ACC FOR MINIMAL LOG DISCREPANCIES IN DIMENSION THREE, II

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Abstract. We prove Shokurov's ACC conjecture for minimal log discrepancies of threefold pairs with coefficients in an arbitrary finite set of real numbers. The conjecture predicts that the minimal log discrepancies at closed points of log canonical pairs of a fixed dimension with coefficients in a DCC set satisfy the ACC, and this paper is the second of a series of three papers which together prove it in dimension three.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Shokurov asked in [Sho88, Problem 5(c)] whether the minimal log discrepancies of log terminal singularities of a fixed dimension satisfy the ascending chain condition (ACC). For pairs with coefficients in a DCC set, the question takes the following form [Sho96, Conjecture 4.2] (see also [Sho04, p. 316]).

Conjecture 1.1 (cf. [Sho88, Problem 5]). *Let d be a positive integer and let $\Gamma \subset [0,1]$ be a DCC set. Then the set of minimal log discrepancies $\mathrm{mld}(X \ni x, B)$, where (X, B) is an lc pair of dimension d, $x \in X$ is a closed point, and $B \in \Gamma$, satisfies the ACC.*

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Shokurov showed that Conjecture 1.1, together with Ambro's lower semicontinuity conjecture [Amb99, Conjecture 2.4], implies the termination of flips [Sho04, Theorem, p. 317]. In Shokurov's implication, it suffices to know Conjecture 1.1 for finite sets Γ [Nak16a, Section 1].

Conjecture 1.1 is known in the following cases.

(1) $d = 2$: [Ale93, Theorem 3.8] for $\Gamma \subset \mathbb{Q}$ (see also [Sho94]), and [HL24, Theorem 1.2] for any DCC set $\Gamma \subset [0, 1]$.
(2) X is toric and B is torus-invariant [Amb06, Theorem 1.1(2)]. See also [Bor97, Corollary 2.1] for toric singularities without boundary, where the minimal discrepancy is taken over all exceptional divisors [Bor97, Definition 1.2].
(3) $(X \ni x, B)$ is an exceptional singularity [HLS26, Theorem 1.2].
(4) $d = 3$ and X is smooth at x [Kaw26, Theorem 1.1]. The values in $(1, 3]$ were treated earlier in [Kaw15a, Theorem 1.3].
(5) $d = 3$ and $x \in X$ is a terminal quotient singularity of index at most a fixed positive integer r [Kaw26, Corollary 1.6], where, for the index $r' \leq r$ of $x \in X$, $B = \sum_j b_j B_j$, with distinct prime divisors B_j and $b_j > 0$, is regarded as an $\mathbb{R}$-ideal as in [Nak16a, Remark 2.2], with exponents b_j/r' in the DCC set $\bigcup_{q=1}^{r} \frac{1}{q}(\Gamma\setminus\{0\})$ of positive real numbers, since $r'B_j$ is Cartier near x for any j (cf. [HLL25, Proof of Lemma 4.2]).
(6) $d = 3$ and X is a fixed normal threefold [Kaw24, Theorem 1.1].
(7) $d = 3$, (X, B) is canonical, and Γ is finite [Nak16a, Corollary 1.5]. Moreover, 1 is the only accumulation point of the set of such values.
(8) $d = 3$, X is terminal near x, and $\mathrm{mld}(X \ni x, B) \geq 1$ [HLL25, Theorem 4.1].
(9) $d = 3$ and $\mathrm{mld}(X \ni x, B) \geq 1$ [Liu26, Theorem 1.2].
(10) $d = 3$ and Γ is the set of standard coefficients

$$\Gamma_{\mathrm{std}} := \{1\} \cup \left\{1 - \frac{1}{m} \;\middle|\; m \in \mathbb{Z}_{\geq 1}\right\},$$

which contains the case $B = 0$ [Liu26, Theorem 1.3].
(11) $d \leq 3$, $\mathrm{mld}(X \ni x, B) \leq 1$, and, for a fixed positive number ϵ and a fixed positive integer N, at most N exceptional prime divisors over X whose centers are closed points have log discrepancy less than $1 + \epsilon$ with respect to (X, B) [Sho96, Corollary 4.5].
(12) The normalized volume $\widehat{\mathrm{vol}}(X \ni x, B)$ is bounded from below by a fixed positive number [HLQ24, Theorem 1.3]. Since $\widehat{\mathrm{vol}}(X \ni x, B) \leq d^{d-1}\mathrm{mld}(X \ni x, B)$ for any klt germ $(X \ni x, B)$ of dimension $d \geq 2$ [Han26, Theorem 1.1], the lower bound on $\widehat{\mathrm{vol}}(X \ni x, B)$ implies that $\mathrm{mld}(X \ni x, B)$ is bounded from below by a positive number.
(13) $B = 0$, X is $\mathbb{Q}$-factorial and klt of regularity one at x, and the values belong to $(0, \epsilon_d)$, where ϵ_d is a positive number depending only on d [Mor23a, Theorem 3].
(14) $B = 0$, $x \in X$ is a quotient singularity, and the values belong to $(0, \epsilon_d)$, where ϵ_d is a positive number depending only on d [Mor23b, Theorem 1].

For $d = 3$, Conjecture 1.1 is still open in general.

This paper is the second of a series of three papers which together prove Conjecture 1.1 for $d = 3$, that is, for any DCC set of real coefficients, any lc pair, and any closed point. The first paper of the series, [Liu26], proves the following results.

Theorem 1.2 ([Liu26, Theorems 1.2, 1.3 and 1.5])**.** *The following statements hold.*

(1) Let $\Gamma \subset [0, 1]$ be a DCC set. Then the set of minimal log discrepancies $\mathrm{mld}(X \ni x, B)$, where (X, B) is an lc pair of dimension three, $x \in X$ is a closed point, $B \in \Gamma$, and $\mathrm{mld}(X \ni x, B) \geq 1$, satisfies the ACC.
(2) Conjecture 1.1 holds for $d = 3$ and $\Gamma = \Gamma_{\mathrm{std}}$.
(3) Let $\Gamma \subset [0, 1]$ be a DCC set. Then the set of positive numbers $\mathrm{mld}((X \ni x, B)/G)$ (see Definition 2.7), where $(X \ni x)$ is a smooth threefold germ, G is a finite abelian group acting faithfully on X and fixing x, and $B = \sum_{j=1}^{s} c_j \,\mathrm{div}(h_j)$ with $c_j \in \Gamma$ and non-zero

regular G-semi-invariant functions h_j near x such that (X, B) is klt near x, satisfies the ACC.

In particular, the minimal log discrepancies in $(0, 1)$ at closed points of $\mathbb{Q}$-factorial klt threefolds without boundary satisfy the ACC [Liu26, Corollary 1.4].

Two further kinds of results fix part of the singularity. The first kind fixes the Cartier index of K_X. Let d and r be positive integers and $I \subset [0, +\infty)$ a finite set. Then the set of minimal log discrepancies $\mathrm{mld}(X \ni x, \mathfrak{a})$, where $(X, \mathfrak{a})$ is an lc pair of dimension d with rK_X Cartier, $x \in X$ is a closed point, and $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ is an $\mathbb{R}$-ideal, that is, a formal product of coherent ideal sheaves $\mathfrak{a}_j$ with exponents $r_j \in I$, has no accumulation point in $\mathbb{R}$ [Nak16a, Corollary 1.3]. Indeed, the set of all log discrepancies of prime divisors over such pairs has no accumulation point in $\mathbb{R}$ [Nak16a, Theorem 1.2]. By [Nak16a, Remark 2.2], boundaries whose components are Cartier are special cases of such $\mathbb{R}$-ideals. Thus [Nak16a, Corollary 1.3 and Remark 2.2] cover the pairs (X, B) such that rK_X is Cartier for a fixed r and B has Cartier components with coefficients in a finite set.

The second kind fixes the germ. On a fixed pair, the minimal log discrepancies of $\mathbb{R}$-ideals with exponents in a fixed finite set form a finite set [Kaw14, Theorem 1.2]. On a fixed klt germ, the ACC for minimal log discrepancies is equivalent to the ACC for a-lc thresholds, to Mustaţă's conjecture on uniform ideal-adic semicontinuity, and to Nakamura's conjecture on the boundedness of a divisor computing the minimal log discrepancy [Kaw21, Theorem 4.6]. We refer the reader to [Kaw21, Conjecture 4.2] for the precise statements. For germs which are not fixed, [Liu18, Theorem 1.6] shows that, in a fixed dimension, the ACC for a-lc thresholds for all $0 \leq a < 1$ is equivalent to the statement that, for any DCC set $\Gamma \subset [0, 1)$, the minimal log discrepancies of lc pairs with coefficients in Γ have no accumulation point from below in $[0, 1)$.

Nakamura's boundedness conjecture, in the form [MN18, Conjecture 1.1], implies the ACC for minimal log discrepancies of $\mathbb{R}$-ideals with exponents in a DCC set on a fixed germ [MN18, Theorem 1.5]. A prime divisor E computing the minimal log discrepancy, with $a(E, X, 0)$ bounded from above in terms of the coefficient set, exists for lc surface germs $(X \ni x, B)$ with $B \in \Gamma$ for a DCC set Γ [HL23, Theorem 1.2], on a fixed klt threefold for boundaries with coefficients in a DCC set [Kaw24, Theorem 1.4], and for threefold germs $(X \ni x, B)$ with X terminal and $\mathrm{mld}(X \ni x, B) \geq 1$ [HLL25, Theorem 1.2].

For threefolds, results are also known on the minimal log discrepancy $\mathrm{mld}(X, B)$ of a pair (X, B), the infimum of $a(E, X, B)$ over all exceptional prime divisors E over X. The number $\mathrm{mld}(X, B)$ is a different invariant from the minimal log discrepancies at closed points in Conjecture 1.1. For any DCC set $\Gamma \subset [0, 1]$, there exists a positive number δ such that the numbers $\mathrm{mld}(X, B)$, where (X, B) is a threefold pair with $B \in \Gamma$, satisfy the ACC in $[1 - \delta, +\infty)$ [HLL25, Theorem 1.1].

Without boundary, the gap theorem [Jia21, Theorem 1.3] shows that there exists a positive number δ_0 such that $\mathrm{mld}(X) < 1$ implies $\mathrm{mld}(X) \leq 1 - \delta_0$ for any threefold X. The optimal bound is $\frac{12}{13}$, attained by the quotient of $\mathbb{A}^3$ by the group μ_{13} of 13th roots of unity ζ acting by $\zeta \cdot (x_1, x_2, x_3) = (\zeta^3 x_1, \zeta^4 x_2, \zeta^5 x_3)$ [LX21, Theorem 1.4 and Example 1.5]. The largest accumulation point of the numbers $\mathrm{mld}(X)$ is 1 (see the remark after [LL22, Theorem 1.3]), and the second largest is $\frac{5}{6}$ [LL22, Theorem 1.3]. The numbers $\mathrm{mld}(X)$ satisfy the ACC in $[\frac{5}{6}, +\infty)$ [LL22, Theorem 1.5]. With coefficients in a DCC set, the numbers $\mathrm{mld}(X, B)$ satisfy the ACC for exceptionally non-canonical threefold pairs [HL25, Theorem 1.4] and for klt threefold pairs with a bounded number of distinct exceptional log discrepancies in $[0, 1]$ [HL25, Theorem 1.6]. For a DCC set $\Gamma \subset [0, 1]$ and lc threefold pairs (X, B) with $B \in \Gamma$ such that X is $\mathbb{Q}$-factorial and polarized by a very ample divisor H with H^3 bounded, the minima of $a(E, X, B)$ over all prime divisors E over X, including the components of B, satisfy the ACC [CL26, Theorem 1.3 and the conventions at the beginning of Section 2].

We now explain why the results on a fixed index or on a fixed germ do not extend to Conjecture 1.1 with $d = 3$ and Γ finite. Let (X_i, B_i) be lc threefold pairs with $B_i \in \Gamma$ and $x_i \in X_i$ closed points such that $a_i := \mathrm{mld}(X_i \ni x_i, B_i)$ is strictly increasing with a finite limit.

Assume first that $\Gamma \subset \mathbb{Q}$, so that $K_{X_i} + B_i$ is $\mathbb{Q}$-Cartier (an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor is $\mathbb{Q}$-Cartier). If $r(K_{X_i} + B_i)$ is Cartier near x_i, then $a(E, X_i, B_i) \in \frac{1}{r}\mathbb{Z}$ for any prime divisor E over X_i with center x_i, and hence $a_i \in \frac{1}{r}\mathbb{Z}$. If the Cartier indices of $K_{X_i} + B_i$ at x_i were bounded by an integer r_0, then all a_i would belong to $\frac{1}{r_0!}\mathbb{Z}$, which contains no strictly increasing sequence with a finite limit. Thus the Cartier indices of $K_{X_i} + B_i$ at x_i are unbounded.

For real coefficients, the values a_i are in general irrational, and denominators are not available. Instead, for a fixed positive integer r, if rK_{X_i} is Cartier near x_i and the components of B_i are Cartier near x_i for infinitely many i, then the corresponding a_i belong to the set of [Nak16a, Corollary 1.3 and Remark 2.2], which contains no strictly increasing sequence with a finite limit. Similarly, the results on a fixed germ consider one germ $(X \ni x)$, while the germs $(X_i \ni x_i)$ vary along the sequence.

The Cartier index of K_X itself is bounded only in some cases. If $x \in X$ is a terminal threefold point at which K_X has Cartier index $n \geq 2$, then there exists a prime divisor E over X with center x and $a(E, X, 0) = 1 + \frac{1}{n}$ [Kaw93, Theorem]. Since $B \geq 0$, we have $1 + \frac{1}{n} = a(E, X, 0) \geq a(E, X, B) \geq \mathrm{mld}(X \ni x, B)$, and [HLL25, Proof of Theorem 4.1, Step 1] bounds n in this way when $\mathrm{mld}(X \ni x, B) > 1$. At canonical threefold points, the Cartier index of K_X is still bounded in terms of $\mathrm{mld}(X \ni x)$ [Kaw15b, Theorem 5.3], while at klt threefold points it is unbounded when $\mathrm{mld}(X \ni x)$ is allowed to vary. We refer the reader to [Liu26, Section 1] for examples. Shokurov's index conjecture [NS22, Conjecture 1.1] predicts that the Cartier index of K_X at x is bounded in terms of $\dim X$ and $\mathrm{mld}(X \ni x)$, and it holds for quotient singularities [NS22, Theorem 1.2]. At non-klt points the bound fails: for any positive integer m, there exists a non-klt $\mathbb{Q}$-Gorenstein threefold point $x \in X$ with $\mathrm{mld}(X \ni x) = 1$ at which K_X has Cartier index m [NS26a, Theorem 1.2 and the paragraph before the acknowledgements in Section 1].

By [Liu26, Construction-Theorem 17.6 and Corollary 17.3], the case of a finite set Γ reduces to the minimal log discrepancies, at the image of o, of quotient pairs $(W/G, D_{W/G})$, where $(W \ni o)$ is a Gorenstein terminal threefold germ [Liu26, Construction-Theorem 5.5(1)], G is a finite cyclic group acting on W and fixing o, $D = \sum_j c_j \,\mathrm{div}(h_j)$ is a G-invariant boundary on W with regular G-semi-invariant functions h_j and non-negative coefficients c_j in a finite set, and $K_W + D = \pi^*(K_{W/G} + D_{W/G})$ for the quotient morphism $\pi : W \to W/G$. Before the quotient, the log discrepancies of the pairs (W, D) belong to the set of [Nak16a, Theorem 1.2 and Remark 2.2], and along a strictly increasing sequence of values only the groups are unbounded: the orders of the groups tend to infinity [Liu26, Construction-Theorem 17.6(2)]. The simplest example has $W = \mathbb{A}^3$ and $D = 0$. For the group μ_n of n-th roots of unity ζ acting by $\zeta \cdot (x_1, x_2, x_3) = (\zeta x_1, \zeta x_2, \zeta x_3)$, the quotient $\mathbb{A}^3/\mu_n$ has minimal log discrepancy $\frac{3}{n}$ at the image of the origin [Liu26, Example 4.2(1)], so the value $\frac{3}{n}$ varies with n while $W = \mathbb{A}^3$ is fixed.

The methods of [Liu26] stop at the following two places. First, standard coefficients $1 - \frac{1}{m}$ can be absorbed into ramification of index m [Liu26, Construction-Theorem 5.3]. The values in $(0, 1)$ then either are minima of piecewise linear functions over the points of a lattice in an open orthant, for which [Liu26, Theorem 4.5] proves the ACC, or arise from groups of bounded order [Liu26, Theorem 13.1 and the beginning of Section 4]. Irrational coefficients cannot be absorbed in this way.

Second, for values at least $1+\epsilon$, where ϵ is a fixed positive number, [Liu26, Proposition 14.12(1)] shows, in the setting of [Liu26, Set-up 14.1] for r and N sufficiently large, that the pair stays lc near the point after the boundary is multiplied by a number $\lambda > 1$ depending only on ϵ, and the lower bounds of [Liu26] for values at least $1 + \epsilon$ use λ. In $(0, 1)$, [Liu26, Proposition 14.12(1)] has no uniform analogue [Liu26, beginning of Subsection 14.3], and with real coefficients in $(0, 1)$, [Liu26] treats only quotients of smooth threefold germs (Theorem 1.2(3)). The germs $(W \ni o)$ of [Liu26, Construction-Theorem 17.6 and Corollary 17.3] are in general isolated cDV points, and for their quotients [Liu26] proves the ACC only in $[1, +\infty)$ [Liu26, Proposition 15.1].

The goal of this paper is Conjecture 1.1 for $d = 3$ and finite sets Γ, which is Theorem 1.3 below.

In Theorem 1.3, no bound is assumed on the Cartier index of K_X or of $K_X + B$, K_X need not be $\mathbb{Q}$-Cartier, X need not be $\mathbb{Q}$-factorial, the germ $(X \ni x)$ is not fixed, (X, B) need not be klt near x, and the coefficients may be irrational. By Theorem 1.2(1), the new content of Theorem 1.3 is the range $(0, 1)$. Moreover, Theorem 1.2(2) covers the infinite set Γ_{std}, and hence any finite subset of Γ_{std}, while Theorem 1.3 covers any finite set of real numbers, including irrational ones. Thus neither Theorem 1.3 nor Theorem 1.2(2) contains the other.

Theorem 1.3. *Let $\Gamma \subset [0, 1]$ be a finite set of real numbers. Then the set of minimal log discrepancies* $\mathrm{mld}(X \ni x, B)$*, where (X, B) is an lc pair of dimension three, $x \in X$ is a closed point, and $B \in \Gamma$, satisfies the ACC.*

The third paper of the series proves that for any positive rational number a and any finite set $S \subset [0, 1] \cap \mathbb{Q}$, there exists a positive real number $\eta(a, S)$ with the following property: for any $\mathbb{Q}$-factorial klt threefold X, any closed point $x \in X$ such that $\mathrm{mld}(X \ni x) \geq a$, and any $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \in S$ and (X, D) is lc near x,

$$\sup\{t \in [0, 1] \mid \mathrm{mld}(X \ni x, tD) \geq a\} \notin (1 - \eta(a, S), 1).$$

The third paper then deduces Conjecture 1.1 for $d = 3$ from Theorem 1.3 and the existence of $\eta(a, S)$.

Sketch of the proof. By [Liu26, Lemma 6.1(3) and Proposition 20.1], it suffices to exclude a strictly increasing sequence in $(0, 1)$ of minimal log discrepancies of $\mathbb{Q}$-factorial klt threefold pairs with coefficients in a fixed finite set.

(1) By [Liu26, Construction-Theorem 17.6(5) and Corollary 17.3], the terms are quotient values α_i of $D_i = \sum_k \mu_k C_{i,k}$ on Gorenstein terminal threefold germs W_i by cyclic groups G_i (Definition 2.7), with fixed μ_k and divisors $C_{i,k}$, which we call *proxies* (Set-up 11.1), such that the pairs $(W_i, C_{i,k})$ are terminal.

(2) Theorem 10.1 excludes sources (Definition 2.7) of type cE, and Theorem 11.2 the others, sorted by the quadratic rank [Liu26, Definition-Lemma 11.1]: smooth sources and sources of quadratic rank at least three by Theorem 7.1, and sources of quadratic rank one or two by Propositions 11.7 and 11.11–11.16.

(3) In each case, bounded groups (Proposition 11.7) and Hausdorff limits H of the G_i with $\dim H \geq 2$ (Lemmas 4.6, 4.7, and 5.8) are excluded, leaving a primitive circle $\mathbf{T}$ (Set-up 5.1) with $|G_i \cap \mathbf{T}| \to +\infty$.

(4) Along $\mathbf{T}$, $\alpha_i \leq \beta_i + c_0/|G_i \cap \mathbf{T}|$ for the minimum β_i of a reference pair (Section 5), and the β_i belong to an ACC set, so a subsequence is non-increasing with limit $L \geq \lim_i \alpha_i$. A compatible generic limit of the reference pairs (Construction-Proposition 4.1) either is klt, and then a margin (Definition 5.2) implies $\alpha_i \geq \beta_i \geq L$, or has a smooth curve with coefficient one, and then an lc augmentation (Lemma 5.8) implies $\alpha_i \geq L$. Since $L \geq \lim_j \alpha_j$, this contradicts the strict increase of the α_j.

In Steps (1)–(4), a source is a germ $(W_i \ni o_i)$ with the action of a finite group G_i (Definition 2.7), and the quotient value of a boundary D_i on it is $\mathrm{mld}((W_i \ni o_i, D_i)/G_i)$, the minimal log discrepancy of the crepant quotient. The quadratic rank is the rank of the quadratic part of a local equation of W_i [Liu26, Definition-Lemma 11.1]. In Step (3), a Hausdorff limit is a closed subgroup $H \subset \mathbb{T}^4$ to which the G_i converge in the Hausdorff distance along a subsequence (Lemma 4.6), and a primitive circle is the one-dimensional subtorus $\mathbf{T} = \mathbf{T}_{(w)}$ of the diagonal torus for a primitive weight vector w (Set-up 5.1). In Step (4), the reference pair is the pair on a fixed surface built from the principal blocks of the equations along $\mathbf{T}$ (Section 5), and a compatible generic limit is a limit of the reference pairs over a field extension (Construction-Proposition 4.1). A margin is a number $\theta > 1$ with θ times the boundary still lc (Definition 5.2),

and an lc augmentation adds $\lambda \operatorname{div}(U)$ for a semi-invariant function U while keeping the pair lc (Lemma 5.8).

Step (1) is quoted from [Liu26], and Steps (2) and (3) are short reductions. The new part is Step (4), whose main tools are described in (a)–(d) below. The proof is by contradiction and does not determine the accumulation points of the set of minimal log discrepancies.

(a) Upper and lower bounds along a circle. Prime divisors computing quotient values need not be given by monomial valuations in the eigenparameters [Liu26, Example 8.10], so upper bounds use explicit prime divisors, and lower bounds hold for any prime divisor. The reference pair is built from the principal blocks along $\mathbf{T}$, the sums of the terms of the least-degree $\mathbf{T}$-weight of the equations, the other blocks having large degree (Set-up 5.1 and Lemma 5.3). A Gauss extension [Liu26, Definition-Proposition 2.11] of a prime divisor computing β with bounded log discrepancy (Theorem 2.9(2)), with value ℓ on the added coordinate satisfying $\ell \equiv q \pmod{e}$ for the q and e of Lemma 3.1, implies $\alpha \leq \beta + c_0/r$ for $r = |G \cap \mathbf{T}|$ (Proposition 5.4).

Under a margin (Definition 5.2), the other blocks can be deleted, and quotient adjunction to a section with coefficient one (Theorem 3.3) implies $\alpha \geq \beta$ (Lemma 5.5). For some circles (Theorem 8.13), the error of a single reference has the wrong sign. It is absorbed by the difference between the minima of the quotients graded by the positive weights and by the negative weights, which is at least a fixed multiple of $|\delta|$, where δ is the adjusted mass of Set-up 5.1.

(b) Margins and floors. Besides the obstruction at [Liu26, beginning of Subsection 14.3], klt reference pairs may have a compatible generic limit which is not klt (Example 4.3), and the sources have no common ambient space. We take compatible generic limits only of the reference pairs, on a fixed germ $\mathbb{A}^n$ or near a fixed proper curve on a fixed surface, and the lc property and the quotient minima pass to the limit (Construction-Proposition 4.1 and Theorem 4.2). The curve is used when the reference at a point does not bound α from below (Remark 7.16). A klt limit provides a margin along a subsequence, which passes to the sources by Lemma 4.14. The non-klt locus of a limit which is not klt consists of smooth curves with coefficient one, called floors in Definition-Proposition 5.7, and an lc augmentation by a semi-invariant function of $\mathbf{T}$-weight zero bounds α_i from below by the limit value (Lemma 5.8).

(c) Radial quotients and crepant models. Terminal crepant models of quotients can have boundary coefficients $\frac{1}{n}$ with n unbounded [Liu26, Section 1]. If $\mathbf{T}$ has no negative weight, then, possibly after a weighted blow-up, the quotient by $G \cap \mathbf{T}$ has a coordinate hyperplane with coefficient tending to one, which the ACC for lc thresholds (Theorem 2.12) raises to one, and adjunction compares α with a value in an ACC set on a fixed ambient (Theorem 8.5).

For a canonical circle with $w_t > 0$ ($w_z + w_t = 0$ in Set-up 7.3), if the mass τ of Set-up 5.1 is non-positive, a weighted blow-up with weights $(1,1,1,0) + \frac{L}{r}w$, for the weight vector w of $\mathbf{T}$ and a fixed L, is a crepant model (Definition-Proposition 4.10) with exceptional coefficients in a DCC set, whose charts are quotients of smooth germs or axial sources ($W_N \ni 0$) (Set-up 7.3) with bounded groups (Theorem 7.9). For mixed circles ($w_x w_y < 0$ and $w_z \geq 0$ in Set-up 7.3), under the conditions of Theorem 7.28, an offset L coprime to r with $L \to +\infty$ and $L/r \to 0$ (Lemma 4.9) leaves at the origin of the only possibly singular chart of the crepant model given by the weighted blow-up with weights $(1,1,1,0) + \frac{L}{r}w$ (for $w_z > 0$ and a fixed L, the model of Theorem 7.27(2)) an axial source with a growing subgroup of the scalar circle $\mathbf{T}_{(-1,-1,-1,1)}$, treated by Theorem 7.15 (Theorem 7.28).

(d) Several labels, and sources of type cE. The quotient value of $D = \sum_k \mu_k C_k$ is not the corresponding combination of the quotient values of the proxies C_k (Example 11.3), and the lattice of Lemma 4.8 needs an integral multiple of $K_W + D$ to be Cartier. We add log discrepancies only at one prime divisor P: $a_D(P) = \sum_k \mu_k a_k(P)$, where $a_k(P) > 0$ and $m|I_G(P)|a_k(P) \in \mathbb{Z}$ (Definition-Lemma 11.4). Here $C_k = \frac{1}{m}\operatorname{div}(h_k)$ for marks h_k and a fixed positive integer m, and $a_k(P) := a(P, W, C_k)/|I_G(P)|$ and $a_D(P) := a(P, W, D)/|I_G(P)|$ are the scores of P. Thus bounded groups lead to finitely many values (Proposition 11.7). The theorems on circles are proved for labels in a DCC set. At a singular point with bounded inertia of a crepant model, if

the limit b_* of the exceptional coefficients b_i is not in the closure of the set of lc thresholds, the pair stays lc after raising the coefficient to $b_* + \eta$, and the quotient values are at least numbers ℓ_i in an ACC set and at most $\ell_i + (b_* - b_i)\ell_i/\eta$, in place of the lattice (Proposition 11.9(2)).

For sources of type cE, Theorem 2.13 implies a gap $\mathrm{mld}(W \ni o, D) \geq 1+\epsilon$, which Theorem 10.5 turns into a margin of the source itself (Lemma D.20), since the gap and Lemma 9.2 leave no component with coefficient one in a generic limit of such pairs in the sense of [Kaw21, Section 3]. Terminality alone does not suffice (Remark 10.6).

Besides [Liu26], the proof uses the ACC and bounded computing divisors for surfaces [HL24, Theorem 1.2], [HL23, Theorem 1.2], results on a fixed threefold [Kaw24, Theorems 1.1, 1.3, and 1.4], the ACC and the $\mathfrak{m}$-adic continuity of lc thresholds [HMX14, Theorem 1.1], [dFEM11, Proposition 2.19(ii)], the absence of accumulation points of log discrepancies for a fixed Gorenstein index and coefficients in a fixed finite set [Nak16a, Theorem 1.2], the lower semicontinuity of minimal log discrepancies [Nak16b, Theorem 1.2(ii)], and the inversion of adjunction for quotients [NS26b, Theorem 7.1].

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce sources, marks, labelled boundaries, and quotient values, and recall the results on surfaces, a fixed threefold, thresholds, and fixed Gorenstein index used repeatedly. In Section 3 we extend several results of the first paper, among them an adjunction formula for quotients of lc pairs (Theorem 3.3) and the replacement of normalized marks by polynomials (Theorem 3.10). In Section 4 we treat compatible generic limits of pairs on fixed ambient spaces (Construction-Proposition 4.1 and Theorem 4.2) and limits of diagonal groups, replace a source and its marks by polynomial data which keep the quotient values and the log discrepancies of finitely many given prime divisors (Construction-Theorem 4.13), and compute minimal log discrepancies on crepant models (Definition-Proposition 4.10). In Section 5 we compare quotient values with those of a reference surface along a primitive circle whose intersection with the group has growing order (Set-up 5.1), from above and from below.

In Section 6 we introduce marked terminal sources (Definition 6.1), classify plane cubics (Definition-Lemma 6.4), and construct from a terminal pair with a cyclic group a Gorenstein terminal cover with a mark and a group, which form a marked source when the support of the boundary contains the point, and bound the orders of the mark (Theorem 6.2). In Section 7 we prove the ACC for the quotient values of smooth sources and sources of quadratic rank at least three with labels in a finite set (Theorem 7.1) and Theorem 7.2 on the axial sources to which isolated sources of quadratic rank three are reduced.

In Section 8 we treat sources of quadratic rank two: circles on split sources, limits of the groups of dimension two (Proposition 8.10), splittings of the binary part (Theorem 8.11), lc augmentations on the toric germs $(xy + z^a t^b = 0)$ (Lemma 8.12), and groups which may exchange the hyperbolic coordinates along circles whose radical weights have opposite signs (Theorem 8.13). In Section 9 we define sources of type cD and of type cE, state a lemma on prime divisors of log discrepancy two over hypersurfaces $(x^2 + g(y, z, t) = 0)$ which contain a smooth surface (Lemma 9.2), and fix the double-plane divisor E_0 of a source of quadratic rank one (Set-up 9.3). In Section 10 we state the ACC for the quotient values of sources of type cE with finitely many real labels (Theorem 10.1) and the results which enter its proof. In Section 11 we treat several labels on sources not of type cE (Theorem 11.2).

In Section 12 we prove Theorem 1.3. Appendices A–E contain the long proofs of the results of Sections 7–11, including the proof of Theorem 10.1.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

The lower and upper bounds in this paper are proved on the source (Definition 2.7), through the formula for quotient values of [Liu26, Lemma 2.6] recalled in Subsection 2.1.

2.1. Pairs and quotients.

In this subsection, we fix the conventions for pairs, $\mathbb{R}$-ideals, minimal log discrepancies, crepant quotients, and formal prime divisors.

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2 ([Liu26, Definitions 2.2, 2.3, and 2.9])**.** Let X be a normal quasi-projective variety and B an $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier. We use the *$\mathbb{R}$-ideals* $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ on X and the log discrepancies $a(E, X, B, \mathfrak{a})$ of prime divisors E over X with respect to $(X, B, \mathfrak{a})$ of [Liu26, Definition 2.2]. We say that $(X, B, \mathfrak{a})$ is *(sub-)lc* (resp. *(sub-)klt*) if $a(E, X, B, \mathfrak{a}) \geq 0$ (resp. > 0) for any prime divisor E over X [Liu26, Definition 2.2], and we write $(X, \mathfrak{a})$ for $(X, 0, \mathfrak{a})$. For a point $\eta \in X$, which need not be closed, we denote by $\operatorname{mld}(X \ni \eta, B, \mathfrak{a})$ the *minimal log discrepancy* of $(X, B, \mathfrak{a})$ at η, computed by the prime divisors over X with center $\overline{\{\eta\}}$, and we write $\operatorname{mld}(X \ni \eta, B)$ if $\mathfrak{a} = \mathcal{O}_X$ [Liu26, Definition 2.2]. We use the sets $\operatorname{MLD}(d, \Gamma)$ and $\operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda)$ of minimal log discrepancies at closed points of [Liu26, Definition 2.3]. For a set $\Gamma \subset [0, 1]$, we denote by Γ_+ the set of finite sums of elements of Γ, where the empty sum is 0 [Liu26, Definition 2.9]. For a set $\Gamma \subset [0, 1]$, we write

$$\Psi(\Gamma) := \left\{ 1 - \frac{1 - c}{e} \;\middle|\; c \in \Gamma,\ e \text{ is a positive integer} \right\}$$

[Liu26, Definition 2.9].

Definition 2.3 ([Liu26, Definitions 2.5 and 2.7, Lemma 2.6])**.** Let X be a normal quasi-projective variety, G a finite group acting faithfully on X, and B a G-invariant $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier. We denote by $(X, B)/G = (X/G, B_{X/G})$ the *crepant quotient* of (X, B) by G, so that $K_X + B$ is the pullback of $K_{X/G} + B_{X/G}$ [Liu26, Definition 2.5 and Lemma 2.6(0)]. For a point $x \in X$ fixed by G, we denote by $(X \ni x, B)/G$ the germ of $(X, B)/G$ at the image of x, and by $\operatorname{mld}((X \ni x, B)/G)$ the minimal log discrepancy of $(X/G, B_{X/G})$ at the image of x. For a prime divisor F over X, we denote by $\operatorname{Stab}_G(F)$ and $I_G(F)$ the *decomposition group* and the *inertia group* of F [Liu26, Definition 2.5]. For a prime divisor F over X, the *prime divisor under* F is the unique prime divisor E over X/G of [Liu26, Definition 2.7 and Lemma 2.6], and F is called a prime divisor *lying over* E. If F is a prime divisor over X lying over E, then

$$a(F, X, B) = |I_G(F)| \cdot a(E, X/G, B_{X/G})$$

[Liu26, Lemma 2.6(2)]. In particular, for a closed point $x \in X$ fixed by G,

$$\operatorname{mld}((X \ni x, B)/G) = \inf_F \frac{a(F, X, B)}{|I_G(F)|},$$

where F runs over the prime divisors over X with center x [Liu26, Lemma 2.6].

Definition 2.4. Let X be a normal quasi-projective variety over a field $k \supset \mathbb{C}$, $x \in X$ a point, G a finite group acting faithfully on X and fixing x, and B a G-invariant $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier near x. Let $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ be an $\mathbb{R}$-ideal on X whose factors $\mathfrak{a}_j$ are G-invariant. We define

$$\operatorname{mld}((X \ni x, B, \mathfrak{a})/G) := \inf_F \frac{a(F, X, B, \mathfrak{a})}{|I_G(F)|},$$

where F runs over the prime divisors over X with $\operatorname{center}_X F = \overline{\{x\}}$, and we write $\operatorname{mld}((X \ni x, \mathfrak{a})/G)$ if $B = 0$.

Let Y be a normal variety over k, $y \in Y$ a closed point, and $\widehat{Y} := \operatorname{Spec} \widehat{\mathcal{O}}_{Y,y}$. An $\mathbb{R}$-*ideal* on $\widehat{Y}$ is a formal product $\widehat{\mathfrak{a}} = \prod_j \widehat{\mathfrak{a}}_j^{r_j}$ of finitely many non-zero ideals $\widehat{\mathfrak{a}}_j \subset \widehat{\mathcal{O}}_{Y,y}$ with real numbers $r_j > 0$. For an $\mathbb{R}$-divisor $\widehat{B}$ on $\widehat{Y}$ such that $K_{\widehat{Y}} + \widehat{B}$ is $\mathbb{R}$-Cartier, an $\mathbb{R}$-ideal $\widehat{\mathfrak{a}} = \prod_j \widehat{\mathfrak{a}}_j^{r_j}$ on $\widehat{Y}$, and a formal prime divisor F over $\widehat{Y}$ [Liu26, Definition 3.2], we define

$$a(F, \widehat{Y}, \widehat{B}, \widehat{\mathfrak{a}}) := a(F, \widehat{Y}, \widehat{B}) - \sum_j r_j \operatorname{mult}_F \widehat{\mathfrak{a}}_j,$$

where $a(F, \widehat{Y}, \widehat{B})$ is the log discrepancy of [Liu26, Definition 3.2], ξ is the generic point of the center of F on $\widehat{Y}$, and $\operatorname{mult}_F \widehat{\mathfrak{a}}_j$ is the minimum of $\operatorname{mult}_F \operatorname{div}(f)$ for non-zero $f \in (\widehat{\mathfrak{a}}_j)_\xi$, as in [Liu26, Definition 2.2].

Suppose that x is a closed point, and let $\widehat{X} := \operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$ with closed point $\hat{x}$. Let $\widehat{B}$ be any G-invariant $\mathbb{R}$-divisor on $\widehat{X}$ such that $K_{\widehat{X}} + \widehat{B}$ is $\mathbb{R}$-Cartier, and let $\widehat{\mathfrak{a}}$ be any $\mathbb{R}$-ideal on $\widehat{X}$ with G-invariant factors, for instance the pullbacks of B and $\mathfrak{a}$. We define

$$\operatorname{mld}((\widehat{X} \ni \hat{x}, \widehat{B}, \widehat{\mathfrak{a}})/G) := \inf_F \frac{a(F, \widehat{X}, \widehat{B}, \widehat{\mathfrak{a}})}{|I_G(F)|},$$

where F runs over the formal prime divisors over $\widehat{X}$ with center $\hat{x}$ of [Liu26, Definition 3.2], and their log discrepancies $a(F, \widehat{X}, \widehat{B}, \widehat{\mathfrak{a}})$ are those of the previous paragraph.

Definition 2.5 ([Liu26, Definition 3.2 and Definition-Lemma 11.1])**.** Let $k \supset \mathbb{C}$ be a field and $(X \ni x)$ a germ of a normal variety over k at a closed point x. We use the *formal prime divisors* over $\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$ and the log discrepancies of formal prime divisors of [Liu26, Definition 3.2]. For a threefold germ $(W \ni o)$ as in [Liu26, Definition-Lemma 11.1], we use the *quadratic rank* of $(W \ni o)$ of [Liu26, Definition-Lemma 11.1]. The *residual cubic* of such a germ $(W \ni o)$ is the cubic of [Liu26, Definition-Lemma 11.1], which is unique up to a linear change of coordinates and a non-zero scalar.

2.2. **Marks and quotient values.** In this subsection, we recall Gauss extensions, semi-invariant functions, and lattice Newton minima from [Liu26], and define marks, labelled boundaries, and quotient values.

Definition 2.6 ([Liu26, Definition-Proposition 2.11 and Definitions 2.14–2.16])**.** For a field L, a non-trivial valuation $v : L^* \to \mathbb{Z}$, an element s transcendental over L, and an integer c, we denote by v_c the *Gauss extension* of v to $L(s)$ with s-value c [Liu26, Definition-Proposition 2.11]. For a finite group G acting on a variety, we use the *G-semi-invariant* regular functions and the *characters* of G attached to them as in [Liu26, Definition 2.14]. For a G-semi-invariant regular function, rational function or power series f, we write χ_f for the character of f, so that $g^* f = \chi_f(g) f$ for any $g \in G$. For coordinates x, y, z, t on which G acts diagonally, we write $\chi_x, \chi_y, \chi_z, \chi_t$ for their characters. For a positive integer d, we write $\mathbb{T}^d := \mathbb{R}^d/\mathbb{Z}^d$, and for a

subgroup $H \subset \mathbb{T}^d$, we denote by N_H the *lattice* of H [Liu26, Definition 2.15]. For a positive integer r and integers $a_1, \dots, a_d$, we denote by $\mu_r(a_1, \dots, a_d)$ the cyclic diagonal group of [Liu26, Definition 2.15]. For a vector $(a_1, \dots, a_d)$, we denote by $\mathbf{T}_{(a_1,\dots,a_d)}$ the one-dimensional subtorus of [Liu26, Definition 2.15]. Let $\rho : \mathbb{R}^d \to \mathbb{T}^d$ be the quotient map of [Liu26, Definition 2.15]. We identify $\mathbb{T}^d$ with $(S^1)^d \subset (\mathbb{C}^*)^d$ through the isomorphism

$$\rho(u) \to (e^{2\pi\sqrt{-1}u_1}, \dots, e^{2\pi\sqrt{-1}u_d}).$$

Under the identification, the elements of finite order of $\mathbb{T}^d$ are the diagonal automorphisms of $\mathbb{A}^d$ of finite order, and a finite diagonal group $G \subset (\mathbb{C}^*)^d$ is regarded as a subgroup of $\mathbb{T}^d$ through its image under the characters of the coordinates [Liu26, Definition 2.15]. For integers $a_1, \dots, a_d$, not all zero, the identification maps the closed subgroup $\rho(\mathbb{R}(a_1, \dots, a_d))$ of $\mathbb{T}^d$ onto $\mathbf{T}_{(a_1,\dots,a_d)} \cap (S^1)^d$, and we also write $\mathbf{T}_{(a_1,\dots,a_d)}$ for $\rho(\mathbb{R}(a_1, \dots, a_d))$. For a closed subgroup $H \subset \mathbb{T}^d$ and a primitive vector $w \in \mathbb{Z}^d$, we write $H^0 = \mathbf{T}_{(w)}$ if the identity component H^0 of H is $\rho(\mathbb{R}w)$. For any finite diagonal group $G \subset (\mathbb{C}^*)^d$ and any $w \in \mathbb{Z}^d \backslash \{0\}$, the image of $G \cap \mathbf{T}_{(w)}$ in $\mathbb{T}^d$ is $G \cap \rho(\mathbb{R}w)$ [Liu26, Lemma 4.3(2)]. In particular, if $H^0 = \mathbf{T}_{(w)}$, then the image of $G \cap \mathbf{T}_{(w)}$ in $\mathbb{T}^d$ is $G \cap H^0$. For a finite abelian group acting on a variety and fixing a closed point o, we use the *eigenparameters* at o of [Liu26, Definition 2.15]. For a weight vector w, we denote by wt_w and in_w the *w-weight* and the *w-initial form* of polynomials and power series in $x_1, \dots, x_d$ [Liu26, Definition 2.16]. We denote by $\psi_{J,c}$ and $\psi_{\min}(H, J, c)$ the functions and the *lattice Newton minima* of [Liu26, Definition 2.16].

Definition 2.7. Let W be a normal variety, $o \in W$ a closed point, and G a finite group acting faithfully on W and fixing o. We call $(W \ni o)$ together with the action of G a *source*. A *mark* on $(W \ni o)$ is a non-zero regular G-semi-invariant function h on a G-invariant open neighborhood of o. It may be a unit, and $\mathrm{div}(h)$ need not be reduced. A *labelled boundary* on $(W \ni o)$ is an $\mathbb{R}$-divisor $D = \sum_{j=1}^{s} c_j \,\mathrm{div}(h_j)$, where the h_j are marks and the *labels* c_j are non-negative real numbers. If $K_W + D$ is $\mathbb{R}$-Cartier near o, choose a G-invariant affine open neighborhood U of o on which all h_j are regular and $K_U + D|_U$ is $\mathbb{R}$-Cartier. The number $\mathrm{mld}((W \ni o, D)/G) := \mathrm{mld}((U \ni o, D|_U)/G)$ is called the *quotient value* of $(W \ni o, D)$ and G, and a prime divisor over W with center o which computes it through the formula at the end of [Liu26, Lemma 2.6] is called a *computing lift*.

In Definition 2.7, such a neighborhood U exists, since o is fixed by G, D is G-invariant, and the intersection of the G-translates of an affine open neighborhood of o on which all h_j are regular and $K_W + D$ is $\mathbb{R}$-Cartier is affine. The quotient value does not depend on U: by [Liu26, Lemma 2.6], $\mathrm{mld}((U \ni o, D|_U)/G)$ is the infimum of $a(F, U, D|_U)/|I_G(F)|$ over the prime divisors F over U with center o, and for another such neighborhood U' of o, the prime divisors over U with center o are the prime divisors over $U \cap U'$ with center o, and for each such F we have $a(F, U, D|_U) = a(F, U \cap U', D|_{U\cap U'})$, and the inertia group $I_G(F)$ computed on U equals the inertia group $I_G(F)$ computed on $U \cap U'$.

We allow marks that are units or have non-reduced divisors, and arbitrary non-negative labels, in Definition 2.7 because they occur in the later constructions. Computing lifts are defined through the formula at the end of [Liu26, Lemma 2.6] because the upper bounds later are obtained by exhibiting such prime divisors.

2.3. **Two results of the first paper.** In the next theorem, $\mathrm{mld}((X \ni x, B)/G)$ is the minimal log discrepancy of the crepant quotient of [Liu26, Definition 2.5], semi-invariant functions are as in [Liu26, Definition 2.14], $\mathbb{T}^d$ and N_H are as in [Liu26, Definition 2.15], and $\psi_{\min}(H, J, c)$ is the lattice Newton minimum of [Liu26, Definition 2.16]. Subsection 2.1 recalls [Liu26, Definition 2.5], and Subsection 2.2 recalls [Liu26, Definitions 2.14, 2.15, and 2.16].

Theorem 2.8 ([Liu26, Theorems 1.5 and 4.5])**.** *The following hold.*

(1) *Let $\Gamma \subset [0,1]$ be a DCC set. Then the set of positive numbers* $\mathrm{mld}((X \ni x, B)/G)$*, where $(X \ni x)$ is a smooth threefold germ, G is a finite abelian group acting faithfully on X and fixing x, and $B = \sum_{j=1}^{s} c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$ and non-zero regular G-semi-invariant functions h_j near x such that (X, B) is klt near x, satisfies the ACC. In particular, the conclusion holds for G-invariant $\mathbb{R}$-divisors $B \geq 0$ with $B \in \Gamma$ and (X, B) klt near x.*

(2) *Let d, s be positive integers and $\Lambda \subset [0, +\infty)$ a DCC set. Then the set of finite numbers $\psi_{\min}(H, J, c)$, where $H \subset \mathbb{T}^d$ is a finite subgroup, $J = (J_1, \dots, J_s)$ is an s-tuple of non-zero monomial ideals of $\mathbb{C}[x_1, \dots, x_d]$, and $c \in \Lambda^s$, satisfies the ACC, and $\psi_{\min}(H, J, c) \geq 0$ whenever $\psi_{\min}(H, J, c)$ is finite. The ACC and the non-negativity in (2) also hold with $\mathbb{C}[x_1, \dots, x_d]$ replaced by $k[x_1, \dots, x_d]$ for any algebraically closed field $k \supset \mathbb{C}$.*

Proof. Part (1) is [Liu26, Theorem 1.5], and the first sentence of (2) is [Liu26, Theorem 4.5]. For a non-zero monomial ideal J_l of $k[x_1, \dots, x_d]$ and $w \in \mathbb{R}^d_{\geq 0}$, $\mathrm{wt}_w(J_l)$ is the minimum of $\langle w, \alpha \rangle$ over the exponents α of the monomials $x^\alpha \in J_l$ [Liu26, Definition 2.16]. Thus $\psi_{\min}(H, J, c)$ depends only on H, c, and the sets of exponents of the monomials in $J_1, \dots, J_s$. For a tuple J of non-zero monomial ideals of $k[x_1, \dots, x_d]$, let $J_{\mathbb{C}}$ be the tuple whose l-th entry is the monomial ideal of $\mathbb{C}[x_1, \dots, x_d]$ generated by the monomials x^α with $x^\alpha \in J_l$. Then $\psi_{\min}(H, J, c) = \psi_{\min}(H, J_{\mathbb{C}}, c)$, and the last sentence of (2) follows from the first. □

2.4. **Surfaces and a fixed threefold.** The reference pairs used later are always on a fixed ambient variety, such as a fixed quotient surface or a fixed $\mathbb{A}^3/H$. The bounded computing divisors of Theorems 2.9(2) and 2.10(3) are the starting points of the Gauss extensions [Liu26, Definition-Proposition 2.11], recalled in Subsection 2.2, in the upper bounds for quotient values. The uniform $\mathfrak{m}$-adic semicontinuity of [Kaw24, Theorem 1.3], which is Theorem 2.10(2), implies that finitely many jets suffice for the generic limits.

Theorem 2.9 ([HL24, Theorem 1.2], [HL23, Theorem 1.2])**.** *Let $\Gamma \subset [0,1]$ be a DCC set and k an algebraically closed field containing $\mathbb{C}$. Then the following hold.*

(1) *The set of $\mathrm{mld}(X \ni x, B)$, where (X, B) is an lc pair over k with $\dim X = 2$, $x \in X$ is a closed point and $B \in \Gamma$, satisfies the ACC.*

(2) *There exists a positive integer l depending only on Γ such that any lc surface germ $(X \ni x, B)$ over k with $B \in \Gamma$ has a prime divisor E over X with $\mathrm{center}_X E = \{x\}$, $a(E, X, B) = \mathrm{mld}(X \ni x, B)$, and $a(E, X, 0) \leq l$.*

Proof. Both [HL24] and [HL23] work over an algebraically closed field of arbitrary characteristic [HL24, Section 1], [HL23, Section 1]. For $k = \mathbb{C}$ and $\Gamma \subset \mathbb{Q}$, part (1) was proved earlier in [Ale93, Theorem 3.8], see also [Sho94]. Write $B = \sum_i b_i B_i$ with distinct prime divisors B_i and $b_i > 0$, and let B' be the sum of the terms $b_i B_i$ with $x \in B_i$. Then $B' \in \Gamma$, and $X^\circ := X \setminus \mathrm{Supp}(B - B')$ is an open neighborhood of x on which $B = B'$. Thus (X°, B') is lc near x, $a(E, X^\circ, B') = a(E, X, B)$ for any prime divisor E over X with $\mathrm{center}_X E = \{x\}$, and $\mathrm{mld}(X^\circ \ni x, B') = \mathrm{mld}(X \ni x, B)$. Replacing (X, B) with (X°, B') in both (1) and (2), we may assume that x is contained in any component of B. Then (1) is [HL24, Theorem 1.2], and (2) is [HL23, Theorem 1.2], which is proved for such germs [HL23, Definitions 2.6 and 2.8]. □

Theorem 2.10 ([Kaw24, Theorems 1.1, 1.3 and 1.4])**.** *Let X be a normal threefold and $\Gamma \subset \mathbb{R}_{>0}$ a DCC set. Then the following hold.*

(1) *The set of $\mathrm{mld}(X \ni \eta, B)$, where $\eta \in X$ is a scheme-theoretic point and (X, B) is a pair with $B \in \Gamma$, satisfies the ACC.*

(2) *There exists a positive integer l depending only on X and Γ such that $\mathrm{mld}(X \ni \eta, B, \mathfrak{a}) = \mathrm{mld}(X \ni \eta, B, \mathfrak{b})$ for any scheme-theoretic point $\eta \in X$, any pair (X, B) with $B \in \Gamma$, and any two $\mathbb{R}$-ideals $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ and $\mathfrak{b} = \prod_j \mathfrak{b}_j^{r_j}$ with $r_j \in \Gamma$ and*

$$\mathfrak{a}_j \mathcal{O}_{X,\eta} + \mathfrak{m}_\eta^l = \mathfrak{b}_j \mathcal{O}_{X,\eta} + \mathfrak{m}_\eta^l \quad \text{for any } j.$$

(3) *If moreover X is klt, then there exists a positive integer l depending only on X and Γ such that for any scheme-theoretic point $\eta \in X$ and any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $B \geq 0$ on X with $B \in \Gamma$ such that (X, B) is lc near η, there exists a prime divisor E over X which computes* $\mathrm{mld}(X \ni \eta, B)$ *and satisfies* $a(E, X, 0) \leq l$.

2.5. **Thresholds and fixed Gorenstein index.** We use lc thresholds of $\mathbb{R}$-divisors and of $\mathbb{R}$-ideals. The local versions follow [dFEM11, Section 2].

Definition 2.11. Let (X, Δ) be an lc pair, $x \in X$ a point, $M \geq 0$ an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X, and $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ an $\mathbb{R}$-ideal on X. The *lc threshold* of M with respect to (X, Δ) is

$$\mathrm{lct}(X, \Delta; M) := \sup\{t \geq 0 \mid (X, \Delta + tM) \text{ is lc}\},$$

and the lc threshold of $\mathfrak{a}$ with respect to (X, Δ) is

$$\mathrm{lct}(X, \Delta; \mathfrak{a}) := \sup\{t \geq 0 \mid (X, \Delta, \mathfrak{a}^t) \text{ is lc}\},$$

where $\mathfrak{a}^t := \prod_j \mathfrak{a}_j^{tr_j}$. The local versions $\mathrm{lct}(X \ni x, \Delta; M)$ and $\mathrm{lct}(X \ni x, \Delta; \mathfrak{a})$ are defined by requiring lc only near x. We omit Δ if $\Delta = 0$.

In the applications, the sources are hypersurface germs $(W \ni o) = (f = 0)$ in $(\mathbb{A}^4 \ni 0)$, and the equations f vary with the pairs, so the lc property cannot be checked on one fixed variety. We check it on the fixed germ $(\mathbb{A}^4 \ni 0)$, where f enters as a factor with exponent one together with the marks. Then the ACC for lc thresholds in Theorem 2.12(1) gives a gap below 1, and the $\mathfrak{m}$-adic continuity in Theorem 2.12(2) gives a truncation order depending only on the number of marks and the set of labels (Lemma 4.14).

Theorem 2.12 ([HMX14, Theorem 1.1], [dFEM11, Propositions 2.11 and 2.19])**.** *The following hold.*

(1) *Let n be a positive integer, and $I \subset [0, 1]$ and $J \subset \mathbb{R}_{>0}$ DCC sets. Then the set of log canonical thresholds* $\mathrm{lct}(X, \Delta; M)$*, where (X, Δ) is an lc pair of dimension n with $\Delta \in I$ and M is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor with $M \in J$, satisfies the ACC.*

(2) *Let $(X \ni x)$ be a klt germ of dimension n, $\epsilon > 0$, and $\mathfrak{a} = \prod_{i=1}^s \mathfrak{a}_i^{q_i}$ and $\mathfrak{b} = \prod_{i=1}^s \mathfrak{b}_i^{q_i}$ two $\mathbb{R}$-ideals with $q_i > 0$ such that* $\mathrm{lct}(X \ni x; \mathfrak{a})$ *is finite. Let l be a positive integer with $l \geq n/(\epsilon \min_i q_i)$, and assume that $\mathfrak{a}_i + \mathfrak{m}_x^l = \mathfrak{b}_i + \mathfrak{m}_x^l$ for all i. Then*

$$|\mathrm{lct}(X \ni x; \mathfrak{a}) - \mathrm{lct}(X \ni x; \mathfrak{b})| \leq \epsilon.$$

Proof. Part (1) is [HMX14, Theorem 1.1]. We prove (2). If $\mathfrak{a}_i \not\subset \mathfrak{m}_x$ for some i, then

$$\mathfrak{b}_i + \mathfrak{m}_x^l = \mathfrak{a}_i + \mathfrak{m}_x^l \not\subset \mathfrak{m}_x,$$

so $\mathfrak{b}_i \not\subset \mathfrak{m}_x$. Thus $\mathfrak{a}_i = \mathfrak{b}_i = \mathcal{O}_X$ near x, and removing the index i changes neither $\mathrm{lct}(X \ni x; \mathfrak{a})$, nor $\mathrm{lct}(X \ni x; \mathfrak{b})$, nor the inequality $l \geq n/(\epsilon \min_i q_i)$. Since $\mathrm{lct}(X \ni x; \mathfrak{a})$ is finite, at least one index remains. Therefore we may assume that $\mathfrak{a}_i, \mathfrak{b}_i \subset \mathfrak{m}_x$ for any i. Let $Z := \operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$. By [dFEM11, Proposition 2.11(ii)], $\mathrm{lct}(X \ni x; \mathfrak{a})$ and $\mathrm{lct}(X \ni x; \mathfrak{b})$ are equal to the lc thresholds on Z of the $\mathbb{R}$-ideals $\prod_i (\mathfrak{a}_i \widehat{\mathcal{O}}_{X,x})^{q_i}$ and $\prod_i (\mathfrak{b}_i \widehat{\mathcal{O}}_{X,x})^{q_i}$, whose factors are proper ideals of $\widehat{\mathcal{O}}_{X,x}$ since $\mathfrak{a}_i, \mathfrak{b}_i \subset \mathfrak{m}_x$. Since $l \geq n/(\epsilon q_i)$ for any i, (2) follows from [dFEM11, Proposition 2.19(ii)]. □

In the applications, the source W is Gorenstein and the labels belong to a finite set, so the log discrepancies of the pair (W, D) itself, before the quotient by G, form a set without accumulation points by Theorem 2.13. For sources of type cE with labels in a finite set S, this implies $\mathrm{mld}(W \ni o, D) \geq 1 + \epsilon_S$ for a real number $\epsilon_S > 0$ depending only on S whenever (W, D) is terminal near o (Lemma D.20), which is used in the proof of Theorem 10.1, and for bounded groups it implies finiteness (Proposition 7.10(2)). Theorem 2.13 is not used for the quotients directly, since the Cartier index of $K_{W/G}$ can be unbounded when $|G|$ is unbounded.

Theorem 2.13 ([Nak16a, Theorem 1.2, Corollary 1.3 and Remark 2.2])**.** *Let d and r be positive integers and $I \subset [0, +\infty)$ a finite set. Then the following hold.*

(1) *The set of log discrepancies* $a(E, X, \mathfrak{a})$*, where* $(X, \mathfrak{a})$ *is an lc pair of dimension* d *with* rK_X *Cartier,* $\mathfrak{a}$ *is an* $\mathbb{R}$*-ideal with exponents in* I*, and* E *is a prime divisor over* X*, has no accumulation point in* $\mathbb{R}$*. In particular, the minimal log discrepancies* $\operatorname{mld}(X \ni x, \mathfrak{a})$ *of such pairs at closed points form a set without accumulation points in* $\mathbb{R}$*.*

(2) *The set of log discrepancies* $a(E, X, \sum_i b_i D_i)$*, where* X *is a normal variety of dimension* d *with* rK_X *Cartier, the* $D_i \geq 0$ *are Cartier divisors on* X*,* $b_i \in I$*,* $(X, \sum_i b_i D_i)$ *is lc, and* E *is a prime divisor over* X*, has no accumulation point in* $\mathbb{R}$*. In particular, the minimal log discrepancies of such pairs at closed points form a set without accumulation points in* $\mathbb{R}$*.*

Proof. Part (1) is [Nak16a, Theorem 1.2 and Corollary 1.3]. Since rK_X is Cartier and the D_i are Cartier, [Nak16a, Remark 2.2] implies

$$a\Big(E, X, \sum_i b_i D_i\Big) = a\Big(E, X, \prod_i \mathcal{O}_X(-D_i)^{b_i}\Big)$$

for any prime divisor E over X. Thus (2) follows from (1). □

For a smooth curve on a smooth surface, the following lemma is [Kaw07, Theorem]. Below, when we cite [Kaw07, Theorem] on such formal germs, we mean Lemma 2.14.

Lemma 2.14. *Let* $k \supset \mathbb{C}$ *be an algebraically closed field and* $(S \ni s)$ *the spectrum of a two-dimensional regular complete local* k*-algebra with residue field* k*, for instance* $\operatorname{Spec} k[[p, q]]$ *or the spectrum of the completion of a smooth surface over* k *at a closed point. Let* $D \subset S$ *be a regular curve through* s *and* $B \geq 0$ *an* $\mathbb{R}$*-divisor on* S *whose support does not contain* D*. Then* $(S, D + B)$ *is lc if and only if* $\operatorname{mult}_s(B|_D) \leq 1$*.*

Proof. Write $B = \sum_i b_i B_i$ with distinct prime divisors $B_i \neq D$ and $b_i > 0$. The regular local ring $\mathcal{O}_{S,s}$ is a unique factorization domain [Mat89, Theorem 20.3], so D and the B_i are Cartier, and

$$\beta := \operatorname{mult}_s(B|_D) = \sum_i b_i (B_i \cdot D)_s,$$

where $(B_i \cdot D)_s \geq 1$, since B_i contains s. Here log discrepancies over a two-dimensional regular excellent local scheme are those of [Kol13, Definition 2.4 and 2.20].

Claim 2.15. *Let* $(T \ni t)$ *be a two-dimensional regular excellent local scheme with residue field* k*,* $C \subset T$ *a regular curve through* t*, and* $\Delta \geq 0$ *an* $\mathbb{R}$*-divisor on* T *whose support does not contain* C*. Assume that* $\operatorname{mult}_t(\Delta|_C) \leq 1$*. Then* $a(E, T, C + \Delta) \geq 0$ *for any prime divisor* E *over* T*.*

Proof. Write $\Delta = \sum_i \delta_i \Delta_i$ with distinct prime divisors Δ_i and $\delta_i > 0$. For prime divisors on T, C has coefficient one, and since $t \in \Delta_i$,

$$\delta_i \leq \delta_i (\Delta_i \cdot C)_t \leq \operatorname{mult}_t(\Delta|_C) \leq 1.$$

An exceptional prime divisor E over T has center t, and by [Kol13, Lemma 2.22] it is the exceptional curve of the last blow-up of a sequence of n blow-ups of closed points, each at the center of E. We induct on n. Let $\pi : T_1 \to T$ be the blow-up of t with exceptional curve $F \cong \mathbb{P}^1_k$, and C_1 and Δ_1 the strict transforms of C and Δ. Let $\mu := \operatorname{mult}_t \Delta$. Since C is regular, $\mu \leq (\Delta \cdot C)_t \leq 1$. A direct calculation in the charts of the blow-up (cf. [Kol13, Theorem 2.29]) gives

$$K_{T_1} + C_1 + \Delta_1 + \mu F = \pi^*(K_T + C + \Delta).$$

Thus $a(F, T, C + \Delta) = 1 - \mu \geq 0$, which is the case $n = 1$.

Assume that $n \geq 2$. Then the center t' of E on T_1 is a closed point of F, and E is reached over $\operatorname{Spec} \mathcal{O}_{T_1,t'}$ after $n - 1$ blow-ups. The scheme $\operatorname{Spec} \mathcal{O}_{T_1,t'}$ is again a two-dimensional regular excellent local scheme with residue field k, since T_1 is regular and of finite type over T [Mat89, p. 260] and t' is a closed point of $F \cong \mathbb{P}^1_k$. If $t' \in C_1$, then t' is the point where $C_1 \cong C$ intersects F transversally, and $(\Delta_1 + \mu F)|_{C_1} = (\pi^*\Delta)|_{C_1}$ corresponds to $\Delta|_C$ under $C_1 \cong C$.

Thus $\operatorname{mult}_{t'}((\Delta_1+\mu F)|_{C_1}) \le 1$, and the induction hypothesis applied to C_1 and $\Delta_1+\mu F$ on $\operatorname{Spec}\mathcal{O}_{T_1,t'}$ gives $a(E,T_1,C_1+\Delta_1+\mu F)\ge 0$. If $t'\notin C_1$, then

$$(\Delta_1\cdot F)_{t'} \le (\Delta_1\cdot F) = \mu \le 1,$$

as $(\pi^*\Delta\cdot F)=0$ and $(F^2)=-1$, so $\operatorname{mult}_{t'}(\Delta_1|_F)\le 1$. Since $\mu\le 1$, the induction hypothesis for F and Δ_1 on $\operatorname{Spec}\mathcal{O}_{T_1,t'}$ implies

$$a(E,T_1,C_1+\Delta_1+\mu F) = a(E,T_1,\Delta_1+\mu F) \ge a(E,T_1,\Delta_1+F)\ge 0.$$

In both cases $a(E,T,C+\Delta)=a(E,T_1,C_1+\Delta_1+\mu F)\ge 0$. □

Proof of Lemma 2.14 continued. If $\beta\le 1$, then the claim applied to $(S\ni s)$ (complete local rings are excellent [Mat89, p. 260]), D, and B shows that $(S,D+B)$ is lc.

Conversely, assume that $\beta>1$. Let $S_0:=S$, $s_0:=s$, $D_0:=D$, $F_0:=0$, and $c_0:=0$. For $j\ge 0$, let S_{j+1} be the blow-up of S_j at s_j with exceptional curve F_{j+1}, D_{j+1} the strict transform of D_j, and $s_{j+1}:=D_{j+1}\cap F_{j+1}$. Let B'_j be the strict transform of B on S_j, and $B'_{i,j}$ the strict transform of B_i. For $j\ge 2$, since D_{j-1} and F_{j-1} meet transversally at s_{j-1}, the earlier exceptional curves do not contain s_j. Applying the equality $K_{T_1}+C_1+\Delta_1+\mu F=\pi^*(K_T+C+\Delta)$ with $\mu=\operatorname{mult}_t\Delta$ to $C:=D_j$ and $\Delta:=B'_j+c_jF_j$ on $\operatorname{Spec}\mathcal{O}_{S_j,s_j}$, we see by induction on j that near s_j the crepant pull-back of $(S,D+B)$ to S_j is $(S_j,D_j+B'_j+c_jF_j)$, where

$$c_{j+1}=c_j+\operatorname{mult}_{s_j}B'_j.$$

Since the strict transform of F_j does not meet D_{j+1}, $(B'_{j+1}+c_{j+1}F_{j+1})|_{D_{j+1}}$ corresponds to $(B'_j+c_jF_j)|_{D_j}$ under $D_{j+1}\cong D_j$. As $(F_j\cdot D_j)_{s_j}=1$ for $j\ge 1$, induction on j gives

$$c_j+\operatorname{mult}_{s_j}(B'_j|_{D_j})=\beta.$$

Since D_j is regular,

$$(B'_{i,j+1}\cdot D_{j+1})_{s_{j+1}} = (B'_{i,j}\cdot D_j)_{s_j}-\operatorname{mult}_{s_j}B'_{i,j},$$

and $\operatorname{mult}_{s_j}B'_{i,j}\ge 1$ whenever $(B'_{i,j}\cdot D_j)_{s_j}>0$. Thus $\operatorname{mult}_{s_j}(B'_j|_{D_j})=0$ for some $j\ge 1$. Then $c_j=\beta$, and

$$a(F_j,S,D+B)=1-c_j=1-\beta<0.$$

Hence $(S,D+B)$ is not lc. □

3. Complements to the first paper

In this section, we prove the versions of results of [Liu26] which the later sections use in a form that [Liu26] does not state. For any result which [Liu26] states, we cite [Liu26].

3.1. **Quotient adjunction.** In this subsection, we prove a precise quotient adjunction for lc pairs with $\mathbb{R}$-ideals (Theorem 3.3), together with a lemma on Gauss extensions with a given inertia group (Lemma 3.1), a lemma on adjoining coordinates with coefficient one (Lemma 3.2), and a version of [Amb99, Proposition 2.1] for curves with $\mathbb{R}$-ideals (Lemma 3.4).

The upper bounds of the later sections lift a prime divisor computing the quotient value of a surface to a prime divisor over the source by a Gauss extension, and they need the inertia group of the lift to be the inertia group of the prime divisor on the surface. For this, the w-value ℓ of the Gauss extension has to satisfy $\ell\equiv q \pmod e$ in the notation of Lemma 3.1, since otherwise the inertia group of the lift is smaller and the ratio $a(F,\cdot)/|I_H(F)|$ of the log discrepancy to the order of the inertia group becomes larger. We state Lemma 3.1 over an algebraically closed field $k\supset\mathbb{C}$ since Proposition 5.4 uses it over k, and the proof uses only that k is algebraically closed of characteristic zero.

Lemma 3.1. *Let $k\supset\mathbb{C}$ be an algebraically closed field, V a normal quasi-projective variety over k, $0\in V$ a closed point, B an $\mathbb{R}$-divisor on V with K_V+B $\mathbb{R}$-Cartier, and H a finite group acting faithfully on V, fixing 0 and preserving B. Let χ be a character of H, and let H act on $V\times\mathbb{A}^1_w$ by $g\cdot(p,w)=(g\cdot p,\chi(g)w)$, so that $g^*w=\chi(g)w$, where g^* denotes the pullback of functions.*

Let E be a prime divisor over V with center 0, $I := I_H(E)$, $e := |I|$, λ the character by which I acts on $\mathfrak{m}_E/\mathfrak{m}_E^2$ through the pullback of functions, where $\mathfrak{m}_E$ is the maximal ideal of the valuation ring of mult_E, and $q \in \{1, \dots, e\}$ the integer with $\chi|_I = \lambda^q$. Then for any positive integer $\ell \equiv q \pmod{e}$, the Gauss extension of mult_E with w-value ℓ [Liu26, Definition-Proposition 2.11] *is mult_F for a prime divisor F over $V \times \mathbb{A}^1_w$ with center $(0,0)$, $I_H(F) = I$,*

$$a(F, V \times \mathbb{A}^1_w, \mathrm{pr}_1^* B) = a(E, V, B) + \ell \quad \text{and} \quad a(F, V \times \mathbb{A}^1_w, \mathrm{pr}_1^* B + \mathrm{div}(w)) = a(E, V, B).$$

Proof. Since k has characteristic zero, I is cyclic and λ is faithful [Liu26, Lemma 2.6(1) and Remark 2.17], so q is well defined. Let π_E be a rational function on V such that $\mathrm{mult}_E \pi_E = 1$. By definition [Liu26, Definition-Proposition 2.11], the Gauss extension v of mult_E with w-value ℓ is the valuation determined by

$$v\Big(\sum_i a_i w^i\Big) = \min_i \big(\mathrm{mult}_E a_i + i\ell\big)$$

for any non-zero polynomial $\sum_i a_i w^i$ in w whose coefficients a_i are rational functions on V, the minimum being taken over the indices i with $a_i \neq 0$.

Step 1. In this step, we construct F and show that $\mathrm{mult}_F = v$.

Let $f : Y \to V$ be a higher model such that E is a prime divisor on Y, and write $K_Y + B_Y := f^*(K_V + B)$. Then $\mathrm{mult}_E B_Y = 1 - a(E, V, B)$. Let $U \subset Y$ be a smooth affine open subset satisfying the following: $E \cap U$ is a non-empty prime divisor on U, π_E is regular on U and $\mathrm{div}(\pi_E|_U) = E \cap U$, $B_Y|_U = (1 - a(E, V, B))(E \cap U)$, and there exists a nowhere vanishing regular top form ω on U. Let

$$h : U \times \mathbb{A}^1_{w'} \to U \times \mathbb{A}^1_w, \qquad (u, w') \to (u, \pi_E(u)^\ell w').$$

Then h is birational, with inverse given by $w' = w/\pi_E^\ell$. Let $F := (E \cap U) \times \mathbb{A}^1_{w'}$. Then F is a prime divisor on $U \times \mathbb{A}^1_{w'}$, hence a prime divisor over $V \times \mathbb{A}^1_w$ through $(f \times \mathrm{id}_{\mathbb{A}^1_w}) \circ h$.

Let $P = \sum_{i \in J} a_i w^i$ be a polynomial in w whose coefficients a_i, $i \in J$, are non-zero rational functions on V. For any $i \in J$, write $a_i = \epsilon_i \pi_E^{m_i}$, where $m_i := \mathrm{mult}_E a_i$ and ϵ_i is a rational function on Y which is regular and invertible at the generic point of E. Let $\mu := \min_{i \in J}(m_i + i\ell)$. Then

$$h^* P = \pi_E^\mu \sum_{i \in J} \epsilon_i \pi_E^{m_i + i\ell - \mu} w'^i,$$

and the restriction of the sum on the right-hand side to F is

$$\sum_{i \in J,\ m_i + i\ell = \mu} \bar{\epsilon}_i w'^i,$$

where $\bar{\epsilon}_i \neq 0$ is the restriction of ϵ_i to E. Since the powers w'^i are distinct, the restriction of the sum to F is non-zero, so $\mathrm{mult}_F(h^* P) = \mu = v(P)$. Since any rational function on $V \times \mathbb{A}^1_w$ is a quotient of two such polynomials, $\mathrm{mult}_F = v$.

Step 2. In this step, we compute the center and the log discrepancies of F.

Since $h(F) = (E \cap U) \times \{0\}$ and $f(E) = \{0\}$, the center of F on $V \times \mathbb{A}^1_w$ is $(0,0)$. Since $\omega \wedge d\pi_E$ is a form of degree $\dim U + 1$ on U, it is zero. Writing ω also for $\mathrm{pr}_1^* \omega$ on $U \times \mathbb{A}^1_w$ and on $U \times \mathbb{A}^1_{w'}$, we have

$$h^*(\omega \wedge dw) = \omega \wedge d(\pi_E^\ell w') = \pi_E^\ell \omega \wedge dw',$$

so $K_{U \times \mathbb{A}^1_{w'}} = h^* K_{U \times \mathbb{A}^1_w} + \ell F$. Moreover,

$$(f \times \mathrm{id}_{\mathbb{A}^1_w})^*(K_{V \times \mathbb{A}^1_w} + \mathrm{pr}_1^* B) = K_{Y \times \mathbb{A}^1_w} + \mathrm{pr}_1^* B_Y \quad \text{and} \quad h^* \mathrm{pr}_1^*(B_Y|_U) = (1 - a(E, V, B))F.$$

Thus

$$K_{U \times \mathbb{A}^1_{w'}} + (1 - a(E, V, B) - \ell)F = h^*\big(K_{U \times \mathbb{A}^1_w} + \mathrm{pr}_1^*(B_Y|_U)\big),$$

and so $a(F, V \times \mathbb{A}^1_w, \mathrm{pr}_1^* B) = a(E, V, B) + \ell$. Since $\mathrm{mult}_F w = v(w) = \ell$, we have

$$a(F, V \times \mathbb{A}^1_w, \mathrm{pr}_1^* B + \mathrm{div}(w)) = a(E, V, B).$$

Step 3. In this step, we show that $I_H(F) = I$.

Let $u := w/\pi_E^\ell$. Since $h^* u = w'$ and $F = (E \cap U) \times \mathbb{A}^1_{w'}$, the residue field of mult_F is generated over the residue field of mult_E by the residue $\bar{u}$ of u, where the residue field of mult_E is embedded through pr_1^*.

Let $g \in I$. Since $g \in \mathrm{Stab}_H(E)$, we have $\mathrm{mult}_E(g^* a) = \mathrm{mult}_E a$ for any rational function a on V. Thus for any polynomial $P = \sum_i a_i w^i$ as in **Step 1**,

$$v(g^* P) = v\Big(\sum_i (g^* a_i)\chi(g)^i w^i\Big) = \min_i \big(\mathrm{mult}_E(g^* a_i) + i\ell\big) = v(P),$$

so $g \in \mathrm{Stab}_H(F)$ by **Step 1**. Since g acts on $\mathfrak{m}_E/\mathfrak{m}_E^2$ by $\lambda(g)$ through the pullback of functions, $g^*\pi_E - \lambda(g)\pi_E \in \mathfrak{m}_E^2$, so the rational function $g^*\pi_E/\pi_E$ has residue $\lambda(g)$ with respect to mult_E. Since

$$g^* u = \chi(g)\Big(\frac{\pi_E}{g^*\pi_E}\Big)^\ell u, \qquad \chi(g) = \lambda(g)^q, \qquad \lambda(g)^e = 1,$$

and $\ell \equiv q \pmod e$, the residue of $g^* u$ is $\lambda(g)^{q-\ell}\bar{u} = \bar{u}$. Since g acts trivially on the residue field of mult_E, g acts trivially on the residue field of mult_F. Thus $g \in I_H(F)$, and so $I \subset I_H(F)$.

Conversely, let $g \in I_H(F)$. For any rational function a on V,

$$\mathrm{mult}_E(g^* a) = \mathrm{mult}_F(\mathrm{pr}_1^* g^* a) = \mathrm{mult}_F(g^* \mathrm{pr}_1^* a) = \mathrm{mult}_F(\mathrm{pr}_1^* a) = \mathrm{mult}_E a,$$

so $g \in \mathrm{Stab}_H(E)$. Since g acts trivially on the residue field of mult_F, which contains the residue field of mult_E through pr_1^*, g acts trivially on the residue field of mult_E. Thus $g \in I$, and so $I_H(F) = I$. □

The next lemma extends [Liu26, Lemma 8.1] to several coordinates and to $\mathbb{R}$-ideals.

Lemma 3.2. *Let Z be a normal quasi-projective variety of positive dimension, $z \in Z$ a closed point, B an $\mathbb{R}$-divisor on Z with $K_Z + B$ $\mathbb{R}$-Cartier, H a finite group acting faithfully on Z, fixing z and preserving B, and $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ an $\mathbb{R}$-ideal on Z with H-invariant factors $\mathfrak{a}_j$. Let $\chi_1, \dots, \chi_k$ be characters of H, let H act on $Z \times \mathbb{A}^k_{t_1,\dots,t_k}$ by the given action on Z and by $g^* t_i = \chi_i(g) t_i$ for any $g \in H$ and any i, where g^* denotes the pullback of functions, let $p : Z \times \mathbb{A}^k \to Z$ be the projection, and let*

$$p^*\mathfrak{a} := \prod_j (\mathfrak{a}_j \mathcal{O}_{Z\times\mathbb{A}^k})^{r_j}, \qquad a := \mathrm{mld}((Z \ni z, B, \mathfrak{a})/H),$$

$$a^{\mathrm{c}} := \mathrm{mld}\Big(\Big(Z \times \mathbb{A}^k \ni (z,0), p^* B + \sum_{i=1}^k \mathrm{div}(t_i), p^*\mathfrak{a}\Big)/H\Big).$$

Then the following hold.

1. *For any prime divisor E over Z with center z, there exists a prime divisor F over $Z \times \mathbb{A}^k$ with center $(z,0)$ such that $I_H(F) = I_H(E)$ and*

$$a\Big(F, Z \times \mathbb{A}^k, p^* B + \sum_{i=1}^k \mathrm{div}(t_i), p^*\mathfrak{a}\Big) = a(E, Z, B, \mathfrak{a}).$$

   In particular, $a^{\mathrm{c}} \le a$.
2. *If $a \ge 0$ or $a^{\mathrm{c}} \ge 0$, then $a^{\mathrm{c}} = a$.*

Proof. For $0 \le i \le k$, let $Z_i := Z \times \mathbb{A}^i_{t_1,\dots,t_i}$ with the induced action of H, $p_i : Z_i \to Z$ the projection,

$$B_i := p_i^* B + \sum_{l=1}^i \mathrm{div}(t_l), \quad \text{and} \quad \mathfrak{a}^{(i)} := \prod_j (\mathfrak{a}_j \mathcal{O}_{Z_i})^{r_j}.$$

Then $Z_k = Z \times \mathbb{A}^k$, $B_k = p^*B + \sum_{i=1}^k \operatorname{div}(t_i)$, and $\mathfrak{a}^{(k)} = p^*\mathfrak{a}$. For $1 \le i \le k$, we have $Z_i = Z_{i-1} \times \mathbb{A}^1_{t_i}$ and $B_i = \operatorname{pr}_1^* B_{i-1} + (t_i = 0)$, and H acts on t_i by χ_i. Moreover, Z_{i-1} is a normal quasi-projective variety of positive dimension, $K_{Z_{i-1}} + B_{i-1}$ is $\mathbb{R}$-Cartier, and H acts faithfully on Z_{i-1}, fixing $(z,0)$ and preserving B_{i-1}. Thus Z_{i-1}, $(z,0)$, B_{i-1}, H, and χ_i satisfy the hypotheses of [Liu26, Lemma 8.1].

Let $1 \le i \le k$. By [Liu26, Lemma 8.1(1)], for any prime divisor F' over Z_i with center $(z,0)$, there exist a positive integer ϱ and a prime divisor E' over Z_{i-1} with center $(z,0)$ such that $I_H(F') \subset I_H(E')$, $a(F', Z_i, B_i) \ge \varrho a(E', Z_{i-1}, B_{i-1})$, and

$$\operatorname{mult}_{F'}(\operatorname{pr}_1^* \zeta) = \varrho \operatorname{mult}_{E'} \zeta \quad \text{for any non-zero rational function } \zeta \text{ on } Z_{i-1}. \tag{3.1}$$

By [Liu26, Lemma 8.1(2)], for any prime divisor E' over Z_{i-1} with center $(z,0)$, a Gauss extension of $\operatorname{mult}_{E'}$ is $\operatorname{mult}_{F'}$ for a prime divisor F' over Z_i with center $(z,0)$ such that $I_H(F') = I_H(E')$ and $a(F', Z_i, B_i) = a(E', Z_{i-1}, B_{i-1})$. For this F', (3.1) holds with $\varrho = 1$.

Let F' be a prime divisor over Z_i with center $(z,0)$, let E' be a prime divisor over Z_{i-1} with center $(z,0)$, and let ϱ be a positive integer such that (3.1) holds. For any j, let $\zeta_1, \dots, \zeta_m$ be generators of $\mathfrak{a}_j \mathcal{O}_{Z_{i-1}}$ near $(z,0)$. Then the pullbacks $\operatorname{pr}_1^* \zeta_l$, $1 \le l \le m$, generate $\mathfrak{a}_j \mathcal{O}_{Z_i}$ near $(z,0)$, so

$$\operatorname{mult}_{F'}(\mathfrak{a}_j \mathcal{O}_{Z_i}) = \min_l \operatorname{mult}_{F'}(\operatorname{pr}_1^* \zeta_l) = \varrho \min_l \operatorname{mult}_{E'} \zeta_l = \varrho \operatorname{mult}_{E'}(\mathfrak{a}_j \mathcal{O}_{Z_{i-1}}).$$

Thus

$$a(F', Z_i, B_i, \mathfrak{a}^{(i)}) - \varrho a(E', Z_{i-1}, B_{i-1}, \mathfrak{a}^{(i-1)}) = a(F', Z_i, B_i) - \varrho a(E', Z_{i-1}, B_{i-1}). \tag{3.2}$$

We have the following for any $1 \le i \le k$.

(i) For any prime divisor F' over Z_i with center $(z,0)$, there exist a positive integer ϱ and a prime divisor E' over Z_{i-1} with center $(z,0)$ such that (3.1) holds, $I_H(F') \subset I_H(E')$, and

$$a(F', Z_i, B_i, \mathfrak{a}^{(i)}) \ge \varrho a(E', Z_{i-1}, B_{i-1}, \mathfrak{a}^{(i-1)}).$$

This follows from [Liu26, Lemma 8.1(1)] and (3.2).

(ii) For any prime divisor E' over Z_{i-1} with center $(z,0)$, there exists a prime divisor F' over Z_i with center $(z,0)$ such that (3.1) holds with $\varrho = 1$, $I_H(F') = I_H(E')$, and

$$a(F', Z_i, B_i, \mathfrak{a}^{(i)}) = a(E', Z_{i-1}, B_{i-1}, \mathfrak{a}^{(i-1)}).$$

This follows from [Liu26, Lemma 8.1(2)] and (3.2).

(1) Let $E_0 := E$. For $i = 1, \dots, k$, (ii) applied to $E' = E_{i-1}$ provides a prime divisor E_i over Z_i with center $(z,0)$ such that $I_H(E_i) = I_H(E_{i-1})$ and

$$a(E_i, Z_i, B_i, \mathfrak{a}^{(i)}) = a(E_{i-1}, Z_{i-1}, B_{i-1}, \mathfrak{a}^{(i-1)}).$$

Then $F := E_k$ satisfies (1). By Definition 2.4,

$$a^{\mathrm{c}} \le \frac{a(F, Z_k, B_k, p^*\mathfrak{a})}{|I_H(F)|} = \frac{a(E, Z, B, \mathfrak{a})}{|I_H(E)|}.$$

Since E is an arbitrary prime divisor over Z with center z, $a^{\mathrm{c}} \le a$.

(2) First assume that $a \ge 0$. Let F be a prime divisor over $Z \times \mathbb{A}^k$ with center $(z,0)$, and let $F_k := F$. For $i = k, \dots, 1$, (i) applied to $F' = F_i$ provides a positive integer ϱ_i and a prime divisor F_{i-1} over Z_{i-1} with center $(z,0)$ such that $I_H(F_i) \subset I_H(F_{i-1})$ and

$$a(F_i, Z_i, B_i, \mathfrak{a}^{(i)}) \ge \varrho_i a(F_{i-1}, Z_{i-1}, B_{i-1}, \mathfrak{a}^{(i-1)}).$$

Let $E := F_0$ and $\varrho := \prod_{i=1}^k \varrho_i \ge 1$. Then E is a prime divisor over Z with center z, $I_H(F) \subset I_H(E)$, and

$$a(F, Z_k, B_k, p^*\mathfrak{a}) \ge \varrho a(E, Z, B, \mathfrak{a}).$$

Since $a(E,Z,B,\mathfrak{a}) \geq |I_H(E)|a \geq 0$ and $|I_H(F)| \leq |I_H(E)|$, we have

$$\frac{a(F,Z_k,B_k,p^*\mathfrak{a})}{|I_H(F)|} \geq \frac{\varrho a(E,Z,B,\mathfrak{a})}{|I_H(F)|} \geq \frac{a(E,Z,B,\mathfrak{a})}{|I_H(E)|} \geq a.$$

Since F is an arbitrary prime divisor over $Z \times \mathbb{A}^k$ with center $(z,0)$, $a^{\mathrm{c}} \geq a$, and so $a^{\mathrm{c}} = a$ by (1).

Now assume that $a^{\mathrm{c}} \geq 0$. By (1), $a \geq a^{\mathrm{c}} \geq 0$, and so $a^{\mathrm{c}} = a$ by the case above. □

The lower bounds of the later sections compare the quotient value of a source with the quotient value of a section which has coefficient one in the boundary. There, the pair on the section may be lc and not klt, the boundary may contain $\mathbb{R}$-ideals which are not principal, the section may be a singular klt variety, such as an A_1 surface section of the threefold chart $(UV + Z^2 + v^d = 0)$, and the point may be the generic point of a curve. Theorem 3.3 allows all of them, whereas [Liu26, Theorem 8.3] covers only klt pairs with principal boundaries at closed points, and [NS26b, Theorem 7.1] also needs that the group has no pseudoreflections and that the branch locus intersects the quotient of the complete intersection in codimension at least two [NS26b, Remark 3.3].

Theorem 3.3. *Let n, c be integers with $c \geq 1$ and $n - c \geq 2$, let G be a finite abelian group acting diagonally on $\mathbb{A}^n$, and let $f_1, \ldots, f_c$ be G-semi-invariant polynomials vanishing at 0 whose common zero scheme X is, near 0, a normal integral complete intersection of codimension c with klt singularities, on which G acts faithfully. Let $\mathfrak{a} = \prod_{j=1}^{s} \mathfrak{a}_j^{r_j}$ be an $\mathbb{R}$-ideal on $\mathbb{A}^n$ with $r_j > 0$ and each $\mathfrak{a}_j$ generated by G-semi-invariant polynomials, such that $\mathfrak{a}|_X$ is non-zero near 0. Let $H \subset G$ be a subgroup.*

(1) If $(X, \mathfrak{a}|_X)$ is lc near 0, then

$$\mathrm{mld}((X \ni 0, \mathfrak{a}|_X)/H) = \mathrm{mld}\Big(\Big(\mathbb{A}^n \ni 0, \sum_{i=1}^{c} \mathrm{div}(f_i), \mathfrak{a}\Big)/H\Big).$$

(2) Let f_{c+1} be a G-semi-invariant polynomial such that $S := X \cap (f_{c+1} = 0)$ is, near 0, a normal integral complete intersection of codimension $c+1$ in $\mathbb{A}^n$ with klt singularities, on which H acts faithfully, and such that $\mathfrak{a}|_S$ is non-zero near 0. If $(S, \mathfrak{a}|_S)$ is lc near 0, then

$$\mathrm{mld}((S \ni 0, \mathfrak{a}|_S)/H) = \mathrm{mld}((X \ni 0, S, \mathfrak{a}|_X)/H) = \mathrm{mld}\Big(\Big(\mathbb{A}^n \ni 0, \sum_{i=1}^{c+1} \mathrm{div}(f_i), \mathfrak{a}\Big)/H\Big),$$

where S has coefficient one in the middle term.

(3) The equalities of (1) and (2) also hold with 0 replaced by the generic point ξ of an H-invariant integral curve through 0 contained in X (in S for (2), in which case we assume in addition that $n - c \geq 3$), the minimal log discrepancies being taken at the image of ξ in the quotients, provided that $(X, \mathfrak{a}|_X)$ (resp. $(S, \mathfrak{a}|_S)$) is lc near ξ.

Proof of Theorem 3.3(1). The proof is long, so we divide it into several steps.

By [Liu26, Theorem 2.22], if a finite group H' acts faithfully and linearly on $\mathbb{A}^m$ without pseudoreflections, $Y \subset \mathbb{A}^m$ is the closed subscheme defined by H'-semi-invariant polynomials $\psi_1, \ldots, \psi_c$ which, near 0, is a normal complete intersection of codimension c with klt singularities and $m - c \geq 2$, and, near the image y' of 0, the image in $\mathbb{A}^m/H'$ of the locus where $\mathbb{A}^m \to \mathbb{A}^m/H'$ is not étale intersects Y/H' in codimension at least two, then for any $\mathbb{R}$-ideal $\mathfrak{b}'$ on $\mathbb{A}^m/H'$ whose restriction to Y/H' is non-zero near y', we have

$$\mathrm{mld}(Y/H' \ni y', \mathfrak{b}'|_{Y/H'}) = \mathrm{mld}\big(\mathbb{A}^m/H' \ni y', (\psi_1^r \cdots \psi_c^r)^{1/r}\mathfrak{b}'\big),$$

where $r := |H'|$. The group H acting on $\mathbb{A}^n$ may have pseudoreflections, and the last condition may fail for X. After adjoining coordinates $t_1, \ldots, t_n$ on which H acts by the characters of the diagonal coordinates $x_1, \ldots, x_n$ of $\mathbb{A}^n$ fixed below, both conditions hold (**Step 1**), and Lemma 3.2 shows that adjoining $t_1, \ldots, t_n$ with coefficient one does not change the quotient values (**Step 3**).

Let $x_1, \dots, x_n$ be coordinates of $\mathbb{A}^n$ with respect to which G acts diagonally, let χ_i be the character of H such that $g^* x_i = \chi_i(g) x_i$ for any $g \in H$, and let $N := |H|$. Let $V^+ := \mathbb{A}^n \times \mathbb{A}^n_{t_1,\dots,t_n}$, and let H act on V^+ by the given action on $\mathbb{A}^n$ and by $g^* t_i = \chi_i(g) t_i$ for any $g \in H$ and any i. Let $p : V^+ \to \mathbb{A}^n$ be the first projection, $X^+ := X \times \mathbb{A}^n$, which is the common zero scheme of $p^* f_1, \dots, p^* f_c$, 0^+ the origin of V^+, $\pi : V^+ \to V^+/H$ the quotient morphism, and $y^+ := \pi(0^+)$. Let

$$a_X := \mathrm{mld}((X \ni 0, \mathfrak{a}|_X)/H), \qquad a_V := \mathrm{mld}\Big(\Big(\mathbb{A}^n \ni 0, \sum_{i=1}^{c} \mathrm{div}(f_i), \mathfrak{a}\Big)/H\Big),$$

$$a_X^+ := \mathrm{mld}\Big(\Big(X^+ \ni 0^+, \sum_{i=1}^{n} \mathrm{div}(t_i|_{X^+}), p^*\mathfrak{a}|_{X^+}\Big)/H\Big),$$

and

$$a_V^+ := \mathrm{mld}\Big(\Big(V^+ \ni 0^+, \sum_{i=1}^{c} \mathrm{div}(p^* f_i) + \sum_{i=1}^{n} \mathrm{div}(t_i), p^*\mathfrak{a}\Big)/H\Big).$$

We need to show that $a_X = a_V$.

Step 1. In this step, we show that H and $X^+ \subset V^+$ satisfy the hypotheses of [Liu26, Theorem 2.22], and that the crepant quotient boundaries of $(V^+, 0)$ and of $(X^+, 0)$ by H are zero near y^+.

The group H acts faithfully and linearly on V^+. Let $g \in H$ be a non-identity element. Since G acts faithfully on X near 0, g acts non-trivially on $\mathbb{A}^n$, so there exists i such that $\chi_i(g) \neq 1$. Thus the fixed locus $\mathrm{Fix}(g)$ of g on V^+ is contained in $(x_i = t_i = 0)$, which has codimension two. In particular, H has no pseudoreflections. Moreover,

$$\mathrm{Fix}(g) \cap X^+ = (\mathrm{Fix}(g) \cap X) \times L_g,$$

where $\mathrm{Fix}(g) \cap X$ is the fixed locus of g on X and $L_g \subset \mathbb{A}^n_{t_1,\dots,t_n}$ is the linear subspace ($t_l = 0$ for any l with $\chi_l(g) \neq 1$). Since X is integral near 0 and g acts non-trivially on any neighborhood of 0 in X, $\mathrm{Fix}(g) \cap X$ has codimension at least one in X near 0. Since $\dim L_g \leq n - 1$,

$$\mathrm{codim}_{X^+}(\mathrm{Fix}(g) \cap X^+) \geq 2 \quad \text{near } 0^+. \tag{3.3}$$

The locus where π is not étale is the union of $\mathrm{Fix}(g)$ over the non-identity elements $g \in H$. Since X^+ is H-invariant and $\pi|_{X^+} : X^+ \to X^+/H$ is finite and surjective, (3.3) implies that, near y^+, the image in V^+/H of the locus where π is not étale intersects X^+/H in codimension at least two.

Near 0^+, X^+ is a normal integral complete intersection of codimension c in V^+, defined by the H-semi-invariant polynomials $p^* f_1, \dots, p^* f_c$, and $\dim X^+ = 2n - c \geq 2$. If $h : X'' \to X$ is a log resolution of X near 0, then $h \times \mathrm{id}_{\mathbb{A}^n}$ is a log resolution of X^+ near 0^+ whose discrepancy coefficients are those of h, so X^+ is klt near 0^+ [KM98, Lemma 2.30 and Corollary 2.31(3)].

Finally, by (3.3) and since $\mathrm{Fix}(g)$ has codimension at least two in V^+ for any non-identity element $g \in H$, no prime divisor on V^+, and no prime divisor on X^+ near 0^+, is fixed pointwise by a non-identity element of H. Thus any such prime divisor has trivial inertia group, and [Liu26, Lemma 2.6(1)(4)] implies that the crepant quotient boundaries of $(V^+, 0)$ and of $(X^+, 0)$ by H are zero near y^+.

Step 2. In this step, we construct an $\mathbb{R}$-ideal $\mathfrak{b}$ on V^+/H and prove (3.4) by [Liu26, Theorem 2.22].

For any j, let $h_{j1}, \dots, h_{jm_j}$ be G-semi-invariant polynomials generating $\mathfrak{a}_j$. Since the order of any character of H divides N, the polynomials $p^* h_{jl}^N$, $p^* f_i^N$, and t_i^N are H-invariant. Let $\mathfrak{b}_j$ be the ideal on V^+/H generated by $p^* h_{j1}^N, \dots, p^* h_{jm_j}^N$, let $\mathfrak{t}_i$ be the ideal generated by t_i^N, and let

$$\mathfrak{b} := \prod_{j=1}^{s} \mathfrak{b}_j^{r_j/N} \cdot \prod_{i=1}^{n} \mathfrak{t}_i^{1/N}.$$

For any prime divisor F over V^+, since $\operatorname{mult}_F(\phi^N) = N \operatorname{mult}_F \phi$ for any non-zero polynomial ϕ,

$$\operatorname{mult}_F(\pi^*\mathfrak{b}_j) = \min_l \operatorname{mult}_F(p^* h_{jl}^N) = N \operatorname{mult}_F(p^*\mathfrak{a}_j) \quad \text{and} \quad \operatorname{mult}_F(\pi^*\mathfrak{t}_i) = N \operatorname{mult}_F t_i.$$

The equalities $\operatorname{mult}_F(\pi^*\mathfrak{b}_j) = N \operatorname{mult}_F(p^*\mathfrak{a}_j)$ and $\operatorname{mult}_F(\pi^*\mathfrak{t}_i) = N \operatorname{mult}_F t_i$ also hold for any prime divisor F over X^+, with $\mathfrak{b}_j$, $\mathfrak{t}_i$, $\mathfrak{a}_j$, and t_i restricted to X^+. Thus

$$a\big(F, V^+, 0, \pi^*\big((f_1^N \cdots f_c^N)^{\frac{1}{N}}\mathfrak{b}\big)\big) = a\Big(F, V^+, \sum_{i=1}^c \operatorname{div}(p^* f_i) + \sum_{i=1}^n \operatorname{div}(t_i), p^*\mathfrak{a}\Big)$$

for any prime divisor F over V^+, and

$$a(F, X^+, 0, \pi^*\mathfrak{b}|_{X^+}) = a\Big(F, X^+, \sum_{i=1}^n \operatorname{div}(t_i|_{X^+}), p^*\mathfrak{a}|_{X^+}\Big)$$

for any prime divisor F over X^+. By **Step 1** and [Liu26, Lemma 2.6(2)(3)], any prime divisor over V^+/H (resp. X^+/H) with center y^+ is the prime divisor under a prime divisor F over V^+ (resp. X^+) with center 0^+, the prime divisor under any such F has center y^+, and the log discrepancy of the prime divisor under F is the log discrepancy of F in the corresponding equality above divided by $|I_H(F)|$. Thus, by Definition 2.4, the minimal log discrepancy at y^+ of the pair

$$\big(V^+/H, (f_1^N \cdots f_c^N)^{\frac{1}{N}}\mathfrak{b}\big) \quad (\text{resp. } (X^+/H, \mathfrak{b}|_{X^+/H}))$$

is a_V^+ (resp. a_X^+).

Since $\mathfrak{a}|_X$ is non-zero near 0 and X^+ is integral near 0^+, $\mathfrak{b}|_{X^+/H}$ is non-zero near y^+. By **Step 1**, H, $X^+ \subset V^+$, and $\mathfrak{b}$ satisfy the hypotheses of [Liu26, Theorem 2.22]. Thus, by [Liu26, Theorem 2.22] (cf. [NS26b, Theorem 7.1]),

$$\operatorname{mld}\big(X^+/H \ni y^+, \mathfrak{b}|_{X^+/H}\big) = \operatorname{mld}\big(V^+/H \ni y^+, (f_1^N \cdots f_c^N)^{\frac{1}{N}}\mathfrak{b}\big). \tag{3.4}$$

Step 3. In this step, we remove the coordinates $t_1, \ldots, t_n$ and conclude the proof.

Since $(X, \mathfrak{a}|_X)$ is lc near 0, $a_X \geq 0$. Let $X^\circ \subset X$ be an H-invariant open neighborhood of 0 such that X° is a normal variety with klt singularities and $(X^\circ, \mathfrak{a}|_{X^\circ})$ is lc. Since a_X and a_X^+ depend only on neighborhoods of 0 and of 0^+, Lemma 3.2(2), applied to $Z = X^\circ$, $z = 0$, $B = 0$, the $\mathbb{R}$-ideal $\mathfrak{a}|_{X^\circ}$, and the characters $\chi_1, \ldots, \chi_n$, implies that $a_X^+ = a_X$.

By **Step 2**, the left-hand side of (3.4) is a_X^+ and the right-hand side of (3.4) is a_V^+. Therefore,

$$a_V^+ = a_X^+ = a_X \geq 0.$$

The group H acts faithfully on $\mathbb{A}^n$, fixing 0 and preserving $\sum_{i=1}^c \operatorname{div}(f_i)$, and the factors $\mathfrak{a}_j$ are H-invariant since they are generated by G-semi-invariant polynomials. Since $a_V^+ \geq 0$, Lemma 3.2(2), applied to $Z = \mathbb{A}^n$, $z = 0$, $B = \sum_{i=1}^c \operatorname{div}(f_i)$, $\mathfrak{a}$, and the characters $\chi_1, \ldots, \chi_n$, implies that $a_V = a_V^+$. Lemma 3.2(2) does not need $(\mathbb{A}^n, \sum_{i=1}^c \operatorname{div}(f_i), \mathfrak{a})$ to be lc. Thus $a_V = a_X$. □

We now prove parts (2) and (3) of Theorem 3.3. For part (3), we use the following version of [Amb99, Proposition 2.1] for curves, with $\mathbb{R}$-ideals. Since [Amb99] treats $\mathbb{R}$-divisors only, we include the proof.

Lemma 3.4. *Let X be a normal quasi-projective variety of dimension at least two, $\mathfrak{a} = \prod_{j=1}^s \mathfrak{a}_j^{r_j}$ an $\mathbb{R}$-ideal on X, and $C \subset X$ an integral curve with generic point η. Let B be an $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier. Assume that $(X, B, \mathfrak{a})$ is lc near η. Then there exists a non-empty open subset $C^\circ \subset C$ such that*

$$\operatorname{mld}(X \ni q, B, \mathfrak{a}) = \operatorname{mld}(X \ni \eta, B, \mathfrak{a}) + 1$$

for any closed point $q \in C^\circ$.

Proof. Since the minimal log discrepancy at a point only depends on an open neighborhood of the point, possibly shrinking X to an open neighborhood of η, we may assume that $(X, B, \mathfrak{a})$ is lc.

Let $h : X' \to X$ be a log resolution of (X, B) such that, for any j, $\mathfrak{a}_j \mathcal{O}_{X'} = \mathcal{O}_{X'}(-L_j)$ for a divisor L_j, such that $h^{-1}(C)$ is a divisor, and such that

$$\Sigma := \operatorname{Exc}(h) \cup \operatorname{Supp} h_*^{-1} B \cup \bigcup_j \operatorname{Supp} L_j \cup h^{-1}(C)$$

is an snc divisor. Let $P_1, \dots, P_l$ be the irreducible components of Σ. Define B' by

$$K_{X'} + B' = h^*(K_X + B) + \sum_j r_j L_j,$$

and write $B' = \sum_{k=1}^{l} b_k P_k$. For any prime divisor E over X' and any j, we have $\operatorname{mult}_E \mathfrak{a}_j = \operatorname{mult}_E L_j$, so

$$a(E, X, B, \mathfrak{a}) = a(E, X', B'). \tag{3.5}$$

In particular, $a_k := 1 - b_k = a(P_k, X, B, \mathfrak{a}) \geq 0$ for any k. Let J_C be the set of indices k such that $h(P_k) = C$. The irreducible components of the non-empty divisor $h^{-1}(C)$ are among $P_1, \dots, P_l$, and $h^{-1}(C) \to C$ is surjective, so $J_C \neq \emptyset$.

We first bound log discrepancies from below. Let E be a prime divisor over X', $W := \operatorname{center}_{X'} E$, and J the set of indices k such that $W \subset P_k$. Since Σ is snc, there exists a regular system of parameters $(z_k)_{k \in J}, z'_1, \dots, z'_m$ of the local ring of X' at the generic point of W such that $P_k = (z_k = 0)$ near the generic point of W for any $k \in J$. The pair $(X', \sum_{k \in J}(z_k = 0) + \sum_{i=1}^{m}(z'_i = 0))$ is snc near the generic point of W, hence lc there [KM98, Corollary 2.31(3)]. Moreover, $B' = \sum_{k \in J} b_k P_k$ near the generic point of W. Thus

$$a(E, X', B') \geq \sum_{k \in J} (1 - b_k) \operatorname{mult}_E z_k + \sum_{i=1}^{m} \operatorname{mult}_E z'_i = \sum_{k \in J} a_k \operatorname{mult}_E z_k + \sum_{i=1}^{m} \operatorname{mult}_E z'_i. \tag{3.6}$$

Since the center of E on X' is W, we have $\operatorname{mult}_E z_k \geq 1$ and $\operatorname{mult}_E z'_i \geq 1$ for any $k \in J$ and any i.

Next we show that

$$\operatorname{mld}(X \ni \eta, B, \mathfrak{a}) = \min_{k \in J_C} a_k. \tag{3.7}$$

For any $k \in J_C$, the center of P_k on X is C, so the left side is at most a_k. Conversely, let E be a prime divisor over X with $\operatorname{center}_X E = C$, and let $W := \operatorname{center}_{X'} E$. Then $h(W) = C$, so W is contained in an irreducible component P_{k_0} of $h^{-1}(C)$. Since $C = h(W) \subset h(P_{k_0}) \subset C$, we have $k_0 \in J_C$, and (3.5) and (3.6) imply $a(E, X, B, \mathfrak{a}) \geq a_{k_0}$. Thus (3.7) holds.

Let $\mathcal{W}$ be the set of irreducible components of the intersections $\bigcap_{k \in J} P_k$, where J runs over the non-empty subsets of $\{1, \dots, l\}$. The set $\mathcal{W}$ is finite, and for any $W \in \mathcal{W}$ with $h(W) \subset C$, the image $h(W)$ is either C or a closed point of C. For $W \in \mathcal{W}$ with $h(W) = C$, the morphism $W \to C$ induced by h is proper and surjective, it is flat over the regular locus of C [Har77, Proposition III.9.7], and hence its fibers over that locus are of pure dimension $\dim W - 1$ [Har77, Corollary III.9.6]. Thus there exists a non-empty open subset C° of the regular locus of C such that the following hold.

(a) $h(W) \cap C^\circ = \emptyset$ for any $W \in \mathcal{W}$ such that $h(W)$ is a closed point of C.
(b) For any $W \in \mathcal{W}$ with $h(W) = C$ and any closed point $q \in C^\circ$, the fiber $W \cap h^{-1}(q)$ is non-empty and of pure dimension $\dim W - 1$.

Let $q \in C^\circ$ be a closed point.

We show that

$$\operatorname{mld}(X \ni q, B, \mathfrak{a}) \geq \operatorname{mld}(X \ni \eta, B, \mathfrak{a}) + 1.$$

Let E be a prime divisor over X with $\operatorname{center}_X E = \{q\}$, let $W := \operatorname{center}_{X'} E$, and let J and $z'_1, \dots, z'_m$ be as in (3.6) for E. Since $W \subset h^{-1}(q) \subset h^{-1}(C)$, W is contained in an irreducible component P_{k_0} of $h^{-1}(C)$. The closed subset $h(P_{k_0}) \subset C$ contains q, so $h(P_{k_0}) = C$ by (a). Hence $k_0 \in J \cap J_C$. Let $W_J \in \mathcal{W}$ be the irreducible component of $\bigcap_{k\in J} P_k$ which contains W. Since $h(W_J) \subset h(P_{k_0}) = C$ and $q \in h(W_J)$, (a) implies $h(W_J) = C$. Since $h(W) = \{q\}$, $W \subsetneq W_J$, so $\operatorname{codim}_{X'} W > |J|$, that is, $m \geq 1$. By (3.5), (3.6), and (3.7),

$$a(E, X, B, \mathfrak{a}) \geq a_{k_0} + 1 \geq \operatorname{mld}(X \ni \eta, B, \mathfrak{a}) + 1.$$

We show that

$$\operatorname{mld}(X \ni q, B, \mathfrak{a}) \leq \operatorname{mld}(X \ni \eta, B, \mathfrak{a}) + 1.$$

By (3.7), there exists $k_1 \in J_C$ such that $a_{k_1} = \operatorname{mld}(X \ni \eta, B, \mathfrak{a})$. By (b), $P_{k_1} \cap h^{-1}(q)$ is non-empty and of pure dimension $\dim X - 2$. For any $k \neq k_1$, any irreducible component W of $P_{k_1} \cap P_k$ belongs to $\mathcal{W}$, $\dim W = \dim X - 2$, and $h(W) \subset C$. If $h(W) = C$, then $\dim(W \cap h^{-1}(q)) = \dim X - 3$ by (b), and otherwise $W \cap h^{-1}(q) = \emptyset$ by (a). Thus there exists an irreducible component V of $P_{k_1} \cap h^{-1}(q)$ which is not contained in P_k for any $k \neq k_1$, and $\operatorname{codim}_{X'} V = 2$. Let $U' \subset X'$ be an open subset such that $U' \cap V$ is non-empty and smooth and $U' \cap P_k = \emptyset$ for any $k \neq k_1$, and let E_V be the exceptional divisor of the blow-up of U' along $U' \cap V$. Since $B'|_{U'} = b_{k_1} P_{k_1}|_{U'}$ and P_{k_1} is smooth and contains V,

$$a(E_V, X', B') = 2 - b_{k_1} = a_{k_1} + 1.$$

Since $\operatorname{center}_X E_V = h(V) = \{q\}$, (3.5) implies

$$\operatorname{mld}(X \ni q, B, \mathfrak{a}) \leq a_{k_1} + 1 = \operatorname{mld}(X \ni \eta, B, \mathfrak{a}) + 1.$$

□

Proof of Theorem 3.3(2) and (3). The proof is long, so we divide it into several steps. Let $x_1, \dots, x_n$ be the coordinates of $\mathbb{A}^n$, and let $\chi_1, \dots, \chi_n$ be the characters of G such that $g^* x_i = \chi_i(g) x_i$ for any $g \in G$ and any i, where g^* denotes the pullback of functions. Let $N := |H|$. The characters of H take values in the N-th roots of unity, so φ^N is H-invariant for any H-semi-invariant regular function φ.

Step 1. In this step, we write quotient values as minimal log discrepancies of quotient pairs, at any point and with the subgroup fixing the point.

Let Z be a normal affine variety with a faithful action of H. Let B_Z be an H-invariant $\mathbb{R}$-divisor on Z such that $K_Z + B_Z$ is $\mathbb{R}$-Cartier. Let $\mathfrak{d} = \prod_j \mathfrak{d}_j^{s_j}$ be an $\mathbb{R}$-ideal on Z such that each $\mathfrak{d}_j$ is generated by H-semi-invariant regular functions $g_{j1}, \dots, g_{jm_j}$. Let $\pi_Z : Z \to Z/H$ be the quotient morphism, and let $(B_Z)_{Z/H}$ be the $\mathbb{R}$-divisor on Z/H such that $K_Z + B_Z = \pi_Z^*(K_{Z/H} + (B_Z)_{Z/H})$ [Liu26, Lemma 2.6(0)]. Let $\mathfrak{d}_j^{(N)}$ be the ideal on Z/H generated by $g_{j1}^N, \dots, g_{jm_j}^N$, and let $\mathfrak{d}_{Z/H} := \prod_j (\mathfrak{d}_j^{(N)})^{s_j/N}$. For a point $\zeta \in Z$, let $H_\zeta := \{g \in H \mid g(\zeta) = \zeta\}$. We show the following.

(i) For any point $\zeta \in Z$,

$$\operatorname{mld}\big(Z/H \ni \pi_Z(\zeta), (B_Z)_{Z/H}, \mathfrak{d}_{Z/H}\big) = \operatorname{mld}((Z \ni \zeta, B_Z, \mathfrak{d})/H_\zeta).$$

(ii) If $B_Z = 0$ and $\operatorname{codim}_Z Z^g \geq 2$ for any $g \in H \backslash \{1\}$, where Z^g is the fixed locus of g, then $(B_Z)_{Z/H} = 0$.

For any prime divisor F over Z and any j, we have

$$\operatorname{mult}_F(\mathfrak{d}_j^{(N)} \mathcal{O}_Z) = \min_k N \operatorname{mult}_F g_{jk} = N \operatorname{mult}_F \mathfrak{d}_j.$$

Thus $a(F, Z, B_Z, \pi_Z^* \mathfrak{d}_{Z/H}) = a(F, Z, B_Z, \mathfrak{d})$, and [Liu26, Lemma 2.6(2)] implies

$$a(F, Z, B_Z, \mathfrak{d}) = |I_H(F)| \cdot a\big(E, Z/H, (B_Z)_{Z/H}, \mathfrak{d}_{Z/H}\big), \tag{3.8}$$

where E is the prime divisor under F, and $\operatorname{center}_{Z/H} E = \pi_Z(\operatorname{center}_Z F)$. By [Liu26, Lemma 2.6(3)] and the uniqueness of E in [Liu26, Lemma 2.6], any prime divisor E over Z/H is

the prime divisor under some prime divisor F over Z. For $g \in H$, the prime divisor F_g over Z with $\mathrm{mult}_{F_g} = \mathrm{mult}_F \circ g^*$ is also a prime divisor over Z lying over E, $\mathrm{center}_Z F_g = g(\mathrm{center}_Z F)$, and $a(F_g, Z, B_Z, \mathfrak{d}) = a(F, Z, B_Z, \mathfrak{d})$, since B_Z and the $\mathfrak{d}_j$ are H-invariant.

We prove (i). Let E be a prime divisor over Z/H with center $\overline{\{\pi_Z(\zeta)\}}$, and let F be a prime divisor over Z lying over E. Since π_Z is finite and the fibers of π_Z are H-orbits, $\mathrm{center}_Z F = \overline{\{g(\zeta)\}}$ for some $g \in H$. Possibly replacing F with $F_{g^{-1}}$, we may assume that $\mathrm{center}_Z F = \overline{\{\zeta\}}$. Conversely, the prime divisor under any prime divisor F over Z with center $\overline{\{\zeta\}}$ has center $\overline{\{\pi_Z(\zeta)\}}$. Any element of $I_H(F)$ preserves mult_F, hence preserves $\mathrm{center}_Z F$, so it fixes ζ. Thus $I_H(F) = I_{H_\zeta}(F)$. Since H_ζ acts faithfully on Z, fixes ζ, and preserves B_Z and the $\mathfrak{d}_j$, (i) follows from (3.8) and Definition 2.4.

We prove (ii). Let P be a prime divisor on Z. By [Liu26, Lemma 2.6(1)(4)], the coefficient of $\pi_Z(P)$ in $(B_Z)_{Z/H}$ is $1 - 1/|I_H(P)|$. Any element of $I_H(P)$ acts trivially on the residue field of mult_P, which is the function field of P, so it fixes P pointwise. Since $\mathrm{codim}_Z Z^g \geq 2$ for any $g \neq 1$, we have $I_H(P) = \{1\}$. Thus $(B_Z)_{Z/H} = 0$.

Finally, for any H-semi-invariant regular function φ on Z and any prime divisor F over Z, [Liu26, Definition 2.2] implies

$$a(F, Z, B_Z + \mathrm{div}(\varphi), \mathfrak{d}) = a(F, Z, B_Z, (\varphi) \cdot \mathfrak{d}).$$

So a principal divisor with coefficient one in the boundary may be moved into the $\mathbb{R}$-ideal as a factor with exponent one, without changing log discrepancies or quotient values.

Step 2. In this step, we assume the conditions of (2), adjoin n coordinates, and use [Liu26, Theorem 2.22] twice.

Let $V^+ := \mathbb{A}^n_{x_1,\dots,x_n} \times \mathbb{A}^n_{t_1,\dots,t_n}$, and let H act on V^+ by the given action on the first factor and by $g^* t_i = \chi_i(g) t_i$ for any $g \in H$ and any i. Let $p : V^+ \to \mathbb{A}^n$ be the first projection, $X^+ := X \times \mathbb{A}^n$, $S^+ := S \times \mathbb{A}^n$, and $0^+ := (0,0) \in V^+$. Since S is integral near 0 and $S = (f_{c+1}|_X = 0)$, we have $\mathrm{div}(f_{c+1}|_X) = S$ near 0. Let

$$\begin{aligned} a_S &:= \mathrm{mld}((S \ni 0, \mathfrak{a}|_S)/H), \\ a_X &:= \mathrm{mld}((X \ni 0, S, \mathfrak{a}|_X)/H), \\ a_V &:= \mathrm{mld}\Big(\Big(\mathbb{A}^n \ni 0, \sum_{i=1}^{c+1} \mathrm{div}(f_i), \mathfrak{a}\Big)/H\Big), \end{aligned}$$

and let a_S^+, a_X^+, and a_V^+ be the numbers a^{c} of Lemma 3.2 for $k = n$, the characters $\chi_1|_H, \dots, \chi_n|_H$, $z = 0$, and $(Z, B, \mathfrak{a})$ equal to $(S, 0, \mathfrak{a}|_S)$, $(X, S, \mathfrak{a}|_X)$, and $(\mathbb{A}^n, \sum_{i=1}^{c+1} \mathrm{div}(f_i), \mathfrak{a})$, respectively. That is,

$$\begin{aligned} a_S^+ &= \mathrm{mld}\Big(\Big(S^+ \ni 0^+, \sum_{i=1}^{n} \mathrm{div}(t_i|_{S^+}), (p^*\mathfrak{a})|_{S^+}\Big)/H\Big), \\ a_X^+ &= \mathrm{mld}\Big(\Big(X^+ \ni 0^+, S^+ + \sum_{i=1}^{n} \mathrm{div}(t_i|_{X^+}), (p^*\mathfrak{a})|_{X^+}\Big)/H\Big), \\ a_V^+ &= \mathrm{mld}\Big(\Big(V^+ \ni 0^+, \sum_{i=1}^{c+1} \mathrm{div}(f_i) + \sum_{i=1}^{n} \mathrm{div}(t_i), p^*\mathfrak{a}\Big)/H\Big). \end{aligned}$$

Quotient values at 0 and 0^+ only depend on H-invariant open neighborhoods of 0 and 0^+, so **Step 1** and Lemma 3.2 are applied to H-invariant affine open neighborhoods of 0 in X and S, and of 0^+ in X^+ and S^+, on which X, S, X^+, and S^+ are normal.

We check the conditions of [Liu26, Theorem 2.22] for the group H acting on V^+, once for S^+ and once for X^+. Let $g \in H \backslash \{1\}$. Since H acts faithfully on S, g acts non-trivially on $\mathbb{A}^n$, so $\chi_i(g) \neq 1$ for some i. Thus the fixed locus $(V^+)^g$ is contained in $(x_i = t_i = 0)$, which is of codimension two in V^+. In particular, H acts faithfully and linearly on V^+ without

pseudoreflections. Let Y be S or X, and $Y^+ := Y \times \mathbb{A}^n$. Since Y is integral near 0 and g acts non-trivially on Y, the fixed locus Y^g is a proper closed subset of Y near 0. Moreover, $(\mathbb{A}^n_{t_1,\dots,t_n})^g \subset (t_i = 0)$. Thus, near 0^+,

$$\operatorname{codim}_{Y^+}(Y^+)^g = \operatorname{codim}_{Y^+}\left(Y^g \times (\mathbb{A}^n_{t_1,\dots,t_n})^g\right) \geq 2. \tag{3.9}$$

The quotient morphism $V^+ \to V^+/H$ is étale at any point with trivial stabilizer, so the locus where it is not étale is contained in $\bigcup_{g\neq 1}(V^+)^g$, whose intersection with Y^+ is $\bigcup_{g\neq 1}(Y^+)^g$. Since the quotient morphism is finite, the image in V^+/H of the locus where $V^+ \to V^+/H$ is not étale intersects Y^+/H in codimension at least two near the image y^+ of 0^+. Near 0^+, S^+ (resp. X^+) is the closed subscheme of V^+ defined by the H-semi-invariant polynomials $f_1, \dots, f_{c+1}$ (resp. $f_1, \dots, f_c$), and it is a normal integral complete intersection of codimension $c+1$ (resp. c). It has klt singularities, since the product of a log resolution of Y with $\mathbb{A}^n$ is a log resolution of Y^+, and the discrepancy of $P \times \mathbb{A}^n$ equals the discrepancy of P for any exceptional prime divisor P of the log resolution of Y. Moreover, $\dim S^+ = 2n - c - 1 \geq n + 1 \geq 2$ and $\dim X^+ = 2n - c \geq 2$. Since $\dim S = n - c - 1$ may be one, [Liu26, Theorem 2.22] is applied to S^+, not to S.

For any j, let $\mathfrak{a}_j^{(N)}$ be the ideal on V^+/H generated by the N-th powers of the G-semi-invariant polynomials generating $\mathfrak{a}_j$, and let

$$\mathfrak{b} := \prod_{i=1}^{n}(t_i^N)^{\frac{1}{N}} \cdot \prod_{j=1}^{s}\left(\mathfrak{a}_j^{(N)}\right)^{\frac{r_j}{N}}.$$

Since $\mathfrak{a}|_S$ is non-zero near 0, the restrictions of $\mathfrak{b}$ to S^+/H and to X^+/H are non-zero near y^+. Since $S \neq X$, the restriction of $(f_{c+1}^N)^{\frac{1}{N}}\mathfrak{b}$ to X^+/H is non-zero near y^+. Thus [Liu26, Theorem 2.22] (see also [NS26b, Theorem 7.1 and Remark 6.5]), applied to S^+ with the $\mathbb{R}$-ideal $\mathfrak{b}$ and to X^+ with the $\mathbb{R}$-ideal $(f_{c+1}^N)^{\frac{1}{N}}\mathfrak{b}$, implies

$$\operatorname{mld}\big(S^+/H \ni y^+, \mathfrak{b}|_{S^+/H}\big) = \operatorname{mld}\big(V^+/H \ni y^+, (f_1^N \cdots f_{c+1}^N)^{\frac{1}{N}}\mathfrak{b}\big),$$

$$\operatorname{mld}\big(X^+/H \ni y^+, ((f_{c+1}^N)^{\frac{1}{N}}\mathfrak{b})|_{X^+/H}\big) = \operatorname{mld}\big(V^+/H \ni y^+, (f_1^N \cdots f_c^N)^{\frac{1}{N}}(f_{c+1}^N)^{\frac{1}{N}}\mathfrak{b}\big).$$

We apply **Step 1** with $\zeta = 0^+$, so that $H_\zeta = H$, and with $B_Z = 0$. Let

$$\mathfrak{d}_S := \prod_{i=1}^{n}(t_i|_{S^+}) \cdot (p^*\mathfrak{a})|_{S^+},$$

$$\mathfrak{d}_X := (f_{c+1}|_{X^+}) \cdot \prod_{i=1}^{n}(t_i|_{X^+}) \cdot (p^*\mathfrak{a})|_{X^+},$$

$$\mathfrak{d}_V := (f_1 \cdots f_{c+1}) \cdot \prod_{i=1}^{n}(t_i) \cdot p^*\mathfrak{a}.$$

For $Z = S^+$ and $\mathfrak{d} = \mathfrak{d}_S$, we have $\mathfrak{d}_{Z/H} = \mathfrak{b}|_{S^+/H}$, and $(B_Z)_{Z/H} = 0$ by **Step 1**(ii), since $B_Z = 0$ and $\operatorname{codim}_{Y^+}(Y^+)^g \geq 2$ for any $g \in H\backslash\{1\}$ by (3.9). By **Step 1**(i) and the last paragraph of **Step 1**, the left side of the first equality is a_S^+. For $Z = X^+$ and $\mathfrak{d} = \mathfrak{d}_X$, we have

$$\mathfrak{d}_{Z/H} = ((f_{c+1}^N)^{\frac{1}{N}}\mathfrak{b})|_{X^+/H},$$

and $(B_Z)_{Z/H} = 0$ by **Step 1**(ii), since $B_Z = 0$ and $\operatorname{codim}_{Y^+}(Y^+)^g \geq 2$ for any $g \in H\backslash\{1\}$ by (3.9), and **Step 1**(i) and the last paragraph of **Step 1** imply that the left side of the second equality is a_X^+, since $\operatorname{div}(f_{c+1}|_{X^+}) = S^+$ near 0^+. For $Z = V^+$, the $\mathbb{R}$-ideals in both right sides are of the form $\mathfrak{d}_{Z/H}$ for $\mathbb{R}$-ideals $\mathfrak{d}$ on V^+ with $\operatorname{mult}_F \mathfrak{d} = \operatorname{mult}_F \mathfrak{d}_V$ for any prime divisor F over V^+, and $(B_Z)_{Z/H} = 0$ by **Step 1**(ii), since, for any $g \in H\backslash\{1\}$ and an index i with $\chi_i(g) \neq 1$,

$(V^+)^g$ is contained in $(x_i = t_i = 0)$, which has codimension two in V^+, so both right sides are equal to a_V^+. Hence

$$a_S^+ = a_V^+ = a_X^+. \tag{3.10}$$

Step 3. In this step, we remove the coordinates $t_1, \dots, t_n$ and prove (2).

Since $(S, \mathfrak{a}|_S)$ is lc near 0, we have $a(F, S, \mathfrak{a}|_S) \geq 0$ for any prime divisor F over S with center 0, so $a_S \geq 0$. By Lemma 3.2(2) for $(S, 0, \mathfrak{a}|_S)$, $a_S^+ = a_S$. By (3.10), $a_X^+ = a_V^+ = a_S \geq 0$. Thus Lemma 3.2(2) for $(X, S, \mathfrak{a}|_X)$ and for $(\mathbb{A}^n, \sum_{i=1}^{c+1} \operatorname{div}(f_i), \mathfrak{a})$, whose numbers a^{c} are a_X^+ and a_V^+, implies $a_X = a_X^+$ and $a_V = a_V^+$. Hence $a_S = a_X = a_V$, which is (2).

Step 4. In this step, we start the proof of (3) and use (1) and (2) at general closed points of the curve.

Let C be an H-invariant integral curve through 0 contained in X (in S for (2)), with generic point ξ. In the setting of (2), $n - c \geq 3$ by assumption, that is, $\dim S \geq 2$.

Let $(Z, B_Z, \mathfrak{a}_Z)$ be one of the triples

$$(X, 0, \mathfrak{a}|_X) \quad \text{and} \quad \Big(\mathbb{A}^n, \sum_{i=1}^{c} \operatorname{div}(f_i), \mathfrak{a}\Big)$$

for (1), and one of the triples

$$(S, 0, \mathfrak{a}|_S), \quad (X, S, \mathfrak{a}|_X), \quad \text{and} \quad \Big(\mathbb{A}^n, \sum_{i=1}^{c+1} \operatorname{div}(f_i), \mathfrak{a}\Big)$$

for (2). Let $Z_0 := X$ for (1) and $Z_0 := S$ for (2), so that $(Z_0, \mathfrak{a}|_{Z_0})$ is lc near ξ by assumption. As in **Step 2**, we work on an H-invariant affine open neighborhood of 0 in Z on which Z is normal, and such a neighborhood contains ξ. With the notation of **Step 1** for $\mathfrak{d} = \mathfrak{a}_Z$, let $\Delta_Z := (B_Z)_{Z/H}$, $\mathfrak{c}_Z := (\mathfrak{a}_Z)_{Z/H}$, and

$$\alpha_Z(\zeta) := \operatorname{mld}\big(Z/H \ni \pi_Z(\zeta), \Delta_Z, \mathfrak{c}_Z\big)$$

for any point $\zeta \in Z$. Since C is H-invariant, $H_\xi = H$, and **Step 1**(i) implies

$$\alpha_Z(\xi) = \operatorname{mld}\big((Z \ni \xi, B_Z, \mathfrak{a}_Z)/H\big).$$

Thus (3) says that $\alpha_Z(\xi)$ does not depend on the choice of $(Z, B_Z, \mathfrak{a}_Z)$ among the triples of (1), resp. (2).

Let $H_1 \subset H$ be the subgroup of the elements which act trivially on C. For any $g \in H \backslash H_1$, the fixed locus of g on C is a proper closed subset of the curve C, hence a finite set. Moreover, X (and S for (2)) is a normal integral complete intersection with klt singularities on an open neighborhood of 0, and $(Z_0, \mathfrak{a}|_{Z_0})$ is lc on an open neighborhood of ξ. Thus there exists a non-empty open subset $C_1 \subset C$ such that, for any closed point $q \in C_1$, $H_q = H_1$, X (and S for (2)) is a normal integral complete intersection with klt singularities near q, and $(Z_0, \mathfrak{a}|_{Z_0})$ is lc near q.

Let $q = (q_1, \dots, q_n) \in C_1$ be a closed point. By **Step 1**(i), $\alpha_Z(q) = \operatorname{mld}((Z \ni q, B_Z, \mathfrak{a}_Z)/H_1)$. Let $y_i := x_i - q_i$ for any i. For $g \in H_1$, we have $g(q) = q$, that is, $\chi_i(g) q_i = q_i$ for any i. Thus $g^* y_i = \chi_i(g) x_i - q_i = \chi_i(g) y_i$ for any i, and H_1 acts diagonally on $\mathbb{A}^n$ in the coordinates $y_1, \dots, y_n$, with $q = (y_1 = \cdots = y_n = 0)$. The polynomials $f_1, \dots, f_{c+1}$ and the generators of the $\mathfrak{a}_j$ are H_1-semi-invariant polynomials in $y_1, \dots, y_n$, and $f_i(q) = 0$ for $i \leq c$ (for $i \leq c+1$ in the setting of (2)), since $q \in C$. Since H acts faithfully on X (on S for (2)), so does H_1. Moreover, H_1 acts faithfully on any open neighborhood of q in X (in S for (2)), and $\mathfrak{a}|_X$ (and $\mathfrak{a}|_S$ for (2)) is non-zero near q, since C is contained in the irreducible component of X (resp. S) through 0, on any non-empty open subset of which each non-identity element of H_1 acts non-trivially and no factor of $\mathfrak{a}$ vanishes identically. Thus the conditions of Theorem 3.3(1) (resp. (2)) hold at q in the coordinates $y_1, \dots, y_n$, with G and H replaced with H_1. By part (1) (resp. part (2), proved

in **Step 3**), the number $\mathrm{mld}((Z \ni q, B_Z, \mathfrak{a}_Z)/H_1)$ does not depend on the choice of $(Z, B_Z, \mathfrak{a}_Z)$ among the triples of (1) (resp. (2)). Since $(Z_0, \mathfrak{a}|_{Z_0})$ is lc near q, we have $\alpha_{Z_0}(q) \geq 0$. Hence

$$\alpha_Z(q) = \alpha_{Z_0}(q) \geq 0 \tag{3.11}$$

for any triple $(Z, B_Z, \mathfrak{a}_Z)$ and any closed point $q \in C_1$.

Step 5. In this step, we pass from the closed points of C to ξ and prove (3).

Let $(Z, B_Z, \mathfrak{a}_Z)$ be one of the triples of **Step 4**. Then $C_Z := \pi_Z(C)$ is an integral curve on Z/H with generic point $\pi_Z(\xi)$, and $\dim Z/H = \dim Z \geq 2$, since $n - c \geq 2$, $n - c \geq 3$ in the setting of (2), and $n \geq 3$. Moreover, $K_{Z/H} + \Delta_Z$ is $\mathbb{R}$-Cartier [Liu26, Lemma 2.6(0)].

We show that $(Z/H, \Delta_Z, \mathfrak{c}_Z)$ is lc near $\pi_Z(\xi)$. Suppose that $(Z/H, \Delta_Z, \mathfrak{c}_Z)$ is not lc near $\pi_Z(\xi)$. The set of points of Z/H near which $(Z/H, \Delta_Z, \mathfrak{c}_Z)$ is lc is open and does not contain $\pi_Z(\xi)$, so it does not intersect C_Z. Since $\dim Z/H \geq 2$, [KM98, Corollary 2.31(1)] and its local form at a closed point [Amb99, Section 1, the second paragraph after Definition 1.1], applied to the sub-pair induced on a log resolution of $(Z/H, \Delta_Z)$ which principalizes the factors of $\mathfrak{c}_Z$, imply $\alpha_Z(q) = -\infty$ for any closed point $q \in C$. This contradicts (3.11).

By Lemma 3.4 for $(Z/H, \Delta_Z, \mathfrak{c}_Z)$ and C_Z, there exists a non-empty open subset $C_Z^\circ \subset C_Z$ such that $\alpha_Z(q) = \alpha_Z(\xi) + 1$ for any closed point $q \in C$ with $\pi_Z(q) \in C_Z^\circ$. The morphism $C \to C_Z$ induced by π_Z is finite and surjective, and there are at most three triples. Thus there exists a closed point $q \in C_1$ such that $\pi_Z(q) \in C_Z^\circ$ for any triple $(Z, B_Z, \mathfrak{a}_Z)$. By (3.11),

$$\alpha_Z(\xi) = \alpha_Z(q) - 1 = \alpha_{Z_0}(q) - 1 = \alpha_{Z_0}(\xi)$$

for any triple $(Z, B_Z, \mathfrak{a}_Z)$. This proves (3). □

Remark 3.5. Theorem 3.3 covers the uses of quotient adjunction in Sections 5–11 and Appendices A–E. There, a coefficient-one section enters as a principal ideal factor (f) with exponent $r_j = 1$, also on singular klt charts such as $(UV + Z^2 + v^d = 0) \subset \mathbb{A}^4$. The $\mathbb{R}$-ideals there may contain real powers of ideals generated by monomials, as in the proofs of Propositions A.14 and A.35. Marks (Definition 2.7) are regular semi-invariant functions on neighborhoods of a closed point. By Theorem 3.10, the later sections replace marks with the H-semi-invariant polynomials $h_{k,0}$ and f_0 of that theorem, and use Theorem 3.3 for $h_{k,0}$ and f_0.

3.2. **Completions and normalized marks.** In this subsection, we show that quotient values of pairs with an $\mathbb{R}$-ideal can be computed on the completion, we lift equivariant isomorphisms of completed hypersurface germs to the ambient power series rings, and we replace marks in a normal form, which are formal power series, by polynomials without changing the quotient values (Theorem 3.10).

Lemma 3.6. *Let $(X \ni x)$ be a normal klt germ, H a finite group acting faithfully on X and fixing x, B an H-invariant $\mathbb{R}$-divisor with $K_X + B$ $\mathbb{R}$-Cartier near x, and $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ an $\mathbb{R}$-ideal on X whose factors $\mathfrak{a}_j$ are H-invariant ideals. Let $\widehat{X} := \operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$, $\hat{x}$ the closed point of $\widehat{X}$, and $\widehat{B}$, $\widehat{\mathfrak{a}}$ the pullbacks. Then*

$$\mathrm{mld}((X \ni x, B, \mathfrak{a})/H) = \mathrm{mld}((\widehat{X} \ni \hat{x}, \widehat{B}, \widehat{\mathfrak{a}})/H).$$

Moreover, any formal prime divisor over $\widehat{X}$ with center $\hat{x}$ restricts to a prime divisor over X with center x with isomorphic residue field, equal log discrepancy with respect to $(B, \mathfrak{a})$ and $(\widehat{B}, \widehat{\mathfrak{a}})$, and equal decomposition and inertia groups, and any prime divisor over X with center x is the restriction of a formal prime divisor over $\widehat{X}$ with center $\hat{x}$.

Proof. Let P be a formal prime divisor over $\widehat{X}$ with center $\hat{x}$. By [Liu26, Lemma 3.3(3)], P restricts to a prime divisor E over X with center x such that the residue fields of P and E are isomorphic,

$$a(P, \widehat{X}, \widehat{B}) = a(E, X, B), \quad \mathrm{Stab}_H(P) = \mathrm{Stab}_H(E), \quad \text{and} \quad I_H(P) = I_H(E),$$

and any prime divisor over X with center x arises in this way. Since P restricts to E, we have $\operatorname{mult}_P g = \operatorname{mult}_E g$ for any $g \in \mathcal{O}_{X,x}$.

Fix j, and let $g_1, \dots, g_m \in \mathcal{O}_{X,x}$ be generators of the stalk of $\mathfrak{a}_j$ at x. Then $g_1, \dots, g_m$ generate $\widehat{\mathfrak{a}}_j$. Since the center of E on X is x, we have

$$\operatorname{mult}_P \widehat{\mathfrak{a}}_j = \min_{1\le i\le m} \operatorname{mult}_P g_i = \min_{1\le i\le m} \operatorname{mult}_E g_i = \operatorname{mult}_E \mathfrak{a}_j.$$

Thus, by the definitions of log discrepancies with respect to $\mathbb{R}$-ideals on $\widehat{X}$ (Definition 2.4) and on X [Liu26, Definition 2.2],

$$a(P, \widehat{X}, \widehat{B}, \widehat{\mathfrak{a}}) = a(P, \widehat{X}, \widehat{B}) - \sum_j r_j \operatorname{mult}_P \widehat{\mathfrak{a}}_j = a(E, X, B) - \sum_j r_j \operatorname{mult}_E \mathfrak{a}_j = a(E, X, B, \mathfrak{a}).$$

This proves the second part of the lemma.

Hence any number $a(P, \widehat{X}, \widehat{B}, \widehat{\mathfrak{a}})/|I_H(P)|$, where P is a formal prime divisor over $\widehat{X}$ with center $\hat{x}$, is of the form $a(E, X, B, \mathfrak{a})/|I_H(E)|$ for a prime divisor E over X with center x, and conversely. By Definition 2.4, both sides of the equality in the lemma equal the infimum of the numbers $a(E, X, B, \mathfrak{a})/|I_H(E)|$, where E runs through the prime divisors over X with center x. □

The next lemma is the formal version of [Liu26, Lemma 2.6] for boundaries which are not pulled back from the variety, such as the limit boundaries of Construction-Proposition 4.1.

Lemma 3.7. *Let $k \supset \mathbb{C}$ be a field, $(X \ni x)$ a germ of a normal variety over k at a closed point x, and H a finite group acting faithfully on X and fixing x. Let $\widehat{X} := \operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$ with the induced action of H, $\widehat{X}/H := \operatorname{Spec} \widehat{\mathcal{O}}^H_{X,x}$, which is the completion of X/H at the image of x* [Liu26, Lemma 3.3(1)]*, and $\pi : \widehat{X} \to \widehat{X}/H$ the quotient morphism. Let $\widehat{B}$ be an H-invariant $\mathbb{R}$-divisor on $\widehat{X}$ such that $K_{\widehat{X}} + \widehat{B}$ is $\mathbb{R}$-Cartier, and*

$$\widehat{B}_{\widehat{X}/H} := \frac{1}{|H|}\pi_*(\widehat{B} + R(\pi)),$$

where $R(\pi)$ is the ramification divisor of π. Then

$$K_{\widehat{X}} + \widehat{B} = \pi^*(K_{\widehat{X}/H} + \widehat{B}_{\widehat{X}/H}).$$

Moreover, for any formal prime divisor $\tilde{P}$ over $\widehat{X}$ [Liu26, Definition 3.2]*, there exist a unique formal prime divisor P over $\widehat{X}/H$ and a unique positive integer e such that $\operatorname{mult}_{\tilde{P}} g = e \operatorname{mult}_P g$ for any non-zero g in the fraction field of $\widehat{\mathcal{O}}^H_{X,x}$, and the following hold.*

(1) $e = |I_H(\tilde{P})|$, the group $I_H(\tilde{P})$ is cyclic, and $I_H(\tilde{P})$ acts faithfully on $\mathfrak{m}_{\tilde{P}}/\mathfrak{m}^2_{\tilde{P}}$, where $\mathfrak{m}_{\tilde{P}}$ is the maximal ideal of the valuation ring of $\operatorname{mult}_{\tilde{P}}$.

(2) $\operatorname{center}_{\widehat{X}/H} P = \pi(\operatorname{center}_{\widehat{X}} \tilde{P})$ and

$$a(\tilde{P}, \widehat{X}, \widehat{B}) = |I_H(\tilde{P})| \cdot a(P, \widehat{X}/H, \widehat{B}_{\widehat{X}/H}).$$

Moreover, let $\mathfrak{b} = \prod_j \mathfrak{b}_j^{r_j}$ be an $\mathbb{R}$-ideal on $\widehat{X}/H$, and

$$\pi^*\mathfrak{b} := \prod_j (\mathfrak{b}_j \mathcal{O}_{\widehat{X}})^{r_j}.$$

Then

$$a(\tilde{P}, \widehat{X}, \widehat{B}, \pi^*\mathfrak{b}) = |I_H(\tilde{P})| \cdot a(P, \widehat{X}/H, \widehat{B}_{\widehat{X}/H}, \mathfrak{b}).$$

(3) For any formal prime divisor P' over $\widehat{X}/H$, there exist a formal prime divisor F' over $\widehat{X}$ and a positive integer e' such that $\operatorname{mult}_{F'} g = e' \operatorname{mult}_{P'} g$ for any non-zero g in the fraction field of $\widehat{\mathcal{O}}^H_{X,x}$.

(4) *If Q is a prime divisor on $\widehat{X}$ with $\operatorname{mult}_Q \widehat{B} = b$ and ramification index e_Q over $\pi(Q)$, then $\operatorname{mult}_{\pi(Q)} \widehat{B}_{\widehat{X}/H} = 1 - \frac{1-b}{e_Q}$.*

In particular, if $\widehat{B} = \sum_j c_j \operatorname{div}(\varphi_j)$ with real numbers c_j and non-zero H-semi-invariant elements $\varphi_j \in \widehat{\mathcal{O}}_{X,x}$ whose characters have orders e_j, then

$$\widehat{B}_{\widehat{X}/H} = \widehat{B}_{\mathrm{br}} + \sum_j \frac{c_j}{e_j} \operatorname{div}(\varphi_j^{e_j}),$$

where $\pi_X : X \to X/H$ is the quotient morphism and $\widehat{B}_{\mathrm{br}}$ is the pullback to $\widehat{X}/H$ of $\frac{1}{|H|}\pi_{X*}R(\pi_X)$, which is the boundary of $(X,0)/H$ when K_X is $\mathbb{R}$-Cartier near x.

Proof. Possibly shrinking X to an H-invariant affine neighborhood of x, we may assume that X is affine. Let $\mathcal{O} := \mathcal{O}_{X,x}$, $\mathcal{O}'$ the local ring of X/H at the image of x, and $\pi_X : X \to X/H$ the quotient morphism. We denote the completions of $\mathcal{O}$ and $\mathcal{O}'$ by $\widehat{\mathcal{O}}$ and $\widehat{\mathcal{O}'}$. Since X and X/H are normal, $\widehat{\mathcal{O}}$ and $\widehat{\mathcal{O}'} = \widehat{\mathcal{O}}^H$ are normal [dFEM11, Remark 2.10], and they are excellent, since they are complete Noetherian local rings. Since $\mathcal{O}$ is finite over $\mathcal{O}'$,

$$\widehat{\mathcal{O}} = \mathcal{O} \otimes_{\mathcal{O}'} \widehat{\mathcal{O}'}$$

[Mat89, Theorem 8.7], so π is finite and is the base change of π_X. Since $\mathcal{O} \subset \widehat{\mathcal{O}}$ and H acts faithfully on X, H acts faithfully on $\widehat{\mathcal{O}}$, so the extension of the fraction fields of $\widehat{\mathcal{O}}$ and $\widehat{\mathcal{O}}^H$ is Galois with group H [AM69, Chapter 5, Exercise 12]. The canonical divisors $K_{\widehat{X}}$ and $K_{\widehat{X}/H}$ are those of [dFEM11, Appendix A].

The first equality and (4). Since π is a finite tame morphism of normal excellent schemes, the Hurwitz formula gives $K_{\widehat{X}} = \pi^* K_{\widehat{X}/H} + R(\pi)$ [Kol13, 2.41]. Let Q be a prime divisor on $\widehat{X}$ with $\operatorname{mult}_Q \widehat{B} = b$ and ramification index e_Q over $\pi(Q)$. The prime divisors on $\widehat{X}$ over $\pi(Q)$ form one H-orbit, and their number m and the degree f of $Q \to \pi(Q)$ satisfy $e_Q f m = |H|$ [Ser79, Chapter I, §7, Proposition 19 and its Corollary]. Since $\operatorname{mult}_Q R(\pi) = e_Q - 1$,

$$\operatorname{mult}_{\pi(Q)} \widehat{B}_{\widehat{X}/H} = \frac{mf(e_Q - 1 + b)}{|H|} = 1 - \frac{1-b}{e_Q},$$

so

$$\operatorname{mult}_Q \pi^* \widehat{B}_{\widehat{X}/H} = e_Q - 1 + b = \operatorname{mult}_Q(R(\pi) + \widehat{B}).$$

Thus $\pi^* \widehat{B}_{\widehat{X}/H} = \widehat{B} + R(\pi)$, and the Hurwitz formula gives the first equality of the lemma. This proves (4). Moreover, $\pi_* \pi^* D = |H| D$ for any $\mathbb{R}$-divisor D on $\widehat{X}/H$, so

$$|H|(K_{\widehat{X}/H} + \widehat{B}_{\widehat{X}/H}) = \pi_*(K_{\widehat{X}} + \widehat{B})$$

is $\mathbb{R}$-Cartier, as the norm of a Cartier divisor is Cartier [Kol13, 2.40], so the log discrepancies with respect to $(\widehat{X}/H, \widehat{B}_{\widehat{X}/H})$ in (2) are defined.

Existence and uniqueness of P and e, and (1). Let $\tilde{P}$ be a formal prime divisor over $\widehat{X}$, a prime divisor on a normal scheme W' which is proper and birational over $\widehat{X}$. Since W' is normal at the generic point of $\tilde{P}$, $\tilde{P}$ remains a prime divisor on any normal scheme proper and birational over W'. Thus, by Chow's lemma [Har77, Exercise II.4.10] and normalization, which is finite since $\widehat{X}$ is excellent, we may assume that W' is projective over $\widehat{X}$. Let W be the normalization of the closure of the image of the generic point of $\widehat{X}$ in the fiber product over $\widehat{X}$ of the translates $\sigma(W')$, $\sigma \in H$. Then W is normal, projective and birational over $\widehat{X}$, and H-equivariant, and $W \to W'$ is an isomorphism over the generic point of $\tilde{P}$, so $\tilde{P}$ is a prime divisor on W. The quotient W/H exists since W is projective over the affine scheme $\widehat{X}$, it is normal, and $W/H \to \widehat{X}/H$ is proper and birational. Let $p : W \to W/H$ be the quotient morphism. Then $P := p(\tilde{P})$ is a formal prime divisor over $\widehat{X}/H$, and $\operatorname{mult}_{\tilde{P}}(p^* g) = e \operatorname{mult}_P g$ for any non-zero g in the fraction field of

$\widehat{\mathcal{O}}^H$, where e is the ramification index of p along $\tilde{P}$. In particular, P and e are determined by $\tilde{P}$. Since the local ring of W/H at P is a discrete valuation ring whose integral closure in the fraction field of $\widehat{\mathcal{O}}$ is the semi-local ring of W at the H-orbit of $\tilde{P}$, and the residue fields have characteristic zero, $e = |I_H(\tilde{P})|$, $I_H(\tilde{P})$ is cyclic, and $I_H(\tilde{P})$ acts faithfully on $\mathfrak{m}_{\tilde{P}}/\mathfrak{m}_{\tilde{P}}^2$ [Ser79, Chapter I, §7, Propositions 19–21 and the Corollary to Proposition 21, Chapter IV, §1, Remark 2, and Chapter IV, §2, Proposition 7 and Corollary 2]. This proves the existence and uniqueness of P and e, and (1).

Proof of (2). Let $h : W \to \widehat{X}$ and $g : W/H \to \widehat{X}/H$ be the induced morphisms, and write

$$K_W + B_W := h^*(K_{\widehat{X}} + \widehat{B}) \quad \text{and} \quad K_{W/H} + B_{W/H} := g^*(K_{\widehat{X}/H} + \widehat{B}_{\widehat{X}/H}).$$

By the first equality of the lemma, $K_W + B_W = p^*(K_{W/H} + B_{W/H})$, and by the Hurwitz formula [Kol13, 2.41], $K_W = p^*K_{W/H} + (e-1)\tilde{P}$ near the generic point of $\tilde{P}$. Comparing the coefficients of $\tilde{P}$, we obtain $1 - \operatorname{mult}_{\tilde{P}} B_W = e(1 - \operatorname{mult}_P B_{W/H})$, that is,

$$a(\tilde{P}, \widehat{X}, \widehat{B}) = |I_H(\tilde{P})| \cdot a(P, \widehat{X}/H, \widehat{B}_{\widehat{X}/H}), \quad \text{and} \quad \operatorname{center}_{\widehat{X}/H} P = g(p(\tilde{P})) = \pi(\operatorname{center}_{\widehat{X}} \tilde{P}).$$

For an $\mathbb{R}$-ideal $\mathfrak{b} = \prod_j \mathfrak{b}_j^{r_j}$ on $\widehat{X}/H$,

$$\operatorname{mult}_{\tilde{P}}(\mathfrak{b}_j \mathcal{O}_{\widehat{X}}) = e \operatorname{mult}_P \mathfrak{b}_j,$$

so

$$\begin{aligned} a(\tilde{P}, \widehat{X}, \widehat{B}, \pi^*\mathfrak{b}) &= a(\tilde{P}, \widehat{X}, \widehat{B}) - \sum_j r_j \operatorname{mult}_{\tilde{P}}(\mathfrak{b}_j \mathcal{O}_{\widehat{X}}) \\ &= e \cdot a(P, \widehat{X}/H, \widehat{B}_{\widehat{X}/H}) - e \sum_j r_j \operatorname{mult}_P \mathfrak{b}_j \\ &= |I_H(\tilde{P})| \cdot a(P, \widehat{X}/H, \widehat{B}_{\widehat{X}/H}, \mathfrak{b}). \end{aligned}$$

This proves (2).

Proof of (3). Let P' be a prime divisor on a normal scheme Y' proper and birational over $\widehat{X}/H$. The normalization X' of Y' in the fraction field of $\widehat{\mathcal{O}}$ is finite over Y', and proper and birational over $\widehat{X}$, since $\widehat{X}$ is the normalization of $\widehat{X}/H$ in the fraction field of $\widehat{\mathcal{O}}$. An irreducible component F' of the preimage of P' in X' which dominates P' is a formal prime divisor over $\widehat{X}$, and $\operatorname{mult}_{F'}(\pi^*g) = e' \operatorname{mult}_{P'} g$ for a positive integer e' and any non-zero g in the fraction field of $\widehat{\mathcal{O}}^H$. This proves (3).

The last formula. Let $\widehat{B} = \sum_j c_j \operatorname{div}(\varphi_j)$ as in the lemma. Since the character of φ_j has order e_j, $\varphi_j^{e_j} \in \widehat{\mathcal{O}}^H$, and

$$\pi^* \operatorname{div}(\varphi_j^{e_j}) = e_j \operatorname{div}(\varphi_j).$$

Since $\pi_*\pi^*D = |H|D$,

$$\frac{1}{|H|}\pi_* \operatorname{div}(\varphi_j) = \frac{1}{e_j} \operatorname{div}(\varphi_j^{e_j}).$$

Since π is the flat base change of π_X, and the canonical divisors of $\widehat{X}$ and $\widehat{X}/H$ are the pullbacks of those of X and X/H [dFEM11, Proposition A.14], the Hurwitz formula shows that $R(\pi)$ is the pullback of $R(\pi_X)$, and $\frac{1}{|H|}\pi_* R(\pi)$ is the pullback of $\frac{1}{|H|}\pi_{X*} R(\pi_X)$. Thus

$$\widehat{B}_{\widehat{X}/H} = \widehat{B}_{\mathrm{br}} + \sum_j \frac{c_j}{e_j} \operatorname{div}(\varphi_j^{e_j}). \qquad \square$$

Proposition 3.15 connects two hypersurface germs by G-equivariant morphisms from a third germ which are étale at the marked points, and thus induces a G-equivariant isomorphism between the completed local rings of the hypersurface germs. The following lemma turns it into

an equivariant isomorphism of the ambient power series rings and an invariant unit, which are the ψ and the v of Theorem 3.10 and Construction-Theorem 4.13.

Lemma 3.8. *Let G be a finite abelian group acting diagonally on $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_n]]$ and on $\mathfrak{R}' := \mathbb{C}[[y_1, \dots, y_n]]$, let $f \in (x_1, \dots, x_n)^2$ and $f' \in (y_1, \dots, y_n)^2$ be G-semi-invariant power series, and let $\phi : \mathfrak{R}/(f) \to \mathfrak{R}'/(f')$ be a G-equivariant isomorphism of $\mathbb{C}$-algebras. Then there exist a G-equivariant isomorphism $\Phi : \mathfrak{R} \to \mathfrak{R}'$ which induces ϕ and a G-invariant unit v of $\mathfrak{R}'$ such that $\Phi(f) = vf'$. In particular, the characters of f and f' are equal, and the characters of $x_1, \dots, x_n$ are those of $y_1, \dots, y_n$ up to order.*

Proof. Let $\mathfrak{m} := (x_1, \dots, x_n)$ and $\mathfrak{m}' := (y_1, \dots, y_n)$ be the maximal ideals, $q : \mathfrak{R} \to \mathfrak{R}/(f)$ and $q' : \mathfrak{R}' \to \mathfrak{R}'/(f')$ the quotient maps, and χ_i the character of x_i. Since f and f' are semi-invariant, the ideals (f) and (f') are G-invariant, so q and q' are G-equivariant. For a character χ of G and an element g of $\mathfrak{R}'$ or of $\mathfrak{R}'/(f')$, let

$$p_\chi(g) := \frac{1}{|G|} \sum_{\sigma \in G} \chi(\sigma)^{-1} \sigma(g).$$

Then $p_\chi(g)$ is semi-invariant with character χ, $p_\chi(g) = g$ if g is semi-invariant with character χ, $p_\chi(\mathfrak{m}') \subset \mathfrak{m}'$, and $q' \circ p_\chi = p_\chi \circ q'$.

Since ϕ is an isomorphism of local rings, $\phi(q(x_i))$ belongs to the maximal ideal $q'(\mathfrak{m}')$ of $\mathfrak{R}'/(f')$, and $\phi(q(x_i))$ is semi-invariant with character χ_i. Pick $g_i' \in \mathfrak{m}'$ with $q'(g_i') = \phi(q(x_i))$, and let $g_i := p_{\chi_i}(g_i')$. Then $g_i \in \mathfrak{m}'$ is semi-invariant with character χ_i, and

$$q'(g_i) = p_{\chi_i}(q'(g_i')) = p_{\chi_i}(\phi(q(x_i))) = \phi(q(x_i)).$$

Let $\Phi : \mathfrak{R} \to \mathfrak{R}'$ be the continuous homomorphism of $\mathbb{C}$-algebras with $\Phi(x_i) = g_i$ for any i. For $\sigma \in G$, the continuous homomorphisms $\sigma \circ \Phi$ and $\Phi \circ \sigma$ coincide, since

$$\sigma(\Phi(x_i)) = \chi_i(\sigma) g_i = \Phi(\sigma(x_i))$$

for any i. Thus Φ is G-equivariant. Since ϕ is an isomorphism of local rings, it is continuous, so $q' \circ \Phi$ and $\phi \circ q$ are continuous homomorphisms which coincide at $x_1, \dots, x_n$. Thus $q' \circ \Phi = \phi \circ q$, that is, Φ induces ϕ.

Since $f \in \mathfrak{m}^2$ and $f' \in \mathfrak{m}'^2$, the maps q and q' induce isomorphisms of the cotangent spaces $\mathfrak{m}/\mathfrak{m}^2$ and $\mathfrak{m}'/\mathfrak{m}'^2$ with the cotangent spaces of $\mathfrak{R}/(f)$ and $\mathfrak{R}'/(f')$. Since ϕ induces an isomorphism between the cotangent spaces of $\mathfrak{R}/(f)$ and $\mathfrak{R}'/(f')$, the classes of $g_1, \dots, g_n$ form a basis of $\mathfrak{m}'/\mathfrak{m}'^2$. Hence Φ is an isomorphism by the formal inverse function theorem [GLS07, Chapter I, Theorem 1.21] (with the trivial valuation on $\mathbb{C}$, the convergent power series ring there is $\mathbb{C}[[x_1, \dots, x_n]]$).

Since ϕ is injective and $q' \circ \Phi = \phi \circ q$, we have

$$\Phi^{-1}((f')) = \mathrm{Ker}(q' \circ \Phi) = \mathrm{Ker}(\phi \circ q) = \mathrm{Ker}(q) = (f).$$

Thus $(\Phi(f)) = (f')$. If $f = 0$, then $f' = 0$, and $v := 1$ satisfies $\Phi(f) = vf'$. Assume that $f \neq 0$. Since $\mathfrak{R}'$ is a domain, $\Phi(f) = vf'$ for a unit v of $\mathfrak{R}'$. Let χ and χ' be the characters of f and f'. For $\sigma \in G$,

$$\chi'(\sigma)\sigma(v)f' = \sigma(vf') = \sigma(\Phi(f)) = \Phi(\sigma(f)) = \chi(\sigma)vf',$$

so $\sigma(v) = \chi(\sigma)\chi'(\sigma)^{-1}v$. Since G acts diagonally, σ preserves the constant term of v, which is non-zero. Thus $\chi(\sigma) = \chi'(\sigma)$ and $\sigma(v) = v$. Therefore v is G-invariant, and $\chi = \chi'$.

Finally, Φ induces a G-equivariant isomorphism

$$\mathfrak{m}/\mathfrak{m}^2 \to \mathfrak{m}'/\mathfrak{m}'^2.$$

The classes of $x_1, \dots, x_n$ and of $y_1, \dots, y_n$ are eigenbases of the representations of the abelian group G on $\mathfrak{m}/\mathfrak{m}^2$ and on $\mathfrak{m}'/\mathfrak{m}'^2$, so the characters of $x_1, \dots, x_n$ are those of $y_1, \dots, y_n$ up to order. □

Normal forms of semi-invariant power series, which use equivariant formal changes of coordinates, the Weierstrass preparation theorem, and division by invariant units, exist only in the completion, whereas the arguments of the later sections, after a numerical admission (Construction-Theorem 4.13), take place on polynomials on a fixed $\mathbb{A}^n$. [Liu26, Construction-Theorem 7.2] replaces each mark by a degree truncation of a formal transform of it, and Theorem 3.10 allows any polynomial congruent to a mark in a normal form, while keeping the quotient values for all subgroups and the data of finitely many prime divisors. The equation f_0 is controlled by the equivariant contact determinacy of [Liu26, Lemma 7.1], which is an equivariant form of the finite contact determinacy of [GP19, Theorem 4.8].

Set-up 3.9. We recall the data of [Liu26, Construction-Theorem 7.2]. There H is a finite abelian group, $n \in \{3,4\}$, and the following hold: (i) $(X \ni x)$ is a smooth threefold germ or an isolated Gorenstein terminal threefold germ, and H acts faithfully on X and fixes x; (ii) X has an H-equivariant closed embedding into a smooth germ $(Y \ni x)$ of dimension n, with $Y = X$ if $n = 3$ and $X = (f = 0)$ for a regular H-semi-invariant function f if $n = 4$, and $x_1, \dots, x_n$ are H-semi-invariant regular parameters of Y at x; (iii) the marks $h_1, \dots, h_s$ are regular H-semi-invariant functions on Y, non-zero on X, their labels $\beta_1, \dots, \beta_s$ are positive real numbers, and (X, B) is klt near x for $B := \sum_k \beta_k \operatorname{div}(h_k|_X)$; (iv) $\beta^{(1)}, \dots, \beta^{(r)} \in \mathbb{R}^s_{\geq 0}$ are finitely many vectors such that $(X, \sum_k \beta_k^{(j)} \operatorname{div}(h_k|_X))$ is klt near x for any j; (v) $P_1, \dots, P_l$ are finitely many prime divisors over X with center x. The power series $\hat{h}_k$, and $\hat{f}$ if $n = 4$, are the expansions of h_k and f in $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_n]]$, that is, their images under $\mathcal{O}_{Y,x} \to \mathfrak{R}$. For a germ $(X_0 \ni 0)$ with semi-invariant polynomials $h_{k,0}$ and $B_0 := \sum_k \beta_k \operatorname{div}(h_{k,0}|_{X_0})$, the conclusions are: (1) (X_0, B_0) is klt near 0, H acts faithfully on X_0, and $\operatorname{mld}((X_0 \ni 0, B_0)/H') = \operatorname{mld}((X \ni x, B)/H')$ for any subgroup $H' \subset H$; (2) there exists an H-equivariant isomorphism between the completions of the local rings of X at x and of X_0 at 0, under which each P_i corresponds to a prime divisor $P_{i,0}$ over X_0 with center 0 having isomorphic residue field and equal decomposition group, inertia group, log discrepancy $a(\cdot, \cdot, 0)$, and orders of h_k and $h_{k,0}$; (3) $(X_0, \sum_k \beta_k^{(j)} \operatorname{div}(h_{k,0}|_{X_0}))$ is klt near 0 for any j.

Theorem 3.10. *Notation and conditions as in* [Liu26, Construction-Theorem 7.2]*, with* $\mathfrak{R} = \mathbb{C}[[x_1, \dots, x_n]]$*, and let* H *act diagonally on* $\mathbb{A}^n$ *so that the coordinates have the characters of the* x_i*. Let* ψ *be an* H*-equivariant automorphism of* $\mathfrak{R}$*, and let* $v, u_1, \dots, u_s$ *be* H*-invariant units of* $\mathfrak{R}$*. For* H*-semi-invariant polynomials* $h_{k,0}$ *with the characters of the* h_k *and, if* $n = 4$*, an* H*-semi-invariant polynomial* f_0 *with the character of* f*, let* $X_0 := \mathbb{A}^3$ *if* $n = 3$*,* $X_0 := (f_0 = 0) \subset \mathbb{A}^4$ *if* $n = 4$*, and* $B_0 := \sum_k \beta_k \operatorname{div}(h_{k,0}|_{X_0})$*. Then there exists a positive integer* N_1 *such that for any* $N \geq N_1$ *there exists* L_1 *such that for any* $L \geq L_1$ *and any such polynomials with*

$$h_{k,0} \equiv u_k \psi(\hat{h}_k) \pmod{(x_1, \dots, x_n)^N} \quad \text{and} \quad f_0 \equiv v\psi(\hat{f}) \pmod{(x_1, \dots, x_n)^L},$$

the action of H *on* $\mathbb{A}^n$ *is faithful,* X_0 *is an isolated Gorenstein terminal germ at* 0 *if* $n = 4$*, each* $h_{k,0}|_{X_0}$ *is non-zero, and the conclusions (1), (2) and (3) of* [Liu26, Construction-Theorem 7.2] *hold for* X_0 *and* B_0*. In (2), each* P_i *corresponds to a prime divisor* $P_{i,0}$ *over* X_0 *with center* 0*, and* $\operatorname{mult}_{P_{i,0}} h_{k,0} = \operatorname{mult}_{P_i} h_k$ *for any* k*.*

Proof of Theorem 3.10. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix the notation and choose N_1.

Let $\mathfrak{m} := (x_1, \dots, x_n) \subset \mathfrak{R}$, and $\partial_i := \partial/\partial x_i$. Let $\widehat{\mathcal{O}} := \mathfrak{R}/(\hat{f})$ if $n = 4$ and $\widehat{\mathcal{O}} := \mathfrak{R}$ if $n = 3$. Then $\widehat{\mathcal{O}}$ is the completion of $\mathcal{O}_{X,x}$ with the induced action of H, and, since X is normal near x, $\widehat{\mathcal{O}}$ is normal [dFEM11, Remark 2.10], hence a domain, as a normal local ring is a domain. Let $\widehat{X} := \operatorname{Spec} \widehat{\mathcal{O}}$, $\hat{x}$ the closed point of $\widehat{X}$, and $\widehat{\mathfrak{m}}$ the maximal ideal of $\widehat{\mathcal{O}}$. We denote the image of $\hat{h}_k$ in $\widehat{\mathcal{O}}$ again by $\hat{h}_k$. Since $h_k|_X \neq 0$ and $\mathcal{O}_{X,x} \to \widehat{\mathcal{O}}$ is injective, $\hat{h}_k \neq 0$. The pullback of B to $\widehat{X}$ is

$$\widehat{B} = \sum_k \beta_k \operatorname{div}(\hat{h}_k).$$

For a formal prime divisor P over $\widehat{X}$ with center $\hat{x}$, we write $a(P,\widehat{X})$ for the log discrepancy of P with respect to $\widehat{X}$ [Liu26, Definition 3.2]. Since $\widehat{X}$ is Gorenstein, for non-zero elements $e_1,\dots,e_s$ of $\widehat{\mathcal{O}}$ the log discrepancy of P with respect to $(\widehat{X},\sum_k \beta_k \operatorname{div}(e_k))$ is

$$a\Big(P,\widehat{X},\sum_k \beta_k \operatorname{div}(e_k)\Big) = a(P,\widehat{X}) - \sum_k \beta_k \operatorname{mult}_P e_k.$$

If $n=4$, then X has an isolated singularity at x, so, by the Jacobian criterion [Har77, Theorem I.5.1] (cf. [Mat89, Theorem 14.2] and [Har77, Proposition II.8.7]), the ideal $(\hat{f},\partial_1\hat{f},\dots,\partial_4\hat{f})$ contains a power $\mathfrak{m}^p$ of $\mathfrak{m}$. Thus $b := p+1$ satisfies

$$\mathfrak{m}^b \subset T(\hat{f}), \quad \text{where} \quad T(g) := (g) + \mathfrak{m}(\partial_1 g,\dots,\partial_4 g) \text{ for } g \in \mathfrak{R}. \tag{$*$}$$

If $n=3$, let $b:=1$.

Let $\beta^{(0)} := (\beta_1,\dots,\beta_s)$. For $0 \le j \le r$, let K_j be the set of indices k with $\beta_k^{(j)} > 0$ and $\hat{h}_k \in \widehat{\mathfrak{m}}$, and let

$$\widehat{\mathfrak{c}}_j := \prod_{k\in K_j} (\hat{h}_k)^{\beta_k^{(j)}}.$$

By (iii) and (iv) of [Liu26, Construction-Theorem 7.2], $(X,\sum_k \beta_k^{(j)} \operatorname{div}(h_k|_X))$ is klt near x. The marks h_k with $k \notin K_j$ and $\beta_k^{(j)} > 0$ are units near x. Thus, by [dFEM11, Proposition 2.11], $(\widehat{X},\widehat{\mathfrak{c}}_j)$ is klt, that is, $\operatorname{lct}(\widehat{X};\widehat{\mathfrak{c}}_j) > 1$ if $K_j \neq \emptyset$. Fix a real number $\delta > 0$ such that $\operatorname{lct}(\widehat{X};\widehat{\mathfrak{c}}_j) > 1+2\delta$ for any j with $K_j \neq \emptyset$. By [dFEM11, Proposition 2.19(ii)], applied to the klt germ $(X \ni x)$, the finitely many $\mathbb{R}$-ideals $\widehat{\mathfrak{c}}_j$ on $\widehat{X}$, and $\epsilon = \delta$, there exists a positive integer N_2 with the following property: for any $N \ge N_2$, any j with $K_j \neq \emptyset$, and any non-zero ideals $\mathfrak{b}_k$ of $\widehat{\mathcal{O}}$, $k \in K_j$, with

$$\mathfrak{b}_k + \widehat{\mathfrak{m}}^N = (\hat{h}_k) + \widehat{\mathfrak{m}}^N,$$

we have

$$\operatorname{lct}\Big(\widehat{X};\prod_{k\in K_j} \mathfrak{b}_k^{\beta_k^{(j)}}\Big) > 1+\delta. \tag{$**$}$$

Fix a prime divisor P_* over X with center x, for example an irreducible component of the exceptional divisor of the normalized blow-up of X at x. By Lemma 3.6, there exist formal prime divisors $\widehat{P}_*,\widehat{P}_1,\dots,\widehat{P}_l$ over $\widehat{X}$ with center $\hat{x}$ which restrict to $P_*,P_1,\dots,P_l$ respectively. Since (X,B) is klt near x, $a(P_*,X,B)$ is a positive real number. Fix a real number $C > a(P_*,X,B)$, let $\beta_{\min} := \min_k \beta_k$ (and $\beta_{\min} := 1$ if $s=0$), which is positive, since $\beta_1,\dots,\beta_s > 0$ by (iii) of [Liu26, Construction-Theorem 7.2], and let $N_1 \ge N_2$ be an integer such that

$$\delta\beta_{\min} N_1 > C|H| \quad \text{and} \quad N_1 > \operatorname{mult}_{\widehat{P}} \hat{h}_k \text{ for any } k \text{ and any } \widehat{P} \in \{\widehat{P}_*,\widehat{P}_1,\dots,\widehat{P}_l\}.$$

Step 2. In this step, we prove the estimate on the fixed germ $\widehat{X}$. Let $N \ge N_1$, and let $e_1,\dots,e_s \in \widehat{\mathcal{O}}$ be H-semi-invariant elements such that

$$e_k - w_k \hat{h}_k \in \widehat{\mathfrak{m}}^N$$

for units $w_1,\dots,w_s$ of $\widehat{\mathcal{O}}$. Let $\widehat{B}_e := \sum_k \beta_k \operatorname{div}(e_k)$. We show the following.

(1) For any formal prime divisor P over $\widehat{X}$ with center $\hat{x}$, either $\operatorname{mult}_P e_k = \operatorname{mult}_P \hat{h}_k$ for any k, or
$$a(P,\widehat{X},\widehat{B}) > C|H| \quad \text{and} \quad a(P,\widehat{X},\widehat{B}_e) > C|H|.$$
(2) $\operatorname{mult}_{\widehat{P}} e_k = \operatorname{mult}_{\widehat{P}} \hat{h}_k$ for any k and for $\widehat{P} = \widehat{P}_*,\widehat{P}_1,\dots,\widehat{P}_l$. In particular, $e_k \neq 0$ for any k.
(3) $(\widehat{X},\sum_k \beta_k^{(j)} \operatorname{div}(e_k))$ is klt for any $0 \le j \le r$.

(4) For any subgroup $H' \subset H$,

$$\inf_P \frac{a(P, \widehat{X}, \widehat{B}_e)}{|I_{H'}(P)|} = \inf_P \frac{a(P, \widehat{X}, \widehat{B})}{|I_{H'}(P)|} = \mathrm{mld}((X \ni x, B)/H'),$$

where P runs through the formal prime divisors over $\widehat{X}$ with center $\hat{x}$.

Let P be a formal prime divisor over $\widehat{X}$ with center $\hat{x}$. Then $\mathrm{mult}_P \widehat{\mathfrak{m}}$ is a positive integer, and

$$\mathrm{mult}_P(e_k - w_k \hat{h}_k) \geq N \,\mathrm{mult}_P \widehat{\mathfrak{m}} \geq N.$$

Since w_k is a unit, $\mathrm{mult}_P(w_k \hat{h}_k) = \mathrm{mult}_P \hat{h}_k$. Thus, if one of $\mathrm{mult}_P e_k$ and $\mathrm{mult}_P \hat{h}_k$ is smaller than $N \,\mathrm{mult}_P \widehat{\mathfrak{m}}$, then $\mathrm{mult}_P e_k = \mathrm{mult}_P \hat{h}_k$. In other words, if $\mathrm{mult}_P e_k \neq \mathrm{mult}_P \hat{h}_k$, then

$$\mathrm{mult}_P e_k \geq N \quad \text{and} \quad \mathrm{mult}_P \hat{h}_k \geq N. \tag{$\dagger$}$$

For $\widehat{P} = \widehat{P}_*, \widehat{P}_1, \ldots, \widehat{P}_l$, we have $\mathrm{mult}_{\widehat{P}} \hat{h}_k < N_1 \leq N$ by the choice of N_1, so ($\dagger$) fails and $\mathrm{mult}_{\widehat{P}} e_k = \mathrm{mult}_{\widehat{P}} \hat{h}_k$. Since $\mathrm{mult}_{\widehat{P}_*} \hat{h}_k$ is finite, $e_k \neq 0$. This proves (2).

We prove (3). Fix j. If $K_j = \emptyset$, then any e_k with $\beta_k^{(j)} > 0$ is congruent to the unit $w_k \hat{h}_k$ modulo $\widehat{\mathfrak{m}}$, hence is a unit. Thus $\sum_k \beta_k^{(j)} \,\mathrm{div}(e_k) = 0$, and $(\widehat{X}, 0)$ is klt. Assume that $K_j \neq \emptyset$. For $k \notin K_j$ with $\beta_k^{(j)} > 0$, $\hat{h}_k$ is a unit, so $e_k \in w_k \hat{h}_k + \widehat{\mathfrak{m}}$ is a unit. For $k \in K_j$, we have $e_k \in (\hat{h}_k) + \widehat{\mathfrak{m}}^N$ and

$$\hat{h}_k = w_k^{-1}(e_k - (e_k - w_k \hat{h}_k)) \in (e_k) + \widehat{\mathfrak{m}}^N,$$

so

$$(e_k) + \widehat{\mathfrak{m}}^N = (\hat{h}_k) + \widehat{\mathfrak{m}}^N.$$

Since $N \geq N_2$, ($**$) implies that

$$\mathrm{lct}\left(\widehat{X}; \prod_{k \in K_j} (e_k)^{\beta_k^{(j)}}\right) > 1 + \delta.$$

Thus the pair

$$\left(\widehat{X}, (1+\delta) \sum_k \beta_k^{(j)} \,\mathrm{div}(e_k)\right)$$

is klt, which implies (3). Moreover, log canonicity on $\widehat{X}$ can be checked on one log resolution, and any formal prime divisor over $\widehat{X}$ is a divisor on some log resolution [dFEM11, Section 2 and Theorems 2.3 and 2.4]. Thus, taking $j = 0$, for any formal prime divisor P over $\widehat{X}$ with center $\hat{x}$,

$$a(P, \widehat{X}) > (1+\delta) \sum_k \beta_k \,\mathrm{mult}_P e_k \quad \text{and} \quad a(P, \widehat{X}) > (1+\delta) \sum_k \beta_k \,\mathrm{mult}_P \hat{h}_k, \tag{$\ddagger$}$$

where the second inequality holds since $\mathrm{lct}(\widehat{X}; \widehat{\mathfrak{c}}_0) > 1 + 2\delta$. If $K_0 = \emptyset$, then ($\ddagger$) holds since both sums vanish and $a(P, \widehat{X}) > 0$.

We prove (1). Let P be a formal prime divisor over $\widehat{X}$ with center $\hat{x}$, and assume that $\mathrm{mult}_P e_k \neq \mathrm{mult}_P \hat{h}_k$ for some k. By ($\ddagger$) and ($\dagger$),

$$\begin{aligned} a(P, \widehat{X}, \widehat{B}) &= a(P, \widehat{X}) - \sum_{k'} \beta_{k'} \,\mathrm{mult}_P \hat{h}_{k'} \\ &> \delta \sum_{k'} \beta_{k'} \,\mathrm{mult}_P \hat{h}_{k'} \geq \delta \beta_k \,\mathrm{mult}_P \hat{h}_k \geq \delta \beta_{\min} N \geq \delta \beta_{\min} N_1 > C|H|. \end{aligned}$$

By the first inequality of ($\ddagger$) and by the bound $\mathrm{mult}_P e_k \geq N$ of ($\dagger$),

$$\begin{aligned} a(P, \widehat{X}, \widehat{B}_e) &= a(P, \widehat{X}) - \sum_{k'} \beta_{k'} \,\mathrm{mult}_P e_{k'} \\ &> \delta \sum_{k'} \beta_{k'} \,\mathrm{mult}_P e_{k'} \geq \delta \beta_k \,\mathrm{mult}_P e_k \geq \delta \beta_{\min} N \geq \delta \beta_{\min} N_1 > C|H|. \end{aligned}$$

This proves (1).

We prove (4). Fix a subgroup $H' \subset H$, and let μ and μ_e be the infima of $a(P, \widehat{X}, \widehat{B})/|I_{H'}(P)|$ and of $a(P, \widehat{X}, \widehat{B}_e)/|I_{H'}(P)|$ respectively, where P runs through the formal prime divisors over $\widehat{X}$ with center $\hat{x}$. By Definition 2.4 and Lemma 3.6 with $\mathfrak{a} = \mathcal{O}_X$, $\mu = \mathrm{mld}((X \ni x, B)/H')$. The inertia group $I_{H'}(P)$ depends only on P and on the action of H' on $\widehat{X}$, so it does not depend on the boundary. Let P be a formal prime divisor over $\widehat{X}$ with center $\hat{x}$. If $\mathrm{mult}_P e_k = \mathrm{mult}_P \hat{h}_k$ for any k, then

$$\frac{a(P, \widehat{X}, \widehat{B}_e)}{|I_{H'}(P)|} = \frac{a(P, \widehat{X}, \widehat{B})}{|I_{H'}(P)|}. \tag{♮}$$

Otherwise, since $|I_{H'}(P)| \le |H|$, (1) implies that both numbers are larger than C. By (2), the first case holds for $\widehat{P}_*$, and by Lemma 3.6,

$$\frac{a(\widehat{P}_*, \widehat{X}, \widehat{B})}{|I_{H'}(\widehat{P}_*)|} \le a(\widehat{P}_*, \widehat{X}, \widehat{B}) = a(P_*, X, B) < C.$$

Thus $\mu < C$ and $\mu_e < C$, so in the definitions of μ and μ_e it suffices to let P run through the formal prime divisors with $\mathrm{mult}_P e_k = \mathrm{mult}_P \hat{h}_k$ for any k, and (♮) holds for such P. Hence $\mu_e = \mu$. This proves (4).

Step 3. In this step, we fix $N \ge N_1$, choose L_1, and, for any $L \ge L_1$ and f_0 as in the theorem, construct an H-equivariant automorphism Θ of $\mathfrak{R}$ which maps (f_0) onto $(\hat{f})$ if $n = 4$.

If $n = 3$, let $L_1 := N$, $\varphi := \mathrm{id}_{\mathfrak{R}}$, and $\Theta := \psi^{-1} = \psi^{-1} \circ \varphi^{-1}$. Then (§) below holds for any $L \ge L_1$, since $\varphi^{-1}(g) = g$ for any $g \in \mathfrak{R}$ and $\mathfrak{m}^{L-b+1} = \mathfrak{m}^L \subset \mathfrak{m}^N$. Assume that $n = 4$. Let $L_1 := N + 2b$, let $L \ge L_1$, let f_0 be an H-semi-invariant polynomial with the character of f such that $f_0 \equiv F' \pmod{\mathfrak{m}^L}$, where

$$F' := v\psi(\hat{f}).$$

Then $L > 2b$ and $L - b + 1 > N$.

We first show that $\mathfrak{m}^b \subset T(F')$. Since v is a unit, $(F') = (\psi(\hat{f}))$. By the chain rule,

$$\partial_i F' = (\partial_i v)\psi(\hat{f}) + v\partial_i(\psi(\hat{f})), \qquad \partial_i(\psi(\hat{f})) = \sum_{i'=1}^{4} \psi(\partial_{i'}\hat{f})\partial_i(\psi(x_{i'})).$$

Since ψ is an automorphism, the matrix $(\partial_i(\psi(x_{i'})))_{i,i'}$ is invertible over $\mathfrak{R}$. Thus

$$T(F') = (\psi(\hat{f})) + \mathfrak{m}(\psi(\partial_1\hat{f}), \dots, \psi(\partial_4\hat{f})).$$

Since $\psi(\mathfrak{m}) = \mathfrak{m}$, we obtain by $(*)$

$$T(F') = \psi(T(\hat{f})) \supset \psi(\mathfrak{m}^b) = \mathfrak{m}^b.$$

We next show that $\mathfrak{m}^b \subset T(f_0)$. We have $f_0 - F' \in \mathfrak{m}^L$ and $\partial_i f_0 - \partial_i F' \in \mathfrak{m}^{L-1}$, so $T(f_0) + \mathfrak{m}^L = T(F') + \mathfrak{m}^L$. Since $L \ge b + 1$, we obtain

$$\mathfrak{m}^b \subset T(F') \subset T(f_0) + \mathfrak{m}^{b+1} = T(f_0) + \mathfrak{m} \cdot \mathfrak{m}^b.$$

By Nakayama's lemma applied to the finitely generated $\mathfrak{R}$-module $(\mathfrak{m}^b + T(f_0))/T(f_0)$, $\mathfrak{m}^b \subset T(f_0)$.

By [Liu26, Lemma 7.1], applied to $G = H$, the set $S = \{1, 2, 3, 4\}$ of the indices of all four coordinates (so that its hypothesis $\mathfrak{m}^b \subset (F) + \mathfrak{m}(\partial_i F \mid i \in S)$ is the inclusion $\mathfrak{m}^b \subset T(F)$ shown above), and $L > 2b$, once to F' and once to f_0, there exist H-equivariant automorphisms φ_1, φ_2 of $\mathfrak{R}$ and H-invariant units v_1, v_2 of $\mathfrak{R}$ such that

$$v_1\varphi_1(F') = F'_{<L}, \quad v_2\varphi_2(f_0) = (f_0)_{<L}, \quad \text{and} \quad \varphi_1(x_i) - x_i,\ \varphi_2(x_i) - x_i \in \mathfrak{m}^{L-b+1} \text{ for any } i,$$

where, for $g \in \mathfrak{R}$, $g_{<L}$ denotes the sum of the terms of g of degree less than L. Since $f_0 \equiv F' \pmod{\mathfrak{m}^L}$, $(f_0)_{<L} = F'_{<L}$. Let $\varphi := \varphi_2^{-1} \circ \varphi_1$. Then

$$\varphi(F') = \varphi_2^{-1}(v_1^{-1}F'_{<L}) = \varphi_2^{-1}(v_1^{-1})\varphi_2^{-1}(v_2\varphi_2(f_0)) = \varphi_2^{-1}(v_1^{-1}v_2)f_0,$$

so $\varphi((F')) = (f_0)$.

Let $c := L - b + 1$. If α is an automorphism of $\mathfrak{R}$ with $\alpha(x_i) - x_i \in \mathfrak{m}^c$ for any i, then $\alpha(x^a) - x^a \in \mathfrak{m}^c$ for any monomial $x^a \neq 1$, so $\alpha(g) - g \in \mathfrak{m}^c$ for any $g \in \mathfrak{R}$, since $\mathfrak{m}^c$ is closed. Moreover, $\alpha^{-1}(g) - g = -(\alpha(g') - g') \in \mathfrak{m}^c$ with $g' := \alpha^{-1}(g)$. Applying the inclusions $\alpha(g) - g \in \mathfrak{m}^c$ and $\alpha^{-1}(g) - g \in \mathfrak{m}^c$ to $\alpha = \varphi_1$ and $\alpha = \varphi_2$, we obtain

$$\varphi^{-1}(g) - g \in \mathfrak{m}^{L-b+1} \subset \mathfrak{m}^N \quad \text{for any } g \in \mathfrak{R}. \tag{§}$$

Let $\Theta := \psi^{-1} \circ \varphi^{-1}$. Then Θ is an H-equivariant automorphism of $\mathfrak{R}$ with $\Theta(\mathfrak{m}^N) = \mathfrak{m}^N$, and

$$\Theta((f_0)) = \psi^{-1}((F')) = \psi^{-1}((\psi(\hat{f}))) = (\hat{f}).$$

In both cases, Θ induces an H-equivariant isomorphism

$$\bar{\Theta} : \widehat{\mathcal{O}}_{X_0,0} \to \widehat{\mathcal{O}},$$

since $\widehat{\mathcal{O}}_{X_0,0}$ is $\mathfrak{R}/(f_0)$ if $n = 4$ and $\mathfrak{R}$ if $n = 3$.

Step 4. In this step, we show that the action of H on $\mathbb{A}^n$ is faithful and that X_0 is an isolated Gorenstein terminal germ at 0 if $n = 4$.

Let $\sigma \in H$ act trivially on $\widehat{\mathcal{O}}$. Since $\mathcal{O}_{X,x} \subset \widehat{\mathcal{O}}$, σ acts trivially on $\mathcal{O}_{X,x}$, so σ is the identity on a neighborhood of x, hence on X, as X is integral and separated. Since H acts faithfully on X, $\sigma = 1$. Any element of H acting trivially on $\mathbb{A}^n$ acts trivially on the x_i, hence on $\mathfrak{R}$ and on $\widehat{\mathcal{O}}$. Any element of H acting trivially on X_0 acts trivially on $\widehat{\mathcal{O}}_{X_0,0}$, hence on $\widehat{\mathcal{O}}$ since $\bar{\Theta}$ is H-equivariant. Thus H acts faithfully on $\mathbb{A}^n$ and on X_0.

Assume that $n = 4$. Let $J_0 := (f_0, \partial_1 f_0, \dots, \partial_4 f_0)$ be the ideal of the local ring $\mathcal{O}_{\mathbb{A}^4,0}$. By **Step 3**, $J_0\mathfrak{R} \supset T(f_0) \supset \mathfrak{m}^b$. Since $\mathfrak{R}$ is faithfully flat over $\mathcal{O}_{\mathbb{A}^4,0}$, J_0 contains the b-th power of the maximal ideal of $\mathcal{O}_{\mathbb{A}^4,0}$. Thus, by the Jacobian criterion [Har77, Theorem I.5.1], X_0 is smooth on $U\backslash\{0\}$ for a neighborhood U of 0. Since X_0 is a hypersurface, it is Gorenstein, and by [Har77, Proposition II.8.23(b)], it is normal near 0, since it is regular in codimension one.

Let E be an exceptional prime divisor over X_0 whose center contains 0. Since K_{X_0} is Cartier, $a(E, X_0, 0)$ is an integer. If the center of E is 0, then by [Liu26, Lemma 3.3(3)], E is the restriction of a formal prime divisor $\widehat{E}$ over $\operatorname{Spec} \widehat{\mathcal{O}}_{X_0,0}$ with log discrepancy $a(E, X_0, 0)$ with respect to $\operatorname{Spec} \widehat{\mathcal{O}}_{X_0,0}$. The isomorphism $\bar{\Theta}$ maps $\widehat{E}$ to a formal prime divisor over $\widehat{X}$ with center $\hat{x}$, which, by [Liu26, Lemma 3.3(3)] again, restricts to a prime divisor E' over X with center x. The isomorphism $\bar{\Theta}$ carries the canonical divisor of [dFEM11, Appendix A] to the canonical divisor, which does not depend on the presentation [dFEM11, Remark 2.12]. Thus

$$a(E, X_0, 0) = a(\widehat{E}, \operatorname{Spec} \widehat{\mathcal{O}}_{X_0,0}, 0) = a(E', X, 0).$$

Since X is terminal and $\dim X = 3$, $a(E', X, 0) > 1$. If the center of E has positive dimension, then the center intersects $U\backslash\{0\}$, which is smooth, hence terminal [KM98, Corollary 2.31(2)], so $a(E, X_0, 0) > 1$. Therefore $a(E, X_0, 0) \geq 2$ for any such E, and X_0 is an isolated Gorenstein terminal germ at 0.

Step 5. In this step, we conclude the proof. Let $L \geq L_1$, and let f_0 and the $h_{k,0}$ be polynomials as in the theorem. Recall that Θ is the automorphism of **Step 3**, with $\Theta = \psi^{-1}$ if $n = 3$. Let e_k be the image of $\Theta(h_{k,0})$ in $\widehat{\mathcal{O}}$, and w_k the image of $\psi^{-1}(u_k)$ in $\widehat{\mathcal{O}}$. Then e_k is the image under $\bar{\Theta}$ of the class of $h_{k,0}$, e_k is H-semi-invariant with the character of h_k, and w_k is a unit. By (§), $\varphi^{-1}(h_{k,0}) - h_{k,0} \in \mathfrak{m}^N$, so

$$\Theta(h_{k,0}) - \psi^{-1}(u_k)\hat{h}_k = \psi^{-1}\big(\varphi^{-1}(h_{k,0}) - h_{k,0}\big) + \psi^{-1}\big(h_{k,0} - u_k\psi(\hat{h}_k)\big) \in \mathfrak{m}^N.$$

Thus $e_k - w_k\hat{h}_k \in \widehat{\mathfrak{m}}^N$, and we may apply **Step 2** to $e_1, \dots, e_s$.

By **Step 2**(2), $e_k \neq 0$. Since $\mathcal{O}_{X_0,0} \to \widehat{\mathcal{O}}_{X_0,0}$ is injective and $\bar{\Theta}$ is an isomorphism, $h_{k,0}|_{X_0} \neq 0$. Let $\widehat{X}_0 := \operatorname{Spec} \widehat{\mathcal{O}}_{X_0,0}$, and let $\widehat{B}_0$ be the pullback of B_0 to $\widehat{X}_0$. Then $\bar{\Theta}$ is an H-equivariant isomorphism of complete local rings which maps the class of $h_{k,0}$ to e_k. Thus $\bar{\Theta}$ induces a bijection between the formal prime divisors over $\widehat{X}_0$ with center the closed point and the formal prime

divisors over $\widehat{X}$ with center $\hat{x}$, which preserves residue fields, decomposition and inertia groups for any subgroup of H, log discrepancies with respect to $\widehat{X}_0$ and $\widehat{X}$ (by [dFEM11, Remark 2.12], as in **Step 4**), and the orders of corresponding elements. In particular, for any j, it preserves log discrepancies with respect to

$$\Big(\widehat{X}_0, \sum_k \beta_k^{(j)} \operatorname{div}(h_{k,0})\Big) \quad \text{and} \quad \Big(\widehat{X}, \sum_k \beta_k^{(j)} \operatorname{div}(e_k)\Big),$$

and it maps $\widehat{B}_0$ to $\widehat{B}_e$.

We prove (3) of [Liu26, Construction-Theorem 7.2]. By **Step 2**(3) and $\bar{\Theta}$, the pair

$$\Big(\widehat{X}_0, \sum_k \beta_k^{(j)} \operatorname{div}(h_{k,0})\Big)$$

is klt for any j. By [dFEM11, Proposition 2.11], $(X_0, \sum_k \beta_k^{(j)} \operatorname{div}(h_{k,0}|_{X_0}))$ is klt near 0 for any j.

We prove (1) of [Liu26, Construction-Theorem 7.2]. The case $j = 0$ of (3) shows that (X_0, B_0) is klt near 0, and H acts faithfully on X_0 by **Step 4**. Let $H' \subset H$ be a subgroup. By **Step 4**, $(X_0 \ni 0)$ is a normal klt germ. By Lemma 3.6 with $\mathfrak{a} = \mathcal{O}_{X_0}$, Definition 2.4, the bijection induced by $\bar{\Theta}$, and **Step 2**(4),

$$\operatorname{mld}((X_0 \ni 0, B_0)/H') = \inf_{P_0} \frac{a(P_0, \widehat{X}_0, \widehat{B}_0)}{|I_{H'}(P_0)|} = \inf_{P} \frac{a(P, \widehat{X}, \widehat{B}_e)}{|I_{H'}(P)|} = \operatorname{mld}((X \ni x, B)/H'),$$

where P_0 and P run through the formal prime divisors with center the closed point over $\widehat{X}_0$ and over $\widehat{X}$ respectively.

We prove (2) of [Liu26, Construction-Theorem 7.2]. The isomorphism $\bar{\Theta}^{-1}$ is an H-equivariant isomorphism between the completions of the local rings of X at x and of X_0 at 0. Fix $1 \le i \le l$. Let $\widehat{P}_{i,0}$ be the formal prime divisor over $\widehat{X}_0$ which corresponds to $\widehat{P}_i$ under $\bar{\Theta}$, and let $P_{i,0}$ be the restriction of $\widehat{P}_{i,0}$, which is a prime divisor over X_0 with center 0 by [Liu26, Lemma 3.3(3)]. By [Liu26, Lemma 3.3(3)] for X and for X_0 with the zero boundary, and by the bijection induced by $\bar{\Theta}$, the prime divisors P_i and $P_{i,0}$ have isomorphic residue fields, equal decomposition groups, equal inertia groups, and

$$a(P_{i,0}, X_0, 0) = a(\widehat{P}_{i,0}, \widehat{X}_0, 0) = a(\widehat{P}_i, \widehat{X}, 0) = a(P_i, X, 0).$$

Finally, by **Step 2**(2),

$$\operatorname{mult}_{P_{i,0}} h_{k,0} = \operatorname{mult}_{\widehat{P}_{i,0}} h_{k,0} = \operatorname{mult}_{\widehat{P}_i} e_k = \operatorname{mult}_{\widehat{P}_i} \hat{h}_k = \operatorname{mult}_{P_i} h_k$$

for any k. This proves (2), and the proof is complete. □

Remark 3.11. The later sections use Theorem 3.10 through Construction-Theorem 4.13, with ψ an equivariant formal change of coordinates and with polynomials f_0 such as $xy + g_0(z,t)$, $x^2 + y^2 + g_0(z,t)$, and $x^2 + g_0(y,z,t)$, where g_0 is a polynomial in the indicated coordinates.

3.3. **Gauss extensions on split hypersurfaces.** In this subsection, we compute log discrepancies and inertia groups of Gauss extensions on the hypersurfaces $(xy \pm g(z,t) = 0) \subset \mathbb{A}^4$, and we prove an upper bound for the quotient minimal log discrepancy of $(xy - g = 0)$ when the group $G \cap \mathbf{T}_{(1,-1,0,0)} = \mu_r(1,-1,0,0)$ is large (Proposition 3.14), which is used in the proof of Theorem 8.3 for the circles of Set-up 5.1. The substitution $y \to -y$ maps $(xy - g = 0)$ isomorphically onto $(xy + g = 0)$ and, for a diagonal group, changes no character. We use the equation $xy+g$ in this subsection, except in Proposition 3.14, which extends [Liu26, Theorem 15.9] and keeps the equation $xy - g$ of [Liu26, Theorem 15.9].

As in [Liu26, Lemma 2.6(1)], $\mathfrak{m}_F$ denotes the maximal ideal of the valuation ring of mult_F, and the residue field of mult_F is the quotient of the valuation ring of mult_F by $\mathfrak{m}_F$. The inertia groups $I_G(F)$ and $I_G(\nu)$ are those of [Liu26, Definition 2.5]. They are defined for any action, faithful or not, such as the action of G on T in Lemma 3.12(2).

Lemma 3.12. *Let V be a normal variety, I a finite group acting on V, F an I-invariant prime divisor over V such that I acts trivially on the residue field of* mult_F. *Let λ be the character by which I acts on $\mathfrak{m}_F/\mathfrak{m}_F^2$.*

(1) If ω is an I-semi-invariant rational section of ω_V with character χ_ω which generates ω_V at the generic point of $\mathrm{center}_V F$, *then $\chi_\omega = \lambda^{a(F,V,0)}$.*

(2) Let $T := \mathbb{A}^2_{z,t}$, $0 \neq g \in (z,t)^2$, $X := (xy + g(z,t) = 0) \subset \mathbb{A}^4$, G a finite diagonal abelian group preserving X and acting faithfully on X, ϱ a positive integer, and ν a prime divisor over T with center 0. Assume that V is a G-invariant open neighborhood of 0 in X, that $I = I_G(F)$, that F has center 0 on V, and that $\mathrm{mult}_F \zeta = \varrho\,\mathrm{mult}_\nu \zeta$ *for any non-zero $\zeta \in \mathbb{C}[z,t]$. Then $|I|$ divides $d_F := a(F,X,0) - \varrho a(\nu,T,0)$.*

(3) Let $(S' \ni p)$ be a smooth surface germ with an action of I fixing p, u, v I-eigenparameters at p, ϱ a positive integer, and ν a prime divisor over S' with center p. Assume that I acts faithfully on V, that there is a dominant I-equivariant rational map $\phi : V \dashrightarrow S'$, and that $\mathrm{mult}_F(\phi^*\zeta) = \varrho\,\mathrm{mult}_\nu \zeta$ *for any non-zero rational function ζ on S'. Then λ is faithful and $|I|$ divides $\varrho(a(\nu,S',0) - \mathrm{mult}_\nu u - \mathrm{mult}_\nu v)$. In particular, if $0 \leq \varrho(a(\nu,S',0) - \mathrm{mult}_\nu u - \mathrm{mult}_\nu v) < |I|$, then ν is the monomial valuation in u, v with weights* $(\mathrm{mult}_\nu u, \mathrm{mult}_\nu v)$.

(4) Let S be a normal variety with an action of I, $\phi : V \dashrightarrow S$ a dominant I-equivariant rational map, ν a prime divisor over S, and ϱ a positive integer such that $\mathrm{mult}_F(\phi^*\zeta) = \varrho\,\mathrm{mult}_\nu \zeta$ *for any non-zero rational function ζ on S. Then ν is I-invariant, I acts trivially on the residue field of* mult_ν *and acts on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ by the character λ^ϱ, and $\chi_\zeta = \lambda^{\varrho\,\mathrm{mult}_\nu \zeta}$ for any non-zero rational function ζ on S with $\gamma^*\zeta = \chi_\zeta(\gamma)\zeta$ for any $\gamma \in I$.*

Proof. For $\gamma \in I$, we write γ^* for the pullback of rational functions and rational differential forms. Let $n := \dim V$ and let $\mathcal{O}_F$ be the valuation ring of mult_F. Since F is I-invariant, $\gamma^*\mathcal{O}_F = \mathcal{O}_F$ and $\gamma^*\mathfrak{m}_F = \mathfrak{m}_F$ for any $\gamma \in I$, and $\gamma^* f - f \in \mathfrak{m}_F$ for any $f \in \mathcal{O}_F$ because I acts trivially on $\mathcal{O}_F/\mathfrak{m}_F$.

We use the following average. Let a finite group K act on a field preserving a valuation ring $\mathcal{O}$ with maximal ideal $\mathfrak{m}$, let χ be a character of K, $\ell \geq 0$ an integer, and $f \in \mathcal{O}$ with $\gamma^* f - \chi(\gamma) f \in \mathfrak{m}^{\ell+1}$ for any $\gamma \in K$. Then

$$f' := \frac{1}{|K|} \sum_{\gamma \in K} \chi(\gamma)^{-1} \gamma^* f \tag{3.12}$$

satisfies $\gamma^* f' = \chi(\gamma) f'$ for any $\gamma \in K$ and $f' - f \in \mathfrak{m}^{\ell+1}$, since reindexing the sum gives the semi-invariance and

$$f' - f = \frac{1}{|K|} \sum_{\gamma \in K} \chi(\gamma)^{-1} (\gamma^* f - \chi(\gamma) f).$$

We first construct semi-invariant generators. Let $h : Y \to V$ be a projective birational morphism from a normal variety Y such that F is a prime divisor on Y. Pick a closed point $q \in F$ at which Y and F are smooth, and local parameters $\pi, \zeta_1, \dots, \zeta_{n-1}$ of Y at q such that $F = (\pi = 0)$ near q. Then π generates $\mathfrak{m}_F$, and $\theta := d\pi \wedge d\zeta_1 \wedge \cdots \wedge d\zeta_{n-1}$ generates the free $\mathcal{O}_F$-module $\Omega^n_{\mathcal{O}_F}$ of rank one, since Ω^1_Y is free near q with basis $d\pi, d\zeta_1, \dots, d\zeta_{n-1}$ by [Har77, Proposition II.8.7 and Theorem II.8.8] and Nakayama's lemma, and $\mathcal{O}_F$ is the localization of $\mathcal{O}_{Y,q}$ at the prime ideal of F. Since $\gamma^*\pi - \lambda(\gamma)\pi \in \mathfrak{m}_F^2$ and $\gamma^*\zeta_i - \zeta_i \in \mathfrak{m}_F$ for any $\gamma \in I$, the average (3.12), applied to I and $\mathcal{O}_F$, gives π' from π with $\chi = \lambda$ and $\ell = 1$, and ζ_i' from ζ_i with $\chi = 1$ and $\ell = 0$. Let

$$\theta' := d\pi' \wedge d\zeta_1' \wedge \cdots \wedge d\zeta_{n-1}'.$$

Then $\pi' - \pi \in \mathfrak{m}_F^2 = \pi^2 \mathcal{O}_F$ and $\zeta_i' - \zeta_i \in \mathfrak{m}_F = \pi \mathcal{O}_F$. Thus

$$d\pi' - d\pi \in \mathfrak{m}_F \Omega^1_{\mathcal{O}_F}, \qquad d\zeta_i' - d\zeta_i \in \mathcal{O}_F d\pi + \mathfrak{m}_F \Omega^1_{\mathcal{O}_F}.$$

Expanding θ' and using $d\pi \wedge d\pi = 0$, we obtain $\theta' - \theta \in \mathfrak{m}_F \Omega^n_{\mathcal{O}_F}$. Thus π' generates $\mathfrak{m}_F$ and θ' generates $\Omega^n_{\mathcal{O}_F}$. Moreover, for any $\gamma \in I$,

$$\gamma^*\pi' = \lambda(\gamma)\pi', \qquad \gamma^*\zeta_i' = \zeta_i', \qquad \gamma^*\theta' = \lambda(\gamma)\theta'.$$

Let f be a non-zero rational function on V such that $\gamma^* f = \chi_f(\gamma) f$ for any $\gamma \in I$, where χ_f is a character of I. We show that

$$\chi_f = \lambda^{\operatorname{mult}_F f}. \tag{3.13}$$

Let $\gamma \in I$. The function $\epsilon := f\pi'^{-\operatorname{mult}_F f}$ is a unit of $\mathcal{O}_F$, and

$$\gamma^*\epsilon = \chi_f(\gamma)\lambda(\gamma)^{-\operatorname{mult}_F f}\epsilon.$$

Since $\gamma^*\epsilon - \epsilon \in \mathfrak{m}_F$ and $\epsilon \notin \mathfrak{m}_F$, we have $\chi_f(\gamma)\lambda(\gamma)^{-\operatorname{mult}_F f} = 1$. When I is abelian, (3.13) is [Liu26, Lemma 9.2(1)] applied to the action of I on V, since $I_I(F) = I$.

(1) Since ω generates ω_V at the generic point of $\operatorname{center}_V F$, ω vanishes to order $a(F,V,0)-1$ along F by the definition of log discrepancies. Thus $\omega = \epsilon\pi'^{a(F,V,0)-1}\theta'$ for a unit ϵ of $\mathcal{O}_F$. For any $\gamma \in I$, we have $\gamma^*\omega = \chi_\omega(\gamma)\omega$ and

$$\gamma^*(\pi'^{a(F,V,0)-1}\theta') = \lambda(\gamma)^{a(F,V,0)}\pi'^{a(F,V,0)-1}\theta',$$

hence

$$\gamma^*\epsilon = \chi_\omega(\gamma)\lambda(\gamma)^{-a(F,V,0)}\epsilon.$$

Since $\gamma^*\epsilon - \epsilon \in \mathfrak{m}_F$ and $\epsilon \notin \mathfrak{m}_F$, we obtain $\chi_\omega(\gamma) = \lambda(\gamma)^{a(F,V,0)}$. This proves (1).

(4) The proofs of (2) and (3) use (4), so we prove (4) first. Since ϕ is I-equivariant, $\gamma^*\phi^* = \phi^*\gamma^*$ for any $\gamma \in I$. Hence

$$\operatorname{mult}_\nu(\gamma^*\zeta) = \frac{1}{\varrho}\operatorname{mult}_F(\gamma^*\phi^*\zeta) = \frac{1}{\varrho}\operatorname{mult}_F(\phi^*\zeta) = \operatorname{mult}_\nu\zeta$$

for any non-zero rational function ζ on S. If moreover $\operatorname{mult}_\nu\zeta \geq 0$, then $\phi^*\zeta \in \mathcal{O}_F$, so

$$\operatorname{mult}_F(\phi^*(\gamma^*\zeta - \zeta)) = \operatorname{mult}_F(\gamma^*\phi^*\zeta - \phi^*\zeta) > 0$$

and $\operatorname{mult}_\nu(\gamma^*\zeta - \zeta) > 0$. Thus ν is I-invariant, and I acts trivially on the residue field of mult_ν. If ζ is a non-zero rational function on S with $\gamma^*\zeta = \chi_\zeta(\gamma)\zeta$ for any $\gamma \in I$, then $\gamma^*\phi^*\zeta = \chi_\zeta(\gamma)\phi^*\zeta$ and $\operatorname{mult}_F(\phi^*\zeta) = \varrho\operatorname{mult}_\nu\zeta$, so $\chi_\zeta = \lambda^{\varrho\operatorname{mult}_\nu\zeta}$ by (3.13). Let α be the character of I on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ and τ_0 a uniformizer of mult_ν. Since I acts trivially on the residue field of mult_ν, $\gamma^*\tau_0 - \alpha(\gamma)\tau_0 \in \mathfrak{m}_\nu^2$ for any $\gamma \in I$. By the average (3.12), applied to I, α, $\ell = 1$, and τ_0, there is a uniformizer τ of mult_ν with $\gamma^*\tau = \alpha(\gamma)\tau$ for any $\gamma \in I$. Therefore $\alpha = \lambda^{\varrho\operatorname{mult}_\nu\tau} = \lambda^\varrho$. This proves (4).

(2) The map $(z,t) : V \dashrightarrow T$ is dominant and G-equivariant, and $\operatorname{mult}_F\zeta = \varrho\operatorname{mult}_\nu\zeta$ holds for any non-zero rational function ζ on T, since ζ is a quotient of two polynomials. Let χ_z, χ_t be the characters of z, t. Since G acts faithfully on V and $I = I_G(F)$, λ is faithful by [Liu26, Lemma 2.6(1)]. Let

$$\omega_0 := \frac{dx \wedge dz \wedge dt}{x}. \tag{3.14}$$

Since $ydx + xdy + g_z dz + g_t dt = 0$ on X, ω_0 is equal to $\frac{dx\wedge dz\wedge dt}{x}$, $-\frac{dy\wedge dz\wedge dt}{y}$, $-\frac{dx\wedge dy\wedge dt}{g_z}$, and $\frac{dx\wedge dy\wedge dz}{g_t}$ on the open subsets $(x \neq 0)$, $(y \neq 0)$, $(g_z \neq 0)$, and $(g_t \neq 0)$ of X, respectively, and on each of the four open subsets the corresponding expression generates Ω^3_X. By the Jacobian criterion, the smooth locus of X is the union of the four open subsets. Since V is normal, ω_0 generates ω_X near 0. Moreover, $\gamma^*\omega_0 = \chi_z(\gamma)\chi_t(\gamma)\omega_0$ for any $\gamma \in G$. Since F has center 0 on V, (1), applied to ω_0, implies

$$(\chi_z\chi_t)|_I = \lambda^{a(F,X,0)}.$$

On the other hand, $dz \wedge dt$ generates ω_T and has the character $\chi_z\chi_t$. By (4), applied to $S = T$, we may apply (1) to the action of I on T and to ν, with conormal character λ^ϱ. Thus

$$(\chi_z\chi_t)|_I = \lambda^{\varrho a(\nu,T,0)}.$$

Therefore $\lambda^{d_F}=1$. Since λ is faithful, $|I|$ divides d_F.
(3) Since I acts faithfully on V and acts trivially on the residue field of mult_F, $I=I_I(F)$, and λ is faithful by [Liu26, Lemma 2.6(1)], after replacing V by an I-invariant affine open subset containing the generic point of $\mathrm{center}_V F$. Let χ_u,χ_v be the characters of u,v. Possibly shrinking S' to an I-invariant neighborhood of p, $du\wedge dv$ generates $\omega_{S'}$, and $du\wedge dv$ has the character $\chi_u\chi_v$. By (4), applied to $S=S'$, we may apply (1) to the action of I on S' and to ν, with conormal character λ^{ϱ}, so $\chi_u\chi_v=\lambda^{\varrho a(\nu,S',0)}$. By (4) again, $\chi_u=\lambda^{\varrho\,\mathrm{mult}_\nu u}$ and $\chi_v=\lambda^{\varrho\,\mathrm{mult}_\nu v}$. Hence

$$\lambda^{\varrho(a(\nu,S',0)-\mathrm{mult}_\nu u-\mathrm{mult}_\nu v)}=1,$$

and $|I|$ divides $\varrho(a(\nu,S',0)-\mathrm{mult}_\nu u-\mathrm{mult}_\nu v)$ because λ is faithful.

Assume now that $0\le\varrho(a(\nu,S',0)-\mathrm{mult}_\nu u-\mathrm{mult}_\nu v)<|I|$. By the divisibility, $a(\nu,S',0)=\mathrm{mult}_\nu u+\mathrm{mult}_\nu v$, since $\varrho>0$. By [Liu26, Lemma 2.13], $a(\nu,S',0)\ge\mathrm{mult}_\nu u+\mathrm{mult}_\nu v$ always holds. By [JM12, Proposition 5.1 and Corollary 5.4], applied to the simple normal crossing divisor $\mathrm{div}(u)+\mathrm{div}(v)$ near p, the equality implies that ν is the monomial valuation in u,v with weights $(\mathrm{mult}_\nu u,\mathrm{mult}_\nu v)$, since the retraction of mult_ν is the monomial valuation in u,v with weights $(\mathrm{mult}_\nu u,\mathrm{mult}_\nu v)$ [JM12, §4.3]. □

On $X=(xy+g=0)$, a prime divisor with center 0 may be computed by a Gauss extension in a fiber coordinate $u=x+\varphi$ rather than in x. Since φ depends only on z,t, the generator $\frac{dx\wedge dz\wedge dt}{x}$ of ω_X keeps the form $\frac{du\wedge dz\wedge dt}{x}$ in the coordinates z,t,u, and the log discrepancy formula of [Liu26, Lemma 18.2] only gains the term $\mathrm{mult}_F u-\mathrm{mult}_F x$.

Proposition 3.13. *Let $T:=\mathbb{A}^2_{z,t}$, $0\ne g\in(z,t)^2$, and $X:=(xy+g(z,t)=0)\subset\mathbb{A}^4$, normal near 0. Let G be a finite diagonal abelian group preserving X and acting faithfully on X, and β the character of x. Let ν be a prime divisor over T with center 0, $J_\nu:=I_G(\nu)$ for the action of G on T, and α_ν the character of J_ν on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$. Let $\varphi\in(z,t)$ be a polynomial which is zero or G-semi-invariant with character β, and $u:=x+\varphi$, so that (z,t,u) is a birational map $X\dashrightarrow T\times\mathbb{A}^1_u$.*

(1) Let ϱ,w be coprime positive integers with $k:=\min\{w,\varrho\,\mathrm{mult}_\nu\varphi\}<\varrho\,\mathrm{mult}_\nu g$. Then the Gauss extension of $\varrho\,\mathrm{mult}_\nu$ with u-value w is mult_F for a prime divisor F over X with center 0, and

$$\mathrm{mult}_F x=k,\quad \mathrm{mult}_F y=\varrho\,\mathrm{mult}_\nu g-k,\quad a(F,X,0)=\varrho a(\nu,T,0)+w-k,$$
$$I_G(F)=\mathrm{Ker}(\beta^{\varrho}\alpha_\nu^{-w}:J_\nu\to\mathbb{C}^*).$$

(2) Let ϱ be a positive integer and F a prime divisor over X with center 0 such that $\mathrm{mult}_F\zeta=\varrho\,\mathrm{mult}_\nu\zeta$ for any non-zero $\zeta\in\mathbb{C}[z,t]$. Then $a(F,X,0)\ge\varrho a(\nu,T,0)+\mathrm{mult}_F u-\mathrm{mult}_F x$, with equality if and only if mult_F is the Gauss extension of $\varrho\,\mathrm{mult}_\nu$ with u-value $\mathrm{mult}_F u$.

Statements (1) and (2) hold with x and $u=x+\varphi$ replaced by y and $u=y+\varphi$, and with β replaced by the character of y, where φ is zero or has the character of y.

Proof. Since $y=-g/x$ on X and $x=u-\varphi$, the map (z,t,u) restricts to an isomorphism from $X\cap(x\ne0)$ onto the open subset $(u\ne\varphi(z,t))$ of $T\times\mathbb{A}^1_u$. It is G-equivariant if G acts on u by β, since x has the character β and φ is zero or has the character β. Through the birational morphism $(z,t,u):X\to T\times\mathbb{A}^1_u$, any prime divisor over X is a prime divisor over $T\times\mathbb{A}^1_u$. A prime divisor F over $T\times\mathbb{A}^1_u$ is a prime divisor over X if $\mathrm{mult}_F y\ge0$ for $y=-g/(u-\varphi)$, since z, t, and $x=u-\varphi$ are regular on $T\times\mathbb{A}^1_u$ and generate the coordinate ring of X together with y. The Gauss extension of [Liu26, Definition-Proposition 2.11] is taken with respect to u and the rational functions on T. Since φ is a polynomial in z,t, we have $dx\wedge dz\wedge dt=du\wedge dz\wedge dt$. Since X is normal near 0, the form ω_0 of (3.14) generates ω_X near 0 by the computation following (3.14), and $du\wedge dz\wedge dt$ generates $\omega_{T\times\mathbb{A}^1_u}$. Let F be a prime divisor over X with center 0. Then $\mathrm{mult}_F z,\mathrm{mult}_F t,\mathrm{mult}_F u>0$, so F has center $(0,0)$ on $T\times\mathbb{A}^1_u$. On a higher model on which

F is a divisor, the divisors of $\frac{du\wedge dz\wedge dt}{x}$ and of $du\wedge dz\wedge dt$ have multiplicities $a(F,X,0)-1$ and $a(F,T\times\mathbb{A}^1_u,0)-1$ along F. Thus

$$a(F,X,0)=a(F,T\times\mathbb{A}^1_u,0)-\operatorname{mult}_F x. \tag{3.15}$$

(1) By [Liu26, Lemma 18.1], applied to T, the boundary 0, the prime divisor ν, and the coprime integers ϱ,w, the Gauss extension of $\varrho\operatorname{mult}_\nu$ with u-value w is mult_F for a prime divisor F over $T\times\mathbb{A}^1_u$, and

$$a(F,T\times\mathbb{A}^1_u,0)=\varrho a(\nu,T,0)+w.$$

Moreover, for any rational function τ on T with $\operatorname{mult}_\nu\tau=1$, the residue R_τ of u^ϱ/τ^w is transcendental over the residue field κ_ν of mult_ν, and the residue field κ_F of mult_F is $\kappa_\nu(R_\tau)$.

We have $\operatorname{mult}_F z,\operatorname{mult}_F t\geq\varrho$ and $\operatorname{mult}_F\varphi=\varrho\operatorname{mult}_\nu\varphi\geq\varrho$, where $\operatorname{mult}_\nu 0=+\infty$. Since $x=u-\varphi$, the definition of the Gauss extension implies $\operatorname{mult}_F x=\min\{w,\varrho\operatorname{mult}_\nu\varphi\}=k\geq 1$. Since $xy=-g$, $\operatorname{mult}_F y=\varrho\operatorname{mult}_\nu g-k>0$. Thus mult_F is non-negative on the coordinate ring of X and positive on x,y,z,t, so F has center 0 on X. By (3.15), $a(F,X,0)=\varrho a(\nu,T,0)+w-k$.

We compute $I_G(F)$. Let $\gamma\in I_G(F)$. Since $\operatorname{mult}_F\zeta=\varrho\operatorname{mult}_\nu\zeta$ for any non-zero rational function ζ on T, γ preserves mult_ν. Since $\kappa_\nu\subset\kappa_F$ and γ acts trivially on κ_F, γ acts trivially on κ_ν. Hence $\gamma\in J_\nu$. Conversely, let $\gamma\in J_\nu$. For rational functions a_i on T,

$$\gamma^*\Big(\sum_i a_iu^i\Big)=\sum_i\gamma^*(a_i)\beta(\gamma)^iu^i$$

and $\operatorname{mult}_\nu\gamma^*(a_i)=\operatorname{mult}_\nu a_i$, so γ preserves mult_F by the definition of the Gauss extension. Let τ_0 be a uniformizer of mult_ν. Since J_ν acts trivially on the residue field of mult_ν by the definition of $I_G(\nu)$, $\delta^*\tau_0-\alpha_\nu(\delta)\tau_0\in\mathfrak{m}_\nu^2$ for any $\delta\in J_\nu$. By the average (3.12), applied to J_ν, α_ν, $\ell=1$, and τ_0, there is a uniformizer τ of mult_ν with $\delta^*\tau=\alpha_\nu(\delta)\tau$ for any $\delta\in J_\nu$. Then

$$\gamma^*\Big(\frac{u^\varrho}{\tau^w}\Big)=\beta(\gamma)^\varrho\alpha_\nu(\gamma)^{-w}\frac{u^\varrho}{\tau^w},$$

so γ maps R_τ to $\beta(\gamma)^\varrho\alpha_\nu(\gamma)^{-w}R_\tau$. Since γ acts trivially on κ_ν, $R_\tau\neq 0$, and $\kappa_F=\kappa_\nu(R_\tau)$, γ acts trivially on κ_F if and only if $\beta(\gamma)^\varrho\alpha_\nu(\gamma)^{-w}=1$. Therefore

$$I_G(F)=\operatorname{Ker}(\beta^\varrho\alpha_\nu^{-w}:J_\nu\to\mathbb{C}^*).$$

(2) Let $c:=\operatorname{mult}_F u$. Since $u\in(x,z,t)$ and F has center 0 on X, $c>0$. Let $h:T'\to T$ be a higher model on which ν is a prime divisor E_ν, and let Z be the center of F on $T'\times\mathbb{A}^1_u$, which exists because $T'\times\mathbb{A}^1_u\to T\times\mathbb{A}^1_u$ is proper. Since $\operatorname{mult}_F\zeta=\varrho\operatorname{mult}_\nu\zeta$ for any non-zero rational function ζ on T, the projection $T'\times\mathbb{A}^1_u\to T'$ maps Z onto the center E_ν of ν on T'. Since $c>0$, $Z\subset T'\times\{0\}$. Thus $Z=E_\nu\times\{0\}$. Let η be the generic point of $E_\nu\times\{0\}$ and τ a local equation of E_ν at the generic point of E_ν. Possibly shrinking T' to a neighborhood of the generic point of E_ν, T' is smooth, $E_\nu=\operatorname{div}(\tau)$, and τ,u are regular parameters of $T'\times\mathbb{A}^1_u$ at η. Since $K_{T'}=h^*K_T+(a(\nu,T,0)-1)E_\nu$ and $\operatorname{mult}_F\tau=\varrho$, we have

$$a(F,T\times\mathbb{A}^1_u,0)=a(F,T'\times\mathbb{A}^1_u,0)+\varrho(a(\nu,T,0)-1).$$

By [Liu26, Lemma 2.13], $a(F,T'\times\mathbb{A}^1_u,0)\geq\operatorname{mult}_F\tau+\operatorname{mult}_F u=\varrho+c$. By [JM12, Proposition 5.1 and Corollary 5.4], applied to the simple normal crossing divisor $\operatorname{div}(\tau)+\operatorname{div}(u)$ near η, equality holds if and only if mult_F is the monomial valuation $v_{\varrho,c}$ in τ,u at η with weights (ϱ,c), since the retraction of mult_F is $v_{\varrho,c}$ [JM12, §4.3]. By (3.15),

$$a(F,X,0)=a(F,T'\times\mathbb{A}^1_u,0)+\varrho(a(\nu,T,0)-1)-\operatorname{mult}_F x\geq\varrho a(\nu,T,0)+c-\operatorname{mult}_F x,$$

with equality if and only if $\operatorname{mult}_F=v_{\varrho,c}$.

It remains to show that $v_{\varrho,c}$ is the Gauss extension of $\varrho\operatorname{mult}_\nu$ with u-value c. Let $a_0,\dots,a_m$ be rational functions on T, not all zero. For any j with $a_j\neq 0$, write $a_j=\tau^{\operatorname{mult}_\nu a_j}\epsilon_j$, where ϵ_j is a unit at the generic point of E_ν, hence at η. In the expansion of $\sum_j a_ju^j$ in the completion of

the local ring at η with respect to τ, u, the monomials $\tau^{\mathrm{mult}_\nu a_j} u^j$ have distinct u-degrees, and any other monomial occurring in $\epsilon_j \tau^{\mathrm{mult}_\nu a_j} u^j$ has weight larger than $\varrho\,\mathrm{mult}_\nu a_j + cj$. Thus

$$v_{\varrho,c}\Big(\sum_j a_j u^j\Big) = \min_j(\varrho\,\mathrm{mult}_\nu a_j + cj),$$

and $v_{\varrho,c}$ is the Gauss extension of $\varrho\,\mathrm{mult}_\nu$ with u-value c. This proves that equality implies that mult_F is the Gauss extension of $\varrho\,\mathrm{mult}_\nu$ with u-value c. Conversely, assume that mult_F is the Gauss extension of $\varrho\,\mathrm{mult}_\nu$ with u-value c. Since the values of the Gauss extension of $\varrho\,\mathrm{mult}_\nu$ with u-value c belong to $\varrho\mathbb{Z} + c\mathbb{Z}$ and mult_F takes the value 1, ϱ and c are coprime. By [Liu26, Lemma 18.1], $a(F, T \times \mathbb{A}^1_u, 0) = \varrho a(\nu, T, 0) + c$, and equality follows from (3.15).

The last statement follows from (1) and (2) applied after interchanging x and y, which interchanges the characters of x and y, since X is symmetric in x and y and G is diagonal. □

[Liu26, Theorem 15.9(3)] proves the upper bound $\alpha \le b_H + \frac{2}{\sqrt{r}}$ under the assumption $\alpha \ge 1+\epsilon$, which is used there to bound the degrees of the equations in **Step 1** of the proof and to make the pair (T, B_T) klt in **Step 2** [Liu26, Theorem 15.9(2)]. We need the upper bound for all positive values, since it is used for quotient values in $(0,1)$ in the proof of Theorem 8.3. In Proposition 3.14, the bound on the equations is the normal form hypothesis with the constant C, and the pair (T, B_T) is only assumed to be lc.

Proposition 3.14. *Fix an integer $e_0 \ge 1$ and a DCC set $\Gamma \subset [0,1]$. Let $X := (xy - g(z,t) = 0) \subset \mathbb{A}^4$ with $0 \ne g \in (z,t)^2$ be normal and klt near 0, let G be a finite diagonal abelian group preserving $xy - g$ up to a character and acting faithfully on X, and let $R := G \cap \mathbf{T}_{(1,-1,0,0)} = \mu_r(1,-1,0,0)$ and $H := G/R$ with $|H| \le e_0$. Let $B := \sum_j c_j \operatorname{div}_X(h_j)$ with positive $c_j \in \Gamma$ and G-semi-invariant polynomials h_j with non-zero restrictions to X, such that (X, B) is klt near 0, and let $\alpha := \mathrm{mld}((X \ni 0, B)/G)$. Write each h_j modulo $xy - g$ as $A_{j,0} + \sum_{k\ge1} x^k A_{j,k} + \sum_{l\ge1} y^l A_{j,-l}$ with $A_{j,k} \in \mathbb{C}[z,t]$. Let $C \ge 0$ be an integer, and assume that for each j there are integers $a_j, b_j \ge 0$ with $\min\{a_j, b_j\} = 0$ and $|a_j - b_j| \le C$ such that $F_j := A_{j,a_j-b_j} \ne 0$ and every integer $k \ne a_j - b_j$ with $A_{j,k} \ne 0$ satisfies $k \equiv a_j - b_j \pmod r$ and $|k| \ge r - C$. Let $c_x := \sum_j c_j a_j$, $c_y := \sum_j c_j b_j$, $c_g := \max\{c_x, c_y\}$, $T := \mathbb{A}^2_{z,t}$, and*

$$B_T := \sum_j c_j \operatorname{div}_T(F_j) + c_g \operatorname{div}_T(g).$$

Assume that (T, B_T) is lc near 0. Then there exists r_0, depending only on e_0, Γ, and C, such that if $r \ge r_0$, then

$$\alpha \le b_H + \frac{2}{\sqrt{r}}, \qquad b_H := \mathrm{mld}((T \ni 0, B_T)/H).$$

Proof. We fix the constants first. Let $\epsilon := \min(\Gamma\backslash\{0\})$ if $\Gamma\backslash\{0\} \ne \emptyset$, and $\epsilon := 1$ otherwise. Let Γ_+ and $\Psi(\cdot)$ be as in [Liu26, Definition 2.9], and let $\Gamma' := \Psi(\Gamma_+ \cap [0,1])$. By [Liu26, Lemma 2.10(1)], applied to Γ and then to $\Gamma_+ \cap [0,1]$, Γ' is a DCC set depending only on Γ. Let L_0 be the positive integer of Theorem 2.9(2) for the DCC set Γ'. Let r_0 be a positive integer such that any integer $r \ge r_0$ satisfies

$$r > C, \qquad 4r \ge 9e_0^2, \qquad \frac{r - C}{\sqrt{r}} > \frac{(C+1)L_0}{\epsilon}.$$

Then r_0 depends only on e_0, Γ, and C. Assume that $r \ge r_0$, and let $\eta := \frac{1}{\sqrt{r}}$.

Step 1. In this step, we pick a prime divisor ν over T which computes b_H and has bounded log discrepancy. Let $\sigma \in G$ act trivially on T. Since σ preserves $xy - g$ up to a character and $\sigma^* g = g \ne 0$, we have $\sigma^*(xy) = xy$, so $\sigma \in G \cap \mathbf{T}_{(1,-1,0,0)} = R$. Thus H acts faithfully on T.

The polynomials F_j and g are G-semi-invariant, since replacing xy by g preserves characters, so B_T is H-invariant. Possibly shrinking T to an H-invariant neighborhood of 0, we may assume that (T, B_T) is lc. For a prime divisor D on T, the coefficient of B_T along D is $\sum_j c_j \mathrm{mult}_D F_j +$

$c_g \operatorname{mult}_D g$, which is a finite sum of elements of Γ because c_g is one of c_x and c_y, and which is at most one because (T, B_T) is lc. Thus $B_T \in \Gamma_+ \cap [0,1]$. Let $f : T \to T/H$ be the quotient map, $\bar{0} := f(0)$, and $B_{T/H}$ the $\mathbb{R}$-divisor with $K_T + B_T = f^*(K_{T/H} + B_{T/H})$. By [Liu26, Lemma 2.6(4)], $B_{T/H} \in \Gamma'$, and $(T/H, B_{T/H})$ is lc near $\bar{0}$ [Liu26, Lemma 2.6(2)(3)]. By Theorem 2.9(2), there exists a prime divisor P over T/H with $\operatorname{center}_{T/H} P = \{\bar{0}\}$, $a(P, T/H, B_{T/H}) = b_H$, and $a(P, T/H, 0) \le L_0$.

By [Liu26, Lemma 2.6(3)], there exists a prime divisor ν over T over P. Then ν has center 0, since $f^{-1}(\bar{0}) = \{0\}$. Let $e_\nu := |I_H(\nu)| \le e_0$. Let $B^0 \ge 0$ be the $\mathbb{R}$-divisor with $K_T = f^*(K_{T/H} + B^0)$. By [Liu26, Lemma 2.6(2)], applied to B_T and to 0,

$$\begin{aligned} \frac{a(\nu, T, B_T)}{e_\nu} &= a(P, T/H, B_{T/H}) = b_H \ge 0, \\ \frac{a(\nu, T, 0)}{e_\nu} &= a(P, T/H, B^0) \le a(P, T/H, 0) \le L_0. \end{aligned} \tag{3.16}$$

Step 2. In this step, we bound the orders of g and of the F_j along ν. By (3.16),

$$\frac{\operatorname{mult}_\nu B_T}{e_\nu} = \frac{a(\nu, T, 0) - a(\nu, T, B_T)}{e_\nu} \le L_0.$$

Since $B_T \ge c_j \operatorname{div}_T(F_j)$ and $c_j \ge \epsilon$, we have $\operatorname{mult}_\nu F_j / e_\nu \le L_0/\epsilon$ for any j. If $a_j + b_j > 0$ for some j, then $c_g \ge c_j \ge \epsilon$, and $B_T \ge c_g \operatorname{div}_T(g)$ implies $\operatorname{mult}_\nu g / e_\nu \le L_0/\epsilon$. Since $g \in (z,t)^2$ and ν has center 0, $\operatorname{mult}_\nu g \ge 2$, so

$$\frac{\operatorname{mult}_\nu g}{e_\nu} \ge \frac{2}{e_0}. \tag{3.17}$$

Let E_0 be the exceptional divisor of the blow-up of $0 \in T$. Since (T, B_T) is lc,

$$0 \le a(E_0, T, B_T) = 2 - \operatorname{mult}_0 B_T \le 2 - c_g \operatorname{mult}_0 g \le 2 - 2c_g.$$

Thus $c_g \le 1$, and $|c_x - c_y| \le c_g \le 1$ because $c_x, c_y \ge 0$.

Step 3. In this step, we pick a lattice vector. Let $J := I_G(\nu)$ for the action of G on T, which is the inverse image of $I_H(\nu)$ in G, and let α_ν be the character of J on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ and $\beta : G \to \mathbb{C}^*$ the character of x. Since H acts faithfully on T, $I_H(\nu)$ acts faithfully on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ [Liu26, Lemma 2.6(1)]. Thus α_ν has order e_ν on J and kernel R. Since β is faithful on R, the homomorphism $(\alpha_\nu, \beta) : J \to (\mathbb{C}^*)^2$ is injective. Identify the roots of unity in $\mathbb{C}^*$ with $\mathbb{Q}/\mathbb{Z} \subset \mathbb{R}/\mathbb{Z}$ through $s \to \exp(2\pi\sqrt{-1}s)$, let $\bar{J} \subset \mathbb{T}^2$ be the image of J under (α_ν, β), and let $N_J := N_{\bar{J}} \subset \mathbb{R}^2$ be the lattice of [Liu26, Definition 2.15], the inverse image of $\bar{J}$ under $\mathbb{R}^2 \to \mathbb{T}^2$. The image of N_J under the first projection is $\frac{1}{e_\nu}\mathbb{Z}$, because $\alpha_\nu(J)$ is the group of e_ν-th roots of unity. A vector $(0, b)$ belongs to N_J if and only if $\exp(2\pi\sqrt{-1}b) = \beta(\sigma)$ for some $\sigma \in J$ with $\alpha_\nu(\sigma) = 1$, that is, for some $\sigma \in R$. Since $\beta(R)$ is the group of r-th roots of unity, the kernel of the first projection on N_J is $\{0\} \times \frac{1}{r}\mathbb{Z}$. Thus there exists $v_0 \in \mathbb{Q}$ such that $(\frac{1}{e_\nu}, v) \in N_J$ if and only if $v \in v_0 + \frac{1}{r}\mathbb{Z}$.

Any half-open interval of length $\frac{1}{r}$ contains exactly one element of $v_0 + \frac{1}{r}\mathbb{Z}$. If $c_x \ge c_y$, let $v_x \in v_0 + \frac{1}{r}\mathbb{Z}$ be the element with $\frac{\operatorname{mult}_\nu g}{e_\nu} - v_x \in [\eta, \eta + \frac{1}{r})$. If $c_x < c_y$, let $v_x \in v_0 + \frac{1}{r}\mathbb{Z}$ be the element with $v_x \in [\eta, \eta + \frac{1}{r})$. Let

$$v_y := \frac{\operatorname{mult}_\nu g}{e_\nu} - v_x, \qquad \theta := \begin{cases} v_y & \text{if } c_x \ge c_y, \\ v_x & \text{if } c_x < c_y. \end{cases}$$

Then $\eta \le \theta < \eta + \frac{1}{r} \le 2\eta$. Since $4r \ge 9e_0^2$, we have $3\eta \le \frac{2}{e_0}$, and (3.17) implies

$$\frac{\operatorname{mult}_\nu g}{e_\nu} - \theta \ge \frac{2}{e_0} - 2\eta \ge \eta.$$

Thus

$$v_x \geq \eta, \qquad v_y \geq \eta, \qquad v_x + v_y = \frac{\operatorname{mult}_\nu g}{e_\nu}. \tag{3.18}$$

Since $N_J \subset \mathbb{Q}^2$, v_x is a positive rational number. Let (ϱ, w) be the primitive vector of $\mathbb{Z}^2$ in $\mathbb{R}_{>0}(\frac{1}{e_\nu}, v_x)$. Since (ϱ, w) is primitive, ϱ and w are coprime. Let

$$e' := |\operatorname{Ker}(\beta^\varrho \alpha_\nu^{-w} : J \to \mathbb{C}^*)|.$$

We show that $N_J \cap \mathbb{R}(\varrho, w) = \frac{1}{e'}\mathbb{Z}(\varrho, w)$. For $(s_1, s_2) \in \mathbb{R}^2$, we have $(s_1, s_2) \in \mathbb{R}(\varrho, w) + \mathbb{Z}^2$ if and only if $ws_1 - \varrho s_2 \in \mathbb{Z}$, because ϱ and w are coprime. Let $(s_1, s_2) \in N_J$ and let $\sigma \in J$ be the element with

$$(\alpha_\nu(\sigma), \beta(\sigma)) = (\exp(2\pi\sqrt{-1}s_1), \exp(2\pi\sqrt{-1}s_2)).$$

Then $ws_1 - \varrho s_2 \in \mathbb{Z}$ if and only if $\beta(\sigma)^\varrho \alpha_\nu(\sigma)^{-w} = 1$. Since (α_ν, β) is injective on J, the group $(N_J \cap (\mathbb{R}(\varrho, w) + \mathbb{Z}^2))/\mathbb{Z}^2$ has order e'. The map

$$\mathbb{R} \to (\mathbb{R}(\varrho, w) + \mathbb{Z}^2)/\mathbb{Z}^2, \qquad s \to s(\varrho, w),$$

is surjective with kernel $\mathbb{Z}$, because $\mathbb{R}(\varrho, w) \cap \mathbb{Z}^2 = \mathbb{Z}(\varrho, w)$. Since $\mathbb{Z}^2 \subset N_J$, $s(\varrho, w) \in N_J$ if and only if the class of $s(\varrho, w)$ belongs to $(N_J \cap (\mathbb{R}(\varrho, w) + \mathbb{Z}^2))/\mathbb{Z}^2$, the subgroup of order e' of

$$\mathbb{R}(\varrho, w)/\mathbb{Z}(\varrho, w) \cong \mathbb{R}/\mathbb{Z}.$$

Thus $s(\varrho, w) \in N_J$ if and only if $s \in \frac{1}{e'}\mathbb{Z}$. Since $(\frac{1}{e_\nu}, v_x) \in N_J \cap \mathbb{R}_{>0}(\varrho, w)$, there exists a positive integer q such that

$$\left(\frac{1}{e_\nu}, v_x\right) = \frac{q}{e'}(\varrho, w). \tag{3.19}$$

By (3.18), $v_x < \frac{\operatorname{mult}_\nu g}{e_\nu}$, so $w < \varrho \operatorname{mult}_\nu g$. With $\varphi = 0$, ϱ and w satisfy the hypotheses of Proposition 3.13(1), where $k = w$.

Step 4. In this step, we construct a prime divisor over X and conclude. After the substitution $y \to -y$ of the beginning of this subsection, X becomes $(xy + g = 0)$. By Proposition 3.13(1), applied with $\varphi = 0$, $u = x$, and $k = w$ (here $J = J_\nu$), the Gauss extension of $\varrho \operatorname{mult}_\nu$ with x-value w is mult_E for a prime divisor E over X with center 0, and

$$\operatorname{mult}_E x = w, \qquad \operatorname{mult}_E y = \varrho \operatorname{mult}_\nu g - w, \qquad a(E, X, 0) = \varrho a(\nu, T, 0),$$
$$|I_G(E)| = |\operatorname{Ker}(\beta^\varrho \alpha_\nu^{-w} : J \to \mathbb{C}^*)| = e'.$$

Moreover, $\operatorname{mult}_E \zeta = \varrho \operatorname{mult}_\nu \zeta$ for any non-zero $\zeta \in \mathbb{C}[z, t]$ by the definition of the Gauss extension. By (3.18) and (3.19), $\frac{q}{e'}(\varrho \operatorname{mult}_\nu g - w) = v_y$. Thus, for any non-zero $\zeta \in \mathbb{C}[z, t]$ and integers $k, l \geq 0$,

$$\frac{q}{e'} \operatorname{mult}_E(x^k y^l \zeta) = kv_x + lv_y + \frac{\operatorname{mult}_\nu \zeta}{e_\nu}. \tag{3.20}$$

Fix j. By (3.20), the term $x^{a_j} y^{b_j} F_j$ of the normal form of h_j satisfies

$$\frac{q}{e'} \operatorname{mult}_E(x^{a_j} y^{b_j} F_j) = a_j v_x + b_j v_y + \frac{\operatorname{mult}_\nu F_j}{e_\nu} =: P_j.$$

Since $\min\{a_j, b_j\} = 0$, $\max\{a_j, b_j\} = |a_j - b_j| \leq C$, and $v_x, v_y \leq \frac{\operatorname{mult}_\nu g}{e_\nu}$, **Step 2** implies

$$P_j \leq |a_j - b_j| \frac{\operatorname{mult}_\nu g}{e_\nu} + \frac{\operatorname{mult}_\nu F_j}{e_\nu} \leq \frac{CL_0}{\epsilon} + \frac{L_0}{\epsilon} = \frac{(C+1)L_0}{\epsilon},$$

where the first term is zero if $a_j = b_j = 0$. Any other term of the normal form of h_j is $x^k A_{j,k}$ with $k \geq r - C$ or $y^l A_{j,-l}$ with $l \geq r - C$, and $r - C > 0$. By (3.20) and (3.18), the product

of $\frac{q}{e'}$ and the order along E of such a term is at least $(r-C)\eta = \frac{r-C}{\sqrt{r}}$, which is larger than $\frac{(C+1)L_0}{\epsilon} \geq P_j$. Thus

$$\frac{q}{e'} \operatorname{mult}_E h_j = P_j.$$

We have $\sum_j c_j(a_j v_x + b_j v_y) = c_x v_x + c_y v_y$. If $c_x \geq c_y$, then $c_g = c_x$, $v_y = \theta$, and $v_x = \frac{\operatorname{mult}_\nu g}{e_\nu} - \theta$. If $c_x < c_y$, then $c_g = c_y$, $v_x = \theta$, and $v_y = \frac{\operatorname{mult}_\nu g}{e_\nu} - \theta$. In both cases,

$$c_x v_x + c_y v_y = c_g \frac{\operatorname{mult}_\nu g}{e_\nu} - |c_x - c_y|\theta.$$

Since $\operatorname{mult}_\nu B_T = \sum_j c_j \operatorname{mult}_\nu F_j + c_g \operatorname{mult}_\nu g$, we obtain $\sum_j c_j P_j = \frac{\operatorname{mult}_\nu B_T}{e_\nu} - |c_x - c_y|\theta$. By (3.19), $\frac{q}{e'} a(E, X, 0) = \frac{a(\nu, T, 0)}{e_\nu}$. Therefore, by (3.16) and **Step 2**,

$$\frac{q}{e'} a(E, X, B) = \frac{a(\nu, T, 0)}{e_\nu} - \sum_j c_j P_j = \frac{a(\nu, T, B_T)}{e_\nu} + |c_x - c_y|\theta = b_H + |c_x - c_y|\theta \leq b_H + 2\eta.$$

Since (X, B) is klt near 0 and E has center 0, $a(E, X, B) > 0$. By the formula for the quotient minimal log discrepancy at the end of [Liu26, Lemma 2.6],

$$\alpha \leq \frac{a(E, X, B)}{|I_G(E)|} = \frac{a(E, X, B)}{e'} \leq \frac{q}{e'} a(E, X, B) \leq b_H + \frac{2}{\sqrt{r}}. \qquad \square$$

3.4. **Morse splittings and normal forms.** In this subsection, we split off a non-degenerate binary quadratic form from the equation of a hypersurface germ in a smooth fourfold germ by equivariant morphisms which are étale at a point, prove that hypersurface germs whose quadratic part has rank at least three are normal and canonical, and construct equivariant normal forms for residual cubics with a cusp or with a tangency.

In the proofs of this subsection, all shrinkings are to G-invariant open neighborhoods of the marked points. Such neighborhoods exist since G is finite and fixes the marked points.

For quadratic rank at least two, Proposition 3.15 below provides the G-equivariant normal forms $Q_2(x, y) + g$ with $Q_2 \in \{xy, x^2 + y^2\}$ used later. Both forms occur: for G-semi-invariant linear forms u_1, u_2 with $\chi_{u_1} \neq \chi_{u_2}$, the form $u_1 u_2$ is not a sum of squares of two G-semi-invariant linear combinations of u_1, u_2, and $u_1^2 + u_2^2$ is not a product of two such combinations. Unlike [Liu26, Theorem 9.1(3)], the proposition does not assume that the germ is normal and klt, and by Proposition 3.15(2) the germ is normal and klt if and only if the normal form is.

Proposition 3.15. *Let $(Y \ni o)$ be a smooth fourfold germ with an action of a finite abelian group G fixing o, f a G-semi-invariant regular function on Y with $f \in \mathfrak{m}_{Y,o}^2$, $X := (f = 0)$, and u_1, u_2, z, t G-eigenparameters at o. Assume that the restriction of the quadratic part of f to the (u_1, u_2)-plane is non-degenerate. Then there exist a smooth surface germ $(V \ni v)$ with an action of G fixing v and G-eigenparameters z, t of V at v, a diagonal action of G on $\mathbb{A}^2_{x,y}$, $Q_2 \in \{xy, x^2 + y^2\}$, a regular G-semi-invariant function g on V vanishing to order at least two at v, and a germ $(X' \ni o')$ with an action of G and G-equivariant morphisms $(X' \ni o') \to (X \ni o)$ and $(X' \ni o') \to (X_0 \ni o_0)$ étale at o', where*

$$X_0 := (Q_2(x, y) + g = 0) \subset \mathbb{A}^2_{x,y} \times V \quad \text{and} \quad o_0 := (0, 0, v),$$

such that the pullbacks to X' of z and t on X coincide with those of z and t on X_0. Moreover the following hold.

1. *If X is normal near o, then $g \neq 0$.*
2. *X is normal and klt near o if and only if X_0 is normal and klt near o_0.*
3. *If X is normal near o, then for any real numbers b_z, b_t the sub-pair $(X, b_z \operatorname{div}(z) + b_t \operatorname{div}(t))$ is sub-klt near o if and only if $(X_0, b_z \operatorname{div}(z) + b_t \operatorname{div}(t))$ is sub-klt near o_0.*

(4) *Under the diagonal action of G on $\mathbb{A}^2_{x,y}$, the coordinates x and y have the characters of u_1 and u_2 respectively, and there exists a G-equivariant isomorphism*

$$\widehat{\mathcal{O}}_{X,o} \cong \mathbb{C}[[x,y,z,t]]/(Q_2(x,y)+\hat{g}(z,t)),$$

where $\hat{g} \in \mathbb{C}[[z,t]]$ is the expansion of g at v, G acts diagonally on $\mathbb{C}[[x,y,z,t]]$ with the characters of u_1,u_2,z,t on x,y,z,t, and z,t on the left correspond to z,t on the right.

Proof. Let Q be the symmetric 2×2 matrix such that u^TQu is the restriction of the quadratic part of f to the (u_1,u_2)-plane, where $u := (u_1,u_2)^T$. For $\gamma \in G$, let

$$\rho(\gamma) := \operatorname{diag}(\chi_{u_1}(\gamma), \chi_{u_2}(\gamma)).$$

Since u^TQu is G-semi-invariant with character χ_f,

$$\rho(\gamma)^TQ\rho(\gamma) = \chi_f(\gamma)Q \quad \text{for any } \gamma \in G. \tag{$*$}$$

Step 1. We construct V. Possibly shrinking Y, we may assume that $(u_1,u_2,z,t) : Y \to \mathbb{A}^4$ is étale and that o is the only point of Y over 0. Let $\partial_{u_1}, \partial_{u_2}$ be the derivations of $\mathcal{O}_Y$ with $\partial_{u_i}(u_j) = \delta_{ij}$ and $\partial_{u_i}(z) = \partial_{u_i}(t) = 0$. Since

$$\partial_{u_i}(\gamma^*k) = \chi_{u_i}(\gamma)\gamma^*(\partial_{u_i}k)$$

for any $\gamma \in G$ and any regular function k, the function $\partial_{u_i}f$ is G-semi-invariant with character $\chi_f\chi_{u_i}^{-1}$. Modulo $\mathfrak{m}^2_{Y,o}$, the vector $(\partial_{u_1}f, \partial_{u_2}f)^T$ is $2Qu$ plus a vector of linear forms in z,t. Since Q is invertible,

$$V := (\partial_{u_1}f = \partial_{u_2}f = 0)$$

is a G-invariant smooth surface germ at $v := o$, and $(z,t)|_V : V \to \mathbb{A}^2_{z,t}$ is étale at v and G-equivariant for the diagonal action of G on $\mathbb{A}^2_{z,t}$ with the characters of z,t. In particular, $z|_V, t|_V$ are G-eigenparameters of V at v, which we denote by z,t. Possibly shrinking V, we may assume that $V \to \mathbb{A}^2_{z,t}$ is étale and that v is the only point of V over 0. Let $g := f|_V$. Then g is G-semi-invariant with character χ_f, and g vanishes to order at least two at v as $f \in \mathfrak{m}^2_{Y,o}$.

Step 2. We write $f-g$ as an equivariant quadratic form on a fiber product. Let

$$Y_1 := Y \times_{\mathbb{A}^2_{z,t}} V,$$

with projections $p_1 : Y_1 \to Y$ and $p_2 : Y_1 \to V$, and let $o_1 := (o,v)$. The group G acts diagonally on Y_1 and fixes o_1, the projections are G-equivariant, and p_1 is étale. Let

$$\delta_i := p_1^*u_i - p_2^*(u_i|_V) \quad \text{for } i=1,2, \quad \text{and} \quad \delta := (\delta_1,\delta_2)^T.$$

Then δ_i is G-semi-invariant with character χ_{u_i}, that is, $\gamma^*\delta = \rho(\gamma)\delta$, and $\delta(o_1) = 0$.

Since the morphisms (u_1,u_2,z,t), $V \to \mathbb{A}^2_{z,t}$, and p_1 are étale at the marked points, we have $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[u_1,u_2,z,t]]$ and $\widehat{\mathcal{O}}_{V,v} = \mathbb{C}[[z,t]]$, p_1^* induces an isomorphism from $\widehat{\mathcal{O}}_{Y,o}$ to $\widehat{\mathcal{O}}_{Y_1,o_1}$, and under the isomorphism p_2^* is the inclusion $\mathbb{C}[[z,t]] \subset \mathbb{C}[[u_1,u_2,z,t]]$. In $\mathbb{C}[[u_1,u_2,z,t]]$, $\partial_{u_i}f = \partial f/\partial u_i$, and the Jacobian matrix of $(\partial f/\partial u_1, \partial f/\partial u_2)$ with respect to u_1,u_2 is $2Q$ at 0. By the formal implicit function theorem, the ideal generated by $\partial f/\partial u_1$ and $\partial f/\partial u_2$ is $(u_1-\xi_1, u_2-\xi_2)$ for some $\xi_1,\xi_2 \in (z,t)\mathbb{C}[[z,t]]$. Hence $\delta_i = u_i - \xi_i$, $g = f(\xi_1,\xi_2,z,t)$, and the Taylor expansion of f in $u_1-\xi_1, u_2-\xi_2$ over $\mathbb{C}[[z,t]]$ together with $(\partial f/\partial u_i)(\xi_1,\xi_2,z,t) = 0$ implies

$$p_1^*f - p_2^*g \in (\delta_1,\delta_2)^2\widehat{\mathcal{O}}_{Y_1,o_1}.$$

Since $\widehat{\mathcal{O}}_{Y_1,o_1}$ is faithfully flat over $\mathcal{O}_{Y_1,o_1}$ [Mat89, Theorems 8.14 and 7.5(ii)], we have

$$p_1^*f - p_2^*g \in (\delta_1,\delta_2)^2\mathcal{O}_{Y_1,o_1}.$$

Thus, possibly shrinking Y_1, there exists a symmetric 2×2 matrix A of regular functions on Y_1 with $p_1^*f - p_2^*g = \delta^T A\delta$. Let

$$\bar{A} := \frac{1}{|G|}\sum_{\gamma\in G}\chi_f(\gamma)^{-1}\rho(\gamma)^T(\gamma^*A)\rho(\gamma),$$

where γ^*A is the matrix of the pullbacks of the entries of A. Then $\bar{A}$ is symmetric, and

$$\delta^T\bar{A}\delta = \frac{1}{|G|}\sum_{\gamma\in G}\chi_f(\gamma)^{-1}\gamma^*(\delta^TA\delta) = \frac{1}{|G|}\sum_{\gamma\in G}\chi_f(\gamma)^{-1}\gamma^*(p_1^*f-p_2^*g) = p_1^*f - p_2^*g.$$

For $\gamma'\in G$, replacing γ by $\gamma\gamma'^{-1}$ in the sum and using that G is abelian and ρ is diagonal, we obtain

$$\gamma'^*\bar{A} = \chi_f(\gamma')\rho(\gamma')^{-1}\bar{A}\rho(\gamma')^{-1}.$$

Moreover $\bar{A}(o_1) = Q$, since on the plane $(dz = dt = 0)$ of the tangent space of Y_1 at o_1, the quadratic part of p_1^*f restricts to u^TQu, the quadratic part of p_2^*g restricts to zero since $g\in(z,t)^2\mathbb{C}[[z,t]]$, and $d\delta_i$ restricts to du_i since $\xi_i\in(z,t)$.

Step 3. We take an equivariant square root. Let $\Lambda := Q^{-1}\bar{A}$. Then $\Lambda(o_1)$ is the identity matrix Id, $\Lambda^T = Q\Lambda Q^{-1}$, and **Step 2** and $(*)$ imply

$$\gamma^*\Lambda = \rho(\gamma)\Lambda\rho(\gamma)^{-1}\quad\text{for any }\gamma\in G.$$

Let $M_2\cong\mathbb{A}^4$ be the space of 2×2 matrices, on which G acts by $\gamma\cdot M := \rho(\gamma)M\rho(\gamma)^{-1}$, and let $\mathrm{sq}: M_2\to M_2$ be the morphism $M\to M^2$. Then $\Lambda: Y_1\to M_2$ and sq are G-equivariant, and sq is étale at Id, since the differential of sq at Id is the multiplication by 2 on M_2. Let

$$Y_2 := Y_1\times_{M_2}M_2,\quad o_2 := (o_1,\mathrm{Id}),$$

the fiber product of Λ and sq, with the diagonal action of G, and let $p: Y_2\to Y_1$ be the first projection. Then p is G-equivariant and étale at o_2, and the second projection is a matrix R of regular functions on Y_2 with

$$R^2 = p^*\Lambda,\quad R(o_2)=\mathrm{Id},\quad \gamma^*R = \rho(\gamma)R\rho(\gamma)^{-1}\quad\text{for any }\gamma\in G.$$

We claim that $R^TQ = QR$. Let $\widehat{\mathfrak{m}}$ be the maximal ideal of $\widehat{\mathcal{O}}_{Y_2,o_2}$. If M, M' are 2×2 matrices over $\widehat{\mathcal{O}}_{Y_2,o_2}$ with $M^2 = M'^2$ and $M\equiv M'\equiv\mathrm{Id}\mod\widehat{\mathfrak{m}}$, then $N := M - M'$ satisfies

$$2N = -(M-\mathrm{Id})N - N(M'-\mathrm{Id}),$$

so the entries of N belong to $\widehat{\mathfrak{m}}^{k+1}$ whenever they belong to $\widehat{\mathfrak{m}}^k$, and $N=0$. The matrix $R' := Q^{-1}R^TQ$ satisfies $R'^2 = Q^{-1}(p^*\Lambda)^TQ = p^*\Lambda$ and $R'(o_2)=\mathrm{Id}$. Thus $R' = R$ in $\widehat{\mathcal{O}}_{Y_2,o_2}$, hence in $\mathcal{O}_{Y_2,o_2}$, which proves the claim. Hence $R^TQR = QR^2 = p^*\bar{A}$. Let $w := Rp^*\delta = (w_1,w_2)^T$. Then

$$p^*(p_1^*f - p_2^*g) = (p^*\delta)^TR^TQR(p^*\delta) = w^TQw,$$

$\gamma^*w = \rho(\gamma)w$ for any $\gamma\in G$, that is, w_i is G-semi-invariant with character χ_{u_i}, and $dw_i = d(p^*\delta_i)$ at o_2.

Step 4. We reach Q_2. Let $Q = (Q_{ij})$. By $(*)$, $Q_{ij}\chi_{u_i}\chi_{u_j} = Q_{ij}\chi_f$ for $i,j\in\{1,2\}$. If $Q_{12}=0$, then $Q_{11}Q_{22}\neq 0$, and we let $x := Q_{11}^{1/2}w_1$, $y := Q_{22}^{1/2}w_2$, and $Q_2 := x^2+y^2$. If $Q_{11}=Q_{22}=0$, then $Q_{12}\neq 0$, and we let $x := 2Q_{12}w_1$, $y := w_2$, and $Q_2 := xy$. Otherwise $Q_{12}\neq 0$ and $Q_{ii}\neq 0$ for some i, so $\chi_{u_1}\chi_{u_2} = \chi_f = \chi_{u_i}^2$, that is, $\chi_{u_1} = \chi_{u_2}$. Then any linear combination of w_1, w_2 is G-semi-invariant, and we take linear combinations x, y of w_1, w_2 with $w^TQw = x^2+y^2$ and let $Q_2 := x^2+y^2$. In each case, x, y are G-semi-invariant, $(x,y)^T = Lw$ for an invertible constant matrix L, and $w^TQw = Q_2(x,y)$. Let G act on $\mathbb{A}^2_{x,y}$ diagonally with the characters of x, y. By $(*)$, each monomial of $Q_2(x,y)$ then has the character χ_f, so $Q_2(x,y)+g$ is G-semi-invariant and X_0 is G-invariant. Let

$$Y' := Y_2,\quad o' := o_2,\quad \psi := p_1\circ p,\quad \varphi := p_2\circ p.$$

Then

$$\psi^* f = Q_2(x,y) + \varphi^* g.$$

The morphism ψ is étale at o'. The functions $x, y, \psi^* z, \psi^* t$ are regular parameters at o', since at o' the span of dx, dy equals the span of $dw_1 = d(p^*\delta_1)$ and $dw_2 = d(p^*\delta_2)$, and $d(p^*\delta_i) - d(\psi^* u_i)$ is a linear combination of $d(\psi^* z), d(\psi^* t)$. Hence the morphism $Y' \to \mathbb{A}^4$ defined by $x, y, \psi^* z, \psi^* t$ is étale at o'. It is the composite of the G-equivariant morphism $(x, y, \varphi) : Y' \to \mathbb{A}^2_{x,y} \times V$ and the étale morphism $\mathbb{A}^2_{x,y} \times V \to \mathbb{A}^4$, so (x, y, φ) is étale at o'. Let

$$X' := \psi^{-1}(X) = (x, y, \varphi)^{-1}(X_0) \quad \text{and} \quad o_0 := (0, 0, v).$$

The restrictions $X' \to X$ of ψ and $X' \to X_0$ of (x, y, φ) are G-equivariant and étale at o', and $\psi^* z = \varphi^* z$ and $\psi^* t = \varphi^* t$ by the definition of Y_1. This proves the first part of the proposition.

Step 5. We prove (1)–(3). Since normality is local in the étale topology [Mat89, Corollary to Theorem 23.9] and the normal locus of a variety is open, X is normal near o if and only if X' is normal near o', if and only if X_0 is normal near o_0.

(1) Suppose that $g = 0$. Then $X_0 = (Q_2(x, y) = 0)$ is the union of two smooth hypersurfaces of $\mathbb{A}^2_{x,y} \times V$ through o_0, namely $(x = 0)$ and $(y = 0)$ if $Q_2 = xy$, and $(x + \sqrt{-1}y = 0)$ and $(x - \sqrt{-1}y = 0)$ if $Q_2 = x^2 + y^2$. Thus the local ring of X_0 at o_0 is not a domain, and X_0 is not normal at o_0. By the previous paragraph, X is not normal near o.

For (2) and (3), we show the following. Let $h : X_1 \to X_2$ be a morphism between normal varieties, étale on an open subset $U_1 \subset X_1$, and B_2 an $\mathbb{R}$-divisor on X_2 with $K_{X_2} + B_2$ $\mathbb{R}$-Cartier. Then $(U_1, h^* B_2)$ is sub-klt if and only if $(h(U_1), B_2)$ is sub-klt. The set $h(U_1)$ is open, and $K_{U_1} + h^* B_2 = h^*(K_{X_2} + B_2)$ on U_1, since canonical forms pull back to canonical forms under étale morphisms. Let $\mu : W \to X_2$ be a log resolution of (X_2, B_2), and write $K_W + B_W = \mu^*(K_{X_2} + B_2)$. Let $W_1 := W \times_{X_2} U_1$ and $\mu_1 : W_1 \to U_1$ the projection. Since $W_1 \to W$ is étale, μ_1 is a log resolution of $(U_1, h^* B_2)$, and $K_{W_1} + B_{W_1} = \mu_1^*(K_{U_1} + h^* B_2)$ for the pullback B_{W_1} of B_W. Any prime divisor D_1 on W_1 maps onto a dense open subset of a prime divisor D on W which intersects $\mu^{-1}(h(U_1))$, and $\operatorname{mult}_{D_1} B_{W_1} = \operatorname{mult}_D B_W$ as $W_1 \to W$ is étale. Conversely, any prime divisor D on W which intersects $\mu^{-1}(h(U_1))$ arises in this way. By [KM98, Lemma 2.30 and Corollary 2.31(3)], both $(U_1, h^* B_2)$ and $(h(U_1), B_2)$ are sub-klt if and only if $\operatorname{mult}_D B_W < 1$ for any prime divisor D on W which intersects $\mu^{-1}(h(U_1))$. This proves the claim (cf. [KM98, Proposition 5.20]). Since h is étale on a neighborhood of a point $q_1 \in X_1$, any open neighborhood of q_1 contains such a U_1, and $h(U_1)$ is an open neighborhood of $h(q_1)$, it follows that (X_2, B_2) is sub-klt near $h(q_1)$ if and only if $(X_1, h^* B_2)$ is sub-klt near q_1.

(2) By the first paragraph of **Step 5**, we may assume that X, X', and X_0 are normal near o, o', and o_0 respectively. Applying the claim with $B_2 = 0$ to $X' \to X$ and to $X' \to X_0$, we see that X is klt near o if and only if X' is klt near o', if and only if X_0 is klt near o_0.

(3) Assume that X is normal near o. Then X' and X_0 are normal near o' and o_0. Since X is integral near o, the functions z and t do not vanish identically on X near o: otherwise f would be a unit times z or t, which contradicts $f \in \mathfrak{m}^2_{Y,o}$. As $X' \to X$ and $X' \to X_0$ are open near o', the pullbacks of z and t do not vanish identically on X' near o' or on X_0 near o_0 either. Since $\psi^* z = \varphi^* z$ and $\psi^* t = \varphi^* t$, the pullbacks of $b_z \operatorname{div}(z) + b_t \operatorname{div}(t)$ from X and from X_0 to X' coincide, and the divisors $K_X + b_z \operatorname{div}(z) + b_t \operatorname{div}(t)$ and $K_{X_0} + b_z \operatorname{div}(z) + b_t \operatorname{div}(t)$ are $\mathbb{R}$-Cartier. Applying the claim to $X' \to X$ and to $X' \to X_0$ implies (3).

(4) In each case of **Step 4**, x and y are non-zero constant multiples of w_1 and w_2, or $\chi_{u_1} = \chi_{u_2}$ and x, y are linear combinations of w_1, w_2, so x and y have the characters χ_{u_1} and χ_{u_2} of w_1 and w_2. The morphisms $X' \to X$ and $X' \to X_0$ are G-equivariant and étale at o', and all residue fields at the marked points are $\mathbb{C}$, so they induce G-equivariant isomorphisms

$$\widehat{\mathcal{O}}_{X,o} \cong \widehat{\mathcal{O}}_{X',o'} \cong \widehat{\mathcal{O}}_{X_0,o_0}.$$

Since $\mathbb{A}^2_{x,y} \times V \to \mathbb{A}^4$ is étale at o_0, we have

$$\widehat{\mathcal{O}}_{\mathbb{A}^2_{x,y}\times V,o_0} = \mathbb{C}[[x,y,z,t]], \quad \text{hence} \quad \widehat{\mathcal{O}}_{X_0,o_0} = \mathbb{C}[[x,y,z,t]]/(Q_2(x,y) + \hat{g}(z,t)).$$

Both isomorphisms of completions follow from [Har77, Exercise III.10.4]. The group G acts on $\mathbb{C}[[x,y,z,t]]$ by continuous automorphisms which multiply x,y,z,t by their characters, so the action is diagonal. Finally z and t correspond to z and t, since $\psi^*z = \varphi^*z$ and $\psi^*t = \varphi^*t$. □

The next lemma, in which the singularity need not be isolated, is used in the proof of Theorem 7.1 for the hypersurface $(xy + z^2 = 0) \subset \mathbb{A}^4$, whose singular locus is a line.

Lemma 3.16. *Let $(X \ni o)$ be a hypersurface germ in a smooth fourfold germ whose quadratic part at o has rank at least three. Then X is normal and canonical near o. In particular, it is klt.*

Proof. Let $(Y \ni o)$ be the smooth fourfold germ and f a local equation of X in Y at o. If $f \notin \mathfrak{m}^2_{Y,o}$, then X is smooth near o. Thus we may assume that $f \in \mathfrak{m}^2_{Y,o}$, and that $r := \operatorname{rank} q \geq 3$ for the quadratic part q of f. Choose regular parameters x_1, x_2, x_3, x_4 at o in which $q = \sum_{i=1}^r x_i^2$. Possibly shrinking Y, we may assume that $(x_1, \dots, x_4) : Y \to \mathbb{A}^4$ is étale and o is the only point of Y over 0. Let $\partial_1, \dots, \partial_4$ be the derivations of $\mathcal{O}_Y$ dual to $dx_1, \dots, dx_4$. Then $\partial_i f \equiv 2x_i$ mod $\mathfrak{m}^2_{Y,o}$ for $i = 1,2,3$, so

$$Z := (\partial_1 f = \partial_2 f = \partial_3 f = 0)$$

is a smooth curve near o. By the Jacobian criterion, $\operatorname{Sing} X \subset Z$, so $\dim \operatorname{Sing} X \leq 1$, while $\dim X = 3$. As X is a hypersurface, it is Cohen–Macaulay, so X is normal near o by Serre's criterion.

Let l be a general linear combination of $x_1, \dots, x_4$ and $H := X \cap (l = 0)$. Since $dl(o)$ does not vanish on the tangent line of Z at o, near o the curve Z intersects $(l = 0)$ only at o. By the Bertini theorem [Har77, Corollary III.10.9 and Remark III.10.9.2] applied to the morphism $X \setminus \operatorname{Sing} X \to \mathbb{P}^3$ defined by $(x_1 : x_2 : x_3 : x_4)$ near o, the surface H is smooth near o outside o. Thus H is a hypersurface in the smooth threefold germ $(l = 0)$ with at most an isolated singularity at o, and H is normal by Serre's criterion.

The restriction of q to a general hyperplane $(l = 0)$ of the tangent space at o has rank at least two: if $r = 3$, then a general hyperplane does not contain the kernel of q and the restriction has rank three, and if $r = 4$, then restricting to a hyperplane lowers the rank by at most two. By the splitting lemma [GLS07, Chapter I, Theorem 2.47], H is either smooth at o or analytically isomorphic near o to $(y_1^2 + y_2^2 + y_3^{n+1} = 0) \subset \mathbb{A}^3$ for some integer $n \geq 1$, as o is an isolated singular point of H. In both cases H is canonical near o [KM98, Theorem 4.20], in particular klt. Since H is a Cartier divisor on X and K_X is Cartier, [KM98, Theorem 5.50(1)] implies that (X, H) is plt near H. Removing H does not decrease log discrepancies, so X is klt near o. Finally, since K_X is Cartier, the log discrepancy of any prime divisor over X is an integer. Thus the log discrepancies of the prime divisors over X whose centers intersect an open neighborhood of o on which X is klt are at least one, that is, X is canonical near o. □

The data in the next proposition are those of the conclusion of [Liu26, Lemma 11.3(1)]: a smooth fourfold germ $(Y \ni o)$, the function f, and G-semi-invariant generators x, y, z, t of the maximal ideal of $\widehat{\mathcal{O}}_{Y,o}$ such that f is a unit times $x^2 + g$ with $g \in (y,z,t)^3$. The cubic part g_3 of g is then the residual cubic of $(X \ni o)$ up to a non-zero scalar [Liu26, Definition-Lemma 11.1].

Proposition 3.17. *Let $(X \ni o)$ be an isolated cDV threefold germ of quadratic rank one with a faithful action of a finite abelian group G fixing o, $(Y \ni o)$ a smooth fourfold germ with an action of G containing X as $(f = 0)$ for a G-semi-invariant regular function f on Y, and x, y, z, t G-semi-invariant generators of the maximal ideal of $\widehat{\mathcal{O}}_{Y,o}$. Assume that f is a unit times $x^2 + g(y,z,t)$ with $g \in (y,z,t)^3$. Let g_3 be the cubic part of g and $C := (g_3 = 0) \subset \mathbb{P}^2$, where y, z, t are homogeneous coordinates of $\mathbb{P}^2$. Then the following hold.*

(1) *If C is an irreducible cuspidal cubic, then after a G-equivariant linear change of y,z,t in $\widehat{\mathcal{O}}_{Y,o}$ we have $g_3=y^2z+t^3$, and there exists a G-equivariant automorphism of $\widehat{\mathcal{O}}_{Y,o}$ transforming f into a unit times $x^2+y^2z+\lambda yt^\ell+b(z,t)$ with $\ell\geq 2$, $\lambda\in\mathbb{C}$, and $b\in(z,t)^3$ whose part independent of z is t^3 times a G-invariant unit.*

(2) *If C is the union of a smooth conic and a line tangent to it, then after a G-equivariant linear change of y,z,t in $\widehat{\mathcal{O}}_{Y,o}$ we have $g_3=y^2z+yt^2$, and there exists a G-equivariant automorphism of $\widehat{\mathcal{O}}_{Y,o}$ transforming f into a unit times $x^2+y^2z+yt^2+b(z,t)$ with $b\in(z,t)^4$.*

In (1) and (2), the series $x^2+y^2z+\lambda yt^\ell+b(z,t)$, resp. $x^2+y^2z+yt^2+b(z,t)$, is G-semi-invariant with the character of x^2, which is the character of f, and the unit is G-invariant.

Proof. We have $\widehat{\mathcal{O}}_{Y,o}=\mathbb{C}[[x,y,z,t]]$, on which G acts diagonally. A series is G-semi-invariant with character χ if and only if any monomial occurring in it has character χ, and an endomorphism Φ of $\mathbb{C}[[x,y,z,t]]$ is G-equivariant if and only if $\Phi(x),\Phi(y),\Phi(z),\Phi(t)$ are G-semi-invariant with characters $\chi_x,\chi_y,\chi_z,\chi_t$ respectively. Write $f=uF$ with $F:=x^2+g$ and u a unit.

Step 1. We show that F is G-semi-invariant with character χ_x^2. The quadratic part of f is $u(0)x^2$, so $\chi_f=\chi_x^2$. For $\gamma\in G$, let $F_\gamma:=x^2+\chi_x(\gamma)^{-2}\gamma^*g$. Then $uF=f=\chi_f(\gamma)^{-1}\gamma^*f=(\gamma^*u)F_\gamma$, so

$$F_\gamma-F=\big(u(\gamma^*u)^{-1}-1\big)F.$$

Since $F_\gamma-F\in\mathbb{C}[[y,z,t]]$ has degree zero in x and F is monic of degree two in x with coefficients in (y,z,t), the uniqueness of the Weierstrass division by F [GLS07, Chapter I, Theorem 1.8] (with the trivial valuation on $\mathbb{C}$, for which $K\langle x\rangle=K[[x]]$ [GLS07, Chapter I, Remark 1.1.1(1)]) implies $F_\gamma=F$. Thus g and g_3 are G-semi-invariant with character χ_x^2.

Let $\Lambda:=\mathbb{C}y+\mathbb{C}z+\mathbb{C}t$, on which G acts linearly. Since $\gamma^*g_3=\chi_x(\gamma)^2g_3$ for any $\gamma\in G$, the curve C is G-stable, and a linear form $l\in\Lambda$ is G-semi-invariant if and only if the line $(l=0)$ is G-stable. A G-equivariant linear change of y,z,t replaces y,z,t by a basis of Λ consisting of G-semi-invariant linear forms. After such a change, f is still a unit times x^2+g with $g\in(y,z,t)^3$, and g_3 is the cubic part of g in the new coordinates.

Step 2. We prove the first part of (1). Assume that C is an irreducible cuspidal cubic, and let p_0 be the cusp of C. We first find a basis y_1,z_1,t_1 of Λ, not necessarily consisting of G-semi-invariant forms, with $g_3=y_1^2z_1+t_1^3$. Choose a basis y_1,z_1,t_1 of Λ with $p_0=(0:1:0)$ such that the tangent cone of C at p_0 is the line $(y_1=0)$ with multiplicity two. Since C has multiplicity two at p_0, we have $g_3=\alpha y_1^2z_1+k(y_1,t_1)$ for some $\alpha\in\mathbb{C}$ and a binary cubic form k. Since C is irreducible, $\alpha\neq 0$ and the coefficient β of t_1^3 in k is non-zero. Replacing t_1 by $t_1+\frac{\beta'}{3\beta}y_1$, where β' is the coefficient of $y_1t_1^2$ in k, we may assume that $\beta'=0$. Then $g_3=y_1^2(\alpha z_1+\alpha' y_1+\alpha'' t_1)+\beta t_1^3$ for some $\alpha',\alpha''\in\mathbb{C}$, and replacing z_1 by $\alpha z_1+\alpha' y_1+\alpha'' t_1$ and t_1 by $\beta^{1/3}t_1$ yields $g_3=y_1^2z_1+t_1^3$.

Let $\nu:\mathbb{P}^1\to C$ be the morphism $\nu(s:r):=(s^3:r^3:-s^2r)$ in the coordinates $(y_1:z_1:t_1)$. Then ν is the normalization of C, ν is bijective, $\nu(0:1)=p_0$, and the smooth points of C are $\nu(1:\mu)$ for $\mu\in\mathbb{C}$ (on the chart $y_1\neq 0$, ν is inverse to $\mu=-t_1/y_1$, and p_0 is the only singular point of C). For a line $(\alpha_1y_1+\alpha_2z_1+\alpha_3t_1=0)$ and $\mu\in\mathbb{C}$, the intersection multiplicity of the line and C at $\nu(1:\mu)$ is the order of vanishing at $(1:\mu)$ of the binary cubic form $\alpha_1s^3-\alpha_3s^2r+\alpha_2r^3$. The order is three if and only if the form is a non-zero multiple of

$$(r-\mu s)^3=r^3-3\mu sr^2+3\mu^2s^2r-\mu^3s^3,$$

which forces $\mu=0$ and $\alpha_1=\alpha_3=0$, as the form has no sr^2 term. Thus $p_1:=\nu(1:0)=(1:0:0)$ is the only smooth point of C at which a line intersects C with multiplicity three, and the line is $(z_1=0)$. Therefore p_0, the line $(y_1=0)$ of the tangent cone at p_0, p_1, the line $(z_1=0)$, and the line $(t_1=0)$ through p_0 and p_1 are determined by C, hence G-stable. By **Step 1**, y_1,z_1,t_1 are G-semi-invariant, which proves the first part of (1).

Step 3. We prove the first part of (2). Assume that $C=C_1\cup L_0$ for a smooth conic C_1 and a line L_0 tangent to C_1 at a point p_0. Since C_1 and L_0 have different degrees, both are G-stable,

and so is p_0, the only point of $C_1 \cap L_0$. The group G acts on C_1 through a finite group G' of automorphisms fixing p_0. Fixing an isomorphism $C_1 \cong \mathbb{P}^1$ which sends p_0 to ∞, the group G' acts on $\mathbb{A}^1 = C_1 \backslash \{p_0\}$ by affine transformations. For a point $p \in C_1 \backslash \{p_0\}$, the point

$$p_1 := \frac{1}{|G'|} \sum_{\gamma \in G'} \gamma(p) \in \mathbb{A}^1 = C_1 \backslash \{p_0\}$$

is fixed by G', since affine transformations preserve barycenters. Let L_1 be the tangent line to C_1 at p_1 and L_2 the line through p_0 and p_1. Then L_1 and L_2 are G-stable. The lines L_0, L_1, L_2 are pairwise distinct, since $L_0 \cap C_1 = \{p_0\}$, $L_1 \cap C_1 = \{p_1\}$, and $p_0 \neq p_1$. They do not pass through a common point, since $L_0 \cap L_2 = \{p_0\}$ and $p_0 \notin L_1$. Thus linear forms y_1, z_1, t_1 defining L_0, L_1, L_2 form a basis of Λ, and they are G-semi-invariant by **Step 1**. In the coordinates $(y_1 : z_1 : t_1)$ we have $p_0 = (0 : 1 : 0)$ and $p_1 = (1 : 0 : 0)$, and a direct calculation shows that the conics through p_0 and p_1 which are tangent to L_0 at p_0 and to L_1 at p_1 are defined by $\alpha y_1 z_1 + \beta t_1^2$ with $(\alpha, \beta) \neq (0, 0)$. Since C_1 is smooth, $\alpha\beta \neq 0$. Thus $g_3 = \epsilon y_1(\alpha y_1 z_1 + \beta t_1^2)$ for some $\epsilon \in \mathbb{C}^*$, and replacing z_1 by $\epsilon\alpha z_1$ and t_1 by $(\epsilon\beta)^{1/2} t_1$ yields $g_3 = y_1^2 z_1 + y_1 t_1^2$. This proves the first part of (2).

Step 4. We eliminate the higher terms (cf. [Liu26, Lemma 11.5]). By **Steps 2 and 3**, we may assume that x, y, z, t are G-semi-invariant and

$$g_3 = y^2 z + cyt^2 + dt^3,$$

where $(c, d) = (0, 1)$ in the case of (1) and $(c, d) = (1, 0)$ in the case of (2). As g_3 has character χ_x^2, we have $\chi_y^2 \chi_z = \chi_x^2$. All automorphisms below fix x.

First, let $F_3 := F$. For $N \geq 4$, let q_N be the sum of the terms of degree N of F_{N-1} which are divisible by y^2, let ϕ_N be the automorphism of $\mathbb{C}[[x, y, z, t]]$ with

$$\phi_N(z) = z - q_N/y^2, \quad \phi_N(x) = x, \quad \phi_N(y) = y, \quad \phi_N(t) = t,$$

and let $F_N := \phi_N(F_{N-1})$. The series q_N/y^2 is a form of degree $N-2$ with character $\chi_x^2 \chi_y^{-2} = \chi_z$, so ϕ_N is G-equivariant. We have $\phi_N(y^2 z) = y^2 z - q_N$, the terms $cyt^2 + dt^3$ do not involve z and are fixed by ϕ_N, and a monomial of degree $k \geq 4$ changes only by terms of degree at least $k + N - 3 \geq N + 1$. Hence, by induction on N, the series $F_N - x^2$ has cubic part g_3 and no monomial of degree between 4 and N divisible by y^2. Since $\phi_N(k) - k \in \mathfrak{m}^{N-2}$ for any $k \in \mathbb{C}[[x, y, z, t]]$, where $\mathfrak{m} := (x, y, z, t)$, the composites $\phi_N \circ \cdots \circ \phi_4$ converge in the $\mathfrak{m}$-adic topology to a G-equivariant automorphism Φ_1, and

$$\Phi_1(F) = x^2 + y^2 z + ya(z, t) + e(z, t), \quad a \in ct^2 + (z, t)^3, \quad e \in dt^3 + (z, t)^4.$$

Second, write $a = a_0(t) + z a_1(z, t)$. Since ct^2 does not involve z, we have $a_1 \in (z, t)^2$ and $a_0 \in (t^2)$. The terms of a have character $\chi_x^2 \chi_y^{-1}$, so a_1 has character $\chi_x^2 \chi_y^{-1} \chi_z^{-1} = \chi_y$. Let Φ_2 be the G-equivariant automorphism with $\Phi_2(y) = y - a_1/2$ fixing x, z, t. A direct calculation shows

$$\Phi_2(\Phi_1(F)) = x^2 + y^2 z + ya_0(t) + e_1(z, t), \quad e_1 := e - \frac{a_0 a_1}{2} - \frac{z a_1^2}{4} \in dt^3 + (z, t)^4.$$

Third, if $a_0 = 0$, let $\lambda := 0$, $\ell := 2$, and $\Phi_3 := \mathrm{id}_{\mathbb{C}[[x,y,z,t]]}$. Otherwise write $a_0 = \lambda t^\ell U(t)$ with $\lambda \neq 0$, $\ell \geq 2$, and $U(0) = 1$. Any monomial t^k of a_0 has character $\chi_t^k = \chi_x^2 \chi_y^{-1} = \chi_t^\ell$, so U is G-invariant. Let Φ_3 be the G-equivariant automorphism with

$$\Phi_3(y) = y/U(t), \quad \Phi_3(z) = U(t)^2 z, \quad \Phi_3(x) = x, \quad \Phi_3(t) = t.$$

Then $\Phi_3(y^2 z) = y^2 z$, $\Phi_3(ya_0) = \lambda y t^\ell$, and $\Phi_3(e_1(z, t)) = e_1(U^2 z, t)$. Let $\Phi := \Phi_3 \circ \Phi_2 \circ \Phi_1$ and $b(z, t) := e_1(U(t)^2 z, t)$. Then

$$\Phi(f) = \Phi(u)\big(x^2 + y^2 z + \lambda y t^\ell + b(z, t)\big),$$

$\Phi(u)$ is a unit, $b \in (z, t)^3$, and $b(0, t) = e_1(0, t) \in dt^3 + (t^4)$.

In the case of (1), we have $d = 1$, so $b(0, t) = t^3 V(t)$ for a unit V with $V(0) = 1$. Any monomial t^k of $b(0, t)$ has character $\chi_x^2 = \chi_t^3$, as t^3 occurs in g_3. Thus V is G-invariant, which proves (1).

In the case of (2), we have $c = 1$ and $d = 0$. Then $a_0 \in t^2 + (t^3)$, so $\lambda = 1$ and $\ell = 2$, and $e_1 \in (z,t)^4$, so $b \in (z,t)^4$. This proves (2).

Finally, we prove the last sentence of the proposition. Let $N := x^2 + y^2 z + \lambda y t^\ell + b(z,t)$ in the case of (1), $N := x^2 + y^2 z + y t^2 + b(z,t)$ in the case of (2), and $u' := \Phi(u)$, so that $\Phi(f) = u'N$. The quadratic part of f is a non-zero multiple of x^2, so $\chi_f = \chi_x^2$. For $\gamma \in G$, since Φ is G-equivariant,

$$\gamma^*(u'N) = \Phi(\gamma^* f) = \chi_x(\gamma)^2 u' N.$$

Let

$$N_\gamma := \chi_x(\gamma)^{-2} \gamma^* N = x^2 + \chi_x(\gamma)^{-2} \gamma^*(N - x^2).$$

Then $u'N = (\gamma^* u') N_\gamma$, both N and N_γ are monic of degree two in x with coefficients in (y,z,t), and

$$N_\gamma - N = \big(u'(\gamma^* u')^{-1} - 1\big) N \in \mathbb{C}[[y,z,t]].$$

As in **Step 1**, the uniqueness of the Weierstrass division by N [GLS07, Chapter I, Theorem 1.8] implies $N_\gamma = N$. Thus $\gamma^* N = \chi_x(\gamma)^2 N$ and $\gamma^* u' = u'$ for any $\gamma \in G$. □

3.5. **Generic limits of plane pairs and transverse surfaces.** In this subsection, we take the generic limits of plane quotient pairs of [Liu26, Section 14] together with a fixed $\mathbb{R}$-ideal factor, we deduce the lc hypotheses of [Liu26, Proposition 14.15] for $p = q = 1$ from the residual plane pair, and we reduce quotient minimal log discrepancies at generic points of curves on threefolds to closed points of generic fibers of invariant functions on the threefolds.

Theorem 4.2(2) allows a fixed $\mathbb{R}$-ideal $\mathfrak{c}$ on the plane when the group is diagonal, and the proof of this case uses Proposition 3.19(2). So in Proposition 3.19 we take generic limits with a fixed $\mathbb{R}$-ideal $\mathfrak{c}$, and the finite-jet statements of [Liu26, Section 14] remain true with $\mathfrak{c}$. In Proposition 3.19, (1) and (2) are [Liu26, Lemma 14.14(1)(2)] with the fixed $\mathbb{R}$-ideal $\mathfrak{c}$ added, and (3) is [Liu26, Definition-Lemma 14.17(2)] with the exponent θ_0 of $\mathfrak{c}$ added (cf. [dFEM11, Proposition 3.3(ii) and Corollary 3.4], where any non-negative exponents are allowed). In (2) and (3), we state the equality for all but finitely many i and the convergence without passing to a subsequence, as proved below for the tuples of [Liu26, Construction 14.13].

Definition 3.18. Let $k \supset \mathbb{C}$ be an algebraically closed field, H a finite group acting on $T_k = \operatorname{Spec} k[[u,t]]$, D an H-invariant $\mathbb{R}$-divisor on T_k, and $\mathfrak{c}_k$ an $\mathbb{R}$-ideal on T_k with H-invariant factors. As in Definition 2.4, we denote by $\operatorname{mld}((T_k \ni 0, D, \mathfrak{c}_k)/H)$ the infimum of $a(F, T_k, D, \mathfrak{c}_k)/|I_H(F)|$ over the formal prime divisors F over T_k with center the closed point [Liu26, Definitions 2.2 and 3.2]. If $k = \mathbb{C}$, H acts faithfully on $\mathbb{A}^2 = \mathbb{A}^2_{u,t}$ fixing 0, and D and $\mathfrak{c}_k$ are the pullbacks to $T_{\mathbb{C}}$ of an H-invariant $\mathbb{R}$-divisor D' and of an $\mathbb{R}$-ideal $\mathfrak{c}'$ with H-invariant factors on $\mathbb{A}^2$, then

$$\operatorname{mld}((T_{\mathbb{C}} \ni 0, D, \mathfrak{c}_{\mathbb{C}})/H) = \operatorname{mld}((\mathbb{A}^2 \ni 0, D', \mathfrak{c}')/H)$$

by Lemma 3.6.

Proposition 3.19. *Let H be a finite diagonal subgroup of $(\mathbb{C}^*)^2$ acting on $\mathbb{A}^2 = \mathbb{A}^2_{u,t}$, $\Lambda \subset \mathbb{R}_{>0}$ a finite set, and $\mathfrak{c} = \prod_{l=1}^m \mathfrak{c}_l^{e_l}$ an $\mathbb{R}$-ideal on $\mathbb{A}^2$ with $e_l > 0$ and each $\mathfrak{c}_l \subset \mathbb{C}[u,t]$ generated by H-semi-invariant polynomials. For an algebraically closed field $k \supset \mathbb{C}$ let $T_k := \operatorname{Spec} k[[u,t]]$ and $\mathfrak{c}_k := \prod_l (\mathfrak{c}_l k[[u,t]])^{e_l}$. Then the following hold.*

(1) There exists a positive integer ℓ, depending only on H, Λ, and $\mathfrak{c}$, such that for any algebraically closed field $k \supset \mathbb{C}$, any two tuples $(f_1, \dots, f_s)$ and $(f'_1, \dots, f'_s)$ of non-zero H-semi-invariant power series in $k[[u,t]]$ with $f_j - f'_j \in (u,t)^\ell$ for any j, and any $c_1, \dots, c_s \in \Lambda$,

$$\operatorname{mld}\Big(\Big(T_k \ni 0, \sum_j c_j \operatorname{div}(f_j), \mathfrak{c}_k\Big)/H\Big) = \operatorname{mld}\Big(\Big(T_k \ni 0, \sum_j c_j \operatorname{div}(f'_j), \mathfrak{c}_k\Big)/H\Big).$$

(2) *Let* $(f_{i,j})_j$, $i \geq 1$, *be tuples as in* [Liu26, Construction 14.13] *with generic limit* $(f_{\infty,j})_j$ *over* k, *and let* $c_1, \dots, c_s \in \Lambda$. *Then for all but finitely many* i

$$\operatorname{mld}\Big(\Big(\mathbb{A}^2 \ni 0, \sum_j c_j \operatorname{div}(f_{i,j}), \mathfrak{c}\Big)/H\Big) = \operatorname{mld}\Big(\Big(T_k \ni 0, \sum_j c_j \operatorname{div}(f_{\infty,j}), \mathfrak{c}_k\Big)/H\Big).$$

(3) *In* (2) *assume moreover that* $f_{i,j}(0) = 0$ *for all* i, j *and* $\mathfrak{c}_l \subset (u,t)$ *for all* l. *Then for any real numbers* $\theta_0, \theta_1, \dots, \theta_s \geq 0$,

$$\operatorname{lct}_0\Big(\mathbb{A}^2; \prod_j (f_{i,j})^{\theta_j} \cdot \mathfrak{c}^{\theta_0}\Big) \to \operatorname{lct}_0\Big(T_k; \prod_j (f_{\infty,j})^{\theta_j} \cdot \mathfrak{c}_k^{\theta_0}\Big).$$

Proof. In the proof, K is an algebraically closed field containing $\mathbb{C}$, $\mathbb{A}^2_K := \operatorname{Spec} K[u,t]$, $T_K := \operatorname{Spec} K[[u,t]]$, $\mathfrak{m} := (u,t)$, and $\mathfrak{c}_K$ is the pullback of $\mathfrak{c}$ to $\mathbb{A}^2_K$ and to T_K. For an $\mathbb{R}$-ideal $\mathfrak{a}$ on $\mathbb{A}^2_K$, we write $\hat{\mathfrak{a}}$ for the pullback of $\mathfrak{a}$ to T_K. For a power series f and a positive integer n, $f^{<n}$ denotes the sum of the terms of f of degree less than n. Since H is diagonal, $f^{<n}$ is H-semi-invariant if f is H-semi-invariant. For any formal prime divisor F over T_K,

$$a\Big(F, T_K, \sum_j c_j \operatorname{div}(f_j), \mathfrak{c}_K\Big) = a\Big(F, T_K, 0, \prod_j (f_j)^{c_j} \cdot \mathfrak{c}_K\Big)$$

[Liu26, Definition 2.2], and the analogous equality holds for prime divisors over $\mathbb{A}^2_K$. Thus all the quotient values in the proposition are quotient values of $\mathbb{R}$-ideals with zero boundary, whose factors are generated by H-semi-invariant elements.

Step 1. In this step, we show that

$$\operatorname{mld}((\mathbb{A}^2_K \ni 0, \mathfrak{a})/H) = \operatorname{mld}((T_K \ni 0, \hat{\mathfrak{a}})/H)$$

for any $\mathbb{R}$-ideal $\mathfrak{a}$ on $\mathbb{A}^2_K$ whose factors are generated by H-semi-invariant polynomials. For $K = \mathbb{C}$, this is Lemma 3.6. The following argument works for any K.

Let P be a formal prime divisor over T_K with center the closed point. Since $\operatorname{mult}_P \mathfrak{m} \geq 1$, we have $\operatorname{mult}_P(f - f^{<n}) \geq n$, hence $\operatorname{mult}_P f = \operatorname{mult}_P(f^{<n})$ for any $f \in K[[u,t]]$ and any integer $n > \operatorname{mult}_P f$. Thus mult_P is determined by $\operatorname{mult}_P|_{K(u,t)}$, and $\operatorname{mult}_P|_{K(u,t)}$ has value group $\mathbb{Z}$. By [dFEM11, Lemma 2.21], $\operatorname{mult}_P|_{K(u,t)} = \operatorname{mult}_{P^{\mathrm{a}}}$ for a prime divisor P^{a} over $\mathbb{A}^2_K$ with center 0, and $a(P, T_K, 0) = a(P^{\mathrm{a}}, \mathbb{A}^2_K, 0)$ by [dFEM11, Propositions 2.11 and A.14]. Conversely, let E be a prime divisor over $\mathbb{A}^2_K$ with center 0, and $Y \to \mathbb{A}^2_K$ a higher model on which E is a divisor. Since $T_K \to \mathbb{A}^2_K$ is flat with regular fibers, $\hat{Y} := Y \times_{\mathbb{A}^2_K} T_K$ is normal, proper and birational over T_K, and the fiber of $\hat{Y}$ over the closed point is the fiber of Y over 0. Hence E is a prime divisor on $\hat{Y}$, and the formal prime divisor P_E defined by E on $\hat{Y}$ satisfies $P_E^{\mathrm{a}} = E$ and has residue field equal to the residue field of E (see [dFEM11, Proposition A.14 and Remark A.15]). Thus the map sending P to P^{a} is a bijection from the set of formal prime divisors over T_K with center the closed point onto the set of prime divisors over $\mathbb{A}^2_K$ with center 0, which commutes with the action of H and preserves residue fields. In particular $I_H(P) = I_H(P^{\mathrm{a}})$. Since moreover $\operatorname{mult}_P \hat{\mathfrak{a}} = \operatorname{mult}_{P^{\mathrm{a}}} \mathfrak{a}$, we have

$$\frac{a(P, T_K, 0, \hat{\mathfrak{a}})}{|I_H(P)|} = \frac{a(P^{\mathrm{a}}, \mathbb{A}^2_K, 0, \mathfrak{a})}{|I_H(P^{\mathrm{a}})|}$$

for any P, and **Step 1** follows.

Step 2. Let $\mathfrak{a} = \prod_{j=1}^s \mathfrak{a}_j^{r_j}$ be an $\mathbb{R}$-ideal on $\mathbb{A}^2_K$ with $r_j > 0$, where $\mathfrak{a}_j$ is generated by H-semi-invariant polynomials $g_{j,1}, \dots, g_{j,n_j}$, some of which may be zero. Let $(Q_K, \Delta_{Q_K}) := (\mathbb{A}^2_K, 0)/H$ be the crepant quotient, $\pi : \mathbb{A}^2_K \to Q_K$ the quotient morphism, and $q := \pi(0)$. Since the order of the character of an H-semi-invariant polynomial divides $|H|$, the polynomials $g_{j,i}^{|H|}$ are H-invariant.

Let $\mathfrak{a}'_j \subset K[u,t]^H$ be the ideal generated by $g_{j,1}^{|H|}, \dots, g_{j,n_j}^{|H|}$, and $\mathfrak{a}' := \prod_j \mathfrak{a}_j'^{r_j/|H|}$, an $\mathbb{R}$-ideal on Q_K. In this step, we show that

$$\operatorname{mld}((\mathbb{A}^2_K \ni 0, \mathfrak{a})/H) = \operatorname{mld}(Q_K \ni q, \Delta_{Q_K}, \mathfrak{a}').$$

For any prime divisor F over $\mathbb{A}^2_K$,

$$\operatorname{mult}_F(\mathfrak{a}'_j \mathcal{O}_{\mathbb{A}^2_K}) = \min_i \operatorname{mult}_F \left(g_{j,i}^{|H|}\right) = |H| \min_i \operatorname{mult}_F g_{j,i} = |H| \operatorname{mult}_F \mathfrak{a}_j,$$

so

$$\operatorname{mult}_F(\mathfrak{a}' \mathcal{O}_{\mathbb{A}^2_K}) = \operatorname{mult}_F \mathfrak{a}.$$

Let F be a prime divisor over $\mathbb{A}^2_K$ with center 0, and E the prime divisor over Q_K with $\operatorname{mult}_F|_{K(Q_K)} = |I_H(F)| \operatorname{mult}_E$. By [Liu26, Lemma 2.6 and Remark 2.17],

$$a(F, \mathbb{A}^2_K, 0) = |I_H(F)| a(E, Q_K, \Delta_{Q_K}), \qquad \operatorname{mult}_F(\mathfrak{a}' \mathcal{O}_{\mathbb{A}^2_K}) = |I_H(F)| \operatorname{mult}_E \mathfrak{a}'.$$

Thus

$$\frac{a(F, \mathbb{A}^2_K, 0, \mathfrak{a})}{|I_H(F)|} = a(E, Q_K, \Delta_{Q_K}) - \operatorname{mult}_E \mathfrak{a}' = a(E, Q_K, \Delta_{Q_K}, \mathfrak{a}').$$

Since $\pi^{-1}(q) = \{0\}$, the center of E on Q_K is q if and only if the center of F on $\mathbb{A}^2_K$ is 0, and for any prime divisor E over Q_K there exists a prime divisor F over $\mathbb{A}^2_K$ with $\operatorname{mult}_F|_{K(Q_K)} = |I_H(F)| \operatorname{mult}_E$ [Liu26, Lemma 2.6]. **Step 2** follows.

Step 3. In this step, we show that quotient values and lc thresholds of a family of $\mathbb{R}$-ideals do not change over a non-empty open subset of the parameter space.

We first compute minimal log discrepancies on a log resolution. Let $\mathfrak{b} = \prod_k \mathfrak{b}_k^{r_k}$ be an $\mathbb{R}$-ideal on Q_K with $r_k > 0$, and $\psi : Y \to Q_K$ a log resolution of $\operatorname{Supp} \Delta_{Q_K}$, the ideals $\mathfrak{b}_k$, and the ideal of q. This means that Y is smooth, $\mathfrak{b}_k \mathcal{O}_Y = \mathcal{O}_Y(-M_k)$ for divisors M_k, and $\operatorname{Exc}(\psi) \cup \psi^{-1}(q)$, the strict transform of $\operatorname{Supp} \Delta_{Q_K}$, and $\bigcup_k \operatorname{Supp} M_k$ form a divisor with simple normal crossings. Let B_Y be the $\mathbb{R}$-divisor with

$$K_Y + B_Y = \psi^*(K_{Q_K} + \Delta_{Q_K}) + \sum_k r_k M_k.$$

Then $a(P, Q_K, \Delta_{Q_K}, \mathfrak{b}) = a(P, Y, B_Y)$ for any prime divisor P over Y. Let $E_1, \dots, E_n$ be the components of $\psi^{-1}(q)$. We show that

$$\operatorname{mld}(Q_K \ni q, \Delta_{Q_K}, \mathfrak{b}) = \min_i (1 - \operatorname{mult}_{E_i} B_Y) \tag{$\dagger$}$$

if $\operatorname{mult}_D B_Y \le 1$ for any prime divisor $D \subset \operatorname{Supp} B_Y \cup \psi^{-1}(q)$ with $D \cap \psi^{-1}(q) \neq \emptyset$, and that $\operatorname{mld}(Q_K \ni q, \Delta_{Q_K}, \mathfrak{b}) = -\infty$ otherwise. Each E_i has center q on Q_K. Let P be a prime divisor over Y whose center on Y is a closed point $y \in \psi^{-1}(q)$. Choose local coordinates z_1, z_2 of Y at y such that $\operatorname{Supp} B_Y \cup \psi^{-1}(q) \subset (z_1 z_2 = 0)$ near y and $(z_1 = 0) \subset \psi^{-1}(q)$, and let b_1, b_2 be the coefficients of $(z_1 = 0)$ and $(z_2 = 0)$ in B_Y. Since $(Y, \operatorname{div}(z_1 z_2))$ is lc near y,

$$a(P, Y, B_Y) \ge (1 - b_1) \operatorname{mult}_P z_1 + (1 - b_2) \operatorname{mult}_P z_2.$$

If $b_1, b_2 \le 1$, then the right hand side is at least $1 - b_1 \ge \min_i (1 - \operatorname{mult}_{E_i} B_Y)$, which proves ($\dagger$). If $\operatorname{mult}_D B_Y > 1$ for some prime divisor $D \subset \operatorname{Supp} B_Y \cup \psi^{-1}(q)$ with $D \cap \psi^{-1}(q) \neq \emptyset$, choose $y \in D \cap \psi^{-1}(q)$ and local coordinates z_1, z_2 of Y at y such that $\operatorname{Supp} B_Y \cup \psi^{-1}(q) \subset (z_1 z_2 = 0)$ near y and $D = (z_1 = 0)$ or $D = (z_2 = 0)$. The exceptional divisor F_w of the weighted blow-up of y with weight w on the equation of D and weight 1 on the other coordinate has center q on Q_K and log discrepancy $w(1 - \operatorname{mult}_D B_Y) + 1 - b$ with respect to (Y, B_Y), where b is the coefficient in B_Y of the other coordinate divisor ($a(F_w, Y, 0) = w + 1$, and the equation of D and the other coordinate have multiplicities w and 1 along F_w, by a toric computation). The log discrepancy $w(1 - \operatorname{mult}_D B_Y) + 1 - b$ tends to $-\infty$ as $w \to +\infty$. Similarly, for an $\mathbb{R}$-ideal $\mathfrak{b}$ on $\mathbb{A}^2_K$ and a log resolution $h : Y' \to \mathbb{A}^2_K$ of the factors of $\mathfrak{b}$ and of $\mathfrak{m}$, $\operatorname{lct}_0(\mathbb{A}^2_K; \mathfrak{b})$ is the minimum of

$a(D,\mathbb{A}^2_K,0)/\operatorname{mult}_D\mathfrak{b}$ over the prime divisors D on Y' with $0\in h(D)$ and $\operatorname{mult}_D\mathfrak{b}>0$ [dFEM11, Section 2].

Now let V be an integral affine scheme of finite type over $\mathbb{C}$ with generic point ξ, and let $g_{j,i}\in\mathcal{O}(V)[u,t]$, $1\le j\le s$, $1\le i\le n_j$. For a point $\bar v$ of V with values in an algebraically closed field $K\supset\mathbb{C}$, let $g_{j,i}(\bar v)\in K[u,t]$ be the polynomial obtained by evaluating the coefficients at $\bar v$, and $\mathfrak{a}_j(\bar v)\subset K[u,t]$ the ideal generated by $g_{j,1}(\bar v),\dots,g_{j,n_j}(\bar v)$. Assume that the $g_{j,i}(\bar v)$ are H-semi-invariant and $\mathfrak{a}_j(\bar v)\neq 0$ for any $\bar v$. We show that there exists a non-empty open subset $V^\circ\subset V$ such that, for any real numbers $c'_1,\dots,c'_s>0$ and $\theta_0,\dots,\theta_s\ge 0$, the numbers

$$\operatorname{mld}\Big(\Big(\mathbb{A}^2_K\ni 0,\prod_j\mathfrak{a}_j(\bar v)^{c'_j}\cdot\mathfrak{c}_K\Big)/H\Big)\quad\text{and}\quad\operatorname{lct}_0\Big(\mathbb{A}^2_K;\prod_j\mathfrak{a}_j(\bar v)^{\theta_j}\cdot\mathfrak{c}_K^{\theta_0}\Big)$$

do not depend on the field K or on the point $\bar v$ of V with values in K mapping into V°.

Fix H-semi-invariant generators of the $\mathfrak{c}_l$. Let $Q:=\operatorname{Spec}\mathbb{C}[u,t]^H$, $q\in Q$ the image of 0, and Δ_Q the boundary of the crepant quotient $(\mathbb{A}^2,0)/H$. Then (Q_K,Δ_{Q_K}) is the base change of (Q,Δ_Q) to K, since taking H-invariants and forming the branch divisor commute with the flat base change $\mathbb{C}\subset K$ (cf. [Liu26, Remark 14.18]). Let $\mathfrak{a}'_{j,V}\subset\mathcal{O}(V)[u,t]^H$ be the ideal generated by the $g_{j,i}^{|H|}$, and $\mathfrak{c}'_l$ the ideal of **Step 2** for $\mathfrak{c}_l$. Take a log resolution of $\operatorname{Supp}\Delta_Q$, the ideals $\mathfrak{a}'_{j,V}$ and $\mathfrak{c}'_l$, and the ideal of q on $Q\times\{\xi\}$ over the field $\kappa(\xi)$, and extend it to a projective morphism $\psi_V:Y_V\to Q\times V^\circ$ over a non-empty open subset $V^\circ\subset V$. By generic flatness and generic smoothness, after shrinking V°, we may assume that Y_V is smooth over V°, that the pullbacks of the $\mathfrak{a}'_{j,V}$ and the $\mathfrak{c}'_l$ are invertible, that the exceptional divisors, $\psi_V^{-1}(q\times V^\circ)$, the strict transforms of the components of $\operatorname{Supp}\Delta_Q\times V^\circ$, and the divisors of the pullbacks form a divisor with simple normal crossings relative to V°, and that any intersection of components of the divisor is either empty or smooth and surjective over V° (cf. [Har77, Corollary III.10.7], applied to Y_V and to the irreducible components of the intersections of components of the divisor, which are non-singular after shrinking V°, since their local rings at the points over ξ are those of their fibers over ξ, which are smooth over $\kappa(\xi)$, and the singular locus is closed). Then, for any $\bar v$ mapping into V°, the fiber of ψ_V over $\bar v$ is a log resolution of $\operatorname{Supp}\Delta_{Q_K}$, the factors of the $\mathbb{R}$-ideal

$$\mathfrak{b}=\prod_j\mathfrak{a}'_j(\bar v)^{c'_j/|H|}\cdot\prod_l\mathfrak{c}_l'^{e_l/|H|}$$

on Q_K, and the ideal of q, where $\mathfrak{a}'_j(\bar v)$ is generated by the $g_{j,i}(\bar v)^{|H|}$, and the relative canonical divisor of Y_V over $Q\times V^\circ$ restricts to the relative canonical divisor of the fiber. Write $\psi_{\bar v}:Y_{\bar v}\to Q_K$ for the fiber of ψ_V over $\bar v$, $\psi_\xi:Y_\xi\to Q_{\kappa(\xi)}$ for the fiber over ξ, and $B_{Y_{\bar v}}$, B_{Y_ξ} for the $\mathbb{R}$-divisors B_Y of (†) on them. Let D be a component of the divisor with simple normal crossings on Y_V above. Then (a) any component of the fiber $D_{\bar v}$ has coefficient in $B_{Y_{\bar v}}$ equal to the coefficient of D_ξ in B_{Y_ξ}, and (b) $D_{\bar v}$ intersects $\psi_{\bar v}^{-1}(q)$ if and only if D intersects $\psi_V^{-1}(q\times V^\circ)$, if and only if D_ξ intersects $\psi_\xi^{-1}(q)$, since the intersection of D with a component of $\psi_V^{-1}(q\times V^\circ)$ is empty or surjective over V°. Thus, by (†) and **Step 2**, the quotient value does not depend on $\bar v$. Replacing $Q\times V^\circ$ with $\mathbb{A}^2\times V^\circ$, the ideals $\mathfrak{a}'_{j,V}$ and $\mathfrak{c}'_l$ with the ideals $\mathfrak{a}_{j,V}\subset\mathcal{O}(V)[u,t]$ generated by the $g_{j,i}$ and the ideals $\mathfrak{c}_l$, the ideal of $q\times V^\circ$ with the ideal of $0\times V^\circ$, and (†) with the description of $\operatorname{lct}_0(\mathbb{A}^2_K;\mathfrak{b})$ by the prime divisors on a log resolution [dFEM11, Section 2], the proof for quotient values also proves the statement for lc thresholds.

Step 4. In this step, $K=\mathbb{C}$. Since the germs in Theorem 2.10(2) are threefold germs, we pass from the surface Q to the threefold $X:=Q\times\mathbb{A}^1_\tau$, and the equality of minimal log discrepancies on X and on Q shown after $(*)$ recovers the minimal log discrepancies on Q from those on X. We write Q, Δ_Q, and q as in **Step 3**, and $\mathfrak{m}_q:=\mathfrak{m}\cap\mathbb{C}[u,t]^H$, the ideal of q in $\mathbb{C}[u,t]^H$. Let $x:=(q,0)$, $p:X\to Q$ the projection, and $\Delta_X:=p^*\Delta_Q+\operatorname{div}(\tau)$. Let $\Gamma\subset\mathbb{R}_{>0}$ be the finite set consisting of the non-zero coefficients of Δ_X and the numbers $c/|H|$, $c\in\Lambda$, and $e_l/|H|$, $1\le l\le m$. Let ℓ_0 be the positive integer of Theorem 2.10(2) for X and Γ, and $\ell:=|H|\ell_0$. Then

ℓ depends only on H, Λ, and $\mathfrak{c}$. We show that

$$\operatorname{mld}\Big(\Big(\mathbb{A}^2 \ni 0, \prod_j \mathfrak{a}_j^{c_j} \cdot \mathfrak{c}\Big)/H\Big) = \operatorname{mld}\Big(\Big(\mathbb{A}^2 \ni 0, \prod_j \mathfrak{d}_j^{c_j} \cdot \mathfrak{c}\Big)/H\Big) \tag{$*$}$$

for any $c_1, \dots, c_s \in \Lambda$ and any non-zero ideals $\mathfrak{a}_j$ and $\mathfrak{d}_j$ of $\mathbb{C}[u,t]$ generated by H-semi-invariant polynomials $g_{j,1}, \dots, g_{j,n_j}$ and $g'_{j,1}, \dots, g'_{j,n_j}$ respectively, some of which may be zero, with $g_{j,i} - g'_{j,i} \in \mathfrak{m}^\ell$ for any j and i.

First, we show that

$$\operatorname{mld}(X \ni x, \Delta_X, \mathfrak{b}\mathcal{O}_X) = \operatorname{mld}(Q \ni q, \Delta_Q, \mathfrak{b})$$

for any $\mathbb{R}$-ideal $\mathfrak{b} = \prod_k \mathfrak{b}_k^{r_k}$ on Q with $r_k > 0$, where $\mathfrak{b}\mathcal{O}_X$ denotes the $\mathbb{R}$-ideal

$$\mathfrak{b}\mathcal{O}_X := \prod_k (\mathfrak{b}_k \mathcal{O}_X)^{r_k}$$

on X (cf. [Liu26, Lemma 8.1]). Let F be a prime divisor over X with center x. By [Liu26, Lemma 8.1(1)], applied with the trivial group, there exist a positive integer ϱ and a prime divisor E over Q such that $\operatorname{mult}_F(\phi \circ p) = \varrho \operatorname{mult}_E \phi$ for any rational function ϕ on Q, and the center of E on Q is q, since $\operatorname{mult}_F(\phi \circ p) > 0$ for any $\phi \in \mathfrak{m}_q$. Let $Q' \to Q$ be a higher model on which E is a divisor and which is smooth at the generic point of E, and $X' := Q' \times \mathbb{A}^1_\tau$. Since $\varrho \operatorname{mult}_E$ has center the generic point of E on Q' and $\operatorname{mult}_F \tau > 0$, the center of F on X' is $E \times \{0\}$. Let z be the pullback of a local equation of E at the generic point of E. Near the generic point of $E \times \{0\}$, the crepant pullback of $K_X + \Delta_X$ to X' is

$$K_{X'} + (1 - a(E, Q, \Delta_Q)) \operatorname{div}(z) + \operatorname{div}(\tau).$$

Since $(X', \operatorname{div}(z\tau))$ is lc near the generic point of $E \times \{0\}$ and $\operatorname{mult}_F z = \varrho$, we have

$$a(F, X, \Delta_X) \geq a(E, Q, \Delta_Q) \operatorname{mult}_F z + (1-1) \operatorname{mult}_F \tau = \varrho a(E, Q, \Delta_Q).$$

Moreover $\operatorname{mult}_F(\mathfrak{b}\mathcal{O}_X) = \varrho \operatorname{mult}_E \mathfrak{b}$, so

$$a(F, X, \Delta_X, \mathfrak{b}\mathcal{O}_X) \geq \varrho a(E, Q, \Delta_Q, \mathfrak{b}).$$

Conversely, for a prime divisor E over Q with center q, let $Q' \to Q$ be a higher model on which E is a divisor and which is smooth at the generic point of E, $X' := Q' \times \mathbb{A}^1_\tau$, and F_E the exceptional divisor of the blow-up of X' along $E \times \{0\}$ near the generic point of $E \times \{0\}$. The valuation $\operatorname{mult}_{F_E}$ is the monomial valuation with weights $(1,1)$ in the parameters z and τ, where z is the pullback of a local equation of E at the generic point of E, hence the Gauss extension of mult_E with τ-value 1, and F_E is a prime divisor over X (cf. [Liu26, Lemma 8.1(2)]). Then F_E has center x on X, $\operatorname{mult}_{F_E}(\phi \circ p) = \operatorname{mult}_E \phi$ for any rational function ϕ on Q, and

$$a(F_E, X, \Delta_X, \mathfrak{b}\mathcal{O}_X) = 2 - (1 - a(E, Q, \Delta_Q)) - 1 - \operatorname{mult}_E \mathfrak{b} = a(E, Q, \Delta_Q, \mathfrak{b}).$$

If $\operatorname{mld}(Q \ni q, \Delta_Q, \mathfrak{b}) \geq 0$, then $a(E, Q, \Delta_Q, \mathfrak{b}) \geq 0$ for any E with center q, and $\varrho \geq 1$ implies

$$a(F, X, \Delta_X, \mathfrak{b}\mathcal{O}_X) \geq a(E, Q, \Delta_Q, \mathfrak{b}) \geq \operatorname{mld}(Q \ni q, \Delta_Q, \mathfrak{b}).$$

Together with the prime divisors F_E, this implies the equality. If $a(E, Q, \Delta_Q, \mathfrak{b}) < 0$ for some E with center q, then $\operatorname{mld}(Q \ni q, \Delta_Q, \mathfrak{b}) = -\infty$ by **Step 3**, applied to a log resolution on which E is a divisor, and the prime divisors F_E imply $\operatorname{mld}(X \ni x, \Delta_X, \mathfrak{b}\mathcal{O}_X) = -\infty$.

Next, we show that

$$\mathfrak{m}^\ell \cap \mathbb{C}[u,t]^H \subset \mathfrak{m}_q^{\ell_0}.$$

Since H is diagonal, $\mathbb{C}[u,t]^H$ is spanned by the H-invariant monomials, so it suffices to show that an H-invariant monomial ϕ of degree at least $\ell = |H|\ell_0$ is a product of ℓ_0 non-constant H-invariant monomials. Write $\phi = y_1 \cdots y_n$ with $y_i \in \{u, t\}$. If $n \geq |H|$, then two of the $|H|+1$ monomials $y_1 \cdots y_i$, $0 \leq i \leq |H|$, have equal characters, so $y_{i+1} \cdots y_{i'}$ is H-invariant for some $0 \leq i < i' \leq |H|$. Removing $\ell_0 - 1$ such H-invariant factors of degree at most $|H|$ one after

another from ϕ, we obtain the claim, since the remaining monomial is H-invariant of degree at least $|H|$.

Now let $\mathfrak{a}'_j$, $\mathfrak{d}'_j$, and $\mathfrak{c}'_l$ be the ideals of **Step 2** for the given generators of $\mathfrak{a}_j$, $\mathfrak{d}_j$, and $\mathfrak{c}_l$. For any j and i,

$$g_{j,i}^{|H|} - g_{j,i}'^{|H|} = (g_{j,i} - g'_{j,i}) \sum_{b=0}^{|H|-1} g_{j,i}^{b} g_{j,i}'^{|H|-1-b}$$

is an H-invariant element of $\mathfrak{m}^{\ell}$, hence an element of $\mathfrak{m}_q^{\ell_0}$. Thus

$$\mathfrak{a}'_j + \mathfrak{m}_q^{\ell_0} = \mathfrak{d}'_j + \mathfrak{m}_q^{\ell_0}, \quad \text{hence} \quad \mathfrak{a}'_j \mathcal{O}_{X,x} + \mathfrak{m}_x^{\ell_0} = \mathfrak{d}'_j \mathcal{O}_{X,x} + \mathfrak{m}_x^{\ell_0}$$

for the maximal ideal $\mathfrak{m}_x$ of $\mathcal{O}_{X,x}$. The boundary Δ_X and the exponents $c_j/|H|$ and $e_l/|H|$ belong to Γ. By Theorem 2.10(2),

$$\begin{aligned} &\operatorname{mld}\Big(X \ni x, \Delta_X, \prod_j (\mathfrak{a}'_j \mathcal{O}_X)^{c_j/|H|} \prod_l (\mathfrak{c}'_l \mathcal{O}_X)^{e_l/|H|}\Big) \\ &\quad = \operatorname{mld}\Big(X \ni x, \Delta_X, \prod_j (\mathfrak{d}'_j \mathcal{O}_X)^{c_j/|H|} \prod_l (\mathfrak{c}'_l \mathcal{O}_X)^{e_l/|H|}\Big). \end{aligned}$$

By the equality of the minimal log discrepancies on X and on Q and by **Step 2**, $(*)$ follows.

Step 5. In this step, we prove (1), with ℓ as in **Step 4**.

We first show $(*)$ over any K. Let $g_{j,i}, g'_{j,i} \in K[u,t]$ be H-semi-invariant polynomials, some of which may be zero, with $g_{j,i} - g'_{j,i} \in \mathfrak{m}^{\ell}$, generating non-zero ideals $\mathfrak{a}_j$ and $\mathfrak{d}_j$, and let $c_1, \dots, c_s \in \Lambda$. Let $A \subset K$ be the $\mathbb{C}$-subalgebra generated by the coefficients of the $g_{j,i}$ and $g'_{j,i}$ and by the inverses of one non-zero coefficient of each family $(g_{j,i})_i$ and $(g'_{j,i})_i$, and $V := \operatorname{Spec} A$. Then both families satisfy the hypotheses of **Step 3** for V, and the inclusion $A \subset K$ defines a point of V with values in K mapping to ξ. For any $\mathbb{C}$-point v of V, $g_{j,i}(v) - g'_{j,i}(v) \in \mathfrak{m}^{\ell}$, since the coefficients of $g_{j,i} - g'_{j,i}$ in degrees less than ℓ are zero. Applying **Step 3** to both families with a common V°, and **Step 4** at a $\mathbb{C}$-point of V°, we obtain $(*)$ over K.

Now let k, (f_j), (f'_j), and $c_1, \dots, c_s$ be as in (1). For $n \geq \ell$, let

$$\mu_n(f) := \operatorname{mld}\Big(\Big(T_k \ni 0, \prod_j ((f_j) + \mathfrak{m}^n)^{c_j} \cdot \mathfrak{c}_k\Big)/H\Big),$$

and let $\mu_n(f')$ be defined by the formula for $\mu_n(f)$ with f'_j in place of f_j. The ideal $(f_j) + \mathfrak{m}^n$ is the pullback to T_k of the ideal of $k[u,t]$ generated by $f_j^{<n}$ and the monomials of degree n, the ideal $(f'_j) + \mathfrak{m}^n$ is the pullback of the ideal generated by $f_j'^{<n}$ and the monomials of degree n, and $f_j^{<n} - f_j'^{<n} \in \mathfrak{m}^{\ell}$. Thus **Step 1** and $(*)$ over k imply $\mu_n(f) = \mu_n(f')$. Let P be a formal prime divisor over T_k with center the closed point. Since

$$(f_j) \subset (f_j) + \mathfrak{m}^{n+1} \subset (f_j) + \mathfrak{m}^n,$$

we have

$$\operatorname{mld}\Big(\Big(T_k \ni 0, \prod_j (f_j)^{c_j} \cdot \mathfrak{c}_k\Big)/H\Big) \leq \mu_{n+1}(f) \leq \mu_n(f).$$

Moreover,

$$\operatorname{mult}_P((f_j) + \mathfrak{m}^n) = \min(\operatorname{mult}_P f_j, n \operatorname{mult}_P \mathfrak{m}) = \operatorname{mult}_P f_j \quad \text{for } n > \max_j \operatorname{mult}_P f_j,$$

so

$$\frac{a(P, T_k, 0, \prod_j (f_j)^{c_j} \cdot \mathfrak{c}_k)}{|I_H(P)|} \geq \mu_n(f) \geq \lim_{n' \to +\infty} \mu_{n'}(f).$$

Hence the left hand side of (1) is equal to $\lim_n \mu_n(f)$, and the right hand side of (1) is equal to $\lim_n \mu_n(f')$. Since $\mu_n(f) = \mu_n(f')$ for any $n \geq \ell$, (1) follows.

Step 6. In this step, we prove (2). Recall from [Liu26, Construction 14.13] that the $f_{i,j}$ are H-semi-invariant polynomials with $\operatorname{mult}_0 f_{i,j} \leq N$ for a fixed positive integer N, that the character of $f_{i,j}$ depends only on j and is the character of $f_{\infty,j}$, and that, for any positive integer d, with $\mathcal{J}_d$ the affine space of s-tuples of polynomials of degree less than d in u,t, the tuple $(f_{\infty,j}^{<d})_j$ is a k-point of $\mathcal{J}_d$ mapping to the generic point of an irreducible closed subset $Z_d \subset \mathcal{J}_d$, and any non-empty open subset of Z_d contains $(f_{i,j}^{<d})_j$ for all but finitely many i.

Fix an integer $d \geq \max(\ell, N+1)$, with ℓ as in (1), and regard Z_d as an integral scheme. The coordinates of $\mathcal{J}_d$ define polynomials $g_j \in \mathcal{O}(Z_d)[u,t]$ of degree less than d whose specializations at the points $(f_{i,j}^{<d})_j$ and $(f_{\infty,j}^{<d})_j$ are $f_{i,j}^{<d}$ and $f_{\infty,j}^{<d}$. The vanishing of the coefficients of g_j at the monomials whose character is not the character of $f_{1,j}$ is a closed condition which holds at the points $(f_{i,j}^{<d})_j$, $i \gg 0$. The vanishing of the coefficients of g_j in degrees at most N is a closed condition which fails at the points $(f_{i,j}^{<d})_j$, since $\operatorname{mult}_0 f_{i,j} \leq N < d$. As the points $(f_{i,j}^{<d})_j$, $i \gg 0$, are dense in Z_d, the first condition holds on Z_d, and the second condition fails at the generic point of Z_d. Let $V \subset Z_d$ be a non-empty affine open subset on which, for any j, the coefficients of g_j in degrees at most N have no common zero. Then V and the g_j satisfy the hypotheses of **Step 3** with $n_j = 1$, and the open subset $V^\circ \subset V$ of **Step 3** contains the image of $(f_{\infty,j}^{<d})_j$ and the points $(f_{i,j}^{<d})_j$ for all but finitely many i. For such i, since $f_{i,j} - f_{i,j}^{<d} \in \mathfrak{m}^d \subset \mathfrak{m}^\ell$ and $f_{\infty,j} - f_{\infty,j}^{<d} \in \mathfrak{m}^\ell$, **Step 1**, (1), and **Step 3** imply

$$\begin{aligned}\operatorname{mld}\Big(\Big(\mathbb{A}^2 \ni 0, \prod_j (f_{i,j})^{c_j} \cdot \mathfrak{c}\Big)/H\Big) &= \operatorname{mld}\Big(\Big(\mathbb{A}^2 \ni 0, \prod_j (f_{i,j}^{<d})^{c_j} \cdot \mathfrak{c}\Big)/H\Big)\\ &= \operatorname{mld}\Big(\Big(\mathbb{A}^2_k \ni 0, \prod_j (f_{\infty,j}^{<d})^{c_j} \cdot \mathfrak{c}_k\Big)/H\Big)\\ &= \operatorname{mld}\Big(\Big(T_k \ni 0, \prod_j (f_{\infty,j})^{c_j} \cdot \mathfrak{c}_k\Big)/H\Big).\end{aligned}$$

Thus (2) holds for all but finitely many i. Multiplying $f_{i,j}$ by a non-zero constant does not change $\operatorname{div}(f_{i,j})$, so (2) also holds for the tuples of [Liu26, Construction 14.13] before their normalization by non-zero constants.

Step 7. In this step, we prove (3). Let $\theta_0, \dots, \theta_s \geq 0$ and $S := \{j \mid \theta_j > 0\}$. For $i \geq 1$, let

$$\mathfrak{a}_i := \prod_{j \in S} (f_{i,j})^{\theta_j} \cdot \mathfrak{c}^{\theta_0} \text{ on } \mathbb{A}^2, \qquad \mathfrak{a}_\infty := \prod_{j \in S} (f_{\infty,j})^{\theta_j} \cdot \mathfrak{c}_k^{\theta_0} \text{ on } T_k,$$

and for a positive integer d let $\mathfrak{a}_i^{<d}$ and $\mathfrak{a}_\infty^{<d}$ be obtained by replacing $f_{i,j}$ and $f_{\infty,j}$ with $f_{i,j}^{<d}$ and $f_{\infty,j}^{<d}$. If $S = \emptyset$ and $\theta_0 = 0$, then both sides of (3) are $+\infty$. Otherwise, the factors of $\mathfrak{a}_i$ with positive exponents are proper ideals, since $f_{i,j}(0) = 0$ and $\mathfrak{c}_l \subset \mathfrak{m}$. The vanishing of the constant term of g_j is a closed condition on Z_d which holds at the points $(f_{i,j}^{<d})_j$, $i \gg 0$, so $f_{\infty,j}(0) = 0$ as well. Thus all the thresholds below are finite. Let $q_{\min} > 0$ be the minimum of the numbers θ_j, $j \in S$, and, if $\theta_0 > 0$, $\theta_0 e_l$, $1 \leq l \leq m$.

Let $\epsilon > 0$ and $d \geq \max(\ell, N+1, 2/(\epsilon q_{\min}))$. Since $(f_{i,j}) + \mathfrak{m}^d = (f_{i,j}^{<d}) + \mathfrak{m}^d$, Theorem 2.12(2), applied to the klt germ $(\mathbb{A}^2 \ni 0)$ with $l = d$, implies

$$\left|\operatorname{lct}_0(\mathbb{A}^2; \mathfrak{a}_i) - \operatorname{lct}_0(\mathbb{A}^2; \mathfrak{a}_i^{<d})\right| \leq \epsilon,$$

and [dFEM11, Proposition 2.19(ii)] on T_k implies

$$\left|\operatorname{lct}_0(T_k; \mathfrak{a}_\infty) - \operatorname{lct}_0(T_k; \mathfrak{a}_\infty^{<d})\right| \leq \epsilon.$$

By [dFEM11, Proposition 2.11], $\operatorname{lct}_0(T_k; \mathfrak{a}_\infty^{<d})$ is equal to the lc threshold at 0 of the corresponding $\mathbb{R}$-ideal on $\mathbb{A}^2_k$. By **Step 3** for the open subset $V^\circ \subset Z_d$ of **Step 6** for this d,

$$\operatorname{lct}_0(\mathbb{A}^2; \mathfrak{a}_i^{<d}) = \operatorname{lct}_0(T_k; \mathfrak{a}_\infty^{<d})$$

for all but finitely many i. Thus

$$\left|\operatorname{lct}_0(\mathbb{A}^2;\mathfrak{a}_i)-\operatorname{lct}_0(T_k;\mathfrak{a}_\infty)\right|\le 2\epsilon$$

for all but finitely many i. Since $\epsilon>0$ is arbitrary,

$$\operatorname{lct}_0(\mathbb{A}^2;\mathfrak{a}_i)\to\operatorname{lct}_0(T_k;\mathfrak{a}_\infty),$$

which proves (3). □

Set-up 3.20. We recall the data of [Liu26, Set-up 14.1] for $p=q=1$. Let $\Gamma\subset[0,1]$ be a DCC set with $\Gamma\backslash\{0\}\neq\emptyset$ and $\gamma:=\gamma(\Gamma)=\inf(\Gamma\backslash\{0\})$ [Liu26, Definition 2.9], and P, e_0 positive integers. Let $X:=\mathbb{A}^3_{x,y,t}$, $u:=xy$, $\mathbf{T}:=\mathbf{T}_{(1,-1,0)}$, and G a finite diagonal abelian group acting faithfully on X and containing $\mu_r(1,-1,0)$ for a positive integer r. Let $c_1,\dots,c_s\in\Gamma$ be positive, and $h_j=h_j^0+R_j$ non-zero G-semi-invariant polynomials, where $h_j^0=x^{a_j}y^{b_j}F_j(u,t)$, with non-negative integers a_j,b_j and non-zero polynomials F_j, has the character of h_j, and $R_j\in(x,y)^N$ for a positive integer N. Let $B:=\sum_j c_j\operatorname{div}(h_j)$ with (X,B) klt near 0, $B^0:=\sum_j c_j\operatorname{div}(h_j^0)$, $c_x:=\sum_j c_ja_j$, $c_y:=\sum_j c_jb_j$, $c_u:=\max\{c_x,c_y\}$, $T:=\mathbb{A}^2_{u,t}$, and

$$B_T:=\sum_j c_j\operatorname{div}_T(F_j)+c_u\operatorname{div}_T(u),$$

so that (T,B_T) is the residual plane pair. When $G\cap\mathbf{T}=\mu_r(1,-1,0)$, $H\subset(\mathbb{C}^*)^2$ is the image of G under the characters of u and t, acting diagonally on T, and $b_H:=\operatorname{mld}((T\ni 0,B_T)/H)$ when (T,B_T) is lc near 0.

The next lemma deduces, for $p=q=1$, the lc hypotheses of [Liu26, Proposition 14.15] on the threefold from the lc property of $(T,\theta B_T)$ for one real number $\theta>1$.

Lemma 3.21. *Notation and conditions as in* [Liu26, Set-up 14.1] *with $p=q=1$, and let $1<\theta'<\theta$. Then the following hold.*

1. *There exists a positive integer N_0, depending only on $\theta-\theta'$ and γ, such that if $(T,\theta B_T)$ is lc near 0 and $N\ge N_0$, then $(X,\theta' B)$ and $(X,\theta' B^0)$ are lc near 0.*
2. *Assume moreover that h_j^0 is the sum of all terms of h_j of $\mathbf{T}$-weight a_j-b_j for any j, that $G\cap\mathbf{T}=\mu_r(1,-1,0)$, and that $|H|\le e_0$. Then there exist positive integers r_0 and N_1, depending only on θ, θ', P, e_0, and Γ, such that if $(T,\theta B_T)$ is lc near 0, $r\ge r_0$, and $N\ge N_1$, then $\operatorname{mld}((X\ni 0,B)/G)\ge b_H$.*

Proof. (1) Let $N_0:=\lceil 3/((\theta-\theta')\gamma)\rceil$, and assume that $(T,\theta B_T)$ is lc near 0 and $N\ge N_0$. For $B':=\theta B^0$, the data of [Liu26, Lemma 14.5] are $c_j'=\theta c_j$, $a_j'=a_j$, $b_j'=b_j$, $F_j'=F_j$, $c_x'=\theta c_x$, and $c_y'=\theta c_y$, and, since $p=q=1$,

$$c_u'=\max\{\theta c_x,\theta c_y\}=\theta c_u,\qquad B_T'=\theta B_T.$$

Thus [Liu26, Lemma 14.5(2)] implies that $(X,\theta B^0)$ is lc near 0. In particular, $(X,\theta' B^0)$ is lc near 0, as $\theta'<\theta$.

Since $N\ge 1$, $h_j(0)=0$ if and only if $h_j^0(0)=0$. Let J be the set of indices j with $h_j(0)=0$. If $J=\emptyset$, then $B=B^0=0$ near 0. Otherwise, let $\mathfrak{m}$ be the maximal ideal of $0\in X=\mathbb{A}^3$, and consider the $\mathbb{R}$-ideals

$$\mathfrak{a}:=\prod_{j\in J}(h_j^0)^{c_j},\qquad \mathfrak{b}:=\prod_{j\in J}(h_j)^{c_j}.$$

Then $\theta\le\operatorname{lct}_0(X;\mathfrak{a})<+\infty$, $c_j\ge\gamma$ for any j, since $c_j\in\Gamma\backslash\{0\}$, and $(h_j^0)+\mathfrak{m}^N=(h_j)+\mathfrak{m}^N$ as $R_j\in(x,y)^N$. Since

$$N\ge N_0\ge\frac{3}{(\theta-\theta')\min_j c_j},$$

Theorem 2.12(2), applied to the klt germ $(X\ni 0)$ with $n=3$, $\epsilon=\theta-\theta'$, and $l=N$, implies

$$\operatorname{lct}_0(X;\mathfrak{b})\ge\operatorname{lct}_0(X;\mathfrak{a})-(\theta-\theta')\ge\theta'.$$

Thus $(X, \theta' B)$ is lc near 0. The integer N_0 depends only on $\theta - \theta'$ and γ, and not on the number of the h_j.

(2) Let r_0 and N_0' be the integers of [Liu26, Proposition 14.15] for $\lambda = \theta'$ and the given P, e_0, and Γ, and let $N_1 := \max(N_0, N_0')$, with N_0 as in (1). Then r_0 and N_1 depend only on θ, θ', P, e_0, and Γ. Assume that $(T, \theta B_T)$ is lc near 0, $r \geq r_0$, and $N \geq N_1$. By (1), $(X, \theta' B^0)$ and $(X, \theta' B)$ are lc near 0, and $(T, \theta' B_T)$ is lc near 0 since $\theta' < \theta$ and $B_T \geq 0$. Thus the hypotheses of [Liu26, Proposition 14.15] hold for $\lambda = \theta' > 1$, and $\mathrm{mld}((X \ni 0, B)/G) \geq b_H$. □

After passing to a crepant model in the sense of Definition-Proposition 4.10 below, the center of a lift of a computing prime divisor can be a curve. The next lemma takes an invariant transverse function t and turns the quotient minimal log discrepancy at the generic point of the curve into the quotient minimal log discrepancy at a closed point of the generic fiber of t. A prime divisor computing the first one computes the second one, with equal inertia groups over the threefold and over the generic fiber, so Proposition 5.4 can be applied after the base change of the generic fiber of t to an algebraic closure Ω of the base field of the generic fiber.

Lemma 3.22. *Let (X, B) be a threefold sub-pair, $C \subset X$ an integral curve with generic point η, and H a finite group acting faithfully on X, preserving B and fixing C pointwise. Let $(X/H, B_{X/H})$ be the crepant quotient and $\bar{\eta}$ the image of η. Then the following hold.*

(1) There exists an H-invariant rational function t on X which is regular at η and whose restriction to C is not constant.

(2) Let t be as in (1), $U \subset X$ the open subset on which t is regular, $k := \mathbb{C}(t)$, X_k the generic fiber of $t : U \to \mathbb{A}^1$, $c \in X_k$ the closed point corresponding to η, and B_k the restriction of B. Then $\mathcal{O}_{X_k,c} = \mathcal{O}_{X,\eta}$ and $\mathrm{mld}((X_k \ni c, B_k)/H) = \mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H})$. Moreover, any prime divisor F over X lying over a prime divisor over X/H which computes $\mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H})$ is a prime divisor over X_k with center c and inertia group $I_H(F)$, and it computes $\mathrm{mld}((X_k \ni c, B_k)/H)$.

(3) In (2), let Ω be an algebraic closure of k, c' a closed point of $X_k \otimes_k \Omega$ over c, X_Ω the connected component of $X_k \otimes_k \Omega$ containing c', and B_Ω the restriction of $B_k \otimes_k \Omega$ to X_Ω. Then H acts on X_Ω fixing c', and

$$\mathrm{mld}((X_\Omega \ni c', B_\Omega)/H) = \mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H}).$$

Moreover, for F as in (2), any prime divisor F' over X_Ω with center c' such that $\mathrm{mult}_{F'}(p^*\zeta) = \mathrm{mult}_F \zeta$ for any non-zero rational function ζ on X, where $p : X_\Omega \to X$ is the induced morphism, has inertia group $I_H(F)$ and computes $\mathrm{mld}((X_\Omega \ni c', B_\Omega)/H)$.

Proof. Let $\pi : X \to X/H$ be the quotient morphism.

(1) Since C is an integral curve, there exists a non-constant rational function $\bar{v}$ on C. As the residue field of $\mathcal{O}_{X,\eta}$ is the field of rational functions on C, there exists $v \in \mathcal{O}_{X,\eta}$ with $v|_C = \bar{v}$. Since H fixes C pointwise, $\sigma(\eta) = \eta$ and $(\sigma^* v)|_C = v|_C$ for any $\sigma \in H$. Thus

$$t := \frac{1}{|H|} \sum_{\sigma \in H} \sigma^* v$$

is an H-invariant rational function on X which is regular at η, and $t|_C = v|_C$ is not constant.

(2) Since $t|_C$ is not constant, any non-zero polynomial in t with coefficients in $\mathbb{C}$ has non-zero restriction to C, and hence is a unit of $\mathcal{O}_{X,\eta}$. Thus η maps to the generic point of $\mathbb{A}^1$, so η is a point of X_k, and $\mathcal{O}_{X_k,\eta}$, the localization of $\mathcal{O}_{X,\eta}$ at $\mathbb{C}[t]\backslash\{0\}$, is equal to $\mathcal{O}_{X,\eta}$. Since $t|_C$ is not constant, the morphism $C \cap U \to \mathbb{A}^1$ is dominant with finite generic fiber, so η is a closed point c of X_k, and $\mathcal{O}_{X_k,c} = \mathcal{O}_{X,\eta}$.

Let F be a prime divisor over X with center C, and $Y \to U$ a projective birational morphism from a normal variety Y on which F is a prime divisor. Then F maps onto C and $t|_C$ is not constant, so F dominates $\mathbb{A}^1$. Thus the generic fiber F_k of F over $\mathbb{A}^1$ is a prime divisor on the generic fiber Y_k of Y over $\mathbb{A}^1$, Y_k is projective and birational over X_k, and the center of F_k on X_k

is c. Conversely, let D be a prime divisor on a normal variety Z which is projective and birational over X_k, with center c on X_k. Spreading out Z and D over an open subset $V \subset \mathbb{A}^1$ [Gro66, §8], and replacing the model so obtained with its normalization, which does not change the generic fiber Z since normalization commutes with localization and Z is normal, we obtain a projective birational morphism $Y \to U \cap t^{-1}(V)$ from a normal variety Y and a prime divisor F on Y with generic fibers Z and D over $\mathbb{A}^1$. The center of F on X is the closure of c, that is, C. Thus taking generic fibers over $\mathbb{A}^1$ gives a one-to-one correspondence between the prime divisors over X with center C and the prime divisors over X_k with center c. In both cases $a(F, X_k, B_k) = a(F, X, B)$, since on a higher model of U on which F is a divisor, the canonical divisor restricts to the relative canonical divisor of the generic fiber over k (cf. [Liu26, Lemma 3.6(1)]).

By [Liu26, Lemma 3.6(3)],

$$\mathrm{mld}((X_k \ni c, B_k)/H) = \mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H}).$$

If $\mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H}) = -\infty$, then no prime divisor computes it, and the remaining statement of (2) is empty. Otherwise, let E be a prime divisor over X/H which computes $\mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H})$, and F a prime divisor over X lying over E [Liu26, Definition 2.7]. Since the fibers of π are H-orbits and H fixes C pointwise, $\pi^{-1}(\pi(C)) = C$. As $\pi(\mathrm{center}_X F) = \mathrm{center}_{X/H} E = \pi(C)$, we have $\mathrm{center}_X F = C$. Hence F is a prime divisor over X_k with center c. Since t is H-invariant, H acts on U over $\mathbb{A}^1$, hence acts k-linearly on X_k, fixing c. Let $Y \to U$ be an H-equivariant projective birational morphism from a normal variety Y on which F is a prime divisor (for example, the normalization of the main component of the fiber product over U of the H-translates of a projective birational model of U on which F is a divisor). Then $I_H(F)$ consists of the elements of H which map F to itself and restrict to the identity on F. Since F_k is dense in F, this is the inertia group of F_k on Y_k, so $I_H(F)$ does not change when F is regarded as a prime divisor over X_k. By [Liu26, Lemma 2.6(2)] and [Liu26, Lemma 3.6(3)],

$$\begin{aligned}\frac{a(F, X_k, B_k)}{|I_H(F)|} = \frac{a(F, X, B)}{|I_H(F)|} &= a(E, X/H, B_{X/H}) \\ &= \mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H}) = \mathrm{mld}((X_k \ni c, B_k)/H).\end{aligned}$$

Thus F computes $\mathrm{mld}((X_k \ni c, B_k)/H)$.

(3) Since H fixes C pointwise, H acts trivially on the point c of X_k, hence on the fiber of $X_k \otimes_k \Omega \to X_k$ over c, as H acts Ω-linearly on $X_k \otimes_k \Omega$. In particular, H fixes c' and preserves X_Ω. The action of H on X_Ω is faithful, since p is dominant and H acts faithfully on U.

Let F be as in (2), $Y \to U$ an H-equivariant projective birational morphism from a normal variety Y on which F is a prime divisor (for example, the normalization of the main component of the fiber product over U of the H-translates of a projective birational model of U on which F is a divisor), Y_k and F_k the generic fibers of Y and F over $\mathbb{A}^1$, and Y_Ω the union of the components of $Y_k \otimes_k \Omega$ over X_Ω. Since Ω is a filtered union of finite separable extensions of k, Y_Ω is normal and $F_k \otimes_k \Omega$ is reduced. A local equation of F at the generic point of F pulls back to a local equation of each irreducible component of $(F_k \otimes_k \Omega) \cap Y_\Omega$ at the generic point of the component. Conversely, let F' be a prime divisor over X_Ω with center c' and $\mathrm{mult}_{F'}(p^*\zeta) = \mathrm{mult}_F \zeta$ for any non-zero rational function ζ on X. Since $Y_\Omega \to X_\Omega$ is proper, F' has a center y' on Y_Ω, and y' maps to the center of mult_F on Y_k, the generic point of F_k. As Ω is algebraic over k, the points of Y_Ω over the generic point of F_k are the generic points of the irreducible components of $(F_k \otimes_k \Omega) \cap Y_\Omega$. Thus y' is the generic point of such a component F''. Since Y_Ω is normal, $\mathcal{O}_{Y_\Omega, F''}$ is a discrete valuation ring dominated by the valuation ring of $\mathrm{mult}_{F'}$, hence equal to it, and $F' = F''$. Thus the prime divisors F' of (3) are the irreducible components of $(F_k \otimes_k \Omega) \cap Y_\Omega$ with center c' on X_Ω.

We show that $I_H(F') = I_H(F)$ for such F'. Since H acts on Y over $\mathbb{A}^1$, H acts Ω-linearly on $Y_k \otimes_k \Omega$ and preserves Y_Ω. Let $\sigma \in I_H(F)$. Then σ acts trivially on F_k, hence on $F_k \otimes_k \Omega$, so σ maps each F' to itself and acts trivially on it. Thus $\sigma \in I_H(F')$. Conversely, let $\sigma \in I_H(F')$.

Then σ maps F to itself, since the image of F' in Y is contained in F and in $\sigma(F)$, and σ acts trivially on F_k, since $F' \to F_k$ is dominant and σ-equivariant. As F_k is dense in F, $\sigma \in I_H(F)$.

We show that $a(F', X_\Omega, B_\Omega) = a(F, X_k, B_k)$. The crepant pullback of $K_{X_\Omega} + B_\Omega$ to Y_Ω is the pullback of the crepant pullback of $K_{X_k} + B_k$ to Y_k, F' is a component of $(F_k \otimes_k \Omega) \cap Y_\Omega$, and $F_k \otimes_k \Omega$ is reduced. Thus $a(F', X_\Omega, B_\Omega) = a(F, X_k, B_k)$.

Finally, let $\bar{t}$ be the rational function on X/H with $t = \pi^*\bar{t}$. Then $\bar{t}$ is regular at $\bar{\eta}$, the restriction of $\bar{t}$ to $\pi(C)$ is not constant, and the generic fiber of $\bar{t}$ on $(X, B)/H$ is $(X_k, B_k)/H$ [Liu26, Lemma 3.6(3)]. Since taking H-invariants commutes with the flat base change from k to Ω, we have $(X_k/H) \otimes_k \Omega = (X_k \otimes_k \Omega)/H$. Hence the image $\bar{c}'$ of c' is a closed point over the image of c, the connected component of $(X_k/H) \otimes_k \Omega$ containing $\bar{c}'$ is X_Ω/H, and the restriction of $B_{X_k/H} \otimes_k \Omega$ to X_Ω/H is the boundary $B_{X_\Omega/H}$ of the crepant quotient $(X_\Omega, B_\Omega)/H$, since the equality $K_{X_k} + B_k = \pi_k^*(K_{X_k/H} + B_{X_k/H})$ [Liu26, Lemma 2.6(0)] for the quotient morphism $\pi_k : X_k \to X_k/H$ is preserved by the base change to Ω, and the pullback of $\mathbb{R}$-divisors under the quotient morphism $X_\Omega \to X_\Omega/H$ is injective, as its composition with the pushforward is multiplication by $|H|$. By [Liu26, Lemma 2.6] and by [Liu26, Lemma 3.6(2)], applied to the threefold sub-pair $(X/H, B_{X/H})$, the curve $\pi(C)$, and the function $\bar{t}$,

$$\mathrm{mld}((X_\Omega \ni c', B_\Omega)/H) = \mathrm{mld}(X_\Omega/H \ni \bar{c}', B_{X_\Omega/H}) = \mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H}).$$

If $\mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H}) = -\infty$, then no F as in (2) exists, and the remaining statement of (3) is empty. Otherwise, together with (2), $I_H(F') = I_H(F)$, and $a(F', X_\Omega, B_\Omega) = a(F, X_k, B_k)$, we obtain

$$\begin{aligned}\frac{a(F', X_\Omega, B_\Omega)}{|I_H(F')|} = \frac{a(F, X_k, B_k)}{|I_H(F)|} &= \mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H}) \\ &= \mathrm{mld}((X_\Omega \ni c', B_\Omega)/H).\end{aligned}$$

Thus F' computes $\mathrm{mld}((X_\Omega \ni c', B_\Omega)/H)$. □

4. Limits, admissions and crepant models

In this section, we prove the statements on limits, numerical admissions, and crepant models used repeatedly in Sections 7–11.

4.1. **Compatible generic limits.**

In this subsection, we construct compatible generic limits of tuples of semi-invariant functions on a fixed germ and near a fixed proper curve or point (Construction-Proposition 4.1), and we show that lc, lc thresholds, and quotient minimal log discrepancies are constant on them (Theorem 4.2).

Case (b) of Construction-Proposition 4.1 is used for reference pairs on a fixed surface near a proper curve or a point E. There, minimal log discrepancies are taken over all centers contained in the image of E, the generic point of E among them, so the jets on the charts have to be compatible through the gluing maps for the limit to be a pair near E.

Construction-Proposition 4.1. Let s be a positive integer and let $\mathcal{A}$ be one of the following fixed ambient data.

(a) $\mathbb{A}^n$ with a fixed finite subgroup $H \subset GL_n(\mathbb{C})$ acting linearly, and the point 0.

(b) A fixed normal surface Y with a fixed connected closed subset $E \subset Y$ which is a proper curve, possibly reducible, or a closed point, and the following fixed data.

(b1) Finitely many indices α, and for each α a normal affine surface $S_\alpha \subset \mathbb{A}^{N_\alpha}$ defined by polynomials, a finite subgroup $K_\alpha \subset GL_{N_\alpha}(\mathbb{C})$ acting linearly on $\mathbb{A}^{N_\alpha}$, preserving S_α and acting faithfully on it, a K_α-invariant open subset $U_\alpha \subset S_\alpha$ with at worst quotient singularities, and a K_α-invariant ideal $\mathfrak{n}_\alpha \subset \mathbb{C}[S_\alpha]$.

(b2) Open subsets $V_\alpha \subset Y$ covering E with isomorphisms $V_\alpha \cong U_\alpha/K_\alpha$, under which $E \cap V_\alpha$ is the image of $U_\alpha \cap V(\mathfrak{n}_\alpha)$, such that the crepant quotients $(U_\alpha, 0)/K_\alpha$ glue to a pair $(\bigcup_\alpha V_\alpha, B_{\mathrm{br}})$.

(b3) A finite group H acting on Y and preserving E.

For instance, S_α may be $\mathbb{A}^2_{p,q}$ with $K_\alpha \subset GL_2(\mathbb{C})$ and $\mathfrak{n}_\alpha = (e_\alpha)$ for a reduced K_α-semi-invariant polynomial e_α such as q.

For $i \geq 1$ let $(f_{i,1}, \dots, f_{i,s})$ be s-tuples of the following form. In case (a), the $f_{i,j}$ are non-zero H-semi-invariant regular functions on $\mathbb{A}^n$ whose characters depend only on j and whose orders at 0 are bounded. In case (b), $f_{i,j} = (f_{i,j,\alpha})_\alpha$, where each $f_{i,j,\alpha}$ is a non-zero K_α-semi-invariant element of the $\mathfrak{n}_\alpha$-adic completion of $\mathbb{C}[S_\alpha]$ (for instance a polynomial in p, q) whose character depends only on j and α, and the following hold.

(i) The divisors of the $f_{i,j,\alpha}$ on the formal completions of the U_α along $V(\mathfrak{n}_\alpha)$ descend to an H-invariant $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D_{i,j}$ on the formal completion $\widehat{Y}$ of $\bigcup_\alpha V_\alpha$ along E.

(ii) For any α and β, the quotient of the pullbacks of $f_{i,j,\alpha}$ and $f_{i,j,\beta}$ to the formal completion, along the preimage of E, of the normalization of the fiber product of U_α and U_β over $V_\alpha \cap V_\beta$ is a unit independent of i.

(iii) The orders of the $f_{i,j,\alpha}$ along the irreducible components of $U_\alpha \cap V(\mathfrak{n}_\alpha)$ are bounded, where the order along a component which is a point is the order with respect to the maximal ideal of that point.

(iv) For any positive integer d, the classes of the $f_{i,j,\alpha}$ in $\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^d$ are represented by restrictions of polynomials whose degrees are bounded independently of i.

For a positive integer d, the *d-jet* of such a tuple is the tuple of the parts of degree less than d in case (a), and of the classes in $\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^d$ in case (b) (the parts of q-degree less than d if $S_\alpha = \mathbb{A}^2_{p,q}$ and $\mathfrak{n}_\alpha = (q)$). The d-jets of the tuples belong to a fixed affine space $\mathcal{J}_d$ of finite dimension.

Then, after passing to a subsequence, there exist an algebraically closed field $k \supset \mathbb{C}$ and s-tuples $(f_{\infty,j})_j$ with the following properties. In case (a), the $f_{\infty,j}$ are non-zero H-semi-invariant power series in $k[[x_1, \dots, x_n]]$ with the characters of the $f_{i,j}$. In case (b), $f_{\infty,j} = (f_{\infty,j,\alpha})_\alpha$ with non-zero K_α-semi-invariant elements $f_{\infty,j,\alpha}$ of the $\mathfrak{n}_\alpha$-adic completion of $k[S_\alpha] := \mathbb{C}[S_\alpha] \otimes_{\mathbb{C}} k$ (that is, of $k[p][[q]]$ if $S_\alpha = \mathbb{A}^2_{p,q}$ and $\mathfrak{n}_\alpha = (q)$) with the characters of the $f_{i,j,\alpha}$, whose divisors define an H-invariant $\mathbb{Q}$-divisor on the formal completion of $Y_k := Y \times_{\mathbb{C}} \operatorname{Spec} k$ along $E_k := E \times_{\mathbb{C}} \operatorname{Spec} k$. In both cases, for any positive integer d the d-jet of $(f_{\infty,j})_j$ maps to the generic point of an irreducible closed subset $Z_d \subset \mathcal{J}_d$, and any non-empty open subset of Z_d contains the d-jets of $(f_{i,j})_j$ for all but finitely many i. We call $(f_{\infty,j})_j$ a *compatible generic limit* of the tuples. Moreover, for any subset $J \subset \{1, \dots, s\}$, the tuple $(f_{\infty,j})_{j \in J}$ is a compatible generic limit of the tuples $(f_{i,j})_{j \in J}$ along the subsequence above.

We write $\mathcal{A}_k$ for $\operatorname{Spec} k[[x_1, \dots, x_n]]$ with the induced action of H in case (a). In case (b), we write $\mathcal{A}_k$ for the formal completion of Y_k along E_k with the induced action of H, together with the formal completions of the $U_{\alpha,k} := U_\alpha \times_{\mathbb{C}} \operatorname{Spec} k$ along the preimages of E_k and the induced actions of the K_α on them. In case (b), a pair on $\mathcal{A}_k$, or on $\widehat{Y}$ (the case $k = \mathbb{C}$, on which the $D_{i,j}$ live), is lc (resp. klt) if the pullback of the pair to $\operatorname{Spec} \widehat{\mathcal{O}}_{Y_k,y}$ is lc (resp. klt) in the sense of [dFEM11, Section 2] for any closed point $y \in E_k$, and the prime divisors over $\mathcal{A}_k$ with center contained in E_k are the irreducible components of E_k if E is a curve, and the formal prime divisors over $\operatorname{Spec} \widehat{\mathcal{O}}_{Y_k,y}$ with center y for the closed points $y \in E_k$. For an algebraically closed field $k \supset \mathbb{C}$, possibly $k = \mathbb{C}$, a *$\mathbb{Q}$-divisor on the formal completion of Y_k along E_k* is a family $(D_y)_y$ of $\mathbb{Q}$-divisors D_y on $\operatorname{Spec} \widehat{\mathcal{O}}_{Y_k,y}$, where y runs over the closed points of E_k. Such a family is *defined by the divisors of* the $f_{\infty,j,\alpha}$ (resp. of the $f_{i,j,\alpha}$ if $k = \mathbb{C}$) if, for any α with $y \in V_{\alpha,k} := V_\alpha \times_{\mathbb{C}} \operatorname{Spec} k$ and any closed point $\tilde{y} \in U_{\alpha,k}$ over y, the pullback of D_y to $\operatorname{Spec} \widehat{\mathcal{O}}_{S_{\alpha,k},\tilde{y}}$, where $S_{\alpha,k} := S_\alpha \times_{\mathbb{C}} \operatorname{Spec} k$, is $\operatorname{div}(f_{\infty,j,\alpha})$ (resp. $\operatorname{div}(f_{i,j,\alpha})$), the element $f_{\infty,j,\alpha}$ (resp. $f_{i,j,\alpha}$) being regarded as an element of $\widehat{\mathcal{O}}_{S_{\alpha,k},\tilde{y}}$ through $\tilde{y} \in V(\mathfrak{n}_\alpha)$. Such a family is *$H$-invariant* if, for any $h \in H$ and any closed point $y \in E_k$, D_y is the pullback of $D_{h(y)}$ under the isomorphism of the completed local rings of Y_k at $h(y)$ and at y induced by h.

In dimension two with diagonal H, case (a) is [Liu26, Construction 14.13], whose d-jets are also the parts of degree less than d. Construction-Proposition 4.1 follows [Kaw24, Section 4] and

[dFEM11, Section 3]. In case (b), the groups K_α need not be diagonal, E may be a point or a reducible curve, and H may permute the charts. We use the properness of E to show that the limit divisors are H-invariant. The proof of Construction-Proposition 4.1 only passes to a subsequence and multiplies none of the $f_{i,j}$ by a constant. A rescaling in case (b) would have to multiply the $f_{i,j,\alpha}$, for fixed i and j, by constants whose ratios for different α do not depend on i, since otherwise the quotients in (ii) would depend on i.

Proof of Construction-Proposition 4.1. The proof is long, so we divide it into several steps.

Step 1. In this step, we define the jet spaces and, in case (b), sections of fixed invertible sheaves near E.

In case (a), let $\mathcal{J}_d$ be the affine space of s-tuples of polynomials of degree less than d in $x_1, \dots, x_n$. In case (b), for any j, α, and d, let $W_{j,\alpha,d}$ be the linear span of the classes of the $f_{i,j,\alpha}$, $i \geq 1$, in $\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^d$. By (iv), the classes of the $f_{i,j,\alpha}$ are represented by restrictions of polynomials of bounded degree, so $W_{j,\alpha,d}$ is spanned by the images of the polynomials of bounded degree in N_α variables, so $\dim W_{j,\alpha,d} < +\infty$. Let $\mathcal{J}_d := \prod_{j,\alpha} W_{j,\alpha,d}$. In both cases, let $x_{i,d} \in \mathcal{J}_d$ be the d-jet of $(f_{i,j})_j$, and let $\tau_d : \mathcal{J}_{d+1} \to \mathcal{J}_d$ be the truncation in case (a) and the map induced by the surjections

$$\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^{d+1} \to \mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^d$$

in case (b). Then τ_d is linear and $\tau_d(x_{i,d+1}) = x_{i,d}$ for any i and d.

In case (a), let $\mathcal{J}_d^+ := \mathcal{J}_d$, $x_{i,d}^+ := x_{i,d}$, and $\tau_d^+ := \tau_d$.

In case (b), let $V_E := \bigcup_\alpha V_\alpha$, so that $\widehat{Y}$ is the formal completion of V_E along E, let $\mathcal{I}_E \subset \mathcal{O}_{V_E}$ be the ideal sheaf of E, let $\pi_\alpha : U_\alpha \to V_\alpha$ be the quotient morphisms, and let $r := \prod_\alpha |K_\alpha|$. Since $\mathfrak{n}_\alpha$ is K_α-invariant, so is $U_\alpha \cap V(\mathfrak{n}_\alpha)$, and hence

$$\pi_\alpha^{-1}(E \cap V_\alpha) = U_\alpha \cap V(\mathfrak{n}_\alpha).$$

Thus, as U_α is quasi-compact, there exists a positive integer c such that

$$\mathfrak{n}_\alpha^c \mathcal{O}_{U_\alpha} \subset \mathcal{I}_E \mathcal{O}_{U_\alpha} \quad \text{and} \quad \mathcal{I}_E^c \mathcal{O}_{U_\alpha} \subset \mathfrak{n}_\alpha \mathcal{O}_{U_\alpha}$$

for any α. Hence the formal completion $\widehat{U}_\alpha$ of U_α along $U_\alpha \cap V(\mathfrak{n}_\alpha)$ is the formal completion along $\pi_\alpha^{-1}(E \cap V_\alpha)$, the $f_{i,j,\alpha}$ restrict to sections of $\mathcal{O}_{\widehat{U}_\alpha}$, and π_α induces a finite surjective morphism from $\widehat{U}_\alpha$ onto the formal completion $\widehat{V}_\alpha$ of V_α along $E \cap V_\alpha$. The Reynolds operator of K_α splits the inclusion of $\mathcal{O}_{V_\alpha}$ into the $\mathcal{O}_{V_\alpha}$-module $\pi_{\alpha *}\mathcal{O}_{U_\alpha}$, so

$$\mathcal{I}_E^m \pi_{\alpha *}\mathcal{O}_{U_\alpha} \cap \mathcal{O}_{V_\alpha} = \mathcal{I}_E^m \mathcal{O}_{V_\alpha}$$

for any m, and $\mathcal{O}_{\widehat{V}_\alpha}$ is the sheaf of K_α-invariant sections of $\pi_{\alpha *}\mathcal{O}_{\widehat{U}_\alpha}$.

Fix j. The order of a character of K_α divides $|K_\alpha|$, so $f_{i,j,\alpha}^r$ is K_α-invariant, and

$$f_{i,j,\alpha}^r|_{\widehat{U}_\alpha} = \pi_\alpha^* g_{i,j,\alpha}$$

for a section $g_{i,j,\alpha}$ of $\mathcal{O}_{\widehat{V}_\alpha}$. The formal completions of U_α, V_α, and V_E along the given closed subsets are normal [Gro65, Scholie 7.8.3(v)], so finite surjective morphisms between them induce injective pullbacks of functions and divisors, and two sections with equal divisors differ by a unit. Since

$$\pi_\alpha^* \operatorname{div}(g_{i,j,\alpha}) = r \operatorname{div}(f_{i,j,\alpha}|_{\widehat{U}_\alpha}) = r\pi_\alpha^* D_{i,j}$$

by (i), and π_α is finite and surjective, we have

$$\operatorname{div}(g_{i,j,\alpha}) = r D_{i,j}|_{\widehat{V}_\alpha}.$$

The normalization $\widetilde{U}_{\alpha\beta}$ of the fiber product of U_α and U_β over $V_\alpha \cap V_\beta$ maps onto $V_\alpha \cap V_\beta$, and the pullback of $g_{i,j,\alpha}/g_{i,j,\beta}$ to the formal completion of $\widetilde{U}_{\alpha\beta}$ along the preimage of E is the r-th power of the unit of (ii), which is independent of i. Thus

$$v_{j,\alpha\beta} := g_{i,j,\alpha}/g_{i,j,\beta}$$

is a unit on the formal completion of $V_\alpha \cap V_\beta$ along $E \cap V_\alpha \cap V_\beta$, independent of i. Since $v_{j,\alpha\beta} = g_{1,j,\alpha}/g_{1,j,\beta}$, the $v_{j,\alpha\beta}$ satisfy the cocycle condition. Hence they are the transition functions of an invertible sheaf L_j on $\widehat{Y}$ with trivializations $L_j|_{\widehat{V}_\alpha} \cong \mathcal{O}_{\widehat{V}_\alpha}$, and the $g_{i,j,\alpha}$ glue to a section $\sigma_{i,j} \in H^0(\widehat{Y}, L_j)$ with

$$\operatorname{div}(\sigma_{i,j}) = rD_{i,j}.$$

For any integer $m \geq 0$, let $E^{(m)} \subset V_E$ be the closed subscheme defined by $\mathcal{I}_E^m$, which is also a closed subscheme of $\widehat{Y}$. Since E is proper, $E^{(m)}$ is empty, or finite over $\mathbb{C}$ if E is a point, or a one-dimensional proper scheme over $\mathbb{C}$ if E is a curve, so it is projective ([Har77, Exercise III.5.8] in the last case), and hence

$$\mathcal{S}_{j,m} := H^0\big(E^{(m)}, L_j|_{E^{(m)}}\big)$$

satisfies $\dim \mathcal{S}_{j,m} < +\infty$ [Har77, Theorem III.5.2(a)]. Let

$$\mathcal{J}_d^+ := \mathcal{J}_d \times \prod_j \mathcal{S}_{j,\lfloor d/c \rfloor} \quad \text{and} \quad x_{i,d}^+ := \big(x_{i,d}, (\sigma_{i,j}|_{E^{(\lfloor d/c \rfloor)}})_j\big),$$

and let $\tau_d^+ : \mathcal{J}_{d+1}^+ \to \mathcal{J}_d^+$ be the product of τ_d and the restriction maps

$$\mathcal{S}_{j,\lfloor (d+1)/c \rfloor} \to \mathcal{S}_{j,\lfloor d/c \rfloor}.$$

Then τ_d^+ is linear and $\tau_d^+(x_{i,d+1}^+) = x_{i,d}^+$ for any i and d.

Step 2. In this step, we pass to a subsequence, and we construct irreducible closed subsets $Z_d^+ \subset \mathcal{J}_d^+$ and the field k.

As in [dFEM11, Section 3], we choose infinite subsets $\mathbb{Z}_{\geq 1} = I_0 \supset I_1 \supset I_2 \supset \cdots$ and closed subsets $Z_d^+ \subset \mathcal{J}_d^+$ inductively. Let $d \geq 1$. Since closed subsets of $\mathcal{J}_d^+$ satisfy the DCC, there exists a closed subset $Z_d^+ \subset \mathcal{J}_d^+$ which is minimal among the closed subsets $Z \subset \mathcal{J}_d^+$ with $\{i \in I_{d-1} \mid x_{i,d}^+ \in Z\}$ infinite. Let $I_d := \{i \in I_{d-1} \mid x_{i,d}^+ \in Z_d^+\}$. By minimality, Z_d^+ is irreducible, and

(P1) any proper closed subset of Z_d^+ contains $x_{i,d}^+$ for only finitely many $i \in I_d$.

The intersection $Z_{d+1}^+ \cap (\tau_d^+)^{-1}(Z_d^+)$ is a closed subset of Z_{d+1}^+ containing $x_{i,d+1}^+$ for any $i \in I_{d+1} \subset I_d$, so it equals Z_{d+1}^+ by (P1) for $d+1$. The closure of $\tau_d^+(Z_{d+1}^+)$ is a closed subset of Z_d^+ containing $x_{i,d}^+$ for any $i \in I_{d+1}$, so it equals Z_d^+ by (P1). Thus τ_d^+ restricts to a dominant morphism $Z_{d+1}^+ \to Z_d^+$.

Choose integers $i_1 < i_2 < \cdots$ with $i_d \in I_d$, and replace the tuples $(f_{i,j})_j$, $i \geq 1$, with the tuples $(f_{i_d,j})_j$, $d \geq 1$. Since $i_{d'} \in I_{d'} \subset I_d$ for any $d' \geq d$, (P1) implies the following after the replacement.

(P2) For any d, $x_{i,d}^+ \in Z_d^+$ for all but finitely many i, and any non-empty open subset of Z_d^+ contains $x_{i,d}^+$ for all but finitely many i.

Let η_d^+ be the generic point of Z_d^+. The dominant morphisms $Z_{d+1}^+ \to Z_d^+$ induce inclusions $\kappa(\eta_d^+) \subset \kappa(\eta_{d+1}^+)$ of residue fields. Let k be an algebraic closure of $\bigcup_d \kappa(\eta_d^+)$, and let $\gamma_d^+ \in \mathcal{J}_d^+(k)$ be the k-point defined by η_d^+ and $\kappa(\eta_d^+) \subset k$. Then $\tau_d^+(\gamma_{d+1}^+) = \gamma_d^+$ for any d. We show the following.

(P3) Let $C \subset \mathcal{J}_d^+$ be a constructible subset containing $x_{i,d}^+$ for any i. Then $\eta_d^+ \in C$. If moreover C is the image of a closed subset $\widehat{C} \subset \mathcal{J}_d^+ \times \mathbb{A}^M$ for some M, then $\widehat{C}$ has a k-point over γ_d^+.

By (P2), $x_{i,d}^+ \in Z_d^+$ for all but finitely many i, so the closure of $C \cap Z_d^+$ contains $x_{i,d}^+$ for infinitely many i. By (P2), a proper closed subset of Z_d^+, whose complement in Z_d^+ is a non-empty open subset, contains $x_{i,d}^+$ for only finitely many i, so the closure of $C \cap Z_d^+$ equals Z_d^+. A constructible dense subset of an irreducible Noetherian topological space contains the generic point [Har77, Exercise II.3.18(b)], so $\eta_d^+ \in C$. Then the fiber of $\widehat{C}$ over η_d^+ is a non-empty scheme of finite type

over $\kappa(\eta_d^+)$, and it has a k-point by Hilbert's Nullstellensatz since k is an algebraically closed extension of $\kappa(\eta_d^+)$. This proves (P3).

Step 3. In this step, we construct $(f_{\infty,j})_j$ and the subsets Z_d, and we show that the $f_{\infty,j}$ are non-zero and semi-invariant.

Let $Z_d \subset \mathcal{J}_d$ be the closure of the image of Z_d^+, and let $\gamma_d \in \mathcal{J}_d(k)$ be the image of γ_d^+. Then Z_d is irreducible, γ_d maps to the generic point of Z_d, and $\tau_d(\gamma_{d+1}) = \gamma_d$. Let $Z_d^\circ \subset Z_d$ be a non-empty open subset. Then the preimage of Z_d° in Z_d^+ is a non-empty open subset, so by (P2), Z_d° contains $x_{i,d}$ for all but finitely many i.

In case (a), the entries of the γ_d are compatible under truncation, so there exist unique power series $f_{\infty,j} \in k[[x_1, \dots, x_n]]$ whose parts of degree less than d are the entries of γ_d for any d. In case (b),

$$W_{j,\alpha,d} \otimes_{\mathbb{C}} k \subset k[S_\alpha]/\mathfrak{n}_\alpha^d k[S_\alpha],$$

and the entries of the γ_d define elements $f_{\infty,j,\alpha}$ of $\varprojlim_d k[S_\alpha]/\mathfrak{n}_\alpha^d k[S_\alpha]$, the $\mathfrak{n}_\alpha$-adic completion of $k[S_\alpha]$. In both cases, the d-jet of $(f_{\infty,j})_j$ is γ_d, which maps to the generic point of Z_d.

Next, we show that the $f_{\infty,j}$ are non-zero. In case (a), let N be an upper bound of the orders of the $f_{i,j}$ at 0. In case (b), fix α, let Γ be an irreducible component of $U_\alpha \cap V(\mathfrak{n}_\alpha)$, and by (iii), let N be an upper bound of the orders of the $f_{i,j,\alpha}$ along Γ. Since $\mathfrak{n}_\alpha$ is contained in the ideal of Γ, the elements of the $\mathfrak{n}_\alpha$-adic completion of $\mathbb{C}[S_\alpha]$ whose classes in $\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^{N+1}$ vanish have order at least $N+1$ along Γ, so the class of $f_{i,j,\alpha}$ in $\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^{N+1}$ is non-zero. In both cases, let $d := N+1$. The subset of $\mathcal{J}_d^+$ where the entry of index j (resp. (j,α)) is non-zero is open and contains $x_{i,d}^+$ for any i, so it contains η_d^+ by (P3). Thus $f_{\infty,j} \neq 0$ (resp. $f_{\infty,j,\alpha} \neq 0$). Moreover, in case (b), for any irreducible component Γ of $U_\alpha \cap V(\mathfrak{n}_\alpha)$, let $I_\Gamma \subset \mathbb{C}[S_\alpha]$ be the prime ideal of the closure of Γ in S_α, let N be a bound of the orders of the $f_{i,j,\alpha}$ along Γ, and let $d := N+1$. Then the points of $\mathcal{J}_d^+$ whose entry of index (j,α) has non-zero image in $(\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^d)_{I_\Gamma}$ form an open subset containing every $x_{i,d}^+$, hence η_d^+ by (P3). Since the map

$$(\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^d)_{I_\Gamma} \otimes_{\mathbb{C}} k \to (k[S_\alpha]/\mathfrak{n}_\alpha^d)_{I_\Gamma k[S_\alpha]}$$

is injective ($I_\Gamma k[S_\alpha]$ is prime as $\mathbb{C}$ is algebraically closed, and its complement acts injectively on the factors $\kappa(I_\Gamma) \otimes_{\mathbb{C}} k$ of a composition series of the Artinian ring $(\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^d)_{I_\Gamma}$ tensored with k), the class of $f_{\infty,j,\alpha}$ in $(k[S_\alpha]/\mathfrak{n}_\alpha^d)_{I_\Gamma k[S_\alpha]}$ is non-zero.

Next, we show the semi-invariance. In case (a), let χ_j be the character of the $f_{i,j}$. Since H acts linearly on $\mathbb{A}^n$, the polynomials of degree less than d which are semi-invariant with character χ_j form a linear subspace. Thus the points of $\mathcal{J}_d^+$ whose entry of index j is semi-invariant with character χ_j form a closed subset containing $x_{i,d}^+$ for any i, and it contains η_d^+ by (P3). Hence the part of degree less than d of $f_{\infty,j}$ is semi-invariant with character χ_j for any d, and so is $f_{\infty,j}$. In case (b), let $\chi_{j,\alpha}$ be the character of the $f_{i,j,\alpha}$. Since K_α acts linearly on $\mathbb{C}[S_\alpha]$ and preserves $\mathfrak{n}_\alpha^d$, as $\mathfrak{n}_\alpha$ is K_α-invariant, the elements of $W_{j,\alpha,d}$ which are semi-invariant with character $\chi_{j,\alpha}$ form a linear subspace. Thus the points of $\mathcal{J}_d^+$ whose entry of index (j,α) is semi-invariant with character $\chi_{j,\alpha}$ form a closed subset containing $x_{i,d}^+$ for any i, and it contains η_d^+ by (P3). Hence the class of $f_{\infty,j,\alpha}$ in $k[S_\alpha]/\mathfrak{n}_\alpha^d k[S_\alpha]$ is semi-invariant with character $\chi_{j,\alpha}$ for any d, and so is $f_{\infty,j,\alpha}$.

It remains to show the statement on divisors in case (b) and the statement on subsets J.

Step 4. In this step, we show that, in case (b), the divisors of the $f_{\infty,j,\alpha}$ define an H-invariant $\mathbb{Q}$-divisor on the formal completion of Y_k along E_k.

For a closed point $y \in E_k$, let $\mathfrak{m}_y$ be the maximal ideal of $\widehat{\mathcal{O}}_{Y_k,y}$, and for $h \in H$, let

$$h^\sharp : \widehat{\mathcal{O}}_{Y_k,h(y)} \to \widehat{\mathcal{O}}_{Y_k,y}$$

be the isomorphism induced by h. For any j and any closed point $y \in E_k$, we construct a $\mathbb{Q}$-divisor $D_{\infty,j,y} \geq 0$ on $\operatorname{Spec} \widehat{\mathcal{O}}_{Y_k,y}$ with the following properties.

(D1) For any α with $y \in V_{\alpha,k}$ and any closed point $\tilde{y} \in U_{\alpha,k}$ over y, the pullback of $D_{\infty,j,y}$ to $\operatorname{Spec} \widehat{\mathcal{O}}_{S_{\alpha,k},\tilde{y}}$ is $\operatorname{div}(f_{\infty,j,\alpha})$.

(D2) $D_{\infty,j,y}$ is the pullback of $D_{\infty,j,h(y)}$ under $h^\sharp$ for any $h \in H$.

Here $f_{\infty,j,\alpha}$ is regarded as an element of $\widehat{\mathcal{O}}_{S_{\alpha,k},\tilde{y}}$ through $\tilde{y} \in V(\mathfrak{n}_\alpha)$. By (D1) and (D2), the family $(D_{\infty,j,y})_{y\in E_k}$ is an H-invariant $\mathbb{Q}$-divisor on the formal completion of Y_k along E_k defined by the divisors of the $f_{\infty,j,\alpha}$ in the sense of Construction-Proposition 4.1.

Fix j. For $m \geq 0$, let t_m be the entry of γ^+_{cm} in

$$\mathcal{S}_{j,m} \otimes_{\mathbb{C}} k = H^0(E_k^{(m)}, L_j).$$

Since $\tau_d^+(\gamma^+_{d+1}) = \gamma^+_d$, the t_m are compatible under restriction.

For any α and m, let $\mathcal{R}_{\alpha,m} \subset \mathcal{J}^+_{cm}$ be the subset of points (P,t) with

$$\pi_\alpha^* t_\alpha = P_{j,\alpha}^r \quad \text{in } H^0\big(U_\alpha, \mathcal{O}_{U_\alpha}/\mathcal{I}_E^m \mathcal{O}_{U_\alpha}\big),$$

where $P_{j,\alpha}$ is the entry of P in $W_{j,\alpha,cm}$, t is the entry in $\mathcal{S}_{j,m}$, and $t_\alpha \in H^0(V_\alpha, \mathcal{O}_{V_\alpha}/\mathcal{I}_E^m)$ corresponds to $t|_{V_\alpha}$ under the trivialization of L_j on $\widehat{V}_\alpha$. The right-hand side is well defined by the choice of c in **Step 1**. Both sides are polynomial in the coordinates of (P,t) with values in a subspace of finite dimension, so $\mathcal{R}_{\alpha,m}$ is closed. By the definition of the $g_{i,j,\alpha}$ in **Step 1**, $\mathcal{R}_{\alpha,m}$ contains $x^+_{i,cm}$ for any i, so it contains η^+_{cm} by (P3). Hence γ^+_{cm} satisfies the defining equation of $\mathcal{R}_{\alpha,m}$ over k.

Let $y \in E_k$ be a closed point, and let α be an index with $y \in V_{\alpha,k}$. Since $\mathcal{I}_E \widehat{\mathcal{O}}_{Y_k,y} \subset \mathfrak{m}_y$, the images of the $t_m|_{V_{\alpha,k}}$ under the trivialization of L_j on $\widehat{V}_\alpha$ define an element

$$a_{\alpha,y} \in \varprojlim_m \widehat{\mathcal{O}}_{Y_k,y}/\mathfrak{m}_y^m = \widehat{\mathcal{O}}_{Y_k,y}.$$

If also $y \in V_{\beta,k}$, then $a_{\alpha,y} = v_{j,\alpha\beta} a_{\beta,y}$, so $\operatorname{div}(a_{\alpha,y})$ does not depend on α. Let $\tilde{y} \in U_{\alpha,k}$ be a closed point over y. Since

$$\mathcal{I}_E \widehat{\mathcal{O}}_{S_{\alpha,k},\tilde{y}} \subset \mathfrak{m}_{\tilde{y}},$$

the defining equation of $\mathcal{R}_{\alpha,m}$ for γ^+_{cm} implies that

$$\pi_\alpha^* a_{\alpha,y} - f_{\infty,j,\alpha}^r \in \mathfrak{m}_{\tilde{y}}^m$$

for any m, so $\pi_\alpha^* a_{\alpha,y} = f^r_{\infty,j,\alpha}$. We have $\tilde{y} \in \Gamma_k$ for some Γ as in **Step 3**, with d as there. If $f_{\infty,j,\alpha}$ were zero in $\widehat{\mathcal{O}}_{S_{\alpha,k},\tilde{y}}$, its class in $(k[S_\alpha]/\mathfrak{n}_\alpha^d)_{\mathfrak{m}_{\tilde{y}}}$ would vanish, as $(k[S_\alpha]/\mathfrak{n}_\alpha^d)_{\mathfrak{m}_{\tilde{y}}}$ injects into its completion [Mat89, Theorem 8.14], and hence so would its class in $(k[S_\alpha]/\mathfrak{n}_\alpha^d)_{I_\Gamma k[S_\alpha]}$, a contradiction. Since $\widehat{\mathcal{O}}_{S_{\alpha,k},\tilde{y}}$ is a domain ($S_{\alpha,k}$ is normal, as S_α is normal and $\mathbb{C}$ is algebraically closed) [Mat89, §32 and Corollary to Theorem 23.9], $f^r_{\infty,j,\alpha} \neq 0$, so $a_{\alpha,y} \neq 0$. Let

$$D_{\infty,j,y} := \frac{1}{r} \operatorname{div}(a_{\alpha,y}).$$

Then $D_{\infty,j,y} \geq 0$ is a $\mathbb{Q}$-divisor which does not depend on α, and the pullback of $D_{\infty,j,y}$ to $\operatorname{Spec} \widehat{\mathcal{O}}_{S_{\alpha,k},\tilde{y}}$ is

$$\frac{1}{r} \operatorname{div}(f^r_{\infty,j,\alpha}) = \operatorname{div}(f_{\infty,j,\alpha}).$$

Thus (D1) holds.

Let $h \in H$. Since $h(E) = E$, h maps $E^{(m)}$ isomorphically onto $E^{(m)}$ for any m. Since $\widehat{Y}$ is also the formal completion of $V_E \cap h^{-1}(V_E)$ along E, h induces an automorphism $\hat{h}$ of $\widehat{Y}$. Let h^*L_j be the pullback of L_j under $\hat{h}$, let $M_h := L_j \otimes (h^*L_j)^{-1}$ on $\widehat{Y}$, and let

$$A_m := H^0\big(E^{(m)}, M_h|_{E^{(m)}}\big) \quad \text{and} \quad A'_m := H^0\big(E^{(m)}, M_h^{-1}|_{E^{(m)}}\big).$$

Since $E^{(m)}$ is projective, $\dim A_m < +\infty$ and $\dim A'_m < +\infty$ [Har77, Theorem III.5.2(a)]. Let

$$\widehat{C}_{h,m} \subset \mathcal{J}^+_{cm} \times A_m \times A'_m$$

be the closed subset of points (P, t, ω, ω') with

$$\omega \cdot h^* t = t \quad \text{and} \quad \omega' \cdot t = h^* t,$$

where t is the entry of (P, t) in $\mathcal{S}_{j,m}$. Since $D_{i,j}$ is H-invariant,

$$\operatorname{div}(h^* \sigma_{i,j}) = r h^* D_{i,j} = r D_{i,j} = \operatorname{div}(\sigma_{i,j}) \quad \text{on } \widehat{Y}.$$

Thus there exists a nowhere vanishing section $\omega_i \in H^0(\widehat{Y}, M_h)$ with $\omega_i \cdot h^* \sigma_{i,j} = \sigma_{i,j}$, and

$$(x^+_{i,cm}, \omega_i|_{E^{(m)}}, \omega_i^{-1}|_{E^{(m)}}) \in \widehat{C}_{h,m}$$

for any i. By Chevalley's theorem [Har77, Exercise II.3.19], the image of $\widehat{C}_{h,m}$ in $\mathcal{J}^+_{cm}$ is constructible, so by (P3), there exist $\omega_m \in A_m \otimes_{\mathbb{C}} k$ and $\omega'_m \in A'_m \otimes_{\mathbb{C}} k$ with

$$\omega_m \cdot h^* t_m = t_m \quad \text{and} \quad \omega'_m \cdot t_m = h^* t_m.$$

Let $y \in E_k$ be a closed point, let $y' := h(y)$, and let α and β be indices with $y \in V_{\alpha,k}$ and $y' \in V_{\beta,k}$. Writing ω_m and ω'_m in the trivializations of L_j on $\widehat{V}_\alpha$ and of $h^* L_j$ over $h^{-1}(V_\beta)$, we obtain $w_m, w'_m \in \widehat{\mathcal{O}}_{Y_k,y}$ with

$$a_{\alpha,y} - w_m h^\sharp(a_{\beta,y'}) \in \mathfrak{m}_y^m \quad \text{and} \quad h^\sharp(a_{\beta,y'}) - w'_m a_{\alpha,y} \in \mathfrak{m}_y^m$$

for any m. By the Krull intersection theorem [Mat89, Theorem 8.10(i)] applied to the finitely generated $\widehat{\mathcal{O}}_{Y_k,y}$-modules $\widehat{\mathcal{O}}_{Y_k,y}/(h^\sharp(a_{\beta,y'}))$ and $\widehat{\mathcal{O}}_{Y_k,y}/(a_{\alpha,y})$, we have

$$a_{\alpha,y} \in (h^\sharp(a_{\beta,y'})) \quad \text{and} \quad h^\sharp(a_{\beta,y'}) \in (a_{\alpha,y}).$$

Thus

$$\operatorname{div}(a_{\alpha,y}) = \operatorname{div}(h^\sharp(a_{\beta,y'})),$$

that is, $D_{\infty,j,y}$ is the pullback of $D_{\infty,j,y'}$ under $h^\sharp$. Thus (D2) holds.

Step 5. In this step, we show the statement on subsets J, in both cases.

Let $J \subset \{1, \dots, s\}$, let $\mathcal{J}_d^J$ be the product of the factors of $\mathcal{J}_d$ with index $j \in J$, and let Z_d^J be the closure of the image of Z_d under the projection $\mathcal{J}_d \to \mathcal{J}_d^J$, which is linear. The image of the generic point of Z_d is the generic point of Z_d^J, and the preimage in Z_d of a non-empty open subset of Z_d^J is a non-empty open subset of Z_d, so the d-jets of the $(f_{i,j})_{j\in J}$ and of $(f_{\infty,j})_{j\in J}$ satisfy the properties of the statement concerning Z_d (the generic point and the non-empty open subsets), with Z_d^J in place of Z_d. The other properties of $(f_{\infty,j})_{j\in J}$ are properties of the individual $f_{\infty,j}$, so $(f_{\infty,j})_{j\in J}$ is a compatible generic limit of the $(f_{i,j})_{j\in J}$, and the proof is finished. □

We now prove that the lc thresholds and the quotient minimal log discrepancies of the tuples of Construction-Proposition 4.1 pass to a compatible generic limit.

Theorem 4.2. *Notation as in Construction-Proposition 4.1. Let $c_1, \dots, c_s$ be positive real numbers. In case (a), let $\mathfrak{c} = \prod_l \mathfrak{c}_l^{e_l}$ be an $\mathbb{R}$-ideal on $\mathbb{A}^n$ with $e_l > 0$ and H-invariant ideals $\mathfrak{c}_l$ such that $(\mathbb{A}^n, \mathfrak{c})$ is klt near 0, write $\mathfrak{c}$ also for the pullback of $\mathfrak{c}$ to $\mathcal{A}_k$, and let $C_i := \sum_j c_j \operatorname{div}(f_{i,j})$ and $C_\infty := \sum_j c_j \operatorname{div}(f_{\infty,j})$. In case (b), let $\mathfrak{c} := \mathcal{O}$, $C_i := \sum_j c_j D_{i,j}$, and C_∞ the $\mathbb{R}$-divisor on $\mathcal{A}_k$ given by $\sum_j c_j \operatorname{div}(f_{\infty,j,\alpha})$ on the charts, and write $(\mathcal{A}, \theta C_i)$ and $(\mathcal{A}_k, \theta C_\infty)$ for the pairs with boundaries $B_{\mathrm{br}} + \theta C_i$ and $B_{\mathrm{br}} + \theta C_\infty$ for any $\theta \geq 0$. In case (a) "near the base" means near 0, and in case (b) it means near E, resp. near E_k. Then, after passing to an infinite subsequence, the following hold.*

(1) The lc thresholds of C_i with respect to $(\mathcal{A}, \mathfrak{c})$ near the base converge to the lc threshold of C_∞ with respect to $(\mathcal{A}_k, \mathfrak{c})$. For any real number $\theta > 0$, $(\mathcal{A}_k, \theta C_\infty, \mathfrak{c})$ is lc near the base if and only if $(\mathcal{A}, \theta C_i, \mathfrak{c})$ is lc near the base for all i. If $(\mathcal{A}_k, \theta C_\infty, \mathfrak{c})$ is klt near the base, then there exists $\theta' > \theta$ such that $(\mathcal{A}, \theta' C_i, \mathfrak{c})$ is lc near the base for all i.

(2) *Assume that $n = 2$ in case (a), that H is diagonal if $\mathfrak{c} \neq \mathcal{O}$, and that $(\mathcal{A}_k, C_\infty, \mathfrak{c})$ is lc near the base. In case (a),*

$$\mathrm{mld}((\mathcal{A} \ni 0, C_i, \mathfrak{c})/H) = \mathrm{mld}((\mathcal{A}_k \ni 0, C_\infty, \mathfrak{c})/H)$$

for any i. In case (b), for any i, the infimum of $a(F, \mathcal{A}, B_{\mathrm{br}} + C_i)/|I_H(F)|$ over the prime divisors F over $\mathcal{A}$ with center contained in E is equal to the infimum of $a(F, \mathcal{A}_k, B_{\mathrm{br}} + C_\infty)/|I_H(F)|$ over the prime divisors F over $\mathcal{A}_k$ with center contained in E_k. By [Liu26, Lemma 2.6(2)] *and Lemma 3.7, the infimum for C_i (resp. C_∞) is the minimum of the log discrepancies of the crepant quotient of $(\mathcal{A}, C_i)$ (resp. $(\mathcal{A}_k, C_\infty)$) by H over the prime divisors with center contained in E/H (resp. E_k/H).*

(3) *In case (a), for any sufficiently large integer d there exist a non-empty open subset $Z_d^\circ \subset Z_d$ containing the d-jets of $(f_{i,j})_j$ for all but finitely many i (the finitely many exceptions depend on d), and an H-equivariant log resolution near the base of the family of pairs defined by the d-jets in Z_d°, with constant dual complex, constant discrepancies, constant multiplicities along the exceptional divisors of the divisor of each entry of the d-jets and of each ideal $\mathfrak{c}_l$, and constant decomposition and inertia groups. Moreover, for the points z of Z_d° with values in algebraically closed fields $k' \supset \mathbb{C}$, the irreducible components of the inverse image of 0, of the exceptional locus, and of the strict transforms of the zero loci of the entries of z and of the $\mathfrak{c}_l$ in the fiber over z of the log resolution can be indexed by one finite set, independent of z, so that the sets of indices of the components with non-empty intersection, the indices of the components mapping to 0, the log discrepancies of the components over $\mathbb{A}^n_{k'}$, and the multiplicities along the components of the divisor of each entry of z and of each $\mathfrak{c}_l$ do not depend on z.*

When the limit pair is klt, part (1) provides a real number larger than 1 by which the reference boundaries may be multiplied, remaining lc, for all members of a subsequence, and part (2) turns the limit of the minima of the reference pairs into the minimum of the limit pair. Part (1) does not turn klt members into a klt limit (Example 4.3). Non-klt limits are treated in Definition-Proposition 5.7.

Example 4.3. For $i \geq 2$, let $f_i := z^2 + t^i$ on $\mathbb{A}^2_{z,t}$, and let $c_1 := \frac{1}{2}$. For any $i \geq 2$, $(\mathbb{A}^2, \frac{1}{2}\operatorname{div}(f_i))$ is klt with minimal log discrepancy 1 at the origin, and the lc threshold of $\frac{1}{2}\operatorname{div}(f_i)$ is $1 + \frac{2}{i}$. The compatible generic limit has $C_\infty = \operatorname{div}(z)$, so $(\mathbb{A}^2_k, C_\infty)$ is lc but not klt.

Proof of Theorem 4.2. The proof is long, so we divide it into several steps. Since (1), (2), and (3) are preserved when we pass to a further subsequence, we pass to subsequences freely.

Step 1. In this step, we fix charts and jets for both cases. In case (a), the index α takes one value, $S_\alpha = U_\alpha := \mathbb{A}^n$, K_α is trivial, $\mathfrak{n}_\alpha := (x_1, \ldots, x_n)$, $Y := \mathbb{A}^n$, $E := \{0\}$, $B_{\mathrm{br}} := 0$, and $f_{i,j,\alpha} := f_{i,j}$. Let $\Sigma_\alpha := U_\alpha \cap V(\mathfrak{n}_\alpha)$, so $\Sigma_\alpha = \{0\}$ in case (a). For an algebraically closed field $k' \supset \mathbb{C}$, let $R_{\alpha,k'}$ be the $\mathfrak{n}_\alpha$-adic completion of $k'[S_\alpha]$, so that $f_{i,j,\alpha} \in R_{\alpha,\mathbb{C}}$ and $f_{\infty,j,\alpha} \in R_{\alpha,k}$. For a closed point $u \in \Sigma_\alpha(k')$, let $\mathfrak{m}_u$ be the maximal ideal of $\widehat{\mathcal{O}}_{U_{\alpha,k'},u}$. Since $\mathfrak{n}_\alpha \subset \mathfrak{m}_u$, the ring $R_{\alpha,k'}$ maps to $\widehat{\mathcal{O}}_{U_{\alpha,k'},u}$. In case (a), $\mathcal{J}_d$ is already a space of polynomials. In case (b), $\mathcal{J}_d = \prod_{j,\alpha} W_{j,\alpha,d}$, where $W_{j,\alpha,d}$ is the linear span of the classes of the $f_{i,j,\alpha}$, $i \geq 1$, in $\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^d$. The classes of the $f_{i,j,\alpha}$ are represented by restrictions of polynomials of bounded degree in N_α variables (Construction-Proposition 4.1), and they are semi-invariant with the character $\chi_{j,\alpha}$ of the $f_{i,j,\alpha}$, so any element of $W_{j,\alpha,d}$ is $\chi_{j,\alpha}$-semi-invariant. For any j and α fix a $\mathbb{C}$-linear map from $W_{j,\alpha,d}$ to the space of polynomials of bounded degree in N_α variables whose composite with the quotient map to $\mathbb{C}[S_\alpha]/\mathfrak{n}_\alpha^d$ is the inclusion, and let $\sigma_{j,\alpha}$ be its composite with the projection

$$p_{j,\alpha} := |K_\alpha|^{-1} \sum_{h \in K_\alpha} \chi_{j,\alpha}(h)^{-1} h^*$$

onto the $\chi_{j,\alpha}$-semi-invariant polynomials. Since K_α acts linearly on $\mathbb{A}^{N_\alpha}$ and preserves S_α and $\mathfrak{n}_\alpha$, the quotient map is K_α-equivariant, so the composite of $\sigma_{j,\alpha}$ with the quotient map is still the

inclusion, and the values of $\sigma_{j,\alpha}$ are $\chi_{j,\alpha}$-semi-invariant. Through the $\sigma_{j,\alpha}$ there exist polynomials $G_{j,\alpha}$ on $\mathbb{A}^{N_\alpha}$ ($N_\alpha := n$ in case (a)) with coefficients in $\mathbb{C}[\mathcal{J}_d]$ such that the tuple of the restrictions of the $G_{j,\alpha}(z)$ to S_α is the tuple of representatives of a k'-point z of $\mathcal{J}_d$, and in case (b) the $G_{j,\alpha}(z)$ are $\chi_{j,\alpha}$-semi-invariant. Let $z_i^{(d)} \in \mathcal{J}_d(\mathbb{C})$ and $z_\infty^{(d)} \in \mathcal{J}_d(k)$ be the d-jets of $(f_{i,j})_j$ and $(f_{\infty,j})_j$. By the property of Construction-Proposition 4.1 on the non-empty open subsets of Z_d, applied to Z_d itself, $z_i^{(d)} \in Z_d$ for all but finitely many i, and $z_\infty^{(d)}$ maps to the generic point of Z_d. Since the d-jets are the classes modulo $\mathfrak{n}_\alpha^d$,

$$f_{i,j,\alpha} - G_{j,\alpha}(z_i^{(d)}) \in \mathfrak{n}_\alpha^d R_{\alpha,\mathbb{C}} \quad \text{and} \quad f_{\infty,j,\alpha} - G_{j,\alpha}(z_\infty^{(d)}) \in \mathfrak{n}_\alpha^d R_{\alpha,k}, \tag{4.1}$$

and since $\mathfrak{n}_\alpha \subset \mathfrak{m}_u$, the images of $f_{i,j,\alpha} - G_{j,\alpha}(z_i^{(d)})$ and of $f_{\infty,j,\alpha} - G_{j,\alpha}(z_\infty^{(d)})$ in $\widehat{\mathcal{O}}_{U_{\alpha,k'},u}$ belong to $\mathfrak{m}_u^d$ for any closed point $u \in \Sigma_\alpha(k')$.

In case (b), the surface U_α has quotient singularities, so it has finitely many singular points. For a closed point $u \in \Sigma_\alpha(k')$, let π_u be the morphism from a scheme $\mathcal{S}_u$ to $\operatorname{Spec} \widehat{\mathcal{O}}_{U_{\alpha,k'},u}$ which is the identity if U_α is smooth at u (in particular in case (a)), and otherwise the smooth formal cover $\mathcal{S}_u = \operatorname{Spec} k'[[\xi_1,\xi_2]]$ whose quotient by the local fundamental group G_u of the quotient singularity, acting linearly and without pseudo-reflections, is $\operatorname{Spec} \widehat{\mathcal{O}}_{U_{\alpha,k'},u}$. Since π_u is étale in codimension one, Lemma 3.7 applied to G_u implies that a pair on $\operatorname{Spec} \widehat{\mathcal{O}}_{U_{\alpha,k'},u}$ is lc, resp. klt, if and only if its pullback to $\mathcal{S}_u$ is lc, resp. klt. Let $\mu_\alpha : \widetilde{U}_\alpha \to U_\alpha$ be a K_α-equivariant log resolution, which is the identity if U_α is smooth, and let $B_{\widetilde{U}_\alpha}$ be the crepant pullback of $(U_\alpha, 0)$, that is,

$$K_{\widetilde{U}_\alpha} + B_{\widetilde{U}_\alpha} = \mu_\alpha^* K_{U_\alpha}.$$

Since quotient singularities are klt [KM98, Proposition 5.20], the coefficients of $B_{\widetilde{U}_\alpha}$ are less than one.

Let $C_i^\alpha := \sum_j c_j \operatorname{div}(f_{i,j,\alpha})$, on U_α in case (a) and on the formal completion of U_α along Σ_α in case (b). Let F be a prime divisor over Y with $\operatorname{center}_Y F \subset E$, and let F'' be a prime divisor over U_α lying over F, for an index α with $\operatorname{center}_Y F \cap V_\alpha \neq \emptyset$. In case (b), if the center of F is a closed point y, then F and F'' are formal prime divisors over $\operatorname{Spec} \widehat{\mathcal{O}}_{Y,y}$ and over $\operatorname{Spec} \widehat{\mathcal{O}}_{U_\alpha,u}$ for a closed point $u \in \Sigma_\alpha$ over y. Set $\iota(F'') := |I_{K_\alpha}(F'')| \cdot |I_H(F)|$. Since $(U_\alpha, C_i^\alpha)/K_\alpha = (V_\alpha, B_{\mathrm{br}} + C_i)$, [Liu26, Lemma 2.6(2)] in case (a), and Lemma 3.7 at the closed points of Σ_α in case (b), applied to K_α and to H, imply

$$\frac{a(F, Y, B_{\mathrm{br}} + C_i)}{|I_H(F)|} = \frac{a(F'', U_\alpha, C_i^\alpha)}{\iota(F'')}. \tag{4.2}$$

In case (b), Lemma 3.7 is applied to the stabilizers $K_u \subset K_\alpha$ and $H_y \subset H$ of u and y, since the quotient by the stabilizer is étale over the quotient by K_α, resp. H, at the image point [Gro71, Exposé V, Proposition 2.2], and for formal prime divisors, I_{K_α} and I_H denote the inertia groups for K_u and H_y. By [Liu26, Lemma 2.6(2)] in case (a), and Lemma 3.7 in case (b), $(\mathcal{A}, \theta C_i, \mathfrak{c})$ is lc near the base if and only if $(U_\alpha, \theta C_i^\alpha, \mathfrak{c})$ is lc near Σ_α for any α, where in case (b) this means lc on the completions at the closed points of Σ_α. By [Liu26, Lemma 3.3(1)] and Lemma 3.7, (4.2) and the lc criterion of the previous sentence also hold for C_∞, with the completions $\widehat{\mathcal{O}}_{U_{\alpha,k},u}$ at the closed points $u \in \Sigma_\alpha(k)$ and formal prime divisors. Thus all statements of the theorem may be checked on the charts U_α near Σ_α.

Step 2. In this step, we spread out log resolutions over a dense open subset of a parameter variety.

Claim 4.4. *Let X be a smooth quasi-projective variety with an action of a finite group G, $V \subset X$ a G-invariant closed subset, T an integral variety on which G acts trivially, $\mathfrak{b}_1, \dots, \mathfrak{b}_m$ ideals of $\mathcal{O}_{X\times T}$ which G permutes and whose restrictions to the generic fiber of $X \times T \to T$ are non-zero, and $P_1, \dots, P_q$ prime divisors over $X \times T$ whose centers dominate T. Then there exist a non-empty open subset $T^\circ \subset T$, a G-equivariant projective birational morphism $\varphi : \widetilde{X} \to X \times T^\circ$*

from a variety $\widetilde{X}$ on which the G-orbits of $P_1, \dots, P_q$ are divisors, a finite surjective morphism $T' \to T^\circ$, and prime divisors $Q_1, \dots, Q_N$ on $\widetilde{X}' := \widetilde{X} \times_{T^\circ} T'$ with the following properties, where $\varphi' : \widetilde{X}' \to X \times T'$ is the base change of φ. Let $k' \supset \mathbb{C}$ be an algebraically closed field and τ a k'-point of T'.

(1) The fiber $\varphi'_\tau : \widetilde{X}'_\tau \to X_{k'}$ is a G-equivariant log resolution of $(X_{k'}, V_{k'})$ and of the ideals $\mathfrak{b}_{r,\tau}$, and the fibers $Q_{1,\tau}, \dots, Q_{N,\tau}$ are irreducible and are the irreducible components of $\mathrm{Exc}(\varphi'_\tau)$, of $\varphi'^{-1}_\tau(V_{k'})$, of the strict transforms of the zero loci of the $\mathfrak{b}_{r,\tau}$, and of the fibers of the strict transforms of $P_1, \dots, P_q$; for $1 \le \nu \le q$, the indices l of the components of the fiber of the strict transform of P_ν form a set depending only on ν. For any non-empty subset $J \subset \{1, \dots, N\}$, the fibers over τ of the irreducible components of $\bigcap_{l\in J} Q_l$ are pairwise disjoint and are the irreducible components of $\bigcap_{l\in J} Q_{l,\tau}$.

(2) The numbers $a(Q_{l,\tau}, X_{k'}, 0)$, $\mathrm{mult}_{Q_{l,\tau}} \mathfrak{b}_{r,\tau}$, and $\dim \varphi'_\tau(Q_{l,\tau})$, whether $\varphi'_\tau(Q_{l,\tau})$ intersects $V_{k'}$, the sets of indices l with $\bigcap_l Q_{l,\tau} \neq \emptyset$, and the groups $\mathrm{Stab}_G(Q_{l,\tau})$ and $I_G(Q_{l,\tau})$ do not depend on τ.

Proof. Let η be the generic point of T and κ the residue field at η. Let φ_η be a G-equivariant log resolution of (X_κ, V_κ) and of the ideals $\mathfrak{b}_{r,\eta}$ on which the G-orbits of the prime divisors P_ν are divisors, which exists by functorial, hence G-equivariant, principalization over any field of characteristic zero [Kol07, Theorem 3.35], applied to the product of the $\mathfrak{b}_{r,\eta}$, of the ideal of V_κ, and of G-invariant ideals with positive multiplicity along the P_ν whose normalized blow-ups have the G-orbits of the P_ν as divisors, which is G-invariant since G permutes the $\mathfrak{b}_{r,\eta}$ and V is G-invariant. For P_ν a divisor on X_κ we take the ideal of the G-orbit of P_ν, and otherwise the product of the G-translates of an ideal whose blow-up has P_ν as a divisor [KM98, Lemma 2.45], [Liu02, Theorem 8.1.24]. The morphism φ_η factors through the normalized blow-ups of the ideals taken for the P_ν [Har77, Proposition II.7.14]. Let Υ be the snc divisor formed by $\mathrm{Exc}(\varphi_\eta)$, $\varphi_\eta^{-1}(V_\kappa)$, the strict transforms of the zero loci of the $\mathfrak{b}_{r,\eta}$, and the strict transforms of the G-orbits of the P_ν. Choose a finite extension κ' of κ such that the irreducible components of all strata of Υ become geometrically irreducible over κ', and let T' be the normalization of T in κ'. Write $\varphi_\eta : \widetilde{X}_\eta \to X_\kappa$, and let $S_{1,\eta}, \dots, S_{p,\eta}$ be the irreducible components of the strata of Υ. Since φ_η, the action of G on $\widetilde{X}_\eta$, and the $S_{b,\eta}$ are given by finitely many equations with coefficients in κ, after shrinking T the morphism φ_η is the generic fiber of a G-equivariant projective birational morphism $\varphi : \widetilde{X} \to X \times T$, and the $S_{b,\eta}$ are the generic fibers of their closures S_b in $\widetilde{X}$. The closed subset of $\widetilde{X}$ where $\widetilde{X}$ is singular or some $\mathfrak{b}_r\mathcal{O}_{\widetilde{X}}$ is not invertible, and the singular loci of the S_b, do not meet the generic fiber. Their images in T are constructible by Chevalley's theorem and do not contain η, so after shrinking T they are empty. By generic smoothness [Har77, Corollary III.10.7] applied to $\widetilde{X} \to T$ and to the $S_b \to T$, after shrinking T the variety $\widetilde{X}$ and the S_b are smooth over T, hence flat with open image, and surjective onto T after shrinking T to the intersection of the images. Hence, after removing the images of the finitely many components that do not dominate T of the zero loci of the $\mathfrak{b}_r\mathcal{O}_{\widetilde{X}}$, of $\varphi^{-1}(V \times T)$, and of the intersections of the components of the closure of Υ, and the image of the closed subset of $\widetilde{X}$ where the closure of Υ does not have simple normal crossings relative to T, which is constructible by Chevalley's theorem and does not contain η since Υ is snc, the closure of Υ has simple normal crossings relative to T, any prime component of it is flat and surjective over T, any $\mathfrak{b}_r\mathcal{O}_{\widetilde{X}}$ is invertible, the zero loci of the $\mathfrak{b}_r\mathcal{O}_{\widetilde{X}}$ and $\varphi^{-1}(V \times T)$ are unions of prime components of the closure of Υ, and, since $\widetilde{X}$ and $X \times T$ are smooth over T, $K_{\widetilde{X}/X\times T}$ restricts to $K_{\widetilde{X}_\tau/X_\tau}$ on any fiber [Har77, Proposition II.8.10]. After shrinking T again, $T' \to T$ is finite étale. Let $Q_1, \dots, Q_N$ be the prime components of the base change of the closure of Υ to T'. Any Q_l and any irreducible component of any stratum of $\sum_l Q_l$ is smooth over T' with geometrically irreducible generic fiber, so after shrinking T the fibers over T' of the Q_l and of the irreducible components of the strata of $\sum_l Q_l$ are irreducible [Gro66, Proposition 9.7.8], and distinct Q_l have distinct fibers. Any

intersection of the Q_l is smooth over the normal variety T', hence normal [Mat89, Corollary to Theorem 23.9]. Thus its irreducible components are pairwise disjoint, and so are their fibers over T'. This implies (1) and the first five items of (2): the log discrepancies and the multiplicities are those of the generic fiber because $K_{\widetilde{X}/X\times T}$ and $\mathfrak{b}_r\mathcal{O}_{\widetilde{X}}$ restrict to the fibers, and the dimensions and the incidences are constant on a dense open subset by Chevalley's theorem. For $g \in G$, $g(Q_l)$ is one of the $Q_{l'}$, so $g(Q_{l,\tau}) = Q_{l,\tau}$ if and only if $g(Q_l) = Q_l$. Moreover, $Q_l \setminus \mathrm{Fix}(g)$ is open in Q_l, and $Q_l \to T'$ is smooth, hence open, so the set of points τ with $Q_{l,\tau} \subset \mathrm{Fix}(g)$ is closed in T'. The set is T' if g acts trivially on Q_l, and a proper closed subset of T' otherwise. Since G is finite, after shrinking T we obtain the last item of (2). We let T° be the open subset of T thus obtained. □

Proof of Theorem 4.2 continued. Any k'-point of T° lifts to a k'-point of T' because k' is algebraically closed, so the data of Claim 4.4(2) do not depend on the k'-point of T°, for all algebraically closed fields $k' \supset \mathbb{C}$.

Step 3. In this step, we prove (3). Let d be larger than the orders of the $f_{\infty,j}$ at 0, so that the $G_j(z_\infty^{(d)})$ are non-zero. Since H acts linearly, the points of $\mathcal{J}_d$ whose entry of index j is H-semi-invariant with the character χ_j of the $f_{i,j}$ for any j form a linear subspace, and it contains the image of $z_\infty^{(d)}$ (Construction-Proposition 4.1), which is the generic point of Z_d, hence it contains Z_d. Thus the ideals (G_j) on $\mathbb{A}^n \times Z_d$ are H-invariant. Apply Claim 4.4 to $X = \mathbb{A}^n$, $G = H$, $V = \{0\}$, $T = Z_d$, the ideals (G_j) and $\mathfrak{c}_l$, and $q = 0$. Let $Z_d^\circ := T^\circ$. By Construction-Proposition 4.1, $z_i^{(d)} \in Z_d^\circ$ for all but finitely many i, and the exceptional indices depend on d. The fiber of φ over a point $z \in Z_d^\circ$ is a log resolution of $(\mathbb{A}^n, \sum_j c_j \operatorname{div}(G_j(z)), \mathfrak{c})$ near 0.

Let z be a point of Z_d° with values in an algebraically closed field $k' \supset \mathbb{C}$, and let $\tau \in T'(k')$ be a lift of z. The fiber of φ over z is the fiber of φ' over τ, and by Claim 4.4(1) the irreducible components of the inverse image of 0, of the exceptional locus, and of the strict transforms of the zero loci of the $G_j(z)$ and of the $\mathfrak{c}_l$ in the fiber of φ over z are $Q_{1,\tau}, \dots, Q_{N,\tau}$. By Claim 4.4(2) with $V = \{0\}$, the sets of indices l with $\bigcap_l Q_{l,\tau} \neq \emptyset$, whether $\varphi'_\tau(Q_{l,\tau}) = \{0\}$ (that is, whether $\dim \varphi'_\tau(Q_{l,\tau}) = 0$ and $\varphi'_\tau(Q_{l,\tau})$ intersects $\{0\}$), the log discrepancies $a(Q_{l,\tau}, \mathbb{A}^n_{k'}, 0)$, the multiplicities along $Q_{l,\tau}$ of the divisor of each $G_j(z)$ and of each $\mathfrak{c}_{l'}$, and the groups $\mathrm{Stab}_H(Q_{l,\tau})$ and $I_H(Q_{l,\tau})$ do not depend on z, which proves the last sentence of (3) and gives constant discrepancies, constant multiplicities along the exceptional divisors, and constant decomposition and inertia groups.

For a non-empty subset $J \subset \{1, \dots, N\}$ and an irreducible component S of $\bigcap_{l\in J} Q_l$, let S_τ be the fiber of S over τ. By Claim 4.4(1), for any such J the S_τ are pairwise disjoint and are the irreducible components of $\bigcap_{l\in J} Q_{l,\tau}$, and $\varphi'^{-1}_\tau(0)$ is the union of the $Q_{l_1,\tau}$ with $\varphi'_\tau(Q_{l_1,\tau}) = \{0\}$. Hence, for non-empty subsets $J \subset J' \subset \{1, \dots, N\}$ and irreducible components S of $\bigcap_{l\in J} Q_l$ and S' of $\bigcap_{l\in J'} Q_l$, we have $S'_\tau \subset S_\tau$ if and only if $S' \subset S$, and S_τ intersects $\varphi'^{-1}_\tau(0)$ if and only if S contains an irreducible component of $\bigcap_{l\in J\cup\{l_1\}} Q_l$ for an index l_1 with $\varphi'_\tau(Q_{l_1,\tau}) = \{0\}$. The dual complex of the snc divisor $\sum_l Q_{l,\tau}$ near $\varphi'^{-1}_\tau(0)$, defined as in [Kol13, Definition 3.62] with k' in place of $\mathbb{C}$, has a $(|J|-1)$-dimensional cell for each S_τ intersecting $\varphi'^{-1}_\tau(0)$, and, for non-empty subsets $J \subset J' \subset \{1, \dots, N\}$ and irreducible components S of $\bigcap_{l\in J} Q_l$ and S' of $\bigcap_{l\in J'} Q_l$ with S_τ and S'_τ intersecting $\varphi'^{-1}_\tau(0)$, the cell of S_τ is a face of the cell of S'_τ if and only if $S'_\tau \subset S_\tau$. Since the irreducible components S of the intersections of the Q_l, the inclusions between them, and the indices l_1 with $\varphi'_\tau(Q_{l_1,\tau}) = \{0\}$ do not depend on z, the dual complex, with the cell of S_τ labeled by S, does not depend on z. This proves (3).

Step 4. In this step, we prove (1). By **Step 1**, we work on the charts. For a tuple t, which is one of $(f_{i,j,\alpha})_{j,\alpha}$, $(f_{\infty,j,\alpha})_{j,\alpha}$, or $(G_{j,\alpha}(z))_{j,\alpha}$ for a point z of $\mathcal{J}_d$, a real number $\theta > 0$, and a

closed point $u \in \Sigma_\alpha$, let

$$\mathfrak{a}_{t,u}(\theta) := \mathfrak{c} \cdot \prod_j (t_{j,\alpha})^{\theta c_j}$$

on $\mathcal{S}_u$ (the pullback by π_u), let $\mathrm{lct}_u(\mathfrak{a}_{t,u}(\theta)) \in (0, +\infty]$ be the lc threshold, which is $+\infty$ if all factors are units at u, and let $\ell_t(\theta)$ be the infimum of $\mathrm{lct}_u(\mathfrak{a}_{t,u}(\theta))$ over all α and u. By [dFEM11, Proposition 2.11], and by Lemma 3.7 applied to G_u at the singular points u of U_α, the pairs

$$\Big(U_\alpha, \theta \sum_j c_j \operatorname{div}(t_{j,\alpha}), \mathfrak{c}\Big)$$

are lc near Σ_α for all α if and only if $\ell_t(\theta) \geq 1$. By **Step 1**, for the tuples of C_i and of C_∞, this is the condition that $(\mathcal{A}, \theta C_i, \mathfrak{c})$, resp. $(\mathcal{A}_k, \theta C_\infty, \mathfrak{c})$, is lc near the base. For a fixed u, the set of real numbers $\theta \geq 0$ with $\mathrm{lct}_u(\mathfrak{a}_{t,u}(\theta)) \geq 1$ is a closed interval $[0, \lambda_{t,u}]$, because the log discrepancy of a fixed prime divisor is an affine function of θ. Let $\lambda_t := \inf_{\alpha,u} \lambda_{t,u}$. Then $\ell_t(\theta) \geq 1$ if and only if $\theta \leq \lambda_t$, and λ_t is the lc threshold in (1). We write $\mathfrak{a}_{i,u}(\theta)$, ℓ_i, and λ_i for the tuple $t = (f_{i,j,\alpha})_{j,\alpha}$ of C_i, and $\mathfrak{a}_{\infty,u}(\theta)$, ℓ_∞, λ_∞, and $\lambda_{\infty,u}$ for the tuple $t = (f_{\infty,j,\alpha})_{j,\alpha}$ of C_∞.

Fix $\theta > 0$ and $\epsilon > 0$, and let $d(\theta, \epsilon)$ be an integer with

$$d(\theta, \epsilon) \geq \frac{n}{\epsilon \min(\{e_l\}_l \cup \{\theta c_j\}_j)},$$

where $n = 2$ in case (b). For $d \geq d(\theta, \epsilon)$, (4.1) and Theorem 2.12(2), in the form [dFEM11, Proposition 2.19(ii)] on $\mathcal{S}_u$, imply

$$|\ell_t(\theta) - \ell_{G(z)}(\theta)| \leq \epsilon \tag{4.3}$$

for any tuple t with d-jet z. Here a factor of $\mathfrak{a}_{t,u}(\theta)$ is a unit at u if and only if the corresponding factor of $\mathfrak{a}_{G(z),u}(\theta)$ is, so $\mathrm{lct}_u(\mathfrak{a}_{t,u}(\theta)) = +\infty$ if and only if $\mathrm{lct}_u(\mathfrak{a}_{G(z),u}(\theta)) = +\infty$.

For d so large that the $G_{j,\alpha}(z_\infty^{(d)})$ are non-zero, apply Claim 4.4 to $X = \widetilde{U}_\alpha$, G trivial, $V = \mu_\alpha^{-1}(\Sigma_\alpha)$, $T = Z_d$, the ideals $(G_{j,\alpha})$, the $\mathfrak{c}_l$, and the ideal sheaves of the components of $\mathrm{Exc}(\mu_\alpha)$, and $q = 0$, for any α, and let $Z'_d \subset Z_d$ be the intersection of the open subsets thus obtained. For a point z of Z'_d, let ψ_z be the composite of φ_z and μ_α, which is a log resolution of U_α and of the $G_{j,\alpha}(z)$ near Σ_α. By [dFEM11, Section 2] and Lemma 3.7, $\mathrm{lct}_u(\mathfrak{a}_{G(z),u}(\theta))$ is the minimum of

$$\frac{a(Q_{l,z}, U_\alpha, 0)}{\sum_{l'} e_{l'} \operatorname{mult}_{Q_{l,z}} \mathfrak{c}_{l'} + \theta \sum_j c_j \operatorname{mult}_{Q_{l,z}} G_{j,\alpha}(z)}$$

over the indices l with $u \in \psi_z(Q_{l,z})$. Here

$$a(Q_{l,z}, U_\alpha, 0) = a(Q_{l,z}, \widetilde{U}_\alpha, B_{\widetilde{U}_\alpha})$$

is determined by $a(Q_{l,z}, \widetilde{U}_\alpha, 0)$ and the multiplicities along $Q_{l,z}$ of the ideal sheaves of the components of $\mathrm{Exc}(\mu_\alpha)$. Thus $\ell_{G(z)}(\theta)$ is the minimum of the quotients above with numerator $a(Q_{l,z}, U_\alpha, 0)$ over the indices l with $\psi_z(Q_{l,z}) \cap \Sigma_\alpha \neq \emptyset$, and by Claim 4.4(2) it does not depend on $z \in Z'_d$, for all θ at once. Denote the common value by $\bar{\ell}_d(\theta)$. By Construction-Proposition 4.1, $z_\infty^{(d)}$ is a k-point of Z'_d, and $z_i^{(d)} \in Z'_d$ for all but finitely many i.

For $k' = \mathbb{C}$, $\mathcal{S}_u \cong \operatorname{Spec} \mathbb{C}[[\xi_1, \ldots, \xi_n]]$, where $n = 2$ in case (b). Since [dFEM11, Theorem 4.2] treats $\mathbb{R}$-ideals on the completions of germs with bounded singularities, it covers the $\mathfrak{a}_{i,u}(\theta)$ on $\mathcal{S}_u$, which in case (b) need not come from a variety. With the finite set of exponents $\{e_l\}_l \cup \{\theta c_j\}_j$, it gives $\gamma_\theta > 0$ such that

$$\mathrm{lct}_u(\mathfrak{a}_{i,u}(\theta)) \notin (1 - \gamma_\theta, 1)$$

for all i, α, and $u \in \Sigma_\alpha(\mathbb{C})$. Otherwise the thresholds $\mathrm{lct}_u(\mathfrak{a}_{i,u}(\theta))$ would contain a strictly increasing sequence converging to 1.

Assume that $(\mathcal{A}_k, \theta C_\infty, \mathfrak{c})$ is lc near the base. Let $\epsilon := \gamma_\theta/3$ and $d \geq d(\theta, \epsilon)$. Then $\ell_\infty(\theta) \geq 1$, so (4.3) implies $\bar{\ell}_d(\theta) \geq 1 - \epsilon$ and $\ell_i(\theta) \geq 1 - 2\epsilon > 1 - \gamma_\theta$ for all but finitely many i. By the choice of γ_θ, $\mathrm{lct}_u(\mathfrak{a}_{i,u}(\theta)) \geq 1$ for any u, so $(\mathcal{A}, \theta C_i, \mathfrak{c})$ is lc near the base for all but finitely many

i. Assume instead that $(\mathcal{A}_k, \theta C_\infty, \mathfrak{c})$ is not lc near the base. Then $\ell_\infty(\theta) < 1$, and for ϵ with $\ell_\infty(\theta) + 2\epsilon < 1$ and $d \geq d(\theta, \epsilon)$, (4.3) implies $\ell_i(\theta) < 1$ for all but finitely many i.

If $\lambda_\infty = +\infty$, then any $f_{\infty,j,\alpha}$ is a unit at any closed point of Σ_α, since $\lambda_{\infty,u} < +\infty$ whenever some $f_{\infty,j,\alpha}$ vanishes at u. Then $\mathrm{mult}_{Q_{l,z}} G_{j,\alpha}(z) = 0$ for any l with $\psi_z(Q_{l,z}) \cap \Sigma_\alpha \neq \emptyset$ at $z = z_\infty^{(d)}$, hence at any $z \in Z'_d$, and by (4.1) the $f_{i,j,\alpha}$ are units along Σ_α for all but finitely many i. Thus $C_i = 0$ near the base and $\lambda_i = +\infty$ for all but finitely many i. If $\lambda_\infty < +\infty$, then $(\mathcal{A}_k, \lambda_\infty C_\infty, \mathfrak{c})$ is lc near the base, so the previous paragraph implies $\lambda_i \geq \lambda_\infty$ for all but finitely many i, and $\lambda_i < \theta$ for all but finitely many i for any $\theta > \lambda_\infty$. Possibly removing finitely many members, we obtain $\lambda_i \geq \lambda_\infty$ for any i and $\lambda_i \to \lambda_\infty$. The equivalence in (1) follows: if $\theta \leq \lambda_\infty$, then $\theta \leq \lambda_i$ for all i, and if $\theta \leq \lambda_i$ for all i, then $\theta \leq \lim_i \lambda_i = \lambda_\infty$.

Finally assume that $(\mathcal{A}_k, \theta C_\infty, \mathfrak{c})$ is klt near the base. We show $\lambda_\infty > \theta$. Then (1) holds with $\theta' = \min\{\lambda_\infty, \theta + 1\}$ by the equivalence. For a closed point u, the pullback to $\mathcal{S}_u$ of the klt pair at u is klt by **Step 1**, and it has a log resolution [dFEM11, Section 2], on which finitely many prime divisors test lc and klt, so $\lambda_{\infty,u} > \theta$. This proves the claim in case (a). In case (b), if E is a point, then each Σ_α, the preimage of $E \cap V_\alpha$ under the quotient morphism by K_α, is finite, so λ_∞ is the minimum of the finitely many $\lambda_{\infty,u}$, each larger than θ. If E is a curve, let Γ run through the irreducible components of $V(\mathfrak{n}_\alpha) \subset S_{\alpha,k}$ meeting $U_{\alpha,k}$, and let $m_{j,\Gamma}$ be the multiplicity of $\mathrm{div}(f_{\infty,j,\alpha})$ along Γ on $\operatorname{Spec} R_{\alpha,k}$. The divisor

$$\Omega_j := \mathrm{div}(f_{\infty,j,\alpha}) - \sum_\Gamma m_{j,\Gamma}\Gamma \geq 0$$

on $\operatorname{Spec} R_{\alpha,k}$ has no component Γ, so the support of Ω_j intersects Σ_α in finitely many closed points. Let Ξ_α be the union of the finite sets of such points for all j, of the singular points of the reduced curve Σ_α, and of the singular points of $U_{\alpha,k}$. For $u \in \Sigma_\alpha \backslash \Xi_\alpha$ on Γ, the surface $U_{\alpha,k}$ and the curve Γ are smooth at u, and θC_∞^α is $\theta(\sum_j c_j m_{j,\Gamma})\Gamma$ near u, so $\lambda_{\infty,u} = (\sum_j c_j m_{j,\Gamma})^{-1}$. Thus λ_∞ is the minimum of finitely many $\lambda_{\infty,u}$, each larger than θ. This proves (1).

Step 5. In this step, we prove (2) in case (a) when H is diagonal. Since H is diagonal and each $\mathfrak{c}_l$ is H-invariant, $\mathfrak{c}_l$ is generated by H-semi-invariant polynomials. Since $n = 2$, H is diagonal, the orders of the $f_{i,j}$ at 0 are bounded, and the characters of the $f_{i,j}$ depend only on j, the tuple $(f_{\infty,j})_j$ of Construction-Proposition 4.1 is a generic limit in the sense of [Liu26, Construction 14.13]. Proposition 3.19(2), with $\Lambda := \{c_1, \dots, c_s\}$, implies that the equality of (2) holds for infinitely many i, and we pass to the subsequence of these i.

Step 6. In this step, we prove (2) when $\mathfrak{c} = \mathcal{O}$, in case (b) and in case (a) with $n = 2$. For a member or for the limit, let β_i, resp. β_∞, be the infimum of $a(F, Y, B_{\mathrm{br}} + C_i)/|I_H(F)|$, resp. of $a(F, \mathcal{A}_k, B_{\mathrm{br}} + C_\infty)/|I_H(F)|$, over the prime divisors F with center contained in E, resp. E_k. In case (b), the infima in (2) are β_i and β_∞ by definition. In case (a), the minimal log discrepancies in (2) are β_i and β_∞ by [Liu26, Lemma 2.6(2)] for the members, Lemma 3.7 for the limit, and Definition 2.4. Thus part (2) is the equality $\beta_i = \beta_\infty$ for any i. By (1) with $\theta = 1$, $(\mathcal{A}, C_i)$ is lc near the base for any i.

The values β_i belong to a finite set. Let Λ_+ be the finite set of sums $\sum_j m_j c_j \leq 1$ with non-negative integers m_j, and let Θ be the finite set of numbers $1 - (1 - \sigma)/(ww')$ with $\sigma \in \Lambda_+$, $1 \leq w \leq \max_\alpha |K_\alpha|$, and $1 \leq w' \leq |H|$. Let Δ_0 be the crepant quotient boundary of (Y, B_{br}) by H. By [Liu26, Lemma 2.6(4)] applied twice, with Lemma 3.7 in case (b), the crepant quotient $(Y/H, \Delta_i)$ of $(Y, B_{\mathrm{br}} + C_i)$ by H has $\Delta_i \in \Theta$ near E/H, and $\Delta_0 \in \Theta$. In case (b), Δ_i is an $\mathbb{R}$-divisor on the formal completion of Y/H along E/H. Let Θ' be the finite set of numbers $\theta_0 + \sum_j c_j m_j/w \leq 1$ with $\theta_0 \in \Theta \cup \{0\}$, non-negative integers m_j, and $1 \leq w \leq \max_\alpha |K_\alpha| \cdot |H|$, and let l_0 be the integer of Theorem 2.9(2) for Θ'. We show that for any i, any closed point $\bar{y} \in E/H$, and any prime divisor Q over Y/H with $\mathrm{center}_{Y/H} Q = \{\bar{y}\}$ (a formal prime divisor over $\operatorname{Spec} \widehat{\mathcal{O}}_{Y/H,\bar{y}}$ with center the closed point in case (b)), some prime divisor P over Y/H with

$\operatorname{center}_{Y/H} P = \{\bar{y}\}$ satisfies

$$a(P, Y/H, \Delta_i) \le a(Q, Y/H, \Delta_i) \quad \text{and} \quad a(P, Y/H, 0) \le l_0.$$

In case (a), this is Theorem 2.9(2), since $\Theta \subset \Theta'$.

In case (b), choose a closed point $y \in E$ over $\bar{y}$, an index α with $y \in V_\alpha$, and a closed point $u \in \Sigma_\alpha$ over y, let $K_u \subset K_\alpha$ and $H_y \subset H$ be the stabilizers of u and y, and let $w := |K_u| \cdot |H_y|$. Let $\widehat{R} := \widehat{\mathcal{O}}_{Y/H,\bar{y}}$, which is the ring of invariants of H_y in $\widehat{\mathcal{O}}_{Y,y}$, which is in turn the ring of invariants of K_u in $\widehat{\mathcal{O}}_{U_\alpha,u}$ [Liu26, Lemma 3.3(1)], since $Y/H_y \to Y/H$ and $U_\alpha/K_u \to U_\alpha/K_\alpha$ induce isomorphisms of the completed local rings at the images of y and u [Gro71, Exposé V, Proposition 2.2(i)], and let $s_{i,j} \in \widehat{R}$ be the norm of $f_{i,j,\alpha}$ from $\widehat{\mathcal{O}}_{U_\alpha,u}$ to $\widehat{R}$. Since the pushforward of the divisor of a function under a finite morphism is the divisor of the norm of the function, Lemma 3.7 applied to K_u and to H_y gives

$$\Delta_i = \Delta_0 + \sum_j \frac{c_j}{w} \operatorname{div}(s_{i,j})$$

on $\operatorname{Spec} \widehat{R}$. Let $\widehat{\mathfrak{m}}$ be the maximal ideal of $\widehat{R}$, and let $g : W \to \operatorname{Spec} \widehat{R}$ be a log resolution of $(\operatorname{Spec} \widehat{R}, \Delta_i)$ and of $\widehat{\mathfrak{m}}$ [dFEM11, Section 2], so that $\widehat{\mathfrak{m}}\mathcal{O}_W$ is invertible and $g^{-1}(\bar{y})$ is the union of the exceptional curves of g. Let M be an integer larger than $l_0 w / \min_j c_j$, than the multiplicities of the $s_{i,j}$ along Q, and than their multiplicities along the exceptional curves of g. Choose $s'_{i,j} \in \mathcal{O}_{Y/H,\bar{y}}$ with $s_{i,j} - s'_{i,j} \in \widehat{\mathfrak{m}}^M$, and let

$$\Delta'_i := \Delta_0 + \sum_j \frac{c_j}{w} \operatorname{div}(s'_{i,j})$$

near $\bar{y}$.

We show that $(Y/H \ni \bar{y}, \Delta'_i)$ is lc. By [Liu26, Lemma 3.3(1)(3)], applied with the trivial group, the prime divisors over Y/H with center $\bar{y}$ correspond to the formal prime divisors over $\operatorname{Spec} \widehat{R}$ with center the closed point, with equal log discrepancies with respect to Δ'_i and its pullback. A curve through $\bar{y}$ with coefficient b in Δ'_i pulls back to a divisor $\sum_k m_k \widehat{C}_k$ on $\operatorname{Spec} \widehat{R}$ with integers $m_k \ge 1$, so the coefficient $m_k b$ of $\widehat{C}_k$ in the pullback of Δ'_i is at least b. Thus it suffices to show that $(\operatorname{Spec} \widehat{R}, \Delta'_i)$ is lc near the closed point.

Any exceptional curve E_l of g has center $\bar{y}$, so $\operatorname{mult}_{E_l} s_{i,j} = \operatorname{mult}_{E_l} s'_{i,j}$, and the log pullbacks of Δ_i and Δ'_i to W have equal coefficients, at most one, along the exceptional curves. Near $g^{-1}(\bar{y})$, write $g^* s_{i,j} = \zeta_j \widetilde{s}_{i,j}$ and $g^* s'_{i,j} = \zeta_j \widetilde{s}'_{i,j}$, where ζ_j is the product of the $\varpi_l^{\operatorname{mult}_{E_l} s_{i,j}}$ for local equations ϖ_l of the E_l. Since $M > \operatorname{mult}_{E_l} s_{i,j}$ for any l, $\widetilde{s}_{i,j} - \widetilde{s}'_{i,j}$ vanishes along the exceptional curves. Thus $\widetilde{s}'_{i,j}$ is a unit wherever $\widetilde{s}_{i,j}$ is. By the snc condition, any point $p \in g^{-1}(\bar{y})$ at which some $\widetilde{s}_{i,j}$ is not a unit is on exactly one exceptional curve E_p and on exactly one component C_p of the strict transform of $\operatorname{Supp} \Delta_i$, which is smooth and transversal to E_p at p, and the restrictions of $\widetilde{s}_{i,j}$ and $\widetilde{s}'_{i,j}$ to E_p coincide and vanish at p to the order $r_j := \operatorname{mult}_{C_p} s_{i,j}$. Let δ_p be the coefficient of C_p in Δ_0, and let B'_p be the sum of δ_p times the strict transform of C_p and of the $(c_j/w) \operatorname{div}(\widetilde{s}'_{i,j})$. The support of B'_p does not contain E_p, and the restriction of B'_p to E_p has coefficient $\delta_p + \sum_j c_j r_j / w$ at p, which is the coefficient of C_p in Δ_i, hence at most one.

By [Kaw07, Theorem] on formal germs (Lemma 2.14), applied to $\operatorname{Spec} \widehat{\mathcal{O}}_{W,p}$, the regular curve given by E_p, and the pullback of B'_p, the pullback of $(W, E_p + B'_p)$ to $\operatorname{Spec} \widehat{\mathcal{O}}_{W,p}$ is lc. Every prime divisor over $\operatorname{Spec} \mathcal{O}_{W,p}$ with center p is the exceptional curve of the last of a sequence of blow-ups of closed points [Kol13, Lemma 2.22]. Blow-ups of closed points commute with the base change to $\operatorname{Spec} \widehat{\mathcal{O}}_{W,p}$ and preserve log discrepancies, and, as for Δ'_i above, the coefficient of a curve through p is at most that of any component of the pullback of the curve, so $(W, E_p + B'_p)$ is lc near p. Since the log pullback of Δ'_i to W is at most $E_p + B'_p$ near p, and is an snc divisor

with coefficients at most one near the other points of $g^{-1}(\bar{y})$, the pair $(\operatorname{Spec}\widehat{R}, \Delta_i')$ is lc near the closed point.

Since $(Y/H \ni \bar{y}, \Delta_i')$ is lc, $\Delta_i' \in \Theta'$. By Theorem 2.9(2), some prime divisor P over Y/H with $\operatorname{center}_{Y/H} P = \{\bar{y}\}$ satisfies $a(P, Y/H, \Delta_i') = \operatorname{mld}(Y/H \ni \bar{y}, \Delta_i')$ and $a(P, Y/H, 0) \le l_0$. We identify P with the formal prime divisor over $\operatorname{Spec}\widehat{R}$ that it defines [dFEM11, Lemma 2.21, Proposition A.14, and Remark A.15]. Since $(Y/H \ni \bar{y}, \Delta_i')$ is lc, $(c_j/w)\operatorname{mult}_P s_{i,j}' \le a(P, Y/H, 0) \le l_0$, so $\operatorname{mult}_P s_{i,j}' < M$, and $\operatorname{mult}_P s_{i,j} = \operatorname{mult}_P s_{i,j}'$ because $\operatorname{mult}_P \widehat{\mathfrak{m}}^M \ge M$. Similarly, $\operatorname{mult}_Q s_{i,j} = \operatorname{mult}_Q s_{i,j}'$ by the choice of M. Thus

$$a(P, Y/H, \Delta_i) = a(P, Y/H, \Delta_i') \le a(Q, Y/H, \Delta_i') = a(Q, Y/H, \Delta_i).$$

Now let P be a prime divisor over Y/H with $\operatorname{center}_{Y/H} P = \{\bar{y}\}$ and $a(P, Y/H, 0) \le l_0$, let F be a prime divisor over Y over P, and let F'' be a prime divisor over U_α as in **Step 1**. By [Liu26, Lemma 2.6(2)] applied to K_α and to H, and since $\Delta_0 \ge 0$,

$$a(F'', U_\alpha, 0) = \iota(F'')a(P, Y/H, \Delta_0) \le N := l_0 \max_\alpha |K_\alpha| \cdot |H|.$$

Since (U_α, C_i^α) is lc, $c_j \operatorname{mult}_{F''} f_{i,j,\alpha} \le N$ for any j. Let r_K be a positive integer such that $r_K K_{U_\alpha}$ is Cartier for any α, so that $a(F'', U_\alpha, 0) \in \frac{1}{r_K}\mathbb{Z}$. By (4.2) and [Liu26, Lemma 2.6(2)], $a(P, Y/H, \Delta_i)$ belongs to the finite set $\mathcal{V}$ of numbers $(a - \sum_j c_j m_j)/w'$ with $a \in \frac{1}{r_K}\mathbb{Z} \cap (0, N]$, integers $0 \le m_j \le N/c_j$, and $1 \le w' \le N$. Thus $\operatorname{mld}(Y/H \ni \bar{y}, \Delta_i)$ is attained by a prime divisor P with $a(P, Y/H, 0) \le l_0$, and it belongs to $\mathcal{V}$. If E is a curve, then for a one-dimensional irreducible component E' of E,

$$\frac{a(E', Y, B_{\mathrm{br}} + C_i)}{|I_H(E')|} = \frac{1 - \operatorname{mult}_{E'}(B_{\mathrm{br}} + C_i)}{|I_H(E')|},$$

and the right-hand side belongs to the finite set of numbers $(1 - \theta_0)/w'$ with $\theta_0 \in \Theta$ and $1 \le w' \le |H|$. Thus β_i is a minimum and belongs to a finite set. Possibly passing to a subsequence, $\beta_i = L$ for a real number L and all i. Fix an integer $d_1 > N/\min_j c_j$ such that the $G_{j,\alpha}(z_\infty^{(d_1)})$ are non-zero.

Components of E. In case (b) with E a curve, let E' be a one-dimensional irreducible component of E, α an index with $E' \cap V_\alpha \ne \emptyset$, and Γ a component of Σ_α over E'. For a non-negative integer m', let $I_\Gamma^{(m')} \subset \mathbb{C}[S_\alpha]$ be the ideal of the functions with multiplicity at least m' along Γ. By (4.2), it suffices to show that

$$\operatorname{mult}_\Gamma f_{i,j,\alpha} = \operatorname{mult}_\Gamma f_{\infty,j,\alpha}$$

for all but finitely many i. Let $m := \operatorname{mult}_\Gamma f_{\infty,j,\alpha}$ and $d > \max\{d_1, m+1\}$. For an integer $m' < d$ and a point z of $\mathcal{J}_d$, $\operatorname{mult}_\Gamma G_{j,\alpha}(z) \ge m'$ if and only if the restriction of $G_{j,\alpha}(z)$ to S_α belongs to $I_\Gamma^{(m')}$, and the points z with $\operatorname{mult}_\Gamma G_{j,\alpha}(z) \ge m'$ form a linear subspace of $\mathcal{J}_d$. Since $\Gamma \subset V(\mathfrak{n}_\alpha)$, (4.1) implies $\operatorname{mult}_\Gamma G_{j,\alpha}(z_\infty^{(d)}) = m$. Since $z_\infty^{(d)}$ maps to the generic point of Z_d, the subspace for $m' = m$ contains Z_d, and the subspace for $m' = m+1$ intersects Z_d in a proper closed subset. Thus $\operatorname{mult}_\Gamma G_{j,\alpha}(z_i^{(d)}) = m$ for all but finitely many i, and since $\operatorname{mult}_\Gamma f_{i,j,\alpha} \le N/c_j < d$, (4.1) implies $\operatorname{mult}_\Gamma f_{i,j,\alpha} = m$ for all but finitely many i. Possibly removing finitely many members, the number $a(E', Y, B_{\mathrm{br}} + C_i)/|I_H(E')|$ does not depend on i and is equal to the corresponding number for C_∞.

The inequality $\beta_\infty \ge L$. We first compare inertia groups of formal prime divisors and of their restrictions.

Claim 4.5. *Let $k' \supset \mathbb{C}$ be an algebraically closed field, X a normal variety over k' with an action of a finite group G, $x \in X$ a closed point, $\widehat{F}$ a formal prime divisor over $\operatorname{Spec}\widehat{\mathcal{O}}_{X,x}$ with center the closed point, and F_0 the prime divisor over X with center x to which $\widehat{F}$ restricts* [dFEM11, Lemma 2.21 and Proposition A.14]. *Then $I_G(\widehat{F}) = I_G(F_0)$, where $I_G(\widehat{F})$ is the inertia group of $\widehat{F}$ for the stabilizer of x in G.*

Proof. Since F_0 has center x, it is a prime divisor on a log resolution $\widetilde{X}_x \to X$ that is equivariant for the stabilizer of x in G. The base change of $\widetilde{X}_x$ to $\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$ is a log resolution on which $\widehat{F}$ is the prime divisor given by F_0 on the common fiber over x [dFEM11, Lemma 2.21 and Remark A.15]. Hence the residue fields of $\widehat{F}$ and of F_0 are equal, an element of G fixes $\widehat{F}$ if and only if it fixes F_0 (both force it to fix x), and it acts trivially on one residue field if and only if on the other. Thus $I_G(\widehat{F}) = I_G(F_0)$. □

Proof of Theorem 4.2 continued. By the paragraph on components of E, it suffices to consider a formal prime divisor F over $\operatorname{Spec} \widehat{\mathcal{O}}_{Y_k,y}$ with center a closed point $y \in E_k$. Choose α with $y \in V_{\alpha,k}$, a closed point $u \in \Sigma_\alpha(k)$ over y, and a formal prime divisor F'' over $\operatorname{Spec} \widehat{\mathcal{O}}_{U_{\alpha,k},u}$ lying over F ([Liu26, Lemma 3.3(1)] and Lemma 3.7, applied to the stabilizers as for (4.2)). Choose $d > \max\{d_1, \max_j \operatorname{mult}_{F''} f_{\infty,j,\alpha}\}$. By (4.1),

$$m_j := \operatorname{mult}_{F''} G_{j,\alpha}(z_\infty^{(d)}) = \operatorname{mult}_{F''} f_{\infty,j,\alpha}.$$

By [dFEM11, Lemma 2.21 and Proposition A.14], F'' restricts to a prime divisor F_0'' over $U_{\alpha,k}$ with center u, with log discrepancy and multiplicities of polynomials equal to those of F''. Let F_0 be the prime divisor over Y_k under F_0'', which is the restriction of F. Claim 4.5 applied to $(U_{\alpha,k}, u, K_\alpha, F'', F_0'')$ and to (Y_k, y, H, F, F_0) gives $I_{K_\alpha}(F'') = I_{K_\alpha}(F_0'')$ and $I_H(F) = I_H(F_0)$, so $\iota(F_0'') = \iota(F'')$, and

$$\frac{a(F, \mathcal{A}_k, B_{\mathrm{br}} + C_\infty)}{|I_H(F)|} = v := \frac{a(F_0'', U_{\alpha,k}, 0) - \sum_j c_j m_j}{\iota(F_0'')}.$$

There exists a finitely generated $\mathbb{C}$-subalgebra $A \subset k$ over which $z_\infty^{(d)}$, u, and F_0'' are defined, and the morphism $T := \operatorname{Spec} A \to Z_d$ is dominant because $z_\infty^{(d)}$ maps to the generic point of Z_d. Fix an H-equivariant resolution $\widetilde{Y} \to Y$, and let X_α be a K_α-equivariant resolution of the normalization of the irreducible component of $U_\alpha \times_Y \widetilde{Y}$ dominating U_α. Then X_α is smooth and quasi-projective, the projection $X_\alpha \to U_\alpha$ is K_α-equivariant, proper, and birational, and the projection $X_\alpha \to \widetilde{Y}$ is K_α-invariant and proper over the preimage $\widetilde{Y}_{V_\alpha}$ of V_α. Let $\mathcal{F}$ be a prime divisor over $U_\alpha \times T$ whose base change to k has F_0'' as a component. Its center dominates T, since the center of F_0'' is the point u, which is defined over A, and $\mathcal{F}$ is a prime divisor over $X_\alpha \times T$, since $X_\alpha \to U_\alpha$ is birational. The ideals $(G_{j,\alpha})$ are K_α-invariant by the choice of the $G_{j,\alpha}$ in **Step 1**, and K_α permutes the ideal sheaves of the components of the exceptional locus of $X_\alpha \to U_\alpha$. Apply Claim 4.4 over T to $X = X_\alpha$, $G = K_\alpha$, V the preimage of Σ_α, the ideals $(G_{j,\alpha})$ and the ideal sheaves of the components of the exceptional locus of $X_\alpha \to U_\alpha$, and $\mathcal{F}$. Let $T^\circ \subset T$, T', $\widetilde{X}'$, and the prime divisors Q_l on $\widetilde{X}'$ be the output of Claim 4.4. The composite morphism from $\widetilde{X}'$ to $\widetilde{Y} \times T'$ maps each Q_l onto a closed subset of $\widetilde{Y}_{V_\alpha} \times T'$, and there is a prime divisor P_l over $\widetilde{Y} \times T'$ under Q_l, whose center dominates T'.

Apply Claim 4.4 over T' also to $X = \widetilde{Y}$, $G = H$, V the preimage of E, no ideals, and the P_l, and let $T_1' \subset T'$, T'', the resolution $\widetilde{Y}''$ over T'', and the prime divisors on $\widetilde{Y}''$ given by Claim 4.4 be the output, where T_1' is the open subset that Claim 4.4 gives in place of T°. We replace T° by the complement of the image of $T' \backslash T_1'$, so that any point of T° lifts to T''. Let P_l'' be the center of P_l on the resolution of $\widetilde{Y} \times T_1'$ given by Claim 4.4. The generic fiber of Q_l over T' is geometrically irreducible and dominates the generic fiber of P_l'' over T_1'. Hence the base change of P_l'' to T'' is irreducible, so it is one of the prime divisors on $\widetilde{Y}''$ given by Claim 4.4, which we denote by $R_{\bar{l}}$. After shrinking T°, by Chevalley's theorem and generic flatness, the closure of the graph of the rational map from the base change of $\widetilde{X}'$ to T'' to $\widetilde{Y}''$ has as fibers over T'' the closures of the graphs of the fibers, the strict transform of Q_l on it has irreducible fibers which map dominantly onto the fibers of $R_{\bar{l}}$, and the dimension of the image of $Q_{l,\tau}$ in $U_{\alpha,k'}$ does not depend on τ.

Thus, for any algebraically closed field $k' \supset \mathbb{C}$ and any k'-point τ of T'', the prime divisor over $Y_{k'}$ under $Q_{l,\tau}$ is $R_{\bar{l},\tau}$, and by Claim 4.4(2) for both applications, $I_{K_\alpha}(Q_{l,\tau})$ and $I_H(R_{\bar{l},\tau})$ do not depend on τ. Therefore

$$\iota(Q_{l,\tau}) = |I_{K_\alpha}(Q_{l,\tau})| \cdot |I_H(R_{\bar{l},\tau})|$$

does not depend on τ. Moreover, the number $a(Q_{l,\tau}, U_\alpha, 0)$ is determined by $a(Q_{l,\tau}, X_\alpha, 0)$ and the multiplicities along $Q_{l,\tau}$ of the ideal sheaves of the components of the exceptional locus of $X_\alpha \to U_\alpha$, so it does not depend on τ. The k-point of T given by $A \subset k$ maps to the generic point of T, so it is a k-point of T°, and F_0'' is a component of the fiber over it of $\mathcal{F}$. By Claim 4.4(1), $F_0'' = Q_{\hat{l},\tau_k}$ for a k-point τ_k of T'' over it and an index $\hat{l}$.

By Chevalley's theorem, the image of T° in Z_d contains a non-empty open subset of Z_d, and it contains $z_i^{(d)}$ for all but finitely many i. For such i, choose a $\mathbb{C}$-point τ of T'' over $z_i^{(d)}$, and let $Q := Q_{\hat{l},\tau}$, a component of the fiber over τ of $\mathcal{F}$. By Claim 4.4(2) and the previous paragraph, Q is a prime divisor over U_α with center a closed point of Σ_α, $a(Q, U_\alpha, 0) = a(F_0'', U_{\alpha,k}, 0)$, $\mathrm{mult}_Q G_{j,\alpha}(z_i^{(d)}) = m_j$, and $\iota(Q) = \iota(F_0'')$. Since $m_j < d$, (4.1) implies $\mathrm{mult}_Q f_{i,j,\alpha} = m_j$. By (4.2), $L = \beta_i \le v$. Since F is arbitrary, $\beta_\infty \ge L$.

The inequality $\beta_\infty \le L$. If β_i is attained at a component of E for infinitely many i, then $\beta_\infty \le L$ by the paragraph on components of E. Otherwise, possibly passing to a subsequence, for any i there exist a closed point $u_i \in \Sigma_\alpha$ for a fixed α and a prime divisor F_i'' over U_α with center u_i such that $a(F_i'', U_\alpha, C_i^\alpha)/\iota(F_i'') = L$, $a(F_i'', U_\alpha, 0) \le N$, and $\mathrm{mult}_{F_i''} f_{i,j,\alpha} \le N/c_j < d_1$.

Let δ_α be the minimum of 1 and of the numbers $a(\Gamma', U_\alpha, 0)$ for the components Γ' of $\mathrm{Exc}(\mu_\alpha)$. Then $0 < \delta_\alpha \le 1$ since U_α is klt, and $\delta_\alpha = 1$ if U_α is smooth. The center of F_i'' on $\widetilde{U}_\alpha$ is either a component of $\mathrm{Exc}(\mu_\alpha)$ or a closed point. In the second case, F_i'' is the exceptional curve of the last blow-up in a sequence of blow-ups of closed points starting on $\widetilde{U}_\alpha$, each on the exceptional curve of the previous blow-up [KM98, Lemma 2.45]. On each surface U' of the sequence, let $\Upsilon_{U'}$ be the snc divisor formed by the strict transforms of the components of $\mathrm{Exc}(\mu_\alpha)$ and the exceptional curves of the previous blow-ups. The support of the crepant pullback of $(U_\alpha, 0)$ to U' is contained in $\Upsilon_{U'}$, and the coefficient of any curve of $\Upsilon_{U'}$ in the crepant pullback is $1 - a(\cdot, U_\alpha, 0)$. Since $\Upsilon_{U'}$ is snc, a blown-up point is on the last exceptional curve and on at most one other curve of $\Upsilon_{U'}$, so by [KM98, Lemma 2.29] the new exceptional curve has $a(\cdot, U_\alpha, 0)$ equal to the log discrepancy of the last curve plus either 1 or the log discrepancy of the other curve. By induction, since the coefficients in the crepant pullback are at most $1 - \delta_\alpha$, any curve of the sequence has $a(\cdot, U_\alpha, 0) \ge \delta_\alpha$, the first exceptional curve has $a(\cdot, U_\alpha, 0) \ge 2\delta_\alpha$, and $a(\cdot, U_\alpha, 0)$ increases by at least δ_α at any later blow-up. Thus F_i'' is obtained by a sequence of $\rho \le N/\delta_\alpha$ blow-ups, where $\rho = 0$ means that F_i'' is a component of $\mathrm{Exc}(\mu_\alpha)$, and possibly passing to a subsequence, ρ does not depend on i.

The sequences of ρ blow-ups of closed points starting on $\widetilde{U}_\alpha$, each on the exceptional curve of the previous blow-up, are parametrized by a variety $\mathcal{B}_\rho$ of finite type, with a universal family over $\widetilde{U}_\alpha \times \mathcal{B}_\rho$. They are constructed inductively as follows. Let $\mathcal{B}_1 := \widetilde{U}_\alpha$, with universal surface the blow-up of $\widetilde{U}_\alpha \times \mathcal{B}_1$ along the diagonal. For $\rho \ge 1$, let $\mathcal{B}_{\rho+1}$ be the exceptional divisor of the universal blow-up over $\mathcal{B}_\rho$, and let the universal surface over $\mathcal{B}_{\rho+1}$ be the blow-up of the pullback of the universal surface over $\mathcal{B}_\rho$ along the tautological section. Each universal surface is smooth over its base, since the blow-up of a smooth family of surfaces along a section is smooth over the base, and the last exceptional divisor of the universal family is the universal divisor. For $\rho = 0$, $\mathcal{B}_0$ is the finite set of components of $\mathrm{Exc}(\mu_\alpha)$.

Let $b_i \in \mathcal{B}_\rho$ be the point of F_i'', and let $\Pi := \{(z_i^{(d_1)}, b_i)\}_i$, a subset of $Z_{d_1} \times \mathcal{B}_\rho$. If no irreducible component of the closure of Π dominates Z_{d_1}, then all $z_i^{(d_1)}$ belong to a proper closed subset of Z_{d_1}, which contradicts Construction-Proposition 4.1. Let T be an irreducible component of the closure of Π which dominates Z_{d_1}. Then $\Pi \cap T$ is dense in T. The universal divisor restricted to T is a prime divisor over $U_\alpha \times T$ whose center dominates T. Repeat the applications

of Claim 4.4 in the paragraph on $\beta_\infty \geq L$, and the shrinking after them, with T in place of $\operatorname{Spec} A$, the projection $T \to Z_{d_1}$ in place of $\operatorname{Spec} A \to Z_d$, d_1 in place of d, and the universal divisor in place of $\mathcal{F}$. After shrinking T°, the fibers over T° of the strict transform on $\widetilde{X}$ of the universal divisor are the fibers of the universal divisor, as prime divisors over U_α, since $\widetilde{X}$ and the pullback to T° of the universal surface over $\mathcal{B}_\rho$ are isomorphic near the generic point of the universal divisor (Chevalley's theorem). We obtain a non-empty open subset $T^\circ \subset T$ such that the value $(a(Q, U_\alpha, 0) - \sum_j c_j \operatorname{mult}_Q G_{j,\alpha}(z))/\iota(Q)$ for the components Q of the fibers of the universal divisor over the points $(z, b) \in T^\circ$ does not depend on the point. The value is L at a point of $\Pi \cap T^\circ$, since $\operatorname{mult}_{F_i''} f_{i,j,\alpha} < d_1$ and by (4.1).

Since $T^\circ \to Z_{d_1}$ is dominant, $T^\circ \times_{Z_{d_1}} \operatorname{Spec} k$, formed with $z_\infty^{(d_1)}$, is a non-empty scheme of finite type over k, so it has a k-point. Let Q be a component of the fiber of the universal divisor over the k-point. Then Q is a prime divisor over $U_{\alpha,k}$ with center a closed point $u \in \Sigma_\alpha(k)$, $\operatorname{mult}_Q G_{j,\alpha}(z_\infty^{(d_1)}) < d_1$, and the value of Q is L. By [dFEM11, Remark A.15], a component $\widehat{Q}$ of the pullback of Q to $\operatorname{Spec} \widehat{\mathcal{O}}_{U_{\alpha,k},u}$ is a formal prime divisor with log discrepancy and multiplicities equal to those of Q, and $\iota(\widehat{Q}) = \iota(Q)$ by Claim 4.5 applied twice, as for F''. By (4.1),

$$\operatorname{mult}_{\widehat{Q}} f_{\infty,j,\alpha} = \operatorname{mult}_Q G_{j,\alpha}(z_\infty^{(d_1)}).$$

By (4.2), the image of $\widehat{Q}$ over $\mathcal{A}_k$ has value L, so $\beta_\infty \leq L$.

Therefore $\beta_\infty = L = \beta_i$ for any i, and the minimum for C_∞ is attained. This proves (2). □

4.2. **Limits of diagonal groups and bounded groups.** In this subsection, we collect the facts on Hausdorff limits of finite diagonal groups and on bounded groups that are used in the later sections. Along the sequences of the later sections the orders of the groups are unbounded. The groups preserve the equations, and we regard them as subgroups of the compact torus $\mathbb{T}^n$ and take their Hausdorff limits. Let $\rho : \mathbb{R}^n \to \mathbb{T}^n$ be the quotient map of [Liu26, Definition 2.15], and equip $\mathbb{T}^n$ with the metric induced by the Euclidean metric of $\mathbb{R}^n$. This subsection treats bounded groups and limits whose Lie algebra contains a vector with all coordinates positive, and Lemma 4.6 describes the circles. Two-dimensional limits are excluded in the later sections, separately for each type of source.

Lemma 4.6. *Let $(G_i)_{i\geq 1}$ be finite subgroups of $(\mathbb{C}^*)^n$ acting diagonally on $\mathbb{A}^n$, regarded as subgroups of $\mathbb{T}^n$. After passing to a subsequence there exists a closed subgroup $H \subset \mathbb{T}^n$ such that the G_i converge to H in the Hausdorff distance, $G_i \subset H$ and G_i intersects every connected component of H for $i \gg 0$. If $|G_i| \to +\infty$, then $\dim H \geq 1$. If moreover $\dim H = 1$, then $H^0 = \rho(\mathbb{R}w)$ for a primitive $w \in \mathbb{Z}^n$, where $\rho : \mathbb{R}^n \to \mathbb{T}^n$ is the quotient map of* [Liu26, Definition 2.15]*, $G_i \cap \mathbf{T}_{(w)}$ is cyclic of order $r_i \to +\infty$, and $[G_i : G_i \cap \mathbf{T}_{(w)}] = |H/H^0|$ for $i \gg 0$.*

Proof. The first sentence is [Liu26, Lemma 4.4] applied to the closed subgroups $G_i \subset \mathbb{T}^n$, with the metric fixed above.

Assume that $|G_i| \to +\infty$. If $\dim H = 0$, then H is finite, and $G_i \subset H$ for $i \gg 0$ implies $|G_i| \leq |H|$ for $i \gg 0$, a contradiction. Thus $\dim H \geq 1$.

Assume moreover that $\dim H = 1$. By [Liu26, Lemma 4.3(1)], H/H^0 is finite and $H^0 = \rho(\mathbb{R}w)$ for a primitive vector $w \in \mathbb{Z}^n$. By [Liu26, Lemma 4.3(2)], the image of $G_i \cap \mathbf{T}_{(w)}$ in $\mathbb{T}^n$ is $G_i \cap H^0$. Since w is primitive, $\mathbb{R}w \cap \mathbb{Z}^n = \mathbb{Z}w$, so ρ induces an isomorphism $\mathbb{R}w/\mathbb{Z}w \to H^0$ of groups. Hence $G_i \cap H^0$ is a finite subgroup of a group isomorphic to $\mathbb{R}/\mathbb{Z}$, and it is cyclic.

For $i \gg 0$, we have $G_i \subset H$, and G_i intersects every coset of H^0 in H. Thus the homomorphism $G_i \to H/H^0$ is surjective with kernel $G_i \cap H^0$. Therefore

$$[G_i : G_i \cap \mathbf{T}_{(w)}] = [G_i : G_i \cap H^0] = |H/H^0|$$

for $i \gg 0$, and

$$r_i := |G_i \cap \mathbf{T}_{(w)}| = \frac{|G_i|}{|H/H^0|} \to +\infty.$$

□

If the Lie algebra of the limit contains a vector with all coordinates positive, the quotient values tend to 0 (Lemma 4.7). For example, the minimal log discrepancy of $\mathbb{A}^3/\mu_n(1,1,1)$ at the image of 0 is $3/n$ [Liu26, Example 4.2(1)], and the groups $\mu_n(1,1,1)$ converge to the circle $\rho(\mathbb{R}(1,1,1))$. A positive lower bound for the quotient values therefore excludes such limits.

Lemma 4.7. *Let $n \in \{3,4\}$, and let $W = \mathbb{A}^3$ if $n = 3$ and $W = (f = 0) \subset \mathbb{A}^4$ for a polynomial f if $n = 4$, such that W is normal and klt near 0. Let G be a finite diagonal abelian group preserving W and acting faithfully on W, and $D = \sum_j c_j \operatorname{div}_W(h_j)$ with $c_j \geq 0$ and G-semi-invariant polynomials h_j such that (W, D) is lc near 0. Then for any $w \in N_G \cap \mathbb{R}^n_{>0}$, with $\operatorname{wt}_w(f) := 0$ for $n = 3$,*

$$\operatorname{mld}((W \ni 0, D)/G) \leq \sum_{k=1}^{n} w_k - \operatorname{wt}_w(f) - \sum_j c_j \operatorname{wt}_w(h_j).$$

Let $\rho : \mathbb{R}^n \to \mathbb{T}^n$ be the quotient map of [Liu26, Definition 2.15]. *If $(W_i, D_i, G_i)_{i\geq 1}$ is a sequence of such data with n fixed, the G_i have a Hausdorff limit $H \subset \mathbb{T}^n$, and the linear subspace $V \subset \mathbb{R}^n$ with $\rho(V) = H^0$* [Liu26, Lemma 4.3(1)] *contains a vector with all coordinates positive, then $\operatorname{mld}((W_i \ni 0, D_i)/G_i) \to 0$.*

Proof. For $v \in \mathbb{R}^n_{\geq 0}$, write

$$\varphi(v) := \sum_{k=1}^{n} v_k - \operatorname{wt}_v(f) - \sum_j c_j \operatorname{wt}_v(h_j).$$

Assume that $n = 4$. Since $W = (f = 0)$ is a closed subscheme of $\mathbb{A}^4$ preserved by the linear diagonal action of G, for any $g \in G$ the ideals (g^*f) and (f) of $\mathbb{C}[x_1, \dots, x_4]$ are equal. Thus $g^*f = \lambda_g f$ for some $\lambda_g \in \mathbb{C}^*$, that is, f is G-semi-invariant. Moreover, $0 \in W$ implies $f(0) = 0$, and W is integral near 0 because W is normal near 0.

The terms of D with $c_j = 0$ or $h_j(0) \neq 0$ do not contribute to D near 0. They do not contribute to φ either, because $\operatorname{wt}_v(h_j) = 0$ when $h_j(0) \neq 0$. Hence we may assume that $c_j > 0$ and $h_j(0) = 0$ for any j. Let $\mathfrak{a} := \prod_j (h_j)^{c_j}$, the unit ideal if there are no terms. Then $\mathfrak{a}$ is an $\mathbb{R}$-ideal on $\mathbb{A}^n$ with positive exponents, and each factor is generated by a G-semi-invariant polynomial. Since (W, D) is lc near 0, so is $(W, \mathfrak{a}|_W)$. If $n = 4$, Theorem 3.3(1), applied with $c = 1$, $f_1 = f$, $X = W$, and the subgroup equal to G, implies

$$\operatorname{mld}((W \ni 0, D)/G) = \operatorname{mld}\Big(\Big(\mathbb{A}^4 \ni 0, \operatorname{div}(f) + \sum_j c_j \operatorname{div}(h_j)\Big)/G\Big),$$

where we use that all factors of $\mathfrak{a}$ are principal. If $n = 3$, then $W = \mathbb{A}^3$ and the left-hand side is already a quotient value on $\mathbb{A}^3$. In both cases, let B be $\operatorname{div}(f) + \sum_j c_j \operatorname{div}(h_j)$ if $n = 4$ and $\sum_j c_j \operatorname{div}(h_j)$ if $n = 3$, and let $B_{\mathbb{A}^n/G}$ be the $\mathbb{R}$-divisor on $\mathbb{A}^n/G$ with $K_{\mathbb{A}^n} + B$ the pullback of $K_{\mathbb{A}^n/G} + B_{\mathbb{A}^n/G}$.

Let $v \in N_G \cap \mathbb{R}^n_{>0}$ be a primitive vector of N_G. By [Liu26, Set-up 8.4 and Lemma 8.5], applied with $d = n$, with the polynomials f (coefficient 1) and h_j (coefficient c_j) if $n = 4$, and with the polynomials h_j (coefficient c_j) if $n = 3$, the toric prime divisor E_v over $\mathbb{A}^n/G$ corresponding to v satisfies

$$a(E_v, \mathbb{A}^n/G, B_{\mathbb{A}^n/G}) = \varphi(v).$$

Since all coordinates of v are positive, v belongs to the interior of the cone $\mathbb{R}^n_{\geq 0}$, so the smallest cone of the fan of $\mathbb{A}^n/G$ containing v is $\mathbb{R}^n_{\geq 0}$. Hence the toric morphism to $\mathbb{A}^n/G$ from a toric variety on which E_v is a divisor maps E_v into the orbit of $\mathbb{R}^n_{\geq 0}$, which is the image of 0 [CLS11, Theorem 3.2.6 and Lemma 3.3.21(b)]. Thus the center of E_v on $\mathbb{A}^n/G$ is the image of 0, and

$$\operatorname{mld}((W \ni 0, D)/G) \leq \varphi(v).$$

Let $w \in N_G \cap \mathbb{R}^n_{>0}$. Then $w = kv$ for a primitive vector $v \in N_G \cap \mathbb{R}^n_{>0}$ of N_G and an integer $k \geq 1$. Since $\mathrm{wt}_{kv} = k\,\mathrm{wt}_v$, we have $\varphi(w) = k\varphi(v)$. By the formula at the end of [Liu26, Lemma 2.6], $\mathrm{mld}((W \ni 0, D)/G) = \inf_F a(F, W, D)/|I_G(F)|$, where F runs over the prime divisors over W with center 0. Since (W, D) is lc near 0, each $a(F, W, D) \geq 0$, so $\mathrm{mld}((W \ni 0, D)/G) \geq 0$, hence $\varphi(v) \geq 0$. Therefore

$$\varphi(w) = k\varphi(v) \geq \varphi(v) \geq \mathrm{mld}((W \ni 0, D)/G).$$

This proves the inequality.

We prove the second part. Let $u \in V$ be a vector with all coordinates positive, and let $\epsilon > 0$. Replacing u with a positive multiple, we may assume that $\sum_k u_k < \epsilon$. Since $\rho(u) \in H^0 \subset H$ and the G_i converge to H in the Hausdorff distance, there exist $g_i \in G_i$ whose distance to $\rho(u)$ tends to 0. Since the metric of $\mathbb{T}^n$ is induced by the Euclidean metric of $\mathbb{R}^n$, there exist $w_i \in \rho^{-1}(g_i)$ with $|w_i - u| \to 0$. Then $w_i \in \rho^{-1}(G_i) = N_{G_i}$, and for $i \gg 0$ we have $w_i \in \mathbb{R}^n_{>0}$ and $\sum_k (w_i)_k < \epsilon$. Since (W_i, D_i) is lc near 0, the formula at the end of [Liu26, Lemma 2.6] gives $\mathrm{mld}((W_i \ni 0, D_i)/G_i) \geq 0$. Let f_i, c_{ij}, and h_{ij} be the polynomial f, the coefficients c_j, and the polynomials h_j of the data (W_i, D_i, G_i), and let φ_i be the function φ for the data (W_i, D_i, G_i). Then for $i \gg 0$,

$$0 \leq \mathrm{mld}((W_i \ni 0, D_i)/G_i) \leq \varphi_i(w_i) \leq \sum_{k=1}^{n} (w_i)_k < \epsilon,$$

where the second inequality is the first part, and the third holds since $c_{ij} \geq 0$, $\mathrm{wt}_{w_i}(f_i) \geq 0$, and $\mathrm{wt}_{w_i}(h_{ij}) \geq 0$ because $w_i \in \mathbb{R}^n_{>0}$. Since $\epsilon > 0$ is arbitrary, $\mathrm{mld}((W_i \ni 0, D_i)/G_i) \to 0$. □

Lemma 4.8. *Fix positive integers N, M. Let $(Y \ni y)$ be a normal germ with $\dim Y \geq 1$, C an $\mathbb{R}$-divisor on Y with $N(K_Y + C)$ Cartier near y, G a finite group acting faithfully on Y, fixing y and preserving C, and $(Z \ni z, \Delta) := (Y \ni y, C)/G$. Assume that $|I_G(F)| \leq M$ for any prime divisor F over Y with center y (for example, if $|G| \leq M$). Then $a(E, Z, \Delta) \in (N \operatorname{lcm}(1, 2, \ldots, M))^{-1}\mathbb{Z}$ for any prime divisor E over Z with center z. If (Y, C) is lc near y, then $\mathrm{mld}((Y \ni y, C)/G)$ is attained and belongs to $(N \operatorname{lcm}(1, 2, \ldots, M))^{-1}\mathbb{Z}_{\geq 0}$. In particular, the set of such minimal log discrepancies in a bounded interval is finite.*

Proof. Let $\ell := \operatorname{lcm}(1, 2, \ldots, M)$ and let $\pi : Y \to Z$ be the quotient morphism. Since G fixes y, we have $\pi^{-1}(z) = \{y\}$.

Let F be a prime divisor over Y with center y, and let $h : Y' \to Y$ be a projective birational morphism from a normal variety Y' on which F is a divisor. Then

$$Na(F, Y, C) = N + \mathrm{mult}_F \left(NK_{Y'} - h^*(N(K_Y + C))\right) \in \mathbb{Z},$$

because $NK_{Y'}$ is a divisor with integer coefficients and $h^*(N(K_Y + C))$ is a Cartier divisor near the generic point of F.

Let E be a prime divisor over Z with center z, and let $Z' \to Z$ be a projective birational morphism from a normal variety Z' on which E is a divisor. Let Y'' be the normalization of the irreducible component of $Y \times_Z Z'$ dominating Y. Then $Y'' \to Y$ is projective and birational, and $Y'' \to Z'$ is finite and surjective, so there exists a prime divisor F on Y'' mapping onto E. The center of F on Y is contained in $\pi^{-1}(z) = \{y\}$. By [Liu26, Lemma 2.6(2)], $a(F, Y, C) = |I_G(F)| a(E, Z, \Delta)$. Since $|I_G(F)| \leq M$, $|I_G(F)|$ divides ℓ, and

$$a(E, Z, \Delta) = \frac{Na(F, Y, C)}{N|I_G(F)|} = \frac{Na(F, Y, C) \cdot \ell/|I_G(F)|}{N\ell} \in \frac{1}{N\ell}\mathbb{Z}.$$

Assume that (Y, C) is lc near y. Then $a(F, Y, C) \geq 0$ for any prime divisor F over Y with center y. For any prime divisor E over Z with center z, by the previous paragraph there exists a prime divisor F over Y with center y and $a(E, Z, \Delta) = a(F, Y, C)/|I_G(F)|$, so $a(E, Z, \Delta) \geq 0$. Thus the set $\mathcal{A}$ of the numbers $a(E, Z, \Delta)$, where E runs over the prime divisors over Z with center z, is contained in $(N\ell)^{-1}\mathbb{Z}_{\geq 0}$. The set $\mathcal{A}$ is non-empty: if $\dim Z = 1$, then z is a prime divisor on Z, and if $\dim Z \geq 2$, any irreducible component of the exceptional divisor of the normalized blow-up of Z at z has center z. A non-empty subset of $(N\ell)^{-1}\mathbb{Z}_{\geq 0}$ has a least element.

Thus $\mathrm{mld}((Y \ni y, C)/G) = \mathrm{mld}(Z \ni z, \Delta) = \min \mathcal{A}$ is attained and belongs to $(N\ell)^{-1}\mathbb{Z}_{\geq 0}$. The last sentence follows because the intersection of $(N\ell)^{-1}\mathbb{Z}_{\geq 0}$ with a bounded interval is finite. □

When C has irrational coefficients, no multiple of $K_Y + C$ is Cartier, and the case of bounded groups is treated by Proposition 11.7(1) instead.

The following elementary lemma is used to choose integers coprime to the order of a cyclic group in short intervals.

Lemma 4.9. *There exists a positive integer r_0 such that for any integer $r \geq r_0$, any real number x, and any real number $y \geq \sqrt{r}$, there exists an integer $L \in (x, x+y]$ with $\gcd(L, r) = 1$.*

Proof. For any integer $k \geq 1$,

$$\frac{(k+1)!}{16^{k+1}} = \frac{k+1}{16} \cdot \frac{k!}{16^k},$$

so the sequence $k!/16^k$ is increasing for $k \geq 16$ and tends to $+\infty$. Thus there exists a positive integer K such that $k! > 16^k$ for any integer $k \geq K$. Let $r_0 := 16^K$.

Let $r \geq r_0$ be an integer, x a real number, and $y \geq \sqrt{r}$ a real number. Let $p_1 < \cdots < p_\omega$ be the distinct prime factors of r. We first show that $16^\omega < r$. If $\omega \geq K$, then since $p_k \geq k+1$ for any k,

$$r \geq \prod_{k=1}^{\omega} p_k \geq \omega! > 16^\omega.$$

If $\omega < K$, then $16^\omega < 16^K \leq r$. Thus $16^\omega < r$, that is, $4^\omega < \sqrt{r}$.

For a positive integer d, the number of integers in $(x, x+y]$ divisible by d is

$$\left\lfloor \frac{x+y}{d} \right\rfloor - \left\lfloor \frac{x}{d} \right\rfloor = \frac{y}{d} + \theta_d, \quad |\theta_d| < 1.$$

By the inclusion–exclusion principle, the number of integers $L \in (x, x+y]$ with $\gcd(L, r) = 1$ is

$$\sum_{d \mid p_1 \cdots p_\omega} \mu(d)\Big(\frac{y}{d} + \theta_d\Big) > y \prod_{k=1}^{\omega} \Big(1 - \frac{1}{p_k}\Big) - 2^\omega \geq y 2^{-\omega} - 2^\omega \geq \sqrt{r} 2^{-\omega} - 2^\omega,$$

where μ is the Möbius function and we use that $p_1 \cdots p_\omega$ has 2^ω positive divisors and $1 - \frac{1}{p_k} \geq \frac{1}{2}$. Since $4^\omega < \sqrt{r}$, the right-hand side is positive. Thus there exists an integer $L \in (x, x+y]$ with $\gcd(L, r) = 1$. □

4.3. **Crepant models.** In this subsection, we show that the minimal log discrepancy of a threefold germ $(X \ni x, B)$ is computed on a crepant model of $(X \ni x, B)$, and that the ACC holds for $\mathrm{mld}(X \ni x, B)$ when the crepant model is locally a finite abelian quotient of a smooth threefold with a boundary $B_V \geq 0$ whose coefficients belong to a DCC set (Definition-Proposition 4.10). We also compute the exceptional prime divisors of weighted blow-ups of hypersurfaces (Lemma 4.12).

Definition-Proposition 4.10. Let $\Gamma \subset [0, 1)$ be a DCC set, (X, B) a threefold pair, and $x \in X$ a closed point. A projective birational morphism $f : Y \to X$ from a normal variety Y together with the $\mathbb{R}$-divisor B_Y on Y such that $K_Y + B_Y = f^*(K_X + B)$ is called a *crepant model* of $(X \ni x, B)$. Let (f, B_Y) be a crepant model of $(X \ni x, B)$, and assume the following.

(i) (Y, B_Y) is klt over a neighborhood of x.
(ii) Any closed point of $f^{-1}(x)$ has a Zariski open neighborhood $U \subset Y$ with $U \cong V/H$, where V is a smooth threefold and H is a finite abelian group acting on V, such that the $\mathbb{R}$-divisor B_V on V with $K_V + B_V = \pi^*(K_U + B_Y|_U)$ for the quotient morphism $\pi : V \to U$ satisfies $B_V \geq 0$ and $B_V \in \Gamma$.

Then

$$\mathrm{mld}(X \ni x, B) = \min\left\{\mathrm{mld}(Y \ni \eta, B_Y) \;\middle|\; \eta \in f^{-1}(x)\right\},$$

where η runs over the scheme-theoretic points of $f^{-1}(x)$. Moreover, the set of positive numbers $\mathrm{mld}(X \ni x, B)$, where $(X \ni x, B)$ runs over the germs which have a crepant model satisfying (i) and (ii) for the fixed Γ, satisfies the ACC.

Proof. **Step 1**. In this step, we show that the coefficients of B_Y near $f^{-1}(x)$ belong to the DCC set $\Psi(\Gamma \cup \{0\}) \subset [0, 1)$.

Let $y \in f^{-1}(x)$ be a closed point, and let $U \cong V/H$, $\pi : V \to U$, and B_V be as in (ii). Let $H_0 \subset H$ be the kernel of the action of H on V. The group H/H_0 acts faithfully on V, and the H/H_0-orbits on V are the H-orbits, hence π is also the quotient morphism by H/H_0. Possibly replacing H with H/H_0, we may assume that H acts faithfully on V. Since $K_V = \pi^* K_U + R_\pi$ for the ramification divisor R_π of π, which is H-invariant, $B_V = \pi^*(B_Y|_U) - R_\pi$ is H-invariant.

Let P be a prime divisor on U, P_V a prime divisor on V with $\pi(P_V) = P$, and e_P the ramification index of π along P_V. Since $(V, B_V)/H = (U, B_Y|_U)$ by (ii) and [Liu26, Lemma 2.6(0)], we have, by [Liu26, Lemma 2.6(4)],

$$\mathrm{mult}_P(B_Y|_U) = 1 - \frac{1 - \mathrm{mult}_{P_V} B_V}{e_P}.$$

Since $B_V \geq 0$ and $B_V \in \Gamma$, we have $\mathrm{mult}_{P_V} B_V \in \Gamma \cup \{0\}$. Thus $\mathrm{mult}_P(B_Y|_U) \in \Psi(\Gamma \cup \{0\})$.

Let $Y^\circ \subset Y$ be the union of the open subsets U of (ii) for all closed points of $f^{-1}(x)$. Any point of $f^{-1}(x)$ specializes to a closed point of $f^{-1}(x)$, hence $f^{-1}(x) \subset Y^\circ$. Thus $B_Y|_{Y^\circ} \in \Psi(\Gamma \cup \{0\})$. Since $\Gamma \cup \{0\} \subset [0, 1)$ is a DCC set, $\Psi(\Gamma \cup \{0\})$ is a DCC set [Liu26, Lemma 2.10(1)], and $\Psi(\Gamma \cup \{0\}) \subset [0, 1)$.

Step 2. In this step, we prove the formula for $\mathrm{mld}(X \ni x, B)$.

Let $X_0 \subset X$ be an open neighborhood of x such that (Y, B_Y) is klt over X_0. For any prime divisor F over X with $\mathrm{center}_X F \cap X_0 \neq \emptyset$, we have $a(F, X, B) = a(F, Y, B_Y) > 0$, since $K_Y + B_Y = f^*(K_X + B)$. Thus (X, B) is klt near x. Let $\mathfrak{m}_x$ be the maximal ideal of $\mathcal{O}_{X,x}$. By [Liu26, Lemma 3.5], applied to a log resolution of (X, B) which principalizes $\mathfrak{m}_x$, there exists a prime divisor F_0 over X such that $\mathrm{center}_X F_0 = \{x\}$ and $a(F_0, X, B) = \mathrm{mld}(X \ni x, B)$.

Let $\eta \in f^{-1}(x)$, and let F be a prime divisor over Y with $\mathrm{center}_Y F = \overline{\{\eta\}}$. Since f is proper and $f(\eta) = x$, we have $f(\overline{\{\eta\}}) = \{x\}$, hence $\mathrm{center}_X F = \{x\}$ and

$$a(F, Y, B_Y) = a(F, X, B) \geq \mathrm{mld}(X \ni x, B).$$

Thus $\mathrm{mld}(Y \ni \eta, B_Y) \geq \mathrm{mld}(X \ni x, B)$ for any $\eta \in f^{-1}(x)$. Conversely, since f is proper, F_0 has a center on Y. Let η_0 be the generic point of $\mathrm{center}_Y F_0$. Then $f(\eta_0) = x$, and

$$\mathrm{mld}(Y \ni \eta_0, B_Y) \leq a(F_0, Y, B_Y) = a(F_0, X, B) = \mathrm{mld}(X \ni x, B).$$

Therefore the minimum over $f^{-1}(x)$ exists, it is attained at η_0, and it is equal to $\mathrm{mld}(X \ni x, B)$.

Step 3. In this step, we prove the ACC.

Suppose that the ACC fails. Then there exist threefold pairs (X_i, B_i), closed points $x_i \in X_i$, and crepant models $(f_i : Y_i \to X_i, B_{Y_i})$ of $(X_i \ni x_i, B_i)$ satisfying (i) and (ii) for the fixed Γ, such that the numbers $a_i := \mathrm{mld}(X_i \ni x_i, B_i)$ satisfy $0 < a_1 < a_2 < \cdots$. By **Step 2**, there exist points $\eta_i \in f_i^{-1}(x_i)$ such that

$$a_i = \mathrm{mld}(Y_i \ni \eta_i, B_{Y_i}).$$

Since f_i is birational, $\dim \overline{\{\eta_i\}} \leq 2$. Possibly passing to a subsequence, we may assume that $\dim \overline{\{\eta_i\}} = d$ for any i, where $d \in \{0, 1, 2\}$. Let $Y_i^\circ \subset Y_i$ be the open subset constructed in **Step 1** for (f_i, B_{Y_i}). Then $\eta_i \in Y_i^\circ$ and $B_{Y_i}|_{Y_i^\circ} \in \Psi(\Gamma \cup \{0\})$.

Case 1. $d = 2$. Then $E_i := \overline{\{\eta_i\}}$ is a prime divisor on Y_i. Since Y_i is normal, E_i is the only prime divisor over Y_i with center E_i. Thus

$$a_i = a(E_i, Y_i, B_{Y_i}) = 1 - \mathrm{mult}_{E_i} B_{Y_i},$$

and $\mathrm{mult}_{E_i} B_{Y_i} \in \Psi(\Gamma \cup \{0\})$ since $\eta_i \in Y_i^\circ$. Since $\Psi(\Gamma \cup \{0\})$ is a DCC set, the set $\{1 - c \mid c \in \Psi(\Gamma \cup \{0\})\}$ satisfies the ACC. This contradicts the assumption that a_i is strictly increasing.

Case 2. $d = 1$. Then η_i is the generic point of an integral curve on Y_i°. The $\mathbb{R}$-divisor $K_{Y_i^\circ} + B_{Y_i}|_{Y_i^\circ}$ is $\mathbb{R}$-Cartier, since it is the pullback of $K_{X_i} + B_i$, and $B_{Y_i}|_{Y_i^\circ} \geq 0$ by **Step 1**. Thus $(Y_i^\circ, B_{Y_i}|_{Y_i^\circ})$ is a threefold pair with $B_{Y_i}|_{Y_i^\circ} \in \Psi(\Gamma \cup \{0\})$, and

$$a_i = \mathrm{mld}(Y_i^\circ \ni \eta_i, B_{Y_i}|_{Y_i^\circ}) > 0.$$

By [Liu26, Lemma 3.7], applied to the DCC set $\Psi(\Gamma \cup \{0\}) \subset [0, 1]$, the numbers a_i satisfy the ACC, a contradiction.

Case 3. $d = 0$. Fix i and drop the index i. Then $y := \eta$ is a closed point of $f^{-1}(x)$. Let $U \cong V/H$, $\pi : V \to U$, and B_V be as in **Step 1**, where H acts faithfully on V and B_V is H-invariant. Let $v \in V$ be a point with $\pi(v) = y$, and $H_v \subset H$ the stabilizer of v. An element of H_v which acts trivially on a neighborhood of v acts trivially on the function field of V, hence on V. Thus H_v acts faithfully on the germ $(V \ni v)$.

We show that $\mathrm{mld}(Y \ni y, B_Y) = \mathrm{mld}((V \ni v, B_V)/H_v)$. Let F be a prime divisor over V with $\mathrm{center}_V F = \{v\}$, and E the prime divisor over U induced by F [Liu26, Definition 2.7]. Then $\mathrm{center}_U E = \{y\}$, and

$$a(F, V, B_V) = |I_H(F)| \cdot a(E, U, B_Y|_U) \tag{4.4}$$

by [Liu26, Lemma 2.6(2)]. Conversely, let E be a prime divisor over U with $\mathrm{center}_U E = \{y\}$, and F a prime divisor over V inducing E. Then $\mathrm{center}_V F$ is a point of $\pi^{-1}(y) = Hv$. Since B_V is H-invariant, after replacing F with $h(F)$ for some $h \in H$, we may assume that $\mathrm{center}_V F = \{v\}$. Any element of $I_H(F)$ preserves $\mathrm{center}_V F$, hence $I_H(F) \subset H_v$ and $I_H(F) = I_{H_v}(F)$. Therefore,

$$\mathrm{mld}(Y \ni y, B_Y) = \mathrm{mld}(U \ni y, B_Y|_U) = \inf_F \frac{a(F, V, B_V)}{|I_{H_v}(F)|} = \mathrm{mld}((V \ni v, B_V)/H_v),$$

where F runs over the prime divisors over V with $\mathrm{center}_V F = \{v\}$, and the last equality follows from [Liu26, Lemma 2.6].

The pair $(U, B_Y|_U)$ is klt near y by (i). Thus (V, B_V) is klt near v, since (4.4) holds for any prime divisor F over V and the prime divisor E over U induced by F [Liu26, Lemma 2.6(2)]. Moreover, $B_V \geq 0$, $B_V \in \Gamma$, B_V is H_v-invariant, and H_v is a finite abelian group which acts faithfully on the smooth threefold germ $(V \ni v)$ and fixes v. By the last sentence of [Liu26, Theorem 1.5], the positive numbers $\mathrm{mld}((V \ni v, B_V)/H_v)$ for all such data satisfy the ACC. Since each a_i is such a number, this contradicts the assumption that a_i is strictly increasing. □

Remark 4.11. Without the conditions $B_V \geq 0$ and $B_V \in \Gamma$ in Definition-Proposition 4.10(ii), the coefficients of a crepant boundary need not belong to a DCC set. For example, for odd $r \geq 3$, the crepant boundary on a terminal crepant model of $(\mathbb{A}^2/\mu_r(1, r-2)) \times \mathbb{A}^1$ has a component with coefficient $\frac{1}{r}$ [Liu26, Section 1].

The crepant models in this paper are constructed from weighted blow-ups. For a primitive vector w, Lemma 4.12 computes the exceptional prime divisor of the strict transform of a hypersurface in the w-weighted blow-up, together with the log discrepancy and the multiplicities of functions along it.

Lemma 4.12. *Let $k \supset \mathbb{C}$ be a field, $n \geq 3$ an integer, $x_1, \dots, x_n$ the coordinates of $\mathbb{A}^n_k$ (we write x, y, z, t if $k = \mathbb{C}$ and $n = 4$), $W = (f = 0) \subset \mathbb{A}^n_k$ a hypersurface through 0 which is normal near 0, and $w \in \mathbb{Z}^n_{>0}$ a primitive vector. Let $f_w := \mathrm{in}_w(f)$. Assume that $(f_w = 0) \subset \mathbb{A}^n_k$ is integral and not contained in any coordinate hyperplane. Then the strict transform of W in the w-weighted blow-up of $\mathbb{A}^n_k$ intersects the exceptional divisor in a single prime divisor E_W. For any $H \in k[x_1, \dots, x_n]$,*

$$\mathrm{mult}_{E_W}(H|_W) \geq \mathrm{wt}_w(H),$$

with equality if f_w does not divide $\mathrm{in}_w(H)$. Moreover,

$$a(E_W, W, 0) = \sum_{i=1}^{n} w_i - \mathrm{wt}_w(f).$$

The lemma also holds for $k = \mathbb{C}$ and a hypersurface $W = (f = 0)$ through o, normal near o, in a smooth germ $(\mathcal{X} \ni o)$ with regular parameters $x_1, \dots, x_n$ at o and $f, H \in \mathcal{O}_{\mathcal{X},o}$, where wt_w and in_w are taken on the expansions in $\mathbb{C}[[x_1, \dots, x_n]]$ and the w-weighted blow-up of $\mathcal{X}$ is $\operatorname{Proj} \bigoplus_{j\geq 0} I_j$ for the ideals $I_j \subset \mathcal{O}_{\mathcal{X}}$ generated by the monomials in $x_1, \dots, x_n$ of w-weight at least j.

Proof. Let $g : Y \to \mathbb{A}^n_k$ be the w-weighted blow-up, E the exceptional divisor of g, and W_Y the strict transform of W. Since $g(E) = \{0\}$, replacing W with a normal open neighborhood of 0 in W changes neither W_Y near E nor E_W, $\mathrm{mult}_{E_W}(H|_W)$, and $a(E_W, W, 0)$. Thus only the normality of W near 0 is used below. Write $|w| := \sum_{j=1}^{n} w_j$. We use the following description of g [CLS11, Proposition 1.3.18], [Kaw05, the paragraph after Lemma 6.1]. The divisor E is isomorphic to $\mathbb{P}(w)$. For $1 \leq i \leq n$, let $u_1, \dots, u_n$ be coordinates on a copy of $\mathbb{A}^n_k$, on which $\zeta \in \mu_{w_i}$ acts by $u_i \to \zeta u_i$ and $u_j \to \zeta^{-w_j} u_j$ for $j \neq i$. Then there exists a morphism $q_i : \mathbb{A}^n_k \to Y$, which is the quotient morphism by μ_{w_i} onto an open subset $Y_i \subset Y$, such that $g \circ q_i$ is given by

$$x_i = u_i^{w_i}, \quad x_j = u_j u_i^{w_j} \quad (j \neq i),$$

$q_i^{-1}(E) = (u_i = 0)$, and the restriction of q_i to $(u_i = 0)$ is the quotient morphism onto the open subset $(x_i \neq 0)$ of $E \cong \mathbb{P}(w)$ given by $u \to [u_1 : \cdots : u_{i-1} : 1 : u_{i+1} : \cdots : u_n]$. Moreover, $Y = \bigcup_{i=1}^{n} Y_i$. For $\phi \in k[x_1, \dots, x_n]$, we write

$$(g \circ q_i)^* \phi = u_i^{\mathrm{wt}_w(\phi)} \phi_i, \quad \phi_i \in k[u_1, \dots, u_n].$$

Then

$$\phi_i|_{(u_i=0)} = \mathrm{in}_w(\phi)(u_1, \dots, u_{i-1}, 1, u_{i+1}, \dots, u_n).$$

Since w is primitive, μ_{w_i} acts freely on

$$O_i := \{u \in \mathbb{A}^n_k \mid u_j \neq 0 \text{ for any } j \neq i\}$$

and on $(u_i \neq 0)$. Thus q_i is étale on $O_i \cup (u_i \neq 0)$.

First we show that $E \cap W_Y$ is a single prime divisor. Let $\bar{f}_i := f_i|_{(u_i=0)}$, which is a non-zero polynomial in the u_j, $j \neq i$. Since u_i does not divide f_i, no irreducible component of $(f_i = 0)$ is contained in $(u_i = 0)$. Hence $q_i^{-1}(W_Y) = (f_i = 0)$ as sets, and

$$E \cap W_Y \cap Y_i = q_i\big((u_i = \bar{f}_i = 0)\big) = (f_w = 0) \cap (x_i \neq 0) \subset \mathbb{P}(w).$$

Thus $E \cap W_Y = (f_w = 0) \subset \mathbb{P}(w)$ as sets. Since $(f_w = 0) \subset \mathbb{A}^n_k$ is integral, f_w is irreducible and $(f_w = 0) \subset \mathbb{P}(w)$ is integral of dimension $n - 2$. Therefore $E \cap W_Y$ is a single prime divisor E_W on W_Y.

Let ξ be the generic point of E_W. Since $(f_w = 0)$ is not contained in any coordinate hyperplane, $\xi \in q_i(O_i)$ for any i. Fix i, let $\xi' \in O_i$ be a point with $q_i(\xi') = \xi$, $\widetilde{W} := (f_i = 0) \cap O_i$, and D' the closure of ξ' in $\widetilde{W}$. On $(u_i \neq 0)$, we have $(g \circ q_i)^* f_w = u_i^{\mathrm{wt}_w(f)} \bar{f}_i$, the morphism q_i is étale, and $(f_w = 0)$ is reduced. Thus $\bar{f}_i$ has no multiple factor, and

$$\widetilde{W} \cap (u_i = 0) = (u_i = \bar{f}_i = 0) \cap O_i$$

is reduced. Hence $\mathcal{O}_{\widetilde{W},\xi'}/(u_i)$ is a field, $\mathcal{O}_{\widetilde{W},\xi'}$ is a discrete valuation ring, and u_i is a uniformizer of $\mathcal{O}_{\widetilde{W},\xi'}$. Since q_i is étale at ξ', for any $\phi \in k[x_1, \dots, x_n]$,

$$\mathrm{mult}_{E_W}(\phi|_W) = \mathrm{mult}_{D'}\left((g \circ q_i)^* \phi|_{\widetilde{W}}\right) = \mathrm{wt}_w(\phi) + \mathrm{mult}_{D'}(\phi_i|_{\widetilde{W}}).$$

In particular, $\mathrm{mult}_{E_W}(H|_W) \geq \mathrm{wt}_w(H)$. Suppose that $\mathrm{mult}_{D'}(H_i|_{\widetilde{W}}) > 0$. Then the polynomial

$$h := H_i|_{(u_i=0)} = \mathrm{in}_w(H)(u_1, \dots, 1, \dots, u_n)$$

vanishes on D'. Let D_0 be the closure of the image of D' under the projection $\mathbb{A}^n_k \to \mathbb{A}^{n-1}_k$ forgetting u_i. Since h and $\bar{f}_i$ do not depend on u_i, $D_0 \times \mathbb{A}^1$ is an irreducible component of $(\bar{f}_i = 0)$, and the pullback

$$(g \circ q_i)^* \operatorname{in}_w(H) = u_i^{\operatorname{wt}_w(H)} h$$

vanishes on $D_0 \times (\mathbb{A}^1 \backslash \{0\})$. The restriction of q_i to $(u_i \neq 0)$ is a finite étale morphism onto $(x_i \neq 0) \subset \mathbb{A}^n_k$, so it maps $D_0 \times (\mathbb{A}^1 \backslash \{0\})$ onto a closed subset of $(f_w = 0) \cap (x_i \neq 0)$ of dimension $n-1$, which is dense in the integral hypersurface $(f_w = 0)$. Thus $\operatorname{in}_w(H)$ vanishes on $(f_w = 0)$, and f_w divides $\operatorname{in}_w(H)$. This proves the equality when f_w does not divide $\operatorname{in}_w(H)$.

Finally, we compute $a(E_W, W, 0)$. Let σ be the residue of $dx_1 \wedge \cdots \wedge dx_n / f$ along W, and $\tilde{\sigma}$ the residue of $du_1 \wedge \cdots \wedge du_n / f_i$ along $\widetilde{W}$. By adjunction [KM98, Proposition 5.73], σ generates the dualizing sheaf of W, and $\tilde{\sigma}$ generates the dualizing sheaf of $\widetilde{W}$. A direct calculation shows

$$(g \circ q_i)^*(dx_1 \wedge \cdots \wedge dx_n) = w_i u_i^{|w|-1} du_1 \wedge \cdots \wedge du_n$$

(cf. [Rei87, (4.8)], [Kaw05, Lemma 6.1]), hence

$$(g \circ q_i)^* \frac{dx_1 \wedge \cdots \wedge dx_n}{f} = w_i u_i^{|w|-1-\operatorname{wt}_w(f)} \frac{du_1 \wedge \cdots \wedge du_n}{f_i},$$

and

$$(g \circ q_i)^* \sigma = w_i u_i^{|w|-1-\operatorname{wt}_w(f)} \tilde{\sigma}$$

as rational forms on $\widetilde{W}$. Since q_i is étale at ξ' and u_i is a uniformizer of $\mathcal{O}_{\widetilde{W},\xi'}$, the coefficient of E_W in $K_{W_Y} - (g|_{W_Y})^* K_W$ is $|w| - 1 - \operatorname{wt}_w(f)$. Thus

$$a(E_W, W, 0) = |w| - \operatorname{wt}_w(f) = \sum_{i=1}^{n} w_i - \operatorname{wt}_w(f).$$

Now let $(\mathcal{X} \ni o)$, $W = (f = 0)$, H, and I_j be as in the last sentence of the lemma. Possibly shrinking $\mathcal{X}$ to a neighborhood of o, we may assume that $f, H \in \mathcal{O}(\mathcal{X})$ and that $(x_1, \ldots, x_n) : \mathcal{X} \to \mathbb{A}^n$ is étale. Then I_j is the pullback of the ideal of $\mathbb{C}[x_1, \ldots, x_n]$ generated by the monomials of w-weight at least j. Thus the w-weighted blow-up $g_{\mathcal{X}} : Y_{\mathcal{X}} \to \mathcal{X}$ is the base change of g, and the exceptional divisor of $g_{\mathcal{X}}$ is $\mathbb{P}(w)$. Moreover, possibly shrinking $\mathcal{X}$ further, $f \in I_{\operatorname{wt}_w(f)}$ and $H \in I_{\operatorname{wt}_w(H)}$, with images f_w in $I_{\operatorname{wt}_w(f)}/I_{\operatorname{wt}_w(f)+1}$ and $\operatorname{in}_w(H)$ in $I_{\operatorname{wt}_w(H)}/I_{\operatorname{wt}_w(H)+1}$, since the expansions of $f - f_w$ and $H - \operatorname{in}_w(H)$ in $\mathbb{C}[[x_1, \ldots, x_n]]$ have only monomials of larger w-weight and $\widehat{\mathcal{O}}_{\mathcal{X},o}$ is faithfully flat over $\mathcal{O}_{\mathcal{X},o}$ [Mat89, Theorems 8.14 and 7.5(ii)]. Hence, on the base change of each Y_i and of q_i, there exist regular functions f_i and H_i such that

$$(g_{\mathcal{X}} \circ q_i)^* f = u_i^{\operatorname{wt}_w(f)} f_i, \quad (g_{\mathcal{X}} \circ q_i)^* H = u_i^{\operatorname{wt}_w(H)} H_i,$$

$$f_i|_{(u_i=0)} = f_w(u_1, \ldots, u_{i-1}, 1, u_{i+1}, \ldots, u_n), \quad H_i|_{(u_i=0)} = \operatorname{in}_w(H)(u_1, \ldots, u_{i-1}, 1, u_{i+1}, \ldots, u_n).$$

The computations of E_W, $\operatorname{mult}_{E_W}(H|_W)$, and $a(E_W, W, 0)$ in the polynomial case use only the description of q_i, the irreducibility of f_w, and the formulas for $(g \circ q_i)^* f$, $(g \circ q_i)^* H$, $f_i|_{(u_i=0)}$, and $H_i|_{(u_i=0)}$, so they hold without change. □

4.4. **Numerical admission and threshold cutoff.** In this subsection, we prove two tools used in the later sections: a numerical admission of a germ with several marks into a germ given by polynomials (Construction-Theorem 4.13), and a threshold cutoff on a fixed smooth germ (Lemma 4.14).

The later families are treated on germs given by polynomial equations in normal form. Construction-Theorem 4.13 replaces a germ with marks by such a polynomial germ simultaneously for finitely many label vectors and finitely many prime divisors, and keeps the quotient minimal log discrepancies of any subgroup. The new pairs are only klt and need not be terminal, so a prime divisor whose log discrepancy is used later to test terminality has to be included among the prime divisors $P_1, \ldots, P_l$ of Construction-Theorem 4.13.

Construction-Theorem 4.13. Let G be a finite abelian group and $n \in \{3,4\}$. Assume that

(i) $(W \ni o)$ is a smooth threefold germ or an isolated Gorenstein terminal threefold germ, and G acts faithfully on W and fixes o;
(ii) W has a G-equivariant closed embedding into a smooth germ $(Y \ni o)$ of dimension n, with $Y = W$ if $n = 3$ and $W = (f = 0)$ for a regular G-semi-invariant function f on Y if $n = 4$, and $x_1, \dots, x_n$ are G-eigenparameters of Y at o;
(iii) $h_1, \dots, h_s$ are regular G-semi-invariant functions on Y whose restrictions to W are marks, $c_1, \dots, c_s$ are positive real numbers, and (W, D) is klt near o for $D := \sum_k c_k \operatorname{div}(h_k|_W)$;
(iv) $c^{(1)}, \dots, c^{(q)} \in \mathbb{R}^s_{\geq 0}$ are finitely many vectors such that $(W, \sum_k c_k^{(j)} \operatorname{div}(h_k|_W))$ is klt near o for any j, and $c^{(0)} := (c_1, \dots, c_s)$;
(v) $P_1, \dots, P_l$ are finitely many prime divisors over W with center o;
(vi) ψ is a G-equivariant automorphism of $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_n]]$, v is a G-invariant unit of $\mathfrak{R}$, and $u_1, \dots, u_s$ are G-invariant units of $\mathfrak{R}$.

Let $\hat{h}_k$, and $\hat{f}$ if $n = 4$, be the expansions of h_k and f in $\mathfrak{R}$. Then there exists a positive integer N_1 such that for any $N \geq N_1$ there exists L_1 such that for any $L \geq L_1$ the following holds. Let $h_{k,0}$ be G-semi-invariant polynomials with the characters of h_k, and, if $n = 4$, f_0 a G-semi-invariant polynomial with the character of f, such that

$$h_{k,0} \equiv u_k \psi(\hat{h}_k) \pmod{(x_1, \dots, x_n)^N} \quad \text{and} \quad f_0 \equiv v\psi(\hat{f}) \pmod{(x_1, \dots, x_n)^L}.$$

Let $W_0 := \mathbb{A}^3$ if $n = 3$ and $W_0 := (f_0 = 0) \subset \mathbb{A}^4$ if $n = 4$, with the diagonal action of G in which the coordinates have the characters of the x_i. Then

(1) G acts faithfully on W_0, and for any $j \in \{0, \dots, q\}$ the pair

$$\Big(W_0, \sum_k c_k^{(j)} \operatorname{div}(h_{k,0}|_{W_0})\Big)$$

is klt near 0 and, for any subgroup $G' \subset G$,

$$\operatorname{mld}\Big(\big(W_0 \ni 0, \textstyle\sum_k c_k^{(j)} \operatorname{div}(h_{k,0}|_{W_0})\big)/G'\Big) = \operatorname{mld}\Big(\big(W \ni o, \textstyle\sum_k c_k^{(j)} \operatorname{div}(h_k|_W)\big)/G'\Big);$$

(2) there is a G-equivariant isomorphism between the completions of the local rings of W at o and of W_0 at 0 under which each P_i corresponds to a prime divisor $P_{i,0}$ over W_0 with center 0 having isomorphic residue field, equal log discrepancy $a(\cdot, \cdot, 0)$, equal orders of h_k and $h_{k,0}$ for any k, and equal decomposition and inertia groups in G;
(3) if $n = 4$, then W_0 is an isolated Gorenstein terminal germ at 0, and, as in (2), the completions of the local rings of W at o and of W_0 at 0 are G-equivariantly isomorphic.

We call $(W_0, (h_{k,0})_k)$ a *numerical admission* of $(W, (h_k)_k)$ with *admitted marks* $h_{k,0}$.

Proof. For any $j \in \{0, \dots, q\}$, let

$$S_j := \{k \in \{1, \dots, s\} \mid c_k^{(j)} > 0\}.$$

For each j, we use Theorem 3.10 with the entries $c_k^{(j)}$, $k \in S_j$, as the labels of [Liu26, Construction-Theorem 7.2(iii)], since the labels there have to be positive and the equality of quotient minimal log discrepancies in [Liu26, Construction-Theorem 7.2(1)] involves only the labels $c_k^{(j)}$ with $k \in S_j$. For an additional vector of [Liu26, Construction-Theorem 7.2(iv)], conclusion (3) there gives only the klt property. By (i), (ii), (iii), (iv), and (v), the germ $(W \ni o)$, the group G, the embedding into Y, the function f if $n = 4$, the parameters $x_1, \dots, x_n$, the functions h_k with $k \in S_j$, the labels $c_k^{(j)}$ with $k \in S_j$, and the prime divisors $P_1, \dots, P_l$ satisfy the conditions of [Liu26, Construction-Theorem 7.2] with $H = G$ and with no additional vector in [Liu26, Construction-Theorem 7.2(iv)]. If $S_j = \emptyset$, then no function h_k is used and the boundary is zero, which is allowed there. By Theorem 3.10, applied to $(W \ni o)$, G, Y, f if $n = 4$, $x_1, \dots, x_n$, the functions h_k and the labels $c_k^{(j)}$ with $k \in S_j$, the prime divisors $P_1, \dots, P_l$, the automorphism

ψ, the unit v, and the units u_k with $k \in S_j$, there exist a positive integer $N_1^{(j)}$, and for any $N \geq N_1^{(j)}$ an integer $L_1^{(j)}(N)$, such that the conclusions of Theorem 3.10 hold for any $N \geq N_1^{(j)}$, any $L \geq L_1^{(j)}(N)$, and any polynomials $h_{k,0}$ with $k \in S_j$ and f_0 satisfying the congruences there.

Let

$$N_1 := \max_{0 \leq j \leq q} N_1^{(j)},$$

and for $N \geq N_1$ let $L_1 := \max_{0 \leq j \leq q} L_1^{(j)}(N)$. Let $L \geq L_1$, and let $h_{k,0}$ and f_0 be polynomials as in the statement. For any j, the polynomials $h_{k,0}$ with $k \in S_j$ and f_0 satisfy the congruences of the j-th application of Theorem 3.10, and the polynomial germ of that application is W_0 for any j, since f_0 does not depend on j. Thus for any j, the conclusions of Theorem 3.10 hold for W_0 and

$$B_0^{(j)} := \sum_{k \in S_j} c_k^{(j)} \operatorname{div}(h_{k,0}|_{W_0}) = \sum_{k=1}^{s} c_k^{(j)} \operatorname{div}(h_{k,0}|_{W_0}).$$

Since $c_k > 0$ for any k, we have $S_0 = \{1, \dots, s\}$. In particular, the application with $j = 0$ shows that $h_{k,0}|_{W_0}$ is non-zero for any k.

(1) For any j, conclusion (1) of [Liu26, Construction-Theorem 7.2], which holds for W_0 and $B_0^{(j)}$ by the j-th application of Theorem 3.10, shows that G acts faithfully on W_0, that $(W_0, B_0^{(j)})$ is klt near 0, and that

$$\operatorname{mld}\big((W_0 \ni 0, B_0^{(j)})/G'\big) = \operatorname{mld}\big((W \ni o, \textstyle\sum_{k \in S_j} c_k^{(j)} \operatorname{div}(h_k|_W)\big)/G'\big)$$

for any subgroup $G' \subset G$. Since $c_k^{(j)} = 0$ for $k \notin S_j$, we obtain (1).

(2) Since $S_0 = \{1, \dots, s\}$, conclusion (2) of [Liu26, Construction-Theorem 7.2], which holds by the application of Theorem 3.10 with $j = 0$, together with the last clause of Theorem 3.10, gives the correspondence of (2) through the G-equivariant isomorphism of completions that it constructs. For $n = 3$ both completions are $\mathbb{C}[[x_1, x_2, x_3]]$, and the isomorphism is induced by ψ.

(3) If $n = 4$, then the application with $j = 0$ shows that W_0 is an isolated Gorenstein terminal germ at 0, and the G-equivariant isomorphism of completions is the one of (2). □

The normal forms used in the later sections include those of Propositions 3.15, 3.17, and 7.31. When a normal form is connected to the germ by G-equivariant morphisms that are étale at a point, as Proposition 3.15 provides, the automorphism ψ and the unit v of Construction-Theorem 4.13(vi) are given by Lemma 3.8, applied to the induced G-equivariant isomorphism of completed local rings.

In the applications, the equation of a hypersurface germ enters as a factor with exponent one on a fixed smooth ambient germ, together with the marks and their labels. Lemma 4.14 shows that the lc property is then decided by a jet whose order depends only on the dimension, the number of factors, and the set of labels.

Lemma 4.14. *Let n and s be positive integers and $\Lambda \subset \mathbb{R}_{>0}$ a DCC set. Then there exists a positive integer L, depending only on n, s, and Λ, such that the following holds. Let $k \supset \mathbb{C}$ be an algebraically closed field, and $f_1, \dots, f_s$ and $f_1', \dots, f_s'$ non-zero elements of $k[[x_1, \dots, x_n]]$ with $f_j - f_j' \in (x_1, \dots, x_n)^L$, and $c_1, \dots, c_s \in \Lambda$. If $(\operatorname{Spec} k[[x_1, \dots, x_n]], \prod_j (f_j)^{c_j})$ is lc, then so is $(\operatorname{Spec} k[[x_1, \dots, x_n]], \prod_j (f_j')^{c_j})$.*

Proof. Let $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_n]]$. Pairs on $\operatorname{Spec} \mathfrak{R}$ and the lc property are taken in the sense of [dFEM11, Section 2], and lc thresholds on $\operatorname{Spec} \mathfrak{R}$ are defined from them as in Definition 2.11. Pairs, the lc property, and lc thresholds on $\operatorname{Spec} k[[x_1, \dots, x_n]]$ for a field $k \supset \mathbb{C}$ are defined in this way as well. We may assume that $\Lambda \neq \emptyset$. Since Λ is a DCC set, $\lambda_0 := \min \Lambda$ exists, and $\lambda_0 > 0$. Let

$$T := \Big\{ \operatorname{lct}\Big(\operatorname{Spec} \mathfrak{R}; \prod_{j=1}^{r} (g_j)^{b_j}\Big) \;\Big|\; r \in \mathbb{Z}_{\geq 1},\ g_j \in (x_1, \dots, x_n) \backslash \{0\},\ b_j \in \Lambda \Big\}.$$

Step 1. *T satisfies the ACC.* Suppose that $t_1 < t_2 < \cdots$ is a strictly increasing sequence in T. For any i, write

$$t_i = \operatorname{lct}\Big(\operatorname{Spec}\mathfrak{R}; \prod_{j=1}^{r_i}(g_{ij})^{b_{ij}}\Big)$$

with $g_{ij} \in (x_1, \dots, x_n)\backslash\{0\}$ and $b_{ij} \in \Lambda$. Since $\prod_j (g_{ij})^{b_{ij}}$ is a non-zero proper $\mathbb{R}$-ideal, $0 < t_i < +\infty$. Let $t_0 := 0$. For any $i \geq 1$, choose a real number $\epsilon_i > 0$ such that

$$3\epsilon_i < \min\{t_i - t_{i-1}, t_{i+1} - t_i\},$$

and an integer d_i such that $d_i \geq n/(\epsilon_i \lambda_0)$ and $g_{ij} \notin (x_1, \dots, x_n)^{d_i}$ for any j. Let g'_{ij} be the sum of the terms of g_{ij} of degree less than d_i. Then g'_{ij} is a non-zero polynomial in $(x_1, \dots, x_n)$, and $g_{ij} - g'_{ij} \in (x_1, \dots, x_n)^{d_i}$. Let

$$t'_i := \operatorname{lct}\Big(\operatorname{Spec}\mathfrak{R}; \prod_{j=1}^{r_i}(g'_{ij})^{b_{ij}}\Big).$$

Since

$$(g_{ij}) + (x_1, \dots, x_n)^{d_i} = (g'_{ij}) + (x_1, \dots, x_n)^{d_i}$$

for any j and $d_i \geq n/(\epsilon_i \min_j b_{ij})$, [dFEM11, Proposition 2.19(ii)], applied to the completion of the smooth germ $(\mathbb{A}^n \ni 0)$, implies that $|t'_i - t_i| \leq \epsilon_i$. Thus

$$t'_i \leq t_i + \epsilon_i < t_{i+1} - \epsilon_{i+1} \leq t'_{i+1}$$

for any $i \geq 1$, because $\epsilon_i + \epsilon_{i+1} < t_{i+1} - t_i$. Hence $t'_1 < t'_2 < \cdots$ is strictly increasing. Since the lc threshold of a pair on an algebraic germ coincides with the lc threshold of the pullback of the pair to the completion [dFEM11, Proposition 2.11], t'_i is the lc threshold at 0 of the $\mathbb{R}$-ideal $\prod_j (g'_{ij}\mathcal{O}_{\mathbb{A}^n})^{b_{ij}}$ on $\mathbb{A}^n$, whose exponents belong to Λ. The lc threshold at 0 is the lc threshold on a sufficiently small open neighborhood of 0 in $\mathbb{A}^n$, and such neighborhoods are smooth varieties of dimension n. By [dFEM11, Theorem 1.1], applied with the DCC set Λ of exponents, the lc thresholds at 0 of $\mathbb{R}$-ideals on $\mathbb{A}^n$ with exponents in Λ satisfy the ACC. This contradicts the strict increase of t'_i.

Step 2. *A gap below one.* By **Step 1**, T contains no strictly increasing sequence converging to 1. Hence there exists a real number $\delta \in (0, 1)$ such that

$$T \cap (1 - \delta, 1) = \emptyset.$$

Step 3. *The integer L.* Let $\epsilon := \delta/2$, and let L be a positive integer such that $L \geq n/(\epsilon\lambda_0)$. Since δ and λ_0 depend only on n and Λ, so does L. In particular, L depends only on n, s, and Λ. Let k, f_j, f'_j, and c_j be as in the statement, and let $\mathfrak{R}_k := k[[x_1, \dots, x_n]]$.

The gap of **Step 2** holds over k:

$$T_k \cap (1 - \delta, 1) = \emptyset,$$

where T_k is defined as T with $\mathfrak{R}_k$ in place of $\mathfrak{R}$. Suppose that $t = \operatorname{lct}(\operatorname{Spec}\mathfrak{R}_k; \prod_{j=1}^{r}(g_j)^{b_j}) \in T_k$ and $t \in (1-\delta, 1)$, and let $\epsilon' > 0$ with $\epsilon' < \min\{t-1+\delta, 1-t\}$. As in **Step 1**, there exist polynomial truncations $g'_j \in (x_1, \dots, x_n)\backslash\{0\}$ of the g_j such that, by [dFEM11, Proposition 2.19(ii)],

$$\Big|\operatorname{lct}\Big(\operatorname{Spec}\mathfrak{R}_k; \prod_{j=1}^{r}(g'_j)^{b_j}\Big) - t\Big| \leq \epsilon'.$$

The coefficients of the g'_j generate a subfield $K_0 \subset k$ finitely generated over $\mathbb{Q}$, and the algebraic closure K of K_0 in k embeds into $\mathbb{C}$, since K has finite transcendence degree over $\mathbb{Q}$ and $\mathbb{C}$ is algebraically closed of uncountable transcendence degree over $\mathbb{Q}$. Since lc thresholds do not change under extension of the algebraically closed ground field [dFEM11, Lemma 2.14(iii)], applied to $\mathbb{A}^n_K$ and the extensions $K \subset k$ and $K \subset \mathbb{C}$, and coincide with the lc thresholds

on the completion [dFEM11, Proposition 2.11], the threshold $\mathrm{lct}(\operatorname{Spec}\mathfrak{R}_k;\prod_j(g'_j)^{b_j})$ belongs to $T\cap(1-\delta,1)$. This contradicts **Step 2**.

Since $L\geq 1$ and $f_j-f'_j\in(x_1,\dots,x_n)$, the element f_j is a unit if and only if f'_j is a unit. Let

$$J:=\{j\in\{1,\dots,s\}\mid f_j\in(x_1,\dots,x_n)\}.$$

A unit generates the unit ideal, so the pairs $(\operatorname{Spec}\mathfrak{R}_k,\prod_j(f_j)^{c_j})$ and $(\operatorname{Spec}\mathfrak{R}_k,\prod_j(f'_j)^{c_j})$ are $(\operatorname{Spec}\mathfrak{R}_k,\mathfrak{a})$ and $(\operatorname{Spec}\mathfrak{R}_k,\mathfrak{a}')$, where

$$\mathfrak{a}:=\prod_{j\in J}(f_j)^{c_j}\quad\text{and}\quad\mathfrak{a}':=\prod_{j\in J}(f'_j)^{c_j}.$$

If $J=\emptyset$, then $\mathfrak{a}'$ is the unit ideal and $(\operatorname{Spec}\mathfrak{R}_k,\mathfrak{a}')$ is lc because $\mathfrak{R}_k$ is regular. Assume that $J\neq\emptyset$. Then $\mathrm{lct}(\operatorname{Spec}\mathfrak{R}_k;\mathfrak{a})$ and $\mathrm{lct}(\operatorname{Spec}\mathfrak{R}_k;\mathfrak{a}')$ belong to T_k. Since $(\operatorname{Spec}\mathfrak{R}_k,\mathfrak{a})$ is lc,

$$\mathrm{lct}(\operatorname{Spec}\mathfrak{R}_k;\mathfrak{a})\geq 1.$$

For any $j\in J$, we have $(f_j)+(x_1,\dots,x_n)^L=(f'_j)+(x_1,\dots,x_n)^L$, and

$$L\geq\frac{n}{\epsilon\lambda_0}\geq\frac{n}{\epsilon\min_{j\in J}c_j}.$$

By [dFEM11, Proposition 2.19(ii)],

$$\mathrm{lct}(\operatorname{Spec}\mathfrak{R}_k;\mathfrak{a}')\geq\mathrm{lct}(\operatorname{Spec}\mathfrak{R}_k;\mathfrak{a})-\epsilon\geq 1-\epsilon>1-\delta.$$

Since $T_k\cap(1-\delta,1)=\emptyset$, we have $\mathrm{lct}(\operatorname{Spec}\mathfrak{R}_k;\mathfrak{a}')\geq 1$. Thus $(\operatorname{Spec}\mathfrak{R}_k,\mathfrak{a}')$ is lc. □

5. Upper and lower bounds along circles

In this section, we bound a quotient value along a primitive circle $\mathbf{T}$ with growing $r=|G\cap\mathbf{T}|$ from above and from below by the minimum of a reference surface pair built from the principal blocks, and the case theorems in later sections provide the reference pair and the charts. The upper bound holds in general, while the lower bound needs a margin of the reference pair, and when a generic limit is not klt, Subsection 5.2 replaces the margin by an augmentation (Lemma 5.8).

5.1. Principal blocks, Gauss extensions and sections.

In this subsection, we define principal blocks, masses, and margins, and we prove the upper bound (Proposition 5.4) and the lower bound (Lemma 5.5).

Set-up 5.1. Let n, W, f, G, w, $\mathbf{T}$, R, r, e_0, Γ, and D be as follows.

(1) $n\in\{3,4\}$, $W=\mathbb{A}^3$ if $n=3$, and $W=(f=0)\subset\mathbb{A}^4$ for a polynomial f if $n=4$, and W is normal and klt near 0.
(2) G is a finite diagonal abelian group preserving W and acting faithfully on W, and f is G-semi-invariant if $n=4$.
(3) $w\in\mathbb{Z}^n$ is a primitive vector, $\mathbf{T}:=\mathbf{T}_{(w)}$, $R:=G\cap\mathbf{T}$, $r:=|R|$, and e_0 is a positive integer with $[G:R]\leq e_0$.
(4) $\Gamma\subset[0,1]$ is a DCC set, and $D=\sum_j c_j\operatorname{div}_W(h_j)$ with positive $c_j\in\Gamma$ and G-semi-invariant polynomials h_j vanishing at 0, such that (W,D) is klt near 0.

We call $\mathbf{T}$ a *primitive circle*. The $\mathbf{T}$-*weight* of a monomial x^α is $\langle w,\alpha\rangle$. For a non-zero polynomial h, let $\kappa(h)$ be the $\mathbf{T}$-weight of a fixed monomial of least degree of h. The *principal block* h^0 of h is the sum of all terms of h of $\mathbf{T}$-weight $\kappa(h)$. Let $\kappa_j:=\kappa(h_j)$, let f^0 be the principal block of f if $n=4$, and let $\kappa(f):=0$ if $n=3$. The *mass* of D along $\mathbf{T}$ is $\tau:=\sum_j c_j\kappa_j$. Let $\epsilon_{\mathbf{T}}:=\sum_{i=1}^n w_i-\kappa(f)$. The *adjusted mass* of D along $\mathbf{T}$ is $\delta:=\tau-\epsilon_{\mathbf{T}}$.

Lemma 5.3 shows that, for r large, κ_j, $\kappa(f)$, h_j^0, and f^0 do not depend on the choice of the monomial of least degree. We use monomials of least degree because, for r large, they have one common $\mathbf{T}$-weight, and the other blocks of h_j have degree at least $(r-|\kappa_j|)/\max_i|w_i|$, which tends to infinity with r. The number $\epsilon_{\mathbf{T}}$ is the $\mathbf{T}$-weight of the residue form of $(f^0=0)$ (of $dx_1\wedge dx_2\wedge dx_3$ if $n=3$).

Definition 5.2. Let (S,C) be a pair such that C is $\mathbb{R}$-Cartier. Let $Z\subset S$ be a closed subset. A *margin* of (S,C) near Z is a real number $\theta>1$ such that $(S,\theta C)$ is lc near Z.

Lemma 5.3. *Notation as in Set-up 5.1. Any two monomials of a G-semi-invariant polynomial have $\mathbf{T}$-weights congruent modulo r. Let M be a positive integer such that each h_j, and f if $n=4$, has a monomial of total degree at most M, and assume that $r>2M\max_i|w_i|$. Then $|\kappa_j|\le M\max_i|w_i|$ and $|\kappa(f)|\le M\max_i|w_i|$, the numbers κ_j and $\kappa(f)$ and the blocks h_j^0 and f^0 do not depend on the choice of the monomial of least degree, and any monomial of $h_j-h_j^0$ (resp. $f-f^0$) has total degree at least $(r-|\kappa_j|)/\max_i|w_i|$ (resp. $(r-|\kappa(f)|)/\max_i|w_i|$). If $W=\mathbb{A}^3$, or $W=(xy+z^2+t^N=0)$ with $N\ge 2$ and no h_j has a monomial divisible by xy, then one can take $M=\lceil 3/\gamma\rceil$, resp. $M=\lceil 2/\gamma\rceil$, where γ is the least positive element of Γ, and $\gamma:=1$ if $\Gamma\subset\{0\}$. In particular, in both cases there exists a positive integer r_0, depending only on w and Γ, such that $r\ge r_0$ implies $r>2M\max_i|w_i|$ for this M.*

Proof. Since w is primitive, R is the cyclic group $\mu_r(w)$ [Liu26, Definition 2.15], generated by the diagonal element g with entries $\zeta^{w_1},\dots,\zeta^{w_n}$ for a primitive r-th root of unity ζ. The pullback by g multiplies a monomial x^α by $\zeta^{\langle w,\alpha\rangle}$. If h is G-semi-invariant with character χ, then $\zeta^{\langle w,\alpha\rangle}=\chi(g)$ for any monomial x^α of h. Thus the $\mathbf{T}$-weights of any two monomials of h are congruent modulo r.

Let $\mu:=\max_i|w_i|$. Then $\mu\ge 1$, and a monomial of total degree m has $\mathbf{T}$-weight in $[-\mu m,\mu m]$. Let h be one of the h_j, or f if $n=4$. By assumption, $\operatorname{mult}_0 h\le M$, so any monomial of h of least degree has $\mathbf{T}$-weight in $[-\mu M,\mu M]$. The $\mathbf{T}$-weights of two such monomials are congruent modulo r and differ by at most $2\mu M<r$, hence they are equal. Thus $\kappa(h)$ and h^0 do not depend on the choice of the monomial of least degree, and $|\kappa(h)|\le\mu M$. If x^α is a monomial of $h-h^0$, then $\langle w,\alpha\rangle\ne\kappa(h)$ and $\langle w,\alpha\rangle\equiv\kappa(h)\pmod r$, so

$$\mu\deg x^\alpha\ge|\langle w,\alpha\rangle|\ge r-|\kappa(h)|.$$

This proves the lemma for a given M. If $n=3$, then $\kappa(f)=0$ and there is nothing to prove for f.

It remains to prove the values of M for $W=\mathbb{A}^3$ and for $W=(xy+z^2+t^N=0)$. If $\Gamma\subset\{0\}$, then there is no h_j and $\gamma=1$, so $M=3$, resp. $M=2$, works, since f has the monomial xy of total degree 2 in the second case. Thus we may assume that $\Gamma\not\subset\{0\}$, so that γ exists and $c_j\ge\gamma$ for any j.

Assume that $W=\mathbb{A}^3$. Let E be the exceptional divisor of the blow-up of $\mathbb{A}^3$ at 0. Then $a(E,\mathbb{A}^3,0)=3$ and $\operatorname{mult}_E h_j=\operatorname{mult}_0 h_j$ for any j [KM98, Lemma 2.29]. Since (W,D) is klt near 0 and $\operatorname{mult}_0 h_i\ge 1$ for any i,

$$\gamma\operatorname{mult}_0 h_j\le\sum_i c_i\operatorname{mult}_0 h_i<a(E,\mathbb{A}^3,0)=3.$$

Hence $\operatorname{mult}_0 h_j<3/\gamma$, and h_j has a monomial of total degree at most $\lceil 3/\gamma\rceil$.

Assume that $W=(xy+z^2+t^N=0)$ with $N\ge 2$, and that no h_j has a monomial divisible by xy. Let $Q:=xy+z^2$ if $N\ge 3$ and $Q:=xy+z^2+t^2$ if $N=2$, and let E be the exceptional divisor of the blow-up of W at 0. Then E is the projectivized tangent cone of W at 0, that is, the quadric $(Q=0)\subset\mathbb{P}^3$, which is integral since Q has rank at least three. Since E is reduced and a Cartier divisor on the blow-up, E is a prime divisor over W. Since the exceptional divisor of the blow-up of $\mathbb{A}^4$ at 0 has log discrepancy 4 and $\operatorname{mult}_0(xy+z^2+t^N)=2$, adjunction implies $a(E,W,0)=4-2=2$. For a polynomial h with lowest-degree form h_{low}, $\operatorname{mult}_E h=\operatorname{mult}_0 h$ if h_{low} is not divisible by Q, because E is defined by Q in $\mathbb{P}^3$. In the lexicographic order with

$x > y > z > t$, the leading monomial of Q is xy, so the leading monomial of any non-zero multiple of Q is divisible by xy. Since h_j has no monomial divisible by xy, neither does the lowest-degree form of h_j. Thus the lowest-degree form of h_j is not divisible by Q, and $\operatorname{mult}_E h_j = \operatorname{mult}_0 h_j$. Since (W, D) is klt near 0,

$$\gamma \operatorname{mult}_0 h_j \le \sum_i c_i \operatorname{mult}_0 h_i = \sum_i c_i \operatorname{mult}_E h_i < a(E, W, 0) = 2.$$

Hence $\operatorname{mult}_0 h_j < 2/\gamma$, and h_j has a monomial of total degree at most $\lceil 2/\gamma \rceil$. Moreover, $f = xy + z^2 + t^N$ has the monomial xy of total degree $2 \le \lceil 2/\gamma \rceil$, as $\gamma \le 1$.

In both cases, $r_0 := 2\lceil 3/\gamma \rceil \mu + 1$, resp. $r_0 := 2\lceil 2/\gamma \rceil \mu + 1$, depends only on w and Γ, and $r \ge r_0$ implies $r > 2M\mu$. □

Proposition 5.4 is the upper bound. A prime divisor ν computing the minimum of the reference pair, with log discrepancy bounded by Theorem 2.9(2), is extended along the coordinate u by a Gauss extension whose u-value ℓ is in the residue class q modulo $e = |I_{\tilde{H}}(\nu)|$ of Lemma 3.1 and exceeds the orders $\operatorname{mult}_\nu \varphi_j$, so that the inertia group does not change and the terms of Φ_j divisible by u do not change the values.

Proposition 5.4. *Let S be a normal variety of dimension at least two over an algebraically closed field $k \supset \mathbb{C}$, $s \in S$ a closed point, $V := S \times \mathbb{A}^1_u$, and $\tilde{H}$ a finite group acting on S faithfully and fixing s. Let χ be a character of $\tilde{H}$, and let $\tilde{H}$ act on V by $g \cdot (p, u) = (g \cdot p, \chi(g)u)$. Let*

$$C_V := b \operatorname{div}(u) + \sum_j c_j \operatorname{div}(\Phi_j)$$

with $b \le 1$, $c_j \ge 0$, and $\tilde{H}$-semi-invariant regular functions Φ_j on V with $\varphi_j := \Phi_j|_{u=0} \ne 0$, and let $C_S := \sum_j c_j \operatorname{div}(\varphi_j)$. Assume that (S, C_S) is lc near s, and let ν be a prime divisor over S with center s computing $\beta := \operatorname{mld}((S \ni s, C_S)/\tilde{H})$, with $e := |I_{\tilde{H}}(\nu)|$. Let q be the integer of Lemma 3.1 for ν and χ. Then for any positive integer $\ell \equiv q \pmod e$ with $\ell > \operatorname{mult}_\nu \varphi_j$ for any j, the Gauss extension F of mult_ν to $k(S)(u)$ with u-value ℓ is a prime divisor over V with center $(s, 0)$, $I_{\tilde{H}}(F) = I_{\tilde{H}}(\nu)$, $\operatorname{mult}_F \Phi_j = \operatorname{mult}_\nu \varphi_j$ for any j, and

$$\frac{a(F, V, C_V)}{|I_{\tilde{H}}(F)|} = \beta + \frac{(1-b)\ell}{e}.$$

Proof. Let U be an affine open neighborhood of s such that $(U, C_S|_U)$ is lc. Then $S' := \bigcap_{g \in \tilde{H}} gU$ is a $\tilde{H}$-invariant affine open neighborhood of s [Har77, Exercise II.4.3], and $\tilde{H}$ acts faithfully on S'. Since β and the log discrepancies of prime divisors over V with center $(s, 0)$ only depend on a neighborhood of s, possibly replacing S with S', we may assume that S is affine and (S, C_S) is lc. In particular, S is quasi-projective and $K_S + C_S$ is $\mathbb{R}$-Cartier. Since each Φ_j is $\tilde{H}$-semi-invariant and $(u = 0)$ is $\tilde{H}$-invariant, each φ_j is $\tilde{H}$-semi-invariant, hence $\tilde{H}$ preserves C_S.

The action $g \cdot (p, u) = (g \cdot p, \chi(g)u)$ is the action of Lemma 3.1, so q is the integer of that lemma. By Lemma 3.1 with $H = \tilde{H}$, the variety S in place of V, $B = C_S$, $E = \nu$, $w = u$, and the character χ of $\tilde{H}$, F is a prime divisor over V with center $(s, 0)$, $I_{\tilde{H}}(F) = I_{\tilde{H}}(\nu)$, and

$$a(F, V, \operatorname{pr}_1^* C_S) = a(\nu, S, C_S) + \ell,$$

where $\operatorname{pr}_1 : V \to S$ is the projection.

Since S is affine, the coordinate ring of V is the polynomial ring over the coordinate ring of S in u. Write $\Phi_j = \sum_{i \ge 0} \varphi_{j,i} u^i$ with regular functions $\varphi_{j,i}$ on S and $\varphi_{j,0} = \varphi_j$. Since the center of ν on S is s, $\operatorname{mult}_\nu \varphi_{j,i} \ge 0$ for any i. By the definition of the Gauss extension [Liu26, Definition-Proposition 2.11],

$$\operatorname{mult}_F \Phi_j = \min_i \big(\operatorname{mult}_\nu \varphi_{j,i} + i\ell \big),$$

and for $i \ge 1$, $\operatorname{mult}_\nu \varphi_{j,i} + i\ell \ge \ell > \operatorname{mult}_\nu \varphi_j$. Hence $\operatorname{mult}_F \Phi_j = \operatorname{mult}_\nu \varphi_j$. Moreover, $\operatorname{mult}_F u = \ell$ and $\operatorname{mult}_F \operatorname{pr}_1^* \varphi_j = \operatorname{mult}_\nu \varphi_j$.

The $\mathbb{R}$-divisor

$$C_V - \mathrm{pr}_1^* C_S = b\,\mathrm{div}(u) + \sum_j c_j\big(\mathrm{div}(\Phi_j) - \mathrm{div}(\mathrm{pr}_1^*\,\varphi_j)\big)$$

is $\mathbb{R}$-Cartier, hence $K_V + C_V$ is $\mathbb{R}$-Cartier and

$$\begin{aligned} a(F, V, C_V) &= a(F, V, \mathrm{pr}_1^* C_S) - b\,\mathrm{mult}_F\, u - \textstyle\sum_j c_j\big(\mathrm{mult}_F\,\Phi_j - \mathrm{mult}_F\,\mathrm{pr}_1^*\,\varphi_j\big) \\ &= a(\nu, S, C_S) + (1-b)\ell. \end{aligned}$$

Since ν computes β, $a(\nu, S, C_S) = e\beta$ [Liu26, Lemma 2.6]. Since $|I_{\tilde{H}}(F)| = e$, dividing by e implies the proposition. □

Lemma 5.5 is the lower bound: for a section S_0 with coefficient $b \le 1$, the quotient log discrepancy of any prime divisor is at least the quotient minimal log discrepancy of the section pair plus $1-b$ times the order of the equation of S_0 divided by the order of the inertia group. In the case sections, a margin first deletes the blocks that a given prime divisor does not see, and then the lemma is applied to the section. The lemma bounds the quotient log discrepancy from below by the quotient minimal log discrepancy of the section pair only when $b \le 1$, and this is why the sign of the adjusted mass δ decides on which side of $\mathbf{T}$ the case sections take the reference pair: there the coefficient of the section is $1 + \delta/r$ on one side and $1 - \delta/r$ on the other.

Lemma 5.5. *Let G be a finite abelian group acting diagonally on $\mathbb{A}^n$, $n \ge 3$, and let X be either $\mathbb{A}^n$, or the common zero scheme of G-semi-invariant polynomials $f_1, \dots, f_c$ vanishing at 0, $c \ge 1$, which is, near 0, a normal integral complete intersection of codimension c with klt singularities, with G acting faithfully on X. Let u be a G-semi-invariant polynomial such that $S_0 := X \cap (u = 0)$ is, near 0, a normal integral complete intersection in $\mathbb{A}^n$ with klt singularities, of dimension at least one, and let $H \subset G$ be a subgroup acting faithfully on S_0. Let $\mathfrak{a} = \prod_{j=1}^s \mathfrak{a}_j^{r_j}$ be an $\mathbb{R}$-ideal on $\mathbb{A}^n$ with $r_j > 0$ and each $\mathfrak{a}_j$ generated by G-semi-invariant polynomials, such that $\mathfrak{a}|_{S_0}$ is non-zero near 0 and $(S_0, \mathfrak{a}|_{S_0})$ is lc near 0, and let $\beta_0 := \mathrm{mld}((S_0 \ni 0, \mathfrak{a}|_{S_0})/H)$. Then for any real number $b \le 1$, any prime divisor P over X with center 0 satisfies*

$$\frac{a(P, X, bS_0, \mathfrak{a}|_X)}{|I_H(P)|} \ge \beta_0 + \frac{(1-b)\,\mathrm{mult}_P\, u}{|I_H(P)|}.$$

If moreover $\dim S_0 \ge 2$ and $C \ni 0$ is an H-invariant integral curve contained in S_0 such that $(S_0, \mathfrak{a}|_{S_0})$ is lc near the generic point ξ of C, then, for any real number $b \le 1$, any prime divisor P over X with center C satisfies

$$\frac{a(P, X, bS_0, \mathfrak{a}|_X)}{|I_H(P)|} \ge \beta_0(\xi) + \frac{(1-b)\,\mathrm{mult}_P\, u}{|I_H(P)|},$$

where $\beta_0(\xi)$ is the minimal log discrepancy of the quotient of $(S_0, \mathfrak{a}|_{S_0})$ by H at the image of ξ.

Proof. Since $\mathfrak{a}|_{S_0}$ is non-zero near 0, so is $\mathfrak{a}|_X$. Since $X \cap (u = 0) = S_0$ is integral near 0, $\mathrm{div}(u|_X) = S_0$ near 0. Hence for any real number b and any prime divisor P over X whose center contains 0,

$$a(P, X, bS_0, \mathfrak{a}|_X) = a(P, X, S_0, \mathfrak{a}|_X) + (1-b)\,\mathrm{mult}_P\, u. \tag{5.1}$$

By Definition 2.4, any prime divisor P over X with center 0 satisfies

$$a(P, X, S_0, \mathfrak{a}|_X) \ge |I_H(P)|\mathrm{mld}((X \ni 0, S_0, \mathfrak{a}|_X)/H),$$

and, for a curve C as in the lemma, any prime divisor P over X with center C satisfies

$$a(P, X, S_0, \mathfrak{a}|_X) \ge |I_H(P)|\mathrm{mld}((X \ni \xi, S_0, \mathfrak{a}|_X)/H).$$

By (5.1), it suffices to show that

$$\beta_0 = \mathrm{mld}((X \ni 0, S_0, \mathfrak{a}|_X)/H),$$

and, in the second case, that $\beta_0(\xi) = \mathrm{mld}((X \ni \xi, S_0, \mathfrak{a}|_X)/H)$.

Assume that $X = \mathbb{A}^n$. The hypotheses of Theorem 3.3(1) with $c = 1$, $f_1 = u$, and the group H in place of G hold: H acts diagonally on $\mathbb{A}^n$, u is H-semi-invariant and vanishes at 0 since $0 \in S_0$, $S_0 = (u = 0)$ is, near 0, a normal integral hypersurface with klt singularities on which H acts faithfully, $\dim S_0 = n - 1 \geq 2$, $\mathfrak{a}|_{S_0}$ is non-zero near 0, and the factors of $\mathfrak{a}$ are generated by H-semi-invariant polynomials. Since $(S_0, \mathfrak{a}|_{S_0})$ is lc near 0, Theorem 3.3(1) implies

$$\beta_0 = \mathrm{mld}((\mathbb{A}^n \ni 0, \mathrm{div}(u), \mathfrak{a})/H) = \mathrm{mld}((\mathbb{A}^n \ni 0, S_0, \mathfrak{a})/H).$$

Assume that $X = (f_1 = \cdots = f_c = 0)$. The hypotheses of Theorem 3.3(2) with $f_{c+1} = u$ hold: S_0 is, near 0, a normal integral complete intersection of codimension $c + 1$ with klt singularities on which H acts faithfully, $\mathfrak{a}|_{S_0}$ is non-zero near 0, and $(S_0, \mathfrak{a}|_{S_0})$ is lc near 0. Thus, by Theorem 3.3(2), $\beta_0 = \mathrm{mld}((X \ni 0, S_0, \mathfrak{a}|_X)/H)$.

For a curve C as in the lemma, we use Theorem 3.3(3) in place of Theorem 3.3(1) and (2), whose hypothesis $n - c \geq 3$ in the setting of (2) is $\dim S_0 \geq 2$: C is an H-invariant integral curve through 0 contained in S_0, and $(S_0, \mathfrak{a}|_{S_0})$ is lc near ξ. Thus, by Theorem 3.3(3), $\beta_0(\xi) = \mathrm{mld}((X \ni \xi, S_0, \mathfrak{a}|_X)/H)$, where for $X = \mathbb{A}^n$ the boundary S_0 is $\mathrm{div}(u)$ near ξ. □

5.2. **Margins, floors and augmentations.** In this subsection, we study compatible generic limits which are not klt, and we bound quotient values from below by lc augmentations.

Near a smooth curve with coefficient one, the quotient value is determined by the order of the restriction of the remaining boundary to the curve and by the order of the image of the group in the automorphism group of the tangent line of the curve. Lemma 5.6 computes the limit β in Definition-Proposition 5.7.

Lemma 5.6. *Let $(S \ni p)$ be a smooth surface germ over an algebraically closed field $k \supset \mathbb{C}$, D a smooth curve through p, $C' \geq 0$ an $\mathbb{R}$-divisor not containing D with $b := \mathrm{mult}_p(C'|_D) \leq 1$, and H a finite abelian group acting faithfully on S, fixing p and preserving D and C'. Let e be the order of the image of H in the automorphism group of the tangent line T_pD. Then*

$$\mathrm{mld}((S \ni p, D + C')/H) = \frac{1 - b}{e}.$$

Proof. We use [Liu26, Lemma 2.6] and [KM98, Lemma 2.29] over k and for $\mathbb{R}$-divisors, where they hold by [Liu26, Remark 2.17]. The proof of Definition-Proposition 5.7 also uses them over k and for $\mathbb{R}$-divisors.

Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{S,p}$ and $I_D \subset \mathcal{O}_{S,p}$ the ideal of D. Since H is abelian, H acts on the lines $I_D/\mathfrak{m}I_D$ and $\mathfrak{m}/(\mathfrak{m}^2 + I_D)$ by characters χ and ψ respectively. Let f_0 be a generator of I_D, and $z_0 \in \mathfrak{m}$ an element whose image generates $\mathfrak{m}/(\mathfrak{m}^2 + I_D)$. Then

$$f := \frac{1}{|H|} \sum_{g \in H} \chi(g)^{-1} g^* f_0 \quad \text{and} \quad z := \frac{1}{|H|} \sum_{g \in H} \psi(g)^{-1} g^* z_0$$

are H-semi-invariant with characters χ and ψ, $f - f_0 \in \mathfrak{m}I_D$, and $z - z_0 \in \mathfrak{m}^2 + I_D$. Thus f generates I_D by Nakayama's lemma. Since D is smooth, $f \notin \mathfrak{m}^2$, so (f, z) is a regular system of parameters of $\mathcal{O}_{S,p}$ and $\widehat{\mathcal{O}}_{S,p} = k[[f, z]]$.

Since $\mathfrak{m}/(\mathfrak{m}^2 + I_D)$ is the cotangent line of D at p, the kernel K of $H \to \mathrm{Aut}(T_pD)$ is $\mathrm{Ker}\,\psi$, and $e = |\psi(H)|$. Any $g \in K$ satisfies $g^*z = z$ and $g^*f = \chi(g)f$. An element of H which fixes f and z acts trivially on $k[[f, z]]$, hence trivially on S, so it is the identity. Thus $\chi|_K$ is injective, K is cyclic of order $m := |K|$, and $D = (f = 0)$ is the only prime divisor through p which is fixed pointwise by a non-trivial element of K. Let $\pi : S \to S' := S/K$ be the quotient morphism, $p' := \pi(p)$, and $D' := \pi(D)$. The functions $u := f^m$ and z are K-invariant, and the completion of $\mathcal{O}_{S',p'}$ is $k[[f, z]]^K = k[[u, z]]$ (by [Liu26, Lemma 3.3(1)]). Hence S' is smooth at p', (u, z) is a regular system of parameters of $\mathcal{O}_{S',p'}$, and $D' = (u = 0)$ is smooth. By [Liu26, Lemma 2.6(4)], the crepant quotient of $(S, D + C')$ by K is $(S', D' + C'')$, where D' has coefficient $1 - \frac{1-1}{m} = 1$ and C'' is the $\mathbb{R}$-divisor with $\pi^*C'' = C'$. Since $\pi^*z = z$ and z restricts to a parameter of D at p

and to a parameter of D' at p', we have

$$\mathrm{mult}_{p'}(C''|_{D'}) = \mathrm{mult}_p\left((\pi^*C'')|_D\right) = b.$$

An element of H which acts trivially on S' fixes z, so it belongs to K. Thus H/K acts faithfully on S', and by [Liu26, Lemma 2.6(5)],

$$\mathrm{mld}((S \ni p, D + C')/H) = \mathrm{mld}((S' \ni p', D' + C'')/(H/K)).$$

The character ψ induces an injective character $\bar{\psi} : H/K \to k^*$, by which H/K acts on z. In particular, H/K is cyclic of order e. The function u is H/K-semi-invariant, since the character χ^m of u is trivial on K. Let ρ be an integer such that H/K acts on u by $\bar{\psi}^\rho$.

We first prove the lower bound. The germ S', the curve D', and the $\mathbb{R}$-divisor C'' are defined over an algebraically closed subfield $k_1 \subset k$ isomorphic to $\mathbb{C}$, and the minimal log discrepancy at a closed point does not change under the base change from k_1 to k [Liu26, Remark 14.18]. Thus [EM04, Theorem 1.1] holds over k, and since $b \le 1$, it implies

$$\mathrm{mld}(S' \ni p', D' + C'') = \mathrm{mld}(D' \ni p', C''|_{D'}) = 1 - b \ge 0.$$

Thus any prime divisor F over S' with $\mathrm{center}_{S'} F = \{p'\}$ satisfies $a(F, S', D' + C'') \ge 1 - b$ and $|I_{H/K}(F)| \le |H/K| = e$. By the formula at the end of [Liu26, Lemma 2.6],

$$\mathrm{mld}\big((S' \ni p', D' + C'')/(H/K)\big) \ge \frac{1-b}{e}.$$

We prove the upper bound. Near p', write $C'' = \sum_l c_l \,\mathrm{div}(g_l)$ with $c_l > 0$ and irreducible $g_l \in \mathfrak{m}_{S',p'}$, where $\mathfrak{m}_{S',p'}$ is the maximal ideal of $\mathcal{O}_{S',p'}$. Since D' is not a component of C'', the integers $n_l := \mathrm{mult}_{p'}(g_l|_{D'})$ are finite, and $b = \sum_l c_l n_l$. Let N be an integer such that $N > n_l$ for any l and $N \equiv \rho \pmod e$, and let F_N be the prime divisor over S' whose valuation v_N is the monomial valuation in (u, z) with $v_N(u) = N$ and $v_N(z) = 1$. The prime divisor F_N is the exceptional curve E_N of the last of N point blow-ups, where E_1 is the exceptional curve of the blow-up of S' at p', and E_i for $i \ge 2$ is the exceptional curve of the blow-up at the intersection point of E_{i-1} and the strict transform of D'. For $i \ge 2$, the intersection point of E_{i-1} and the strict transform of D' is not on the strict transforms of $E_1, \dots, E_{i-2}$ (in the chart $u = u_i z^i$ of the i-th blow-up, E_i is $(z = 0)$, the strict transform of D' is $(u_i = 0)$, and the strict transforms of $E_1, \dots, E_{i-1}$ are disjoint from the chart), so [KM98, Lemma 2.29] implies $a(E_1, S', 0) = 2$ and $a(E_i, S', 0) = a(E_{i-1}, S', 0) + 1$ for $i \ge 2$. Hence $a(F_N, S', 0) = N + 1$. Any monomial in u, z divisible by u has value at least $N > n_l$, so $v_N(g_l) = n_l$. Therefore

$$a(F_N, S', D' + C'') = N + 1 - N - \sum_l c_l n_l = 1 - b.$$

The group H/K acts on the rational function u/z^N by $\bar{\psi}^{\rho - N}$, which is trivial since $\bar{\psi}$ has order e and $N \equiv \rho \pmod e$. Since H/K acts diagonally on (u, z), it preserves v_N, and since the residue field of v_N is generated over k by the residue of u/z^N (a polynomial in u, z which is homogeneous of weight d is $z^d P(u/z^N)$ for a polynomial P), the group H/K acts trivially on the residue field of v_N. Hence $I_{H/K}(F_N) = H/K$, and

$$\frac{a(F_N, S', D' + C'')}{|I_{H/K}(F_N)|} = \frac{1-b}{e}.$$

Together with the lower bound and the formula at the end of [Liu26, Lemma 2.6], this proves the lemma. □

The pairs $(\mathcal{A}, C_i)$ of Theorem 4.2 need not have a margin (Definition 5.2) independent of i (Example 4.3), so there is no analogue of [Liu26, Proposition 14.12(1)], which multiplies the boundary by a fixed real number larger than one. When the generic limit is not klt, the positivity of the limit β of the quotient values forces the non-klt locus of the generic limit to consist of smooth, pairwise disjoint curves with coefficient one.

Definition-Proposition 5.7. Notation as in Construction-Proposition 4.1 and Theorem 4.2, in case (a) with $n = 2$ or in case (b), with $\mathfrak{c} = \mathcal{O}$. Assume that $(\mathcal{A}, C_i)$ is lc near the base for any i. In case (a), let $\beta_i := \mathrm{mld}((\mathcal{A} \ni 0, C_i)/H)$. In case (b), let β_i be the minimum of the log discrepancies of the crepant quotient of $(\mathcal{A}, C_i)$ by H over the prime divisors with center contained in E/H. Assume that the β_i converge to a real number $\beta > 0$, and that $(\mathcal{A}_k, C_\infty)$ is not klt near the base. Then $(\mathcal{A}_k, C_\infty)$ is lc near the base, and any lc center of $(\mathcal{A}_k, C_\infty)$ meeting the base is a curve with coefficient one in the boundary of $(\mathcal{A}_k, C_\infty)$. In case (a) such a curve is smooth at 0. In case (b), for any closed point $\tilde{y}$ over E_k of a chart cover $U_{\alpha,k} := U_\alpha \times_{\mathbb{C}} \operatorname{Spec} k$, let

$$\mathcal{S}_{\tilde{y}} = \operatorname{Spec} k[[\xi_1, \xi_2]] \to \operatorname{Spec} \widehat{\mathcal{O}}_{U_{\alpha,k},\tilde{y}}$$

be the smooth formal cover, which is the identity if U_α is smooth at $\tilde{y}$ and the quotient by the local fundamental group of the quotient singularity otherwise. Then the preimage of such a curve in $\mathcal{S}_{\tilde{y}}$ is smooth. No two such curves meet at 0 in case (a), and no two such preimages meet at the closed point of $\mathcal{S}_{\tilde{y}}$ in case (b). Moreover $(\mathcal{A}_k, C_\infty - D)$ is klt near the base, where D is the sum of all such curves.

Such curves are called the *floors* of the limit. In case (a) there exists exactly one floor D, and $\beta = (1-b)/e$, where $b := \mathrm{mult}_0((C_\infty - D)|_D)$ and e is the order of the image of H in the automorphism group of the tangent line $T_0 D$.

Proof. The hypotheses are preserved when we pass to an infinite subsequence of the tuples, and $(f_{\infty,j})_j$ is a compatible generic limit of any infinite subsequence, since the property of Z_d in Construction-Proposition 4.1 is preserved. Thus, possibly passing to an infinite subsequence, we may assume that (1) and (2) of Theorem 4.2 hold for all i. In case (b), log discrepancies with respect to $(\mathcal{A}_k, C_\infty)$ are taken with respect to the boundary $B_{\mathrm{br}} + C_\infty$.

Step 1. In this step we prove that $(\mathcal{A}_k, C_\infty)$ is lc near the base and that

$$a(P, \mathcal{A}_k, C_\infty) \geq \beta \tag{5.2}$$

for any prime divisor P over $\mathcal{A}_k$ with center 0 in case (a), and with center contained in E_k in case (b).

Since $(\mathcal{A}, C_i)$ is lc near the base for any i, Theorem 4.2(1) with $\theta = 1$ implies that $(\mathcal{A}_k, C_\infty)$ is lc near the base. Since $\mathfrak{c} = \mathcal{O}$, and $n = 2$ in case (a), Theorem 4.2(2) implies that β_i is equal to $\mathrm{mld}((\mathcal{A}_k \ni 0, C_\infty)/H)$ in case (a), and to the minimum of the log discrepancies of the crepant quotient of $(\mathcal{A}_k, C_\infty)$ by H over the prime divisors with center contained in E_k/H in case (b), for any i. Thus β is equal to the corresponding value for $(\mathcal{A}_k, C_\infty)$. For a prime divisor P as in (5.2), the corresponding log discrepancy of the crepant quotient is $a(P, \mathcal{A}_k, C_\infty)/|I_H(P)|$ ([Liu26, Definition 3.2] in case (a), Lemma 3.7 in case (b)). Hence $a(P, \mathcal{A}_k, C_\infty) \geq |I_H(P)|\beta \geq \beta$.

Step 2. In this step we introduce a regular local model $(\mathcal{S} \ni \tilde{y}, \Theta)$ at any closed point of the base.

In case (a), let $(\mathcal{S} \ni \tilde{y}, \Theta) := (\mathcal{A}_k \ni 0, C_\infty)$. In case (b), let $y \in E_k$ be a closed point, α an index with $y \in V_\alpha \times_{\mathbb{C}} \operatorname{Spec} k$, $\tilde{y}$ a closed point of $U_{\alpha,k}$ over y, and $K_{\tilde{y}} \subset K_\alpha$ the stabilizer of $\tilde{y}$. Since $\tilde{y}$ is a closed point with residue field k, its decomposition group in K_α is $K_{\tilde{y}}$, so the morphism $U_{\alpha,k}/K_{\tilde{y}} \to U_{\alpha,k}/K_\alpha$ induces an isomorphism of the completed local rings at the image of $\tilde{y}$ and at y [Gro71, Exposé V, Proposition 2.2(i)]. Hence, by [Liu26, Lemma 3.3(1)] for the action of $K_{\tilde{y}}$ on $U_{\alpha,k}$, we have

$$\widehat{\mathcal{O}}_{Y_k,y} = \widehat{\mathcal{O}}^{K_{\tilde{y}}}_{U_{\alpha,k},\tilde{y}}.$$

Let $\mathcal{S} := \mathcal{S}_{\tilde{y}}$, let $\pi_{\tilde{y}}$ be the local fundamental group of $U_{\alpha,k}$ at $\tilde{y}$, which is trivial if U_α is smooth at $\tilde{y}$, and let $G_{\tilde{y}}$ be the group of automorphisms of $\mathcal{S}$ which lift elements of $K_{\tilde{y}}$. Any element of $K_{\tilde{y}}$ lifts to $\mathcal{S}$, since $\mathcal{S}_{\tilde{y}}$ is the local universal cover (the punctured spectrum of $k[[\xi_1, \xi_2]]$ is simply connected by purity [Gro71, Exposé X, Théorème 3.1]), and two lifts of an element of $K_{\tilde{y}}$ differ by an element of $\pi_{\tilde{y}}$. Thus $G_{\tilde{y}}$ is a finite group generated by $\pi_{\tilde{y}}$ and the lifts of the elements of

$K_{\tilde{y}}$, and

$$\widehat{\mathcal{O}}_{\mathcal{S}}^{G_{\tilde{y}}} = \left(\widehat{\mathcal{O}}_{\mathcal{S}}^{\pi_{\tilde{y}}}\right)^{K_{\tilde{y}}} = \widehat{\mathcal{O}}_{U_{\alpha,k},\tilde{y}}^{K_{\tilde{y}}} = \widehat{\mathcal{O}}_{Y_k,y},$$

where $\widehat{\mathcal{O}}_{\mathcal{S}} := k[[\xi_1,\xi_2]]$. Let $\mathfrak{m}$ be the maximal ideal of $\widehat{\mathcal{O}}_{\mathcal{S}}$. Since $G_{\tilde{y}}$ is finite and k has characteristic zero, the surjection $\mathfrak{m} \to \mathfrak{m}/\mathfrak{m}^2$ has a $G_{\tilde{y}}$-equivariant section, so we may choose ξ_1,ξ_2 on which $G_{\tilde{y}}$ acts linearly. Thus $\mathcal{S}$ with the action of $G_{\tilde{y}}$ is the completion of $\mathbb{A}^2_k$ at 0 with a linear action. We denote the closed point of $\mathcal{S}$ again by $\tilde{y}$, and let

$$\Theta := \sum_j c_j \operatorname{div}(f_{\infty,j,\alpha}) \text{ on } \mathcal{S},$$

where Θ is defined since $\tilde{y}$ is a point over E_k. The pullbacks of the $f_{\infty,j,\alpha}$ to $\mathcal{S}$ are $G_{\tilde{y}}$-semi-invariant, since they are $\pi_{\tilde{y}}$-invariant and K_α-semi-invariant on $U_{\alpha,k}$. The group $\pi_{\tilde{y}}$ contains no pseudo-reflection, so the quotient morphism of $\mathcal{S}$ by $\pi_{\tilde{y}}$ is étale in codimension one, and the crepant quotient of $(\mathcal{S},0)$ by $\pi_{\tilde{y}}$ is $(\operatorname{Spec}\widehat{\mathcal{O}}_{U_{\alpha,k},\tilde{y}},0)$. Since the crepant quotient of $(U_\alpha,0)$ by K_α is $(V_\alpha,B_{\mathrm{br}})$ and C_∞ is given by $\sum_j c_j \operatorname{div}(f_{\infty,j,\alpha})$ on the chart, the crepant quotient of $(\mathcal{S},\Theta)$ by $G_{\tilde{y}}$ is $(\operatorname{Spec}\widehat{\mathcal{O}}_{Y_k,y}, B_{\mathrm{br}}+C_\infty)$ (Lemma 3.7).

In both cases, $\mathcal{S} = \operatorname{Spec} k[[\xi_1,\xi_2]]$ for parameters ξ_1,ξ_2 at $\tilde{y}$, and we claim that $(\mathcal{S},\Theta)$ is lc and that

$$a(\tilde{P},\mathcal{S},\Theta) \geq \beta \tag{5.3}$$

for any formal prime divisor $\tilde{P}$ over $\mathcal{S}$ with center $\tilde{y}$. In case (a), the claim is **Step 1**. In case (b), by Lemma 3.7, any formal prime divisor $\tilde{P}$ over $\mathcal{S}$ satisfies

$$a(\tilde{P},\mathcal{S},\Theta) = |I_{G_{\tilde{y}}}(\tilde{P})| \cdot a(P, \operatorname{Spec}\widehat{\mathcal{O}}_{Y_k,y}, B_{\mathrm{br}}+C_\infty)$$

for a prime divisor P over $\operatorname{Spec}\widehat{\mathcal{O}}_{Y_k,y}$ whose center is the image of the center of $\tilde{P}$. The right side is non-negative since $(\mathcal{A}_k,C_\infty)$ is lc near the base, and it is at least β if $\operatorname{center}_{\mathcal{S}}\tilde{P} = \{\tilde{y}\}$, since then $\operatorname{center} P = \{y\}$, and (5.2) holds for P.

Step 3. In this step we prove that $\operatorname{mult}_{\tilde{y}}\Theta < 2$.

Write $\Theta = \sum_j c_j \operatorname{div}(\varphi_j)$ with non-zero $\varphi_j \in k[[\xi_1,\xi_2]]$ (the images of the $f_{\infty,j}$, resp. the pullbacks of the $f_{\infty,j,\alpha}$). Let ℓ be an integer which is at least the integer of [Liu26, Lemma 14.14(1)] for the trivial group and the finite set $\{c_1,\dots,c_s\}$ (two tuples of semi-invariant power series whose differences have multiplicity at least ℓ give crepant quotients with equal minimal log discrepancies at the closed point), and which is larger than $\operatorname{mult}_{\tilde{y}}\varphi_j$ for any j. Let $\varphi_j^{<\ell} \in k[\xi_1,\xi_2]$ be the part of φ_j of degree less than ℓ, and

$$\Theta^{<\ell} := \sum_j c_j \operatorname{div}(\varphi_j^{<\ell}) \quad \text{on } \mathbb{A}^2_k = \operatorname{Spec} k[\xi_1,\xi_2].$$

Then $\operatorname{mult}_0\Theta^{<\ell} = \operatorname{mult}_{\tilde{y}}\Theta$. By [Liu26, Lemma 14.14(1)] for the trivial group, [Liu26, Lemma 3.3(2)], and (5.3),

$$\operatorname{mld}(\mathbb{A}^2_k \ni 0, \Theta^{<\ell}) = \operatorname{mld}(\mathcal{S} \ni \tilde{y}, \Theta) \geq \beta > 0.$$

By [KM98, Lemma 2.29], the exceptional curve F_0 of the blow-up of $\mathbb{A}^2_k$ at 0 satisfies $a(F_0,\mathbb{A}^2_k,\Theta^{<\ell}) = 2 - \operatorname{mult}_0\Theta^{<\ell}$. Thus $2-\operatorname{mult}_{\tilde{y}}\Theta > 0$.

Step 4. In this step we prove the statements on the lc centers and on $C_\infty - D$.

Let P be a prime divisor with $a(P,\mathcal{A}_k,C_\infty) = 0$ whose center intersects the base, namely a prime divisor over $\mathcal{A}_k$ in case (a), and a prime divisor over $\operatorname{Spec}\widehat{\mathcal{O}}_{Y_k,y}$ for a closed point $y \in E_k$ in case (b). By (5.2), the center of P is not the closed point. Since $\mathcal{A}_k$, resp. $\operatorname{Spec}\widehat{\mathcal{O}}_{Y_k,y}$, is normal of dimension two, the center Σ of P is a curve, $P=\Sigma$, and Σ has coefficient $1-a(P,\mathcal{A}_k,C_\infty)=1$ in the boundary. In case (b), the branches of E_k at y have coefficients at most $1-\beta<1$ by (5.2), so Σ is not a branch of E_k. Hence any lc center of $(\mathcal{A}_k,C_\infty)$ meeting the base is a curve with coefficient one in the boundary.

Conversely, the curves with coefficient one in the boundary are lc centers, and in case (a) all of them pass through 0. Let $\tilde{y}$ and $(\mathcal{S}, \Theta)$ be as in **Step 2**, and let $\tilde{D}$ be the sum of the curves through $\tilde{y}$ with coefficient one in Θ. In case (a), $\tilde{D} = D$. In case (b), the preimage in $\mathcal{S}$ of any curve with coefficient one in $B_{\mathrm{br}} + C_\infty$ through y is contained in $\tilde{D}$, since by Lemma 3.7 (that is, [Liu26, Lemma 2.6(4)] on the completion) a prime divisor of $\mathcal{S}$ with coefficient c in Θ and ramification index e' maps to a prime divisor with coefficient $1 - \frac{1-c}{e'}$, which is equal to 1 only if $c = 1$. By **Step 3**,

$$\operatorname{mult}_{\tilde{y}} \tilde{D} \le \operatorname{mult}_{\tilde{y}} \Theta < 2.$$

Thus at most one curve with coefficient one in Θ passes through $\tilde{y}$, and it is smooth at $\tilde{y}$. Since $\tilde{y}$ is the closed point 0 of $\mathcal{A}_k$ in case (a), and an arbitrary closed point of $U_{\alpha,k}$ over E_k in case (b), the floor is smooth at 0 in case (a), the preimages of the floors in $\mathcal{S}_{\tilde{y}}$ are smooth in case (b), and no two floors, resp. no two preimages of floors in $\mathcal{S}_{\tilde{y}}$, pass through 0, resp. through the closed point of $\mathcal{S}_{\tilde{y}}$.

Suppose that $a(P, \mathcal{A}_k, C_\infty - D) \le 0$ for a prime divisor P whose center intersects the base, over $\mathcal{A}_k$ in case (a), and over $\operatorname{Spec} \widehat{\mathcal{O}}_{Y_k,y}$ for a closed point $y \in E_k$ in case (b). Since

$$a(P, \mathcal{A}_k, C_\infty) = a(P, \mathcal{A}_k, C_\infty - D) - \operatorname{mult}_P D \le -\operatorname{mult}_P D \le 0,$$

and $(\mathcal{A}_k, C_\infty)$ is lc near the base, we have $a(P, \mathcal{A}_k, C_\infty) = 0$ and $\operatorname{mult}_P D = 0$. By the first paragraph of this step, P is a floor, so $\operatorname{mult}_P D = 1$, a contradiction. Thus $(\mathcal{A}_k, C_\infty - D)$ is klt near the base.

Step 5. In this step we assume case (a) and prove that there exists exactly one floor D and that $\beta = (1-b)/e$.

Since $(\mathcal{A}_k, C_\infty)$ is lc and not klt, there exists a prime divisor P over $\mathcal{A}_k$ with $a(P, \mathcal{A}_k, C_\infty) = 0$. By **Step 4**, the center of P is a floor. All floors pass through 0, and no two of them meet at 0, so there exists exactly one floor D. Since the $f_{\infty,j}$ are H-semi-invariant, H preserves C_∞, hence permutes the floors, and so D is H-invariant.

The group $H \subset GL_2(\mathbb{C})$ preserves the line $T_0 D$ in the tangent space k^2 of $\mathcal{A}_k$ at 0, so H is contained in a Borel subgroup $\mathcal{B} \subset GL_2(k)$. The unipotent radical of $\mathcal{B}$ is isomorphic to the additive group of k, which has no non-trivial finite subgroup. Thus H injects into the quotient of $\mathcal{B}$ by the unipotent radical, which is isomorphic to $(k^*)^2$, so H is abelian. Since H is abelian, as in the second paragraph of the proof of Lemma 5.6, there exist H-semi-invariant $f, z \in k[[x_1, x_2]]$, where x_1, x_2 are the coordinates of Construction-Proposition 4.1(a), with $D = (f = 0)$ and $k[[x_1, x_2]] = k[[f, z]]$. Let χ_f and χ_z be the characters of f and z. Since H acts faithfully on $k[[f, z]]$, the homomorphism $(\chi_f, \chi_z) : H \to (\mathbb{C}^*)^2$ is injective. Let H' be the image of (χ_f, χ_z), a finite diagonal subgroup of $(\mathbb{C}^*)^2$ acting on $\mathbb{A}^2_{u,t}$. Then the isomorphism $k[[u, t]] \to k[[x_1, x_2]]$ given by $u \to f$ and $t \to z$ is H-equivariant.

For any j, let m_j be the largest integer such that f^{m_j} divides $f_{\infty,j}$, and let $G_j \in k[[u, t]]$ be the power series with $G_j(f, z) = f_{\infty,j}/f^{m_j}$. Since D has coefficient one in C_∞, we have $\sum_j c_j m_j = 1$, and

$$C_\infty - D = \sum_j c_j \operatorname{div}\big(G_j(f, z)\big).$$

Each G_j is H'-semi-invariant and $G_j(0, t) \neq 0$. Let ℓ be an integer which is at least the integer of [Liu26, Lemma 14.14(1)] for H' and the finite set $\{1, c_1, \dots, c_s\}$, and which is larger than $\operatorname{mult}_0 G_j(0, t)$ for any j. Let $G_j^{<\ell}$ be the part of G_j of degree less than ℓ. Since H' is diagonal, $G_j^{<\ell}$ is an H'-semi-invariant polynomial with the character of G_j, and $\operatorname{mult}_0 G_j^{<\ell}(0, t) = \operatorname{mult}_0 G_j(0, t)$. Let

$$C' := \sum_j c_j \operatorname{div}(G_j^{<\ell}) \quad \text{on } \mathbb{A}^2_k = \operatorname{Spec} k[u, t].$$

By [Liu26, Lemma 14.14(1)] for the tuples $(u, G_1, \dots, G_s)$ and $(u, G_1^{<\ell}, \dots, G_s^{<\ell})$ with labels $1, c_1, \dots, c_s$, and by [Liu26, Lemma 3.3(2)],

$$\operatorname{mld}((\mathbb{A}^2_k \ni 0, \operatorname{div}(u) + C')/H') = \operatorname{mld}((\mathcal{A}_k \ni 0, C_\infty)/H) = \beta.$$

Moreover

$$\operatorname{mult}_0 \left(C'|_{(u=0)}\right) = \sum_j c_j \operatorname{mult}_0 G_j(0,t) = \operatorname{mult}_0 \left((C_\infty - D)|_D\right) = b,$$

and the image of H' in $\operatorname{Aut}(T_0(u=0))$ has order $|\chi_z(H)|$, which is the order e of the image of H in $\operatorname{Aut}(T_0 D)$.

Let N be an integer larger than $\operatorname{mult}_0 G_j(0,t)$ for any j. Let F_N be the prime divisor over $\mathbb{A}^2_k$ whose valuation is the monomial valuation in (u,t) with weights $(N,1)$. Then $\operatorname{center}_{\mathbb{A}^2_k} F_N = \{0\}$, and F_N is the exceptional curve of the last of N point blow-ups, the first at 0 and each later one at the intersection point of the last exceptional curve and the strict transform of $(u=0)$, which is not on the strict transforms of the earlier exceptional curves (in the chart $u = u_i t^i$ of the i-th blow-up, the i-th exceptional curve is $(t=0)$, the strict transform of $(u=0)$ is $(u_i = 0)$, and the strict transforms of the earlier exceptional curves are disjoint from the chart). Thus [KM98, Lemma 2.29] gives $a(F_N, \mathbb{A}^2_k, 0) = N+1$. Any monomial in u,t divisible by u has value at least $N > \operatorname{mult}_0 G_j^{<\ell}(0,t)$, so $\operatorname{mult}_{F_N} G_j^{<\ell} = \operatorname{mult}_0 G_j(0,t)$, and $a(F_N, \mathbb{A}^2_k, \operatorname{div}(u)+C') = N+1-N-b = 1-b$. By the formula at the end of [Liu26, Lemma 2.6], $1-b \geq |I_{H'}(F_N)|\beta > 0$, so $b < 1$. Lemma 5.6, applied to the germ of $\mathbb{A}^2_k$ at 0, the curve $(u=0)$, the $\mathbb{R}$-divisor C' and the group H', implies $\beta = (1-b)/e$. □

For a pair (W, D) with an action of a finite group G, a G-semi-invariant regular function U with $(W, D + \lambda \operatorname{div}(U))$ lc bounds the quotient log discrepancy of any prime divisor with center in the zero locus of U from below by λ/h, where h is the order of the character of U. In the applications, λ/h equals the limit β of Definition-Proposition 5.7.

Lemma 5.8. *Let (W, D) be a pair, G a finite group acting faithfully on W and preserving D, $Z \subset W$ a G-invariant closed subset, and U a G-semi-invariant regular function on a G-invariant open neighborhood of Z which vanishes on Z and whose character has order h. If $\lambda > 0$ and $(W, D + \lambda \operatorname{div}(U))$ is lc near Z, then*

$$\frac{a(P, W, D)}{|I_G(P)|} \geq \frac{\lambda}{h}$$

for any prime divisor P over W with $\operatorname{center}_W P \subset Z$. In particular, if $Z = \{o\}$ for a closed point o fixed by G, then $\operatorname{mld}((W \ni o, D)/G) \geq \lambda/h$.

Proof. Let V be the G-invariant open neighborhood of Z on which U is defined, $\pi : W \to W/G$ the quotient morphism, and P a prime divisor over W with $\operatorname{center}_W P \subset Z$. Since $(W, D + \lambda \operatorname{div}(U))$ is lc near Z,

$$a(P, W, D) - \lambda \operatorname{mult}_P U = a(P, W, D + \lambda \operatorname{div}(U)) \geq 0.$$

Since the character of U has order h, U^h is a G-invariant regular function on V, so $U^h = \pi^* \bar{U}$ for a regular function $\bar{U}$ on V/G, and $\bar{U}$ vanishes on $\pi(Z)$. By [Liu26, Lemma 2.6, the statement before (0), (1), and (2)], there exists a prime divisor P' over W/G with $\operatorname{center}_{W/G} P' = \pi(\operatorname{center}_W P) \subset \pi(Z)$ such that $\operatorname{mult}_P = |I_G(P)| \operatorname{mult}_{P'}$ on the rational functions on W/G. Since $\bar{U}$ is regular near $\operatorname{center}_{W/G} P'$ and vanishes on it, $\operatorname{mult}_{P'} \bar{U}$ is a positive integer. Thus

$$h \operatorname{mult}_P U = \operatorname{mult}_P(U^h) = |I_G(P)| \operatorname{mult}_{P'} \bar{U} \geq |I_G(P)|,$$

and hence

$$\frac{a(P, W, D)}{|I_G(P)|} \geq \frac{\lambda \operatorname{mult}_P U}{|I_G(P)|} \geq \frac{\lambda}{h}.$$

If $Z = \{o\}$ for a closed point o fixed by G, the inequality $\operatorname{mld}((W \ni o, D)/G) \geq \lambda/h$ follows from the formula at the end of [Liu26, Lemma 2.6]. □

6. Marked terminal sources

In this section, we introduce marked sources (Definition 6.1), together with their quadratic ranks and residual cubics. Theorem 6.2 constructs from a terminal pair a Gorenstein terminal cover with a mark and a group, which form a marked source when the support of the boundary contains the point, and proves the bounds on the orders of the mark which follow from terminality. Definition-Lemma 6.4 classifies plane cubics. Sources with several marks and real labels are treated in Sections 7, 8, 10, and 11.

Definition 6.1. Let m be a positive integer, $(W \ni o)$ a smooth threefold germ or an isolated Gorenstein terminal threefold germ, G a finite abelian group acting faithfully on W and fixing o, and h a mark on $(W \ni o)$ which vanishes at o. We say that $(W \ni o, h, G)$ is a *marked source of level* m if $(W, \frac{1}{m}\operatorname{div}(h))$ is terminal near o. The *quotient value* of a marked source $(W \ni o, h, G)$ of level m is $\operatorname{mld}((W \ni o, \frac{1}{m}\operatorname{div}(h))/G)$. If W is singular at o, the *quadratic rank* and the *residual cubic* of $(W \ni o, h, G)$ are those of $(W \ni o)$ in the sense of [Liu26, Definition-Lemma 11.1].

We require $(W, \frac{1}{m}\operatorname{div}(h))$ to be terminal, and not only klt, because terminality bounds the orders of h along fixed prime divisors over W (Theorem 6.2(2), whose bounds hold for any marked source of level m with W singular at o, as we note after its proof). Construction-Theorem 4.13(1) only states that the pairs on a numerical admission are klt, but by Construction-Theorem 4.13(2) a numerical admission keeps the orders of the marks along finitely many given prime divisors, so the bounds of Theorem 6.2(2) pass to it.

Theorem 6.2. *Let m be a positive integer, and $(T \ni c, C)$ a terminal threefold germ such that $C \in \{\frac{1}{m}, \dots, \frac{m-1}{m}\}$, $m(K_T + C)$ is Cartier, and the Cartier index r of K_T at c is at most m. Let H be a finite cyclic group acting faithfully on T, fixing c and preserving C, which acts on $\mathcal{O}_T(m(K_T + C))$ near c through a faithful character. Assume that $a := \operatorname{mld}((T \ni c, C)/H) > 0$, and let F be a prime divisor over T with center c, $I_H(F) = H$, and $a(F, T, C) = |H|a$. Then the following hold.*

(1) There exist a quasi-étale morphism $p : (W \ni o) \to (T \ni c)$ of degree r with $p^{-1}(c) = \{o\}$ and a finite abelian group $\tilde{H}$ acting faithfully on W, with an exact sequence

$$1 \to \mu_r \to \tilde{H} \to H \to 1$$

and $W/\mu_r = T$, such that W is Gorenstein terminal, $p^*C = \frac{1}{m}\operatorname{div}(h)$ for an $\tilde{H}$-semi-invariant regular function h, and $(W, \frac{1}{m}\operatorname{div}(h))$ is terminal near o. For a prime divisor $F^\#$ over W lying over F and $G := I_{\tilde{H}}(F^\#)$, the group G is cyclic, the character of an $\tilde{H}$-semi-invariant generator of ω_W at o and the character of h are jointly faithful on G, and

$$\operatorname{mld}\big((W \ni o, \tfrac{1}{m}\operatorname{div}(h))/G\big) = a.$$

Moreover, let $\Delta_{W/G}$ and $C_{T/H}$ be the boundaries of the crepant quotients of $(W, \frac{1}{m}\operatorname{div}(h))$ by G and of (T, C) by H. Then the log discrepancy with respect to $\Delta_{W/G}$ of any prime divisor over W/G with center the image of o is a positive integer multiple of the log discrepancy with respect to $C_{T/H}$ of a prime divisor over T/H with center the image of c.

(2) Assume that W is singular at o, and let $\rho \in \{1, 2, 3, 4\}$ be the quadratic rank of $(W \ni o)$. If $\rho \geq 2$, then for each irreducible component Q of the exceptional locus of the blow-up of W at o, the prime divisor E over W given by Q satisfies $a(E, W, 0) = 2$ and $\operatorname{mult}_E h \leq m - 1$. If $\rho = 1$, let $W = (f = 0) \subset (Y \ni o)$ be an $\tilde{H}$-equivariant closed embedding into a smooth fourfold germ given by [Liu26, Construction-Theorem 5.5(3)], *let x, y, z, t be $\tilde{H}$-eigenparameters of $\widehat{\mathcal{O}}_{Y,o}$ in which the quadratic part of f is a non-zero multiple of x^2, and let $w := (3, 2, 2, 2)$. Then there exists a unique prime divisor E_0 over W such that $\operatorname{mult}_{E_0}(\Phi|_W) = \operatorname{wt}_w(\Phi)$ for any $\Phi \in \mathbb{C}[[x, y, z, t]]$ such that $\operatorname{in}_w(\Phi)$ is not divisible by $\operatorname{in}_w(f)$; we say that E_0 is* given by the weights w in x, y, z, t. *Moreover, E_0*

does not depend on the choice of x, y, z, t, $a(E_0, W, 0) = 3$, *and* $\operatorname{mult}_{E_0} h \le 2m-1$. *In all cases* $\operatorname{mult}_o h \le m-1$.

Proof of Theorem 6.2. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct p, $\tilde{H}$, h, and G, and we prove the statements of (1) on quotient values.

Since (T, C) is terminal and $C \in \{\frac{1}{m}, \dots, \frac{m-1}{m}\}$, the pair (T, C) is klt. By [Liu26, Construction-Theorem 5.5(1)] for $(T \ni c)$, H, the boundary C, and F, after shrinking H-invariant neighborhoods of c, there exist a quasi-étale Galois morphism $p : (W \ni o) \to (T \ni c)$ of degree r with group μ_r and a finite abelian group $\tilde{H}$ acting faithfully on W and fixing o such that $p^{-1}(c) = \{o\}$, W is Gorenstein and terminal, $W/\mu_r = T$, there exists an exact sequence

$$1 \to \mu_r \to \tilde{H} \to H \to 1$$

compatible with p, and the character κ of an $\tilde{H}$-semi-invariant generator ω of ω_W at o is faithful on μ_r.

Since p is quasi-étale, $K_W = p^*K_T$ by the Hurwitz formula, so $K_W + p^*C = p^*(K_T + C)$ and $m(K_W + p^*C)$ is Cartier near o. Since K_W is Cartier, $mp^*C \ge 0$ is Cartier near o, and it is $\tilde{H}$-invariant because C is H-invariant. Let h' be a generator of $\mathcal{O}_W(-mp^*C)$ at o, and let ψ be the character through which $\tilde{H}$ acts on the line $\mathcal{O}_W(-mp^*C) \otimes k(o)$. Then

$$\sigma^* h' - \psi(\sigma) h' \in \mathfrak{m}_o \mathcal{O}_W(-mp^*C)$$

for any $\sigma \in \tilde{H}$, so

$$h := \frac{1}{|\tilde{H}|} \sum_{\sigma \in \tilde{H}} \psi(\sigma)^{-1} \sigma^* h'$$

satisfies $h - h' \in \mathfrak{m}_o \mathcal{O}_W(-mp^*C)$, and h generates $\mathcal{O}_W(-mp^*C)$ at o by Nakayama's lemma. Possibly shrinking W to an $\tilde{H}$-invariant neighborhood of o, we may assume that h is a regular $\tilde{H}$-semi-invariant function on W with character ψ and $\operatorname{div}(h) = mp^*C$. Thus $p^*C = \frac{1}{m}\operatorname{div}(h)$.

Let $F^{\#}$ be a prime divisor over W lying over F and $G := I_{\tilde{H}}(F^{\#})$. By [Liu26, Construction-Theorem 5.5(2)(c)], G is cyclic, and by [Liu26, Construction-Theorem 5.5(2)(f)], applied with $p^*C = \frac{1}{m}\operatorname{div}(h)$,

$$\operatorname{mld}\big((W \ni o, \tfrac{1}{m}\operatorname{div}(h))/G\big) = a.$$

Let P be a prime divisor over W/G with center the image of o, and P' a prime divisor over W lying over P. Since o is fixed by G, the center of P' on W is o, so the prime divisor P'' over T/H under P' has center the image of c. By [Liu26, Construction-Theorem 5.5(2)(h)], the number $|I_{\tilde{H}}(P')|/|G \cap I_{\tilde{H}}(P')|$ is a positive integer and

$$a\big(P, W/G, \Delta_{W/G}\big) = \frac{|I_{\tilde{H}}(P')|}{|G \cap I_{\tilde{H}}(P')|} a\big(P'', T/H, C_{T/H}\big),$$

where $\Delta_{W/G}$ and $C_{T/H}$ are as in (1). This proves the last sentence of (1).

Step 2. In this step, we prove that $(W, \frac{1}{m}\operatorname{div}(h))$ is terminal near o and that κ and ψ are jointly faithful on G.

Let R be an exceptional prime divisor over W, and R_T the prime divisor over T under R. Let Z be the center of R on W. Since p is finite, the center of R_T on T is $p(Z)$ and $\dim p(Z) = \dim Z \le 1$, so R_T is exceptional over T. Since $K_W + p^*C = p^*(K_T + C)$, the pair (T, C) is the crepant quotient of (W, p^*C) by μ_r, and [Liu26, Lemma 2.6(2)] implies

$$a\big(R, W, \tfrac{1}{m}\operatorname{div}(h)\big) = |I_{\mu_r}(R)| a(R_T, T, C) \ge a(R_T, T, C) > 1.$$

Thus $(W, \frac{1}{m}\operatorname{div}(h))$ is terminal near o.

Let s be an H-semi-invariant generator of $\mathcal{O}_T(m(K_T + C))$ at c whose character χ is faithful on H. Since p is quasi-étale and $\tilde{H}$ acts compatibly with p, the pullback p^*s is an $\tilde{H}$-semi-invariant

generator of $\mathcal{O}_W(m(K_W+p^*C))$ at o whose character $\tilde{\chi}$ is the composition of $\tilde{H}\to H$ with χ. The section $\omega^{\otimes m}h^{-1}$ also generates

$$\mathcal{O}_W(m(K_W+p^*C))=\mathcal{O}_W(mK_W+\mathrm{div}(h))$$

at o, so $e:=p^*s/(\omega^{\otimes m}h^{-1})$ is an $\tilde{H}$-semi-invariant unit at o. Since o is fixed by $\tilde{H}$, we have $(\sigma^*e)(o)=e(o)\neq 0$ for any $\sigma\in\tilde{H}$, so e is $\tilde{H}$-invariant, and

$$\tilde{\chi}=\kappa^m\psi^{-1}.$$

Since χ is faithful on H, $\mathrm{Ker}(\kappa^m\psi^{-1})=\mu_r$. Let $\sigma\in\tilde{H}$ with $\kappa(\sigma)=\psi(\sigma)=1$. Then $\sigma\in\mu_r$, and $\sigma=1$ since κ is faithful on μ_r. Thus κ and ψ are jointly faithful on $\tilde{H}$, and in particular on G. This finishes the proof of (1).

Step 3. In this step, we assume that W is singular at o, and we show that W is a hypersurface of multiplicity two in a smooth fourfold germ and that the residual cubic is non-zero if $\rho=1$.

By [Liu26, Construction-Theorem 5.5(3)], $(W\ni o)$ is an isolated cDV singularity, and there exist an $\tilde{H}$-equivariant closed embedding of W into a smooth fourfold germ $(Y\ni o)$ with $W=(f=0)$ for a regular $\tilde{H}$-semi-invariant function f, and $\tilde{H}$-semi-invariant regular parameters of Y at o. By [Rei83, (0.5)], there exist parameters x',y',z',t' of $\widehat{\mathcal{O}}_{Y,o}$ such that $f=e'(f_0(x',y',z')+t'g')$, where e' is a unit, $g'\in\widehat{\mathcal{O}}_{Y,o}$, and f_0 defines a Du Val singularity. The parameters x',y',z',t' exist in the completion of $\mathcal{O}_{Y,o}$ since the embedding $W\subset Y$ and the embedding of [Rei83, (0.5)] are minimal, as W is singular at o: lifting the images of the four parameters under an isomorphism of completions of $\mathcal{O}_{W,o}$ gives a homomorphism of power series rings in four variables that is bijective on cotangent spaces, hence an isomorphism, and it maps the equation of [Rei83, (0.5)] to a unit times f. A Du Val singularity is analytically isomorphic to one of $x^2+y^2+z^{n+1}$, $x^2+y^2z+z^{n-1}$, $x^2+y^3+z^4$, $x^2+y^3+yz^3$, and $x^2+y^3+z^5$ [Ste05, Section 2], so $\mathrm{mult}_0 f_0=2$. Hence $\mathrm{mult}_o f\leq 2$, and $\mathrm{mult}_o f=2$ as W is singular at o. Let q be the quadratic part of f. Then ρ is the rank of q [Liu26, Definition-Lemma 11.1].

Assume that $\rho=1$, and write $q=\lambda\ell^2$ for a non-zero constant λ and a linear form ℓ. We use the following observation for a power series ϕ whose quadratic part is a non-zero multiple of ℓ^2: multiplying ϕ by a unit multiplies the restriction of the cubic part of ϕ to $(\ell=0)$ by a non-zero constant, and composing ϕ with an automorphism of the power series ring transforms the restriction by the linear part of the automorphism, since all other contributions to the cubic part are multiples of ℓ. In the parameters x',y',z',t', the quadratic part of $e'(0)f_0$ is the restriction of q to $(t'=0)$, which is non-zero as $\mathrm{mult}_0 f_0=2$. Thus the quadratic part of f_0 has rank one, and f_0 is of type D or E, since the forms $x^2+y^2+z^{n+1}$ have quadratic rank at least two. The cubic part of $x^2+y^2z+z^{n-1}$ restricted to $(x=0)$ is y^2z or y^2z+z^3, and that of the three forms of type E is y^3. By the observation, the cubic part of f in x',y',z',t' is non-zero on $(\ell=0)\cap(t'=0)$, hence on $(\ell=0)$. The residual cubic of $(W\ni o)$ is, up to a linear change of variables and a non-zero scalar, the restriction of the cubic part of f to $(\ell=0)$ in any parameters [Liu26, Definition-Lemma 11.1], so it is non-zero.

Step 4. In this step, we compute log discrepancies on the blow-up of o.

Let $\pi:Y'\to Y$ be the blow-up of o, $V:=\pi^{-1}(o)\cong\mathbb{P}^3$, and W' the strict transform of W, so $W'\to W$ is the blow-up of W at o [Har77, Corollary II.7.15]. By [KM98, Lemma 2.29], $K_{Y'}=\pi^*K_Y+3V$, and $\pi^*W=W'+2V$ as $\mathrm{mult}_o f=2$. Hence $K_{Y'}+W'=\pi^*(K_Y+W)+V$, and by adjunction [KM98, Proposition 5.73],

$$K_{W'}=(\pi|_{W'})^*K_W+V|_{W'}$$

near any point of codimension one of W' at which W' is regular. The ideal of $V|_{W'}$ is $\mathfrak{m}_o\mathcal{O}_{W'}$. Therefore, for any prime divisor E on W' contained in V such that W' is regular at the generic point of E,

$$a(E,W,0)=1+\mathrm{mult}_E\,\mathfrak{m}_o.$$

The scheme $W' \cap V$ is the projectivized tangent cone $(q = 0) \subset V$ (the exceptional divisor of the blow-up is Proj of the tangent cone, defined by the initial form q of f).

Step 5. In this step, we prove (2) for $\rho \geq 2$.

Since $\rho \geq 2$, the quadric $(q = 0) \subset V$ is reduced: it is irreducible if $\rho \geq 3$ and the union of two distinct planes if $\rho = 2$. Let Q be an irreducible component of $(q = 0)$, η the generic point of Q, and $x_1, \dots, x_4$ parameters of Y at o such that the linear part of x_1 does not vanish on Q. Near η, the function x_1 generates $\mathfrak{m}_o \mathcal{O}_{Y'}$, the functions $u_j := x_j/x_1$ are regular, and the strict transform of f is

$$\frac{f}{x_1^2} = q(1, u_2, u_3, u_4) + x_1 \beta$$

for a function β regular at η. Since $V = (x_1 = 0)$ near η and Q is a reduced component of $(q = 0)$, the elements x_1 and $q(1, u_2, u_3, u_4)$ form a regular system of parameters of the regular local ring $\mathcal{O}_{Y',\eta}$ of dimension two, and so do x_1 and f/x_1^2. Thus $\mathcal{O}_{W',\eta}$ is a discrete valuation ring with uniformizer x_1. Let E be the prime divisor over W given by Q. Then $\operatorname{mult}_E \mathfrak{m}_o = \operatorname{mult}_E x_1 = 1$, so $a(E, W, 0) = 2$ by **Step 4**. Since $(W, \frac{1}{m} \operatorname{div}(h))$ is terminal near o (**Step 2**) and E has center o,

$$1 < a\big(E, W, \tfrac{1}{m} \operatorname{div}(h)\big) = 2 - \frac{1}{m} \operatorname{mult}_E h.$$

Hence $\operatorname{mult}_E h < m$, and $\operatorname{mult}_E h \leq m - 1$ as $\operatorname{mult}_E h$ is an integer.

Step 6. In this step, we prove (2) for $\rho = 1$.

By [Liu26, Lemma 11.3(1)] and its proof, which is valid for the embedding $W = (f = 0) \subset (Y \ni o)$ of **Step 3**, there exist $\tilde{H}$-eigenparameters x, y, z, t of $\widehat{\mathcal{O}}_{Y,o}$ in which the expansion of f is $x^2 + g(y, z, t)$ with $\operatorname{mult}_0 g \geq 3$, so

$$\widehat{\mathcal{O}}_{W,o} = \mathbb{C}[[x, y, z, t]]/(x^2 + g(y, z, t)).$$

The quadratic part of f is x^2, so x, y, z, t are eigenparameters as in (2). Let g_3 be the cubic part of g, so $\operatorname{in}_w(f) = x^2 + g_3$. Then g_3 is the residual cubic of $(W \ni o)$, and $g_3 \neq 0$ by **Step 3**. The quadric $(q = 0)$ is the plane $\Pi := (x = 0) \subset V$ counted twice. Let η be the generic point of Π. Since the linear parts of y, z, t restrict to coordinates on Π, the image of t generates $\mathfrak{m}_o \widehat{\mathcal{O}}_{Y',\eta}$, the elements $\tilde{x} := x/t$, $u := y/t$, and $v := z/t$ belong to $\widehat{\mathcal{O}}_{Y',\eta}$, the elements $t, \tilde{x}$ form a regular system of parameters of $\widehat{\mathcal{O}}_{Y',\eta}$, and the residues of u, v are algebraically independent over $\mathbb{C}$. We have

$$x^2 + g(y, z, t) = t^2\big(\tilde{x}^2 + t(g_3(u, v, 1) + t\gamma)\big)$$

for some $\gamma \in \widehat{\mathcal{O}}_{Y',\eta}$, and $g_3(u, v, 1)$ is a unit as $g_3 \neq 0$. Hence the completion of $\mathcal{O}_{W',\eta}$ is

$$\widehat{\mathcal{O}}_{Y',\eta}/\big(\tilde{x}^2 + t(g_3(u, v, 1) + t\gamma)\big),$$

in which t is $\tilde{x}^2$ times a unit. Thus the completion of $\mathcal{O}_{W',\eta}$ is a discrete valuation ring with uniformizer $\tilde{x}$, so $\mathcal{O}_{W',\eta}$ is a discrete valuation ring, as $\mathcal{O}_{W',\eta}$ and its completion have equal dimensions and equal embedding dimensions [Mat89, p. 157], and its valuation extends to the completion. Let E_0 be the prime divisor over W given by Π. Then E_0 is the unique prime divisor over W whose center on the blow-up W' of W at o is the support of $W' \cap V$, so E_0 depends only on $(W \ni o)$.

We extend $\operatorname{mult}_{E_0}$ to $\widehat{\mathcal{O}}_{W,o}$ through the continuous homomorphism

$$\widehat{\mathcal{O}}_{W,o} \to \widehat{\mathcal{O}}_{W',\eta}$$

(cf. [Liu26, Lemma 3.3(3)]). Since $\operatorname{mult}_{E_0} t = 2$ and $\operatorname{mult}_{E_0} \tilde{x} = 1$,

$$(\operatorname{mult}_{E_0} x, \operatorname{mult}_{E_0} y, \operatorname{mult}_{E_0} z, \operatorname{mult}_{E_0} t) = (3, 2, 2, 2), \quad \operatorname{mult}_{E_0} \mathfrak{m}_o = 2,$$

and $a(E_0, W, 0) = 3$ by **Step 4**. Any element of $\widehat{\mathcal{O}}_{W,o}$ is uniquely of the form $a_0(y, z, t) + x b_0(y, z, t)$ with $a_0, b_0 \in \mathbb{C}[[y, z, t]]$. If $a_0 \neq 0$ and $d := \operatorname{mult}_0 a_0$, then $a_0 = t^d(a_{0,d}(u, v, 1) + t\alpha)$ for some

$\alpha \in \widehat{\mathcal{O}}_{Y',\eta}$, where $a_{0,d}$ is the homogeneous part of degree d of a_0, and $a_{0,d}(u,v,1)$ is a unit. Hence $\operatorname{mult}_{E_0} a_0 = 2\operatorname{mult}_0 a_0$, and similarly $\operatorname{mult}_{E_0}(xb_0) = 3 + 2\operatorname{mult}_0 b_0$ if $b_0 \neq 0$. The integers $2\operatorname{mult}_0 a_0$ and $3 + 2\operatorname{mult}_0 b_0$ have different parities, so

$$\operatorname{mult}_{E_0}(a_0 + xb_0) = \min\{2\operatorname{mult}_0 a_0, 3 + 2\operatorname{mult}_0 b_0\},$$

where $\operatorname{mult}_0 0 := +\infty$.

We show that E_0 is given by the weights w in x, y, z, t in the sense of (2). Let $\Phi \in \mathbb{C}[[x,y,z,t]]$ such that $\operatorname{in}_w(\Phi)$ is not divisible by $\operatorname{in}_w(f) = x^2 + g_3$. By the Weierstrass division theorem, $\Phi = \Theta \cdot (x^2 + g) + a_0 + xb_0$ with $\Theta \in \mathbb{C}[[x,y,z,t]]$ and $a_0, b_0 \in \mathbb{C}[[y,z,t]]$. Here $a_0 + xb_0 \neq 0$, since otherwise $\Phi = \Theta \cdot (x^2 + g)$, so $\operatorname{in}_w(\Phi)$ is 0 or $\operatorname{in}_w(\Theta)(x^2 + g_3)$, which is divisible by $x^2 + g_3$. Monomials in y, z, t have even weight and monomials divisible by x but not by x^2 have odd weight, so $\nu_2 := \operatorname{wt}_w(a_0 + xb_0)$ is $\min\{2\operatorname{mult}_0 a_0, 3 + 2\operatorname{mult}_0 b_0\}$, and $\operatorname{in}_w(a_0 + xb_0)$ is non-zero of degree at most one in x, hence not divisible by $x^2 + g_3$. Let $\nu_1 := \operatorname{wt}_w(\Theta \cdot (x^2 + g))$. If $\nu_1 < \nu_2$, then

$$\operatorname{in}_w(\Phi) = \operatorname{in}_w(\Theta)(x^2 + g_3),$$

which is excluded. Otherwise

$$\operatorname{wt}_w(\Phi) = \nu_2 = \operatorname{mult}_{E_0}(a_0 + xb_0) = \operatorname{mult}_{E_0}(\Phi|_W),$$

because if $\nu_1 = \nu_2$, then

$$\operatorname{in}_w(\Theta)(x^2 + g_3) + \operatorname{in}_w(a_0 + xb_0) \neq 0.$$

Next, we show that E_0 is the unique prime divisor over W given by the weights w in x, y, z, t. Any element of $\widehat{\mathcal{O}}_{W,o}$ is $(a_0 + xb_0)|_W$ with $a_0, b_0 \in \mathbb{C}[[y,z,t]]$, and if $a_0 + xb_0 \neq 0$, then $\operatorname{in}_w(a_0 + xb_0)$ is not divisible by $x^2 + g_3$. Hence a prime divisor E over W given by the weights w in x, y, z, t satisfies $\operatorname{mult}_E = \operatorname{mult}_{E_0}$ on $\widehat{\mathcal{O}}_{W,o}$, hence on the function field of W, so $E = E_0$.

Finally, we show that E_0 does not depend on the eigenparameters. Let x', y', z', t' be $\tilde{H}$-eigenparameters of $\widehat{\mathcal{O}}_{Y,o}$ in which the quadratic part of f is a non-zero multiple of x'^2. Then $x' - \lambda x \in \mathfrak{m}_o^2$ for some $\lambda \neq 0$, and y', z', t' are linear combinations of y, z, t modulo $(x) + \mathfrak{m}_o^2$. Since x has weight 3 and any element of $\mathfrak{m}_o^2$ has weight at least 4, the change of parameters preserves the weights and induces a graded linear automorphism σ on the w-homogeneous parts. The w-initial form of f in x', y', z', t' is a non-zero multiple of x'^2 plus the cubic part of $f(0, y', z', t')$, and it is $\sigma(x^2 + g_3)$. Hence a prime divisor over W is given by the weights w in x', y', z', t' if and only if it is given by the weights w in x, y, z, t. Thus E_0 is the unique prime divisor over W given by the weights w in x', y', z', t', and it does not depend on the choice of eigenparameters.

Since $(W, \frac{1}{m}\operatorname{div}(h))$ is terminal near o and E_0 has center o,

$$1 < a\big(E_0, W, \tfrac{1}{m}\operatorname{div}(h)\big) = 3 - \frac{1}{m}\operatorname{mult}_{E_0} h.$$

Hence $\operatorname{mult}_{E_0} h \leq 2m - 1$.

Step 7. In this step, we prove that $\operatorname{mult}_o h \leq m - 1$.

Let $k := \operatorname{mult}_o h$, so $h \in \mathfrak{m}_o^k$. If $\rho \geq 2$ and E is as in **Step 5**, then

$$k = k \operatorname{mult}_E \mathfrak{m}_o \leq \operatorname{mult}_E h \leq m - 1.$$

If $\rho = 1$, then

$$2k = k \operatorname{mult}_{E_0} \mathfrak{m}_o \leq \operatorname{mult}_{E_0} h \leq 2m - 1,$$

so $k \leq m - 1$. □

Remark 6.3. The bound $r \leq m$ and the faithful character of H on $\mathcal{O}_T(m(K_T + C))$ are properties of the terminal models of [Liu26, Construction-Theorem 17.2(3)]: there the Cartier index of K_T at any closed point is at most m, and the character is that of the invertible sheaf $\mathcal{M}_{X'}$ of [Liu26, Construction-Theorem 17.2]. The faithful character gives the joint faithfulness in (1) (**Step 2** of the proof).

The proof of Theorem 6.2(2) uses only that $(W \ni o)$ is an isolated cDV singularity with an equivariant closed embedding as in **Step 3**, and, for the bounds on $\mathrm{mult}_E h$, $\mathrm{mult}_{E_0} h$, and $\mathrm{mult}_o h$, that $(W, \frac{1}{m}\mathrm{div}(h))$ is terminal near o. Hence Theorem 6.2(2) holds for any marked source $(W \ni o, h, G)$ of level m with W singular at o, with G in place of $\tilde{H}$ and the G-equivariant closed embedding of [Liu26, Lemma 5.4] in place of that of [Liu26, Construction-Theorem 5.5(3)], since $(W \ni o)$ is an isolated cDV singularity by [Rei83, Theorem 1.1]. The statements of Theorem 6.2(2) on $a(E, W, 0)$, on $a(E_0, W, 0)$, and on E_0 hold for any isolated cDV threefold germ $(W \ni o)$ singular at o, with the trivial group.

Sources of quadratic rank one are divided into families according to their residual cubics. The list of Definition-Lemma 6.4 is the list of cases of Sections 10 and 11.

Definition-Lemma 6.4. Let $C \subset \mathbb{P}^2$ be the zero scheme of a non-zero ternary cubic form. Then C is one of the following.

(1) A smooth cubic.
(2) An irreducible cubic with one node; we call C a *nodal cubic*.
(3) An irreducible cubic with one cusp; we call C a *cuspidal cubic*.
(4) The union of a smooth conic and a line meeting it transversally; we call C a *secant conic*.
(5) The union of a smooth conic and a tangent line; we call C a *tangent conic*.
(6) The union of three non-concurrent lines; we call C a *triangle*.
(7) The union of three concurrent lines.
(8) The zero scheme of L^2M for linearly independent linear forms L, M, that is, the union of a line counted twice and a different line; we call C a *double line*.
(9) The zero scheme of L^3 for a non-zero linear form L, that is, a line counted three times.

Proof. Let ϕ be the cubic form. If ϕ is irreducible, then C is smooth, or C has exactly one singular point, which is a node or a cusp [BK86, Section 7.3, Propositions 12 and 13]. Assume that ϕ is reducible and has no repeated factor. Then either $\phi = \phi_2 L$ for an irreducible quadratic form ϕ_2 and a linear form L, or ϕ is the product of three pairwise non-proportional linear forms. In the first case, ϕ_2 has rank three, so the conic $(\phi_2 = 0)$ is smooth, and by Bézout's theorem the line $(L = 0)$ intersects $(\phi_2 = 0)$ in two distinct points or in one point with multiplicity two, so C is a secant conic or a tangent conic. In the second case, the three lines are concurrent or form a triangle. Assume that ϕ has a repeated irreducible factor φ. Then φ^2 divides ϕ, so $2 \deg \varphi \leq 3$ and φ is linear. Thus ϕ is a non-zero multiple of L^2M for linear forms L, M which are not proportional, or of L^3 for a linear form L. □

7. Sources of quadratic rank at least three

In this section, we prove Theorems 7.1 and 7.2, reducing smooth sources and non-isolated sources of quadratic rank three to Theorem 2.8(1). By Construction-Theorem 4.13 and Proposition 7.31, any other source of quadratic rank at least three may be replaced by a node or by a hypersurface $W_N = (xy + z^2 + t^N = 0)$ with a diagonal group (an axial source, Set-up 7.3), where neither N nor the order of the group is bounded. The crepant models used for the other families have axial sources at their vertices, with groups growing along a circle, so Theorem 7.1 and the theorems on circles of this section are used repeatedly in the later sections.

Theorem 7.1. *For any finite set $\Gamma \subset [0, 1]$, the set of positive numbers*

$$\mathrm{mld}((W \ni o, D)/G),$$

where $(W \ni o)$ is a smooth threefold germ or a hypersurface threefold germ whose quadratic part has rank at least three, G is a finite group acting faithfully on W and fixing o, and $D = \sum_j c_j \mathrm{div}(h_j)$ with $c_j \in \Gamma$ and G-semi-invariant regular functions h_j such that (W, D) is klt near o, satisfies the ACC. For any DCC set $\Gamma \subset [0, 1]$, the set of positive numbers $\mathrm{mld}((W \ni o, D)/G)$, where $(W \ni o)$ is a smooth germ, a germ of quadratic rank four, or a non-isolated germ of quadratic rank three, and G and D are as in the first sentence with $c_j \in \Gamma$, satisfies the ACC.

Theorem 7.2. *For an integer $N \geq 2$, let $W_N := (xy + z^2 + t^N = 0) \subset \mathbb{A}^4$. Let $\Gamma \subset [0,1]$ be a DCC set. For any positive integer i, let $N_i \geq 3$ be an integer, let G_i be a finite diagonal abelian group preserving the equation of W_{N_i} up to a character and acting faithfully on W_{N_i}, and let D_i be a labelled boundary on $(W_{N_i} \ni 0)$ with labels in Γ such that (W_{N_i}, D_i) is klt near 0. Assume that the numbers*

$$a_i := \mathrm{mld}((W_{N_i} \ni 0, D_i)/G_i)$$

are positive. Then the following hold.

(1) If the a_i are strictly increasing, then, possibly passing to a subsequence, there exist a positive integer M and cyclic subgroups $I_i \subset G_i$ of order at most M such that I_i is the inertia group of a computing lift and

$$a_i = \mathrm{mld}((W_{N_i} \ni 0, D_i)/I_i)$$

for any i.

(2) If Γ is finite, then the set of positive numbers $\mathrm{mld}((W_N \ni 0, D)/G)$, where $N \geq 3$ is an integer, G is a finite diagonal abelian group preserving the equation of W_N up to a character and acting faithfully on W_N, and D is a labelled boundary on $(W_N \ni 0)$ with labels in Γ such that (W_N, D) is klt near 0, satisfies the ACC.

Theorem 7.2(2) needs finitely many labels only for the groups of bounded order left by Theorem 7.2(1), where Proposition 7.10(2) obtains finiteness from Theorem 2.13.

We prove Theorems 7.1 and 7.2 in Subsection 7.4, from the case theorems of Subsections 7.1, 7.2, and 7.3. The circles along which the groups of an axial source grow are named in the following set-up.

Set-up 7.3. Let Γ, e_0, w, $\mathbf{T}$, N, G, R, r, d, D, κ_j, h_j^0, I_j, τ, A, and α be as follows.

(1) $\Gamma \subset [0,1]$ is a DCC set and e_0 is a positive integer.
(2) $w = (w_x, w_y, w_z, w_t) \in \mathbb{Z}^4$ is a primitive vector with $w_x + w_y = 2w_z$, and $\mathbf{T} := \mathbf{T}_{(w)}$.
(3) $N \geq 3$ is an integer, and G is a finite diagonal abelian group preserving the equation of W_N up to a character and acting faithfully on W_N.
(4) $R := G \cap \mathbf{T}$, $r := |R|$, $d := (Nw_t - 2w_z)/r \in \mathbb{Z}$, and $[G : R] \leq e_0$.
(5) $D = \sum_j c_j \operatorname{div}(h_j)$ is a labelled boundary on $(W_N \ni 0)$ with $c_j \in \Gamma$ such that (W_N, D) is klt near 0, where, after multiplication by invariant units, each h_j is a polynomial without monomials divisible by xy.
(6) For the non-units h_j with $c_j > 0$, κ_j and h_j^0 are as in Set-up 5.1 for $\mathbf{T}$, and I_j is the least total degree in x, y, z of the monomials of h_j^0.
(7) $\tau := \sum_j c_j \kappa_j$, $A := \sum_j c_j I_j$, and $\alpha := \mathrm{mld}((W_N \ni 0, D)/G)$.

We call $(W_N \ni 0)$ with the action of G an *axial source*, and w_z the *square weight* of $\mathbf{T}$. The circle $\mathbf{T}$ is *canonical* if $w_z + w_t = 0$. If $w_x w_y \neq 0$ and $w_z < 0$, then $\mathbf{T}$ is *interior* if $w_x w_y > 0$, and *exterior* if $w_x w_y < 0$. If $w_x w_y < 0$ and $w_z \geq 0$, then $\mathbf{T}$ is *mixed*. The circle $\mathbf{T}_{(-1,-1,-1,1)}$ is the *scalar circle*. We say that α *cannot increase along* $\mathbf{T}$ (under a condition on the data) if there exists no sequence of such data, with Γ, e_0, and w fixed and $r \to +\infty$ (satisfying the condition), whose values α are positive and strictly increasing. The integers N and d may vary along such a sequence.

In Set-up 7.3(4), r divides $Nw_t - 2w_z$ because z^2 and t^N have the character of the equation on R. For interior canonical circles, d is the exponent of the coordinate u in the chart $X_d = (XY + Z^2 + u^d = 0)$ of Set-up A.17. The crepant models of Theorems 7.9 and 7.27 have the axial sources W_{dL-2} at their vertices, and for bounded d the values α satisfy the ACC under the sign conditions of Theorems 7.9(1), 7.27(3), and 7.11(2), so the theorems on circles do not fix d.

The object with which the quotient value α is compared depends on the name of the circle $\mathbf{T}$. Interior canonical circles are compared with the toric partial resolution of Construction-Proposition 7.14 along the exceptional curve of the resolution for non-negative τ and with the

plane $\mathbb{A}^2_{P,Q}$ of Theorem 7.15 for non-positive τ, exterior canonical circles with the y-root plane of Theorem 7.17 for non-negative τ and with a toric surface constructed in Appendix A for non-positive τ, and circles with non-negative square weight with a root plane or a crepant model.

Remark 7.4. In the proof of Theorem 7.2, only circles with $w_t \neq 0$ occur, since the character of t is non-trivial on the limit H^0 of Proposition 7.19(2). Replacing w by $-w$, which does not change $\mathbf{T}$, we may assume $w_t > 0$. Up to the exchange of x and y, the circles $\mathbf{T}$ with $w_t > 0$ are treated by the following results.

condition on w	result
$w_z + w_t = 0$	Corollary 7.18
$w_z < 0$, $w_x w_y > 0$, and $w_z + w_t \neq 0$	Theorem 7.21
$w_z < 0$, $w_x w_y < 0$, and $w_z + w_t \neq 0$	Theorem 7.22
$w_z < 0$, $w_x w_y = 0$, and $w_z + w_t \neq 0$	Theorem 7.23
$w_z \geq 0$ and $w \neq (0,0,0,1)$	Corollary 7.29
$w = (0,0,0,1)$	excluded by Proposition 7.19(2)

Connected limits of dimension two of the groups G are excluded by Propositions 7.19 and 7.20 as well.

7.1. **Nodes and axial sources: fixed exponents, extractions and bounded groups.** In this subsection, we state the results on nodes, on axial sources with a fixed exponent, on canonical circles with non-positive mass, on bounded groups, and on the circles $\mathbf{T}_{(0,2s,s,v)}$. The results of this subsection are proved in Appendix A.

Proposition 7.5. *Let $\Gamma \subset [0,1]$ be a DCC set. For any positive integer i, let $W_i = (q_i = 0) \subset \mathbb{A}^4$ be a node with q_i a non-degenerate quadratic form, let G_i be a finite diagonal abelian group preserving q_i up to a character and acting faithfully on W_i, and let D_i be a labelled boundary on $(W_i \ni 0)$ with labels in Γ such that (W_i, D_i) is klt near 0. Assume that the numbers*

$$a_i := \mathrm{mld}((W_i \ni 0, D_i)/G_i)$$

are positive and strictly increasing. Then, possibly passing to a subsequence and replacing G_i with the inertia group of a computing lift, which does not change a_i, the following hold after a linear change of eigenparameters.

(1) $W_i = W := (xy = z^2 + t^2)$ for any i.
(2) There exist coprime positive integers p, s and a positive integer e_0 such that, for any i, the group $R_i := G_i \cap \mathbf{T}_{(-p,p+2s,s,s)}$ has order n_i and $[G_i : R_i] = e_0$, and $n_i \to +\infty$.

Theorem 7.6. *Let p, s be coprime positive integers, e_0 a positive integer, and $\Gamma \subset [0,1]$ a DCC set. Let $W := (xy = z^2 + t^2) \subset \mathbb{A}^4$ and $\mathbf{T} := \mathbf{T}_{(-p,p+2s,s,s)}$. Then the set of positive numbers $\mathrm{mld}((W \ni 0, D)/G)$, where G is a finite diagonal abelian group preserving W and acting faithfully on W such that $[G : G \cap \mathbf{T}] \leq e_0$, and D is a labelled boundary on $(W \ni 0)$ with labels in Γ such that (W, D) is klt near 0, satisfies the ACC.*

In the proof of Theorem 7.6, the quotient value is compared with the crepant boundary on a G/R-equivariant projective toric birational morphism $Y \to W/R$, where $R = G \cap \mathbf{T}$ and Y has only simplicial toric singularities.

Theorem 7.7. *Let $\Gamma \subset [0,1]$ be a DCC set. Then the set of positive numbers $\mathrm{mld}((W \ni o, D)/G)$, where $(W \ni o)$ is a threefold node, that is, a hypersurface threefold germ whose quadratic part has rank four, G is a finite abelian group acting faithfully on W and fixing o, and D is a labelled boundary on $(W \ni o)$ with labels in Γ such that (W, D) is klt near o, satisfies the ACC.*

For a fixed N, the proof of the next theorem removes the coordinate reflections by induction on N, and along each remaining circle it passes to a crepant model at a ray of fixed lattice depth. As $N \to +\infty$, the exponent of the axial source at the vertex of the crepant model varies with N, so the later results treat one circle at a time.

Theorem 7.8. *Let $N \geq 2$ be an integer and $\Gamma \subset [0,1]$ a DCC set. Then the set of positive numbers $\mathrm{mld}((W_N \ni 0, D)/G)$, where G is a finite diagonal abelian group preserving the equation of W_N up to a character and acting faithfully on W_N, and D is a labelled boundary on $(W_N \ni 0)$ with labels in Γ such that (W_N, D) is klt near 0, satisfies the ACC.*

Theorem 7.9. *Notation as in Set-up 7.3. Assume that $w = (u, -2v-u, -v, v)$, where u, v are integers with $\gcd(u,v) = 1$ and $v > 0$, and that $\tau \leq 0$. Let $M := \lceil 2/\gamma \rceil$ for $\gamma := \min(\Gamma \backslash \{0\})$, and $M := 1$ if $\Gamma \subset \{0\}$. Let $\bar{0}$ be the image of 0 in W_N/R. Then for any even integer $L > 4M$ there exist an integer r_0 and a DCC set $\Gamma' \subset [0,1)$, depending only on Γ, u, v, and L, such that if $r \geq r_0$, then there exists a G/R-equivariant projective birational morphism $g : Y \to W_N/R$, which is an isomorphism over $(W_N/R)\backslash\{\bar{0}\}$, such that $\mathrm{Exc}(g)$ is one prime divisor E of weight*

$$\left(1 + \frac{uL}{r},\ 1 - \frac{(2v+u)L}{r},\ 1 - \frac{vL}{r},\ \frac{vL}{r}\right)$$

in x, y, z, t, and such that the crepant boundary B_Y of $(W_N, D)/R$ on Y satisfies $B_Y \geq 0$, (Y, B_Y) is klt, and

$$\mathrm{mult}_E B_Y = A + \frac{\tau L}{r} \in \Gamma'.$$

If $r \geq r_0$, then any point of $g^{-1}(\bar{0})$ has a smooth quasi-étale quotient chart, that is, an open neighborhood $U \subset Y$ with $U \cong \widetilde{U}/H$ for a smooth threefold $\widetilde{U}$ and a finite abelian group H such that the quotient morphism $\widetilde{U} \to U$ is quasi-étale, except possibly the origin of one chart, whose underlying germ is W_{dL-2} with a group of order $vL[G:R]$ and a labelled boundary with labels in Γ'. The chart of the origin is

$$V := (x_1 y_1 + z_1^2 + \lambda^{dL-2} = 0) \subset \mathbb{A}^4_{x_1, y_1, z_1, \lambda}$$

with $x = x_1 \lambda^{1+uL/r}$, $y = y_1 \lambda^{1-(2v+u)L/r}$, $z = z_1 \lambda^{1-vL/r}$, and $t = \lambda^{vL/r}$, and the group acting on V is the group of all lifts to V of the action of G/R, which contains the deck transformations of V over the chart. On V, each non-unit h_j with $c_j > 0$ equals $\lambda^{I_j + \kappa_j L/r} \mathcal{H}_j$, where $\mathcal{H}_j$ is a regular semi-invariant function on V whose monomials have non-negative integral exponents of λ. The labelled boundary on V is

$$\sum_j c_j \,\mathrm{div}(\mathcal{H}_j) + \left(A + \frac{\tau L}{r}\right) \mathrm{div}(\lambda),$$

where j runs over the non-units h_j with $c_j > 0$. Moreover, the following hold.

(1) For any integer $d_0 \geq 1$, the set of positive numbers α with $\tau \leq 0$ and $d \leq d_0$, for data as in Set-up 7.3 with $w = (u, -2v-u, -v, v)$ but without the bound $[G:R] \leq e_0$, satisfies the ACC.

(2) Any strictly increasing sequence of positive numbers α_i with $\tau_i \leq 0$ and $r_i \to +\infty$ has, possibly passing to a subsequence, $d_i \to +\infty$, and each α_i is the quotient value of a labelled boundary with labels in Γ' on $W_{d_i L - 2}$ by a group of order at most vLe_0.

Theorem 7.9 is one weighted blow-up at a fixed lattice depth L along the circle. Away from the chart of t, it is a crepant model in the sense of Definition-Proposition 4.10, and the sign condition on τ puts its exceptional coefficient in a DCC set.

For bounded groups, the quotient value on W_N is compared with the quotient value on the hypersurface W_0 of Proposition 7.10: a prime divisor of bounded log discrepancy does not see t^N when N is large.

Proposition 7.10. *Let M be a positive integer and $\Gamma \subset [0,1]$ a DCC set. Let $N \geq 2$ be an integer, G a finite diagonal abelian group of order at most M preserving the equation of W_N up to a character and acting faithfully on W_N, and D_N a labelled boundary on $(W_N \ni 0)$ with labels in Γ such that (W_N, D_N) is klt near 0. Let $W_0 := (xy + z^2 = 0) \subset \mathbb{A}^4$ with the restriction of the diagonal action of G on $\mathbb{A}^4$. Write $D_N = \sum_j c_j \,\mathrm{div}(h_j)$ with each mark h_j a polynomial without monomials divisible by xy, and let $D_0 := \sum_j c_j \,\mathrm{div}(h_j|_{W_0})$ on W_0. Then the following hold.*

(1) *There exists* $N_1 = N_1(\Gamma)$ *such that* (W_0, D_0) *is lc near* 0 *if* $N \geq N_1$, *and for any* $\lambda > 0$ *there exists* $N_0 = N_0(M, \Gamma, \lambda)$ *such that if* $N \geq N_0$, *then any prime divisor* E *over* W_N/G *or* W_0/G *with center the image of* 0 *and log discrepancy at most* λ *with respect to* $(W_N, 0)/G$ *or* $(W_0, 0)/G$ *corresponds to a prime divisor* E' *over the other quotient such that* E *and* E' *have equal log discrepancies, one computed with respect to* $(W_N, D_N)/G$ *and the other with respect to* $(W_0, D_0)/G$. *In particular,*

$$0 < \operatorname{mld}((W_N \ni 0, D_N)/G) \leq \operatorname{mld}((W_0 \ni 0, D_0)/G)$$

for N *sufficiently large depending only on* M *and* Γ.

(2) *If* Γ *is finite, then the set of positive numbers* $\operatorname{mld}((W_N \ni 0, D_N)/G)$ *for all such data and all* N *is finite.*

Theorem 7.11. *Let* $\Gamma \subset [0, 1]$ *be a DCC set. Then the following hold.*

(1) *There exist* $\eta > 0$ *and* $C > 0$, *depending only on* Γ, *such that the following holds. Let* $m \geq 2$ *be an integer,* $S_0 := (t = 0) \subset W_m$, G *a finite diagonal abelian group preserving the equation of* W_m *up to a character and acting faithfully on* S_0, D *a labelled boundary on* $(W_m \ni 0)$ *with labels in* Γ *whose marks do not vanish identically on* S_0, *and* β *a real number with* $1 - \eta < \beta < 1$ *such that* $(W_m, D + \beta \operatorname{div}(t))$ *is klt near* 0. *Then* $(W_m, D + \operatorname{div}(t))$ *and* $(S_0, D|_{S_0})$ *are lc near* 0, *and*

$$b_0 \leq \operatorname{mld}((W_m \ni 0, D + \beta \operatorname{div}(t))/G) \leq b_0 + C(1 - \beta),$$

where $b_0 := \operatorname{mld}((S_0 \ni 0, D|_{S_0})/G)$. *Moreover, the set of the numbers* b_0 *for all such data satisfies the ACC.*

(2) *Notation as in Set-up 7.3, with* $w = (0, 2s, s, v)$ *for coprime positive integers* s, v, *but without the bound* $[G : R] \leq e_0$. *Then* α *cannot increase along* $\mathbf{T}$, *and for any integer* $d_0 \geq 1$ *the set of positive numbers* α *with* $0 < d \leq d_0$ *satisfies the ACC.*

In Theorem 7.11(2), the quotient value α is compared, on a crepant model of W_N/G, with quotients of W_d and of $\mathbb{A}^3$ whose boundaries contain a coordinate hyperplane with coefficient close to one, and Theorem 7.11(1) treats the quotients of W_d.

7.2. **Canonical circles.** In this subsection, we show that α cannot increase along any canonical circle, for any sign of τ. The scalar circle $\mathbf{T}_{(-1,-1,-1,1)}$ is the canonical circle with $(a, b, v) = (1, 1, 1)$ in Construction-Proposition 7.14. The results of this subsection are proved in Appendix A.

For canonical circles the adjusted mass equals the mass, so the statements below are written in terms of τ.

Theorem 7.12. *Notation as in Set-up 7.3 with* $w = (0, -2, -1, 1)$ *or* $w = (-2, 0, -1, 1)$. *If* $w = (0, -2, -1, 1)$, *let*

$$C := \sum_j c_j \operatorname{div}\left(h_j^0(-p^2, 1, p, q)\right) \quad \text{on } \mathbb{A}^2_{p,q},$$

let $\tilde{H}$ *be the quotient of the group of pairs* (g, ξ) *with* $g \in G$ *and* $\xi^2 = g_y$ *by the subgroup*

$$\{((1, \lambda^{-2}, \lambda^{-1}, \lambda), \lambda^{-1}) : \lambda^r = 1\},$$

with $\tilde{H}$ *acting on* $\mathbb{A}^2_{p,q}$ *by* $(p, q) \to (g_z \xi^{-1} p, g_t \xi q)$, *and let* $b := \operatorname{mld}((\mathbb{A}^2 \ni 0, C)/\tilde{H})$. *Then the following hold.*

(1) α *cannot increase along* $\mathbf{T}$.

(2) *Assume that* $w = (0, -2, -1, 1)$. *Then* $|\tilde{H}| = 2[G : R]$, $\tilde{H}$ *acts faithfully on* $\mathbb{A}^2_{p,q}$, *and there exists* C_0, *depending only on* Γ *and* e_0, *such that for* $r \gg 0$ *the pair* $(\mathbb{A}^2, C)$ *is lc near* 0 *and*

$$\alpha \leq b + \frac{C_0}{r}.$$

(3) *Assume that* $w = (0, -2, -1, 1)$, $\tau \geq 0$, *and* $(\mathbb{A}^2, \theta C)$ *is lc near* 0 *for a fixed* $\theta > 1$. *Then* $\alpha \geq b$ *for* $r \gg 0$.

The upper bound in Theorem 7.12(2) is the case of the following proposition in which x has weight zero, and the proposition holds for any canonical circle on which t has non-zero weight and for any sign of τ. The name of the y-root plane in the proposition refers to the chart of W_N on which a root of y is adjoined, since p and q are functions on that chart.

Proposition 7.13. *Notation as in Set-up 7.3 with $w = (-a, -b, -v, v)$, where $v > 0$, $b > 0$, $a + b = 2v$, and $\gcd(a, v) = 1$. Let*

$$\varphi_j(p, q) := h_j^0(-p^2, 1, p, q) \quad \text{and} \quad C_{\text{plane}} := \sum_j c_j \operatorname{div}(\varphi_j) \quad \text{on } \mathbb{A}^2_{p,q},$$

let $\tilde{H}$ be the quotient of the group of pairs (g, ξ) with $g \in G$ and $\xi^b = g_y$ by the subgroup

$$\tilde{R} := \{((\lambda^{-a}, \lambda^{-b}, \lambda^{-v}, \lambda^v), \lambda^{-1}) : \lambda^r = 1\},$$

acting on $\mathbb{A}^2_{p,q}$ by $(p, q) \to (g_z \xi^{-v} p, g_t \xi^v q)$, and let

$$\beta := \operatorname{mld}\big((\mathbb{A}^2 \ni 0, C_{\text{plane}})/\tilde{H}\big).$$

We call the plane $\mathbb{A}^2_{p,q}$ with the action of $\tilde{H}$ the y-root plane. *Then the following hold.*

(1) $|\tilde{H}| = b[G : R]$, and $\tilde{H}$ acts faithfully on $\mathbb{A}^2_{p,q}$.

(2) There exist r_0 and C_0, depending only on a, b, v, Γ, e_0, such that if $r \geq r_0$, then, for any d and any sign of τ, the pair $(\mathbb{A}^2, C_{\text{plane}})$ is lc near 0,

$$\alpha \leq \beta + \frac{C_0}{r},$$

and β belongs to an ACC set depending only on a, b, v, Γ, e_0. More precisely, there exists a prime divisor over W_N/G with center the image of 0 whose log discrepancy with respect to $(W_N, D)/G$ is

$$\beta + \frac{\tau k}{re}$$

for a positive integer $e \leq be_0$ and a positive integer k bounded in terms of a, b, v, Γ, e_0.

(3) Exchanging x and y, (1) and (2) cover all canonical circles $\mathbf{T}_{(u,-2v-u,-v,v)}$ with $v > 0$.

By Remark 7.16 below, for interior canonical circles the minimum at the origin of the plane quotient of Proposition 7.13 does not bound α from below, even with a fixed margin, so for $\tau \geq 0$ the lower bound uses the minimum along the exceptional curve of a partial resolution.

Construction-Proposition 7.14. Notation as in Set-up 7.3 with $w = (-a, -b, -v, v)$, where $a, b, v > 0$, $a + b = 2v$, and $\gcd(a, v) = 1$. Let

$$S_v := (UV = (-1)^v Z^{2v}) \subset \mathbb{A}^3_{U,V,Z},$$

which is the image of $(xy + z^2 = 0) \subset \mathbb{A}^4$ under the $\mathbf{T}$-invariant morphism $(x, y, z, t) \to (x^v t^a, y^v t^b, zt)$, since $xy = -z^2$ there and $a + b = 2v$, and already the slice $t = 1$ maps onto S_v. The group $H := G/R$ acts on S_v through the characters of $x^v t^a$, $y^v t^b$, and zt. The surface S_v is the toric surface of the cone $\mathbb{R}^2_{\geq 0}$ in the lattice $\mathbb{Z}^2 + \frac{1}{2v}(1, -1)\mathbb{Z}$. Let $Y \to S_v$ be the toric partial resolution inserting the ray $(\frac{a}{2v}, \frac{b}{2v})$ in the lattice $\mathbb{Z}^2 + \frac{1}{2v}(1, -1)\mathbb{Z}$, with exceptional curve E and with charts

$$\mathbb{A}^2_{p,q}/\mu_b(-v, v), \quad U = (-1)^v p^{2v} q^a,\ V = q^b,\ Z = pq,$$

and

$$\mathbb{A}^2_{p',q'}/\mu_a(-v, v), \quad U = q'^a,\ V = (-1)^v p'^{2v} q'^b,\ Z = p'q'.$$

Then the following hold.

(1) The divisors with chart equations $h_j^0(-p^2, 1, p, q)$ and $h_j^0(1, -p'^2, p', q')$ glue to $\mathbb{Q}$-Cartier divisors $D_{Y,j} \geq 0$ on Y, and H acts faithfully on Y, preserving E and

$$\Delta_Y := \sum_j c_j D_{Y,j}.$$

(2) There exist r_0 and C, depending only on a, b, v, Γ, e_0, such that if $r \geq r_0$, then, for any d, the pair (Y, Δ_Y) is lc near E, the minimum β_Y of the log discrepancies of $(Y, \Delta_Y)/H$ over all prime divisors with center contained in E/H is attained and belongs to an ACC set depending only on a, b, v, Γ, e_0, and

$$\alpha \leq \beta_Y + \frac{C}{r}.$$

Theorem 7.15. *Notation as in Construction-Proposition 7.14. Let*

$$C_t := \sum_j c_j \operatorname{div}\left(h_j^0(P^2, -Q^2, PQ, 1)\right) \quad \text{on } \mathbb{A}^2_{P,Q},$$

let $\tilde{H}_t$ be the quotient of the group of triples (g, ξ, η) with $g \in G$, $\xi^v = g_t$, and $\eta^2 = g_x \xi^a$ by the subgroup

$$\{((\lambda^{-a}, \lambda^{-b}, \lambda^{-v}, \lambda^v), \lambda, 1) : \lambda^r = 1\},$$

with $\tilde{H}_t$ acting on $\mathbb{A}^2_{P,Q}$ by $(P, Q) \to (\eta P, g_z \xi^v Q/\eta)$, and let $\beta_t := \operatorname{mld}((\mathbb{A}^2 \ni 0, C_t)/\tilde{H}_t)$. Then the following hold.

(1) α cannot increase along $\mathbf{T}$.
(2) $|\tilde{H}_t| = 2v[G : R]$, and $\tilde{H}_t$ acts faithfully on $\mathbb{A}^2_{P,Q}$.
(3) For any $\theta > 1$ there exist r_1, depending only on $a, b, v, \Gamma, e_0, \theta$, and C', depending only on a, b, v, Γ, e_0, such that if $r \geq r_1$, then, for any d, the following hold.
  (a) If $\tau \geq 0$ and $(Y, \theta\Delta_Y)$ is lc near E, then

$$\beta_Y \leq \alpha \leq \beta_Y + \frac{C'}{r}.$$

  (b) If $\tau \leq 0$ and $(\mathbb{A}^2, \theta C_t)$ is lc near 0, then

$$\beta_t \leq \alpha \leq \beta_t + \frac{C'}{r}.$$

Remark 7.16. Let $r \geq 7$, $N := r - 2$, $G := \mu_r(-1, -3, -2, 2)$ acting on W_N, and $D := c \operatorname{div}(t)$ with $0 < c < 1$. The data (W_N, G, D) satisfy Set-up 7.3 with $w = (-1, -3, -2, 2)$, $R = G$, $d = 2$, and $\tau = 2c$, and

$$\alpha = 1 - c + \frac{2c}{r},$$

computed by the exceptional prime divisor of the weighted blow-up of W_N with weights $(1, 3, 2, r-2)$ (only the upper bound $\alpha \leq 1 - c + \frac{2c}{r}$ is used below, and equality follows from the concavity of α in c, since W_N/G is Gorenstein canonical and $\operatorname{mld}((W_N \ni 0, \operatorname{div}(t))/G) = \frac{2}{r}$ by a lattice computation). On the root chart $\eta^3 = y$, $p = z/\eta^2$, $q = t\eta^2$, the exceptional prime divisor of the weighted blow-up induces on $\mathbb{A}^2_{p,q}$ the valuation $r \operatorname{mult}_{(q=0)}$, whose center is the curve $(q = 0)$. In the notation of Proposition 7.13 with $(a, b, v) = (1, 3, 2)$, we have $C_{\text{plane}} = c \operatorname{div}(q)$, $\tilde{H} = \mu_3(1, 2)$, and $\beta = 1 - \frac{2c}{3}$, so

$$\beta - \alpha = c\left(\frac{1}{3} - \frac{2}{r}\right) > 0.$$

For $c = \frac{1}{2}$ the pair $(\mathbb{A}^2, \frac{3}{2}C_{\text{plane}})$ is klt near 0, so $\frac{3}{2}$ is a margin of $(\mathbb{A}^2, C_{\text{plane}})$ near 0 (Definition 5.2) for any r. Thus the minimum at the origin of the plane quotient of Proposition 7.13 does not bound α from below, even with a fixed margin.

For exterior canonical circles with $\tau \geq 0$, the reference is the y-root plane of Theorem 7.17. For $\tau \leq 0$, the proof in Appendix A works on a toric surface along the exceptional curve of a partial resolution. That surface does not appear in this section.

Theorem 7.17. *Notation as in Set-up 7.3 with $w = (a, -b, -v, v)$, where $a, v > 0$, $\gcd(a, v) = 1$, and $b = a + 2v$. For $\lambda \in \mathbb{C}^*$ let $\mathbf{T}(\lambda) \in \mathbf{T}$ be the element with entries λ^a, λ^{-b}, λ^{-v}, λ^v. Let*

$$C_y := \sum_j c_j \operatorname{div}\left(h_j^0(-p^2, 1, p, q)\right) \quad \text{on } \mathbb{A}^2_{p,q},$$

let

$$\tilde{H} := \{(g, \zeta) : g \in G,\ \zeta^b = g_y\}/\{(\mathbf{T}(\lambda), \lambda^{-1}) : \lambda^r = 1\},$$

acting on $\mathbb{A}^2_{p,q}$ by $(p, q) \to (g_z\zeta^{-v}p, g_t\zeta^v q)$, and let

$$\beta := \operatorname{mld}\big((\mathbb{A}^2 \ni 0, C_y)/\tilde{H}\big).$$

Then the following hold.

(1) α cannot increase along $\mathbf{T}$.
(2) $|\tilde{H}| = b[G : R]$, and $\tilde{H}$ acts faithfully on $\mathbb{A}^2_{p,q}$.
(3) For any $\theta > 1$ there exist r_0, depending only on $a, v, \Gamma, e_0, \theta$, and C_0, depending only on a, v, Γ, e_0, such that if $\tau \geq 0$, $(\mathbb{A}^2, \theta C_y)$ is lc near 0, and $r \geq r_0$, then, for any d,

$$\beta \leq \alpha \leq \beta + \frac{C_0}{r}.$$

Since $\mathbf{T}_{(w)} = \mathbf{T}_{(-w)}$, we may assume $w_t \geq 0$. After exchanging x and y, any canonical circle with $w_t > 0$ is one of the circles of Theorems 7.12, 7.15, and 7.17. The only canonical circle with $w_t = 0$ is $\mathbf{T}_{(1,-1,0,0)}$, along which α cannot increase by Theorem 8.3.

Corollary 7.18. *Notation as in Set-up 7.3 with $w = (u, -2v - u, -v, v)$, where $v > 0$ and $\gcd(u, v) = 1$. Then α cannot increase along $\mathbf{T}$. Therefore any strictly increasing sequence of positive α with w fixed and $[G : R] \leq e_0$ has bounded r, hence bounded $|G|$.*

7.3. **Limits of groups and the remaining circles.** In this subsection, we reduce strictly increasing sequences on axial sources to bounded groups or to one fixed circle, and we show that α cannot increase along the circles that are not canonical. The results of this subsection are proved in Appendix A.

Proposition 7.19. *Let $\Gamma \subset [0, 1]$ be a DCC set. Let $a_i = \operatorname{mld}((W_{N_i} \ni 0, D_i)/G_i)$ be a strictly increasing sequence of positive numbers, where $N_i \geq 3$, G_i is a finite diagonal abelian group preserving the equation of W_{N_i} up to a character and acting faithfully on W_{N_i}, and D_i is a labelled boundary with labels in Γ such that (W_{N_i}, D_i) is klt near 0. Then, after passing to a subsequence and replacing G_i by the inertia group of a computing lift, which is cyclic and keeps a_i, we have $N_i \to +\infty$ and one of the following holds.*

(1) The orders $|G_i|$ are bounded.
(2) $|G_i| \to +\infty$, the G_i converge to a closed subgroup $H \subset \{w \in \mathbb{T}^4 : w_x + w_y = 2w_z\}$ with $G_i \subset H$, $\dim H^0 \in \{1, 2\}$, $[G_i : G_i \cap H^0] = |H/H^0|$, the Lie algebra of H^0 contains no vector with all coordinates positive, and the character of t is non-trivial on H^0. If $\dim H^0 = 1$, then $H^0 = \mathbf{T}_{(u,2s-u,s,v)}$ with $u, s \in \mathbb{Z}$, $v > 0$, $\gcd(u, s, v) = 1$, and $(u, s) \neq (0, 0)$, it is not the case that $s > 0$ and $0 < u < 2s$, and $|G_i \cap H^0| \to +\infty$.

When the limit H^0 is two-dimensional, the lower bound $a_i \geq \epsilon$ forces the primitive character annihilating H^0 to have entries of one sign, and the character defines the invariant monomial Φ of Proposition 7.20. The lc of $(W_{N_i}, D_i + \lambda \operatorname{div}(\Phi))$ implies the lower bound λ/q, and a toric prime divisor over W_{N_i} implies the upper bound λ/q in the limit, so the sequence is squeezed.

Proposition 7.20. *Let $\Gamma \subset [0, 1]$ be a DCC set and $\epsilon > 0$. Let $N_i \geq 3$ be integers with $N_i \to +\infty$, G_i finite diagonal abelian groups preserving the equation of W_{N_i} up to a character and acting faithfully on W_{N_i}, and D_i labelled boundaries with labels in Γ such that (W_{N_i}, D_i) is klt near 0, and let $a_i := \operatorname{mld}((W_{N_i} \ni 0, D_i)/G_i)$. Assume that $a_i \geq \epsilon$ and that the G_i converge to a closed subgroup $H \subset \{w \in \mathbb{T}^4 : w_x + w_y = 2w_z\}$ with $\dim H^0 = 2$. Then, after passing to a subsequence,*

there exist a positive integer q, a non-constant monomial Φ in x, y, z, t, and a real number $\lambda > 0$ such that for $i \gg 0$, Φ^q is G_i-invariant, $(W_{N_i}, D_i + \lambda \operatorname{div}(\Phi))$ is lc near 0,

$$a_i \geq \frac{\lambda}{q}, \quad \text{and} \quad \limsup_i a_i \leq \frac{\lambda}{q}.$$

In particular, the a_i are not strictly increasing.

We now turn to the circles with negative square weight.

Theorem 7.21. *Notation as in Set-up 7.3 with $w = (-a, -b, -c, v)$, where $a, b, c, v > 0$, $a + b = 2c$, and $\gcd(a, c, v) = 1$. Then α cannot increase along $\mathbf{T}$, for any sign of the adjusted mass $\delta = \tau + c - v$ of D along $\mathbf{T}$ (Set-up 5.1).*

Theorem 7.22. *Notation as in Set-up 7.3 with $w = (a, -b, -c, v)$, where $a, c, v > 0$, $b = a + 2c$, and $\gcd(a, c, v) = 1$. Then α cannot increase along $\mathbf{T}$, for any sign of $v - c$ and of the adjusted mass $\delta = \tau + c - v$ of D along $\mathbf{T}$ (Set-up 5.1).*

Theorem 7.23. *Notation as in Set-up 7.3 with $w = (0, -2c, -c, v)$ or $w = (-2c, 0, -c, v)$, where $c, v > 0$ are coprime. Then α cannot increase along $\mathbf{T}$.*

The remaining circles have non-negative square weight. The mixed circles are divided by the signs of the adjusted mass δ and of A into the cases of Theorems 7.24–7.28, and Corollary 7.29 assembles all cases. Theorem 7.24 is the case $p = v = 1$ of Theorem 7.25, and the proof of Theorem 7.27 uses it on a chart.

Theorem 7.24. *Notation as in Set-up 7.3 with $w = (-1, 1, 0, 1)$, and let e be the order of the character of z on G. Assume that, for each j, the block h_j^0 contains the monomial t^{κ_j} with a non-zero coefficient, and that $\tau \leq 1$. Then the following hold.*

(1) α cannot increase along $\mathbf{T}$ for data satisfying the assumptions above.
(2) $e \leq e_0$, and for $r \gg 0$ the pair $(W_N, D + \operatorname{div}(z))$ is lc near 0, so that $\alpha \geq 1/e$.

Theorem 7.25. *Notation as in Set-up 7.3 with $w = (-p, p, 0, v)$, where $p, v > 0$ are coprime. Then α cannot increase along $\mathbf{T}$ under the condition that, for each j, the block h_j^0 contains a power of t with a non-zero coefficient (so v divides κ_j), and $\tau \leq v$.*

Theorem 7.26. *Notation as in Set-up 7.3 with $w = (-p, p + 2s, s, v)$, where $p, v > 0$, $s \geq 0$, and $\gcd(p, s, v) = 1$. Let $\delta = \tau - (s + v)$ be the adjusted mass of D along $\mathbf{T}$ (Set-up 5.1). Then α cannot increase along $\mathbf{T}$ under the condition $\delta \geq 0$.*

Theorem 7.27. *Notation as in Set-up 7.3 with $w = (-p, p + 2s, s, v)$, where $p, s, v > 0$ and $\gcd(p, s, v) = 1$. Let $\delta = \tau - (s + v)$ be the adjusted mass of D along $\mathbf{T}$ (Set-up 5.1). Then the following hold.*

(1) α cannot increase along $\mathbf{T}$ under the conditions $\delta \leq 0$ and $A = 0$. If $v > s$, then α cannot increase along $\mathbf{T}$ under the conditions $\delta \leq 0$ and $A = 0$ also without the bound $[G : R] \leq e_0$.
(2) There exists a positive integer L, depending only on Γ, p, s, v, such that if $\delta \leq 0$, $A > 0$, $Nv > 2s$, and $r \gg 0$, then W_N/G has a klt projective crepant model with boundary ≥ 0 whose only possibly singular chart has underlying germ W_{dL-2}, with a group of order $vL[G : R]$, and the crepant pullback of the boundary of the model to W_{dL-2} has coefficient $A + \delta L/r$ along the preimage of the exceptional divisor.
(3) For each integer $d_0 \geq 1$, the set of values α of the data with Γ, e_0, and w fixed, $\delta \leq 0$, and $0 < d \leq d_0$ satisfies the ACC.

In the proof of Theorem 7.28 (Appendix A), the crepant model of Theorem 7.27(2) is taken with an offset L growing with r. The coefficient of Theorem 7.27(2) stays bounded below by a positive constant, and the vertex of the model carries a growing subgroup of the scalar circle, which is the circle of Construction-Proposition 7.14 with $(a, b, v) = (1, 1, 1)$, so Theorem 7.15(1) holds for it with no condition on the sign of the mass.

condition	weight vector	result
$s \geq 0$	$(u, 2s-u, s, v)$	Corollary 7.29
$s < 0$, $v = c$	$(u, -2c-u, -c, c)$	Corollary 7.18
$s < 0$, $v \neq c$, $-2c < u < 0$	$(-a, -b, -c, v)$	Theorem 7.21
$s < 0$, $v \neq c$, $u \in \{0, -2c\}$	$(0, -2c, -c, v)$, $(-2c, 0, -c, v)$	Theorem 7.23
$s < 0$, $v \neq c$, $u > 0$	$(a, -a-2c, -c, v)$	Theorem 7.22
$s < 0$, $v \neq c$, $u < -2c$	$(u, -2c-u, -c, v)$; after exchanging x, y: $(a, -a-2c, -c, v)$	Theorem 7.22

TABLE 1. The circles $\mathbf{T}$ for $s < 0$ and $s \geq 0$ in the proof of Theorem 7.2.

Theorem 7.28. *Notation as in Set-up 7.3 with $w = (-p, p+2s, s, v)$, where $p, v > 0$, $s \geq 0$, and $\gcd(p, s, v) = 1$. Let $\delta = \tau - (s+v)$ be the adjusted mass of D along $\mathbf{T}$ (Set-up 5.1). Then α cannot increase along $\mathbf{T}$ under the conditions $A > 0$ and $\delta \leq 0$.*

Corollary 7.29. *Notation as in Set-up 7.3 with $w = (u, 2s-u, s, v)$, where $s \geq 0$, $v > 0$, $\gcd(u, s, v) = 1$, and $(u, s) \neq (0, 0)$. Then α cannot increase along $\mathbf{T}$.*

7.4. **Proofs of the class theorems.** In this subsection, we prove Theorems 7.2 and 7.1.

Proof of Theorem 7.2. (1) Suppose that the a_i are strictly increasing. By Proposition 7.19, possibly passing to a subsequence and replacing each G_i with the inertia group of a computing lift, which is cyclic and keeps a_i, we may assume that $N_i \to +\infty$ and that case (1) or case (2) of Proposition 7.19 holds. In the first case, $I_i := G_i$ and $M := \max_i |G_i|$ satisfy (1). Thus we may assume that $|G_i| \to +\infty$, and that the G_i converge to a closed subgroup $H \subset \{w \in \mathbb{T}^4 \mid w_x + w_y = 2w_z\}$ with $G_i \subset H$, $\dim H^0 \in \{1, 2\}$, and $[G_i : G_i \cap H^0] = |H/H^0|$. If $\dim H^0 = 2$, then Proposition 7.20, applied with $\epsilon := a_1$, implies that the a_i are not strictly increasing, a contradiction. Hence $\dim H^0 = 1$,

$$H^0 = \mathbf{T} := \mathbf{T}_{(u,2s-u,s,v)}, \quad u, s \in \mathbb{Z}, \quad v > 0, \quad \gcd(u, s, v) = 1, \quad (u, s) \neq (0, 0),$$

and $r_i := |G_i \cap \mathbf{T}| \to +\infty$.

We show that this case does not occur. Let $w := (u, 2s-u, s, v)$, $e_0 := |H/\mathbf{T}|$, and $R_i := G_i \cap \mathbf{T}$, so that $[G_i : R_i] = e_0$. Then the data (W_{N_i}, G_i, D_i) satisfy the conditions of Set-up 7.3 for Γ, e_0, and w. The group R_i is the subgroup of order r_i of $\mathbf{T}$, that is, $R_i = \mu_{r_i}(w)$. Since z^2 and t^{N_i} have the character of the equation $xy + z^2 + t^{N_i}$, the characters of z^2 and t^{N_i} coincide on $\mu_{r_i}(w)$, so $r_i \mid N_i v - 2s$ and $d_i := (N_i v - 2s)/r_i$ is an integer. For the normal form of the marks, write a mark h of D_i as $h = p/q$ with polynomials p, q and $q(0) \neq 0$, multiply h by the G_i-invariant unit $\prod_{g \in G_i} g^* q$, and replace xy by $-z^2 - t^{N_i}$ repeatedly. We obtain a polynomial without monomials divisible by xy, with the character of h and with divisor $\operatorname{div}(h)$ near 0, because xy, z^2, and t^{N_i} all have the character of the equation $xy + z^2 + t^{N_i}$. Since $0 < a_1 < a_2 < \cdots$ and $r_i \to +\infty$, it suffices to show that α cannot increase along $\mathbf{T}$.

If $s < 0$, let $c := -s > 0$, so that $w = (u, -2c-u, -c, v)$ and $\gcd(u, c, v) = 1$. Table 1 lists, for each value of (u, s, v), the result which shows that α cannot increase along $\mathbf{T}$. Here $(a, b) := (-u, 2c+u)$ in the third row, $a := u$ in the fifth row, and $a := -2c-u$ in the sixth row. In the first row, $(u, s) \neq (0, 0)$. In the second row, $w = (u, -2v-u, -v, v)$ and $\gcd(u, v) = \gcd(u, c, v) = 1$. In the third, fifth and sixth rows, $a, c, v > 0$ and $\gcd(a, c, v) = \gcd(u, c, v) = 1$, and in the third row $a, b > 0$ and $a + b = 2c$. In the fourth row, $\gcd(c, v) = \gcd(u, c, v) = 1$. In the last row, we first exchange x and y. The exchange is an automorphism of W_{N_i} fixing 0. It conjugates G_i to a diagonal group, keeps the normal form of the marks, the labels, r_i, $[G_i : R_i]$, d_i, and a_i, and sends $\mathbf{T}$ to $\mathbf{T}_{(-2c-u,u,-c,v)}$, whose weight vector is $(a, -a-2c, -c, v)$ with $a = -2c-u > 0$. Thus in any case α cannot increase along $\mathbf{T}$, a contradiction. This proves (1).

(2) Assume that Γ is finite, and suppose that the set in (2) does not satisfy the ACC. Then there exist integers $N_i \geq 3$, finite diagonal abelian groups G_i preserving the equation of W_{N_i} up

to a character and acting faithfully on W_{N_i}, and labelled boundaries D_i on $(W_{N_i} \ni 0)$ with labels in Γ such that (W_{N_i}, D_i) is klt near 0 and the numbers $a_i := \operatorname{mld}((W_{N_i} \ni 0, D_i)/G_i)$ satisfy $0 < a_1 < a_2 < \cdots$. By (1), possibly passing to a subsequence, there exist a positive integer M and cyclic subgroups $I_i \subset G_i$ such that $|I_i| \leq M$ and $a_i = \operatorname{mld}((W_{N_i} \ni 0, D_i)/I_i)$. Each I_i is a finite diagonal abelian group of order at most M which preserves the equation of W_{N_i} up to a character and acts faithfully on W_{N_i}, and the marks of D_i are I_i-semi-invariant. Since $N_i \geq 3$ and the labels of D_i belong to the finite set Γ, Proposition 7.10(2) implies that the a_i belong to a finite set, which contradicts the strict increase of the a_i. □

The next lemma diagonalizes an abelian action on a source of quadratic rank three and splits off the quadratic part of the equation equivariantly.

Lemma 7.30. *Let $(W \ni o)$ be a singular hypersurface threefold germ whose quadratic part has rank three, and G a finite abelian group acting faithfully on W and fixing o. Assume that G is cyclic, or that there exist ambient G-eigenparameters x, y, z, t at o in which the quadratic part of a local equation of W is $xy + z^2$. Then there exists a G-equivariant isomorphism*

$$\widehat{\mathcal{O}}_{W,o} \cong \mathbb{C}[[x, y, z, t]]/(xy + z^2 + g(t)),$$

where G acts diagonally on x, y, z, t with $\chi_x\chi_y = \chi_z^2$, and $g \in \mathbb{C}[[t]]$ is either 0 or of order at least three. Moreover, $g \neq 0$ if and only if $(W \ni o)$ is an isolated singularity, and in this case one may take $g = t^N$ with $N := \operatorname{mult}_0 g$.

Proof. **Step 1.** In this step we find G-eigencoordinates in which the quadratic part of the equation is $xy + z^2$.

Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{W,o}$. Since W is a singular hypersurface threefold germ, $\dim \mathfrak{m}/\mathfrak{m}^2 = 4$. In the second case of the assumption, let $x_1, \ldots, x_4$ be the restrictions of x, y, z, t to W. Otherwise, lift a basis of $\mathfrak{m}/\mathfrak{m}^2$ consisting of G-eigenvectors to regular functions, and project each lift onto the character of the corresponding basis vector. In both cases we obtain G-semi-invariant regular functions $x_1, \ldots, x_4$ whose classes form a basis of $\mathfrak{m}/\mathfrak{m}^2$. Let G act diagonally on $\mathfrak{R} := \mathbb{C}[[X_1, \ldots, X_4]]$, where X_i has the character of x_i. Then $X_i \to x_i$ defines a G-equivariant surjection $\mathfrak{R} \to \widehat{\mathcal{O}}_{W,o}$, whose kernel is a principal ideal (F) since W is a hypersurface germ of embedding dimension four. The group G acts on the line $(F)/\mathfrak{m}_{\mathfrak{R}}(F)$ by a character χ_f. By Nakayama's lemma, the projection of F onto χ_f is again a generator of (F), so we may assume that F is χ_f-semi-invariant. The quadratic part q of F has rank three and is χ_f-semi-invariant.

In the second case of the assumption, F is the product of a unit e of $\mathfrak{R}$ and the expansion of a local equation of W whose quadratic part is $X_1X_2 + X_3^2$. Possibly replacing F with $e(0)^{-1}F$, we may assume that $q = X_1X_2 + X_3^2$, and we let $x := X_1$, $y := X_2$, $z := X_3$, and $T := X_4$. Since xy and z^2 are monomials of the χ_f-semi-invariant series F, we have $\chi_x\chi_y = \chi_z^2 = \chi_f$.

Assume now that G is cyclic. Let $U \subset \mathfrak{R}$ be the space of linear forms, and $U' \subset U$ the image of the linear map $U^\vee \to U$ associated with q. Then $\dim U' = 3$, q is a non-degenerate quadratic form in the elements of U', and U' is G-stable since q is semi-invariant. Choose a G-eigenvector $T \in U$ spanning a G-stable complement of U', and write $U' = \bigoplus_\beta U'_\beta$, where U'_β consists of the linear forms of character β. A product $\ell\ell'$ with $\ell \in U'_\beta$ and $\ell' \in U'_\gamma$ has character $\beta\gamma$. Since q is χ_f-semi-invariant, we have

$$q = \sum_{\beta^2 = \chi_f} q_\beta + \sum_{\{\beta,\gamma\},\ \beta \neq \gamma,\ \beta\gamma = \chi_f} q_{\beta,\gamma},$$

where q_β is a quadratic form in the elements of U'_β, and $q_{\beta,\gamma}$ is a bilinear form in the elements of U'_β and of U'_γ. Since q is non-degenerate on U', each q_β is non-degenerate, each $q_{\beta,\gamma}$ is a perfect pairing, so $\dim U'_\beta = \dim U'_\gamma$, and any character β with $U'_\beta \neq 0$ satisfies $\beta^2 = \chi_f$ or $U'_{\chi_f\beta^{-1}} \neq 0$. There are two cases.

Case 1. Some character β with $U'_\beta \neq 0$ satisfies $\beta^2 \neq \chi_f$. Let $\gamma := \chi_f\beta^{-1} \neq \beta$. Then $1 \leq \dim U'_\beta = \dim U'_\gamma$ and $\dim U'_\beta + \dim U'_\gamma \leq 3$, so U'_β and U'_γ are spanned by eigenvectors x and y respectively, and $U' = U'_\beta \oplus U'_\gamma \oplus U'_{\beta'}$ for a character β' with $\dim U'_{\beta'} = 1$. As a second pair would occupy two more dimensions, $\beta'^2 = \chi_f$. Let z span $U'_{\beta'}$. Then $q = \lambda xy + \mu z^2$ with $\lambda\mu \neq 0$, and after rescaling x and z we obtain $q = xy + z^2$. Moreover $\chi_x\chi_y = \beta\gamma = \chi_f$ and $\chi_z^2 = \beta'^2 = \chi_f$.

Case 2. Any character β with $U'_\beta \neq 0$ satisfies $\beta^2 = \chi_f$. The character group of the cyclic group G is cyclic, so the equation $\beta^2 = \chi_f$ has at most two solutions. Since $\dim U' = 3$, there exists β with $\dim U'_\beta \geq 2$. Any element of U'_β is an eigenvector of character β, and q_β is a non-degenerate quadratic form in at least two variables. Hence, after a linear change of coordinates in U'_β, we have $q_\beta = xy + q'_\beta$, where $x, y \in U'_\beta$ and q'_β is a non-degenerate quadratic form in the remaining $\dim U'_\beta - 2$ coordinates. If $\dim U'_\beta = 3$, then $U' = U'_\beta$, $q = q_\beta$, and $q'_\beta = \mu z^2$ for an element $z \in U'_\beta$ and some $\mu \neq 0$. If $\dim U'_\beta = 2$, then $q'_\beta = 0$ and $U' = U'_\beta \oplus U'_{\beta'}$, where $\beta' \neq \beta$ is the other solution of $\beta'^2 = \chi_f$ and $\dim U'_{\beta'} = 1$. As $\beta\beta' \neq \beta^2 = \chi_f$, the form q has no mixed terms, so $q = q_\beta + q_{\beta'}$ and $q_{\beta'} = \mu z^2$ for a generator z of $U'_{\beta'}$ and some $\mu \neq 0$. In either subcase $q = xy + \mu z^2$ with z an eigenvector, and after rescaling z we obtain $q = xy + z^2$. Moreover $\chi_x\chi_y = \beta^2 = \chi_f$ and $\chi_z^2 = \chi_f$.

In both cases x, y, z, T are G-eigencoordinates of $\mathfrak{R}$, obtained by a G-equivariant linear change of coordinates, with

$$q = xy + z^2, \qquad \chi_x\chi_y = \chi_z^2 = \chi_f.$$

The cyclicity of G is used only for the bound on the number of solutions of $\beta^2 = \chi_f$.

Step 2. In this step we split off q by a G-equivariant automorphism of $\mathfrak{R}$ fixing T.

Let $k \geq 3$ be an integer, and assume that there exists a G-equivariant automorphism Θ_k of $\mathfrak{R}$ with identity linear part, fixing T, such that

$$\Theta_k(F) = xy + z^2 + g_k(T) + F_k,$$

where g_k is a polynomial in T of degree less than k and $F_k \in \mathfrak{m}_{\mathfrak{R}}^k$. For $k = 3$, we take $\Theta_3 := \mathrm{id}_{\mathfrak{R}}$ and $g_3 := 0$. Write the homogeneous part of degree k of F_k as $\gamma_k T^k + R_k$ with $\gamma_k \in \mathbb{C}$ and $R_k \in (x, y, z)$. Since $\partial_x q = y$, $\partial_y q = x$ and $\partial_z q = 2z$ generate the ideal (x, y, z), we may write

$$R_k = b_x\partial_x q + b_y\partial_y q + b_z\partial_z q$$

with b_x, b_y, b_z homogeneous of degree $k-1$. As Θ_k is G-equivariant, R_k is χ_f-semi-invariant. Since $\partial_i q$ has character $\chi_f\chi_i^{-1}$ for $i \in \{x, y, z\}$, the equality is preserved when each b_i is replaced with the projection of b_i onto χ_i. Let σ_k be the G-equivariant automorphism of $\mathfrak{R}$ with

$$\sigma_k(x) = x - b_x, \quad \sigma_k(y) = y - b_y, \quad \sigma_k(z) = z - b_z, \quad \sigma_k(T) = T.$$

Then

$$\sigma_k(xy + z^2) = xy + z^2 - R_k + b_xb_y + b_z^2,$$

where

$$b_xb_y + b_z^2 \in \mathfrak{m}_{\mathfrak{R}}^{2k-2} \subset \mathfrak{m}_{\mathfrak{R}}^{k+1}.$$

Moreover σ_k fixes $g_k(T)$, and

$$\sigma_k(F_k) - F_k \in \mathfrak{m}_{\mathfrak{R}}^{2k-2} \subset \mathfrak{m}_{\mathfrak{R}}^{k+1}.$$

Hence $\Theta_{k+1} := \sigma_k \circ \Theta_k$ and $g_{k+1} := g_k + \gamma_k T^k$ satisfy the assumption for $k+1$. Since σ_k is the identity modulo $\mathfrak{m}_{\mathfrak{R}}^{k-1}$, the automorphisms Θ_k converge in the $\mathfrak{m}_{\mathfrak{R}}$-adic topology to a G-equivariant automorphism Θ of $\mathfrak{R}$ with identity linear part, and

$$\Theta(F) = xy + z^2 + g(T), \qquad g := \sum_{k\geq 3} \gamma_k T^k.$$

Thus Θ induces the G-equivariant isomorphism in the lemma, with $t := T$ and $\chi_x\chi_y = \chi_z^2$.

Step 3. In this step we prove the last statement.

By the Jacobian criterion, $(W \ni o)$ is an isolated singularity if and only if the ideal generated by F and $\partial F/\partial X_1, \dots, \partial F/\partial X_4$ has finite colength in $\mathfrak{R}$. The colength of the ideal does not change under automorphisms of $\mathfrak{R}$ and under multiplication of F by units. For $xy + z^2 + g(T)$ the ideal is $(x, y, z, g(T), g'(T))$. If $g \neq 0$ has order N, it is (x, y, z, T^{N-1}), of finite colength. If $g = 0$, it is (x, y, z), of infinite colength.

Assume that $g \neq 0$, let $N := \operatorname{mult}_0 g \geq 3$, and write $g = \gamma T^N u(T)$ with $\gamma \neq 0$ and $u(0) = 1$. Since $xy + z^2 + g(T)$ is χ_f-semi-invariant and the monomials in the eigencoordinates are eigenvectors, any monomial T^k appearing in g has character χ_f. Hence $\chi_T^N = \chi_f$ and u is G-invariant. Then the power series

$$u^{1/N} := \sum_{k \geq 0} \binom{1/N}{k} (u-1)^k$$

is G-invariant, and $t := \gamma^{1/N} T u^{1/N}$, for a fixed N-th root $\gamma^{1/N}$, is a χ_T-semi-invariant power series with non-zero linear part. Thus x, y, z, t are G-eigencoordinates of $\mathfrak{R}$, and $xy + z^2 + g(T) = xy + z^2 + t^N$. □

By Lemma 7.30, an isolated source of quadratic rank three with a cyclic group action is formally the axial source W_N with a diagonal action, so Construction-Theorem 4.13 replaces it with W_N without changing the quotient values.

Proposition 7.31. *Let $(W \ni o)$ be an isolated Gorenstein terminal threefold germ of quadratic rank three, G a finite abelian group acting faithfully on W and fixing o, and $h_1, \dots, h_m$ marks. Let $c_1, \dots, c_m \geq 0$ be real numbers such that $(W, \sum_j c_j \operatorname{div}(h_j))$ is klt near o, let $c^{(1)}, \dots, c^{(q)} \in \mathbb{R}^m_{\geq 0}$ be vectors such that $(W, \sum_j c_j^{(i)} \operatorname{div}(h_j))$ is klt near o for any i, and let $P_1, \dots, P_l$ be prime divisors over W with center o. Assume that G is cyclic, or that there exist ambient G-eigenparameters x, y, z, t at o in which the quadratic part of a local equation of W is $xy + z^2$. Then there exist an integer $N \geq 3$, a faithful diagonal action of G on W_N whose tangent representation at 0, character of the equation, and character of the canonical form coincide with those of the action of G on W at o, and G-semi-invariant polynomials p_j with the characters of h_j, such that $(W_N, \sum_j c_j \operatorname{div}(p_j))$ is klt near 0 and $(W_N, (p_j)_j)$ is a numerical admission of $(W, (h_j)_j)$ in the sense of Construction-Theorem 4.13 for the vectors $(c_j)_j, c^{(1)}, \dots, c^{(q)}$ and the prime divisors $P_1, \dots, P_l$. If moreover x', y', z', t' are ambient G-eigenparameters at o in which the quadratic part of a local equation of W is $x'y' + z'^2$, then, possibly after exchanging x and y, the coordinates x, y, z, t of W_N have the characters of x', y', z', t', respectively.*

Proof. Since G is abelian, there exist G-semi-invariant regular functions $x_1, \dots, x_4$ whose classes form a basis of $\mathfrak{m}/\mathfrak{m}^2$ (lift a basis of G-eigenvectors of $\mathfrak{m}/\mathfrak{m}^2$ and project each lift onto the character of the corresponding basis vector), where $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}_{W,o}$. The functions $x_1, \dots, x_4$ define a G-equivariant closed embedding of W into the smooth fourfold germ $(Y \ni o) := (\mathbb{A}^4 \ni 0)$, with the diagonal action of G in which the coordinates have the characters of the x_i. The ideal of W in $\mathcal{O}_{Y,o}$ is principal, and by Nakayama's lemma, the projection of a generator onto the character of G on the ideal modulo $\mathfrak{m}_{Y,o}$ times the ideal is again a generator. Thus $W = (f = 0)$ for a regular G-semi-invariant function f on Y, of character χ_f. Lifting each h_j to a regular function on Y and projecting the lift onto the character of h_j, we may regard $h_1, \dots, h_m$ as G-semi-invariant regular functions on Y.

Let $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_4]]$, and let $\hat{f}$ and $\hat{h}_j$ be the expansions of f and h_j in $\mathfrak{R}$, so $\widehat{\mathcal{O}}_{W,o} = \mathfrak{R}/(\hat{f})$. Since $(W \ni o)$ is an isolated singularity, Lemma 7.30 implies that there exist an integer $N \geq 3$ and a G-equivariant isomorphism

$$\phi : \mathfrak{R}/(\hat{f}) \to \mathbb{C}[[x, y, z, t]]/(f_N), \qquad f_N := xy + z^2 + t^N,$$

where G acts diagonally on x, y, z, t. As W is singular, $\hat{f} \in (x_1, \dots, x_4)^2$, and $f_N \in (x, y, z, t)^2$. By Lemma 3.8, there exist a G-equivariant isomorphism $\Phi : \mathfrak{R} \to \mathbb{C}[[x, y, z, t]]$ inducing ϕ and a

G-invariant unit v_0 of $\mathbb{C}[[x,y,z,t]]$ such that $\Phi(\hat{f}) = v_0 f_N$. Moreover, f_N has the character χ_f, and the characters of $x_1,\dots,x_4$ are those of x,y,z,t up to order. Possibly renumbering the x_i, we may assume that x_1,x_2,x_3,x_4 have the characters of x,y,z,t respectively, and we identify $\mathbb{C}[[x,y,z,t]]$ with $\mathfrak{R}$ by $x = x_1$, $y = x_2$, $z = x_3$, and $t = x_4$. Here x,y,z,t are the coordinates given by Lemma 7.30. The eigenparameters in the assumption of the proposition enter only through that lemma. Then Φ is a G-equivariant automorphism of $\mathfrak{R}$, and $W_N = (f_N = 0) \subset \mathbb{A}^4$ carries the diagonal action of G in which the coordinates have the characters of the x_i.

Since W is klt near o and the h_j are non-zero, there exists a real number $\epsilon > 0$ such that $(W, \epsilon \sum_j \operatorname{div}(h_j))$ is klt near o. We apply Construction-Theorem 4.13 with $n = 4$, the embedding $W \subset Y$ and the eigenparameters $x_1,\dots,x_4$ in (i) and (ii), the marks $h_1,\dots,h_m$ with the positive vector $(\epsilon,\dots,\epsilon)$ in (iii), the vectors $(c_1,\dots,c_m), c^{(1)},\dots,c^{(q)}$ in (iv), the prime divisors $P_1,\dots,P_l$ in (v), and, in (vi), the automorphism Φ, the invariant unit v_0^{-1} and the units $u_j := 1$. For sufficiently large truncation orders of the marks, let p_j be the sum of the monomials of $\Phi(\hat{h}_j)$ of degree less than the truncation order. Since Φ is G-equivariant, p_j is a G-semi-invariant polynomial with the character of h_j, and p_j satisfies the congruence of Construction-Theorem 4.13 with $u_j = 1$. Let $f_0 := v_0^{-1}\Phi(\hat{f}) = f_N$. Then f_0 satisfies the congruence of Construction-Theorem 4.13 for any truncation order of the equation, and $W_0 = W_N$. By Construction-Theorem 4.13(1) and (2), G acts faithfully on W_N, $(W_N, \sum_j c_j \operatorname{div}(p_j))$ is klt near 0, and $(W_N, (p_j)_j)$ is a numerical admission of $(W, (h_j)_j)$: for the vector $(c_j)_j$ and for each vector $c^{(i)}$, the quotient minimal log discrepancies by any subgroup of G coincide, and the prime divisors $P_1,\dots,P_l$ correspond.

The coordinates of $\mathbb{A}^4$ have the characters of $x_1,\dots,x_4$, so the diagonal action of G on W_N has the tangent representation of W at o, and f_N has the character χ_f of f. The canonical forms of W at o and of W_N at 0 are the residues of $dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4/f$ and of $dx \wedge dy \wedge dz \wedge dt/f_N$. The characters of both residues are determined by the characters of the coordinates and of the equations, hence coincide.

Finally, let x',y',z',t' be as in the last statement. By Construction-Theorem 4.13(3), there is a G-equivariant isomorphism of the completed local rings of W at o and of W_N at 0. It induces a G-equivariant isomorphism Λ of the cotangent spaces and an isomorphism of the tangent cones, so $\operatorname{Sym}^2 \Lambda$ maps $x'y' + z'^2$ to a non-zero multiple of $xy + z^2$. The span of x',y',z' is the smallest subspace whose second symmetric power contains $x'y' + z'^2$, and the span of x,y,z is the one for $xy + z^2$, so Λ maps the first span onto the second. Hence t has the character of t', and the characters ψ_x, ψ_y, ψ_z of x,y,z are those of x',y',z' up to order. The characters of the equations coincide, so

$$\chi_{x'}\chi_{y'} = \chi_{z'}^2 = \psi_x\psi_y = \psi_z^2.$$

If $\psi_z = \chi_{x'}$, then $\{\psi_x,\psi_y\} = \{\chi_{y'},\chi_{z'}\}$, and $\chi_{y'}\chi_{z'} = \chi_{x'}\chi_{y'}$ gives $\chi_{z'} = \chi_{x'} = \psi_z$. The case $\psi_z = \chi_{y'}$ is symmetric. Thus $\psi_z = \chi_{z'}$ and $\{\psi_x,\psi_y\} = \{\chi_{x'},\chi_{y'}\}$. The exchange of x and y preserves f_N and the diagonal action, so possibly after it, x,y,z,t have the characters of x',y',z',t'. □

Proof of Theorem 7.1. Let $\Gamma \subset [0,1]$ be a DCC set, $(W \ni o)$ a smooth threefold germ or a hypersurface threefold germ whose quadratic part has rank at least three, G a finite group acting faithfully on W and fixing o, and $D = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$ and G-semi-invariant regular functions h_j such that (W,D) is klt near o. Assume that $a := \operatorname{mld}((W \ni o, D)/G) > 0$.

Step 1. In this step we reduce to the case when G is cyclic.

Let F be a computing lift for $(W \ni o, D)/G$ (a computing lift exists, since the minimal log discrepancy of the klt pair $(W,D)/G$ at the image of o is attained by a prime divisor E [Liu26, Lemma 3.5], and a prime divisor over W lying over E exists, has center o, and computes the quotient value [Liu26, Lemma 2.6(2) and (3)]), and $I := I_G(F)$. Then I is cyclic [Liu26, Lemma 2.6(1)]. The image of the center of F on W/I is the image of o, so [Liu26, Lemma 3.1(1)], applied with $X' = Y = W$, f the identity and $H = I$, implies that $\operatorname{mld}((W \ni o, D)/I) = a$. The

group I acts faithfully on W and fixes o, and the h_j are I-semi-invariant. Possibly replacing G with I, we may assume that G is cyclic. In particular, G is abelian.

Step 2. In this step we treat smooth germs and germs of quadratic rank four.

If $(W \ni o)$ is smooth, then a belongs to the set of Theorem 2.8(1). If the quadratic part of $(W \ni o)$ has rank four, then $(W \ni o)$ is a node, D is a labelled boundary on $(W \ni o)$ with labels in Γ, and a belongs to the set of Theorem 7.7. Both sets satisfy the ACC for any DCC set $\Gamma \subset [0,1]$.

Step 3. In this step we treat isolated singularities of quadratic rank three.

Assume that the quadratic part of $(W \ni o)$ has rank three and that $(W \ni o)$ is an isolated singularity. Choose coordinates x, y, z, t of a smooth fourfold germ containing W and a local equation F' of W whose quadratic part does not involve t, and write $F' = f_3(x,y,z) + tF_1$ with $f_3 := F'(x,y,z,0)$. The quadratic part of f_3 is non-degenerate, so $(f_3 = 0) \subset \mathbb{A}^3$ is an A_1 point and $(W \ni o)$ is a compound Du Val point [Rei83, (0.5)]. As $(W \ni o)$ is isolated, it is Gorenstein terminal [Rei83, Theorem 1.1]. By Proposition 7.31, applied to W, the cyclic group G, the marks h_j and the labels c_j, there exist an integer $N \geq 3$, a faithful diagonal action of G on W_N which preserves the equation of W_N up to the character of a G-semi-invariant local equation of W at o, and G-semi-invariant polynomials p_j such that $(W_N, \sum_j c_j \operatorname{div}(p_j))$ is klt near 0 and

$$\operatorname{mld}\Big(\big(W_N \ni 0, \textstyle\sum_j c_j \operatorname{div}(p_j)\big)/G\Big) = \operatorname{mld}((W \ni o, D)/G) = a.$$

Since $c_j \in \Gamma$, a belongs to the set of Theorem 7.2(2), which satisfies the ACC when Γ is finite.

Step 4. In this step we assume that the quadratic part of $(W \ni o)$ has rank three and that $(W \ni o)$ is not an isolated singularity, and we replace (W, D) with a polynomial pair on $V_0 := (xy + z^2 = 0) \subset \mathbb{A}^4$ without changing the quotient minimal log discrepancies.

Since G is cyclic by **Step 1** and $(W \ni o)$ is a singular hypersurface threefold germ of quadratic rank three, the hypotheses of Lemma 7.30 hold, and its last statement gives $g = 0$ since $(W \ni o)$ is not an isolated singularity. Hence there exists a G-equivariant isomorphism

$$\phi : \widehat{\mathcal{O}}_{W,o} \to \mathbb{C}[[x,y,z,t]]/(xy + z^2) = \widehat{\mathcal{O}}_{V_0,0},$$

where G acts diagonally on x, y, z, t with $\chi_x \chi_y = \chi_z^2$. We let G act diagonally on $\mathbb{A}^4$ with the characters $\chi_x, \chi_y, \chi_z, \chi_t$. Then $xy + z^2$ is G-semi-invariant of character $\chi_f := \chi_z^2$, and G acts on V_0. The action of G on V_0 is faithful: an element acting trivially on V_0 acts trivially on $\widehat{\mathcal{O}}_{V_0,0}$, hence on $\widehat{\mathcal{O}}_{W,o}$ by ϕ and on $\mathcal{O}_{W,o} \subset \widehat{\mathcal{O}}_{W,o}$, so it is the identity. By Lemma 3.16, V_0 is klt near 0. Let $\mathfrak{m}_0$ be the maximal ideal of $\widehat{\mathcal{O}}_{V_0,0}$ and $\widehat{V}_0 := \operatorname{Spec} \widehat{\mathcal{O}}_{V_0,0}$.

Construction-Theorem 4.13 does not apply here, since its condition (i) requires $(W \ni o)$ to be smooth or an isolated Gorenstein terminal germ, and $(W \ni o)$ is not isolated. We run its truncation argument on V_0 directly.

We compute on completions. By Lemma 3.6 and [Liu26, Lemma 2.6], for a normal klt germ $(X \ni x)$ with a faithful action of a subgroup $K \subset G$ fixing x and a K-invariant $\mathbb{R}$-divisor $B = \sum_j c_j \operatorname{div}(f_j)$ with (X, B) klt near x, $\operatorname{mld}((X \ni x, B)/K)$ is the infimum of $a(E, X, B)/|I_K(E)|$ over the prime divisors E over $\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$ with center the closed point, and $\operatorname{lct}(X \ni x; \prod_j (f_j)^{c_j})$ is the lc threshold of the pullback to $\operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$ [dFEM11, Proposition 2.11(ii)]. For a prime divisor E over $\widehat{V}_0$ with center the closed point, a subgroup $K \subset G$, and non-zero elements $f_1, \dots, f_m$ of $\widehat{\mathcal{O}}_{V_0,0}$, let

$$A(E) := a(E, \widehat{V}_0, 0), \qquad s_K(E; f_1, \dots, f_m) := \frac{A(E) - \sum_j c_j E(f_j)}{|I_K(E)|},$$

where $E(f_j)$ is the order of f_j along E. Let $b_j := \phi(\hat{h}_j)$, where $\hat{h}_j$ is the image of h_j in $\widehat{\mathcal{O}}_{W,o}$. Then b_j is a non-zero G-semi-invariant element with the character of h_j, and, since ϕ is G-equivariant,

for any subgroup $K \subset G$,

$$\mathrm{mld}((W \ni o, D)/K) = \inf_E s_K(E; b_1, \dots, b_m).$$

Let $\tilde{b}_j \in \mathbb{C}[[x,y,z,t]]$ be a lift of b_j with the character of h_j, obtained by projecting any lift onto the character of h_j. For a positive integer L, let $p_{j,L}$ be the sum of the monomials of $\tilde{b}_j$ of degree less than L. Then $p_{j,L}$ is a G-semi-invariant polynomial with the character of h_j, and

$$p_{j,L} \equiv b_j \pmod{\mathfrak{m}_0^L}. \tag{7.1}$$

If h_j is a unit, then $p_{j,L}$ has the non-zero constant term $b_j(0)$, hence is a unit near 0, for any $L \geq 1$. Since $b_j \neq 0$ and $\bigcap_L \mathfrak{m}_0^L = 0$, we have $p_{j,L}|_{V_0} \neq 0$ for any $L \gg 0$.

Let J be the set of indices j with $c_j > 0$ and $h_j(o) = 0$. For $j \notin J$, either $c_j = 0$, or h_j is a unit, in which case b_j is a unit and $p_{j,L}$ is a unit near 0. Hence

$$\sum_j c_j E(b_j) = \sum_{j \in J} c_j E(b_j) \quad \text{and} \quad \sum_j c_j E(p_{j,L}) = \sum_{j \in J} c_j E(p_{j,L})$$

for any prime divisor E over $\widehat{V}_0$. If $J = \emptyset$, then $D = 0$ near o and $\sum_j c_j \operatorname{div}(p_{j,L}) = 0$ near 0, so for $L \gg 0$ the minimal log discrepancies of the quotients by any subgroup coincide by ϕ. Assume that $J \neq \emptyset$, and let $c_{\min} := \min_{j \in J} c_j > 0$. Since (W, D) is klt near o, the log canonical threshold $\mathrm{lct}(W \ni o; \prod_{j \in J}(h_j)^{c_j})$ is computed by finitely many prime divisors on a log resolution, and is greater than 1. Hence there exists a real number $\delta > 0$ such that

$$\mathrm{lct}\Big(\widehat{V}_0; \prod_{j \in J}(b_j)^{c_j}\Big) > 1 + 2\delta.$$

By (7.1), $(b_j) + \mathfrak{m}_0^L = (p_{j,L}) + \mathfrak{m}_0^L$ for any j. Since V_0 is klt near 0 and the ideals (b_j) and $(p_{j,L})$, $j \in J$, of $\widehat{\mathcal{O}}_{V_0,0}$ are proper, [dFEM11, Proposition 2.19(ii)], applied to V_0 at 0, whose spectrum of the completed local ring is $\widehat{V}_0$, with $n = 3$, $\epsilon := \delta$, the exponents c_j, $j \in J$, and $d := L$, implies that for any $L \geq 3/(\delta c_{\min})$ such that $p_{j,L}|_{V_0} \neq 0$ for $j \in J$,

$$\mathrm{lct}\Big(\widehat{V}_0; \prod_{j \in J}(p_{j,L})^{c_j}\Big) > 1 + \delta.$$

Since $A(E) > 0$ for any E, we obtain, for any prime divisor E over $\widehat{V}_0$ with center the closed point,

$$A(E) > (1+\delta)\sum_{j \in J} c_j E(b_j) \quad \text{and} \quad A(E) > (1+\delta)\sum_{j \in J} c_j E(p_{j,L}). \tag{7.2}$$

In particular, $(V_0, \sum_j c_j \operatorname{div}(p_{j,L}))$ is klt near 0.

Fix a prime divisor E_* over $\widehat{V}_0$ with center the closed point, and a real number

$$C > A(E_*) - \sum_j c_j E_*(b_j).$$

Choose $L \geq 3/(\delta c_{\min})$ such that $p_{j,L}|_{V_0} \neq 0$ for any j and

$$\delta c_{\min} L > C|G|, \qquad L \cdot E_*(\mathfrak{m}_0) > E_*(b_j) \text{ for any } j \text{ with } h_j(o) = 0.$$

Let $p_j := p_{j,L}$ and $D_0 := \sum_j c_j \operatorname{div}(p_j)$. Let $K \subset G$ be a subgroup, and let S_0 be the set of prime divisors E over $\widehat{V}_0$ with center the closed point such that $E(b_j) \neq E(p_j)$ for some $j \in J$. If $E \notin S_0$, then $s_K(E; b_1, \dots, b_m) = s_K(E; p_1, \dots, p_m)$. If $E \in S_0$ and $j \in J$ satisfies $E(b_j) \neq E(p_j)$, then (7.1) implies that $E(b_j) \geq L \cdot E(\mathfrak{m}_0) \geq L$ and $E(p_j) \geq L$. By (7.2),

$$A(E) - \sum_j c_j E(b_j) > \delta \sum_{j \in J} c_j E(b_j) \geq \delta c_{\min} L > C|G|,$$

and, by the second inequality of (7.2) and $E(p_j) \geq L$,

$$A(E) - \sum_j c_j E(p_j) > \delta \sum_{j \in J} c_j E(p_j) \geq \delta c_{\min} L > C|G|.$$

Since $|I_K(E)| \leq |G|$, we have $s_K(E; b_1, \dots, b_m) > C$ and $s_K(E; p_1, \dots, p_m) > C$. By the choice of L, $E_*(p_j) = E_*(b_j)$ for any j, so $E_* \notin S_0$ and

$$s_K(E_*; b_1, \dots, b_m) = s_K(E_*; p_1, \dots, p_m) \leq A(E_*) - \sum_j c_j E_*(b_j) < C.$$

Hence both infima $\inf_E s_K(E; b_1, \dots, b_m)$ and $\inf_E s_K(E; p_1, \dots, p_m)$ are less than C, so they are the infima over the prime divisors $E \notin S_0$, where $s_K(E; b_1, \dots, b_m) = s_K(E; p_1, \dots, p_m)$. Therefore

$$\operatorname{mld}((W \ni o, D)/K) = \operatorname{mld}((V_0 \ni 0, D_0)/K)$$

for any subgroup $K \subset G$. In particular, $a = \operatorname{mld}((V_0 \ni 0, D_0)/G)$.

Step 5. In this step we pass from (V_0, D_0) to a smooth threefold.

Let $S := \mathbb{A}^3_{P,Q,t}$ and

$$\rho : S \to V_0, \qquad (P, Q, t) \to (P^2, -Q^2, PQ, t).$$

Then ρ is the quotient of S by μ_2 acting by $(P, Q, t) \to (-P, -Q, t)$, since the ring of invariants $\mathbb{C}[P^2, Q^2, PQ, t]$ is $\mathbb{C}[x, y, z, t]/(xy + z^2)$ by $x = P^2$, $y = -Q^2$, and $z = PQ$. The fixed locus of μ_2 is the t-axis, so ρ is étale in codimension one. Let

$$\widetilde{G} := \{(g, \lambda) \in G \times \mathbb{C}^* \mid \lambda^2 = \chi_x(g)\},$$

an abelian group of order $2|G|$. Let $(g, \lambda) \in \widetilde{G}$ act on S by multiplying the coordinates P, Q, and t by λ, $\chi_z(g)\lambda^{-1}$, and $\chi_t(g)$, respectively. Since $\chi_x\chi_y = \chi_z^2$, it multiplies P^2, $-Q^2$, PQ, and t by $\chi_x(g)$, $\chi_y(g)$, $\chi_z(g)$, and $\chi_t(g)$, so ρ is equivariant with respect to the projection $\widetilde{G} \to G$. The action of $\widetilde{G}$ on S is faithful: if (g, λ) acts trivially on S, then g acts trivially on V_0, so $g = 1$, and then $\lambda = 1$. The projection $\widetilde{G} \to G$ is surjective with kernel $\{(1, \pm 1)\}$, which acts on S as μ_2. Hence $S/\widetilde{G} = V_0/G$.

Let $H_j := \rho^* p_j = p_j(P^2, -Q^2, PQ, t)$ and $B_S := \sum_j c_j \operatorname{div}(H_j)$. The H_j are non-zero regular $\widetilde{G}$-semi-invariant functions. Since ρ is étale in codimension one, $K_S + B_S = \rho^*(K_{V_0} + D_0)$, that is, $(S \ni 0, B_S)/\mu_2 = (V_0 \ni 0, D_0)$. Thus (S, B_S) is klt near 0, since

$$a(F, S, B_S) = |I_{\mu_2}(F)| \cdot a(E, V_0, D_0)$$

for any prime divisor F over S and the prime divisor E over V_0 under it [Liu26, Lemma 2.6(0) and (2)]. Composing the crepant pullbacks by $S \to S/\mu_2 = V_0$ and by $V_0 \to V_0/G$, and using $\rho^{-1}(0) = \{0\}$, we obtain

$$(S \ni 0, B_S)/\widetilde{G} = (V_0 \ni 0, D_0)/G,$$

so $a = \operatorname{mld}((S \ni 0, B_S)/\widetilde{G})$. Since $c_j \in \Gamma$, a belongs to the set of Theorem 2.8(1), which satisfies the ACC for any DCC set $\Gamma \subset [0, 1]$.

Step 6. In this step we conclude the proof.

A singular hypersurface threefold germ whose quadratic part has rank at least three has quadratic rank four, or quadratic rank three and an isolated singularity, or quadratic rank three and a non-isolated singularity. By **Steps 1–5**, any positive value $\operatorname{mld}((W \ni o, D)/G)$ in Theorem 7.1 belongs to the union of the sets of Theorem 2.8(1), Theorem 7.7 and Theorem 7.2(2). If Γ is finite, each of the three sets satisfies the ACC, hence so does the union. If $(W \ni o)$ is smooth, of quadratic rank four, or of quadratic rank three with a non-isolated singularity, then **Steps 2, 4, and 5** show that the value belongs to the union of the sets of Theorem 2.8(1) and Theorem 7.7, which satisfies the ACC for any DCC set $\Gamma \subset [0, 1]$. □

8. Sources of quadratic rank two

In this section, we treat sources of quadratic rank two, as a complement to Section 7 on quadratic rank at least three. By [Liu26, Theorem 9.1(3)] and Construction-Theorem 4.13, a source of quadratic rank two becomes, after numerical admission, a polynomial hypersurface in $\mathbb{A}^4$ defined by $xy + g(z,t)$ or by $x^2 + y^2 + g(z,t)$, on which G acts diagonally (Theorem 8.11(1)), and the difficulties are that the split radical order can be unbounded, that the group can exchange the hyperbolic coordinates, and that along a circle whose weights on z and t have opposite signs, no single reference pair bounds the quotient value from below (Theorems 8.6 and 8.13). We formulate the theorems on circles in this section for labelled boundaries with labels in a DCC set, since Section 11 uses them in that generality.

We use the following invariant of a source of quadratic rank two.

Definition 8.1. Let $(W \ni o, h, G)$ be a marked source of level m of quadratic rank two, G-equivariantly embedded in a smooth fourfold germ $(M \ni o)$. Since W is klt, by [Liu26, Theorem 9.1(3)] with $B = 0$ (a local equation of W in M can be taken G-semi-invariant by character projection) there exist formal coordinates x, y, z, t of M at o on which G acts diagonally and in which the completed local equation of W is $xy + \hat{g}(z,t)$ or $x^2 + y^2 + \hat{g}(z,t)$.

In the second case we replace x and y by $x + \sqrt{-1}y$ and $x - \sqrt{-1}y$. Here the monomials x^2 and y^2 have the character of the equation, so the characters of x and y have equal squares, and G acts monomially on the pair $\{x + \sqrt{-1}y, x - \sqrt{-1}y\}$. In both cases the equation is then $xy + \hat{g}(z,t)$, G acts diagonally on z, t, and G acts monomially on the pair $\{x, y\}$.

The multiplicity $\operatorname{mult}_0 \hat{g}$ is the *split radical order* of W. It does not depend on the choice of the coordinates (see below).

Proof of the claims in Definition 8.1. We prove that the coordinates x, y, z, t exist, that the split radical order n satisfies $n \geq 3$, and that a general hyperplane section of W through o is a Du Val singularity of type A_{n-1}. In particular, the split radical order does not depend on the choice of the coordinates.

The G-equivariant morphisms of [Liu26, Theorem 9.1(3)] are étale at the base point, so they induce G-equivariant isomorphisms of the completed local rings. Since $T_oW = T_oM$ and the restriction

$$\widehat{\mathcal{O}}_{M,o} \to \widehat{\mathcal{O}}_{W,o}$$

is a G-equivariant surjection, G-semi-invariant lifts of x, y, z, t to $\widehat{\mathcal{O}}_{M,o}$ with the characters of x, y, z, t form a regular system of parameters.

Let $n := \operatorname{mult}_0 \hat{g}$. Then $n \geq 3$, since the quadratic rank is two. Let $\eta \in \mathfrak{m}_o$ be an element with general linear part. Solving $\eta = 0$ for t, we write $t = \varphi(x, y, z)$ on $(\eta = 0)$, where the linear part of $\varphi(0, 0, z)$ is cz for a general $c \in \mathbb{C}$. In the coordinates x, y, z, the equation of $(\eta = 0)$ is

$$f := xy + \hat{g}(z, \varphi(x, y, z)) = xy + xA + yB + C(z),$$

where A and B are power series of order at least $n - 1$, and $C(z) := \hat{g}(z, \varphi(0, 0, z))$ has order exactly n, since c is general. Since $xA + yB$ has order at least $n \geq 3$, by the implicit function theorem the partial derivatives of f in x and y define a curve Σ with equations $x = \xi(z)$ and $y = \upsilon(z)$, and ξ and υ have order at least $n - 1$, since A and B do. The function $f - f(\xi, \upsilon, z)$ and its partial derivatives in x and y vanish on Σ, so $f - f(\xi, \upsilon, z)$ is a quadratic form Φ in $x - \xi$ and $y - \upsilon$ with coefficients in $\mathbb{C}[[x, y, z]]$ whose value at 0 is $(x - \xi)(y - \upsilon)$. The discriminant of Φ is a unit, so $\Phi = L_1 L_2$ for linear forms L_1, L_2 in $x - \xi$ and $y - \upsilon$ with coefficients in $\mathbb{C}[[x, y, z]]$. Taking $x'' := L_1$ and $y'' := L_2$, we obtain formal coordinates x'', y'', z with $f = x''y'' + f(\xi, \upsilon, z)$, as in the splitting lemma (cf. [GLS07, Chapter I, Theorem 2.47]). The difference

$$f(\xi, \upsilon, z) - C(z) = \xi\upsilon + \xi A(\xi, \upsilon, z) + \upsilon B(\xi, \upsilon, z)$$

has order at least $2n - 2 > n$. Thus $(\eta = 0)$ is a Du Val singularity of type A_{n-1}. Since the analytic type of a general hyperplane section through o is an invariant of the germ $(W \ni o)$, so is n. □

Subsection 8.1 formulates the theorems on circles on split sources. Subsection 8.2 treats split sources whose groups converge to a subgroup of dimension two, Subsection 8.3 the equivariant splitting of the binary part, and Subsection 8.4 lc augmentations on the toric germs $(xy+z^at^b=0)$ and groups which may exchange x and y along circles whose radical weights have opposite signs. Since the split radical order is an invariant of $(W \ni o)$, Section 11 treats sequences of sources of quadratic rank two with split radical orders tending to $+\infty$ and with a fixed split radical order separately (Propositions 11.11 and 11.12).

8.1. **Circles on split sources.** In this subsection, we formulate the theorems on circles on split sources. They are proved in Appendix B.

Set-up 8.2. Let Γ, e_0, $\mathbf{T}$, W, G, R, r, D, and α be as follows.

(1) $\Gamma \subset [0,1]$ is a DCC set and e_0 is a positive integer.
(2) $\mathbf{T}=\mathbf{T}_{(w)}$ is a primitive circle, where $w=(w_x,w_y,w_z,w_t)\in\mathbb{Z}^4$.
(3) $W=(xy+g(z,t)=0)\subset\mathbb{A}^4$, where $0\neq g\in(z,t)^2$ is a polynomial, and W is normal and klt near 0.
(4) G is a finite abelian group acting faithfully on W, linearly and monomially on the pair $\{x,y\}$ and diagonally on z,t, and preserving the equation of W up to a character.
(5) $R:=G\cap\mathbf{T}$, $r:=|R|$, and $[G:R]\leq e_0$.
(6) D is a labelled boundary on $(W\ni 0)$ with labels in Γ such that (W,D) is klt near 0, and $\alpha:=\mathrm{mld}((W\ni 0,D)/G)$.

We call $(W\ni 0)$ with the action of G a *split source*, w_x,w_y the *hyperbolic weights* of $\mathbf{T}$, and w_z,w_t the *radical weights* of $\mathbf{T}$. The phrase "α cannot increase along $\mathbf{T}$" is used as in Set-up 7.3.

The prototype of two-sided bounds along circles is [Liu26, Theorem 15.9(3)], which bounds the quotient value of a split source along a growing hyperbolic circle from below by the quotient value b of a plane pair on $\mathbb{A}^2_{z,t}$ with the action of G/R, and from above by $b+2/\sqrt{r}$, but only when the quotient value is at least $1+\epsilon$ and r is large in terms of ϵ. For values in $(0,1)$, the lower bound depends on a margin (Definition 5.2) of the generic limit of the plane pairs, the floor (Definition-Proposition 5.7) of a non-klt limit is replaced by an lc augmentation (Lemma 5.8) by a fixed multiple of $\mathrm{div}(z)$ or $\mathrm{div}(t)$, and log canonicity is detected on $\mathbb{A}^4$ by inversion of adjunction [Kaw07, Theorem] and after truncation by Lemma 4.14. Theorem 8.3 and Proposition 8.4(2) are written with $xy-g$ in place of $xy+g$, as in their proofs. As in Subsection 3.3, the change of sign of y carries one equation to the other and changes no character of a diagonal group.

Theorem 8.3. *Fix a positive integer e_0 and a DCC set $\Gamma\subset[0,1]$. Let $W_i=(xy-g_i(z,t)=0)\subset\mathbb{A}^4$ be normal klt germs at 0, where $0\neq g_i\in(z,t)^2$, let G_i be finite diagonal abelian groups preserving the equation of W_i up to a character and acting faithfully on W_i, and let D_i be labelled boundaries on $(W_i\ni 0)$ with labels in Γ such that (W_i,D_i) is klt near 0. Assume that*

$$R_i:=G_i\cap\mathbf{T}_{(1,-1,0,0)}=\mu_{r_i}(1,-1,0,0),\qquad r_i\to+\infty,\qquad [G_i:R_i]\leq e_0.$$

Then the numbers $\mathrm{mld}((W_i\ni 0,D_i)/G_i)$ do not form a strictly increasing sequence of positive numbers.

Proposition 8.4. *Fix a DCC set $\Gamma\subset[0,1]$.*

(1) Let $(W\ni o)$ be a normal $\mathbb{Q}$-Gorenstein threefold germ, G a finite abelian group acting faithfully on W and fixing o, and D a labelled boundary on $(W\ni o)$ with labels in Γ such that (W,D) is klt near o. Assume that there exists a G-equivariant projective crepant resolution $Y\to W$. Then the set of positive numbers $\mathrm{mld}((W\ni o,D)/G)$ for all such data satisfies the ACC.

(2) Let $W=(xy-g(z,t)=0)\subset\mathbb{A}^4$ be an isolated Gorenstein terminal germ at 0, where $0\neq g\in(z,t)^2$ is reduced, G a finite diagonal abelian group preserving the equation of W up to a character and acting faithfully on W, and D a labelled boundary on $(W\ni 0)$ with labels in Γ such that (W,D) is klt near 0. After a G-equivariant change of x by an

invariant unit, write $g = F_1 \cdots F_k$, *where each* F_i *is a* G*-semi-invariant polynomial that is the product of one* G*-orbit of irreducible factors of* g *in the local ring of* $\mathbb{A}^2_{z,t}$ *at* 0. *Then, after shrinking* W *to a* G*-stable neighborhood of* 0, *there exists a* G*-equivariant small crepant modification* $Y \to W$, *isomorphic away from* 0, *covered near the fiber over* 0 *by* G*-stable charts* U_i, *each an open subset of the hypersurface* $(u_i v_i = F_i(z,t)) \subset \mathbb{A}^4_{u_i,v_i,z,t}$ *containing the origin of* $\mathbb{A}^4_{u_i,v_i,z,t}$, *with*

$$x = u_i \prod_{l<i} F_l, \qquad y = v_i \prod_{l>i} F_l,$$

on which G *acts diagonally, such that for any subgroup* $G' \subset G$, *the number* $\operatorname{mld}((W \ni 0, D)/G')$ *is the minimum of the minimal log discrepancies of* $(Y, D_Y)/G'$ *at the scheme-theoretic points of the fiber of* $Y/G' \to W/G'$ *over the image of* 0, *where* D_Y *is the pull-back of* D. *Moreover, if the numbers* $\operatorname{mld}((W \ni 0, D)/G)$ *of a sequence of such data are positive and strictly increasing, then, after passing to a subsequence, each of them is equal to* $\operatorname{mld}((U \ni 0, D_U)/G)$ *for one of the charts* $U = U_i$ *of the corresponding data such that* 0 *is a singular point of* U, *where* D_U *is the pull-back of* D. *In particular, the set of positive numbers* $\operatorname{mld}((W \ni 0, D)/G)$ *for all such data such that any irreducible factor of* g *is smooth at* 0 *and preserved by* G *satisfies the ACC.*

In Theorem 8.5, the subgroup $\mu_r(1,0,0,0)$ is generated by reflections, and the branch coefficient of the quotient along the image of $(x = 0)$ tends to 1 as $r \to +\infty$. A gap of lc thresholds below 1 allows us to replace the branch coefficient by 1 for $r \geq r_0$, and the restriction to $(x = 0)$ is then a pair on the fixed quotient of $\mathbb{A}^3$ by $G/\mu_r(1,0,0,0)$ with a component of coefficient one. The quotient values b of such pairs satisfy the ACC and are computed by prime divisors of bounded log discrepancy, and Gauss extensions of the computing prime divisors bound the quotient value from above by $b + M/r$.

Theorem 8.5. *Fix a positive integer* e_0 *and a DCC set* $\Gamma \subset [0,1]$. *Then there exist an integer* $r_0 = r_0(\Gamma) \geq 3$ *and a real number* $M = M(\Gamma, e_0)$ *with the following property. Let* $W = (f(x,u_1,u_2,u_3) = 0) \subset \mathbb{A}^4$ *be an isolated singular cDV germ at* 0, G *a finite diagonal abelian group preserving* f *up to a character and acting faithfully on* W, *and* $D = \sum_j c_j \operatorname{div}(h_j)$, *where* $c_j \in \Gamma$ *and the* h_j *are* G*-semi-invariant polynomials not vanishing identically on* W. *Assume that*

$$R := G \cap \mathbf{T}_{(1,0,0,0)} = \mu_r(1,0,0,0), \qquad r \geq r_0, \qquad [G:R] \leq e_0,$$

and that (W, D) *is klt near* 0. *Then the following hold.*

(1) $f = F(x^r, u)$ *and* $h_j = x^{a_j} H_j(x^r, u)$ *for polynomials* F *and* H_j *with* $H_j(0,u) \neq 0$, *where* a_j *is the least* x*-exponent of* h_j.

(2) $A := \sum_j c_j a_j$ *satisfies* $0 \leq A < 1$, *and the pair* (V, C_0) *is lc near* 0, *where* $V := \mathbb{A}^3_u$ *and*

$$C_0 := \operatorname{div}(F(0,u)) + \sum_j c_j \operatorname{div}(H_j(0,u)).$$

(3) We have

$$b \leq \operatorname{mld}((W \ni 0, D)/G) \leq b + \frac{M}{r}, \qquad b := \operatorname{mld}((V \ni 0, C_0)/(G/R)),$$

and the numbers b *for all such data belong to an ACC set depending only on* Γ *and* e_0.

In particular, the numbers $\operatorname{mld}((W \ni 0, D)/G)$ *for such data do not form a strictly increasing sequence with* $r \to +\infty$.

Theorem 8.6 is proved on the smooth cover $x = p^n$, $y = -q^n$, $z = pq$ of $(xy + z^n = 0)$, where the terms $a_j(t)z^j$ have orders tending to infinity with r. If a generic limit (Construction-Proposition 4.1) of the reference pairs that bound α from below, on the surface $(xy + z^n = 0) \subset \mathbb{A}^3$ and on a weighted blow-up of a quotient of it, is not klt along the exceptional curve of a weighted

blow-up of the cover, then the total intersection number of the boundary with the exceptional curve is smaller than 2, independently of n, and the deck group μ_n of the cover $x = p^n$, $y = -q^n$, $z = pq$ acts non-trivially on it. Thus the unique floor (Definition-Proposition 5.7) passes through a fixed point of μ_n on the exceptional curve and defines an lc augmentation by an invariant function.

Theorem 8.6. *Fix an integer $n \geq 3$, coprime positive integers c, v, a positive integer e_0, and a DCC set $\Gamma \subset [0,1]$. Let $d := \gcd(n,2)$, $A := nc/d$, and $B := 2c/d$. Notation and conditions as in Set-up 8.2, where G may exchange x and y. Assume that $\mathbf{T} = \mathbf{T}_{(-A,-A,-B,v)}$ and that*

$$W = \Big(xy + z^n + \sum_{j \leq n-2} a_j(t) z^j = 0\Big)$$

is an isolated Gorenstein terminal germ of quadratic rank two, where the a_j are polynomials with $a_j(0) = 0$. Then α cannot increase along $\mathbf{T}$. The conclusion that α cannot increase along $\mathbf{T}$ also holds for isolated Gorenstein terminal germs $(W \ni o)$ of quadratic rank two with a finite abelian group G acting faithfully on W and fixing o and a G-equivariant isomorphism

$$\widehat{\mathcal{O}}_{W,o} \cong \mathbb{C}[[x,y,z,t]]/(Q(x,y) + \hat{g}(z,t)),$$

where Q is a non-degenerate quadratic form, G acts linearly on the span of x, y and diagonally on z, t, $\operatorname{mult}_0 \hat{g}(z,0) = n$, and $\mathbf{T}$ acts by λ^{-A} on x and y, by λ^{-B} on z, and by λ^v on t, and with a labelled boundary D on $(W \ni o)$ with labels in Γ such that (W, D) is klt near o, where $\alpha := \operatorname{mld}((W \ni o, D)/G)$. Here $R := G \cap \mathbf{T}$ and $r := |R|$ are taken in the group of linear automorphisms of $\mathbb{C}[[x,y,z,t]]$, and $[G : R] \leq e_0$.

The circles $\mathbf{T}_{(1,1,2,0)}$ and $\mathbf{T}_{(A,B,A+B,0)}$ of Theorems 8.7 and 8.8 act trivially on t, and after the weighted blow-up with the weights of the circle the group R acts on each chart $x_l = \rho^{w_l}$, $x_j = \rho^{w_j} x'_j$ $(j \neq l)$, at a coordinate $x_l \in \{x, y, z\}$ with $w_l > 0$, through the exceptional parameter ρ alone, so we run the argument of Theorem 8.5 on the charts: the branch coefficient of the exceptional divisor tends to 1 and is raised to 1 by Theorem 2.12, and adjunction to the exceptional divisor puts α between a value β in an ACC set and $\beta + M/r$ for a constant M.

Theorem 8.7. *Fix a positive integer e_0 and a DCC set $\Gamma \subset [0,1]$. Notation and conditions as in Set-up 8.2, where G may exchange x and y. Assume that $\mathbf{T} = \mathbf{T}_{(1,1,2,0)}$ and that $W = (xy + zf(z,t) = 0)$ is an isolated Gorenstein terminal germ of quadratic rank two, where $f(0,0) = 0$ and $f(0,t) \neq 0$. Then α cannot increase along $\mathbf{T}$. The conclusion that α cannot increase along $\mathbf{T}$ also holds for isolated Gorenstein terminal germs $(W \ni o)$ of quadratic rank two with a finite abelian group G acting faithfully on W and fixing o and a G-equivariant isomorphism*

$$\widehat{\mathcal{O}}_{W,o} \cong \mathbb{C}[[x,y,z,t]]/(Q(x,y) + \hat{g}(z,t)),$$

where Q is a non-degenerate quadratic form, G acts linearly on the span of x, y and diagonally on z, t, and $\mathbf{T}$ acts by λ on x and y, by λ^2 on z, and trivially on t, and with a labelled boundary D on $(W \ni o)$ with labels in Γ such that (W, D) is klt near o, where $\alpha := \operatorname{mld}((W \ni o, D)/G)$. Here $R := G \cap \mathbf{T}$ and $r := |R|$ are taken in the group of linear automorphisms of $\mathbb{C}[[x,y,z,t]]$, and $[G : R] \leq e_0$.

Theorem 8.8. *Fix coprime non-negative integers A, B with $C := A + B > 0$, a positive integer e_0, and a DCC set $\Gamma \subset [0,1]$. Notation and conditions as in Set-up 8.2. Assume that G is diagonal, $\mathbf{T} = \mathbf{T}_{(A,B,C,0)}$, and $W = (xy + zf(z,t) = 0)$ is an isolated Gorenstein terminal germ of quadratic rank two, where $f(0,0) = 0$ and $f(0,t) \neq 0$. Then α cannot increase along $\mathbf{T}$.*

Theorem 8.7 allows G to exchange x and y. For diagonal G, it is the case $A = B = 1$ of Theorem 8.8.

Theorem 8.9. *Fix positive integers N, v, s with $\gcd(N, v, s) = 1$, a positive integer e_0, and a DCC set $\Gamma \subset [0,1]$. Notation and conditions as in Set-up 8.2. Assume that G is diagonal,*

$\mathbf{T} = \mathbf{T}_{(0,N,v,s)}$ or $\mathbf{T} = \mathbf{T}_{(N,0,v,s)}$, $0 \neq g \in (z,t)^3$, and the sum of the terms of g of $\mathbf{T}$-weight N is non-zero. Then α cannot increase along $\mathbf{T}$.

8.2. **Two-dimensional limits.** In this subsection, we consider split sources whose groups converge to a subgroup of dimension two. The result of this subsection is proved in Appendix B.

If the coefficient of $z^a t^b$ in g_i is non-zero for any i, then the semi-invariance of $xy + g_i$ places G_i in the subtorus $w_x + w_y = aw_z + bw_t$ of $\mathbb{T}^4$. When the limit H of the G_i satisfies $\dim H^0 = 2$, the lower bound ϵ forces the primitive generator of the annihilator of H^0 to be non-negative on the cone $\{w \in \mathbb{R}^4_{\geq 0} : w_x + w_y = aw_z + bw_t\}$ of weights of x, y, z, t, and the regular monomial Φ whose character is the generator bounds the quotient values from below and from above.

Proposition 8.10. *Let $a, b \geq 0$ be integers with $n := a + b \geq 2$, $\Gamma \subset [0,1]$ a DCC set, and $\epsilon > 0$ a real number. Let $W_i = (xy + g_i(z,t) = 0) \subset \mathbb{A}^4$ be germs at 0, where the g_i are polynomials, G_i finite abelian groups acting faithfully and diagonally on W_i and preserving the equations up to characters, and D_i labelled boundaries on $(W_i \ni 0)$ with labels in Γ. Assume that*

(i) W_i is normal and klt near 0, $\operatorname{mult}_0 g_i = n$, and the coefficient of $z^a t^b$ in g_i is non-zero,
(ii) (W_i, D_i) is klt near 0 and $\alpha_i := \operatorname{mld}((W_i \ni 0, D_i)/G_i) \geq \epsilon$, and
(iii) the G_i converge to a closed subgroup $H \subset \mathbb{T}^4$ with $\dim H^0 = 2$.

Then

(1) we have

$$G_i \subset \{w \in \mathbb{T}^4 : w_x + w_y = aw_z + bw_t\} \text{ for any } i, \quad \text{and hence}$$
$$H \subset \{w \in \mathbb{T}^4 : w_x + w_y = aw_z + bw_t\},$$

and

(2) after passing to a subsequence, there exist a non-constant monomial Φ in x, y, z, t whose character has order $q := |H/H^0|$ on each G_i, and a real number $s > 0$, such that $(W_i, D_i + s \operatorname{div}(\Phi))$ is lc near 0 and $\alpha_i \geq \frac{s}{q}$ for any i, and

$$\limsup_{i \to +\infty} \alpha_i \leq \frac{s}{q}.$$

In particular, the α_i do not form a strictly increasing sequence.

8.3. **Binary splitting.** In this subsection, we split the binary part of a source of quadratic rank two equivariantly (Theorem 8.11). The theorem is proved in Appendix B.

An equivariant splitting of the binary part of a source of quadratic rank two is of one of two types: the quadratic part becomes XY, or it becomes $X^2 + Y^2$ with $\chi_X \neq \chi_Y$ and $\chi_X^2 = \chi_Y^2$. If, in the second type, the group contains the reflection $X \to -X$, then each mark becomes $X^\epsilon A_j(Y, z, t)$ with $\epsilon \in \{0, 1\}$, and the quotient by the reflection is the smooth germ $\mathbb{A}^3_{Y,z,t}$, and we use [Liu26, Theorem 1.5] on $\mathbb{A}^3_{Y,z,t}$.

Theorem 8.11. *The following hold.*

(1) Let $(W \ni o)$ be an isolated Gorenstein terminal singular threefold germ embedded as $(f = 0)$ in a smooth fourfold germ $(M \ni o)$, G a finite cyclic group acting faithfully on W and on M and fixing o, $h_1, \dots, h_s$ marks on $(W \ni o)$, $\mathcal{C} \subset \mathbb{R}^s_{\geq 0}$ a finite set, and $P_1, \dots, P_l$ finitely many prime divisors over W with center o. Assume that the embedding $W \subset M$ is G-equivariant, that f is G-semi-invariant, that the quadratic part of f has rank two, and that $(W, \sum_k c_k \operatorname{div}(h_k))$ is klt near o for any $(c_1, \dots, c_s) \in \mathcal{C}$. Then there exists a numerical admission (Construction-Theorem 4.13) of $(W, (h_k)_k)$ for $\mathcal{C}$ and $P_1, \dots, P_l$ with group G to a polynomial isolated Gorenstein terminal germ $V \subset \mathbb{A}^4_{X,Y,z,t}$, on which G acts diagonally in X, Y, z, t, such that either
(i) $V = (XY + g_0(z,t) = 0)$, or
(ii) $V = (X^2 + Y^2 + g_0(z,t) = 0)$ and $\chi_X \neq \chi_Y$,

where $0 \neq g_0 \in (z,t)^3$ is a polynomial. If moreover x,y,z,t are G-eigen formal coordinates of W at o as in Definition 8.1 before the replacement of x,y by $x \pm \sqrt{-1}y$, in which the completed equation of W is $xy+\hat{g}(z,t)$, resp. $x^2+y^2+\hat{g}(z,t)$ with $\chi_x \neq \chi_y$, then for any positive integer L' the admission can be taken such that the G-equivariant isomorphism of completions of Construction-Theorem 4.13(2) maps x,y,z,t to X,Y,z,t modulo the L'-th power of the maximal ideal. In particular, if $w \in \mathbb{Z}^4_{>0}$ is a primitive vector with $\max_j w_j < L'$ such that the lowest w-weighted part of the completed equation is irreducible and not divisible by any coordinate, and the prime divisor over W defined by the weights w in x,y,z,t is one of $P_1,\dots,P_l$, then it corresponds to the prime divisor over V defined by the weights w in X,Y,z,t. If the completed equation is $x^2+y^2+\hat{g}(z,t)$ with $\chi_x = \chi_y$, then $x' := x+\sqrt{-1}y$, $y' := x-\sqrt{-1}y$, z, t are G-eigen formal coordinates in which the completed equation is $x'y'+\hat{g}(z,t)$, and the first case holds for x',y',z,t.

(2) Let $\Gamma \subset [0,1]$ be a DCC set. Then the set of positive numbers

$$\operatorname{mld}\Big((W \ni o, \textstyle\sum_k c_k \operatorname{div}(h_k))/G\Big),$$

where $(W \ni o)$, G, and $h_1,\dots,h_s$ are as in (1) with s arbitrary such that, for some numerical admission V as in (1), case (ii) holds and G contains an element acting on V by $(X,Y,z,t) \to (-X,Y,z,t)$, and $c_1,\dots,c_s \in \Gamma$ with $(W, \sum_k c_k \operatorname{div}(h_k))$ klt near o, satisfies the ACC.

8.4. **Fixed split radical order.** In this subsection, we state Lemma 8.12 on lc augmentations on the toric germs $(xy+z^at^b=0)$ with a circle action, and Theorem 8.13 on sources of quadratic rank two with completed equation $xy+\hat{g}(z,t)$, where the lowest homogeneous part of $\hat{g}$ is z^at^b, along the circles $\mathbf{T}_{(-A,-A,-C,D)}$ and with groups which may exchange x and y. Both are used in Appendix E for sources of quadratic rank two with a fixed split radical order. The proofs of Lemma 8.12 and Theorem 8.13 are given in Appendix B.

Lemma 8.12 is the form that Lemma 5.8 takes on the toric germ $(xy+z^at^b=0)$ with a circle action. When the pair (Y,B_Y) of Lemma 8.12 is lc but not klt with positive minimal log discrepancy β over Σ, the pull-back of (Y,B_Y) to a smooth toric cover of Y has a unique coefficient-one curve through the origin, a smooth floor as in Definition-Proposition 5.7. A semi-invariant parameter transverse to the floor descends to a regular $\mathbf{T}_{(w)}$-invariant function U on $(xy+z^at^b=0)$, and the lc augmentation by U yields $\lambda/q=\beta$.

Lemma 8.12. *Let $a,b \geq 0$ be integers with $a+b \geq 2$, $V := (xy+z^at^b=0) \subset \mathbb{A}^4$, and $w=(u,v,c,d) \in \mathbb{Z}^4$ a primitive vector with $u+v=ac+bd$ such that $\mathbb{R}w$ intersects the cone*

$$\{(\xi_x,\xi_y,\xi_z,\xi_t) \in \mathbb{R}^4_{\geq 0} : \xi_x+\xi_y = a\xi_z+b\xi_t\}$$

only at 0. Let G be a finite abelian group acting faithfully on V, monomially on $\{x,y\}$ and diagonally on z,t, commuting with $\mathbf{T}_{(w)}$ and acting torically on the invariant torus, and let $H := G/(G \cap \mathbf{T}_{(w)})$. Let $D_V = \sum_j e_j \operatorname{div}(h_j)$ with $e_j > 0$ and non-zero regular functions h_j on V, both $\mathbf{T}_{(w)}$-homogeneous of weight κ_j and G-semi-invariant, and $\delta := \sum_j e_j\kappa_j - c - d$. Let $\mathbb{C}[V]_k$ be the space of regular functions on V of $\mathbf{T}_{(w)}$-weight k, $Q_0 := \operatorname{Spec}\mathbb{C}[V]_0$, and $Y^{\pm} := \operatorname{Proj}\bigoplus_{k\geq 0}\mathbb{C}[V]_{\pm k}$, which are projective and birational over Q_0. For $\ell \in \{x,y,z,t\}$ whose $\mathbf{T}_{(w)}$-weight w_ℓ satisfies $\pm w_\ell > 0$, the open subset of $Y^{\pm}$ where ℓ does not vanish is the quotient of $V_\ell := V \cap (\ell = 1)$ by $\mu_{|w_\ell|} \subset \mathbf{T}_{(w)}$. Let $B^{\pm}$ be the $\mathbb{R}$-divisor on $Y^{\pm}$ whose restriction to each such open subset is the boundary of the crepant quotient of $(V_\ell, D_V|_{V_\ell})$ by $\mu_{|w_\ell|}$. Let $\epsilon := +$ if $\delta < 0$, $\epsilon := -$ if $\delta > 0$, and $\epsilon \in \{+,-\}$ arbitrary if $\delta = 0$. Let (Y,B_Y) be the crepant quotient of (Y^ϵ, B^ϵ) by H, and Σ the fiber of Y over the image of the origin of Q_0. Let β be the minimum of the log discrepancies of (Y,B_Y) over the prime divisors with center in Σ. Then the following hold.

(1) If (Y,B_Y) is lc but not klt near Σ and $\beta > 0$, then there exist a non-constant regular $\mathbf{T}_{(w)}$-invariant G-semi-invariant function U on V with $U(0)=0$ and a real number

$\lambda > 0$ such that $(V, D_V + \lambda \operatorname{div}(U))$ is lc near 0 and $\lambda/q = \beta$, where q is the order of the character of U on G.

(2) If (Y, B_Y) is lc but not klt near Σ and $\beta > 0$, then any prime divisor F over V with center 0 satisfies $a(F, V, D_V)/|I_G(F)| \geq \beta$.

(3) Let $B_0 \geq 0$ be the push-forward of B^ϵ to Q_0. There exist an H-invariant monomial ideal I on Q_0, equal to (1) or primary at the origin, and a positive integer N, both depending only on a, b, w, and the sign of δ, such that $a(P, Y^\epsilon, B^\epsilon) = a(P, Q_0, B_0, I^{|\delta|/N})$ for any prime divisor P over Q_0.

The condition on $\mathbb{R}w$ in Lemma 8.12 means that $\mathbf{T}_{(w)}$ has monomial functions on V of positive and of negative weight. It excludes in particular the circles $\mathbf{T}_{(w)}$ with all entries of w positive, along which the quotient values tend to zero by Lemma 4.7.

Along the circle $\mathbf{T}_{(-A,-A,-C,D)}$ of Theorem 8.13, the coordinates x and y both have weight $-A$, and an element of G_i exchanging x and y permutes the conditions given by x and y on the varieties Y^+ and Y^- of Lemma 8.12. By Hensel's lemma and the Weierstrass preparation theorem, $\hat{g}_i$ is a unit times the product of a monic polynomial in z and a monic polynomial in t, with lowest homogeneous parts z^a and t^b. Dividing both x and y by one square root of the unit removes it and commutes with the exchange of x and y. In the comparison with $Y^\pm$, the monic factor in t, resp. in z, is replaced by t^b, resp. z^a, also by dividing x and y by a square root: when 1 minus the quotient of the monic factor by t^b, resp. z^a, has positive order along the prime divisor P in question, the quotient has a square root ς in the completion of the function field along P, and replacing x, y by $x/\varsigma, y/\varsigma$ keeps the comparison compatible with the exchange of x and y.

Theorem 8.13. *Fix integers $a, b \geq 1$ with $a + b \geq 3$, integers $A \geq 0$ and $C, D > 0$ with $\gcd(A, C, D) = 1$ and $aC - bD = 2A$, a positive integer e_0 and a DCC set $\Gamma \subset [0, 1]$, and let $\mathbf{T} := \mathbf{T}_{(-A,-A,-C,D)}$. Let $(W_i \ni o_i)$ be isolated Gorenstein terminal threefold germs of quadratic rank two, G_i finite abelian groups acting faithfully on W_i and fixing o_i, and B_i labelled boundaries with labels in Γ such that (W_i, B_i) is klt near o_i. Assume that there exist G_i-equivariant isomorphisms*

$$\widehat{\mathcal{O}}_{W_i,o_i} \cong \mathbb{C}[[x, y, z, t]]/(xy + \hat{g}_i(z, t)),$$

with G_i acting linearly and monomially on $\{x, y\}$ and diagonally on z, t, $\operatorname{mult}_0 \hat{g}_i = a + b$ and the homogeneous part of degree $a+b$ of $\hat{g}_i$ equal to $z^a t^b$, and that $|G_i \cap \mathbf{T}| \to +\infty$ and $[G_i : G_i \cap \mathbf{T}] \leq e_0$. Then the numbers $\operatorname{mld}((W_i \ni o_i, B_i)/G_i)$ do not form a strictly increasing sequence of positive numbers.

9. Sources of quadratic rank one

In this section, we define sources of type cD and of type cE, state a lemma on prime divisors of log discrepancy two over hypersurfaces $(x^2 + g(y, z, t) = 0)$ which contain a smooth surface, and fix the eigenparameters and the double-plane divisor E_0 of a source of quadratic rank one. A source of quadratic rank one has a local equation $x^2 + g(y, z, t)$ with $\operatorname{mult}_0 g \geq 3$, and it is of type cD unless the residual cubic is the cube of a linear form, in which case it is of type cE and is treated in Section 10. Sources of type cD are treated, with several labels, in Section 11.

Definition 9.1. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one, and g_3 the residual cubic of $(W \ni o)$ [Liu26, Definition-Lemma 11.1]. We say that $(W \ni o)$ is *of type cE* if g_3 is the cube of a linear form, and *of type cD* otherwise.

9.1. **A divisor of log discrepancy two.** In this subsection, we state a lemma on prime divisors of log discrepancy two over a hypersurface $(x^2 + g(y, z, t) = 0)$ which contains a smooth surface. It is proved in Appendix C.

Lemma 9.2, including the persistence of E under changes of high order jets in its second paragraph, is used in Section 11 to exclude a smooth prime divisor with coefficient one in a compatible generic limit (Construction-Proposition 4.1).

Lemma 9.2. *Let k be an algebraically closed field containing $\mathbb{C}$, $R = k[[x,y,z,t]]/(x^2+g(y,z,t))$ a normal complete local domain with $\operatorname{mult}_0 g \geq 3$ and non-zero cubic part g_3 of g, D a prime divisor of $\operatorname{Spec} R$ through the closed point o such that R/I_D is regular of dimension two, and $B = \sum_l b_l \operatorname{div}(\phi_l)$ with $\phi_l \in R\backslash\{0\}$ and real numbers $b_l \geq 0$. Let $c_D := \sum_l b_l \operatorname{mult}_D \phi_l$. Then there exists a prime divisor E over $\operatorname{Spec} R$ with center o such that*

$$a(E, \operatorname{Spec} R, 0) = 2, \qquad \sum_l b_l \operatorname{mult}_E \phi_l \geq c_D.$$

In particular, $\operatorname{mld}(\operatorname{Spec} R \ni o, B) \leq 2 - c_D$, and $\operatorname{mld}(\operatorname{Spec} R \ni o, \frac{1}{m}\operatorname{div}(h)) \leq 1$ for any integer $m \geq 1$ and any $h \in R\backslash\{0\}$ with $\operatorname{mult}_D h \geq m$.

Moreover, there exist formal coordinates X, Y, z', t' of $k[[x,y,z,t]]$ with $D = (X = Y = 0)$, where z', t' are linear forms in y, z, t, and a weight vector $\mathbf{w} \in \{(2,2,1,1),(3,3,1,1),(3,2,2,1)\}$ such that the lowest $\mathbf{w}$-weighted part of the equation of $\operatorname{Spec} R$ in X, Y, z', t' is reduced and E is an exceptional prime divisor of the weighted blow-up with weights $\mathbf{w}$ in X, Y, z', t'. There exists an integer N such that, for any g' with $g' - g \in (x,y,z,t)^N$ for which $R' := k[[x,y,z,t]]/(x^2+g')$ is normal, and any $\phi_l' \in R'$ for which ϕ_l and ϕ_l' have representatives in $k[[x,y,z,t]]$ with difference in $(x,y,z,t)^N$, the weighted blow-up of $\operatorname{Spec} R'$ with weights $\mathbf{w}$ in X, Y, z', t' has an exceptional prime divisor E' with $a(E', \operatorname{Spec} R', 0) = 2$ and $\operatorname{mult}_{E'} \phi_l' \geq \operatorname{mult}_E \phi_l$ for all l.

Furthermore, let $m \geq 1$ be an integer and, for $i \geq 1$, let $F_i, h_i \in \mathbb{C}[x,y,z,t]$ be polynomials such that $W_i := (F_i = 0) \subset \mathbb{A}^4$ is normal near $0 \in W_i$ and h_i does not vanish identically on W_i. Let (F_*, h_*) be a compatible generic limit of the pairs (F_i, h_i) over k in case (a) of Construction-Proposition 4.1 with trivial group. Assume that $k[[x,y,z,t]]/(F_*)$ is isomorphic as a k-algebra to a normal ring $R = k[[x,y,z,t]]/(x^2+g(y,z,t))$ with $\operatorname{mult}_0 g \geq 3$ and $g_3 \neq 0$, and that $\operatorname{mult}_D h_* \geq m$ for a prime divisor D of $\operatorname{Spec} R$ through o with R/I_D regular of dimension two. Then for infinitely many i there exists a prime divisor E_i over W_i with center 0 such that $a(E_i, W_i, 0) = 2$ and $\operatorname{mult}_{E_i} h_i \geq m$.

9.2. **The double-plane divisor.** In this subsection, we fix the eigenparameters and the double-plane divisor E_0 of a source of quadratic rank one (Set-up 9.3), which are used in Section 11 and Appendix E.

Set-up 9.3. Let m, $(W \ni o)$, G, h, x, y, z, t, f, g_3, Y, E_0, s, P, and α be as follows.

(1) $m \geq 2$ is an integer.
(2) $(W \ni o)$ is an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one in a smooth fourfold germ, and G is a finite abelian group acting faithfully on W and fixing o.
(3) h is a G-semi-invariant regular function with $h(o) = 0$ such that $(W, \frac{1}{m}\operatorname{div}(h))$ is terminal near o.
(4) x, y, z, t are ambient G-eigenparameters in which the quadratic part of a local equation f of W is x^2, and $g_3(y,z,t)$ is the residual cubic.
(5) $Y \to W$ is the normalized strict transform of W in the weighted blow-up of the ambient germ with weights $(3,2,2,2)$ in x, y, z, t, and E_0 is the exceptional prime divisor of $Y \to W$ (it is unique by Lemma C.2(2)).
(6) $s := \operatorname{mult}_{E_0} h$, and the image of h in the graded ring of E_0, a quotient of $\mathbb{C}[x,y,z,t]$ by $x^2 + g_3$ (Lemma C.2(3)), is $P(y,z,t)$ or $xP(y,z,t)$ with P homogeneous, and $s = 2\deg P$, resp. $s = 3 + 2\deg P$.
(7) For $u \in \{y, z, t\}$, the *u-root chart* is the base change of the morphism $\mathbb{A}^4_{\sigma,X,u',u''} \to \mathbb{A}^4$ given by $u = \sigma^2$, $x = \sigma^3 X$, and the other two of y, z, t equal to $\sigma^2 u'$ and $\sigma^2 u''$, along the morphism (x,y,z,t) from the ambient smooth fourfold germ to $\mathbb{A}^4$, with the coordinates σ, X, u', u''.
(8) On the u-root chart, the exceptional divisor is $(\sigma = 0)$ and the strict transform of W is $(\sigma^{-6} f = 0)$, since the $(3,2,2,2)$-initial form of f is the irreducible form $x^2 + g_3$ of weight 6 and the root charts are quasi-étale over the weighted blow-up (Lemma C.2(1) and (2)).

(9) $g_3 \neq 0$: by [KM98, Corollary 5.38] and [Liu26, Lemma 11.3(3)], $(W \ni o)$ is an isolated cDV point of type cD or cE, so a general hyperplane section of W through o is a Du Val point of type D or E [Liu26, Definition-Lemma 11.1]; the residual cubic of the section is the restriction of g_3 to a general plane (substitute the equation of the hyperplane into f, whose quadratic part is x^2), and it is non-zero, since by [Liu26, Definition-Lemma 11.1] it is determined up to a linear change of coordinates and a non-zero scalar, and the normal forms of types D and E in [KM98, Theorem 4.20] have non-zero residual cubics.

(10) $\alpha := \mathrm{mld}((W \ni o, \frac{1}{m}\operatorname{div}(h))/G)$.

The prime divisor E_0 does not depend on the eigenparameters. Indeed, if x', y', z', t' are other ambient G-eigenparameters in which the quadratic part of a local equation of W is x'^2, then the linear parts of x and x' are proportional, so $x' \in \mathbb{C}^* x + \mathfrak{m}_o^2$, and every element of $\mathfrak{m}_o^2$ has weight at least 4 for the weights $(3,2,2,2)$. Thus every monomial in x', y', z', t' of weight at least n belongs to the ideal generated by the monomials in x, y, z, t of weight at least n, and conversely, so the weighted blow-ups of the ambient germ with weights $(3,2,2,2)$ in x, y, z, t and in x', y', z', t' coincide. The argument above remains valid for formal G-eigenparameters, so E_0 depends only on the completion $\widehat{\mathcal{O}}_{W,o}$ with its G-action, and under a G-equivariant isomorphism of completions the prime divisors E_0 correspond to each other [dFEM11, Lemma 2.21].

10. Sources of type cE

In this section, we state the ACC for quotient values on sources of type cE with finitely many real labels, together with the results that enter the proof of the ACC. All proofs are in Appendix D. An isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic [Liu26, Definition-Lemma 11.1] is the cube of a linear form is of type cE in the sense of Definition 9.1. Such a germ is an isolated cDV point [KM98, Corollary 5.38], so it is of type cE in the sense of [Liu26, Definition-Lemma 11.1 and Lemma 11.3(3)], that is, a general hyperplane section through o is a Du Val singularity of type E_6, E_7, or E_8. Thus the germ is a point of type cE in the sense of [Kol98, Definition 2.3]. Along a circle of the form in Construction-Proposition 10.3(2), the proof of Theorem 10.1 lifts prime divisors over W to prime divisors over $\mathbb{A}^4$ and discards the blocks of the equations whose orders along the lifted divisor exceed a fixed bound. To discard the blocks, the proof needs a bound on the log discrepancies of the prime divisors over W with center o whose quotient log discrepancies are bounded by a fixed constant, which for sources of type cE follows from the margin of Theorem 10.5.

Theorem 10.1. *Fix a finite set $S \subset [0,1]$. Then the set of numbers $\mathrm{mld}((W \ni o, \Delta)/G)$ in $(0,1)$, where $(W \ni o)$ is an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form, G is a finite cyclic group acting faithfully on W and fixing o, and $\Delta = \sum_j c_j \operatorname{div}(h_j)$ is a labelled boundary (Definition 2.7) with $c_j \in S$ such that (W, Δ) is terminal near o, satisfies the ACC. Moreover there exists $\eta_S > 0$ such that $a(P, W, \Delta) \geq \eta_S a(P, W, 0)$ for any such data and any prime divisor P over W whose center contains o.*

The proof of Theorem 10.1 in Appendix D uses Theorems 2.13(2), 10.2, 10.4, 8.5, and 10.5, and its case division follows the proof of Construction-Proposition 10.3. Theorem 10.1 is used for the finite set $\{\mu_1/m, \dots, \mu_s/m\}$ of real labels of Set-up 11.1 in Section 12.

10.1. **Normal forms and circles.** In this subsection, we state the bound for regular functions in $\mathfrak{m}_o \backslash \mathfrak{m}_o^2$, the normal forms of sources of type cE, and the theorem on negative scalar circles. The proofs are in Appendix D.

As recalled at the beginning of this section, a general hyperplane section of a source of type cE through o is a Du Val singularity of type E_6, E_7, or E_8. The proof of the bound $\frac{1}{15}$ in Theorem 10.2 uses a bound on the orders of the pullbacks of functions in $\mathfrak{m}_o \backslash \mathfrak{m}_o^2$ to the binary polyhedral cover of such a section.

Theorem 10.2. *Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form, and $h \in \mathfrak{m}_o \backslash \mathfrak{m}_o^2$ a regular function. Then $(W, \frac{1}{15}\operatorname{div}(h))$ is lc near o.*

The terms of low degree of the equation of a source of type cE confine a diagonal group to a compact two-dimensional torus whose Lie algebra contains a vector with positive entries. A two-dimensional limit of the groups along a sequence with strictly increasing quotient values would imply the existence of prime divisors with arbitrarily small log discrepancy over the quotients, so the limit is a circle $\mathbf{T}_{(3s,2s,C,D)}$ with integers s, C, D as in Construction-Proposition 10.3(2). When the first non-zero part among B_4, A_3, B_5 of the normal form in Construction-Proposition 10.3(1)(d) is a power of a linear form, the source has a G-stable Du Val section, the circle does not act trivially on the coordinate defining it since the singularity is isolated, and Theorem 10.4, or Theorem 8.5 when the circle acts trivially on x, y, and z, excludes the sequence, so the non-zero part is a single monomial $z^a t^b$ with positive exponents a and b.

Construction-Proposition 10.3. Fix $m \geq 3$. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form, G a finite cyclic group acting faithfully on W and fixing o, and $h \in \mathfrak{m}_o^2$ a G-semi-invariant regular function with $(W, \frac{1}{m}\operatorname{div}(h))$ terminal near o.

(1) There exists a numerical admission (Construction-Theorem 4.13) with the group G to a germ W' and a mark h' with the following properties.
  (a) $W' = (x^2 + y^3 + yA(z,t) + B(z,t) = 0)$ with $\operatorname{mult} A \geq 3$ and $\operatorname{mult} B \geq 4$.
  (b) W' is isolated Gorenstein terminal.
  (c) G acts diagonally on x, y, z, t.
  (d) With A_k, B_k the homogeneous parts of degree k, exactly one of the following holds: (E_6) $B_4 \neq 0$, (E_7) $B_4 = 0$ and $A_3 \neq 0$, or (E_8) $B_4 = A_3 = 0$ and $B_5 \neq 0$.
  (e) The admitted mark $h' \in \mathfrak{m}^2$ has the character of h.
  (f) $(W', \frac{1}{m}\operatorname{div}(h'))$ is klt.

(2) Let $(W_i \ni o_i, G_i, h_i)$ be a sequence of such data, and let

$$\alpha_i := \operatorname{mld}((W_i \ni o_i, \frac{1}{m}\operatorname{div}(h_i))/G_i).$$

Assume that (α_i) is a strictly increasing sequence in $(0,1)$. Then, after passing to a subsequence and admitting as in (1), there exist integers s, C, D with $\gcd(s, C, D) = 1$ and $e_0 \geq 1$ with the following properties.
  (a) The order of $G_i \cap \mathbf{T}_{(3s,2s,C,D)}$ tends to $+\infty$.
  (b) $[G_i : G_i \cap \mathbf{T}_{(3s,2s,C,D)}] \leq e_0$.
  (c) The four weights $3s, 2s, C, D$ are neither all positive nor all negative.
  (d) The non-zero homogeneous part B_4, A_3, or B_5 of (1) is $\lambda_i z^a t^b$ with $\lambda_i \in \mathbb{C}^*$ and $a, b \geq 1$, and

$$\begin{aligned} &a + b = 4,\ 6s = aC + bD && \text{in case } (\mathrm{E}_6),\\ &a + b = 3,\ 4s = aC + bD && \text{in case } (\mathrm{E}_7),\\ &a + b = 5,\ 6s = aC + bD && \text{in case } (\mathrm{E}_8). \end{aligned}$$

In particular $C \neq D$.

Theorem 10.4 treats the case, excluded in Construction-Proposition 10.3(2)(d), in which the first non-zero part among B_4, A_3, B_5 is a power of a linear form, except for the circle acting trivially on x, y, and z, which Theorem 8.5 treats. Then the source has a G-stable Du Val section. A circle acting trivially on the coordinate that defines this section does not occur, since the singularity is isolated. With type D_4, Theorem 10.4 is also used in Section 11.

Theorem 10.4. *Fix one of the Du Val types D_k ($k \geq 4$), E_6, E_7, E_8, with the weighted homogeneous polynomial $f_0 = x^2 + g_0(y,z)$ and weights $q = (q_x, q_y, q_z)$: $x^2 + y^2 z + z^{k-1}$ with*

$q=(k-1,k-2,2)$, $x^2+y^3+z^4$ with $q=(6,4,3)$, $x^2+y^3+yz^3$ with $q=(9,6,4)$, and $x^2+y^3+z^5$ with $q=(15,10,6)$, respectively, or the image of such a polynomial under a fixed change of the coordinates y,z preserving the weights q_y,q_z, such as $x^2+y^3+z^3$ with $q=(3,2,2)$ for D_4. Fix a weighted homogeneous basis $M_1,\dots,M_\mu$ of $\mathbb{C}[y,z]/(\partial_y g_0,\partial_z g_0)$ (for $x^2+y^3+z^3$, for instance, $1,y,z,yz$), coprime positive integers c,v, a positive integer e_0 and a DCC set $\Gamma\subset[0,1]$, and let $\mathbf{T}:=\mathbf{T}_{(-cq_x,-cq_y,-cq_z,v)}$, acting on x,y,z,t with the weights $(-cq_x,-cq_y,-cq_z,v)$. Consider data $(W\ni o,G,D)$ of one of the following two kinds. In both kinds, G is a finite abelian group acting faithfully on W and diagonally on x,y,z,t with $[G:G\cap\mathbf{T}]\le e_0$, and D is a labelled boundary with labels in Γ such that (W,D) is klt near o.

(1) $W=(f_0+\sum_j a_j(t)M_j=0)\subset\mathbb{A}^4$ with $a_j\in t\mathbb{C}[t]$ is an isolated Gorenstein terminal germ at the origin o, and G acts on f_0 and on each non-zero summand $a_j(t)M_j$ by one character of G.

(2) $(W\ni o)$ is an isolated Gorenstein terminal germ with a G-equivariant isomorphism

$$\widehat{\mathcal{O}}_{W,o}\cong\mathbb{C}[[x,y,z,t]]/(\hat f),$$

where the quadratic part of $\hat f$ is x^2 and $\hat f(x,y,z,0)=f_0$.

Then, for each of the kinds (1) and (2), there exists no sequence of data of that kind with $|G\cap\mathbf{T}|\to+\infty$ whose numbers $\operatorname{mld}((W\ni o,D)/G)$ are positive and strictly increasing.

10.2. **A uniform margin.** In this subsection, we state the uniform margin for boundaries on sources of type cE. The proof of Theorem 10.5 is in Appendix D. The proof uses a generic limit in the sense of [Kaw21, Section 3], Lemma D.18(2), and Lemma 9.2.

Theorem 10.5. *Let $\gamma\in(0,1]$ and $\epsilon\in(0,2)$. Then there exists a real number $\theta=\theta(\gamma,\epsilon)>1$ with the following property. Let $(W\ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form, and $\Delta=\sum_j c_j\operatorname{div}(h_j)$ with non-zero $h_j\in\mathfrak{m}_o$ and $\gamma\le c_j\le 1$. Assume that $\operatorname{mld}(W\ni o,\Delta)\ge 1+\epsilon$. Then $(W,\theta\Delta)$ is lc near o. Therefore, with $\eta:=(\theta-1)/\theta$, any prime divisor P over W whose center contains o satisfies $a(P,W,\Delta)\ge\eta a(P,W,0)$, and if a finite group G fixes o and preserves Δ, then for any real number $L>0$, any prime divisor P over W with center o and $0<a(P,W,\Delta)\le L|I_G(P)|$ satisfies $a(P,W,0)\le(L/\eta)|I_G(P)|$ and $\sum_j c_j\operatorname{mult}_P h_j\le((1-\eta)L/\eta)|I_G(P)|$.*

Remark 10.6. For $n\ge 226$ let $W_n=(x^2+y^3+z^5+t^n=0)\subset\mathbb{A}^4$ and $c_n=\frac{1}{15}-\frac{1}{n}$. Then $(W_n,c_n\operatorname{div}(x))$ is terminal and $\operatorname{mld}(W_n\ni 0,c_n\operatorname{div}(x))=1+\frac{15}{n}$, while $\operatorname{lct}_0(W_n;\operatorname{div}(x))=\frac{1}{15}+\frac{2}{n}$. The prime divisors F_n with weights $(15n,10n,6n,30)/\gcd(n,30)$ on x,y,z,t satisfy $a(F_n,W_n,c_n\operatorname{div}(x))/a(F_n,W_n,0)=45/(n+30)$. Since $c_n>\frac{1}{30}$ and $45/(n+30)\to 0$, in Theorem 10.5 with $\gamma=\frac{1}{30}$ the number ϵ cannot be replaced by the condition that the pair is terminal.

The polynomial $x^2+y^3+z^5+t^n$ is Newton non-degenerate [IP01, Definition 2.6]. For $q\in[0,1]$ and $w=(w_1,w_2,w_3,w_4)\in\mathbb{R}^4_{\ge 0}$ let

$$\Xi_q(w):=(1-q)w_1+w_2+w_3+w_4-\min(2w_1,3w_2,5w_3,nw_4).$$

The non-degeneracy of [IP01, Definition 2.6] is a condition on any face of the Newton polyhedron, the non-compact faces included. Thus, on the toric embedded resolution π of W_n in [IP01, Proposition 2.7], the strict transform of W_n intersects any torus orbit in $\pi^{-1}(0)$ transversally [Var76, Lemma 10.3(3)], so near $\pi^{-1}(0)$ it forms a simple normal crossing divisor with the toric boundary, which contains the strict transforms of the coordinate hyperplanes. A calculation on π shows that $a(F,W_n,q\operatorname{div}(x))\ge\Xi_q(w)$ for any prime divisor F over W_n whose center contains 0, where $w=(\operatorname{mult}_F x,\operatorname{mult}_F y,\operatorname{mult}_F z,\operatorname{mult}_F t)$, with equality for the exceptional prime divisors of the weighted blow-ups with weights $(15,10,6,1)$ and $(15n,10n,6n,30)/\gcd(n,30)$.

For the minimal log discrepancy, let F have center 0, so that $w\in(\mathbb{N}^+)^4$. The bounds $w_1\ge\frac{1}{2}k$, $w_2\ge\frac{1}{3}k$, $w_3\ge\frac{1}{5}k$ for $k:=\min(2w_1,3w_2,5w_3,nw_4)$ imply $\Xi_{1/15}(w)\ge w_4\ge 1$, and

$\Xi_{1/15}(w) \in \frac{1}{15}\mathbb{Z}$, with $\Xi_{1/15}(w) = 1$ only if $w = (15j, 10j, 6j, 1)$ and $30j \le n$ for a positive integer j. Since $\Xi_{c_n}(w) = \Xi_{1/15}(w) + w_1/n$, if $\Xi_{1/15}(w) > 1$, then $\Xi_{1/15}(w) \ge \frac{16}{15}$ and $\Xi_{c_n}(w) \ge \frac{16}{15} + \frac{w_1}{n} > 1 + \frac{15}{n}$, as $w_1 \ge 1$ and $n \ge 226$. If $\Xi_{1/15}(w) = 1$, then $w = (15j, 10j, 6j, 1)$ and $\Xi_{c_n}(w) = 1 + \frac{15j}{n} \ge 1 + \frac{15}{n}$, with equality for $w = (15, 10, 6, 1)$, where $a(F, W_n, 0) = 2$ and $\operatorname{mult}_F x = 15$. Away from 0, both W_n and $\operatorname{div}(x)$ are smooth and $c_n < 1$, so $(W_n, c_n \operatorname{div}(x))$ is terminal near 0 and $\operatorname{mld}(W_n \ni 0, c_n \operatorname{div}(x)) = 1 + \frac{15}{n}$.

For the lc threshold, let $\lambda_n := \frac{1}{15} + \frac{2}{n}$. The bounds $w_1 \ge \frac{1}{2}k$, $w_2 \ge \frac{1}{3}k$, $w_3 \ge \frac{1}{5}k$ together with $w_4 \ge \frac{1}{n}k$ imply

$$\Xi_{\lambda_n}(w) \ge k\Big(\frac{1-\lambda_n}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{n} - 1\Big) = 0$$

for any $w \in \mathbb{R}^4_{\ge 0}$, so $(W_n, \lambda_n \operatorname{div}(x))$ is lc near 0. With $g_n := \gcd(n, 30)$, the prime divisor F_n has $a(F_n, W_n, 0) = (n+30)/g_n$ and $\operatorname{mult}_{F_n} x = 15n/g_n$, so $a(F_n, W_n, \lambda_n \operatorname{div}(x)) = 0$ and $a(F_n, W_n, c_n \operatorname{div}(x)) = 45/g_n$, which implies the lc threshold and the ratio $45/(n+30)$.

Remark 10.7. The hypothesis that W is of type cE cannot be dropped from Theorem 10.5. For $i \ge 4$ the pair $(\mathbb{A}^3, \frac{1}{2}\operatorname{div}(z^2 + x^i + t^i))$ is terminal and $\operatorname{mld}(\mathbb{A}^3 \ni 0, \frac{1}{2}\operatorname{div}(z^2 + x^i + t^i)) = 2$, while $\operatorname{lct}_0(\mathbb{A}^3; z^2 + x^i + t^i) = \frac{1}{2} + \frac{2}{i}$ (by the toric computation of Remark 10.6 for the non-degenerate polynomial $z^2 + x^i + t^i$ [IP01, Proposition 2.7]). Hence $(\mathbb{A}^3, \theta \cdot \frac{1}{2}\operatorname{div}(z^2 + x^i + t^i))$ is lc near 0 only for $\theta \le 1 + \frac{4}{i}$, which tends to 1, and the generic limit $\frac{1}{2}\operatorname{div}(z^2) = \operatorname{div}(z)$ of the boundaries has a component with coefficient one.

11. Several labels

In this section, we prove Theorem 11.2. By [Liu26, Construction-Theorem 17.6(5)] (see Section 12), real coefficients lead to a convex combination of rational proxies $\frac{1}{m}\operatorname{div}(h_k)$ on one source. We prove Theorem 11.2 for such boundaries, using the circle theorems for labelled boundaries in the form of Section 5. Log discrepancies of the proxies are added only at one prime divisor at a time (Definition-Lemma 11.4).

Set-up 11.1. Let s, m, $\mu_1, \dots, \mu_s$, $(W \ni o)$, G, and $h_1, \dots, h_s$ be as follows.

(1) s and m are positive integers, and $\mu_1, \dots, \mu_s$ are positive real numbers with $\sum_k \mu_k = 1$.
(2) $(W \ni o)$ is a smooth threefold germ or an isolated Gorenstein terminal threefold germ.
(3) G is a finite cyclic group acting faithfully on W and fixing o.
(4) $h_1, \dots, h_s$ are marks on $(W \ni o)$ (Definition 2.7) such that (W, C_k) is terminal near o for $C_k := \frac{1}{m}\operatorname{div}(h_k)$, $k = 1, \dots, s$.

Let $c_k := \mu_k/m$ for each k. Let $D := \sum_k \mu_k C_k = \sum_k c_k \operatorname{div}(h_k)$, the labelled boundary with labels $c_1, \dots, c_s$ (Definition 2.7). Let $\alpha := \operatorname{mld}((W \ni o, D)/G)$. For a prime divisor P over W with center o, let $a_k(P) := a(P, W, C_k)/|I_G(P)|$ and $a_D(P) := a(P, W, D)/|I_G(P)|$. We call the C_k the *proxies* of D, and $a_k(P)$ and $a_D(P)$ the *scores* of P.

Some results below allow, in place of (4), the following weaker condition, under which we keep the definitions of C_k, c_k, D, α, $a_k(P)$, and $a_D(P)$:

(4') $h_1, \dots, h_s$ are marks on $(W \ni o)$ such that (W, D) is klt near o and $a(P, W, 0) - \frac{1}{m}\operatorname{mult}_P h_k > 0$ for any k and any prime divisor P over W with center o.

Condition (4) implies (4'): for a prime divisor P over W with center o, $a(P, W, 0) - \frac{1}{m}\operatorname{mult}_P h_k = a(P, W, C_k) > 1$, and (W, D) is klt near o, since each (W, C_k) is terminal, hence klt, near o (the coefficients are less than one by [KM98, Corollary 2.31(3)]) and $a(P, W, D) = \sum_k \mu_k a(P, W, C_k)$ for any prime divisor P over W. Unless (4') is allowed explicitly, data as in Set-up 11.1 satisfy (1)–(4).

Each proxy (W, C_k) is assumed terminal, and not only the pair (W, D): the bounds of Definition-Lemma 11.5(3) are the sums of the bounds that the terminality of each proxy gives, and for bounded groups the score of each proxy is positive and satisfies an integrality condition

(Definition-Lemma 11.4(1)). For bounded groups, the integrality of $m|I_G(P)|a_k(P)$ replaces the lattice of Lemma 4.8 (Proposition 11.7). Condition (4') holds for the numerical admissions of Proposition 11.8, whose proxies need not be terminal (Lemma 11.10).

Theorem 11.2. *Notation as in Set-up 11.1, with s, m and $\mu_1, \dots, \mu_s$ fixed. Then the set of numbers $\alpha \in (0,1)$, for all such data for which $(W \ni o)$ is not of type cE, satisfies the ACC.*

Sources of type cE are left out of Theorem 11.2 because Theorem 10.1 treats them for any finite set of real labels without proxies, and there the margin is given by the source itself (Theorem 10.5). For the other sources, the margins are given by the reference pairs of Section 5. We prove Theorem 11.2 in Subsection 11.3.

11.1. **Scores and blocks.** In this subsection, we show that the scores of one prime divisor are linear in the proxies, compute the boundary on a quotient chart, and define the normal forms of the marks on a double cover together with their blocks along a circle.

Example 11.3. Let $r \geq 3$ be an integer, $\lambda \in (0,1)$ a real number, and $G = \mu_r(1,-1,0)$ acting on $\mathbb{A}^3$ with coordinates x, y, z. Let $C_1 := \lambda \operatorname{div}(x)$ and $C_2 := \lambda \operatorname{div}(y)$. Since the boundaries are torus invariant, the quotient values below are minima of linear functions over the vectors of N_G with positive coordinates [Liu26, Lemma 3.4], and a direct calculation shows

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, C_1)/G) = \operatorname{mld}((\mathbb{A}^3 \ni 0, C_2)/G) = 2 - \lambda + \frac{\lambda}{r},$$

attained at the vectors $(\frac{r-1}{r}, \frac{1}{r}, 1)$ and $(\frac{1}{r}, \frac{r-1}{r}, 1)$ respectively, whereas

$$\operatorname{mld}\Big(\Big(\mathbb{A}^3 \ni 0, \frac{1}{2}(C_1 + C_2)\Big)/G\Big) = 2 - \frac{\lambda}{2} > 2 - \lambda + \frac{\lambda}{r}.$$

Minimal log discrepancies of quotients are not additive in the proxies (Example 11.3), so we only add log discrepancies at one prime divisor.

Definition-Lemma 11.4. Notation as in Set-up 11.1, where (4) may be replaced by (4').

(1) For any prime divisor P over W with center o, $a_D(P) = \sum_k \mu_k a_k(P)$ and $a_k(P) > 0$ for each k. Moreover, $m|I_G(P)|a_k(P)$ is a positive integer.

(2) Let $x_1, \dots, x_4$ be ambient G-eigenparameters at o, f a G-semi-invariant local equation of W in them ($f = x_4$ if W is smooth), $v \in \mathbb{Z}^4_{>0}$ a primitive vector such that the lowest v-weighted part f_v of f is reduced and irreducible, E_v the exceptional prime divisor of the normalized strict transform $Y \to W$ of W in the weighted blow-up with weights v. Let l be an index such that f_v is not a multiple of x_l. The *root chart* of v at x_l is the affine space with coordinates ρ and x'_j, $j \neq l$, mapping to the ambient germ by $x_l = \rho^{v_l}$ and $x_j = \rho^{v_j} x'_j$. It carries the action of the group K generated by μ_{v_l} and the lifts of the elements of G: an element $\zeta \in \mu_{v_l}$ acts by $\rho \to \zeta\rho$ and $x'_j \to \zeta^{-v_j} x'_j$, and an element of G acting by $x_j \to g_j x_j$, $j = 1, \dots, 4$, lifts to $\rho \to \epsilon\rho$ and $x'_j \to g_j \epsilon^{-v_j} x'_j$, $j \neq l$, for any ϵ with $\epsilon^{v_l} = g_l$. Let W' be the strict transform of W in the root chart, d the order of the subgroup of μ_{v_l} fixing an irreducible component of $(\rho = 0) \cap W'$ pointwise, $A := da(E_v, W, 0)$, and $s_k := d \operatorname{mult}_{E_v} h_k$, and write $h_k = \rho^{s_k} H_k$ on W'. Let $R' \subset K$ be a subgroup of order r which acts on ρ alone. We call W'/R', with the parameter $q := \rho^r$ and the action of K/R', the *quotient chart*. Let Δ' be the $\mathbb{R}$-divisor on W'/R' such that $K_{W'/R'} + \Delta'$ is the pullback of $K_{W/G} + D_{W/G}$, where $(W/G, D_{W/G})$ is the crepant quotient of (W, D). Then the following hold.

(a) $d = 1$ unless f_v is a linear monomial.

(b) Each H_k is regular on W' and descends to a regular function on W'/R', and

$$\Delta' = \sum_k c_k \operatorname{div}(H_k) + b \operatorname{div}(q), \qquad b = 1 - \frac{A - \sum_k c_k s_k}{r}.$$

(c) For prime divisors P' over W'/R' and P over W with center o lying over a common prime divisor over W/G,
$$\frac{a(P', W'/R', \Delta')}{|I_{K/R'}(P')|} = a_D(P).$$

(d) W' is normal, W'/μ_{v_l} is the open subset Y_l of Y lying over the chart of the weighted blow-up corresponding to x_l, $(W'/R')/(K/R') = Y_l/G$ is an open subset of Y/G, and K/R' acts faithfully on W'/R'.

Proof. (1) Let P be a prime divisor over W with center o. Since $\operatorname{div}(h_k)$ is Cartier, $a(P, W, C_k) = a(P, W, 0) - \frac{1}{m}\operatorname{mult}_P h_k$ for each k. Since $\sum_k \mu_k = 1$ and $c_k = \mu_k/m$,
$$\sum_k \mu_k a(P, W, C_k) = a(P, W, 0) - \sum_k c_k \operatorname{mult}_P h_k = a(P, W, D).$$
Dividing by $|I_G(P)|$, we obtain $a_D(P) = \sum_k \mu_k a_k(P)$. Under (4), $a(P, W, C_k) > 1$, since (W, C_k) is terminal near o and P is exceptional over W. Under (4'), $a(P, W, C_k) = a(P, W, 0) - \frac{1}{m}\operatorname{mult}_P h_k > 0$ by assumption. Hence $a_k(P) > 0$. Moreover,
$$m|I_G(P)|a_k(P) = ma(P, W, C_k) = ma(P, W, 0) - \operatorname{mult}_P h_k,$$
and $a(P, W, 0) \in \mathbb{Z}$ because K_W is Cartier. Thus $m|I_G(P)|a_k(P)$ is a positive integer.

(2) We first describe W'. Away from $(\rho = 0)$, the root chart maps onto $(x_l \neq 0)$ as the quotient by μ_{v_l}, which acts freely there. Thus $W'\backslash(\rho = 0) \to W \cap (x_l \neq 0)$ is étale. Let $v(f)$ be the least v-weight of a monomial of f. Then $W' = (f' = 0)$, where $f(\rho^{v_1}x'_1, \dots, \rho^{v_4}x'_4) = \rho^{v(f)}f'$ with $x'_l := 1$, and $f'|_{\rho=0} = f_v(x'_1, \dots, x'_4)$. Since f_v is reduced and is not a multiple of x_l, the Cartier divisor $(\rho = 0) \cap W'$ of W' is non-empty and reduced. Hence the hypersurface W' is regular at the generic points of $(\rho = 0) \cap W'$. Since $W'\backslash(\rho = 0)$ is étale over the normal variety W, W' is normal by Serre's criterion. The quotient W'/μ_{v_l} is normal and maps finitely and birationally onto the strict transform of W in the chart $\mathbb{A}^4/\mu_{v_l}$ of the weighted blow-up. Thus W'/μ_{v_l} is an open subset Y_l of Y, and $(\rho = 0) \cap W'$ maps onto $E_v \cap Y_l$, which is irreducible because f_v is irreducible.

(a) Since $E_v \cap Y_l$ is irreducible, μ_{v_l} permutes the irreducible components of $(\rho = 0) \cap W'$ transitively. As μ_{v_l} is abelian, the pointwise stabilizers in μ_{v_l} of the irreducible components of $(\rho = 0) \cap W'$ are all equal, so d does not depend on the component. Suppose that $\zeta \in \mu_{v_l}$, $\zeta \neq 1$, fixes a component S pointwise. Since $\zeta^{v_l} = 1$ and v is primitive, $\zeta^{v_j} \neq 1$ for some $j \neq l$, so $S \subset (\rho = x'_j = 0)$. As S is a surface in $(\rho = 0) \cong \mathbb{A}^3$, $S = (\rho = x'_j = 0)$. Thus x'_j divides $f'|_{\rho=0}$, that is, any monomial of f_v is divisible by x_j. Since f_v is irreducible, f_v is a non-zero constant multiple of x_j. This proves (a).

(b) By [Liu26, Lemma 2.6(1)] applied to $W' \to Y_l$, d is the ramification index of any irreducible component S of $(\rho = 0) \cap W'$ over E_v, so $\operatorname{mult}_S \varphi = d \operatorname{mult}_{E_v} \varphi$ for any regular function φ on W near o. In particular, $\operatorname{mult}_S h_k = s_k$. Since ρ is a unit away from $(\rho = 0)$, the rational function $H_k = \rho^{-s_k} h_k$ has no pole on the normal variety W', so H_k is regular on W', and H_k does not vanish identically on any component of $(\rho = 0) \cap W'$.

The group K acts diagonally in ρ and the x'_j, so K is abelian and ρ is K-semi-invariant. Moreover, $W' \to W$ is equivariant for the surjection $K \to G$ whose kernel is μ_{v_l}, so h_k is K-semi-invariant, and so is H_k. Let $\gamma \in R'$. Then γ fixes $(\rho = 0)$ pointwise and $H_k \circ \gamma = \chi(\gamma) H_k$ for a constant $\chi(\gamma)$. Restricting to a component of $(\rho = 0) \cap W'$ on which H_k is not identically zero, we obtain $\chi(\gamma) = 1$. Thus H_k is R'-invariant and descends to a regular function on W'/R', again denoted by H_k.

Let $\Delta_{W'}$ be the $\mathbb{R}$-divisor on W' such that $K_{W'} + \Delta_{W'}$ is the pullback of $K_W + D$. Away from $(\rho = 0)$, the morphism $W' \to W$ is étale, so on $W'\backslash(\rho = 0)$ we have
$$\Delta_{W'} = \sum_k c_k \operatorname{div}(h_k) = \sum_k c_k \operatorname{div}(H_k).$$

Let S be an irreducible component of $(\rho = 0) \cap W'$, and D_{Y_l} the $\mathbb{R}$-divisor on Y_l such that $K_{Y_l} + D_{Y_l}$ is the pullback of $K_W + D$. Then (Y_l, D_{Y_l}) is the crepant quotient of $(W', \Delta_{W'})$ by μ_{v_l}, and [Liu26, Lemma 2.6(2)] implies

$$\begin{aligned} a(S, W', \Delta_{W'}) &= da(E_v, Y_l, D_{Y_l}) = da(E_v, W, D) \\ &= d\Big(a(E_v, W, 0) - \sum_k c_k \operatorname{mult}_{E_v} h_k\Big) = A - \sum_k c_k s_k. \end{aligned}$$

Since $(\rho = 0) \cap W'$ is reduced, we obtain

$$\Delta_{W'} = \sum_k c_k \operatorname{div}(H_k) + \Big(1 - A + \sum_k c_k s_k\Big) \operatorname{div}(\rho).$$

An element $\gamma \neq 1$ of R' multiplies ρ by a non-trivial root of unity and fixes the x'_j, so the fixed locus of γ on W' is $(\rho = 0) \cap W'$, and R' is the inertia group of any component of $(\rho = 0) \cap W'$. The image of $(\rho = 0) \cap W'$ in W'/R' is $\operatorname{div}(q)$, and $\operatorname{div}(H_k)$ on W' is the pullback of $\operatorname{div}(H_k)$ on W'/R'. By [Liu26, Lemma 2.6(4)], the crepant quotient of $(W', \Delta_{W'})$ by R' is the pair

$$\Big(W'/R',\ \sum_k c_k \operatorname{div}(H_k) + b \operatorname{div}(q)\Big),$$

where

$$b = 1 - \frac{1 - (1 - A + \sum_k c_k s_k)}{r} = 1 - \frac{A - \sum_k c_k s_k}{r}.$$

Since $K_{W'} + \Delta_{W'}$ is the pullback of $K_{W/G} + D_{W/G}$ and pullback of divisors along $W' \to W'/R'$ is injective,

$$\Delta' = \sum_k c_k \operatorname{div}(H_k) + b \operatorname{div}(q).$$

This proves (b).

(c) Since μ_{v_l} is the kernel of $K \to G$, $(W'/R')/(K/R') = W'/K = Y_l/G$, which is an open subset of Y/G, and $Y/G \to W/G$ is birational. Moreover, K acts faithfully on W', because an element acting trivially on W' maps to the identity of G and μ_{v_l} acts freely away from $(\rho = 0)$. Hence K/R' acts faithfully on W'/R' by [Liu26, Lemma 2.6(5)], as W' is irreducible ($x_l|_W$ is not in $\mathfrak{m}^2_{W,o}$, hence for any prime p it is not the p-th power of a rational function on W). Let P_0 be the prime divisor over W/G under P'. By assumption, P_0 is also under P. The crepant quotient of $(W'/R', \Delta')$ by K/R' is $(Y_l/G, D'')$, where $K_{Y_l/G} + D''$ is the pullback of $K_{W/G} + D_{W/G}$ under the birational morphism $Y_l/G \to W/G$, so [Liu26, Lemma 2.6(2)] implies

$$a(P', W'/R', \Delta') = |I_{K/R'}(P')| a(P_0, W/G, D_{W/G}).$$

Again by [Liu26, Lemma 2.6(2)], $a(P, W, D) = |I_G(P)| a(P_0, W/G, D_{W/G})$. Thus

$$\frac{a(P', W'/R', \Delta')}{|I_{K/R'}(P')|} = a(P_0, W/G, D_{W/G}) = \frac{a(P, W, D)}{|I_G(P)|} = a_D(P).$$

The first paragraph of the proof of (2) and the proof of (c) prove (d). □

The marks h_k may have different characters, so each mark has its own weight κ_k and block h_k^0 along a circle (Definition-Lemma 11.5(2)), and Definition-Lemma 11.5(3) adds up the bounds that the terminality of each proxy gives along o and E_0.

Definition-Lemma 11.5. Notation as in Set-up 11.1, where (4) may be replaced by (4') in (1) and (2), with $W = (x^2 + g(y, z, t) = 0)$ in ambient G-eigenparameters x, y, z, t, and let $\mathbf{T} = \mathbf{T}_{(w)}$ be a primitive circle with $r := |G \cap \mathbf{T}|$.

(1) Each h_k is equal in $\widehat{\mathcal{O}}_{W,o}$ to a unique G-semi-invariant element $\phi_k(y, z, t) + x\xi_k(y, z, t)$ with $\phi_k, \xi_k \in \mathbb{C}[[y, z, t]]$, the *normal form* of h_k.

(2) The $\mathbf{T}$-weights of the monomials of $\phi_k + x\xi_k$ belong to one residue class modulo r. If $r > 2(3m-1)\max_j |w_j|$, then the residue class contains a unique integer κ_k with $|\kappa_k| < r/2$, and any monomial of $\phi_k + x\xi_k - h_k^0$ has total degree at least $(r - |\kappa_k|)/\max_j |w_j|$, where h_k^0 is the sum of the terms of $\phi_k + x\xi_k$ of $\mathbf{T}$-weight κ_k.

(3) If $(W \ni o)$ is of type cD, then, with E_0 the prime divisor over W given by the weights $(3,2,2,2)$ in x, y, z, t,

$$\sum_k c_k \operatorname{mult}_o h_k \le 1 - \frac{1}{m}, \qquad \sum_k c_k \operatorname{mult}_{E_0} h_k \le 2 - \frac{1}{m}.$$

Proof. We first bound the orders of the marks, for use in (2) and (3). Let E be a prime divisor over W with center o. Then $\operatorname{mult}_E h_k \ge \operatorname{mult}_o h_k$, and

$$a(E, W, 0) - \frac{\operatorname{mult}_E h_k}{m} = a(E, W, C_k).$$

If W is smooth, we take for E the exceptional divisor of the blow-up of o, so that $a(E, W, 0) = 3$. If W is singular and the quadratic part $x^2 + g_2$ of $x^2 + g$ has rank at least two, we take for E an irreducible component of the exceptional divisor of the blow-up of W at o, so that $a(E, W, 0) = 2$. If $(W \ni o)$ has quadratic rank one, we take $E := E_0$, so that $a(E_0, W, 0) = 3$ and $2\operatorname{mult}_o h_k \le \operatorname{mult}_{E_0} h_k$. If W is singular and $x^2 + g_2$ has rank at least two, then $x^2 + g_2$ is reduced, and adjunction implies $a(E, W, 0) = 4 - 2 = 2$.

Assume that $(W \ni o)$ is of quadratic rank one. Then $g_2 = 0$, and the cubic part g_3 of g is the residual cubic of $(W \ni o)$. We show that $g_3 \neq 0$. The germ $(W \ni o)$ is an isolated cDV point [KM98, Corollary 5.38]. If $g_3 = 0$, a general hyperplane section of W through o has quadratic rank one and zero residual cubic, and by [Liu26, Definition-Lemma 11.1] (with $d = 3$) both properties are invariant under analytic isomorphism, whereas every Du Val point in the list of [KM98, Theorem 4.20] has quadratic rank at least two or a non-zero residual cubic. Since g_3 is a non-zero form of odd degree, $-g_3$ is not a square, so $x^2 + g_3$ is reduced and irreducible. The lowest $(3,2,2,2)$-weighted part of $x^2 + g$ is $x^2 + g_3$, of weight 6, and it is not a linear monomial. Hence E_0 is the exceptional prime divisor of the normalized strict transform of W in the weighted blow-up with weights $(3,2,2,2)$, $\operatorname{mult}_{E_0}$ of a regular function on W is at least the $(3,2,2,2)$-weighted order of any lift (Definition-Lemma 11.4(2)(a)), and adjunction implies

$$a(E_0, W, 0) = (3 + 2 + 2 + 2) - 6 = 3.$$

Let $\tilde{h}_k$ be a lift of h_k to the ambient germ with $\tilde{h}_k \in \mathfrak{m}^{\operatorname{mult}_o h_k}$, where $\mathfrak{m}$ is the maximal ideal of the ambient germ. All weights are at least 2, so the $(3,2,2,2)$-weighted order of $\tilde{h}_k$ is at least $2\operatorname{mult}_o h_k$. Hence $2\operatorname{mult}_o h_k \le \operatorname{mult}_{E_0} h_k$.

Under (4'), $a(E, W, C_k) > 0$, so $\operatorname{mult}_E h_k \le m a(E, W, 0) - 1 \le 3m - 1$, and $\operatorname{mult}_o h_k \le 3m - 1$ in all cases. Under (4), $a(E, W, C_k) > 1$ since (W, C_k) is terminal near o, so $\operatorname{mult}_E h_k \le m(a(E, W, 0) - 1) - 1$. Thus $\operatorname{mult}_o h_k \le \operatorname{mult}_E h_k \le 2m - 1$ if W is smooth, and $\operatorname{mult}_o h_k \le \operatorname{mult}_E h_k \le m - 1$ if W is singular and $x^2 + g_2$ has rank at least two. If $(W \ni o)$ is of quadratic rank one, then $\operatorname{mult}_{E_0} h_k \le 2m - 1$ and $2\operatorname{mult}_o h_k \le 2m - 1$, so

$$\operatorname{mult}_o h_k \le m - 1, \qquad \operatorname{mult}_{E_0} h_k \le 2m - 1. \tag{11.1}$$

(1) Since $(x^2 + g)(x, 0, 0, 0) = x^2$, the Weierstrass division theorem [GLS07, Chapter I, Theorem 1.8] in $\mathbb{C}[[x, y, z, t]]$ (with the trivial valuation on $\mathbb{C}$, for which $K\langle x\rangle = K[[x]]$ by [GLS07, Chapter I, Remark 1.1.1(1)]) implies that any element of $\mathbb{C}[[x, y, z, t]]$ is congruent modulo $x^2 + g$ to a unique element of the form $\phi(y, z, t) + x\xi(y, z, t)$. As $\widehat{\mathcal{O}}_{W,o} = \mathbb{C}[[x, y, z, t]]/(x^2 + g)$, this proves existence and uniqueness. Since G acts diagonally, any $\gamma \in G$ maps $\phi_k + x\xi_k$ to an element of the form $\phi(y, z, t) + x\xi(y, z, t)$, which is equal to $\chi(\gamma)h_k$ in $\widehat{\mathcal{O}}_{W,o}$, where χ is the character of h_k. By uniqueness, γ maps $\phi_k + x\xi_k$ to $\chi(\gamma)(\phi_k + x\xi_k)$, so the normal form is G-semi-invariant.

(2) Since w is primitive, $\lambda \to (\lambda^{w_1}, \dots, \lambda^{w_4})$ is an isomorphism from $\mathbb{C}^*$ onto $\mathbf{T}$. Thus $G \cap \mathbf{T}$ is the cyclic group of order r generated by $(e^{2\pi\sqrt{-1}w_1/r}, \dots, e^{2\pi\sqrt{-1}w_4/r})$, which multiplies a

monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ by $e^{2\pi\sqrt{-1}\sum_j w_j a_j/r}$. Since the normal form is G-semi-invariant by (1), the $\mathbf{T}$-weights $\sum_j w_j a_j$ of the monomials of $\phi_k + x\xi_k$ (Set-up 5.1) are congruent modulo r.

The normal form maps to h_k in $\widehat{\mathcal{O}}_{W,o}$, so it has a monomial of total degree at most $\operatorname{mult}_o h_k \le 3m-1$. Assume that $r > 2(3m-1)\max_j |w_j|$. The $\mathbf{T}$-weight κ_k of that monomial satisfies $|\kappa_k| \le (3m-1)\max_j |w_j| < r/2$, so κ_k is the unique integer of the residue class with $|\kappa_k| < r/2$. A monomial of the normal form whose $\mathbf{T}$-weight is not κ_k has $\mathbf{T}$-weight $\kappa_k + \ell r$ with $\ell \neq 0$, so $|\kappa_k + \ell r| \ge r - |\kappa_k|$. On the other hand, $|\kappa_k + \ell r|$ is at most the total degree of the monomial times $\max_j |w_j|$. Thus any monomial of $\phi_k + x\xi_k - h_k^0$ has total degree at least $(r - |\kappa_k|)/\max_j |w_j|$.

(3) Under (4), (11.1) gives $\operatorname{mult}_o h_k \le m-1$ and $\operatorname{mult}_{E_0} h_k \le 2m-1$ for each k, as a source of type cD is of quadratic rank one (Definition 9.1). Multiplying by $c_k = \mu_k/m$, summing over k, and using $\sum_k \mu_k = 1$, we obtain

$$\sum_k c_k \operatorname{mult}_o h_k \le \frac{m-1}{m} = 1 - \frac{1}{m}, \qquad \sum_k c_k \operatorname{mult}_{E_0} h_k \le \frac{2m-1}{m} = 2 - \frac{1}{m}.$$

□

Remark 11.6. In Set-up 11.1 all h_k may be units. Then $D = 0$ near o, $\alpha = \operatorname{mld}((W \ni o, 0)/G)$, and the boundary of the crepant quotient $(W, 0)/G$ consists of branch divisors with coefficients $1 - \frac{1}{e}$ [Liu26, Lemma 2.6(4)], where the ramification indices e need not be bounded, so the coefficients $1 - \frac{1}{e}$ need not belong to a fixed finite set. For example, take $h_1 = \cdots = h_s = 1$, and let n be a positive integer, $\omega := e^{2\pi\sqrt{-1}/3}$, $\zeta := e^{2\pi\sqrt{-1}/(6n)}$, $W = (x^2 + y^3 + z^3 + t^{6n} = 0) \subset \mathbb{A}^4$, and G the cyclic group of order $6n$ generated by $\operatorname{diag}(1, \omega, \omega, \zeta)$. The partial derivatives of $x^2 + y^3 + z^3 + t^{6n}$ vanish only at 0, and a general hyperplane section of W through 0 is a Du Val point of type D_4, since its quadratic part is x^2 and its cubic part $y^3 + z^3$ restricted to the plane has three distinct linear factors. Thus $(W \ni 0)$ is an isolated cDV point, hence Gorenstein terminal [KM98, Corollary 5.38], of quadratic rank one with residual cubic $y^3 + z^3$, which is not a cube, so it is of type cD. The inertia group of $(t = 0)$ is generated by $\operatorname{diag}(1, 1, 1, \zeta^3)$, so $(t = 0)$ is a branch divisor with coefficient $1 - \frac{1}{2n}$. The prime divisor over W given by the weights $(3n, 2n, 2n, 1)$ has log discrepancy $(3n + 2n + 2n + 1) - 6n = n + 1$ and inertia group generated by $\operatorname{diag}(1, \omega^2, \omega^2, \zeta^2)$, of order $3n$. Thus

$$\alpha \le \frac{n+1}{3n} = \frac{1}{3} + \frac{1}{3n} < 1.$$

The numbers α of such data are minimal log discrepancies of threefold pairs with coefficients in $\{1 - \frac{1}{e} \mid e \in \mathbb{Z}_{\ge 1}\}$, and they satisfy the ACC by [Liu26, Theorem 1.3].

11.2. **Admission, bounded groups and crepant models.** In this subsection, we prove four results on the data of Set-up 11.1: the case of bounded groups together with the sources covered by Theorem 7.1 and [Liu26, Proposition 11.10] (Proposition 11.7), a numerical admission which keeps the scores of the proxies positive (Proposition 11.8), the weighted blow-ups over sources of quadratic rank one (Proposition 11.9), and the properties of the numerical admissions of Proposition 11.8 (Lemma 11.10).

Proposition 11.7. *Notation as in Set-up 11.1, where (4) may be replaced by (4') in (0), and fix s, m, and $\mu_1, \dots, \mu_s$. Then the following hold.*

(0) There exists a prime divisor F over W with center o such that $a_D(F) = \alpha$.

(1) For any positive integer M and any real number $\lambda > 0$, the numbers $\alpha \in (0, \lambda]$ for all data as in Set-up 11.1 with the fixed s, m, $\mu_1, \dots, \mu_s$ and with $|G| \le M$ belong to a finite set.

(2) The numbers $\alpha \in (0, 1)$ for all data as in Set-up 11.1 with the fixed s, m, $\mu_1, \dots, \mu_s$ such that W is smooth, or W has quadratic rank at least three, or W has quadratic rank one and a smooth or an irreducible nodal residual cubic, satisfy the ACC.

Proof. Let $\lambda > 0$ be a real number and P a prime divisor over W with center o such that $a_D(P) \le \lambda$. By Definition-Lemma 11.4(1), $a_D(P) = \sum_k \mu_k a_k(P)$ and $a_k(P) > 0$ for any k, hence $a_k(P) \le \lambda/\mu_k$ for any k. Moreover, $m|I_G(P)|a_k(P)$ is a positive integer, and $|I_G(P)| \le |G|$. For a positive integer M, let $S(M,\lambda)$ be the set of numbers $\sum_k \mu_k t_k$, where $t_k \in \frac{1}{m\cdot M!}\mathbb{Z}$ and $0 < t_k \le \lambda/\mu_k$ for any k. The set $S(M,\lambda)$ is finite and depends only on s, m, $\mu_1,\dots,\mu_s$, M, and λ. Since $|I_G(P)|$ divides $M!$ when $|G| \le M$, we have shown

$$a_D(P) \in S(M,\lambda) \quad \text{if } |G| \le M \text{ and } a_D(P) \le \lambda. \tag{11.2}$$

(0) Recall the formula at the end of [Liu26, Lemma 2.6]:

$$\operatorname{mld}((X \ni x, B)/G) = \inf_F \frac{a(F,X,B)}{|I_G(F)|},$$

where G is a finite group acting faithfully on a normal quasi-projective variety X and fixing a closed point x, B is a G-invariant $\mathbb{R}$-divisor such that $K_X + B$ is $\mathbb{R}$-Cartier, and F runs over the prime divisors over X with center x. For $(W \ni o, D)$ and G, it gives $\alpha = \inf_P a_D(P)$, where P runs over the prime divisors over W with center o. By (11.2) with $M = |G|$ and $\lambda = \alpha + 1$, the numbers $a_D(P) \le \alpha + 1$ belong to the finite set $S(|G|, \alpha+1)$, and at least one of them is smaller than $\alpha + 1$. Thus the infimum is a minimum.

(1) Assume that $|G| \le M$ and $\alpha \le \lambda$. By (0), $\alpha = a_D(F)$ for a prime divisor F over W with center o, so $\alpha \in S(M,\lambda)$ by (11.2).

(2) For any prime divisor P over W, $a(P,W,D) = \sum_k \mu_k a(P,W,C_k)$, and each (W,C_k) is terminal near o. Thus (W,D) is klt near o [KM98, Corollary 2.31(1) and (3)]. If W is smooth or has quadratic rank at least three, then α is one of the positive numbers of Theorem 7.1 for the finite set $\Gamma := \{c_1,\dots,c_s\} \subset [0,1]$: G is a finite group acting faithfully on W and fixing o, $D = \sum_k c_k \operatorname{div}(h_k)$, and the h_k are G-semi-invariant regular functions (Definition 2.7). Thus the numbers α with W smooth or of quadratic rank at least three satisfy the ACC.

Suppose that there exist data $(W_i \ni o_i)$, G_i, $h_{i,k}$, $i \ge 1$, as in Set-up 11.1, such that W_i has quadratic rank one and a smooth or an irreducible nodal residual cubic, and such that the corresponding numbers $\alpha_i \in (0,1)$ are strictly increasing. Let $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$ and $a_i := \operatorname{mld}((W_i \ni o_i, 0)/G_i)$. By the formula at the end of [Liu26, Lemma 2.6],

$$a_i = \inf_P \frac{a(P,W_i,0)}{|I_{G_i}(P)|} \ge \inf_P \frac{a(P,W_i,D_i)}{|I_{G_i}(P)|} = \alpha_i,$$

where P runs over the prime divisors over W_i with center o_i. The inequality holds since $a(P,W_i,0) - a(P,W_i,D_i) = \sum_k c_k \operatorname{mult}_P h_{i,k} \ge 0$. Thus $a_i \ge \alpha_1 > 0$. The germ $(W_i \ni o_i)$ is an isolated Gorenstein terminal threefold singularity of quadratic rank one, and G_i is cyclic, hence abelian. By [Liu26, Proposition 11.10], $|G_i| \le 162/a_i \le 162/\alpha_1$. By (1) with $M = \lfloor 162/\alpha_1 \rfloor$ and $\lambda = 1$, the numbers α_i belong to a finite set, a contradiction. Since a union of three sets satisfying the ACC satisfies the ACC, we obtain (2). □

A numerical admission of $(W,(h_k)_k)$ has to keep a prime divisor F computing α together with $I_G(F)$, and the order bounds which the terminality of the proxies implies at fixed prime divisors, such as E_0 and the prime divisors given by the weights $(2m+2, 2m+1, 2, 1)$ in Subsection E.10. Terminality of the proxies need not survive the admission, so we include, for each k, the vector with k-th entry $\frac{1}{m}$ and other entries 0 among the vectors in Construction-Theorem 4.13(iv), and F, the prime divisors $P_1,\dots,P_l$ of Proposition 11.8 below, and E_0 if W has quadratic rank one, among the prime divisors in Construction-Theorem 4.13(v).

Proposition 11.8. *Notation as in Set-up 11.1. Let F be a prime divisor over W with center o such that $a_D(F) = \alpha$ (Proposition 11.7(0)), and $P_1,\dots,P_l$ finitely many prime divisors over W with center o. Assume that W is singular. Let $(W_0,(h_{k,0})_k)$ be a numerical admission of $(W,(h_k)_k)$ (Construction-Theorem 4.13) for the labels $c_1,\dots,c_s$, such that*

(i) the vectors in Construction-Theorem 4.13(iv) include, for each k, the vector in $\mathbb{R}^s_{\geq 0}$ whose k-th entry is $\frac{1}{m}$ and whose other entries are 0, and
(ii) the prime divisors in Construction-Theorem 4.13(v) include F, $P_1, \dots, P_l$, and E_0 if W has quadratic rank one.

Such numerical admissions exist. Let $D_0 := \sum_k c_k \operatorname{div}(h_{k,0}|_{W_0})$, and F_0 the prime divisor over W_0 corresponding to F. Then

(1) $\alpha = \operatorname{mld}((W_0 \ni 0, D_0)/G)$, $I_G(F_0) = I_G(F)$, and $a(F_0, W_0, D_0) = |I_G(F_0)|\alpha$, and
(2) $a(P, W_0, 0) - \frac{1}{m}\operatorname{mult}_P(h_{k,0}|_{W_0}) > 0$ for any k and any prime divisor P over W_0 with center 0.

Proof. We show that such numerical admissions exist. Each (W, C_k) is terminal, hence klt, near o [KM98, Corollary 2.31(1) and (3)], so the vectors in (i) satisfy Construction-Theorem 4.13(iv). Ambient G-eigenparameters $x_1, \dots, x_4$ at o with a G-semi-invariant local equation of W, as in Definition-Lemma 11.4(2), satisfy Construction-Theorem 4.13(ii), and projecting any lift of h_k to the ambient germ to the character of h_k provides the regular G-semi-invariant functions of Construction-Theorem 4.13(iii).

(1) Construction-Theorem 4.13(1) for the vector $(c_1, \dots, c_s)$ and the subgroup G implies $\alpha = \operatorname{mld}((W \ni o, D)/G) = \operatorname{mld}((W_0 \ni 0, D_0)/G)$. By Construction-Theorem 4.13(2) for F, we have $I_G(F_0) = I_G(F)$, $a(F_0, W_0, 0) = a(F, W, 0)$, and $\operatorname{mult}_{F_0}(h_{k,0}|_{W_0}) = \operatorname{mult}_F h_k$ for any k. Thus

$$\begin{aligned} a(F_0, W_0, D_0) &= a(F_0, W_0, 0) - \sum_k c_k \operatorname{mult}_{F_0}(h_{k,0}|_{W_0}) \\ &= a(F, W, 0) - \sum_k c_k \operatorname{mult}_F h_k = a(F, W, D) = |I_G(F)|\alpha. \end{aligned}$$

(2) Fix k, and let $C_{k,0} := \frac{1}{m}\operatorname{div}(h_{k,0}|_{W_0})$. Construction-Theorem 4.13(1) for the vector in (i) with k-th entry $\frac{1}{m}$ and the trivial subgroup implies

$$\operatorname{mld}(W_0 \ni 0, C_{k,0}) = \operatorname{mld}(W \ni o, C_k).$$

Since (W, C_k) is terminal near o, $a(P', W, C_k) > 1$ for any prime divisor P' over W with center o, hence $\operatorname{mld}(W \ni o, C_k) \geq 1$. Thus, for any prime divisor P over W_0 with center 0,

$$a(P, W_0, 0) - \frac{1}{m}\operatorname{mult}_P(h_{k,0}|_{W_0}) = a(P, W_0, C_{k,0}) \geq \operatorname{mld}(W_0 \ni 0, C_{k,0}) \geq 1 > 0. \quad \square$$

Over a source of quadratic rank one, the weighted blow-ups of Definition-Lemma 11.4(2) induce proper birational morphisms onto W/G with crepant boundaries, and the coefficient of the exceptional divisor is $1 - (A - \sum_k c_k s_k)/|I_G(E_v)|$. At a singular point with bounded inertia, the values with real labels do not belong to the fixed lattice of Lemma 4.8, and Proposition 11.9(2) replaces Lemma 4.8 by the bounds $\ell_i \leq \alpha_i \leq \ell_i + (b_* - b_i)\ell_i/\eta$, where b_* is the limit of the b_i. The assumption of Proposition 11.9(2) on b_* is checked in **Step 8** of the proof of Proposition 11.13, where b_* is taken outside the closure of the ACC set of Theorem 2.12(1).

Proposition 11.9. *Notation as in Set-up 11.1, where (4) may be replaced by (4'). Assume that W has quadratic rank one and that the residual cubic of W is not a cube. Then the following hold.*

(0) Let $x_1, \dots, x_4$ be ambient G-eigenparameters at o, f a G-semi-invariant local equation of W in them, v a primitive vector of positive integers such that the lowest v-weighted part f_v of f is reduced and irreducible, $g : Y' \to W$ the normalized strict transform of W in the weighted blow-up with weights v, E_v the exceptional prime divisor of g, $A := a(E_v, W, 0)$, and $s_k := \operatorname{mult}_{E_v} h_k$. Let $\bar{o}$ be the image of o in W/G, $\phi : Y := Y'/G \to W/G$ the induced morphism, $(W/G, D_{W/G})$ the crepant quotient of (W, D), and

$K_Y + \Delta_Y := \phi^*(K_{W/G} + D_{W/G})$. *Then the coefficient of the image of* E_v *in* Δ_Y *is*

$$1 - \frac{A - \sum_k c_k s_k}{|I_G(E_v)|},$$

and

$$\alpha = \min\{\mathrm{mld}(Y \ni \xi, \Delta_Y) \mid \xi \in \phi^{-1}(\bar{o})\},$$

where ξ *runs over the scheme-theoretic points of* $\phi^{-1}(\bar{o})$. *Moreover, there exists a prime divisor* E *over* W/G *with center* $\bar{o}$ *such that* $a(E, W/G, D_{W/G}) = \alpha$, *and for any such* E, *the minimum is attained at the generic point of the center of* E *on* Y.

(1) *Let* $(W_i \ni o_i)$, G_i, $h_{i,k}$, $x_{i,1}, \dots, x_{i,4}$, f_i, *and* v_i, $i \geq 1$, *be a sequence of data as in (0) with the fixed* s, m, $\mu_1, \dots, \mu_s$, *and denote by* α_i, $\bar{o}_i$, $\phi_i : Y_i \to W_i/G_i$, Δ_{Y_i}, *and* E_{v_i} *the corresponding objects. Let* b_i *be the coefficient of the image of* E_{v_i} *in* Δ_{Y_i}, *and assume that* $b_i \in [0, 1)$ *and* $b_i \leq b_{i+1}$ *for any* i. *For any* i, *let* ξ_i *be the generic point of the center on* Y_i *of a prime divisor* E_i *over* W_i/G_i *with center* $\bar{o}_i$ *such that* $a(E_i, W_i/G_i, D_{W_i/G_i}) = \alpha_i$. *Then there exists a DCC set* $\Lambda \subset [0, 1]$ *such that* $\Delta_{Y_i} \in \Lambda$ *for any* i, *and the set* $\{\alpha_i \mid \dim \overline{\{\xi_i\}} \geq 1\}$ *satisfies the ACC. Moreover, the set of the* α_i *with the following property satisfies the ACC:* ξ_i *is a closed point, and for some* l, *the quotient chart* W_i'/R_i' *of* v_i *at* $x_{i,l}$ *(Definition-Lemma 11.4(2)), where* R_i' *is the subgroup of the elements of* K_i *acting on* ρ *alone, has a closed point* y_i'' *over* ξ_i *such that* $(W_i'/R_i' \ni y_i'')$ *is a smooth germ, a hypersurface germ of quadratic rank four, or a non-isolated hypersurface germ of quadratic rank three.*

(2) *Let* $(Y_i \ni y_i)$, $i \geq 1$, *be smooth threefold germs or hypersurface threefold germs of quadratic rank at least three, and* J_i *finite cyclic groups of bounded order acting faithfully on* Y_i *and fixing* y_i. *Let* $H_{i,k}$ *and* q_i *be non-zero* J_i*-semi-invariant regular functions on* Y_i, *and* $b_i \in [0, 1)$ *non-decreasing real numbers with limit* $b_* \in (0, 1)$, *such that* (Y_i, Δ_i) *is klt near* y_i, *where*

$$\Delta_i := \sum_k c_k \operatorname{div}(H_{i,k}) + b_i \operatorname{div}(q_i).$$

Let $\alpha_i := \mathrm{mld}((Y_i \ni y_i, \Delta_i)/J_i)$. *Assume that* b_* *is not in the closure of the set of the finite log canonical thresholds of* $\operatorname{div}(q_i)$ *with respect to* $(Y_i, \sum_k c_k \operatorname{div}(H_{i,k}))$ *near* y_i, $i \geq 1$. *Then there exists a real number* $\eta > 0$ *such that, for* $i \gg 0$, *the pair*

$$\big(Y_i, \textstyle\sum_k c_k \operatorname{div}(H_{i,k}) + (b_* + \eta) \operatorname{div}(q_i)\big)$$

is lc near y_i, *the positive numbers*

$$\ell_i := \mathrm{mld}\Big(\big(Y_i \ni y_i, \textstyle\sum_k c_k \operatorname{div}(H_{i,k}) + b_* \operatorname{div}(q_i)\big)/J_i\Big)$$

satisfy the ACC, and

$$\ell_i \leq \alpha_i \leq \ell_i + \frac{(b_* - b_i)\ell_i}{\eta}.$$

Proof. (0) Since $x_1, \dots, x_4$ are G-eigenparameters, G acts on the weighted blow-up with weights v, hence on Y', and g is G-equivariant. Let $p : W \to W/G$ and $p' : Y' \to Y$ be the quotient morphisms, so that $p \circ g = \phi \circ p'$, and let $K_{Y'} + D_{Y'} := g^*(K_W + D)$. Then

$$p'^*(K_Y + \Delta_Y) = g^* p^*(K_{W/G} + D_{W/G}) = g^*(K_W + D) = K_{Y'} + D_{Y'},$$

so (Y, Δ_Y) is the crepant quotient of $(Y', D_{Y'})$ by G. The coefficient of E_v in $D_{Y'}$ is

$$1 - a(E_v, W, D) = 1 - A + \sum_k c_k s_k.$$

By [Liu26, Lemma 2.6(1)(4)], the ramification index of p' along E_v is $|I_G(E_v)|$, and the coefficient of $p'(E_v)$ in Δ_Y is

$$1 - \frac{1 - (1 - A + \sum_k c_k s_k)}{|I_G(E_v)|} = 1 - \frac{A - \sum_k c_k s_k}{|I_G(E_v)|}.$$

By definition, $\alpha = \mathrm{mld}(W/G \ni \bar{o}, D_{W/G})$. Since ϕ is proper and birational and $K_Y + \Delta_Y = \phi^*(K_{W/G} + D_{W/G})$, we have $a(E, Y, \Delta_Y) = a(E, W/G, D_{W/G})$ for any prime divisor E over Y. A prime divisor E over W/G has center $\bar{o}$ if and only if the center of E on Y is the closure of a point $\xi \in \phi^{-1}(\bar{o})$. Thus

$$\alpha = \inf\{\mathrm{mld}(Y \ni \xi, \Delta_Y) \mid \xi \in \phi^{-1}(\bar{o})\}.$$

By Proposition 11.7(0), there exists a prime divisor F over W with center o such that $a_D(F) = \alpha$. By [Liu26, Lemma 2.6(2)], there exists a prime divisor E over W/G with center $p(o) = \bar{o}$ such that $a(F, W, D) = |I_G(F)|a(E, W/G, D_{W/G})$, hence $a(E, W/G, D_{W/G}) = \alpha$. For any such E, let ξ_E be the generic point of the center of E on Y. Then $\xi_E \in \phi^{-1}(\bar{o})$ and

$$\mathrm{mld}(Y \ni \xi_E, \Delta_Y) \le a(E, Y, \Delta_Y) = \alpha \le \mathrm{mld}(Y \ni \xi_E, \Delta_Y).$$

Thus the infimum is a minimum, attained at ξ_E.

(1) Let $\Gamma := \{c_1, \dots, c_s\}$, $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$, $g_i : Y_i' \to W_i$ the morphisms of (0), $p_i : W_i \to W_i/G_i$ and $p_i' : Y_i' \to Y_i$ the quotient morphisms, and $E_i' := p_i'(E_{v_i})$. The morphism g_i is an isomorphism over $W_i \backslash \{o_i\}$, and $g_i^{-1}(o_i)$ is the support of the pullback of the exceptional divisor of the weighted blow-up, which is E_{v_i}. Since $p_i^{-1}(\bar{o}_i) = \{o_i\}$, we have $\phi_i^{-1}(\bar{o}_i) = p_i'(g_i^{-1}(o_i)) = E_i'$ as sets, and ϕ_i is an isomorphism over $(W_i/G_i)\backslash\{\bar{o}_i\}$. Thus any prime divisor B' on Y_i other than E_i' is the strict transform of a prime divisor B on W_i/G_i, and the coefficient of B' in Δ_{Y_i} equals the coefficient of B in D_{W_i/G_i}. Since (W_i, D_i) is klt near o_i by Set-up 11.1(4'), which (4) implies, $D_i \in \Gamma_+ \cap [0, 1]$, hence $D_{W_i/G_i} \in \Psi(\Gamma_+ \cap [0, 1])$ by [Liu26, Lemma 2.10(5)]. Let

$$\Lambda := \Psi(\Gamma_+ \cap [0, 1]) \cup \{b_i \mid i \ge 1\} \subset [0, 1].$$

Then $\Delta_{Y_i} \in \Lambda$ for any i. The set $\Psi(\Gamma_+ \cap [0, 1])$ satisfies the DCC by [Liu26, Lemma 2.10(1)], and so does $\{b_i \mid i \ge 1\}$, since b_i is non-decreasing. Thus Λ satisfies the DCC.

Let i be an index such that $\overline{\{\xi_i\}}$ is a divisor. Since $\overline{\{\xi_i\}} \subset \phi_i^{-1}(\bar{o}_i) = E_i'$, the center of E_i on Y_i is E_i', so $E_i = E_i'$ and $\alpha_i = a(E_i', Y_i, \Delta_{Y_i}) = 1 - b_i$. If $\alpha_{i_1} < \alpha_{i_2} < \cdots$ for such indices, then $b_{i_1} > b_{i_2} > \cdots$, hence $i_1 > i_2 > \cdots$. Thus the set of the α_i for such indices satisfies the ACC.

Let i be an index such that $\overline{\{\xi_i\}}$ is a curve. By (0), $\alpha_i = \mathrm{mld}(Y_i \ni \xi_i, \Delta_{Y_i}) > 0$, where (Y_i, Δ_{Y_i}) is a threefold pair with $\Delta_{Y_i} \in \Lambda$. By [Liu26, Lemma 3.7], the set of the α_i for such indices satisfies the ACC. Thus the set $\{\alpha_i \mid \dim \overline{\{\xi_i\}} \ge 1\}$ satisfies the ACC.

For the last statement of (1), fix an index i with the property of (1), and drop the index i. Let W', ρ, x_j' $(j \ne l)$, K, and H_k be as in Definition-Lemma 11.4(2) for v at x_l, $R' \subset K$ the subgroup of the elements acting on ρ alone, $r := |R'|$, $q := \rho^r$, and $y := \xi$.

Step 1. In this step, we show that y has an open neighborhood U in Y such that $W'/R' \to U$ is the quotient by K/R', that K/R' acts faithfully on W'/R', and that the inertia group in K/R' of any prime divisor on W'/R' contained in $(q = 0)$ is trivial. By Definition-Lemma 11.4(2)(d), applied with the normalized strict transform Y' of (0) in the role of Y there, $U := Y_l'/G = W'/K$, where $Y_l' \subset Y'$ is the open subset Y_l there, is an open subset of $Y = Y'/G$, $W'/R' \to U$ is the quotient by K/R', and K/R' acts faithfully on W'/R'. The open subset U contains y, since the point y'' of W'/R' maps to y.

Since W is singular, $f \in \mathfrak{m}_o^2$, so no x_j divides f_v: otherwise $f_v = ax_j$ for some $a \in \mathbb{C}^*$, as f_v is irreducible. Since distinct monomials of f_v remain distinct after substituting 1 for x_l, no x_j' vanishes on any irreducible component of $W' \cap (\rho = 0)$. As K is generated by diagonal maps, an element of K acting trivially on an irreducible component of $W' \cap (\rho = 0)$ acts trivially on all x_j', hence acts on ρ alone and belongs to R'. Thus, for any prime divisor T on W'/R' contained in $(q = 0)$, the inertia group $I_{K/R'}(T)$ is trivial, since a lift to K of an element of $I_{K/R'}(T)$ fixes pointwise the irreducible components of $W' \cap (\rho = 0)$ over T, as R' acts trivially on $(\rho = 0)$.

Step 2. In this step, we show that

$$\Delta' = \sum_k c_k \operatorname{div}(H_k) + b \operatorname{div}(q),$$

where b is the coefficient of $E' := p'(E_v)$ in Δ_Y and Δ' is the $\mathbb{R}$-divisor of Definition-Lemma 11.4(2), that $(W'/R', \Delta')$ is klt, and that the H_k and q are K/R'-semi-invariant regular functions on W'/R'. By Definition-Lemma 11.4(2), $K_{W'/R'} + \Delta'$ is the pullback of $K_U + \Delta_Y|_U$, and

$$\Delta' = \sum_k c_k \operatorname{div}(H_k) + b' \operatorname{div}(q),$$

where $b' = 1 - (A - \sum_k c_k s_k)/r$. Since W is singular, f_v is not a linear monomial, so $d = 1$ in Definition-Lemma 11.4(2) by its part (a), and its A and s_k are those of (0). Any prime divisor contained in $(q = 0)$ maps onto E' with ramification index one (**Step 1**), so [Liu26, Lemma 2.6(1)(4)] implies $b = 1 - (1 - b')/1 = b'$. By Definition-Lemma 11.4(2)(b), each H_k descends to a regular function on W'/R', which is K/R'-semi-invariant since $H_k = \rho^{-s_k} h_k$ with h_k and ρ K-semi-invariant, and so is q. The pair $(W/G, D_{W/G})$ is klt near $\bar{o}$ by [Liu26, Lemma 2.6(2)(3)], since (W, D) is klt near o by Set-up 11.1(4'), which (4) implies. Since $K_Y + \Delta_Y = \phi^*(K_{W/G} + D_{W/G})$, the pair (Y, Δ_Y) is klt, and so is $(W'/R', \Delta')$ by [Liu26, Lemma 2.6(2)].

Step 3. In this step, we conclude. Let $J \subset K/R'$ be the stabilizer of y''. By [Liu26, Lemma 2.6(2)(3)], any prime divisor over U with center y is the image of a prime divisor P'' over W'/R' whose center is a point over y. Since K/R' acts transitively on the points over y, we may assume that the center of P'' is y''. Then $I_{K/R'}(P'')$ preserves y'', so $I_{K/R'}(P'') = I_J(P'')$. By [Liu26, Lemma 2.6(2)] and the formula at the end of [Liu26, Lemma 2.6], applied to J acting on W'/R' and fixing y'',

$$\operatorname{mld}(Y \ni y, \Delta_Y) = \inf_{P''} \frac{a(P'', W'/R', \Delta')}{|I_J(P'')|} = \operatorname{mld}((W'/R' \ni y'', \Delta')/J),$$

where P'' runs over the prime divisors over W'/R' with center y''.

Restoring the index i, (0) and **Steps 1–3** imply that

$$\alpha_i = \operatorname{mld}\Big(\big(W_i'/R_i' \ni y_i'', \textstyle\sum_k c_k \operatorname{div}(H_{i,k}) + b_i \operatorname{div}(q_i)\big)/J_i \Big),$$

where $(W_i'/R_i' \ni y_i'')$ is a smooth germ, a germ of quadratic rank four, or a non-isolated germ of quadratic rank three, J_i is a finite group acting faithfully on W_i'/R_i' and fixing y_i'', the $H_{i,k}$ and q_i are J_i-semi-invariant regular functions, the pair is klt near y_i'', and the labels belong to the DCC set $\Gamma \cup \{b_i \mid i \geq 1\} \subset [0, 1]$. By Theorem 7.1, the set of the α_i for such indices satisfies the ACC.

(2) Let $\Delta_{0,i} := \sum_k c_k \operatorname{div}(H_{i,k})$, and $t_i \in (0, +\infty]$ the log canonical threshold of $\operatorname{div}(q_i)$ with respect to $(Y_i, \Delta_{0,i})$ near y_i. Let $\mathcal{T}$ be the set of the finite t_i. Since b_* is not in the closure of $\mathcal{T}$, there exists a real number $\eta > 0$ such that $(b_* - 2\eta, b_* + 2\eta) \cap \mathcal{T} = \emptyset$. Since (Y_i, Δ_i) is klt near y_i, $t_i \geq b_i$, and $b_i > b_* - 2\eta$ for $i \gg 0$. Thus $t_i \geq b_* + 2\eta$ for $i \gg 0$, and $(Y_i, \Delta_{0,i} + (b_* + \eta) \operatorname{div}(q_i))$ is lc near y_i.

Fix $i \gg 0$, and let $\Delta_{*,i} := \Delta_{0,i} + b_* \operatorname{div}(q_i)$ and $\theta := b_*/(b_* + \eta) \in (0, 1)$. For any prime divisor P over a neighborhood of y_i,

$$a(P, Y_i, \Delta_{*,i}) = \theta a\big(P, Y_i, \Delta_{0,i} + (b_* + \eta) \operatorname{div}(q_i)\big) + (1 - \theta) a(P, Y_i, \Delta_{0,i}),$$

where the first term is non-negative and $a(P, Y_i, \Delta_{0,i}) \geq a(P, Y_i, \Delta_i) > 0$, as $b_i \geq 0$. Thus $(Y_i, \Delta_{*,i})$ is klt near y_i, and $\ell_i > 0$. By Theorem 7.1 for the finite set $\{c_1, \ldots, c_s, b_*\} \subset [0, 1]$, the numbers ℓ_i satisfy the ACC.

Let P be a prime divisor over Y_i with center y_i. Since $(Y_i, \Delta_{*,i} + \eta \operatorname{div}(q_i))$ is lc near y_i,

$$\eta \operatorname{mult}_P q_i \leq a(P, Y_i, \Delta_{*,i}).$$

Since $b_i \leq b_*$,

$$a(P, Y_i, \Delta_{*,i}) \leq a(P, Y_i, \Delta_i) = a(P, Y_i, \Delta_{*,i}) + (b_* - b_i) \operatorname{mult}_P q_i \leq \Big(1 + \frac{b_* - b_i}{\eta}\Big) a(P, Y_i, \Delta_{*,i}).$$

Dividing by $|I_{J_i}(P)|$ and taking the infimum over P, we obtain $\ell_i \le \alpha_i \le \ell_i + (b_* - b_i)\ell_i/\eta$ by the formula at the end of [Liu26, Lemma 2.6]. □

Lemma 11.10. *Notation as in Set-up 11.1, with W singular, and let $(W_0, (h_{k,0})_k)$ be a numerical admission of $(W, (h_k)_k)$ satisfying (i) and (ii) of Proposition 11.8. Then the following hold.*

(1) The germ $(W_0 \ni 0)$, the group G, and the marks $h_{1,0}|_{W_0}, \dots, h_{s,0}|_{W_0}$ satisfy Set-up 11.1(1)–(3) and (4') for s, m, and $\mu_1, \dots, \mu_s$. In particular, Definition-Lemmas 11.4 and 11.5(1)(2) and Propositions 11.7(0) and 11.9 apply to the admitted data whenever their remaining hypotheses hold.

(2) If $(W \ni o)$ is of type cD and the integers N and L of the admission in Construction-Theorem 4.13 are larger than $\max_k \operatorname{mult}_o h_k$, then $\operatorname{mult}_0(h_{k,0}|_{W_0}) = \operatorname{mult}_o h_k$ and $\operatorname{mult}_{E_{0,0}}(h_{k,0}|_{W_0}) = \operatorname{mult}_{E_0} h_k$ for any k, where $E_{0,0}$ is the prime divisor over W_0 corresponding to E_0 (Construction-Theorem 4.13(2)). In particular,

$$\sum_k c_k \operatorname{mult}_0(h_{k,0}|_{W_0}) \le 1 - \frac{1}{m}, \qquad \sum_k c_k \operatorname{mult}_{E_{0,0}}(h_{k,0}|_{W_0}) \le 2 - \frac{1}{m}.$$

Proof of Lemma 11.10. (1) Since W is singular, $W_0 = (f_0 = 0) \subset \mathbb{A}^4$ is an isolated Gorenstein terminal germ at 0 by Construction-Theorem 4.13(3), and G, the cyclic group of Set-up 11.1(3), acts faithfully on W_0 and fixes 0 by Construction-Theorem 4.13(1). Each $h_{k,0}|_{W_0}$ is G-semi-invariant and non-zero, since $\operatorname{mult}_{F_0}(h_{k,0}|_{W_0}) = \operatorname{mult}_F h_k$ by Construction-Theorem 4.13(2). The pair (W_0, D_0) is klt near 0 by Construction-Theorem 4.13(1) for the vector $(c_1, \dots, c_s)$, and the inequality of Set-up 11.1(4') holds by Proposition 11.8(2). The last sentence of (1) holds since Definition-Lemmas 11.4 and 11.5(1)(2) and Propositions 11.7(0) and 11.9 allow (4') in their statements.

(2) Recall that the admission of Proposition 11.8 is taken in ambient G-eigenparameters $x_1, \dots, x_4$ at o with a G-semi-invariant local equation f of W (Construction-Theorem 4.13(ii)). Let $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_4]]$, $\mathfrak{m} := (x_1, \dots, x_4) \subset \mathfrak{R}$, $\hat{h}_k$ the expansion of the G-semi-invariant lift of h_k in Construction-Theorem 4.13(iii), and $\hat{f}$ the expansion of f. Then ψ is a G-equivariant automorphism of $\mathfrak{R}$, each of $u_0, u_1, \dots, u_s$ is a unit of $\mathfrak{R}$, where u_0 is the unit called v in Construction-Theorem 4.13(vi), and $h_{k,0} \equiv u_k \psi(\hat{h}_k) \pmod{\mathfrak{m}^N}$ and $f_0 \equiv u_0 \psi(\hat{f}) \pmod{\mathfrak{m}^L}$ (Construction-Theorem 4.13(vi) and the congruences after it). Let $e \le \min(N, L)$ be a non-negative integer. Since ψ is an automorphism of $\mathfrak{R}$ and u_k and u_0 are units, the congruences $h_{k,0} \equiv u_k \psi(\hat{h}_k)$ and $f_0 \equiv u_0 \psi(\hat{f})$ give

$$\mathfrak{m}^e + (f_0) = \mathfrak{m}^e + (u_0 \psi(\hat{f})) = \psi\big(\mathfrak{m}^e + (\hat{f})\big),$$

and $h_{k,0} \in \mathfrak{m}^e + (f_0)$ if and only if $\hat{h}_k \in \mathfrak{m}^e + (\hat{f})$. Taking $e = \operatorname{mult}_o h_k$ and $e = \operatorname{mult}_o h_k + 1$, we obtain $\operatorname{mult}_0(h_{k,0}|_{W_0}) = \operatorname{mult}_o h_k$. A source of type cD has quadratic rank one, so E_0 is among the prime divisors in Construction-Theorem 4.13(v) by Proposition 11.8(ii), and $\operatorname{mult}_{E_{0,0}}(h_{k,0}|_{W_0}) = \operatorname{mult}_{E_0} h_k$ by Construction-Theorem 4.13(2). Thus

$$\sum_k c_k \operatorname{mult}_0(h_{k,0}|_{W_0}) = \sum_k c_k \operatorname{mult}_o h_k, \qquad \sum_k c_k \operatorname{mult}_{E_{0,0}}(h_{k,0}|_{W_0}) = \sum_k c_k \operatorname{mult}_{E_0} h_k,$$

and the right-hand sides are at most $1 - \frac{1}{m}$ and $2 - \frac{1}{m}$ respectively by Definition-Lemma 11.5(3) for $(W \ni o)$, G, $h_1, \dots, h_s$, which satisfy (4). □

11.3. **The cases and the proof of the theorem.** In this subsection, we state six propositions that exclude strictly increasing sequences of quotient values for the classes of sources not covered by Proposition 11.7, and we prove Theorem 11.2.

For sources of quadratic rank two we use the split radical order of Definition 8.1. Since a general hyperplane section of W through o is a Du Val singularity of type A_{n-1} with n the split radical order, the split radical order depends only on $(W \ni o)$, and it is defined for any W of quadratic rank two in Set-up 11.1.

Propositions 11.11–11.16 treat several labels on sources of quadratic rank two and of type cD. Their proofs are given in Appendix E. From Sections 8 and 9 they use only the theorems on circles, Definitions 8.1 and 9.1, Set-up 8.2, Proposition 8.10, Theorem 8.11, Lemma 8.12, Set-up 9.3 and the paragraph on E_0 that follows it, and Lemma 9.2, and from the corresponding appendices they use only Lemmas B.16, C.2, and C.3.

Three points are specific to several labels. First, the proofs use, in place of the lattice of Lemma 4.8, finitely many score vectors (Proposition 11.7), finitely many real reference values (Lemma E.6), or a real number outside the closure of a set of log canonical thresholds (Proposition 11.9(2)). Second, the lower bounds are given by the block h_k^0 and the weight κ_k of each mark h_k (Set-up 5.1, and Definition-Lemma 11.5(2) for the normal forms in quadratic rank one) together with the mass $\sum_k c_k\kappa_k$ of D along the circle (Set-up 5.1). Third, the marks may be units (Remark 11.6). The case when D vanishes near o is **Step 1** of the proof of Theorem 11.2.

Proposition 11.11. *Fix s, m, and $\mu_1, \dots, \mu_s$ as in Set-up 11.1. Then there exists no sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with W_i of quadratic rank two with split radical orders (Definition 8.1) tending to $+\infty$, whose numbers*

$$\alpha_i := \mathrm{mld}((W_i \ni o_i, \sum_k c_k \operatorname{div}(h_{i,k}))/G_i)$$

form a strictly increasing sequence in $(0, 1)$.

Proposition 11.12. *Fix s, m, and $\mu_1, \dots, \mu_s$ as in Set-up 11.1. Then there exists no sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with W_i of quadratic rank two with a fixed split radical order (Definition 8.1), whose numbers*

$$\alpha_i := \mathrm{mld}((W_i \ni o_i, \sum_k c_k \operatorname{div}(h_{i,k}))/G_i)$$

form a strictly increasing sequence in $(0, 1)$.

Proposition 11.13. *Fix s, m, and $\mu_1, \dots, \mu_s$ as in Set-up 11.1. Then there exists no sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with W_i of quadratic rank one whose residual cubic defines a triangle or a secant conic (Definition-Lemma 6.4), whose numbers*

$$\alpha_i := \mathrm{mld}((W_i \ni o_i, \sum_k c_k \operatorname{div}(h_{i,k}))/G_i)$$

form a strictly increasing sequence in $(0, 1)$.

Proposition 11.14. *Fix s, m, and $\mu_1, \dots, \mu_s$ as in Set-up 11.1. Then there exists no sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with W_i of quadratic rank one whose residual cubic defines the union of three concurrent lines (Definition-Lemma 6.4), whose numbers*

$$\alpha_i := \mathrm{mld}((W_i \ni o_i, \sum_k c_k \operatorname{div}(h_{i,k}))/G_i)$$

form a strictly increasing sequence in $(0, 1)$.

Proposition 11.15. *Fix s, m, and $\mu_1, \dots, \mu_s$ as in Set-up 11.1. Then there exists no sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with W_i of quadratic rank one whose residual cubic defines a cuspidal cubic or a tangent conic (Definition-Lemma 6.4), whose numbers*

$$\alpha_i := \mathrm{mld}((W_i \ni o_i, \sum_k c_k \operatorname{div}(h_{i,k}))/G_i)$$

form a strictly increasing sequence in $(0, 1)$.

Proposition 11.16. *Fix s, m, and $\mu_1, \dots, \mu_s$ as in Set-up 11.1. Then there exists no sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with W_i of quadratic rank one whose residual cubic defines a double line (Definition-Lemma 6.4), whose numbers*

$$\alpha_i := \mathrm{mld}((W_i \ni o_i, \sum_k c_k \operatorname{div}(h_{i,k}))/G_i)$$

form a strictly increasing sequence in $(0,1)$.

Proof of Theorem 11.2. Suppose that the theorem does not hold. Then there exists a sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with s, m, and $\mu_1, \dots, \mu_s$ fixed and $(W_i \ni o_i)$ not of type cE (Definition 9.1), such that the numbers

$$\alpha_i := \mathrm{mld}((W_i \ni o_i, D_i)/G_i), \quad D_i := \sum_{k=1}^{s} c_k \operatorname{div}(h_{i,k}),$$

form a strictly increasing sequence in $(0,1)$. Any subsequence of $(\alpha_i)_i$ is again strictly increasing in $(0,1)$, so we may pass to subsequences. We show that any class of sources leads to a contradiction.

Step 1. In this step we treat the case when all the marks are units.

Possibly passing to a subsequence, we may assume that either $h_{i,k}$ is a unit near o_i for any i and k, or for any i there exists k such that $h_{i,k}$ is not a unit near o_i.

Suppose that we are in the first case. Then $D_i = 0$ near o_i, so $\alpha_i = \mathrm{mld}((W_i \ni o_i, 0)/G_i)$. Let $(W_i/G_i, B_i)$ be the crepant quotient of $(W_i, 0)$ by G_i [Liu26, Definition 2.5]. Any coefficient of B_i is of the form $1 - \frac{1}{e}$ with e a ramification index of $W_i \to W_i/G_i$ [Liu26, Lemma 2.6(4)], and $(W_i/G_i, B_i)$ is klt since W_i is terminal [KM98, Proposition 5.20(4)]. Then $B_i \in \Gamma_{\mathrm{std}}$, where Γ_{std} is the set of [Liu26, Theorem 1.3], and α_i is the minimal log discrepancy of $(W_i/G_i, B_i)$ at the image of o_i. Thus $\alpha_i \in \mathrm{MLD}(3, \Gamma_{\mathrm{std}})$ [Liu26, Definition 2.3] for any i. Since $\mathrm{MLD}(3, \Gamma_{\mathrm{std}})$ satisfies the ACC [Liu26, Theorem 1.3], this is not possible. Therefore, in the rest of the proof we may assume that for any i there exists k such that $h_{i,k}$ is not a unit near o_i.

Step 2. In this step we treat smooth sources and sources of quadratic rank at least three.

If W_i is singular, then $(W_i \ni o_i)$ is an isolated cDV hypersurface germ [KM98, Corollary 5.38]. Thus the quadratic rank and the residual cubic of $(W_i \ni o_i)$ are defined [Liu26, Definition-Lemma 11.1], and the quadratic rank belongs to $\{1, 2, 3, 4\}$. Possibly passing to a subsequence, we may assume that either W_i is smooth for any i, or W_i is singular for any i and the quadratic rank of $(W_i \ni o_i)$ does not depend on i. If W_i is smooth for any i, or if the quadratic rank is at least three, then the α_i satisfy the ACC by Proposition 11.7(2), which is not possible. Therefore, W_i is singular for any i and the quadratic rank is one or two.

Step 3. In this step we treat the case of quadratic rank two.

Suppose that the quadratic rank is two. Let n_i be the split radical order of W_i (Definition 8.1). The n_i are positive integers, so possibly passing to a subsequence, we may assume that either n_i does not depend on i, or $n_i \to +\infty$. In the first case Proposition 11.12 implies a contradiction, and in the second case Proposition 11.11 implies a contradiction. Therefore, the quadratic rank is one.

Step 4. In this step we treat the case of quadratic rank one and conclude the proof.

Since the quadratic rank is one, by [Liu26, Lemma 11.3(1)] the completion of $\mathcal{O}_{W_i, o_i}$ is

$$\mathbb{C}[[x, y, z, t]]/(x^2 + g_i(y, z, t)), \quad g_i \in (y, z, t)^3,$$

and the residual cubic of $(W_i \ni o_i)$ is the cubic part $g_{3,i}$ of g_i. Since $(W_i \ni o_i)$ is an isolated Gorenstein terminal germ of quadratic rank one, $g_{3,i} \neq 0$ for any i by the argument of Set-up 9.3(9). By Definition-Lemma 6.4, the zero scheme of $g_{3,i}$ in $\mathbb{P}^2$ is a smooth cubic, a nodal cubic, a cuspidal cubic, a secant conic, a tangent conic, a triangle, the union of three concurrent lines, a double line, or a line counted three times. Since $g_{3,i}$ is unique up to a linear change of coordinates and a non-zero scalar [Liu26, Definition-Lemma 11.1], the class of the zero scheme of

$g_{3,i}$ depends only on $(W_i \ni o_i)$. Possibly passing to a subsequence, we may assume that the class does not depend on i.

Since $(W_i \ni o_i)$ is not of type cE, $g_{3,i}$ is not the cube of a linear form (Definition 9.1), so the zero scheme of $g_{3,i}$ is not a line counted three times. We consider the remaining eight classes.

- If the zero scheme of $g_{3,i}$ is a smooth cubic or a nodal cubic, then the α_i satisfy the ACC by Proposition 11.7(2), which is not possible.
- If the zero scheme of $g_{3,i}$ is a triangle or a secant conic, then Proposition 11.13 implies a contradiction.
- If the zero scheme of $g_{3,i}$ is the union of three concurrent lines, then Proposition 11.14 implies a contradiction.
- If the zero scheme of $g_{3,i}$ is a cuspidal cubic or a tangent conic, then Proposition 11.15 implies a contradiction.
- If the zero scheme of $g_{3,i}$ is a double line, then Proposition 11.16 implies a contradiction.

Thus no class of sources is possible, and the theorem is proved. □

12. Proof of the main theorem

In this section, we prove Theorem 1.3 by combining the reduction of [Liu26] with Theorems 11.2 and 10.1.

Proof of Theorem 1.3. The proof is long, so we divide it into several steps.

Step 1. In this step, we reduce Theorem 1.3 to the case of a strictly increasing sequence in $(0,1)$ of minimal log discrepancies of $\mathbb{Q}$-factorial klt threefold pairs with coefficients in a fixed finite set.

The set in Theorem 1.3 is the set $\mathrm{MLD}(3,\Gamma)$ of [Liu26, Definition 2.3]. Since Γ is finite, by [Liu26, Lemma 6.1(3)], it suffices to show that $\mathrm{MLD}_{\mathrm{Qf}}(3,\Lambda')$ satisfies the ACC for any finite set $\Lambda' \subset [0,1]$.

Fix a finite set $\Lambda' \subset [0,1]$, and assume, to the contrary, that there exists a strictly increasing sequence $a_1 < a_2 < \cdots$ in $\mathrm{MLD}_{\mathrm{Qf}}(3,\Lambda')$. Since Λ' is a DCC set, $\mathrm{MLD}_{\mathrm{Qf}}(3,\Lambda') \cap [1,+\infty)$ satisfies the ACC by [Liu26, Proposition 20.1]. If $a_{i_0} \geq 1$ for some i_0, then $a_{i_0} < a_{i_0+1} < \cdots$ is a strictly increasing sequence in $\mathrm{MLD}_{\mathrm{Qf}}(3,\Lambda') \cap [1,+\infty)$, which is impossible. Thus $a_i < 1$ for any i. For each i, write $a_i = \mathrm{mld}(X_i \ni x_i, B_i)$, where (X_i, B_i) is a klt pair, X_i is a $\mathbb{Q}$-factorial threefold, $x_i \in X_i$ is a closed point, and $B_i \in \Lambda'$. Since (X_i, B_i) is klt, $a_i > 0$. Hence

$$0 < a_1 \leq a_i < 1$$

for any i.

Step 2. In this step, we fix the numbers s, m, and $\mu_1, \ldots, \mu_s$, and attach to each pair (X_i, B_i) the data of [Liu26, Construction-Theorem 17.6].

Let s and q be as in [Liu26, Construction-Theorem 5.2], and let L, $\mu_1, \ldots, \mu_s$, and $\mathbf{R} = (r_{kj})$ be as in [Liu26, Construction-Proposition 17.1], both applied with $\Gamma := \Lambda'$. Let $m := qL$. By [Liu26, Construction-Theorem 5.2] and [Liu26, Construction-Proposition 17.1], $\mu_k > 0$ for any k, $\sum_k \mu_k = 1$, and s, q, L, and $\mu_1, \ldots, \mu_s$ depend only on Λ'. Thus s, m, and $\mu_1, \ldots, \mu_s$ do not depend on i.

Let $\epsilon := a_1$ and $A_0 := 1$. By **Step 1**, the assumptions (i)–(iii) of [Liu26, Construction-Theorem 17.6] hold for $\Gamma := \Lambda'$, ϵ, A_0, and the pairs (X_i, B_i), the points x_i, and the numbers a_i. By [Liu26, Construction-Theorem 17.6], for each i there exists a prime divisor E_i over X_i computing a_i, and, after deleting finitely many terms, (1)–(5) of [Liu26, Construction-Theorem 17.6] hold for the E_i. Since X_i is $\mathbb{Q}$-factorial, the morphism f_i of [Liu26, Construction-Theorem 17.6(1)] can be taken to be the identity. Possibly shrinking X_i to the affine neighborhood U_i of x_i of [Liu26, Construction-Theorem 17.6(1)], we may assume that X_i is affine and that the pair (X_i'', B_i'') and the point x_i'' of [Liu26, Construction-Theorem 17.6(1)] are (X_i, B_i) and x_i.

Let $\rho_i : X_i' \to X_i$ be the cover of [Liu26, Construction-Theorem 5.2] at x_i, with group G_i, and let $B_i^{(1)}, \ldots, B_i^{(s)}$ be the $\mathbb{Q}$-divisors $B^{(j)}$ of [Liu26, Construction-Theorem 5.2] for (X_i, B_i)

at x_i. Since ρ_i is finite, X_i' is affine. Let F_i be a prime divisor over X_i' lying over E_i [Liu26, Definition 2.7], and let $I_i := I_{G_i}(F_i)$. By [Liu26, Construction-Theorem 17.6(2)(c), (2)(f)], I_i is cyclic and

$$a(F_i, X_i', \rho_i^* B_i) = |I_i| a_i > 1.$$

Let $g_i : Y_i \to X_i'$ be the I_i-equivariant projective birational morphism of [Liu26, Construction-Theorem 17.6(3)], and let B_{Y_i} be the $\mathbb{R}$-divisor defined by

$$K_{Y_i} + B_{Y_i} = g_i^*(K_{X_i'} + \rho_i^* B_i).$$

By [Liu26, Construction-Theorem 17.6(3)(a), (3)(d), (3)(g)], $B_{Y_i} \geq 0$, Y_i is terminal, and $\mathrm{center}_{Y_i} F_i$ is a closed point y_i fixed by I_i. By the definition of B_{Y_i},

$$a(F_i, Y_i, B_{Y_i}) = a(F_i, X_i', \rho_i^* B_i) = |I_i| a_i.$$

By [Liu26, Construction-Theorem 17.6(4)(c)],

$$\mathrm{mld}((Y_i \ni y_i, B_{Y_i})/I_i) = a_i.$$

For each k, let $C_i^{(k)} := \sum_j r_{kj} B_i^{(j)}$, and let $C_{Y_i}^{(k)}$ be the $\mathbb{R}$-divisor defined by

$$K_{Y_i} + C_{Y_i}^{(k)} = g_i^*(K_{X_i'} + \rho_i^* C_i^{(k)}).$$

By [Liu26, Construction-Theorem 17.6(5)(f)], $(Y_i, C_{Y_i}^{(k)})$ is terminal for any k.

Step 3. In this step, we construct, for each i, data as in Set-up 11.1 with the numbers s, m, and $\mu_1, \dots, \mu_s$ of **Step 2** and with $\alpha = a_i$.

Since Y_i is terminal, Y_i is normal. Since g_i is projective and X_i' is affine, Y_i is quasi-projective. In [Liu26, Corollary 17.3], let $\Gamma := \Lambda'$, $(X, B) := (X_i, B_i)$, $x := x_i$, $\pi := \rho_i$, $G := G_i$, $E := E_i$, $F := F_i$, $T := Y_i$, and $g := g_i$. The assumptions of [Liu26, Corollary 17.3] that $B_{Y_i} \geq 0$, that Y_i is terminal, and that $\mathrm{center}_{Y_i} F_i$ is a closed point hold by **Step 2**. The $\mathbb{R}$-divisors $C'^{(k)}$ of [Liu26, Construction-Proposition 17.1] for $X = X_i$ and $\pi = \rho_i$ are $\rho_i^* C_i^{(k)}$. Hence the $\mathbb{R}$-divisors B_T and $C_T^{(k)}$ of [Liu26, Corollary 17.3] are B_{Y_i} and $C_{Y_i}^{(k)}$.

Let r_i be the Cartier index of K_{Y_i} at y_i. By [Liu26, Corollary 17.3], the conclusions of [Liu26, Construction-Theorem 5.5] hold for $(Y_i \ni y_i, B_{Y_i})$, $H = I_i$, and $F = F_i$. Thus, after shrinking I_i-invariant neighborhoods of y_i, the following hold.

(a) There exists a finite quasi-étale morphism $\tau_i : W_i \to Y_i$, the canonical cover of [Liu26, Construction-Theorem 5.5(1)], such that $\tau_i^{-1}(y_i)$ is a single point o_i, W_i is Gorenstein and terminal, and τ_i is Galois with group μ_{r_i} and $W_i/\mu_{r_i} = Y_i$ [Liu26, Construction-Theorem 5.5(1)(a), (1)(b), (1)(d)].

(b) There exist a finite abelian group $\hat{I}_i$ acting faithfully on W_i and fixing o_i, and an exact sequence

$$1 \to \mu_{r_i} \to \hat{I}_i \to I_i \to 1$$

compatible with τ_i, such that $W_i/\hat{I}_i = Y_i/I_i$ [Liu26, Construction-Theorem 5.5(1)(e), (1)(f)].

(c) For a prime divisor F_i' over W_i lying over F_i, the group $H_i' := I_{\hat{I}_i}(F_i')$ is cyclic and

$$\mathrm{mld}((W_i \ni o_i, \tau_i^* B_{Y_i})/H_i') = \mathrm{mld}((Y_i \ni y_i, B_{Y_i})/I_i) = a_i$$

[Liu26, Construction-Theorem 5.5(2)(c), (2)(f)].

(d) Either W_i is smooth at o_i, or $(W_i \ni o_i)$ is an isolated cDV singularity and there exists an $\hat{I}_i$-equivariant closed embedding of W_i into a smooth fourfold germ $(V_i \ni o_i)$ such that W_i is the zero locus of a regular $\hat{I}_i$-semi-invariant function on V_i [Liu26, Construction-Theorem 5.5(3)]. Let $V_i := W_i$ in the first case.

(e) There exist regular $\hat{I}_i$-semi-invariant functions on V_i whose restrictions $h_{i,1}, \dots, h_{i,s}$ to W_i are non-zero and satisfy

$$\operatorname{div}(h_{i,k}) = m\tau_i^* C_{Y_i}^{(k)} \text{ for any } k \quad \text{and} \quad \tau_i^* B_{Y_i} = \sum_k \frac{\mu_k}{m} \operatorname{div}(h_{i,k})$$

[Liu26, Corollary 17.3(3)].

For each k, let $C_{i,k} := \frac{1}{m}\operatorname{div}(h_{i,k})$. By (e), $C_{i,k} = \tau_i^* C_{Y_i}^{(k)}$. We show that $(W_i, C_{i,k})$ is terminal near o_i. Since τ_i is quasi-étale,

$$K_{W_i} + C_{i,k} = \tau_i^*(K_{Y_i} + C_{Y_i}^{(k)}),$$

so $(Y_i, C_{Y_i}^{(k)}) = (W_i, C_{i,k})/\mu_{r_i}$ by (a) and [Liu26, Definition 2.5]. Let P' be an exceptional prime divisor over W_i, and let P be the prime divisor over Y_i under P'. Since τ_i is finite, $\operatorname{center}_{Y_i} P = \tau_i(\operatorname{center}_{W_i} P')$ has codimension at least two in Y_i, so P is exceptional over Y_i. By [Liu26, Lemma 2.6(2)] and **Step 2**,

$$a(P', W_i, C_{i,k}) = |I_{\mu_{r_i}}(P')| \cdot a(P, Y_i, C_{Y_i}^{(k)}) > 1.$$

Thus $(W_i, C_{i,k})$ is terminal near o_i for any k.

Let $D_i := \sum_k \mu_k C_{i,k}$. By (e),

$$D_i = \sum_k \frac{\mu_k}{m} \operatorname{div}(h_{i,k}) = \tau_i^* B_{Y_i}.$$

We show that s, m, $\mu_1, \dots, \mu_s$, $(W_i \ni o_i)$, $G := H_i'$, and $h_{i,1}, \dots, h_{i,s}$ satisfy (1)–(4) of Set-up 11.1. Condition (1) holds by **Step 2**. By (a) and (d), $(W_i \ni o_i)$ is a smooth threefold germ or an isolated Gorenstein terminal threefold germ, so (2) holds. By (b) and (c), H_i' is a finite cyclic subgroup of $\hat{I}_i$, and H_i' acts faithfully on W_i and fixes o_i, so (3) holds. By (e), each $h_{i,k}$ is a non-zero regular $\hat{I}_i$-semi-invariant function on W_i near o_i, hence a non-zero regular H_i'-semi-invariant function on W_i near o_i. Thus $h_{i,1}, \dots, h_{i,s}$ are marks on $(W_i \ni o_i)$ (Definition 2.7), and (4) holds since $(W_i, C_{i,k})$ is terminal near o_i for any k. The proxies of Set-up 11.1 are the $C_{i,k}$, the labelled boundary D of Set-up 11.1 is D_i, with labels $\mu_1/m, \dots, \mu_s/m$, and, by (c),

$$\alpha = \operatorname{mld}((W_i \ni o_i, D_i)/H_i') = \operatorname{mld}((W_i \ni o_i, \tau_i^* B_{Y_i})/H_i') = a_i.$$

Step 4. In this step, we derive a contradiction and conclude the proof.

Possibly passing to a subsequence, we may assume that one of the following two cases occurs.

Case 1. For any i, $(W_i \ni o_i)$ is of type cE (Definition 9.1), that is, $(W_i \ni o_i)$ is an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form. Let

$$S := \left\{ \frac{\mu_1}{m}, \dots, \frac{\mu_s}{m} \right\}.$$

Then $S \subset [0,1]$ is a finite set, and S does not depend on i by **Step 2**. By **Step 3**, H_i' is a finite cyclic group acting faithfully on W_i and fixing o_i, and $D_i = \sum_k \frac{\mu_k}{m} \operatorname{div}(h_{i,k})$ is a labelled boundary with labels in S. For any exceptional prime divisor P over W_i, since $\sum_k \mu_k = 1$, **Step 3** implies

$$a(P, W_i, D_i) = \sum_k \mu_k a(P, W_i, C_{i,k}) > \sum_k \mu_k = 1.$$

Thus (W_i, D_i) is terminal near o_i. Therefore, by **Step 3**, each $a_i = \operatorname{mld}((W_i \ni o_i, D_i)/H_i') \in (0,1)$ belongs to the set of Theorem 10.1 for S. Since $a_1 < a_2 < \cdots$ is strictly increasing, this contradicts Theorem 10.1.

Case 2. For any i, $(W_i \ni o_i)$ is not of type cE (Definition 9.1), including the case that W_i is smooth at o_i. By **Step 3**, for each i the number $a_i \in (0,1)$ is the number α of data as in Set-up 11.1 for the numbers s, m, and $\mu_1, \dots, \mu_s$ of **Step 2**, which do not depend on i,

and $(W_i \ni o_i)$ is not of type cE. Since $a_1 < a_2 < \cdots$ is strictly increasing, this contradicts Theorem 11.2.

In both cases we obtain a contradiction. Thus $\mathrm{MLD}_{\mathrm{Qf}}(3, \Lambda')$ satisfies the ACC for any finite set $\Lambda' \subset [0,1]$, and Theorem 1.3 follows from **Step 1**. □

Appendix A. Proofs for sources of quadratic rank at least three

In this section, we prove the results of Section 7. We first prove a lemma on the hypersurfaces W_N of Set-up 7.3, which is used throughout this section.

Lemma A.1. *Let $N \geq 2$ be an integer, let $W_N := (xy + z^2 + t^N = 0) \subset \mathbb{A}^4$, and let E_1 be the exceptional divisor of the blow-up of W_N at 0. For a non-zero polynomial h in x, y, z, t, let $m(h)$ be the least total degree of a monomial of h.*

(1) A non-zero polynomial without monomials divisible by xy is divisible neither by $xy + z^2$ nor by $xy + z^2 + t^2$.

(2) The divisor E_1 is a prime divisor over W_N with center 0, $a(E_1, W_N, 0) = 2$, and $\mathrm{mult}_{E_1} h = m(h)$ for any non-zero polynomial h without monomials divisible by xy.

(3) Let $c_1, \ldots, c_s$ be positive real numbers and $h_1, \ldots, h_s$ non-zero polynomials without monomials divisible by xy and vanishing at 0. If $(W_N, \sum_j c_j \operatorname{div}(h_j))$ is klt near 0, then $\sum_j c_j m(h_j) < 2$. In particular, if $c_j \geq \gamma$ for a positive real number γ and any j, then $s < 2/\gamma$ and $m(h_j) < 2/\gamma$ for any j.

Proof. (1) Let F be a non-zero polynomial without monomials divisible by xy. Assume that $F = (xy + z^2)F'$ or $F = (xy + z^2 + t^2)F'$ for a polynomial F'. Then $F' \neq 0$. Let ℓ be the largest x-exponent of a monomial of F', and μ' a monomial of F' with x-exponent ℓ. The monomials of z^2F' and t^2F' have x-exponent at most ℓ, and $xy\mu'$ has x-exponent $\ell + 1$, so $xy\mu'$ is a monomial of F, a contradiction.

(2) The hypersurface W_N is normal, since its singular locus is $\{0\}$ [Har77, Proposition II.8.23(b)]. Let $f := xy + z^2 + t^N$ and $w := (1,1,1,1)$. The initial form f_w is $xy + z^2$ if $N \geq 3$ and $xy + z^2 + t^2$ if $N = 2$, so $(f_w = 0)$ is an integral quadric not contained in a coordinate hyperplane. The $(1,1,1,1)$-weighted blow-up of $\mathbb{A}^4$ is the blow-up of 0. Thus by Lemma 4.12 with $n = 4$ and $W = W_N$, $E_{W_N} = E_1$ is a prime divisor over W_N with center 0 (the blow-up of W_N at 0 is the strict transform of W_N in the blow-up of $\mathbb{A}^4$ at 0 [Har77, Corollary II.7.15]), and

$$a(E_1, W_N, 0) = 4 - 2 = 2.$$

Let h be a non-zero polynomial without monomials divisible by xy. Then $\mathrm{in}_w(h)$ is the lowest-degree part of h, which has no monomial divisible by xy, so f_w does not divide $\mathrm{in}_w(h)$ by (1). By Lemma 4.12, $\mathrm{mult}_{E_1} h = \mathrm{wt}_w(h) = m(h)$.

(3) Since $(W_N, \sum_j c_j \operatorname{div}(h_j))$ is klt near 0 and E_1 has center 0, (2) gives

$$0 < a\Big(E_1, W_N, \sum_j c_j \operatorname{div}(h_j)\Big) = 2 - \sum_j c_j m(h_j).$$

Since the h_j vanish at 0, $m(h_j) \geq 1$ for any j. Thus if $c_j \geq \gamma$ for any j, then $s\gamma \leq \sum_j c_j m(h_j) < 2$ and $\gamma m(h_j) \leq c_j m(h_j) < 2$ for any j. □

A.1. Nodes: reduction.

In this subsection, we prove Proposition 7.5.

Lemma A.2. *Let $\Gamma \subset [0,1]$ be a DCC set, N a positive integer, and $W := (xy = z^2 + t^2) \subset \mathbb{A}^4$. Then the set of positive numbers $\mathrm{mld}((W \ni 0, D)/G)$, where G is a finite diagonal abelian group preserving W and acting faithfully on W with $|G| \leq N$, and $D = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$ and G-semi-invariant polynomials h_j such that (W, D) is klt near 0, satisfies the ACC.*

Proof. Any element of such a group G has order at most N, so $G \subset \mu_{N!}^4$. Thus there exist only finitely many such groups G, and it suffices to fix G. Let $X := W/G$, $\pi : W \to X$ the quotient morphism, $\bar{0} := \pi(0)$, and D_X the $\mathbb{R}$-divisor on X with $K_W + D = \pi^*(K_X + D_X)$. Then X is a

fixed normal threefold. Since the h_j are polynomials, D is an $\mathbb{R}$-divisor on W and $K_W + D$ is $\mathbb{R}$-Cartier, so $K_X + D_X$ is $\mathbb{R}$-Cartier and $D_X \geq 0$ [Liu26, Lemma 2.6(0)]. For a prime divisor P on W, the coefficient $b_P := \operatorname{mult}_P D = \sum_j c_j \operatorname{mult}_P h_j$ belongs to Γ_+. By [Liu26, Lemma 2.6(4)],

$$\operatorname{mult}_{\pi(P)} D_X = 1 - \frac{1 - b_P}{e_P},$$

where e_P is the ramification index of π along P, and $1 \leq e_P \leq |G| \leq N$. Thus the coefficients of D_X belong to

$$\Psi_N := \left\{ 1 - \frac{1-b}{e} \;\middle|\; b \in \Gamma_+,\ 1 \leq e \leq N \right\} \setminus \{0\}.$$

Since Γ_+ is a DCC set [Liu26, Lemma 2.10(1)], the set $\{1 - 1/e + b/e \mid b \in \Gamma_+\}$ is a DCC set for any e, and Ψ_N is a DCC set of positive real numbers as a finite union of DCC sets of non-negative real numbers with 0 removed. Moreover,

$$\operatorname{mld}((W \ni 0, D)/G) = \operatorname{mld}(X \ni \bar{0}, D_X).$$

By [Kaw24, Theorem 1.1] on the fixed normal threefold X with the DCC set Ψ_N, the numbers $\operatorname{mld}(X \ni \bar{0}, D_X)$ satisfy the ACC. □

Proof of Proposition 7.5. Write $D_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$, where the $c_{ij} \in \Gamma$ are the labels and the h_{ij} are the marks (Definition 2.7), and let χ_{ij} be the character of h_{ij}. Near 0 write $h_{ij} = \varphi_{ij}/\psi_{ij}$ with polynomials φ_{ij}, ψ_{ij} on $\mathbb{A}^4$ and $\psi_{ij}(0) \neq 0$, and let $u_{ij} := \prod_{\omega \in G_i} \omega^* \psi_{ij}$, a G_i-invariant polynomial with $u_{ij}(0) \neq 0$. Then $h_{ij} u_{ij}$ is the restriction of the polynomial $\varphi_{ij} \prod_{\omega \neq \mathrm{id}_{\mathbb{A}^4}} \omega^* \psi_{ij}$, the component of character χ_{ij} of $\varphi_{ij} \prod_{\omega \neq \mathrm{id}_{\mathbb{A}^4}} \omega^* \psi_{ij}$ is a G_i-semi-invariant polynomial whose restriction to W_i is $h_{ij} u_{ij}$, and $\operatorname{div}(h_{ij} u_{ij}) = \operatorname{div}(h_{ij})$ near 0. Thus we may assume that the h_{ij} are G_i-semi-invariant polynomials.

The proof is long, so we divide it into several steps.

Step 1. In this step we replace G_i with the inertia group of a computing lift, which is cyclic and keeps a_i.

Fix i and write $W = W_i$, $G = G_i$, $D = D_i$, and $a = a_i$. Let $\bar{0}$ be the image of 0 in W/G and $(W/G, D_{W/G})$ the crepant quotient. By [Liu26, Lemma 2.6(2)(3)], $(W/G, D_{W/G})$ is klt near $\bar{0}$, so there exists a prime divisor over W/G with center $\bar{0}$ which computes $a = \operatorname{mld}(W/G \ni \bar{0}, D_{W/G})$ [Liu26, Lemma 3.5]. Let F be a prime divisor over W lying over it, a computing lift (Definition 2.7). Then $\operatorname{center}_W F = \{0\}$ and $a = a(F, W, D)/|I_G(F)|$ by [Liu26, Lemma 2.6(2)]. Let $I := I_G(F)$, which is cyclic by [Liu26, Lemma 2.6(1)]. For any prime divisor P over W with center 0, we have $I_I(P) = I_G(P) \cap I$ [Liu26, Definition 2.5], and

$$\frac{a(P, W, D)}{|I_G(P) \cap I|} = \frac{|I_G(P)|}{|I_G(P) \cap I|} \cdot \frac{a(P, W, D)}{|I_G(P)|} \geq a,$$

since $|I_G(P)|/|I_G(P) \cap I|$ is a positive integer, $a(P, W, D)/|I_G(P)| \geq a$ by the formula at the end of [Liu26, Lemma 2.6], and $a > 0$. Equality holds for $P = F$, since $I_G(F) \cap I = I$. By the formula at the end of [Liu26, Lemma 2.6] again,

$$\operatorname{mld}((W \ni 0, D)/I) = \inf_P \frac{a(P, W, D)}{|I_I(P)|} = a,$$

where P runs over the prime divisors over W with center 0, and F computes $\operatorname{mld}((W \ni 0, D)/I)$. This is [Liu26, Lemma 3.1(1)] with $X' = W$, $H = I$, $Y = W$, and f the identity. The group $I \subset G$ is diagonal, acts faithfully on W, and preserves q_i up to a character, and the h_{ij} are I-semi-invariant polynomials. Possibly replacing G_i with I for any i, we may assume that G_i is cyclic.

Step 2. In this step we classify the characters of G_i on the eigenparameters, reduce to the fixed node $W = (xy = z^2 + t^2)$, and show that $|G_i| \to +\infty$.

Fix i and write $G = G_i$ and $q = q_i$. Let $x_1, \dots, x_4$ be eigenparameters, $\omega^* x_k = \chi_k(\omega) x_k$ and $\omega^* q = \chi_q(\omega) q$ for $\omega \in G$. For a character α of G, let V_α be the span of the x_k with $\chi_k = \alpha$, and $m_\alpha := \dim V_\alpha$.

Write $q = \sum_{k,l} q_{kl} x_k x_l$ with $q_{kl} = q_{lk}$. Then $q_{kl} \neq 0$ only if $\chi_k \chi_l = \chi_q$. Since the matrix (q_{kl}) is invertible, for any α the block of (q_{kl}) with rows indexed by V_α and columns indexed by $V_{\chi_q/\alpha}$ is invertible. In particular, $m_\alpha = m_{\chi_q/\alpha}$. Linear changes of basis inside the spaces V_α keep eigenparameters, and they lead to the following normal forms.

(1) If $\alpha \neq \chi_q/\alpha$, then in a basis $u_1, \dots, u_{m_\alpha}$ of V_α and the dual basis $v_1, \dots, v_{m_\alpha}$ of $V_{\chi_q/\alpha}$ with respect to the block, the part of q in $V_\alpha \cdot V_{\chi_q/\alpha}$ is $\sum_{k=1}^{m_\alpha} u_k v_k$.

(2) If $\alpha^2 = \chi_q$, then the restriction of q to V_α is a non-degenerate quadratic form, equal to $u_1^2 + \cdots + u_{m_\alpha}^2$ in a basis $u_1, \dots, u_{m_\alpha}$ of V_α. Moreover, $u_1^2 + u_2^2 = (u_1 + \sqrt{-1}u_2)(u_1 - \sqrt{-1}u_2)$ is a product of two eigenparameters of character α.

Since G is cyclic, the character group of G is cyclic and contains at most one element ϵ of order two. Hence if $\alpha_0^2 = \chi_q$, then the characters α with $\alpha^2 = \chi_q$ are α_0 and $\alpha_0\epsilon$. Let $m_{\alpha_0\epsilon} := 0$ if $|G|$ is odd, and $m_{\alpha_0} = m_{\alpha_0\epsilon} := 0$ if no character α satisfies $\alpha^2 = \chi_q$. By (1), the characters α with $\alpha^2 \neq \chi_q$ contribute an even number to $4 = \sum_\alpha m_\alpha$, so $m_{\alpha_0} + m_{\alpha_0\epsilon} \in \{0, 2, 4\}$ and $m_{\alpha_0} \equiv m_{\alpha_0\epsilon} \pmod 2$. Thus one of the following holds.

(a) m_{α_0} and $m_{\alpha_0\epsilon}$ are even, that is,

$$(m_{\alpha_0}, m_{\alpha_0\epsilon}) \in \{(0,0), (2,0), (0,2), (4,0), (0,4), (2,2)\}.$$

Then (1) and (2) write q as the sum of two products of eigenparameters, and after renaming the eigenparameters and replacing t with $-t$, we have $q = xy - zt$.

(b) m_{α_0} and $m_{\alpha_0\epsilon}$ are odd, that is, $(m_{\alpha_0}, m_{\alpha_0\epsilon}) \in \{(1,1), (1,3), (3,1)\}$. Then ϵ exists and $|G|$ is even. Exchanging α_0 and $\alpha_0\epsilon$, we may assume that $m_{\alpha_0\epsilon} = 1$, and let t span $V_{\alpha_0\epsilon}$ with $q|_{V_{\alpha_0\epsilon}} = t^2$. If $m_{\alpha_0} = 1$, then the remaining two eigenparameters span $V_\alpha \oplus V_{\chi_q/\alpha}$ for a character $\alpha \neq \chi_q/\alpha$ with $m_\alpha = 1$, and (1) and (2) imply $q = xy + z^2 + t^2$ with $x \in V_\alpha$, $y \in V_{\chi_q/\alpha}$, and $z \in V_{\alpha_0}$. If $m_{\alpha_0} = 3$, then $q|_{V_{\alpha_0}} = u_1^2 + u_2^2 + u_3^2$, and with $x := u_1 + \sqrt{-1}u_2$, $y := u_1 - \sqrt{-1}u_2$, and $z := u_3$ we again have $q = xy + z^2 + t^2$.

We exclude case (a). The germ $(xy - zt = 0) \ni 0$ is an isolated Gorenstein terminal germ: the blow-up of 0 is a resolution whose exceptional divisor P is a smooth quadric surface, and $a(P, (xy - zt = 0), 0) = 4 - 2 = 2$. The polynomial $g = zt$ is reduced, the irreducible factors z and t of g are smooth at 0, and z and t are preserved by G since they are eigenparameters. By Proposition 8.4(2), the positive values $\mathrm{mld}((xy - zt = 0) \ni 0, D)/G)$, where G is a finite diagonal abelian group preserving $xy - zt$ up to a character and acting faithfully on $(xy - zt = 0)$ and D is a labelled boundary with labels in Γ such that $((xy - zt = 0), D)$ is klt near 0, satisfy the ACC. Since the a_i are strictly increasing, case (a) occurs only for finitely many i. Possibly passing to a subsequence, we may assume that case (b) holds for any i.

In case (b), replacing y with $-y$, we have $W_i = W := (xy = z^2 + t^2)$ for any i, and

$$\chi_x \chi_y = \chi_z^2 = \chi_t^2 = \chi_q, \qquad \chi_t = \chi_z \epsilon.$$

The labels are kept, and the marks are still G_i-semi-invariant polynomials. Let $|G_i| = 2\ell_i$, let ω_i be a generator of G_i, and let $\zeta_i := \exp(2\pi\sqrt{-1}/(2\ell_i))$. Since ω_i generates G_i, $\epsilon(\omega_i) = -1 = \zeta_i^{\ell_i}$. Thus there exist integers β_1, β_3 such that the values of $\chi_x, \chi_y, \chi_z, \chi_t$ at ω_i are ζ_i to the powers

$$(\beta_1, 2\beta_3 - \beta_1, \beta_3, \beta_3 + \ell_i) \mod 2\ell_i. \tag{A.1}$$

If $|G_i| \leq N$ for infinitely many i and some N, then Lemma A.2, applied to the DCC set Γ of the proposition, contradicts the strict increase of the a_i. Thus, possibly passing to a subsequence, we may assume that $|G_i| = 2\ell_i \to +\infty$.

Step 3. In this step we prove the ACC for the circles $\mathbf{T}_{(0,2,1,1)}$ and $\mathbf{T}_{(2,0,1,1)}$ on W with bounded index.

Claim A.3. *Let e_0 be a positive integer. Then the set of positive numbers* $\mathrm{mld}((W \ni 0, D)/G)$, *where G is a finite diagonal abelian group preserving W and acting faithfully on W with $[G : G \cap \mathbf{T}_{(0,2,1,1)}] \le e_0$, and $D = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$ and G-semi-invariant polynomials h_j such that (W, D) is klt near 0, satisfies the ACC. The set obtained by replacing $\mathbf{T}_{(0,2,1,1)}$ with $\mathbf{T}_{(2,0,1,1)}$ also satisfies the ACC.*

Proof. Exchanging x and y preserves W and maps $\mathbf{T}_{(0,2,1,1)}$ onto $\mathbf{T}_{(2,0,1,1)}$, so it suffices to prove the first statement. Fix data G and D of the claim, and let $R := G \cap \mathbf{T}_{(0,2,1,1)}$ and $r := |R|$. Since

$$\mathbf{T}_{(0,2,1,1)} = \{(1, \lambda^2, \lambda, \lambda) \mid \lambda \in \mathbb{C}^*\},$$

we have

$$R = \{(1, \zeta^2, \zeta, \zeta) \mid \zeta^r = 1\}.$$

If $r = 1$, then $|G| \le e_0$, and Lemma A.2, applied to the DCC set Γ of the proposition and $N = e_0$, implies the ACC for the corresponding values. So we may assume that $r \ge 2$. For $\omega \in G$ write $\omega = (\omega_x, \omega_y, \omega_z, \omega_t)$, so that $\omega_x \omega_y = \omega_z^2 = \omega_t^2$.

Let $f : W' \to W$ be the weighted blow-up with weights $(2, 1, 1)$ on (y, z, t) along the x-axis $(y = z = t = 0) \subset W$, that is, the normalized blow-up of the ideal (y, z^2, zt, t^2), and let E be the exceptional divisor. Then W' is covered by three charts U_y, U_z, and U_t (a direct computation in the charts, where the y-chart is $\mathbb{A}^3/\mu_2$ since its coordinate ring is generated by the μ_2-invariant monomials of degree two):

- on U_z, $z = \lambda$, $t = \lambda t_1$, $y = \lambda^2 y_1$, and $U_z \cong (xy_1 = 1 + t_1^2) \times \mathbb{A}^1_\lambda$;
- on U_t, $t = \lambda$, $z = \lambda z_2$, $y = \lambda^2 y_2$, and $U_t \cong (xy_2 = 1 + z_2^2) \times \mathbb{A}^1_\lambda$;
- on U_y, $y = \lambda^2$, $z = \lambda Z$, $t = \lambda T$, $x = Z^2 + T^2$, and $U_y \cong \mathbb{A}^3_{\lambda,Z,T}/\mu_2$, where μ_2 acts by $(\lambda, Z, T) \to (-\lambda, -Z, -T)$.

The surface $(xy_1 = 1 + t_1^2) \subset \mathbb{A}^3_{x,y_1,t_1}$ is smooth, since $x = y_1 = t_1 = 0$ does not satisfy $xy_1 = 1 + t_1^2$. Similarly, the surface $(xy_2 = 1 + z_2^2) \subset \mathbb{A}^3_{x,y_2,z_2}$ is smooth. On U_z and U_t, $E = (\lambda = 0)$ is irreducible, and the preimage of $E \cap U_y$ in $\mathbb{A}^3_{\lambda,Z,T}$ is $(\lambda = 0)$. The generator $x^{-1} dx \wedge dz \wedge dt$ of ω_W pulls back on U_z to

$$\lambda \cdot x^{-1} dx \wedge d\lambda \wedge dt_1,$$

and $x^{-1} dx \wedge dt_1$ is a nowhere vanishing 2-form on $(xy_1 = 1 + t_1^2)$. Thus $a(E, W, 0) = 2$. Let $m_j := \operatorname{mult}_E h_j$ and

$$S := \sum_j c_j m_j = \operatorname{mult}_E D.$$

Then $a(E, W, D) = 2 - S$, and $0 \le S < 2$ since (W, D) is klt near 0. On each chart we write the pull-back of h_j as $\lambda^{m_j} \tilde{h}_j$, where $\tilde{h}_j$ is not divisible by λ.

Since y, z, t are G-semi-invariant, G acts on W' and preserves U_y, U_z, and U_t. An element $(1, \zeta^2, \zeta, \zeta) \in R$ acts on U_z and on U_t by $\lambda \to \zeta\lambda$ and trivially on the other coordinates. On U_y, let

$$\widetilde{G} := \{(\omega, \alpha) \in G \times \mathbb{C}^* \mid \alpha^2 = \omega_y\},$$

an abelian group acting on $\mathbb{A}^3_{\lambda,Z,T}$ by

$$(\lambda, Z, T) \to (\alpha\lambda, \omega_z \alpha^{-1} Z, \omega_t \alpha^{-1} T).$$

The action of $\widetilde{G}$ lifts the action of G on U_y: it maps $y = \lambda^2$, $z = \lambda Z$, and $t = \lambda T$ to $\omega_y y$, $\omega_z z$, and $\omega_t t$, and $x = Z^2 + T^2$ to $\omega_z^2 \omega_y^{-1} x = \omega_x x$. The kernel of $\widetilde{G} \to G$ is $\{(1, \pm 1)\}$, which acts as μ_2, so $U_y/G = \mathbb{A}^3_{\lambda,Z,T}/\widetilde{G}$. The subgroup

$$\widetilde{R} := \{((1, \zeta^2, \zeta, \zeta), \zeta) \mid \zeta^r = 1\} \subset \widetilde{G}$$

has order r and acts by $\lambda \to \zeta\lambda$ and trivially on Z, T.

Let $W^\circ \subset W$ be a G-invariant open neighborhood of 0 such that $(W^\circ, D|_{W^\circ})$ is klt. Then $D|_{W^\circ} \in \Gamma_+ \cap [0,1)$. Let $X := W/G$, $\bar{0}$ the image of 0, (X, D_X) the crepant quotient of (W, D), $Y := W'/G$, $\phi : Y \to X$ the induced projective birational morphism, and B_Y the $\mathbb{R}$-divisor with $K_Y + B_Y = \phi^*(K_X + D_X)$. Let Θ be the union of $\Gamma_+ \cap [0,1)$ and

$$\left\{1 - \frac{2-b}{r'} \;\middle|\; b \in \Gamma_+ \cap [0,2),\ r' \in \mathbb{Z}_{\geq 2}\right\}.$$

Then $\Theta \subset [0,1)$, and Θ depends only on Γ. We show that Θ is a DCC set. The set Γ_+ is a DCC set [Liu26, Lemma 2.10(1)]. Assume that $\theta_k = 1 - (2-b_k)/r_k$ is a strictly decreasing sequence with $b_k \in \Gamma_+ \cap [0,2)$ and $r_k \geq 2$. Possibly passing to a subsequence, we may assume that the b_k are non-decreasing. Then $(2-b_{k+1})/r_{k+1} > (2-b_k)/r_k \geq (2-b_{k+1})/r_k$, and $2 - b_{k+1} > 0$, so $r_{k+1} < r_k$ for any k, which is impossible. Thus Θ is a DCC set.

We check the conditions of Definition-Proposition 4.10 for $(X \ni \bar{0}, D_X)$, (ϕ, B_Y), and Θ.

(i) Let $X^\circ := W^\circ/G$. The pair $(X^\circ, D_X|_{X^\circ})$ is klt [Liu26, Lemma 2.6(2)(3)], so

$$a(P, Y, B_Y) = a(P, X, D_X) > 0$$

for any prime divisor P over Y whose center intersects $\phi^{-1}(X^\circ)$. The components of B_Y over X° other than the image E_Y of E are images of components of $f_*^{-1}D$, with non-negative coefficients. By [Liu26, Lemma 2.6(2)],

$$\operatorname{mult}_{E_Y} B_Y = 1 - \frac{a(E, W, D)}{|I_G(E)|} = 1 - \frac{2-S}{|I_G(E)|}.$$

Since R acts trivially on E, $R \subset I_G(E)$, so $|I_G(E)| \geq r \geq 2$ and $\operatorname{mult}_{E_Y} B_Y \in [0,1)$. Thus $B_Y \geq 0$ over X°, and (Y, B_Y) is klt over X°.

(ii) Let $U_z^\circ := U_z \cap f^{-1}(W^\circ)$, $V_z := U_z^\circ/R$, and $H_z := G/R$. Since R acts only on λ, V_z is an open subset of $(xy_1 = 1 + t_1^2) \times \mathbb{A}^1_v$ with $v := \lambda^r$, hence a smooth threefold, and $U_z^\circ/G \cong V_z/H_z$ is a Zariski open subset of Y. Let $K_{W'} + D_{W'} = f^*(K_W + D)$, so that $D_{W'} = f_*^{-1}D + (S-1)E$. The $\mathbb{R}$-divisor B_{V_z} of Definition-Proposition 4.10 satisfies $K_{U_z^\circ} + D_{W'}|_{U_z^\circ} = \pi_z^*(K_{V_z} + B_{V_z})$ for the quotient morphism $\pi_z : U_z^\circ \to V_z$, so (V_z, B_{V_z}) is the crepant quotient of $(U_z^\circ, D_{W'}|_{U_z^\circ})$ by R. The only prime divisor on U_z° fixed pointwise by a non-trivial element of R is $E \cap U_z^\circ = (\lambda = 0)$, with ramification index r. By [Liu26, Lemma 2.6(4)], the coefficient of the image of E in B_{V_z} is

$$1 - \frac{1-(S-1)}{r} = 1 - \frac{2-S}{r},$$

and the other coefficients of B_{V_z} are coefficients of $D|_{W^\circ}$. Thus $B_{V_z} \geq 0$ and $B_{V_z} \in \Theta$. Similarly, with $U_t^\circ := U_t \cap f^{-1}(W^\circ)$, $V_t := U_t^\circ/R$, and $H_t := G/R$, we have $B_{V_t} \geq 0$ and $B_{V_t} \in \Theta$.

On U_y, let $U_y^\circ := U_y \cap f^{-1}(W^\circ)$, $\widetilde{U}$ the preimage of U_y° in $\mathbb{A}^3_{\lambda,Z,T}$, $V_y := \widetilde{U}/\widetilde{R}$, an open subset of $\mathbb{A}^3_{v,Z,T}$ with $v := \lambda^r$, and $H_y := \widetilde{G}/\widetilde{R}$, a finite abelian group of order $2[G:R]$. Then $U_y^\circ/G \cong \widetilde{U}/\widetilde{G} \cong V_y/H_y$ is a Zariski open subset of Y. The morphism $\widetilde{U} \to U_y$ is the quotient by μ_2, which fixes only the origin, so it is étale in codimension one, and the crepant pull-back of $D_{W'}$ to $\widetilde{U}$ has coefficient $S-1$ along $(\lambda = 0)$ and coefficients of $D|_{W^\circ}$ along the other prime divisors. The subgroup $\widetilde{R}$ fixes pointwise only the prime divisor $(\lambda = 0)$, with ramification index r. As on U_z, [Liu26, Lemma 2.6(4)] implies that the crepant pull-back B_{V_y} of $K_Y + B_Y$ to V_y has coefficient $1 - (2-S)/r$ along $(v = 0)$ and coefficients of $D|_{W^\circ}$ elsewhere. Thus $B_{V_y} \geq 0$ and $B_{V_y} \in \Theta$.

Since U_y, U_z, and U_t cover W', the open subsets V_y/H_y, V_z/H_z, and V_t/H_t cover $\phi^{-1}(\bar{0})$, and (ii) holds. By Definition-Proposition 4.10, the positive numbers

$$\operatorname{mld}(X \ni \bar{0}, D_X) = \operatorname{mld}((W \ni 0, D)/G)$$

with $r \geq 2$ satisfy the ACC. Together with the case $r = 1$, this proves the claim. □

Proof of Proposition 7.5 continued.

Step 4. In this step we take the Hausdorff limit of the groups G_i and show that the identity component of the limit is a circle or a torus of dimension two.

Regard the G_i as finite subgroups of $\mathbb{T}^4 = \mathbb{R}^4/\mathbb{Z}^4$. By (A.1), the generator ω_i corresponds to

$$\frac{1}{2\ell_i}(\beta_1, 2\beta_3 - \beta_1, \beta_3, \beta_3 + \ell_i) \mod \mathbb{Z}^4,$$

so G_i is contained in the closed subgroup

$$\Sigma := \{\theta \in \mathbb{T}^4 \mid \theta_1 + \theta_2 = 2\theta_3,\ 2\theta_3 = 2\theta_4\}.$$

Let

$$\Sigma^0 := \{\theta \in \mathbb{T}^4 \mid \theta_1 + \theta_2 = 2\theta_3,\ \theta_3 = \theta_4\},$$

the image of $\mathbb{T}^2$ under

$$(\theta_1, \theta_3) \to (\theta_1, 2\theta_3 - \theta_1, \theta_3, \theta_3).$$

Then Σ^0 is a torus of dimension two, and Σ is the union of Σ^0 and the translate of Σ^0 by $(0, 0, 0, \frac{1}{2})$.

By Lemma 4.6, possibly passing to a subsequence, there exists a closed subgroup $H \subset \mathbb{T}^4$ such that the G_i converge to H in the Hausdorff distance, and $G_i \subset H$ and G_i intersects any connected component of H for $i \gg 0$. Since $|G_i| \to +\infty$, $\dim H \geq 1$. Since the closed subgroup Σ contains any G_i, it contains H, and the identity component H^0 is contained in Σ^0. Thus $\dim H \in \{1, 2\}$, and the linear subspace $\mathfrak{h} \subset \mathbb{R}^4$ with $\rho(\mathfrak{h}) = H^0$ [Liu26, Lemma 4.3(1)], where $\rho : \mathbb{R}^4 \to \mathbb{T}^4$ is the quotient map, is contained in $\{w \in \mathbb{R}^4 \mid w_1 + w_2 = 2w_3,\ w_3 = w_4\}$.

Step 5. In this step we exclude all limits except the circle $\mathbf{T}_{(-p,p+2s,s,s)}$ by the toric upper bound, Theorem 8.3, and Claim A.3, and finish the proof.

The data (W, D_i, G_i) satisfy the hypotheses of Lemma 4.7 with $n = 4$ and $f = xy - z^2 - t^2$: W is normal and klt near 0, G_i is a finite diagonal abelian group preserving W and acting faithfully on W, the h_{ij} are G_i-semi-invariant polynomials, and (W, D_i) is lc near 0. If $\mathfrak{h}$ contains a vector with all coordinates positive, then Lemma 4.7 implies $a_i \to 0$, which contradicts $a_i \geq a_1 > 0$. Hence no vector of $\mathfrak{h}$ has all coordinates positive.

If $\dim H = 2$, then $\mathfrak{h} = \{w_1 + w_2 = 2w_3,\ w_3 = w_4\}$ contains $(1, 1, 1, 1)$, which is impossible.

Thus $\dim H = 1$. By Lemma 4.6, $\mathfrak{h} = \mathbb{R}w$ for a primitive $w \in \mathbb{Z}^4$, and $w = (w_1, w_2, w_3, w_3)$ with $w_1 + w_2 = 2w_3$. Moreover, $G_i \cap \mathbf{T}_{(w)}$ is cyclic of order $n_i \to +\infty$, and $[G_i : G_i \cap \mathbf{T}_{(w)}] = |H/H^0|$ for $i \gg 0$. Since $\mathbf{T}_{(w)} = \mathbf{T}_{(-w)}$, we may replace w with $-w$.

- If $w_3 = 0$, then $w_1 = -w_2$, and since w is primitive, $w = \pm(1, -1, 0, 0)$. Then $R_i' := G_i \cap \mathbf{T}_{(1,-1,0,0)}$ is a cyclic subgroup of the circle $\mathbf{T}_{(1,-1,0,0)}$ of order n_i, so $R_i' = \mu_{n_i}(1, -1, 0, 0)$ with $n_i \to +\infty$, and $[G_i : R_i'] = |H/H^0|$ for $i \gg 0$. Theorem 8.3, applied with $z^2 + t^2 \in (z, t)^2$ in place of g_i for any i and with $e_0 = |H/H^0|$, implies that the a_i do not form a strictly increasing sequence of positive numbers, a contradiction.
- If $w_3 \neq 0$, replacing w with $-w$, we may assume that $w_3 > 0$. If $w_1 > 0$ and $w_2 > 0$, then $w \in \mathfrak{h}$ has all coordinates positive, which is impossible. If $w_1 = 0$, then $w_2 = 2w_3$, and since w is primitive, $w = (0, 2, 1, 1)$. Then $[G_i : G_i \cap \mathbf{T}_{(0,2,1,1)}] = |H/H^0|$ for $i \gg 0$, and Claim A.3 with $e_0 = |H/H^0|$ contradicts the strict increase of the a_i. Similarly, $w_2 = 0$ implies $w = (2, 0, 1, 1)$, which is impossible by the second statement of Claim A.3. In the remaining case, w_1 and w_2 are non-zero and not both positive. Since $w_1 + w_2 = 2w_3 > 0$, exactly one of them is negative, and exchanging x and y, which preserves W, we may assume that $w_1 < 0 < w_2$.

Thus, possibly exchanging x and y, we have $w_1 < 0 < w_3$. Let $p := -w_1$ and $s := w_3$. Then p, s are positive integers, $w = (-p, p + 2s, s, s)$, and $\gcd(p, s) = \gcd(p, p + 2s, s) = 1$ since w is primitive. Let $e_0 := |H/H^0| \geq 1$ and $R_i := G_i \cap \mathbf{T}_{(-p,p+2s,s,s)}$. Then p, s, and e_0 depend only on H, $|R_i| = n_i \to +\infty$, and $[G_i : R_i] = e_0$ for $i \gg 0$. Possibly passing to a subsequence, we may assume that $|R_i| = n_i$ and $[G_i : R_i] = e_0$ for any i. Here G_i is the inertia group of a computing

lift (**Step 1**), the values a_i are kept, and the data are on $W = (xy = z^2 + t^2)$ after a linear change of eigenparameters (**Step 2**). This finishes the proof. □

A.2. **Nodes: circles.** In this subsection, we prove Theorem 7.6.

Proof of Theorem 7.6. The proof is long, so we divide it into several steps. Let $R := G \cap \mathbf{T}$ and $r := |R|$. Let (Q, Δ) be the crepant quotient of (W, D) by G, and $\bar{0} \in Q$ the image of 0, so that

$$\alpha := \mathrm{mld}((W \ni 0, D)/G) = \mathrm{mld}(Q \ni \bar{0}, \Delta).$$

We assume that $\alpha > 0$. Let $\gamma := \min(\Gamma \backslash \{0\})$ if $\Gamma \not\subset \{0\}$, and $\gamma := 1$ otherwise. Let

$$m_0 := \left\lceil \frac{2}{\gamma} \right\rceil, \quad m_1 := (p+2s)m_0, \quad m_2 := m_0 \max\left\{1, \frac{2s}{p}\right\}, \quad m_3 := 2s + 2(p+2s),$$

let k_0 be a positive integer such that $\frac{k_0 - 1}{2} > m_2$, and let $\epsilon_0 := \frac{1}{p} - \frac{1}{p+2s} > 0$. The constants $m_0, m_1, m_2, m_3, k_0, \epsilon_0$ depend only on p, s, and Γ.

Step 1. In this step we normalize the marks and bound the data of the marks which vanish at 0.

By Definition 2.7, $D = \sum_j c_j \, \mathrm{div}_W(h_j)$, where $c_j \in \Gamma$ and the h_j are non-zero regular G-semi-invariant functions on a G-invariant neighborhood of 0. For any j, there exists a polynomial g_j such that $g_j(0) \neq 0$ and $g_j h_j$ is regular on W. Then

$$h_j \prod_{\omega \in G} \omega^* g_j = (g_j h_j) \prod_{\omega \neq \mathrm{id}_W} \omega^* g_j$$

is a G-semi-invariant element of $\mathbb{C}[W]$ which differs from h_j by a G-invariant unit near 0. Thus, possibly replacing h_j with $h_j \prod_{\omega \in G} \omega^* g_j$, we may assume that $h_j \in \mathbb{C}[W]$ for any j. Since a mark h_j with $c_j = 0$ or $h_j(0) \neq 0$ does not contribute to D near 0, possibly removing such marks, we may assume that $c_j > 0$ and $h_j(0) = 0$ for any j.

For $f \in \mathbb{C}[W] = \mathbb{C}[x, y, z, t]/(xy - z^2 - t^2)$, the *normal form* of f is the unique linear combination of monomials $x^{a_1} y^{a_2} z^{a_3} t^{a_4}$ with $\min\{a_1, a_2\} = 0$ whose image in $\mathbb{C}[W]$ is f, and a *monomial of* f is a monomial with non-zero coefficient in the normal form of f. Since G is diagonal and preserves W, G acts on the monomials xy, z^2, and t^2 by one character. Thus G acts on any monomial of a G-semi-invariant f by the character of f. Since $xy - z^2 - t^2$ is homogeneous, the monomials of degree m in normal form induce a basis of $\mathfrak{m}_0^m / \mathfrak{m}_0^{m+1}$, where $\mathfrak{m}_0$ is the maximal ideal of $\mathcal{O}_{W,0}$. Hence $\mathrm{mult}_0 h_j$ is the least degree of a monomial of h_j, and $\mathrm{mult}_0 h_j \geq 1$. Let $z_\pm := z \pm \sqrt{-1} t$. Then $W = (xy = z_+ z_-)$ is the affine toric variety whose characters $\chi^{(1,0,0)}$, $\chi^{(0,1,0)}$, $\chi^{(0,0,1)}$ are x, z_+, z_-, and the exceptional prime divisor of the blow-up of W at 0 is the toric prime divisor of the ray of $(1, 1, 1)$, since x, y, z_+, z_- all have value 1 on it. The log discrepancy of the toric prime divisor of $(1, 1, 1)$ is the value 2 at $(1, 1, 1)$ of the support function of K_W, which is $u_2 + u_3$ in the coordinates (u_1, u_2, u_3) dual to the characters $\chi^{(1,0,0)}$, $\chi^{(0,1,0)}$, $\chi^{(0,0,1)}$ [CLS11, Theorem 8.2.3 and Lemma 11.4.10], and the order of h_j along it is the least degree of a character with non-zero coefficient in h_j, which is $\mathrm{mult}_0 h_j$. Since (W, D) is klt near 0,

$$\sum_j c_j \, \mathrm{mult}_0 h_j < 2. \tag{A.2}$$

Since $c_j \geq \gamma$ for any j, the number of marks is less than $\frac{2}{\gamma} \leq m_0$, and $\mathrm{mult}_0 h_j < \frac{2}{c_j} \leq m_0$ for any j.

The $\mathbf{T}$-*weight* of a monomial $x^{a_1} y^{a_2} z^{a_3} t^{a_4}$ is $-pa_1 + (p+2s)a_2 + s(a_3 + a_4)$. Since $\gcd(p, s) = 1$, $\mathbf{T}$ is the image of the injective homomorphism

$$\mathbb{C}^* \to (\mathbb{C}^*)^4, \quad \lambda \to (\lambda^{-p}, \lambda^{p+2s}, \lambda^s, \lambda^s),$$

and R is the image of the group of r-th roots of unity. A monomial of $\mathbf{T}$-weight κ is R-semi-invariant with character $\zeta \to \zeta^\kappa$.

For any j, fix a monomial $x^{a_{j1}}y^{a_{j2}}z^{a_{j3}}t^{a_{j4}}$ of h_j of degree $\operatorname{mult}_0 h_j$, and let κ_j be the $\mathbf{T}$-weight of $x^{a_{j1}}y^{a_{j2}}z^{a_{j3}}t^{a_{j4}}$. Then $|\kappa_j| \le (p+2s)\operatorname{mult}_0 h_j \le m_1$. Let h_j^0 be the sum of the terms of the normal form of h_j whose monomials have $\mathbf{T}$-weight κ_j. Since any monomial of the normal form of h_j has the character of h_j, the normal form of h_j is a G-semi-invariant polynomial on $\mathbb{A}^4$. For it and the primitive vector $(-p, p+2s, s, s)$, the weight $\kappa(h_j)$ and the principal block of Set-up 5.1 are κ_j and h_j^0. Since h_j is R-semi-invariant, the $\mathbf{T}$-weight of any monomial of $h_j - h_j^0$ is $\kappa_j + lr$ for a non-zero integer l. Let

$$d_j := \min\{a_1 - a_2 \mid x^{a_1}y^{a_2}z^{a_3}t^{a_4} \text{ is a monomial of } h_j^0\}.$$

If $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ is a monomial of h_j^0 with $a_2 > 0$, then $a_1 = 0$ and $(p+2s)a_2 + s(a_3+a_4) = \kappa_j \le m_1$, so $a_2 \le m_0$. Moreover, $d_j \le a_{j1} - a_{j2} \le \operatorname{mult}_0 h_j \le m_0$. Thus $-m_0 \le d_j \le m_0$.

Step 2. In this step we compute the orders of the marks along two weight vectors close to the ends of the interval $[\frac{1}{p+2s}, \frac{1}{p}]$.

For $\theta \in [\frac{1}{p+2s}, \frac{1}{p}]$, let

$$w(\theta) := \big(1 - p\theta, (p+2s)\theta - 1, s\theta, s\theta\big).$$

The entries of $w(\theta)$ are non-negative, positive if $\frac{1}{p+2s} < \theta < \frac{1}{p}$, and at most $\max\{1, \frac{2s}{p}\}$. Since $w(\theta)_x + w(\theta)_y = 2w(\theta)_z = 2w(\theta)_t$, the polynomial $xy - z^2 - t^2$ is homogeneous for the weights $w(\theta)$ of x, y, z, t. Hence $\mathbb{C}[W]$ is graded by $w(\theta)$-weight, and the monomials in normal form are homogeneous. The $w(\theta)$-weight of a monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of $\mathbf{T}$-weight κ is

$$a_1(1 - p\theta) + a_2((p+2s)\theta - 1) + s\theta(a_3 + a_4) = (a_1 - a_2) + \kappa\theta. \tag{A.3}$$

For a non-zero $f \in \mathbb{C}[W]$, let $\nu_\theta(f)$ be the least $w(\theta)$-weight of a monomial of f. Define the affine function

$$\rho_j(\theta) := d_j + \kappa_j\theta, \quad \theta \in \left[\frac{1}{p+2s}, \frac{1}{p}\right].$$

By (A.3), $\rho_j(\theta)$ is the least $w(\theta)$-weight of a monomial of h_j^0, and

$$\rho_j(\theta) \le \nu_\theta(x^{a_{j1}}y^{a_{j2}}z^{a_{j3}}t^{a_{j4}}) \le \operatorname{mult}_0 h_j \max\left\{1, \frac{2s}{p}\right\} \le m_2.$$

Since all entries of $w(\theta)$ are non-negative on the closed interval, the $w(\theta)$-weight of any monomial is non-negative. Thus

$$0 \le \rho_j(\theta) \le m_2 \quad \text{for any } \theta \in \left[\frac{1}{p+2s}, \frac{1}{p}\right]. \tag{A.4}$$

Let n_0 be a positive integer such that any integer $r' \ge n_0$ satisfies

$$r' > 2m_1, \quad \frac{k_0}{r'} < \frac{\epsilon_0}{2}, \quad \frac{\epsilon_0(r' - m_1)}{2} > m_2, \quad \frac{2s}{p+2s} + \frac{m_3 k_0}{r'} < 1. \tag{A.5}$$

Then n_0 depends only on p, s, and Γ. We assume that $r \ge n_0$ until we treat the case $r < n_0$. Let

$$\theta_{\text{left}} := \frac{1}{r}\left(\left\lfloor \frac{r}{p+2s} \right\rfloor + k_0\right), \quad \theta_{\text{right}} := \frac{1}{r}\left(\left\lceil \frac{r}{p} \right\rceil - k_0\right).$$

Then

$$\frac{k_0 - 1}{r} < \theta_{\text{left}} - \frac{1}{p+2s} \le \frac{k_0}{r}, \quad \frac{k_0 - 1}{r} < \frac{1}{p} - \theta_{\text{right}} \le \frac{k_0}{r}.$$

Since $\frac{k_0 - 1}{2} > m_2 \ge m_0 \ge 1$, we have $k_0 \ge 2$. Since $k_0 \ge 2$ and $\frac{k_0}{r} < \frac{\epsilon_0}{2}$, both θ_{left} and θ_{right} belong to the open interval $(\frac{1}{p+2s}, \frac{1}{p})$, and for $\theta \in \{\theta_{\text{left}}, \theta_{\text{right}}\}$,

$$\left(\theta - \frac{1}{p+2s}\right)(r - m_1) > m_2 \quad \text{and} \quad \left(\frac{1}{p} - \theta\right)(r - m_1) > m_2. \tag{A.6}$$

If $\theta = \theta_{\text{left}}$, then

$$\left(\theta - \frac{1}{p+2s}\right)(r-m_1) > (k_0-1)\left(1-\frac{m_1}{r}\right) > \frac{k_0-1}{2} > m_2$$

and $\frac{1}{p} - \theta \geq \epsilon_0 - \frac{k_0}{r} > \frac{\epsilon_0}{2}$, so $(\frac{1}{p}-\theta)(r-m_1) > \frac{\epsilon_0(r-m_1)}{2} > m_2$ by (A.5). The case $\theta = \theta_{\text{right}}$ is symmetric.

We show that for $\theta \in \{\theta_{\text{left}}, \theta_{\text{right}}\}$ and any j,

$$\nu_\theta(h_j) = \rho_j(\theta). \tag{A.7}$$

Let $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ be a monomial of $h_j - h_j^0$, with $\mathbf{T}$-weight $\kappa' = \kappa_j + lr$ and $l \neq 0$. Since $r > 2m_1 \geq 2|\kappa_j|$, either $\kappa' \geq r - m_1 > 0$ or $\kappa' \leq -(r-m_1) < 0$.

- If $\kappa' > 0$ and $a_2 = 0$, then the $w(\theta)$-weight is $a_1 + \kappa'\theta \geq (\theta - \frac{1}{p+2s})\kappa'$. If $\kappa' > 0$ and $a_1 = 0$, then $(p+2s)a_2 \leq \kappa'$, and the $w(\theta)$-weight is $-a_2 + \kappa'\theta \geq (\theta - \frac{1}{p+2s})\kappa'$. In both cases the $w(\theta)$-weight is at least $(\theta - \frac{1}{p+2s})(r-m_1)$.
- If $\kappa' < 0$, then $a_2 = 0$, since $a_2 > 0$ implies $a_1 = 0$ and $\kappa' = (p+2s)a_2 + s(a_3+a_4) > 0$. Thus $pa_1 = s(a_3+a_4) - \kappa' \geq -\kappa'$, and the $w(\theta)$-weight is

$$a_1 + \kappa'\theta \geq \left(\frac{1}{p} - \theta\right)(-\kappa') \geq \left(\frac{1}{p} - \theta\right)(r-m_1).$$

By (A.6) and (A.4), any monomial of $h_j - h_j^0$ has $w(\theta)$-weight larger than $m_2 \geq \rho_j(\theta)$. Since $h_j^0 \neq 0$, (A.7) follows.

Step 3. In this step we fix $\theta_0 \in \{\theta_{\text{left}}, \theta_{\text{right}}\}$ and bound the value at θ_0 of an affine function φ_0.

Let $\tau := \sum_j c_j\kappa_j$, the mass of D along $\mathbf{T}$ (Set-up 5.1). By (A.2),

$$|\tau| \leq (p+2s)\sum_j c_j \operatorname{mult}_0 h_j < 2(p+2s),$$

so $|2s - \tau| \leq m_3$. Let $\theta_0 := \theta_{\text{left}}$ if $2s - \tau \geq 0$, and $\theta_0 := \theta_{\text{right}}$ if $2s - \tau < 0$. On the closed interval $[\frac{1}{p+2s}, \frac{1}{p}]$, let

$$\varphi_0(\theta) := 2s\theta - \sum_j c_j\rho_j(\theta) = (2s-\tau)\theta - \sum_j c_j d_j, \quad \text{and} \quad \varphi := \varphi_0(\theta_0).$$

By (A.4), $\varphi_0(\frac{1}{p+2s}) \leq \frac{2s}{p+2s}$. If $2s - \tau < 0$, then φ_0 is decreasing, so $\varphi_0(\frac{1}{p}) \leq \varphi_0(\frac{1}{p+2s})$. In both cases θ_0 has distance at most $\frac{k_0}{r}$ from an end θ_e of the interval with $\varphi_0(\theta_e) \leq \frac{2s}{p+2s}$. Since the slope of φ_0 is $2s - \tau$, (A.5) implies that

$$\varphi \leq \frac{2s}{p+2s} + \frac{m_3k_0}{r} < 1. \tag{A.8}$$

Step 4. In this step we construct a toric variety Y over W/R with one exceptional prime divisor E and compute the crepant boundary D_Y on Y.

Recall from **Step 1** that $W = (xy = z_+z_-)$. Let $N := \mathbb{Z}^3$ with coordinates (u_1, u_2, u_3), and $M := \mathbb{Z}^3$ the dual lattice, where the characters $\chi^{(1,0,0)}$, $\chi^{(0,1,0)}$, $\chi^{(0,0,1)}$ are x, z_+, z_-, so that $y = \chi^{(-1,1,1)}$. Then W is the affine toric variety of the cone

$$\sigma := \{u \in \mathbb{R}^3 \mid u_1 \geq 0,\ u_2 \geq 0,\ u_3 \geq 0,\ u_2 + u_3 - u_1 \geq 0\}$$

with rays generated by $(0,1,0)$, $(0,0,1)$, $(1,1,0)$, and $(1,0,1)$. The group $\mathbf{T}$ acts on (x, z_+, z_-) by $(\lambda^{-p}, \lambda^s, \lambda^s)$, so R is generated by the torus element corresponding to $\frac{1}{r}(-p, s, s)$. Let

$$N_R := N + \mathbb{Z}\frac{1}{r}(-p, s, s),$$

which is the lattice N_R of [Liu26, Definition 2.15] for the R-semi-invariant functions x, z_+, z_-, and let $M_R \subset M$ be the dual lattice. Since the R-invariant characters are χ^m with $m \in M_R$,

W/R is the affine toric variety of σ in N_R. If $u \in N_R$ is a multiple of $(0,1,0)$ or $(0,0,1)$, then $u - j\frac{1}{r}(-p,s,s) \in N$ for an integer j with $jp \equiv js \equiv 0 \pmod r$. If $u \in N_R$ is a multiple of $(1,1,0)$ or $(1,0,1)$, then $u - j\frac{1}{r}(-p,s,s) \in N$ for an integer j with $js \equiv j(p+s) \equiv 0 \pmod r$. In both cases $j \equiv 0 \pmod r$ since $\gcd(p,s) = 1$, so $u \in N$. Thus the four generators above are primitive in N_R. Since R acts freely on the torus of W, the quotient map $W \to W/R$ is quasi-étale. Let D_R be the crepant boundary of $(W,D)/R$. For any j, h_j^r is R-invariant, and we let $\bar{h}_j \in \mathbb{C}[W/R]$ be the function with pullback h_j^r. By [Liu26, Lemma 2.6(4)], D is the pullback of D_R, so

$$D_R = \sum_j \frac{c_j}{r} \operatorname{div}(\bar{h}_j).$$

For $u \in \sigma$ and a non-zero $f \in \mathbb{C}[W]$, let $\operatorname{val}_u(f)$ be the least value $\langle m, u\rangle$ over the characters χ^m with non-zero coefficient in f. For a primitive $u \in N_R \cap \sigma\backslash\{0\}$, let E_u be the toric prime divisor over W/R of the ray $\mathbb{R}_{\ge 0}u$. Then $\operatorname{mult}_{E_u} g = \operatorname{val}_u(g)$ for $g \in \mathbb{C}[W/R]$. Since $K_{W/R} = -\sum_\varrho D_\varrho$ over the four toric prime divisors D_ϱ and $u_2 + u_3 = 1$ at the four primitive generators, $a(E_u, W/R, 0) = u_2 + u_3$ by [CLS11, Theorem 8.2.3 and Lemma 11.4.10]. As $\operatorname{val}_u(h_j^r) = r \operatorname{val}_u(h_j)$, we obtain

$$a(E_u, W/R, D_R) = u_2 + u_3 - \sum_j c_j \operatorname{val}_u(h_j). \tag{A.9}$$

Let $v := (1 - p\theta_0, s\theta_0, s\theta_0)$. Since $r\theta_0 \in \mathbb{Z}$, $v = (1,0,0) + r\theta_0 \cdot \frac{1}{r}(-p,s,s) \in N_R$. Since $\frac{1}{p+2s} < \theta_0 < \frac{1}{p}$, v belongs to the interior of σ. The values $\langle m, v\rangle$ for the characters x, y, z_+, z_- are $1 - p\theta_0$, $(p+2s)\theta_0 - 1$, $s\theta_0$, $s\theta_0$, which are the $w(\theta_0)$-weights of x, y, z, t. Thus any character χ^m is homogeneous of $w(\theta_0)$-weight $\langle m, v\rangle$. Since the characters χ^m, $m \in \sigma^\vee \cap M$, and the monomials in normal form are two bases of $\mathbb{C}[W]$ consisting of homogeneous elements, both $\operatorname{val}_v(f)$ and $\nu_{\theta_0}(f)$ are the least weight of a non-zero homogeneous component of f. By (A.7),

$$\operatorname{val}_v(h_j) = \nu_{\theta_0}(h_j) = \rho_j(\theta_0) \quad \text{for any } j.$$

Write $v = qv'$ with $v' \in N_R$ primitive and q a positive integer. The monomial $x^{p+2s}y^p = \chi^{(2s,p,p)}$ is $\mathbf{T}$-invariant, so $(2s,p,p) \in M_R$, and

$$\langle (2s,p,p), v\rangle = 2s(1 - p\theta_0) + 2ps\theta_0 = 2s.$$

Since $\langle (2s,p,p), v'\rangle$ is a positive integer, q divides $2s$. Let $E := E_{v'}$. By (A.9),

$$a(E, W/R, D_R) = \frac{1}{q}\left(2s\theta_0 - \sum_j c_j\rho_j(\theta_0)\right) = \frac{\varphi}{q}.$$

Let F be a prime divisor over W lying over E. Then the center of F on W is 0, and $a(F,W,D) = |I_R(F)| \cdot a(E, W/R, D_R)$ by [Liu26, Lemma 2.6(2)]. Since (W,D) is klt near 0, $\varphi > 0$. By (A.8),

$$\beta := 1 - \frac{\varphi}{q} \in (0,1). \tag{A.10}$$

Let Y be the toric variety of the star subdivision of σ at v', whose maximal cones are the four cones generated by v' and the generators of a two-dimensional face of σ, and $\mu : Y \to W/R$ the induced toric morphism. Each maximal cone is simplicial. The morphism μ is projective by [CLS11, Proposition 11.1.6(c)]. Since v' is interior to σ, μ is an isomorphism over $(W/R)\backslash\{0/R\}$, and $\mu^{-1}(0/R) = E$. Let

$$D_Y := \mu_*^{-1} D_R + \beta E.$$

Then $K_Y + D_Y = \mu^*(K_{W/R} + D_R)$ and $D_Y \ge 0$. Let $Y^\circ := Y\backslash\mu^{-1}(\operatorname{Nklt}(W/R, D_R))$. Since $(W/R, D_R)$ is klt near $0/R$ by [Liu26, Lemma 2.6(2)(3)], Y° is an open neighborhood of E, and (Y°, D_Y) is klt. A component of D_R intersecting $\mu(Y^\circ)$ has coefficient less than 1, equal to the coefficient of a component of D by [Liu26, Lemma 2.6(4)], which belongs to Γ_+. Thus the coefficients of the components of D_Y intersecting Y° belong to $(\Gamma_+ \cap [0,1)) \cup \{\beta\}$.

Let $\omega \in G$ act by $(x,y,z,t) \to (a_x x, a_y y, a_z z, a_t t)$. Since ω preserves W, $a_x a_y = a_z^2 = a_t^2$, so $a_t = \pm a_z$. If $a_t = a_z$, then ω multiplies z_+ and z_- by a_z. If $a_t = -a_z$, then $\omega^* z_+ = a_z z_-$ and $\omega^* z_- = a_z z_+$. Thus any element of G acts on W as an element of the torus of W, possibly composed with the toric automorphism induced by the lattice involution $\iota(u_1,u_2,u_3) := (u_1,u_3,u_2)$. The involution ι fixes $\frac{1}{r}(-p,s,s)$ and v, so ι preserves N_R, v', and the fan of Y. Hence the action of G/R on W/R lifts to Y, μ is G/R-equivariant, and D_Y and Y° are G/R-invariant. If $\omega \in G$ acts trivially on W/R, then W is the union of the fixed loci of $\omega'^{-1}\omega$ for $\omega' \in R$, which are closed, so $\omega'^{-1}\omega = \mathrm{id}_W$ for some $\omega' \in R$ as W is irreducible, and $\omega \in R$. Thus G/R acts faithfully on W/R and on Y.

Step 5. In this step we show that the set Λ_E of the numbers β of (A.10), for all data of Theorem 7.6 with p, s, Γ fixed and $r \geq n_0$, satisfies the DCC.

Suppose that there exist data (G_i, D_i) with $r_i \geq n_0$ such that β_i is strictly decreasing. We use the index i for the objects of **Steps 1–4** of the i-th data. The number of marks is less than m_0, $|\kappa_{ij}| \leq m_1$, $|d_{ij}| \leq m_0$, and q_i divides $2s$. Possibly passing to a subsequence, we may assume that the number of marks, the integers $\kappa_j := \kappa_{ij}$ and $d_j := d_{ij}$, $q := q_i$, and the choice between θ_{left} and θ_{right} do not depend on i. Then ρ_j does not depend on i. Since Γ satisfies the DCC, possibly passing to a subsequence, we may assume that $c_{ij} \leq c_{i+1,j}$ for any i and j. Possibly passing to a subsequence again, we may assume that either r_i is constant, or r_i is strictly increasing with limit $+\infty$. In the first case, $\theta_{0,i}$ is constant. In the second case, $\theta_{0,i}$ converges to $\frac{1}{p+2s}$ from above if $\theta_{0,i} = \theta_{\mathrm{left},i}$ and to $\frac{1}{p}$ from below if $\theta_{0,i} = \theta_{\mathrm{right},i}$, so possibly passing to a subsequence, we may assume that $\theta_{0,i}$ is non-increasing if $\theta_{0,i} = \theta_{\mathrm{left},i}$ and non-decreasing if $\theta_{0,i} = \theta_{\mathrm{right},i}$. Since the choice between θ_{left} and θ_{right} is determined by the sign of $2s - \tau_i$, in either case

$$\theta_{0,i+1} \leq \theta_{0,i} \text{ if } 2s - \tau_{i+1} \geq 0, \qquad \theta_{0,i+1} \geq \theta_{0,i} \text{ if } 2s - \tau_{i+1} < 0.$$

Since $\varphi_{0,i+1}$ is affine with slope $2s - \tau_{i+1}$,

$$\varphi_{i+1} - \varphi_i = (2s - \tau_{i+1})(\theta_{0,i+1} - \theta_{0,i}) - \sum_j (c_{i+1,j} - c_{ij})\rho_j(\theta_{0,i}). \tag{A.11}$$

The first term of (A.11) is non-positive, and the second sum is non-negative by (A.4). Thus $\varphi_{i+1} \leq \varphi_i$, and $\beta_{i+1} = 1 - \frac{\varphi_{i+1}}{q} \geq \beta_i$, a contradiction.

By [Liu26, Lemma 2.10(1)], Γ_+ satisfies the DCC. Hence

$$\Lambda := (\Gamma_+ \cap [0,1)) \cup \Lambda_E \cup \{0\}$$

is a DCC subset of $[0,1)$ which depends only on p, s, and Γ, and the coefficients of the components of D_Y intersecting Y° belong to Λ by **Step 4**.

Step 6. In this step we show that α is the minimal log discrepancy of a pair (Z, D_Z) at a scheme-theoretic point over $\bar{0}$, where $D_Z \in \Psi(\Lambda)$. We do not apply Definition-Proposition 4.10 to the morphism $\bar{\mu} : Z \to Q^\circ$ constructed below, since condition (ii) there asks for local presentations by finite abelian groups, while the groups $\widetilde{H}$ of the local presentations of Z constructed below may be non-abelian. We prove directly the part of the first statement of Definition-Proposition 4.10 that we need.

Let $Z := Y^\circ/(G/R)$, $\pi_Y : Y^\circ \to Z$ the quotient map, and D_Z the crepant boundary of $(Y^\circ, D_Y)/(G/R)$. By [Liu26, Lemma 2.6(4)], $D_Z \in \Psi(\Lambda)$. By [Liu26, Lemma 2.6(5)], (Q, Δ) is the crepant quotient of $(W/R, D_R)$ by G/R. Let $\pi : W/R \to Q$ be the quotient map, and $Q^\circ := Q \backslash \pi(\mathrm{Nklt}(W/R, D_R))$, an open neighborhood of $\bar{0}$ with $\pi^{-1}(Q^\circ) = \mu(Y^\circ)$. Since μ is proper, birational, and G/R-equivariant, it induces a proper birational morphism $\bar{\mu} : Z \to Q^\circ$ with $\bar{\mu} \circ \pi_Y = \pi \circ \mu$. By [Liu26, Lemma 2.6(0)],

$$\pi_Y^*(K_Z + D_Z) = K_Y + D_Y = \mu^*(K_{W/R} + D_R) = \mu^*\pi^*(K_Q + \Delta) = \pi_Y^*\bar{\mu}^*(K_Q + \Delta)$$

over Y°. Since π_Y is finite and surjective,

$$K_Z + D_Z = \bar{\mu}^*(K_Q + \Delta). \tag{A.12}$$

Thus $a(P, Z, D_Z) = a(P, Q, \Delta)$ for any prime divisor P over Q°.

By [Liu26, Lemma 3.5], there exists a prime divisor P over Q with center $\bar{0}$ such that $a(P, Q, \Delta) = \alpha$. Since $\bar{\mu}$ is proper, the center of P on Z is the closure of a point $\eta \in \bar{\mu}^{-1}(\bar{0})$. Any prime divisor P' over Z with center $\overline{\{\eta\}}$ has center $\bar{0}$ on Q, so $a(P', Z, D_Z) = a(P', Q, \Delta) \geq \alpha$, with equality for $P' = P$. Hence

$$\alpha = \mathrm{mld}(Z \ni \eta, D_Z).$$

Since $\pi^{-1}(\bar{0}) = \{0/R\}$, $\mu^{-1}(0/R) = E$, and π_Y is surjective, $\bar{\mu}^{-1}(\bar{0}) = \pi_Y(E)$.

Step 7. In this step we show that the numbers α with $r \geq n_0$ belong to a finite union of ACC sets depending only on p, s, and Γ, according to the dimension of $\overline{\{\eta\}}$.

If $\overline{\{\eta\}}$ is a prime divisor on Z, then $\alpha = 1 - b$, where $b \in \Psi(\Lambda)$ is the coefficient of $\overline{\{\eta\}}$ in D_Z. By [Liu26, Lemma 2.10(1)], $\Psi(\Lambda)$ satisfies the DCC, so $\{1 - b \mid b \in \Psi(\Lambda)\}$ satisfies the ACC.

If $\overline{\{\eta\}}$ is a curve, then $\alpha = \mathrm{mld}(Z \ni \eta, D_Z) > 0$, where (Z, D_Z) is a threefold pair with $D_Z \in \Psi(\Lambda)$ and $\Psi(\Lambda) \subset [0, 1]$ is a DCC set, so α belongs to the ACC set of [Liu26, Lemma 3.7] for $\Psi(\Lambda)$.

Suppose that η is a closed point. Let $\xi \in E \cap Y^\circ$ be a point over η, $H_\xi \subset G/R$ the stabilizer of ξ, and σ_ξ the cone of the fan of Y whose torus orbit contains ξ. For any prime divisor F' over Y° with center ξ, $I_{G/R}(F') \subset H_\xi$, so $I_{H_\xi}(F') = I_{G/R}(F')$. By [Liu26, Lemma 2.6(2)(3)], any prime divisor over Z with center η is the image of a prime divisor F' over Y° whose center is a point of $\pi_Y^{-1}(\eta)$, and after translating F' by an element of G/R we may take the center of F' to be ξ. Hence, by [Liu26, Lemma 2.6(2)] for G/R and by the formula at the end of [Liu26, Lemma 2.6] for H_ξ, which fixes ξ, both $\mathrm{mld}(Z \ni \eta, D_Z)$ and $\mathrm{mld}((Y^\circ \ni \xi, D_Y)/H_\xi)$ are the infimum of $a(F', Y^\circ, D_Y)/|I_{G/R}(F')|$ over the prime divisors F' over Y° with center ξ, so they are equal. Since $\xi \in E$, $v' \in \sigma_\xi$, and σ_ξ is generated by v' and $n_\xi - 1$ of the four generators of σ, where $n_\xi := \dim \sigma_\xi \in \{1, 2, 3\}$. The involution ι fixes none of the four generators of σ. Since any element of H_ξ preserves σ_ξ, if $n_\xi = 2$, then H_ξ consists of torus elements.

Let $N_\xi := N_R \cap \mathbb{R}\sigma_\xi$, choose a sublattice $N'' \subset N_R$ with $N_R = N_\xi \oplus N''$, and let N' be the sum of N'' and the sublattice generated by the primitive generators of σ_ξ. If $n_\xi = 1$, then $N' = N_R$ since v' is primitive. If $n_\xi = 3$, then $N'' = 0$. Thus if H_ξ contains an element which is not a torus element, then $n_\xi \in \{1, 3\}$, $\iota(\sigma_\xi) = \sigma_\xi$, and N' is ι-invariant. The affine toric variety $U_{\sigma_\xi, N'}$ of σ_ξ in N' is isomorphic to $\mathbb{A}^{n_\xi} \times (\mathbb{C}^*)^{3-n_\xi}$. Let $V \subset U_{\sigma_\xi, N'}$ be the preimage of Y° under the toric morphism $U_{\sigma_\xi, N'} \to U_{\sigma_\xi}$, which is the quotient by the finite abelian group $H_0 := \mathrm{Ker}(T_{N'} \to T_{N_R})$ [CLS11, Proposition 1.3.18]. Since the primitive generators of σ_ξ are primitive in N' and in N_R, and H_0 acts freely on the torus, $V \to U_{\sigma_\xi} \cap Y^\circ$ is quasi-étale. Let $\widetilde{H}$ be the group of automorphisms of V which are torus elements of $T_{N'}$, possibly composed with the automorphism induced by ι on N', and which induce elements of H_ξ on U_{σ_ξ}. Then $\widetilde{H}$ is finite, $H_0 \subset \widetilde{H}$ is normal, and $\widetilde{H}/H_0 \cong H_\xi$. Let B' be the pullback of D_Y to V. Since $V \to U_{\sigma_\xi} \cap Y^\circ$ is quasi-étale, $B' \geq 0$, $B' \in \Lambda$, and (V, B') is klt by [Liu26, Lemma 2.6(2)(4)]. By [Liu26, Lemma 2.6(5)], the crepant quotient of (V, B') by $\widetilde{H}$ is the crepant quotient of $(U_{\sigma_\xi} \cap Y^\circ, D_Y)$ by H_ξ. Thus

$$\alpha = \mathrm{mld}(Z \ni \eta, D_Z) = \mathrm{mld}(V/\widetilde{H} \ni \bar{\xi}, B'_{V/\widetilde{H}}),$$

where $\bar{\xi}$ is the image of ξ and $B'_{V/\widetilde{H}}$ is the crepant boundary.

Since $\widetilde{H}$ may be non-abelian, we reduce to the inertia group of a computing prime divisor. Let P be a prime divisor over $V/\widetilde{H}$ with center $\bar{\xi}$ such that $a(P, V/\widetilde{H}, B'_{V/\widetilde{H}}) = \alpha$, which exists by [Liu26, Lemma 3.5], as $(V/\widetilde{H}, B'_{V/\widetilde{H}})$ is klt near $\bar{\xi}$ by [Liu26, Lemma 2.6(2)(3)]. Let F be a prime divisor over V lying over P, x' the center of F on V, which is a closed point, and $I := I_{\widetilde{H}}(F)$. By [Liu26, Lemma 2.6(1)], I is cyclic. Since I preserves F, I fixes x'. By [Liu26, Lemma 3.1(1)], applied with $H = I$ and the identity morphism of V,

$$\mathrm{mld}((V \ni x', B')/I) = \alpha.$$

Since $(V \ni x')$ is a smooth threefold germ, I is abelian and acts faithfully on V fixing x', $B' \geq 0$ is I-invariant with $B' \in \Lambda$, $\Lambda \subset [0,1]$ is a DCC set, and (V, B') is klt near x', the last sentence of Theorem 2.8(1), applied to $(V \ni x')$, I, and B', implies that α belongs to an ACC set depending only on Λ.

Step 8. In this step we treat the case $r < n_0$ and conclude the proof.

If $r < n_0$, then $|G| = [G : R] \cdot r < e_0 n_0$, the normal forms of the h_j are G-semi-invariant polynomials (**Step 1**), and Lemma A.2 with $e_0 n_0$ in place of N implies that the numbers α with $r < n_0$ satisfy the ACC.

By **Steps 6 and 7**, the numbers α with $r \geq n_0$ belong to the union of $\{1 - b \mid b \in \Psi(\Lambda)\}$, the ACC set of [Liu26, Lemma 3.7] for $\Psi(\Lambda)$, and the set of Theorem 2.8(1) for Λ, each of which satisfies the ACC. By the previous paragraph, the numbers α with $r < n_0$ satisfy the ACC. Since Λ and n_0 depend only on p, s, and Γ, the positive numbers $\mathrm{mld}((W \ni 0, D)/G)$ satisfy the ACC. □

A.3. **Nodes and fixed axial exponents.** In this subsection, we prove Theorems 7.7 and 7.8.

Proof of Theorem 7.7. A threefold node is an isolated cDV point, hence an isolated Gorenstein terminal point [KM98, Corollary 5.38], so Construction-Theorem 4.13 can be used.

Suppose that there exist threefold nodes $(W_i \ni o_i)$, finite abelian groups G_i acting faithfully on W_i and fixing o_i, and labelled boundaries D_i on $(W_i \ni o_i)$ with labels in Γ such that (W_i, D_i) is klt near o_i and the numbers $a_i := \mathrm{mld}((W_i \ni o_i, D_i)/G_i)$ are positive and strictly increasing. Removing the marks with label 0 and the units does not change D_i, so we may assume that all labels of D_i are positive and that no mark of D_i is a unit (possibly with no marks). Fix i, and write W, o, G, $D = \sum_j c_j \,\mathrm{div}(h_j)$, and a for W_i, o_i, G_i, D_i, and a_i.

Since $(W \ni o)$ is a hypersurface singularity of embedding dimension four and G is abelian, there exist a smooth fourfold germ $(Y \ni o)$ with an action of G, a G-equivariant closed embedding $W \subset Y$, a regular G-semi-invariant function f on Y with $W = (f = 0)$, and G-eigenparameters $x_1, \dots, x_4$ of Y at o, of characters $\chi_1, \dots, \chi_4$. We embed a G-invariant affine neighborhood of o in W equivariantly into an affine space $\mathbb{A}^M$ with a diagonal action of G, mapping o to 0. The projection onto a character of an element of the ideal I_W of W belongs to I_W, so there exist G-semi-invariant $g_5, \dots, g_M \in I_W$ whose differentials at o form a basis of the image of I_W in the cotangent space of $\mathbb{A}^M$ at o. We let Y be the germ at o of $(g_5 = \cdots = g_M = 0)$, $x_1, \dots, x_4$ four coordinates of $\mathbb{A}^M$ whose restrictions form a regular system of parameters of Y at o, and f the projection onto χ_0 of a generator of the ideal I of W in $\mathcal{O}_{Y,o}$, where χ_0 is the character of G on the line $I/\mathfrak{m}_{Y,o} I$. We lift the marks h_j to regular G-semi-invariant functions on Y, denoted again by h_j, by projecting lifts onto the characters of the h_j.

Let $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_4]]$, the completion of $\mathcal{O}_{Y,o}$, on which G acts diagonally through $\chi_1, \dots, \chi_4$, let $\mathfrak{m} := (x_1, \dots, x_4)$, and let $\hat{f}$ and $\hat{h}_j$ be the expansions of f and h_j. Since $(W \ni o)$ is a node, the quadratic part q of $\hat{f}$ is non-degenerate, and q is a semi-invariant of character χ_0.

We construct a G-equivariant automorphism ϕ of $\mathfrak{R}$ with identity linear part and $\phi(\hat{f}) = q$. Let $F \in q + \mathfrak{m}^3$ be a semi-invariant of character χ_0, and write $F = q + F_d + F'$ with F_d homogeneous of degree $d \geq 3$ and $F' \in \mathfrak{m}^{d+1}$. Since q is non-degenerate, the linear forms $\partial q/\partial x_1, \dots, \partial q/\partial x_4$ form a basis of the space of linear forms, so $F_d = \sum_i b_i \partial q/\partial x_i$ for homogeneous polynomials b_i of degree $d-1$. The form $\partial q/\partial x_i$ is a semi-invariant of character $\chi_0 \chi_i^{-1}$ and F_d is a semi-invariant of character χ_0, so replacing each b_i with the projection of b_i onto the character χ_i keeps the equality $F_d = \sum_i b_i \partial q/\partial x_i$. The substitution $x_i \to x_i - b_i$ defines a G-equivariant automorphism ϕ_d of $\mathfrak{R}$ with identity linear part, and

$$\phi_d(F) \in q - \sum_i b_i \frac{\partial q}{\partial x_i} + q(b_1, \dots, b_4) + F_d + \mathfrak{m}^{d+1} = q + \mathfrak{m}^{d+1},$$

since $q(b_1,\dots,b_4)\in\mathfrak{m}^{2d-2}$ and $2d-2>d$. Starting from $F=\hat{f}$ and repeating the substitution for $d=3,4,\dots$, the automorphisms $\phi_d\circ\cdots\circ\phi_3$ converge $\mathfrak{m}$-adically, as ϕ_d is the identity modulo $\mathfrak{m}^{d-1}$, and the limit ϕ satisfies $\phi(\hat{f})=q$.

We use Construction-Theorem 4.13 for $n=4$, the embedding $W\subset Y$, the eigenparameters $x_1,\dots,x_4$, the marks h_j with the labels c_j, no vectors in (iv), no prime divisors in (v), $\psi:=\phi$, $v:=1$, and $u_j:=1$. For sufficiently large truncation orders, let $h_{j,0}$ be the sum of the terms of $\phi(\hat{h}_j)$ of degree less than the truncation order, a G-semi-invariant polynomial with the character of h_j, and let $f_0:=q=\phi(\hat{f})$. Then G acts faithfully on $W_0:=(q=0)\subset\mathbb{A}^4$ through the characters $\chi_1,\dots,\chi_4$, the pair (W_0,D_0) with $D_0:=\sum_j c_j\operatorname{div}(h_{j,0}|_{W_0})$ is klt near 0, and

$$\operatorname{mld}((W_0\ni 0,D_0)/G)=\operatorname{mld}((W\ni o,D)/G)=a.$$

Thus W_0, with the diagonal action of G and the labelled boundary D_0, satisfies the hypotheses of Proposition 7.5: q is a non-degenerate quadratic form, G preserves q up to the character χ_0 and acts faithfully on W_0, and the labels of D_0 belong to Γ.

Writing q_i and $D_{i,0}$ for q and D_0 when W, o, G, and D are W_i, o_i, G_i, and D_i, we obtain nodes $W_{i,0}=(q_i=0)\subset\mathbb{A}^4$, finite diagonal abelian groups G_i, and labelled boundaries $D_{i,0}$ with labels in Γ such that $(W_{i,0},D_{i,0})$ is klt near 0 and $\operatorname{mld}((W_{i,0}\ni 0,D_{i,0})/G_i)=a_i$. By Proposition 7.5, after passing to a subsequence and replacing G_i by the inertia group of a computing lift, which keeps a_i, the data are, after a linear change of eigenparameters, on the fixed node $W=(xy=z^2+t^2)$, and there exist fixed coprime positive integers p,s and a fixed positive integer e_0 such that $R_i:=G_i\cap\mathbf{T}_{(-p,p+2s,s,s)}$ has order $n_i\to+\infty$ and $[G_i:R_i]=e_0$. By Theorem 7.6, the positive numbers a_i satisfy the ACC, which contradicts the strict increase of a_i. □

Before the proof of Theorem 7.8, we construct the weighted extractions of quotients of W_N used in that proof.

Lemma A.4. *Let $N\geq 3$ be an integer, $k:=\gcd(2,N)$, and $R\subset G$ finite diagonal abelian groups preserving the equation of W_N up to a character and acting faithfully on W_N, with no non-identity element of G acting non-trivially on exactly one of the coordinates x,y,z,t. Let $r:=|R|$, let e_x,e_y,e_z,e_t be the standard basis of $\mathbb{Z}^4$, let $M_R\subset\mathbb{Z}^4$ be the dual lattice of N_R, and let $v_*=(v_{*x},v_{*y},v_{*z},v_{*t})\in N_R$ be a primitive vector such that one of the following holds:*

(a) $v_{*x},v_{*y},v_{*z},v_{*t}>0$ and $v_{*x}+v_{*y}=2v_{*z}=Nv_{*t}$;
(b) $R=\mu_r(0,\frac{2N}{k},\frac{N}{k},\frac{2}{k})$ and $v_*=\frac{1}{kr}(0,2N,N,2)$.

Let Σ_* be the star subdivision of $\mathbb{R}^4_{\geq 0}$ at v_*, X_{Σ_*} the toric variety of Σ_* with respect to N_R, $\varphi:Y\to W_N/R$ the normalization of the strict transform of W_N/R in X_{Σ_*}, and $\pi:W_N\to W_N/R$ the quotient morphism. Let $\Xi:=\{x,y,z,t\}$ in case (a) and $\Xi:=\{y,z,t\}$ in case (b). For $\xi\in\Xi$, let $N_\xi\subset N_R$ be the sublattice generated by v_* and e_l, $l\in\{x,y,z,t\}\backslash\{\xi\}$, let $K_\xi:=N_R/N_\xi$, and let $V_\xi\subset\mathbb{A}^4$ be the hypersurface, in the coordinates λ and u_l, $l\neq\xi$, defined by $u_xu_y+u_z^2+u_t^N=0$ with $u_\xi:=1$, that is,

$$\begin{aligned} V_x&=(u_y+u_z^2+u_t^N=0),\quad & V_y&=(u_x+u_z^2+u_t^N=0),\\ V_z&=(u_xu_y+1+u_t^N=0),\quad & V_t&=(u_xu_y+u_z^2+1=0).\end{aligned}\tag{A.13}$$

Then the following hold.

(1) π is quasi-étale, and φ is projective, birational, and G/R-equivariant, an isomorphism over the complement of the image of 0 in case (a) and of the image of the x-axis in case (b), with exactly one exceptional prime divisor E.
(2) For any $\xi\in\Xi$, V_ξ is a smooth threefold, K_ξ is a finite abelian group acting diagonally on V_ξ and freely in codimension one, and Y is covered by open subsets $Y_\xi\cong V_\xi/K_\xi$, $\xi\in\Xi$. Under $V_\xi\to Y_\xi\to W_N/R$, the Laurent monomial $x^\alpha:=x^{\alpha_x}y^{\alpha_y}z^{\alpha_z}t^{\alpha_t}$, $\alpha\in M_R$, pulls

back to

$$\lambda^{\langle \alpha, v_* \rangle} \prod_{l \neq \xi} u_l^{\alpha_l},$$

and $E \cap Y_\xi$ *is the image of* $(\lambda = 0)$.

(3) $a(E, W_N/R, 0) = v_{*z} + v_{*t}$, *and for any* R*-semi-invariant polynomial* H *not divisible by* $xy + z^2 + t^N$, *the* R*-invariant function* H^r *on* W_N/R *satisfies*

$$\frac{1}{r} \operatorname{mult}_E(H^r) \geq \operatorname{wt}_{v_*}(H),$$

with equality if $xy + z^2 + t^N$ *does not divide* $\operatorname{in}_{v_*}(H)$.

(4) *For any* $\xi \in \Xi$, *the inverse image* $\tilde{G}_\xi$ *of* G/R *under the isogeny*

$$T_{N_\xi} := N_\xi \otimes_{\mathbb{Z}} \mathbb{C}^* \to T_{N_R} := N_R \otimes_{\mathbb{Z}} \mathbb{C}^* = (\mathbb{C}^*)^4/R$$

is a finite abelian group acting diagonally on $\mathbb{A}^4$ *and preserving* V_ξ, *the open subset* Y_ξ *is* G/R*-invariant, and* $V_\xi/\tilde{G}_\xi \cong Y_\xi/(G/R)$, *an open subset of* $Y/(G/R)$.

Proof. Let $f := xy + z^2 + t^N$. If $\gamma e_l \in N_R$ for some l and some $\gamma \in \mathbb{Q}\backslash\mathbb{Z}$, then the image of γe_l in $\mathbb{T}^4$ is a non-identity element of R acting non-trivially only on the coordinate l, which is excluded. Thus e_x, e_y, e_z, e_t are primitive in N_R. Since no non-identity element of R acts non-trivially on exactly one coordinate, the fixed locus in $\mathbb{A}^4$ of any non-identity element of R is contained in a coordinate two-plane. The restrictions of f to the six coordinate two-planes are $z^2 + t^N$, t^N, z^2, t^N, z^2, and xy, so W_N contains no coordinate two-plane. Thus the fixed locus on W_N of any non-identity element of R has codimension at least two, and π is quasi-étale.

The maximal cones of Σ_* are the cones σ_ξ spanned by v_* and e_l, $l \neq \xi$, for $\xi \in \Xi$. In case (b), v_* belongs to the relative interior of the cone spanned by e_y, e_z, e_t, and any σ_ξ contains e_x. Since Σ_* is the star subdivision at the primitive vector $v_* \in N_R$ of the fan of $\mathbb{A}^4/R$, the toric morphism $X_{\Sigma_*} \to \mathbb{A}^4/R$ is projective [CLS11, Proposition 11.1.6(c)]. It is an isomorphism over the complement of the closure Z of the torus orbit corresponding to the cone spanned by e_l, $l \in \Xi$, which is the image of 0 in case (a) and of the x-axis in case (b), and the preimage of Z is the toric divisor D_{v_*} of the ray $\mathbb{R}_{\geq 0} v_*$. Since W_N/R is normal, φ is projective, birational, and an isomorphism over $(W_N/R)\backslash Z$, and the exceptional locus of φ is the preimage of D_{v_*}. The group $G/R \subset T_{N_R}$ acts on X_{Σ_*} by translations and preserves the strict transform of the G/R-invariant subvariety W_N/R, so φ is G/R-equivariant.

Fix $\xi \in \Xi$. Since $v_{*\xi} > 0$, the vectors v_* and e_l, $l \neq \xi$, form a basis of N_ξ, consisting of primitive vectors of N_R. Let λ and u_l, $l \neq \xi$, be the dual coordinates on $U_{\sigma_\xi, N_\xi} = \mathbb{A}^4$. Then $U_{\sigma_\xi, N_R} = \mathbb{A}^4/K_\xi$, where $K_\xi \subset T_{N_\xi}$ acts diagonally, and a Laurent monomial x^α with $\alpha \in M_R$ pulls back to

$$\lambda^{\langle \alpha, v_* \rangle} \prod_{l \neq \xi} u_l^{\alpha_l},$$

as the exponents are the coordinates of α in the dual basis. An element of K_ξ fixing a coordinate hyperplane of $\mathbb{A}^4$ pointwise is the image of $\gamma b \in N_R$, where b is the corresponding basis vector and $\gamma \in \mathbb{Q}$. As b is primitive in N_R, $\gamma \in \mathbb{Z}$ and the element is trivial. Thus the fixed locus in $\mathbb{A}^4$ of any non-identity element of K_ξ is contained in a coordinate two-plane.

Since R preserves f up to a character, the Laurent monomials $z^2/(xy)$ and $t^N/(xy)$ are R-invariant. In both cases (a) and (b), $2v_{*z} - v_{*x} - v_{*y} = N v_{*t} - v_{*x} - v_{*y} = 0$, so $z^2/(xy)$ and $t^N/(xy)$ pull back to $u_z^2/(u_x u_y)$ and $u_t^N/(u_x u_y)$ with $u_\xi := 1$. As $W_N \cap (\mathbb{C}^*)^4 = (1 + z^2/(xy) + t^N/(xy) = 0)$, the preimage in T_{N_ξ} of $(W_N/R) \cap T_{N_R}$ is $V_\xi \cap T_{N_\xi}$. Let g_ξ be the polynomial defining V_ξ in (A.13). If $\xi = x$, it has degree one in u_y with coefficient 1. If $\xi = y$, it has degree one in u_x with coefficient 1. If $\xi \in \{z, t\}$, it has degree one in u_x with coefficient u_y, which is coprime to the remaining terms $1 + u_t^N$, resp. $u_z^2 + 1$. Hence V_ξ is integral, and it is not contained in a coordinate hyperplane. Thus V_ξ is the closure of $V_\xi \cap T_{N_\xi}$, and the image of V_ξ in U_{σ_ξ, N_R} is the closure S_ξ of $(W_N/R) \cap T_{N_R}$ there, and V_ξ is the preimage of S_ξ. The morphism $V_\xi/K_\xi \to S_\xi$

is finite and birational and V_ξ/K_ξ is normal, so Y_ξ, the preimage of U_{σ_ξ,N_R} in Y, is isomorphic to V_ξ/K_ξ. The threefold V_ξ is smooth: if $\xi \in \{x,y\}$, then $\partial g_\xi/\partial u_y = 1$ or $\partial g_\xi/\partial u_x = 1$, and if $\xi = z$ (resp. $\xi = t$), then the partial derivatives of g_ξ in u_x, u_y, u_t (resp. u_x, u_y, u_z) vanish only at $u_x = u_y = u_t = 0$ (resp. $u_x = u_y = u_z = 0$), where $g_\xi = 1$. Moreover, g_ξ does not vanish identically on any coordinate two-plane of $\mathbb{A}^4$: if $\xi \in \{x,y\}$, then setting two of the four coordinates to zero leaves a non-zero term among the three terms of g_ξ, which involve distinct variables, and if $\xi \in \{z,t\}$, then the constant term 1 remains. Thus the fixed locus in V_ξ of any non-identity element of K_ξ has codimension at least two, and K_ξ acts on V_ξ freely in codimension one.

The threefold V_ξ is $\mathbb{A}^1_\lambda \times (g_\xi = 0)$, so $(\lambda = 0) \subset V_\xi$ is a prime divisor, and the image of $(\lambda = 0)$ in Y_ξ is the preimage of D_{v_*} in Y_ξ. For any $\xi \in \Xi$, the image of $(\lambda = 0)$ in Y_ξ contains the preimage in Y of the torus orbit $O(v_*)$ of the ray $\mathbb{R}_{\geq 0}v_*$ as a dense open subset, namely the image of $(\lambda = 0) \cap (\prod_{l\neq\xi} u_l \neq 0)$, which is irreducible. Thus the exceptional locus of φ is a prime divisor E, and $E \cap Y_\xi$ is the image of $(\lambda = 0)$. Since K_ξ acts freely in codimension one, $V_\xi \to Y_\xi$ is unramified along $(\lambda = 0)$, and since g_ξ is not divisible by any u_l,

$$\operatorname{mult}_E(x^\alpha) = \langle \alpha, v_* \rangle \quad \text{for any } \alpha \in M_R. \tag{A.14}$$

This proves (1) and (2).

We prove (3). Assume first (a). Let w be the primitive vector of $\mathbb{Z}^4$ on the ray $\mathbb{R}_{\geq 0}v_*$. Since $\mathbb{Z}^4 \subset N_R$ and v_* is primitive in N_R, $w = mv_*$ for a positive integer m. As $w_x + w_y = 2w_z = Nw_t$, $\operatorname{in}_w(f) = f$, and $(f = 0) \subset \mathbb{A}^4$ is integral and not contained in any coordinate hyperplane. Let E_W be the prime divisor over W_N of Lemma 4.12 for W_N and w. Then $\operatorname{mult}_{E_W}(H|_{W_N}) \geq \operatorname{wt}_w(H)$ for any polynomial H not divisible by f, with equality if f does not divide $\operatorname{in}_w(H)$, and $a(E_W, W_N, 0) = \sum_l w_l - \operatorname{wt}_w(f) = w_z + w_t$. The weighted blow-up of $\mathbb{A}^4$ with weights w is the toric variety of Σ_* with respect to $\mathbb{Z}^4$, and the inclusion $\mathbb{Z}^4 \subset N_R$ induces a finite morphism from it to X_{Σ_*}, mapping the exceptional divisor onto D_{v_*} and the strict transform of W_N onto that of W_N/R. Thus the induced rational map from the strict transform of W_N to Y maps E_W onto a surface in the exceptional locus of φ, that is, onto E, and $\operatorname{mult}_{E_W}(\pi^* g) = |I_R(E_W)| \operatorname{mult}_E g$ for any rational function g on W_N/R [Liu26, Lemma 2.6(1)]. Since v_* is primitive in N_R, there exists $\alpha \in M_R$ with $\langle \alpha, v_* \rangle = 1$. Writing x^α as a quotient of two monomials, for which the equality holds, we obtain $\operatorname{mult}_{E_W}(x^\alpha) = \langle \alpha, w \rangle = m$, while $\operatorname{mult}_E(x^\alpha) = 1$ by (A.14). Thus $|I_R(E_W)| = m$. As π is quasi-étale, $a(E_W, W_N, 0) = ma(E, W_N/R, 0)$ [Liu26, Lemma 2.6(2)], so $a(E, W_N/R, 0) = (w_z + w_t)/m = v_{*z} + v_{*t}$, and for any R-semi-invariant polynomial H not divisible by f,

$$\frac{1}{r}\operatorname{mult}_E(H^r) = \frac{1}{mr}\operatorname{mult}_{E_W}(H^r) = \frac{1}{m}\operatorname{mult}_{E_W} H \geq \frac{1}{m}\operatorname{wt}_w(H) = \operatorname{wt}_{v_*}(H),$$

with equality if f does not divide $\operatorname{in}_{v_*}(H) = \operatorname{in}_w(H)$.

Assume next (b), and let $w := rv_* = (0, \frac{2N}{k}, \frac{N}{k}, \frac{2}{k}) \in \mathbb{Z}^4$. Let $\eta : \mathbb{A}^4 \to \mathbb{A}^4$ be the morphism

$$\eta(\mu, u_x, u_y, u_z) = \big(u_x,\ \mu^{2N/k}u_y,\ \mu^{N/k}u_z,\ \mu^{2/k}\big).$$

Then $f \circ \eta = \mu^{2N/k}(u_x u_y + u_z^2 + 1)$. Let $V_t' := (u_x u_y + u_z^2 + 1 = 0) \subset \mathbb{A}^4$ with the coordinates μ, u_x, u_y, u_z. Then η maps V_t' into W_N, and $\eta(V_t' \cap (\mu \neq 0)) = W_N \cap (t \neq 0)$. Let $\psi : V_t' \to Y_t$ be the composite of $V_t' \to V_t$, $(\mu, u_x, u_y, u_z) \to (\mu^r, u_x, u_y, u_z)$ (the coordinate $\lambda = \mu^r$ on V_t), and $V_t \to Y_t$. For $\alpha \in M_R$, both $\eta^* x^\alpha$ and $\psi^* x^\alpha$ are

$$\mu^{r\langle \alpha, v_* \rangle} u_x^{\alpha_x} u_y^{\alpha_y} u_z^{\alpha_z}$$

by (2), so $\pi \circ \eta|_{V_t'} = \varphi \circ \psi$. Let $F := (\mu = 0) \subset V_t'$. Since $V_t \to Y_t$ is unramified along $(\lambda = 0)$, ψ maps F onto $E \cap Y_t$ with ramification index r, and $\operatorname{mult}_F(\psi^* g) = r \operatorname{mult}_E g$ for any rational function g on Y.

The form $\omega := \mathrm{Res}(dx \wedge dy \wedge dz \wedge dt/f)$ generates ω_{W_N} and is R-semi-invariant, so $\omega^{\otimes r}$ is the pullback of a generator of $\omega^{[r]}_{W_N/R}$, as π is quasi-étale. Since

$$\eta^*(dx \wedge dy \wedge dz \wedge dt) = \pm\frac{2}{k}\mu^{\frac{3N+2}{k}-1} du_x \wedge du_y \wedge du_z \wedge d\mu$$

and $f \circ \eta = \mu^{2N/k}(u_x u_y + u_z^2 + 1)$, and since η is étale over $(t \neq 0)$, where the residue commutes with the pullback, we obtain on $V_t' \cap (\mu \neq 0)$, hence as rational forms on V_t',

$$\eta^*\omega = \pm\frac{2}{k}\mu^{\frac{N+2}{k}-1}\,\mathrm{Res}\Big(\frac{du_x \wedge du_y \wedge du_z \wedge d\mu}{u_x u_y + u_z^2 + 1}\Big),$$

and the residue on the right generates $\omega_{V_t'}$. Thus $K_{V_t'} = (\pi \circ \eta)^* K_{W_N/R} + (\frac{N+2}{k} - 1)F$ near the generic point of F. On the other hand, $K_Y = \varphi^* K_{W_N/R} + (a(E, W_N/R, 0) - 1)E$, and $K_{V_t'} = \psi^* K_Y + (r-1)F$ near the generic point of F by the Hurwitz formula. Thus

$$\frac{N+2}{k} - 1 = r\big(a(E, W_N/R, 0) - 1\big) + r - 1, \quad \text{that is,} \quad a(E, W_N/R, 0) = \frac{N+2}{kr} = v_{*z} + v_{*t}.$$

Let H be an R-semi-invariant polynomial not divisible by f. Then

$$\mathrm{mult}_E(H^r) = \frac{1}{r}\,\mathrm{mult}_F(H^r \circ \eta) = \mathrm{mult}_F(H \circ \eta),$$

and

$$H \circ \eta = \mu^{\mathrm{wt}_w(H)}\big(\,\mathrm{in}_w(H)(u_x, u_y, u_z, 1) + \mu H'\big)$$

for a polynomial H'. Thus

$$\mathrm{mult}_F(H \circ \eta) \geq \mathrm{wt}_w(H) = r\,\mathrm{wt}_{v_*}(H),$$

with equality if $\mathrm{in}_w(H)(u_x, u_y, u_z, 1)$ does not vanish identically on $(u_x u_y + u_z^2 + 1 = 0) \subset \mathbb{A}^3$. If it vanishes there, then

$$\mathrm{in}_w(H) \circ \eta = \mu^{\mathrm{wt}_w(H)}\,\mathrm{in}_w(H)(u_x, u_y, u_z, 1)$$

vanishes on V_t', so $\mathrm{in}_w(H)$ vanishes on $W_N \cap (t \neq 0)$, which is dense in W_N, and f divides $\mathrm{in}_w(H) = \mathrm{in}_{v_*}(H)$. This proves (3).

We prove (4). Fix $\xi \in \Xi$. The group $G/R \subset T_{N_R}$ acts on U_{σ_ξ, N_R} by translations, so Y_ξ is G/R-invariant. The group $\tilde{G}_\xi$ is an extension of G/R by K_ξ, hence finite abelian, and acts on $U_{\sigma_\xi, N_\xi} = \mathbb{A}^4$ by translations of T_{N_ξ}, that is, diagonally. Since V_ξ is the preimage of the G/R-invariant subset S_ξ, V_ξ is $\tilde{G}_\xi$-invariant, and

$$V_\xi/\tilde{G}_\xi = (V_\xi/K_\xi)/(G/R) \cong Y_\xi/(G/R),$$

which is an open subset of $Y/(G/R)$ as Y_ξ is a G/R-invariant open subset of Y. □

Proof of Theorem 7.8. The proof is long, so we divide it into several steps. Let $f := xy + z^2 + t^N$ and $k := \gcd(2, N)$.

Step 1. In this step we reduce to marks that are polynomials with positive labels, and we set up an induction on N.

The partial derivatives $y, x, 2z, Nt^{N-1}$ of f vanish only at 0, so W_N has an isolated singularity at 0 and is normal by Serre's criterion. For general $a, b, c \in \mathbb{C}$, the hyperplane section $t = ax + by + cz$ of W_N is $(xy + z^2 + (ax + by + cz)^N = 0) \subset \mathbb{A}^3$, an A_1 point at 0 as its quadratic part, $xy + z^2$ if $N \geq 3$ and $xy + z^2 + (ax + by + cz)^2$ if $N = 2$, is non-degenerate for general a, b, c. Thus $(W_N \ni 0)$ is an isolated cDV point, and W_N is terminal near 0 [KM98, Corollary 5.38]. Let h be a mark on $(W_N \ni 0)$. Then $h = A/A'$ for regular functions A, A' on W_N with $A'(0) \neq 0$. The function $U := \prod_{g \in G} g^* A'$ is G-invariant with $U(0) \neq 0$, and $hU = A \prod_{g \neq 1} g^* A'$ is a regular function on W_N with the character of h, hence the restriction of a G-semi-invariant polynomial of that character. Since $\mathrm{div}(hU) = \mathrm{div}(h)$ near 0, replacing each mark h with hU does not change D near 0. Removing the marks with label 0 and the units does not change D either. Thus we may

assume that $D = \sum_j c_j \operatorname{div}(h_j)$, where the $c_j \in \Gamma$ are positive and the h_j are G-semi-invariant polynomials with $h_j(0) = 0$ (possibly with no marks).

We prove the theorem by strong induction on N. If $N = 2$, then $(W_2 \ni 0)$ is a node, and the theorem is Theorem 7.7. Let $N \geq 3$, and assume the theorem for all exponents in $\{2, \dots, N-1\}$ and all DCC sets. Suppose that there exist finite diagonal abelian groups G_i preserving the equation of W_N up to a character and acting faithfully on W_N, and labelled boundaries $D_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$ on $(W_N \ni 0)$ with labels in Γ, where the $c_{ij} \in \Gamma$ are positive and the h_{ij} are G_i-semi-invariant polynomials with $h_{ij}(0) = 0$, such that (W_N, D_i) is klt near 0 and the numbers $a_i := \operatorname{mld}((W_N \ni 0, D_i)/G_i)$ are positive and strictly increasing. We derive a contradiction.

Step 2. In this step we treat the groups G_i containing a non-identity element that acts non-trivially on exactly one coordinate.

For $g \in G_i$, write

$$g \cdot (x, y, z, t) = (\chi_x(g)x, \chi_y(g)y, \chi_z(g)z, \chi_t(g)t).$$

Since g preserves f up to a character,

$$\chi_x(g)\chi_y(g) = \chi_z(g)^2 = \chi_t(g)^N. \tag{A.15}$$

Thus a non-identity element acting non-trivially on exactly one coordinate acts non-trivially on z or on t.

Assume that G_i contains an element ι acting non-trivially only on z. Then $\chi_z(\iota) = -1$ by (A.15). Since $z^2 = -xy - t^N$ on W_N, the quotient of W_N by $\langle\iota\rangle$ is $\mathbb{A}^3$ with the coordinates x, y, t. Let Δ_i be the $\mathbb{R}$-divisor on $\mathbb{A}^3$ such that $K_{W_N} + D_i$ is the pullback of $K_{\mathbb{A}^3} + \Delta_i$. Then $\Delta_i \in \Psi(\Gamma_+ \cap [0,1])$ [Liu26, Lemma 2.10(5)], $(\mathbb{A}^3, \Delta_i)$ is klt near 0 [KM98, Proposition 5.20], $G_i/\langle\iota\rangle$ acts faithfully on $\mathbb{A}^3$, as an element of G_i acting trivially on x, y, t belongs to $\langle\iota\rangle$ by (A.15), and $a_i = \operatorname{mld}((\mathbb{A}^3 \ni 0, \Delta_i)/(G_i/\langle\iota\rangle))$ [Liu26, Lemma 2.6(5)]. Since $\Psi(\Gamma_+ \cap [0,1])$ is a DCC set [Liu26, Lemma 2.10(1)], the numbers a_i for such G_i satisfy the ACC by the last sentence of Theorem 2.8(1).

Assume that G_i contains a non-identity element acting non-trivially only on t. By (A.15), the subgroup of elements of G_i acting trivially on x, y, z is the group μ_e acting on t, for an integer $e \geq 2$ dividing N. The morphism

$$W_N \to W_{N/e} = (xy + z^2 + t'^{N/e} = 0), \quad (x, y, z, t) \to (x, y, z, t^e),$$

is the quotient by μ_e, and the coordinate ring of W_N is the direct sum of t^l times the coordinate ring of $W_{N/e}$, $0 \leq l < e$, which are the eigenspaces of μ_e. Thus $h_{ij} = t^{l_j} H_j$ with $0 \leq l_j < e$ and H_j a regular function on $W_{N/e}$, semi-invariant for G_i/μ_e. Let $\sigma := \sum_j c_{ij} l_j$. The divisor $\operatorname{div}(t)$ on W_N is the prime divisor $P := (t = xy + z^2 = 0)$ and $\operatorname{mult}_P h_{ij} \geq l_j$, so $\sigma < 1$ as (W_N, D_i) is klt. The quotient morphism is branched only along P, with ramification index e, and $\operatorname{div}(t')$ pulls back to eP. By the Hurwitz formula, $K_{W_N} + D_i$ is the pullback of $K_{W_{N/e}} + \Delta_i$, where

$$\Delta_i := \sum_j c_{ij} \operatorname{div}(H_j) + \left(1 - \frac{1-\sigma}{e}\right) \operatorname{div}(t').$$

As $\sigma \in \Gamma_+ \cap [0,1)$, the labels of Δ_i belong to the DCC set $\Gamma_1 := \Gamma \cup \Psi(\Gamma_+ \cap [0,1])$ [Liu26, Lemma 2.10(1)], $(W_{N/e}, \Delta_i)$ is klt near 0 [KM98, Proposition 5.20], G_i/μ_e is a finite diagonal abelian group preserving the equation of $W_{N/e}$ up to a character and acting faithfully on $W_{N/e}$, and $a_i = \operatorname{mld}((W_{N/e} \ni 0, \Delta_i)/(G_i/\mu_e))$ [Liu26, Lemma 2.6(5)]. If $N/e = 1$, then $W_1 \cong \mathbb{A}^3$ and $\Delta_i \in (\Gamma_1)_+ \cap [0,1]$, a DCC set [Liu26, Lemma 2.10(1)(5)], so the ACC for such a_i follows from the last sentence of Theorem 2.8(1). If $N/e = 2$, it follows from Theorem 7.7 with Γ_1, and if $3 \leq N/e \leq N-1$, from the induction hypothesis with Γ_1. Since N has finitely many divisors, the numbers a_i for such G_i belong to a finite union of ACC sets.

Thus, passing to a subsequence, we may assume that no non-identity element of G_i acts non-trivially on exactly one coordinate.

Step 3. In this step we show that $|G_i| \to +\infty$ after passing to a subsequence.

Otherwise, passing to a subsequence, there exists m with $|G_i| \le m$ for any i, so the G_i are subgroups of the finite group $\mu_{m!}^4$, and passing to a subsequence again, $G_i = G$ is fixed. Let $Q := W_N/G$, let $o \in Q$ be the image of 0, and let B_i be the $\mathbb{R}$-divisor on Q such that $K_{W_N} + D_i$ is the pullback of $K_Q + B_i$. Then $a_i = \mathrm{mld}(Q \ni o, B_i)$ [Liu26, Definition 2.5]. Since the h_{ij} are polynomials, $K_Q + B_i$ is $\mathbb{R}$-Cartier and $B_i \ge 0$ on all of Q [Liu26, Lemma 2.6(0)], and the coefficients of B_i belong to

$$\left\{1 - \frac{1-b}{e} \;\middle|\; b \in \Gamma_+,\ 1 \le e \le |G|\right\} \setminus \{0\}$$

[Liu26, Lemma 2.6(4)], a DCC set of positive real numbers as a finite union of images of the DCC set Γ_+ [Liu26, Lemma 2.10(1)] under increasing affine maps. Thus (Q, B_i) is a pair in the sense of [Kaw24], whose definition contains no lc condition, and this contradicts [Kaw24, Theorem 1.1] on the fixed normal threefold Q.

Step 4. In this step we determine the Hausdorff limit of the groups G_i.

By **Step 3** and Lemma 4.6, passing to a subsequence, the G_i converge to a closed subgroup $H \subset \mathbb{T}^4$ with $\dim H \ge 1$ in the Hausdorff distance, and $G_i \subset H$ for $i \gg 0$. Let $\mathfrak{h} \subset \mathbb{R}^4$ be the linear subspace with $\rho(\mathfrak{h}) = H^0$, where $\rho : \mathbb{R}^4 \to \mathbb{T}^4$ is the quotient map [Liu26, Definition 2.15], [Liu26, Lemma 4.3(1)]. By (A.15), the G_i are contained in the closed subgroup

$$\{\gamma \in \mathbb{T}^4 : \gamma_x + \gamma_y = 2\gamma_z = N\gamma_t\},$$

so H is contained in it, and

$$\mathfrak{h} \subset \{v \in \mathbb{R}^4 : v_x + v_y = 2v_z = Nv_t\} = \mathbb{R}(1,-1,0,0) + \mathbb{R}(N,N,N,2).$$

If $\dim H^0 = 2$, then $(N,N,N,2) \in \mathfrak{h}$, and Lemma 4.7 implies $a_i \to 0$, which contradicts $a_i \ge a_1 > 0$. Thus $\dim H^0 = 1$. By Lemma 4.6, $H^0 = \rho(\mathbb{R}w)$ for a primitive $w \in \mathbb{Z}^4$, $R_i := G_i \cap \mathbf{T}_{(w)}$ is cyclic of order $r_i \to +\infty$, so $R_i = \mu_{r_i}(w)$, and $[G_i : R_i] = |H/H^0| =: e_0$ for $i \gg 0$. Since $w \in \mathfrak{h}$, $w_x + w_y = 2w_z = Nw_t$. If N is odd, then w_t is even. Thus $s := kw_t/2$ is an integer, and

$$w = \left(u,\ \frac{2Ns}{k} - u,\ \frac{Ns}{k},\ \frac{2s}{k}\right), \quad u := w_x \in \mathbb{Z}.$$

If $s = 0$, then $w = \pm(1,-1,0,0)$. Since $f = xy - g$ with $g := -z^2 - t^N \in (z,t)^2$, and $R_i = G_i \cap \mathbf{T}_{(1,-1,0,0)} = \mu_{r_i}(1,-1,0,0)$ with $r_i \to +\infty$ and $[G_i : R_i] = |H/H^0|$ for $i \gg 0$ by Lemma 4.6, Theorem 8.3 shows that the a_i do not form a strictly increasing sequence of positive numbers, a contradiction.

Thus $s \ne 0$, and replacing w by $-w$ we may assume $s > 0$. If $u > 0$ and $\frac{2Ns}{k} - u > 0$, then all entries of $w \in \mathfrak{h}$ are positive, and Lemma 4.7 implies $a_i \to 0$, a contradiction. If $u = 0$, then $w = s(0, \frac{2N}{k}, \frac{N}{k}, \frac{2}{k})$, and $s = 1$ since w is primitive and $\gcd(\frac{N}{k}, \frac{2}{k}) = 1$. Otherwise, one of u and $\frac{2Ns}{k} - u$ is not positive. The automorphism $(x,y,z,t) \to (y,x,z,t)$ preserves f, and replacing G_i and the h_{ij} by their images under it does not change the a_i. Thus, possibly interchanging x and y, it remains to treat

$$w = \left(0, \frac{2N}{k}, \frac{N}{k}, \frac{2}{k}\right) \quad \text{and} \quad w = \left(-p,\ q,\ \frac{Ns}{k},\ \frac{2s}{k}\right), \quad q := p + \frac{2Ns}{k},$$

where p and s are positive integers, coprime since w is primitive. In both cases, $R_i = \mu_{r_i}(w)$, $r_i \to +\infty$, and $N_{R_i} = \mathbb{Z}^4 + \mathbb{Z}\frac{w}{r_i}$.

Step 5. In this step we show the following. Let $\Lambda \subset [0,1)$ be a DCC set containing $\Gamma_+ \cap [0,1)$. Assume that, for any i, there exists a vector $v_{*,i} \in N_{R_i}$, where R_i and $r_i = |R_i|$ are as in **Step 4**, such that $R_i \subset G_i$ and $v_{*,i}$ satisfy the hypotheses of Lemma A.4, and let $\varphi_i : Y_i \to W_N/R_i$ and E_i be the morphism and the exceptional prime divisor given by Lemma A.4. Let $\pi_i : W_N \to W_N/R_i$ be the quotient morphism and B_i' the $\mathbb{R}$-divisor on W_N/R_i with $K_{W_N} + D_i = \pi_i^*(K_{W_N/R_i} + B_i')$. If

$$\beta_i := 1 - a(E_i, W_N/R_i, B_i') \in \Lambda$$

for any i, then we obtain a contradiction.

Since π_i is quasi-étale by Lemma A.4(1), $K_{W_N} = \pi_i^* K_{W_N/R_i}$, so

$$B_i' = \sum_j \frac{c_{ij}}{r_i} \operatorname{div}(h_{ij}^{r_i}),$$

and, after replacing W_N by a G_i-invariant open neighborhood of 0 on which (W_N, D_i) is klt (Definition-Proposition 4.10 is a statement on a neighborhood of the image of 0 in W_N/G_i), the coefficients of B_i' are coefficients of D_i, which belong to $\Gamma_+ \cap [0,1]$ [Liu26, Lemma 2.10(5)] and are less than one since (W_N, D_i) is klt. Let $B_{Y_i} := \varphi_{i*}^{-1} B_i' + \beta_i E_i$. As E_i is the only exceptional prime divisor of φ_i, $K_{Y_i} + B_{Y_i} = \varphi_i^*(K_{W_N/R_i} + B_i')$, $B_{Y_i} \geq 0$, and $B_{Y_i} \in \Lambda$. Moreover, (Y_i, B_{Y_i}) is klt, since $(W_N/R_i, B_i')$ is klt [KM98, Proposition 5.20] and log discrepancies do not change under crepant pullback [KM98, Lemma 2.30].

Let $Q_i := W_N/G_i$, let $o_i \in Q_i$ be the image of 0, and let B_{Q_i} be the $\mathbb{R}$-divisor on Q_i such that $K_{W_N} + D_i$ is the pullback of $K_{Q_i} + B_{Q_i}$, so that $a_i = \operatorname{mld}(Q_i \ni o_i, B_{Q_i})$ [Liu26, Definition 2.5]. Let $Z_i := Y_i/(G_i/R_i)$, let B_{Z_i} be the $\mathbb{R}$-divisor on Z_i such that $K_{Y_i} + B_{Y_i}$ is the pullback of $K_{Z_i} + B_{Z_i}$, and let $\bar{\varphi}_i : Z_i \to Q_i$ be the morphism induced by φ_i. Since $(W_N, D_i)/G_i = ((W_N, D_i)/R_i)/(G_i/R_i)$ [Liu26, Lemma 2.6(5)] and φ_i is G_i/R_i-equivariant, $K_{Z_i} + B_{Z_i} = \bar{\varphi}_i^*(K_{Q_i} + B_{Q_i})$. The morphism $\bar{\varphi}_i$ is birational, and it is projective since φ_i is projective and G_i/R_i-equivariant: the norm of a φ_i-ample line bundle is a G_i/R_i-linearized φ_i-ample line bundle, and a power of it descends to Z_i. Thus $(\bar{\varphi}_i, B_{Z_i})$ is a crepant model of $(Q_i \ni o_i, B_{Q_i})$, and (Z_i, B_{Z_i}) is klt [KM98, Proposition 5.20].

By Lemma A.4(4), Z_i is covered by open subsets $U \cong V_\xi/\tilde{G}_{i,\xi}$, where V_ξ is a smooth threefold of (A.13) and $\tilde{G}_{i,\xi}$ is a finite abelian group, and the quotient morphism $V_\xi \to U$ is the composite of $V_\xi \to Y_{i,\xi}$ and $Y_{i,\xi} \to U$. Since $V_\xi \to Y_{i,\xi}$ is quasi-étale by Lemma A.4(2), the $\mathbb{R}$-divisor B_V such that $K_{V_\xi} + B_V$ is the pullback of $K_U + B_{Z_i}|_U$ is the pullback of B_{Y_i}, so $B_V \geq 0$ and $B_V \in \Lambda$. Thus, by Definition-Proposition 4.10 for the DCC set Λ, the positive numbers a_i belong to a set satisfying the ACC, which contradicts the strict increase of a_i.

Step 6. In this step we exclude $w = (0, \frac{2N}{k}, \frac{N}{k}, \frac{2}{k})$.

Passing to a subsequence, we may assume $r_i \geq \frac{N+2}{k}$. Let $v_{*,i} := \frac{1}{kr_i}(0, 2N, N, 2) = \frac{w}{r_i}$, which is primitive in $N_{R_i} = \mathbb{Z}^4 + \mathbb{Z}\frac{w}{r_i}$ as w is primitive in $\mathbb{Z}^4$. By **Steps 2 and 4**, $R_i \subset G_i$ and $v_{*,i}$ satisfy the hypotheses of Lemma A.4 in case (b). With the notation of **Step 5**, Lemma A.4(3) implies

$$a(E_i, W_N/R_i, B_i') = \frac{N+2}{kr_i} - \sum_j \frac{c_{ij}}{r_i} \operatorname{mult}_{E_i}(h_{ij}^{r_i}) = \frac{1}{r_i}\Big(\frac{N+2}{k} - \sigma_i\Big),$$

where $\sigma_i := \sum_j c_{ij} \operatorname{mult}_{E_i}(h_{ij}^{r_i})$. Since the $\operatorname{mult}_{E_i}(h_{ij}^{r_i})$ are non-negative integers, $\sigma_i \in \Gamma_+$, and $\sigma_i < \frac{N+2}{k}$ since $(W_N/R_i, B_i')$ is klt. Thus $\beta_i = 1 - \frac{1}{r_i}(\frac{N+2}{k} - \sigma_i)$ belongs to the set

$$\Lambda_0 := \Big\{ 1 - \frac{1}{r}\Big(\frac{N+2}{k} - \sigma\Big) \;\Big|\; \sigma \in \Gamma_+,\ \sigma < \frac{N+2}{k},\ r \in \mathbb{Z},\ r \geq \frac{N+2}{k} \Big\} \subset [0,1).$$

If $\beta^{(1)} > \beta^{(2)} > \cdots$ belong to Λ_0, with integers $r^{(n)}$ and $\sigma^{(n)} \in \Gamma_+$, then $\frac{1}{r^{(n)}} \cdot \frac{N+2}{k} \geq 1 - \beta^{(1)} > 0$ bounds $r^{(n)}$, and for a fixed value of $r^{(n)}$ the $\sigma^{(n)}$ strictly decrease, which contradicts the DCC of Γ_+ [Liu26, Lemma 2.10(1)]. Thus Λ_0 is a DCC set, and **Step 5** with $\Lambda := (\Gamma_+ \cap [0,1)) \cup \Lambda_0$ yields a contradiction.

Step 7. In this step, for $w = (-p, q, \frac{Ns}{k}, \frac{2s}{k})$, we construct a vector of N_{R_i} at which the orders of all marks are computed by the principal blocks.

Fix i, and write G, R, r, D, c_j, and h_j for G_i, R_i, r_i, D_i, c_{ij}, and h_{ij}. The monomials not divisible by xy form a basis of $\mathbb{C}[x,y,z,t]/(f)$, and the representative of a polynomial in terms of the monomials not divisible by xy is obtained by replacing xy with $-z^2 - t^N$. Since xy, z^2, and t^N are G-semi-invariant of the character of f and have $\mathbf{T}_{(w)}$-weight $\frac{2Ns}{k}$, the representative of h_j is a G-semi-invariant polynomial with the character of h_j and with restriction $h_j|_{W_N}$ to

W_N. Possibly replacing each h_j with the representative of h_j, we may assume that no monomial of any h_j is divisible by xy. Let m_j be the least degree of a monomial of h_j. Then $m_j \geq 1$, as $h_j(0) = 0$.

Since $N \geq 3$, the h_j have no monomial divisible by xy and vanish at 0, and (W_N, D) is klt near 0, Lemma A.1(3) implies

$$\sum_j c_j m_j < 2. \tag{A.16}$$

If $\Gamma \not\subset \{0\}$, let $\gamma := \min(\Gamma\backslash\{0\})$ and $m_0 := \lceil 2/\gamma \rceil$. Otherwise no marks exist, and we let $m_0 := 1$. Since $c_j \geq \gamma$, the last sentence of Lemma A.1(3) implies that there exist fewer than m_0 marks and that $m_j < m_0$ for any j.

We use Set-up 5.1 for $n = 4$, W_N, G, w, R, r, e_0, Γ, and D. Let $\kappa_j := \kappa(h_j)$, the $\mathbf{T}_{(w)}$-weight of a fixed monomial of least degree of h_j, and h_j^0 the principal block of h_j. As $\sum_l w_l = \frac{(3N+2)s}{k}$ and $\kappa(f) = \frac{2Ns}{k}$, the adjusted mass of D along $\mathbf{T}_{(w)}$ is

$$\delta = \sum_j c_j \kappa_j - \frac{(N+2)s}{k}.$$

Since the entries of w have absolute value at most q, the $\mathbf{T}_{(w)}$-weight of a monomial of degree m has absolute value at most qm, so $|\kappa_j| \leq qm_j < qm_0$. As h_j is R-semi-invariant and $R = \mu_r(w)$, the $\mathbf{T}_{(w)}$-weights of the monomials of h_j are congruent modulo r, so any monomial of $h_j - h_j^0$ has weight $\kappa_j + \ell r$ for a non-zero integer ℓ.

For $\frac{1}{q} \leq \theta \leq \frac{1}{p}$, let

$$v(\theta) := (1, -1, 0, 0) + \theta w = \Big(1 - p\theta,\ q\theta - 1,\ \frac{Ns\theta}{k},\ \frac{2s\theta}{k}\Big).$$

The entries of $v(\theta)$ are non-negative, positive for $\frac{1}{q} < \theta < \frac{1}{p}$, and at most $\max(1, \frac{2Ns}{kp})$, and $v_x(\theta) + v_y(\theta) = 2v_z(\theta) = Nv_t(\theta)$. The $v(\theta)$-weight of a monomial x^α of $\mathbf{T}_{(w)}$-weight κ is $\alpha_x - \alpha_y + \kappa\theta$. Let d_j be the minimum of $\alpha_x - \alpha_y$ over the monomials x^α of h_j^0, and let $\rho_j(\theta) := d_j + \kappa_j\theta$, so that $\mathrm{wt}_{v(\theta)}(h_j^0) = \rho_j(\theta)$. If $\alpha_y > 0$ for a monomial x^α of h_j^0, then $\alpha_x = 0$ and $q\alpha_y \leq \kappa_j < qm_0$, and the fixed monomial of least degree has $\alpha_x - \alpha_y \leq m_j$. Thus $|d_j| < m_0$. Moreover, evaluating the weight of the fixed monomial of least degree,

$$0 \leq \rho_j(\theta) \leq \rho_0 := m_0 \max\Big(1, \frac{2Ns}{kp}\Big) \quad \text{for} \quad \frac{1}{q} \leq \theta \leq \frac{1}{p}. \tag{A.17}$$

Let $\delta_0 := \frac{1}{p} - \frac{1}{q} > 0$, fix an integer $\ell_0 > 2\rho_0 + 1$, and fix an integer r_0 such that any $r \geq r_0$ satisfies

$$r \geq 2qm_0, \quad \frac{\ell_0}{r} < \frac{\delta_0}{2}, \quad \frac{\delta_0}{2}(r - qm_0) > \rho_0, \quad \frac{(N+2)s}{kq} + \Big(2q + \frac{(N+2)s}{k}\Big)\frac{\ell_0}{r} < 1, \tag{A.18}$$

which is possible since $q > \frac{2Ns}{k} \geq \frac{(N+2)s}{k}$. The numbers m_0, ρ_0, ℓ_0, and r_0 depend only on N, p, s, and Γ. Assume $r \geq r_0$, and let

$$\theta_- := \frac{1}{r}\Big(\Big\lfloor \frac{r}{q} \Big\rfloor + \ell_0\Big), \qquad \theta_+ := \frac{1}{r}\Big(\Big\lceil \frac{r}{p} \Big\rceil - \ell_0\Big).$$

Then $\frac{\ell_0 - 1}{r} \leq \theta_- - \frac{1}{q} \leq \frac{\ell_0}{r}$ and $\frac{\ell_0 - 1}{r} \leq \frac{1}{p} - \theta_+ \leq \frac{\ell_0}{r}$, while $\frac{1}{p} - \theta_-$ and $\theta_+ - \frac{1}{q}$ are at least $\delta_0 - \frac{\ell_0}{r} > \frac{\delta_0}{2}$. In particular, $\frac{1}{q} < \theta_\pm < \frac{1}{p}$.

We show that for $\theta \in \{\theta_-, \theta_+\}$ and any j,

$$\mathrm{wt}_{v(\theta)}(h_j) = \rho_j(\theta), \quad \text{and } f \text{ does not divide } \mathrm{in}_{v(\theta)}(h_j). \tag{A.19}$$

Let x^α be a monomial of $h_j - h_j^0$, of $\mathbf{T}_{(w)}$-weight $\kappa' = \kappa_j + \ell r$ with $\ell \neq 0$. Assume $\ell > 0$. Then $\kappa' \geq r - qm_0 > 0$. If $\alpha_y = 0$, then the $v(\theta)$-weight of x^α is $\alpha_x + \kappa'\theta \geq (\theta - \frac{1}{q})\kappa'$, and if $\alpha_x = 0$,

then $q\alpha_y \le \kappa'$ and the weight is $\kappa'\theta - \alpha_y \ge (\theta - \frac{1}{q})\kappa'$. Assume $\ell < 0$. Then $\kappa' \le -(r - qm_0) < 0$, so $\alpha_x > 0$, $\alpha_y = 0$, and $p\alpha_x \ge -\kappa'$, and the weight is $\alpha_x + \kappa'\theta \ge (\frac{1}{p} - \theta)(-\kappa')$. Since $r - qm_0 \ge \frac{r}{2}$ by (A.18), in all cases the weight is at least

$$\min\Big(\frac{\ell_0 - 1}{2},\ \frac{\delta_0}{2}(r - qm_0)\Big) > \rho_0 \ge \rho_j(\theta)$$

by (A.17) and (A.18). Thus $\mathrm{wt}_{v(\theta)}(h_j) = \rho_j(\theta)$, and $\mathrm{in}_{v(\theta)}(h_j)$ is a non-zero sum of terms of h_j^0. No monomial of $\mathrm{in}_{v(\theta)}(h_j)$ is divisible by xy, so the representative of $\mathrm{in}_{v(\theta)}(h_j)$ in terms of the monomials not divisible by xy is non-zero, and f does not divide $\mathrm{in}_{v(\theta)}(h_j)$. This proves (A.19).

Step 8. In this step we exclude $w = (-p, q, \frac{Ns}{k}, \frac{2s}{k})$.

Keep the notation of **Step 7** and assume $r \ge r_0$. Let $\theta := \theta_-$ if $\delta \le 0$ and $\theta := \theta_+$ if $\delta > 0$. Since $r\theta \in \mathbb{Z}$ and $N_R = \mathbb{Z}^4 + \mathbb{Z}\frac{w}{r}$, $v(\theta) \in N_R$. Write $v(\theta) = ev_*$ with v_* primitive in N_R and e a positive integer. The monomial $x^q y^p$ has $\mathbf{T}_{(w)}$-weight 0, so $(q, p, 0, 0) \in M_R$, and $\langle (q, p, 0, 0), v(\theta)\rangle = q - p = \frac{2Ns}{k}$. Thus e divides $\frac{2Ns}{k}$. By **Step 2** and since $\frac{1}{q} < \theta < \frac{1}{p}$, $R \subset G$ and v_* satisfy the hypotheses of Lemma A.4 in case (a). With the notation of **Step 5** without the index i, Lemma A.4(3) and (A.19) imply $\frac{1}{r}\,\mathrm{mult}_E(h_j^r) = \frac{1}{e}\rho_j(\theta)$ and

$$a(E, W_N/R, 0) = \frac{1}{e}(v_z(\theta) + v_t(\theta)) = \frac{(N+2)s\theta}{ke},$$

so

$$a(E, W_N/R, B') = \frac{1}{e}\Big(\frac{(N+2)s\theta}{k} - \sum_j c_j\rho_j(\theta)\Big) = -\frac{1}{e}\Big(\delta\theta + \sum_j c_j d_j\Big). \tag{A.20}$$

The number

$$-\delta\theta' - \sum_j c_j d_j = \frac{(N+2)s\theta'}{k} - \sum_j c_j\rho_j(\theta')$$

is affine in $\theta' \in [\frac{1}{q}, \frac{1}{p}]$, and by (A.17) it is at most $\frac{(N+2)s}{kq}$ for $\theta' = \frac{1}{q}$. If $\delta \le 0$, it is non-decreasing in θ', so it is at most $\frac{(N+2)s}{kq} + |\delta|\frac{\ell_0}{r}$ for $\theta' = \theta_-$. If $\delta > 0$, it is decreasing in θ', so for $\theta' = \theta_+$ it exceeds the number for $\theta' = \frac{1}{p}$ by at most $\delta\frac{\ell_0}{r}$, and it is at most $\frac{(N+2)s}{kq} + \delta\frac{\ell_0}{r}$. By (A.16),

$$|\delta| \le q\sum_j c_j m_j + \frac{(N+2)s}{k} < 2q + \frac{(N+2)s}{k}.$$

Thus (A.18) and (A.20) imply $a(E, W_N/R, B') < \frac{1}{e} \le 1$, and $a(E, W_N/R, B') > 0$ since $(W_N/R, B')$ is klt. Hence

$$\beta = 1 - \frac{1}{e}\Big(\frac{(N+2)s\theta}{k} - \sum_j c_j\rho_j(\theta)\Big) \in (0, 1).$$

Let $\Lambda_{p,s}$ be the set of the numbers β defined by (A.20) and $\beta = 1 - a(E, W_N/R, B')$ for all G, R, and D as in **Step 7** with $r \ge r_0$, for the fixed N, p, s, and Γ. We show that $\Lambda_{p,s}$ is a DCC set. Suppose that $\beta^{(1)} > \beta^{(2)} > \cdots$ belong to $\Lambda_{p,s}$. Each $\beta^{(n)}$ is determined by $r^{(n)}$, the sign of $\delta^{(n)}$, $e^{(n)}$, a divisor of $\frac{2Ns}{k}$, fewer than m_0 labels $c_j^{(n)} \in \Gamma$, and integers $\kappa_j^{(n)}, d_j^{(n)}$ with $|\kappa_j^{(n)}| < qm_0$ and $|d_j^{(n)}| < m_0$. Passing to a subsequence, we may assume that the sign of $\delta^{(n)}$, $e^{(n)} = e$, the number of labels, and the integers $\kappa_j^{(n)} = \kappa_j$ and $d_j^{(n)} = d_j$ do not depend on n, and, as Γ is a DCC set, that $c_j^{(n)}$ is non-decreasing in n for any j. If $r^{(n)}$ is bounded, then we may also assume that $r^{(n)}$, hence $\theta^{(n)}$, does not depend on n. Otherwise, we may assume $r^{(n)} \to +\infty$. Then $\theta^{(n)} \to \frac{1}{q}$ from

above if $\delta^{(n)} \le 0$, and $\theta^{(n)} \to \frac{1}{p}$ from below if $\delta^{(n)} > 0$, and passing to a subsequence, $\theta^{(n)}$ is decreasing in the first case and increasing in the second. In all cases, (A.20) implies

$$e\big(\beta^{(n)} - \beta^{(n+1)}\big) = -\delta^{(n+1)}\big(\theta^{(n+1)} - \theta^{(n)}\big) - \sum_j \big(c_j^{(n+1)} - c_j^{(n)}\big)\rho_j(\theta^{(n)}) \le 0,$$

as $\delta^{(n+1)}(\theta^{(n+1)} - \theta^{(n)}) \ge 0$ and $\rho_j(\theta^{(n)}) \ge 0$. This contradicts $\beta^{(n)} > \beta^{(n+1)}$, so $\Lambda_{p,s}$ is a DCC set.

Passing to the subsequence with $r_i \ge r_0$, **Step 5** with $\Lambda := (\Gamma_+ \cap [0,1)) \cup \Lambda_{p,s}$ yields a contradiction. By **Step 4**, this completes the induction and the proof. □

A.4. **Canonical extractions and bounded groups.** In this subsection, we prove Theorem 7.9 and Proposition 7.10.

Proof of Theorem 7.9. The proof is long, so we divide it into several steps. Let

$$C_{u,v} := \max(1, |u|, |2v+u|, v), \qquad w_L := \Big(1+\frac{uL}{r}, 1-\frac{(2v+u)L}{r}, 1-\frac{vL}{r}, \frac{vL}{r}\Big) = (1,1,1,0)+\frac{L}{r}w,$$

and write $w_{L,x}, w_{L,y}, w_{L,z}, w_{L,t}$ for the entries of w_L. Marks that are units or have label zero do not contribute to D near 0, so we omit them. Thus $c_j > 0$ and $h_j(0) = 0$ for any j, and no mark is left if $\Gamma \subset \{0\}$.

Step 1. In this step we bound the number of marks, their degrees, and their $\mathbf{T}$-weights.

Since $N \ge 3$, the lowest-degree part of $xy + z^2 + t^N$ is $xy + z^2$, which defines an integral quadric not contained in any coordinate hyperplane. By Lemma A.1(1), a non-zero polynomial without monomials divisible by xy is not divisible by $xy + z^2$. Let m_j be the least total degree of a monomial of h_j. Since (W_N, D) is klt near 0, Lemma A.1(3) gives

$$\sum_j c_j m_j < 2. \tag{A.21}$$

Since $c_j \ge \gamma$, Lemma A.1(3) also shows that the number of marks and each m_j are less than $2/\gamma \le M$. A monomial of least degree of h_j belongs to h_j^0, so $I_j \le m_j \le M$ and $|\kappa_j| \le C_{u,v} m_j \le C_{u,v} M$.

The group R consists of the elements $(\zeta^u, \zeta^{-2v-u}, \zeta^{-v}, \zeta^v)$ with $\zeta^r = 1$. Since $\max_i |w_i| \le C_{u,v}$ and each h_j has a monomial of degree $m_j \le M$, Lemma 5.3 shows that the $\mathbf{T}$-weights of the monomials of h_j are congruent to κ_j modulo r, and that if $r > 2C_{u,v}M$, then κ_j and h_j^0 do not depend on the choice of the monomial of least degree. Multiplication by an invariant unit multiplies the lowest-degree part of h_j by a non-zero constant, since the representative modulo $xy + z^2 + t^N$ in $\mathbb{C}[[x,y,z,t]]$ without monomials divisible by xy is unique. Hence if $r > 2C_{u,v}M$, then κ_j does not depend on the invariant unit either.

Step 2. In this step we choose r_0 and compute the w_L-weights of the marks.

Choose r_0, depending only on Γ, u, v, and L, such that any integer $r \ge r_0$ satisfies

$$\begin{aligned} &r > 2C_{u,v}M, \quad \frac{C_{u,v}L}{r} < \frac{1}{2}, \quad \frac{C_{u,v}ML}{r} < M, \\ &\frac{r - C_{u,v}M}{2C_{u,v}} > 2M, \quad L\Big(1 - \frac{C_{u,v}M}{r}\Big) > 2M. \end{aligned} \tag{A.22}$$

Assume that $r \ge r_0$. Then $w_{L,t} > 0$, and $w_{L,x}, w_{L,y}, w_{L,z} > 1/2$. For a monomial x^α, let

$$x^\alpha = x^{\alpha_x} y^{\alpha_y} z^{\alpha_z} t^{\alpha_t}, \qquad I(\alpha) := \alpha_x + \alpha_y + \alpha_z, \qquad \kappa(\alpha) := \langle w, \alpha \rangle.$$

Then

$$\langle w_L, \alpha \rangle = I(\alpha) + \kappa(\alpha)\frac{L}{r}.$$

Let $\omega_j := I_j + \kappa_j L/r$, the least w_L-weight of a monomial of h_j^0. Then $\omega_j \geq 0$, and a monomial of least degree of h_j has w_L-weight at most $M + C_{u,v}ML/r < 2M$, so $\omega_j < 2M$. Let x^α be a monomial of h_j with $\kappa(\alpha) = \kappa_j + \ell r$ and $\ell \neq 0$. If $\ell > 0$, then

$$\langle w_L, \alpha\rangle \geq \kappa(\alpha)\frac{L}{r} \geq L\Big(1 - \frac{C_{u,v}M}{r}\Big) > 2M.$$

If $\ell < 0$, then $\kappa(\alpha) \leq C_{u,v}M - r < 0$. Since $\kappa(\alpha) \geq -C_{u,v}I(\alpha)$, we have $I(\alpha) \geq (r - C_{u,v}M)/C_{u,v}$, and $\langle w_L, \alpha\rangle \geq I(\alpha)/2 > 2M$. Thus

$$\mathrm{wt}_{w_L}(h_j) = \omega_j = I_j + \kappa_j\frac{L}{r}, \tag{A.23}$$

and $\mathrm{in}_{w_L}(h_j)$ is the sum of the terms of h_j^0 with $I(\alpha) = I_j$, a non-zero polynomial without monomials divisible by xy. Moreover

$$w_{L,x} + w_{L,y} = 2w_{L,z} = 2 - \frac{2vL}{r}, \qquad Nw_{L,t} - 2w_{L,z} = \frac{v(N+2)L}{r} - 2 = dL - 2 > 0, \tag{A.24}$$

so $\mathrm{in}_{w_L}(xy + z^2 + t^N) = xy + z^2$.

Let e_W be the positive integer such that $e_W w_L$ is a primitive vector of $\mathbb{Z}^4$. It exists since $rw_L \in \mathbb{Z}^4$. By Lemma 4.12 with the weights $e_W w_L$ and **Step 1**, there exists a prime divisor E_W over $W_N \ni 0$ such that $\mathrm{mult}_{E_W} h = e_W \,\mathrm{wt}_{w_L}(h)$ for any polynomial h without monomials divisible by xy, and

$$a(E_W, W_N, 0) = e_W(w_{L,x} + w_{L,y} + w_{L,z} + w_{L,t} - 2w_{L,z}) = e_W(w_{L,z} + w_{L,t}) = e_W. \tag{A.25}$$

In particular, $\mathrm{mult}_{E_W} h_j = e_W\omega_j$ for any j.

Step 3. In this step we construct Y and the quasi-étale charts V_k, and we compute the order of the group on the singular chart.

Let $M_R \subset \mathbb{Z}^4$ be the lattice of the vectors α with $\langle w, \alpha\rangle \in r\mathbb{Z}$, that is, of the exponents of R-invariant Laurent monomials, and let $N_R := \mathbb{Z}^4 + \mathbb{Z}\frac{w}{r}$ be the dual lattice. Then $\mathbb{A}^4/R$ is the affine toric variety of the positive orthant σ with respect to N_R [CLS11, Proposition 1.3.18]. The vector w_L belongs to N_R, and w_L is primitive in N_R since zt is R-invariant and $\langle w_L, (0,0,1,1)\rangle = w_{L,z} + w_{L,t} = 1$. The coordinate vectors e_x, e_y, e_z, e_t are primitive in N_R as well. If one of them is not primitive in N_R, then an element $(\zeta^u, \zeta^{-2v-u}, \zeta^{-v}, \zeta^v) \neq 1$ of R has exactly one eigenvalue different from one. If the moved coordinate is x or y, then the eigenvalues on z and on the other one of x, y imply $\zeta^v = \zeta^u = 1$. If the moved coordinate is z or t, then the eigenvalues on the other one of z, t and on x imply $\zeta^v = \zeta^u = 1$. Since $\gcd(u, v) = 1$, $\zeta = 1$, a contradiction. Thus R contains no pseudoreflection of $\mathbb{A}^4$. No coordinate subspace of codimension two of $\mathbb{A}^4$ is contained in W_N, since the restrictions of $xy + z^2 + t^N$ to the six such subspaces are $z^2 + t^N$, t^N, z^2, t^N, z^2, and xy. Hence no non-identity element of R fixes a divisor of W_N, and the quotient morphism $\pi_R : W_N \to W_N/R$ is quasi-étale.

Since $w_{L,l} > 0$ for any $l \in \{x, y, z, t\}$, w_L is in the interior of σ. Let Σ be the star subdivision of σ at the ray $\mathbb{R}_{\geq 0}w_L$. The maximal cones of Σ are $\sigma_k := \mathrm{Cone}(w_L, e_l \mid l \neq k)$ for $k \in \{x, y, z, t\}$, and the toric morphism $X_\Sigma \to \mathbb{A}^4/R$ is projective [CLS11, Proposition 11.1.6(c)] and an isomorphism over the complement of the origin. Let Y be the normalization of the strict transform of W_N/R in X_Σ, $\bar{0} \in W_N/R$ the image of 0, and $g : Y \to W_N/R$ the induced projective birational morphism, an isomorphism over $W_N/R\backslash\{\bar{0}\}$. The group G/R acts on $\mathbb{A}^4/R$ through the torus T_{N_R}, hence on X_Σ and on Y, and g is G/R-equivariant.

Let $N_k := \mathbb{Z}w_L + \sum_{l\neq k}\mathbb{Z}e_l \subset N_R$. The affine toric variety U_{σ_k, N_k} is $\mathbb{A}^4$, whose coordinates are λ and the three of x_1, y_1, z_1, t_1 other than the one indexed by k. They are the characters of the basis of the dual lattice of N_k dual to w_L and the e_l with $l \neq k$. Moreover $U_{\sigma_k, N_R} = U_{\sigma_k, N_k}/(N_R/N_k)$ [CLS11, Proposition 1.3.18]. For $l \in \{x, y, z, t\}$ we write l_1 for the coordinate among x_1, y_1, z_1, t_1 indexed by l. For $\alpha \in M_R$, the monomial x^α pulls back to

$$\lambda^{\langle w_L, \alpha\rangle}\prod_{l\neq k} l_1^{\alpha_l}. \tag{A.26}$$

For a non-zero R-semi-invariant polynomial $h = \sum_\alpha a_\alpha x^\alpha$ with $\mathrm{wt}_{w_L}(h) = \omega$, let

$$h^{(k)} := \sum_\alpha a_\alpha \lambda^{\langle w_L, \alpha\rangle - \omega} \prod_{l \neq k} l_1^{\alpha_l}. \tag{A.27}$$

The exponents of λ in (A.27) are non-negative integers, since the difference of two exponents of h belongs to M_R and $\langle w_L, M_R\rangle \subset \mathbb{Z}$. By (A.26), h^r pulls back to $\lambda^{r\omega}(h^{(k)})^r$, and h pulls back to $\lambda^\omega h^{(k)}$ if h is R-invariant.

By (A.24), $\mathrm{wt}_{w_L}(xy + z^2 + t^N) = 2w_{L,z}$, and the polynomials $(xy + z^2 + t^N)^{(k)}$ for $k = x, y, z, t$ are

$$\begin{aligned} &y_1 + z_1^2 + \lambda^{dL-2} t_1^N, \qquad && x_1 + z_1^2 + \lambda^{dL-2} t_1^N, \\ &x_1 y_1 + 1 + \lambda^{dL-2} t_1^N, \qquad && x_1 y_1 + z_1^2 + \lambda^{dL-2}. \end{aligned} \tag{A.28}$$

Let $V_k \subset U_{\sigma_k, N_k}$ be the hypersurface defined by the polynomial of (A.28) indexed by k. By (A.26), the R-invariant function $(xy + z^2 + t^N)/z^2$ on the torus of $\mathbb{A}^4/R$ pulls back to $(xy + z^2 + t^N)^{(k)}$ times a Laurent monomial in the coordinates other than λ. Each polynomial of (A.28) has degree one in one variable, namely y_1, x_1, x_1, and x_1 for $k = x, y, z, t$, with coefficient 1 or y_1, and the coefficient is coprime to the rest of the polynomial. Hence V_k is irreducible. Since V_k is irreducible and not contained in a coordinate hyperplane, V_k is the inverse image of the strict transform of W_N/R in U_{σ_k, N_R}. The hypersurfaces V_x and V_y are smooth, V_z is smooth near $(\lambda = 0)$ since $x_1 y_1 = -1$ there, and V_t is W_{dL-2} in the coordinates x_1, y_1, z_1, λ, which is normal with an isolated singular point at the origin. Possibly shrinking W_N/R to a neighborhood of $\bar{0}$, we may assume that V_z is smooth. Then $V_k/(N_R/N_k)$ is normal, since V_x, V_y, and V_z are smooth and V_t is normal, and $V_k/(N_R/N_k)$ maps finitely and birationally onto the strict transform of W_N/R in U_{σ_k, N_R}. Hence $V_k/(N_R/N_k)$ is the normalization of the strict transform of W_N/R in U_{σ_k, N_R}, that is, the inverse image Y_k of U_{σ_k, N_R} in Y.

The generators w_L and e_l of σ_k are primitive in N_R. An element of N_R/N_k acting on exactly one coordinate of $\mathbb{A}^4$ is the class of qb for a generator b of σ_k and a rational number $0 < q < 1$ with $qb \in N_R$, so N_R/N_k contains no pseudoreflection. Setting two coordinates equal to zero in any polynomial of (A.28) leaves a non-zero polynomial, so no coordinate subspace of codimension two of $\mathbb{A}^4$ is contained in V_k. Hence $V_k \to Y_k$ is quasi-étale.

The fiber $g^{-1}(\bar{0})$ is the image of $(\lambda = 0)$, and $V_k \cap (\lambda = 0)$ is defined in $\mathbb{A}^3$ by $y_1 + z_1^2$, $x_1 + z_1^2$, $x_1 y_1 + 1$, and $x_1 y_1 + z_1^2$ for $k = x, y, z, t$, respectively. Each of them is integral, and their images in Y contain a common dense open subset, the image of the locus where all coordinates other than λ are non-zero. Hence $g^{-1}(\bar{0})$ is a prime divisor E, the divisor of the ray $\mathbb{R}_{\geq 0} w_L$, and E is the only exceptional prime divisor of g. Moreover, $\mathrm{div}(\lambda)$ on V_k is reduced, and $V_k \to Y_k$ is unramified along $\mathrm{div}(\lambda)$.

Let G_k be the inverse image of G/R under the isogeny $T_{N_k} \to T_{N_R}$. Then G_k is a finite diagonal abelian group acting on $\mathbb{A}^4$ and preserving V_k, and $V_k/G_k = Y_k/(G/R)$. The kernel of $T_{N_k} \to T_{N_R}$ is N_R/N_k, so G_k contains the deck transformations of V_k over Y_k. Conversely, an automorphism of V_k lifting the action of an element of G/R on Y_k differs from an element of G_k by an automorphism of V_k over Y_k, which is an element of N_R/N_k since $V_k \to Y_k$ is the quotient of the integral variety V_k by the faithful action of N_R/N_k. Hence G_k is the group of all lifts to V_k of the action of G/R. The index $[N_R : N_t]$ is $\det(e_x, e_y, e_z, w_L) = w_{L,t}$ divided by the covolume $1/r$ of N_R, so

$$[N_R : N_t] = vL, \qquad |G_t| = vL[G : R]. \tag{A.29}$$

The group G_t fixes the origin of V_t and preserves $x_1 y_1 + z_1^2 + \lambda^{dL-2}$ up to a character. It acts faithfully on V_t, since a diagonal element acting trivially on V_t fixes the non-zero functions x_1, y_1, z_1, and λ on V_t.

Step 4. In this step we compute the crepant boundary on Y and on the charts V_k, and we construct Γ'.

Let h be an R-semi-invariant polynomial without monomials divisible by xy, and $\omega := \mathrm{wt}_{w_L}(h)$. The restriction of $h^{(t)}$ to $(\lambda = 0)$ is $\sum a_\alpha x_1^{\alpha_x} y_1^{\alpha_y} z_1^{\alpha_z}$, the sum over the exponents with $\langle w_L, \alpha\rangle = \omega$. Two such exponents have distinct $(\alpha_x, \alpha_y, \alpha_z)$ since $w_{L,t} > 0$. Thus $h^{(t)}|_{(\lambda=0)}$ is a non-zero polynomial without monomials divisible by $x_1 y_1$, and by **Step 1** it does not vanish identically on $V_t \cap (\lambda = 0) = (x_1 y_1 + z_1^2 = 0)$. Since h^r pulls back to $\lambda^{r\omega}(h^{(t)})^r$ and $V_t \to Y_t$ is unramified along $\mathrm{div}(\lambda)$, we have $\mathrm{mult}_E(h^r) = r\omega$, and $\mathrm{mult}_E h = \omega$ if h is R-invariant. Any R-invariant regular function on W_N is the restriction of such a polynomial h, since the reduction by $xy = -z^2 - t^N$ preserves R-characters. By **Step 2**, $\mathrm{mult}_{E_W} = e_W \mathrm{mult}_E$ on R-invariant regular functions. Thus E_W is a prime divisor over W_N lying over E with $|I_R(E_W)| = e_W$ [Liu26, Lemma 2.6(1)]. Since π_R is quasi-étale, [Liu26, Lemma 2.6(2)] and (A.25) imply

$$a(E, W_N/R, 0) = \frac{a(E_W, W_N, 0)}{e_W} = 1.$$

Let $D_{W_N/R}$ be the $\mathbb{R}$-divisor with $\pi_R^*(K_{W_N/R} + D_{W_N/R}) = K_{W_N} + D$. Since π_R is quasi-étale and h_j^r is R-invariant, $D_{W_N/R} = \sum_j \frac{c_j}{r}\mathrm{div}(h_j^r)$, and $\mathrm{mult}_E(h_j^r) = r\omega_j$ by (A.23). Let B_Y be the $\mathbb{R}$-divisor with $K_Y + B_Y = g^*(K_{W_N/R} + D_{W_N/R})$. Then $B_Y = g_*^{-1} D_{W_N/R} + \beta E$ with

$$\beta = 1 - a(E, W_N/R, D_{W_N/R}) = \sum_j c_j\omega_j = A + \tau\frac{L}{r}. \tag{A.30}$$

We have $\beta \geq 0$ since $\omega_j \geq 0$. The pair $(W_N/R, D_{W_N/R})$ is klt near $\bar{0}$ [Liu26, Lemma 2.6(2)(3)], so $\beta < 1$, $B_Y \geq 0$, and (Y, B_Y) is klt.

By the remark after (A.27), the exponents of λ in $h_j^{(k)}$ are non-negative integers. Since $V_k \to Y_k$ is unramified along $\mathrm{div}(\lambda)$ and h_j^r pulls back to $\lambda^{r\omega_j}(h_j^{(k)})^r$, the polynomial $h_j^{(k)}$ does not vanish identically on $(\lambda = 0)$. Hence the $\mathbb{R}$-divisor B_{V_k} on V_k such that $K_{V_k} + B_{V_k}$ is the pullback of $K_Y + B_Y$ is

$$B_{V_k} = \sum_j c_j \,\mathrm{div}(h_j^{(k)}) + \beta\,\mathrm{div}(\lambda), \tag{A.31}$$

and (V_k, B_{V_k}) is klt [Liu26, Lemma 2.6(2)]. The ratio of two monomials of $h_j^{(k)}$ is the pullback of the ratio of two monomials of h_j, which is a G-invariant character of T_{N_R}. Hence $h_j^{(k)}$ is G_k-semi-invariant, and so is λ. Let $Z := Y/(G/R)$, and let (Z, D_Z) be the crepant quotient of (Y, B_Y) by G/R. Over $Y_k/(G/R)$, (Z, D_Z) is the crepant quotient of (V_k, B_{V_k}) by G_k [Liu26, Lemma 2.6(5)]. In particular, the germ of (Z, D_Z) at the image z_0 of the origin of V_t is the crepant quotient germ $(W_{dL-2} \ni 0, B_{V_t})/G_t$, and B_{V_t} is a labelled boundary (Definition 2.7) with labels c_j and β. The chart V of the statement is V_t. Let $\mathcal{H}_j := h_j^{(t)}$. By (A.26) with $k = t$, the pullback of R-invariant monomials is the substitution $x = x_1\lambda^{w_{L,x}}$, $y = y_1\lambda^{w_{L,y}}$, $z = z_1\lambda^{w_{L,z}}$, $t = \lambda^{w_{L,t}}$, and the substitution turns h_j into $\lambda^{\omega_j}\mathcal{H}_j$ with $\omega_j = I_j + \kappa_j L/r$ by (A.23). By (A.31) and (A.30),

$$B_{V_t} = \sum_j c_j\,\mathrm{div}(\mathcal{H}_j) + \left(A + \frac{\tau L}{r}\right)\mathrm{div}(\lambda).$$

We now construct Γ'. Since (W_N, D) is klt and $h_j(0) = 0$, $c_j < 1$. Let $\mathfrak{B}$ be the set of real numbers $b < 1$ of the form

$$b = \sum_{j=1}^{s} c_j\left(I_j + \kappa_j\frac{L}{r}\right),$$

where $s \leq M$, $c_j \in \Gamma \cap (0,1)$, $I_j \in \{0, 1, \ldots, M\}$, $\kappa_j \in \mathbb{Z} \cap [-C_{u,v}M, C_{u,v}M]$, $r \geq r_0$, $\sum_j c_j\kappa_j \leq 0$, and $I_j + \kappa_j L/r \geq 0$ for any j. Let $\Gamma' := (\Gamma \cap [0,1)) \cup \mathfrak{B}$. It depends only on Γ, u, v, and L. In particular, Γ' does not depend on N, d, or G. By **Steps 1 and 2** and (A.30), $\beta \in \mathfrak{B}$, and $c_j \in \Gamma'$ for any j.

We show that $\mathfrak{B}$ satisfies the DCC, hence so does Γ'. Suppose that $b_i \in \mathfrak{B}$ is strictly decreasing. Since s, I_j, and κ_j range over finite sets, after passing to a subsequence, they do not depend on i, the labels c_{ij} are non-decreasing in i for any j, and r_i is either constant or strictly increasing. If r_i is constant, then b_i is non-decreasing since $I_j + \kappa_j L/r_i \geq 0$, a contradiction. Otherwise, $\eta_i := L/r_i$ is strictly decreasing, and $\tau_i := \sum_j c_{ij}\kappa_j \leq 0$. Then

$$b_{i+1} - b_i = \tau_{i+1}(\eta_{i+1} - \eta_i) + \sum_j (c_{i+1,j} - c_{ij})(I_j + \kappa_j \eta_i) \geq 0,$$

a contradiction. The sign $\tau \leq 0$ is used only here.

The prime coefficients of B_Y and of B_{V_k} are finite sums of elements of Γ' and less than one, so they belong to $\Gamma'_+ \cap [0,1)$. By [Liu26, Lemma 2.6(4)], $D_Z \in \Theta := \Psi(\Gamma'_+ \cap [0,1))$, which is a DCC set [Liu26, Lemma 2.10(1)].

Step 5. In this step we determine where the centers of the computing prime divisors are, and we prove (1) and (2).

Let $Q := W_N/G$, $\bar{0}_Q \in Q$ the image of 0, and $(Q, D_Q) := (W_N, D)/G$. Then $(Q, D_Q) = ((W_N, D)/R)/(G/R)$ [Liu26, Lemma 2.6(5)], so the induced birational morphism $\bar{g} : Z \to Q$ satisfies $K_Z + D_Z = \bar{g}^*(K_Q + D_Q)$. The morphism $\bar{g}$ is projective since g is projective and G/R-equivariant: the norm of a g-ample line bundle is a G/R-linearized g-ample line bundle, and a power of it descends to Z. Thus $(\bar{g}, D_Z)$ is a crepant model of $(Q \ni \bar{0}_Q, D_Q)$ in the sense of Definition-Proposition 4.10. Since (Q, D_Q) is klt near $\bar{0}_Q$, there exists a prime divisor F over Q with center $\bar{0}_Q$ such that $a(F, Q, D_Q) = \alpha$ [Liu26, Lemma 3.5]. Let C be the center of F on Z, and η_C the generic point of C. Then $C \subset \bar{g}^{-1}(\bar{0}_Q)$, so any prime divisor over Z with center C has center $\bar{0}_Q$ on Q, so the log discrepancy of such a prime divisor with respect to (Z, D_Z) is at least α. Since F has center C,

$$\mathrm{mld}(Z \ni \eta_C, D_Z) = \alpha. \tag{A.32}$$

We consider the possible centers C.

(a) If C is a prime divisor, then $\alpha = 1 - \mathrm{mult}_C D_Z \in \{1 - b \mid b \in \Theta\}$, which satisfies the ACC since Θ is a DCC set.

(b) If C is a curve, then by (A.32) and [Liu26, Lemma 3.7] applied to the threefold pair (Z, D_Z) with $D_Z \in \Theta$, α belongs to a set satisfying the ACC and depending only on Θ.

(c) If C is a closed point $z \neq z_0$, then z is the image of a point $p \in V_k$ for some k such that V_k is smooth at p. Let $H_p \subset G_k$ be the stabilizer of p. Since (Z, D_Z) is klt near z, there is a prime divisor E_z over Z with center z such that $a(E_z, Z, D_Z) = \mathrm{mld}(Z \ni z, D_Z)$ [Liu26, Lemma 3.5]. By [Liu26, Lemma 2.6(3)] there is a prime divisor F_z over V_k lying over E_z [Liu26, Definition 2.7]. Its center on V_k is a point of the G_k-orbit of p, so after replacing F_z by a G_k-translate, the center of F_z on V_k is p. The inertia group $I_{G_k}(F_z)$ fixes the center of F_z, hence $I_{G_k}(F_z) \subset H_p$. By [Liu26, Lemma 3.1(1)] applied to the quotient $V_k \to V_k/G_k$ near z, the identity morphism of V_k, and the subgroup H_p of G_k, we have $\mathrm{mld}(Z \ni z, D_Z) = \mathrm{mld}((V_k \ni p, B_{V_k})/H_p)$. The group H_p is a finite abelian group acting faithfully on the smooth threefold germ $(V_k \ni p)$, possibly with pseudoreflections, and $B_{V_k} \in \Gamma'_+ \cap [0,1)$. By the last sentence of Theorem 2.8(1), α belongs to a set satisfying the ACC and depending only on Γ'.

(d) If $C = \{z_0\}$, then $\alpha = \mathrm{mld}((W_{dL-2} \ni 0, B_{V_t})/G_t)$, where B_{V_t} is a labelled boundary with labels in Γ', (W_{dL-2}, B_{V_t}) is klt near 0, and $|G_t| = vL[G : R]$ by (A.29).

By **Step 3**, any point of $g^{-1}(\bar{0})$ has a smooth quasi-étale quotient chart $V_k \to Y_k$ except possibly the origin of V_t, which has the properties in (d). None of the arguments in **Steps 1–5** uses the bound $[G : R] \leq e_0$.

We prove (1). Fix an even integer $L > 4M$, and let r_0 and Γ' be the integer of (A.22) and the set of **Step 4** for this L. The parity of L is not used in this proof. Fix d_0, and suppose that α_i is a strictly increasing sequence of positive numbers α as in (1). If $r_i < r_0$ for infinitely

many i, then $N_i+2=d_ir_i/v\le d_0r_0$ for these i, so after passing to a subsequence, N_i is constant, and Theorem 7.8 with labels in Γ implies a contradiction. Otherwise, $r_i\ge r_0$ for $i\gg 0$, and by (a)–(d), α_i belongs to the union of the three ACC sets of (a)–(c) and the sets of Theorem 7.8 for the finitely many exponents $dL-2$ with $d\le d_0$ and labels in Γ'. A finite union of sets satisfying the ACC satisfies the ACC, a contradiction. This proves (1).

We prove (2). Let α_i be a strictly increasing sequence of positive numbers α with $\tau_i\le 0$ and $r_i\to+\infty$. We may assume that $r_i\ge r_0$ for any i. By (1), d_i is unbounded, so after passing to a subsequence, $d_i\to+\infty$. The sets of (a)–(c) satisfy the ACC, so they contain only finitely many α_i. After passing to a subsequence, for any i, the center on Z_i of any prime divisor computing α_i is $z_{0,i}$, and by (d),

$$\alpha_i=\operatorname{mld}((W_{d_iL-2}\ni 0,B_{V_t,i})/G_{t,i}),\qquad |G_{t,i}|=vL[G_i:R_i]\le vLe_0,$$

where $B_{V_t,i}$ has labels in Γ' for any i, and Γ' does not depend on d_i. This proves (2). We treat such values when proving Theorems 7.12 and 7.15 for non-positive mass, which together with Theorem 7.17 imply Corollary 7.18. □

Proof of Proposition 7.10. The proof is long, so we divide it into several steps. Let

$$\gamma:=\min(\Gamma\backslash\{0\}),\qquad m_0:=\lceil 2/\gamma\rceil,$$

and $\gamma:=m_0:=1$ if $\Gamma\subset\{0\}$. Let $f_N:=xy+z^2+t^N$, $f_0:=xy+z^2$, $Q_N:=W_N/G$, and $Q_0:=W_0/G$. For $g\in G$ we write $g(x,y,z,t)=(\chi_x(g)x,\chi_y(g)y,\chi_z(g)z,\chi_t(g)t)$. Since G preserves f_N up to a character, $\chi_x\chi_y=\chi_z^2=\chi_t^N$.

Step 1. In this step we bound the number of marks and prove that (W_0,D_0) is lc near 0 for $N\ge N_1$.

Marks that are units or have label zero do not contribute to D_N or D_0 near 0, so we omit them. By Lemma A.1(1), each h_j is not divisible by f_0, hence non-zero on W_0, and D_0 is defined. Let E_1 be the exceptional divisor of the blow-up of W_N at 0, so that $a(E_1,W_N,0)=2$ by Lemma A.1(2). Since (W_N,D_N) is klt near 0 and $c_j\ge\gamma$, Lemma A.1(3) gives $\sum_j c_jm(h_j)<2$, so at most m_0 marks with positive labels remain.

Let $\Lambda:=(\Gamma\backslash\{0\})\cup\{1\}$, a DCC subset of $\mathbb{R}_{>0}$, and let k_0 be the maximum over $s\in\{1,\dots,m_0+1\}$ of the integers of Lemma 4.14 with $n=4$, s, and Λ. Let $N_1:=k_0$, which depends only on Γ. Assume that $N\ge N_1$. By inversion of adjunction [Kaw07, Theorem] along the normal Cartier divisor $W_N\subset\mathbb{A}^4$, the pair $(\mathbb{A}^4,W_N+\sum_j c_j\operatorname{div}(h_j))$ is lc near 0. Hence the pair $(\operatorname{Spec}\mathbb{C}[[x,y,z,t]],(f_N)\prod_j(h_j)^{c_j})$ is lc [dFEM11, Proposition 2.11]. Since $f_N-f_0=t^N\in(x,y,z,t)^{k_0}$, Lemma 4.14 implies that the pair $(\operatorname{Spec}\mathbb{C}[[x,y,z,t]],(f_0)\prod_j(h_j)^{c_j})$ is lc. By [dFEM11, Proposition 2.11], $(\mathbb{A}^4,W_0+\sum_j c_j\operatorname{div}(h_j))$ is lc near 0, and by inversion of adjunction [Kaw07, Theorem] along the normal Cartier divisor W_0, (W_0,D_0) is lc near 0. This proves the first statement of (1).

Step 2. In this step we prove the correspondence of prime divisors in (1).

Let $\phi:W_N\dashrightarrow W_0$ be the birational map

$$\phi(x,y,z,t)=\Big(x,y+\frac{t^N}{x},z,t\Big),$$

an isomorphism over $(x\ne 0)$ with inverse $(x,y_0,z,t)\to(x,y_0-t^N/x,z,t)$, where x,y_0,z,t are the coordinates on W_0. Since $\chi_x\chi_y=\chi_t^N$, we have $g^*(t^N/x)=\chi_y(g)t^N/x$ for any $g\in G$, so ϕ is G-equivariant. Hence ϕ induces a birational map $Q_N\dashrightarrow Q_0$, any prime divisor over W_N is a prime divisor over W_0 through ϕ, and the inertia and decomposition groups in G of a prime divisor over W_N and over W_0 coincide.

We show that $(W_N,\operatorname{div}(z)+\operatorname{div}(t))$ and $(W_0,\operatorname{div}(z)+\operatorname{div}(t))$ are lc near 0. The Cartier divisor $(t=0)$ of W_N and of W_0 is the surface $S_1:=(xy+z^2=0)\subset\mathbb{A}^3$, and $\mathbb{A}^2\to S_1$, $(p,q_1)\to(p^2,-q_1^2,pq_1)$, is the quasi-étale quotient by $(p,q_1)\to(-p,-q_1)$. Since $\operatorname{div}_{S_1}(z)$ pulls

back to $\operatorname{div}(p)+\operatorname{div}(q_1)$, the pair $(S_1, \operatorname{div}_{S_1}(z))$ is lc [Liu26, Lemma 2.6(2)(3)]. Inversion of adjunction [Kaw07, Theorem] along $(t=0)$ implies the claim.

Fix $\lambda>0$ and let

$$N_0(M,\Gamma,\lambda) := \max\bigl(N_1, \lfloor M\lambda(2+1/\gamma)\rfloor+1\bigr).$$

Assume that $N \ge N_0(M,\Gamma,\lambda)$. Let W_* be one of W_N and W_0, D_* the corresponding one of D_N and D_0, and y_* the corresponding one of y and y_0. Let E be a prime divisor over W_*/G with center the image of 0 such that the log discrepancy of E with respect to $(W_*,0)/G$ is at most λ, and let $\tilde{E}$ be a prime divisor over W_* lying over E [Liu26, Lemma 2.6(3)], so that the center of $\tilde{E}$ on W_* is 0. Then $|I_G(\tilde{E})| \le |G| \le M$, so $a(\tilde{E}, W_*, 0) \le M\lambda$ [Liu26, Lemma 2.6(2)]. Since $(W_*, \operatorname{div}(z)+\operatorname{div}(t))$ and (W_*, D_*) are lc near 0 (the previous paragraph, the hypothesis, and **Step 1**, since $N \ge N_1$),

$$\operatorname{mult}_{\tilde{E}} z + \operatorname{mult}_{\tilde{E}} t \le M\lambda, \qquad \sum_j c_j \operatorname{mult}_{\tilde{E}} h_j \le M\lambda. \tag{A.33}$$

Let $T_{\tilde{E}} := N\operatorname{mult}_{\tilde{E}} t - 2\operatorname{mult}_{\tilde{E}} z$. Since $\operatorname{mult}_{\tilde{E}} t \ge 1$ and $N > M\lambda(2+1/\gamma)$, (A.33) implies

$$T_{\tilde{E}} \ge N - 2M\lambda > \frac{M\lambda}{\gamma} \ge \operatorname{mult}_{\tilde{E}} h_j \quad \text{for any } j. \tag{A.34}$$

Since $T_{\tilde{E}} > 0$, $\operatorname{mult}_{\tilde{E}}(z^2+t^N) = 2\operatorname{mult}_{\tilde{E}} z$, so $\operatorname{mult}_{\tilde{E}} x + \operatorname{mult}_{\tilde{E}} y_* = 2\operatorname{mult}_{\tilde{E}} z$ on both hypersurfaces. Thus

$$\operatorname{mult}_{\tilde{E}}(t^N/x) = T_{\tilde{E}} + \operatorname{mult}_{\tilde{E}} y_* > \operatorname{mult}_{\tilde{E}} y_*,$$

and the other one of y, y_0, which is $y_* \pm t^N/x$, has order $\operatorname{mult}_{\tilde{E}} y_* > 0$ along $\tilde{E}$. Hence the center of $\tilde{E}$ on the other hypersurface is also 0.

Write a polynomial h without monomials divisible by xy as $h = P_0(x,z,t) + \sum_{b\ge1} y^b P_b(z,t)$. Replacing y_* by $y_* \pm t^N/x$ changes h by the sum of the terms $\binom{b}{i}(\pm t^N/x)^i y_*^{b-i} P_b$ with $1 \le i \le b$, whose orders along $\tilde{E}$ are at least $b\operatorname{mult}_{\tilde{E}} y_* + T_{\tilde{E}} > T_{\tilde{E}}$. By (A.34), for any j, the restrictions of h_j to W_N and to W_0 have equal orders along $\tilde{E}$. The residue form of both hypersurfaces is $\Omega := dx\wedge dz\wedge dt/x$, which generates the canonical sheaf near 0 [KM98, Proposition 5.73], and $\phi^*\Omega = \Omega$. Hence $a(\tilde{E}, W_N, 0) = a(\tilde{E}, W_0, 0)$, and so $a(\tilde{E}, W_N, D_N) = a(\tilde{E}, W_0, D_0)$. Let E' be the prime divisor over the other one of Q_N and Q_0 under $\tilde{E}$. Since the inertia groups coincide, [Liu26, Lemma 2.6(2)] implies that E and E' have equal log discrepancies with respect to $(W_N,0)/G$ and $(W_0,0)/G$, and with respect to $(W_N,D_N)/G$ and $(W_0,D_0)/G$. This proves the correspondence in (1).

Step 3. In this step we prove $0 < \operatorname{mld}((W_N \ni 0, D_N)/G) \le \operatorname{mld}((W_0 \ni 0, D_0)/G)$ for N sufficiently large.

Let $S := \mathbb{A}^3_{p,q_1,t}$ and $S \to W_0$, $(p,q_1,t) \to (p^2, -q_1^2, pq_1, t)$, the quasi-étale quotient by $(p,q_1,t) \to (-p,-q_1,t)$. Let

$$\tilde{G} := \{(g,\rho) \mid g\in G,\ \rho^2 = \chi_x(g)\}, \qquad (g,\rho)(p,q_1,t) = (\rho p, \chi_z(g)\rho^{-1}q_1, \chi_t(g)t).$$

Since $\chi_x\chi_y = \chi_z^2$, the morphism $S \to W_0$ is equivariant with respect to $\tilde{G} \to G$, whose kernel is generated by $(1,-1)$. Hence $\tilde{G}$ is a finite diagonal abelian group of order $2|G| \le 2M$ acting faithfully on S, and $S/\tilde{G} = Q_0$. Since $\tilde{G}$ is contained in the group of diagonal matrices whose entries are $(2M)!$-th roots of unity, only finitely many varieties Q_0 occur, and they are klt and $\mathbb{Q}$-factorial [KM98, Proposition 5.20], [KM98, Lemma 5.16].

Let Δ_0 be the crepant boundary of $(W_0,D_0)/G$ on Q_0, assume that $N \ge N_1$, and let Δ_0' be the sum of the prime components of Δ_0 through the image of 0, with their coefficients in Δ_0. A prime divisor P of W_0 over such a component passes through 0, since 0 is the only point over the image of 0. Thus $\operatorname{mult}_P D_0$ is a finite sum of elements of Γ and at most one since (W_0,D_0)

is lc near 0. Since the inertia group of P has order at most M, [Liu26, Lemma 2.6(4)] implies that $\Delta_0' \in \Theta_M \backslash \{0\}$, where

$$\Theta_M := \bigcup_{e \le M} \left\{ 1 - \frac{1-s}{e} \;\middle|\; s \in \Gamma_+ \cap [0,1] \right\}$$

is a DCC set [Liu26, Lemma 2.10(1)]. Since Q_0 is $\mathbb{Q}$-factorial, Δ_0' is $\mathbb{R}$-Cartier. Near the image of 0 we have $\Delta_0 = \Delta_0'$, so any prime divisor over Q_0 with center the image of 0 has equal log discrepancies with respect to (Q_0, Δ_0) and (Q_0, Δ_0'). By [Kaw24, Theorem 1.4] applied to the finitely many Q_0 and $\Theta_M \backslash \{0\}$, there exist a real number $L_{\mathrm{ref}} = L_{\mathrm{ref}}(M, \Gamma)$ and a prime divisor F_0 over Q_0 with center the image of 0 such that $a(F_0, Q_0, \Delta_0') = \mathrm{mld}((W_0 \ni 0, D_0)/G)$ and $a(F_0, Q_0, 0) \le L_{\mathrm{ref}}$. The log discrepancy of F_0 with respect to $(W_0, 0)/G$ is at most $a(F_0, Q_0, 0) \le L_{\mathrm{ref}}$, since the crepant boundary of $(W_0, 0)/G$ on Q_0 is ≥ 0. If $N \ge N_0(M, \Gamma, L_{\mathrm{ref}})$, then by **Step 2**, F_0 corresponds to a prime divisor over Q_N with center the image of 0 whose log discrepancy with respect to $(W_N, D_N)/G$ is $\mathrm{mld}((W_0 \ni 0, D_0)/G)$. Thus

$$\mathrm{mld}((W_N \ni 0, D_N)/G) \le \mathrm{mld}((W_0 \ni 0, D_0)/G),$$

and $\mathrm{mld}((W_N \ni 0, D_N)/G) > 0$ since (W_N, D_N) is klt near 0 [Liu26, Lemma 2.6(2)(3)]. This proves (1).

Step 4. In this step we compute the branch divisor of $W_N \to Q_N$ and the Cartier index of K_{Q_N}.

Let $\epsilon_z := 1$ if $(1, 1, -1, 1) \in G$ and $\epsilon_z := 0$ otherwise, and let m be the order of the subgroup of G of the elements of the form $(1, 1, 1, \zeta)$. Let $g \in G$ be a non-identity element fixing a prime divisor P of W_N pointwise. Then P is contained in the coordinate subspace defined by the vanishing of the coordinates moved by g. The restrictions of f_N to the six coordinate subspaces of codimension two of $\mathbb{A}^4$ are $z^2 + t^N$, t^N, z^2, t^N, z^2, and xy, so none of them is contained in W_N. If g moves at least two coordinates, then P is contained in the intersection of W_N with a coordinate subspace of codimension at least two, which is of dimension at most one, a contradiction. Hence g moves exactly one coordinate. Since $\chi_x \chi_y = \chi_z^2 = \chi_t^N$, g moves neither only x nor only y. Hence either $g = (1, 1, -1, 1)$ and $P = (z = 0)$, or $g = (1, 1, 1, \zeta)$ and $P = (t = 0)$. The divisors $\mathrm{div}(z) = (xy + t^N = 0)$ and $\mathrm{div}(t) = (xy + z^2 = 0)$ on W_N are prime, and their inertia groups have orders $1 + \epsilon_z$ and m. Thus the ramification divisor of $\pi_N : W_N \to Q_N$ is

$$R_N = \epsilon_z \,\mathrm{div}(z) + (m-1)\,\mathrm{div}(t), \tag{A.35}$$

and $K_{W_N} = \pi_N^* K_{Q_N} + R_N$.

The residue form $\Omega = dx \wedge dz \wedge dt/x$ generates the canonical sheaf of W_N near 0 [KM98, Proposition 5.73], and $g^*\Omega = \chi_z(g)\chi_t(g)\Omega$. Let $\psi := \chi_z^{1+\epsilon_z}\chi_t^m$ and $l := M!$. Since the order of any element of G divides l, the l-pluricanonical form $\Omega_l := (z^{\epsilon_z} t^{m-1}\Omega)^{\otimes l}$ satisfies $g^*\Omega_l = \psi(g)^l \Omega_l = \Omega_l$. Thus Ω_l descends to a rational section of $\mathcal{O}_{Q_N}(lK_{Q_N})$, whose divisor Ξ satisfies $\pi_N^*\Xi = \mathrm{div}(\Omega_l) - lR_N = 0$ near 0 by (A.35). Hence $M!K_{Q_N}$ is Cartier near the image of 0.

Step 5. In this step we prove (2).

Assume that Γ is finite, and let $q := |G|$. The functions z^q, t^q, and h_j^q are G-invariant. Let Δ_N be the $\mathbb{R}$-divisor with $\pi_N^*(K_{Q_N} + \Delta_N) = K_{W_N} + D_N$. Then $\pi_N^*\Delta_N = D_N + R_N$, and by (A.35),

$$\Delta_N = \frac{\epsilon_z}{q}\,\mathrm{div}(z^q) + \frac{m-1}{q}\,\mathrm{div}(t^q) + \sum_j \frac{c_j}{q}\,\mathrm{div}(h_j^q), \tag{A.36}$$

since both sides pull back to $D_N + R_N$. The divisors $\mathrm{div}(z^q)$, $\mathrm{div}(t^q)$, and $\mathrm{div}(h_j^q)$ are Cartier divisors ≥ 0 on Q_N, and their coefficients ϵ_z/q, $(m-1)/q$, c_j/q in (A.36) belong to the finite set

$$\left\{ \frac{c}{q'}, \frac{1}{q'}, \frac{m'-1}{q'} \;\middle|\; c \in \Gamma,\ 1 \le m' \le q' \le M \right\}.$$

The pair (Q_N, Δ_N) is klt near the image of 0 [Liu26, Lemma 2.6(2)(3)], and $M!K_{Q_N}$ is Cartier near the image of 0 by **Step 4**. Possibly shrinking Q_N to a neighborhood of the image of 0,

Theorem 2.13(2) with dimension 3 and the integer $M!$ implies that the numbers $\mathrm{mld}((W_N \ni 0, D_N)/G) = \mathrm{mld}(Q_N \ni \pi_N(0), \Delta_N)$, for all such data and all N, form a subset $\mathcal{S}_M$ of $\mathbb{R}$ without accumulation points in $\mathbb{R}$. Moreover, with E_1 as in **Step 1**,

$$\mathrm{mld}((W_N \ni 0, D_N)/G) \le \frac{a(E_1, W_N, D_N)}{|I_G(E_1)|} \le a(E_1, W_N, 0) = 2$$

[Liu26, Lemma 2.6]. Since $\mathcal{S}_M$ has no accumulation point in $\mathbb{R}$, the bounded set $\mathcal{S}_M \cap (0, 2]$ is finite. This proves (2). □

A.5. **Circles $(0, 2s, s, v)$.** In this subsection, we prove Theorem 7.11.

For the circles $(0, 2s, s, v)$, the weighted blow-up of W_N with weights $(0, 2s, s, v)$ is covered over 0 by the quotients of two hypersurfaces, at most one of them singular, the crepant boundary on the chart of the possibly singular hypersurface contains a coordinate hyperplane whose coefficient tends to 1 as $r \to +\infty$, and the ACC for log canonical thresholds allows us to replace the coefficient of the hyperplane by 1, which reduces (2) to (1).

Proof of Theorem 7.11. The proof is long, so we divide it into several steps.

Step 1. In this step we bring the marks of (1) into polynomial form.

Let $f := xy + z^2 + t^m$, so that $W_m = (f = 0) \subset \mathbb{A}^4$. For $g \in G$ we write $g^*x = \chi_x(g)x$, $g^*y = \chi_y(g)y$, $g^*z = \chi_z(g)z$, and $g^*t = \chi_t(g)t$. Write $D = \sum_j c_j \operatorname{div}(h_j)$. Marks with $c_j = 0$ do not change D, and marks which are units at 0 do not change D near 0. Hence we may assume that $c_j > 0$ and $h_j(0) = 0$ for any j. Write $h_j = a_j/u_j$ with regular functions a_j, u_j on W_m such that $u_j(0) \neq 0$. The function $\prod_{g\in G} g^*u_j$ is G-invariant and is a unit near 0. Multiplying h_j by it changes neither $\operatorname{div}(h_j)$ near 0 nor $\operatorname{mult}_F h_j$ for any prime divisor F over W_m with center 0, and keeps $h_j|_{S_0} \neq 0$. Thus we may assume that h_j is regular on W_m. Since G is diagonal and W_m is G-invariant, the sum of the monomials of a polynomial lift of h_j whose character is that of h_j is again a lift of h_j. Thus $h_j = P_j|_{W_m}$ for a G-semi-invariant polynomial P_j on $\mathbb{A}^4$, and $P_j|_{S_0} \neq 0$. Let γ be the least element of $\Gamma\backslash\{0\}$, and $\gamma := 1$ if $\Gamma \subset \{0\}$.

Step 2. In this step we define η and show that

$$\Big(\mathbb{A}^4,\ \Delta := \operatorname{div}(f) + \operatorname{div}(t) + \sum_j c_j \operatorname{div}(P_j)\Big) \text{ is lc near } 0 \tag{A.37}$$

whenever $1 - \eta < \beta < 1$ and $(W_m, D + \beta \operatorname{div}(t))$ is klt near 0.

Let $\Lambda := (\Gamma\backslash\{0\}) \cup \{1\}$, a DCC set. By [dFEM11, Theorem 1.1], the log canonical thresholds at 0 of $\mathbb{R}$-ideals on $\mathbb{A}^4$ whose exponents belong to Λ form an ACC set $\mathcal{T}$. Hence there exists $\eta \in (0, 1)$, depending only on Γ, such that $\mathcal{T} \cap (1 - \eta, 1) = \emptyset$: otherwise elements of $\mathcal{T} \cap (1 - \frac{1}{k}, 1)$, $k = 1, 2, \dots$, contain a strictly increasing sequence.

Assume that $1 - \eta < \beta < 1$ and that $(W_m, D + \beta \operatorname{div}(t))$ is klt near 0. The hypersurface W_m has an isolated cDV singularity at 0, hence is terminal near 0 [KM98, Corollary 5.38], and G acts faithfully on W_m since it acts faithfully on S_0. Theorem 3.3(1), applied with $n = 4$, $c = 1$, $f_1 = f$, the trivial subgroup, and $\mathfrak{a} = (t)^\beta \prod_j (P_j)^{c_j}$, implies

$$\mathrm{mld}\Big(\mathbb{A}^4 \ni 0, \operatorname{div}(f) + \beta \operatorname{div}(t) + \sum_j c_j \operatorname{div}(P_j)\Big) = \mathrm{mld}(W_m \ni 0, D + \beta \operatorname{div}(t)) > 0.$$

By [KM98, Corollary 2.31(1)] and its local form at a closed point [Amb99, Section 1, the second paragraph after Definition 1.1], a pair of dimension at least two which is not lc near a closed point has minimal log discrepancy $-\infty$ at the point. Hence the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(f) + \beta \operatorname{div}(t) + \sum_j c_j \operatorname{div}(P_j)\Big)$$

is lc near 0. Let μ be the log canonical threshold at 0 of the $\mathbb{R}$-ideal $(f)(t)\prod_j (P_j)^{c_j}$, that is, the largest real number $c \geq 0$ such that $(\mathbb{A}^4, c\Delta)$ is lc near 0. Since $\beta < 1$, we have

$$\beta\Delta \leq \operatorname{div}(f) + \beta \operatorname{div}(t) + \sum_j c_j \operatorname{div}(P_j),$$

so $\beta \leq \mu$. Since $\operatorname{div}(f) = W_m$ is a prime divisor through 0, $\mu \leq 1$. As $\mu \in \mathcal{T}$ and $\mu \geq \beta > 1 - \eta$, we obtain $\mu = 1$, which is (A.37).

Step 3. In this step we show that $(W_m, D + \operatorname{div}(t))$ and $(S_0, D|_{S_0})$ are lc near 0.

The surface $S_0 = (xy + z^2 = 0) \subset \mathbb{A}^3_{x,y,z}$ has a Du Val singularity of type A_1 at 0, hence is klt [KM98, Theorem 4.20]. The divisor W_m on $\mathbb{A}^4$ and the divisor $S_0 = \operatorname{div}_{W_m}(t)$ on W_m are normal and Cartier. By [KM98, Theorem 5.50(1)] with zero boundary, $(\mathbb{A}^4, W_m)$ is plt near W_m, and (W_m, S_0) is plt near S_0.

Let $0 < \epsilon < 1$. Log discrepancies are affine functions of the boundary. Hence for any prime divisor F over $\mathbb{A}^4$ other than W_m whose center contains a point near 0,

$$a\big(F, \mathbb{A}^4, W_m + (1-\epsilon)(\Delta - W_m)\big) = \epsilon a(F, \mathbb{A}^4, W_m) + (1-\epsilon)a(F, \mathbb{A}^4, \Delta) > 0$$

by (A.37), so $(\mathbb{A}^4, W_m + (1-\epsilon)(\Delta - W_m))$ is plt near 0. Since $(\Delta - W_m)|_{W_m} = S_0 + D$, [KM98, Theorem 5.50(1)] implies that $(W_m, (1-\epsilon)(S_0 + D))$ is klt near 0. The boundaries in [KM98, Theorem 5.50(1)] are $\mathbb{Q}$-divisors, and we pass to our $\mathbb{R}$-divisors as follows. All boundaries in this step are real combinations of principal divisors on $\mathbb{A}^4$ or on the hypersurface W_m, whose canonical divisor is Cartier. Being plt near 0 is a finite set of strict inequalities on a fixed log resolution, affine in the coefficients. Thus each real coefficient may be raised to a nearby rational number without changing the support, keeping the pair plt near 0. Then [KM98, Theorem 5.50(1)] shows that the restriction with the raised coefficients is klt, and lowering the coefficients back keeps it klt. Letting $\epsilon \to 0$, $(W_m, D + \operatorname{div}(t)) = (W_m, S_0 + D)$ is lc near 0. Since log discrepancies are affine functions of the boundary and (W_m, S_0) is plt near S_0,

$$a(F, W_m, S_0 + (1-\epsilon)D) = \epsilon a(F, W_m, S_0) + (1-\epsilon)a(F, W_m, S_0 + D) > 0$$

for any prime divisor F over W_m other than S_0 whose center contains a point near 0. Hence $(W_m, S_0 + (1-\epsilon)D)$ is plt near 0, $(S_0, (1-\epsilon)D|_{S_0})$ is klt near 0 by [KM98, Theorem 5.50(1)], applied after raising the real coefficients to nearby rational numbers and lowering them back, and $(S_0, D|_{S_0})$ is lc near 0.

Step 4. In this step we prove the lower bound of (1) and the ACC for b_0.

By **Step 3**, the hypotheses of Theorem 3.3(2) hold for $n = 4$, $c = 1$, $f_1 = f$, $f_2 = t$, the subgroup G, and $\mathfrak{a} = \prod_j (P_j)^{c_j}$: the surface $S_0 = W_m \cap (t = 0)$ is a normal integral complete intersection with klt singularities, and G acts faithfully on S_0. Hence, by Theorem 3.3(2),

$$b_0 = \operatorname{mld}((W_m \ni 0, D + \operatorname{div}(t))/G). \tag{A.38}$$

For any prime divisor F over W_m with center 0,

$$a(F, W_m, D + \beta \operatorname{div}(t)) = a(F, W_m, D + \operatorname{div}(t)) + (1-\beta)\operatorname{mult}_F t \geq a(F, W_m, D + \operatorname{div}(t)).$$

Dividing by $|I_G(F)|$ and taking the infimum over F in the formula at the end of [Liu26, Lemma 2.6], (A.38) implies

$$\operatorname{mld}((W_m \ni 0, D + \beta \operatorname{div}(t))/G) \geq b_0.$$

Since $(S_0, D|_{S_0})$ is lc, $D|_{S_0} \in \Gamma_+ \cap [0,1]$. Let $(S_0/G, B_0)$ be the crepant quotient of $(S_0, D|_{S_0})$ and $\bar{0} \in S_0/G$ the image of 0. Then $B_0 \in \Theta_0 := \Psi(\Gamma_+ \cap [0,1])$, and $\Gamma_+ \cap [0,1]$ and Θ_0 are DCC sets [Liu26, Lemma 2.10(1)(5)]. Since $b_0 = \operatorname{mld}(S_0/G \ni \bar{0}, B_0)$ [Liu26, Definition 2.5], Theorem 2.9(1) implies that the numbers b_0 belong to the ACC set of minimal log discrepancies of lc surface germs with coefficients in Θ_0, and the ACC set depends only on Γ.

Step 5. In this step we prove the upper bound of (1).

Let n_0 be the positive integer of Theorem 2.9(2) for the DCC set Θ_0, fix an integer $k_0 > 4n_0 + n_0/\gamma + 1$, and let $C := k_0 + 2$. Then η and C depend only on Γ. By Theorem 2.9(2), there exists a prime divisor E over S_0/G with center $\bar{0}$ such that $a(E, S_0/G, B_0) = b_0$ and $a(E, S_0/G, 0) \le n_0$. Let ν be a prime divisor over S_0 lying over E (it exists by [Liu26, Lemma 2.6(3)]). Since 0 is the only preimage of $\bar{0}$, $\operatorname{center}_{S_0} \nu = \{0\}$ [Liu26, Lemma 2.6(2)]. Let $e := |I_G(\nu)|$. Let $B_{\mathrm{br}} \ge 0$ be the boundary of the crepant quotient of $(S_0, 0)$. By [Liu26, Lemma 2.6(2)],

$$a(\nu, S_0, D|_{S_0}) = eb_0 \quad \text{and} \quad a(\nu, S_0, 0) = ea(E, S_0/G, B_{\mathrm{br}}) \le ea(E, S_0/G, 0) \le en_0.$$

In particular, ν computes b_0. Since $a(\nu, S_0, D|_{S_0}) \ge 0$, we have

$$\sum_j c_j \operatorname{mult}_\nu(P_j|_{S_0}) \le a(\nu, S_0, 0) \le en_0,$$

hence

$$\operatorname{mult}_\nu(P_j|_{S_0}) \le en_0/\gamma \quad \text{for any } j. \tag{A.39}$$

The morphism $\mathbb{A}^2_{u_1,u_2} \to S_0$, $(u_1, u_2) \to (u_1^2, -u_2^2, u_1u_2)$, is the quotient by μ_2 acting by -1, is étale in codimension one, and pulls $\operatorname{div}_{S_0}(x)$ back to $2\operatorname{div}(u_1)$. Since $(\mathbb{A}^2, \operatorname{div}(u_1))$ is lc, $(S_0, \frac{1}{2}\operatorname{div}(x))$ is lc [KM98, Proposition 5.20]. Thus

$$\operatorname{mult}_\nu x \le 2a(\nu, S_0, 0) \le 2en_0. \tag{A.40}$$

Let $V := (xy + z^2 = 0) \subset \mathbb{A}^4$, so that $V = S_0 \times \mathbb{A}^1_t$, and G acts on V by $g \cdot (p, t) = (g \cdot p, \chi_t(g)t)$. Let $i_\nu \in \{1, \dots, e\}$ be the integer q of Lemma 3.1 for ν and χ_t, and choose an integer $k \equiv i_\nu \pmod e$ with $ek_0 < k \le e(k_0 + 2)$. The hypotheses of Proposition 5.4 with $S = S_0$, $s = 0$, $u = t$, the group G, the character χ_t, the coefficient β of $\operatorname{div}(t)$, $\Phi_j = P_j|_V$, and the integer k hold: G acts faithfully on S_0 by assumption, $\beta < 1$, $P_j|_V$ is a G-semi-invariant polynomial with $(P_j|_V)|_{t=0} = P_j|_{S_0} \ne 0$ by **Step 1**, $(S_0, D|_{S_0})$ is lc near 0 by **Step 3**, ν has center 0 and computes b_0, and $k > ek_0 > en_0/\gamma \ge \operatorname{mult}_\nu(P_j|_{S_0})$ for any j by (A.39). Thus, by Proposition 5.4, the Gauss extension F of mult_ν with t-value k is a prime divisor over V with center $(0, 0)$, $I_G(F) = I_G(\nu)$, $\operatorname{mult}_F(P_j|_V) = \operatorname{mult}_\nu(P_j|_{S_0})$ for any j, and

$$\frac{a\big(F, V, \beta \operatorname{div}(t) + \sum_j c_j \operatorname{div}(P_j|_V)\big)}{|I_G(F)|} = b_0 + \frac{(1-\beta)k}{e}. \tag{A.41}$$

The birational map $\psi : V \dashrightarrow W_m$, $(x, y, z, t) \to (x, y - t^m/x, z, t)$, is G-equivariant, since $g^*(t^m/x) = \chi_t(g)^m \chi_x(g)^{-1} t^m/x = \chi_y(g)t^m/x$. Through ψ we regard F as a prime divisor over W_m. As ψ is G-equivariant and birational, the inertia group $I_G(F)$ of F over V equals the inertia group of F over W_m. Since F is a Gauss extension, $\operatorname{mult}_F x = \operatorname{mult}_\nu x$, and (A.40) and $m \ge 2$ imply

$$\operatorname{mult}_F(t^m/x) = mk - \operatorname{mult}_\nu x \ge 2k - 2en_0 > 2ek_0 - 2en_0 > en_0/\gamma.$$

We show that F has center 0 on W_m, that $a(F, W_m, 0) = a(F, V, 0)$, and that

$$\operatorname{mult}_F(P_j|_{W_m}) = \operatorname{mult}_\nu(P_j|_{S_0}).$$

(i) The pullbacks by ψ of the coordinates x, y, z, t of W_m are x, $y - t^m/x$, z, t. They have positive order along F, since F has center $(0, 0)$ on V and $\operatorname{mult}_F(t^m/x) > 0$. Hence $\operatorname{center}_{W_m} F = 0$.

(ii) For a rational 3-form θ, let $\operatorname{mult}_F \theta$ be the order of θ along F on a higher model on which F is a divisor. The form $\frac{dx}{x} \wedge dz \wedge dt$ generates the canonical sheaves of V and of W_m near 0 [KM98, Proposition 5.73], and ψ pulls it back to itself. Hence

$$a(F, W_m, 0) = 1 + \operatorname{mult}_F\left(\frac{dx}{x} \wedge dz \wedge dt\right) = a(F, V, 0).$$

(iii) On V, $\psi^*(P_j|_{W_m}) - P_j|_V = P_j(x, y - t^m/x, z, t) - P_j(x, y, z, t)$ is the product of t^m/x and a polynomial in $x, y, z, t, t^m/x$, all of which have non-negative order along F. Hence

$$\mathrm{mult}_F\left(\psi^*(P_j|_{W_m}) - P_j|_V\right) \geq \mathrm{mult}_F(t^m/x) > en_0/\gamma \geq \mathrm{mult}_F(P_j|_V),$$

so $\mathrm{mult}_F(P_j|_{W_m}) = \mathrm{mult}_F(P_j|_V) = \mathrm{mult}_\nu(P_j|_{S_0})$.

As $\psi^* t = t$, (ii) and (iii) imply

$$a(F, W_m, D + \beta \operatorname{div}(t)) = a\Big(F, V, \beta \operatorname{div}(t) + \sum_j c_j \operatorname{div}(P_j|_V)\Big).$$

By (i), (A.41), and the formula at the end of [Liu26, Lemma 2.6],

$$\mathrm{mld}((W_m \ni 0, D + \beta \operatorname{div}(t))/G) \leq \frac{a(F, W_m, D + \beta \operatorname{div}(t))}{|I_G(F)|} = b_0 + \frac{(1-\beta)k}{e} \leq b_0 + C(1-\beta).$$

Together with **Steps 3 and 4**, this proves (1).

Step 6. In this step we set up (2) and reduce it to the case $Nv > 2s$.

Notation as in Set-up 7.3 with $w = (0, 2s, s, v)$, without the bound $[G : R] \leq e_0$. For $g \in G$ we write $\chi_x(g), \chi_y(g), \chi_z(g), \chi_t(g)$ as in **Step 1**. Since G preserves the equation of W_N up to a character, $\chi_x\chi_y = \chi_z^2 = \chi_t^N$. As w is primitive, $R = \mu_r(0, 2s, s, v)$. For $\zeta \in \mu_r$ let $g_\zeta \in R$ be the element acting by $(x, y, z, t) \to (x, \zeta^{2s}y, \zeta^s z, \zeta^v t)$. If $g \in G$ acts by $(x, y, z, t) \to (x, \zeta^{2s}y, \zeta^s z, \zeta^v t)$ for some $\zeta \in \mathbb{C}^*$, then $g \in G \cap \mathbf{T} = R$, so $g = g_\xi$ for some $\xi \in \mu_r$, and $(\zeta/\xi)^s = (\zeta/\xi)^v = 1$ implies $\zeta = \xi \in \mu_r$. By Set-up 7.3(5), after multiplying each h_j by an invariant unit, which changes neither D near 0 nor the log discrepancies of prime divisors with center 0, we may assume that each h_j is a polynomial without monomials divisible by xy. From now on we work over a neighborhood of $0 \in W_N$ on which (W_N, D) is klt.

For each N with $3 \leq N \leq 2s/v$, the positive values α of all data with N fixed form an ACC set by Theorem 7.8. The union $\mathcal{A}_0$ of the finitely many ACC sets satisfies the ACC [Liu26, Lemma 2.10(4)] and depends only on Γ, s, and v. Thus a strictly increasing sequence of values α contains only finitely many terms with $Nv \leq 2s$, and from now on we assume that $Nv > 2s$, that is, $d \geq 1$.

Step 7. In this step we construct the weighted blow-up $\pi : W' \to W_N$ and compute the exceptional divisor E of π.

Let $\mathbb{A}' \to \mathbb{A}^4$ be the weighted blow-up with weights $(0, 2s, s, v)$ in x, y, z, t, that is, the product of $\mathbb{A}^1_x$ and the weighted blow-up of $\mathbb{A}^3_{y,z,t}$ at 0 with weights $(2s, s, v)$. It is projective and G-equivariant. Let $\pi : W' \to W_N$ be the strict transform of W_N. The t-, y-, and z-charts of $\mathbb{A}'$ are the quotients of $\mathbb{A}^4_{x,Y,Z,\lambda}$, $\mathbb{A}^4_{x,\lambda,Z,T_0}$, and $\mathbb{A}^4_{x,Y,\lambda,T_0}$ by μ_v, μ_{2s}, and μ_s (cf. [CLS11, Example 1.3.20]), mapped to $\mathbb{A}^4$ by

$$(x, \lambda^{2s}Y, \lambda^s Z, \lambda^v), \qquad (x, \lambda^{2s}, \lambda^s Z, \lambda^v T_0), \qquad (x, \lambda^{2s}Y, \lambda^s, \lambda^v T_0),$$

where $\zeta \in \mu_v$ (resp. μ_{2s}, μ_s) multiplies λ by ζ and the other coordinates by the inverse powers of ζ for which the maps are invariant. Since $Nv - 2s = rd$, f pulls back to $\lambda^{2s}(xY + Z^2 + \lambda^{rd})$, $\lambda^{2s}(x + Z^2 + \lambda^{rd}T_0^N)$, and $\lambda^{2s}(xY + 1 + \lambda^{rd}T_0^N)$, respectively. Hence W' is covered by the quotients of

$$\tilde{X}_t := (xY + Z^2 + \lambda^{rd} = 0) \subset \mathbb{A}^4, \qquad \tilde{X}_y := (x + Z^2 + \lambda^{rd}T_0^N = 0) \cong \mathbb{A}^3_{\lambda,Z,T_0},$$

and of $(xY + 1 + \lambda^{rd}T_0^N = 0)$. The last hypersurface is smooth, $\tilde{X}_t$ has at most one singular point, and so W' is normal. The preimage of 0 is empty in the z-chart, since $x = \lambda = 0$ contradicts $xY + 1 + \lambda^{rd}T_0^N = 0$. It is the Y-axis in $\tilde{X}_t$ and the T_0-axis in $\tilde{X}_y$. Thus $\pi^{-1}(0)$ is contained in the t- and y-charts, and it is the proper irreducible curve with weighted homogeneous coordinates $[y : t]$.

Let E be the exceptional divisor of π. The preimages of E in $\tilde{X}_t$ and $\tilde{X}_y$ are the prime divisors $(\lambda = 0)$. An element $\zeta \in \mu_v$ acting trivially on $(\lambda = 0) \subset \tilde{X}_t$ satisfies $\zeta^s = 1$, since Z does not

vanish identically on $(\lambda = 0)$, and an element $\zeta \in \mu_{2s}$ acting trivially on $(\lambda = 0) \subset \tilde{X}_y$ satisfies $\zeta^s = \zeta^v = 1$, since Z and T_0 do not vanish identically on $(\lambda = 0)$. As $\gcd(s, v) = 1$, in both cases $\zeta = 1$, so the inertia groups of $(\lambda = 0)$ for the actions of μ_v and μ_{2s} are trivial. The t- and y-charts of W' are $\tilde{X}_t/\mu_v$ and $\tilde{X}_y/\mu_{2s}$, and the intersection of E with each of them is the image of $(\lambda = 0)$. Thus, by [Liu26, Lemma 2.6, the statement before (0), (1), and (2)], the restriction of the order along $(\lambda = 0)$ to rational functions on W_N is mult_E. For a polynomial h and an integer k, let $h_{[k]}$ be the sum of the terms of h of weight k, where $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ has weight $2sa_2 + sa_3 + va_4$. If $h \neq 0$ has no monomial divisible by xy and k_1 is the least integer with $h_{[k_1]} \neq 0$, then

$$\mathrm{mult}_E h = k_1. \tag{A.42}$$

The pullback of h to $\tilde{X}_t$ is

$$h(x, \lambda^{2s}Y, \lambda^s Z, \lambda^v) = \sum_{k \geq k_1} \lambda^k h_{[k]}(x, Y, Z, 1).$$

A monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of weight k_1 is determined by a_1, a_2, a_3, since $va_4 = k_1 - 2sa_2 - sa_3$, so $h_{[k_1]}(x, Y, Z, 1)$ is non-zero and has no monomial divisible by xY. By Lemma A.1(1), in the variables x, Y, Z, it is not divisible by $xY + Z^2$. Thus $h_{[k_1]}(x, Y, Z, 1)$ does not vanish identically on $(\lambda = 0) \cong (xY + Z^2 = 0)$, which implies (A.42).

Let $q_j := \mathrm{mult}_E h_j$ and $\sigma := \sum_j c_j q_j$. The form $\omega := \frac{dx}{x} \wedge dz \wedge dt$ generates the canonical sheaf of W_N [KM98, Proposition 5.73], and the pullback of ω to $\tilde{X}_t$ is

$$v\lambda^{s+v-1}\frac{dx}{x} \wedge dZ \wedge d\lambda, \tag{A.43}$$

where $\frac{dx}{x} \wedge dZ \wedge d\lambda$ generates the canonical sheaf of $\tilde{X}_t$. Since the order along $(\lambda = 0)$ restricts to mult_E, the Hurwitz formula and (A.43) imply $a(E, W_N, 0) = s + v$. As (W_N, D) is klt near 0 and $\mathrm{center}_{W_N} E$ is the x-axis,

$$a(E, W_N, D) = s + v - \sigma > 0. \tag{A.44}$$

Step 8. In this step we describe W'/G near the fiber over 0 and compute the crepant boundary on the charts of W'/G.

Let $\bar{0} \in W_N/G$ be the image of 0, $B_{W_N/G}$ the boundary of the crepant quotient of (W_N, D), and $\pi_G : W'/G \to W_N/G$ the induced projective birational morphism. Let B' be the $\mathbb{R}$-divisor with $K_{W'/G} + B' = \pi_G^*(K_{W_N/G} + B_{W_N/G})$, so that (π_G, B') is a crepant model of $(W_N/G \ni \bar{0}, B_{W_N/G})$ (Definition-Proposition 4.10).

Let

$$\tilde{G}_t := \{(g, \zeta) \in G \times \mathbb{C}^* \mid \zeta^v = \chi_t(g)\}.$$

The group $\tilde{G}_t$, of order $v|G|$, acts on $\tilde{X}_t$ by

$$x \to \chi_x(g)x, \quad Y \to \chi_y(g)\zeta^{-2s}Y, \quad Z \to \chi_z(g)\zeta^{-s}Z, \quad \lambda \to \zeta\lambda$$

(as pullbacks). It contains the covering group $\{1\} \times \mu_v$, and the map $\tilde{X}_t \to W'$ is equivariant for $\tilde{G}_t \to G$. Hence the t-chart of W'/G is $\tilde{X}_t/\tilde{G}_t$. The subgroup $\tilde{R}_t := \{(g_\zeta, \zeta) \mid \zeta \in \mu_r\}$ fixes x, Y, Z and acts on λ through μ_r. Thus $\tilde{X}_t/\tilde{R}_t$ is

$$X_d := (xY + Z^2 + b^d = 0) \subset \mathbb{A}^4_{x,Y,Z,b}, \qquad b = \lambda^r,$$

and $H_t := \tilde{G}_t/\tilde{R}_t$, of order $v[G : R]$, acts diagonally on X_d, preserves the equation up to a character, and X_d/H_t is the t-chart of W'/G. If (g, ζ) acts trivially on x, Y, Z, then g acts by $(x, y, z, t) \to (x, \zeta^{2s}y, \zeta^s z, \zeta^v t)$, so $\zeta \in \mu_r$ and $g = g_\zeta$ by **Step 6**. Thus $(g, \zeta) \in \tilde{R}_t$. Since x, Y, Z do not vanish identically on $S_d := (b = 0) \subset X_d$, H_t acts faithfully on S_d. For the y-chart, $\tilde{G}_y := \{(g, \zeta) \mid \zeta^{2s} = \chi_y(g)\}$ acts on $\tilde{X}_y$ by $\lambda \to \zeta\lambda$, $Z \to \chi_z(g)\zeta^{-s}Z$, $T_0 \to \chi_t(g)\zeta^{-v}T_0$, the subgroup $\tilde{R}_y := \{(g_\zeta, \zeta) \mid \zeta \in \mu_r\}$ acts on λ only, and $\tilde{X}_y/\tilde{R}_y = \mathbb{A}^3_{b,Z,T_0}$ with $b = \lambda^r$. As for

H_t, if $(g,\zeta)\in\tilde{G}_y$ acts trivially on Z and T_0, then $\chi_z(g)=\zeta^s$, $\chi_t(g)=\zeta^v$, and $\chi_y(g)=\zeta^{2s}$, so $\chi_x(g)=\chi_z(g)^2\chi_y(g)^{-1}=1$ by **Step 6**. Thus g acts by $(x,y,z,t)\to(x,\zeta^{2s}y,\zeta^s z,\zeta^v t)$, so $\zeta\in\mu_r$ and $g=g_\zeta$ by **Step 6**, that is, $(g,\zeta)\in\tilde{R}_y$. Since Z and T_0 do not vanish identically on $(b=0)$, $H_y:=\tilde{G}_y/\tilde{R}_y$, of order $2s[G:R]$, acts faithfully on $(b=0)\subset\mathbb{A}^3_{b,Z,T_0}$, hence on $\mathbb{A}^3_{b,Z,T_0}$, with quotient the y-chart of W'/G. By **Step 7**, X_d/H_t and $\mathbb{A}^3_{b,Z,T_0}/H_y$ cover $\pi_G^{-1}(\bar{0})$.

As h_j is semi-invariant under R and g_ζ multiplies a monomial of weight k by ζ^k, all monomials of h_j have weights congruent to q_j modulo r, and q_j is the least of them by (A.42). Let

$$\mathcal{P}_j:=\sum_{i\geq 0}b^i(h_j)_{[q_j+ir]}(x,Y,Z,1),\qquad \mathcal{P}_j^y:=\sum_{i\geq 0}b^i(h_j)_{[q_j+ir]}(-Z^2-b^dT_0^N,1,Z,T_0).$$

Then h_j pulls back to $\lambda^{q_j}\mathcal{P}_j(x,Y,Z,\lambda^r)$ on $\tilde{X}_t$ and to $\lambda^{q_j}\mathcal{P}_j^y(\lambda^r,Z,T_0)$ on $\tilde{X}_y$. Since the order along $(\lambda=0)$ restricts to mult_E, $\mathcal{P}_j|_{S_d}\neq 0$ and $\mathcal{P}_j^y|_{(b=0)}\neq 0$. The functions $\mathcal{P}_j$ and $\mathcal{P}_j^y$ are semi-invariant under H_t and H_y, since h_j and λ^{q_j} are semi-invariant under $\tilde{G}_t$ and $\tilde{G}_y$. Since h_j pulls back to $\lambda^{q_j}\mathcal{P}_j(x,Y,Z,\lambda^r)$, (A.43) shows that $K_{W'/G}+B'$ pulls back to

$$K_{\tilde{X}_t}+(1-s-v+\sigma)\operatorname{div}(\lambda)+\sum_j c_j\operatorname{div}(\mathcal{P}_j(x,Y,Z,\lambda^r))$$

on $\tilde{X}_t$. The quotient map $\tilde{X}_t\to X_d$, $b=\lambda^r$, is ramified of index r along $(\lambda=0)$ and étale elsewhere over a neighborhood of 0 in codimension one, so by [Liu26, Lemma 2.6(4)] the coefficient β of $\operatorname{div}(b)$ satisfies $1-\beta=(s+v-\sigma)/r$, and B' pulls back to

$$B_d:=\beta\operatorname{div}(b)+\sum_j c_j\operatorname{div}(\mathcal{P}_j)\text{ on }X_d,\qquad \beta:=1-\frac{s+v-\sigma}{r}. \tag{A.45}$$

Similarly, $\omega=-\frac{dy}{y}\wedge dz\wedge dt$ (on W_N, $ydx\wedge dz\wedge dt=-xdy\wedge dz\wedge dt$ by differentiating $xy+z^2+t^N=0$) pulls back to $-2s\lambda^{s+v-1}d\lambda\wedge dZ\wedge dT_0$ on $\tilde{X}_y$, with exponent $s+v-1$ as on $\tilde{X}_t$, so B' pulls back to

$$B_y:=\beta\operatorname{div}(b)+\sum_j c_j\operatorname{div}(\mathcal{P}_j^y)\quad\text{on }\mathbb{A}^3_{b,Z,T_0}.$$

Assume that $r\geq s+v$. Then $0\leq\beta<1$ by (A.44), so $B_d,B_y\geq 0$, and hence $B'\geq 0$ over a neighborhood of $\bar{0}$. The pair $(W_N/G,B_{W_N/G})$ is klt near $\bar{0}$ [Liu26, Lemma 2.6(2)(3)], hence $(W'/G,B')$ is klt over a neighborhood of $\bar{0}$ [KM98, Lemma 2.30], and (X_d,B_d) and $(\mathbb{A}^3_{b,Z,T_0},B_y)$ are klt near the preimages of $\pi_G^{-1}(\bar{0})$ [Liu26, Lemma 2.6(0)(2)]. Let

$$\Theta_1:=\left\{1-\frac{s+v-\sigma'}{r'}\;\middle|\;\sigma'\in\Gamma_+,\ \sigma'<s+v,\ r'\in\mathbb{Z},\ r'\geq s+v\right\},\qquad \Theta:=\Theta_1\cup(\Gamma_+\cap[0,1]).$$

The set $\Theta\subset[0,1]$ is a DCC set depending only on Γ, s, and v, since $\Gamma_+\cap[0,1]$ is a DCC set [Liu26, Lemma 2.10(1)] and Θ_1 is a DCC set: a strictly decreasing sequence in Θ_1 with bounded r' has a subsequence with constant r', whose σ' strictly decrease in the DCC set Γ_+. With unbounded r' it has a subsequence tending to 1, which is impossible for a strictly decreasing sequence of numbers less than 1. Since $\mathcal{P}_j$ and $\mathcal{P}_j^y$ do not vanish identically on $(b=0)$, the coefficient of $(b=0)$ in B_d and B_y is $\beta\in\Theta_1$, and the other coefficients belong to $\Gamma_+\cap[0,1]$. Thus $B_d,B_y\in\Theta$, and the coefficients of B' over a neighborhood of $\bar{0}$ belong to the DCC set $\Psi(\Theta)$ [Liu26, Lemma 2.10(1)(5)].

Step 9. In this step we show, for $r\geq s+v$, that α is the least minimal log discrepancy at the points of $\pi_G^{-1}(\bar{0})$, and that only the image of $0\in X_d$ with $d\geq 2$ can have minimal log discrepancy outside two fixed ACC sets.

Any prime divisor over W_N/G with center $\bar{0}$ has center on W'/G the closure of a scheme-theoretic point of $\pi_G^{-1}(\bar{0})$, and conversely any prime divisor over W'/G whose center is contained

in $\pi_G^{-1}(\bar{0})$ has center $\bar{0}$ on W_N/G. The log discrepancies with respect to $B_{W_N/G}$ and B' are equal [KM98, Lemma 2.30]. As $\alpha = \mathrm{mld}(W_N/G \ni \bar{0}, B_{W_N/G})$ [Liu26, Definition 2.5],

$$\alpha = \min\left\{\mathrm{mld}(W'/G \ni \xi, B') \;\middle|\; \xi \in \pi_G^{-1}(\bar{0})\right\}. \tag{A.46}$$

Let $\mathcal{A}_1$ be the ACC set of Theorem 2.8(1) for the DCC set Θ, and $\mathcal{A}_2$ the ACC set of [Liu26, Lemma 3.7] for the DCC set $\Psi(\Theta)$. Both depend only on Γ, s, and v. Let $o_d \in X_d/H_t$ be the image of $0 \in X_d$.

(i) If ξ is the generic point of the curve $\pi_G^{-1}(\bar{0})$, then $\mathrm{mld}(W'/G \ni \xi, B') \in \mathcal{A}_2$ by **Step 8**.

(ii) If ξ is a closed point of $\mathbb{A}^3_{b,Z,T_0}/H_y$, or of X_d/H_t other than o_d when $d \geq 2$, let p be a preimage of ξ in $\mathbb{A}^3_{b,Z,T_0}$ or X_d, and let H_p be the stabilizer of p. The variety $\mathbb{A}^3_{b,Z,T_0}$ is smooth, and X_d is smooth outside 0, and smooth if $d = 1$. A prime divisor F with center p has $I_{H_y}(F)$ (resp. $I_{H_t}(F)$) contained in H_p, so $I_{H_p}(F) = I_{H_y}(F)$ (resp. $I_{H_p}(F) = I_{H_t}(F)$), and the prime divisors with center ξ lift, after a translation by an element of H_y (resp. H_t), to prime divisors with center p. Hence, by [Liu26, Lemma 2.6(2)(3)] and the formula at the end of [Liu26, Lemma 2.6], $\mathrm{mld}(W'/G \ni \xi, B')$ is $\mathrm{mld}((\mathbb{A}^3_{b,Z,T_0} \ni p, B_y)/H_p)$ or $\mathrm{mld}((X_d \ni p, B_d)/H_p)$, which belongs to $\mathcal{A}_1$ by **Step 8**.

(iii) If $d \geq 2$, then, by [Liu26, Lemma 2.6(0)] and the formula at the end of [Liu26, Lemma 2.6], $\mathrm{mld}(W'/G \ni o_d, B') = \mathrm{mld}((X_d \ni 0, B_d)/H_t)$.

Step 10. In this step we prove (2).

Assume that there exists a sequence of data as in Set-up 7.3 with $w = (0, 2s, s, v)$, indexed by i, with $r_i \to +\infty$ and α_i positive and strictly increasing. By **Step 6**, after dropping finitely many terms, $N_i v > 2s$ and $r_i \geq s + v$ for any i. By (A.46), $\alpha_i = \mathrm{mld}(W_i'/G_i \ni \xi_i, B_i')$ for some $\xi_i \in \pi_{G_i}^{-1}(\bar{0})$. Since $\mathcal{A}_1 \cup \mathcal{A}_2$ satisfies the ACC [Liu26, Lemma 2.10(4)], **Step 9** implies that for $i \gg 0$, $d_i \geq 2$ and

$$\alpha_i = \mathrm{mld}((X_{d_i} \ni 0, B_{d_i})/H_{t,i}).$$

Since $0 < s + v - \sigma_i \leq s + v$ and $r_i \to +\infty$, $\beta_i \to 1$, so $\beta_i > 1 - \eta$ for $i \gg 0$. We apply (1) with $m := d_i$, the coordinates (x, Y, Z, b) in place of (x, y, z, t), the group $H_{t,i}$ in place of G, the labelled boundary $\sum_j c_j \operatorname{div}(\mathcal{P}_j)$ in place of D, and $\beta := \beta_i$. The hypotheses of (1) hold by **Step 8**: $H_{t,i}$ is a finite diagonal abelian group preserving the equation of X_{d_i} up to a character and acting faithfully on S_{d_i}, $c_j \in \Gamma$, $\mathcal{P}_j|_{S_{d_i}} \neq 0$, and (X_{d_i}, B_{d_i}) is klt near 0. Thus

$$b_{0,i} \leq \alpha_i \leq b_{0,i} + C(1 - \beta_i),$$

where the $b_{0,i}$ belong to an ACC set depending only on Γ. Passing to a subsequence, we may assume that the $b_{0,i}$ are non-increasing. Then $\alpha_i \leq b_{0,1} + C$, so α_i converges to a real number A, and $\alpha_i < A$ for any i. Since $1 - \beta_i \to 0$, $b_{0,i} \to A$, and since the $b_{0,i}$ are non-increasing, $b_{0,i} \geq A > \alpha_i \geq b_{0,i}$, a contradiction. Hence α cannot increase along $\mathbf{T}$.

Fix an integer $d_0 \geq 1$, and let α_i be a strictly increasing sequence of positive values with $0 < d_i \leq d_0$. If the r_i are unbounded, a subsequence with $r_i \to +\infty$ contradicts the previous paragraph. Otherwise $N_i = (d_i r_i + 2s)/v$ is bounded, so infinitely many N_i are equal, which contradicts Theorem 7.8. This proves (2). □

A.6. **The circle** $(0, -2, -1, 1)$**.** In this subsection, we prove Proposition A.5 and Theorem 7.12. Proposition A.5 compares α with the minimal log discrepancy of the quotient of a pair on a fixed plane.

Proposition A.5. *Notation as in Set-up 7.3 with* $w = (0, -2, -1, 1)$*. Let*

$$\varphi_j(p, q) := h_j^0(-p^2, 1, p, q) \qquad \text{and} \qquad C := \sum_j c_j \operatorname{div}(\varphi_j) \quad \text{on } \mathbb{A}^2_{p,q}.$$

For $g \in G$ write $g = (g_x, g_y, g_z, g_t)$, where g acts on $\mathbb{A}^4$ by $(x, y, z, t) \to (g_x x, g_y y, g_z z, g_t t)$. Let $\tilde{H}$ be the quotient of the group of pairs (g, ξ) with $g \in G$ and $\xi^2 = g_y$ by the subgroup

$$\tilde{R} := \{((1, \lambda^{-2}, \lambda^{-1}, \lambda), \lambda^{-1}) : \lambda^r = 1\},$$

acting on $\mathbb{A}^2_{p,q}$ by $(p, q) \to (g_z \xi^{-1} p, g_t \xi q)$, and let $b := \mathrm{mld}((\mathbb{A}^2 \ni 0, C)/\tilde{H})$. Then the following hold.

(1) The group $\tilde{H}$ has order $2[G : R]$ and acts faithfully on $\mathbb{A}^2_{p,q}$.
(2) There exist r_0 and C_0, depending only on Γ and e_0, such that if $\tau \geq 0$ and $r \geq r_0$, then $(\mathbb{A}^2, C)$ is lc near 0, b belongs to an ACC set depending only on Γ and e_0, and $\alpha \leq b + C_0/r$.
(3) For any $\theta > 1$ there exists r_1, depending only on Γ, e_0, and θ, such that if $\tau \geq 0$, $r \geq r_1$, and $(\mathbb{A}^2, \theta C)$ is lc near 0, then $\alpha \geq b$.
(4) Under the condition $\tau > 0$, α cannot increase along $\mathbf{T}$.

Proof of Proposition A.5(1) and (2). The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1).

Since g preserves the equation $xy + z^2 + t^N$ of W_N up to a character,

$$g_x g_y = g_z^2 = g_t^N \quad \text{for any } g \in G. \tag{A.47}$$

Let $\tilde{G}$ be the group of pairs (g, ξ) with $g \in G$ and $\xi^2 = g_y$, with the componentwise product. The homomorphism $\tilde{G} \to G$, $(g, \xi) \to g$, is surjective with kernel $\{(1, 1), (1, -1)\}$, so $|\tilde{G}| = 2|G|$. Since

$$\mathbf{T} = \{(1, \lambda^{-2}, \lambda^{-1}, \lambda) : \lambda \in \mathbb{C}^*\}$$

and $R = G \cap \mathbf{T}$ has order r,

$$R = \{(1, \lambda^{-2}, \lambda^{-1}, \lambda) : \lambda^r = 1\}.$$

Thus $\tilde{R}$ is a subgroup of $\tilde{G}$ of order r, and

$$|\tilde{H}| = \frac{2|G|}{r} = 2[G : R].$$

For the element (g, ξ) of $\tilde{R}$ given by λ, we have $g_z = \xi = \lambda^{-1}$ and $g_t = \lambda$, so $g_z \xi^{-1} = 1$ and $g_t \xi = 1$. Thus the action of $\tilde{G}$ on $\mathbb{A}^2_{p,q}$ by $(p, q) \to (g_z \xi^{-1} p, g_t \xi q)$ factors through $\tilde{H}$.

Assume that $(g, \xi) \in \tilde{G}$ satisfies $g_z \xi^{-1} = g_t \xi = 1$. Then $\xi = g_z = g_t^{-1}$ and $g_y = \xi^2 = g_t^{-2}$. By (A.47), $g_t^{-2} = g_z^2 = g_t^N$, and

$$g_x = \frac{g_t^N}{g_y} = g_t^{N+2} = 1.$$

Thus $g = (1, \lambda^{-2}, \lambda^{-1}, \lambda)$ with $\lambda := g_t$, so $g \in G \cap \mathbf{T} = R$, $\lambda^r = 1$, and $\xi = \lambda^{-1}$. Therefore $(g, \xi) \in \tilde{R}$, and $\tilde{H}$ acts faithfully on $\mathbb{A}^2_{p,q}$. This proves (1).

Step 2. In this step we normalize the functions h_j and bound the mass τ. From now on we assume that $\tau \geq 0$.

The functions h_j are the polynomials without monomials divisible by xy of Set-up 7.3(5), and κ_j, h_j^0, τ, C, and b are defined from them. Removing the terms with $c_j = 0$ or $h_j(0) \neq 0$ changes none of D near 0, κ_j, h_j^0, τ, C, and b (Set-up 7.3(6) and (7)). Thus we may assume that $c_j > 0$ for any j and that each h_j vanishes at 0.

Let $\gamma := \min(\Gamma \backslash \{0\})$ if $\Gamma \not\subset \{0\}$, and $\gamma := 1$ otherwise, and let $m_0 := \lceil 2/\gamma \rceil$. For any j, let m_j be the least total degree of a monomial of h_j. By Lemma A.1(3), since (W_N, D) is klt near 0 and the h_j are polynomials without monomials divisible by xy vanishing at 0,

$$\sum_j c_j m_j < 2. \tag{A.48}$$

Since $c_j \geq \gamma$, (A.48) gives $m_j < 2/\gamma \leq m_0$ for any j. By Lemma 5.3, applied with $M = m_0$ and $\max_i |w_i| = 2$, if $r > 4m_0$, then any two monomials of h_j have $\mathbf{T}$-weights congruent modulo r, $|\kappa_j| \leq 2m_0$, and every monomial of $h_j - h_j^0$ has total degree at least $(r - 2m_0)/2$ for any j. We assume that $r > 4m_0$ from now on. The $\mathbf{T}$-weight of $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ is $a_4 - 2a_2 - a_3 \leq a_1 + a_2 + a_3 + a_4$, so $\kappa_j \leq m_j$. By (A.48),

$$0 \leq \tau = \sum_j c_j\kappa_j \leq \sum_j c_j m_j < 2. \tag{A.49}$$

For $l \in \mathbb{Z}$, let $h_j^{(l)}$ be the sum of the terms of h_j of $\mathbf{T}$-weight $\kappa_j + lr$. Then $h_j = \sum_l h_j^{(l)}$, a finite sum, and $h_j^{(0)} = h_j^0$. For any monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of $h_j^{(l)}$,

$$a_4 = \kappa_j + lr + 2a_2 + a_3 \geq \kappa_j + lr. \tag{A.50}$$

Step 3. In this step we express the quotient log discrepancies of prime divisors over W_N by log discrepancies over $\mathbb{A}^3_{u,p,q}$.

Let $\tilde{G}$ act on $\mathbb{A}^3_{\sigma,p,q}$ by $(\sigma, p, q) \to (\xi\sigma, g_z\xi^{-1}p, g_t\xi q)$ for $(g, \xi) \in \tilde{G}$, and let

$$\psi : \mathbb{A}^3_{\sigma,p,q} \dashrightarrow W_N, \quad (\sigma, p, q) \to \Big(-p^2 - \frac{q^N}{\sigma^{N+2}},\ \sigma^2,\ p\sigma,\ \frac{q}{\sigma}\Big),$$

which is well defined because $(-p^2 - q^N\sigma^{-N-2})\sigma^2 + (p\sigma)^2 + (q/\sigma)^N = 0$. By (A.47),

$$\frac{g_z^2}{\xi^2} = \frac{g_z^2}{g_y} = g_x \quad \text{and} \quad \frac{g_t^N\xi^N}{\xi^{N+2}} = \frac{g_t^N}{g_y} = g_x,$$

so $\psi((g, \xi) \cdot (\sigma, p, q)) = g \cdot \psi(\sigma, p, q)$. The fiber of ψ over a general point $(x, y, z, t) \in W_N$ consists of the points $(\sigma_0, z/\sigma_0, t\sigma_0)$ and $(-\sigma_0, -z/\sigma_0, -t\sigma_0)$, where $\sigma_0^2 = y$, and they form an orbit of $\iota := (1, -1) \in \tilde{G}$, which acts by $(\sigma, p, q) \to (-\sigma, -p, -q)$. Thus ψ induces a birational map $\mathbb{A}^3_{\sigma,p,q}/\langle\iota\rangle \dashrightarrow W_N$. Since $\tilde{G}/\langle\iota\rangle \cong G$ compatibly with ψ, ψ induces a birational map $\mathbb{A}^3_{\sigma,p,q}/\tilde{G} \dashrightarrow W_N/G$.

Let $\pi : \mathbb{A}^3_{\sigma,p,q} \to \mathbb{A}^3_{u,p,q}$, $(\sigma, p, q) \to (\sigma^r, p, q)$. The element $((1, \lambda^{-2}, \lambda^{-1}, \lambda), \lambda^{-1})$ of $\tilde{R}$ acts by $(\sigma, p, q) \to (\lambda^{-1}\sigma, p, q)$, so π is the quotient by $\tilde{R}$. The group $\tilde{H}$ acts on $\mathbb{A}^3_{u,p,q}$ by

$$(u, p, q) \to (\xi^r u, g_z\xi^{-1}p, g_t\xi q),$$

which is well defined because $\xi^r = \lambda^{-r} = 1$ on $\tilde{R}$, and π is equivariant. Thus $\mathbb{A}^3_{u,p,q}/\tilde{H} = \mathbb{A}^3_{\sigma,p,q}/\tilde{G}$, which is birational to W_N/G. By (1), $\tilde{H}$ acts faithfully on $\mathbb{A}^3_{u,p,q}$.

Let $\bar{x} := -p^2 - q^N/u^d$. Since $N + 2 = dr$ (Set-up 7.3(4)), $\psi^*x = \pi^*\bar{x}$. For a monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of $h_j^{(l)}$, (A.50) implies $2a_2 + a_3 - a_4 = -\kappa_j - lr$, hence

$$\psi^*(x^{a_1}y^{a_2}z^{a_3}t^{a_4}) = (\psi^*x)^{a_1}\sigma^{2a_2+a_3-a_4}p^{a_3}q^{a_4} = \sigma^{-\kappa_j}\pi^*(u^{-l}\bar{x}^{a_1}p^{a_3}q^{a_4}).$$

Thus

$$\psi^*h_j = \sigma^{-\kappa_j}\pi^*\tilde{h}_j, \qquad \text{where} \quad \tilde{h}_j := \sum_l u^{-l}h_j^{(l)}(\bar{x}, 1, p, q) \tag{A.51}$$

is a rational function on $\mathbb{A}^3_{u,p,q}$. By definition, $\varphi_j = h_j^{(0)}(-p^2, 1, p, q)$. Any monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of $h_j^{(0)}$ satisfies $a_4 - 2a_2 - a_3 = \kappa_j$ and $a_1a_2 = 0$, and the substitution sends it to $(-1)^{a_1}p^{2a_1+a_3}q^{a_4}$. The exponents $2a_1 + a_3$ and a_4 determine $2(a_1 - a_2) = (2a_1 + a_3) - (a_4 - \kappa_j)$, hence a_1 and a_2 because $a_1a_2 = 0$, and then a_3. Thus distinct monomials of $h_j^{(0)}$ are sent to distinct monomials in p, q. Since $h_j^{(0)} = h_j^0$ contains the fixed monomial of least degree of h_j, $h_j^{(0)} \neq 0$, and therefore $\varphi_j \neq 0$. The proof of $\varphi_j \neq 0$ uses neither $\tau \geq 0$ nor $r > 4m_0$, and holds for any h_j of Set-up 7.3(5),

so $\varphi_j \neq 0$ for any data as in the proposition, and C in the statement is defined. For $(g,\xi) \in \tilde{G}$ and a monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of $h_j^{(0)}$, we have $a_4 - 2a_2 - a_3 = \kappa_j$, hence

$$g_x^{a_1}(g_z\xi^{-1})^{a_3}(g_t\xi)^{a_4} = g_x^{a_1}g_y^{a_2}g_z^{a_3}g_t^{a_4}\xi^{\kappa_j}.$$

Since p^2 has the character $g_z^2\xi^{-2} = g_x$ of x by (A.47), the left-hand side is the value at (g,ξ) of the character of $(-p^2)^{a_1}p^{a_3}q^{a_4}$. Since h_j is G-semi-invariant, $g_x^{a_1}g_y^{a_2}g_z^{a_3}g_t^{a_4}$ is the value at g of the character of h_j for any monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of h_j, so the right-hand side does not depend on the monomial of $h_j^{(0)}$. Thus φ_j is $\tilde{H}$-semi-invariant.

Let ω be the Poincaré residue of $dx\wedge dy\wedge dz\wedge dt/(xy+z^2+t^N)$ along W_N. Then ω generates ω_{W_N}, and possibly replacing ω with $-\omega$, we may assume that $\omega = dy\wedge dz\wedge dt/y$ on $W_N \cap (y \neq 0)$. A direct calculation shows that

$$\psi^*\omega = \frac{2d\sigma\wedge dp\wedge dq}{\sigma} = \frac{2}{r}\pi^*\Big(\frac{du\wedge dp\wedge dq}{u}\Big). \tag{A.52}$$

Let

$$\Delta := \Big(1-\frac{\tau}{r}\Big)\operatorname{div}(u) + \sum_j c_j \operatorname{div}(\tilde{h}_j) \quad \text{on } \mathbb{A}^3_{u,p,q}.$$

Let P, F, and $\tilde{\nu}$ be prime divisors over $\mathbb{A}^3_{\sigma,p,q}$, W_N, and $\mathbb{A}^3_{u,p,q}$, respectively, and e_1, e_2 positive integers with

$$\operatorname{mult}_P(\psi^*f) = e_1 \operatorname{mult}_F f \quad \text{and} \quad \operatorname{mult}_P(\pi^*f') = e_2 \operatorname{mult}_{\tilde{\nu}} f'$$

for any non-zero rational functions f on W_N and f' on $\mathbb{A}^3_{u,p,q}$. We show that

$$\frac{a(F,W_N,D)}{|I_G(F)|} = \frac{a(\tilde{\nu},\mathbb{A}^3_{u,p,q},\Delta)}{|I_{\tilde{H}}(\tilde{\nu})|}. \tag{A.53}$$

For a non-zero rational 3-form η, we write $\operatorname{mult}_F\eta$ for the order along F of the pullback of η to a higher model on which F is a divisor, and similarly for $\tilde{\nu}$ and P. Since ω generates ω_{W_N} and $du\wedge dp\wedge dq$ generates $\omega_{\mathbb{A}^3}$,

$$a(F,W_N,D) = 1 + \operatorname{mult}_F\omega - \sum_j c_j \operatorname{mult}_F h_j,$$

$$a(\tilde{\nu},\mathbb{A}^3_{u,p,q},\Delta) = 1 + \operatorname{mult}_{\tilde{\nu}}\Big(\frac{du\wedge dp\wedge dq}{u}\Big) + \frac{\tau}{r}\operatorname{mult}_{\tilde{\nu}} u - \sum_j c_j \operatorname{mult}_{\tilde{\nu}}\tilde{h}_j.$$

Let η and η' be non-zero rational 3-forms on W_N and $\mathbb{A}^3_{u,p,q}$, respectively. Let $B_1 := -\operatorname{div}(\psi^*\eta)$, which is $\langle\iota\rangle$-invariant, and $B_2 := -\operatorname{div}(\pi^*\eta')$, which is $\tilde{R}$-invariant. Taking the canonical divisor of $\mathbb{A}^3_{\sigma,p,q}$ to be $\operatorname{div}(\psi^*\eta)$, resp. $\operatorname{div}(\pi^*\eta')$, we have $K+B_1 = 0$, resp. $K+B_2 = 0$. Thus the boundary of the crepant quotient of $(\mathbb{A}^3_{\sigma,p,q}, B_1)$ by $\langle\iota\rangle$ is $-\operatorname{div}(\eta)$, through the birational map $\mathbb{A}^3_{\sigma,p,q}/\langle\iota\rangle \dashrightarrow W_N$, and the boundary of the crepant quotient of $(\mathbb{A}^3_{\sigma,p,q}, B_2)$ by $\tilde{R}$ is $-\operatorname{div}(\eta')$. Moreover, $a(F,W_N,-\operatorname{div}(\eta)) = 1+\operatorname{mult}_F\eta$ does not depend on the birational model of W_N, and

$$a(\tilde{\nu},\mathbb{A}^3_{u,p,q},-\operatorname{div}(\eta')) = 1+\operatorname{mult}_{\tilde{\nu}}\eta'.$$

By [Liu26, Lemma 2.6(1)], $e_1 = |I_{\langle\iota\rangle}(P)|$ and $e_2 = |I_{\tilde{R}}(P)|$. Thus [Liu26, Lemma 2.6(2)], for the quotient by $\langle\iota\rangle$ with B_1 and for the quotient π by $\tilde{R}$ with B_2, implies

$$1+\operatorname{mult}_P(\psi^*\eta) = e_1(1+\operatorname{mult}_F\eta) \quad \text{and} \quad 1+\operatorname{mult}_P(\pi^*\eta') = e_2(1+\operatorname{mult}_{\tilde{\nu}}\eta').$$

Since $\sigma^r = \pi^* u$, $\operatorname{mult}_P \sigma = e_2 \operatorname{mult}_{\tilde{\nu}} u/r$. Thus (A.51) and (A.52) imply

$$\begin{aligned} e_1 a(F, W_N, D) &= 1 + \operatorname{mult}_P(\psi^*\omega) - \sum_j c_j \operatorname{mult}_P(\psi^* h_j) \\ &= 1 + \operatorname{mult}_P \pi^*\Big(\frac{du \wedge dp \wedge dq}{u}\Big) - \sum_j c_j\big(\operatorname{mult}_P(\pi^*\tilde{h}_j) - \kappa_j \operatorname{mult}_P \sigma\big) \\ &= e_2 a(\tilde{\nu}, \mathbb{A}^3_{u,p,q}, \Delta). \end{aligned}$$

By [Liu26, Lemma 2.6, the statement before (0), and (1)], the restriction of mult_F to the rational functions on W_N/G is $|I_G(F)| \operatorname{mult}_{\bar{F}}$ for a prime divisor $\bar{F}$ over W_N/G, and the restriction of $\operatorname{mult}_{\tilde{\nu}}$ to the rational functions on $\mathbb{A}^3_{u,p,q}/\tilde{H}$ is $|I_{\tilde{H}}(\tilde{\nu})| \operatorname{mult}_{\bar{F}'}$ for a prime divisor $\bar{F}'$ over $\mathbb{A}^3_{u,p,q}/\tilde{H}$. Under the birational map $\mathbb{A}^3_{u,p,q}/\tilde{H} \dashrightarrow W_N/G$, the restriction of mult_P to the rational functions on the quotient is both $e_1|I_G(F)| \operatorname{mult}_{\bar{F}}$ and $e_2|I_{\tilde{H}}(\tilde{\nu})| \operatorname{mult}_{\bar{F}'}$. Thus $\bar{F} = \bar{F}'$ and $e_1|I_G(F)| = e_2|I_{\tilde{H}}(\tilde{\nu})|$, and (A.53) follows.

Step 4. In this step we construct prime divisors over W_N with center 0 from prime divisors over $\mathbb{A}^2_{p,q}$.

Let ν be a prime divisor over $\mathbb{A}^2_{p,q}$ with center 0, let $m := \min\{\operatorname{mult}_\nu p, \operatorname{mult}_\nu q\} \geq 1$, and let k be a positive integer with

$$k < (r - 2m_0)m. \tag{A.54}$$

By Lemma 3.1, applied to the trivial group, $V = \mathbb{A}^2_{p,q}$, $B = 0$, the coordinate u in place of w, and $\ell = k$, the Gauss extension of mult_ν with u-value k is $\operatorname{mult}_{\tilde{\nu}}$ for a prime divisor $\tilde{\nu}$ over $\mathbb{A}^3_{u,p,q}$ with center 0, and

$$a(\tilde{\nu}, \mathbb{A}^3_{u,p,q}, 0) = a(\nu, \mathbb{A}^2_{p,q}, 0) + k. \tag{A.55}$$

Since π is the quotient of $\mathbb{A}^3_{\sigma,p,q}$ by the group $\tilde{R} \cong \mu_r$ acting on σ (**Step 3**), by [Liu26, Lemma 2.6(3)] there exist a prime divisor P over $\mathbb{A}^3_{\sigma,p,q}$ lying over $\tilde{\nu}$ [Liu26, Definition 2.7] and a positive integer e_2 with $\operatorname{mult}_P(\pi^* f') = e_2 \operatorname{mult}_{\tilde{\nu}} f'$ for any non-zero rational function f' on $\mathbb{A}^3_{u,p,q}$. Let $v := \operatorname{mult}_P / e_2$, and write $v(f)$ for $v(\psi^* f)$ and $v(f')$ for $v(\pi^* f')$. In particular, $v(f') = \operatorname{mult}_{\tilde{\nu}} f'$ for any non-zero rational function f' on $\mathbb{A}^3_{u,p,q}$, by the choice of e_2, and $\operatorname{mult}_{\tilde{\nu}} f'' = \operatorname{mult}_\nu f''$ for any non-zero rational function f'' on $\mathbb{A}^2_{p,q}$, by the definition of the Gauss extension. Then

$$v(\sigma) = \frac{k}{r}, \quad v(p) = \operatorname{mult}_\nu p \geq m, \quad v(q) = \operatorname{mult}_\nu q \geq m.$$

By (A.54), $r > 2m_0 \geq 2$ and $rv(q) - k \geq rm - k > 0$. Since $d \geq 1$ and $m_0 \geq 1$,

$$v\Big(\frac{q^N}{u^d}\Big) = Nv(q) - dk = d\big(rv(q) - k\big) - 2v(q) \geq (r-2)v(q) - k \geq (r - 2m_0)m - k > 0. \tag{A.56}$$

Thus

$$v(y) = \frac{2k}{r} > 0, \quad v(z) = v(p) + \frac{k}{r} > 0, \quad v(t) = v(q) - \frac{k}{r} > 0,$$

$$v(x) = v(\bar{x}) \geq \min\Big\{2v(p), v\Big(\frac{q^N}{u^d}\Big)\Big\} > 0.$$

By **Step 3**, ψ induces a birational map $\mathbb{A}^3_{\sigma,p,q}/\langle\iota\rangle \dashrightarrow W_N$, so the functions $\psi^* f$, for the rational functions f on W_N, are exactly the ι-invariant rational functions on $\mathbb{A}^3_{\sigma,p,q}$. By [Liu26, Lemma 2.6, the statement before (0)] for the group $\langle\iota\rangle$, there exist a positive integer e_1 and a prime divisor F over $\mathbb{A}^3_{\sigma,p,q}/\langle\iota\rangle$, that is, a prime divisor F over W_N, with $\operatorname{mult}_P(\psi^* f) = e_1 \operatorname{mult}_F f$ for any non-zero rational function f on W_N. Since x, y, z, t have positive values, $\operatorname{center}_{W_N} F = \{0\}$. In particular, (A.53) holds for F and $\tilde{\nu}$.

We show that

$$\operatorname{mult}_{\tilde\nu}(\tilde h_j-\varphi_j)\ge \min\{k,(r-2m_0)m-k\}\quad\text{for any } j. \tag{A.57}$$

By (A.51),

$$\tilde h_j-\varphi_j=\sum_{l\ne 0}u^{-l}h_j^{(l)}(\bar x,1,p,q)+\Big(h_j^{(0)}(\bar x,1,p,q)-h_j^{(0)}(-p^2,1,p,q)\Big).$$

Since $\bar x$, p, and q have positive values, any term $u^n\bar x^{a_1}p^{a_3}q^{a_4}$ with $n\ge 1$ has value at least k. For $l\ge 1$ and a monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of $h_j^{(l)}$, (A.50) and $\kappa_j\ge -2m_0$ imply

$$v(u^{-l}\bar x^{a_1}p^{a_3}q^{a_4})\ge -lk+(\kappa_j+lr)v(q)=l\big(rv(q)-k\big)+\kappa_j v(q)\ge (r-2m_0)v(q)-k\ge (r-2m_0)m-k.$$

The difference $h_j^{(0)}(\bar x,1,p,q)-h_j^{(0)}(-p^2,1,p,q)$ is the product of $\bar x+p^2=-q^N/u^d$ and a polynomial in $\bar x,p,q$. By (A.56), it has value at least $(r-2m_0)m-k$. This proves (A.57).

Step 5. In this step we show that there exists a positive integer r_2, depending only on Γ, such that $(\mathbb{A}^2,C)$ is lc near 0 if $r\ge r_2$.

By [dFEM11, Theorem 1.1], the numbers $\operatorname{lct}(\mathbb{A}^2\ni 0;\prod_j\mathfrak{a}_j^{c_j})$ with $c_j\in\Gamma$ satisfy the ACC. Thus there exists a real number $\delta\in(0,1)$, depending only on Γ, such that any such number less than one is at most $1-\delta$. Let $\ell:=\lceil 8/(\delta\gamma)\rceil$, which depends only on Γ. By Theorem 2.12(2) (that is, [dFEM11, Propositions 2.11 and 2.19(ii)]) with $n=2$ and $\epsilon=\delta/4$, if $J\ne\emptyset$ is a set of indices with $\varphi_j(0)=0$ for $j\in J$, $c_j\ge\gamma$ for $j\in J$, and $\mathfrak{b}_j:=(\varphi_j)+(p,q)^\ell$, then $\operatorname{lct}(\mathbb{A}^2\ni 0;\prod_{j\in J}(\varphi_j)^{c_j})$ and $\operatorname{lct}(\mathbb{A}^2\ni 0;\prod_{j\in J}\mathfrak{b}_j^{c_j})$ differ by at most $\delta/4$. Let $k_0:=\ell+1$ and

$$r_2:=\max\Big\{4m_0+1,2m_0+2k_0,\Big\lfloor\frac{4k_0}{\delta\gamma}\Big\rfloor+1\Big\}.$$

Assume that $r\ge r_2$ and that $(\mathbb{A}^2,C)$ is not lc near 0. Let J_0 be the set of indices j with $\varphi_j(0)=0$, $\mathfrak{b}:=\prod_{j\in J_0}\mathfrak{b}_j^{c_j}$,

$$\mu:=\operatorname{lct}\Big(\mathbb{A}^2\ni 0;\prod_{j\in J_0}(\varphi_j)^{c_j}\Big),\quad\text{and}\quad \mu_0:=\operatorname{lct}(\mathbb{A}^2\ni 0;\mathfrak{b}).$$

Then $J_0\ne\emptyset$ and $\mu<1$, hence $\mu\le 1-\delta$, and

$$\mu_0\le\mu+\frac{\delta}{4}\le 1-\frac{3\delta}{4}<1-\frac{\delta}{2}.$$

Let ν be a prime divisor over $\mathbb{A}^2_{p,q}$ computing μ_0. Since each $\mathfrak{b}_j$ with $j\in J_0$ is (p,q)-primary, $\operatorname{center}_{\mathbb{A}^2}\nu=\{0\}$. Let $m:=\min\{\operatorname{mult}_\nu p,\operatorname{mult}_\nu q\}$. Then

$$\operatorname{mult}_\nu\mathfrak{b}_j=\min\{\operatorname{mult}_\nu\varphi_j,\ell m\}\ge m\quad\text{for } j\in J_0,$$

and

$$a(\nu,\mathbb{A}^2_{p,q},0)-\sum_{j\in J_0}c_j\operatorname{mult}_\nu\mathfrak{b}_j=(\mu_0-1)\sum_{j\in J_0}c_j\operatorname{mult}_\nu\mathfrak{b}_j\le-\frac{\delta\gamma}{2}m. \tag{A.58}$$

Since $r\ge 2m_0+2k_0$, the integer $k:=k_0m$ satisfies (A.54) and $(r-2m_0)m-k\ge k_0m$. Let $\tilde\nu$ and F be the prime divisors constructed in **Step 4** for ν and k. By (A.57), $\operatorname{mult}_{\tilde\nu}(\tilde h_j-\varphi_j)\ge k_0m>\ell m$. Hence

$$\operatorname{mult}_{\tilde\nu}\tilde h_j\ge\min\{\operatorname{mult}_\nu\varphi_j,\ell m\}=\operatorname{mult}_\nu\mathfrak{b}_j\quad\text{for } j\in J_0,$$

and $\operatorname{mult}_{\tilde\nu}\tilde h_j=0$ for $j\notin J_0$. By (A.55), (A.58), and (A.49),

$$a(\tilde\nu,\mathbb{A}^3_{u,p,q},\Delta)\le a(\nu,\mathbb{A}^2_{p,q},0)+k_0m-\Big(1-\frac{\tau}{r}\Big)k_0m-\sum_{j\in J_0}c_j\operatorname{mult}_\nu\mathfrak{b}_j\le-\frac{\delta\gamma}{2}m+\frac{2k_0m}{r}<0,$$

where the last inequality follows from $r>4k_0/(\delta\gamma)$. By (A.53), $a(F,W_N,D)<0$. Since $\operatorname{center}_{W_N}F=\{0\}$, this contradicts the klt property of (W_N,D) near 0. Therefore $(\mathbb{A}^2,C)$ is lc near 0 if $r\ge r_2$.

Step 6. In this step we show that b belongs to an ACC set as in (2), and we choose a prime divisor computing b with bounded log discrepancy.

Assume that $r \geq r_2$. Let $(\mathbb{A}^2/\tilde{H}, C_{\tilde{H}})$ be the crepant quotient of $(\mathbb{A}^2, C)$ by $\tilde{H}$, and let $\bar{0}$ be the image of 0. Any prime divisor Q on $\mathbb{A}^2$ through 0 satisfies $\operatorname{mult}_Q C = \sum_j c_j \operatorname{mult}_Q \varphi_j \in \Gamma_+$, and $\operatorname{mult}_Q C \leq 1$ because $(\mathbb{A}^2, C)$ is lc near 0. By [Liu26, Lemma 2.6(4)], $C_{\tilde{H}} \in \Gamma'$ near $\bar{0}$, where $\Gamma' := \Psi(\Gamma_+ \cap [0,1])$, and Γ' is a DCC set depending only on Γ by [Liu26, Lemma 2.10(1)]. By [Liu26, Lemma 2.6], $(\mathbb{A}^2/\tilde{H} \ni \bar{0}, C_{\tilde{H}})$ is an lc surface germ, and $b = \operatorname{mld}(\mathbb{A}^2/\tilde{H} \ni \bar{0}, C_{\tilde{H}})$. By Theorem 2.9(1), b belongs to an ACC set depending only on Γ', hence only on Γ, and we take it as the ACC set of (2).

By Theorem 2.9(2), there exist a positive integer n_0, depending only on Γ', and a prime divisor E over $\mathbb{A}^2/\tilde{H}$ with center $\bar{0}$ such that $a(E, \mathbb{A}^2/\tilde{H}, C_{\tilde{H}}) = b$ and $a(E, \mathbb{A}^2/\tilde{H}, 0) \leq n_0$. Let ν be a prime divisor over $\mathbb{A}^2_{p,q}$ lying over E, and let $e := |I_{\tilde{H}}(\nu)|$. Since the preimage of $\bar{0}$ is 0, $\operatorname{center}_{\mathbb{A}^2} \nu = \{0\}$. Let $B_{\mathrm{br}} \geq 0$ be the boundary of the crepant quotient of $(\mathbb{A}^2, 0)$ by $\tilde{H}$. By [Liu26, Lemma 2.6(2)], and since $e \leq |\tilde{H}| \leq 2e_0$ by (1),

$$a(\nu, \mathbb{A}^2_{p,q}, C) = eb \quad \text{and} \quad a(\nu, \mathbb{A}^2_{p,q}, 0) = ea(E, \mathbb{A}^2/\tilde{H}, B_{\mathrm{br}}) \leq ea(E, \mathbb{A}^2/\tilde{H}, 0) \leq 2e_0 n_0.$$

In particular, ν computes b. Since $b \geq 0$ and $c_j \geq \gamma$,

$$\gamma \operatorname{mult}_\nu \varphi_j \leq \sum_i c_i \operatorname{mult}_\nu \varphi_i = a(\nu, \mathbb{A}^2_{p,q}, 0) - eb \leq 2e_0 n_0 \quad \text{for any } j.$$

Let $k_1 := \lceil 2e_0 n_0/\gamma \rceil$, which depends only on Γ and e_0. Then

$$\operatorname{mult}_\nu \varphi_j \leq k_1 \quad \text{for any } j. \tag{A.59}$$

Step 7. In this step we prove the upper bound and finish the proof of (2).

Let

$$r_0 := \max\{r_2, 2m_0 + 2k_1 + 2e_0 + 1\} \quad \text{and} \quad C_0 := 2(k_1 + 2e_0),$$

which depend only on Γ and e_0, and assume that $r \geq r_0$. Let ν and e be the prime divisor and the integer of **Step 6**. Let χ be the character of $\tilde{H}$ given by $(g, \xi) \to \xi^r$, so that $\tilde{H}$ acts on $\mathbb{A}^3_{u,p,q} = \mathbb{A}^2_{p,q} \times \mathbb{A}^1_u$ by $(g, \xi) \cdot (p, q, u) = ((g, \xi) \cdot (p, q), \chi(g, \xi)u)$. Let $i_\nu \in \{1, \dots, e\}$ be the integer q of Lemma 3.1 for ν and χ, and let k be the integer with $k_1 < k \leq k_1 + e$ and $k \equiv i_\nu \pmod{e}$. By (1), $e \leq |\tilde{H}| \leq 2e_0$, so $k \leq k_1 + 2e_0$. Let

$$\Delta_0 := \left(1 - \frac{\tau}{r}\right) \operatorname{div}(u) + \sum_j c_j \operatorname{div}(\varphi_j) \quad \text{on } \mathbb{A}^3_{u,p,q}.$$

The hypotheses of Proposition 5.4 with $S = \mathbb{A}^2_{p,q}$, $s = 0$, the group $\tilde{H}$, the character χ, the coefficient $1 - \tau/r$ of $\operatorname{div}(u)$, and $\Phi_j = \varphi_j$ hold: $\tilde{H}$ acts faithfully on $\mathbb{A}^2_{p,q}$ by (1), $1 - \tau/r \leq 1$ because $\tau \geq 0$, φ_j is $\tilde{H}$-semi-invariant and non-zero by **Step 3**, $(\mathbb{A}^2, C)$ is lc near 0 by **Step 5**, ν computes b by **Step 6**, and $k > k_1 \geq \operatorname{mult}_\nu \varphi_j$ for any j by (A.59). By Proposition 5.4, the Gauss extension $\tilde{\nu}$ of mult_ν with u-value k is a prime divisor over $\mathbb{A}^3_{u,p,q}$ with center 0, $I_{\tilde{H}}(\tilde{\nu}) = I_{\tilde{H}}(\nu)$, and

$$\frac{a(\tilde{\nu}, \mathbb{A}^3_{u,p,q}, \Delta_0)}{|I_{\tilde{H}}(\tilde{\nu})|} = b + \frac{\tau k}{re}. \tag{A.60}$$

Let $m := \min\{\operatorname{mult}_\nu p, \operatorname{mult}_\nu q\} \geq 1$. Since $r > 2m_0 + 2k_1 + 2e_0$,

$$k \leq k_1 + 2e_0 < r - 2m_0 \leq (r - 2m_0)m \quad \text{and} \quad (r - 2m_0)m - k \geq r - 2m_0 - k_1 - 2e_0 > k_1.$$

Thus k satisfies (A.54), and the prime divisor F over W_N constructed in **Step 4** for ν and k has center 0 and satisfies (A.53). By (A.57) and (A.59),

$$\operatorname{mult}_{\tilde{\nu}}(\tilde{h}_j - \varphi_j) > k_1 \geq \operatorname{mult}_\nu \varphi_j = \operatorname{mult}_{\tilde{\nu}} \varphi_j \quad \text{for any } j,$$

hence $\operatorname{mult}_{\tilde\nu}\tilde h_j = \operatorname{mult}_{\tilde\nu}\varphi_j$ for any j. Since $\Delta-\Delta_0$ is the sum of $c_j\operatorname{div}(\tilde h_j/\varphi_j)$ over j, this implies

$$a(\tilde\nu,\mathbb{A}^3_{u,p,q},\Delta)=a(\tilde\nu,\mathbb{A}^3_{u,p,q},\Delta_0).$$

By (A.53), (A.60), and (A.49),

$$\frac{a(F,W_N,D)}{|I_G(F)|}=b+\frac{\tau k}{re}\le b+\frac{2(k_1+2e_0)}{r}=b+\frac{C_0}{r}.$$

Since $\operatorname{center}_{W_N}F=\{0\}$, [Liu26, Lemma 2.6] implies

$$\alpha=\operatorname{mld}((W_N\ni 0,D)/G)\le\frac{a(F,W_N,D)}{|I_G(F)|}\le b+\frac{C_0}{r}.$$

The integers r_0 and C_0 do not depend on N or d. Together with **Steps 5 and 6**, this proves (2). □

We now prove parts (3) and (4) of Proposition A.5.

Proof of Proposition A.5(3) and (4). The proof is long, so we divide it into several steps.

Step 1. In this step we fix notation for the weight blocks of the h_j and for two varieties birational to W_N/G, and we compare log discrepancies on them.

We keep the normalization of the h_j and the notation γ, m_0, m_j, and $h_j^{(l)}$ of **Step 2** of the proof of (1) and (2). In particular, for any j, $c_j>0$, $h_j(0)=0$, and h_j is a polynomial without monomials divisible by xy. By (A.48), $\sum_j c_j\le\sum_j c_jm_j<2$, so the number of indices j is at most m_0.

From now on we assume that $r>4m_0$. By Lemma 5.3, applied with $M=m_0$ and $\max_i|w_i|=2$, we have $|\kappa_j|\le 2m_0$, the $\mathbf{T}$-weight of any monomial of h_j is κ_j+lr for some $l\in\mathbb{Z}$, and any monomial of $h_j-h_j^0$ has total degree at least $(r-2m_0)/2$. Let

$$h_j^+:=\sum_{l\ge0}h_j^{(l)},\qquad h_j^-:=\sum_{l\le0}h_j^{(l)},\qquad D^0:=\sum_j c_j\operatorname{div}(h_j^0).$$

Since the $\mathbf{T}$-weight of $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ is $a_4-2a_2-a_3$, any monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of $h_j^{(l)}$ satisfies

$$a_4\ge r-2m_0\ \text{ if } l\ge1,\qquad\text{and}\qquad 2a_2+a_3\ge r-2m_0\ \text{ if } l\le-1. \tag{A.61}$$

We describe the first chart. Let $X_d:=(X_1X_2+X_3^2+X_4^d=0)\subset\mathbb{A}^4_{X_1,X_2,X_3,X_4}$, let $S:=X_d\cap(X_4=0)$, which is the surface $(X_1X_2+X_3^2=0)\subset\mathbb{A}^3_{X_1,X_2,X_3}$, and let $H:=G/R$. Since $N+2=dr$, the polynomial $X_1X_2+X_3^2+X_4^d$ pulls back to $t^2(xy+z^2+t^N)$ under

$$\psi_1:W_N\to X_d,\qquad (x,y,z,t)\to(X_1,X_2,X_3,X_4)=(x,yt^2,zt,t^r).$$

An element $g\in G$ acts on X_1,X_2,X_3,X_4 by $(g_xg_y^{-1}g_t^{-2},g_yg_t^2,g_zg_t,g_t^r)$, and R acts trivially on them. Thus, by (A.47), H acts diagonally on $\mathbb{A}^4$ and preserves X_d, and ψ_1 is G-equivariant. The points of W_N over a point of X_d with $X_4\ne0$ are the points $(X_1,X_2t^{-2},X_3t^{-1},t)$ with $t^r=X_4$, which form one R-orbit. Hence the morphism $(W_N\cap(t\ne0))/R\to X_d\cap(X_4\ne0)$ induced by ψ_1 is bijective and birational onto the smooth variety $X_d\cap(X_4\ne0)$, hence an isomorphism by Zariski's main theorem. If $g\in G$ acts trivially on X_1,X_2,X_3, then $g_x=1$, $g_y=g_t^{-2}$, and $g_z=g_t^{-1}$, so $g\in G\cap\mathbf{T}=R$. Since X_1,X_2,X_3 restrict to non-zero functions on S, H acts faithfully on S and on X_d. The threefold X_d is smooth if $d=1$, and has an isolated cDV point at 0 if $d\ge2$, because the section S of X_d by $X_4=0$ is an A_1 surface singularity. Thus X_d is terminal near 0 [Kol98, Theorem 2.4], and S is klt.

We describe the second chart. Since y is not the square of a rational function on W_N, the subvariety $(\sigma^2=y)\subset W_N\times\mathbb{A}^1_\sigma$ is irreducible. Let W' be the normalization of $(\sigma^2=y)$. Let $\tilde G$ be the group of pairs (g,ξ) with $g\in G$ and $\xi^2=g_y$. It acts on W' by g on x,y,z,t and

by $\sigma \to \xi\sigma$, and $W'/\tilde{G} = W_N/G$, because the kernel $\{(1, \pm 1)\}$ of $\tilde{G} \to G$ is the group of deck transformations of $W' \to W_N$. Let

$$u := \sigma^r, \qquad p := z/\sigma, \qquad q := t\sigma.$$

Since $x = -(z^2 + t^N)/y$ and $N + 2 = dr$, we have

$$x = -p^2 - \frac{q^N}{u^d} = -p^2 - \frac{t^N}{y}. \tag{A.62}$$

On $W' \cap (\sigma \neq 0)$ the functions σ, z, t are coordinates, and $\tilde{R}$ acts on them by $(\lambda^{-1}, \lambda^{-1}, \lambda)$ with $\lambda^r = 1$. Hence $(\sigma, z, t) \to (u, p, q)$ is the quotient of $W' \cap (\sigma \neq 0)$ by $\tilde{R}$ onto $\mathbb{A}^3_{u,p,q} \cap (u \neq 0)$. Thus $\tilde{H} = \tilde{G}/\tilde{R}$ acts diagonally on $\mathbb{A}^3_{u,p,q}$ by $(\xi^r, g_z\xi^{-1}, g_t\xi)$, and W_N/G is birational to $\mathbb{A}^3/\tilde{H}$. Let $\iota \in \tilde{H}$ be the image of $(1, -1) \in \tilde{G}$. The element $(1, -1)$ itself is also denoted ι in the proof of (1) and (2). It acts on (p, q) by $(-1, -1)$, so $\iota \neq 1$. The homomorphism $\tilde{H} \to H$, $(g, \xi) \to gR$, is surjective, and $|\tilde{H}| = 2|H|$ by Proposition A.5(1), so the kernel of $\tilde{H} \to H$ is $\{1, \iota\}$.

We define normalized orders. Let F be a prime divisor over W_N, and let E be the prime divisor over W_N/G which is the image of F. Let $\tilde{H}$ act on $\mathbb{P}^3 \supset \mathbb{A}^3_{u,p,q}$ by the diagonal action on (u, p, q). Let F', F_1, and $\tilde{F}$ be prime divisors over W', X_d, and $\mathbb{P}^3$ whose images over $W'/\tilde{G}$, X_d/H, and $\mathbb{P}^3/\tilde{H}$ correspond to E under the birational identifications of $W'/\tilde{G}$, X_d/H, and $\mathbb{P}^3/\tilde{H}$ with W_N/G induced by $W' \to W_N$, by ψ_1, and by $(\sigma, z, t) \to (u, p, q)$. For a semi-invariant rational function f on W_N, W', X_d, or $\mathbb{A}^3_{u,p,q}$, let $[f]$ be $\operatorname{mult}_F f/|I_G(F)|$, $\operatorname{mult}_{F'} f/|I_{\tilde{G}}(F')|$, $\operatorname{mult}_{F_1} f/|I_H(F_1)|$, or $\operatorname{mult}_{\tilde{F}} f/|I_{\tilde{H}}(\tilde{F})|$, respectively. By [Liu26, Lemma 2.6, the statement before (0), and (1)], applied to the quotients by G, $\tilde{G}$, H, and $\tilde{H}$, the inertia orders $|I_G(F)|$, $|I_{\tilde{G}}(F')|$, $|I_H(F_1)|$, and $|I_{\tilde{H}}(\tilde{F})|$ are the ramification indices over E. Thus $[f] = \operatorname{mult}_E(f^m)/m$ for any positive integer m such that f^m is invariant. Thus $[f]$ does not depend on the choices of F', F_1, and $\tilde{F}$, and it is compatible with the pullbacks by $W' \to W_N$, by ψ_1, and by $W' \dashrightarrow \mathbb{A}^3$. In particular,

$$\begin{aligned} &[X_1] = [x], \quad [X_2] = [y] + 2[t], \quad [X_3] = [z] + [t], \quad [X_4] = r[t], \\ &[\sigma] = \frac{1}{2}[y], \quad [u] = \frac{r}{2}[y], \quad [p] = [z] - \frac{1}{2}[y], \quad [q] = [t] + \frac{1}{2}[y]. \end{aligned} \tag{A.63}$$

Moreover $[x], [y], [z], [t] \geq 0$, and $\operatorname{center}_{W_N} F = \{0\}$ if and only if $[x], [y], [z], [t] > 0$. We have $\operatorname{center}_{\mathbb{P}^3} \tilde{F} \cap \mathbb{A}^3 \neq \emptyset$, that is, $\tilde{F}$ is a prime divisor over $\mathbb{A}^3$, if and only if $[u], [p], [q] \geq 0$.

We compare log discrepancies. The residue forms

$$\omega_W := \frac{dx \wedge dz \wedge dt}{x} = -\frac{dy \wedge dz \wedge dt}{y} \qquad \text{and} \qquad \omega_X := \frac{dX_1 \wedge dX_3 \wedge dX_4}{X_1}$$

generate ω_{W_N} and ω_{X_d}, and

$$\psi_1^*\omega_X = rt^r\omega_W \qquad \text{and} \qquad \omega_W = -\frac{2}{r} \cdot \frac{du \wedge dp \wedge dq}{u} \quad \text{on } W',$$

where the first identity follows from a direct calculation, and the second follows from (A.52), since $\omega_W = -dy \wedge dz \wedge dt/y$ and $u = \sigma^r$. Since $\psi_1^* X_4 = t^r$, the first identity says that the form ω_X/X_4, which trivializes $K_{X_d} + S$, pulls back to $r\omega_W$. The second identity says that ω_W is a constant multiple of the form $du \wedge dp \wedge dq/u$, which trivializes $K_{\mathbb{A}^3} + \operatorname{div}(u)$. Thus [Liu26, Lemma 2.6(2)], applied to the quotients by G, $\tilde{G}$, H, and $\tilde{H}$, implies that, for any prime divisor F over W_N, the first equality in

$$\frac{a(F, W_N, 0)}{|I_G(F)|} = \frac{a(F_1, X_d, S)}{|I_H(F_1)|} = \frac{a(\tilde{F}, \mathbb{A}^3, \operatorname{div}(u))}{|I_{\tilde{H}}(\tilde{F})|} \tag{A.64}$$

holds, and the second holds if $[u], [p], [q] \geq 0$. For an $\mathbb{R}$-divisor $B = \sum_k b_k \operatorname{div}(f_k)$ with semi-invariant rational functions f_k on W_N, we have $a(F, W_N, B)/|I_G(F)| = a(F, W_N, 0)/|I_G(F)| - \sum_k b_k[f_k]$, and similarly on X_d and on $\mathbb{A}^3$.

We write the blocks on X_d and on W'. For any monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of $h_j^{(l)}$, the equality $a_4 - 2a_2 - a_3 = \kappa_j + lr$ implies

$$x^{a_1}y^{a_2}z^{a_3}t^{a_4} = t^{\kappa_j}\psi_1^*\big(X_4^l X_1^{a_1} X_2^{a_2} X_3^{a_3}\big) \qquad \text{and} \qquad x^{a_1}y^{a_2}z^{a_3}t^{a_4} = \sigma^{-\kappa_j}u^{-l}x^{a_1}p^{a_3}q^{a_4} \ \text{ on } W'.$$

Hence there exist polynomials $P_j^{(l)} \in \mathbb{C}[X_1, X_2, X_3]$ and $\hat{h}_j \in \mathbb{C}[u, x, p, q]$ such that

$$h_j^{(l)} = t^{\kappa_j}\psi_1^*\big(X_4^l P_j^{(l)}\big) \qquad \text{and} \qquad h_j^- = \sigma^{-\kappa_j}\hat{h}_j(u, x, p, q).$$

Let $P_j^+ := \sum_{l\geq 0} X_4^l P_j^{(l)}$, so that $h_j^+ = t^{\kappa_j}\psi_1^* P_j^+$ and $P_j^+|_S = P_j^{(0)}|_S$, and let

$$\check{h}_j := \hat{h}_j(u, -p^2, p, q) \in \mathbb{C}[u, p, q].$$

Only the terms with $l = 0$ survive at $u = 0$, so $\check{h}_j|_{u=0} = h_j^0(-p^2, 1, p, q) = \varphi_j$, and

$$\varphi_j = q^{\kappa_j} P_j^{(0)}(-p^2, q^2, pq), \tag{A.65}$$

which follows by substituting $(x, y, z, t) = (-p^2, 1, p, q)$ in

$$h_j^0 = t^{\kappa_j}\psi_1^* P_j^{(0)} = t^{\kappa_j} P_j^{(0)}(x, yt^2, zt).$$

Equivalently, each monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of h_j^0 has $a_4 = \kappa_j + 2a_2 + a_3$, and

$$(-p^2)^{a_1}p^{a_3}q^{\kappa_j+2a_2+a_3} = q^{\kappa_j}(-p^2)^{a_1}(q^2)^{a_2}(pq)^{a_3}.$$

For a monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of h_j^0 we have $a_1a_2 = 0$ and $a_4 - 2a_2 - a_3 = \kappa_j$, so $2a_1 + a_3$ and a_4 determine a_1, a_2, a_3, and the substitution $(x, y, z, t) = (-p^2, 1, p, q)$ sends distinct monomials of h_j^0 to distinct monomials in p, q. Since h_j^0 contains a monomial of least degree of h_j (Set-up 7.3), $h_j^0 \neq 0$, so $\varphi_j \neq 0$, and (A.65) implies $P_j^{(0)}|_S \neq 0$. Each monomial of h_j has the G-character of h_j. Moreover, R and $\tilde{R}$ act trivially on $X_1, \dots, X_4$ and on u, x, p, q, and $g_x = (g_z\xi^{-1})^2$ for any $(g, \xi) \in \tilde{G}$ by (A.47) and $\xi^2 = g_y$. Thus $P_j^{(l)}$ and P_j^+ are H-semi-invariant, and $\hat{h}_j$ and $\check{h}_j$ are $\tilde{H}$-semi-invariant, and so is $\varphi_j = \check{h}_j|_{u=0}$, since $\tilde{H}$ acts diagonally on (u, p, q).

Step 2. In this step we relate pairs on W_N to pairs on S and on $\mathbb{A}^2_{p,q}$ by adjunction, and we prove a cutoff for lc pairs on W_N.

Let $\psi_2 : \mathbb{A}^2_{p,q} \to S$ be the morphism $(p, q) \to (X_1, X_2, X_3) = (-p^2, q^2, pq)$. Since $g_x = (g_z\xi^{-1})^2$, $g_y g_t^2 = (g_t\xi)^2$, and $g_z g_t = (g_z\xi^{-1})(g_t\xi)$ for any $(g, \xi) \in \tilde{G}$, the morphism ψ_2 is equivariant with respect to $\tilde{H} \to H$. Moreover ψ_2 is the quotient by $\{1, \iota\}$, which acts freely outside 0.

(a) For any H-invariant $\mathbb{R}$-divisor $B'_S \geq 0$ on S, the pair (S, B'_S) is lc near 0 if and only if $(\mathbb{A}^2, \psi_2^* B'_S)$ is lc near 0, and in this case

$$\operatorname{mld}((S \ni 0, B'_S)/H) = \operatorname{mld}((\mathbb{A}^2 \ni 0, \psi_2^* B'_S)/\tilde{H}).$$

This follows from [Liu26, Lemma 2.6], since $K_{\mathbb{A}^2} + \psi_2^* B'_S = \psi_2^*(K_S + B'_S)$ and $\mathbb{A}^2/\tilde{H} = S/H$. If $\tau \geq 0$, let

$$B_S := \sum_j c_j \operatorname{div}\big(P_j^{(0)}|_S\big) + \frac{\tau}{2}\operatorname{div}(X_2|_S) \geq 0.$$

Here $P_j^{(0)}|_S \neq 0$ by **Step 1**. By (A.65), $\psi_2^* X_2 = q^2$, and $\tau = \sum_j c_j\kappa_j$, we have

$$\psi_2^* B_S = \sum_j c_j\big(\operatorname{div}(\varphi_j) - \kappa_j \operatorname{div}(q)\big) + \tau \operatorname{div}(q) = C.$$

In particular, if $\tau \geq 0$ and $(\mathbb{A}^2, C)$ is lc near 0, then (S, B_S) is lc near 0 and $\operatorname{mld}((S \ni 0, B_S)/H) = b$.

(b) Let $Q_1, \dots, Q_{s_1}$ be H-semi-invariant polynomials in $X_1, \dots, X_4$ with $Q_k|_S \neq 0$, let $b_1, \dots, b_{s_1}$ be positive real numbers, and let $\rho \geq 0$. Let

$$B'_S := \sum_k b_k \operatorname{div}(Q_k|_S) + \rho \operatorname{div}(X_2|_S) \qquad \text{and} \qquad \Delta := S + \sum_k b_k \operatorname{div}(Q_k|_{X_d}) + \rho \operatorname{div}(X_2|_{X_d}),$$

and assume that (S, B'_S) is lc near 0. Then the following hold.

(b1) The pair (X_d, Δ) is lc near 0.

(b2) If $\operatorname{center}_{W_N} F = \{0\}$, then $a(F_1, X_d, \Delta)/|I_H(F_1)| \geq \operatorname{mld}((S \ni 0, B'_S)/H)$.

(b3) Let $\kappa'_1, \dots, \kappa'_{s_1}$ be integers with $\sum_k b_k \kappa'_k = 2\rho$ such that $f_k := t^{\kappa'_k} \psi_1^* Q_k$ are polynomials. Then $(W_N, \sum_k b_k \operatorname{div}(f_k))$ is lc near 0, and if $\operatorname{center}_{W_N} F = \{0\}$, then

$$\frac{a(F, W_N, \sum_k b_k \operatorname{div}(f_k))}{|I_G(F)|} \geq \operatorname{mld}((S \ni 0, B'_S)/H) + \rho[y].$$

Since X_d is Gorenstein and S is a normal Cartier divisor on X_d, $(K_{X_d} + \Delta)|_S = K_S + B'_S$, so (b1) follows from inversion of adjunction [Kaw07, Theorem]. If $\operatorname{center}_{W_N} F = \{0\}$, then $\operatorname{center}_{X_d} F_1 = \{\psi_1(0)\} = \{0\}$, and (b2) is Lemma 5.5 for the diagonal action of H on $\mathbb{A}^4$, the complete intersection X_d, the section $S = X_d \cap (X_4 = 0)$ on which H acts faithfully (**Step 1**), the $\mathbb{R}$-ideal $\prod_k (Q_k)^{b_k} \cdot (X_2)^\rho$ (without the last factor if $\rho = 0$), and the coefficient 1 of S. For (b3), the equalities (A.64), (A.63), $[f_k] = \kappa'_k[t] + [Q_k]$, and $\sum_k b_k \kappa'_k = 2\rho$ imply that

$$\frac{a(F, W_N, \sum_k b_k \operatorname{div}(f_k))}{|I_G(F)|} = \frac{a(F_1, X_d, \Delta)}{|I_H(F_1)|} + \rho[y] \tag{A.66}$$

for any prime divisor F over W_N. Let $V \ni 0$ be an open subset of X_d on which (X_d, Δ) is lc. Since ψ_1 is a morphism, $\operatorname{center}_{X_d} F_1$ is the closure of $\psi_1(\operatorname{center}_{W_N} F)$, so it intersects V whenever $\operatorname{center}_{W_N} F$ intersects $\psi_1^{-1}(V)$. As $[y] \geq 0$, (A.66) and (b1) imply that $(W_N, \sum_k b_k \operatorname{div}(f_k))$ is lc on $\psi_1^{-1}(V)$, and (A.66) and (b2) imply the inequality of (b3).

(c) If $\operatorname{center}_{W_N} F = \{0\}$, then

$$\frac{a(F, W_N, 0)}{|I_G(F)|} \geq \max\left\{\frac{[x]}{2}, \frac{[yt^2]}{2}, [zt]\right\}. \tag{A.67}$$

To prove (c), take $s_1 = 1$, $\rho = 0$, $\kappa'_1 = 0$, and (Q_1, b_1) equal to $(X_1, \frac{1}{2})$, $(X_2, \frac{1}{2})$, or $(X_3, 1)$, so that f_1 is x, yt^2, or zt. Then $\psi_2^* B'_S$ is $\operatorname{div}(p)$, $\operatorname{div}(q)$, or $\operatorname{div}(p) + \operatorname{div}(q)$, so (S, B'_S) is lc near 0 by (a), and $\operatorname{mld}((S \ni 0, B'_S)/H) \geq 0$. By (b3), $a(F, W_N, b_1 \operatorname{div}(f_1)) \geq 0$, which is (A.67).

(d) Let $\Lambda \subset \mathbb{R}_{>0}$ be a DCC set containing 1, and let s' be a positive integer. Then there exists a positive integer ℓ, depending only on Λ and s', with the following property. Let $f_1, \dots, f_{s'}$ and $f'_1, \dots, f'_{s'}$ be polynomials on $\mathbb{A}^4$ not divisible by $xy + z^2 + t^N$, with $f_k - f'_k \in (x, y, z, t)^\ell$, and let $b_1, \dots, b_{s'} \in \Lambda$. If $(W_N, \sum_k b_k \operatorname{div}(f_k|_{W_N}))$ is lc near 0, then so is $(W_N, \sum_k b_k \operatorname{div}(f'_k|_{W_N}))$. To prove (d), let ℓ be the integer of Lemma 4.14 for $n = 4$, $s' + 1$ functions, and Λ. By inversion of adjunction [Kaw07, Theorem], $(W_N, \sum_k b_k \operatorname{div}(f_k|_{W_N}))$ is lc near 0 if and only if the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^N) + \sum_k b_k \operatorname{div}(f_k)\Big)$$

is lc near 0, and by [dFEM11, Proposition 2.11] this holds if and only if

$$\Big(\operatorname{Spec} \mathbb{C}[[x, y, z, t]],\ (xy + z^2 + t^N) \cdot \prod_k f_k^{b_k}\Big)$$

is lc. Both equivalences also hold with f'_k in place of f_k. Since the tuples $(xy + z^2 + t^N, f_1, \dots, f_{s'})$ and $(xy + z^2 + t^N, f'_1, \dots, f'_{s'})$ have the common first entry $xy + z^2 + t^N$, Lemma 4.14 implies (d), and ℓ does not depend on N.

Step 3. In this step we prove (3).

Fix $\theta > 1$. Let $\Theta_1 := \theta/(\theta-1)$ and $\Theta_2 := 1/(\gamma(\theta-1))$, let ℓ be the integer of **Step 2**(d) for $\Lambda := \{1\} \cup \{\theta c : c \in \Gamma,\ c > 0\}$ and $s' = m_0$, and let r_1 be the least integer such that

$$r_1 > \max\big\{4m_0,\ 2\ell + 2m_0,\ 2m_0 + 4e_0\Theta_2,\ 2 + 4e_0\big(\Theta_2 + 2(m_0+1)\Theta_1\big)\big\}.$$

Then r_1 depends only on Γ, e_0, and θ. Assume that $\tau \geq 0$, $r \geq r_1$, and $(\mathbb{A}^2, \theta C)$ is lc near 0.

We first show that $(W_N, \theta D)$ is lc near 0. Since $\psi_2^*(\theta B_S) = \theta C$ and $(\mathbb{A}^2, \theta C)$ is lc near 0, the pair $(S, \theta B_S)$ is lc near 0 by **Step 2**(a). **Step 2**(b3) with $Q_j = P_j^{(0)}$, $b_j = \theta c_j$, $\kappa_j' = \kappa_j$, and $\rho = \theta\tau/2$, so that $f_j = h_j^0$, implies that $(W_N, \theta D^0)$ is lc near 0. Any monomial of $h_j - h_j^0$ has total degree at least $(r - 2m_0)/2 \geq \ell$ (**Step 1**), and the number of indices j is at most m_0. The h_j and the h_j^0 have no monomial divisible by xy, hence are not divisible by $xy + z^2 + t^N$ (see the proof of Lemma 5.3). Thus **Step 2**(d), with $f_k = f_k' = 1$ and $b_k = 1$ for the remaining indices $k \leq m_0$, implies that $(W_N, \theta D)$ is lc near 0.

We next show that $b \leq 1$. Let E_1 be the exceptional divisor of the blow-up of $\mathbb{A}^2$ at 0. Since ι acts on $\mathbb{A}^2$ by -1, it fixes E_1 pointwise, so $|I_{\tilde{H}}(E_1)| \geq 2$ and

$$b \leq \frac{a(E_1, \mathbb{A}^2, C)}{|I_{\tilde{H}}(E_1)|} \leq \frac{2 - \operatorname{mult}_0 C}{2} \leq 1.$$

By [Liu26, Lemma 2.6], α is the minimum of $a(F, W_N, D)/|I_G(F)|$ over the prime divisors F over W_N with $\operatorname{center}_{W_N} F = \{0\}$. Fix such F. Since $b \leq 1$, we may assume that $a(F, W_N, D)/|I_G(F)| \leq 1$, and it suffices to show $a(F, W_N, D)/|I_G(F)| \geq b$. Let $A_F := a(F, W_N, 0)/|I_G(F)|$. Then

$$A_F - \sum_j c_j[h_j] \leq 1 \qquad \text{and} \qquad A_F - \theta\sum_j c_j[h_j] \geq 0,$$

where the second inequality holds because $(W_N, \theta D)$ is lc near 0. Hence $(\theta - 1)\sum_j c_j[h_j] \leq 1$, and

$$A_F \leq \Theta_1 \qquad \text{and} \qquad [h_j] \leq \Theta_2 \ \text{ for any } j. \tag{A.68}$$

By (A.67), $[x], [yt^2] \leq 2\Theta_1$ and $[zt] \leq \Theta_1$. The functions x, yt^2, and zt are R-invariant, so their G-characters factor through H and have order at most $|H| \leq e_0$. Hence for some positive integer $m \leq e_0$, x^m is a G-invariant regular function vanishing at 0, so $[x] = \operatorname{mult}_E(x^m)/m \geq 1/e_0$. Likewise, for some positive integers $m', m'' \leq e_0$, the functions $(yt^2)^{m'}$ and $(zt)^{m''}$ are G-invariant regular functions vanishing at 0, so $[yt^2] \geq 1/e_0$ and $[zt] \geq 1/e_0$. Hence

$$[x] \geq \frac{1}{e_0}, \qquad [yt^2] \geq \frac{1}{e_0}, \qquad [zt] \geq \frac{1}{e_0}. \tag{A.69}$$

We distinguish two cases.

Case I: $[t] \leq 1/(4e_0)$. By (A.63) and (A.69),

$$\frac{1}{2}[y] = \frac{1}{2}[yt^2] - [t] \geq \frac{1}{4e_0} \qquad \text{and} \qquad [z] = [zt] - [t] \geq \frac{3}{4e_0}.$$

By (A.61), any monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of $h_j^{(l)}$ with $l \leq -1$ satisfies

$$[x^{a_1}y^{a_2}z^{a_3}t^{a_4}] \geq a_2[y] + a_3[z] \geq \frac{2a_2 + a_3}{4e_0} \geq \frac{r - 2m_0}{4e_0} > \Theta_2.$$

Since $h_j - h_j^+ = \sum_{l \leq -1} h_j^{(l)}$ and $[h_j] \leq \Theta_2$ by (A.68), we obtain $[h_j^+] = [h_j]$, so $a(F, W_N, D) = a(F, W_N, \sum_j c_j \operatorname{div}(h_j^+))$. We use **Step 2**(b3) with $Q_j = P_j^+$, $b_j = c_j$, $\kappa_j' = \kappa_j$, and $\rho = \tau/2$, so that $f_j = h_j^+$. As $P_j^+|_S = P_j^{(0)}|_S$, the $\mathbb{R}$-divisor B_S' is B_S, and by **Step 2**(a), (S, B_S) is lc near 0 with $\operatorname{mld}((S \ni 0, B_S)/H) = b$. Therefore **Step 2**(b3) implies

$$\frac{a(F, W_N, D)}{|I_G(F)|} \geq b + \frac{\tau}{2}[y] \geq b.$$

Case II: $[t] > 1/(4e_0)$. By (A.61), any monomial $x^{a_1}y^{a_2}z^{a_3}t^{a_4}$ of $h_j^{(l)}$ with $l \geq 1$ satisfies

$$[x^{a_1}y^{a_2}z^{a_3}t^{a_4}] \geq a_4[t] > \frac{r-2m_0}{4e_0} > \Theta_2.$$

Since $h_j - h_j^- = \sum_{l\geq 1} h_j^{(l)}$, (A.68) implies $[h_j^-] = [h_j]$. Since $N+2 = dr \geq r$,

$$[t^N] = N[t] > \frac{r-2}{4e_0} > 2\Theta_1 \geq 2[zt] \geq [z^2].$$

As $xy = -z^2 - t^N$ on W_N, we obtain $[x] + [y] = [z^2] = 2[z]$, so $[y] \leq 2[z] \leq 2\Theta_1$. By (A.63), $[p] = [x]/2 > 0$, $[q] = [t] + [y]/2 > 0$, and $[u] = r[y]/2 > 0$, so $\operatorname{center}_{\mathbb{A}^3} \tilde{F} = \{0\}$. By (A.62), $x + p^2 = -t^N/y$ on W', so $\hat{h}_j(u,x,p,q) - \check{h}_j$ is the product of t^N/y and a polynomial η_j in u,x,p,q, and $[\eta_j] \geq 0$ since $[u],[x],[p],[q] > 0$. Hence

$$\left[\hat{h}_j(u,x,p,q) - \check{h}_j\right] \geq N[t] - [y] > \frac{r-2}{4e_0} - 2\Theta_1 > \Theta_2 + 2m_0\Theta_1.$$

On the other hand, by $h_j^- = \sigma^{-\kappa_j}\hat{h}_j(u,x,p,q)$, (A.63), and $|\kappa_j| \leq 2m_0$,

$$\left[\hat{h}_j(u,x,p,q)\right] = [h_j] + \frac{\kappa_j}{2}[y] \leq \Theta_2 + 2m_0\Theta_1.$$

Thus $[\check{h}_j] = [\hat{h}_j(u,x,p,q)]$, that is, $[h_j] = [\check{h}_j] - \kappa_j[u]/r$. By (A.64),

$$\frac{a(F,W_N,D)}{|I_G(F)|} = \frac{a(\tilde{F},\mathbb{A}^3,\operatorname{div}(u))}{|I_{\tilde{H}}(\tilde{F})|} - \sum_j c_j[h_j] = \frac{a\left(\tilde{F},\mathbb{A}^3,(1-\frac{\tau}{r})\operatorname{div}(u) + \sum_j c_j \operatorname{div}(\check{h}_j)\right)}{|I_{\tilde{H}}(\tilde{F})|}.$$

We use Lemma 5.5 for the diagonal action of $\tilde{H}$ on $\mathbb{A}^3_{u,p,q}$, the section $(u=0) = \mathbb{A}^2_{p,q}$ on which $\tilde{H}$ acts faithfully (Proposition A.5(1)), the $\mathbb{R}$-ideal $\prod_j(\check{h}_j)^{c_j}$, whose restriction to $(u=0)$ is $\prod_j(\varphi_j)^{c_j}$, and the coefficient $1-\tau/r \leq 1$. Since $(\mathbb{A}^2, C)$ is lc near 0 and $b = \operatorname{mld}((\mathbb{A}^2 \ni 0, C)/\tilde{H})$, we obtain

$$\frac{a(F,W_N,D)}{|I_G(F)|} \geq b + \frac{\tau[u]}{r} \geq b.$$

In both cases $a(F,W_N,D)/|I_G(F)| \geq b$. Hence $\alpha \geq b$, which proves (3).

Step 4. In this step we assume that (4) fails, and we reduce to fixed plane data with fixed coefficients.

Suppose that there exist data as in Set-up 7.3, indexed by $i \geq 1$, with Γ, e_0, and $w = (0,-2,-1,1)$ fixed, $r_i \to +\infty$, $\tau_i > 0$, and positive and strictly increasing values α_i. We add the index i to N, d, G, R, D, c_j, h_j, κ_j, φ_j, C, $\tilde{H}$, b, D^0, and B_S, and we normalize the h_{ij} as at the beginning of **Step 1**. Let r_0 and C_0 be the constants of (2). Passing to a subsequence, we may assume that $r_i \geq r_0$ for any i. Then $(\mathbb{A}^2, C_i)$ is lc near 0, b_i belongs to an ACC set depending only on Γ and e_0, and $\alpha_i \leq b_i + C_0/r_i$. Passing to a subsequence, we may assume that $(b_i)_i$ is non-increasing, and we let $\beta := \lim_i b_i$. Since $\alpha_i \leq b_1 + C_0$, the limit $\alpha_\infty := \lim_i \alpha_i$ exists, and $\alpha_\infty \leq \lim_i(b_i + C_0/r_i) = \beta$. Thus

$$0 < \alpha_i < \alpha_\infty \leq \beta \leq b_i \qquad \text{for any } i. \tag{A.70}$$

The number of indices j is at most m_0 (**Step 1**), and it is positive since $\tau_i > 0$. By (1), $\tilde{H}_i$ is a diagonal subgroup of $GL_2(\mathbb{C})$ of order $2[G_i : R_i] \leq 2e_0$, so it is contained in $\mu_{(2e_0)!} \times \mu_{(2e_0)!}$, and there exist finitely many such subgroups. Since Γ is a DCC set, passing to a subsequence, we may assume that the number s of indices j does not depend on i, that $(c_{ij})_i$ is non-decreasing for any j with limit $c_j^* \in [\gamma, 1]$, that $\tilde{H}_i = \tilde{H}$ does not depend on i, and that the $\tilde{H}$-character of φ_{ij} depends only on j. Since $(\mathbb{A}^2, C_i)$ is lc near 0, $2 - \sum_j c_{ij} \operatorname{mult}_0 \varphi_{ij} \geq 0$, so $\operatorname{mult}_0 \varphi_{ij} \leq 2/\gamma$. Let

$$C_i^* := \sum_j c_j^* \operatorname{div}(\varphi_{ij}).$$

We show that $(\mathbb{A}^2, C_i^*)$ is lc near 0 for $i \gg 0$. Let $\vartheta_i := \mathrm{lct}_0(\mathbb{A}^2; C_i^*)$ (Definition 2.11), and let $\lambda_i := \min_j c_{ij}/c_j^*$. Then $C_i \geq \lambda_i C_i^*$, so $\vartheta_i \geq \lambda_i$, and $\lambda_i \to 1$. By [dFEM11, Theorem 1.1], the numbers ϑ_i belong to an ACC set, because they are lc thresholds of the $\mathbb{R}$-ideals $\prod_j (\varphi_{ij})^{c_j^*}$ on $\mathbb{A}^2$ with exponents in the finite set $\{c_1^*, \dots, c_s^*\}$. If $\vartheta_i < 1$ for infinitely many i, then, since $\vartheta_i \geq \lambda_i \to 1$, the set $\{\vartheta_i : \vartheta_i < 1\}$ contains a strictly increasing sequence, a contradiction. Passing to a subsequence, we may assume that $(\mathbb{A}^2, C_i^*)$ is lc near 0 for any i, and we let $\beta_i^* := \mathrm{mld}((\mathbb{A}^2 \ni 0, C_i^*)/\tilde{H})$.

We show that $\lim_i \beta_i^* = \beta$. Since $C_i \leq C_i^*$, $\beta_i^* \leq b_i$. Let $\Gamma^* := \{c_1^*, \dots, c_s^*\}$. The coefficients of the crepant quotient of $(\mathbb{A}^2, C_i^*)$ by $\tilde{H}$ belong, near the image of 0, to the DCC set $\Psi(\Gamma_+^* \cap [0,1])$ [Liu26, Lemma 2.10(1)(5)]. Let n_0 be the integer of Theorem 2.9(2) for $\Psi(\Gamma_+^* \cap [0,1])$. There exists a prime divisor E_i over $\mathbb{A}^2/\tilde{H}$ with center the image of 0, computing β_i^*, with $a(E_i, \mathbb{A}^2/\tilde{H}, 0) \leq n_0$. Let P_i be a prime divisor over $\mathbb{A}^2$ whose image over $\mathbb{A}^2/\tilde{H}$ is E_i. Since the branch divisor of $\mathbb{A}^2 \to \mathbb{A}^2/\tilde{H}$ is effective, [Liu26, Lemma 2.6] implies

$$\frac{a(P_i, \mathbb{A}^2, 0)}{|I_{\tilde{H}}(P_i)|} \leq a(E_i, \mathbb{A}^2/\tilde{H}, 0) \leq n_0,$$

so $a(P_i, \mathbb{A}^2, 0) \leq 2e_0 n_0$. Since $a(P_i, \mathbb{A}^2, C_i^*) \geq 0$,

$$\sum_j c_j^* \operatorname{mult}_{P_i} \varphi_{ij} \leq 2e_0 n_0 \quad \text{and} \quad \operatorname{mult}_{P_i} \varphi_{ij} \leq \frac{2e_0 n_0}{\gamma}.$$

Therefore

$$b_i \leq \frac{a(P_i, \mathbb{A}^2, C_i)}{|I_{\tilde{H}}(P_i)|} = \beta_i^* + \sum_j \frac{(c_j^* - c_{ij}) \operatorname{mult}_{P_i} \varphi_{ij}}{|I_{\tilde{H}}(P_i)|} \leq \beta_i^* + \frac{2e_0 n_0}{\gamma} \sum_j (c_j^* - c_{ij}).$$

As $c_{ij} \to c_j^*$ and $\beta_i^* \leq b_i \to \beta$, we obtain $\lim_i \beta_i^* = \beta$.

Step 5. In this step we take a generic limit of the plane pairs $(\mathbb{A}^2, C_i^*)$ and exclude the case when the limit is klt.

We use Construction-Proposition 4.1 in case (a) with $n = 2$, the fixed group $\tilde{H} \subset GL_2(\mathbb{C})$, and the s-tuples $(\varphi_{i1}, \dots, \varphi_{is})$. By **Steps 1 and 4**, the φ_{ij} are non-zero $\tilde{H}$-semi-invariant polynomials whose characters depend only on j and whose orders at 0 are at most $2/\gamma$. Passing to a subsequence and multiplying the φ_{ij} by non-zero constants, which changes none of C_i, C_i^*, b_i, and β_i^*, we obtain an algebraically closed field $k \supset \mathbb{C}$ and a compatible generic limit $(\varphi_{\infty 1}, \dots, \varphi_{\infty s})$ in $k[[p,q]]$. Let $\mathcal{A}_k = \operatorname{Spec} k[[p,q]]$ and

$$C_\infty := \sum_j c_j^* \operatorname{div}(\varphi_{\infty j}).$$

We use Theorem 4.2 with the coefficients $c_1^*, \dots, c_s^*$ and $\mathfrak{c} = \mathcal{O}$, so that the $\mathbb{R}$-divisors C_i of Theorem 4.2 are our C_i^*. Passing to a subsequence, Theorem 4.2(1) holds.

Assume that $(\mathcal{A}_k, C_\infty)$ is klt near 0. By Theorem 4.2(1), there exists $\theta' > 1$ such that $(\mathbb{A}^2, \theta' C_i^*)$ is lc near 0 for any i. Since $C_i \leq C_i^*$, $(\mathbb{A}^2, \theta' C_i)$ is lc near 0 for any i. As $r_i \to +\infty$ and $\tau_i > 0$, (3) with $\theta = \theta'$ implies $\alpha_i \geq b_i$ for $i \gg 0$, which contradicts (A.70). Hence $(\mathcal{A}_k, C_\infty)$ is not klt near 0.

Step 6. In this step we exclude the case when $(\mathcal{A}_k, C_\infty)$ is not klt near 0, which finishes the proof of (4).

By **Steps 4 and 5**, $(\mathbb{A}^2, C_i^*)$ is lc near 0 for any i, $\lim_i \beta_i^* = \beta > 0$ by (A.70), and $(\mathcal{A}_k, C_\infty)$ is not klt near 0. Thus the hypotheses of Definition-Proposition 5.7 hold, and $(\mathcal{A}_k, C_\infty)$ is lc near 0 with exactly one floor Σ, which is smooth at 0. Moreover, by Definition-Proposition 5.7 and Lemma 5.6,

$$\beta = \frac{1 - b'}{e}, \qquad \text{where} \qquad b' := \operatorname{mult}_0 \left((C_\infty - \Sigma)|_\Sigma \right) \leq 1,$$

and e is the order of the image of $\tilde{H}$ in the automorphism group of the tangent line $T_0\Sigma$. Since $\tilde{H}$ preserves C_∞ and Σ is the only floor, Σ is $\tilde{H}$-invariant. Since Σ is smooth at 0, there exists $z_0 \in \{p, q\}$ with $\mathrm{mult}_0(\mathrm{div}(z_0)|_\Sigma) = 1$. Let χ be the character of $\tilde{H}$ on z_0. The restriction of z_0 to the $\tilde{H}$-invariant line $T_0\Sigma$ is a non-zero linear form, so $h \in \tilde{H}$ acts trivially on $T_0\Sigma$ if and only if $\chi(h) = 1$, and e is the order of χ. Since $\chi(\iota) = -1$, e is even. Let

$$\mu := 1 - b' = e\beta > 0.$$

We show that $(\mathbb{A}^2, C_i + \mu\,\mathrm{div}(z_0))$ is lc near 0 for $i \gg 0$. Since Σ is smooth at 0, has coefficient one in C_∞, and is not contained in $\mathrm{div}(z_0)$,

$$\mathrm{mult}_0\left((C_\infty - \Sigma + \mu\,\mathrm{div}(z_0))|_\Sigma\right) = b' + \mu = 1.$$

Since $C_\infty - \Sigma \geq 0$ does not contain Σ, Lemma 2.14, applied to $D = \Sigma$ and $B = C_\infty - \Sigma + \mu\,\mathrm{div}(z_0)$, implies that $(\mathcal{A}_k, C_\infty + \mu\,\mathrm{div}(z_0))$ is lc. Since z_0 does not depend on i, $(\varphi_{\infty 1}, \dots, \varphi_{\infty s}, z_0)$ is a compatible generic limit of the $(s+1)$-tuples $(\varphi_{i1}, \dots, \varphi_{is}, z_0)$, because for each jet order d the irreducible closed subset of d-jets of Construction-Proposition 4.1 for the $(s+1)$-tuples is the product of Z_d and the d-jet of z_0. Here $Z_d \subset \mathcal{J}_d$ is the irreducible closed subset of that construction for the s-tuples, whose generic point is the d-jet of $(\varphi_{\infty 1}, \dots, \varphi_{\infty s})$, and d denotes the jet order, not the integer d_i. Theorem 4.2(1), with the coefficients $c_1^*, \dots, c_s^*, \mu$ and $\theta = 1$, implies that, after passing to a subsequence, $(\mathbb{A}^2, C_i^* + \mu\,\mathrm{div}(z_0))$ is lc near 0 for any i. Since $C_i \leq C_i^*$, $(\mathbb{A}^2, C_i + \mu\,\mathrm{div}(z_0))$ is lc near 0 for any i.

We lift the augmentation to W_{N_i}. Let $(Y_0, f_0) := (X_1, x)$ if $z_0 = p$, and $(Y_0, f_0) := (X_2, yt^2)$ if $z_0 = q$. Then $f_0 = \psi_1^* Y_0$, and $\psi_2^* Y_0$ is $-p^2$ or q^2, so

$$\psi_2^*\Big(B_{S,i} + \frac{\mu}{2}\,\mathrm{div}(Y_0|_S)\Big) = C_i + \mu\,\mathrm{div}(z_0).$$

By **Step 2**(a), $(S, B_{S,i} + \frac{\mu}{2}\,\mathrm{div}(Y_0|_S))$ is lc near 0. **Step 2**(b3) on W_{N_i}, with the polynomials $P_{ij}^{(0)}$ and Y_0, the coefficients c_{ij} and $\mu/2$, the integers κ_{ij} and 0, and $\rho = \tau_i/2$, implies that $(W_{N_i}, D_i^0 + \frac{\mu}{2}\,\mathrm{div}(f_0))$ is lc near 0. Let ℓ' be the integer of **Step 2**(d) for

$$\Lambda' := \{1, \mu/2\} \cup \{c \in \Gamma : c > 0\}$$

and $s' = m_0 + 1$. For $i \gg 0$, $(r_i - 2m_0)/2 \geq \ell'$, so any monomial of $h_{ij} - h_{ij}^0$ has total degree at least ℓ'. The h_{ij}, the h_{ij}^0, and f_0 have no monomial divisible by xy, hence are not divisible by $xy + z^2 + t^{N_i}$ (see the proof of Lemma 5.3). **Step 2**(d), with the tuples $(h_{ij}^0)_j, f_0$ and $(h_{ij})_j, f_0$ completed by the constant 1 with coefficient 1, implies that

$$\Big(W_{N_i}, D_i + \frac{\mu}{2}\,\mathrm{div}(f_0)\Big) \text{ is lc near } 0 \quad \text{for } i \gg 0.$$

We compute the character of f_0. For $(g, \xi) \in \tilde{G}_i$, the character of x at g is $g_x = (g_z\xi^{-1})^2$, and the character of yt^2 at g is $g_y g_t^2 = (g_t\xi)^2$. Thus the G_i-character of f_0, pulled back to $\tilde{G}_i$, is the pullback of χ^2. Since $\tilde{G}_i \to G_i$ and $\tilde{G}_i \to \tilde{H}$ are surjective, the order of the G_i-character of f_0 is the order of χ^2, which is $e/2$. The group G_i acts faithfully on W_{N_i}, preserves D_i, and fixes 0, and f_0 vanishes at 0. By Lemma 5.8 with $Z = \{0\}$, $U = f_0$, and $\lambda = \mu/2$,

$$\alpha_i = \mathrm{mld}((W_{N_i} \ni 0, D_i)/G_i) \geq \frac{\mu/2}{e/2} = \frac{\mu}{e} = \beta \qquad \text{for } i \gg 0,$$

which contradicts (A.70).

Thus both alternatives of **Steps 5 and 6** are impossible, and α cannot increase along **T** under the condition $\tau > 0$. None of r_0, r_1, C_0, ℓ, ℓ', and n_0 depends on N or d, and the sequence in **Step 4** has arbitrary N_i and d_i. This proves (4). □

Proof of Theorem 7.12. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce to the circle $w = (0, -2, -1, 1)$ with $\tau \leq 0$.

Exchanging x and y maps W_N to itself, maps the circle $\mathbf{T}_{(-2,0,-1,1)}$ onto $\mathbf{T}_{(0,-2,-1,1)}$, and preserves the conditions of Set-up 7.3. Thus we may assume that $w = (0,-2,-1,1)$. For $\tau \geq 0$, the log canonicity of $(\mathbb{A}^2, C)$ near 0 and the inequality $\alpha \leq b + C_0/r$ are Proposition A.5(2), and the lower bound $\alpha \geq b$ under a fixed $\theta > 1$ with $(\mathbb{A}^2, \theta C)$ lc near 0 is Proposition A.5(3). A strictly increasing sequence of positive values α with $r \to +\infty$ has a subsequence with $\tau > 0$ for any member or a subsequence with $\tau \leq 0$ for any member, and Proposition A.5(4) excludes the first one. Therefore it remains to prove the following two statements for data with $\tau \leq 0$. First, for $r \gg 0$, $(\mathbb{A}^2, C)$ is lc near 0 and $\alpha \leq b + C_0/r$. Second, there exists no strictly increasing sequence of positive values α with $r \to +\infty$ and with $\tau \leq 0$ for any member. From now on we assume $\tau \leq 0$.

Let $\gamma := \min(\Gamma \backslash \{0\})$ if $\Gamma \not\subset \{0\}$, and $\gamma := 1$ if $\Gamma \subset \{0\}$, in which case $D = 0$. Let $m_0 := \lceil 2/\gamma \rceil$. We only consider the non-units h_j with $c_j > 0$, and we write each of them as a polynomial without monomials divisible by xy. The exceptional divisor of the blow-up of W_N at 0 has log discrepancy 2 with respect to $(W_N, 0)$, and (W_N, D) is klt near 0, so $\sum_j c_j \operatorname{mult}_0 h_j < 2$. Thus the number of the h_j is at most m_0. By Lemma 5.3, applied with $M = m_0$ and $\max_i |w_i| = 2$, for $r \geq r_0'$, where r_0' is the integer r_0 of Lemma 5.3 and depends only on Γ, $|\kappa_j| \leq 2m_0$ and any monomial of $h_j - h_j^0$ has $\mathbf{T}$-weight $\kappa_j + lr$ for a non-zero integer l and total degree at least $(r - 2m_0)/2$. Since κ_j is the $\mathbf{T}$-weight of a monomial of h_j of least degree and $|e' - 2b - c| \leq 2(a + b + c + e')$ for any monomial $x^a y^b z^c t^{e'}$, $|\kappa_j| \leq 2 \operatorname{mult}_0 h_j$. Hence $|\tau| \leq 2 \sum_j c_j \operatorname{mult}_0 h_j < 4 \leq 2m_0$ if $\Gamma \not\subset \{0\}$, and $\tau = 0$ otherwise. Since G preserves $xy + z^2 + t^N$ up to a character, for any $g = (g_x, g_y, g_z, g_t) \in G$,

$$g_x g_y = g_z^2 = g_t^N. \tag{A.71}$$

Step 2. In this step we pass from C to the plane pair $(\mathbb{A}^2, C_t)$.

Let

$$\Phi_j(P,Q) := h_j^0(P^2, -Q^2, PQ, 1) \quad \text{and} \quad C_t := \sum_j c_j \operatorname{div}(\Phi_j) \quad \text{on } \mathbb{A}^2_{P,Q}.$$

Since h_j^0 has no monomial divisible by xy and, by the $\mathbf{T}$-homogeneity of h_j^0, the exponent of t in a monomial of h_j^0 is determined by its exponents in x, y, z, the substitution $(x, y, z) \to (P^2, -Q^2, PQ)$ maps distinct monomials of h_j^0 to distinct monomials, so $\Phi_j \neq 0$. Since h_j^0 is $\mathbf{T}$-homogeneous of weight κ_j,

$$h_j^0(x, \lambda^{-2} y, \lambda^{-1} z, \lambda t) = \lambda^{\kappa_j} h_j^0(x, y, z, t) \quad \text{for any } \lambda \in \mathbb{C}^*.$$

Taking $\lambda = q$, we obtain

$$h_j^0(-p^2, 1, p, q) = q^{\kappa_j} h_j^0(-p^2, q^2, pq, 1) = q^{\kappa_j} \Phi_j(ip, -iq). \tag{A.72}$$

Hence, under the linear change of coordinates $(P, Q) = (ip, -iq)$,

$$C = C_t + \tau \operatorname{div}(Q). \tag{A.73}$$

For $(g, \xi) \in \tilde{H}$ let $\epsilon := g_z \xi^{-1}$. By (A.71), $\epsilon^2 = g_z^2 / g_y = g_x$ and $g_t \xi = g_z g_t \epsilon^{-1}$, so $\tilde{H}$ acts on $\mathbb{A}^2_{P,Q}$ by

$$(P, Q) \to (\epsilon P, g_z g_t \epsilon^{-1} Q), \qquad \epsilon^2 = g_x. \tag{A.74}$$

The action (A.74) commutes with the linear change of coordinates, and the element $\iota \in \tilde{H}$ given by $(1, -1)$, the image of the element $(1, -1)$ of the group of pairs (g, ξ), acts by $(P, Q) \to (-P, -Q)$. The group $\tilde{H}$ has order $2[G : R]$ and acts faithfully on $\mathbb{A}^2_{P,Q}$ (Proposition A.5(1)). Let

$$\beta := \operatorname{mld}((\mathbb{A}^2 \ni 0, C_t)/\tilde{H})$$

whenever $(\mathbb{A}^2, C_t)$ is lc near 0. Since $\tau \leq 0$, (A.73) implies $C \leq C_t$. Thus, if $(\mathbb{A}^2, C_t)$ is lc near 0, then $(\mathbb{A}^2, C)$ is lc near 0 and $b \geq \beta$. If $\tau = 0$, then $C = C_t$ and $b = \beta$.

Step 3. In this step we compare log discrepancies over W_N with log discrepancies over two charts.

Let $X_d := (XY + Z^2 + v^d = 0) \subset \mathbb{A}^4_{X,Y,Z,v}$ and $S_0 := X_d \cap (v = 0) = (XY + Z^2 = 0) \subset \mathbb{A}^3_{X,Y,Z}$. The functions

$$X = x, \quad Y = yt^2, \quad Z = zt, \quad v = t^r$$

have $\mathbf{T}$-weight 0 except v, which is invariant under $R = \mu_r$, and

$$XY + Z^2 + v^d = t^2(xy + z^2 + t^N)$$

since $rd = N + 2$. Since t is determined by v up to μ_r and $y = Y/t^2$, $z = Z/t$, they define a birational map $W_N/R \dashrightarrow X_d$. The group G/R acts on X_d by

$$(X, Y, Z, v) \to (g_x X, g_y g_t^2 Y, g_z g_t Z, g_t^r v),$$

and the birational map is G/R-equivariant. The action of G/R on S_0 is faithful: if $g_x = g_y g_t^2 = g_z g_t = 1$, then $g = (1, \lambda^{-2}, \lambda^{-1}, \lambda)$ with $\lambda = g_t$, so $g \in G \cap \mathbf{T} = R$.

Next, let P be a square root of x and $Q := zt/P$. Then

$$X = P^2, \qquad Z = PQ, \qquad Y = -Q^2 - \frac{v^d}{P^2}, \tag{A.75}$$

and X_d is birational to $V := \mathbb{A}^3_{P,Q,v}$ through the double cover (A.75), whose deck transformation is ι. In other words, W_N/G is birational to $V/\tilde{H}$ through the double cover $x = P^2$ of W_N, followed by the quotient by R. Here the bijection $(g, \xi) \to (g, \epsilon)$, $\epsilon = g_z \xi^{-1}$, of **Step 2** induces an isomorphism from $\tilde{H}$ to the quotient of the group of pairs (g, ϵ) with $g \in G$ and $\epsilon^2 = g_x$ by the subgroup $\{(g, 1) : g \in R\}$, which is the image of $\{((1, \lambda^{-2}, \lambda^{-1}, \lambda), \lambda^{-1})\}$. The pair (g, ϵ) acts on the double cover by $P \to \epsilon P$. Here $\tilde{H}$ acts on V by (A.74) and $v \to g_t^r v$, which is trivial on $\{((1, \lambda^{-2}, \lambda^{-1}, \lambda), \lambda^{-1})\}$. The section $(v = 0) \subset V$ is the plane $\mathbb{A}^2_{P,Q}$ of **Step 2**. Since $\frac{dx}{x} \wedge dz \wedge dt$ generates ω_{W_N} near 0, $\frac{dX}{X} \wedge dZ \wedge dv$ generates ω_{X_d} near 0, and $dP \wedge dQ \wedge dv$ generates ω_V, a direct calculation implies

$$\frac{dx}{x} \wedge dz \wedge dt = \frac{1}{r} \cdot \frac{dX}{X} \wedge dZ \wedge \frac{dv}{v} = \frac{2}{r} dP \wedge dQ \wedge \frac{dv}{v}. \tag{A.76}$$

By Lemma 5.3, any monomial $x^a y^b z^c t^{e'}$ of h_j has $\mathbf{T}$-weight $e' - 2b - c = \kappa_j + lr$ for an integer l, and $x^a y^b z^c t^{e'} = t^{\kappa_j} X^a Y^b Z^c v^l$. Hence

$$h_j = t^{\kappa_j} B_j, \qquad B_j = \sum_{l \in \mathbb{Z}} v^l B_j^l(X, Y, Z), \qquad B_j^0 = h_j^0(X, Y, Z, 1), \tag{A.77}$$

where $v^l B_j^l$ collects the terms of $\mathbf{T}$-weight $\kappa_j + lr$, and $B_j^0(P^2, -Q^2, PQ) = \Phi_j(P, Q)$. Each B_j^l is a G/R-semi-invariant polynomial, and $B_j^l = 0$ for $l \neq 0$ if $h_j = h_j^0$. Since $\operatorname{div}(t)$ corresponds to $\frac{1}{r}\operatorname{div}(v)$, (A.76) and (A.77) imply that $K_{W_N} + D$ corresponds to $K_{X_d} + B_{X_d}$ and to $K_V + B_V$, where

$$B_{X_d} := \Big(1 + \frac{\tau}{r}\Big) S_0 + \sum_j c_j \operatorname{div}(B_j), \qquad B_V := \Big(1 + \frac{\tau}{r}\Big) \operatorname{div}(v) + \sum_j c_j \operatorname{div}(B_j).$$

Together with [Liu26, Lemma 2.6(2)(5)], this implies the following. Let E be a prime divisor over W_N, and let E_d and E_V be prime divisors over X_d and over V whose images on the quotients correspond to the image of E on W_N/G. Then

$$\frac{a(E, W_N, D)}{|I_G(E)|} = \frac{a(E_d, X_d, B_{X_d})}{|I_{G/R}(E_d)|} = \frac{a(E_V, V, B_V)}{|I_{\tilde{H}}(E_V)|}. \tag{A.78}$$

If E has center 0, then X, Y, Z, v vanish along E, so E_d has center 0. Conversely, if E_V has center 0 and $\operatorname{mult}_{E_V} v > 2 \operatorname{mult}_{E_V} P + 2 \operatorname{mult}_{E_V} Q$, then by (A.75) the orders of x, y, z, t along E_V are

$$\begin{gathered} 2 \operatorname{mult}_{E_V} P, \qquad 2 \operatorname{mult}_{E_V} Q - \frac{2}{r} \operatorname{mult}_{E_V} v, \\ \operatorname{mult}_{E_V} P + \operatorname{mult}_{E_V} Q - \frac{1}{r} \operatorname{mult}_{E_V} v, \qquad \frac{1}{r} \operatorname{mult}_{E_V} v, \end{gathered} \tag{A.79}$$

and E has center 0 once the four numbers in (A.79) are positive.

Finally, $\mathbb{A}^2_{P,Q} \to S_0$, $(P,Q) \to (P^2, -Q^2, PQ)$, is the quotient by $\langle\iota\rangle$, which is étale in codimension one, and it is equivariant for $\tilde{H} \to G/R$ with kernel $\langle\iota\rangle$. Thus $(\mathbb{A}^2, C_t)$ is the crepant pullback of $(S_0, \sum_j c_j \operatorname{div}(B_j^0|_{S_0}))$, the pair $(S_0, \sum_j c_j \operatorname{div}(B_j^0|_{S_0}))$ is lc near 0 if and only if $(\mathbb{A}^2, C_t)$ is lc near 0 [KM98, Proposition 5.20(4)], and in this case

$$\operatorname{mld}\Big(\Big(S_0 \ni 0, \sum_j c_j \operatorname{div}(B_j^0|_{S_0})\Big)/(G/R)\Big) = \beta \tag{A.80}$$

by [Liu26, Lemma 2.6(5)].

Step 4. In this step we show that the log canonicity of a plane pair implies the log canonicity of the corresponding homogeneous pairs.

Claim A.6. *Let $c'_1, \dots, c'_s$ be positive real numbers, and let $h'_1, \dots, h'_s$ be $\mathbf{T}$-homogeneous polynomials of $\mathbf{T}$-weights $\kappa'_1, \dots, \kappa'_s$ without monomials divisible by xy, such that $\sum_j c'_j\kappa'_j \le 0$. Let $W_0 := (xy + z^2 = 0) \subset \mathbb{A}^4$. Let*

$$C'_t := \sum_j c'_j \operatorname{div}(h'_j(P^2, -Q^2, PQ, 1)) \quad \text{on } \mathbb{A}^2_{P,Q}.$$

If $(\mathbb{A}^2, C'_t)$ is lc near 0, then $(W_{N'}, \sum_j c'_j \operatorname{div}(h'_j))$ is lc near 0 for any integer $N' \ge 1$, and $(W_0, \sum_j c'_j \operatorname{div}(h'_j))$ is lc near 0.

Proof. Let $\tau' := \sum_j c'_j\kappa'_j$, $D' := \sum_j c'_j \operatorname{div}(h'_j)$, and $B'_j := h'_j(X, Y, Z, 1)$. We apply **Step 3** to $W_{N'}$ with G and R replaced by the trivial group, r by 1, and d by $N' + 2$. For W_0 we apply **Step 3** with the term v^d deleted, so that X_d is replaced by $X_\infty := (XY + Z^2 = 0) \subset \mathbb{A}^4_{X,Y,Z,v}$. Since h'_j is $\mathbf{T}$-homogeneous, $h'_j = t^{\kappa'_j} B'_j$. Let X' be $X_{N'+2}$, resp. X_∞. By (A.78), for any prime divisor E over $W_{N'}$, resp. W_0, with center 0, the corresponding prime divisor E' over X' has center 0 and

$$\begin{aligned} a(E, W_{N'}, D') &= a\Big(E', X', (1+\tau')S_0 + \sum_j c'_j \operatorname{div}(B'_j)\Big) \\ &\ge a\Big(E', X', S_0 + \sum_j c'_j \operatorname{div}(B'_j)\Big), \end{aligned}$$

since $\tau' \le 0$ and v vanishes along E' (resp. with W_0 in place of $W_{N'}$). The variety X' is a normal klt hypersurface in $\mathbb{A}^4$, and $S_0 = X' \cap (v = 0)$ is the normal klt surface $(XY + Z^2 = 0)$. By the last paragraph of **Step 3** applied to C'_t, $(S_0, \sum_j c'_j \operatorname{div}(B'_j|_{S_0}))$ is lc near 0. By Lemma 5.5, applied with the trivial group, $u = v$, $b = 1$, and $\mathfrak{a} = \prod_j (B'_j)^{c'_j}$, the right-hand side is non-negative. Thus any prime divisor over $W_{N'}$, resp. W_0, with center 0 has non-negative log discrepancy. If a pair is not lc near 0, then it has a prime divisor with center 0 and negative log discrepancy (cf. [KM98, Corollary 2.31(1)]). This proves the Claim. □

Proof of Theorem 7.12 continued.

Step 5. In this step we show that $(\mathbb{A}^2, C_t)$ is lc near 0 and $\alpha \le \beta + C_0/r$ for $r \gg 0$.

Recall W_0 from the Claim of **Step 4**, and let $D^0 := \sum_j c_j \operatorname{div}(h_j^0|_{W_0})$. Since (W_N, D) is lc near 0, $(\mathbb{A}^4, W_N + \sum_j c_j \operatorname{div}(h_j))$ is lc near 0 [Kaw07, Theorem], that is, the $\mathbb{R}$-ideal $\mathfrak{a}_N := (xy + z^2 + t^N) \prod_j (h_j)^{c_j}$ has lc threshold at least 1 at 0. Let $\mathfrak{a}^0 := (xy + z^2) \prod_j (h_j^0)^{c_j}$. The exponents of $\mathfrak{a}_N$ and $\mathfrak{a}^0$ belong to the DCC set $\Gamma \cup \{1\}$. By [dFEM11, Theorem 1.1], there exists $\epsilon_0 > 0$, depending only on Γ, such that no such lc threshold at 0 belongs to $[1 - \epsilon_0, 1)$. The corresponding factors of $\mathfrak{a}_N$ and $\mathfrak{a}^0$ are congruent modulo $(x, y, z, t)^n$ with $n = \min\{N, \lceil (r - 2m_0)/2 \rceil\}$ (**Step 1**), and $N = rd - 2 \ge r - 2$. By Theorem 2.12(2), applied on $\mathbb{A}^4$ with $\epsilon_0/2$ in place of ϵ, there exists r_2, depending only on Γ, such that if $r \ge r_2$, then the lc

thresholds of $\mathfrak{a}_N$ and $\mathfrak{a}^0$ at 0 differ by less than ϵ_0. Thus the lc threshold of $\mathfrak{a}^0$ at 0 is at least 1, and (W_0, D^0) is lc near 0 [Kaw07, Theorem].

Let $\mathbb{A}^3_{\sigma_1,\sigma_2,t} \to W_0$ be the morphism $(x,y,z,t) = (\sigma_1^2, -\sigma_2^2, \sigma_1\sigma_2, t)$, which is the quotient by $(\sigma_1,\sigma_2,t) \to (-\sigma_1,-\sigma_2,t)$ and is étale in codimension one. By the $\mathbf{T}$-homogeneity of h^0_j with $\lambda = t$,

$$h^0_j(\sigma_1^2, -\sigma_2^2, \sigma_1\sigma_2, t) = t^{\kappa_j}\Phi_j(\sigma_1, \sigma_2 t). \tag{A.81}$$

Let

$$B^0 := \sum_j c_j \operatorname{div}(h^0_j(\sigma_1^2, -\sigma_2^2, \sigma_1\sigma_2, t)).$$

Then $(\mathbb{A}^3, B^0)$ is the crepant pullback of (W_0, D^0), so it is lc near 0 [KM98, Proposition 5.20(4)]. Suppose that $(\mathbb{A}^2, C_t)$ is not lc near 0. Then there exists a prime divisor ν over $\mathbb{A}^2_{P,Q}$ with center 0 and $a(\nu, \mathbb{A}^2, C_t) < 0$ (cf. [KM98, Corollary 2.31(1)]). Choose a positive integer m such that $ma(\nu, \mathbb{A}^2, C_t) - \tau < 0$ and $m \operatorname{mult}_\nu Q > 1$, and let E be the Gauss extension of $m \operatorname{mult}_\nu$ with t-value 1 to $\mathbb{A}^2_{P,Q} \times \mathbb{A}^1_t$, which is mult_E for a prime divisor E [Liu26, Definition-Proposition 2.11 and Lemma 18.1], with

$$a(E, \mathbb{A}^2 \times \mathbb{A}^1_t, \operatorname{pr}_1^* C_t) = ma(\nu, \mathbb{A}^2, C_t) + 1$$

[Liu26, Lemma 18.1]. The birational map

$$\mathbb{A}^3_{\sigma_1,\sigma_2,t} \dashrightarrow \mathbb{A}^2_{P,Q} \times \mathbb{A}^1_t, \quad (P,Q,t) = (\sigma_1, \sigma_2 t, t),$$

satisfies $dP \wedge dQ \wedge dt = t d\sigma_1 \wedge d\sigma_2 \wedge dt$. Together with (A.81), this implies that $K_{\mathbb{A}^3} + B^0$ corresponds to

$$K_{\mathbb{A}^2 \times \mathbb{A}^1} + \operatorname{pr}_1^* C_t + (1+\tau)\operatorname{div}(t).$$

The orders of σ_1, σ_2, t along E are $m \operatorname{mult}_\nu P > 0$, $m \operatorname{mult}_\nu Q - 1 > 0$, and 1, so E is a prime divisor over $\mathbb{A}^3_{\sigma_1,\sigma_2,t}$ with center 0 and

$$a(E, \mathbb{A}^3, B^0) = ma(\nu, \mathbb{A}^2, C_t) + 1 - (1+\tau) = ma(\nu, \mathbb{A}^2, C_t) - \tau < 0,$$

a contradiction. Thus $(\mathbb{A}^2, C_t)$ is lc near 0 for $r \geq \max\{r_0', r_2\}$, and so is $(\mathbb{A}^2, C)$ (**Step 2**).

We prove the upper bound. The coefficients of the crepant quotient of $(\mathbb{A}^2, C_t)$ by $\tilde{H}$ belong to the DCC set $\Psi(\Gamma_+ \cap [0,1])$ [Liu26, Lemma 2.10(1)(5)] (cf. [HMX14, Proposition 3.4.1(3)]). By Theorem 2.9(2), applied to the crepant quotient germ, there exists m_1, depending only on Γ, and a prime divisor ν over $\mathbb{A}^2_{P,Q}$ with center 0 computing β whose image $\bar{\nu}$ over $\mathbb{A}^2/\tilde{H}$ satisfies $a(\bar{\nu}, \mathbb{A}^2/\tilde{H}, 0) \leq m_1$. Let $e := |I_{\tilde{H}}(\nu)| \leq 2e_0$. Since the coefficients of the branch boundary are non-negative, [Liu26, Lemma 2.6(2)] implies $a(\nu, \mathbb{A}^2, 0) \leq em_1 \leq 2e_0m_1$. Since $(\mathbb{A}^2, C_t)$ is lc near 0, $c_j \operatorname{mult}_\nu \Phi_j \leq a(\nu, \mathbb{A}^2, 0)$, and $\operatorname{mult}_\nu P + \operatorname{mult}_\nu Q \leq a(\nu, \mathbb{A}^2, 0)$ [Liu26, Lemma 2.13]. Let $m_2 := 2e_0m_1$ and $m_3 := 2m_2 + m_2/\gamma$. Then

$$\operatorname{mult}_\nu P + \operatorname{mult}_\nu Q \leq m_2 \quad \text{and} \quad \operatorname{mult}_\nu \Phi_j \leq m_3 - 2m_2 \ \text{ for any } j. \tag{A.82}$$

Let $\chi_v : (g, \xi) \to g_t^r$ be the character of $\tilde{H}$ on v, let n_ν be the integer of Lemma 3.1 for ν and χ_v, and let ℓ be the integer with $\ell \equiv n_\nu \pmod{e}$ and $m_3 < \ell \leq m_3 + e$. We check the hypotheses of Proposition 5.4.

- $S = \mathbb{A}^2_{P,Q}$ is a normal surface, and $s = 0$.
- $\tilde{H}$ is finite, fixes 0, and acts faithfully on $\mathbb{A}^2_{P,Q}$ (**Step 2**, Proposition A.5(1)).
- $\tilde{H}$ acts on $V = \mathbb{A}^2_{P,Q} \times \mathbb{A}^1_v$ by (A.74) and the character χ_v on v (well defined on $\tilde{H}$, **Step 3**).
- The coefficient of $\operatorname{div}(v)$ is $1 + \tau/r \leq 1$ since $\tau \leq 0$, and $c_j \geq 0$.
- Each Φ_j, viewed as a function on V independent of v, is $\tilde{H}$-semi-invariant, since $\Phi_j(P,Q) = B^0_j(P^2, -Q^2, PQ)$ with B^0_j a G/R-semi-invariant polynomial and $\tilde{H} \to G/R$ equivariant (**Step 3**), and $\Phi_j|_{v=0} = \Phi_j \neq 0$ (**Step 2**).
- $(\mathbb{A}^2, C_t)$ is lc near 0 (first half of this Step).

- ν computes β (by its choice above), and $e = |I_{\tilde{H}}(\nu)|$.
- $\ell \equiv n_\nu \pmod e$ with $\ell > m_3 \geq \operatorname{mult}_\nu \Phi_j$ by (A.82).

By Proposition 5.4, the Gauss extension of mult_ν with v-value ℓ is $\operatorname{mult}_{E_V}$ for a prime divisor E_V over V with center 0, $I_{\tilde{H}}(E_V) = I_{\tilde{H}}(\nu)$, $\operatorname{mult}_{E_V} \Phi_j = \operatorname{mult}_\nu \Phi_j$, and

$$\frac{a(E_V, V, (1+\frac{\tau}{r})\operatorname{div}(v) + \sum_j c_j \operatorname{div}(\Phi_j))}{|I_{\tilde{H}}(E_V)|} = \beta - \frac{\tau\ell}{re}. \tag{A.83}$$

We show that $\operatorname{mult}_{E_V} B_j = \operatorname{mult}_\nu \Phi_j$ for $r \gg 0$. By (A.75), $Y = -Q^2 - v^d/P^2$, and by (A.82),

$$\operatorname{mult}_{E_V}(v^d/P^2) \geq \ell - 2\operatorname{mult}_\nu P > m_3 - 2m_2 \geq \operatorname{mult}_\nu \Phi_j.$$

In particular, since $\ell > m_3 \geq 2\operatorname{mult}_\nu P + 2\operatorname{mult}_\nu Q$ by (A.82), $\operatorname{mult}_{E_V} Y = 2\operatorname{mult}_\nu Q \geq 2$ and $\operatorname{mult}_{E_V} Z \geq 2$. Thus $B_j^0(P^2, Y, PQ) - \Phi_j(P, Q)$, which is divisible by v^d/P^2 with a quotient of non-negative order along E_V, has order larger than $\operatorname{mult}_\nu \Phi_j$. For $l \geq 1$, $v^l B_j^l$ has order at least $\ell > \operatorname{mult}_\nu \Phi_j$. For $l \leq -1$, a monomial $x^a y^b z^c t^{e'}$ of h_j of $\mathbf{T}$-weight $\kappa_j + lr$ satisfies $2b + c = e' - \kappa_j - lr \geq -lr - 2m_0$, so the order of $X^a Y^b Z^c v^l$ along E_V is at least $2b + c + l\ell \geq (-l)(r-\ell) - 2m_0 \geq r - m_3 - e - 2m_0$, which is larger than m_3 for $r > 2m_3 + 2e_0 + 2m_0$. Therefore $\operatorname{mult}_{E_V} B_j = \operatorname{mult}_\nu \Phi_j = \operatorname{mult}_{E_V} \Phi_j$, and the left-hand side of (A.83) is $a(E_V, V, B_V)/|I_{\tilde{H}}(E_V)|$.

By (A.82), $\ell > m_3 \geq 2\operatorname{mult}_\nu P + 2\operatorname{mult}_\nu Q$, and for $r > \ell$ the four numbers in (A.79) are positive. Thus the prime divisor E over W_N corresponding to E_V has center 0, and by (A.78) and (A.83),

$$\alpha \leq \frac{a(E, W_N, D)}{|I_G(E)|} = \beta - \frac{\tau\ell}{re} \leq \beta + \frac{2m_0(m_3 + 2e_0)}{r}.$$

Let $C_0^- := 2m_0(m_3 + 2e_0)$, which depends only on Γ and e_0. Since $b \geq \beta$ (**Step 2**), we obtain $\alpha \leq \beta + C_0^-/r \leq b + C_0^-/r$ for $r \gg 0$. Together with Proposition A.5(2) for $\tau \geq 0$, the inequality $\alpha \leq b + C_0/r$ of the theorem holds with $C_0 := \max\{C_0^+, C_0^-\}$, where C_0^+ is the constant C_0 of Proposition A.5(2), for r at least the maximum of the thresholds of that proposition and of this step.

Step 6. In this step we prove the lower bound under a fixed margin of $(\mathbb{A}^2, C_t)$.

Claim A.7. *Let $\theta > 1$ be a real number. Then there exists r_1, depending only on Γ, e_0, and θ, such that if $\tau \leq 0$, $r \geq r_1$, and $(\mathbb{A}^2, \theta C_t)$ is lc near 0, then $\alpha \geq \beta$.*

Proof. The proof of the Claim is divided into four parts.

Margins. By the Claim of **Step 4** applied to the labels θc_j, $(W_0, \theta D^0)$ is lc near 0. As in **Step 5**, with the DCC set $\theta\Gamma \cup \{1\}$ in place of $\Gamma \cup \{1\}$, [Kaw07, Theorem], [dFEM11, Theorem 1.1], and Theorem 2.12(2) imply that $(W_N, \theta D)$ is lc near 0 for $r \geq r_3$, where r_3 depends only on Γ and θ. Let

$$B := \sum_j c_j \operatorname{div}(h_j(\sigma_1^2, -\sigma_2^2, \sigma_1\sigma_2, t)) \quad \text{on } \mathbb{A}^3_{\sigma_1,\sigma_2,t},$$

and recall B^0 from **Step 5**. Since $(\mathbb{A}^3, \theta B^0)$ is the crepant pullback of $(W_0, \theta D^0)$, it is lc near 0 [KM98, Proposition 5.20(4)]. A monomial $x^a y^b z^c t^{e'}$ of $h_j - h_j^0$ maps to $\pm\sigma_1^{2a+c}\sigma_2^{2b+c}t^{e'}$ with $e' - (2b + c) = \kappa_j + lr$ and $l \neq 0$, so

$$h_j(\sigma_1^2, -\sigma_2^2, \sigma_1\sigma_2, t) - h_j^0(\sigma_1^2, -\sigma_2^2, \sigma_1\sigma_2, t) \in (\sigma_2, t)^{r-2m_0}. \tag{A.84}$$

By [dFEM11, Theorem 1.1] and Theorem 2.12(2) on the fixed $\mathbb{A}^3$, $(\mathbb{A}^3, \theta B)$ is lc near 0 for $r \geq r_4$, where r_4 depends only on Γ and θ. Since $\theta > 1$, $(\mathbb{A}^3, B)$ is klt near 0.

Bounds. Since β is at most the log discrepancy of the exceptional divisor of the blow-up of $\mathbb{A}^2$ at 0, $\beta \leq 2$. Thus it suffices to show $a(E, W_N, D)/|I_G(E)| \geq \beta$ for any prime divisor E over W_N with center 0 and $a(E, W_N, D)/|I_G(E)| \leq 2$. Fix such E, and write $[f] := \operatorname{mult}_E f/|I_G(E)|$ for a regular

function f near 0. Since $(W_N, \theta D)$ is lc near 0, $(\theta-1)\sum_j c_j[h_j] \le a(E, W_N, D)/|I_G(E)| \le 2$. Let $\Theta_1 := \theta/(\theta-1)$ and $\Theta_2 := 1/(\gamma(\theta-1))$. Thus

$$[h_j] \le 2\Theta_2 \quad \text{and} \quad \frac{a(E, W_N, 0)}{|I_G(E)|} \le 2\Theta_1.$$

The surface $W_N \cap (t=0)$ is $(xy+z^2=0) \subset \mathbb{A}^3$, whose quotient morphism $(P,Q) \to (P^2, -Q^2, PQ)$ from $\mathbb{A}^2$ is étale in codimension one and pulls back the restrictions of $\frac{1}{2}\operatorname{div}(x) + \frac{1}{2}\operatorname{div}(y)$ and $\operatorname{div}(z)$ to $\operatorname{div}(P) + \operatorname{div}(Q)$. Thus by [KM98, Proposition 5.20(4)] and [Kaw07, Theorem], the pairs

$$(W_N, \frac{1}{2}\operatorname{div}(x) + \frac{1}{2}\operatorname{div}(y) + \operatorname{div}(t)) \quad \text{and} \quad (W_N, \operatorname{div}(z) + \operatorname{div}(t))$$

are lc near 0. Evaluating the log discrepancies of both pairs at E, we obtain $[x], [y] \le 4\Theta_1$ and $[z], [t] \le 2\Theta_1$. The functions x, yt^2, and zt are G-semi-invariant, invariant under R, and vanish at 0, so their $[G:R]$-th powers are G-invariant regular functions vanishing at 0, whose orders along E are positive multiples of $|I_G(E)|$ [Liu26, Lemma 2.6]. Since $[G:R] \le e_0$,

$$[x] \ge \frac{1}{e_0}, \qquad [y] + 2[t] \ge \frac{1}{e_0}, \qquad [z] + [t] \ge \frac{1}{e_0}. \tag{A.85}$$

Let $\rho_- := \min\{[y]/2, [z]\}$. By (A.85), $[t] + \rho_- \ge 1/(2e_0)$. We call a monomial of h_j of $\mathbf{T}$-weight $\kappa_j + lr$ with $l \le -1$ a *negative monomial*. A negative monomial $x^a y^b z^c t^{e'}$ satisfies $2b + c \ge r - 2m_0$, so $[x^a y^b z^c t^{e'}] \ge (r - 2m_0)\rho_-$.

Case A: any negative monomial $\mathfrak{n}$ of any h_j satisfies $[\mathfrak{n}] > 2\Theta_2$. Let h_j^+ be the sum of the terms of h_j of $\mathbf{T}$-weight $\kappa_j + lr$ with $l \ge 0$. Then $[h_j - h_j^+] > 2\Theta_2 \ge [h_j]$, so $[h_j^+] = [h_j]$. By (A.77), $h_j^+ = t^{\kappa_j} B_j^+$ with the G/R-semi-invariant polynomial $B_j^+ := \sum_{l \ge 0} v^l B_j^l$, and $B_j^+|_{v=0} = B_j^0$. Let E_d be the prime divisor over X_d corresponding to E, which has center 0 (**Step 3**). Since $\operatorname{mult}_{E_d} B_j = \operatorname{mult}_{E_d} B_j^+$, $\tau \le 0$, and v vanishes along E_d, (A.78) implies

$$\begin{aligned} \frac{a(E, W_N, D)}{|I_G(E)|} &= \frac{a(E_d, X_d, S_0 + \sum_j c_j \operatorname{div}(B_j^+)) - \frac{\tau}{r}\operatorname{mult}_{E_d} v}{|I_{G/R}(E_d)|} \\ &\ge \frac{a(E_d, X_d, S_0 + \sum_j c_j \operatorname{div}(B_j^+))}{|I_{G/R}(E_d)|}. \end{aligned}$$

The group G/R acts diagonally on $\mathbb{A}^4_{X,Y,Z,v}$ and faithfully on X_d and on S_0 (**Step 3**), X_d is a normal klt hypersurface, $S_0 = X_d \cap (v=0)$ is a normal klt complete intersection, and $(S_0, \sum_j c_j \operatorname{div}(B_j^0|_{S_0}))$ is lc near 0 (**Steps 3 and 5**). By Lemma 5.5, applied with $u = v$, $b = 1$, $H = G/R$, and $\mathfrak{a} = \prod_j (B_j^+)^{c_j}$, and by (A.80), the right-hand side is at least β.

Case B: some negative monomial $\mathfrak{n}$ of some h_j satisfies $[\mathfrak{n}] \le 2\Theta_2$. Then $\rho_- \le 2\Theta_2/(r-2m_0)$, so $\rho_- \le 1/(4e_0)$ and $[t] \ge 1/(4e_0)$ for $r \ge 2m_0 + 8e_0\Theta_2$. Let $y_0 := -z^2/x = y + t^N/x$. The map $(x,y,z,t) \to (x, y_0, z, t)$ is a G-equivariant birational map $W_N \dashrightarrow W_0$, which is the identity on (x,z,t), and $\frac{dx}{x} \wedge dz \wedge dt$ generates both ω_{W_N} and ω_{W_0} near 0. Since $N \ge r-2$,

$$\frac{\operatorname{mult}_E(t^N/x)}{|I_G(E)|} \ge \frac{r-2}{4e_0} - 4\Theta_1 > 2\Theta_2 + 4\Theta_1$$

for $r \ge r_5$, where r_5 depends only on Γ, e_0, and θ. Since $[y] \le 4\Theta_1$, the order of y_0 along E equals that of y, so E is a prime divisor over W_0 with center 0 and with inertia group $I_G(E)$. For any j, $h_j(x,y,z,t) - h_j(x,y_0,z,t)$ is divisible by t^N/x with a quotient of non-negative order along E, and $[h_j] \le 2\Theta_2$, so the orders of $h_j(x,y,z,t)$ and $h_j(x,y_0,z,t)$ along E are equal. Thus, with $\hat{D} := \sum_j c_j \operatorname{div}(h_j|_{W_0})$ (with h_j in place of the blocks h_j^0 in D^0 of **Step 5**),

$$\frac{a(E, W_N, D)}{|I_G(E)|} = \frac{a(E, W_0, \hat{D})}{|I_G(E)|} \ge \operatorname{mld}((W_0 \ni 0, \hat{D})/G).$$

Let G^+ be the group of pairs (g, ϵ) with $g \in G$ and $\epsilon^2 = g_x$, acting on $\mathbb{A}^3_{\sigma_1,\sigma_2,t}$ by

$$(\sigma_1, \sigma_2, t) \to (\epsilon\sigma_1, g_z\epsilon^{-1}\sigma_2, g_t t).$$

By (A.71), the morphism $\mathbb{A}^3 \to W_0$ of **Step 5** is $G^+ \to G$-equivariant with deck transformation $(1, -1)$. Thus $\mathrm{mld}((W_0 \ni 0, \hat{D})/G) = \mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G^+)$ [Liu26, Lemma 2.6(5)].

Lemma 3.21(2) is used with $\theta' := (1 + \theta)/2$. In the following table, the left column lists the letters of [Liu26, Set-up 14.1] and the right column their values here. In the rest of this case, the letters $a_j, b_j, F_j, c_x, c_y, c_u, T, B_T, H, b_H$ stand for their values in the right column, and all other letters keep their meaning in this proof.

[Liu26, Set-up 14.1]	this proof
p, q	$1, 1$
P	1
x, y, t	t, σ_2, σ_1
u	$t\sigma_2 = Q$
T	$\mathbb{A}^2_{Q,P}$
r	r
N	$r - 2m_0$
G	G^+
H	$\tilde{H}$
e_0	$2e_0$
a_j, b_j, F_j	as defined below the table
c_x, c_y, c_u	$\sum_{\kappa_j>0} c_j\kappa_j$, $\sum_{\kappa_j<0} c_j(-\kappa_j)$, $\max\{c_x, c_y\}$
B_T	C_t
b_H	β

In the right column, P and Q are the coordinates of the plane $\mathbb{A}^2_{P,Q}$ of **Step 2**, identified with σ_1 and $\sigma_2 t$ by the map of **Step 5**. The symbol e_0 of [Liu26, Set-up 14.1] differs from the e_0 of Set-up 7.3. By (A.81),

$$h^0_j(\sigma_1^2, -\sigma_2^2, \sigma_1\sigma_2, t) = t^{a_j}\sigma_2^{b_j}F_j(Q, P)$$

with $(a_j, b_j, F_j) = (\kappa_j, 0, \Phi_j(P, Q))$ if $\kappa_j \geq 0$ and $(a_j, b_j, F_j) = (0, -\kappa_j, Q^{\kappa_j}\Phi_j(P, Q))$ if $\kappa_j < 0$. Here $Q^{-\kappa_j}$ divides Φ_j since the left-hand side of (A.81) is a polynomial. We check the conditions.

- G^+ acts diagonally and faithfully on (σ_1, σ_2, t): if $\epsilon = g_z\epsilon^{-1} = g_t = 1$, then $g_x = g_z = g_t = 1$ and $g_y = g_z^2/g_x = 1$ by (A.71).
- $G^+ \cap \mathbf{T}_{(1,-1,0)} = \mu_r(1, -1, 0)$ in the coordinates (t, σ_2, σ_1): an element of the intersection has $\epsilon = 1$, $g_t = \lambda$, $g_z = \lambda^{-1}$, $g_x = 1$, and $g_y = \lambda^{-2}$, so $g \in G \cap \mathbf{T} = R = \mu_r$. Conversely, $(g, 1)$ with $g \in R$ belongs to the intersection.
- The terms of $h_j(\sigma_1^2, -\sigma_2^2, \sigma_1\sigma_2, t)$ of $\mathbf{T}_{(1,-1,0)}$-weight $a_j - b_j = \kappa_j$ are exactly

  $$h^0_j(\sigma_1^2, -\sigma_2^2, \sigma_1\sigma_2, t),$$

  since the substitution preserves the $\mathbf{T}$-weight of monomials, and the remainder belongs to $(t, \sigma_2)^{r-2m_0}$ by (A.84).
- $(\mathbb{A}^3, B)$ is klt near 0 (the part on margins).
- $c_x = \sum_{\kappa_j>0} c_j\kappa_j$ and $c_y = \sum_{\kappa_j<0} c_j(-\kappa_j)$, so $c_x - c_y = \tau \leq 0$ and $c_u = \max\{c_x, c_y\} = c_y$. Thus

  $$B_T = \sum_j c_j \operatorname{div}(F_j) + c_y \operatorname{div}(Q) = C_t,$$

  and $(T, \theta B_T)$ is lc near 0.
- The image of G^+ under the characters $g_z g_t \epsilon^{-1}$ of Q and ϵ of P is $\tilde{H}$ acting by (A.74), so $|H| = 2[G : R] \leq 2e_0$ and $b_H = \beta$.

By Lemma 3.21(2), there exists r_6, depending only on Γ, e_0, and θ, such that for $r \geq r_6$,

$$\mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G^+) \geq \beta.$$

In both cases $a(E, W_N, D)/|I_G(E)| \geq \beta$. Hence the Claim holds with

$$r_1 := \max\{r_0', r_2, \dots, r_6,\ 2m_0 + 8e_0\Theta_2\}. \qquad \square$$

Proof of Theorem 7.12 continued.

Step 7. In this step we show that α cannot increase along $\mathbf{T}$ under the condition $\tau \leq 0$.

Suppose that there exists a sequence of data as in Set-up 7.3, indexed by i, with Γ, e_0, and $w = (0, -2, -1, 1)$ fixed, $\tau_i \leq 0$, $r_i \to +\infty$, and α_i positive and strictly increasing. We add the index i to the objects of **Steps 1–6**, write $D_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$, $C_{t,i} = \sum_j c_{ij} \operatorname{div}(\Phi_{ij})$, and $\beta_i = \mathrm{mld}((\mathbb{A}^2 \ni 0, C_{t,i})/\tilde{H}_i)$, and we pass to subsequences without mentioning it. We may assume $r_i \geq \max\{r_0', r_2\}$ and that r_i is large enough for **Step 5**. Let $\alpha_\infty := \lim_i \alpha_i$. Then $0 < \alpha_i < \alpha_\infty$ for any i.

The coefficients of the crepant quotient of $(\mathbb{A}^2, C_{t,i})$ by $\tilde{H}_i$ belong to the DCC set $\Psi(\Gamma_+ \cap [0,1])$ [Liu26, Lemma 2.10(1)(5)] (cf. [HMX14, Proposition 3.4.1(3)]), so the β_i belong to an ACC set (Theorem 2.9(1)). Thus we may assume that β_i is non-increasing with limit L. By **Step 5**, $\alpha_i \leq \beta_i + C_0^-/r_i$, so $L \geq \alpha_\infty$, and

$$0 < \alpha_i < \alpha_\infty \leq L \quad \text{for any } i. \tag{A.86}$$

The number s of the labels is at most m_0, and $|\kappa_{ij}| \leq 2m_0$ (**Step 1**), so we may assume that s and $\kappa_j := \kappa_{ij}$ do not depend on i. The group $\tilde{H}_i$ is a diagonal subgroup of $GL_2(\mathbb{C})$ of order at most $2e_0$, so any element of $\tilde{H}_i$ is a pair of $(2e_0)!$-th roots of unity. Thus we may assume that $\tilde{H} := \tilde{H}_i$ and the characters of the Φ_{ij} do not depend on i. Since Γ is a DCC set, we may assume that c_{ij} is non-decreasing in i with limit $c_j^* > 0$. Then

$$\tau^* := \sum_j c_j^* \kappa_j = \lim_i \tau_i \leq 0.$$

Since $(\mathbb{A}^2, C_{t,i})$ is lc near 0, the blow-up of $\mathbb{A}^2$ at 0 implies $\operatorname{mult}_0 \Phi_{ij} \leq 2/\gamma$.

Let $C_i^* := \sum_j c_j^* \operatorname{div}(\Phi_{ij})$. Let $\lambda_i := \min_j c_{ij}/c_j^*$. Then $\lambda_i \to 1$ and $\lambda_i C_i^* \leq C_{t,i}$, so the lc threshold of C_i^* with respect to $\mathbb{A}^2$ at 0 is at least λ_i. Since the coefficients of C_i^* belong to the finite set $\{c_j^*\}$, the lc thresholds of the C_i^* at 0 satisfy the ACC [dFEM11, Theorem 1.1], so they are at least 1 for $i \gg 0$. Thus $(\mathbb{A}^2, C_i^*)$ is lc near 0. Let $\beta_i^* := \mathrm{mld}((\mathbb{A}^2 \ni 0, C_i^*)/\tilde{H})$. Near the image of 0, the coefficients of the crepant quotient of $(\mathbb{A}^2, C_i^*)$ by $\tilde{H}$ belong to a finite, hence DCC, set (the components of $\operatorname{div}(\Phi_{ij})$ through 0 have multiplicity at most $\operatorname{mult}_0 \Phi_{ij} \leq 2/\gamma$, and the ramification indices are at most $|\tilde{H}| \leq 2e_0$, see [Liu26, Lemma 2.6(4)]), so by Theorem 2.9(2), applied to the crepant quotient germ, there exist m_1', depending only on Γ, e_0, and the c_j^*, and prime divisors ν_i over $\mathbb{A}^2$ with center 0 computing β_i^* with $a(\nu_i, \mathbb{A}^2, 0) \leq m_1'$, as in **Step 5**. Since $(\mathbb{A}^2, C_i^*)$ is lc near 0, $c_j^* \operatorname{mult}_{\nu_i} \Phi_{ij} \leq m_1'$. Since $C_{t,i} \leq C_i^*$,

$$0 \leq \beta_i - \beta_i^* \leq \frac{a(\nu_i, \mathbb{A}^2, C_{t,i}) - a(\nu_i, \mathbb{A}^2, C_i^*)}{|I_{\tilde{H}}(\nu_i)|} \leq m_1' \sum_j \frac{c_j^* - c_{ij}}{c_j^*}.$$

Thus $\beta_i^* \to L$. Moreover

$$\beta_i^* = \frac{a(\nu_i, \mathbb{A}^2, 0) - \sum_j c_j^* \operatorname{mult}_{\nu_i} \Phi_{ij}}{|I_{\tilde{H}}(\nu_i)|},$$

where $a(\nu_i, \mathbb{A}^2, 0)$ and $|I_{\tilde{H}}(\nu_i)|$ are bounded positive integers and $\operatorname{mult}_{\nu_i} \Phi_{ij}$ are bounded non-negative integers. Hence the β_i^* belong to a finite set, and we may assume that $\beta_i^* = L$ for any i.

We apply Construction-Proposition 4.1, case (a) with $n = 2$ and the group $\tilde{H}$, to the tuples $(\Phi_{i1}, \dots, \Phi_{is}, P, Q)$. The entries P and Q do not depend on i, so their limits are P and Q. We

obtain an algebraically closed field $k \supset \mathbb{C}$ and $\tilde{H}$-semi-invariant power series $\Phi_{\infty,j} \in k[[P,Q]]$. For any subset of the entries, the corresponding subtuple of the limit is a compatible generic limit of the corresponding subtuples by the statement of Construction-Proposition 4.1 on subsets $J \subset \{1,\dots,s+2\}$. Let $\mathcal{A}_k = \operatorname{Spec} k[[P,Q]]$ and

$$C_\infty := \sum_j c_j^* \operatorname{div}(\Phi_{\infty,j}).$$

By Theorem 4.2(1), applied with $\mathfrak{c} = \mathcal{O}$, $(\mathcal{A}_k, C_\infty)$ is lc, since $(\mathbb{A}^2, C_i^*)$ is lc near 0 for any i. By Theorem 4.2(2),

$$\operatorname{mld}((\mathcal{A}_k \ni 0, C_\infty)/\tilde{H}) = \beta_i^* = L > 0.$$

Suppose that $(\mathcal{A}_k, C_\infty)$ is klt. By Theorem 4.2(1), there exists $\theta > 1$ such that $(\mathbb{A}^2, \theta C_i^*)$, hence $(\mathbb{A}^2, \theta C_{t,i})$, is lc near 0 for any i. By the Claim of **Step 6**, $\alpha_i \geq \beta_i \geq L$ for $i \gg 0$, which contradicts (A.86).

Suppose that $(\mathcal{A}_k, C_\infty)$ is not klt. By Definition-Proposition 5.7, $(\mathcal{A}_k, C_\infty)$ has exactly one floor Σ, Σ is smooth at 0, and $L = (1-b')/e$, where $b' := \operatorname{mult}_0((C_\infty - \Sigma)|_\Sigma)$ and e is the order of the image of $\tilde{H}$ in the automorphism group of $T_0\Sigma$. Since Σ is the only floor, Σ is $\tilde{H}$-invariant. Let $\mu := 1 - b' = eL > 0$. If Σ is not tangent to $(P = 0)$, let $z_0 := P$. Otherwise, let $z_0 := Q$. Then $z_0|_\Sigma$ is a parameter of Σ at 0. Since $z_0|_\Sigma$ generates the cotangent space of Σ at 0, e is the order of the character χ of z_0. Since ι acts by $z_0 \to -z_0$, e is even.

Since $z_0|_\Sigma$ is a parameter of Σ at 0, Σ is not contained in $\operatorname{div}(z_0)$. Thus the floor Σ is a regular curve through 0 with coefficient one in $C_\infty + \mu \operatorname{div}(z_0)$, and

$$\operatorname{mult}_0((C_\infty - \Sigma + \mu \operatorname{div}(z_0))|_\Sigma) = b' + \mu = 1.$$

Hence the pair $(\mathcal{A}_k, C_\infty + \mu \operatorname{div}(z_0))$ is lc near 0 by [Kaw07, Theorem] on the formal germ $\mathcal{A}_k$ (Lemma 2.14). By Theorem 4.2(1), applied to the subtuples $(\Phi_{i1},\dots,\Phi_{is},z_0)$ with the labels $c_1^*,\dots,c_s^*,\mu$,

$$\left(\mathbb{A}^2, C_i^* + \mu \operatorname{div}(z_0)\right) \text{ is lc near 0 for any } i. \tag{A.87}$$

Let $U_0 := x$ if $z_0 = P$, and $U_0 := -yt^2$ if $z_0 = Q$. The polynomials x and yt^2 are $\mathbf{T}$-homogeneous of weight 0 and have no monomial divisible by xy, and

$$x(P^2, -Q^2, PQ, 1) = P^2, \qquad -yt^2(P^2, -Q^2, PQ, 1) = Q^2;$$

so $U_0(P^2, -Q^2, PQ, 1) = z_0^2$, and U_0 satisfies the conditions on the h_j' in the Claim of **Step 4**. By the Claim of **Step 4**, applied to the polynomials h_{ij}^0 with the labels c_j^* and to U_0 with the label $\mu/2$, whose mass is $\tau^* \leq 0$, (A.87) implies that the pair

$$(W_0, \sum_j c_j^* \operatorname{div}(h_{ij}^0) + \frac{\mu}{2} \operatorname{div}(U_0))$$

is lc near 0. As in **Step 5**, with the finite set $\{1, \mu/2\} \cup \{c_j^*\}$ of exponents and the factor U_0 on both W_0 and W_{N_i}, [Kaw07, Theorem], [dFEM11, Theorem 1.1], and Theorem 2.12(2) imply that the pair

$$(W_{N_i}, \sum_j c_j^* \operatorname{div}(h_{ij}) + \frac{\mu}{2} \operatorname{div}(U_0))$$

is lc near 0 for $i \gg 0$. Since $c_{ij} \leq c_j^*$,

$$\left(W_{N_i}, D_i + \frac{\mu}{2} \operatorname{div}(U_0)\right) \text{ is lc near 0 for } i \gg 0.$$

The function U_0 is a G_i-semi-invariant regular function vanishing at 0. The character of U_0 is $g \to g_x$ if $z_0 = P$ and $g \to g_y g_t^2$ if $z_0 = Q$. By (A.74), the characters $g \to g_x$ and $g \to g_y g_t^2$ pull

back to $\tilde{H}$ as $\epsilon^2 = \chi^2$ and $(g_t\xi)^2 = \chi^2$, respectively, and $\tilde{H} \to G_i/R_i$ is surjective. Since e is even, the character of U_0 has order $e/2$. By Lemma 5.8, applied with $Z = \{0\}$, $U = U_0$, and $\lambda = \mu/2$,

$$\alpha_i \geq \frac{\mu/2}{e/2} = \frac{\mu}{e} = L$$

for $i \gg 0$, which contradicts (A.86). This proves **Step 7**, and the proof of Theorem 7.12 is complete. □

A.7. **Canonical circles.** Here we prove Proposition 7.13, Construction-Proposition 7.14, and the claims of Remark 7.16.

We first construct a root chart along the circle $\mathbf{T}$. On the root chart, the quotient W_N/G is birational to a quotient of $\mathbb{A}^3$, and the principal blocks of the equations h_j become the plane equations φ_j.

Construction A.8. Notation as in Set-up 7.3 with $w = (-a, -b, -v, v)$, where $v > 0$, $b > 0$, $a + b = 2v$, and $\gcd(a, v) = 1$. Let γ be the least positive element of Γ, with $\gamma := 1$ if $\Gamma \subset \{0\}$, let $M := \lceil 2/\gamma \rceil$, and let $C_1 := \max\{1, v, -\frac{a}{2}\}$. Then

$$\max_i |w_i| = \max\{|a|, b, v\} \geq C_1.$$

Assume that $r > 2M \max_i |w_i|$, and that for any j, $c_j > 0$ and h_j is a G-semi-invariant polynomial with $h_j(0) = 0$ and without monomials divisible by xy.

(1) For any j and any $L \in \mathbb{Z}$, let $h_{j,L}$ be the sum of the terms of h_j of $\mathbf{T}$-weight $\kappa_j + Lr$. By Lemma 5.3, $|\kappa_j| \leq M \max_i |w_i|$ and $h_j = \sum_{L\in\mathbb{Z}} h_{j,L}$ with $h_{j,0} = h_j^0$. Let $\varphi_j(p, q) := h_j^0(-p^2, 1, p, q)$ and $C_{\text{plane}} := \sum_j c_j \operatorname{div}(\varphi_j)$ on $\mathbb{A}^2_{p,q}$.

(2) Let $\tilde{G}$ be the group of pairs (g, ξ) with $g \in G$ and $\xi^b = g_y$, let

$$\tilde{R} := \{((\lambda^{-a}, \lambda^{-b}, \lambda^{-v}, \lambda^v), \lambda^{-1}) : \lambda^r = 1\} \subset \tilde{G},$$

let $\tilde{H} := \tilde{G}/\tilde{R}$, and let $\tilde{G}_0 := \{(1, \xi) : \xi^b = 1\} \subset \tilde{G}$.

(3) Let $X := \mathbb{A}^1_\eta \times \mathbb{A}^2_{p,q}$ and $\mathbb{A}^3 := \mathbb{A}^1_u \times \mathbb{A}^2_{p,q}$. Let $\tilde{G}$ act on X by

$$(g, \xi) \cdot (\eta, p, q) := (\xi\eta, g_z\xi^{-v}p, g_t\xi^v q),$$

and let $\rho : X \to \mathbb{A}^3$ be the morphism $(\eta, p, q) \to (\eta^r, p, q)$. Since $\tilde{R}$ acts on X by $(\eta, p, q) \to (\lambda^{-1}\eta, p, q)$, ρ is the quotient of X by $\tilde{R}$, and $\tilde{H}$ acts on $\mathbb{A}^3$ by

$$(g, \xi) \cdot (u, p, q) = (\chi(g, \xi)u, g_z\xi^{-v}p, g_t\xi^v q), \qquad \chi(g, \xi) := \xi^r.$$

(4) Let $x' := -p^2 - q^N u^{-d}$, a rational function on $\mathbb{A}^3$, and let $\psi : X \dashrightarrow \mathbb{A}^4$ be the rational map $(\eta, p, q) \to (\eta^a x', \eta^b, \eta^v p, \eta^{-v} q)$. Let

$$\Phi_j := \sum_{L\in\mathbb{Z}} u^{-L} h_{j,L}(x', 1, p, q) \in \mathbb{C}[u, u^{-1}, p, q] \quad \text{and} \quad \Delta_3 := \left(1 - \frac{\tau}{r}\right) \operatorname{div}(u) + \sum_j c_j \operatorname{div}(\Phi_j)$$

on $\mathbb{A}^3$.

(5) For a prime divisor F over $\mathbb{A}^3$, a non-zero rational function θ on $\mathbb{A}^3$, and an integer i, let

$$\operatorname{mult}_F(\eta^i\theta) := \frac{i}{r} \operatorname{mult}_F u + \operatorname{mult}_F \theta.$$

This is well defined since $\eta^r = u$. For instance, $\operatorname{mult}_F \psi^* x = \frac{a}{r} \operatorname{mult}_F u + \operatorname{mult}_F x'$ and $\operatorname{mult}_F \psi^* t = \operatorname{mult}_F q - \frac{v}{r} \operatorname{mult}_F u$.

The next lemma compares log discrepancies over W_N/G with log discrepancies over $\mathbb{A}^3$.

Lemma A.9. *Notation as in Construction A.8. Then the following hold.*

(1) The rational map ψ is a $\tilde{G}$-equivariant dominant rational map $X \dashrightarrow W_N$ of degree b, the group $\tilde{G}_0$ preserves the fibers of ψ, and $\tilde{G}$ acts faithfully on X.

(2) The group $\tilde{H}$ has order $b[G : R]$ and acts faithfully on $\mathbb{A}^2_{p,q}$.

(3) *For any j, $\psi^*h_j = \eta^{-\kappa_j}\Phi_j$ and $\varphi_j \neq 0$. The functions u, φ_j, and Φ_j are $\tilde{H}$-semi-invariant.*

(4) *Let F be a prime divisor over $\mathbb{A}^3$ such that $\mathrm{mult}_F\,\psi^*x$, $\mathrm{mult}_F\,\psi^*y$, $\mathrm{mult}_F\,\psi^*z$, and $\mathrm{mult}_F\,\psi^*t$ are positive. Then there exists a prime divisor E over W_N/G with center the image of 0 such that*

$$a(E, W_N/G, D_{W_N/G}) = \frac{a(F, \mathbb{A}^3, \Delta_3)}{|I_{\tilde{H}}(F)|},$$

where $(W_N/G, D_{W_N/G})$ is the crepant quotient of (W_N, D) [Liu26, Definition 2.5].

Proof. (1) Since G is diagonal and preserves the equation of W_N up to a character, $g_xg_y = g_z^2 = g_t^N$ for any $g \in G$. By Set-up 7.3(4), $rd = v(N+2)$. Since $a + b = 2v$,

$$\psi^*(xy + z^2 + t^N) = \eta^{2v}x' + \eta^{2v}p^2 + \eta^{-vN}q^N = -\eta^{2v-rd}q^N + \eta^{-vN}q^N = 0,$$

so ψ maps X into W_N. The projection $W_N \dashrightarrow \mathbb{A}^3_{y,z,t}$ is birational and $(\psi^*y, \psi^*z, \psi^*t) = (\eta^b, \eta^v p, \eta^{-v}q)$. Thus ψ is dominant of degree b. The group $\tilde{G}_0$ preserves the fibers of ψ, because $(1, \xi)$ preserves η^b, $\eta^v p$, and $\eta^{-v}q$. Let $(g, \xi) \in \tilde{G}$. Since $\xi^{a-2v} = \xi^{-b} = g_y^{-1}$, we have $g_z^2\xi^{a-2v} = g_t^N\xi^{a-2v} = g_x$. Hence

$$(\xi\eta)^a\Big(- (g_z\xi^{-v}p)^2 - \frac{(g_t\xi^v q)^N}{(\xi\eta)^{v(N+2)}}\Big) = g_x\eta^a x',$$

and

$$(\xi\eta)^b = g_y\eta^b, \qquad (\xi\eta)^v \cdot g_z\xi^{-v}p = g_z\eta^v p, \qquad (\xi\eta)^{-v} \cdot g_t\xi^v q = g_t\eta^{-v}q.$$

Thus $\psi \circ (g, \xi) = g \circ \psi$ as rational maps $X \dashrightarrow W_N$. If (g, ξ) acts trivially on X, then $\xi = 1$ and $g_z = g_t = 1$, so $g_y = \xi^b = 1$ and $g_x = g_z^2g_y^{-1} = 1$. Hence $\tilde{G}$ acts faithfully on X.

(2) For any $g \in G$ there exist exactly b numbers ξ with $\xi^b = g_y$, so $|\tilde{G}| = b|G|$. Since $\gcd(a, b, v) = \gcd(a, v) = 1$, the homomorphism $\iota : \mathbb{C}^* \to \mathbf{T}$ given by

$$\iota(\lambda) := (\lambda^{-a}, \lambda^{-b}, \lambda^{-v}, \lambda^v)$$

is injective. Hence $|\tilde{R}| = r$ and $|\tilde{H}| = b|G|/r = b[G : R]$. Assume that $(g, \xi) \in \tilde{G}$ acts trivially on $\mathbb{A}^2_{p,q}$. Then $g_z = \xi^v$, $g_t = \xi^{-v}$, and $g_y = \xi^b$, so $g_x = g_z^2g_y^{-1} = \xi^{2v-b} = \xi^a$. With $\lambda := \xi^{-1}$, we obtain $g = \iota(\lambda) \in G \cap \mathbf{T} = R$. Since ι is injective and $|R| = r$, $\lambda^r = 1$. Thus $(g, \xi) \in \tilde{R}$, and $\tilde{H}$ acts faithfully on $\mathbb{A}^2_{p,q}$.

(3) For any j and L,

$$h_{j,L}(\lambda^{-a}x, \lambda^{-b}y, \lambda^{-v}z, \lambda^v t) = \lambda^{\kappa_j + Lr}h_{j,L}(x, y, z, t).$$

Taking $\lambda = \eta^{-1}$ and $(x, y, z, t) = (x', 1, p, q)$, we obtain

$$\psi^*h_{j,L} = \eta^{-\kappa_j}u^{-L}h_{j,L}(x', 1, p, q),$$

and the sum over L is $\psi^*h_j = \eta^{-\kappa_j}\Phi_j$. Let $\psi_0 : X \dashrightarrow W_0 := (xy + z^2 = 0) \subset \mathbb{A}^4$ be the rational map $(\eta, p, q) \to (-\eta^a p^2, \eta^b, \eta^v p, \eta^{-v}q)$. The argument of (1) with the terms q^N removed shows that ψ_0 is $\tilde{G}$-equivariant, and the identity of (3) for $L = 0$, $\lambda = \eta^{-1}$, and $(x, y, z, t) = (-p^2, 1, p, q)$ implies $\psi_0^*h_j^0 = \eta^{-\kappa_j}\varphi_j$. Since G is diagonal, all terms of h_j have the character of h_j, so h_j^0 is G-semi-invariant. Since η is $\tilde{G}$-semi-invariant with character $(g, \xi) \to \xi$, the functions $\Phi_j = \eta^{\kappa_j}\psi^*h_j$, $\varphi_j = \eta^{\kappa_j}\psi_0^*h_j^0$, and $u = \eta^r$ are $\tilde{G}$-semi-invariant. They are functions on $\mathbb{A}^3$, on which the action of $\tilde{G}$ factors through $\tilde{H}$. Hence they are $\tilde{H}$-semi-invariant.

It remains to show that $\varphi_j \neq 0$. A monomial $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ with $n_1n_2 = 0$ is mapped to $(-1)^{n_1}p^{n_p}q^{n_4}$, where $n_p := 2n_1 + n_3$. Since $a = 2v - b$, the $\mathbf{T}$-weight of $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ is

$$-an_1 - bn_2 - vn_3 + vn_4 = b(n_1 - n_2) - vn_p + vn_4.$$

For monomials of $\mathbf{T}$-weight κ_j, the pair (n_p, n_4) determines $n_1 - n_2$ as $b > 0$, hence n_1 and n_2 as $n_1n_2 = 0$, hence n_3. Thus distinct monomials of h_j^0 are mapped to distinct monomials, and $\varphi_j \neq 0$ since $h_j^0 \neq 0$.

(4) Let F_X be a prime divisor over X lying over F [Liu26, Definition 2.7], and let $e_X := |I_{\tilde{R}}(F_X)|$. By [Liu26, Lemma 2.6(1)], $\operatorname{mult}_{F_X}\theta = e_X \operatorname{mult}_F\theta$ for any non-zero rational function θ on $\mathbb{A}^3$. Since $r\operatorname{mult}_{F_X}\eta = \operatorname{mult}_{F_X}u$, $\operatorname{mult}_{F_X}(\eta^i\theta) = e_X\operatorname{mult}_F(\eta^i\theta)$ for any integer i. Let $\pi_W : W' \to W_N$ be the normalization of W_N in the function field of X. By (1), W' is birational to X, $\tilde{G}$ acts faithfully on W', π_W is $\tilde{G}$-equivariant, and π_W is the quotient of W' by $\tilde{G}_0$. By assumption, $\operatorname{mult}_{F_X}$ is positive on ψ^*x, ψ^*y, ψ^*z, and ψ^*t. Hence F_X is a prime divisor over W', and the prime divisor F_W over W_N under F_X satisfies $\operatorname{center}_{W_N}F_W = \{0\}$. Let E be the prime divisor over W_N/G under F_W. Then the center of E is the image of 0.

We use the following description of log discrepancies. Let Y be a normal threefold, Ω a non-zero rational 3-form on Y, $\theta_1,\dots,\theta_l$ non-zero rational functions on Y, and $s_1,\dots,s_l$ real numbers. Let $K_Y := \operatorname{div}_Y(\Omega)$ and $B_Y := \sum_i s_i\operatorname{div}_Y(\theta_i) - K_Y$. Then $K_Y + B_Y = \sum_i s_i\operatorname{div}_Y(\theta_i)$ is $\mathbb{R}$-Cartier. For a prime divisor P over Y, let $f : Y' \to Y$ be a higher model on which P is a divisor. With $K_{Y'} := \operatorname{div}_{Y'}(\Omega)$, we have $f_*K_{Y'} = K_Y$ and $f^*(K_Y + B_Y) = \sum_i s_i\operatorname{div}_{Y'}(\theta_i)$. Hence

$$a(P,Y,B_Y) = 1 + \operatorname{mult}_P\Omega - \sum_i s_i\operatorname{mult}_P\theta_i, \tag{A.88}$$

where $\operatorname{mult}_P\Omega$ is the coefficient of P in $\operatorname{div}_{Y'}(\Omega)$.

The Poincaré residue $\Omega_W := \frac{dy\wedge dz\wedge dt}{y}$ of $\frac{dx\wedge dy\wedge dz\wedge dt}{xy+z^2+t^N}$ generates the dualizing sheaf of W_N, so $K_{W_N} := \operatorname{div}_{W_N}(\Omega_W) = 0$ and $K_{W_N} + D = \sum_j c_j\operatorname{div}(h_j)$. Let

$$K_{W'} := \operatorname{div}_{W'}(\pi_W^*\Omega_W) \quad\text{and}\quad D' := \sum_j c_j\operatorname{div}_{W'}(h_j) - K_{W'}.$$

Then $K_{W'} + D' = \pi_W^*(K_{W_N} + D)$, so (W_N, D) is the crepant quotient of (W', D') by $\tilde{G}_0$. Similarly, $K_{\mathbb{A}^3} = \operatorname{div}(du\wedge dp\wedge dq) = 0$. Let $K_X := \operatorname{div}_X(\rho^*(du\wedge dp\wedge dq))$ and

$$\Delta_X := \Big(1 - \frac{\tau}{r}\Big)\operatorname{div}_X(u) + \sum_j c_j\operatorname{div}_X(\Phi_j) - K_X.$$

Then $K_X + \Delta_X = \rho^*(K_{\mathbb{A}^3} + \Delta_3)$, so $(\mathbb{A}^3, \Delta_3)$ is the crepant quotient of (X, Δ_X) by $\tilde{R}$. Thus (W', D') with $\Omega = \pi_W^*\Omega_W$ and (X, Δ_X) with $\Omega = \rho^*(du\wedge dp\wedge dq)$ are of the form considered in (A.88), where $\Delta_X \geq 0$ need not hold.

A direct calculation from $\psi^*y = \eta^b$, $\psi^*z = \eta^v p$, $\psi^*t = \eta^{-v}q$, and $\rho^*u = \eta^r$ shows

$$\psi^*\Omega_W = b\frac{d\eta}{\eta}\wedge dp\wedge dq = \frac{b}{r}\rho^*\Big(\frac{du}{u}\wedge dp\wedge dq\Big). \tag{A.89}$$

By (A.89) and (3),

$$\operatorname{mult}_{F_X}\Omega_W = \operatorname{mult}_{F_X}(du\wedge dp\wedge dq) - \operatorname{mult}_{F_X}u \quad\text{and}\quad \operatorname{mult}_{F_X}h_j = \operatorname{mult}_{F_X}\Phi_j - \frac{\kappa_j}{r}\operatorname{mult}_{F_X}u.$$

Since $\tau = \sum_j c_j\kappa_j$, (A.88) implies

$$\begin{aligned} a(F_X, W', D') &= 1 + \operatorname{mult}_{F_X}\Omega_W - \sum_j c_j\operatorname{mult}_{F_X}h_j \\ &= 1 + \operatorname{mult}_{F_X}(du\wedge dp\wedge dq) - \Big(1-\frac{\tau}{r}\Big)\operatorname{mult}_{F_X}u - \sum_j c_j\operatorname{mult}_{F_X}\Phi_j \\ &= a(F_X, X, \Delta_X). \end{aligned}$$

By [Liu26, Lemma 2.6(1)], the order of an inertia group is the ramification index of the corresponding prime divisor. Ramification indices are multiplicative in the towers $W' \to W_N \to W_N/G$ and $X \to \mathbb{A}^3 \to \mathbb{A}^3/\tilde{H}$, and both composites are birational to the quotient by $\tilde{G}$. Hence

$$|I_{\tilde{G}_0}(F_X)|\cdot|I_G(F_W)| = |I_{\tilde{G}}(F_X)| = |I_{\tilde{R}}(F_X)|\cdot|I_{\tilde{H}}(F)|.$$

By [Liu26, Lemma 2.6(2)], applied to $\tilde{G}_0$ on W', to G on W_N, and to $\tilde{R}$ on X,

$$a(F_X, W', D') = |I_{\tilde{G}_0}(F_X)| \cdot |I_G(F_W)| \cdot a(E, W_N/G, D_{W_N/G}),$$
$$a(F_X, X, \Delta_X) = |I_{\tilde{R}}(F_X)| \cdot a(F, \mathbb{A}^3, \Delta_3).$$

Dividing the equality $a(F_X, W', D') = a(F_X, X, \Delta_X)$ by $|I_{\tilde{G}}(F_X)|$ implies (4). □

The next lemma bounds $\operatorname{mult}_F(\Phi_j - \varphi_j)$ from below for the Gauss extensions F of plane prime divisors with center 0.

Lemma A.10. *Notation as in Construction A.8. Let ν be a prime divisor over $\mathbb{A}^2_{p,q}$ with center 0, let $m := \min\{\operatorname{mult}_\nu p, \operatorname{mult}_\nu q\}$, and let c be a positive integer. Let F be the prime divisor over $\mathbb{A}^3$ such that mult_F is the Gauss extension of mult_ν with u-value c* [Liu26, Definition-Proposition 2.11]. *Assume that $r \geq (\frac{c}{m} + 4) \max_i |w_i|$. Then $\operatorname{mult}_F \psi^* x$, $\operatorname{mult}_F \psi^* y$, $\operatorname{mult}_F \psi^* z$, and $\operatorname{mult}_F \psi^* t$ are positive, and for any j,*

$$\operatorname{mult}_F(\Phi_j - \varphi_j) \geq \min\Big\{c,\ \frac{(r - M \max_i |w_i|)m}{C_1} - c,\ \Big(\frac{r}{v} - 2\Big)m - c\Big\}.$$

Proof. By [Liu26, Lemma 18.1] with $\varrho = 1$, the Gauss extension is mult_F for a prime divisor F over $\mathbb{A}^3$ with center the origin of $\mathbb{A}^3$. For $\theta = \sum_i \theta_i u^i$ with rational functions θ_i on $\mathbb{A}^2_{p,q}$, $\operatorname{mult}_F \theta = \min_i\{\operatorname{mult}_\nu \theta_i + ic\}$. Since $d \geq 1$, the terms of $x' = -p^2 - q^N u^{-d}$ have distinct powers of u, so

$$\operatorname{mult}_F x' = \min\{2 \operatorname{mult}_\nu p,\ N \operatorname{mult}_\nu q - dc\}.$$

The assumption implies $rm \geq \max_i |w_i|(c+4m)$, hence $rm \geq v(c+4m)$, $rm > |a|c$, and $rm \geq C_1 c$. Since $N = \frac{dr}{v} - 2$, $\frac{r}{v} \operatorname{mult}_\nu q \geq \frac{r}{v} m \geq c$, and $d \geq 1$,

$$N \operatorname{mult}_\nu q - dc = d\Big(\frac{r}{v} \operatorname{mult}_\nu q - c\Big) - 2 \operatorname{mult}_\nu q \geq \Big(\frac{r}{v} - 2\Big) \operatorname{mult}_\nu q - c \geq \Big(\frac{r}{v} - 2\Big)m - c \geq 2m. \tag{A.90}$$

Hence $\operatorname{mult}_F x' \geq 2m$. Thus

$$\operatorname{mult}_F \psi^* y = \frac{bc}{r} > 0, \quad \operatorname{mult}_F \psi^* z = \operatorname{mult}_\nu p + \frac{vc}{r} > 0,$$
$$\operatorname{mult}_F \psi^* t \geq m - \frac{vc}{r} > 0, \quad \operatorname{mult}_F \psi^* x \geq 2m - \frac{|a|c}{r} > 0.$$

By Construction A.8(4),

$$\Phi_j - \varphi_j = \sum_{L \neq 0} u^{-L} h_{j,L}(x', 1, p, q) + \big(h^0_j(x', 1, p, q) - h^0_j(-p^2, 1, p, q)\big).$$

We bound mult_F of each summand from below. Let $x^{n_1} y^{n_2} z^{n_3} t^{n_4}$ be a monomial of $h_{j,L}$, so $n_1 n_2 = 0$. The image of $x^{n_1} y^{n_2} z^{n_3} t^{n_4}$ in Φ_j is a multiple of $u^{-L} x'^{n_1} p^{n_3} q^{n_4}$, and $\operatorname{mult}_F x' \geq 2m$, $\operatorname{mult}_F p \geq m$, and $\operatorname{mult}_F q \geq m$.

If $L \leq -1$, then $\operatorname{mult}_F(u^{-L} x'^{n_1} p^{n_3} q^{n_4}) \geq |L|c \geq c$.

If $L \geq 1$, let $n_p := 2n_1 + n_3$. Since $a = 2v - b$, the $\mathbf{T}$-weight of $x^{n_1} y^{n_2} z^{n_3} t^{n_4}$ is $-an_1 - bn_2 - vn_3 + vn_4 = b(n_1 - n_2) - vn_p + vn_4$. Since $n_1 n_2 = 0$ and $n_1 \leq \frac{n_p}{2}$, $b(n_1 - n_2) \leq \frac{b}{2} n_p$. Since $b - 2v = -a$ and $|\kappa_j| \leq M \max_i |w_i|$,

$$Lr - M \max_i |w_i| \leq \kappa_j + Lr \leq -\frac{a}{2} n_p + vn_4 \leq C_1(n_p + n_4).$$

Since $rm \geq C_1 c$ and $L \geq 1$,

$$\begin{aligned} \operatorname{mult}_F(u^{-L} x'^{n_1} p^{n_3} q^{n_4}) &\geq (n_p + n_4)m - Lc \\ &\geq L\Big(\frac{rm}{C_1} - c\Big) - \frac{M \max_i |w_i| m}{C_1} \geq \frac{(r - M \max_i |w_i|)m}{C_1} - c. \end{aligned}$$

For $L = 0$, the difference $h_j^0(x', 1, p, q) - h_j^0(-p^2, 1, p, q)$ is the product of $x' + p^2 = -q^N u^{-d}$ with a polynomial in x', p, and q. Since $\operatorname{mult}_F x'$, $\operatorname{mult}_F p$, and $\operatorname{mult}_F q$ are non-negative, (A.90) implies

$$\operatorname{mult}_F \left(h_j^0(x', 1, p, q) - h_j^0(-p^2, 1, p, q)\right) \geq N \operatorname{mult}_\nu q - dc \geq \left(\frac{r}{v} - 2\right)m - c.$$

The three bounds imply the lemma. □

Proof of Proposition 7.13. If $c_j = 0$ or $h_j(0) \neq 0$, then $c_j \operatorname{div}(h_j) = 0$ near 0, and the index j does not enter τ or C_{plane} (Set-up 7.3(6)). Multiplying h_j by an invariant unit does not change $\operatorname{div}(h_j)$ near 0. Possibly removing the terms $c_j \operatorname{div}(h_j)$ with $c_j = 0$ or $h_j(0) \neq 0$ from D and replacing each h_j by the product of h_j with an invariant unit, we may assume that for any j, $c_j > 0$ and h_j is a G-semi-invariant polynomial with $h_j(0) = 0$ and without monomials divisible by xy (Set-up 7.3(5)). Let γ, M, and C_1 be as in Construction A.8, and assume that $r > 2M \max_i |w_i|$. Then the objects of Construction A.8 are defined, and φ_j, C_{plane}, and $\tilde{H}$ are those of Proposition 7.13. By Lemma A.9(2), $\tilde{H}$ has order $b[G : R] \leq be_0$ and acts faithfully on $\mathbb{A}^2_{p,q}$. The proof of Lemma A.9(2) does not use $r > 2M \max_i |w_i|$, so Proposition 7.13(1) holds for any r. If no index j is left, then $C_{\text{plane}} = 0$, $\tau = 0$, and all sums over j below are zero. By [Liu26, Definition 2.5], $\alpha = \operatorname{mld}(W_N/G \ni \bar{0}, D_{W_N/G})$, where $\bar{0}$ is the image of 0.

The proof is long, so we divide it into several steps.

Step 1. In this step we show that $|\tau| < 2 \max_i |w_i|$ and that there exist at most M indices j.

For any j, let m_j be the least total degree of a monomial of h_j. By Lemma A.1(3), $\sum_j c_j m_j < 2$, so there are at most $\frac{2}{\gamma} \leq M$ indices j. Since κ_j is the $\mathbf{T}$-weight of a monomial of degree m_j, $|\kappa_j| \leq m_j \max_i |w_i|$. Therefore $|\tau| \leq \max_i |w_i| \sum_j c_j m_j < 2 \max_i |w_i|$.

Step 2. In this step we show that $(\mathbb{A}^2, C_{\text{plane}})$ is lc near 0 when $r \geq r_1$, where r_1 is defined in (A.91) below.

The coefficients of C_{plane} belong to Γ_+, a DCC set [Liu26, Lemma 2.10(1)]. By Theorem 2.12(1), applied with $n = 2$ and the DCC sets $\{0\}$ and $\Gamma_+ \backslash \{0\}$, there exists $\eta_0 \in (0, 1)$, depending only on Γ, such that $\operatorname{lct}(U, 0; B') \leq 1 - \eta_0$ for any smooth surface U and any $\mathbb{R}$-divisor $B' \geq 0$ on U with coefficients in Γ_+ and $\operatorname{lct}(U, 0; B') < 1$. Let $B \geq 0$ be an $\mathbb{R}$-divisor on $\mathbb{A}^2$ with coefficients in Γ_+. Since $\operatorname{lct}_0(\mathbb{A}^2; B) = \operatorname{lct}(U, 0; B|_U)$ for any small enough open neighborhood U of 0, the choice of η_0 applied to the $\mathbb{R}$-divisor $B|_U$ shows that $\operatorname{lct}_0(\mathbb{A}^2; B) < 1$ implies $\operatorname{lct}_0(\mathbb{A}^2; B) \leq 1 - \eta_0$. Let $\mathfrak{m} := (p, q)$, and let L_0 be an integer with $L_0 > \frac{4}{\eta_0 \gamma}$, depending only on Γ. By Theorem 2.12(2) with $n = 2$, $\epsilon = \frac{\eta_0}{2}$, and the exponents $c_j \geq \gamma$, L_0 has the following property: for any $\mathbb{R}$-ideals $\prod_{j \in J} \mathfrak{b}_j^{c_j}$ and $\prod_{j \in J} \mathfrak{b}_j'^{c_j}$ on $\mathbb{A}^2$ near 0 with $c_j \in \Gamma \backslash \{0\}$, the first of which has finite log canonical threshold at 0, and

$$\mathfrak{b}_j + \mathfrak{m}^{L_0} = \mathfrak{b}'_j + \mathfrak{m}^{L_0},$$

the log canonical thresholds at 0 differ by at most $\frac{\eta_0}{2}$. Let

$$r_1 := \left(\frac{4L_0}{\eta_0 \gamma} + 2L_0 + 2M + 4\right) \max_i |w_i| + 1, \tag{A.91}$$

and assume that $r \geq r_1$. Then $r > 2M \max_i |w_i|$.

Let J_1 be the set of indices j with $\varphi_j(0) = 0$. If J_1 is empty, then $C_{\text{plane}} = 0$ near 0. Assume that J_1 is non-empty and that $(\mathbb{A}^2, C_{\text{plane}})$ is not lc near 0. Then the log canonical threshold $\operatorname{lct}_0\left(\prod_{j \in J_1} (\varphi_j)^{c_j}\right)$ at 0 is smaller than 1, hence at most $1 - \eta_0$. For $j \in J_1$, let $\mathfrak{a}_j := (\varphi_j) + \mathfrak{m}^{L_0}$, and let $\mathfrak{a} := \prod_{j \in J_1} \mathfrak{a}_j^{c_j}$. Then $\operatorname{lct}_0(\mathfrak{a}) \leq 1 - \frac{\eta_0}{2}$. Let ν be a prime divisor over $\mathbb{A}^2$ computing $\operatorname{lct}_0(\mathfrak{a})$ on a log resolution of $\mathfrak{m} \prod_{j \in J_1} \mathfrak{a}_j$. Since each $\mathfrak{a}_j$ is $\mathfrak{m}$-primary, ν has center 0. Let $m := \min\{\operatorname{mult}_\nu p, \operatorname{mult}_\nu q\} \geq 1$ and $s_j := \operatorname{mult}_\nu \mathfrak{a}_j = \min\{\operatorname{mult}_\nu \varphi_j, L_0 m\}$ for $j \in J_1$. Since

$\mathfrak{a}_j \subset \mathfrak{m}$, $s_j \geq m$, so $\sum_{j\in J_1} c_j s_j \geq \gamma m$. Therefore

$$a(\nu, \mathbb{A}^2, 0) - \sum_{j\in J_1} c_j s_j = \big(\operatorname{lct}_0(\mathfrak{a}) - 1\big) \sum_{j\in J_1} c_j s_j \leq -\frac{\eta_0}{2} \sum_{j\in J_1} c_j s_j \leq -\frac{\eta_0\gamma}{2} m. \tag{A.92}$$

Let F be the prime divisor over $\mathbb{A}^3$ such that mult_F is the Gauss extension of mult_ν with u-value $L_0 m$. By [Liu26, Lemma 18.1] with $\varrho = 1$, $a(F, \mathbb{A}^3, 0) = a(\nu, \mathbb{A}^2, 0) + L_0 m$. Since $r \geq r_1 \geq (L_0 + 4)\max_i |w_i|$, Lemma A.10 with $c = L_0 m$ implies that $\operatorname{mult}_F \psi^* x$, $\operatorname{mult}_F \psi^* y$, $\operatorname{mult}_F \psi^* z$, and $\operatorname{mult}_F \psi^* t$ are positive, and that

$$\operatorname{mult}_F(\Phi_j - \varphi_j) \geq \min\Big\{L_0 m,\ \Big(\frac{r - M\max_i |w_i|}{C_1} - L_0\Big)m,\ \Big(\frac{r}{v} - 2 - L_0\Big)m\Big\} \geq L_0 m,$$

where the last inequality holds since

$$C_1 \leq \max_i |w_i|, \qquad v \leq \max_i |w_i|, \qquad r \geq (2L_0 + 2M + 4)\max_i |w_i|.$$

Since $\operatorname{mult}_F \varphi_j = \operatorname{mult}_\nu \varphi_j$, we obtain $\operatorname{mult}_F \Phi_j \geq \min\{\operatorname{mult}_\nu \varphi_j, L_0 m\}$, and the right-hand side equals s_j for $j \in J_1$ and 0 for $j \notin J_1$. By (A.92) and **Step 1**,

$$\begin{aligned} a(F, \mathbb{A}^3, \Delta_3) &= a(F, \mathbb{A}^3, 0) - \Big(1 - \frac{\tau}{r}\Big) L_0 m - \sum_j c_j \operatorname{mult}_F \Phi_j \\ &\leq a(\nu, \mathbb{A}^2, 0) - \sum_{j\in J_1} c_j s_j + \frac{|\tau| L_0 m}{r} < -\frac{\eta_0\gamma}{2} m + \frac{2\max_i |w_i| L_0 m}{r} < 0, \end{aligned}$$

where the last inequality holds since $r > \frac{4L_0 \max_i |w_i|}{\eta_0\gamma}$. By Lemma A.9(4), there exists a prime divisor E over W_N/G with center $\bar{0}$ and $a(E, W_N/G, D_{W_N/G}) < 0$. Since (W_N, D) is klt near 0, $(W_N/G, D_{W_N/G})$ is klt near $\bar{0}$ [Liu26, Lemma 2.6(2)(3)], a contradiction. Thus $(\mathbb{A}^2, C_{\text{plane}})$ is lc near 0.

Step 3. In this step we show that β belongs to an ACC set depending only on Γ, and we fix a prime divisor ν over $\mathbb{A}^2$ computing β with bounded multiplicities of the φ_j.

Let $\Sigma := \Gamma_+ \cap [0, 1]$. Since $(\mathbb{A}^2, C_{\text{plane}})$ is lc near 0, the coefficients of C_{plane} near 0 belong to Σ. Let $(\mathbb{A}^2/\tilde{H}, C_{\mathbb{A}^2/\tilde{H}})$ be the crepant quotient of $(\mathbb{A}^2, C_{\text{plane}})$ and $\bar{0} \in \mathbb{A}^2/\tilde{H}$ the image of 0. By [Liu26, Lemma 2.6(4)], the coefficients of $C_{\mathbb{A}^2/\tilde{H}}$ near $\bar{0}$ belong to $\Psi(\Sigma)$, which is a DCC set contained in $[0, 1]$ [Liu26, Lemma 2.10(1)]. By [Liu26, Lemma 2.6(2)(3)], $(\mathbb{A}^2/\tilde{H}, C_{\mathbb{A}^2/\tilde{H}})$ is lc near $\bar{0}$, and $\beta = \operatorname{mld}(\mathbb{A}^2/\tilde{H} \ni \bar{0}, C_{\mathbb{A}^2/\tilde{H}})$. By Theorem 2.9(1), applied over a $\tilde{H}$-invariant affine neighborhood of 0 given by [Liu26, Lemma 2.10(5)], β belongs to an ACC set depending only on Γ, in particular only on a, b, v, Γ, e_0, which is the dependence in Proposition 7.13(2).

Let l be the positive integer of Theorem 2.9(2) for the DCC set $\Psi(\Sigma)$. Then there exists a prime divisor $\bar{\nu}$ over $\mathbb{A}^2/\tilde{H}$ with center $\bar{0}$ such that

$$a(\bar{\nu}, \mathbb{A}^2/\tilde{H}, C_{\mathbb{A}^2/\tilde{H}}) = \beta \quad \text{and} \quad a(\bar{\nu}, \mathbb{A}^2/\tilde{H}, 0) \leq l.$$

Let ν be a prime divisor over $\mathbb{A}^2$ lying over $\bar{\nu}$, and let $e := |I_{\tilde{H}}(\nu)| \leq |\tilde{H}| \leq b e_0$. Since $\tilde{H}$ fixes 0, ν has center 0. By [Liu26, Lemma 2.6(2)], $a(\nu, \mathbb{A}^2, C_{\text{plane}}) = e\beta$. Let $(\mathbb{A}^2/\tilde{H}, B_0)$ be the crepant quotient of $(\mathbb{A}^2, 0)$. Then $B_0 \geq 0$ by [Liu26, Lemma 2.6(4)], and [Liu26, Lemma 2.6(2)] implies

$$a(\nu, \mathbb{A}^2, 0) = e \cdot a(\bar{\nu}, \mathbb{A}^2/\tilde{H}, B_0) \leq e \cdot a(\bar{\nu}, \mathbb{A}^2/\tilde{H}, 0) \leq b e_0 l =: L_1.$$

Since $(\mathbb{A}^2, C_{\text{plane}})$ is lc near 0, $c_j \operatorname{mult}_\nu \varphi_j \leq a(\nu, \mathbb{A}^2, 0) \leq L_1$, so $\operatorname{mult}_\nu \varphi_j \leq \frac{L_1}{\gamma}$ for any j.

Step 4. In this step we construct the prime divisor E over W_N/G of the proposition and conclude the proof.

Let $I := I_{\tilde{H}}(\nu)$, and let $i_0 \in \{1, \dots, e\}$ be the integer of Lemma 3.1 for ν and the character χ of Construction A.8(3). Let $k_0 := \lfloor \frac{L_1}{\gamma} \rfloor$, and let k be the integer with $k_0 < k \leq k_0 + e$ and

$k \equiv i_0 \pmod e$. Then $k \le k_0 + be_0$. By Lemma A.9(2)(3), $\tilde H$ acts faithfully on $\mathbb{A}^2_{p,q}$, fixes 0, and preserves C_{plane}, and $\tilde H$ acts on u by χ. By Lemma 3.1, applied with $V = \mathbb{A}^2_{p,q}$, $H = \tilde H$, $B = C_{\text{plane}}$, $E = \nu$, the coordinate u in place of w, and $\ell = k$, the Gauss extension of mult_ν with u-value k is mult_F for a prime divisor F over $\mathbb{A}^3$ with center the origin of $\mathbb{A}^3$, $I_{\tilde H}(F) = I$, and

$$a(F, \mathbb{A}^3, \operatorname{pr}_1^* C_{\text{plane}}) = a(\nu, \mathbb{A}^2, C_{\text{plane}}) + k = e\beta + k.$$

Let

$$r_2 := (2k_0 + be_0 + M + 4) \max_i |w_i| + 1,$$

and assume that $r \ge r_2$. Since $m := \min\{\operatorname{mult}_\nu p, \operatorname{mult}_\nu q\} \ge 1$ and $k \le k_0 + be_0$, $r \ge (\frac{k}{m} + 4) \max_i |w_i|$. By Lemma A.10 with $c = k$, $\operatorname{mult}_F \psi^* x$, $\operatorname{mult}_F \psi^* y$, $\operatorname{mult}_F \psi^* z$, and $\operatorname{mult}_F \psi^* t$ are positive, and since $C_1 \le \max_i |w_i|$ and $v \le \max_i |w_i|$,

$$\operatorname{mult}_F(\Phi_j - \varphi_j) \ge \min\left\{k,\ \frac{r - M \max_i |w_i|}{C_1} - k,\ \frac{r}{v} - 2 - k\right\} > k_0 \ge \operatorname{mult}_\nu \varphi_j = \operatorname{mult}_F \varphi_j,$$

since $\operatorname{mult}_\nu \varphi_j$ is an integer at most $\frac{L_1}{\gamma}$. Hence $\operatorname{mult}_F \Phi_j = \operatorname{mult}_\nu \varphi_j$ for any j. Since $\operatorname{mult}_F u = k$,

$$a(F, \mathbb{A}^3, \Delta_3) = a(F, \mathbb{A}^3, \operatorname{pr}_1^* C_{\text{plane}}) - \Big(1 - \frac{\tau}{r}\Big)k - \sum_j c_j\big(\operatorname{mult}_F \Phi_j - \operatorname{mult}_F \varphi_j\big) = e\beta + \frac{\tau k}{r}.$$

By Lemma A.9(4), there exists a prime divisor E over W_N/G with center $\bar 0$ such that

$$a(E, W_N/G, D_{W_N/G}) = \frac{a(F, \mathbb{A}^3, \Delta_3)}{|I_{\tilde H}(F)|} = \beta + \frac{\tau k}{re},$$

where $e \le be_0$ and $k \le k_0 + be_0$. Let $r_0 := \max\{r_1, r_2\}$ and $C_0 := 2 \max_i |w_i|(k_0 + be_0)$. They depend only on a, b, v, Γ, and e_0, and not on d, N, or the degrees of the h_j. If $r \ge r_0$, then **Step 1** implies

$$\alpha \le a(E, W_N/G, D_{W_N/G}) \le \beta + \frac{|\tau| k}{r} \le \beta + \frac{C_0}{r}.$$

No sign condition on τ was used.

Finally, let $\mathbf{T} = \mathbf{T}_{(w)}$ be a canonical circle with $w = (w_x, -2v - w_x, -v, v)$ and $v > 0$. Since $w_x + w_y = -2v < 0$, one of w_x and w_y is negative. The automorphism $(x, y, z, t) \to (y, x, z, t)$ of $\mathbb{A}^4$ preserves W_N, conjugates diagonal groups to diagonal groups, preserves the polynomials without monomials divisible by xy, and exchanges w_x and w_y. Thus we may assume that $w_y < 0$. Then $w = (-a, -b, -v, v)$ with $b := -w_y > 0$ and $a + b = 2v$. Since $\gcd(a, v)$ divides $\gcd(a, b, v)$ as $b = 2v - a$, and w is primitive, $\gcd(a, v) = 1$. Thus w is as in Construction A.8, and **Steps 1**–**4** apply to any canonical circle. □

We now prove Construction-Proposition 7.14 and Remark 7.16, using the root chart of Construction A.8.

Proof of Construction-Proposition 7.14. The proof is long, so we divide it into several steps. Units h_j do not change D near 0 and an invariant unit factor does not change $\operatorname{div}(h_j)$, so we consider only the non-units h_j with $c_j > 0$, without monomials divisible by xy.

Step 1. In this step we show that $Y = Y_y \cup Y_x$ for the toric charts Y_y and Y_x defined below, that the $\mathbb{Q}$-divisors $D_{Y,j}$ defining Δ_Y are well defined, and that H acts faithfully on Y and preserves E and Δ_Y. This step holds for any r. Let φ_j, C_{plane}, $\tilde R$, and $\tilde H$ be as in Construction A.8(1)–(3). Their definitions do not use the assumption $r > 2M \max_i |w_i|$ made there.

Let Y_y and Y_x be the images of the charts $\mathbb{A}^2_{p,q}/\mu_b(-v, v)$ and $\mathbb{A}^2_{p',q'}/\mu_a(-v, v)$, let o_y and o_x be the images of the origins, and let $n_E := (\frac{a}{2v}, \frac{b}{2v})$. The characters of the torus of S_v form the lattice $\{m \in \mathbb{Z}^2 : m_1 \equiv m_2 \pmod{2v}\}$ dual to $\mathbb{Z}^2 + \frac{1}{2v}(1, -1)\mathbb{Z}$, and U, V, Z are, up to signs, the characters $(2v, 0)$, $(0, 2v)$, $(1, 1)$. Since $b = 2v - a$ and $\gcd(a, v) = 1$, we have $\gcd(b, v) = 1$. Hence the monomial $p^{n_1} q^{n_2}$ is $\mu_b(-v, v)$-invariant if and only if $n_1 \equiv n_2 \pmod b$, and $p'^{n_1} q'^{n_2}$ is $\mu_a(-v, v)$-invariant if and only if $n_1 \equiv n_2 \pmod a$. A direct calculation shows that, up to

non-zero constants, the first chart sends the character m to $p^{\langle m,(1,0)\rangle}q^{\langle m,n_E\rangle}$ and the second chart sends it to $p'^{\langle m,(0,1)\rangle}q'^{\langle m,n_E\rangle}$. Thus Y_y and Y_x are the affine toric surfaces of the cones spanned by $(1,0), n_E$ and by $n_E, (0,1)$, $Y = Y_y \cup Y_x$, and $E \cap Y_y$ and $E \cap Y_x$ are the images of $(q = 0)$ and $(q' = 0)$. In particular $E\backslash Y_y = \{o_x\}$. Since $\gcd(b, v) = 1$, no non-identity element of $\mu_b(-v, v)$ fixes a point of $\mathbb{A}^2_{p,q}\backslash\{0\}$, so $\mathbb{A}^2_{p,q} \to Y_y$ is quasi-étale. Similarly, $\mathbb{A}^2_{p',q'} \to Y_x$ is quasi-étale.

Let G act on S_v through the characters of the $\mathbf{T}$-invariant monomials $U = x^v t^a$, $V = y^v t^b$, and $Z = zt$. The group G acts by diagonal matrices preserving $UV = (-1)^v Z^{2v}$, hence through the torus of S_v, and R acts trivially. Thus the action of H on S_v lifts to Y and preserves Y_y, Y_x, and E. Since G is diagonal and preserves $xy + z^2 + t^N$ up to a character, $g_x g_y = g_z^2$ for any $g \in G$. Using $g_x g_y = g_z^2$ and $\xi^b = g_y$, a direct calculation shows that the first chart $\mathbb{A}^2_{p,q} \to S_v$ is equivariant with respect to $\tilde{H} \to H$, $(g, \xi) \to gR$. The kernel of $\tilde{H} \to H$ is the image of $\{(1, \xi) : \xi^b = 1\}$, which acts on $\mathbb{A}^2_{p,q}$ as $\mu_b(-v, v)$, because a pair (g, ξ) with $g = (\lambda^{-a}, \lambda^{-b}, \lambda^{-v}, \lambda^v) \in R$ and $\xi^b = g_y$ is congruent to $(1, \lambda\xi)$ modulo $\tilde{R}$. Thus the action of H on $Y_y = \mathbb{A}^2_{p,q}/\mu_b(-v, v)$ is induced by $\tilde{H}$. By Proposition 7.13, $\tilde{H}$ acts faithfully on $\mathbb{A}^2_{p,q}$. If $h \in \tilde{H}$ acts trivially on Y_y, then $\mathbb{A}^2_{p,q}$ is the union of the closed subsets $\{s : h \cdot s = \zeta \cdot s\}$ for $\zeta \in \mu_b(-v, v)$, so $h \in \mu_b(-v, v)$. Thus $\tilde{H}$ acts faithfully on Y.

Let χ_j be the character of h_j. For a monomial $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ of h_j^0, we have $n_1 n_2 = 0$ and $-an_1 - bn_2 - vn_3 + vn_4 = \kappa_j$, so

$$v(n_4 - 2n_1 - n_3) = \kappa_j + b(n_2 - n_1). \tag{A.93}$$

The monomial $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ gives the term $(-1)^{n_1}p^{2n_1+n_3}q^{n_4}$ of φ_j, whose pullback by (g, ξ) is $(-1)^{n_1}p^{2n_1+n_3}q^{n_4}$ multiplied by

$$g_z^{2n_1+n_3}g_t^{n_4}\xi^{v(n_4-2n_1-n_3)} = g_z^{2n_1+n_3}g_t^{n_4}g_y^{n_2-n_1}\xi^{\kappa_j} = g_x^{n_1}g_y^{n_2}g_z^{n_3}g_t^{n_4}\xi^{\kappa_j} = \chi_j(g)\xi^{\kappa_j},$$

by (A.93), $\xi^b = g_y$, and $g_z^2 = g_x g_y$. Therefore the pullback of φ_j by (g, ξ) is $\chi_j(g)\xi^{\kappa_j}\varphi_j$. Thus $\tilde{H}$ preserves C_{plane}, and φ_j^b is $\mu_b(-v, v)$-invariant. Hence $\operatorname{div}(\varphi_j)$ is the pullback of a $\mathbb{Q}$-divisor $D^{(1)}_{Y,j} \geq 0$ on Y_y, $bD^{(1)}_{Y,j} = \operatorname{div}(\varphi_j^b)$ on Y_y, and H preserves $D^{(1)}_{Y,j}$. Exchanging x and y replaces w with $(-b, -a, -v, v)$, where $\gcd(b, v) = \gcd(2v - a, v) = 1$, interchanges Y_y and Y_x, and preserves Set-up 7.3(5). Thus any statement on Y_y in this proof has a counterpart on Y_x, with φ_j replaced by $\varphi'_j(p', q') := h_j^0(1, -p'^2, p', q')$, C_{plane} replaced by $C'_{\text{plane}} := \sum_j c_j \operatorname{div}(\varphi'_j)$, and $\tilde{H}$ replaced by the group $\tilde{H}'$ of order $a|H|$ given by Proposition 7.13 with x and y exchanged. In particular, $\operatorname{div}(\varphi'_j)$ is the pullback of a $\mathbb{Q}$-divisor $D^{(2)}_{Y,j} \geq 0$ on Y_x such that $aD^{(2)}_{Y,j}$ is Cartier, and H preserves $D^{(2)}_{Y,j}$.

We show that $D^{(1)}_{Y,j}$ and $D^{(2)}_{Y,j}$ coincide on $Y_y \cap Y_x$, which is the image of $(p \neq 0) \subset \mathbb{A}^2_{p,q}$. Let

$$Q := \{(p, q, \vartheta) : p \neq 0,\ \vartheta^a = -p^2\} \subset \mathbb{A}^3, \qquad f : Q \to \mathbb{A}^2_{p',q'}, \quad (p, q, \vartheta) \to (\vartheta^{-v}p, \vartheta^v q).$$

Since $a + b = 2v$ and $\vartheta^a = -p^2$, a direct calculation shows that the compositions $Q \to \mathbb{A}^2_{p,q} \to S_v$ and $Q \to \mathbb{A}^2_{p',q'} \to S_v$ coincide. As $Y \to S_v$ is birational, the induced maps from Q to Y_y and to Y_x coincide, and the common map $Q \to Y_y \cap Y_x$ is finite and surjective. For a monomial $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ of h_j^0, the ratio of the corresponding terms of $\varphi'_j \circ f$ and φ_j is

$$(-p^2)^{n_2-n_1}\vartheta^{v(n_4-2n_2-n_3)} = \vartheta^{-an_1-bn_2-vn_3+vn_4} = \vartheta^{\kappa_j},$$

so $\varphi'_j \circ f = \vartheta^{\kappa_j}\varphi_j$ on Q. Since ϑ is a unit on Q, the pullbacks of $D^{(1)}_{Y,j}$ and $D^{(2)}_{Y,j}$ to Q coincide, and hence so do $D^{(1)}_{Y,j}$ and $D^{(2)}_{Y,j}$ on $Y_y \cap Y_x$. Thus they glue to a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D_{Y,j} \geq 0$ on Y, and H preserves E and $\Delta_Y = \sum_j c_j D_{Y,j}$.

Since $\mathbb{A}^2_{p,q} \to Y_y$ is quasi-étale, C_{plane} is the pullback of Δ_Y. Since $\mu_b(-v, v)$ is a normal subgroup of $\tilde{H}$ with quotient H, [Liu26, Lemma 2.6(5)] implies that

$$(\mathbb{A}^2_{p,q}, C_{\text{plane}})/\tilde{H} = (Y_y, \Delta_Y)/H, \tag{A.94}$$

and similarly $(\mathbb{A}^2_{p',q'}, C'_{\mathrm{plane}})/\tilde{H}' = (Y_x, \Delta_Y)/H$. Since $\mathbb{A}^2_{p,q} \to Y_y$ is unramified in codimension one, $\Delta_Y \in \Gamma_+$.

Step 2. In this step we estimate $\Phi_j - \varphi_j$ along Gauss extensions. From now on, let γ and M be as in Construction A.8. We assume $r > 2M \max_i |w_i|$ and use the notation of Construction A.8. In particular, $|\kappa_j| \le M \max_i |w_i|$ for any j (Lemma 5.3). Lemma A.10 does not apply here, since $\min\{\mathrm{mult}_\nu p, \mathrm{mult}_\nu q\} = 0$ for the prime divisors ν below. Instead, we use that $\mathrm{mult}_F p = 0$ and $\mathrm{mult}_F x' = 0$ for the prime divisors F below, so only the exponent of q contributes.

Let ν be a prime divisor over $\mathbb{A}^2_{p,q}$ with $\mathrm{mult}_\nu p = 0$ and $n_q := \mathrm{mult}_\nu q \ge 1$, let $s > s_1 \ge 0$ be integers such that

$$(r - M \max_i |w_i| - 2v) n_q > v(s + s_1), \tag{A.95}$$

and let F be the prime divisor over $\mathbb{A}^3_{u,p,q}$ such that mult_F is the Gauss extension of mult_ν with u-value s [Liu26, Definition-Proposition 2.11] (it exists by [Liu26, Lemma 18.1]). We show that F satisfies the hypothesis of Lemma A.9(4), that $\mathrm{mult}_F x' = 0$, and that

$$\mathrm{mult}_F(\Phi_j - \varphi_j) > s_1 \quad \text{for any } j.$$

We have $\mathrm{mult}_F u = s$, $\mathrm{mult}_F p = 0$, $\mathrm{mult}_F q = n_q$, and $\mathrm{mult}_F \varphi_j = \mathrm{mult}_\nu \varphi_j$. By (A.95), $rn_q/v > s$. Since $d \ge 1$ and $dr = v(N+2)$,

$$\mathrm{mult}_F(q^N u^{-d}) = Nn_q - ds = d\Big(\frac{rn_q}{v} - s\Big) - 2n_q \ge \Big(\frac{r}{v} - 2\Big)n_q - s > s_1 \ge 0,$$

where the strict inequality follows from (A.95). As $\mathrm{mult}_F p^2 = 0$, we obtain $\mathrm{mult}_F x' = 0$. Thus, by Construction A.8(5),

$$\mathrm{mult}_F \psi^* x = \frac{a}{r}\mathrm{mult}_F u + \mathrm{mult}_F x' = \frac{as}{r},$$
$$\mathrm{mult}_F \psi^* y = \frac{b}{r}\mathrm{mult}_F u = \frac{bs}{r},$$
$$\mathrm{mult}_F \psi^* z = \frac{v}{r}\mathrm{mult}_F u + \mathrm{mult}_F p = \frac{vs}{r},$$

and $\mathrm{mult}_F \psi^* t = \mathrm{mult}_F q - \frac{v}{r}\mathrm{mult}_F u = n_q - \frac{vs}{r}$ are positive, since $a, b > 0$ and $rn_q > vs$.

For a monomial $x^{n_1} y^{n_2} z^{n_3} t^{n_4}$ of $h_{j,L}$, we have $n_1 n_2 = 0$, and its image in Φ_j is a multiple of $u^{-L} x'^{n_1} p^{n_3} q^{n_4}$. By Construction A.8(1) and (4), $\Phi_j - \varphi_j$ is the sum of the following terms.

(1) The terms $u^{-L} x'^{n_1} p^{n_3} q^{n_4}$ with $L < 0$. Each has order at least $-Ls \ge s > s_1$ along F.

(2) The terms $u^{-L} x'^{n_1} p^{n_3} q^{n_4}$ with $L > 0$. Since $a, b, v > 0$,

$$\kappa_j + Lr = -an_1 - bn_2 - vn_3 + vn_4 \le vn_4,$$

so $n_4 \ge (Lr - M \max_i |w_i|)/v$. Each such term has order

$$-Ls + n_4 n_q \ge L\Big(\frac{rn_q}{v} - s\Big) - \frac{M \max_i |w_i|}{v} n_q \ge \frac{r - M \max_i |w_i|}{v} n_q - s > s_1$$

along F, by (A.95).

(3) For $L = 0$, the difference $h^0_j(x', 1, p, q) - h^0_j(-p^2, 1, p, q)$, which is the product of $x' + p^2 = -q^N u^{-d}$ and a polynomial in x', p, and q. It has order at least $\mathrm{mult}_F(q^N u^{-d}) > s_1$ along F.

Thus $\mathrm{mult}_F(\Phi_j - \varphi_j) > s_1$.

Step 3. In this step we show that (Y, Δ_Y) is lc near E when r is large.

Let r_y, C_y be the constants of Proposition 7.13 and r_x, C_x those of Proposition 7.13 with x and y exchanged, and assume $r \ge \max\{r_x, r_y\}$. Then $(\mathbb{A}^2_{p,q}, C_{\mathrm{plane}})$ and $(\mathbb{A}^2_{p',q'}, C'_{\mathrm{plane}})$ are lc near the origins, so (Y, Δ_Y) is lc near o_y and o_x by (A.94) and [Liu26, Lemma 2.6(2)(3)]. By **Step 1**, it suffices to show that $(\mathbb{A}^2_{p,q}, C_{\mathrm{plane}})$ is lc near $(p_0, 0)$ for any $p_0 \ne 0$.

We first bound τ. By Lemma A.1(3), $\sum_j c_j m_j < 2$, where m_j is the least total degree of a monomial of h_j. Since $|\kappa_j| \le m_j \max_i |w_i|$,

$$|\tau| < 2 \max_i |w_i|. \tag{A.96}$$

By [Liu26, Lemma 2.10(1)], Γ_+ is a DCC set. For a closed point $x_1 \in \mathbb{A}^2$ and a non-zero $\mathbb{R}$-divisor $B \ge 0$ on $\mathbb{A}^2$, $\mathrm{lct}_{x_1}(\mathbb{A}^2; B) = \mathrm{lct}(U, 0; B|_U)$ for any small enough open neighborhood $U \subset \mathbb{A}^2$ of x_1. Thus Theorem 2.12(1), applied with $n = 2$ and the DCC sets $\{0\}$ and $\Gamma_+\backslash\{0\}$ to the pair $(U, 0)$ and the $\mathbb{R}$-divisor $B|_U$, implies that the thresholds $\mathrm{lct}_{x_1}(\mathbb{A}^2; B)$, for closed points $x_1 \in \mathbb{A}^2$ and non-zero $\mathbb{R}$-divisors $B \ge 0$ on $\mathbb{A}^2$ with coefficients in Γ_+, satisfy the ACC. In particular, those smaller than 1 have a maximum if they exist, so there exists $\eta_0 \in (0, 1)$ depending only on Γ such that $\mathrm{lct}_{x_1}(\mathbb{A}^2; B) < 1$ implies $\mathrm{lct}_{x_1}(\mathbb{A}^2; B) \le 1 - \eta_0$ for any such x_1 and B. Fix an integer $L_0 > 4/(\eta_0\gamma)$, let $k_1 := L_0 + 1$, and assume that

$$r - M \max_i |w_i| - 2v > v(k_1 + L_0) \qquad \text{and} \qquad r > \frac{4k_1 \max_i |w_i|}{\eta_0\gamma}.$$

Suppose that $(\mathbb{A}^2_{p,q}, C_{\mathrm{plane}})$ is not lc near $(p_0, 0)$ for some $p_0 \ne 0$. Let $\mathfrak{m} := (p - p_0, q)$, and let J be the set of indices j with $\varphi_j(p_0, 0) = 0$. Then $J \ne \emptyset$ and

$$\mathrm{lct}_{(p_0,0)}\left(\mathbb{A}^2; \sum_{j\in J} c_j \operatorname{div}(\varphi_j)\right) \le 1 - \eta_0.$$

By Theorem 2.12(2), applied at $(p_0, 0)$ with $n = 2$, $\epsilon = \eta_0/2$, $l = L_0$, and the exponents $c_j \ge \gamma$, so that $L_0 > 4/(\eta_0\gamma) \ge 2/(\epsilon \min_{j\in J} c_j)$, the $\mathbb{R}$-ideal $\mathfrak{a} := \prod_{j\in J}((\varphi_j) + \mathfrak{m}^{L_0})^{c_j}$ satisfies

$$\lambda_1 := \mathrm{lct}_{(p_0,0)}(\mathbb{A}^2; \mathfrak{a}) \le 1 - \frac{\eta_0}{2}.$$

Let ν be a prime divisor over $\mathbb{A}^2_{p,q}$ computing λ_1, and let $w'_j := \mathrm{mult}_\nu((\varphi_j) + \mathfrak{m}^{L_0})$ for $j \in J$. Then $a(\nu, \mathbb{A}^2, 0) = \lambda_1 \sum_{j\in J} c_j w'_j$. Since the ideals $(\varphi_j) + \mathfrak{m}^{L_0}$, $j \in J$, are $\mathfrak{m}$-primary near $(p_0, 0)$, $\mathrm{center}_{\mathbb{A}^2} \nu = \{(p_0, 0)\}$. Let $m := \mathrm{mult}_\nu \mathfrak{m} \ge 1$. Then $m \le w'_j \le L_0 m$ for $j \in J$, $\mathrm{mult}_\nu p = 0$, $\mathrm{mult}_\nu q \ge m$, and

$$a(\nu, \mathbb{A}^2, 0) - \sum_{j\in J} c_j w'_j = (\lambda_1 - 1) \sum_{j\in J} c_j w'_j \le -\frac{\eta_0\gamma}{2} m. \tag{A.97}$$

Let F be the prime divisor over $\mathbb{A}^3_{u,p,q}$ such that mult_F is the Gauss extension of mult_ν with u-value $k_1 m$ [Liu26, Definition-Proposition 2.11] (it exists by [Liu26, Lemma 18.1]). By the choice of r, (A.95) holds with $s = k_1 m$ and $s_1 = L_0 m$, so **Step 2** implies that F satisfies the hypothesis of Lemma A.9(4) and that $\mathrm{mult}_F(\Phi_j - \varphi_j) > L_0 m$. Hence $\mathrm{mult}_F \Phi_j \ge \min\{\mathrm{mult}_\nu \varphi_j, L_0 m\} = w'_j$ for $j \in J$, and $\mathrm{mult}_F \Phi_j = 0$ for $j \notin J$, since then $\mathrm{mult}_\nu \varphi_j = 0$. By [Liu26, Lemma 18.1] with $\varrho = 1$, $a(F, \mathbb{A}^3, 0) = a(\nu, \mathbb{A}^2, 0) + k_1 m$. By (A.96) and (A.97),

$$a(F, \mathbb{A}^3, \Delta_3) = a(\nu, \mathbb{A}^2, 0) + \frac{\tau k_1 m}{r} - \sum_j c_j \mathrm{mult}_F \Phi_j \le -\frac{\eta_0\gamma}{2} m + \frac{2k_1 \max_i |w_i|}{r} m < 0.$$

By Lemma A.9(4), there exists a prime divisor over W_N/G with center the image of 0 and negative log discrepancy with respect to $(W_N, D)/G$, which contradicts [Liu26, Lemma 2.6(2)(3)], since (W_N, D) is klt near 0. Thus $(\mathbb{A}^2_{p,q}, C_{\mathrm{plane}})$ is lc near $(q = 0)$, and (Y, Δ_Y) is lc near E by (A.94).

Step 4. In this step we show that β_Y is attained and belongs to an ACC set depending only on Γ.

Let $(Y/H, \Delta_{Y/H})$ be the crepant quotient of (Y, Δ_Y) by H, and let $\Gamma' := \Psi(\Gamma_+ \cap [0, 1])$. By **Steps 1 and 3**, the coefficients of Δ_Y near E belong to $\Gamma_+ \cap [0, 1]$, so the coefficients of $\Delta_{Y/H}$ near E/H belong to Γ' by [Liu26, Lemma 2.6(4)], and Γ' is a DCC set [Liu26, Lemma 2.10(1)]. If the center of a prime divisor over Y/H is contained in E/H, then it is E/H or a closed point of

E/H. Let c_H be the coefficient of E/H in $\Delta_{Y/H}$, so that $a(E/H, Y/H, \Delta_{Y/H}) = 1 - c_H$. There exists a finite set Σ of closed points of E/H such that, near any closed point of $(E/H)\backslash\Sigma$, the surface Y/H and the curve E/H are smooth and $\Delta_{Y/H} = c_H(E/H)$. Let $\varpi \in (E/H)\backslash\Sigma$ be a closed point, let $V \subset Y/H$ be a smooth open neighborhood of ϖ on which E/H is smooth and $\Delta_{Y/H}|_V = c_H(E/H)|_V$, let $f : Y' \to V$ be the blow-up at ϖ, let E_ϖ be the exceptional curve of f, and let $\tilde{E}$ be the strict transform of E/H. Then $E_\varpi + \tilde{E}$ is an snc divisor and

$$K_{Y'} + c_H\tilde{E} + (c_H - 1)E_\varpi = f^*(K_V + \Delta_{Y/H}|_V).$$

Since $c_H \in \Gamma' \subset [0, 1]$, the coefficients c_H and $c_H - 1$ are at most 1, so [Liu26, Lemma 3.5], applied to $(V, \Delta_{Y/H}|_V)$ and f, implies

$$\mathrm{mld}(Y/H \ni \varpi, \Delta_{Y/H}) = a(E_\varpi, Y/H, \Delta_{Y/H}) = 2 - c_H > 1 - c_H.$$

Since $(Y/H, \Delta_{Y/H})$ is lc near E/H by **Step 3** and [Liu26, Lemma 2.6(2)(3)], each closed-point minimal log discrepancy is attained [Liu26, Lemma 3.5]. Thus

$$\beta_Y = \min\big(\{1 - c_H\} \cup \{\mathrm{mld}(Y/H \ni \varpi, \Delta_{Y/H}) : \varpi \in \Sigma\}\big)$$

is attained, and $\beta_Y \geq 0$ by **Step 3**. Since $c_H \in \Gamma'$, β_Y belongs to the union of $\{1 - c : c \in \Gamma'\}$ and the ACC set of Theorem 2.9(1) for Γ'. Both sets satisfy the ACC and depend only on Γ.

Step 5. In this step we show that $\alpha \leq \beta_Y + C/r$.

Let l be the integer of Theorem 2.9(2) for Γ', let $L_1 := be_0l$, let $k_0 := \lfloor L_1/\gamma \rfloor$, and assume that

$$r - M \max_i |w_i| - 2v > v(2k_0 + be_0).$$

Let

$$C := \max\{C_x, C_y, 2(k_0 + be_0) \max_i |w_i|\}.$$

By **Step 4**, $\beta_Y = a(P_Y, Y/H, \Delta_{Y/H})$ for a prime divisor P_Y over Y/H whose center is E/H or a closed point of E/H. If the center of P_Y is a closed point ϖ, then $\beta_Y = \mathrm{mld}(Y/H \ni \varpi, \Delta_{Y/H})$, since

$$\beta_Y \leq \mathrm{mld}(Y/H \ni \varpi, \Delta_{Y/H}) \leq a(P_Y, Y/H, \Delta_{Y/H}).$$

We distinguish three cases.

Case 1: the center of P_Y is the image $\bar{o}_y$ of o_y or the image $\bar{o}_x$ of o_x. If it is $\bar{o}_y$, then, by (A.94),

$$\beta_Y = \mathrm{mld}(Y/H \ni \bar{o}_y, \Delta_{Y/H}) = \mathrm{mld}((\mathbb{A}^2 \ni 0, C_{\mathrm{plane}})/\tilde{H}),$$

and Proposition 7.13 implies $\alpha \leq \beta_Y + C_y/r$. If it is $\bar{o}_x$, then Proposition 7.13 with x and y exchanged implies $\alpha \leq \beta_Y + C_x/r$.

Case 2: the center of P_Y is a closed point $\varpi \neq \bar{o}_x, \bar{o}_y$. By **Step 1**, ϖ is the image of a point $(p_0, 0) \in \mathbb{A}^2_{p,q}$ with $p_0 \neq 0$. Let $\tilde{H}_0$ be the stabilizer of $(p_0, 0)$ in $\tilde{H}$, and let $(\mathbb{A}^2/\tilde{H}_0, C_0)$ be the crepant quotient of $(\mathbb{A}^2_{p,q}, C_{\mathrm{plane}})$ by $\tilde{H}_0$. The inertia group in $\tilde{H}$ of a prime divisor ν over $\mathbb{A}^2_{p,q}$ with center $(p_0, 0)$ fixes $(p_0, 0)$, so it is a subgroup of $\tilde{H}_0$ and equals $I_{\tilde{H}_0}(\nu)$. Hence $\mathrm{mld}(Y/H \ni \varpi, \Delta_{Y/H})$ and $\mathrm{mld}((\mathbb{A}^2 \ni (p_0, 0), C_{\mathrm{plane}})/\tilde{H}_0)$ both equal the infimum of $a(\nu, \mathbb{A}^2, C_{\mathrm{plane}})/|I_{\tilde{H}}(\nu)|$ over these ν. Thus, by (A.94) and [Liu26, Lemma 2.6(2)(3)(5)],

$$\beta_Y = \mathrm{mld}(Y/H \ni \varpi, \Delta_{Y/H}) = \mathrm{mld}\big((\mathbb{A}^2 \ni (p_0, 0), C_{\mathrm{plane}})/\tilde{H}_0\big).$$

We have $C_0 \in \Gamma'$ [Liu26, Lemma 2.10(5)]. By Theorem 2.9(2), there exists a prime divisor E_1 over $\mathbb{A}^2/\tilde{H}_0$ with center the image of $(p_0, 0)$, $a(E_1, \mathbb{A}^2/\tilde{H}_0, C_0) = \beta_Y$, and $a(E_1, \mathbb{A}^2/\tilde{H}_0, 0) \leq l$. Let ν be a prime divisor over $\mathbb{A}^2_{p,q}$ over E_1. Then $\mathrm{center}_{\mathbb{A}^2}\nu = \{(p_0, 0)\}$. Let $e := |I_{\tilde{H}_0}(\nu)| \leq |\tilde{H}| \leq be_0$. By [Liu26, Lemma 2.6(2)], $a(\nu, \mathbb{A}^2, C_{\mathrm{plane}}) = e\beta_Y$, and $a(\nu, \mathbb{A}^2, 0) = ea(E_1, \mathbb{A}^2/\tilde{H}_0, B_0)$, where $B_0 \geq 0$ is the boundary of the crepant quotient of $(\mathbb{A}^2_{p,q}, 0)$ by $\tilde{H}_0$. Hence $a(\nu, \mathbb{A}^2, 0) \leq el \leq L_1$. Since $(\mathbb{A}^2_{p,q}, C_{\mathrm{plane}})$ is lc near $(p_0, 0)$ and $c_j \geq \gamma$, we obtain $\mathrm{mult}_\nu \varphi_j \leq L_1/\gamma$ for any j. Let χ be the character of $\tilde{H}_0$ on u, $(g, \xi) \to \xi^r$, and let $i_0 \in \{1, \ldots, e\}$ be the integer of Lemma 3.1 for ν and χ. Fix an integer $k \equiv i_0 \pmod{e}$ with $k_0 < k \leq k_0 + e$. Lemma 3.1, applied with

$V = \mathbb{A}^2_{p,q}$, the closed point $(p_0, 0)$, the group $\tilde{H}_0$, which acts faithfully on $\mathbb{A}^2_{p,q}$ by Proposition 7.13, $B = C_{\text{plane}}$, $E = \nu$, and $\ell = k$, implies that the Gauss extension of mult_ν with u-value k is mult_F for a prime divisor F over $\mathbb{A}^3_{u,p,q}$ with center $((p_0, 0), 0)$, $I_{\tilde{H}_0}(F) = I_{\tilde{H}_0}(\nu)$, and

$$a(F, \mathbb{A}^3, \text{pr}_1^* C_{\text{plane}}) = a(\nu, \mathbb{A}^2, C_{\text{plane}}) + k.$$

An element of $I_{\tilde{H}}(F)$ or of $I_{\tilde{H}}(\nu)$ fixes the closed point $((p_0, 0), 0)$ or $(p_0, 0)$, so $I_{\tilde{H}}(F) = I_{\tilde{H}_0}(F)$ and $I_{\tilde{H}}(\nu) = I_{\tilde{H}_0}(\nu)$. Thus $|I_{\tilde{H}}(F)| = e$.

Case 3: the center of P_Y is E/H. Let $\nu := \text{mult}_{(q=0)}$ on $\mathbb{A}^2_{p,q}$, $I := I_{\tilde{H}}(\nu)$, and $e := |I| \le be_0$. By **Step 1** and [Liu26, Lemma 2.6(2)], $a(\nu, \mathbb{A}^2, C_{\text{plane}}) = e\beta_Y$. Since $(\mathbb{A}^2_{p,q}, C_{\text{plane}})$ is lc near $(q = 0)$, $\text{mult}_\nu \varphi_j \le 1/\gamma \le L_1/\gamma$ for any j. By [Liu26, Lemma 2.6(1)], I is cyclic and acts faithfully on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ by a character λ_ν. Since q is a semi-invariant uniformizer of ν, λ_ν is the character of q restricted to I. Let χ be the character of $\tilde{H}$ on u, $(g, \xi) \to \xi^r$, and let $i_0 \in \{1, \dots, e\}$ be the integer with $\chi|_I = \lambda_\nu^{i_0}$. Fix an integer $k \equiv i_0 \pmod e$ with $k_0 < k \le k_0 + e$, and let F be the prime divisor over $\mathbb{A}^3_{u,p,q}$ such that mult_F is the Gauss extension of mult_ν with u-value k [Liu26, Definition-Proposition 2.11] (it exists by [Liu26, Lemma 18.1]). By [Liu26, Lemma 18.1] with $\varrho = 1$, the center of F on $\mathbb{A}^3_{u,p,q}$ is $(q = u = 0)$, the residue field of F is generated over the residue field of ν by the residue of u/q^k, and

$$a(F, \mathbb{A}^3, \text{pr}_1^* C_{\text{plane}}) = a(\nu, \mathbb{A}^2, C_{\text{plane}}) + k.$$

We show that $I_{\tilde{H}}(F) = I$. An element of $I_{\tilde{H}}(F)$ preserves F, hence ν, and acts trivially on the residue field of F, which contains the residue field of ν. Thus $I_{\tilde{H}}(F) \subset I$. Conversely, an element $\sigma \in I$ preserves mult_ν and multiplies u by a constant, so σ preserves F. Moreover, σ acts trivially on the residue field of ν, and the pullback of u/q^k by σ is $\chi(\sigma)\lambda_\nu(\sigma)^{-k} u/q^k = u/q^k$ by the choice of k. Thus $\sigma \in I_{\tilde{H}}(F)$, and $|I_{\tilde{H}}(F)| = e$.

End of the proof in Cases 2 and 3. In both cases, $\text{mult}_\nu p = 0$, $n_q := \text{mult}_\nu q \ge 1$, $\text{mult}_\nu \varphi_j \le k_0$ since $\text{mult}_\nu \varphi_j$ is an integer, and $k_0 < k \le k_0 + be_0$. By the choice of r, (A.95) holds with $s = k$ and $s_1 = k_0$. By **Step 2**, F satisfies the hypothesis of Lemma A.9(4) and

$$\text{mult}_F(\Phi_j - \varphi_j) > k_0 \ge \text{mult}_\nu \varphi_j = \text{mult}_F \varphi_j,$$

so $\text{mult}_F \Phi_j = \text{mult}_\nu \varphi_j$ for any j. Hence

$$a(F, \mathbb{A}^3, \Delta_3) = a(F, \mathbb{A}^3, \text{pr}_1^* C_{\text{plane}}) - \Big(1 - \frac{\tau}{r}\Big)k = a(\nu, \mathbb{A}^2, C_{\text{plane}}) + \frac{\tau k}{r} = e\beta_Y + \frac{\tau k}{r}.$$

By Lemma A.9(4), as α is the minimum of the log discrepancies over the prime divisors with center the image of 0, and by (A.96),

$$\alpha \le \frac{a(F, \mathbb{A}^3, \Delta_3)}{e} = \beta_Y + \frac{\tau k}{re} \le \beta_Y + \frac{2(k_0 + be_0)\max_i |w_i|}{r} \le \beta_Y + \frac{C}{r}.$$

Let r_0 be the maximum of $2M \max_i |w_i| + 1$, r_x, r_y, and the least integers satisfying the inequalities assumed in **Steps 3 and 5**. Then r_0 and C depend only on a, b, v, Γ, e_0, **Step 1** proves part (1) of Construction-Proposition 7.14 for any r, and **Steps 1–5** prove part (2) for any $r \ge r_0$ and any d. □

Proof of Remark 7.16. An element $\lambda \in \mu_r$ acts on (x, y, z, t) by $(\lambda^{-1}, \lambda^{-3}, \lambda^{-2}, \lambda^2)$. Since $N = r - 2$, the monomials xy, z^2, t^N have weights -4, -4, $2r - 4$, so G preserves the equation of W_N up to a character, and G acts faithfully on W_N, since x has weight -1. The vector $w = (-1, -3, -2, 2)$ is primitive with $w_x + w_y = 2w_z$, and $G \subset \mathbf{T}_{(w)}$. Thus $R = G$, $[G : R] = 1$, and $d = (Nw_t - 2w_z)/r = (2r - 4 + 4)/r = 2$. The function t has no monomial divisible by xy, $\kappa(t) = 2$, $h^0 = t$, and $\tau = 2c$.

Let $S := (t = 0) \subset W_N$, so that $S \cong (xy + z^2 = 0) \subset \mathbb{A}^3$ is a normal klt surface and $D = cS$. The hypersurface W_N is smooth outside 0, so the different of 0 on S is zero, and $K_{W_N} + S$ is

Cartier. By [Kaw07, Theorem], (W_N, S) is lc near S. For any prime divisor P over W_N with center 0, $a(P, W_N, S)$ is a non-negative integer, so

$$a(P, W_N, 0) = a(P, W_N, S) + \operatorname{mult}_P t \geq 1.$$

Thus W_N is klt near 0. Since $a(P, W_N, cS) = (1-c)a(P, W_N, 0) + ca(P, W_N, S)$ for any prime divisor P over W_N, (W_N, cS) is klt near 0. Thus the data satisfy Set-up 7.3 with $\Gamma = \{c\}$, $e_0 = 1$, $w = (-1, -3, -2, 2)$, $R = G$, $d = 2$, and $\tau = 2c$.

We compute $\operatorname{mld}((W_N \ni 0, S)/G)$. The group G acts faithfully on S, since $x|_S$ has weight -1. By Theorem 3.3(2) with $n = 4$, $c = 1$, $f_1 = xy + z^2 + t^N$, $f_2 = t$, the trivial ideal, and $H = G$,

$$\operatorname{mld}((W_N \ni 0, S)/G) = \operatorname{mld}((S \ni 0)/G).$$

The map

$$\mathbb{A}^2_{\varsigma_1,\varsigma_2} \to S, \quad (\varsigma_1, \varsigma_2) \to (-\varsigma_1^2, \varsigma_2^2, \varsigma_1\varsigma_2)$$

is the quotient by ± 1 and is quasi-étale, and $\mu_{2r}(-1, -3)$ acts on $\mathbb{A}^2_{\varsigma_1,\varsigma_2}$ compatibly with G through $\zeta \to \zeta^2$, with kernel ± 1. By [Liu26, Lemma 2.6(4)(5)] and [Liu26, Lemma 3.4],

$$\operatorname{mld}((S \ni 0)/G) = \min\left\{w_1 + w_2 : (w_1, w_2) \in \left(\mathbb{Z}^2 + \frac{1}{2r}(1, 3)\mathbb{Z}\right) \cap \mathbb{R}^2_{>0}\right\}.$$

On the coset of $\frac{k}{2r}(1, 3)$ with $1 \leq k \leq 2r - 1$, the value $2r(w_1 + w_2)$ is at least $k + k'$, where $k' \in \{1, \ldots, 2r\}$ and $k' \equiv 3k \pmod{2r}$. Since $2r \geq 14$, $k + k'$ equals 4, 8, 12 for $k = 1, 2, 3$ and is at least 5 for $k \geq 4$, and on the zero coset $w_1 + w_2 \geq 2$. Thus

$$\operatorname{mld}((W_N \ni 0, S)/G) = \frac{2}{r}. \tag{A.98}$$

We compute α. Let F be a prime divisor over W_N with center 0, $e_F := |I_G(F)|$, and λ_F the character of $I_G(F)$ on $\mathfrak{m}_F/\mathfrak{m}_F^2$, which is faithful [Liu26, Lemma 2.6(1)]. By [Liu26, Lemma 9.2(2)], applied to $W_N = (f = 0) \subset \mathbb{A}^4$ with $f = xy + z^2 + t^N$ and the eigenparameters x, y, z, t of weights $-1, -3, -2, 2$, $\lambda_F^{a(F,W_N,0)}$ is the restriction to $I_G(F)$ of the character of weight $-1-3-2+2-(-4) = 0$, since f has weight -4 ($N = r-2$). Thus $\lambda_F^{a(F,W_N,0)} = 1$, so e_F divides the positive integer $a(F, W_N, 0)$. By (A.98), [Liu26, Lemma 2.6(2)], and the formula at the end of [Liu26, Lemma 2.6], $a(F, W_N, S)/e_F \geq 2/r$. Hence

$$\frac{a(F, W_N, cS)}{e_F} = (1-c)\frac{a(F, W_N, 0)}{e_F} + c\frac{a(F, W_N, S)}{e_F} \geq 1 - c + \frac{2c}{r}.$$

Let F_0 be the exceptional prime divisor of the weighted blow-up of W_N with weights $(1, 3, 2, r-2)$. It is a prime divisor, since xy and z^2 have weight 4, while t^N has weight $(r-2)N = (r-2)^2 > 4$, and $(xy + z^2 = 0) \subset \mathbb{P}(1, 3, 2, r-2)$ is irreducible and reduced. Let F_4 be the exceptional divisor of the weighted blow-up of $\mathbb{A}^4$ with weights $(1, 3, 2, r-2)$. Then $a(F_4, \mathbb{A}^4, 0) = 1+3+2+(r-2)$ [JM12, Proposition 5.1 and (5.1)] and $\operatorname{mult}_{F_4}(xy + z^2 + t^N) = 4$. Since $(xy + z^2 = 0) \subset \mathbb{P}(1, 3, 2, r-2)$ is not contained in a coordinate hyperplane and $(1, 3, 2, r-2)$ is primitive, the strict transform of W_N is smooth and intersects F_4 transversally at the generic point of F_0, and the adjunction formula implies $a(F_0, W_N, 0) = r$. Moreover, $\operatorname{mult}_{F_0} t = r - 2$. Since $\lambda^2 = \lambda^{-(r-2)}$, the element $\lambda \in G$ acts as $(\mu, \mu^3, \mu^2, \mu^{r-2})$ with $\mu = \lambda^{-1}$, which preserves F_0 and acts trivially on the function field of F_0. Thus $I_G(F_0) = G$, and

$$\frac{a(F_0, W_N, cS)}{r} = \frac{r - c(r-2)}{r} = 1 - c + \frac{2c}{r}.$$

By [Liu26, Lemma 2.6(2)] and the formula at the end of [Liu26, Lemma 2.6], $\alpha = 1 - c + \frac{2c}{r}$.

We compute β. The vector w is $(-a, -b, -v, v)$ with $(a, b, v) = (1, 3, 2)$, which satisfies $a+b = 2v$, $b > 0$, and $\gcd(a, v) = 1$. As $h^0 = t$, $\varphi(p, q) = h^0(-p^2, 1, p, q) = q$ and $C_{\text{plane}} = c\operatorname{div}(q)$. Since $R = G$, a pair (g, ξ) with $g \in G$ and $\xi^3 = g_y$ has $g = (\lambda^{-1}, \lambda^{-3}, \lambda^{-2}, \lambda^2)$ and $\xi = \lambda^{-1}\zeta$ with

$\zeta^3 = 1$, so it is congruent to $(1, \zeta)$ modulo $\tilde{R}$, and $(1, \zeta)$ acts on (p, q) by $(\zeta^{-2}p, \zeta^2 q) = (\zeta p, \zeta^2 q)$. Thus $\tilde{H} = \mu_3(1, 2)$. By [Liu26, Lemma 3.4],

$$\beta = \min\Big\{w_1 + (1-c)w_2 : (w_1, w_2) \in \Big(\mathbb{Z}^2 + \frac{1}{3}(1,2)\mathbb{Z}\Big) \cap \mathbb{R}^2_{>0}\Big\}.$$

The minima on the three cosets are $2 - c$, $1 - \frac{2c}{3}$, and $1 - \frac{c}{3}$, attained at $(1, 1)$, $(\frac{1}{3}, \frac{2}{3})$, and $(\frac{2}{3}, \frac{1}{3})$. Since $0 < c < 1$, $\beta = 1 - \frac{2c}{3}$, and

$$\beta - \alpha = \frac{c}{3} - \frac{2c}{r} = c\Big(\frac{1}{3} - \frac{2}{r}\Big) > 0,$$

since $r \geq 7$. For $c = \frac{1}{2}$, the pair

$$\Big(\mathbb{A}^2, \frac{3}{2}C_{\text{plane}}\Big) = \Big(\mathbb{A}^2, \frac{3}{4}\operatorname{div}(q)\Big)$$

is klt, so $\frac{3}{2}$ is a margin of $(\mathbb{A}^2, C_{\text{plane}})$ near 0 for any $r \geq 7$, while $\alpha < \beta$.

Finally, we show that F_0 induces on $\mathbb{A}^2_{p,q}$ a valuation whose center is the curve $(q = 0)$. On the root chart $\eta^3 = y$, $p = z/\eta^2$, $q = t\eta^2$, normalize the extension of $\operatorname{mult}_{F_0}$ so that y has value 3. Then η, p, q have values 1, 0, r, and $p^3 = z^3/y^2$ has weighted degree 0 and is non-constant on F_0. Thus F_0 induces on $\mathbb{A}^2_{p,q}$ the valuation $r \operatorname{mult}_{(q=0)}$, whose center is the curve $(q = 0)$ and not the origin. □

A.8. **Interior canonical circles.** In this subsection, we prove Proposition A.14 on the interior canonical circles $\mathbf{T}_{(-a,-b,-v,v)}$ with non-negative mass. We first pass from W_N to a threefold X_d by adjoining a root s of t of order v, and then compare X_d with the reference surface Y of Construction-Proposition 7.14 by quotient adjunction (Theorem 3.3).

Construction-Proposition A.11. Let a, b, v be positive integers with $a+b = 2v$ and $\gcd(a, v) = 1$, let $\mathbf{T} := \mathbf{T}_{(-a,-b,-v,v)}$, let $N \geq 3$ be an integer, and let G be a finite diagonal abelian group preserving the equation of W_N up to a character and acting faithfully on W_N. Let $R := G \cap \mathbf{T}$ and $r := |R|$. Then r divides $v(N + 2)$. Let $d := v(N + 2)/r$, let

$$\widetilde{W} := (xy + z^2 + s^{vN} = 0) \subset \mathbb{A}^4_{x,y,z,s}, \qquad \widetilde{X} := (x_1y_1 + z_1^2 + s^{v(N+2)} = 0) \subset \mathbb{A}^4_{x_1,y_1,z_1,s},$$

$$X_d := (x_1y_1 + z_1^2 + \omega^d = 0) \subset \mathbb{A}^4_{x_1,y_1,z_1,\omega},$$

and let $S_0 := X_d \cap (\omega = 0)$. Write each $g \in G$ as $g = \operatorname{diag}(g_x, g_y, g_z, g_t)$ in the coordinates x, y, z, t. Let

$$\widetilde{G} := \{(g, \sigma) \in G \times \mathbb{C}^* \mid \sigma^v = g_t\}$$

act on $\widetilde{W}$, on $\widetilde{X}$ and on $\mathbb{A}^4_{x_1,y_1,z_1,\omega}$ by

$$(x, y, z, s) \to (g_x x, g_y y, g_z z, \sigma s), \qquad (x_1, y_1, z_1, s) \to (g_x\sigma^a x_1, g_y\sigma^b y_1, g_z\sigma^v z_1, \sigma s),$$

$$(x_1, y_1, z_1, \omega) \to (g_x\sigma^a x_1, g_y\sigma^b y_1, g_z\sigma^v z_1, \sigma^r\omega),$$

let $G_d \subset GL_4(\mathbb{C})$ be the image of $\widetilde{G}$ under the last action, and let

$$\mu_v := \{(1, \zeta) \mid \zeta^v = 1\} \subset \widetilde{G}, \qquad \widetilde{R} := \{((\sigma^{-a}, \sigma^{-b}, \sigma^{-v}, \sigma^v), \sigma) \mid \sigma^r = 1\} \subset \widetilde{G}.$$

Let P_W be a prime divisor over W_N with center 0, let $\widetilde{P}$ be a prime divisor over $\widetilde{W}$ lying over P_W, and let P be the prime divisor over X_d with $\widetilde{P}$ lying over P, where $\widetilde{W}$ and $\widetilde{X}$ are identified birationally through $(x, y, z, s) \to (xs^a, ys^b, zs^v, s)$. For a $\widetilde{G}$-semi-invariant rational function u on $\widetilde{W}$, let $[u] := \operatorname{mult}_{\widetilde{P}} u/|I_{\widetilde{G}}(\widetilde{P})|$. Let $D = \sum_j c_j \operatorname{div}(h_j)$, where c_j are real numbers and h_j are non-zero G-semi-invariant polynomials. For each j let κ_j be the $\mathbf{T}$-weight of a monomial of h_j, let $\tau := \sum_j c_j\kappa_j$, and let

$$\check{h}_j := s^{-\kappa_j} h_j(x_1 s^{-a}, y_1 s^{-b}, z_1 s^{-v}, s^v).$$

Then the following hold.

(1) $\widetilde{W}$, $\widetilde{X}$, and X_d are normal integral hypersurfaces, X_d is klt, and S_0 is a normal klt surface.

(2) The group G_d is a finite diagonal abelian group of order $v[G:R]$, preserving the equation of X_d up to a character and acting faithfully on X_d and on S_0, and μ_v maps isomorphically onto a subgroup of G_d, again denoted by μ_v.
(3) The morphism $\widetilde{W} \to W_N$ given by $t = s^v$ is the quotient by μ_v, the morphism $\widetilde{X} \to X_d$ given by $\omega = s^r$ is the quotient by $\widetilde{R}$, and $(x, y, z, s) \to (xs^a, ys^b, zs^v, s)$ is a $\widetilde{G}$-equivariant isomorphism $\widetilde{W} \cap (s \neq 0) \to \widetilde{X} \cap (s \neq 0)$.
(4) The group $\widetilde{G}/\mu_v$ is G through the projection, $\widetilde{G}/\widetilde{R} = G_d$, and $G_d/\mu_v \cong G/R$.
(5) The center of $\widetilde{P}$ on $\widetilde{X}$ and the center of P on X_d are 0.
(6) The number $[u]$ equals $\operatorname{mult}_{P_W} u/|I_G(P_W)|$ for any G-semi-invariant rational function u on W_N, and equals $\operatorname{mult}_P u/|I_{G_d}(P)|$ for any G_d-semi-invariant rational function u on X_d.
(7) The inequalities $[x], [y], [z], [s] > 0$ and the equalities

$$[x_1] = [x] + a[s], \quad [y_1] = [y] + b[s], \quad [z_1] = [z] + v[s], \quad [t] = v[s], \quad [\omega] = r[s]$$

hold, so that

$$\min\left(\frac{[x_1]}{a}, \frac{[y_1]}{b}, \frac{[z_1]}{v}\right) - \frac{[\omega]}{r} = \min\left(\frac{[x]}{a}, \frac{[y]}{b}, \frac{[z]}{v}\right) > 0.$$

(8) The function $\check{h}_j$ is a G_d-semi-invariant element of $\mathbb{C}[x_1, y_1, z_1, \omega, \omega^{-1}]$, $\check{h}_j = h_j(x_1, y_1, z_1, 1)$ if h_j is homogeneous for $\mathbf{T}$, and

$$\frac{a(P_W, W_N, D)}{|I_G(P_W)|} = \frac{a(P, X_d, S_0 + \sum_j c_j \operatorname{div}(\check{h}_j))}{|I_{G_d}(P)|} - \frac{\tau}{r}[\omega].$$

Proof. The groups $\widetilde{G}$ and $\widetilde{R}$ are defined by a root of t. They are not the groups $\tilde{G}$ and $\tilde{R}$ of Construction A.8, which are defined by a root of y.

We prove (1) and (2). Since $\gcd(a, v) = 1$, the weights $-a, -b, -v, v$ have greatest common divisor one. Thus

$$\mathbb{C}^* \to \mathbf{T}, \qquad \sigma \to (\sigma^{-a}, \sigma^{-b}, \sigma^{-v}, \sigma^v),$$

is an isomorphism, and R is the image of the group μ_r of r-th roots of unity. For a primitive r-th root of unity σ, the corresponding element of R multiplies $xy + z^2$ by σ^{-2v} and t^N by σ^{vN}. Since G preserves $xy + z^2 + t^N$ up to a character, $\sigma^{v(N+2)} = 1$, so r divides $v(N+2)$.

The equations of $\widetilde{W}$, $\widetilde{X}$, and X_d are of degree one in x, resp. x_1, with coprime coefficients, so they are irreducible. Their partial derivatives vanish simultaneously on them only at 0, because $vN \geq 2$, $v(N+2) \geq 2$, and X_d is smooth if $d = 1$. Thus the three hypersurfaces are smooth outside 0, and, being hypersurfaces in $\mathbb{A}^4$ regular in codimension one, they are normal [Har77, Proposition II.8.23(b)]. The morphism $\mathbb{A}^2_{u,u'} \to S_0$, $(u, u') \to (u^2, -u'^2, uu')$, is the quotient by $\mu_2(1, 1)$, so S_0 is a normal klt surface [Liu26, Lemma 2.6(2)(3)]. Since X_d is a hypersurface and $S_0 = (\omega = 0)$ is a normal Cartier divisor on X_d, we have $K_{S_0} = (K_{X_d} + S_0)|_{S_0}$, so (X_d, S_0) is lc near S_0 by [Kaw07, Theorem]. Hence any prime divisor P' over X_d with center 0 satisfies

$$a(P', X_d, 0) = a(P', X_d, S_0) + \operatorname{mult}_{P'} \omega \geq \operatorname{mult}_{P'} \omega > 0.$$

As X_d is smooth outside 0, X_d is klt.

Let $(g, \sigma) \in \widetilde{G}$. Since G preserves $xy + z^2 + t^N$ up to a character, $g_x g_y = g_z^2 = g_t^N$. As $a + b = 2v$, $rd = v(N+2)$ and $\sigma^v = g_t$, the element (g, σ) multiplies $x_1 y_1$, z_1^2, and ω^d by $g_x g_y \sigma^{2v}$, $g_z^2 \sigma^{2v}$, and $g_t^N \sigma^{2v}$, which are equal. Thus G_d preserves the equation of X_d up to a character. If (g, σ) acts trivially on S_0, then $g_x \sigma^a = g_y \sigma^b = g_z \sigma^v = 1$, because x_1, y_1 and z_1 do not vanish identically on S_0. Together with $g_t = \sigma^v$, this implies

$$g = (\sigma^{-a}, \sigma^{-b}, \sigma^{-v}, \sigma^v) \in G \cap \mathbf{T} = R,$$

so $\sigma^r = 1$ and $(g, \sigma) \in \widetilde{R}$. Conversely $\widetilde{R}$ acts trivially on $\mathbb{A}^4_{x_1,y_1,z_1,\omega}$. Hence the kernel of $\widetilde{G} \to G_d$ is $\widetilde{R}$, G_d acts faithfully on S_0 and on X_d, and, since any $g \in G$ has exactly v lifts to $\widetilde{G}$,

$$|G_d| = \frac{|\widetilde{G}|}{|\widetilde{R}|} = \frac{v|G|}{r} = v[G : R].$$

An element $(1, \zeta) \in \mu_v \cap \widetilde{R}$ satisfies $(\zeta^{-a}, \zeta^{-b}, \zeta^{-v}, \zeta^v) = 1$, so $\zeta = 1$. This proves (1) and (2).

We prove (3) and (4). The group μ_v acts by $s \to \zeta s$ and fixes x, y, z, so

$$\mathbb{C}[x, y, z, s]^{\mu_v} = \mathbb{C}[x, y, z, s^v],$$

and $xy + z^2 + s^{vN}$ is the pullback of $xy + z^2 + t^N$ under $t = s^v$. Thus $\widetilde{W}/\mu_v = W_N$. The element $((\sigma^{-a}, \sigma^{-b}, \sigma^{-v}, \sigma^v), \sigma)$ of $\widetilde{R}$ acts on $\widetilde{X}$ by $(x_1, y_1, z_1, s) \to (x_1, y_1, z_1, \sigma s)$, and $x_1y_1 + z_1^2 + s^{v(N+2)}$ is the pullback of $x_1y_1 + z_1^2 + \omega^d$ under $\omega = s^r$. Thus $\widetilde{X}/\widetilde{R} = X_d$. Since $a + b = 2v$, for $(x_1, y_1, z_1) = (xs^a, ys^b, zs^v)$ we have

$$x_1y_1 + z_1^2 + s^{v(N+2)} = s^{2v}(xy + z^2 + s^{vN}),$$

so $(x, y, z, s) \to (xs^a, ys^b, zs^v, s)$ maps $\widetilde{W} \cap (s \neq 0)$ isomorphically onto $\widetilde{X} \cap (s \neq 0)$, with inverse $(x_1, y_1, z_1, s) \to (x_1s^{-a}, y_1s^{-b}, z_1s^{-v}, s)$, and a direct calculation shows that it is $\widetilde{G}$-equivariant. The projection $\widetilde{G} \to G$ is surjective with kernel μ_v and commutes with the actions on $\widetilde{W}$ and W_N, so $\widetilde{G}/\mu_v = G$. The kernel of $\widetilde{G} \to G_d$ is $\widetilde{R}$, as shown above, so $\widetilde{G}/\widetilde{R} = G_d$. Since the image of $\widetilde{R}$ in G is R, we obtain $G_d/\mu_v = \widetilde{G}/(\widetilde{R}\mu_v) \cong G/R$.

We prove (5), (6), and (7). Since $\widetilde{W} \to W_N$ is finite and the preimage of 0 is 0, the center of $\widetilde{P}$ on $\widetilde{W}$ is 0. Thus x, y, z, and s have positive multiplicity along $\widetilde{P}$. The identities $x_1 = xs^a$, $y_1 = ys^b$, $z_1 = zs^v$, $t = s^v$, and $\omega = s^r$ imply the formulas for $[x_1]$, $[y_1]$, $[z_1]$, $[t]$, and $[\omega]$, and x_1, y_1, z_1, and s have positive multiplicity along $\widetilde{P}$. As x_1, y_1, z_1, s are the coordinates of the affine variety $\widetilde{X}$, $\widetilde{P}$ is a prime divisor over $\widetilde{X}$ with center 0, and P has center 0. By [Liu26, Lemma 2.6(1)(5)], the order of an inertia group is the ramification index, and ramification indices are multiplicative in the towers $\widetilde{W} \to W_N \to W_N/G$ and $\widetilde{X} \to X_d \to X_d/G_d$, whose composites are the quotients by $\widetilde{G}$. Thus

$$\operatorname{mult}_{\widetilde{P}} u = |I_{\mu_v}(\widetilde{P})| \operatorname{mult}_{P_W} u, \qquad |I_{\widetilde{G}}(\widetilde{P})| = |I_{\mu_v}(\widetilde{P})| \cdot |I_G(P_W)|$$

for any rational function u on W_N, and likewise with $\widetilde{R}$, P, and G_d in place of μ_v, P_W, and G. This implies both expressions for $[u]$. Finally $[x_1]/a - [\omega]/r = [x]/a$, $[y_1]/b - [\omega]/r = [y]/b$, and $[z_1]/v - [\omega]/r = [z]/v$.

We prove (8). By Lemma 5.3, the $\mathbf{T}$-weight of any monomial $x^{e_1}y^{e_2}z^{e_3}t^{e_4}$ of h_j is $\kappa_j + rl$ for an integer l, and

$$s^{-\kappa_j}x^{e_1}y^{e_2}z^{e_3}t^{e_4} = x_1^{e_1}y_1^{e_2}z_1^{e_3}\omega^l.$$

Thus $\check{h}_j \in \mathbb{C}[x_1, y_1, z_1, \omega, \omega^{-1}]$, and $l = 0$ for all monomials if h_j is homogeneous for $\mathbf{T}$, in which case $\check{h}_j = h_j(x_1, y_1, z_1, 1)$. As s and h_j are $\widetilde{G}$-semi-invariant, $\check{h}_j$ is G_d-semi-invariant.

Let $\Omega_W := dx \wedge dz \wedge dt/x$ and $\Omega_X := dx_1 \wedge dz_1 \wedge d\omega/x_1$ be the Poincaré residues of

$$\frac{dx \wedge dy \wedge dz \wedge dt}{xy + z^2 + t^N} \qquad \text{and} \qquad \frac{dx_1 \wedge dy_1 \wedge dz_1 \wedge d\omega}{x_1y_1 + z_1^2 + \omega^d},$$

respectively. They generate the dualizing sheaves of W_N and X_d, respectively. Let $(\widetilde{W}, \widetilde{D})$ be the crepant pullback of (W_N, D), and $(\widetilde{X}, \widetilde{B})$ the crepant pullback of $(X_d, S_0 + \sum_j c_j \operatorname{div}(\check{h}_j))$. Since $\operatorname{div}_{W_N}(\Omega_W) = 0$ and $\operatorname{div}_{X_d}(\Omega_X/\omega) = -S_0$, (A.88), applied to $\widetilde{W}$ with the pullback of Ω_W

and to $\widetilde{X}$ with the pullback of Ω_X/ω, gives

$$a(\widetilde{P},\widetilde{W},\widetilde{D}) = 1 + \operatorname{mult}_{\widetilde{P}}\Omega_W - \sum_j c_j \operatorname{mult}_{\widetilde{P}} h_j,$$
$$a(\widetilde{P},\widetilde{X},\widetilde{B}) = 1 + \operatorname{mult}_{\widetilde{P}}(\Omega_X/\omega) - \sum_j c_j \operatorname{mult}_{\widetilde{P}} \check{h}_j,$$

where the forms and functions are pulled back to $\widetilde{W}$ and $\widetilde{X}$. By [Liu26, Lemma 2.6(2)],

$$a(\widetilde{P},\widetilde{W},\widetilde{D}) = |I_{\mu_v}(\widetilde{P})|a(P_W,W_N,D), \qquad a(\widetilde{P},\widetilde{X},\widetilde{B}) = |I_{\widetilde{R}}(\widetilde{P})|a\Big(P,X_d,S_0+\sum_j c_j \operatorname{div}(\check{h}_j)\Big).$$

A direct calculation with $x = x_1 s^{-a}$, $z = z_1 s^{-v}$, $t = s^v$, and $\omega = s^r$ shows that on $\widetilde{W}\cap(s\neq 0) = \widetilde{X}\cap(s\neq 0)$,

$$\Omega_W = v\frac{dx_1\wedge dz_1\wedge ds}{x_1 s} = \frac{v}{r}\cdot\frac{\Omega_X}{\omega}.$$

Since $h_j = s^{\kappa_j}\check{h}_j$ and $\tau = \sum_j c_j\kappa_j$, we obtain

$$a(\widetilde{P},\widetilde{W},\widetilde{D}) = a(\widetilde{P},\widetilde{X},\widetilde{B}) - \tau \operatorname{mult}_{\widetilde{P}} s.$$

Dividing by

$$|I_{\widetilde{G}}(\widetilde{P})| = |I_{\mu_v}(\widetilde{P})|\cdot|I_G(P_W)| = |I_{\widetilde{R}}(\widetilde{P})|\cdot|I_{G_d}(P)|$$

and using $[s] = [\omega]/r$, we obtain (8). □

Lemma A.12. *Notation as in Construction-Proposition A.11. Let Q be a positive integer divisible by a, b, and v, and let $\mathfrak{a} := (x_1^{Q/a}, y_1^{Q/b}, z_1^{Q/v})$. Let $c_1,\dots,c_s$ be positive real numbers, let $B_1,\dots,B_s$ be G_d-semi-invariant polynomials in x_1,y_1,z_1,ω with $B_j|_{S_0}\neq 0$, let $\tau\geq 0$ be a real number, and let $\mathfrak{b} := \mathfrak{a}^{\tau/Q}\prod_j B_j^{c_j}$, where the factor $\mathfrak{a}^{\tau/Q}$ is omitted if $\tau = 0$.*

(1) If $(S_0,\mathfrak{b}|_{S_0})$ is lc near 0, then any prime divisor P over X_d with center 0 satisfies

$$\frac{a(P,X_d,S_0+\sum_j c_j\operatorname{div}(B_j))}{|I_{G_d}(P)|}$$
$$\geq \operatorname{mld}((S_0\ni 0,\mathfrak{b}|_{S_0})/G_d) + \frac{\tau}{|I_{G_d}(P)|}\min\Big(\frac{\operatorname{mult}_P x_1}{a},\frac{\operatorname{mult}_P y_1}{b},\frac{\operatorname{mult}_P z_1}{v}\Big).$$

(2) We use the surface S_v, the toric partial resolution $Y\to S_v$, the curve E and the charts $\mathbb{A}^2_{p,q}/\mu_b(-v,v)$ and $\mathbb{A}^2_{p',q'}/\mu_a(-v,v)$ of Y of Construction-Proposition 7.14. Let $f_1,\dots,f_s$ be non-zero G-semi-invariant polynomials in x,y,z,t without monomials divisible by xy, such that f_j is homogeneous for $\mathbf{T}$ of weight κ_j, $\tau = \sum_j c_j\kappa_j$, and $B_j|_{S_0} = f_j(x_1,y_1,z_1,1)|_{S_0}$ for each j. Let $D_j\geq 0$ be the $\mathbb{Q}$-Cartier divisor on Y whose pullbacks to $\mathbb{A}^2_{p,q}$ and $\mathbb{A}^2_{p',q'}$ are $\operatorname{div}(f_j(-p^2,1,p,q))$ and $\operatorname{div}(f_j(1,-p'^2,p',q'))$. Then $H := G_d/\mu_v \cong G/R$ acts on Y as in Construction-Proposition 7.14, preserving E and each D_j. Moreover $(S_0,\mathfrak{b}|_{S_0})$ is lc near 0 if and only if $(Y,\sum_j c_jD_j)$ is lc near E, and in this case $\operatorname{mld}((S_0\ni 0,\mathfrak{b}|_{S_0})/G_d)$ is the infimum of $a(P',Y,\sum_j c_jD_j)/|I_H(P')|$ over all prime divisors P' over Y with center contained in E.

Proof. We prove (1). By Construction-Proposition A.11(1)(2), $X_d = (x_1y_1+z_1^2+\omega^d = 0)$ and $S_0 = X_d\cap(\omega = 0)$ are normal integral complete intersections in $\mathbb{A}^4$ with klt singularities, the diagonal group G_d acts faithfully on both, and $x_1y_1+z_1^2+\omega^d$ and ω are G_d-semi-invariant. The factors of $\mathfrak{b}$ are generated by G_d-semi-invariant polynomials, and $\mathfrak{b}|_{S_0}\neq 0$. By Theorem 3.3(2), applied with $n=4$, $c=1$, the polynomials $x_1y_1+z_1^2+\omega^d$ and ω in place of f_1 and f_2, G_d in place of both G and H, and $\mathfrak{b}$ in place of $\mathfrak{a}$ of Theorem 3.3,

$$\operatorname{mld}((S_0\ni 0,\mathfrak{b}|_{S_0})/G_d) = \operatorname{mld}((X_d\ni 0,S_0,\mathfrak{b}|_{X_d})/G_d),$$

where S_0 has coefficient one. The right side is the infimum of $a(P, X_d, S_0, \mathfrak{b}|_{X_d})/|I_{G_d}(P)|$ over the prime divisors P over X_d with center 0 by Definition 2.4, as the factors of $\mathfrak{b}$ are G_d-invariant ideals, being generated by G_d-semi-invariant polynomials. For such P,

$$a(P, X_d, S_0, \mathfrak{b}|_{X_d}) = a\Big(P, X_d, S_0 + \sum_j c_j \operatorname{div}(B_j)\Big) - \frac{\tau}{Q} \operatorname{mult}_P \mathfrak{a},$$

$$\operatorname{mult}_P \mathfrak{a} = Q \min\Big(\frac{\operatorname{mult}_P x_1}{a}, \frac{\operatorname{mult}_P y_1}{b}, \frac{\operatorname{mult}_P z_1}{v}\Big).$$

This proves (1).

We prove (2). Let $\pi_0 : \widetilde{S}_0 \to S_0$ be the strict transform of $S_0 \subset \mathbb{A}^3_{x_1,y_1,z_1}$ in the weighted blow-up of $\mathbb{A}^3$ with weights (a, b, v), and let $\widetilde{E} := \pi_0^{-1}(0)$. As G_d is diagonal, G_d acts on $\widetilde{S}_0$. Since $x_1 = y_1 = 0$ implies $z_1 = 0$ on S_0, $\widetilde{S}_0$ is covered by the strict transforms V_y and V_x of S_0 in the y_1-chart and in the x_1-chart. Since $x_1y_1 + z_1^2$ is homogeneous of weight $2v$ for the weights (a, b, v), a direct calculation shows

$$V_y = \mathbb{A}^2_{\lambda,p}/\mu_b(1, -v), \quad (x_1, y_1, z_1) = (-\lambda^a p^2, \lambda^b, \lambda^v p),$$

$$V_x = \mathbb{A}^2_{\lambda',p'}/\mu_a(1, -v), \quad (x_1, y_1, z_1) = (\lambda'^a, -\lambda'^b p'^2, \lambda'^v p'),$$

and $\widetilde{E}$ is the image of $(\lambda = 0)$ and of $(\lambda' = 0)$. The subgroup $\mu_v \subset G_d$ acts on S_0 by $(x_1, y_1, z_1) \to (\zeta^a x_1, \zeta^b y_1, z_1)$, and the action of μ_v lifts to $(\lambda, p) \to (\zeta\lambda, p)$ and $(\lambda', p') \to (\zeta\lambda', p')$.

Since $a \equiv -b$ modulo v, $\gcd(a, v) = 1$ and $x_1y_1 = -z_1^2$ on S_0, the μ_v-invariant functions on S_0 are generated by $U := x_1^v$, $V := y_1^v$, and $Z := z_1$, so $S_0/\mu_v = S_v$. Both V_y/μ_v and $\mathbb{A}^2_{p,q}/\mu_b(-v, v)$ are the quotient of $\mathbb{A}^2_{\lambda,p}$ by the group generated by $\mu_b(1, -v)$ and $\mu_v(1, 0)$, through $q = \lambda^v$, and both morphisms to S_v are given by

$$U = (-1)^v p^{2v} q^a, \qquad V = q^b, \qquad Z = pq.$$

Likewise, both V_x/μ_v and $\mathbb{A}^2_{p',q'}/\mu_a(-v, v)$ are the quotient of $\mathbb{A}^2_{\lambda',p'}$ by the group generated by $\mu_a(1, -v)$ and $\mu_v(1, 0)$, through $q' = \lambda'^v$, and both morphisms to S_v are given by

$$U = q'^a, \qquad V = (-1)^v p'^{2v} q'^b, \qquad Z = p'q'.$$

Since $\widetilde{S}_0/\mu_v \to S_v$ and $Y \to S_v$ are birational, the isomorphisms of charts above glue to an isomorphism $\widetilde{S}_0/\mu_v \cong Y$ over S_v, under which $\widetilde{E}/\mu_v = E$. Thus $H = G_d/\mu_v$ acts on Y preserving E, and $H \cong G/R$ by Construction-Proposition A.11(4). Since $(g, \sigma) \in \widetilde{G}$ multiplies x_1^v, y_1^v, and z_1 by the values at g of the characters of $x^v t^a$, $y^v t^b$, and zt, under the isomorphism $H \cong G/R$ the action of H on S_v, hence on Y, is that of Construction-Proposition 7.14.

On $\mathbb{A}^2_{\lambda,p}$, the ideal $\mathfrak{a}$ pulls back to (λ^Q), since $y_1^{Q/b} = \lambda^Q$ and λ^Q divides $x_1^{Q/a}$ and $z_1^{Q/v}$. As b divides Q and $\gcd(b, v) = 1$, λ^Q is a regular function on V_y, and the prime divisor $(\lambda = 0)$ has trivial inertia under $\mu_b(1, -v)$. Likewise, $\mathfrak{a}$ pulls back to (λ'^Q) on $\mathbb{A}^2_{\lambda',p'}$, λ'^Q is a regular function on V_x, and $(\lambda' = 0)$ has trivial inertia under $\mu_a(1, -v)$. Hence $\operatorname{mult}_P \mathfrak{a} = Q \operatorname{mult}_P \widetilde{E}$ for any prime divisor P over $\widetilde{S}_0$. Let $\widetilde{B}$ be the $\mathbb{R}$-divisor on $\widetilde{S}_0$ with

$$K_{\widetilde{S}_0} + \widetilde{B} = \pi_0^*\Big(K_{S_0} + \sum_j c_j \operatorname{div}(B_j|_{S_0})\Big) + \tau\widetilde{E}.$$

Then

$$a(P, S_0, \mathfrak{b}|_{S_0}) = a(P, \widetilde{S}_0, \widetilde{B})$$

for any prime divisor P over $\widetilde{S}_0$, and $\widetilde{B}$ is G_d-invariant.

The residue form $dy_1 \wedge dz_1/y_1$ generates the dualizing sheaf of S_0 and pulls back to $b\lambda^{v-1} d\lambda \wedge dp$ on $\mathbb{A}^2_{\lambda,p}$. Since f_j is homogeneous for $\mathbf{T}$ of weight κ_j, $f_j(x_1, y_1, z_1, 1)$ pulls back to

$\lambda^{-\kappa_j} f_j(-p^2, 1, p, \lambda^v)$. As $\tau = \sum_j c_j\kappa_j$, the crepant pullback of $(\widetilde{S}_0, \widetilde{B})$ to $\mathbb{A}^2_{\lambda,p}$ is

$$(1-v)\operatorname{div}(\lambda) + \sum_j c_j\Big(\operatorname{div}(f_j(-p^2,1,p,\lambda^v)) - \kappa_j \operatorname{div}(\lambda)\Big) + \tau\operatorname{div}(\lambda)$$
$$= (1-v)\operatorname{div}(\lambda) + \sum_j c_j \operatorname{div}(f_j(-p^2,1,p,\lambda^v)).$$

On the other hand, $\mathbb{A}^2_{p,q} \to \mathbb{A}^2_{p,q}/\mu_b(-v,v)$ is quasi-étale since $\gcd(b,v)=1$, and $\mathbb{A}^2_{\lambda,p} \to \mathbb{A}^2_{p,q}$, $q=\lambda^v$, is ramified only along $(\lambda=0)$, with index v. By the definition of D_j and the Hurwitz formula, the crepant pullback of $(Y, \sum_j c_jD_j)$ to $\mathbb{A}^2_{\lambda,p}$ is also the right side of the last display. Likewise, on $\mathbb{A}^2_{\lambda',p'}$ the crepant pullbacks of $(\widetilde{S}_0, \widetilde{B})$ and of $(Y, \sum_j c_jD_j)$ are both

$$(1-v)\operatorname{div}(\lambda') + \sum_j c_j \operatorname{div}(f_j(1,-p'^2,p',\lambda'^v)).$$

By [Liu26, Lemma 2.6(4)], the boundary of a crepant quotient by a finite group is determined by the crepant pullback of the quotient pair, so

$$(\widetilde{S}_0, \widetilde{B})/\mu_v = \Big(Y, \sum_j c_jD_j\Big).$$

On $\mathbb{A}^2_{\lambda,p}$, D_j pulls back to $\operatorname{div}(f_j(-p^2,1,p,\lambda^v))$, $f_j(x_1,y_1,z_1,1)$ pulls back to $\lambda^{-\kappa_j}f_j(-p^2,1,p,\lambda^v)$, and $\widetilde{E}$ pulls back to $\operatorname{div}(\lambda)$, and likewise on $\mathbb{A}^2_{\lambda',p'}$. Hence the quotient map $\widetilde{S}_0 \to Y$ pulls D_j back to

$$\pi_0^*\operatorname{div}(f_j(x_1,y_1,z_1,1)|_{S_0}) + \kappa_j\widetilde{E},$$

which is G_d-invariant, because by Construction-Proposition A.11(8) the function $f_j(x_1,y_1,z_1,1)$ is G_d-semi-invariant. Thus H preserves each D_j.

Let P be a prime divisor over $\widetilde{S}_0$, and let P' be the prime divisor over Y with P lying over P'. By [Liu26, Lemma 2.6(1)(2)(5)],

$$a(P, S_0, \mathfrak{b}|_{S_0}) = a(P, \widetilde{S}_0, \widetilde{B}) = |I_{\mu_v}(P)|a\Big(P', Y, \sum_j c_jD_j\Big),$$

$$|I_{G_d}(P)| = |I_{\mu_v}(P)| \cdot |I_H(P')|.$$

Since π_0 is proper with $\pi_0^{-1}(0) = \widetilde{E}$, and the preimage of E in $\widetilde{S}_0$ is $\widetilde{E}$, the preimages of the neighborhoods of 0 in S_0 and of the neighborhoods of E in Y are both fundamental systems of neighborhoods of $\widetilde{E}$ in $\widetilde{S}_0$. Moreover the center of P on S_0 is 0 if and only if the center of P' on Y is contained in E, and any prime divisor over Y is P' for some P [Liu26, Lemma 2.6(3)]. Thus $(S_0, \mathfrak{b}|_{S_0})$ is lc near 0 if and only if $(Y, \sum_j c_jD_j)$ is lc near E, and in this case

$$\operatorname{mld}((S_0 \ni 0, \mathfrak{b}|_{S_0})/G_d) = \inf_P \frac{a(P, S_0, \mathfrak{b}|_{S_0})}{|I_{G_d}(P)|} = \inf_{P'} \frac{a(P', Y, \sum_j c_jD_j)}{|I_H(P')|},$$

where P runs over the prime divisors over S_0 with center 0 and P' over the prime divisors over Y with center contained in E. □

In the proof of Proposition A.14, Lemma A.13 is applied to the principal blocks with all coefficients multiplied by a margin, and to the principal blocks together with one more homogeneous polynomial of $\mathbf{T}$-weight zero.

Lemma A.13. *Let a, b, v be positive integers with $a+b=2v$ and $\gcd(a,v)=1$, let $\mathbf{T} := \mathbf{T}_{(-a,-b,-v,v)}$, and let S_v, the toric partial resolution $Y \to S_v$, its exceptional curve E, and the charts $\mathbb{A}^2_{p,q}/\mu_b(-v,v)$ and $\mathbb{A}^2_{p',q'}/\mu_a(-v,v)$ of Y be as in Construction-Proposition 7.14. They depend only on a, b, and v. Let $c_1, \dots, c_s$ be positive real numbers and $f_1, \dots, f_s$ non-zero polynomials in x, y, z, t without monomials divisible by xy such that f_j is homogeneous for $\mathbf{T}$ of weight κ_j, and let $D_j \geq 0$ be the $\mathbb{Q}$-Cartier divisor on Y whose pullbacks to $\mathbb{A}^2_{p,q}$ and $\mathbb{A}^2_{p',q'}$*

are $\operatorname{div}(f_j(-p^2,1,p,q))$ *and* $\operatorname{div}(f_j(1,-p'^2,p',q'))$. *If* $\sum_j c_j\kappa_j \geq 0$ *and* $(Y,\sum_j c_jD_j)$ *is lc near* E, *then* $(W_N,\sum_j c_j\operatorname{div}(f_j))$ *is lc near* 0 *for any integer* $N\geq 3$.

Proof. Let $N\geq 3$ be an integer, let $D:=\sum_j c_j\operatorname{div}(f_j)$ on W_N, and let $\tau:=\sum_j c_j\kappa_j\geq 0$. We apply Construction-Proposition A.11 to the trivial group $G=\{1\}$, so that $R=\{1\}$, $r=1$, $d=v(N+2)$, and $H=G/R$ is trivial. Let $\check{f}_j:=f_j(x_1,y_1,z_1,1)$, which is the function $\check{h}_j$ of Construction-Proposition A.11(8) for $h_j=f_j$, so $\check{f}_j$ is G_d-semi-invariant. The polynomial $\check{f}_j$ is non-zero and has no monomial divisible by x_1y_1, while comparing the terms of highest degree in x_1 shows that any non-zero multiple of $x_1y_1+z_1^2$ has a monomial divisible by x_1y_1. Thus $\check{f}_j|_{S_0}\neq 0$. Let $Q:=abv$, let $\mathfrak{a}:=(x_1^{Q/a},y_1^{Q/b},z_1^{Q/v})$, and let $\mathfrak{b}:=\mathfrak{a}^{\tau/Q}\prod_j\check{f}_j^{c_j}$, where the factor $\mathfrak{a}^{\tau/Q}$ is omitted if $\tau=0$. Since $(Y,\sum_j c_jD_j)$ is lc near E, Lemma A.12(2), applied with $B_j:=\check{f}_j$, implies that $(S_0,\mathfrak{b}|_{S_0})$ is lc near 0. In particular

$$\beta_0:=\operatorname{mld}((S_0\ni 0,\mathfrak{b}|_{S_0})/G_d)\geq 0.$$

Let P_W be a prime divisor over W_N with center 0, and take $\widetilde{P}$, P, and $[\cdot]$ from Construction-Proposition A.11. Since G is trivial, $|I_G(P_W)|=1$, and $[u]=\operatorname{mult}_P u/|I_{G_d}(P)|$ for the coordinates u of X_d. By Construction-Proposition A.11(8), Lemma A.12(1), and Construction-Proposition A.11(7),

$$\begin{aligned}a(P_W,W_N,D)&=\frac{a(P,X_d,S_0+\sum_j c_j\operatorname{div}(\check{f}_j))}{|I_{G_d}(P)|}-\tau[\omega]\\&\geq\beta_0+\tau\Big(\min\Big(\frac{[x_1]}{a},\frac{[y_1]}{b},\frac{[z_1]}{v}\Big)-[\omega]\Big)\\&=\beta_0+\tau\min\Big(\frac{[x]}{a},\frac{[y]}{b},\frac{[z]}{v}\Big).\end{aligned}$$

As $\beta_0\geq 0$ and $\tau\geq 0$, $a(P_W,W_N,D)\geq 0$ for any prime divisor P_W over W_N with center 0.

Suppose that (W_N,D) is not lc near 0. Let $f:W'\to W_N$ be a log resolution of (W_N,D), and let $K_{W'}+B'=f^*(K_{W_N}+D)$. By [KM98, Lemma 2.30 and Corollary 2.31(3)], some component P_1 of B' with coefficient $c_1'>1$ satisfies $0\in f(P_1)$. Let $o'\in P_1\cap f^{-1}(0)$ be a closed point, and let u_1,u_2,u_3 be local coordinates at o' such that $P_1=(u_1=0)$ and $B'=\sum_{i=1}^3 c_i'\operatorname{div}(u_i)$ near o' for real numbers c_2',c_3'. For a positive integer k, let P_k' be the exceptional divisor of the weighted blow-up of W' at o' with weights $(k,1,1)$ in u_1,u_2,u_3. A direct calculation shows

$$a(P_k',W_N,D)=a(P_k',W',B')=k(1-c_1')+(1-c_2')+(1-c_3'),$$

which is negative for $k\gg 0$. Since the center of P_k' on W_N is $f(o')=0$, this contradicts the previous paragraph. Thus (W_N,D) is lc near 0. □

We now prove the lower bound by β_Y for non-negative mass, and the non-increase of α for positive mass, along interior canonical circles.

Proposition A.14. *Notation as in Construction-Proposition 7.14, and assume that* $\tau\geq 0$.

1. *For any real number* $\theta>1$ *there exists a positive integer* r_0 *depending only on* a,b,v,Γ,e_0, *and* θ *such that the following holds. If* $r\geq r_0$ *and* $(Y,\theta\Delta_Y)$ *is lc near* E, *then, for any* d, *any prime divisor over* W_N/G *with center the image of* 0 *has log discrepancy at least* β_Y *with respect to the crepant quotient of* (W_N,D) *by* G. *In particular,* $\beta_Y\leq\alpha\leq\beta_Y+C/r$, *where* C *is as in Construction-Proposition 7.14.*
2. *If* $\tau>0$, *then* α *cannot increase along* $\mathbf{T}$.

Proof. The proof is long, so we divide it into several steps. Since a unit h_j does not change D near 0, and multiplication by an invariant unit does not change $\operatorname{div}(h_j)$, we may assume that any h_j is a non-unit without monomials divisible by xy, and that $c_j>0$ for any j. Let γ be the least positive element of Γ, $M_0:=\lceil 2/\gamma\rceil$, and $M_1:=\max(a,b)$. Then $\max(a,b,v)=M_1$. Since (W_N,D) is klt near 0 and the blow-up of 0 has log discrepancy 2 with respect to $(W_N,0)$ (its

exceptional divisor is the quadric cone ($xy + z^2 = 0$) in $\mathbb{P}^3$, and the log discrepancy is $4 - 2$ by adjunction, cf. [KM98, Lemma 2.29]), $\sum_j c_j < 2$, so the number of indices j is less than M_0. By Lemma 5.3, after enlarging r_0 (depending only on a, b, v, Γ) we may assume that, for $r \geq r_0$, $|\kappa_j| \leq M_0 M_1$, and any monomial of $h_j - h_j^0$ has total degree at least $(r - M_0 M_1)/M_1$. Moreover, any two monomials of h_j have $\mathbf{T}$-weights congruent modulo r, so

$$h_j = \sum_{l \in \mathbb{Z}} h_j^{(l)}, \tag{A.99}$$

where $h_j^{(l)}$ is the sum of the terms of h_j of $\mathbf{T}$-weight $\kappa_j + lr$, and $h_j^{(0)} = h_j^0$. Since G is diagonal, any $h_j^{(l)}$ is G-semi-invariant with the character of h_j.

Step 1. In this step we write the crepant quotient of (W_N, D) by G on the cover X_d given by $s^v = t$.

We use Construction-Proposition A.11 and Lemma A.12 and their notation ($\widetilde{W}$, $\widetilde{G}$, $\widetilde{R}$, G_d, μ_v, X_d, S_0, x_1, y_1, z_1, ω, $[\cdot]$, $\check{h}_j$). In particular, $|G_d| = v[G : R] \leq ve_0$. A monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ of $\mathbf{T}$-weight κ is equal to $x_1^{i_1} y_1^{i_2} z_1^{i_3} s^\kappa$ on $\widetilde{W}$. By (A.99),

$$\begin{gathered} \check{h}_j = \sum_{l \in \mathbb{Z}} \omega^l \Phi_j^l(x_1, y_1, z_1), \qquad \Phi_j^l(x_1, y_1, z_1) := s^{-\kappa_j - lr} h_j^{(l)}, \\ \Phi_j^0(x_1, y_1, z_1) = h_j^0(x_1, y_1, z_1, 1), \end{gathered} \tag{A.100}$$

where Φ_j^l is a polynomial without monomials divisible by $x_1 y_1$.

Let P_W be a prime divisor over W_N with center 0, and let P be the prime divisor over X_d given by P_W in Construction-Proposition A.11. We say that P corresponds to P_W. Since $\operatorname{div}(\omega) = S_0$ on X_d, Construction-Proposition A.11(6) and (8) imply

$$\frac{a(P_W, W_N, D)}{|I_G(P_W)|} = \frac{a(P, X_d, 0) - \operatorname{mult}_P \omega - \sum_j c_j \operatorname{mult}_P \check{h}_j - \frac{\tau}{r} \operatorname{mult}_P \omega}{|I_{G_d}(P)|}. \tag{A.101}$$

For a prime divisor P over X_d, let

$$m_P := \min\left(\frac{\operatorname{mult}_P x_1}{a}, \frac{\operatorname{mult}_P y_1}{b}, \frac{\operatorname{mult}_P z_1}{v}\right).$$

By Construction-Proposition A.11(5), the center of P on X_d is 0. By Construction-Proposition A.11(6) and (7), and since $[t] = v[s]$,

$$[x_1] = [x] + \frac{a}{v}[t], \quad [y_1] = [y] + \frac{b}{v}[t], \quad [z_1] = [z] + [t], \quad [\omega] = \frac{r}{v}[t], \tag{A.102}$$

$m_P / |I_{G_d}(P)| = \min([x]/a, [y]/b, [z]/v) + [t]/v$, and

$$\operatorname{mult}_P \omega > 0 \quad \text{and} \quad m_P > \frac{\operatorname{mult}_P \omega}{r}. \tag{A.103}$$

Step 2. In this step we introduce a monomial ideal $\mathfrak{a}$ on X_d and show that $\mathfrak{a}$ and the coordinate functions define lc pairs on S_0.

Let Q be a positive integer divisible by a, b, and v, and let

$$\mathfrak{a} := (x_1^{Q/a}, y_1^{Q/b}, z_1^{Q/v}) \subset \mathbb{C}[x_1, y_1, z_1, \omega].$$

Then $\operatorname{mult}_P \mathfrak{a} = Q m_P$ for any prime divisor P over X_d with center 0. The double cover $\mathbb{A}^2_{\xi_1, \xi_2} \to S_0$, $(\xi_1, \xi_2) \to (\xi_1^2, -\xi_2^2, \xi_1 \xi_2)$, the quotient by $\mu_2(1, 1)$, is quasi-étale, and the pullbacks of $\frac{1}{2}\operatorname{div}(x_1)$, $\frac{1}{2}\operatorname{div}(y_1)$, and $\operatorname{div}(z_1)$ are $\operatorname{div}(\xi_1)$, $\operatorname{div}(\xi_2)$, and $\operatorname{div}(\xi_1 \xi_2)$. Hence $(S_0, \frac{1}{2}\operatorname{div}(x_1))$, $(S_0, \frac{1}{2}\operatorname{div}(y_1))$, and $(S_0, \operatorname{div}(z_1))$ are lc near 0. Since $z_1^{Q/v} \in \mathfrak{a}$, we have $\frac{v}{Q} \operatorname{mult}_P \mathfrak{a} \leq \operatorname{mult}_P z_1$ for any prime divisor P over S_0, so $(S_0, \mathfrak{a}^{v/Q})$ is lc near 0 because $(S_0, \operatorname{div}(z_1))$ is. Thus

$$(S_0, \mathfrak{a}^{v/Q}), \quad \left(S_0, \frac{1}{2}\operatorname{div}(x_1)\right), \quad \left(S_0, \frac{1}{2}\operatorname{div}(y_1)\right), \quad (S_0, \operatorname{div}(z_1)) \quad \text{are lc near 0.} \tag{A.104}$$

Step 3. In this step we prove the lower bound by β_Y at the prime divisors P such that $\operatorname{mult}_P \check{h}_j = \operatorname{mult}_P B_j$ for polynomials B_j with $B_j|_{S_0} = \Phi_j^0|_{S_0}$ (Claim A.15). We assume that (Y, Δ_Y) is lc near E. This holds if $(Y, \theta\Delta_Y)$ is lc near E for some $\theta > 1$, as assumed in (1).

Claim A.15. *Let $B_j \in \mathbb{C}[x_1, y_1, z_1, \omega]$ be G_d-semi-invariant polynomials with $B_j|_{S_0} = \Phi_j^0|_{S_0}$, and let P be a prime divisor over X_d with center 0. Then*

$$\frac{a\big(P, X_d, S_0 + \sum_j c_j \operatorname{div}(B_j)\big) - \tau m_P}{|I_{G_d}(P)|} \geq \beta_Y.$$

If moreover P corresponds to a prime divisor P_W over W_N with center 0 and $\operatorname{mult}_P \check{h}_j = \operatorname{mult}_P B_j$ for every j, then $a(P_W, W_N, D)/|I_G(P_W)| \geq \beta_Y$.

Proof. Since Φ_j^0 is a non-zero polynomial without monomials divisible by $x_1 y_1$, $B_j|_{S_0} = \Phi_j^0|_{S_0} \neq 0$. Let $\mathfrak{b} := \mathfrak{a}^{\tau/Q} \prod_j B_j^{c_j}$, where the factor $\mathfrak{a}^{\tau/Q}$ is omitted if $\tau = 0$. The polynomials $f_j := h_j^0$ are G-semi-invariant, have no monomials divisible by xy, and are homogeneous for $\mathbf{T}$ of weight κ_j, $\tau = \sum_j c_j \kappa_j$, and $B_j|_{S_0} = h_j^0(x_1, y_1, z_1, 1)|_{S_0}$. For these f_j, we have $D_j = D_{Y,j}$ in Lemma A.12(2), so $\sum_j c_j D_j = \Delta_Y$. Since (Y, Δ_Y) is lc near E, Lemma A.12(2) shows that $(S_0, \mathfrak{b}|_{S_0})$ is lc near 0 and that $\operatorname{mld}((S_0 \ni 0, \mathfrak{b}|_{S_0})/G_d) = \beta_Y$, since $a(P', Y, \Delta_Y)/|I_H(P')|$ is the log discrepancy of the prime divisor over Y/H under P' with respect to $(Y, \Delta_Y)/H$ [Liu26, Lemma 2.6(2)(3)]. Thus the first inequality follows from Lemma A.12(1). Under the second set of assumptions, $\operatorname{div}(\omega) = S_0$ on X_d, so by (A.101) and the first inequality,

$$\begin{aligned}\frac{a(P_W, W_N, D)}{|I_G(P_W)|} &= \frac{a\big(P, X_d, S_0 + \sum_j c_j \operatorname{div}(B_j)\big) - \frac{\tau}{r}\operatorname{mult}_P \omega}{|I_{G_d}(P)|} \\ &\geq \beta_Y + \frac{\tau\big(m_P - \frac{1}{r}\operatorname{mult}_P \omega\big)}{|I_{G_d}(P)|} \geq \beta_Y,\end{aligned}$$

where the last inequality follows from (A.103) and $\tau \geq 0$. □

Proof of Proposition A.14 continued. Moreover $\beta_Y \leq 1$: the prime divisor E/H has log discrepancy $(1 - \operatorname{mult}_E \Delta_Y)/|I_H(E)| \leq 1$ with respect to $(Y, \Delta_Y)/H$. Hence it suffices to bound $a(P_W, W_N, D)/|I_G(P_W)|$ from below for the prime divisors P_W over W_N with center 0 such that $a(P_W, W_N, D)/|I_G(P_W)| \leq 1$.

Step 4. In this step we assume the margin θ of (1), show that $(W_N, \theta D)$ is lc near 0, and bound the multiplicities of the coordinates. Assume that $(Y, \theta\Delta_Y)$ is lc near E.

Let $D^0 := \sum_j c_j \operatorname{div}(h_j^0)$. Since $\sum_j \theta c_j \kappa_j = \theta\tau \geq 0$, Lemma A.13 for the labels θc_j and the polynomials h_j^0 implies that $(W_N, \theta D^0)$ is lc near 0. By inversion of adjunction [Kaw07, Theorem], $(\mathbb{A}^4, W_N + \theta D^0_{\mathbb{A}^4})$ is lc near 0, where $D^0_{\mathbb{A}^4} := \sum_j c_j \operatorname{div}_{\mathbb{A}^4}(h_j^0)$, and the pullback of $(\mathbb{A}^4, W_N + \theta D^0_{\mathbb{A}^4})$ to $\operatorname{Spec} \mathbb{C}[[x, y, z, t]]$ is lc near 0 [dFEM11, Proposition 2.11]. Let L be the maximum of the integers of Lemma 4.14 for $n = 4$, the DCC set $\Lambda := \{1\} \cup \{\theta c \mid c \in \Gamma, c > 0\}$, and at most M_0 functions. We assume that r is so large that $(r - M_0 M_1)/M_1 \geq L$. Then $h_j - h_j^0 \in (x, y, z, t)^L$ for any j, and Lemma 4.14 for the tuples $(xy + z^2 + t^N, h_1^0, h_2^0, \dots)$ and $(xy + z^2 + t^N, h_1, h_2, \dots)$ with the exponents $(1, \theta c_1, \theta c_2, \dots)$ implies that $(\mathbb{A}^4, W_N + \theta \sum_j c_j \operatorname{div}_{\mathbb{A}^4}(h_j))$ is lc near 0. By inversion of adjunction [Kaw07, Theorem] again, $(W_N, \theta D)$ is lc near 0.

Let P_W be a prime divisor over W_N with center 0 and $a(P_W, W_N, D)/|I_G(P_W)| \leq 1$, and let P be a prime divisor over X_d corresponding to P_W as in **Step 1**. Let

$$M_2 := \frac{1}{\gamma(\theta - 1)}, \qquad M_3 := \frac{\theta}{\theta - 1}, \qquad M_4 := \frac{1}{M_1 v e_0}.$$

Since $a(P_W, W_N, \theta D) \geq 0$ and $a(P_W, W_N, D) \leq |I_G(P_W)|$, we have $(\theta - 1)\sum_j c_j [h_j] \leq 1$ and $a(P_W, W_N, 0)/|I_G(P_W)| \leq M_3$. Since $a(P_W, W_N, \theta D^0) \geq 0$, we have $\theta \sum_j c_j [h_j^0] \leq M_3$. Hence $[h_j] \leq M_2$ and $[h_j^0] \leq M_2$ for any j. By (A.104) and Theorem 3.3(2) with the trivial subgroup,

the minimal log discrepancies at 0 of (X_d, S_0) with the $\mathbb{R}$-ideals $\mathfrak{a}^{v/Q}$, $(x_1)^{1/2}$, $(y_1)^{1/2}$, and (z_1) are non-negative. Thus

$$a(P, X_d, 0) - \operatorname{mult}_P \omega \geq \max\Big(vm_P,\ \frac{\operatorname{mult}_P x_1}{2},\ \frac{\operatorname{mult}_P y_1}{2},\ \operatorname{mult}_P z_1\Big).$$

By (A.101) for the boundary 0,

$$\frac{a(P, X_d, 0) - \operatorname{mult}_P \omega}{|I_{G_d}(P)|} = \frac{a(P_W, W_N, 0)}{|I_G(P_W)|} \leq M_3.$$

Since $\operatorname{mult}_P x_1$, $\operatorname{mult}_P y_1$, and $\operatorname{mult}_P z_1$ are positive integers and $|I_{G_d}(P)| \leq |G_d| \leq ve_0$, we have $m_P/|I_{G_d}(P)| \geq M_4$. Together with (A.103) and (A.102), we obtain

$$\text{(A.105)} \qquad \begin{gathered} M_4 \leq \min\Big(\frac{[x]}{a}, \frac{[y]}{b}, \frac{[z]}{v}\Big) + \frac{[t]}{v} \leq \frac{M_3}{v}, \qquad [x], [y] \leq 2M_3, \qquad [z], [t] \leq M_3, \\ [h_j] \leq M_2, \qquad [h_j^0] \leq M_2. \end{gathered}$$

Step 5. In this step we separate two cases by the monomials of $h_j - h_j^0$, and we prove the lower bound when $[t]$ is small.

Let $\epsilon_r := M_2/(r - M_0M_1)$. If $\operatorname{mult}_{P_W} h_j = \operatorname{mult}_{P_W} h_j^0$ for every j, then, since $\check{h}_j = s^{-\kappa_j} h_j$ and $\Phi_j^0 = s^{-\kappa_j} h_j^0$ on $\widetilde{W}$ ((A.100) with $l = 0$), $[\check{h}_j] = [\Phi_j^0]$, that is, $\operatorname{mult}_P \check{h}_j = \operatorname{mult}_P \Phi_j^0$ by Construction-Proposition A.11(6), and Claim A.15 for $B_j := \Phi_j^0$ implies $a(P_W, W_N, D)/|I_G(P_W)| \geq \beta_Y$. Otherwise there exists j with $\operatorname{mult}_{P_W} h_j \neq \operatorname{mult}_{P_W} h_j^0$, so

$$[h_j - h_j^0] = \min([h_j], [h_j^0]) \leq M_2,$$

and there exists a monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of $h_j - h_j^0$ with $i_1[x] + i_2[y] + i_3[z] + i_4[t] \leq M_2$, and the $\mathbf{T}$-weight of $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ is $\kappa_j + lr$ for an integer $l \neq 0$. If $l > 0$, then $vi_4 \geq r + \kappa_j \geq r - M_0M_1$. If $l < 0$, then $ai_1 + bi_2 + vi_3 = vi_4 - \kappa_j - lr \geq r - M_0M_1$. Thus

$$\text{(A.106)} \qquad \frac{[t]}{v} \leq \epsilon_r \qquad \text{or} \qquad \min\Big(\frac{[x]}{a}, \frac{[y]}{b}, \frac{[z]}{v}\Big) \leq \epsilon_r.$$

We assume that r is so large that $\epsilon_r \leq M_4/4$.

Assume that $[t]/v \leq \epsilon_r$. By (A.105), $\min([x]/a, [y]/b, [z]/v) \geq M_4/2$, so any monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of $h_j^{(l)}$ with $l < 0$ satisfies

$$i_1[x] + i_2[y] + i_3[z] + i_4[t] \geq (ai_1 + bi_2 + vi_3)\frac{M_4}{2} \geq \frac{(r - M_0M_1)M_4}{2} > M_2 \geq [h_j].$$

Hence $h_j^+ := \sum_{l \geq 0} h_j^{(l)}$ satisfies $[h_j^+] = [h_j]$. Let $B_j := \sum_{l \geq 0} \omega^l \Phi_j^l$. Then B_j is a polynomial, $h_j^+ = s^{\kappa_j} B_j$ on $\widetilde{W}$, so B_j is G_d-semi-invariant, $B_j|_{S_0} = \Phi_j^0|_{S_0}$, and $\operatorname{mult}_P B_j = \operatorname{mult}_P \check{h}_j$. By Claim A.15, $a(P_W, W_N, D)/|I_G(P_W)| \geq \beta_Y$.

Step 6. In this step we prove the lower bound when $\min([x]/a, [y]/b, [z]/v) \leq \epsilon_r$, which completes the proof of (1).

By (A.105), $[t]/v \geq M_4/2$. Any monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of $h_j^{(l)}$ with $l > 0$ satisfies $i_4[t] \geq ((r - M_0M_1)/v) \cdot (vM_4/2) > M_2$, so $h_j^- := \sum_{l \leq 0} h_j^{(l)}$ satisfies $[h_j^-] = [h_j]$. On W_N, $xy + z^2 = -t^N$, and $N = rd/v - 2 \geq r/v - 2$, so

$$[xy + z^2] = N[t] \geq \Big(\frac{r}{v} - 2\Big)\frac{vM_4}{2}.$$

We assume that r is so large that the right hand side is greater than $4M_3$. If $[x] + [y] \neq 2[z]$, then $[xy + z^2] = \min([x] + [y], 2[z]) \leq 4M_3$ by (A.105), a contradiction. Thus $[x] + [y] = 2[z]$, and

$$\frac{[z]}{v} = \frac{a \cdot \frac{[x]}{a} + b \cdot \frac{[y]}{b}}{a + b} \geq \min\Big(\frac{[x]}{a}, \frac{[y]}{b}\Big).$$

Exchanging (a,x) with (b,y), and the charts $\mathbb{A}^2_{p,q}/\mu_b(-v,v)$ and $\mathbb{A}^2_{p',q'}/\mu_a(-v,v)$ of Y, if necessary, we may assume that $[y]/b \le [x]/a$, so $[y]/b \le \epsilon_r$.

Let G'' be the group of pairs (g,ζ) with $g \in G$ and $\zeta^b = g_y$, acting diagonally on $\mathbb{A}^3_{\eta,p,q}$ by $(\eta,p,q) \to (\zeta\eta, g_z\zeta^{-v}p, g_t\zeta^v q)$. Since $g_x g_y = g_z^2 = g_t^N$ for $g \in G$, the rational map

$$\mathbb{A}^3_{\eta,p,q} \dashrightarrow W_N, \qquad (\eta,p,q) \to \Big(\eta^a\big(-p^2 - q^N\eta^{-v(N+2)}\big),\ \eta^b,\ \eta^v p,\ \eta^{-v}q\Big),$$

is G''-equivariant, and it is generically the quotient by $\{(1,\zeta) \mid \zeta^b = 1\}$. Hence $\mathbb{A}^3_{\eta,p,q}/G''$ is birational to W_N/G. The lift

$$\{((\zeta^a,\zeta^b,\zeta^v,\zeta^{-v}),\zeta) \mid \zeta^r = 1\} \cong \mu_r$$

of R acts on η only, and the quotient is $\mathbb{A}^3_{u,p,q}$ with $u := \eta^r$. Let $G_\eta := G''/\mu_r$, acting diagonally on $\mathbb{A}^3_{u,p,q}$. If (g,ζ) acts trivially on p and q, then $g_z = \zeta^v$, $g_t = \zeta^{-v}$, $g_y = \zeta^b$, and $g_x = g_z^2/g_y = \zeta^a$, so $g \in R$ and $(g,\zeta) \in \mu_r$. Thus G_η acts faithfully on the plane $(u=0) = \mathbb{A}^2_{p,q}$. The subgroup $\{(1,\zeta) \mid \zeta^b = 1\}$ acts on $\mathbb{A}^2_{p,q}$ as $\mu_b(-v,v)$, and the functions $(-1)^v p^{2v} q^a$, q^b, and pq have the characters of $x^v t^a$, $y^v t^b$, and zt. Hence the action of G_η on $\mathbb{A}^2_{p,q}$ lifts the action of H on the chart $\mathbb{A}^2_{p,q}/\mu_b(-v,v)$ of Y, and $\mathbb{A}^2_{p,q}/G_\eta = (\mathbb{A}^2_{p,q}/\mu_b(-v,v))/H$.

Let $x' := -p^2 - q^N/u^d$, so that $x = \eta^a x'$. A monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of $\mathbf{T}$-weight $\kappa_j + lr$ is equal to $\eta^{-\kappa_j}u^{-l}x'^{i_1}p^{i_3}q^{i_4}$. Hence

$$\eta^{\kappa_j}h_j^- = B_j^-(u,x',p,q) := \sum_{l\le 0} u^{-l}h_j^{(l)}(x',1,p,q), \qquad h_j^\eta := B_j^-(u,-p^2,p,q) \in \mathbb{C}[u,p,q],$$

where h_j^η is G_η-semi-invariant, since the characters of x' and $-p^2$ are equal. Moreover $h_j^\eta(0,p,q) = h_j^0(-p^2,1,p,q)$ is the equation of the pullback of $D_{Y,j}$ to $\mathbb{A}^2_{p,q}$. The form $dy\wedge dz\wedge dt/y$ generates the dualizing sheaf of W_N near 0, and a direct calculation shows that it is equal to $\frac{b}{r}du\wedge dp\wedge dq/u$. Since $\operatorname{div}(\eta) = \frac{1}{r}\operatorname{div}(u)$, the crepant quotient of (W_N,D) by G is crepant birational to the crepant quotient of

$$\Big(\mathbb{A}^3_{u,p,q},\ \Big(1-\frac{\tau}{r}\Big)\operatorname{div}(u) + \sum_j c_j \operatorname{div}(\eta^{\kappa_j}h_j)\Big)$$

by G_η. Let P_η be a prime divisor over $\mathbb{A}^3_{u,p,q}$ lying over the prime divisor over $\mathbb{A}^3_{u,p,q}/G_\eta$ that corresponds to the prime divisor over W_N/G under P_W through the birational map $\mathbb{A}^3_{u,p,q}/G_\eta \dashrightarrow W_N/G$ above. For a semi-invariant rational function f on W_N, $[f] = \operatorname{mult}_{P_W} f/|I_G(P_W)|$ by Construction-Proposition A.11(6), and if f^l is G-invariant for a positive integer l, then $[f]$ is $\frac{1}{l}$ times the multiplicity of f^l along the prime divisor over W_N/G under P_W [Liu26, Lemma 2.6(1)]. Hence $[f] = \operatorname{mult}_{P_\eta} f/|I_{G_\eta}(P_\eta)|$ when f is regarded as a function on $\mathbb{A}^3_{u,p,q}$, and we set $[f] := \operatorname{mult}_{P_\eta} f/|I_{G_\eta}(P_\eta)|$ also for the semi-invariant rational functions f on $\mathbb{A}^3_{u,p,q}$. Since $u^b = y^r$, $p^b = z^b/y^v$, $q^b = t^b y^v$, and $x'^b = x^b/y^a$,

$$\begin{aligned} &[u] = \frac{r}{b}[y] \le r\epsilon_r, \quad [p] = [z] - \frac{v}{b}[y] = \frac{1}{2}\Big([x] - \frac{a}{b}[y]\Big) \ge 0,\\ &[q] = [t] + \frac{v}{b}[y] \ge \frac{vM_4}{2}, \quad [x'] = 2[p] \ge 0. \end{aligned} \tag{A.107}$$

Thus the center of P_η on $\mathbb{A}^3_{u,p,q}$ is contained in the curve $C_\eta := (u = q = 0)$.

We compare $\eta^{\kappa_j}h_j$ with h_j^η at P_η. Since $[h_j - h_j^-] > [h_j]$, we have

$$[\eta^{\kappa_j}h_j] = [B_j^-] = \frac{\kappa_j}{b}[y] + [h_j] \le M_0M_1\epsilon_r + M_2.$$

The polynomial $B_j^-(u,x',p,q) - h_j^\eta$ is the product of $x' + p^2 = -q^N/u^d$ and a polynomial in u,x',p,q, whose bracket is non-negative by (A.107). Since $rd = v(N+2)$, $d \ge 1$, and

$\frac{r}{v}[q]-[u]=\frac{r}{v}[t]$,

$$[B_j^- - h_j^\eta] \geq N[q]-d[u]=d\cdot\frac{r}{v}[t]-2[q]\geq\frac{rM_4}{2}-2(M_3+v\epsilon_r).$$

We assume that r is so large that the right hand side is greater than $M_0M_1\epsilon_r+M_2$. Then $[h_j^\eta]=[\eta^{\kappa_j}h_j]$ for any j. Since $\tau\geq 0$, we obtain

$$\begin{aligned}\frac{a(P_W,W_N,D)}{|I_G(P_W)|}&=\frac{a\big(P_\eta,\mathbb{A}^3_{u,p,q},(1-\frac{\tau}{r})\operatorname{div}(u)+\sum_j c_j\operatorname{div}(h_j^\eta)\big)}{|I_{G_\eta}(P_\eta)|}\\&\geq\frac{a\big(P_\eta,\mathbb{A}^3_{u,p,q},\operatorname{div}(u)+\sum_j c_j\operatorname{div}(h_j^\eta)\big)}{|I_{G_\eta}(P_\eta)|}.\end{aligned}$$

The pair $(\mathbb{A}^2_{p,q},\sum_j c_j\operatorname{div}(h_j^\eta(0,p,q)))$ is the pullback of (Y,Δ_Y) under the quasi-étale morphism $\mathbb{A}^2_{p,q}\to\mathbb{A}^2_{p,q}/\mu_b(-v,v)$ (note $\gcd(b,v)=1$), so it is lc near $(q=0)$, and the crepant quotient of $(\mathbb{A}^2_{p,q},\sum_j c_j\operatorname{div}(h_j^\eta(0,p,q)))$ by G_η is the crepant quotient of (Y,Δ_Y) by H over the chart. Assume first that the center of P_η is a closed point $o=(0,p_0,0)$. Let $K:=\operatorname{Stab}_{G_\eta}(o)$. Then $I_{G_\eta}(P_\eta)=I_K(P_\eta)$, and if $p_0\neq 0$, then K acts trivially on p, so after the translation $p\to p-p_0$ the group K still acts diagonally and o becomes the origin. By Lemma 5.5 for $\mathbb{A}^3_{u,p,q}$, the group K, the function u with $(u=0)$ in place of S_0, the $\mathbb{R}$-ideal $\prod_j(h_j^\eta)^{c_j}$, and the coefficient 1 of $(u=0)$ in place of the number b of Lemma 5.5,

$$\frac{a\big(P_\eta,\mathbb{A}^3_{u,p,q},\operatorname{div}(u)+\sum_j c_j\operatorname{div}(h_j^\eta)\big)}{|I_K(P_\eta)|}\geq\operatorname{mld}\Big(\Big(\mathbb{A}^2_{p,q}\ni o,\sum_j c_j\operatorname{div}(h_j^\eta(0,p,q))\Big)\Big/K\Big)\geq\beta_Y,$$

since any prime divisor P' over $\mathbb{A}^2_{p,q}$ with center o corresponds to a prime divisor P_Y over Y with center contained in E, and the quotient log discrepancies of P' and P_Y are equal. Assume next that the center of P_η is C_η. The curve $C_\eta\ni 0$ is G_η-invariant and contained in $(u=0)$, and the second part of Lemma 5.5 for G_η bounds the left hand side from below by the quotient log discrepancy of the prime divisor $(q=0)$ of $\mathbb{A}^2_{p,q}$, which is the log discrepancy of E/H with respect to $(Y,\Delta_Y)/H$, hence at least β_Y.

We have proved that $a(P_W,W_N,D)/|I_G(P_W)|\geq\beta_Y$ for any prime divisor P_W over W_N with center 0 once r is at least an integer r_0 determined by the conditions on r in **Steps 4–6**, Lemma 5.3, and Construction-Proposition 7.14. The conditions depend only on a,b,v,Γ,e_0, and θ. Thus $\alpha\geq\beta_Y$, and $\alpha\leq\beta_Y+C/r$ by Construction-Proposition 7.14. This proves (1).

Step 7. In this step we start the proof of (2): we take a sequence contradicting (2) and fix the discrete data of the sequence.

Assume that there exists a sequence of data as in Set-up 7.3, indexed by $i\geq 1$, with Γ, e_0, and $w=(-a,-b,-v,v)$ fixed, $r_i\to+\infty$, and $\tau_i>0$, whose values satisfy $0<\alpha_1<\alpha_2<\cdots$. We add the index i to the notation: N_i, d_i, G_i, H_i, $D_i=\sum_j c_{ij}\operatorname{div}(h_{ij})$, κ_{ij}, $\Delta_i:=\Delta_{Y,i}=\sum_j c_{ij}D_{Y,ij}$, and $\beta_i:=\beta_{Y,i}$. The surface Y, the curve E, and the charts of Y depend only on a,b,v. As at the beginning of the proof, we may assume that the h_{ij} are non-units without monomials divisible by xy and that $c_{ij}>0$. Passing to a subsequence, we may assume that r_i is at least the integer r_0 of Construction-Proposition 7.14, so $\alpha_i\leq\beta_i+C/r_i$. Let $\alpha_\infty:=\lim_i\alpha_i$. Then $0<\alpha_i<\alpha_\infty$ for any i.

Let $\nu_{ij}:=\operatorname{mult}_E D_{Y,ij}$, a non-negative integer since the quotient map of $\mathbb{A}^2_{p,q}$ by $\mu_b(-v,v)$ is quasi-étale, and let $\varrho_i:=\operatorname{mult}_E\Delta_i=\sum_j c_{ij}\nu_{ij}$. Since $\beta_i\leq(1-\varrho_i)/|I_{H_i}(E)|\leq 1-\varrho_i$ and $\beta_i\geq\alpha_i-C/r_i>\alpha_1-C/r_i\geq\alpha_1/2$ for $i\gg 0$, we have $\varrho_i\leq 1-\alpha_1/2$ for $i\gg 0$. Passing to a subsequence, we may assume the following.

- The number n_h of indices j $(n_h<M_0)$, the integers $\kappa_j:=\kappa_{ij}$ $(|\kappa_j|\leq M_0M_1)$, and the integers $\nu_j:=\nu_{ij}$ $(\nu_j\leq 1/\gamma)$ do not depend on i.

- The image of G_i in $(\mathbb{C}^*)^3$ under the characters of $x^v t^a$, $y^v t^b$, and zt, which is the image of H_i acting on Y, is a fixed subgroup H of order at most e_0; there exist only finitely many subgroups of $(\mathbb{C}^*)^3$ of order at most e_0. Let $\chi_y : (\mathbb{C}^*)^2 \to (\mathbb{C}^*)^3$ be the homomorphism given by the characters of $(-1)^v p^{2v} q^a$, q^b, and pq on $\mathbb{A}^2_{p,q}$, and $K_y := \chi_y^{-1}(H)$. The kernel of χ_y is $\mu_b(-v, v)$, and K_y is the image of the group G_η of **Step 6** acting on $\mathbb{A}^2_{p,q}$. Let $\chi_x : (\mathbb{C}^*)^2 \to (\mathbb{C}^*)^3$ be the homomorphism given by the characters of q'^a, $(-1)^v p'^{2v} q'^b$, and $p'q'$ on $\mathbb{A}^2_{p',q'}$, and $K_x := \chi_x^{-1}(H)$.
- The characters of $h^0_{ij}(-p^2, 1, p, q)$ under K_y and of $h^0_{ij}(1, -p'^2, p', q')$ under K_x do not depend on i; there exist only finitely many characters of the bounded groups K_y and K_x.
- Since Γ is a DCC set, the sequence $(c_{ij})_i$ is non-decreasing with limit $c^*_j \in [\gamma, 1]$ for any j.
- Since β_i belongs to an ACC set (Construction-Proposition 7.14), the sequence $(\beta_i)_i$ is non-increasing with limit β.

Let $\tau^* := \sum_j c^*_j \kappa_j = \lim_i \tau_i \geq 0$ and $\varrho^* := \sum_j c^*_j \nu_j = \lim_i \varrho_i \leq 1 - \alpha_1/2$. Since $\alpha_i \leq \beta_i + C/r_i$, we have $0 < \alpha_\infty \leq \beta \leq \beta_i$.

Step 8. In this step we replace the coefficients c_{ij} with c^*_j. Let $\Delta^*_i := \sum_j c^*_j D_{Y,ij} \geq \Delta_i$.

First, we show that (Y, Δ^*_i) is lc near E for $i \gg 0$. It suffices to check that the pullbacks of Δ^*_i to the charts $\mathbb{A}^2_{p,q}$ and $\mathbb{A}^2_{p',q'}$ are lc near $(q = 0)$ and $(q' = 0)$. The lc thresholds of the pullbacks at the points of $(q = 0)$ and $(q' = 0)$ are at least $\min_j c_{ij}/c^*_j$, which converges to 1. The lc thresholds of $\mathbb{R}$-divisors on smooth surfaces with coefficients in the DCC set $\{c^*_j\}_+ \backslash \{0\}$ satisfy the ACC [HMX14, Theorem 1.1], so they are at least 1 for $i \gg 0$.

Let β^*_i be the minimum of the log discrepancies of $(Y, \Delta^*_i)/H$ over the prime divisors with center contained in E/H. Let $(Y/H, \Delta_{Y/H,i})$ and $(Y/H, \Delta^*_{Y/H,i})$ be the crepant quotients of (Y, Δ_i) and (Y, Δ^*_i) by H, and let $\bar{D}_{ij}$ be the $\mathbb{R}$-divisor on Y/H whose pullback to Y is $D_{Y,ij}$. The coefficients of $\Delta_{Y/H,i}$ and $\Delta^*_{Y/H,i}$ belong to the DCC set $\Psi(\Gamma'_+ \cap [0, 1])$ of [Liu26, Definition 2.9] for the DCC set $\Gamma' := \Gamma \cup \{c^*_j\}_j$, since $\Delta_i \in \Gamma'_+ \cap [0, 1]$ and $\Delta^*_i \in \Gamma'_+ \cap [0, 1]$ near E: the $D_{Y,ij}$ have integral multiplicities along prime divisors (the quotient maps of the charts are quasi-étale), and the pairs are lc near E. The set $\Psi(\Gamma'_+ \cap [0, 1])$ is $D(\Gamma')$ in the notation of [HMX14, Section 3.4], so it is a DCC set [HMX14, Proposition 3.4.1(3)]. Let l_0 be the integer of Theorem 2.9(2) for $\Psi(\Gamma'_+ \cap [0, 1])$, and $l'_0 := \max(l_0, 1)$. For any closed point $y \in E/H$, Theorem 2.9(2) for the lc germ $(Y/H \ni y, \Delta^*_{Y/H,i})$ implies that $\beta^*_i = a(P_i, Y/H, \Delta^*_{Y/H,i})$ for a prime divisor P_i over Y/H which is E/H or satisfies $a(P_i, Y/H, 0) \leq l_0$. In both cases $a(P_i, Y/H, 0) \leq l'_0$, and $a(P_i, Y/H, \Delta^*_{Y/H,i}) \geq 0$ implies $\operatorname{mult}_{P_i} \bar{D}_{ij} \leq l'_0/\gamma$. Hence

$$\beta_i \geq \beta^*_i = a(P_i, Y/H, \Delta_{Y/H,i}) - \sum_j (c^*_j - c_{ij}) \operatorname{mult}_{P_i} \bar{D}_{ij} \geq \beta_i - \frac{l'_0}{\gamma} \sum_j (c^*_j - c_{ij}),$$

so $\beta^*_i \to \beta$. Moreover $\beta^*_i = a(P_i, Y/H, \Delta^*_{Y/H,i})$ is determined by $a(P_i, Y/H, 0)$, the multiplicities of the branch divisor and of the $\bar{D}_{ij}$ along P_i, and the fixed numbers c^*_j. Since Y has cyclic quotient singularities of orders a and b and $|H| \leq e_0$, the numbers $a(P_i, Y/H, 0)$, $\operatorname{mult}_{P_i} \bar{D}_{ij}$, and the multiplicity of the branch divisor along P_i belong to $\frac{1}{ab \cdot e_0!}\mathbb{Z} \cap [0, l'_0/\gamma]$, a finite set. Thus β^*_i belongs to a finite set, and passing to a subsequence, $\beta^*_i = \beta$ for any i.

Step 9. In this step we take a compatible generic limit near E and treat the case when the limit is klt.

We apply Construction-Proposition 4.1 in case (b) to the surface Y, the curve E, the charts $\mathbb{A}^2_{p,q}/\mu_b(-v, v)$ and $\mathbb{A}^2_{p',q'}/\mu_a(-v, v)$ with $e_\alpha = q$ and q' (the branch divisor is zero, since both quotient maps are quasi-étale), the group H, and the tuples whose members are

$$f_{i,j} := \left(h^0_{ij}(-p^2, 1, p, q),\ h^0_{ij}(1, -p'^2, p', q')\right),\ 1 \leq j \leq n_h,$$

together with the three constant members given by the chart functions of x^vt^a, y^vt^b, and zt. We check the hypotheses. The characters are fixed by **Step 7**, and the divisors descend to the H-invariant $\mathbb{Q}$-Cartier divisors $D_{Y,ij}$ and $\mathrm{div}_Y(x^vt^a)$, $\mathrm{div}_Y(y^vt^b)$, $\mathrm{div}_Y(zt)$. The points $(-p^2,1,p,q)$ and $(1,-p'^2,p',q')$ of $(xy+z^2=0)\subset\mathbb{A}^4$ with a common image in S_v belong to one $\mathbf{T}$-orbit, so over the normalization of the fiber product of $\mathbb{A}^2_{p,q}$ and $\mathbb{A}^2_{p',q'}$ over the intersection of the charts, $(1,-p'^2,p',q')=\varpi\cdot(-p^2,1,p,q)$ for a unit ϖ independent of i, and the quotient of the chart functions on $\mathbb{A}^2_{p,q}$ and $\mathbb{A}^2_{p',q'}$ of a $\mathbf{T}$-homogeneous polynomial of weight κ is ϖ^κ, independent of i. The orders along $(q=0)$ and $(q'=0)$ are the fixed ν_j, and the coefficient of q^{i_4} in $h^0_{ij}(-p^2,1,p,q)$ is a linear combination of the $p^{2i_1+i_3}$ with $ai_1+bi_2+vi_3=vi_4-\kappa_j$, $i_1i_2=0$, whose degree is bounded in terms of a,b,v,i_4,κ_j, and the coefficient of q'^{i_4} in $h^0_{ij}(1,-p'^2,p',q')$ is a polynomial in p' whose degree is bounded in terms of a,b,v,i_4,κ_j.

Passing to a subsequence, we obtain an algebraically closed field $k\supset\mathbb{C}$ and a compatible generic limit. Here $Y_k:=Y\times_{\mathbb{C}}\operatorname{Spec}k$, $E_k:=E\times_{\mathbb{C}}\operatorname{Spec}k$, $\mathcal{A}_k$ is the formal completion of Y_k along E_k, and for a positive integer l' we write $Z_{l'}\subset\mathcal{J}_{l'}$ for the irreducible closed subset of l'-jets of Construction-Proposition 4.1. Since the jets of a constant member do not depend on i, the corresponding members of the limit are the chart functions of x^vt^a, y^vt^b, and zt up to non-zero constants. Let $f_{\infty,j}$, $1\le j\le n_h$, be the first n_h members of the limit. By Construction-Proposition 4.1 for the subset $J=\{1,\dots,n_h\}$, $(f_{\infty,j})_{j\le n_h}$ is a compatible generic limit of the tuples $(f_{i,j})_{j\le n_h}$. Let

$$\Delta_\infty:=\sum_j c^*_j\,\mathrm{div}(f_{\infty,j})\quad\text{on }\mathcal{A}_k.$$

By **Step 8** and Theorem 4.2(1) for the labels c^*_j and $\theta=1$, $(\mathcal{A}_k,\Delta_\infty)$ is lc near E_k, and by Theorem 4.2(2), the minimum of the log discrepancies of $(\mathcal{A}_k,\Delta_\infty)/H$ over the prime divisors with center contained in E_k/H is equal to $\beta^*_i=\beta$.

Assume that $(\mathcal{A}_k,\Delta_\infty)$ is klt near E_k. By Theorem 4.2(1) there exists $\theta'>1$ such that $(Y,\theta'\Delta^*_i)$, hence $(Y,\theta'\Delta_i)$, is lc near E for any i. By (1) for θ', $\alpha_i\ge\beta_i\ge\beta\ge\alpha_\infty$ for $i\gg0$, which contradicts $\alpha_i<\alpha_\infty$. Thus we may assume that $(\mathcal{A}_k,\Delta_\infty)$ is not klt near E_k. By Definition-Proposition 5.7 for the pairs (Y,Δ^*_i), whose minima are $\beta^*_i=\beta>0$, $(\mathcal{A}_k,\Delta_\infty)$ has at least one floor, any floor is a curve with coefficient one in Δ_∞ whose preimages in the chart covers are smooth at the points over E_k, no two such preimages intersect at a point over E_k, and $\Delta_\infty-F$ is klt near E_k, where F is the sum of the floors.

Step 10. In this step we show that Δ_∞ has exactly one floor, that it intersects E_k at exactly one point, and that it is transverse to E_k there. The key input is the bound (A.109) on the intersection number of $\Delta_\infty-\varrho^*E_k$ with E_k.

Let o_y and o_x be the images of the origins of $\mathbb{A}^2_{p,q}$ and $\mathbb{A}^2_{p',q'}$. For an $\mathbb{R}$-divisor $T\ge0$ on $\mathcal{A}_k$ not containing E_k and a closed point $o'\in E_k$, let $(T\cdot E_k)_{o'}$ be $\frac{1}{b}(\tilde T\cdot(q=0))_0$ if $o'=o_y$, where $\tilde T$ is the pullback of T to $\operatorname{Spec}k[[p,q]]$, $\frac{1}{a}(\tilde T\cdot(q'=0))_0$ if $o'=o_x$, and the local intersection number at the smooth point o' of Y_k otherwise. Let $(T\cdot E_k):=\sum_{o'}(T\cdot E_k)_{o'}$. For divisors on Y, $(T\cdot E_k)_{o'}$ and $(T\cdot E_k)$ are the usual intersection numbers (projection formula for the quotient maps of the charts, which are quasi-étale of degrees b and a).

Let $\Sigma_x\subset Y$ be the closure of $(p=0)$ in the chart $\mathbb{A}^2_{p,q}/\mu_b(-v,v)$, and $\Sigma_y\subset Y$ the closure of $(p'=0)$ in the other chart. A direct calculation on $\mathbb{A}^2_{p,q}$ and $\mathbb{A}^2_{p',q'}$ shows that

$$\begin{gathered}(E^2)=-\frac{2v}{ab},\qquad(\Sigma_x\cdot E)=\frac1b,\qquad(\Sigma_y\cdot E)=\frac1a,\\ \mathrm{div}_Y(x^vt^a)=aE+2v\Sigma_x,\qquad\mathrm{div}_Y(y^vt^b)=bE+2v\Sigma_y,\qquad\mathrm{div}_Y(zt)=E+\Sigma_x+\Sigma_y,\end{gathered}\tag{A.108}$$

and that the divisors on Y with chart equations x,y,z,t evaluated at $(-p^2,1,p,q)$ and $(1,-p'^2,p',q')$ are $2\Sigma_x$, $2\Sigma_y$, $\Sigma_x+\Sigma_y$, and E. Hence the Weil divisor D_M on Y with the

chart equations of a monomial $M = x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of $\mathbf{T}$-weight κ satisfies

$$(D_M \cdot E) = \frac{2ai_1 + 2bi_2 + 2vi_3 - 2vi_4}{ab} = -\frac{2\kappa}{ab}.$$

If h is $\mathbf{T}$-homogeneous of weight κ and D_h is defined by the chart equations of h, then h/M is a $\mathbf{T}$-invariant rational function, hence a rational function on S_v, and $D_h - D_M$ is the divisor of h/M on Y, so $(D_h \cdot E) = -2\kappa/(ab)$. In particular $(D_{Y,ij} \cdot E) = -2\kappa_j/(ab)$.

Since being divisible by q^{ν_j} is a closed condition and having a non-zero coefficient of q^{ν_j} is an open condition on the (ν_j+1)-jets, $\operatorname{mult}_{E_k} \operatorname{div}(f_{\infty,j}) = \nu_j$, so $\operatorname{mult}_{E_k} \Delta_\infty = \varrho^*$. Let $D'_{\infty,j} := \operatorname{div}(f_{\infty,j}) - \nu_j E_k$ and $D'_{ij} := D_{Y,ij} - \nu_j E$. Let A_j and A'_j be the coefficients of q^{ν_j} and q'^{ν_j} in the chart functions of $f_{\infty,j}$ on $\mathbb{A}^2_{p,q}$ and $\mathbb{A}^2_{p',q'}$, which are polynomials in p and p'. Since $\mu_b(-v,v)$ acts freely on $(q=0)\backslash\{0\}$,

$$(D'_{\infty,j} \cdot E_k) = \frac{1}{b}\deg A_j + \frac{1}{a}\operatorname{mult}_{(p'=0)} \operatorname{div}(A'_j),$$

and

$$(D'_{ij} \cdot E) = \frac{1}{b}\deg A_{ij} + \frac{1}{a}\operatorname{mult}_{(p'=0)} \operatorname{div}(A'_{ij}),$$

where A_{ij} and A'_{ij} are the coefficients of q^{ν_j} and q'^{ν_j} in the chart functions of $f_{i,j}$. The numbers $\deg A_j$ and $\operatorname{mult}_{(p'=0)} \operatorname{div}(A'_j)$ are constant on a non-empty open subset of Z_{ν_j+1} (the degree is lower semicontinuous and the order at $p'=0$ is upper semicontinuous on Z_{ν_j+1}), which contains the jets of $f_{i,j}$ for $i \gg 0$. Hence

$$(D'_{\infty,j} \cdot E_k) = (D'_{ij} \cdot E) = \frac{2v\nu_j - 2\kappa_j}{ab}.$$

Write $\Delta_\infty = \varrho^* E_k + \Delta'$. Since $\tau^* \geq 0$ and $\varrho^* < 1$,

$$(\Delta' \cdot E_k) = \frac{2v\varrho^* - 2\tau^*}{ab} \leq \Big(\frac{1}{a} + \frac{1}{b}\Big)\varrho^* < \frac{1}{a} + \frac{1}{b}. \tag{A.109}$$

Since $\varrho^* < 1$, any floor is a curve other than E_k, and it is contained in the support of Δ' with coefficient one. The local intersection number of a floor with E_k at a point of intersection is at least $\frac{1}{b}$ at o_y, at least $\frac{1}{a}$ at o_x, and at least 1 at any other point. If the floors intersect E_k at two distinct points, then $(\Delta' \cdot E_k) \geq \frac{1}{a} + \frac{1}{b}$, since $\frac{1}{b} + 1 \geq \frac{1}{a} + \frac{1}{b}$, $\frac{1}{a} + 1 \geq \frac{1}{a} + \frac{1}{b}$, and $2 \geq \frac{1}{a} + \frac{1}{b}$, which contradicts (A.109). If two distinct floors intersect E_k at one point o', then their preimages intersect at a point over o', which is excluded by **Step 9**. Hence Δ_∞ has exactly one floor F, and $F \cap E_k = \{o\}$ for a closed point o. Since H preserves Δ_∞, it preserves F and fixes o.

We show that F is transverse to E_k at o. If $o \neq o_x, o_y$ and F is tangent to E_k, then $(F \cdot E_k)_o \geq 2 \geq \frac{1}{a} + \frac{1}{b}$, a contradiction. Assume that $o = o_y$, let $\tilde{F}$ be the preimage of F in $\operatorname{Spec} k[[p,q]]$, smooth at 0, and let $k_0 := (\tilde{F} \cdot (q=0))_0$. Assume that $k_0 \geq 2$. Then $\tilde{F} = (q = \varphi(p))$ with $\operatorname{mult}_0 \operatorname{div}(\varphi) = k_0$. Since $q - \varphi(p)$ is semi-invariant under $(p,q) \to (\zeta^{-v}p, \zeta^v q)$, $\zeta^b = 1$, we have $\zeta^{v(k_0+1)} = 1$, so $b \mid k_0 + 1$ since $\gcd(b,v) = 1$. By (A.109),

$$\frac{k_0}{b} \leq (\Delta' \cdot E_k) \leq \Big(\frac{1}{a} + \frac{1}{b}\Big)\varrho^*,$$

so $\varrho^* \geq \frac{ak_0}{a+b}$. Let

$$\sigma_0 := \operatorname{mult}_0\big((\tilde{\Delta}_\infty - \tilde{F})|_{\tilde{F}}\big),$$

where $\tilde{\Delta}_\infty$ is the pullback of Δ_∞. Since $(\mathcal{A}_k, \Delta_\infty)$ is lc near o, $\sigma_0 \leq 1$ by inversion of adjunction [Kaw07, Theorem]. By Lemma 5.6 for the curve $\tilde{F}$ and $C' = \tilde{\Delta}_\infty - \tilde{F}$, with the group K_y, the minimal log discrepancy at o of $(\mathcal{A}_k, \Delta_\infty)/H$ is equal to $(1-\sigma_0)/e'$ for a positive integer e', and it is at least $\beta > 0$. Thus

$$\varrho^* k_0 \leq \sigma_0 < 1, \qquad \text{so} \qquad k_0^2 < \frac{a+b}{a} = 1 + \frac{b}{a} \leq 1 + b.$$

If $b = 1$, then $k_0^2 \geq 4 > 2$. If $b = 2$, then k_0 is odd, so $k_0^2 \geq 9 > 3$. If $b \geq 3$, then $k_0 \geq b - 1$ and $(b-1)^2 \geq b + 1$. This is a contradiction, so $k_0 = 1$. If $o = o_x$, then exchanging (a, p, q) with (b, p', q') in the argument for $o = o_y$ shows $k_0 = 1$. Thus E_k restricts to a local parameter of F at o (on the chart cover if $o \in \{o_x, o_y\}$).

Step 11. In this step we augment Δ_∞ by a $\mathbf{T}$-invariant monomial so that the pair stays lc near E_k, and we compute β.

Let $\tilde{o}$ be a preimage of o in the chart cover $\mathbb{A}^2_{p,q} \times_{\mathbb{C}} \operatorname{Spec} k$ (resp. $\mathbb{A}^2_{p',q'} \times_{\mathbb{C}} \operatorname{Spec} k$ if $o = o_x$), $S_{\tilde{o}}$ the spectrum of the completed local ring of the chart cover at $\tilde{o}$ (so $S_{\tilde{o}} = \operatorname{Spec} k[[p, q]]$ if $o = o_y$, and $S_{\tilde{o}} = \operatorname{Spec} k[[p', q']]$ if $o = o_x$), $\tilde{F}$ the preimage of F in $S_{\tilde{o}}$, $\tilde{\Delta}_\infty$ the pullback of Δ_∞ to $S_{\tilde{o}}$, and $K := \operatorname{Stab}_{K_y}(\tilde{o})$ (resp. $\operatorname{Stab}_{K_x}(\tilde{o})$), which acts faithfully. Let

$$\sigma := \operatorname{mult}_{\tilde{o}} \left((\tilde{\Delta}_\infty - \tilde{F})|_{\tilde{F}} \right),$$

and let e be the order of the image of K in the automorphism group of the tangent line $T_{\tilde{o}}\tilde{F}$. As in **Step 10**, $\sigma < 1$, and Lemma 5.6 for the curve $\tilde{F}$ and $C' = \tilde{\Delta}_\infty - \tilde{F}$ implies

$$\beta \leq \frac{1 - \sigma}{e}. \tag{A.110}$$

Since $\tilde{F}$ is transverse to the preimage of E_k, $\sigma \geq \varrho^*$. Let

$$(\Upsilon, \lambda) := \left(y^v t^b, \frac{1-\sigma}{b} \right) \text{ if } o = o_y, \qquad \left(x^v t^a, \frac{1-\sigma}{a} \right) \text{ if } o = o_x, \qquad (zt, 1-\sigma) \text{ otherwise,}$$

and let e_Υ be the order of the character of Υ under G_i, which is the order of the character of Υ under H since Υ is $\mathbf{T}$-invariant. In particular e_Υ does not depend on i. By (A.108),

$$\lambda \operatorname{mult}_E \operatorname{div}_Y(\Upsilon) = 1 - \sigma \leq 1 - \varrho^*.$$

Let

$$\Delta_\infty^\Upsilon := \Delta_\infty + \lambda \operatorname{div}(\Upsilon) \quad \text{on } \mathcal{A}_k.$$

Claim A.16. *The pair $(\mathcal{A}_k, \Delta_\infty^\Upsilon)$ is lc near E_k, and $\beta = \lambda / e_\Upsilon$.*

Proof. Assume that $o = o_y$. The coefficient of E_k in Δ_∞^Υ is $\varrho^* + 1 - \sigma \leq 1$. On $\operatorname{Spec} k[[p, q]]$, $\Upsilon = q^b$, so near 0 the pullback of Δ_∞^Υ is

$$\tilde{F} + (\tilde{\Delta}_\infty - \tilde{F}) + (1 - \sigma) \operatorname{div}(q),$$

whose restriction to $\tilde{F}$ has multiplicity $\sigma + 1 - \sigma = 1$ at 0. By inversion of adjunction [Kaw07, Theorem], it is lc near 0. Near a point of E_k other than o, by (A.108),

$$\Delta_\infty^\Upsilon = (\varrho^* + 1 - \sigma) E_k + (\Delta' - F) + 2v\lambda\Sigma_y.$$

Since $(F \cdot E_k) = \frac{1}{b}$, (A.109), (A.108), and $1 - \sigma \leq 1 - \varrho^*$ imply

$$\left((\Delta' - F + 2v\lambda\Sigma_y) \cdot E_k \right) \leq \left(\frac{1}{a} + \frac{1}{b} \right) \varrho^* - \frac{1}{b} + \frac{2v}{ab}(1 - \varrho^*) = \frac{1}{a}.$$

Hence the restriction of the pullback of $\Delta' - F + 2v\lambda\Sigma_y$ to $(q' = 0)$ has multiplicity at most $a \cdot \frac{1}{a} = 1$ at the origin of $\operatorname{Spec} k[[p', q']]$, and the restriction to E_k has multiplicity at most $\frac{1}{a} \leq 1$ at any smooth point of Y_k on E_k other than o. By inversion of adjunction [Kaw07, Theorem] on the chart covers, where E_k is the smooth curve $(q = 0)$ or $(q' = 0)$, the pair with E_k of coefficient one is lc near the points of E_k other than o, and so is $(\mathcal{A}_k, \Delta_\infty^\Upsilon)$ since the quotient maps of the charts are quasi-étale. The case $o = o_x$ follows from the argument for $o = o_y$ after exchanging (a, x, Σ_x) with (b, y, Σ_y). Assume that $o \neq o_x, o_y$. Near o, $\operatorname{div}(zt) = E_k$, and the argument at o in the case $o = o_y$ shows lc near o. Near a point of E_k other than o, the boundary is

$$(\varrho^* + 1 - \sigma) E_k + (\Delta' - F) + (1 - \sigma)(\Sigma_x + \Sigma_y),$$

and since $(F \cdot E_k) = 1$,

$$\left((\Delta' - F + (1-\sigma)(\Sigma_x + \Sigma_y)) \cdot E_k \right) \leq \left(\frac{1}{a} + \frac{1}{b} \right) \varrho^* - 1 + \left(\frac{1}{a} + \frac{1}{b} \right)(1 - \varrho^*) = \frac{1}{a} + \frac{1}{b} - 1 \leq \min\left(\frac{1}{a}, \frac{1}{b} \right),$$

where the last inequality follows from $a, b \geq 1$. Thus the restrictions have multiplicity at most $b \cdot \frac{1}{b} = 1$ at o_y, at most $a \cdot \frac{1}{a} = 1$ at o_x, and at most 1 elsewhere, and inversion of adjunction [Kaw07, Theorem] on the chart covers implies that $(\mathcal{A}_k, \Delta_\infty^\Upsilon)$ is lc near the points of E_k other than o.

We show that $\beta \geq \lambda/e_\Upsilon$. The function Υ is a regular function on Y, semi-invariant under H with a character of order e_Υ, and $\operatorname{div}_Y(\Upsilon) \geq E$. Let $o' \in E_k$ be a closed point, $\tilde{o}'$ a preimage of o' in a chart cover, and K' the stabilizer of $\tilde{o}'$ in K_y or K_x. The pullback of Υ is K'-semi-invariant, with a character of order dividing e_Υ, since Υ is invariant under $\mu_b(-v, v)$ and $\mu_a(-v, v)$. By Lemma 5.8 for the germ at $\tilde{o}'$, the group K', $Z = \{\tilde{o}'\}$, and the formal prime divisors of [Liu26, Definition 3.2], any prime divisor with center o' has quotient log discrepancy at least λ/e_Υ. For the prime divisor E_k itself, let I_E be the inertia group of $(q = 0)$ in K_y. Since Υ^{e_Υ} is K_y-invariant and vanishes on $(q = 0)$, $|I_E| \leq e_\Upsilon \operatorname{mult}_{(q=0)} \Upsilon$, and since $\varrho^* + \lambda \operatorname{mult}_{(q=0)} \Upsilon \leq 1$,

$$\frac{1 - \varrho^*}{|I_E|} \geq \frac{\lambda \operatorname{mult}_{(q=0)} \Upsilon}{e_\Upsilon \operatorname{mult}_{(q=0)} \Upsilon} = \frac{\lambda}{e_\Upsilon}.$$

Hence $\beta \geq \lambda/e_\Upsilon$.

We show that $(1 - \sigma)/e = \lambda/e_\Upsilon$, which implies $\beta = \lambda/e_\Upsilon$ by (A.110). Assume that $o = o_y$. Since $\tilde{F}$ is transverse to $(q = 0)$, $q|_{\tilde{F}}$ is a local parameter of $\tilde{F}$ at 0, so e is the order of the character χ_q of q under $K = K_y$. Since $\mu_b(-v, v) \subset K_y$ acts on q by ζ^v and $\gcd(b, v) = 1$, $b \mid e$. Since $\Upsilon = q^b$, the character of Υ is χ_q^b, of order e/b. Thus $e_\Upsilon = e/b$ and $\lambda/e_\Upsilon = (1 - \sigma)/e$. If $o = o_x$, then exchanging (b, q, K_y) with (a, q', K_x) in the argument for $o = o_y$ shows $\lambda/e_\Upsilon = (1 - \sigma)/e$. Assume that $o \neq o_x, o_y$. Since $\mu_b(-v, v)$ acts freely on $(q = 0)\backslash\{0\}$ and H fixes o, the group K maps isomorphically onto H. The function $\Upsilon = pq$ restricts to a local parameter of $\tilde{F}$ at $\tilde{o}$, since p is a unit near $\tilde{o}$ and $\tilde{F}$ is transverse to $(q = 0)$. Thus $e = e_\Upsilon$ and $\lambda/e_\Upsilon = (1 - \sigma)/e$. □

Proof of Proposition A.14 continued. **Step 12.** In this step we specialize the augmentation to W_{N_i} and conclude.

The chart functions of Υ are members of the tuples of **Step 9**, and the corresponding member of the limit is the pair of chart functions of Υ up to a non-zero constant. By Theorem 4.2(1) for the tuples with members $f_{i,1}, \dots, f_{i,n_h}$ and Υ, the labels $c_1^*, \dots, c_{n_h}^*, \lambda$, and $\theta = 1$, and by Claim A.16, after passing to a subsequence $(Y, \Delta_i^* + \lambda \operatorname{div}_Y(\Upsilon))$ is lc near E for any i, and so is $(Y, \Delta_i + \lambda \operatorname{div}_Y(\Upsilon))$. The polynomial Υ is $\mathbf{T}$-homogeneous of weight 0 without monomials divisible by xy, and $\operatorname{div}_Y(\Upsilon)$ is the divisor with chart equations $\Upsilon(-p^2, 1, p, q)$ and $\Upsilon(1, -p'^2, p', q')$. Since $\sum_j c_{ij}\kappa_j + \lambda \cdot 0 = \tau_i \geq 0$, Lemma A.13 for the labels c_{ij}, λ and the polynomials h_{ij}^0, Υ implies that the pair

$$\Big(W_{N_i}, \sum_j c_{ij} \operatorname{div}(h_{ij}^0) + \lambda \operatorname{div}(\Upsilon)\Big)$$

is lc near 0. Let L' be the maximum of the integers of Lemma 4.14 for $n = 4$, the DCC set $\{1, \lambda\} \cup (\Gamma\backslash\{0\})$, and at most $M_0 + 1$ functions. For $i \gg 0$, $(r_i - M_0 M_1)/M_1 \geq L'$, so $h_{ij} - h_{ij}^0 \in (x, y, z, t)^{L'}$, and by inversion of adjunction [Kaw07, Theorem], [dFEM11, Proposition 2.11], and Lemma 4.14 for the tuples $(xy + z^2 + t^{N_i}, h_{i1}^0, \dots, h_{in_h}^0, \Upsilon)$ and $(xy + z^2 + t^{N_i}, h_{i1}, \dots, h_{in_h}, \Upsilon)$ with the exponents $(1, c_{i1}, \dots, c_{in_h}, \lambda)$, as in **Step 4**, $(W_{N_i}, D_i + \lambda \operatorname{div}(\Upsilon))$ is lc near 0. The function Υ is G_i-semi-invariant with a character of order e_Υ and vanishes at 0. By Lemma 5.8 for (W_{N_i}, D_i), the group G_i, $Z = \{0\}$, the function Υ, and $h = e_\Upsilon$, and by Claim A.16,

$$\alpha_i = \operatorname{mld}((W_{N_i} \ni 0, D_i)/G_i) \geq \frac{\lambda}{e_\Upsilon} = \beta \geq \alpha_\infty$$

for $i \gg 0$, which contradicts $\alpha_i < \alpha_\infty$. This proves (2). □

Proof of Theorem 7.15(2). This is the case $c = v$ of Construction-Proposition A.33(1). □

We now prove Theorem 7.15(1) and (3). For positive mass τ, part (1) is Proposition A.14, and for non-negative mass τ, part (3)(a) is Proposition A.14 together with Construction-Proposition 7.14. For non-positive mass τ we pass to the chart of the v-th root of t, on which the quotient of (W_N, D) by G becomes the quotient of a pair on $\mathbb{A}^3_{P,Q,u}$ whose zero locus of u carries C_t, and at the prime divisors where the terms of h_j of $\mathbf{T}$-weight less than κ_j matter, we replace W_N by W_0.

Set-up A.17. Notation as in Construction-Proposition 7.14. Since units and zero labels do not change D near 0, we assume that $c_j > 0$ and $h_j(0) = 0$ for any j. Let $\gamma := \min(\Gamma\backslash\{0\})$ ($\gamma := 1$ if $\Gamma \subset \{0\}$), $M := \lceil 2/\gamma \rceil$, $w_{\max} := \max(a, b, v)$, and $M_1 := M w_{\max}$. By Lemma 5.3, applied with $w = (-a, -b, -v, v)$ and this M, there exists a positive integer r_0, depending only on a, b, v, Γ, such that if $r \geq r_0$, then $|\kappa_j| \leq M_1$ for any j and any monomial of $h_j - h_j^0$ has total degree at least $(r - M_1)/w_{\max}$. For $\lambda \in \mathbb{C}^*$ let

$$\mathbf{T}(\lambda) := (\lambda^{-a}, \lambda^{-b}, \lambda^{-v}, \lambda^v) \in \mathbf{T}.$$

We use C_t, $\tilde{H}_t$, and β_t of Theorem 7.15. There $\tilde{H}_t = L_t/R_t$, where L_t is the group of triples (g, ξ, η) and $R_t = \{(\mathbf{T}(\lambda), \lambda, 1) : \lambda^r = 1\}$. The group L_t acts on $\mathbb{A}^3_{P,Q,s}$ by

$$(P, Q, s) \to (\eta P, g_z \xi^v Q/\eta, \xi s),$$

and $\tilde{H}_t$ acts on $\mathbb{A}^3_{P,Q,u}$ by $(P, Q, u) \to (\eta P, g_z \xi^v Q/\eta, \xi^r u)$. Let H_t be the quotient of the group of pairs (g, ξ) with $g \in G$ and $\xi^v = g_t$ by $\{(\mathbf{T}(\lambda), \lambda) : \lambda^r = 1\}$, acting on $\mathbb{A}^4_{X,Y,Z,u}$ by

$$(X, Y, Z, u) \to (g_x \xi^a X, g_y \xi^b Y, g_z \xi^v Z, \xi^r u).$$

Let

$$X_d := (XY + Z^2 + u^d = 0), \qquad X_\infty := (XY + Z^2 = 0) \quad \text{in } \mathbb{A}^4_{X,Y,Z,u},$$
$$S_0 := X_d \cap (u = 0) = X_\infty \cap (u = 0), \qquad W_0 := (xy + z^2 = 0) \subset \mathbb{A}^4_{x,y,z,t}.$$

In the coordinates $(X, Y, Z, u) = (x_1, y_1, z_1, \omega)$, X_d, S_0, and H_t are the threefold X_d, the surface S_0, and the group G_d of Construction-Proposition A.11.

For a G-semi-invariant polynomial g without monomials divisible by xy and an integer ℓ, let $g^{(\ell)}$ be the sum of the terms of g of $\mathbf{T}$-weight $\kappa(g) + \ell r$, so that $g = \sum_\ell g^{(\ell)}$ by Lemma 5.3, and $g^{(0)} = g^0$. Let

$$\hat{g} := \sum_\ell u^\ell g^{(\ell)}(X, Y, Z, 1), \qquad g^{(\geq 0)} := \sum_{\ell \geq 0} g^{(\ell)}.$$

For $D' = \sum_k c'_k \operatorname{div}(g_k)$ with real numbers c'_k and G-semi-invariant polynomials g_k without monomials divisible by xy, let $\tau(D') := \sum_k c'_k \kappa(g_k)$ and

$$\Delta(D') := \Big(1 + \frac{\tau(D')}{r}\Big) \operatorname{div}(u) + \sum_k c'_k \operatorname{div}(\hat{g}_k),$$

regarded on X_d and on X_∞ in the coordinates X, Y, Z, u, and on $\mathbb{A}^3_{P,Q,u}$ through $X = P^2$, $Y = -Q^2 - u^d P^{-2}$, $Z = PQ$. The $\hat{g}_k$ are rational functions (the blocks with $\ell < 0$ give negative powers of u, and on $\mathbb{A}^3_{P,Q,u}$ the substitution gives negative powers of P), so $\Delta(D')$ is an $\mathbb{R}$-divisor that need not be effective. Log discrepancies with respect to it are defined as usual. We write $\Delta_t := \Delta(D)$ on $\mathbb{A}^3_{P,Q,u}$ and $\Delta_d := \Delta(D)$ on X_d. For a finite group K acting on a normal variety V and preserving an $\mathbb{R}$-divisor B with $K_V + B$ $\mathbb{R}$-Cartier, $(V/K, B_{V/K})$ denotes the crepant quotient.

Lemma A.18. *Notation as in Set-up A.17. Let $D' = \sum_k c'_k \operatorname{div}(g_k)$ be as there, with g_k vanishing at 0, and let $D'_0 := \sum_k c'_k \operatorname{div}_{W_0}(g_k)$.*

(1) *H_t has order $v[G:R]$, preserves X_d and X_∞, and acts faithfully on S_0. The morphism $\mathbb{A}^2_{P,Q} \to S_0$, $(P,Q) \to (P^2, -Q^2, PQ, 0)$, is the quotient by the class of $(1,1,-1)$ in $\tilde{H}_t$, it is equivariant along the surjection $\tilde{H}_t \to H_t$ sending the class of (g,ξ,η) to the class of (g,ξ), and*

$$\beta_t = \mathrm{mld}\Big(\Big(S_0 \ni 0, \sum_j c_j \operatorname{div}(h_j^0(X,Y,Z,1))\Big)/H_t\Big).$$

(2) *The rational map $\mathbb{A}^3_{P,Q,s} \dashrightarrow W_N$ given by*

$$(P,Q,s) \to \big(P^2 s^{-a}, -Q^2 s^{-b} - P^{-2} s^{vN+a}, PQs^{-v}, s^v\big)$$

is equivariant along the projection $L_t \to G$ and induces birational maps

$$\mathbb{A}^3_{P,Q,u}/\tilde{H}_t \dashrightarrow W_N/G, \qquad X_d/H_t \dashrightarrow W_N/G.$$

The rational map $\mathbb{A}^3_{P,Q,s} \dashrightarrow W_0$ given by

$$(P,Q,s) \to (P^2 s^{-a}, -Q^2 s^{-b}, PQs^{-v}, s^v)$$

is equivariant along $L_t \to G$ and induces birational maps

$$\mathbb{A}^3_{P,Q,u}/\tilde{H}_t \dashrightarrow W_0/G, \qquad X_\infty/H_t \dashrightarrow W_0/G.$$

(3) *Let F be a prime divisor over $\mathbb{A}^3_{P,Q,u}$, and let E be the prime divisor over W_N/G induced by F through (2). Then*

$$a(E, W_N/G, D'_{W_N/G}) = \frac{a(F, \mathbb{A}^3_{P,Q,u}, \Delta(D'))}{|I_{\tilde{H}_t}(F)|}.$$

The formula for $a(E, W_N/G, D'_{W_N/G})$ also holds with $\mathbb{A}^3_{P,Q,u}$ and $\tilde{H}_t$ replaced by X_d and H_t, and with W_N, D', X_d replaced by W_0, D'_0, X_∞.

(4) *If E is a prime divisor over W_N/G with center the image of 0, then the prime divisor over X_d inducing E has center 0. If E is a prime divisor over W_0/G with center the image of 0, then the prime divisor over X_∞ inducing E has center 0.*

Proof. (1) Since $\gcd(a,v)=1$, the map $\lambda \to \mathbf{T}(\lambda)$ is injective, so $R = \mathbf{T}(\mu_r)$ and the subgroup $\{(\mathbf{T}(\lambda),\lambda) : \lambda^r = 1\}$ has order r. Thus $|H_t| = v|G|/r = v[G:R]$. The elements $(\mathbf{T}(\lambda),\lambda)$ act trivially on $\mathbb{A}^4_{X,Y,Z,u}$, so the action of H_t is well defined. Since G preserves $xy + z^2 + t^N$ up to a character, $g_x g_y = g_z^2 = g_t^N$. Hence XY, Z^2, and u^d are multiplied by $g_z^2 \xi^{2v}$, where for u^d we use $\xi^{rd} = \xi^{v(N+2)} = g_t^N g_t^2 = g_z^2 \xi^{2v}$. Thus H_t preserves X_d and X_∞. If (g,ξ) acts trivially on S_0, then $g_x = \xi^{-a}$, $g_z = \xi^{-v}$, $g_y = g_z^2/g_x = \xi^{-b}$, and $g_t = \xi^v$, so $g = \mathbf{T}(\xi) \in G \cap \mathbf{T} = R$ and $\xi^r = 1$. Thus H_t acts faithfully on S_0.

The map $\tilde{H}_t \to H_t$ is surjective, since for any (g,ξ) there exists η with $\eta^2 = g_x \xi^a$. The class of (g,ξ,η) maps to the trivial class if and only if $g = \mathbf{T}(\lambda)$ and $\xi = \lambda$ with $\lambda^r = 1$, in which case $\eta^2 = 1$. Modulo R_t the kernel is generated by the class of $(1,1,-1)$, which acts on $\mathbb{A}^2_{P,Q}$ by -1, and $\mathbb{A}^2_{P,Q} \to S_0$ is the quotient by -1. Equivariance follows from $\eta^2 = g_x \xi^a$ and $(g_z \xi^v/\eta)^2 = g_z^2 \xi^{2v}/(g_x \xi^a) = g_y \xi^b$. Since -1 fixes only the origin, the quotient by it is quasi-étale, and $\Phi_j := h_j^0(P^2, -Q^2, PQ, 1)$ is the pullback of $h_j^0(X,Y,Z,1)$. The formula for β_t follows from the quotient in stages [Liu26, Lemma 2.6(5)].

(2) The rational map, which we denote by ϕ, is equivariant: $(\xi s)^v = g_t s^v$, $(\eta P)^2 (\xi s)^{-a} = g_x P^2 s^{-a}$, and for the other coordinates equivariance follows from $g_x g_y = g_z^2 = g_t^N$. Let (P,Q,s) and (P',Q',s') be general points with

$$\phi(P,Q,s) = \phi(P',Q',s').$$

Then $s' = \xi s$ with $\xi^v = 1$, $P' = \eta P$ with $\eta^2 = \xi^a$, and $Q' = Q/\eta$. Thus the general fibers of ϕ are the orbits of the kernel

$$K_0 := \{(1,\xi,\eta) : \xi^v = 1,\ \eta^2 = \xi^a\}$$

of $L_t \to G$, which has order $2v$. Since $L_t \to G$ is surjective, ϕ induces a birational map $\mathbb{A}^3_{P,Q,s}/L_t \dashrightarrow W_N/G$. The morphism $\mathbb{A}^3_{P,Q,s} \to \mathbb{A}^3_{P,Q,u}$, $u = s^r$, is the quotient by R_t, and $(P, Q, s) \to (P^2, -Q^2, PQ, s^r)$ is the quotient onto X_∞ by the subgroup generated by R_t and $(1, 1, -1)$. Finally $(X, Y, Z, u) \to (X, Y - u^d X^{-1}, Z, u)$ is an H_t-equivariant birational map $X_\infty \dashrightarrow X_d$, and on X_d we have $X = xs^a$, $Y = ys^b$, $Z = zs^v$, $u = s^r$.

(3) The rational 3-form $\Omega := dx \wedge dz \wedge dt/x$ generates ω_{W_N} and ω_{W_0} near 0, as the residue of the hypersurface equation. Since $dX/X \wedge dZ = 2dP \wedge dQ$, a direct calculation shows

$$\phi^*\Omega = 2v dP \wedge dQ \wedge \frac{ds}{s} = \frac{2v}{r} dP \wedge dQ \wedge \frac{du}{u} = \frac{v}{r} \cdot \frac{dX \wedge dZ \wedge du}{Xu}. \tag{A.111}$$

Any monomial of $g_k^{(\ell)}$ pulls back to $s^{\kappa(g_k)+\ell r}$ times a monomial in X, Y, Z, so

$$g_k \circ \phi = s^{\kappa(g_k)} \hat{g}_k. \tag{A.112}$$

Let F_V be a prime divisor over $\mathbb{A}^3_{P,Q,s}$ inducing F, and for F_V let

$$A(F_V) := 1 + \operatorname{mult}_{F_V}(\phi^*\Omega) - \sum_k c'_k \operatorname{mult}_{F_V}(g_k \circ \phi).$$

Let F_W be the prime divisor over W_N induced by F_V, with ramification index e_W. By the Hurwitz formula,

$$1 + \operatorname{mult}_{F_V}(\phi^*\Omega) = e_W(1 + \operatorname{mult}_{F_W} \Omega), \qquad \operatorname{mult}_{F_V}(g_k \circ \phi) = e_W \operatorname{mult}_{F_W} g_k,$$

so $A(F_V) = e_W a(F_W, W_N, D')$. By [Liu26, Lemma 2.6(1)(2)], $a(F_W, W_N, D') = |I_G(F_W)| a(E, W_N/G, D'_{W_N/G})$ and $|I_{L_t}(F_V)| = e_W |I_G(F_W)|$. Thus

$$a(E, W_N/G, D'_{W_N/G}) = \frac{A(F_V)}{|I_{L_t}(F_V)|}.$$

On the other hand $\operatorname{mult}_{F_V} u = r \operatorname{mult}_{F_V} s$, and by (A.111) and (A.112),

$$A(F_V) = 1 + \operatorname{mult}_{F_V}(dP \wedge dQ \wedge du) - \Big(1 + \frac{\tau(D')}{r}\Big) \operatorname{mult}_{F_V} u - \sum_k c'_k \operatorname{mult}_{F_V} \hat{g}_k.$$

Since $dP \wedge dQ \wedge du$ generates $\omega_{\mathbb{A}^3}$, the Hurwitz formula and [Liu26, Lemma 2.6(1)] applied to $\mathbb{A}^3_{P,Q,s} \to \mathbb{A}^3_{P,Q,u}$ imply $A(F_V) = e_A a(F, \mathbb{A}^3_{P,Q,u}, \Delta(D'))$ and $|I_{L_t}(F_V)| = e_A |I_{\tilde{H}_t}(F)|$, where e_A is the ramification index of F_V over F. This proves the first formula. For X_d and X_∞ we use the generator $dX \wedge dZ \wedge du/X$ of ω_{X_d} and of ω_{X_∞} near 0 and the last expression in (A.111). For W_0 the pullback of g_k is again $s^{\kappa(g_k)}\hat{g}_k$, now in the coordinates of X_∞.

(4) The prime divisor F_W over W_N inducing E has center 0, so $\operatorname{mult}_{F_W}$ is positive on x, y, z, t. Since $X = xs^a$, $Y = ys^b$, $Z = zs^v$, $u = s^r$ on X_d and $v \operatorname{mult} s = \operatorname{mult} t > 0$, the orders of X, Y, Z, u are positive, and the center on $X_d \subset \mathbb{A}^4$ is 0. For W_0, on X_∞ we have $X = xs^a$, $Y = ys^b$, $Z = zs^v$, $u = s^r$ with $y = -z^2/x$ on W_0, so the orders of X, Y, Z, u along the prime divisor over X_∞ inducing E are positive, and its center on $X_\infty \subset \mathbb{A}^4$ is 0. □

The next lemma is the lower bound on the root chart. Part (2) is used with labels outside Γ and with extra equations of $\mathbf{T}$-weight zero.

Lemma A.19. *Notation as in Set-up A.17, and assume that $\tau \le 0$.*

(1) Assume that $(\mathbb{A}^2, C_t)$ is lc near 0. Let E be a prime divisor over W_N/G with center the image of 0, and F_W a prime divisor over W_N inducing E. If $\operatorname{mult}_{F_W}(h_j - h_j^{(\ge 0)}) > \operatorname{mult}_{F_W} h_j$ for any j, then $a(E, W_N/G, D_{W_N/G}) \ge \beta_t$.

(2) Let c'_k be finitely many positive real numbers and g_k G-semi-invariant polynomials vanishing at 0 and without monomials divisible by xy, such that any monomial of g_k has $\mathbf{T}$-weight κ'_k and $\sum_k c'_k \kappa'_k \le 0$. If $(\mathbb{A}^2, \sum_k c'_k \operatorname{div}(g_k(P^2, -Q^2, PQ, 1)))$ is lc near 0, then $(W_N, \sum_k c'_k \operatorname{div}(g_k))$ and $(W_0, \sum_k c'_k \operatorname{div}(g_k))$ are lc near 0.

Proof. (1) Let F be the prime divisor over X_d inducing E. By Lemma A.18(4), $\operatorname{center}_{X_d} F = \{0\}$. By (A.112), $\hat{h}_j - \hat{h}_j^{(\geq 0)}$ and $\hat{h}_j$ are $s^{-\kappa_j}(h_j - h_j^{(\geq 0)})$ and $s^{-\kappa_j} h_j$, so the hypothesis implies $\operatorname{mult}_F \hat{h}_j = \operatorname{mult}_F B_j^+$, where

$$B_j^+ := \sum_{\ell \geq 0} u^\ell h_j^{(\ell)}(X, Y, Z, 1)$$

is an H_t-semi-invariant polynomial on $\mathbb{A}^4$ with $B_j^+|_{u=0} = h_j^0(X, Y, Z, 1)$. Let $\mathfrak{a} := \prod_j (B_j^+)^{c_j}$. Since $\operatorname{div}_{X_d}(u) = S_0$, the log discrepancy of F depends only on the orders of u and of the equations, and Lemma A.18(3) implies

$$a(E, W_N/G, D_{W_N/G}) = \frac{a(F, X_d, \Delta_d)}{|I_{H_t}(F)|} = \frac{a(F, X_d, (1 + \tau/r)S_0, \mathfrak{a}|_{X_d})}{|I_{H_t}(F)|}.$$

The variety X_d is smooth for $d = 1$. For $d \geq 2$, X_d is the axial source $W_d = (xy + z^2 + t^d = 0)$, an isolated cDV point, so it is terminal of index one by [KM98, Corollary 5.38]. Thus X_d is normal and klt near 0. The section $S_0 = X_d \cap (u = 0)$ is a normal integral complete intersection with an A_1 singularity. By Lemma A.18(1), the diagonal group H_t acts faithfully on X_d and on S_0, $(S_0, \mathfrak{a}|_{S_0})$ is the quasi-étale quotient of $(\mathbb{A}^2, C_t)$ by -1, hence it is lc near 0, and $\operatorname{mld}((S_0 \ni 0, \mathfrak{a}|_{S_0})/H_t) = \beta_t$. By Lemma 5.5, applied with $n = 4$, the complete intersection X_d, the group H_t, the coordinate u, the section S_0, the $\mathbb{R}$-ideal $\mathfrak{a}$, and $b = 1 + \tau/r \leq 1$,

$$\frac{a(F, X_d, (1 + \tau/r)S_0, \mathfrak{a}|_{X_d})}{|I_{H_t}(F)|} \geq \beta_t - \frac{\tau \operatorname{mult}_F u}{r|I_{H_t}(F)|} \geq \beta_t.$$

(2) Let $D' := \sum_k c'_k \operatorname{div}(g_k)$. Since g_k is homogeneous for $\mathbf{T}$, $\hat{g}_k = g_k(X, Y, Z, 1)$ is a polynomial, $\tau(D') = \sum_k c'_k \kappa'_k \leq 0$, and the pair $(S_0, \sum_k c'_k \operatorname{div}(\hat{g}_k|_{S_0}))$ is the quasi-étale quotient of the plane pair of the hypothesis by -1, hence lc near 0. Let β' be the quotient value of $(S_0 \ni 0, \sum_k c'_k \operatorname{div}(\hat{g}_k|_{S_0}))$ by H_t, so $\beta' \geq 0$. Let F_W be a prime divisor over W_N with center 0, and E and F the prime divisors over W_N/G and over X_d induced by F_W. Then $\operatorname{center}_{X_d} F = \{0\}$, and Lemma A.18(3) together with Lemma 5.5, applied as in (1) with $\mathfrak{a} := \prod_k \hat{g}_k^{c'_k}$ and $b = 1 + \tau(D')/r$, implies

$$\frac{a(F_W, W_N, D')}{|I_G(F_W)|} = a(E, W_N/G, D'_{W_N/G}) = \frac{a(F, X_d, \Delta(D'))}{|I_{H_t}(F)|} \geq \beta' \geq 0. \tag{A.113}$$

Suppose that (W_N, D') is not lc near 0. Then (W_N, D') has a prime divisor with center 0 and negative log discrepancy (cf. [KM98, Corollary 2.31(1)], on a log resolution of (W_N, D') with [KM98, Lemma 2.30 and Corollary 2.31(3)]), which contradicts (A.113). The hypersurface $X_\infty = (XY + Z^2 = 0) \subset \mathbb{A}^4$ is the product of an A_1 surface singularity and $\mathbb{A}^1_u$, hence normal and klt. The group H_t preserves it and acts faithfully on $S_0 = X_\infty \cap (u = 0)$ by Lemma A.18(1), hence faithfully on X_∞. Replacing W_N and X_d with W_0 and X_∞, and using the last parts of Lemma A.18(3) and (4), we obtain (A.113) for W_0 and X_∞, and [KM98, Corollary 2.31(1)] shows as for W_N that $(W_0, \sum_k c'_k \operatorname{div}(g_k))$ is lc near 0. □

The upper bound uses one Gauss extension of a divisor computing β_t, and only needs $(\mathbb{A}^2, C_t)$ to be lc.

Lemma A.20. *Notation as in Set-up A.17.*

(1) Let m_j be the least total degree of a monomial of h_j. Then $\sum_j c_j m_j < 2$. In particular the number of indices j is at most M, and $|\tau| < 2w_{\max}$.

(2) There exist positive integers r_2 and k_0, depending only on a, b, v, Γ, e_0, such that if $\tau \leq 0$, $r \geq r_2$, and $(\mathbb{A}^2, C_t)$ is lc near 0, then there exists a prime divisor E over W_N/G with center the image of 0 and

$$a(E, W_N/G, D_{W_N/G}) = \beta_t - \frac{\tau k}{re}$$

for some integers $1 \leq k \leq k_0$ and $1 \leq e \leq 2ve_0$. In particular $\alpha \leq \beta_t + 2w_{\max} k_0/r$.

Proof. (1) By Lemma A.1(3), $\sum_j c_j m_j < 2$. A monomial of degree m_j has $\mathbf{T}$-weight at most $w_{\max} m_j$ in absolute value, so $|\tau| \le w_{\max} \sum_j c_j m_j < 2w_{\max}$. Since $c_j \ge \gamma$ and $m_j \ge 1$, the number of indices j is less than $2/\gamma \le M$.

(2) Let $\Theta := \Psi(\Gamma_+ \cap [0,1])$, a DCC set by [Liu26, Lemma 2.10(1)], and let l be the integer of Theorem 2.9(2) for Θ. Let $M_3 := 2ve_0 l/\gamma$, $k_0 := \lfloor 3M_3 \rfloor + 2ve_0$, and let $r_2 \ge r_0$, with r_0 as in Set-up A.17, be an integer with $r_2 > w_{\max}(k_0 + M_3) + M_1$.

By [Liu26, Lemma 2.6(4)], the coefficients of the crepant quotient of $(\mathbb{A}^2, C_t)$ by $\tilde{H}_t$ belong to Θ. By Theorem 2.9(2) there exists a prime divisor E_0 over $\mathbb{A}^2/\tilde{H}_t$ computing β_t with $a(E_0, \mathbb{A}^2/\tilde{H}_t, 0) \le l$. Let ν be a prime divisor over $\mathbb{A}^2$ inducing E_0, and $e := |I_{\tilde{H}_t}(\nu)| \le |\tilde{H}_t| = 2v[G:R] \le 2ve_0$ (Theorem 7.15(2)). Since the branch $\mathbb{Q}$-divisor $B_{\tilde{H}_t}$ of $\mathbb{A}^2 \to \mathbb{A}^2/\tilde{H}_t$ satisfies $B_{\tilde{H}_t} \ge 0$, [Liu26, Lemma 2.6(2)] implies $a(\nu, \mathbb{A}^2, 0) \le e a(E_0, \mathbb{A}^2/\tilde{H}_t, 0) \le 2ve_0 l$. Since $(\mathbb{A}^2, \operatorname{div}(P) + \operatorname{div}(Q))$ and $(\mathbb{A}^2, C_t)$ are lc and $c_j \ge \gamma$,

$$1 \le \operatorname{mult}_\nu P,\ \operatorname{mult}_\nu Q, \qquad \operatorname{mult}_\nu P + \operatorname{mult}_\nu Q \le M_3, \qquad \operatorname{mult}_\nu \Phi_j \le M_3,$$

where $\Phi_j = h^0_j(P^2, -Q^2, PQ, 1)$. Let χ be the character of $\tilde{H}_t$ sending the class of (g, ξ, η) to ξ^r, so that $\tilde{H}_t$ acts on $\mathbb{A}^3_{P,Q,u} = \mathbb{A}^2_{P,Q} \times \mathbb{A}^1_u$ by $(p,u) \to (\sigma \cdot p, \chi(\sigma)u)$ for $\sigma \in \tilde{H}_t$. Let q be the integer of Lemma 3.1 for ν and χ, and let k be the least integer with $k > 3M_3$ and $k \equiv q \pmod e$. Then $k \le \lfloor 3M_3 \rfloor + e \le k_0$.

Let

$$C_V := \left(1 + \frac{\tau}{r}\right) \operatorname{div}(u) + \sum_j c_j \operatorname{div}(\Phi_j),$$

an $\mathbb{R}$-divisor on $\mathbb{A}^3_{P,Q,u}$, where the Φ_j are regular and do not depend on u. By Proposition 5.4, applied with $S = \mathbb{A}^2_{P,Q}$, $s = 0$, the group $\tilde{H}_t$, which acts faithfully on S (Theorem 7.15(2)), the coordinate u, the $\mathbb{R}$-divisor C_V, the coefficient $b = 1 + \tau/r \le 1$, the prime divisor ν, and the integer k, there exists a prime divisor F over $\mathbb{A}^3_{P,Q,u}$ with center 0 such that mult_F is the Gauss extension of mult_ν with u-value k, $I_{\tilde{H}_t}(F) = I_{\tilde{H}_t}(\nu)$, $\operatorname{mult}_F \Phi_j = \operatorname{mult}_\nu \Phi_j$, and

$$\frac{a(F, \mathbb{A}^3_{P,Q,u}, C_V)}{e} = \beta_t - \frac{\tau k}{re}. \tag{A.114}$$

We show that $\operatorname{mult}_F \hat{h}_j = \operatorname{mult}_F \Phi_j$ for any j. On $\mathbb{A}^3_{P,Q,u}$ we have $X = P^2$, $Z = PQ$, and $Y = -Q^2 + \epsilon$ with $\epsilon := -u^d P^{-2}$. Since mult_F is a Gauss extension [Liu26, Definition-Proposition 2.11],

$$\operatorname{mult}_F \epsilon = dk - 2\operatorname{mult}_\nu P \ge k - 2\operatorname{mult}_\nu P > M_3 + 2\operatorname{mult}_\nu Q,$$

so $\operatorname{mult}_F Y = 2\operatorname{mult}_\nu Q$, and X, Y, Z, ϵ have non-negative orders along F. First, $h^0_j(X,Y,Z,1) - \Phi_j$ is ϵ times a polynomial in X, Q^2, ϵ, Z, so $\operatorname{mult}_F(h^0_j(X,Y,Z,1) - \Phi_j) > M_3 \ge \operatorname{mult}_F \Phi_j$. Second, for $\ell > 0$, $\operatorname{mult}_F(u^\ell h^{(\ell)}_j(X,Y,Z,1)) \ge \ell k > M_3$. Third, let $\ell = -m < 0$ and let $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ be a monomial of $h^{(\ell)}_j$. The $\mathbf{T}$-weight of $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ is $\kappa_j - mr$, so $ai_1 + bi_2 + vi_3 = vi_4 - \kappa_j + mr \ge mr - M_1$, and since $2\operatorname{mult}_\nu P/a$, $2\operatorname{mult}_\nu Q/b$, and $(\operatorname{mult}_\nu P + \operatorname{mult}_\nu Q)/v$ are at least $1/w_{\max}$,

$$\operatorname{mult}_F(u^{-m} X^{i_1} Y^{i_2} Z^{i_3}) \ge \frac{mr - M_1}{w_{\max}} - mk \ge \frac{r}{w_{\max}} - k - \frac{M_1}{w_{\max}} > M_3.$$

Thus $\operatorname{mult}_F \hat{h}_j = \operatorname{mult}_F \Phi_j$.

Since the log discrepancy of F only depends on the orders of u and of the equations along F, $a(F, \mathbb{A}^3_{P,Q,u}, \Delta_t) = a(F, \mathbb{A}^3_{P,Q,u}, C_V)$. Let E be the prime divisor over W_N/G induced by F. By Lemma A.18(3) and (A.114), $a(E, W_N/G, D_{W_N/G}) = \beta_t - \tau k/(re)$. Normalizing the extension of mult_F to $\mathbb{A}^3_{P,Q,s}$ so that $\operatorname{mult} u = k$, the orders of x, y, z, t are

$$2\operatorname{mult}_\nu P - \frac{ak}{r}, \quad 2\operatorname{mult}_\nu Q - \frac{bk}{r}, \quad \operatorname{mult}_\nu P + \operatorname{mult}_\nu Q - \frac{vk}{r}, \quad \frac{vk}{r},$$

which are positive since $r > w_{\max}k_0$. Hence the center of E is the image of 0, and $\alpha \le \beta_t + |\tau|k/r \le \beta_t + 2w_{\max}k_0/r$. □

The lower bound needs a margin $\theta > 1$ for C_t. The proof separates the prime divisors at which the blocks $h_j^{(\ell)}$ with $\ell < 0$ are negligible from those at which the term t^N is negligible.

Lemma A.21. *Notation as in Set-up A.17, and let $D_0 := \sum_j c_j \operatorname{div}_{W_0}(h_j)$. For any $\theta > 1$ there exists a positive integer r_3, depending only on $a, b, v, \Gamma, e_0, \theta$, such that if $\tau \le 0$, $r \ge r_3$, and $(\mathbb{A}^2, \theta C_t)$ is lc near 0, then the following hold.*

(1) $(W_N, \theta D)$ and $(W_0, \theta D_0)$ are lc near 0.
(2) $a(E, W_N/G, D_{W_N/G}) \ge \beta_t$ for any prime divisor E over W_N/G with center the image of 0. In particular $\beta_t \le \alpha$.

Proof. The proof is long, so we divide it into several steps. By Lemma A.20(1), $|\tau| < 2w_{\max}$ and the number of indices j is at most M. Let

$$\theta_0 := \frac{2\theta}{\theta - 1}, \qquad M_2 := \frac{2}{\gamma(\theta - 1)}, \qquad c_0 := \frac{1}{4abve_0}.$$

Step 1. In this step we prove (1). By Lemma A.19(2) with the labels θc_j and the polynomials h_j^0, whose $\mathbf{T}$-weights satisfy $\sum_j \theta c_j \kappa_j = \theta\tau \le 0$, the pairs $(W_N, \theta \sum_j c_j \operatorname{div}(h_j^0))$ and $(W_0, \theta \sum_j c_j \operatorname{div}(h_j^0))$ are lc near 0. By inversion of adjunction [Kaw07, Theorem], the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^N) + \theta \sum_j c_j \operatorname{div}(h_j^0)\Big)$$

is lc near 0. Let $\mathcal{H}_j := h_j(p^2, -q^2, pq, t)$ and $\mathcal{H}_j^0 := h_j^0(p^2, -q^2, pq, t)$. The morphism $\mathbb{A}^3_{p,q,t} \to W_0$, $(p, q, t) \to (p^2, -q^2, pq, t)$, is the quasi-étale quotient by $(p, q) \to (-p, -q)$, so $(\mathbb{A}^3, \theta \sum_j c_j \operatorname{div}(\mathcal{H}_j^0))$ is lc near 0 by [Liu26, Lemma 2.6(2)].

Let $\Lambda_\theta := \{\theta c : c \in \Gamma,\ c > 0\} \cup \{1\}$, a DCC subset of $\mathbb{R}_{>0}$, and let L_1 be the maximum of the integers of Lemma 4.14 for $n \in \{3, 4\}$, at most $M + 1$ functions, and Λ_θ. We take $r_3 \ge \max(r_0, w_{\max}L_1 + M_1)$. By Set-up A.17, any monomial of $h_j - h_j^0$ has total degree at least $(r - M_1)/w_{\max} \ge L_1$, and so has any monomial of $\mathcal{H}_j - \mathcal{H}_j^0$ in p, q, t. Since lc near 0 is equivalent to lc of the completion [dFEM11, Proposition 2.11], Lemma 4.14, applied on $\mathbb{A}^4$ to the tuples $(xy + z^2 + t^N, h_1^0, h_2^0, \dots)$ and $(xy + z^2 + t^N, h_1, h_2, \dots)$ with the exponents $(1, \theta c_1, \theta c_2, \dots)$, and on $\mathbb{A}^3_{p,q,t}$ to the tuples $(\mathcal{H}_1^0, \mathcal{H}_2^0, \dots)$ and $(\mathcal{H}_1, \mathcal{H}_2, \dots)$ with the exponents $(\theta c_1, \theta c_2, \dots)$, implies that the pairs

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^N) + \theta \sum_j c_j \operatorname{div}(h_j)\Big) \quad \text{and} \quad \Big(\mathbb{A}^3, \theta \sum_j c_j \operatorname{div}(\mathcal{H}_j)\Big)$$

are lc near 0. By [Kaw07, Theorem] and [Liu26, Lemma 2.6(2)], $(W_N, \theta D)$ and $(W_0, \theta D_0)$ are lc near 0. Since $\theta > 1$, (W_0, D_0) is klt near 0.

Step 2. In this step we bound the orders at a prime divisor of small log discrepancy. Let F_1' be the prime divisor over $\mathbb{A}^2$ induced by the blow-up of 0. Then $\beta_t \le a(F_1', \mathbb{A}^2, C_t)/|I_{\tilde{H}_t}(F_1')| \le 2$. Thus it suffices to prove (2) for a prime divisor E with $a(E, W_N/G, D_{W_N/G}) \le 2$. Let F be a prime divisor over W_N inducing E, $e := |I_G(F)|$, and $[f] := \operatorname{mult}_F f/e$ for a regular function f near 0. By [Liu26, Lemma 2.6(2)], $a(F, W_N, D) \le 2e$, and since $(W_N, \theta D)$ is lc, $\theta \sum_j c_j \operatorname{mult}_F h_j \le a(F, W_N, 0)$. Thus $(1 - 1/\theta)a(F, W_N, 0) \le 2e$, and

$$\frac{a(F, W_N, 0)}{e} \le \theta_0, \qquad [h_j] \le \frac{\theta_0}{\theta\gamma} = M_2. \tag{A.115}$$

The prime divisor $S' := \mathrm{div}_{W_N}(t) = (xy + z^2 = t = 0)$ is the quotient of $\mathbb{A}^2_{p,q}$ by -1 through $x = p^2$, $y = -q^2$, $z = pq$, and

$$\frac{1}{2}\,\mathrm{div}(x)|_{S'} + \frac{1}{2}\,\mathrm{div}(y)|_{S'} = \mathrm{div}(z)|_{S'}$$

is the image of $(pq = 0)$. Thus the restrictions are lc, and by [Kaw07, Theorem] the pairs

$$\Big(W_N, \frac{1}{2}\,\mathrm{div}(x) + \frac{1}{2}\,\mathrm{div}(y) + \mathrm{div}(t)\Big) \quad \text{and} \quad \big(W_N, \mathrm{div}(z) + \mathrm{div}(t)\big)$$

are lc near 0. By (A.115),

$$[x], [y] \le 2\theta_0, \qquad [z], [t] \le \theta_0. \tag{A.116}$$

The polynomials $x^v t^a$, $y^v t^b$, and zt have $\mathbf{T}$-weight 0, so R acts trivially on them and their characters have order at most $[G : R] \le e_0$. If U is one of them and m_U is the order of the character of U, then U^{m_U} descends to a regular function on W_N/G vanishing at the image of 0, so $m_U[U] = \mathrm{mult}_E(U^{m_U}) \ge 1$ by [Liu26, Lemma 2.6(1)]. Thus

$$v[x] + a[t] \ge \frac{1}{e_0}, \qquad v[y] + b[t] \ge \frac{1}{e_0}, \qquad [z] + [t] \ge \frac{1}{e_0}. \tag{A.117}$$

Let $\rho_t := [t]/v$ and $\rho_- := \min([x]/a, [y]/b, [z]/v)$. If $\rho_t \le c_0$ and $\rho_- \le c_0$, then applying (A.117) to a coordinate attaining ρ_- implies $2abvc_0 \ge 1/e_0$, a contradiction. Thus $\rho_t > c_0$ or $\rho_- > c_0$.

Step 3. In this step we treat the case when $[f] > M_2$ for any monomial f of any $h_j^{(\ell)}$ with $\ell < 0$. Then $[h_j - h_j^{(\ge 0)}] > M_2 \ge [h_j]$ by (A.115). Since $(\mathbb{A}^2, C_t)$ is lc, Lemma A.19(1) implies $a(E, W_N/G, D_{W_N/G}) \ge \beta_t$.

Step 4. In this step we treat the remaining case and show that t^N is negligible at F. Let $f = x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ be a monomial of some $h_j^{(\ell)}$ with $\ell < 0$ and $[f] \le M_2$. The $\mathbf{T}$-weight of f is at most $\kappa_j - r$, so $ai_1 + bi_2 + vi_3 \ge r - M_1$, and

$$M_2 \ge [f] \ge i_1[x] + i_2[y] + i_3[z] \ge \rho_-(r - M_1).$$

We take $r_3 > M_1 + M_2/c_0$. Then $\rho_- < c_0$, so $\rho_t > c_0$ by **Step 2**, and $[t] > vc_0$. Since $N + 2 = rd/v \ge r/v$, we have $N[t] \ge (r - 2v)c_0$. We take $r_3 > 2v + (4\theta_0 + M_2)/c_0$. By (A.116),

$$N[t] > [x] + [y] + M_2, \qquad N[t] > 2[z].$$

Since $xy + z^2 + t^N = 0$ on W_N, the minimum of $[x] + [y]$, $2[z]$, $N[t]$ is attained at least twice, so $[x] + [y] = 2[z]$. Let $y_0 := -z^2/x$. Then $[y_0] = 2[z] - [x] = [y] > 0$ and $y - y_0 = -t^N/x$, so $[y - y_0] = N[t] - [x] > M_2$. For any j, $h_j(x, y, z, t) - h_j(x, y_0, z, t)$ is $y - y_0$ times a polynomial in x, y, y_0, z, t, so

$$[h_j(x, y, z, t) - h_j(x, y_0, z, t)] > M_2 \ge [h_j], \quad \text{hence} \quad [h_j(x, y_0, z, t)] = [h_j].$$

The rational map $W_N \dashrightarrow W_0$, $(x, y, z, t) \to (x, -z^2/x, z, t)$, is birational and G-equivariant. Let F_0 and E_0 be the prime divisors over W_0 and W_0/G induced by F and E. Since $[x], [y_0], [z], [t] > 0$, $\mathrm{center}_{W_0} F_0 = \{0\}$. The form $dx \wedge dz \wedge dt/x$ generates ω_{W_N} and ω_{W_0} near 0, the inertia groups $I_G(F)$ and $I_G(F_0)$ coincide, and the orders of the equations coincide. Thus

$$a(E, W_N/G, D_{W_N/G}) = \frac{a(F, W_N, D)}{e} = \frac{a(F_0, W_0, D_0)}{e} = a(E_0, W_0/G, D_{0,W_0/G}).$$

Step 5. In this step we bound $a(E_0, W_0/G, D_{0,W_0/G})$ from below by [Liu26, Theorem 16.21]. Let $\tilde{G}$ be the group of pairs (g, ς) with $g \in G$ and $\varsigma^2 = g_x$, acting on $\mathbb{A}^3_{p,q,t}$ by $(p, q, t) \to (\varsigma p, g_z \varsigma^{-1} q, g_t t)$. Then $\tilde{G}$ is a finite diagonal group of order $2|G|$ acting faithfully on $\mathbb{A}^3$, and

$\mathbb{A}^3_{p,q,t} \to W_0$ is the quasi-étale quotient by $\{(1,\pm 1)\}$. With $\mathcal{H}_j = h_j(p^2, -q^2, pq, t)$ from **Step 1**, [Liu26, Lemma 2.6(5)] implies

$$\mathrm{mld}((W_0 \ni 0, D_0)/G) = \mathrm{mld}\Big(\Big(\mathbb{A}^3 \ni 0, \sum_j c_j \operatorname{div}(\mathcal{H}_j)\Big)/\tilde{G}\Big).$$

Let $\delta := \gcd(a, 2v) \in \{1, 2\}$ and $\mathbf{T}' := \mathbf{T}_{(a/\delta, b/\delta, -2v/\delta)}$, so $\gcd(a/\delta, b/\delta, 2v/\delta) = 1$. An element (g, ς) of $\tilde{G}$ lying in $\mathbf{T}'$ has $g \in G \cap \mathbf{T} = R$, and $\tilde{G} \cap \mathbf{T}'$ is the image of μ_{2r} under $\zeta \to (\mathbf{T}(\zeta^2), \zeta^{-a})$, which acts on (p, q, t) by $(\zeta^{-a}, \zeta^{-b}, \zeta^{2v})$. The map $\zeta \to (\mathbf{T}(\zeta^2), \zeta^{-a})$ has kernel μ_δ, so $|\tilde{G} \cap \mathbf{T}'| = 2r/\delta$, and since $|\tilde{G}| = 2|G|$,

$$|\tilde{G}/(\tilde{G} \cap \mathbf{T}')| = \delta[G : R] \le 2e_0.$$

The monomial $p^{2i_1+i_3} q^{2i_2+i_3} t^{i_4}$ of $\mathcal{H}_j$ coming from $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ has $\mathbf{T}'$-weight $-2\kappa/\delta$, where κ is the $\mathbf{T}$-weight of $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$. The monomials of $\mathcal{H}_j - \mathcal{H}_j^0$ have total degree at least $(r - M_1)/w_{\max}$, and $\mathcal{H}_j^0$ has a monomial of total degree at most $2M$. We take $r_3 > 2Mw_{\max} + M_1$. Thus the principal block of $\mathcal{H}_j$ for $\mathbf{T}'$ is $\mathcal{H}_j^0$, and the mass of $\sum_j c_j \operatorname{div}(\mathcal{H}_j)$ along $\mathbf{T}'$ is

$$\sum_j c_j \Big(-\frac{2\kappa_j}{\delta}\Big) = -\frac{2\tau}{\delta} \ge 0.$$

By **Step 1**, $(\mathbb{A}^3, \sum_j c_j \operatorname{div}(\mathcal{H}_j))$ is klt near 0. Let n_1 be the threshold of [Liu26, Theorem 16.21] for the circle $\mathbf{T}'$ on $\mathbb{A}^3_{p,q,t}$, the residual order bound $2e_0$, and the labels in Γ. We take $r_3 > \delta n_1/2$, so that $|\tilde{G} \cap \mathbf{T}'| = 2r/\delta > n_1$. By [Liu26, Theorem 16.21], applied to the group $\tilde{G}$, the minimal log discrepancy above is at least the number b_0 of [Liu26, Construction-Theorem 16.9] for the principal blocks.

Since the mass is non-negative (here the number $r-p-q$ of [Liu26, Construction-Theorem 16.9] for $\mathbf{T}'$ is $(2v - a - b)/\delta = 0$, so the number τ there equals $-2\tau/\delta \ge 0$, and we are in case (1) there), b_0 is the quotient value at the origin of the plane pair on the cover $t = \varsigma_1^{2v/\delta}$, $U = p\varsigma_1^{a/\delta}$, $V = q\varsigma_1^{b/\delta}$ by the group $\tilde{H}$ of [Liu26, Construction-Theorem 16.9], the image in $\mathrm{Aut}(\mathbb{A}^2_{U,V})$ of the group of the cover. On the cover, $U^2 = xs^a$ and $UV = zs^v$ with $s := \varsigma_1^{2/\delta}$, so (U, V) are the coordinates (P, Q), and the plane pair is $(\mathbb{A}^2_{P,Q}, C_t)$. The group of the cover consists of the triples (g, ς, ψ) with $(g, \varsigma) \in \tilde{G}$ and $\psi^{2v/\delta} = g_t$, acting by

$$(U, V, \varsigma_1) \to (\varsigma\psi^{a/\delta} U, g_z \varsigma^{-1} \psi^{b/\delta} V, \psi\varsigma_1).$$

The map sending (g, ς, ψ) to the class of $(g, \psi^{2/\delta}, \varsigma\psi^{a/\delta})$ in $\tilde{H}_t$ is a homomorphism, since $\xi := \psi^{2/\delta}$ and $\eta := \varsigma\psi^{a/\delta}$ satisfy $\xi^v = g_t$ and $\eta^2 = g_x \xi^a$. It is compatible with the actions on $\mathbb{A}^2_{U,V} = \mathbb{A}^2_{P,Q}$, since $g_z \xi^v/\eta = g_z \varsigma^{-1} \psi^{b/\delta}$. It is surjective: given (g, ξ, η), take ψ with $\psi^{2/\delta} = \xi$ and $\varsigma := \eta\psi^{-a/\delta}$. Since $\tilde{H}_t$ acts faithfully on $\mathbb{A}^2_{P,Q}$ (Theorem 7.15(2)), the group $\tilde{H}$ is $\tilde{H}_t$, of order $2v[G : R] = (2v/\delta) \cdot \delta[G : R]$. Thus $b_0 = \beta_t$, and

$$a(E, W_N/G, D_{W_N/G}) = a(E_0, W_0/G, D_{0,W_0/G}) \ge \beta_t.$$

□

Proof of Theorem 7.15(1) and (3). The proof is long, so we divide it into several steps.

Step 1. In this step we fix the constants and prove Theorem 7.15(3). Let r_Y and C be the numbers r_0 and C of Construction-Proposition 7.14, r_θ the integer r_0 of Proposition A.14 for θ, and r_2, k_0, $r_3 = r_3(\theta)$ the integers of Lemmas A.20 and A.21. Let

$$C' := \max(C, 2w_{\max} k_0), \qquad r_1 := \max(r_Y, r_\theta, r_2, r_3(\theta)).$$

Since $r_2 \ge r_0$ for the integer r_0 of Set-up A.17, also $r_1 \ge r_0$. Let $r \ge r_1$. If $\tau \ge 0$ and $(Y, \theta\Delta_Y)$ is lc near E, then $\beta_Y \le \alpha$ by Proposition A.14 and $\alpha \le \beta_Y + C'/r$ by Construction-Proposition 7.14. If $\tau \le 0$ and $(\mathbb{A}^2, \theta C_t)$ is lc near 0, then $\beta_t \le \alpha$ by Lemma A.21(2) and $\alpha \le \beta_t + C'/r$ by Lemma A.20(2).

Suppose that α can increase along $\mathbf{T}$. Then there exists a sequence of data as in Set-up A.17, indexed by i, with $r_i \to +\infty$ and strictly increasing positive values α_i. Passing to a subsequence, either $\tau_i > 0$ for any i, which contradicts Proposition A.14, or $\tau_i \le 0$ for any i. In the remaining steps we assume $\tau_i \le 0$ and derive a contradiction. We write N_i, d_i, G_i, R_i, $D_i = \sum_j c_{ij}\operatorname{div}(h_{ij})$, κ_{ij}, $C_{t,i}$, $\tilde{H}_{t,i}$, and $\beta_{t,i}$ for the data, and $\Phi_{ij} := h^0_{ij}(P^2, -Q^2, PQ, 1)$. Let $\alpha_\infty := \lim_i \alpha_i$, so $\alpha_i < \alpha_\infty$.

Step 2. In this step we fix the discrete data and an extraction depth. By Lemma A.20(1) and Lemma 5.3, the number n_h of indices, the weights κ_{ij}, and the least total degrees I_j in x, y, z of the monomials of h^0_{ij} belong to finite sets. Passing to a subsequence, they do not depend on i, we write $\kappa_j := \kappa_{ij}$, and c_{ij} is non-decreasing with limit c^*_j for any j. Let $A_* := \sum_j c^*_j I_j$ and $\tau_* := \sum_j c^*_j \kappa_j = \lim_i \tau_i \le 0$.

If $I_j = 0$, then h^0_{ij} has a monomial t^{i_4} with $i_4 \ge 1$, so $\kappa_j = vi_4 > 0$. Since $\tau_i \le 0$, if $n_h \ge 1$, then $I_j > 0$ for some j, and $A_* > 0$.

By Theorem 7.9, applied with $-a$ in place of u at a fixed even depth $L' > 4M$, for i large the coefficient $\sum_j c_{ij} I_j + \tau_i L'/r_i$ of the exceptional divisor belongs to $[0, 1)$. Since $|\tau_i| < 2w_{\max}$ and $r_i \to +\infty$, $\sum_j c_{ij} I_j + \tau_i L'/r_i \to A_*$, so $A_* \le 1$. Let

$$\Lambda := \Big\{ \Big(1 - \frac{1}{n}\Big) c^*_j, \Big(1 - \frac{1}{n}\Big) A_* : n \ge 2,\ 1 \le j \le n_h \Big\},$$

a DCC subset of $[0, 1]$, since it is the union of finitely many increasing sequences. Let n_0 be the threshold of [Liu26, Theorem 16.21] for the circle $\mathbf{T}_{(1,1,-2)}$ on $\mathbb{A}^3$, the bound ve_0 for the residual order, and the labels in Λ. We fix an even integer $L > 4M$ with $2L \ge n_0$.

Step 3. In this step we show that $(\mathbb{A}^2, C_{t,i})$ is lc near 0 for i large. If $n_h = 0$, then $C_{t,i} = 0$. Assume $n_h \ge 1$, so $A_* > 0$ by **Step 2**. Consider Theorem 7.9 with $-a$ in place of u, so that $w = (-a, -2v + a, -v, v) = (-a, -b, -v, v)$, at the depth L. Passing to a subsequence, by Theorem 7.9(2), $d_i \to +\infty$, and for i large W_{N_i}/R_i has the model of Theorem 7.9. The chart at the possibly singular point is the hypersurface $V_i \subset \mathbb{A}^4_{x_1,y_1,z_1,\lambda}$ given by $x_1 y_1 + z_1^2 + \lambda^{d_i L - 2} = 0$, with

$$x = x_1 \lambda^{1 - aL/r_i}, \quad y = y_1 \lambda^{1 - bL/r_i}, \quad z = z_1 \lambda^{1 - vL/r_i}, \quad t = \lambda^{vL/r_i},$$

and with a finite diagonal group G'_i of order $vL[G_i : R_i] \le vLe_0$. A monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ of $h^{(\ell)}_{ij}$ becomes

$$x_1^{i_1} y_1^{i_2} z_1^{i_3} \lambda^{i_1 + i_2 + i_3 + \kappa_j L/r_i + \ell L},$$

so h_{ij} becomes $\lambda^{I_j + \kappa_j L/r_i} \mathcal{H}_{ij}$, where

$$\mathcal{H}_{ij} := \lambda^{-I_j - \kappa_j L/r_i} h_{ij}\big(x_1 \lambda^{1 - aL/r_i}, y_1 \lambda^{1 - bL/r_i}, z_1 \lambda^{1 - vL/r_i}, \lambda^{vL/r_i}\big)$$

is the polynomial with the coefficients of h_{ij} in which each monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ of $h^{(\ell)}_{ij}$, for every ℓ, is replaced by $x_1^{i_1} y_1^{i_2} z_1^{i_3} \lambda^{i_1 + i_2 + i_3 - I_j + \ell L}$. The labelled boundary of Theorem 7.9 on V_i is

$$\sum_j c_{ij} \operatorname{div}(\mathcal{H}_{ij}) + \beta'_i \operatorname{div}(\lambda),$$

where

$$\beta'_i := \sum_j c_{ij} I_j + \frac{L}{r_i} \tau_i \in [0, 1) \tag{A.118}$$

is the coefficient of the exceptional divisor and the exponents of λ in $\mathcal{H}_{ij}$ are non-negative integers. The $\mathcal{H}_{ij}$ have no monomials divisible by $x_1 y_1$.

We may assume that r_i is strictly increasing. Since $I_j + \kappa_j L/r_i \geq 0$ is the order of h_{ij} along the exceptional divisor,

$$\beta'_{i+1} - \beta'_i = \tau_{i+1}\Big(\frac{L}{r_{i+1}} - \frac{L}{r_i}\Big) + \sum_j (c_{i+1,j} - c_{ij})\Big(I_j + \frac{\kappa_j L}{r_i}\Big) \geq 0.$$

Thus β'_i is non-decreasing, and $\{\beta'_i\}$ is a DCC set. Moreover $\beta'_i \leq A_*$ and $\beta'_i \to A_*$. By Proposition 7.10(1), applied with the group bound vLe_0 and the DCC label set $\Gamma \cup \{\beta'_i\}$, the pair

$$\Big(x_1y_1 + z_1^2 = 0, \sum_j c_{ij}\operatorname{div}(\mathcal{H}_{ij}) + \beta'_i \operatorname{div}(\lambda)\Big)$$

is lc near 0 for $d_iL - 2 \geq N_1(\Gamma \cup \{\beta'_i\})$. The quasi-étale double cover $\mathbb{A}^3_{p,q,\lambda}$ of $(x_1y_1 + z_1^2 = 0)$, with $x_1 = p^2$, $y_1 = -q^2$, $z_1 = pq$, carries the lc pair

$$\Big(\mathbb{A}^3_{p,q,\lambda}, \sum_j c_{ij}\operatorname{div}(\tilde{\mathcal{H}}_{ij}) + \beta'_i \operatorname{div}(\lambda)\Big), \qquad \tilde{\mathcal{H}}_{ij} := \mathcal{H}_{ij}(p^2, -q^2, pq, \lambda), \tag{A.119}$$

with the group $\tilde{G}'_i$ of the lifts of G'_i, of order $2vL[G_i : R_i]$.

Let

$$\Theta_i := \sum_j c_j^* \operatorname{div}(\tilde{\mathcal{H}}_{ij}) + A_* \operatorname{div}(\lambda).$$

Since (A.119) is lc, the lc threshold of Θ_i at 0 is at least $\min(\beta'_i/A_*, \min_j c_{ij}/c_j^*)$, which tends to 1. By the ACC for lc thresholds with exponents in the finite set $\{c_j^*, A_*\}$ on $\mathbb{A}^3$ [dFEM11, Theorem 1.1], $(\mathbb{A}^3, \Theta_i)$ is lc near 0 for i large. Hence $(\mathbb{A}^3, (1-1/n)\Theta_i)$ is klt near 0 for any $n \geq 2$, with labels in Λ.

For $\zeta \in \mu_{2L}$, the element $(\zeta^{-2}, \zeta^{-2}, \zeta^{-2}, \zeta^2)$ of $\mu_L(-1,-1,-1,1)$ preserves V_i, since $\zeta^{2(d_iL-2)} = \zeta^{-4}$, and it fixes every R_i-invariant monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$, whose pullback is

$$x_1^{i_1}y_1^{i_2}z_1^{i_3}\lambda^{i_1+i_2+i_3+L\kappa/r_i}$$

with $\kappa/r_i \in \mathbb{Z}$, κ its $\mathbf{T}$-weight. Thus $\mu_L(-1,-1,-1,1)$ consists of lifts of the identity of G_i/R_i and is contained in G'_i. The element acting on $\mathbb{A}^3_{p,q,\lambda}$ by $(\zeta^{-1}, \zeta^{-1}, \zeta^2)$ lifts $(\zeta^{-2}, \zeta^{-2}, \zeta^{-2}, \zeta^2)$, so $\tilde{G}'_i$ contains $\mu_{2L}(-1,-1,2)$. Thus

$$|\tilde{G}'_i \cap \mathbf{T}_{(1,1,-2)}| \geq 2L \geq n_0, \qquad |\tilde{G}'_i/(\tilde{G}'_i \cap \mathbf{T}_{(1,1,-2)})| \leq v[G_i : R_i] \leq ve_0.$$

The monomial of $\tilde{\mathcal{H}}_{ij}$ coming from $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ in $h_{ij}^{(\ell)}$ is a multiple of the monomial

$$p^{2i_1+i_3}q^{2i_2+i_3}\lambda^{i_1+i_2+i_3-I_j+\ell L},$$

of $\mathbf{T}_{(1,1,-2)}$-weight $2I_j - 2\ell L$ and total degree $3(i_1+i_2+i_3) - I_j + \ell L$. For $\ell = 0$ the least total degree is $2I_j \leq 2M$. For $\ell > 0$ the total degree is at least $L - I_j > 2M$. For $\ell < 0$, $ai_1 + bi_2 + vi_3 \geq r_i - M_1$, so the total degree is at least $2(r_i - M_1)/w_{\max} > 2M$ for i large. Thus the principal block of $\tilde{\mathcal{H}}_{ij}$ for $\mathbf{T}_{(1,1,-2)}$ is

$$\tilde{\mathcal{H}}^0_{ij} = \lambda^{-I_j} h^0_{ij}(p^2\lambda, -q^2\lambda, pq\lambda, 1), \tag{A.120}$$

in which each monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of h^0_{ij} is replaced by $p^{2i_1+i_3}(-q^2)^{i_2}q^{i_3}\lambda^{i_1+i_2+i_3-I_j}$. Let

$$\Theta^0_i := \sum_j c_j^* \operatorname{div}(\tilde{\mathcal{H}}^0_{ij}) + A_* \operatorname{div}(\lambda).$$

By [Liu26, Theorem 16.21(2)], applied to $\tilde{G}'_i$ and $(1-1/n)\Theta_i$, the pair $(\mathbb{A}^3, (1-1/n)\Theta^0_i)$ is lc near 0 for any $n \geq 2$, hence so is $(\mathbb{A}^3, \Theta^0_i)$ (lc near 0 is checked on one log resolution of $(\mathbb{A}^3, \Theta^0_i)$ [KM98, Corollary 2.32(2)], so it is a closed condition on the coefficient). The mass of

$(\mathbb{A}^3, \Theta_i^0)$ along $\mathbf{T}_{(1,1,-2)}$ is $2\sum_j c_j^* I_j - 2A_* = 0$. On the cover $\lambda = \varsigma_1^2$, $U = p\varsigma_1$, $V = q\varsigma_1$ of [Liu26, Construction-Theorem 16.9],

$$\tilde{\mathcal{H}}_{ij}^0 = \varsigma_1^{-2I_j} h_{ij}^0(U^2, -V^2, UV, 1),$$

so by [Liu26, Construction-Theorem 16.9] the zero-mass homogeneous pair above is lc only if $(\mathbb{A}^2_{U,V}, \sum_j c_j^* \operatorname{div}(\Phi_{ij}(U,V)))$ is lc near 0. Let $C_{t,i}^* := \sum_j c_j^* \operatorname{div}(\Phi_{ij})$. Then $(\mathbb{A}^2, C_{t,i}^*)$ is lc near 0, and since $c_{ij} \le c_j^*$, so is $(\mathbb{A}^2, C_{t,i})$, for i large.

Step 4. In this step we pass to a compatible generic limit of the plane pairs. By **Step 3** and Lemma A.20(2), $\alpha_i \le \beta_{t,i} + C'/r_i$ for i large. By [Liu26, Lemma 2.6(4)] the coefficients of the quotients of $(\mathbb{A}^2, C_{t,i})$ by $\tilde{H}_{t,i}$ belong to the DCC set $\Psi(\Gamma_+ \cap [0,1])$, so the $\beta_{t,i}$ belong to an ACC set by Theorem 2.9(1). Passing to a subsequence, $\beta_{t,i}$ is non-increasing with limit β_∞, and $\beta_\infty \ge \alpha_\infty > 0$.

The groups $\tilde{H}_{t,i}$ are diagonal of order $2v[G_i : R_i] \le 2ve_0$ (Theorem 7.15(2)), so after passing to a subsequence they are equal to one group $\tilde{H}_\infty$, and the characters of Φ_{ij}, P^{2v}, and Q^{2v} do not depend on i. Let $\beta_{t,i}^* := \operatorname{mld}((\mathbb{A}^2 \ni 0, C_{t,i}^*)/\tilde{H}_\infty)$. The coefficients of the quotients of $(\mathbb{A}^2, C_{t,i})$ and $(\mathbb{A}^2, C_{t,i}^*)$ by $\tilde{H}_\infty$ belong to the DCC set $\Psi(\Gamma_+^* \cap [0,1])$, where $\Gamma^* := \Gamma \cup \{c_1^*, \dots, c_{n_h}^*\}$. Let l^* be the integer of Theorem 2.9(2) for $\Psi(\Gamma_+^* \cap [0,1])$. There exist prime divisors ν_i and ν_i^* over $\mathbb{A}^2$ computing $\beta_{t,i}$ and $\beta_{t,i}^*$ with $|I_{\tilde{H}_\infty}(\nu_i)| \le 2ve_0$, $|I_{\tilde{H}_\infty}(\nu_i^*)| \le 2ve_0$, and $a(\nu_i, \mathbb{A}^2, 0), a(\nu_i^*, \mathbb{A}^2, 0) \le 2ve_0 l^*$, since the branch $\mathbb{Q}$-divisor $B_{\tilde{H}_\infty}$ of $\mathbb{A}^2 \to \mathbb{A}^2/\tilde{H}_\infty$ satisfies $B_{\tilde{H}_\infty} \ge 0$ [Liu26, Lemma 2.6(2)]. Since both pairs are lc and $c_j^* \ge c_{ij} \ge \gamma$, the orders of the Φ_{ij} along ν_i and ν_i^* are at most $2ve_0 l^*/\gamma$. Since $C_{t,i} \le C_{t,i}^*$, we have $\beta_{t,i}^* \le \beta_{t,i}$. Testing ν_i^* in $C_{t,i}$ gives

$$\beta_{t,i} \le \frac{a(\nu_i^*, \mathbb{A}^2, C_{t,i})}{|I_{\tilde{H}_\infty}(\nu_i^*)|} = \beta_{t,i}^* + \frac{\sum_j (c_j^* - c_{ij}) \operatorname{mult}_{\nu_i^*} \Phi_{ij}}{|I_{\tilde{H}_\infty}(\nu_i^*)|},$$

so

$$0 \le \beta_{t,i} - \beta_{t,i}^* \le \sum_j (c_j^* - c_{ij}) \operatorname{mult}_{\nu_i^*} \Phi_{ij} \to 0.$$

Since $a(\nu_i^*, \mathbb{A}^2, 0) \le 2ve_0 l^*$, $\operatorname{mult}_{\nu_i^*} \Phi_{ij} \le 2ve_0 l^*/\gamma$, and $|I_{\tilde{H}_\infty}(\nu_i^*)| \le 2ve_0$ are integers bounded independently of i, and the labels c_j^* are fixed, the values

$$\beta_{t,i}^* = \frac{a(\nu_i^*, \mathbb{A}^2, C_{t,i}^*)}{|I_{\tilde{H}_\infty}(\nu_i^*)|}$$

belong to a finite set. Thus $\beta_{t,i}^* = \beta_\infty$ after passing to a subsequence.

Each Φ_{ij} is non-zero, since $(x,y,z) \to (P^2, -Q^2, PQ)$ maps distinct monomials of $\mathbf{T}$-weight κ_j without xy to distinct monomials, and has order at most $2M$ at 0. We apply Construction-Proposition 4.1 in case (a) with $n = 2$ and the group $\tilde{H}_\infty$ to the tuples $(\Phi_{i1}, \dots, \Phi_{in_h}, P^{2v}, Q^{2v})$. After passing to a subsequence, we obtain a compatible generic limit $(\Phi_{\infty 1}, \dots, \Phi_{\infty n_h}, P^{2v}, Q^{2v})$ over an algebraically closed field $k \supset \mathbb{C}$, where the last two entries can be taken to be P^{2v} and Q^{2v} since the corresponding entries of the tuples do not depend on i. Any subtuple of a compatible generic limit is a compatible generic limit of the subtuples, since the projections of the jet spaces map the generic points of the Z_d to the generic points of the closures of their images. Let $C_\infty := \sum_j c_j^* \operatorname{div}(\Phi_{\infty j})$ on $\mathcal{A}_k = \operatorname{Spec} k[[P,Q]]$. By Theorem 4.2(1), $(\mathcal{A}_k, C_\infty)$ is lc near 0, and by Theorem 4.2(2),

$$\operatorname{mld}((\mathcal{A}_k \ni 0, C_\infty)/\tilde{H}_\infty) = \beta_\infty.$$

Step 5. In this step we treat the case when $(\mathcal{A}_k, C_\infty)$ is klt near 0. By Theorem 4.2(1) there exists $\theta > 1$ such that $(\mathbb{A}^2, \theta C_{t,i}^*)$ is lc near 0 for any i, hence so is $(\mathbb{A}^2, \theta C_{t,i})$. By **Step 1** with this θ, for $r_i \ge r_1(a,b,v,\Gamma,e_0,\theta)$,

$$\alpha_i \ge \beta_{t,i} \ge \beta_\infty \ge \alpha_\infty > \alpha_i,$$

a contradiction.

Step 6. In this step we treat the case when $(\mathcal{A}_k, C_\infty)$ is not klt near 0. By Definition-Proposition 5.7 in case (a) with $n = 2$, applied to the pairs $(\mathbb{A}^2, C^*_{t,i})$, whose values satisfy $\beta^*_{t,i} = \beta_\infty > 0$, $(\mathcal{A}_k, C_\infty)$ has exactly one floor F_∞, which is smooth at 0, and

$$\beta_\infty = \frac{1 - b_\infty}{e_\infty},$$

where $b_\infty := \mathrm{mult}_0((C_\infty - F_\infty)|_{F_\infty}) \leq 1$ and e_∞ is the order of the image of $\tilde{H}_\infty$ in the automorphism group of the tangent line T_0F_∞. Let $\mu := 1 - b_\infty$, so $\mu = e_\infty\beta_\infty > 0$. Since F_∞ is smooth at 0, there exists $z_0 \in \{P, Q\}$ whose restriction to F_∞ is a parameter at 0. Then $\tilde{H}_\infty$ acts on T_0F_∞ through the character χ_{z_0} of z_0, so e_∞ is the order of χ_{z_0} and $\beta_\infty = \mu/e_\infty$.

Let $B_\infty := C_\infty - F_\infty + \mu\,\mathrm{div}(z_0)$. Since the coefficient of F_∞ in C_∞ is 1 and $\mu > 0$, $B_\infty \geq 0$. Since $z_0|_{F_\infty}$ is a parameter at 0, B_∞ does not contain F_∞, and $\mathrm{mult}_0(B_\infty|_{F_\infty}) = b_\infty + \mu = 1$. Thus Lemma 2.14 implies that the pair

$$\Big(\mathcal{A}_k, C_\infty + \frac{\mu}{2v}\,\mathrm{div}(z_0^{2v})\Big) = \big(\mathcal{A}_k, F_\infty + B_\infty\big)$$

is lc near 0. By Theorem 4.2(1), applied to the subtuple $(\Phi_{i1}, \dots, \Phi_{in_h}, z_0^{2v})$ with the coefficients $c^*_1, \dots, c^*_{n_h}, \mu/(2v)$,

$$\Big(\mathbb{A}^2, C^*_{t,i} + \frac{\mu}{2v}\,\mathrm{div}(z_0^{2v})\Big) \text{ is lc near 0 for any } i. \tag{A.121}$$

Let $\Xi_0 := x^v t^a$ if $z_0 = P$, and $\Xi_0 := (-1)^v y^v t^b$ if $z_0 = Q$. Then Ξ_0 is a G_i-semi-invariant monomial vanishing at 0, of $\mathbf{T}$-weight 0, and $\Xi_0(P^2, -Q^2, PQ, 1) = z_0^{2v}$. By Lemma A.19(2), applied with the labels $c^*_1, \dots, c^*_{n_h}, \mu/(2v)$, the polynomials $h^0_{i1}, \dots, h^0_{in_h}, \Xi_0$, and the mass $\tau_* \leq 0$, the pair $(W_{N_i}, D_i^{+,0})$ is lc near 0, where

$$D_i^{+,0} := \sum_j c^*_j\,\mathrm{div}(h^0_{ij}) + \frac{\mu}{2v}\,\mathrm{div}(\Xi_0), \qquad D_i^{+} := \sum_j c^*_j\,\mathrm{div}(h_{ij}) + \frac{\mu}{2v}\,\mathrm{div}(\Xi_0).$$

Let L_2 be the integer of Lemma 4.14 for $n = 4$, $n_h + 2$ functions, and the finite set $\{c^*_1, \dots, c^*_{n_h}, \mu/(2v), 1\}$. For i large, $h_{ij} - h^0_{ij} \in (x, y, z, t)^{L_2}$ by Lemma 5.3. By inversion of adjunction [Kaw07, Theorem], $(\mathbb{A}^4, \mathrm{div}(xy + z^2 + t^{N_i}) + D_i^{+,0})$ is lc near 0. Since lc near 0 is equivalent to lc of the completion [dFEM11, Proposition 2.11], Lemma 4.14, applied on $\mathbb{A}^4$ to the tuples $(xy + z^2 + t^{N_i}, h^0_{i1}, \dots, h^0_{in_h}, \Xi_0)$ and $(xy + z^2 + t^{N_i}, h_{i1}, \dots, h_{in_h}, \Xi_0)$ with the exponents $(1, c^*_1, \dots, c^*_{n_h}, \mu/(2v))$, implies that $(\mathbb{A}^4, \mathrm{div}(xy + z^2 + t^{N_i}) + D_i^{+})$ is lc near 0. By [Kaw07, Theorem] again, (W_{N_i}, D_i^{+}) is lc near 0. Since $c_{ij} \leq c^*_j$,

$$\Big(W_{N_i}, D_i + \frac{\mu}{2v}\,\mathrm{div}(\Xi_0)\Big) \text{ is lc near 0.} \tag{A.122}$$

Let m be the order of the G_i-character of Ξ_0. The class of (g, ξ, η) in $\tilde{H}_\infty = \tilde{H}_{t,i}$ acts on P^{2v} by $\eta^{2v} = g_x^v g_t^a$ and on Q^{2v} by $(g_z\xi^v/\eta)^{2v} = g_y^v g_t^b$, which are the values at g of the characters of $x^v t^a$ and $y^v t^b$. Since $L_t \to G_i$ is surjective, the character $\chi_{z_0^{2v}} = \chi_{z_0}^{2v}$ of $\tilde{H}_\infty$ has order m. The subgroup of $\tilde{H}_\infty$ given by the classes of $(1, \xi, \eta)$ with $\xi^v = 1$ and $\eta^2 = \xi^a$ has order $2v$, and it acts on P by η and on Q by η^{-1}. If $\eta = 1$, then $\xi^a = \xi^v = 1$, so $\xi = 1$ since $\gcd(a, v) = 1$. Thus the subgroup acts faithfully on z_0, $2v$ divides the order e_∞ of χ_{z_0}, and $m = e_\infty/(2v)$. By (A.122) and Lemma 5.8 with $Z = \{0\}$, $U = \Xi_0$, and the coefficient $\mu/(2v)$,

$$\alpha_i \geq \frac{\mu}{2vm} = \frac{\mu}{e_\infty} = \beta_\infty \geq \alpha_\infty > \alpha_i,$$

a contradiction. Thus α cannot increase along $\mathbf{T}$, and the sequence may have any values of N_i and d_i, so the conclusion is uniform in d. □

A.9. **Exterior canonical circles.** In this subsection, we prove Proposition A.22 and, for $\tau \le 0$, Proposition A.32. Proposition A.22 contains the bounds $\beta \le \alpha \le \beta + C_0/r$ of Theorem 7.17 for $\tau \ge 0$ and the non-increase of α along $\mathbf{T}$ for $\tau > 0$. The $\mathbb{R}$-divisor C_y and the group $\tilde{H}$ below are those of Theorem 7.17.

Proposition A.22. *Notation as in Set-up 7.3 with $w = (a, -b, -v, v)$, where $a, v > 0$, $\gcd(a, v) = 1$, and $b = a + 2v$, and for $\lambda \in \mathbb{C}^*$ let $\mathbf{T}(\lambda) \in \mathbf{T}$ be the element with entries λ^a, λ^{-b}, λ^{-v}, λ^v. Let $C_y := \sum_j c_j \operatorname{div}(h_j^0(-p^2, 1, p, q))$ on $\mathbb{A}^2_{p,q}$, let*

$$\tilde{H} := \{(g, \zeta) : g \in G,\ \zeta^b = g_y\}/\{(\mathbf{T}(\lambda), \lambda^{-1}) : \lambda^r = 1\},$$

acting on $\mathbb{A}^2_{p,q}$ by $(p, q) \to (g_z\zeta^{-v}p, g_t\zeta^v q)$, and let $\beta := \operatorname{mld}((\mathbb{A}^2 \ni 0, C_y)/\tilde{H})$ when $(\mathbb{A}^2, C_y)$ is lc near 0.

(1) Let $\theta > 1$. There exist r_0, depending only on $a, v, \Gamma, e_0, \theta$, and C_0, depending only on a, v, Γ, e_0, such that if $\tau \ge 0$, $(\mathbb{A}^2, \theta C_y)$ is lc near 0, and $r \ge r_0$, then $\beta \le \alpha \le \beta + C_0/r$ for any d.

(2) If $\tau > 0$, then α cannot increase along $\mathbf{T}$.

The upper bound is Proposition 7.13. For the lower bound we compare W_N/G with three threefold charts. Two of them contain the affine charts of a toric crepant partial resolution of the plane quotient as sections, and the third contains the plane $\mathbb{A}^2_{p,q}$ itself.

Construction-Proposition A.23. Let a, v be coprime positive integers, $b := a + 2v$, $w := (a, -b, -v, v)$, $\mathbf{T} := \mathbf{T}_{(w)}$, and for $\lambda \in \mathbb{C}^*$ let $\mathbf{T}(\lambda) \in \mathbf{T}$ be the element with entries λ^a, λ^{-b}, λ^{-v}, λ^v. Let $N \ge 3$ and G be as in Set-up 7.3(3), $R := G \cap \mathbf{T}$, $r := |R|$, and $d := v(N + 2)/r$. Let

$$W^{(t)} := (xy + z^2 + s^{vN} = 0) \subset \mathbb{A}^4_{x,y,z,s},$$
$$W^{(x)} := (\eta_x^a y + z^2 + t^N = 0) \subset \mathbb{A}^4_{\eta_x,y,z,t},$$
$$W^{(y)} := (x\eta^b + z^2 + t^N = 0) \subset \mathbb{A}^4_{x,\eta,z,t},$$

with the finite morphisms $W^{(t)} \to W_N$, $W^{(x)} \to W_N$, and $W^{(y)} \to W_N$ given by $t = s^v$, $x = \eta_x^a$, and $y = \eta^b$. We regard

$$x_1 := \frac{x}{s^a}, \quad y_1 := ys^b, \quad z_1 := zs^v, \quad u_1 := s^r,$$
$$u_2 := \eta_x^r, \quad p_x := z\eta_x^v, \quad q_x := \frac{t}{\eta_x^v}, \qquad u := \eta^r, \quad p := \frac{z}{\eta^v}, \quad q := t\eta^v$$

as rational functions on $W^{(t)}$, $W^{(x)}$, and $W^{(y)}$, respectively. The coordinates on $W^{(y)}$ carry no subscript, because $\mathbb{A}^2_{p,q} = (u = 0)$ is the plane of Proposition A.22 that carries C_y and $\tilde{H}$. Let $X_d := (x_1y_1 + z_1^2 + u_1^d = 0) \subset \mathbb{A}^4$, $S_0 := X_d \cap (u_1 = 0)$, and $\ell_1 := (y_1 = z_1 = u_1 = 0) \subset S_0$. Let

$$\tilde{H}_1 := \{(g, \zeta) : g \in G,\ \zeta^v = g_t\}/\{(\mathbf{T}(\lambda), \lambda) : \lambda^r = 1\},$$
$$\tilde{H}_2 := \{(g, \zeta) : g \in G,\ \zeta^a = g_x\}/\{(\mathbf{T}(\lambda), \lambda) : \lambda^r = 1\},$$

acting on $\mathbb{A}^4_{x_1,y_1,z_1,u_1}$ and on $\mathbb{A}^3_{u_2,p_x,q_x}$, respectively, by

$$(x_1, y_1, z_1, u_1) \to (g_x\zeta^{-a}x_1, g_y\zeta^b y_1, g_z\zeta^v z_1, \zeta^r u_1),$$
$$(u_2, p_x, q_x) \to (\zeta^r u_2, g_z\zeta^v p_x, g_t\zeta^{-v} q_x),$$

and let the group $\tilde{H}$ of Proposition A.22 act on $\mathbb{A}^3_{u,p,q}$ by $(u, p, q) \to (\zeta^r u, g_z\zeta^{-v}p, g_t\zeta^v q)$. Let $\bar{G} := G/R$ and

$$\Phi_+ := (-1)^v x^v z^a, \qquad \Phi_- := y^v t^b, \qquad \Phi_0 := zt.$$

Then the following hold.

(1) The following hold.
  (a) The number d is a positive integer.

(b) The functions above define dominant rational maps $W^{(t)} \dashrightarrow X_d$, $W^{(x)} \dashrightarrow \mathbb{A}^3_{u_2,p_x,q_x}$, and $W^{(y)} \dashrightarrow \mathbb{A}^3_{u,p,q}$, and there exist birational maps $X_d/\tilde{H}_1 \dashrightarrow W_N/G$, $\mathbb{A}^3_{u_2,p_x,q_x}/\tilde{H}_2 \dashrightarrow W_N/G$, and $\mathbb{A}^3_{u,p,q}/\tilde{H} \dashrightarrow W_N/G$ such that the morphisms from $W^{(t)}$, $W^{(x)}$, and $W^{(y)}$ to W_N/G are the composites

$$W^{(t)} \dashrightarrow X_d \to X_d/\tilde{H}_1 \dashrightarrow W_N/G,$$
$$W^{(x)} \dashrightarrow \mathbb{A}^3_{u_2,p_x,q_x} \to \mathbb{A}^3_{u_2,p_x,q_x}/\tilde{H}_2 \dashrightarrow W_N/G,$$
$$W^{(y)} \dashrightarrow \mathbb{A}^3_{u,p,q} \to \mathbb{A}^3_{u,p,q}/\tilde{H} \dashrightarrow W_N/G.$$

(c) The orders of $\tilde{H}_1$, $\tilde{H}_2$, and $\tilde{H}$ are $v[G:R]$, $a[G:R]$, and $b[G:R]$.
(d) The group $\tilde{H}_1$ preserves X_d and S_0 and acts faithfully on S_0, and $\tilde{H}_2$ and $\tilde{H}$ act faithfully on $(u_2 = 0)$ and $(u = 0)$.
(e) Near 0, X_d is normal and klt and $S_0 = (x_1y_1 + z_1^2 = 0) \subset \mathbb{A}^3_{x_1,y_1,z_1}$ is a normal klt surface.

(2) Let P be a prime divisor over W_N with center 0, and, for a rational function f on $W^{(t)}$, $W^{(x)}$, or $W^{(y)}$ (for instance the pullback of a rational function on X_d, $\mathbb{A}^3_{u_2,p_x,q_x}$, or $\mathbb{A}^3_{u,p,q}$) such that f^m is the pullback of a G-semi-invariant rational function on W_N for some positive integer m, let $[f] := \mathrm{mult}_P(f^m)/(m|I_G(P)|)$, so that $[f] = \mathrm{mult}_P f/|I_G(P)|$ if f is a G-semi-invariant regular function near $0 \in W_N$.
(a) If $[x]/a \geq [t]/v$, then $[x_1] = [x] - a[t]/v$, $[y_1] = [y] + b[t]/v$, $[z_1] = [z] + [t]$, $[u_1] = r[t]/v$, and there exists a prime divisor Q_1 over X_d with $\mathrm{center}_{X_d} Q_1 \subset \ell_1$ such that

$$\frac{a(P, W_N, 0)}{|I_G(P)|} = \frac{a(Q_1, X_d, S_0)}{|I_{\tilde{H}_1}(Q_1)|}, \qquad [f] = \frac{\mathrm{mult}_{Q_1} f}{|I_{\tilde{H}_1}(Q_1)|}$$

for any $\tilde{H}_1$-semi-invariant rational function f on X_d.
(b) If $[x]/a < [t]/v$, then $[u_2] = r[x]/a$, $[p_x] = [z] + v[x]/a$, $[q_x] = [t] - v[x]/a$, and there exists a prime divisor Q_2 over $\mathbb{A}^3_{u_2,p_x,q_x}$ with center 0 such that

$$\frac{a(P, W_N, 0)}{|I_G(P)|} = \frac{a(Q_2, \mathbb{A}^3, (u_2 = 0))}{|I_{\tilde{H}_2}(Q_2)|}, \qquad [f] = \frac{\mathrm{mult}_{Q_2} f}{|I_{\tilde{H}_2}(Q_2)|}$$

for any $\tilde{H}_2$-semi-invariant rational function f on $\mathbb{A}^3_{u_2,p_x,q_x}$, and if moreover $Nv > a$, then $2[z] \geq [x]$.
(c) If $v[y] < b[z]$, then $[u] = r[y]/b$, $[p] = [z] - v[y]/b$, $[q] = [t] + v[y]/b$, and there exists a prime divisor Q_3 over $\mathbb{A}^3_{u,p,q}$ with center 0 such that

$$\frac{a(P, W_N, 0)}{|I_G(P)|} = \frac{a(Q_3, \mathbb{A}^3, (u = 0))}{|I_{\tilde{H}}(Q_3)|}, \qquad [f] = \frac{\mathrm{mult}_{Q_3} f}{|I_{\tilde{H}}(Q_3)|}$$

for any $\tilde{H}$-semi-invariant rational function f on $\mathbb{A}^3_{u,p,q}$.

(3) The functions Φ_+, Φ_-, and Φ_0 have $\mathbf{T}$-weight zero, their characters on G factor through $\bar{G}$, and $\Phi_+(-p^2, 1, p, q) = p^b$, $\Phi_-(-p^2, 1, p, q) = q^b$, and $\Phi_0(-p^2, 1, p, q) = pq$.

(4) Let $x^i y^l z^m t^n$ be a monomial with $il = 0$ and $\mathbf{T}$-weight κ. Then, on $W^{(t)}$, $W^{(x)}$, and $W^{(y)}$, respectively,

$$x^i y^l z^m t^n = s^\kappa x_1^i y_1^l z_1^m = \eta_x^\kappa (-p_x^2 - u_2^d q_x^N)^l p_x^m q_x^n$$
$$= \eta^{-\kappa} (x\eta^a)^i p^m q^n, \qquad \text{where } x\eta^a = -p^2 - \frac{q^N}{u^d}.$$

Proof. For $\lambda^r = 1$ the element $\mathbf{T}(\lambda) \in R$ multiplies xy by λ^{-2v} and t^N by λ^{Nv}, and it preserves the equation of W_N up to a character. Thus $\lambda^{v(N+2)} = 1$ for any λ with $\lambda^r = 1$, and d is a positive integer. Part (3) is a direct calculation, as $R \subset \mathbf{T}$ acts trivially on functions of $\mathbf{T}$-weight

zero. The identities of (4), $x_1y_1+z_1^2+u_1^d=s^{2v}(xy+z^2+t^N)=0$, and $y\eta_x^b=-p_x^2-u_2^dq_x^N$ are direct calculations with $v(N+2)=rd$.

We prove the statements of (1) on X_d, S_0, and $\tilde{H}_1$, and (2)(a). The group $\hat{G}_1:=\{(g,\zeta):g\in G,\ \zeta^v=g_t\}$ of order $v|G|$ acts on $W^{(t)}$, with g acting on x,y,z and $s\to\zeta s$. The morphism $W^{(t)}\to W_N$ is the quotient by $\{(1,\zeta):\zeta^v=1\}$, the kernel of $\hat{G}_1\to G$, so $W^{(t)}/\hat{G}_1=W_N/G$. Let

$$\widetilde{X}:=(x_1y_1+z_1^2+s^{v(N+2)}=0)\subset\mathbb{A}^4_{x_1,y_1,z_1,s}.$$

Since $x=x_1s^a$, $y=y_1s^{-b}$, and $z=z_1s^{-v}$, the morphism $(x,y,z,s)\to(x_1,y_1,z_1,s)$ is an isomorphism of $W^{(t)}\cap(s\neq0)$ onto $\widetilde{X}\cap(s\neq0)$. The subgroup $R_1:=\{(\mathbf{T}(\lambda),\lambda):\lambda^r=1\}$ of order r fixes x_1, y_1, and z_1 and multiplies s by λ, and $\widetilde{X}\to X_d$, $u_1=s^r$, is the quotient by R_1. Hence $\tilde{H}_1=\hat{G}_1/R_1$, of order $v|G|/r=v[G:R]$, acts on X_d by the formula defining the action of $\tilde{H}_1$ on $\mathbb{A}^4_{x_1,y_1,z_1,u_1}$, and $X_d/\tilde{H}_1$ is birational to $W^{(t)}/\hat{G}_1=W_N/G$. If (g,ζ) fixes x_1, y_1, and z_1, then $g_x=\zeta^a$, $g_y=\zeta^{-b}$, $g_z=\zeta^{-v}$, and $g_t=\zeta^v$, so $g=\mathbf{T}(\zeta)\in R$ and $\zeta^r=1$. As S_0 is not contained in a coordinate hyperplane, $\tilde{H}_1$ acts faithfully on S_0. The hypersurface X_d is smooth outside 0, so it is normal. The surface S_0 is the quotient $\mathbb{A}^2/\mu_2$ by $(x_1,y_1,z_1)=(U^2,-V^2,UV)$, so it is klt, and (X_d,S_0) is lc near 0 by [Kaw07, Theorem]. Thus $a(F,X_d,0)=a(F,X_d,S_0)+\operatorname{mult}_F u_1>0$ for any prime divisor F over X_d with center 0, and X_d is klt near 0.

The 3-forms $\Omega:=dx\wedge dz\wedge dt/x$ and $\Omega_1:=dx_1\wedge dz_1\wedge du_1/x_1$ are the residues of $dx\wedge dy\wedge dz\wedge dt/(xy+z^2+t^N)$ and $dx_1\wedge dy_1\wedge dz_1\wedge du_1/(x_1y_1+z_1^2+u_1^d)$, so they generate ω_{W_N} and ω_{X_d} near 0, and a direct calculation shows

$$\Omega=\frac{v}{r}\cdot\frac{\Omega_1}{u_1}.$$

Let P' be a prime divisor over $W^{(t)}$ lying over P, and let Q_1 be the prime divisor over X_d that is the image of P' under the dominant rational map $W^{(t)}\dashrightarrow X_d$. The crepant pullback of $(W_N,0)$ under $t=s^v$, which is the quotient by $\{(1,\zeta):\zeta^v=1\}$ ramified along $(s=0)$ with index v, is $(W^{(t)},(1-v)\operatorname{div}(s))$, and the crepant pullback of (X_d,S_0) under $u_1=s^r$ is $(\widetilde{X},\operatorname{div}(s))$. Both are given by the pullbacks of the forms Ω and $\frac{v}{r}\Omega_1/u_1$, which are equal on $W^{(t)}\cap(s\neq0)\cong\widetilde{X}\cap(s\neq0)$. Thus

$$a(P',W^{(t)},(1-v)\operatorname{div}(s))=a(P',\widetilde{X},\operatorname{div}(s)).$$

By [Liu26, Lemma 2.6(1)(2)(5)], applied to $\hat{G}_1$ and to the image $\tilde{H}_1$ of $\hat{G}_1$,

$$\frac{a(P,W_N,0)}{|I_G(P)|}=\frac{a(P',W^{(t)},(1-v)\operatorname{div}(s))}{|I_{\hat{G}_1}(P')|}=\frac{a(Q_1,X_d,S_0)}{|I_{\tilde{H}_1}(Q_1)|}.$$

Let f be a rational function on $W^{(t)}$ with f^m the pullback of a G-semi-invariant rational function g on W_N as in (2). Replacing m by $m|G|$, which does not change $\operatorname{mult}_P(f^m)/m$, we may assume that g is G-invariant, as $g^{|G|}$ is. Then f^m is $\hat{G}_1$-invariant, hence a rational function on $\widetilde{X}/R_1$, which is birational to X_d, and

$$\frac{\operatorname{mult}_P(f^m)}{|I_G(P)|}=\frac{\operatorname{mult}_{P'}(f^m)}{|I_{\hat{G}_1}(P')|}=\frac{\operatorname{mult}_{Q_1}(f^m)}{|I_{\tilde{H}_1}(Q_1)|}.$$

This proves the formulas of (2)(a). The values $[x_1],[y_1],[z_1],[u_1]$ follow from $x_1^v=x^vt^{-a}$, $y_1^v=y^vt^b$, $z_1=zt$, and $u_1^v=t^r$. Since $[y_1],[z_1],[u_1]>0$ and $[x_1]\geq0$, the valuation of Q_1 is non-negative on the coordinate ring of X_d, and $\operatorname{center}_{X_d}Q_1\subset\ell_1$.

We prove the statements of (1) on $\tilde{H}_2$ and (2)(b). The group $\hat{G}_2:=\{(g,\zeta):g\in G,\ \zeta^a=g_x\}$ of order $a|G|$ acts on $W^{(x)}$, with g acting on y,z,t and $\eta_x\to\zeta\eta_x$. The morphism $W^{(x)}\to W_N$ is the quotient by $\{(1,\zeta):\zeta^a=1\}$, the kernel of $\hat{G}_2\to G$, so $W^{(x)}/\hat{G}_2=W_N/G$. Since $y\eta_x^b=-p_x^2-u_2^dq_x^N$, $z=p_x\eta_x^{-v}$, and $t=q_x\eta_x^v$, the morphism $(\eta_x,y,z,t)\to(\eta_x,p_x,q_x)$ is an

isomorphism of $W^{(x)}\cap(\eta_x\neq 0)$ onto $\mathbb{A}^3_{\eta_x,p_x,q_x}\cap(\eta_x\neq 0)$. The subgroup $R_2:=\{(\mathbf{T}(\lambda),\lambda):\lambda^r=1\}$ of order r fixes p_x and q_x and multiplies η_x by λ, and $\mathbb{A}^3_{\eta_x,p_x,q_x}\to\mathbb{A}^3_{u_2,p_x,q_x}$, $u_2=\eta_x^r$, is the quotient by R_2. Hence $\tilde{H}_2=\hat{G}_2/R_2$, of order $a|G|/r=a[G:R]$, acts on $\mathbb{A}^3_{u_2,p_x,q_x}$ by the formula defining its action, and $\mathbb{A}^3_{u_2,p_x,q_x}/\tilde{H}_2$ is birational to $W^{(x)}/\hat{G}_2=W_N/G$. If (g,ζ) fixes p_x and q_x, then $g_z=\zeta^{-v}$, $g_t=\zeta^v$, $g_x=\zeta^a$, and $g_y=g_z^2/g_x=\zeta^{-b}$, because the characters of xy and z^2 on G are equal. Hence $g=\mathbf{T}(\zeta)\in R$ and $\zeta^r=1$, so $(g,\zeta)\in R_2$, and $\tilde{H}_2$ acts faithfully on $(u_2=0)$.

On $W^{(x)}\cap(\eta_x\neq 0)$, a direct calculation shows

$$\Omega=\frac{a}{r}\cdot\frac{du_2\wedge dp_x\wedge dq_x}{u_2}.$$

Let P'' be a prime divisor over $W^{(x)}$ lying over P, and let Q_2 be the prime divisor over $\mathbb{A}^3_{u_2,p_x,q_x}$ that is the image of P'' under the dominant rational map $W^{(x)}\dashrightarrow\mathbb{A}^3_{u_2,p_x,q_x}$. The crepant pullback of $(W_N,0)$ under $x=\eta_x^a$, which is the quotient by $\{(1,\zeta):\zeta^a=1\}$ ramified along $(\eta_x=0)$ with index a, is $(W^{(x)},(1-a)\operatorname{div}(\eta_x))$, and the crepant pullback of $(\mathbb{A}^3_{u_2,p_x,q_x},(u_2=0))$ under $u_2=\eta_x^r$ is $(\mathbb{A}^3_{\eta_x,p_x,q_x},\operatorname{div}(\eta_x))$. Both are given by the pullbacks of the forms Ω and $\frac{a}{r}du_2\wedge dp_x\wedge dq_x/u_2$, which are equal on $(\eta_x\neq 0)$, where $W^{(x)}$ and $\mathbb{A}^3_{\eta_x,p_x,q_x}$ are identified. Thus

$$a(P'',W^{(x)},(1-a)\operatorname{div}(\eta_x))=a(P'',\mathbb{A}^3_{\eta_x,p_x,q_x},\operatorname{div}(\eta_x)).$$

By [Liu26, Lemma 2.6(1)(2)(5)], applied to $\hat{G}_2$ and to the image $\tilde{H}_2$ of $\hat{G}_2$,

$$\frac{a(P,W_N,0)}{|I_G(P)|}=\frac{a(P'',W^{(x)},(1-a)\operatorname{div}(\eta_x))}{|I_{\hat{G}_2}(P'')|}=\frac{a(Q_2,\mathbb{A}^3,(u_2=0))}{|I_{\tilde{H}_2}(Q_2)|}.$$

Let f be a rational function on $W^{(x)}$ with f^m the pullback of a G-semi-invariant rational function g on W_N. Replacing m by $m|G|$, we may assume that g is G-invariant. Then f^m is $\hat{G}_2$-invariant, hence a rational function on $\mathbb{A}^3_{\eta_x,p_x,q_x}/R_2=\mathbb{A}^3_{u_2,p_x,q_x}$, and

$$\frac{\operatorname{mult}_P(f^m)}{|I_G(P)|}=\frac{\operatorname{mult}_{P''}(f^m)}{|I_{\hat{G}_2}(P'')|}=\frac{\operatorname{mult}_{Q_2}(f^m)}{|I_{\tilde{H}_2}(Q_2)|}.$$

This proves the formulas of (2)(b). The values $[u_2],[p_x],[q_x]$ follow from $u_2^a=x^r$, $p_x^a=z^ax^v$, and $q_x^a=t^ax^{-v}$. They are positive, as $[x]>0$ and $[x]/a<[t]/v$, so Q_2 has center 0.

We prove the statements of (1) on $\tilde{H}$ and (2)(c). The group $\hat{G}_3:=\{(g,\zeta):g\in G,\ \zeta^b=g_y\}$ of order $b|G|$ acts on $W^{(y)}$, with g acting on x,z,t and $\eta\to\zeta\eta$. The morphism $W^{(y)}\to W_N$ is the quotient by $\{(1,\zeta):\zeta^b=1\}$, the kernel of $\hat{G}_3\to G$, so $W^{(y)}/\hat{G}_3=W_N/G$. Since $x\eta^a=-p^2-q^N/u^d$, $z=p\eta^v$, and $t=q\eta^{-v}$, the morphism $(x,\eta,z,t)\to(\eta,p,q)$ is an isomorphism of $W^{(y)}\cap(\eta\neq 0)$ onto $\mathbb{A}^3_{\eta,p,q}\cap(\eta\neq 0)$. The subgroup $R_3:=\{(\mathbf{T}(\lambda),\lambda^{-1}):\lambda^r=1\}$ of order r fixes p and q and multiplies η by λ^{-1}, and $\mathbb{A}^3_{\eta,p,q}\to\mathbb{A}^3_{u,p,q}$, $u=\eta^r$, is the quotient by R_3. Hence $\tilde{H}=\hat{G}_3/R_3$, of order $b|G|/r=b[G:R]$, acts on $\mathbb{A}^3_{u,p,q}$ by the formula defining its action, and $\mathbb{A}^3_{u,p,q}/\tilde{H}$ is birational to $W^{(y)}/\hat{G}_3=W_N/G$. The faithfulness of $\tilde{H}$ on $(u=0)$ is part of Proposition 7.13, applied with $-a$ in place of a.

The generator $dy\wedge dz\wedge dt/y$ of ω_{W_N} near 0 equals $-\Omega$, and on $W^{(y)}\cap(\eta\neq 0)$ a direct calculation shows

$$-\Omega=\frac{b}{r}\cdot\frac{du\wedge dp\wedge dq}{u}.$$

Let P''' be a prime divisor over $W^{(y)}$ lying over P, and let Q_3 be the prime divisor over $\mathbb{A}^3_{u,p,q}$ that is the image of P''' under the dominant rational map $W^{(y)}\dashrightarrow\mathbb{A}^3_{u,p,q}$. The crepant pullback of $(W_N,0)$ under $y=\eta^b$, which is the quotient by $\{(1,\zeta):\zeta^b=1\}$ ramified along $(\eta=0)$ with index b, is $(W^{(y)},(1-b)\operatorname{div}(\eta))$, and the crepant pullback of $(\mathbb{A}^3_{u,p,q},(u=0))$ under $u=\eta^r$ is

$(\mathbb{A}^3_{\eta,p,q}, \operatorname{div}(\eta))$. Both are given by the pullbacks of the forms $-\Omega$ and $\frac{b}{r} du \wedge dp \wedge dq/u$, which are equal on $(\eta \neq 0)$, where $W^{(y)}$ and $\mathbb{A}^3_{\eta,p,q}$ are identified. Thus

$$a(P''', W^{(y)}, (1-b)\operatorname{div}(\eta)) = a(P''', \mathbb{A}^3_{\eta,p,q}, \operatorname{div}(\eta)).$$

By [Liu26, Lemma 2.6(1)(2)(5)], applied to $\hat{G}_3$ and to the image $\tilde{H}$ of $\hat{G}_3$,

$$\frac{a(P, W_N, 0)}{|I_G(P)|} = \frac{a(P''', W^{(y)}, (1-b)\operatorname{div}(\eta))}{|I_{\hat{G}_3}(P''')|} = \frac{a(Q_3, \mathbb{A}^3, (u=0))}{|I_{\tilde{H}}(Q_3)|}.$$

Let f be a rational function on $W^{(y)}$ with f^m the pullback of a G-semi-invariant rational function g on W_N. Replacing m by $m|G|$, we may assume that g is G-invariant. Then f^m is $\hat{G}_3$-invariant, hence a rational function on $\mathbb{A}^3_{\eta,p,q}/R_3 = \mathbb{A}^3_{u,p,q}$, and

$$\frac{\operatorname{mult}_P(f^m)}{|I_G(P)|} = \frac{\operatorname{mult}_{P'''}(f^m)}{|I_{\hat{G}_3}(P''')|} = \frac{\operatorname{mult}_{Q_3}(f^m)}{|I_{\tilde{H}}(Q_3)|}.$$

This proves the formulas of (2)(c). The values $[u], [p], [q]$ follow from $u^b = y^r$, $p^b = z^b y^{-v}$, and $q^b = t^b y^v$. They are positive, as $[y] > 0$ and $v[y] < b[z]$, so Q_3 has center 0.

It remains to prove $2[z] \geq [x]$ in (2)(b). Since $[x] > 0$ and $Nv > a$, we have $N[t] > Nv[x]/a > [x]$. If $2[z] < [x]$, then $[x] + [y] > 2[z]$ and $N[t] > 2[z]$, and $z^2 = -xy - t^N$ implies $2[z] \geq \min([x] + [y], N[t]) > 2[z]$, a contradiction. □

Construction-Proposition A.24. Notation as in Construction-Proposition A.23. Let c_j be finitely many positive real numbers and h^0_j non-zero G-semi-invariant $\mathbf{T}$-homogeneous polynomials without monomials divisible by xy, of $\mathbf{T}$-weights κ_j, such that $\tau := \sum_j c_j \kappa_j \geq 0$. Let

$$\varphi_j := h^0_j(-p^2, 1, p, q), \qquad H^t_j := h^0_j(x_1, y_1, z_1, 1), \qquad H^x_j := h^0_j(1, -p_x^2, p_x, q_x),$$

$C_y := \sum_j c_j \operatorname{div}(\varphi_j)$ on $\mathbb{A}^2_{p,q}$, and

$$C_t := \sum_j c_j \operatorname{div}(H^t_j|_{S_0}) + \frac{\tau}{b}\operatorname{div}(y_1|_{S_0}) \text{ on } S_0, \qquad C_x := \sum_j c_j \operatorname{div}(H^x_j) + \frac{2\tau}{b}\operatorname{div}(p_x) \text{ on } \mathbb{A}^2_{p_x,q_x}.$$

Assume that $(\mathbb{A}^2, C_y)$ is lc near 0, and let $\beta := \operatorname{mld}((\mathbb{A}^2 \ni 0, C_y)/\tilde{H})$. Then the following hold.

(1) The following hold.
   (a) For any integer κ, distinct monomials of $\mathbf{T}$-weight κ without monomial factor xy have distinct images under each of the substitutions $(x, y, z, t) \to (-p^2, 1, p, q)$, $(x_1, y_1, z_1, 1)$, and $(1, -p_x^2, p_x, q_x)$.
   (b) The functions φ_j, $H^t_j|_{S_0}$, and H^x_j are non-zero, and $C_t \geq 0$ and $C_x \geq 0$.

(2) The pair (S_0, C_t) is lc near ℓ_1, and $a(Q, S_0, C_t)/|I_{\tilde{H}_1}(Q)| \geq \beta$ for any prime divisor Q over S_0 with $\operatorname{center}_{S_0} Q \subset \ell_1$.

(3) The pair $(\mathbb{A}^2, C_x)$ is lc near 0, and $\operatorname{mld}((\mathbb{A}^2 \ni 0, C_x)/\tilde{H}_2) \geq \beta$.

Proof. (1) The exponent n of a monomial $x^i y^l z^m t^n$ of $\mathbf{T}$-weight κ is determined by i, l, m, and κ. If $il = 0$, the three substitutions send the monomial to $(-1)^i p^{2i+m} q^n$, to $x_1^i y_1^l z_1^m$, and to $(-1)^l p_x^{2l+m} q_x^n$. Using $ai - bl - vm + vn = \kappa$, a direct calculation shows that distinct monomials of $\mathbf{T}$-weight κ have distinct images. As the monomials $x_1^i y_1^l z_1^m$ with $il = 0$ are linearly independent on S_0, the three functions are non-zero. Since $\tau \geq 0$, we have $C_t \geq 0$ and $C_x \geq 0$.

(2) and (3) The subgroup $\mu_b := \{(1, \zeta) : \zeta^b = 1\} \subset \tilde{H}$ acts on $\mathbb{A}^2_{p,q}$ by $(\zeta^{-v} p, \zeta^v q)$. As $\gcd(b, v) = 1$, no non-trivial element of μ_b fixes a curve pointwise, and $S_b := \mathbb{A}^2/\mu_b$ is the surface $(\Phi_+\Phi_- = \Phi_0^b) \subset \mathbb{A}^3$, where we use $\Phi_+ = p^b$, $\Phi_- = q^b$, $\Phi_0 = pq$ as coordinates. The group $\tilde{H}/\mu_b$ acts faithfully on S_b through the torus, and it is the group $\bar{G}$ acting through the characters of Φ_+, Φ_-, Φ_0. Let Δ be the descent of C_y to S_b. Then $(\mathbb{A}^2, C_y)/\tilde{H}$ is the crepant quotient $(S_b, \Delta)/\bar{G}$, and $\operatorname{mld}((S_b \ni 0, \Delta)/\bar{G}) = \beta$ by [Liu26, Lemma 2.6(5)] and the formula at the end of [Liu26, Lemma 2.6].

The lattice of S_b is $\mathbb{Z}^2+\mathbb{Z}\frac{1}{b}(1,-1)$ and the cone of S_b is $\mathbb{R}^2_{\geq 0}$, in the coordinates dual to the exponents of p and q. The vector $(a/b, 2v/b) = (0,1) + a\cdot\frac{1}{b}(1,-1)$ belongs to the lattice and $a/b + 2v/b = 1$. Let $\pi : Y \to S_b$ be the toric morphism defined by inserting the ray through $(a/b, 2v/b)$. Then π is crepant, $\bar{G}$ acts on Y, and $E := \pi^{-1}(0)$ is a prime divisor. The fan of Y has two maximal cones $\sigma_1 := \mathrm{cone}((1,0),(a/b,2v/b))$ and $\sigma_2 := \mathrm{cone}((a/b,2v/b),(0,1))$, of index $2v$ and a in the lattice $\mathbb{Z}^2+\mathbb{Z}\frac{1}{b}(1,-1)$. The corresponding affine charts are $V_1 := S_0/\mu_v$, of type A_{2v-1}, and $V_2 := \mathbb{A}^2_{p_x,q_x}/\mu_a$, of type A_{a-1}, where $\mu_v = \{(1,\zeta) : \zeta^v = 1\} \subset \tilde{H}_1$ and $\mu_a = \{(1,\zeta) : \zeta^a = 1\} \subset \tilde{H}_2$, with

$$(\Phi_+,\Phi_-,\Phi_0) = ((-1)^v x_1^v z_1^a, y_1^v, z_1) \text{ on } V_1,$$
$$(\Phi_+,\Phi_-,\Phi_0) = ((-1)^v p_x^a, (-1)^v p_x^{2v} q_x^b, p_x q_x) \text{ on } V_2.$$

The characters of the right-hand sides under $\tilde{H}_1$ and $\tilde{H}_2$ are those of Φ_+,Φ_-,Φ_0 on G, so the quotient maps $S_0 \to V_1$ and $\mathbb{A}^2_{p_x,q_x} \to V_2$ are equivariant for $\tilde{H}_1 \to \bar{G}$ and $\tilde{H}_2 \to \bar{G}$. As $\gcd(a,v) = \gcd(b,v) = 1$, no non-trivial element of μ_v or μ_a fixes a curve pointwise. The curve E is the image of ℓ_1 in V_1 and of $(p_x = 0)$ in V_2. The formulas for (Φ_+,Φ_-,Φ_0) identify the charts: on V_1, the image of the curve $\ell_1 = (y_1 = z_1 = 0)$ gives the ray $(a/b, 2v/b)$, and the image of the curve $(x_1 = z_1 = 0)$ gives the ray $(1,0)$.

Let Δ_Y be the crepant pullback of Δ, so $K_Y + \Delta_Y = \pi^*(K_{S_b} + \Delta)$. We show that the pullbacks of Δ_Y to S_0 and to $\mathbb{A}^2_{p_x,q_x}$ are C_t and C_x. Since π is toric and crepant, $\Delta_Y - \partial Y = \pi^*(\Delta - \partial S_b)$, where ∂ denotes the toric boundary, and

$$\Delta - \partial S_b = \sum_j \frac{c_j}{b}\,\mathrm{div}_{S_b}(\varphi_j^b) - \mathrm{div}_{S_b}(\Phi_0)$$

is an $\mathbb{R}$-linear combination of principal divisors. On a finite cover of S_0 on which y_1 has a b-th root $y_1^{1/b}$, we have $q = y_1^{v/b}$ and $p = z_1 y_1^{-v/b}$, and the $\mathbf{T}$-homogeneity of h_j^0 implies $\varphi_j = y_1^{\kappa_j/b} H_j^t$, that is, $\varphi_j^b = y_1^{\kappa_j}(H_j^t)^b$. The toric boundary of V_1 pulls back to $\mathrm{div}(z_1|_{S_0})$ and $\Phi_0 = z_1$. Thus the pullback of Δ_Y to S_0 is

$$\sum_j c_j\,\mathrm{div}(H_j^t|_{S_0}) + \frac{\tau}{b}\,\mathrm{div}(y_1|_{S_0}) - \mathrm{div}(z_1|_{S_0}) + \mathrm{div}(z_1|_{S_0}) = C_t.$$

Replacing S_0, $\varphi_j^b = y_1^{\kappa_j}(H_j^t)^b$, and $\Phi_0 = z_1$ by $\mathbb{A}^2_{p_x,q_x}$, $\varphi_j^b = (-p_x^2)^{\kappa_j}(H_j^x)^b$, and $\Phi_0 = p_x q_x$, the argument of the last paragraph shows that the pullback of Δ_Y to $\mathbb{A}^2_{p_x,q_x}$ is C_x.

The pair (Y,Δ_Y) is lc near E, because (S_b,Δ) is lc near 0, and so (S_0,C_t) and $(\mathbb{A}^2_{p_x,q_x},C_x)$ are lc near ℓ_1 and near $(p_x = 0)$. Let Q be a prime divisor over S_0 with $\mathrm{center}_{S_0} Q \subset \ell_1$. By [Liu26, Lemma 2.6(2)(5)] applied to $S_0 \to V_1$ and to $\bar{G}$, the number $a(Q,S_0,C_t)/|I_{\tilde{H}_1}(Q)|$ is the log discrepancy of the image $\bar{Q}$ of Q with respect to the crepant quotient $(Y,\Delta_Y)/\bar{G}$. The center of $\bar{Q}$ is contained in $E/\bar{G}$, and $(Y,\Delta_Y)/\bar{G}$ is crepant over $(S_b,\Delta)/\bar{G}$. Hence the log discrepancy is at least $\mathrm{mld}((S_b \ni 0,\Delta)/\bar{G}) = \beta$. Now let Q' be a prime divisor over $\mathbb{A}^2_{p_x,q_x}$ with center 0, so that $\mathrm{center}_{\mathbb{A}^2} Q' \subset (p_x = 0)$. By [Liu26, Lemma 2.6(2)(5)] applied to $\mathbb{A}^2_{p_x,q_x} \to V_2$ and to $\bar{G}$, the number $a(Q',\mathbb{A}^2,C_x)/|I_{\tilde{H}_2}(Q')|$ is the log discrepancy of the image $\bar{Q}'$ of Q' with respect to $(Y,\Delta_Y)/\bar{G}$. The center of $\bar{Q}'$ is contained in $E/\bar{G}$, so

$$\frac{a(Q',\mathbb{A}^2,C_x)}{|I_{\tilde{H}_2}(Q')|} \geq \mathrm{mld}((S_b \ni 0,\Delta)/\bar{G}) = \beta.$$

Taking the minimum over Q', we obtain $\mathrm{mld}((\mathbb{A}^2 \ni 0, C_x)/\tilde{H}_2) \geq \beta$. □

Lemma A.25. *Notation and assumptions as in Construction-Proposition A.24. Let P be a prime divisor over W_N with center 0, $[\cdot]$ as in Construction-Proposition A.23(2), and h_j G-semi-invariant regular functions near $0 \in W_N$, one for each j. Assume that one of the following holds.*

(1) $[x]/a \geq [t]/v$, *and there exist* $\tilde{H}_1$*-semi-invariant polynomials* $B_j(x_1, y_1, z_1, u_1)$ *with* $B_j(x_1, y_1, z_1, 0) = H_j^t$ *and* $[h_j] = \kappa_j[t]/v + [B_j]$ *for any* j.

(2) $[x]/a < [t]/v$, $Nv > a$, *and there exist* $\tilde{H}_2$*-semi-invariant polynomials* $B_j(u_2, p_x, q_x)$ *with* $B_j(0, p_x, q_x) = H_j^x$ *and* $[h_j] = \kappa_j[x]/a + [B_j]$ *for any* j.

(3) $v[y] < b[z]$, *and there exist* $\tilde{H}$*-semi-invariant polynomials* $B_j(u, p, q)$ *with* $B_j(0, p, q) = \varphi_j$ *and* $[h_j] = -\kappa_j[y]/b + [B_j]$ *for any* j.

Then

$$\frac{a(P, W_N, \sum_j c_j \operatorname{div}(h_j))}{|I_G(P)|} \geq \beta.$$

Proof. Let $D' := \sum_j c_j \operatorname{div}(h_j)$, so that $a(P, W_N, D')/|I_G(P)| = a(P, W_N, 0)/|I_G(P)| - \sum_j c_j[h_j]$.

Assume (1), and let Q_1 be as in Construction-Proposition A.23(2)(a). Since $[y_1] = [y] + b[t]/v$,

$$\frac{a(P, W_N, D')}{|I_G(P)|} = \frac{a\big(Q_1, X_d, S_0 + \sum_j c_j \operatorname{div}(B_j) + \frac{\tau}{b}\operatorname{div}(y_1)\big)}{|I_{\tilde{H}_1}(Q_1)|} + \frac{\tau[y]}{b}. \tag{A.123}$$

As $\tau \geq 0$, it suffices to bound the first term from below by β. We take the data of Lemma 5.5 to be $X := X_d \subset \mathbb{A}^4$, defined by the $\tilde{H}_1$-semi-invariant polynomial $x_1y_1 + z_1^2 + u_1^d$, the group $\tilde{H}_1$ in place of both G and H, $u := u_1$ with the coefficient of S_0 equal to one, and the $\mathbb{R}$-ideal $\mathfrak{a} := \prod_j B_j^{c_j} \cdot y_1^{\tau/b}$ (without the last factor if $\tau = 0$). The restriction of $\mathfrak{a}$ to S_0 defines C_t. The hypotheses of Lemma 5.5 hold by Construction-Propositions A.23(1) and A.24. We distinguish three cases.

(i) If $\operatorname{center}_{X_d} Q_1 = \{0\}$, Lemma 5.5 bounds the first term of (A.123) from below by $\operatorname{mld}((S_0 \ni 0, C_t)/\tilde{H}_1)$, which is at least β by Construction-Proposition A.24(2).

(ii) If $\operatorname{center}_{X_d} Q_1$ is a closed point $o = (x_o, 0, 0, 0)$ with $x_o \neq 0$, let $H_o \subset \tilde{H}_1$ be the stabilizer of o. Then H_o acts trivially on x_1, so it acts diagonally in the coordinates $(x_1 - x_o, y_1, z_1, u_1)$, in which $x_1y_1 + z_1^2 + u_1^d$, u_1, and the B_j and y_1 are H_o-semi-invariant polynomials. The projection

$$X_d \cap (x_1 \neq 0) \to (x_1 \neq 0) \subset \mathbb{A}^3_{x_1, z_1, u_1}$$

is an $\tilde{H}_1$-equivariant isomorphism with inverse $y_1 = -(z_1^2 + u_1^d)/x_1$, so X_d and S_0 are smooth near o. An element of H_o acting trivially on S_0 near o acts trivially on the irreducible surface S_0, so H_o acts faithfully on S_0 by Construction-Proposition A.23(1)(d). The inertia group of a prime divisor with center o is contained in H_o, so $I_{H_o}(Q_1) = I_{\tilde{H}_1}(Q_1)$. Hence Lemma 5.5 in the coordinates $(x_1 - x_o, y_1, z_1, u_1)$ centered at o, with H_o in place of both G and H, bounds the first term of (A.123) from below by $\operatorname{mld}((S_0 \ni o, C_t)/H_o)$, and $\operatorname{mld}((S_0 \ni o, C_t)/H_o)$ is the minimum of $a(Q, S_0, C_t)/|I_{\tilde{H}_1}(Q)|$ over the prime divisors Q over S_0 with center o. It is at least β by Construction-Proposition A.24(2).

(iii) If $\operatorname{center}_{X_d} Q_1 = \ell_1$, the last sentence of Lemma 5.5, with the $\tilde{H}_1$-invariant curve $\ell_1 \ni 0$, bounds the first term of (A.123) from below by the minimal log discrepancy of $(S_0, C_t)/\tilde{H}_1$ at the image of the generic point of ℓ_1, that is, by the left-hand side of

$$\frac{a(\ell_1, S_0, C_t)}{|I_{\tilde{H}_1}(\ell_1)|} \geq \beta,$$

and the inequality holds by Construction-Proposition A.24(2).

Assume (2), and let Q_2 be as in Construction-Proposition A.23(2)(b). Since $[p_x] = [z] + v[x]/a$,

$$\frac{a(P, W_N, D')}{|I_G(P)|} = \frac{a\big(Q_2, \mathbb{A}^3, (u_2 = 0) + \sum_j c_j \operatorname{div}(B_j) + \frac{2\tau}{b}\operatorname{div}(p_x)\big)}{|I_{\tilde{H}_2}(Q_2)|} + \frac{\tau(2[z] - [x])}{b}. \tag{A.124}$$

The second term is non-negative by Construction-Proposition A.23(2)(b). Lemma 5.5 with $X := \mathbb{A}^3_{u_2,p_x,q_x}$, the group $\tilde{H}_2$, $u := u_2$ with coefficient one, the section $\mathbb{A}^2_{p_x,q_x} = (u_2 = 0)$ in place of S_0, and $\mathfrak{a} := \prod_j B_j^{c_j} \cdot p_x^{2\tau/b}$ bounds the first term from below by $\mathrm{mld}((\mathbb{A}^2 \ni 0, C_x)/\tilde{H}_2) \geq \beta$.

Assume (3), and let Q_3 be as in Construction-Proposition A.23(2)(c). Since $[u] = r[y]/b$,

$$\frac{a(P, W_N, D')}{|I_G(P)|} = \frac{a\big(Q_3, \mathbb{A}^3, (u=0) + \sum_j c_j \operatorname{div}(B_j)\big)}{|I_{\tilde{H}}(Q_3)|} + \frac{\tau[y]}{b},$$

and Lemma 5.5 with $X := \mathbb{A}^3_{u,p,q}$, the group $\tilde{H}$, u with coefficient one, the section $\mathbb{A}^2_{p,q} = (u = 0)$, and $\mathfrak{a} := \prod_j B_j^{c_j}$ bounds the first term from below by β. □

Lemma A.26. *Let a, v be coprime positive integers, $b := a + 2v$, $w := (a, -b, -v, v)$, $\mathbf{T} := \mathbf{T}_{(w)}$, and $N \geq 3$ an integer with $Nv > a$. Let c_j be finitely many positive real numbers and h_j^0 non-zero $\mathbf{T}$-homogeneous polynomials without monomials divisible by xy, of $\mathbf{T}$-weights κ_j, such that $\tau := \sum_j c_j\kappa_j \geq 0$. If $(\mathbb{A}^2, \sum_j c_j \operatorname{div}(h_j^0(-p^2, 1, p, q)))$ is lc near 0, then $(W_N, \sum_j c_j \operatorname{div}(h_j^0))$ is lc near 0.*

Proof. We use Construction-Propositions A.23 and A.24 with the trivial group $G = \{1\}$, so that $r = 1$, $d = v(N+2)$, and any $\mathbf{T}$-homogeneous polynomial is G-semi-invariant. Let $D^0 := \sum_j c_j \operatorname{div}(h_j^0)$, β as in Construction-Proposition A.24, and P a prime divisor over W_N with center 0. If $[x]/a \geq [t]/v$, then $h_j^0 = s^{\kappa_j} H_j^t$ by Construction-Proposition A.23(4), so $[h_j^0] = \kappa_j[t]/v + [H_j^t]$, and Lemma A.25(1) with $B_j := H_j^t$ shows $a(P, W_N, D^0) \geq \beta \geq 0$. If $[x]/a < [t]/v$, then $h_j^0 = \eta_x^{\kappa_j} B_j$ with $B_j := h_j^0(1, -p_x^2 - u_2^d q_x^N, p_x, q_x)$ by Construction-Proposition A.23(4), and Lemma A.25(2) shows $a(P, W_N, D^0) \geq \beta \geq 0$.

If (W_N, D^0) is not lc near 0, then it has a prime divisor with center 0 and negative log discrepancy (cf. [KM98, Corollary 2.31(1)]), which contradicts the first paragraph. □

Lemma A.27. *Notation as in Set-up 7.3 with $w = (a, -b, -v, v)$, where $a, v > 0$, $\gcd(a, v) = 1$, and $b = a + 2v$, and as in Construction-Proposition A.23, and let $\epsilon_0 := 1/(8b^2ve_0)$. Let P be a prime divisor over W_N with center 0, and let $[\cdot]$ be as in Construction-Proposition A.23(2). Then*

$$v[x] + a[z] \geq \frac{1}{e_0}, \qquad v[y] + b[t] \geq \frac{1}{e_0}, \qquad [z] + [t] \geq \frac{1}{e_0},$$

and $\min([x]/a, [t]/v)$ and $\min([y]/b, [z]/v)$ are not both at most ϵ_0.

Proof. Let $\Phi \in \{\Phi_+, \Phi_-, \Phi_0\}$ and $m := [G : R] \leq e_0$. The character of Φ factors through $\bar{G}$, so Φ^m is G-invariant and descends to a regular function on W_N/G vanishing at the image of 0. The ramification index of P over the image of P in W_N/G is $|I_G(P)|$ [Liu26, Lemma 2.6(1)], so $m[\Phi]$ is a positive integer, and $[\Phi] \geq 1/e_0$. This is the display, as $[\Phi_+] = v[x] + a[z]$, $[\Phi_-] = v[y] + b[t]$, and $[\Phi_0] = [z] + [t]$.

Assume that both minima are at most ϵ_0. If $[x] \leq a\epsilon_0$ and $[z] \leq v\epsilon_0$, then $v[x] + a[z] \leq 2av\epsilon_0 < 1/e_0$. If $[t] \leq v\epsilon_0$ and $[y] \leq b\epsilon_0$, then $v[y] + b[t] \leq 2bv\epsilon_0 < 1/e_0$. If $[t] \leq v\epsilon_0$ and $[z] \leq v\epsilon_0$, then $[z] + [t] \leq 2v\epsilon_0 < 1/e_0$. If $[x] \leq a\epsilon_0$ and $[y] \leq b\epsilon_0$, then the display implies $a[z] \geq 1/e_0 - av\epsilon_0 \geq 1/(2e_0)$ and $b[t] \geq 1/e_0 - bv\epsilon_0 \geq 1/(2e_0)$, so

$$[x] + [y] \leq (a+b)\epsilon_0 < \frac{1}{4be_0} < \min\left(\frac{1}{ae_0}, \frac{3}{2be_0}\right) \leq \min(2[z], N[t]),$$

and $xy = -z^2 - t^N$ implies $[x] + [y] \geq \min(2[z], N[t])$. Each case is a contradiction. □

Proof of Proposition A.22. The proof is long, so we divide it into several steps. Let γ be the least positive element of Γ, $M := \lceil 2/\gamma \rceil$, $D^0 := \sum_j c_j \operatorname{div}(h_j^0)$, and $\epsilon_0 := 1/(8b^2ve_0)$. Only the h_j with $c_j > 0$ which are not units near 0 are considered.

Step 1. In this step we bound the equations h_j and show that $(W_N, \theta D)$ is lc near 0 for large r.

By Lemma 5.3 with M and $\max_i|w_i| = b$, if $r > 2bM$ then $\operatorname{mult}_0 h_j \le M$, $|\kappa_j| \le bM$, and any monomial of $h_j - h_j^0$ has total degree at least $(r - bM)/b$ and $\mathbf{T}$-weight κ with $|\kappa - \kappa_j| \ge r$, so $\kappa \ge r - bM$ or $\kappa \le -(r - bM)$. Since (W_N, D) is klt near 0 by Set-up 7.3(5), Lemma A.1(3) gives $\sum_j c_j \operatorname{mult}_0 h_j < 2$, and the number of indices j is less than $2/\gamma \le M$.

In Proposition 7.13, applied with $-a$ in place of a, the group and the $\mathbb{R}$-divisor are $\tilde{H}$ and C_y of Proposition A.22. It provides r_0' and C_0, depending only on a, v, Γ, e_0, such that if $r \ge r_0'$ then $(\mathbb{A}^2, C_y)$ is lc near 0 and $\alpha \le \beta + C_0/r$, and the values β belong to an ACC set depending only on a, v, Γ, e_0.

Assume that $\tau \ge 0$, that $(\mathbb{A}^2, \theta C_y)$ is lc near 0, and that $r > 2bM$. Then $Nv > a$, as $Nv = rd - 2v$ and $rd - 2v \ge r - 2v > a$. Lemma A.26 with the labels θc_j shows that $(W_N, \theta D^0)$ is lc near 0. By [Kaw07, Theorem] and [dFEM11, Proposition 2.11], the pair

$$\Big(\operatorname{Spec}\mathbb{C}[[x,y,z,t]], (xy + z^2 + t^N)\cdot\prod_j (h_j^0)^{\theta c_j}\Big)$$

is lc. Let L_1 be the maximum of the integers L of Lemma 4.14 for $n = 4$, at most $M+1$ functions, and the DCC set $\theta(\Gamma\backslash\{0\}) \cup \{1\}$. The integer L_1 depends only on Γ and θ. If $(r - bM)/b \ge L_1$, then $h_j - h_j^0 \in (x,y,z,t)^{L_1}$, and Lemma 4.14, with the equation $xy + z^2 + t^N$ kept with label one, shows that

$$\Big(\operatorname{Spec}\mathbb{C}[[x,y,z,t]], (xy + z^2 + t^N)\cdot\prod_j h_j^{\theta c_j}\Big)$$

is lc. By [dFEM11, Proposition 2.11] and [Kaw07, Theorem] again, $(W_N, \theta D)$ is lc near 0.

Let $\Theta_1 := 2\theta/(\theta - 1)$, $\Theta_2 := 2/(\gamma(\theta - 1))$, and

$$r_0 := \max\Big\{r_0',\ 2bM + 1,\ bM + bL_1,\ bM + \frac{\Theta_2}{\epsilon_0} + 1,\ bM + \frac{2v\Theta_2}{a\epsilon_0} + 1,\ 2v + \frac{4(\Theta_1 + \Theta_2)}{\epsilon_0} + 1\Big\}.$$

It depends only on $a, v, \Gamma, e_0, \theta$. If $r \ge r_0$, then $r \ge r_0'$, $r > 2bM$, $(r - bM)/b \ge L_1$, and

$$\frac{\Theta_2}{r - bM} < \epsilon_0, \qquad \frac{2v\Theta_2}{r - bM} < a\epsilon_0, \qquad (r - 2v)\epsilon_0 > 4\Theta_1, \qquad (r - 2v)\epsilon_0 > 4\Theta_2.$$

Until the end of the proof of (1), we assume $r \ge r_0$.

Step 2. In this step we bound the values $[f]$ at a prime divisor P over W_N with center 0 and $a(P, W_N, D) \le 2|I_G(P)|$, with $[\cdot]$ as in Construction-Proposition A.23(2).

Since $(W_N, \theta D)$ is lc near 0, we have $a(P, W_N, 0) \ge \theta\sum_j c_j \operatorname{mult}_P h_j$. Combined with $a(P, W_N, 0) - \sum_j c_j \operatorname{mult}_P h_j \le 2|I_G(P)|$ and $c_j \ge \gamma$, this implies

$$\frac{a(P, W_N, 0)}{|I_G(P)|} \le \Theta_1, \qquad [h_j] \le \Theta_2. \tag{A.125}$$

As $(W_N, \theta D^0)$ is lc near 0, the first bound of (A.125) implies $\theta\sum_j c_j[h_j^0] \le \Theta_1$, and $[h_j^0] \le \Theta_2$.

Let $S := W_N \cap (t = 0) = (xy + z^2 = 0) \subset \mathbb{A}^3$ and

$$\Delta_S := \frac{1}{2}\operatorname{div}(x|_S) + \frac{1}{2}\operatorname{div}(y|_S).$$

Under $S \cong \mathbb{A}^2/\mu_2$, $(x,y,z) = (U^2, -V^2, UV)$, both $\operatorname{div}(z|_S)$ and Δ_S pull back to $\operatorname{div}(UV)$, so the pairs $(S, \operatorname{div}(z|_S))$ and (S, Δ_S) are lc [KM98, Proposition 5.20(4)]. By [Kaw07, Theorem], the pairs

$$\big(W_N, \operatorname{div}(t) + \operatorname{div}(z)\big), \qquad \Big(W_N, \operatorname{div}(t) + \frac{1}{2}\operatorname{div}(x) + \frac{1}{2}\operatorname{div}(y)\Big)$$

are lc near 0, for any N. Evaluating their log discrepancies at P and using (A.125),

$$[x], [y] \le 2\Theta_1, \qquad [z], [t] \le \Theta_1. \tag{A.126}$$

Step 3. In this step we reduce the lower bound $\alpha \geq \beta$ to two cases.

The blow-up of $0 \in \mathbb{A}^2$ shows $\beta \leq 2$. By [Liu26, Lemma 2.6], α is the minimum of $a(P, W_N, D)/|I_G(P)|$ over the prime divisors P over W_N with center 0, so it suffices to show $a(P, W_N, D)/|I_G(P)| \geq \beta$ for P as in **Step 2**. Let

$$\rho_+ := \min\Big(\frac{[x]}{a}, \frac{[t]}{v}\Big), \qquad \rho_- := \min\Big(\frac{[y]}{b}, \frac{[z]}{v}\Big).$$

Assume first that $[h_j] = [h_j^0]$ for all j. By Construction-Proposition A.23(4), $h_j^0 = s^{\kappa_j} H_j^t$ and $h_j^0 = \eta_x^{\kappa_j} h_j^0(1, -p_x^2 - u_2^d q_x^N, p_x, q_x)$. Hence Lemma A.25(1) with $B_j := H_j^t$ if $[x]/a \geq [t]/v$, and Lemma A.25(2) with $B_j := h_j^0(1, -p_x^2 - u_2^d q_x^N, p_x, q_x)$ otherwise, imply $a(P, W_N, D)/|I_G(P)| \geq \beta$, as $Nv > a$.

Otherwise there exists j with $[h_j] \neq [h_j^0]$. As $[h_j], [h_j^0] \leq \Theta_2$, some monomial $x^i y^l z^m t^n$ of $h_j - h_j^0$ has $i[x] + l[y] + m[z] + n[t] \leq \Theta_2$. If $\kappa := ai - bl - vm + vn$ satisfies $\kappa \geq r - bM$, then $ai + vn \geq r - bM$ and $\Theta_2 \geq i[x] + n[t] \geq (r - bM)\rho_+$. If $\kappa \leq -(r - bM)$, then $bl + vm \geq r - bM$ and $\Theta_2 \geq (r - bM)\rho_-$. Thus

$$\rho_+ \leq \frac{\Theta_2}{r - bM} \qquad \text{or} \qquad \rho_- \leq \frac{\Theta_2}{r - bM}. \tag{A.127}$$

Since $\Theta_2/(r - bM) < \epsilon_0$, Lemma A.27 shows that exactly one of the inequalities (A.127) holds, and then the other minimum is greater than ϵ_0.

Step 4. In this step we treat the case $\rho_+ \leq \Theta_2/(r - bM)$, so that $\rho_- > \epsilon_0$.

Let $h_j^{\geq}$ be the sum of the terms of h_j of $\mathbf{T}$-weight at least κ_j. A monomial of $h_j - h_j^{\geq}$ has weight $\kappa \leq -(r - bM)$, so, as in **Step 3**, the value $[\cdot]$ of such a monomial is at least $(r - bM)\rho_- > (r - bM)\epsilon_0 > \Theta_2 \geq [h_j]$. Hence $[h_j^{\geq}] = [h_j]$. The weights of the monomials of $h_j^{\geq}$ belong to $\kappa_j + r\mathbb{Z}_{\geq 0}$ by Lemma 5.3. By Construction-Proposition A.23(4) and the identities $s^{\kappa} = s^{\kappa_j} u_1^{(\kappa - \kappa_j)/r}$ and $\eta_x^{\kappa} = \eta_x^{\kappa_j} u_2^{(\kappa - \kappa_j)/r}$, we can write

$$h_j^{\geq} = s^{\kappa_j} B_j(x_1, y_1, z_1, u_1) = \eta_x^{\kappa_j} B_j'(u_2, p_x, q_x)$$

with polynomials B_j, B_j', semi-invariant under $\tilde{H}_1$ and $\tilde{H}_2$, such that $B_j(x_1, y_1, z_1, 0) = H_j^t$ and $B_j'(0, p_x, q_x) = H_j^x$, because the terms of weight κ_j form h_j^0. Thus

$$[h_j] = \frac{\kappa_j [t]}{v} + [B_j] = \frac{\kappa_j [x]}{a} + [B_j'].$$

As $Nv > a$, Lemma A.25(1) if $[x]/a \geq [t]/v$, and Lemma A.25(2) otherwise, imply $a(P, W_N, D)/|I_G(P)| \geq \beta$.

Step 5. In this step we treat the case $\rho_- \leq \Theta_2/(r - bM)$, so that $\rho_+ > \epsilon_0$, that is, $[x] > a\epsilon_0$ and $[t] > v\epsilon_0$.

As in **Step 4**, the sum $h_j^{\leq}$ of the terms of h_j of weight at most κ_j satisfies $[h_j^{\leq}] = [h_j]$. If $[z]/v \leq \Theta_2/(r - bM)$, then $2[z] \leq 2v\Theta_2/(r - bM) < a\epsilon_0 < \min([x] + [y], N[t])$, because $[x] > a\epsilon_0$ and $N[t] > Nv\epsilon_0 > a\epsilon_0$, which contradicts $z^2 = -xy - t^N$. Hence $[y]/b \leq \Theta_2/(r - bM)$. By (A.126), $[x] + [y] \leq 4\Theta_1$ and $2[z] \leq 2\Theta_1$, whereas $N[t] > Nv\epsilon_0 \geq (r - 2v)\epsilon_0 > 4\Theta_1$. Thus $[x] + [y] = 2[z]$, and we may use Construction-Proposition A.23(2)(c), with

$$[u] = \frac{r[y]}{b} \leq \frac{r\Theta_2}{r - bM} \leq 2\Theta_2, \qquad [q] = [t] + \frac{v[y]}{b} > v\epsilon_0,$$

$$[p] = [z] - \frac{v[y]}{b} = \frac{[x]}{2} + \frac{a[y]}{2b} > 0.$$

Let $x' := x\eta^a = -p^2 - q^N/u^d$, so $[x'] = [x] + a[y]/b > 0$. By Construction-Proposition A.23(4) and $\eta^{-\kappa} = \eta^{-\kappa_j} u^{(\kappa_j - \kappa)/r}$ for $\kappa \leq \kappa_j$, we can write $h_j^{\leq} = \eta^{-\kappa_j} B_j^-(u, x', p, q)$ with a polynomial $B_j^-(u, X', p, q)$ such that $B_j^-(0, X', p, q) = h_j^0(X', 1, p, q)$. Let $B_j := B_j^-(u, -p^2, p, q)$. Then B_j

is $\tilde{H}$-semi-invariant, as the characters of x' and $-p^2$ under $\tilde{H}$ are equal, and $B_j(0,p,q)=\varphi_j$. Moreover $B_j^-(u,x',p,q)-B_j=(x'+p^2)\cdot Q'$ for a polynomial Q' in u,x',p,q, so $[Q']\geq 0$ and

$$\begin{aligned}[B_j^-(u,x',p,q)-B_j]&\geq N[q]-d[u]=d\Big(\frac{r}{v}[q]-[u]\Big)-2[q]\\&\geq\Big(\frac{r}{v}-2\Big)[q]-[u]>(r-2v)\epsilon_0-2\Theta_2>2\Theta_2,\end{aligned}$$

where we use $N=rd/v-2$, $d\geq 1$, and $r[q]/v-[u]>r\epsilon_0-2\Theta_2>0$. On the other hand

$$[B_j^-(u,x',p,q)]=[h_j]+\frac{\kappa_j[y]}{b}\leq\Theta_2+\frac{bM\Theta_2}{r-bM}\leq 2\Theta_2.$$

Hence $[B_j]=[h_j]+\kappa_j[y]/b$, that is, $[h_j]=-\kappa_j[y]/b+[B_j]$. Since $v[y]<b[z]$, Lemma A.25(3) implies $a(P,W_N,D)/|I_G(P)|\geq\beta$.

Step 6. In this step we prove (1). By **Steps 3–5**, $a(P,W_N,D)/|I_G(P)|\geq\beta$ for any prime divisor P over W_N with center 0, so $\alpha\geq\beta$. The bound $\alpha\leq\beta+C_0/r$ is in **Step 1**. The numbers r_0 and C_0 do not depend on d, on the degrees of the h_j, or on the terms of $h_j-h_j^0$.

Step 7. In this step we start the proof of (2) and take a compatible generic limit of the plane divisors.

Assume that there exists a sequence of data as in Set-up 7.3, indexed by i, with Γ, e_0, and w fixed, $r_i\to+\infty$, $\tau_i>0$, and positive and strictly increasing values α_i. We denote the data by $N_i,d_i,G_i,D_i,h_{i,j},c_{i,j}$, and so on. We may assume $r_i\geq r_0'$ and $r_i>2bM$. By **Step 1**, $\alpha_i\leq\beta_i+C_0/r_i\leq 2+C_0$, as $\beta_i\leq 2$ (**Step 3**), so $\alpha_\infty:=\lim_i\alpha_i$ is a positive real number with $\alpha_i<\alpha_\infty$. As the β_i belong to an ACC set, after passing to a subsequence the β_i are non-increasing with limit $L\geq\lim_i(\alpha_i-C_0/r_i)=\alpha_\infty>0$.

By **Step 1** the number of indices j is less than M and $|\kappa_{i,j}|\leq bM$. After passing to a subsequence, the number of indices and the $\kappa_{i,j}=\kappa_j$ do not depend on i, and for any j the sequence $c_{i,j}$ is non-decreasing with limit $c_j^*\in[\gamma,1]$. The group $\tilde{H}_i$ is a diagonal subgroup of $(\mathbb{C}^*)^2$ of order $b[G_i:R_i]\leq be_0$, so it is contained in $\mu^2_{(be_0)!}$. After passing to a subsequence, $\tilde{H}_i=\tilde{H}$ and the characters of the $\varphi_{i,j}:=h^0_{i,j}(-p^2,1,p,q)$ do not depend on i. A monomial of least total degree of $h_{i,j}$ has degree at most M, and it belongs to $h^0_{i,j}$, since the monomials of $h_{i,j}-h^0_{i,j}$ have degree at least $(r_i-bM)/b>M$ by **Step 1**. Its image has degree at most $2M$ and is not cancelled, by Construction-Proposition A.24(1). Thus $\operatorname{mult}_0\varphi_{i,j}\leq 2M$. Construction-Proposition 4.1, in case (a) with $n=2$ and the group $\tilde{H}\subset GL_2(\mathbb{C})$, applied to the tuples $(\varphi_{i,j})_j$, provides, after passing to a subsequence, a compatible generic limit $(\varphi_{\infty,j})_j$ over a field k. Let

$$C_i^*:=\sum_j c_j^*\operatorname{div}(\varphi_{i,j}),\qquad C_\infty:=\sum_j c_j^*\operatorname{div}(\varphi_{\infty,j}),$$
$$\beta_i^*:=\operatorname{mld}((\mathbb{A}^2\ni 0,C_i^*)/\tilde{H}).$$

We compare $C_{y,i}$ with C_i^*, which has the limit labels c_j^*. First, we show that $(\mathbb{A}^2,C_i^*)$ is lc near 0 for $i\gg 0$. Since $C_{y,i}\geq(\min_j c_{i,j}/c_j^*)C_i^*$, the lc threshold of C_i^* near 0 is at least $\min_j c_{i,j}/c_j^*$, which tends to 1. The lc thresholds on $\mathbb{A}^2$ of divisors with labels in the finite set $\{c_j^*\}$ satisfy the ACC [dFEM11, Theorem 1.1], so they cannot be less than 1 for infinitely many i. Second, we show that $\beta_i^*=L$ for $i\gg 0$. The inequality $C_i^*\geq C_{y,i}$ implies $\beta_i^*\leq\beta_i$. Let $\Gamma^*:=\Gamma\cup\{c_j^*\}_j\subset[0,1]$, a DCC set. The crepant quotients of the $(\mathbb{A}^2,C_{y,i})$ and $(\mathbb{A}^2,C_i^*)$ by $\tilde{H}$ have coefficients in $\Psi(\Gamma^*_+\cap[0,1])$, a DCC set [Liu26, Lemma 2.10(1)(5)]. Let l be the integer of Theorem 2.9(2) for $\Psi(\Gamma^*_+\cap[0,1])$. For any i there exists a prime divisor E_i' over $\mathbb{A}^2$ with center 0 computing β_i^* such that $a(E_i',\mathbb{A}^2,0)\leq be_0l$, since the branch divisor has non-negative coefficients and $|\tilde{H}|\leq be_0$. As $(\mathbb{A}^2,C_i^*)$ is lc and $c_j^*\geq\gamma$, we have $\operatorname{mult}_{E_i'}\varphi_{i,j}\leq be_0l/\gamma$, and evaluating E_i' on

$(\mathbb{A}^2, C_{y,i})$ shows

$$\beta_i \le \beta_i^* + \sum_j (c_j^* - c_{i,j}) \frac{be_0 l}{\gamma}.$$

Thus $\beta_i^* \to L$. The numbers β_i^* belong to a finite set, because $a(E_i', \mathbb{A}^2, 0)$ and the $\mathrm{mult}_{E_i'} \varphi_{i,j}$ are bounded non-negative integers and $|I_{\tilde{H}}(E_i')| \le be_0$. Hence $\beta_i^* = L$ for $i \gg 0$.

By Theorem 4.2(1) with $\theta = 1$ and $\mathfrak{c} = \mathcal{O}$, the pair $(\mathcal{A}_k, C_\infty)$ is lc near 0, and by Theorem 4.2(2), $\mathrm{mld}((\mathcal{A}_k \ni 0, C_\infty)/\tilde{H}) = L$.

Step 8. In this step we treat the case where $(\mathcal{A}_k, C_\infty)$ is klt near 0.

By Theorem 4.2(1) there exists $\theta' > 1$ such that $(\mathbb{A}^2, \theta' C_i^*)$ is lc near 0 for any i, hence so is $(\mathbb{A}^2, \theta' C_{y,i})$. Since $\tau_i \ge 0$ and r_i is at least the number r_0 of (1) for θ' when $i \gg 0$, part (1) with θ' in place of θ implies $\alpha_i \ge \beta_i \ge L \ge \alpha_\infty$ for $i \gg 0$, which contradicts $\alpha_i < \alpha_\infty$.

Step 9. In this step we treat the case where $(\mathcal{A}_k, C_\infty)$ is not klt near 0, and finish the proof of (2).

By Definition-Proposition 5.7, C_∞ has a unique floor F, and

$$L = \frac{\sigma}{e}, \qquad \sigma := 1 - \mathrm{mult}_0((C_\infty - F)|_F),$$

where e is the order of the image of $\tilde{H}$ in the automorphism group of $T_0 F$. In particular $\sigma > 0$. The curve F is $\tilde{H}$-invariant, as it is the unique floor. The element $(1, \zeta)$ of $\mu_b \subset \tilde{H}$ with ζ a primitive b-th root of unity acts on $\mathbb{A}^2$ by $(\zeta^{-v} p, \zeta^v q)$ with $\zeta^{-v} \ne \zeta^v$, because b does not divide $2v$, as $b > 2v$. Hence the only $\tilde{H}$-invariant lines in $T_0 \mathbb{A}^2$ are the coordinate axes, and one of p, q restricts to a parameter of F at 0. Let $z_0 \in \{p, q\}$ be such a coordinate, let $\Phi := \Phi_+$ if $z_0 = p$ and $\Phi := \Phi_-$ if $z_0 = q$, so that $\Phi(-p^2, 1, p, q) = z_0^b$ by Construction-Proposition A.23(3). Then e is the order of the character of z_0 on $\tilde{H}$. Since μ_b acts faithfully on z_0, b divides e.

We show that $(\mathbb{A}^2, C_i^* + \sigma \operatorname{div}(z_0))$ is lc near 0 for infinitely many i. The restriction of the $\mathbb{R}$-divisor $(C_\infty - F) + \sigma \operatorname{div}(z_0)$ to the regular curve F has multiplicity $(1 - \sigma) + \sigma = 1$ at 0, so $(\mathcal{A}_k, C_\infty + \sigma \operatorname{div}(z_0))$ is lc near 0 by Lemma 2.14. If $\sigma < 1$, then $(\mathbb{A}^2, \sigma \operatorname{div}(z_0))$ is klt and the ideal (z_0) is $\tilde{H}$-invariant, and Theorem 4.2(1) with $\theta = 1$ and $\mathfrak{c} := (z_0)^\sigma$ shows the claim. If $\sigma = 1$, then $0 \notin \mathrm{Supp}(C_\infty - F)$, and $m_j := \mathrm{mult}_0(\varphi_{\infty,j}|_{(z_0=0)})$ satisfies $\sum_j c_j^* m_j = \mathrm{mult}_0(F|_{(z_0=0)}) = 1$. The condition $\mathrm{mult}_0(\varphi|_{(z_0=0)}) \le m_j$ on the $(m_j + 1)$-jets of φ is open and holds at the jet of $\varphi_{\infty,j}$, so it holds for $\varphi_{i,j}$ for all but finitely many i by Construction-Proposition 4.1. Thus $\mathrm{mult}_0(C_i^*|_{(z_0=0)}) \le 1$, and $(\mathbb{A}^2, C_i^* + \operatorname{div}(z_0))$ is lc near 0 by [Kaw07, Theorem].

As $C_{y,i} \le C_i^*$ and $\Phi(-p^2, 1, p, q) = z_0^b$, the pair

$$\left(\mathbb{A}^2, C_{y,i} + \frac{\sigma}{b} \operatorname{div}\left(\Phi(-p^2, 1, p, q)\right)\right)$$

is lc near 0 for infinitely many i. The polynomial Φ is $\mathbf{T}$-homogeneous of weight zero by Construction-Proposition A.23(3), without monomials divisible by xy, so the mass of the list $(h_{i,j}^0, \Phi)$ with labels $(c_{i,j}, \sigma/b)$ is $\tau_i > 0$, and $N_i v > a$ for $i \gg 0$. Lemma A.26 shows that $(W_{N_i}, D_i^0 + \frac{\sigma}{b} \operatorname{div}(\Phi))$ is lc near 0. The argument of **Step 1**, with the DCC set $(\Gamma \backslash \{0\}) \cup \{\sigma/b, 1\}$, at most $M + 2$ functions, and Φ appended both to the $h_{i,j}^0$ and to the $h_{i,j}$, shows that $(W_{N_i}, D_i + \frac{\sigma}{b} \operatorname{div}(\Phi))$ is lc near 0 for infinitely many i.

Let $g \in G_i$ and ζ with $\zeta^b = g_y$. Since $g_x g_y = g_z^2$, we have $g_x^v g_z^a = g_z^{2v+a} g_y^{-v} = (g_z \zeta^{-v})^b$ and $g_y^v g_t^b = (g_t \zeta^v)^b$. Hence the multiplier of Φ under g is the b-th power of the multiplier of z_0 under the class of (g, ζ) in $\tilde{H}$. As b divides e, the character of Φ on G_i has order e/b. Since Φ vanishes at 0, Lemma 5.8 with $U := \Phi$ and $\lambda := \sigma/b$ implies

$$\alpha_i \ge \frac{\sigma/b}{e/b} = \frac{\sigma}{e} = L \ge \alpha_\infty$$

for infinitely many i, which contradicts $\alpha_i < \alpha_\infty$. This proves (2). □

We now treat exterior canonical circles with non-positive mass. The reference object is a pair on a crepant partial resolution of the A_{b-1} plane quotient, whose exceptional curve E has the coefficient β_E of Construction-Proposition A.28(3).

Construction-Proposition A.28. Notation as in Set-up 7.3 and Construction-Proposition A.23, with $w = (a, -b, -v, v)$, where $a, v > 0$, $\gcd(a, v) = 1$, and $b = a + 2v$. Let $\tilde{H}$ be the group of Proposition A.22, acting on $\mathbb{A}^2_{p,q}$, and let $\mu_b \subset \tilde{H}$ be the image of the pairs $(1, \zeta)$ with $\zeta^b = 1$, which acts by $(p, q) \to (\zeta^{-v}p, \zeta^v q)$. Then the following hold.

(1) μ_b is central in $\tilde{H}$, $\tilde{H}/\mu_b \cong \bar{G}$, and $S_b := \mathbb{A}^2_{p,q}/\mu_b$ is the A_{b-1} singularity, embedded by (p^b, q^b, pq) into $\mathbb{A}^3$ with image $(\Phi_+\Phi_- = \Phi_0^b)$. Let $\pi : Y \to S_b$ be the toric partial resolution given by the ray $(a/b, 2v/b)$, E the exceptional curve, and C_p, C_q the strict transforms of the images of $(p = 0)$ and $(q = 0)$. Then π is crepant, $\bar{G}$ acts on Y preserving E, and Y is covered by two open subsets $V_1 \ni E \cap C_p$ and $V_2 \ni E \cap C_q$ with

$$V_1 \cong S_0/\mu_v, \quad S_0 := (x_1y_1 + z_1^2 = 0) \subset \mathbb{A}^3, \qquad V_2 \cong \mathbb{A}^2_{p_x,q_x}/\mu_a,$$

where $\varsigma \in \mu_v$ acts by $(\varsigma^{-a}x_1, \varsigma^b y_1, z_1)$ and $\xi \in \mu_a$ acts by $(\xi^v p_x, \xi^{-v} q_x)$, and on which $(\Phi_+, \Phi_-, \Phi_0) \circ \pi$ is $((-1)^v x_1^v z_1^a, y_1^v, z_1)$ and $((-1)^v p_x^a, (-1)^v p_x^{2v} q_x^b, p_x q_x)$, respectively. On S_0 the preimages of E and C_p are $(y_1 = z_1 = 0)$ and $(x_1 = z_1 = 0)$, and on $\mathbb{A}^2_{p_x,q_x}$ the preimages of E and C_q are $(p_x = 0)$ and $(q_x = 0)$. Moreover

$$(E^2) = -\frac{b}{2av}, \qquad (C_p \cdot E) = \frac{1}{2v}, \qquad (C_q \cdot E) = \frac{1}{a}, \tag{A.128}$$
$$\operatorname{div}_Y(\Phi_+) = aE + bC_p, \qquad \operatorname{div}_Y(\Phi_-) = 2vE + bC_q, \qquad \operatorname{div}_Y(\Phi_0) = E + C_p + C_q.$$

(2) Let h be a non-zero G-semi-invariant polynomial, $\mathbf{T}$-homogeneous of $\mathbf{T}$-weight κ, without monomials divisible by xy. Then there exists a $\mathbb{Q}$-divisor $D_h \geq 0$ on Y, invariant under $\bar{G}$, whose pullbacks to S_0 and to $\mathbb{A}^2_{p_x,q_x}$ are

$$\operatorname{div}_{S_0}\big(h(x_1, y_1, z_1, 1)\big) \quad \text{and} \quad \operatorname{div}\big(h(1, -p_x^2, p_x, q_x)\big), \tag{A.129}$$

and $D_h = \pi^*\Delta_h - \frac{2\kappa}{b}E$, where Δ_h is the image on S_b of $\operatorname{div}(h(-p^2, 1, p, q))$. The coefficient of E in D_h is the least value $\iota(h)$ of $2l + m$ over the monomials $x^i y^l z^m t^n$ of h, and $(\pi^*\Delta_h \cdot E) = 0$. Moreover, with $\varphi_h(p, q) := h(-p^2, 1, p, q)$, the function φ_h^b descends to S_b, and the pullbacks of φ_h^b to S_0 and to $\mathbb{A}^2_{p_x,q_x}$ are $y_1^\kappa h(x_1, y_1, z_1, 1)^b$ and $(-p_x^2)^\kappa h(1, -p_x^2, p_x, q_x)^b$.

(3) Assume that $\tau \leq 0$, write h_j^0 for the principal blocks of Set-up 7.3, $\iota_j := \iota(h_j^0)$, $\beta_E := \sum_j c_j\iota_j$, and

$$B_- := \sum_j c_j D_{h_j^0} = \pi^*\Delta - \frac{2\tau}{b}E, \qquad \Delta := \sum_j c_j \Delta_{h_j^0}.$$

Let γ be the least positive element of Γ, and $M := \lceil 2/\gamma \rceil$. Then there are at most M non-units h_j with $c_j > 0$, and $\sum_j c_j < 2$ over them. There exists r_-, depending only on a, v, Γ, e_0, such that if $r \geq r_-$, then $|\kappa_j| \leq bM$, every monomial of $h_j - h_j^0$ has degree at least $(r - bM)/b$, $(W_0, \sum_j c_j \operatorname{div}(h_j^0))$ is lc near 0, where $W_0 := (xy + z^2 = 0) \subset \mathbb{A}^4$, the pair (Y, B_-) is lc near E, and the minimum β_- of the log discrepancies of the crepant quotient $(Y, B_-)/\bar{G}$ over the prime divisors with center contained in $E/\bar{G}$ is attained and non-negative. The coefficients of $(Y, B_-)/\bar{G}$ belong to a DCC set depending only on a, v, Γ, e_0, and the numbers β_- belong to an ACC set depending only on a, v, Γ, e_0. If $\tau = 0$, then $B_- = \pi^*\Delta$ and $\beta_- = \beta$.

(4) Let $\mathbb{A}^3_{p_0,t,q_0} \to W_0$ be the double cover given by $x = p_0^2$, $y = -q_0^2$, $z = p_0q_0$, let G' be the group of triples $(\epsilon_1, g_t, \epsilon_2)$ with $g \in G$, $\epsilon_1^2 = g_x$, and $\epsilon_1\epsilon_2 = g_z$, acting diagonally on (p_0, t, q_0), let $g_1 := \gcd(a, 2v)$, $\mathbf{T}' := \mathbf{T}_{(a/g_1, 2v/g_1, -b/g_1)}$ acting on (p_0, t, q_0), and $h_j' := h_j^0(p_0^2, -q_0^2, p_0q_0, t)$. Then G' has order $2|G|$ and acts faithfully on $\mathbb{A}^3$, $\mathbb{A}^3/G' = W_0/G$,

$|G' \cap \mathbf{T}'| = 2r/g_1$, and $|G'/(G' \cap \mathbf{T}')| = g_1[G : R]$. If $\tau \le 0$ and $r \ge r_-$, then the minimum b' of [Liu26, Construction-Theorem 16.9] (denoted b there) for the weights $(a/g_1, 2v/g_1, b/g_1)$, the group G', the polynomials h'_j, and the labels c_j is β_-.

Proof. (1) The subgroup μ_b intersects the subgroup $\{(\mathbf{T}(\lambda), \lambda^{-1})\}$ only in the identity, so μ_b injects into $\tilde{H}$. The group μ_b is central in $\tilde{H}$ since $\tilde{H}$ is abelian. The map $(g, \zeta) \to gR$ is surjective with kernel the pairs $(\mathbf{T}(\lambda), \zeta)$ with $\zeta^b = \lambda^{-b}$, which is the product of μ_b and $\{(\mathbf{T}(\lambda), \lambda^{-1})\}$, so $\tilde{H}/\mu_b \cong \bar{G}$. Since $\gcd(b, v) = \gcd(a, v) = 1$, the group μ_b acts with weights $(1, -1)$ up to a generator, the invariants of μ_b are generated by p^b, q^b, pq, and an element of $\tilde{H}$ acts trivially on S_b if and only if it belongs to μ_b. Thus $\bar{G}$ acts faithfully on S_b, through the characters of Φ_+, Φ_-, Φ_0. The ray $(a/b, 2v/b)$ belongs to the lattice $\mathbb{Z}^2 + \mathbb{Z}\frac{1}{b}(1, -1)$ of S_b, as $(a/b, 2v/b) = (0, 1) + a \cdot \frac{1}{b}(1, -1)$, and belongs to the segment joining $(1, 0)$ and $(0, 1)$ since $a + 2v = b$, so π is crepant. The maximal cones of Y are of type A_{2v-1} and A_{a-1}, and a direct toric calculation shows that the corresponding charts are V_1 and V_2, with π given by the maps above. For instance, $(-1)^v x_1^v z_1^a \cdot y_1^v = z_1^b$ on S_0 and $(-1)^v p_x^a \cdot (-1)^v p_x^{2v} q_x^b = (p_x q_x)^b$. The actions of μ_v on S_0 and of μ_a on $\mathbb{A}^2$ are free outside the origins, since $\gcd(a, v) = \gcd(b, v) = 1$. In particular, they do not ramify along the curves above. Thus $\operatorname{div}_Y(\Phi_0) = E + C_p + C_q$ and the other two divisors follow from $\operatorname{div}_{S_0}(x_1) = 2(x_1 = z_1 = 0)$, $\operatorname{div}_{S_0}(y_1) = 2(y_1 = z_1 = 0)$, and $\operatorname{div}_{S_0}(z_1) = (x_1 = z_1 = 0) + (y_1 = z_1 = 0)$. The curves C_p and E intersect at a cyclic quotient point of order $2v$ transversally on the cover, so $(C_p \cdot E) = \frac{1}{2v}$. Likewise, $(C_q \cdot E) = \frac{1}{a}$, and $(\operatorname{div}_Y(\Phi_0) \cdot E) = 0$ implies $(E^2) = -\frac{1}{2v} - \frac{1}{a} = -\frac{b}{2av}$.

(2) Each monomial $x^i y^l z^m t^n$ of h maps to $(-1)^i p^{2i+m} q^n$, and has $\mathbf{T}$-weight

$$ai - bl - vm + vn \equiv -v(2i + m) + vn \pmod{b},$$

since $a \equiv -2v \pmod{b}$. Hence $\varphi_h(p, q) := h(-p^2, 1, p, q)$ is μ_b-semi-invariant with character $\zeta \to \zeta^\kappa$, φ_h^b descends to S_b, and $\Delta_h = \frac{1}{b} \operatorname{div}_{S_b}(\varphi_h^b)$. Two distinct monomials of h have distinct images, since they have $\mathbf{T}$-weight κ and no factor xy. Hence $\varphi_h \ne 0$, and likewise $h(x_1, y_1, z_1, 1)$ and $h(1, -p_x^2, p_x, q_x)$ are non-zero. Let ϑ be a b-th root of y_1 on a cover of S_0, and let $p := \vartheta^{-v} z_1$ and $q := \vartheta^v$. Then $p^b = (-1)^v x_1^v z_1^a$, $q^b = y_1^v$, $pq = z_1$, and $\mathbf{T}(\vartheta) \cdot (x_1, y_1, z_1, 1) = (-p^2, 1, p, q)$ on S_0, so

$$\varphi_h(p, q) = \vartheta^\kappa h(x_1, y_1, z_1, 1), \qquad \varphi_h^b = y_1^\kappa h(x_1, y_1, z_1, 1)^b.$$

Similarly, with $\lambda^b = -p_x^2$ on a cover of $\mathbb{A}^2_{p_x, q_x}$, $p := \lambda^{-v} p_x$ and $q := \lambda^v q_x$, we have $\mathbf{T}(\lambda) \cdot (1, -p_x^2, p_x, q_x) = (-p^2, 1, p, q)$ and $\varphi_h^b = (-p_x^2)^\kappa h(1, -p_x^2, p_x, q_x)^b$. Therefore the pullbacks of $\pi^* \Delta_h$ to S_0 and to $\mathbb{A}^2_{p_x, q_x}$ are the divisors (A.129) plus $\frac{\kappa}{b} \operatorname{div}(y_1)$ and $\frac{2\kappa}{b} \operatorname{div}(p_x)$, respectively. As the pullbacks of E are $\frac{1}{2} \operatorname{div}(y_1)$ and $\operatorname{div}(p_x)$, the $\mathbb{Q}$-divisor $D_h := \pi^* \Delta_h - \frac{2\kappa}{b} E$ has the pullbacks (A.129), so $D_h \ge 0$ and D_h is invariant under $\bar{G}$. The coefficient of $(p_x = 0)$ in $\operatorname{div}(h(1, -p_x^2, p_x, q_x))$ is $\iota(h)$, since distinct monomials of h have distinct images. Finally $(\pi^* \Delta_h \cdot E) = 0$ by the projection formula.

(3) and (4). Since (W_N, D) is klt near 0, Lemma A.1(3) gives $\sum_j c_j \operatorname{mult}_0 h_j < 2$ over the non-units h_j with $c_j > 0$, so there are at most M of them. By Lemma 5.3, if $r \ge r_-$, then $|\kappa_j| \le bM$ and every monomial of $h_j - h_j^0$ has degree at least $(r - bM)/b$. By [Kaw07, Theorem], the pair

$$(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^N) + \sum_j c_j \operatorname{div}(h_j))$$

is lc near 0. The exponents belong to the DCC set $\Lambda := \{1\} \cup (\Gamma \backslash \{0\})$, and $t^N \in \mathfrak{m}_0^N$ with $N \ge r/v - 2$ because $d \ge 1$. By Lemma 4.14 and [dFEM11, Proposition 2.11], after enlarging r_-, the pair

$$(\mathbb{A}^4, \operatorname{div}(xy + z^2) + \sum_j c_j \operatorname{div}(h_j^0))$$

is lc near 0, and [Kaw07, Theorem] again implies that $(W_0, \sum_j c_j \operatorname{div}(h_j^0))$ is lc near 0.

We now recall [Liu26, Construction-Theorem 16.9] in the letters of this paper. With the notation of (4), given $g \in G$ there are two choices of ϵ_1, and then $\epsilon_2^2 = g_z^2/g_x = g_y$. Hence G' has order $2|G|$, acts faithfully, and $\mathbb{A}^3/G' = W_0/G$. Note that $g_1 \in \{1, 2\}$ as $\gcd(a, v) = 1$. The image of $\mathbf{T}'$ in $\mathrm{Aut}(W_0)$ is $\mathbf{T}$ and $\mathbf{T}' \to \mathbf{T}$ has degree $2/g_1$, so $n' := |G' \cap \mathbf{T}'| = 2r/g_1$ and $H' := G'/(G' \cap \mathbf{T}')$ has order $g_1[G : R] \le 2e_0$. The lifts h'_j are $\mathbf{T}'$-homogeneous of weight $2\kappa_j/g_1$. Thus the numbers (p, q, r, n, τ) of [Liu26, Construction-Theorem 16.9] are $(p', q', r', n', \tau') := (a/g_1, 2v/g_1, b/g_1, 2r/g_1, 2\tau/g_1)$, since $r' - p' - q' = 0$, and the group H there is H'.

The pair $(\mathbb{A}^3, \sum_j c_j \operatorname{div}(h'_j))$ is lc near 0 by [Liu26, Lemma 2.6]. Construction-Theorem 16.9 of [Liu26] uses the cover $(p_0, t, \eta') \to (p_0, t, \eta'^{r'})$, the plane coordinates $u' := p_0\eta'^{p'}$, $v' := t\eta'^{q'}$, the $\mathbb{R}$-divisor $C' := \sum_j c_j \operatorname{div}(h'_j(u', v', 1))$ on $\mathbb{A}^2_{u',v'}$, and the image $\tilde{H}'$ of the pairs (g', ϵ) with $g' \in G'$ and $\epsilon^{r'} = g'_{q_0}$ in $\mathrm{Aut}(\mathbb{A}^2_{u',v'})$. We write u', v', b', Z' for the plane coordinates, the minimum and the quotient surface there. By [Liu26, Construction-Theorem 16.9], the following hold.

(i) If $\tau' = 0$, then $(Z', B_{Z'}) := (\mathbb{A}^2, C')/\tilde{H}'$ is lc, and $b' := \mathrm{mld}(Z' \ni \bar{0}, B_{Z'})$ satisfies
$$b' \le \mathrm{mld}((\mathbb{A}^3 \ni 0, \sum_j c_j \operatorname{div}(h'_j))/G').$$

(ii) If $\tau' < 0$, then $\gcd(p', q') = 1$; let $f : V' \to \mathbb{A}^2_{u',v'}$ be the weighted blow-up with weights (p', q') and exceptional curve E', let $\beta_z := \sum_j c_j \operatorname{mult}_{(q_0=0)} \operatorname{div}(h'_j)$, and $B_{V'} := f_*^{-1}C' + (1 - r'(1 - \beta_z))E'$. Then, near $f^{-1}(0)$, the crepant quotient $(Z', B_{Z'}) := (V', B_{V'})/\tilde{H}'$ satisfies $B_{Z'} \ge 0$ and is lc, the minimum b' of $a(F, Z', B_{Z'})$ over the prime divisors F over Z' with center contained in the image $\bar{E}'$ of E' is attained and satisfies
$$0 \le b' \le \mathrm{mld}((\mathbb{A}^3 \ni 0, \sum_j c_j \operatorname{div}(h'_j))/G'),$$
the order of the inertia group of E' in $\tilde{H}'$ is $r'e_{\mathrm{res}}$ with $1 \le e_{\mathrm{res}} \le |H'|$, and $\operatorname{mult}_{\bar{E}'} B_{Z'} = 1 - \frac{1-\beta_z}{e_{\mathrm{res}}}$.

In both cases the coefficients of $B_{Z'}$ belong to a DCC set depending only on $p', q', r', 2e_0, \Gamma$, and the numbers b' satisfy the ACC.

We now match the objects of [Liu26, Construction-Theorem 16.9] with ours. Let λ be a root of $\lambda^b = -1$. Since $\mathbf{T}(\lambda) \cdot (-p^2, 1, p, q) = (u'^2, -1, u', v')$ for $(u', v') = (\lambda^{-v}p, \lambda^v q)$, the linear isomorphism $(p, q) \to (\lambda^{-v}p, \lambda^v q)$ maps C_y onto C'. A direct calculation shows that it maps $\tilde{H}$ onto $\tilde{H}'$: for $g_1 = 1$ the element (g', ϵ) corresponds to (g, ϵ^2), and for $g_1 = 2$ to (g, ϵ), where $g \in G$ is the image of g'. In particular $\tilde{H}'$ contains μ_b. The inertia of E' in μ_b consists of the elements (ζ^{-v}, ζ^v) of the one-parameter subgroup $\{(\varrho^{p'}, \varrho^{q'})\}$. Such an element satisfies $\varrho^{b/g_1} = \varrho^{p'+q'} = 1$, and conversely for $\varrho^{b/g_1} = 1$ the element $\zeta := \varrho^{2/g_1}$ satisfies $\zeta^b = 1$ and $(\zeta^{-v}, \zeta^v) = (\varrho^{p'}, \varrho^{q'})$. Since $\gcd(p', q') = 1$, the ramification index of E' under μ_b is exactly $b/g_1 = r'$, in accordance with $|I_{\tilde{H}'}(E')| = r'e_{\mathrm{res}}$ in (ii). When $g_1 = 2$, the elements of μ_b induced by the deck transformation $(p_0, t, q_0) \to (-p_0, t, -q_0)$ satisfy $\zeta^{b/2} = -1$, hence do not fix E' pointwise.

The weighted blow-up V' is the toric variety of the ray $(p', q') = \frac{b}{g_1}(\frac{a}{b}, \frac{2v}{b})$, so $V'/\mu_b = Y$ and E' maps onto E. Moreover $\beta_z = \beta_E$, because $\operatorname{mult}_{(q_0=0)}$ of the lift of $x^i y^l z^m t^n$ is $2l + m$. Since the ramification index is r', the coefficient of E in $(V', B_{V'})/\mu_b$ is $1 - \frac{r'(1-\beta_E)}{r'} = \beta_E$, which is the coefficient of E in B_- by (2). The other components of $B_{V'}$ map to $\pi_*^{-1}\Delta$, as μ_b acts on $\mathbb{A}^2\backslash\{0\}$ freely. Thus $(V', B_{V'})/\mu_b = (Y, B_-)$ near E, $(Z', B_{Z'}) = (Y, B_-)/\bar{G}$, and $b' = \beta_-$ if $\tau < 0$, which is (4) in this case. If $\tau = 0$, then $B_- = \pi^*\Delta$, π is crepant and $\pi^{-1}(\bar{0}) = E$, so $\beta_- = \mathrm{mld}((\mathbb{A}^2 \ni 0, C_y)/\tilde{H}) = \beta = b'$ by (i), which is (4) in this case. In both cases (Y, B_-) is lc near E by [Liu26, Lemma 2.6], and the remaining statements are those of (i) and (ii), except for the coefficient of $E/\bar{G}$ when $\tau = 0$. If $\tau = 0$, then the coefficient of $E/\bar{G}$ in $(Y, B_-)/\bar{G}$ is $1 - (1 - \beta_E)/|I_{\bar{G}}(E)|$ by [Liu26, Lemma 2.6(1)(4)], where $\beta_E = \sum_j c_j\iota_j \in \Gamma_+$ by (2), and $\beta_E \le 1$ as (Y, B_-) is lc near E. Since $(Y, \pi^*\Delta)/\bar{G}$ is isomorphic to $(\mathbb{A}^2, C_y)/\tilde{H}$ outside $E/\bar{G}$, [Liu26,

Lemma 2.10(5)], applied to $(\mathbb{A}^2_{p,q}, C_y)$ and $\tilde{H}$ at 0, shows that the other coefficients belong to $\Psi(\Gamma_+ \cap [0,1])$. Thus the coefficients of $(Y, B_-)/\bar{G}$ belong to $\Psi(\Gamma_+ \cap [0,1])$, which is a DCC set by [Liu26, Lemma 2.10(1)]. □

The next construction-proposition writes the ratios $a(F, W, D_W)/|I_G(F)|$ in three charts over V_1 and V_2. Recall that $[f] := \mathrm{mult}_F f/|I_G(F)|$ for a regular function f near 0 and a prime divisor F over W_N (or W_0) fixed in the argument.

Construction-Proposition A.29. Notation as in Construction-Proposition A.28. Let W be W_N or W_0, let c_j be finitely many real numbers, and let h_j be non-zero G-semi-invariant polynomials without monomials divisible by xy. Let κ_j be integers such that the $\mathbf{T}$-weights of the monomials of h_j are congruent to κ_j modulo r, write $h_j = \sum_k h_{j,k}$, where $h_{j,k}$ is the sum of the terms of h_j of $\mathbf{T}$-weight $\kappa_j + kr$, and let $\tau := \sum_j c_j\kappa_j$ and $D_W := \sum_j c_j \operatorname{div}_W(h_j)$. Let $X := X_d = (x_1y_1 + z_1^2 + u_1^d = 0) \subset \mathbb{A}^4$ if $W = W_N$, and $X := S_0 \times \mathbb{A}^1_{u_1}$ if $W = W_0$. If $W = W_0$, the terms u_1^d, $u_2^d q_x^N$, s^{rd}, and η_x^{rd} below are omitted. Consider the following data.

(a) The group $\hat{G}_1$ of pairs (g, ζ) with $g \in G$ and $\zeta^v = g_t$, acting on

$$\bar{X} := (x_1y_1 + z_1^2 + s^{rd} = 0) \subset \mathbb{A}^4_{x_1,y_1,z_1,s}$$

by $(g_x\zeta^{-a}x_1, g_y\zeta^b y_1, g_z\zeta^v z_1, \zeta s)$, the rational map $\psi_1(x_1, y_1, z_1, s) := \mathbf{T}(s) \cdot (x_1, y_1, z_1, 1)$ to W, the morphism $\bar{X} \to X$ given by $u_1 = s^r$, the image $\tilde{H}_1$ of $\hat{G}_1$ in $\mathrm{Aut}(X)$, and $B_j^{(1)} := \sum_k u_1^k h_{j,k}(x_1, y_1, z_1, 1)$.

(b) The group $\hat{G}_2$ of triples (g, ζ, ϵ) with $g \in G$, $\zeta^v = g_t$, and $\epsilon^2 = g_x\zeta^{-a}$, acting on $\mathbb{A}^3_{P,Q,s}$ by $(\epsilon P, g_z\zeta^v\epsilon^{-1}Q, \zeta s)$, the rational map

$$\psi_2(P, Q, s) := \mathbf{T}(s) \cdot (P^2, -Q^2 - s^{rd}P^{-2}, PQ, 1)$$

to W, the morphism $\mathbb{A}^3_{P,Q,s} \to \mathbb{A}^3_{P,Q,u_1}$ given by $u_1 = s^r$, the image K_2 of $\hat{G}_2$ in $\mathrm{Aut}(\mathbb{A}^3_{P,Q,u_1})$, and $B_j^{(2)} := \sum_k u_1^k h_{j,k}(P^2, -Q^2 - u_1^d P^{-2}, PQ, 1)$.

(c) The group $\hat{G}_3$ of pairs (g, ξ) with $g \in G$ and $\xi^a = g_x$, acting on $\mathbb{A}^3_{\eta_x,p_x,q_x}$ by $(\xi\eta_x, g_z\xi^v p_x, g_t\xi^{-v}q_x)$, the rational map

$$\psi_3(\eta_x, p_x, q_x) := \mathbf{T}(\eta_x) \cdot (1, -p_x^2 - \eta_x^{rd}q_x^N, p_x, q_x)$$

to W, the morphism $\mathbb{A}^3_{\eta_x,p_x,q_x} \to \mathbb{A}^3_{u_2,p_x,q_x}$ given by $u_2 = \eta_x^r$, the image $\tilde{H}_2$ of $\hat{G}_3$ in $\mathrm{Aut}(\mathbb{A}^3_{u_2,p_x,q_x})$, and $B_j^{(3)} := \sum_k u_2^k h_{j,k}(1, -p_x^2 - u_2^d q_x^N, p_x, q_x)$.

For $W = W_N$, $X = X_d$, S_0, ℓ_1, $\tilde{H}_1$, and $\tilde{H}_2$ are the objects of Construction-Proposition A.23. Write $V^{(1)} := X$, $V^{(2)} := \mathbb{A}^3_{P,Q,u_1}$, $V^{(3)} := \mathbb{A}^3_{u_2,p_x,q_x}$, $K_1 := \tilde{H}_1$, $K_3 := \tilde{H}_2$, and $w_1 = w_2 := u_1$, $w_3 := u_2$. Then the following hold.

(1) The groups K_1, K_2, K_3 are diagonal of orders $v[G : R]$, $2v[G : R]$, $a[G : R]$, and act faithfully on $S_0 = X \cap (u_1 = 0)$, on $\mathbb{A}^2_{P,Q}$ and on $\mathbb{A}^2_{p_x,q_x}$, respectively. The quotients W/G and $V^{(i)}/K_i$ are birational for $i = 1, 2, 3$. Moreover $S_0/K_1 = V_1/\bar{G} = \mathbb{A}^2_{P,Q}/K_2$ and $\mathbb{A}^2_{p_x,q_x}/K_3 = V_2/\bar{G}$, where $x_1 = P^2$, $y_1 = -Q^2$, $z_1 = PQ$.

(2) Let F be a prime divisor over W, and let ν be a prime divisor over $V^{(i)}$ inducing over the common quotient the prime divisor induced by F. Then

$$\frac{a(F, W, D_W)}{|I_G(F)|} = \frac{a\big(\nu, V^{(i)}, (1 + \frac{\tau}{r})\operatorname{div}(w_i) + \sum_j c_j \operatorname{div}(B_j^{(i)})\big)}{|I_{K_i}(\nu)|}. \tag{A.130}$$

(3) Let F have center 0 on W. If $[x]/a \geq [t]/v$, then the center of ν on X for $i = 1$ is a closed point or the generic point of the curve $\ell_1 := (y_1 = z_1 = u_1 = 0)$. If $[x]/a < [t]/v$, then the center of ν on $\mathbb{A}^3_{u_2,p_x,q_x}$ for $i = 3$ is the origin.

Proof. (1) Since $g_x g_y = g_z^2 = g_t^N$ for $g \in G$ and $\zeta^{rd} = \zeta^{v(N+2)}$, the group $\hat{G}_1$ preserves $\bar{X}$, and ψ_1 is $\hat{G}_1$-equivariant over the surjection $\hat{G}_1 \to G$: for instance $x_1 s^a \to g_x \zeta^{-a} x_1 \cdot \zeta^a s^a$. The map ψ_1 is dominant of degree v, since s is a v-th root of t and then $(x_1, y_1, z_1) = \mathbf{T}(s)^{-1} \cdot (x, y, z)$, and the kernel $\{(1,\zeta) : \zeta^v = 1\}$ acts simply transitively on the general fibers. Thus $\bar{X}/\hat{G}_1$ is birational to W/G. The subgroup $\{(\mathbf{T}(\lambda), \lambda) : \lambda^r = 1\}$ acts by $s \to \lambda s$ only, with quotient X, so $X/\tilde{H}_1 = \bar{X}/\hat{G}_1$ and $|\tilde{H}_1| = v|G|/r$. If (g,ζ) acts trivially on S_0, then $g = \mathbf{T}(\zeta) \in R$, so $\zeta^r = 1$ and (g,ζ) belongs to that subgroup. Hence $\tilde{H}_1$ acts faithfully on S_0. The image of $\{(1,\zeta)\}$ in $\tilde{H}_1$ is the group μ_v of Construction-Proposition A.28(1), and the characters of $(-1)^v x_1^v z_1^a$, y_1^v, z_1 under (g,ζ) are those of Φ_+, Φ_-, Φ_0 under g. Thus $S_0/\tilde{H}_1 = V_1/\bar{G}$. In (b) and (c), the kernels of $\hat{G}_2 \to G$ and $\hat{G}_3 \to G$ have orders $2v$ and a and act simply transitively on the general fibers of ψ_2 and ψ_3. The image of $\{(1,\zeta,\epsilon)\}$ in K_2 is μ_{2v} acting by $(\epsilon, \epsilon^{-1})$ because $\gcd(a,v) = 1$. An element acting trivially on $\mathbb{A}^2_{P,Q}$ or on $\mathbb{A}^2_{p_x,q_x}$ is the image of some $(\mathbf{T}(\lambda), \lambda, 1)$ or $(\mathbf{T}(\lambda), \lambda)$ with $\lambda^r = 1$, since $g_y = g_z^2/g_x$. Finally, the characters of $(-1)^v P^b Q^a$, $(-1)^v Q^{2v}$, PQ and of $(-1)^v p_x^a$, $(-1)^v p_x^{2v} q_x^b$, $p_x q_x$ are those of Φ_+, Φ_-, Φ_0.

(2) The form $\Omega_W := dx \wedge dz \wedge dt/x$ generates ω_W near 0, and $\Omega_X := dx_1 \wedge dz_1 \wedge du_1/x_1$ generates ω_X. A direct calculation shows

$$\psi_1^*\Omega_W = \frac{v}{r}\frac{\Omega_X}{u_1}, \qquad \psi_2^*\Omega_W = \frac{2v}{r}\frac{dP \wedge dQ \wedge du_1}{u_1}, \qquad \psi_3^*\Omega_W = \frac{a}{r}\frac{du_2 \wedge dp_x \wedge dq_x}{u_2}$$

on $\bar{X}$, $\mathbb{A}^3_{P,Q,s}$, and $\mathbb{A}^3_{\eta_x,p_x,q_x}$, and, by the $\mathbf{T}$-weights of the $h_{j,k}$,

$$\psi_1^* h_j = s^{\kappa_j} B_j^{(1)}, \qquad \psi_2^* h_j = s^{\kappa_j} B_j^{(2)}, \qquad \psi_3^* h_j = \eta_x^{\kappa_j} B_j^{(3)},$$

with $s^{\kappa_j} = u_1^{\kappa_j/r}$ and $\eta_x^{\kappa_j} = u_2^{\kappa_j/r}$. Hence, for a prime divisor $\bar{F}$ over the common quotient, the log discrepancies of $\bar{F}$ with respect to the crepant quotients of (W, D_W) and of

$$(V^{(i)}, (1 + \frac{\tau}{r})\operatorname{div}(w_i) + \sum_j c_j \operatorname{div}(B_j^{(i)}))$$

are equal by the Hurwitz formula [Kol13, 2.41], and (A.130) follows from [Liu26, Lemma 2.6(2)].

(3) Let F_1 be a prime divisor over $\bar{X}$ inducing F and ν. Then $\operatorname{mult}_{F_1}$ of s, x_1, y_1, z_1 is a positive multiple of $[t]/v$, $[x] - a[t]/v$, $[y] + b[t]/v$, $[z] + [t]$, respectively. If $[x]/a \geq [t]/v$, the four numbers are non-negative, and positive for s, y_1, z_1, which proves the first statement. If $[x]/a < [t]/v$, then mult of η_x, p_x, q_x along a prime divisor over $\mathbb{A}^3_{\eta_x,p_x,q_x}$ inducing F are positive multiples of $[x]/a$, $[z] + v[x]/a$, $[t] - v[x]/a$, which are positive. □

Lemma A.30. *Notation as in Construction-Proposition A.29. Assume that $c_j > 0$, $\tau \leq 0$, and that each h_j is a sum of terms of $\mathbf{T}$-weights $\kappa_j + kr$ with $k \geq 0$, whose principal part $h_{j,0}$ is non-zero. Let $B' := \sum_j c_j D_{h_{j,0}}$ (Construction-Proposition A.28(2)), assume that (Y, B') is lc near E, and let β' be the minimum of the log discrepancies of $(Y, B')/\bar{G}$ over the prime divisors with center contained in $E/\bar{G}$. Then any prime divisor F over W with center 0 satisfies*

$$\frac{a(F, W, D_W)}{|I_G(F)|} \geq \beta' \geq 0.$$

If moreover each h_j is $\mathbf{T}$-homogeneous, then (W, D_W) is lc near 0.

Proof. The functions $B_j^{(1)}$ and $B_j^{(3)}$ are polynomials, since $k \geq 0$, and their restrictions to $(u_1 = 0)$ and $(u_2 = 0)$ are $h_{j,0}(x_1, y_1, z_1, 1)$ and $h_{j,0}(1, -p_x^2, p_x, q_x)$. By Construction-Proposition A.28(2), the pairs

$$(S_0, \sum_j c_j \operatorname{div}(h_{j,0}(x_1, y_1, z_1, 1))) \quad \text{and} \quad (\mathbb{A}^2, \sum_j c_j \operatorname{div}(h_{j,0}(1, -p_x^2, p_x, q_x)))$$

are the pullbacks of (Y, B') to the quasi-étale covers of V_1 and V_2, hence lc near the preimages of E, and by Construction-Proposition A.29(1) their quotient values by $\tilde{H}_1$ and $\tilde{H}_2$, respectively, at

the origins, which are points over E fixed by $\tilde{H}_1$ and $\tilde{H}_2$, are at least β'. Let F have center 0 and let $c_u := 1 + \tau/r \le 1$.

If $[x]/a < [t]/v$, let $\mathfrak{a} := \prod_j (B_j^{(3)})^{c_j}$. By Lemma 5.5, with $\mathbb{A}^3_{u_2,p_x,q_x}$ in place of X, the section $(u_2 = 0)$, and the group $\tilde{H}_2$, and by Construction-Proposition A.29(2)(3), the prime divisor ν over $\mathbb{A}^3_{u_2,p_x,q_x}$ with center 0 satisfies

$$\frac{a(F,W,D_W)}{|I_G(F)|} = \frac{a(\nu,\mathbb{A}^3,c_u\operatorname{div}(u_2),\mathfrak{a})}{|I_{\tilde{H}_2}(\nu)|} \ge \beta' + \frac{(1-c_u)\operatorname{mult}_\nu u_2}{|I_{\tilde{H}_2}(\nu)|} \ge \beta'.$$

If $[x]/a \ge [t]/v$, let ν be over X as in Construction-Proposition A.29(3). We check the hypotheses of Lemma 5.5 for $n = 4$, $c = 1$, $f_1 = x_1y_1 + z_1^2 + u_1^d$ (or $x_1y_1 + z_1^2$) and $u = u_1$. Near 0, the variety X is a normal hypersurface with a cA singularity, hence klt. The surface $S_0 = X \cap (u_1 = 0)$ is the normal klt surface $(x_1y_1 + z_1^2 = 0)$. The group $\tilde{H}_1$ is diagonal and acts faithfully on S_0 by Construction-Proposition A.29(1). The factors of the $\mathbb{R}$-ideal $\mathfrak{a} := \prod_j (B_j^{(1)})^{c_j}$ are generated by $\tilde{H}_1$-semi-invariant polynomials, and $\mathfrak{a}|_{S_0} \ne 0$. By the first paragraph, $\operatorname{mld}((S_0 \ni 0, \mathfrak{a}|_{S_0})/\tilde{H}_1) \ge \beta'$. If the center of ν is 0, Lemma 5.5 with the section $(u_1 = 0)$ and the group $\tilde{H}_1$ implies that the left side of (A.130) is at least β'.

If the center of ν is a closed point $o = (x_o, 0, 0, 0)$ of ℓ_1 with $x_o \ne 0$, let H_o be the stabilizer of o in $\tilde{H}_1$. As $\tilde{H}_1$ is diagonal and $x_o \ne 0$, the group H_o acts trivially on x_1, so it acts diagonally on the coordinates $(x_1 - x_o, z_1, u_1)$. The projection

$$X \cap (x_1 \ne 0) \to (x_1 \ne 0) \subset \mathbb{A}^3_{x_1,z_1,u_1}$$

is an isomorphism with inverse given by $y_1 = -(z_1^2 + u_1^d)/x_1$, and it is $\tilde{H}_1$-equivariant, hence H_o-equivariant. After multiplying each $B_j^{(1)}$ by a power of the invariant unit x_1, which changes neither its divisor near o nor its character, it becomes a polynomial in $(x_1 - x_o, z_1, u_1)$. The group H_o acts faithfully on $(u_1 = 0)$ near o: an element acting trivially near o acts trivially on a dense open subset of the irreducible surface S_0, hence on S_0, and $\tilde{H}_1$ acts faithfully on S_0 by Construction-Proposition A.29(1). The conclusion of Lemma 5.5 involves only the prime divisors with center 0, and a prime divisor over $(x_1 \ne 0)$ with center o is a prime divisor over $\mathbb{A}^3$ with center o, and conversely. By the first paragraph, $(S_0, \mathfrak{a}|_{S_0})$ is lc near o, as $o \in (y_1 = z_1 = 0)$, the preimage of E in S_0 (Construction-Proposition A.28(1)). Thus the hypotheses of Lemma 5.5 hold with $\mathbb{A}^3$ in place of X, the point o as origin, the section $(u_1 = 0)$ and the group H_o, and, through the projection, the number β_0 of Lemma 5.5 is $\operatorname{mld}((S_0 \ni o, \mathfrak{a}|_{S_0})/H_o)$.

Let ν' be a prime divisor over S_0 with center o, and let

$$C_{S_0} := \sum_j c_j \operatorname{div}(h_{j,0}(x_1, y_1, z_1, 1))$$

be the boundary on S_0 in the first paragraph. The inertia group $I_{\tilde{H}_1}(\nu')$ fixes the center o of ν', so $I_{\tilde{H}_1}(\nu') = I_{H_o}(\nu')$. As the factors of $\mathfrak{a}|_{S_0}$ are the principal ideals generated by the $h_{j,0}(x_1, y_1, z_1, 1)$, we have $a(\nu', S_0, \mathfrak{a}|_{S_0}) = a(\nu', S_0, C_{S_0})$. By the first paragraph and Construction-Proposition A.29(1), the crepant quotient $(S_0, C_{S_0})/\tilde{H}_1$ is the restriction of $(Y, B')/\bar{G}$ to $S_0/\tilde{H}_1 = V_1/\bar{G}$ [Liu26, Lemma 2.6(0)]. Thus, by [Liu26, Lemma 2.6(2)], $a(\nu', S_0, C_{S_0})/|I_{\tilde{H}_1}(\nu')|$ is the log discrepancy, with respect to $(Y, B')/\bar{G}$, of the prime divisor under ν', whose center is the image of o. The image of o is a closed point of $E/\bar{G}$, since o belongs to the preimage of E, so the log discrepancy of the prime divisor under ν' is at least β'. Hence, by Definition 2.4 for the group H_o, which fixes o and preserves the factors of $\mathfrak{a}|_{S_0}$, the number $\operatorname{mld}((S_0 \ni o, \mathfrak{a}|_{S_0})/H_o)$, the infimum of $a(\nu', S_0, \mathfrak{a}|_{S_0})/|I_{H_o}(\nu')|$ over the prime divisors ν' over S_0 with center o, is at least β'. Together with the equality $I_{H_o}(\nu) = I_{\tilde{H}_1}(\nu)$, as the inertia group of ν fixes the center o of ν, Lemma 5.5 implies that the left side of (A.130) is at least β'.

If the center of ν is the generic point of ℓ_1, we use the second part of Lemma 5.5 with X, the section $(u_1 = 0)$, the $\mathbb{R}$-ideal $\mathfrak{a}$ and the group $\tilde{H}_1$ as in the case where the center of ν is 0, and

with $C = \ell_1$, which is a $\tilde{H}_1$-invariant integral curve in S_0 through 0. As ℓ_1 is the preimage of E in S_0, the pair $(S_0, \mathfrak{a}|_{S_0})$ is lc near the generic point of ℓ_1 by the first paragraph, and the minimal log discrepancy of the quotient of $(S_0, \mathfrak{a}|_{S_0})$ by $\tilde{H}_1$ at the image of the generic point of ℓ_1 is the log discrepancy of the prime divisor $E/\bar{G}$ with respect to $(Y, B')/\bar{G}$, which is at least β'. Thus the left side of (A.130) is at least β'. Finally $\beta' \geq 0$ since (Y, B') is lc near E.

Assume that the h_j are $\mathbf{T}$-homogeneous. If (W, D_W) is not lc near 0, then it has a prime divisor with center 0 and negative log discrepancy (cf. [KM98, Corollary 2.31(1)]), which contradicts the first part of this proof. □

Lemma A.31. *Notation as in Set-up 7.3 with $w = (a, -b, -v, v)$, where $a, v > 0$, $\gcd(a, v) = 1$, and $b = a + 2v$. Let F be a prime divisor over W_N with center 0 and $[f] := \operatorname{mult}_F f/|I_G(F)|$. Then $[x], [y] \leq 2a(F, W_N, 0)/|I_G(F)|$ and $[z], [t] \leq a(F, W_N, 0)/|I_G(F)|$.*

Proof. The hypersurface $(t = 0)$ of W_N is the normal surface $S' := (xy + z^2 = 0) \subset \mathbb{A}^3$, and

$$\frac{1}{2}\operatorname{div}_{S'}(x) + \frac{1}{2}\operatorname{div}_{S'}(y) = \operatorname{div}_{S'}(z) = (x = z = 0) + (y = z = 0),$$

whose pullback under $(P, Q) \to (P^2, -Q^2, PQ)$ is $\operatorname{div}(P) + \operatorname{div}(Q)$. Hence, by [Kaw07, Theorem], the pairs

$$(W_N, \frac{1}{2}\operatorname{div}(x) + \frac{1}{2}\operatorname{div}(y) + \operatorname{div}(t)) \quad \text{and} \quad (W_N, \operatorname{div}(z) + \operatorname{div}(t))$$

are lc near 0. Thus

$$a(F, W_N, 0) \geq \tfrac{1}{2}\operatorname{mult}_F x + \tfrac{1}{2}\operatorname{mult}_F y + \operatorname{mult}_F t, \qquad a(F, W_N, 0) \geq \operatorname{mult}_F z + \operatorname{mult}_F t.$$

Dividing by $|I_G(F)|$ gives the lemma. □

Proposition A.32. *Notation as in Construction-Proposition A.28, with $\tau \leq 0$.*

(1) There exist r_1 and C_0, depending only on a, v, Γ, e_0, such that if $r \geq r_1$, then, for every d, there exists a prime divisor F over W_N with center 0 and

$$\frac{a(F, W_N, D)}{|I_G(F)|} = \beta_- - \frac{\tau k}{re}$$

for positive integers e and k with $|\tau|k \leq C_0$. In particular $\alpha \leq \beta_- + C_0/r$.

(2) For any $\theta > 1$ there exists r_2, depending only on $a, v, \Gamma, e_0, \theta$, such that if $(Y, \theta B_-)$ is lc near E and $r \geq r_2$, then $\beta_- \leq \alpha$ for every d.

Proof. Let γ and M be as in Construction-Proposition A.28(3). We assume that $r \geq r_-$, and consider only the non-units h_j with $c_j > 0$. Then $|\tau| < 2bM$.

(1) Let $\bar{\nu}$ be a prime divisor over $Y/\bar{G}$ with center contained in $E/\bar{G}$ computing β_-. By Construction-Proposition A.28(3), the coefficients of $(Y, B_-)/\bar{G}$ belong to a DCC set depending only on a, v, Γ, e_0, so by Theorem 2.9(2), if the center of $\bar{\nu}$ is a closed point, we may assume that $a(\bar{\nu}, Y/\bar{G}, 0) \leq l$ for some l depending only on a, v, Γ, e_0. Let ν' be a prime divisor over $\mathbb{A}^2_{P,Q}$ (if the center of $\bar{\nu}$ is the image of $E \cap C_p$) or over $\mathbb{A}^2_{p_x,q_x}$ (otherwise) inducing $\bar{\nu}$, through Construction-Proposition A.29(1). Let T be $\mathbb{A}^2_{P,Q}$ or $\mathbb{A}^2_{p_x,q_x}$, according to the plane over which ν' is taken, and let $f_j := h^0_j(P^2, -Q^2, PQ, 1)$ on $\mathbb{A}^2_{P,Q}$ and $f_j := h^0_j(1, -p_x^2, p_x, q_x)$ on $\mathbb{A}^2_{p_x,q_x}$. The groups K_2 and $\tilde{H}_2$ have orders at most be_0, so $a(\nu', \mathbb{A}^2, 0) \leq be_0 l$. Since the $\mathbb{R}$-divisor $C_T := \sum_j c_j \operatorname{div}(f_j)$ on T, which is the pullback of B_- (Construction-Propositions A.28(2) and A.29(1)), is lc and $c_j \geq \gamma$, we have $\operatorname{mult}_{\nu'} f_j \leq be_0 l/\gamma$. If the center of $\bar{\nu}$ is the generic point of $E/\bar{G}$, then ν' is $(p_x = 0)$, with $\operatorname{mult}_{\nu'} f_j = \iota_j \leq 2M$. In all cases $\operatorname{mult}_{\nu'}$ of the plane coordinates and of the f_j are at most $m_1 := be_0 l/\gamma + 2M$. Let $c_u := 1 + \tau/r \leq 1$, let $C_0 := 2bM(5m_1 + be_0)$, and let k be a positive integer with $k > 5m_1$ in the congruence class modulo $|I(\nu')|$ fixed in each case below. As $|I(\nu')| \leq be_0$, we may take $k \leq 5m_1 + be_0 \leq C_0$, so that $|\tau|k < 2bM(5m_1 + be_0) = C_0$.

Assume first that ν' is over $\mathbb{A}^2_{p_x,q_x}$ with center a closed point $\tilde{y}$ of $(p_x=0)$, and let $K_{\tilde{y}}$ be the stabilizer of $\tilde{y}$ in $\tilde{H}_2$. Let $\Phi_j := h^0_j(1,-p_x^2,p_x,q_x)$ and

$$C_V := c_u \operatorname{div}(u_2) + \sum_j c_j \operatorname{div}(\Phi_j).$$

By [Liu26, Lemma 3.1], $\operatorname{mld}((T\ni\tilde{y}, C_T)/K_{\tilde{y}}) = \beta_-$, where $T=\mathbb{A}^2_{p_x,q_x}$. Hence, by Proposition 5.4, applied to $S=\mathbb{A}^2_{p_x,q_x}$, the point $\tilde{y}$, the group $K_{\tilde{y}}$, the character of u_2, the Φ_j, C_V, and $\ell := k$ in the class of Lemma 3.1, the Gauss extension F of $\operatorname{mult}_{\nu'}$ with u_2-value k satisfies $I(F)=I(\nu')$, $\operatorname{mult}_F \Phi_j = \operatorname{mult}_{\nu'} f_j$, and

$$\frac{a(F,\mathbb{A}^3,C_V)}{e} = \beta_- + \frac{(1-c_u)k}{e} = \beta_- - \frac{\tau k}{re}, \qquad e := |I(\nu')|.$$

If the center of $\bar{\nu}$ is the generic point of $E/\bar{G}$, let I_E be the inertia group of $(p_x=0)$ in $\tilde{H}_2$, which is cyclic with a faithful character on p_x, let $k \equiv q \pmod{|I_E|}$ for the integer q with $\chi_{u_2}|_{I_E} = \chi_{p_x}|^q_{I_E}$, and let F be the monomial valuation with weights $(k,1,0)$ on (u_2,p_x,q_x). A direct calculation shows that $I(F)=I_E$, $\operatorname{mult}_F \Phi_j = \iota_j$, and

$$\frac{a(F,\mathbb{A}^3,C_V)}{|I_E|} = \frac{1-\beta_E+(1-c_u)k}{|I_E|} = \beta_- - \frac{\tau k}{r|I_E|}.$$

In both cases, the prime divisor F has, through ψ_3, the orders

$$\operatorname{mult}_F x = \frac{ak}{r}, \qquad \operatorname{mult}_F y = 2\operatorname{mult}_F p_x - \frac{bk}{r},$$
$$\operatorname{mult}_F z = \operatorname{mult}_F p_x - \frac{vk}{r}, \qquad \operatorname{mult}_F t = \operatorname{mult}_F q_x + \frac{vk}{r},$$

since $dk > 2\operatorname{mult}_F p_x$. As $\operatorname{mult}_F p_x \geq 1$, for $r \geq 2bC_0$ the four orders are positive with $\operatorname{mult}_F y \geq 1$ and $\operatorname{mult}_F z \geq \frac{1}{2}$. Hence F induces a prime divisor over W_N with center 0. A monomial $x^i y^l z^m t^n$ of h_j of $\mathbf{T}$-weight at most $\kappa_j - r$ satisfies $bl + vm \geq r - bM$, so the order of the monomial along F is at least $\frac{r-bM}{2b}$, which exceeds $\frac{bMk}{r} + m_1 \geq \operatorname{mult}_F(\eta_x^{\kappa_j}\Phi_j)$ for r large. The other terms of $B^{(3)}_j - \Phi_j$ are divisible by u_2, except $h^0_j(1,-p_x^2-u_2^d q_x^N,p_x,q_x)-\Phi_j$, which is divisible by u_2^d. So all of them have order at least $k > m_1$. Hence $\operatorname{mult}_F B^{(3)}_j = \operatorname{mult}_F \Phi_j$ for every j, and (A.130) implies that the induced prime divisor over W_N has $a(\cdot,W_N,D)/|I_G(\cdot)| = \beta_- - \tau k/(re)$.

If ν' is over $\mathbb{A}^2_{P,Q}$ with center 0, let $\Phi_j := h^0_j(P^2,-Q^2,PQ,1)$ and

$$C_V := c_u \operatorname{div}(u_1) + \sum_j c_j \operatorname{div}(\Phi_j).$$

By Proposition 5.4, applied to $S=\mathbb{A}^2_{P,Q}$, the point 0, the group K_2, the character of u_1, the Φ_j, C_V, and $\ell := k$ in the class of Lemma 3.1 for ν' and the character of u_1, with $e := |I_{K_2}(\nu')|$, there exists a prime divisor F over $\mathbb{A}^3_{P,Q,u_1}$ with $a(F,\mathbb{A}^3,C_V)/e = \beta_- - \tau k/(re)$ and $\operatorname{mult}_F \Phi_j = \operatorname{mult}_{\nu'} f_j$. Now $\operatorname{mult}_F Q \geq 1$, $k > 5m_1 \geq 2\operatorname{mult}_F P + 2\operatorname{mult}_F Q + m_1$, and through ψ_2

$$\operatorname{mult}_F x = 2\operatorname{mult}_F P + \frac{ak}{r}, \qquad \operatorname{mult}_F y = 2\operatorname{mult}_F Q - \frac{bk}{r},$$
$$\operatorname{mult}_F z = \operatorname{mult}_F P + \operatorname{mult}_F Q - \frac{vk}{r}, \qquad \operatorname{mult}_F t = \frac{vk}{r},$$

because $\operatorname{mult}_F(u_1^d P^{-2}) \geq k - 2\operatorname{mult}_F P > 2\operatorname{mult}_F Q$. The difference

$$h^0_j(P^2,-Q^2-u_1^d P^{-2},PQ,1) - \Phi_j$$

is $u_1^d P^{-2}$ times a polynomial in $P, Q, u_1^d P^{-2}$, so the order of the difference is at least $k - 2\operatorname{mult}_F P > m_1$. The terms of $B^{(2)}_j$ divisible by u_1 have order at least k, and the monomials of h_j of $\mathbf{T}$-weight at most $\kappa_j - r$ have order at least $\frac{r-bM}{2b}$ along F by the orders of y and z

above. Hence $\mathrm{mult}_F B_j^{(2)} = \mathrm{mult}_F \Phi_j$, and (A.130) implies the first statement. Since $|\tau|k \le C_0$, it implies $\alpha \le \beta_- + C_0/r$.

(2) Let $\Theta_1 := \frac{2\theta}{\theta-1}$, and $\Theta_2 := \frac{2}{\gamma(\theta-1)}$. The pair $(Y, \theta B_-)$ is the pair (Y, B') of Lemma A.30 for the labels θc_j and the h_j^0, with mass $\theta\tau \le 0$. As the h_j^0 are $\mathbf{T}$-homogeneous, the last statement of Lemma A.30, applied with $W = W_N$ and with $W = W_0$, shows that $(W_N, \theta \sum_j c_j \operatorname{div}(h_j^0))$ and $(W_0, \theta \sum_j c_j \operatorname{div}(h_j^0))$ are lc near 0. By [Kaw07, Theorem] and Lemma 4.14 on $\mathbb{A}^4$ with the equation $xy + z^2 + t^N$ and exponents in $\{1\} \cup \theta(\Gamma\backslash\{0\})$, the pair $(W_N, \theta D)$ is lc near 0 for r large. Likewise, by [Liu26, Lemma 2.6] and Lemma 4.14 on the double cover $\mathbb{A}^3_{p_0,t,q_0}$ of W_0, the pair $(\mathbb{A}^3, \theta B_{\mathrm{ref}})$ is lc near 0 for r large, where

$$B_{\mathrm{ref}} := \sum_j c_j \operatorname{div}(h_j(p_0^2, -q_0^2, p_0 q_0, t)).$$

Thus $(\mathbb{A}^3, B_{\mathrm{ref}})$ is klt near 0.

Let F be a prime divisor over W_N with center 0, $e := |I_G(F)|$, and $A_F := a(F, W_N, 0)$. Since β_- is at most the log discrepancy of the prime divisor $E/\bar{G}$ with respect to $(Y, B_-)/\bar{G}$, which is at most one, we may assume $a(F, W_N, D) \le 2e$. As $(W_N, \theta D)$ is lc, $A_F \ge \theta \sum_j c_j \,\mathrm{mult}_F h_j$, so

$$\sum_j c_j[h_j] \le \frac{2}{\theta - 1}, \qquad [h_j] \le \Theta_2, \qquad \frac{A_F}{e} \le \Theta_1,$$

and Lemma A.31 implies $[x], [y] \le 2\Theta_1$ and $[z], [t] \le \Theta_1$. A monomial $x^i y^l z^m t^n$ of h_j of $\mathbf{T}$-weight $\kappa_j + k'r$ satisfies $ai + vn \ge r - bM$ if $k' \ge 1$, and $bl + vm \ge r - bM$ if $k' \le -1$. Hence, with $\rho_+ := \min([x]/a, [t]/v)$ and $\rho_- := \min([y]/b, [z]/v)$, we have $[x^i y^l z^m t^n] \ge (r - bM)\rho_+$, resp. $\ge (r - bM)\rho_-$.

Case (i): every monomial of h_j of $\mathbf{T}$-weight at most $\kappa_j - r$ has $[\cdot] > \Theta_2$, for every j. Let $\hat{h}_j$ be the sum of the terms of h_j of $\mathbf{T}$-weights at least κ_j. Since $[h_j - \hat{h}_j] > \Theta_2 \ge [h_j]$, we have $[\hat{h}_j] = [h_j]$, so $a(F, W_N, D) = a(F, W_N, \sum_j c_j \operatorname{div}(\hat{h}_j))$. The $\hat{h}_j$ satisfy the hypotheses of Lemma A.30 with $h_{j,0} = h_j^0$ and $B' = B_-$, which is lc near E by Construction-Proposition A.28(3). Hence $a(F, W_N, D)/e \ge \beta_-$.

Case (ii): some monomial of some h_j of $\mathbf{T}$-weight at most $\kappa_j - r$ has $[\cdot] \le \Theta_2$. With $\epsilon_0 := 1/(8b^2 v e_0)$, we have $\rho_- \le \Theta_2/(r - bM) \le \epsilon_0$ for r large, so $\rho_+ > \epsilon_0$ by Lemma A.27, and $[t] > v\epsilon_0$. As $N + 2 = rd/v \ge r/v$, we have $N[t] \ge (r - 2v)\epsilon_0 > 4\Theta_1 + \Theta_2$ for r large. Thus $N[t] > [x] + [y]$ and $N[t] > 2[z]$, and $xy + z^2 = -t^N$ implies $[x] + [y] = 2[z]$. The projections $W_N \to \mathbb{A}^3_{x,z,t}$ and $W_0 \to \mathbb{A}^3_{x,z,t}$ are birational and G-equivariant, and $dx \wedge dz \wedge dt/x$ generates ω_{W_N} and ω_{W_0} near 0. Hence F is also a prime divisor F_0 over W_0, with $a(F_0, W_0, 0) = A_F$ and $I_G(F_0) = I_G(F)$. On W_0 the function y is $y_0 := -z^2/x$, and $[y_0] = 2[z] - [x] = [y] > 0$, so F_0 has center 0 on W_0. Since $y - y_0 = -t^N/x$, the difference $h_j(x, y, z, t) - h_j(x, y_0, z, t)$ is $-t^N/x$ times a polynomial in x, y, y_0, z, t, so

$$[h_j(x, y, z, t) - h_j(x, y_0, z, t)] \ge N[t] - [x] > \Theta_2 \ge [h_j].$$

Therefore, with $D_{\mathrm{ref}} := \sum_j c_j \operatorname{div}_{W_0}(h_j)$,

$$\frac{a(F, W_N, D)}{e} = \frac{a(F_0, W_0, D_{\mathrm{ref}})}{e} \ge \mathrm{mld}((W_0 \ni 0, D_{\mathrm{ref}})/G) = \mathrm{mld}((\mathbb{A}^3 \ni 0, B_{\mathrm{ref}})/G'),$$

the last equality by [Liu26, Lemma 2.6(5)], with G' as in Construction-Proposition A.28(4). We check the hypotheses of [Liu26, Theorem 16.21(2)] with $(p', q', r') = (a/g_1, 2v/g_1, b/g_1)$, the bound $2e_0$ on $|G'/(G' \cap \mathbf{T}')| = g_1[G : R]$, and $n' = 2r/g_1$. For r large, n' is at least the n_0 there, and the pair $(\mathbb{A}^3, B_{\mathrm{ref}})$ is klt near 0. The monomials of least degree of h_j are monomials of h_j^0 and have degree at most M, so their lifts have degree at most $2M$, while the lifts of the monomials of $h_j - h_j^0$ have degree at least $(r - bM)/b$. Hence, for r large, the principal blocks of the lifted equations in the sense of [Liu26, Theorem 16.21] are the h_j' of Construction-Proposition A.28(4), and the

minimum b' of [Liu26, Theorem 16.21(2)] for them is β_- by Construction-Proposition A.28(4). Thus, by [Liu26, Theorem 16.21(2)], $a(F, W_N, D)/e \geq \beta_-$.

In both cases $a(F, W_N, D)/e \geq \beta_-$, so $\alpha \geq \beta_-$. □

Proof of Theorem 7.17. The group $\tilde{H}$ of Theorem 7.17 is the group $\tilde{H}$ of Proposition A.22, which is the group of Proposition 7.13 with $-a$ in place of a. The hypotheses of Proposition 7.13 hold since $-a + b = 2v$, $b > 0$, and $\gcd(a, v) = 1$. It is the image of the pairs (g, ζ) with $\zeta^b = g_y$, of order $b[G : R]$, and on the y-root plane it corresponds to $\eta^b = y$, $p = z/\eta^v$, $q = t\eta^v$. Thus (2) is Proposition 7.13(1). Statement (3) is Proposition A.22(1). We prove (1).

Assume that $(\alpha_i)_i$ is a strictly increasing sequence of positive numbers, where α_i is the value α of data $(N_i, G_i, D_i = \sum_j c_{ij} \operatorname{div}(h_{ij}))$ as in Set-up 7.3 with Γ, e_0, w fixed and $r_i \to +\infty$. If $\tau_i > 0$ for infinitely many i, this contradicts Proposition A.22(2). Thus we may assume that $\tau_i \leq 0$ and $r_i \geq \max(r_-, r_1)$ for every i, where r_- is the constant of Construction-Proposition A.28(3) and r_1 that of Proposition A.32(1). All constants below depend only on a, v, Γ, e_0 and the subsequences, not on the d_i. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the discrete data. Passing to a subsequence, we may assume that the number of non-units h_{ij} with $c_{ij} > 0$, which is at most M, the κ_j, and the ι_j do not depend on i, since $|\kappa_j| \leq bM$ and $\iota_j \leq 2M$. The group $\bar{G}_i := G_i/R_i$ acts on Y as a subgroup of order at most e_0 of the torus of Y, and $K_{2,i}$ and $\tilde{H}_{2,i}$ are diagonal groups of orders at most be_0. There are finitely many such subgroups, so we may assume that they and the characters of the chart functions of (A.129) for the h^0_{ij} do not depend on i, and we write $\bar{G}$, K_2, $\tilde{H}_2$. Since Γ is a DCC set, we may assume that c_{ij} is non-decreasing in i with limit $c^*_j > 0$. Let β_i be the number β_- of Construction-Proposition A.28 for the i-th data. By the ACC in Construction-Proposition A.28(3), we may assume that β_i is non-increasing with limit L. Proposition A.32(1) implies

$$\alpha_i < \lim_{i'} \alpha_{i'} \leq L.$$

Let $\tau_* := \sum_j c^*_j \kappa_j = \lim_i \tau_i \leq 0$ and $\beta^*_E := \sum_j c^*_j \iota_j = \lim_i \beta_{E,i}$. As β_i is at most the log discrepancy $(1 - \beta_{E,i})/|I_{\bar{G}}(E)| \leq 1 - \beta_{E,i}$ of the prime divisor $E/\bar{G}$, we have $\beta^*_E \leq 1 - \lim_i \alpha_i < 1$.

Step 2. In this step we raise the labels to c^*_j. Let $B^*_i := \sum_j c^*_j D_{h^0_{ij}} \geq B_{-,i}$, and let β^*_i be the minimum of the log discrepancies of $(Y, B^*_i)/\bar{G}$ over the prime divisors with center contained in $E/\bar{G}$. We show that, for i large, (Y, B^*_i) is lc near E and $\beta^*_i = L$.

The pair (Y, B^*_i) is lc near E if and only if the pullbacks $(\mathbb{A}^2, \sum_j c^*_j \operatorname{div}(\varphi_{ij}))$ to $\mathbb{A}^2_{P,Q}$ and $\mathbb{A}^2_{p_x,q_x}$ are lc near the preimages of E, where φ_{ij} denote the chart functions. Since $(Y, B_{-,i})$ is lc near E, the lc thresholds of $\sum_j c^*_j \operatorname{div}(\varphi_{ij})$ near the preimages of E are at least $\min_j c_{ij}/c^*_j$, which tends to one. By [dFEM11, Theorem 1.1], the lc thresholds belong to an ACC set, so they are at least one for i large. The coefficients of $(Y, B^*_i)/\bar{G}$ belong to the DCC set of [Liu26, Lemma 2.10(1)(5)] for $\Gamma \cup \{c^*_j\}_j$, so by Theorem 2.9(2) we may choose prime divisors ν'_i over the planes computing β^*_i with $a(\nu'_i, \mathbb{A}^2, 0)$, $\operatorname{mult}_{\nu'_i} \varphi_{ij}$, and $|I(\nu'_i)|$ bounded independently of i. For the generic point of E, we take $(Q = 0)$ or $(p_x = 0)$, with $\operatorname{mult} \varphi_{ij} = \iota_j$. Then $\beta^*_i \leq \beta_i$ and

$$\beta_i \leq \frac{a(\nu'_i, \mathbb{A}^2, \sum_j c_{ij} \operatorname{div}(\varphi_{ij}))}{|I(\nu'_i)|} = \beta^*_i + \sum_j (c^*_j - c_{ij}) \frac{\operatorname{mult}_{\nu'_i} \varphi_{ij}}{|I(\nu'_i)|},$$

so $\beta^*_i \to L$. The numbers β^*_i belong to the finite set of numbers $(n_1 - \sum_j c^*_j m_j)/e$ with bounded positive integers n_1, e and bounded non-negative integers m_j, hence $\beta^*_i = L$ for i large.

Step 3. In this step we construct the limit. We apply Construction-Proposition 4.1(b) to the fixed surface Y, the curve E, the charts $V_1 \cong \mathbb{A}^2_{P,Q}/\mu_{2v}(1, -1)$ and $V_2 \cong \mathbb{A}^2_{p_x,q_x}/\mu_a(v, -v)$ with $e_1 = Q$ and $e_2 = p_x$, the branch divisor $B_{\mathrm{br}} = 0$ (the charts are quasi-étale), the group $\bar{G}$, and the tuples formed by the chart functions of the h^0_{ij} for the indices j of **Step 1**, and of Φ_+, Φ_-, Φ_0. The hypotheses hold. The characters are fixed by **Step 1**. The divisors descend to the $\bar{G}$-invariant

divisors $D_{h^0_{ij}}$ and $\mathrm{div}_Y(\Phi_\pm)$, $\mathrm{div}_Y(\Phi_0)$ by Construction-Proposition A.28(2). On the overlap of the charts, the b-th powers of the chart functions of h^0_{ij} have the fixed quotient $(-p_x^2)^{\kappa_j}/y_1^{\kappa_j}$ by Construction-Proposition A.28(2), so after passing to a subsequence their quotients are units independent of i. The orders along $(e_\alpha = 0)$ are $\iota_j \le 2M$. For a $\mathbf{T}$-homogeneous h of weight κ_j, the coefficient of p_x^l in $h(1, -p_x^2, p_x, q_x)$, resp. of Q^l in $h(P^2, -Q^2, PQ, 1)$, is a polynomial of degree bounded in terms of l, κ_j, a, v, as the $\mathbf{T}$-weight bounds the remaining exponents. We obtain an algebraically closed field $k \supset \mathbb{C}$, the formal completion $\mathcal{A}_k$ of Y_k along E_k, and limits of the chart functions. Since Φ_+, Φ_-, Φ_0 do not depend on i, their limits are themselves, and the limit of any subtuple is the corresponding subtuple of the limit: the projection of the jet spaces onto the jets of the subtuple maps the generic point of each set Z_d of Construction-Proposition 4.1 to the generic point of the closure of the image of Z_d, and the preimage in Z_d of a non-empty open subset of that closure is a non-empty open subset of Z_d. Let $C_\infty := \sum_j c_j^* \,\mathrm{div}(f_{\infty,j})$. By Theorem 4.2(1)(2) and **Step 2**, $(\mathcal{A}_k, C_\infty)$ is lc near E_k, and the minimum of the log discrepancies of the quotient of $(\mathcal{A}_k, C_\infty)$ by $\bar G$ over the prime divisors with center contained in $E_k/\bar G$ is L.

If $(\mathcal{A}_k, C_\infty)$ is klt near E_k, then by Theorem 4.2(1) there exists $\theta > 1$ such that $(Y, \theta B_i^*)$, hence $(Y, \theta B_{-,i})$, is lc near E for every i. Proposition A.32(2) then implies $\alpha_i \ge \beta_i \ge L$ for i large, which contradicts $\alpha_i < L$. We assume from now on that $(\mathcal{A}_k, C_\infty)$ is not klt near E_k.

Step 4. In this step we show that there is exactly one floor, that it intersects E_k at one point fixed by $\bar G$, and that the preimage of the floor is transverse to the preimage of E_k. Write $C_\infty = \beta_E^* E_k + T'$ with $T' \ge 0$ not containing E_k. For a closed point $\varpi \in E_k$ let $(T' \cdot E_k)_\varpi$ be the local intersection number on the chart cover divided by the order of the chart group at ϖ, which is $2v$ at $E \cap C_p$, a at $E \cap C_q$, and one elsewhere. Fix j and an integer $d > \iota_j$, and let $\mathcal{J}_d$ and Z_d be as in Construction-Proposition 4.1. The condition that the quotient of the chart functions on the overlap of V_1 and V_2 is the unit of **Step 3**, together with the vanishing of the coefficients of e_α^l for $l < \iota_j$, is a closed condition on $\mathcal{J}_d$, and so is the vanishing of the coefficient of $e_\alpha^{\iota_j}$. The d-jet of the i-th tuple satisfies the first condition and does not satisfy the second for every i, since the order of the chart functions along $(e_\alpha = 0)$ is ι_j. As the d-jet of the limit maps to the generic point of Z_d, and every non-empty open subset of Z_d contains the d-jets of all but finitely many i, the limit satisfies the first condition and does not satisfy the second. Thus the restrictions $(f_{\infty,j,\alpha}/e_\alpha^{\iota_j})|_{(e_\alpha=0)}$ glue to a non-zero section of the orbifold line bundle on E_k given by the transition factors of $\mathcal{O}_Y(D_{h^0_{ij}} - \iota_j E)|_E$, which do not depend on i. The sum of the orders of the zeros of this section, each divided by the order of the chart group at the zero, is the degree of the bundle. By Construction-Proposition A.28(2), $D_{h^0_{ij}} = \pi^*\Delta_{h^0_{ij}} - \frac{2\kappa_j}{b}E$ with $(\pi^*\Delta_{h^0_{ij}} \cdot E) = 0$, so

$$\sum_\varpi \Big(\big(\mathrm{div}(f_{\infty,j}) - \iota_j E_k \big) \cdot E_k \Big)_\varpi = \big((D_{h^0_{ij}} - \iota_j E) \cdot E \big) = \Big(-\frac{2\kappa_j}{b} - \iota_j \Big)(E^2).$$

As $T' = \sum_j c_j^* (\mathrm{div}(f_{\infty,j}) - \iota_j E_k)$, summing over j with the weights c_j^* and using (A.128) gives

$$\sum_\varpi (T' \cdot E_k)_\varpi = \sum_j c_j^* \Big(-\frac{2\kappa_j}{b} - \iota_j \Big)(E^2) = \frac{b\beta_E^* + 2\tau_*}{2av} \le \Big(\frac{1}{a} + \frac{1}{2v} \Big)\beta_E^* < \frac{1}{a} + \frac{1}{2v}. \tag{A.131}$$

By Definition-Proposition 5.7, applied with the limit value $L > 0$, there exists a floor. Floors have coefficient one, so they are different from E_k as $\beta_E^* < 1$, and their preimages in the chart covers are smooth and pairwise disjoint over E_k. A floor through $E \cap C_p$, through $E \cap C_q$, or through another point of E_k contributes at least $\frac{1}{2v}$, $\frac{1}{a}$, or 1 to (A.131). Hence two distinct points of E_k on floors are impossible, and a point other than $E \cap C_p$ and $E \cap C_q$ is possible only if $a = 1$, since

$$\sum_\varpi (T' \cdot E_k)_\varpi < \frac{1}{a} + \frac{1}{2v} \le 1 \quad \text{if } a \ge 2.$$

Thus there exists a unique floor F, and $F \cap E_k$ is one point ϖ, which is fixed by $\bar{G}$.

Let $\tilde{F}$ and $\tilde{E}$ be the preimages of F and E_k in the chart cover at ϖ, let $\hat{K}$ be K_2 if $\varpi = E \cap C_p$ and $\tilde{H}_2$ otherwise, which fixes the preimage of ϖ, let $\rho := \mathrm{mult}_{\varpi}((C_\infty - F)|_{\tilde{F}})$ on the cover, $\mu := 1 - \rho$, and n_F the order of the image of $\hat{K}$ in $\mathrm{Aut}(T_{\varpi}\tilde{F})$. By Lemma 5.6 (here $\rho \le 1$ by Lemma 2.14, as $(\mathcal{A}_k, C_\infty)$ is lc near E_k), the minimum of the log discrepancies over the prime divisors with center ϖ is at most μ/n_F (we use only the upper bound in the proof of Lemma 5.6, which is given by the monomial valuation in semi-invariant parameters, uses only [KM98, Lemma 2.29] and [Liu26, Lemma 2.6], and holds on the formal germ of the chart cover at ϖ, whose boundary is given by power series, with Lemma 3.7 in place of [Liu26, Lemma 2.6]), so $0 < L \le \mu/n_F$ and $\rho < 1$. If $\tilde{F}$ is tangent to $\tilde{E}$ with contact $k_1 \ge 2$ at a point which is not torus-fixed, then (A.131) would be at least $2 > \frac{1}{a} + \frac{1}{2v}$. Let ϖ be torus-fixed with chart index n, and m' the other index, so $\{n, m'\} = \{2v, a\}$. In coordinates $(\tilde{u}, \tilde{e})$ with $\tilde{E} = (\tilde{e} = 0)$ on which $\mu_n \subset \hat{K}$ acts by (ζ, ζ^{-1}), write $\tilde{F} = (\tilde{e} = f(\tilde{u}))$ with $\mathrm{mult}_0 f = k_1 \ge 2$. The invariance of $\tilde{F}$ under μ_n forces $n \mid k_1 + 1$. By (A.131), $\frac{k_1}{n} \le \frac{m'+n}{m'n}\beta_E^*$, while $\beta_E^* k_1 \le \rho < 1$. Thus $k_1^2 < 1 + \frac{n}{m'} \le 1 + n$, which is impossible: for $n = 1$ since $k_1 \ge 2$, for $n = 2$ since $k_1 \ge 3$ is odd, and for $n \ge 3$ since $k_1 \ge n - 1$ and $(n-1)^2 \ge n + 1$. Hence $\tilde{F}$ is transverse to $\tilde{E}$, and $\beta_E^* \le \rho$, that is, $\mu \le 1 - \beta_E^*$.

Step 5. In this step we construct an augmentation on Y. Let

$$(U_0, \lambda_0) := \Big(\Phi_-, \frac{\mu}{2v}\Big), \quad \Big(\Phi_+, \frac{\mu}{a}\Big), \quad (\Phi_0, \mu)$$

if $\varpi = E \cap C_p$, if $\varpi = E \cap C_q$, and otherwise, respectively. We show that $(\mathcal{A}_k, C_\infty + \lambda_0 \operatorname{div}(U_0))$ is lc near E_k.

Let $\varpi = E \cap C_p$. By (A.128),

$$C_\infty + \lambda_0 \operatorname{div}(U_0) = (\beta_E^* + \mu)E_k + T' + \frac{b\mu}{2v} C_q,$$

and $\beta_E^* + \mu \le 1$. On the cover at ϖ, the pullback of U_0 is $(-1)^v Q^{2v}$, so the restriction of the boundary minus $\tilde{F}$ to $\tilde{F}$ has coefficient $\rho + \mu = 1$ at ϖ, as $(\tilde{F} \cdot \tilde{E})_{\varpi} = 1$. By [Kaw07, Theorem], the pair is lc near ϖ. We raise the coefficient of E_k to one near the other points of E_k, so that [Kaw07, Theorem] can be used along $\tilde{E}$ on the chart covers. This is allowed since $\beta_E^* + \mu \le 1$ and a larger boundary being lc implies the claim. It then suffices to bound the restricted coefficients there. Since $(F \cdot E_k)_{\varpi} = \frac{1}{2v}$, $(C_q \cdot E) = \frac{1}{a}$ and $\frac{b}{2av} = \frac{1}{a} + \frac{1}{2v}$,

$$\sum_{\varpi' \ne \varpi} \Big((T' - F + \tfrac{b\mu}{2v} C_q) \cdot E_k\Big)_{\varpi'} \le \Big(\frac{1}{a} + \frac{1}{2v}\Big)\beta_E^* - \frac{1}{2v} + \Big(\frac{1}{a} + \frac{1}{2v}\Big)(1 - \beta_E^*) = \frac{1}{a}. \tag{A.132}$$

So the restricted coefficient on the cover is at most $a \cdot \frac{1}{a} = 1$ at $E \cap C_q$ and at most $\frac{1}{a} \le 1$ at the other points, and the pair is lc near E_k. If $\varpi = E \cap C_q$, then $\operatorname{div}_Y(\Phi_+) = aE + bC_p$, so near ϖ we have $\lambda_0 \operatorname{div}_Y(U_0) = \mu E$, whose pullback to the cover $\mathbb{A}^2_{p_x, q_x}$ is $\mu \operatorname{div}(p_x)$. Thus the restriction of the boundary minus $\tilde{F}$ to $\tilde{F}$ has coefficient $\rho + \mu = 1$ at ϖ, as $\tilde{F}$ is transverse to $\tilde{E}$, and the pair is lc near ϖ by [Kaw07, Theorem]. Since $(F \cdot E_k)_{\varpi} = \frac{1}{a}$ and $(C_p \cdot E) = \frac{1}{2v}$,

$$\sum_{\varpi' \ne \varpi} \Big((T' - F + \tfrac{b\mu}{a} C_p) \cdot E_k\Big)_{\varpi'} \le \Big(\frac{1}{a} + \frac{1}{2v}\Big)\beta_E^* - \frac{1}{a} + \Big(\frac{1}{a} + \frac{1}{2v}\Big)(1 - \beta_E^*) = \frac{1}{2v}. \tag{A.133}$$

So the restricted coefficient on the cover is at most $2v \cdot \frac{1}{2v} = 1$ at $E \cap C_p$ and at most $\frac{1}{2v} \le 1$ at the other points, and the pair is lc near E_k. If ϖ is not torus-fixed, then $a = 1$, Φ_0 is a local equation of E at ϖ, the coefficient at ϖ is again $\rho + \mu = 1$, and, as $\beta_E^* + \mu \le 1$,

$$\sum_{\varpi' \ne \varpi} \Big((T' - F + \mu(C_p + C_q)) \cdot E_k\Big)_{\varpi'} \le \Big(\frac{1}{a} + \frac{1}{2v}\Big)(\beta_E^* + \mu) - 1 \le \frac{1}{a} + \frac{1}{2v} - 1 \le \min\Big(\frac{1}{a}, \frac{1}{2v}\Big). \tag{A.134}$$

So the restricted coefficients on the covers are at most $2v \cdot \frac{1}{2v} = 1$ at $E \cap C_p$, at most $a \cdot \frac{1}{a} = 1$ at $E \cap C_q$, and at most 1 elsewhere, and the pair is lc near E_k.

Let m be the order of the character of U_0 under G_i, which is the order of the character of U_0 under $\bar{G}$ and does not depend on i. We show $\lambda_0/m = \mu/n_F$. If $\varpi = E \cap C_p$, then Q is semi-invariant under K_2 and $Q|_{\tilde{F}}$ is a parameter at ϖ, so n_F is the order of the character χ_Q of Q. As $\mu_{2v} \subset K_2$ acts on Q faithfully, $2v \mid n_F$, and the character of U_0 is χ_Q^{2v} by Construction-Proposition A.29(1), of order $n_F/(2v)$. Likewise $m = n_F/a$ if $\varpi = E \cap C_q$. If $\varpi = (0, q_x^0)$ with $q_x^0 \neq 0$, then $\bar{G}$ fixes ϖ, so it acts trivially on q_x, and the character of $U_0 = p_x q_x$ is that of p_x, whose restriction to $\tilde{F}$ is a parameter. Thus $m = n_F$. In all cases

$$L \leq \frac{\mu}{n_F} = \frac{\lambda_0}{m}. \tag{A.135}$$

Step 6. In this step we conclude. By Theorem 4.2(1) applied to the tuples with U_0 added and the labels c_j^* and λ_0, the pair $(Y, B_i^* + \lambda_0 \operatorname{div}_Y(U_0))$ is lc near E for every i. As U_0 is a G_i-semi-invariant monomial of $\mathbf{T}$-weight zero without factor xy, $\operatorname{div}_Y(U_0) = D_{U_0}$ and the mass of the labelled equations $(h_{ij}^0)_j, U_0$ with labels $(c_j^*)_j, \lambda_0$ is $\tau_* \leq 0$. By Lemma A.30, the pair

$$(W_{N_i}, \sum_j c_j^* \operatorname{div}(h_{ij}^0) + \lambda_0 \operatorname{div}(U_0))$$

is lc near 0. By [Kaw07, Theorem] and Lemma 4.14 on $\mathbb{A}^4$ with the equations $xy + z^2 + t^{N_i}$ and U_0, and exponents in the finite set $\{1, \lambda_0\} \cup \{c_j^*\}_j$, the pair

$$(W_{N_i}, \sum_j c_j^* \operatorname{div}(h_{ij}) + \lambda_0 \operatorname{div}(U_0))$$

is lc near 0 for i large, and so is $(W_{N_i}, D_i + \lambda_0 \operatorname{div}(U_0))$ since $c_{ij} \leq c_j^*$. Lemma 5.8 with $Z = \{0\}$ and $U = U_0$, together with (A.135), implies

$$\alpha_i \geq \frac{\lambda_0}{m} \geq L,$$

which contradicts $\alpha_i < L$. □

Proof of Corollary 7.18. Write $w = (u, -2v - u, -v, v)$. The exchange $(x, y, z, t) \to (y, x, z, t)$ is an automorphism of W_N which conjugates G to a finite diagonal group, maps $\mathbf{T}$ to $\mathbf{T}_{(-2v-u,u,-v,v)}$, preserves the labels, the absence of monomials divisible by xy, the $\mathbf{T}$-weights of monomials, τ, r, $[G : R]$, d, and α. There are three cases.

If $-2v < u < 0$, let $a := -u$ and $b := 2v + u$. Then $a, b > 0$, $a + b = 2v$, and $\gcd(a, v) = 1$, so $w = (-a, -b, -v, v)$ is an interior canonical circle as in Construction-Proposition 7.14. The scalar circle $u = -1$, $v = 1$ is the case $a = b = v = 1$. Theorem 7.15 implies that α cannot increase along $\mathbf{T}$ for every sign of τ.

If $u \in \{0, -2v\}$, then $\gcd(u, v) = 1$ forces $v = 1$, and w is $(0, -2, -1, 1)$ or $(-2, 0, -1, 1)$, and Theorem 7.12 treats w.

If $u > 0$, this case is Theorem 7.17 with $a := u$. If $u < -2v$, the exchange of x and y reduces to the circle $(-2v - u, u, -v, v)$ with $-2v - u > 0$ and $\gcd(-2v - u, v) = 1$, which is Theorem 7.17 with $a := -2v - u$.

Finally, let $(\alpha_i)_i$ be a strictly increasing sequence of positive values with w fixed and $[G_i : R_i] \leq e_0$. If r_i were unbounded, a subsequence with $r_i \to +\infty$ would still be strictly increasing, which is impossible. Hence r_i is bounded, and $|G_i| = r_i[G_i : R_i] \leq e_0 \sup_i r_i$. □

A.10. **Limits of groups on axial sources.** In this subsection, we prove Proposition 7.19. The exponents N_i are forced to grow by Theorem 7.8, and the groups are controlled by Hausdorff limits in the compact torus $\mathbb{T}^4$ (Lemma 4.6). The circles $\mathbf{T}_{(1,-1,0,0)}$ and $\mathbf{T}_{(0,0,0,1)}$ are excluded by Theorems 8.3 and 8.5.

Proof of Proposition 7.19. The proof is long, so we divide it into several steps.

Step 1. In this step we replace G_i by the inertia group of a computing lift, and we show that $N_i \to +\infty$.

Fix i, and let $\bar{0}$ be the image of 0 in W_{N_i}/G_i. Since (W_{N_i}, D_i) is klt near 0, the crepant quotient $(W_{N_i}, D_i)/G_i$ is klt near $\bar{0}$ [Liu26, Lemma 2.6(2)(3)], so a_i is attained by a prime divisor over W_{N_i}/G_i with center $\bar{0}$ [Amb99, Section 1, the second paragraph after Definition 1.1], which is induced by a computing lift F_i with center 0 (Definition 2.7). Let $I_i := I_{G_i}(F_i)$, which is cyclic [Liu26, Lemma 2.6(1)]. By [Liu26, Lemma 3.1(1)], applied with $X' = Y = W_{N_i}$, f the identity and $H = I_i$, we have $\mathrm{mld}((W_{N_i} \ni 0, D_i)/I_i) = a_i$, and F_i is a computing lift for the action of I_i, since $I_{I_i}(F_i) = I_i$.

Since $I_i \subset G_i$, the group I_i is a finite diagonal abelian group preserving the equation of W_{N_i} up to a character and acting faithfully on W_{N_i}. Any G_i-semi-invariant function is I_i-semi-invariant, and any G_i-invariant open subset is I_i-invariant, so D_i is a labelled boundary on $(W_{N_i} \ni 0)$ together with the action of I_i. Possibly replacing G_i with I_i, we may assume that G_i is cyclic and that $G_i = I_{G_i}(F_i)$.

Suppose that the sequence (N_i) has a bounded subsequence. Since the N_i are integers, there exists an integer $N \geq 3$ with $N_i = N$ for infinitely many i. The corresponding a_i form a strictly increasing infinite sequence of positive numbers in the set of Theorem 7.8 for the exponent N, which contradicts Theorem 7.8. Thus, after passing to a subsequence, $N_i \to +\infty$.

After passing to a further subsequence, either the orders $|G_i|$ are bounded, which is (1), or $|G_i| \to +\infty$. From now on we assume that $|G_i| \to +\infty$, and we prove (2).

Step 2. In this step we show that we may assume that any mark of D_i is the restriction of a G_i-semi-invariant polynomial.

Fix i and a mark h of D_i, and let χ be the character of h. Since h is regular near 0, there exist polynomials P and Q with $Q(0) \neq 0$ and a G_i-invariant open neighborhood U of 0 in W_{N_i} such that $h = P/Q$ on U and g^*Q is invertible on U for any $g \in G_i$. The function $q := \prod_{g \in G_i} g^*Q$ is G_i-invariant and invertible on U. Thus hq is a G_i-semi-invariant regular function on U of character χ, $\mathrm{div}(hq) = \mathrm{div}(h)$ on U, and hq is the restriction to U of the polynomial

$$P \prod_{g \in G_i, g \neq 1} g^*Q.$$

Since G_i is diagonal, any monomial is G_i-semi-invariant. Let h' be the sum of the monomials of $P \prod_{g \neq 1} g^*Q$ of character χ. The restriction of polynomials to U is G_i-equivariant, and the ring of regular functions on U is the direct sum of the G_i-isotypic components. Since hq has character χ, $h'|_U = hq$. Possibly replacing each mark h of D_i with the restriction of h' to W_{N_i}, we may assume that

$$D_i = \sum_j c_{ij} \, \mathrm{div}_{W_{N_i}}(h_{ij})$$

near 0, where $c_{ij} \in \Gamma$ and the h_{ij} are G_i-semi-invariant polynomials not vanishing identically on W_{N_i}. The replacement does not change D_i near 0, so (W_{N_i}, D_i) is still klt near 0 and a_i does not change.

Step 3. In this step we construct the limit H and show that $H \subset \{w \in \mathbb{T}^4 : w_x + w_y = 2w_z\}$ and $[G_i : G_i \cap H^0] = |H/H^0|$.

Let $\rho : \mathbb{R}^4 \to \mathbb{T}^4$ be the quotient map onto $\mathbb{T}^4 = \mathbb{R}^4/\mathbb{Z}^4$ [Liu26, Definition 2.15]. We regard G_i as a subgroup of $\mathbb{T}^4$: an element $w = (w_x, w_y, w_z, w_t) \in \mathbb{T}^4$ acts on $\mathbb{A}^4$ by multiplying x, y, z, and t by $e^{2\pi\sqrt{-1}w_x}$, $e^{2\pi\sqrt{-1}w_y}$, $e^{2\pi\sqrt{-1}w_z}$, and $e^{2\pi\sqrt{-1}w_t}$ respectively. The character of a monomial $x^{m_1}y^{m_2}z^{m_3}t^{m_4}$ is then the homomorphism $w \to m_1w_x + m_2w_y + m_3w_z + m_4w_t$ to $\mathbb{R}/\mathbb{Z}$, and the character of t is $w \to w_t$. In the additive notation of [Liu26, Definition 2.15], the subgroup $\{\lambda_x\lambda_y = \lambda_z^2\}$ of $(\mathbb{C}^*)^4$ is $\{w \in \mathbb{T}^4 : w_x + w_y = 2w_z\}$.

By Lemma 4.6, after passing to a subsequence and discarding finitely many terms, there exists a closed subgroup $H \subset \mathbb{T}^4$ such that the G_i converge to H in the Hausdorff distance, $G_i \subset H$, and G_i intersects each connected component of H. Moreover, $\dim H \geq 1$ since $|G_i| \to +\infty$. The group H/H^0 is finite [Liu26, Lemma 4.3(1)]. The homomorphism $G_i \to H/H^0$ is surjective with kernel $G_i \cap H^0$, so

$$[G_i : G_i \cap H^0] = |H/H^0|.$$

The characters on G_i of the monomials xy, z^2, and t^{N_i} of the equation $xy + z^2 + t^{N_i}$ of W_{N_i} coincide, since G_i preserves the equation up to a character. Thus

$$w_x + w_y = 2w_z \quad \text{and} \quad 2w_z = N_i w_t \quad \text{in } \mathbb{R}/\mathbb{Z} \text{ for any } w \in G_i. \tag{A.136}$$

The closed subgroup $\{w \in \mathbb{T}^4 : w_x + w_y = 2w_z\}$ contains G_i for any i. Since any point of H is a limit of points of the G_i, $H \subset \{w \in \mathbb{T}^4 : w_x + w_y = 2w_z\}$.

Let $\mathfrak{h} \subset \mathbb{R}^4$ be the linear subspace with $\rho(\mathfrak{h}) = H^0$ [Liu26, Lemma 4.3(1)], that is, the Lie algebra of H^0. For $\eta = (\eta_x, \eta_y, \eta_z, \eta_t) \in \mathfrak{h}$ and $\lambda \in \mathbb{R}$, $\rho(\lambda\eta) \in H$, so $\lambda(\eta_x + \eta_y - 2\eta_z) \in \mathbb{Z}$ for any $\lambda \in \mathbb{R}$. Thus

$$\mathfrak{h} \subset \{\eta \in \mathbb{R}^4 : \eta_x + \eta_y = 2\eta_z\},$$

and in particular $\dim H^0 \leq 3$.

Step 4. In this step we show that $\mathfrak{h}$ contains no vector with all coordinates positive, and that $\dim H^0 \in \{1, 2\}$.

Since (W_{N_i}, D_i) is klt near 0 and $D_i \geq 0$, W_{N_i} is normal and klt near 0. By **Step 2**, $D_i = \sum_j c_{ij} \operatorname{div}_{W_{N_i}}(h_{ij})$ near 0 with $c_{ij} \geq 0$ and G_i-semi-invariant polynomials h_{ij}, and (W_{N_i}, D_i) is lc near 0. Suppose that $\mathfrak{h}$ contains a vector with all coordinates positive. Then Lemma 4.7, applied with $n = 4$, $W = W_{N_i}$, and $f = xy + z^2 + t^{N_i}$, implies that $a_i \to 0$. This contradicts $a_i \geq a_1 > 0$. Thus $\mathfrak{h}$ contains no vector with all coordinates positive.

The vector $(1, 1, 1, 1)$ satisfies $1 + 1 = 2 \cdot 1$, so $\mathfrak{h} \neq \{\eta \in \mathbb{R}^4 : \eta_x + \eta_y = 2\eta_z\}$ and $\dim H^0 \leq 2$ by **Step 3**. Since $\dim H \geq 1$ and H/H^0 is finite, $\dim H^0 \geq 1$. Thus $\dim H^0 \in \{1, 2\}$.

Step 5. In this step we show that the character of t is non-trivial on H^0.

Suppose that $w_t = 0$ for any $w \in H^0$. Let $J := |H/H^0|$. The character $w \to w_t$ of H factors through H/H^0, so $Jw_t = 0$ for any $w \in H$. After passing to a subsequence, there exists an integer N_0 with $N_i \equiv N_0 \pmod{J}$ for any i. For $w \in G_i$, (A.136) and $G_i \subset H$ imply

$$2w_z - N_0 w_t = (2w_z - N_i w_t) + \frac{N_i - N_0}{J} \cdot Jw_t = 0 \quad \text{in } \mathbb{R}/\mathbb{Z}.$$

Thus the closed subgroup $\{w \in \mathbb{T}^4 : 2w_z = N_0 w_t\}$ contains G_i for any i, hence contains H. For $w \in H^0$ we obtain $2w_z = 0$. The image of the connected group H^0 under $w \to w_z$ is a connected subgroup of $\{0, \frac{1}{2}\} \subset \mathbb{R}/\mathbb{Z}$, so $w_z = 0$ for any $w \in H^0$. By **Step 3**, $w_x + w_y = 0$ for any $w \in H^0$. Therefore

$$H^0 \subset \{w \in \mathbb{T}^4 : w_x + w_y = 0,\ w_z = w_t = 0\} = \mathbf{T}_{(1,-1,0,0)}.$$

Since $\dim H^0 \geq 1$ (**Step 4**), $H^0 = \mathbf{T}_{(1,-1,0,0)}$ and $\dim H = 1$.

By Lemma 4.6, $R_i := G_i \cap \mathbf{T}_{(1,-1,0,0)}$ is cyclic of order $r_i \to +\infty$ and $[G_i : R_i] = J$ for $i \gg 0$. Since R_i is a finite subgroup of order r_i of the circle $\mathbf{T}_{(1,-1,0,0)}$, $R_i = \mu_{r_i}(1, -1, 0, 0)$. Let $g_i := -z^2 - t^{N_i}$. Then $W_{N_i} = (xy - g_i = 0)$ and $0 \neq g_i \in (z, t)^2$. Theorem 8.3, applied with $e_0 := J$ and the data (W_{N_i}, G_i, D_i) for $i \gg 0$, implies that the a_i do not form a strictly increasing sequence of positive numbers. This is a contradiction. Thus the character of t is non-trivial on H^0.

Step 6. In this step we assume that $\dim H^0 = 1$, and we prove the rest of (2).

By Lemma 4.6, $H^0 = \mathbf{T}_{(\omega)}$ for a primitive vector $\omega = (\omega_x, \omega_y, \omega_z, \omega_t) \in \mathbb{Z}^4$, and $r_i := |G_i \cap H^0| \to +\infty$. Since $\omega \in \mathfrak{h}$, **Step 3** implies $\omega_x + \omega_y = 2\omega_z$. Let $u := \omega_x$, $s := \omega_z$, and $v := \omega_t$. Then

$$\omega = (u, 2s - u, s, v), \qquad \gcd(u, s, v) = \gcd(u, 2s - u, s, v) = 1.$$

The character of t takes the value λv modulo $\mathbb{Z}$ at $\rho(\lambda\omega)$, so $v \neq 0$ by **Step 5**. Since $\mathbf{T}_{(\omega)} = \mathbf{T}_{(-\omega)}$, possibly replacing ω with $-\omega$, we may assume that $v > 0$. If $s > 0$ and $0 < u < 2s$, then $\omega_x, \omega_y, \omega_z, \omega_t > 0$, which contradicts **Step 4**. Thus it is not the case that $s > 0$ and $0 < u < 2s$.

We show that $(u, s) \neq (0, 0)$. Suppose that $u = s = 0$. Since $\gcd(u, s, v) = 1$ and $v > 0$, we have $\omega = (0, 0, 0, 1)$ and $H^0 = \mathbf{T}_{(0,0,0,1)}$. By Lemma 4.6, for $i \gg 0$,

$$R_i := G_i \cap \mathbf{T}_{(0,0,0,1)} = \mu_{r_i}(0, 0, 0, 1), \qquad [G_i : R_i] = |H/H^0|.$$

We check the hypotheses of Theorem 8.5 with t in the role of x, (x, y, z) in the role of (u_1, u_2, u_3), $f = t^{N_i} + xy + z^2$, and $e_0 := |H/H^0|$. The germ of W_{N_i} at 0 has an isolated singularity, since the partial derivatives y, x, $2z$, and $N_i t^{N_i - 1}$ of $xy + z^2 + t^{N_i}$ vanish simultaneously only at 0. Moreover, the germ of W_{N_i} at 0 is cDV: a general hyperplane through 0 is $(t = \alpha x + \beta y + \gamma z)$ for general $\alpha, \beta, \gamma \in \mathbb{C}$, and the restriction of the equation to it is

$$xy + z^2 + (\alpha x + \beta y + \gamma z)^{N_i},$$

whose quadratic part $xy + z^2$ is non-degenerate since $N_i \geq 3$. Thus a general hyperplane section of W_{N_i} through 0 has an A_1 singularity at 0. Moreover, $r_i \geq r_0(\Gamma)$ for $i \gg 0$, and by **Step 2**, $D_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$ near 0 with $c_{ij} \in \Gamma$ and G_i-semi-invariant polynomials h_{ij} not vanishing identically on W_{N_i}. Thus Theorem 8.5 implies that the a_i with $i \gg 0$ do not form a strictly increasing sequence, since $r_i \to +\infty$. This is a contradiction. Thus $(u, s) \neq (0, 0)$.

Finally, $|G_i \cap H^0| = r_i \to +\infty$.

Steps 3–6 prove (2). □

A.11. **Limits of dimension two on axial sources.** In this subsection, we prove Proposition 7.20.

Proof of Proposition 7.20. The proof is long, so we divide it into several steps. Let

$$f_i := xy + z^2 + t^{N_i}, \qquad W_{N_i} = (f_i = 0) \subset \mathbb{A}^4,$$

and write

$$D_i = \sum_{j=1}^{s_i} c_{ij} \operatorname{div}_{W_{N_i}}(h_{ij})$$

with marks h_{ij} and labels $c_{ij} \in \Gamma$ (Definition 2.7). The marks that are units near 0 and the marks with label 0 contribute zero to D_i near 0, so we may remove them from the list and assume that $c_{ij} > 0$ and $h_{ij}(0) = 0$ for any i and j. We identify finite diagonal groups with subgroups of $\mathbb{T}^4 = \mathbb{R}^4/\mathbb{Z}^4$ as in [Liu26, Definition 2.15], with the quotient maps $\rho : \mathbb{R}^d \to \mathbb{T}^d$, so that $\rho(w)$ acts on a monomial M in x, y, z, t by the character value $\rho(\operatorname{wt}_w(M))$. We write $w = (w_x, w_y, w_z, w_t)$ for the coordinates of $\mathbb{R}^4$ and $\mathbb{T}^4$, and $\deg M$ for the degree of a monomial M.

Step 1. In this step we replace the marks by polynomials of bounded least degree and fix their monomials of least degree.

Let h be a mark of D_i with G_i-character χ. Write $h = p/u$ near 0, where p and u are restrictions of polynomials and $u(0) \neq 0$. The function $u' := \prod_{g \in G_i} g^* u$ is G_i-invariant, and it is a unit near 0 since each $g \in G_i$ fixes 0 and $u(0) \neq 0$. Moreover $hu' = p \prod_{g \neq 1} g^* u$ is the restriction to W_{N_i} of a polynomial P. Since G_i preserves W_{N_i}, the restriction map $\mathbb{C}[x, y, z, t] \to \mathbb{C}[W_{N_i}]$ is G_i-equivariant, so the χ-isotypic component of P is a G_i-semi-invariant polynomial with restriction hu'. Thus, multiplying each h_{ij} by a G_i-invariant unit, which does not change D_i near 0, we may assume that each h_{ij} is a G_i-semi-invariant polynomial. Since G_i preserves f_i up to a character, the monomials xy, z^2, and t^{N_i} all have the G_i-character of f_i. Hence replacing xy by $-z^2 - t^{N_i}$ repeatedly changes h_{ij} by a multiple of f_i and keeps the G_i-character. Possibly replacing h_{ij} in this way, we may assume that no monomial of h_{ij} is divisible by xy. Moreover $h_{ij}(0) = 0$, so any monomial of h_{ij} has positive degree.

Since (W_{N_i}, D_i) is klt near 0, Lemma 4.7, applied to the trivial group and $w = (1,1,1,1)$, implies that

$$0 < \operatorname{mld}(W_{N_i} \ni 0, D_i) \le 4 - \operatorname{wt}_{(1,1,1,1)}(f_i) - \sum_j c_{ij} \operatorname{wt}_{(1,1,1,1)}(h_{ij}) = 2 - \sum_j c_{ij} \operatorname{wt}_{(1,1,1,1)}(h_{ij}). \tag{A.137}$$

Here $\operatorname{wt}_{(1,1,1,1)}(f_i) = 2$ as $N_i \ge 3$, and $\operatorname{wt}_{(1,1,1,1)}(h_{ij}) \ge 1$ is the least degree of a monomial of h_{ij}. If $\Gamma \subset \{0\}$, then $s_i = 0$ for any i. Otherwise let γ be the minimum of $\Gamma \backslash \{0\}$, which exists since Γ is a DCC set. By (A.137), $s_i\gamma < 2$, and any h_{ij} has a monomial of degree less than $2/\gamma$. There are only finitely many monomials of degree less than $2/\gamma$, and any sequence in a DCC set has a non-decreasing subsequence. Thus, after passing to a subsequence, there exist a non-negative integer s, real numbers $c_1^*, \dots, c_s^*$, and monomials $M_1, \dots, M_s$ such that for any i and j:

- $s_i = s$,
- the sequence $(c_{ij})_{i \ge 1}$ is non-decreasing with limit c_j^*, so $0 < c_{ij} \le c_j^* \le 1$, and
- M_j is a monomial of h_{ij} of least degree.

Step 2. In this step we describe the subgroup $\{w \in \mathbb{T}^4 : w_x + w_y = 2w_z\}$ through the cover $\mathbb{A}^3_{\xi_1,\xi_2,\xi_3} \to V$, and we introduce cover exponents.

Let $\iota : \mathbb{R}^3 \to \mathbb{R}^4$ be the linear map $\iota(v) = (2v_1, 2v_2, v_1 + v_2, v_3)$. Since $\iota(\mathbb{Z}^3) \subset \mathbb{Z}^4$, it induces a homomorphism $\iota : \mathbb{T}^3 \to \mathbb{T}^4$ with $\iota \circ \rho = \rho \circ \iota$. A direct calculation shows that the kernel of $\iota : \mathbb{T}^3 \to \mathbb{T}^4$ is $\{0, \rho(\frac{1}{2}, \frac{1}{2}, 0)\}$. The image $\iota(\mathbb{T}^3)$ is a connected closed subgroup of dimension three contained in $\{w \in \mathbb{T}^4 : w_x + w_y = 2w_z\}$. The subgroup $\{w \in \mathbb{T}^4 : w_x + w_y = 2w_z\}$ is connected of dimension three, since $(1,1,-2,0)$ is primitive in $\mathbb{Z}^4$. Hence

$$\{w \in \mathbb{T}^4 : w_x + w_y = 2w_z\} = \iota(\mathbb{T}^3),$$

and $\iota : \mathbb{T}^3 \to \iota(\mathbb{T}^3)$ is a covering map of degree two.

Let $\mathcal{M} := \{(a,b,c) \in \mathbb{Z}^3 : a + b \in 2\mathbb{Z}\}$. For $\beta \in \mathcal{M}$, the homomorphism

$$\chi_\beta : \iota(\mathbb{T}^3) \to \mathbb{T}^1, \qquad \chi_\beta(\iota(\rho(v))) = \rho(\langle \beta, v \rangle),$$

is well defined, since $\langle \beta, (\frac{1}{2}, \frac{1}{2}, 0) \rangle \in \mathbb{Z}$. The characters of $\mathbb{T}^3$ are the maps $\rho(v) \to \rho(\langle \beta, v \rangle)$ with $\beta \in \mathbb{Z}^3$, and such a character is trivial on $\{0, \rho(\frac{1}{2}, \frac{1}{2}, 0)\}$ if and only if $\beta \in \mathcal{M}$. Thus $\beta \to \chi_\beta$ is an isomorphism from $\mathcal{M}$ onto the character group of $\iota(\mathbb{T}^3)$.

Let $V := (xy + z^2 = 0) \subset \mathbb{A}^4$, and let $\pi : \mathbb{A}^3_{\xi_1,\xi_2,\xi_3} \to V$ be the morphism $x = \xi_1^2$, $y = -\xi_2^2$, $z = \xi_1\xi_2$, $t = \xi_3$. It is the quotient by the group of order two acting by $(\xi_1, \xi_2, \xi_3) \to (-\xi_1, -\xi_2, \xi_3)$, which is étale in codimension one. For a monomial $M = x^a y^b z^c t^e$, let $\alpha_M := (2a + c, 2b + c, e) \in \mathcal{M} \cap \mathbb{Z}^3_{\ge 0}$ be the cover exponent of M, so that $\pi^* M = (-1)^b \xi_1^{2a+c} \xi_2^{2b+c} \xi_3^e$. A direct calculation shows that for any $v \in \mathbb{R}^3$,

$$\operatorname{wt}_{\iota(v)}(M) = \langle \alpha_M, v \rangle, \qquad \deg M = \left\langle \alpha_M, \left(\frac{1}{2}, \frac{1}{2}, 1\right) \right\rangle, \tag{A.138}$$

and that the character of M restricted to $\iota(\mathbb{T}^3)$ is χ_{α_M}. Moreover, if M is not divisible by xy, then $\min\{a,b\} = 0$ and $c = \min\{2a + c, 2b + c\}$, so M is determined by α_M. Conversely, any $\beta \in \mathcal{M} \cap \mathbb{Z}^3_{\ge 0}$ is the cover exponent of a monomial not divisible by xy. For $j = 1, \dots, s$, let $\alpha_j := \alpha_{M_j}$.

Step 3. In this step we construct the integer q, the character χ_m, and the monomial Φ, and we show that m has no negative coordinate.

By Lemma 4.6 and the uniqueness of Hausdorff limits, after passing to a subsequence, $G_i \subset H$ and G_i intersects any connected component of H for any i. Let $\mathbf{T} := H^0$ and $q := |H/\mathbf{T}|$, which is finite by [Liu26, Lemma 4.3(1)]. The quotient $\iota(\mathbb{T}^3)/\mathbf{T}$ is a compact connected abelian Lie

group of dimension one, that is, a circle. Hence there exists $m \in \mathcal{M}$ such that χ_m induces an isomorphism $\iota(\mathbb{T}^3)/\mathbf{T} \to \mathbb{T}^1$, and

$$\{\beta \in \mathcal{M} : \chi_\beta(\mathbf{T}) = 0\} = \mathbb{Z}m.$$

In particular, $\ker \chi_m = \mathbf{T}$, so χ_m maps H onto a subgroup of $\mathbb{T}^1$ of order q. Since $G_i \subset H$ intersects any connected component of H,

$$H = \chi_m^{-1}\Big(\rho\Big(\frac{1}{q}\mathbb{Z}\Big)\Big), \qquad \chi_m(G_i) = \rho\Big(\frac{1}{q}\mathbb{Z}\Big). \tag{A.139}$$

Let $m^\perp := \{v \in \mathbb{R}^3 : \langle m, v\rangle = 0\}$. Since m is integral, $\rho(\iota(m^\perp))$ is a closed connected subgroup of dimension two contained in $\ker \chi_m = \mathbf{T}$. Hence $\mathbf{T} = \rho(\iota(m^\perp))$, so $\iota(m^\perp)$ is the Lie algebra of H, that is, the linear subspace of $\mathbb{R}^4$ whose image under ρ is H^0 [Liu26, Lemma 4.3(1)].

Assume that $m_k > 0 > m_l$ for some $k, l \in \{1, 2, 3\}$, and let n be the third index. Since $m_k v_k + m_l v_l$ takes any real value for $(v_k, v_l) \in \mathbb{R}^2_{>0}$, there exists $v \in m^\perp \cap \mathbb{R}^3_{>0}$ with $v_n = 1$. Then $\iota(v)$ is a vector of the Lie algebra of H with all coordinates positive. By **Step 1** the marks h_{ij} are G_i-semi-invariant polynomials, so Lemma 4.7 implies that $a_i \to 0$, which contradicts $a_i \geq \epsilon$. Therefore, possibly replacing m with $-m$, we may assume that $m \in \mathbb{Z}^3_{\geq 0}$. Moreover $m \neq 0$, and $m_1 + m_2$ is even.

Let Φ be the monomial not divisible by xy with $\alpha_\Phi = m$, that is,

$$\Phi = x^{(m_1-m_2)/2} z^{m_2} t^{m_3} \ \text{ if } m_1 \geq m_2, \qquad \Phi = y^{(m_2-m_1)/2} z^{m_1} t^{m_3} \ \text{ if } m_2 > m_1.$$

Then Φ is non-constant, and $\pi^*\Phi = \pm\xi_1^{m_1}\xi_2^{m_2}\xi_3^{m_3}$. By **Step 2** and (A.139), the G_i-character of Φ is $\chi_m|_{G_i}$, whose image is $\rho(\frac{1}{q}\mathbb{Z})$. Thus the G_i-character of Φ has order q, and Φ^q is G_i-invariant for any i.

Step 4. In this step we show that each mark coincides on W_{N_i}, up to an invariant unit, with a function whose difference from M_j has arbitrarily high order.

For any i and j, let $h_{ij}^{\mathbf{T}}$ be the sum of the terms of h_{ij} whose monomial M satisfies $\chi_{\alpha_M}|_{\mathbf{T}} = \chi_{\alpha_j}|_{\mathbf{T}}$, equivalently $\alpha_M - \alpha_j \in \mathbb{Z}m$ (**Step 3**). We show that for any positive integer L,

$$h_{ij} - h_{ij}^{\mathbf{T}} \in (x, y, z, t)^L \quad \text{for any } j \text{ and any } i \gg 0. \tag{A.140}$$

We may assume that $L > \deg M_j$ for any j. Let P_L be the finite set of vectors $\alpha_M - \alpha_{M'} \notin \mathbb{Z}m$, where M and M' are monomials of degree less than L. For $\beta \in P_L$, the character χ_β is non-trivial on $\mathbf{T}$, so there exists $u_\beta \in \mathbf{T}$ with $\chi_\beta(u_\beta) \neq 0$. Since the G_i converge to $H \supset \mathbf{T}$, there exist $g_i \in G_i$ converging to u_β, and then $\chi_\beta(g_i) \neq 0$ for $i \gg 0$. Hence for $i \gg 0$, $\chi_\beta|_{G_i}$ is non-trivial for any $\beta \in P_L$. On the other hand, any monomial of h_{ij} has the G_i-character of h_{ij}, so for any monomial M of h_{ij}, the character $\chi_{\alpha_M - \alpha_j}$ is trivial on G_i. Therefore, for $i \gg 0$, any monomial M of h_{ij} of degree less than L satisfies $\alpha_M - \alpha_j \notin P_L$, that is, $\alpha_M - \alpha_j \in \mathbb{Z}m$, and (A.140) holds.

Let M be a monomial of $h_{ij}^{\mathbf{T}}$, and write $\alpha_M = \alpha_j + \nu m$ with $\nu \in \mathbb{Z}$. By (A.138),

$$\deg M = \deg M_j + \nu\Big\langle m, \Big(\frac{1}{2}, \frac{1}{2}, 1\Big)\Big\rangle, \qquad \Big\langle m, \Big(\frac{1}{2}, \frac{1}{2}, 1\Big)\Big\rangle > 0,$$

and since M_j is a monomial of h_{ij} of least degree, $\nu \geq 0$. Since monomials not divisible by xy are determined by their cover exponents, M is determined by ν, and $\nu = 0$ if and only if $M = M_j$. Moreover $\pi^*M = \pm\xi^{\alpha_j + \nu m} = \pm\pi^*(M_j\Phi^\nu)$, where $\xi^\beta := \xi_1^{\beta_1}\xi_2^{\beta_2}\xi_3^{\beta_3}$. Since π is surjective and V is reduced, $M|_V = \pm(M_j\Phi^\nu)|_V$. Finally M and M_j both have the G_i-character of h_{ij}, so Φ^ν is G_i-invariant, and $q \mid \nu$ by **Step 3**. Thus there exists a polynomial ϕ_{ij} in one variable, in which only exponents divisible by q occur, such that

$$h_{ij}^{\mathbf{T}}|_V = \big(M_j\phi_{ij}(\Phi)\big)|_V, \qquad \phi_{ij}(0) \neq 0, \tag{A.141}$$

where $\phi_{ij}(0)$ is the coefficient of M_j in h_{ij}. In particular $\phi_{ij}(\Phi)$ is a G_i-invariant polynomial and a unit near 0, as $\Phi(0) = 0$.

Since $xy+z^2$ is irreducible, (A.141) implies that there exists a polynomial g_{ij} with

$$h_{ij}^{\mathbf{T}} - M_j\phi_{ij}(\Phi) = (xy+z^2)g_{ij}.$$

The polynomials $h_{ij}^{\mathbf{T}}$ and $M_j\phi_{ij}(\Phi)$ are G_i-semi-invariant with the G_i-character of h_{ij}, and $xy+z^2$ is G_i-semi-invariant with the G_i-character of f_i. Hence g_{ij} is G_i-semi-invariant, and $h_{ij}-f_ig_{ij}$ is a G_i-semi-invariant polynomial with the G_i-character of h_{ij} and whose restriction to W_{N_i} equals $h_{ij}|_{W_{N_i}}$. Since $f_i = xy+z^2+t^{N_i}$,

$$h_{ij} - f_ig_{ij} = M_j\phi_{ij}(\Phi) + \big(h_{ij} - h_{ij}^{\mathbf{T}}\big) - t^{N_i}g_{ij}. \tag{A.142}$$

Let $\tilde{h}_{ij} := (h_{ij}-f_ig_{ij})/\phi_{ij}(\Phi)$, a regular function near 0 on $\mathbb{A}^4$, which we also regard as an element of $\mathbb{C}[[x,y,z,t]]$. Then

$$\tilde{h}_{ij} = M_j + \frac{h_{ij} - h_{ij}^{\mathbf{T}} - t^{N_i}g_{ij}}{\phi_{ij}(\Phi)},$$

and $\tilde{h}_{ij}|_{W_{N_i}}$ and $h_{ij}|_{W_{N_i}}$ differ by the unit $\phi_{ij}(\Phi)|_{W_{N_i}}$ near 0. Since $N_i\to+\infty$, (A.140) implies that for any positive integer L,

$$\tilde{h}_{ij} - M_j \in (x,y,z,t)^L\mathbb{C}[[x,y,z,t]] \quad \text{for any } j \text{ and any } i \gg 0. \tag{A.143}$$

Step 5. In this step we prove that $\limsup_i a_i \le \lambda/q$ for a real number $\lambda \ge q\epsilon$ defined by the monomials M_j, the limits c_j^*, and m.

Let $v\in\mathbb{R}^3_{>0}$ with $\langle m,v\rangle = 1/q$. Then $\chi_m(\iota(\rho(v))) = \rho(1/q)$, so $\iota(\rho(v))\in H$ by (A.139), that is, $\iota(v)\in N_H$. Since the G_i converge to H, there exist $g_i\in G_i$ converging to $\iota(\rho(v))$. Since $\iota:\mathbb{T}^3\to\iota(\mathbb{T}^3)$ is a covering map and $G_i\subset\iota(\mathbb{T}^3)$, there exist $v_i\in\mathbb{R}^3$ with $\iota(\rho(v_i)) = g_i$ and $v_i\to v$. Let $w_i := \iota(v_i)$. Then $\rho(w_i) = g_i$, so $w_i\in N_{G_i}$. Moreover, by (A.139),

$$\rho(\langle m,v_i\rangle) = \chi_m(g_i) \in \rho\Big(\frac{1}{q}\mathbb{Z}\Big),$$

so $\langle m,v_i\rangle\in\frac{1}{q}\mathbb{Z}$. As $\langle m,v_i\rangle\to 1/q$, we obtain $\langle m,v_i\rangle = 1/q$ for $i\gg 0$. Since $v>0$, $w_i\in N_{G_i}\cap\mathbb{R}^4_{>0}$ for $i\gg 0$.

Fix a positive integer L_0 such that

$$L_0 \min_{1\le k\le 4}\iota(v)_k > 2\max_j\langle\alpha_j,v\rangle.$$

Since $v_i\to v$ and $N_i\to+\infty$, for $i\gg 0$ we have

$$L_0\min_{1\le k\le 4}(w_i)_k > \max_j\langle\alpha_j,v_i\rangle, \qquad N_i\ge L_0, \qquad N_i(v_i)_3 > 2\big((v_i)_1+(v_i)_2\big),$$

and (A.140) holds with $L=L_0$. The monomials xy, z^2, and t^{N_i} of f_i have w_i-weights $2((v_i)_1+(v_i)_2)$, $2((v_i)_1+(v_i)_2)$, and $N_i(v_i)_3$, so

$$\mathrm{wt}_{w_i}(f_i) = 2\big((v_i)_1+(v_i)_2\big).$$

In (A.142), any monomial of $(h_{ij}-h_{ij}^{\mathbf{T}}) - t^{N_i}g_{ij}$ has degree at least L_0, hence w_i-weight at least $L_0\min_{1\le k\le 4}(w_i)_k > \langle\alpha_j,v_i\rangle$. The monomials of $M_j\phi_{ij}(\Phi)$ other than M_j are $M_j\Phi^\nu$ with $\nu\ge q$, of w_i-weight $\langle\alpha_j,v_i\rangle+\nu/q$ by (A.138). Hence M_j occurs in $h_{ij}-f_ig_{ij}$ with coefficient $\phi_{ij}(0)\ne 0$, and any other monomial of $h_{ij}-f_ig_{ij}$ has larger w_i-weight. Since $\tilde{h}_{ij} = (h_{ij}-f_ig_{ij})/\phi_{ij}(\Phi)$ and $\phi_{ij}(0)\ne 0$, we obtain

$$\mathrm{wt}_{w_i}(h_{ij}-f_ig_{ij}) = \mathrm{wt}_{w_i}(\tilde{h}_{ij}) = \langle\alpha_j,v_i\rangle \quad \text{for any } j \text{ and any } i\gg 0. \tag{A.144}$$

By Lemma 4.7 with $W = W_{N_i}$, $f = f_i$, $G = G_i$,

$$D_i = \sum_j c_{ij}\,\mathrm{div}_{W_{N_i}}(h_{ij}-f_ig_{ij}),$$

and $w = w_i$, by (A.144), and since $\sum_{k=1}^4 (w_i)_k = 3((v_i)_1 + (v_i)_2) + (v_i)_3$, for $i \gg 0$ we have

$$\epsilon \le a_i \le \sum_{k=1}^4 (w_i)_k - \mathrm{wt}_{w_i}(f_i) - \sum_j c_{ij}\langle \alpha_j, v_i\rangle = \langle \ell_i, v_i\rangle, \qquad \ell_i := (1,1,1) - \sum_j c_{ij}\alpha_j.$$

Let $\ell := (1,1,1) - \sum_j c_j^*\alpha_j = \lim_i \ell_i$. Letting $i \to +\infty$, we obtain

$$\epsilon \le \langle \ell, v\rangle \quad \text{and} \quad \limsup_i a_i \le \langle \ell, v\rangle \quad \text{for any } v \in \mathbb{R}^3_{>0} \text{ with } \langle m, v\rangle = \frac{1}{q}. \tag{A.145}$$

We draw two consequences of (A.145). First, let $k \in \{1,2,3\}$ be an index with $m_k = 0$. Fix $v \in \mathbb{R}^3_{>0}$ with $\langle m, v\rangle = 1/q$, which exists since $m \ge 0$ and $m \ne 0$. For any $r > 0$, the vector $v + re_k$ satisfies the condition of (A.145), so $\epsilon \le \langle \ell, v\rangle + r\ell_k$. Hence $\ell_k \ge 0$. Second, let $k \in \{1,2,3\}$ be an index with $m_k > 0$. For $0 < \delta < 1/(q\sum_l m_l)$, the vector

$$v^{(\delta)} := \delta \sum_{l \ne k} e_l + \frac{1 - q\delta \sum_{l\ne k} m_l}{qm_k} e_k$$

is positive with $\langle m, v^{(\delta)}\rangle = 1/q$, and $v^{(\delta)} \to e_k/(qm_k)$ as $\delta \to 0$. Hence (A.145) implies that

$$\epsilon \le \frac{\ell_k}{qm_k} \quad \text{and} \quad \limsup_i a_i \le \frac{\ell_k}{qm_k}.$$

Let

$$\lambda := \min_{k : m_k > 0} \frac{\ell_k}{m_k}.$$

Then $\lambda \ge q\epsilon > 0$, $\limsup_i a_i \le \lambda/q$, and

$$\ell_k \ge \lambda m_k \quad \text{for } k = 1,2,3, \tag{A.146}$$

where for $m_k = 0$ this is $\ell_k \ge 0$.

Step 6. In this step we prove that $(W_{N_i}, D_i + \lambda \operatorname{div}_{W_{N_i}}(\Phi))$ is lc near 0 and that $a_i \ge \lambda/q$ for $i \gg 0$.

Let

$$\Delta_V := \sum_j c_j^* \operatorname{div}_V(M_j) + \lambda \operatorname{div}_V(\Phi).$$

Since $\pi^* M_j = \pm \xi^{\alpha_j}$ and $\pi^*\Phi = \pm\xi^m$,

$$\pi^*\Delta_V = \sum_{k=1}^3 \Big(\sum_j c_j^* \alpha_{j,k} + \lambda m_k \Big) \operatorname{div}(\xi_k) = \sum_{k=1}^3 (1 - \ell_k + \lambda m_k) \operatorname{div}(\xi_k),$$

where $\alpha_{j,k}$ is the k-th coordinate of α_j. The coefficients are non-negative since c_j^*, α_j, λ, and m are non-negative, and they are at most one by (A.146). Hence $(\mathbb{A}^3, \pi^*\Delta_V)$ is lc. The map π is the quotient by a group of order two which is étale in codimension one, so the ramification index of π along any prime divisor is one. By [Liu26, Lemma 2.6(0)(4)], the crepant boundary of $(\mathbb{A}^3, \pi^*\Delta_V)$ on V is Δ_V. By [Liu26, Lemma 2.6(3)], any prime divisor F over V is dominated by a prime divisor over $\mathbb{A}^3$, and if F' is such a prime divisor over $\mathbb{A}^3$, then by [Liu26, Lemma 2.6(2)],

$$a(F, V, \Delta_V) = \frac{a(F', \mathbb{A}^3, \pi^*\Delta_V)}{|I(F')|} \ge 0.$$

Hence (V, Δ_V) is lc. The variety V is a normal Cartier divisor on $\mathbb{A}^4$, and the monomials M_j and Φ do not vanish identically on V. By [Kaw07, Theorem],

$$\Big(\mathbb{A}^4, V + \sum_j c_j^* \operatorname{div}(M_j) + \lambda \operatorname{div}(\Phi)\Big)$$

is lc near V, in particular near 0. By [dFEM11, Proposition 2.11], the pair

$$\Big(\operatorname{Spec}\mathbb{C}[[x,y,z,t]],(xy+z^2)\prod_j(M_j)^{c_j^*}(\Phi)^{\lambda}\Big)$$

is lc.

Let L be the positive integer of Lemma 4.14 for $n=4$, for $s+2$ functions, and for the finite set

$$\Lambda:=\{1,\lambda,c_1^*,\dots,c_s^*\}\subset\mathbb{R}_{>0}.$$

For $i\gg 0$, we have $f_i-(xy+z^2)=t^{N_i}\in(x,y,z,t)^L$ and, by (A.143), $\tilde{h}_{ij}-M_j\in(x,y,z,t)^L$ for any j. We compare $xy+z^2$ with f_i, each M_j with $\tilde{h}_{ij}$, and Φ with itself, with exponents 1, c_j^*, and λ. Lemma 4.14 implies that

$$\Big(\operatorname{Spec}\mathbb{C}[[x,y,z,t]],(f_i)\prod_j(\tilde{h}_{ij})^{c_j^*}(\Phi)^{\lambda}\Big)$$

is lc for $i\gg 0$. By [dFEM11, Proposition 2.11],

$$\Big(\mathbb{A}^4,W_{N_i}+\sum_j c_j^*\operatorname{div}(\tilde{h}_{ij})+\lambda\operatorname{div}(\Phi)\Big)$$

is lc near 0 for $i\gg 0$. The hypersurface W_{N_i} is a normal Cartier divisor on $\mathbb{A}^4$, and neither $\tilde{h}_{ij}$ nor Φ vanishes identically on W_{N_i}, since

$$\tilde{h}_{ij}|_{W_{N_i}}=h_{ij}|_{W_{N_i}}/\phi_{ij}(\Phi)|_{W_{N_i}}\neq 0.$$

By [Kaw07, Theorem],

$$\Big(W_{N_i},\sum_j c_j^*\operatorname{div}_{W_{N_i}}(\tilde{h}_{ij})+\lambda\operatorname{div}_{W_{N_i}}(\Phi)\Big)$$

is lc near 0 for $i\gg 0$. Since $\tilde{h}_{ij}|_{W_{N_i}}$ and $h_{ij}|_{W_{N_i}}$ differ by a unit near 0, we have

$$\operatorname{div}_{W_{N_i}}(\tilde{h}_{ij})=\operatorname{div}_{W_{N_i}}(h_{ij})\quad\text{near }0.$$

Since $0<c_{ij}\le c_j^*$ and $\operatorname{div}_{W_{N_i}}(h_{ij})\ge 0$ near 0, we obtain that

$$\big(W_{N_i},D_i+\lambda\operatorname{div}_{W_{N_i}}(\Phi)\big)\text{ is lc near }0\text{ for }i\gg 0. \tag{A.147}$$

By **Step 3**, $\Phi|_{W_{N_i}}$ is a G_i-semi-invariant regular function which vanishes at 0 and whose G_i-character has order q. By (A.147) and Lemma 5.8, applied with $Z=\{0\}$ and $U=\Phi|_{W_{N_i}}$, we obtain $a_i\ge\lambda/q$ for $i\gg 0$.

Step 7. In this step we conclude the proof.

By **Steps 3, 5, and 6**, after passing to a subsequence, the positive integer q, the non-constant monomial Φ, and the real number $\lambda>0$ satisfy the following for $i\gg 0$: Φ^q is G_i-invariant, $(W_{N_i},D_i+\lambda\operatorname{div}(\Phi))$ is lc near 0, $a_i\ge\lambda/q$, and $\limsup_i a_i\le\lambda/q$. If the a_i were strictly increasing, then for $i\gg 0$,

$$a_i<\lim_{i'\to+\infty}a_{i'}=\limsup_{i'}a_{i'}\le\frac{\lambda}{q}\le a_i,$$

a contradiction. Hence the a_i are not strictly increasing. □

A.12. **Interior circles.** In this subsection, we prove Theorem 7.21. The proof of Theorem 7.21 does not use $v\neq c$, and in Appendix E the theorem is applied without the assumption $v\neq c$. We first construct a plane pair and a surface pair attached to an interior circle, and we bound α from above and from below by their minimal log discrepancies.

Construction-Proposition A.33. Notation as in Set-up 7.3 with $w = (-a, -b, -c, v)$, where $a, b, c, v > 0$, $a + b = 2c$, and $\gcd(a, c, v) = 1$, let $\delta := \tau + c - v$ be the adjusted mass of Set-up 5.1, let $H := G/R$, and for $\lambda \in \mathbb{C}^*$ let $\mathbf{T}(\lambda) \in \mathbf{T}$ be the element with entries λ^{-a}, λ^{-b}, λ^{-c}, λ^v. Let $e_E := \gcd(a, c)$, $e_a := \gcd(a, v)$, $e_b := \gcd(b, v)$, $S_0 := (\bar{x}\bar{y} + \bar{z}^2 = 0) \subset \mathbb{A}^3$, and $S := S_0/\mu_v(a, b, c)$. Let $\pi : Y \to S$ be the toric partial resolution with one exceptional curve E such that Y is covered near E by the charts $\mathbb{A}^2_{p,q}/\mu_b(-c, v)$ and $\mathbb{A}^2_{p',q'}/\mu_a(-c, v)$, where the first chart is induced by $\bar{x} = -\tilde{q}^a p^2$, $\bar{y} = \tilde{q}^b$, and $\bar{z} = \tilde{q}^c p$ for a coordinate $\tilde{q}$ on a cover of $\mathbb{A}^2_{p,q}$ with $\tilde{q}^v = q$, and the second by exchanging $(a, \bar{x})$ with $(b, \bar{y})$. Let Σ_x and Σ_y be the strict transforms of the images in S of the lines $(\bar{x} = \bar{z} = 0)$ and $(\bar{y} = \bar{z} = 0)$ of S_0. Let $\mathcal{P}_{\mathbf{T}}$ be the set of non-zero polynomials in x, y, z, t without monomials divisible by xy and homogeneous for $\mathbf{T}$. For $f \in \mathcal{P}_{\mathbf{T}}$, let $m(f)$ be the least exponent of t in the monomials of f, let

$$f_t := f(P^2, -Q^2, PQ, 1), \qquad f_y := f(-p^2, 1, p, q), \qquad f_x := f(1, -p'^2, p', q'),$$

and let $D_Y(f) \geq 0$ be the $\mathbb{Q}$-Cartier divisor on Y whose pullbacks to $\mathbb{A}^2_{p,q}$ and $\mathbb{A}^2_{p',q'}$ are $\operatorname{div}(f_y)$ and $\operatorname{div}(f_x)$. Let $D_{Y,j} := D_Y(h^0_j)$, $\Delta_Y := \sum_j c_j D_{Y,j}$,

$$B_{\mathrm{br}} := \Big(1 - \frac{1}{e_E}\Big)E + \Big(1 - \frac{1}{e_b}\Big)\Sigma_x + \Big(1 - \frac{1}{e_a}\Big)\Sigma_y, \qquad B_Y := B_{\mathrm{br}} + \Delta_Y,$$

let $m_j := m(h^0_j)$, and let $m_E := \sum_j c_j m_j$. Let $\tilde{H}_t$, $\tilde{H}_y$, and $\tilde{H}_x$ be the quotients

$$\{(g, s_g, p_g) : g \in G,\ s_g^v = g_t,\ p_g^2 = g_x s_g^a\}/\{(\mathbf{T}(\lambda), \lambda, 1) : \lambda^r = 1\},$$

$$\{(g, \eta_g) : g \in G,\ \eta_g^b = g_y\}/\{(\mathbf{T}(\lambda), \lambda^{-1}) : \lambda^r = 1\},$$

$$\{(g, \xi_g) : g \in G,\ \xi_g^a = g_x\}/\{(\mathbf{T}(\lambda), \lambda^{-1}) : \lambda^r = 1\},$$

acting on $\mathbb{A}^2_{P,Q}$, $\mathbb{A}^2_{p,q}$, and $\mathbb{A}^2_{p',q'}$ by $(P, Q) \to (p_g P, g_z s_g^c Q/p_g)$, $(p, q) \to (g_z \eta_g^{-c} p, g_t \eta_g^v q)$, and $(p', q') \to (g_z \xi_g^{-c} p', g_t \xi_g^v q')$, and let $C_t := \sum_j c_j \operatorname{div}((h^0_j)_t)$ on $\mathbb{A}^2_{P,Q}$. Let $(c'_j, f_j)_j$ be a finite family of pairs of a positive real number c'_j and an element $f_j \in \mathcal{P}_{\mathbf{T}}$, and let

$$C' := \sum_j c'_j \operatorname{div}((f_j)_t) \text{ on } \mathbb{A}^2_{P,Q}, \qquad B'_Y := B_{\mathrm{br}} + \sum_j c'_j D_Y(f_j), \qquad \delta' := \sum_j c'_j \kappa(f_j) + c - v.$$

Then the following hold.

(1) The following hold.
   (a) The groups $\tilde{H}_t$, $\tilde{H}_y$, and $\tilde{H}_x$ have orders $2v[G : R]$, $b[G : R]$, and $a[G : R]$ and act faithfully.
   (b) The groups $\tilde{H}_y$ and $\tilde{H}_x$ contain the groups $\mu_b(-c, v)$ and $\mu_a(-c, v)$ of the charts $\mathbb{A}^2_{p,q}/\mu_b(-c, v)$ and $\mathbb{A}^2_{p',q'}/\mu_a(-c, v)$ and induce one action of H on Y, which preserves E, Σ_x, Σ_y, and each $D_{Y,j}$.
   (c) If $f \in \mathcal{P}_{\mathbf{T}}$ is G-semi-invariant, then f_t, f_y, and f_x are semi-invariant under $\tilde{H}_t$, $\tilde{H}_y$, and $\tilde{H}_x$, respectively.

(2) The following hold.
   (a) For $f \in \mathcal{P}_{\mathbf{T}}$, distinct monomials of f have distinct images under each of the three substitutions defining f_t, f_y, and f_x.
   (b) For $f \in \mathcal{P}_{\mathbf{T}}$, $f_y = q^{m(f)}(\varphi(p) + q\varphi'(p, q))$ for polynomials $\varphi \neq 0$ and φ'.
   (c) For $f \in \mathcal{P}_{\mathbf{T}}$, the coefficient of E in $D_Y(f)$ is $m(f)/e_E$.
   (d) The chart morphisms $\mathbb{A}^2_{p,q} \to Y$ and $\mathbb{A}^2_{p',q'} \to Y$ pull $K_Y + B'_Y$ back to $K_{\mathbb{A}^2} + \sum_j c'_j \operatorname{div}((f_j)_y)$ and $K_{\mathbb{A}^2} + \sum_j c'_j \operatorname{div}((f_j)_x)$, respectively.
   (e) The intersection numbers on Y are $(E^2) = -\frac{2ve_E^2}{ab}$, $(\Sigma_x \cdot E) = \frac{e_E e_b}{b}$, $(\Sigma_y \cdot E) = \frac{e_E e_a}{a}$, and $(D_Y(f) \cdot E) = -\frac{2e_E \kappa(f)}{ab}$ for $f \in \mathcal{P}_{\mathbf{T}}$.
   (f) The coefficient of E in B_Y is $1 - \frac{1 - m_E}{e_E}$.

(g) We have
$$K_Y + B'_Y = \pi^*(K_S + B'_S) + \frac{\delta'}{ve_E}E,$$
where B'_S is the $\mathbb{R}$-divisor with $K_{\mathbb{A}^2} + C' = \phi^*(K_S + B'_S)$ for the quotient morphism $\phi : \mathbb{A}^2_{P,Q} \to S$ given by $\bar{x} = P^2$, $\bar{y} = -Q^2$, $\bar{z} = PQ$.

(h) For $c'_j = c_j$ and $f_j = h^0_j$, the formula of (g) gives $K_Y + B_Y = \pi^*(K_S + B_S) + \frac{\delta}{ve_E}E$, where B_S is the $\mathbb{R}$-divisor with $K_{\mathbb{A}^2} + C_t = \phi^*(K_S + B_S)$.

(3) If $\delta' \le 0$ and $(\mathbb{A}^2, C')$ is lc near 0, or if $\delta' \ge 0$ and (Y, B'_Y) is lc near E, then $(W_N, \sum_j c'_j \operatorname{div}(f_j))$ is lc near 0 for every $N \ge 3$.

Proof. The proof is long, so we divide it into several steps. For any $g \in G$ we have
$$g_x g_y = g_z^2 = g_t^N,$$
since G is diagonal and preserves $xy + z^2 + t^N$ up to a character. Since $b = 2c - a$, we have $\gcd(a, b, c, v) = \gcd(a, c, v) = 1$, so $\lambda \to \mathbf{T}(\lambda)$ is injective and $R = \{\mathbf{T}(\lambda) : \lambda^r = 1\}$.

Step 1. In this step we prove (1).

For $g \in G$ there are v choices of s_g and then two choices of p_g, and the triples $(\mathbf{T}(\lambda), \lambda, 1)$ with $\lambda^r = 1$ form a subgroup of order r. Thus $|\tilde{H}_t| = 2v|G|/r = 2v[G : R]$, and, counting the b choices of η_g and the a choices of ξ_g, we obtain $|\tilde{H}_y| = b[G : R]$ and $|\tilde{H}_x| = a[G : R]$. The triple $(\mathbf{T}(\lambda), \lambda, 1)$ maps (P, Q) to $(P, \lambda^{-c}\lambda^c Q) = (P, Q)$, so $\tilde{H}_t$ acts on $\mathbb{A}^2_{P,Q}$. The pair $(\mathbf{T}(\lambda), \lambda^{-1})$ acts trivially on $\mathbb{A}^2_{p,q}$, since $g_z \eta_g^{-c} = g_t \eta_g^v = 1$ for $(g, \eta_g) = (\mathbf{T}(\lambda), \lambda^{-1})$, so $\tilde{H}_y$ acts on $\mathbb{A}^2_{p,q}$, and similarly $\tilde{H}_x$ acts on $\mathbb{A}^2_{p',q'}$. Assume that (g, s_g, p_g) acts trivially. Then $p_g = 1$ and $g_z = s_g^{-c}$, hence $g_x = s_g^{-a}$, $g_y = g_z^2/g_x = s_g^{a-2c} = s_g^{-b}$, and $g_t = s_g^v$. Thus $g = \mathbf{T}(s_g) \in R$, so $s_g^r = 1$ and $(g, s_g, p_g) = (\mathbf{T}(s_g), s_g, 1)$ is trivial in $\tilde{H}_t$. If (g, η_g) acts trivially on $\mathbb{A}^2_{p,q}$, then $g_z = \eta_g^c$, $g_t = \eta_g^{-v}$, $g_y = \eta_g^b$, and $g_x = g_z^2/g_y = \eta_g^a$, so $g = \mathbf{T}(\eta_g^{-1}) \in R$ and $(g, \eta_g) = (\mathbf{T}(\lambda), \lambda^{-1})$ with $\lambda = \eta_g^{-1}$, $\lambda^r = 1$. Exchanging (a, x) with (b, y), we obtain that $\tilde{H}_x$ acts faithfully on $\mathbb{A}^2_{p',q'}$.

The pairs $(1, \zeta)$ with $\zeta^b = 1$ act by $(p, q) \to (\zeta^{-c}p, \zeta^v q)$, and $(1, \zeta)$ is trivial in $\tilde{H}_y$ only if $\zeta = 1$, since $\lambda \to \mathbf{T}(\lambda)$ is injective. Thus $\mu_b(-c, v) \subset \tilde{H}_y$. The homomorphism $\tilde{H}_y \to H$, $(g, \eta_g) \to gR$, is surjective, and if $g = \mathbf{T}(\lambda) \in R$, then $\eta_g = \lambda^{-1}\zeta$ with $\zeta^b = 1$, so $(g, \eta_g) = (\mathbf{T}(\lambda), \lambda^{-1}) \cdot (1, \zeta)$. Hence $\tilde{H}_y/\mu_b(-c, v) \cong H$, and, exchanging (a, x) with (b, y), we obtain $\mu_a(-c, v) \subset \tilde{H}_x$ and $\tilde{H}_x/\mu_a(-c, v) \cong H$.

The pairs (g, s_g) with $s_g^v = g_t$ act on S_0 by
$$(\bar{x}, \bar{y}, \bar{z}) \to (g_x s_g^a \bar{x}, g_y s_g^b \bar{y}, g_z s_g^c \bar{z}),$$
since $g_x g_y = g_z^2$. Another choice of s_g changes the automorphism by an element of $\mu_v(a, b, c)$, and $(\mathbf{T}(\lambda), \lambda)$ with $\lambda^r = 1$ acts trivially. Thus H acts on S. On the first chart, the substitution $\tilde{q} \to \eta_g s_g \tilde{q}$, $p \to g_z \eta_g^{-c} p$ satisfies
$$(\eta_g s_g \tilde{q})^b = g_y s_g^b \tilde{q}^b, \quad (\eta_g s_g \tilde{q})^c g_z \eta_g^{-c} p = g_z s_g^c \tilde{q}^c p, \quad -(\eta_g s_g \tilde{q})^a (g_z \eta_g^{-c} p)^2 = g_x s_g^a (-\tilde{q}^a p^2),$$
where the last equality uses $g_z^2 \eta_g^{a-2c} = g_z^2/g_y = g_x$, and it maps $q = \tilde{q}^v$ to $g_t \eta_g^v q$. Hence the action of (g, η_g) on $\mathbb{A}^2_{p,q}$ covers the action of g on S, and, exchanging $(a, \bar{x})$ with $(b, \bar{y})$, the action of (g, ξ_g) on $\mathbb{A}^2_{p',q'}$ covers the action of g on S. Let U_1 and U_2 be the images of the charts $\mathbb{A}^2_{p,q}/\mu_b(-c, v)$ and $\mathbb{A}^2_{p',q'}/\mu_a(-c, v)$ in Y. Since π is an isomorphism over $S\backslash\{0\}$, the actions of H on U_1 and U_2 coincide on the dense open subset $\pi^{-1}(S\backslash\{0\}) \cap U_1 \cap U_2$, hence on $U_1 \cap U_2$, since Y is separated, so they define one action of H on Y. The action of H on Y is diagonal on the charts, so it preserves the toric curves E, Σ_x, and Σ_y. The identity
$$f_y = \tilde{q}^{\kappa(f)} f(\bar{x}, \bar{y}, \bar{z}, 1) = \Big(\frac{\tilde{q}}{s}\Big)^{\kappa(f)} f(x, y, z, t) \quad \text{for } f \in \mathcal{P}_{\mathbf{T}}, \tag{A.148}$$

where $s^v = t$ and $(\bar{x}, \bar{y}, \bar{z}) = (xs^a, ys^b, zs^c)$, follows from

$$(-\tilde{q}^a p^2, \tilde{q}^b, \tilde{q}^c p, 1) = \mathbf{T}(\tilde{q}^{-1}) \cdot (-p^2, 1, p, q), \qquad (\bar{x}, \bar{y}, \bar{z}, 1) = \mathbf{T}(s^{-1}) \cdot (x, y, z, t).$$

By (A.148), f_y is $\tilde{H}_y$-semi-invariant for any G-semi-invariant $f \in \mathcal{P}_{\mathbf{T}}$, and f_x is $\tilde{H}_x$-semi-invariant by exchanging (a, x) with (b, y). Since $p_g^2 = g_x s_g^a$ and $(g_z s_g^c / p_g)^2 = g_y s_g^b$, the element (g, s_g, p_g) maps $(P^2, -Q^2, PQ, 1)$ to $g\mathbf{T}(s_g^{-1}) \cdot (P^2, -Q^2, PQ, 1)$, so f_t is $\tilde{H}_t$-semi-invariant for any G-semi-invariant $f \in \mathcal{P}_{\mathbf{T}}$. Since G is diagonal and h_j is G-semi-invariant, h_j^0 is G-semi-invariant. Thus H preserves each $D_{Y,j}$.

Step 2. In this step we describe Y and prove (2) except the formula for $K_Y + B'_Y$.

Let

$$\Lambda := \mathbb{Z}^2 + \mathbb{Z}\Big(\frac{1}{2}, \frac{1}{2}\Big) + \mathbb{Z}\Big(\frac{a}{2v}, \frac{b}{2v}\Big).$$

The lattice $\Lambda_0^\vee := \{(i, j) \in \mathbb{Z}^2 : i + j \in 2\mathbb{Z}\}$ is the lattice of exponents of the Laurent monomials $P^i Q^j$ in $\bar{x} = P^2$, $\bar{y} = -Q^2$, $\bar{z} = PQ$, and the dual lattice

$$\Lambda^\vee = \Big\{(i, j) \in \Lambda_0^\vee : \frac{ai + bj}{2} \in v\mathbb{Z}\Big\}$$

is the lattice of exponents of the monomials invariant under $\mu_v(a, b, c)$. Thus S is the affine toric surface of the cone $\mathbb{R}^2_{\geq 0}$ in Λ. For $(i, j) \in \Lambda_0^\vee$ we have $\frac{ai+bj}{2} = a\frac{i-j}{2} + cj$, so the numbers $\frac{ai+bj}{2}$ with $(i, j) \in \Lambda_0^\vee$ form $e_E\mathbb{Z}$, and since $\gcd(e_E, v) = \gcd(a, c, v) = 1$, the numbers $\frac{ai+bj}{2}$ for $(i, j) \in \Lambda^\vee$ form $ve_E\mathbb{Z}$. Hence

$$\rho_E := \frac{1}{2ve_E}(a, b)$$

is a primitive vector of Λ with $\langle \rho_E, \Lambda^\vee \rangle = \mathbb{Z}$, and Y is the toric surface obtained by inserting the ray $\mathbb{R}_{\geq 0}\rho_E$. On the first chart $P^2 = \bar{x} = -\tilde{q}^a p^2$ and $Q^2 = -\bar{y} = -\tilde{q}^b$, so

$$P^i Q^j = \pm\sqrt{-1}^{i+j} p^i q^{(ai+bj)/(2v)} \quad \text{for } (i, j) \in \Lambda^\vee.$$

The map $(i, j) \to (i, (ai + bj)/(2v))$ is a bijection from $\Lambda^\vee$ onto the set of exponents (i, k) of the $\mu_b(-c, v)$-invariant Laurent monomials $p^i q^k$, that is, of the pairs with $vk \equiv ci \pmod b$, since for such (i, k) the number $j = (2vk - ai)/b$ is an integer and $i + j = 2(vk + (c - a)i)/b$ is even. The bijection maps the dual cone $\{i \geq 0,\ ai + bj \geq 0\}$ of the cone spanned by $(1, 0)$ and ρ_E onto $\{i \geq 0,\ k \geq 0\}$. Hence the chart $\mathbb{A}^2_{p,q}/\mu_b(-c, v)$ is the affine chart of the cone spanned by $(1, 0)$ and ρ_E, with $E = (q = 0)$ (the ray ρ_E) and $\Sigma_x = (p = 0)$ (the ray $(1, 0)$). Exchanging $(a, \bar{x}, p, q)$ with $(b, \bar{y}, p', q')$, the chart $\mathbb{A}^2_{p',q'}/\mu_a(-c, v)$ is the affine chart of the cone spanned by ρ_E and $(0, 1)$. The element $\zeta \in \mu_b$ fixes the curve $(q = 0)$ pointwise if and only if $\zeta^c = 1$, and the curve $(p = 0)$ if and only if $\zeta^v = 1$. Since $\gcd(b, c) = \gcd(a, c) = e_E$, the chart morphism $\varsigma : \mathbb{A}^2_{p,q} \to Y$ has ramification indices e_E along E and e_b along Σ_x, and by the Hurwitz formula

$$K_{\mathbb{A}^2_{p,q}} + \sum_j c'_j \operatorname{div}((f_j)_y) = \varsigma^*(K_Y + B'_Y). \tag{A.149}$$

Exchanging $(a, \bar{x}, p, q)$ with $(b, \bar{y}, p', q')$, we obtain the analogous identity for the chart morphism $\mathbb{A}^2_{p',q'} \to Y$ and $(f_j)_x$.

We check that $D_Y(f)$ is well defined for $f \in \mathcal{P}_{\mathbf{T}}$, and we prove the claims of (2) on the images of monomials. The function f_y is $\mu_b(-c, v)$-semi-invariant by (A.148). Off E, by (A.148), both charts define the $\mathbb{Q}$-divisor whose pullback to $S_0\backslash\{0\}$ is $\operatorname{div}(f(\bar{x}, \bar{y}, \bar{z}, 1))$. Let $x^{i_1} y^{i_2} z^{i_3} t^l$ be a monomial of f. Then $i_1 i_2 = 0$ and

$$b(i_2 - i_1) = vl - \kappa(f) - c(2i_1 + i_3),$$

since $-ai_1 - bi_2 - ci_3 + vl = \kappa(f)$ and $a = 2c - b$. The monomials of f with a fixed exponent l of t have distinct images $\pm p^{2i_1+i_3} q^l$ under the substitution defining f_y, since $2i_1 + i_3$ determines $i_2 - i_1$, hence (i_1, i_2, i_3). Since l is determined by (i_1, i_2, i_3) and $\kappa(f)$, distinct monomials of f have distinct images under the substitution defining f_y, and, exchanging (a, x) with (b, y),

under the substitution defining f_x. The monomial $x^{i_1}y^{i_2}z^{i_3}t^l$ maps to $\pm P^{2i_1+i_3}Q^{2i_2+i_3}$ under the substitution defining f_t, and $(2i_1+i_3, 2i_2+i_3)$ determines i_1-i_2, hence (i_1,i_2,i_3) since $i_1i_2=0$, hence l. Thus distinct monomials of f have distinct images under the substitution defining f_t. In particular $f_y = q^{m(f)}(\varphi(p)+q\varphi'(p,q))$ with $\varphi\neq 0$, and since the chart morphism has ramification index e_E along E, both charts assign to E the coefficient $m(f)/e_E$ in $D_Y(f)$. Hence the $\mathbb{Q}$-divisors defined by the charts $\mathbb{A}^2_{p,q}/\mu_b(-c,v)$ and $\mathbb{A}^2_{p',q'}/\mu_a(-c,v)$ coincide near E. In particular the coefficient of E in B_Y is $1-\frac{1}{e_E}+\frac{m_E}{e_E}=1-\frac{1-m_E}{e_E}$.

The curves $(p=0)$ and $(q=0)$ intersect transversally at the origin of $\mathbb{A}^2_{p,q}$, the chart morphism has degree b, and the pullbacks of Σ_x and E are $e_b\operatorname{div}(p)$ and $e_E\operatorname{div}(q)$. Thus $b(\Sigma_x\cdot E)=e_be_E$, and the computation on the second chart shows $a(\Sigma_y\cdot E)=e_ae_E$. By the definition of D_Y,

$$D_Y(x)=\frac{2}{e_b}\Sigma_x,\quad D_Y(y)=\frac{2}{e_a}\Sigma_y,\quad D_Y(z)=\frac{1}{e_b}\Sigma_x+\frac{1}{e_a}\Sigma_y,\quad D_Y(t)=\frac{1}{e_E}E. \tag{A.150}$$

If $f,f'\in\mathcal{P}_{\mathbf{T}}$ and $\kappa(f)=\kappa(f')$, then the monomials of $f(\bar{x},\bar{y},\bar{z},1)$ and $f'(\bar{x},\bar{y},\bar{z},1)$ have weights congruent to $-\kappa(f)$ modulo v for $\mu_v(a,b,c)$, so $f(\bar{x},\bar{y},\bar{z},1)/f'(\bar{x},\bar{y},\bar{z},1)$ is a rational function on S, and by (A.148) the pullback of $f(\bar{x},\bar{y},\bar{z},1)/f'(\bar{x},\bar{y},\bar{z},1)$ to Y has divisor $D_Y(f)-D_Y(f')$. By the projection formula,

$$((D_Y(f)-D_Y(f'))\cdot E)=0.$$

For $f=x^vt^a$ and $f'=1$, the last equality and (A.150) imply $\frac{2v}{e_b}\cdot\frac{e_Ee_b}{b}+\frac{a}{e_E}(E^2)=0$, that is, $(E^2)=-\frac{2ve_E^2}{ab}$. Since $\gcd(a,c,v)=1$, the integers $vl-ai_1-ci_3$ with $i_1,i_3,l\geq 0$ form $\mathbb{Z}$, so any $f\in\mathcal{P}_{\mathbf{T}}$ has the weight of a monomial $x^{i_1}z^{i_3}t^l$, and by (A.150)

$$\begin{aligned}(D_Y(f)\cdot E)=(D_Y(x^{i_1}z^{i_3}t^l)\cdot E)&=\frac{2i_1e_E}{b}+i_3\Big(\frac{e_E}{b}+\frac{e_E}{a}\Big)-\frac{2lve_E}{ab}\\&=\frac{2e_E(ai_1+ci_3-vl)}{ab}=-\frac{2e_E\kappa(f)}{ab}.\end{aligned}$$

Step 3. In this step we prove (2) for any finitely many positive c'_j and $f_j\in\mathcal{P}_{\mathbf{T}}$, with C', B'_S, B'_Y, and δ' of the statement, that is, the formula for $K_Y+B'_Y$.

Let $m':=\sum_j c'_jm(f_j)$. We show that

$$K_Y+B'_Y=\pi^*(K_S+B'_S)+\frac{\delta'}{ve_E}E. \tag{A.151}$$

Then (A.151) is the formula of (2), and for $c'_j=c_j$ and $f_j=h_j^0$ it is the formula for K_Y+B_Y.

The morphism ϕ is the quotient by the group $\{(\zeta,\varrho):\zeta^v=1,\ \varrho^2=\zeta^a\}$ of order $2v$ acting by $(P,Q)\to(\varrho P,\zeta^cQ/\varrho)$. The elements fixing $(P=0)$ pointwise are those with $\varrho=\zeta^c$, that is, $\zeta^v=\zeta^b=1$, and the elements fixing $(Q=0)$ pointwise are those with $\varrho=1$, that is, $\zeta^v=\zeta^a=1$. Thus ϕ has ramification indices e_b and e_a along the lines $(P=0)$ and $(Q=0)$, which map onto the images of $(\bar{x}=\bar{z}=0)$ and $(\bar{y}=\bar{z}=0)$. Since $\mathbb{A}^2_{P,Q}\to S_0$ is étale in codimension one and $(f_j)_t$ is the pullback of $f_j(\bar{x},\bar{y},\bar{z},1)$, the Hurwitz formula and **Step 2** imply that B'_Y and $\pi_*^{-1}B'_S$ coincide off E. Hence

$$K_Y+B'_Y=\pi^*(K_S+B'_S)+\Big(a(E,S,B'_S)-\frac{1-m'}{e_E}\Big)E.$$

We compute $a(E,S,B'_S)$. Let F be the exceptional divisor of the weighted blow-up of $\mathbb{A}^2_{P,Q}$ with weights $(a,b)/\gcd(a,b)$, so that $a(F,\mathbb{A}^2,0)=(a+b)/\gcd(a,b)$ and $\operatorname{mult}_F(P^iQ^{i'})=(ai+bi')/\gcd(a,b)$. Since $(a,b)/\gcd(a,b)=n_F\rho_E$ with $n_F:=2ve_E/\gcd(a,b)$, the prime divisor F maps onto E with ramification index n_F, so $a(F,\mathbb{A}^2,C')=n_Fa(E,S,B'_S)$ by [Liu26, Lemma 2.6(1)(2)]. The monomial $x^{i_1}y^{i_2}z^{i_3}t^l$ maps to $\pm P^{2i_1+i_3}Q^{2i_2+i_3}$, distinct monomials of f_j map to distinct monomials by the part of (2) proved in **Step 2**, and

$$\operatorname{mult}_F(P^{2i_1+i_3}Q^{2i_2+i_3})=\frac{2(ai_1+bi_2+ci_3)}{\gcd(a,b)}=\frac{2(vl-\kappa(f_j))}{\gcd(a,b)}.$$

Thus

$$\operatorname{mult}_F((f_j)_t) = \frac{2(vm(f_j) - \kappa(f_j))}{\gcd(a,b)},$$

and

$$a(E,S,B'_S) = \frac{\gcd(a,b)}{2ve_E} \cdot \frac{a+b-2vm'+2\sum_j c'_j\kappa(f_j)}{\gcd(a,b)} = \frac{c-vm'+\sum_j c'_j\kappa(f_j)}{ve_E}.$$

Hence $a(E,S,B'_S) - \frac{1-m'}{e_E} = \frac{\delta'}{ve_E}$, which proves (A.151).

Step 4. In this step we prove (3).

Fix $N \geq 3$, let $D' := \sum_j c'_j \operatorname{div}(f_j)$ on W_N, let $\tau' := \sum_j c'_j\kappa(f_j)$, and let

$$Z_X := (xy + z^2 + s^{vN} = 0) \subset \mathbb{A}^4_{x,y,z,s}, \qquad X' := (\bar{x}\bar{y} + \bar{z}^2 + s^{vN+2c} = 0) \subset \mathbb{A}^4_{\bar{x},\bar{y},\bar{z},s},$$

with the morphisms $\pi_Z : Z_X \to W_N$, $(x,y,z,s) \to (x,y,z,s^v)$, and $g : Z_X \to X'$, $(x,y,z,s) \to (xs^a, ys^b, zs^c, s)$. The morphism g is well defined since

$$(xs^a)(ys^b) + (zs^c)^2 + s^{vN+2c} = s^{2c}(xy + z^2 + s^{vN}),$$

it is birational, and it is an isomorphism over $X'\backslash(s=0)$. The varieties W_N, Z_X, and X' are normal hypersurfaces, the surface $S_0 = X' \cap (s=0)$ is normal, and $\operatorname{div}_{Z_X}(s)$ is the prime divisor $(xy + z^2 = 0) \times \{0\}$. Since π_Z is a cyclic cover of degree v totally ramified along $\operatorname{div}_{Z_X}(s)$, the Hurwitz formula implies $K_{Z_X} = \pi_Z^* K_{W_N} + (v-1)\operatorname{div}_{Z_X}(s)$. The forms $d\bar{x} \wedge d\bar{z} \wedge ds/\bar{x}$ and $dx \wedge dz \wedge ds/x$ generate $\omega_{X'}$ and ω_{Z_X}. Since $d\bar{x} \wedge d\bar{z} \wedge ds = s^{a+c} dx \wedge dz \wedge ds$ (the terms with ds in $d\bar{x}$ and $d\bar{z}$ vanish after the wedge product with ds), dividing by $\bar{x} = xs^a$ shows that g^* of the first form is $s^c dx \wedge dz \wedge ds/x$, so $K_{Z_X} = g^* K_{X'} + c\operatorname{div}_{Z_X}(s)$. Let $\Psi'_j := f_j(\bar{x},\bar{y},\bar{z},1)$. Since $(xs^a, ys^b, zs^c, 1) = \mathbf{T}(s^{-1}) \cdot (x,y,z,s^v)$, we have $\Psi'_j \circ g = s^{-\kappa(f_j)} f_j \circ \pi_Z$. Thus for any real number θ_s,

$$g^*\Big(K_{X'} + \theta_s \operatorname{div}(s) + \sum_j c'_j \operatorname{div}(\Psi'_j)\Big) = \pi_Z^*(K_{W_N} + D') + (\theta_s - 1 - \delta')\operatorname{div}_{Z_X}(s), \tag{A.152}$$

since $v - 1 - c - \tau' = -1 - \delta'$. The polynomial Ψ'_j does not involve s, and $\Psi'_j|_{S_0} \neq 0$, since the pullback of $\Psi'_j|_{S_0}$ to $\mathbb{A}^2_{P,Q}$ is $(f_j)_t \neq 0$ by (2). Since X' is a hypersurface and S_0 is a normal Cartier divisor on X', [Kaw07, Theorem] implies that, for any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $\Theta \geq 0$ on X' whose support does not contain S_0, the pair $(X', S_0 + \Theta)$ is lc near 0 if $(S_0, \Theta|_{S_0})$ is lc near 0. By (A.152) and [KM98, Proposition 5.20], applied to π_Z, with $\pi_Z^{-1}(0) = \{0\}$, the pair (W_N, D') is lc near 0 as soon as

$$g^*\Big(K_{X'} + \theta_s \operatorname{div}(s) + \sum_j c'_j \operatorname{div}(\Psi'_j) + \Theta'\Big) \geq \pi_Z^*(K_{W_N} + D')$$

for some θ_s and some $\Theta' \geq 0$ such that the pair

$$\Big(X', \theta_s \operatorname{div}(s) + \sum_j c'_j \operatorname{div}(\Psi'_j) + \Theta'\Big)$$

has non-negative log discrepancies over all prime divisors whose center contains 0.

Assume that $\delta' \leq 0$ and that $(\mathbb{A}^2, C')$ is lc near 0. Since $\mathbb{A}^2_{P,Q} \to S_0$ is étale in codimension one and $(f_j)_t$ is the pullback of Ψ'_j, the pair $(S_0, \sum_j c'_j \operatorname{div}(\Psi'_j|_{S_0}))$ is lc near 0 by [KM98, Proposition 5.20]. Hence $(X', S_0 + \sum_j c'_j \operatorname{div}(\Psi'_j))$ is lc near 0. Take $\theta_s := 1 + \delta' \leq 1$ and $\Theta' := 0$. Since $\theta_s \leq 1$, the log discrepancies of the pair

$$\Big(X', \theta_s \operatorname{div}(s) + \sum_j c'_j \operatorname{div}(\Psi'_j)\Big)$$

are at least those of $(X', S_0 + \sum_j c'_j \operatorname{div}(\Psi'_j))$. By (A.152), (W_N, D') is lc near 0.

Assume that $\delta' \geq 0$ and that (Y, B'_Y) is lc near E. Let k be a positive integer divisible by a, b, and c with $k \geq \delta'$, and let

$$\Xi := \alpha_1 \bar{x}^{k/a} + \alpha_2 \bar{y}^{k/b} + \alpha_3 \bar{z}^{k/c}$$

for general $(\alpha_1, \alpha_2, \alpha_3) \in \mathbb{C}^3$. The monomials of Ξ have weight k for $\mu_v(a, b, c)$, so Ξ^v is a function on S. Let $\Xi_S := \frac{1}{v} \operatorname{div}_S(\Xi^v)$. On the first chart,

$$\Xi = \tilde{q}^k(\alpha_1(-p^2)^{k/a} + \alpha_2 + \alpha_3 p^{k/c}),$$

so the pullbacks of $\pi^*\Xi_S$ to $\mathbb{A}^2_{p,q}$ and to $\mathbb{A}^2_{p',q'}$ are

$$\frac{k}{v}\operatorname{div}(q) + \operatorname{div}(\alpha_1(-p^2)^{k/a} + \alpha_2 + \alpha_3 p^{k/c}), \qquad \frac{k}{v}\operatorname{div}(q') + \operatorname{div}(\alpha_1 + \alpha_2(-p'^2)^{k/b} + \alpha_3 p'^{k/c}).$$

Since the pullback of E is $e_E \operatorname{div}(q)$, we obtain $\pi^*\Xi_S = \frac{k}{ve_E}E + \Xi_Y$, where the pullbacks of Ξ_Y to the charts are general members of base point free linear systems. By (A.151),

$$\pi^*\Big(K_S + B'_S + \frac{\delta'}{k}\Xi_S\Big) = K_Y + B'_Y + \frac{\delta'}{k}\Xi_Y.$$

Since $\delta'/k \leq 1$, [KM98, Lemma 5.17] on the charts, (A.149), and [KM98, Proposition 5.20] imply that $(Y, B'_Y + \frac{\delta'}{k}\Xi_Y)$ is lc near E. Since $\pi^{-1}(0) = E$, the pair $(S, B'_S + \frac{\delta'}{k}\Xi_S)$ is lc near 0, and since the pullback of Ξ_S to S_0 is $\operatorname{div}(\Xi)$, the pair

$$\Big(S_0, \sum_j c'_j \operatorname{div}(\Psi'_j|_{S_0}) + \frac{\delta'}{k}\operatorname{div}(\Xi|_{S_0})\Big)$$

is lc near 0 by [KM98, Proposition 5.20]. Hence the pair

$$\Big(X', S_0 + \sum_j c'_j \operatorname{div}(\Psi'_j) + \frac{\delta'}{k}\operatorname{div}(\Xi)\Big)$$

is lc near 0. Take $\theta_s := 1$ and $\Theta' := \frac{\delta'}{k}\operatorname{div}(\Xi)$. Since

$$\Xi \circ g = s^k \Xi_0, \qquad \Xi_0 := \alpha_1 x^{k/a} + \alpha_2 y^{k/b} + \alpha_3 z^{k/c},$$

(A.152) implies

$$g^*\Big(K_{X'} + S_0 + \sum_j c'_j \operatorname{div}(\Psi'_j) + \frac{\delta'}{k}\operatorname{div}(\Xi)\Big) = \pi_Z^*(K_{W_N} + D') + \frac{\delta'}{k}\operatorname{div}_{Z_X}(\Xi_0) \geq \pi_Z^*(K_{W_N} + D').$$

Thus (W_N, D') is lc near 0. □

The following lemma expresses the log discrepancies over W_N on the threefold X_d of the t-root chart and on the threefolds $\mathbb{A}^2_{P,Q} \times \mathbb{A}^1$ and $\mathbb{A}^2_{p,q} \times \mathbb{A}^1$.

Lemma A.34. *Notation as in Construction-Proposition A.33. Let*

$$Z_X := (xy + z^2 + s^{vN} = 0) \subset \mathbb{A}^4_{x,y,z,s}$$

with the finite morphism $\pi_Z : Z_X \to W_N$, $(x, y, z, s) \to (x, y, z, s^v)$, let Z_t be the normalization of the double cover $(P^2 = xs^a) \subset Z_X \times \mathbb{A}^1_P$, and let Z_y be the normalization of $(\eta^b = y) \subset W_N \times \mathbb{A}^1_\eta$. Let $\bar{x} := xs^a$, $\bar{y} := ys^b$, $\bar{z} := zs^c$, $u := s^r$, $Q := \bar{z}/P$, $p := z\eta^{-c}$, $q := t\eta^v$, $u_y := \eta^r$, $\tilde{x} := x\eta^{-a}$, and

$$X_d := (\bar{x}\bar{y} + \bar{z}^2 + u^d = 0) \subset \mathbb{A}^4.$$

Let $\tilde{H}_d$ be the quotient of $\{(g, s_g) : g \in G,\ s_g^v = g_t\}$ by $\{(\mathbf{T}(\lambda), \lambda) : \lambda^r = 1\}$, acting on X_d by

$$(\bar{x}, \bar{y}, \bar{z}, u) \to (g_x s_g^a \bar{x}, g_y s_g^b \bar{y}, g_z s_g^c \bar{z}, s_g^r u),$$

and let $\tilde{H}_t$ *and* $\tilde{H}_y$ *act on* $\mathbb{A}^3_{P,Q,u}$ *and* $\mathbb{A}^3_{p,q,u_y}$ *by their actions on* $\mathbb{A}^2_{P,Q}$ *and* $\mathbb{A}^2_{p,q}$ *and by* $u \to s_g^r u$ *and* $u_y \to \eta_g^r u_y$. *For any* j *and* $\ell \in \mathbb{Z}$ *let* $h_{j,\ell}$ *be the sum of the terms of* h_j *of* $\mathbf{T}$*-weight* $\kappa_j + \ell r$, *and let*

$$\Psi_j := \sum_\ell u^\ell h_{j,\ell}(\bar{x}, \bar{y}, \bar{z}, 1), \qquad \Psi_j^y := \sum_\ell u_y^{-\ell} h_{j,\ell}(\tilde{x}, 1, p, q).$$

Let $(V, \tilde{H}, C_V)$ *be one of the triples*

$$\Big(X_d, \tilde{H}_d, \Big(1 + \frac{\delta}{r}\Big)\operatorname{div}(u) + \sum_j c_j \operatorname{div}(\Psi_j)\Big), \quad \Big(\mathbb{A}^3_{P,Q,u}, \tilde{H}_t, \Big(1 + \frac{\delta}{r}\Big)\operatorname{div}(u) + \sum_j c_j \operatorname{div}(\Psi_j)\Big),$$

$$\Big(\mathbb{A}^3_{p,q,u_y}, \tilde{H}_y, \Big(1 - \frac{\delta}{r}\Big)\operatorname{div}(u_y) + \sum_j c_j \operatorname{div}(\Psi_j^y)\Big),$$

and let $Z_V := Z_X$, Z_t, *resp.* Z_y, *and let* $\tilde{L}_V$ *be the group of the pairs* (g, s_g), *of the triples* (g, s_g, p_g), *resp. of the pairs* (g, η_g) *of Construction-Proposition A.33 and of the definition of* $\tilde{H}_d$, *acting on* Z_V *by* g *on* x, y, z, t *and by* $s \to s_g s$, $P \to p_g P$, *resp.* $\eta \to \eta_g \eta$. *Here the* μ*-invariant functions* Ψ_j *and* Ψ_j^y *are regarded as rational functions on* V *through the birational map* $Z_V/\mu \dashrightarrow V$ *of (1), that is,* $\bar{x} = P^2$, $\bar{y} = -Q^2 - u^d/P^2$, $\bar{z} = PQ$ *on* $\mathbb{A}^3_{P,Q,u}$ *and* $\tilde{x} = -p^2 - q^N/u_y^d$ *on* $\mathbb{A}^3_{p,q,u_y}$. *Then the following hold.*

(1) $h_j = \sum_\ell h_{j,\ell}$, $h_{j,0} = h_j^0$, *and* $h_j = s^{\kappa_j}\Psi_j = \eta^{-\kappa_j}\Psi_j^y$. *The group* $\tilde{H}_d$ *has order* $v[G : R]$ *and acts faithfully on* X_d *and on* $S_0 = X_d \cap (u = 0)$. *The group* $\tilde{L}_V$ *acts faithfully on* Z_V, *the morphism* $Z_V \to W_N/G$ *is the quotient by* $\tilde{L}_V$, *and the formulas for* $\bar{x}, \bar{y}, \bar{z}, u$, *resp.* P, Q, u, *resp.* p, q, u_y *define an* $\tilde{L}_V$*-equivariant dominant rational map* $\varphi_V : Z_V \dashrightarrow V$ *which induces a birational map* $Z_V/\mu \dashrightarrow V$ *for the subgroup* μ *of the classes* $(\mathbf{T}(\lambda), \lambda)$, $(\mathbf{T}(\lambda), \lambda, 1)$, *resp.* $(\mathbf{T}(\lambda), \lambda^{-1})$ *with* $\lambda^r = 1$; *in particular* $V/\tilde{H}$ *and* W_N/G *are birational.*

(2) Let F_Z *be a prime divisor over* Z_V, *let* F *and* F_W *be the prime divisors over* V *and over* W_N *induced by* F_Z *through* φ_V *and* $Z_V \to W_N$, *and for a rational function* f *on* Z_V *let* $[f] := \operatorname{mult}_{F_Z} f/|I(F_Z)|$, *where* $I(F_Z)$ *is the inertia group of* F_Z *in* $\tilde{L}_V$; *for a rational function* f *on* V *or on* W_N, $[f]$ *denotes* $[\cdot]$ *of the pullback of* f *to* Z_V. *Then* $[f] = \operatorname{mult}_F f/|I_{\tilde{H}}(F)|$ *for a rational function* f *on* V, $[f] = \operatorname{mult}_{F_W} f/|I_G(F_W)|$ *for a rational function* f *on* W_N, *and*

$$\frac{a(F_W, W_N, D)}{|I_G(F_W)|} = \frac{a(F, V, C_V)}{|I_{\tilde{H}}(F)|}.$$

For any prime divisor F' *over* V, *resp. over* W_N, *there exists a prime divisor* F_Z *over* Z_V *such that* $F = F'$, *resp.* $F_W = F'$.

Proof. By Lemma 5.3, any two monomials of h_j have $\mathbf{T}$-weights congruent modulo r, so $h_j = \sum_\ell h_{j,\ell}$, and $h_{j,0} = h_j^0$ by definition. Since $(\bar{x}, \bar{y}, \bar{z}, 1) = \mathbf{T}(s^{-1}) \cdot (x, y, z, t)$ and $(\tilde{x}, 1, p, q) = \mathbf{T}(\eta) \cdot (x, y, z, t)$, we have

$$h_{j,\ell} = s^{\kappa_j + \ell r} h_{j,\ell}(\bar{x}, \bar{y}, \bar{z}, 1) = \eta^{-\kappa_j - \ell r} h_{j,\ell}(\tilde{x}, 1, p, q),$$

which implies $h_j = s^{\kappa_j}\Psi_j = \eta^{-\kappa_j}\Psi_j^y$.

The equation $xy + z^2 + t^N = 0$ implies $\bar{x}\bar{y} + \bar{z}^2 + s^{Nv+2c} = 0$, and $s^{Nv+2c} = u^d$. The varieties Z_V are normal: Z_X is an irreducible hypersurface with an isolated singularity. The divisor of xs^a on Z_X has a prime component with odd coefficient and the divisor of y on W_N has a prime component with coefficient one, so both covers are irreducible, and Z_t and Z_y are their normalizations. The morphism $W_N \to \mathbb{A}^3_{x,z,t}$ is birational, $Z_V \to W_N$ has degree v, $2v$, resp. b, and $(\bar{x}, \bar{z}, s)$, (P, Q, s), resp. (η, p, q) give birational maps from Z_V to $\mathbb{A}^3$, while $(\bar{x}, \bar{z}, u)$, (P, Q, u), resp. (p, q, u_y) give birational maps from V to $\mathbb{A}^3$. Since $u = s^r$ and $u_y = \eta^r$, the map φ_V is dominant of degree r. The action of $\tilde{L}_V$ on Z_X, $Z_X \times \mathbb{A}^1_P$, resp. $W_N \times \mathbb{A}^1_\eta$ preserves the equations and lifts to the normalizations, φ_V is $\tilde{L}_V$-equivariant, $|\tilde{L}_V| = \deg(Z_V \to W_N) \cdot |G|$, and the

action is faithful, so the finite birational morphism $Z_V/\tilde{L}_V \to W_N/G$ of normal varieties is an isomorphism, that is, $Z_V \to W_N/G$ is the quotient by $\tilde{L}_V$. The subgroup μ fixes $\bar{x}$, $\bar{y}$, $\bar{z}$, u, P, Q, resp. p, q, u_y, and has order r, so φ_V factors through Z_V/μ and $Z_V/\mu \dashrightarrow V$ is birational. Since $\tilde{H} = \tilde{L}_V/\mu$, the varieties $V/\tilde{H}$ and $Z_V/\tilde{L}_V = W_N/G$ are birational. An element of $\tilde{L}_V$ acting trivially on V acts trivially on Z_V/μ, so it belongs to μ, the Galois group of $Z_V \to Z_V/\mu$. In particular $\tilde{H}_d$ acts faithfully on X_d, and $|\tilde{H}_d| = v|G|/r$. If (g, s_g) acts trivially on S_0, then $g_x = s_g^{-a}$, $g_y = s_g^{-b}$, and $g_z = s_g^{-c}$, and $g_t = s_g^v$ by definition. Thus $g = \mathbf{T}(s_g) \in R$, so $s_g^r = 1$ and (g, s_g) is trivial in $\tilde{H}_d$.

For (2), the ramification index of F_Z over the image of F_Z on W_N/G is $|I(F_Z)|$, and it is the product of the ramification index of F_Z over F_W, resp. over F, and $|I_G(F_W)|$, resp. $|I_{\tilde{H}}(F)|$. Hence the formulas for $[f]$ hold. The forms

$$\omega_W := \frac{dx \wedge dz \wedge dt}{x}, \quad \omega_d := \frac{d\bar{x} \wedge d\bar{z} \wedge du}{\bar{x}}, \quad dP \wedge dQ \wedge du, \quad dp \wedge dq \wedge du_y$$

generate ω_{W_N}, ω_{X_d}, $\omega_{\mathbb{A}^3_{P,Q,u}}$, and $\omega_{\mathbb{A}^3_{p,q,u_y}}$. Since $d\bar{x} \wedge d\bar{z} = s^{a+c} dx \wedge dz$ modulo ds, $du = rs^{r-1}ds$, and $dt = vs^{v-1}ds$, we have

$$\omega_d = rs^{r-1+c}\frac{dx \wedge dz \wedge ds}{x}, \qquad \omega_W = vs^{v-1}\frac{dx \wedge dz \wedge ds}{x},$$

which imply the first identity below. For $\bar{x} = P^2$ and $\bar{z} = PQ$ we have $d\bar{x} \wedge d\bar{z} = 2P^2 dP \wedge dQ$, which implies the second. On W_N we have $\omega_W = -dy \wedge dz \wedge dt/y$, and $y = \eta^b$, $z = p\eta^c$, $t = q\eta^{-v}$, $u_y = \eta^r$ imply

$$\frac{dy \wedge dz \wedge dt}{y} = b\eta^{c-v-1} d\eta \wedge dp \wedge dq = \frac{b}{r}\eta^{c-v}\frac{dp \wedge dq \wedge du_y}{u_y},$$

which implies the third. Thus

$$\omega_W = \frac{v}{r}s^{v-c}\frac{\omega_d}{u}, \qquad \omega_d = 2dP \wedge dQ \wedge du, \qquad \omega_W = -\frac{b}{r}\eta^{c-v}\frac{dp \wedge dq \wedge du_y}{u_y}.$$

The crepant pullback (Z_V, B_Z) of (W_N, D) satisfies

$$a(F_Z, Z_V, B_Z) = 1 + \operatorname{mult}_{F_Z}(\omega_W) - \sum_j c_j \operatorname{mult}_{F_Z} h_j,$$

which is the ramification index of F_Z over F_W times $a(F_W, W_N, D)$ by [Liu26, Lemma 2.6(1)(2)]. Let $\omega_V^0 := \omega_d$, $dP \wedge dQ \wedge du$, resp. $dp \wedge dq \wedge du_y$, the generator of ω_V above. Let Z'' and V'' be normal models of Z_V and V on which φ_V is a finite morphism near the generic point of F_Z. Near the generic point of F_Z, Z'' is the normalization of V'' in Z_V, so $Z'' \to V''$ is the quotient by μ there, since $Z_V/\mu \dashrightarrow V$ is birational by (1). The crepant pullback (Z_V, B'_Z) of (V, C_V) by φ_V, computed on Z'' and V'', satisfies $a(F_Z, Z_V, B'_Z) = 1 + \operatorname{mult}_{F_Z}(\omega_V^0) - \operatorname{mult}_{F_Z} C_V$, which is the ramification index of F_Z over F times $a(F, V, C_V)$ by [Liu26, Lemma 2.6(1)(2)] for $Z'' \to V''$. Since $\operatorname{mult}_{F_Z} s = \operatorname{mult}_{F_Z} u/r$, $\operatorname{mult}_{F_Z}(\eta) = \operatorname{mult}_{F_Z} u_y/r$, and $\tau - v + c = \delta$, the three identities and (1) imply

$$\operatorname{mult}_{F_Z}(\omega_W) - \sum_j c_j \operatorname{mult}_{F_Z} h_j = \operatorname{mult}_{F_Z}(\omega_V^0) - \operatorname{mult}_{F_Z} C_V.$$

Dividing by $|I(F_Z)|$ proves the formula for the log discrepancies. The last sentence of (2) holds since a prime divisor F' over W_N, resp. over V, is a divisor on a normal model W' of W_N, resp. V' of V, and a component of the preimage of F' in the normalization of the main component of $W' \times_{W_N} Z_V$, resp. in the normalization of V' in Z_V through the birational map $Z_V/\mu \dashrightarrow V$, is a prime divisor F_Z over Z_V inducing it. □

Proposition A.35. *Notation as in Construction-Proposition A.33. Let*

$$\beta_t := \operatorname{mld}\big((\mathbb{A}^2 \ni 0, C_t)/\tilde{H}_t\big)$$

when $(\mathbb{A}^2, C_t)$ is lc near 0, and let β_Y be the infimum of the log discrepancies of $(Y, B_Y)/H$ over the prime divisors with center contained in E/H when (Y, B_Y) is lc near E.

(1) There exist r_0 and C, depending only on a, c, v, Γ, e_0, such that for $r \geq r_0$ and any d the following hold. If $\delta \leq 0$, then $(\mathbb{A}^2, C_t)$ is lc near 0 and $\alpha \leq \beta_t + C/r$. If $\delta \geq 0$, then (Y, B_Y) is lc near E, the infimum β_Y is a minimum, and $\alpha \leq \beta_Y + C/r$. If $\delta = 0$, then $\beta_t = \beta_Y$. The numbers β_t and β_Y obtained in this way belong to an ACC set depending only on a, c, v, Γ, e_0.

(2) Let $\theta > 1$. There exists r_1, depending only on $a, c, v, \Gamma, e_0, \theta$, such that for $r \geq r_1$ the following hold. If $\delta \leq 0$, $\theta\tau + c - v \leq 0$, and $(\mathbb{A}^2, \theta C_t)$ is lc near 0, then $\alpha \geq \beta_t$. If $\delta \geq 0$, $\theta\tau + c - v \geq 0$, and $(Y, B_{\mathrm{br}} + \theta\Delta_Y)$ is lc near E, then $\alpha \geq \beta_Y$.

Proof. The proof is long, so we divide it into several steps. We use the notation of Lemma A.34. Let γ be the least positive element of Γ, $M := \lceil 2/\gamma \rceil$, and $C_1 := M \max_i |w_i|$. Since W_N has multiplicity two at 0, the exceptional divisor of the blow-up of $0 \in W_N$ has log discrepancy 2 with respect to $(W_N, 0)$, so $\sum_j c_j \operatorname{mult}_0 h_j < 2$ because (W_N, D) is klt near 0. We only consider the indices j with $c_j > 0$ and $h_j(0) = 0$, since the other terms of D are zero near 0. There are at most M such indices, $\sum_j c_j < 2$, and by Lemma 5.3 there exists r_2, depending only on w and Γ, such that for $r \geq r_2$ we have $|\kappa_j| \leq C_1$ and any monomial of $h_j - h_j^0$ has total degree at least $(r - C_1)/\max_i |w_i|$. In particular $|\delta| \leq 2C_1 + c + v$. Let Γ_+ and $\Psi(\cdot)$ be as in [Liu26, Definition 2.9], and let $\Gamma' := \Psi(\Gamma_+ \cap [0, 1])$, a DCC set by [Liu26, Lemma 2.10(1)]. Let l be the integer of Theorem 2.9(2) for Γ'.

Step 1. In this step we prove that the pairs of (1) are lc for r large, that β_Y is a minimum, the ACC, and $\beta_t = \beta_Y$ if $\delta = 0$.

Let $f_N := xy + z^2 + t^N$. By [Kaw07, Theorem], the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(f_N) + \sum_j c_j \operatorname{div}(h_j)\Big)$$

is lc near 0. Let L_0 be the integer of Lemma 4.14 for $n = 4$ and $M + 1$ functions, and the DCC set $(\Gamma \cup \{1\})\backslash\{0\}$. Since $N = (dr - 2c)/v \geq (r - 2c)/v$, there exists $r_3 \geq r_2$, depending only on a, c, v, Γ, such that t^N and all $h_j - h_j^0$ belong to $(x, y, z, t)^{L_0}$ for $r \geq r_3$. By Lemma 4.14 and [dFEM11, Proposition 2.11], the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2) + \sum_j c_j \operatorname{div}(h_j^0)\Big)$$

is lc near 0, so $(W_0, \sum_j c_j \operatorname{div}(h_j^0))$ is lc near 0 by [Kaw07, Theorem], where $W_0 := (xy + z^2 = 0)$. The morphism $\mathbb{A}^3_{P_0,Q_0,t} \to W_0$, $(x, y, z) = (P_0^2, -Q_0^2, P_0Q_0)$, is étale in codimension one, so

$$\Big(\mathbb{A}^3, \sum_j c_j \operatorname{div}(h_j^0(P_0^2, -Q_0^2, P_0Q_0, t))\Big)$$

is lc near 0 by [KM98, Proposition 5.20]. The group $\hat{G} := \{(g, \varrho_0) : g \in G,\ \varrho_0^2 = g_x\}$ of order $2|G|$ acts faithfully and diagonally on $\mathbb{A}^3$ by

$$(P_0, Q_0, t) \to (\varrho_0 P_0, g_z \varrho_0^{-1} Q_0, g_t t),$$

and $\mathbb{A}^3/\hat{G} = W_0/G$. Since $b = 2c - a$, we have $\gcd(a, b, 2v) = \gcd(a, 2c, 2v)$. An odd prime dividing $\gcd(a, b, 2v)$ divides a, c, and v, and if 4 divided $\gcd(a, b, 2v)$, then 2 would divide a, c, and v. Since $\gcd(a, c, v) = 1$, $\gcd(a, b, 2v)$ divides 2. Let $\mathbf{T}'$ be the primitive circle of weights $(a, b, -2v)/\gcd(a, b, 2v)$ on $\mathbb{A}^3$. The homomorphism $\mathbf{T}' \to \mathbf{T}$ with $\lambda' \to \mathbf{T}(\lambda'^{-2/\gcd(a,b,2v)})$ is compatible with $\mathbb{A}^3 \to W_0$ and has a kernel of order $2/\gcd(a, b, 2v)$ contained in $\hat{G}$, and any element of R lifts to $\hat{G} \cap \mathbf{T}'$. Thus $|\hat{G} \cap \mathbf{T}'| = 2r/\gcd(a, b, 2v)$ and $[\hat{G} : \hat{G} \cap \mathbf{T}'] = \gcd(a, b, 2v)[G : R] \leq 2e_0$. The pullback of h_j^0 is $\mathbf{T}'$-homogeneous of weight $-2\kappa_j/\gcd(a, b, 2v)$, and the number denoted τ in [Liu26, Construction-Theorem 16.9] for the pullbacks of the h_j^0 is $-2\delta/\gcd(a, b, 2v)$.

We compare the group and the reference pair of [Liu26, Construction-Theorem 16.9] with $\tilde{H}_t$, $(\mathbb{A}^2, C_t)$, and (Y, B_Y). [Liu26, Construction-Theorem 16.9] uses a root ϑ with $\vartheta^{2v/\gcd(a,b,2v)} = t$ and the plane coordinates $P_0\vartheta^{a/\gcd(a,b,2v)}$ and $Q_0\vartheta^{b/\gcd(a,b,2v)}$. Choose the root ϑ with $\vartheta^{2/\gcd(a,b,2v)} = s$, which is possible since $s^v = t$. Then the plane coordinates are P and Q, and the pullback of h_j^0 is $s^{\kappa_j}(h_j^0)_t$. An element $((g, \varrho_0), \vartheta_g)$ of the group of [Liu26, Construction-Theorem 16.9] maps to

$$(g, \vartheta_g^{2/\gcd(a,b,2v)}, \varrho_0\vartheta_g^{a/\gcd(a,b,2v)}) \in \tilde{H}_t,$$

which defines an isomorphism compatible with the actions on $\mathbb{A}^2_{P,Q}$, since both groups have order $2v[G : R]$ and act faithfully. Hence, if $\delta \le 0$, the reference pair of [Liu26, Construction-Theorem 16.9] is $(\mathbb{A}^2, C_t)$ with the group $\tilde{H}_t$. If $\delta > 0$, it is the pullback of (Y, B_Y) to the weighted blow-up $\hat{Y}$ of $\mathbb{A}^2_{P,Q}$ with weights $(a, b)/\gcd(a, b)$, with the group $\tilde{H}_t$, as we now check. The surface Y is the quotient of $\hat{Y}$ by the kernel of $\tilde{H}_t \to H$. Off the exceptional curves both pairs consist of the strict transforms of the curves $\operatorname{div}((h_j^0)_t)$ with coefficients c_j. Let $\hat{E}$ be the exceptional curve of $\hat{Y}$. Then $\hat{E}$ has ramification index $2ve_E/\gcd(a, b)$ over E, so by Construction-Proposition A.33(2) the coefficient of $\hat{E}$ in the pullback of B_Y is $1 - 2v(1 - m_E)/\gcd(a, b)$. In [Liu26, Construction-Theorem 16.9(2)] for the weights $(a, b, -2v)/\gcd(a, b, 2v)$, the weighted blow-up has weights $(a, b)/\gcd(a, b)$, the ratio of the third weight to the greatest common divisor of the first two is $2v/\gcd(a, b)$, and the coefficient of $(t = 0)$ in $\sum_j c_j \operatorname{div}(h_j^0(P_0^2, -Q_0^2, P_0Q_0, t))$ is m_E, since distinct monomials of h_j^0 have distinct images under the substitution defining $(h_j^0)_t$ by Construction-Proposition A.33(2). Hence the coefficient of the exceptional curve in [Liu26, Construction-Theorem 16.9(2)] is also $1-2v(1-m_E)/\gcd(a, b)$. By [Liu26, Construction-Theorem 16.9], $(\mathbb{A}^2, C_t)$ is lc near 0 if $\delta \le 0$, and (Y, B_Y) is lc near E if $\delta > 0$. If $\delta = 0$, then $K_Y + B_Y = \pi^*(K_S + B_S)$ by Construction-Proposition A.33(2), and (S, B_S) is lc near 0 by [KM98, Proposition 5.20], so (Y, B_Y) is lc near E.

If $\delta = 0$, the pair $(Y, B_Y)/H$ is crepant over $(S, B_S)/H$, and $\mathbb{A}^2_{P,Q}/\tilde{H}_t = S/H$ with $(\mathbb{A}^2, C_t)/\tilde{H}_t = (S, B_S)/H$, because the kernel of $\tilde{H}_t \to H$, $(g, s_g, p_g) \to gR$, is

$$\{(1, \zeta, \varrho) : \zeta^v = 1,\ \varrho^2 = \zeta^a\},$$

the group of ϕ. The prime divisors over S/H with center the image of 0 are the prime divisors over Y/H with center contained in E/H. Thus $\beta_t = \beta_Y$.

Assume that (Y, B_Y) is lc near E. Near any point of E/H, the pair $(Y, B_Y)/H$ is the crepant quotient of

$$\Big(\mathbb{A}^2_{p,q}, \sum_j c_j \operatorname{div}((h_j^0)_y)\Big) \text{ by } \tilde{H}_y, \quad \text{or of} \quad \Big(\mathbb{A}^2_{p',q'}, \sum_j c_j \operatorname{div}((h_j^0)_x)\Big) \text{ by } \tilde{H}_x,$$

by Construction-Proposition A.33(1)(2). Since (Y, B_Y) is lc near E, the pairs $(\mathbb{A}^2_{p,q}, \sum_j c_j \operatorname{div}((h_j^0)_y))$ and $(\mathbb{A}^2_{p',q'}, \sum_j c_j \operatorname{div}((h_j^0)_x))$ are lc near the preimage of E by [KM98, Proposition 5.20], and since their coefficients c_j belong to Γ, the coefficients of the boundary of $(Y, B_Y)/H$ belong to $\Gamma' = \Psi(\Gamma_+ \cap [0, 1])$ [Liu26, Lemma 2.10(5)]. Let e_I be the order of the inertia group of $(q = 0)$ in $\tilde{H}_y$. The curve E/H has coefficient $1 - \frac{1-m_E}{e_I}$, so the log discrepancy of E/H is $\frac{1-m_E}{e_I}$. Except at finitely many closed points of E/H, the surface Y/H is smooth and the boundary is $(1 - \frac{1-m_E}{e_I})E/H$, so the minimal log discrepancy at such a point is $1 + \frac{1-m_E}{e_I}$. Hence β_Y is the minimum of $\frac{1-m_E}{e_I} \le 1$ and of the minimal log discrepancies at the remaining closed points. Since $e_I \le |\tilde{H}_y| \le be_0$ and $|\tilde{H}_x| \le ae_0$, the numbers $\frac{1-\sigma}{e}$ with $\sigma \in \Gamma_+$ and $1 \le e \le \max(a, b)e_0$ form an ACC set, and the minimal log discrepancies at closed points belong to an ACC set by Theorem 2.9(1) applied to Γ'. Theorem 2.9(1) also implies the ACC for the numbers β_t, since the boundary of $(\mathbb{A}^2, C_t)/\tilde{H}_t$ has coefficients in Γ'. We have $\beta_t \le 2$, since the minimal log discrepancy of a surface germ at a closed point is at most 2.

Step 2. In this step we prove the upper bound of (1) for $\delta \le 0$.

Let $C_3 := \max(a,b,2v)e_0 l$ and $C_4 := 2C_3(1+1/\gamma)$. By Theorem 2.9(2) there exists a prime divisor over $\mathbb{A}^2_{P,Q}/\tilde{H}_t$ with center the image of 0 which computes β_t and whose log discrepancy with respect to the quotient surface is at most l. Let ν be a prime divisor over $\mathbb{A}^2_{P,Q}$ with center 0 above it, and let $e := |I_{\tilde{H}_t}(\nu)| \le 2ve_0$. By [Liu26, Lemma 2.6(2)], applied to the quotient morphism $\mathbb{A}^2_{P,Q} \to \mathbb{A}^2_{P,Q}/\tilde{H}_t$, whose branch $\mathbb{Q}$-divisor is ≥ 0, we have $a(\nu, \mathbb{A}^2, 0) \le el \le C_3$. Since $(\mathbb{A}^2, C_t)$ and $(\mathbb{A}^2, \operatorname{div}(P) + \operatorname{div}(Q))$ are lc near 0, we obtain

$$\operatorname{mult}_\nu (h^0_j)_t \le \frac{C_3}{\gamma}, \qquad 1 \le \operatorname{mult}_\nu P,\ \operatorname{mult}_\nu Q \le C_3.$$

The hypotheses of Proposition 5.4 hold for the plane $\mathbb{A}^2_{P,Q}$, the point 0, $\mathbb{A}^3_{P,Q,u} = \mathbb{A}^2_{P,Q} \times \mathbb{A}^1_u$, the group $\tilde{H}_t$, which acts faithfully on $\mathbb{A}^2_{P,Q}$ by Construction-Proposition A.33(1), the character $(g, s_g, p_g) \to s_g^r$, the functions $(h^0_j)_t$, which do not involve u, the coefficient $1 + \delta/r \le 1$ of $\operatorname{div}(u)$, and ν. Let q_ν be the integer of Lemma 3.1 for ν and the character $(g, s_g, p_g) \to s_g^r$, and let k_0 be the integer with $k_0 \equiv q_\nu \pmod e$ and $C_4 < k_0 \le C_4 + e$. Proposition 5.4 implies that the Gauss extension F of mult_ν with u-value k_0 is a prime divisor over $\mathbb{A}^3_{P,Q,u}$ with center 0, $I_{\tilde{H}_t}(F) = I_{\tilde{H}_t}(\nu)$, $\operatorname{mult}_F (h^0_j)_t = \operatorname{mult}_\nu (h^0_j)_t$, and

$$\frac{a(F, \mathbb{A}^3, (1+\frac{\delta}{r})\operatorname{div}(u) + \sum_j c_j \operatorname{div}((h^0_j)_t))}{|I_{\tilde{H}_t}(F)|} = \beta_t - \frac{\delta k_0}{re}. \tag{A.153}$$

Moreover $\operatorname{mult}_F P = \operatorname{mult}_\nu P$, $\operatorname{mult}_F Q = \operatorname{mult}_\nu Q$, and $\operatorname{mult}_F u = k_0$.

We show that $\operatorname{mult}_F \Psi_j = \operatorname{mult}_F (h^0_j)_t$ for r large, where Ψ_j is regarded as a rational function on $\mathbb{A}^3_{P,Q,u}$ via $\bar{x} = P^2$, $\bar{y} = -Q^2 - u^d/P^2$, $\bar{z} = PQ$. Since $\operatorname{mult}_F (u^d/P^2) \ge k_0 - 2C_3 > 2C_3$, we have $\operatorname{mult}_F \bar{y} = 2\operatorname{mult}_F Q$, and all of P^2, $\bar{y}$, PQ have non-negative value. The difference $h^0_j(P^2, \bar{y}, PQ, 1) - (h^0_j)_t$ is divisible by $\bar{y} + Q^2 = -u^d/P^2$, so the value of the difference is at least $k_0 - 2C_3 > C_3/\gamma$. For $\ell \ge 1$ the value of $u^\ell h_{j,\ell}(\bar{x}, \bar{y}, \bar{z}, 1)$ is at least $k_0 > C_3/\gamma$. For $\ell \le -1$, any monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ of $h_{j,\ell}$ satisfies $ai_1 + bi_2 + ci_3 = vi_4 - \kappa_j - \ell r \ge |\ell| r - C_1$, so the value of $u^\ell P^{2i_1} \bar{y}^{i_2} (PQ)^{i_3}$ is at least

$$\frac{|\ell| r - C_1}{\max(a,b,c)} - |\ell| k_0 \ge \frac{r}{\max(a,b,c)} - C_4 - 2ve_0 - C_1$$

for $r \ge \max(a,b,c)k_0$. Hence there exists $r_4 \ge r_3$ with $r_4 \ge \max(a,b,c)(C_4 + 2ve_0)$, depending only on a, c, v, Γ, e_0, such that for $r \ge r_4$ the terms of $\Psi_j - (h^0_j)_t$ have value larger than $\operatorname{mult}_F (h^0_j)_t$, and $\operatorname{mult}_F \Psi_j = \operatorname{mult}_F (h^0_j)_t$.

By Lemma A.34(2) for $V = \mathbb{A}^3_{P,Q,u}$, there is a prime divisor F_Z over Z_V inducing F. Let F_W be the prime divisor over W_N induced by F_Z. Since $x = P^2 s^{-a}$, $y = \bar{y} s^{-b}$, $z = PQ s^{-c}$, and $t = s^v$, Lemma A.34(2) implies

$$e[x] = 2\operatorname{mult}_\nu P - \frac{ak_0}{r}, \quad e[y] = 2\operatorname{mult}_\nu Q - \frac{bk_0}{r}, \quad e[z] = \operatorname{mult}_\nu P + \operatorname{mult}_\nu Q - \frac{ck_0}{r}, \quad e[t] = \frac{vk_0}{r},$$

which are positive for $r > \max(a,b,c)k_0$. Thus the center of F_W is 0. By Lemma A.34(2), $\operatorname{mult}_F \Psi_j = \operatorname{mult}_F (h^0_j)_t$, and (A.153),

$$\alpha \le \frac{a(F_W, W_N, D)}{|I_G(F_W)|} = \beta_t - \frac{\delta k_0}{re} \le \beta_t + \frac{C_0}{r}, \qquad C_0 := (2C_1 + c + v)(C_4 + \max(a,b,2v)e_0).$$

Step 3. In this step we prove the upper bound of (1) for $\delta \ge 0$.

By **Step 1**, β_Y is the log discrepancy of a prime divisor E'' over Y/H whose center is either a closed point $\bar{o}$ of E/H or E/H itself. Exchanging (a, x) with (b, y) if necessary, we may assume that the center intersects the image of the chart $\mathbb{A}^2_{p,q}$.

Assume first that the center of E'' is a closed point $\bar o$. Let $o \in \mathbb{A}^2_{p,q}$ be a point over $\bar o$ and $\tilde H_o$ the stabilizer of o in $\tilde H_y$. By Construction-Proposition A.33(2), the germ of $(Y, B_Y)/H$ at $\bar o$ is the crepant quotient of $(\mathbb{A}^2_{p,q} \ni o, \sum_j c_j \operatorname{div}((h^0_j)_y))$ by $\tilde H_o$, and the boundary of the quotient has coefficients in Γ'. As in **Step 2**, by Theorem 2.9(2) there exists a prime divisor ν over $\mathbb{A}^2_{p,q}$ with center o such that mult_ν induces a divisor computing β_Y, $e := |I_{\tilde H_y}(\nu)| \le be_0$, $a(\nu, \mathbb{A}^2, 0) \le C_3$, and

$$\operatorname{mult}_\nu (h^0_j)_y \le \frac{C_3}{\gamma}, \qquad 0 \le \operatorname{mult}_\nu p \le C_3, \qquad 1 \le \operatorname{mult}_\nu q \le C_3,$$

where $\operatorname{mult}_\nu q \ge 1$ since $q(o) = 0$. The inertia group of ν in $\tilde H_y$ fixes o, so it is the inertia group in $\tilde H_o$. The hypotheses of Proposition 5.4 hold for the plane $\mathbb{A}^2_{p,q}$, the point o, $\mathbb{A}^3_{p,q,u_y}$, the group $\tilde H_o$, which acts faithfully by Construction-Proposition A.33(1), the character $(g, \eta_g) \to \eta_g^r$, the functions $(h^0_j)_y$, the coefficient $1 - \delta/r \le 1$ of $\operatorname{div}(u_y)$, and ν. Let q_ν be the integer of Lemma 3.1 for ν, the group $\tilde H_o$, and the character $(g, \eta_g) \to \eta_g^r$, and let k_0 be the integer with $k_0 \equiv q_\nu \pmod e$ and $C_4 < k_0 \le C_4 + e$. By Proposition 5.4 there exists a prime divisor F over $\mathbb{A}^3_{p,q,u_y}$ with center $(o, 0)$, $I_{\tilde H_y}(F) = I_{\tilde H_y}(\nu)$, $\operatorname{mult}_F (h^0_j)_y = \operatorname{mult}_\nu (h^0_j)_y$, $\operatorname{mult}_F p = \operatorname{mult}_\nu p$, $\operatorname{mult}_F q = \operatorname{mult}_\nu q$, $\operatorname{mult}_F u_y = k_0$, and

$$\frac{a(F, \mathbb{A}^3, (1 - \frac{\delta}{r}) \operatorname{div}(u_y) + \sum_j c_j \operatorname{div}((h^0_j)_y))}{|I_{\tilde H_y}(F)|} = \beta_Y + \frac{\delta k_0}{re}. \tag{A.154}$$

Assume next that $E'' = E/H$, so that $\beta_Y = \frac{1 - m_E}{e_I}$ by **Step 1**. Let $\nu := \operatorname{mult}_{(q=0)}$ on $\mathbb{A}^2_{p,q}$, let I be the inertia group of $(q = 0)$ in $\tilde H_y$, so that $e := |I| = e_I$, let χ_I be the character by which I acts on q, which has order e, and let $q_E \in \{1, \dots, e\}$ be the integer with $\eta_g^r = \chi_I(g, \eta_g)^{q_E}$ on I. Let k_0 be the integer with $k_0 \equiv q_E \pmod e$ and $C_4 < k_0 \le C_4 + e$, and let F be the exceptional divisor of the weighted blow-up of $\mathbb{A}^3_{p,q,u_y}$ along the line $(q = u_y = 0)$ with weights 1 and k_0 for q and u_y. Then $a(F, \mathbb{A}^3, 0) = 1 + k_0$, $\operatorname{mult}_F p = 0$, $\operatorname{mult}_F q = 1$, $\operatorname{mult}_F u_y = k_0$, and $\operatorname{mult}_F (h^0_j)_y = m_j$, since $(h^0_j)_y = q^{m_j}(\varphi_j(p) + q\varphi'_j(p, q))$ with $\varphi_j \ne 0$ by Construction-Proposition A.33(2). An element of $I_{\tilde H_y}(F)$ fixes the center $(q = u_y = 0)$ of F pointwise, so it belongs to I. Conversely, I acts trivially on F, since F is birational to $\mathbb{A}^2$ with coordinates p and u_y/q^{k_0}, I fixes p, and $\eta_g^r \chi_I(g, \eta_g)^{-k_0} = 1$ on I. Thus $I_{\tilde H_y}(F) = I$ and

$$\frac{a(F, \mathbb{A}^3, (1 - \frac{\delta}{r}) \operatorname{div}(u_y) + \sum_j c_j \operatorname{div}((h^0_j)_y))}{|I_{\tilde H_y}(F)|} = \frac{1 + k_0 - (1 - \frac{\delta}{r})k_0 - m_E}{e} = \beta_Y + \frac{\delta k_0}{re},$$

which is (A.154) in this case.

In both cases we show that $\operatorname{mult}_F \Psi^y_j = \operatorname{mult}_F (h^0_j)_y$ for r large. Since $N = (dr - 2c)/v$ and $\operatorname{mult}_F q \ge 1$,

$$\operatorname{mult}_F (q^N/u^d_y) \ge d\Big(\frac{r}{v} - k_0\Big) - \frac{2c}{v} \ge \frac{r}{v} - k_0 - \frac{2c}{v}$$

for $r \ge vk_0$. Thus for r large, $\operatorname{mult}_F \tilde x = 2 \operatorname{mult}_F p \ge 0$ for $\tilde x = -p^2 - q^N/u^d_y$, and the difference $h^0_j(\tilde x, 1, p, q) - (h^0_j)_y$, which is divisible by q^N/u^d_y, has value larger than C_3/γ. For $\ell \ge 1$, any monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ of $h_{j,\ell}$ satisfies $vi_4 = \kappa_j + \ell r + ai_1 + bi_2 + ci_3 \ge \ell r - C_1$, so the value of $u_y^{-\ell} \tilde x^{i_1} p^{i_3} q^{i_4}$ is at least $\frac{\ell r - C_1}{v} - \ell k_0 \ge \frac{r}{v} - k_0 - \frac{C_1}{v}$. For $\ell \le -1$ the value of $u_y^{-\ell} h_{j,\ell}(\tilde x, 1, p, q)$ is at least $k_0 > C_3/\gamma$. We have $\operatorname{mult}_F (h^0_j)_y \le C_3/\gamma$ (in the second case $\operatorname{mult}_F (h^0_j)_y = m_j \le 1/\gamma \le C_3/\gamma$, since $c_j m_j \le m_E \le 1$ by the lc property of (Y, B_Y)). Hence there exists $r_5 \ge r_4$, depending only on a, c, v, Γ, e_0, such that $\operatorname{mult}_F \Psi^y_j = \operatorname{mult}_F (h^0_j)_y$ for $r \ge r_5$.

By Lemma A.34(2) for $V = \mathbb{A}^3_{p,q,u_y}$, there is a prime divisor F_Z over Z_V inducing F. Let F_W be the prime divisor over W_N induced by F_Z. Since $x = \tilde{x}\eta^a$, $y = \eta^b$, $z = p\eta^c$, and $t = q\eta^{-v}$,

$$e[x] = 2\,\mathrm{mult}_F\, p + \frac{ak_0}{r}, \quad e[y] = \frac{bk_0}{r}, \quad e[z] = \mathrm{mult}_F\, p + \frac{ck_0}{r}, \quad e[t] = \mathrm{mult}_F\, q - \frac{vk_0}{r},$$

which are positive for $r > vk_0$. Thus the center of F_W is 0, and by Lemma A.34(2) and (A.154),

$$\alpha \le \frac{a(F_W, W_N, D)}{|I_G(F_W)|} = \beta_Y + \frac{\delta k_0}{re} \le \beta_Y + \frac{C_0}{r}.$$

Taking $r_0 \ge r_5$ larger than $\max(a,b,c,v)(C_4 + \max(a,b,2v)e_0)$ and $C := C_0$ proves (1).

Step 4. In this step we fix $\theta > 1$ and prove that $(W_N, \theta D)$ is lc and the bounds on a prime divisor over W_N used in (2).

Assume the hypotheses of the first or of the second case of (2), and let $\beta := \beta_t$ in the first case and $\beta := \beta_Y$ in the second. By Construction-Proposition A.33(3), applied to $c'_j = \theta c_j$ and $f_j = h^0_j$, with $\delta' = \theta\tau + c - v$, the pair $(W_{N'}, \theta\sum_j c_j \,\mathrm{div}(h^0_j))$ is lc near 0 for any $N' \ge 3$, so the pair

$$\Big(\mathbb{A}^4, \mathrm{div}(xy + z^2 + t^{N'}) + \theta\sum_j c_j\,\mathrm{div}(h^0_j)\Big)$$

is lc near 0 by [Kaw07, Theorem]. Let L_1 be the integer of Lemma 4.14 for $n = 4$ and $M + 1$ functions, and the DCC set $(\Gamma \cup \theta\Gamma \cup \{1\})\backslash\{0\}$. There exists $r_6 \ge r_0$, depending only on $a, c, v, \Gamma, e_0, \theta$, such that all $h_j - h^0_j$ belong to $(x,y,z,t)^{L_1}$ for $r \ge r_6$. Let $D_0 := \sum_j c_j\,\mathrm{div}(h_j|_{W_0})$. Lemma 4.14, applied with $N' = N$ to the tuple formed by $xy + z^2 + t^N$ and the h^0_j and the tuple formed by $xy + z^2 + t^N$ and the h_j, together with [dFEM11, Proposition 2.11] and [Kaw07, Theorem], shows that $(W_N, \theta D)$ is lc near 0. Lemma 4.14, applied with a fixed $N' \ge \max(3, L_1)$ to the tuple formed by $xy + z^2 + t^{N'}$ and the h^0_j and the tuple formed by $xy + z^2$ and the h_j, together with [dFEM11, Proposition 2.11] and [Kaw07, Theorem], shows that $(W_0, \theta D_0)$ is lc near 0. In both applications the equation has coefficient one and the j-th function has coefficient θc_j. Let

$$B_3 := \sum_j c_j\,\mathrm{div}(h_j(P_0^2, -Q_0^2, P_0Q_0, t)).$$

By [KM98, Proposition 5.20], $(\mathbb{A}^3, \theta B_3)$ is lc near 0, so $(\mathbb{A}^3, B_3)$ is klt near 0.

Let F_W be a prime divisor over W_N with center 0, let $[f] := \mathrm{mult}_{F_W} f/|I_G(F_W)|$, and let $a_W := a(F_W, W_N, 0)/|I_G(F_W)|$. Since $\beta_t \le 2$ and $\beta_Y \le 1$ by **Step 1**, we may assume that $a(F_W, W_N, D)/|I_G(F_W)| < 2$. Since $(W_N, \theta D)$ is lc near 0, we have $\theta\sum_j c_j[h_j] \le a_W$, and $a_W - \sum_j c_j[h_j] < 2$. Hence

$$[h_j] \le C_5 := \frac{2}{\gamma(\theta - 1)}, \qquad a_W \le C_6 := \frac{2\theta}{\theta - 1}.$$

The hypersurface W_N is Newton non-degenerate, so it has a toric embedded resolution π_W [IP01, Proposition 2.7], [Var76, §10, Lemmas 10.2(2) and 10.3]. The non-degeneracy condition holds for any face of the Newton polyhedron, the non-compact faces included [IP01, Definition 2.6], so the strict transform of W_N is transversal to any torus orbit in $\pi_W^{-1}(0)$ [Var76, Lemma 10.3(3)]. Hence near 0 the strict transform of W_N and the toric boundary form a simple normal crossing divisor, and the strict transforms of the coordinate hyperplanes are contained in the toric boundary. The log discrepancies of the pairs

$$\Big(W_N, \frac{1}{2}(\mathrm{div}(x) + \mathrm{div}(y)) + \mathrm{div}(t)\Big), \qquad (W_N, \mathrm{div}(z) + \mathrm{div}(t))$$

along the toric divisor of weights (w'_1, w'_2, w'_3, w'_4) are

$$\frac{w'_1 + w'_2}{2} + w'_3 - \min(w'_1 + w'_2, 2w'_3, Nw'_4) \ge 0, \qquad w'_1 + w'_2 - \min(w'_1 + w'_2, 2w'_3, Nw'_4) \ge 0,$$

so both pairs are lc near 0. Thus $\frac{1}{2}([x]+[y])+[t] \le a_W$ and $[z]+[t] \le a_W$, which imply $[x],[y] \le 2C_6$ and $[z],[t] \le C_6$.

The monomials $x^v t^a$, $y^v t^b$, and $z^v t^c$ have $\mathbf{T}$-weight zero, so R acts trivially on them and their characters factor through H. Thus $U^{|H|}$ is G-invariant for $U = x^v t^a$, $y^v t^b$, $z^v t^c$, and since $U^{|H|}$ vanishes at 0, $[U^{|H|}]$ is a positive integer. Hence

$$v[x]+a[t] \ge \frac{1}{e_0}, \qquad v[y]+b[t] \ge \frac{1}{e_0}, \qquad v[z]+c[t] \ge \frac{1}{e_0}. \tag{A.155}$$

Let $\rho_t := [t]/v$, $\rho_- := \min([x]/a,[y]/b,[z]/c)$, and $c_0 := 1/(4v\max(a,b,c)e_0)$. If $\rho_t \le c_0$ and $\rho_- \le c_0$, then the inequality of (A.155) for the coordinate attaining ρ_- fails: for example, if $\rho_- = [x]/a$, then $v[x]+a[t] \le 2vac_0 < 1/e_0$. Thus $\rho_t > c_0$ or $\rho_- > c_0$.

For $\ell \ne 0$, any monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of $h_{j,\ell}$ satisfies $ai_1+bi_2+ci_3 \ge |\ell|r - C_1$ if $\ell < 0$, and $vi_4 \ge \ell r - C_1$ if $\ell > 0$. Thus

$$[h_{j,\ell}] \ge \rho_-(r-C_1) \ \text{ if } \ell<0, \qquad [h_{j,\ell}] \ge \rho_t(r-C_1) \ \text{ if } \ell>0. \tag{A.156}$$

We distinguish the cases $\rho_- > C_5/(r-C_1)$ and $\rho_- \le C_5/(r-C_1)$.

Step 5. In this step we prove $a(F_W,W_N,D)/|I_G(F_W)| \ge \beta$ if $\rho_- > C_5/(r-C_1)$.

By (A.156), $[h_{j,\ell}] > C_5 \ge [h_j]$ for $\ell < 0$, so $h_j^{\ge 0} := \sum_{\ell \ge 0} h_{j,\ell}$ satisfies $[h_j^{\ge 0}] = [h_j]$. By Lemma A.34(2) for $V = X_d$, there is a prime divisor F_Z over Z_V inducing F_W. Let F be the prime divisor over X_d induced by F_Z. Let

$$\Psi_j^{\ge 0} := \sum_{\ell \ge 0} u^\ell h_{j,\ell}(\bar{x},\bar{y},\bar{z},1).$$

Then $\Psi_j^{\ge 0}$ is a $\tilde{H}_d$-semi-invariant polynomial with $\Psi_j^{\ge 0}|_{S_0} = h_j^0(\bar{x},\bar{y},\bar{z},1)|_{S_0}$, and $h_j^{\ge 0} = s^{\kappa_j}\Psi_j^{\ge 0}$, so $\operatorname{mult}_F \Psi_j^{\ge 0} = \operatorname{mult}_F \Psi_j$ by Lemma A.34(1)(2). Let $\mathfrak{b} := \prod_j (\Psi_j^{\ge 0})^{c_j}$. Since $\operatorname{div}(u) = S_0$ on X_d,

$$a(F,X_d,C_V) = a\Big(F,X_d,\Big(1+\frac{\delta}{r}\Big)S_0,\mathfrak{b}\Big)$$

for

$$C_V = \Big(1+\frac{\delta}{r}\Big)\operatorname{div}(u) + \sum_j c_j \operatorname{div}(\Psi_j).$$

Since $\bar{x} = xs^a$, $\bar{y} = ys^b$, $\bar{z} = zs^c$, and $u = s^r$ have positive value at F_Z, the center of F is 0. The hypersurface X_d is normal with klt singularities near 0, $\tilde{H}_d$ acts diagonally on $\mathbb{A}^4$ and faithfully on X_d and on the normal klt surface S_0 by Lemma A.34(1), and u is $\tilde{H}_d$-semi-invariant. The morphism $\mathbb{A}^2_{P,Q} \to S_0$ is étale in codimension one, and $\tilde{H}_t \to \tilde{H}_d$, $(g,s_g,p_g) \to (g,s_g)$, is surjective with kernel generated by $(P,Q) \to (-P,-Q)$. Thus $(\mathbb{A}^2,C_t)/\tilde{H}_t = (S_0,\mathfrak{b}|_{S_0})/\tilde{H}_d$.

Assume that $\delta \le 0$. Then $(S_0,\mathfrak{b}|_{S_0})$ is lc near 0, and $\operatorname{mld}((S_0 \ni 0,\mathfrak{b}|_{S_0})/\tilde{H}_d) = \beta_t$. By Lemma 5.5 with the coefficient $1+\delta/r \le 1$ and the prime divisor F,

$$\frac{a(F_W,W_N,D)}{|I_G(F_W)|} = \frac{a(F,X_d,(1+\frac{\delta}{r})S_0,\mathfrak{b})}{|I_{\tilde{H}_d}(F)|} \ge \beta_t - \frac{\delta \operatorname{mult}_F u}{r|I_{\tilde{H}_d}(F)|} \ge \beta_t.$$

Assume that $\delta \ge 0$. Let k be a positive integer divisible by a, b, and c, let $I_w := (\bar{x}^{k/a},\bar{y}^{k/b},\bar{z}^{k/c})$, and let $\mathfrak{a} := \mathfrak{b}\cdot I_w^{\delta/k}$. The ideal I_w is generated by $\tilde{H}_d$-semi-invariant monomials. By Lemma A.34(2),

$$\begin{aligned}\frac{\operatorname{mult}_F I_w}{k} - \frac{\operatorname{mult}_F u}{r} &= |I_{\tilde{H}_d}(F)| \min\Big(\frac{[\bar{x}]}{a},\frac{[\bar{y}]}{b},\frac{[\bar{z}]}{c}\Big) - \frac{|I_{\tilde{H}_d}(F)|[u]}{r} \\ &= |I_{\tilde{H}_d}(F)| \min\Big(\frac{[x]}{a},\frac{[y]}{b},\frac{[z]}{c}\Big) > 0,\end{aligned}$$

where $[\cdot]$ is taken at F_Z. Thus

$$a(F,X_d,C_V)=a(F,X_d,S_0,\mathfrak{a})+\delta\Big(\frac{\operatorname{mult}_F I_w}{k}-\frac{\operatorname{mult}_F u}{r}\Big)\geq a(F,X_d,S_0,\mathfrak{a}).$$

We show that $(S_0,\mathfrak{a}|_{S_0})$ is lc near 0 and $\operatorname{mld}((S_0\ni 0,\mathfrak{a}|_{S_0})/\tilde{H}_d)=\beta_Y$. Let F_0 be a prime divisor over S_0 whose center contains 0, let $\bar{F}$ be the induced prime divisor over S/H, and let $[f]_0:=\operatorname{mult}_{F_0} f/|I_{\tilde{H}_d}(F_0)|$. The center of $\bar{F}$ on Y/H intersects E/H, and we may assume that it intersects the image of the first chart. We extend $[\cdot]_0$ to rational functions on the cover $\mathbb{A}^2_{p,\tilde{q}}$ of the first chart, with $\tilde{q}^v=q$, as follows. Let T be a normal surface with a finite morphism $T\to S/H$ that is Galois and dominates both S_0 and $\mathbb{A}^2_{p,\tilde{q}}$, let F_T be a prime divisor over T inducing F_0, and let $[f]_0:=\operatorname{mult}_{F_T} f/|I(F_T)|$, where $I(F_T)$ is the inertia group of F_T in the Galois group of $T\to S/H$. Such T exists: for an irreducible component T' of $S_0\times_{S/H}\mathbb{A}^2_{p,\tilde{q}}$ dominating $\mathbb{A}^2_{p,\tilde{q}}$, the normalization of S/H in the Galois closure of $T'\to S/H$ is such a surface. The prime divisor F_T exists by [Liu26, Lemma 2.6(3)] applied to $T\to S_0$. The values $[\tilde{q}]_0$, $[q]_0$, and $[I_w]_0$ below do not depend on the choice of F_T, since the prime divisors over T inducing F_0 are conjugate under the Galois group of $T\to S/H$, and $\tilde{q}$, q, and the generators of I_w are semi-invariant. Since ramification indices are multiplicative, the second definition of $[f]_0$ coincides with the first for rational functions on S_0. On $\mathbb{A}^2_{p,\tilde{q}}$ the ideal I_w is generated by $\tilde{q}^k$, so $[I_w]_0/k=[\tilde{q}]_0=[q]_0/v$. The pullback of E to $\mathbb{A}^2_{p,q}$ is $e_E\operatorname{div}(q)$, so we set $[E]_0:=e_E[q]_0$. Hence $\frac{\delta}{k}[I_w]_0=\frac{\delta}{ve_E}[E]_0$. Write $a(\bar{F},(S,B_S)/H)$ and $a(\bar{F},(Y,B_Y)/H)$ for the log discrepancies of $\bar{F}$ with respect to $(S,B_S)/H$ and $(Y,B_Y)/H$. Then Construction-Proposition A.33(2) implies

$$\frac{a(F_0,S_0,\mathfrak{a}|_{S_0})}{|I_{\tilde{H}_d}(F_0)|}=a(\bar{F},(S,B_S)/H)-\frac{\delta}{ve_E}[E]_0=a(\bar{F},(Y,B_Y)/H).$$

Since (Y,B_Y) is lc near E and the prime divisors over S/H with center the image of 0 are those over Y/H with center contained in E/H, we obtain that $(S_0,\mathfrak{a}|_{S_0})$ is lc near 0 and $\operatorname{mld}((S_0\ni 0,\mathfrak{a}|_{S_0})/\tilde{H}_d)=\beta_Y$. By Lemma 5.5 with the coefficient 1 and the prime divisor F,

$$\frac{a(F_W,W_N,D)}{|I_G(F_W)|}=\frac{a(F,X_d,C_V)}{|I_{\tilde{H}_d}(F)|}\geq\frac{a(F,X_d,S_0,\mathfrak{a})}{|I_{\tilde{H}_d}(F)|}\geq\beta_Y.$$

Step 6. In this step we prove $a(F_W,W_N,D)/|I_G(F_W)|\geq\beta$ if $\rho_-\leq C_5/(r-C_1)$.

There exists $r_7\geq r_6$, depending only on a,c,v,Γ,e_0,θ, such that for $r\geq r_7$ we have

$$C_5/(r-C_1)\leq c_0,\qquad (r-2c)c_0>2C_6+C_5.$$

Then $\rho_t>c_0$ by **Step 4**, so $[t]>vc_0$, and $N\geq(r-2c)/v$ implies $N[t]>2C_6+C_5\geq 2[z]$. Since $xy=-(z^2+t^N)$ on W_N, we obtain $[x]+[y]=2[z]$. The projections of W_N and W_0 to $\mathbb{A}^3_{x,z,t}$ are birational, with $y=-(z^2+t^N)/x$ on W_N and $y_0:=-z^2/x$ on W_0, and they are G-equivariant for the action of G on $\mathbb{A}^3_{x,z,t}$ by the characters g_x, g_z, g_t. Let F_0 be the prime divisor over W_0 corresponding to F_W under the birational map $W_N\dashrightarrow W_0$ over $\mathbb{A}^3_{x,z,t}$. Then $I_G(F_0)=I_G(F_W)$, and $[y_0]=2[z]-[x]=[y]>0$, so the center of F_0 is 0. The form $dx\wedge dz\wedge dt/x$ generates both ω_{W_N} and ω_{W_0}, so $a(F_W,W_N,0)=a(F_0,W_0,0)$. The difference $h_j(x,y,z,t)-h_j(x,y_0,z,t)$ is divisible by $y-y_0=-t^N/x$, with a quotient of non-negative value, so the value of the difference is at least $N[t]-[x]>C_5\geq[h_j]$. Thus $[h_j(x,y_0,z,t)]=[h_j]$ and

$$\frac{a(F_W,W_N,D)}{|I_G(F_W)|}=\frac{a(F_0,W_0,D_0)}{|I_G(F_0)|}\geq\operatorname{mld}((W_0\ni 0,D_0)/G)=\operatorname{mld}((\mathbb{A}^3\ni 0,B_3)/\hat{G}),$$

where the last equality holds since $\mathbb{A}^3\to W_0$ is étale in codimension one and $\hat{G}\to G$ is surjective with kernel generated by $(P_0,Q_0)\to(-P_0,-Q_0)$.

The pair $(\mathbb{A}^3,B_3)$ is klt near 0 by **Step 4**, $c_j\in\Gamma$, and by **Step 1** the group $\hat{G}$ is diagonal and faithful with $|\hat{G}\cap\mathbf{T}'|=2r/\gcd(a,b,2v)$ and $[\hat{G}:\hat{G}\cap\mathbf{T}']\leq 2e_0$. The pullback of a monomial of $\mathbf{T}$-weight κ and total degree n is a monomial of $\mathbf{T}'$-weight $-2\kappa/\gcd(a,b,2v)$ and total degree

between n and $2n$. By the choice of r_2, for r large the pullback of h_j^0 contains a monomial of degree at most $2M$, the monomials of the pullback of $h_j - h_j^0$ have larger degree, and the principal block of the pullback of h_j for $\mathbf{T}'$ is the pullback of h_j^0. By [Liu26, Theorem 16.21(2)], there exists $r_8 \geq r_7$, depending only on $a, c, v, \Gamma, e_0, \theta$, such that for $r \geq r_8$ the number of [Liu26, Construction-Theorem 16.9] for the pullbacks of the h_j^0, which we denote by β_{ref}, satisfies $\beta_{\mathrm{ref}} \leq \mathrm{mld}((\mathbb{A}^3 \ni 0, B_3)/\hat{G})$. By **Step 1**, $\beta_{\mathrm{ref}} = \beta_t$ if $\delta \leq 0$ and $\beta_{\mathrm{ref}} = \beta_Y$ if $\delta > 0$, and $\beta_t = \beta_Y$ if $\delta = 0$. Hence $a(F_W, W_N, D)/|I_G(F_W)| \geq \beta$.

Steps 5 and 6 hold for any prime divisor F_W over W_N with center 0 such that

$$a(F_W, W_N, D)/|I_G(F_W)| < 2.$$

Taking $r_1 := r_8$, we obtain $\alpha \geq \beta$, which proves (2). □

Proof of Theorem 7.21. The proof is long, so we divide it into several steps.

Step 1. In this step we fix a sequence violating the theorem, pass to a subsequence on which the discrete data are constant, and define the limit value L.

Assume that the theorem fails. Then there exist data as in Set-up 7.3, with Γ, e_0, and $w = (-a, -b, -c, v)$ fixed and indexed by $i \geq 1$, such that $r_i \to +\infty$ and $0 < \alpha_1 < \alpha_2 < \cdots$. We add the index i to the notation of Set-up 7.3 and Construction-Proposition A.33: N_i, G_i, R_i, $H_i := G_i/R_i$, $D_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$, κ_{ij}, h_{ij}^0, m_{ij}, τ_i, δ_i, $\tilde{H}_{t,i}$, $\tilde{H}_{y,i}$, $\tilde{H}_{x,i}$, and so on. Let $\alpha_\infty := \lim_i \alpha_i$. Then $0 < \alpha_i < \alpha_\infty$ for any i. Removing the units among the h_{ij} and the terms with $c_{ij} = 0$ changes neither D_i near 0 nor τ_i, so we may assume that each h_{ij} is a non-unit polynomial without monomials divisible by xy and that $c_{ij} > 0$.

Let γ be the least positive element of Γ and $M := \lceil 2/\gamma \rceil$. Let E' be the exceptional divisor of the blow-up of the origin of $\mathbb{A}^4$, so that $a(E', \mathbb{A}^4, 0) = 4$. Since W_{N_i} has multiplicity two at 0, its strict transform is the pullback of W_{N_i} minus $2E'$, and its intersection with $E' \cong \mathbb{P}^3$ is the reduced irreducible quadric cone $E_0 := (xy + z^2 = 0)$, since $N_i \geq 3$. Thus E_0 is the exceptional divisor of the blow-up of the origin of W_{N_i}, and $a(E_0, W_{N_i}, 0) = 4 - 2 = 2$ by adjunction. Moreover $\mathrm{mult}_{E_0} h_{ij} \geq 1$, so $\sum_j c_{ij} < 2$ because (W_{N_i}, D_i) is klt near 0. Thus the number s_i of the h_{ij} is less than $2/\gamma$. By Lemma 5.3, for i large, $|\kappa_{ij}| \leq M \max(a, b, c, v)$ and h_{ij}^0 contains a monomial $\mathfrak{n}_{ij}$ of total degree at most M. In particular $m_{ij} \leq M$.

For $f \in \mathcal{P}_{\mathbf{T}}$, let f_t, f_y, f_x, $m(f)$, and $D_Y(f)$ be as in Construction-Proposition A.33, so that $C_{t,i} = \sum_j c_{ij} \operatorname{div}((h_{ij}^0)_t)$. By Construction-Proposition A.33(2), distinct monomials of f have distinct images under each of the three substitutions defining f_t, f_y, and f_x, $f_y = q^{m(f)}(\varphi(p) + q\varphi'(p, q))$ for polynomials $\varphi \neq 0$ and φ', and the coefficient of E in $D_Y(f)$ is $m(f)/e_E$. Thus $\mathrm{mult}_0(h_{ij}^0)_t \leq 2M$, and the least exponent of q in $(h_{ij}^0)_y$ and of q' in $(h_{ij}^0)_x$ is m_{ij}. If f is moreover G_i-semi-invariant, then f_t, f_y, and f_x are semi-invariant under $\tilde{H}_{t,i}$, $\tilde{H}_{y,i}$, and $\tilde{H}_{x,i}$ by Construction-Proposition A.33(1). Combining Construction-Proposition A.33(2) with the coefficient $m(f)/e_E$ and with $(E^2) = -\frac{2ve_E^2}{ab}$, we obtain, for $f \in \mathcal{P}_{\mathbf{T}}$,

$$\left(\left(D_Y(f) - \frac{m(f)}{e_E} E\right) \cdot E\right) = \frac{2e_E(vm(f) - \kappa(f))}{ab}. \tag{A.157}$$

By Construction-Proposition A.33(1), the groups $\tilde{H}_{t,i}$, $\tilde{H}_{y,i}$, and $\tilde{H}_{x,i}$ act faithfully and diagonally on $\mathbb{A}^2$, with orders at most $2ve_0$, be_0, and ae_0, and $\tilde{H}_{y,i}$ induces the action of H_i on Y. A diagonal torus contains finitely many subgroups of bounded order, and a finite group has finitely many characters. Thus, possibly passing to a subsequence, we may assume the following.

- $s_i = s$, $\kappa_{ij} = \kappa_j$, $m_{ij} = m_j$, and $\mathfrak{n}_{ij} = \mathfrak{n}_j$ do not depend on i.
- The images of $\tilde{H}_{t,i}$, $\tilde{H}_{y,i}$, and $\tilde{H}_{x,i}$ in $GL_2(\mathbb{C})$ and the action of H_i on Y do not depend on i. We denote them by $\tilde{H}_t$, $\tilde{H}_y$, $\tilde{H}_x$, and H.
- The characters of $(h_{ij}^0)_t$, $(h_{ij}^0)_y$, and $(h_{ij}^0)_x$ do not depend on i.
- Either $\delta_i \leq 0$ for any i, or $\delta_i \geq 0$ for any i.

- Since Γ is a DCC set, for any j, the sequence $(c_{ij})_i$ is non-decreasing with limit $c_j^* \in (0,1]$.

Let $\tau^* := \sum_j c_j^* \kappa_j$, $\delta_* := \tau^* + c - v$, and $m_E^* := \sum_j c_j^* m_j$. Then $\delta_i \to \delta_*$, and δ_* has the weak sign of the δ_i.

Let $\beta_i := \beta_{t,i}$ if $\delta_i \le 0$ for any i, and $\beta_i := \beta_{Y,i}$ otherwise. By Proposition A.35(1), for i large, $\alpha_i \le \beta_i + C/r_i$, and the β_i belong to an ACC set. Possibly passing to a subsequence, the β_i are non-increasing with limit L, and

$$\beta_i \ge L \ge \alpha_\infty > 0.$$

We obtain a contradiction by showing that $\alpha_i \ge L$ for some i.

Step 2. In this step we replace the labels c_{ij} by their limits c_j^*, show that, for i large, $(\mathbb{A}^2, C_{t,i}^*)$ if $\delta_* \le 0$, resp. $(Y, B_{Y,i}^*)$ if $\delta_* \ge 0$, defined below, is lc with minimum L, and reduce the lc property of the pairs on W_{N_i} of Claim A.37 to that of pairs on $\mathbb{A}^2$ and on Y.

Let

$$D_i^* := \sum_j c_j^* \operatorname{div}(h_{ij}), \quad C_{t,i}^* := \sum_j c_j^* \operatorname{div}((h_{ij}^0)_t), \quad \Delta_{Y,i}^* := \sum_j c_j^* D_Y(h_{ij}^0), \quad B_{Y,i}^* := B_{\mathrm{br}} + \Delta_{Y,i}^*.$$

When $(\mathbb{A}^2, C_{t,i}^*)$ is lc near 0, let $\beta_{t,i}^* := \operatorname{mld}((\mathbb{A}^2 \ni 0, C_{t,i}^*)/\tilde{H}_t)$. When $(Y, B_{Y,i}^*)$ is lc near E, let $\beta_{Y,i}^*$ be the infimum of the log discrepancies of $(Y, B_{Y,i}^*)/H$ over the prime divisors with center contained in E/H.

For a boundary $B_{\mathrm{br}} + \Theta$ on Y, or on the formal completion of $Y_{\mathbb{K}}$ along $E_{\mathbb{K}}$ for an extension field $\mathbb{K}$ of the ground field, which is lc near E, resp. $E_{\mathbb{K}}$, such an infimum is a minimum, since it is the least of the value at E and of finitely many closed-point minimal log discrepancies of the quotient: at a closed point of E off the singular points of Y, off $\Sigma_x \cup \Sigma_y$, off the other components of the boundary, and off the finitely many points whose stabilizer in H is larger than the pointwise stabilizer of E, the quotient germ is smooth and the boundary is a multiple of the image of E, so the value there exceeds the value at E by one [Liu26, Lemma 2.6(2)(4)].

By Construction-Proposition A.33(2) with no labels c_j', the chart morphisms $\mathbb{A}_{p,q}^2 \to Y$ and $\mathbb{A}_{p',q'}^2 \to Y$ pull $K_Y + B_{\mathrm{br}}$ back to $K_{\mathbb{A}^2}$. Thus the crepant quotients of $(\mathbb{A}_{p,q}^2, 0)$ and $(\mathbb{A}_{p',q'}^2, 0)$ glue to B_{br} near E, and the crepant pullback of B_{br} to either root chart is zero.

Claim A.36. *For i large, the following hold. If $\delta_* \le 0$, then $(\mathbb{A}^2, C_{t,i}^*)$ is lc near 0 and $\beta_{t,i}^* = L$. If $\delta_* \ge 0$, then $(Y, B_{Y,i}^*)$ is lc near E and $\beta_{Y,i}^* = L$.*

Proof. By **Step 1**, if $\delta_* > 0$, then $\delta_i \ge 0$ for any i, and if $\delta_* < 0$, then $\delta_i \le 0$ for any i. Assume first that $\delta_i \ge 0$ for any i. Let θ_i be the largest real number such that $(Y, B_{\mathrm{br}} + \theta_i \Delta_{Y,i}^*)$ is lc near E. By Proposition A.35(1), $(Y, B_{Y,i})$ is lc near E. Since $\Delta_{Y,i} \ge (\min_j c_{ij}/c_j^*)\Delta_{Y,i}^*$,

$$\min_j \frac{c_{ij}}{c_j^*} \le \theta_i.$$

Since the crepant pullback of B_{br} to the root charts is zero, θ_i is the lc threshold of $\sum_j c_j^* \operatorname{div}((h_{ij}^0)_y)$ or of $\sum_j c_j^* \operatorname{div}((h_{ij}^0)_x)$ at a closed point of $\mathbb{A}^2$ over E. By [dFEM11, Theorem 1.1], such lc thresholds belong to an ACC set depending only on the c_j^*. If $\theta_i < 1$ for infinitely many i, then the θ_i with $\theta_i < 1$ form an infinite sequence converging to 1 from below, which contains a strictly increasing subsequence, a contradiction. Thus $\theta_i \ge 1$ for i large, that is, $(Y, B_{Y,i}^*)$ is lc near E.

Since $\Delta_{Y,i}^* \ge \Delta_{Y,i}$, $\beta_{Y,i}^* \le \beta_{Y,i}$. Let P_i be a prime divisor computing $\beta_{Y,i}^*$. If $P_i \ne E$, then P_i computes the minimal log discrepancy of the quotient of $(Y, B_{Y,i}^*)$ by H at a closed point, and the coefficients of the quotient boundary belong to the DCC set $\Psi(\Gamma_+^* \cap [0,1])$, $\Gamma^* := \{c_1^*, \dots, c_s^*\}$ [Liu26, Lemma 2.6(4) and Lemma 2.10(1)]. Thus by Theorem 2.9(2), we may choose P_i with $a(P_i, Y/H, 0) \le l_{\Gamma^*}$ for a positive integer l_{Γ^*} depending only on Γ^*. Let $\tilde{P}_i$ be a prime divisor over a root chart $\mathbb{A}^2$ lying over P_i, with inertia group I_i in $\tilde{H}_y$ or $\tilde{H}_x$. Since the boundary of the crepant quotient of $(\mathbb{A}^2, 0)$ is ≥ 0, [Liu26, Lemma 2.6(2)] implies $a(\tilde{P}_i, \mathbb{A}^2, 0) \le |I_i| l_{\Gamma^*}$. If $P_i = E$, let $\tilde{P}_i$

be the line $(q=0)$, so that $a(\tilde{P}_i,\mathbb{A}^2,0)=1$. In both cases $a(\tilde{P}_i,\mathbb{A}^2,0)\le l':=\max(a,b)e_0 l_{\Gamma^*}$, and the lc property implies

$$\sum_j c_j^*\operatorname{mult}_{\tilde{P}_i}(h_{ij}^0)_y\le l',$$

where $(h_{ij}^0)_y$ is replaced by $(h_{ij}^0)_x$ on the second root chart. Let $\gamma^*:=\min_j c_j^*$. By [Liu26, Lemma 2.6(2)],

$$\beta_{Y,i}^*=\frac{a(\tilde{P}_i,\mathbb{A}^2,0)-\sum_j c_j^*\operatorname{mult}_{\tilde{P}_i}(h_{ij}^0)_y}{|I_i|}$$

belongs to a finite set depending only on l', γ^*, and the c_j^*, and

$$\beta_{Y,i}\le\frac{a(\tilde{P}_i,\mathbb{A}^2,0)-\sum_j c_{ij}\operatorname{mult}_{\tilde{P}_i}(h_{ij}^0)_y}{|I_i|}\le\beta_{Y,i}^*+\frac{l'}{\gamma^*}\sum_j(c_j^*-c_{ij}).$$

Thus $\beta_{Y,i}^*\to L$, and $\beta_{Y,i}^*=L$ for i large. If $\delta_i\le 0$ for any i, then $(\mathbb{A}^2,C_{t,i}^*)$ is lc near 0 and $\beta_{t,i}^*=L$ for i large, by the argument for $(Y,B_{Y,i}^*)$ with (Y,B_{br},E,H) replaced by $(\mathbb{A}^2,0,\{0\},\tilde{H}_t)$, the root charts replaced by $\mathbb{A}^2_{P,Q}$, and $\max(a,b)$ replaced by $2v$. This proves the claim when $\delta_*\ne 0$.

Assume that $\delta_*=0$. Let $B_{S,i}^*$ be the $\mathbb{R}$-divisor on S with

$$K_{\mathbb{A}^2}+C_{t,i}^*=\phi^*(K_S+B_{S,i}^*),$$

as B_S in Construction-Proposition A.33(2). By Construction-Proposition A.33(2) with $c_j'=c_j^*$ and $f_j=h_{ij}^0$, for which $\delta'=\delta_*=0$,

$$K_Y+B_{Y,i}^*=\pi^*(K_S+B_{S,i}^*).$$

Since ϕ is the quotient morphism of Construction-Proposition A.33(2), $\tilde{H}_t$ acts on S through H, and E is the fiber of π over the image of 0, $(\mathbb{A}^2,C_{t,i}^*)$ is lc near 0 if and only if $(Y,B_{Y,i}^*)$ is lc near E, and then $\beta_{t,i}^*=\beta_{Y,i}^*$ [Liu26, Lemma 2.6(2)(5)]. Thus the claim holds for both pairs. □

Proof of Theorem 7.21 continued. The following claim reduces the lc property of $(W_{N_i},\theta D_i^*+\lambda\operatorname{div}(U))$ to that of $(\mathbb{A}^2,\theta C_{t,i}^*+\lambda\operatorname{div}(U_t))$ or of $(Y,B_{\mathrm{br}}+\theta\Delta_{Y,i}^*+\lambda D_Y(U))$.

Claim A.37. *Let $\theta\ge 1$ and $\lambda\ge 0$ be real numbers, let $U\in\mathcal{P}_{\mathbf{T}}$ have $\mathbf{T}$-weight zero, and let $\delta_\theta:=\theta\tau^*+c-v$. Assume that for any i, either $\delta_\theta\le 0$ and $(\mathbb{A}^2,\theta C_{t,i}^*+\lambda\operatorname{div}(U_t))$ is lc near 0, or $\delta_\theta\ge 0$ and $(Y,B_{\mathrm{br}}+\theta\Delta_{Y,i}^*+\lambda D_Y(U))$ is lc near E. Then $(W_{N_i},\theta D_i^*+\lambda\operatorname{div}(U))$ is lc near 0 for i large. If $\lambda=0$, the terms with U are omitted.*

Proof. The polynomials h_{ij}^0 and U belong to $\mathcal{P}_{\mathbf{T}}$, and $\kappa(U)=0$. Thus Construction-Proposition A.33(3), with labels $\theta c_1^*,\dots,\theta c_s^*,\lambda$ (without λ if $\lambda=0$), implies that the pair

$$\Big(W_N,\theta\sum_j c_j^*\operatorname{div}(h_{ij}^0)+\lambda\operatorname{div}(U)\Big)$$

is lc near 0 for any $N\ge 3$, since

$$\sum_j\theta c_j^*\kappa_j+\lambda\kappa(U)+c-v=\delta_\theta.$$

Let $f_i:=xy+z^2+t^{N_i}$. By [Kaw07, Theorem], the pair

$$\Big(\mathbb{A}^4,\operatorname{div}(f_i)+\theta\sum_j c_j^*\operatorname{div}(h_{ij}^0)+\lambda\operatorname{div}(U)\Big)$$

is lc near 0. A pair on $\mathbb{A}^4$ is lc near 0 if and only if the corresponding pair on $\operatorname{Spec}\mathbb{C}[[x,y,z,t]]$ is lc [dFEM11, Proposition 2.11(i)]. Let L_0 be the integer of Lemma 4.14 for $n=4$, $s+2$ functions (resp. $s+1$ if $\lambda=0$), and the finite set

$$\{1,\theta c_1^*,\dots,\theta c_s^*\}\cup(\{\lambda\}\backslash\{0\}).$$

By Lemma 5.3, any monomial of $h_{ij}-h_{ij}^0$ has total degree at least $(r_i-|\kappa_j|)/\max(a,b,c,v)$, so $h_{ij}-h_{ij}^0\in(x,y,z,t)^{L_0}$ for i large. Thus by Lemma 4.14, applied to the tuples $(f_i,U,h_{i1}^0,\dots,h_{is}^0)$ and $(f_i,U,h_{i1},\dots,h_{is})$, resp. to $(f_i,h_{i1}^0,\dots,h_{is}^0)$ and $(f_i,h_{i1},\dots,h_{is})$ if $\lambda=0$, the pair

$$\Big(\mathbb{A}^4,\operatorname{div}(f_i)+\theta\sum_j c_j^*\operatorname{div}(h_{ij})+\lambda\operatorname{div}(U)\Big)$$

is lc near 0 for i large. By [Kaw07, Theorem], $(W_{N_i},\theta D_i^*+\lambda\operatorname{div}(U))$ is lc near 0. □

Proof of Theorem 7.21 continued.

Step 3. In this step we take compatible generic limits of the tuples formed by the $(h_{ij}^0)_t$, resp. by the pairs $((h_{ij}^0)_y,(h_{ij}^0)_x)$, and we compute the intersection numbers with E of the limit divisors.

Let

$$U_1:=x^{v/e_a}t^{a/e_a},\qquad U_2:=y^{v/e_b}t^{b/e_b},\qquad U_3:=x^vt^a,\qquad U_4:=y^vt^b.$$

They belong to $\mathcal{P}_{\mathbf{T}}$ and have $\mathbf{T}$-weight zero.

If $\delta_*\le 0$, we use case (a) of Construction-Proposition 4.1 with $n=2$, the group $\tilde{H}_t$, and the tuples formed by the $(h_{ij}^0)_t$ and the $(U_k)_t$. The hypotheses hold by **Step 1**: the characters do not depend on i, and the orders at 0 are bounded, by $2M$ for the $(h_{ij}^0)_t$. We write $f_{\infty,j}$ for the limit of $(h_{ij}^0)_t$ and $C_{t,\infty}:=\sum_j c_j^*\operatorname{div}(f_{\infty,j})$ on $\operatorname{Spec}\mathbb{K}[[P,Q]]$.

If $\delta_*\ge 0$, we use case (b) with Y, E, the root charts $V_1:=\mathbb{A}^2_{p,q}/\mu_b(-c,v)$ and $V_2:=\mathbb{A}^2_{p',q'}/\mu_a(-c,v)$ of Construction-Proposition A.33, e_α equal to q and q', the group H acting on Y, and the tuples formed by the pairs $((h_{ij}^0)_y,(h_{ij}^0)_x)$ and $((U_k)_y,(U_k)_x)$. We check the hypotheses. The branch divisor of the charts is B_{br} by **Step 2**. The chart functions are semi-invariant with characters independent of i by **Step 1**, and the $\mathbb{Q}$-divisors $D_Y(h_{ij}^0)$ and $D_Y(U_k)$ are $\mathbb{Q}$-Cartier [Liu26, Lemma 2.6] and H-invariant because their chart functions are semi-invariant.

For $f\in\mathcal{P}_{\mathbf{T}}$ and a monomial $\mathfrak{n}$ with $\kappa(\mathfrak{n})=\kappa(f)$, the homogeneity of f and $\mathfrak{n}$ shows that $f_y/\mathfrak{n}_y$ and $f_x/\mathfrak{n}_x$ are the pullbacks of the rational function

$$\frac{f(\bar{x},\bar{y},\bar{z},1)}{\mathfrak{n}(\bar{x},\bar{y},\bar{z},1)}$$

on S. Thus on the normalization of the fiber product of the root charts over $V_1\cap V_2$,

$$f_y\mathfrak{n}_x=f_x\mathfrak{n}_y, \tag{A.158}$$

and $\mathfrak{n}_y/\mathfrak{n}_x$ is a unit there, since p and p' are units over $V_1\cap V_2$ and the pullbacks of q and q' differ by a unit. Applied to $f=h_{ij}^0$ and $\mathfrak{n}=\mathfrak{n}_j$, the quotient of the chart functions is the unit $(\mathfrak{n}_j)_y/(\mathfrak{n}_j)_x$, which does not depend on i. The orders of $(h_{ij}^0)_y$ along $(q=0)$ are m_j. Finally, for any l, the monomials of h_{ij}^0 with exponent l of t are of the form $x^{\nu_1}y^{\nu_2}z^{\nu_3}t^l$ with $a\nu_1+b\nu_2+c\nu_3=vl-\kappa_j$, so there exist finitely many such monomials, and the coefficients of q^l in $(h_{ij}^0)_y$ have bounded degrees in p. Similarly on the second root chart. We write $f_{\infty,j,1}$ and $f_{\infty,j,2}$ for the limits of $(h_{ij}^0)_y$ and $(h_{ij}^0)_x$, $D_{\infty,j}$ for the divisor that they define on the formal completion $\mathcal{A}_{\mathbb{K}}$ of $Y_{\mathbb{K}}$ along $E_{\mathbb{K}}$, and $C_{Y,\infty}:=\sum_j c_j^*D_{\infty,j}$.

If $\delta_*=0$, we take both limits along one subsequence. Each of them is defined over an algebraically closed field, and we write $\mathbb{K}$ for the field of the limit under consideration. The entries $(U_k)_t$, resp. $((U_k)_y,(U_k)_x)$, do not depend on i, so the corresponding entries of the limits are non-zero constant multiples of them: the d-jets of the corresponding entries of the tuples belong to a line in $\mathcal{J}_d$, which is closed. Moreover, after deleting corresponding entries of the tuples and of a compatible generic limit, the remaining limit is a compatible generic limit of the

remaining tuples, because the closure of the image of Z_d in the corresponding factor of $\mathcal{J}_d$ is irreducible, and the preimage in Z_d of a non-empty open subset of it is non-empty and open. Thus Theorem 4.2 is available, with any positive labels, for the tuples of the h^0_{ij} and for the tuples of the h^0_{ij} and one U_k, with the limits above and U_k.

Claim A.38. *Assume that $\delta_* \geq 0$. Then for any j, the functions $f_{\infty,j,1}$ and $f_{\infty,j,2}$ have the characters of $(h^0_{ij})_y$ and $(h^0_{ij})_x$, satisfy*

$$f_{\infty,j,1}(\mathfrak{n}_j)_x = f_{\infty,j,2}(\mathfrak{n}_j)_y$$

on the base change of the normalization of the fiber product above, and have least exponents of q and q' equal to m_j. Thus the coefficient of $E_{\mathbb{K}}$ in $D_{\infty,j}$ is m_j/e_E, and

$$(D_{\infty,j} \cdot E_{\mathbb{K}}) = -\frac{2e_E\kappa_j}{ab}.$$

Proof. The characters are those of the $(h^0_{ij})_y$ and $(h^0_{ij})_x$ by Construction-Proposition 4.1. By (A.158), $(h^0_{ij})_y(\mathfrak{n}_j)_x = (h^0_{ij})_x(\mathfrak{n}_j)_y$ for any i. Since the pullbacks of q and q' to the fiber product differ by a unit, the identity modulo q^d depends only on the d-jets, and it is a closed condition on $\mathcal{J}_d$, preserved by multiplying both chart functions by a common non-zero constant. Let $\mathcal{C} \subset \mathcal{J}_d$ be a closed subset containing the d-jets of the tuples with $i \geq i_0$. If $Z_d \not\subset \mathcal{C}$, then $Z_d \backslash \mathcal{C}$ is a non-empty open subset of Z_d containing the d-jet of no tuple with $i \geq i_0$, which contradicts Construction-Proposition 4.1. Thus $Z_d \subset \mathcal{C}$, and the identity holds for the limit modulo q^d for any d, hence holds. The vanishing of the coefficients of q^l for $l < m_j$ is a closed condition, so it holds for the limit. The non-vanishing of the coefficient of q^{m_j} is an open condition $\mathcal{V}$, preserved by non-zero constants and satisfied by all d-jets for $d > m_j$. Since all but finitely many d-jets belong to Z_d, $Z_d \cap \mathcal{V}$ is non-empty, so it contains the generic point of Z_d. Thus the least exponent of q in $f_{\infty,j,1}$ is m_j, and similarly for q'. As the pullback of E to either root chart is $e_E(q = 0)$, resp. $e_E(q' = 0)$, the coefficient of $E_{\mathbb{K}}$ in $D_{\infty,j}$ is m_j/e_E.

The functions $f_{\infty,j,1}/(\mathfrak{n}_j)_y$ and $f_{\infty,j,2}/(\mathfrak{n}_j)_x$ are invariant under the groups of the charts and satisfy

$$f_{\infty,j,1}(\mathfrak{n}_j)_x = f_{\infty,j,2}(\mathfrak{n}_j)_y,$$

so they glue to a meromorphic function φ_j on $\mathcal{A}_{\mathbb{K}}$ with

$$D_{\infty,j} = D_Y(\mathfrak{n}_j)_{\mathbb{K}} + \operatorname{div}(\varphi_j).$$

Since

$$\mathcal{O}(\operatorname{div}(\varphi_j))|_{E_{\mathbb{K}}} \cong \mathcal{O}_{E_{\mathbb{K}}},$$

we have $(\operatorname{div}(\varphi_j) \cdot E_{\mathbb{K}}) = 0$. By Construction-Proposition A.33(2), $(D_Y(\mathfrak{n}_j) \cdot E) = -\frac{2e_E\kappa_j}{ab}$. □

Proof of Theorem 7.21 continued. Assume that $\delta_* \geq 0$. By Claim A.38, the coefficient of $E_{\mathbb{K}}$ in $C_{Y,\infty}$ is m^*_E/e_E, and the coefficient of $E_{\mathbb{K}}$ in $B_{\mathrm{br}} + C_{Y,\infty}$ is $1 - \frac{1-m^*_E}{e_E}$. Let $T_\infty := C_{Y,\infty} - \frac{m^*_E}{e_E}E_{\mathbb{K}}$. By Claim A.38 and $(E^2) = -\frac{2ve^2_E}{ab}$,

$$\iota := \frac{(T_\infty \cdot E_{\mathbb{K}})}{e_E} = \frac{2(vm^*_E - \tau^*)}{ab}. \tag{A.159}$$

After discarding finitely many i, by Claim A.36, $(\mathbb{A}^2, C^*_{t,i})$ is lc near 0 with $\beta^*_{t,i} = L$ for any i if $\delta_* \leq 0$, and $(Y, B^*_{Y,i})$ is lc near E with $\beta^*_{Y,i} = L$ for any i if $\delta_* \geq 0$. Thus by Theorem 4.2(1), (2), possibly passing to a subsequence, the limit pairs $(\operatorname{Spec}\mathbb{K}[[P,Q]], C_{t,\infty})$ (if $\delta_* \leq 0$) and $(\mathcal{A}_{\mathbb{K}}, B_{\mathrm{br}} + C_{Y,\infty})$ (if $\delta_* \geq 0$) are lc near the base, and the minimum of the log discrepancies of the quotient by $\tilde{H}_t$, resp. H, over the prime divisors with center the origin, resp. contained in $E_{\mathbb{K}}/H$, is L. The minimum along $E_{\mathbb{K}}$ is attained, as shown in **Step 2**.

Step 4. In this step we treat the case when the limits of **Step 3** are klt near the base.

Thus we assume that the limit on $\operatorname{Spec}\mathbb{K}[[P,Q]]$ is klt near 0 if $\delta_* \leq 0$, and that the limit on $\mathcal{A}_{\mathbb{K}}$ is klt near $E_{\mathbb{K}}$ if $\delta_* \geq 0$.

By Theorem 4.2(1) with $\theta = 1$, there exists $\theta_0 > 1$ such that $(\mathbb{A}^2, \theta_0 C^*_{t,i})$ is lc near 0 for any i if $\delta_* \le 0$, and $(Y, B_{\mathrm{br}} + \theta_0 \Delta^*_{Y,i})$ is lc near E for any i if $\delta_* \ge 0$. We fix $\theta \in (1, \theta_0]$ and one reference as follows. If $\delta_* < 0$, we take the plane and θ close to 1 with $\theta\tau^* + c - v < 0$. If $\delta_* > 0$, we take Y and θ close to 1 with $\theta\tau^* + c - v > 0$. If $\delta_* = 0$, then $\tau^* = v - c$ and

$$\theta\tau^* + c - v = (\theta - 1)(v - c),$$

and we take the plane if $v \le c$ and Y if $v > c$. In all cases, Claim A.37 with $\lambda = 0$ implies that $(W_{N_i}, \theta D^*_i)$ is lc near 0 for i large. As W_{N_i} is klt and $\theta > 1$, (W_{N_i}, D^*_i) is klt near 0. Thus W_{N_i}, G_i, and D^*_i satisfy Set-up 7.3 with Γ replaced by the DCC set $\Gamma \cup \Gamma^*$, with the κ_j and h^0_{ij} of **Step 1**, with $\tau = \tau^*$ and $\delta = \delta_*$, and with the reference pairs $(\mathbb{A}^2, C^*_{t,i})$ and $(Y, B^*_{Y,i})$. By Proposition A.35(2) for $\Gamma \cup \Gamma^*$ and θ, and by Claim A.36, for i large,

$$\alpha^*_i := \mathrm{mld}((W_{N_i} \ni 0, D^*_i)/G_i) \ge L.$$

Since $D_i \le D^*_i$, $\alpha_i \ge \alpha^*_i \ge L$, a contradiction.

Step 5. In this step we assume that $\delta_* \le 0$ and that $(\operatorname{Spec} \mathbb{K}[[P, Q]], C_{t,\infty})$ is not klt near 0, and we obtain a contradiction by Lemma 5.8 applied to a function U of $\mathbf{T}$-weight zero.

By Claim A.36 and **Step 3**, Definition-Proposition 5.7 in case (a) with $n = 2$ implies that $(\operatorname{Spec} \mathbb{K}[[P, Q]], C_{t,\infty})$ is lc near 0 and has exactly one floor F, which is the curve denoted D in Definition-Proposition 5.7. The curve F is smooth at 0 and $\tilde{H}_t$-invariant. Let

$$\rho_F := 1 - \mathrm{mult}_0((C_{t,\infty} - F)|_F),$$

and let o_F be the order of the image of $\tilde{H}_t$ in the automorphism group of $T_0 F$. By Definition-Proposition 5.7 and Lemma 5.6 with $e = o_F$,

$$L = \frac{\rho_F}{o_F}.$$

In particular $\rho_F > 0$. Since F is smooth at 0, one of P and Q, denoted z_0, restricts to a parameter of F at 0. As z_0 is $\tilde{H}_t$-semi-invariant, o_F is the order of the character χ_{z_0} of z_0.

If $z_0 = P$, let $U := U_1$ and $\ell := 2v/e_a$. If $z_0 = Q$, let $U := U_2$ and $\ell := 2v/e_b$. Then $U_t = \pm z_0^\ell$. For the class of (g, s_g, p_g) in $\tilde{H}_t$,

$$p_g^\ell = (g_x s_g^a)^{v/e_a} = g_x^{v/e_a} g_t^{a/e_a}, \qquad \left(\frac{g_z s_g^c}{p_g}\right)^{2v/e_b} = g_z^{2v/e_b} g_x^{-v/e_b} g_t^{b/e_b} = g_y^{v/e_b} g_t^{b/e_b},$$

using $s_g^v = g_t$, $p_g^2 = g_x s_g^a$, $2c - a = b$, and $g_z^2 = g_x g_y$. Thus the character of U under G_i, which is trivial on R_i, pulls back to $\chi^\ell_{z_0}$ under $\tilde{H}_t \to H_i$. The kernel of $\tilde{H}_t \to H_i$ consists of the classes of $(1, s, p)$ with $s^v = 1$ and $p^2 = s^a$, and on it χ_{z_0} takes exactly the ℓ-th roots of unity as values, because p^2 and $(s^c/p)^2 = s^b$ run through the (v/e_a)-th, resp. (v/e_b)-th, roots of unity with both signs of p. Hence ℓ divides o_F, and the order o_U of the character of U is

$$o_U = \frac{o_F}{\ell}.$$

Let $\lambda := \rho_F/\ell$. Then

$$\lambda \operatorname{div}(U_t) = \rho_F \operatorname{div}(z_0), \qquad \mathrm{mult}_0\left((C_{t,\infty} - F + \rho_F \operatorname{div}(z_0))|_F\right) = (1 - \rho_F) + \rho_F = 1.$$

By [Kaw07, Theorem], the pair

$$\left(\operatorname{Spec} \mathbb{K}[[P, Q]], C_{t,\infty} + \lambda \operatorname{div}(U_t)\right)$$

is lc near 0. By **Step 3** and Theorem 4.2(1), $(\mathbb{A}^2, C^*_{t,i} + \lambda \operatorname{div}(U_t))$ is lc near 0 for any i. Since $\delta_* \le 0$, Claim A.37 with $\theta = 1$ implies that $(W_{N_i}, D^*_i + \lambda \operatorname{div}(U))$ is lc near 0 for i large, hence so is $(W_{N_i}, D_i + \lambda \operatorname{div}(U))$. The function U is G_i-semi-invariant and vanishes at 0. By Lemma 5.8 with $Z = \{0\}$ and $h = o_U$,

$$\alpha_i \ge \frac{\lambda}{o_U} = \frac{\rho_F}{o_F} = L,$$

a contradiction.

Step 6. In this step we assume that $\delta_* \geq 0$ and that $(\mathcal{A}_{\mathbb{K}}, B_{\mathrm{br}} + C_{Y,\infty})$ is not klt near $E_{\mathbb{K}}$, and we show that there is exactly one floor and determine its position.

By Claim A.36, $\beta^*_{Y,i} = L$, and the value at E is at least $\beta^*_{Y,i}$, so $\frac{1-m^*_E}{e_E} \geq L > 0$ and $m^*_E < 1$. Since $\delta_* \geq 0$, $\tau^* \geq v - c$, and (A.159) implies

$$\iota \leq \frac{2(c - v(1-m^*_E))}{ab} < \frac{2c}{ab} = \frac{1}{a} + \frac{1}{b}. \tag{A.160}$$

Recall that $m^*_E = \sum_j c^*_j m_j$ and $\tau^* = \sum_j c^*_j \kappa_j$. We compute on the root chart covers $\mathbb{A}^2_{p,q}$ and $\mathbb{A}^2_{p',q'}$ over $\mathbb{K}$, completed along $(q=0)$ and $(q'=0)$. Let $\varpi_x := E \cap \Sigma_x$ and $\varpi_y := E \cap \Sigma_y$, the images of the origins. For a closed point $\varpi \in E_{\mathbb{K}}$, let $\tilde{\varpi}$ be a point over ϖ of one of the chart covers, and let n_ϖ be the order of the stabilizer of $\tilde{\varpi}$ in $\mu_b(-c,v)$, resp. $\mu_a(-c,v)$. At a point with $p \neq 0$, resp. $p' \neq 0$, the stabilizer of $\tilde{\varpi}$ consists of the ζ with $\zeta^c = 1$, so

$$n_\varpi = b \text{ if } \varpi = \varpi_x, \qquad n_\varpi = a \text{ if } \varpi = \varpi_y, \qquad n_\varpi = e_E \text{ otherwise.}$$

Let $\tilde{C}_\infty$ be the pullback of $B_{\mathrm{br}} + C_{Y,\infty}$ to the chart cover. By **Step 2** and Claim A.38,

$$\tilde{C}_\infty = \sum_j c^*_j \operatorname{div}(f_{\infty,j,\alpha}) = m^*_E(q=0) + \tilde{T}_\infty,$$

where $\tilde{T}_\infty$ is the pullback of T_∞. The coefficients c^*_j are real, so we argue for each summand. Let Θ be one of the $\mathbb{Q}$-divisors $D_{\infty,j} - \frac{m_j}{e_E}E_{\mathbb{K}}$, or $D_Y(U_k)_{\mathbb{K}} - \frac{m(U_k)}{e_E}E_{\mathbb{K}}$ for $k = 1,\dots,4$, and let $\tilde{\Theta}$ be its pullback to the chart cover. On the first chart cover $\tilde{\Theta}$ is the divisor of $f_{\infty,j,1}q^{-m_j}$, resp. $(U_k)_y q^{-m(U_k)}$, which is semi-invariant under $\mu_b(-c,v)$ by Claim A.38, resp. because $\mu_b(-c,v)$ acts diagonally on p,q and $(U_k)_y$ is a monomial in p,q up to sign, and similarly on the second chart cover. Thus Θ is $\mathbb{Q}$-Cartier. Let φ be a local equation at ϖ of a positive integer multiple $k_\Theta\Theta$ that is Cartier. The group $\mu_b(-c,v)$ acts on $(q=0) \cong \mathbb{A}^1_p$ by $p \to \zeta^{-c}p$, with kernel of order $\gcd(b,c) = e_E$, so the morphism from $(q=0)$ to $E_{\mathbb{K}}$ near $\tilde{\varpi}$ is the quotient by a cyclic group of order b/e_E, and its ramification index at $\tilde{\varpi}$ is n_ϖ/e_E. Thus the order at $\tilde{\varpi}$ of the restriction to $(q=0)$ of the pullback of φ is $\frac{n_\varpi}{e_E}$ times the order at ϖ of $\varphi|_{E_{\mathbb{K}}}$, that is,

$$\frac{(\Theta \cdot E_{\mathbb{K}})_\varpi}{e_E} = \frac{(\tilde{\Theta} \cdot (q=0))_{\tilde{\varpi}}}{n_\varpi}$$

for the local intersection numbers. Only $\varphi|_{E_{\mathbb{K}}}$ enters, so this holds on the formal completion along $E_{\mathbb{K}}$. Both sides are $\mathbb{R}$-linear in Θ, and

$$T_\infty = \sum_j c^*_j \Big(D_{\infty,j} - \frac{m_j}{e_E} E_{\mathbb{K}} \Big).$$

Thus

$$\frac{(T_\infty \cdot E_{\mathbb{K}})_\varpi}{e_E} = \frac{(\tilde{T}_\infty \cdot (q=0))_{\tilde{\varpi}}}{n_\varpi}, \tag{A.161}$$

with q replaced by q' on the second chart cover. For any real number λ and any k, (A.161) also holds with T_∞ replaced by

$$T_\infty + \lambda \Big(D_Y(U_k)_{\mathbb{K}} - \frac{m(U_k)}{e_E} E_{\mathbb{K}} \Big)$$

and $\tilde{T}_\infty$ replaced by its pullback. The sum of (A.161) over ϖ is ι.

By Claim A.36 and **Step 3**, Definition-Proposition 5.7 in case (b) implies that any lc center of $(\mathcal{A}_{\mathbb{K}}, B_{\mathrm{br}} + C_{Y,\infty})$ intersecting $E_{\mathbb{K}}$ is a floor, whose preimage in the chart covers is smooth at the points over $E_{\mathbb{K}}$, and that no two preimages of floors intersect at a point over $E_{\mathbb{K}}$. The curve $E_{\mathbb{K}}$ is not a floor, because the coefficient of $E_{\mathbb{K}}$ in $B_{\mathrm{br}} + C_{Y,\infty}$ is $1 - \frac{1-m^*_E}{e_E} < 1$. As the limit is lc and not klt near $E_{\mathbb{K}}$, there exists a floor. Let F be a floor, which is a curve denoted D in

Definition-Proposition 5.7, and let $\varpi \in F \cap E_{\mathbb{K}}$. The preimage $\tilde{F}$ of F at $\tilde{\varpi}$ is a smooth curve with coefficient one in $\tilde{C}_\infty$ [Liu26, Lemma 2.6(4)], and $\tilde{F} \neq (q=0)$. Thus $\tilde{T}_\infty \geq \tilde{F}$, and with $\sigma_\varpi := (\tilde{F} \cdot (q=0))_{\tilde{\varpi}} \geq 1$,

$$\frac{(T_\infty \cdot E_{\mathbb{K}})_\varpi}{e_E} \geq \frac{\sigma_\varpi}{n_\varpi}. \tag{A.162}$$

Since e_E divides a and b, two distinct points of $E_{\mathbb{K}}$ on floors would contribute at least $\frac{1}{a}+\frac{1}{b}$ to ι by (A.162), which contradicts (A.160). Thus there exists exactly one floor F, and $F \cap E_{\mathbb{K}}$ is one closed point ϖ_F. As H preserves $E_{\mathbb{K}}$ and permutes the floors, F and ϖ_F are H-invariant. Let $\sigma := \sigma_{\varpi_F}$.

Let $\tilde{H}_F \subset \tilde{H}_y$ or $\tilde{H}_x$ be the stabilizer of $\tilde{\varpi}_F$. It maps onto H because ϖ_F is H-fixed, and the quotient of $(\mathbb{A}^2, \tilde{C}_\infty)$ by $\tilde{H}_F$ near $\tilde{\varpi}_F$ is the germ of $(\mathcal{A}_{\mathbb{K}}, B_{\mathrm{br}} + C_{Y,\infty})/H$ at ϖ_F [Liu26, Lemma 2.6(5)] (the completion of a quotient X/G at the image of a closed point x is $\hat{\mathcal{O}}_{X,x}^{\mathrm{Stab}_G(x)}$. When $\tilde{\varpi}_F$ is the origin, $\tilde{H}_F = \tilde{H}_y$). Let

$$\rho_F := 1 - \mathrm{mult}_{\tilde{\varpi}_F}((\tilde{C}_\infty - \tilde{F})|_{\tilde{F}}),$$

and let o_F be the order of the image of $\tilde{H}_F$ in the automorphism group of $T_{\tilde{\varpi}_F}\tilde{F}$. By Lemma 5.6 with $e = o_F$, the minimal log discrepancy of the quotient at ϖ_F is ρ_F/o_F. Thus $\rho_F/o_F \geq L > 0$. Since $\tilde{C}_\infty - \tilde{F} \geq m_E^*(q=0)$,

$$m_E^* \sigma \leq 1 - \rho_F < 1. \tag{A.163}$$

We distinguish the position of ϖ_F.

Case 1: $\varpi_F = \varpi_x$ and $\sigma = 1$. The floor F is transverse to $E_{\mathbb{K}}$ at ϖ_x.

Case 2: $\varpi_F \notin \{\varpi_x, \varpi_y\}$. Then $n_{\varpi_F} = e_E$. If $\sigma \geq 2$, then $\iota \geq \frac{2}{e_E} \geq \frac{1}{a}+\frac{1}{b}$, which contradicts (A.160). Thus $\sigma = 1$ and $\frac{1}{e_E} < \frac{1}{a}+\frac{1}{b}$. Writing $a = e_E a'$ and $b = e_E b'$, we obtain $1 < \frac{1}{a'}+\frac{1}{b'}$, so $\min(a', b') = 1$, that is, $\min(a,b) = e_E$.

Case 3: $\varpi_F = \varpi_x$ and $\sigma \geq 2$. Then $\tilde{F} = (q = f(p))$ for a power series f with $\mathrm{mult}_0 f = \sigma$. Since $\tilde{H}_F = \tilde{H}_y$ preserves $\tilde{F}$, comparing the leading coefficients on both sides of $g^*(q - f(p)) \in (q - f(p))$ for $g \in \tilde{H}_y$ implies

$$\chi_q = \chi_p^\sigma \quad \text{on } \tilde{H}_y, \tag{A.164}$$

where χ_p and χ_q are the characters of p and q. On $\mu_b(-c, v)$, (A.164) reads $\zeta^v = \zeta^{-c\sigma}$, so $v + c\sigma = \omega b$ for a positive integer ω. By (A.162) at ϖ_x and (A.159),

$$\frac{\sigma}{b} \leq \frac{2(vm_E^* - \tau^*)}{ab}, \quad \text{that is,} \quad \delta_* \leq vm_E^* + c - v - \frac{a\sigma}{2}.$$

Since $\delta_* \geq 0$, $m_E^* < \frac{1}{\sigma}$ by (A.163), $v + c\sigma = \omega b$, and $a = 2c - b$,

$$0 \leq 2\sigma\delta_* < 2v + 2c\sigma - 2v\sigma - a\sigma^2 = b\big(\sigma^2 - 2\omega(\sigma - 1)\big). \tag{A.165}$$

Thus $2\omega < \frac{\sigma^2}{\sigma-1} = \sigma + 1 + \frac{1}{\sigma-1}$. On the other hand, $2\omega b = 2v + 2c\sigma > (a+b)\sigma > b\sigma$, so $2\omega > \sigma$. If $\sigma = 2$, then $2 < 2\omega < 4$, which is impossible. If $\sigma \geq 3$, then $\sigma < 2\omega < \sigma + \frac{3}{2}$, so $2\omega = \sigma + 1$. Thus σ is odd, $\sigma \geq 3$, and $v + c\sigma = \frac{(\sigma+1)b}{2}$. With $2c = a + b$, we obtain

$$b = a\sigma + 2v, \qquad c = \frac{a(\sigma+1)}{2} + v.$$

Then $c \equiv v \pmod{a}$, so $\gcd(a,c) = \gcd(a,v) = \gcd(a,c,v) = 1$, that is, $e_E = e_a = 1$.

Case 4: $\varpi_F = \varpi_y$. Exchanging (a, p, q, x, Σ_x) with (b, p', q', y, Σ_y) in Cases 1 and 3, either $\sigma = 1$, or σ is odd, $\sigma \geq 3$, $a = b\sigma + 2v$, and $e_E = e_b = 1$.

Thus the floor F is in one of the following positions:

- transverse to $E_{\mathbb{K}}$ at ϖ_x or ϖ_y;

- transverse to $E_{\mathbb{K}}$ at a closed point other than ϖ_x, ϖ_y, in which case $\min(a,b) = e_E$;
- tangent to $E_{\mathbb{K}}$ at ϖ_x with σ odd, $\sigma \geq 3$, and $b = a\sigma + 2v$, resp. at ϖ_y with $a = b\sigma + 2v$.

Step 7. In this step we close each case of **Step 6** by Lemma 5.8, applied with $Z = E$ and with $Z = \{0\}$ to a function U of **T**-weight zero.

In each case we fix $U \in \{U_1, U_2, U_3, U_4\}$, a positive integer ℓ, and a parameter $z_0 \in \{p, q\}$ of $\tilde{F}$ at $\tilde{\varpi}_F$ (with p, q replaced by p', q' on the second chart cover). Let $\lambda := \rho_F/\ell$. We choose them such that the following hold:

(i) o_F is the order of the character of z_0 on $\tilde{H}_F$, ℓ divides o_F, and the order o_U of the character of U under H is o_F/ℓ;
(ii) the pair $(\mathbb{A}^2, \tilde{C}_\infty + \lambda \operatorname{div}(U_y))$, resp. with U_x, is lc near $\tilde{\varpi}_F$;
(iii) the pair
$$(\mathcal{A}_{\mathbb{K}}, B_{\mathrm{br}} + C_{Y,\infty} + \lambda D_Y(U))$$
is lc near any closed point of $E_{\mathbb{K}}$ other than ϖ_F.

The first part of (i) holds whenever z_0 restricts to a parameter of $\tilde{F}$ at $\tilde{\varpi}_F$, since z_0 is semi-invariant under $\tilde{H}_F$, as in **Step 5**. For (ii), it suffices that
$$\operatorname{mult}_{\tilde{\varpi}_F}(\lambda \operatorname{div}(U_y)|_{\tilde{F}}) = \rho_F,$$
since then
$$\operatorname{mult}_{\tilde{\varpi}_F}((\tilde{C}_\infty - \tilde{F} + \lambda \operatorname{div}(U_y))|_{\tilde{F}}) = 1,$$
and we use [Kaw07, Theorem] along $\tilde{F}$. For (iii), let m_λ be the coefficient of $(q = 0)$ in $\tilde{C}_\infty + \lambda \operatorname{div}(U_y)$, and let ι_{rest} be the sum over the closed points $\varpi \neq \varpi_F$ of $E_{\mathbb{K}}$ of
$$\frac{\left(\left(T_\infty + \lambda(D_Y(U) - \frac{m(U)}{e_E}E)\right) \cdot E_{\mathbb{K}}\right)_{\varpi}}{e_E}.$$
It suffices that $m_\lambda \leq 1$ and that $n_\varpi \iota_{\mathrm{rest}} \leq 1$ for any $\varpi \neq \varpi_F$, because then by (A.161), the pair on the chart cover obtained by setting the coefficient of $(q = 0)$ equal to 1 has restriction to $(q = 0)$ of multiplicity at most one at $\tilde{\varpi}$, so it is lc near $\tilde{\varpi}$ by [Kaw07, Theorem], and lowering the coefficient of $(q = 0)$ back to m_λ preserves the lc property. By (A.157), for U of weight zero,
$$\frac{1}{e_E}\left(\lambda\left(D_Y(U) - \frac{m(U)}{e_E}E\right) \cdot E\right) = \frac{2v\lambda m(U)}{ab},$$
so the sum over all closed points ϖ of $E_{\mathbb{K}}$ of the local terms of ι_{rest}, together with the term at ϖ_F, is $\iota + \frac{2v\lambda m(U)}{ab}$.

Case 1. Let $U := U_2 = y^{v/e_b} t^{b/e_b}$, $\ell := b/e_b$, and $z_0 := q$. As $\tilde{F}$ is transverse to $(q = 0)$, z_0 restricts to a parameter of $\tilde{F}$ at $\tilde{\varpi}_F$. Then $U_y = q^\ell$ and $U_x = (-p'^2)^{v/e_b} q'^\ell$. For the class of (g, η_g) in $\tilde{H}_y$, $\chi_q = g_t \eta_g^v$, so $\chi_q^\ell = g_t^{b/e_b} g_y^{v/e_b}$ is the character of U. On $\mu_b(-c, v)$, χ_q is $\zeta \to \zeta^v$, of order $b/e_b = \ell$. As $\tilde{H}_F = \tilde{H}_y$ maps onto H, (i) holds. Since $\tilde{F}$ is transverse to $(q = 0)$,
$$\operatorname{mult}_{\tilde{\varpi}_F}(\lambda \operatorname{div}(q^\ell)|_{\tilde{F}}) = \rho_F,$$
so (ii) holds. By (A.163) with $\sigma = 1$, $m_\lambda = m_E^* + \rho_F \leq 1$. The strict part of $\lambda D_Y(U)$ is supported on Σ_y, and $\lambda m(U) = \rho_F$. Removing the contribution at ϖ_x, which is at least $\frac{1}{b}$ by (A.162), and using $\tau^* \geq v - c$, we obtain

$$\iota_{\mathrm{rest}} \leq \frac{2(vm_E^* - \tau^*)}{ab} + \frac{2v\rho_F}{ab} - \frac{1}{b} = \frac{2(v(m_E^* + \rho_F) - \tau^*)}{ab} - \frac{1}{b} \leq \frac{2c}{ab} - \frac{1}{b} = \frac{1}{a}. \tag{A.166}$$

For $\varpi \neq \varpi_x$, $n_\varpi \in \{a, e_E\}$ and $e_E \leq a$, so $n_\varpi \iota_{\mathrm{rest}} \leq 1$, and (iii) holds. The transverse case at ϖ_y in Case 4 is treated by the argument for Case 1 with $U := U_1$, $\ell := a/e_a$, and $z_0 := q'$.

Case 2. Assume that $a = e_E \leq b$. Then a divides c, so $\gcd(a, v) = \gcd(a, c, v) = 1$. Let $U := U_3 = x^v t^a$, $\ell := a$, and $z_0 := q$, and choose $\tilde{\varpi}_F = (p_0, 0)$ with $p_0 \neq 0$ on $\mathbb{A}^2_{p,q}$. As $\tilde{F}$ is transverse to $(q = 0)$, z_0 restricts to a parameter of $\tilde{F}$ at $\tilde{\varpi}_F$. Then $U_y = (-p^2)^v q^a$ is a unit

times q^a near $\tilde{\varpi}_F$, and $U_x = q'^a$. The kernel of $\tilde{H}_F \to H$ is the stabilizer of $\tilde{\varpi}_F$ in $\mu_b(-c, v)$, of order $e_E = a$, on which χ_q is $\zeta \to \zeta^v$, of order a. Thus a divides o_F. A direct calculation implies that the character of U pulls back to $\chi_p^{2v}\chi_q^a$ on $\tilde{H}_y$, and χ_p is trivial on $\tilde{H}_F$ because $\tilde{H}_F$ fixes $p_0 \neq 0$. As $\tilde{H}_F$ maps onto H, (i) holds. Since $\tilde{F}$ is transverse to $(q = 0)$,

$$\mathrm{mult}_{\tilde{\varpi}_F}(\lambda \operatorname{div}(U_y)|_{\tilde{F}}) = \lambda a = \rho_F,$$

so (ii) holds, and $m_\lambda = m_E^* + \rho_F \leq 1$ by (A.163). The strict part of $\lambda D_Y(U)$ is supported on Σ_x, and $\lambda m(U) = \rho_F$. Removing the contribution at ϖ_F, which is at least $\frac{1}{e_E} = \frac{1}{a}$, we obtain as in (A.166)

$$\iota_{\mathrm{rest}} \leq \frac{2(v(m_E^* + \rho_F) - \tau^*)}{ab} - \frac{1}{a} \leq \frac{2c}{ab} - \frac{1}{a} = \frac{1}{b}.$$

Since $n_\varpi \in \{a, b, e_E\}$ and $a = e_E \leq b$, $n_\varpi \iota_{\mathrm{rest}} \leq 1$, and (iii) holds. If $b = e_E \leq a$, we use $U := U_4 = y^v t^b$ and $\ell := b$ with the roles of the charts exchanged.

Case 3. Here σ is odd, $\sigma \geq 3$, $b = a\sigma + 2v$, and $e_E = 1$. Let $U := U_3 = x^v t^a$, $\ell := b$, and $z_0 := p$. As $\tilde{F} = (q = f(p))$, z_0 restricts to a parameter of $\tilde{F}$ at $\tilde{\varpi}_F$. Then $U_y = (-p^2)^v q^a$ and $U_x = q'^a$. The character of U pulls back to $\chi_p^{2v}\chi_q^a$, which is $\chi_p^{2v+a\sigma} = \chi_p^b$ on $\tilde{H}_y$ by (A.164). On $\mu_b(-c, v)$, χ_p is $\zeta \to \zeta^{-c}$, of order $b/\gcd(b, c) = b/e_E = b$. As $\tilde{H}_F = \tilde{H}_y$ maps onto H, (i) holds. On $\tilde{F} = (q = f(p))$, $U_y|_{\tilde{F}} = (-p^2)^v f(p)^a$ is a unit times $p^{2v+a\sigma} = p^b$, so

$$\mathrm{mult}_{\tilde{\varpi}_F}(\lambda \operatorname{div}(U_y)|_{\tilde{F}}) = \lambda b = \rho_F,$$

and (ii) holds. By (A.163), $\rho_F \leq 1 - m_E^*\sigma$, so

$$m_\lambda = m_E^* + \frac{a\rho_F}{b} \leq \frac{bm_E^* + a - am_E^*\sigma}{b} = \frac{a + 2vm_E^*}{b} < \frac{a\sigma + 2v}{b} = 1.$$

The strict part of $\lambda D_Y(U)$ is supported on Σ_x, which intersects E only at $\varpi_F = \varpi_x$. Removing the contribution at ϖ_x, which is at least $\frac{\sigma}{b}$, and using $\tau^* \geq v - c$, $2c = a + b$, and $b = a\sigma + 2v$,

$$\iota_{\mathrm{rest}} \leq \frac{2(vm_E^* - v + c)}{ab} - \frac{\sigma}{b} = \frac{a + 2vm_E^*}{ab} < \frac{a + 2v}{ab} \leq \frac{1}{a}.$$

For $\varpi \neq \varpi_x$, $n_\varpi \in \{a, e_E\} = \{a, 1\}$, so $n_\varpi \iota_{\mathrm{rest}} < 1$, and (iii) holds. The tangent case at ϖ_y in Case 4 is treated by the argument for Case 3 with $U := U_4$, $\ell := a$, and $z_0 := p'$.

In each case, (ii), (iii), and [Liu26, Lemma 2.6] imply that

$$(\mathcal{A}_{\mathbb{K}}, B_{\mathrm{br}} + C_{Y,\infty} + \lambda D_Y(U))$$

is lc near $E_{\mathbb{K}}$. By **Step 3** and Theorem 4.2(1), $(Y, B_{Y,i}^* + \lambda D_Y(U))$ is lc near E for any i. The chart functions U_y and U_x are regular, invariant under the groups of the charts, and equal on the overlap by (A.158) with $\mathfrak{n} = 1$, so U is a regular H-semi-invariant function on a neighborhood of E which vanishes on E, with divisor $D_Y(U)$ and character of order o_U. By Lemma 5.8 with $Z = E$, $\beta_{Y,i}^* \geq \lambda/o_U = \rho_F/o_F$. Together with $L \leq \rho_F/o_F$ from **Step 6** and $\beta_{Y,i}^* = L$, we obtain

$$L = \frac{\rho_F}{o_F} = \frac{\lambda}{o_U}.$$

Since $\delta_* \geq 0$, Claim A.37 with $\theta = 1$ implies that $(W_{N_i}, D_i^* + \lambda \operatorname{div}(U))$ is lc near 0 for i large, hence so is $(W_{N_i}, D_i + \lambda \operatorname{div}(U))$. The function U is G_i-semi-invariant, vanishes at 0, and has character of order o_U, since the character of U is trivial on R_i. By Lemma 5.8 with $Z = \{0\}$ and $h = o_U$,

$$\alpha_i \geq \frac{\lambda}{o_U} = L,$$

a contradiction.

Steps 4, 5, and 7 cover all cases: either some limit of **Step 3** is not klt near the base, and **Step 5** or **Steps 6 and 7** apply to it, or all of them are klt, and **Step 4** is used. Thus no sequence as in **Step 1** exists. □

A.13. **Exterior circles.** In this subsection, we prove Theorem 7.22. We first construct a plane pair and a surface pair attached to the circle, and then bound α from above and from below by their minimal log discrepancies.

Construction-Proposition A.39. Notation as in Set-up 7.3 with $w = (a, -b, -c, v)$, where $a, c, v > 0$, $b = a + 2c$, and $\gcd(a, c, v) = 1$, let $\delta := \tau + c - v$ be the adjusted mass of Set-up 5.1, let $H := G/R$, and for $\lambda \in \mathbb{C}^*$ let $\mathbf{T}(\lambda) \in \mathbf{T}$ be the element with entries λ^a, λ^{-b}, λ^{-c}, λ^v. Let $e_E := \gcd(a, v)$, $e_p := \gcd(b, v)$, and $e_q := \gcd(a, c)$. Let

$$\tilde{H}_y := \{(g, \eta_g) : g \in G,\ \eta_g^b = g_y\}/\{(\mathbf{T}(\lambda), \lambda^{-1}) : \lambda^r = 1\},$$

acting on $\mathbb{A}^2_{p,q}$ by $(p, q) \to (g_z\eta_g^{-c}p, g_t\eta_g^v q)$, and let $C_y := \sum_j c_j \operatorname{div}(h_j^0(-p^2, 1, p, q))$ on $\mathbb{A}^2_{p,q}$. Let $S := \mathbb{A}^2_{p,q}/\mu_b(-c, v)$, let $\pi : Y \to S$ be the toric partial resolution inserting the primitive ray $\frac{1}{be_E}(a, 2v)$ in the lattice $\mathbb{Z}^2 + \frac{1}{b}(-c, v)\mathbb{Z}$, with exceptional curve E, and let C_p and C_q be the strict transforms of the images of $(p = 0)$ and $(q = 0)$. Near E the surface Y is covered by the charts $S_0/\mu_v(-a, b, c)$, where $S_0 := (\bar{x}\bar{y} + \bar{z}^2 = 0) \subset \mathbb{A}^3$, and $\mathbb{A}^2_{p_x,q_x}/\mu_a(c, -v)$, induced by $\bar{x} = x/s^a$, $\bar{y} = ys^b$, $\bar{z} = zs^c$ with $s^v = t$, and by $p_x = z\eta^c$, $q_x = t/\eta^v$ with $\eta^a = x$, respectively. On the overlap, with $q_x = \xi^v$, the transition is $\bar{x} = \xi^{-a}$, $\bar{y} = -p_x^2\xi^b$, $\bar{z} = p_x\xi^c$. Let $u_t := s^r$ and $u_x := \eta^r$, let

$$X_d := (\bar{x}\bar{y} + \bar{z}^2 + u_t^d = 0) \subset \mathbb{A}^4_{\bar{x},\bar{y},\bar{z},u_t},$$

so that $S_0 = X_d \cap (u_t = 0)$, and let $\mathbb{A}^3_{u_x,p_x,q_x}$ be the x-root chart of W_N, on which $y\eta^b = -p_x^2 - u_x^d q_x^N$. Let $\tilde{H}_t^{(2)}$ be the group of the triples (g, s_g, ς_g) with $g \in G$, $s_g^v = g_t$, and $\varsigma_g^2 = g_x s_g^{-a}$, modulo $\{(\mathbf{T}(\lambda), \lambda, 1) : \lambda^r = 1\}$, acting on $\mathbb{A}^3_{P,Q,u_t}$ by

$$(P, Q, u_t) \to (\varsigma_g P, g_z s_g^c \varsigma_g^{-1} Q, s_g^r u_t),$$

where $\mathbb{A}^3_{P,Q,u_t} \dashrightarrow X_d$ is given by $\bar{x} = P^2$, $\bar{z} = PQ$, and $\bar{y} = -Q^2 - u_t^d/P^2$, so that $\mathbb{A}^2_{P,Q} = (u_t = 0)$ is the double cover of S_0 given by $\bar{x} = P^2$, $\bar{y} = -Q^2$, $\bar{z} = PQ$. Let $\tilde{H}_t$ be the group of the pairs (g, s_g) with $g \in G$ and $s_g^v = g_t$, modulo $\{(\mathbf{T}(\lambda), \lambda) : \lambda^r = 1\}$, acting on $\mathbb{A}^4_{\bar{x},\bar{y},\bar{z},u_t}$ by

$$(\bar{x}, \bar{y}, \bar{z}, u_t) \to (g_x s_g^{-a}\bar{x}, g_y s_g^b \bar{y}, g_z s_g^c \bar{z}, s_g^r u_t),$$

and let $\tilde{H}_x$ be the group of the pairs (g, ϵ) with $g \in G$ and $\epsilon^a = g_x$, modulo $\{(\mathbf{T}(\lambda), \lambda) : \lambda^r = 1\}$, acting on $\mathbb{A}^3_{u_x,p_x,q_x}$ by

$$(u_x, p_x, q_x) \to (\epsilon^r u_x, g_z\epsilon^c p_x, g_t\epsilon^{-v} q_x).$$

Let $\mathcal{P}_{\mathbf{T}}$ be the set of non-zero polynomials in x, y, z, t without monomials divisible by xy and homogeneous for $\mathbf{T}$, and for $f \in \mathcal{P}_{\mathbf{T}}$ let $D_Y(f) \geq 0$ be the $\mathbb{Q}$-Cartier divisor on Y whose pullbacks to S_0 and $\mathbb{A}^2_{p_x,q_x}$ are $\operatorname{div}(f(\bar{x}, \bar{y}, \bar{z}, 1))$ and $\operatorname{div}(f(1, -p_x^2, p_x, q_x))$. Let $D_{Y,j} := D_Y(h_j^0)$, $\Delta_Y := \sum_j c_j D_{Y,j}$,

$$B_{\mathrm{br}} := \Big(1 - \frac{1}{e_E}\Big)E + \Big(1 - \frac{1}{e_p}\Big)C_p + \Big(1 - \frac{1}{e_q}\Big)C_q, \qquad B_Y := B_{\mathrm{br}} + \Delta_Y,$$

let n_j be the least value of $2i_2 + i_3$ over the monomials $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of h_j^0, and let $m_E := \sum_j c_j n_j$. Then the following hold.

(1) The following hold.
  (a) The following hold.
    (i) The groups $\tilde{H}_y$, $\tilde{H}_t^{(2)}$, $\tilde{H}_t$, and $\tilde{H}_x$ are diagonal of orders $b[G : R]$, $2v[G : R]$, $v[G : R]$, and $a[G : R]$.
    (ii) The group $\tilde{H}_y$ contains $\mu_b(-c, v)$ and acts faithfully on $\mathbb{A}^2_{p,q}$.
    (iii) The group $\tilde{H}_t^{(2)}$ acts faithfully on $\mathbb{A}^2_{P,Q}$, the class of $(1, 1, -1)$ acts as the deck involution of $\mathbb{A}^2_{P,Q} \to S_0$, and $(g, s_g, \varsigma_g) \to (g, s_g)$ induces an isomorphism from

the quotient of $\tilde{H}_t^{(2)}$ by the subgroup generated by the class of $(1,1,-1)$ onto $\tilde{H}_t$.

(iv) The group $\tilde{H}_t$ preserves X_d and acts faithfully on S_0 and on X_d, and $\tilde{H}_x$ contains $\mu_a(c,-v)$ and acts faithfully on $\mathbb{A}^2_{p_x,q_x}$ and on $\mathbb{A}^3_{u_x,p_x,q_x}$.

(v) The G-invariant polynomials in x,y,z,t induce birational maps $\mathbb{A}^3_{P,Q,u_t}/\tilde{H}_t^{(2)} \dashrightarrow W_N/G$, $X_d/\tilde{H}_t \dashrightarrow W_N/G$, and $\mathbb{A}^3_{u_x,p_x,q_x}/\tilde{H}_x \dashrightarrow W_N/G$.

(b) The following hold.

(i) The group $\tilde{H}_y$ induces an action of H on Y which preserves E, C_p, C_q, and each $D_{Y,j}$.

(ii) The maps $(g,s_g,\varsigma_g)\to gR$ and $(g,\epsilon)\to gR$ are surjective homomorphisms onto H whose kernels are the group K_t of the automorphisms $(P,Q)\to(\varsigma P,\zeta^c\varsigma^{-1}Q)$ with $\zeta^v=1$ and $\varsigma^2=\zeta^{-a}$, of order $2v$, and the group $K_x:=\mu_a(c,-v)$, and the actions of $\tilde{H}_t^{(2)}$ and $\tilde{H}_x$ induce the action of H on the charts $S_0/\mu_v(-a,b,c)=\mathbb{A}^2_{P,Q}/K_t$ and $\mathbb{A}^2_{p_x,q_x}/K_x$ of Y.

(c) If $f\in\mathcal{P}_{\mathbf{T}}$ is G-semi-invariant of $\mathbf{T}$-weight zero, then the character of f is trivial on R, and $f(-p^2,1,p,q)$, $f(P^2,-Q^2,PQ,1)$, and $f(1,-p_x^2,p_x,q_x)$ are semi-invariant under $\tilde{H}_y$, $\tilde{H}_t^{(2)}$, and $\tilde{H}_x$, with the pullbacks of the character of f under the maps $(g,\eta_g)\to gR$, $(g,s_g,\varsigma_g)\to gR$, and $(g,\epsilon)\to gR$.

(2) The following hold.

(a) The intersection numbers on Y are $(E^2)=-\frac{be_E^2}{2av}$, $(C_p\cdot E)=\frac{e_Ee_p}{2v}$, $(C_q\cdot E)=\frac{e_Ee_q}{a}$, and $(D_Y(f)\cdot E)=\frac{e_E\kappa(f)}{av}$ for $f\in\mathcal{P}_{\mathbf{T}}$.

(b) The pullbacks of E to S_0 and to $\mathbb{A}^2_{p_x,q_x}$ are $e_E(\bar{y}=\bar{z}=0)$ and $e_E\operatorname{div}(p_x)$, and $\operatorname{div}(\bar{y})=2(\bar{y}=\bar{z}=0)$ on S_0 near the generic point of $(\bar{y}=\bar{z}=0)$.

(c) For a finite family $(c'_j,f_j)_j$ of positive real numbers c'_j and elements $f_j\in\mathcal{P}_{\mathbf{T}}$, the chart morphisms $S_0\to Y$ and $\mathbb{A}^2_{p_x,q_x}\to Y$ pull $K_Y+B_{\mathrm{br}}+\sum_j c'_jD_Y(f_j)$ back to

$$K_{S_0}+\sum_j c'_j\operatorname{div}(f_j(\bar{x},\bar{y},\bar{z},1)) \quad\text{and}\quad K_{\mathbb{A}^2}+\sum_j c'_j\operatorname{div}(f_j(1,-p_x^2,p_x,q_x)).$$

(d) The coefficient of E in B_Y is $1-\frac{1-m_E}{e_E}$, and

$$K_Y+B_Y=\pi^*(K_S+B_S)-\frac{2\delta}{be_E}E,$$

where B_S is the $\mathbb{R}$-divisor with $K_{\mathbb{A}^2}+C_y=\varphi^*(K_S+B_S)$ for the quotient morphism $\varphi:\mathbb{A}^2_{p,q}\to S$.

(3) For a finite family $(c'_j,f_j)_j$ of positive real numbers c'_j and elements $f_j\in\mathcal{P}_{\mathbf{T}}$, with $\delta':=\sum_j c'_j\kappa(f_j)+c-v$: if $\delta'\le 0$ and $(Y,B_{\mathrm{br}}+\sum_j c'_jD_Y(f_j))$ is lc near E, or if $\delta'\ge 0$ and $(\mathbb{A}^2,\sum_j c'_j\operatorname{div}(f_j(-p^2,1,p,q)))$ is lc near 0, then $(W_N,\sum_j c'_j\operatorname{div}(f_j))$ is lc near 0 for every $N\ge 3$, where in the second case $Nv>a$.

If $c=v$ and $\gcd(a,v)=1$, then $e_E=e_p=e_q=1$, and Y, E, C_p, C_q, the charts $S_0/\mu_v(-a,b,c)$ and $\mathbb{A}^2_{p_x,q_x}/\mu_a(c,-v)$, and $D_Y(f)$ are those of Construction-Proposition A.28, with $\bar{x},\bar{y},\bar{z}$ in place of x_1,y_1,z_1 and $D_Y(f)$ in place of D_f.

Proof of Construction-Proposition A.39. The proof is long, so we divide it into several steps. Here and below, only the non-units h_j with $c_j>0$ enter D near 0, and the index j runs only over such h_j. The weight a is the x-weight of the circle, not the number A of Set-up 7.3.

Step 1. In this step we prove the statements of (1) on $\tilde{H}_y$. Since w is primitive, $\lambda\to\mathbf{T}(\lambda)$ is injective, so $R=\{\mathbf{T}(\lambda):\lambda^r=1\}$. For $\lambda^r=1$ we have $(\lambda^{-1})^b=\mathbf{T}(\lambda)_y$, so $\{(\mathbf{T}(\lambda),\lambda^{-1}):\lambda^r=1\}$ is a subgroup of order r of the group of pairs (g,η_g), which has order $b|G|$. Thus $|\tilde{H}_y|=b|G|/r=b[G:R]$. The element $(\mathbf{T}(\lambda),\lambda^{-1})$ multiplies p by $\lambda^{-c}\lambda^c=1$ and q by $\lambda^v\lambda^{-v}=1$, so the action

on $\mathbb{A}^2_{p,q}$ is well defined. The pairs $(1,\eta)$ with $\eta^b=1$ act by $(\eta^{-c}p,\eta^v q)$, and $(1,\eta)$ belongs to the subgroup only if $\eta=1$. Hence $\mu_b(-c,v)\subset\tilde{H}_y$.

Assume that (g,η) acts trivially on $\mathbb{A}^2_{p,q}$. Then $g_z=\eta^c$ and $g_t=\eta^{-v}$. Since g preserves $xy+z^2+t^N$ up to a character, $g_xg_y=g_z^2$, so $g_x=\eta^{2c-b}=\eta^{-a}$, and $g=\mathbf{T}(\eta^{-1})$. Thus $g\in G\cap\mathbf{T}=R$, so $\eta^r=1$ and $(g,\eta)=(\mathbf{T}(\lambda),\lambda^{-1})$ with $\lambda=\eta^{-1}$. Hence $\tilde{H}_y$ acts faithfully.

The map $(g,\eta_g)\to gR$ is a surjective homomorphism $\tilde{H}_y\to H$ whose kernel consists of the classes of $(\mathbf{T}(\lambda),\eta)$, and

$$(\mathbf{T}(\lambda),\eta)=(\mathbf{T}(\lambda),\lambda^{-1})\cdot(1,\lambda\eta).$$

Hence the kernel is $\mu_b(-c,v)$ and $\tilde{H}_y/\mu_b(-c,v)\cong H$. The diagonal action of H on the toric surface S commutes with the torus, hence lifts to the toric partial resolution Y and preserves the torus-invariant curves E, C_p, and C_q. For a monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of $\mathbf{T}$-weight κ, using $g_z^2=g_xg_y$ and $\eta_g^b=g_y$, a direct calculation shows that (g,η_g) multiplies $(-p^2)^{i_1}p^{i_3}q^{i_4}$ by

$$g_x^{i_1}g_y^{i_1}g_z^{i_3}g_t^{i_4}\eta_g^{-c(2i_1+i_3)+vi_4}=g_x^{i_1}g_y^{i_2}g_z^{i_3}g_t^{i_4}\cdot\eta_g^{ai_1-bi_2-ci_3+vi_4}=g_x^{i_1}g_y^{i_2}g_z^{i_3}g_t^{i_4}\cdot\eta_g^{\kappa}.$$

Since G is diagonal and h_j is G-semi-invariant, all monomials of h_j have the G-character of h_j, and all monomials of h_j^0 have $\mathbf{T}$-weight κ_j. Hence $h_j^0(-p^2,1,p,q)$ is $\tilde{H}_y$-semi-invariant. Away from E, $D_{Y,j}$ is the strict transform of the image of $\mathrm{div}(h_j^0(-p^2,1,p,q))$ on S (by the homogeneity $h_j^0(\mathbf{T}(\lambda)\cdot u)=\lambda^{\kappa_j}h_j^0(u)$), which is H-invariant as $h_j^0(-p^2,1,p,q)$ is $\tilde{H}_y$-semi-invariant, and E is H-invariant, so $D_{Y,j}$ is H-invariant.

Step 2. In this step we prove the chart description, the remaining statements of (1), and the statements of (2) on the charts. On W_N with $s^v=t$ we have

$$\bar{x}\bar{y}+\bar{z}^2=s^{2c}(xy+z^2)=-s^{2c+vN}=-u_t^d,$$

since $vN+2c=rd$. Thus X_d is the t-root chart of W_N, and $S_0=X_d\cap(u_t=0)$. With $\eta^a=x$, $u_x=\eta^r$, $p_x=z\eta^c$, and $q_x=t/\eta^v$,

$$y\eta^b=-(z^2+t^N)\eta^{2c}=-p_x^2-u_x^dq_x^N,$$

so the x-root chart of W_N is $\mathbb{A}^3_{u_x,p_x,q_x}$, with section $(u_x=0)=\mathbb{A}^2_{p_x,q_x}$. On $W_0=(xy+z^2=0)$ the substitutions $\bar{x}=x/s^a$, $\bar{y}=ys^b$, $\bar{z}=zs^c$, $p_x=z\eta^c$, and $q_x=t/\eta^v$ yield S_0 and $y\eta^b=-p_x^2$. If $q_x=\xi^v$, then $\xi=s/\eta$, and $\bar{x}=\eta^a/s^a=\xi^{-a}$, $\bar{z}=p_x\eta^{-c}s^c=p_x\xi^c$, and $\bar{y}=-p_x^2\eta^{-b}s^b=-p_x^2\xi^b$ on W_0. The charts of Y near E are the quotients of S_0 and $\mathbb{A}^2_{p_x,q_x}$ by the deck groups μ_v of s and μ_a of η, acting with weights $(-a,b,c)$ on $(\bar{x},\bar{y},\bar{z})$ and $(c,-v)$ on (p_x,q_x) (the exponents of $\bar{x},\bar{y}$, resp. p_x,q_x, in p,q span the dual cones of the cones spanned by $(1,0),(a,2v)$ and by $(a,2v),(0,1)$, and the invariant monomials have the character lattice of S, by a direct toric calculation), and the point $E\cap C_p$ is the image of $0\in S_0$ while $E\cap C_q$ is the image of $0\in\mathbb{A}^2_{p_x,q_x}$. In the charts, E is the image of $(\bar{y}=\bar{z}=0)$ and of $(p_x=0)$, C_p is the image of $(\bar{x}=\bar{z}=0)$, and C_q is the image of $(q_x=0)$.

The group $\tilde{H}_t$ acts on $\mathbb{A}^4$ by

$$(\bar{x},\bar{y},\bar{z},u_t)\to(g_xs_g^{-a}\bar{x},g_ys_g^b\bar{y},g_zs_g^c\bar{z},s_g^ru_t).$$

As in **Step 1**, $|\tilde{H}_t|=v[G:R]$, $\tilde{H}_t$ is diagonal, preserves X_d up to a character, and contains $\mu_v(-a,b,c)$ as the classes of $(1,s_g)$. If (g,s_g) acts trivially on S_0, then $g_x=s_g^a$, $g_y=s_g^{-b}$, and $g_z=s_g^{-c}$. Since $g_t=s_g^v$, we get $g=\mathbf{T}(s_g)\in R$, so $s_g^r=1$ and (g,s_g) is trivial in $\tilde{H}_t$. Hence $\tilde{H}_t$ acts faithfully on S_0, and hence on X_d. The group $\tilde{H}_t^{(2)}$ acts on $\mathbb{A}^2_{P,Q}$ by $(P,Q)\to(\varsigma_gP,g_zs_g^c\varsigma_g^{-1}Q)$. Since

$$(g_zs_g^c\varsigma_g^{-1})^2=\frac{g_xg_ys_g^{2c}}{g_xs_g^{-a}}=g_ys_g^b,$$

the action on $\mathbb{A}^2_{P,Q}$ lifts the action of $\tilde{H}_t$ on S_0. The group $\tilde{H}_t^{(2)}$ has order $2v[G:R]$, and $(1,1,-1)$ is the deck involution of $\mathbb{A}^2_{P,Q}\to S_0$. The map $(g,s_g,\varsigma_g)\to(g,s_g)$ from $\tilde{H}_t^{(2)}$ to $\tilde{H}_t$ is

surjective, and its kernel consists of the classes of $(\mathbf{T}(\lambda), \lambda, \pm 1)$ with $\lambda^r = 1$, that is, of $(1, 1, \pm 1)$, since $\varsigma_g^2 = \lambda^a \lambda^{-a} = 1$ for $(g, s_g) = (\mathbf{T}(\lambda), \lambda)$. If (g, s_g, ς_g) acts trivially on $\mathbb{A}^2_{P,Q}$, then $\varsigma_g = 1$ and $g_z s_g^c = 1$, so $g_x = s_g^a$, $g_z = s_g^{-c}$, $g_y = g_z^2/g_x = s_g^{-b}$, and $g_t = s_g^v$. Thus $g = \mathbf{T}(s_g) \in R$, so $s_g^r = 1$, and the triple is trivial. The group $\tilde{H}_t^{(2)}$ also acts on $\mathbb{A}^3_{P,Q,u_t}$ by $u_t \to s_g^r u_t$, and the rational map $\mathbb{A}^3_{P,Q,u_t} \dashrightarrow X_d$ given by $\bar{x} = P^2$, $\bar{z} = PQ$, and $\bar{y} = -Q^2 - u_t^d/P^2$ is equivariant, as $s_g^{rd+a}/g_x = s_g^{Nv+b}/g_x = g_y s_g^b$ by $s_g^{Nv} = g_t^N = g_x g_y$. Similarly the group $\tilde{H}_x$ acts on $\mathbb{A}^3_{u_x,p_x,q_x}$ by

$$(u_x, p_x, q_x) \to (\epsilon^r u_x, g_z \epsilon^c p_x, g_t \epsilon^{-v} q_x).$$

It has order $a[G : R]$ and contains $\mu_a(c, -v)$. If (g, ϵ) acts trivially on $\mathbb{A}^2_{p_x,q_x}$, then $g_z = \epsilon^{-c}$, $g_t = \epsilon^v$, $g_x = \epsilon^a$, and $g_y = g_z^2/g_x = \epsilon^{-b}$, so $g = \mathbf{T}(\epsilon) \in R$ and (g, ϵ) is trivial. Hence $\tilde{H}_x$ acts faithfully on $\mathbb{A}^2_{p_x,q_x}$, and also on $\mathbb{A}^3_{u_x,p_x,q_x}$. A G-invariant polynomial in x, y, z, t is a sum of monomials whose $\mathbf{T}$-weights are divisible by r, so, written in the coordinates of the charts, it involves s and η only through $u_t = s^r$ and $u_x = \eta^r$. Thus the G-invariant polynomials define rational maps $\mathbb{A}^3_{P,Q,u_t} \dashrightarrow W_N/G$, $X_d \dashrightarrow W_N/G$, and $\mathbb{A}^3_{u_x,p_x,q_x} \dashrightarrow W_N/G$, which are invariant under $\tilde{H}_t^{(2)}$, $\tilde{H}_t$, and $\tilde{H}_x$, and the induced rational maps $\mathbb{A}^3_{P,Q,u_t}/\tilde{H}_t^{(2)} \dashrightarrow W_N/G$, $X_d/\tilde{H}_t \dashrightarrow W_N/G$, and $\mathbb{A}^3_{u_x,p_x,q_x}/\tilde{H}_x \dashrightarrow W_N/G$ are birational (X_d is the quotient of the cover $(s^v = t)$ of W_N by the classes $(\mathbf{T}(\lambda), \lambda)$, $\lambda^r = 1$, so the degrees match).

The maps $(g, s_g, \varsigma_g) \to gR$ and $(g, \epsilon) \to gR$ are surjective homomorphisms onto H, as s_g, ς_g, and ϵ exist for every $g \in G$. If $gR = R$, then $g = \mathbf{T}(\lambda)$ with $\lambda^r = 1$, and multiplying (g, s_g, ς_g) by the inverse of $(\mathbf{T}(\lambda), \lambda, 1)$ we see that the class of (g, s_g, ς_g) is the class of $(1, \zeta, \varsigma)$ with $\zeta := s_g/\lambda$ and $\varsigma := \varsigma_g$, where $\zeta^v = g_t/\lambda^v = 1$ and $\varsigma^2 = g_x s_g^{-a} = \zeta^{-a}$. The triple $(1, \zeta, \varsigma)$ acts by $(P, Q) \to (\varsigma P, \zeta^c \varsigma^{-1} Q)$, and for any ζ, ς with $\zeta^v = 1$ and $\varsigma^2 = \zeta^{-a}$ the automorphism $(P, Q) \to (\varsigma P, \zeta^c \varsigma^{-1} Q)$ is the action of $(1, \zeta, \varsigma) \in \tilde{H}_t^{(2)}$. Hence the kernel of $(g, s_g, \varsigma_g) \to gR$ is K_t, of order $2v$. It contains the deck involution, and $K_t/\{\pm 1\}$ is $\mu_v(-a, b, c)$ acting on S_0, so $\mathbb{A}^2_{P,Q}/K_t = S_0/\mu_v(-a, b, c)$. Likewise,

$$(\mathbf{T}(\lambda), \epsilon) = (\mathbf{T}(\lambda), \lambda) \cdot (1, \epsilon/\lambda), \qquad (\epsilon/\lambda)^a = 1,$$

so the kernel of $(g, \epsilon) \to gR$ is $\mu_a(c, -v) = K_x$, and the kernel of $(g, \eta_g) \to gR$ is $\mu_b(-c, v)$ by **Step 1**. Thus the actions of $\tilde{H}_t^{(2)}$ and $\tilde{H}_x$ induce on the charts $\mathbb{A}^2_{P,Q}/K_t$ and $\mathbb{A}^2_{p_x,q_x}/K_x$ of Y the action of H of **Step 1** (all three are induced by the action of g on the $\mathbf{T}$-quotient of W_0, normalized by $\mathbf{T}(\eta_g)$, $\mathbf{T}(s_g^{-1})$, and $\mathbf{T}(\epsilon^{-1})$, respectively).

Let $f \in \mathcal{P}_{\mathbf{T}}$ be G-semi-invariant of $\mathbf{T}$-weight zero, with character χ_f. Since $\mathbf{T}$ acts trivially on f, χ_f is trivial on $R \subset \mathbf{T}$. Since G is diagonal, each monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ of f satisfies $\chi_f(g) = g_x^{i_1} g_y^{i_2} g_z^{i_3} g_t^{i_4}$ and $-ai_1 + bi_2 + ci_3 = vi_4$. The element (g, s_g, ς_g) multiplies $\bar{x}$, $\bar{y}$, and $\bar{z}$ by $g_x s_g^{-a}$, $g_y s_g^b$, and $g_z s_g^c$, hence multiplies $\bar{x}^{i_1} \bar{y}^{i_2} \bar{z}^{i_3}$ by

$$g_x^{i_1} g_y^{i_2} g_z^{i_3} s_g^{-ai_1+bi_2+ci_3} = g_x^{i_1} g_y^{i_2} g_z^{i_3} s_g^{vi_4} = g_x^{i_1} g_y^{i_2} g_z^{i_3} g_t^{i_4}.$$

Using $g_z^2 = g_x g_y$, $b = a + 2c$, and $\epsilon^a = g_x$, a direct calculation shows that (g, ϵ) multiplies $(-p_x^2)^{i_2} p_x^{i_3} q_x^{i_4}$ by

$$g_x^{i_1} g_y^{i_2} g_z^{i_3} g_t^{i_4} \cdot \epsilon^{-ai_1+bi_2+ci_3-vi_4} = g_x^{i_1} g_y^{i_2} g_z^{i_3} g_t^{i_4},$$

and the formula of **Step 1** with $\kappa = 0$, which uses $\eta_g^b = g_y$, shows that (g, η_g) multiplies $(-p^2)^{i_1} p^{i_3} q^{i_4}$ by $g_x^{i_1} g_y^{i_2} g_z^{i_3} g_t^{i_4}$. Hence $f(-p^2, 1, p, q)$, $f(P^2, -Q^2, PQ, 1)$, and $f(1, -p_x^2, p_x, q_x)$ are semi-invariant under $\tilde{H}_y$, $\tilde{H}_t^{(2)}$, and $\tilde{H}_x$, with the pullbacks of the character of H induced by χ_f. This proves (1).

For $f \in \mathcal{P}_{\mathbf{T}}$ of weight $\kappa = \kappa(f)$, the monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ becomes $\pm P^{2i_1+i_3} Q^{2i_2+i_3}$ in $f(\bar{x}, \bar{y}, \bar{z}, 1)$ and $\pm p_x^{2i_2+i_3} q_x^{i_4}$ in $f(1, -p_x^2, p_x, q_x)$. Since $i_1 i_2 = 0$, the pair $(2i_1 + i_3, 2i_2 + i_3)$ determines (i_1, i_2, i_3), and i_4 is determined by the weight. Similarly, $(2i_2 + i_3, i_4)$ and the weight

determine the monomial. Hence both substitutions are non-zero, and the coefficient of $(Q=0)$, resp. $(p_x=0)$, in their divisors is

$$n(f) := \min\{2i_2 + i_3\}$$

over the monomials of f. On the overlap, since $-ai_1 + bi_2 + ci_3 = -\kappa + vi_4$ for a monomial of weight κ, a direct calculation shows

$$f(\bar{x}, \bar{y}, \bar{z}, 1) = \xi^{-\kappa} f(1, -p_x^2, p_x, q_x).$$

Since ξ is invertible on the overlap, $\operatorname{div}(f(\bar{x}, \bar{y}, \bar{z}, 1))$ and $\operatorname{div}(f(1, -p_x^2, p_x, q_x))$ glue to the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D_Y(f)$. The generic inertia of $\mu_v(-a, b, c)$ along $(\bar{y} = \bar{z} = 0)$ and of $\mu_a(c, -v)$ along $(p_x = 0)$ has order $e_E = \gcd(a, v)$, along $(\bar{x} = \bar{z} = 0)$ it has order $\gcd(b, v) = e_p$, and along $(q_x = 0)$ it has order $\gcd(a, c) = e_q$. Since K_t and $\mu_a(c, -v)$ act diagonally on $\mathbb{A}^2_{P,Q}$ and $\mathbb{A}^2_{p_x,q_x}$, only the coordinate axes can have non-trivial generic inertia, and $\mathbb{A}^2_{P,Q} \to S_0$ is étale in codimension one. Thus no other curve on the charts has non-trivial generic inertia. Hence B_{br} is the branch divisor of the charts $S_0/\mu_v(-a, b, c)$ and $\mathbb{A}^2_{p_x,q_x}/\mu_a(c, -v)$, the coefficient of E in $D_Y(f)$ is $n(f)/e_E$, and the coefficient of E in B_Y is

$$1 - \frac{1}{e_E} + \frac{m_E}{e_E} = 1 - \frac{1 - m_E}{e_E}.$$

By the generic inertia orders above, the pullbacks of E to S_0 and to $\mathbb{A}^2_{p_x,q_x}$ are $e_E(\bar{y} = \bar{z} = 0)$ and $e_E \operatorname{div}(p_x)$, and $\operatorname{div}(\bar{y}) = 2(\bar{y} = \bar{z} = 0)$ on S_0 near the generic point of $(\bar{y} = \bar{z} = 0)$, as $\bar{y} = -\bar{z}^2/\bar{x}$ there. Since B_{br} is the branch divisor of the charts, for finitely many pairs of a positive real number c'_j and $f_j \in \mathcal{P}_{\mathbf{T}}$, the chart morphisms $S_0 \to Y$ and $\mathbb{A}^2_{p_x,q_x} \to Y$ pull $K_Y + B_{\mathrm{br}} + \sum_j c'_j D_Y(f_j)$ back to

$$K_{S_0} + \sum_j c'_j \operatorname{div}(f_j(\bar{x}, \bar{y}, \bar{z}, 1)) \qquad \text{and} \qquad K_{\mathbb{A}^2} + \sum_j c'_j \operatorname{div}(f_j(1, -p_x^2, p_x, q_x)).$$

Step 3. In this step we prove the remaining statements of (2). Let $k_E := \gcd(a, 2v)$, and let $\bar{\pi} : \bar{Y} \to \mathbb{A}^2_{p,q}$ be the weighted blow-up with weights $(a/k_E, 2v/k_E)$ on (p, q), with exceptional curve $\bar{E}$ and with strict transforms $\bar{C}_p$ and $\bar{C}_q$ of $(p = 0)$ and $(q = 0)$. The surface $\bar{Y}$ is the toric surface of the fan in $\mathbb{Z}^2$ obtained by inserting the ray through $(a, 2v)$, which is the fan of Y, so $Y = \bar{Y}/\mu_b(-c, v)$. Let $\psi : \bar{Y} \to Y$ be the quotient morphism, of degree b. Since $\bar{Y}$ is a weighted blow-up of a smooth surface,

$$(\bar{E}^2) = -\frac{k_E^2}{2av}, \qquad (\bar{E} \cdot \bar{C}_p) = \frac{k_E}{2v}, \qquad (\bar{E} \cdot \bar{C}_q) = \frac{k_E}{a}.$$

The function field of $\bar{E}$ is generated by the restriction of $q^{a/k_E}/p^{2v/k_E}$, on which $\zeta \in \mu_b$ acts by ζ^{vb/k_E}, and a direct calculation shows that the inertia group of $\bar{E}$ in $\mu_b(-c, v)$ has order be_E/k_E. The inertia groups of $\bar{C}_p$ and $\bar{C}_q$ have orders e_p and e_q. Hence $\psi^* E = \frac{be_E}{k_E}\bar{E}$, $\psi^* C_p = e_p \bar{C}_p$, $\psi^* C_q = e_q \bar{C}_q$, and the projection formula $(\psi^* D_1 \cdot \psi^* D_2) = b(D_1 \cdot D_2)$ implies

$$(E^2) = -\frac{be_E^2}{2av}, \qquad (C_p \cdot E) = \frac{e_E e_p}{2v}, \qquad (C_q \cdot E) = \frac{e_E e_q}{a}.$$

For $f \in \mathcal{P}_{\mathbf{T}}$, let Δ_f be the $\mathbb{Q}$-divisor on S with $\varphi^* \Delta_f = \operatorname{div}(f(-p^2, 1, p, q))$. The monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ of weight κ becomes $\pm p^{2i_1 + i_3} q^{i_4}$, and

$$a(2i_1 + i_3) + 2vi_4 = 2(ai_1 - bi_2 - ci_3 + vi_4) + b(2i_2 + i_3) = 2\kappa + b(2i_2 + i_3).$$

Hence the toric valuation of the ray $\frac{1}{be_E}(a, 2v)$ takes the value $\frac{2\kappa(f) + bn(f)}{be_E}$ on $f(-p^2, 1, p, q)$, and since π is an isomorphism over $S \backslash \{\pi(E)\}$ under which the strict transform of Δ_f corresponds to

$D_Y(f) - \frac{n(f)}{e_E}E$ by **Step 2** (by the homogeneity $f(\mathbf{T}(\lambda)\cdot u) = \lambda^{\kappa(f)}f(u)$, as for the transition formula of **Step 2**),

$$\pi^*\Delta_f = D_Y(f) + \frac{2\kappa(f)}{be_E}E. \tag{A.167}$$

Since $(\pi^*\Delta_f \cdot E) = 0$, (A.167) implies

$$(D_Y(f)\cdot E) = -\frac{2\kappa(f)}{be_E}(E^2) = \frac{e_E\kappa(f)}{av}.$$

Let

$$B_S^0 := \Big(1-\frac{1}{e_p}\Big)\varphi(p=0) + \Big(1-\frac{1}{e_q}\Big)\varphi(q=0),$$

so that $K_{\mathbb{A}^2} = \varphi^*(K_S + B_S^0)$ and $B_S = B_S^0 + \sum_j c_j\Delta_{h_j^0}$. The log discrepancy function of the toric pair (S, B_S^0) is the linear function on $\mathbb{Z}^2 + \frac{1}{b}(-c, v)\mathbb{Z}$ with value 1 at $(1,0)$ and $(0,1)$, as $(\mathbb{A}^2, 0)$ is crepant over (S, B_S^0). Hence

$$a(E,S,B_S^0) = \frac{a+2v}{be_E}, \qquad a(E,S,B_S) = \frac{a+2v-2\tau-bm_E}{be_E}.$$

The strict transform of B_S is $B_Y - (1-\frac{1-m_E}{e_E})E$ by **Step 2**, so the coefficient of E in $K_Y + B_Y - \pi^*(K_S+B_S)$ is

$$1 - \frac{1-m_E}{e_E} - 1 + \frac{a+2v-2\tau-bm_E}{be_E} = \frac{a+2v-b-2\tau}{be_E} = \frac{2(v-c-\tau)}{be_E} = -\frac{2\delta}{be_E}.$$

This proves (2).

Step 4. In this step we prove (3). Let $D' := \sum_j c_j' \operatorname{div}(f_j)$ on W_N, let $\tau' := \sum_j c_j'\kappa(f_j)$, so $\delta' = \tau' + c - v$, and let F be a prime divisor over W_N with center 0. We use the charts of **Step 2** for the trivial group, that is, with $u_t = s$, $u_x = \eta$, and $d = Nv + 2c$, and we write

$$X' := (\bar{x}\bar{y} + \bar{z}^2 + s^{Nv+2c} = 0) \subset \mathbb{A}^4_{\bar{x},\bar{y},\bar{z},s}.$$

Let

$$C_t' := \sum_j c_j' \operatorname{div}(f_j(\bar{x},\bar{y},\bar{z},1)), \qquad C_x' := \sum_j c_j' \operatorname{div}(f_j(1, y\eta^b, p_x, q_x))$$

on X' and on $\mathbb{A}^3_{\eta,p_x,q_x}$, respectively.

The rational 3-form $\omega := \frac{dx}{x}\wedge dz\wedge dt$ generates ω_{W_N} near 0, since $\partial(xy+z^2+t^N)/\partial y = x$. A direct calculation shows

$$\omega = vs^{v-c}\frac{d\bar{x}}{\bar{x}}\wedge d\bar{z}\wedge\frac{ds}{s} = a\eta^{v-c}\frac{d\eta}{\eta}\wedge dp_x\wedge dq_x,$$

and $\frac{d\bar{x}}{\bar{x}}\wedge d\bar{z}\wedge ds$ generates $\omega_{X'}$. Moreover $f_j = s^{\kappa(f_j)}f_j(\bar{x},\bar{y},\bar{z},1)$ and $f_j = \eta^{\kappa(f_j)}f_j(1, y\eta^b, p_x, q_x)$, where $y\eta^b = -p_x^2 - \eta^d q_x^N$. Let

$$\Delta_t := (1+\delta')\operatorname{div}(s) + C_t', \qquad \Delta_x := (1+\delta')\operatorname{div}(\eta) + C_x'.$$

The rational maps $X' \dashrightarrow W_N$ and $\mathbb{A}^3_{\eta,p_x,q_x} \dashrightarrow W_N$ of the charts are generically Galois with groups μ_v, acting on s, and μ_a, acting on η, and by the expressions of ω above the pairs (X', Δ_t) and $(\mathbb{A}^3, \Delta_x)$ are crepant over (W_N, D'). Let ν be a prime divisor over X', resp. over $\mathbb{A}^3$, whose valuation extends mult_F with ramification index e. Then

$$a(F, W_N, D') = \frac{1}{e}a(\nu, X', \Delta_t), \qquad \text{resp.} \qquad a(F, W_N, D') = \frac{1}{e}a(\nu, \mathbb{A}^3, \Delta_x), \tag{A.168}$$

by [Liu26, Lemma 2.6]. Write $\nu(\cdot)$ for the values of ν.

If $\nu(x)/a \geq \nu(t)/v$, we use X': then $\nu(\bar{x}) = \nu(x) - \frac{a}{v}\nu(t) \geq 0$ and $\nu(\bar{y}), \nu(\bar{z}), \nu(s) > 0$, so the center of ν on X' is contained in the curve $(\bar{y} = \bar{z} = s = 0)$ of $S_0 = X' \cap (s = 0)$. If

$\nu(x)/a < \nu(t)/v$, we use $\mathbb{A}^3$: then $\nu(\eta), \nu(p_x), \nu(q_x) > 0$, so the center of ν is the origin of $\mathbb{A}^3_{\eta,p_x,q_x}$.

The variety X' is singular only at 0, since $Nv+2c \geq 2$, so X' is normal, and the Cartier divisor S_0 is a normal surface. The pullback of $(Y, B_{\mathrm{br}} + \sum_j c'_j D_Y(f_j))$ to the chart S_0, resp. $\mathbb{A}^2_{p_x,q_x}$, is

$$\Big(S_0, \sum_j c'_j \operatorname{div}(f_j(\bar{x}, \bar{y}, \bar{z}, 1))\Big), \qquad \text{resp.} \qquad \Big(\mathbb{A}^2, \sum_j c'_j \operatorname{div}(f_j(1, -p_x^2, p_x, q_x))\Big),$$

by **Step 2**. By (2) for the labels c'_j and the polynomials f_j (**Step 3** uses no property of c_j and h^0_j other than $h^0_j \in \mathcal{P}_{\mathbf{T}}$),

$$K_Y + B_{\mathrm{br}} + \sum_j c'_j D_Y(f_j) + \frac{2\delta'}{be_E} E = \pi^*(K_S + B'_S), \tag{A.169}$$

where

$$\varphi^*(K_S + B'_S) = K_{\mathbb{A}^2} + \sum_j c'_j \operatorname{div}(f_j(-p^2, 1, p, q)).$$

By **Step 2**, the pullbacks of E to S_0 and to $\mathbb{A}^2_{p_x,q_x}$ are $e_E(\bar{y} = \bar{z} = 0)$ and $e_E \operatorname{div}(p_x)$, and $\operatorname{div}(\bar{y}) = 2(\bar{y} = \bar{z} = 0)$ on S_0, as $\bar{y}$ vanishes on S_0 only along $(\bar{y} = \bar{z} = 0)$. Hence the pullbacks of $\frac{2\delta'}{be_E}E$ to the charts are $\frac{\delta'}{b}\operatorname{div}(\bar{y})$ and $\frac{2\delta'}{b}\operatorname{div}(p_x)$.

Assume first that $\delta' \leq 0$ and $(Y, B_{\mathrm{br}} + \sum_j c'_j D_Y(f_j))$ is lc near E. Then $(S_0, C'_t|_{S_0})$ is lc near $(\bar{y} = \bar{z} = 0)$ and $(\mathbb{A}^2, \sum_j c'_j \operatorname{div}(f_j(1, -p_x^2, p_x, q_x)))$ is lc near 0, as crepant finite pullbacks of an lc pair are lc [Liu26, Lemma 2.6]. Since S_0 and $(\eta = 0)$ are reduced Cartier divisors on X' and on $\mathbb{A}^3$ which are not components of C'_t and C'_x, [Kaw07, Theorem] implies that $(X', S_0 + C'_t)$ and $(\mathbb{A}^3, (\eta = 0) + C'_x)$ are lc near the curve $(\bar{y} = \bar{z} = s = 0)$ and near 0, respectively. As $\delta' \leq 0$,

$$a(\nu, X', \Delta_t) = a(\nu, X', S_0 + C'_t) - \delta'\nu(s) \geq 0,$$
$$a(\nu, \mathbb{A}^3, \Delta_x) = a(\nu, \mathbb{A}^3, (\eta = 0) + C'_x) - \delta'\nu(\eta) \geq 0.$$

Hence $a(F, W_N, D') \geq 0$ by (A.168).

Assume next that $\delta' \geq 0$, $Nv > a$, and $(\mathbb{A}^2, \sum_j c'_j \operatorname{div}(f_j(-p^2, 1, p, q)))$ is lc near 0. Then (S, B'_S) is lc near the image of 0, so by (A.169) the pair $(Y, B_{\mathrm{br}} + \sum_j c'_j D_Y(f_j) + \frac{2\delta'}{be_E}E)$ is lc near E. By [Kaw07, Theorem], applied to the reduced Cartier divisors $S_0 \subset X'$ and $(\eta = 0) \subset \mathbb{A}^3$ as in the case $\delta' \leq 0$, the pairs

$$\Big(X', S_0 + C'_t + \frac{\delta'}{b}\operatorname{div}(\bar{y})\Big) \qquad \text{and} \qquad \Big(\mathbb{A}^3, (\eta = 0) + C'_x + \frac{2\delta'}{b}\operatorname{div}(p_x)\Big)$$

are lc near $(\bar{y} = \bar{z} = s = 0)$ and near 0. Since $\nu(\bar{y}) = \nu(y) + b\nu(s)$,

$$a(\nu, X', \Delta_t) = a\Big(\nu, X', S_0 + C'_t + \frac{\delta'}{b}\operatorname{div}(\bar{y})\Big) + \frac{\delta'}{b}\nu(y) \geq 0.$$

On $\mathbb{A}^3$, since $\nu(p_x) = \nu(z) + c\nu(\eta)$ and $\nu(x) = a\nu(\eta)$,

$$a(\nu, \mathbb{A}^3, \Delta_x) = a\Big(\nu, \mathbb{A}^3, (\eta = 0) + C'_x + \frac{2\delta'}{b}\operatorname{div}(p_x)\Big) + \frac{\delta'}{b}\big(2\nu(z) - \nu(x)\big).$$

Since we use $\mathbb{A}^3$ only when $\nu(x)/a < \nu(t)/v$, we have $N\nu(t) > \frac{Nv}{a}\nu(x) > \nu(x)$. If $2\nu(z) < \nu(x)$, then $\nu(xy) \geq \nu(x) > 2\nu(z)$ and $\nu(t^N) > 2\nu(z)$, which contradicts $xy + z^2 + t^N = 0$. Hence $2\nu(z) \geq \nu(x)$ and $a(\nu, \mathbb{A}^3, \Delta_x) \geq 0$. Thus $a(F, W_N, D') \geq 0$ by (A.168).

In both cases $a(F, W_N, D') \geq 0$ for every prime divisor F over W_N with center 0. Hence (W_N, D') is lc near 0 (cf. [KM98, Corollary 2.31(1)]). This proves (3). □

Proposition A.40. *Notation as in Construction-Proposition A.39. Let*

$$\beta := \operatorname{mld}((\mathbb{A}^2 \ni 0, C_y)/\tilde{H}_y)$$

when $(\mathbb{A}^2, C_y)$ is lc near 0, and let β_Y be the infimum of the log discrepancies of $(Y, B_Y)/H$ over the prime divisors with center contained in E/H when (Y, B_Y) is lc near E.

(1) There exist r_0 and C, depending only on a, c, v, Γ, e_0, such that for $r \geq r_0$ and every d the following hold. If $\delta \leq 0$, then (Y, B_Y) is lc near E, the infimum β_Y is a minimum, and $\alpha \leq \beta_Y + C/r$. If $\delta \geq 0$, then $(\mathbb{A}^2, C_y)$ is lc near 0 and $\alpha \leq \beta + C/r$. If $\delta = 0$, then $\beta = \beta_Y$. The numbers β and β_Y obtained in this way belong to an ACC set depending only on a, c, v, Γ, e_0.

(2) Let $\theta > 1$. There exists r_1, depending only on $a, c, v, \Gamma, e_0, \theta$, such that for $r \geq r_1$ the following hold. If $\delta \leq 0$, $\theta\tau + c - v \leq 0$, and $(Y, B_{\mathrm{br}} + \theta\Delta_Y)$ is lc near E, then $\alpha \geq \beta_Y$. If $\delta \geq 0$, $\theta\tau + c - v \geq 0$, and $(\mathbb{A}^2, \theta C_y)$ is lc near 0, then $\alpha \geq \beta$.

In (2) the factor θ multiplies only the labelled part Δ_Y, resp. C_y, and not the branch divisor B_{br}.

Proof of Proposition A.40. The proof is long, so we divide it into several steps. We first fix the charts. Let X_d, u_t, u_x, $\tilde{H}_t^{(2)}$, $\tilde{H}_t$, and $\tilde{H}_x$ be as in Construction-Proposition A.39. By Construction-Proposition A.39(1)(a), the groups $\tilde{H}_t^{(2)}$, $\tilde{H}_t$, and $\tilde{H}_x$ are diagonal, $\tilde{H}_t^{(2)}$ acts faithfully on $\mathbb{A}^2_{P,Q}$, $\tilde{H}_t$ preserves X_d and acts faithfully on S_0 and on X_d, $\tilde{H}_x$ acts faithfully on $\mathbb{A}^2_{p_x,q_x}$ and on $\mathbb{A}^3_{u_x,p_x,q_x}$, and the G-invariant polynomials in x, y, z, t induce birational maps $\mathbb{A}^3_{P,Q,u_t}/\tilde{H}_t^{(2)} \dashrightarrow W_N/G$, $X_d/\tilde{H}_t \dashrightarrow W_N/G$, and $\mathbb{A}^3_{u_x,p_x,q_x}/\tilde{H}_x \dashrightarrow W_N/G$. Recall that $\mathbb{A}^3_{P,Q,u_t}$ maps to X_d by $\bar{x} = P^2$, $\bar{z} = PQ$, and $\bar{y} = -Q^2 - u_t^d/P^2$, and that (g, s_g, ς_g) and (g, ϵ) act on $\mathbb{A}^3_{P,Q,u_t}$ and $\mathbb{A}^3_{u_x,p_x,q_x}$ by

$$(P, Q, u_t) \to (\varsigma_g P, g_z s_g^c \varsigma_g^{-1} Q, s_g^r u_t), \qquad (u_x, p_x, q_x) \to (\epsilon^r u_x, g_z \epsilon^c p_x, g_t \epsilon^{-v} q_x).$$

By Construction-Proposition A.39(2) with $c'_j := c_j$ and $f_j := h_j^0$, the pullbacks of (Y, B_Y) to S_0 and to $\mathbb{A}^2_{p_x,q_x}$ are

$$\big(S_0, C_t\big), \quad C_t := \sum_j c_j \operatorname{div}(h_j^0(\bar{x}, \bar{y}, \bar{z}, 1)), \qquad \big(\mathbb{A}^2, C_x\big), \quad C_x := \sum_j c_j \operatorname{div}(h_j^0(1, -p_x^2, p_x, q_x)).$$

For a prime divisor F over W_N and a regular function f near 0 we write $[f] := \operatorname{mult}_F f/|I_G(F)|$.

Step 1. In this step we fix constants and prove that the pairs (Y, B_Y) and $(\mathbb{A}^2, C_y)$ are lc near E and near 0 for $r \geq r_0$. Let γ be the least positive element of Γ ($\gamma := 1$ if $\Gamma \subset \{0\}$), let $M := \lceil 2/\gamma \rceil$, and let $C_1 := M \max_i |w_i|$. The blow-up of $0 \in W_N$ has exceptional divisor E_0 with $a(E_0, W_N, 0) = 2$, so $\sum_j c_j \operatorname{mult}_0 h_j < 2$ as (W_N, D) is klt near 0. Hence the number of indices j is at most M, $\sum_j c_j \leq 2$, and $|\tau| \leq 2 \max_i |w_i|$. By Lemma 5.3, for r larger than a constant depending only on w and Γ, we have $|\kappa_j| \leq C_1$, the blocks h_j^0 are well defined, and every monomial of $h_j - h_j^0$ has $\mathbf{T}$-weight $\kappa_j + \ell r$ with an integer $\ell \neq 0$ and total degree at least $(r - C_1)/\max_i |w_i|$. Since $\mathbf{T}(\lambda)$ preserves $xy + z^2 + t^N$ up to a character exactly when $\lambda^{Nv+2c} = 1$, and R preserves W_N, we have $rd = Nv + 2c$, so $N \geq (r - 2c)/v$.

Since (W_N, D) is klt near 0, [Kaw07, Theorem] implies that the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^N) + \sum_j c_j \operatorname{div}(h_j)\Big)$$

is lc near 0. The differences t^N and $h_j - h_j^0$ belong to $(x, y, z, t)^L$ with

$$L \geq \min\{N, (r - C_1)/\max_i |w_i|\}.$$

By Lemma 4.14 with $n = 4$, applied, for each number at most $M + 1$ of functions, with the largest of the integers it provides, and the DCC set $(\Gamma \backslash \{0\}) \cup \{1\} \subset \mathbb{R}_{>0}$, together with [dFEM11,

Proposition 2.11], there exists r_0, depending only on a, c, v, Γ, such that for $r \geq r_0$ the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy+z^2) + \sum_j c_j \operatorname{div}(h_j^0)\Big)$$

is lc near 0. Hence (W_0, D^0) is lc near 0 by [Kaw07, Theorem], where $W_0 := (xy+z^2=0)$ and $D^0 := \sum_j c_j \operatorname{div}(h_j^0|_{W_0})$.

On W_0 the substitutions $\bar{x} = x/s^a$, $\bar{y} = ys^b$, $\bar{z} = zs^c$, $p_x = z\eta^c$, and $q_x = t/\eta^v$, without the terms u_t^d and $u_x^d q_x^N$, yield the charts $S_0 \times \mathbb{A}^1_s$ and $\mathbb{A}^2_{p_x,q_x} \times \mathbb{A}^1_\eta$, and, with $\eta^b = y$, $p = z\eta^{-c}$, $q = t\eta^v$, and $x = -p^2\eta^{-a}$, the chart $\mathbb{A}^2_{p,q} \times \mathbb{A}^1_\eta$. The rational 3-form $\omega := \frac{dx}{x} \wedge dz \wedge dt$ generates ω_{W_0} and ω_{W_N} near 0, and a direct calculation shows

$$\omega = vs^{v-c}\frac{d\bar{x}}{\bar{x}} \wedge d\bar{z} \wedge \frac{ds}{s} = a\eta^{v-c}\frac{d\eta}{\eta} \wedge dp_x \wedge dq_x = \pm b\eta^{c-v}\frac{d\eta}{\eta} \wedge dp \wedge dq.$$

Since a monomial of weight κ equals s^κ, η^κ, resp. $\eta^{-\kappa}$, times the substituted monomial, the pairs

$$\begin{gathered}\big(S_0 \times \mathbb{A}^1_s, (1+\delta)\operatorname{div}(s) + \operatorname{pr}_1^* C_t\big),\\ \big(\mathbb{A}^2_{p_x,q_x} \times \mathbb{A}^1_\eta, (1+\delta)\operatorname{div}(\eta) + \operatorname{pr}_1^* C_x\big),\\ \big(\mathbb{A}^2_{p,q} \times \mathbb{A}^1_\eta, (1-\delta)\operatorname{div}(\eta) + \operatorname{pr}_1^* C_y\big)\end{gathered}$$

are crepant over (W_0, D^0). We show that (S_0, C_t) is lc near the curve $L_t := (\bar{y} = \bar{z} = 0)$, and at the end of **Step 1** we list the changes for $(\mathbb{A}^2, C_x)$ and $(\mathbb{A}^2, C_y)$. Let Θ be a prime divisor over S_0 with center a point $o \in L_t$, let $g : S' \to S_0$ be a resolution on which Θ is a divisor, and for a positive integer m let F_m be the exceptional divisor of the weighted blow-up of $S' \times \mathbb{A}^1_s$ along $\Theta \times \{0\}$ with weights $(m, 1)$ on a local equation of Θ and on s. Then $\operatorname{mult}_{F_m}$ equals $m \operatorname{mult}_\Theta$ on functions of S_0 and $\operatorname{mult}_{F_m} s = 1$, and

$$a\big(F_m, S_0 \times \mathbb{A}^1_s, (1+\delta)\operatorname{div}(s) + \operatorname{pr}_1^* C_t\big) = ma(\Theta, S_0, C_t) - \delta.$$

The values of $\operatorname{mult}_{F_m}$ on $x = s^a\bar{x}$, $y = s^{-b}\bar{y}$, $z = s^{-c}\bar{z}$, and $t = s^v$ are $m \operatorname{mult}_\Theta \bar{x} + a$, $m \operatorname{mult}_\Theta \bar{y} - b$, $m \operatorname{mult}_\Theta \bar{z} - c$, and v, which are positive for m large since $\operatorname{mult}_\Theta \bar{y}, \operatorname{mult}_\Theta \bar{z} > 0$. Hence F_m induces a prime divisor over W_0 with center 0 whose log discrepancy with respect to (W_0, D^0) is a positive multiple of $ma(\Theta, S_0, C_t) - \delta$. As (W_0, D^0) is lc near 0, $a(\Theta, S_0, C_t) \geq 0$. On $\mathbb{A}^2_{p_x,q_x} \times \mathbb{A}^1_\eta$ the values on x, y, z, t are a, $2m \operatorname{mult}_\Theta p_x - b$, $m \operatorname{mult}_\Theta p_x - c$, $m \operatorname{mult}_\Theta q_x + v$ for Θ with center on $(p_x = 0)$, and on $\mathbb{A}^2_{p,q} \times \mathbb{A}^1_\eta$ they are $2m \operatorname{mult}_\Theta p - a$, b, $m \operatorname{mult}_\Theta p + c$, $m \operatorname{mult}_\Theta q - v$ for Θ with center 0, with log discrepancy $ma(\Theta) - \delta$, resp. $ma(\Theta) + \delta$. Since the non-lc locus is closed, it suffices to test prime divisors with center a closed point of L_t, resp. of $(p_x = 0)$ (cf. [KM98, Corollary 2.31(1)]). Hence (S_0, C_t) is lc near L_t, $(\mathbb{A}^2, C_x)$ is lc near $(p_x = 0)$, and $(\mathbb{A}^2, C_y)$ is lc near 0, for $r \geq r_0$ and for any sign of δ. Since (S_0, C_t) and $(\mathbb{A}^2, C_x)$ are the crepant pullbacks of (Y, B_Y) to the charts, (Y, B_Y) is lc near E [Liu26, Lemma 2.6].

Step 2. In this step we prove that β_Y is a minimum and that β and β_Y belong to an ACC set. The crepant quotient of (Y, B_Y) by H is a pair $(Y/H, B_{Y/H})$, and the crepant quotient of $(\mathbb{A}^2, C_y)$ by $\tilde{H}_y$ is a pair $(\mathbb{A}^2/\tilde{H}_y, B_y)$. The coefficients of $B_{Y/H}$ and B_y have the form $1 - \frac{1-\gamma'}{m'}$ with $\gamma' \in \Gamma_+ \cap [0,1]$ and positive integers m' [Liu26, Lemma 2.6(4)], so they belong to the set $\Psi(\Gamma_+ \cap [0,1])$, which is a DCC set by [Liu26, Lemma 2.10(1)], since $\Gamma_+ \cap [0,1]$ is a DCC set by [Liu26, Lemma 2.10(1)] applied to Γ. Hence β belongs to the ACC set of Theorem 2.9(1) for $\Psi(\Gamma_+ \cap [0,1])$ [Liu26, Lemma 2.6]. Let b_E be the coefficient of E/H in $B_{Y/H}$. The log discrepancy of E/H is $1 - b_E$, and $\{1 - b_E\}$ is an ACC set. Near a closed point y' of E/H at which Y/H and E/H are smooth and $B_{Y/H} = b_E E/H$, the pair $(Y/H, E/H)$ is log smooth, so $a(\Theta', Y/H, E/H) \geq 1$ for any prime divisor Θ' with center y', and since $b_E \leq 1$,

$$\begin{aligned}a(\Theta', Y/H, b_E E/H) &= a(\Theta', Y/H, E/H) + (1 - b_E)\operatorname{mult}_{\Theta'}(E/H)\\ &\geq 1 + (1 - b_E)\operatorname{mult}_{\Theta'}(E/H) \geq 2 - b_E > 1 - b_E.\end{aligned}$$

Hence β_Y is the minimum of $1-b_E$ and of the finitely many numbers $\mathrm{mld}(Y/H \ni y', B_{Y/H})$ at the remaining closed points y' of E/H, each of which is attained by Theorem 2.9(2). Thus β_Y is a minimum, and it belongs to the union of $\{1-b_E\}$ and the ACC set of Theorem 2.9(1), which is an ACC set depending only on a, c, v, Γ, e_0.

Step 3. In this step we prove that $\beta = \beta_Y$ if $\delta = 0$. By Construction-Proposition A.39(2), if $\delta = 0$, then $K_Y + B_Y = \pi^*(K_S + B_S)$, so $a(\Theta, Y, B_Y) = a(\Theta, S, B_S)$ for every prime divisor Θ over Y. The prime divisors over S with center the image o of 0 are exactly the prime divisors over Y with center contained in $E = \pi^{-1}(o)$, and both sets are preserved by H. Since $(\mathbb{A}^2, C_y)/\tilde{H}_y = (S, B_S)/H$ [Liu26, Lemma 2.6], we obtain $\beta = \mathrm{mld}((S \ni o, B_S)/H) = \beta_Y$.

Step 4. In this step we write the log discrepancies of prime divisors over W_N on the charts. By Lemma 5.3, each h_j is the sum of the blocks of h_j of $\mathbf{T}$-weight $\kappa_j + \ell r$, $\ell \in \mathbb{Z}$, so

$$h_j = s^{\kappa_j} A_j, \qquad h_j = \eta^{\kappa_j} A'_j, \qquad h_j = \eta_y^{-\kappa_j} A''_j,$$

where $A_j = \sum_\ell u_t^\ell P_j^\ell(\bar{x}, \bar{y}, \bar{z})$, $A'_j = \sum_\ell u_x^\ell Q_j^\ell(p_x, q_x, y\eta^b)$ with $y\eta^b = -p_x^2 - u_x^d q_x^N$, and, with $\eta_y^b = y$, $u_y := \eta_y^r$, $p = z\eta_y^{-c}$, $q = t\eta_y^v$, and $x = \eta_y^{-a}\tilde{x}$ for $\tilde{x} := -p^2 - q^N/u_y^d$, $A''_j = \sum_\ell u_y^{-\ell} T_j^\ell(\tilde{x}, p, q)$. Here P_j^ℓ, Q_j^ℓ, T_j^ℓ are polynomials, and the terms with $\ell = 0$ are the principal blocks:

$$P_j^0(\bar{x}, \bar{y}, \bar{z}) = h_j^0(\bar{x}, \bar{y}, \bar{z}, 1), \qquad Q_j^0(p_x, q_x, y\eta^b) = h_j^0(1, y\eta^b, p_x, q_x), \qquad T_j^0(\tilde{x}, p, q) = h_j^0(\tilde{x}, 1, p, q).$$

The group $\tilde{H}_y$ acts on $\mathbb{A}^3_{u_y,p,q}$ through $u_y \to \eta_g^r u_y$, the classes $(\mathbf{T}(\lambda), \lambda^{-1})$ with $\lambda^r = 1$ fix u_y, p, and q, and the quotient $\mathbb{A}^3_{u_y,p,q}/\tilde{H}_y$ is birational to W_N/G. A direct calculation shows

$$\omega = \frac{v}{r} s^{v-c} \frac{d\bar{x}}{\bar{x}} \wedge d\bar{z} \wedge \frac{du_t}{u_t} = \frac{a}{r} \eta^{v-c} \frac{du_x}{u_x} \wedge dp_x \wedge dq_x = \pm \frac{b}{r} \eta_y^{c-v} \frac{du_y}{u_y} \wedge dp \wedge dq. \tag{A.170}$$

Let

$$\Delta_t := \left(1 + \frac{\delta}{r}\right) \mathrm{div}(u_t) + \sum_j c_j \,\mathrm{div}(A_j),$$

$$\Delta_x := \left(1 + \frac{\delta}{r}\right) \mathrm{div}(u_x) + \sum_j c_j \,\mathrm{div}(A'_j),$$

$$\Delta_y := \left(1 - \frac{\delta}{r}\right) \mathrm{div}(u_y) + \sum_j c_j \,\mathrm{div}(A''_j),$$

where for a rational function A the log discrepancy with respect to $\mathrm{div}(A)$ is defined by linearity in the value of A. Let F be a prime divisor over W_N with center 0, and let ν be a prime divisor over X_d, resp. $\mathbb{A}^3_{u_x,p_x,q_x}$, resp. $\mathbb{A}^3_{u_y,p,q}$, lying over F, that is, $\mathrm{mult}_\nu(\phi^*\zeta)$ is a positive multiple of $\mathrm{mult}_F \zeta$ for the rational map ϕ of the chart to W_N/G above and any non-zero rational function ζ on W_N/G, regarded as a G-invariant rational function on W_N. The charts induce birational maps from $X_d/\tilde{H}_t$, $\mathbb{A}^3_{u_x,p_x,q_x}/\tilde{H}_x$, and $\mathbb{A}^3_{u_y,p,q}/\tilde{H}_y$ to W_N/G, under which the forms in (A.170) correspond to ω up to a constant factor, and $s^{\kappa_j}A_j$, $\eta^{\kappa_j}A'_j$, and $\eta_y^{-\kappa_j}A''_j$ correspond to h_j. Thus the crepant quotients of (W_N, D) by G, of (X_d, Δ_t) by $\tilde{H}_t$, of $(\mathbb{A}^3, \Delta_x)$ by $\tilde{H}_x$, and of $(\mathbb{A}^3, \Delta_y)$ by $\tilde{H}_y$ are crepant birational. Let F_G be the prime divisor over W_N/G induced by F, which is also induced by ν. The log discrepancies of F_G with respect to the four crepant quotients coincide, as they can be computed on one model on which F_G appears [KM98, Lemma 2.30]. By [Liu26, Lemma 2.6(2)], each of the four ratios in (A.171) below equals the log discrepancy of F_G, so

$$\frac{a(F, W_N, D)}{|I_G(F)|} = \frac{a(\nu, X_d, \Delta_t)}{|I_{\tilde{H}_t}(\nu)|} = \frac{a(\nu, \mathbb{A}^3, \Delta_x)}{|I_{\tilde{H}_x}(\nu)|} = \frac{a(\nu, \mathbb{A}^3, \Delta_y)}{|I_{\tilde{H}_y}(\nu)|} \tag{A.171}$$

for the corresponding ν. The equalities (A.171) also hold with X_d, Δ_t, and $\tilde{H}_t$ replaced by $\mathbb{A}^3_{P,Q,u_t}$, the pullback of Δ_t, and $\tilde{H}_t^{(2)}$, where $\bar{x} = P^2$, $\bar{z} = PQ$, $\bar{y} = -Q^2 - u_t^d/P^2$, and $\frac{d\bar{x}}{\bar{x}} \wedge d\bar{z} = 2dP \wedge dQ$:

the crepant quotient of $\mathbb{A}^3_{P,Q,u_t}$ with the pullback of Δ_t by $\tilde{H}_t^{(2)}$ is crepant birational to the crepant quotient of (W_N, D) by G, and we use [Liu26, Lemma 2.6(2)] for the group $\tilde{H}_t^{(2)}$.

Step 5. In this step we prove $\alpha \le \beta_Y + C_0/r$ when $\delta \le 0$. By **Step 2** and Theorem 2.9(2), there exist a positive integer l, depending only on Γ, and a prime divisor ν_0 over Y/H with center contained in E/H, $a(\nu_0, Y/H, B_{Y/H}) = \beta_Y$, and $a(\nu_0, Y/H, 0) \le l$ (if the center of ν_0 is E/H, then $\nu_0 = E/H$). Let $l_1 := \max(2v, a, b)e_0 l$. Lift ν_0 to a prime divisor ν over $\mathbb{A}^2_{P,Q}$ with the group $\tilde{H}_t^{(2)}$ if the center of ν_0 is a closed point of the image of S_0, and over $\mathbb{A}^2_{p_x,q_x}$ with the group $\tilde{H}_x$ otherwise. Then the center of ν is a point s_0 of $(Q = 0)$, resp. of $(p_x = 0)$, or ν is the curve $(p_x = 0)$. Let $\tilde{H}$ be the stabilizer of s_0 (resp. $\tilde{H} := \tilde{H}_x$ in the last case), let $e := |I_{\tilde{H}}(\nu)| \le \max(2v, a)e_0$, and let C_ν be $\sum_j c_j \operatorname{div}(h_j^0(P^2, -Q^2, PQ, 1))$, resp. C_x. Since $(\mathbb{A}^2, C_\nu)$ and $(\mathbb{A}^2, 0)$ are crepant over $(Y/H, B_{Y/H})$ and over $(Y/H, B')$ for a boundary $B' \ge 0$,

$$\frac{a(\nu, \mathbb{A}^2, C_\nu)}{e} = \beta_Y, \qquad a(\nu, \mathbb{A}^2, 0) \le el \le l_1.$$

Hence the values of ν on the coordinates are at most l_1, and since $(\mathbb{A}^2, C_\nu)$ is lc, $\sum_j c_j \operatorname{mult}_\nu \varphi_j \le l_1$, so $\operatorname{mult}_\nu \varphi_j \le l_1/\gamma$ for the components φ_j of C_ν. Let χ be the character of $\tilde{H}$ on u_t, resp. u_x, and let q_ν be the integer of Lemma 3.1 for ν and χ (in the last case, the integer $q_\nu \in \{1, \dots, e\}$ with $\chi|_I = \lambda_I^{q_\nu}$, where λ_I is the faithful character of $I := I_{\tilde{H}}(\nu)$ on p_x). Let k_0 be the integer with $k_0 \equiv q_\nu \pmod e$ and $4l_1 + l_1/\gamma < k_0 \le 4l_1 + l_1/\gamma + e$, so

$$k_0 \le k_1 := 4l_1 + l_1/\gamma + \max(2v, a, b)e_0,$$

and k_1 depends only on a, c, v, Γ, e_0.

Assume that ν has center a closed point s_0. By Proposition 5.4, applied to $S := \mathbb{A}^2$, the point s_0, the group $\tilde{H}$, which acts faithfully on S by Construction-Proposition A.39(1), the character χ, $u := u_t$ (resp. u_x), $1 + \delta/r = 1 - |\delta|/r$ in place of b, $\Phi_j := \varphi_j \circ \mathrm{pr}_1$, and $\ell := k_0$, the Gauss extension F' of mult_ν with u-value k_0 is a prime divisor over $V := S \times \mathbb{A}^1_u$ with center $(s_0, 0)$, $I_{\tilde{H}}(F') = I_{\tilde{H}}(\nu)$, $\operatorname{mult}_{F'} \Phi_j = \operatorname{mult}_\nu \varphi_j$, and

$$\frac{a(F', V, (1 - |\delta|/r)\operatorname{div}(u) + \sum_j c_j \operatorname{div}(\Phi_j))}{e} = \beta_Y + \frac{|\delta| k_0}{re}. \tag{A.172}$$

If ν is the curve $(p_x = 0)$, let F' be the exceptional divisor of the weighted blow-up of $V = \mathbb{A}^3_{u_x,p_x,q_x}$ along $(p_x = u_x = 0)$ with weights $(1, k_0)$ on (p_x, u_x). Then $\operatorname{mult}_{F'} p_x = 1$, $\operatorname{mult}_{F'} u_x = k_0$, $a(F', V, 0) = 1 + k_0$, and F' is birational to $\mathbb{A}^2$ with coordinates q_x and $u_x/p_x^{k_0}$. An element $g \in \tilde{H}_x$ acts trivially on F' if and only if $g \in I$ and $\chi(g) = \lambda_I(g)^{k_0}$, which holds for every $g \in I$ as $k_0 \equiv q_\nu \pmod e$. Hence $I_{\tilde{H}_x}(F') = I$. Since $\Phi_j = \varphi_j \circ \mathrm{pr}_1$ is a function of (p_x, q_x), $\operatorname{mult}_{F'} \Phi_j = \operatorname{mult}_\nu \varphi_j$, so

$$\begin{aligned} a\Big(F', V, \Big(1 - \frac{|\delta|}{r}\Big)\operatorname{div}(u_x) + \sum_j c_j \operatorname{div}(\Phi_j)\Big) &= 1 + k_0 - \Big(1 - \frac{|\delta|}{r}\Big)k_0 - \sum_j c_j \operatorname{mult}_\nu \varphi_j \\ &= a(\nu, \mathbb{A}^2, C_x) + \frac{|\delta| k_0}{r}, \end{aligned}$$

and dividing by $e = |I|$ and using $a(\nu, \mathbb{A}^2, C_x)/e = \beta_Y$, we obtain (A.172).

Proposition 5.4 does not compare F' with W_N, and we now compare them. In the t-chart, $V = \mathbb{A}^3_{P,Q,u_t}$ and $\operatorname{mult}_{F'} Q = \operatorname{mult}_\nu Q \ge 1$. The value of u_t^d/P^2 is at least $dk_0 - 2\operatorname{mult}_\nu P > 2l_1 + l_1/\gamma$, so $\operatorname{mult}_{F'} \bar{y} = 2\operatorname{mult}_\nu Q$ and $\operatorname{mult}_{F'}$ of $P_j^0(\bar{x}, \bar{y}, \bar{z}) - \varphi_j$ is larger than $\operatorname{mult}_\nu \varphi_j$. A block with $\ell > 0$ has value at least $\ell k_0 > \operatorname{mult}_\nu \varphi_j$. A monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ of a block with $\ell = -m < 0$ satisfies $bi_2 + ci_3 \ge mr - C_1$, and $2c < b$, so the value of $u_t^{-m} \bar{x}^{i_1} \bar{y}^{i_2} \bar{z}^{i_3}$ is at least

$$(2i_2 + i_3) - mk_0 \ge \frac{2(mr - C_1)}{b} - mk_0,$$

which, possibly enlarging r_0, is larger than l_1/γ for $r \geq r_0$. Hence $\operatorname{mult}_{F'} A_j = \operatorname{mult}_\nu \varphi_j$ for every j. In the x-chart, the estimates for $\bar{y}$ and for the blocks with $\ell \neq 0$ hold with $\operatorname{mult}_{F'}(y\eta^b) = 2\operatorname{mult}_\nu p_x$ in place of $\operatorname{mult}_{F'} \bar{y} = 2\operatorname{mult}_\nu Q$, the error $u_x^d q_x^N$ of value at least dk_0 in place of u_t^d/P^2, and the p_x-exponent $2i_2 + i_3$ of a monomial of a block with $\ell < 0$. Since V is $\mathbb{A}^3_{P,Q,u_t}$ (resp. the chart $\mathbb{A}^3_{u_x,p_x,q_x}$ itself), the log discrepancy of F' in **Step 4** depends only on the values of F' on u and on the A_j (resp. A'_j), so it equals the left-hand side of (A.172). The values of F' on s (resp. η) are k_0/r, so the values on x, y, z, t are

$$2\operatorname{mult}_\nu P + \frac{ak_0}{r},\ 2\operatorname{mult}_\nu Q - \frac{bk_0}{r},\ \operatorname{mult}_\nu P + \operatorname{mult}_\nu Q - \frac{ck_0}{r},\ \frac{vk_0}{r},$$
$$\text{resp.}\quad \frac{ak_0}{r},\ 2\operatorname{mult}_\nu p_x - \frac{bk_0}{r},\ \operatorname{mult}_\nu p_x - \frac{ck_0}{r},\ \operatorname{mult}_\nu q_x + \frac{vk_0}{r},$$

and, possibly enlarging r_0 so that $r_0 > \max(a, b, v)k_1$, they are all positive for $r \geq r_0$. Thus F' induces a prime divisor F over W_N with center 0, and since the inertia group of F' in $\tilde{H}_t^{(2)}$ (resp. $\tilde{H}_x$) fixes the center of F', it equals $I_{\tilde{H}}(F')$. By (A.171),

$$\alpha \leq \frac{a(F, W_N, D)}{|I_G(F)|} = \beta_Y + \frac{|\delta|k_0}{re} \leq \beta_Y + \frac{C_0}{r},$$

where $C_0 := (2\max_i |w_i| + c + v)k_1$ depends only on a, c, v, Γ, e_0, as $|\delta| \leq |\tau| + |c - v|$. This holds for every d.

Step 6. In this step we prove $\alpha \leq \beta + C_0/r$ when $\delta \geq 0$. By **Step 2** and Theorem 2.9(2), with l as in **Step 5**, there exists a prime divisor ν_0 over $\mathbb{A}^2/\tilde{H}_y$ with center the image of 0, $a(\nu_0, \mathbb{A}^2/\tilde{H}_y, B_y) = \beta$, and $a(\nu_0, \mathbb{A}^2/\tilde{H}_y, 0) \leq l$. Let ν be a prime divisor over $\mathbb{A}^2_{p,q}$ lying over ν_0, with center 0, and let $e := |I_{\tilde{H}_y}(\nu)| \leq be_0$. Since $(\mathbb{A}^2, 0)$ is crepant over $(\mathbb{A}^2/\tilde{H}_y, B')$ for a boundary $B' \geq 0$, $a(\nu, \mathbb{A}^2, 0) \leq el \leq l_1$. Since $(\mathbb{A}^2, C_y)$ is lc near 0 by **Step 1**, as in **Step 5** the values of ν on p and q are at most l_1 and $\operatorname{mult}_\nu h_j^0(-p^2, 1, p, q) \leq l_1/\gamma$. Let χ be the character of $\tilde{H}_y$ on u_y, and choose k_0 as in **Step 5**, so $k_0 \leq k_1$. By Proposition 5.4, applied to $S = \mathbb{A}^2_{p,q}$, the point 0, the group $\tilde{H}_y$, which acts faithfully on S by Construction-Proposition A.39(1), $1 - \delta/r$ in place of b, and $\Phi_j := h_j^0(-p^2, 1, p, q)$, there exists a prime divisor F' over $\mathbb{A}^3_{u_y,p,q}$ with

$$\frac{a(F', \mathbb{A}^3, (1 - \delta/r)\operatorname{div}(u_y) + \sum_j c_j \operatorname{div}(\Phi_j))}{|I_{\tilde{H}_y}(F')|} = \beta + \frac{\delta k_0}{re}.$$

The value of q^N/u_y^d is

$$N\operatorname{mult}_\nu q - dk_0 = d\Big(\frac{r}{v}\operatorname{mult}_\nu q - k_0\Big) - \frac{2c}{v}\operatorname{mult}_\nu q,$$

which, possibly enlarging r_0, is larger than $2l_1 + l_1/\gamma$ for $r \geq r_0$ and every $d \geq 1$. Hence $\operatorname{mult}_{F'} \tilde{x} = 2\operatorname{mult}_\nu p$ and $\operatorname{mult}_{F'}(T_j^0(\tilde{x}, p, q) - \Phi_j) > \operatorname{mult}_\nu \Phi_j$. A block with $\ell < 0$ has the factor $u_y^{-\ell}$ and value at least k_0. A monomial of a block with $\ell > 0$ satisfies $ai_1 + vi_4 \geq \ell r - C_1$, so the value of $u_y^{-\ell}\tilde{x}^{i_1}p^{i_3}q^{i_4}$ is at least $(\ell r - C_1)/\max(a, v) - \ell k_0 > l_1/\gamma$ for $r \geq r_0$, possibly enlarging r_0. Hence $\operatorname{mult}_{F'} A''_j = \operatorname{mult}_\nu \Phi_j$. The values of F' on x, y, z, t are

$$2\operatorname{mult}_\nu p - \frac{ak_0}{r},\qquad \frac{bk_0}{r},\qquad \operatorname{mult}_\nu p + \frac{ck_0}{r},\qquad \operatorname{mult}_\nu q - \frac{vk_0}{r},$$

and, possibly enlarging r_0 so that $r_0 > \max(a, b, v)k_1$, they are all positive for $r \geq r_0$. By (A.171), $\alpha \leq \beta + \delta k_0/(re) \leq \beta + C_0/r$ for every d. Together with **Steps 1–3**, this proves (1) with $C := C_0$.

Step 7. In this step we start the proof of (2) and bound the orders at a prime divisor of small log discrepancy. Assume that either

(I) $\delta \leq 0$, $\theta\tau + c - v \leq 0$, and $(Y, B_{\mathrm{br}} + \theta\Delta_Y)$ is lc near E, or
(II) $\delta \geq 0$, $\theta\tau + c - v \geq 0$, and $(\mathbb{A}^2, \theta C_y)$ is lc near 0.

Since α is the minimum of $a(F,W_N,D)/|I_G(F)|$ over the prime divisors F over W_N with center 0, it suffices to prove that $a(F,W_N,D)/|I_G(F)| \geq \beta_Y$ in case (I) and $a(F,W_N,D)/|I_G(F)| \geq \beta$ in case (II) for any such F. In case (II) we have $Nv \geq r-2c > a$ for $r > a+2c$. By Construction-Proposition A.39(3) with $c'_j := \theta c_j$ and $f_j := h_j^0$, so that $\delta' = \theta\tau + c - v$, the pair $(W_N, \theta\sum_j c_j \operatorname{div}(h_j^0))$ is lc near 0. By [Kaw07, Theorem], Lemma 4.14 for the DCC set $\theta(\Gamma\backslash\{0\})\cup\{1\}$, and [dFEM11, Proposition 2.11], as in **Step 1**, there exists $r_1 \geq r_0$, depending only on $a, c, v, \Gamma, e_0, \theta$, such that for $r \geq r_1$ the pairs

$$(W_N, \theta D) \qquad \text{and} \qquad \Big(W_0, \theta\sum_j c_j \operatorname{div}(h_j|_{W_0})\Big)$$

are lc near 0: for the second pair we pass from W_N to W_0 with the blocks h_j^0, and then from h_j^0 to h_j on W_0. Let $D_0 := \sum_j c_j \operatorname{div}(h_j|_{W_0})$. Since $\theta > 1$ and W_0 is klt, (W_0, D_0) is klt near 0.

The coefficient of E in B_Y is $1 - \frac{1-m_E}{e_E} \geq 0$, so the log discrepancy of E/H with respect to the crepant quotient of (Y, B_Y) by H is at most one, and $\beta_Y \leq 1$. If $\delta \geq 0$, then the coefficient of E in $B_Y + \frac{2\delta}{be_E}E$ is $1 - \frac{1-m_E}{e_E} + \frac{2\delta}{be_E} \geq 0$, so the log discrepancy of E/H with respect to the crepant quotient of $(Y, B_Y + \frac{2\delta}{be_E}E)$ by H is at most one. Since $(Y, B_Y + \frac{2\delta}{be_E}E)$ is crepant over (S, B_S) by Construction-Proposition A.39(2), $\beta = \operatorname{mld}((S \ni o, B_S)/H)$ is at most that log discrepancy, so $\beta \leq 1$. Let F be a prime divisor over W_N with center 0. Hence, to prove the bound for F, we may assume that $a(F,W_N,D)/|I_G(F)| \leq 2$. Let $C_2 := 2\theta/(\theta-1)$ and $C_3 := 2/(\gamma(\theta-1))$. Since $a(F,W_N,D) = a(F,W_N,\theta D) + (\theta-1)\sum_j c_j \operatorname{mult}_F h_j$,

$$[h_j] \leq C_3, \qquad \frac{a(F,W_N,0)}{|I_G(F)|} \leq 2 + \frac{2}{\theta-1} = C_2.$$

The surface $W_N \cap (t=0) = (xy+z^2=0) \subset \mathbb{A}^3$ is the quotient of $\mathbb{A}^2_{P,Q}$ by ± 1 through $x = P^2$, $y = -Q^2$, $z = PQ$, and the pullbacks of $\frac{1}{2}(\operatorname{div}(x) + \operatorname{div}(y))$ and of $\operatorname{div}(z)$ are $\operatorname{div}(P) + \operatorname{div}(Q)$. Hence the pairs

$$\Big(W_N, \frac{1}{2}\operatorname{div}(x) + \frac{1}{2}\operatorname{div}(y) + \operatorname{div}(t)\Big) \qquad \text{and} \qquad \big(W_N, \operatorname{div}(z) + \operatorname{div}(t)\big)$$

are lc near 0 by [Kaw07, Theorem], and

$$[x], [y] \leq 2C_2, \qquad [z], [t] \leq C_2.$$

The monomials $x^b y^a$, $x^c z^a$, $y^v t^b$, and $z^v t^c$ have $\mathbf{T}$-weight zero, so G acts on each of them through H, by a character of order at most e_0. A G-invariant regular function U vanishing at 0 satisfies $[U] = \operatorname{mult}_{F_G} U \in \mathbb{N}^+$, where F_G is the prime divisor over W_N/G induced by F [Liu26, Definition 2.7 and Lemma 2.6(1)]. Hence

$$b[x]+a[y], \quad c[x]+a[z], \quad v[y]+b[t], \quad v[z]+c[t] \;\geq\; \frac{1}{e_0}.$$

Let $\rho_+ := \min([x]/a, [t]/v)$, $\rho_- := \min([y]/b, [z]/c)$, and $C_4 := 1/(2e_0 \max_i |w_i|^2)$. If $\rho_+ < C_4$ and $\rho_- < C_4$, then one of the four sums is smaller than $2\max_i |w_i|^2 C_4 = 1/e_0$, a contradiction. Hence $\max(\rho_+, \rho_-) \geq C_4$. A monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of h_j of weight $\kappa_j + \ell r$ satisfies $ai_1 + vi_4 \geq \ell r - C_1$ if $\ell > 0$ and $bi_2 + ci_3 \geq |\ell| r - C_1$ if $\ell < 0$, so the value $[\cdot]$ of the monomial is at least $(\ell r - C_1)\rho_+$, resp. $(|\ell| r - C_1)\rho_-$.

Step 8. In this step we prove the lower bound when $\rho_- > C_3/(r - C_1)$. Let h_j^+ be the sum of the blocks of h_j with $\ell \geq 0$. By **Step 7**, every monomial of $h_j - h_j^+$ has value larger than $C_3 \geq [h_j]$, so $[h_j^+] = [h_j]$. On the charts, $h_j^+ = s^{\kappa_j} A_j^+$ and $h_j^+ = \eta^{\kappa_j} A_j'^+$ with the polynomials

$$A_j^+ := \sum_{\ell \geq 0} u_t^\ell P_j^\ell(\bar{x}, \bar{y}, \bar{z}), \qquad A_j'^+ := \sum_{\ell \geq 0} u_x^\ell Q_j^\ell(p_x, q_x, -p_x^2 - u_x^d q_x^N),$$

which are semi-invariant for the diagonal groups $\tilde{H}_t$ on $\mathbb{A}^4_{\bar{x},\bar{y},\bar{z},u_t}$ and $\tilde{H}_x$ on $\mathbb{A}^3_{u_x,p_x,q_x}$, and whose restrictions to $(u_t=0)$ and $(u_x=0)$ are $h^0_j(\bar{x},\bar{y},\bar{z},1)$ and $h^0_j(1,-p_x^2,p_x,q_x)$. For the prime divisor ν of **Step 4**, the values of A_j and A_j^+ (resp. A'_j and $A_j'^+$) are equal, so we may replace A_j by A_j^+ in (A.171). Let

$$\mathfrak{a}_t := \prod_j (A_j^+)^{c_j}\cdot \bar{y}^{\max(\delta,0)/b}, \qquad \mathfrak{a}_x := \prod_j (A_j'^+)^{c_j}\cdot p_x^{2\max(\delta,0)/b}.$$

By Construction-Proposition A.39(2), $K_Y + B_Y + \frac{2\delta}{be_E}E = \pi^*(K_S+B_S)$, the pullbacks of E to S_0 and to $\mathbb{A}^2_{p_x,q_x}$ are $e_E(\bar{y}=\bar{z}=0)$ and $e_E\operatorname{div}(p_x)$, and $\operatorname{div}(\bar{y}) = 2(\bar{y}=\bar{z}=0)$ on S_0 near the generic point of $(\bar{y}=\bar{z}=0)$. Since $\bar{y}$ vanishes on the normal surface S_0 only along $(\bar{y}=\bar{z}=0)$, $\operatorname{div}(\bar{y}) = 2(\bar{y}=\bar{z}=0)$ on S_0. Let $B_* := B_Y + \frac{2\max(\delta,0)}{be_E}E$. Hence the restrictions $(S_0, \mathfrak{a}_t|_{S_0})$ and $(\mathbb{A}^2_{p_x,q_x}, \mathfrak{a}_x|_{\mathbb{A}^2})$ are the pullbacks of (Y,B_*) to the charts. The pair (Y,B_*) is lc near E by **Step 1** and Construction-Proposition A.39(2). Let β_* be the minimum of the log discrepancies of $(Y,B_*)/H$ over the prime divisors with center contained in E/H. In case (I), $\delta\le 0$, so $B_*=B_Y$ and $\beta_*=\beta_Y$. In case (II), (Y,B_*) is crepant over (S,B_S), and the argument of **Step 3** shows $\beta_*=\beta$.

Assume that $[x]/a \ge [t]/v$. Then the center of ν on X_d is contained in the curve $L_t = (\bar{y}=\bar{z}=u_t=0)$. The variety X_d is normal and singular only at 0, and (X_d, S_0) is lc near S_0 by [Kaw07, Theorem], as S_0 is a klt surface. Hence $a(\Theta, X_d, 0) \ge \operatorname{mult}_\Theta u_t > 0$ for every prime divisor Θ over X_d with center 0, and X_d is a klt complete intersection near 0, on which $\tilde{H}_t$ acts faithfully. In case (I),

$$a(\nu, X_d, \Delta_t) = a\Big(\nu, X_d, \Big(1+\frac{\delta}{r}\Big)S_0, \mathfrak{a}_t|_{X_d}\Big),$$

and $1+\delta/r\le 1$. In case (II), since $\nu(\bar{y}) = \nu(y) + b\nu(s)$ and $\nu(u_t) = r\nu(s)$,

$$a(\nu, X_d, \Delta_t) = a(\nu, X_d, S_0, \mathfrak{a}_t|_{X_d}) + \frac{\delta}{b}\nu(y) \ge a(\nu, X_d, S_0, \mathfrak{a}_t|_{X_d}).$$

If the center of ν is 0, Lemma 5.5, applied to the complete intersection $X_d\subset\mathbb{A}^4$ with the diagonal group $\tilde{H}_t$, the function u_t, the section S_0, the subgroup $\tilde{H}_t$, which acts faithfully on S_0, and the $\mathbb{R}$-ideal $\mathfrak{a}_t$, with $1+\delta/r$ (resp. 1) in place of b, implies

$$\frac{a(\nu, X_d, \Delta_t)}{|I_{\tilde{H}_t}(\nu)|} \ge \operatorname{mld}((S_0\ni 0, \mathfrak{a}_t|_{S_0})/\tilde{H}_t) \ge \beta_*.$$

If the center of ν is L_t, the second part of Lemma 5.5 with $C := L_t$ implies that $a(\nu, X_d, \Delta_t)/|I_{\tilde{H}_t}(\nu)|$ is at least the minimal log discrepancy of $(Y,B_*)/H$ at the generic point of E/H, which is at least β_*. If the center of ν is a closed point $o' = (\bar{x}_0, 0, 0, 0)$ with $\bar{x}_0\ne 0$, then the stabilizer $\tilde{H}_{o'}$ of o' acts trivially on $\bar{x}$, X_d is smooth near o' with coordinates $(\bar{x}-\bar{x}_0, \bar{z}, u_t)$ in which $\tilde{H}_{o'}$ acts diagonally, and $\bar{x}^{m'}A_j^+$ is, for m' large, a semi-invariant polynomial in $(\bar{x}-\bar{x}_0,\bar{z},u_t)$ which differs from A_j^+ by a unit near o'. Similarly, for m' large, $\bar{x}^{m'}\bar{y}$ is a semi-invariant polynomial in $(\bar{x}-\bar{x}_0,\bar{z},u_t)$ which differs from $\bar{y}$ by a unit near o'. Lemma 5.5 with $X=\mathbb{A}^3$ and the group $\tilde{H}_{o'}$, which acts faithfully on S_0, implies the bound with the minimal log discrepancy of the quotient at the image of o', which is at least β_*.

Assume that $[x]/a < [t]/v$. Then the center of ν is the origin of $\mathbb{A}^3_{u_x,p_x,q_x}$. In case (I),

$$a(\nu, \mathbb{A}^3, \Delta_x) = a\Big(\nu, \mathbb{A}^3, \Big(1+\frac{\delta}{r}\Big)(u_x=0), \mathfrak{a}_x\Big).$$

In case (II), since $\nu(p_x) = \nu(z) + c\nu(\eta)$, $\nu(u_x) = r\nu(\eta)$, and $\nu(x) = a\nu(\eta)$,

$$a(\nu, \mathbb{A}^3, \Delta_x) = a(\nu, \mathbb{A}^3, (u_x=0), \mathfrak{a}_x) + \frac{\delta}{b}\big(2\nu(z)-\nu(x)\big),$$

and $2\nu(z) \geq \nu(x)$: indeed

$$N\nu(t) > \frac{Nv}{a}\nu(x) > \nu(x),$$

so if $2\nu(z) < \nu(x)$, then

$$\nu(z^2) < \min(\nu(xy), \nu(t^N)),$$

which contradicts $xy + z^2 + t^N = 0$. Lemma 5.5, applied to $X = \mathbb{A}^3_{u_x,p_x,q_x}$ with the diagonal group $\tilde{H}_x$, the function u_x, the section $\mathbb{A}^2_{p_x,q_x}$, on which $\tilde{H}_x$ acts faithfully by Construction-Proposition A.39(1), and the $\mathbb{R}$-ideal $\mathfrak{a}_x$, implies $a(\nu, \mathbb{A}^3, \Delta_x)/|I_{\tilde{H}_x}(\nu)| \geq \beta_*$.

In both sectors (A.171) implies $a(F, W_N, D)/|I_G(F)| \geq \beta_*$.

Step 9. In this step we prove the lower bound when $\rho_- \leq C_3/(r - C_1)$. Possibly enlarging r_1, we may assume that $C_3/(r - C_1) < C_4$, so $\rho_+ \geq C_4$ by **Step 7**, and $[x] \geq aC_4$, $[t] \geq vC_4$. Since $N \geq (r - 2c)/v$ and $[x], [y], [z]$ are bounded by **Step 7**, possibly enlarging r_1 we have

$$N[t] > \max\big([x] + [y],\ 2[z],\ [x] + C_3\big).$$

Hence the equation $xy + z^2 + t^N = 0$ implies $[x] + [y] = 2[z]$. The projection $(x, y, z, t) \to (x, z, t)$ restricts to G-equivariant birational morphisms $W_N \to \mathbb{A}^3_{x,z,t}$ and $W_0 \to \mathbb{A}^3_{x,z,t}$, with inverses given by $y = -(z^2 + t^N)/x$ and $y = -z^2/x$, where G acts diagonally on $\mathbb{A}^3_{x,z,t}$ and $g_x g_y = g_z^2$ for $g \in G$. Let F_0 be the prime divisor over W_0 whose valuation corresponds to mult_F under the projections to $\mathbb{A}^3_{x,z,t}$. The center of F_0 is 0, as the value of $-z^2/x$ is $2[z] - [x] = [y] > 0$, and $I_G(F_0) = I_G(F)$. For each j, the difference $h_j(x, -(z^2+t^N)/x, z, t) - h_j(x, -z^2/x, z, t)$ is divisible by t^N/x with a cofactor of non-negative value, so it has value at least $N[t] - [x] > C_3 \geq [h_j]$. Hence $\mathrm{mult}_F(h_j|_{W_N}) = \mathrm{mult}_{F_0}(h_j|_{W_0})$. The form $\omega = \frac{dx}{x} \wedge dz \wedge dt$ generates ω_{W_N} and ω_{W_0} near 0, so

$$\frac{a(F, W_N, D)}{|I_G(F)|} = \frac{a(F_0, W_0, D_0)}{|I_G(F_0)|} \geq \mathrm{mld}((W_0 \ni 0, D_0)/G).$$

Let $g_0 := \gcd(a, b, 2v)$, which is 1 or 2 as $\gcd(a, c, v) = 1$. The quotient of $\mathbb{A}^3_{p_0,t,q_0}$ by ± 1 through $x = p_0^2$, $y = -q_0^2$, $z = p_0 q_0$ is W_0 and is étale in codimension one, so G lifts to a finite diagonal group G_0 of order $2|G|$ on $\mathbb{A}^3_{p_0,t,q_0}$, and

$$\mathrm{mld}((W_0 \ni 0, D_0)/G) = \mathrm{mld}((\mathbb{A}^3 \ni 0, B_0)/G_0), \qquad B_0 := \sum_j c_j \,\mathrm{div}(h_j(p_0^2, -q_0^2, p_0 q_0, t)),$$

where $(\mathbb{A}^3, B_0)$ is klt near 0 by **Step 7**. The circle $\mathbf{T}' := \mathbf{T}_{(a/g_0, 2v/g_0, -b/g_0)}$ on (p_0, t, q_0) maps onto $\mathbf{T}$ by $\mu \to \mu^{2/g_0}$, with kernel generated by -1 if $g_0 = 1$. Hence $|G_0 \cap \mathbf{T}'| = 2r/g_0$ and $[G_0 : G_0 \cap \mathbf{T}'] = g_0[G : R] \leq 2e_0$. A monomial of h_j of weight κ lifts to a monomial of $\mathbf{T}'$-weight $2\kappa/g_0$, the least degree of the lifts is at most $2M$, and the other monomials have degrees tending to infinity with r. Hence, by Lemma 5.3, the principal blocks of the lifts are the lifts of the h_j^0. The lifted labels have $\mathbf{T}'$-weights $2\kappa_j/g_0$, so the lifted τ is $2\tau/g_0$, and the ambient space $\mathbb{A}^3_{p_0,t,q_0}$ has no equation, so its $\epsilon_{\mathbf{T}'}$ is the sum $(a + 2v - b)/g_0$ of the weights. Hence the adjusted mass of the lifted data along $\mathbf{T}'$ is $(2\tau + b - a - 2v)/g_0 = 2(\tau + c - v)/g_0 = 2\delta/g_0$ by $b = a + 2c$.

We now match [Liu26, Construction-Theorem 16.9] for $\mathbb{A}^3_{x',y',z'}$ with $(x', y', z') := (p_0, t, q_0)$ and weights $(a/g_0, 2v/g_0, b/g_0)$ with the plane $\mathbb{A}^2_{p,q}$. That construction uses the cover $q_0 = \vartheta^{b/g_0}$, $u' = p_0 \vartheta^{a/g_0}$, $v' = t\vartheta^{2v/g_0}$. Let $\zeta^b = -1$ and $\eta_y := \zeta\vartheta^{2/g_0}$. Then $\eta_y^b = -q_0^2 = y$, and

$$p = z\eta_y^{-c} = \zeta^{-c} p_0 \vartheta^{(b-2c)/g_0} = \zeta^{-c} u', \qquad q = t\eta_y^v = \zeta^v v',$$

so the plane of [Liu26, Construction-Theorem 16.9] is $\mathbb{A}^2_{p,q}$ up to constants, the group of order $(b/g_0) \cdot g_0[G : R]$ in [Liu26, Construction-Theorem 16.9] is $\tilde{H}_y$, and the boundary on $\mathbb{A}^2_{p,q}$ is C_y. If $2\delta/g_0 < 0$, that construction uses the weighted blow-up of $\mathbb{A}^2_{u',v'}$ with weights $(a/k_E, 2v/k_E)$, that is, the weighted blow-up $\bar{Y} \to \mathbb{A}^2_{p,q}$ with weights $(a/k_E, 2v/k_E)$, $k_E := \gcd(a, 2v)$, whose quotient by $\mu_b(-c, v)$ is Y, with the strict transform of C_y and the coefficient $1 - b(1 - m_E)/k_E$ along $\bar{E}$.

Since a direct calculation shows that the inertia group of the exceptional curve $\bar{E}$ in $\mu_b(-c,v)$ has order be_E/k_E, the crepant quotient by $\mu_b(-c,v)$ has coefficient $1-(1-m_E)/e_E$ along E, and the quotient pair is (Y,B_Y). Hence the minimum of [Liu26, Construction-Theorem 16.9] for the lifted principal blocks is β_Y if $\delta<0$ and β if $\delta\geq 0$, and $\beta=\beta_Y$ if $\delta=0$ by **Step 3**. Let r_2 be the threshold of [Liu26, Theorem 16.21] for the circle $\mathbf{T}'$ on $\mathbb{A}^3_{p_0,t,q_0}$, the residual order bound $2e_0$, and the labels in Γ. Possibly enlarging r_1, we may assume $2r/g_0>r_2$, so $|G_0\cap\mathbf{T}'|>r_2$, and [Liu26, Theorem 16.21], applied to the group G_0 and the klt pair $(\mathbb{A}^3,B_0)$, implies that $\mathrm{mld}((\mathbb{A}^3\ni 0,B_0)/G_0)$ is at least the minimum of [Liu26, Construction-Theorem 16.9] for the lifted principal blocks, that is,

$$\mathrm{mld}((\mathbb{A}^3\ni 0,B_0)/G_0)\geq\beta_*.$$

By **Steps 8 and 9**, $a(F,W_N,D)/|I_G(F)|\geq\beta_*$ for every prime divisor F over W_N with center 0, so $\alpha\geq\beta_Y$ in case (I) and $\alpha\geq\beta$ in case (II). This proves (2). □

Proof of Theorem 7.22. The proof is long, so we divide it into several steps.

Step 1. In this step we fix a sequence contradicting the theorem, normalize it, and describe the charts of Y near E.

Assume that there exists a sequence of data of Set-up 7.3, with Γ, e_0 and $w=(a,-b,-c,v)$ fixed, consisting of N_i, G_i, R_i, r_i, d_i, $D_i=\sum_j c_{ij}\operatorname{div}(h_{ij})$, τ_i, and α_i, such that $r_i\to+\infty$ and $0<\alpha_1<\alpha_2<\cdots$. Let $\delta_i:=\tau_i+c-v$. For the i-th data we use the notation of Construction-Proposition A.39, adding the index i to the objects which depend on the data. The terms with $c_{ij}=0$ or $h_{ij}(0)\neq 0$ do not change D_i near 0, so we may assume that $c_{ij}>0$ and $h_{ij}(0)=0$ for any i and j. Let $\gamma:=\min(\Gamma\backslash\{0\})$ ($\gamma:=1$ if $\Gamma\subset\{0\}$). The blow-up of W_{N_i} at 0 has one exceptional prime divisor E_1, isomorphic to the quadric $(xy+z^2=0)\subset\mathbb{P}^3$, with $a(E_1,W_{N_i},0)=2$ and $\operatorname{mult}_{E_1}h_{ij}\geq\operatorname{mult}_0 h_{ij}$. Since (W_{N_i},D_i) is klt near 0,

$$\sum_j c_{ij}\operatorname{mult}_0 h_{ij}<2.$$

Thus the number of the h_{ij}, the multiplicities $\operatorname{mult}_0 h_{ij}$, the weights $\kappa(h_{ij})$, and the integers n_{ij} of Construction-Proposition A.39 are bounded in terms of a, c, v, and γ, since

$$|\kappa(h_{ij})|\leq\max(a,b,c,v)\operatorname{mult}_0 h_{ij},\qquad n_{ij}\leq 2\operatorname{mult}_0 h_{ij}.$$

By Construction-Proposition A.39(1), the groups $\tilde{H}_{y,i}$, $\tilde{H}^{(2)}_{t,i}$, and $\tilde{H}_{x,i}$, acting on $\mathbb{A}^2_{p,q}$, $\mathbb{A}^2_{P,Q}$, and $\mathbb{A}^2_{p_x,q_x}$, are diagonal subgroups of $GL_2(\mathbb{C})$ of orders $b[G_i:R_i]\leq be_0$, $2v[G_i:R_i]\leq 2ve_0$, and $a[G_i:R_i]\leq ae_0$. There exist only finitely many diagonal subgroups of $GL_2(\mathbb{C})$ of bounded order. Since Γ is a DCC set, possibly passing to a subsequence, we may assume the following:

- the number s of the h_{ij}, and the integers $\kappa_j:=\kappa(h_{ij})$ and $n_j:=n_{ij}$, do not depend on i;
- the groups $\tilde{H}_y:=\tilde{H}_{y,i}$, $\tilde{H}^{(2)}_t:=\tilde{H}^{(2)}_{t,i}$, and $\tilde{H}_x:=\tilde{H}_{x,i}$ do not depend on i, and neither do the characters under them of the semi-invariant functions $h^0_{ij}(-p^2,1,p,q)$, $h^0_{ij}(P^2,-Q^2,PQ,1)$, and $h^0_{ij}(1,-p_x^2,p_x,q_x)$;
- for any j, the label c_{ij} is non-decreasing in i, with limit $c_j^*\geq\gamma$.

In particular the action of $H=\tilde{H}_y/\mu_b(-c,v)$ on Y does not depend on i. The limits c_j^* of the labels belong to the DCC set $\Gamma^*:=\Gamma\cup\{c_1^*,\dots,c_s^*\}$. Let

$$\tau^*:=\sum_j c_j^*\kappa_j,\qquad \delta_*:=\tau^*+c-v,\qquad m_E^*:=\sum_j c_j^* n_j.$$

Then $\tau_i\to\tau^*$ and $\delta_i\to\delta_*$. Possibly passing to a subsequence, we may assume that $\delta_i\leq 0$ for any i, or that $\delta_i\geq 0$ for any i. In the first case let β_i be the number β_Y of Proposition A.40 for the i-th data, and in the second case the number β. By Proposition A.40(1), after removing

finitely many i, β_i is defined, $\alpha_i \le \beta_i + C/r_i$, and the β_i belong to an ACC set. Possibly passing to a subsequence, we may assume that β_i is non-increasing, with limit L. Thus

$$0 < \alpha_i < \lim_{k\to+\infty} \alpha_k \le L \le \beta_i \quad \text{for any } i. \tag{A.173}$$

Let K_t and $K_x = \mu_a(c,-v)$ be the groups of Construction-Proposition A.39(1). We write (ψ,ς) for the element $(P,Q) \to (\varsigma P, \psi^c\varsigma^{-1}Q)$ of K_t, where $\psi^v = 1$ and $\varsigma^2 = \psi^{-a}$ (the parameter ψ is written ζ in Construction-Proposition A.39(1)). We write $\zeta \in \mu_a$ for the element $(p_x,q_x) \to (\zeta^c p_x, \zeta^{-v} q_x)$ of K_x. As $\bar{x} = P^2$, $\bar{y} = -Q^2$ and $\bar{z} = PQ$, the charts of Y near E of Construction-Proposition A.39 are $V_t := S_0/\mu_v(-a,b,c) = \mathbb{A}^2_{P,Q}/K_t$ and $V_x := \mathbb{A}^2_{p_x,q_x}/K_x$, and $E\backslash\{\mathfrak{p}_t\} \subset V_x$, where $\mathfrak{p}_t := E \cap C_p$ and $\mathfrak{p}_x := E \cap C_q$ are the images of the origins. The preimages of E are $E'_t := (Q=0)$ and $E'_x := (p_x = 0)$, and the preimages of C_p and C_q are $(P=0)$ and $(q_x=0)$. An element (ψ,ς) of K_t fixes $E'_t = (Q=0)$ pointwise if and only if $\varsigma = 1$, that is, it is $(\psi,1)$ with $\psi \in \mu_{\gcd(a,v)}$. It fixes $(P=0)$ pointwise if and only if $\psi^c\varsigma^{-1} = 1$, which with $\varsigma^2 = \psi^{-a}$ gives $\varsigma = \psi^c$ and $\psi^b = 1$, that is, it is (ψ,ψ^c) with $\psi \in \mu_{\gcd(b,v)}$. An element $\zeta \in \mu_a$ of K_x fixes $E'_x = (p_x = 0)$ pointwise if and only if $\zeta^{-v} = 1$, and it fixes $(q_x = 0)$ pointwise if and only if $\zeta^c = 1$. Thus the subgroups of K_t and K_x fixing E'_t, $(P=0)$, E'_x, and $(q_x=0)$ pointwise are cyclic of orders $e_E = \gcd(a,v)$, $e_p = \gcd(b,v)$, e_E, and $e_q = \gcd(a,c)$. A non-identity diagonal element fixes a line pointwise only if the line is a coordinate axis, so K_t and K_x contain no other quasi-reflections. Thus the crepant quotients of $(\mathbb{A}^2_{P,Q},0)$ and $(\mathbb{A}^2_{p_x,q_x},0)$ by K_t and K_x glue to $(Y,B_{\rm br})$ near E, the pullbacks of E to the charts are $e_E E'_t$ and $e_E E'_x$, and for any $\mathbb{R}$-divisor Θ on Y the pullback of $(Y,B_{\rm br}+\Theta)$ to a chart is $(\mathbb{A}^2,\Theta')$, where Θ' is the pullback of Θ. By Construction-Proposition A.39(1), K_t and K_x are the kernels of $\tilde{H}^{(2)}_t \to H$ and $\tilde{H}_x \to H$, and $\tilde{H}^{(2)}_t$ and $\tilde{H}_x$ induce the action of H on V_t and V_x.

Step 2. In this step we replace the labels c_{ij} by their limits c^*_j, and we show that the pairs $(Y, B_{\rm br} + \Delta^*_{Y,i})$ near E, resp. $(\mathbb{A}^2, C^*_{y,i})$ near 0, stay lc, with minima tending to L.

Let

$$\Delta^*_{Y,i} := \sum_j c^*_j D_Y(h^0_{ij}), \qquad C^*_{y,i} := \sum_j c^*_j \operatorname{div}(h^0_{ij}(-p^2,1,p,q)), \qquad D^*_i := \sum_j c^*_j \operatorname{div}(h_{ij}).$$

When $(Y, B_{\rm br} + \Delta^*_{Y,i})$ is lc near E, let $\beta^*_{Y,i}$ be the infimum of the log discrepancies of $(Y, B_{\rm br} + \Delta^*_{Y,i})/H$ over the prime divisors with center contained in E/H. When $(\mathbb{A}^2, C^*_{y,i})$ is lc near 0, let $\beta^*_{y,i} := \operatorname{mld}((\mathbb{A}^2 \ni 0, C^*_{y,i})/\tilde{H}_y)$.

First assume that $\delta_i \le 0$ for any i. Let t_i be the lc threshold of $\Delta^*_{Y,i}$ with respect to $(Y,B_{\rm br})$ near E. By **Step 1**, t_i is the infimum of the lc thresholds of the pullbacks of $\Delta^*_{Y,i}$ to $\mathbb{A}^2_{P,Q}$ and $\mathbb{A}^2_{p_x,q_x}$ at the closed points over E. The lc threshold at a closed point is a lower semicontinuous function taking finitely many values and E is proper, so t_i is the lc threshold of the $\mathbb{R}$-ideal $\prod_j f_j^{c^*_j}$ at a closed point of one of the fixed smooth surfaces $\mathbb{A}^2_{P,Q}$ and $\mathbb{A}^2_{p_x,q_x}$, where f_j is $h^0_{ij}(P^2,-Q^2,PQ,1)$, resp. $h^0_{ij}(1,-p_x^2,p_x,q_x)$, the pullback of h^0_{ij} to that chart. Thus the t_i belong to an ACC set [dFEM11, Theorem 1.1]. By Proposition A.40(1), after removing finitely many i, $(Y, B_{\rm br} + \Delta_{Y,i})$ is lc near E, so $t_i \ge \min_j c_{ij}/c^*_j$, which tends to 1. Thus $t_i \ge 1$ for $i \gg 0$, that is, $(Y, B_{\rm br} + \Delta^*_{Y,i})$ is lc near E for $i \gg 0$. Next assume that $\delta_i \ge 0$ for any i. Let t'_i be the lc threshold of $C^*_{y,i}$ with respect to $\mathbb{A}^2$ near 0. Then t'_i is the lc threshold of the $\mathbb{R}$-ideal $\prod_j h^0_{ij}(-p^2,1,p,q)^{c^*_j}$ at the origin of the fixed smooth surface $\mathbb{A}^2_{p,q}$, so the t'_i belong to an ACC set [dFEM11, Theorem 1.1]. By Proposition A.40(1), after removing finitely many i, $(\mathbb{A}^2, C_{y,i})$ is lc near 0, so $t'_i \ge \min_j c_{ij}/c^*_j$, which tends to 1. Thus $t'_i \ge 1$ for $i \gg 0$, that is, $(\mathbb{A}^2, C^*_{y,i})$ is lc near 0 for $i \gg 0$.

Next we show that $\beta^*_{Y,i} - \beta_i \to 0$ if $\delta_i \le 0$ for any i, and $\beta^*_{y,i} - \beta_i \to 0$ if $\delta_i \ge 0$ for any i. We treat the first case. For the second case, replace $(Y, B_{\rm br} + \Delta^*_{Y,i})$, H, and E/H with $(\mathbb{A}^2, C^*_{y,i})$, $\tilde{H}_y$, and the image of 0 in the following argument. For non-negative real numbers $c'_1,\dots,c'_s$, the

crepant quotient of $(Y, B_{\mathrm{br}} + \sum_j c'_j D_Y(h^0_{ij}))$ by H has boundary $B_{i,0} + \sum_j c'_j B_{i,j}$, with $B_{i,j} \geq 0$ not depending on the c'_j. For $c'_j = c^*_j$ the coefficients belong to a DCC set depending only on Γ^* [Liu26, Lemma 2.10]. At all but finitely many closed points of E/H, the minimal log discrepancy is the log discrepancy of E/H plus one. Thus, by Theorem 2.9(2), there exist a positive integer l depending only on Γ^* and, for any i, a prime divisor P_i over Y/H with

$$\beta^*_{Y,i} = a\Big(P_i, Y/H, B_{i,0} + \sum_j c^*_j B_{i,j}\Big),$$

such that either $P_i = E/H$, or $\mathrm{center}_{Y/H} P_i$ is a closed point of E/H and $a(P_i, Y/H, 0) \leq l$. In the second case $a(P_i, Y/H, B_{i,0}) \leq l$ and $a(P_i, Y/H, B_{i,0} + \sum_j c^*_j B_{i,j}) \geq 0$, so $\mathrm{mult}_{P_i} B_{i,j} \leq l/\gamma$. In the first case $\mathrm{mult}_{E/H} B_{i,j} \leq n_j$. Thus

$$\beta^*_{Y,i} \leq \beta_i \leq \beta^*_{Y,i} + \max(l/\gamma, n_1, \dots, n_s) \sum_j (c^*_j - c_{ij}),$$

and $\beta^*_{Y,i} \to L$.

Finally assume that $\delta_* = 0$. Both sides of the identity for $K_Y + B_Y$ in Construction-Proposition A.39(2) are affine functions of the labels. For a fixed i and $0 < \epsilon \ll 1$, $(W_{N_i}, \epsilon D^*_i)$ is klt near 0, so N_i, G_i, and ϵD^*_i are data of Set-up 7.3 for the finite set $\{\epsilon c^*_1, \dots, \epsilon c^*_s\}$ in place of Γ, with e_0 and w of **Step 1**. Applying Construction-Proposition A.39(2) to two such values of ϵ, we obtain

$$K_Y + B_{\mathrm{br}} + \Delta^*_{Y,i} = \pi^*(K_S + B^*_{S,i}) - \frac{2\delta_*}{b e_E} E = \pi^*(K_S + B^*_{S,i}), \tag{A.174}$$

where $K_{\mathbb{A}^2} + C^*_{y,i} = \varphi^*(K_S + B^*_{S,i})$. As $\pi^{-1}(\varphi(0)) = E$ and $(\mathbb{A}^2, C^*_{y,i})/\tilde{H}_y = (S, B^*_{S,i})/H$, (A.174) and [Liu26, Lemma 2.6(2)(5)] imply that $(Y, B_{\mathrm{br}} + \Delta^*_{Y,i})$ is lc near E if and only if $(\mathbb{A}^2, C^*_{y,i})$ is lc near 0, and that in this case $\beta^*_{Y,i} = \beta^*_{y,i}$.

Claim A.41. *Let $\theta \geq 1$ and $\lambda_1, \dots, \lambda_k$ be positive real numbers and $U_1, \dots, U_k$ polynomials in x, y, z, t, all independent of i. If*

$$(W_{N_i}, \theta \sum_j c^*_j \operatorname{div}(h^0_{ij}) + \sum_l \lambda_l \operatorname{div}(U_l))$$

is lc near 0 for any i, then

$$(W_{N_i}, \theta D^*_i + \sum_l \lambda_l \operatorname{div}(U_l))$$

is lc near 0 for any $i \gg 0$.

Proof. Let L_0 be the integer of Lemma 4.14 for $n = 4$, $s + k + 1$ functions, and the finite set

$$\{1, \theta c^*_1, \dots, \theta c^*_s, \lambda_1, \dots, \lambda_k\}.$$

Since h_{ij} is semi-invariant under $R_i \cong \mu_{r_i}$, the $\mathbf{T}$-weights of the monomials of h_{ij} are congruent to κ_j modulo r_i. Thus any monomial of $h_{ij} - h^0_{ij}$ has degree at least $(r_i - |\kappa_j|)/\max(a, b, c, v)$, and $h_{ij} - h^0_{ij} \in (x, y, z, t)^{L_0}$ for $i \gg 0$. By [Kaw07, Theorem], the pair

$$(\mathbb{A}^4, W_{N_i} + \theta \sum_j c^*_j \operatorname{div}(h^0_{ij}) + \sum_l \lambda_l \operatorname{div}(U_l))$$

is lc near 0. Lemma 4.14, applied on the completion at 0 to the families of functions $(xy + z^2 + t^{N_i}, (h^0_{ij})_j, (U_l)_l)$ and $(xy + z^2 + t^{N_i}, (h_{ij})_j, (U_l)_l)$, shows that

$$(\mathbb{A}^4, W_{N_i} + \theta D^*_i + \sum_l \lambda_l \operatorname{div}(U_l))$$

is lc near 0 for $i \gg 0$ (lc near 0 is detected on the completion [dFEM11, Proposition 2.11 and Lemma 2.21]). By [Kaw07, Theorem] again,

$$(W_{N_i}, \theta D_i^* + \sum_l \lambda_l \operatorname{div}(U_l))$$

is lc near 0. □

Both [Kaw07, Theorem] and Lemma 5.6 are results on varieties. In the rest of the proof, we will use [Kaw07, Theorem] in **Steps 5, 6, and 8**, and we will use Lemma 5.6 in **Step 6**, for pairs whose boundaries are given by power series over an algebraically closed field $\mathbb{K} \supset \mathbb{C}$, where lc is meant in the sense of [dFEM11, Section 2] (Construction-Proposition 4.1). We reduce to polynomial truncations.

Claim A.42. *Let $\mathbb{K} \supset \mathbb{C}$ be an algebraically closed field, $\mathfrak{p}'$ a closed point of $\mathbb{A}^2_{\mathbb{K}}$, J a finite abelian group acting faithfully on $\mathbb{A}^2_{\mathbb{K}}$ and fixing $\mathfrak{p}'$, Λ a finite set of positive real numbers containing 1, and (u, ν) coordinates centered at $\mathfrak{p}'$ and semi-invariant under J, where u is a generator of the ideal of a J-invariant curve smooth at $\mathfrak{p}'$ and $\nu|_{(u=0)}$ is a parameter of $(u = 0)$ at $\mathfrak{p}'$. Let*

$$\operatorname{div}(u) + \sum_l b_l \operatorname{div}(g_l), \qquad b_l \in \Lambda,$$

be a boundary on $\operatorname{Spec} \mathbb{K}[[u, \nu]]$, with J-semi-invariant power series $g_l \in \mathbb{K}[[u, \nu]]$ not divisible by u. Let M be an integer at least the integer of [Liu26, Lemma 14.14(1)] *for J and Λ and larger than $\operatorname{mult}_0 g_l(0, \nu)$ for every l, and let $g_l^{<M}$ be the part of g_l of degree less than M. Then the pair above and the truncated pair*

$$\Big(\mathbb{A}^2_{u,\nu}, \operatorname{div}(u) + \sum_l b_l \operatorname{div}(g_l^{<M})\Big)$$

on $\mathbb{A}^2_{u,\nu} := \operatorname{Spec} \mathbb{K}[u, \nu]$, on which J acts through the characters of u and ν, have equal minimal log discrepancies for the quotients by J at the closed point of $\operatorname{Spec} \mathbb{K}[[u, \nu]]$ and at the origin of $\mathbb{A}^2_{u,\nu}$ respectively, the pair above is lc near its closed point if and only if the truncated pair is lc near the origin, and the multiplicities at the origin of their restrictions to $(u = 0)$ are equal. In particular, [Kaw07, Theorem], *which holds over $\mathbb{K}$ by* [Liu26, Remark 14.18], *and Lemma 5.6 apply to the pair above through the truncated pair.*

Proof. The coordinates (u, ν) exist, as J is abelian and preserves the curve. Thus J acts diagonally on $\mathbb{K}[[u, \nu]]$, and each $g_l^{<M}$ is a J-semi-invariant polynomial with the character of g_l. The completion of $\mathbb{A}^2_{u,\nu}$ at the origin is $\mathbb{K}[[u, \nu]]$, on which J acts through the characters of u and ν, faithfully, since J acts faithfully on the local ring $\mathcal{O}_{\mathbb{A}^2_{\mathbb{K}}, \mathfrak{p}'}$, a subring of $\mathbb{K}[[u, \nu]]$. Thus J is a finite diagonal subgroup of $(\mathbb{C}^*)^2$ acting on $\mathbb{A}^2_{u,\nu}$. By [Liu26, Lemma 14.14(1)] and [Liu26, Lemma 3.3(2)], applied to $\mathbb{K}[[u, \nu]]$, the pair of the claim and the truncated pair have equal minimal log discrepancies for the quotients by J at the closed point of $\operatorname{Spec} \mathbb{K}[[u, \nu]]$ and at the origin of $\mathbb{A}^2_{u,\nu}$ respectively. A surface pair is lc near a closed point if and only if its minimal log discrepancy there is non-negative [Amb99, Section 1, the second paragraph after Definition 1.1]. On $\operatorname{Spec} \mathbb{K}[[u, \nu]]$, with the formal prime divisors of [Liu26, Definition 3.2], a pair is lc near the closed point if and only if its minimal log discrepancy there is non-negative, by computing the minimal log discrepancy on a log resolution. The sign does not change under the quotient by J, by the last display of [Liu26, Lemma 2.6], which holds over $\mathbb{K}$ [Liu26, Remark 2.17], and by [Liu26, Definition 3.2]. Thus the pair of the claim is lc near the closed point of $\operatorname{Spec} \mathbb{K}[[u, \nu]]$ if and only if the truncated pair is lc near the origin. As $M > \operatorname{mult}_0 g_l(0, \nu)$, $\operatorname{mult}_0 g_l^{<M}(0, \nu) = \operatorname{mult}_0 g_l(0, \nu)$, so the multiplicities at the origin of the restrictions to $(u = 0)$ are equal. □

Proof of Theorem 7.22 continued.

Step 3. In this step we construct a compatible generic limit $(f_{\infty,j})_j$ and show that the minimum of the limit pair is L.

We say that we are in Case (Y) if $\delta_* < 0$, or $\delta_* = 0$ and $v \le c$. We say that we are in Case (P) if $\delta_* > 0$, or $\delta_* = 0$ and $v > c$. If $\delta_* \neq 0$, then δ_i has the sign of δ_* for $i \gg 0$. Thus, by **Step 2** (with (A.174) when $\delta_* = 0$), after removing finitely many i, $(Y, B_{\mathrm{br}} + \Delta^*_{Y,i})$ is lc near E for any i and $\beta^*_{Y,i} \to L$ in Case (Y), and $(\mathbb{A}^2, C^*_{y,i})$ is lc near 0 for any i and $\beta^*_{y,i} \to L$ in Case (P).

If $s = 0$, then we do not construct a limit, and we will treat this case in **Step 4**. In the rest of this step we assume $s \ge 1$.

In Case (Y) we apply Construction-Proposition 4.1 in case (b) to the surface Y, the curve E, the charts $V_t = \mathbb{A}^2_{P,Q}/K_t$ and $V_x = \mathbb{A}^2_{p_x,q_x}/K_x$ of **Step 1** with the equations Q and p_x of the preimages of E, the group H acting on Y, and the tuples $(f_{i,1}, \dots, f_{i,s})$ with

$$f_{i,j} := \big(h^0_{ij}(P^2, -Q^2, PQ, 1), h^0_{ij}(1, -p_x^2, p_x, q_x)\big), \qquad D_{i,j} = D_Y(h^0_{ij}).$$

The hypotheses hold. By **Step 1**, (Y, B_{br}) is the glued crepant quotient of the charts, and the characters are fixed. The $\mathbb{Q}$-divisors $D_Y(h^0_{ij})$ are H-invariant (Construction-Proposition A.39(1)). On the overlap, where $q_x = \xi^v$, the transition of Construction-Proposition A.39 and the $\mathbf{T}$-homogeneity of h^0_{ij} imply

$$h^0_{ij}(\bar{x}, \bar{y}, \bar{z}, 1) = \xi^{-\kappa_j} h^0_{ij}(1, -p_x^2, p_x, q_x), \tag{A.175}$$

with a factor independent of i. A monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of h^0_{ij} becomes $\pm P^{2i_1+i_3}Q^{2i_2+i_3}$ and $\pm p_x^{2i_2+i_3}q_x^{i_4}$, and distinct monomials of h^0_{ij} have distinct images, because $ai_1 - bi_2 - ci_3 + vi_4 = \kappa_j$ and $i_1 i_2 = 0$. Thus the orders of the chart functions along E'_t and E'_x are n_j, and if $2i_2 + i_3 < d'$, then i_2, i_3, and $ai_1 + vi_4 = \kappa_j + bi_2 + ci_3$ are bounded, which bounds the degrees of the classes modulo $Q^{d'}$ and $p_x^{d'}$. We obtain an algebraically closed field $\mathbb{K} \supset \mathbb{C}$ and a compatible generic limit $(f_{\infty,j})_j$, and we let C_∞ be the divisor given by $\sum_j c^*_j \operatorname{div}(f_{\infty,j})$ on the charts. With the labels c^*_j, the pairs $(\mathcal{A}, C_i)$ of Theorem 4.2 are the pairs $(Y, B_{\mathrm{br}} + \Delta^*_{Y,i})$ near E.

In Case (P) we apply Construction-Proposition 4.1 in case (a) with $n = 2$, the group $\tilde{H}_y$, and $f_{i,j} := h^0_{ij}(-p^2, 1, p, q)$, whose multiplicities at 0 are at most $2/\gamma$ since $(\mathbb{A}^2, C^*_{y,i})$ is lc near 0. Then $C_i = C^*_{y,i}$, and we let $C_\infty := \sum_j c^*_j \operatorname{div}(f_{\infty,j})$.

In both cases, let $U \in \mathcal{P}_{\mathbf{T}}$ be a monomial of $\mathbf{T}$-weight zero, and let $f_U := U(-p^2, 1, p, q)$ in Case (P), resp. $f_U := (U(P^2, -Q^2, PQ, 1), U(1, -p_x^2, p_x, q_x))$ in Case (Y). Then $(f_{\infty,j})_j$ extended by f_U is a compatible generic limit of the tuples extended by f_U. The entry f_U is a non-zero semi-invariant with a fixed character, and in case (b) its divisor $D_Y(U)$ is H-invariant, as U is a rational function on Y on which H acts by a character. For every d', let $U^{(d')}$ be the d'-jet of f_U. The d'-jets of the extended tuples belong to $Z_{d'} \times \{U^{(d')}\}$, where $Z_{d'}$ is the irreducible closed subset of Construction-Proposition 4.1, and the d'-jet of the extended limit maps to its generic point. A non-empty open subset of $Z_{d'} \times \{U^{(d')}\}$ is $V \times \{U^{(d')}\}$ for a non-empty open subset $V \subset Z_{d'}$, which contains the d'-jets of $(f_{i,j})_j$ for all but finitely many i. When we add an augmentation, Theorem 4.2 is used for the extended compatible generic limit, with a positive label on the new entry.

By Theorem 4.2(1), $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ is lc near the base, and by Theorem 4.2(2), after passing to a subsequence, $\beta^*_{Y,i}$, resp. $\beta^*_{y,i}$, does not depend on i. Since it tends to L,

$$\beta^*_{Y,i} = L \text{ in Case (Y)}, \qquad \beta^*_{y,i} = L \text{ in Case (P)}, \qquad \text{for any } i, \tag{A.176}$$

and L is the minimum of the log discrepancies of the crepant quotient of $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ by H over the prime divisors with center contained in $E_{\mathbb{K}}/H$, resp. $L = \operatorname{mld}((\mathcal{A}_{\mathbb{K}} \ni 0, C_\infty)/\tilde{H}_y)$. If $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ is klt near the base, we will conclude in **Step 4**. Otherwise we will conclude in **Step 5** in Case (P), and in **Steps 6–8** in Case (Y).

Step 4. In this step we treat the case where $s = 0$, and the case where $s \ge 1$ and $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ is klt near the base.

If $s \ge 1$, then by Theorem 4.2(1), there exists $\theta_0 > 1$ such that $(Y, B_{\mathrm{br}} + \theta_0 \Delta^*_{Y,i})$ is lc near E for any i in Case (Y), and $(\mathbb{A}^2, \theta_0 C^*_{y,i})$ is lc near 0 for any i in Case (P). If $s = 0$, then $\Delta^*_{Y,i} = 0$

and $C^*_{y,i}=0$, the pairs (Y,B_{br}) near E and $(\mathbb{A}^2,0)$ near 0 are klt and do not depend on i, so $\beta^*_{Y,i}$, resp. $\beta^*_{y,i}$, does not depend on i and equals L by **Step 2**. Thus (A.176) holds when $s=0$, and we let $\theta_0>1$ be arbitrary. Then $(Y,B_{\mathrm{br}}+\theta_0\Delta^*_{Y,i})=(Y,B_{\mathrm{br}})$ is lc near E and $(\mathbb{A}^2,\theta_0C^*_{y,i})=(\mathbb{A}^2,0)$ is lc near 0.

First assume that $\delta_*\neq 0$. Choose $\theta\in(1,\theta_0]$ such that $\theta\tau^*+c-v$ has the sign of $\delta_*=\tau^*+c-v$. As $\tau_i\to\tau^*$, for $i\gg 0$ both $\theta\tau_i+c-v$ and δ_i have the sign of δ_*. Since $c_{ij}\le c^*_j$, $(Y,B_{\mathrm{br}}+\theta\Delta_{Y,i})$ is lc near E in Case (Y), and $(\mathbb{A}^2,\theta C_{y,i})$ is lc near 0 in Case (P). Proposition A.40(2) implies $\alpha_i\ge\beta_i$ for $i\gg 0$, which contradicts (A.173).

Next assume that $\delta_*=0$, and let $\theta:=\theta_0$. Then

$$\theta\tau^*+c-v=(\theta-1)(v-c),$$

which is non-positive in Case (Y) and positive in Case (P). Construction-Proposition A.39(3), applied to $(h^0_{ij})_j$ with the labels $(\theta c^*_j)_j$, implies that $(W_N,\theta\sum_j c^*_j\operatorname{div}(h^0_{ij}))$ is lc near 0 for any $N\ge 3$ (with $Nv>a$ in Case (P), which holds for $N=N_i$ and $i\gg 0$, as $N_iv+2c=r_id_i\ge r_i$). By Claim A.41, $(W_{N_i},\theta D^*_i)$ is lc near 0 for $i\gg 0$. Since W_{N_i} is klt near 0 and $\theta>1$, (W_{N_i},D^*_i) is klt near 0. Thus N_i, G_i, and D^*_i are data of Set-up 7.3 for the DCC set Γ^*, with e_0 and w of **Step 1**, principal blocks h^0_{ij}, and adjusted mass $\tau^*+c-v=0$. Let $\alpha^*_i:=\operatorname{mld}((W_{N_i}\ni 0,D^*_i)/G_i)$. Since $D_i\le D^*_i$, $\alpha_i\ge\alpha^*_i$. Proposition A.40(2), for Γ^* and θ, implies $\alpha^*_i\ge\beta^*_{Y,i}$ in Case (Y) and $\alpha^*_i\ge\beta^*_{y,i}$ in Case (P), for $i\gg 0$. By (A.176), $\alpha_i\ge\alpha^*_i\ge L$, which contradicts (A.173).

Step 5. In this step we treat Case (P) when $(\mathcal{A}_{\mathbb{K}},C_\infty)$ is not klt near 0.

By (A.176) and Definition-Proposition 5.7, in case (a) with $n=2$, the group $\tilde H_y$, and $\beta=L$, the pair $(\mathcal{A}_{\mathbb{K}},C_\infty)$ has exactly one floor F, F is smooth at 0, and

$$L=\frac{\rho_F}{o_F},\qquad \rho_F:=1-\operatorname{mult}_0((C_\infty-F)|_F),$$

where o_F is the order of the character of $\tilde H_y$ on T_0F. As $L>0$, $\rho_F>0$. Choose $\varpi\in\{p,q\}$ such that $\varpi|_F$ is a parameter of F at 0, and let

$$(U,\ell):=\left(x^{c/e_q}z^{a/e_q},\frac{b}{e_q}\right)\text{ if }\varpi=p,\qquad (U,\ell):=\left(y^{v/e_p}t^{b/e_p},\frac{b}{e_p}\right)\text{ if }\varpi=q.$$

Then $U\in\mathcal{P}_{\mathbf{T}}$ has $\mathbf{T}$-weight zero, $U(0)=0$, and $U(-p^2,1,p,q)=\pm\varpi^\ell$, as $b=a+2c$.

Since $\varpi|_F$ is a semi-invariant parameter, o_F is the order of the character χ_ϖ of ϖ under $\tilde H_y$. The subgroup $\mu_b(-c,v)\subset\tilde H_y$ acts on p and q by characters of orders $b/\gcd(b,c)=b/e_q$ and $b/\gcd(b,v)=b/e_p$, so ℓ divides o_F. By Construction-Proposition A.39(1), the character of U under G_i is trivial on R_i, and the character χ^ℓ_ϖ of $U(-p^2,1,p,q)=\pm\varpi^\ell$ under $\tilde H_y$ is its pullback under the surjection $(g,\eta_g)\to gR_i$ onto H. For a G_i-semi-invariant monomial U, let o_U be the order of the character of U under G_i. Thus $o_U=o_F/\ell$.

Let $\lambda:=\rho_F/\ell$. Then

$$\lambda\operatorname{div}(U(-p^2,1,p,q))=\rho_F\operatorname{div}(\varpi),\qquad \operatorname{mult}_0\left((C_\infty-F+\rho_F\operatorname{div}(\varpi))|_F\right)=1-\rho_F+\rho_F=1.$$

By [Kaw07, Theorem] and Claim A.42, for the point 0, the group $\tilde H_y$, an equation u of F, and $\Lambda=\{1,c^*_1,\dots,c^*_s,\lambda\}$, the pair

$$(\mathcal{A}_{\mathbb{K}},C_\infty+\lambda\operatorname{div}(U(-p^2,1,p,q)))$$

is lc near 0. By Theorem 4.2(1) for the tuples extended by $U(-p^2,1,p,q)$ with the label λ (**Step 3**), after passing to a subsequence, $(\mathbb{A}^2,C^*_{y,i}+\lambda\operatorname{div}(U(-p^2,1,p,q)))$ is lc near 0 for any i. Since $\delta_*\ge 0$ and $N_iv>a$ for $i\gg 0$, Construction-Proposition A.39(3), applied to $(h^0_{ij})_j$ and U with the labels $(c^*_j)_j$ and λ, implies that

$$(W_{N_i},\sum_j c^*_j\operatorname{div}(h^0_{ij})+\lambda\operatorname{div}(U))$$

is lc near 0. By Claim A.41 and $D_i \le D_i^*$, the pair $(W_{N_i}, D_i + \lambda \operatorname{div}(U))$ is lc near 0 for $i \gg 0$. Lemma 5.8, applied to $(W_{N_i} \ni 0)$, D_i, G_i, $Z = \{0\}$, and U with $h = o_U$, implies

$$\alpha_i \ge \frac{\lambda}{o_U} = \frac{\rho_F}{o_F} = L,$$

which contradicts (A.173).

Step 6. In this step we assume Case (Y) and that $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ is not klt near $E_{\mathbb{K}}$, and we show that there exists exactly one floor and that it intersects $E_{\mathbb{K}}$ at exactly one point.

For a closed point $\mathfrak{p} \in E_{\mathbb{K}}$, let $(\mathbb{A}^2, K, \tilde{G}, E')$ be $(\mathbb{A}^2_{P,Q}, K_t, \tilde{H}_t^{(2)}, E'_t)$ if $\mathfrak{p} = \mathfrak{p}_t$ and $(\mathbb{A}^2_{p_x,q_x}, K_x, \tilde{H}_x, E'_x)$ otherwise, over $\mathbb{K}$, let $\mathfrak{p}' \in \mathbb{A}^2$ be a point over $\mathfrak{p}$, and let $K_{\mathfrak{p}'}$ be the stabilizer of $\mathfrak{p}'$ in K. As K_x acts on q_x by ζ^{-v},

$$|K_{\mathfrak{p}'}| = 2v \text{ if } \mathfrak{p} = \mathfrak{p}_t, \qquad |K_{\mathfrak{p}'}| = a \text{ if } \mathfrak{p} = \mathfrak{p}_x, \qquad |K_{\mathfrak{p}'}| = e_E \text{ otherwise.}$$

For an $\mathbb{R}$-divisor Θ on $\mathcal{A}_{\mathbb{K}}$ not containing $E_{\mathbb{K}}$, with pullback Θ' to $\mathbb{A}^2$, the projection formula for the quotient of the germ of $\mathbb{A}^2$ at $\mathfrak{p}'$ by $K_{\mathfrak{p}'}$ (the morphism $\mathbb{A}^2/K_{\mathfrak{p}'} \to \mathbb{A}^2/K$ induces an isomorphism of the completed local rings at the images of $\mathfrak{p}'$ [Gro71, Exposé V, Proposition 2.2(i)]) and the pullback $e_E E'$ of $E_{\mathbb{K}}$ imply

$$\frac{(\Theta \cdot E_{\mathbb{K}})_{\mathfrak{p}}}{e_E} = \frac{(\Theta' \cdot E')_{\mathfrak{p}'}}{|K_{\mathfrak{p}'}|}. \tag{A.177}$$

By **Step 3**, the orders of $h^0_{ij}(1, -p_x^2, p_x, q_x)$ along E'_x are n_j. The jets of the $f_{i,j}$ belong to the closed subset of jets of order at least n_j along E'_t and E'_x, and to the open subset of jets of order at most n_j, and (A.175) is a closed linear condition on jets. Thus $f_{\infty,j}$ has order n_j along E'_t and E'_x and satisfies (A.175). Hence the coefficient of E in $D_Y(h^0_{ij})$ is n_j/e_E, the coefficient of $E_{\mathbb{K}}$ in C_∞ is m^*_E/e_E, and the coefficient of E' in the pullback C'_∞ of C_∞ is m^*_E. The quotients of the chart functions of $f_{\infty,j}$ by those of h^0_{ij} are invariant under K_t and K_x and coincide on the overlap, so $\operatorname{div}(f_{\infty,j}) - D_Y(h^0_{ij})$ is principal on $\mathcal{A}_{\mathbb{K}}$. Let $T := C_\infty - (m^*_E/e_E)E_{\mathbb{K}}$. By Construction-Proposition A.39(2), $\tau^* \le v - c$ (as $\delta_* \le 0$ in Case (Y)), and $b = a + 2c$,

$$\frac{(T \cdot E_{\mathbb{K}})}{e_E} = \frac{1}{e_E} \sum_j c_j^* \Big(\frac{e_E \kappa_j}{av} + \frac{n_j}{e_E} \cdot \frac{b e_E^2}{2av} \Big) = \frac{b m^*_E + 2\tau^*}{2av} \le \frac{1}{a} + \frac{1}{2v} - \frac{b(1 - m^*_E)}{2av}. \tag{A.178}$$

By (A.176) and [Liu26, Lemma 2.6(2)],

$$0 < L = \beta^*_{Y,i} \le a(E, Y, B_{\mathrm{br}} + \Delta^*_{Y,i}) = \frac{1 - m^*_E}{e_E}.$$

Thus $m^*_E < 1$, and

$$\frac{(T \cdot E_{\mathbb{K}})}{e_E} < \frac{1}{a} + \frac{1}{2v}. \tag{A.179}$$

By (A.176) and Definition-Proposition 5.7, in case (b) with $\beta = L$, the floors of $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ are curves with coefficient one in $B_{\mathrm{br}} + C_\infty$ whose preimages in the charts are smooth at the points over $E_{\mathbb{K}}$, and no two such preimages intersect at a point over $E_{\mathbb{K}}$. Since $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ is not klt near $E_{\mathbb{K}}$, a floor exists. The coefficient of $E_{\mathbb{K}}$ in $B_{\mathrm{br}} + C_\infty$ is $1 - (1 - m^*_E)/e_E < 1$, so a floor F is not $E_{\mathbb{K}}$. Let $\mathfrak{p} \in F \cap E_{\mathbb{K}}$, let F' be the preimage of F near $\mathfrak{p}'$, let $\sigma := (F' \cdot E')_{\mathfrak{p}'} \ge 1$ be the contact order, and let e_F be the order of the subgroup of K fixing F' pointwise. Then the pullback of F is $e_F F'$, the coefficient of F in B_{br} is $1 - 1/e_F$ (**Step 1**), and the coefficient of F in T is $1/e_F$. Since $T \ge 0$, (A.177) implies

$$\frac{(T \cdot E_{\mathbb{K}})_{\mathfrak{p}}}{e_E} \ge \frac{\sigma}{|K_{\mathfrak{p}'}|} \ge \frac{1}{|K_{\mathfrak{p}'}|}. \tag{A.180}$$

If two distinct floors pass through a point $\mathfrak{p}$, then their preimages pass through $\mathfrak{p}'$, as the preimages of floors are K-invariant, which is impossible. If two distinct points $\mathfrak{p}_1, \mathfrak{p}_2 \in E_{\mathbb{K}}$ belong

to floors, then, as $|K_{\mathfrak{p}'_k}| \in \{2v, a, e_E\}$, the values $2v$ and a occur only at $\mathfrak{p}_t$ and $\mathfrak{p}_x$, and e_E divides a and v, (A.180) implies

$$\frac{(T \cdot E_{\mathbb{K}})}{e_E} \geq \frac{1}{|K_{\mathfrak{p}'_1}|} + \frac{1}{|K_{\mathfrak{p}'_2}|} \geq \frac{1}{a} + \frac{1}{2v},$$

which contradicts (A.179). Thus there exists exactly one floor F, and $F \cap E_{\mathbb{K}} = \{\mathfrak{p}_F\}$ for a closed point $\mathfrak{p}_F$. Since H permutes the floors, $\mathfrak{p}_F$ is fixed by H.

From now on let $\mathfrak{p} := \mathfrak{p}_F$, choose $\mathfrak{p}'$ to be the origin if $\mathfrak{p}_F \in \{\mathfrak{p}_t, \mathfrak{p}_x\}$, and let $\tilde{G}_{\mathfrak{p}'}$ be the stabilizer of $\mathfrak{p}'$ in $\tilde{G}$. Let

$$\rho_F := 1 - \mathrm{mult}_{\mathfrak{p}'}((C'_\infty - F')|_{F'}),$$

and let o_F be the order of the character of $\tilde{G}_{\mathfrak{p}'}$ on $T_{\mathfrak{p}'}F'$. Since $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ is lc near $E_{\mathbb{K}}$ (so $(\mathrm{Spec}\,\widehat{\mathcal{O}}_{\mathbb{A}^2,\mathfrak{p}'}, C'_\infty)$ is lc by Lemma 3.7(2) for $K_{\mathfrak{p}'}$, as the completion of $\mathcal{A}_{\mathbb{K}}$ at $\mathfrak{p}_F$ is the ring of invariants of $K_{\mathfrak{p}'}$ in $\widehat{\mathcal{O}}_{\mathbb{A}^2,\mathfrak{p}'}$ [Liu26, Lemma 3.3(1)], since $\mathbb{A}^2/K_{\mathfrak{p}'} \to \mathbb{A}^2/K$ induces an isomorphism of the completed local rings at the images of $\mathfrak{p}'$, $K_{\mathfrak{p}'}$ being the stabilizer of $\mathfrak{p}'$ in K [Gro71, Exposé V, Proposition 2.2(i)]), $\rho_F \geq 0$ by [Kaw07, Theorem] and Claim A.42, for the point $\mathfrak{p}'$, the group $\tilde{G}_{\mathfrak{p}'}$, an equation u of F', and $\Lambda = \{1, c_1^*, \ldots, c_s^*\}$. Lemma 5.6 and Claim A.42, for these $\mathfrak{p}'$, $\tilde{G}_{\mathfrak{p}'}$, u, and Λ, the curve F', and the $\mathbb{R}$-divisor $C'_\infty - F'$, compute

$$\mathrm{mld}((\mathbb{A}^2 \ni \mathfrak{p}', C'_\infty)/\tilde{G}_{\mathfrak{p}'}) = \frac{\rho_F}{o_F}.$$

We show that this is the minimal log discrepancy of the crepant quotient of $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ by H at the image of $\mathfrak{p}_F$. By Construction-Proposition A.39(1), $\tilde{G} \to H$ is surjective with kernel K, and $\mathfrak{p}_F$ is fixed by H, so $\tilde{G} = K\tilde{G}_{\mathfrak{p}'}$ and $\tilde{G}_{\mathfrak{p}'} \to H$ is surjective with kernel $K_{\mathfrak{p}'}$. As $K_{\mathfrak{p}'}$ and $\tilde{G}_{\mathfrak{p}'}$ are the stabilizers of $\mathfrak{p}'$ in K and in $\tilde{G}$, the morphisms $\mathbb{A}^2/K_{\mathfrak{p}'} \to \mathbb{A}^2/K$ and $\mathbb{A}^2/\tilde{G}_{\mathfrak{p}'} \to \mathbb{A}^2/\tilde{G}$ induce isomorphisms of the completed local rings at the images of $\mathfrak{p}'$ [Gro71, Exposé V, Proposition 2.2(i)], so the completions of $\mathcal{A}_{\mathbb{K}}$ at $\mathfrak{p}_F$ and of $\mathcal{A}_{\mathbb{K}}/H$ at the image of $\mathfrak{p}_F$ are the rings of invariants of $K_{\mathfrak{p}'}$ and of $\tilde{G}_{\mathfrak{p}'}$ in $\widehat{\mathcal{O}}_{\mathbb{A}^2,\mathfrak{p}'}$ [Liu26, Lemma 3.3(1)]. The inertia group in $\tilde{G}$ of a prime divisor over $\mathbb{A}^2$ with center $\mathfrak{p}'$ fixes $\mathfrak{p}'$, so it is contained in $\tilde{G}_{\mathfrak{p}'}$. Lemma 3.7, the version of [Liu26, Lemma 2.6(2)(3)] for completions, applied to $\tilde{G}_{\mathfrak{p}'}$, $K_{\mathfrak{p}'}$, and H, gives the equality. Let $\tilde{P}$ be a formal prime divisor over $\mathrm{Spec}\,\widehat{\mathcal{O}}_{\mathbb{A}^2,\mathfrak{p}'}$, and P' its image over the spectrum of the ring of invariants of $K_{\mathfrak{p}'}$. By Lemma 3.7(1) and the multiplicativity of ramification indices,

$$|I_{\tilde{G}_{\mathfrak{p}'}}(\tilde{P})| = |I_{K_{\mathfrak{p}'}}(\tilde{P})| \cdot |I_H(P')|,$$

Lemma 3.7(2) gives the corresponding equalities of log discrepancies and centers, and by Lemma 3.7(3) every formal prime divisor over the quotients is such an image. By (A.176),

$$L \leq \frac{\rho_F}{o_F}, \tag{A.181}$$

so $\rho_F > 0$. As $C'_\infty - F' \geq m_E^* E'$ and $(F' \cdot E')_{\mathfrak{p}'} = \sigma$,

$$0 < \rho_F \leq 1 - m_E^* \sigma. \tag{A.182}$$

Step 7. In this step we show that F' is transverse to E' at $\mathfrak{p}'$, with one exception at $\mathfrak{p}_t$, and we compute the numerical constraints at $\mathfrak{p}_F$.

By (A.180), (A.178), (A.182), and $\tau^* \leq v - c$,

$$\frac{\sigma}{|K_{\mathfrak{p}'}|} \leq \frac{bm_E^* + 2\tau^*}{2av} < \frac{b\sigma^{-1} + 2v - 2c}{2av}. \tag{A.183}$$

First assume that $\mathfrak{p}_F \in E_{\mathbb{K}} \backslash (C_p \cup C_q)$, so $|K_{\mathfrak{p}'}| = e_E$. If $\sigma \geq 2$, then $\sigma/e_E \geq 1/a + 1/v$, which contradicts (A.179). Thus $\sigma = 1$ and $1/e_E < 1/a + 1/(2v)$. If $a \geq 2e_E$, then $1/a + 1/(2v) \leq 1/(2e_E) + 1/(2e_E)$, as $e_E \leq v$, a contradiction. As e_E divides a, we obtain $a = e_E$, that is, a divides v, and $e_q = \gcd(a, c) = \gcd(a, c, v) = 1$.

Next assume that $\mathfrak{p}_F = \mathfrak{p}_x$ and $\sigma \geq 2$. Then $F' = (p_x = f(q_x))$ with $\mathrm{mult}_0 f = \sigma$. As F' is K_x-invariant, the leading term of f has the character of p_x, that is, $\zeta^{-v\sigma} = \zeta^c$ on K_x. Thus $c + v\sigma = a\omega$ for a positive integer ω. As $|K_{\mathfrak{p}'}| = a$, (A.183) and $b = a + 2c$ imply $2v\sigma^2 < a + 2c + 2\sigma(v - c)$, that is,

$$2(\sigma - 1)(v\sigma + c) < a, \qquad \text{that is,} \qquad 2(\sigma - 1)\omega < 1,$$

which is impossible. Thus $\sigma = 1$.

Finally assume that $\mathfrak{p}_F = \mathfrak{p}_t$ and $\sigma \geq 2$. Then $F' = (Q = f(P))$ with $\mathrm{mult}_0 f = \sigma$. As F' is invariant under $\tilde{G}_{\mathfrak{p}'} = \tilde{H}_t^{(2)}$, the function $Q - f(P)$ is semi-invariant and the leading term of f has the character of Q, so

$$\chi_Q = \chi_P^\sigma \quad \text{on } \tilde{H}_t^{(2)}, \tag{A.184}$$

where χ_P and χ_Q are the characters of P and Q. The element $(P, Q) \to (-P, -Q)$ of K_t ($\psi = 1$, $\varsigma = -1$) and (A.184) show that σ is odd, so $\sigma \geq 3$. For $(\psi, \varsigma) \in K_t$, (A.184) reads $\psi^c \varsigma^{-1} = \varsigma^\sigma$, so $\psi^c = (\varsigma^2)^{(\sigma+1)/2} = \psi^{-a(\sigma+1)/2}$. As ψ runs over μ_v, v divides $c + a(\sigma + 1)/2$, that is, $a\sigma + b = 2v\omega$ for a positive integer ω. As $|K_{\mathfrak{p}'}| = 2v$, (A.183) implies $a\sigma^2 + 2\sigma c < b + 2\sigma v$. Since $\sigma(a\sigma + 2c) = \sigma(2v\omega - a)$ and $b + \sigma a = 2v\omega$, we obtain $\sigma(\omega - 1) < \omega$. As $\sigma \geq 3$, $\omega = 1$. Thus

$$2v = a\sigma + b, \qquad v = c + \frac{a(\sigma + 1)}{2} > c, \qquad e_E = e_q = 1, \tag{A.185}$$

where $e_q = \gcd(a, c)$ divides v, hence $e_q = \gcd(a, c, v) = 1$, and $e_E = \gcd(a, v) = \gcd(a, c)$.

Step 8. In this step we construct an augmentation lc near $E_{\mathbb{K}}$ and conclude.

By **Step 7**, exactly one of the following cases occurs, and we define U, ϖ, the chart function U' of U at $\mathfrak{p}'$, and ℓ as follows:

(i) $\mathfrak{p}_F = \mathfrak{p}_t$ and $\sigma = 1$: $U := y^{v/e_p} t^{b/e_p}$, $\varpi := Q$, $U' := \bar{y}^{v/e_p} = \pm Q^{2v/e_p}$, $\ell := 2v/e_p$;
(ii) $\mathfrak{p}_F = \mathfrak{p}_x$: $U := x^{c/e_q} z^{a/e_q}$, $\varpi := p_x$, $U' := p_x^{a/e_q}$, $\ell := a/e_q$;
(iii) $\mathfrak{p}_F \in E_{\mathbb{K}} \backslash (C_p \cup C_q)$: $U := x^c z^a$, $\varpi := p_x$, $U' := p_x^a$, $\ell := a$;
(iv) $\mathfrak{p}_F = \mathfrak{p}_t$ and $\sigma \geq 3$: $U := x^c z^a$, $\varpi := P$, $U' := \bar{x}^c \bar{z}^a = P^b Q^a$, $\ell := b + a\sigma = 2v$.

In any case $U \in \mathcal{P}_{\mathbf{T}}$ has $\mathbf{T}$-weight zero and $U(0) = 0$. The restriction $\varpi|_{F'}$ is a parameter of F' at $\mathfrak{p}'$, as F' is transverse to E' in (i)–(iii) and $F' = (Q = f(P))$ in (iv). The restriction $U'|_{F'}$ is a unit times $(\varpi|_{F'})^\ell$. Moreover U' is U written in the coordinates of the root cover ($\bar{y}^{v/e_p} = y^{v/e_p} t^{b/e_p}$ as $\bar{y} = y s^b$ and $s^v = t$, $\bar{x}^c \bar{z}^a = x^c z^a$, and $p_x^{a/e_q} = x^{c/e_q} z^{a/e_q}$ as $\eta^a = x$).

We show that ℓ divides o_F and that $o_U = o_F/\ell$. In (i), (ii), and (iv), $\mathfrak{p}'$ is the origin, so $\tilde{G}_{\mathfrak{p}'} = \tilde{G}$. In (iii), a divides v, so K_x fixes q_x and $\mathfrak{p}'$, and since $\mathfrak{p}_F$ is fixed by H, any element of $\tilde{H}_x$ maps $\mathfrak{p}'$ into $K_x \mathfrak{p}' = \{\mathfrak{p}'\}$, so again $\tilde{G}_{\mathfrak{p}'} = \tilde{G}$. As $\varpi|_{F'}$ is a semi-invariant parameter, o_F is the order of the character χ_ϖ of ϖ under $\tilde{G}$. The image of $K_{\mathfrak{p}'} = K$ under χ_ϖ has order ℓ. In (i), the kernel of $(\psi, \varsigma) \to \psi^c \varsigma^{-1}$ is the subgroup fixing $(P = 0)$ pointwise, of order e_p by **Step 1**, hence the image has order $2v/e_p$. In (ii) and (iii), $\zeta \to \zeta^c$ on μ_a has image of order a/e_q. In (iv), $(\psi, \varsigma) \to \varsigma$ has kernel in $\{\psi^a = 1\}$, which is trivial as $e_E = 1$. Thus ℓ divides o_F. The character of U' under $\tilde{G}$ is χ_ϖ^ℓ, by (A.184) in (iv), and has order o_F/ℓ. By the last sentence of Construction-Proposition A.39(1), the character of U' under $\tilde{G}$ is the pullback of the character of U under G_i, which is trivial on R_i, under the surjection $\tilde{G} \to H$. Thus $o_U = o_F/\ell$.

Let $\lambda := \rho_F/\ell$. We show that

$$(\mathcal{A}_{\mathbb{K}}, C_\infty + \lambda D_Y(U)_{\mathbb{K}})$$

is lc near $E_{\mathbb{K}}$, where $D_Y(U)_{\mathbb{K}}$ is the base change of $D_Y(U)$. Let $n(U)$ be the least value of $2i_2 + i_3$ over the monomials of U, that is, $n(U) = \ell$ in (i)–(iii) and $n(U) = a$ in (iv). By the computation of **Step 3**, applied to U, the chart functions of U have order $n(U)$ along E'_t and E'_x. As the

pullbacks of E are $e_E E'_t$ and $e_E E'_x$ (**Step 1**), the coefficient of E in $D_Y(U)$ is $n(U)/e_E$. By Construction-Proposition A.39(2), $(D_Y(U)\cdot E)=0$, so $\Theta_U := D_Y(U)-(n(U)/e_E)E$ satisfies

$$\frac{(\Theta_U\cdot E)}{e_E}=\frac{n(U)b}{2av}.$$

The $\mathbb{Q}$-divisor Θ_U is supported on C_q in (i), as $U(1,-p_x^2,p_x,q_x)=\pm p_x^{2v/e_p}q_x^{b/e_p}$, and on C_p in (ii)–(iv), as $U(P^2,-Q^2,PQ,1)=\pm P^{b/e_q}Q^{a/e_q}$, resp. P^bQ^a. Thus $\mathfrak{p}_F\notin\Theta_U$ in (i)–(iii), and $\Theta_U\cap E=\{\mathfrak{p}_t\}=\{\mathfrak{p}_F\}$ in (iv).

Near $\mathfrak{p}'$, the pullback of $B_{\mathrm{br}}+C_\infty+\lambda D_Y(U)_{\mathbb{K}}$ is $C'_\infty+\lambda\operatorname{div}(U')$, and

$$\operatorname{mult}_{\mathfrak{p}'}\left((C'_\infty-F'+\lambda\operatorname{div}(U'))|_{F'}\right)=1-\rho_F+\lambda\ell=1.$$

By [Kaw07, Theorem] and Claim A.42, for the point $\mathfrak{p}'$, the group $\tilde{G}_{\mathfrak{p}'}$, an equation u of F', and $\Lambda=\{1,c_1^*,\dots,c_s^*,\lambda\}$, the pair

$$\left(\operatorname{Spec}\widehat{\mathcal{O}}_{\mathbb{A}^2,\mathfrak{p}'},C'_\infty+\lambda\operatorname{div}(U')\right)$$

is lc. As the completion of $\mathcal{A}_{\mathbb{K}}$ at $\mathfrak{p}_F$ is the ring of invariants of $K_{\mathfrak{p}'}$ in $\widehat{\mathcal{O}}_{\mathbb{A}^2,\mathfrak{p}'}$ (**Step 6**), Lemma 3.7(2)(3) for $K_{\mathfrak{p}'}$ shows that the pair

$$(\mathcal{A}_{\mathbb{K}},C_\infty+\lambda D_Y(U)_{\mathbb{K}})$$

is lc near $\mathfrak{p}_F$.

Let $\mathfrak{p}\in E_{\mathbb{K}}\backslash\{\mathfrak{p}_F\}$. Near $\mathfrak{p}'$, the pullback is $(m_E^*+\lambda n(U))E'+\Theta'$, where Θ' is the pullback of $\Theta:=T+\lambda(\Theta_U)_{\mathbb{K}}$. In (i)–(iii), $m_E^*+\lambda n(U)=m_E^*+\rho_F\le 1$ by (A.182) with $\sigma=1$. In (iv), (A.182) and (A.185) imply

$$m_E^*+\frac{a\rho_F}{2v}\le m_E^*+\frac{a(1-\sigma m_E^*)}{2v}=\frac{a+bm_E^*}{2v}<\frac{a\sigma+b}{2v}=1.$$

We bound $(\Theta\cdot E_{\mathbb{K}})_{\mathfrak{p}}$ by (A.178) and (A.180) at $\mathfrak{p}_F$. In (i)–(iii), using $m_E^*+\rho_F\le 1$, $\tau^*\le v-c$, and $b=a+2c$,

$$\sum_{\mathfrak{p}\neq\mathfrak{p}_F}\frac{(\Theta\cdot E_{\mathbb{K}})_{\mathfrak{p}}}{e_E}\le\frac{b(m_E^*+\rho_F)+2\tau^*}{2av}-\frac{1}{|K_{\mathfrak{p}'_F}|}\le\frac{1}{a}+\frac{1}{2v}-\frac{1}{|K_{\mathfrak{p}'_F}|},$$

where $\mathfrak{p}'_F$ is the point over $\mathfrak{p}_F$ fixed above. The right-hand side is $1/a$ in (i), and $1/(2v)$ in (ii) and (iii), as $|K_{\mathfrak{p}'_F}|=e_E=a$ in (iii). For $\mathfrak{p}\neq\mathfrak{p}_F$, $|K_{\mathfrak{p}'}|\in\{a,e_E\}$ in (i), $|K_{\mathfrak{p}'}|\in\{2v,e_E\}$ in (ii), and $|K_{\mathfrak{p}'}|\in\{2v,a\}$ in (iii), where $e_E\le a$, $e_E\le v$, and $a=e_E\le v$ in (iii). Thus $|K_{\mathfrak{p}'}|$ times the right-hand side is at most one. In (iv), $\Theta=T$ near $\mathfrak{p}\neq\mathfrak{p}_t$, and (A.185) implies

$$\sum_{\mathfrak{p}\neq\mathfrak{p}_t}\frac{(T\cdot E_{\mathbb{K}})_{\mathfrak{p}}}{e_E}\le\frac{bm_E^*+2v-2c}{2av}-\frac{\sigma}{2v}=\frac{bm_E^*+a}{2av},$$

while $|K_{\mathfrak{p}'}|\in\{a,1\}$ and $a(bm_E^*+a)/(2av)<(a\sigma+b)/(2v)=1$. In all cases (A.177) implies $(\Theta'\cdot E')_{\mathfrak{p}'}\le 1$. By [Kaw07, Theorem] and Claim A.42, for the point $\mathfrak{p}'$, the group $K_{\mathfrak{p}'}$, an equation u of E', and $\Lambda=\{1,c_1^*,\dots,c_s^*,\lambda\}$, $(\mathbb{A}^2,E'+\Theta')$ is lc near $\mathfrak{p}'$, and so is $(\mathbb{A}^2,(m_E^*+\lambda n(U))E'+\Theta')$. As $K_{\mathfrak{p}'}$ is the stabilizer of $\mathfrak{p}'$ in K, the morphism $\mathbb{A}^2/K_{\mathfrak{p}'}\to\mathbb{A}^2/K$ induces an isomorphism of the completed local rings at the images of $\mathfrak{p}'$ [Gro71, Exposé V, Proposition 2.2(i)], so the completion of $\mathcal{A}_{\mathbb{K}}$ at $\mathfrak{p}$ is the ring of invariants of $K_{\mathfrak{p}'}$ in $\widehat{\mathcal{O}}_{\mathbb{A}^2,\mathfrak{p}'}$ [Liu26, Lemma 3.3(1)]. By Lemma 3.7(2)(3) for $K_{\mathfrak{p}'}$, the version of [Liu26, Lemma 2.6(2)(3)] for completions, the pair is lc near $\mathfrak{p}$. Thus

$$(\mathcal{A}_{\mathbb{K}},C_\infty+\lambda D_Y(U)_{\mathbb{K}})$$

is lc near $E_{\mathbb{K}}$.

By Theorem 4.2(1) for the tuples extended by the chart functions of U with the label λ (**Step 3**), after passing to a subsequence, $(Y,B_{\mathrm{br}}+\Delta_{Y,i}^*+\lambda D_Y(U))$ is lc near E for any i. Since

$\delta_* \le 0$, Construction-Proposition A.39(3), applied to $(h^0_{ij})_j$ and U with the labels $(c^*_j)_j$ and λ, implies that

$$(W_{N_i}, \sum_j c^*_j \operatorname{div}(h^0_{ij}) + \lambda \operatorname{div}(U))$$

is lc near 0. By Claim A.41 and $D_i \le D^*_i$, the pair $(W_{N_i}, D_i + \lambda \operatorname{div}(U))$ is lc near 0 for $i \gg 0$. Lemma 5.8, applied to $(W_{N_i} \ni 0)$, D_i, G_i, $Z = \{0\}$, and U with $h = o_U$, and (A.181) imply

$$\alpha_i \ge \frac{\lambda}{o_U} = \frac{\rho_F}{o_F} \ge L,$$

which contradicts (A.173).

Steps 4, 5, and 8 cover all cases, so no sequence as in **Step 1** exists. Thus α cannot increase along $\mathbf{T}$. □

A.14. **Circles** $(0, -2c, -c, v)$**.** In this subsection, we prove Theorem 7.23.

Proof of Theorem 7.23. Exchanging the coordinates x and y is an automorphism of W_N which carries the circle $\mathbf{T}_{(-2c,0,-c,v)}$ to $\mathbf{T}_{(0,-2c,-c,v)}$, so we may assume that $w = (0, -2c, -c, v)$. The integer c is unrelated to the labels c_j of D. We write $g = (g_x, g_y, g_z, g_t)$ for the element of G acting by $x \to g_x x$, $y \to g_y y$, $z \to g_z z$, $t \to g_t t$. Since G preserves the equation of W_N up to a character, $g_x g_y = g_z^2 = g_t^N$. For $\lambda \in \mathbb{C}^*$ let

$$\mathbf{T}(\lambda) := (1, \lambda^{-2c}, \lambda^{-c}, \lambda^v).$$

Since $\gcd(c, v) = 1$, $\lambda \to \mathbf{T}(\lambda)$ is injective, and $R = \{\mathbf{T}(\lambda) \mid \lambda^r = 1\}$. We discard the h_j which are units or have label $c_j = 0$, which does not change D near 0. The proof is long, so we divide it into several steps.

Step 1. In this step we bound the equations h_j and their principal blocks.

Let $\gamma := \min(\Gamma \backslash \{0\})$. If no h_j is left, all sums below are empty. Since (W_N, D) is klt near 0, Lemma A.1(3) gives

$$\sum_j c_j \operatorname{mult}_0 h_j < 2.$$

Let $C_1 := \lceil 2/\gamma \rceil$, $C_2 := \max\{2c, v\}$, and $C_3 := C_1 C_2$. Then there are fewer than C_1 indices j, $\operatorname{mult}_0 h_j < C_1$, $|\kappa_j| \le C_3$, and $|\tau| < 2C_2$. Since $\mathbf{T}(\lambda)$ multiplies a monomial of $\mathbf{T}$-weight k' by $\lambda^{k'}$ and all monomials of h_j have the R-character of h_j, every monomial of h_j has $\mathbf{T}$-weight $\kappa_j + \ell r$ for an integer ℓ. We write $h_j = \sum_\ell h^{(\ell)}_j$, where $h^{(\ell)}_j$ is the sum of the terms of weight $\kappa_j + \ell r$, so $h^{(0)}_j = h^0_j$. A monomial of $h^{(\ell)}_j$ with $\ell \ne 0$ has weight of absolute value at least $r - C_3$, hence degree at least $(r - C_3)/C_2$. In particular, if $r > 2C_3$, then $\operatorname{mult}_0 h^0_j = \operatorname{mult}_0 h_j$. Moreover, $Nv + 2c = rd \ge r$, so

$$N \ge \frac{r - 2c}{v}. \tag{A.186}$$

We let

$$\Phi_j(x_1, z_1) := h^0_j(x_1^2, -z_1^2, x_1 z_1, 1) \quad \text{and} \quad \varphi_j(p, q) := h^0_j(-p^2, 1, p, q).$$

Since h^0_j has no monomial divisible by xy and all monomials of h^0_j have $\mathbf{T}$-weight κ_j, a direct calculation shows that distinct monomials of h^0_j are sent to distinct monomials of Φ_j, and to distinct monomials of φ_j. Thus $\Phi_j \ne 0$, $\varphi_j \ne 0$, and $\operatorname{mult}_0 \Phi_j, \operatorname{mult}_0 \varphi_j \le 2C_1$.

Claim A.43. *Let $\Lambda \subset \mathbb{R}_{>0}$ be a DCC set. There exists r_Λ depending only on Γ, Λ, c, and v with the following property. Let $r \ge r_\Lambda$, let $c'_j \in \Lambda$ for the indices j of the h_j, let U be a non-zero polynomial in x, y, z, t vanishing at 0 and $\lambda_U \in \Lambda \cup \{0\}$, and let $\epsilon \in \{0, 1\}$. Then*

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^N) + \sum_j c'_j \operatorname{div}(h_j) + \lambda_U \operatorname{div}(U)\Big)$$

is lc near 0 *if and only if*

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2 + \epsilon t^N) + \sum_j c'_j \operatorname{div}(h_j^0) + \lambda_U \operatorname{div}(U)\Big)$$

is lc near 0.

Proof. There are fewer than C_1 indices j, so each pair in the claim is given by at most $C_1 + 1$ non-zero functions with labels in the DCC set $\Lambda \cup \{1\} \subset \mathbb{R}_{>0}$, where U is omitted if $\lambda_U = 0$. Let L_0 be the largest of the integers of Lemma 4.14 for $n = 4$, the numbers $s \le C_1 + 1$ of functions, and the DCC set $\Lambda \cup \{1\}$, and let $r_\Lambda := \max\{C_2 L_0 + C_3, vL_0 + 2c\}$. If $r \ge r_\Lambda$, then $\min\{N, (r - C_3)/C_2\} \ge L_0$ by (A.186), so the functions $(1-\epsilon)t^N$ and $h_j - h_j^0$ are in $(x, y, z, t)^{L_0}$. By [dFEM11, Proposition 2.11(i)], each pair in the claim is lc near 0 if and only if its pullback to $\operatorname{Spec} \mathbb{C}[[x, y, z, t]]$ is lc. Thus Lemma 4.14, applied in both directions, gives the claim. □

Step 2. In this step we introduce root models of W_N/G and compute the crepant boundaries on them.

Let $\tilde{G}'_t$ be the group of pairs (g, ξ) with $g \in G$ and $\xi^v = g_t$, and let (g, ξ) act on $\mathbb{A}^4_{X,Y,Z,w}$ by

$$(X, Y, Z, w) \to (g_x X, g_y \xi^{2c} Y, g_z \xi^c Z, \xi^r w).$$

Let

$$X_d := (XY + Z^2 + w^d = 0) \subset \mathbb{A}^4_{X,Y,Z,w} \quad \text{and} \quad S_t := X_d \cap (w = 0) = (XY + Z^2 = 0) \subset \mathbb{A}^3_{X,Y,Z}.$$

Since $\xi^{rd} = \xi^{Nv+2c} = g_t^N \xi^{2c}$ and $g_x g_y = g_z^2 = g_t^N$, the action preserves X_d. The subgroup $\tilde{R}_t := \{(\mathbf{T}(\lambda), \lambda) \mid \lambda^r = 1\}$ acts trivially, and $G'_t := \tilde{G}'_t/\tilde{R}_t$ is a diagonal group of order $v[G : R]$. If (g, ξ) fixes X, Y, and Z, then $g_x = 1$, $g_y = \xi^{-2c}$, $g_z = \xi^{-c}$, and $g_t = \xi^v$, so $g = \mathbf{T}(\xi) \in R$, $\xi^r = 1$, and $(g, \xi) \in \tilde{R}_t$. Thus G'_t acts faithfully on S_t, and hence on X_d. Let $W'_N \to W_N$ be the normalization of the base change of W_N by $\mathbb{A}^1_s \to \mathbb{A}^1_t$, $s \to s^v$, with the induced action of $\tilde{G}'_t$, where (g, ξ) multiplies s by ξ. Then the morphism

$$W'_N \to X_d, \qquad (x, y, z, s) \to (X, Y, Z, w) = (x, y s^{2c}, z s^c, s^r) \tag{A.187}$$

is $\tilde{G}'_t$-equivariant and generically the quotient by $\tilde{R}_t$, since $xy + z^2 + t^N = s^{-2c}(XY + Z^2 + w^d)$. Hence we have the finite map $X_d \to X_d/G'_t$ and the birational map $X_d/G'_t \dashrightarrow W_N/G$ given by (A.187). The threefold X_d is klt near 0: for $d = 1$ it is smooth, and for $d \ge 2$ the blow-up of 0 has an exceptional prime divisor of log discrepancy 2 with respect to X_d, the charts of the blow-up other than the w-chart are smooth along the exceptional divisor, and the w-chart is X_{d-2}, where X_0 is smooth.

Let $\tilde{G}_t$ be the group of triples (g, ξ, η) with $(g, \xi) \in \tilde{G}'_t$ and $\eta^2 = g_x$, acting on $V_t := \mathbb{A}^2_{x_1,z_1} \times \mathbb{A}^1_w$ by

$$(x_1, z_1, w) \to (\eta x_1, g_z \xi^c \eta^{-1} z_1, \xi^r w),$$

and let

$$G_t := \tilde{G}_t/\{(\mathbf{T}(\lambda), \lambda, 1) \mid \lambda^r = 1\},$$

a diagonal group of order $2v[G : R]$. The rational map

$$V_t \dashrightarrow X_d, \qquad (x_1, z_1, w) \to (x_1^2, -z_1^2 - w^d x_1^{-2}, x_1 z_1, w) \tag{A.188}$$

is equivariant with respect to $G_t \to G'_t$, $(g, \xi, \eta) \to (g, \xi)$, and generically the quotient by the involution $\iota := (1, 1, -1)$ of G_t. On $(w = 0)$ it restricts to the quotient $\mathbb{A}^2_{x_1,z_1} \to S_t$ by ι, which is étale in codimension one. An element of G_t acting trivially on $\mathbb{A}^2_{x_1,z_1}$ maps to the identity of G'_t and has $\eta = 1$, so G_t acts faithfully on $\mathbb{A}^2_{x_1,z_1}$.

Let $\tilde{G}_y$ be the group of pairs (g, ζ) with $g \in G$ and $\zeta^{2c} = g_y$, acting on $V_y := \mathbb{A}^3_{u,p,q}$ by

$$(u, p, q) \to (\zeta^r u, g_z \zeta^{-c} p, g_t \zeta^v q),$$

and let

$$G_y := \tilde{G}_y/\{(\mathbf{T}(\lambda), \lambda^{-1}) \mid \lambda^r = 1\},$$

a diagonal group of order $2c[G : R]$. If (g, ζ) fixes p and q, then $g_z = \zeta^c$, $g_t = \zeta^{-v}$, and $g_x = g_z^2/g_y = 1$, so $g = \mathbf{T}(\zeta^{-1}) \in R$ and $\zeta^r = 1$. Thus G_y acts faithfully on $\mathbb{A}^2_{p,q} = (u = 0)$. Writing $y = \varsigma^{2c}$, $u = \varsigma^r$, $p = z\varsigma^{-c}$, and $q = t\varsigma^v$, the equation of W_N becomes

$$x = -p^2 - u^{-d}q^N, \tag{A.189}$$

and (A.189) defines the birational map $V_y/G_y \dashrightarrow W_N/G$ sending (u, p, q) to

$$(x, y, z, t) = (-p^2 - u^{-d}q^N, \varsigma^{2c}, p\varsigma^c, q\varsigma^{-v}), \quad \text{where } \varsigma^r = u.$$

We now write the crepant boundaries on X_d, V_t, and V_y. A monomial $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ of $h_j^{(\ell)}$ equals $s^{\kappa_j}w^\ell X^{n_1}Y^{n_2}Z^{n_3}$ on W'_N, and equals $\varsigma^{-\kappa_j}u^{-\ell}x^{n_1}p^{n_3}q^{n_4}$ in the coordinates of (A.189). Thus

$$h_j = s^{\kappa_j}\sum_\ell w^\ell P_j^{(\ell)}(X, Y, Z) = \varsigma^{-\kappa_j}\sum_\ell u^{-\ell}Q_j^{(\ell)}(x, p, q)$$

for polynomials $P_j^{(\ell)}$ and $Q_j^{(\ell)}$ with $P_j^{(0)}(X, Y, Z) = h_j^0(X, Y, Z, 1)$ and $Q_j^{(0)}(-p^2, p, q) = \varphi_j$. Let

$$\tilde{P}_j := \sum_\ell w^\ell P_j^{(\ell)}(X, Y, Z) \quad \text{and} \quad H_j := \sum_\ell u^{-\ell}Q_j^{(\ell)}(-p^2 - u^{-d}q^N, p, q),$$

rational functions on X_d and V_y. Up to non-zero constants, the generator $\omega := dx \wedge dz \wedge dt/x$ of the canonical sheaf of W_N near 0 satisfies

$$\omega = s^{v-c}\frac{dX \wedge dZ \wedge dw}{Xw} = s^{v-c}\frac{dx_1 \wedge dz_1 \wedge dw}{w} = \varsigma^{c-v}\frac{du \wedge dp \wedge dq}{u},$$

where $dX \wedge dZ \wedge dw/X$ generates the canonical sheaf of X_d near 0. Let

$$\begin{aligned}
B_{X_d} &:= \Big(1 + \frac{\delta}{r}\Big)\operatorname{div}(w) + \sum_j c_j \operatorname{div}(\tilde{P}_j),\\
B_{V_t} &:= \Big(1 + \frac{\delta}{r}\Big)\operatorname{div}(w) + \sum_j c_j \operatorname{div}\big(\tilde{P}_j(x_1^2, -z_1^2 - w^d x_1^{-2}, x_1 z_1, w)\big),\\
B_{V_y} &:= \Big(1 - \frac{\delta}{r}\Big)\operatorname{div}(u) + \sum_j c_j \operatorname{div}(H_j).
\end{aligned}$$

Here $\delta = \tau - \epsilon_{\mathbf{T}} = \tau + c - v$, since $\epsilon_{\mathbf{T}} = (v - 3c) - \kappa(xy + z^2 + t^N) = v - c$. Let $(Y', B_{Y'}, G')$ be one of (X_d, B_{X_d}, G'_t), (V_t, B_{V_t}, G_t), and (V_y, B_{V_y}, G_y), and let $(W_N/G, D_G)$ be the crepant quotient of (W_N, D). We show that for any prime divisor E over W_N/G and any prime divisor F' over Y' which is mapped onto E by the maps above,

$$a(E, W_N/G, D_G) = \frac{a(F', Y', B_{Y'})}{|I_{G'}(F')|}. \tag{A.190}$$

Let $\Omega := \omega\prod_j h_j^{-c_j}$, and let

$$\operatorname{div}(\Omega) := \operatorname{div}(\omega) - \sum_j c_j \operatorname{div}(h_j),$$

where $\operatorname{div}(s) = \operatorname{div}(w)/r$ and $\operatorname{div}(\varsigma) = \operatorname{div}(u)/r$. Since $s^r = w$, $\varsigma^r = u$, and $\sum_j c_j\kappa_j = \tau$, by the expressions of h_j through s and $\tilde{P}_j$ and through ς and H_j, and of ω through generators of the canonical sheaves, near 0 we have $D = -\operatorname{div}(\Omega)$ on W_N and $B_{Y'} = -\operatorname{div}(\Omega)$ on Y'. Thus, by the Hurwitz formula and [Liu26, Lemma 2.6(2)], applied to the finite quotients $W_N \to W_N/G$,

$X_d \to X_d/G'_t$, $V_t \to V_t/G_t$, and $V_y \to V_y/G_y$, the crepant boundaries on W_N/G, X_d/G'_t, V_t/G_t, and V_y/G_y are all $-\operatorname{div}(\Omega)$, and

$$\frac{a(F', Y', B_{Y'})}{|I_{G'}(F')|} = a(E, Y'/G', -\operatorname{div}(\Omega)).$$

Let $\widetilde{W}$ be a common resolution of Y'/G' and W_N/G for the birational map $Y'/G' \dashrightarrow W_N/G$, with morphisms $\pi_1 : \widetilde{W} \to Y'/G'$ and $\pi_2 : \widetilde{W} \to W_N/G$. The crepant pullbacks of $(Y'/G', -\operatorname{div}(\Omega))$ by π_1 and of $(W_N/G, D_G)$ by π_2 both equal $(\widetilde{W}, -\operatorname{div}_{\widetilde{W}}(\Omega))$, so [KM98, Lemma 2.30], applied to π_1 and to π_2, gives $a(E, Y'/G', -\operatorname{div}(\Omega)) = a(E, W_N/G, D_G)$. This proves (A.190).

Let E be a prime divisor over W_N/G with center the image of 0, let F be a prime divisor over W_N mapped onto E, and let F' be as in (A.190). Recall that $[f] = \operatorname{mult}_F f/|I_G(F)|$ for a regular function f near 0. For a rational function ϕ on Y' we write likewise $[\phi] := \operatorname{mult}_{F'} \phi/|I_{G'}(F')|$. If a power ϕ^m is a G-invariant rational function on W_N and a G'-invariant rational function on Y', then both values of $[\phi]$ equal $\operatorname{mult}_E(\phi^m)/m$ [Liu26, Lemma 2.6(1)]. Taking for ϕ the functions in the equalities $x = X$, $y^r = Y^r w^{-2c}$, $z^r = Z^r w^{-c}$, $t^r = w^v$, $y^r = u^{2c}$, $z^2 = p^2 y$, and $t^{2c} = q^{2c} y^{-v}$, we obtain

$$\begin{aligned} &[Y] = [y] + \frac{2c}{r}[w], \quad [Z] = [z] + \frac{c}{r}[w], \quad [w] = \frac{r}{v}[t], \\ &[u] = \frac{r[y]}{2c}, \quad [p] = [z] - \frac{[y]}{2}, \quad [q] = [t] + \frac{v[y]}{2c}. \end{aligned} \tag{A.191}$$

Since $[x], [y], [z], [t] > 0$, the functions X, Y, Z, w have positive values $[\cdot]$, so the center of any prime divisor over X_d mapped onto E is 0.

Step 3. In this step we compare the plane pairs of a $\mathbf{T}$-homogeneous boundary through a pair on a residual plane.

Let $D' = \sum_j c'_j \operatorname{div}(h'_j)$, where $c'_j > 0$ and the h'_j are non-zero G-semi-invariant $\mathbf{T}$-homogeneous polynomials vanishing at 0 without monomials divisible by xy, with $\mathbf{T}$-weights κ'_j. Let $\tau(D') := \sum_j c'_j \kappa'_j$, $\delta(D') := \tau(D') + c - v$,

$$C_t(D') := \sum_j c'_j \operatorname{div}\left(h'_j(x_1^2, -z_1^2, x_1 z_1, 1)\right) \text{ on } \mathbb{A}^2_{x_1,z_1},$$

$$C_y(D') := \sum_j c'_j \operatorname{div}\left(h'_j(-p^2, 1, p, q)\right) \text{ on } \mathbb{A}^2_{p,q},$$

and, when the pairs are lc near 0,

$$\beta_t(D') := \operatorname{mld}((\mathbb{A}^2 \ni 0, C_t(D'))/G_t) \quad \text{and} \quad \beta_y(D') := \operatorname{mld}((\mathbb{A}^2 \ni 0, C_y(D'))/G_y).$$

As in **Step 1**, $h'_j(x_1^2, -z_1^2, x_1 z_1, 1)$ and $h'_j(-p^2, 1, p, q)$ are non-zero. We write

$$D^0 := \sum_j c_j \operatorname{div}(h^0_j), \qquad \beta_t := \beta_t(D^0), \qquad \beta_y := \beta_y(D^0),$$

$$C_t := C_t(D^0) = \sum_j c_j \operatorname{div}(\Phi_j), \qquad C_y := C_y(D^0) = \sum_j c_j \operatorname{div}(\varphi_j).$$

Then $\tau(D^0) = \tau$ and $\delta(D^0) = \delta$.

Let $\mathbb{A}^3_{x_1,q_0,t} \to W_0 = (xy + z^2 = 0) \subset \mathbb{A}^4$ be the morphism $(x_1, q_0, t) \to (x_1^2, -q_0^2, x_1 q_0, t)$, the quotient by $(x_1, q_0, t) \to (-x_1, -q_0, t)$, which is étale in codimension one. The $\mathbf{T}$-weights of x_1, q_0, t are $0, -c, v$. The monomials of $h'_j(x_1^2, -q_0^2, x_1 q_0, t)$ have weight κ'_j, and since $\gcd(c, v) = 1$, any two solutions (n, n') of $vn - cn' = \kappa'_j$ differ by an integer multiple of (c, v). Let (a'_j, b'_j) be the pair with the least n among the exponent pairs (n, n') of $t^n q_0^{n'}$ in the monomials of $h'_j(x_1^2, -q_0^2, x_1 q_0, t)$. Then the exponent pairs of $t^n q_0^{n'}$ in the monomials of $h'_j(x_1^2, -q_0^2, x_1 q_0, t)$ are of the form $(a'_j + ck, b'_j + vk)$ with $k \geq 0$. A monomial $x^{n_1} y^{n_2} z^{n_3} t^{n_4}$ with $n_1 n_2 = 0$ is sent to

$\pm x_1^{2n_1+n_3} q_0^{2n_2+n_3} t^{n_4}$, and distinct such monomials have distinct images, so no terms cancel. Thus, with $M_0 := t^c q_0^v$,

$$h'_j(x_1^2, -q_0^2, x_1 q_0, t) = t^{a'_j} q_0^{b'_j} F_j(M_0, x_1), \qquad \kappa'_j = v a'_j - c b'_j,$$

for polynomials F_j with $F_j(0, x_1) \neq 0$. Let $A_t := \sum_j c'_j a'_j$ and $A_q := \sum_j c'_j b'_j$, so $\tau(D') = vA_t - cA_q$. Let $\Xi := \mathbb{A}^2_{M_0, x_1}$,

$$\rho_t := 1 - \frac{1 - A_q}{v}, \qquad \rho_y := 1 - \frac{1 - A_t}{c}, \qquad \text{so that} \qquad \rho_y - \rho_t = \frac{c - v + vA_t - cA_q}{cv} = \frac{\delta(D')}{cv},$$

and let

$$B_\Xi := \sum_j c'_j \operatorname{div}_\Xi(F_j) + \max\{\rho_t, \rho_y\} \operatorname{div}_\Xi(M_0). \tag{A.192}$$

(i) The morphism $\mathbb{A}^2_{x_1,z_1} \to \Xi$, $(x_1, z_1) \to (z_1^v, x_1)$, is the quotient by the subgroup $\mu^{(t)} := \{(1, \xi, 1) \mid \xi^v = 1\}$ of G_t, which acts by $z_1 \to \xi^c z_1$, that is, by μ_v on z_1, since $\gcd(c, v) = 1$. Since G_t acts faithfully on $\mathbb{A}^2_{x_1,z_1}$, $\mu^{(t)}$ is the kernel of the action of G_t on Ξ. Moreover $h'_j(x_1^2, -z_1^2, x_1 z_1, 1) = z_1^{b'_j} F_j(z_1^v, x_1)$. Thus, by the Hurwitz formula [Liu26, Lemma 2.6(4)], for any real number e' the crepant quotient of $(\mathbb{A}^2, C_t(D') + e' \operatorname{div}(z_1))$ by $\mu^{(t)}$ is

$$\Big(\Xi, \sum_j c'_j \operatorname{div}_\Xi(F_j) + \Big(\rho_t + \frac{e'}{v}\Big) \operatorname{div}_\Xi(M_0)\Big).$$

Similarly, let $\mathrm{i} := \sqrt{-1}$. The morphism $\mathbb{A}^2_{p,q} \to \Xi$, $(p, q) \to (\mathrm{i}^v q^c, -\mathrm{i}p)$, is the quotient by the subgroup $\mu^{(y)} := \{(1, \zeta) \mid \zeta^c = 1\}$ of G_y, which acts by μ_c on q and is the kernel of the action of G_y on Ξ, and

$$h'_j(-p^2, 1, p, q) = \mathrm{i}^{b'_j} q^{a'_j} F_j(\mathrm{i}^v q^c, -\mathrm{i}p).$$

Thus the crepant quotient of $(\mathbb{A}^2, C_y(D') + e' \operatorname{div}(q))$ by $\mu^{(y)}$ is

$$\Big(\Xi, \sum_j c'_j \operatorname{div}_\Xi(F_j) + \Big(\rho_y + \frac{e'}{c}\Big) \operatorname{div}_\Xi(M_0)\Big).$$

A direct calculation shows that $(g, \xi, \eta) \in G_t$ and $(g, \zeta) \in G_y$ act on Ξ by

$$(M_0, x_1) \to (g_z^v g_t^c \eta^{-v} M_0, \eta x_1),$$

with $\eta = g_z \zeta^{-c}$ in the second case, which satisfies $\eta^2 = g_x$. Hence G_t and G_y both act on Ξ through the group H_Ξ, the image of the group of pairs (g, η) with $g \in G$ and $\eta^2 = g_x$.

(ii) By (i) and [Liu26, Lemma 2.6(5)], taking $e' = \delta(D')/c$ on $\mathbb{A}^2_{x_1,z_1}$ and $e' = -\delta(D')/v$ on $\mathbb{A}^2_{p,q}$,

$$\begin{aligned} \beta_y(D') &= \operatorname{mld}\Big(\Big(\mathbb{A}^2 \ni 0, C_t(D') + \frac{\delta(D')}{c} \operatorname{div}(z_1)\Big)/G_t\Big), \\ \beta_t(D') &= \operatorname{mld}\Big(\Big(\mathbb{A}^2 \ni 0, C_y(D') - \frac{\delta(D')}{v} \operatorname{div}(q)\Big)/G_y\Big), \end{aligned} \tag{A.193}$$

where in each line one side is defined if and only if the other side is, both pairs being crepant over the quotient by H_Ξ of the pair

$$\Big(\Xi, \sum_j c'_j \operatorname{div}_\Xi(F_j) + \rho \operatorname{div}_\Xi(M_0)\Big),$$

where $\rho = \rho_y$ in the first line and $\rho = \rho_t$ in the second line. If $\delta(D') \leq 0$, then $(\mathbb{A}^2, C_t(D'))$ is crepant over (Ξ, B_Ξ), and if $\delta(D') \geq 0$, then $(\mathbb{A}^2, C_y(D'))$ is crepant over (Ξ, B_Ξ). In particular, if $\delta(D') = 0$, then $\beta_t(D') = \beta_y(D')$.

(iii) The variables and integers $(x, y, t; p, q)$ of [Liu26, Set-up 14.1] correspond to $(t, q_0, x_1; v, c)$ here, so $u = M_0$, and the primed quantities of [Liu26, Lemma 14.5] are $c'_x = A_t$, $c'_y = A_q$,

$c'_u = \max\{\rho_t, \rho_y\}$, and $B'_T = B_\Xi$. The data of [Liu26, Set-up 14.1] are the group $\mu_r(v, -c, 0)$, which acts faithfully since $\gcd(c, v) = 1$, the set $\{1/2\}$, the integer $\max\{c, v\}$, any positive integer as its e_0, and one label $1/2$ with the polynomial x_1, and the primed data of [Liu26, Lemma 14.5(3)] are $c'_j \geq 0$, a'_j, b'_j, and $F'_j = F_j$, which that lemma allows to be arbitrary non-negative real numbers, non-negative integers, and non-zero polynomials in u, t. The data of [Liu26, Set-up 14.1] above only serve to fix X, u, and T there, which are $\mathbb{A}^3_{x_1,q_0,t}$, M_0, and Ξ here. Thus, by [Liu26, Lemma 14.5(3)], if the pair

$$\Big(\mathbb{A}^3_{x_1,q_0,t}, \sum_j c'_j \operatorname{div}\big(t^{a'_j} q_0^{b'_j} F_j(M_0, x_1)\big)\Big)$$

is lc near 0, then (Ξ, B_Ξ) is lc near 0. Since the morphism $\mathbb{A}^3_{x_1,q_0,t} \to W_0$ is étale in codimension one, by (ii) and [Liu26, Lemma 2.6(2)] we deduce: if the pair $(W_0, \sum_j c'_j \operatorname{div}(h'_j|_{W_0}))$ is lc near 0, then $(\mathbb{A}^2, C_t(D'))$ is lc near 0 when $\delta(D') \leq 0$, and $(\mathbb{A}^2, C_y(D'))$ is lc near 0 when $\delta(D') \geq 0$.

(iv) We show that there exists r_1 depending only on Γ, c, and v such that if $r \geq r_1$, then $(W_0, \sum_j c_j \operatorname{div}(h^0_j|_{W_0}))$ is lc near 0. Let $r_1 := r_\Lambda$ be given by Claim A.43 for $\Lambda = (\Gamma\backslash\{0\}) \cup \{1\}$. Since (W_N, D) is klt near 0, the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^N) + \sum_j c_j \operatorname{div}(h_j)\Big)$$

is lc near 0 [Kaw07, Theorem]. If $r \geq r_1$, then by Claim A.43 with $c'_j = c_j$, $\lambda_U = 0$, and $\epsilon = 0$, the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2) + \sum_j c_j \operatorname{div}(h^0_j)\Big)$$

is lc near 0, so $(W_0, \sum_j c_j \operatorname{div}(h^0_j|_{W_0}))$ is lc near 0 [Kaw07, Theorem]. By (iii), if $r \geq r_1$, then $(\mathbb{A}^2, C_t)$ is lc near 0 when $\delta \leq 0$, and $(\mathbb{A}^2, C_y)$ is lc near 0 when $\delta \geq 0$. We let $\beta := \beta_t$ if $\delta \leq 0$ and $\beta := \beta_y$ if $\delta \geq 0$. If $\delta = 0$, then $\beta_t = \beta_y$ by (ii).

(v) Let $(\mathbb{A}^2, C_t)$ be lc near 0. Then $C_t \in \Gamma_+ \cap [0, 1]$, and the coefficients of the crepant quotient of $(\mathbb{A}^2, C_t)$ by G_t are of the form $1 - (1 - b)/e$ with $b \in \Gamma_+ \cap [0, 1]$ and a positive integer e. They belong to a DCC set Γ' depending only on Γ [HMX14, Proposition 3.4.1(3)]. Likewise, if $(\mathbb{A}^2, C_y)$ is lc near 0, then the crepant quotient of $(\mathbb{A}^2, C_y)$ by G_y has coefficients in Γ'.

Step 4. In this step we show the following. Let D' be as in **Step 3**. If $\delta(D') \leq 0$ and $(\mathbb{A}^2, C_t(D'))$ is lc near 0, then $a(E, W_N/G, D'_G) \geq \beta_t(D')$ for any prime divisor E over W_N/G with center the image of 0, where $(W_N/G, D'_G)$ is the crepant quotient of (W_N, D'), and (W_N, D') is lc near 0. If $\delta(D') \geq 0$ and $(\mathbb{A}^2, C_y(D'))$ is lc near 0, then $a(E, W_N/G, D'_G) \geq \beta_y(D')$ for any such E, and (W_N, D') is lc near 0.

Let E be such a prime divisor and F' a prime divisor over X_d mapped onto E. By **Step 2**, the center of F' on X_d is 0. Let $P'_j := h'_j(X, Y, Z, 1)$, a G'_t-semi-invariant polynomial with $h'_j = s^{\kappa'_j} P'_j$ on W'_N. By (A.190) for D' in place of D,

$$|I_{G'_t}(F')| \cdot a(E, W_N/G, D'_G) = a\Big(F', X_d, \Big(1 + \frac{\delta(D')}{r}\Big) \operatorname{div}(w) + \sum_j c'_j \operatorname{div}(P'_j)\Big).$$

The hypotheses of Lemma 5.5 hold with $n = 4$, the group G'_t acting diagonally on $\mathbb{A}^4_{X,Y,Z,w}$ in place of G, the hypersurface $X_d = (XY + Z^2 + w^d = 0)$ in place of X, which near 0 is normal, integral, and klt by **Step 2** and on which G'_t acts faithfully, the function w in place of u, the section $S_0 = S_t$, which is an A_1 surface singularity, hence normal and klt, the group $H = G'_t$, which acts faithfully on S_t by **Step 2**, and $b = 1$.

If $\delta(D') \leq 0$, let $\mathfrak{a} := \prod_j (P'_j)^{c'_j}$. The pullback of $\mathfrak{a}|_{S_t}$ to $\mathbb{A}^2_{x_1,z_1}$ is the ideal of $C_t(D')$. Since $\mathbb{A}^2_{x_1,z_1} \to S_t$ is the quotient by $\iota \in G_t$ and is étale in codimension one, $(S_t, \mathfrak{a}|_{S_t})$ is lc near 0 and

$\operatorname{mld}((S_t \ni 0, \mathfrak{a}|_{S_t})/G'_t) = \beta_t(D')$ [Liu26, Lemma 2.6(5)]. By Lemma 5.5,

$$\frac{a(F', X_d, S_t + \sum_j c'_j \operatorname{div}(P'_j))}{|I_{G'_t}(F')|} \geq \beta_t(D'),$$

and since $-\delta(D') \operatorname{mult}_{F'} w/r \geq 0$, we obtain $a(E, W_N/G, D'_G) \geq \beta_t(D')$.

If $\delta(D') \geq 0$, let $\mathfrak{a} := Y^{\delta(D')/(2c)} \prod_j (P'_j)^{c'_j}$, the factor $Y^{\delta(D')/(2c)}$ being omitted if $\delta(D') = 0$. Since $Y|_{S_t}$ pulls back to $-z_1^2$, the pullback of $\mathfrak{a}|_{S_t}$ to $\mathbb{A}^2_{x_1,z_1}$ is the $\mathbb{R}$-ideal of $C_t(D') + (\delta(D')/c)\operatorname{div}(z_1)$, so $(S_t, \mathfrak{a}|_{S_t})$ is lc near 0 and $\operatorname{mld}((S_t \ni 0, \mathfrak{a}|_{S_t})/G'_t) = \beta_y(D')$ by (A.193) and [Liu26, Lemma 2.6(5)]. By Lemma 5.5 and (A.191),

$$\begin{aligned} a(E, W_N/G, D'_G) &= \frac{a(F', X_d, S_t, \mathfrak{a}|_{X_d})}{|I_{G'_t}(F')|} + \delta(D')\Big(\frac{[Y]}{2c} - \frac{[w]}{r}\Big) \\ &\geq \beta_y(D') + \frac{\delta(D')[y]}{2c} \geq \beta_y(D'). \end{aligned} \tag{A.194}$$

In both cases $\operatorname{mld}((W_N \ni 0, D')/G) \geq 0$, so $(W_N/G, D'_G)$ is lc near the image of 0 [Amb99, Section 1, the second paragraph after Definition 1.1], and (W_N, D') is lc near 0 [Liu26, Lemma 2.6(2)].

Step 5. In this step we show that there exist r_2 and C_6 depending only on Γ, e_0, c, and v such that if $r \geq r_2$, then there exist positive integers k and e with $k \leq C_6$ and

$$\alpha \leq \beta_t - \frac{\delta k}{re} \text{ if } \delta \leq 0, \qquad \alpha \leq \beta_y + \frac{\delta k}{re} \text{ if } \delta \geq 0; \qquad \text{in particular} \qquad \alpha \leq \beta + \frac{3C_2C_6}{r}. \tag{A.195}$$

Let $r \geq \max\{r_1, 2C_3 + 1\}$. Assume first that $\delta \leq 0$. By **Step 3**(iv) and (v), the crepant quotient of $(\mathbb{A}^2, C_t)$ by G_t is lc near the image of 0 with coefficients in Γ'. By Theorem 2.9(2) for the DCC set Γ', there exist C_4 depending only on Γ and a prime divisor E_0 over $\mathbb{A}^2/G_t$ with center the image of 0 computing β_t such that $a(E_0, \mathbb{A}^2/G_t, 0) \leq C_4$. Let ν be a prime divisor over $\mathbb{A}^2_{x_1,z_1}$ mapped onto E_0, and $e := |I_{G_t}(\nu)| \leq |G_t| \leq 2ve_0$. Since the branch divisor Δ_t of $\mathbb{A}^2 \to \mathbb{A}^2/G_t$ satisfies $\Delta_t \geq 0$, by [Liu26, Lemma 2.6(2)],

$$a(\nu, \mathbb{A}^2, 0) \leq e \cdot a(E_0, \mathbb{A}^2/G_t, 0) \leq C_5 := 2C_2e_0C_4.$$

Since $(\mathbb{A}^2, \operatorname{div}(x_1) + \operatorname{div}(z_1))$ and $(\mathbb{A}^2, C_t)$ are lc and $c_j \geq \gamma$,

$$\operatorname{mult}_\nu x_1 + \operatorname{mult}_\nu z_1 \leq C_5 \quad \text{and} \quad \operatorname{mult}_\nu \Phi_j \leq \frac{C_5}{\gamma}.$$

Let χ be the character of w under G_t, and q_ν the integer of Lemma 3.1 for ν and χ. Let k be the least integer with $k \equiv q_\nu \pmod e$ and $k > 4C_5 + C_5/\gamma$, so $k \leq C_6 := 4C_5 + C_5/\gamma + 2C_2e_0 + 1$. Let F be the Gauss extension of mult_ν with w-value k. By Proposition 5.4 for $S = \mathbb{A}^2_{x_1,z_1}$, $V = V_t$, $\tilde{H} = G_t$, which acts faithfully on S, the character χ, the boundary $\operatorname{div}(w) + \sum_j c_j \operatorname{div}(\Phi_j)$ with the functions Φ_j independent of w, and the prime divisor ν, F is a prime divisor over V_t with center 0, $|I_{G_t}(F)| = e$, $\operatorname{mult}_F \Phi_j = \operatorname{mult}_\nu \Phi_j$, and

$$\frac{a(F, V_t, \operatorname{div}(w) + \sum_j c_j \operatorname{div}(\Phi_j))}{e} = \beta_t.$$

We show that mult_F of $\tilde{P}_j(x_1^2, \tilde{Y}, x_1z_1, w)$ is $\operatorname{mult}_\nu \Phi_j$, where $\tilde{Y} := -z_1^2 - w^d x_1^{-2}$. Since $\operatorname{mult}_F(w^d x_1^{-2}) \geq k - 2C_5 > 2C_5 + C_5/\gamma$, $\operatorname{mult}_F \tilde{Y} = 2\operatorname{mult}_\nu z_1 \geq 0$ and $\operatorname{mult}_F(\tilde{Y} + z_1^2) > C_5/\gamma$. Since $P_j^{(0)}(x_1^2, \tilde{Y}, x_1z_1) - \Phi_j$ is divisible by $\tilde{Y} + z_1^2$ with a quotient which is a polynomial in $x_1, z_1, \tilde{Y}$, $\operatorname{mult}_F(P_j^{(0)}(x_1^2, \tilde{Y}, x_1z_1) - \Phi_j) > C_5/\gamma$. For $\ell \geq 1$,

$$\operatorname{mult}_F\left(w^\ell P_j^{(\ell)}(x_1^2, \tilde{Y}, x_1z_1)\right) \geq \ell k > \frac{C_5}{\gamma}.$$

For $\ell = -m \le -1$, a monomial $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ of $h_j^{(\ell)}$ satisfies $2cn_2+cn_3 = vn_4-\kappa_j+mr \ge mr-C_3$, so $n_2+n_3 \ge (mr-C_3)/(2c)$, and the corresponding term of $w^{-m}P_j^{(-m)}(x_1^2, \tilde{Y}, x_1z_1)$ has value at least

$$-mk+n_2+n_3 \ge m\Big(\frac{r}{2c}-k\Big)-\frac{C_3}{2c} > \frac{C_5}{\gamma}$$

once $r > 2c(C_6+C_3+C_5/\gamma)$. Since $\operatorname{mult}_F \Phi_j = \operatorname{mult}_\nu \Phi_j \le C_5/\gamma$, the claim follows. Hence

$$\frac{a(F,V_t,B_{V_t})}{e} = \frac{a(F,V_t,\operatorname{div}(w)+\sum_j c_j \operatorname{div}(\Phi_j))}{e} - \frac{\delta k}{re} = \beta_t - \frac{\delta k}{re}.$$

On V_t, $x = x_1^2$, $y^r = \tilde{Y}^r w^{-2c}$, $z^r = (x_1z_1)^r w^{-c}$, and $t^r = w^v$, so the values of x, y, z, t along F divided by e are

$$\frac{2\operatorname{mult}_\nu x_1}{e},\quad \frac{2\operatorname{mult}_\nu z_1}{e}-\frac{2ck}{re},\quad \frac{\operatorname{mult}_\nu x_1+\operatorname{mult}_\nu z_1}{e}-\frac{ck}{re},\quad \frac{vk}{re},$$

which are positive for $r > 2ck$. Thus F is mapped onto a prime divisor E over W_N/G with center the image of 0, and (A.190) implies $\alpha \le a(E, W_N/G, D_G) = \beta_t - \delta k/(re)$.

Assume that $\delta \ge 0$. By **Step 3**(iv) and (v) and Theorem 2.9(2), as in the case $\delta \le 0$, $(\mathbb{A}^2, C_y)/G_y$ has a prime divisor computing β_y whose lift ν over $\mathbb{A}^2_{p,q}$ satisfies $e := |I_{G_y}(\nu)| \le 2C_2e_0$, $\operatorname{mult}_\nu p+\operatorname{mult}_\nu q \le C_5$, and $\operatorname{mult}_\nu \varphi_j \le C_5/\gamma$. Let χ be the character of u under G_y, q_ν the integer of Lemma 3.1 for ν and χ, and k the least integer with $k \equiv q_\nu \pmod e$ and $k > 4C_5 + C_5/\gamma$, so $k \le C_6$, and let F be the Gauss extension of mult_ν with u-value k. By Proposition 5.4 for $S = \mathbb{A}^2_{p,q}$, $V = V_y$, $\tilde{H} = G_y$, the character χ, the boundary

$$\Theta_y := \operatorname{div}(u) + \sum_j c_j \operatorname{div}(\varphi_j),$$

and the prime divisor ν, F is a prime divisor over V_y with center 0, $|I_{G_y}(F)| = e$, $\operatorname{mult}_F \varphi_j = \operatorname{mult}_\nu \varphi_j$, and $a(F, V_y, \Theta_y)/e = \beta_y$. Let $\tilde{x} := -p^2 - u^{-d}q^N$. Since $Nv+2c = rd$ (**Step 1**) and $d \ge 1$,

$$\operatorname{mult}_F(u^{-d}q^N) = N\operatorname{mult}_\nu q - dk = d\Big(\frac{r}{v}\operatorname{mult}_\nu q - k\Big) - \frac{2c}{v}\operatorname{mult}_\nu q \ge \frac{r}{v} - k - \frac{2c}{v}C_5, \tag{A.196}$$

which exceeds $2C_5+C_5/\gamma$ for r large depending only on Γ, e_0, c, and v. Thus $\operatorname{mult}_F \tilde{x} = 2\operatorname{mult}_\nu p$, and $Q_j^{(0)}(\tilde{x}, p, q) - \varphi_j$ has value larger than C_5/γ. For $\ell = -m \le -1$, $u^m Q_j^{(-m)}(\tilde{x}, p, q)$ has value at least $mk > C_5/\gamma$. For $\ell = m \ge 1$, a monomial of $h_j^{(m)}$ satisfies

$$vn_4 = \kappa_j + mr + 2cn_2 + cn_3 \ge mr - C_3,$$

so the corresponding term of $u^{-m}Q_j^{(m)}(\tilde{x}, p, q)$ has value at least $-mk + (mr-C_3)/v > C_5/\gamma$ for $r > v(C_6+C_3+C_5/\gamma)$. Hence $\operatorname{mult}_F H_j = \operatorname{mult}_\nu \varphi_j$ and

$$\frac{a(F,V_y,B_{V_y})}{e} = \beta_y + \frac{\delta k}{re}.$$

Since $x = \tilde{x}$, $y^r = u^{2c}$, $z^2 = p^2y$, and $t^{2c} = q^{2c}y^{-v}$, the values of x, y, z, t along F divided by e are

$$\frac{2\operatorname{mult}_\nu p}{e},\quad \frac{2ck}{re},\quad \frac{\operatorname{mult}_\nu p}{e}+\frac{ck}{re},\quad \frac{\operatorname{mult}_\nu q}{e}-\frac{vk}{re},$$

which are positive for $r > vk$. Thus F is mapped onto a prime divisor E over W_N/G with center the image of 0, and (A.190) implies $\alpha \le a(E, W_N/G, D_G) = \beta_y + \delta k/(re)$. Since $|\delta| \le |\tau| + |c-v| < 3C_2$, the last inequality of (A.195) follows.

Step 6. In this step we show the lower bound under a margin. Let $\Lambda \subset [0,1]$ be a DCC set containing Γ, and let $\theta > 1$. We show that there exists r_3 depending only on Γ, Λ, e_0, c, v, and θ with the following property. Let $r \ge r_3$, let $c'_j \in \Lambda$ with $c'_j \ge c_j$, let $D' := \sum_j c'_j \operatorname{div}(h_j)$ and $D'^0 := \sum_j c'_j \operatorname{div}(h_j^0)$, let $\delta' := \delta(D'^0)$, and assume that either

(1) $\delta' \le 0$, $\delta(\theta D'^0) \le 0$, and $(\mathbb{A}^2, \theta C_t(D'^0))$ is lc near 0, in which case let $\beta' := \beta_t(D'^0)$, or
(2) $\delta' \ge 0$, $\delta(\theta D'^0) \ge 0$, and $(\mathbb{A}^2, \theta C_y(D'^0))$ is lc near 0, in which case let $\beta' := \beta_y(D'^0)$.

Then $(W_N, \theta D')$ is lc near 0 and $\mathrm{mld}((W_N \ni 0, D')/G) \ge \beta'$. By definition,

$$\delta(\theta D'^0) = \theta\tau(D'^0) + c - v \quad \text{and} \quad C_t(\theta D'^0) = \theta C_t(D'^0).$$

By **Step 4** for $\theta D'^0$, the pair $(W_N, \theta D'^0)$ is lc near 0, so $(\mathbb{A}^4, \mathrm{div}(xy + z^2 + t^N) + \theta D'^0)$ is lc near 0 [Kaw07, Theorem]. We take r_3 at least the integer r_Λ of Claim A.43 for the DCC set $(\theta\Lambda \cup \{1\})\setminus\{0\}$. By Claim A.43 with the labels $\theta c'_j$, $\lambda_U = 0$, and $\epsilon = 1$, $(\mathbb{A}^4, \mathrm{div}(xy+z^2+t^N)+\theta D')$ is lc near 0, and hence $(W_N, \theta D')$ is lc near 0 [Kaw07, Theorem].

Let E be a prime divisor over W_N/G with center the image of 0, let $a_E := a(E, W_N/G, D'_G)$, and assume that $a_E \le 2$. Let F be a prime divisor over W_N mapped onto E, and $a_0 := a(F, W_N, 0)/|I_G(F)|$. Since

$$a(F, W_N, \theta D') = \theta a(F, W_N, D') - (\theta - 1)a(F, W_N, 0) \ge 0 \quad \text{and}$$
$$a(F, W_N, \theta D') = a(F, W_N, 0) - \theta \sum_j c'_j \,\mathrm{mult}_F\, h_j,$$

we obtain

$$a_0 \le C_7 := \frac{2\theta}{\theta - 1} \quad \text{and} \quad [h_j] \le C_8 := \frac{2}{\gamma(\theta - 1)}. \tag{A.197}$$

Let $S' := W_N \cap (t = 0) = (xy + z^2 = 0) \subset \mathbb{A}^3$ and $L_x := (x = z = 0)$, $L_y := (y = z = 0)$. Then $\mathrm{div}(x)|_{S'} = 2L_x$, $\mathrm{div}(y)|_{S'} = 2L_y$, and $\mathrm{div}(z)|_{S'} = L_x + L_y$, and $(S', L_x + L_y)$ is the toric A_1 surface singularity with toric boundary, which is lc. By [Kaw07, Theorem], the pairs

$$\Big(W_N, \frac{1}{2}\,\mathrm{div}(x) + \frac{1}{2}\,\mathrm{div}(y) + \mathrm{div}(t)\Big) \quad \text{and} \quad \big(W_N, \mathrm{div}(z) + \mathrm{div}(t)\big)$$

are lc near 0. Thus $\frac{1}{2}([x] + [y]) + [t] \le a_0$ and $[z] + [t] \le a_0$, so

$$[x], [y] \le 2C_7 \quad \text{and} \quad [z], [t] \le C_7.$$

Let $\sigma := \gcd(2, v)$ and $\psi := y^{v/\sigma} t^{2c/\sigma}$. The monomials x, ψ, and $z^v t^c$ have $\mathbf{T}$-weight zero, so R acts trivially on them, and their characters have order at most $[G : R] \le e_0$. Since an invariant power of each of them vanishes at the image of 0,

$$[x] \ge \frac{1}{e_0}, \qquad \frac{v}{\sigma}[y] + \frac{2c}{\sigma}[t] \ge \frac{1}{e_0}, \qquad v[z] + c[t] \ge \frac{1}{e_0}. \tag{A.198}$$

For $\ell \ge 1$, a monomial of $h_j^{(\ell)}$ has $n_4 \ge (r - C_3)/v$, so the value $[\cdot]$ of such a monomial is at least $(r - C_3)[t]/v$. For $\ell \le -1$, a monomial of $h_j^{(\ell)}$ has $2cn_2 + cn_3 \ge r - C_3$, so the value of such a monomial is at least $(r - C_3)\min\{[y]/(2c), [z]/c\}$. Consider the conditions

$$\text{(A)} \quad [t] \le \frac{vC_8}{r - C_3}, \qquad \text{(B)} \quad \min\left\{\frac{[y]}{2c}, \frac{[z]}{c}\right\} \le \frac{C_8}{r - C_3}.$$

If (A) fails, every monomial of $h_j^{(\ell)}$ with $\ell \ge 1$ has value larger than C_8, and if (B) fails, every monomial of $h_j^{(\ell)}$ with $\ell \le -1$ has value larger than C_8. By (A.198), (A) and (B) do not hold together once r is large depending only on Γ, Λ, e_0, c, v, and θ, and we assume that r is so large. We show $a_E \ge \beta'$ in the three remaining cases.

If neither (A) nor (B) holds, then $h_j - h_j^0$ is a sum of monomials of value larger than $C_8 \ge [h_j]$, so $[h_j^0] = [h_j]$ for any j, and $a_E = a(E, W_N/G, D'^0_G) \ge \beta'$ by **Step 4**, where $(W_N/G, D'^0_G)$ is the crepant quotient of (W_N, D'^0).

If (A) holds, then (B) fails. Let $\tilde{P}_j^+ := \sum_{\ell \ge 0} w^\ell P_j^{(\ell)}(X, Y, Z)$, a G'_t-semi-invariant polynomial with $\tilde{P}_j^+|_{w=0} = P_j^{(0)}(X, Y, Z) = h_j^0(X, Y, Z, 1)$. Since $s^{\kappa_j}\tilde{P}_j^+ = \sum_{\ell \ge 0} h_j^{(\ell)}$ and the monomials of

$h_j^{(\ell)}$ with $\ell \le -1$ have value larger than $C_8 \ge [h_j]$, we have $[\tilde{P}_j^+] = [\tilde{P}_j]$ for any prime divisor F' over X_d mapped onto E. By (A.190) for D',

$$a_E = \frac{a(F', X_d, (1+\delta'/r)\operatorname{div}(w) + \sum_j c'_j \operatorname{div}(\tilde{P}_j^+))}{|I_{G'_t}(F')|}.$$

Since $\tilde{P}_j^+|_{S_t} = P_j^{(0)}|_{S_t}$, the hypotheses of Lemma 5.5 hold for the data of **Step 4** with $\mathfrak{a} := \prod_j (\tilde{P}_j^+)^{c'_j}$ if $\delta' \le 0$ and $\mathfrak{a} := Y^{\delta'/(2c)} \prod_j (\tilde{P}_j^+)^{c'_j}$ if $\delta' \ge 0$, and the section pairs are those of **Step 4**. By Lemma 5.5, as in (A.194), $a_E \ge \beta_t(D'^0)$ if $\delta' \le 0$ and $a_E \ge \beta_y(D'^0) + \delta'[y]/(2c)$ if $\delta' \ge 0$. Thus $a_E \ge \beta'$.

If (B) holds, then (A) fails and the monomials of $h_j^{(\ell)}$ with $\ell \le -1$ may have small values. By (B) and (A.198), $[t] \ge \eta_0 := 1/(4ce_0)$ once r is large. By (A.186), for r large,

$$N[t] \ge \frac{(r-2c)\eta_0}{v} > 4C_7 \ge \max\{[x]+[y], 2[z]\}.$$

Since $xy + z^2 = -t^N$ on W_N, it follows that $[x] + [y] = 2[z]$. If $[z] \le cC_8/(r - C_3)$, then $[x] \le 2[z] < 1/e_0$ for r large, contradicting (A.198). Thus $[y] \le 2cC_8/(r - C_3)$. Let F'' be a prime divisor over V_y mapped onto E. By (A.191),

$$\begin{aligned} [p] &= [z] - \frac{[y]}{2} = \frac{[x]}{2} \ge \frac{1}{2e_0}, \\ [q] &= [t] + \frac{v[y]}{2c} \in [\eta_0, C_7 + vC_8], \\ [u] &= \frac{r[y]}{2c} \le \frac{rC_8}{r - C_3} \le 2C_8, \end{aligned} \tag{A.199}$$

for $r \ge 2C_3$. Thus the center of F'' on V_y is 0. The monomials of $h_j^{(\ell)}$ with $\ell \ge 1$ have value at least $(r - C_3)\eta_0/v > C_8$ for r large. Let

$$H'_j := \sum_{\ell \le 0} u^{-\ell} Q_j^{(\ell)}(-p^2, p, q).$$

Since the characters of x and $-p^2$ under G_y are equal, H'_j is a G_y-semi-invariant polynomial on V_y with $H'_j|_{u=0} = \varphi_j$. The function $H_j - H'_j$ is the sum of $\sum_{\ell \ge 1} u^{-\ell} Q_j^{(\ell)}(\tilde{x}, p, q)$, whose terms have value $[\cdot]$ larger than $C_8 + \kappa_j[u]/r$, and of

$$\sum_{\ell \le 0} u^{-\ell} \big(Q_j^{(\ell)}(\tilde{x}, p, q) - Q_j^{(\ell)}(-p^2, p, q)\big),$$

which is divisible by $\tilde{x} + p^2 = -u^{-d} q^N$ with a quotient of non-negative value. Since $N = (rd - 2c)/v$, by (A.199),

$$[u^{-d} q^N] = d\Big(\frac{r}{v}[q] - [u]\Big) - \frac{2c}{v}[q] \ge \frac{r\eta_0}{v} - 2C_8 - \frac{2c}{v}(C_7 + vC_8),$$

which tends to $+\infty$ uniformly in $d \ge 1$. Since $[H_j] = [h_j] + \kappa_j[u]/r \le C_8 + \kappa_j[u]/r$, we obtain $[H'_j] = [H_j]$ for r large. By (A.190) for D',

$$a_E \cdot |I_{G_y}(F'')| = a\Big(F'', V_y, \Big(1 - \frac{\delta'}{r}\Big)\operatorname{div}(u) + \sum_j c'_j \operatorname{div}(H'_j)\Big).$$

The hypotheses of Lemma 5.5 hold for $X = \mathbb{A}^3_{u,p,q}$ with the diagonal group G_y, the function u, the section $S_0 = \mathbb{A}^2_{p,q}$, the group $H = G_y$, which acts faithfully on S_0, and $b = 1$. If $\delta' \ge 0$, let $\mathfrak{a} := \prod_j (H'_j)^{c'_j}$. Then $\mathfrak{a}|_{S_0}$ is the $\mathbb{R}$-ideal of $C_y(D'^0)$, and by Lemma 5.5,

$$a_E \ge \beta_y(D'^0) + \frac{\delta'[u]}{r} \ge \beta'.$$

If $\delta' \le 0$, let $\mathfrak{a} := q^{-\delta'/v}\prod_j (H'_j)^{c'_j}$. Then $\mathfrak{a}|_{S_0}$ is the $\mathbb{R}$-ideal of $C_y(D'^0) - (\delta'/v)\operatorname{div}(q)$, whose quotient by G_y has minimal log discrepancy $\beta_t(D'^0)$ by (A.193), and by (A.191),

$$a_E \ge \beta_t(D'^0) + (-\delta')\Big(\frac{[q]}{v} - \frac{[u]}{r}\Big) = \beta_t(D'^0) + \frac{(-\delta')[t]}{v} \ge \beta'. \tag{A.200}$$

Finally, the exceptional divisor ν_0 of the blow-up of $0 \in \mathbb{A}^2$ is invariant under the diagonal groups G_t and G_y and satisfies $a(\nu_0, \mathbb{A}^2, 0) = 2$, so $\beta' \le 2$. Hence $a_E \ge \beta'$ also when $a_E > 2$, and $\operatorname{mld}((W_N \ni 0, D')/G) \ge \beta'$.

Step 7. In this step we set up a sequence contradicting the theorem, take a compatible generic limit of plane pairs, and exclude the case when the limit is klt.

Assume that there exists a sequence of data as in the theorem, indexed by i, with Γ, e_0, and w fixed, $r_i \to +\infty$, and values α_i positive and strictly increasing. We add the index i to the notation. All constants $C_1, \dots, C_8$, r_1, r_2, and r_3 above depend only on Γ, e_0, c, v, and the indicated Λ and θ, and not on d. Possibly passing to a subsequence, we may assume the following.

(1) $r_i \ge \max\{r_1, r_2, 2C_3 + 1\}$.
(2) The number s of the h_{ij} and the weights $\kappa_j := \kappa_{ij}$ do not depend on i.
(3) For any j, c_{ij} is non-decreasing in i with limit $c^*_j \in (0, 1]$.
(4) Either $\delta_i \le 0$ for any i or $\delta_i \ge 0$ for any i.
(5) The groups $G_{t,i}$ and $G_{y,i}$, which are diagonal subgroups of $(\mathbb{C}^*)^2$ of order at most $2C_2e_0$, do not depend on i, and neither do the characters of Φ_{ij} and φ_{ij}.

Let $\tau^* := \sum_j c^*_j\kappa_j$ and $\delta_* := \tau^* + c - v = \lim_i \delta_i$.

The values β_i are minimal log discrepancies of lc surface germs with coefficients in Γ' (**Step 3**(v)), so they satisfy the ACC by Theorem 2.9(1). Possibly passing to a subsequence, β_i is non-increasing with limit L. By (A.195), $\alpha_i < \lim_{i'} \alpha_{i'} \le L$ for any i, and $L > 0$.

Let $D^*_i := \sum_j c^*_j \operatorname{div}(h_{ij})$ and $D^{*0}_i := \sum_j c^*_j \operatorname{div}(h^0_{ij})$, so $\delta(D^{*0}_i) = \delta_*$. Let (C_i, G_β) be $(C_{t,i}, G_t)$ if $\delta_i \le 0$ and $(C_{y,i}, G_y)$ if $\delta_i \ge 0$, and let C^*_i be the corresponding plane boundary of D^{*0}_i. Since $(\mathbb{A}^2, C_i)$ is lc near 0 and $c_{ij}/c^*_j \to 1$, the lc threshold of C^*_i on $\mathbb{A}^2$ at 0 tends to a number at least one. By the ACC for lc thresholds on $\mathbb{A}^2$ with exponents in $\{c^*_j\}_j$ [dFEM11, Theorem 1.1], $(\mathbb{A}^2, C^*_i)$ is lc near 0 for $i \gg 0$. Let $\beta^*_i := \operatorname{mld}((\mathbb{A}^2 \ni 0, C^*_i)/G_\beta) \le \beta_i$. By Theorem 2.9(2) for the DCC set obtained as in **Step 3**(v) from $\Gamma \cup \{c^*_j\}_j$ in place of Γ, as in **Step 5**, β^*_i is computed by a prime divisor whose lift ν_i over $\mathbb{A}^2$ has $a(\nu_i, \mathbb{A}^2, 0)$ and the orders of the plane equations of h^0_{ij} along ν_i bounded independently of i. Thus, writing Ψ_{ij} for the plane equation of h^0_{ij} on the plane of C_i,

$$\beta^*_i \le \beta_i \le \frac{a(\nu_i, \mathbb{A}^2, C_i)}{|I_{G_\beta}(\nu_i)|} \le \beta^*_i + \sum_j (c^*_j - c_{ij}) \operatorname{mult}_{\nu_i} \Psi_{ij},$$

and the last sum tends to 0, so $\beta^*_i \to L$. Since β^*_i is determined by bounded integers and the fixed labels c^*_j, it takes finitely many values, so $\beta^*_i = L$ for $i \gg 0$.

Let $\mathbb{A}^2_P$ with group G_P be the t-plane $\mathbb{A}^2_{x_1,z_1}$ with G_t if $\delta_* < 0$, or $\delta_* = 0$ and $v \le c$, and the y-plane $\mathbb{A}^2_{p,q}$ with G_y if $\delta_* > 0$, or $\delta_* = 0$ and $v > c$. Let $D^0_i := \sum_j c_{ij} \operatorname{div}(h^0_{ij})$, and for a boundary D' as in **Step 3** let $C_P(D') := C_t(D')$, resp. $C_y(D')$, according as $\mathbb{A}^2_P = \mathbb{A}^2_{x_1,z_1}$, resp. $\mathbb{A}^2_{p,q}$. Let $C^*_{P,i} := C_P(D^{*0}_i)$. If $\delta_* \ne 0$, then $C^*_{P,i} = C^*_i$. If $\delta_* = 0$, then $\delta(D^{*0}_i) = 0$, and by **Step 3**(ii) the pairs $(\mathbb{A}^2_{x_1,z_1}, C_t(D^{*0}_i))$ and $(\mathbb{A}^2_{p,q}, C_y(D^{*0}_i))$ are both crepant over the pair (Ξ, B_Ξ) of (A.192) for $D' = D^{*0}_i$. In both cases, possibly passing to a subsequence, $(\mathbb{A}^2_P, C^*_{P,i})$ is lc near 0 and $\operatorname{mld}((\mathbb{A}^2_P \ni 0, C^*_{P,i})/G_P) = L$ for any i.

We apply Construction-Proposition 4.1(a) with $n = 2$ and $H = G_P$ to the tuples consisting of the plane equations of $h^0_{i1}, \dots, h^0_{is}$ on $\mathbb{A}^2_P$ followed by the fixed polynomials $x_1^2, z_1^{2v/\sigma}$ if $\mathbb{A}^2_P = \mathbb{A}^2_{x_1,z_1}$, and $p^2, q^{2c/\sigma}$ if $\mathbb{A}^2_P = \mathbb{A}^2_{p,q}$. The entries are non-zero G_P-semi-invariant polynomials with characters independent of i and with bounded orders at 0. Since the last two entries do not

depend on i, their compatible generic limits are the fixed polynomials up to non-zero constants. By the statement of Construction-Proposition 4.1 on subsets $J \subset \{1, \dots, s+2\}$, the sub-tuple of the first s entries, and that sub-tuple followed by one of the last two entries, are compatible generic limits of the corresponding sub-tuples. Let $\mathbb{K} \supset \mathbb{C}$ be the algebraically closed field given by Construction-Proposition 4.1, and let $\mathcal{A}_{\mathbb{K}}$ be $\operatorname{Spec} \mathbb{K}[[x_1, z_1]]$, resp. $\operatorname{Spec} \mathbb{K}[[p, q]]$, with the induced action of G_P (the germ $\mathcal{A}_k$ of Construction-Proposition 4.1 for $k = \mathbb{K}$). Let

$$C_\infty := \sum_{j \le s} c_j^* \operatorname{div}(f_{\infty,j}).$$

By Theorem 4.2(1) and (2) with $\mathfrak{c} = \mathcal{O}$, possibly passing to a subsequence, $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ is lc near 0 and $\operatorname{mld}((\mathcal{A}_{\mathbb{K}} \ni 0, C_\infty)/G_P) = L$.

Assume that $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ is klt near 0. By Theorem 4.2(1), there exists $\theta_1 > 1$ such that $(\mathbb{A}^2_P, \theta_1 C^*_{P,i})$ is lc near 0 for any i.

If $\delta_* \neq 0$, choose $\theta \in (1, \theta_1]$ such that $\delta_* + (\theta - 1)\tau^*$ has the sign of δ_*. Since

$$\delta(\theta D_i^0) = \theta\tau_i + c - v \to \delta_* + (\theta - 1)\tau^*$$

and $\delta_i \to \delta_*$, for $i \gg 0$ both δ_i and $\delta(\theta D_i^0)$ have the sign of δ_*, and $(\mathbb{A}^2_P, \theta C_P(D_i^0))$ is lc near 0 as $c_{ij} \le c_j^*$, where $C_P(D_i^0) = C_i$. By **Step 6** with $\Lambda = \Gamma$, $c'_j = c_{ij}$, and θ, $\alpha_i \ge \beta_i \ge L$ for $i \gg 0$, a contradiction.

If $\delta_* = 0$, let $\theta := \theta_1$. Then $\tau^* = v - c$, $\delta(D_i^{*0}) = 0$, and

$$\delta(\theta D_i^{*0}) = \theta\tau^* + c - v = (\theta - 1)(v - c),$$

which is non-positive if $\mathbb{A}^2_P = \mathbb{A}^2_{x_1,z_1}$ and positive if $\mathbb{A}^2_P = \mathbb{A}^2_{p,q}$. By **Step 6** with $\Lambda = \Gamma \cup \{c_j^*\}_j$, $c'_j = c_j^*$, and θ, for $i \gg 0$ the pair $(W_{N_i}, \theta D_i^*)$ is lc near 0 and $\operatorname{mld}((W_{N_i} \ni 0, D_i^*)/G_i) \ge L$. Since $D_i \le D_i^*$, $\alpha_i \ge L$, a contradiction.

Step 8. In this step we exclude the case when $(\mathcal{A}_{\mathbb{K}}, C_\infty)$ is not klt near 0, which finishes the proof.

By Definition-Proposition 5.7 in case (a) with $n = 2$ and $\mathfrak{c} = \mathcal{O}$, applied to the pairs $(\mathbb{A}^2_P, C^*_{P,i})$, whose quotient values are all equal to $L > 0$, there is exactly one floor D_∞, it is smooth at 0 with coefficient one in C_∞, $(\mathcal{A}_{\mathbb{K}}, C_\infty - D_\infty)$ is klt near 0, and

$$L = \frac{1 - b}{h}, \qquad b := \operatorname{mult}_0((C_\infty - D_\infty)|_{D_\infty}),$$

where h is the order of the image of G_P in the automorphism group of $T_0 D_\infty$. Since the floor is unique, D_∞ is G_P-invariant. Let $\lambda := 1 - b = hL > 0$. Since D_∞ is smooth at 0, one of the coordinates of $\mathbb{A}^2_P$, say z_0, restricts to a parameter of D_∞ at 0. Then G_P acts on $T_0 D_\infty$ through the character of z_0, so h is the order of the character of z_0. Let

$$(m_0, U_0) := \begin{cases} (2, x) & \text{if } z_0 \in \{x_1, p\}, \\ (2v/\sigma, \psi) & \text{if } z_0 = z_1, \\ (2c/\sigma, \psi) & \text{if } z_0 = q, \end{cases}$$

where $\psi = y^{v/\sigma} t^{2c/\sigma}$. Then U_0 is a G-semi-invariant monomial of $\mathbf{T}$-weight zero which is not divisible by xy, and a direct calculation shows that the plane equation of U_0 is $z_0^{m_0}$ up to a non-zero constant: the plane equations of x are x_1^2 and $-p^2$, and those of ψ are $(-1)^{v/\sigma} z_1^{2v/\sigma}$ and $q^{2c/\sigma}$.

We compute the order of the character of U_0. The kernel Σ_t of the surjection $G_t \to G/R$, $(g, \xi, \eta) \to gR$, consists of the classes of $(1, \xi, \eta)$ with $\xi^v = 1$ and $\eta^2 = 1$, acting by $(x_1, z_1) \to (\eta x_1, \xi^c \eta^{-1} z_1)$. The images of Σ_t under the characters of x_1 and z_1 have orders 2 and $\operatorname{lcm}(2, v) = 2v/\sigma$. The kernel Σ_y of the surjection $G_y \to G/R$ consists of the classes of $(1, \zeta)$ with $\zeta^{2c} = 1$, acting by $(p, q) \to (\zeta^{-c} p, \zeta^v q)$. The images of Σ_y under the characters of p and q have orders 2 and $2c/\gcd(2c, v) = 2c/\sigma$. Thus the restriction of the character of z_0 to Σ_t or Σ_y has order m_0, so m_0 divides h, and the character of $z_0^{m_0}$ is trivial on Σ_t or Σ_y and has order h/m_0. It is

the pullback of the character of U_0 through the surjection onto G/R, as R acts trivially on U_0. Hence the character of U_0 under G has order $m := h/m_0$.

The restriction of $C_\infty - D_\infty + \lambda \operatorname{div}(z_0)$ to D_∞ has multiplicity $b + \lambda = 1$ at 0, so the pair

$$(\mathcal{A}_{\mathbb{K}}, C_\infty + \lambda \operatorname{div}(z_0))$$

is lc near 0 [Kaw07, Theorem]. Since

$$\lambda \operatorname{div}(z_0) = \frac{\lambda}{m_0} \operatorname{div}(z_0^{m_0})$$

and $z_0^{m_0}$ is, up to a non-zero constant, one of the last two entries of the tuples, Theorem 4.2(1) applied to the sub-tuple of the first s entries and the entry $z_0^{m_0}$, a compatible generic limit of the corresponding sub-tuples by Construction-Proposition 4.1, with the labels $c_1^*, \dots, c_s^*, \lambda/m_0$, shows that, possibly passing to a subsequence, the pair

$$\left(\mathbb{A}^2_P, C^*_{P,i} + \frac{\lambda}{m_0} \operatorname{div}(z_0^{m_0})\right)$$

is lc near 0 for any i, and it is the plane pair on $\mathbb{A}^2_P$ of the $\mathbf{T}$-homogeneous boundary $D_i^{*0} + \frac{\lambda}{m_0} \operatorname{div}(U_0)$, whose adjusted mass is δ_*, since U_0 has weight zero. By the choice of $\mathbb{A}^2_P$, $\delta_* \le 0$ if $\mathbb{A}^2_P = \mathbb{A}^2_{x_1,z_1}$ and $\delta_* \ge 0$ if $\mathbb{A}^2_P = \mathbb{A}^2_{p,q}$. By **Step 4**, $(W_{N_i}, D_i^{*0} + \frac{\lambda}{m_0} \operatorname{div}(U_0))$ is lc near 0, so the pair

$$\left(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^{N_i}) + D_i^{*0} + \frac{\lambda}{m_0} \operatorname{div}(U_0)\right)$$

is lc near 0 [Kaw07, Theorem]. Let

$$\Lambda := (\Gamma \cup \{c_j^*\}_j \cup \{\lambda/m_0\}) \backslash \{0\},$$

which does not depend on i. For each i with $r_i \ge r_\Lambda$, Claim A.43 with $c'_j = c_j^*$, $U = U_0$, $\lambda_U = \lambda/m_0$, and $\epsilon = 1$ implies that the pair

$$\left(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^{N_i}) + D_i^* + \frac{\lambda}{m_0} \operatorname{div}(U_0)\right)$$

is lc near 0. Thus $(W_{N_i}, D_i^* + \frac{\lambda}{m_0} \operatorname{div}(U_0))$ is lc near 0 for $i \gg 0$ [Kaw07, Theorem]. By Lemma 5.8 with $Z = \{0\}$, $U = U_0$, whose character has order m, and the coefficient λ/m_0,

$$\alpha_i \ge \operatorname{mld}((W_{N_i} \ni 0, D_i^*)/G_i) \ge \frac{\lambda}{m_0 m} = \frac{\lambda}{h} = L$$

for $i \gg 0$, a contradiction.

Therefore no such sequence exists, that is, α cannot increase along $\mathbf{T}$. None of the constants above depends on d, so the sequence may have varying d. □

A.15. **The circle $(-1,1,0,1)$.** In this subsection, we prove Theorem 7.24 by comparing α with the minimum $\beta = \min(\beta_t, \beta_y)$ of quotient log discrepancies on two reference surfaces S_t and S_y, which are sections of charts of W_N/R. Lemma A.52 gives $\alpha \le \beta + m_3/r$, and Lemma A.53 gives $\alpha \ge \beta$ when the reference pairs have a margin (Definition 5.2). In the proof of the theorem, the case $\chi_y = \chi_t$ is Lemma A.54, and otherwise a generic limit of the pairs on S_y either provides such a margin or is not klt, in which case the limit of β is $1/e$ and Lemma A.47 gives a contradiction.

Set-up A.44. Let Γ, e_0, N, G, R, r, d, $D = \sum_j c_j \operatorname{div}(h_j)$, κ_j, h_j^0, τ, and α be as in Set-up 7.3 with $w = (-1,1,0,1)$, so that $\mathbf{T}$ acts on (x,y,z,t) with weights $(-1,1,0,1)$. Assume that for each j the block h_j^0 contains a non-zero monomial t^{κ_j}, and that $\tau \le 1$. Removing the h_j that are units near 0 or have $c_j = 0$, and multiplying each h_j by a non-zero constant, we assume that each h_j is a polynomial vanishing at 0 without monomials divisible by xy, that $c_j > 0$, and that the coefficient of t^{κ_j} in h_j is one. Let $f := xy + z^2 + t^N$, so that $W_N = (f = 0)$, and let $D^0 := \sum_j c_j \operatorname{div}(h_j^0)$. Let γ be the least positive element of Γ ($\gamma := 1$ if $\Gamma \subset \{0\}$), and $m_0 := \lfloor 2/\gamma \rfloor$. For $g \in G$ let $g^*x = \chi_x(g)x$, $g^*y = \chi_y(g)y$, $g^*z = \chi_z(g)z$, and $g^*t = \chi_t(g)t$. Let

$H := G/R$, $\zeta := \chi_z$, and $\eta := \chi_y/\chi_t$. Since ζ and η are trivial on R, we regard them as characters of H. Let e be the order of ζ, and e_1 the order of the restriction of ζ to $\ker(\eta)$.

Since f is G-semi-invariant, $\chi_x\chi_y = \zeta^2 = \chi_t^N$. In particular $\chi_x = \zeta^2/\chi_y$, and an element $(\lambda^{-1}, \lambda, 1, \lambda)$ of R with λ a primitive r-th root of unity satisfies $\lambda^N = 1$, so $r \mid N$ and $d = N/r$. Moreover $e \le |H| = [G : R] \le e_0$.

Construction-Proposition A.45. Notation as in Set-up A.44, and assume that $r > 2m_0$.

(1) The number of marks h_j is at most m_0, $\operatorname{mult}_0 h_j \le m_0$, and $1 \le \kappa_j \le m_0$.

(2) For $m \in \mathbb{Z}$ let $h_j^{(m)}$ be the sum of the terms of h_j of $\mathbf{T}$-weight $\kappa_j + mr$.
   (a) $h_j = \sum_m h_j^{(m)}$ and $h_j^{(0)} = h_j^0$.
   (b) For $m < 0$, any monomial of $h_j^{(m)}$ has the form $x^a z^c t^{a+\kappa_j+mr}$ with $a \ge |m|r - \kappa_j$.
   (c) For $m > 0$, the sum of the y-exponent and the t-exponent of any monomial of $h_j^{(m)}$ is at least r.
   (d) In particular any monomial of $h_j - h_j^0$ has degree at least $r - m_0$.

(3) There exist polynomials $f_{j,b}(z)$, $0 \le b \le \kappa_j$, and $g_{j,a}(z)$, $a \ge 1$, almost all zero, with $f_{j,0}(0) = 1$ and

$$h_j^0 = \sum_{b=0}^{\kappa_j} y^b t^{\kappa_j - b} f_{j,b}(z) + \sum_{a \ge 1} x^a t^{a+\kappa_j} g_{j,a}(z).$$

(4) Let $U := xt$, $V := y/t$, $Z := z$, $s := t^r$, $T_0 := t/y$, $s' := y^r$, and $\Xi := Z^2 + s'^d T_0^N$, so that $xy = -\Xi$ on W_N.
   (a) For $m \ge 0$ there exist polynomials $P_{j,m} \in \mathbb{C}[U, V, Z]$ and $Q_{j,m} \in \mathbb{C}[Z, T_0, s']$ such that
   $$h_j^{(m)} = t^{\kappa_j} s^m P_{j,m}(U, V, Z) = y^{\kappa_j} s'^m Q_{j,m}(Z, T_0, s')$$
   on W_N.
   (b) The polynomials $P_j := P_{j,0}$ and $Q_j(Z, T_0) := Q_{j,0}(Z, T_0, 0)$ satisfy
   $$P_j = \sum_{b=0}^{\kappa_j} V^b f_{j,b}(Z) + \sum_{a \ge 1} U^a g_{j,a}(Z) \quad \text{and} \quad Q_j = T_0^{\kappa_j} P_j(-Z^2 T_0, T_0^{-1}, Z).$$

(5) (a) $P_j(0, 0, 0) = 1$.
   (b) $P_j(0, V, 0)$ is a polynomial of degree at most κ_j with constant term one.
   (c) $Q_j(0, T_0)$ is a polynomial of degree κ_j with leading coefficient one.
   (d) The polynomial P_j is G-invariant.
   (e) The polynomial Q_j is G-semi-invariant with character $\eta^{-\kappa_j}$.

Proof. Since (W_N, D) is klt near 0, $\sum_j c_j \operatorname{mult}_0 h_j < 2$ by Lemma A.1(3). Since $c_j \ge \gamma$ and $\operatorname{mult}_0 h_j \ge 1$, the number of marks and each $\operatorname{mult}_0 h_j$ are at most m_0. The $\mathbf{T}$-weight of a monomial has absolute value at most the degree, so the weight κ_j of a monomial of least degree satisfies $|\kappa_j| \le m_0$. Since t^{κ_j} is a monomial of h_j and $h_j(0) = 0$, $\kappa_j \ge 1$. This proves (1).

The group R acts on a monomial of $\mathbf{T}$-weight k by the k-th power of the character $(\lambda^{-1}, \lambda, 1, \lambda) \to \lambda$. Since h_j is R-semi-invariant, the $\mathbf{T}$-weights of the monomials of h_j are congruent modulo r, so $h_j = \sum_m h_j^{(m)}$. The monomials of least degree have weights in $[-m_0, m_0]$, and $r > 2m_0$, so they all have weight κ_j, and $h_j^{(0)} = h_j^0$. Let $x^a y^b z^c t^k$ be a monomial of $h_j^{(m)}$, so that $ab = 0$ and $-a + b + k = \kappa_j + mr$. If $m < 0$, then $\kappa_j + mr < 0$, hence $a > 0$, $b = 0$, and $a = k - \kappa_j - mr \ge |m|r - \kappa_j$. If $m > 0$ and $a = 0$, then $b + k = \kappa_j + mr \ge r$. If $m > 0$ and $b = 0$, then $k = \kappa_j + mr + a \ge r$. Since $\kappa_j \le m_0$, any monomial of $h_j^{(m)}$ with $m \ne 0$ has degree at least $r - m_0$. This proves (2).

For (3), a monomial $x^a y^b z^c t^k$ of weight κ_j with $ab = 0$ is $y^b z^c t^{\kappa_j - b}$ with $0 \le b \le \kappa_j$ if $a = 0$, and $x^a z^c t^{a+\kappa_j}$ if $a \ge 1$. The coefficient of t^{κ_j} is one, so $f_{j,0}(0) = 1$.

For (4), take a monomial of $h_j^{(m)}$ with $m \geq 0$. By (2) and the computation of (3) with $\kappa_j + mr$ in place of κ_j, it is $y^b z^c t^{\kappa_j+mr-b}$ with $b \leq \kappa_j + mr$, or $x^a z^c t^{a+\kappa_j+mr}$ with $a \geq 1$. Then

$$y^b z^c t^{\kappa_j+mr-b} = t^{\kappa_j} s^m V^b Z^c = y^{\kappa_j} s'^m Z^c T_0^{\kappa_j+mr-b},$$

$$x^a z^c t^{a+\kappa_j+mr} = t^{\kappa_j} s^m U^a Z^c = y^{\kappa_j} s'^m (-\Xi)^a Z^c T_0^{a+\kappa_j+mr},$$

where we use $x = -\Xi/y$ and $t = yT_0$. All exponents are non-negative, which proves the existence of $P_{j,m}$ and $Q_{j,m}$. For $m = 0$ the first identities imply the formula for P_j, and the second ones together with $\Xi|_{s'=0} = Z^2$ imply

$$Q_j(Z, T_0) = \sum_b T_0^{\kappa_j - b} f_{j,b}(Z) + \sum_{a \geq 1} (-Z^2)^a T_0^{a+\kappa_j} g_{j,a}(Z) = T_0^{\kappa_j} P_j(-Z^2 T_0, T_0^{-1}, Z).$$

For (5), the first three statements follow from (3) and (4), because $f_{j,0}(0) = 1$ and the terms with $a \geq 1$ of $Q_j(0, T_0)$ vanish. The monomials of h_j^0 have the character of t^{κ_j}, hence $P_j = h_j^0/t^{\kappa_j}$ is G-invariant, and $Q_j = T_0^{\kappa_j} P_j(-Z^2 T_0, T_0^{-1}, Z)$ has the character of $T_0^{\kappa_j}$, which is $\eta^{-\kappa_j}$. □

The following lemma passes from a principal tuple to the marks h_j on the fixed ambient space $\mathbb{A}^4$.

Lemma A.46. *Notation as in Set-up A.44. Let $\theta_0 \geq 1$ be a real number and $b_0 \in \{0, 1\}$. Then there exists a positive integer $r(\theta_0)$ depending only on Γ and θ_0 such that if $r \geq r(\theta_0)$ and $(W_N, \theta_0 D^0 + b_0 \operatorname{div}(z))$ is lc near 0, then $(W_N, \theta_0 D + b_0 \operatorname{div}(z))$ is lc near 0.*

Proof. Let $\Lambda := \{1\} \cup \theta_0(\Gamma \backslash \{0\})$, a DCC subset of $\mathbb{R}_{>0}$ depending only on Γ and θ_0. By Lemma A.1(3), there are at most m_0 marks h_j. Let L be the largest of the integers given by Lemma 4.14 with $n = 4$, the DCC set Λ, and at most $m_0 + 2$ functions, and let

$$r(\theta_0) := \max(2m_0 + 1, L + m_0),$$

which depends only on Γ and θ_0. Assume that $r \geq r(\theta_0)$ and that $(W_N, \theta_0 D^0 + b_0 \operatorname{div}(z))$ is lc near 0.

Let $D_{\mathbb{A}}^0 := \sum_j c_j \operatorname{div}_{\mathbb{A}^4}(h_j^0)$ and $D_{\mathbb{A}} := \sum_j c_j \operatorname{div}_{\mathbb{A}^4}(h_j)$. Since W_N is normal, by [Kaw07, Theorem], $(\mathbb{A}^4, W_N + \theta_0 D_{\mathbb{A}}^0 + b_0 \operatorname{div}(z))$ is lc near 0. Let $\widehat{\mathbb{A}}^4 := \operatorname{Spec} \mathbb{C}[[x, y, z, t]]$. By [dFEM11, Proposition 2.11], the pair $(\widehat{\mathbb{A}}^4, \mathfrak{a}^0)$ is lc, where

$$\mathfrak{a}^0 := (f) \cdot (z)^{b_0} \cdot \prod_j (h_j^0)^{\theta_0 c_j}.$$

The exponents of $\mathfrak{a}^0$ belong to Λ, because each c_j is positive. Since $r > 2m_0$, Construction-Proposition A.45(2) gives $h_j - h_j^0 \in \mathfrak{m}_0^{r-m_0} \subset \mathfrak{m}_0^L$, where $\mathfrak{m}_0$ is the maximal ideal of $\mathbb{C}[[x, y, z, t]]$. By Lemma 4.14 with $n = 4$ and $k = \mathbb{C}$, applied to the functions f, z (if $b_0 = 1$), and h_j^0, replaced by f, z, and h_j, the pair $(\widehat{\mathbb{A}}^4, \mathfrak{a})$ is lc, where

$$\mathfrak{a} := (f) \cdot (z)^{b_0} \cdot \prod_j (h_j)^{\theta_0 c_j}.$$

By [dFEM11, Proposition 2.11], $(\mathbb{A}^4, W_N + \theta_0 D_{\mathbb{A}} + b_0 \operatorname{div}(z))$ is lc near 0, and by [Kaw07, Theorem], $(W_N, \theta_0 D + b_0 \operatorname{div}(z))$ is lc near 0. □

Lemma A.47. *Notation as in Set-up A.44. There exists a positive integer r_0 depending only on Γ such that if $r \geq r_0$, then $(W_N, D + \operatorname{div}(z))$ is lc near 0. In particular, if $r \geq r_0$, then*

$$\frac{a(P, W_N, D)}{|I_G(P)|} \geq \frac{1}{e}$$

for any prime divisor P over W_N with center 0, and $\alpha \geq 1/e$, where $e \leq e_0$.

Proof. Let $r_0 := \max(2m_0+1, r(1))$, where $r(1)$ is the integer of Lemma A.46 for $\theta_0 = 1$, and assume that $r \geq r_0$. Let $S_N := W_N \cap (z=0) = (xy + t^N = 0)$, a Du Val surface. Let ϖ_1, ϖ_2 be coordinates on $\mathbb{A}^2$, and let $\mathbb{A}^2 \to S_N$ be the morphism $x = \varpi_1^N$, $y = -\varpi_2^N$, $t = \varpi_1\varpi_2$. It is the quotient by the group μ_N acting by

$$(\varpi_1, \varpi_2) \to (\epsilon\varpi_1, \epsilon^{-1}\varpi_2),$$

which contains no reflection, so it is quasi-étale. By Construction-Proposition A.45(3),

$$y^b t^{\kappa_j - b} = (-1)^b \varpi_1^{\kappa_j - b} \varpi_2^{\kappa_j + b(N-1)} \quad \text{and} \quad x^a t^{a+\kappa_j} = \varpi_1^{(N+1)a+\kappa_j} \varpi_2^{a+\kappa_j},$$

hence the pullback of $h_j^0|_{S_N}$ is $\varpi_2^{\kappa_j} F_j(\varpi_1, \varpi_2)$ for a polynomial F_j with $F_j(\varpi_1, 0) = \varpi_1^{\kappa_j}$, because $b(N-1) > 0$ for $b \geq 1$ and $f_{j,0}(0) = 1$. Let $\Delta_N := \sum_j c_j \operatorname{div}(F_j)$. Thus the pullback of $(S_N, D^0|_{S_N})$ is $(\mathbb{A}^2, \tau \operatorname{div}(\varpi_2) + \Delta_N)$. Since

$$(\Delta_N \cdot \operatorname{div}(\varpi_2))_0 = \sum_j c_j \kappa_j = \tau \leq 1,$$

[Kaw07, Theorem] implies that $(\mathbb{A}^2, \operatorname{div}(\varpi_2) + \Delta_N)$ is lc near 0. Since $\tau \leq 1$, the pair $(\mathbb{A}^2, \tau \operatorname{div}(\varpi_2) + \Delta_N)$ is lc near 0, and by [KM98, Proposition 5.20] and [Liu26, Lemma 2.6(2)(3)], $(S_N, D^0|_{S_N})$ is lc near 0. Since $S_N = (z=0)$ is a normal Cartier divisor on W_N, by [Kaw07, Theorem], $(W_N, D^0 + \operatorname{div}(z))$ is lc near 0. By Lemma A.46 with $\theta_0 = 1$ and $b_0 = 1$, $(W_N, D + \operatorname{div}(z))$ is lc near 0.

The function z is G-semi-invariant with character ζ of order e, and $e \leq [G : R] \leq e_0$. By Lemma 5.8 applied to the closed subset $\{0\}$, the function z, whose character has order e, and $\lambda = 1$, the last two statements hold. □

We now define the reference surfaces S_t and S_y. They are the sections $(s=0) \subset Y_t$ and $(s'=0) \subset Y_y$ of the charts of Construction-Proposition A.49 below. The chart Y_t is used for the prime divisors F over W_N with center 0 and $\operatorname{mult}_F t \leq \operatorname{mult}_F y$, and Y_y for the others.

Construction-Proposition A.48. Notation as in Construction-Proposition A.45, and assume that $r > 2m_0$. Let

$$S_t := (UV + Z^2 = 0) \subset \mathbb{A}^3_{(U,V,Z)}, \quad L_t := (U = Z = 0) \subset S_t, \quad C_t := \sum_j c_j \operatorname{div}(P_j|_{S_t}),$$

$$S_y := \mathbb{A}^2_{(Z,T_0)}, \quad C_y := \sum_j c_j \operatorname{div}(Q_j),$$

and let H act on $\mathbb{A}^3_{(U,V,Z)}$ and on S_y diagonally, through the pullback of functions, by the characters

$$(U, V, Z) : \left(\frac{\zeta^2}{\eta}, \eta, \zeta\right), \qquad (Z, T_0) : (\zeta, \eta^{-1}). \tag{A.201}$$

Let β_t be the infimum of $a(P, S_t, C_t)/|I_H(P)|$ over the prime divisors P over S_t with center contained in L_t, let $\beta_y := \operatorname{mld}((S_y \ni 0, C_y)/H)$, $\beta := \min(\beta_t, \beta_y)$, and $\beta_H := \operatorname{mld}((S_t \ni 0, 0)/H)$. Then the following hold.

(1) H acts faithfully on S_t and on S_y, preserving L_t, C_t, and C_y.
(2) $(S_t, C_t + \operatorname{div}(Z))$ is lc near L_t, and $(S_y, C_y + \operatorname{div}(Z))$ is lc near 0.
(3) (a) $K := \ker(\eta)$ is cyclic of order e_1, and $\zeta|_K$ is faithful.
    (b) $\beta_t = \min(\beta_H, 1/e_1)$.
    (c) There exist finitely many isomorphism classes of H with the characters ζ and η and $|H| \leq e_0$, so β_t belongs to a finite set depending only on e_0.
(4) $1/e \leq \beta \leq \beta_t \leq 1$.
(5) β_y belongs to a set satisfying the ACC, depending only on Γ and e_0.
(6) If η is non-trivial, then $(S_t, \theta C_t)$ is lc near L_t for any real number $\theta \in [1, 2]$.

Proof. (1) Since $UV + Z^2$ has character ζ^2, H preserves S_t. Let $g \in G$ act trivially on $\mathbb{A}^3_{(U,V,Z)}$ or on S_y. In both cases $\chi_z(g) = 1$ and $\chi_y(g) = \chi_t(g)$, hence $\chi_x(g) = \zeta(g)^2/\chi_y(g) = \chi_t(g)^{-1}$, so $g \in G \cap \mathbf{T} = R$. Since none of U, V, Z vanishes identically on S_t, an element of H acting trivially on S_t acts trivially on $\mathbb{A}^3_{(U,V,Z)}$. Thus H acts faithfully on S_t and on S_y. By Construction-Proposition A.45(5), H preserves C_t and C_y, and L_t is a coordinate axis.

(2) Since $P_j(0,0,0) = 1$, $C_t = 0$ near the vertex 0 of S_t. Let $\mathbb{A}^2_{(\varpi_1,\varpi_2)} \to S_t$ be the quotient by μ_2 acting by -1, given by $U = \varpi_1^2$, $V = -\varpi_2^2$, $Z = \varpi_1\varpi_2$. It is quasi-étale, and the pullback of $\operatorname{div}(Z)$ is $\operatorname{div}(\varpi_1) + \operatorname{div}(\varpi_2)$, so by [KM98, Proposition 5.20] and [Liu26, Lemma 2.6(2)(3)], $(S_t, \operatorname{div}(Z))$ is lc near 0. Let $p = (0, v_0, 0) \in L_t$ with $v_0 \neq 0$. Near p, S_t is smooth with coordinates (V, Z), $U = -Z^2/V$, and $\operatorname{div}(Z) = L_t$. By Construction-Proposition A.45(5), $P_j(0, V, 0)$ is a non-zero polynomial of degree at most κ_j, so $L_t \not\subset \operatorname{Supp} C_t$ and

$$(C_t \cdot L_t)_p = \sum_j c_j \operatorname{mult}_{v_0} P_j(0, V, 0) \leq \sum_j c_j \kappa_j = \tau \leq 1.$$

By [Kaw07, Theorem], $(S_t, C_t + L_t)$ is lc near p. Thus $(S_t, C_t + \operatorname{div}(Z))$ is lc near L_t. Similarly, $Q_j(0, T_0)$ is a non-zero polynomial of degree κ_j, so

$$(C_y \cdot \operatorname{div}(Z))_0 = \sum_j c_j \operatorname{mult}_0 Q_j(0, T_0) \leq \tau \leq 1,$$

and by [Kaw07, Theorem], $(S_y, C_y + \operatorname{div}(Z))$ is lc near 0.

(3) Let $K := \ker(\eta) \subset H$. The group K fixes V, and the stabilizer in H of a point $p = (0, v_0, 0)$ with $v_0 \neq 0$ is K. Near p, K acts on the coordinates (V, Z) of S_t by $(1, \zeta)$, and the action is faithful by (1), so K is cyclic of order e_1. The prime divisors P over S_t with center contained in L_t are of three kinds.

- P has center 0. Since $C_t = 0$ near 0, the infimum of $a(P, S_t, C_t)/|I_H(P)|$ over such P is β_H, and it is attained, because the minimal log discrepancy of the crepant quotient of the klt germ $(S_t \ni 0)$ by H is attained on a log resolution [Liu26, Lemma 3.5], and a prime divisor attaining it lifts to a prime divisor over S_t with center 0 [Liu26, Lemma 2.6(2)(3)].
- P has center a point $p = (0, v_0, 0)$ with $v_0 \neq 0$. Then $I_H(P) \subset K$, so $I_H(P) = I_K(P)$. By (2) and Lemma 5.8 applied to the group K acting on $S_t \cap (V \neq 0)$, the closed subset $\{p\}$, the function Z, whose character on K has order e_1, and $\lambda = 1$, we have $a(P, S_t, C_t)/|I_H(P)| \geq 1/e_1$.
- $P = L_t$. Then $a(L_t, S_t, C_t) = 1$ because $L_t \not\subset \operatorname{Supp} C_t$, and $I_H(L_t) = K$ because K is the subgroup of H fixing L_t pointwise. Thus $a(L_t, S_t, C_t)/|I_H(L_t)| = 1/e_1$.

Hence $\beta_t = \min(\beta_H, 1/e_1)$. The number β_H and the integer e_1 depend only on the isomorphism class of H with the characters ζ and η, and there exist finitely many such classes with $|H| \leq e_0$.

(4) The function Z has character ζ of order e and vanishes on L_t. By (2) and Lemma 5.8 applied on S_t to the closed subset L_t, the function Z, whose character has order e, and $\lambda = 1$, and on S_y to the closed subset $\{0\}$, the function Z, and $\lambda = 1$, we have $\beta_t \geq 1/e$ and $\beta_y \geq 1/e$. By (3), $\beta \leq \beta_t \leq 1/e_1 \leq 1$.

(5) By (2), (S_y, C_y) is lc near 0. We have $C_y \in \Gamma_+ \cap [0, 1]$. By [Liu26, Lemma 2.10(1)(5)], the coefficients of the crepant quotient $(S_y/H, C_{y,H})$ belong to the DCC set $\Psi(\Gamma_+ \cap [0, 1])$, and by [Liu26, Lemma 2.6], β_y is the minimal log discrepancy of the lc surface germ $(S_y/H \ni 0, C_{y,H})$. By [HL24, Theorem 1.2], such numbers satisfy the ACC.

(6) Let $n_\eta \geq 2$ be the order of η. Since P_j is H-invariant and H acts on V by η, $P_j(0, V, 0)$ is a polynomial in V^{n_η}. Hence for $v_0 \neq 0$ the n_η points ϵv_0 with $\epsilon^{n_\eta} = 1$ have equal multiplicities in $P_j(0, V, 0)$, and $\operatorname{mult}_{v_0} P_j(0, V, 0) \leq \kappa_j/n_\eta$. Thus, near $p = (0, v_0, 0)$,

$$(\theta C_t \cdot L_t)_p \leq \frac{\theta\tau}{n_\eta} \leq 1,$$

since $\theta \le 2$, $\tau \le 1$, and $n_\eta \ge 2$, and by [Kaw07, Theorem], $(S_t, L_t + \theta C_t)$ is lc near p. Since $C_t = 0$ near the vertex, $(S_t, \theta C_t)$ is lc near L_t. □

The next Construction-Proposition describes three charts of W_N/R and the formula for log discrepancies on them.

Construction-Proposition A.49. Notation as in Construction-Propositions A.45 and A.48. Let $s'' := x^r$, and let

$$Y_t := (UV + Z^2 + s^d = 0) \subset \mathbb{A}^4_{(U,V,Z,s)}, \quad Y_y := \mathbb{A}^3_{(Z,T_0,s')}, \quad Y_x := \mathbb{A}^3_{(s'',Z,U)}.$$

For $c \in \{t, y, x\}$ let $\xi_t := t$, $\xi_y := y$, $\xi_x := x$, let $s_t := s$, $s_y := s'$, $s_x := s''$, let $\epsilon_t = \epsilon_y := 1$ and $\epsilon_x := -1$ be the $\mathbf{T}$-weights of t, y, and x, and let $\omega_t := dU \wedge dZ \wedge ds/U$, $\omega_y := dZ \wedge dT_0 \wedge ds'$, and $\omega_x := ds'' \wedge dZ \wedge dU$.

(1) (a) The coordinates of Y_c are R-invariant functions on W_N satisfying the equation of Y_c.
(b) The rational map $\pi_c : W_N \dashrightarrow Y_c$ defined by the coordinates of Y_c is the quotient $W_N \to W_N/R$ followed by a birational map $W_N/R \dashrightarrow Y_c$.
(c) The group H acts on Y_c diagonally, with the characters (A.201) on U, V, Z, T_0, and with the characters χ_t^r, χ_y^r, χ_x^r on s, s', s''.
(d) The group H acts faithfully on $Y_t \cap (s = 0) = S_t$, on $Y_y \cap (s' = 0) = S_y$, and on $S_x := Y_x \cap (s'' = 0) = \mathbb{A}^2_{(Z,U)}$.
(e) The variety Y_t is klt near 0.

(2) Let F be a prime divisor over W_N along which the coordinates of Y_c have non-negative orders, and let F' be the image of F under π_c, a prime divisor over Y_c.
(a) $\operatorname{mult}_F g = |I_R(F)| \operatorname{mult}_{F'} g$ for any rational function g on Y_c, regarded as a function on W_N.
(b) Let P be a prime divisor over W_N with center 0. Then the coordinates of Y_x have positive orders along P. If $\operatorname{mult}_P t \le \operatorname{mult}_P y$, then the coordinates of Y_t have non-negative orders along P. If $\operatorname{mult}_P y < \operatorname{mult}_P t$, then the coordinates of Y_y have positive orders along P.
(c) If F has center 0, $\operatorname{mult}_F t \le \operatorname{mult}_F y$, and $c = t$, then the center of F' is contained in $L_t = (U = Z = s = 0)$.
(d) If F has center 0 and either $\operatorname{mult}_F y < \operatorname{mult}_F t$ and $c = y$, or $c = x$, then the coordinates of Y_c have positive orders along F, and the center of F' is 0.
(e) Conversely, any prime divisor over Y_c along which the regular functions on W_N/R have non-negative orders is the image under π_c of a prime divisor over W_N.

(3) Let $B = \sum_k b_k \operatorname{div}(g_k)$, where b_k are real numbers and g_k are G-semi-invariant regular functions on W_N near 0, on which an element $(\lambda^{-1}, \lambda, 1, \lambda)$ of R acts by $\lambda^{\kappa'_k}$, let $\tau' := \sum_k b_k \kappa'_k$, and let F and F' be as in (2).
(a) $|I_G(F)| = |I_R(F)| \cdot |I_H(F')|$.
(b) We have

$$\frac{a(F, W_N, B)}{|I_G(F)|} = \frac{1}{|I_H(F')|} a\Big(F', Y_c, \Big(1 - \frac{\epsilon_c(1-\tau')}{r}\Big) \operatorname{div}(s_c) + \sum_k b_k \operatorname{div}(\xi_c^{-\epsilon_c \kappa'_k} g_k)\Big). \tag{A.202}$$

Proof. (1) The group R acts on (x, y, z, t) by $(\lambda^{-1}, \lambda, 1, \lambda)$, so U, V, Z, s, T_0, s', s'' are R-invariant. On W_N we have $UV = xy = -Z^2 - s^d$. Let $p \in W_N$ with $xyt \ne 0$ at p. Then p is determined by $\pi_t(p) = (U, V, Z, s)$ and the value of t, an r-th root of s, through $x = U/t$, $y = Vt$, and $z = Z$. Similarly p is determined by $\pi_y(p) = (Z, T_0, s')$ and the value of y, an r-th root of s', through $t = yT_0$, $x = -(Z^2 + t^N)/y$, and $z = Z$. Finally p is determined by $\pi_x(p) = (s'', Z, U)$ and the value of x, an r-th root of s'', through $t = U/x$, $y = -(Z^2 + t^N)/x$, and $z = Z$. Thus a general fiber of π_c has at most r points. Since π_c is R-invariant and R has order r and acts faithfully on W_N, π_c is the quotient by R followed by a birational map $W_N/R \dashrightarrow Y_c$. The characters follow from $\chi_x = \zeta^2/\chi_y$. The faithfulness on S_t and S_y is Construction-Proposition A.48(1). If

$g \in G$ acts trivially on S_x, then $\chi_z(g) = 1$ and $\chi_x(g)\chi_t(g) = 1$, hence $\chi_y(g) = \chi_t(g)$, and as in Construction-Proposition A.48(1), $g \in R$. The surface S_t is a klt Cartier divisor on the normal hypersurface Y_t, so by [KM98, Theorem 5.50], (Y_t, S_t) is plt near 0, and Y_t is klt near 0.

(2) Let P be a prime divisor over W_N with center 0, so that x, y, z, t have positive orders along P. Then s'', Z, U have positive orders along P. If $\operatorname{mult}_P t \le \operatorname{mult}_P y$, then U, Z, s have positive orders and $V = y/t$ has non-negative order along P, so the center of the image of P under π_t is contained in $Y_t \cap (U = Z = s = 0) = L_t$. If $\operatorname{mult}_P y < \operatorname{mult}_P t$, then Z, T_0, s' have positive orders along P. This proves (b), and for $P = F$ it proves (c) and (d). Since π_c is the quotient by R followed by a birational map, the relation $\operatorname{mult}_F g = |I_R(F)| \operatorname{mult}_{F'} g$ is [Liu26, Lemma 2.6(1)]. Conversely, let F' be a prime divisor over Y_c along which the regular functions on W_N/R have non-negative orders. Then F' has a center on the affine variety W_N/R, and we let $Y' \to W_N/R$ be a proper birational morphism on which F' is a prime divisor. Any component of the preimage of F' on the normalization of the main component of $Y' \times_{W_N/R} W_N$ is a prime divisor over W_N with image F'.

(3) Since $W_N/R \dashrightarrow Y_c$ is an H-equivariant birational map, we regard F' also as the image of F on W_N/R, a prime divisor over W_N/R. Let E be the image of F on W_N/G. By the statement before (0) and (1) of [Liu26, Lemma 2.6] for the quotients $W_N \to W_N/G$ and $W_N \to W_N/R$, and for the quotient of W_N/R by H, which is W_N/G by [Liu26, Lemma 2.6(5)],

$$\operatorname{mult}_F g = |I_G(F)| \operatorname{mult}_E g, \quad \operatorname{mult}_F g' = |I_R(F)| \operatorname{mult}_{F'} g', \quad \operatorname{mult}_{F'} g = |I_H(F')| \operatorname{mult}_E g$$

for any rational functions g on W_N/G and g' on W_N/R, where the prime divisor over W_N/G in the third equality is E by the uniqueness in the statement before (0) of [Liu26, Lemma 2.6], since its valuation is proportional to mult_E. Comparing the three equalities, $|I_G(F)| = |I_R(F)| \cdot |I_H(F')|$. Let $\omega_W := dx \wedge dz \wedge dt/x$, a generator of ω_{W_N}. A direct calculation shows that, up to sign,

$$\omega_W = \frac{1}{r} t^{1-r} \omega_t = \frac{1}{r} y^{1-r} \omega_y = \frac{1}{r} x^{-1-r} \omega_x,$$

that is, $\omega_W = \pm \frac{1}{r} \xi_c^{\epsilon_c - r} \omega_c$. For example, $dU \wedge dZ \wedge ds = r t^r dx \wedge dz \wedge dt$, so $\omega_t = r t^{r-1} \omega_W$. The form ω_c generates ω_{Y_c} near the center of F'. Write $e_R := |I_R(F)|$, and regard ω_c as a form on W_N. By [Liu26, Lemma 2.6(0)(2)] for the quotient $W_N \to W_N/R$ and the R-invariant divisor $-\operatorname{div}(\omega_c)$ on W_N, with $K_{W_N} := \operatorname{div}(\omega_c)$, the crepant quotient boundary on W_N/R is $-\operatorname{div}(\omega_c)$ with $K_{W_N/R} := \operatorname{div}(\omega_c)$. Since log discrepancies do not change under the birational map $W_N/R \dashrightarrow Y_c$, we have

$$a(F, W_N, -\operatorname{div}(\omega_c)) = 1 + \operatorname{mult}_F \operatorname{div}(\omega_c) \quad \text{and} \quad a(F', Y_c, -\operatorname{div}(\omega_c)) = 1 + \operatorname{mult}_{F'} \operatorname{div}(\omega_c).$$

By [Liu26, Lemma 2.6(2)],

$$a(F, W_N, -\operatorname{div}(\omega_c)) = e_R a(F', Y_c, -\operatorname{div}(\omega_c)).$$

Thus

$$\operatorname{mult}_F \operatorname{div}(\omega_c) = e_R(\operatorname{mult}_{F'} \operatorname{div}(\omega_c) + 1) - 1.$$

Moreover $\operatorname{mult}_F \xi_c = e_R \operatorname{mult}_{F'} s_c / r$ and

$$\operatorname{mult}_F g_k = \epsilon_c \kappa'_k \operatorname{mult}_F \xi_c + e_R \operatorname{mult}_{F'}(\xi_c^{-\epsilon_c \kappa'_k} g_k),$$

because $\xi_c^{-\epsilon_c \kappa'_k} g_k$ is R-invariant. Hence

$$\begin{aligned} a(F, W_N, B) &= 1 + \operatorname{mult}_F \operatorname{div}(\omega_W) - \sum_k b_k \operatorname{mult}_F g_k \\ &= e_R\Big(1 + \operatorname{mult}_{F'} \operatorname{div}(\omega_c) + \frac{\epsilon_c - r - \epsilon_c \tau'}{r} \operatorname{mult}_{F'} s_c - \sum_k b_k \operatorname{mult}_{F'}(\xi_c^{-\epsilon_c \kappa'_k} g_k)\Big) \\ &= e_R a\Big(F', Y_c, \Big(1 - \frac{\epsilon_c(1 - \tau')}{r}\Big) \operatorname{div}(s_c) + \sum_k b_k \operatorname{div}(\xi_c^{-\epsilon_c \kappa'_k} g_k)\Big). \end{aligned}$$

Dividing by $|I_G(F)| = e_R |I_H(F')|$ proves (A.202). □

The next two lemmas bound the quotient log discrepancies over W_N from below by β when each mark h_j is replaced by a sum of blocks $h_j^{(m)}$ of h_j with non-negative m, among them h_j^0, which is the block with m zero by Construction-Proposition A.45(2), through the sections S_t and S_y of the charts. We write $\mathfrak{a}|_Y$ for the restriction of an $\mathbb{R}$-ideal to a closed subvariety Y.

Lemma A.50. *Notation as in Construction-Proposition A.49. Let $c \in \{t, y\}$, let $\mathbb{A}_t := \mathbb{A}^4_{(U,V,Z,s)}$ and $\mathbb{A}_y := Y_y$, and let $\mathfrak{a} = \prod_k \mathfrak{a}_k^{b_k}$ be an $\mathbb{R}$-ideal on $\mathbb{A}_c$ with $b_k > 0$ and $\mathfrak{a}_k$ generated by H-semi-invariant polynomials, such that $\mathfrak{a}|_{S_c}$ is non-zero, where S_c is S_t or S_y. If $c = t$, assume that $(S_t, \mathfrak{a}|_{S_t})$ is lc near L_t, and let $\beta(\mathfrak{a})$ be the infimum of $a(P, S_t, \mathfrak{a}|_{S_t})/|I_H(P)|$ over the prime divisors P over S_t with center contained in L_t. If $c = y$, assume that $(S_y, \mathfrak{a}|_{S_y})$ is lc near 0, and let $\beta(\mathfrak{a}) := \operatorname{mld}((S_y \ni 0, \mathfrak{a}|_{S_y})/H)$. Then any prime divisor F' over Y_c with center contained in L_t if $c = t$, and with center 0 if $c = y$, satisfies*

$$\frac{a(F', Y_c, \operatorname{div}(s_c), \mathfrak{a}|_{Y_c})}{|I_H(F')|} \geq \beta(\mathfrak{a}).$$

Proof. If $c = y$, the inequality is Lemma 5.5 with $X = \mathbb{A}^3$, the group H, $u = s'$, $S_0 = S_y$, and $b = 1$, since H acts faithfully on S_y by Construction-Proposition A.49(1).

Let $c = t$. The chart Y_t is the complete intersection $(UV + Z^2 + s^d = 0)$ in $\mathbb{A}^4$ with the diagonal action of H, it is normal and klt near 0, and H acts faithfully on Y_t and on the klt complete intersection $S_t = Y_t \cap (s = 0)$ by Construction-Proposition A.49(1). The center of F' is 0, a point $p = (0, v_0, 0, 0)$ with $v_0 \neq 0$, or L_t.

If the center is 0, Lemma 5.5 with $X = Y_t$, $u = s$, $S_0 = S_t$, and $b = 1$ implies that the left hand side is at least $\operatorname{mld}((S_t \ni 0, \mathfrak{a}|_{S_t})/H) \geq \beta(\mathfrak{a})$.

If the center is L_t, the statement of Lemma 5.5 for curve centers, applied to the H-invariant curve $L_t \subset S_t$ (see also Theorem 3.3(3)), implies that the left hand side is at least the minimal log discrepancy of $(S_t, \mathfrak{a}|_{S_t})/H$ at the image of the generic point of L_t. The only prime divisor over S_t with center L_t is L_t, so the minimal log discrepancy at the image of the generic point of L_t is

$$\frac{a(L_t, S_t, \mathfrak{a}|_{S_t})}{|I_H(L_t)|} \geq \beta(\mathfrak{a}).$$

If the center is $p = (0, v_0, 0, 0)$ with $v_0 \neq 0$, then the open subset $Y_t \cap (V \neq 0)$ is isomorphic to $(\mathbb{A}^1 \backslash \{0\})_V \times \mathbb{A}^2_{(Z,s)}$ through $U = -(Z^2 + s^d)/V$. Thus $X_p := Y_t \cap (V \neq 0)$ is smooth with coordinates $(V - v_0, Z, s)$ at p, and the stabilizer $K = \ker(\eta)$ of p in H acts on $(V - v_0, Z, s)$ diagonally by $(1, \zeta, \chi_t^r)$. The section $X_p \cap (s = 0)$ is $S_t \cap (V \neq 0)$, with coordinates $(V - v_0, Z)$, on which K acts faithfully by Construction-Proposition A.48(1). Since $I_H(F') \subset K$, $I_H(F') = I_K(F')$. Each $\mathfrak{a}_k$ is generated by H-semi-invariant polynomials in (U, V, Z, s). On X_p we substitute $U = -(Z^2 + s^d)/V$ and multiply each generator by a power of V. Since V is a unit near p and $K = \ker(\eta)$ acts trivially on V, whose character is η, we obtain ideals $\mathfrak{a}'_k$ generated by K-semi-invariant polynomials in $(V - v_0, Z, s)$ with $\mathfrak{a}'_k = \mathfrak{a}_k$ near p, and the translation $V \to V - v_0$ is K-equivariant. Since X_p is an open subset of $\mathbb{A}^3_{(V,Z,s)}$, and the log discrepancies of prime divisors with center p, as well as the lc of $(S_t, \mathfrak{a}|_{S_t})$ near p, depend only on a neighborhood of p, we use Lemma 5.5 with $X = \mathbb{A}^3_{(V-v_0,Z,s)}$, the diagonal group K with characters $(1, \zeta, \chi_t^r)$, $u = s$, $S_0 = \mathbb{A}^2_{(V-v_0,Z)}$, $b = 1$, and the $\mathbb{R}$-ideal $\prod_k \mathfrak{a}_k'^{b_k}$. It implies that the left hand side is at least $\operatorname{mld}((S_t \ni p, \mathfrak{a}|_{S_t})/K)$. For prime divisors P over S_t with center p we have $I_H(P) = I_K(P)$, so $\operatorname{mld}((S_t \ni p, \mathfrak{a}|_{S_t})/K) \geq \beta(\mathfrak{a})$. □

Lemma A.51. *Notation as in Construction-Proposition A.49. For each j let $g_j := \sum_{m \in M_j} h_j^{(m)}$ for a finite set $M_j \subset \mathbb{Z}_{\geq 0}$ containing 0. Then*

$$\frac{a(F, W_N, \sum_j c_j \operatorname{div}(g_j))}{|I_G(F)|} \geq \beta$$

for any prime divisor F over W_N with center 0. In particular $a(F, W_N, D^0)/|I_G(F)| \geq \beta$.

Proof. Assume first that $\mathrm{mult}_F t \leq \mathrm{mult}_F y$, and let F' be the prime divisor over Y_t of Construction-Proposition A.49(2). Each g_j is G-semi-invariant, and R acts on it by λ^{κ_j}. By Construction-Proposition A.45(4), $\Phi_j := t^{-\kappa_j} g_j = \sum_{m \in M_j} s^m P_{j,m}(U, V, Z)$ is a polynomial with $\Phi_j|_{S_t} = P_j|_{S_t}$. By (A.202) with $\tau' = \tau$,

$$\frac{a(F, W_N, \sum_j c_j \operatorname{div}(g_j))}{|I_G(F)|} = \frac{a(F', Y_t, \operatorname{div}(s) + \sum_j c_j \operatorname{div}(\Phi_j))}{|I_H(F')|} + \frac{(1-\tau)\,\mathrm{mult}_{F'} s}{r|I_H(F')|}.$$

The last term is non-negative because $\tau \leq 1$. Let $\mathfrak{a} := \prod_j (\Phi_j)^{c_j}$. Then $\mathfrak{a}|_{S_t}$ defines C_t, so $(S_t, \mathfrak{a}|_{S_t})$ is lc near L_t by Construction-Proposition A.48(2), and $\beta(\mathfrak{a}) = \beta_t$. By Lemma A.50, the first term is at least $\beta_t \geq \beta$.

If $\mathrm{mult}_F y < \mathrm{mult}_F t$, let F' be the prime divisor over Y_y of Construction-Proposition A.49(2). Then $\Phi_j := y^{-\kappa_j} g_j = \sum_{m \in M_j} s'^m Q_{j,m}$ is a polynomial with $\Phi_j|_{S_y} = Q_j$. By (A.202) with $c = y$ and $\tau' = \tau$,

$$\frac{a(F, W_N, \sum_j c_j \operatorname{div}(g_j))}{|I_G(F)|} = \frac{a(F', Y_y, \operatorname{div}(s') + \sum_j c_j \operatorname{div}(\Phi_j))}{|I_H(F')|} + \frac{(1-\tau)\,\mathrm{mult}_{F'} s'}{r|I_H(F')|}.$$

The last term is non-negative. By Construction-Proposition A.48(2) and Lemma A.50 with $c = y$ and $\mathfrak{a} := \prod_j (\Phi_j)^{c_j}$, for which $\beta(\mathfrak{a}) = \beta_y$, the first term is at least $\beta_y \geq \beta$.

□

For the upper bound we use Gauss extensions (Proposition 5.4) of prime divisors computing β.

Lemma A.52. *Notation as in Set-up A.44 and Construction-Proposition A.48. There exist positive integers r_1 and m_3 depending only on Γ and e_0 such that if $r \geq r_1$, then*

$$\alpha \leq \beta + \frac{m_3}{r}.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix prime divisors computing β_H and β_y with bounded invariants. By Construction-Proposition A.48(3), there exist finitely many isomorphism classes of H with the action (A.201) on S_t, and for each class we fix a prime divisor ν_H over S_t with center 0 computing β_H, which exists because the minimal log discrepancy of the crepant quotient of the klt germ $(S_t \ni 0)$ by H is attained on a log resolution [Liu26, Lemma 3.5] and a prime divisor attaining it lifts to S_t [Liu26, Lemma 2.6(2)(3)]. By Construction-Proposition A.48(2) and [Liu26, Lemmas 2.6(2)(3) and 2.10(1)(5)], the crepant quotient $(S_y/H \ni 0, C_{y,H})$ is an lc surface germ with coefficients in the DCC set $\Psi(\Gamma_+ \cap [0,1])$. By Theorem 2.9(2), there exist an integer n_1 depending only on Γ and a prime divisor E over S_y/H with center 0 computing β_y such that $a(E, S_y/H, 0) \leq n_1$. By [Liu26, Lemma 2.6(3)], there exists a prime divisor ν_y over S_y over E, and by [Liu26, Lemma 2.6(2)], ν_y computes β_y, and since the branch divisor B_H of $S_y \to S_y/H$ satisfies $B_H \geq 0$,

$$a(\nu_y, S_y, 0) \leq |I_H(\nu_y)| a(E, S_y/H, 0) \leq e_0 n_1.$$

Since $(S_y, \operatorname{div}(Z) + \operatorname{div}(T_0))$ and (S_y, C_y) are lc, $\mathrm{mult}_{\nu_y} Z + \mathrm{mult}_{\nu_y} T_0 \leq a(\nu_y, S_y, 0)$ and $\gamma\,\mathrm{mult}_{\nu_y} Q_j \leq a(\nu_y, S_y, 0)$. Hence there exists a positive integer m_1 depending only on Γ and e_0 such that $a(\nu_H, S_t, 0)$, $\mathrm{mult}_{\nu_H} U$, $\mathrm{mult}_{\nu_H} V$, $\mathrm{mult}_{\nu_H} Z$, $\mathrm{mult}_{\nu_y} Z$, $\mathrm{mult}_{\nu_y} T_0$, and $\mathrm{mult}_{\nu_y} Q_j$ are at most m_1. Let

$$m_3 := 2m_1 + e_0 + 2 \quad \text{and} \quad r_1 := 2m_0 + 2m_1 + m_3 + 1,$$

and assume that $r \geq r_1$. Let $b := 1 - (1-\tau)/r$. Since $0 \leq \tau \leq 1$, $b \leq 1$ and $1 - b \leq 1/r$. By Construction-Proposition A.48(3), $\beta = \min(\beta_H, 1/e_1, \beta_y)$. We show that α is at most each of $\beta_H + m_3/r$, $1/e_1 + m_3/r$, and $\beta_y + m_3/r$.

Step 2. In this step we show that $\alpha \le \beta_y + m_3/r$. Let $e_y := |I_H(\nu_y)|$, let q_y be the integer of Lemma 3.1 for ν_y and the character χ_y^r of s', and let ℓ be an integer with $\ell \equiv q_y \pmod{e_y}$ and $2m_1 < \ell \le 2m_1 + e_0$. Let

$$C_V := b\,\mathrm{div}(s') + \sum_j c_j\,\mathrm{div}(Q_{j,0}), \qquad Q_{j,0}|_{s'=0} = Q_j.$$

Since $\ell > \mathrm{mult}_{\nu_y} Q_j$, by Proposition 5.4 with $S = S_y$, the point 0, $u = s'$, so that $S \times \mathbb{A}^1_u = Y_y$, the group H, which acts faithfully on S_y by Construction-Proposition A.49(1), the character χ_y^r, and C_V, the Gauss extension F' of mult_{ν_y} with s'-value ℓ is a prime divisor over Y_y with center 0, $I_H(F') = I_H(\nu_y)$, $\mathrm{mult}_{F'} Q_{j,0} = \mathrm{mult}_{\nu_y} Q_j$, and

$$\frac{a(F', Y_y, C_V)}{|I_H(F')|} = \beta_y + \frac{(1-b)\ell}{e_y} \le \beta_y + \frac{\ell}{r}.$$

We show that $\mathrm{mult}_{F'}(y^{-\kappa_j} h_j) = \mathrm{mult}_{\nu_y} Q_j$. Recall that $\Xi = Z^2 + s'^d T_0^N$. Since $N \ge r > 2m_1$ and $\mathrm{mult}_{\nu_y} T_0 \ge 1$,

$$\mathrm{mult}_{F'}(s'^d T_0^N) \ge N > 2\,\mathrm{mult}_{\nu_y} Z, \quad \text{so} \quad \mathrm{mult}_{F'}\,\Xi = 2\,\mathrm{mult}_{\nu_y} Z.$$

By Construction-Proposition A.45(4), for $m > 0$, $\mathrm{mult}_{F'}(y^{-\kappa_j} h_j^{(m)}) \ge m\ell > m_1$. For $m < 0$, by Construction-Proposition A.45(2), any monomial of $h_j^{(m)}$ is $x^a z^c t^{a+\kappa_j+mr}$ with $a \ge |m|r - \kappa_j$, and

$$y^{-\kappa_j} x^a z^c t^{a+\kappa_j+mr} = (-\Xi)^a Z^c T_0^{a+\kappa_j+mr} s'^m.$$

The order of $(-\Xi)^a Z^c T_0^{a+\kappa_j+mr} s'^m$ along F' is at least

$$2a - |m|\ell \ge |m|(2r-\ell) - 2m_0 \ge 2r - \ell - 2m_0 > m_1.$$

Thus

$$\mathrm{mult}_{F'}(y^{-\kappa_j} h_j) = \mathrm{mult}_{F'} Q_{j,0} = \mathrm{mult}_{\nu_y} Q_j.$$

The rational functions $y^r = s'$, $t^r = s' T_0^r$, $z^r = Z^r$, and $x^r = (-\Xi)^r/s'$ on Y_y have orders ℓ, $r\,\mathrm{mult}_{\nu_y} T_0 + \ell$, $r\,\mathrm{mult}_{\nu_y} Z$, and $2r\,\mathrm{mult}_{\nu_y} Z - \ell$ along F', which are positive since $r > \ell$. The regular functions on W_N/R are generated by R-invariant monomials in x, y, z, t, whose r-th powers are monomials in x^r, y^r, z^r, t^r, so they have non-negative orders along F'. By Construction-Proposition A.49(2), there exists a prime divisor F over W_N with image F' under π_y, and the orders of x, y, z, t along F are $|I_R(F)|/r$ times those of x^r, y^r, z^r, t^r along F', hence positive. Thus F has center 0. By (A.202) with $B = D$ and $c = y$,

$$\begin{aligned}\alpha \le \frac{a(F, W_N, D)}{|I_G(F)|} &= \frac{a(F', Y_y, b\,\mathrm{div}(s') + \sum_j c_j\,\mathrm{div}(y^{-\kappa_j} h_j))}{|I_H(F')|} \\ &= \frac{a(F', Y_y, C_V)}{|I_H(F')|} \le \beta_y + \frac{\ell}{r} \le \beta_y + \frac{m_3}{r},\end{aligned}$$

since $\ell \le 2m_1 + e_0 < m_3$.

Step 3. In this step we show that $\alpha \le \beta_H + m_3/r$. Let $e_H := |I_H(\nu_H)|$, let q_H be the integer of Lemma 3.1 for ν_H and the character χ_t^r of s, and let ℓ be an integer with $\ell \equiv q_H \pmod{e_H}$ and $2m_1 < \ell \le 2m_1 + e_0$. The functions V, Z are coordinates of the dense open subset $S_t \cap (V \ne 0)$ of S_t, with $U = -Z^2/V$, and V, Z, s are coordinates of the dense open subset $Y_t \cap (V \ne 0)$ of Y_t, with $U = -(Z^2 + s^d)/V$. Thus Y_t is birational to $S_t \times \mathbb{A}^1_s$. A direct calculation shows that, up to sign, $\omega_t = dV \wedge dZ \wedge ds/V$ and that $dV \wedge dZ/V$ generates ω_{S_t}. Thus ω_t generates $\omega_{S_t \times \mathbb{A}^1_s}$, and for any prime divisor over both Y_t and $S_t \times \mathbb{A}^1_s$ and any $\mathbb{R}$-linear combination Δ of divisors of rational functions, the log discrepancies of Δ on Y_t and on $S_t \times \mathbb{A}^1_s$ are equal.

Let $\Phi_j := \mathrm{pr}_1^*(P_j|_{S_t})$, and let

$$C_V := b\,\mathrm{div}(s) + \sum_j c_j\,\mathrm{div}(\Phi_j) \quad \text{on } S_t \times \mathbb{A}^1_s.$$

Since $C_t = 0$ near 0, ν_H computes $\mathrm{mld}((S_t \ni 0, C_t)/H) = \beta_H$. By Proposition 5.4 with $S = S_t$, the vertex 0, $u = s$, the group H, and the character χ_t^r, the Gauss extension F' of mult_{ν_H} with s-value ℓ is a prime divisor over $S_t \times \mathbb{A}^1_s$ with center $(0,0)$, $I_H(F') = I_H(\nu_H)$, and

$$\frac{a(F', S_t \times \mathbb{A}^1_s, C_V)}{|I_H(F')|} = \beta_H + \frac{(1-b)\ell}{e_H} \le \beta_H + \frac{\ell}{r}.$$

Since $\mathrm{mult}_{F'}(s^d) \ge \ell > 2m_1 \ge 2\,\mathrm{mult}_{\nu_H} Z$, we have

$$\mathrm{mult}_{F'} U = 2\,\mathrm{mult}_{\nu_H} Z - \mathrm{mult}_{\nu_H} V = \mathrm{mult}_{\nu_H}(U|_{S_t}) \ge 1.$$

Thus U, V, Z, s have positive orders along F', and F' is a prime divisor over Y_t with center 0.

We show that $\mathrm{mult}_{F'}(t^{-\kappa_j} h_j) = 0 = \mathrm{mult}_{F'} \Phi_j$. Since $P_j(0,0,0) = 1$, both $P_j(U,V,Z)$ and Φ_j are units at the centers of F'. For $m > 0$,

$$\mathrm{mult}_{F'}(t^{-\kappa_j} h_j^{(m)}) = \mathrm{mult}_{F'}(s^m P_{j,m}) \ge m\ell > 0.$$

For $m < 0$, any monomial $x^a z^c t^{a+\kappa_j+mr}$ of $h_j^{(m)}$ satisfies $t^{-\kappa_j} x^a z^c t^{a+\kappa_j+mr} = U^a Z^c s^m$ with $a \ge |m|r - \kappa_j$, so the order of $U^a Z^c s^m$ along F' is at least $a - |m|\ell \ge |m|(r-\ell) - m_0 > 0$. Thus $\mathrm{mult}_{F'}(t^{-\kappa_j} h_j) = 0$.

The rational functions $t^r = s$, $y^r = sV^r$, $z^r = Z^r$, and $x^r = U^r/s$ on Y_t have orders ℓ, $r\,\mathrm{mult}_{\nu_H} V + \ell$, $r\,\mathrm{mult}_{\nu_H} Z$, and $r\,\mathrm{mult}_{F'} U - \ell$ along F', which are positive since $\mathrm{mult}_{F'} U \ge 1$ and $r > \ell$. As in **Step 2**, by Construction-Proposition A.49(2), there exists a prime divisor F over W_N with image F' under π_t, and F has center 0. By (A.202) with $B = D$ and $c = t$, and by the equality of log discrepancies on Y_t and on $S_t \times \mathbb{A}^1_s$,

$$\begin{aligned}\alpha &\le \frac{a(F, W_N, D)}{|I_G(F)|} = \frac{a(F', Y_t, b\,\mathrm{div}(s) + \sum_j c_j\,\mathrm{div}(t^{-\kappa_j} h_j))}{|I_H(F')|} \\ &= \frac{a(F', S_t \times \mathbb{A}^1_s, C_V)}{|I_H(F')|} \le \beta_H + \frac{\ell}{r} \le \beta_H + \frac{m_3}{r},\end{aligned}$$

since $\ell \le 2m_1 + e_0 < m_3$.

Step 4. In this step we show that $\alpha \le 1/e_1 + m_3/r$. By Construction-Proposition A.48(3), K is cyclic of order e_1 and $\zeta|_K$ is faithful, so $\chi_t^r|_K = (\zeta|_K)^{q_1}$ for an integer q_1. Let ℓ be an integer with $\ell \equiv q_1 \pmod{e_1}$ and $3 \le \ell \le e_1 + 2$. The open subset $Y_t \cap (V \ne 0)$ is isomorphic to $(\mathbb{A}^1 \backslash \{0\})_V \times \mathbb{A}^2_{(Z,s)}$ through $U = -(Z^2 + s^d)/V$. Let F' be the exceptional divisor of the weighted blow-up of $Y_t \cap (V \ne 0)$ along $(Z = s = 0)$ with weights 1 on Z and ℓ on s. The center of F' is L_t. Since $Y_t \cap (V \ne 0)$ is smooth with coordinates (V, Z, s), $a(F', Y_t, 0) = 1 + \ell$, the sum of the weights. The restrictions of V and s/Z^ℓ to F' are coordinates of a dense open subset of F'. Any element of H acting trivially on F' fixes V, so $I_H(F') \subset K$. Since K acts on s/Z^ℓ by $\chi_t^r \zeta^{-\ell}$, which is trivial on K, we have $I_H(F') = K$ and $|I_H(F')| = e_1$.

Since $\ell \ge 3$ and $d \ge 1$, $\mathrm{mult}_{F'} U = \mathrm{mult}_{F'}(Z^2 + s^d) = 2$. The polynomial $P_j(U,V,Z)$ is congruent to $P_j(0,V,0) \ne 0$ modulo the ideal (Z, s) of $Y_t \cap (V \ne 0)$, so $\mathrm{mult}_{F'} P_j(U,V,Z) = 0$. By Construction-Proposition A.45(2)(4), for $m > 0$ the order of $t^{-\kappa_j} h_j^{(m)} = s^m P_{j,m}$ along F' is at least $m\ell > 0$, and for $m < 0$ any monomial of $t^{-\kappa_j} h_j^{(m)}$ is $U^a Z^c s^m$ with $a \ge |m|r - \kappa_j$, of order at least $2a - |m|\ell \ge |m|(2r - \ell) - 2m_0 > 0$. Thus $\mathrm{mult}_{F'}(t^{-\kappa_j} h_j) = 0$, and

$$\frac{a(F', Y_t, b\,\mathrm{div}(s) + \sum_j c_j\,\mathrm{div}(t^{-\kappa_j} h_j))}{|I_H(F')|} = \frac{1 + \ell - b\ell}{e_1} \le \frac{1}{e_1} + \frac{\ell}{r}.$$

The rational functions $t^r = s$, $y^r = sV^r$, $z^r = Z^r$, and $x^r = U^r/s$ on Y_t have orders ℓ, ℓ, r, and $2r - \ell$ along F', which are positive since $r > \ell$. As in **Step 2**, by Construction-Proposition A.49(2), there exists a prime divisor F over W_N with image F' under π_t, and F has center 0. By (A.202)

with $B = D$ and $c = t$,

$$\alpha \le \frac{a(F, W_N, D)}{|I_G(F)|} = \frac{a(F', Y_t, b\,\mathrm{div}(s) + \sum_j c_j \,\mathrm{div}(t^{-\kappa_j} h_j))}{|I_H(F')|} \le \frac{1}{e_1} + \frac{\ell}{r} \le \frac{1}{e_1} + \frac{m_3}{r},$$

since $\ell \le e_1 + 2 \le e_0 + 2 \le m_3$. □

The lower bound $\alpha \ge \beta$ holds when (S_t, C_t) and (S_y, C_y) have a common margin θ near L_t and near 0 respectively (Definition 5.2).

Lemma A.53. *Notation as in Set-up A.44 and Construction-Proposition A.48. Let $\theta > 1$ be a real number such that $(S_t, \theta C_t)$ is lc near L_t and $(S_y, \theta C_y)$ is lc near 0. Then there exists a positive integer r_2 depending only on Γ, e_0, and θ such that if $r \ge r_2$, then $a(F, W_N, D)/|I_G(F)| \ge \beta$ for any prime divisor F over W_N with center 0. In particular $\alpha \ge \beta$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the constants. Since the hypotheses hold for any smaller $\theta > 1$, possibly replacing θ with $\min(\theta, 5/4)$, we may assume that $\theta \le 5/4$. Let $\theta' := (1+\theta)/2 \in (1, 9/8]$, and let m_2 be the least integer with $m_2 \ge 2\theta'/((\theta'-1)\gamma)$. Let $r(\theta')$ be the integer of Lemma A.46 for $\theta_0 = \theta'$, r_0 the integer of Lemma A.47, and

$$r_2 := \max\big(r_0, r(\theta'), m_0 + 4e_0 m_2 + 1\big).$$

Assume that $r \ge r_2$. For a prime divisor F over W_N with center 0 we write $\mathrm{mult}_F x$, $\mathrm{mult}_F y$, $\mathrm{mult}_F z$, $\mathrm{mult}_F t$ for the orders of the coordinates and $e_G := |I_G(F)|$.

Step 2. In this step we show that $(W_N, \theta' D)$ is lc near 0. Since

$$\theta' C_t + \frac{1}{2}\,\mathrm{div}(Z) = \frac{1}{2}\theta C_t + \frac{1}{2}(C_t + \mathrm{div}(Z)),$$

Construction-Proposition A.48(2) implies that $(S_t, \theta' C_t + \frac{1}{2}\,\mathrm{div}(Z))$ is lc near L_t, and similarly $(S_y, \theta' C_y + \frac{1}{2}\,\mathrm{div}(Z))$ is lc near 0.

Let F be a prime divisor over W_N with center 0, and let $m_F := \min(\mathrm{mult}_F t, \mathrm{mult}_F y)$. If $\mathrm{mult}_F t > 2\,\mathrm{mult}_F z$ and $\mathrm{mult}_F y > 2\,\mathrm{mult}_F z$, then z^2 has smaller order along F than xy and t^N, which contradicts $xy + z^2 + t^N = 0$. Thus

$$m_F \le 2\,\mathrm{mult}_F z. \tag{A.203}$$

Let $B := \theta' D^0 + \frac{1}{2}\,\mathrm{div}(z)$. The function z has $\mathbf{T}$-weight 0, so we may use (A.202) with $\tau' = \theta'\tau$. Assume that $\mathrm{mult}_F t \le \mathrm{mult}_F y$, and let F' be the prime divisor over Y_t of Construction-Proposition A.49(2). Then

$$\frac{a(F, W_N, B)}{e_G} = \frac{a(F', Y_t, \mathrm{div}(s), \mathfrak{a}|_{Y_t})}{|I_H(F')|} + \frac{(1-\theta'\tau)\,\mathrm{mult}_{F'} s}{r|I_H(F')|}, \qquad \mathfrak{a} := \prod_j P_j^{\theta' c_j} \cdot (Z)^{1/2}.$$

Since $(S_t, \mathfrak{a}|_{S_t})$ is lc near L_t, $\beta(\mathfrak{a}) \ge 0$, and Lemma A.50 implies that the first term is non-negative. Since $\mathrm{mult}_F t = |I_R(F)|\,\mathrm{mult}_{F'} s/r$ and $e_G = |I_R(F)| \cdot |I_H(F')|$, we obtain

$$a(F, W_N, B) \ge (1 - \theta'\tau)\,\mathrm{mult}_F t.$$

If $\mathrm{mult}_F y < \mathrm{mult}_F t$, then (A.202) on Y_y and Lemma A.50 with $\mathfrak{a} := \prod_j Q_{j,0}^{\theta' c_j} \cdot (Z)^{1/2}$ imply $a(F, W_N, B) \ge (1 - \theta'\tau)\,\mathrm{mult}_F y$. Since $\tau \le 1$, $1 - \theta'\tau \ge 1 - \theta'$. By (A.203) and $\theta' \le 9/8$,

$$\begin{aligned} a(F, W_N, \theta' D^0) &= a(F, W_N, B) + \frac{1}{2}\,\mathrm{mult}_F z \ge \frac{1}{2}\,\mathrm{mult}_F z + (1-\theta'\tau) m_F \\ &\ge \frac{1}{2}\,\mathrm{mult}_F z - 2(\theta'-1)\,\mathrm{mult}_F z \ge 0. \end{aligned}$$

Thus $\mathrm{mld}(W_N \ni 0, \theta' D^0) \ge 0$, so $(W_N, \theta' D^0)$ is lc near 0 [Amb99, Section 1, the second paragraph after Definition 1.1]. By Lemma A.46 with $\theta_0 = \theta'$ and $b_0 = 0$, $(W_N, \theta' D)$ is lc near 0.

Step 3. In this step we bound the invariants of a prime divisor F over W_N with center 0 and $a(F, W_N, D) \le 2e_G$. We show that

$$a(F, W_N, 0),\ \operatorname{mult}_F h_j,\ \operatorname{mult}_F h_j^0,\ \operatorname{mult}_F z\ \le m_2 e_G. \tag{A.204}$$

Let $\delta_F := \sum_j c_j \operatorname{mult}_F h_j$. Since $(W_N, \theta' D)$ is lc near 0, $a(F, W_N, 0) - \theta'\delta_F \ge 0$, so

$$(\theta' - 1)\delta_F \le a(F, W_N, D) \le 2e_G,$$

$$a(F, W_N, 0) = a(F, W_N, D) + \delta_F \le \frac{2\theta' e_G}{\theta' - 1},$$

where the second line uses $\delta_F \le 2e_G/(\theta' - 1)$ from the first. Moreover $\operatorname{mult}_F h_j \le \delta_F/\gamma$. Since $(W_N, \theta' D^0)$ is lc near 0, $\theta'\gamma \operatorname{mult}_F h_j^0 \le a(F, W_N, 0)$. Since $(W_N, \operatorname{div}(z))$ is lc near 0 by Lemma A.47, $\operatorname{mult}_F z \le a(F, W_N, 0)$. Hence (A.204) holds.

Step 4. In this step we prove two inequalities for a prime divisor F over W_N with center 0. The functions xy, xt, and z are R-invariant and G-semi-invariant, so their characters are characters of H. For $u \in \{xy, xt, z\}$ let n_u be the order of the character of u. Then $n_u \le |H| \le e_0$, and u^{n_u} is a G-invariant regular function vanishing at 0. By [Liu26, Lemma 2.6(1)], the order along F of a G-invariant regular function vanishing at 0 is at least e_G. Hence

$$\operatorname{mult}_F x + \operatorname{mult}_F y \ge \frac{e_G}{e_0},\quad \operatorname{mult}_F x + \operatorname{mult}_F t \ge \frac{e_G}{e_0},\quad \operatorname{mult}_F z \ge \frac{e_G}{e_0}. \tag{A.205}$$

Assume moreover that $a(F, W_N, D) \le 2e_G$ and that $\operatorname{mult}_F h_j \ne \operatorname{mult}_F h_j^0$ for some j. Then by (A.204),

$$\operatorname{mult}_F(h_j - h_j^0) \le \max(\operatorname{mult}_F h_j, \operatorname{mult}_F h_j^0) \le m_2 e_G,$$

so $h_j - h_j^0$ has a monomial of order at most $m_2 e_G$ along F. By Construction-Proposition A.45(2), any monomial of $h_j - h_j^0$ has x-exponent at least $r - m_0$, or the sum of the y-exponent and the t-exponent at least r. Thus

$$\min(\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F t) \le \frac{m_2 e_G}{r - m_0} \le \frac{e_G}{4e_0}. \tag{A.206}$$

Step 5. In this step we treat a prime divisor F over W_N with center 0, $a(F, W_N, D) \le 2e_G$, and $\operatorname{mult}_F x \le m_2 e_G/(r - m_0)$. By (A.205) and (A.206), $\operatorname{mult}_F y \ge e_G/(2e_0)$ and $\operatorname{mult}_F t \ge e_G/(2e_0)$. By Construction-Proposition A.45(2), any monomial of $h_j^{(m)}$ with $m > 0$ has order at least $re_G/(2e_0) > m_2 e_G$ along F. By (A.204), $\operatorname{mult}_F h_j = \operatorname{mult}_F h_j^{\le 0}$, where $h_j^{\le 0} := \sum_{m \le 0} h_j^{(m)}$.

We use the chart Y_x of Construction-Proposition A.49, with coordinates $s'' = x^r$, Z, $U = xt$. A monomial of $h_j^{(m)}$ with $m \le 0$ is

$$x^a z^c t^{a + \kappa_j + mr} = x^{-\kappa_j} s''^{-m} U^{a + \kappa_j + mr} Z^c,$$

or, if $m = 0$,

$$y^b z^c t^{\kappa_j - b} = x^{-\kappa_j} (xy)^b U^{\kappa_j - b} Z^c.$$

Thus there exists a polynomial $\hat{h}_j$ in four variables such that $h_j^{\le 0} = x^{-\kappa_j} \hat{h}_j(s'', Z, U, xy)$. Let $\tilde{h}_j(s'', Z, U) := \hat{h}_j(s'', Z, U, -Z^2)$ and $\bar{h}_j(Z, U) := \tilde{h}_j(0, Z, U)$. Only the block $m = 0$ contributes to $\bar{h}_j$, and by Construction-Proposition A.45(3),

$$\bar{h}_j = U^{\kappa_j} P_j(U, -Z^2/U, Z).$$

The polynomial $\hat{h}_j(s'', Z, U, xy) - \tilde{h}_j(s'', Z, U)$ is divisible by $xy + Z^2 = -t^N$, with quotient a polynomial in s'', Z, U, xy. Hence

$$\operatorname{mult}_F \left(h_j^{\le 0} - x^{-\kappa_j} \tilde{h}_j\right) \ge N \operatorname{mult}_F t - \kappa_j \operatorname{mult}_F x \ge \frac{re_G}{2e_0} - \frac{m_0 m_2 e_G}{r - m_0} > m_2 e_G,$$

and by (A.204),

$$\operatorname{mult}_F h_j = \operatorname{mult}_F(x^{-\kappa_j} \tilde{h}_j). \tag{A.207}$$

Let $S_x = \mathbb{A}^2_{(Z,U)}$ as in Construction-Proposition A.49(1), $C_x := \sum_j c_j \operatorname{div}(\bar{h}_j)$, and $\mathfrak{b} := (U, Z^2)$. We show that

$$\frac{a(\nu, S_x, C_x, \mathfrak{b}^{1-\tau})}{|I_H(\nu)|} \geq \beta \tag{A.208}$$

for any prime divisor ν over S_x with center 0, where $\mathfrak{b}^{1-\tau}$ is omitted if $\tau = 1$. Assume first that $\operatorname{mult}_\nu U \leq 2 \operatorname{mult}_\nu Z$. The H-equivariant birational map $S_x \dashrightarrow S_t$ given by $(U, V, Z) = (U, -Z^2/U, Z)$ sends ν to a prime divisor over S_t with center contained in L_t, because U and Z have positive orders and V has non-negative order. Since $dZ \wedge dU = \pm U \cdot dU \wedge dZ/U$ and $\bar{h}_j = U^{\kappa_j} P_j$, we have

$$a(\nu, S_x, C_x) = a(\nu, S_t, C_t) + (1-\tau) \operatorname{mult}_\nu U,$$

and $\operatorname{mult}_\nu \mathfrak{b} = \operatorname{mult}_\nu U$. Hence the left hand side of (A.208) is $a(\nu, S_t, C_t)/|I_H(\nu)| \geq \beta_t$. Assume next that $\operatorname{mult}_\nu U > 2 \operatorname{mult}_\nu Z$. The H-equivariant birational map $S_x \dashrightarrow S_y$ given by $T_0 = -U/Z^2$ sends ν to a prime divisor over S_y with center 0. Since $dZ \wedge dU = -Z^2 dZ \wedge dT_0$ and $\bar{h}_j = (-Z^2)^{\kappa_j} Q_j$, we have

$$a(\nu, S_x, C_x) = a(\nu, S_y, C_y) + 2(1-\tau) \operatorname{mult}_\nu Z,$$

and $\operatorname{mult}_\nu \mathfrak{b} = 2 \operatorname{mult}_\nu Z$. Hence the left hand side of (A.208) is $a(\nu, S_y, C_y)/|I_H(\nu)| \geq \beta_y$. This proves (A.208). Since $\beta \geq 1/e > 0$ by Construction-Proposition A.48(4), (A.208) implies $a(\nu, S_x, C_x, \mathfrak{b}^{1-\tau}) \geq 0$ for any prime divisor ν over S_x with center 0. Let $\psi : \tilde{S} \to S_x$ be a log resolution of (S_x, C_x) such that $\mathfrak{b}\mathcal{O}_{\tilde{S}} = \mathcal{O}_{\tilde{S}}(-M)$ for a divisor M and $\operatorname{Exc}(\psi) \cup \operatorname{Supp}(\psi_*^{-1} C_x) \cup \operatorname{Supp} M$ is a simple normal crossing divisor, and let $\tilde{B}$ be the $\mathbb{R}$-divisor with $K_{\tilde{S}} + \tilde{B} = \psi^*(K_{S_x} + C_x) + (1-\tau)M$. Then

$$a(\nu, \tilde{S}, \tilde{B}) = a(\nu, S_x, C_x, \mathfrak{b}^{1-\tau})$$

for any prime divisor ν over S_x, and any prime divisor over $\tilde{S}$ whose center is a point of $\psi^{-1}(0)$ has center 0 on S_x. Hence $\operatorname{mld}(\tilde{S} \ni p, \tilde{B}) \geq 0$ for any closed point $p \in \psi^{-1}(0)$, so $(\tilde{S}, \tilde{B})$ is sub-lc near p [Amb99, Section 1, the second paragraph after Definition 1.1]. Since ψ is proper, $(\tilde{S}, \tilde{B})$ is sub-lc over a neighborhood of 0, that is, $(S_x, C_x, \mathfrak{b}^{1-\tau})$ is lc near 0.

Let F' be the prime divisor over Y_x of Construction-Proposition A.49(2). The polynomials $\tilde{h}_j$ are H-semi-invariant, $\mathfrak{b}$ is generated by H-semi-invariant polynomials, and H acts faithfully on S_x. Let

$$\mathfrak{a} := \prod_j \tilde{h}_j^{c_j} \cdot \mathfrak{b}^{1-\tau}.$$

By Lemma 5.5 with $X = Y_x = \mathbb{A}^3$, $u = s''$, $S_0 = S_x$, $b = 1$, and $\mathfrak{a}$, and by (A.208),

$$\frac{a(F', Y_x, \operatorname{div}(s''), \mathfrak{a})}{|I_H(F')|} \geq \beta.$$

By (A.202) with $B = D$, $c = x$, $\epsilon_x = -1$, and by (A.207),

$$\frac{a(F, W_N, D)}{e_G} = \frac{a(F', Y_x, \operatorname{div}(s''), \mathfrak{a})}{|I_H(F')|} + \frac{(1-\tau)}{|I_H(F')|}\Big(\operatorname{mult}_{F'} \mathfrak{b} - \frac{\operatorname{mult}_{F'} s''}{r}\Big).$$

It remains to show that $\operatorname{mult}_{F'} \mathfrak{b} \geq \operatorname{mult}_{F'} s''/r$, that is, $\operatorname{mult}_F x + \operatorname{mult}_F t \geq \operatorname{mult}_F x$ and $2 \operatorname{mult}_F z \geq \operatorname{mult}_F x$. The first inequality holds because $\operatorname{mult}_F t \geq 0$. Since $N \operatorname{mult}_F t \geq re_G/(2e_0) > 2m_2 e_G \geq 2 \operatorname{mult}_F z$ by (A.204), the equation $xy = -z^2 - t^N$ implies $\operatorname{mult}_F x + \operatorname{mult}_F y = 2 \operatorname{mult}_F z$, so $2 \operatorname{mult}_F z \geq \operatorname{mult}_F x$. Thus $a(F, W_N, D)/e_G \geq \beta$.

Step 6. In this step we treat a prime divisor F over W_N with center 0, $a(F, W_N, D) \leq 2e_G$, and $\min(\operatorname{mult}_F y, \operatorname{mult}_F t) \leq m_2 e_G/(r - m_0)$. By (A.205) and (A.206), $\operatorname{mult}_F x \geq e_G/(2e_0)$. By Construction-Proposition A.45(2), any monomial of $h_j^{(m)}$ with $m < 0$ has x-exponent at least

$r - m_0$, so the order along F of such a monomial is at least $(r-m_0)e_G/(2e_0) > m_2 e_G$. By (A.204), $\operatorname{mult}_F h_j = \operatorname{mult}_F h_j^{\geq 0}$, where $h_j^{\geq 0} := \sum_{m\geq 0} h_j^{(m)}$. Hence

$$a(F, W_N, D) = a\Big(F, W_N, \sum_j c_j \operatorname{div}(h_j^{\geq 0})\Big),$$

and Lemma A.51 implies $a(F, W_N, D)/e_G \geq \beta$.

Step 7. In this step we conclude. Let F be a prime divisor over W_N with center 0. If $a(F, W_N, D) > 2e_G$, then $a(F, W_N, D)/e_G > 2 > \beta$ by Construction-Proposition A.48(4). If $a(F, W_N, D) \leq 2e_G$ and $\operatorname{mult}_F h_j = \operatorname{mult}_F h_j^0$ for any j, then $a(F, W_N, D) = a(F, W_N, D^0)$, and Lemma A.51 implies $a(F, W_N, D)/e_G \geq \beta$. Otherwise (A.206) holds, and **Step 5** or **Step 6** implies $a(F, W_N, D)/e_G \geq \beta$. Since α is the infimum of $a(F, W_N, D)/|I_G(F)|$ over the prime divisors F over W_N with center 0 [Liu26, Lemma 2.6], $\alpha \geq \beta$. □

If $\chi_y = \chi_t$, the upper bound is given by a toric prime divisor on the ambient space $\mathbb{A}^4$.

Lemma A.54. *Notation as in Set-up A.44, and assume that η is trivial. Then $\ker(\zeta) = R$, so that $e = [G : R]$. Moreover there exists a positive integer r_3 depending only on Γ and e_0 such that if $r \geq r_3$, then*

$$\alpha \leq \frac{1}{e} + \frac{4}{r}.$$

Proof. If $g \in G$ and $\zeta(g) = 1$, then $\chi_y(g) = \chi_t(g)$ and $\chi_x(g) = \chi_y(g)^{-1}$, so $g \in G \cap \mathbf{T} = R$. Thus $\ker(\zeta) = R$, and H is cyclic of order e.

Let $r_3 := \max(r_0, 2m_0 + 4e_0 + 4)$, where r_0 is the integer of Lemma A.47, and assume that $r \geq r_3$. We regard G as a subgroup of $\mathbb{T}^4$ through $g \to (\chi_x(g), \chi_y(g), \chi_z(g), \chi_t(g))$, and let $N_G \subset \mathbb{R}^4$ be the lattice of G [Liu26, Definition 2.15]. Let $g_0 \in G$ with $\zeta(g_0) = \exp(2\pi\sqrt{-1}/e)$, and let $v \in \mathbb{R}$ with $\chi_t(g_0) = \exp(2\pi\sqrt{-1}v)$. Multiplying g_0 by elements of R changes v by the elements of $\frac{1}{r}\mathbb{Z}$, so we may assume that $3/r \leq v \leq 4/r$. Since $\chi_y = \chi_t$ and $\chi_x = \zeta^2/\chi_y$, the vector

$$w' := \Big(\frac{2}{e} - v,\ v,\ \frac{1}{e},\ v\Big)$$

belongs to N_G, and $w' \in \mathbb{R}^4_{>0}$ because $r > 2e_0 \geq 2e$.

We compute the weights $\operatorname{wt}_{w'}$ of f and h_j. Since $r \mid N$, $Nv \geq 3 > 2/e$, so $\operatorname{wt}_{w'}(f) = 2/e$, attained by xy and z^2. A monomial $x^a y^b z^c t^k$ of $\mathbf{T}$-weight $\kappa' = -a + b + k$ has w'-weight

$$\kappa' v + \frac{2a + c}{e}.$$

The monomials of h_j^0 have weight at least $\kappa_j v$, attained by t^{κ_j}. For $m > 0$, the monomials of $h_j^{(m)}$ have weight at least $(\kappa_j + mr)v \geq \kappa_j v + 3$. For $m < 0$, by Construction-Proposition A.45(2), the monomials of $h_j^{(m)}$ have $b = 0$ and $a \geq r - m_0$, so their weight is at least

$$a\Big(\frac{2}{e} - v\Big) \geq (r - m_0)\Big(\frac{2}{e_0} - \frac{4}{r}\Big) \geq \frac{r - m_0}{e_0} > 4 \geq \frac{4m_0}{r} \geq \kappa_j v.$$

Thus $\operatorname{wt}_{w'}(h_j) = \kappa_j v$.

The hypersurface $W_N = (f = 0)$ is normal and klt near 0, G is diagonal and acts faithfully on W_N, and (W_N, D) is klt near 0. By Lemma 4.7 with $n = 4$, $W = W_N$, the group G, D, and $w = w' \in N_G \cap \mathbb{R}^4_{>0}$,

$$\alpha \leq \sum_i w_i' - \operatorname{wt}_{w'}(f) - \sum_j c_j \operatorname{wt}_{w'}(h_j) = \frac{3}{e} + v - \frac{2}{e} - \tau v = \frac{1}{e} + (1 - \tau)v \leq \frac{1}{e} + \frac{4}{r},$$

because $0 \leq \tau \leq 1$. □

Proof of Theorem 7.24. Removing units and zero labels and rescaling the marks change neither D near 0 nor the hypotheses, so we may use the notation of Set-up A.44. Part (2) is Lemma A.47, where $r \gg 0$ means $r \geq r_0$ for the integer r_0 of Lemma A.47, which depends only on Γ. We prove part (1).

Assume that there exists a sequence of data as in Set-up A.44, indexed by $i \geq 1$, with Γ, e_0, and w fixed, $r_i \to +\infty$, satisfying the hypotheses of the theorem, with $0 < \alpha_1 < \alpha_2 < \cdots$. We add the index i to the notation. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the discrete data. Passing to a subsequence, we may assume that $r_i \geq \max(2m_0 + 1, r_0, r_1, r_3)$ for any i, where r_0, r_1, r_3 are the integers of Lemmas A.47, A.52, and A.54. By Construction-Proposition A.45(1), passing to a subsequence, we may assume that the number J of marks and the integers $\kappa_j = \kappa_{ij}$ do not depend on i. Since Γ is a DCC set, passing to a subsequence, we may assume that c_{ij} is non-decreasing in i with limit $c_j^* \geq \gamma > 0$ for each j. Since $|H_i| \leq e_0$, passing to a subsequence, we may assume that H_i with the characters ζ_i and η_i is isomorphic to a fixed group H with characters ζ and η, so that H acts on $S_y = \mathbb{A}^2_{(Z,T_0)}$ and on S_t by (A.201) for any i. Then e, e_1, and, by Construction-Proposition A.48(3), β_t do not depend on i. By Lemma A.52 and Construction-Proposition A.48(4), $\alpha_i \leq 1 + m_3/r_i$, so $\alpha_\infty := \lim_i \alpha_i$ exists and $\alpha_i < \alpha_\infty$ for any i. By Lemma A.47, $\alpha_i \geq 1/e$ for any i.

Step 2. In this step we treat the case when η is trivial. By Lemma A.54, $\alpha_i \leq 1/e + 4/r_i$, so $\alpha_\infty \leq 1/e \leq \alpha_1 < \alpha_\infty$, a contradiction. From now on we assume that η is non-trivial. By Construction-Proposition A.48(6), $(S_t, \theta C_{t,i})$ is lc near L_t for any i and any $\theta \in [1, 2]$.

Step 3. In this step we reduce the theorem to the plane S_y. By Construction-Proposition A.48(4) and (5), the numbers $\beta_{y,i}$ are at least $1/e$ and belong to a set satisfying the ACC. Passing to a subsequence, we may assume that $\beta_{y,i}$ is non-increasing with limit $\bar{\beta}_y \geq 1/e$. Then $\beta_i = \min(\beta_t, \beta_{y,i})$ is non-increasing with limit $\bar{\beta} := \min(\beta_t, \bar{\beta}_y)$, and by Lemma A.52,

$$\alpha_\infty \leq \bar{\beta}. \tag{A.209}$$

Let $C^*_{y,i} := \sum_j c_j^* \operatorname{div}(Q_{ij})$. Since $\tau_i = \sum_j c_{ij}\kappa_j \leq 1$ for any i, $\sum_j c_j^*\kappa_j \leq 1$. By Construction-Proposition A.45(5), $\operatorname{mult}_0 Q_{ij}(0, T_0) \leq \kappa_j$, so

$$(C^*_{y,i} \cdot \operatorname{div}(Z))_0 \leq \sum_j c_j^*\kappa_j \leq 1,$$

and by [Kaw07, Theorem], $(S_y, C^*_{y,i} + \operatorname{div}(Z))$ is lc near 0 for any i.

Step 4. In this step we take a compatible generic limit on S_y. The group H acts on $\mathbb{A}^2_{(Z,T_0)}$ through a finite diagonal subgroup of $GL_2(\mathbb{C})$. The functions Z and Q_{ij} are non-zero H-semi-invariant polynomials with characters ζ and $\eta^{-\kappa_j}$, independent of i, and $\operatorname{mult}_0 Q_{ij} \leq \kappa_j$. By Construction-Proposition 4.1 in case (a) with $n = 2$, passing to a subsequence and multiplying the functions by non-zero constants, there exists a compatible generic limit $(Z_\infty, Q_1^*, \ldots, Q_J^*)$ over an algebraically closed field $k \supset \mathbb{C}$ of the tuples $(Z, Q_{i1}, \ldots, Q_{iJ})$.

For a positive integer l, let $\mathcal{J}_l$ and $Z_l \subset \mathcal{J}_l$ be the space of l-jets of the tuples and the closed subset of Construction-Proposition 4.1 (there written with d in place of l). The set Z_l has nothing to do with the coordinate Z. For any l, the l-jets of the first entries of the tuples are multiples of the l-jet of Z. Any non-empty open subset of Z_l contains such jets, so Z_l is contained in the closed subset of $\mathcal{J}_l$ whose first entry is a multiple of the l-jet of Z. Thus $Z_\infty = \lambda Z$ for some $\lambda \in k^*$. By Construction-Proposition 4.1 for the subset of the last J entries, $(Q_1^*, \ldots, Q_J^*)$ is a compatible generic limit of the tuples $(Q_{i1}, \ldots, Q_{iJ})$.

Let $C_\infty := \sum_j c_j^* \operatorname{div}(Q_j^*)$ on $\mathcal{A}_k = \operatorname{Spec} k[[Z, T_0]]$. By **Step 3** and Theorem 4.2(1) with $\mathfrak{c} = \mathcal{O}$ and coefficients $(1, c_1^*, \ldots, c_J^*)$, passing to a subsequence, $(\mathcal{A}_k, \operatorname{div}(Z) + C_\infty)$ is lc near 0. In particular $(\mathcal{A}_k, C_\infty)$ is lc near 0. By Theorem 4.2(2) with $n = 2$ and $\mathfrak{c} = \mathcal{O}$, passing to a subsequence,

$$\operatorname{mld}((S_y \ni 0, C^*_{y,i})/H) = \operatorname{mld}((\mathcal{A}_k \ni 0, C_\infty)/H) =: \beta^*$$

for any i.

Step 5. In this step we show that $\bar{\beta}_y = \beta^*$. Since $c_{ij} \le c_j^*$, $C_{y,i} \le C_{y,i}^*$ and $\beta_{y,i} \ge \beta^*$. The coefficients of $C_{y,i}^*$ are finite sums of the numbers c_j^* and are at most one, so the coefficients of the crepant quotient of $(S_y, C_{y,i}^*)$ by H belong to a DCC set depending only on $(c_j^*)_j$ [Liu26, Lemma 2.10(1)(5)]. By Theorem 2.9(2) and [Liu26, Lemma 2.6(0)(2)(3)], there exist an integer n_2 depending only on $(c_j^*)_j$ and prime divisors ν_i over S_y with center 0 computing $\mathrm{mld}((S_y \ni 0, C_{y,i}^*)/H)$ such that $a(\nu_i, S_y, 0) \le e_0 n_2$. Since $(S_y, C_{y,i}^*)$ is lc, $c_j^* \operatorname{mult}_{\nu_i} Q_{ij} \le e_0 n_2$. Hence

$$\beta^* \le \beta_{y,i} \le \frac{a(\nu_i, S_y, C_{y,i})}{|I_H(\nu_i)|} = \beta^* + \sum_j \frac{(c_j^* - c_{ij}) \operatorname{mult}_{\nu_i} Q_{ij}}{|I_H(\nu_i)|} \le \beta^* + \sum_j \frac{(c_j^* - c_{ij}) e_0 n_2}{c_j^*},$$

and the right hand side tends to β^*. Thus $\bar{\beta}_y = \beta^*$.

Step 6. In this step we treat the case when $(\mathcal{A}_k, C_\infty)$ is klt near 0. By Theorem 4.2(1), there exists $\theta > 1$ such that, passing to a subsequence, $(S_y, \theta C_{y,i}^*)$ is lc near 0 for any i. Since lc is preserved when θ decreases, possibly replacing θ with $\min(\theta, 2)$, we may assume that $\theta \le 2$. Since $C_{y,i} \le C_{y,i}^*$, $(S_y, \theta C_{y,i})$ is lc near 0, and by **Step 2**, $(S_t, \theta C_{t,i})$ is lc near L_t. By Lemma A.53, $\alpha_i \ge \beta_i$ for $r_i \ge r_2$, where r_2 depends only on Γ, e_0, and θ. Since β_i is non-increasing with limit $\bar{\beta}$, (A.209) implies that for $r_i \ge r_2$,

$$\alpha_i \ge \beta_i \ge \bar{\beta} \ge \alpha_\infty > \alpha_i,$$

a contradiction.

Step 7. In this step we treat the case when $(\mathcal{A}_k, C_\infty)$ is not klt near 0. By **Step 5**, $\beta^* = \bar{\beta}_y \ge 1/e > 0$. By Definition-Proposition 5.7 in case (a) with $n = 2$, applied to the pairs $(S_y, C_{y,i}^*)$ and the group H, there exists exactly one floor D_∞, a curve through 0, smooth at 0, with coefficient one in C_∞, and

$$\beta^* = \frac{1 - b}{e'},$$

where $b := \operatorname{mult}_0((C_\infty - D_\infty)|_{D_\infty})$ and e' is the order of the image of H in the automorphism group of the tangent line of D_∞ at 0 (Lemma 5.6).

Since $(\mathcal{A}_k, \operatorname{div}(Z) + C_\infty)$ is lc near 0, $D_\infty \ne (Z = 0)$ and $(Z = 0) \not\subset \operatorname{Supp} C_\infty$. For each j let $o_j := \operatorname{mult}_0(Q_j^*|_{(Z=0)})$, which is finite because $(Z = 0) \not\subset \operatorname{Supp} C_\infty$, so that $(C_\infty \cdot \operatorname{div}(Z))_0 = \sum_j c_j^* o_j$. Let $m > \max_j o_j$ be an integer, and let E_m be the exceptional prime divisor of the weighted blow-up of $\mathcal{A}_k$ at 0 with weights m on Z and 1 on T_0. Then $\operatorname{mult}_{E_m}$ is the monomial valuation with $\operatorname{mult}_{E_m} Z = m$ and $\operatorname{mult}_{E_m} T_0 = 1$, the center of E_m is 0, and $a(E_m, \mathcal{A}_k, 0) = m+1$ [JM12, Proposition 5.1 and (5.1)]. Writing $Q_j^* = Q_j^*(0, T_0) + Z \cdot R_j$ with $R_j \in k[[Z, T_0]]$, we have $\operatorname{mult}_{E_m} Q_j^*(0, T_0) = o_j < m \le \operatorname{mult}_{E_m}(ZR_j)$, so $\operatorname{mult}_{E_m} Q_j^* = o_j$. Hence

$$a(E_m, \mathcal{A}_k, \operatorname{div}(Z) + C_\infty) = m + 1 - m - \sum_j c_j^* o_j = 1 - \sum_j c_j^* o_j.$$

Since $(\mathcal{A}_k, \operatorname{div}(Z) + C_\infty)$ is lc near 0, $(C_\infty \cdot \operatorname{div}(Z))_0 = \sum_j c_j^* o_j \le 1$. Since D_∞ passes through 0, $(D_\infty \cdot \operatorname{div}(Z))_0 \ge 1$, and since $C_\infty - D_\infty \ge 0$,

$$(D_\infty \cdot \operatorname{div}(Z))_0 = 1 \quad \text{and} \quad ((C_\infty - D_\infty) \cdot \operatorname{div}(Z))_0 = 0.$$

Thus $C_\infty - D_\infty$ does not pass through 0, so $b = 0$. Moreover $Z|_{D_\infty}$ is a parameter of D_∞ at 0, and H acts on it by ζ, so $e' = e$. Hence $\bar{\beta}_y = \beta^* = 1/e$. Since $\beta_t \ge 1/e$ by Construction-Proposition A.48(4), $\bar{\beta} = 1/e$, and (A.209) and Lemma A.47 imply

$$\alpha_\infty \le \frac{1}{e} \le \alpha_1 < \alpha_\infty,$$

a contradiction. This proves the theorem. □

A.16. **Circles with square weight zero.** In this subsection, we prove Theorem 7.25. For $p = v = 1$ it is Theorem 7.24. For general coprime p and v, the charts of W_N given by a v-th root of t and by a p-th root of y have different degrees, and the higher terms of the marks are controlled by a third chart given by a p-th root of x.

Proof of Theorem 7.25. The proof is long, so we divide it into several steps. The constants M, M_0, M_1, M_2, M_3 below depend only on Γ, e_0, p, v, and, in **Step 7**, on the number θ fixed there. Write $\chi_x, \chi_y, \chi_z, \chi_t$ for the characters of G with $g^*x = \chi_x(g)x$, $g^*y = \chi_y(g)y$, $g^*z = \chi_z(g)z$, and $g^*t = \chi_t(g)t$, where g^* is the pullback of functions. Since G preserves the equation $xy+z^2+t^N$ up to a character, $\chi_x\chi_y = \chi_z^2 = \chi_t^N$. For $\epsilon \in \mathbb{C}^*$ let

$$g_\epsilon := \operatorname{diag}(\epsilon^{-p}, \epsilon^p, 1, \epsilon^v) \in \mathbf{T}.$$

Since w is primitive, $\epsilon \to g_\epsilon$ is injective, so $R = \{g_\epsilon \mid \epsilon^r = 1\}$. Removing from D the h_j that are units near 0 or have $c_j = 0$ changes neither D near 0 nor τ. Thus we may assume that $c_j > 0$ for any j and that each h_j is a non-unit polynomial without monomials divisible by xy. Let $\gamma := \min(\Gamma \cap (0,1])$ ($\gamma := 1$ if $\Gamma \cap (0,1]$ is empty).

Step 1. In this step we bound the data of the marks and determine the shape of the principal blocks.

Let F_0 be the exceptional divisor of the blow-up of W_N at 0. The tangent cone of W_N at 0 is the integral quadric $(xy + z^2 = 0)$, so $a(F_0, W_N, 0) = 2$. Any non-zero multiple of $xy + z^2$ contains a monomial divisible by xy, so $\operatorname{mult}_{F_0} h_j = \operatorname{mult}_0 h_j$, the least degree of a monomial of h_j. Since (W_N, D) is klt near 0,

$$\sum_j c_j \operatorname{mult}_0 h_j < a(F_0, W_N, 0) = 2.$$

Thus the number of the h_j is less than $2/\gamma$, $\operatorname{mult}_0 h_j < 2/\gamma$, and $|\kappa_j| < 2\max(p,v)/\gamma$. The block h_j^0 contains a non-zero monomial t^{B_j}, so $\kappa_j = vB_j$ for an integer $B_j \ge 0$. If $B_j = 0$, then h_j^0, hence h_j, contains a non-zero constant, which is impossible since h_j is a non-unit. Hence

$$1 \le B_j \le M := \lfloor 2\max(p,v)/(v\gamma) \rfloor, \qquad \sigma := \sum_j c_j B_j = \frac{\tau}{v} \le 1.$$

Since h_j contains t^{B_j}, the character of h_j is $\chi_t^{B_j}$. The element $g_\epsilon \in R$ acts on a monomial of $\mathbf{T}$-weight a by ϵ^a, so the $\mathbf{T}$-weights of the monomials of h_j are contained in $vB_j + r\mathbb{Z}$. For $\ell \in \mathbb{Z}$, we call the sum of the terms of h_j of $\mathbf{T}$-weight $vB_j + \ell r$ the *ℓ-th block* of h_j. Thus h_j^0 is the 0-th block. Let $x^{m_1}y^{m_2}z^{m_3}t^{m_4}$ be a monomial of the ℓ-th block, so $m_1m_2 = 0$ and $-pm_1 + pm_2 + vm_4 = vB_j + \ell r$. If $\ell < 0$ and $r > vM$, then $m_1 \ne 0$, since otherwise the left-hand side is non-negative. Thus

$$m_2 = 0 \ \text{ and } \ pm_1 \ge |\ell| r - vM \ \text{ if } \ell < 0, \qquad pm_2 + vm_4 \ge \ell r \ \text{ if } \ell > 0. \tag{A.210}$$

In particular, for $r \gg 0$, any monomial of $h_j - h_j^0$ has degree at least $k_r := \lceil (r - vM)/\max(p,v) \rceil$, and $k_r \to +\infty$ as $r \to +\infty$. If $\ell = 0$ and $m_1 = 0$, then $pm_2 + vm_4 = vB_j$, so $v \mid m_2$ since $\gcd(p,v) = 1$. If $\ell = 0$ and $m_2 = 0$, then $vm_4 - pm_1 = vB_j$, so $v \mid m_1$. Multiplying h_j by a non-zero constant, we obtain polynomials $f_{j,k}$ and $g_{j,k}$ such that

$$h_j^0 = \sum_{0 \le k \le \lfloor B_j/p \rfloor} y^{vk} t^{B_j - pk} f_{j,k}(z) + \sum_{k \ge 1} x^{vk} t^{B_j + pk} g_{j,k}(z), \qquad f_{j,0}(0) = 1, \tag{A.211}$$

where the second sum is finite. No bound on the degrees of the $f_{j,k}$ and $g_{j,k}$ is used below.

Step 2. In this step we prove that $(W_N, D + \operatorname{div}(z))$ is lc near 0 for $r \gg 0$, and hence $\alpha \ge 1/h$, where h is the order of χ_z.

Let $D^0 := \sum_j c_j \operatorname{div}(h_j^0)$ and $S_z := W_N \cap (z = 0) = (xy + t^N = 0) \subset \mathbb{A}^3$. The morphism $\mathbb{A}^2_{(P,Q)} \to S_z$, $(P,Q) \to (P^N, -Q^N, PQ)$, is the quotient by μ_N acting by $(\zeta P, \zeta^{-1}Q)$, and it is

étale in codimension one. By (A.211), the pullback of $h_j^0|_{S_z}$ is $Q^{B_j}F_j(P,Q)$, where the term $y^{vk}t^{B_j-pk}$ contributes $Q^{B_j+k(Nv-p)}P^{B_j-pk}$ up to a constant and the term $x^{vk}t^{B_j+pk}$ contributes $Q^{B_j+pk}P^{Nvk+B_j+pk}$. By Set-up 7.3, $Nv = rd \geq r > p$ for $r \gg 0$, so F_j is a polynomial with $F_j(P,0) = P^{B_j}$. The restriction of $\sum_j c_j \operatorname{div}(F_j)$ to $(Q=0)$ is σ times the origin, and $\sigma \leq 1$. Let $\Theta := \sum_j c_j \operatorname{div}(F_j)$. By [Kaw07, Theorem], $(\mathbb{A}^2, \operatorname{div}(Q)+\Theta)$ is lc near 0, hence so is $(\mathbb{A}^2, \sigma\operatorname{div}(Q)+\Theta)$, which is the pullback of $(S_z, D^0|_{S_z})$. By [Liu26, Lemma 2.6(2)(3)], $(S_z, D^0|_{S_z})$ is lc near 0. Since $F_j(P,0) \neq 0$, S_z is not a component of D^0, and $S_z = \operatorname{div}(z)$ is Cartier on W_N and normal, so the different of D^0 on S_z is $D^0|_{S_z}$ [Kol13, Proposition 4.5(4) and (4.2.10)], applied to $\Delta = 0$, and extended to $\mathbb{R}$-divisors by linearity, since each $\operatorname{div}(h_j^0)$ is Cartier. By [Kaw07, Theorem],

$$(W_N, D^0 + \operatorname{div}(z)) \text{ is lc near } 0. \tag{A.212}$$

Let $f_N := xy + z^2 + t^N$. By (A.212) and [Kaw07, Theorem], the pair

$$(\mathbb{A}^4, \operatorname{div}(f_N) + \operatorname{div}(z) + \sum_j c_j \operatorname{div}(h_j^0))$$

is lc near 0. Let $s_0 := \lfloor 2/\gamma \rfloor + 2$ and $\Lambda := (\Gamma \cap (0,1]) \cup \{1\}$, and let k_Λ be the largest of the integers L of Lemma 4.14 for $n = 4$, the set Λ, and the numbers $s \leq s_0$ of functions. The polynomials f_N, z, h_j^0, and likewise f_N, z, h_j, are at most s_0 in number, and $h_j - h_j^0 \in \mathfrak{m}^{k_r}$, where $\mathfrak{m}$ is the maximal ideal of 0, with $k_r \geq k_\Lambda$ for $r \gg 0$. By Lemma 4.14 with $n = 4$, at most s_0 functions and Λ, together with [dFEM11, Proposition 2.11] to pass between $\mathbb{A}^4$ near 0 and its completion at 0, for $r \gg 0$ the pair

$$(\mathbb{A}^4, \operatorname{div}(f_N) + \operatorname{div}(z) + \sum_j c_j \operatorname{div}(h_j))$$

is lc near 0, and by [Kaw07, Theorem],

$$(W_N, D + \operatorname{div}(z)) \text{ is lc near } 0. \tag{A.213}$$

The function z has $\mathbf{T}$-weight 0, so χ_z is trivial on R and $h \leq [G:R] \leq e_0$. By Lemma 5.8, applied with the closed point 0, the function z, and the coefficient 1 of $\operatorname{div}(z)$, we obtain $\alpha \geq 1/h$.

Step 3. In this step we introduce three root charts of W_N and compare log discrepancies on them.

For the first chart, adjoin λ with $\lambda^v = t$ and let

$$x = \lambda^{-p}U, \quad y = \lambda^p V, \quad z = Z, \quad s_t := \lambda^r, \qquad X_d := (UV + Z^2 + s_t^d = 0) \subset \mathbb{A}^4_{(U,V,Z,s_t)}, \tag{A.214}$$

where $t^N = \lambda^{Nv} = s_t^d$. Let

$$\widetilde{G}_t := \{(g,\epsilon) \in G \times \mathbb{C}^* \mid \epsilon^v = \chi_t(g)\}, \qquad R_t := \{(g_\epsilon, \epsilon) \mid \epsilon^r = 1\} \subset \widetilde{G}_t,$$

where $\widetilde{G}_t$ acts by g on x, y, z, t and by $(g,\epsilon)^*\lambda = \epsilon\lambda$. The functions U, V, Z, s_t are R_t-invariant, X_d is birational to the quotient by R_t of the normalization of $(\lambda^v = t) \subset W_N \times \mathbb{A}^1_\lambda$ (as in Construction-Proposition A.49(1), the R_t-invariant rational functions on the normalization are the rational functions of U, Z, s_t), and $G_t := \widetilde{G}_t/R_t$ has order $v[G:R]$ and acts diagonally on $\mathbb{A}^4_{(U,V,Z,s_t)}$ with the characters $(\chi_x\epsilon^p, \chi_y\epsilon^{-p}, \chi_z, \epsilon^r)$, preserving X_d. If (g,ϵ) acts trivially on $S_t := X_d \cap (s_t = 0) = (UV + Z^2 = 0)$, then $\chi_z(g) = 1$, $\chi_x(g) = \epsilon^{-p}$, $\chi_y(g) = \epsilon^p$, and $\chi_t(g) = \epsilon^v$, so $g = g_\epsilon \in R$, $\epsilon^r = 1$, and $(g,\epsilon) \in R_t$. Thus G_t acts faithfully on S_t, hence on X_d. The hypersurface X_d is smooth if $d = 1$ and has an isolated cA singularity at 0 otherwise. Thus X_d is normal near 0, since its singular locus is a point of a hypersurface of dimension three. An isolated cA point is an isolated cDV point, hence terminal of index one [KM98, Corollary 5.38], in particular klt. Thus X_d is klt near 0, and S_t is an A_1 surface singularity. Let $L_t := (U = Z = 0) \subset S_t$, the V-axis.

For the second chart, adjoin ρ with $\rho^p = y$ and let

$$t = \rho^v T_0, \quad z = Z, \quad s_y := \rho^r, \quad x = -\frac{Z^2 + s_y^d T_0^N}{\rho^p}. \tag{A.215}$$

Let $\widetilde{G}_y := \{(g,\epsilon) \mid \epsilon^p = \chi_y(g)\}$ with $(g,\epsilon)^*\rho = \epsilon\rho$, $R_y := \{(g_\epsilon, \epsilon) \mid \epsilon^r = 1\}$, and $G_y := \widetilde{G}_y/R_y$, of order $p[G : R]$, acting diagonally on $\mathbb{A}^3_{(Z,T_0,s_y)}$ with the characters $(\chi_z, \chi_t\epsilon^{-v}, \epsilon^r)$. If (g,ϵ) acts trivially on $S_y := (s_y = 0) = \mathbb{A}^2_{(Z,T_0)}$, then $\chi_z(g) = 1$, $\chi_t(g) = \epsilon^v$, $\chi_y(g) = \epsilon^p$, and $\chi_x(g) = \chi_z(g)^2\chi_y(g)^{-1} = \epsilon^{-p}$, so $(g,\epsilon) \in R_y$. Thus G_y acts faithfully on S_y.

For the third chart, adjoin ϱ with $\varrho^p = x$ and let

$$t = \varrho^{-v} V_0, \quad z = Z, \quad s_x := \varrho^r, \quad xy = -Z^2 - \frac{V_0^N}{s_x^d}. \tag{A.216}$$

Let $\widetilde{G}_x := \{(g,\epsilon) \mid \epsilon^p = \chi_x(g)\}$ with $(g,\epsilon)^*\varrho = \epsilon\varrho$, $R_x := \{(g_\epsilon, \epsilon^{-1}) \mid \epsilon^r = 1\}$, and $G_x := \widetilde{G}_x/R_x$, of order $p[G : R]$, acting diagonally on $\mathbb{A}^3_{(s_x,Z,V_0)}$ with the characters $(\epsilon^r, \chi_z, \chi_t\epsilon^v)$. If (g,ϵ) acts trivially on $S_x := (s_x = 0) = \mathbb{A}^2_{(Z,V_0)}$, then $\chi_z(g) = 1$, $\chi_t(g) = \epsilon^{-v}$, and $\chi_x(g) = \epsilon^p$, so $g = g_{\epsilon^{-1}} \in R$ and $(g,\epsilon) \in R_x$. Thus G_x acts faithfully on S_x. Each of G_t, G_y, G_x maps onto G/R, so χ_z has order h on each of them.

Let F be a prime divisor over W_N, and E the prime divisor over W_N/G induced by F. For f one of λ, ρ, ϱ, or a semi-invariant rational function on W_N or on one of the charts, some power f^n with $n \geq 1$ is a rational function on W_N/G. Let $[f] := \mathrm{mult}_E(f^n)/n$. By [Liu26, Lemma 2.6], $[f] = \mathrm{mult}_{F'} f/|I(F')|$ for any prime divisor F' inducing E over W_N, over one of the charts, or over the normalization of W_N in the root extension, whichever carries f, where the inertia group is taken in the corresponding group. For regular f on W_N this is the notation $[f] = \mathrm{mult}_F f/|I_G(F)|$ of this section, and $[\lambda] = [t]/v$, $[\rho] = [y]/p$, $[\varrho] = [x]/p$.

For a G-semi-invariant rational function f on W_N with character χ_t^B, the rational functions

$$f_t := \lambda^{-vB} f, \qquad f_y := \rho^{-vB} f, \qquad f_x := \varrho^{vB} f$$

are invariant under R_t, R_y, R_x respectively, and are rational functions on X_d, $\mathbb{A}^3_{(Z,T_0,s_y)}$, $\mathbb{A}^3_{(s_x,Z,V_0)}$. For $D' = \sum_j c'_j \operatorname{div}(f_j)$ with real numbers c'_j and such f_j of characters $\chi_t^{B'_j}$, let $\sigma' := \sum_j c'_j B'_j$ and

$$\begin{aligned}
\Delta_t(D') &:= \Big(1 - \frac{v(1-\sigma')}{r}\Big) S_t + \sum_j c'_j \operatorname{div}(f_{j,t}) \ \text{ on } X_d, \\
\Delta_y(D') &:= \Big(1 - \frac{v(1-\sigma')}{r}\Big) S_y + \sum_j c'_j \operatorname{div}(f_{j,y}) \ \text{ on } \mathbb{A}^3_{(Z,T_0,s_y)}, \\
\Delta_x(D') &:= \Big(1 + \frac{v(1-\sigma')}{r}\Big) S_x + \sum_j c'_j \operatorname{div}(f_{j,x}) \ \text{ on } \mathbb{A}^3_{(s_x,Z,V_0)},
\end{aligned} \tag{A.217}$$

where $S_t = \operatorname{div}(s_t)$, $S_y = \operatorname{div}(s_y)$, and $S_x = \operatorname{div}(s_x)$.

Claim A.55. *Let F be a prime divisor over W_N, E the prime divisor over W_N/G induced by F, and F_t, F_y, F_x be prime divisors over X_d, $\mathbb{A}^3_{(Z,T_0,s_y)}$, $\mathbb{A}^3_{(s_x,Z,V_0)}$ inducing E. Then*

$$\frac{a(F, W_N, D')}{|I_G(F)|} = \frac{a(F_t, X_d, \Delta_t(D'))}{|I_{G_t}(F_t)|} = \frac{a(F_y, \mathbb{A}^3, \Delta_y(D'))}{|I_{G_y}(F_y)|} = \frac{a(F_x, \mathbb{A}^3, \Delta_x(D'))}{|I_{G_x}(F_x)|}.$$

Proof. We prove the first equality. Replacing $(\lambda, s_t, X_d, \omega_{X_d}, R_t, G_t)$ with

$$(\rho, s_y, \mathbb{A}^3, ds_y \wedge dZ \wedge dT_0, R_y, G_y) \quad \text{or} \quad (\varrho, s_x, \mathbb{A}^3, ds_x \wedge dZ \wedge dV_0, R_x, G_x),$$

and the equation $\lambda^v = t$ with $\rho^p = y$ or with $\varrho^p = x$, the argument below proves the second and the third equalities, with the identities at the end of the proof (similarly, the R_y-invariant rational functions on the normalization of $(\rho^p = y)$ are the rational functions of Z, T_0, s_y, and

the R_x-invariant ones on the normalization of $(\varrho^p = x)$ are those of s_x, Z, V_0). Let $\widehat{W}$ be the normalization of $(\lambda^v = t) \subset W_N \times \mathbb{A}^1_\lambda$ and $\widehat{D}'$ the crepant pullback of D', so that $K_{\widehat{W}} + \widehat{D}'$ is the pullback of $K_{W_N} + D'$. By [Liu26, Lemma 2.6(5)], $(W_N, D')/G$ is the quotient of $(\widehat{W}, \widehat{D}')$ by $\widetilde{G}_t$, which is the quotient by G_t of $(\widehat{W}, \widehat{D}')/R_t$. Let $\omega_W := dx \wedge dz \wedge dt/x$ and $\omega_{X_d} := dU \wedge dZ \wedge ds_t/U$, which generate ω_{W_N} and ω_{X_d} near 0. A direct calculation with (A.214) shows

$$\omega_W = \frac{v}{r} \lambda^{v-r} \omega_{X_d}.$$

Let P be a prime divisor over $\widehat{W}$, P' the induced prime divisor over X_d, and e the ramification index of P over P'. Then $\operatorname{mult}_P f_j = vB'_j \operatorname{mult}_P \lambda + e \operatorname{mult}_{P'} f_{j,t}$ and $\operatorname{mult}_P \lambda = e \operatorname{mult}_{P'} s_t/r$. With the Hurwitz formula

$$\operatorname{mult}_P \operatorname{div}(\omega_{X_d}) = e \operatorname{mult}_{P'} \operatorname{div}(\omega_{X_d}) + e - 1,$$

we obtain

$$a(P, \widehat{W}, \widehat{D}') = 1 + \operatorname{mult}_P \operatorname{div}(\omega_W) - \sum_j c'_j \operatorname{mult}_P f_j = e \cdot a(P', X_d, \Delta_t(D')).$$

Since the ramification indices multiply in the tower $\widehat{W} \to X_d \to X_d/G_t$, [Liu26, Lemma 2.6(2)] applied to the quotients by $\widetilde{G}_t$ and by G_t implies the first equality. For the second and the third equalities, the identities are

$$\omega_W = -\frac{dy \wedge dz \wedge dt}{y} = -\frac{p}{r} \rho^{v-r} ds_y \wedge dZ \wedge dT_0, \qquad \omega_W = \frac{p}{r} \varrho^{-v-r} ds_x \wedge dZ \wedge dV_0,$$

$f_j = \rho^{vB'_j} f_{j,y}$, $s_y = \rho^r$, $f_j = \varrho^{-vB'_j} f_{j,x}$, and $s_x = \varrho^r$. □

Proof of Theorem 7.25 continued. Let F be a prime divisor over W_N with center 0, so $[x], [y], [z], [t] > 0$, and let $\mu_F := \min([t]/v, [y]/p)$. If $[t]/v \le [y]/p$, then $[U] = [x] + p[t]/v > 0$, $[V] = [y] - p[t]/v \ge 0$, $[Z] > 0$, and $[s_t] = r[t]/v = r\mu_F > 0$, so the center of F_t on X_d is contained in L_t. If $[y]/p < [t]/v$, then $[Z] > 0$, $[T_0] = [t] - v[y]/p > 0$, and $[s_y] = r[y]/p = r\mu_F > 0$, so the center of F_y is the origin. Moreover $[s_x] = r[x]/p > 0$, $[Z] > 0$, and $[V_0] = v[x]/p + [t] > 0$, so the center of F_x is the origin.

Step 4. In this step we define the section pairs and the number β, and prove that $1/h \le \beta \le 1$ and that β belongs to an ACC set depending only on Γ, e_0, and v.

By (A.211), (A.214), and (A.215), $(h_j^0)_t = P_j$ and $(h_j^0)_y|_{S_y} = Q_j$, where

$$\begin{aligned} P_j(U, V, Z) &:= \sum_{0 \le k \le \lfloor B_j/p \rfloor} V^{vk} f_{j,k}(Z) + \sum_{k \ge 1} U^{vk} g_{j,k}(Z), \\ Q_j(Z, T_0) &:= \sum_{0 \le k \le \lfloor B_j/p \rfloor} T_0^{B_j - pk} f_{j,k}(Z) + \sum_{k \ge 1} (-Z^2)^{vk} T_0^{B_j + pk} g_{j,k}(Z), \end{aligned} \tag{A.218}$$

and $(h_j^0)_y$ is the polynomial Q_j with $-Z^2$ replaced by $-(Z^2 + s_y^d T_0^N)$. Thus $P_j(0,0,0) = 1$, and $Q_j(0, T_0)$ is a polynomial of degree B_j with leading coefficient 1. Since h_j^0 and λ^{vB_j} both have character $\chi_t^{B_j}$ on $\widetilde{G}_t$, P_j is G_t-invariant. Let $C_t := \sum_j c_j \operatorname{div}(P_j|_{S_t})$ on S_t and $C_y := \sum_j c_j \operatorname{div}(Q_j)$ on S_y. Since $\operatorname{mult}_0 Q_j(0, T_0) \le B_j$, the restriction of C_y to $(Z = 0)$ has multiplicity at most $\sigma \le 1$ at 0, so by [Kaw07, Theorem],

$$(S_y, C_y + \operatorname{div}(Z)) \text{ is lc near } 0. \tag{A.219}$$

Let χ_V be the character of V on G_t, that is, $(g, \epsilon) \to \chi_y(g)\epsilon^{-p}$, let $\psi := \chi_y^v \chi_t^{-p}$, a character of G/R, and let m be the order of ψ. Then χ_V^v is the pullback of ψ, since $\epsilon^v = \chi_t(g)$. The kernel of $G_t \to G/R$ is the image of $\{(1, \epsilon) \mid \epsilon^v = 1\}$, on which $\chi_V(1, \epsilon) = \epsilon^{-p}$ has order v since $\gcd(p, v) = 1$. Thus the order of χ_V is vo for a positive integer o. Since $\psi^o = \chi_V^{vo} = 1$, $m \mid o$, and since $\chi_V^{vm} = \psi^m = 1$, $o \mid m$. Hence χ_V has order vm. The polynomial $P_j(0, V, 0)$ is G_t-invariant,

of degree at most $v\lfloor B_j/p\rfloor$, with constant term 1. So the non-zero roots of $P_j(0,V,0)$ form orbits of length vm under $V\to\zeta V$, $\zeta\in\mu_{vm}$, with equal multiplicities, and each non-zero root has multiplicity at most $B_j/(pm)$. Near a point $v_0\neq 0$ of L_t, S_t is smooth with coordinates $(V-v_0,Z)$, $\operatorname{div}(Z)=L_t$, and

$$(C_t\cdot L_t)_{v_0}\le\frac{\sigma}{pm}\le\frac{1}{m}.$$

Near the vertex 0 of S_t, $C_t=0$, and $(S_t,\operatorname{div}(Z))$ is lc, since the pullback of $(S_t,\operatorname{div}(Z))$ under the quotient $\mathbb{A}^2_{(a,b)}\to S_t$, $(a,b)\to(a^2,-b^2,ab)$, by ±1, which is étale in codimension one, is $(\mathbb{A}^2,\operatorname{div}(ab))$ ([Liu26, Lemma 2.6(2)(3)]). By [Kaw07, Theorem], we obtain

$$(S_t,C_t+\operatorname{div}(Z))\text{ is lc near }L_t,\quad\text{and }(S_t,2C_t)\text{ is lc near }L_t\text{ if }m\ge 2,\tag{A.220}$$

where for the second part we use $(2C_t\cdot L_t)_{v_0}\le 2/m\le 1$.

Let β_t be the infimum of the log discrepancies of $(S_t,C_t)/G_t$ over the prime divisors with center contained in the image of L_t, $\beta_y:=\operatorname{mld}((S_y\ni 0,C_y)/G_y)$, and $\beta:=\min(\beta_t,\beta_y)$. The stabilizer in G_t of a point $v_0\neq 0$ of L_t is $\ker\chi_V$. Let h' be the order of χ_z on $\ker\chi_V$. An element of $\ker\chi_V$ fixing Z fixes $U=-Z^2/V$ on S_t, so it is the identity, and $|\ker\chi_V|=h'$. The inertia group of the prime divisor L_t is $\ker\chi_V$, and L_t is not a component of C_t since $P_j(0,V,0)\neq 0$. Thus L_t has log discrepancy $1/h'$ on $(S_t,C_t)/G_t$. For a prime divisor P over S_t with center $v_0\in L_t\backslash\{0\}$, $I_{G_t}(P)$ is contained in the stabilizer $\ker\chi_V$, so Lemma 5.8, applied to the group $\ker\chi_V$ and the function Z with (A.220), implies $a(P,S_t,C_t)/|I_{G_t}(P)|\ge 1/h'$. Hence

$$\beta_t=\min\{\operatorname{mld}((S_t\ni 0,0)/G_t),1/h'\}.\tag{A.221}$$

By Lemma 5.8, applied to G_t and to G_y with the function Z, together with (A.219) and (A.220), we have $\beta_t\ge 1/h$ and $\beta_y\ge 1/h$. Thus $1/h\le\beta\le\beta_t\le 1/h'\le 1$. By (A.221), β_t depends only on the diagonal action of G_t on $\mathbb{A}^3_{(U,V,Z)}$, and $|G_t|\le ve_0$, so β_t belongs to a finite set depending only on v and e_0.

Assume that $\psi=1$. The kernel of $\ker\chi_V\to G/R$ is contained in the image of $\{(1,\epsilon)\mid\epsilon^v=1\}$, on which χ_V is injective, so $\ker\chi_V\to G/R$ is injective. For $g\in G$ choose ϵ with $\epsilon^v=\chi_t(g)$. Then $\chi_V(g,\epsilon)^v=\psi(g)=1$, and multiplying ϵ by a v-th root of unity ζ multiplies $\chi_V(g,\epsilon)$ by ζ^{-p}, which runs over μ_v, so $\ker\chi_V\to G/R$ is surjective. Thus $\ker\chi_V\to G/R$ is an isomorphism, $h'=h$, and

$$\beta=\beta_t=1/h\quad\text{if }\psi=1.\tag{A.222}$$

Since (S_y,C_y) is lc, $C_y\in\Gamma_+\cap[0,1]$, and by [Liu26, Lemma 2.10(1)(5)] the coefficients of the quotient of (S_y,C_y) by G_y belong to the DCC set $\Psi(\Gamma_+\cap[0,1])$. By [HL24, Theorem 1.2], β_y belongs to an ACC set depending only on Γ. Since β_t belongs to a finite set depending only on v and e_0, $\beta=\min(\beta_t,\beta_y)$ belongs to an ACC set depending only on Γ, e_0, and v.

Step 5. In this step we bound from below the log discrepancies of the boundaries made of blocks with $\ell\ge 0$.

Claim A.56. *For any j let g_j be the sum of h_j^0 and of some ℓ-th blocks of h_j with $\ell\ge 1$, and let $D_g:=\sum_j c_j\operatorname{div}(g_j)$. Then for $r\gg 0$, any prime divisor F over W_N with center 0 satisfies*

$$\frac{a(F,W_N,D_g)}{|I_G(F)|}\ge\beta+v(1-\sigma)\mu_F\ge\beta.$$

Proof. The polynomial g_j has character $\chi_t^{B_j}$. By (A.214) and (A.215), a monomial $x^{m_1}y^{m_2}z^{m_3}t^{m_4}$ of $\mathbf{T}$-weight $vB_j+\ell r$ is equal to $\lambda^{vB_j}s_t^\ell U^{m_1}V^{m_2}Z^{m_3}$ and to $\rho^{vB_j}s_y^\ell(-Z^2-s_y^dT_0^N)^{m_1}Z^{m_3}T_0^{m_4}$. Hence $(g_j)_t$ and $(g_j)_y$ are polynomials in (U,V,Z,s_t) and in (Z,T_0,s_y), and by (A.218) their restrictions to S_t and S_y are $P_j|_{S_t}$ and Q_j. Let $b:=1-v(1-\sigma)/r\le 1$. By Claim A.55, with F_t and F_y as there,

$$\frac{a(F,W_N,D_g)}{|I_G(F)|}=\frac{a(F_t,X_d,bS_t+\sum_j c_j\operatorname{div}((g_j)_t))}{|I_{G_t}(F_t)|}=\frac{a(F_y,\mathbb{A}^3,bS_y+\sum_j c_j\operatorname{div}((g_j)_y))}{|I_{G_y}(F_y)|}.$$

Assume that $[t]/v \le [y]/p$, so the center of F_t is contained in L_t and $(1-b)[s_t] = v(1-\sigma)\mu_F$ by **Step 3**. The hypersurface $X_d \subset \mathbb{A}^4$ with the diagonal action of G_t satisfies the hypotheses of Theorem 3.3 with $n=4$ and $c=1$, the section $S_t = X_d \cap (s_t = 0)$ is a normal integral klt complete intersection of dimension two on which G_t acts faithfully, and (S_t, C_t) is lc near L_t by (A.220). If the center of F_t is 0, Lemma 5.5 with $X := X_d$, $u := s_t$, $S_0 := S_t$, $H := G_t$, $\mathfrak{a} := \prod_j ((g_j)_t)^{c_j}$, and the coefficient b implies

$$\begin{aligned} \frac{a(F_t, X_d, bS_t + \sum_j c_j \operatorname{div}((g_j)_t))}{|I_{G_t}(F_t)|} &\ge \operatorname{mld}((S_t \ni 0, C_t)/G_t) + (1-b)[s_t] \\ &\ge \beta_t + v(1-\sigma)\mu_F. \end{aligned} \tag{A.223}$$

If the center of F_t is L_t, the second part of Lemma 5.5 with $C := L_t$, that is, Theorem 3.3(3) at the generic point of L_t, implies (A.223) with $1/h'$ in place of $\operatorname{mld}((S_t \ni 0, C_t)/G_t)$, and $1/h' \ge \beta_t$. If the center of F_t is a closed point $v_0 \ne 0$ of L_t, then $X_d \cap (V \ne 0)$ is isomorphic to the open subset $(V \ne 0)$ of $\mathbb{A}^3_{(V,Z,s_t)}$ through $U = -(Z^2 + s_t^d)/V$. The stabilizer $\ker \chi_V$ of v_0 acts diagonally and faithfully on $\mathbb{A}^3_{(V-v_0,Z,s_t)}$, trivially on $V - v_0$, and faithfully on $(s_t = 0)$, and for a large integer n_0 the functions $V^{n_0}(g_j)_t$ are semi-invariant polynomials in $(V - v_0, Z, s_t)$ with

$$\operatorname{div}(V^{n_0}(g_j)_t) = \operatorname{div}((g_j)_t) \quad \text{near } v_0.$$

Since $I_{G_t}(F_t)$ fixes the center v_0, $I_{G_t}(F_t) = I_{\ker \chi_V}(F_t)$. Lemma 5.5 with $X := \mathbb{A}^3_{(V-v_0,Z,s_t)}$, $u := s_t$, $S_0 := X \cap (s_t = 0)$, $H := \ker \chi_V$, $\mathfrak{a} := \prod_j (V^{n_0}(g_j)_t)^{c_j}$, and the coefficient b, where S_0 is S_t near v_0 and $(S_0, \mathfrak{a}|_{S_0})$ is lc near v_0 by (A.220), together with $\operatorname{mld}((S_t \ni v_0, C_t)/\ker \chi_V) \ge 1/h'$ from **Step 4**, implies (A.223) with $1/h'$ in place of $\operatorname{mld}((S_t \ni 0, C_t)/G_t)$.

Assume that $[y]/p < [t]/v$, so the center of F_y is 0 and $(1-b)[s_y] = v(1-\sigma)\mu_F$. Lemma 5.5 with $X := \mathbb{A}^3_{(Z,T_0,s_y)}$, $u := s_y$, $S_0 := S_y$, $H := G_y$, $\mathfrak{a} := \prod_j ((g_j)_y)^{c_j}$, and the coefficient b, together with (A.219), implies the lower bound $\beta_y + v(1-\sigma)\mu_F$. □

Proof of Theorem 7.25 continued.

Step 6. In this step we prove that $\alpha \le \beta + M_3/r$ for $r \gg 0$.

By (A.221), β is equal to $1/h'$, to $\operatorname{mld}((S_t \ni 0, 0)/G_t)$, or to β_y. In the last two cases let S be S_t or S_y, and C_S be 0 or C_y. The coefficients of the crepant quotient of (S, C_S) belong to $\Psi(\Gamma_+ \cap [0,1])$ (**Step 4**), so by [HL23, Theorem 1.2] there exists a prime divisor E_S over the quotient germ computing β with $a(E_S, S', 0) \le N_\Gamma$, where N_Γ is the constant of [HL23, Theorem 1.2] for the DCC set $\Psi(\Gamma_+ \cap [0,1])$ and $(S', B_{S'})$ is the crepant quotient of $(S, 0)$. By [Liu26, Lemma 2.6(2)(3)], there exists a prime divisor F_S over S with center 0 inducing E_S. Let $e_S := |I(F_S)| \le \max(p, v)e_0$. Since $B_{S'} \ge 0$, [Liu26, Lemma 2.6(2)] gives

$$a(F_S, S, 0) = e_S a(E_S, S', B_{S'}) \le e_S N_\Gamma \le \max(p, v)e_0 N_\Gamma.$$

On S_t let $L_t' := (V = Z = 0)$, so $\operatorname{div}(U) = 2L_t$, $\operatorname{div}(V) = 2L_t'$, and $\operatorname{div}(Z) = L_t + L_t'$. The pairs $(S_t, L_t + L_t')$, $(S_y, \operatorname{div}(ZT_0))$, and (S_y, C_y) are lc, so the multiplicities along F_S of U, V, Z if $S = S_t$, and of Z, T_0 and the Q_j if $S = S_y$, are at most $2a(F_S, S, 0)/\gamma$. If $S = S_t$, then $P_j(0,0,0) = 1$, so P_j is a unit at the center 0 of F_S and $\operatorname{mult}_{F_S} P_j = 0$. Let $M_0 := \max(p, v)e_0 \max(2N_\Gamma/\gamma, 1)$, so $M_0 \ge 2a(F_S, S, 0)/\gamma$ and $M_0 \ge e_S$. Let $q_S \in \{1, \dots, e_S\}$ be the integer of Lemma 3.1 for F_S and the character of s_t (resp. s_y), and let q be the integer with $2M_0 < q \le 2M_0 + e_S$ and $q \equiv q_S \pmod{e_S}$.

Assume that $\beta = \beta_y$. Then $\mathbb{A}^3_{(Z,T_0,s_y)} = S_y \times \mathbb{A}^1_{s_y}$ with the action of G_y of Lemma 3.1. By Lemma 3.1 applied to the variety S_y, the boundary C_y, the group G_y, the coordinate s_y with its character, the prime divisor F_S, and the s_y-value q (so the integer of that lemma is q_S), the Gauss extension F of F_S with s_y-value q is a prime divisor over $\mathbb{A}^3$ with center 0, $I_{G_y}(F) = I_{G_y}(F_S)$, $\operatorname{mult}_F s_y = q$, and

$$a(F, \mathbb{A}^3, \operatorname{pr}_1^* C_y + bS_y) = a(F_S, S_y, C_y) + (1-b)q,$$

where $b := 1 - v(1-\sigma)/r$. By [Liu26, Definition-Proposition 2.11], the multiplicity along F of $\sum_i a_i s_y^i$, with a_i polynomials in (Z, T_0), is $\min_i(\operatorname{mult}_{F_S} a_i + iq)$. We show $\operatorname{mult}_F (h_j)_y = \operatorname{mult}_{F_S} Q_j$. The contribution of h_j^0 differs from Q_j by a multiple of $s_y^d T_0^N$, of multiplicity at least $dq > 2M_0$. The contribution of an ℓ-th block with $\ell \geq 1$ is divisible by s_y^ℓ, of multiplicity at least q. Since $\operatorname{mult}_F(Z^2 + s_y^d T_0^N) = 2 \operatorname{mult}_{F_S} Z$, by (A.210) a monomial of an ℓ-th block with $\ell < 0$ contributes multiplicity at least

$$2m_1 - |\ell| q \geq 2m_1 - \frac{q(pm_1 + vM)}{r} \geq m_1 - \frac{qvM}{r} \geq \frac{r - vM}{p} - \frac{qvM}{r}$$

for $r \geq pq$, which is larger than M_0 for $r \gg 0$. Since $\operatorname{mult}_{F_S} Q_j \leq M_0$, $\operatorname{mult}_F (h_j)_y = \operatorname{mult}_{F_S} Q_j$ for any j. The prime divisor over W_N/G induced by F has center the image of 0, since

$$[y] = p[s_y]/r > 0, \quad [t] = [T_0] + v[s_y]/r > 0,$$
$$[z] = [Z] \geq 1/e_S, \quad [x] = 2[Z] - p[s_y]/r \geq (2 - pq/r)/e_S > 0$$

for $r > pq$. By Claim A.55 with $D' := D$,

$$\alpha \leq \frac{a(F, \mathbb{A}^3, \Delta_y(D))}{|I_{G_y}(F)|} = \frac{a(F_S, S_y, C_y) + (1-b)q}{e_S} = \beta + \frac{v(1-\sigma)q}{re_S}.$$

Assume that $\beta = \operatorname{mld}((S_t \ni 0, 0)/G_t)$. The map

$$X_d \dashrightarrow S_t \times \mathbb{A}^1_{s_t}, \qquad (U, V, Z, s_t) \to ((-Z^2/V, V, Z), s_t),$$

is birational and G_t-equivariant, and it is the identity on V, Z, s_t. Recall that $C_t = 0$ near 0 (**Step 4**), so $a(F_S, S_t, C_t) = a(F_S, S_t, 0)$ and $\beta = \operatorname{mld}((S_t \ni 0, C_t)/G_t)$. Both X_d and $S_t \times \mathbb{A}^1_{s_t}$ have canonical sheaves generated near the centers below by $dV \wedge dZ \wedge ds_t/V$, so $a(P, X_d, B_1) = a(P, S_t \times \mathbb{A}^1_{s_t}, B_2)$ for any prime divisor P over both and any boundaries B_1, B_2 with $\operatorname{mult}_P B_1 = \operatorname{mult}_P B_2$, and P has equal inertia groups in G_t over both. Lemma 3.1 applied to the variety S_t, the boundary C_t, the group G_t, the coordinate s_t with its character, the prime divisor F_S, and the s_t-value q (so the integer of that lemma is q_S) implies that the Gauss extension F of F_S with s_t-value q is a prime divisor over $S_t \times \mathbb{A}^1_{s_t}$ with center $(0, 0)$, $I_{G_t}(F) = I_{G_t}(F_S)$, $\operatorname{mult}_F s_t = q$, and

$$a(F, S_t \times \mathbb{A}^1_{s_t}, \operatorname{pr}_1^* C_t + bS_t) = a(F_S, S_t, C_t) + (1-b)q.$$

On X_d, $U = -(Z^2 + s_t^d)/V$ and $\operatorname{mult}_F(Z^2 + s_t^d) = 2 \operatorname{mult}_{F_S} Z$ since $dq > 2M_0$, so $\operatorname{mult}_F U = \operatorname{mult}_{F_S} U \geq 1$. The difference $P_j(U, V, Z) - P_j(-Z^2/V, V, Z)$ is divisible by s_t^d/V with a cofactor of non-negative multiplicity, so $\operatorname{mult}_F(P_j(U, V, Z) - P_j(-Z^2/V, V, Z)) \geq dq - M_0 > M_0$. The contribution of an ℓ-th block with $\ell \geq 1$ has multiplicity at least q, and that of a monomial of an ℓ-th block with $\ell < 0$ has multiplicity at least $m_1 - |\ell| q \geq m_1(1 - pq/r) - qvM/r$, which is larger than M_0 for $r \gg 0$. Thus $\operatorname{mult}_F (h_j)_t = \operatorname{mult}_{F_S} P_j$ for any j. The center on W_N is 0, since $[t] = v[s_t]/r > 0$, $[x] = [U] - p[s_t]/r \geq (1 - pq/r)/e_S > 0$ for $r > pq$, $[y] = [V] + p[s_t]/r > 0$, and $[z] > 0$. By Claim A.55 with $D' := D$,

$$\alpha \leq \frac{a(F, X_d, \Delta_t(D))}{|I_{G_t}(F)|} = \frac{a(F_S, S_t, C_t) + (1-b)q}{e_S} = \beta + \frac{v(1-\sigma)q}{re_S}.$$

Assume that $\beta = 1/h'$. The group $\ker \chi_V$ is cyclic, since χ_z is injective on it, so there exists $q' \in \{1, \dots, h'\}$ such that $\chi_{s_t} = \chi_z^{q'}$ on $\ker \chi_V$, where χ_{s_t} is the character of s_t. Let q be the integer with $2 < q \leq 2 + h'$ and $q \equiv q' \pmod{h'}$. Near the generic point of L_t, V is a unit and (V, Z, s_t) are coordinates on X_d. Let F be the monomial valuation with weights $(0, 1, q)$ on (V, Z, s_t). The residue field of F is generated by V and s_t/Z^q, so

$$I_{G_t}(F) = \{g \in \ker \chi_V \mid \chi_{s_t}(g) = \chi_z(g)^q\} = \ker \chi_V,$$

and $a(F, X_d, 0) = 1 + q$ [JM12, Proposition 5.1 and (5.1)], $\operatorname{mult}_F s_t = q$. Moreover $\operatorname{mult}_F U = 2$, $P_j(U, V, Z) - P_j(0, V, 0) \in (U, Z)$ with $P_j(0, V, 0) \neq 0$, the ℓ-th blocks with $\ell \geq 1$ contribute

multiplicity at least q, and those with $\ell < 0$ contribute at least $2m_1 - |\ell|q$, which is positive for $r \gg 0$. Thus $\operatorname{mult}_F (h_j)_t = 0$, and by Claim A.55,

$$\alpha \le \frac{a(F, X_d, \Delta_t(D))}{|I_{G_t}(F)|} = \frac{1 + q - bq}{h'} = \frac{1}{h'} + \frac{v(1-\sigma)q}{rh'},$$

where the center on W_N is 0 since $[Z] = 1/h'$, $[U] = 2/h'$, $[V] = 0$, $[s_t] = q/h'$, hence $[t] > 0$, $[y] > 0$, $[z] > 0$, and $[x] = (2 - pq/r)/h' > 0$ for $r > pq$.

Since $1 - \sigma \le 1$, $q \le 2M_0 + \max(p, v)e_0 + 2$, and $h' \le ve_0$, we obtain

$$\alpha \le \beta + \frac{M_3}{r}, \qquad M_3 := v(2M_0 + \max(p, v)e_0 + 2). \tag{A.224}$$

Step 7. In this step we fix $\theta > 1$, assume that $(S_t, \theta C_t)$ is lc near L_t and $(S_y, \theta C_y)$ is lc near 0, and prove that $\alpha \ge \beta$ for $r \gg 0$.

Claim A.57. *Let $(Y, B_Y, \mathfrak{b})$ be a pair with an $\mathbb{R}$-ideal $\mathfrak{b}$ and $y \in Y$ a closed point such that $a(P, Y, B_Y, \mathfrak{b}) \ge 0$ for any prime divisor P over Y with center y. Then $(Y, B_Y, \mathfrak{b})$ is lc near y.*

Proof. Otherwise there exist a log resolution $\varphi : Y' \to Y$ of $(Y, B_Y, \mathfrak{b})$ near y and a prime divisor P on Y' with $a(P, Y, B_Y, \mathfrak{b}) < 0$ and $y \in \varphi(P)$. Choose $y' \in P \cap \varphi^{-1}(y)$ and coordinates $(y_1, \dots, y_n)$ at y' with $P = (y_1 = 0)$ in which the exceptional locus and the support of the pullback of the boundary and of $\mathfrak{b}$ are coordinate hyperplanes. For $k \ge 1$ the monomial valuation P_k with weights $(k, 1, \dots, 1)$ has center y' on Y', hence center y on Y, and the log discrepancy of P_k is $k \cdot a(P, Y, B_Y, \mathfrak{b}) + \sum_{i \ge 2} a_i$ [JM12, Proposition 5.1 and (5.1)], where a_i is the log discrepancy of $(y_i = 0)$. Thus $a(P_k, Y, B_Y, \mathfrak{b}) < 0$ for $k \gg 0$. □

Proof of Theorem 7.25 continued. Decreasing θ, we may assume that $\theta - 1 \le p/(4v)$. Let $\theta' := (1 + \theta)/2$. Since

$$\theta' C_t + \frac{1}{2}\operatorname{div}(Z) = \frac{1}{2}\theta C_t + \frac{1}{2}(C_t + \operatorname{div}(Z)),$$

(A.220) implies that $(S_t, \theta' C_t + \frac{1}{2}\operatorname{div}(Z))$ is lc near L_t, and (A.219) implies that $(S_y, \theta' C_y + \frac{1}{2}\operatorname{div}(Z))$ is lc near 0.

Let F be a prime divisor over W_N with center 0. Assume that $[t]/v \le [y]/p$. Claim A.55 with $D' := \theta' D^0$, so $\sigma' = \theta'\sigma$, implies

$$\frac{a(F, W_N, \theta' D^0)}{|I_G(F)|} = \frac{a(F_t, X_d, S_t + \theta' \sum_j c_j \operatorname{div}(P_j) + \frac{1}{2}\operatorname{div}(Z))}{|I_{G_t}(F_t)|} + \frac{1}{2}[z] + v(1 - \theta'\sigma)\mu_F. \tag{A.225}$$

Since X_d is a hypersurface and $S_t = \operatorname{div}(s_t)$ is Cartier and normal, as an A_1 surface singularity, the different on S_t of an $\mathbb{R}$-linear combination of Cartier divisors on X_d not containing S_t, such as $\operatorname{div}(P_j)$ and $\operatorname{div}(Z)$, is its restriction [Kol13, Proposition 4.5(4) and (4.2.10)], applied to $\Delta = 0$ and extended to $\mathbb{R}$-divisors by linearity. Thus [Kaw07, Theorem] shows that

$$(X_d, S_t + \theta' \sum_j c_j \operatorname{div}(P_j) + \frac{1}{2}\operatorname{div}(Z))$$

is lc near L_t. The center of F_t is contained in L_t, so the first term is non-negative. If $[y]/p < [t]/v$, then (A.225) holds with X_d, F_t, G_t, S_t, P_j replaced by $\mathbb{A}^3_{(Z, T_0, s_y)}$, F_y, G_y, S_y, $(h_j^0)_y$, and with $\mu_F = [y]/p$, and the first term is non-negative by [Kaw07, Theorem]. Moreover $\mu_F \le 2[z]/p$, since otherwise $[y] > 2[z]$ and $[t] > 2v[z]/p$, so $[xy] > 2[z]$ and $N[t] > 2[z]$ as $Nv \ge r > p$, and z^2 would be the only term of $xy + z^2 + t^N$ of least multiplicity along F. Since $\sigma \le 1$, $v(1 - \theta'\sigma)\mu_F \ge -v(\theta' - 1)\mu_F$, and since $2v(\theta' - 1)/p = v(\theta - 1)/p \le 1/4$,

$$\frac{a(F, W_N, \theta' D^0)}{|I_G(F)|} \ge \frac{1}{2}[z] - v(\theta' - 1)\mu_F \ge \Big(\frac{1}{2} - \frac{2v}{p}(\theta' - 1)\Big)[z] \ge \frac{1}{4}[z] > 0. \tag{A.226}$$

By Claim A.57, $(W_N, \theta' D^0)$ is lc near 0. By [Kaw07, Theorem], Lemma 4.14 with $n = 4$, at most s_0 functions, and the DCC set $\Lambda_{\theta'} := \theta'(\Gamma \cap (0,1]) \cup \{1\}$ (for $r \gg 0$, k_r is at least the integers L of Lemma 4.14 for $\Lambda_{\theta'}$), and [dFEM11, Proposition 2.11], $(W_N, \theta' D)$ is lc near 0 for $r \gg 0$.

Fix a prime divisor F over W_N with center 0. If $a(F, W_N, D)/|I_G(F)| > 2$, then it is larger than β. Assume that $a(F, W_N, D) \le 2|I_G(F)|$. Since $(W_N, \theta' D)$ is lc, $a(F, W_N, 0) \ge \theta' \sum_j c_j \operatorname{mult}_F h_j$, so $(1 - 1/\theta') a(F, W_N, 0) \le a(F, W_N, D) \le 2|I_G(F)|$. Using also that $(W_N, \theta' D^0)$ is lc and that $(W_N, \operatorname{div}(z))$ is lc by (A.213), we obtain

$$[h_j],\ [h_j^0],\ [z] \le M_1 := \frac{2\theta'}{(\theta' - 1)\gamma}.$$

The regular functions xy, $\varpi := x^v t^p$, and z have $\mathbf{T}$-weight 0, so their characters are trivial on R and have orders at most e_0. The powers of xy, ϖ, and z by the orders of their characters are G-invariant regular functions vanishing at 0, with multiplicity at least one along the prime divisor over W_N/G induced by F. Hence

$$[x] + [y] \ge \frac{1}{e_0}, \qquad v[x] + p[t] \ge \frac{1}{e_0}, \qquad [z] \ge \frac{1}{e_0}. \tag{A.227}$$

If $[h_j] = [h_j^0]$ for any j, then $a(F, W_N, D) = a(F, W_N, D^0)$, and Claim A.56 with $g_j := h_j^0$ implies $a(F, W_N, D)/|I_G(F)| \ge \beta$. Otherwise $[h_j - h_j^0] = \min([h_j], [h_j^0]) \le M_1$ for some j, so some monomial $x^{m_1} y^{m_2} z^{m_3} t^{m_4}$ of an ℓ-th block of h_j with $\ell \ne 0$ satisfies $m_1[x] + m_2[y] + m_4[t] \le M_1$. By (A.210), with $M_2 := \max(p, v) M_1$,

$$[x] \le \frac{M_2}{r - vM} \quad \text{or} \quad \min([y], [t]) \le \frac{M_2}{r}.$$

Assume that $[x] \le M_2/(r - vM)$. By (A.227), $[y] \ge 1/(2e_0)$ and $[t] \ge 1/(2pe_0)$ for $r \gg 0$. By (A.210), a monomial of an ℓ-th block with $\ell \ge 1$ has $[\cdot] \ge r \min([y]/p, [t]/v) \ge r/(2pve_0) > M_1$ for $r \gg 0$. Let h_j^- be the sum of the ℓ-th blocks of h_j with $\ell \le 0$. Then $[h_j^-] = [h_j]$ for any j. Since $m_1 m_2 = 0$, by (A.216) a monomial of an ℓ-th block equals $\varrho^{-vB_j} s_x^{-\ell} (xy)^{m_2} Z^{m_3} V_0^{m_4}$. Thus there exist polynomials $\Phi_{j,\ell}$ with

$$(h_j^-)_x = \sum_{\ell \le 0} s_x^{-\ell} \Phi_{j,\ell}(xy, Z, V_0).$$

Let

$$\Phi_j(s_x, Z, V_0) := \sum_{\ell \le 0} s_x^{-\ell} \Phi_{j,\ell}(-Z^2, Z, V_0), \qquad A_j(Z, V_0) := \Phi_j(0, Z, V_0).$$

By (A.211),

$$A_j(Z, V_0) = \sum_{0 \le k \le \lfloor B_j/p \rfloor} (-Z^2)^{vk} V_0^{B_j - pk} f_{j,k}(Z) + \sum_{k \ge 1} V_0^{B_j + pk} g_{j,k}(Z). \tag{A.228}$$

By (A.216), $(h_j^-)_x - \Phi_j$ is the product of $xy + Z^2 = -t^N$ and a polynomial in xy, Z, V_0, s_x, which has non-negative multiplicity along F_x. Since $[t^N] \ge N/(2pe_0) \ge r/(2pve_0)$ while

$$[(h_j^-)_x] = [\varrho^{vB_j}] + [h_j^-] \le \frac{vMM_2}{p(r - vM)} + M_1,$$

we obtain for $r \gg 0$

$$[(h_j)_x] = [\varrho^{vB_j}] + [h_j^-] = [\Phi_j] \quad \text{for any } j. \tag{A.229}$$

Claim A.58. *Let $C_x := \sum_j c_j \operatorname{div}(A_j)$ on S_x and $\mathfrak{q} := (V_0^p, Z^{2v})$. Then*

$$\operatorname{mld}((S_x \ni 0, C_x, \mathfrak{q}^{(1-\sigma)/p})/G_x) \ge \beta,$$

and $(S_x, C_x, \mathfrak{q}^{(1-\sigma)/p})$ is lc near 0, where the factor $\mathfrak{q}^{(1-\sigma)/p}$ is omitted if $\sigma = 1$.

Proof. By (A.214), (A.215), and (A.216), ϖ is equal to U^v, to $(-Z^2)^v T_0^p$ on S_y, and to V_0^p. The rational maps $S_t \dashrightarrow \mathbb{A}^2_{(Z,\varpi)}$, $(U,V,Z) \to (Z, U^v)$, $S_y \dashrightarrow \mathbb{A}^2_{(Z,\varpi)}$, $(Z,T_0) \to (Z,(-Z^2)^v T_0^p)$, and $S_x \to \mathbb{A}^2_{(Z,\varpi)}$, $(Z,V_0) \to (Z, V_0^p)$, are the quotients by the kernels of G_t, G_y, $G_x \to G/R$, which act by $U \to \zeta^p U$ ($\zeta^v = 1$), $T_0 \to \zeta^{-v} T_0$ ($\zeta^p = 1$), and $V_0 \to \zeta^v V_0$ ($\zeta^p = 1$). The three rational maps to $\mathbb{A}^2_{(Z,\varpi)}$ are equivariant for the action of G/R on $\mathbb{A}^2_{(Z,\varpi)}$ through $(\chi_z, \chi_x^v \chi_t^p)$. Thus S_t/G_t, S_y/G_y, and S_x/G_x are birational to $\overline{S} := \mathbb{A}^2_{(Z,\varpi)}/(G/R)$. Let E be a prime divisor over $\overline{S}$ with center the image of $0 \in S_x$, and define $[f]$ for semi-invariant rational functions f on S_t, S_y, S_x as in **Step 3**. By [Liu26, Lemma 2.6(2)(5)], the log discrepancies at E of the three crepant quotient pairs can be computed on a Galois cover of $\overline{S}$ dominating S_t, S_y, S_x.

Assume that $p[V_0] \le 2v[Z]$. On the cover $U^v = V_0^p$, and by (A.218) and (A.228), $A_j = V_0^{B_j} P_j(U, -Z^2/U, Z)$, and $dU \wedge dZ/U = \frac{p}{v} dV_0 \wedge dZ/V_0$, where $dU \wedge dZ/U$ generates the canonical sheaf of S_t. Hence

$$a(E,(S_x,C_x)/G_x) = a(E,(S_t,C_t)/G_t) + (1-\sigma)[V_0].$$

Moreover $[U] = p[V_0]/v > 0$, $[Z] > 0$, and $[V] = 2[Z] - [U] \ge 0$, so the center of E on S_t/G_t is contained in the image of L_t, and $a(E,(S_t,C_t)/G_t) \ge \beta_t$. Since

$$\frac{1-\sigma}{p}[\mathfrak{q}] = \frac{1-\sigma}{p} \min(p[V_0], 2v[Z]) = (1-\sigma)[V_0],$$

we obtain

$$a(E,(S_x,C_x,\mathfrak{q}^{(1-\sigma)/p})/G_x) \ge \beta_t \ge \beta.$$

Assume that $p[V_0] > 2v[Z]$. On the cover let $\eta := V_0/T_0$, so $\eta^p = (-Z^2)^v$, $[\eta] = 2v[Z]/p$, and $[T_0] = [V_0] - 2v[Z]/p > 0$. As $[Z] > 0$, the center of E on S_y/G_y is the image of 0. By (A.218) and (A.228), $A_j = \eta^{B_j} Q_j$, and $dZ \wedge dV_0 = \eta dZ \wedge dT_0$. Hence

$$a(E,(S_x,C_x)/G_x) = a(E,(S_y,C_y)/G_y) + \frac{2v}{p}(1-\sigma)[Z],$$

and $(1-\sigma)[\mathfrak{q}]/p = 2v(1-\sigma)[Z]/p$, so

$$a(E,(S_x,C_x,\mathfrak{q}^{(1-\sigma)/p})/G_x) \ge \beta_y \ge \beta.$$

Thus the minimal log discrepancy is at least $\beta > 0$. Any prime divisor over S_x with center 0 has non-negative log discrepancy by [Liu26, Lemma 2.6(2)], and Claim A.57 implies the last statement. □

Proof of Theorem 7.25 continued. The group G_x acts diagonally and faithfully on $\mathbb{A}^3_{(s_x,Z,V_0)}$ and on S_x, and $\mathfrak{q}$ is generated by semi-invariant monomials. By Lemma 5.5 with $X := \mathbb{A}^3_{(s_x,Z,V_0)}$, $u := s_x$, $S_0 := S_x$, $H := G_x$, $\mathfrak{a} := \prod_j \Phi_j^{c_j} \mathfrak{q}^{(1-\sigma)/p}$, the coefficient $b := 1$, and Claim A.58,

$$\frac{a(F_x, \mathbb{A}^3, S_x + \sum_j c_j \operatorname{div}(\Phi_j), \mathfrak{q}^{(1-\sigma)/p})}{|I_{G_x}(F_x)|} \ge \beta.$$

By Claim A.55 with $D' := D$ and (A.229),

$$\frac{a(F,W_N,D)}{|I_G(F)|} = \frac{a(F_x,\mathbb{A}^3,\Delta_x(D))}{|I_{G_x}(F_x)|} \ge \beta + (1-\sigma)\Big(\min\Big([V_0], \frac{2v}{p}[Z]\Big) - \frac{v}{r}[s_x]\Big).$$

Since $[t] \ge 1/(2pe_0)$ and $[z] \le M_1$, $N[t] > 2[z]$ for $r \gg 0$, so $[x] + [y] = 2[z]$ by $xy = -z^2 - t^N$. As $[V_0] = v[x]/p + [t]$, $2v[Z]/p = v([x]+[y])/p$, and $v[s_x]/r = v[x]/p$, the bracket is $\min([t], v[y]/p) > 0$. Hence $a(F,W_N,D)/|I_G(F)| \ge \beta$.

Assume that $\min([y],[t]) \le M_2/r$. By (A.227), $[x] \ge 1/(2ve_0)$ for $r \gg 0$. By (A.210), a monomial of an ℓ-th block with $\ell < 0$ has $[\cdot] \ge m_1[x] \ge (r - vM)/(2pve_0) > M_1$ for $r \gg 0$. Let h_j^+ be the sum of the ℓ-th blocks of h_j with $\ell \ge 0$. Then $[h_j^+] = [h_j]$ for

any j, so $a(F, W_N, D) = a(F, W_N, \sum_j c_j \operatorname{div}(h_j^+))$, and Claim A.56 with $g_j := h_j^+$ implies $a(F, W_N, D)/|I_G(F)| \geq \beta$.

Therefore $\alpha \geq \beta$ for $r \gg 0$.

Step 8. In this step we prove that α cannot increase along $\mathbf{T}$.

Suppose that there exists a sequence of data as in Set-up 7.3 with Γ, e_0, w fixed, $r_i \to +\infty$, and positive strictly increasing values α_i. We add the index i to the data. By **Step 1**, possibly passing to a subsequence, we may assume that the number s of the h_{ij} and the integers B_j do not depend on i, and that each sequence $(c_{ij})_i$ is non-decreasing with limit $c_j^* \in (0, 1]$. Since $|G_{t,i}| \leq ve_0$ and $|G_{y,i}| \leq pe_0$, possibly passing to a subsequence, we may assume that the diagonal actions of $G_{t,i}$ on $\mathbb{A}^3_{(U,V,Z)}$ and of $G_{y,i}$ on $\mathbb{A}^2_{(Z,T_0)}$ do not depend on i, since such a subgroup is contained in $\mu^3_{(ve_0)!}$ (resp. $\mu^2_{(pe_0)!}$), and we write G_t and G_y. Thus h, m, and β_t do not depend on i.

If $m = 1$, then by (A.222), **Step 2**, and (A.224), $1/h \leq \alpha_i \leq 1/h + M_3/r_i$ for $i \gg 0$. Thus $\alpha_i \to 1/h \leq \alpha_i$, which is impossible for a strictly increasing sequence.

Assume that $m \geq 2$. By (A.220), $(S_t, 2C_{t,i})$ is lc near L_t for any i. By **Step 4**, the $\beta_{y,i}$ belong to an ACC set and $\beta_{y,i} \geq 1/h$, so possibly passing to a subsequence, $(\beta_{y,i})_i$ is non-increasing with limit $\bar{\beta}_y \geq 1/h$. Let $\bar{\beta} := \min(\beta_t, \bar{\beta}_y)$. Then $\beta_i \geq \bar{\beta}$ and $\beta_i \to \bar{\beta}$, so $\lim_i \alpha_i \leq \bar{\beta}$ by (A.224).

Let $C^*_{y,i} := \sum_j c_j^* \operatorname{div}(Q_{ij})$ on S_y and $\delta_i := \min_j c_{ij}/c_j^*$. Then $\delta_i \leq 1$, $\delta_i \to 1$, and

$$\delta_i(C^*_{y,i} + \operatorname{div}(Z)) \leq C_{y,i} + \operatorname{div}(Z),$$

so by (A.219) the lc threshold of $C^*_{y,i} + \operatorname{div}(Z)$ at 0 is at least δ_i. The coefficients of $C^*_{y,i} + \operatorname{div}(Z)$ belong to the set J^* of the non-zero sums $m_0 + \sum_j m_j c_j^*$ with non-negative integers $m_0, \dots, m_s$, a DCC set, since it has finitely many elements below any bound. By Theorem 2.12(1) with $n = 2$, $I = \{0\}$, $J = J^*$, X a small enough Zariski neighborhood of 0 in S_y, and $\Delta = 0$, the lc thresholds at 0 of $C^*_{y,i} + \operatorname{div}(Z)$ with respect to S_y satisfy the ACC. As they are at least δ_i and $\delta_i \to 1$, $(S_y, C^*_{y,i} + \operatorname{div}(Z))$ is lc near 0 for $i \gg 0$. Let $\beta^*_{y,i} := \operatorname{mld}((S_y \ni 0, C^*_{y,i})/G_y)$. Since $C_{y,i} \leq C^*_{y,i}$, $\beta_{y,i} \geq \beta^*_{y,i}$. By [Liu26, Lemma 2.10(1)(5)], the coefficients of the crepant quotient of $(S_y, C^*_{y,i})$ by G_y belong to the DCC set $\Psi(\Lambda^*_+ \cap [0, 1])$, where $\Lambda^* := \{c_1^*, \dots, c_s^*\}$. By [HL23, Theorem 1.2], there exist prime divisors E_i over S_y/G_y with center the image of 0 computing $\beta^*_{y,i}$ with $a(E_i, S_y/G_y, 0) \leq N^*$, where N^* is the integer of [HL23, Theorem 1.2] for $\Psi(\Lambda^*_+ \cap [0, 1])$. By [Liu26, Lemma 2.6(2)(3)], there exist prime divisors F_i over S_y with center 0 inducing E_i. They compute $\beta^*_{y,i}$, and

$$a(F_i, S_y, 0) = |I_{G_y}(F_i)| a(E_i, S_y/G_y, B_{\mathrm{br}}) \leq pe_0 N^*,$$

where $B_{\mathrm{br}} \geq 0$ is the boundary of the crepant quotient of $(S_y, 0)$ by G_y. Thus $a(F_i, S_y, 0)$ is bounded independently of i. Since $(S_y, C^*_{y,i})$ is lc, $\operatorname{mult}_{F_i} Q_{ij} \leq a(F_i, S_y, 0)/c_j^*$ is bounded independently of i as well. Hence

$$0 \leq \beta_{y,i} - \beta^*_{y,i} \leq \sum_j (c_j^* - c_{ij}) \frac{\operatorname{mult}_{F_i} Q_{ij}}{|I_{G_y}(F_i)|} \to 0,$$

and $\beta^*_{y,i} \to \bar{\beta}_y$.

Each Q_{ij} is G_y-semi-invariant with the character of $T_0^{B_j}$, and $\operatorname{mult}_0 Q_{ij} \leq B_j$. Construction-Proposition 4.1 in case (a) with $n = 2$, the group G_y, and the tuples $(Q_{i1}, \dots, Q_{is})$ yields, after passing to a subsequence and multiplying the Q_{ij} by non-zero constants, a compatible generic limit $(Q_{\infty,1}, \dots, Q_{\infty,s})$ over a field $k \supset \mathbb{C}$. Let $C_\infty := \sum_j c_j^* \operatorname{div}(Q_{\infty,j})$ on $\mathcal{A}_k = \operatorname{Spec} k[[Z, T_0]]$. For $d' > \max_j B_j$, in the notation of Construction-Proposition 4.1, the coefficients of T_0^l with $B_j < l < d'$ in the d'-jets of the Q_{ij} restricted to $(Z = 0)$ vanish. The vanishing of a coefficient is a linear condition on d'-jets, so it holds on $Z_{d'}$ and for $Q_{\infty,j}$. The coefficient of $T_0^{B_j}$ is non-zero on the d'-jets of the Q_{ij}, so it vanishes on a proper closed subset of $Z_{d'}$, which does not contain

the generic point. Thus $Q_{\infty,j}(0,T_0)$ is a polynomial of degree B_j, and

$$((Z=0)\cdot C_\infty)_0 = \sum_j c_j^* \operatorname{mult}_0 Q_{\infty,j}(0,T_0) \le \sum_j c_j^* B_j = \lim_i \sum_j c_{ij} B_j \le 1. \tag{A.230}$$

Assume that $(\mathcal{A}_k, C_\infty)$ is klt near 0. By Theorem 4.2(1) with $\mathfrak{c} = \mathcal{O}$, there exists $\theta_\infty > 1$ such that $(S_y, \theta_\infty C^*_{y,i})$ is lc near 0 for any i, hence so is $(S_y, \theta_\infty C_{y,i})$. Let $\theta := \min(\theta_\infty, 2) > 1$. Since $\theta \le 2$ and $(S_t, 2C_{t,i})$ is lc near L_t by (A.220) ($m \ge 2$), and $(S_y, \theta C_{y,i})$ is lc near 0, both hypotheses of **Step 7** hold, so its conclusion holds for the data with index i for $i \gg 0$. Thus $\alpha_i \ge \beta_i \ge \bar\beta \ge \lim_i \alpha_i$, which contradicts the strict increase.

Assume that $(\mathcal{A}_k, C_\infty)$ is not klt near 0. Since $(S_y, C^*_{y,i})$ is lc near 0 for $i \gg 0$ and $\beta^*_{y,i} \to \bar\beta_y > 0$, Definition-Proposition 5.7 implies that the limit has exactly one floor E_∞, which is a curve through 0, smooth at 0, with coefficient one in C_∞, and that $\bar\beta_y = (1 - b_\infty)/e$ for the numbers b_∞ and e of Lemma 5.6 for the point 0, the curve E_∞, the $\mathbb{R}$-divisor $C_\infty - E_\infty$, and the group G_y. Since $Q_{\infty,j}(0,T_0) \ne 0$, $(Z=0)$ is not a component of C_∞, so $((Z=0)\cdot E_\infty)_0 \ge 1$, and (A.230) implies

$$((Z=0)\cdot E_\infty)_0 = 1, \qquad ((Z=0)\cdot (C_\infty - E_\infty))_0 = 0.$$

Thus $0 \notin \operatorname{Supp}(C_\infty - E_\infty)$, so $b_\infty = \operatorname{mult}_0((C_\infty - E_\infty)|_{E_\infty}) = 0$, and $Z|_{E_\infty}$ is a parameter of E_∞ at 0. Since G_y acts on Z through χ_z, of order h, $e = h$ and $\bar\beta_y = 1/h$. Since $\beta_t \ge 1/h$, $\bar\beta = 1/h$ and $\lim_i \alpha_i \le 1/h$. But $\alpha_i \ge 1/h$ for $i \gg 0$ by **Step 2**, which contradicts the strict increase. □

A.17. **Mixed circles with non-negative adjusted mass.** The proof of Theorem 7.26 adjoins the p-th root ξ of x, which gives a cover $\widetilde{W} \to W_N$. The quotient of $\widetilde{W}$ by the subgroup of order r lying over R that fixes $U := \xi^s z$ and $V := \xi^v t$ is birational to $\mathbb{A}^3_{(u_x,U,V)}$ with $u_x = \xi^r$, and its section $(u_x = 0)$ is the plane $\mathbb{A}^2_{(U,V)}$ with a group of order $p[G:R]$. Prime divisors over W_N with center 0 are compared with $\mathbb{A}^2_{(U,V)}$ through the t-root and y-root sections in the proof of Claim A.59, whose crepant boundaries differ from the pullback of the boundary of the plane by a non-negative multiple of one coordinate divisor.

Proof of Theorem 7.26. The proof is long, so we divide it into several steps.

Let $w_y := p + 2s$, so that $w = (-p, w_y, s, v)$, and let $\bar w := \max(w_y, v)$, the largest absolute value of an entry of w. Since $N \ge 3$, the monomials of least degree of $xy + z^2 + t^N$ are xy and z^2, of $\mathbf{T}$-weight $2s$, so $\kappa(xy + z^2 + t^N) = 2s$ and $\epsilon_{\mathbf{T}} = -p + w_y + s + v - 2s = s + v$. Thus $\delta = \tau - (s+v)$ is the adjusted mass of Set-up 5.1. Assume that the theorem fails. Then there exists a sequence of data (N_i, G_i, D_i) of Set-up 7.3 with Γ, e_0, and w fixed, $r_i := |G_i \cap \mathbf{T}| \to +\infty$, and $\delta_i \ge 0$, such that $0 < \alpha_1 < \alpha_2 < \cdots$. Let $\alpha_\infty := \lim_i \alpha_i$. In **Steps 1–4** we fix i and drop the index: N, G, $R = \{\mathbf{T}(\lambda) : \lambda^r = 1\}$, r, $d = (Nv - 2s)/r$, $D = \sum_j c_j \operatorname{div}(h_j)$, where the h_j are the non-units with $c_j > 0$, written without monomials divisible by xy. Let $\gamma := \min(\Gamma\backslash\{0\})$, and $\gamma := 1$ if $\Gamma \subset \{0\}$. The constants M_0, M_1, M_2 depend only on Γ, p, s, v, e_0, and M_3 and the bound for r in **Step 4** depend also on θ. We write "$r \gg 0$" if r is larger than such a constant. For $g \in G$ we write $g^*x = \chi_x(g)x$, and similarly χ_y, χ_z, χ_t.

Step 1. In this step we bound the data, construct the root plane $\mathbb{A}^2_{(U,V)}$ with the group G_x, and prove that the plane pair is lc.

Let $\deg_j$ be the least degree of a monomial of h_j. By Lemma A.1(2), the exceptional prime divisor E_1 of the blow-up of 0 satisfies $a(E_1, W_N, 0) = 2$ and $\operatorname{mult}_{E_1} h_j = \deg_j$. Since (W_N, D) is klt near 0, Lemma A.1(3) gives

$$\sum_j c_j \deg_j < 2. \tag{A.231}$$

Thus the number of indices j is at most $2/\gamma$, $\deg_j < 2/\gamma$, and $|\kappa_j| \le \bar w \deg_j < M_0 := 2\bar w/\gamma$. The group R acts on a monomial of $\mathbf{T}$-weight κ by λ^κ, and the monomials of h_j have one character,

so their $\mathbf{T}$-weights are congruent modulo r. For $r > 2M_0$, all the monomials of least degree of h_j have weight κ_j, and we write

$$h_j = \sum_{m\in\mathbb{Z}} h_{j,m},$$

where $h_{j,m}$ is the sum of the terms of h_j of weight $\kappa_j + mr$, so $h_{j,0} = h_j^0$. A monomial of $h_j - h_j^0$ has weight of absolute value at least $r - M_0$, hence degree at least $(r - M_0)/\bar{w}$.

The equation of W_N is semi-invariant under R, and xy, z^2, t^N have weights $2s, 2s, Nv$, so $d \in \mathbb{Z}$. If $d \le 0$, then $N \le 2s/v$. If $d_i \le 0$ for infinitely many i, then there exists $N' \le 2s/v$ such that the set $\{i : N_i = N'\}$ is infinite, and Theorem 7.8, applied to the indices i with $N_i = N'$, contradicts the strict increase of the α_i. Possibly passing to a subsequence, we may assume that $d \ge 1$ for any i. In particular $N = (dr + 2s)/v \ge r/v$.

Let $W_0 := (xy + z^2 = 0) \subset \mathbb{A}^4$ and $D^0 := \sum_j c_j \operatorname{div}(h_j^0|_{W_0})$. Since (W_N, D) is klt near 0, the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^2 + t^N) + \sum_j c_j \operatorname{div}(h_j)\Big)$$

is lc near 0 by inversion of adjunction [Kaw07, Theorem]. The differences t^N and $h_j - h_j^0$ belong to $\mathfrak{m}_0^{\lceil (r-M_0)/\bar{w}\rceil}$. By Lemma 4.14 with $n = 4$ and $\Lambda = \{1\} \cup (\Gamma\backslash\{0\})$, taking the largest of the integers for the numbers of functions at most $1 + 2/\gamma$ (**Step 1**), together with [dFEM11, Proposition 2.11], and by [Kaw07, Theorem] again, (W_0, D^0) is lc near 0 for $r \gg 0$.

Let $\widetilde{W}$ be the normalization of $W_N \times_{\mathbb{A}^1_x} \mathbb{A}^1_\xi$, where $\xi^p = x$, and let

$$U := \xi^s z, \quad V := \xi^v t, \quad y_\xi := \xi^{w_y} y, \quad u_x := \xi^r.$$

Since $\xi^{2s}(xy + z^2 + t^N) = y_\xi + U^2 + \xi^{2s-Nv}V^N$ and $Nv - 2s = dr$,

$$y_\xi = -U^2 - V^N u_x^{-d} \quad \text{on } \widetilde{W}. \tag{A.232}$$

The rational map $(\xi, U, V) : \widetilde{W} \dashrightarrow \mathbb{A}^3$ is birational, with inverse $x = \xi^p$, $z = \xi^{-s}U$, $t = \xi^{-v}V$, $y = -(z^2 + t^N)/x$. Let

$$\hat{G}_x := \{(g, \zeta) \in G \times \mathbb{C}^* : \zeta^p = \chi_x(g)\}, \qquad \hat{R}_x := \{(\mathbf{T}(\lambda), \lambda^{-1}) : \lambda^r = 1\},$$

where (g, ζ) acts on $\widetilde{W}$ by g on W_N and by $\xi \to \zeta\xi$. Then $|\hat{G}_x| = p|G|$, $|\hat{R}_x| = r$, and (g, ζ) multiplies U, V, u_x by $\chi_z(g)\zeta^s$, $\chi_t(g)\zeta^v$, ζ^r. Thus $\hat{R}_x$ fixes U and V and acts on ξ by μ_r, so (u_x, U, V) induces a birational map $\widetilde{W}/\hat{R}_x \dashrightarrow \mathbb{A}^3_{(u_x,U,V)}$, on which $G_x := \hat{G}_x/\hat{R}_x$, of order $p[G : R]$, acts diagonally. Since G preserves the equation of W_N up to a character, $\chi_x\chi_y = \chi_z^2$. If (g, ζ) fixes U and V, then $\chi_z(g) = \zeta^{-s}$, $\chi_t(g) = \zeta^{-v}$, $\chi_x(g) = \zeta^p$, and $\chi_y(g) = \zeta^{-w_y}$, so $g = \mathbf{T}(\zeta^{-1}) \in R$ and $(g, \zeta) \in \hat{R}_x$. Thus G_x acts faithfully on $\mathbb{A}^2_{(U,V)}$. The deck group $\{(1, \zeta) : \zeta^p = 1\}$ has trivial intersection with $\hat{R}_x$, since $\mathbf{T}(\lambda) = 1$ forces $\lambda = 1$, so G_x contains a subgroup of order p acting by $(U, V) \to (\zeta^s U, \zeta^v V)$. Since $\hat{G}_x \to G$ is surjective, $W_N/G = \widetilde{W}/\hat{G}_x$, and we obtain a birational map $W_N/G \dashrightarrow \mathbb{A}^3_{(u_x,U,V)}/G_x$.

The 3-form $\omega := x^{-1}dx \wedge dz \wedge dt$ generates the canonical sheaf of W_N near 0, and on $\widetilde{W}$,

$$\omega = p\xi^{-s-v-1}d\xi \wedge dU \wedge dV.$$

A monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of $\mathbf{T}$-weight κ equals $\xi^{-\kappa}y_\xi^{i_2}U^{i_3}V^{i_4}$. Thus $\Phi_j := \xi^{\kappa_j}h_j$ satisfies

$$\Phi_j = \sum_m u_x^{-m} Q_{j,m}(y_\xi, U, V), \qquad Q_{j,m}(y_\xi, U, V) := \xi^{\kappa_j + mr} h_{j,m}, \tag{A.233}$$

with polynomials $Q_{j,m}$, and $Q_{j,0}(y_\xi, U, V) = h_j^0(1, y_\xi, U, V)$. By (A.232), Φ_j is a rational function of (u_x, U, V). Pulling back ω and D, the crepant pullback of (W_N, D) to $\mathbb{A}^3_{(\xi,U,V)}$ is $(\mathbb{A}^3_{(\xi,U,V)}, B_\xi)$

with

$$B_\xi := (1-\delta)\operatorname{div}(\xi) + \sum_j c_j \operatorname{div}(\Phi_j),$$

and the crepant quotient of $(\mathbb{A}^3_{(\xi,U,V)}, B_\xi)$ by μ_r is $(\mathbb{A}^3_{(u_x,U,V)}, B_x)$ with

$$B_x := \Big(1-\frac{\delta}{r}\Big)\operatorname{div}(u_x) + \sum_j c_j \operatorname{div}(\Phi_j).$$

By [Liu26, Lemma 2.6(2)(5)] and the formula at the end of [Liu26, Lemma 2.6], for a prime divisor F over W_N and a prime divisor F_x over $\mathbb{A}^3_{(u_x,U,V)}$ whose image over W_N/G is the image of F,

$$\frac{a(F, W_N, D)}{|I_G(F)|} = \frac{a(F_x, \mathbb{A}^3, B_x)}{|I_{G_x}(F_x)|}, \tag{A.234}$$

both sides being the log discrepancy of the image of F over W_N/G with respect to the crepant quotient. The center of F is 0 if and only if x, y, z, t have positive multiplicity along a prime divisor over $\widetilde{W}$ lying over F.

On W_0 the computation of (A.232) and (A.233) holds with $V^N u_x^{-d}$ removed, so on the normalization of $W_0 \times_{\mathbb{A}^1_x} \mathbb{A}^1_\xi$ we have

$$\xi^{\kappa_j} h^0_j = P_j(U,V), \qquad P_j(U,V) := h^0_j(1, -U^2, U, V),$$

and the crepant pullback of (W_0, D^0) to $\mathbb{A}^3_{(\xi,U,V)}$ is

$$(1-\delta)\operatorname{div}(\xi) + \sum_j c_j \operatorname{div}(P_j).$$

A monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ of h^0_j maps to $(-1)^{i_2} U^{2i_2+i_3} V^{i_4}$, and $p(i_2 - i_1) = \kappa_j - s(2i_2 + i_3) - v i_4$. Since $i_1 i_2 = 0$, the exponents (i_1, i_2, i_3, i_4) are determined by $(2i_2 + i_3, i_4)$. Thus no two monomials cancel, $P_j \neq 0$, $\operatorname{mult}_0 P_j \le 2\deg_j < 4/\gamma$, and P_j is G_x-semi-invariant. Let

$$\Delta := \sum_j c_j \operatorname{div}(P_j) \quad \text{on } \mathbb{A}^2_{(U,V)}.$$

We show that $(\mathbb{A}^2, \Delta)$ is lc near 0. Let P be a prime divisor over $\mathbb{A}^2$ with center 0, k a positive integer, and F_k the Gauss extension of $k \operatorname{mult}_P$ with ξ-value 1 [Liu26, Definition-Proposition 2.11]. Then F_k is a prime divisor over $\mathbb{A}^3_{(\xi,U,V)}$, $\operatorname{mult}_{F_k} P_j = k \operatorname{mult}_P P_j$, and $a(F_k, \mathbb{A}^3, 0) = k a(P, \mathbb{A}^2, 0) + 1$ [Liu26, Lemma 18.1]. Hence

$$a\Big(F_k, \mathbb{A}^3, (1-\delta)\operatorname{div}(\xi) + \sum_j c_j \operatorname{div}(P_j)\Big) = k a(P, \mathbb{A}^2, \Delta) + \delta.$$

The multiplicities along F_k of x, z, t, $y = -\xi^{-w_y} U^2$ are p, $k \operatorname{mult}_P U - s$, $k \operatorname{mult}_P V - v$, $2k \operatorname{mult}_P U - w_y$, which are positive for $k > \bar{w}$. Thus F_k is over a prime divisor over W_0 with center 0, and since (W_0, D^0) is lc near 0, the left-hand side is non-negative [Liu26, Lemma 2.6(2)]. Letting $k \to +\infty$ we obtain $a(P, \mathbb{A}^2, \Delta) \ge 0$. If $(\mathbb{A}^2, \Delta)$ were not lc near 0, let E be a prime divisor over $\mathbb{A}^2$ with $a(E, \mathbb{A}^2, \Delta) < 0$ whose center contains 0, and take a log resolution of $(\mathbb{A}^2, \Delta)$ on which E is a prime divisor E' and the preimage of 0 is a divisor. If E' is contained in the preimage of 0, its center is 0. Otherwise E' intersects a component Σ of the preimage of 0, and blowing up a point of $E' \cap \Sigma$ and then repeatedly the intersection of E' with the newest exceptional divisor yields prime divisors with center 0 and log discrepancies $a(\Sigma, \mathbb{A}^2, \Delta) + k a(E, \mathbb{A}^2, \Delta)$, $k = 1, 2, \dots$, which tend to $-\infty$ [KM98, Lemma 2.29 and Corollary 2.31(1)]. Thus $(\mathbb{A}^2, \Delta)$ is lc near 0.

Let $\beta := \operatorname{mld}((\mathbb{A}^2 \ni 0, \Delta)/G_x)$. The coefficients of the crepant quotient of $(\mathbb{A}^2, \Delta)$ by G_x belong to the DCC set $\Psi(\Gamma_+ \cap [0,1])$ [Liu26, Lemma 2.10(1)(5)]. Thus $\beta \in [0, 2]$ (the blow-up of 0, [KM98, Lemma 2.29]) belongs to an ACC set depending only on Γ [HL24, Theorem 1.2], and by [HL23, Theorem 1.2] there exists a prime divisor E over $\mathbb{A}^2/G_x$ computing β with $a(E, \mathbb{A}^2/G_x, 0) \le N_\Gamma$,

where N_Γ is the integer of [HL23, Theorem 1.2] for the DCC set $\Psi(\Gamma_+ \cap [0,1])$. By [Liu26, Lemma 2.6(2)(3)], there exists a prime divisor P_β over $\mathbb{A}^2$ with center 0 lying over E, P_β computes β, and

$$a(P_\beta, \mathbb{A}^2, 0) = |I_{G_x}(P_\beta)| a(E, \mathbb{A}^2/G_x, B_{\mathrm{br}}),$$

where $B_{\mathrm{br}} \geq 0$ is the boundary of the crepant quotient of $(\mathbb{A}^2, 0)$ by G_x. Since

$$a(E, \mathbb{A}^2/G_x, B_{\mathrm{br}}) \leq a(E, \mathbb{A}^2/G_x, 0) \leq N_\Gamma, \qquad |I_{G_x}(P_\beta)| \leq |G_x| = p[G:R] \leq pe_0,$$

we obtain

$$a(P_\beta, \mathbb{A}^2, 0) \leq pe_0 N_\Gamma.$$

Since P_β computes β,

$$a(P_\beta, \mathbb{A}^2, 0) - \sum_k c_k \operatorname{mult}_{P_\beta} P_k = a(P_\beta, \mathbb{A}^2, \Delta) = |I_{G_x}(P_\beta)|\beta \geq 0. \tag{A.235}$$

Let $M_1 := pe_0 N_\Gamma/\gamma$. Since $a(P_\beta, \mathbb{A}^2, 0) \geq \operatorname{mult}_{P_\beta} U + \operatorname{mult}_{P_\beta} V$ [Liu26, Lemma 2.13] and

$$\gamma \operatorname{mult}_{P_\beta} P_j \leq c_j \operatorname{mult}_{P_\beta} P_j \leq a(P_\beta, \mathbb{A}^2, 0),$$

where the second inequality holds by (A.235), the multiplicities $\operatorname{mult}_{P_\beta} U$, $\operatorname{mult}_{P_\beta} V$, and $\operatorname{mult}_{P_\beta} P_j$ are at most M_1.

Step 2. In this step we prove that $\alpha \leq \beta + M_2/r$ for $r \gg 0$.

By (A.231), $0 \leq \delta \leq \tau \leq \bar{w} \sum_j c_j \deg_j < 2\bar{w}$. Let $e := |I_{G_x}(P_\beta)| \leq pe_0$, let χ be the character of u_x on G_x, let q be the integer of Lemma 3.1 for P_β and χ, and let ℓ be the integer with $M_1 < \ell \leq M_1 + e$ and $\ell \equiv q \pmod e$. By Proposition 5.4 for $S = \mathbb{A}^2_{(U,V)}$, the point 0, $u = u_x$, $\tilde{H} = G_x$, which acts faithfully on S by **Step 1**, $b = 1 - \delta/r \leq 1$, and the regular functions P_j, which do not depend on u_x, the Gauss extension F of $\operatorname{mult}_{P_\beta}$ with u_x-value ℓ is a prime divisor over $\mathbb{A}^3_{(u_x,U,V)}$ with center 0, $|I_{G_x}(F)| = e$, $\operatorname{mult}_F P_j = \operatorname{mult}_{P_\beta} P_j \leq M_1$, and

$$\frac{a\big(F, \mathbb{A}^3, (1 - \frac{\delta}{r}) \operatorname{div}(u_x) + \sum_j c_j \operatorname{div}(P_j)\big)}{e} = \beta + \frac{\delta \ell}{re}. \tag{A.236}$$

Recall that the multiplicity along F of $\sum_i u_x^i g_i(U,V)$ is $\min_i(i\ell + \operatorname{mult}_{P_\beta} g_i)$.

We show that $\operatorname{mult}_F \Phi_j = \operatorname{mult}_F P_j$ for $r \gg 0$, which is the multiplicity along F of $Q_{j,0}(-U^2, U, V)$. Since $\operatorname{mult}_{P_\beta} V \geq 1$, $\ell \leq M_1 + pe_0$ and $d \geq 1$,

$$\operatorname{mult}_F(V^N u_x^{-d}) = N \operatorname{mult}_{P_\beta} V - d\ell \geq d\Big(\frac{r}{v} - \ell\Big) + \frac{2s}{v} \geq \frac{r}{v} - M_1 - pe_0.$$

For $r \gg 0$ the right-hand side is larger than $2M_1 + 2\operatorname{mult}_{P_\beta} U$, so $\operatorname{mult}_F y_\xi = 2\operatorname{mult}_{P_\beta} U \geq 2$ by (A.232). The difference $Q_{j,0}(y_\xi, U, V) - Q_{j,0}(-U^2, U, V)$ is $y_\xi + U^2 = -V^N u_x^{-d}$ times a polynomial in y_ξ, U, V, so the multiplicity of the difference along F is larger than M_1. For $m < 0$, $\operatorname{mult}_F(u_x^{-m} Q_{j,m}) \geq \ell > M_1$. For $m > 0$, by (A.233) a monomial $y_\xi^{i_2} U^{i_3} V^{i_4}$ of $Q_{j,m}$ is $\xi^{\kappa_j + mr} x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ for a monomial $x^{i_1} y^{i_2} z^{i_3} t^{i_4}$ of $h_{j,m}$, of weight $\kappa_j + mr$, so $w_y i_2 + s i_3 + v i_4 = \kappa_j + mr + p i_1 \geq \kappa_j + mr$. Let $\eta := \min(2/w_y, 1/v)$ if $s = 0$, and $\eta := \min(2/w_y, 1/v, 1/s)$ if $s > 0$. Then $2i_2 + i_3 + i_4 \geq \eta(w_y i_2 + s i_3 + v i_4)$, and since $\operatorname{mult}_F y_\xi \geq 2$ and $\operatorname{mult}_F U, \operatorname{mult}_F V \geq 1$,

$$\operatorname{mult}_F\left(u_x^{-m} y_\xi^{i_2} U^{i_3} V^{i_4}\right) \geq \eta(\kappa_j + mr) - m\ell \geq m(\eta r - \ell) - \eta M_0 \geq \eta r - \ell - \eta M_0$$

once $\eta r \geq \ell$, and $\eta r - \ell - \eta M_0 > M_1$ for $r \gg 0$. By (A.233), $\Phi_j - P_j$ has multiplicity larger than $M_1 \geq \operatorname{mult}_F P_j$ along F, so $\operatorname{mult}_F \Phi_j = \operatorname{mult}_F P_j$, and the left-hand side of (A.236) equals $a(F, \mathbb{A}^3, B_x)/e$.

Let F_W be a prime divisor over W_N whose image over W_N/G is the image of F. Normalizing by $\operatorname{mult} \xi = \ell/r$, the multiplicities of $x, z, t, y = \xi^{-w_y} y_\xi$ are

$$\frac{p\ell}{r}, \quad \operatorname{mult}_{P_\beta} U - \frac{s\ell}{r}, \quad \operatorname{mult}_{P_\beta} V - \frac{v\ell}{r}, \quad 2\operatorname{mult}_{P_\beta} U - \frac{w_y \ell}{r},$$

which are positive for $r \gg 0$. Thus the center of F_W is 0, and by (A.234) and (A.236),

$$\alpha \le \frac{a(F_W, W_N, D)}{|I_G(F_W)|} = \beta + \frac{\delta\ell}{re} \le \beta + \frac{M_2}{r}, \qquad M_2 := 2\bar{w}(M_1 + pe_0). \tag{A.237}$$

Step 3. In this step we prove the following lower bound.

Claim A.59. *Let $h'_1, \dots, h'_n$ be G-semi-invariant polynomials vanishing at 0, $c'_1, \dots, c'_n$ positive real numbers, κ'_k the least $\mathbf{T}$-weight of a monomial of h'_k, h'^0_k the sum of the terms of h'_k of weight κ'_k, and $P'_k := h'^0_k(1, -U^2, U, V)$. Assume that $P'_k \neq 0$ for any k, that*

$$\delta' := \sum_k c'_k \kappa'_k - (s+v) \ge 0,$$

*and that $(\mathbb{A}^2, \Delta')$ is lc near 0, where $\Delta' := \sum_k c'_k \operatorname{div}(P'_k)$ (the P'_k are G_x-semi-invariant, as the P_j in **Step 1**). Let $\beta' := \operatorname{mld}((\mathbb{A}^2 \ni 0, \Delta')/G_x)$ and $D' := \sum_k c'_k \operatorname{div}(h'_k|_{W_N})$, where $h'_k|_{W_N} \neq 0$ since $P'_k \neq 0$. Then (W_N, D') is lc near 0, and*

$$\frac{a(F, W_N, D')}{|I_G(F)|} \ge \beta' \tag{A.238}$$

for any prime divisor F over W_N with center 0.

Proof. Since the weights of the monomials of h'_k are congruent modulo r, they belong to $\kappa'_k + r\mathbb{Z}_{\ge 0}$.

Let F be a prime divisor over W_N with center 0, and $\rho_F := \min(\operatorname{mult}_F y/w_y, \operatorname{mult}_F t/v) > 0$. We claim that $\operatorname{mult}_F z > s\rho_F$. If $s = 0$, the inequality holds since the center of F is 0. If $s > 0$ and $\operatorname{mult}_F z \le s\rho_F$, then

$$2\operatorname{mult}_F z \le 2s\rho_F < \min\left(\operatorname{mult}_F x + w_y\rho_F, Nv\rho_F\right) \le \min(\operatorname{mult}_F(xy), \operatorname{mult}_F(t^N)),$$

because $\operatorname{mult}_F x > 0$, $w_y \ge 2s$, and $Nv = dr + 2s > 2s$. Then z^2 is the only term of $xy + z^2 + t^N$ of least multiplicity along F, which is impossible. If $\rho_F = \operatorname{mult}_F t/v$, we use the t-root chart, and otherwise the y-root chart.

The t-root chart. Let $\sigma_t^v = t$, $x = \sigma_t^{-p} X$, $y = \sigma_t^{w_y} Y$, $z = \sigma_t^s Z$, and $u_t := \sigma_t^r$. Then

$$xy + z^2 + t^N = \sigma_t^{2s}(XY + Z^2 + u_t^d),$$

and the normalization of $W_N \times_{\mathbb{A}^1_t} \mathbb{A}^1_{\sigma_t}$ is birational to

$$(XY + Z^2 + \sigma_t^{rd} = 0) \subset \mathbb{A}^4_{(X,Y,Z,\sigma_t)},$$

whose quotient by $\sigma_t \to \lambda\sigma_t$, $\lambda \in \mu_r$, is

$$W_t := (XY + Z^2 + u_t^d = 0) \subset \mathbb{A}^4_{(X,Y,Z,u_t)}.$$

The variety W_t is normal, klt, and integral near 0, since $d \ge 1$ [KM98, Corollary 5.38]. Let

$$\hat{G}_t := \{(g, \zeta) : \zeta^v = \chi_t(g)\}, \qquad G_t := \hat{G}_t / \{(\mathbf{T}(\lambda), \lambda) : \lambda^r = 1\},$$

where $\hat{G}_t$ acts by g and $\sigma_t \to \zeta\sigma_t$. The group G_t has order $v[G : R]$. Then G_t acts diagonally on $\mathbb{A}^4_{(X,Y,Z,u_t)}$, preserving W_t, with (g, ζ) multiplying X, Y, Z by $\chi_x(g)\zeta^p, \chi_y(g)\zeta^{-w_y}, \chi_z(g)\zeta^{-s}$. If (g, ζ) fixes X, Y, Z, then $g = \mathbf{T}(\zeta) \in R$, so G_t acts faithfully on $S_t := W_t \cap (u_t = 0) = (XY + Z^2 = 0) \subset \mathbb{A}^3_{(X,Y,Z)}$. On the root cover, $\omega = v\sigma_t^{s+v-1} X^{-1} dX \wedge dZ \wedge d\sigma_t$, and $h'_k = \sigma_t^{\kappa'_k} P^t_k$, where P^t_k is a polynomial in X, Y, Z, u_t with $P^t_k|_{u_t=0} = h'^0_k(X, Y, Z, 1)$, since a monomial of weight $\kappa'_k + mr$ equals $\sigma_t^{\kappa'_k} u_t^m X^{i_1} Y^{i_2} Z^{i_3}$. As in **Step 1**, the crepant quotient of (W_N, D') by G corresponds to that of (W_t, B_t) by G_t, where

$$B_t := \left(1 + \frac{\delta'}{r}\right) \operatorname{div}(u_t) + \sum_k c'_k \operatorname{div}(P^t_k),$$

and (A.234) holds with (W_t, B_t, G_t) in place of $(\mathbb{A}^3, B_x, G_x)$. Normalizing by $\operatorname{mult} \sigma_t = \rho_F$, the multiplicities of X, Z, u_t along a prime divisor F_t over W_t corresponding to F are $\operatorname{mult}_F x + p\rho_F$,

$\mathrm{mult}_F z - s\rho_F$, $r\rho_F$, which are positive, and that of Y is $\mathrm{mult}_F y - w_y\rho_F \geq 0$. Thus the center of F_t is contained in the axis $C_t := (X = Z = u_t = 0)$, which is a G_t-invariant curve contained in S_t.

The y-root chart. Let $\sigma_y^{w_y} = y$, $x = \sigma_y^{-p}X$, $z = \sigma_y^s Z$, $t = \sigma_y^v T_0$, and $u_y := \sigma_y^r$. Then $X = -Z^2 - u_y^d T_0^N$, and the chart is $\mathbb{A}^3_{(u_y,Z,T_0)}$. Let G_y be the quotient of $\{(g,\zeta) : \zeta^{w_y} = \chi_y(g)\}$ by $\{(\mathbf{T}(\lambda),\lambda) : \lambda^r = 1\}$, of order $w_y[G:R]$. It acts diagonally on $\mathbb{A}^3_{(u_y,Z,T_0)}$, and if (g,ζ) fixes Z and T_0, then $\chi_x(g) = \chi_z(g)^2\chi_y(g)^{-1} = \zeta^{-p}$ and $g = \mathbf{T}(\zeta) \in R$, so G_y acts faithfully on $S_y := (u_y = 0) = \mathbb{A}^2_{(Z,T_0)}$. Since

$$\omega = -y^{-1}dy \wedge dz \wedge dt = -w_y\sigma_y^{s+v-1}d\sigma_y \wedge dZ \wedge dT_0,$$

the crepant quotient of (W_N, D') by G corresponds to that of $(\mathbb{A}^3, B_y)$ by G_y, where

$$B_y := \Big(1 + \frac{\delta'}{r}\Big)\operatorname{div}(u_y) + \sum_k c'_k \operatorname{div}(P_k^y), \qquad P_k^y := \sigma_y^{-\kappa'_k}h'_k, \qquad P_k^y|_{u_y=0} = h_k'^0(-Z^2, 1, Z, T_0),$$

where P_k^y is a polynomial in u_y, Z, T_0. Along a prime divisor F_y corresponding to F, normalized by $\mathrm{mult}\,\sigma_y = \rho_F$, we have $\mathrm{mult}_{F_y} Z > 0$, $\mathrm{mult}_{F_y} u_y > 0$, and $\mathrm{mult}_{F_y} T_0 = \mathrm{mult}_F t - v\rho_F \geq 0$. Thus the center of F_y is contained in the axis $C_y := (u_y = Z = 0) \subset S_y$.

The sections and the plane. On S_t and on S_y, let

$$B_{S_t} := \frac{\delta'}{p}\operatorname{div}(X) + \sum_k c'_k \operatorname{div}(P_k^t|_{S_t}), \qquad B_{S_y} := \frac{2\delta'}{p}\operatorname{div}(Z) + \sum_k c'_k \operatorname{div}(P_k^y|_{S_y}).$$

We show that for any prime divisor P over S_t with center contained in C_t (resp. over S_y with center contained in C_y),

$$\frac{a(P, S_t, B_{S_t})}{|I_{G_t}(P)|} \geq \beta' \qquad \Big(\text{resp. } \frac{a(P, S_y, B_{S_y})}{|I_{G_y}(P)|} \geq \beta'\Big). \tag{A.239}$$

Let $\widetilde{S}_t$ be the normalization of $S_t \times_{\mathbb{A}^1_X} \mathbb{A}^1_\Xi$, where $\Xi^p = X$, and let

$$\varphi_t := (\Xi^s Z, \Xi^v) : \widetilde{S}_t \to \mathbb{A}^2_{(U,V)}.$$

By the $\mathbf{T}$-homogeneity of $h_k'^0$ with $\lambda = \Xi$, $\varphi_t^* P'_k = \Xi^{\kappa'_k} P_k^t|_{S_t}$, and a direct calculation shows

$$\varphi_t^*(dU \wedge dV) = -\frac{v}{p}\Xi^{s+v}\frac{dX \wedge dZ}{X}.$$

Let $\pi_t : \widetilde{S}_t \to S_t$ be the projection. Since $\operatorname{div}(X) = p\operatorname{div}(\Xi)$ on $\widetilde{S}_t$ and $\sum_k c'_k\kappa'_k = \delta' + s + v$,

$$\varphi_t^*\Delta' - \operatorname{div}\big(\varphi_t^*(dU \wedge dV)\big) = \pi_t^* B_{S_t} - \operatorname{div}\big(\pi_t^*(X^{-1}dX \wedge dZ)\big),$$

so the crepant pullbacks of $(\mathbb{A}^2, \Delta')$ by φ_t and of (S_t, B_{S_t}) by π_t coincide. Let χ_X be the character of X on G_t. The group

$$\check{G}_t := \{(g', \zeta) \in G_t \times \mathbb{C}^* : \zeta^p = \chi_X(g')\}$$

acts on $\widetilde{S}_t$, with quotient S_t/G_t, and acts on U and V by scalars. For g' the class of (g, ζ_t), the element (g', ζ) multiplies U and V by $\chi_z(g)\zeta_x^s$ and $\chi_t(g)\zeta_x^v$, where $\zeta_x := \zeta/\zeta_t$ satisfies $\zeta_x^p = \chi_x(g)$, which is the action of the class of $(g, \zeta_x) \in \hat{G}_x$. Conversely any $(g, \zeta_x) \in \hat{G}_x$ arises from ζ_t with $\zeta_t^v = \chi_t(g)$ and $\zeta = \zeta_x\zeta_t$. Thus the image of $\check{G}_t$ in $\mathrm{Aut}(\mathbb{A}^2)$ is G_x, and the kernel K_t has order $|\check{G}_t|/|G_x| = v$. Since φ_t has degree v and $\check{G}_t$ acts faithfully, $\widetilde{S}_t/K_t \to \mathbb{A}^2$ is birational. By [Liu26, Lemma 2.6(2)(5)], the crepant quotients of (S_t, B_{S_t}) by G_t and of $(\mathbb{A}^2, \Delta')$ by G_x are crepant birational, compatibly with the log discrepancies divided by the orders of the inertia groups. If the center of P is contained in C_t, then on $\widetilde{S}_t$ the center of the pullback of P is over $X = Z = 0$, where $\Xi = 0$, so U and V have positive multiplicity and the corresponding prime divisor over $\mathbb{A}^2$ has center 0. This implies (A.239) for S_t.

On S_y, let $\widetilde{S}_y$ be the normalization of $(\Xi^p = -Z^2) \subset \mathbb{A}^3_{(\Xi,Z,T_0)}$ and $\varphi_y := (\Xi^s Z, \Xi^v T_0)$. Then $\varphi_y^* P'_k = \Xi^{\kappa'_k} P^y_k|_{S_y}$ and, since $p\Xi^{p-1} d\Xi = -2Z dZ$,

$$\varphi_y^*(dU \wedge dV) = \frac{w_y}{p} \Xi^{s+v} dZ \wedge dT_0.$$

Since

$$\frac{2\delta'}{p} \operatorname{div}(Z) = \delta' \operatorname{div}(\Xi) \quad \text{on } \widetilde{S}_y,$$

the identity of crepant pullbacks holds with

$$(\varphi_y, \pi_y, B_{S_y}, dZ \wedge dT_0) \quad \text{in place of} \quad (\varphi_t, \pi_t, B_{S_t}, X^{-1} dX \wedge dZ),$$

where $\pi_y : \widetilde{S}_y \to S_y$ is the projection. Let χ_Z be the character of Z on G_y, and let

$$\check{G}_y := \{(g', \zeta) \in G_y \times \mathbb{C}^* : \zeta^p = \chi_Z(g')^2\},$$

which acts on $\widetilde{S}_y$ with quotient S_y/G_y. The image of $\check{G}_y$ in $\operatorname{Aut}(\mathbb{A}^2)$ is G_x by the computation for $\check{G}_t$, and its kernel K_y has order $|\check{G}_y|/|G_x| = w_y$. An element $(g', \zeta) \in \check{G}_y$ acting trivially on $\widetilde{S}_y$ fixes Ξ, Z, and T_0, so $\zeta = 1$ and $g' = 1$, since G_y acts faithfully on S_y. An element $(g', \zeta) \in K_y$ fixes U, so $\zeta^{w_y} = 1$ by $\Xi^{w_y} = -U^2$, and it is determined by ζ, since $Z = \Xi^{-s} U$ and $T_0 = \Xi^{-v} V$ on $\widetilde{S}_y$. Thus K_y acts simply transitively on a general fiber of φ_y, which consists of the w_y points with $\Xi^{w_y} = -U^2$, and since each component of $\widetilde{S}_y$ dominates $\mathbb{A}^2$, K_y permutes the components transitively. Hence for a component $\widetilde{S}^1_y$ with stabilizer $\check{G}^1_y$, we have $\check{G}_y = K_y \check{G}^1_y$, so the image of $\check{G}^1_y$ in $\operatorname{Aut}(\mathbb{A}^2)$ is G_x, the order of $K_y \cap \check{G}^1_y$ is the degree of $\varphi_y|_{\widetilde{S}^1_y}$, $\widetilde{S}^1_y/(K_y \cap \check{G}^1_y) \to \mathbb{A}^2$ is birational, and $\widetilde{S}^1_y/\check{G}^1_y = S_y/G_y$. If $\widetilde{S}_y$ is reducible, we use $\widetilde{S}^1_y$ and $\check{G}^1_y$ in place of $\widetilde{S}_y$ and $\check{G}_y$. On $\widetilde{S}_y$, $Z = 0$ implies $\Xi = 0$, so U and V have positive multiplicity along any prime divisor with center over C_y. This implies (A.239) for S_y.

Since $\beta' \geq 0$, by (A.239) any prime divisor over S_t with center a point of C_t has non-negative log discrepancy with respect to (S_t, B_{S_t}). If (S_t, B_{S_t}) were not lc near C_t, let E be a prime divisor over S_t with $a(E, S_t, B_{S_t}) < 0$ whose center contains a point o of C_t, and take a log resolution of (S_t, B_{S_t}) on which E is a prime divisor E' and the preimage of o is a divisor. If E' is contained in the preimage of o, its center is o. Otherwise E' intersects a component F_0 of the preimage of o, and blowing up a point of $E' \cap F_0$ and then repeatedly the intersection of E' with the newest exceptional divisor yields prime divisors with center o and log discrepancies $a(F_0, S_t, B_{S_t}) + k a(E, S_t, B_{S_t})$, $k = 1, 2, \ldots$, which tend to $-\infty$ [KM98, Lemma 2.29 and Corollary 2.31(1)]. Thus (S_t, B_{S_t}) is lc near C_t. Replacing S_t, B_{S_t}, and C_t with S_y, B_{S_y}, and C_y, we obtain that (S_y, B_{S_y}) is lc near C_y.

The lower bound. Assume that $\rho_F = \operatorname{mult}_F t/v$, and let $\mathfrak{a}_t := X^{\delta'/p} \prod_k (P^t_k)^{c'_k}$. Then $\mathfrak{a}_t|_{S_t}$ defines B_{S_t}, and

$$a(F_t, W_t, B_t) = a(F_t, W_t, S_t, \mathfrak{a}_t) + \frac{\delta'}{p}\Big(\operatorname{mult}_{F_t} X - \frac{p}{r} \operatorname{mult}_{F_t} u_t \Big).$$

Since $x = \sigma_t^{-p} X$ and $u_t = \sigma_t^r$, the bracket is a positive multiple of $\operatorname{mult}_F x > 0$, so

$$a(F_t, W_t, B_t) \geq a(F_t, W_t, S_t, \mathfrak{a}_t). \tag{A.240}$$

The hypotheses of Lemma 5.5 hold for $b = 1$, the complete intersection $W_t \subset \mathbb{A}^4$ with the diagonal group G_t, $u = u_t$, and $S_0 = S_t$, which is a normal integral klt complete intersection on which G_t acts faithfully. The center of F_t is 0, the generic point of C_t, or a closed point $o' \in C_t \backslash \{0\}$. If it is 0, we take $H = G_t$, and $\operatorname{mld}((S_t \ni 0, B_{S_t})/G_t) \geq \beta'$ by (A.239). If it is the generic point of C_t, we use the last sentence of Lemma 5.5 with $C = C_t$ and $H = G_t$. If it is $o' = (0, y_0, 0, 0)$ with $y_0 \neq 0$, we translate Y by y_0. The stabilizer $H := \operatorname{Stab}_{G_t}(o')$ acts trivially on Y, hence diagonally in the coordinates $(X, Y - y_0, Z, u_t)$, the equation of W_t is H-semi-invariant, W_t and S_t are smooth at o', and H acts faithfully on S_t. Since $I_{G_t}(F_t) \subset H$, $I_H(F_t) = I_{G_t}(F_t)$, and the

minimal log discrepancy at o' of the quotient of (S_t, B_{S_t}) by H is at least β' by (A.239). In the three cases Lemma 5.5 implies $a(F_t, W_t, S_t, \mathfrak{a}_t)/|I_{G_t}(F_t)| \geq \beta'$, and (A.238) follows from (A.240).

Assume that $\rho_F = \operatorname{mult}_F y/w_y$, and let $\mathfrak{a}_y := X^{\delta'/p}\prod_k (P_k^y)^{c'_k}$ on $\mathbb{A}^3_{(u_y,Z,T_0)}$, where $X = -Z^2 - u_y^d T_0^N$. Since $X|_{S_y} = -Z^2$, $\mathfrak{a}_y|_{S_y}$ defines B_{S_y}, and

$$a(F_y, \mathbb{A}^3, B_y) = a(F_y, \mathbb{A}^3, S_y, \mathfrak{a}_y) + \frac{\delta'}{p}\Big(\operatorname{mult}_{F_y} X - \frac{p}{r}\operatorname{mult}_{F_y} u_y\Big).$$

Since $x = \sigma_y^{-p} X$ and $u_y = \sigma_y^r$, the bracket is a positive multiple of $\operatorname{mult}_F x > 0$, so $a(F_y, \mathbb{A}^3, B_y) \geq a(F_y, \mathbb{A}^3, S_y, \mathfrak{a}_y)$. The hypotheses of Lemma 5.5 hold for $b = 1$, the affine space $\mathbb{A}^3_{(u_y,Z,T_0)}$ with the diagonal group G_y, $u = u_y$, and $S_0 = S_y = \mathbb{A}^2_{(Z,T_0)}$, on which G_y acts faithfully. The center of F_y is 0, the generic point of C_y, or a closed point $o' = (0, 0, t_0) \in C_y$ with $t_0 \neq 0$. If it is 0, we take $H = G_y$, and $\operatorname{mld}((S_y \ni 0, B_{S_y})/G_y) \geq \beta'$ by (A.239). If it is the generic point of C_y, we use the last sentence of Lemma 5.5 with $C = C_y$ and $H = G_y$. If it is o', we translate T_0 by t_0. The stabilizer $H := \operatorname{Stab}_{G_y}(o')$ acts trivially on T_0, hence diagonally in the coordinates $(u_y, Z, T_0 - t_0)$, and H acts faithfully on S_y. Since $I_{G_y}(F_y) \subset H$, $I_H(F_y) = I_{G_y}(F_y)$, and the minimal log discrepancy at o' of the quotient of (S_y, B_{S_y}) by H is at least β' by (A.239). In the three cases Lemma 5.5 implies $a(F_y, \mathbb{A}^3, S_y, \mathfrak{a}_y)/|I_{G_y}(F_y)| \geq \beta'$. Since the crepant quotient of (W_N, D') by G corresponds to that of $(\mathbb{A}^3, B_y)$ by G_y, (A.234) holds with $(\mathbb{A}^3, B_y, G_y)$ in place of $(\mathbb{A}^3, B_x, G_x)$, and (A.238) follows.

Since $\beta' \geq 0$, by (A.238) any prime divisor over W_N with center 0 has non-negative log discrepancy with respect to (W_N, D'). If (W_N, D') were not lc near 0, let E be a prime divisor over W_N with $a(E, W_N, D') < 0$ whose center contains 0, and take a log resolution of (W_N, D') on which E is a prime divisor E' and the preimage of 0 is a divisor. If E' is contained in the preimage of 0, its center is 0. Otherwise E' intersects a component F_0 of the preimage of 0, and blowing up a component of $E' \cap F_0$ and then repeatedly the intersection of E' with the newest exceptional divisor yields prime divisors with center 0 and log discrepancies $a(F_0, W_N, D') + ka(E, W_N, D')$, $k = 1, 2, \dots$, which tend to $-\infty$ [KM98, Lemma 2.29 and Corollary 2.31(1)]. Thus (W_N, D') is lc near 0. □

Proof of Theorem 7.26 continued.

Step 4. In this step we prove that if $(\mathbb{A}^2, \theta\Delta)$ is lc near 0 for a real number $\theta > 1$, then $\alpha \geq \beta$ for $r \gg 0$, where the bound for r also depends on θ.

We apply Claim A.59 to $h'_j = h_j^0$ and $c'_j = \theta c_j$, since $\theta\tau - (s+v) = \theta\delta + (\theta-1)(s+v) \geq 0$. Thus $(W_N, \theta D_N^0)$ is lc near 0, where $D_N^0 = \sum_j c_j \operatorname{div}(h_j^0|_{W_N})$. By [Kaw07, Theorem], Lemma 4.14 with $n = 4$ and $\Lambda = \{1\} \cup \theta(\Gamma\backslash\{0\})$, taking the largest of the integers for the numbers of functions at most $1 + 2/\gamma$ (**Step 1**), together with [dFEM11, Proposition 2.11], and [Kaw07, Theorem] again, $(W_N, \theta D)$ is lc near 0 for $r \gg 0$.

Let F be a prime divisor over W_N with center 0, $e_G := |I_G(F)|$, and $[f] := \operatorname{mult}_F f/e_G$. Since $\beta \leq 2$ (the blow-up of 0, [KM98, Lemma 2.29]), we may assume that $a(F, W_N, D) \leq 2e_G$. Since $a(F, W_N, \theta D) \geq 0$ and $a(F, W_N, \theta D_N^0) \geq 0$, we obtain $a(F, W_N, 0) \leq 2\theta e_G/(\theta - 1)$ and

$$[h_j] \leq M_3, \qquad [h_j^0] \leq M_3, \qquad M_3 := \frac{2\theta}{(\theta-1)\gamma}.$$

The monomials $x^s z^p$, $x^v t^p$, $x^{w_y} y^p$ have $\mathbf{T}$-weight 0, so their characters are trivial on R and have order at most e_0. A power of each of them of exponent at most e_0 is a G-invariant regular function vanishing at 0, hence has multiplicity at least e_G along F [Liu26, Lemma 2.6(1)]. Thus

$$s[x] + p[z] \geq \frac{1}{e_0}, \qquad v[x] + p[t] \geq \frac{1}{e_0}, \qquad w_y[x] + p[y] \geq \frac{1}{e_0}. \tag{A.241}$$

If $\operatorname{mult}_F h_j = \operatorname{mult}_F h_j^0$ for each j, then $a(F, W_N, D) = a(F, W_N, D_N^0) \geq \beta e_G$ by Claim A.59 with $h'_j = h_j^0$ and $c'_j = c_j$, for which $\delta' = \delta$ and $\beta' = \beta$. Otherwise, for some j some monomial

$x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ of $h_j - h_j^0$ satisfies $[x^{i_1}y^{i_2}z^{i_3}t^{i_4}] \le M_3$. If the monomial has weight $\kappa_j + mr$ with $m < 0$, then $pi_1 \ge r - M_0$, and if $m > 0$, then $w_y i_2 + si_3 + vi_4 \ge r - M_0$. Thus

$$\min\big([x], [y]/w_y, [t]/v, [z]/s\big) \le \frac{pM_3}{r - M_0}, \tag{A.242}$$

where the term $[z]/s$ is omitted if $s = 0$.

Assume first that $[x] \le pM_3/(r - M_0)$. By (A.241), for $r \gg 0$, $[y]/w_y$, $[t]/v$ and, if $s > 0$, $[z]/s$ are at least $\mu_0 := 1/(2p\bar{w}e_0)$. Thus any monomial of weight $\kappa_j + mr$ with $m > 0$ has $[\cdot] \ge \mu_0(r - M_0) > M_3$, and $\operatorname{mult}_F h_j = \operatorname{mult}_F h_j^{\le 0}$, where $h_j^{\le 0} := \sum_{m \le 0} h_{j,m}$. By (A.233),

$$\xi^{\kappa_j} h_j^{\le 0} = \sum_{m \le 0} u_x^{-m} Q_{j,m}(y_\xi, U, V).$$

Let P_j^x be the polynomial in u_x, U, V obtained by replacing y_ξ with $-U^2$. Then P_j^x is G_x-semi-invariant, since the characters of y_ξ and U^2 on G_x are equal, and $P_j^x|_{u_x=0} = P_j$. By (A.232), $\xi^{\kappa_j} h_j^{\le 0} - P_j^x$ is divisible by $V^N u_x^{-d} = \xi^{2s} t^N$, with a quotient that is a polynomial in u_x, y_ξ, U, V, which have non-negative multiplicity along F. Since $[\xi^{-\kappa_j}] \ge -M_0[x]/p$ and $[t^N] \ge Nv\mu_0 \ge r\mu_0$, we obtain $\operatorname{mult}_F(h_j^{\le 0} - \xi^{-\kappa_j} P_j^x) > M_3 e_G$ for $r \gg 0$. Thus $\operatorname{mult}_{F_x} \Phi_j = \operatorname{mult}_{F_x} P_j^x$ for a prime divisor F_x over $\mathbb{A}^3_{(u_x,U,V)}$ corresponding to F, whose center is 0. By (A.234) and Lemma 5.5 with $X = \mathbb{A}^3$, $u = u_x$, $S_0 = \mathbb{A}^2_{(U,V)}$, $H = G_x$, $\mathfrak{a} = \prod_j (P_j^x)^{c_j}$, and $b = 1 - \delta/r \le 1$,

$$\frac{a(F, W_N, D)}{e_G} = \frac{a\big(F_x, \mathbb{A}^3, (1 - \frac{\delta}{r}) \operatorname{div}(u_x) + \sum_j c_j \operatorname{div}(P_j^x)\big)}{|I_{G_x}(F_x)|} \ge \beta + \frac{\delta \operatorname{mult}_{F_x} u_x}{r|I_{G_x}(F_x)|} \ge \beta.$$

Assume next that $[x] > pM_3/(r - M_0)$. Then one of $[y]/w_y$, $[t]/v$, $[z]/s$ is at most $pM_3/(r - M_0)$, and by (A.241), $[x] \ge 1/(2\bar{w}e_0)$ for $r \gg 0$. Thus any monomial of weight $\kappa_j + mr$ with $m < 0$ has $[\cdot] \ge (r - M_0)/(2p\bar{w}e_0) > M_3$, and $\operatorname{mult}_F h_j = \operatorname{mult}_F h_j^{\ge 0}$, where $h_j^{\ge 0} := \sum_{m \ge 0} h_{j,m}$. The least weight of a monomial of $h_j^{\ge 0}$ is κ_j, with block h_j^0, so we may apply Claim A.59 to $h'_j = h_j^{\ge 0}$ and $c'_j = c_j$, with $P'_j = P_j$, $\delta' = \delta$ and $\beta' = \beta$. Thus

$$a(F, W_N, D) = a\Big(F, W_N, \sum_j c_j \operatorname{div}(h_j^{\ge 0})\Big) \ge \beta e_G.$$

Since F is arbitrary, $\alpha \ge \beta$.

Step 5. In this step we pass to a compatible generic limit of the plane pairs and reach a contradiction.

We restore the index i. Since the number of indices j and the $|\kappa_{ij}|$ are bounded, Γ is a DCC set, the β_i belong to an ACC set, and a finite diagonal subgroup of $(\mathbb{C}^*)^2$ of order at most pe_0 is contained in $\mu^2_{(pe_0)!}$, possibly passing to a subsequence we may assume the following: the number l of indices j and the weights $\kappa_j = \kappa_{ij}$ do not depend on i; c_{ij} is non-decreasing in i with limit $c_j^* > 0$; $G_{x,i} = H$ is a fixed subgroup of $(\mathbb{C}^*)^2$ and the character of P_{ij} on H depends only on j; $r_i \gg 0$ in the sense of **Steps 1**–**3**; and β_i is non-increasing with limit $\bar{\beta}$. By (A.237), $\bar{\beta} \ge \alpha_\infty > 0$. Since $\delta_i = \sum_j c_{ij}\kappa_j - (s + v) \ge 0$, also $\delta^* := \sum_j c_j^* \kappa_j - (s + v) \ge 0$.

Let $\Delta_i^* := \sum_j c_j^* \operatorname{div}(P_{ij})$. Since $(\mathbb{A}^2, \Delta_i)$ is lc near 0 and $c_{ij} \ge t_i c_j^*$ for $t_i := \min_j c_{ij}/c_j^* \to 1$, the lc threshold of Δ_i^* at 0 is at least t_i. The lc threshold at 0 of an $\mathbb{R}$-divisor on $\mathbb{A}^2$ is equal to its lc threshold on a small enough Zariski neighborhood of 0, since on a fixed log resolution only finitely many prime divisors occur and we may take the neighborhood disjoint from the centers of those whose center does not contain 0. The coefficients of Δ_i^* belong to the DCC set J^* of the non-zero sums $\sum_j m_j c_j^*$ with non-negative integers m_j. By Theorem 2.12(1) with $n = 2$, $I = \{0\}$, $J = J^*$, X such a neighborhood, $\Delta = 0$, and $M = \Delta_i^*$, the lc thresholds of the Δ_i^* at 0 belong to an ACC set. Since they are at least $t_i \to 1$, they are at least 1 for $i \gg 0$, that is, $(\mathbb{A}^2, \Delta_i^*)$ is lc near 0 for $i \gg 0$. Let $\beta_i^* := \operatorname{mld}((\mathbb{A}^2 \ni 0, \Delta_i^*)/H) \le \beta_i$, and let P_i^* be a prime divisor computing

β_i^* as in **Step 1**, with $\mathrm{mult}_{P_i^*} P_{ij} \le M_1^*$, where M_1^* is the constant M_1 of **Step 1** for the DCC set $\Gamma \cup \{c_j^*\}_j$ in place of Γ. Then

$$\beta_i \le \frac{a(P_i^*, \mathbb{A}^2, \Delta_i)}{|I_H(P_i^*)|} \le \beta_i^* + M_1^* \sum_j (c_j^* - c_{ij}),$$

so $\beta_i^* \to \bar\beta$.

We apply Construction-Proposition 4.1 in case (a), with $\mathbb{A}^2$, the group H, and the tuples $(P_{i1}, \dots, P_{il}, U, V)$, which consist of non-zero H-semi-invariant regular functions whose characters depend only on the position in the tuple and whose orders at 0 are at most $4/\gamma$. Possibly passing to a subsequence, we obtain a compatible generic limit $(P_{\infty,1}, \dots, P_{\infty,l}, U, V)$ over an algebraically closed field $k \supset \mathbb{C}$. The last two entries are U and V, since the jets of a constant entry are constant and the functions may be multiplied by non-zero constants. By Construction-Proposition 4.1, any sub-tuple of the limit is a compatible generic limit of the corresponding sub-tuples. Let $\Delta_\infty := \sum_j c_j^* \operatorname{div}(P_{\infty,j})$ on $\mathcal{A}_k = \operatorname{Spec} k[[U, V]]$. By Theorem 4.2(1) with $\mathfrak{c} = \mathcal{O}$, $(\mathcal{A}_k, \Delta_\infty)$ is lc near 0.

Assume that $(\mathcal{A}_k, \Delta_\infty)$ is klt near 0. By Theorem 4.2(1), there exists $\theta > 1$ such that $(\mathbb{A}^2, \theta\Delta_i^*)$ is lc near 0 for any i, hence so is $(\mathbb{A}^2, \theta\Delta_i)$, since $\Delta_i \le \Delta_i^*$. By **Step 4**, $\alpha_i \ge \beta_i \ge \bar\beta \ge \alpha_\infty$ for $i \gg 0$, which contradicts the strict increase of the α_i.

Assume that $(\mathcal{A}_k, \Delta_\infty)$ is not klt near 0. By Definition-Proposition 5.7 in case (a) with $n = 2$, applied to Δ_i^* and $\beta_i^* \to \bar\beta > 0$, there exists exactly one floor E, which is smooth at 0, $(\mathcal{A}_k, \Delta_\infty - E)$ is klt near 0, and

$$\bar\beta = \frac{1 - b_E}{e_E}, \qquad b_E := \mathrm{mult}_0\left((\Delta_\infty - E)|_E\right),$$

where e_E is the order of the image of H in the automorphism group of the tangent line T_0E. The floor E is H-invariant, since it is unique. Let $\varpi \in \{U, V\}$ be a coordinate whose restriction to E is a parameter at 0. Since $\varpi|_E$ is H-semi-invariant, e_E is the order e_ϖ of the character χ_ϖ of ϖ on H. Let $c_E := 1 - b_E$, so that $c_E = e_\varpi \bar\beta > 0$. Then

$$\mathrm{mult}_0\left((\Delta_\infty - E + c_E \operatorname{div}(\varpi))|_E\right) = b_E + c_E = 1.$$

Since the floor E is a regular curve through 0 with coefficient one, not contained in $\operatorname{div}(\varpi)$, and the multiplicity at 0 of the restriction to E of the rest of the boundary is $b_E + c_E = 1$, the pair $(\mathcal{A}_k, \Delta_\infty + c_E \operatorname{div}(\varpi))$ is lc near 0 by [Kaw07, Theorem] on the formal germ $\mathcal{A}_k$ (Lemma 2.14). By Theorem 4.2(1) for the sub-tuple $(P_{i1}, \dots, P_{il}, \varpi)$ with the coefficients $(c_1^*, \dots, c_l^*, c_E)$, possibly passing to a subsequence, $(\mathbb{A}^2, \Delta_i^* + c_E \operatorname{div}(\varpi))$ is lc near 0 for any i.

If $\varpi = U$, let $n_\varpi := p/\gcd(p, s)$ and $h_\varpi := x^{sn_\varpi/p} z^{n_\varpi}$. If $\varpi = V$, let $n_\varpi := p/\gcd(p, v)$ and $h_\varpi := x^{vn_\varpi/p} t^{n_\varpi}$. Then $h_\varpi = \varpi^{n_\varpi}$ on $\widetilde{W}$, h_ϖ has $\mathbf{T}$-weight 0, and $h_\varpi(1, -U^2, U, V) = \varpi^{n_\varpi}$. If $s = 0$ and $\varpi = U$, then $n_\varpi = 1$ and $h_\varpi = z$. The subgroup of $H = G_x$ of order p constructed in **Step 1** acts on U by ζ^s and on V by ζ^v, $\zeta \in \mu_p$, so the restriction of χ_ϖ to that subgroup has order n_ϖ. Thus n_ϖ divides e_ϖ. The function ϖ^{e_ϖ} is $\hat G_{x,i}$-invariant on $\widetilde{W}_i$ and equals $h_\varpi^{e_\varpi/n_\varpi}$, so $h_\varpi^{e_\varpi/n_\varpi}$ is a G_i-invariant regular function on W_{N_i} vanishing at 0.

We apply Claim A.59 to the polynomials h_{ij}^0 with the coefficients c_j^* and to h_ϖ with the coefficient c_E/n_ϖ. The adjusted mass is $\delta^* \ge 0$, since h_ϖ has weight 0, and the plane pair is $(\mathbb{A}^2, \Delta_i^* + c_E \operatorname{div}(\varpi))$, which is lc near 0. Thus the pair

$$\Big(W_{N_i}, \sum_j c_j^* \operatorname{div}(h_{ij}^0) + \frac{c_E}{n_\varpi} \operatorname{div}(h_\varpi)\Big)$$

is lc near 0. By [Kaw07, Theorem], Lemma 4.14 with $n = 4$ and $\Lambda = \{1\} \cup \{c_j^*\}_j \cup \{c_E/n_\varpi\}$, together with [dFEM11, Proposition 2.11], and [Kaw07, Theorem] again, the pair

$$\Big(W_{N_i}, \sum_j c_j^* \operatorname{div}(h_{ij}) + \frac{c_E}{n_\varpi} \operatorname{div}(h_\varpi)\Big)$$

is lc near 0 for $i \gg 0$. Since $c_{ij} \le c_j^*$ and

$$\frac{c_E}{n_\varpi}\operatorname{div}(h_\varpi) = \frac{c_E}{e_\varpi}\operatorname{div}\left(h_\varpi^{e_\varpi/n_\varpi}\right),$$

the pair

$$\left(W_{N_i}, D_i + \frac{c_E}{e_\varpi}\operatorname{div}\left(h_\varpi^{e_\varpi/n_\varpi}\right)\right)$$

is lc near 0. By Lemma 5.8 with $Z = \{0\}$ and the G_i-invariant function $h_\varpi^{e_\varpi/n_\varpi}$, whose character has order 1,

$$\alpha_i \ge \frac{c_E}{e_\varpi} = \bar\beta \ge \alpha_\infty$$

for $i \gg 0$, which contradicts the strict increase of the α_i. This proves the theorem. □

A.18. **Mixed circles: walls.** In this subsection, we prove Theorem 7.27. The crepant model of W_N/G in the proof is the strict transform in one weighted blow-up of $\mathbb{A}^4/G$, for three families of weight vectors (Table 2), and the proof studies the vertex of the t-chart of the weighted blow-up.

Set-up A.60. We use the notation of Set-up 7.3 with $w = (-p, q, s, v)$, where p, s, v are positive integers with $\gcd(p, s, v) = 1$ and $q := p + 2s$. Thus $d = (Nv - 2s)/r$, $R = \{w(\zeta) \mid \zeta^r = 1\}$ with $w(\zeta) := (\zeta^{-p}, \zeta^q, \zeta^s, \zeta^v)$, and the $\mathbf{T}$-weight of a monomial $\mu = x^{i_x}y^{i_y}z^{i_z}t^{i_t}$ is

$$\kappa(\mu) = -pi_x + qi_y + si_z + vi_t.$$

The index j runs over the indices with $c_j > 0$ for which h_j is not a unit at 0. Let $(Q, \Delta) := (W_N, D)/G$ be the crepant quotient [Liu26, Definition 2.5] and $o \in Q$ the image of 0, so that $\alpha = \operatorname{mld}(Q \ni o, \Delta)$. Possibly shrinking Q to a neighborhood of o, we may assume that the marks which are units at 0 do not vanish on the preimage of Q and that (Q, Δ) is klt, since (W_N, D) is klt near 0 [Liu26, Lemma 2.6(2)(3)]. If $\Gamma \not\subset \{0\}$, let $\gamma := \min(\Gamma\backslash\{0\})$ and $M := \lceil 2\max(q, v)/\gamma \rceil$. If $\Gamma \subset \{0\}$, let $M := 1$.

Lemma A.61. *Notation as in Set-up A.60.*

(1) A non-zero polynomial in x, y, z, t without monomials divisible by xy is not divisible by $xy + z^2$.

(2) $\sum_j c_j \operatorname{mult}_0 h_j < 2$, where $\operatorname{mult}_0 h_j$ is the least degree of a monomial of h_j. Moreover, the number of indices j is at most M, $|\kappa_j| \le M$, $I_j \le M$, and $\delta > -(2p + s + v)$.

(3) The $\mathbf{T}$-weights of the monomials of h_j are congruent modulo r. If $r > 2M$, then all monomials of least degree of h_j have $\mathbf{T}$-weight κ_j.

(4) Assume that $A = 0$. Let ν_j be the largest y-exponent of a monomial of h_j^0, and $\nu_0 := \sum_j c_j\nu_j$. Then h_j^0 contains a non-zero multiple of $t^{\kappa_j/v}$, κ_j is a positive multiple of v, $\delta \ge -(s + v)$, $q\nu_j \le \kappa_j \le M$, and $-\nu_j$ is the least value of $i_x - i_y$ over the monomials $x^{i_x}y^{i_y}z^{i_z}t^{i_t}$ of h_j^0.

Proof. (1) This is Lemma A.1(1).

(2) By Lemma A.1(3), $\sum_j c_j \operatorname{mult}_0 h_j < 2$, and the number of indices j and each $\operatorname{mult}_0 h_j$ are smaller than $2/\gamma$, which is at most M. Let μ_j be a monomial of h_j^0 of degree $\operatorname{mult}_0 h_j$. Then $|\kappa_j| = |\kappa(\mu_j)| \le \max(q, v)\operatorname{mult}_0 h_j < M$, $I_j \le \operatorname{mult}_0 h_j < M$, and $\kappa_j \ge -pi_x(\mu_j) \ge -p\operatorname{mult}_0 h_j$, where $i_x(\mu_j)$ is the x-exponent of μ_j. Thus $\tau = \sum_j c_j\kappa_j > -2p$ and $\delta > -(2p + s + v)$.

(3) Since $\gcd(p, s, v) = 1$, $\zeta \to w(\zeta)$ is injective, so $R = w(\mu_r)$, and $w(\zeta)$ multiplies a monomial μ by $\zeta^{\pm\kappa(\mu)}$, with a sign independent of μ. Since h_j is R-semi-invariant, $\zeta^{\kappa(\mu)} = \zeta^{\kappa(\mu')}$ for any monomials μ, μ' of h_j and any $\zeta \in \mu_r$, that is, r divides $\kappa(\mu) - \kappa(\mu')$. If μ, μ' have least degree, then $|\kappa(\mu) - \kappa(\mu')| < 2M < r$ by (2), so $\kappa(\mu) = \kappa(\mu')$. Since h_j^0 contains the monomials of least degree of h_j, their common weight is κ_j.

(4) Since $A = 0$, $I_j = 0$ for any j, so h_j^0 contains a non-zero multiple of a monomial t^{i_t} with $vi_t = \kappa_j$, and $i_t \ge 1$ because h_j is not a unit. Thus $\kappa_j \in v\mathbb{Z}_{>0}$, $\tau \ge 0$, and $\delta \ge -(s + v)$. Let $\mu = x^{i_x}y^{i_y}z^{i_z}t^{i_t}$ be a monomial of h_j^0. If $i_y > 0$, then $i_x = 0$ because μ is not divisible by xy, so

$i_x - i_y = -i_y \geq -\nu_j$, and $\kappa_j = qi_y + si_z + vi_t \geq qi_y$. If $i_y = 0$, then $i_x - i_y \geq 0 \geq -\nu_j$. Thus $q\nu_j \leq \kappa_j$, and $-\nu_j$ is attained by a monomial with y-exponent ν_j if $\nu_j > 0$, and by $t^{\kappa_j/v}$ if $\nu_j = 0$. Finally $\kappa_j \leq M$ by (2). □

The following construction is the weighted blow-up used in the three cases of the proof. The weight vector need not be primitive.

Construction-Proposition A.62. Notation as in Set-up A.60. Let $\mathbf{a} = (a_x, a_y, a_z, a_t) \in \mathbb{Z}^4_{>0}$, $u \in \mathbb{Z}^4$, and let k and $N' \geq 3$ be integers with

$$\mathbf{a} = ru + kw, \qquad a_x + a_y = 2a_z, \qquad Na_t - 2a_z = rN'. \tag{A.243}$$

For $g \in G$ let g_x, g_y, g_z, g_t be the diagonal entries of g. For $i \in \{x, y, z, t\}$, let

$$\widehat{G}_i := \{(g, \zeta) \in G \times \mathbb{C}^* \mid \zeta^{a_i} = g_i\}$$

act on $\mathbb{A}^4$ with coordinates λ and X_l, $l \in \{x, y, z, t\}\backslash\{i\}$, by $\lambda \to \zeta\lambda$ and $X_l \to g_l\zeta^{-a_l}X_l$. Write x_x, x_y, x_z, x_t for x, y, z, t. Let ψ_i be the $\widehat{G}_i$-equivariant morphism to $\mathbb{A}^4$ given by $x_i = \lambda^{a_i}$ and $x_l = \lambda^{a_l}X_l$ for $l \neq i$, and let $b := \lambda^r$. Let $(x', y', z', t') := (X_x, X_y, X_z, X_t)$, and let V_i be the hypersurface in $\mathbb{A}^4$ with coordinates b and X_l, $l \neq i$, defined by

$$y' + z'^2 + b^{N'}t'^N, \qquad x' + z'^2 + b^{N'}t'^N, \qquad x'y' + 1 + b^{N'}t'^N, \qquad x'y' + z'^2 + b^{N'} \tag{A.244}$$

for $i = x, y, z, t$ respectively, and let $\widetilde{V}_i$ be the hypersurface defined by the polynomial of (A.244) for i with b replaced by λ^r. Let $\widehat{R} := \{(w(\zeta^k), \zeta) \mid \zeta^r = 1\}$, $H_i := \widehat{G}_i/\widehat{R}$, $C_* := a_z + a_t$, $\omega_j := \mathrm{wt}_{\mathbf{a}}(h_j)$, and

$$\beta := 1 - \frac{C_*}{r} + \sum_j \frac{c_j\omega_j}{r}. \tag{A.245}$$

Then there exists a projective birational morphism $\varphi : Y \to Q$ from a normal variety Y, which is an isomorphism over $Q\backslash\{o\}$, with the following properties.

(1) $\widehat{R} \subset \widehat{G}_i$, and the group H_i has order $a_i[G : R]$, acts diagonally and faithfully on V_i, and preserves the equation of V_i up to a character.
(2) Y is covered by open subsets $U_i \cong V_i/H_i$, $i \in \{x, y, z, t\}$, and $\varphi^{-1}(o) \cap U_i$ is the image of $(b = 0) \cap V_i$.
(3) V_x, V_y, and V_z are smooth, $V_t = W_{N'}$ is smooth outside 0, and $\varphi^{-1}(o)$ is a prime divisor E.
(4) For any i and j, $h_j \circ \psi_i = \lambda^{\omega_j}P_j(\lambda^r, X_l)$ for an H_i-semi-invariant polynomial P_j in b and the X_l (depending on i), which does not vanish identically on $(b = 0) \cap V_i$.
(5) For any i and j, the monomials of P_j are the images of the monomials μ of h_j under the injective map
$$\mu \to b^{(\mathrm{wt}_{\mathbf{a}}(\mu) - \omega_j)/r} \prod_{l \neq i} X_l^{i_l}.$$
(6) For any i, the crepant pullback of $K_Y + \Delta_Y := \varphi^*(K_Q + \Delta)$ to V_i is $K_{V_i} + B_{V_i}$, where
$$B_{V_i} := \beta\,\mathrm{div}(b) + \sum_j c_j\,\mathrm{div}(P_j).$$
(7) The coefficient of E in Δ_Y is $1 - (1 - \beta)/e$, where $e := |I_{H_t}((b = 0))|$ is the order of the inertia group of the prime divisor $(b = 0)$ of V_t.
(8) (Y, Δ_Y) is klt over a neighborhood of o, and $\beta < 1$.
(9) If $\beta \geq 0$, then $\Delta_Y \geq 0$.

Proof. Since $\mathbf{a} \in \mathbb{Z}^4_{>0}$, the star subdivision of $\mathbb{R}^4_{\geq 0}$ at the ray $\mathbb{R}_{\geq 0}\mathbf{a}$, with respect to the lattice N_G, defines a projective birational toric morphism $\mathcal{Y} \to \mathbb{A}^4/G$ [CLS11, Proposition 11.1.6(c)], which is an isomorphism over the complement of the image of 0. A direct calculation shows that, for $i \in \{x, y, z, t\}$, the chart of $\mathcal{Y}$ of the cone spanned by $\mathbf{a}$ and the e_l with $l \neq i$ is

$\mathbb{A}^4/\widehat{G}_i$, with the morphism to $\mathbb{A}^4/G$ induced by ψ_i [CLS11, Proposition 1.3.18]. By (A.243), $\psi_i^*(xy+z^2+t^N) = \lambda^{2a_z}\widetilde{F}_i$, where $\widetilde{F}_i$ is the equation of $\widetilde{V}_i$. For instance,

$$\psi_t^*(xy+z^2+t^N) = \lambda^{a_x+a_y}x'y' + \lambda^{2a_z}z'^2 + \lambda^{Na_t} = \lambda^{2a_z}(x'y' + z'^2 + \lambda^{rN'}).$$

Since $\widetilde{F}_i$ does not vanish identically on $(\lambda=0)$, the strict transform of W_N in the chart is $\widetilde{V}_i$. Let Y be the strict transform of Q in $\mathcal{Y}$ and $\varphi : Y \to Q$ the induced morphism. Then Y is covered by the open subsets $U_i := \widetilde{V}_i/\widehat{G}_i$.

(1) and (2) For $\zeta \in \mu_r$, $\zeta^{a_i} = \zeta^{ru_i+kw_i} = (\zeta^k)^{w_i}$, so $(w(\zeta^k), \zeta) \in \widehat{G}_i$, and it multiplies X_l by $(\zeta^k)^{w_l}\zeta^{-a_l} = \zeta^{-ru_l} = 1$ and λ by ζ. Thus $\widehat{R}$ acts trivially on b and the X_l, and $\widetilde{V}_i \to V_i$, $\lambda \to \lambda^r$, is the quotient by $\widehat{R} \cong \mu_r$. Hence $U_i \cong V_i/H_i$. Since $|\widehat{G}_i| = a_i|G|$ and $|\widehat{R}| = r$, $|H_i| = a_i[G:R]$. If (g,ζ) acts trivially on b and the X_l, then $\zeta^r = 1$ and $g_l = \zeta^{a_l} = \zeta^{kw_l}$ for any l, so $(g,\zeta) \in \widehat{R}$. Thus H_i acts faithfully on $\mathbb{A}^4$, hence on V_i, which is contained in no coordinate hyperplane. Since $xy+z^2+t^N$ and λ are $\widehat{G}_i$-semi-invariant, so is $\widetilde{F}_i$, and the equation of V_i is H_i-semi-invariant. Since $a_l > 0$ for any l, $\psi_i^{-1}(0) = (\lambda = 0)$, so $\varphi^{-1}(o) \cap U_i$ is the image of $(\lambda=0) \cap \widetilde{V}_i$, that is, of $(b=0) \cap V_i$.

(3) The derivative of the equation of V_x (resp. V_y) with respect to y' (resp. x') is 1. At a singular point of V_z, $x' = y' = 0$, so $b^{N'}t'^N = -1$, and the derivative $N'b^{N'-1}t'^N$ with respect to b is non-zero, a contradiction. The equation $x'y' + z'^2 + b^{N'}$ of V_t has non-zero gradient outside 0. Let $C := (xy+z^2=0) \subset \mathbb{A}^4$, which is irreducible. Then $(\lambda=0) \cap \widetilde{V}_i$ is the slice $C \cap (x_i = 1)$. Hence $\varphi^{-1}(o)$ is the image of $C\backslash\{0\}$ under the quotient by G and by the $\mathbb{C}^*$-action of weight $\mathbf{a}$, and it is irreducible of dimension two. Each V_i is a hypersurface with at most one singular point, hence normal [Har77, Proposition II.8.23(b)], and Y is normal since $U_i \cong V_i/H_i$ with V_i normal.

(4) and (5) For a monomial μ with exponents i_x, i_y, i_z, i_t,

$$\psi_i^*\mu = \lambda^{\mathrm{wt}_{\mathbf{a}}(\mu)} \prod_{l\neq i} X_l^{i_l}, \qquad \mathrm{wt}_{\mathbf{a}}(\mu) = r\sum_l u_l i_l + k\kappa(\mu). \tag{A.246}$$

By Lemma A.61(3), $\mathrm{wt}_{\mathbf{a}}(\mu) - \omega_j \in r\mathbb{Z}_{\geq 0}$ for any monomial μ of h_j, which defines P_j. The exponents i_l, $l \neq i$, together with $\mathrm{wt}_{\mathbf{a}}(\mu)$ determine i_i, so the map of monomials is injective. Since $h_j \circ \psi_i$ and λ are $\widehat{G}_i$-semi-invariant, P_j is H_i-semi-invariant. The restriction of P_j to $(b=0) \cap V_i = C \cap (x_i = 1)$ is the restriction of $\mathrm{in}_{\mathbf{a}}(h_j)$. If it vanished, then, since $\mathrm{in}_{\mathbf{a}}(h_j)$ is $\mathbf{a}$-homogeneous and any $\mathbb{C}^*$-orbit of weight $\mathbf{a}$ in $C \cap (x_i \neq 0)$ intersects $(x_i = 1)$, $\mathrm{in}_{\mathbf{a}}(h_j)$ would vanish on the dense open subset $C \cap (x_i \neq 0)$ of C. Thus $xy + z^2$ would divide $\mathrm{in}_{\mathbf{a}}(h_j)$, which is a non-zero polynomial without monomials divisible by xy, contradicting Lemma A.61(1).

(6) and (7) The form $\omega_W := dx \wedge dz \wedge dt/x$ generates ω_{W_N} near 0. A direct calculation shows that

$$\psi_i^*\omega_W = \pm a_i \lambda^{C_*-1}\omega_{\widetilde{V}_i},$$

where $\omega_{\widetilde{V}_i}$ is the Poincaré residue of the standard 4-form along $\widetilde{F}_i$. For instance,

$$\psi_t^*\omega_W = a_t\lambda^{a_z+a_t-1}dx' \wedge dz' \wedge d\lambda/x'.$$

By (4), the crepant pullback of $K_{W_N} + D$, hence of $K_Q + \Delta$, to $\widetilde{V}_i$ is

$$K_{\widetilde{V}_i} + \Big(1 - C_* + \sum_j c_j\omega_j\Big)\,\mathrm{div}(\lambda) + \sum_j c_j\,\mathrm{div}(P_j).$$

The quotient $\widetilde{V}_i \to V_i$ is ramified only along $(\lambda=0)$, with index r. By [Liu26, Lemma 2.6(4)], the coefficient of $\mathrm{div}(b)$ in the crepant pullback to V_i is $1 - (C_* - \sum_j c_j\omega_j)/r = \beta$, and the coefficient of E in Δ_Y is $1 - (1-\beta)/e$.

(8) and (9) Since (W_N, D) is klt near 0, (Q, Δ) is klt near o [Liu26, Lemma 2.6(2)(3)]. The log discrepancies of (Y, Δ_Y) over a neighborhood of o are log discrepancies of (Q, Δ) [KM98, Lemma 2.30], hence positive. In particular $0 < a(E, Q, \Delta) = (1-\beta)/e$, so $\beta < 1$. If $\beta \geq 0$, then $B_{V_i} \geq 0$, so $\Delta_Y \geq 0$ by [Liu26, Lemma 2.6(4)]. □

Construction A.63. Notation as in Set-up A.60, and assume that $Nv > 2s$, so that $d \geq 1$. Fix the following integers.

(R1) $L := M + 5$.

(R2) $L_0 := q(M+3)+1$, L the integer in $\{L_0, \dots, L_0 + q - 1\}$ with $L \equiv -r \pmod q$, and $k := (r+L)/q$.

(R3) $n := \lceil v/(2s) \rceil$, $m := \lfloor pn/v \rfloor + 1$, $L_0 := 2sn + 3v + vM + 1$, L the integer in $\{L_0, \dots, L_0 + v - 1\}$ with $L \equiv -nr \pmod v$, and $k := (nr+L)/v$.

The weight vector $\mathbf{a}$, the vector u, and the integers k, N', C_* of Construction-Proposition A.62 for the three rows are listed in Table 2, with the number β of (A.245) computed in Lemma A.64.

row	$\mathbf{a}$	u	k
(R1)	$(r - pL, r + qL, r + sL, vL)$	$(1,1,1,0)$	L
(R2)	$(r - pk, L, sk, vk)$	$(1,-1,0,0)$	$(r+L)/q$
(R3)	$(mr - pk, qk - mr, sk, L)$	$(m,-m,0,-n)$	$(nr+L)/v$

row	N'	C_*	β
(R1)	$dL - 2$	$r + (s+v)L$	$A + \delta L/r$
(R2)	kd	$(s+v)k$	$1 - \nu_0 + \delta k/r$
(R3)	$(dL - 2sn)/v$	$sk + L$	$1 - sn/v + \delta L/(vr)$

TABLE 2. The three weight vectors of Construction A.63. Row (R1) is used when $A > 0$, row (R2) when $A = 0$ and $v \leq s$, and row (R3) when $A = 0$ and $v > s$. The group H_t has order $a_t[G : R]$, that is, $vL[G : R]$, $vk[G : R]$, and $L[G : R]$, respectively.

Lemma A.64. *Notation as in Construction A.63. There exists a positive integer $r_1 > 2M$, depending only on Γ, p, s, v, such that if $r \geq r_1$ and $\delta \leq 0$, then the following hold for row (R1) if $A > 0$, for row (R2) if $A = 0$ and $v \leq s$, and for row (R3) if $A = 0$ and $v > s$.*

(1) $\mathbf{a} \in \mathbb{Z}^4_{>0}$, and $\mathbf{a}, u, k, N'$ satisfy (A.243) with $N' \geq 3$.

(2) $\omega_j = rI_j + L\kappa_j$, $\omega_j = k\kappa_j - r\nu_j$, and $\omega_j = L\kappa_j/v$ in rows (R1), (R2), and (R3) respectively.

(3) β is as in Table 2, and $\beta \geq \gamma/2$, $\beta \geq (p+s-v)/(2q)$, and $\beta \geq (1 - sn/v)/2$ in rows (R1), (R2), and (R3) respectively. In particular $0 < \beta < 1$.

Proof. In row (R3), $1 \leq n < v/s$: if $v \leq 2s$ then $n = 1 < v/s$, and if $v > 2s$ then $\lceil v/(2s) \rceil < v/(2s) + 1 < v/s$. Moreover,

$$pn < vm \leq qn,$$

since $\lfloor pn/v \rfloor > pn/v - 1$, $vm \leq pn + v$, and $v \leq 2sn$. In each row L is bounded in terms of p, s, v, M, and k is an integer.

(1) Row (R1): $\mathbf{a} = r(1,1,1,0) + Lw$, $r - pL > 0$ for $r > pL$, $a_x + a_y = 2r + 2sL = 2a_z$, and $Na_t - 2a_z = L(Nv - 2s) - 2r = r(dL - 2)$ with $dL - 2 \geq L - 2 \geq 3$.

Row (R2):

$$r(1,-1,0,0) + kw = (r - pk, qk - r, sk, vk),$$

and $qk - r = L$, $r - pk = (2sr - pL)/q > 0$ for $r > pL$, $a_x + a_y = (q-p)k = 2a_z$, and $Na_t - 2a_z = k(Nv - 2s) = rkd$ with $kd \geq k \geq r/q \geq 3$ for $r \geq 3q$.

Row (R3):

$$r(m,-m,0,-n) + kw = (mr - pk, qk - mr, sk, vk - nr),$$

$$mr - pk = \frac{r(vm - pn) - pL}{v} > 0, \qquad qk - mr = \frac{r(qn - vm) + qL}{v} > 0,$$

where $vk - nr = L$, the first inequality holds for $r > pL$, $a_x + a_y = 2sk = 2a_z$, and $Na_t - 2a_z = NL - 2sk = r(dL - 2sn)/v$. Since $dL \equiv -dnr = -n(Nv - 2s) \equiv 2sn \pmod v$, $N' = (dL - 2sn)/v$ is an integer, and $N' \geq (L - 2sn)/v > 3$.

(2) Let μ be a monomial with exponents i_x, i_y, i_z, i_t. By (A.246),

$$\frac{\mathrm{wt}_{\mathbf{a}}(\mu)}{r} = \pi(\mu) + \frac{L\kappa(\mu)}{\rho_0 r}, \tag{A.247}$$

where $\rho_0 = 1, q, v$ in rows (R1), (R2), (R3), and

$$\pi(\mu) = i_x + i_y + i_z \qquad \text{in (R1)},$$

$$\pi(\mu) = i_x - i_y + \frac{\kappa(\mu)}{q} = \frac{2si_x + si_z + vi_t}{q} \qquad \text{in (R2)},$$

$$\pi(\mu) = m(i_x - i_y) - ni_t + \frac{n\kappa(\mu)}{v} = \frac{(vm - pn)i_x + (qn - vm)i_y + sni_z}{v} \qquad \text{in (R3)}.$$

Thus $\pi(\mu) \geq \rho_1 i_x \geq 0$, where $\rho_1 = 1$, $2s/q$, $(vm - pn)/v$ respectively, so $\rho_1 > 0$. Let π_j be the least value of π on the monomials of h_j^0. Then $\pi_j = I_j$ in (R1), $\pi_j = \kappa_j/q - \nu_j$ in (R2) by Lemma A.61(4), and $\pi_j = 0$ in (R3), attained by $t^{\kappa_j/v}$. In each case $0 \leq \pi_j \leq M$ by Lemma A.61(2)(4).

Let μ be a monomial of h_j which is not a monomial of h_j^0. By Lemma A.61(3), $\kappa(\mu) = \kappa_j + lr$ for a non-zero integer l. We show that, for $r \gg 0$,

$$\frac{\mathrm{wt}_{\mathbf{a}}(\mu)}{r} > \pi_j + \frac{L\kappa_j}{\rho_0 r}. \tag{A.248}$$

If $l \geq 1$, then by (A.247),

$$\frac{\mathrm{wt}_{\mathbf{a}}(\mu)}{r} \geq \frac{L\kappa_j}{\rho_0 r} + \frac{L}{\rho_0},$$

and $L/\rho_0 > M \geq \pi_j$ in each row. If $l \leq -1$, then $\kappa(\mu) \leq M - r < 0$, and $-pi_x \leq \kappa(\mu)$, so $i_x \geq -\kappa(\mu)/p$ and

$$\frac{\mathrm{wt}_{\mathbf{a}}(\mu)}{r} \geq \rho_1 i_x + \frac{L\kappa(\mu)}{\rho_0 r} \geq (-\kappa(\mu))\Big(\frac{\rho_1}{p} - \frac{L}{\rho_0 r}\Big) \geq \frac{(r - M)\rho_1}{2p}$$

for $r \geq 2pL/(\rho_0\rho_1)$, which is larger than $2M \geq \pi_j + L\kappa_j/(\rho_0 r)$ for $r \gg 0$. Thus (A.248) holds, and $\omega_j = r\pi_j + L\kappa_j/\rho_0$, which is $rI_j + L\kappa_j$, $r(\kappa_j/q - \nu_j) + L\kappa_j/q = k\kappa_j - r\nu_j$, and $L\kappa_j/v$ in the three rows.

(3) Substituting (2) and C_* into (A.245):

$$\beta = 1 - \frac{r + (s + v)L}{r} + \sum_j c_j\Big(I_j + \frac{L\kappa_j}{r}\Big) = A + \frac{\delta L}{r} \qquad \text{in (R1)},$$

$$\beta = 1 - \frac{(s + v)k}{r} + \sum_j c_j\Big(\frac{k\kappa_j}{r} - \nu_j\Big) = 1 - \nu_0 + \frac{\delta k}{r} \qquad \text{in (R2)}$$

$$= \frac{p + s - v}{q} + \sum_j c_j\Big(\frac{\kappa_j}{q} - \nu_j\Big) + \frac{\delta L}{qr},$$

$$\beta = 1 - \frac{sk + L}{r} + \sum_j c_j \frac{L\kappa_j}{vr} = 1 - \frac{sn}{v} + \frac{\delta L}{vr} \qquad \text{in (R3)},$$

where we used $k/r = 1/q + L/(qr)$ in (R2) and $k/r = n/v + L/(vr)$ in (R3). In (R1), $A \geq \gamma$ since $A > 0$, and $\delta > -(2p + s + v)$ by Lemma A.61(2), so $\beta \geq \gamma - (2p + s + v)L/r \geq \gamma/2$ for $r \gg 0$. In (R2), $\kappa_j/q - \nu_j \geq 0$, $p + s - v \geq p > 0$, and $\delta \geq -(s + v)$ by Lemma A.61(4), so $\beta \geq (p + s - v)/q - (s + v)L/(qr) \geq (p + s - v)/(2q)$ for $r \gg 0$. In (R3), $sn < v$ and $\delta \geq -(s + v)$, so $\beta \geq 1 - sn/v - (s + v)L/(vr) \geq (1 - sn/v)/2$ for $r \gg 0$. Finally $\beta < 1$ by Construction-Proposition A.62(8). All the bounds on r above depend only on Γ, p, s, v. □

Lemma A.65. *Let $\Gamma \subset [0,1]$ be a DCC set, M a positive integer, β_0, σ real numbers, and $\Lambda \subset (0,+\infty)$ a set satisfying the ACC. Then the set of real numbers*

$$\beta_0 + \sum_{j=1}^{l} c_j(\epsilon_j + \kappa_j\vartheta) - \sigma\vartheta,$$

where $l \le M$, $c_j \in \Gamma$, ϵ_j, κ_j are integers with $|\epsilon_j|, |\kappa_j| \le M$, and $\vartheta \in \Lambda$, with $\epsilon_j + \kappa_j\vartheta \ge 0$ for any j and $\sum_j c_j\kappa_j \le \sigma$, satisfies the DCC.

In particular, fix Γ, p, s, v, and let $\mathcal{B}_1$, $\mathcal{B}_2$, $\mathcal{B}_3$ be the sets of the numbers β of rows (R1), (R2), (R3), for all data of Set-up A.60 with the fixed Γ, p, s, v and with $Nv > 2s$, $r \ge r_1$, $\delta \le 0$, and the condition on A and v of the row in Lemma A.64. Then $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3 \subset (0,1)$ are DCC sets.

Proof. Assume that the set contains a strictly decreasing sequence. Possibly passing to a subsequence, we may assume that l and the integers ϵ_j, κ_j do not depend on the member, that each c_j is non-decreasing, and that ϑ is non-increasing. For two members with labels $c_j \le c'_j$ and parameters $\vartheta \ge \vartheta'$, the second one minus the first one is

$$\sum_j (c'_j - c_j)(\epsilon_j + \kappa_j\vartheta') + (\vartheta' - \vartheta)\Big(\sum_j c_j\kappa_j - \sigma\Big) \ge 0,$$

a contradiction.

For the second statement, by Lemma A.64(2)(3),

$$\beta = \sum_j c_j\Big(I_j + \kappa_j\frac{L}{r}\Big) - (s+v)\frac{L}{r} \qquad \text{in (R1)},$$

$$\beta = 1 + \sum_j c_j\Big(-\nu_j + \kappa_j\frac{k}{r}\Big) - (s+v)\frac{k}{r} \qquad \text{in (R2)},$$

$$\beta = 1 - \frac{sn}{v} + \sum_j c_j\kappa_j\frac{L}{vr} - (s+v)\frac{L}{vr} \qquad \text{in (R3)},$$

with $\epsilon_j + \kappa_j\vartheta = \omega_j/r \ge 0$, $\sum_j c_j\kappa_j = \tau \le s+v$ since $\delta \le 0$, and $l, |I_j|, |\nu_j|, |\kappa_j| \le M$ by Lemma A.61. The parameters L/r, $k/r = 1/q + L/(qr)$, and $L/(vr)$ belong to the sets $\{L/r' \mid r' \ge 1\}$, $\{1/q + L'/(qr') \mid r' \ge 1,\ L_0 \le L' < L_0 + q\}$, and $\{L'/(vr') \mid r' \ge 1,\ L_0 \le L' < L_0 + v\}$, which satisfy the ACC. Finally $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3 \subset (0,1)$ by Lemma A.64(3). □

Lemma A.66. *Let $\Theta \subset [0,1]$ be a DCC set. Consider data as in Set-up A.60, without the bound $[G:R] \le e_0$, together with a vector $\mathbf{a}$ as in Construction-Proposition A.62 such that $\Gamma \subset \Theta$ and $\beta \in \Theta$, and let $o' \in Y$ be the image of $0 \in V_t$. Then the following hold.*

(1) The minimal log discrepancy α satisfies

$$\alpha = \min\{\mathrm{mld}(Y \ni \eta, \Delta_Y) \mid \eta \in \varphi^{-1}(o)\},$$

where η runs over the scheme-theoretic points of $\varphi^{-1}(o)$.

(2) The set of positive numbers $\mathrm{mld}(Y \ni \eta, \Delta_Y)$, where the data vary and $\eta \ne o'$ is a scheme-theoretic point of $\varphi^{-1}(o)$, satisfies the ACC.

Proof. (1) Since φ is proper and birational and $K_Y + \Delta_Y = \varphi^*(K_Q + \Delta)$, the prime divisors over Q with center o are the prime divisors over Y whose center is contained in $\varphi^{-1}(o)$, and $a(F, Y, \Delta_Y) = a(F, Q, \Delta)$ for any prime divisor F over Y [KM98, Lemma 2.30]. Hence $\mathrm{mld}(Y \ni \eta, \Delta_Y) \ge \alpha$ for any $\eta \in \varphi^{-1}(o)$. Since (Q, Δ) is klt near o, $\alpha = a(F, Q, \Delta)$ for a prime divisor F over Q with center o [Amb99, Section 1, the second paragraph after Definition 1.1], and equality holds for the generic point η of the center of F on Y.

(2) By Construction-Proposition A.62(6)(8), $B_{V_i} \in \Theta_+ \cap [0,1]$, so $\Delta_Y \in \Psi(\Theta_+ \cap [0,1])$ [Liu26, Lemma 2.6(4)], which is a DCC set [Liu26, Lemma 2.10(1)]. Let $\eta \in \varphi^{-1}(o)$ be a scheme-theoretic point different from o'. If η is the generic point of E, then $\mathrm{mld}(Y \ni \eta, \Delta_Y)$ is one minus the

coefficient of E in Δ_Y, which belongs to the ACC set $1-\Psi(\Theta_+\cap[0,1])$. If η is the generic point of an integral curve, the ACC follows from [Liu26, Lemma 3.7] applied to (Y,Δ_Y) and the DCC set $\Psi(\Theta_+\cap[0,1])$. If η is a closed point, let $\tilde\eta\in V_i$ be a point over η for some i. Then $b(\tilde\eta)=0$, and $\tilde\eta\neq 0$ if $i=t$, so V_i is smooth at $\tilde\eta$ by Construction-Proposition A.62(3). Let $H_{\tilde\eta}\subset H_i$ be the stabilizer of $\tilde\eta$. For a prime divisor F over V_i with center $\tilde\eta$, any element of H_i that maps F to itself fixes the center $\tilde\eta$ of F, so $I_{H_i}(F)=I_{H_{\tilde\eta}}(F)$. Any prime divisor over V_i/H_i with center the image of $\tilde\eta$ is the image of such a prime divisor F, after a translation by an element of H_i. Since $U_i\cong V_i/H_i$ is open in Y, by [Liu26, Lemma 2.6(2)(3)] applied to H_i and to $H_{\tilde\eta}$,

$$\operatorname{mld}(Y\ni\eta,\Delta_Y)=\operatorname{mld}((V_i\ni\tilde\eta,B_{V_i})/H_{\tilde\eta}).$$

Here $H_{\tilde\eta}$ is a finite abelian group acting faithfully on the smooth threefold germ $(V_i\ni\tilde\eta)$ and fixing $\tilde\eta$,

$$B_{V_i}=\beta\operatorname{div}(b)+\sum_j c_j\operatorname{div}(P_j)$$

with $H_{\tilde\eta}$-semi-invariant functions b,P_j and $\beta,c_j\in\Theta$, and (V_i,B_{V_i}) is klt near $\tilde\eta$ [Liu26, Lemma 2.6(2)]. Thus the ACC follows from Theorem 2.8(1) applied to Θ. The set of (2) is the union of the three sets of values at the generic point of E, at the generic points of integral curves, and at closed points, so it satisfies the ACC [Liu26, Lemma 2.10(4)]. □

Proof of Theorem 7.27. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce to the case $Nv>2s$ for (1) and (3). The element $w(\zeta)\in R$ multiplies xy, z^2, t^N by $\zeta^{2s},\zeta^{2s},\zeta^{Nv}$ (up to a common sign of the exponents), and $xy+z^2+t^N$ is R-semi-invariant, so $r\mid Nv-2s$. If $Nv<2s$, then $r\le 2s-Nv$ is bounded. The exponents N with $Nv\le 2s$ form a finite set, so a sequence of data with $Nv\le 2s$ may be assumed to have a constant N, and then the positive values α of the sequence satisfy the ACC by Theorem 7.8. Data with $Nv\le 2s$ have $d\le 0$, so they are excluded by the hypothesis $Nv>2s$ of (2) and by the hypothesis $d>0$ of (3). Thus we may assume that $Nv>2s$, so that $d\ge 1$. Let r_1 be the integer given by Lemma A.64.

Step 2. In this step we construct the crepant model for $A>0$. Assume that $\delta\le 0$, $A>0$, and $r\ge r_1$, and use row (R1) of Table 2, with $L=M+5$, which depends only on Γ,p,s,v. By Lemma A.64, $\mathbf{a}$ satisfies (A.243) with $N'=dL-2\ge 3$, and $0<\beta=A+\delta L/r<1$. By Construction-Proposition A.62, $\varphi:Y\to Q$ is a projective crepant model with $\Delta_Y\ge 0$, (Y,Δ_Y) is klt over a neighborhood of o, Y is covered by the charts V_i/H_i, where V_x,V_y,V_z are smooth and $V_t=W_{dL-2}$, the group H_t has order $a_t[G:R]=vL[G:R]$, and the coefficient of the exceptional divisor $(b=0)$ in B_{V_t} is $\beta=A+\delta L/r$. This is the coefficient of (2), since B_{V_t} is the crepant pullback of Δ_Y to $V_t=W_{dL-2}$ (Construction-Proposition A.62(6)) and $(b=0)$ is the preimage of the exceptional divisor E in V_t (Construction-Proposition A.62(2) and (3)). The exceptional prime divisor E of φ has coefficient $1-(1-\beta)/e$ in Δ_Y, where $e=|I_{H_t}((b=0))|$. This proves (2).

Step 3. In this step we show that α cannot increase along $\mathbf{T}$ if $A=0$ and $v\le s$. Assume that a sequence of such data with $\delta\le 0$, $Nv>2s$, and $r\to+\infty$ has strictly increasing positive values α. Possibly removing finitely many members, we may assume that $r\ge r_1$, and we use row (R2), with $k=(r+L)/q$. By Lemma A.64, $N'=kd\ge 3$ and $\beta=1-\nu_0+\delta k/r\in\mathcal{B}_2$. Let $\Theta:=\Gamma\cup\mathcal{B}_2$, a DCC set by Lemma A.65. By Lemma A.66, possibly passing to a subsequence, the minimum of Lemma A.66(1) is attained at o' for any member, since the values $\operatorname{mld}(Y\ni\eta,\Delta_Y)$ with $\eta\neq o'$ belong to the ACC set of Lemma A.66(2), which contains only finitely many members of the strictly increasing sequence α. Hence

$$\alpha=\operatorname{mld}((V_t\ni 0,B_{V_t})/H_t).$$

By Lemma A.61(4), h_j^0 contains a non-zero multiple of $t^{\kappa_j/v}$, and

$$\psi_t^*t^{\kappa_j/v}=\lambda^{a_t\kappa_j/v}=\lambda^{k\kappa_j}=\lambda^{\omega_j}b^{\nu_j}.$$

By Construction-Proposition A.62(5), P_j contains a non-zero multiple of b^{ν_j}, and P_j has no monomial divisible by $x'y'$. In particular P_j is a unit at 0 if $\nu_j = 0$.

For $\zeta \in \mu_k$, $(1,\zeta) \in \widehat{G}_t$ since $a_t = vk$, and it acts on (x', y', z', b) by

$$(\zeta^{-a_x}, \zeta^{-a_y}, \zeta^{-a_z}, \zeta^r) = (\zeta^{-r}, \zeta^r, 1, \zeta^r),$$

because $a_x = r - pk$, $a_y = qk - r$, and $a_z = sk$. Let $w' := (-1,1,0,1)$, $\mathbf{T}' := \mathbf{T}_{(w')}$, $G' := H_t$, and $R' := G' \cap \mathbf{T}'$. The images of the elements $(1,\zeta)$, $\zeta \in \mu_k$, form the subgroup of $\mathbf{T}'$ of order $k/\gcd(k,r)$. Since $\gcd(k,r)$ divides $qk - r = L$,

$$|R'| \geq \frac{k}{L} \geq \frac{r}{qL} \to +\infty, \qquad [G' : R'] \leq \frac{vk[G:R]L}{k} \leq v(L_0 + q - 1)e_0.$$

Since $|R'| \to +\infty$, removing finitely many members, we may assume that $|R'| > 2M$. The $\mathbf{T}'$-weight $-i_{x'} + i_{y'} + i_b$ of a monomial $x'^{i_{x'}} y'^{i_{y'}} z'^{i_{z'}} b^{i_b}$ has absolute value at most $i_{x'} + i_{y'} + i_{z'} + i_b$, and the $\mathbf{T}'$-weights of the monomials of the R'-semi-invariant polynomial P_j are congruent modulo $|R'|$. If $\nu_j > 0$, then any monomial of P_j of degree at most $\nu_j \leq M$ has $\mathbf{T}'$-weight congruent to the weight ν_j of b^{ν_j} and of absolute value at most ν_j, hence equal to ν_j, so the degree of the monomial is ν_j. Thus $\operatorname{mult}_0 P_j = \nu_j$, and the principal block of P_j along $\mathbf{T}'$ has weight ν_j and contains b^{ν_j}. The mark b has weight 1. Near 0, B_{V_t} is the boundary

$$D' := \beta \operatorname{div}(b) + \sum_{\nu_j > 0} c_j \operatorname{div}(P_j),$$

and the mass of D' along $\mathbf{T}'$ is

$$\tau' = \beta + \sum_j c_j \nu_j = 1 + \frac{\delta k}{r} \leq 1. \tag{A.249}$$

The following data are as in Set-up 7.3 with $w = w'$.

(1) The DCC set Θ, and the integer $v(L_0 + q - 1)e_0$ in place of e_0.
(2) The weight vector $w' = (-1,1,0,1)$, which is primitive with $w'_x + w'_y = 2w'_z$.
(3) The exponent $N' = kd \geq 3$ and the group $G' = H_t$, which is finite, diagonal, abelian, faithful on $W_{N'}$, and preserves the equation up to a character (Construction-Proposition A.62(1)), with coordinates (x', y', z', b) in place of (x, y, z, t).
(4) The subgroup $R' = G' \cap \mathbf{T}'$, with $[G' : R'] \leq v(L_0 + q - 1)e_0$.
(5) The boundary D', which is klt near 0 by Construction-Proposition A.62(8) and [Liu26, Lemma 2.6(2)], and whose P_j have no monomial divisible by $x'y'$.

For the marks b and P_j, $\nu_j > 0$, the principal blocks contain b and b^{ν_j}, of weights 1 and ν_j, and $\tau' \leq 1$ by (A.249), so the data satisfy the condition of Theorem 7.24(1). Hence the values α form a strictly increasing sequence of values of data as in Set-up 7.3 that satisfy the condition of Theorem 7.24(1), with $|R'| \to +\infty$, which contradicts Theorem 7.24(1).

Step 4. In this step we show that α cannot increase along $\mathbf{T}$ if $A = 0$ and $v > s$, without the bound $[G : R] \leq e_0$. Assume that a sequence of such data with $\delta \leq 0$, $Nv > 2s$, and $r \to +\infty$ has strictly increasing positive values α. As in **Step 3**, we may assume that $r \geq r_1$, and we use row (R3), with $n = \lceil v/(2s) \rceil$, $m = \lfloor pn/v \rfloor + 1$, and $k = (nr + L)/v$. By Lemma A.64, $N' = (dL - 2sn)/v \geq 3$ and $\beta = 1 - sn/v + \delta L/(vr) \in \mathcal{B}_3$. Let $\Theta := \Gamma \cup \mathcal{B}_3$, a DCC set by Lemma A.65. By Lemma A.66, possibly passing to a subsequence as in **Step 3**, $\alpha = \operatorname{mld}((V_t \ni 0, B_{V_t})/H_t)$ for any member.

By Lemma A.61(4), h_j^0 contains a non-zero multiple of $t^{\kappa_j/v}$, and

$$\psi_t^* t^{\kappa_j/v} = \lambda^{L\kappa_j/v} = \lambda^{\omega_j}.$$

By Construction-Proposition A.62(5), P_j has a non-zero constant term, so P_j is a unit at 0 for any j. Hence $B_{V_t} = \beta \operatorname{div}(b)$ near 0. Moreover, E is a prime divisor over Q with center o, so

$$0 < \alpha \leq a(E, Q, \Delta) = \frac{1 - \beta}{e} \leq 1 - \beta < 1.$$

In [Liu26, Theorem 18.8(1)], consider a single mark $h_1 = b$ (the case $s = 1$ in the notation there), with label $c_1 = \beta$ in the DCC set $\mathcal{B}_3$, the coordinates (x', y', z', b) in place of (x, y, z, t), so that (z, t) corresponds to (z', b), the polynomial $g = z'^2 + b^{N'} \in (z', b)^2$ with $N' \geq 3$, and the group H_t with no bound on $|H_t|$. The germ $W_{N'}$ is a normal klt threefold germ at 0, and $(W_{N'}, \beta \operatorname{div}(b))$ is klt near 0 by Construction-Proposition A.62(8) and [Liu26, Lemma 2.6(2)], and it is klt outside 0, since $W_{N'}$ and $(b = 0)$ are smooth there and $\beta < 1$. Thus, by [Liu26, Theorem 18.8(1)], the values $\alpha \in (0, 1)$ satisfy the ACC, a contradiction. Lemmas A.61, A.64, A.65, A.66, Construction-Proposition A.62, Theorem 7.8 in **Step 1**, and [Liu26, Theorem 18.8(1)] do not use the bound $[G : R] \leq e_0$. Together with **Steps 1 and 3**, this proves (1).

Step 5. In this step we prove the ACC for the subclass $\delta \leq 0$, $0 < d \leq d_0$. Assume that a sequence of such data has strictly increasing positive values α. If r is bounded along a subsequence, then $Nv - 2s = dr \leq d_0 r$ is bounded, so N is bounded, and possibly passing to a subsequence, N is constant, which contradicts Theorem 7.8. Hence we may assume that $r \to +\infty$. By **Steps 3 and 4**, $A > 0$ for all but finitely many members. Removing them, we may assume that $A > 0$ and $r \geq r_1$, and we use the crepant model of **Step 2**. Let $\Theta := \Gamma \cup \mathcal{B}_1$, a DCC set by Lemma A.65. By Lemma A.66, possibly passing to a subsequence as in **Step 3**,

$$\alpha = \operatorname{mld}((W_{dL-2} \ni 0, B_{V_t})/H_t)$$

for any member, where B_{V_t}, defined in Construction-Proposition A.62(6), is a labelled boundary with labels in Θ, klt near 0, and H_t is a finite diagonal abelian group acting faithfully on W_{dL-2} and preserving the equation of W_{dL-2} up to a character. Since $dL - 2 \in \{L - 2, \dots, d_0 L - 2\}$, possibly passing to a subsequence, $dL - 2$ is constant, which contradicts Theorem 7.8. This proves (3). □

A.19. **Mixed circles: the growing offset.** In this subsection, we prove Theorem 7.28 and Corollary 7.29.

Proof of Theorem 7.28. The argument below holds uniformly for $s \geq 0$. For $s = 0$ the circle is $\mathbf{T}_{(-p,p,0,v)}$. The proof is long, so we divide it into several steps.

Assume, to the contrary, that there exists a sequence of data of Set-up 7.3 with $w = (-p, p + 2s, s, v)$, with Γ, e_0 and w fixed and $r \to +\infty$, such that $\alpha > 0$, $A > 0$, and $\delta \leq 0$ for any term, and such that the values α are strictly increasing. We write (W_{N_i}, G_i, D_i) for the i-th term, $D_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$, and r_i, R_i, α_i, A_i, τ_i, δ_i, d_i for the corresponding groups and numbers of Set-up 7.3. When we work with one term, we drop the index i. For $\epsilon \in \mu_r$ let

$$g_\epsilon := (\epsilon^{-p}, \epsilon^{p+2s}, \epsilon^s, \epsilon^v);$$

since w is primitive, $R = \{g_\epsilon \mid \epsilon \in \mu_r\}$. Recall that

$$\operatorname{wt}_w(x^{m_1} y^{m_2} z^{m_3} t^{m_4}) = -pm_1 + (p + 2s)m_2 + sm_3 + vm_4.$$

Step 1. In this step we bound the numerical data of the marks and fix the lists (κ_j) and (I_j).

Since $v > 0$, only finitely many N satisfy $Nv \leq 2s$, and for each of them Theorem 7.8 implies that the positive numbers α_i with $N_i = N$ satisfy the ACC. Possibly passing to a subsequence, we may assume that $N_i v > 2s$, hence $d_i \geq 1$, for any i.

Removing from D the marks h_j with $c_j = 0$ and the marks h_j with $h_j(0) \neq 0$ changes neither D near 0 nor α, A, and τ (a mark h_j with $h_j(0) \neq 0$ has $\kappa_j = I_j = 0$). Thus we may assume that $c_j > 0$ and $h_j(0) = 0$ for any j. By Set-up 7.3(5), we may assume that each h_j is a polynomial without monomials divisible by xy.

Since $A > 0$, the set $\Gamma \backslash \{0\}$ is non-empty, and we let $\gamma := \min(\Gamma \backslash \{0\})$, which exists as Γ is a DCC set. For a polynomial h, let $\operatorname{mult}_0 h$ be the least total degree of the monomials of h. Since (W_N, D) is klt near 0, Lemma A.1(2)(3) gives

$$\sum_j c_j \operatorname{mult}_0 h_j < 2,$$

and, as $c_j \geq \gamma$, the number of marks and each $\operatorname{mult}_0 h_j$ are smaller than $2/\gamma$. Let

$$M := \left\lceil \frac{2(p+2s+v)}{\gamma} \right\rceil.$$

Since h_j is semi-invariant under R, the numbers wt_w of the monomials of h_j are congruent modulo r. A monomial of h_j of degree $\operatorname{mult}_0 h_j$ has $|\operatorname{wt}_w| \leq (p+2s+v)\operatorname{mult}_0 h_j \leq M$. Hence, if $r > 2M$, then the monomials of h_j of degree $\operatorname{mult}_0 h_j$ have the common weight κ_j, any monomial of h_j has $\operatorname{wt}_w \in \kappa_j + r\mathbb{Z}$, and h_j^0 is the sum of the terms of h_j whose monomials have $\operatorname{wt}_w = \kappa_j$. Moreover,

$$|\kappa_j| \leq M, \qquad 0 \leq I_j \leq \operatorname{mult}_0 h_j \leq M, \qquad \kappa_j \geq -p \operatorname{mult}_0 h_j.$$

The last inequality and $\sum_j c_j \operatorname{mult}_0 h_j < 2$ imply $\tau > -2p$, and hence

$$\delta > -(2p+s+v). \tag{A.250}$$

Possibly passing to a subsequence, we may assume that $r_i > 2M$ for any i, that the number of marks and the lists (κ_j) and (I_j) do not depend on i, and that the sequence $(c_{ij})_i$ is non-decreasing for any j, which is possible because Γ is a DCC set. Since $A_i > 0$, we have $I_j > 0$ for some j, so $A_i \geq \gamma$ for any i.

Step 2. In this step we choose the growing offset L and compute the orders of the marks along a weight vector depending on L.

By Lemma 4.9, applied to the interval $(\sqrt{r_i}, 2\sqrt{r_i}]$, for any $r_i \geq r_0$ there exists an integer L_i such that

$$\sqrt{r_i} < L_i \leq 2\sqrt{r_i} \quad \text{and} \quad \gcd(L_i, r_i) = 1.$$

Then $L_i \to +\infty$ and $L_i/r_i \leq 2/\sqrt{r_i} \to 0$. Possibly passing to a subsequence, we may assume that the sequence $(L_i/r_i)_i$ is strictly decreasing and, since $\sqrt{r_i} < L_i \leq 2\sqrt{r_i}$, that

$$L > 2M+5, \qquad r > p(L+2M), \qquad (2p+s+v)L \leq \frac{\gamma r}{2} \tag{A.251}$$

for any term. Let

$$\omega = (\omega_x, \omega_y, \omega_z, \omega_t) := (r - pL,\ r + (p+2s)L,\ r + sL,\ vL) \in \mathbb{Z}^4_{>0}.$$

Since $\omega/r = (1,1,1,0) + (L/r)w$, the image of ω/r in $\mathbb{T}^4$ is g_ϵ with $\epsilon = e^{2\pi\sqrt{-1}L/r}$, so $\omega/r \in N_R \subset N_G$. Moreover

$$\omega \equiv Lw \pmod{r}. \tag{A.252}$$

Let $x^{m_1}y^{m_2}z^{m_3}t^{m_4}$ be a monomial of h_j, let $n \in \mathbb{Z}$ with

$$\operatorname{wt}_w(x^{m_1}y^{m_2}z^{m_3}t^{m_4}) = \kappa_j + nr,$$

and let $I := m_1 + m_2 + m_3$. Then

$$\operatorname{wt}_\omega(x^{m_1}y^{m_2}z^{m_3}t^{m_4}) = rI + L\kappa_j + nrL = rI_j + L\kappa_j + r(I - I_j + nL). \tag{A.253}$$

We show that

$$I - I_j + nL \geq 0, \tag{A.254}$$

with equality if and only if $n = 0$ and $I = I_j$. If $n = 0$, then the monomial belongs to h_j^0 and $I \geq I_j$ by the definition of I_j. If $n \geq 1$, then $I - I_j + nL \geq L - M > 0$ by (A.251). If $n \leq -1$, then $pm_1 \geq -nr - \kappa_j \geq -nr - M$ because $(p+2s)m_2 + sm_3 + vm_4 \geq 0$, so $I \geq m_1 \geq (-nr - M)/p$ and

$$I - I_j + nL \geq (-n)\left(\frac{r}{p} - L\right) - \frac{M}{p} - M \geq \frac{r}{p} - L - 2M > 0$$

by (A.251). Thus $\operatorname{wt}_\omega(h_j) = rI_j + L\kappa_j > 0$, where the inequality holds because $h_j(0) = 0$ and the entries of ω are positive. Finally,

$$\operatorname{wt}_\omega(xy) = \operatorname{wt}_\omega(z^2) = 2r + 2sL, \qquad \operatorname{wt}_\omega(t^N) = NvL = 2r + 2sL + r(dL - 2),$$

and $dL - 2 \geq L - 2 \geq 3$, since $d \geq 1$ and $L > 2M + 5$.

Step 3. In this step we construct a projective birational morphism $f : Y \to W_N/G$ and describe the charts of Y. For $s > 0$, this is the model of Construction-Proposition A.62 with $\mathbf{a} = \omega$, $u = (1, 1, 1, 0)$, $k = L$, and $N' = dL - 2$, which is row (R1) of Table 2 when L is the integer fixed in Construction A.63. Here $s \geq 0$ and L grows with r.

Let $X := W_N/G$, let $o \in X$ be the image of 0, and let B be the $\mathbb{R}$-divisor on X with $K_{W_N} + D = \pi^*(K_X + B)$ for the quotient morphism $\pi : W_N \to X$, so that $\alpha = \mathrm{mld}(X \ni o, B)$. Possibly shrinking X to a neighborhood of o, we may assume that (W_N, D) is klt, hence (X, B) is klt by [Liu26, Lemma 2.6(2)(3)].

Let Σ be the star subdivision of the cone $\mathbb{R}^4_{\geq 0}$ at the ray $\mathbb{R}_{\geq 0}\omega$ with respect to the lattice N_G, and let $Y_\Sigma \to \mathbb{A}^4/G$ be the corresponding projective birational toric morphism [CLS11, Proposition 11.1.6(c)]. Since the entries of ω are positive, $Y_\Sigma \to \mathbb{A}^4/G$ is an isomorphism over the complement of the image of 0. Let Y be the strict transform of X in Y_Σ and $f : Y \to X$ the induced morphism. Then f is a projective birational morphism and an isomorphism over $X\backslash\{o\}$.

Let $q \in \{x, y, z, t\}$, and let e_x, e_y, e_z, e_t be the standard basis vectors of $\mathbb{Z}^4$. The vectors ω and e_l, $l \neq q$, form a basis of a sublattice $N'_q \subset N_G$. Hence the affine chart of Y_Σ corresponding to the cone spanned by ω and e_l, $l \neq q$, is $\mathbb{A}^4/\tilde{G}_q$, where $\mathbb{A}^4$ has the coordinates λ and l', $l \neq q$, dual to the basis ω, e_l of N'_q, and

$$\tilde{G}_q := \{(g, \zeta) \in G \times \mathbb{C}^* \mid \zeta^{\omega_q} = g_q\}, \qquad g = (g_x, g_y, g_z, g_t),$$

is identified, through its action below, with the image of N_G/N'_q in the torus $N'_q \otimes \mathbb{C}^*$ of this $\mathbb{A}^4$ [CLS11, Proposition 1.3.18]. The group $\tilde{G}_q$ has order $\omega_q|G|$ and acts by $\lambda \to \zeta\lambda$ and $l' \to \zeta^{-\omega_l} g_l l'$ for $l \neq q$. Explicitly, the morphism $\psi_q : \mathbb{A}^4 \to \mathbb{A}^4$ is given by

$$x = \lambda^{r-pL} x', \qquad y = \lambda^{r+(p+2s)L} y', \qquad z = \lambda^{r+sL} z', \qquad t = \lambda^{vL} t', \tag{A.255}$$

where the factor q' is omitted. The morphism ψ_q is equivariant with respect to the projection $\tilde{G}_q \to G$. Let $b := \lambda^r$. By **Step 2**,

$$\psi_q^*(xy + z^2 + t^N) = \lambda^{2r+2sL} F_q, \qquad F_q := x'y' + z'^2 + b^{dL-2} t'^N,$$

where the factor q' in F_q is replaced by 1. Thus Y is covered by the four open subsets $\tilde{V}_q/\tilde{G}_q$, where $\tilde{V}_q := (F_q = 0) \subset \mathbb{A}^4$.

Let $\tilde{R}_q := \{(g_{\zeta^L}, \zeta) \mid \zeta \in \mu_r\}$. By (A.252), $\zeta^{\omega_q} = \zeta^{Lw_q} = (g_{\zeta^L})_q$, so $\tilde{R}_q \subset \tilde{G}_q$, and $\zeta^{-\omega_l}(g_{\zeta^L})_l = \zeta^{Lw_l - \omega_l} = 1$ for any l. Hence $\tilde{R}_q \cong \mu_r$ acts trivially on the coordinates l', $l \neq q$, and by $\lambda \to \zeta\lambda$. Thus $V_q := \tilde{V}_q/\tilde{R}_q$ is the hypersurface $(F_q = 0)$ in the affine space $\mathbb{A}^4$ with the coordinates l', $l \neq q$, and b. Let $G'_q := \tilde{G}_q/\tilde{R}_q$, a group of order $\omega_q[G : R]$ acting diagonally on b and l', $l \neq q$. The action of G'_q is faithful: if (g, ζ) acts trivially on b and on l', $l \neq q$, then $\zeta^r = 1$ and $g_l = \zeta^{\omega_l}$ for any l (for $l = q$ by the definition of $\tilde{G}_q$), so $g = g_{\zeta^L}$ by (A.252), and $(g, \zeta) \in \tilde{R}_q$. Let $U_q := V_q/G'_q \subset Y$ and let $\pi_q : V_q \to U_q$ be the quotient morphism. The four hypersurfaces are

$$V_x = (y' + z'^2 + b^{dL-2} t'^N = 0), \qquad V_y = (x' + z'^2 + b^{dL-2} t'^N = 0),$$
$$V_z = (x'y' + 1 + b^{dL-2} t'^N = 0), \qquad V_t = (x'y' + z'^2 + b^{dL-2} = 0).$$

The hypersurfaces V_x and V_y are smooth. The hypersurface V_z is smooth, because $x' = y' = 0$ on V_z implies $bt' \neq 0$, hence $\partial F_z/\partial t' = Nb^{dL-2} t'^{N-1} \neq 0$. Each V_q is integral and is not contained in any coordinate hyperplane, so G'_q acts faithfully on V_q. In the t-chart we write t' for b, so that

$$V_t = W_{dL-2} = (x'y' + z'^2 + t'^{dL-2} = 0), \qquad dL - 2 \geq 3,$$

which is smooth outside 0, and we write $G' := G'_t$, a finite diagonal abelian group of order $vL[G : R]$. Each V_q is normal [Har77, Proposition II.8.23(b)], as it is a hypersurface smooth outside 0, so Y is normal, as $U_q = V_q/G'_q$. Since λ and $\psi_t^*(xy + z^2 + t^N)$ are $\tilde{G}_t$-semi-invariant, G' preserves $x'y' + z'^2 + t'^{dL-2}$ up to a character.

Since the entries of ω are positive, a point of $\tilde{V}_q$ maps to $0 \in W_N$ if and only if $\lambda = 0$ there. Thus $f^{-1}(o) \cap U_q = \pi_q(E_q)$, where $E_q := (b = 0) \subset V_q$ is the integral surface

$$(y' + z'^2 = 0), \quad (x' + z'^2 = 0), \quad (x'y' + 1 = 0), \quad (x'y' + z'^2 = 0)$$

in $\mathbb{A}^3$ for $q = x, y, z, t$ respectively, and $\operatorname{div}(b) = E_q$ on V_q. Hence $f^{-1}(o)$ is a prime divisor E_Y on Y, the image of the surface $(xy + z^2 = 0)$ of the weighted projective space with weights ω.

Step 4. In this step we compute the crepant boundaries on the charts V_q and the coefficient β of the exceptional divisor.

Recall that ψ_q is given by (A.255), with the factor q' omitted. By (A.253) and (A.254), $\psi_q^* h_j = \lambda^{rI_j + L\kappa_j} P_j$, where P_j is h_j with each monomial $x^{m_1} y^{m_2} z^{m_3} t^{m_4}$ of h_j replaced, keeping its coefficient, by

$$x'^{m_1} y'^{m_2} z'^{m_3} t'^{m_4} b^{I - I_j + nL}, \tag{A.256}$$

with n and I as in (A.253), and with the factor q' replaced by 1. In the t-chart, where t' stands for b, the monomial (A.256) reads

$$x'^{m_1} y'^{m_2} z'^{m_3} t'^{I - I_j + nL}.$$

By (A.254), P_j is a polynomial in b and l', $l \neq q$. Since h_j and λ are $\tilde{G}_q$-semi-invariant and $\tilde{R}_q$ acts trivially on b and l', P_j is a G'_q-semi-invariant regular function on V_q. We show that P_j does not vanish identically on E_q. By (A.254), $P_j|_{b=0}$ is the polynomial $h_j^\flat$, the sum of the terms of h_j^0 with $I = I_j$, evaluated at (x', y', z', t') with $q' = 1$. The polynomial $h_j^\flat$ is ω-homogeneous, so if P_j vanished on $E_q = (x'y' + z'^2 = 0,\ q' = 1)$, then $h_j^\flat$ would vanish on $(xy + z^2 = 0) \cap (q \neq 0)$, which is dense in the integral hypersurface $(xy + z^2 = 0) \subset \mathbb{A}^4$. Then $xy + z^2$ would divide $h_j^\flat$, and the lexicographically largest monomial of $h_j^\flat$, with $x > y > z > t$, would be divisible by xy, a contradiction.

Let $\Omega := dx \wedge dz \wedge dt/x = -dy \wedge dz \wedge dt/y$, the residue form generating $\mathcal{O}_{W_N}(K_{W_N})$ near 0. On $\tilde{V}_q$ near $(\lambda = 0)$, the residue forms

$$\Omega_x = dz' \wedge dt' \wedge d\lambda, \quad \Omega_y = dz' \wedge dt' \wedge d\lambda, \quad \Omega_z = \frac{dx' \wedge dt' \wedge d\lambda}{x'}, \quad \Omega_t = \frac{dx' \wedge dz' \wedge d\lambda}{x'}$$

are generators, as $x' \neq 0$ on $\tilde{V}_z$ near $(\lambda = 0)$. A direct calculation shows that

$$\psi_q^* \Omega = \vartheta_q \lambda^{r + (s+v)L - 1} \Omega_q \tag{A.257}$$

for a non-zero constant ϑ_q. Here $r + (s + v)L$ is the sum of the entries of ω minus $2r + 2sL$. For instance, in the t-chart $dt = vL\lambda^{vL-1} d\lambda$, so

$$dx \wedge dz \wedge dt = vL\lambda^{(r - pL) + (r + sL) + vL - 1} dx' \wedge dz' \wedge d\lambda,$$

and dividing by $x = \lambda^{r - pL} x'$ we obtain $\psi_t^* \Omega = vL\lambda^{r + (s+v)L - 1} \Omega_t$. In the y-chart we use $\Omega = -dy \wedge dz \wedge dt/y$. The divisor $\operatorname{div}(\lambda)$ on $\tilde{V}_q$ is the prime divisor $(\lambda = 0) \cap \tilde{V}_q$, since $F_q|_{\lambda=0}$ is irreducible. By (A.257),

$$K_{\tilde{V}_q} + \tilde{D}_q = \psi_q^*(K_{W_N} + D), \qquad \tilde{D}_q := \sum_j c_j \operatorname{div}(P_j) + \Big(\sum_j c_j (rI_j + L\kappa_j) - r - (s+v)L + 1 \Big) \operatorname{div}(\lambda).$$

The morphism $\tilde{V}_q \to V_q$ is the quotient by $\tilde{R}_q \cong \mu_r$, which acts by $\lambda \to \zeta\lambda$ and trivially on the other coordinates, so it is ramified only along $\operatorname{div}(\lambda)$, with index r. By [Liu26, Lemma 2.6(4)], $K_{\tilde{V}_q} + \tilde{D}_q$ is the pullback of $K_{V_q} + D_q$, where

$$D_q := \beta \operatorname{div}(b) + \sum_j c_j \operatorname{div}(P_j)$$

and, since $\sum_j c_j(rI_j + L\kappa_j) = rA + L\tau$ and $\delta = \tau - (s+v)$,

$$\beta = 1 - \frac{r + (s+v)L - \sum_j c_j(rI_j + L\kappa_j)}{r} = A + \frac{\delta L}{r}. \tag{A.258}$$

In particular, β does not depend on q.

Let B_Y be the $\mathbb{R}$-divisor on Y with $K_Y + B_Y = f^*(K_X + B)$. The composite

$$\tilde{V}_q \to V_q \xrightarrow{\pi_q} U_q \subset Y \xrightarrow{f} X$$

coincides with the composite of $\psi_q|_{\tilde{V}_q} : \tilde{V}_q \to W_N$ and π. Hence $K_{V_q} + D_q$ and $\pi_q^*(K_{U_q} + B_Y|_{U_q})$ both pull back to $K_{\tilde{V}_q} + \tilde{D}_q$ on $\tilde{V}_q$, and since $\tilde{V}_q \to V_q$ is finite and surjective,

$$K_{V_q} + D_q = \pi_q^*(K_{U_q} + B_Y|_{U_q}). \tag{A.259}$$

By [Liu26, Lemma 2.6(4)], $\mathrm{mult}_{E_Y} B_Y = 1 - (1-\beta)/e$, where e is the ramification index of π_t along E_t.

Step 5. In this step we show that the numbers β_i form a non-decreasing sequence in $(0,1)$, and we find one DCC set containing the coefficients of all the boundaries D_q and B_Y.

By (A.258), (A.250), $A \geq \gamma$, and (A.251),

$$\beta > \gamma - \frac{(2p+s+v)L}{r} \geq \frac{\gamma}{2}.$$

Hence $\mathrm{mult}_{E_Y} B_Y \geq 0$. The other components of B_Y are strict transforms of components of B, so $B_Y \geq 0$. Since $K_Y + B_Y = f^*(K_X + B)$ and (X, B) is klt, (Y, B_Y) is klt. By (A.259) and [Liu26, Lemma 2.6(2)], (V_q, D_q) is klt for any q, and in particular $\beta < 1$.

Let $i < i'$, so that $L_{i'}/r_{i'} < L_i/r_i$ by **Step 2**, and let $c_j := c_{ij}$ and $c'_j := c_{i'j}$, so that $c'_j \geq c_j$ by **Step 1**. Since the lists (κ_j) and (I_j) do not depend on i, (A.258) implies

$$\beta_{i'} - \beta_i = \sum_j (c'_j - c_j)(I_j + \kappa_j L_{i'}/r_{i'}) + \delta_i(L_{i'}/r_{i'} - L_i/r_i) \geq 0, \tag{A.260}$$

because $I_j + \kappa_j L_{i'}/r_{i'} = (rI_j + L\kappa_j)/r$ for the i'-th term, which is positive by **Step 2**, $c'_j - c_j \geq 0$, $\delta_i \leq 0$, and $L_{i'}/r_{i'} - L_i/r_i < 0$. Thus $(\beta_i)_i$ is a non-decreasing sequence in $[\gamma/2, 1)$, and

$$\Lambda := \Gamma \cup \{\beta_i \mid i \geq 1\} \subset [0,1]$$

is a DCC set. For any i, any q and any prime divisor P on V_q, $\mathrm{mult}_P D_q$ is a finite sum of elements of Λ, and $\mathrm{mult}_P D_q < 1$ as (V_q, D_q) is klt. Thus $D_q \in \Lambda_+ \cap [0,1]$. By (A.259) and [Liu26, Lemma 2.6(4) and Lemma 2.10(1)], $B_Y \in \Theta$ for the DCC set

$$\Theta := \Psi(\Lambda_+ \cap [0,1]) \subset [0,1],$$

which depends neither on i nor on q.

Step 6. In this step we find a subgroup of order L of G' in the scalar circle $\mathbf{T}_{(-1,-1,-1,1)}$.

Recall that $G' = \tilde{G}_t/\tilde{R}_t$ acts on $V_t = W_{dL-2}$ with the coordinates x', y', z' and $t' = b$. For $\zeta \in \mu_L$, the element $(1, \zeta)$ belongs to $\tilde{G}_t$, since $\zeta^{\omega_t} = \zeta^{vL} = 1$. As $\omega_x = r - pL$, $\omega_y = r + (p+2s)L$ and $\omega_z = r + sL$ are congruent to r modulo L, $(1, \zeta)$ acts by

$$(x', y', z', t') \to (\zeta^{-r}x', \zeta^{-r}y', \zeta^{-r}z', \zeta^r t').$$

Since $\gcd(L, r) = 1$, ζ^r runs over μ_L when ζ runs over μ_L. Since G' acts faithfully, the image of $\{(1,\zeta) \mid \zeta \in \mu_L\}$ in G' is the subgroup $\mu_L(-1,-1,-1,1)$ of order L of $\mathbf{T}_{(-1,-1,-1,1)}$. Let $R' := G' \cap \mathbf{T}_{(-1,-1,-1,1)}$. Then

$$|R'| \geq L, \qquad [G' : R'] \leq \frac{|G'|}{L} = v[G : R] \leq ve_0. \tag{A.261}$$

Step 7. In this step we treat the centers over o and conclude the proof.

Since $K_Y + B_Y = f^*(K_X + B)$ and f is an isomorphism over $X\backslash\{o\}$, a prime divisor over X has center o if and only if the center on Y is contained in $f^{-1}(o) = E_Y$, and the log discrepancies with respect to (X, B) and (Y, B_Y) are equal [KM98, Lemma 2.30]. Hence

$$\alpha = \min\{\mathrm{mld}(Y \ni \eta, B_Y) \mid \eta \in E_Y\}, \tag{A.262}$$

where η runs over the scheme-theoretic points of E_Y, and the minimum is attained since (X, B) is klt [Amb99, Section 1, the second paragraph after Definition 1.1]. For any i, let $\eta_i \in E_{Y_i}$ be a point attaining the minimum in (A.262), and let $o'_i \in U_t \subset Y_i$ be the image of $0 \in V_t = W_{d_iL_i-2}$. Possibly passing to a subsequence, we may assume that one of the following cases holds for any i. Recall that Λ and Θ are the DCC sets of **Step 5**.

(1) η_i is the generic point of E_{Y_i}. Then

$$\alpha_i = 1 - \mathrm{mult}_{E_{Y_i}} B_{Y_i} \in \{1 - \theta \mid \theta \in \Theta\},$$

which satisfies the ACC.

(2) η_i is the generic point of a curve. By [Liu26, Lemma 3.7], for the DCC set $\Theta \subset [0, 1]$, the positive numbers $\mathrm{mld}(Y' \ni \xi, B')$, where (Y', B') is a threefold pair with $B' \in \Theta$ and ξ is the generic point of an integral curve on Y', satisfy the ACC. Since $B_{Y_i} \in \Theta$, the numbers $\alpha_i = \mathrm{mld}(Y_i \ni \eta_i, B_{Y_i})$ satisfy the ACC.

(3) η_i is a closed point different from o'_i. Let $q \in \{x, y, z, t\}$ with $\eta_i \in U_q$, let $\tilde{v} \in V_q$ be a closed point over η_i, and let $H \subset G'_q$ be the stabilizer of $\tilde{v}$. Then $b(\tilde{v}) = 0$, and $\tilde{v} \neq 0$ if $q = t$, since $\eta_i \neq o'_i$. Hence V_q is smooth near $\tilde{v}$ by **Step 3**. Since G'_q acts faithfully on the integral variety V_q (**Step 3**), so does its subgroup H on any non-empty open subset, and in particular H acts faithfully on the germ $(V_q \ni \tilde{v})$. By (A.259), [Liu26, Lemma 2.6(2)(3)], and the formula at the end of [Liu26, Lemma 2.6], $\alpha_i = \mathrm{mld}((V_q \ni \tilde{v}, D_q)/H)$. Here an element of G'_q mapping a prime divisor F over V_q with center $\tilde{v}$ to itself fixes $\tilde{v}$, so $I_{G'_q}(F) = I_H(F)$, and for every prime divisor E over U_q with center η_i there is such an F over E by [Liu26, Lemma 2.6(3)], after a translation by an element of G'_q. The boundary

$$D_q = \beta \,\mathrm{div}(b) + \sum_j c_j \,\mathrm{div}(P_j)$$

has labels in Λ and H-semi-invariant marks b and P_j (**Step 4**), and (V_q, D_q) is klt near $\tilde{v}$ (**Step 5**). By Theorem 2.8(1), with Γ replaced by Λ, the numbers α_i satisfy the ACC.

(4) $\eta_i = o'_i$. Then $\alpha_i = \mathrm{mld}((W_{d_iL_i-2} \ni 0, D_t)/G'_i)$. By **Steps 3, 5, and 6**, $(W_{d_iL_i-2}, G'_i, D_t)$ are data of Set-up 7.3 for the scalar circle $\mathbf{T}_{(-1,-1,-1,1)}$, which is the canonical circle of Construction-Proposition 7.14 whose three parameters are equal to 1. For $(W_{d_iL_i-2}, G'_i, D_t)$, Γ is replaced by Λ, e_0 by ve_0, N by $d_iL_i - 2 \geq 3$, and r by $|R'_i| \geq L_i$: the group G'_i is a finite diagonal abelian group preserving $x'y' + z'^2 + t'^{d_iL_i-2}$ up to a character and acting faithfully on $W_{d_iL_i-2}$, D_t has labels in Λ and is klt near 0, by (A.256) no mark of D_t has a monomial divisible by $x'y'$, and $[G'_i : R'_i] \leq ve_0$ by (A.261). Since $|R'_i| \geq L_i \to +\infty$ and Λ and ve_0 do not depend on i, Theorem 7.15(1) implies that the positive numbers α_i are not strictly increasing.

In any case, $(\alpha_i)_i$ is not a strictly increasing sequence of positive numbers, a contradiction. □

Proof of Corollary 7.29. Assume, to the contrary, that there exists a sequence of data of Set-up 7.3 with $w = (u, 2s - u, s, v)$, with Γ, e_0 and w fixed and $r \to +\infty$, whose values α are strictly increasing and satisfy $\alpha > 0$.

The automorphism of $\mathbb{A}^4$ exchanging x and y preserves $xy + z^2 + t^N$. It conjugates G to a finite diagonal abelian group, maps $\mathbf{T}_{(u,2s-u,s,v)}$ onto $\mathbf{T}_{(2s-u,u,s,v)}$, and preserves r, $[G : R]$, and α. Hence, replacing u by $2s - u$ if $u > s$, we may assume that $u \leq s$. If $s = 0$, then $u \neq 0$ because $(u, s) \neq (0, 0)$, so $u < 0$. Removing from D the marks h_j with $c_j = 0$ and the marks h_j with $h_j(0) \neq 0$ changes neither D near 0 nor α, A, and τ, so we may assume that $c_j > 0$ and $h_j(0) = 0$ for any j. Then $A = \sum_j c_j I_j \geq 0$, since $c_j > 0$ and $I_j \geq 0$ for any j. Possibly passing

to a subsequence, we may assume that all the terms belong to one row of Table 3, and we show that each row leads to a contradiction.

circle	condition on the terms	reason
$u<0,\ s\geq 0$	$\delta\geq 0$	Theorem 7.26
$u<0,\ s\geq 0$	$A>0,\ \delta<0$	Theorem 7.28
$u<0,\ s>0$	$A=0,\ \delta<0$	Theorem 7.27
$u<0,\ s=0$	$A=0,\ \delta<0$	Theorem 7.25
$u=0,\ s>0$	none	Theorem 7.11(2)
$0<u\leq s$	$Nv\geq 2s$	$\alpha\leq(s+v)/r$ by Lemma 4.7
$0<u\leq s$	$Nv<2s$	Theorem 7.8

TABLE 3. The cases in the proof of Corollary 7.29.

The rows with $u<0$. Let $p:=-u>0$. Then $w=(-p,p+2s,s,v)$, and $\gcd(p,s,v)=1$ since $\gcd(u,s,v)=1$. If $\delta\geq 0$ for any term, then Theorem 7.26 yields a contradiction. If $A>0$ and $\delta<0$ for any term, then Theorem 7.28 yields a contradiction. Assume that $A=0$ and $\delta<0$ for any term. If $s>0$, then $p,s,v>0$, and Theorem 7.27 yields a contradiction. If $s=0$, then $w=(-p,p,0,v)$ with $\gcd(p,v)=1$. Since $A=\sum_j c_jI_j=0$ and $c_j>0$, we have $I_j=0$, so h_j^0 contains a non-zero monomial that is a power of t, for any j. Moreover, $\tau=\delta+v<v$. Thus Theorem 7.25 yields a contradiction.
The row with $u=0$. Then $s>0$ and $w=(0,2s,s,v)$, and $\gcd(s,v)=1$ since $\gcd(u,s,v)=1$. Theorem 7.11(2) yields a contradiction.
The rows with $0<u\leq s$. Then $s>0$ and $0<u<2s$. Since $v>0$, only finitely many N satisfy $Nv<2s$, and for each of them Theorem 7.8 implies that the positive numbers α with exponent N satisfy the ACC. Assume that $Nv\geq 2s$ for any term, and let

$$w':=\frac{1}{r}(u,2s-u,s,v).$$

The image of w' in $\mathbb{T}^4$ is $(\epsilon^u,\epsilon^{2s-u},\epsilon^s,\epsilon^v)$ with $\epsilon=e^{2\pi\sqrt{-1}/r}$, a generator of R, so $w'\in N_R\subset N_G$. Since $0<u<2s$ and $s,v>0$, the entries of w' are positive. Since $Nv\geq 2s$,

$$\mathrm{wt}_{w'}(xy+z^2+t^N)=\frac{\min\{2s,Nv\}}{r}=\frac{2s}{r}.$$

As (W_N,D) is klt near 0, W_N is normal and klt near 0. By Lemma 4.7 with $n=4$, and since $\mathrm{wt}_{w'}(h_j)\geq 0$ for any j,

$$\alpha\leq\frac{u+(2s-u)+s+v}{r}-\frac{2s}{r}-\sum_j c_j\,\mathrm{wt}_{w'}(h_j)\leq\frac{s+v}{r}.$$

Thus $\alpha\to 0$ as $r\to+\infty$, which contradicts the assumption that the positive values α are strictly increasing. □

APPENDIX B. PROOFS FOR SOURCES OF QUADRATIC RANK TWO

In this section, we prove the results of Section 8. We begin with the circle $(1,-1,0,0)$, on which the quotient values are bounded from above and from below by quotient values on the plane $\mathbb{A}^2_{z,t}$. We then construct small resolutions of split sources, and treat the circle $(1,0,0,0)$, the families of scalar circles $(-A,-A,-B,v)$, the axis circles $(1,1,2,0)$ and $(A,B,A+B,0)$, and the circles with a zero hyperbolic weight. Finally we treat two-dimensional limits, inertia groups exchanging x and y, the splitting of the binary part, augmentations from lc toric quotients, and exchanging groups on circles with radical weights of opposite signs.

B.1. **The circle** $(1,-1,0,0)$**.** In this subsection, we prove Theorem 8.3.

Proof of Theorem 8.3. The proof is long, so we divide it into several steps.

Step 1. In this step we normalize the marks and bound their orders at 0.

Let $T := \mathbb{A}^2_{z,t}$. Let $\gamma := \min(\Gamma\backslash\{0\})$ if $\Gamma \not\subset \{0\}$, which exists as Γ is a DCC set, and let $\gamma := 1$ otherwise. Let $N_0 := \lceil 2/\gamma \rceil + 1$ and $s_0 := \lfloor 2/\gamma \rfloor + 1$. In **Steps 1–4** we fix i and drop it from the notation: $W = (xy - g = 0)$, $R = \mu_r(1,-1,0,0)$, $H := G/R$, and $\alpha := \mathrm{mld}((W \ni 0, D)/G)$.

Write $D = \sum_j c_j \operatorname{div}(h_j)$ with marks h_j and labels $c_j \in \Gamma$ (Definition 2.7). Since α only depends on the germ of (W, D) at 0, we may remove the marks with $c_j = 0$. Write a mark h_j as p/q with p and q in the coordinate ring of W and $q(0) \neq 0$. The function $\prod_{\sigma\in G} \sigma^* q$ is G-invariant and a unit near 0, and $h_j \prod_{\sigma\in G} \sigma^* q$ is a regular function on W with the character of h_j. Removing the marks that are units near 0, we may assume that each h_j is a G-semi-invariant element of the coordinate ring of W with $h_j(0) = 0$, and that $c_j \in \Gamma\backslash\{0\}$. Since G acts diagonally on $\mathbb{A}^4$, the sum of the monomials of a lift of h_j to $\mathbb{C}[x,y,z,t]$ with the character of h_j is again a lift of h_j. Since $xy - g$ is G-semi-invariant, replacing xy with g in a monomial does not change the character. Thus we may assume that each h_j is a G-semi-invariant polynomial of the form

$$h_j = A_{j,0} + \sum_{k\geq 1} x^k A_{j,k} + \sum_{l\geq 1} y^l A_{j,-l}, \qquad A_{j,k} \in \mathbb{C}[z,t].$$

The polynomials $A_{j,k}$ are determined by $h_j|_W$, as $1, x, x^2, \ldots, y, y^2, \ldots$ form a basis of $\mathbb{C}[x,y,z,t]/(xy-g)$ over $\mathbb{C}[z,t]$. For $k \geq 0$ (resp. $k < 0$), *the term of index* k of h_j is $x^k A_{j,k}$ (resp. $y^{-k} A_{j,k}$), and we write $\mathrm{mult}_0 h_j$ for the order of the polynomial h_j at 0.

Let E be an irreducible component of the exceptional divisor of the blow-up of W at 0. If $g \in (z,t)^3$, there are two such components, over $(x = 0)$ and over $(y = 0)$ in the tangent cone $(xy = 0)$, and otherwise there is one. Since W is a hypersurface of multiplicity two at 0 with reduced tangent cone, the adjunction formula on the blow-up of $\mathbb{A}^4$ at 0 yields $a(E, W, 0) = 2$, and $\mathrm{mult}_E h_j \geq \mathrm{mult}_0 h_j$ for any j. As (W, D) is klt near 0, $a(E, W, D) > 0$, and by the formula at the end of [Liu26, Lemma 2.6],

$$0 < \alpha \leq \frac{a(E,W,D)}{|I_G(E)|} \leq 2 - \sum_j c_j \,\mathrm{mult}_0 h_j. \tag{B.1}$$

Since $c_j \geq \gamma$ and $\mathrm{mult}_0 h_j \geq 1$ for any j, there are at most $s_0 - 1$ marks, and $\mathrm{mult}_0 h_j < 2/\gamma$ for any j. In particular, each h_j has a non-zero term of index k with $|k| < 2/\gamma < N_0$.

Let ϵ be a primitive r-th root of unity. The element $(\epsilon, \epsilon^{-1}, 1, 1)$ of R multiplies the term of index k by ϵ^k, so the indices of the non-zero terms of h_j are congruent modulo r. Assume that $r > 2N_0$. Then there exists exactly one index κ_j with $|\kappa_j| < N_0$ and $A_{j,\kappa_j} \neq 0$, and any other index k with $A_{j,k} \neq 0$ satisfies $|k| \geq r - |\kappa_j| > r - N_0$. Let $a_j := \max\{\kappa_j, 0\}$, $b_j := \max\{-\kappa_j, 0\}$, $F_j := A_{j,\kappa_j}$, and $h_j^0 := x^{a_j} y^{b_j} F_j$. As h_j has no monomial divisible by xy and a monomial of least degree of h_j has degree $\mathrm{mult}_0 h_j < 2/\gamma$, we have $\kappa_j = \kappa(h_j)$, and h_j^0 is the principal block of h_j along $\mathbf{T}_{(1,-1,0,0)}$ in the sense of Set-up 5.1. Then h_j^0 is a G-semi-invariant polynomial with the character of h_j and with non-zero restriction to W, and

$$h_j - h_j^0 \in (x,y)^{r-N_0}. \tag{B.2}$$

Hence the hypotheses of Proposition 3.14 on the marks hold for the polynomials h_j and the integers a_j and b_j, with $C := N_0$. Let $c_x := \sum_j c_j a_j$, $c_y := \sum_j c_j b_j$, $c_g := \max\{c_x, c_y\}$,

$$B_T := \sum_j c_j \operatorname{div}_T(F_j) + c_g \operatorname{div}_T(g), \qquad b_H := \mathrm{mld}((T \ni 0, B_T)/H),$$

and let $D^0 := \sum_j c_j \operatorname{div}_W(h_j^0)$.

Let $\sigma \in G$ act trivially on T. Since σ preserves $xy - g$ up to a character and $\sigma^* g = g \neq 0$, we have $\sigma^*(xy) = xy$, so $\sigma \in G \cap \mathbf{T}_{(1,-1,0,0)} = R$. Thus H acts faithfully on T.

Step 2. In this step we show that the log canonicity of pairs on W near 0 is determined by a fixed jet of the marks, uniformly in g.

Claim B.1. *Let $\Lambda \subset \mathbb{R}_{>0}$ be a DCC set with $1 \in \Lambda$. Then there exists a positive integer L_Λ, depending only on Λ and Γ, with the following property. Let $s \le s_0$, let $f_1, \dots, f_s, f_1', \dots, f_s'$ be polynomials on $\mathbb{A}^4$ with non-zero restrictions to W and $f_j - f_j' \in (x,y,z,t)^{L_\Lambda}$ for any j, and let $c_1', \dots, c_s' \in \Lambda$. Then $(W, \sum_j c_j' \operatorname{div}_W(f_j))$ is lc near 0 if and only if $(W, \sum_j c_j' \operatorname{div}_W(f_j'))$ is lc near 0.*

Proof. Let L_Λ be the maximum of the integers of Lemma 4.14 for $n = 4$, the set Λ, and s' factors, $1 \le s' \le s_0 + 1$. Since W is a normal Cartier hypersurface in $\mathbb{A}^4$, the different on W of an $\mathbb{R}$-divisor on $\mathbb{A}^4$ without W as a component is the restriction. Hence by [Kaw07, Theorem], $(W, \sum_j c_j' \operatorname{div}_W(f_j))$ is lc near 0 if and only if $(\mathbb{A}^4, W + \sum_j c_j' \operatorname{div}(f_j))$ is lc near 0, and by [dFEM11, Proposition 2.11], $(\mathbb{A}^4, W + \sum_j c_j' \operatorname{div}(f_j))$ is lc near 0 if and only if $(\operatorname{Spec} \mathbb{C}[[x,y,z,t]], (xy-g)\prod_j (f_j)^{c_j'})$ is lc. Both equivalences also hold with f_j' in place of f_j for any j. Lemma 4.14, for the tuples $(xy-g, f_1, \dots, f_s)$ and $(xy-g, f_1', \dots, f_s')$ with the exponents $(1, c_1', \dots, c_s')$, implies the claim in both directions. The integer L_Λ does not depend on g, hence not on i. □

Proof of Theorem 8.3 continued.

Step 3. In this step we show that (T, B_T) is lc near 0 and bound α from above.

By [Liu26, Lemma 18.2(2)] and [Liu26, Lemma 18.2(1)], applied after replacing y with $-y$, the following hold, respectively.

(i) For any prime divisor P over W with center 0, there exist an integer $\varrho \ge 1$ and a prime divisor ν over T with center 0 such that $\operatorname{mult}_P f = \varrho \operatorname{mult}_\nu f$ for any rational function f on T, and $a(P, W, 0) \ge \varrho a(\nu, T, 0)$.

(ii) For any prime divisor ν over T with center 0 and any coprime positive integers ϱ and ℓ with $\ell < \varrho \operatorname{mult}_\nu g$, the Gauss extension of $\varrho \operatorname{mult}_\nu$ with x-value ℓ is mult_P for a prime divisor P over W with center 0, and $\operatorname{mult}_P y = \varrho \operatorname{mult}_\nu g - \ell$, $\operatorname{mult}_P f = \varrho \operatorname{mult}_\nu f$ for any rational function f on T, and $a(P, W, 0) = \varrho a(\nu, T, 0)$.

In (i) and (ii), $\operatorname{mult}_P x + \operatorname{mult}_P y = \operatorname{mult}_P g = \varrho \operatorname{mult}_\nu g$ as $xy = g$ on W, and $\operatorname{mult}_\nu g \ge 2$ as $g \in (z,t)^2$.

Claim B.2. *Let $\theta > 0$ and $\lambda \ge 0$ be real numbers, and let $\zeta \in \{z, t\}$. Then $(W, \theta D^0 + \lambda \operatorname{div}_W(\zeta))$ is lc near 0 if and only if $(T, \theta B_T + \lambda \operatorname{div}_T(\zeta))$ is lc near 0.*

Proof. Let $B_W := \theta D^0 + \lambda \operatorname{div}_W(\zeta)$ and $B_T' := \theta B_T + \lambda \operatorname{div}_T(\zeta)$. If P is a prime divisor over W and ν a prime divisor over T with $\operatorname{mult}_P f = \varrho \operatorname{mult}_\nu f$ for any rational function f on T, then

$$a(P, W, B_W) = a(P, W, 0) - \theta c_x \operatorname{mult}_P x - \theta c_y \operatorname{mult}_P y - \varrho\Big(\theta \sum_j c_j \operatorname{mult}_\nu F_j + \lambda \operatorname{mult}_\nu \zeta\Big).$$

Assume that (T, B_T') is lc near 0. Let P be a prime divisor over W with center 0, and let ϱ and ν be as in (i). Since $c_g = \max\{c_x, c_y\}$ and $\operatorname{mult}_P x + \operatorname{mult}_P y = \varrho \operatorname{mult}_\nu g$,

$$a(P, W, B_W) \ge \varrho a(\nu, T, 0) - \theta c_g \varrho \operatorname{mult}_\nu g - \varrho\Big(\theta \sum_j c_j \operatorname{mult}_\nu F_j + \lambda \operatorname{mult}_\nu \zeta\Big) = \varrho a(\nu, T, B_T') \ge 0.$$

Suppose that (W, B_W) is not lc near 0. Let $f : Y \to W$ be a log resolution of (W, B_W) such that $f^{-1}(0)$ is a divisor and $f^{-1}(0) \cup \operatorname{Exc}(f) \cup f_*^{-1} B_W$ is a simple normal crossing divisor. Then there exists a prime divisor E on Y with $a(E, W, B_W) < 0$ and $0 \in f(E)$ [KM98, Lemma 2.30 and Corollary 2.31(3)]. Since $a(P, W, B_W) \ge 0$ for any prime divisor P over W with center 0, E is not a component of $f^{-1}(0)$, so there exists a component F of $f^{-1}(0)$ with $E \cap F \ne \emptyset$. Blowing up a component of the curve $E \cap F$, and then successively the intersection of the new exceptional divisor with the strict transform of E, we obtain prime divisors G_k over W with center 0 and,

by [KM98, Lemma 2.29], as each blown-up curve is contained in exactly two components of the simple normal crossing divisor,

$$a(G_k, W, B_W) = a(F, W, B_W) + ka(E, W, B_W),$$

which is negative for k large, although G_k has center 0. Thus (W, B_W) is lc near 0.

Conversely, assume that (W, B_W) is lc near 0. Let ν be a prime divisor over T with center 0 and $m := \mathrm{mult}_\nu g \geq 2$. For an integer $\varrho \geq 1$, let $\ell := \varrho m - 1$ if $c_x \geq c_y$, and $\ell := 1$ otherwise. Then ℓ and ϱ are coprime and $1 \leq \ell < \varrho m$. Let P be the prime divisor of (ii). By (ii), $a(P, W, 0) = \varrho a(\nu, T, 0)$ and $(\mathrm{mult}_P x, \mathrm{mult}_P y) = (\ell, \varrho m - \ell)$, which is $(\varrho m - 1, 1)$ if $c_x \geq c_y$ and $(1, \varrho m - 1)$ otherwise. Substituting into the formula for $a(P, W, B_W)$ above and using $c_g = \max\{c_x, c_y\}$, we obtain

$$a(P, W, B_W) = \varrho a(\nu, T, B'_T) + \theta |c_x - c_y|,$$

which is non-negative since (W, B_W) is lc near 0 and P has center 0. Letting $\varrho \to +\infty$, we obtain $a(\nu, T, B'_T) \geq 0$. A prime divisor Q over T whose center is a curve through 0 and with $a(Q, T, B'_T) < 0$ would yield, after blowing up points, a prime divisor Q' over T with center 0 and $a(Q', T, B'_T) < 0$. Thus (T, B'_T) is lc near 0. □

Proof of Theorem 8.3 continued. Let $\Lambda_0 := \{1\} \cup (\Gamma \backslash \{0\})$, and assume that $r > 2N_0$ and $r - N_0 \geq L_{\Lambda_0}$. By (B.2), $h_j - h_j^0 \in (x, y, z, t)^{L_{\Lambda_0}}$ for any j. As (W, D) is klt near 0 and there are at most $s_0 - 1$ marks, Claim B.1 implies that (W, D^0) is lc near 0, and Claim B.2 with $\theta = 1$ and $\lambda = 0$ implies that (T, B_T) is lc near 0. Thus the hypotheses of Proposition 3.14 hold for $X = W$ and $B = D$, and there exists r_0, depending only on e_0, Γ, and the integer C of Proposition 3.14, which is $N_0 = \lceil 2/\gamma \rceil + 1$, such that if $r \geq r_0$, then

$$\alpha \leq b_H + \frac{2}{\sqrt{r}}. \tag{B.3}$$

As N_0 depends only on Γ, the integer r_0 depends only on e_0 and Γ. Let E_T be the exceptional curve of the blow-up of T at 0. Since (T, B_T) is lc near 0, $a(E_T, T, B_T) \geq 0$, and by the formula at the end of [Liu26, Lemma 2.6],

$$c_g \,\mathrm{mult}_0 g + \sum_j c_j \,\mathrm{mult}_0 F_j \leq 2, \qquad b_H \leq \frac{a(E_T, T, B_T)}{|I_H(E_T)|} \leq 2. \tag{B.4}$$

Step 4. In this step we bound α from below by b_H when (T, B_T) has a margin.

Claim B.3. *Let $\theta > 1$. Then there exists a positive integer r_1, depending only on θ, e_0, and Γ, such that if $r \geq r_1$ and $(T, \theta B_T)$ is lc near 0, then $\alpha \geq b_H$.*

Proof. Let $\Lambda_\theta := \{1\} \cup \theta(\Gamma \backslash \{0\})$ and $M := 2/(\gamma(\theta - 1))$, and let r_1 be a positive integer with $r_1 > 2N_0$, $r_1 - N_0 \geq L_{\Lambda_\theta}$, and $(r_1 - N_0)/e_0 > M$. Assume that $r \geq r_1$ and that $(T, \theta B_T)$ is lc near 0. Then (T, B_T) is lc near 0. By Claim B.2, $(W, \theta D^0)$ is lc near 0, and by (B.2) and Claim B.1 for Λ_θ, $(W, \theta D)$ is lc near 0.

By the formula at the end of [Liu26, Lemma 2.6], α is the infimum of $a(P, W, D)/|I_G(P)|$ over the prime divisors P over W with center 0. Let P be such a prime divisor and $e := |I_G(P)|$. We show that $a(P, W, D) \geq eb_H$. By (B.4), we may assume that $a(P, W, D) \leq 2e$. Since $(W, \theta D)$ is lc near 0,

$$0 \leq a(P, W, \theta D) = a(P, W, D) - (\theta - 1) \sum_j c_j \,\mathrm{mult}_P h_j.$$

As $c_j \geq \gamma$ for any j,

$$\mathrm{mult}_P h_j \leq Me \quad \text{for any } j. \tag{B.5}$$

By (i) in **Step 3**, there exist an integer $\varrho \geq 1$ and a prime divisor ν over T with center 0 such that $\mathrm{mult}_P f = \varrho \,\mathrm{mult}_\nu f$ for any rational function f on T. Let $e_\nu := |I_H(\nu)| \leq e_0$. Let f be a rational function on T/H whose order along the prime divisor over T/H induced by ν is one.

By [Liu26, Lemma 2.6(1)], the pullback of f to T has order e_ν along ν, so the pullback of f to W has order ϱe_ν along P. The pullback of f to W is G-invariant, so by [Liu26, Lemma 2.6(1)] the integer e divides ϱe_ν. Thus $\varrho/e \geq 1/e_\nu \geq 1/e_0$, and

$$\frac{\operatorname{mult}_P x + \operatorname{mult}_P y}{e} = \frac{\varrho \operatorname{mult}_\nu g}{e} \geq \frac{2}{e_0}. \tag{B.6}$$

The automorphism $(x, y, z, t) \to (y, x, z, t)$ of $\mathbb{A}^4$ preserves W and R, conjugates G to a finite diagonal group whose image in the automorphism group of T is H, and carries D, D^0, and P to the corresponding objects for the marks $h_j(y, x, z, t)$. It interchanges a_j with b_j, c_x with c_y, and $\operatorname{mult}_P x$ with $\operatorname{mult}_P y$, and it changes none of the F_j, B_T, b_H, e, $a(P, W, D)$, and the hypotheses of the claim. Thus we may assume that $\operatorname{mult}_P x \leq \operatorname{mult}_P y$, and then $\operatorname{mult}_P y \geq e/e_0$ by (B.6).

Let h_j^+ be the sum of the terms of h_j of index at least κ_j. Since the indices of the non-zero terms of h_j are congruent to κ_j modulo r, $|\kappa_j| < N_0$, and $r > 2N_0$, any other non-zero term of h_j has index $k \leq \kappa_j - r < 0$ with $|k| > r - N_0$, so it is $y^l A_{j,-l}$ with $l > r - N_0$, and the order of $y^l A_{j,-l}$ along P is at least $(r - N_0) \operatorname{mult}_P y \geq (r - N_0)e/e_0 > Me$. By (B.5),

$$\operatorname{mult}_P h_j^+ = \operatorname{mult}_P h_j \quad \text{for any } j. \tag{B.7}$$

Let $\xi := x^r$, and let $\phi : W \to \mathbb{A}^3_{\xi,z,t}$ be the morphism $(x, y, z, t) \to (x^r, z, t)$. The indices of the non-zero terms of h_j^+ are $\kappa_j + rl$ with $l \geq 0$, and $\kappa_j + rl > 0$ for $l \geq 1$. If $\kappa_j < 0$, then $y^{b_j} F_j = x^{-b_j} g^{b_j} F_j$ on W. Hence there exist polynomials F_j^+ in ξ, z, t with

$$h_j^+ = x^{\kappa_j} \phi^* F_j^+ \text{ on } W, \qquad F_j^+(0, z, t) = g^{b_j} F_j. \tag{B.8}$$

For $\sigma \in G$ with diagonal entries $\sigma_x, \sigma_y, \sigma_z, \sigma_t$, let the image of σ in H act on $\mathbb{A}^3_{\xi,z,t}$ by $(\xi, z, t) \to (\sigma_x^r \xi, \sigma_z z, \sigma_t t)$. The action is well defined as R acts trivially on x^r, z, and t, and ϕ is equivariant. Since H acts faithfully on T, it acts diagonally and faithfully on $\mathbb{A}^3_{\xi,z,t}$ and on $S_0 := (\xi = 0) = T$. The polynomials ξ, g, and F_j^+ are H-semi-invariant, as $\phi^* F_j^+ = x^{-\kappa_j} h_j^+$ is G-semi-invariant.

Over $\mathbb{A}^3 \backslash (\xi = 0)$, the morphism ϕ is the quotient of $W \backslash (x = 0) \cong (\mathbb{A}^1 \backslash \{0\}) \times T$ by R. Hence ϕ induces birational maps $W/R \dashrightarrow \mathbb{A}^3$ and $W/G \dashrightarrow \mathbb{A}^3/H$. Let P' be the prime divisor over $\mathbb{A}^3$ induced by P, and e_R the positive integer with $\operatorname{mult}_P(\phi^* f) = e_R \operatorname{mult}_{P'} f$ for any rational function f on $\mathbb{A}^3$. Then $\operatorname{center}_{\mathbb{A}^3} P' = \phi(0) = 0$. Let f be a rational function on W/G whose order along the prime divisor over W/G induced by P is one. By [Liu26, Lemma 2.6(1)] for G on W and for H on $\mathbb{A}^3$, the pullbacks of f have order e along P and order $|I_H(P')|$ along P'. Thus $e = e_R |I_H(P')|$.

Up to sign, $x^{-1} dx \wedge dz \wedge dt$ is the residue of $(xy - g)^{-1} dx \wedge dy \wedge dz \wedge dt$ along W, so it generates ω_W near 0, and

$$\phi^*(\xi^{-1} d\xi \wedge dz \wedge dt) = r x^{-1} dx \wedge dz \wedge dt.$$

Thus $K_W = \phi^*(K_{\mathbb{A}^3} + S_0)$. Since ϕ is generically finite of degree r, the Hurwitz formula [Kol13, 2.41], applied at the generic point of P to the normalization, in the function field of W, of a model of $\mathbb{A}^3$ on which P' is a prime divisor, gives $a(P, W, 0) = e_R a(P', \mathbb{A}^3, S_0)$. Since $\operatorname{mult}_P x = e_R \operatorname{mult}_{P'} \xi / r$, $\operatorname{mult}_P(\phi^* F_j^+) = e_R \operatorname{mult}_{P'} F_j^+$, and $\sum_j c_j \kappa_j = c_x - c_y$, it follows from (B.7) and (B.8) that

$$a(P, W, D) = e_R a\Big(P', \mathbb{A}^3, \Big(1 + \frac{c_x - c_y}{r}\Big) S_0 + \sum_j c_j \operatorname{div}(F_j^+)\Big). \tag{B.9}$$

Let

$$\mathfrak{a} := \prod_j (F_j^+)^{c_j} \cdot (g)^{\max\{c_x - c_y, 0\}},$$

where the last factor is omitted if $c_x \le c_y$. Since $c_y + \max\{c_x - c_y, 0\} = c_g$, by (B.8) the restriction $\mathfrak{a}|_{S_0}$ corresponds to

$$\sum_j c_j \operatorname{div}_T(g^{b_j} F_j) + \max\{c_x - c_y, 0\} \operatorname{div}_T(g) = B_T,$$

and (T, B_T) is lc near 0. Lemma 5.5, for $X = \mathbb{A}^3_{\xi,z,t}$, $n = 3$, the group H in place of both groups, $u = \xi$, the $\mathbb{R}$-ideal $\mathfrak{a}$, $\beta_0 = b_H$, and $b = 1$, yields

$$\frac{a(P', \mathbb{A}^3, S_0, \mathfrak{a})}{|I_H(P')|} \ge b_H. \tag{B.10}$$

The difference

$$a\Big(P', \mathbb{A}^3, \Big(1 + \frac{c_x - c_y}{r}\Big) S_0 + \sum_j c_j \operatorname{div}(F_j^+)\Big) - a(P', \mathbb{A}^3, S_0, \mathfrak{a})$$

equals

$$-\frac{c_x - c_y}{r} \operatorname{mult}_{P'} \xi + \max\{c_x - c_y, 0\} \operatorname{mult}_{P'} g.$$

If $c_x \le c_y$, it is non-negative. If $c_x > c_y$, then since $\operatorname{mult}_{P'} g = (\operatorname{mult}_P x + \operatorname{mult}_P y)/e_R$ and $\operatorname{mult}_{P'} \xi / r = \operatorname{mult}_P x / e_R$, it is equal to $(c_x - c_y) \operatorname{mult}_P y / e_R \ge 0$. Hence by (B.9), (B.10), and $e = e_R |I_H(P')|$, we obtain $a(P, W, D)/e \ge b_H$. Thus $\alpha \ge b_H$. □

Proof of Theorem 8.3 continued.

Step 5. In this step we assume that the theorem fails, and pass to subsequences on which the plane data have fixed groups, characters, and limits of labels.

From now on we keep the index i: we write $W_i = (xy - g_i = 0)$, $D_i = \sum_j c_{i,j} \operatorname{div}(h_{i,j})$, and $\kappa_{i,j}$, $F_{i,j}$, $h^0_{i,j}$, D^0_i, $c_{x,i}$, $c_{y,i}$, $c_{g,i}$, $B_{T,i}$, $H_i := G_i/R_i$, and $b_{H,i}$ for the objects of **Steps 1–4**, and $\alpha_i := \operatorname{mld}((W_i \ni 0, D_i)/G_i)$. Suppose that the α_i form a strictly increasing sequence. Since $r_i \to +\infty$, removing finitely many i, we may assume that $r_i > 2N_0$, $r_i - N_0 \ge L_{\Lambda_0}$, and $r_i \ge r_0$ for any i. By (B.1) and (B.3), we have $0 < \alpha_i \le 2$ and $\alpha_i \le b_{H,i} + 2/\sqrt{r_i}$ for any i. Let $\alpha_\infty := \lim_i \alpha_i \in (0, 2]$.

The group H_i acts diagonally and faithfully on T, and $|H_i| \le e_0$, so $H_i \subset \mu_{e_0!} \times \mu_{e_0!}$. The polynomials $F_{i,j}$ and g_i are H_i-semi-invariant. Possibly passing to a subsequence, we may assume that $H_i = H$ for a fixed group H, that the number s of marks and the integers $\kappa_{i,j} = \kappa_j$ do not depend on i (recall that $|\kappa_{i,j}| < N_0$), that the characters of $F_{i,j}$ and of g_i under H do not depend on i, and, as Γ is a DCC set, that $c_{i,j}$ is non-decreasing in i with limit $c^*_j \in (0, 1]$ for any j. Then a_j and b_j do not depend on i, and $c_{x,i}$, $c_{y,i}$, and $c_{g,i}$ are non-decreasing in i. Let $c^*_g := \lim_i c_{g,i}$. If $c^*_g = 0$, then $c_{g,i} = 0$ for any i. If $c^*_g > 0$, removing finitely many i, we may assume that $c_{g,i} > 0$ for any i.

The coefficient of $B_{T,i}$ along any prime divisor on T through 0 is a finite sum of elements of Γ, as $c_{g,i} = \sum_j c_{i,j} a_j$ or $c_{g,i} = \sum_j c_{i,j} b_j$, and is at most one, as $(T, B_{T,i})$ is lc near 0 (**Step 3**). By [Liu26, Lemma 2.6(4)], the crepant quotient of $(T, B_{T,i})$ by H has coefficients in $\Psi(\Gamma_+ \cap [0, 1])$ near the image of 0, and $\Psi(\Gamma_+ \cap [0, 1]) \subset [0, 1]$ is a DCC set by [Liu26, Lemma 2.10]. As $b_{H,i}$ is the minimal log discrepancy at the image of 0 of the crepant quotient, which is an lc surface pair, Theorem 2.9(1) shows that the $b_{H,i}$ belong to an ACC set. Possibly passing to a subsequence, we may assume that $b_{H,i}$ is non-increasing in i, and we let $\beta := \lim_i b_{H,i}$. As $\alpha_i \le b_{H,i} + 2/\sqrt{r_i}$,

$$\alpha_i < \alpha_\infty \le \beta \le b_{H,i} \quad \text{for any } i. \tag{B.11}$$

Let

$$C_i := \sum_j c^*_j \operatorname{div}_T(F_{i,j}) + c^*_g \operatorname{div}_T(g_i), \qquad \beta_i := \operatorname{mld}((T \ni 0, C_i)/H),$$

where the last term of C_i is omitted if $c^*_g = 0$. If $s = 0$, then $c^*_g = 0$ and $C_i = B_{T,i} = 0$, so $(T, \theta B_{T,i})$ is lc near 0 for any $\theta > 1$, and Claim B.3 gives $\alpha_i \ge b_{H,i}$ for i large, which contradicts (B.11). Thus we may assume $s \ge 1$. Let ς_i be the minimum of the ratios $c_{i,j}/c^*_j$ and, if $c^*_g > 0$,

of $c_{g,i}/c_g^*$. Then $\varsigma_i \le 1$, $\varsigma_i \to 1$, and $\varsigma_i C_i \le B_{T,i}$, so $(T, \varsigma_i C_i)$ is lc near 0. The lc thresholds at 0 of the $\mathbb{R}$-ideals $\prod_j (F_{i,j})^{c_j^*} \cdot (g_i)^{c_g^*}$ on T have exponents in a fixed finite set, so they satisfy the ACC by [dFEM11, Theorem 1.1]. As the thresholds are at least $\varsigma_i \to 1$, they are at least one for i large. Removing finitely many i, we may assume that (T, C_i) is lc near 0 for any i.

The coefficients of C_i along the prime divisors on T through 0 belong to the finite set Σ^* of finite sums of elements of $\{c_1^*, \dots, c_s^*, c_g^*\} \backslash \{0\}$ that are at most one. By [Liu26, Lemma 2.6(4)], the coefficients of the crepant quotient of (T, C_i) by H belong to the finite set

$$\Sigma' := \Big\{ 1 - \frac{1-c}{e} \;\Big|\; c \in \Sigma^* \cup \{0\},\ 1 \le e \le e_0 \Big\}.$$

Let L_1 be the integer of Theorem 2.9(2) for the finite set Σ'. Then for any i, there exists a prime divisor over T/H with center the image of 0, computing the minimal log discrepancy at the image of 0 of the crepant quotient of (T, C_i) by H, with log discrepancy at most L_1 with respect to $(T/H, 0)$. Let ν_i be a prime divisor over T inducing the prime divisor over T/H of Theorem 2.9(2) ([Liu26, Lemma 2.6(3)]), and $e_i := |I_H(\nu_i)|$. By [Liu26, Lemma 2.6(2)], $a(\nu_i, T, C_i) = e_i \beta_i$, and $a(\nu_i, T, 0) \le e_i L_1$, as the crepant quotient of $(T, 0)$ by H is a pair $(T/H, \Delta)$ with $\Delta \ge 0$. Since $a(\nu_i, T, C_i) \ge 0$, we have $c_j^* \operatorname{mult}_{\nu_i} F_{i,j} \le e_i L_1$ for any j and $c_g^* \operatorname{mult}_{\nu_i} g_i \le e_i L_1$. As $B_{T,i} \le C_i$ and $b_{H,i} \le a(\nu_i, T, B_{T,i})/e_i$ by the formula at the end of [Liu26, Lemma 2.6],

$$0 \le b_{H,i} - \beta_i \le \frac{a(\nu_i, T, B_{T,i}) - a(\nu_i, T, C_i)}{e_i} \le L_1 \Big(\sum_j \frac{c_j^* - c_{i,j}}{c_j^*} + \frac{c_g^* - c_{g,i}}{c_g^*} \Big), \tag{B.12}$$

where the last term is omitted if $c_g^* = 0$. Thus $\lim_i \beta_i = \beta$.

Step 6. In this step we take a compatible generic limit of the plane data and compute the quotient value of the limit.

By (B.4) for $B_{T,i}$, we have $\operatorname{mult}_0 F_{i,j} \le 2/\gamma$ for any i and j, and, if $c_g^* > 0$, $\operatorname{mult}_0 g_i \le 2/c_{g,1}$ for any i. Thus the hypotheses of Construction-Proposition 4.1 in case (a) hold for $n = 2$, the plane T, the group H, and the tuples $(F_{i,1}, \dots, F_{i,s}, g_i)$, where g_i is omitted if $c_g^* = 0$. Possibly passing to a subsequence, we obtain an algebraically closed field $k \supset \mathbb{C}$ and a compatible generic limit $(F_{\infty,1}, \dots, F_{\infty,s}, g_\infty)$ of the tuples $(F_{i,1}, \dots, F_{i,s}, g_i)$ in $k[[z,t]]$. Then C_i is the $\mathbb{R}$-divisor C_i of Theorem 4.2 for the labels $(c_1^*, \dots, c_s^*, c_g^*)$ and $\mathfrak{c} = \mathcal{O}$, and we let

$$C_\infty := \sum_j c_j^* \operatorname{div}(F_{\infty,j}) + c_g^* \operatorname{div}(g_\infty)$$

on $\operatorname{Spec} k[[z,t]]$, as there. Possibly passing to a subsequence, since (T, C_i) is lc near 0 for any i, Theorem 4.2(1) shows that $(\operatorname{Spec} k[[z,t]], C_\infty)$ is lc, and Theorem 4.2(2) shows that $\beta_i = \operatorname{mld}((\operatorname{Spec} k[[z,t]] \ni 0, C_\infty)/H)$ for any i. By (B.12),

$$\operatorname{mld}((\operatorname{Spec} k[[z,t]] \ni 0, C_\infty)/H) = \beta_i = \beta > 0 \quad \text{for any } i. \tag{B.13}$$

Step 7. In this step we treat the case where $(\operatorname{Spec} k[[z,t]], C_\infty)$ is klt.

By Theorem 4.2(1) with $\theta = 1$, there exists $\theta' > 1$ such that $(T, \theta' C_i)$ is lc near 0 for any i. As $B_{T,i} \le C_i$, the pair $(T, \theta' B_{T,i})$ is lc near 0 for any i. Let r_1 be the integer of Claim B.3 for θ'. By Claim B.3, $\alpha_i \ge b_{H,i}$ for any i with $r_i \ge r_1$, which contradicts (B.11) as $r_i \to +\infty$.

Step 8. In this step we treat the case where $(\operatorname{Spec} k[[z,t]], C_\infty)$ is not klt.

Since (T, C_i) is lc near 0 for any i and $\beta_i = \beta > 0$, Definition-Proposition 5.7 in case (a) with $n = 2$ and $\mathfrak{c} = \mathcal{O}$ shows that there exists exactly one floor Φ, that Φ is smooth at 0 and has coefficient one in C_∞, and that $\beta = \lambda/e$, where $\lambda := 1 - \operatorname{mult}_0((C_\infty - \Phi)|_\Phi)$ and e is the order of the image of H in the automorphism group of the tangent line $T_0\Phi$ (Lemma 5.6). As H preserves C_∞, it permutes the floors, so Φ is H-invariant. Let $u \in k[[z,t]]$ be a generator of the ideal of Φ. As Φ is smooth at 0, $u \notin (z,t)^2$. Let $\zeta := t$ if $\partial u/\partial z(0) \ne 0$, and $\zeta := z$ otherwise, and let ζ' be the other coordinate, so that $\{\zeta, \zeta'\} = \{z, t\}$. Then (u, ζ) is a regular system of parameters of $k[[z,t]]$, and $\zeta|_\Phi$ is a uniformizer of Φ at 0. Let χ_ζ be the character of ζ under H and m_ζ the

order of $\chi_\zeta(H)$. Since $\sigma^*\zeta = \chi_\zeta(\sigma)\zeta$ for $\sigma \in H$, an element of H acts trivially on $T_0\Phi$ if and only if it belongs to $\operatorname{Ker}\chi_\zeta$, so $e = m_\zeta$. As $\beta > 0$ and $C_\infty - \Phi \geq 0$,

$$\beta = \frac{\lambda}{m_\zeta}, \qquad 0 < \lambda \leq 1. \tag{B.14}$$

We show that, possibly passing to a subsequence, $(T, C_i + \lambda \operatorname{div}_T(\zeta))$ is lc near 0 for any i.

First assume that $\lambda < 1$. Since $\zeta|_\Phi$ is a uniformizer of Φ at 0, Φ is not a component of $C_\infty - \Phi + \lambda \operatorname{div}(\zeta)$ and

$$\operatorname{mult}_0\left((C_\infty - \Phi + \lambda \operatorname{div}(\zeta))|_\Phi\right) = (1-\lambda) + \lambda = 1.$$

By Lemma 2.14, $(\operatorname{Spec} k[[z,t]], C_\infty + \lambda \operatorname{div}(\zeta))$ is lc. As (ζ) is an H-invariant ideal and $(\mathbb{A}^2, (\zeta)^\lambda)$ is klt near 0, Theorem 4.2(1) with $\mathfrak{c} = (\zeta)^\lambda$ shows that, possibly passing to a subsequence, $(T, C_i + \lambda \operatorname{div}_T(\zeta))$ is lc near 0 for any i.

Next assume that $\lambda = 1$. Then $\operatorname{mult}_0((C_\infty - \Phi)|_\Phi) = 0$, and as any prime divisor on $\operatorname{Spec} k[[z,t]]$ contains the closed point, $C_\infty = \Phi$. As (u, ζ) is a regular system of parameters, $\operatorname{mult}_0(C_\infty|_{(\zeta=0)}) = 1$. Let m_j and m_g be the orders of $F_{\infty,j}|_{(\zeta=0)}$ and $g_\infty|_{(\zeta=0)}$ in $k[[\zeta']]$. Then

$$\sum_j c_j^* m_j + c_g^* m_g = 1.$$

Fix an integer $d > \max\{m_1, \dots, m_s, m_g\}$. Let $\mathcal{J}_d$ be the affine space of d-jets and $Z_d \subset \mathcal{J}_d$ the irreducible closed subset of Construction-Proposition 4.1 for the tuples $(F_{i,1}, \dots, F_{i,s}, g_i)$ and this d. For $l < d$, the coefficient of $(\zeta')^l$ in the restriction to $(\zeta = 0)$ of a member of a d-jet is a linear form on $\mathcal{J}_d$ defined over $\mathbb{C}$. Since the d-jet of $(F_{\infty,1}, \dots, F_{\infty,s}, g_\infty)$ maps to the generic point of Z_d, the forms for $l < m_j$ (resp. $l < m_g$) vanish on Z_d, and the form for $l = m_j$ (resp. $l = m_g$) is non-zero on a non-empty open subset of Z_d. By Construction-Proposition 4.1, Z_d and the open subset contain the d-jets of $(F_{i,1}, \dots, F_{i,s}, g_i)$ for all but finitely many i. Removing finitely many i, the restrictions $F_{i,j}|_{(\zeta=0)}$ and $g_i|_{(\zeta=0)}$ have orders m_j and m_g at 0 for any i, so $(\zeta = 0)$ is not a component of C_i and $\operatorname{mult}_0(C_i|_{(\zeta=0)}) = 1$. By [Kaw07, Theorem], $(T, C_i + \operatorname{div}_T(\zeta))$ is lc near 0 for any i.

In both cases, as $B_{T,i} \leq C_i$, the pair $(T, B_{T,i} + \lambda \operatorname{div}_T(\zeta))$ is lc near 0 for any i, and Claim B.2 with $\theta = 1$ implies that $(W_i, D_i^0 + \lambda \operatorname{div}_{W_i}(\zeta))$ is lc near 0. Let $\Lambda_\lambda := \{1, \lambda\} \cup (\Gamma \backslash \{0\})$. For any i with $r_i - N_0 \geq L_{\Lambda_\lambda}$, (B.2) and Claim B.1 for the tuples $(h_{i,1}, \dots, h_{i,s}, \zeta)$ and $(h_{i,1}^0, \dots, h_{i,s}^0, \zeta)$, which have at most s_0 members, imply that

$$(W_i, D_i + \lambda \operatorname{div}_{W_i}(\zeta)) \text{ is lc near } 0. \tag{B.15}$$

The function ζ is G_i-semi-invariant and vanishes at 0. As R_i acts trivially on ζ and the image of G_i in the automorphism group of T is H, the character of ζ under G_i has order m_ζ. By Lemma 5.8 for (W_i, D_i), the group G_i, $Z = \{0\}$, and $U = \zeta$, and by (B.15) and (B.14),

$$\alpha_i \geq \frac{\lambda}{m_\zeta} = \beta$$

for any i with $r_i - N_0 \geq L_{\Lambda_\lambda}$. As $r_i \to +\infty$, this contradicts (B.11). Thus both cases contradict (B.11), and the α_i do not form a strictly increasing sequence of positive numbers. □

B.2. **Small resolutions of split sources.** In this subsection, we prove Proposition 8.4.

Proof of Proposition 8.4. The proof is long, so we divide it into several steps. In both (1) and (2), the labelled boundary has the form $D = \sum_j c_j \operatorname{div}(h_j)$ of Definition 2.7, where $c_j \in \Gamma$ and the marks h_j are non-zero regular G-semi-invariant functions near the closed point.

Step 1. In this step, for a G-equivariant crepant model of $(W \ni o, D)$ and a subgroup $G' \subset G$, we show that the quotient value $\operatorname{mld}((W \ni o, D)/G')$ is the minimum of the minimal log discrepancies at the scheme-theoretic points of the fiber of the quotient, and we control the coefficients of the quotient boundary.

We consider the data of (1), and the data of (2) with $o = 0$. Let $f : Y \to W$ be a G-equivariant projective birational morphism from a normal variety Y such that $K_Y = f^*K_W$. Possibly shrinking W to a G-stable neighborhood of o on which all h_j are regular, and replacing Y by its inverse image, we may assume that (W, D) is klt and that the h_j are regular on W. Let

$$D_Y := \sum_j c_j \operatorname{div}(f^*h_j).$$

Any f^*h_j is a mark on Y, that is, a non-zero regular G-semi-invariant function, whose character is the character of h_j, so $D_Y \geq 0$ and $K_Y + D_Y = f^*(K_W + D)$. Thus (f, D_Y) is a crepant model of $(W \ni o, D)$ in the sense of Definition-Proposition 4.10, and $a(P, Y, D_Y) = a(P, W, D) > 0$ for any prime divisor P over Y. Since $D_Y = \sum_j c_j \operatorname{div}(f^*h_j)$ with $c_j \in \Gamma$, $D_Y \in \Gamma_+$, and since (Y, D_Y) is klt, $D_Y \in \Gamma_+ \cap [0, 1)$.

The group G, and hence any subgroup of G, acts faithfully on Y, since f is G-equivariant and birational and G acts faithfully on W. Let $G' \subset G$ be a subgroup, $Z := Y/G'$, $Q := W/G'$, $\pi_Y : Y \to Z$ and $\pi_W : W \to Q$ the quotient morphisms, $\bar{f} : Z \to Q$ the induced projective birational morphism, $\bar{o} := \pi_W(o)$, and $(Z, D_Z) := (Y, D_Y)/G'$ and $(Q, \Delta) := (W, D)/G'$ the crepant quotients [Liu26, Definition 2.5]. If P is a prime divisor on Y and e is the ramification index of π_Y along P, then

$$\operatorname{mult}_{\pi_Y(P)} D_Z = 1 - \frac{1 - \operatorname{mult}_P D_Y}{e}$$

by [Liu26, Lemma 2.6(4)]. Let $\Theta := \Psi(\Gamma_+ \cap [0, 1))$. Then $D_Z \in \Theta$, and Θ is a DCC set: by [Liu26, Lemma 2.10(1)], $\Gamma_+ \cap [0, 1]$ is a DCC set, and so is $\Psi(\Gamma_+ \cap [0, 1]) \supset \Theta$. In particular, $D_Z \geq 0$. Moreover,

$$\pi_Y^*(K_Z + D_Z) = K_Y + D_Y = f^*(K_W + D) = f^*\pi_W^*(K_Q + \Delta) = \pi_Y^*\bar{f}^*(K_Q + \Delta).$$

Since the pull-back of $\mathbb{R}$-divisors under the finite surjective morphism π_Y of normal varieties is injective,

$$K_Z + D_Z = \bar{f}^*(K_Q + \Delta). \tag{B.16}$$

By [Liu26, Lemma 2.6(2)(3)], (Z, D_Z) and (Q, Δ) are klt.

Let $\alpha := \operatorname{mld}((W \ni o, D)/G') = \operatorname{mld}(Q \ni \bar{o}, \Delta)$. Then $\alpha > 0$, since (Q, Δ) is klt. By [Liu26, Lemma 3.5], α is the minimum of the log discrepancies of the components of the inverse image of $\bar{o}$ on a log resolution of (Q, Δ) on which the ideal of $\bar{o}$ is invertible. In particular, there exists a prime divisor E over Q such that $\operatorname{center}_Q E = \bar{o}$ and $a(E, Q, \Delta) = \alpha$. Since $\bar{f}$ is proper, $\operatorname{center}_Z E$ exists, and we let η be the generic point of $\operatorname{center}_Z E$. Then $\eta \in \bar{f}^{-1}(\bar{o})$. Let η' be a scheme-theoretic point of $\bar{f}^{-1}(\bar{o})$ and P a prime divisor over Z such that $\operatorname{center}_Z P = \overline{\{\eta'\}}$. Since $\overline{\{\eta'\}} \subset \bar{f}^{-1}(\bar{o})$, we have $\operatorname{center}_Q P = \bar{o}$, and (B.16) implies $a(P, Z, D_Z) = a(P, Q, \Delta) \geq \alpha$. For $\eta' = \eta$ and $P = E$, equality holds. Thus

$$\alpha = \operatorname{mld}(Z \ni \eta, D_Z) = \min\left\{\operatorname{mld}(Z \ni \eta', D_Z) \;\middle|\; \eta' \in \bar{f}^{-1}(\bar{o})\right\}, \tag{B.17}$$

where η' runs over the scheme-theoretic points of $\bar{f}^{-1}(\bar{o})$, and E computes $\operatorname{mld}(Z \ni \eta, D_Z)$.

Step 2. In this step we show that, in the notation of **Step 1**, α belongs to one of three ACC sets $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$, depending only on Γ, unless η is a closed point and Y is singular at the points over η.

Let $\mathcal{A}_1 := \{1 - \theta \mid \theta \in \Theta\}$, which is an ACC set since Θ is a DCC set. Let $\mathcal{A}_2$ be the ACC set of [Liu26, Lemma 3.7] for the DCC set $\Theta \subset [0, 1)$. Let $\mathcal{A}_3$ be the set of positive numbers of Theorem 2.8(1) for the DCC set Γ, which satisfies the ACC.

Assume that $\overline{\{\eta\}}$ is a prime divisor P on Z. Since Z is normal, P is the only prime divisor over Z with center P. Thus $\alpha = a(P, Z, D_Z) = 1 - \operatorname{mult}_P D_Z \in \mathcal{A}_1$ by (B.17).

Assume that $\overline{\{\eta\}}$ is a curve. Since (Z, D_Z) is a threefold pair with $D_Z \in \Theta$, and $\alpha = \operatorname{mld}(Z \ni \eta, D_Z) > 0$ by (B.17), we have $\alpha \in \mathcal{A}_2$.

Assume that η is a closed point and that Y is smooth at a point $y \in Y$ over η. Let $H \subset G'$ be the stabilizer of y, $\rho : Y/H \to Z$ the induced finite morphism, and $\bar{y} \in Y/H$ the image of y. Since completion is flat over $\mathcal{O}_{Z,\eta}$, taking G'-invariants commutes with completion. Since G' permutes the points of the orbit $G'y$ transitively and H is the stabilizer of y,

$$\widehat{\mathcal{O}}_{Z,\eta} = \Big(\prod_{y' \in G'y} \widehat{\mathcal{O}}_{Y,y'} \Big)^{G'} = \big(\widehat{\mathcal{O}}_{Y,y}\big)^{H} = \widehat{\mathcal{O}}_{Y/H,\bar{y}},$$

where the first equality uses the decomposition of the completion of a semi-local ring [Mat89, Theorem 8.15], and the last equality is [Liu26, Lemma 3.3(1)]. Thus ρ induces an isomorphism of the completed local rings at $\bar{y}$ and η, and ρ is étale at $\bar{y}$ [Har77, Exercise III.10.4]. Let $(Y/H, D_{Y/H}) := (Y, D_Y)/H$. Over a neighborhood of $\bar{y}$ on which ρ is étale, $\rho^*(K_Z + D_Z) = K_{Y/H} + \rho^* D_Z$, and the pull-backs of $K_{Y/H} + \rho^* D_Z$ and of $K_{Y/H} + D_{Y/H}$ to Y are both $K_Y + D_Y$. Hence $\rho^* D_Z = D_{Y/H}$ near $\bar{y}$. The base change by ρ of a log resolution of (Z, D_Z) on which the ideal of η is invertible is, over a neighborhood of $\bar{y}$, a log resolution of $(Y/H, D_{Y/H})$ on which the ideal of $\bar{y}$ is invertible. The fiber over $\bar{y}$ is isomorphic to the fiber over η, since both residue fields are $\mathbb{C}$, and the log discrepancy with respect to $(Y/H, D_{Y/H})$ of a component E' over $\bar{y}$ equals the log discrepancy with respect to (Z, D_Z) of the image of E' over η. By [Liu26, Lemma 3.5] and (B.17),

$$\alpha = \mathrm{mld}(Y/H \ni \bar{y}, D_{Y/H}) = \mathrm{mld}((Y \ni y, D_Y)/H).$$

By **Step 1**, G acts faithfully on Y. An element of H acting trivially on a neighborhood of y acts trivially on Y, so H acts faithfully on the smooth threefold germ $(Y \ni y)$, and H is abelian and fixes y. The functions $f^* h_j$ are non-zero regular H-semi-invariant functions near y, $D_Y = \sum_j c_j \operatorname{div}(f^* h_j)$ with $c_j \in \Gamma$, and (Y, D_Y) is klt. Since Theorem 2.8(1) does not assume that H contains no pseudo-reflections, $\alpha \in \mathcal{A}_3$.

Step 3. In this step we prove (1).

Let $f : Y \to W$ be a G-equivariant projective crepant resolution. We apply **Steps 1 and 2** with $G' = G$. Since Y is smooth, η is the generic point of a prime divisor or of a curve, or η is a closed point over which Y is smooth. Thus

$$\mathrm{mld}((W \ni o, D)/G) \in \mathcal{A}_1 \cup \mathcal{A}_2 \cup \mathcal{A}_3.$$

Since $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$ satisfy the ACC and depend only on Γ, (1) holds.

Step 4. In this step we prove the factorization $g = F_1 \cdots F_k$ of (2) after a G-equivariant change of x by an invariant unit, and we shrink the (z,t)-plane.

Let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of x, y, z, t. Since G acts diagonally and preserves $xy - g(z,t)$ up to a character, comparing the terms of $\sigma^*(xy - g)$ for $\sigma \in G$ shows that g is G-semi-invariant with character $\chi_x \chi_y$. Any $\sigma \in G$ acts on $\mathbb{C}[z,t]$ by the linear substitution $(z,t) \to (\chi_z(\sigma) z, \chi_t(\sigma) t)$, which fixes 0. Hence σ^* maps any irreducible factor of g to a non-zero constant multiple of an irreducible factor of g, and preserves the factors vanishing at 0. Since g is reduced, the irreducible factors of g are pairwise non-associate. Thus, up to non-zero constants, G permutes the irreducible factors of g vanishing at 0. Let $F_1, \ldots, F_k$ be the products of the G-orbits of the irreducible factors of g vanishing at 0, one product for each orbit, with one representative for each element of the orbit. Since $0 \neq g \in (z,t)^2$, $k \geq 1$. For $\sigma \in G$, $\sigma^* F_i$ is a non-zero constant multiple of F_i, so F_i is a G-semi-invariant polynomial. Let χ_i be the character of F_i.

Let $u_0 := g/(F_1 \cdots F_k) \in \mathbb{C}[z,t]$. Then $u_0(0) \neq 0$, and u_0 is G-semi-invariant with a character χ. Taking the values at 0 of both sides of $\sigma^* u_0 = \chi(\sigma) u_0$ shows $\chi(\sigma) = 1$, so u_0 is G-invariant. On the G-stable open subset $(u_0 \neq 0) \subset \mathbb{A}^4$, the substitution of $u_0 x$ for x is a G-equivariant isomorphism, since u_0 is a G-invariant unit there, and it transforms $xy - g$ into $u_0(xy - F_1 \cdots F_k)$. The marks h_j become regular functions of the new coordinates, their characters do not change, and D does not change. Possibly replacing x with x/u_0, we may assume that $g = F_1 \cdots F_k$ near 0.

Distinct F_i have no common irreducible factor, and each F_i is reduced. Hence the set Σ of points $p \neq 0$ of $\mathbb{A}^2_{z,t}$ at which two of the F_i vanish or at which some curve $(F_i = 0)$ is singular is finite, and Σ is G-stable. Let ϵ_1 be a polynomial vanishing on Σ with $\epsilon_1(0) \neq 0$, and let $\epsilon := u_0 \prod_{\sigma \in G} \sigma^* \epsilon_1$. Then ϵ is G-invariant, and $\epsilon(0) \neq 0$ since any $\sigma \in G$ fixes 0. Let $S := (\epsilon \neq 0) \subset \mathbb{A}^2_{z,t}$, a G-stable affine open neighborhood of 0, with coordinate ring $\mathcal{O}(S) = \mathbb{C}[z,t][\epsilon^{-1}]$, a unique factorization domain. Possibly shrinking W, we may assume that

$$W = (xy = F_1 \cdots F_k) \subset \mathbb{A}^2_{x,y} \times S.$$

On S, any curve $(F_i = 0)$ and any curve $(\prod_{l \in L} F_l = 0)$ for a non-empty set L of indices are smooth outside 0, and $(F_i = \prod_{l \in L} F_l = 0) = \{0\}$ if $i \notin L$.

Step 5. In this step we construct one small modification of a hypersurface $(wy = ab)$ over S.

Let $a, b \in \mathcal{O}(S)$ be reduced G-semi-invariant functions with characters χ_a, χ_b such that $a(0) = b(0) = 0$, the curves $(a = 0)$ and $(b = 0)$ are smooth outside 0, and $(a = b = 0) = \{0\}$. Let

$$X := (wy = ab) \subset \mathbb{A}^2_{w,y} \times S,$$

where G acts diagonally on the coordinates w, y, z, t with characters $\chi_w, \chi_y, \chi_z, \chi_t$, and $\chi_w \chi_y = \chi_a \chi_b$, so that $wy - ab$ is G-semi-invariant and X is G-stable. Let $X' \subset X \times \mathbb{P}^1$ be the closure of the graph of the rational map $X \dashrightarrow \mathbb{P}^1$ defined by $[w : a]$. We show the following.

(a) $X' = X'_a \cup X'_b$, where

$$X'_a = (wv = a) \subset \mathbb{A}^2_{w,v} \times S, \quad y = bv, \qquad X'_b = (uy = b) \subset \mathbb{A}^2_{u,y} \times S, \quad w = au,$$

glued along $uv = 1$.

(b) The morphism $X' \to X$ is projective and an isomorphism over $X \backslash \{0\}$, and the fiber over 0 is a curve isomorphic to $\mathbb{P}^1$.

(c) X'_a and X'_b are normal, X'_a is smooth outside the origin $w = v = z = t = 0$, and X'_a is singular at the origin if and only if $da(0) = 0$. Similarly, X'_b is smooth outside the origin $u = y = z = t = 0$, and X'_b is singular at the origin if and only if $db(0) = 0$.

(d) X' is G-stable for the action of G on $X \times \mathbb{P}^1$ by

$$\sigma \cdot (p, [s_0 : s_1]) = (\sigma \cdot p, [\chi_w(\sigma) s_0 : \chi_a(\sigma) s_1]),$$

and G acts diagonally on the coordinates w, v, z, t of X'_a and u, y, z, t of X'_b with $\chi_v = \chi_a / \chi_w$ and $\chi_u = \chi_w / \chi_a$.

For (a), let v be the coordinate of $\mathbb{A}^1_v = \mathbb{P}^1 \backslash \{[0 : 1]\}$, so that $v = a/w$ on the graph over $(w \neq 0)$. The closed subscheme $V_a := (y = bv,\ wv = a) \subset X \times \mathbb{A}^1_v$ is isomorphic to the hypersurface $(wv = a) \subset \mathbb{A}^2_{w,v} \times S$, since the equation $wy = ab$ follows from $y = bv$ and $wv = a$. Thus V_a is integral, as $wv - a$ is irreducible in the polynomial ring $\mathcal{O}(S)[w, v]$. Over $(w \neq 0)$, V_a is the graph of $v = a/w$, and $V_a \cap (w = 0) = \mathbb{A}^1_v \times (a = 0)$, with $\dim(\mathbb{A}^1_v \times (a = 0)) = 2$. Thus V_a is the closure of the graph in $X \times \mathbb{A}^1_v$, that is, $V_a = X'_a$. With $u = w/a$ on $\mathbb{A}^1_u = \mathbb{P}^1 \backslash \{[1 : 0]\}$, the argument for V_a with (y, w, b, a, u) in place of (w, y, a, b, v) shows that the closure of the graph in $X \times \mathbb{A}^1_u$ is $(w = au,\ uy = b)$, which is isomorphic to $(uy = b) \subset \mathbb{A}^2_{u,y} \times S$, since $wy = ab$ follows from $w = au$ and $uy = b$. As $\mathbb{P}^1 = \mathbb{A}^1_v \cup \mathbb{A}^1_u$ and $uv = 1$ on $\mathbb{A}^1_v \cap \mathbb{A}^1_u$, (a) holds.

For (b), $X' \to X$ is projective as X' is closed in $X \times \mathbb{P}^1$. Since $wy = ab$, $[w : a] = [b : y]$ wherever both are defined, so the rational map is a morphism outside $(w = y = a = b = 0)$, which is the point 0 since $(a = b = 0) = \{0\}$. The graph of the morphism $X \backslash \{0\} \to \mathbb{P}^1$ is closed in $(X \backslash \{0\}) \times \mathbb{P}^1$ and isomorphic to $X \backslash \{0\}$, so $X' \to X$ is an isomorphism over $X \backslash \{0\}$. Over 0, the points of X'_a are the points with $w = z = t = 0$, at which $y = bv = 0$ and $wv = a$ hold since $a(0) = b(0) = 0$. Thus they form the v-line. Similarly, the points of X'_b over 0 form the u-line. Thus the fiber over 0 is isomorphic to $\mathbb{P}^1$, and (b) holds.

For (c), the partial derivatives of $wv - a$ with respect to w, v, z, t are v, w, $-\partial a/\partial z$, $-\partial a/\partial t$. Hence X'_a is singular exactly at the points with $w = v = 0$ such that $a = 0$ and $da = 0$ at (z, t), that is, at the singular points of the reduced curve $(a = 0)$. The only possible one is the origin, which is singular if and only if $da(0) = 0$. As a hypersurface in a smooth fourfold whose singular

locus has codimension at least two, X'_a is normal by Serre's criterion. The partial derivatives of $uy-b$ with respect to u,y,z,t are y, u, $-\partial b/\partial z$, $-\partial b/\partial t$. Hence X'_b is singular exactly at the points with $u=y=0$ lying over the singular points of the reduced curve ($b=0$), that is, at most at the origin, which is singular if and only if $db(0)=0$, and X'_b is normal by Serre's criterion. Thus (c) holds.

For (d), $v=a/w$ and $u=w/a$ are G-semi-invariant rational functions with characters χ_a/χ_w and χ_w/χ_a. Hence the rational map $[w:a]$ is G-equivariant for the action on $\mathbb{P}^1$ in (d), and the graph and the closure X' of the graph are G-stable. The charts X'_a and X'_b are G-stable, and G acts on them diagonally in the coordinates w,v,z,t and u,y,z,t. This proves (d).

Step 6. In this step we construct the modification $f:Y\to W$ of (2) and the charts U_i. In the complete local setting, the blow-ups $Y_{i+1}\to Y_i$, $1\le i\le k-1$, below are the blow-ups of [Wem21, Section 6.2] with the factorization $F_1\cdots F_k$ of g in place of the factorization into irreducible factors there. We construct the algebraic charts and check the gluing and the equivariance directly.

Let $Y_1:=W$, $U'_1:=W$, and $w_1:=x$. Let $1\le i\le k-1$, and assume that we have constructed a G-equivariant projective birational morphism $Y_i\to W$, which is an isomorphism over $W\setminus\{0\}$, such that Y_i is covered by G-stable open subsets $U_1,\dots,U_{i-1},U'_i$ with

$$U_l=(u_lv_l=F_l)\subset\mathbb{A}^2_{u_l,v_l}\times S,\quad x=u_l\prod_{m<l}F_m,\quad y=v_l\prod_{m>l}F_m\quad(l<i),$$

$$U'_i=\Big(w_iy=\prod_{m\ge i}F_m\Big)\subset\mathbb{A}^2_{w_i,y}\times S,\quad x=w_i\prod_{m<i}F_m.$$

For $i=1$, this holds. Let $Y_{i+1}\subset Y_i\times\mathbb{P}^1$ be the closure of the graph of the rational map $Y_i\dashrightarrow\mathbb{P}^1$ defined by $[w_i:F_i]$.

On U_l with $l<i$, the expressions of x on U_l and on U'_i imply $w_i=u_l/\prod_{l\le m<i}F_m$. Since $F_l=u_lv_l$,

$$\frac{F_i}{w_i}=\frac{F_i\prod_{l\le m<i}F_m}{u_l}=v_lF_i\prod_{l<m<i}F_m,$$

which is regular on U_l. Hence the rational map $[w_i:F_i]=[1:F_i/w_i]$ is a morphism on U_l, and $Y_{i+1}\to Y_i$ is an isomorphism over U_l. Over U'_i, we apply **Step 5** with $(w,y,a,b)=(w_i,y,F_i,\prod_{m>i}F_m)$. The hypotheses of **Step 5** hold by **Step 4**. For the character condition, $x=w_i\prod_{m<i}F_m$ and the character $\chi_x\chi_y$ of $F_1\cdots F_k$ (**Step 4**) imply

$$\chi_{w_i}\chi_y=\frac{\chi_x\chi_y}{\prod_{m<i}\chi_m}=\prod_{m\ge i}\chi_m=\chi_a\chi_b.$$

Let $u_i:=w_i$ and $v_i:=F_i/w_i$. The chart X'_a is

$$U_i=(u_iv_i=F_i)\subset\mathbb{A}^2_{u_i,v_i}\times S,\quad x=u_i\prod_{m<i}F_m,\quad y=v_i\prod_{m>i}F_m.$$

Let $w_{i+1}:=w_i/F_i$. The chart X'_b is

$$U'_{i+1}=\Big(w_{i+1}y=\prod_{m\ge i+1}F_m\Big)\subset\mathbb{A}^2_{w_{i+1},y}\times S,\quad x=w_{i+1}\prod_{m<i+1}F_m.$$

By **Step 5**, $Y_{i+1}\to Y_i$ is G-equivariant, projective, and an isomorphism outside the origin of U'_i, which maps to $0\in W$. Thus Y_{i+1} satisfies the inductive assumption with $i+1$ in place of i.

Let $Y:=Y_k$ and $f:Y\to W$ the composite morphism, so that $Y=W$ and f is the identity if $k=1$. Let $u_k:=w_k$ and $v_k:=y$. Then Y is covered by the G-stable charts

$$U_i=(u_iv_i=F_i)\subset\mathbb{A}^2_{u_i,v_i}\times S,\quad x=u_i\prod_{l<i}F_l,\quad y=v_i\prod_{l>i}F_l\quad(1\le i\le k),$$

where U_i is an open subset of the hypersurface $(u_iv_i = F_i) \subset \mathbb{A}^4_{u_i,v_i,z,t}$. The morphism f is G-equivariant, projective, birational, and an isomorphism over $W\backslash\{0\}$. Each step adds one curve isomorphic to $\mathbb{P}^1$ over 0, so $\dim f^{-1}(0) \leq 1$ and f is small. If $k \geq 2$, then by **Step 5**(c), any U_i is normal and smooth outside the origin $u_i = v_i = z = t = 0$, and the origin of U_i is a singular point of U_i if and only if $dF_i(0) = 0$. If $k = 1$, then $U_1 = W$, and the argument of **Step 5**(c) with (x, y, F_1) in place of (w, v, a) shows that U_1 is normal and smooth outside the origin and that the origin is a singular point of U_1 if and only if $dF_1(0) = 0$, as the curve $(F_1 = 0)$ is smooth outside 0 by **Step 4**. The functions

$$u_i = \frac{x}{\prod_{l<i} F_l}, \qquad v_i = \frac{y}{\prod_{l>i} F_l}$$

are G-semi-invariant with characters

$$\chi_{u_i} = \frac{\chi_x}{\prod_{l<i} \chi_l}, \qquad \chi_{v_i} = \frac{\chi_y}{\prod_{l>i} \chi_l}.$$

Thus G acts diagonally on the coordinates u_i, v_i, z, t of U_i, and G fixes the origin of U_i. The group G acts faithfully on Y, since f is birational and G-equivariant.

Step 7. In this step we show that f is crepant.

Possibly shrinking W to a G-stable neighborhood of 0 on which all h_j are regular and (W, D) is klt, and replacing Y and the U_i by their inverse images, we may assume that the h_j are regular on W. By adjunction, ω_W is generated near 0 by the residue form

$$\omega := \frac{dx}{x} \wedge dz \wedge dt$$

of $xy - F_1 \cdots F_k$, and ω_{U_i} is generated by the residue form $\frac{du_i}{u_i} \wedge dz \wedge dt$ of $u_iv_i - F_i$. Since $x = u_i \prod_{l<i} F_l$ on U_i and $d(\prod_{l<i} F_l)$ is an $\mathcal{O}(S)$-linear combination of dz and dt,

$$f^*\omega = \frac{du_i}{u_i} \wedge dz \wedge dt + \frac{d(\prod_{l<i} F_l)}{\prod_{l<i} F_l} \wedge dz \wedge dt = \frac{du_i}{u_i} \wedge dz \wedge dt.$$

Thus $f^*\omega$ generates ω_Y on any chart U_i, and $K_Y = f^*K_W$. Thus the hypotheses of **Step 1** hold for f. Let $D_{U_i} := D_Y|_{U_i}$, the pull-back of D to U_i. Since f is small, $D_Y = f_*^{-1}D$.

Step 8. In this step we finish the proof of (2).

By **Steps 6 and 7**, $f : Y \to W$ is a G-equivariant small crepant modification of W, isomorphic away from 0, covered by the G-stable charts U_i with the chart formulas of (2), on which G acts diagonally. For any subgroup $G' \subset G$, **Step 1** applied to f and G' implies that the quotient value $\mathrm{mld}((W \ni 0, D)/G')$ is the minimum of the minimal log discrepancies at the scheme-theoretic points of the fiber of the quotient $Z = Y/G' \to Q = W/G'$ over the image of 0, by (B.17).

Let (W_n, G_n, D_n), $n \geq 1$, be data as in (2) such that the numbers $\alpha_n := \mathrm{mld}((W_n \ni 0, D_n)/G_n)$ are positive and strictly increasing. Let $f_n : Y_n \to W_n$ be the modification of **Step 6** for (W_n, G_n, D_n), with charts $U_{n,i}$ and boundaries $D_{U_{n,i}}$, and let $Z_n := Y_n/G_n$ with the crepant quotient boundary D_{Z_n} and the induced morphism $\bar{f}_n : Z_n \to W_n/G_n$. By **Step 1** with $G' = G_n$, there exists a scheme-theoretic point η_n of $\bar{f}_n^{-1}(\bar{0})$, where $\bar{0}$ is the image of 0, such that $\alpha_n = \mathrm{mld}(Z_n \ni \eta_n, D_{Z_n})$ and $\overline{\{\eta_n\}}$ is the center of a prime divisor computing α_n. Since $\dim f_n^{-1}(0) \leq 1$, η_n is the generic point of a curve or a closed point. Since $\mathcal{A}_2 \cup \mathcal{A}_3$ satisfies the ACC, only finitely many α_n belong to $\mathcal{A}_2 \cup \mathcal{A}_3$. Thus by **Step 2**, after passing to a subsequence, η_n is a closed point and Y_n is singular at any point over η_n for any n. Let $y_n \in f_n^{-1}(0)$ be a point over η_n, and $U_{n,i}$ a chart containing y_n. By **Step 6**, y_n is the origin of $U_{n,i}$, which is a singular point of $U_{n,i}$, and G_n fixes y_n. Thus y_n is the only point of Y_n over η_n, $U_{n,i}/G_n$ is an open neighborhood of η_n in Z_n, and the restriction of D_{Z_n} to $U_{n,i}/G_n$ is the crepant quotient boundary of $(U_{n,i}, D_{U_{n,i}})/G_n$. Hence

$$\alpha_n = \mathrm{mld}((U_{n,i} \ni 0, D_{U_{n,i}})/G_n).$$

Finally, assume that any irreducible factor of g is smooth at 0 and preserved by G. Then any G-orbit of irreducible factors of g vanishing at 0 has one element, so any F_i is a non-zero constant multiple of an irreducible factor of g smooth at 0, and $dF_i(0) \neq 0$. By **Step 6**, no chart origin is a singular point. Thus there exists no strictly increasing sequence of positive quotient values $\mathrm{mld}((W \ni 0, D)/G)$ for such data, that is, the positive quotient values satisfy the ACC. □

B.3. **The circle** $(1,0,0,0)$**.** In this subsection, we prove Theorem 8.5. The reflection subgroup R is removed by passing to the quotient $\mathbb{A}^4_{X,u}$ with $X = x^r$, and the minimal log discrepancy is then bounded from both sides by a minimal log discrepancy on one of finitely many fixed threefolds $\mathbb{A}^3/(G/R)$.

Proof of Theorem 8.5. The proof is long, so we divide it into several steps. Write $u = (u_1, u_2, u_3)$.

Step 1. In this step we define the constants r_0 and M, and an ACC set $\mathcal{A}$ which will contain the numbers b.

Let $\gamma := \min(\Gamma\backslash\{0\})$, which exists since Γ is a DCC set, and let $\gamma := 1$ if $\Gamma \subset \{0\}$. Let

$$\Sigma := (\Gamma \cup \{1\})_+ \cap (0,1] \quad \text{and} \quad \Xi := \left\{1 - \frac{1-s}{e'} \;\middle|\; s \in \Sigma \cup \{0\},\ e' \in \{1, \dots, e_0\}\right\}\backslash\{0\}.$$

Then Σ and Ξ are DCC sets [Liu26, Lemma 2.10(1)], and $\Xi \subset (0,1]$.

By [HMX14, Theorem 1.1], the set $\mathrm{LCT}_4(\Sigma, \{1\})$ of the log canonical thresholds $\mathrm{lct}(Y', B'; M')$, where (Y', B') is an lc pair with $\dim Y' = 4$ and $B' \in \Sigma$, and M' is a non-zero reduced $\mathbb{R}$-Cartier divisor on Y', satisfies the ACC. Hence there exists a real number $\eta \in (0,1)$, depending only on Γ, such that

$$\mathrm{LCT}_4(\Sigma, \{1\}) \cap (1-\eta, 1) = \emptyset,$$

since otherwise $\mathrm{LCT}_4(\Sigma, \{1\})$ would contain a strictly increasing sequence converging to 1. Let $r_0 \geq 3$ be an integer such that $1/r_0 < \eta$.

Let $\mathcal{S}$ be the set of diagonal subgroups $\Pi \subset (\mathbb{C}^*)^3$ with $|\Pi| \leq e_0$. Any element of such a group Π has order dividing $|\Pi|$, hence belongs to $\mu^3_{e_0!}$. Thus $\mathcal{S}$ is finite. For any $\Pi \in \mathcal{S}$, V/Π is a $\mathbb{Q}$-factorial klt toric threefold: the quotient morphism $V \to V/\Pi$ is finite and surjective and $V = \mathbb{A}^3$ is smooth, so V/Π is $\mathbb{Q}$-factorial [KM98, Lemma 5.16], and $(V, 0)$ is the crepant pullback of the crepant quotient of $(V, 0)$ by Π, whose boundary has non-negative coefficients, so V/Π is klt [KM98, Proposition 5.20(4)]. Parts (1) and (3) of Theorem 2.10 allow any DCC subset of $\mathbb{R}_{>0}$ and any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $B \geq 0$, so they hold for boundaries on V/Π with real coefficients in the positive DCC set Ξ, which may contain 1. For any $\Pi \in \mathcal{S}$, let $\mathcal{A}_\Pi$ be the ACC set of Theorem 2.10(1) and L_Π the positive integer of Theorem 2.10(3), for the threefold V/Π and the DCC set Ξ. Let

$$\mathcal{A} := \bigcup_{\Pi \in \mathcal{S}} \mathcal{A}_\Pi, \qquad L := \max_{\Pi \in \mathcal{S}} L_\Pi, \qquad M := \frac{L}{\gamma} + 1.$$

Since $\mathcal{S}$ is finite, $\mathcal{A}$ satisfies the ACC. The set $\mathcal{A}$ and the numbers L and M depend only on Γ and e_0, and r_0 depends only on Γ.

Now consider W, f, G, R, h_j, c_j, and D satisfying the hypotheses of Theorem 8.5.

Step 2. In this step we prove $f = F(x^r, u)$ and $h_j = x^{a_j} H_j(x^r, u)$, and we describe the action of G/R on $Y := \mathbb{A}^4_{X,u}$, where X is a coordinate.

Near 0, W is normal by Serre's criterion, since W is a hypersurface and 0 is an isolated singular point of the threefold W. Thus W is integral near 0, that is, f is a prime element of $\mathcal{O}_{\mathbb{A}^4,0}$. Moreover, W is klt near 0, since (W, D) is klt near 0 and $D \geq 0$.

Let ζ be a primitive r-th root of unity. Then R is generated by $(\zeta, 1, 1, 1)$, which multiplies a monomial $x^i u_1^{m_1} u_2^{m_2} u_3^{m_3}$ by ζ^i. Let h be a non-zero G-semi-invariant polynomial and a the least x-exponent of h. Since h is semi-invariant under $(\zeta, 1, 1, 1)$, the x-exponents of the monomials of h are congruent to a modulo r. Hence

$$h = x^a H(x^r, u)$$

for a polynomial H with $H(0,u)\neq 0$. Applied to h_j, this proves $h_j = x^{a_j}H_j(x^r,u)$ with $H_j(0,u)\neq 0$. Applied to f, this implies $f = x^a F(x^r,u)$ with $F(0,u)\neq 0$. If $a\geq 1$, then x divides f, and since f is a prime element of $\mathcal{O}_{\mathbb{A}^4,0}$, f is x times a unit of $\mathcal{O}_{\mathbb{A}^4,0}$, so W is smooth at 0, a contradiction. Thus $a=0$, that is, $f=F(x^r,u)$, and the R-character of f is trivial. Write

$$f_0 := F(0,u) \quad \text{and} \quad H_{j,0} := H_j(0,u),$$

which are non-zero polynomials on $V=\mathbb{A}^3_u$.

Since G acts diagonally, G acts on x^r, u_1, u_2, and u_3 by characters, which are trivial on R. Thus G/R acts diagonally on $Y=\mathbb{A}^4_{X,u}$, with X pulled back to x^r, and on V. Let χ be the character of G/R such that $g^*X=\chi(g)X$ for any $g\in G/R$. The polynomials $F(X,u)$ and $H_j(X,u)$ on Y are G/R-semi-invariant, since they pull back to f and $x^{-a_j}h_j$, which are G-semi-invariant with characters trivial on R. The group G/R acts faithfully on V, since an element of G acting trivially on u has the form $(s,1,1,1)$, hence belongs to $G\cap \mathbf{T}_{(1,0,0,0)}=R$.

Removing the terms with $c_j=0$ changes neither D, A, nor C_0. Thus we may assume that $c_j>0$, hence $c_j\geq\gamma$, for any j.

Step 3. In this step we prove $0\leq A<1$ and reduce the minimal log discrepancy of $(W\ni 0, D)/G$ to a crepant quotient of a pair on Y.

Since $c_j\geq 0$ and $a_j\geq 0$, $A\geq 0$. If $x|_W=0$ near 0, then $W=(x=0)$ near 0 and W is smooth at 0, a contradiction. Thus $\operatorname{div}(x|_W)\geq 0$ is a non-zero Cartier divisor on W through 0 (as W is normal near 0, [Har77, Proposition II.6.3A]). Let P be a prime divisor on W through 0 with $\operatorname{mult}_P\operatorname{div}(x|_W)\geq 1$. Since $H_j(x^r,u)|_W$ is a non-zero regular function,

$$D = A\operatorname{div}(x|_W) + \sum_j c_j \operatorname{div}(H_j(x^r,u)|_W) \geq A\operatorname{div}(x|_W).$$

Since (W,D) is klt near 0, $1>\operatorname{mult}_P D\geq A\operatorname{mult}_P\operatorname{div}(x|_W)\geq A$. Thus $0\leq A<1$.

Let $\pi:\mathbb{A}^4\to Y$ be the morphism $(x,u)\to(x^r,u)$, which is the quotient by R. Let

$$\Delta := \operatorname{div}(f) + \sum_j c_j\operatorname{div}(h_j) \quad \text{on } \mathbb{A}^4, \qquad \beta := 1-\frac{1-A}{r},$$

and on Y let

$$B_{\text{base}} := \operatorname{div}(F) + \sum_j c_j\operatorname{div}(H_j), \qquad B_\beta := \beta\operatorname{div}(X) + B_{\text{base}}, \qquad B_1 := \operatorname{div}(X) + B_{\text{base}}.$$

Since $0\leq A<1$, $1-\frac{1}{r}\leq\beta<1$. The morphism π is ramified exactly along $(x=0)$, with index r. By the Hurwitz formula, $K_{\mathbb{A}^4}=\pi^*(K_Y+(1-\frac{1}{r})\operatorname{div}(X))$. Moreover,

$$\pi^*\operatorname{div}(X)=r\operatorname{div}(x), \qquad \pi^*\operatorname{div}(F)=\operatorname{div}(f), \qquad \pi^*\operatorname{div}(H_j)=\operatorname{div}(h_j)-a_j\operatorname{div}(x).$$

Since $-(r-1)+r\beta-A=0$, we obtain

$$K_{\mathbb{A}^4}+\Delta=\pi^*(K_Y+B_\beta).$$

Thus (Y,B_β) is the crepant quotient of $(\mathbb{A}^4,\Delta)$ by R, and the crepant quotient of $(\mathbb{A}^4,\Delta)$ by G is the crepant quotient of (Y,B_β) by G/R [Liu26, Lemma 2.6(5)]. In particular,

$$\operatorname{mld}((\mathbb{A}^4\ni 0,\Delta)/G) = \operatorname{mld}((Y\ni 0, B_\beta)/(G/R)). \tag{B.18}$$

Since W is a normal Cartier divisor on the smooth variety $\mathbb{A}^4$ near 0, the different of $\sum_j c_j\operatorname{div}(h_j)$ on W is D. Applying inversion of adjunction [Kaw07, Theorem] on an open neighborhood of 0 in $\mathbb{A}^4$ over which W is normal and (W,D) is klt, we see that $(\mathbb{A}^4,\Delta)$ is lc near 0. By [Liu26, Lemma 2.6(2)(3)], a log discrepancy of (Y,B_β) is a log discrepancy of $(\mathbb{A}^4,\Delta)$ divided by the order of an inertia group, so (Y,B_β) is lc near 0.

Let $\mathfrak{a}:=\prod_j(h_j)^{c_j}$, and let $\mathfrak{a}:=(1)$ if no term remains. The hypotheses of Theorem 3.3(1) with ambient dimension 4, $c=1$, $f_1=f$, the group G, the subgroup G, and the $\mathbb{R}$-ideal $\mathfrak{a}$ hold: f is G-semi-invariant and vanishes at 0, W is near 0 a normal integral hypersurface

with klt singularities on which G acts faithfully (**Step 2**), the ideals (h_j) are generated by G-semi-invariant polynomials, $\mathfrak{a}|_W$ is non-zero near 0, and $(W, \mathfrak{a}|_W)$ is the klt pair (W, D). Since $\operatorname{mult}_E h_j = \operatorname{mult}_E \operatorname{div}(h_j)$ for any prime divisor E over $\mathbb{A}^4$ and any j, we have

$$a(E, \mathbb{A}^4, \operatorname{div}(f), \mathfrak{a}) = a(E, \mathbb{A}^4, \Delta)$$

for any such E, also when $\mathfrak{a} = (1)$, in which case $\Delta = \operatorname{div}(f)$. Thus Definition 2.4 gives

$$\operatorname{mld}((\mathbb{A}^4 \ni 0, \operatorname{div}(f), \mathfrak{a})/G) = \operatorname{mld}((\mathbb{A}^4 \ni 0, \Delta)/G).$$

Thus Theorem 3.3(1) and (B.18) imply

$$\operatorname{mld}((W \ni 0, D)/G) = \operatorname{mld}((Y \ni 0, B_\beta)/(G/R)). \tag{B.19}$$

Step 4. In this step we prove that (Y, B_1) is lc near 0.

Let Y° be an open neighborhood of 0 in Y on which (Y, B_β) is lc. Since $f_0 \neq 0$ and $H_{j,0} \neq 0$ for any j, X divides neither F nor H_j, so the hyperplane $(X = 0)$ is not a component of B_{base}. Thus any prime component Q of $B_{\text{base}}|_{Y^\circ}$ has coefficient

$$\mu_Q := \operatorname{mult}_Q F + \sum_j c_j \operatorname{mult}_Q H_j$$

both in B_{base} and in B_β. The number μ_Q is a positive finite sum of elements of $\Gamma \cup \{1\}$, and $\mu_Q \leq 1$ since $(Y^\circ, B_\beta|_{Y^\circ})$ is lc. Hence $B_{\text{base}}|_{Y^\circ} \in \Sigma$. Moreover, $(Y^\circ, B_{\text{base}}|_{Y^\circ})$ is lc since $B_{\text{base}} \leq B_\beta$. Let

$$t := \operatorname{lct}(Y^\circ, B_{\text{base}}|_{Y^\circ}; \operatorname{div}(X)|_{Y^\circ}) \in \operatorname{LCT}_4(\Sigma, \{1\}).$$

Since $(Y^\circ, B_\beta|_{Y^\circ})$ is lc, $t \geq \beta$. Since the hyperplane $(X = 0)$ has coefficient t in $B_{\text{base}} + t \operatorname{div}(X)$, $t \leq 1$. Since $r \geq r_0$ by assumption,

$$1 - \eta < 1 - \frac{1}{r_0} \leq 1 - \frac{1}{r} \leq \beta \leq t \leq 1,$$

hence $t = 1$ by the choice of η in **Step 1**. Since the log discrepancies of the pair

$$(Y^\circ, B_{\text{base}}|_{Y^\circ} + s \operatorname{div}(X)|_{Y^\circ})$$

are affine functions of s, $(Y^\circ, B_1|_{Y^\circ})$ is lc. Thus (Y, B_1) is lc near 0.

Step 5. In this step we prove that (V, C_0) is lc near 0, that $b \leq \operatorname{mld}((W \ni 0, D)/G)$, and that $b \in \mathcal{A}$, and we find a prime divisor ν over V with bounded orders of f_0 and $H_{j,0}$.

Under the isomorphism $V \to (X = 0) \subset Y$, $u \to (0, u)$, the hyperplane $(X = 0)$ is a smooth Cartier divisor which is not a component of B_{base}, and

$$B_{\text{base}}|_{(X=0)} = \operatorname{div}(f_0) + \sum_j c_j \operatorname{div}(H_{j,0}) = C_0.$$

Since $(Y^\circ, B_1|_{Y^\circ})$ is lc by **Step 4**, inversion of adjunction [Kaw07, Theorem] implies that (V, C_0) is lc near 0.

The hypotheses of Theorem 3.3(1) for Y with ambient dimension 4, $c = 1$, $f_1 = X$, the group G/R, the subgroup G/R, and the $\mathbb{R}$-ideal $(F) \prod_j (H_j)^{c_j}$ hold: X is G/R-semi-invariant and vanishes at 0, the zero scheme $(X = 0) \cong V$ is smooth, G/R acts faithfully on V (**Step 2**), F and H_j are G/R-semi-invariant (**Step 2**), the restriction $(f_0) \prod_j (H_{j,0})^{c_j}$ is non-zero, and the pair (V, C_0) is lc near 0. Likewise

$$a(E, Y, \operatorname{div}(X), (F) \prod_j (H_j)^{c_j}) = a(E, Y, B_1)$$

for any prime divisor E over Y, and $a(E, V, (f_0) \prod_j (H_{j,0})^{c_j}) = a(E, V, C_0)$ for any prime divisor E over V, so Theorem 3.3(1) and Definition 2.4 give

$$b = \operatorname{mld}((V \ni 0, C_0)/(G/R)) = \operatorname{mld}((Y \ni 0, B_1)/(G/R)). \tag{B.20}$$

Since $\beta < 1$, $B_\beta \le B_1$, so $a(E, Y, B_\beta) \ge a(E, Y, B_1)$ for any prime divisor E over Y. By [Liu26, Lemma 2.6], $\mathrm{mld}((Y \ni 0, B_\beta)/(G/R)) \ge \mathrm{mld}((Y \ni 0, B_1)/(G/R))$. Thus (B.19) and (B.20) imply

$$\mathrm{mld}((W \ni 0, D)/G) \ge b.$$

Let C_0° be the sum of the components of C_0 through 0, with the coefficients they have in C_0. Any coefficient of C_0° has the form $\mathrm{mult}_Q f_0 + \sum_j c_j \mathrm{mult}_Q H_{j,0}$, which is a positive finite sum of elements of $\Gamma \cup \{1\}$, and which is at most 1 since (V, C_0) is lc near 0. Thus $C_0^\circ \in \Sigma$. Since G/R fixes 0 and f_0 and $H_{j,0}$ are G/R-semi-invariant, C_0° is G/R-invariant. Let $\bar 0 \in V/(G/R)$ be the image of 0, and let $\bar C$ be the boundary of the crepant quotient of (V, C_0°) by G/R. By [Liu26, Lemma 2.6(4)], any coefficient of $\bar C$ has the form $1 - \frac{1-s}{e'}$, where $s \in \Sigma \cup \{0\}$ and e' is a ramification index, hence $e' \le |G/R| \le e_0$ [Liu26, Lemma 2.6(1)]. Thus $\bar C \in \Xi$. Since C_0 and C_0° coincide near 0,

$$b = \mathrm{mld}(V/(G/R) \ni \bar 0, \bar C),$$

and $(V/(G/R), \bar C)$ is lc near $\bar 0$ [Liu26, Lemma 2.6(2)(3)]. Through the faithful action on V, G/R is a diagonal subgroup of $(\mathbb{C}^*)^3$ of order $[G : R] \le e_0$, so $G/R \in \mathcal{S}$. Moreover, $K_{V/(G/R)} + \bar C$ is $\mathbb{R}$-Cartier, since $V/(G/R)$ is $\mathbb{Q}$-factorial. By **Step 1**, $b \in \mathcal{A}_{G/R} \subset \mathcal{A}$, and there exists a prime divisor $\bar\nu$ over $V/(G/R)$ which computes $\mathrm{mld}(V/(G/R) \ni \bar 0, \bar C)$ and satisfies $a(\bar\nu, V/(G/R), 0) \le L$.

Let ν be a prime divisor over V lying over $\bar\nu$ [Liu26, Lemma 2.6(3) and Definition 2.7], and let $e := |I_{G/R}(\nu)|$. Since the preimage of $\bar 0$ in V is $\{0\}$, the center of ν on V is 0. Since C_0° and C_0 coincide near 0, [Liu26, Lemma 2.6(2)] implies $a(\nu, V, C_0) = eb$. Let $(V/(G/R), \bar B_0)$ be the crepant quotient of $(V, 0)$ by G/R. Since $\bar B_0 \ge 0$ [Liu26, Lemma 2.6(0)], [Liu26, Lemma 2.6(2)] also implies $a(\nu, V, 0) \le e a(\bar\nu, V/(G/R), 0) \le eL$. Since $b \ge 0$,

$$\mathrm{mult}_\nu f_0 + \sum_j c_j \mathrm{mult}_\nu H_{j,0} = a(\nu, V, 0) - a(\nu, V, C_0) \le eL.$$

Since $c_j \ge \gamma$ for any j (**Step 2**),

$$\mathrm{mult}_\nu f_0 \le eL \quad \text{and} \quad \mathrm{mult}_\nu H_{j,0} \le \frac{eL}{\gamma} \quad \text{for any } j. \tag{B.21}$$

Step 6. In this step we prove $\mathrm{mld}((W \ni 0, D)/G) \le b + \frac{M}{r}$.

Recall that ν is the prime divisor over V with center $\bar 0$ of **Step 5**, $e = |I_{G/R}(\nu)|$, L is the integer of **Step 1**, and χ is the character of G/R on X. The hypotheses of Lemma 3.1 with $k = \mathbb{C}$, the variety V, the point 0, $B = 0$, the group G/R, the character χ, the coordinate $w = X$, and the prime divisor ν hold: the group G/R acts faithfully on V and fixes 0 (**Step 2**), and the action of G/R on $V \times \mathbb{A}^1_X = Y$ is $g \cdot (p, X) = (g \cdot p, \chi(g)X)$. Let $q \in \{1, \dots, e\}$ be the integer of Lemma 3.1 for ν and χ. Since the interval $(\frac{eL}{\gamma}, \frac{eL}{\gamma} + e]$ contains e consecutive integers, there exists an integer ℓ such that

$$\ell \equiv q \pmod e \quad \text{and} \quad \frac{eL}{\gamma} < \ell \le e\Big(\frac{L}{\gamma} + 1\Big).$$

By Lemma 3.1, the Gauss extension of mult_ν with X-value ℓ is $\mathrm{mult}_{E'}$ for a prime divisor E' over Y with center 0, such that

$$I_{G/R}(E') = I_{G/R}(\nu) \quad \text{and} \quad a(E', Y, 0) = a(\nu, V, 0) + \ell.$$

By the definition of the Gauss extension [Liu26, Definition-Proposition 2.11],

$$\mathrm{mult}_{E'}\Big(\sum_{i \ge 0} \phi_i X^i\Big) = \min_i \big(\mathrm{mult}_\nu \phi_i + i\ell\big)$$

for any regular functions ϕ_i on V. Since the center of ν is 0, $\mathrm{mult}_\nu \phi_i \ge 0$, so any monomial with positive X-exponent has value at least ℓ. Since $\gamma \le 1$, (B.21) implies $\mathrm{mult}_\nu f_0 < \ell$ and

$\operatorname{mult}_\nu H_{j,0} < \ell$. Thus

$$\operatorname{mult}_{E'} X = \ell, \qquad \operatorname{mult}_{E'} F = \operatorname{mult}_\nu f_0, \qquad \operatorname{mult}_{E'} H_j = \operatorname{mult}_\nu H_{j,0}.$$

Hence

$$\begin{aligned} a(E', Y, B_\beta) &= a(\nu, V, 0) + \ell - \beta\ell - \operatorname{mult}_\nu f_0 - \sum_j c_j \operatorname{mult}_\nu H_{j,0} \\ &= a(\nu, V, C_0) + (1-\beta)\ell = eb + \frac{(1-A)\ell}{r}. \end{aligned}$$

Since the center of E' on Y is 0 and $|I_{G/R}(E')| = e$, [Liu26, Lemma 2.6] implies

$$\operatorname{mld}((Y \ni 0, B_\beta)/(G/R)) \le \frac{a(E', Y, B_\beta)}{e} = b + \frac{(1-A)\ell}{re} \le b + \frac{L/\gamma + 1}{r} = b + \frac{M}{r},$$

where we use $0 < 1 - A \le 1$ and $\ell \le e(\frac{L}{\gamma} + 1)$. By (B.19) and **Step 5**,

$$b \le \operatorname{mld}((W \ni 0, D)/G) \le b + \frac{M}{r}.$$

Together with **Steps 1–5**, this proves Theorem 8.5 except the last sentence.

Step 7. In this step we prove the last sentence of Theorem 8.5.

Suppose that (W_i, G_i, D_i) is a sequence of data satisfying the hypotheses of Theorem 8.5, in particular $r_i \ge r_0$, with $r_i \to +\infty$, such that $\alpha_i := \operatorname{mld}((W_i \ni 0, D_i)/G_i)$ is strictly increasing. Let b_i be the number b for (W_i, G_i, D_i). By **Step 6**, $b_i \le \alpha_i \le b_i + \frac{M}{r_i}$, so $\alpha_i - b_i \to 0$. Since (α_i) is strictly increasing, $b_i \le \alpha_i$, and $\alpha_i - b_i \to 0$, the set $\{b_i\}$ does not satisfy the ACC [Liu26, Lemma 14.11]. However, $b_i \in \mathcal{A}$ for any i by **Step 5**, and $\mathcal{A}$ satisfies the ACC by **Step 1**, a contradiction. □

B.4. **The circles $(-A, -A, -B, v)$ on split sources of any order.** In this subsection, we prove Theorem 8.6.

Proof of Theorem 8.6. The proof is long, so we divide it into several steps. **Steps 1–7** treat the germs $W = (xy + g(z,t) = 0)$ of the first part of Theorem 8.6, and the germs with a completed presentation are reduced to them at the end of the proof.

Step 1. In this step, we fix the notation for one such datum, introduce the principal blocks and the cover of degree n, and show that (W^0, D^{pr}) is lc when r is large.

Let $W = (xy + g(z,t) = 0) \subset \mathbb{A}^4$ with $g = z^n + \sum_{j \le n-2} a_j(t) z^j$ and $a_j(0) = 0$, and let G, R, r, and α be as in Set-up 8.2. We write the boundary as $D = \sum_l b_l \operatorname{div}_W(h_l)$, with the letter b since the letter c is the integer of the circle, where $b_l \in \Gamma$ and the h_l are G-semi-invariant polynomials (a mark is regular near 0 by Definition 2.7, and its product with the G-invariant unit $\prod_{\iota \in G} \iota^* Q$, where Q is a denominator with $Q(0) \ne 0$, is the restriction to W of a polynomial, whose component with the character of the mark is a G-semi-invariant polynomial with restriction to W equal to the product, so its divisor near 0 equals that of the mark). In this proof, a constant is called *bounded* if it depends only on n, c, v, e_0, and Γ, and "for large r" means "for r larger than a bounded constant". Since $\gcd(A, B) = c$ and $\gcd(c, v) = 1$, the vector $(-A, -A, -B, v)$ is primitive. Since $nB = 2A$, the polynomial $xy + z^n$ has $\mathbf{T}$-weight $-2A$.

A mark with $b_l = 0$ or $h_l(0) \ne 0$ does not change D near 0, so we may assume that $b_l > 0$ and $h_l(0) = 0$ for any l. For any polynomial m, the restrictions of h_l and $h_l - m(xy + g)$ to W are equal, and $xy + g$ is G-semi-invariant with the character of xy. Substituting $-g$ for xy in the monomials of h_l divisible by xy lowers the degree in (x, y), so possibly replacing each h_l with a G-semi-invariant polynomial whose restriction to W equals that of h_l, we may assume that no monomial of any h_l is divisible by xy. Let $\gamma := \min(\Gamma \setminus \{0\})$, and $\gamma := 1$ if $\Gamma \subset \{0\}$.

Since W has quadratic rank two, $g \in (z,t)^3$, so the exceptional divisor of the blow-up of W at 0 is the reduced cone $(xy = 0) \subset \mathbb{P}^3$, the restriction to the strict transform of W of the exceptional divisor of the blow-up of $\mathbb{A}^4$ at 0, which has log discrepancy 4 [KM98, Lemma 2.29]. Since

$\operatorname{mult}_0(xy+g)=2$, adjunction implies that both components P of the cone satisfy $a(P,W,0)=2$, and they satisfy $\operatorname{mult}_P h \geq \operatorname{mult}_0 h$ for any polynomial h. Since (W,D) is klt,

$$0 < a(P,W,D) \leq 2 - \sum_l b_l \operatorname{mult}_0 h_l, \quad \text{hence} \quad \sum_l b_l \operatorname{mult}_0 h_l < 2. \tag{B.22}$$

Thus the number of marks is at most $2/\gamma$ and $\operatorname{mult}_0 h_l \leq 2/\gamma$ for any l.

Let $t^\sigma z^j$ be a monomial of $a_j(t)z^j$ with non-zero coefficient, $0 \leq j \leq n-2$. The group R is the subgroup $\{\mathbf{T}(\lambda) \mid \lambda^r = 1\}$ of $\mathbf{T}$, and $xy+g$ is R-semi-invariant with the character $\lambda^{-2A} = \lambda^{-nB}$ of xy. Comparing λ^{-nB} with the character $\lambda^{\sigma v - jB}$ of $t^\sigma z^j$, we obtain an integer $N_{j,\sigma}$ with

$$\sigma v + (n-j)B = N_{j,\sigma} r. \tag{B.23}$$

The left side is positive because $\sigma \geq 1$, so $N_{j,\sigma} \geq 1$ and $\sigma \geq (r-nB)/v$. Thus the orders at 0 of the monomials of $g - z^n$ are at least $(r-nB)/v$, and $\operatorname{mult}_0 g = n$ for large r.

For any l, let κ_l be the $\mathbf{T}$-weight of a fixed monomial of least degree of h_l, and let h_l^0 be the principal block of h_l (Set-up 5.1). Since $\operatorname{mult}_0 h_l \leq 2/\gamma$, the integers κ_l are bounded. Since h_l is R-semi-invariant, the $\mathbf{T}$-weights of the monomials of h_l are congruent to κ_l modulo r, so $h_l = \sum_{\ell \in \mathbb{Z}} h_l^{(\ell)}$, where $h_l^{(\ell)}$ is the sum of the terms of weight $\kappa_l + \ell r$ and $h_l^{(0)} = h_l^0$. A monomial $x^{k_1}y^{k_2}z^{k_3}t^{k_4}$ of weight $\kappa_l + \ell r$ satisfies

$$A(k_1+k_2) + Bk_3 \geq -\kappa_l - \ell r \quad \text{and} \quad vk_4 \geq \kappa_l + \ell r, \tag{B.24}$$

so for $\ell \neq 0$ the degree of the monomial is at least $(|\ell| r - |\kappa_l|)/\max(A,v)$. In particular, for large r all the monomials of h_l of degree at most $2/\gamma$ belong to h_l^0, and $h_l^0 \neq 0$. Since G commutes with $\mathbf{T}$ and an exchange of x and y preserves the $\mathbf{T}$-weights, any $h_l^{(\ell)}$ is G-semi-invariant with the character of h_l. Let $n_l := \operatorname{mult}_{(t=0)} h_l^0$ be the least exponent of t in the monomials of h_l^0, and let

$$\tau = \sum_l b_l \kappa_l, \qquad \delta = \tau + B - v, \qquad b_t := \sum_l b_l n_l. \tag{B.25}$$

The numbers τ, δ, and b_t are bounded, and δ is the adjusted mass of D along $\mathbf{T}$, since $\epsilon_{\mathbf{T}} = v - B$.

Let $W^0 := (xy + z^n = 0) \subset \mathbb{A}^4$ and $D^0 := \sum_l b_l \operatorname{div}_{W^0}(h_l)$, $D^{\mathrm{pr}} := \sum_l b_l \operatorname{div}_{W^0}(h_l^0)$. No non-zero multiple of $xy + z^n$ is free of monomials divisible by xy, so the restrictions of h_l and h_l^0 to W^0 are non-zero. Each monomial of g has the character of z^n, so G preserves W^0. Consider the morphism

$$\mathbb{A}^3_{p,q,t} \to W^0, \qquad (p,q,t) \to (p^n, -q^n, pq, t). \tag{B.26}$$

It is the quotient by $\mu_n(1,-1,0)$, which contains no pseudo-reflection, so W^0 is klt and (B.26) is quasi-étale. Let $\tilde{G}$ be the group of linear automorphisms of $\mathbb{A}^3_{p,q,t}$ which lift elements of G. An element of G acting by

$$(x,y,z,t) \to (\alpha_1 x, \alpha_2 y, \alpha_3 z, \alpha_4 t)$$

lifts exactly to $(p,q,t) \to (\omega_1 p, \omega_2 q, \alpha_4 t)$ with $\omega_1^n = \alpha_1$ and $\omega_1\omega_2 = \alpha_3$, and an element acting by $(x,y) \to (\alpha_1 y, \alpha_2 x)$ lifts exactly to $(p,q) \to (\omega_1 q, \omega_2 p)$ with $\omega_1^n = -\alpha_1$ and $\omega_1\omega_2 = \alpha_3$. Thus $\tilde{G}$ is a finite monomial group of order $n|G|$, the kernel of $\tilde{G} \to G$ is $\mu_n(1,-1,0)$, and $(W^0, D^0)/G = (\mathbb{A}^3, \tilde{D}^0)/\tilde{G}$, where $\tilde{D}^0$ is the pullback of D^0. The group $\tilde{G}$ need not be abelian.

Let $g_0 := \gcd(c,d)$, $e_1 := d/g_0$, $c' := c/g_0$, and $v' := e_1 v$. Then $\gcd(c',v') = 1$, so $\mathbf{T}' := \mathbf{T}_{(-c',-c',v')}$ is a primitive circle on $\mathbb{A}^3_{p,q,t}$. Since $nc' = e_1 A$ and $2c' = e_1 B$, the element $\mathbf{T}'(s)$ lifts $\mathbf{T}(s^{e_1})$. The kernel of $\mathbf{T}' \to \mathbf{T}$ is $\{\mathbf{T}'(s) \mid s^{e_1} = 1\}$, and $\mathbf{T}'(s)$ belongs to $\mu_n(1,-1,0)$ if and only if $s^{2c'} = s^{nc'} = s^{v'} = 1$. Since $\gcd(2c', nc', v') = \gcd(dc', v') = e_1 \gcd(c,v) = e_1$, the condition $s^{2c'} = s^{nc'} = s^{v'} = 1$ for $\mathbf{T}'(s) \in \mu_n(1,-1,0)$ is equivalent to the condition $s^{e_1} = 1$ defining the kernel of $\mathbf{T}' \to \mathbf{T}$. As $\tilde{G}$ contains all lifts of R, we obtain

$$|\tilde{G} \cap \mathbf{T}'| = e_1 r \quad \text{and} \quad [\tilde{G} : \tilde{G} \cap \mathbf{T}'] = \frac{n}{e_1}[G:R] \leq \frac{n e_0}{e_1}. \tag{B.27}$$

The group $\tilde{G} \cap \mathbf{T}'$ acts by scalars on (p,q), so it is central in $\tilde{G}$.

Claim B.4. *Let $\Lambda \subset \mathbb{R}_{>0}$ be a DCC set and s_0 a positive integer. Then there exists a positive integer r_0, depending only on $n, c, v, e_0, \Gamma, \Lambda$, and s_0, such that the following holds if $r \geq r_0$. Let $\theta_l, \lambda_k \in \Lambda$ and let $u_1, \dots, u_m$ be $\mathbf{T}$-homogeneous polynomials with $m \leq s_0$ whose restrictions to W and W^0 are non-zero. Then the three pairs*

$$\Big(W, \sum_l \theta_l \operatorname{div}(h_l) + \sum_k \lambda_k \operatorname{div}(u_k)\Big),$$
$$\Big(W^0, \sum_l \theta_l \operatorname{div}(h_l) + \sum_k \lambda_k \operatorname{div}(u_k)\Big),$$
$$\Big(W^0, \sum_l \theta_l \operatorname{div}(h_l^0) + \sum_k \lambda_k \operatorname{div}(u_k)\Big)$$

are either all lc near 0 or all not lc near 0.

Proof. By [Kaw07, Theorem], each pair is lc near 0 if and only if the pair on $\mathbb{A}^4$ with the hypersurface (W or W^0) added with coefficient one is lc near 0. By (B.23) and (B.24), $xy + g \equiv xy + z^n$ and $h_l \equiv h_l^0$ modulo $(x, y, z, t)^{N_r}$, where $N_r \to +\infty$ as $r \to +\infty$ and N_r depends only on n, c, v, Γ, and r. The number of functions is at most $2/\gamma + 1 + s_0$ and the coefficients belong to the DCC set $\Lambda \cup \{1\}$. Since lc near 0 is detected on the completion at 0 [dFEM11, Proposition 2.11(i)], Lemma 4.14, applied in both directions to any two of the three tuples of equations, implies the claim. □

Proof of Theorem 8.6 continued. Applying Claim B.4 with $\Lambda = \Gamma \backslash \{0\}$ and $m = 0$, we see that (W^0, D^{pr}) and (W^0, D^0) are lc near 0 for large r.

Step 2. In this step, we construct the surface references, the plane reference for $\delta \leq 0$ and the reference near a proper curve for $\delta \geq 0$, and compare them with W^0.

Let $S^0 := (XY + Z^n = 0) \subset \mathbb{A}^3_{X,Y,Z}$, let $\bar{p}, \bar{q}$ be the coordinates of the quasi-étale cover $\mathbb{A}^2_{\bar{p},\bar{q}} \to S^0$, $(\bar{p}, \bar{q}) \to (\bar{p}^n, -\bar{q}^n, \bar{p}\bar{q})$, and let $I_0 := (X^2, Y^2, Z^n)$. For a polynomial u on $\mathbb{A}^4$ write $u^S := u(X, Y, Z, 1)$. Let s be a root $s^v = t$ and let

$$X = xs^A, \qquad Y = ys^A, \qquad Z = zs^B. \tag{B.28}$$

Let $\hat{W} := (xy + z^n = 0) \subset \mathbb{A}^4_{x,y,z,s}$, the product of the normal surface $(xy + z^n = 0) \subset \mathbb{A}^3$ with $\mathbb{A}^1_s$, with the finite morphism $\hat{W} \to W^0$, $(x, y, z, s) \to (x, y, z, s^v)$. Since $XY + Z^n = s^{2A}(xy + z^n)$, (B.28) defines a birational morphism $\hat{W} \to S^0 \times \mathbb{A}^1_s$, $(x, y, z, s) \to (xs^A, ys^A, zs^B, s)$, which is an isomorphism over $s \neq 0$. A polynomial u of $\mathbf{T}$-weight κ satisfies $u = s^\kappa u^S$. Let $\hat{G}_s$ be the group of automorphisms of $\mathbb{A}^4_{x,y,z,s}$ which lift elements of G: an element acting on t by α_4 lifts to $s \to as$ with $a^v = \alpha_4$, so $|\hat{G}_s| = v|G|$. Let H_- be the image of $\hat{G}_s$ in $\operatorname{Aut}(S^0)$. An element of $\hat{G}_s$ acting trivially on X, Y, Z does not exchange X and Y, and acts on (x, y, z, t) by $(a^{-A}, a^{-A}, a^{-B}, a^v)$, that is, it lifts $\mathbf{T}(a) \in G$. Thus the kernel of $\hat{G}_s \to H_-$ consists of the elements $s \to \lambda s$ lifting $\mathbf{T}(\lambda) \in R$, which we call the *canonical lifts* of the elements of R. It has order r, and H_- is a group of order $v[G : R]$ acting faithfully on S^0. Let $\hat{H}_-$ be the group of order $nv[G : R]$ of lifts of H_- to $\mathbb{A}^2_{\bar{p},\bar{q}}$.

Let $\Phi_l := h_l^{0,S}(\bar{p}^n, -\bar{q}^n, \bar{p}\bar{q})$. By (B.24), a monomial $x^{k_1}y^{k_2}z^{k_3}t^{k_4}$ of h_l^0 maps to a monomial of Φ_l of degree

$$n(k_1 + k_2) + 2k_3 = \frac{n}{A}\big(A(k_1 + k_2) + Bk_3\big) = \frac{2(vk_4 - \kappa_l)}{B}. \tag{B.29}$$

In particular $\operatorname{mult}_0 \Phi_l = 2(vn_l - \kappa_l)/B$, and $\sum_l b_l \operatorname{mult}_0 \Phi_l = 2(vb_t - \tau)/B$.

Next we describe the curve reference. Let $U_y := \mathbb{A}^2_{p_y,q_y}$, and consider the rational map $\mathbb{A}^2_{\eta,p_y} \dashrightarrow S^0$ and the morphism $\mathbb{A}^2_{\eta,p_y} \to U_y$ given by

$$X = -\eta^A p_y^n, \qquad Y = \eta^A, \qquad Z = \eta^B p_y, \qquad q_y = \eta^v, \tag{B.30}$$

and define $U_x = \mathbb{A}^2_{p_x,q_x}$ and the parameter η' by exchanging X and Y in (B.30). Let Y_+ be the weighted blow-up of S^0 with weights (A, A, B) on (X, Y, Z), divided by the image $\mu_v \subset H_-$ of the lifts $s \to \zeta s$, $\zeta^v = 1$, of the identity of G, which acts on (X, Y, Z) by $(\zeta^A, \zeta^A, \zeta^B)$, and let $E \subset Y_+$ be the exceptional curve of $Y_+ \to S^0/\mu_v$. The weighted blow-up chart where $Y \neq 0$ is $\mathbb{A}^2_{\eta,p_y}/\mu_A$, and μ_v acts on it by $\eta \to \zeta\eta$, $\zeta^v = 1$. Thus by (B.30), Y_+ is covered near E by the images of U_y and U_x, which are the quotients by $K := \mu_A$ acting by $(p_y, q_y) \to (\zeta^B p_y, \zeta^{-v} q_y)$, $\zeta \in \mu_A$, and E is the image of $(q_y = 0)$ and of $(q_x = 0)$. The group H_-/μ_v acts on Y_+ and preserves E. An element exchanging X and Y exchanges U_y and U_x. Let

$$\varphi^y_l := h^0_l(-p_y^n, 1, p_y, q_y), \qquad \varphi^x_l := h^0_l(1, -p_x^n, p_x, q_x). \tag{B.31}$$

By (B.30), $h^{0,S}_l = \eta^{-\kappa_l}\varphi^y_l(p_y, \eta^v)$ and $\operatorname{mult}_{(q_y=0)} \varphi^y_l = n_l$. Let ξ_x and ξ_y be the points of E which are the images of the origins of U_x and U_y.

Claim B.5. *Let $\theta_l \geq 0$ be real numbers, $\delta' := \sum_l \theta_l\kappa_l + B - v$, $b'_t := \sum_l \theta_l n_l$, $\delta'_- := \max(-\delta', 0)$, and $\mathfrak{i}' := I_0^{\delta'/(2A)}$ if $\delta' \geq 0$ and $\mathfrak{i}' := \mathcal{O}$ if $\delta' \leq 0$. Then the crepant pullback of*

$$\Big(S^0, \sum_l \theta_l \operatorname{div}(h^{0,S}_l), \mathfrak{i}'\Big)/\mu_v$$

to Y_+ is a pair whose pullback to U_y, resp. U_x, is

$$\Big(U_y, \sum_l \theta_l \operatorname{div}(\varphi^y_l) + \frac{\delta'_-}{v}\operatorname{div}(q_y)\Big), \quad \text{resp.} \quad \Big(U_x, \sum_l \theta_l \operatorname{div}(\varphi^x_l) + \frac{\delta'_-}{v}\operatorname{div}(q_x)\Big).$$

The inertia group of E in K has order c and acts faithfully on q_y, so the multiplicity of the boundary of the crepant pullback along E is $1 - (1 - b'_t - \delta'_-/v)/c$. Moreover, the conclusions hold for any $\mathbf{T}$-homogeneous polynomials u_l none of whose monomials is divisible by xy in place of the h^0_l, with the $\mathbf{T}$-weight of u_l, $\operatorname{mult}_{(t=0)} u_l$, $u_l(-p_y^n, 1, p_y, q_y)$, and $u_l(1, -p_x^n, p_x, q_x)$ in place of κ_l, n_l, φ^y_l, and φ^x_l. For $\theta_l = b_l$ and $\delta \geq 0$, we denote the crepant pullback by (Y_+, B_+).

Proof. Let $F_\eta := (\eta = 0)$ on $\mathbb{A}^2_{\eta,p_y}$. A direct calculation from (B.30) shows that the residue form $dX \wedge dZ/X$ of S^0 pulls back to $-A\eta^{B-1} d\eta \wedge dp_y$. By (B.30), I_0 pulls back to (η^{2A}) and $h^{0,S}_l$ to $\eta^{-\kappa_l}\varphi^y_l(p_y, \eta^v)$, whose order along F_η is $vn_l - \kappa_l$. Thus the crepant boundary on $\mathbb{A}^2_{\eta,p_y}$ has multiplicity

$$1 - B + \sum_l \theta_l(vn_l - \kappa_l) + \max(\delta', 0) = 1 - v(1 - b'_t) + \delta'_-$$

along F_η, and the other components of the crepant boundary are those of $\sum_l \theta_l \operatorname{div}(\varphi^y_l/q_y^{n_l})$. The quotient by μ_v, $q_y = \eta^v$, is ramified of index v along F_η, so the multiplicity of the crepant boundary along $(q_y = 0)$ is $b'_t + \delta'_-/v$ [Liu26, Lemma 2.6(4)]. Thus the first statement holds for U_y, and U_x is obtained by exchanging X and Y. An element $\zeta \in K$ fixes $(q_y = 0)$ pointwise if and only if $\zeta^B = 1$, that is, $\zeta^{\gcd(A,B)} = \zeta^c = 1$, and it acts on q_y by ζ^{-v}, faithfully since $\gcd(c, v) = 1$. Thus the multiplicity along E is $1 - (1 - b'_t - \delta'_-/v)/c$ by [Liu26, Lemma 2.6(4)]. For polynomials u_l as in the claim, u_l^S pulls back to $\eta^{-\kappa}u_l(-p_y^n, 1, p_y, \eta^v)$ for the $\mathbf{T}$-weight κ of u_l, and the order of $u_l(-p_y^n, 1, p_y, q_y)$ along $(q_y = 0)$ is $\operatorname{mult}_{(t=0)} u_l$, since the coefficient of the least power of t in u_l is $\mathbf{T}$-homogeneous and has no monomial divisible by xy, so it is not a multiple of $xy + z^n$ and does not vanish identically at $(-p_y^n, 1, p_y)$. Thus the computation above holds for the u_l. □

Proof of Theorem 8.6 continued. We define the reference values

$$\beta_- := \operatorname{mld}\big((S^0 \ni 0, \textstyle\sum_l b_l \operatorname{div}(h^{0,S}_l))/H_-\big) \quad (\delta \leq 0),$$
$$\beta_+ := \operatorname{mld}\big((S^0 \ni 0, \textstyle\sum_l b_l \operatorname{div}(h^{0,S}_l), I_0^{\delta/(2A)})/H_-\big) \quad (\delta \geq 0).$$

For $\delta = 0$ they are equal. Since any prime divisor over S^0/μ_v with center the image of 0 has center contained in E on Y_+, Claim B.5 implies that β_+ is the minimum of the log discrepancies of the crepant quotient of (Y_+, B_+) by H_-/μ_v over the prime divisors with center contained

in the image of E. The only such prime divisor with center a curve is the image of E, and a general closed point of E is not the center of a computing divisor, since there the minimal log discrepancy of the quotient is the log discrepancy of the image of E plus one. Let $\beta := \beta_-$ if $\delta \le 0$ and $\beta := \beta_+$ if $\delta \ge 0$.

The following claim compares lc near 0 on W^0 with the references. For a $\mathbf{T}$-homogeneous polynomial u we write $\kappa(u)$ for the $\mathbf{T}$-weight of u.

Claim B.6. *Let $u_1, \dots, u_m$ be $\mathbf{T}$-homogeneous polynomials on $\mathbb{A}^4$ with $u_k|_{W^0} \ne 0$ and $u_k^S|_{S^0} \ne 0$ for any k, let $\theta_k > 0$, $C := \sum_k \theta_k \operatorname{div}(u_k)$ on W^0, $C_S := \sum_k \theta_k \operatorname{div}(u_k^S)$ on S^0, and $\delta_C := \sum_k \theta_k \kappa(u_k) + B - v$. Let $\mathfrak{i}_C := \mathcal{O}$ if $\delta_C \le 0$ and $\mathfrak{i}_C := I_0^{\delta_C/(2A)}$ if $\delta_C \ge 0$. Then (W^0, C) is lc near 0 if and only if $(S^0, C_S, \mathfrak{i}_C)$ is lc near 0. Moreover, if (W^0, C) is lc near 0, then (S^0, C_S) is lc near 0.*

Proof. The morphism $\hat{W} \to W^0$ is finite, x, y, z, s are regular on $\hat{W}$, and the origin is the only point of $\hat{W}$ over 0.

Let P be a prime divisor over W^0, invariant under $\mathbf{T}$, with $0 \in \operatorname{center}_{W^0} P$, and let P_s be a prime divisor over $\hat{W}$ over P. Since the lift of $\mathbf{T}$ acting by $s \to \lambda s$ is connected and permutes the finitely many prime divisors over P, P_s is invariant under it. Through the birational morphism $\hat{W} \to S^0 \times \mathbb{A}^1_s$, P_s is also a prime divisor over $S^0 \times \mathbb{A}^1_s$. Let $\pi : S^0 \times \mathbb{A}^1_s \to S^0$ be the projection. The lifted torus acts trivially on X, Y, Z, so the terms $f_j s^j$ of a polynomial $\sum_j f_j s^j$, $f_j \in \mathbb{C}[X, Y, Z]$, are linear combinations of the translates of $\sum_j f_j s^j$ under the lifted torus, and

$$\operatorname{mult}_{P_s}\Big(\sum_j f_j s^j\Big) = \min_{f_j|_{S^0} \ne 0} \big(\operatorname{mult}_{P_s}(f_j \circ \pi) + j\kappa_s\big), \qquad \kappa_s := \operatorname{mult}_{P_s} s \ge 0. \tag{B.32}$$

The restriction of $\operatorname{mult}_{P_s}$ to the non-zero rational functions pulled back from S^0 is not identically zero: otherwise $\operatorname{mult}_{P_s} x = \operatorname{mult}_{P_s} X - A\kappa_s = -A\kappa_s$ by (B.28), so $\kappa_s = 0$ since x is regular on $\hat{W}$, and then, by (B.32), $\operatorname{mult}_{P_s}$ would vanish on all the polynomials $\sum_j f_j s^j$ with some $f_j|_{S^0} \ne 0$, hence on all the non-zero rational functions on $\hat{W}$, since $\hat{W}$ is birational to $S^0 \times \mathbb{A}^1_s$. Thus there are a positive integer ϱ and a prime divisor ν over S^0 with $\operatorname{mult}_{P_s}(f \circ \pi) = \varrho \operatorname{mult}_\nu f$ for any $f \in \mathbb{C}[X, Y, Z]$ with $f|_{S^0} \ne 0$ [Liu26, Lemma 2.12]. Thus

$$\operatorname{mult}_{P_s}\Big(\sum_j f_j s^j\Big) = \min_{f_j|_{S^0} \ne 0} (\varrho \operatorname{mult}_\nu f_j + j\kappa_s),$$

that is, $\operatorname{mult}_{P_s}$ is the Gauss extension [Liu26, Definition-Proposition 2.11] of $\varrho \operatorname{mult}_\nu$ with s-value κ_s. For any $h \in \mathbb{C}[X, Y, Z]$ with $h(0) \ne 0$, the function $h(xs^A, ys^A, zs^B)$ is a unit on $\hat{W}$ near the origin, and $\operatorname{center}_{\hat{W}} P_s$ contains the origin, so $\operatorname{mult}_{P_s}(h \circ \pi) = 0$. Thus $0 \in \operatorname{center}_{S^0} \nu$.

By (B.28), the residue form of W^0 satisfies

$$\frac{dx \wedge dz \wedge dt}{x} = v s^{v-B} \frac{dX \wedge dZ}{X} \wedge \frac{ds}{s},$$

and $u_k = s^{\kappa(u_k)} u_k^S$. Thus the crepant pullback of (W^0, C) to $\hat{W}$ and $(S^0 \times \mathbb{A}^1_s, (1+\delta_C)\operatorname{div}(s) + C_S)$ are crepant through the birational morphism $\hat{W} \to S^0 \times \mathbb{A}^1_s$. On a model S' of S^0 on which ν is a prime divisor with local equation e at the generic point of ν, $\operatorname{mult}_{P_s}$ is the monomial valuation in (e, s) with weights (ϱ, κ_s), which are coprime as $\operatorname{mult}_{P_s}$ takes the value one. Thus the log discrepancy of P_s over $S' \times \mathbb{A}^1_s$ is $\varrho + \kappa_s$, and the relative canonical divisor of S' over S^0 contributes $\varrho(a(\nu, S^0, 0) - 1)$. Thus the log discrepancy of P_s over $S^0 \times \mathbb{A}^1_s$ is $\varrho a(\nu, S^0, 0) + \kappa_s$. Moreover $\operatorname{mult}_{P_s}(u_k^S \circ \pi) = \varrho \operatorname{mult}_\nu u_k^S$. Let

$$m_P := \min(\operatorname{mult}_{P_s} x/A, \operatorname{mult}_{P_s} y/A, \operatorname{mult}_{P_s} z/B) \ge 0.$$

Since $\varrho \operatorname{mult}_\nu X = \operatorname{mult}_{P_s} x + A\kappa_s$ and similarly for Y, Z, we have $\varrho \operatorname{mult}_\nu I_0/(2A) = m_P + \kappa_s$. Since $a(P_s, \hat{W}, \cdot)$ is a positive multiple of $a(P, W^0, C)$ [Liu26, Lemma 2.6(2)], we obtain that

$a(P, W^0, C)$ is a positive multiple of

$$\varrho a(\nu, S^0, C_S) - \delta_C \kappa_s = \varrho a(\nu, S^0, C_S, \mathfrak{i}_C) + \max(\delta_C, 0) m_P - \min(\delta_C, 0)\kappa_s. \tag{B.33}$$

Assume that $(S^0, C_S, \mathfrak{i}_C)$ is lc near 0 and that (W^0, C) is not lc near 0. On a $\mathbf{T}$-equivariant log resolution of (W^0, C) [Kol07, Proposition 3.9.1 and Theorem 3.35] any prime divisor is invariant under $\mathbf{T}$ since $\mathbf{T}$ is connected, so there exists a $\mathbf{T}$-invariant P with $0 \in \operatorname{center}_{W^0} P$ and $a(P, W^0, C) < 0$. The right side of (B.33) is non-negative, a contradiction.

Conversely, assume that (W^0, C) is lc near 0, and let ν be a prime divisor over S^0 with $0 \in \operatorname{center}_{S^0} \nu$. Let $m_\nu := \operatorname{mult}_\nu I_0/(2A)$, and let ϱ be a positive integer with $\varrho m_\nu \in \mathbb{Z}$. For $\kappa \in \{0, \varrho m_\nu\}$, let P_s be the prime divisor over $S^0 \times \mathbb{A}^1_s$ whose valuation is c_κ times the Gauss extension of $\varrho \operatorname{mult}_\nu$ with s-value κ for a positive rational number c_κ, and let P be the induced prime divisor over W^0. Then $\operatorname{mult}_{P_s}(f \circ \pi) = c_\kappa \varrho \operatorname{mult}_\nu f$ and $\operatorname{mult}_{P_s} s = c_\kappa \kappa$. Thus (B.33) holds for P_s with $c_\kappa \varrho$ and $c_\kappa \kappa$ in place of ϱ and κ_s, and each side is c_κ times the corresponding side of (B.33) with ϱ and $\kappa_s = \kappa$, which we consider below. Moreover $\operatorname{mult}_{P_s} x, \operatorname{mult}_{P_s} y, \operatorname{mult}_{P_s} z, \operatorname{mult}_{P_s} t \geq 0$. Writing a polynomial h in x, y, z, t with $h(0) \neq 0$ as a Laurent polynomial in s with coefficients in $\mathbb{C}[X, Y, Z]$, the coefficient of s^0 has constant term $h(0)$, so $\operatorname{mult}_{P_s} h \leq 0$, and $0 \in \operatorname{center}_{W^0} P$. For $\kappa = 0$ the left side of (B.33) is $\varrho a(\nu, S^0, C_S)$, and for $\kappa = \varrho m_\nu$ and $\delta_C \geq 0$ we have $m_P = 0$ and the right side of (B.33) is $\varrho a(\nu, S^0, C_S, \mathfrak{i}_C)$. Thus both are non-negative. □

Proof of Theorem 8.6 continued. By **Step 1** and Claim B.6 with $u_k = h^0_l$ and $\theta_k = b_l$, for large r the reference pair defining β is lc near 0. By Claim B.5, if $\delta \geq 0$, then (Y_+, B_+) is lc near E, so $b_t \leq 1$, and the log discrepancy of the image of E implies $\beta_+ \leq (1 - b_t)/c \leq 1$. Since S^0 is a surface, $\beta_- \leq 2$.

The crepant quotients defining β are lc surface pairs whose boundary coefficients belong to the DCC set [HMX14, Proposition 3.4.1(3)]

$$\{1 - (1 - s')/m' \mid s' \in \Gamma_+ \cap [0, 1], m' \in \mathbb{Z}_{\geq 1}\},$$

since each coefficient of a crepant quotient is of the form $1 - (1 - s')/m'$ [Liu26, Lemma 2.6(4)]. By Theorem 2.9(1), the values β_- belong to an ACC set depending only on Γ, and so do the values β_+, as each is the minimum of the log discrepancy of the image of E, which is one minus a coefficient in the DCC set above, and of minimal log discrepancies at closed points. By Theorem 2.9(2), β is computed by a prime divisor ν_0 over the reference quotient surface with $a(\nu_0, \cdot, 0) \leq l_0$, l_0 depending only on Γ, unless $\delta \geq 0$ and ν_0 is the image of E. Lifting ν_0 to $\mathbb{A}^2_{\bar{p},\bar{q}}$, resp. to U_y or U_x (or to the normalization of $U_y \times_{Y_+} U_x$ at a closed point of E other than ξ_x and ξ_y), the stabilizer of the center has bounded order, since $|H_-| = v[G : R] \leq v e_0$ and $|K| = A$. Thus the log discrepancy of the lift over the smooth chart, the orders of the coordinates and the orders of Φ_l, resp. φ^y_l, φ^x_l, along the lift are bounded, using $b_l \geq \gamma$.

Step 3. In this step, we compare prime divisors over two hypersurfaces $(x_1 x_2 + g_1 = 0)$ and $(x_1 x_2 + x_3^n = 0)$ in $\mathbb{A}^4$, and we prove the upper bound $\alpha \leq \beta + C_{\mathrm{up}}/r$ for a bounded constant C_{up}.

By the formula at the end of [Liu26, Lemma 2.6], α is the infimum of $a(P, W, D)/|I_G(P)|$ over the prime divisors P over W with center 0.

Claim B.7. *Let H be a finite group acting linearly on $\mathbb{A}^4_{x_1,x_2,x_3,x_4}$, monomially on $\{x_1, x_2\}$ and diagonally on x_3, x_4, let $g_1 \in \mathbb{C}[x_3, x_4]$, and assume that $x_1 x_2 + g_1$ and $x_1 x_2 + x_3^n$ are H-semi-invariant with equal characters and define hypersurfaces V_1 and V_2 which are normal near 0. We replace V_1 and V_2 by H-stable open neighborhoods of 0 which are normal varieties. Let V' be the normalization of the union of the irreducible components of $(\rho^2 g_1 = x_3^n) \subset V_1 \times \mathbb{A}^1_\rho$ dominating V_1, with H acting on V' and fixing ρ, and let $\pi_1 : V' \to V_1$ be the projection. (V' is irreducible unless x_3^n/g_1 is the square of a rational function on V_1, and the argument below does not use irreducibility.) Then $x_1^\circ := \rho x_1$, $x_2^\circ := \rho x_2$, x_3, x_4 satisfy $x_1^\circ x_2^\circ + x_3^n = 0$, and*

$$\phi : V' \to V_2, \qquad (x_1, x_2, x_3, x_4, \rho) \to (x_1^\circ, x_2^\circ, x_3, x_4),$$

is an H-equivariant dominant morphism. Through (ϕ,ρ), V' is birational to the normalization of the union of the irreducible components of $(\rho^2 g_1 = x_3^n) \subset V_2 \times \mathbb{A}^1_\rho$ dominating V_2. Let ν_1 be a prime divisor over V_1 with $\operatorname{mult}_{\nu_1}(g_1 - x_3^n) > n \operatorname{mult}_{\nu_1} x_3$, let ν' be a prime divisor over V' mapping onto ν_1 with $\operatorname{mult}_{\nu'}(\rho-1) > 0$, and let ν_2 be the prime divisor over V_2 induced by ν' through ϕ. Then the following hold.

(1) The prime divisor ν' exists and is unique, and

$$\operatorname{mult}_{\nu'}(f\circ\pi_1) = \operatorname{mult}_{\nu_1} f, \qquad \operatorname{mult}_{\nu'}(f'\circ\phi) = \operatorname{mult}_{\nu_2} f'$$

for any non-zero rational functions f on V_1 and f' on V_2, and

$$\operatorname{mult}_{\nu'}(\rho-1) = \operatorname{mult}_{\nu_1}(g_1 - x_3^n) - n\operatorname{mult}_{\nu_1} x_3.$$

(2) $a(\nu_1, V_1, 0) = a(\nu_2, V_2, 0)$, and the decomposition and inertia groups of ν_1 and ν_2 in H are equal.

(3) Assume that $\operatorname{mult}_{\nu_1} x_i \geq 0$ for any i. Then $\operatorname{mult}_{\nu_2} h(x_1^\circ, x_2^\circ, x_3, x_4) = \operatorname{mult}_{\nu_1} h$ for any polynomial h in $x_1,\dots,x_4$ with $\operatorname{mult}_{\nu_1} h < \operatorname{mult}_{\nu'}(\rho-1)$.

Statements (1)–(3) also hold with the roles of V_1 and V_2 exchanged, with ρ^{-1} in place of ρ in the proof below.

Proof. Since $\rho^2 g_1 = x_3^n$ on V', we have $x_1^\circ x_2^\circ = \rho^2 x_1 x_2 = -\rho^2 g_1 = -x_3^n$, so ϕ maps V' to V_2. Since $x_1 = x_1^\circ/\rho$ and $x_2 = x_2^\circ/\rho$, (ϕ,ρ) is birational onto the union of the irreducible components of $(\rho^2 g_1 = x_3^n) \subset V_2 \times \mathbb{A}^1_\rho$ dominating V_2. Since $x_1x_2 + g_1$ and $x_1x_2 + x_3^n$ have equal characters, g_1 and x_3^n have the character of x_1x_2, and the function $\rho^2 = x_3^n/g_1$ is H-invariant. Thus H acts on V' fixing ρ, x_1°, x_2° have the characters (and the exchange) of x_1, x_2, and ϕ is H-equivariant. Since

$$\operatorname{mult}_{\nu_1}(\rho^2-1) = \operatorname{mult}_{\nu_1}(g_1 - x_3^n) - n\operatorname{mult}_{\nu_1} x_3 > 0,$$

on a model V_1'' of V_1 on which ν_1 is a prime divisor E_1, the function ρ^2 is regular near the generic point of E_1, and $\rho^2|_{E_1} = 1$. Thus there is a neighborhood $N_1 \subset V_1''$ of the generic point of E_1 over which the double cover $\rho^2 = x_3^n/g_1$ of V_1'' is étale, and over N_1 the preimage of E_1 is $(T^2 = 1) \subset E_1 \times \mathbb{A}^1_T$, $T = \rho$, the disjoint union of two prime divisors $(\rho = 1)$ and $(\rho = -1)$, each mapping isomorphically onto an open subset of E_1. The function $\rho - 1$ vanishes along $(\rho = 1)$ and not along $(\rho = -1)$, since $(\rho-1)|_{(\rho=-1)} = -2$. Thus ν' exists and is the prime divisor $(\rho = 1)$, $\operatorname{mult}_{\nu'}(f\circ\pi_1) = \operatorname{mult}_{\nu_1} f$ for any non-zero rational function f on V_1, and, since $\rho+1$ is a unit at the generic point of ν',

$$\operatorname{mult}_{\nu'}(\rho-1) = \operatorname{mult}_{\nu'}(\rho^2-1) = \operatorname{mult}_{\nu_1}(\rho^2-1).$$

Since ρ^2 is also a rational function on V_2 and $\operatorname{mult}_{\nu'}(\rho^2-1) > 0$, we have $\operatorname{mult}_{\nu_2}(\rho^2-1) > 0$, and the argument above, on a model of V_2 on which ν_2 is a prime divisor, shows that ϕ is étale at the generic point of ν', that $\operatorname{mult}_{\nu'}(f'\circ\phi) = \operatorname{mult}_{\nu_2} f'$ for any non-zero rational function f' on V_2, and that ν' is the unique prime divisor over V' mapping onto ν_2 with $\operatorname{mult}_{\nu'}(\rho-1) > 0$. Thus (1) holds. The residue forms satisfy

$$\frac{dx_1^\circ}{x_1^\circ} \wedge dx_3 \wedge dx_4 = \frac{dx_1}{x_1} \wedge dx_3 \wedge dx_4,$$

since $d\rho/\rho$ is a combination of dx_3 and dx_4. Since π_1 and ϕ are étale at the generic point of ν', $a(\nu_i, V_i, 0)$ is one plus the order along ν' of the pullback of the residue form of V_i for $i = 1, 2$, so $a(\nu_1, V_1, 0) = a(\nu_2, V_2, 0)$. Any $\iota \in H$ fixes ρ, so $\iota\nu'$ is the prime divisor over V' mapping onto $\iota\nu_1$, resp. $\iota\nu_2$, along which $\rho - 1$ vanishes. By the uniqueness in (1) over ν_1 and over ν_2, ι fixes ν_1 if and only if it fixes ν', if and only if it fixes ν_2. For ι in the decomposition group of ν_1, which is that of ν' and of ν_2, the centers of ν_1, ν', and ν_2 on the models above are birational to each other through π_1 and ϕ, compatibly with ι, so ι acts trivially on one of them if and only if it acts trivially on the others, and the inertia groups are equal. For (3), each monomial m satisfies $m(x_1^\circ, x_2^\circ, x_3, x_4) - m = m\cdot(\rho^{e_m} - 1)$ for some $e_m \geq 0$, and $\operatorname{mult}_{\nu'}(m\cdot(\rho^{e_m}-1)) \geq \operatorname{mult}_{\nu'}(\rho-1)$ since $\operatorname{mult}_{\nu'} m \geq 0$. □

Proof of Theorem 8.6 continued. Assume first that $\delta \le 0$. Let $w := s^r$, and let

$$V_r := \Big(XY + Z^n + \sum_{j\le n-2} \tilde{a}_j(w) Z^j = 0\Big) \subset \mathbb{A}^4_{X,Y,Z,w}, \tag{B.34}$$

where

$$\tilde{a}_j(w) := \sum_\sigma a_{j\sigma} w^{(\sigma v+(n-j)B)/r} \quad \text{and} \quad a_j(t) = \sum_\sigma a_{j\sigma} t^\sigma.$$

By (B.23), $\tilde{a}_j \in w\mathbb{C}[w]$, and by (B.28), $s^{2A}(xy+g) = XY + Z^n + \sum_j \tilde{a}_j(w) Z^j$. The canonical lifts of the elements of R act trivially on X, Y, Z, w, so H_- acts on V_r, and V_r is birational, through $(x, y, z, s) \to (X, Y, Z, s^r)$, to the quotient by the canonical lifts of the elements of R of the normalization W_s of $(xy + g(z, s^v) = 0) \subset \mathbb{A}^4_{x,y,z,s}$. For any l let

$$\bar{h}_l := \sum_{\ell\ge 0} w^\ell h_l^{(\ell),S},$$

written in X, Y, Z, w. The polynomial $\bar{h}_l$ is H_--semi-invariant, and $\bar{h}_l|_{w=0} = h_l^{0,S}$. The map $(x, y, z, s) \to (X, Y, Z, w)$ is the quotient by the free action of μ_r on $(s \ne 0)$, so a singular point of $V_r \cap (w \ne 0)$ near 0 has $X = Y = 0$ and corresponds to a singular point $(0, 0, z, t)$ of W, where (z, t) is a singular point of $(g = 0)$ and $t = s^v \ne 0$ is small. Since g is monic in z and $a_j(0) = 0$, z is then small, which contradicts the isolatedness of the singularity of W at 0. Thus $V_r \backslash S^0$ is smooth near 0, where $S^0 = V_r \cap (w = 0)$. Since S^0 is a Cartier divisor on V_r which is regular outside 0, V_r is regular along $S^0\backslash\{0\}$. Thus V_r is normal near 0, the different of (V_r, S^0) on S^0 is zero, (V_r, S^0) is lc near 0 by [Kaw07, Theorem], and any prime divisor Q over V_r whose center contains 0 satisfies $a(Q, V_r, 0) \ge \operatorname{mult}_Q S^0 > 0$ if $\operatorname{center} Q \subset S^0$, and $a(Q, V_r, 0) \ge 1$ otherwise. Thus V_r is klt near 0.

Claim B.8. *Let P be a prime divisor over W with center 0 and let P_V be the induced prime divisor over V_r. Assume that $\operatorname{mult}_P(h_l - h_l^{(\ge 0)}) > \operatorname{mult}_P h_l^{(\ge 0)}$ for any l, where $h_l^{(\ge 0)} := \sum_{\ell\ge 0} h_l^{(\ell)}$. Then*

$$\frac{a(P, W, D)}{|I_G(P)|} = \frac{a\big(P_V, V_r, (1+\delta/r)\operatorname{div}(w) + \sum_l b_l \operatorname{div}(\bar{h}_l)\big)}{|I_{H_-}(P_V)|}. \tag{B.35}$$

Proof. By (B.28), the residue forms satisfy

$$\frac{dx \wedge dz \wedge dt}{x} = \frac{v}{r} s^{v-B} \frac{dX \wedge dZ}{X} \wedge \frac{dw}{w},$$

and $h_l = s^{\kappa_l} \sum_\ell w^\ell h_l^{(\ell),S}$. Thus, as crepant quotients [Liu26, Lemma 2.6(2)(5)],

$$(W, D)/G = \Big(V_r, (1+\delta/r)\operatorname{div}(w) + \sum_l b_l \operatorname{div}(s^{-\kappa_l} h_l)\Big)/H_-,$$

and the assumption implies $\operatorname{mult}_{P_V}(s^{-\kappa_l} h_l) = \operatorname{mult}_{P_V} \bar{h}_l$. □

Proof of Theorem 8.6 continued. Let ν be a prime divisor over S^0 with center 0 computing β_- with the bounded data of **Step 2**, $e := |I_{H_-}(\nu)|$, and M_1 a bounded integer larger than $n \operatorname{mult}_\nu Z + \operatorname{mult}_\nu h_l^{0,S}$ for any l. Let k be the least integer larger than M_1 in the congruence class modulo e of Lemma 3.1 for ν and the character of w, so that $k \le M_1 + e$ is bounded. Let $V^0 := S^0 \times \mathbb{A}^1_w = (XY + Z^n = 0) \subset \mathbb{A}^4_{X,Y,Z,w}$. By Proposition 5.4, applied to $S = S^0$, the coordinate w in place of u, the group H_-, which acts faithfully on S^0, the coefficient denoted b there equal to $1 + \delta/r \le 1$, the functions denoted Φ_j there taken to be $\bar{h}_l$ with coefficients b_l, and the integer denoted ℓ there equal to k, the Gauss extension $\hat{\nu}$ of mult_ν with w-value k satisfies $I_{H_-}(\hat{\nu}) = I_{H_-}(\nu)$, $\operatorname{mult}_{\hat{\nu}} \bar{h}_l = \operatorname{mult}_\nu h_l^{0,S}$, and

$$\frac{a\big(\hat{\nu}, V^0, (1+\delta/r)\operatorname{div}(w) + \sum_l b_l \operatorname{div}(\bar{h}_l)\big)}{e} = \beta_- - \frac{\delta k}{re}.$$

Since $\tilde{a}_j \in (w)$, for $g_V := Z^n + \sum_j \tilde{a}_j(w) Z^j$ and any l we have

$$\operatorname{mult}_{\hat{\nu}}(g_V - Z^n) - n \operatorname{mult}_{\hat{\nu}} Z \geq k - n \operatorname{mult}_{\nu} Z > \operatorname{mult}_{\hat{\nu}} \bar{h}_l.$$

Thus Claim B.7, with $V_1 = V_r$ and $V_2 = V^0$ and starting from $\hat{\nu}$ over V_2, yields a prime divisor $\hat{\nu}'$ over V_r with $a(\hat{\nu}', V_r, 0) = a(\hat{\nu}, V^0, 0)$, $\operatorname{mult}_{\hat{\nu}'} \bar{h}_l = \operatorname{mult}_{\hat{\nu}} \bar{h}_l$ for any l, and $I_{H_-}(\hat{\nu}') = I_{H_-}(\hat{\nu})$. Since $\operatorname{mult}_{\hat{\nu}'} X, \operatorname{mult}_{\hat{\nu}'} Y, \operatorname{mult}_{\hat{\nu}'} Z \geq 1$, by (B.24) the terms $w^\ell h_l^{(\ell),S}$ with $\ell < 0$ have order at least $|\ell|(r/\max(A,B) - k) - |\kappa_l|$ along $\hat{\nu}'$. Thus, for large r, the assumption of Claim B.8 holds for the prime divisor P over W induced by $\hat{\nu}'$. Along a prime divisor over W_s mapping onto $\hat{\nu}'$, the orders of x, y, z, t are positive multiples of $\operatorname{mult}_{\hat{\nu}'} X - Ak/r$, $\operatorname{mult}_{\hat{\nu}'} Y - Ak/r$, $\operatorname{mult}_{\hat{\nu}'} Z - Bk/r$, and vk/r, all positive for $r > \max(A,B)k$, so P has center 0. By (B.35), $\alpha \leq \beta_- - \delta k/(re)$.

Assume next that $\delta \geq 0$, and let ν be a prime divisor over Y_+ computing β_+ with the bounded data of **Step 2**. Let $H_y \subset \operatorname{Aut}(U_y)$ be the group generated by K and the lifts of the elements of H_-/μ_v not exchanging X and Y, and define H_x similarly. Let $\tilde{\eta} := \eta/s$ with η as in (B.30), and similarly $\tilde{\eta}' := \eta'/s$ on U_x, so that $x = -\tilde{\eta}^A p_y^n$, $y = \tilde{\eta}^A$, $z = \tilde{\eta}^B p_y$, and $t = \tilde{\eta}^{-v} q_y$ on W^0 by (B.28) and (B.30). The canonical lift $s \to \lambda s$ of $\mathbf{T}(\lambda) \in R$ acts by $\tilde{\eta} \to \lambda^{-1}\tilde{\eta}$ and trivially on U_y and on $u := \tilde{\eta}^r$, and by the residue form of Claim B.5, the crepant quotients of (W^0, D^{pr}) by G and of

$$\left(U_y \times \mathbb{A}^1_u, \left(1 - \frac{\delta}{r}\right) \operatorname{div}(u) + \sum_l b_l \operatorname{div}(\varphi_l^y)\right) \tag{B.36}$$

by H_y have equal log discrepancies over the points over center ν (an element exchanging x and y exchanges ξ_x and ξ_y, and the closed points of E fixed by such an element are treated with the chart replaced below, so such an element does not fix center ν pointwise and belongs to no inertia group in G of a prime divisor used below, whose center maps onto center ν), and similarly for U_x [Liu26, Lemma 2.6(2)(5)]. At a closed point of E other than ξ_x and ξ_y whose stabilizer contains an exchange, we replace U_y with the normalization of $U_y \times_{Y_+} U_x$, which is étale over U_y and U_x there because the transition $(\eta'/\eta)^A = -p_y^n$ is a unit, and u with a generator of the ideal $(\tilde{\eta}^r)$ semi-invariant under the stabilizer K' of the point. Such a generator exists. The ideal $(\tilde{\eta}^r)$ of the local ring $\mathcal{O}'$ at the point is invariant under K', and for the maximal ideal $\mathfrak{m}$ of $\mathcal{O}'$, $(\tilde{\eta}^r)/\mathfrak{m}(\tilde{\eta}^r)$ is a one-dimensional representation of K' with a character χ. Since K' is finite,

$$u' := \frac{1}{|K'|} \sum_{\iota \in K'} \chi(\iota)^{-1} \iota^*(\tilde{\eta}^r) \in (\tilde{\eta}^r)$$

is semi-invariant with the character χ, and $u' - \tilde{\eta}^r \in \mathfrak{m}(\tilde{\eta}^r)$. Thus u' generates $(\tilde{\eta}^r)$ by Nakayama's lemma, and u' differs from $\tilde{\eta}^r$ by a unit, so the orders along prime divisors and the Gauss extension values used below do not change when $\tilde{\eta}^r$ is replaced with u'. Similarly, the ideal (φ_l^y) of $\mathcal{O}'$ is K'-invariant, since an element exchanging x and y carries φ_l^y to a constant multiple of φ_l^x and $\varphi_l^y/\varphi_l^x = (\eta/\eta')^{\kappa_l}$ is a unit there. For the character χ_l of K' on $(\varphi_l^y)/\mathfrak{m}(\varphi_l^y)$, the function

$$\frac{1}{|K'|} \sum_{\iota \in K'} \chi_l(\iota)^{-1} \iota^* \varphi_l^y$$

is a K'-semi-invariant generator of (φ_l^y) which differs from φ_l^y by a unit, and on the replaced chart we replace φ_l^y with it.

Let M_1 be a bounded integer larger than the orders of φ_l^y along the lift of ν for any l. If center ν is a closed point o' of E, let ν also denote a lift over a point of U_y (or of the replaced chart) over o', $H_{o'}$ the stabilizer, and $e := |I_{H_{o'}}(\nu)|$. Let k be the least integer larger than M_1 in the congruence class of Lemma 3.1 for ν and the character of u, so $k \leq M_1 + e$. By Proposition 5.4, applied to the chart (or the replaced chart), the coordinate u, the group $H_{o'}$, which acts faithfully on the chart, the coefficient denoted b there equal to $1 - \delta/r \leq 1$, the functions denoted Φ_j there taken to be φ_l^y (on the replaced chart, their K'-semi-invariant replacements) with coefficients b_l, and the integer denoted ℓ there equal to k, the Gauss extension $\hat{\nu}$ of mult_ν with u-value k

satisfies $I(\hat\nu) = I(\nu)$, $\operatorname{mult}_{\hat\nu} \varphi_l^y = \operatorname{mult}_\nu \varphi_l^y$, and the quotient log discrepancy of $\hat\nu$ for the pair of (B.36) is $\beta_+ + \delta k/(re)$.

If ν is the image of E, let $I(E)$ be the inertia group of $(q_y = 0)$ in H_y and $e := |I(E)|$. The group $I(E)$ is cyclic and acts on q_y by a faithful character λ_E [Liu26, Lemma 2.6(1)], so the character of u restricted to $I(E)$ is $\lambda_E^{k_0}$ for an integer k_0, and we let k be the least integer larger than M_1 with $k \equiv k_0$ modulo e. Let $\hat\nu$ be the Gauss extension of $\operatorname{mult}_{(q_y=0)}$ with u-value k. The prime divisor $\hat\nu$ is birational to $(q_y = 0) \times \mathbb{A}^1$, where the coordinate of $\mathbb{A}^1$ is the restriction of u/q_y^k to $\hat\nu$. The group $I(E)$ fixes $(q_y = 0)$ pointwise and acts trivially on the restriction of u/q_y^k by the choice of k, so $I(E) \subset I(\hat\nu)$. Conversely, an element of $I(\hat\nu)$ fixes pointwise the center $(q_y = u = 0)$ of $\hat\nu$ on $U_y \times \mathbb{A}^1_u$, whose projection to U_y is $(q_y = 0)$, so $I(\hat\nu) = I(E)$. Moreover $\operatorname{mult}_{\hat\nu} \varphi_l^y = n_l$, and $a(\hat\nu) = a((q_y = 0), U_y, 0) + k$, so the quotient log discrepancy of $\hat\nu$ is $(1 - b_t)/e + \delta k/(re) = \beta_+ + \delta k/(re)$.

In both cases $\operatorname{mult}_{\hat\nu} q_y > 0$ and $\operatorname{mult}_{\hat\nu} \tilde\eta = k/r$, so the values of x, y, z, t along $\hat\nu$ are

$$n \operatorname{mult}_{\hat\nu} p_y + \frac{Ak}{r}, \qquad \frac{Ak}{r}, \qquad \operatorname{mult}_{\hat\nu} p_y + \frac{Bk}{r}, \qquad \operatorname{mult}_{\hat\nu} q_y - \frac{vk}{r},$$

all positive for large r, so $\hat\nu$ induces a prime divisor P^0 over W^0 with center 0. By (B.23), for a monomial $t^\sigma z^j$ of $g - z^n$,

$$\operatorname{mult}_{\hat\nu}(t^\sigma z^j) - n \operatorname{mult}_{\hat\nu} z = N_{j,\sigma}\Big(\frac{r}{v} \operatorname{mult}_{\hat\nu} q_y - k\Big) - (n - j)\Big(\frac{B}{v} \operatorname{mult}_{\hat\nu} q_y + \operatorname{mult}_{\hat\nu} p_y\Big), \tag{B.37}$$

which tends to $+\infty$ uniformly as $r \to +\infty$, since $N_{j,\sigma} \geq 1$, $\operatorname{mult}_{\hat\nu} q_y \geq 1$, and the other values are bounded. A term of $h_l^{(\ell)}$ with $\ell > 0$ is $\tilde\eta^{-\kappa_l - \ell r}$ times a monomial in p_y, q_y of q_y-degree at least $(\kappa_l + \ell r)/v$ by (B.24), so the value of the term along $\hat\nu$ exceeds $\operatorname{mult}_{\hat\nu} h_l^0$ by at least

$$\ell\Big(\frac{r}{v} \operatorname{mult}_{\hat\nu} q_y - k\Big) - \frac{|\kappa_l| \operatorname{mult}_{\hat\nu} q_y}{v} - M_1,$$

which is positive for large r. A term with $\ell < 0$ is $u^{|\ell|}\tilde\eta^{-\kappa_l}$ times a polynomial in p_y, q_y, and the value of the term exceeds $\operatorname{mult}_{\hat\nu} h_l^0$ by at least $|\ell|k - M_1 > 0$. Thus Claim B.7, with $V_1 = W$ and $V_2 = W^0$ and starting from P^0 over V_2, yields a prime divisor P over W with center 0, $a(P, W, 0) = a(P^0, W^0, 0)$, $\operatorname{mult}_P h_l = \operatorname{mult}_{P^0} h_l^0$, and $I_G(P) = I_G(P^0)$, and $\alpha \leq \beta_+ + \delta k/(re)$.

Since $|\delta|$ and k are bounded, we obtain a bounded constant C_{up} such that for large r

$$\alpha \leq \beta + \frac{C_{\mathrm{up}}}{r}. \tag{B.38}$$

Step 4. In this step, we prove the lower bound $\alpha \geq \beta$ under a fixed margin of the reference.

For a real number $\theta \geq 1$ let $\delta_\theta := \theta\tau + B - v$, and let the *reference pair for* θ be

$$\Big(S^0, \theta \sum_l b_l \operatorname{div}(h_l^{0,S})\Big) \text{ if } \delta_\theta \leq 0, \qquad \Big(S^0, \theta \sum_l b_l \operatorname{div}(h_l^{0,S}), I_0^{\delta_\theta/(2A)}\Big) \text{ if } \delta_\theta \geq 0.$$

Claim B.9. *Let $\theta > 1$. Then there exists a positive integer r_θ, depending only on n, c, v, e_0, Γ, and θ, such that if $r \geq r_\theta$, either $\delta \leq 0$ and $\delta_\theta \leq 0$ or $\delta \geq 0$ and $\delta_\theta \geq 0$, and the reference pair for θ is lc near 0, then $a(P, W, D)/|I_G(P)| \geq \beta$ for any prime divisor P over W with center 0. In particular $\alpha \geq \beta$.*

Proof. In this proof, "bounded" and "for large r" allow dependence also on θ, that is, "for large r" means $r \geq r_\theta$. By Claim B.6 with $C = \theta D^{\mathrm{pr}}$, $(W^0, \theta D^{\mathrm{pr}})$ is lc near 0, and by Claim B.4 with $\Lambda = \theta(\Gamma \backslash \{0\})$, for large r

$$(W, \theta D) \quad \text{and} \quad (W^0, \theta D^0) \quad \text{are lc near } 0. \tag{B.39}$$

In particular (W, D) and (W^0, D^0) are klt near 0.

Let P be a prime divisor over W with center 0, $e := |I_G(P)|$, and $[f] := \operatorname{mult}_P f/e$ for a regular function f near 0. Since $\beta \le 2$, we may assume that $a(P,W,D) \le 2e$. Let $a_0 := a(P,W,0)$. By (B.39), $\theta \sum_l b_l \operatorname{mult}_P h_l \le a_0$, so $a_0(1-1/\theta) \le 2e$ and

$$\frac{a_0}{e} \le \frac{2\theta}{\theta-1}, \qquad [h_l] \le \frac{2}{(\theta-1)\gamma} \text{ for any } l.$$

Since $g(z,0) = z^n$, the surface $W \cap (t=0)$ is $(xy + z^n = 0) \subset \mathbb{A}^3$, and the pairs

$$\Big(W \cap (t=0), \frac{1}{n}(\operatorname{div}(x) + \operatorname{div}(y))\Big) \quad \text{and} \quad \big(W \cap (t=0), \operatorname{div}(z)\big)$$

pull back under $(p,q) \to (p^n, -q^n, pq)$ to $(\mathbb{A}^2, \operatorname{div}(p) + \operatorname{div}(q))$. By [Kaw07, Theorem],

$$\Big(W, \frac{1}{n}(\operatorname{div}(x) + \operatorname{div}(y)) + \operatorname{div}(t)\Big) \quad \text{and} \quad (W, \operatorname{div}(z) + \operatorname{div}(t)) \quad \text{are lc near 0.} \tag{B.40}$$

Thus $[x] + [y] \le na_0/e$ and $[z] + [t] \le a_0/e$ are bounded.

Let $I_R := I_G(P) \cap R$. The monomials $x^v t^A$, $y^v t^A$, and $z^v t^B$ have $\mathbf{T}$-weight 0, so they are R-invariant. By [Liu26, Lemma 2.6(1)], $I_G(P)$ is cyclic and acts faithfully on $\mathfrak{m}_P/\mathfrak{m}_P^2$, and an I_R-invariant function vanishing along P has order divisible by $|I_R|$. Since $I_G(P)/I_R$ injects into G/R, $e \le e_0|I_R|$, and therefore

$$v[x] + A[t] \ge \frac{1}{e_0}, \qquad v[y] + A[t] \ge \frac{1}{e_0}, \qquad v[z] + B[t] \ge \frac{1}{e_0}. \tag{B.41}$$

Let $M_0 := 2/((\theta-1)\gamma) + 1$, so that $M_0 > [h_l]$ for any l. We distinguish two cases.

Case $[g - z^n] - n[z] \le M_0$. Then some monomial $t^\sigma z^j$ of $g - z^n$ satisfies $\sigma[t] \le n[z] + M_0$. By (B.23), $\sigma \ge (r - nB)/v$, so $[t] \le v(n[z] + M_0)/(r - nB)$, and (B.41) implies $[x], [y], [z] \ge 1/(2ve_0)$ for large r. By (B.24), any term of $h_l^{(\ell)}$ with $\ell < 0$ has normalized order at least $(|\ell|r - |\kappa_l|)/(2ve_0 \max(A,B))$, which exceeds $M_0 > [h_l]$ for large r. Thus P satisfies the assumption of Claim B.8. Let P_V be the induced prime divisor over V_r, $e_V := |I_{H_-}(P_V)|$, $\mathfrak{a} := \prod_l \bar{h}_l^{b_l}$, and

$$m_V := \min(\operatorname{mult}_{P_V} X/A, \operatorname{mult}_{P_V} Y/A, \operatorname{mult}_{P_V} Z/B) = \operatorname{mult}_{P_V} I_0/(2A).$$

Since $\operatorname{mult}_P x > 0$ and $X = xw^{A/r}$, and similarly for y, z, we have $m_V > \operatorname{mult}_{P_V} w/r$. By (B.35),

$$\begin{aligned} \frac{a(P,W,D)}{e} &= \frac{a(P_V, V_r, \operatorname{div}(w), \mathfrak{a}) - \delta \operatorname{mult}_{P_V} w/r}{e_V} \\ &= \frac{a(P_V, V_r, \operatorname{div}(w), \mathfrak{a} I_0^{\delta/(2A)}) + \delta(m_V - \operatorname{mult}_{P_V} w/r)}{e_V}, \end{aligned} \tag{B.42}$$

the second form being used for $\delta \ge 0$. The group H_- is abelian and acts linearly on the span of X, Y, so in an eigenbasis of the span of X, Y it acts diagonally on $\mathbb{A}^4$, and V_r is a hypersurface with an equation semi-invariant under H_-. By **Step 3**, V_r is normal and klt near 0. Moreover H_- acts faithfully on S^0. The ideals $(\bar{h}_l)$ and I_0 are generated by H_--semi-invariant polynomials. By Lemma 5.5 with the polynomial denoted u there equal to w, $S_0 = S^0$, the coefficient denoted b there equal to 1, and $\mathfrak{a}$, resp. $\mathfrak{a} I_0^{\delta/(2A)}$, whose restriction to S^0 is the reference pair for $\theta = 1$, which is lc near 0 by **Step 2**, we obtain

$$a(P_V, V_r, \operatorname{div}(w), \mathfrak{a}) \ge e_V \beta_- \text{ for } \delta \le 0, \qquad a(P_V, V_r, \operatorname{div}(w), \mathfrak{a} I_0^{\delta/(2A)}) \ge e_V \beta_+ \text{ for } \delta \ge 0.$$

Since $-\delta \operatorname{mult}_{P_V} w/r \ge 0$, resp. $\delta(m_V - \operatorname{mult}_{P_V} w/r) \ge 0$, (B.42) implies $a(P,W,D)/e \ge \beta$.

Case $[g - z^n] - n[z] > M_0$. Then $\operatorname{mult}_P g = n \operatorname{mult}_P z$, so $[x] + [y] = n[z]$. By Claim B.7 with $V_1 = W$, $V_2 = W^0$, and $\nu_1 = P$, $\operatorname{mult}_{\nu'}(\rho - 1) > M_0 e > \operatorname{mult}_P h_l$, so there exists a prime divisor P^0 over W^0 with center 0, $a(P^0, W^0, D^0) = a(P,W,D)$, and $I_G(P^0) = I_G(P)$. Let $\tilde{P}$ be a prime divisor over $\mathbb{A}^3_{p,q,t}$ over P^0 for the cover (B.26), $I := I_{\tilde{G}}(\tilde{P})$, which is cyclic [Liu26, Lemma 2.6(1)], and let H be the subgroup of $\tilde{G}$ generated by I and $\tilde{G} \cap \mathbf{T}'$. Since $\tilde{G} \cap \mathbf{T}'$ is central, H is abelian, $|H \cap \mathbf{T}'| = e_1 r$, and $[H : H \cap \mathbf{T}'] \le n e_0/e_1$ by (B.27). After a linear change of (p,q) diagonalizing H, the group $H \cap \mathbf{T}'$ still acts by scalars, so the $\mathbf{T}'$-weights and the principal blocks

do not change in the new coordinates. The pair $(\mathbb{A}^3, \tilde{D}^0)$, $\tilde{D}^0 := \sum_l b_l \operatorname{div}(h_l(p^n, -q^n, pq, t))$, is klt near 0 by (B.39), and a monomial of h_l of degree m' lifts to monomials of degree at most nm' and $\mathbf{T}'$-weight e_1 times the $\mathbf{T}$-weight of the monomial. By (B.24), for large r the principal block of $h_l(p^n, -q^n, pq, t)$ along $\mathbf{T}'$ is $h_l^0(p^n, -q^n, pq, t)$. By [Liu26, Theorem 16.21(2)], in the notation of [Liu26, Construction-Theorem 16.9] with $(p_{\mathrm{L}}, q_{\mathrm{L}}, r_{\mathrm{L}}) = (c', c', v')$, the circle $\mathbf{T}'$ with the orientation $(c', c', -v')$, the group H, and the index bound ne_0/e_1, for large r we have

$$\frac{a(\tilde{P}, \mathbb{A}^3, \tilde{D}^0)}{|I|} \geq \operatorname{mld}((\mathbb{A}^3 \ni 0, \tilde{D}^0)/H) \geq b_{\mathrm{L}}, \tag{B.43}$$

where b_{L} is the number denoted b in [Liu26, Construction-Theorem 16.9] for the principal blocks, and the first inequality holds since $I_H(\tilde{P}) = I$. The left side of (B.43) is $a(P^0, W^0, D^0)/|I_G(P^0)| = a(P, W, D)/e$ by [Liu26, Lemma 2.6(2)(5)].

It remains to show $b_{\mathrm{L}} \geq \beta$. In the orientation $(c', c', -v')$, the lifted principal blocks have weights $-e_1\kappa_l$, and the weights of p, q, t add up to $2c' - v' = e_1(B - v)$. Thus the number τ_{L} of [Liu26, Construction-Theorem 16.9] equals $-e_1\tau - e_1(B - v) = -e_1\delta$, so case (1) there is $\delta \leq 0$ and case (2) is $\delta > 0$.

The circle-homogeneous cover there is $t = \sigma_{\mathrm{L}}^{v'}$ with plane coordinates $p\sigma_{\mathrm{L}}^{c'}$, $q\sigma_{\mathrm{L}}^{c'}$. Since $\sigma_{\mathrm{L}}^{e_1 v} = t$, we may take $s = \sigma_{\mathrm{L}}^{e_1}$, and then $(p\sigma_{\mathrm{L}}^{c'})^n = xs^A = X$, $-(q\sigma_{\mathrm{L}}^{c'})^n = Y$, and $p\sigma_{\mathrm{L}}^{c'} \cdot q\sigma_{\mathrm{L}}^{c'} = Z$. Thus the plane there is $\mathbb{A}^2_{\bar{p},\bar{q}}$ of **Step 2**, the lifted principal blocks restrict to Φ_l up to powers of σ_{L}, and the group there acts on $\mathbb{A}^2_{\bar{p},\bar{q}}$ through a subgroup $\bar{H} \subset \hat{H}_-$, since a lift of $h \in H$ to σ_{L} induces on $(\bar{p}, \bar{q})$ a lift of the image of h in H_-.

In case (1), the last display of [Liu26, Construction-Theorem 16.9] writes b_{L} as the minimal log discrepancy at 0 of the quotient by $\bar{H}$ of the plane pair with boundary $\sum_l b_l \operatorname{div}(\Phi_l)$ and the ideal

$$\mathfrak{b}_{\mathrm{L}}^{\tau_{\mathrm{L}}^-/(p_{\mathrm{L}}q_{\mathrm{L}}r_{\mathrm{L}}|H|)} = \mathcal{O},$$

since $\tau_{\mathrm{L}}^- = 0$. In case (2), b_{L} is the minimum, over the prime divisors with center in the image of the exceptional curve $\bar{E}$ of the blow-up of $\mathbb{A}^2_{\bar{p},\bar{q}}$ at 0 (weights $(1,1)$), of the log discrepancies of the quotient by $\bar{H}$ of the pair with boundary the strict transform of $\sum_l b_l \operatorname{div}(\Phi_l)$ plus $(1 - v'(1-b_t)/c')\bar{E}$. Since $v'/c' = dv/c = 2v/B$, and since the ideal I_0 pulls back to $(\bar{p}^{2n}, \bar{q}^{2n}, \bar{p}^n\bar{q}^n)$, the log discrepancy of $\bar{E}$ is

$$a\big(\bar{E}, \mathbb{A}^2, \textstyle\sum_l b_l \operatorname{div}(\Phi_l), I_0^{\delta/(2A)}\big) = 2 - \sum_l b_l \operatorname{mult}_0 \Phi_l - \frac{2n\delta}{2A} = 2 - \frac{2(vb_t - \tau)}{B} - \frac{2\delta}{B} = \frac{2v}{B}(1 - b_t)$$

by (B.29), so the pair of [Liu26, Construction-Theorem 16.9] on the blow-up is the crepant pullback of

$$\Big(\mathbb{A}^2_{\bar{p},\bar{q}}, \sum_l b_l \operatorname{div}(\Phi_l), I_0^{\delta/(2A)}\Big),$$

whose prime divisors with center 0 are those with center in $\bar{E}$. Hence in both cases

$$b_{\mathrm{L}} = \operatorname{mld}\big((\mathbb{A}^2_{\bar{p},\bar{q}} \ni 0, \textstyle\sum_l b_l \operatorname{div}(\Phi_l), \mathfrak{i})/\bar{H}\big)$$

with $\mathfrak{i} = \mathcal{O}$ for $\delta \leq 0$ and $\mathfrak{i} = I_0^{\delta/(2A)}$ for $\delta > 0$. Since $\bar{H} \subset \hat{H}_-$ and the pair is lc near 0, $a(\nu)/|I_{\bar{H}}(\nu)| \geq a(\nu)/|I_{\hat{H}_-}(\nu)|$ for any prime divisor ν over $\mathbb{A}^2_{\bar{p},\bar{q}}$ with center 0, and by [Liu26, Lemma 2.6(5)] the infimum of the right side is β, since $(\mathbb{A}^2_{\bar{p},\bar{q}}, \cdot)/\hat{H}_- = (S^0, \cdot)/H_-$. Thus $b_{\mathrm{L}} \geq \beta$. □

Proof of Theorem 8.6 continued. **Step 5.** In this step, we assume that the conclusion fails for the germs as in **Step 1**, pass to compatible generic limits of the references, and treat klt limits.

Assume that there exist data (W_i, G_i, D_i) as in **Step 1** with Γ, e_0, and $\mathbf{T}$ fixed, $r_i \to +\infty$, and positive strictly increasing α_i. We add the index i to the objects of **Steps 1–4**. Passing to a subsequence, we may assume that r_i is larger than any bound used below, the number of marks and the integers κ_l and n_l do not depend on i, either $\delta_i \leq 0$ for any i or $\delta_i \geq 0$ for any i, and b_{il} is non-decreasing with limit $b_l^* \geq \gamma$ for any l. If G_i contains an element ι_i

acting by $(x,y)\to(\varpi_1 y,\varpi_2 x)$, then replacing x and y with $\vartheta_i x$ and $\vartheta_i^{-1}y$ for a constant ϑ_i with $\vartheta_i^4=\varpi_2/\varpi_1$ preserves xy, $\mathbf{T}$, R_i, and the form of the marks, and afterwards ι_i acts by $x\to\varpi y$, $y\to\varpi x$ for a constant ϖ. Since G_i is abelian, the elements of G_i not exchanging x and y commute with ι_i, so they act on (x,y) by scalars, and the other elements of G_i exchanging x and y are the products of ι_i with them. Thus possibly replacing x and y in this way, we may assume that any element of G_i exchanging x and y acts by $x\to\varpi y$, $y\to\varpi x$ for a constant ϖ depending on the element. The groups $H_{-,i}$, $\hat H_{-,i}$, $H_{-,i}/\mu_v$, K, and the groups $H_{y,i}$, $H_{x,i}$ of **Step 3** acting on U_y and U_x are monomial with entries roots of unity of bounded order, so we may assume that they and the characters of Φ_{il}, φ^y_{il}, φ^x_{il} do not depend on i. We write $\hat H:=\hat H_{-,i}$ and $H_E:=H_{-,i}/\mu_v$. Let $\tau^*:=\sum_l b_l^*\kappa_l$, $\delta^*:=\tau^*+B-v$, and $b_t^*:=\sum_l b_l^* n_l$, the limits of τ_i, δ_i, and $b_{t,i}$.

By **Step 2**, the values β_i belong to an ACC set, so passing to a subsequence we may assume that β_i is non-increasing with limit β_∞. By (B.38), $0<\lim_i\alpha_i\le\beta_\infty$.

We raise the labels b_{il} to b_l^* and keep the reference equations Φ_{il}, φ^y_{il}, and φ^x_{il}. The *raised reference* is the pair

$$\Big(\mathbb{A}^2_{\bar p,\bar q},\sum_l b_l^*\operatorname{div}(\Phi_{il})\Big)$$

near 0 if $\delta_i\le 0$, and the pairs

$$\Big(U_\omega,\sum_l b_l^*\operatorname{div}(\varphi^\omega_{il})\Big),\qquad \omega\in\{x,y\},$$

near the preimages of E if $\delta_i\ge 0$. Let

$$t_i:=\sup\Big\{t\;\Big|\;\Big(\mathbb{A}^2_{\bar p,\bar q},t\sum_l b_l^*\operatorname{div}(\Phi_{il})\Big)\text{ is lc near }0\Big\}$$

if $\delta_i\le 0$, and

$$t_i:=\sup\Big\{t\;\Big|\;\Big(U_\omega,t\sum_l b_l^*\operatorname{div}(\varphi^\omega_{il})\Big)\text{ is lc near the preimage of }E\text{ for }\omega\in\{x,y\}\Big\}$$

if $\delta_i\ge 0$. Since the reference with labels b_{il} is lc, $t_i\ge\min_l b_{il}/b_l^*\to 1$. Each t_i is the lc threshold at a closed point of a smooth surface of a divisor whose coefficients are sums $\sum_l m_l b_l^*$ with $m_l\in\mathbb{Z}_{\ge 0}$ not all zero, which form a DCC subset of $\mathbb{R}_{>0}$ since the number of marks is fixed and $b_l^*\ge\gamma$, so the t_i belong to an ACC set [HMX14, Theorem 1.1]. Thus $t_i\ge 1$ for large i, and the raised references are lc. Let β_i^* be the minimal log discrepancy defining β_i for the raised reference, that is,

$$\beta_i^*:=\operatorname{mld}\Big(\Big(\mathbb{A}^2\ni 0,\sum_l b_l^*\operatorname{div}(\Phi_{il})\Big)/\hat H\Big)$$

if $\delta_i\le 0$, and the minimum of the log discrepancies of the crepant quotient by H_E over the prime divisors with center contained in the image of E if $\delta_i\ge 0$. Then $\beta_i^*\le\beta_i$. By Theorem 2.9(2) and **Step 2**, β_i^* is computed by a prime divisor whose log discrepancy over the smooth chart, whose inertia group, and whose orders of the marks are bounded, so

$$\beta_i\le\beta_i^*+C_1\sum_l(b_l^*-b_{il})$$

for a constant C_1 independent of i, and $\beta_i^*\to\beta_\infty$. Moreover β_i^* belongs to the finite set of the numbers $\big(N'-\sum_l b_l^* m_l\big)/e'$ with bounded positive integers N' and e' and bounded non-negative integers m_l. Passing to a subsequence, we may assume that $\beta_i^*=\beta_\infty$ for any i.

If $\delta_i\le 0$, we apply Construction-Proposition 4.1(a) on $\mathbb{A}^2_{\bar p,\bar q}$ with the group $\hat H$ to the tuples $(\Phi_{il})_l$, whose orders at 0 are $2(vn_l-\kappa_l)/B$ by (B.29).

If $\delta_i\ge 0$, we apply Construction-Proposition 4.1(b) to the fixed surface Y_+, the curve E, the charts U_y/K and U_x/K with $e_\omega=q_\omega$, the group H_E, and the tuples $(\varphi^y_{il},\varphi^x_{il})_l$. The hypotheses of case (b) hold: the divisors of φ^y_{il} and φ^x_{il} descend to an H_E-invariant $\mathbb{Q}$-divisor D_{il} on Y_+,

the pullback of the divisor of $h^{0,S}_{il}$ plus $\frac{\kappa_l}{cv}E$, which is $\mathbb{Q}$-Cartier near E since Y_+ has quotient singularities there, the quotient $\varphi^y_{il}/\varphi^x_{il} = (\eta/\eta')^{\kappa_l}$ on the normalization of $U_y \times_{Y_+} U_x$ is a unit independent of i, $\operatorname{mult}_{(q_y=0)} \varphi^y_{il} = n_l$, and by (B.24) a monomial of h^0_{il} with t-exponent $k_4 = k'$ contributes to the coefficient of $q_y^{k'}$ in φ^y_{il} a monomial $p_y^{nk_1+k_3}$ and to that of $q_x^{k'}$ in φ^x_{il} a monomial $p_x^{nk_2+k_3}$, with

$$(nk_1 + k_3) + (nk_2 + k_3) = \frac{2(vk' - \kappa_l)}{B}. \tag{B.44}$$

Thus the coefficients of $q_y^{k'}$ have bounded degrees, and, since $\varphi^y_{il} = (\eta/\eta')^{\kappa_l}\varphi^x_{il}$, $q_x = (\eta'/\eta)^v q_y$, and $(\eta'/\eta)^B = p_y/p_x$, the coefficients of $q_y^{k'}$ and $q_x^{k'}$ glue to a form of degree $2(vk' - \kappa_l)/B$ on the smooth rational curve $\hat{E}$ obtained by gluing $(q_y = 0)$ and $(q_x = 0)$ along $p_y p_x = \epsilon_0$, where ϵ_0 is the value of $p_y p_x$ on a fixed irreducible component of the normalization of $U_y \times_{Y_+} U_x$ over E. To compute ϵ_0, note that the rational function X/Y on Y_+ is $-p_y^n$ on U_y by (B.30) and $-p_x^{-n}$ on U_x, so $(p_y p_x)^n = 1$ on the normalization, and $p_y p_x$ is a constant ϵ_0 on each irreducible component. Each component is finite over U_y, so it contains a point over a point of U_y with $q_y \neq 0$ and $p_y = 1$. At such a point, choosing η and η' with $\eta^v = q_y$ and $\eta'^v = q_x$ such that the values of X, Y, and Z given by (B.30) and by the formulas for U_x are equal, we obtain $\eta'^A = X = -\eta^A$ and $\eta'^B p_x = Z = \eta^B$, so

$$\epsilon_0 = p_x = (\eta'/\eta)^{-B}, \qquad (\eta'/\eta)^A = -1.$$

Since $nB = 2A$, $\epsilon_0^{n/d} = (\eta'/\eta)^{-2A/d}$, which is 1 if $d = 1$ and -1 if $d = 2$. In particular $\epsilon_0 \neq 1$ if $d = 2$. We obtain an algebraically closed field $\mathbf{k} \supset \mathbb{C}$ and a compatible generic limit $(\Phi_{\infty,l})_l$, resp. $(\varphi^y_{\infty,l}, \varphi^x_{\infty,l})_l$. In case (a), let $C_i := \sum_l b_l^* \operatorname{div}(\Phi_{il})$ on $\mathbb{A}^2_{\bar{p},\bar{q}}$ and $C_\infty := \sum_l b_l^* \operatorname{div}(\Phi_{\infty,l})$ on $\mathcal{A}_{\mathbf{k}}$. In case (b), let $C_i := \sum_l b_l^* D_{il}$ on Y_+ and let C_∞ be the $\mathbb{R}$-divisor on $\mathcal{A}_{\mathbf{k}}$ given by $\sum_l b_l^* \operatorname{div}(\varphi^\omega_{\infty,l})$ on the charts. Then C_i and C_∞ are the divisors of Theorem 4.2 for $c_l = b_l^*$ and $\mathfrak{c} = \mathcal{O}$. The tuples with non-zero coefficient of $q_y^{n_l}$ form an open subset of the jet space containing the jets of the tuples, so the coefficient of $q_y^{n_l}$ in $\varphi^y_{\infty,l}$ is non-zero, and the restriction of the strict part of C_∞ to $\hat{E}_{\mathbf{k}}$ has degree

$$D_{\mathrm{tot}} := \sum_l b_l^* \frac{2(vn_l - \kappa_l)}{B} = \frac{2(vb_t^* - \tau^*)}{B}. \tag{B.45}$$

By Theorem 4.2(1), $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ is lc near the base, and by Theorem 4.2(2), the quotient minimum of $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ is β_∞: $\operatorname{mld}((\mathcal{A}_{\mathbf{k}} \ni 0, C_\infty)/\hat{H}) = \beta_\infty$ in case (a), and in case (b) the minimum of the log discrepancies of the crepant quotient of $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ by H_E over the prime divisors with center contained in $E_{\mathbf{k}}/H_E$ is β_∞.

Assume that $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ is klt near the base. We fix a real number $\epsilon > 0$ such that $(\mathcal{A}_{\mathbf{k}}, (1+\epsilon)C_\infty, \mathfrak{c}_\epsilon)$ is klt near the base in case (a), where

$$\mathfrak{c}_\epsilon := (\bar{p}^{2n}, \bar{q}^{2n}, \bar{p}^n\bar{q}^n)^\epsilon,$$

and such that

$$\big(\mathcal{A}_{\mathbf{k}}, (1+\epsilon)C_\infty + \epsilon \operatorname{div}(q_\omega)\big)$$

is klt near the base in case (b). Such an ϵ exists. In case (a), $\mathcal{A}_{\mathbf{k}} = \operatorname{Spec} \mathbf{k}[[\bar{p}, \bar{q}]]$, and the pair $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ together with the ideal $(\bar{p}^{2n}, \bar{q}^{2n}, \bar{p}^n\bar{q}^n)$ has a log resolution [dFEM11, Theorems 2.3 and 2.4], which exists for quasi-excellent schemes. The klt property is read off on the finitely many prime divisors of one log resolution, so it persists under small perturbations of the coefficients and of the exponent of the ideal. In case (b), klt near $E_{\mathbf{k}}$ is checked at the closed points of $E_{\mathbf{k}}$, on the charts. The coefficient of $(q_\omega = 0)$ in C_∞ is b_t^*, which is less than one since the limit is klt, and the other components of C_∞ meet $(q_\omega = 0)$ at finitely many points of the charts. At any other point the boundary near the point is $b_t^* \operatorname{div}(q_\omega)$, so any ϵ with $(1+\epsilon)b_t^* + \epsilon < 1$ works there, and at the finitely many remaining points we use a log resolution as in case (a). In case (b), the tuples augmented by (q_y, q_x), which do not depend on i, satisfy the assumptions of

Construction-Proposition 4.1(b), and the compatible generic limit of the augmented tuples is the limit tuple augmented by (q_y, q_x) itself, since the jets of a constant tuple are constant. Thus Theorem 4.2(1) holds for the augmented tuples, and the corresponding pairs for C_i are lc near the base for any i.

Assume first that $\delta^* \neq 0$. Choose $\theta \in (1, 1+\epsilon]$ such that $\theta\tau^* + B - v$ has the sign of δ^*. For large i, $\delta_{\theta,i} = \theta\tau_i + B - v$ has the sign of δ_i, and the reference pair for θ of (W_i, G_i, D_i) is lc near 0: in case (a) it is bounded by $(1+\epsilon)C_i$, and in case (b) the crepant pullback of the reference pair is

$$\Big(U_\omega, \theta \sum_l b_{il} \operatorname{div}(\varphi^\omega_{il})\Big)$$

by Claim B.5, bounded by $(1+\epsilon)C_i$. By Claim B.9, $\alpha_i \geq \beta_i \geq \beta_\infty$ for large i, which contradicts $\alpha_i < \lim_j \alpha_j \leq \beta_\infty$.

Assume next that $\delta^* = 0$, so $\tau^* = v - B$. Consider the raised tuples (W_i, G_i, D_i^*), $D_i^* := \sum_l b_l^* \operatorname{div}(h_{il})$, with labels in the DCC set $\Gamma \cup \{b_l^*\}$ and adjusted mass 0. For them the plane reference and the curve reference coincide: by Claim B.5 with $\theta_l = b_l^*$ and $\delta' = 0$, the crepant pullback of the raised plane reference is the raised curve reference, and both minima are $\beta_i^* = \beta_\infty$. For $\theta > 1$, $\delta_\theta^* = \theta\tau^* + B - v = (\theta-1)(v-B)$, which is non-positive if $v \leq B$ and positive if $v > B$. Thus the sign condition of Claim B.9 holds with $\delta = 0$, in its first alternative if $v \leq B$ and in its second if $v > B$. Let $\theta \in (1, 1+\epsilon]$ with $(\theta-1)|v-B| \leq \min(\epsilon v, 2A\epsilon)$. If $v \leq B$ and the limit is in case (a), or $v > B$ and the limit is in case (b), the reference pair for θ is lc near the base, since its pullback to $\mathbb{A}^2_{\bar p,\bar q}$, resp. U_ω, has boundary $\theta C_i \leq (1+\epsilon)C_i$ (by Claim B.5 in case (b)), as in the case $\delta^* \neq 0$. If $v \leq B$ and the limit is in case (b), by Claim B.5 with $\theta_l = \theta b_l^*$ and $\delta' = (\theta-1)(v-B) \leq 0$, the crepant pullback of $(S^0, \theta\sum_l b_l^* \operatorname{div}(h^{0,S}_{il}))$ to U_ω is

$$\Big(U_\omega, \theta \sum_l b_l^* \operatorname{div}(\varphi^\omega_{il}) + \frac{(\theta-1)(B-v)}{v} \operatorname{div}(q_\omega)\Big),$$

which is lc near the preimage of E because $(\theta-1)(B-v)/v \leq \epsilon$. If $v > B$ and the limit is in case (a), the reference

$$\Big(S^0, \theta \sum_l b_l^* \operatorname{div}(h^{0,S}_{il}), I_0^{(\theta-1)(v-B)/(2A)}\Big)$$

is lc near 0 because $(\theta-1)(v-B)/(2A) \leq \epsilon$. The data (W_i, G_i, D_i^*) satisfy the assumptions of **Steps 1–4** with $\Gamma \cup \{b_l^*\}$ in place of Γ, since the bounded quantities of **Step 1** (the number of marks, $\operatorname{mult}_0 h_{il}$, and κ_l) depend only on the marks, and the least positive element of $\Gamma \cup \{b_l^*\}$ is γ as $b_l^* \geq \gamma$. By Claim B.6 with $C = \theta D_i^{*,\mathrm{pr}}$ and Claim B.4 with $\Lambda = \theta((\Gamma \cup \{b_l^*\}) \setminus \{0\})$, the lc reference pair for θ implies that $(W_i, \theta D_i^*)$ is lc near 0 for large i, so $a(P, W_i, D_i^*) \geq (1 - 1/\theta)a(P, W_i, 0) > 0$ for any prime divisor P over W_i with $0 \in \operatorname{center}_{W_i} P$, and (W_i, D_i^*) is klt near 0. Moreover the value β for D_i^* is $\beta_i^* = \beta_\infty \leq 2$, and the reference pair for $\theta = 1$ is the raised reference, which is lc. In all cases Claim B.9, applied to (W_i, G_i, D_i^*) with $\Gamma \cup \{b_l^*\}$ in place of Γ, implies that $\operatorname{mld}((W_i \ni 0, D_i^*)/G_i) \geq \beta_\infty$ for large i. Since $D_i \leq D_i^*$, $\alpha_i \geq \beta_\infty$, which contradicts $\alpha_i < \lim_j \alpha_j \leq \beta_\infty$.

Thus $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ is lc and not klt near the base. It remains to treat the case where $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ is not klt near the base, which we do without a margin.

Step 6. In this step, we treat a non-klt limit in case (a), that is, $\delta_i \leq 0$, by an augmentation of character type.

Let $g_{Av} := \gcd(A, v)$. By Definition-Proposition 5.7(a), applied with $\mathfrak{c} = \mathcal{O}$ and $\beta_i^* = \beta_\infty > 0$, the limit $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ has exactly one floor F, F is smooth at 0, and $(\mathcal{A}_{\mathbf{k}}, C_\infty - F)$ is klt near 0. Since F is the only floor, it is invariant under $\hat H$, so the tangent line T_0F is invariant under $\mu_n(1,-1) \subset \hat H$, which acts on $\bar p$ and $\bar q$ by the characters ζ and ζ^{-1}. Since $n \geq 3$, $\zeta \neq \zeta^{-1}$ for a generator ζ of μ_n, so T_0F is a coordinate axis. An element exchanging $\bar p$ and $\bar q$ exchanges $(\bar p = 0)$ and $(\bar q = 0)$, so $\hat H$ contains no such element. Thus $\hat H$ is diagonal, in particular abelian,

and G_i contains no element exchanging x and y. Let $Z_0 := \bar{p}$ if $T_0F = (\bar{q} = 0)$ and $Z_0 := \bar{q}$ if $T_0F = (\bar{p} = 0)$, so that $Z_0|_F$ is a parameter of F at 0. Let $b_F := \mathrm{mult}_0((C_\infty - F)|_F)$ and let e_F be the order of the image of $\hat{H}$ in $\mathrm{Aut}(T_0F)$. By Definition-Proposition 5.7(a) and Lemma 5.6,

$$\beta_\infty = \frac{1 - b_F}{e_F}, \qquad 1 - b_F > 0. \tag{B.46}$$

Since F has coefficient one in C_∞ and $Z_0|_F$ is a parameter of F at 0, the $\mathbb{R}$-divisor $C_\infty - F + (1 - b_F)\,\mathrm{div}(Z_0) \geq 0$ does not contain F, and

$$\mathrm{mult}_0\left((C_\infty - F + (1 - b_F)\,\mathrm{div}(Z_0))|_F\right) = b_F + (1 - b_F) = 1.$$

Thus Lemma 2.14, applied to $\mathrm{Spec}\,\mathbf{k}[[\bar{p}, \bar{q}]]$, the regular curve F, and $C_\infty - F + (1 - b_F)\,\mathrm{div}(Z_0)$, implies that $(\mathcal{A}_\mathbf{k}, C_\infty + (1 - b_F)\,\mathrm{div}(Z_0))$ is lc near 0.

Let $U := x^{v/g_{Av}} t^{A/g_{Av}}$ if $Z_0 = \bar{p}$ and $U := y^{v/g_{Av}} t^{A/g_{Av}}$ if $Z_0 = \bar{q}$. Then U has $\mathbf{T}$-weight 0, and by (B.28), $U^S = X^{v/g_{Av}}$, resp. $Y^{v/g_{Av}}$, which is $Z_0^{nv/g_{Av}}$ on $\mathbb{A}^2_{\bar{p},\bar{q}}$ up to sign. Let

$$\lambda := \frac{(1 - b_F)g_{Av}}{nv}, \qquad \text{so that} \qquad \lambda\,\mathrm{div}(U^S) = (1 - b_F)\,\mathrm{div}(Z_0) \text{ on } \mathbb{A}^2_{\bar{p},\bar{q}}.$$

Since U^S does not depend on i, the compatible generic limit of the tuples augmented by U^S is the limit tuple augmented by U^S, so Theorem 4.2(1) holds for the augmented tuples. The lifts to $\mathbb{A}^2_{\bar{p},\bar{q}}$ of the elements $s \to \zeta s$, $\zeta^v = 1$, act on X by ζ^A, and $\{\zeta^A \mid \zeta^v = 1\} = \mu_{v/g_{Av}}$, so they act on Z_0 by all the elements a with $a^n \in \mu_{v/g_{Av}}$, that is, by $\mu_{nv/g_{Av}}$. Thus nv/g_{Av} divides e_F. The character of U^S under $\hat{H}$ is the (nv/g_{Av})-th power of the character of Z_0, whose order on T_0F is e_F, so the character of U^S has order $m_U := e_F g_{Av}/(nv)$. Since U is invariant under R and under the lifts of the identity, and $\hat{H} \to G_i/R_i$ is surjective, m_U is the order of the character of the G_i-semi-invariant function U.

By Theorem 4.2(1) for the tuples augmented by U^S, $(\mathbb{A}^2_{\bar{p},\bar{q}}, C_i + \lambda\,\mathrm{div}(U^S))$ is lc near 0 for any i. By Claim B.6, whose adjusted mass here is $\delta^* \leq 0$ since $\kappa(U) = 0$, the pair

$$\Big(W_i^0, \sum_l b_l^*\,\mathrm{div}(h_{il}^0) + \lambda\,\mathrm{div}(U)\Big)$$

is lc near 0. By Claim B.4 with $\Lambda = (\Gamma\backslash\{0\}) \cup \{b_l^*\} \cup \{\lambda\}$, the pair

$$\Big(W_i, \sum_l b_l^*\,\mathrm{div}(h_{il}) + \lambda\,\mathrm{div}(U)\Big)$$

is lc near 0 for large i, and since $b_{il} \leq b_l^*$, so is $(W_i, D_i + \lambda\,\mathrm{div}(U))$. By Lemma 5.8 with $Z = \{0\}$,

$$\alpha_i \geq \frac{\lambda}{m_U} = \frac{(1 - b_F)g_{Av}}{nv} \cdot \frac{nv}{e_F g_{Av}} = \frac{1 - b_F}{e_F} = \beta_\infty$$

for large i, which contradicts $\alpha_i < \lim_j \alpha_j \leq \beta_\infty$.

Step 7. In this step, we treat a non-klt limit in case (b), that is, $\delta_i \geq 0$, using a bound on the intersection with E which does not depend on n.

The coefficient of $(q_y = 0)$ in C_∞ is b_t^*, since the coefficient of $q_y^{n_l}$ in $\varphi^y_{\infty,l}$ is non-zero. The image $E_\mathbf{k}/H_E$ of $E_\mathbf{k}$ under the quotient map $\mathcal{A}_\mathbf{k} \to \mathcal{A}_\mathbf{k}/H_E$ is a prime divisor on $\mathcal{A}_\mathbf{k}/H_E$ whose center is $E_\mathbf{k}/H_E$ itself, so it is one of the prime divisors in the minimum that defines β_∞ in **Step 5**. Its log discrepancy with respect to the crepant quotient of $(\mathcal{A}_\mathbf{k}, C_\infty)$ by H_E is a positive multiple of $1 - b_t^*$ [Liu26, Lemma 2.6(4)], so $\beta_\infty > 0$ implies $b_t^* < 1$. Since $\delta^* \geq 0$, that is, $-\tau^* \leq B - v$, (B.45) implies

$$D_{\mathrm{tot}} = \frac{2(vb_t^* - \tau^*)}{B} \leq 2 - \frac{2v}{B}(1 - b_t^*) < 2. \tag{B.47}$$

By Definition-Proposition 5.7, applied in case (b) with $\beta_i^* = \beta_\infty > 0$, any lc center of $(\mathcal{A}_\mathbf{k}, C_\infty)$ intersecting $E_\mathbf{k}$ is a floor, the preimages of the floors in the chart covers are smooth at the points over $E_\mathbf{k}$, and no two of them intersect at a point over $E_\mathbf{k}$. Since the limit is not klt near the base,

there exists a floor F, and $F \neq E_{\mathbf{k}}$ since $b_t^* < 1$. Let $\tilde{F}$ be the preimage of F in the chart covers. Any component of $\tilde{F}$ through a point of $\hat{E}_{\mathbf{k}}$ has coefficient one in C_∞, so it contributes at least 1 to the degree D_{tot} of the restriction of the strict part of C_∞ to $\hat{E}_{\mathbf{k}}$, and at least 2 if it is tangent to $\hat{E}_{\mathbf{k}}$. By (B.47), $\tilde{F} \cap \hat{E}_{\mathbf{k}}$ is a single point, $\tilde{F}$ is transversal to $\hat{E}_{\mathbf{k}}$ there, and F is the only floor. The group K preserves $\tilde{F}$ and acts on $\hat{E}_{\mathbf{k}}$ by $p_y \to \zeta^B p_y$, $\zeta \in \mu_A$, through a cyclic group of order $A/\gcd(A,B) = n/d \geq 2$, whose only fixed points are the points $p_y = 0$ and $p_x = 0$ over ξ_y and ξ_x. Thus F intersects $E_{\mathbf{k}}$ only at ξ_y or only at ξ_x. Exchanging x and y if necessary, we may assume that it is ξ_y. Since F is the only floor, H_E fixes ξ_y, so it contains no element exchanging ξ_y and ξ_x, and G_i contains no element exchanging x and y. Thus H_y is the group acting on U_y over a neighborhood of ξ_y, it is diagonal, and $q_y|_{\tilde{F}}$ is a parameter of $\tilde{F}$ at the origin 0_y of U_y.

Let $b_F := \mathrm{mult}_{0_y}((C_\infty - \tilde{F})|_{\tilde{F}})$ and let e_F be the order of the image of H_y in $\mathrm{Aut}(T_{0_y}\tilde{F})$. Since $(q_y = 0)$ has coefficient b_t^* in $C_\infty - \tilde{F}$ and is transversal to $\tilde{F}$, $b_F \geq b_t^*$. Moreover $b_F \leq 1$ by Lemma 2.14, since $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ is lc at 0_y. Let f_F be an H_y-semi-invariant equation of $\tilde{F}$ at 0_y, and m_l the order of $\varphi^y_{\infty,l}$ along $\tilde{F}$. Replacing f_F and the H_y-semi-invariant power series $\varphi^y_{\infty,l}/f_F^{m_l}$ by their parts of degree less than a large integer gives a pair on $\mathbb{A}^2_{\mathbf{k}}$ whose numbers b_F and e_F equal those of $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ at 0_y (Nakayama's lemma) and whose quotient by the diagonal group H_y has minimal log discrepancy at the image of 0 equal to that of the quotient of $(\mathcal{A}_{\mathbf{k}}, C_\infty)$ at the image of ξ_y [Liu26, Lemmas 14.14(1) and 3.3(2)]. Thus Lemma 5.6, applied to the pair on $\mathbb{A}^2_{\mathbf{k}}$, shows that the minimal log discrepancy of the quotient at the image of ξ_y is $(1 - b_F)/e_F$, so

$$\beta_\infty \leq \frac{1 - b_F}{e_F}, \qquad 0 < 1 - b_F \leq 1 - b_t^*. \tag{B.48}$$

Let

$$U := y^{v/g_{Av}} t^{A/g_{Av}}, \qquad \lambda := \frac{(1 - b_F) g_{Av}}{A}.$$

By (B.30) and $t = \tilde{\eta}^{-v} q_y$, $U = q_y^{A/g_{Av}}$ on U_y, and $U = \pm p_x^{nv/g_{Av}} q_x^{A/g_{Av}}$ on U_x. Thus $\lambda \operatorname{div}(U) = (1 - b_F)\operatorname{div}(q_y)$ on U_y. Since the functions U on U_y and U_x do not depend on i, the compatible generic limit of the tuples augmented by them is the limit tuple augmented by them, so Theorem 4.2(1) holds for the augmented tuples. The group K acts on q_y by ζ^{-v}, $\zeta \in \mu_A$, whose image is $\mu_{A/g_{Av}}$, so A/g_{Av} divides e_F, and the character of U under H_y has order $m_U := e_F g_{Av}/A$. The function U is invariant under R since $\kappa(U) = 0$, and the elements of K act trivially on $q_y^{A/g_{Av}}$, so m_U is the order of the character of the G_i-semi-invariant function U.

We show that $(\mathcal{A}_{\mathbf{k}}, C'_\infty)$, $C'_\infty := C_\infty + \lambda \operatorname{div}(U)$, is lc near $E_{\mathbf{k}}$. Near 0_y, the multiplicity of $(C'_\infty - \tilde{F})|_{\tilde{F}}$ at 0_y is $b_F + (1 - b_F) = 1$, so $(\mathcal{A}_{\mathbf{k}}, C'_\infty)$ is lc near ξ_y by Lemma 2.14, applied on the completion $\operatorname{Spec} \mathbf{k}[[p_y, q_y]]$ of the chart $U_{y,\mathbf{k}}$ at 0_y with the regular curve $\tilde{F}$ (lc on $\mathcal{A}_{\mathbf{k}}$ is defined pointwise on the charts in Construction-Proposition 4.1). For $\omega \in \{x, y\}$, the coefficient of $(q_\omega = 0)$ in C'_∞ is $b_t^* + 1 - b_F \leq 1$. The restriction to $\hat{E}_{\mathbf{k}}$ of the strict part of C'_∞ has degree $D_{\mathrm{tot}} + \lambda nv/g_{Av}$, where $\lambda \cdot nv/g_{Av} = (1 - b_F)nv/A = 2v(1 - b_F)/B$ is the contribution of U at the point over ξ_x. Removing the contribution 1 of $\tilde{F}$ at 0_y, the other contributions add up to

$$\frac{2(vb_t^* - \tau^*)}{B} - 1 + \frac{2v(1 - b_F)}{B} \leq \frac{2(v - \tau^*)}{B} - 1 \leq 1, \tag{B.49}$$

by (B.48) and $\delta^* \geq 0$. All the contributions are non-negative, so at any point o'' of $\hat{E}_{\mathbf{k}}$ other than 0_y, the restriction to $\hat{E}_{\mathbf{k}}$ of $C'_\infty - (b_t^* + 1 - b_F)(q_\omega = 0)$ has multiplicity at most 1 at o''. Let C''_∞ be C'_∞ with the coefficient of $(q_\omega = 0)$ raised to 1. By Lemma 2.14 with the regular curve $(q_\omega = 0)$, applied on the completion of the chart $U_{\omega,\mathbf{k}}$ at o'', $(\mathcal{A}_{\mathbf{k}}, C''_\infty)$ is lc near o'', hence so is $(\mathcal{A}_{\mathbf{k}}, C'_\infty)$.

By Theorem 4.2(1) for the tuples augmented by U, the pair

$$\Big(U_\omega, \sum_l b_l^* \operatorname{div}(\varphi^\omega_{il}) + \lambda \operatorname{div}(U)\Big)$$

is lc near the preimage of E for any i. By Claim B.5, applied to the polynomials h^0_{il} and U with the numbers b^*_l and λ, whose adjusted mass is $\delta^* \geq 0$ since $\kappa(U) = 0$, and by Claim B.6, the pair

$$\Big(W^0_i, \sum_l b^*_l \operatorname{div}(h^0_{il}) + \lambda \operatorname{div}(U)\Big)$$

is lc near 0. By Claim B.4 with $\Lambda = (\Gamma\backslash\{0\}) \cup \{b^*_l\} \cup \{\lambda\}$ and $b_{il} \leq b^*_l$, $(W_i, D_i + \lambda \operatorname{div}(U))$ is lc near 0 for large i. By Lemma 5.8 with $Z = \{0\}$ and (B.48),

$$\alpha_i \geq \frac{\lambda}{m_U} = \frac{1 - b_F}{e_F} \geq \beta_\infty$$

for large i, which contradicts $\alpha_i < \lim_j \alpha_j \leq \beta_\infty$. Thus the theorem holds for the germs of **Step 1**.

Step 8. In this step, we reduce the germs with a completed presentation to the germs of **Step 1** by a numerical admission.

Let $(W \ni o)$, G, and $D = \sum_l b_l \operatorname{div}(h_l)$ be as in the second part of Theorem 8.6, with a G-equivariant isomorphism

$$\widehat{\mathcal{O}}_{W,o} \cong \mathbb{C}[[x, y, z, t]]/(Q(x, y) + \hat{g}(z, t)).$$

It suffices to show that for large r there exists a datum (W_0, G, D_0) as in **Step 1** with the group G, the subgroup R, the labels b_l, and the quotient value α of $(W \ni o, G, D)$.

Since W has embedding dimension four at o, G-eigenfunctions $x_1, \dots, x_4$ near o whose classes form a basis of $\mathfrak{m}_o/\mathfrak{m}_o^2$ define a G-equivariant closed embedding of a neighborhood of o into a smooth fourfold germ $(Y \ni o)$ with $W = (f = 0)$ for a G-semi-invariant regular function f. The ideal $(Q + \hat{g})$ is G-stable, so for any $\iota \in G$ we have $\iota^*(Q + \hat{g}) = \upsilon_\iota(Q + \hat{g})$ for a unit υ_ι. Since $\hat{g} \in (z, t)^3$ and $\iota^*\hat{g} \in \mathbb{C}[[z, t]]$, if υ_ι were not a constant, then, for its non-zero homogeneous part $\upsilon_{\iota,m}$ of least positive degree m, comparing the parts of degree $m + 2$ would show that $\upsilon_{\iota,m}Q$, a non-zero element of $(x, y)^2$, is the part of degree $m + 2$ of $\iota^*\hat{g} - \upsilon_\iota(0)\hat{g} \in \mathbb{C}[[z, t]]$, which is impossible. Thus $Q + \hat{g}$ is G-semi-invariant, and so is its part Q of degree two. Taking an eigenbasis of the span of x, y on the right side, Lemma 3.8 yields a G-equivariant isomorphism

$$\psi_1 : \mathbb{C}[[x_1, \dots, x_4]] \to \mathbb{C}[[x, y, z, t]]$$

and a G-invariant unit υ_1 with $\psi_1(\hat{f}) = \upsilon_1(Q + \hat{g})$.

We bring $Q + \hat{g}$ to a normal form. Since G preserves Q up to a character, it permutes the isotropic lines of Q, so in isotropic coordinates G acts monomially on $\{x, y\}$ and $Q = xy$ up to a constant. The coefficient of z^n in $\hat{g}$ is non-zero since $\operatorname{mult}_0 \hat{g}(z, 0) = n$, so $\hat{g}$ has the character of z^n. By the Weierstrass preparation theorem [GLS07, Chapter I, Theorem 1.6], $\hat{g} = u_0 \cdot (z^n + \sum_{j<n} \hat{a}_j(t)z^j)$ with a unit $u_0 \in \mathbb{C}[[z, t]]$ and $\hat{a}_j(0) = 0$. For any $\iota \in G$, $\iota^*\hat{g}$ is the product of ι^*u_0 and $\chi_z(\iota)^n$ times a Weierstrass polynomial, so the uniqueness of the unit and of the coefficients implies that u_0 is G-invariant and $\hat{a}_j(t)z^j$ has the character of z^n. Let $u_1 := u_0(z - \hat{a}_{n-1}(t)/n, t)$, a G-invariant unit, and let $u_1^{1/2}$ be its square root with value $u_1(0)^{1/2}$ at 0, which is G-invariant and commutes with the exchange of x and y. Let ψ_2 be the automorphism of $\mathbb{C}[[x, y, z, t]]$ obtained by composing the linear change of (x, y) to isotropic coordinates of Q, with the constant of Q absorbed in x, and the substitution

$$x \to u_1^{1/2}x, \qquad y \to u_1^{1/2}y, \qquad z \to z - \frac{\hat{a}_{n-1}(t)}{n}, \qquad t \to t.$$

It is G-equivariant, with G acting on the new coordinates by the linear maps by which it acts on the isotropic coordinates of Q and on z, t, that is, monomially on $\{x, y\}$ and diagonally on z, t, since the shift of z is G-equivariant as $\hat{a}_{n-1}(t)$ has the character of z. Since $u_0 \in \mathbb{C}[[z, t]]$, we have $\psi_2(u_0) = u_1$ and

$$\psi_2(Q + \hat{g}) = \psi_2(u_0)f_\infty, \qquad f_\infty := xy + z^n + \sum_{j \leq n-2} \hat{a}_j(t)z^j, \quad \hat{a}_j(0) = 0,$$

where we write $\hat{a}_j(t)$ again for the coefficient of z^j in $\psi_2(z^n + \sum_{j<n} \hat{a}_j(t)z^j)$. Its coefficient of z^{n-1} is zero. Let $\psi := \psi_2 \circ \psi_1$ and $v_2 := \psi_2(v_1 u_0)$, a G-invariant unit, so that

$$\psi(\hat{f}) = v_2 f_\infty.$$

The group G acts on the new coordinates as on the isotropic coordinates of Q and on z, t, and $\mathbf{T}$ acts by scalars on the span of x, y, so $R = G \cap \mathbf{T}$ does not change in the new coordinates. Since f_∞ is R-semi-invariant with the character of xy, each monomial $t^\sigma z^j$ of $\hat{a}_j(t)z^j$ satisfies (B.23), so $\sigma \geq (r - nB)/v$.

We lift the marks to G-semi-invariant regular functions on Y, and the marks with label 0 are dropped, since they do not change D. We apply Construction-Theorem 4.13 with $n = 4$, the marks h_l, $c^{(0)} = (b_l)_l$, no further vectors and prime divisors, ψ, the unit v_2^{-1}, and $u_l = 1$, in an eigenbasis of the span of x, y. For a power series ϕ and a positive integer N, let $\phi_{<N}$ be the sum of the terms of ϕ of degree less than N. Let $N \geq N_1$ and $L \geq L_1$ be the integers there, let $h_{l,0} := \psi(\hat{h}_l)_{<N}$, and let $f_0 := xy + z^n + \sum_{j \leq n-2} a_j(t)z^j$, where $a_j := (\hat{a}_j)_{<L}$. Then $h_{l,0}$ and f_0 are G-semi-invariant polynomials with the characters of h_l and f, $f_0 \equiv v_2^{-1}\psi(\hat{f})$ modulo $(x, y, z, t)^L$, and $a_j(0) = 0$. Let $W_0 := (f_0 = 0) \subset \mathbb{A}^4$ and $D_0 := \sum_l b_l \operatorname{div}(h_{l,0}|_{W_0})$. By Construction-Theorem 4.13(1) and (3), G acts faithfully on W_0, (W_0, D_0) is klt near 0,

$$\operatorname{mld}((W_0 \ni 0, D_0)/G) = \operatorname{mld}((W \ni o, D)/G) = \alpha,$$

and W_0 is an isolated Gorenstein terminal germ at 0. For large r, f_0 has no quadratic monomial other than xy, since $\sigma \geq (r - nB)/v$, so W_0 has quadratic rank two. Thus (W_0, G, D_0) is a datum as in **Step 1** with the group G, the subgroup R, the order r, the labels b_l, and the value α of $(W \ni o, G, D)$. All the root extensions, modifications, and comparisons of **Steps 1–7** are carried out on W_0. Hence a sequence of germs with a completed presentation with $r_i \to +\infty$ and positive strictly increasing α_i yields such a sequence of germs of **Step 1**, which does not exist by **Steps 5–7**. □

B.5. **The circle** $(1, 1, 2, 0)$**.** In this subsection, we prove Theorem 8.7.

Lemma B.10. *Let $\Gamma \subset [0, 1]$ be a DCC set and N a positive integer. Then there exist an ACC set $\mathcal{B} \subset \mathbb{R}_{\geq 0}$ and real numbers $\epsilon \in (0, 1)$ and $M > 0$, depending only on Γ and N, such that the following holds. Let H be a finite abelian group of order at most N acting diagonally on $\mathbb{A}^4$ with coordinates v, y_1, y_2, y_3 and faithfully on $V := (v = 0) \cong \mathbb{A}^3$, let $P_1, \dots, P_s$ be H-semi-invariant polynomials not divisible by v, $c_1, \dots, c_s \in (\Gamma \cup \{1\})\backslash\{0\}$, $B := \sum_j c_j \operatorname{div}(P_j)$, $B|_V := \sum_j c_j \operatorname{div}(P_j|_V)$, and $b \in (1 - \epsilon, 1]$. If $(\mathbb{A}^4, B + b \operatorname{div}(v))$ is lc near 0, then $(V, B|_V)$ is lc near 0, and*

$$\beta := \operatorname{mld}((V \ni 0, B|_V)/H) \in \mathcal{B}, \qquad \beta \leq \operatorname{mld}((\mathbb{A}^4 \ni 0, B + b \operatorname{div}(v))/H) \leq \beta + M(1 - b).$$

Proof. Let $I := (\Gamma \cup \{1\})_+ \cap [0, 1]$, which is a DCC set [Liu26, Lemma 2.10]. By Theorem 2.12(1) with $n = 4$, I, and $J = \{1\}$, the set $\mathcal{T}$ of log canonical thresholds of that theorem satisfies the ACC. Hence there exists $\epsilon \in (0, 1)$, depending only on Γ, with $\mathcal{T} \cap (1 - \epsilon, 1) = \emptyset$.

Let θ be the log canonical threshold of $\operatorname{div}(v)$ with respect to $(\mathbb{A}^4, B)$ near 0. Since $(\mathbb{A}^4, B + b \operatorname{div}(v))$ is lc near 0, and $\operatorname{div}(v)$ is a prime divisor through 0 that is not a component of B, we have $b \leq \theta \leq 1$. The coefficients of B near 0 are sums of the c_j and are at most 1, so they belong to I. Thus $\theta \in \mathcal{T} \cap [b, 1]$, and since $b > 1 - \epsilon$, we obtain $\theta = 1$. Thus $(\mathbb{A}^4, B + \operatorname{div}(v))$ is lc near 0. Since B and $\operatorname{div}(v)$ have no common component, $(V, B|_V)$ is lc near 0 [Kaw07, Theorem].

Since H acts faithfully on V, the restrictions $P_j|_V$ are non-zero as no P_j is divisible by v, and $(V, B|_V)$ is lc near 0, Theorem 3.3(1) with $n = 4$, $c = 1$, $f_1 = v$, $X = V$, the $\mathbb{R}$-ideal $\mathfrak{a} = \prod_j (P_j)^{c_j}$, and H in place of both G and H implies

$$\beta = \operatorname{mld}\big((\mathbb{A}^4 \ni 0, B + \operatorname{div}(v))/H\big).$$

Lowering the coefficient of $\operatorname{div}(v)$ from 1 to b does not decrease any log discrepancy, and the inertia groups do not depend on the boundary. Thus $\beta \leq \operatorname{mld}((\mathbb{A}^4 \ni 0, B + b \operatorname{div}(v))/H)$.

We show that β belongs to an ACC set depending only on Γ and N. Since H is a diagonal group of order at most N, the entries of the elements of H are roots of unity of order at most N. Hence the varieties V/H belong to a finite set $\{V_1, \dots, V_m\}$ depending only on N, and each V_k is klt and $\mathbb{Q}$-factorial [KM98, Lemma 5.16 and Proposition 5.20(4)]. Let $\bar{0}$ be the image of 0 in V/H, and let Δ' be the $\mathbb{R}$-divisor on V/H whose crepant pull-back is $B|_V$. Since $(V, B|_V)$ is lc near 0, the coefficients of $B|_V$ near 0 belong to I. Thus the coefficients of Δ' near $\bar{0}$ belong to the DCC set

$$\Lambda := \left\{1 - \frac{1-c}{e} \;\middle|\; c \in I,\ 1 \le e \le N\right\}\backslash\{0\}$$

[KM98, Proposition 5.20], [Liu26, Lemma 2.6(4) and Lemma 2.10]. The pair $(V/H, \Delta')$ is lc near $\bar{0}$ [KM98, Proposition 5.20], and $\beta = \operatorname{mld}(V/H \ni \bar{0}, \Delta')$ [Liu26, Lemma 2.6]. Let Δ'' be the sum of the components of Δ' through $\bar{0}$, with their coefficients in Δ'. Then $\Delta'' \in \Lambda$, Δ'' is $\mathbb{R}$-Cartier since V/H is $\mathbb{Q}$-factorial, and $\Delta'' = \Delta'$ near $\bar{0}$, so $(V/H, \Delta'')$ is lc near $\bar{0}$ and $\beta = \operatorname{mld}(V/H \ni \bar{0}, \Delta'')$. Thus Theorem 2.10(1) applied to $V_1, \dots, V_m$ and Λ implies that β belongs to an ACC set $\mathcal{B} \subset \mathbb{R}_{\ge 0}$ depending only on Γ and N. By Theorem 2.10(3) applied to $V_1, \dots, V_m$ and Λ, there exists a positive integer l, depending only on Γ and N, such that there exists a prime divisor S over V/H which computes $\operatorname{mld}(V/H \ni \bar{0}, \Delta'') = \beta$ and satisfies $a(S, V/H, 0) \le l$.

Let γ be the least positive element of the DCC set $\Gamma \cup \{1\}$, and let $M := l/\gamma + 1$. Let ν be a prime divisor over V with center 0 whose image over V/H is S [Liu26, Lemma 2.6(3)], and let $e := |I_H(\nu)|$. By [Liu26, Lemma 2.6(2)],

$$a(\nu, V, B|_V) = e\beta \ge 0 \quad \text{and} \quad a(\nu, V, 0) \le e a(S, V/H, 0) \le el,$$

where the inequality $a(\nu, V, 0) \le e a(S, V/H, 0)$ holds since the $\mathbb{R}$-divisor Ξ on V/H whose crepant pull-back to V is 0 satisfies $\Xi \ge 0$. Hence

$$\sum_j c_j \operatorname{mult}_\nu(P_j|_V) \le el.$$

Since $c_j \ge \gamma$ for any j, we obtain

$$\operatorname{mult}_\nu(P_j|_V) \le \frac{el}{\gamma} \quad \text{for any } j.$$

We now construct a prime divisor over $\mathbb{A}^4$ which bounds $\operatorname{mld}((\mathbb{A}^4 \ni 0, B + b\operatorname{div}(v))/H)$ from above. Let q be the integer of Lemma 3.1 applied with V, H, the character of v, $B|_V$, and ν in place of V, H, χ, B, and E. Let ℓ be an integer with $\ell \equiv q \pmod{e}$ and

$$\frac{el}{\gamma} < \ell \le eM,$$

which exists since the interval has length e. Lemma 3.1 implies that there exists a prime divisor F over $\mathbb{A}^4 = V \times \mathbb{A}^1_v$ with center 0, $\operatorname{mult}_F v = \ell$, $I_H(F) = I_H(\nu)$, and

$$a(F, \mathbb{A}^4, \operatorname{pr}_1^*(B|_V)) = a(\nu, V, B|_V) + \ell.$$

For a polynomial $P = \sum_k P_k(y_1, y_2, y_3)v^k$, we have $\operatorname{mult}_F P = \min_k(\operatorname{mult}_\nu P_k + k\ell)$ [Liu26, Definition-Proposition 2.11]. If $\ell > \operatorname{mult}_\nu P_0$, then $\operatorname{mult}_F P = \operatorname{mult}_\nu P_0$. By the last bound on $\operatorname{mult}_\nu(P_j|_V)$, this holds for $P = P_j$, with $P_0 = P_j|_V$, for any j. Hence $\operatorname{mult}_F B = \operatorname{mult}_F \operatorname{pr}_1^*(B|_V)$, and

$$a(F, \mathbb{A}^4, B + b\operatorname{div}(v)) = a(\nu, V, B|_V) + \ell - b\ell = e\beta + (1-b)\ell.$$

The prime divisor over $\mathbb{A}^4/H$ under F has center the image of 0 and log discrepancy $a(F, \mathbb{A}^4, B + b\operatorname{div}(v))/|I_H(F)|$ with respect to the crepant quotient [Liu26, Lemma 2.6(2)], and $|I_H(F)| = e$. Thus

$$\operatorname{mld}((\mathbb{A}^4 \ni 0, B + b\operatorname{div}(v))/H) \le \frac{a(F, \mathbb{A}^4, B + b\operatorname{div}(v))}{e} = \beta + \frac{(1-b)\ell}{e} \le \beta + M(1-b).$$

□

Proof of Theorem 8.7. The proof is long, so we divide it into several steps.

Step 1. In this step we introduce the weighted blow-up of the t-axis and show that $\sigma < 2$.

Consider data as in Set-up 8.2 with $W = (xy + zf(z,t) = 0) \subset \mathbb{A}^4$, where f is a polynomial with $f(0,0) = 0$ and $f(0,t) \neq 0$, and with

$$\mathbf{T} = \mathbf{T}_{(1,1,2,0)} = \{(\lambda, \lambda, \lambda^2, 1)\}.$$

Thus R acts on a monomial $x^a y^b z^c t^d$ by λ^{a+b+2c}, and we call $a+b+2c$ the *weight* of $x^a y^b z^c t^d$. Write $D = \sum_j c_j \operatorname{div}(h_j)$, where $c_j \in \Gamma$ are the labels and h_j are the marks. As in Set-up 5.1(4), we remove the terms with $c_j = 0$ and the terms whose mark h_j is a unit near 0, and this does not change D near 0. Thus $c_j > 0$ for any j, and no h_j is a unit near 0.

If q is a denominator of h_j with $q(0) \neq 0$, then $\prod_{g \in G} g^* q$ is a G-invariant unit near 0, and multiplying h_j by it does not change D near 0. Thus we may assume that each h_j is an element of $\mathbb{C}[W]$. Since $xy = -zf$ on W, any element of $\mathbb{C}[W]$ is the restriction of a unique polynomial with no monomial divisible by xy. Since G acts monomially on $\{x, y\}$ and diagonally on z, t, the polynomial of a G-semi-invariant element is G-semi-invariant, and we denote it again by h_j. Let k_j be the least weight of a monomial of h_j, and let

$$\sigma := \sum_j c_j k_j.$$

Since R acts diagonally and h_j is R-semi-invariant, the weight of any monomial of h_j is congruent to k_j modulo r. Moreover, the equation $xy + zf$ is G-semi-invariant, and xy and z both have the character λ^2 on R. Hence f is R-invariant.

Let $C := (x = y = z = 0)$ be the t-axis, and let $\pi : W' \to W$ be the strict transform of W in the weighted blow-up of $\mathbb{A}^4$ along C with weights $(1,1,2)$ on (x,y,z). Since G acts linearly on the span of x, y and diagonally on z, t, and x, y both have weight 1, G acts on W' and π is G-equivariant. Let U_x, U_y, and U_z be the x-chart, the y-chart, and the z-chart of W'. On U_x we have $x = s$, $y = sY$, and $z = s^2 Z$, and W' is defined by $Y + Zf(s^2 Z, t) = 0$. Eliminating Y, we obtain $U_x \cong \mathbb{A}^3_{(s,Z,t)}$. The chart U_y is obtained by exchanging x and y.

We show that

$$h_j|_{U_x} = s^{k_j} H_j^x(s, Z, t) \quad \text{with} \quad H_j^x(0, Z, t) \neq 0. \tag{B.50}$$

A monomial $x^a z^c t^d$ becomes $s^{a+2c} Z^c t^d$, and a monomial $y^b z^c t^d$ with $b \geq 1$ becomes $s^{b+2c}(-Zf(s^2 Z, t))^b Z^c t^d$. Hence $H_j^x(0, Z, t)$ is the sum of the terms $\varphi(t) Z^c$ coming from the monomials $\varphi(t) x^a z^c$ of weight k_j and of the terms $\psi(t)(-f(0,t))^b Z^{b+c}$ coming from the monomials $\psi(t) y^b z^c$ of weight k_j with $b \geq 1$. In the first family $c \leq k_j/2$, and in the second family $b + c = k_j - c > k_j/2$. Since c determines the monomial of weight k_j in each family, the exponents of Z are pairwise different. Since $f(0,t) \neq 0$, this implies (B.50).

The 3-form $\omega := \frac{dx \wedge dz \wedge dt}{x}$ generates ω_W near 0 [KM98, Proposition 5.73]. On U_x,

$$\pi^* \omega = \frac{ds}{s} \wedge (2sZ\,ds + s^2\,dZ) \wedge dt = s\,ds \wedge dZ \wedge dt.$$

Let E be the closure of $(s = 0) \subset U_x$ in W'. It is an exceptional prime divisor over W with center C, and by (B.50),

$$a(E, W, 0) = 2 \quad \text{and} \quad \operatorname{mult}_E h_j = k_j \text{ for any } j.$$

Since (W, D) is klt near 0 and $0 \in C$,

$$0 < a(E, W, D) = 2 - \sigma. \tag{B.51}$$

Since σ is a sum of $\sum_j k_j$ elements of Γ, we have $\sigma \in \Gamma_+ \cap [0, 2)$.

Step 2. In this step we compute the crepant pull-backs of $K_W + D$ to the charts of W' divided by R, and show that their coefficients belong to one DCC set.

The chart U_z is the quotient of

$$\widetilde{W} := (XY + f(s^2, t) = 0) \subset \mathbb{A}^4_{(s,X,Y,t)}$$

by the involution $(s, X, Y, t) \to (-s, -X, -Y, t)$, where $z = s^2$, $x = sX$, and $y = sY$. The polynomial $XY + f(s^2, t)$ is irreducible, and the singular locus of $\widetilde{W}$ is contained in the curve $(X = Y = f(s^2, t) = 0)$. Thus $\widetilde{W}$ is normal by Serre's criterion. Let $p : \widetilde{W} \to W$ be the induced morphism. As in (B.50), $h_j \circ p = s^{k_j} H_j$ for a polynomial $H_j(s, X, Y, t)$, and $H_j(0, X, Y, t)$ is the non-zero sum of the terms of weight k_j of h_j with x, y, z replaced by $X, Y, 1$. Since $H_j(0, X, Y, t)$ has no monomial divisible by XY, it is not divisible by the irreducible polynomial $XY + f(0, t)$. Hence the prime divisor $(s = 0) = (XY + f(0, t) = 0)$ of $\widetilde{W}$ is not a component of $\operatorname{div}(H_j)$. Moreover,

$$p^*\omega = \Big(\frac{ds}{s} + \frac{dX}{X}\Big) \wedge 2s\,ds \wedge dt = 2s\frac{dX \wedge ds \wedge dt}{X},$$

and $\frac{dX \wedge ds \wedge dt}{X}$ generates $\omega_{\widetilde{W}}$ [KM98, Proposition 5.73]. Pulling back $K_W + D$, we obtain

$$K_{\widetilde{W}} + \sum_j c_j \operatorname{div}(H_j) + (\sigma - 1)\operatorname{div}(s) = p^*(K_W + D). \tag{B.52}$$

For $g \in G$, let $\chi_z(g)$ and $\chi_t(g)$ be the characters of z and t, and let M_g be the matrix of g on the span of x, y. Here M_g may exchange x and y. Let $\widetilde{G}$ be the group of pairs (g, ρ) with $g \in G$ and $\rho^2 = \chi_z(g)$, acting on $\mathbb{A}^4_{(s,X,Y,t)}$ by

$$s \to \rho s, \quad (X, Y) \to \rho^{-1} M_g(X, Y), \quad t \to \chi_t(g)t.$$

Then $\widetilde{G}$ is a finite abelian group of order $2|G|$, it acts on $\widetilde{W}$, and p is equivariant with respect to $\widetilde{G} \to G$, $(g, \rho) \to g$. The kernel of $\widetilde{G} \to G$ is generated by the involution above. Hence $\widetilde{W}/\widetilde{G} = U_z/G$, which is an open subset of W'/G since U_z is G-stable. Since G acts faithfully on W, the group $\widetilde{G}$ acts faithfully on $\mathbb{A}^4_{(s,X,Y,t)}$, hence on $\widetilde{W}$, which is not contained in a hyperplane.

For λ with $\lambda^r = 1$, let $g_\lambda := (\lambda, \lambda, \lambda^2, 1) \in R$, and let

$$\widetilde{R} := \{(g_\lambda, \lambda) \mid \lambda^r = 1\} \subset \widetilde{G}.$$

The element (g_λ, λ) acts by $s \to \lambda s$ and trivially on X, Y, t. Conversely, if $(g, \rho) \in \widetilde{G}$ acts trivially on X, Y, t, then $M_g = \rho \cdot \mathrm{id}$ and $\chi_t(g) = 1$, so g acts on (x, y, z, t) by $(\rho, \rho, \rho^2, 1)$. Thus $g = g_\rho \in R$ and $(g, \rho) \in \widetilde{R}$. Therefore $\widetilde{R}$ is the subgroup of elements acting trivially on X, Y, t, the group $\bar{G} := \widetilde{G}/\widetilde{R}$ acts faithfully on $\mathbb{A}^3_{(X,Y,t)}$, and $|\bar{G}| = 2[G : R] \le 2e_0$.

Since f is R-invariant, any monomial $z^c t^d$ of f satisfies $r \mid 2c$. Hence $f(s^2, t) = \bar{f}(s^r, t)$ for a polynomial $\bar{f}$ with $\bar{f}(0, t) = f(0, t)$. Since h_j and s^{k_j} both have the character λ^{k_j} under (g_λ, λ), the polynomial H_j is $\widetilde{R}$-invariant, and $H_j = \bar{H}_j(s^r, X, Y, t)$ for a polynomial $\bar{H}_j$. Let $u := s^r$. The quotient of $\widetilde{W}$ by $\widetilde{R}$ is

$$\overline{W} := (XY + \bar{f}(u, t) = 0) \subset \mathbb{A}^4_{(u,X,Y,t)},$$

the quotient morphism $\widetilde{W} \to \overline{W}$ is ramified only along $(s = 0)$, with ramification index r, and $\bar{G}$ acts on $\overline{W}$ with $\overline{W}/\bar{G} = U_z/G$. By (B.52) and [KM98, Proposition 5.20] (see also [Liu26, Lemma 2.6(4)]), the crepant pull-back of $K_W + D$ to $\overline{W}$ is $K_{\overline{W}} + B_{\overline{W}}$, where

$$B_{\overline{W}} := \sum_j c_j \operatorname{div}(\bar{H}_j) + b_r \operatorname{div}(u), \qquad b_r := 1 - \frac{2 - \sigma}{r}. \tag{B.53}$$

Here b_r is the solution b of $1 - (\sigma - 1) = r(1 - b)$, since the log discrepancy of $(s = 0)$ with respect to the left-hand side of (B.52) is $1 - (\sigma - 1)$ and $\widetilde{W} \to \overline{W}$ is ramified along $(s = 0)$ with ramification index r.

The group R acts on $U_x = \mathbb{A}^3_{(s,Z,t)}$ by $s \to \lambda s$ and trivially on Z, t, and by (B.50), $H^x_j = \bar{H}^x_j(s^r, Z, t)$ for a polynomial $\bar{H}^x_j$ with $\bar{H}^x_j(0, Z, t) \neq 0$. Thus $U_x/R = \mathbb{A}^3_{(u,Z,t)}$, and the crepant pull-back of $K_W + D$ to U_x/R is $K_{U_x/R} + D''_x$, where

$$D''_x := \sum_j c_j \operatorname{div}(\bar{H}^x_j) + b_r \operatorname{div}(u). \tag{B.54}$$

Exchanging x and y, we obtain $U_y/R \cong \mathbb{A}^3$ and an $\mathbb{R}$-divisor D''_y on U_y/R such that $K_{U_y/R} + D''_y$ is the crepant pull-back of $K_W + D$, given by the formula (B.54) with x and y exchanged.

Assume that $r \geq 3$. By (B.51), $0 < 1 - \frac{2}{r} \leq b_r < 1$. Thus $B_{\overline{W}} \geq 0$ and $D''_x \geq 0$. Since (W, D) is klt near 0, the pairs $(\overline{W}, B_{\overline{W}})$ and $(U_x/R, D''_x)$ are klt over a neighborhood of 0 [KM98, Proposition 5.20].

Let $\mathcal{B}$ be the set of numbers $1 - \frac{2-\sigma'}{r'}$, where $r' \geq 3$ is an integer and $\sigma' \in \Gamma_+ \cap [0, 2)$. We show that $\mathcal{B}$ is a DCC set. Let $b^{(n)} = 1 - \frac{2-\sigma_n}{r_n}$ be a strictly decreasing sequence in $\mathcal{B}$. If r_n is unbounded, then possibly passing to a subsequence, $r_n \to +\infty$ and $b^{(n)} \geq 1 - \frac{2}{r_n} \to 1$, which contradicts $b^{(n)} < b^{(1)} < 1$. If r_n is bounded, then possibly passing to a subsequence, r_n is constant and σ_n is strictly decreasing, which contradicts the DCC of Γ_+ [Liu26, Lemma 2.10]. Thus

$$\Theta := \Gamma \cup \mathcal{B} \subset [0, 1]$$

is a DCC set, and $c_j, b_r \in \Theta$.

Step 3. In this step we show that, along a sequence of strictly increasing values, α is eventually the minimal log discrepancy at the point of W'/G over the origin of $\overline{W}$.

Assume that there exists a sequence of data (W_i, G_i, D_i) as in **Step 1**, with Γ and e_0 fixed and $r_i \to +\infty$, whose values satisfy $0 < \alpha_1 < \alpha_2 < \cdots$. Possibly passing to a subsequence, we may assume that $r_i \geq 3$ for any i. We add the index i to the objects of **Steps 1 and 2**.

Let $(W/G, D_{W/G})$ be the crepant quotient of (W, D), let $\pi_G : W'/G \to W/G$ be the morphism induced by π, and let D' be the $\mathbb{R}$-divisor with $K_{W'/G} + D' = \pi_G^*(K_{W/G} + D_{W/G})$. Let $\bar{0}$ be the image of 0 in W/G. Then $\alpha = \operatorname{mld}(W/G \ni \bar{0}, D_{W/G})$ [Liu26, Lemma 2.6]. Since π_G is proper birational and D' is the crepant pull-back, a prime divisor over W/G with center $\bar{0}$ has center contained in $\pi_G^{-1}(\bar{0})$, and the log discrepancies of such a divisor with respect to $(W'/G, D')$ and to $(W/G, D_{W/G})$ are equal. Moreover, $(W/G, D_{W/G})$ is klt near $\bar{0}$ [KM98, Proposition 5.20], so [Liu26, Lemma 3.5], applied to a log resolution of $(W/G, D_{W/G})$ near $\bar{0}$ on which $\mathfrak{m}_{\bar{0}}$ becomes invertible and the preimage of $\bar{0}$ is part of the snc divisor, implies that there exists a prime divisor computing α. Hence

$$\alpha = \min\bigl\{\operatorname{mld}(W'/G \ni \eta, D') \;\big|\; \eta \in \pi_G^{-1}(\bar{0})\bigr\},$$

where η runs over the scheme-theoretic points of $\pi_G^{-1}(\bar{0})$, and the minimum is attained at the generic point of the center of a prime divisor computing α.

On U_z, the fiber $\pi^{-1}(0)$ is the image of $(s = t = XY = 0) \subset \widetilde{W}$, since $f(0, 0) = 0$. The points of $\pi^{-1}(0) \cap U_z$ with $X \neq 0$ (resp. $Y \neq 0$) belong to U_x (resp. U_y). Let $p_0 \in U_z$ be the image of the origin of $\widetilde{W}$. Thus $\pi^{-1}(0) \subset U_x \cup U_y \cup \{p_0\}$, and $\pi^{-1}(0)$ is a curve. On U_x, the fiber $\pi^{-1}(0)$ is $(s = t = 0)$, and on it $Y = -Zf(0, 0) = 0$. Since $U_x \cap U_y = (Y \neq 0) \subset U_x$, we have $\pi^{-1}(0) \cap U_x \cap U_y = \emptyset$. Let $p \in \pi^{-1}(0) \cap U_x$ be a closed point, and let G_p be the stabilizer of p. An element of G exchanging x and y maps U_x onto U_y, so it does not fix p. Hence any element of G_p acts diagonally on (x, y, z, t), preserves U_x, and acts diagonally on (s, Z, t). Moreover $R \subset G_p$, since R acts trivially on $(s = 0)$.

First, let η be a closed point of $\pi_G^{-1}(\bar{0})$ other than the image of p_0. By symmetry we may assume that η is the image of a closed point $p \in \pi^{-1}(0) \cap U_x$. Since G_p is the stabilizer of p, the morphism $U_x/G_p \to W'/G$ is étale at the image of p [Gro71, Exposé V, Proposition 2.2(ii)], and the crepant pull-back of $K_{W'/G} + D'$ to U_x/G_p is the crepant quotient of $(U_x/R, D''_x)$ by G_p/R.

Hence, after shrinking U_x/G_p around the image of p, [Liu26, Lemma 9.3] implies that the prime divisors with center η correspond bijectively to the prime divisors with center the image of p, with equal log discrepancies. Taking the quotient first by R and then by G_p/R, we obtain

$$\mathrm{mld}(W'/G \ni \eta, D') = \mathrm{mld}\big((U_x/R \ni \bar{p}, D''_x)/(G_p/R)\big)$$

by [Liu26, Lemma 2.6(5)], where $\bar{p}$ is the image of p. Here $U_x/R = \mathbb{A}^3_{(u,Z,t)}$ is smooth, G_p/R is a finite abelian group acting faithfully and diagonally on U_x/R and fixing $\bar{p}$ (an element of G_p acting trivially on (u,Z,t) fixes the function $Y|_{(s=0)} = -Zf(0,t)$, which is non-zero since $f(0,t) \neq 0$, so it acts on (x,y,z,t) by $(\lambda,\lambda,\lambda^2,1)$ for some λ and belongs to $G \cap \mathbf{T} = R$), the functions $\bar{H}^x_j$ and u are G_p/R-semi-invariant, and $(U_x/R, D''_x)$ is klt near $\bar{p}$ with $c_j, b_r \in \Theta$ by (B.54). By Theorem 2.8(1) applied to the DCC set Θ, the positive values $\mathrm{mld}(W'_i/G_i \ni \eta, D'_i)$ at such closed points belong to an ACC set $\mathcal{A}_1$.

Second, let η be the generic point of a curve in $\pi_G^{-1}(\bar{0})$. Then η is the image of the generic point of a curve in $\pi^{-1}(0) \cap (U_x \cup U_y)$. Near η, the coefficients of D' are of the form $1 - \frac{1-c}{e}$, where e is a positive integer and c is a coefficient of D''_x or D''_y [Liu26, Lemma 2.6(4)]. By (B.54), $c \in \Theta_+ \cap [0,1]$. Thus, possibly shrinking W'/G to a neighborhood of η, $D' \in \Psi(\Theta_+ \cap [0,1]) \cap [0,1]$, and $\Psi(\Theta_+ \cap [0,1]) \cap [0,1]$ is a DCC set [Liu26, Lemma 2.10]. By [Liu26, Lemma 3.7], the positive values $\mathrm{mld}(W'_i/G_i \ni \eta, D'_i)$ at such generic points belong to an ACC set $\mathcal{A}_2$.

The fiber $\pi_G^{-1}(\bar{0})$ has no component of dimension two. Since $0 < \alpha_1 < \alpha_2 < \cdots$, $\alpha_i \notin \mathcal{A}_1 \cup \mathcal{A}_2$ for $i \gg 0$. Thus for $i \gg 0$, any prime divisor computing α_i has center the image o'_i of $p_{0,i}$ on W'_i/G_i. Possibly passing to a subsequence, $\alpha_i = \mathrm{mld}(W'_i/G_i \ni o'_i, D'_i)$ for any i. The origin of $\overline{W}_i$ is fixed by $\bar{G}_i$, it maps to o'_i, $\overline{W}_i/\bar{G}_i$ is an open subset of W'_i/G_i, and $B_{\overline{W}_i}$ is the crepant pull-back of D'_i. Hence by [Liu26, Lemma 2.6],

$$\alpha_i = \mathrm{mld}\big((\overline{W}_i \ni 0, B_{\overline{W}_i})/\bar{G}_i\big). \tag{B.55}$$

Step 4. In this step we show that $\beta_i \leq \alpha_i \leq \beta_i + 2M/r_i$ for a number β_i in a fixed ACC set and a fixed constant M, and derive a contradiction.

Recall that $\overline{W}_i = (XY + \bar{f}_i(u,t) = 0) \subset \mathbb{A}^4_{(u,X,Y,t)}$ and that $\bar{G}_i$ acts faithfully on $\mathbb{A}^3_{(X,Y,t)}$ with $|\bar{G}_i| \leq 2e_0$. After a linear change of the coordinates X, Y, we may assume that $\bar{G}_i$ acts diagonally on $\mathbb{A}^4_{(u,X,Y,t)}$. Then $\overline{W}_i = (Q_i(X,Y) + \bar{f}_i(u,t) = 0)$ for a non-degenerate quadratic form Q_i. Since $\bar{G}_i$ acts faithfully on $(u=0) = \mathbb{A}^3_{(X,Y,t)}$, it acts faithfully on $\mathbb{A}^4$. An element of $\bar{G}_i$ acting trivially on $\overline{W}_i$ acts trivially on the linear span of $\overline{W}_i$, which is $\mathbb{A}^4$ since the hypersurface $\overline{W}_i$ is not contained in a hyperplane. Hence $\bar{G}_i$ acts faithfully on $\overline{W}_i$. Let

$$B_i := \mathrm{div}(Q_i + \bar{f}_i) + \sum_j c_{ij} \,\mathrm{div}(\bar{H}_{ij}) \quad \text{on } \mathbb{A}^4,$$

$$\Delta_i := \mathrm{div}(Q_i + f_i(0,t)) + \sum_j c_{ij} \,\mathrm{div}(\bar{H}_{ij}|_{u=0}) \quad \text{on } \mathbb{A}^3_{(X,Y,t)} = (u=0).$$

By **Step 2**, $\bar{H}_{ij}|_{u=0} = H_{ij}(0,X,Y,t) \neq 0$, and $Q_i + f_i(0,t) \neq 0$. Thus neither B_i nor Δ_i contains $(u=0)$, and $\Delta_i = B_i|_{u=0}$.

By Theorem 3.3(1) applied with $c=1$, $f_1 = Q_i + \bar{f}_i$, the $\mathbb{R}$-ideal $\mathfrak{a} = (u)^{b_{r_i}} \prod_j (\bar{H}_{ij})^{c_{ij}}$, and $H = \bar{G}_i$, (B.55) implies

$$\alpha_i = \mathrm{mld}\big((\mathbb{A}^4 \ni 0, B_i + b_{r_i} \,\mathrm{div}(u))/\bar{G}_i\big). \tag{B.56}$$

Since $(\overline{W}_i, B_{\overline{W}_i})$ is klt near 0 and $\overline{W}_i$ is a normal Cartier divisor on $\mathbb{A}^4$, the pair $(\mathbb{A}^4, B_i + b_{r_i} \,\mathrm{div}(u))$ is lc near 0 [Kaw07, Theorem]. Let $\mathcal{A}_3$, ϵ, and M be the ACC set and the numbers given by Lemma B.10 for Γ and the integer $2e_0$ in place of N. For $i \gg 0$, $1 - b_{r_i} = \frac{2-\sigma_i}{r_i} \leq \frac{2}{r_i} < \epsilon$, and the lemma, applied with $H = \bar{G}_i$, $v = u$, the polynomials $Q_i + \bar{f}_i$ (label 1) and $\bar{H}_{ij}$ (labels c_{ij}),

which are not divisible by u by **Step 2**, and $b = b_{r_i}$, shows that $\beta_i := \mathrm{mld}((\mathbb{A}^3 \ni 0, \Delta_i)/\bar{G}_i) \in \mathcal{A}_3$ and, by (B.56), that

$$\beta_i \le \alpha_i \tag{B.57}$$

and

$$\alpha_i \le \beta_i + M(1 - b_{r_i}) \le \beta_i + \frac{2M}{r_i}. \tag{B.58}$$

Possibly passing to a subsequence, we may assume that $\beta_i \in \mathcal{A}_3$, (B.57), and (B.58) hold for any i.

Since the β_i belong to an ACC set, possibly passing to a subsequence, we may assume that β_i is non-increasing. Let $\beta := \lim_{i\to+\infty} \beta_i$. Since $r_i \to +\infty$, (B.57) and (B.58) imply $\lim_{i\to+\infty} \alpha_i = \beta$ and $\alpha_i \ge \beta_i \ge \beta$ for any i. This contradicts the strict increase of α_i, since $\alpha_1 < \alpha_2 \le \beta \le \alpha_1$. Thus α cannot increase along $\mathbf{T}$, which proves the first part of Theorem 8.7.

Step 5. In this step we prove the second part of Theorem 8.7 by reduction to the first part.

Let $(W \ni o)$, G, and $D = \sum_k c_k \operatorname{div}(h_k)$ be as in the second part, with $r > 2$. Since Set-up 7.3 only considers $r \to +\infty$, it suffices to construct data as in the first part with labels in Γ, the bound e_0, and the values of r and α of (W, G, D). Since W is singular at o and

$$\widehat{\mathcal{O}}_{W,o} \cong \mathbb{C}[[x, y, z, t]]/(Q + \hat{g}),$$

we have $Q + \hat{g} \in (x, y, z, t)^2$ and $\dim \mathfrak{m}_o/\mathfrak{m}_o^2 = 4$. Choosing G-semi-invariant lifts of a basis of $\mathfrak{m}_o/\mathfrak{m}_o^2$ consisting of G-eigenvectors, we obtain a G-equivariant closed embedding of W into a smooth fourfold germ $(\mathcal{Y} \ni o)$ with $W = (f = 0)$ for a regular G-semi-invariant function f on $\mathcal{Y}$, and G-eigenparameters $x_1, \dots, x_4$ of $\mathcal{Y}$ at o. We lift the marks h_k to regular G-semi-invariant functions on $\mathcal{Y}$, denoted again by h_k. The marks with label 0 are dropped, since they do not change D. Let $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_4]]$, and let $\hat{f}$ and $\hat{h}_k$ be the expansions of f and h_k.

Since G acts linearly on the span of x, y and is abelian, after a linear change of x, y we may assume that G acts diagonally on $\mathbb{C}[[x, y, z, t]]$. The ideal $(Q + \hat{g})$ is G-stable, so for any $\iota \in G$ we have $\iota^*(Q + \hat{g}) = \upsilon_\iota(Q + \hat{g})$ for a unit υ_ι. As W has quadratic rank two, $\hat{g} \in (z, t)^3$, and $\iota^*\hat{g} \in \mathbb{C}[[z, t]]$. If υ_ι were not a constant, then, for its non-zero homogeneous part $\upsilon_{\iota,m}$ of least positive degree m, comparing the parts of degree $m + 2$ would show that $\upsilon_{\iota,m}Q$, a non-zero element of $(x, y)^2$, is the part of degree $m+2$ of $\iota^*\hat{g} - \upsilon_\iota(0)\hat{g} \in \mathbb{C}[[z, t]]$, which is impossible. Thus $Q + \hat{g}$ is G-semi-invariant, and, comparing the parts of degree two, $Q(x, y)$ is a non-degenerate G-semi-invariant quadratic form whose character is that of $Q + \hat{g}$. By Lemma 3.8, the given G-equivariant isomorphism

$$\mathfrak{R}/(\hat{f}) \cong \widehat{\mathcal{O}}_{W,o} \cong \mathbb{C}[[x, y, z, t]]/(Q + \hat{g})$$

lifts to a G-equivariant isomorphism $\mathfrak{R} \to \mathbb{C}[[x, y, z, t]]$ which maps $\hat{f}$ to a G-invariant unit times $Q + \hat{g}$, and the characters of $x_1, \dots, x_4$ are those of x, y, z, t up to order. Renumbering $x_1, \dots, x_4$ and identifying x, y, z, t with $x_1, \dots, x_4$, we obtain a G-equivariant automorphism ψ of $\mathfrak{R}$ and a G-invariant unit v of $\mathfrak{R}$ such that

$$v\psi(\hat{f}) = Q(x, y) + \hat{g}(z, t).$$

Since $\mathbf{T}$ acts by λ on x and y, the equation $Q + \hat{g}$ has the character λ^2 on R. Since t is R-invariant and $\lambda^2 \ne 1$ for a generator λ of R when $r > 2$, we have $\hat{g}(0, t) = 0$. If z^2 divided $\hat{g}$, then all partial derivatives of $Q + \hat{g}$ would vanish on the formal t-axis ($x = y = z = 0$), which contradicts the fact that W has an isolated singularity at o, since $\mathcal{O}_{W,o}$ is excellent, so that $\operatorname{Spec} \widehat{\mathcal{O}}_{W,o}$ is regular outside its closed point [Gro65, Scholie 7.8.3(v)]. Hence $\hat{g} = z\hat{f}_1$ with $\hat{f}_1(0, t) \ne 0$. Moreover, $\hat{f}_1(0, 0) = 0$, since $Q + \hat{g}$ has no linear term.

For a power series ϕ and a positive integer L, let $\phi_{<L}$ be the sum of the terms of ϕ of degree less than L. Since the coordinates are G-eigenparameters, $\phi_{<L}$ is G-semi-invariant with the character of ϕ for any G-semi-invariant ϕ. We apply Construction-Theorem 4.13 with $n = 4$, the

label vector $c^{(0)} = (c_k)_k$ only in (iv), no prime divisors in (v), the automorphism ψ, the unit v, and $u_k = 1$ for any k. Let N_1 be the integer of Construction-Theorem 4.13, let $N \geq N_1$, let L_1 be the corresponding integer, and let $L \geq \max\{L_1, 3, \mu + 2\}$, where μ is the order of the non-zero series $\hat{f}_1(0,t)$. Let $h_{k,0} := \psi(\hat{h}_k)_{<N}$ and

$$f_0 := (v\psi(\hat{f}))_{<L} = Q(x,y) + z(\hat{f}_1)_{<L-1}(z,t).$$

By Construction-Theorem 4.13(1) and (3), $W_0 := (f_0 = 0) \subset \mathbb{A}^4$ is an isolated Gorenstein terminal germ at 0 whose completion is G-equivariantly isomorphic to $\widehat{\mathcal{O}}_{W,o}$, G acts faithfully on W_0, the pair (W_0, D_0) with $D_0 := \sum_k c_k \operatorname{div}(h_{k,0}|_{W_0})$ is klt near 0, and

$$\operatorname{mld}((W_0 \ni 0, D_0)/G') = \operatorname{mld}((W \ni o, D)/G')$$

for any subgroup $G' \subset G$. In particular, W_0 has quadratic rank two, and the value α of (W_0, G, D_0) is equal to that of (W, G, D). Since $L - 1 > \mu$, $(\hat{f}_1)_{<L-1}(0,t) \neq 0$.

Since G preserves Q up to a character, it permutes the set of isotropic lines of Q. Thus in linear coordinates x, y of the span of x, y with $Q = xy$, the group G acts monomially on $\{x, y\}$, and $\mathbf{T}$ still acts by λ on x and y. Truncation by degree commutes with linear changes of the coordinates x, y. Hence $W_0 = (xy + zf'(z,t) = 0)$ with the polynomial $f' := (\hat{f}_1)_{<L-1}$, which satisfies $f'(0,0) = 0$ and $f'(0,t) \neq 0$. The coordinates of W_0 have the characters of x, y, z, t, so the subgroups $R = G \cap \mathbf{T}$, the numbers r, and the indices $[G : R]$ of (W_0, G, D_0) and (W, G, D) coincide. Thus (W_0, G, D_0) satisfies the hypotheses of the first part, with the labels $c_k \in \Gamma$, the bound e_0, and the values of r and α of (W, G, D). Therefore a sequence of data as in the second part with $r \to +\infty$ and positive strictly increasing values α would imply such a sequence for the first part, which is impossible by **Step 4**. This proves the second part of Theorem 8.7. □

B.6. **Circles** $(A, B, A + B, 0)$**.** In this subsection, we prove Theorem 8.8.

Proof of Theorem 8.8. The proof is long, so we divide it into several steps.

Suppose that the theorem fails. Then there exist data $W_i = (xy + zf_i(z,t) = 0)$, G_i, $R_i = G_i \cap \mathbf{T}$, $r_i = |R_i|$, and $D_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$ as in Set-up 8.2, $i \geq 1$, with G_i diagonal, $\mathbf{T} = \mathbf{T}_{(A,B,C,0)}$, W_i an isolated Gorenstein terminal germ of quadratic rank two, $f_i(0,0) = 0$, and $f_i(0,t) \neq 0$, such that $r_i \to +\infty$ and $0 < \alpha_1 < \alpha_2 < \cdots$, where $\alpha_i := \operatorname{mld}((W_i \ni 0, D_i)/G_i)$. Let γ be the least positive element of Γ, and $\gamma := 1$ if $\Gamma \subset \{0\}$. In **Steps 1** and **2** we fix i and drop the index i.

Step 1. In this step we normalize the marks, define their orders k_j along $\mathbf{T}$, and prove that $\tau := \sum_j c_j k_j < C$.

Removing the marks with $c_j = 0$ and the units does not change D near 0. Thus we may assume that $c_j \geq \gamma$ and $h_j(0) = 0$ for any j. Let Q be a polynomial with $Q(0) \neq 0$ such that Qh_j is the restriction of a polynomial. Multiplying h_j by the G-invariant unit $\prod_{g \in G} g^*Q$ and reducing modulo $xy = -zf$, we may assume that h_j is a polynomial without monomials divisible by xy. Distinct polynomials without monomials divisible by xy define distinct functions on W, and g^*h_j is again such a polynomial for any $g \in G$ as G is diagonal. Hence h_j is a G-semi-invariant polynomial, and any monomial of h_j is G-semi-invariant with the character of h_j.

For $g \in G$ and $v \in \{x, y, z, t\}$ write $g^*v = \chi_v(g)v$. Since $\gcd(A, B, C) = 1$, the elements of R are the elements $\lambda_{\mathbf{T}}$ with $\lambda^r = 1$, where

$$\lambda_{\mathbf{T}}^*(x, y, z, t) = (\lambda^A x, \lambda^B y, \lambda^C z, t).$$

Thus the $\mathbf{T}$-weights $Aa + Bb + Cc$ of the monomials $x^a y^b z^c t^{c'}$ of h_j are congruent modulo r. Let k_j be the least $\mathbf{T}$-weight of a monomial of h_j. Write $h_j = \sum_{(a,b,c)} m_{abc}(t) x^a y^b z^c$ with polynomials m_{abc}, where $ab = 0$ whenever $m_{abc} \neq 0$. We use the identity

$$Aa + Bb + Cc = A(a + c) + B(b + c), \tag{B.59}$$

and the fact that $(a + c, b + c)$ determines (a, b, c) when $ab = 0$.

First let $A, B > 0$. At the generic point η of the t-axis, f is a unit since $f(0,t) \neq 0$. Thus $z = -xy/f$, W is smooth at η, and x, y are regular parameters of $\mathcal{O}_{W,\eta}$. Let E be the prime

divisor over W whose valuation is the monomial valuation in x, y at η with weights (A, B). Since $\gcd(A, B) = 1$, E is the exceptional divisor of the weighted blow-up of (x, y) with weights (A, B) near η, and $a(E, W, 0) = A + B = C$. Modulo elements of larger E-multiplicity, $m_{abc}(t)x^a y^b z^c$ equals $m_{abc}(t)(-f(0,t))^{-c}x^{a+c}y^{b+c}$, whose E-multiplicity is (B.59). Hence the terms of h_j of $\mathbf{T}$-weight k_j correspond to distinct monomials in x, y with coefficients non-zero at η, and $\operatorname{mult}_E h_j = k_j$.

Next let $A = 0$, so $B = C = 1$ and the $\mathbf{T}$-weight of $x^a y^b z^c$ is $b + c$. Then $E := (y = z = 0)$ is a prime divisor on W with $a(E, W, 0) = 1 = C$. At the generic point of E, both x and f are units, $y = -zf/x$, and z is a uniformizer. Thus $m_{abc}(t)x^a y^b z^c = m_{abc}(t)(-f)^b x^{a-b} z^{b+c}$, and modulo z^{k_j+1},

$$h_j \equiv z^{k_j} \sum_{b+c=k_j} m_{abc}(t)(-f(0,t))^b x^{a-b}.$$

Since $ab = 0$, distinct pairs (a, b) have distinct exponents $a - b$, so $\operatorname{mult}_E h_j = k_j$. The case $B = 0$ follows by exchanging x and y.

In both cases the center of E contains 0, and (W, D) is klt near 0. Thus $a(E, W, D) = C - \tau > 0$, that is,

$$\tau = \sum_j c_j k_j < C. \tag{B.60}$$

Since the quadratic rank of $xy + zf$ is two, zf has no term of degree two. Hence $n_W := \operatorname{mult}_0(zf) \geq 3$. Let $(zf)_{n_W} \neq 0$ be the homogeneous part of degree n_W of zf. The part of $xy + zf$ of least weight for the weights $(n_W - 1, 1, 1, 1)$ is $xy + (zf)_{n_W}$, which is irreducible and not divisible by a coordinate. By Lemma 4.12, the weighted blow-up with weights $(n_W - 1, 1, 1, 1)$ defines a prime divisor E' over W with center 0, $a(E', W, 0) = (n_W - 1) + 1 + 1 + 1 - n_W = 2$, and $\operatorname{mult}_{E'} h_j \geq \operatorname{mult}_0 h_j$, since all weights are at least one. As (W, D) is klt near 0,

$$\sum_j c_j \operatorname{mult}_0 h_j \leq \sum_j c_j \operatorname{mult}_{E'} h_j < a(E', W, 0) = 2.$$

Thus $\operatorname{mult}_0 h_j < 2/\gamma \leq M := \lceil 2/\gamma \rceil$ for any j. We use Lemma 5.3 with $n = 4$, the defining polynomial $xy + zf$ of W, which has the monomial xy of degree $2 \leq M$, and each h_j, which has a monomial of degree $\operatorname{mult}_0 h_j < M$. Since the weights $(A, B, C, 0)$ are non-negative, Lemma 5.3 shows that $0 \leq \kappa(h_j) \leq MC$ whenever $r > 2MC$. Let $r_0 := 2MC + 1$, which depends only on M, that is, on Γ, and on $(A, B, C, 0)$. Let $r \geq r_0$. As all $\mathbf{T}$-weights $Aa + Bb + Cc$ of the monomials of h_j are non-negative and congruent to $\kappa(h_j)$ modulo r, and $\kappa(h_j) < r$, the least $\mathbf{T}$-weight k_j of a monomial of h_j equals $\kappa(h_j)$. Possibly discarding finitely many i, we may assume that $r \geq r_0$, so in particular $r > C$. Then τ is the mass of Set-up 5.1, $\epsilon_{\mathbf{T}} = C$ is the $\mathbf{T}$-weight of xy, and $\delta = \tau - C < 0$.

Finally, $xy + zf$ is G-semi-invariant, and xy and z both have $\mathbf{T}$-weight C. Thus f is R-invariant, that is, any monomial $z^a t^{c'}$ of f satisfies $Ca \equiv 0 \pmod{r}$.

Step 2. In this step we construct a proper birational morphism $\varphi : W' \to W$, describe the root charts of φ and their groups, and compute the crepant boundaries on the quotients of the root charts by a subgroup R'.

Let $(q_x, q_y, q_z) := (A, B, C)$, and let $\varphi : W' \to W$ be the normalized strict transform of W in the weighted blow-up of $\mathbb{A}^4$ with weights $(A, B, C, 0)$. If $A, B > 0$, the weighted blow-up is centered at the t-axis. If $A = 0$ (resp. $B = 0$), φ is the blow-up of the ideal (y, z) (resp. (x, z)) of W. For $u \in \{x, y, z\}$ with $q_u > 0$, the root chart U_u with anchor u is given by $u = s^{q_u}$ and $v = s^{q_v} v_1$ for $v \in \{x, y, z\} \backslash \{u\}$:

$$U_z = (x_1 y_1 + f(s^C, t) = 0) \subset \mathbb{A}^4_{x_1, y_1, s, t}, \tag{B.61}$$

$$U_x = (y_1 + z_1 f(s^C z_1, t) = 0) \cong \mathbb{A}^3_{s, z_1, t}, \tag{B.62}$$

$$U_y = (x_1 + z_1 f(s^C z_1, t) = 0) \cong \mathbb{A}^3_{s, z_1, t}. \tag{B.63}$$

The variety W' is covered by the quotients U_u/μ_{q_u}, where $\rho \in \mu_{q_u}$ acts by $\rho^* s = \rho s$ and $\rho^* v_1 = \rho^{-q_v} v_1$. If $A = 0$, then $B = C = 1$, no chart has anchor x, $x_1 = x$, and (B.63) and (B.61) are the charts $(x_1 + z_1 f(sz_1, t) = 0)$ and $(x_1 y_1 + f(s,t) = 0)$ of the blow-up of (y, z). The charts U_x and U_y are smooth. The chart U_z is normal near $s = t = 0$: since W has an isolated singularity at 0, zf is reduced, so $f(s^C, t)$ is reduced as $f(0,t) \neq 0$, and the singular locus of U_z is finite near $s = t = 0$.

The fiber $\varphi^{-1}(0)$ is a curve. On U_x and U_y it is $(s = t = 0)$. On U_z it is $(s = t = 0,\ x_1 y_1 = 0)$ if $A, B > 0$, and $(x_1 = s = t = 0)$ if $A = 0$ (symmetrically $(y_1 = s = t = 0)$ if $B = 0$). If $A, B > 0$, then $\varphi^{-1}(0) = (xy = 0) \subset \mathbb{P}(A, B, C)$ consists of two rational curves meeting at the image of the origin of U_z. If $A = 0$, the ideal (y, z) is generated by z near the points of E where $x \neq 0$, and by y near the points of E where $f(0,t) \neq 0$. Thus, if $A = 0$ (resp. $B = 0$), then, near 0, φ is an isomorphism over $W \backslash \{0\}$, φ is small, and $\varphi^{-1}(0) \cong \mathbb{P}^1$.

For each chart let

$$\widehat{G}_u := \{(g, \rho) \in G \times \mathbb{C}^* \mid \rho^{q_u} = \chi_u(g)\},$$

acting on U_u by $(g,\rho)^* s = \rho s$, $(g,\rho)^* v_1 = \chi_v(g)\rho^{-q_v} v_1$, and $(g,\rho)^* t = \chi_t(g) t$. Then $|\widehat{G}_u| = q_u |G|$, the morphism $U_u \to W$ is equivariant with respect to the projection $\widehat{G}_u \to G$, whose kernel is μ_{q_u}, and $U_u/\widehat{G}_u = (U_u/\mu_{q_u})/G$ is an open subset of W'/G. The subgroup

$$R' := \{(\lambda_{\mathbf{T}}, \lambda) \mid \lambda^r = 1\} \subset \widehat{G}_u$$

has order r, and $(\lambda_{\mathbf{T}}, \lambda)$ acts by $s \to \lambda s$ and trivially on the other coordinates. Let $\widetilde{G}_u := \widehat{G}_u / R'$, a group of order $q_u [G : R]$.

We show that any $(g,\rho) \in \widehat{G}_u$ acting trivially on the divisor $(s = 0)$ of U_u belongs to R'. On U_z, $(s = 0) = (x_1 y_1 + f(0,t) = 0) \subset \mathbb{A}^3_{x_1, y_1, t}$, on which x_1, y_1, and t are non-zero. Hence $\chi_x(g) = \rho^A$, $\chi_y(g) = \rho^B$, and $\chi_t(g) = 1$. On U_x, $(s = 0) = (y_1 + z_1 f(0,t) = 0)$, on which z_1, t, and $y_1 = -z_1 f(0,t)$ are non-zero, and we again obtain $\chi_x(g) = \rho^A$, $\chi_y(g) = \rho^B$, and $\chi_t(g) = 1$. The chart U_y is symmetric. Together with $\chi_u(g) = \rho^{q_u}$, g acts on $\mathbb{A}^4$ as $\rho_{\mathbf{T}}$. Since G is diagonal, $g = \rho_{\mathbf{T}} \in G \cap \mathbf{T} = R$, so $\rho^r = 1$ and $(g,\rho) \in R'$.

Any monomial of $\mathbf{T}$-weight m pulls back to U_u as s^m times a monomial in t and the coordinates v_1. As the $\mathbf{T}$-weights of h_j are congruent to k_j modulo r, on each chart

$$h_j = s^{k_j} H_j$$

with H_j a polynomial in s^r, t, and the coordinates v_1. In particular, H_j is R'-invariant and $\widehat{G}_u$-semi-invariant. The function $H_j|_{(s=0)}$ is the restriction of the sum of the terms $m_{abc}(t) x^a y^b z^c$ of h_j of $\mathbf{T}$-weight k_j, with x, y, z replaced by $x_1, y_1, 1$ on U_z, by $1, y_1, z_1$ on U_x, and by $x_1, 1, z_1$ on U_y. On U_z, the monomials x_1^a $(a \geq 0)$ and y_1^b $(b \geq 1)$ form a basis of $\mathbb{C}[x_1, y_1, t]/(x_1 y_1 + f(0,t))$ over $\mathbb{C}[t]$, and (a, b) determines c for a fixed weight. On U_x (resp. U_y), the term restricts to a non-zero polynomial in t times z_1^{b+c} (resp. z_1^{a+c}), and by (B.59) the weight k_j together with $b + c$ (resp. $a + c$) determines $(a + c, b + c)$, hence (a, b, c). Thus distinct terms restrict to $\mathbb{C}[t]$-linearly independent functions, and hence $H_j|_{(s=0)} \neq 0$.

The form $\Omega := dx \wedge dz \wedge dt / x = -dy \wedge dz \wedge dt / y$ generates ω_W near 0 [KM98, Proposition 5.73]. On U_z, we have $x = s^A x_1$ and $z = s^C$. Thus $dx/x = A ds/s + dx_1/x_1$ and $dz = C s^{C-1} ds$, and the pullback of Ω to U_z is $C s^{C-1} (dx_1/x_1) \wedge ds \wedge dt$. Similarly, the pullback of Ω to U_x is $A s^{C-1} ds \wedge dz_1 \wedge dt$, and to U_y is $-B s^{C-1} ds \wedge dz_1 \wedge dt$. Each pullback is s^{C-1} times a generator of ω_{U_u} [KM98, Proposition 5.73]. For $A = 0$ the exponent is $C - 1 = 0$. Hence, for $\varphi_u : U_u \to W$,

$$\varphi_u^*(K_W + D) = K_{U_u} + \sum_j c_j \operatorname{div}(H_j) + (1 - C + \tau) \operatorname{div}(s). \tag{B.64}$$

Let $w := s^r$. Since r divides Ca for any monomial $z^a t^{c'}$ of f, $f(s^C, t) = \bar{f}(w, t)$ for a polynomial $\bar{f}$ with $\bar{f}(0,t) = f(0,t)$, and $f(s^C z_1, t)$ is a polynomial in w, z_1, and t. Thus the

quotients $W''_u := U_u/R'$ are

$$W''_z = (x_1y_1 + \bar{f}(w,t) = 0) \subset \mathbb{A}^4_{x_1,y_1,w,t}, \qquad W''_x \cong W''_y \cong \mathbb{A}^3_{w,z_1,t}. \tag{B.65}$$

Write $H_j = \bar{H}_j(w,\dots)$ with $\bar{H}_j$ a polynomial on W''_u, not divisible by w. The morphism $U_u \to W''_u$ is ramified only along $(s=0)$, with index r. By (B.64) and [Liu26, Lemma 2.6], the crepant boundary on W''_u is

$$\Delta_{W''_u} = \sum_j c_j \operatorname{div}(\bar{H}_j) + b_r \operatorname{div}(w), \qquad b_r := 1 - \frac{C-\tau}{r} = 1 + \frac{\delta}{r}. \tag{B.66}$$

Since $r > C$ and $0 \le \tau < C$ by (B.60), $0 < b_r < 1$, so $\Delta_{W''_u} \ge 0$. The pair $(W''_u, \Delta_{W''_u})$ is crepant to (W,D) through U_u, hence klt near the preimage of 0 by [KM98, Lemma 2.30] and [Liu26, Lemma 2.6]. In particular, W''_z is klt near 0. The group $\widetilde{G}_u$ preserves $\Delta_{W''_u}$, acts faithfully on $(w=0) \subset W''_u$ and on W''_u, and $\widetilde{G}_z$ acts diagonally on $\mathbb{A}^4_{x_1,y_1,w,t}$, fixes 0, and has order $C[G:R] \le Ce_0$.

Step 3. In this step we show that, after passing to a subsequence, α_i is the minimal log discrepancy of the quotient of $(W''_z, \Delta_{W''_z})$ at the origin. Until the three cases below are settled, we keep i fixed and drop the index i, except in the definition of Γ', which involves all the numbers b_{r_i}.

Let $\Delta_{W'}$ be the $\mathbb{R}$-divisor with $K_{W'} + \Delta_{W'} = \varphi^*(K_W + D)$. For any prime divisor P over W, $a(P,W,D) = a(P,W',\Delta_{W'})$ [KM98, Lemma 2.30], and P has center 0 on W if and only if the center of P on W' is contained in $\varphi^{-1}(0)$. By [Liu26, Lemma 2.6], both statements hold for the quotients $X' := W'/G \to W/G$ with the crepant boundaries $\Delta_{X'}$ and $D_{W/G}$, and on the open subset $W''_u/\widetilde{G}_u = U_u/\widehat{G}_u$ of X' the boundary $\Delta_{X'}$ is the crepant quotient boundary of $(W''_u, \Delta_{W''_u})$. Let $\bar{0}$ be the image of 0 in W/G. As $(W/G, D_{W/G})$ is klt near $\bar{0}$, there exists a prime divisor F over W/G with center $\bar{0}$ and $a(F, W/G, D_{W/G}) = \alpha$, by [Liu26, Lemma 3.5] applied to a log resolution of $(W/G, D_{W/G})$ on which the preimage of $\bar{0}$ is a divisor. Let Z be the center of F on X', and η_Z the generic point of Z. Since Z is contained in the fiber over $\bar{0}$, any prime divisor over X' with center Z has center $\bar{0}$ on W/G. Thus

$$\alpha = \operatorname{mld}(X' \ni \eta_Z, \Delta_{X'}).$$

By **Step 2**, the only point of the preimage of $\varphi^{-1}(0)$ in the charts W''_u which can be a singular point of W''_u is the origin o of W''_z: W''_x and W''_y are smooth, and W''_z is smooth at any point with $x_1 \ne 0$ or $y_1 \ne 0$. Let $\Gamma' := \Gamma \cup \{b_{r_i} \mid i \ge 1\}$. Since $b_{r_i} < 1$ and $b_{r_i} \ge 1 - C/r_i \to 1$, Γ' is a DCC set. Let $\mathcal{A}$ be the set of positive values of Theorem 2.8(1) for Γ', an ACC set. We consider three cases.

(1) Z is a closed point other than the image of o. Let p be a point of W''_u over Z and $\widetilde{G}_{u,p}$ its stabilizer in $\widetilde{G}_u$. Then W''_u is smooth at p, the morphism $W''_u/\widetilde{G}_{u,p} \to X'$ is étale at the image of p [Gro71, Exposé V, Proposition 2.2(ii)], and the crepant pullback of $K_{X'} + \Delta_{X'}$ to $W''_u/\widetilde{G}_{u,p}$ is the crepant quotient of $(W''_u, \Delta_{W''_u})$ by $\widetilde{G}_{u,p}$. Hence, after shrinking around the image of p, [Liu26, Lemma 9.3] and [Liu26, Lemma 2.6(5)] give $\alpha = \operatorname{mld}((W''_u \ni p, \Delta_{W''_u})/\widetilde{G}_{u,p})$, where $\widetilde{G}_{u,p}$ is a finite abelian group acting faithfully on the smooth germ and fixing p. By (B.66), $\Delta_{W''_u}$ is a sum of divisors of non-zero regular $\widetilde{G}_{u,p}$-semi-invariant functions with coefficients in Γ'. Thus $\alpha \in \mathcal{A}$.

(2) Z is a curve. Near η_Z, the coefficients of $\Delta_{X'}$ are $1 - \frac{1-c}{e}$ with e a positive integer and c a coefficient of some $\Delta_{W''_u}$ [Liu26, Lemma 2.6(4)], and $c \in \Gamma'_+ \cap [0,1]$ by (B.66), as $(W''_u, \Delta_{W''_u})$ is klt near the preimage of 0 (**Step 2**), so $\Delta_{X'} \in \Psi(\Gamma'_+ \cap [0,1]) \cap [0,1]$ near η_Z, a DCC set [Liu26, Lemma 2.10]. By [Liu26, Lemma 3.7], the positive values $\operatorname{mld}(X' \ni \eta_Z, \Delta_{X'})$ belong to an ACC set $\mathcal{A}_2$.

(3) Z is the image of o. Since $\widetilde{G}_z$ fixes o, $\alpha = \operatorname{mld}((W''_z \ni 0, \Delta_{W''_z})/\widetilde{G}_z)$ by [Liu26, Lemma 2.6].

The set $\mathcal{A}\cup\mathcal{A}_2$ satisfies the ACC and α_i is strictly increasing, so (1) and (2) hold only for finitely many i. Restoring the index i and passing to a subsequence, we obtain, with $b_i := b_{r_i}$,

$$\alpha_i = \mathrm{mld}\Big(\Big(W''_{z,i} \ni 0, \sum_j c_{ij}\,\mathrm{div}(\bar{H}_{ij}) + b_i\,\mathrm{div}(w)\Big)/\widetilde{G}_{z,i}\Big) \quad \text{for any } i. \tag{B.67}$$

Step 4. In this step we pass to $\mathbb{A}^4_{x_1,y_1,w,t}$, compare α_i with the quotient values β_i of pairs on the hyperplane $(w=0)$, and derive a contradiction.

Recall $W''_{z,i} = (x_1y_1 + \bar{f}_i(w,t) = 0)$ from (B.65), and let

$$\Delta_i := \mathrm{div}(x_1y_1 + \bar{f}_i(w,t)) + \sum_j c_{ij}\,\mathrm{div}(\bar{H}_{ij}) \quad \text{on } \mathbb{A}^4_{x_1,y_1,w,t}.$$

By **Step 2**, $\widetilde{G}_{z,i}$ acts diagonally on $\mathbb{A}^4$ and faithfully on $W''_{z,i}$, and $x_1y_1+\bar{f}_i$, $\bar{H}_{ij}$, w are semi-invariant polynomials. Theorem 3.3(1), applied with $n=4$, $c=1$, $f_1 = x_1y_1+\bar{f}_i$, $H=G=\widetilde{G}_{z,i}$, and $\mathfrak{a} = \prod_j(\bar{H}_{ij})^{c_{ij}}\cdot(w)^{b_i}$, and (B.67) imply

$$\alpha_i = \mathrm{mld}((\mathbb{A}^4 \ni 0, \Delta_i + b_i\,\mathrm{div}(w))/\widetilde{G}_{z,i}). \tag{B.68}$$

Since $(W''_{z,i}, \Delta_{W''_{z,i}})$ is klt, $(\mathbb{A}^4, \Delta_i + b_i\,\mathrm{div}(w))$ is lc near 0 by [Kaw07, Theorem] applied to $\mathbb{A}^4$, the normal Cartier divisor $W''_{z,i}$, and

$$\sum_j c_{ij}\,\mathrm{div}(\bar{H}_{ij}) + b_i\,\mathrm{div}(w).$$

Lemma B.10, applied to Γ and $N = Ce_0$, gives an ACC set $\mathcal{B}$ and real numbers $\epsilon\in(0,1)$ and $M_1>0$. Since $1-b_i \le \frac{C}{r_i}$, possibly discarding finitely many i, we may assume that $b_i > 1-\epsilon$ for any i. The group $\widetilde{G}_{z,i}$ acts diagonally on $\mathbb{A}^4_{x_1,y_1,w,t}$ and faithfully on $V := (w=0)$, since it acts faithfully on $V\cap W''_{z,i}$, and $x_1y_1+\bar{f}_i$ and the $\bar{H}_{ij}$ are semi-invariant polynomials not divisible by w (**Step 2**). By the lemma with $H = \widetilde{G}_{z,i}$, $v=w$, and $b=b_i$, the number $\beta_i := \mathrm{mld}((V\ni 0, \Delta_{V,i})/\widetilde{G}_{z,i})$, where $\Delta_{V,i} = \Delta_i|_V$, belongs to $\mathcal{B}$, and, by (B.68) and $1-b_i \le \frac{C}{r_i}$,

$$\beta_i \le \alpha_i \le \beta_i + \frac{CM_1}{r_i}. \tag{B.69}$$

Since $\{\beta_i\}\subset\mathcal{B}$ satisfies the ACC, possibly passing to a subsequence, β_i is non-increasing, and $\beta_\infty := \lim_{i\to+\infty}\beta_i \le \beta_i$ for any i. Since $r_i\to+\infty$, (B.69) implies $\lim_{i\to+\infty}\alpha_i = \beta_\infty$. As α_i is strictly increasing, $\alpha_i < \beta_\infty \le \beta_i \le \alpha_i$ for $i \gg 0$, a contradiction. □

B.7. **Circles with a zero hyperbolic weight.** In this subsection, we prove Theorem 8.9. The three positive weights force any monomial of a mark outside the principal block to be divisible by a high power of the radial parameter $u=\lambda^r$ of Construction-Proposition B.13, where λ is the root parameter of the weighted blow-up of the x-axis, and the exceptional surface of the blow-up, with the boundary cut out by the principal blocks, controls α up to an error of order $1/r$.

Set-up B.11. Notation as in Set-up 8.2, with G diagonal and $\mathbf{T} = \mathbf{T}_{(0,N,v,s)}$ for positive integers N,v,s with $\gcd(N,v,s)=1$, where $0\neq g\in(z,t)^3$ and the sum g^0 of the terms of g of $\mathbf{T}$-weight N is non-zero. As in Set-up 5.1, the units and the marks with label 0 are removed from $D=\sum_j c_j\,\mathrm{div}(h_j)$. If $\Gamma\not\subset\{0\}$, let γ be the least positive element of Γ, which exists since Γ is a DCC set, and let $\gamma := 1$ if $\Gamma\subset\{0\}$, in which case there are no marks. Writing $h_j = p/q$ near 0 with polynomials p,q and $q(0)\neq 0$, multiplying h_j by the G-invariant polynomial $\prod_{\sigma\in G}\sigma^*q$, which does not vanish at 0 and so does not change D near 0, and replacing the polynomial $p\prod_{\sigma\in G\setminus\{1\}}\sigma^*q$ by its isotypic component for the character of h_j, we may assume that each h_j is the restriction to W of a G-semi-invariant polynomial in x,y,z,t. Replacing repeatedly a monomial $x^ay^bz^ct^d$ with $a,b\ge 1$ by $-gx^{a-1}y^{b-1}z^ct^d$, which agrees with $x^ay^bz^ct^d$ on W and has the character of $x^ay^bz^ct^d$ since $xy+g$ is G-semi-invariant, we may moreover assume that no

monomial of h_j is divisible by xy. Let κ_j be the least $\mathbf{T}$-weight of a monomial of h_j, let h_j^0 be the sum of the terms of h_j of weight κ_j, and let

$$\tau = \sum_j c_j\kappa_j, \qquad \epsilon_{\mathbf{T}} = N + v + s - \kappa(xy + g) = v + s, \qquad \delta = \tau - \epsilon_{\mathbf{T}} = \sum_j c_j\kappa_j - (v + s).$$

Here $\kappa(xy + g) = N$, since xy is the monomial of least degree of $xy + g$. By Lemma B.12(2), if $r > 2\max(N, v, s)/\gamma$, then $\kappa_j = \kappa(h_j)$ and h_j^0 is the principal block of h_j, so τ, $\epsilon_{\mathbf{T}}$, and δ are the mass, $\epsilon_{\mathbf{T}}$, and the adjusted mass of Set-up 5.1. Let

$$W^0 := (xy + g^0(z, t) = 0) \subset \mathbb{A}^4, \qquad D^0 := \sum_j c_j \operatorname{div}(h_j^0|_{W^0}).$$

Lemma B.12. *Notation as in Set-up B.11. Then the following hold.*

(1) $\sum_j c_j \operatorname{mult}_0 h_j < 2$, *where* mult_0 *is the order at* 0 *of a polynomial in* x, y, z, t. *In particular, the number of marks is at most* $2/\gamma$ *and* $\operatorname{mult}_0 h_j < 2/\gamma$ *for any* j.

(2) If $r > 2\max(N, v, s)/\gamma$, *then* $\kappa_j < 2\max(N, v, s)/\gamma$, *any monomial of* h_j *of least degree has weight* κ_j, *and any monomial of* $h_j - h_j^0$ *has weight* $\kappa_j + \ell r$ *for some integer* $\ell \geq 1$, *for any* j.

(3) If $r > N$, *then* $g = g^0 + \sum_{\ell \geq 1} g_\ell$, *where* g_ℓ *is the sum of the terms of* g *of weight* $N + \ell r$, *and* $\deg g^0 \leq \lfloor N/\min(v, s) \rfloor$.

(4) $\gcd(v, s) = 1$ *and* $\delta \geq -(v + s)$. *If* $r > N$, *then moreover* $\delta < 0$.

(5) The polynomial h_j^0 *does not vanish identically on* W^0, *and* W^0 *is normal.*

Proof. Let $n = \operatorname{mult}_0 g \geq 3$ and let P be the exceptional divisor over W of the weighted blow-up of $\mathbb{A}^4$ with weights $(n - 1, 1, 1, 1)$ on (x, y, z, t). The part of $xy + g$ of least weight is $xy + \operatorname{in}(g)$ with $\operatorname{in}(g) \neq 0$, which is irreducible (it has degree one in x with coprime coefficients y and $\operatorname{in}(g)$) and not divisible by a coordinate. By Lemma 4.12, P is a prime divisor over W with center 0, $a(P, W, 0) = (n - 1) + 1 + 1 + 1 - n = 2$, and $\operatorname{mult}_P h_j \geq \operatorname{mult}_0 h_j$, since each coordinate has weight at least one. Since (W, D) is klt near 0,

$$0 < a(P, W, D) \leq 2 - \sum_j c_j \operatorname{mult}_0 h_j,$$

which implies (1).

For (2), since R is the cyclic group generated by $\mathbf{T}(\zeta)$ for a primitive r-th root of unity ζ, and h_j is G-semi-invariant, the weights of all monomials of h_j are congruent modulo r. A monomial of h_j of least degree has weight at most $\max(N, v, s) \operatorname{mult}_0 h_j < 2\max(N, v, s)/\gamma < r$, so $\kappa_j < 2\max(N, v, s)/\gamma$. Since all weights are non-negative and congruent to κ_j modulo r, a monomial of h_j of least degree, which has weight $< r$, has weight κ_j, and any monomial of $h_j - h_j^0$ has weight $\kappa_j + \ell r$ with $\ell \geq 1$. Since $xy + g$ is G-semi-invariant and the monomial xy of $xy + g$ has weight N, the weights of all monomials of $xy + g$ are congruent to N modulo r, so they belong to $N + r\mathbb{Z}_{\geq 0}$ if $r > N$. This implies (3), since a monomial $z^a t^b$ of weight $av + bs = N$ satisfies $a + b \leq N/\min(v, s)$.

For (4), the monomials of g^0 have weights $N = av + bs$, so $\gcd(v, s)$ divides $\gcd(N, v, s) = 1$. Since $\tau \geq 0$, $\delta \geq -(v + s)$. Assume that $r > N$. Via $y = -g/x$, $W \cap (x \neq 0) \cong (\mathbb{A}^1 \backslash \{0\}) \times \mathbb{A}^2_{z,t}$. Let E be the prime divisor over W given by the exceptional divisor of the weighted blow-up of $(\mathbb{A}^1 \backslash \{0\}) \times \mathbb{A}^2_{z,t}$ along $(z = t = 0)$ with weights (v, s) on (z, t). By (3), $g = g^0 + \sum_{\ell \geq 1} g_\ell$, so $\operatorname{mult}_E y = \operatorname{mult}_E(-g/x) = N$, the center of E on W is the x-axis $(y = z = t = 0)$, which passes through 0, and $a(E, W, 0) = v + s$, since $\frac{dx}{x} \wedge dz \wedge dt$ generates ω_W. Since mult_E is the monomial valuation with weights (v, s) in z, t and the terms of $g - g^0$ have weights larger than N, the mult_E-initial form of $h_j|_W$, written with $y = -g/x$, is $h_j^0(x, -g^0/x, z, t)$, which is non-zero by the first part of (5), since W^0 is integral (as $xy + g^0$ has degree one in x with coprime coefficients y and $g^0 \in \mathbb{C}[z, t] \backslash \{0\}$) and $W^0 \cap (x \neq 0)$ is dense in W^0. Hence $\operatorname{mult}_E h_j = \kappa_j$. Since (W, D) is

klt near 0,

$$0 < a(E, W, D) = v + s - \sum_j c_j \kappa_j = -\delta.$$

For (5), the restriction of h_j^0 to W^0, written as a rational function of x, z, t on $W^0 \cap (x \neq 0)$ with $y = -g^0/x$, is a linear combination with non-zero coefficients of the functions $x^a z^c t^d$ and $(-g^0/x)^b z^c t^d$, where $x^a z^c t^d$ runs over the monomials of h_j^0 with $b = 0$, and $y^b z^c t^d$ over those with $b \geq 1$ and $a = 0$. Terms with distinct exponents of x cannot cancel, and for a fixed exponent the polynomials $z^c t^d (g^0)^b$ are linearly independent, so $h_j^0|_{W^0} \neq 0$. Finally, the singular locus of W^0 is contained in

$$\Sigma := (x = y = 0) \cap (g^0 = \partial_z g^0 = \partial_t g^0 = 0)$$

and $\dim \Sigma \leq 1$, so the hypersurface W^0 is normal. □

From now on we assume that $r > \max(N, 2\max(N, v, s)/\gamma)$. Then D^0 is defined by the principal blocks, and the equations F_a and H_j of Construction-Proposition B.13 are polynomials in u.

Construction-Proposition B.13. Notation as in Set-up B.11, and let $w_y := N$, $w_z := v$, and $w_t := s$. Fix an anchor coordinate $a \in \{y, z, t\}$. Let λ be a new coordinate, and substitute

$$x = x', \qquad y = \lambda^N y', \qquad z = \lambda^v z', \qquad t = \lambda^s t',$$

where the primed coordinate of the anchor is set to 1. For any polynomial h all of whose monomials have weights in $\kappa + r\mathbb{Z}_{\geq 0}$, let

$$h^\sharp := \lambda^{-\kappa} h(x', \lambda^N y', \lambda^v z', \lambda^s t'),$$

which is a polynomial in the three remaining primed coordinates and the radial parameter $u := \lambda^r$. Write $\mathbb{A}^4_{\cdot,\lambda}$, resp. $\mathbb{A}^4_{\cdot,u}$, for the affine space with coordinates the three remaining primed coordinates and λ, resp. u. Let

$$F_a := (xy + g)^\sharp, \qquad H_j := h_j^\sharp, \qquad F_a^0 := F_a|_{u=0}, \qquad H_j^0 := H_j|_{u=0}.$$

Let $V_a := (F_a(\cdot, \lambda^r) = 0) \subset \mathbb{A}^4_{\cdot,\lambda}$ and $U_a := (F_a = 0) \subset \mathbb{A}^4_{\cdot,u}$, let $S_a := U_a \cap (u = 0) = (F_a^0 = 0)$, $C_a := S_a \cap (x' = 0)$, and

$$\Delta_a := \sum_j c_j \operatorname{div}(H_j^0|_{S_a}), \qquad B_a := \sum_j c_j \operatorname{div}(H_j|_{U_a}) + \left(1 + \frac{\delta}{r}\right) \operatorname{div}(u|_{U_a}).$$

Let $\widehat{G}_a$ be the group of pairs (σ, ζ) with $\sigma \in G$ and ζ^{w_a} equal to the eigenvalue of σ on the anchor, acting diagonally on $\mathbb{A}^4_{\cdot,\lambda}$ by the unique action for which the substitution is equivariant and $\lambda \to \zeta\lambda$. Let

$$R' = \{(\mathbf{T}(\zeta), \zeta) : \zeta^r = 1\} \subset \widehat{G}_a, \qquad \widetilde{G}_a := \widehat{G}_a / R'.$$

Then the following hold.

(1) (a) R' acts on V_a only through $\lambda \to \zeta\lambda$, and faithfully.
(b) $U_a = V_a/R'$.
(c) The group $\widetilde{G}_a$ has order $w_a[G : R] \leq \max(N, v, s)e_0$.
(d) $\widetilde{G}_a$ acts diagonally on $\mathbb{A}^4_{\cdot,u}$.
(e) F_a, u and the H_j are $\widetilde{G}_a$-semi-invariant.

(2) (a) The section S_a and the curve C_a are

$$S_y = (x' + g^0(z', t') = 0), \quad S_z = (x'y' + g^0(1, t') = 0), \quad S_t = (x'y' + g^0(z', 1) = 0),$$
$$C_a = S_a \cap (x' = 0).$$

(b) The surface S_a is integral and normal.
(c) The surface S_a is smooth or has A_{d-1} points with $d \leq \lfloor N/\min(v, s) \rfloor$.
(d) The group $\widetilde{G}_a$ acts faithfully on S_a.

(e) U_a is normal near S_a.
(f) U_a is klt near S_a.

(3) $H_j^0|_{S_a} \neq 0$ for any j, and $F_a - F_a^0$ and $H_j - H_j^0$ are divisible by u.

(4) (a) The composite

$$V_a \to W, \qquad (x', \cdot, \lambda) \to (x', \lambda^N y', \lambda^v z', \lambda^s t'),$$

is $\widehat{G}_a$-equivariant.
(b) $V_a/\widehat{G}_a = U_a/\widetilde{G}_a$ is an open subset of the quotient by G of the normalized weighted blow-up $W' \to W$ of the x-axis with weights (N, v, s) on (y, z, t).
(c) The pair $(U_a, B_a)/\widetilde{G}_a$ is the crepant pullback of $(W, D)/G$ on $V_a/\widehat{G}_a$.
(d) The preimage of 0 in U_a is C_a.

(5) (a) For any prime divisor P over U_a there exists a prime divisor P_W over W with

$$\frac{a(P, U_a, B_a)}{|I_{\widetilde{G}_a}(P)|} = \frac{a(P_W, W, D)}{|I_G(P_W)|},$$

such that the closure of the center of P_W contains 0 if the closure of the center of P intersects C_a, and the center of P_W is 0 if the center of P is contained in C_a.
(b) We have

$$\alpha = \min_{a \in \{y,z,t\}} \inf_P \frac{a(P, U_a, B_a)}{|I_{\widetilde{G}_a}(P)|},$$

where P runs over the prime divisors over U_a with center contained in C_a.

(6) Replacing (W, D) with (W^0, D^0) in the definitions of F_a, U_a, and B_a yields the chart $U_a^0 := (F_a^0 = 0) = S_a \times \mathbb{A}^1_u$ with the boundary

$$B_a^0 := \mathrm{pr}_1^* \Delta_a + (1 + \delta/r)\,\mathrm{div}(u),$$

and (1), (2), (4), and (5) hold for (W^0, D^0), U_a^0, B_a^0 in place of (W, D), U_a, B_a.

Proof. By Lemma B.12(2) and (3), the monomials of $xy + g$ have weights in $N + r\mathbb{Z}_{\geq 0}$ and the monomials of h_j have weights in $\kappa_j + r\mathbb{Z}_{\geq 0}$, so F_a and H_j are polynomials, and the terms of weight N, resp. κ_j, are exactly the terms not divisible by u. This implies (3) except for the first part, and the formulas for S_a in (2). For $\sigma \in G$ with eigenvalues $(\sigma_x, \sigma_y, \sigma_z, \sigma_t)$ and $(\sigma, \zeta) \in \widehat{G}_a$, the action on $(x', y', z', t', \lambda)$ is

$$(\sigma_x x', \zeta^{-N}\sigma_y y', \zeta^{-v}\sigma_z z', \zeta^{-s}\sigma_t t', \zeta\lambda),$$

which is diagonal. For $(\mathbf{T}(\zeta), \zeta) \in R'$ the action is $(x', y', z', t', \zeta\lambda)$. Since $\mathbf{T}(\zeta) \in G$ if and only if $\zeta^r = 1$ (recall $\gcd(N, v, s) = 1$), R' has order r, and $|\widetilde{G}_a| = w_a|G|/r = w_a[G : R]$. Since V_a is the preimage of U_a under $\lambda \to \lambda^r$, $U_a = V_a/R'$. The polynomials $xy + g$ and h_j are G-semi-invariant and λ is $\widehat{G}_a$-semi-invariant, so F_a, H_j, and u are semi-invariant. This proves (1).

For (2), $g^0(1, t')$ and $g^0(z', 1)$ are non-zero polynomials of degree at most $\lfloor N/\min(v, s) \rfloor$, since distinct monomials in z, t of weight N have distinct exponents of t, resp. of z. Hence $S_y \cong \mathbb{A}^2_{z',t'}$ is smooth, and S_z is integral and is smooth outside the points $(0, 0, \alpha_0)$ with α_0 a root of $g^0(1, t')$ of multiplicity $d \geq 2$, where it is an A_{d-1} point, and similarly for S_t. Let $(\sigma, \zeta) \in \widehat{G}_a$ act trivially on S_a. Since no coordinate function vanishes identically on S_a, the element (σ, ζ) fixes the three primed coordinates, so $\sigma = \mathbf{T}(\zeta)$ on W, hence $(\sigma, \zeta) \in R'$. Thus $\widetilde{G}_a$ acts faithfully on S_a. The section $S_a = \mathrm{div}(u|_{U_a})$ is a normal Cartier divisor with isolated singularities, and the singular locus of U_a intersects S_a in a finite set, so $\dim \mathrm{Sing}(U_a) \leq 1$ near S_a, and the hypersurface U_a is normal near S_a. Since S_a is smooth or has A_{d-1} points, S_a is klt, so (U_a, S_a) is plt near S_a [KM98, Theorem 5.50(1)], and U_a is klt near S_a. The first part of (3) is Lemma B.12(5) after the substitution, since the restriction of h_j^0 to W^0 and the restriction of H_j^0 to S_a differ by a power of λ under the substitution.

For (4), V_a is normal near $(\lambda = 0)$ by the argument of (2) applied to the Cartier divisor $(\lambda = 0) \cap V_a = S_a$, and the substitution is the coordinate chart of the anchor on the weighted

blow-up of the x-axis composed with the quotient by the subgroup $\{(1,\zeta):\zeta^{w_a}=1\}$ of $\widehat{G}_a$, which implies the first two parts of (4). If $a=z$, then ω_W is generated by $\frac{dx}{x}\wedge dz\wedge dt$, and ω_{V_z} by $\frac{dx'}{x'}\wedge dt'\wedge d\lambda$, and a direct calculation shows that $\frac{dx}{x}\wedge dz\wedge dt$ pulls back to $\pm v\lambda^{v+s-1}\frac{dx'}{x'}\wedge dt'\wedge d\lambda$. If $a=t$, then $\frac{dx}{x}\wedge dz\wedge dt$ pulls back to $\pm s\lambda^{v+s-1}\frac{dx'}{x'}\wedge dz'\wedge d\lambda$, and $\frac{dx'}{x'}\wedge dz'\wedge d\lambda$ generates ω_{V_t}. If $a=y$, then the generator $\frac{dy}{y}\wedge dz\wedge dt$ of ω_W pulls back to $\pm N\lambda^{v+s-1}dz'\wedge dt'\wedge d\lambda$, and $dz'\wedge dt'\wedge d\lambda$ generates ω_{V_y}. Since $h_j=\lambda^{\kappa_j}H_j$ on V_a, the crepant pullback of (W,D) to V_a is the $\mathbb{R}$-divisor $B_{V_a}:=$

$$\sum_j c_j\operatorname{div}(H_j)+\Big(\sum_j c_j\kappa_j-(v+s-1)\Big)\operatorname{div}(\lambda)=\sum_j c_j\operatorname{div}(H_j)+(1+\delta)\operatorname{div}(\lambda).$$

The quotient $V_a\to U_a$ is ramified only along $\lambda=0$, with index r, and H_j is R'-invariant, so the crepant quotient boundary on U_a is B_a, whose coefficient of $\operatorname{div}(u)$ is $1-(1-(1+\delta))/r=1+\delta/r$ [Liu26, Lemma 2.6(4)]. The pullback is crepant over W by construction, and by quotients in stages [Liu26, Lemma 2.6(5)], $(V_a,B_{V_a})/\widehat{G}_a=(U_a,B_a)/\widetilde{G}_a$. The fiber of $W'\to W$ over 0 is the subset of the exceptional divisor where $x=0$, whose preimage in U_a is $(u=x'=0)\cap U_a=C_a$.

For (5), let E be the image of P over W/G under the morphism $U_a/\widetilde{G}_a\to W/G$ of (4), and let P_W be a prime divisor over W whose image over W/G is E [Liu26, Lemma 2.6(3)]. By (4) and [KM98, Lemma 2.30], both sides of the equality are equal to $a(E,W/G,D_{W/G})$ [Liu26, Lemma 2.6(2)], where $(W/G,D_{W/G})$ is the crepant quotient of (W,D) [Liu26, Definition 2.5]. Since $W\to W/G$ is finite and 0 is the only point over the image of 0, the statements on centers follow from the last part of (4). Since the charts of the three anchors cover W' and $W'/G\to W/G$ is proper and birational, any prime divisor over W/G with center the image of 0 is the image of a prime divisor over some U_a with center contained in C_a [Liu26, Lemma 2.6(2) and (3)], so the formula at the end of [Liu26, Lemma 2.6] implies the formula for α.

For (6), g^0 and the h_j^0 are G-semi-invariant, G acts faithfully on W^0 since no coordinate vanishes on W^0, and W^0 is normal by Lemma B.12(5). Replacing (W,D) with (W^0,D^0) in the definitions of F_a and H_j yields F_a^0 and H_j^0, which do not involve u, and does not change $\delta=\sum_j c_j\kappa_j-(v+s)$, since $\kappa(h_j^0)=\kappa_j$ and $\kappa(xy+g^0)=N$. Possibly replacing (W,D) with (W^0,D^0), the argument for (1), (2), (4), and (5) above implies (6). □

Lemma B.14. *Notation as in Construction-Proposition B.13, and let $(Y_a,\Delta_{Y_a}):=(S_a,\Delta_a)/\widetilde{G}_a$. There exists a positive integer r_0 depending only on N,v,s and Γ such that, if $r\ge r_0$, then the following hold.*

(1) For any $a\in\{y,z,t\}$, the pairs (S_a,Δ_a) and (Y_a,Δ_{Y_a}) are lc near C_a and near the image of C_a respectively, and $\Delta_{Y_a}\in\Psi(\Gamma_+\cap[0,1])$ near the image of C_a.

(2) Let

$$\beta:=\min_{a\in\{y,z,t\}}\inf_P\frac{a(P,S_a,\Delta_a)}{|I_{\widetilde{G}_a}(P)|},$$

where P runs over the prime divisors over S_a with center contained in C_a. Then one of the following holds:

(a) there exist a and a closed point $p\in C_a$ with stabilizer $G_p\subset\widetilde{G}_a$ such that

$$\beta=\operatorname{mld}((S_a\ni p,\Delta_a)/G_p)=\operatorname{mld}(Y_a\ni\bar{p},\Delta_{Y_a}),$$

where $\bar{p}$ is the image of p, or

(b) there exist a and an integral curve $C'\subset C_a$ such that $\beta=(1-c')/e'$, where $c'=\operatorname{mult}_{C'}\Delta_a$ and $e'=|I_{\widetilde{G}_a}(C')|$.

(3) β belongs to an ACC set depending only on Γ.

Proof. For (1), by [Kaw07, Theorem] applied to $\mathbb{A}^4$ and the normal hypersurface W, the pair

$$\Big(\mathbb{A}^4,\operatorname{div}(xy+g)+\sum_j c_j\operatorname{div}(h_j)\Big)$$

is lc near 0, that is, the $\mathbb{R}$-ideal $\mathfrak{b} := (xy+g)\prod_j (h_j)^{c_j}$ on $\mathbb{A}^4$ satisfies $\mathrm{lct}_0(\mathfrak{b}) \geq 1$. Let $\mathfrak{b}^0 := (xy+g^0)\prod_j (h_j^0)^{c_j}$. By Lemma B.12(1), $\mathfrak{b}$ and $\mathfrak{b}^0$ have $1+m$ factors with $m \leq 2/\gamma$, and their exponents belong to $\{1\} \cup (\Gamma \backslash \{0\})$. By Lemma B.12(2) and (3), $(xy+g)-(xy+g^0)$ and $h_j - h_j^0$ belong to $\mathfrak{m}_0^{\lceil r/\max(N,v,s)\rceil}$. Let L be the maximum over $m \leq 2/\gamma$ of the integers given by Lemma 4.14 for power series in four variables, $1+m$ factors, and the DCC set $\Lambda = \{1\} \cup (\Gamma \backslash \{0\})$, and let r_0 be an integer with $r_0 > \max(N, 2\max(N,v,s)/\gamma)$ and $r_0 \geq \max(N,v,s)L$, which depends only on N, v, s, Γ. For $r \geq r_0$, Lemma 4.14 on $\mathbb{C}[[x,y,z,t]]$, which does not change log canonicity [dFEM11, Proposition 2.11], shows that the pair

$$\Big(\mathbb{A}^4, \mathrm{div}(xy+g^0) + \sum_j c_j \,\mathrm{div}(h_j^0)\Big)$$

is lc near 0, and [Kaw07, Theorem], applied to $\mathbb{A}^4$ and the normal hypersurface W^0, implies that (W^0, D^0) is lc near 0.

Assume that (S_a, Δ_a) is not lc near C_a. Since the non-lc locus of (S_a, Δ_a) is closed, it contains a closed point $p \in C_a$, so $\mathrm{mld}(S_a \ni p, \Delta_a) = -\infty$ [Amb99, Section 1, the second paragraph after Definition 1.1], and there exists a prime divisor P over S_a with center p and $a(P, S_a, \Delta_a) < 0$. Then $P \times \mathbb{A}^1_u$ is a prime divisor over $U_a^0 = S_a \times \mathbb{A}^1_u$ on which u does not vanish, so $a(P \times \mathbb{A}^1_u, U_a^0, B_a^0) = a(P, S_a, \Delta_a) < 0$, and the closure of the center $\{p\} \times \mathbb{A}^1_u$ intersects $C_a \times \{0\}$. By Construction-Proposition B.13(5) and (6), there exists a prime divisor over W^0 with negative log discrepancy with respect to (W^0, D^0) whose center has 0 in the closure, which contradicts the lc property of (W^0, D^0) near 0. Thus (S_a, Δ_a) is lc near C_a. Shrinking S_a to a $\widetilde{G}_a$-invariant neighborhood of C_a, the coefficient $\sum_j c_j \,\mathrm{mult}_{P'}(H_j^0|_{S_a})$ of Δ_a along any prime divisor P' on S_a belongs to Γ_+, and it is at most 1 since (S_a, Δ_a) is lc. Thus $\Delta_a \in \Gamma_+ \cap [0,1]$, the coefficients of Δ_{Y_a} are of the form $1-(1-c)/e_{P'}$ with $c \in \Gamma_+ \cap [0,1]$ [Liu26, Lemma 2.6(4)], so $\Delta_{Y_a} \in \Psi(\Gamma_+ \cap [0,1])$, a DCC set [Liu26, Lemma 2.10(1)], and (Y_a, Δ_{Y_a}) is lc [KM98, Proposition 5.20(4)]. This proves (1).

For (2) and (3), fix a and let $\bar{C}_a$ be the image of C_a. By [Liu26, Lemma 2.6(2) and (3)], the infimum over P in the definition of β is the infimum of $\mathrm{mld}(Y_a \ni \xi, \Delta_{Y_a})$ over the closed points $\xi \in \bar{C}_a$ and the generic points ξ of the components of $\bar{C}_a$. Outside a finite subset of $\bar{C}_a$, Y_a is smooth, $\bar{C}_a$ is smooth, and $\Delta_{Y_a} = c''\bar{C}''$ near ξ for the component $\bar{C}''$ through ξ and some $c'' \in [0,1]$. At such a closed point ξ, with local parameters w_1, w_2 and $\bar{C}'' = (w_1 = 0)$, any prime divisor Q with center ξ satisfies $a(Q, Y_a, c''\bar{C}'') \geq (1-c'')\,\mathrm{mult}_Q w_1 + \mathrm{mult}_Q w_2 > 1 - c''$ [Liu26, Lemma 2.13], which is the minimal log discrepancy at the generic point of $\bar{C}''$. Thus the infimum is a minimum over finitely many points, each of which is attained [Liu26, Lemma 3.5]. If it is attained at a closed point $\bar{p}$, let $p \in C_a$ be a point over $\bar{p}$. For any prime divisor P with center p, $I_{\widetilde{G}_a}(P) \subset G_p$, so $I_{G_p}(P) = I_{\widetilde{G}_a}(P)$, and by [Liu26, Lemma 2.6(2) and (3)] and the formula at the end of [Liu26, Lemma 2.6],

$$\mathrm{mld}(Y_a \ni \bar{p}, \Delta_{Y_a}) = \mathrm{mld}((S_a \ni p, \Delta_a)/G_p).$$

If it is attained at the generic point of $\bar{C}''$, the image of an integral curve $C' \subset C_a$, then the coefficient of $\bar{C}''$ in Δ_{Y_a} is $1-(1-c')/e'$ [Liu26, Lemma 2.6(4)], so the value is $(1-c')/e'$. This proves (2).

For (3), the closed-point values belong to the ACC set of Theorem 2.9(1) for the DCC set $\Psi(\Gamma_+ \cap [0,1])$, and the values at generic points of curves are of the form $1-c''$ with $c'' \in \Psi(\Gamma_+ \cap [0,1])$, which form an ACC set. Both sets depend only on Γ. □

The upper bound uses Gauss extensions on the chart $U_a^0 = S_a \times \mathbb{A}^1_u$ of W^0, and the following lemma moves them to the chart U_a of W.

Lemma B.15. *Notation as in Construction-Proposition B.13, and assume that $r \geq r_0$. Let $a \in \{y,z,t\}$, let $p \in C_a$ be a closed point with stabilizer $G_p \subset \widetilde{G}_a$, and let ν be a prime divisor over S_a with center p. Let $\vartheta := \mathrm{mult}_\nu x'$ if $a \in \{z,t\}$ and $\vartheta := 0$ if $a = y$. Let k be a positive*

integer such that $k > \vartheta$ *and* $k > \mathrm{mult}_\nu(H_j^0|_{S_a}) + \vartheta$ *for any* j*, and let* F^0 *be the Gauss extension of* mult_ν *to* $S_a \times \mathbb{A}^1_u$ *with* u*-value* k [Liu26, Definition-Proposition 2.11]*, and assume that* F^0 *is a prime divisor over* U_a^0 *with center* $(p,0)$*. Then there exists a prime divisor* F *over* U_a *with center* $(p,0)$ *such that*

$$I_{G_p}(F) = I_{G_p}(F^0) \quad \text{and} \quad a(F, U_a, B_a) = a(F^0, U_a^0, B_a^0).$$

Proof. Let $a = z$. The case $a = t$ follows from the argument below for $a = z$, with z' in place of t'. The function x' does not vanish identically on U_z or on U_z^0, and

$$y'|_{U_z} = -\frac{g^0(1,t') + \sum_{\ell\geq 1} u^\ell g_\ell(1,t')}{x'}, \qquad y'|_{U_z^0} = -\frac{g^0(1,t')}{x'}.$$

Thus the projections of U_z and of U_z^0 to the coordinates (x', t', u) are birational, and the induced birational map $\phi : U_z \dashrightarrow U_z^0$ is G_p-equivariant, since G_p acts on x', t', and u on both charts by the restrictions of the characters of G_p on $\mathbb{A}^4_{\cdot,u}$. Since $\partial F_z/\partial y' = \partial F_z^0/\partial y' = x'$, the Poincaré residues

$$\frac{dx' \wedge dt' \wedge du}{x'}$$

generate ω_{U_z} and $\omega_{U_z^0}$ near $(p,0)$, and they correspond under ϕ. Let F be the prime divisor over U_z corresponding to F^0 under ϕ. Then $a(F, U_z, 0) = a(F^0, U_z^0, 0)$, as both are one plus the mult_{F^0}-value of the rational 3-form $\frac{dx'\wedge dt'\wedge du}{x'}$, and $I_{G_p}(F) = I_{G_p}(F^0)$.

Let $y_1 := y'|_{U_z}$ and $y_0 := y'|_{U_z^0}$. Since $g_\ell(1,t')$ is regular and $\mathrm{mult}_{F^0} u = k$,

$$\mathrm{mult}_{F^0}(y_1 - y_0) \geq k - \mathrm{mult}_\nu x' = k - \vartheta > 0.$$

The functions x', $t' - t'(p)$, u, and $y_0 - y'(p)$ have positive values for mult_{F^0}, since F^0 has center $(p,0)$ on U_z^0, so $y_1 - y'(p) = (y_0 - y'(p)) + (y_1 - y_0)$ has positive value, and the center of F on U_z is $(p,0)$. By Construction-Proposition B.13(3), $H_j|_{U_z} = H_j^0(x', y_1, t') + uR_j$ for a polynomial R_j in x', y_1, t', u. The difference $H_j^0(x', y_1, t') - H_j^0(x', y_0, t')$ is $y_1 - y_0$ times a polynomial in x', y_0, y_1, t', all of which have non-negative values. Therefore

$$\mathrm{mult}_F\left(H_j|_{U_z} - H_j^0|_{U_z^0}\right) \geq k - \vartheta > \mathrm{mult}_\nu(H_j^0|_{S_z}) = \mathrm{mult}_{F^0}(H_j^0|_{U_z^0}),$$

so $\mathrm{mult}_F H_j = \mathrm{mult}_{F^0} H_j^0$. Since $\mathrm{mult}_F u = \mathrm{mult}_{F^0} u = k$, the equality of log discrepancies follows from the definitions of B_z and B_z^0.

Let $a = y$. Then the projections of U_y and of U_y^0 to the coordinates (z', t', u) are isomorphisms onto $\mathbb{A}^3_{z',t',u}$, with $x'|_{U_y} = -g^0(z',t') - \sum_{\ell\geq 1} u^\ell g_\ell(z',t')$ and $x'|_{U_y^0} = -g^0(z',t')$, and $dz' \wedge dt' \wedge du$ generates both canonical sheaves. The difference $x'|_{U_y} - x'|_{U_y^0}$ has value at least k, and $x'|_{U_y^0}$ has positive value since $p \in C_y \subset (x' = 0)$. Possibly replacing y_1, y_0 with $x'|_{U_y}, x'|_{U_y^0}$ and ϑ with 0, the argument of the case $a = z$ implies the lemma. □

Proof of Theorem 8.9. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the notation. Assume that there exists a sequence of data of Set-up 8.2 satisfying the hypotheses of Theorem 8.9, with Γ, e_0, N, v, s fixed and $r_i \to +\infty$, whose values α_i are positive and strictly increasing. Possibly passing to a subsequence and exchanging x and y, we may assume that $\mathbf{T} = \mathbf{T}_{(0,N,v,s)}$ for any i, and that $r_i \geq r_0$, where r_0 is the integer of Lemma B.14. We use the notation of Set-up B.11 and Construction-Proposition B.13 for each i, and we drop the index i while we work with a single i. Let β be the number of Lemma B.14(2). Recall that $\delta = \tau - \epsilon_{\mathbf{T}}$ with $\epsilon_{\mathbf{T}} = v + s$, and $-(v+s) \leq \delta < 0$ by Lemma B.12(4). If $\Gamma \subset \{0\}$, there are no marks.

Step 2. In this step we show that $\alpha \geq \beta$. By Construction-Proposition B.13(5), it suffices to show that

$$\frac{a(P, U_a, B_a)}{|I_{\widetilde{G}_a}(P)|} \geq \beta$$

for any a and any prime divisor P over U_a whose center Z is contained in C_a. Let $p \in Z$ be a closed point and $G_p \subset \widetilde{G}_a$ the stabilizer of p. We translate the coordinates of $\mathbb{A}^4_{\cdot,u}$ to p. A coordinate which does not vanish at p has a character which is trivial on G_p, so G_p acts diagonally on the translated coordinates, and F_a, u and the H_j are G_p-semi-invariant polynomials. The group $I_{\widetilde{G}_a}(P)$ acts trivially on the residue field of P, hence fixes Z pointwise. Thus $I_{\widetilde{G}_a}(P) \subset G_p$, and $I_{\widetilde{G}_a}(P) = I_{G_p}(P)$.

By Construction-Proposition B.13(1)–(3) and Lemma B.14(1), the hypotheses of Lemma 5.5 hold for the group G_p acting diagonally on $\mathbb{A}^4$, the hypersurface $X = U_a$, the function u with $S_0 = S_a$, the subgroup $H = G_p$ if $Z = \{p\}$, resp. the stabilizer $H \subset G_p$ of Z if Z is a curve, the $\mathbb{R}$-ideal $\mathfrak{a} = \prod_j (H_j)^{c_j}$, for which $(S_a, \mathfrak{a}|_{S_a}) = (S_a, \Delta_a)$, and $b = 1 + \delta/r \le 1$. Since $I_{\widetilde{G}_a}(P) \subset H$, Lemma 5.5 gives

$$\frac{a(P, U_a, B_a)}{|I_{\widetilde{G}_a}(P)|} \ge \beta_0 - \frac{\delta \operatorname{mult}_P u}{r|I_{\widetilde{G}_a}(P)|} \ge \beta_0,$$

where β_0 is the minimal log discrepancy of the quotient of (S_a, Δ_a) by H at the image of the generic point of Z. If $Z = \{p\}$, then $H = G_p$ fixes p, and by the formula at the end of [Liu26, Lemma 2.6], β_0 is the infimum of $a(P', S_a, \Delta_a)/|I_H(P')|$ over the prime divisors P' over S_a with center Z. If Z is a curve, then Z is the only prime divisor over S_a with center Z, and by [Liu26, Lemma 2.6(1) and (4)],

$$\beta_0 = \frac{1 - \operatorname{mult}_Z \Delta_a}{|I_H(Z)|} = \frac{a(Z, S_a, \Delta_a)}{|I_H(Z)|}.$$

Thus, in both cases, β_0 is the infimum of $a(P', S_a, \Delta_a)/|I_H(P')|$ over the prime divisors P' over S_a with center Z. For any such P', $I_{\widetilde{G}_a}(P')$ acts trivially on the residue field of P', hence fixes Z pointwise, so $I_{\widetilde{G}_a}(P') \subset H$, $I_H(P') = I_{\widetilde{G}_a}(P')$, and $\beta_0 \ge \beta$.

Step 3. In this step we show that

$$\alpha \le \beta + \frac{(v+s)k}{re}$$

for positive integers k and e with $e \le \max(N, v, s)e_0$ and

$$k \le \max(N, v, s)e_0\big(1 + (\lfloor N/\min(v, s)\rfloor + \gamma^{-1})l\big),$$

where l is the integer of Theorem 2.9(2) for the DCC set $\Psi(\Gamma_+ \cap [0, 1])$, which depends only on Γ. We consider the cases (a) and (b) of Lemma B.14(2).

Assume that $\beta = \operatorname{mld}((S_a \ni p, \Delta_a)/G_p)$ for a closed point $p \in C_a$. Let $\bar{p}$ be the image of p in Y_a. By Lemma B.14(1) and (2), (Y_a, Δ_{Y_a}) is lc near $\bar{p}$, $\Delta_{Y_a} \in \Psi(\Gamma_+ \cap [0, 1])$ near $\bar{p}$, and $\operatorname{mld}(Y_a \ni \bar{p}, \Delta_{Y_a}) = \beta$. By Theorem 2.9(2), there exists a prime divisor E over Y_a with center $\bar{p}$ such that $a(E, Y_a, \Delta_{Y_a}) = \beta$ and $a(E, Y_a, 0) \le l$. By [Liu26, Lemma 2.6(3)], there exists a prime divisor ν' over S_a whose image over Y_a is E. By [Liu26, Lemma 2.6(2)], the image of the center of ν' is $\bar{p}$, so the center of ν' is $\sigma(p)$ for some $\sigma \in \widetilde{G}_a$, and $\nu := \sigma^{-1}(\nu')$ is a prime divisor over S_a with center p whose image is E. Let

$$e := |I_{\widetilde{G}_a}(\nu)| = |I_{G_p}(\nu)| \le |\widetilde{G}_a| \le \max(N, v, s)e_0.$$

By [Liu26, Lemma 2.6(2)], $a(\nu, S_a, \Delta_a) = e\beta$, so ν computes $\operatorname{mld}((S_a \ni p, \Delta_a)/G_p)$, and

$$a(\nu, S_a, 0) = ea(E, Y_a, \Delta_{\mathrm{br}}) \le ea(E, Y_a, 0) \le el,$$

where $\Delta_{\mathrm{br}} \ge 0$ is the branch divisor of $S_a \to Y_a$, so that K_{S_a} is the pullback of $K_{Y_a} + \Delta_{\mathrm{br}}$.

We bound the values of ν. Since (S_a, Δ_a) is lc near p and $c_j \ge \gamma$,

$$\gamma \operatorname{mult}_\nu(H_j^0|_{S_a}) \le a(\nu, S_a, 0) - a(\nu, S_a, \Delta_a) \le el.$$

Let $a = z$, so that $p = (0, y'(p), \alpha_0)$ for a root α_0 of $g^0(1, t')$ of multiplicity $d \le \lfloor N/\min(v, s)\rfloor$, and write $g^0(1, t') = (t' - \alpha_0)^d U(t')$ with $U(\alpha_0) \ne 0$. Near p, $\operatorname{div}(x'|_{S_z}) = dL$, where $L = (x' =$

$t' - \alpha_0 = 0$). We show that (S_z, L) is lc near p. If $y'(p) \neq 0$, then S_z and L are smooth at p, so (S_z, L) is lc near p [KM98, Corollary 2.31(3)]. If $y'(p) = 0$, then near 0 the morphism

$$\mathbb{A}^2_{\rho_1,\rho_2} \to S_z, \qquad (\rho_1, \rho_2) \to \big(U(\alpha_0 + \rho_1\rho_2)\rho_1^d, -\rho_2^d, \alpha_0 + \rho_1\rho_2\big)$$

is the quotient by μ_d acting by $(\rho_1, \rho_2) \to (\zeta\rho_1, \zeta^{-1}\rho_2)$, which is étale in codimension one and pulls L back to $\operatorname{div}(\rho_1)$, so (S_z, L) is lc near p [KM98, Proposition 5.20(4)]. Thus

$$\vartheta = \operatorname{mult}_\nu x' = d \operatorname{mult}_\nu L \leq da(\nu, S_z, 0) \leq \lfloor N/\min(v, s)\rfloor el.$$

The case $a = t$ follows from the argument for $a = z$, with z' in place of t', and $\vartheta = 0$ if $a = y$. Let χ be the character of G_p on u and q_ν the integer of Lemma 3.1 for ν and χ. Let k be a positive integer with $k \equiv q_\nu \pmod e$ and

$$\big(\lfloor N/\min(v, s)\rfloor + \gamma^{-1}\big)el < k \leq \big(\lfloor N/\min(v, s)\rfloor + \gamma^{-1}\big)el + e.$$

Then $k > \vartheta$ and $k > \operatorname{mult}_\nu(H_j^0|_{S_a}) + \vartheta$ for any j, and k satisfies the bound of this step.

The hypotheses of Proposition 5.4 hold, in the notation of Proposition 5.4, with S_a in place of S, the point p in place of s, $U_a^0 = S_a \times \mathbb{A}^1_u$ in place of V, G_p in place of $\tilde{H}$, which acts faithfully on S_a by Construction-Proposition B.13(2), $1 + \delta/r$ in place of b, H_j^0 in place of Φ_j, so that B_a^0 and Δ_a are in place of C_V and C_S, and the integer k in place of ℓ, and with the character χ and the prime divisor ν above. By Proposition 5.4, the Gauss extension F^0 of mult_ν with u-value k is a prime divisor over U_a^0 with center $(p, 0)$, $I_{G_p}(F^0) = I_{G_p}(\nu)$, and

$$\frac{a(F^0, U_a^0, B_a^0)}{e} = \beta + \frac{(1-b)k}{e} = \beta - \frac{\delta k}{re}.$$

By Lemma B.15, there exists a prime divisor F over U_a with center $(p, 0) \in C_a$, $I_{G_p}(F) = I_{G_p}(\nu)$, and $a(F, U_a, B_a) = a(F^0, U_a^0, B_a^0)$. Since $I_{\widetilde{G}_a}(F)$ fixes the center of F, $I_{\widetilde{G}_a}(F) = I_{G_p}(F)$ has order e. By Construction-Proposition B.13(5) and $\delta \geq -(v + s)$,

$$\alpha \leq \frac{a(F, U_a, B_a)}{e} = \beta - \frac{\delta k}{re} \leq \beta + \frac{(v+s)k}{re}.$$

Assume that $\beta = (1 - c')/e'$ for an integral curve $C' \subset C_a$, with $c' = \operatorname{mult}_{C'} \Delta_a$ and $e' = |I_{\widetilde{G}_a}(C')| \leq \max(N, v, s)e_0$. Let $I' := I_{\widetilde{G}_a}(C')$, which is cyclic [Liu26, Lemma 2.6(1)], and fix a general closed point $p \in C'$. Then U_a, S_a, and C' are smooth at p, $\operatorname{Supp} \Delta_a \subset C'$ near p, and I' fixes C' pointwise. Since I' is finite abelian, there exist I'-semi-invariant regular functions w_1, w_2 near p such that (w_1, w_2, u) are local parameters of U_a at p and $C' = (w_1 = u = 0)$ near p. Let λ' be the character of I' on w_1, which generates the group of characters of I' [Liu26, Lemma 2.6(1)], and let $q' \in \{1, \dots, e'\}$ be the integer with $\chi|_{I'} = \lambda'^{q'}$, where χ is the character of u. Let $m_j := \operatorname{mult}_{C'}(H_j^0|_{S_a})$, so that $\sum_j c_j m_j = c' \leq 1$ and $m_j \leq \gamma^{-1}$. Let k be a positive integer with $k \equiv q' \pmod{e'}$ and $\gamma^{-1} < k \leq \gamma^{-1} + e'$, which satisfies the bound of this step.

Let F be the monomial valuation with weights $(1, k)$ on (w_1, u) at the generic point of C'. Then F is a prime divisor over U_a with center $C' \subset C_a$, and near the generic point of C', F is the exceptional divisor of the weighted blow-up of the ideal (w_1, u) with weights $(1, k)$. On its chart $w_1 = w_1'$, $u = w_1'^k u'$, we have $dw_1 \wedge dw_2 \wedge du = w_1'^k dw_1' \wedge dw_2 \wedge du'$ and $F = (w_1' = 0)$, so $a(F, U_a, 0) = 1 + k$, and the residue field of F is generated over the function field of C' by the class of $u' = u/w_1^k$, equivalently of w_1^k/u. By Construction-Proposition B.13(3), near the generic point of C', $H_j = w_1^{m_j}\epsilon_j + uR_j$ with ϵ_j a unit and R_j regular, so $\operatorname{mult}_F H_j = m_j$ since $k > m_j$. Thus

$$a(F, U_a, B_a) = 1 + k - \sum_j c_j m_j - \Big(1 + \frac{\delta}{r}\Big)k = 1 - c' - \frac{\delta k}{r} > 0.$$

Any element of I' preserves F, since w_1 and u are semi-invariant and I' fixes C' pointwise, and it acts trivially on the class of w_1^k/u, since $\lambda'^k = \lambda'^{q'} = \chi$ on I'. Thus $I' \subset I_{\widetilde{G}_a}(F)$. By

Construction-Proposition B.13(5),

$$\alpha \le \frac{a(F,U_a,B_a)}{|I_{\widetilde{G}_a}(F)|} \le \frac{1-c'}{e'} - \frac{\delta k}{re'} \le \beta + \frac{(v+s)k}{re'}.$$

Step 4. In this step we conclude the proof. By Lemma B.14(3), the numbers β_i belong to an ACC set depending only on Γ. Possibly passing to a subsequence, we may assume that the sequence β_i is non-increasing, with limit β_∞. By **Steps 2 and 3**,

$$\beta_i \le \alpha_i \le \beta_i + \frac{(v+s)k_i}{r_i e_i},$$

where k_i is bounded in terms of N, v, s, e_0, Γ, and $e_i \ge 1$. Since $r_i \to +\infty$, $\alpha_i \to \beta_\infty$. Since the sequence α_i is strictly increasing, $\alpha_i < \beta_\infty$ for any i, while $\alpha_i \ge \beta_i \ge \beta_\infty$, a contradiction. □

B.8. **Two-dimensional limits.** In this subsection, we prove Proposition 8.10.

Proof of Proposition 8.10. Recall from Definition 2.7 that $D_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$, where $c_{ij} \in \Gamma$ and the h_{ij} are non-zero regular G_i-semi-invariant functions on a G_i-invariant open neighborhood of 0 in W_i. Let $\rho : \mathbb{R}^4 \to \mathbb{T}^4$ be the quotient map. We regard the G_i as subgroups of $\mathbb{T}^4$, where, for $w = (w_x, w_y, w_z, w_t) \in \mathbb{R}^4$, $\rho(w)$ acts on $\mathbb{A}^4$ by

$$(x,y,z,t) \to \left(e^{2\pi\sqrt{-1}w_x}x, e^{2\pi\sqrt{-1}w_y}y, e^{2\pi\sqrt{-1}w_z}z, e^{2\pi\sqrt{-1}w_t}t\right).$$

For $\beta \in \mathbb{Z}^4_{\ge 0}$, write β as

$$\beta = (\beta_x, \beta_y, \beta_z, \beta_t),$$

and let

$$\mathbf{x}^\beta := x^{\beta_x}y^{\beta_y}z^{\beta_z}t^{\beta_t}, \qquad |\beta| := \beta_x + \beta_y + \beta_z + \beta_t.$$

For $\beta \in \mathbb{Z}^4$, let $\chi_\beta : \mathbb{T}^4 \to \mathbb{R}/\mathbb{Z}$ be the character induced by $w \to \langle \beta, w\rangle$, so that $\rho(w)$ multiplies $\mathbf{x}^\beta$ by $\exp(2\pi\sqrt{-1}\langle\beta, w\rangle)$, and the character of $\mathbf{x}^\beta$ on G_i is $\chi_\beta|_{G_i}$. Let

$$\eta := (1,1,-a,-b), \qquad \mathfrak{t}_{a,b} := \{(X,Y,Z,T) \in \mathbb{R}^4 \mid X + Y = aZ + bT\},$$

and $\mathbb{T}_{a,b} := \{w \in \mathbb{T}^4 \mid w_x + w_y = aw_z + bw_t\}$, the kernel of χ_η. Since $\langle \eta, (1,0,0,0)\rangle = 1$, we have $\mathbb{T}_{a,b} = \rho(\mathfrak{t}_{a,b})$.

Since G_i is diagonal and multiplies $xy + g_i$ by a character, and xy and z^at^b are monomials of $xy + g_i$, we have $\chi_{(1,1,0,0)} = \chi_{(0,0,a,b)}$ on G_i, that is, $G_i \subset \mathbb{T}_{a,b}$ for any i. Since $\mathbb{T}_{a,b}$ is closed in $\mathbb{T}^4$, $H \subset \mathbb{T}_{a,b}$. This proves (1).

The proof is long, so we divide it into several steps.

Step 1. In this step we replace the marks by polynomials and bound them along one prime divisor over W_i.

For a regular function h on a neighborhood of 0 in $\mathbb{A}^4$ and a positive integer k, we write $h \in \mathfrak{m}^k$ if h belongs to the k-th power of the maximal ideal of the local ring of $\mathbb{A}^4$ at 0, and for $h \in \mathbb{C}[[x,y,z,t]]$ we write $h \in \mathfrak{m}^k$ if h belongs to the k-th power of the maximal ideal of $\mathbb{C}[[x,y,z,t]]$. For a polynomial h, $h \in \mathfrak{m}^k$ means that any monomial of h has degree at least k.

Fix i and j. Near 0 on W_i we may write $h_{ij} = p/u$, where $p, u \in \mathbb{C}[x,y,z,t]$ and $u(0) \ne 0$. Replacing u with $\prod_{g\in G_i} g^*u$ and p with $p\prod_{g\in G_i\setminus\{1\}} g^*u$, we may assume that u is G_i-invariant. Then $p|_{W_i} = uh_{ij}$ is G_i-semi-invariant with the character of h_{ij}, so replacing p with the isotypic component of p for the character of h_{ij} does not change $p|_{W_i}$, and we may assume that p is G_i-semi-invariant. Since $u(0) \ne 0$, $\operatorname{div}(h_{ij}) = \operatorname{div}(p)$ near 0. Thus, replacing each h_{ij} with the corresponding p, we may assume that all h_{ij} are G_i-semi-invariant polynomials. Removing the marks with $c_{ij} = 0$ or $h_{ij}(0) \ne 0$ does not change D_i near 0, so we may assume that $c_{ij} > 0$ and $h_{ij}(0) = 0$ for any i, j. If no mark remains, all sums over j below are empty. If $\Gamma \not\subset \{0\}$, let γ be the minimum of $\Gamma\setminus\{0\}$, which exists because Γ is a DCC set, and let $\gamma := 1$ if $\Gamma \subset \{0\}$.

Let $(g_i)_n \ne 0$ be the homogeneous part of degree $n = \operatorname{mult}_0 g_i$ of g_i. The part of $xy + g_i$ of least weight for the weights $(n-1,1,1,1)$ is $xy + (g_i)_n$, which is irreducible since it has degree

one in x with coprime coefficients y and $(g_i)_n \in \mathbb{C}[z,t]\backslash\{0\}$, and is not divisible by a coordinate. By Lemma 4.12, the weighted blow-up with the weights $(n-1,1,1,1)$ defines a prime divisor E_i over W_i with center 0 and

$$a(E_i, W_i, 0) = (n-1) + 1 + 1 + 1 - n = 2.$$

Since (W_i, D_i) is klt near 0,

$$0 < a(E_i, W_i, D_i) = 2 - \sum_j c_{ij} \operatorname{mult}_{E_i} h_{ij}. \tag{B.70}$$

Since the center of E_i is 0, $\operatorname{mult}_{E_i} h \geq k$ for any regular function $h \in \mathfrak{m}^k$. In particular, $\operatorname{mult}_{E_i} h_{ij} \geq 1$. Hence, by (B.70), the number of marks of D_i is less than $2/\gamma$, and

$$\operatorname{mult}_{E_i} h_{ij} < \frac{2}{\gamma} \quad \text{for any } i, j. \tag{B.71}$$

Since Γ is a DCC set, possibly passing to a subsequence, we may assume that the number of marks of D_i is a constant l, and that for any $1 \leq j \leq l$ the sequence $(c_{ij})_i$ is non-decreasing with limit $c_j^* \in [\gamma, 1]$.

Step 2. In this step we reduce monomials modulo $f_0 := xy + z^a t^b$ to monomials not divisible by xy.

Let

$$\sigma := \mathfrak{t}_{a,b} \cap \mathbb{R}^4_{\geq 0}, \qquad \sigma^\circ := \mathfrak{t}_{a,b} \cap \mathbb{R}^4_{>0}, \qquad \mathcal{N} := \{\beta \in \mathbb{Z}^4_{\geq 0} : \beta_x \beta_y = 0\},$$

and $w_0 := (n, n, 2, 2)$. Since $2n = 2a + 2b$, $w_0 \in \sigma^\circ$. For $w \in \sigma$ and $\lambda > 0$, $w + \lambda w_0 \in \sigma^\circ$, so σ is the closure of σ°. Moreover $\langle \eta, w \rangle = 0$ for $w \in \mathfrak{t}_{a,b}$, and

$$2|\beta| \leq \langle \beta, w_0 \rangle \leq n|\beta| \quad \text{for any } \beta \in \mathbb{Z}^4_{\geq 0}. \tag{B.72}$$

(i) *Let $\theta \in \mathbb{Z}^4$ with $\langle \theta, w \rangle \geq 0$ for any $w \in \sigma$. Then there exists a unique $\beta \in \mathcal{N}$ with $\beta - \theta \in \mathbb{Z}\eta$.* If $\beta, \beta' \in \mathcal{N}$ and $\beta - \beta' = k\eta$ with $k > 0$, then $\beta_x \geq k$ and $\beta_y \geq k$, which contradicts $\beta_x \beta_y = 0$. The case $k < 0$ is symmetric. Thus β is unique. For existence, let $(r, 0, p, p') := \theta - \theta_y \eta$. The elements of σ are the vectors $(X, aZ + bT - X, Z, T)$ with $Z, T \geq 0$ and $0 \leq X \leq aZ + bT$, and on such a vector $\langle \theta, \cdot \rangle$ takes the value $rX + pZ + p'T$. If $r \geq 0$, then taking $X = 0$ we get $p, p' \geq 0$, and $\beta := (r, 0, p, p') \in \mathcal{N}$. If $r < 0$, then taking $X = aZ + bT$ we get $p + ar \geq 0$ and $p' + br \geq 0$, and $\beta := (r, 0, p, p') - r\eta = (0, -r, p + ar, p' + br) \in \mathcal{N}$.

(ii) *Let $\beta \in \mathbb{Z}^4_{\geq 0}$, and let $\beta' \in \mathcal{N}$ with $\beta - \beta' \in \mathbb{Z}\eta$, which exists by (i). Then $\mathbf{x}^\beta \in \pm \mathbf{x}^{\beta'} + (f_0)$.* If $\beta_x, \beta_y \geq 1$, then

$$\mathbf{x}^\beta = f_0 \mathbf{x}^{\beta - (1,1,0,0)} - \mathbf{x}^{\beta - \eta},$$

and we conclude by induction on $\min(\beta_x, \beta_y)$, using the uniqueness in (i).

(iii) *Any polynomial P can be written as $P = f_0 B + P^{\mathcal{N}}$, where B is a polynomial and $P^{\mathcal{N}}$ is a linear combination of monomials $\mathbf{x}^\beta$ with $\beta \in \mathcal{N}$. If P is G_i-semi-invariant, then any monomial of $P^{\mathcal{N}}$ has the character of P on G_i, and we may choose B to be G_i-semi-invariant.* The first sentence follows from (ii). Since $\chi_\eta = 0$ on $\mathbb{T}_{a,b} \supset G_i$, the monomials $\mathbf{x}^\beta$ and $\mathbf{x}^{\beta - \eta}$ in (ii) satisfy $\chi_\beta = \chi_{\beta - \eta}$ on G_i, which implies that any monomial of $P^{\mathcal{N}}$ has the character of P. Replacing B with the isotypic component of B for the character of P minus $\chi_{(1,1,0,0)}$, the equality $P = f_0 B + P^{\mathcal{N}}$ still holds.

Step 3. In this step we construct q and Φ.

By Lemma 4.6 and the uniqueness of Hausdorff limits, possibly passing to a subsequence, we may assume that $G_i \subset H$ and that G_i intersects any connected component of H for any i. By [Liu26, Lemma 4.3(1)], H/H^0 is finite and $H^0 = \rho(\mathfrak{h})$ for a linear subspace $\mathfrak{h} \subset \mathbb{R}^4$ spanned by $\mathfrak{h} \cap \mathbb{Z}^4$, with $\dim \mathfrak{h} = 2$. Since $\rho(\mathfrak{h}) \subset \mathbb{T}_{a,b}$, the function $\langle \eta, \cdot \rangle$ takes integral values on the

connected set $\mathfrak{h} \ni 0$, so $\mathfrak{h} \subset \mathfrak{t}_{a,b}$. The lattice $\mathfrak{h}^\perp \cap \mathbb{Z}^4$ is saturated of rank two and contains the primitive vector η, so η extends to a basis of $\mathfrak{h}^\perp \cap \mathbb{Z}^4$: there exists $\varphi \in \mathbb{Z}^4$ such that

$$\mathfrak{h}^\perp \cap \mathbb{Z}^4 = \mathbb{Z}\varphi \oplus \mathbb{Z}\eta, \qquad \mathfrak{h} = \mathfrak{t}_{a,b} \cap \varphi^\perp. \tag{B.73}$$

If $\mathfrak{h}$ contained a vector with all coordinates positive, then Lemma 4.7, in dimension 4 and applied to the data (W_i, D_i, G_i) with the polynomial marks of **Step 1**, would imply that $\alpha_i \to 0$, which contradicts $\alpha_i \geq \epsilon$. Hence, by (B.73), $\langle \varphi, w \rangle \neq 0$ for any $w \in \sigma^\circ$. Since σ° is convex, $\langle \varphi, \cdot \rangle$ has constant sign on σ°. Replacing φ with $-\varphi$, we may assume that

$$\langle \varphi, w \rangle > 0 \ \text{ for any } w \in \sigma^\circ, \qquad \langle \varphi, w \rangle \geq 0 \ \text{ for any } w \in \sigma. \tag{B.74}$$

By **Step 2**(i), possibly replacing φ with an element of $\varphi + \mathbb{Z}\eta$, which does not change (B.73) or the values of $\langle \varphi, \cdot \rangle$ on $\mathfrak{t}_{a,b}$, we may assume that $\varphi \in \mathcal{N}$. By (B.74), $\varphi \neq 0$, so

$$\Phi := \mathbf{x}^\varphi$$

is a non-constant monomial in x, y, z, t.

Since $\mathbb{Z}\varphi \oplus \mathbb{Z}\eta$ is saturated in $\mathbb{Z}^4$, there exists $\nu \in \mathbb{Z}^4$ with $\langle \varphi, \nu \rangle = 1$ and $\langle \eta, \nu \rangle = 0$. If $w \in \mathfrak{t}_{a,b}$ and $\langle \varphi, w \rangle = k \in \mathbb{Z}$, then $w - k\nu \in \mathfrak{h}$ by (B.73), so $\rho(w) = \rho(w - k\nu) \in H^0$. Hence the kernel of χ_φ on $\mathbb{T}_{a,b} = \rho(\mathfrak{t}_{a,b})$ is H^0. Let $q := |H/H^0|$. Then χ_φ induces an injective homomorphism $H/H^0 \to \mathbb{R}/\mathbb{Z}$, so $\chi_\varphi(H) = \frac{1}{q}\mathbb{Z}/\mathbb{Z}$. Since $\langle \varphi, \cdot \rangle$ is non-zero on $\mathfrak{t}_{a,b}$, the set $\{w \in \mathbb{T}_{a,b} : q\chi_\varphi(w) = 0\}$ is a union of q cosets of H^0 and contains H, which is also a union of q cosets of H^0. Thus

$$H = \{w \in \mathbb{T}_{a,b} : q\chi_\varphi(w) = 0\}. \tag{B.75}$$

Since $G_i \subset H$ intersects any connected component of H, $\chi_\varphi(G_i) = \frac{1}{q}\mathbb{Z}/\mathbb{Z}$. Thus the character of Φ on G_i takes values in the group of q-th roots of unity and has order q for any i. In particular, Φ^q is G_i-invariant.

Step 4. In this step we normalize the equation of W_i and the marks modulo $\mathfrak{m}^{k_i}$ for integers $k_i \to +\infty$.

We first show that there exist positive integers k_i with $k_i \to +\infty$ such that the following holds: for any i and any $\beta, \beta' \in \mathbb{Z}^4_{\geq 0}$ with $|\beta|, |\beta'| < k_i$ and $\chi_\beta = \chi_{\beta'}$ on G_i,

$$\beta - \beta' \in \mathbb{Z}\varphi \oplus \mathbb{Z}\eta. \tag{B.76}$$

Let

$$\delta \in \mathbb{Z}^4 \backslash (\mathbb{Z}\varphi \oplus \mathbb{Z}\eta).$$

By (B.73), $\langle \delta, \cdot \rangle$ is non-zero on $\mathfrak{h}$, so there exists $v \in \mathfrak{h}$ with $\langle \delta, v \rangle = \frac{1}{2}$. Since the G_i converge to $H \ni \rho(v)$, there exist $\kappa_i \in G_i$ with $\kappa_i \to \rho(v)$, so $\chi_\delta(\kappa_i) \to \frac{1}{2} + \mathbb{Z}$. Hence there exists $i(\delta)$ such that χ_δ is non-trivial on G_i for any $i \geq i(\delta)$. For any positive integer d, the set of differences $\beta - \beta' \notin \mathbb{Z}\varphi \oplus \mathbb{Z}\eta$ with $|\beta|, |\beta'| < d$ is finite, so there exists i_d such that (B.76) holds for any $i \geq i_d$ and any β, β' with $|\beta|, |\beta'| < d$ and $\chi_\beta = \chi_{\beta'}$ on G_i. We may assume that $(i_d)_d$ is non-decreasing. Removing the terms with $i < i_1$, the integers $k_i := \max\{d \leq i : i_d \leq i\}$ satisfy $k_i \to +\infty$ and (B.76). Possibly removing finitely many terms, we may also assume that $k_i > \max(n, n/\gamma)$ for any i.

Replacing x with a non-zero multiple of x and the equation $xy + g_i$ with the corresponding multiple, we may assume that the coefficient of $z^a t^b$ in g_i is one.

Let $z^p t^{p'}$ be a monomial of g_i with $p + p' < k_i$. Since g_i is G_i-semi-invariant, $\chi_{(0,0,p,p')} = \chi_{(0,0,a,b)}$ on G_i, so by (B.76) there exist integers k, k' with

$$(0, 0, p - a, p' - b) = k\varphi + k'\eta.$$

Pairing with w_0, we get $2(p + p' - n) = k\langle \varphi, w_0 \rangle$. Since $p + p' \geq \operatorname{mult}_0 g_i = n$ and $\langle \varphi, w_0 \rangle > 0$ by (B.74), $k \geq 0$. Since $k\chi_\varphi = 0$ on G_i, **Step 3** implies that q divides k. The first two coordinates imply $k\varphi_x + k' = k\varphi_y + k' = 0$. We consider two cases.

- If $\varphi_x = \varphi_y = 0$, then $k' = 0$ and $z^p t^{p'} = z^a t^b \Phi^k$.
- If $(\varphi_x, \varphi_y) \neq (0, 0)$, then $\varphi_x \neq \varphi_y$ because $\varphi \in \mathcal{N}$, so $k = k' = 0$ and $z^p t^{p'} = z^a t^b$.

Thus there exists a polynomial u_i in one variable with $u_i(0) = 1$, equal to 1 in the second case, such that

$$g_i = z^a t^b U_i + K_i, \qquad U_i := u_i(\Phi^q), \qquad K_i \in \mathfrak{m}^{k_i},$$

where K_i is a polynomial in z, t. The polynomial U_i is G_i-invariant, depends only on z, t, and $U_i(0) = 1$. The morphism $\psi_i : (x, y, z, t) \to (U_i x, y, z, t)$ is a G_i-equivariant automorphism of the open set $(U_i \neq 0) \ni 0$, with inverse $(x, y, z, t) \to (x/U_i, y, z, t)$, and $\psi_i^*(xy + g_i) = U_i f_i$, where

$$f_i := xy + z^a t^b + \frac{K_i}{U_i}, \qquad f_i - f_0 = \frac{K_i}{U_i} \in \mathfrak{m}^{k_i}. \tag{B.77}$$

Since Φ is a monomial in z, t when $U_i \neq 1$, $\Phi \circ \psi_i = \Phi$, and the scaling of x above multiplies Φ by a non-zero constant. Thus, replacing W_i, D_i, and h_{ij} with $\psi_i^{-1}(W_i)$, $\psi_i^* D_i$, and $h_{ij} \circ \psi_i$, we may assume that $W_i = (U_i f_i = 0)$ near 0. Then the h_{ij} are still G_i-semi-invariant polynomials, the replacements preserve α_i, the divisor of Φ, and the character of Φ, and (B.71) holds for the prime divisor over the new W_i corresponding to E_i, which we still denote by E_i and whose center is still 0.

Now fix j. By **Step 2**(iii), $h_{ij} = f_0 B'_{ij} + h_{ij}^{\mathcal{N}}$, where B'_{ij} is a polynomial and $h_{ij}^{\mathcal{N}}$ is a linear combination of monomials $\mathbf{x}^\beta$ with $\beta \in \mathcal{N}$, all with the character of h_{ij} on G_i. Near 0 on W_i we have $f_i = 0$, so $f_0 = -K_i/U_i$ by (B.77) and

$$h_{ij}|_{W_i} = \Big(h_{ij}^{\mathcal{N}} - \frac{K_i}{U_i} B'_{ij}\Big)\Big|_{W_i}.$$

If all monomials of $h_{ij}^{\mathcal{N}}$ had degree at least $2/\gamma$, then, since $k_i > 2/\gamma$, **Step 1** would imply $\operatorname{mult}_{E_i} h_{ij} \geq 2/\gamma$, which contradicts (B.71). Hence $h_{ij}^{\mathcal{N}}$ has a monomial of degree less than $2/\gamma$, and by (B.72) a monomial $\mathbf{x}^\beta$ with $\langle \beta, w_0 \rangle < 2n/\gamma$. Let $\mathbf{x}^{\beta_{ij}}$ be a monomial of $h_{ij}^{\mathcal{N}}$ such that $\langle \beta_{ij}, w_0 \rangle$ is minimal among the monomials of $h_{ij}^{\mathcal{N}}$. By (B.72), $|\beta_{ij}| < n/\gamma$, so only finitely many vectors are possible for β_{ij}. Possibly passing to a subsequence, we may assume that $\beta_{ij} = \beta_j$ does not depend on i. Multiplying h_{ij} by a non-zero constant, we may assume that the coefficient of $\mathbf{x}^{\beta_j}$ in $h_{ij}^{\mathcal{N}}$ is one.

Let $\mathbf{x}^{\beta'}$ be a monomial of $h_{ij}^{\mathcal{N}}$ with $|\beta'| < k_i$. Since $|\beta_j| < n/\gamma < k_i$ and $\chi_{\beta'} = \chi_{\beta_j}$ on G_i, (B.76) implies $\beta' - \beta_j = k\varphi + k'\eta$ for some integers k, k'. Since $\langle \beta', w_0 \rangle \geq \langle \beta_j, w_0 \rangle$ and $\langle \varphi, w_0 \rangle > 0$, $k \geq 0$. Since $k\chi_\varphi = 0$ on G_i, q divides k. By **Step 2**(i), β' is the unique element of $\mathcal{N}$ in $\beta_j + k\varphi + \mathbb{Z}\eta$, so $\mathbf{x}^{\beta'} \in \pm \mathbf{x}^{\beta_j} \Phi^k + (f_0)$ by **Step 2**(ii). Collecting the monomials of $h_{ij}^{\mathcal{N}}$ of degree less than k_i and writing each of them as in the previous sentence, we obtain a polynomial v_{ij} in one variable with $v_{ij}(0) = 1$, a polynomial $Q_{ij} \in \mathfrak{m}^{k_i}$ (the sum of the monomials of $h_{ij}^{\mathcal{N}}$ of degree at least k_i), and a polynomial B_{ij} (B'_{ij} plus the corresponding combination of the cofactors of f_0 in the relations $\mathbf{x}^{\beta'} \in \pm \mathbf{x}^{\beta_j} \Phi^k + (f_0)$) such that (B.78) holds. Replacing B_{ij} with its isotypic component as in **Step 2**(iii), we may assume that B_{ij} is G_i-semi-invariant. Thus

$$h_{ij} = f_0 B_{ij} + \mathbf{x}^{\beta_j} v_{ij}(\Phi^q) + Q_{ij}, \tag{B.78}$$

where v_{ij} is a polynomial in one variable with $v_{ij}(0) = 1$, and B_{ij} and Q_{ij} are G_i-semi-invariant polynomials with $Q_{ij} \in \mathfrak{m}^{k_i}$. The polynomial $v_{ij}(\Phi^q)$ is G_i-invariant. On the G_i-invariant open neighborhood $(U_i v_{ij}(\Phi^q) \neq 0)$ of 0, let

$$\tilde{h}_{ij} := \frac{h_{ij} - f_i B_{ij}}{v_{ij}(\Phi^q)}, \qquad P_{ij} := U_i v_{ij}(\Phi^q) \tilde{h}_{ij} = U_i h_{ij} - (U_i f_i) B_{ij}.$$

By (B.77) and (B.78), $U_i(f_i - f_0) = K_i$ and

$$h_{ij} - f_i B_{ij} = \mathbf{x}^{\beta_j} v_{ij}(\Phi^q) + Q_{ij} - (f_i - f_0) B_{ij}.$$

Hence:

(a) P_{ij} is a G_i-semi-invariant polynomial, and

$$P_{ij} = U_i v_{ij}(\Phi^q) \mathbf{x}^{\beta_j} + U_i Q_{ij} - K_i B_{ij}, \qquad U_i Q_{ij} - K_i B_{ij} \in \mathfrak{m}^{k_i};$$

(b) near 0 on W_i, $\tilde{h}_{ij} = h_{ij}/v_{ij}(\Phi^q)$, so

$$\operatorname{div}_{W_i}(\tilde{h}_{ij}) = \operatorname{div}_{W_i}(P_{ij}) = \operatorname{div}_{W_i}(h_{ij}) \quad \text{near } 0;$$

(c) we have

$$\tilde{h}_{ij} - \mathbf{x}^{\beta_j} = \frac{Q_{ij} - (f_i - f_0)B_{ij}}{v_{ij}(\Phi^q)} \in \mathfrak{m}^{k_i}.$$

Step 5. In this step we define s and prove $\limsup_i \alpha_i \le s/q$.

For $w = (X, Y, Z, T) \in \mathbb{R}^4$, let $\ell_j(w) := \langle \beta_j, w \rangle$ and

$$L(w) := Z + T - \sum_{j=1}^{l} c_j^* \ell_j(w), \qquad s := \inf\{L(w) : w \in \sigma^\circ,\ \langle \varphi, w \rangle = 1\}.$$

The set in the definition of s is non-empty by (B.74), since it contains $w_0/\langle \varphi, w_0 \rangle$.

Fix $w \in \sigma^\circ$ with $\langle \varphi, w \rangle = 1$. Since $w/q \in \mathfrak{t}_{a,b}$ and

$$q\chi_\varphi(\rho(w/q)) = \langle \varphi, w \rangle + \mathbb{Z} = 0,$$

(B.75) implies $\rho(w/q) \in H$. Since the G_i converge to H, there exist $w_i \in N_{G_i}$ with $w_i \to w/q$. Since $\langle \eta, w_i \rangle \in \mathbb{Z}$ and $\langle \eta, w/q \rangle = 0$, possibly removing finitely many terms, we may assume that $w_i = (X_i, Y_i, Z_i, T_i) \in \sigma^\circ$, that is, $w_i \in N_{G_i} \cap \mathbb{R}^4_{>0}$ and $X_i + Y_i = aZ_i + bT_i$. Since $k_i \to +\infty$, we may also assume that

$$k_i \min_k (w_i)_k > \max\{X_i + Y_i, \ell_1(w_i), \dots, \ell_l(w_i)\}.$$

For polynomials P, P' we have

$$\operatorname{wt}_{w_i}(PP') = \operatorname{wt}_{w_i}(P) + \operatorname{wt}_{w_i}(P'),$$

because the parts of lowest weight multiply, and $\operatorname{wt}_{w_i}(P + P') = \operatorname{wt}_{w_i}(P)$ if $\operatorname{wt}_{w_i}(P') > \operatorname{wt}_{w_i}(P)$. Moreover, $\operatorname{wt}_{w_i}(P) \ge k_i \min_k (w_i)_k$ if $P \in \mathfrak{m}^{k_i}$. Since U_i and $v_{ij}(\Phi^q)$ are polynomials in Φ^q with constant term 1 and $\langle \varphi, w_i \rangle > 0$, we have

$$\operatorname{wt}_{w_i}(U_i) = \operatorname{wt}_{w_i}(v_{ij}(\Phi^q)) = 0.$$

The monomials xy and $z^a t^b$ of f_0 have weights $X_i + Y_i$ and $aZ_i + bT_i$, and $X_i + Y_i = aZ_i + bT_i$. Hence, by $U_i f_i = U_i f_0 + K_i$ and (a),

$$\operatorname{wt}_{w_i}(U_i f_i) = X_i + Y_i, \qquad \operatorname{wt}_{w_i}(P_{ij}) = \ell_j(w_i).$$

By (a) and (b), $D_i = \sum_j c_{ij} \operatorname{div}_{W_i}(P_{ij})$ near 0, where the P_{ij} are G_i-semi-invariant polynomials, and $W_i = (U_i f_i = 0)$ near 0. By Lemma 4.7 in dimension 4, applied to $w_i \in N_{G_i} \cap \mathbb{R}^4_{>0}$,

$$\alpha_i \le (X_i + Y_i + Z_i + T_i) - (X_i + Y_i) - \sum_j c_{ij} \ell_j(w_i) = Z_i + T_i - \sum_j c_{ij} \ell_j(w_i).$$

Letting $i \to +\infty$ and using $c_{ij} \to c_j^*$, we get $\limsup_i \alpha_i \le L(w/q) = L(w)/q$. Since w is arbitrary,

$$\limsup_i \alpha_i \le \frac{s}{q}. \tag{B.79}$$

Since $\alpha_i \ge \epsilon$, we have $L(w) \ge q\epsilon$ for any $w \in \sigma^\circ$ with $\langle \varphi, w \rangle = 1$, so $s \ge q\epsilon > 0$.

Moreover, for any $w \in \sigma^\circ$, $\langle \varphi, w \rangle > 0$ by (B.74), and applying the definition of s to $w/\langle \varphi, w \rangle$ we get $L(w) \ge s\langle \varphi, w \rangle$. Since σ is the closure of σ°,

$$L(w) - s\langle \varphi, w \rangle \ge 0 \quad \text{for any } w \in \sigma. \tag{B.80}$$

Step 6. In this step we prove that $(W_i, D_i + s \operatorname{div}(\Phi))$ is lc near 0 and that $\alpha_i \ge s/q$.

Let $V := (f_0 = 0) \subset \mathbb{A}^4$. Since f_0 has degree one in x with coprime coefficients y and $z^a t^b$, V is integral, and since the singular locus of V is contained in $(x = y = zt = 0)$, whose dimension is at most one, V is normal by Serre's criterion. Recall that $\Phi = \mathbf{x}^{\varphi}$ and $\ell_j(w) = \langle \beta_j, w \rangle$. Let

$$B_V := \sum_j c_j^* \operatorname{div}_V(\mathbf{x}^{\beta_j}) + s \operatorname{div}_V(\Phi).$$

Let F be a prime divisor over V, and $w_F := (\operatorname{mult}_F x, \operatorname{mult}_F y, \operatorname{mult}_F z, \operatorname{mult}_F t)$. Since x, y, z, t are regular on V and $xy = -z^a t^b$ on V, $w_F \in \sigma$. The canonical sheaf ω_V is generated by

$$\Omega := \frac{dx \wedge dz \wedge dt}{x} = zt \cdot \frac{dx}{x} \wedge \frac{dz}{z} \wedge \frac{dt}{t},$$

the residue of $dx \wedge dy \wedge dz \wedge dt / f_0$. On a higher model of V on which F is a divisor, $\frac{dx}{x} \wedge \frac{dz}{z} \wedge \frac{dt}{t}$ has at most a simple pole along F, since $(\mathbb{A}^3_{x,z,t}, \operatorname{div}(xzt))$ is lc [KM98, Corollary 2.31(3)] and the projection $V \to \mathbb{A}^3_{x,z,t}$, $(x, y, z, t) \to (x, z, t)$, is birational, so $a(F, V, 0) = 1 + \operatorname{mult}_F \Omega \geq \operatorname{mult}_F z + \operatorname{mult}_F t$. Hence, by (B.80),

$$a(F, V, B_V) \geq \operatorname{mult}_F z + \operatorname{mult}_F t - \sum_j c_j^* \ell_j(w_F) - s\langle \varphi, w_F \rangle = L(w_F) - s\langle \varphi, w_F \rangle \geq 0.$$

Thus (V, B_V) is lc, in particular near 0.

Since V is a normal Cartier hypersurface of the smooth variety $\mathbb{A}^4$, the different of

$$\sum_j c_j^* \operatorname{div}(\mathbf{x}^{\beta_j}) + s \operatorname{div}(\Phi)$$

on V is B_V. By [Kaw07, Theorem], applied on a neighborhood of 0 in $\mathbb{A}^4$,

$$\Big(\mathbb{A}^4, V + \sum_j c_j^* \operatorname{div}(\mathbf{x}^{\beta_j}) + s \operatorname{div}(\Phi)\Big)$$

is lc near 0. By [dFEM11, Proposition 2.11], the pair

$$\Big(\operatorname{Spec} \mathbb{C}[[x, y, z, t]], (f_0) \prod_j (\mathbf{x}^{\beta_j})^{c_j^*} (\Phi)^s\Big)$$

is lc. In $\mathbb{C}[[x, y, z, t]]$ the functions U_i and $v_{ij}(\Phi^q)$ are units, and by (B.77) and (c),

$$f_i - f_0 \in \mathfrak{m}^{k_i}, \qquad \tilde{h}_{ij} - \mathbf{x}^{\beta_j} \in \mathfrak{m}^{k_i} \quad \text{for any } j.$$

Since $k_i \to +\infty$, by Lemma 4.14, applied to power series in four variables, with the finite set $\Lambda := \{1, s, c_1^*, \dots, c_l^*\} \subset \mathbb{R}_{>0}$, the $l + 2$ factors $f_0, \mathbf{x}^{\beta_1}, \dots, \mathbf{x}^{\beta_l}, \Phi$ and $f_i, \tilde{h}_{i1}, \dots, \tilde{h}_{il}, \Phi$, and the exponents $1, c_1^*, \dots, c_l^*, s$, possibly removing finitely many terms, we may assume that the pair

$$\Big(\operatorname{Spec} \mathbb{C}[[x, y, z, t]], (f_i) \prod_j (\tilde{h}_{ij})^{c_j^*} (\Phi)^s\Big)$$

is lc for any i. By [dFEM11, Proposition 2.11] again,

$$\Big(\mathbb{A}^4, \operatorname{div}(f_i) + \sum_j c_j^* \operatorname{div}(\tilde{h}_{ij}) + s \operatorname{div}(\Phi)\Big)$$

is lc near 0. Since $U_i(0) = 1$, $\operatorname{div}(f_i) = W_i$ near 0, and W_i is a normal Cartier hypersurface of $\mathbb{A}^4$ near 0, so the different of

$$\sum_j c_j^* \operatorname{div}(\tilde{h}_{ij}) + s \operatorname{div}(\Phi)$$

on W_i is the restriction to W_i. By [Kaw07, Theorem], the pair

$$\Big(W_i, \sum_j c_j^* \operatorname{div}_{W_i}(\tilde{h}_{ij}) + s \operatorname{div}_{W_i}(\Phi)\Big)$$

is lc near 0. By (b),

$$\operatorname{div}_{W_i}(\tilde{h}_{ij}) = \operatorname{div}_{W_i}(h_{ij}) \quad \text{near } 0,$$

and $c_{ij} \le c_j^*$ because $(c_{ij})_i$ is non-decreasing with limit c_j^*. Thus

$$(W_i, D_i + s\operatorname{div}(\Phi)) \text{ is lc near } 0. \tag{B.81}$$

By **Step 3**, Φ is a G_i-semi-invariant regular function vanishing at 0 whose character has order q. By Lemma 5.8, applied with $W = W_i$, $D = D_i$, $G = G_i$, $Z = \{0\}$, $U = \Phi$, $h = q$, and $\lambda = s$ in the notation of Lemma 5.8, (B.81) implies

$$\alpha_i = \operatorname{mld}((W_i \ni 0, D_i)/G_i) \ge \frac{s}{q}.$$

Together with **Step 3** and (B.79), this proves (2).

Finally, if the α_i were strictly increasing, then so would be the terms of the subsequence above, and $\alpha_i < \lim_{i'} \alpha_{i'} \le s/q \le \alpha_i$ for any i in the subsequence by (B.79), a contradiction. □

B.9. **Inertia groups exchanging x and y.** In this subsection, we prove Lemma B.16, which bounds the log discrepancy of a prime divisor whose inertia group exchanges x and y.

Lemma B.16. *Let $g \in \mathbb{C}[z,t]$ be a non-zero polynomial such that $W := (xy + g = 0) \subset \mathbb{A}^4$ is normal. Let G be a finite abelian group acting linearly on $\mathbb{A}^4$, monomially on $\{x, y\}$ and diagonally on z, t, preserving $xy+g$ up to a character, and acting faithfully on W, and let $G^{\mathrm{s}} \subset G$ be the subgroup preserving each of x and y. Let F be a prime divisor over W with center 0, $e := |I_G(F)|$, and*

$$\theta_F := a(F, W, 0) - \operatorname{mult}_F z - \operatorname{mult}_F t.$$

Then $\theta_F \ge 0$. If $I_G(F) \subset G^{\mathrm{s}}$, then $\theta_F \in e\mathbb{Z}$. If $I_G(F) \not\subset G^{\mathrm{s}}$, then e is even and $\theta_F \ge \frac{e}{2}$.

Proof. Let $\omega := (dx/x) \wedge dz \wedge dt$, the Poincaré residue of $dx \wedge dy \wedge dz \wedge dt/(xy + g)$, which generates ω_W [KM98, Proposition 5.73]. The projection $W \to \mathbb{A}^3_{x,z,t}$ is birational, and ω is a generator of $\omega_{\mathbb{A}^3}(\operatorname{div}(x))$, so

$$a(F, W, 0) = 1 + \operatorname{mult}_F \omega = a(F, \mathbb{A}^3_{x,z,t}, \operatorname{div}(x)),$$

where $\operatorname{mult}_F \omega$ is the order along F of the pullback of ω to a smooth model on which F is a divisor. Hence

$$\theta_F = a\big(F, \mathbb{A}^3_{x,z,t}, \operatorname{div}(x) + \operatorname{div}(z) + \operatorname{div}(t)\big) \ge 0,$$

since $(\mathbb{A}^3, \operatorname{div}(xzt))$ is lc.

Let χ_z and χ_t be the characters of z and t, and let $\eta : G \to \{\pm 1\}$ be the character with kernel G^{s}. For $\gamma \in G \backslash G^{\mathrm{s}}$, the pullback $\gamma^* x$ is a multiple of y, so $\gamma^*(dx/x) = dy/y$. On W we have $dx/x + dy/y = dg/g$, and $dg \wedge dz \wedge dt = 0$ since $g \in \mathbb{C}[z,t]$. Hence

$$\gamma^* \omega = \eta(\gamma)\chi_z(\gamma)\chi_t(\gamma)\omega \quad \text{for any } \gamma \in G,$$

whatever the character of $xy+g$ is. Let $I := I_G(F)$. By [Liu26, Lemma 2.6(1)], I is cyclic of order e and acts faithfully on $\mathfrak{m}_F/\mathfrak{m}_F^2$ by a character λ. By [Liu26, Lemma 9.2(1)], $\chi_z|_I = \lambda^{\operatorname{mult}_F z}$ and $\chi_t|_I = \lambda^{\operatorname{mult}_F t}$. Moreover $\gamma^*\omega = \lambda(\gamma)^{a(F,W,0)}\omega$ for $\gamma \in I$ by Lemma 3.12(1), since ω is a G-semi-invariant generator of ω_W, and $I = I_G(F)$ preserves F and acts trivially on the residue field of mult_F [Liu26, Definition 2.5]. Comparing the characters of ω on I, we obtain

$$\lambda^{\theta_F} = \eta|_I.$$

If $I \subset G^{\mathrm{s}}$, then $\eta|_I$ is trivial, so e divides θ_F, as λ is faithful of order e. If $I \not\subset G^{\mathrm{s}}$, then $\eta|_I$ has order two. Since λ is a faithful character of the cyclic group I of order e, e is even and $\theta_F \equiv \frac{e}{2} \pmod{e}$. As $\theta_F \ge 0$, we obtain $\theta_F \ge \frac{e}{2}$. □

B.10. **Binary splitting.** In this subsection, we prove Theorem 8.11.

Proof of Theorem 8.11. The proof is long, so we divide it into several steps. Let $(M \ni o)$ be the smooth fourfold germ of the statement, with $W = (f = 0) \subset M$, and let χ_f be the character of f.

Step 1. In this step we split off the binary part of f and fix the coordinates X, Y.

The group G is finite cyclic and acts on the smooth fourfold germ $(M \ni o)$ fixing o, f is G-semi-invariant, $f \in \mathfrak{m}_{M,o}^2$ since W is singular at o, and the quadratic part q of f has rank two. The partial derivatives of q span a G-stable two-dimensional subspace Λ of $\mathfrak{m}_{M,o}/\mathfrak{m}_{M,o}^2$, and q is a non-degenerate quadratic form in any basis of Λ. Choose G-eigenvectors spanning Λ and G-eigenvectors spanning a G-stable complement of Λ, and lift them by character projection to G-eigenparameters u_1, u_2, z, t of M at o, where u_1, u_2 lift the basis of Λ. Thus the restriction of q to the (u_1, u_2)-plane is non-degenerate, and q is a quadratic form in u_1, u_2 alone.

By Proposition 3.15, applied to $(M \ni o)$, f, W, and u_1, u_2, z, t, there exist a smooth surface germ $(T \ni v)$ with an action of G fixing v and G-eigenparameters z, t of T at v, a diagonal action of G on $\mathbb{A}^2_{x,y}$, $Q_2 \in \{xy, x^2 + y^2\}$, a regular G-semi-invariant function g on T vanishing to order at least two at v, and a germ $(W' \ni o')$ with an action of G and G-equivariant morphisms $(W' \ni o') \to (W \ni o)$ and $(W' \ni o') \to (W_1 \ni o_1)$ étale at o', where

$$W_1 := (Q_2(x, y) + g = 0) \subset \mathbb{A}^2_{x,y} \times T \quad \text{and} \quad o_1 := (0, 0, v).$$

We define G-eigenvectors X, Y of the (x, y)-plane as follows.

- If $Q_2 = xy$, let $X := x$ and $Y := y$.
- If $Q_2 = x^2+y^2$ and $\chi_x = \chi_y$, then G acts on the (x, y)-plane by scalars. Let $X := x+\sqrt{-1}y$ and $Y := x - \sqrt{-1}y$. Then X, Y are G-eigenvectors and $Q_2 = XY$.
- If $Q_2 = x^2 + y^2$ and $\chi_x \neq \chi_y$, let $X := x$ and $Y := y$.

Let $\Theta := XY$ in the first two cases and $\Theta := X^2 + Y^2$ in the third case. Then $Q_2 = \Theta$, G acts diagonally on X, Y, and $\chi_X \neq \chi_Y$ if $\Theta = X^2 + Y^2$. **Step 1** does not involve the marks, the label vectors, or the prime divisors in the statement.

Step 2. In this step we lift the étale maps of **Step 1** to an automorphism of the ambient power series ring.

Let $\hat{f}$ be the expansion of f in $\widehat{\mathcal{O}}_{M,o} = \mathbb{C}[[u_1, u_2, z, t]]$, and $\hat{g}$ the expansion of g in $\widehat{\mathcal{O}}_{T,v} = \mathbb{C}[[z, t]]$. The completion of $\mathbb{A}^2_{x,y} \times T$ at o_1 is $\mathfrak{R}' := \mathbb{C}[[X, Y, z, t]]$, on which G acts diagonally. The morphisms $W' \to W$ and $W' \to W_1$ are G-equivariant and étale at o', and the residue fields at o, o', and o_1 are $\mathbb{C}$. Hence they induce G-equivariant isomorphisms of completed local rings, and we obtain a G-equivariant isomorphism

$$\phi : \mathbb{C}[[u_1, u_2, z, t]]/(\hat{f}) \to \mathfrak{R}'/(\Theta + \hat{g}).$$

Here $\hat{f} \in (u_1, u_2, z, t)^2$ and $\Theta + \hat{g} \in (X, Y, z, t)^2$. We show that $\Theta + \hat{g}$ is G-semi-invariant. Let $\gamma \in G$. Since G acts on W_1 and diagonally on $\mathfrak{R}'$, we have $\gamma^*(\Theta + \hat{g}) = \beta_\gamma(\Theta + \hat{g})$ for a unit β_γ of $\mathfrak{R}'$. By Proposition 3.15 and its part (1), $\hat{g}$ is G-semi-invariant and non-zero. Setting $X = Y = 0$ gives $\gamma^*\hat{g} = \beta_\gamma(0, 0, z, t)\hat{g}$, so $\beta_\gamma(0) = \chi_{\hat{g}}(\gamma)$. Setting $z = t = 0$ and comparing the parts of degree two gives $\gamma^*\Theta = \beta_\gamma(0)\Theta$. Hence

$$\gamma^*(\Theta + \hat{g}) = \chi_{\hat{g}}(\gamma)(\Theta + \hat{g}).$$

By Lemma 3.8, there exist a G-equivariant isomorphism $\Phi : \mathbb{C}[[u_1, u_2, z, t]] \to \mathfrak{R}'$ inducing ϕ and a G-invariant unit v_1 of $\mathfrak{R}'$ such that $\Phi(\hat{f}) = v_1(\Theta+\hat{g})$. In particular, $\Theta+\hat{g}$ has the character χ_f, and the characters of u_1, u_2, z, t are those of X, Y, z, t up to order. By Proposition 3.15, the pullbacks to W' of z, t on W coincide with those of z, t on W_1. The pullbacks of z, t on W are non-zero since $f \in \mathfrak{m}_{M,o}^2$ and $W' \to W$ is étale at o', so z, t on T have the characters of z, t on M. Hence $\{\chi_{u_1}, \chi_{u_2}\} = \{\chi_X, \chi_Y\}$ with multiplicities. Possibly exchanging u_1 and u_2, we may assume that $\chi_{u_1} = \chi_X$ and $\chi_{u_2} = \chi_Y$.

Let $(x_1, x_2, x_3, x_4) := (u_1, u_2, z, t)$ and $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_4]]$. We identify $\mathfrak{R}'$ with $\mathfrak{R}$ by the G-equivariant isomorphism $X, Y, z, t \to x_1, x_2, x_3, x_4$. Then $\psi := \Phi$ is a G-equivariant automorphism of $\mathfrak{R}$, v_1^{-1} is a G-invariant unit of $\mathfrak{R}$, and

$$v_1^{-1}\psi(\hat{f}) = \Theta + \hat{g}. \tag{B.82}$$

The coordinates of $\mathbb{A}^4$ below are written X, Y, z, t, with the diagonal action of G in which they have the characters of $x_1, \dots, x_4$.

Step 3. In this step we construct V by Construction-Theorem 4.13.

The linear part of ψ is invertible and $v_1(0) \neq 0$, so by (B.82) the quadratic part of $\Theta + \hat{g}$ has rank two. Since Θ has rank two in X, Y and $\hat{g} \in \mathbb{C}[[z, t]]$, the quadratic part of $\hat{g}$ is zero, that is, $\hat{g} \in (z, t)^3$.

Lift the marks to regular G-semi-invariant functions on M near o with the characters of the marks, by lifting and character projection, and denote the lifts again by $h_1, \dots, h_s$. Since W is klt, there exists a vector $c^{(0)} \in \mathbb{R}^s_{>0}$ such that $(W, \sum_k c_k^{(0)} \operatorname{div}(h_k|_W))$ is klt near o, and $c^{(0)}$ plays the role of $(c_1, \dots, c_s)$ in Construction-Theorem 4.13(iii). Let $c^{(1)}, \dots, c^{(q)}$ be the elements of the finite set $\mathcal{C}$ of the statement. Let $P_1, \dots, P_l$ be the prime divisors in the statement, and take the units u_k of Construction-Theorem 4.13(vi) to be 1. With $n = 4$, the eigenparameters $x_1, \dots, x_4$, and ψ of **Step 2**, with v_1^{-1} in place of v in (vi), the conditions (i)–(vi) of Construction-Theorem 4.13 hold. Let N_1 be the integer of Construction-Theorem 4.13, fix $N \geq N_1$, and let L_1 be the integer for N. Fix an integer $L \geq L_1$. Let $g_0 \in \mathbb{C}[z, t]$ be the sum of the terms of $\hat{g}$ of degree smaller than L, and let

$$f_0 := \Theta + g_0 \in \mathbb{C}[X, Y, z, t].$$

Let $h_{k,0}$ be the sum of the terms of $\psi(\hat{h}_k)$ of degree smaller than N. Since G acts diagonally, any monomial of a semi-invariant series has the character of the series. Hence $\hat{g}$, g_0, and f_0 are G-semi-invariant with the character χ_f, and $h_{k,0}$ is G-semi-invariant with the character of h_k. Moreover

$$h_{k,0} \equiv \psi(\hat{h}_k) \pmod{(x_1, \dots, x_4)^N} \quad \text{and} \quad f_0 - v_1^{-1}\psi(\hat{f}) = g_0 - \hat{g} \in (x_1, \dots, x_4)^L.$$

Let $V := (f_0 = 0) \subset \mathbb{A}^4$. By Construction-Theorem 4.13, $(V, (h_{k,0})_k)$ is a numerical admission of $(W, (h_k)_k)$ with the group G, G acts faithfully on V, and, since $n = 4$, V is an isolated Gorenstein terminal germ at 0 by Construction-Theorem 4.13(3). Since $\hat{g} \in (z, t)^3$ by the first paragraph of this step, $g_0 \in (z, t)^3$. Moreover $g_0 \neq 0$, since otherwise $V = (\Theta = 0)$ is singular along the (z, t)-plane, whereas V has an isolated singularity at 0. If $\Theta = XY$, then $V = (XY + g_0(z, t) = 0)$. If $\Theta = X^2 + Y^2$, then $V = (X^2 + Y^2 + g_0(z, t) = 0)$ and $\chi_X \neq \chi_Y$ by **Step 1**. In both alternatives G acts diagonally on X, Y, z, t. This proves the existence of the numerical admission in Theorem 8.11(1).

Step 4. In this step we show that the hypothesis of Theorem 8.11(2) does not depend on the numerical admission.

Let $\Gamma \subset [0, 1]$ be a DCC set, and let W, G, $h_1, \dots, h_s$ and labels $c_1, \dots, c_s \in \Gamma$ satisfy the hypotheses of Theorem 8.11(2), with $D := \sum_j c_j \operatorname{div}(h_j)$ klt near o. The hypothesis of Theorem 8.11(2), namely the second alternative and an element of G acting by $(X, Y, z, t) \to (-X, Y, z, t)$, is a condition on a numerical admission, which is not unique. For any numerical admission $\widetilde{V}$ given by Theorem 8.11(1), Construction-Theorem 4.13(2) gives a G-equivariant isomorphism of the completions of W at o and of $\widetilde{V}$ at 0. Hence the characters of the coordinates X, Y, z, t of $\widetilde{V}$ are those of G on the cotangent space of W at o. The isomorphism of completions induces an isomorphism of tangent cones, whose ideals are generated by the quadratic parts of the equations, so the induced isomorphism of cotangent spaces maps the subspace Λ of **Step 1**, spanned by the partial derivatives of the quadratic part of f, onto the span of the classes of X, Y. Thus $\{\chi_X, \chi_Y\}$ and $\{\chi_z, \chi_t\}$ are the characters of G on Λ and on $(\mathfrak{m}_{M,o}/\mathfrak{m}_{M,o}^2)/\Lambda$, and the type of the quadratic part is determined: the second alternative holds if and only if the characters

χ_X, χ_Y of G on Λ satisfy $\chi_X \neq \chi_Y$ and $\chi_X^2 = \chi_Y^2 = \chi_f$ (then $\chi_X\chi_Y \neq \chi_f$, so the quadratic part is not of the form XY). Thus an element of G acting by $(X, Y, z, t) \to (-X, Y, z, t)$ on $\widetilde{V}$ acts by $(X, Y, z, t) \to (-X, Y, z, t)$ on the admission V of **Steps 1–3**, after labelling by X the eigencoordinate on which it acts by -1.

Step 5. In this step we show that the reflection forces each mark on V to be a monomial in X times a polynomial in Y, z, t.

By the hypothesis of Theorem 8.11(2) and **Step 4**, $\Theta = X^2 + Y^2$, and G contains the element τ acting on $\mathbb{A}^4$ by $(X, Y, z, t) \to (-X, Y, z, t)$. Since **Steps 1 and 2** do not involve the labels, Θ and the characters of X, Y, z, t do not depend on the labels. Fix a prime divisor P_* over W with center o. We apply **Step 3** with $\mathcal{C} := \{(c_1, \dots, c_s)\}$, whose entries may vanish, with the vector $c^{(0)} \in \mathbb{R}^s_{>0}$ of **Step 3**, and with the prime divisor P_* in place of $P_1, \dots, P_l$. Let

$$P := Y^2 + g_0(z, t), \qquad V = (X^2 + P = 0), \qquad p_j := h_{j,0}|_V, \qquad D_0 := \sum_j c_j \operatorname{div}(p_j).$$

By Construction-Theorem 4.13(1) for the vector $c^{(1)} = (c_1, \dots, c_s)$, (V, D_0) is klt near 0 and

$$\operatorname{mld}((V \ni 0, D_0)/G) = \operatorname{mld}((W \ni o, D)/G). \tag{B.83}$$

Let $P_{*,0}$ be the prime divisor over V with center 0 corresponding to P_* by Construction-Theorem 4.13(2). Then $\operatorname{mult}_{P_{*,0}} p_j = \operatorname{mult}_{P_*} h_j < +\infty$, so $p_j \neq 0$ in $\mathcal{O}_V$ for any j.

Since $X^2 + P$ is monic of degree two in X,

$$\mathbb{C}[X, Y, z, t]/(X^2 + P) = \mathbb{C}[Y, z, t] \oplus X\mathbb{C}[Y, z, t],$$

and both summands are G-stable. Write $p_j = a_j + Xb_j$ with $a_j, b_j \in \mathbb{C}[Y, z, t]$. The element τ acts trivially on $\mathbb{C}[Y, z, t]$ and multiplies X by -1. Since p_j is G-semi-invariant, $\tau^* p_j = \pm p_j$, so $b_j = 0$ or $a_j = 0$ by the uniqueness of the decomposition. Thus there exist $\epsilon_j \in \{0, 1\}$ and non-zero polynomials $\bar{p}_j \in \mathbb{C}[Y, z, t]$ such that

$$p_j = X^{\epsilon_j} \bar{p}_j(Y, z, t) \quad \text{in } \mathcal{O}_V. \tag{B.84}$$

The polynomial $\bar{p}_j$ is G-semi-invariant with the character $\chi_{h_j}\chi_X^{-\epsilon_j}$, which is trivial on τ since $\chi_{h_j}(\tau) = (-1)^{\epsilon_j}$ and $\chi_X(\tau) = -1$.

Step 6. In this step we descend (V, D_0) to $V' := V/\langle\tau\rangle$.

The invariant ring of τ on $\mathbb{C}[X, Y, z, t]/(X^2 + P)$ is $\mathbb{C}[Y, z, t]$. Hence $V' = \mathbb{A}^3_{Y,z,t}$ is smooth, and the quotient map $\pi : V \to V'$ is $(X, Y, z, t) \to (Y, z, t)$, a double cover branched along $(P = 0)$. Let $H := G/\langle\tau\rangle$. Then H is a finite cyclic group acting diagonally on V' and fixing 0. Since $f_0 = X^2 + Y^2 + g_0$ is G-semi-invariant with the character χ_f, P is G-semi-invariant with the character χ_f, and $\chi_f(\tau) = \chi_Y(\tau)^2 = 1$. Together with **Step 5**, P and $\bar{p}_j$ are H-semi-invariant.

We show that the action of H on V' is faithful. Let $\sigma \in G$ act trivially on Y, z, t. Comparing the coefficients of X^2 and Y^2 in $\sigma^* f_0 = \chi_f(\sigma) f_0$ implies $\chi_X(\sigma)^2 = \chi_f(\sigma) = \chi_Y(\sigma)^2 = 1$. So σ acts on $\mathbb{A}^4$ as the identity or as τ. Since G acts faithfully on V, $\sigma \in \langle\tau\rangle$. Hence H acts faithfully on V'.

We show that the polynomial P is reduced near 0. If R^2 divides P in $\mathcal{O}_{\mathbb{A}^3,0}$ for a non-unit R, then $X^2 + P$ and the partial derivatives of $X^2 + P$ vanish on the surface $(X = R = 0) \subset V$ through 0, which contradicts the isolated singularity of V at 0. Hence P is reduced near 0, and by the Hurwitz formula [Kol13, 2.41],

$$K_V = \pi^*\Big(K_{V'} + \frac{1}{2}\operatorname{div}(P)\Big).$$

Since $X^2 = -P$ on V, $\operatorname{div}_V(X) = \frac{1}{2}\pi^* \operatorname{div}(P)$, and (B.84) implies

$$\operatorname{div}_V(p_j) = \pi^*\Big(\operatorname{div}(\bar{p}_j) + \frac{\epsilon_j}{2}\operatorname{div}(P)\Big).$$

Let

$$C_X := \sum_j c_j \epsilon_j \quad \text{and} \quad B_{V'} := \frac{1+C_X}{2}\operatorname{div}(P) + \sum_j c_j \operatorname{div}(\bar{p}_j).$$

Then $B_{V'} \geq 0$, $B_{V'}$ is H-invariant, and

$$K_V + D_0 = \pi^*(K_{V'} + B_{V'}).$$

Hence $(V', B_{V'})$ is klt near 0 [KM98, Proposition 5.20(4)], which holds for $\mathbb{R}$-divisors [Liu26, Remark 2.17].

Let F be a prime divisor over V with center 0, let ν be the prime divisor over V' with center 0 induced by F, and let $\rho \in \{1, 2\}$ be the ramification index of π along F. Then $a(F, V, D_0) = \rho a(\nu, V', B_{V'})$ by [Liu26, Lemma 2.6(1) and (2)], applied to the action of $\langle\tau\rangle$ on V with the boundary D_0, whose quotient map is π, and whose crepant quotient $(V, D_0)/\langle\tau\rangle$ is $(V', B_{V'})$ by [Liu26, Lemma 2.6(4)]. The quotient map $V \to V/G$ is the composite $V \to V' \to V'/H$, ramification indices along a prime divisor multiply along the composite, and the order of the inertia group equals the ramification index of the quotient map [Liu26, Lemma 2.6(1)]. Thus $|I_G(F)| = \rho |I_H(\nu)|$ and

$$\frac{a(F, V, D_0)}{|I_G(F)|} = \frac{a(\nu, V', B_{V'})}{|I_H(\nu)|}.$$

Since the fiber of π over 0 is $\{0\}$, any prime divisor over V' with center 0 is the prime divisor induced by some prime divisor over V with center 0. Moreover $(V, D_0)/G = (V', B_{V'})/H$ is the quotient in stages [Liu26, Lemma 2.6]. Together with (B.83),

$$\operatorname{mld}((W \ni o, D)/G) = \operatorname{mld}((V' \ni 0, B_{V'})/H). \tag{B.85}$$

Step 7. In this step we bound the labels of $B_{V'}$ and conclude the ACC.

Since $g_0(0) = 0$, $P(0) = 0$. Since P is reduced near 0 and $B_{V'} \geq 0$, the coefficient of any component of $(P = 0)$ through 0 in $B_{V'}$ is at least $\frac{1+C_X}{2}$. Since $(V', B_{V'})$ is klt near 0, $\frac{1+C_X}{2} < 1$, so $C_X < 1$. The number C_X is 0 or a finite sum of elements of Γ. Let

$$\Gamma' := \Gamma \cup \left\{ \frac{1+\gamma}{2} \;\middle|\; \gamma \in \Gamma_+ \cap [0, 1] \right\}.$$

Then $\Gamma' \subset [0, 1]$, and Γ' is a DCC set depending only on Γ, since $\Gamma_+ \cap [0, 1]$ is a DCC set [Liu26, Lemma 2.10(1)]. The boundary $B_{V'}$ is a sum of labels in Γ' times divisors of non-zero regular H-semi-invariant functions near 0. Non-reduced functions and common components are allowed in Theorem 2.8(1), whose boundaries are arbitrary sums $\sum_j c_j \operatorname{div}(h_j)$. Thus $(V' \ni 0)$ is a smooth threefold germ, H is a finite abelian group acting faithfully on V' and fixing 0, $B_{V'}$ has labels in Γ', and $(V', B_{V'})$ is klt near 0. By Theorem 2.8(1) for the DCC set Γ', the positive numbers $\operatorname{mld}((V' \ni 0, B_{V'})/H)$ satisfy the ACC. By (B.85), the positive quotient values $\operatorname{mld}((W \ni o, D)/G)$ in Theorem 8.11(2) form a subset of the set in Theorem 2.8(1) for Γ'. Hence they satisfy the ACC.

Step 8. In this step we prove the part of Theorem 8.11(1) on the formal coordinates x, y, z, t.

To distinguish them from the parameters of **Steps 1–3**, we write $\tilde{x}, \tilde{y}, \tilde{z}, \tilde{t}$ for the G-eigen formal coordinates in the statement, which are elements of $\widehat{\mathcal{O}}_{M,o}$, and $\tilde{g}$ for the series $\hat{g}$ of the statement. Let $\tilde{\Theta} := \tilde{x}\tilde{y}$, resp. $\tilde{\Theta} := \tilde{x}^2 + \tilde{y}^2$, and $e := \tilde{\Theta} + \tilde{g}(\tilde{z}, \tilde{t})$. If $\tilde{\Theta} = \tilde{x}^2 + \tilde{y}^2$ and $\chi_{\tilde{x}} = \chi_{\tilde{y}}$, then $\tilde{x}' := \tilde{x} + \sqrt{-1}\tilde{y}$ and $\tilde{y}' := \tilde{x} - \sqrt{-1}\tilde{y}$ have the character $\chi_{\tilde{x}}$, so $\tilde{x}', \tilde{y}', \tilde{z}, \tilde{t}$ are G-eigen formal coordinates, and $e = \tilde{x}'\tilde{y}' + \tilde{g}(\tilde{z}, \tilde{t})$ since $\tilde{x}^2 + \tilde{y}^2 = \tilde{x}'\tilde{y}'$. Thus the completed equation is $\tilde{x}'\tilde{y}' + \tilde{g}(\tilde{z}, \tilde{t})$ in $\tilde{x}', \tilde{y}', \tilde{z}, \tilde{t}$, and the last sentence of Theorem 8.11(1) follows from the first case for $\tilde{x}', \tilde{y}', \tilde{z}, \tilde{t}$, which is proved below. Replacing $\tilde{x}, \tilde{y}$ by $\tilde{x}', \tilde{y}'$ if $\tilde{\Theta} = \tilde{x}^2 + \tilde{y}^2$ and $\chi_{\tilde{x}} = \chi_{\tilde{y}}$, we may assume that $\tilde{\Theta} = \tilde{x}\tilde{y}$, or that $\tilde{\Theta} = \tilde{x}^2 + \tilde{y}^2$ and $\chi_{\tilde{x}} \neq \chi_{\tilde{y}}$. Fix a positive integer L'. The ideal $(\hat{f})$ of $\widehat{\mathcal{O}}_{M,o}$ is generated by e, so $\hat{f} = v_2 e$ for a unit v_2, and $e \in \mathfrak{m}^2_{M,o}\widehat{\mathcal{O}}_{M,o}$. Hence $\tilde{g} \in (\tilde{z}, \tilde{t})^2$. The quadratic part of e is $\tilde{\Theta}$ plus the quadratic part of $\tilde{g}$, and it has rank two since the quadratic part of f has rank two. Since $\tilde{\Theta}$ has rank two in $\tilde{x}, \tilde{y}$, $\tilde{g} \in (\tilde{z}, \tilde{t})^3$.

Let e' be the sum of the terms of e, as a series in $\tilde{x},\tilde{y},\tilde{z},\tilde{t}$, with the character χ_f, and v_2' the sum of the terms of v_2^{-1} with the trivial character. Since $e = v_2^{-1}\hat{f}$ and $\hat{f}$ is G-semi-invariant with the character χ_f, we have $e' = v_2'\hat{f}$, and v_2' is a G-invariant unit with $v_2'(0) = v_2^{-1}(0)$. Thus the quadratic parts of e and e' are both $v_2^{-1}(0)$ times the quadratic part of $\hat{f}$. Hence all terms of $\tilde{\Theta}$ have the character χ_f, and $e' = \tilde{\Theta} + \tilde{g}'(\tilde{z},\tilde{t})$, where $\tilde{g}' \in (\tilde{z},\tilde{t})^3$ is the sum of the terms of $\tilde{g}$ with the character χ_f. Since e' is a unit multiple of e, for any $w \in \mathbb{Z}^4_{>0}$ the lowest w-weighted parts of e and e' differ by a non-zero constant factor. Replacing $\tilde{g}$ by $\tilde{g}'$ and v_2 by $v_2'^{-1}$, we may assume that e is G-semi-invariant with the character χ_f and that v_2 is a G-invariant unit.

Let $x_1,\dots,x_4$ be G-eigenparameters of M at o lifting the classes of $\tilde{x},\tilde{y},\tilde{z},\tilde{t}$ in $\mathfrak{m}_{M,o}/\mathfrak{m}_{M,o}^2$, obtained by lifting and character projection. Let $\mathfrak{R} := \mathbb{C}[[x_1,\dots,x_4]]$, which is $\widehat{\mathcal{O}}_{M,o}$, and $\mathfrak{m} := (x_1,\dots,x_4)$. The substitution of $\tilde{x},\tilde{y},\tilde{z},\tilde{t}$ for $x_1,\dots,x_4$ is a G-equivariant automorphism of $\mathfrak{R}$, since its linear part is the identity and $x_1,\dots,x_4$ have the characters of $\tilde{x},\tilde{y},\tilde{z},\tilde{t}$. Let $\tilde{\psi}$ be its inverse, so that

$$\tilde{\psi}(\tilde{x}) = x_1, \qquad \tilde{\psi}(\tilde{y}) = x_2, \qquad \tilde{\psi}(\tilde{z}) = x_3, \qquad \tilde{\psi}(\tilde{t}) = x_4.$$

As in **Step 2**, we write X,Y,z,t for $x_1,\dots,x_4$ and for the coordinates of $\mathbb{A}^4$, with the diagonal action of G in which they have the characters of $x_1,\dots,x_4$, and let $\Theta := \tilde{\Theta}(X,Y)$. Then $\tilde{v} := \tilde{\psi}(v_2)^{-1}$ is a G-invariant unit of $\mathfrak{R}$ and

$$\hat{f}' := \tilde{v}\tilde{\psi}(\hat{f}) = \Theta + \tilde{g}(z,t).$$

Since W is smooth outside o near o, the ideal of $\mathfrak{R}$ generated by $\hat{f}$ and its partial derivatives contains a power of $\mathfrak{m}$ by the Jacobian criterion. By the chain rule, the ideal J' generated by $\hat{f}'$ and its partial derivatives is the image of that ideal under $\tilde{\psi}$, since $\tilde{v}$ is a unit. Hence there is a positive integer b with

$$\mathfrak{m}^b \subset \mathfrak{m}J' \subset (\hat{f}') + \mathfrak{m}\Big(\frac{\partial\hat{f}'}{\partial X}, \frac{\partial\hat{f}'}{\partial Y}, \frac{\partial\hat{f}'}{\partial z}, \frac{\partial\hat{f}'}{\partial t}\Big).$$

By [Liu26, Lemma 7.1], if $F \in \mathfrak{R}$ is G-semi-invariant, $S \subset \{1,\dots,4\}$, and

$$\mathfrak{m}^b \subset (F) + \mathfrak{m}\Big(\frac{\partial F}{\partial x_i} \;\Big|\; i \in S\Big),$$

then for any integer $L > 2b$ there exist a G-equivariant automorphism φ of $\mathfrak{R}$ and a G-invariant unit v' of $\mathfrak{R}$ with $v'\varphi(F) = F_{<L}$ and $\varphi(x_i) - x_i \in \mathfrak{m}^{L-b+1}$ for any $i \in S$, where $F_{<L}$ is the sum of the terms of F of degree smaller than L.

We apply Construction-Theorem 4.13 as in **Step 3**, with the eigenparameters $x_1,\dots,x_4$ above and with $\tilde{\psi}$ and $\tilde{v}$ in place of ψ and v in (vi). Let N_1 be its integer, fix an integer $N \geq N_1$ with $N > \operatorname{mult}_{P_i} h_k$ for any i and k, let L_1 be the integer for N, and fix an integer

$$L \geq \max\{L_1, 2b+1, N+b-1, L'+b\}.$$

Here $L \geq L_1$ is the condition of Construction-Theorem 4.13, $L \geq 2b+1$ is the hypothesis $L > 2b$ of [Liu26, Lemma 7.1], $L \geq N+b-1$ gives $L-b+1 \geq N$, which is used for $\Psi(\hat{h}_k) - h_{k,0} \in \mathfrak{m}^N$ below, and $L \geq L'+b$ gives $L-b+1 > L'$, which is used for the congruences of $\Psi(\tilde{x}),\dots,\Psi(\tilde{t})$ below. Let $h_{k,0}$ be the sum of the terms of $\tilde{\psi}(\hat{h}_k)$ of degree smaller than N, g_0 the sum of the terms of $\tilde{g}$ of degree smaller than L, $f_0 := \Theta + g_0$, and $V := (f_0 = 0) \subset \mathbb{A}^4$. Then $(V,(h_{k,0})_k)$ is a numerical admission of $(W,(h_k)_k)$ with the group G, V is an isolated Gorenstein terminal germ at 0 by Construction-Theorem 4.13(3), and $g_0 \in (z,t)^3$. Moreover $g_0 \neq 0$, since otherwise $V = (\Theta = 0)$ is singular along the (z,t)-plane, whereas V has an isolated singularity at 0. If $\tilde{\Theta} = \tilde{x}\tilde{y}$, then $V = (XY + g_0(z,t) = 0)$, which is the alternative (1)(i) of Theorem 8.11. If $\tilde{\Theta} = \tilde{x}^2 + \tilde{y}^2$, then $\chi_X = \chi_{\tilde{x}} \neq \chi_{\tilde{y}} = \chi_Y$ and $V = (X^2 + Y^2 + g_0(z,t) = 0)$, which is the alternative (1)(ii) of Theorem 8.11.

By [Liu26, Lemma 7.1], applied to $F = \hat{f}'$ with $S = \{1,2,3,4\}$, there exist a G-equivariant automorphism φ of $\mathfrak{R}$ and a G-invariant unit v' of $\mathfrak{R}$ such that

$$v'\varphi(\hat{f}') = f_0 \quad \text{and} \quad \varphi(x_i) - x_i \in \mathfrak{m}^{L-b+1} \quad \text{for } i = 1,\dots,4,$$

since $\hat{f}'_{<L} = \Theta + g_0 = f_0$ as $L > 2$. Let $\Psi := \varphi \circ \tilde{\psi}$. Then $\Psi(\hat{f}) = \varphi(\tilde{v})^{-1}v'^{-1}f_0$, so Ψ maps the ideal $(\hat{f})$ onto (f_0) and induces a G-equivariant isomorphism

$$\iota : \widehat{\mathcal{O}}_{W,o} = \mathfrak{R}/(\hat{f}) \to \mathfrak{R}/(f_0) = \widehat{\mathcal{O}}_{V,0}.$$

Since ι is a G-equivariant isomorphism of $\mathbb{C}$-algebras, it maps each formal prime divisor with center the closed point to one with isomorphic residue field, equal decomposition and inertia groups, and equal $a(\cdot,\cdot,0)$. By Lemma 3.6, the formal prime divisors over $\operatorname{Spec}\widehat{\mathcal{O}}_{W,o}$, resp. $\operatorname{Spec}\widehat{\mathcal{O}}_{V,0}$, with center the closed point correspond to the prime divisors over W with center o, resp. over V with center 0. Since $L - b + 1 \geq N$, the identity

$$\varphi(qq') - qq' = \varphi(q)\big(\varphi(q') - q'\big) + \big(\varphi(q) - q\big)q'$$

gives $\varphi(q) - q \in \mathfrak{m}^N$ for any polynomial q in $x_1,\dots,x_4$, hence for any $q \in \mathfrak{R}$, since $\varphi(\mathfrak{m}^N) = \mathfrak{m}^N$. Thus $\Psi(\hat{h}_k) - h_{k,0} \in \mathfrak{m}^N$. Since $N > \operatorname{mult}_{P_i} h_k$, the order of $h_{k,0}$ along the image of P_i under ι is $\operatorname{mult}_{P_i} h_k$. Hence we may take ι as the isomorphism of Construction-Theorem 4.13(2). Since $L - b + 1 > L'$,

$$\Psi(\tilde{x}) - X,\ \Psi(\tilde{y}) - Y,\ \Psi(\tilde{z}) - z,\ \Psi(\tilde{t}) - t \in \mathfrak{m}^{L'}.$$

Thus ι maps $\tilde{x},\tilde{y},\tilde{z},\tilde{t}$ to X,Y,z,t modulo the L'-th power of the maximal ideal.

Let $w \in \mathbb{Z}^4_{>0}$ be a primitive vector as in the statement. For an integer k, let $I_k \subset \mathfrak{R}$ be the ideal generated by the monomials in $\tilde{x},\tilde{y},\tilde{z},\tilde{t}$ of w-weight at least k, and $J_k \subset \mathfrak{R}$ the ideal generated by the monomials in X,Y,z,t of w-weight at least k. Since $w_j \geq 1$ for any j, $\mathfrak{m}^{L'} \subset I_{L'} \cap J_{L'}$. Write $\tilde{x}_1,\dots,\tilde{x}_4$ for $\tilde{x},\tilde{y},\tilde{z},\tilde{t}$, and $\Psi(\tilde{x}_j) = x_j + r_j$ with $r_j \in \mathfrak{m}^{L'}$. For $a \in \mathbb{Z}^4_{\geq 0}$, let $\tilde{x}^a := \prod_j \tilde{x}_j^{a_j}$ and $x^a := \prod_j x_j^{a_j}$. For a non-zero series in $\tilde{x},\tilde{y},\tilde{z},\tilde{t}$, its w-weight wt_w is the minimum of $\operatorname{wt}_w(x^a)$ over the monomials $\tilde{x}^a$ occurring in it. In the expansion of

$$\Psi(\tilde{x}^a) = \prod_j (x_j + r_j)^{a_j},$$

each term other than x^a contains a factor r_j in place of a factor x_j for some j, so it is an element of $J_{\operatorname{wt}_w(x^a) - w_j + L'}$, and $\operatorname{wt}_w(x^a) - w_j + L' > \operatorname{wt}_w(x^a)$ since $\max_j w_j < L'$. Thus $\Psi(\tilde{x}^a) \in x^a + J_{\operatorname{wt}_w(x^a)+1}$, and $\Psi(I_k) \subset J_k$ for any k. Since $\Psi^{-1}(x_j) = \tilde{x}_j - \Psi^{-1}(r_j)$ and $\Psi^{-1}(r_j) \in \mathfrak{m}^{L'}$, the argument above with Ψ^{-1} in place of Ψ gives $\Psi^{-1}(J_k) \subset I_k$. Hence $\Psi(I_k) = J_k$ for any k, and $\Psi(e) \in e(X,Y,z,t) + J_{\operatorname{wt}_w(e)+1}$.

Since f_0 is a unit multiple of $\Psi(\hat{f}) = \Psi(v_2)\Psi(e)$, the lowest w-weighted part of f_0 is a non-zero constant multiple of that of e with $\tilde{x},\tilde{y},\tilde{z},\tilde{t}$ replaced by X,Y,z,t. Hence it is irreducible and not divisible by any coordinate, which is the hypothesis of Lemma 4.12 for $V \subset \mathbb{A}^4$ and w. Let

$$\mathcal{B}_I := \operatorname{Proj} \bigoplus_{k\geq 0} I_k \quad \text{and} \quad \mathcal{B}_J := \operatorname{Proj} \bigoplus_{k\geq 0} J_k.$$

By Lemma 4.12 for $V \subset \mathbb{A}^4$ and w, the strict transform of V in the w-weighted blow-up of $\mathbb{A}^4$ intersects the exceptional divisor in a single prime divisor E_V. We call E_V the *prime divisor over V defined by the weights w in X,Y,z,t*. The ideal J_k is generated by the ideal of $\mathbb{C}[X,Y,z,t]$ generated by the monomials in X,Y,z,t of w-weight at least k, so $\mathcal{B}_J$ is the base change of the w-weighted blow-up of $\mathbb{A}^4$ along the flat morphism $\operatorname{Spec}\mathfrak{R} \to \mathbb{A}^4$. Strict transforms commute with flat base change, the exceptional divisor maps to 0, and the fiber of $\operatorname{Spec}\mathfrak{R} \to \mathbb{A}^4$ over 0 is a single reduced point. Hence the strict transform of $\operatorname{Spec}\mathfrak{R}/(f_0)$ in $\mathcal{B}_J$ is the base change of the strict transform of V, it intersects the exceptional divisor in a single prime divisor $\widehat{E}_V$, the preimage of E_V, and a uniformizer at the generic point of E_V is a uniformizer at the generic point of $\widehat{E}_V$. Thus $\widehat{E}_V$ restricts to E_V. Since Ψ maps I_k onto J_k for any k and $(\hat{f})$ onto (f_0), it induces an

isomorphism $\mathcal{B}_I \to \mathcal{B}_J$ which maps the strict transform of $\operatorname{Spec}\mathfrak{R}/(\hat{f})$ and the exceptional divisor in $\mathcal{B}_I$ to those in $\mathcal{B}_J$. Thus the strict transform of $\operatorname{Spec}\mathfrak{R}/(\hat{f})$ in $\mathcal{B}_I$ intersects the exceptional divisor in a single prime divisor $\widehat{E}_W$, which is a formal prime divisor over $\operatorname{Spec}\widehat{\mathcal{O}}_{W,o}$ with center the closed point, and ι maps $\widehat{E}_W$ to $\widehat{E}_V$. The prime divisor over W defined by the weights w in $\tilde{x}, \tilde{y}, \tilde{z}, \tilde{t}$ is, by definition, the restriction of $\widehat{E}_W$ to W, a prime divisor over W with center o by Lemma 3.6. Hence, if it is one of $P_1, \dots, P_l$, then ι maps it to E_V. □

B.11. **Augmentations from lc toric quotients.** In this subsection, we prove Lemma 8.12. The difference between the log discrepancies of (Y^ϵ, B^ϵ) and of the push-forward (Q_0, B_0) is $|\delta|$ times a non-negative concave piecewise linear function ψ vanishing on both rays of the cone of Q_0. The function ψ is a rational multiple of the order function of a monomial ideal, which is primary at the origin unless the function is zero, in which case the ideal is (1). On the smooth toric cover of Q_0, a non-klt pair with positive minimum has exactly one smooth coefficient-one curve through the origin.

Proof of Lemma 8.12. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the toric notation and write the log discrepancies of (V, D_V) in terms of the torus of Q_0.

Write x_1, x_2, x_3, x_4 for x, y, z, t and w_1, w_2, w_3, w_4 for u, v, c, d, so that w_i is the $\mathbf{T}_{(w)}$-weight of x_i. Let $T_V := V \cap (xyzt \neq 0)$ be the torus of V. The cocharacter lattice of T_V is $\Lambda := \{(X, Y, Z, T) \in \mathbb{Z}^4 \mid X + Y = aZ + bT\}$, the cone of V is $\sigma = \Lambda_{\mathbb{R}} \cap \mathbb{R}^4_{\geq 0}$, and w is a primitive element of Λ. Let $M := \operatorname{Hom}(\Lambda, \mathbb{Z})$, for $m \in M$ let e^m be the corresponding character of T_V, and let $m_i \in M$ be the character of x_i. Then (m_1, m_3, m_4) is a basis of M, and $m_1 + m_2 = am_3 + bm_4$. Let $\mathcal{S} \subset M$ be the semigroup generated by $m_1, \dots, m_4$, so that $\mathbb{C}[V] = \mathbb{C}[\mathcal{S}]$. Since σ is the set of $n \in \Lambda_{\mathbb{R}}$ with $\langle m_i, n\rangle \geq 0$ for $i = 1, \dots, 4$, the cone generated by $\mathcal{S}$ is $\sigma^\vee$, and $\mathcal{S}$ generates the group M. The hypersurface V is Cohen–Macaulay, and by the Jacobian criterion its singular locus is contained in $V \cap (x = y = zt = 0)$, which has codimension two in V, so V is normal by Serre's criterion. Thus $\mathcal{S}$ is saturated in M, that is, $\mathcal{S} = \sigma^\vee \cap M$, and $\mathbb{C}[V] = \mathbb{C}[\sigma^\vee \cap M]$ is the coordinate ring of a normal affine toric variety [CLS11, Theorem 1.3.5].

Let $\Lambda_0 := \Lambda/\mathbb{Z}w$, $M_0 := w^\perp \cap M = \operatorname{Hom}(\Lambda_0, \mathbb{Z})$, and $\bar{\sigma} \subset \Lambda_{0,\mathbb{R}}$ the image of σ. If $\bar{\sigma}$ contains $\pm\bar{n}$ for some $\bar{n} \neq 0$, then there exist $n_1, n_2 \in \sigma$ with images $\bar{n}, -\bar{n}$, so $n_1 + n_2 \in \sigma \cap \mathbb{R}w = \{0\}$. Since $\sigma \subset \mathbb{R}^4_{\geq 0}$, we obtain $n_1 = n_2 = 0$, a contradiction. Thus $\bar{\sigma}$ is a strongly convex two-dimensional rational cone. Let $r_1, r_2 \in \Lambda_0$ be the primitive generators of the rays of $\bar{\sigma}$. Since

$$\mathbb{C}[V]_0 = \mathbb{C}[\sigma^\vee \cap M_0] = \mathbb{C}[\bar{\sigma}^\vee \cap M_0],$$

the surface Q_0 is the normal affine toric surface of $\bar{\sigma}$ [CLS11, Theorem 1.3.5]. Let $\bar{T}$ be the torus of Q_0, $o_0 \in Q_0$ the torus-fixed point, and S_1, S_2 the torus-invariant prime divisors of Q_0 corresponding to r_1, r_2. For any prime divisor P over Q_0, let $n_P \in \bar{\sigma} \cap \Lambda_0$ be the vector with $\langle m, n_P\rangle = \operatorname{mult}_P \operatorname{div} \mathrm{e}^m$ for $m \in M_0$. Then $n_{S_k} = r_k$ for $k = 1, 2$, and $n_S = 0$ for any prime divisor S on Q_0 other than S_1, S_2, since S intersects $\bar{T}$.

The graded rings $\bigoplus_{k \geq 0} \mathbb{C}[V]_{\pm k}$ are finitely generated, so $Y^\pm \to Q_0$ is projective [CLS11, Theorem 7.0.9], and passing to a Veronese subring generated in degree one does not change Proj [CLS11, Lemma 14.2.8]. Any element of $\mathbb{C}[V]_{\pm k}$ with $k > 0$ is a linear combination of monomials, each of which is divisible by some x_i with $\pm w_i > 0$. Thus the open subsets $(x_i \neq 0) \subset Y^\pm$ with $\pm w_i > 0$ cover $Y^\pm$. Both index sets $\{i \mid w_i > 0\}$ and $\{i \mid w_i < 0\}$ are non-empty, since otherwise $w \in \sigma$ or $-w \in \sigma$.

Since w is primitive, there exists $m_\chi \in M$ with $\langle m_\chi, w\rangle = 1$. Let $\chi := \mathrm{e}^{m_\chi}$. Then $M = M_0 \oplus \mathbb{Z}m_\chi$, so the quotient morphism $V \to Q_0$ and χ induce an isomorphism $T_V \to \bar{T} \times \mathbb{G}_m$, and V is birational to $Q_0 \times \mathbb{A}^1$. For $i = 1, \dots, 4$ let $m_i^0 := m_i - w_i m_\chi \in M_0$, so that

$$x_i = \chi^{w_i} \mathrm{e}^{m_i^0}.$$

Under the isomorphism $T_V \to \bar{T} \times \mathbb{G}_m$, the group $\mathbf{T}_{(w)}$ acts trivially on $\bar{T}$ and multiplies χ by the coordinate of $\mathbf{T}_{(w)}$. Thus the $\mathbf{T}_{(w)}$-invariant rational function $\chi^{-\kappa_j} h_j$ on V is the pullback of a rational function $\bar{h}_j$ on Q_0, and $h_j = \chi^{\kappa_j} \bar{h}_j$. We use without further comment the standard facts that the chart $(x_i \neq 0) \subset Y^{\pm}$, the spectrum of the degree-zero part of $\mathbb{C}[V][x_i^{-1}]$, is $(V \cap (x_i = 1))/\mu_{|w_i|}$, that the isomorphism $T_V \to \bar{T} \times \mathbb{G}_m$ gives $(T_V \cap (x_i = 1))/\mu_{|w_i|} \cong \bar{T}$, so $Y^{\pm} \to Q_0$ is birational, and that $B^{\epsilon} \geq 0$ [Liu26, Lemma 2.6(0)].

Choose a basis m', m'' of M_0 such that

$$d\log x \wedge d\log z \wedge d\log t = d\log \chi \wedge d\log \mathrm{e}^{m'} \wedge d\log \mathrm{e}^{m''}.$$

Such a basis exists: since $M = M_0 \oplus \mathbb{Z} m_\chi$, for any basis m', m'' of M_0 the triple (m_χ, m', m'') is a basis of M, as is (m_1, m_3, m_4), so

$$d\log x \wedge d\log z \wedge d\log t = \pm d\log \chi \wedge d\log \mathrm{e}^{m'} \wedge d\log \mathrm{e}^{m''},$$

and exchanging m' and m'' if necessary we obtain the equality. Let

$$\eta := \mathrm{e}^{m_3^0 + m_4^0} d\log \mathrm{e}^{m'} \wedge d\log \mathrm{e}^{m''},$$

a rational 2-form on Q_0. The residue form $\Omega := dx \wedge dz \wedge dt/x$ of $dx \wedge dy \wedge dz \wedge dt/(xy + z^a t^b)$ is a nowhere vanishing section of ω_V. On T_V we have $\Omega = ztd\log x \wedge d\log z \wedge d\log t$, and $zt = \chi^{c+d} \mathrm{e}^{m_3^0 + m_4^0}$. Thus

$$\Omega = \chi^{c+d} d\log \chi \wedge \eta. \tag{B.86}$$

For a prime divisor F over V (resp. Q_0) and a non-zero rational 3-form (resp. 2-form) ϖ on V (resp. Q_0), let $\mathrm{mult}_F \operatorname{div} \varpi$ be the coefficient along F of the divisor of the pullback of ϖ to a higher model on which F is a divisor. Since $K_V = \operatorname{div} \Omega = 0$, for any prime divisor F over V,

$$a(F, V, D_V) = 1 + \mathrm{mult}_F \operatorname{div} \Omega - \sum_j e_j \,\mathrm{mult}_F \operatorname{div} h_j. \tag{B.87}$$

For a prime divisor P over Q_0, let

$$A(P) := 1 + \mathrm{mult}_P \operatorname{div} \eta - \sum_j e_j \,\mathrm{mult}_P \operatorname{div} \bar{h}_j,$$

and for $n \in \Lambda_{0,\mathbb{R}}$ let

$$\varphi_+(n) := \max_{w_i > 0} \Big(- \frac{\langle m_i^0, n \rangle}{w_i} \Big), \qquad \varphi_-(n) := \min_{w_i < 0} \Big(- \frac{\langle m_i^0, n \rangle}{w_i} \Big).$$

The functions $\varphi_\pm$ are rational, piecewise linear, and positively homogeneous, φ_+ is convex, and φ_- is concave.

Step 2. In this step we prove that for any prime divisor P over Q_0,

$$a(P, Y^{\epsilon}, B^{\epsilon}) = A(P) - \delta \varphi_\epsilon(n_P). \tag{B.88}$$

Since $Y^{\epsilon} \to Q_0$ is projective and birational, P is a prime divisor over Y^{ϵ}. By **Step 1**, the center of P on Y^{ϵ} intersects $Y_i^{\epsilon} := (x_i \neq 0) \subset Y^{\epsilon}$ for some i with $\epsilon w_i > 0$. For any k with $\epsilon w_k > 0$, the function $x_k^{|w_i|} x_i^{-|w_k|}$ has $\mathbf{T}_{(w)}$-weight 0, so it is regular on Y_i^{ϵ}, and it equals $\mathrm{e}^{|w_i| m_k^0 - |w_k| m_i^0}$. Thus

$$|w_i| \langle m_k^0, n_P \rangle \geq |w_k| \langle m_i^0, n_P \rangle$$

for any such k, that is,

$$- \frac{\langle m_i^0, n_P \rangle}{w_i} = \varphi_\epsilon(n_P). \tag{B.89}$$

Let $V_i := V \cap (x_i = 1)$, $\tau_i : V_i \to V_i/\mu_{|w_i|} = Y_i^{\epsilon}$ the quotient map, and $\chi_i := \chi|_{V_i}$. Since $d\log x_i = w_i d\log \chi + d\log \mathrm{e}^{m_i^0}$ and $d\log \mathrm{e}^{m_i^0} \wedge \eta = 0$ (a 3-form on the surface $\bar{T}$), (B.86) implies

$$\Omega = \frac{\chi^{c+d}}{w_i x_i} dx_i \wedge \eta.$$

Since V_i is the Cartier divisor $(x_i-1=0)$ on V, by adjunction the Poincaré residue $\Omega_i := \chi_i^{c+d}\eta|_{V_i}$ generates ω_{V_i} up to the constant w_i^{-1}. Each $h_j|_{V_i} = \chi_i^{\kappa_j}\bar{h}_j$ is non-zero, since otherwise h_j vanishes on the dense subset $\mathbf{T}_{(w)}\cdot V_i = V\cap(x_i\neq 0)$. Let P' be a prime divisor over V_i over P, and ϱ' the ramification index of P' over P, so that

$$\operatorname{mult}_{P'}\operatorname{div}\zeta = \varrho'\operatorname{mult}_P\operatorname{div}\zeta$$

for any non-zero rational function ζ on Q_0. Since $c+d-\sum_j e_j\kappa_j = -\delta$,

$$a(P', V_i, D_V|_{V_i}) = 1+\operatorname{mult}_{P'}\operatorname{div}\eta - \sum_j e_j\operatorname{mult}_{P'}\operatorname{div}\bar{h}_j - \delta\operatorname{mult}_{P'}\operatorname{div}\chi_i.$$

By the Hurwitz formula for τ_i along P',

$$1+\operatorname{mult}_{P'}\operatorname{div}\eta = \varrho'(1+\operatorname{mult}_P\operatorname{div}\eta).$$

Since $\chi_i^{w_i}\mathrm{e}^{m_i^0} = x_i|_{V_i} = 1$, (B.89) implies

$$\operatorname{mult}_{P'}\operatorname{div}\chi_i = -\varrho'\langle m_i^0, n_P\rangle/w_i = \varrho'\varphi_\epsilon(n_P).$$

Thus

$$a(P', V_i, D_V|_{V_i}) = \varrho'\big(A(P)-\delta\varphi_\epsilon(n_P)\big).$$

By the definition of B^ϵ and [Liu26, Lemma 2.6(2)], $a(P', V_i, D_V|_{V_i}) = \varrho' a(P, Y^\epsilon, B^\epsilon)$, which implies (B.88).

Step 3. In this step we prove (3). **Steps 1–3** do not use the assumptions that (Y, B_Y) is lc but not klt near Σ and that $\beta>0$. Thus (3) holds without them.

Let φ_{lin} be the linear function on $\Lambda_{0,\mathbb{R}}$ with $\varphi_{\mathrm{lin}}(r_k) = \varphi_\epsilon(r_k)$ for $k=1,2$, and $m_{\mathrm{lin}}\in M_0\otimes\mathbb{Q}$ the element with $\langle m_{\mathrm{lin}}, \cdot\rangle = \varphi_{\mathrm{lin}}$. Let

$$D_{\mathrm{lin}} := \varphi_{\mathrm{lin}}(r_1)S_1 + \varphi_{\mathrm{lin}}(r_2)S_2,$$

so that $lD_{\mathrm{lin}} = \operatorname{div}\mathrm{e}^{lm_{\mathrm{lin}}}$ for any positive integer l with $lm_{\mathrm{lin}}\in M_0$. Then $\operatorname{mult}_P D_{\mathrm{lin}} = \varphi_{\mathrm{lin}}(n_P)$ for any prime divisor P over Q_0. Let

$$B_0' := -\operatorname{div}\eta + \sum_j e_j\operatorname{div}\bar{h}_j + \delta D_{\mathrm{lin}}.$$

Since $K_{Q_0} = \operatorname{div}\eta$ and D_{lin} is $\mathbb{Q}$-Cartier, $K_{Q_0}+B_0'$ is $\mathbb{R}$-Cartier, and

$$a(P, Q_0, B_0') = A(P) - \delta\varphi_{\mathrm{lin}}(n_P)$$

for any prime divisor P over Q_0. For a prime divisor S on Q_0, the coefficient of B_0 along S is the coefficient of B^ϵ along the strict transform of S, which equals $1-A(S)+\delta\varphi_\epsilon(n_S)$ by (B.88). The coefficient of B_0' along S is $1-A(S)+\delta\varphi_{\mathrm{lin}}(n_S)$. Since $n_S\in\{0, r_1, r_2\}$, the coefficients of B_0 and B_0' along S are equal. Thus $B_0 = B_0'$, and by (B.88), for any prime divisor P over Q_0,

$$a(P, Q_0, B_0) - a(P, Y^\epsilon, B^\epsilon) = \delta\big(\varphi_\epsilon(n_P)-\varphi_{\mathrm{lin}}(n_P)\big) = |\delta|\psi(n_P), \tag{B.90}$$

where $\psi := \varphi_{\mathrm{lin}} - \varphi_+$ if $\epsilon=+$, and $\psi := \varphi_- - \varphi_{\mathrm{lin}}$ if $\epsilon=-$. If $\delta=0$, then (B.90) implies (3) with $I=(1)$ and $N=1$. From now on in this step we assume that $\delta\neq 0$, so ϵ is the sign of $-\delta$.

For i with $\epsilon w_i>0$ let

$$\theta_i := \epsilon(m_{\mathrm{lin}} + m_i^0/w_i)\in M_0\otimes\mathbb{Q}.$$

Then

$$\psi(n) = \min_{\epsilon w_i>0}\langle\theta_i, n\rangle.$$

Any $n\in\bar{\sigma}$ can be written as $n = s_1r_1+s_2r_2$ with $s_1, s_2\geq 0$. If $\epsilon=+$, then, since φ_+ is convex and positively homogeneous,

$$\varphi_+(n)\leq s_1\varphi_+(r_1)+s_2\varphi_+(r_2) = \varphi_{\mathrm{lin}}(n),$$

and if $\epsilon=-$, then similarly $\varphi_-(n)\geq\varphi_{\mathrm{lin}}(n)$. Thus $\psi\geq 0$ on $\bar{\sigma}$, and $\psi(r_1)=\psi(r_2)=0$. If m_χ is replaced by $m_\chi+m$ with $m\in M_0$, then m_i^0/w_i is replaced by $m_i^0/w_i - m$, and $\varphi_\pm$ and φ_{lin} are

replaced by $\varphi_\pm + m$ and $\varphi_{\mathrm{lin}} + m$. Hence θ_i and ψ depend only on a, b, w, and ϵ. Let N be the least positive integer with $N\theta_i \in M_0$ for any i with $\epsilon w_i > 0$. Let I be the ideal of $\mathbb{C}[Q_0]$ spanned by the monomials e^m with $m \in M_0$ and $\langle m, n\rangle \geq N\psi(n)$ for any $n \in \bar{\sigma}$. Then I depends only on a, b, w, and the sign of δ. Since $\langle \theta_i, \cdot\rangle \geq \psi \geq 0$ on $\bar{\sigma}$, we have $\mathrm{e}^{N\theta_i} \in I$. For any prime divisor P over Q_0, any non-zero $\zeta \in I$ satisfies $\mathrm{mult}_P \operatorname{div} \zeta \geq N\psi(n_P)$, and

$$\mathrm{mult}_P \operatorname{div} \mathrm{e}^{N\theta_i} = N\langle \theta_i, n_P\rangle.$$

Thus

$$\mathrm{mult}_P I = N\psi(n_P). \tag{B.91}$$

Let $s := |\delta|/N$. By (B.90) and (B.91),

$$a(P, Y^\epsilon, B^\epsilon) = a(P, Q_0, B_0) - s\,\mathrm{mult}_P I = a(P, Q_0, B_0, I^s)$$

for any prime divisor P over Q_0.

If ψ vanishes at some interior point n_0 of $\bar{\sigma}$, let n be in the interior and choose $\nu > 0$ with $n_0 - \nu n \notin \bar{\sigma}$, which is possible since $\bar{\sigma}$ is closed and strongly convex. The ray from νn through n_0 leaves $\bar{\sigma}$ at a point $n' \in \mathbb{R}_{\geq 0} r_1 \cup \mathbb{R}_{\geq 0} r_2$ beyond n_0, so $n_0 = t_0 \nu n + (1 - t_0) n'$ with $t_0 \in (0, 1)$, and the concavity and positive homogeneity of ψ imply $0 = \psi(n_0) \geq t_0 \nu \psi(n) \geq 0$. Thus $\psi \neq 0$ implies $\psi > 0$ in the interior of $\bar{\sigma}$. If $\psi = 0$, then $\mathrm{e}^0 \in I$, so $I = (1)$. If $\psi \neq 0$, then $1 \notin I$, so I vanishes at o_0. Moreover, since $\psi(r_k) = 0 = \min_i \langle \theta_i, r_k\rangle$, some $\mathrm{e}^{N\theta_i} \in I$ satisfies $\langle \theta_i, r_k\rangle = 0$, so it does not vanish at the general point of S_k. Hence the cosupport of I, a closed torus-invariant subset of Q_0, is $\{o_0\}$, and I is primary at o_0.

Since G commutes with $\mathbf{T}_{(w)}$ and preserves D_V, the group H acts on Y^ϵ and on Q_0 and preserves B^ϵ and B_0. Since G acts monomially on $\{x, y\}$ and diagonally on z, t, any $g \in G$ maps each x_i to a non-zero multiple of some $x_{i'}$. Thus g maps each character e^m of T_V to a non-zero multiple of $\mathrm{e}^{A_g(m)}$, where A_g is the lattice automorphism of M that permutes $m_1, \dots, m_4$, fixing m_3, m_4 and fixing or exchanging m_1, m_2. Since g commutes with $\mathbf{T}_{(w)}$, A_g preserves the weight $\langle \cdot, w\rangle$, so it preserves M_0 and $\bar{\sigma}^\vee \cap M_0 = \sigma^\vee \cap M_0$. For the image $\bar{g} \in H$ of g, let $A_{\bar{g}} := A_g|_{M_0}$. Then $\bar{g}$ maps each e^m with $m \in M_0$ to a non-zero multiple of $\mathrm{e}^{A_{\bar{g}}(m)}$, and $n_{\bar{g}(P)} = (A_{\bar{g}}^{-1})^\vee(n_P)$. By (B.90), $\psi(n_{\bar{g}(P)}) = \psi(n_P)$ for any P. Since any rational ray of $\bar{\sigma}$ is spanned by n_P for a torus-invariant prime divisor P over Q_0, the function ψ is invariant under the action of H on $\Lambda_{0,\mathbb{R}}$ by continuity. Thus I is H-invariant. This proves (3).

Step 4. In this step we pass to a smooth toric cover of Q_0, find a coefficient-one curve D through the origin, and compute β by Lemma 5.6.

Let $\Lambda' := \mathbb{Z}r_1 + \mathbb{Z}r_2 \subset \Lambda_0$ and $K_0 := \Lambda_0/\Lambda'$. By [CLS11, Example 1.3.20], the affine toric variety of $\bar{\sigma}$ with respect to Λ' is $\mathbb{A}^2$ with coordinates p_1, p_2, and the toric morphism $\pi_0 : \mathbb{A}^2 \to Q_0$ induced by $\Lambda' \subset \Lambda_0$ is the quotient by K_0. Since r_1, r_2 are primitive in Λ_0, no non-trivial element of K_0 fixes a coordinate axis pointwise. Thus π_0 is étale in codimension one, and $\pi_0^{-1}(o_0) = \{0\}$. For any $\bar{g} \in H$, the dual $A_{\bar{g}}^\vee$ of the lattice automorphism $A_{\bar{g}}$ of M_0 from **Step 3** is a lattice automorphism of Λ_0 preserving $\bar{\sigma}$, so it permutes r_1, r_2 and preserves Λ'. Since $\bar{g}$ maps each e^m with $m \in M_0$ to a non-zero multiple of $\mathrm{e}^{A_{\bar{g}}(m)}$, $\bar{g}$ lifts to a monomial linear automorphism of $\mathbb{A}^2$, and the group $\widetilde{H} \subset GL_2(\mathbb{C})$ of all lifts of elements of H fits into an exact sequence

$$1 \to K_0 \to \widetilde{H} \to H \to 1.$$

Let $B' := \pi_0^* B_0$ and $\mathfrak{b} := I \cdot \mathcal{O}_{\mathbb{A}^2}$. Then $B' \geq 0$, $K_{\mathbb{A}^2} + B' = \pi_0^*(K_{Q_0} + B_0)$, and $\mathfrak{b}$ is an $\widetilde{H}$-invariant monomial ideal equal to (1) or primary at 0.

Let P be a prime divisor over Q_0, and $\widetilde{P}$ a prime divisor over $\mathbb{A}^2$ over P. Then

$$\mathrm{mult}_{\widetilde{P}} \mathfrak{b} = |I_{K_0}(\widetilde{P})|\,\mathrm{mult}_P I$$

by [Liu26, Lemma 2.6(1)(2)]. The group H acts faithfully on Q_0, as [Liu26, Lemma 2.6] requires: if $g \in G$ acts trivially on $\mathbb{C}[V]_0$, then $A_g|_{M_0}$ is the identity, so g does not exchange x and y, since otherwise $w_1 = w_2$ and A_g maps $m_1 - m_2 \in M_0$ to $m_2 - m_1$, and a diagonal element acting

trivially on the characters in M_0 is an element of $\mathbf{T}_{(w)}$. By [Liu26, Lemma 2.6(2)] for the quotient π_0 by K_0 (where $\mathfrak{b} = \pi_0^* I$), by [Liu26, Lemma 2.6(1)(5)] and the multiplicativity of ramification indices in $\mathbb{A}^2 \to Q_0 \to Q_0/H = \mathbb{A}^2/\widetilde{H}$, which imply

$$|I_{\widetilde{H}}(\widetilde{P})| = |I_{K_0}(\widetilde{P})| \cdot |I_H(P)|,$$

and by (3),

$$\frac{a(\widetilde{P}, \mathbb{A}^2, B', \mathfrak{b}^s)}{|I_{\widetilde{H}}(\widetilde{P})|} = \frac{a(P, Q_0, B_0, I^s)}{|I_H(P)|} = \frac{a(P, Y^\epsilon, B^\epsilon)}{|I_H(P)|}. \tag{B.92}$$

By [Liu26, Lemma 2.6(2)], the right hand side is the log discrepancy of (Y, B_Y) along the image of P. The center of $\widetilde{P}$ on $\mathbb{A}^2$ is 0 if and only if the center of P on Y^ϵ is contained in the fiber over o_0, and the preimage of Σ in Y^ϵ is the fiber of Y^ϵ over o_0, since H fixes o_0. Since the non-klt locus of (Y, B_Y) is closed, the assumptions on (Y, B_Y) imply the following.

(a) The pair $(\mathbb{A}^2, B', \mathfrak{b}^s)$ is lc near 0, and there exists a prime divisor $\widetilde{P}_1$ over $\mathbb{A}^2$ whose center contains 0 with $a(\widetilde{P}_1, \mathbb{A}^2, B', \mathfrak{b}^s) = 0$.

(b) β is the minimum of

$$\frac{a(\widetilde{P}, \mathbb{A}^2, B', \mathfrak{b}^s)}{|I_{\widetilde{H}}(\widetilde{P})|}$$

over the prime divisors $\widetilde{P}$ over $\mathbb{A}^2$ with center 0.

By (b) and $\beta > 0$, the center of $\widetilde{P}_1$ is not 0, so $\widetilde{P}_1$ is a prime divisor D on $\mathbb{A}^2$ through 0. Since $\mathfrak{b}$ is (1) or primary at 0, $\operatorname{mult}_D \mathfrak{b} = 0$, so $\operatorname{mult}_D B' = 1$. If $B' \geq D + D''$ for another prime divisor D'' through 0, or if D is singular at 0, then $\operatorname{mult}_0 B' \geq 2$, and the exceptional divisor E_0 of the blow-up of 0 satisfies

$$a(E_0, \mathbb{A}^2, B', \mathfrak{b}^s) \leq 2 - \operatorname{mult}_0 B' \leq 0,$$

which contradicts (b). Thus D is the unique prime divisor through 0 with coefficient one in B', and D is smooth at 0. In particular, D is $\widetilde{H}$-invariant.

We replace $\mathfrak{b}^s$ by an $\mathbb{R}$-divisor (cf. [Laz04, Proposition 9.2.28]). Fix an $\widetilde{H}$-equivariant log resolution $f : A' \to \mathbb{A}^2$ of $(\mathbb{A}^2, B', \mathfrak{b})$, obtained by blowing up $\widetilde{H}$-orbits of points finitely many times (for instance the minimal log resolution of $(\mathbb{A}^2, B', \mathfrak{b})$, which is unique, since any log resolution factors through it (cf. [Har77, Proposition V.5.3]), and hence $\widetilde{H}$-equivariant), with

$$\mathfrak{b} \cdot \mathcal{O}_{A'} = \mathcal{O}_{A'}(-F_\mathfrak{b}),$$

and let $B_{A'}$ be the $\mathbb{R}$-divisor with $K_{A'} + B_{A'} = f^*(K_{\mathbb{A}^2} + B') + sF_\mathfrak{b}$. Then

$$a(\widetilde{P}, \mathbb{A}^2, B', \mathfrak{b}^s) = a(\widetilde{P}, A', B_{A'})$$

for any prime divisor $\widetilde{P}$ over $\mathbb{A}^2$, and by (a) the coefficients of $B_{A'}$ are at most one near $f^{-1}(0)$. Let Ξ be the set of points $p \in \operatorname{Exc}(f)$ such that

$$\operatorname{Stab}_{\widetilde{H}}(p) \neq I_{\widetilde{H}}(E)$$

for some component E of $\operatorname{Exc}(f)$ through p. If $g \in \widetilde{H} \backslash I_{\widetilde{H}}(E)$, then g has finitely many fixed points on E, so Ξ is finite. Let $W_\mathfrak{b} \subset \mathfrak{b}$ be the span of the monomial generators of $\mathfrak{b}$. It is $\widetilde{H}$-invariant, since $\widetilde{H}$ acts by monomial linear maps, and $f^* W_\mathfrak{b}$ generates $\mathcal{O}_{A'}(-F_\mathfrak{b})$. Fix an integer $k > s$, and let $f_1, \dots, f_k \in W_\mathfrak{b}$ be general. Since any $g \in \widetilde{H}$ acts on $W_\mathfrak{b}$ by a linear automorphism, the elements $g^* f_i$ are general as well. By Bertini's theorem applied to each $g^* f_i$, $f^* \operatorname{div}(g^* f_i) = F_\mathfrak{b} + S_{i,g}$, where near $f^{-1}(0)$ each curve $S_{i,g}$ is smooth, intersects $\operatorname{Supp} B_{A'} \cup \operatorname{Exc}(f)$ transversally and only at points contained in exactly one component of $\operatorname{Supp} B_{A'} \cup \operatorname{Exc}(f)$, and does not contain points of Ξ. Since $f^* W_\mathfrak{b}$ is base point free and $(g^* f_i, g'^* f_j)$ is a general pair for $i \neq j$, the curves $S_{i,g}$ and $S_{j,g'}$ with $i \neq j$ do not meet near $f^{-1}(0)$. Curves $S_{i,g}$ and $S_{i,g'}$ may

coincide, for instance $S_{i,g'} = S_{i,g}$ if $g'g^{-1} \in K_0$, since K_0 acts trivially on $W_{\mathfrak{b}}$, or meet on $f^{-1}(0)$. Let

$$\Delta := \frac{s}{k|\widetilde{H}|} \sum_{i=1}^{k} \sum_{g \in \widetilde{H}} \operatorname{div}(g^* f_i), \qquad c_\Delta := \frac{s}{k|\widetilde{H}|} < 1.$$

Then $\Delta \geq 0$ is $\widetilde{H}$-invariant, $D \not\subset \operatorname{Supp} \Delta$, and

$$f^*(K_{\mathbb{A}^2} + B' + \Delta) = K_{A'} + B_{A'} + c_\Delta \sum_{i,g} S_{i,g}.$$

Since $g^* f_i \in \mathfrak{b}$, for any prime divisor $\widetilde{P}$ over $\mathbb{A}^2$,

$$a(\widetilde{P}, \mathbb{A}^2, B' + \Delta) \leq a(\widetilde{P}, \mathbb{A}^2, B', \mathfrak{b}^s). \tag{B.93}$$

For a point $p \in f^{-1}(0)$ on some $S_{i,g}$, let E be the unique component of $\operatorname{Supp} B_{A'} \cup \operatorname{Exc}(f)$ through p, so $E \subset f^{-1}(0)$ and $a_E := a(E, \mathbb{A}^2, B', \mathfrak{b}^s) \geq 0$, and let Θ_p be the sum, with coefficients, of the components of $c_\Delta \sum_{i,g} S_{i,g}$ through p. They come from one index i, so Θ_p is a sum of smooth curves transversal to E with total coefficient at most $c_\Delta |\widetilde{H}| = s/k < 1$, and

$$B_{A'} + c_\Delta \sum_{i,g} S_{i,g} = (1 - a_E)E + \Theta_p$$

near p. Thus $(A', E + \Theta_p)$ is lc near p [Kaw07, Theorem]. Since $(A', B_{A'})$ is log smooth with coefficients at most one near $f^{-1}(0)$, the pair $(A', B_{A'} + c_\Delta \sum_{i,g} S_{i,g})$ is lc near $f^{-1}(0)$, so $(\mathbb{A}^2, B' + \Delta)$ is lc near 0 [KM98, Lemma 2.30 and Corollary 2.31(3)].

We show that the minimum of

$$\frac{a(\widetilde{P}, \mathbb{A}^2, B' + \Delta)}{|I_{\widetilde{H}}(\widetilde{P})|}$$

over the prime divisors $\widetilde{P}$ over $\mathbb{A}^2$ with center 0 is β. By (B.93) and (b), it is at most β. Let $\widetilde{P}$ be a prime divisor over $\mathbb{A}^2$ with center 0, and p the generic point of the center of $\widetilde{P}$ on A'. If $p \notin \bigcup_{i,g} S_{i,g}$, then

$$a(\widetilde{P}, \mathbb{A}^2, B' + \Delta) = a(\widetilde{P}, A', B_{A'}) = a(\widetilde{P}, \mathbb{A}^2, B', \mathfrak{b}^s).$$

Otherwise p is a closed point of some $S_{i,g}$ and of $f^{-1}(0)$, so E, a_E, and Θ_p are defined for p. Near p,

$$a(\widetilde{P}, A', (1 - a_E)E + \Theta_p) = a(\widetilde{P}, A', E + \Theta_p) + a_E \operatorname{mult}_{\widetilde{P}} E \geq a_E,$$

since $(A', E + \Theta_p)$ is lc near p and $\operatorname{mult}_{\widetilde{P}} E \geq 1$. Since $I_{\widetilde{H}}(\widetilde{P})$ fixes p and $p \notin \Xi$, we have

$$I_{\widetilde{H}}(\widetilde{P}) \subset \operatorname{Stab}_{\widetilde{H}}(p) = I_{\widetilde{H}}(E).$$

Thus

$$\frac{a(\widetilde{P}, \mathbb{A}^2, B' + \Delta)}{|I_{\widetilde{H}}(\widetilde{P})|} \geq \frac{a_E}{|I_{\widetilde{H}}(E)|} \geq \beta$$

by (b). This proves that the minimum is β.

Let $C' := B' - D + \Delta$ and $\gamma := 1 - \operatorname{mult}_0(C'|_D)$. Since $(\mathbb{A}^2, D + C')$ is lc near 0, the pair $(D, C'|_D)$ is lc near 0 [Kaw07, Theorem], so $\gamma \geq 0$. Since $\widetilde{H} \subset GL_2(\mathbb{C})$ is finite and preserves the line $T_0 D$, it is contained in a Borel subgroup of $GL_2(\mathbb{C})$, whose unipotent radical has no non-trivial finite subgroup. Thus $\widetilde{H}$ embeds into $(\mathbb{C}^*)^2$, and $\widetilde{H}$ is abelian. Let e be the order of the image of $\widetilde{H}$ in $\operatorname{Aut}(T_0 D)$. By Lemma 5.6 applied to $(\mathbb{A}^2 \ni 0, D + C')$ and $\widetilde{H}$,

$$\operatorname{mld}((\mathbb{A}^2 \ni 0, D + C')/\widetilde{H}) = \frac{\gamma}{e}.$$

By the previous paragraph, $\beta = \gamma/e$. In particular, $\gamma = e\beta > 0$.

Step 5. In this step we construct U and λ, and we prove that $(Y^\epsilon, B^\epsilon + \lambda \operatorname{div} u_0)$ is lc over a neighborhood of o_0, where u_0 is the function on Q_0 whose pullback to V is U.

The tangent line $T_0 D$ is an $\widetilde{H}$-invariant line in $T_0\mathbb{A}^2 = \mathbb{C}^2$. By Maschke's theorem, there exists an $\widetilde{H}$-invariant line $L' \subset \mathbb{C}^2$ with $\mathbb{C}^2 = T_0 D \oplus L'$. Let Z_0 be a linear form on $\mathbb{A}^2$ with kernel L'. Then Z_0 is $\widetilde{H}$-semi-invariant, $D \not\subset \operatorname{div} Z_0$, and $Z_0|_D$ is a local parameter of D at 0. Let r be the order of the character of Z_0 on $\widetilde{H}$. Since $\widetilde{H}$ acts on the cotangent line of D at 0 through the character of Z_0, we have $r = e$, so

$$\beta = \frac{\gamma}{r}. \tag{B.94}$$

Let ℓ be the order of the character of Z_0 on K_0. Then ℓ divides r, and Z_0^ℓ is K_0-invariant. Thus there exists a regular function u_0 on $Q_0 = \mathbb{A}^2/K_0$ with $\pi_0^* u_0 = Z_0^\ell$. The function u_0 is non-constant, vanishes at o_0, and is H-semi-invariant. Since the character of Z_0^ℓ on $\widetilde{H}$ has order r/ℓ and is trivial on K_0, the character of u_0 on H has order $q := r/\ell$. Let $\lambda := \gamma/\ell > 0$. Then

$$\frac{\lambda}{q} = \frac{\gamma/\ell}{r/\ell} = \frac{\gamma}{r} = \beta.$$

Since $\operatorname{mult}_0(C'|_D) = 1 - \gamma$ and $Z_0|_D$ vanishes to order one at 0, after shrinking $\mathbb{A}^2$ to a neighborhood of 0 the $\mathbb{R}$-divisor $(C' + \gamma \operatorname{div} Z_0)|_D$ is $1 \cdot [0]$. Thus $(D, (C' + \gamma \operatorname{div} Z_0)|_D)$ is lc, and $(\mathbb{A}^2, B' + \Delta + \gamma \operatorname{div} Z_0)$ is lc near 0 [Kaw07, Theorem]. By (B.93), $(\mathbb{A}^2, B' + \gamma \operatorname{div} Z_0, \mathfrak{b}^s)$ is lc near 0. Since

$$\gamma \operatorname{div} Z_0 = \lambda \pi_0^* \operatorname{div} u_0,$$

[Liu26, Lemma 2.6(2)] implies that $(Q_0, B_0 + \lambda \operatorname{div} u_0, I^s)$ is lc on a neighborhood Q_0° of o_0. By (3), for any prime divisor P over Q_0,

$$a(P, Y^\epsilon, B^\epsilon + \lambda \operatorname{div} u_0) = a(P, Q_0, B_0 + \lambda \operatorname{div} u_0, I^s), \tag{B.95}$$

where u_0 also denotes the pullback of u_0 to Y^ϵ. Thus $(Y^\epsilon, B^\epsilon + \lambda \operatorname{div} u_0)$ is lc over Q_0°.

Since

$$\mathbb{C}[Q_0] = \mathbb{C}[V]_0 = \mathbb{C}[V]^{\mathbf{T}_{(w)}},$$

the pullback U of u_0 under the quotient morphism $V \to Q_0$ is a regular $\mathbf{T}_{(w)}$-invariant function on V. The action of G on $\mathbb{C}[V]_0$ is the action of H on $\mathbb{C}[Q_0]$ through $G \to H$, so U is G-semi-invariant and the character of U on G has order q. Every monomial e^m with $0 \neq m \in \sigma^\vee \cap M_0$ vanishes at o_0 and, as a function on V, at 0. Since $u_0(o_0) = 0$, the function u_0 is a linear combination of such monomials, so $U(0) = 0$. Since $\mathbb{C}[Q_0] \to \mathbb{C}[V]$ is injective and u_0 is non-constant, U is non-constant.

Step 6. In this step we prove that $(V, D_V + \lambda \operatorname{div} U)$ is lc near 0, and we conclude the proof.

Let $V^\circ \subset V$ be the preimage of Q_0° under $V \to Q_0$, an open neighborhood of 0. Let F be a prime divisor over V whose center intersects V°. On the non-zero rational functions ζ on Q_0, the function $\zeta \to \operatorname{mult}_F \operatorname{div} \zeta$ is either zero or a discrete valuation v_F with value group $\varrho\mathbb{Z}$ for a positive integer ϱ. In the second case, since the general fibers of $V \to Q_0$ are curves and the residue field of F has transcendence degree two, the inequality of Abhyankar [Bou72, Chapter VI, §10, no. 3, Corollary 1 to Theorem 1] shows that the residue field of v_F has transcendence degree at least one (and at most $\dim Q_0 - 1 = 1$, by the inequality of Abhyankar for the restriction of v_F to the rational functions on Q_0). The valuation v_F has a center on Q_0, since $V \to Q_0$ is a morphism. Thus by [KM98, Lemma 2.45] there exists a prime divisor P over Q_0 such that

$$\operatorname{mult}_F \operatorname{div} \zeta = \varrho \operatorname{mult}_P \operatorname{div} \zeta$$

for any non-zero rational function ζ on Q_0. In the first case,

$$\operatorname{mult}_F \operatorname{div} x_i = w_i \operatorname{mult}_F \operatorname{div} \chi$$

for any i. If $\operatorname{mult}_F \operatorname{div} \chi \neq 0$, then $\operatorname{mult}_F \operatorname{div} x_i < 0$ for some i, since both signs of w_i occur, which contradicts the regularity of x_i on V. If $\operatorname{mult}_F \operatorname{div} \chi = 0$, let Z_F be the center of F on V.

If the image of Z_F in Q_0 were contained in a proper closed subset, then some non-zero regular function ζ on Q_0 would satisfy $\mathrm{mult}_F \operatorname{div} \zeta > 0$. Thus Z_F dominates Q_0, so $\dim Z_F \geq 2$. Since $Z_F \neq V$ and V is normal, F is the prime divisor Z_F on V. Then $\mathrm{mult}_F \operatorname{div} \Omega = 0$,

$$\mathrm{mult}_F \operatorname{div} h_j = \kappa_j \mathrm{mult}_F \operatorname{div} \chi + \mathrm{mult}_F \operatorname{div} \bar{h}_j = 0,$$

and $\mathrm{mult}_F \operatorname{div} U = 0$, so $a(F, V, D_V + \lambda \operatorname{div} U) = 1$ by (B.87). From now on we assume that the second case holds. Since $\mathrm{mult}_F \operatorname{div} \zeta \geq 0$ for any function ζ regular at the generic point of the center of F on V, the center of P on Q_0 contains the image of that point, which is contained in Q_0°.

We show that

$$1 + \mathrm{mult}_F \operatorname{div}(\eta \wedge d \log \chi) \geq \varrho(1 + \mathrm{mult}_P \operatorname{div} \eta). \tag{B.96}$$

Let $Q' \to Q_0$ be a resolution on which P is a prime divisor, and $X' := Q' \times \mathbb{P}^1$, where χ is the coordinate of $\mathbb{P}^1$. By **Step 1**, X' is birational to V. The center of F on X' maps to the generic point of P, so near the generic point of the center of F, the divisor of $\eta \wedge d \log \chi$ on X' is

$$(\mathrm{mult}_P \operatorname{div} \eta)(P \times \mathbb{P}^1) - Q' \times \{0\} - Q' \times \{\infty\}.$$

Since $\mathrm{mult}_F(P \times \mathbb{P}^1) = \varrho$,

$$1 + \mathrm{mult}_F \operatorname{div}(\eta \wedge d \log \chi) = a(F, X', P \times \mathbb{P}^1 + Q' \times \{0\} + Q' \times \{\infty\}) + \varrho(1 + \mathrm{mult}_P \operatorname{div} \eta),$$

and $a(F, X', P \times \mathbb{P}^1 + Q' \times \{0\} + Q' \times \{\infty\}) \geq 0$ since $(X', P \times \mathbb{P}^1 + Q' \times \{0\} + Q' \times \{\infty\})$ is log smooth with coefficients one near the center of F [KM98, Corollary 2.31(3)]. This proves (B.96).

Since

$$\mathrm{mult}_F \operatorname{div} h_j = \kappa_j \mathrm{mult}_F \operatorname{div} \chi + \varrho \, \mathrm{mult}_P \operatorname{div} \bar{h}_j, \qquad \mathrm{mult}_F \operatorname{div} U = \varrho \, \mathrm{mult}_P \operatorname{div} u_0,$$

(B.86), (B.87), and (B.96) imply

$$a(F, V, D_V + \lambda \operatorname{div} U) \geq \varrho\big(A(P) - \lambda \, \mathrm{mult}_P \operatorname{div} u_0\big) - \delta \, \mathrm{mult}_F \operatorname{div} \chi.$$

By (B.88),

$$A(P) - \lambda \, \mathrm{mult}_P \operatorname{div} u_0 = a(P, Y^\epsilon, B^\epsilon + \lambda \operatorname{div} u_0) + \delta \varphi_\epsilon(n_P).$$

Thus

$$a(F, V, D_V + \lambda \operatorname{div} U) \geq \varrho a(P, Y^\epsilon, B^\epsilon + \lambda \operatorname{div} u_0) + \delta\big(\varrho \varphi_\epsilon(n_P) - \mathrm{mult}_F \operatorname{div} \chi\big).$$

The first term is non-negative by (B.95), since the center of P on Q_0 intersects Q_0°. For the second term,

$$\mathrm{mult}_F \operatorname{div} x_i = w_i \, \mathrm{mult}_F \operatorname{div} \chi + \varrho \langle m_i^0, n_P \rangle \geq 0$$

for any i. If $\delta < 0$, then $\epsilon = +$, and the inequalities for $w_i > 0$ imply $\mathrm{mult}_F \operatorname{div} \chi \geq \varrho \varphi_+(n_P)$, so the second term is non-negative. If $\delta > 0$, then $\epsilon = -$, and the inequalities for $w_i < 0$ imply $\mathrm{mult}_F \operatorname{div} \chi \leq \varrho \varphi_-(n_P)$, so the second term is non-negative. If $\delta = 0$, the second term is zero. Thus $a(F, V, D_V + \lambda \operatorname{div} U) \geq 0$, and $(V, D_V + \lambda \operatorname{div} U)$ is lc on V°. Together with **Step 5**, this proves (1).

Finally, G acts faithfully on V and preserves D_V, the function U is G-semi-invariant with character of order q and vanishes at the G-fixed point 0, $\lambda > 0$, and $(V, D_V + \lambda \operatorname{div} U)$ is lc near 0. By Lemma 5.8 with $Z = \{0\}$, any prime divisor F over V with center 0 satisfies

$$\frac{a(F, V, D_V)}{|I_G(F)|} \geq \frac{\lambda}{q} = \beta.$$

This proves (2). □

B.12. **Exchanging groups on circles with radical weights of opposite signs.** In this subsection, we prove Theorem 8.13. We first reduce to polynomial split sources ($xy + P_iQ_i = 0$) with P_i monic in z and Q_i monic in t. We then bound the quotient values from above and from below by the minimal log discrepancies of two graded surface quotients along $\mathbf{T}$ (Lemmas B.23 and B.22), and we conclude with compatible generic limits of the surface quotients.

Set-up B.17. Fix integers $a, b \geq 1$ with $a + b \geq 3$, integers $A \geq 0$ and $C, D > 0$ with $\gcd(A, C, D) = 1$ and $aC - bD = 2A$, a positive integer e_0, and a DCC set $\Gamma \subset [0, 1]$, and let $\mathbf{T} := \mathbf{T}_{(-A,-A,-C,D)}$. Let $s \geq 0$ be an integer, and let $\kappa_1, \dots, \kappa_s$ be integers. For any $i \geq 1$, let the following data be given.

(1) $W_i := (xy + P_iQ_i = 0) \subset \mathbb{A}^4$, where
$$P_i = z^a + \sum_{k<a} P_{i,k}(t)z^k \quad \text{and} \quad Q_i = t^b + \sum_{k<b} Q_{i,k}(z)t^k$$
are polynomials with $\mathrm{mult}_0(P_i - z^a) > a$ and $\mathrm{mult}_0(Q_i - t^b) > b$, and $(W_i \ni 0)$ is an isolated Gorenstein terminal germ.
(2) G_i is a finite abelian group acting faithfully on W_i, linearly and monomially on the pair $\{x, y\}$ and diagonally on z, t.
(3) xy, P_i, and Q_i are G_i-semi-invariant with the characters of xy, z^a, and t^b, respectively.
(4) $R_i := G_i \cap \mathbf{T}$, $r_i := |R_i|$, and $H_i := G_i/R_i$ satisfy $[G_i : R_i] \leq e_0$.
(5) $r_i \to +\infty$.
(6) $B_i = \sum_{j=1}^{s} c_{ij} \operatorname{div}(h_{ij}|_{W_i})$, where $c_{ij} \in \Gamma\backslash\{0\}$ and each h_{ij} is a G_i-semi-invariant polynomial vanishing at 0 without monomials divisible by xy.
(7) For any j, the sum h^0_{ij} of the monomials of h_{ij} of $\mathbf{T}$-weight κ_j is non-zero.
(8) The least total degree m_i of a monomial of $h_{ij} - h^0_{ij}$, $1 \leq j \leq s$, satisfies $m_i \to +\infty$.
(9) $\tau_i := \sum_j c_{ij}\kappa_j$, $\delta_i := \tau_i + C - D$, and $\alpha_i := \mathrm{mld}((W_i \ni 0, B_i)/G_i)$.
(10) $V_0 := (xy + z^at^b = 0) \subset \mathbb{A}^4$, and $B^0_i := \sum_j c_{ij} \operatorname{div}(h^0_{ij}|_{V_0})$.
(11) $\Omega := dx \wedge dz \wedge dt/x$, a rational 3-form on W_i and on V_0, and $\mathcal{E}$ is the vector field on V_0 generating the action of $\mathbf{T}$.

The $\mathbf{T}$-weights of x, y, z, t are $-A, -A, -C, D$, and xy and z^at^b have $\mathbf{T}$-weight $-2A$ since $aC - bD = 2A$. If κ_j is the $\mathbf{T}$-weight of a monomial of least degree of h_{ij} for any j, then h^0_{ij} is the principal block of h_{ij}, τ_i is the mass of B_i along $\mathbf{T}$, and δ_i is the adjusted mass of B_i along $\mathbf{T}$. Indeed, in the notation of Set-up 5.1, the monomial xy of least degree of $xy + P_iQ_i$ has $\mathbf{T}$-weight $-2A$, so $\epsilon_{\mathbf{T}} = (-A - A - C + D) - (-2A) = D - C$, and $\delta_i = \tau_i - \epsilon_{\mathbf{T}}$. The circle $\mathbf{T}$ acts by scalars on the span of x and y, so G_i commutes with $\mathbf{T}$ and $G_i\mathbf{T}$ is an abelian group.

Lemma B.18. *Notation as in Set-up B.17.*

(1) Let z^mt^n be a monomial of $P_i - z^a$, resp. of $Q_i - t^b$. Then there exists an integer $l \geq 1$ such that
$$\begin{aligned} -Cm + Dn &= -aC + lr_i \quad \text{and} \quad n \geq \frac{lr_i - aC}{D}, \qquad \text{resp.} \\ -Cm + Dn &= bD - lr_i \quad \text{and} \quad m \geq \frac{lr_i - bD}{C}. \end{aligned} \tag{B.97}$$
Any monomial of $h_{ij} - h^0_{ij}$ has $\mathbf{T}$-weight $\kappa_j + lr_i$ for some integer $l \neq 0$.

(2) V_0 is normal and klt. For any prime divisor P over W_i, resp. over V_0,
$$a(P, W_i, 0) \geq \mathrm{mult}_P z + \mathrm{mult}_P t, \qquad \text{resp.} \qquad a(P, V_0, 0) \geq \mathrm{mult}_P z + \mathrm{mult}_P t.$$

(3) Let $\Gamma' \subset [0, +\infty)$ be a DCC set and $f_1, \dots, f_{s'}$ polynomials. For any i let $c'_{i1}, \dots, c'_{is} \in \Gamma'$ and $c''_1, \dots, c''_{s'} \in \Gamma'$, and let
$$\Delta_i := \sum_j c'_{ij} \operatorname{div}(h_{ij}) + \sum_k c''_k \operatorname{div}(f_k), \qquad \Delta^0_i := \sum_j c'_{ij} \operatorname{div}(h^0_{ij}) + \sum_k c''_k \operatorname{div}(f_k),$$

as $\mathbb{R}$-divisors on $\mathbb{A}^4$ not containing W_i, resp. V_0. Then there exists i_0 such that for any $i \geq i_0$, $(W_i, \Delta_i|_{W_i})$ is lc near 0 if and only if $(V_0, \Delta_i^0|_{V_0})$ is lc near 0.

Proof. (1) The group R_i is a cyclic subgroup of order r_i of $\mathbf{T}$, and an element $\lambda \in R_i$ multiplies a monomial of $\mathbf{T}$-weight w by λ^w. Since P_i is R_i-semi-invariant with the character of z^a, any monomial $z^m t^n$ of $P_i - z^a$ satisfies $-Cm + Dn = -aC + lr_i$ for an integer l. Since $m < a$ and $n \geq 0$, $lr_i = C(a-m) + Dn > 0$, so $l \geq 1$ and $Dn = lr_i - C(a-m) \geq lr_i - aC$. The case of Q_i is symmetric, and the last sentence holds since h_{ij} is R_i-semi-invariant.

(2) The singular locus of V_0 is contained in $(x = y = 0) \cap (zt = 0)$, which has codimension two in V_0. Since V_0 is a hypersurface, it is normal. On V_0 and on W_i, x, z, t are rational coordinates, and Ω is a generator of the canonical module near 0 by adjunction, because $\partial(xy + P_iQ_i)/\partial y = x$. We have

$$\Omega = zt \cdot \frac{dx}{x} \wedge \frac{dz}{z} \wedge \frac{dt}{t}.$$

On a log resolution on which P is a divisor and the divisors of x, z, and t are supported on a simple normal crossing divisor, the form $\frac{dx}{x} \wedge \frac{dz}{z} \wedge \frac{dt}{t}$ has at most a simple pole along any prime divisor P. Thus $a(P, W_i, 0) = 1 + \operatorname{mult}_P \Omega \geq \operatorname{mult}_P z + \operatorname{mult}_P t$, and, for P over V_0, $a(P, V_0, 0) = 1 + \operatorname{mult}_P \Omega \geq \operatorname{mult}_P z + \operatorname{mult}_P t$. The variety V_0 is smooth on $(zt \neq 0)$, and a prime divisor P whose center is contained in $(zt = 0)$ satisfies $\operatorname{mult}_P z + \operatorname{mult}_P t > 0$. Hence V_0 is klt.

(3) We may assume that $\Gamma' \neq \{0\}$, and let γ' be the least positive element of $\Gamma' \cup \{1\}$. By (1) and Set-up B.17(8), $(xy + P_iQ_i) - (xy + z^a t^b)$ and $h_{ij} - h_{ij}^0$ belong to $(x, y, z, t)^{m_i'}$ for integers $m_i' \to +\infty$. By [dFEM11, Theorem 1.1], the lc thresholds at 0 of the $\mathbb{R}$-ideals $(f') \cdot \prod_j (h_j')^{c_j'} \cdot \prod_k (f_k)^{c_k''}$ on $\mathbb{A}^4$, where f', h_j' are polynomials and $c_j', c_k'' \in \Gamma'$, form an ACC set. Hence there exists $\eta_0 > 0$ such that no such threshold belongs to $(1 - \eta_0, 1)$. By Theorem 2.12(2) with $\epsilon = \eta_0/2$, if $m_i' \geq 8/(\eta_0 \gamma')$, then the lc thresholds at 0 of

$$(xy + P_iQ_i) \cdot \prod_j (h_{ij})^{c_{ij}'} \cdot \prod_k (f_k)^{c_k''} \quad \text{and} \quad (xy + z^a t^b) \cdot \prod_j (h_{ij}^0)^{c_{ij}'} \cdot \prod_k (f_k)^{c_k''}$$

differ by at most $\eta_0/2$. Thus one of them is at least 1 if and only if the other one is. Since W_i and V_0 are normal Cartier divisors of $\mathbb{A}^4$, [Kaw07, Theorem] shows that $(W_i, \Delta_i|_{W_i})$ is lc near 0 if and only if $(\mathbb{A}^4, W_i + \Delta_i)$ is lc near 0, and similarly for $(V_0, \Delta_i^0|_{V_0})$ and $(\mathbb{A}^4, V_0 + \Delta_i^0)$. □

Construction-Proposition B.19. Notation as in Set-up B.17.

(1) (a) If G_i is not diagonal, then the diagonal elements of G_i act by scalars on the span of x and y.

(b) If G_i is not diagonal, then there exists $\epsilon_i \in \mathbb{C} \setminus \{0\}$ such that G_i acts diagonally in the coordinates $u_1 := x + \epsilon_i y$, $u_2 := x - \epsilon_i y$, z, t, and $xy = (u_1^2 - u_2^2)/(4\epsilon_i)$.

(c) If G_i is diagonal, let $u_1 := x$ and $u_2 := y$.

Let $\bar{V}_0 := \operatorname{Spec} \mathbb{C}[V_0]^{\mathbf{T}}$, let $\bar{o} \in \bar{V}_0$ be the image of 0, let $\mathbb{C}[V_0] = \bigoplus_{m \in \mathbb{Z}} \mathbb{C}[V_0]_m$ be the grading by $\mathbf{T}$-weight, and let

$$Y_0^{\pm} := \operatorname{Proj}_{\bar{V}_0} \bigoplus_{m \geq 0} \mathbb{C}[V_0]_{\pm m}.$$

By Set-up B.17(2) and (3), $xy + z^a t^b$ is G_i-semi-invariant, since G_i preserves W_i and hence xy and P_iQ_i have equal characters, so G_i preserves V_0. Since G_i commutes with $\mathbf{T}$, it preserves the grading, and $R_i \subset \mathbf{T}$ acts trivially on $\bar{V}_0$ and on $Y_0^{\pm}$. Hence H_i acts on $\bar{V}_0$ and on $Y_0^{\pm}$. Let $Y_i^{\pm} := Y_0^{\pm}/H_i$, with fibers $C_i^{\pm}$ over the image of $\bar{o}$ in $\bar{V}_0/H_i$. For $q \in \{u_1, u_2, z, t\}$ with $\mathbf{T}$-weight $w_q \neq 0$, let $S_q := V_0 \cap (q = 1)$ and $G_{q,i} := \{g \in G_i \mathbf{T} \mid g^* q = q\}$. Let $S := (XY + Z^a = 0) \subset \mathbb{A}^3$, identified with S_t by $X = x$, $Y = y$, $Z = z$, let $G_{S,i} := G_{t,i}$, $\bar{h}_{ij} := h_{ij}^0(X, Y, Z, 1)$, and $\bar{B}_i := \sum_j c_{ij} \operatorname{div}(\bar{h}_{ij})$.

(2) (a) For any q with $w_q \neq 0$, $G_{q,i}$ is a finite abelian group of order $|w_q| \cdot |H_i|$ acting faithfully on S_q and diagonally in the coordinates of $\mathbb{A}^4$ other than q.

(b) For any q with $w_q \neq 0$, $S_q/G_{q,i}$ is an open subset of Y_i^+ if $w_q > 0$ and of Y_i^- if $w_q < 0$.
(c) The open subsets $S_q/G_{q,i}$ cover Y_i^+ and Y_i^-.
(d) The morphisms $Y_i^\pm \to \bar{V}_0/H_i$ are projective and birational.
(e) We have $Y_i^+ = S/G_{S,i}$, the morphism $Y_i^+ \to \bar{V}_0/H_i$ is an isomorphism, and C_i^+ is the image of the origin of S.

(3) Let $B_i^\pm$ be the $\mathbb{R}$-divisor on $Y_i^\pm$ whose restriction to $S_q/G_{q,i}$ is the boundary of the crepant quotient

$$\Big(S_q, \sum_j c_{ij} \operatorname{div}(h_{ij}^0|_{S_q})\Big)/G_{q,i}.$$

If (V_0, B_i^0) is lc near 0, then $(Y_i^\pm, B_i^\pm)$ is lc near $C_i^\pm$, the minimum $\beta_i^\pm$ of $a(\nu, Y_i^\pm, B_i^\pm)$ over the prime divisors ν over $Y_i^\pm$ with center contained in $C_i^\pm$ is attained, $\beta_i^\pm \in [0, 2]$, and $\beta_i^\pm$ belongs to an ACC set $\mathcal{B}$ depending only on Γ.

Proof. (1) Let $g \in G_i$ with $g^*x = \alpha_g y$ and $g^*y = \alpha'_g x$. A diagonal element g' of G_i with $g'^*x = d_1 x$ and $g'^*y = d_2 y$ commutes with g, so $\alpha_g d_2 = d_1 \alpha_g$ and $d_1 = d_2$. Any element of G_i exchanging x and y is the product of g and a diagonal element, so it is the product of g and a scalar on the span of x and y. Let $\epsilon_i^2 = \alpha_g/\alpha'_g$. Then

$$g^*(x \pm \epsilon_i y) = \alpha_g y \pm \epsilon_i \alpha'_g x = \pm\epsilon_i \alpha'_g (x \pm \epsilon_i y),$$

and G_i is diagonal in u_1, u_2, z, t. The formula for xy is a direct calculation.

(2) The scheme $Y_0^\pm$ is the relative Proj of a finitely generated graded $\mathbb{C}[V_0]^{\mathbf{T}}$-algebra, hence projective over $\bar{V}_0$. It is birational to $\bar{V}_0$, since both signs of $\mathbf{T}$-weights occur on V_0 (those of t and of z), so that any $\mathbf{T}$-invariant rational function on V_0 is a quotient of two $\mathbf{T}$-invariant regular functions, and both $Y_0^\pm$ and $\bar{V}_0$ are birational to the quotient of the open subset $V_0 \cap (zt \neq 0)$, on which $\mathbf{T}$ acts with finite stabilizers. Hence $Y_i^\pm \to \bar{V}_0/H_i$ are projective and birational. The coordinates u_1, u_2, z, t are eigenvectors of $G_i\mathbf{T}$. For $g \in G_i\mathbf{T}$ with $g^*q = \chi q$, there exists $\lambda \in \mathbf{T}$ with $\lambda^{w_q} = \chi^{-1}$, so $G_i\mathbf{T} = G_{q,i}\mathbf{T}$. Since $G_{q,i} \cap \mathbf{T} = \mu_{|w_q|}$ and $G_i\mathbf{T}/\mathbf{T} \cong G_i/R_i$, $|G_{q,i}| = |w_q| \cdot |H_i|$. If $g \in G_{q,i}$ acts trivially on S_q, then g acts trivially on $V_0 \cap (q \neq 0) = \mathbf{T} \cdot S_q$ since g commutes with $\mathbf{T}$. Since V_0 is not contained in a hyperplane of $\mathbb{A}^4$, $g = 1$. The open subset $D_+(q) \subset Y_0^\pm$ is $\operatorname{Spec}(\mathbb{C}[V_0]_q)_0 \cong S_q/\mu_{|w_q|}$, it is H_i-invariant, and $D_+(q)/H_i = S_q/G_{q,i}$. A monomial of positive $\mathbf{T}$-weight is divisible by t, and a monomial of negative $\mathbf{T}$-weight is divisible by z, or, if $A > 0$, by x or y. Since $(x \neq 0) \cup (y \neq 0) = (u_1 \neq 0) \cup (u_2 \neq 0)$, the subsets $D_+(q)$ cover $Y_0^\pm$. In particular $Y_0^+ = D_+(t) = S_t/\mu_D$ is affine, so the projective birational morphism $Y_0^+ \to \bar{V}_0$ onto the normal variety $\bar{V}_0$ is an isomorphism. The point $\bar{o}$ is the common zero of the invariants $x^D t^A$, $y^D t^A$, $z^D t^C$ (of x, y, $z^D t^C$ if $A = 0$), so C_i^+ is the image of $X = Y = Z = 0$.

(3) On the overlap of two charts the restrictions of h_{ij}^0 differ by units, so the boundaries $B_i^\pm$ are well defined. Assume that (V_0, B_i^0) is lc near 0. The non-lc locus $\mathcal{N}$ of (V_0, B_i^0) is closed, $\mathbf{T}$-invariant, and does not contain 0. A point $p_0 \in S_q$ over $C_i^\pm$ is a point at which all $\mathbf{T}$-invariant regular functions vanish, so the closure of $\mathbf{T} \cdot p_0$ contains 0, and $p_0 \notin \mathcal{N}$. The morphism $\mathbb{G}_m \times S_q \to V_0 \cap (q \neq 0)$, $(\lambda, p_0) \to \lambda \cdot p_0$, is étale, and $h_{ij}^0(\lambda \cdot p_0) = \lambda^{\kappa_j} h_{ij}^0(p_0)$. Hence $(S_q, \sum_j c_{ij} \operatorname{div}(h_{ij}^0|_{S_q}))$ is lc near the points over $C_i^\pm$, and so is the crepant quotient.

Near the points over $C_i^\pm$, the coefficient of a prime divisor of S_q in $\sum_j c_{ij} \operatorname{div}(h_{ij}^0|_{S_q})$ is $\sum_j n_j c_{ij} \leq 1$ with $n_j \in \mathbb{Z}_{\geq 0}$, since the pair is lc there. So $B_i^\pm \in \Gamma_1$ near $C_i^\pm$, where $\Gamma_1 := \Psi(\Gamma_+ \cap [0, 1])$ is a DCC set [Liu26, Definition 2.9 and Lemma 2.10(1)(5)]. The fiber $C_i^\pm$ is a point or a curve. A prime divisor with center a component C' of C_i^- is C' itself, with log discrepancy $1 - \operatorname{mult}_{C'} B_i^-$, and a closed-point minimal log discrepancy is attained [Liu26, Lemma 3.5]. At a general closed point of C' the minimal log discrepancy is at least $1 - \operatorname{mult}_{C'} B_i^-$, so $\beta_i^\pm$ is the minimum over finitely many closed points and components, and it is attained. Hence, restricting $(Y_i^\pm, B_i^\pm)$ to a neighborhood of $C_i^\pm$ on which $B_i^\pm \in \Gamma_1$, $\beta_i^\pm$ belongs to the

union $\mathcal{B}$ of $\{1-c \mid c \in \Gamma_1\}$ and of the set of minimal log discrepancies at closed points of lc surface pairs with coefficients in Γ_1, which is an ACC set by Theorem 2.9(1). Any section S_q is smooth or has an A_{d-1} point $XY+Z^d=0$ at the points over $C_i^{\pm}$. The blow-up E of the preimage of such a point on the smooth A_{d-1} cover $\mathbb{A}^2$ (the identity if S_q is smooth there) has $a(E,\mathbb{A}^2,0)=2$. The pullback B' of $B_i^{\pm}$ to $\mathbb{A}^2$ is effective, so the image of E on $Y_i^{\pm}$ has log discrepancy $a(E,\mathbb{A}^2,B')/|I(E)| \le 2/|I(E)| \le 2$, where $I(E)$ is the inertia group. Thus $\beta_i^{\pm} \le 2$. □

Lemma B.20. *Notation as in Construction-Proposition B.19. Let ν be a prime divisor over $\bar{V}_0/H_i$ with center the image of $\bar{o}$. By Construction-Proposition B.19(2), ν is also a prime divisor over $Y_i^{\pm}$ with center contained in $C_i^{\pm}$, and any such prime divisor arises in this way. For a $\mathbf{T}$-invariant rational function f on V_0, let $[f]_\nu := \mathrm{mult}_{\nu'} f/|I_{H_i}(\nu')|$ for a prime divisor ν' over $\bar{V}_0$ lying over ν. For $q \in \{u_1,u_2,z\}$ with $w_q<0$ let*

$$\ell_q(\nu) := \frac{[q^D t^{|w_q|}]_\nu}{D|w_q|}, \qquad \ell(\nu) := \min_q \ell_q(\nu), \qquad d_0 := e_0 D \max(A,C).$$

(1) $\ell(\nu) \ge 1/d_0$, and the center of ν on Y_i^- intersects $S_q/G_{q,i}$ if and only if $\ell_q(\nu)=\ell(\nu)$.
(2) $a(\nu,Y_i^-,B_i^-) = a(\nu,Y_i^+,B_i^+) - \delta_i \ell(\nu)$.
(3) Assume that (V_0,B_i^0) is lc near 0. If $\delta_i<0$, then $\beta_i^- \ge \beta_i^+ + |\delta_i|/d_0$. If $\delta_i>0$, then $\beta_i^+ \ge \beta_i^- + \delta_i/d_0$. If $\delta_i=0$, then $\beta_i^+=\beta_i^-$. In particular $\beta_i := \min(\beta_i^+,\beta_i^-)$ is equal to β_i^+ if $\delta_i \le 0$ and to β_i^- if $\delta_i \ge 0$.
(4) Assume that $\delta_i \ge 0$. If $A>0$, let $M := \mathrm{lcm}(A,C)$ and $\mathfrak{b} := (X^{M/A}, Y^{M/A}, Z^{M/C}) \subset \mathbb{C}[S]$. If $A=0$, let $M:=C$ and $\mathfrak{b} := (Z)$. Then $\mathfrak{b}$ is $G_{S,i}$-invariant, and for any prime divisor ν' over S with center 0 lying over ν,

$$a(\nu,Y_i^-,B_i^-) = \frac{a(\nu',S,\bar{B}_i,\mathfrak{b}^{\delta_i/M})}{|I_{G_{S,i}}(\nu')|}.$$

In particular, if (V_0,B_i^0) is lc near 0, then

$$\beta_i^- = \mathrm{mld}\big((S \ni 0, \bar{B}_i, \mathfrak{b}^{\delta_i/M})/G_{S,i}\big).$$

Proof. (1) The function $q^D t^{|w_q|}$ is a $\mathbf{T}$-invariant regular function vanishing at 0, so $\mathrm{mult}_{\nu'}(q^D t^{|w_q|})$ is a positive integer. Since $|I_{H_i}(\nu')| \le |H_i| \le e_0$ and $|w_q| \le \max(A,C)$, $\ell_q(\nu) \ge 1/d_0$. The ring of $D_+(q)$ is generated by the $\mathbf{T}$-invariant Laurent monomials ϖ/q^n, where ϖ is a monomial in the coordinates and $n \ge 0$. Write $\varpi = t^{n_t} q^{n_q} \prod_{q'} q'^{n_{q'}} \cdot \varpi_0$, where q' runs over the coordinates of negative weight other than q and ϖ_0 is a monomial in the coordinates of weight zero. Writing ϖ/q^n as a product of rational powers of the invariants $q'^D t^{|w_{q'}|}$ and $q^D t^{|w_q|}$ and of ϖ_0, and using that ϖ/q^n has weight zero, we obtain

$$[\varpi/q^n]_\nu = \sum_{q' \ne q} n_{q'} |w_{q'}| \big(\ell_{q'}(\nu) - \ell_q(\nu)\big) + n_t D \ell_q(\nu) + [\varpi_0]_\nu.$$

Since $[\varpi_0]_\nu \ge 0$, $[\varpi/q^n]_\nu \ge 0$ for any ϖ/q^n, that is, the center of ν intersects $S_q/G_{q,i}$, if and only if $\ell_q(\nu)=\ell(\nu)$.

(2) The form Ω is $\mathbf{T}$-homogeneous of weight $D-C$. For any q with $w_q \ne 0$, the pullback of Ω under the étale morphism $\mathbb{G}_m \times S_q \to V_0 \cap (q \ne 0)$ of the proof of Construction-Proposition B.19(3) is

$$\lambda^{D-C} \frac{d\lambda}{\lambda} \wedge (\iota_{\mathcal{E}}\Omega)|_{S_q},$$

so $(\iota_{\mathcal{E}}\Omega)|_{S_q}$ generates the canonical module of S_q. The rational $\mathbb{R}$-forms

$$\Theta_+ := \iota_{\mathcal{E}}\Omega \cdot \prod_j (h_{ij}^0)^{-c_{ij}} \cdot t^{\delta_i/D}, \qquad \Theta_-^{(q)} := \iota_{\mathcal{E}}\Omega \cdot \prod_j (h_{ij}^0)^{-c_{ij}} \cdot q^{-\delta_i/|w_q|} \quad (w_q<0)$$

have weight $D-C-\tau_i+\delta_i=0$, are annihilated by $\iota_{\mathcal{E}}$, and restrict to

$$(\iota_{\mathcal{E}}\Omega)|_{S_q}\cdot\prod_j(h_{ij}^0|_{S_q})^{-c_{ij}}$$

on S_t, resp. S_q. Thus Θ_+ and $\Theta_-^{(q)}$ are pullbacks of rational $\mathbb{R}$-forms on $\bar{V}_0$, which is birational to $Y_0^{\pm}$, and for a prime divisor ν' over $\bar{V}_0$ we write $\mathrm{mult}_{\nu'}\Theta_+$ and $\mathrm{mult}_{\nu'}\Theta_-^{(q)}$ for the orders along ν' of the $\mathbb{R}$-forms on $\bar{V}_0$. For ν' as in the definition of $[\cdot]_\nu$, the definition of the crepant quotient and [Liu26, Lemma 2.6(1)(2)] imply

$$a(\nu,Y_i^+,B_i^+)=\frac{1+\mathrm{mult}_{\nu'}\Theta_+}{|I_{H_i}(\nu')|},\qquad a(\nu,Y_i^-,B_i^-)=\frac{1+\mathrm{mult}_{\nu'}\Theta_-^{(q)}}{|I_{H_i}(\nu')|},$$

the second equality if the center of ν intersects $S_q/G_{q,i}$. By (1), choose q with $\ell_q(\nu)=\ell(\nu)$. Then

$$a(\nu,Y_i^-,B_i^-)-a(\nu,Y_i^+,B_i^+)=\left[q^{-\delta_i/|w_q|}t^{-\delta_i/D}\right]_\nu=-\frac{\delta_i}{D|w_q|}[q^Dt^{|w_q|}]_\nu=-\delta_i\ell(\nu).$$

(3) By the first sentence of the lemma, β_i^+ and β_i^- are the minima of $a(\nu,Y_i^+,B_i^+)$ and of $a(\nu,Y_i^-,B_i^-)$ over one set of prime divisors ν, namely the prime divisors over $\bar{V}_0/H_i$ with center the image of $\bar{o}$, so (3) follows from (1) and (2).

(4) The generator $Z^{M/C}$ is semi-invariant, and if $A>0$, $G_{S,i}$ permutes $X^{M/A}$ and $Y^{M/A}$ up to scalars, so $\mathfrak{b}$ is $G_{S,i}$-invariant. Let $e':=|I_{G_{S,i}}(\nu')|$. Since $Y_i^+=S/G_{S,i}=\bar{V}_0/H_i$ and $(x^Dt^A)|_{S_t}=X^D$, $(z^Dt^C)|_{S_t}=Z^D$, we have $\ell_x(\nu)=\mathrm{mult}_{\nu'}X/(Ae')$, $\ell_y(\nu)=\mathrm{mult}_{\nu'}Y/(Ae')$, and $\ell_z(\nu)=\mathrm{mult}_{\nu'}Z/(Ce')$, where ℓ_x,ℓ_y are defined as ℓ_q. Since x,y and u_1,u_2 are linear combinations of each other, $\min(\ell_{u_1},\ell_{u_2})=\min(\ell_x,\ell_y)$. Thus $\ell(\nu)=\mathrm{mult}_{\nu'}\mathfrak{b}/(Me')$, and (4) follows from (2) and $a(\nu,Y_i^+,B_i^+)=a(\nu',S,\bar{B}_i)/e'$. The proofs of (1), (2), and (4) use neither that the labels are at most one nor that the marks have no monomial divisible by xy. □

The next lemma is the passage from the reference surfaces to V_0.

Lemma B.21. *Notation as in Set-up B.17 and Construction-Proposition B.19(1)(2). Let $f_1,\dots,f_n$ be G_i-semi-invariant and $\mathbf{T}$-homogeneous polynomials of $\mathbf{T}$-weights $\kappa_1',\dots,\kappa_n'$ which are non-zero on V_0, let $c_1',\dots,c_n'\geq 0$, $\Delta:=\sum_k c_k'\,\mathrm{div}(f_k|_{V_0})$, and $\delta':=\sum_k c_k'\kappa_k'+C-D$. Assume that either $\delta'\leq 0$ and $(S_t,\Delta|_{S_t})$ is lc near the origin, or $\delta'\geq 0$ and $(S_q,\Delta|_{S_q})$ is lc near the points over C_i^- for any q with $w_q<0$. Then (V_0,Δ) is lc near 0.*

Proof. Let P be a prime divisor over V_0 with center 0. In the first case of the assumption let $q:=t$, and in the second case let $q\in\{u_1,u_2,z\}$ be a coordinate of negative weight with the least value of $\mathrm{mult}_P q/|w_q|$. Consider the dominant rational map

$$\pi_q:\mathbb{A}^1_v\times S_q\dashrightarrow V_0,\qquad (v,p_0)\to\lambda\cdot p_0,\quad \lambda=v^{\pm 1},$$

with $\lambda=v$ if $q=t$ and $\lambda=v^{-1}$ if $w_q<0$. It is the quotient by $\mu_{|w_q|}$ over $v\neq 0$, and $q\circ\pi_q=v^{|w_q|}$. Let $\omega_q:=(\iota_{\mathcal{E}}\Omega)|_{S_q}$. Since Ω has weight $D-C$ and $f_k(\lambda\cdot p_0)=\lambda^{\kappa_k'}f_k(p_0)$,

$$\pi_q^*\Big(\Omega\cdot\prod_k f_k^{-c_k'}\Big)=\pm v^{\mp\delta'}\frac{dv}{v}\wedge\omega_q\cdot\prod_k(f_k|_{S_q})^{-c_k'},$$

with the upper signs for $q=t$. Hence π_q is crepant for the pairs

$$(V_0,\Delta)\quad\text{and}\quad\big(\mathbb{A}^1_v\times S_q,b_v\,\mathrm{div}(v)+\mathrm{pr}^*\Delta|_{S_q}\big),$$

where $b_v:=1+\delta'\leq 1$ if $q=t$ and $b_v:=1-\delta'\leq 1$ if $w_q<0$. Let P' be a prime divisor over $\mathbb{A}^1_v\times S_q$ lying over P. The functions v and $q'\cdot q^{-w_{q'}/w_q}$, $q'\neq q$, have non-negative values at P' by the choice of q, and v has positive value. If $q=t$, the values of v and of the functions $q'\cdot q^{-w_{q'}/w_q}$ at P' are positive, since x,y,z,t have positive values at P, so the center of P' is the origin p_0 of $\{0\}\times S_t$. If $w_q<0$, the center of P' is contained in $\{0\}\times S_q$ and in the zero locus of t and of the coordinates of weight zero, where all non-constant $\mathbf{T}$-invariant monomials vanish. So the

center of P' is a point p_0 over C_i^- or, if $A > 0$ and the least value of $\operatorname{mult}_P q/|w_q|$ is attained by two coordinates, a curve $C \subset \{0\} \times S_q$ over C_i^-. By Lemma 5.5 with the trivial group, applied to the hypersurface $\mathbb{A}^1_v \times S_q \subset \mathbb{A}^4$, the section $S_q = (v = 0)$, and the $\mathbb{R}$-ideal $\prod_k (f_k|_{S_q})^{c'_k}$, at p_0 if the center is a point, and by its second part at a point $p_0 \in C$ translated to the origin if the center is C, where $(S_q, \Delta|_{S_q})$ is lc near p_0 and near the generic point of C with minimal log discrepancy $1 - \operatorname{mult}_C \Delta|_{S_q} \geq 0$ there,

$$a(P', \mathbb{A}^1_v \times S_q, b_v \operatorname{div}(v) + \operatorname{pr}^* \Delta|_{S_q}) \geq (1 - b_v) \operatorname{mult}_{P'} v \geq 0.$$

By [Liu26, Lemma 2.6(2)], $a(P, V_0, \Delta) \geq 0$. If (V_0, Δ) were not lc near 0, there would be a prime divisor E over V_0 with $a(E, V_0, \Delta) < 0$ whose center contains 0. If the center of E is 0, this contradicts the inequality $a(P, V_0, \Delta) \geq 0$ for the prime divisors P over V_0 with center 0. Otherwise take a log resolution $f : V' \to V_0$ on which E is a divisor and $f^{-1}(0)$ is a divisor, with simple normal crossing support, and a component F of $f^{-1}(0)$ meeting E. Then $F \neq E$, since the center of E is not 0. Blowing up $E \cap F$ and then, repeatedly, the intersection of E with the last exceptional divisor gives prime divisors with center 0 and log discrepancies $a(F, V_0, \Delta) + ka(E, V_0, \Delta)$, $k \geq 1$, which are negative for k large. Hence (V_0, Δ) is lc near 0. □

Lemma B.22. *Notation as in Set-up B.17 and Construction-Proposition B.19, and assume that (V_0, B_i^0) is lc near 0. Let $\theta > 1$ and let $\mathcal{I} \subset \mathbb{N}^+$ be an infinite subset. For $i \in \mathcal{I}$ let $\delta_i(\theta) := \theta\tau_i + C - D$ and let $B_i^{\pm}(\theta)$ be defined as $B_i^{\pm}$ with the labels θc_{ij}. Assume that one of the following holds for any $i \in \mathcal{I}$.*

(a) $\delta_i \leq 0$, $\delta_i(\theta) \leq 0$, and $(Y_i^+, B_i^+(\theta))$ is lc near C_i^+.
(b) $\delta_i \geq 0$, $\delta_i(\theta) \geq 0$, and $(Y_i^-, B_i^-(\theta))$ is lc near C_i^-.

Then for all but finitely many $i \in \mathcal{I}$, any prime divisor P over W_i with center 0 satisfies

$$\frac{a(P, W_i, B_i)}{|I_{G_i}(P)|} \geq \beta_i.$$

In particular $\alpha_i \geq \beta_i$.

Proof. The proof is long, so we divide it into several steps. The set $\mathcal{I}$ is the union of the subsets on which (a), resp. (b), holds. It suffices to prove the lemma for each of them that is infinite. We first treat an infinite subset on which (a) holds, and denote it again by $\mathcal{I}$. The case (b) is symmetric and is explained in **Step 5**. If $\Gamma \not\subset \{0\}$, let γ be the least element of $\Gamma \backslash \{0\}$, which exists since Γ is a DCC set, and let $\gamma := 1$ otherwise.

Step 1. In this step we show that $(W_i, \theta B_i)$ is lc near 0 for $i \in \mathcal{I}$, $i \gg 0$. The pair $(S_t, \theta \sum_j c_{ij} \operatorname{div}(h^0_{ij}|_{S_t}))$, whose crepant quotient by $G_{t,i}$ is $(Y_i^+, B_i^+(\theta))$ (Construction-Proposition B.19(2)(3)), is lc near the origin since $(Y_i^+, B_i^+(\theta))$ is lc near C_i^+. By Lemma B.21 with $f_j = h^0_{ij}$, $c'_j = \theta c_{ij}$, and $\delta' = \delta_i(\theta) \leq 0$, $(V_0, \theta B_i^0)$ is lc near 0. By Lemma B.18(3) with $\Gamma' = \theta\Gamma$, $(W_i, \theta B_i)$ is lc near 0 for $i \gg 0$.

Step 2. In this step we bound the values of the coordinates. It suffices to consider the prime divisors P with $a(P, W_i, B_i)/|I_{G_i}(P)| < \beta_i$. Fix $i \in \mathcal{I}$ as in **Step 1** and a prime divisor P over W_i with center 0 such that $a(P, W_i, B_i)/e < \beta_i$, where $e := |I_{G_i}(P)|$. For a rational function f on W_i write $[f] := \operatorname{mult}_P f/e$. Since $\beta_i \leq 2$ and $a(P, W_i, \theta B_i) \geq 0$, we have $(\theta - 1) \sum_j c_{ij}[h_{ij}] \leq a(P, W_i, B_i)/e < 2$. Let $\Lambda' := 2/(\theta - 1)$ and $\Lambda := 2\theta/(\theta - 1)$. Then

$$\frac{a(P, W_i, 0)}{e} \leq \Lambda, \qquad \sum_j c_{ij}[h_{ij}] \leq \Lambda', \qquad [h_{ij}] \leq \Lambda'/\gamma, \qquad [z] + [t] \leq \Lambda, \tag{B.98}$$

where the last inequality follows from Lemma B.18(2). Let $I_R(P) := I_{G_i}(P) \cap R_i$, so $e \leq e_0 |I_R(P)|$. For a coordinate $q \in \{x, y, u_1, u_2, z\}$ of negative weight, $q^D t^{|w_q|}$ is a regular $\mathbf{T}$-invariant, hence $I_R(P)$-invariant, function vanishing at 0. The group $I_R(P)$ is cyclic and acts faithfully on $\mathfrak{m}_P/\mathfrak{m}_P^2$ [Liu26, Lemma 2.6(1)], so $\operatorname{mult}_P(q^D t^{|w_q|})$ is a positive multiple of $|I_R(P)|$ [Liu26, Lemma 9.2(1)].

If $A=0$, then x and y are regular $\mathbf{T}$-invariant functions vanishing at 0, so $\mathrm{mult}_P x$ and $\mathrm{mult}_P y$ are positive multiples of $|I_R(P)|$ by [Liu26, Lemma 9.2(1)]. Hence

$$D[q]+|w_q|[t]\geq\frac{1}{e_0}\quad\text{for any } q \text{ with } w_q<0,\qquad\text{and}\qquad [x],[y]\geq\frac{1}{e_0}\quad\text{if } A=0. \tag{B.99}$$

Let $\epsilon:=1/(2d_0)$, where $d_0=e_0D\max(A,C)$ is the integer of Lemma B.20. Since x,y and u_1,u_2 are linear combinations of each other, $\min([x],[y])=\min([u_1],[u_2])$. If $[q]<\epsilon$ for some q with $w_q<0$, then $D\epsilon\leq 1/(2e_0)$ and (B.99) imply $[t]\geq 1/(2e_0\max(A,C))$.

Step 3. In this step we treat the case when $[q]\geq\epsilon$ for any coordinate q of negative weight. Let $\mathfrak{u}:=Q_i/t^b$, a G_i-invariant rational function. By (B.97), any monomial of $\mathfrak{u}-1$ is z^mt^{n-b} with $m\geq(r_i-bD)/C$, so by (B.98)

$$[\mathfrak{u}-1]\geq\frac{(r_i-bD)\epsilon}{C}-b\Lambda=:\Upsilon_i\to+\infty.$$

Let $\widehat{K}_P$ be the completion of the function field of W_i with respect to mult_P. Assume that $\Upsilon_i>\Lambda'/\gamma$. Then $\mathfrak{u}$ has a square root $\varsigma\in\widehat{K}_P$ with $[\varsigma-1]=[\mathfrak{u}-1]$. Let $x':=x/\varsigma$, $y':=y/\varsigma$, and $W_i':=(x'y'+P_it^b=0)\subset\mathbb{A}^4$. Since $\varsigma\in\widehat{K}_P$, the fields $\mathbb{C}(W_i)$ and $\mathbb{C}(W_i')$ are dense subfields of $\widehat{K}_P$ (since $\mathfrak{u}\in\mathbb{C}(z,t)\subset\mathbb{C}(W_i')$ and $[\mathfrak{u}-1]>0$, Hensel's lemma shows that ς belongs to the closure of $\mathbb{C}(W_i')$, which therefore contains $x=\varsigma x'$ and $y=\varsigma y'$), so P induces a prime divisor P' over W_i' with center 0 whose value group and residue field are those of P. Letting G_i fix ς, any $g\in G_i$ acts on x',y',z,t by the matrix by which it acts on x,y,z,t, and $I_{G_i}(P')=I_{G_i}(P)$. Since $d\varsigma/\varsigma$ is a combination of dz and dt, $dx\wedge dz\wedge dt/x=dx'\wedge dz\wedge dt/x'$, and the orders along P and P' of rational 3-forms on W_i and on W_i' only depend on the images of the forms in the completed module of differentials of $\widehat{K}_P$, so $a(P,W_i,0)=a(P',W_i',0)$. Since h_{ij} has no monomial divisible by xy, $h_{ij}-h_{ij}(x',y',z,t)$ is a sum of terms $(\varsigma^k-1)m$ with $k\geq1$ and m a monomial in x',y',z,t, so $\mathrm{mult}_{P'}h_{ij}(x',y',z,t)=\mathrm{mult}_P h_{ij}$ by (B.98).

Write $h_{ij}(x',y',z,t)=h_{ij}^{\geq}+h_{ij}^{<}$, where $h_{ij}^{\geq}$, resp. $h_{ij}^{<}$, is the sum of the monomials of $\mathbf{T}$-weight κ_j+lr_i with $l\geq0$, resp. $l<0$. A monomial $x'^{n_1}y'^{n_2}z^{n_3}t^{n_4}$ of $h_{ij}^{<}$ satisfies $A(n_1+n_2)+Cn_3\geq r_i-|\kappa_j|$, so the value of the monomial is at least $\epsilon(r_i-|\kappa_j|)/\max(A,C)$, which is larger than Λ'/γ for $i\gg0$. Thus $\mathrm{mult}_{P'}h_{ij}^{\geq}=\mathrm{mult}_P h_{ij}$ for $i\gg0$, and

$$a(P,W_i,B_i)=a\Big(P',W_i',\sum_j c_{ij}\,\mathrm{div}(h_{ij}^{\geq})\Big).$$

The group G_i commutes with $\mathbf{T}$, so $h_{ij}^{\geq}$ is G_i-semi-invariant.

Let ϱ be a D-th root of t, let $v:=\varrho^{r_i}$, and let $X':=x'\varrho^A$, $Y':=y'\varrho^A$, $Z:=z\varrho^C$. By (B.97) and the definition of $h_{ij}^{\geq}$, there exist polynomials $P_{(l)}$ and $\bar{h}_{ijl}$ with

$$P_it^b=\varrho^{-2A}\Big(Z^a+\sum_{l\geq1}v^lP_{(l)}(Z)\Big),$$

$$h_{ij}^{\geq}=\varrho^{\kappa_j}\Big(\bar{h}_{ij}(X',Y',Z)+\sum_{l\geq1}v^l\bar{h}_{ijl}(X',Y',Z)\Big).$$

Let

$$X_i:=\Big(X'Y'+Z^a+\sum_{l\geq1}v^lP_{(l)}(Z)=0\Big)\subset\mathbb{A}^4_{v,X',Y',Z},$$

whose section $(v=0)$ is S. Let G_i act through its lifts on the cover of W_i' given by $t=\varrho^D$. The lift of R_i acts on ϱ by the r_i-th roots of unity and trivially on X',Y',Z, and the lifts of G_i induce an action on X_i of a group $\hat{G}_i$ of order $D|H_i|$, which acts on X',Y',Z as $G_{S,i}$ acts on

X, Y, Z, and on v by a character. If G_i is not diagonal, we replace X', Y' by $u_1' := X' + \epsilon_i Y'$ and $u_2' := X' - \epsilon_i Y'$ as in Construction-Proposition B.19(1). In the coordinates v, u_1', u_2', Z,

$$X_i = \Big(\frac{u_1'^2 - u_2'^2}{4\epsilon_i} + Z^a + \sum_{l\geq 1} v^l P_{(l)}(Z) = 0\Big) \quad \text{and} \quad S = \Big(\frac{u_1'^2 - u_2'^2}{4\epsilon_i} + Z^a = 0\Big)$$

are complete intersections on which $\hat{G}_i$ acts diagonally. The group $\hat{G}_i$ acts faithfully on S by Construction-Proposition B.19(2). A direct calculation shows that

$$\frac{dx' \wedge dz \wedge dt}{x'} = \frac{D}{r_i}\varrho^{D-C} \cdot \frac{dX' \wedge dZ \wedge dv}{vX'},$$

and $dX' \wedge dZ \wedge dv/X'$ generates the canonical module of X_i. Let $h_{ij}^\natural := \bar{h}_{ij} + \sum_{l\geq 1} v^l \bar{h}_{ijl}$, so that $h_{ij}^{\geqq} = \varrho^{\kappa_j} h_{ij}^\natural$. Since

$$\varrho^{D-C} \prod_j (\varrho^{\kappa_j})^{-c_{ij}} = \varrho^{-\delta_i} = v^{-\delta_i/r_i},$$

we obtain

$$\frac{dx' \wedge dz \wedge dt}{x'} \prod_j (h_{ij}^{\geqq})^{-c_{ij}} = \frac{D}{r_i} v^{-1-\delta_i/r_i} \frac{dX' \wedge dZ \wedge dv}{X'} \prod_j (h_{ij}^\natural)^{-c_{ij}}.$$

Hence the pairs

$$\Big(W_i', \sum_j c_{ij} \operatorname{div}(h_{ij}^{\geqq})\Big)/G_i \quad \text{and} \quad \Big(X_i, (1+\delta_i/r_i)\operatorname{div}(v) + \sum_j c_{ij} \operatorname{div}(h_{ij}^\natural)\Big)/\hat{G}_i$$

are crepant birational. Let P'' be a prime divisor over X_i lying over P'. The center of P'' is the origin, since v, X', Y', Z have positive values. Since the Cartier divisor S of X_i is normal and klt, X_i is normal near S. Since moreover $K_{X_i} + S$ is Cartier, (X_i, S) is plt near S by [KM98, Theorem 5.50(1)]. Thus X_i is klt near S, since $a(E, X_i, 0) = a(E, X_i, S) + \operatorname{mult}_E S > 0$ for any exceptional prime divisor E over X_i whose center intersects S. By Construction-Proposition B.19(2), (3), $(S, \bar{B}_i)$ is lc near 0 and $\operatorname{mld}((S \ni 0, \bar{B}_i)/\hat{G}_i) = \beta_i^+$. By Lemma 5.5 with $u = v$, the coefficient $1 + \delta_i/r_i \leq 1$ of S, and the $\mathbb{R}$-ideal $\prod_j (h_{ij}^\natural)^{c_{ij}}$, and by [Liu26, Lemma 2.6(2)],

$$\begin{aligned}\frac{a(P, W_i, B_i)}{e} &= \frac{a(P'', X_i, (1+\delta_i/r_i)\operatorname{div}(v) + \sum_j c_{ij}\operatorname{div}(h_{ij}^\natural))}{|I_{\hat{G}_i}(P'')|} \\ &\geq \beta_i^+ - \frac{\delta_i \operatorname{mult}_{P''} v}{r_i |I_{\hat{G}_i}(P'')|} \geq \beta_i^+ = \beta_i.\end{aligned}$$

Step 4. In this step we treat the case when $[q_1] < \epsilon$ for some coordinate q_1 of negative weight. By **Step 2**, $[t] \geq \epsilon_t := 1/(2e_0 \max(A, C))$. Let $\mathfrak{u}' := P_i/z^a$. By (B.97), any monomial of $\mathfrak{u}' - 1$ is $z^{m-a}t^n$ with $n \geq (r_i - aC)/D$, so

$$[\mathfrak{u}' - 1] \geq \frac{(r_i - aC)\epsilon_t}{D} - a\Lambda =: \Upsilon_i' \to +\infty.$$

For $i \gg 0$, $\Upsilon_i' > \Lambda'/\gamma$, so $\mathfrak{u}'$ has a square root $\varsigma' \in \widehat{K}_P$ with $[\varsigma' - 1] = [\mathfrak{u}' - 1]$. The argument of **Step 3** with ς' in place of ς yields $x'' := x/\varsigma'$, $y'' := y/\varsigma'$, $W_i'' := (x''y'' + z^a Q_i = 0)$ and a prime divisor P' over W_i'' with center 0 such that $I_{G_i}(P') = I_{G_i}(P)$, $a(P', W_i'', 0) = a(P, W_i, 0)$, and $\operatorname{mult}_{P'} h_{ij}(x'', y'', z, t) = \operatorname{mult}_P h_{ij}$ for $i \gg 0$. Write $h_{ij}(x'', y'', z, t) = h_{ij}^{>} + h_{ij}^{\leqq}$, where $h_{ij}^{>}$ is the sum of the monomials of $\mathbf{T}$-weight $\kappa_j + lr_i$ with $l > 0$. A monomial $x''^{n_1}y''^{n_2}z^{n_3}t^{n_4}$ of $h_{ij}^{>}$ satisfies $Dn_4 \geq r_i - |\kappa_j|$, so the value of the monomial is at least $\epsilon_t(r_i - |\kappa_j|)/D$, which is larger than Λ'/γ for $i \gg 0$. Thus $\operatorname{mult}_{P'} h_{ij}^{\leqq} = \operatorname{mult}_P h_{ij}$.

Let $q \in \{u_1, u_2, z\}$ be a coordinate of negative weight with the least value ρ of $[q]/|w_q|$. Since $\min([x], [y]) = \min([u_1], [u_2])$ and $|w_q| \geq 1$, $\rho \leq [q_1] < \epsilon$. Let ϱ be an $|w_q|$-th root of q, which has $\mathbf{T}$-weight -1, let $v := \varrho^{r_i}$, and for any other coordinate q' let $q'\varrho^{w_{q'}}$ be the corresponding

coordinate of S_q. The equation $x''y'' + z^a Q_i$ has $\mathbf{T}$-weight $-2A$, and by (B.97) the monomials of $x''y'' + z^a Q_i$ have weights $-2A - lr_i$ with $l \geq 0$. Hence $x''y'' + z^a Q_i = \varrho^{2A}(\mathcal{F}_0 + \sum_{l\geq 1} v^l \mathcal{F}_l)$, where $\mathcal{F}_0 = 0$ is the equation of S_q in the coordinates of S_q and the $\mathcal{F}_l$ are polynomials in the coordinates of S_q. Since the monomials of $h_{ij}^{\leq}$ have $\mathbf{T}$-weights $\kappa_j + lr_i$ with $l \leq 0$, there exist polynomials $\bar{h}'_{ijl}$ in the coordinates of S_q with

$$h_{ij}^{\leq} = \varrho^{-\kappa_j}\Big(h_{ij}^0|_{S_q} + \sum_{l\geq 1} v^l \bar{h}'_{ijl}\Big).$$

Let $X_i^q := (\mathcal{F}_0 + \sum_{l\geq 1} v^l \mathcal{F}_l = 0)$, a hypersurface in the affine 4-space with coordinate v and the coordinates of S_q, with section $(v = 0) = S_q$. As in **Step 3**, the lifts of G_i act on X_i^q through a group $\hat{G}_i^q$ of order $|w_q| \cdot |H_i|$, which acts on S_q as $G_{q,i}$, diagonally and faithfully, and on v by a character. Let s_1, s_2, s_3 be the coordinates $q'\varrho^{w_{q'}}$, $q' \neq q$, of S_q, and let Ω_i^q be the residue of $dv \wedge ds_1 \wedge ds_2 \wedge ds_3$ along X_i^q, which generates the canonical module of X_i^q. The $\mathbf{T}$-weights of the coordinates of $\mathbb{A}^4$ sum to $-2A - C + D$, so in the coordinates ϱ, s_1, s_2, s_3 we have

$$dx'' \wedge dy'' \wedge dz \wedge dt = c_q \varrho^{2A+C-D} \frac{d\varrho}{\varrho} \wedge ds_1 \wedge ds_2 \wedge ds_3$$

for a constant $c_q \neq 0$. Since $x''y'' + z^a Q_i = \varrho^{2A}(\mathcal{F}_0 + \sum_{l\geq 1} v^l \mathcal{F}_l)$ and $\frac{d\varrho}{\varrho} = \frac{dv}{r_i v}$, taking residues gives

$$\frac{dx'' \wedge dz \wedge dt}{x''} = \pm \frac{c_q}{r_i} \varrho^{C-D} \cdot \frac{\Omega_i^q}{v}.$$

Let $h_{ij}^{\flat} := h_{ij}^0|_{S_q} + \sum_{l\geq 1} v^l \bar{h}'_{ijl}$, so that $h_{ij}^{\leq} = \varrho^{-\kappa_j} h_{ij}^{\flat}$. Since

$$\varrho^{C-D} \prod_j (\varrho^{-\kappa_j})^{-c_{ij}} = \varrho^{\delta_i} = v^{\delta_i/r_i},$$

we obtain

$$\frac{dx'' \wedge dz \wedge dt}{x''} \prod_j (h_{ij}^{\leq})^{-c_{ij}} = \pm \frac{c_q}{r_i} v^{-1+\delta_i/r_i} \Omega_i^q \prod_j (h_{ij}^{\flat})^{-c_{ij}}.$$

Hence the pairs

$$\Big(W_i'', \sum_j c_{ij} \operatorname{div}(h_{ij}^{\leq})\Big)/G_i \quad \text{and} \quad \Big(X_i^q, (1 - \delta_i/r_i)\operatorname{div}(v) + \sum_j c_{ij} \operatorname{div}(h_{ij}^{\flat})\Big)/\hat{G}_i^q$$

are crepant birational.

Let P'' be a prime divisor over X_i^q lying over P'. For any G_i-semi-invariant rational function f on W_i'', $\operatorname{mult}_{P''} f/|I_{\hat{G}_i^q}(P'')| = [f]$, since both sides are the value at the image of P' on the common quotient. In particular $\operatorname{mult}_{P''} v/(r_i|I_{\hat{G}_i^q}(P'')|) = [q]/|w_q| = \rho$. The coordinates $q'\varrho^{w_{q'}}$ of S_q have values $[q'] - |w_{q'}|\rho \geq 0$ if $w_{q'} < 0$, and the coordinate of S_q corresponding to t and the coordinates of weight zero have positive values. Thus the center of P'' is contained in $\{0\} \times S_q$ and in the zero locus of the coordinate of S_q corresponding to t and of the coordinates of weight zero, where all non-constant $\mathbf{T}$-invariant monomials vanish. So it is a point p_0 of S_q over C_i^- or, if $A > 0$ and the least value of $[q]/|w_q|$ is attained by two coordinates, a curve $C'' \subset S_q$ whose image $\bar{C}''$ in Y_i^- is a component of C_i^-. Since the Cartier divisor S_q of X_i^q is normal and klt and $K_{X_i^q} + S_q$ is Cartier, X_i^q is normal and klt near S_q by [KM98, Theorem 5.50(1)], as in **Step 3**. The log discrepancy of P'' with respect to X_i^q and the boundary with coefficient c_v of $\operatorname{div}(v)$ is an affine function of c_v with slope $-\operatorname{mult}_{P''} v$.

If the center of P'' is p_0, the stabilizer K_{p_0} of p_0 in $\hat{G}_i^q$ contains $I_{\hat{G}_i^q}(P'')$ and acts diagonally in the coordinates translated to p_0, since it fixes the non-zero coordinates of p_0. Near the image

of p_0, Y_i^- is the quotient of S_q by K_{p_0}, so

$$\mathrm{mld}\Big((S_q \ni p_0, \sum_j c_{ij}\,\mathrm{div}(h_{ij}^0|_{S_q}))/K_{p_0}\Big) \geq \beta_i^-.$$

If the center of P'' is C'', then $H := I_{\hat{G}_i^q}(P'')$ fixes C'' pointwise, so it acts diagonally in the coordinates translated to a closed point $c_0 \in C''$, and $H \subset I_{G_{q,i}}(C'')$. Thus the minimal log discrepancy of the quotient by H at the image of the generic point of C'' is

$$\frac{a(C'', S_q, \sum_j c_{ij}\,\mathrm{div}(h_{ij}^0|_{S_q}))}{|H|} \geq \frac{a(C'', S_q, \sum_j c_{ij}\,\mathrm{div}(h_{ij}^0|_{S_q}))}{|I_{G_{q,i}}(C'')|} = a(\bar{C}'', Y_i^-, B_i^-) \geq \beta_i^-.$$

By Lemma 5.5 with the coefficient one of S_q and the $\mathbb{R}$-ideal $\prod_j (h_{ij}^\flat)^{c_{ij}}$, applied at p_0 to K_{p_0} if the center of P'' is p_0, and by its second part, applied at c_0 to H if the center of P'' is C'', together with the affine dependence on c_v,

$$\frac{a(P, W_i, B_i)}{e} \geq \beta_i^- + \frac{\delta_i\,\mathrm{mult}_{P''} v}{r_i |I_{\hat{G}_i^q}(P'')|} = \beta_i^- + \delta_i\rho \geq \beta_i^+ + \frac{|\delta_i|}{d_0} - |\delta_i|\epsilon \geq \beta_i,$$

where we use Lemma B.20(3), $\delta_i \leq 0$, $\rho < \epsilon$, and $\epsilon = 1/(2d_0)$.

Step 5. In this step we treat the case (b). Then $\beta_i = \beta_i^-$ by Lemma B.20(3). **Step 1** holds by the second case of Lemma B.21, since for any q with $w_q < 0$ the pair $(S_q, \theta \sum_j c_{ij}\,\mathrm{div}(h_{ij}^0|_{S_q}))$ is lc near the points over C_i^-, and the proof of **Step 2** uses (a) only through **Step 1**, so **Step 2** holds. If $[t] \geq \epsilon$, then the argument of **Step 4** with the bound $[t] \geq \epsilon$ in place of $[t] \geq \epsilon_t$ yields $a(P, W_i, B_i)/e \geq \beta_i^- + \delta_i\rho \geq \beta_i$. If $[t] < \epsilon$, then (B.99) implies $[q] \geq (1/e_0 - \max(A, C)\epsilon)/D \geq 1/(2e_0 D) \geq \epsilon$ for any q of negative weight, and the argument of **Step 3**, with Lemma 5.5 applied with the coefficient one of S and with the log discrepancy of P'' affine in the coefficient of $\mathrm{div}(v)$ with slope $-\mathrm{mult}_{P''} v$ as in **Step 4**, yields

$$\frac{a(P, W_i, B_i)}{e} \geq \beta_i^+ - \frac{\delta_i\,\mathrm{mult}_{P''} v}{r_i |I_{\hat{G}_i}(P'')|} = \beta_i^+ - \frac{\delta_i [t]}{D} \geq \beta_i^- + \frac{\delta_i}{d_0} - \delta_i\epsilon \geq \beta_i,$$

where $\mathrm{mult}_{P''} v/(r_i |I_{\hat{G}_i}(P'')|) = [t]/D$ since $v^D = t^{r_i}$.

The integers i excluded in **Steps 1–5** depend only on θ, Γ, s, the κ_j and the integers of Set-up B.17, and not on P. This proves the lemma. □

Lemma B.23. *Notation as in Set-up B.17 and Construction-Proposition B.19, and assume that (V_0, B_i^0) is lc near 0. There exists a positive real number Λ_0, independent of i, such that $\alpha_i \leq \beta_i + \Lambda_0/r_i$ for $i \gg 0$.*

Proof. Let γ be the least element of $\Gamma\backslash\{0\}$ ($\gamma := 1$ if $\Gamma \subset \{0\}$), $g_1 := e_0 \max(A, C, D)$, and $d_1 := \max(a, b)$. We use the sign $+$ if $\delta_i \leq 0$ and the sign $-$ if $\delta_i > 0$, so $\beta_i = \beta_i^\pm$ by Lemma B.20(3). By Theorem 2.9(2) applied near $C_i^\pm$, where $B_i^\pm \in \Gamma_1$ (Construction-Proposition B.19(3)), there exists a prime divisor ν over $Y_i^\pm$ with center in $C_i^\pm$ computing β_i such that $a(\nu, Y_i^\pm, 0) \leq l_\Gamma$ for an integer l_Γ depending only on Γ. If the center of ν is a curve, then ν is that curve and $a(\nu, Y_i^\pm, 0) = 1$. Let $q := t$ in the first case, and in the second case let q be a coordinate of negative weight such that the center of ν intersects $S_q/G_{q,i}$ (Lemma B.20(1)). Fix a point $p_0 \in S_q$ over a point of the center, and a prime divisor ν' over S_q lying over ν whose center contains p_0. Let K_{p_0} be the stabilizer of p_0 in $G_{q,i}$ and $e' := |I_{K_{p_0}}(\nu')| \leq g_1$. Since $I_{G_{q,i}}(\nu')$ fixes the center of ν' pointwise, it is contained in K_{p_0}, so $a(\nu', S_q, B_{S_q})/e' = \beta_i$, where $B_{S_q} := \sum_j c_{ij}\,\mathrm{div}(h_{ij}^0|_{S_q})$, and if the center of ν' is p_0, then ν' computes $\mathrm{mld}((S_q \ni p_0, B_{S_q})/K_{p_0}) = \beta_i$. Moreover $a(\nu', S_q, 0) \leq e' a(\nu, Y_i^\pm, 0) \leq g_1 l_\Gamma$ by [Liu26, Lemma 2.6(2)]. The surface S_q is smooth at p_0 or p_0 is an A_{d-1} point with $d \leq d_1$. Let ν'' be a prime divisor over the smooth cover $\mathbb{A}^2_{\xi_1,\xi_2} \to (S_q \ni p_0)$ (the identity if S_q is smooth at p_0, with local parameters ξ_1, ξ_2) lying over ν', with ramification

index r''. Then $a(\nu'', \mathbb{A}^2, 0) = r'' a(\nu', S_q, 0)$ by [Liu26, Lemma 2.6(2)], and the value of ν'' on any non-zero linear form in ξ_1, ξ_2 is at most $a(\nu'', \mathbb{A}^2, 0)$, since

$$a\big(\nu'', \mathbb{A}^2, \operatorname{div}(\ell) + \operatorname{div}(\ell')\big) \geq 0$$

for any basis ℓ, ℓ' of the linear forms [KM98, Corollary 2.31(3)]. Near p_0, each coordinate of S_q vanishing at p_0 is, up to a unit, a product of at most $a + b \leq 2d_1$ linear forms in ξ_1, ξ_2: a monomial, or, for u_1 and u_2 at the origin of S_t or of S_z when G_i is not diagonal, a form $c\xi_1^d + c'\xi_2^d$ with $cc' \neq 0$. So, since $d_1 \geq 2$, their values at ν' are at most

$$\frac{(a+b)a(\nu'', \mathbb{A}^2, 0)}{r''} = (a+b)a(\nu', S_q, 0) \leq 2d_1 g_1 l_\Gamma \leq L_1 := \frac{d_1^2 g_1 l_\Gamma}{\gamma}.$$

Since $\sum_j c_{ij} \operatorname{mult}_{\nu'} h^0_{ij}|_{S_q} \leq a(\nu', S_q, 0)$ with $c_{ij} \geq \gamma$, the values of ν' on the $h^0_{ij}|_{S_q}$ are at most $g_1 l_\Gamma/\gamma \leq L_1$. The values of ν' on the coordinates vanishing on the center of ν' are positive integers.

Let X_i^q, v, ϱ, $\hat{G}_i^q$, and $h^\natural_{ij}$, resp. $h^\flat_{ij}$, be as in **Steps 3 and 4** of the proof of Lemma B.22 ($X_i^t = X_i$ and $\hat{G}_i^t = \hat{G}_i$), constructed from $W_i' = (x'y' + P_i t^b = 0)$ if $q = t$ and from $W_i'' = (x''y'' + z^a Q_i = 0)$ if $w_q < 0$, and let $b_v := 1 + \delta_i/r_i$ if $q = t$ and $b_v := 1 - \delta_i/r_i$ if $w_q < 0$, so $b_v \leq 1$. The chart equation is $\mathcal{F}_0 + \Xi = 0$, where $\mathcal{F}_0 = 0$ is the equation of S_q and $\Xi \in v\mathbb{C}[v, \ldots]$. Let $V^\flat := S_q \times \mathbb{A}^1_v$ with the action of $\hat{G}_i^q$, and let $\varkappa$ be an integer with $\varkappa \equiv \varkappa_\nu \pmod{e'}$, where $\varkappa_\nu$ is the integer of Lemma 3.1 for ν' and the character of v used in Proposition 5.4, and $(a+b+1)L_1 < \varkappa \leq (a+b+1)L_1 + g_1$. If the center of ν' is p_0, let F be the Gauss extension of ν' with v-value $\varkappa$. By Proposition 5.4 applied to K_{p_0} and $\Phi_j := \operatorname{pr}^* h^0_{ij}|_{S_q}$, F is a prime divisor over $V^\flat$ with center $(p_0, 0)$, $I_{K_{p_0}}(F) = I_{K_{p_0}}(\nu')$, $\operatorname{mult}_F \Phi_j = \operatorname{mult}_{\nu'} h^0_{ij}|_{S_q}$, and

$$\frac{a(F, V^\flat, b_v \operatorname{div}(v) + \sum_j c_{ij} \operatorname{div}(\Phi_j))}{|I_{K_{p_0}}(F)|} = \beta_i + \frac{(1-b_v)\varkappa}{e'}. \tag{B.100}$$

If the center of ν' is a curve $C_\nu \ni p_0$, let F be the monomial valuation with values 1 and $\varkappa$ on a local equation of C_ν and on v at the generic point of $C_\nu \times \{0\}$. The cyclic group $I_{K_{p_0}}(\nu')$ acts on a local equation of C_ν by a faithful character and on v by the $\varkappa_\nu$-th power of that character, so the residue $v/\pi^\varkappa$ of F, with π the local equation, is $I_{K_{p_0}}(\nu')$-invariant, $I_{K_{p_0}}(F) = I_{K_{p_0}}(\nu')$, and (B.100) holds since $a(F, \cdot) = a(\nu', S_q, B_{S_q}) + (1 - b_v)\varkappa$.

Next we pass from $V^\flat$ to X_i^q. Write $\mathcal{F}_0 = \mathcal{Q} + \mathcal{R}_0$, where $\mathcal{Q}$ is the part of $\mathcal{F}_0$ in the coordinates of S_q coming from x, y (or u_1, u_2) and $\mathcal{R}_0$ is the rest. If $q = t$, resp. $q = z$, let $\tilde{u}_1, \tilde{u}_2$ be the coordinates of S_q coming from x, y (or u_1, u_2). Then $\mathcal{Q}$ is a non-degenerate quadratic form in $\tilde{u}_1, \tilde{u}_2$, $\mathcal{R}_0 = Z^a$, resp. $\mathcal{R}_0 = T^b$ with T the coordinate of S_z corresponding to t, and Ξ is a polynomial in v and the variable of $\mathcal{R}_0$. In this case let ς_1 be the square root of $\mathcal{R}_0/(\mathcal{R}_0 + \Xi)$ with $\operatorname{mult}_F(\varsigma_1 - 1) > 0$, and divide $\tilde{u}_1$ and $\tilde{u}_2$ by ς_1. If $q \in \{u_1, u_2\}$ and G_i is diagonal, then $\mathcal{F}_0 + \Xi$ is the other coordinate coming from x, y (or u_1, u_2) plus a polynomial in the remaining two coordinates and v, so X_i^q is isomorphic to $V^\flat$ by solving for that coordinate. In this case let $\varsigma_1 := 1$. If $q \in \{u_1, u_2\}$ and G_i is not diagonal, let u' be the other coordinate of S_q coming from u_1, u_2, which is ± 1 at p_0. Then $\mathcal{F}_0 = (1 - u'^2)/(4\epsilon_i) + Z^a T^b$ if $q = u_1$, and $\mathcal{F}_0 = (u'^2 - 1)/(4\epsilon_i) + Z^a T^b$ if $q = u_2$. In this case let ς_1 be the square root of $1 + 4\epsilon_i \Xi/u'^2$ if $q = u_1$, resp. of $1 - 4\epsilon_i \Xi/u'^2$ if $q = u_2$, with $\operatorname{mult}_F(\varsigma_1 - 1) > 0$, and multiply u' by ς_1.

In each case ς_1 is $\hat{G}_i^q$-invariant, since $\mathcal{R}_0$ and $\mathcal{R}_0 + \Xi$ have equal characters (if $q \in \{u_1, u_2\}$ and G_i is not diagonal, $\mathcal{F}_0$ contains the non-zero constant $\pm 1/(4\epsilon_i)$, so $\mathcal{F}_0$, u'^2, and Ξ are $\hat{G}_i^q$-invariant), and $\varsigma_1 - 1$ has value at least $\varkappa - (a+b)L_1 > L_1$ at F, since any monomial of Ξ is divisible by v. Thus the substitution embeds $\mathbb{C}(X_i^q)$ $\hat{G}_i^q$-equivariantly into the completion of $\mathbb{C}(V^\flat)$ along F as a dense subfield (since ς_1^2 is a rational function on X_i^q and $\operatorname{mult}_F(\varsigma_1 - 1) > 0$, Hensel's lemma shows that ς_1 belongs to the closure of $\mathbb{C}(X_i^q)$), and F induces a prime divisor F_1 over X_i^q whose residue field, value group, and inertia group are those of F. The canonical

forms $dX \wedge dZ \wedge dv/X$, resp. the residue forms of the hypersurfaces, differ by a unit of value zero. The marks $h_{ij}^{\natural}$, resp. $h_{ij}^{\flat}$, differ from Φ_j by terms divisible by v and by terms $(\varsigma_1^m - 1) \cdot n$ with n a monomial, which have values larger than $L_1 \geq \mathrm{mult}_F \Phi_j$. Hence F_1 satisfies (B.100) on X_i^q with the boundary

$$b_v \operatorname{div}(v) + \sum_j c_{ij} \operatorname{div}(h_{ij}^{\natural}),$$

resp. with $h_{ij}^{\flat}$ in place of $h_{ij}^{\natural}$.

By the crepant birational equivalence of **Step 3**, resp. **Step 4**, of the proof of Lemma B.22, F_1 induces a prime divisor P' over W_i', resp. W_i'', with

$$\frac{a(P', W_i', \sum_j c_{ij} \operatorname{div}(h_{ij}^{\geqslant}))}{|I_{G_i}(P')|} = \beta_i + \frac{(1 - b_v)\varkappa}{e'},$$

resp. with W_i'' and $h_{ij}^{\leqslant}$. Consider first $q = t$. A monomial of $h_{ij}(x', y', z, t) - h_{ij}^{\geqslant}$ of weight $\kappa_j + lr_i$, $l < 0$, is $\varrho^{\kappa_j} v^l$ times a monomial in X', Y', Z of degree at least $(|l|r_i - |\kappa_j|)/\max(A, C)$, and X', Y', Z vanish on the center of ν'. So the value of the monomial at F_1 minus the value of ϱ^{κ_j} is at least $|l|(r_i/\max(A, C) - \varkappa) - |\kappa_j|$, which is larger than L_1 for $i \gg 0$, uniformly in l. Thus $\mathrm{mult}_{P'} h_{ij}(x', y', z, t) = \mathrm{mult}_{P'} h_{ij}^{\geqslant}$. A monomial of $Q_i/t^b - 1$ is v^{-l} times a monomial in Z of degree at least $(lr_i - bD)/C$ with $l \geq 1$, so the value of the monomial is at least $l(r_i/C - \varkappa) - bD/C > L_1$ for $i \gg 0$. As in **Step 3** of the proof of Lemma B.22, the square root ς of Q_i/t^b with $\mathrm{mult}_{F_1}(\varsigma - 1) > 0$ and $x = \varsigma x'$, $y = \varsigma y'$ yield a prime divisor P over W_i with $a(P, W_i, B_i)/|I_{G_i}(P)| = \beta_i + (1 - b_v)\varkappa/e'$. If $w_q < 0$, then a monomial of $h_{ij}(x'', y'', z, t) - h_{ij}^{\leqslant}$, resp. of $P_i/z^a - 1$, is $\varrho^{-\kappa_j} v^{-l}$, resp. v^{-l}, with $l \geq 1$, times a monomial of degree at least $(lr_i - |\kappa_j|)/D$, resp. $(lr_i - aC)/D$, in the coordinate of S_q corresponding to t, which vanishes on the center of ν'. If $q \in \{u_1, u_2\}$, a monomial of $P_i/z^a - 1$ also contains a power of the coordinate of S_q corresponding to z with exponent in $[-a, -1]$, whose value at F_1 is at least $-aL_1$. So the value of such a monomial at F_1 is at least $(lr_i - aC)/D - l\varkappa - aL_1$, which is larger than L_1 for $i \gg 0$, uniformly in l. The estimates of the case $q = t$ and the square root of P_i/z^a yield P. The values of $t = \varrho^D$ and of $x = \varsigma X' \varrho^{-A}$, $y = \varsigma Y' \varrho^{-A}$, $z = Z\varrho^{-C}$ at F_1 are $D\varkappa/r_i$ and at least $1 - \max(A, C)\varkappa/r_i$, which are positive for $i \gg 0$. For $w_q < 0$, the coordinate of S_q corresponding to t and the coordinates of weight zero vanish on the center of ν', and the estimate of the case $q = t$ shows that $\min(\mathrm{mult}_{F_1} x, \mathrm{mult}_{F_1} y, \mathrm{mult}_{F_1} z, \mathrm{mult}_{F_1} t) > 0$ for $i \gg 0$. Thus the center of P on W_i is 0. Since $|1 - b_v| = |\delta_i|/r_i$, $e' \geq 1$, $\varkappa \leq (a + b + 1)L_1 + g_1$, and $|\delta_i| \leq \sum_j |\kappa_j| + C + D$,

$$\alpha_i \leq \beta_i + \frac{|\delta_i|\varkappa}{r_i} \leq \beta_i + \frac{\Lambda_0}{r_i}, \qquad \Lambda_0 := \Big(\sum_j |\kappa_j| + C + D\Big)\big((a + b + 1)L_1 + g_1\big). \tag{B.101}$$

□

Proof of Theorem 8.13. The proof is long, so we divide it into several steps. Assume to the contrary that $\alpha_i := \mathrm{mld}((W_i \ni o_i, B_i)/G_i)$ form a strictly increasing sequence of positive numbers. Let γ be the least element of $\Gamma \backslash \{0\}$, and $\gamma := 1$ if $\Gamma \subset \{0\}$.

Step 1. In this step we reduce to the data of Set-up B.17 with (V_0, B_i^0) lc near 0. The marks with label zero and the marks that are units near o_i do not change B_i, so we omit them. By [Liu26, Lemma 5.4(2) and its last paragraph], $W_i = (f = 0)$ in a smooth fourfold germ $(\mathcal{M}_i \ni o_i)$ with a G_i-semi-invariant f and G_i-eigenparameters $x_1, \ldots, x_4$, and the marks lift to G_i-semi-invariant regular functions h_j on $\mathcal{M}_i$. Let $\mathfrak{R} := \mathbb{C}[[x_1, \ldots, x_4]]$. The composite of $\mathfrak{R} \to \widehat{\mathcal{O}}_{W_i, o_i}$ with the given isomorphism onto $\mathbb{C}[[x, y, z, t]]/(xy + \hat{g}_i)$ is a G_i-equivariant surjection. Since G_i is finite abelian, the projection onto the χ-component for a character χ of G_i commutes with the quotient map from $\mathbb{C}[[x, y, z, t]]$ onto $\mathbb{C}[[x, y, z, t]]/(xy + \hat{g}_i)$, so the image of each x_k has a G_i-semi-invariant lift $\tilde{x}_k \in \mathbb{C}[[x, y, z, t]]$ with the character of x_k. Sending x_k to $\tilde{x}_k$ defines a G_i-equivariant homomorphism $\psi_0 : \mathfrak{R} \to \mathbb{C}[[x, y, z, t]]$. Since $\hat{f} \in (x_1, \ldots, x_4)^2$ and $xy + \hat{g}_i \in (x, y, z, t)^2$, ψ_0

induces a surjection, hence an isomorphism, of the four-dimensional cotangent spaces, so ψ_0 is an isomorphism. Moreover $\psi_0(\hat{f})\mathbb{C}[[x,y,z,t]] = (xy+\hat{g}_i)$, since ψ_0 induces the given isomorphism of the quotients.

The homogeneous part of degree $a+b$ of $\hat{g}_i$ is z^at^b, a product of the coprime forms z^a and t^b. By Hensel's lemma $\hat{g}_i = \hat{P}_1\hat{Q}_1$ with homogeneous parts of lowest degree z^a and t^b, and by the Weierstrass preparation theorem $\hat{P}_1$ and $\hat{Q}_1$ are units times monic polynomials $\hat{P} \in \mathbb{C}[[t]][z]$ of degree a and $\hat{Q} \in \mathbb{C}[[z]][t]$ of degree b. Thus $\hat{g}_i = \mathfrak{u}_0\hat{P}\hat{Q}$ for a unit $\mathfrak{u}_0$. For $g \in G_i$, $g^*(xy+\hat{g}_i) = u_g(xy+\hat{g}_i)$ for a unit u_g, since G_i preserves the ideal $(xy+\hat{g}_i)$, and comparing the coefficients of x^my^n, using that G_i acts monomially on $\{x,y\}$, shows that u_g is a constant. So $\hat{g}_i$ is G_i-semi-invariant with the character of z^at^b. Since G_i is diagonal on z,t, for any $g \in G_i$ the power series $g^*\hat{P}_1$ and $g^*\hat{Q}_1$ have lowest homogeneous parts proportional to z^a and t^b, and their product is a constant multiple of $\hat{g}_i$, so by the uniqueness up to units of the factorization in Hensel's lemma G_i preserves the ideals $(\hat{P}_1)$ and $(\hat{Q}_1)$, and by the uniqueness of monic Weierstrass polynomials $\hat{P}$ and $\hat{Q}$ are semi-invariant with the characters of z^a and t^b. Hence $\mathfrak{u}_0$ is G_i-invariant. Let $\mathfrak{u}_0^{1/2}$ be the square root with constant term a fixed square root of $\mathfrak{u}_0(0)$. The function $\mathfrak{u}_0^{1/2}$ is G_i-invariant. Replacing x,y with $\mathfrak{u}_0^{1/2}x, \mathfrak{u}_0^{1/2}y$, which commutes with the action of G_i on the pair $\{x,y\}$, the equation becomes $\mathfrak{u}_0(xy+\hat{P}\hat{Q})$. Thus there exist a G_i-equivariant isomorphism ψ and a G_i-invariant unit v_0 with $v_0\psi(\hat{f}) = xy+\hat{P}\hat{Q}$.

Fix integers N and L. We take N large and then L large depending on N. For $k<a$, resp. $k<b$, let $P_{i,k}$, resp. $Q_{i,k}$, be the sum of the terms of degree less than L of the coefficient of z^k in $\hat{P}$, resp. of t^k in $\hat{Q}$, and let $P_i := z^a + \sum_{k<a} P_{i,k}z^k$ and $Q_i := t^b + \sum_{k<b} Q_{i,k}t^k$. The polynomials P_i and Q_i are monic, semi-invariant with the characters of z^a, t^b, and $xy+P_iQ_i \equiv v_0\psi(\hat{f}) \pmod{(x,y,z,t)^L}$. Let $h_{j,0}$ be the sum of the terms of degree less than N of $\psi(\hat{h}_j)$. The polynomial $h_{j,0}$ is semi-invariant since G_i acts linearly. For such N and L, Construction-Theorem 4.13 applied to ψ, v_0 in place of v, $u_j = 1$, and G_i (in the eigencoordinates of Construction-Proposition B.19(1)) shows that $(xy+P_iQ_i=0)$ is an isolated Gorenstein terminal germ with a faithful action of G_i, that the pair with the polynomial marks $h_{j,0}$ is klt near 0, and that the quotient value of the pair is α_i. Replacing xy with $-P_iQ_i$ in the monomials of $h_{j,0}$ divisible by xy, which does not change the restriction to $(xy+P_iQ_i=0)$, we may assume that the $h_{j,0}$ have no monomial divisible by xy. We replace W_i with $(xy+P_iQ_i=0)$ and the marks h_j with $h_{j,0}$.

Since $\operatorname{mult}_0(P_iQ_i) = a+b \geq 3$, the exceptional divisor of the blow-up of 0 in W_i contains the prime divisors E_1, E_2 over the planes $x=0$ and $y=0$ of $(xy=0) \subset \mathbb{P}^3$, with $a(E_1,W_i,0)=2$, and $\operatorname{mult}_{E_1} h \geq \operatorname{mult}_0 h \geq 1$ for any mark h. Since (W_i,B_i) is klt, $\sum_j c_{ij}\operatorname{mult}_{E_1} h_{j,0} < 2$. Hence the number of marks is less than $2/\gamma$, and any mark has a monomial of total degree less than $2/\gamma$. Passing to a subsequence, the number s of marks is fixed and, for any j, the $\mathbf{T}$-weight κ_j of a monomial of least degree of the j-th mark is fixed. All monomials of a G_i-semi-invariant polynomial have $\mathbf{T}$-weights congruent modulo r_i, and a monomial of total degree m has weight of absolute value at most $m\max(A,C,D)$. Thus any monomial of the j-th mark of weight different from κ_j has total degree at least $(r_i - |\kappa_j|)/\max(A,C,D) \to +\infty$. The principal parts h_{ij}^0 are non-zero on V_0, since a non-zero polynomial without monomials divisible by xy is not divisible by $xy+z^at^b$. Thus the data satisfy Set-up B.17, and by Lemma B.18(3) with $\Gamma' = \Gamma$, (V_0, B_i^0) is lc near 0 for $i \gg 0$.

Step 2. In this step we take limits of the minima and of the labels. By Lemma B.23, $\alpha_i \leq \beta_i + \Lambda_0/r_i$. The numbers β_i belong to the ACC set $\mathcal{B}$ of Construction-Proposition B.19(3), so after passing to a subsequence they are non-increasing with limit β, and $\beta \geq \lim_i \alpha_i > \alpha_i > 0$ for any i. Since Γ is a DCC set, after passing to a subsequence c_{ij} is non-decreasing in i with limit $c_j^* \in (0,1]$ for any j. Let $\tau^* := \sum_j c_j^*\kappa_j$ and $\delta^* := \tau^* + C - D$. The groups $G_{S,i}$ act on S monomially in X,Y and diagonally in Z, and they have order at most De_0. After replacing x,y with $\vartheta_i x, \vartheta_i^{-1}y$ for constants ϑ_i, which preserves xy, and passing to a subsequence, $G_{S,i} = G_S$ is a fixed group and the characters of the $\bar{h}_{ij}$ do not depend on i: if $G_{S,i}$ contains an element g

with $g^*X = \alpha Y$ and $g^*Y = \alpha' X$, then $\vartheta_i^4 = \alpha'/\alpha$ makes $\alpha = \alpha'$, and then the coefficients of the elements of $G_{S,i}$ are roots of unity of order at most $2De_0$, so there are only finitely many such groups. Let $\mathbb{A}^2_{\xi_1,\xi_2} \to S$, $X = \xi_1^a$, $Y = -\xi_2^a$, $Z = \xi_1\xi_2$, be the quotient by μ_a acting by (ζ, ζ^{-1}), which is quasi-étale, and let $\tilde{G} \subset GL_2(\mathbb{C})$ be the group of the lifts of the elements of G_S, of order $a|G_S| = aD|H_i|$. Here an element with $g^*X = \alpha_X X$, $g^*Y = \alpha_Y Y$, $g^*Z = \alpha_Z Z$ lifts to the linear maps $\tilde{g}$ with $\tilde{g}^*\xi_1 = \beta\xi_1$, $\tilde{g}^*\xi_2 = \alpha_Z\beta^{-1}\xi_2$, $\beta^a = \alpha_X$, and an element with $g^*X = \alpha Y$, $g^*Y = \alpha' X$, $g^*Z = \alpha_Z Z$ lifts to the $\tilde{g}$ with $\tilde{g}^*\xi_1 = \beta\xi_2$, $\tilde{g}^*\xi_2 = \alpha_Z\beta^{-1}\xi_1$, $\beta^a = -\alpha$. The maps $\tilde{g}$ lift g since $\alpha_X\alpha_Y = \alpha_Z^a$, resp. $\alpha\alpha' = \alpha_Z^a$, and the lifts of g form a coset of μ_a. Let $\tilde{h}_{ij}$ be the pullback of $\bar{h}_{ij}$. The order of $\tilde{h}_{ij}$ at 0 is bounded, since h_{ij}^0 has a monomial of degree less than $2/\gamma$, and distinct monomials of weight κ_j without xy restrict to distinct monomials on S, the exponent of t being determined by the weight.

Step 3. In this step we raise the labels and show that the raised minima are equal to β for $i \gg 0$. Let $B_i^* := \sum_j c_j^* \operatorname{div}(h_{ij}|_{W_i})$, and define B_i^{*0}, $\bar{B}_i^*$, and $B_i^{*\pm}$ from the labels c_j^*. Let $\sigma := +$ if $\delta^* < 0$, or if $\delta^* = 0$ and $\tau^* \le 0$, and let $\sigma := -$ otherwise. By Lemma B.20(4) applied to $\delta_i > 0$ for $i \gg 0$ if $\delta^* > 0$, and by Construction-Proposition B.19(3), the pairs $(S, \bar{B}_i)$ and, if $\delta^* > 0$, $(S, \bar{B}_i, \mathfrak{b}^{\delta_i/M})$ are lc near 0. Lemma B.20(4) considers only the prime divisors over S with center 0 and over Y_i^- with center contained in C_i^-, which suffices here and below, since a pair is lc near a closed subset $\mathcal{Z}$ if all prime divisors with center contained in $\mathcal{Z}$ have non-negative log discrepancies: if a prime divisor P with center intersecting $\mathcal{Z}$ had negative log discrepancy, then repeated blow-ups over a closed point $p \in \mathcal{Z}$ of the center of P would yield prime divisors with center p and negative log discrepancies (cf. [KM98, Corollary 2.31(1)]).

The exponents of the raised $\mathbb{R}$-ideals $\prod_j \bar{h}_{ij}^{c_j^*}$ and $\prod_j \bar{h}_{ij}^{c_j^*} \cdot \mathfrak{b}^{\delta^*/M}$ are at most those of $\prod_j \bar{h}_{ij}^{c_{ij}}$ and $\prod_j \bar{h}_{ij}^{c_{ij}} \cdot \mathfrak{b}^{\delta_i/M}$ divided by $\lambda_i' := \min_j(c_{ij}/c_j^*, \delta_i/\delta^*)$ (the last term only if $\delta^* > 0$), with $\lambda_i' := 1$ if $s = 0$ and $\delta^* \le 0$, and $\lambda_i' \to 1$. So the lc thresholds at 0 of the raised $\mathbb{R}$-ideals with respect to S are at least λ_i'. Pulling back to $\mathbb{A}^2_{\xi_1,\xi_2}$, which does not change the lc thresholds since the cover is quasi-étale [KM98, Proposition 5.20], the lc thresholds of the pullbacks belong to an ACC set by [dFEM11, Theorem 1.1], since the exponents c_j^* and δ^*/M are fixed. Hence they are at least one for $i \gg 0$, that is, $(S, \bar{B}_i^*)$ and, if $\delta^* > 0$, $(S, \bar{B}_i^*, \mathfrak{b}^{\delta^*/M})$ are lc near 0. By Lemma B.20(4) if $\sigma = -$, and by Construction-Proposition B.19(2)(3) and [Liu26, Lemma 2.6(2)] if $\sigma = +$, the raised reference $(Y_i^\sigma, B_i^{*\sigma})$ is lc near C_i^σ. By Lemma B.21 with $\delta' = \delta^*$, (V_0, B_i^{*0}) is lc near 0. Thus the raised data satisfy Set-up B.17 with Γ replaced by $\Gamma \cup \{c_1^*, \dots, c_s^*\}$, together with the assumption of Lemmas B.22 and B.23 that (V_0, B_i^{*0}) is lc near 0. Let $\beta_i^{*\pm}$ be the raised minima and $\beta_i^* := \min(\beta_i^{*+}, \beta_i^{*-})$, which is equal to $\beta_i^{*\sigma}$ by Lemma B.20(3).

We have $\beta_i^{*\pm} \le \beta_i^{\pm}$. Conversely, by Theorem 2.9(2) there exists a prime divisor ν computing $\beta_i^{*\pm}$ with $a(\nu, Y_i^\pm, 0)$ bounded independently of i, so the values of ν on the marks are bounded and

$$\beta_i^\pm \le a(\nu, Y_i^\pm, B_i^\pm) \le \beta_i^{*\pm} + \Lambda'' \sum_j (c_j^* - c_{ij})$$

for a constant Λ''. Hence $\beta_i^* - \beta_i \to 0$ and $\beta_i^* \to \beta$. Moreover $\beta_i^* = a(\nu', S_q, B_i')/|I(\nu')|$, where S_q is one of the finitely many sections of Construction-Proposition B.19(2), B_i' is the restriction of $\sum_j c_j^* \operatorname{div}(h_{ij}^0)$ to S_q, $I(\nu')$ is the inertia group in $G_{q,i}$, and ν' is a prime divisor over S_q with $a(\nu', S_q, 0)$ a bounded integer, bounded integral values on the marks, and bounded inertia. Since the labels c_j^* are fixed, β_i^* belongs to a finite set. After passing to a subsequence, $\beta_i^* = \beta$ for any i.

Step 4. In this step we take a compatible generic limit on $\mathbb{A}^2_{\xi_1,\xi_2}$. Let $\tilde{\mathfrak{b}}$ be the pullback of $\mathfrak{b}$ to $\mathbb{A}^2_{\xi_1,\xi_2}$. If $\sigma = -$ and $A > 0$, fix very general members $\varphi_1, \dots, \varphi_{N_0}$ of $\tilde{\mathfrak{b}}$ with N_0 large and let

$$\mathfrak{D}_1 := \frac{1}{MN_0|\tilde{G}|} \sum_{k=1}^{N_0} \operatorname{div}\Big(\prod_{g\in\tilde{G}} g^*\varphi_k\Big).$$

If $\sigma = -$ and $A = 0$, let $\mathfrak{D}_1 := \frac{1}{C}\operatorname{div}(\xi_1\xi_2)$, the pullback of $\frac{1}{M}\operatorname{div}(Z)$. If $\sigma = +$, let $\mathfrak{D}_1 := 0$. Let $\mathcal{L}$ be the finite set of lines through 0 consisting of the axes and the eigenlines of the non-scalar elements of $\tilde{G}$. Let $\mathcal{L}'$ be a union of lines in $\mathcal{L}$, $\Delta' \geq 0$ an $\mathbb{R}$-divisor supported on $\operatorname{Supp}\sum_j \operatorname{div}(\tilde{h}_{ij}) + \mathcal{L}'$ for some i, and $0 \leq \eta \leq \delta^* + 1$. We claim that $(\mathbb{A}^2, \Delta' + \eta\mathfrak{D}_1)$ is lc near 0 if and only if $(\mathbb{A}^2, \Delta', \tilde{\mathfrak{b}}^{\eta/M})$ is lc near 0, and that if they are, then

$$\operatorname{mld}\big((\mathbb{A}^2 \ni 0, \Delta' + \eta\mathfrak{D}_1)/\tilde{G}\big) = \operatorname{mld}\big((\mathbb{A}^2 \ni 0, \Delta', \tilde{\mathfrak{b}}^{\eta/M})/\tilde{G}\big).$$

If $\sigma = +$, only $\eta = 0$ occurs below, and the claim is trivial. If $\sigma = -$ and $A = 0$, then $\tilde{\mathfrak{b}} = (\xi_1\xi_2)$ is principal and $M = C$, so $\eta\mathfrak{D}_1 = \frac{\eta}{M}\operatorname{div}(\xi_1\xi_2)$ and the claim is trivial. Assume that $\sigma = -$ and $A > 0$. On a $\tilde{G}$-equivariant log resolution $Y \to \mathbb{A}^2$ of

$$\Big(\mathbb{A}^2, \operatorname{Supp}\sum_j \operatorname{div}(\tilde{h}_{ij}) + \mathcal{L}', \tilde{\mathfrak{b}}\Big),$$

we have $\operatorname{mult}_E \mathfrak{D}_1 = \frac{1}{M}\operatorname{mult}_E \tilde{\mathfrak{b}}$ for any exceptional divisor E. Since the φ_k are very general and there are countably many such resolutions, near the preimage of 0 the strict transforms of the curves $\operatorname{div}(g^*\varphi_k)$ are smooth, meet the exceptional curves transversally, and intersect the preimage of 0 only at general points y of exceptional curves E, whose stabilizer in $\tilde{G}$ is $I_{\tilde{G}}(E)$ and which are not on the strict transform of Δ', and the strict transforms of curves $\operatorname{div}(g^*\varphi_k)$ with different k do not intersect near the preimage of 0. The coefficients in $\eta\mathfrak{D}_1$ of the curves through such a point y sum to at most $\eta/(MN_0) < 1/2$, since only the translates of one φ_k pass through y. Near y, the boundary on Y is $(1 - a_E)E + \sum_k c_k D_k$, where a_E is the log discrepancy of E and the D_k are the strict transforms through y, with $\sum_k c_k < 1/2$. Any prime divisor F with center y satisfies $a(F, Y, \sum_k c_k D_k) \geq 1$, since $\operatorname{mult}_y \sum_k c_k D_k < 1/2$, and $(Y, E + \sum_k c_k D_k)$ is lc near y by [Kaw07, Theorem], since $\operatorname{mult}_y((\sum_k c_k D_k)|_E) < 1$. If $0 \leq a_E \leq 1$, then by linearity in the boundary,

$$a(F, Y, (1 - a_E)E + \sum_k c_k D_k) = (1 - a_E)a(F, Y, E + \sum_k c_k D_k) + a_E a(F, Y, \sum_k c_k D_k) \geq a_E,$$

and if $a_E > 1$, then

$$a(F, Y, (1 - a_E)E + \sum_k c_k D_k) = a(F, Y, \sum_k c_k D_k) + (a_E - 1)\operatorname{mult}_F E \geq a_E.$$

Moreover $I_{\tilde{G}}(F) \subset I_{\tilde{G}}(E)$. If F is a prime divisor over $\mathbb{A}^2$ with center 0 whose center on Y is not one of the points y, then the crepant pullbacks of $(\mathbb{A}^2, \Delta' + \eta\mathfrak{D}_1)$ and of $(\mathbb{A}^2, \Delta', \tilde{\mathfrak{b}}^{\eta/M})$ to Y coincide near the generic point of the center of F on Y, so

$$a(F, \mathbb{A}^2, \Delta' + \eta\mathfrak{D}_1) = a(F, \mathbb{A}^2, \Delta', \tilde{\mathfrak{b}}^{\eta/M}).$$

Moreover

$$a(F, \mathbb{A}^2, \Delta' + \eta\mathfrak{D}_1) \leq a(F, \mathbb{A}^2, \Delta', \tilde{\mathfrak{b}}^{\eta/M})$$

for any prime divisor F over $\mathbb{A}^2$, since $\operatorname{mult}_F \mathfrak{D}_1 \geq \frac{1}{M}\operatorname{mult}_F \tilde{\mathfrak{b}}$. Near 0 but away from 0, the curves $\operatorname{div}(g^*\varphi_k)$ are smooth and do not intersect $\operatorname{Supp}\Delta'$, and the coefficients in $\eta\mathfrak{D}_1$ of the curves through any point sum to less than $\frac{1}{2}$, which proves the claim.

Let $\Theta_i^* := \sum_j c_j^* \operatorname{div}(\tilde{h}_{ij})$. Let $\Theta_i := \Theta_i^* + \delta^*\mathfrak{D}_1$. Since $\mathbb{A}^2_{\xi_1,\xi_2} \to S$ is quasi-étale, so that lc and quotient minimal log discrepancies do not change under it [KM98, Proposition 5.20], [Liu26, Lemma 2.6(2)(4)], **Step 3**, together with the claim and Lemma B.20(4) if $\sigma = -$,

and with Construction-Proposition B.19(2)(3) if $\sigma = +$, implies that $(\mathbb{A}^2, \Theta_i)$ is lc near 0 and $\mathrm{mld}((\mathbb{A}^2 \ni 0, \Theta_i)/\tilde{G}) = \beta_i^* = \beta$.

If $s = 0$ and $\sigma = +$, then $\Theta_i = 0$ for any i. In this case we take no limit, and we let $k := \mathbb{C}$ and $\Theta_\infty := 0$, so that $(\mathbb{A}^2_k, \Theta_\infty)$ is klt near 0. Otherwise, we apply Construction-Proposition 4.1 in case (a) with $n = 2$, the finite group $\tilde{G} \subset GL_2(\mathbb{C})$, and the tuples

$$\Big(\tilde{h}_{i1}, \dots, \tilde{h}_{is}, \prod_g g^*\varphi_1, \dots, \prod_g g^*\varphi_{N_0}\Big) \text{ if } \sigma = -,\ A > 0, \qquad (\tilde{h}_{i1}, \dots, \tilde{h}_{is}, \xi_1\xi_2) \text{ if } \sigma = -,\ A = 0,$$

and $(\tilde{h}_{i1}, \dots, \tilde{h}_{is})$ if $\sigma = +$, whose orders at 0 are bounded by **Step 2**. Let $k \supset \mathbb{C}$ be the algebraically closed field, $Z_d \subset \mathcal{J}_d$ $(d \geq 1)$ the irreducible closed subsets, and $(f_{\infty,1}, \dots, f_{\infty,s'})$, power series in $k[[\xi_1, \xi_2]]$, the compatible generic limit given by Construction-Proposition 4.1 after passing to a subsequence, where s' is the length of the tuples. The entries after the $\tilde{h}_{ij}$ do not depend on i, so the projection of Z_d to their jets is a point, and we may take the entries $f_{\infty,j}$, $j > s$, of the limit equal to the given functions. In each application of Theorem 4.2 below, the entries with coefficient zero are removed by passing to a subtuple (Construction-Proposition 4.1). Let

$$\Theta_\infty := \sum_j c_j^* \operatorname{div}(f_{\infty,j}) + \delta^*\mathfrak{D}_1 \quad \text{on } \operatorname{Spec} k[[\xi_1, \xi_2]].$$

By Theorem 4.2(1) with $\theta = 1$, $(\mathbb{A}^2_k, \Theta_\infty)$ is lc near 0.

Step 5. In this step we treat the case when $(\mathbb{A}^2_k, \Theta_\infty)$ is klt near 0. If $\sigma = -$, fix $\eta > 0$ such that $(\mathbb{A}^2_k, \Theta_\infty + \eta\mathfrak{D}_1)$ is klt near 0. If $\sigma = +$, let $\eta := 0$. If $s = 0$ and $\sigma = +$, then $\Theta_i = 0$ and $\eta = 0$, and we fix any $\theta' > 1$. Otherwise, by Theorem 4.2(1) applied to the tuples of **Step 4** with the coefficients of $\Theta_i + \eta\mathfrak{D}_1$, there exists $\theta' > 1$ such that $(\mathbb{A}^2, \theta'(\Theta_i + \eta\mathfrak{D}_1))$ is lc near 0 for any i. Let $\delta^*(\theta) := \theta\tau^* + C - D$. By the definition of σ, there exists $\theta \in (1, \theta']$ such that $\delta^*(\theta) \leq 0$ if $\sigma = +$, and

$$0 \leq \delta^*(\theta) \leq \min(\theta'(\delta^* + \eta), \delta^* + 1)$$

if $\sigma = -$, since $\delta^*(\theta) \to \delta^*$ as $\theta \to 1$ and $\delta^*(\theta) = (\theta - 1)\tau^*$ with $\tau^* > 0$ if $\sigma = -$ and $\delta^* = 0$. Then

$$\theta\Theta_i^* + \delta^*(\theta)\mathfrak{D}_1 \leq \theta'(\Theta_i + \eta\mathfrak{D}_1),$$

so $(\mathbb{A}^2, \theta\Theta_i^* + \delta^*(\theta)\mathfrak{D}_1)$ is lc near 0. By **Step 4**, together with Lemma B.20(4) for the labels θc_j^* if $\sigma = -$, and with Construction-Proposition B.19(2)(3) and [Liu26, Lemma 2.6(2)] if $\sigma = +$, the pair $(Y_i^\sigma, B_i^{*\sigma}(\theta))$ is lc near C_i^σ for any i. By Lemma B.22 applied to the raised data, $\alpha_i^* := \mathrm{mld}((W_i \ni 0, B_i^*)/G_i) \geq \beta_i^* = \beta$ for $i \gg 0$. Since $B_i \leq B_i^*$, $\alpha_i \geq \alpha_i^* \geq \beta > \alpha_i$, a contradiction.

Step 6. In this step we treat the case when $(\mathbb{A}^2_k, \Theta_\infty)$ is not klt near 0. By Definition-Proposition 5.7 in case (a) with $H = \tilde{G}$, there exists exactly one floor Σ, it is smooth at 0, $\Theta_\infty - \Sigma$ does not contain Σ, and

$$\beta = \frac{1 - \mathrm{mult}_0((\Theta_\infty - \Sigma)|_\Sigma)}{e_\Sigma} = \frac{\mu}{e_\Sigma}, \qquad \mu := 1 - \mathrm{mult}_0((\Theta_\infty - \Sigma)|_\Sigma) > 0, \tag{B.102}$$

where e_Σ is the order of the image of $\tilde{G}$ in the automorphism group of $T_0\Sigma$. If $\sigma = -$ and $A = 0$, then $\mathfrak{D}_1 = \frac{1}{C}\operatorname{div}(\xi_1\xi_2)$ and the floor may be one of the axes. Definition-Proposition 5.7 and Lemma 5.6 place no condition on the floor besides being a component of coefficient one, so this case needs no separate treatment. The group $\tilde{G}$ preserves Θ_∞, hence Σ and $T_0\Sigma$.

We show that $\tilde{G}$ is abelian. The group $G_S \subset G_i\mathbf{T}$ is abelian. If $a \geq 3$, the deck element (ζ, ζ^{-1}), ζ a primitive a-th root of unity, has the axes as eigenlines with distinct eigenvalues, so $T_0\Sigma$ is an axis. The lift of an element of G_S exchanging X and Y exchanges the axes, so G_S, G_i, and $\tilde{G}$ are diagonal. If $a = 2$ and $T_0\Sigma$ is an axis, then again no element of $\tilde{G}$ exchanges the axes, so $\tilde{G}$ is diagonal. If $a = 2$ and $T_0\Sigma$ is not an axis, then any diagonal element of $\tilde{G}$ preserves $T_0\Sigma$

and is a scalar, and for two anti-diagonal elements g, g' the element $g^{-1}g'$ is diagonal, hence a scalar, so g and g' commute. If $a = 1$, then $\tilde{G} = G_S$. Thus $\tilde{G}$ is a finite abelian group.

Next we choose a linear $\tilde{G}$-eigenform ξ_Σ defined over $\mathbb{C}$ such that $\xi_\Sigma|_\Sigma$ is a parameter at 0, and a G_i-semi-invariant $\mathbf{T}$-homogeneous polynomial ξ_W of weight $-A$ in x, y, z, t. If $a \geq 2$ and either $T_0\Sigma$ is an axis or all elements of $\tilde{G}$ are scalars, let $\xi_\Sigma := \xi_1$ and $\xi_W := x$ if $T_0\Sigma$ is not the ξ_2-axis, and $\xi_\Sigma := \xi_2$ and $\xi_W := y$ otherwise. Then $\xi_\Sigma^a = X$, resp. $\xi_\Sigma^a = -Y$. If $a = 2$, $T_0\Sigma$ is not an axis and $\tilde{G}$ has an element g that is not a scalar, then g is anti-diagonal and the eigenlines of g are $\tilde{G}$-invariant lines defined over $\mathbb{C}$, one of which is $T_0\Sigma$. Let $\xi_\Sigma := \xi_1 + c_\xi \xi_2$ be the linear form vanishing on the other eigenline. An element λ of the subgroup $\mu_D = G_S \cap \mathbf{T}$ acts on X, Y, Z by $\lambda^{-A}, \lambda^{-A}, \lambda^{-C}$, and the lifts of λ are diagonal, hence scalars ζ' with $\zeta'^2 = \lambda^{-A}$ and $\zeta'^2 = \lambda^{-C}$. So D divides $C - A$, and since $2A = 2C - bD$, $b = 2(C - A)/D$ is even. Let

$$\xi_W := x + 2c_\xi z t^{b/2} - c_\xi^2 y,$$

which has $\mathbf{T}$-weight $-A$ since $-C + Db/2 = -A$, and whose restriction to S_t pulls back to $\xi_\Sigma^2 = X + 2c_\xi Z - c_\xi^2 Y$. If $a = 1$, then $S \cong \mathbb{A}^2_{X,Y}$, $\tilde{G} = G_S$ acts linearly on X, Y, and we let ξ_Σ be a linear eigenform in X, Y vanishing on a $\tilde{G}$-invariant complement of $T_0\Sigma$ (the other eigenline of an exchanging element if such an element exists, an axis different from $T_0\Sigma$ otherwise), and $\xi_W := \xi_\Sigma(x, y)$. In any case the restriction of ξ_W to S_t is a semi-invariant whose pullback is $\pm\xi_\Sigma^a$.

For $g \in G_i$, write $g = g_S\lambda$ with $g_S \in G_S$ and $\lambda \in \mathbf{T}$. Since g_S commutes with $\mathbf{T}$, $g_S^*\xi_W$ and ξ_W are $\mathbf{T}$-homogeneous of weight $-A$, and their restrictions to S_t are proportional. A $\mathbf{T}$-homogeneous function is determined by the restriction of the function to S_t on $\mathbf{T} \cdot S_t$, which is dense in V_0, and $g_S^*\xi_W$ and ξ_W are linear combinations of x, y and $zt^{b/2}$, so they are proportional as polynomials. Thus ξ_W is G_i-semi-invariant.

Let $g_0 := \gcd(A, D)$, with $g_0 = D$ if $A = 0$, and let

$$U := \xi_W^{D/g_0} t^{A/g_0}, \qquad \ell_U := aD/g_0. \tag{B.103}$$

Then U is a regular G_i-semi-invariant function of $\mathbf{T}$-weight zero vanishing at 0, and the pullback of $U|_{S_t}$ to $\mathbb{A}^2_{\xi_1,\xi_2}$ is $\pm\xi_\Sigma^{\ell_U}$. The kernel K_0 of the surjection $\tilde{G} \to H_i$ consists of the lifts of μ_D and the deck group μ_a, and has order aD. An element of K_0 multiplies ξ_Σ by a scalar ζ' with $\zeta'^a \in \{\lambda^{-A} \mid \lambda \in \mu_D\} = \mu_{D/g_0}$, and all such ζ' occur. For $a = 2$ and $\xi_\Sigma = \xi_1 + c_\xi\xi_2$ the claim uses that the elements of K_0 are scalars. Thus the character χ_{ξ_Σ} of ξ_Σ has order exactly ℓ_U on K_0, so ℓ_U divides the order e_Σ of χ_{ξ_Σ} on $\tilde{G}$. The character of U on G_i is trivial on R_i, since U has weight zero, and as a character of H_i it is the character $\chi_{\xi_\Sigma}^{\ell_U}$ of $\tilde{G}$, which is trivial on K_0. Hence

$$\text{the character of } U \text{ on } G_i \text{ has order } h_U := e_\Sigma/\ell_U. \tag{B.104}$$

Let $\lambda_U := \mu/\ell_U$ and

$$\Delta_\Sigma := (\Theta_\infty - \Sigma + \mu \operatorname{div}(\xi_\Sigma))|_\Sigma.$$

We show that the pair

$$\big(\mathbb{A}^2_k, \Theta_\infty + \lambda_U \operatorname{div}(U|_{S_t})\big) = \big(\mathbb{A}^2_k, \Theta_\infty + \mu \operatorname{div}(\xi_\Sigma)\big)$$

is lc near 0. Since $\xi_\Sigma|_\Sigma$ is a parameter of Σ at 0, Σ is not a component of $\Theta_\infty - \Sigma + \mu \operatorname{div}(\xi_\Sigma)$, and $\operatorname{mult}_0 \Delta_\Sigma = (1 - \mu) + \mu = 1$. By Lemma 2.14 on $\operatorname{Spec} k[[\xi_1, \xi_2]]$, $(\mathbb{A}^2_k, \Theta_\infty + \mu \operatorname{div}(\xi_\Sigma))$ is lc near 0.

We add the constant entry $U|_{S_t}$ with coefficient λ_U to the tuples of **Step 4**. By Theorem 4.2(1), $(\mathbb{A}^2, \Theta_i + \lambda_U \operatorname{div}(U|_{S_t}))$ is lc near 0 for any i. The curve $(\xi_\Sigma = 0)$ is a line in $\mathcal{L}$, so by **Step 4**,

$$\big(S, \bar{B}_i^* + \lambda_U \operatorname{div}(U|_{S_t}), \mathfrak{b}^{\delta^*/M}\big)$$

is lc near 0, where the ideal is omitted if $\sigma = +$. Since U has weight zero, adding the mark U with label λ_U changes neither τ^* nor δ^*. By Lemma B.20(4) if $\sigma = -$, and by Lemma B.21, $(V_0, B_i^{*0} + \lambda_U \operatorname{div}(U|_{V_0}))$ is lc near 0. By Lemma B.18(3) with $\Gamma' = \{c_1^*, \dots, c_s^*, \lambda_U\}$ and $f_1 = U$,

$(W_i, B_i^* + \lambda_U \operatorname{div}(U|_{W_i}))$ is lc near 0 for $i \gg 0$. By Lemma 5.8 applied to the closed subset $\{0\}$, (B.104), and (B.102),

$$\alpha_i \geq \operatorname{mld}((W_i \ni 0, B_i^*)/G_i) \geq \frac{\lambda_U}{h_U} = \frac{\mu}{\ell_U} \cdot \frac{\ell_U}{e_\Sigma} = \beta > \alpha_i,$$

a contradiction. This proves the theorem. □

Appendix C. Proofs for sources of quadratic rank one

In this section, we prove the results of Section 9. We first prove Lemma 9.2, which holds for any source of quadratic rank one, and describe the prime divisor E_0 on the root charts of Set-up 9.3 (Subsection C.2). Finally, we prove a normal form for the equation of a source whose residual cubic is a double line (Lemma C.3), which is used in Appendix E.

C.1. **A divisor of log discrepancy two.** In this subsection, we prove Lemma 9.2.

Proof of Lemma 9.2. The proof is long, so we divide it into several steps.

Step 1. In this step we straighten D and compute the cubic part of g in the new coordinates.

Let $\mathfrak{m}$ be the maximal ideal of $k[[x, y, z, t]]$, and $\mathfrak{m}_D$ the maximal ideal of R/I_D. Since $\operatorname{mult}_0 g \geq 3$, the tangent cone of $\operatorname{Spec} R$ at o is $(x^2 = 0)$. Since R/I_D is regular, the tangent cone of D at o is the tangent plane T_oD. Since T_oD is contained in the tangent cone of $\operatorname{Spec} R$ at o, $T_oD \subset (x = 0)$. Thus $T_oD = (x = L = 0)$ for a non-zero linear form L in y, z, t. Hence $x|_D \in \mathfrak{m}_D^2$, and $x^2|_D \in \mathfrak{m}_D^4$. Since $g - g_3 \in \mathfrak{m}^4$, we obtain

$$g_3|_D \equiv g|_D = -x^2|_D \equiv 0 \mod \mathfrak{m}_D^4.$$

The class of $g_3|_D$ in $\mathfrak{m}_D^3/\mathfrak{m}_D^4$ is the restriction of the cubic form g_3 to T_oD. Thus g_3 vanishes on $(x = L = 0)$, and L divides g_3.

A linear change of the coordinates y, z, t keeps the form $x^2 + g(y, z, t)$ of the equation of $\operatorname{Spec} R$ and preserves $\mathfrak{m}^n$ for any n. Possibly replacing y, z, t with linear forms in y, z, t, we may assume that $L = y$. Then $y|_D \in \mathfrak{m}_D^2$, and the restrictions of z, t to D form a regular system of parameters of R/I_D. Since R/I_D is complete, $R/I_D = k[[z, t]]$, and there exist $\sigma_0, \beta \in (z, t)^2 k[[z, t]]$ such that $x|_D = \sigma_0$ and $y|_D = \beta$. Let

$$X := x - \sigma_0(z, t), \qquad Y := y - \beta(z, t).$$

Then X, Y, z, t are formal coordinates of $k[[x, y, z, t]]$. Since $k[[x, y, z, t]]/(X, Y) = k[[z, t]] = R/I_D$, the inverse image of I_D in $k[[x, y, z, t]]$ is (X, Y). Thus $D = (X = Y = 0)$ and $x^2 + g \in (X, Y)$. Writing $x^2 + g = (X + \sigma_0)^2 + g(Y + \beta, z, t)$ as a power series in X and Y, we obtain

$$x^2 + g = F := X^2 + 2\sigma_0 X + \sigma_1 Y + \sigma_2 Y^2 + \sigma_3 Y^3 + \sigma_4 Y^4, \tag{C.1}$$

where $\sigma_1, \sigma_2, \sigma_3 \in k[[z, t]]$ and $\sigma_4 \in k[[Y, z, t]]$, and where the term of degree zero in X, Y vanishes because $x^2 + g \in (X, Y)$. Since $\operatorname{mult}_0 g \geq 3$ and $\beta \in (z, t)^2$,

$$\sigma_1 = \frac{\partial g}{\partial y}(\beta, z, t) \in (z, t)^2, \qquad \sigma_2 = \frac{1}{2}\frac{\partial^2 g}{\partial y^2}(\beta, z, t) \in (z, t).$$

For $\sigma \in k[[z, t]]$ and an integer $n \geq 0$, we denote by $\sigma^{(n)}$ the homogeneous part of degree n of σ. Since $\beta \in (z, t)^2$ and $\operatorname{mult}_0 \frac{\partial g}{\partial y} \geq 2$, we have $g(Y + \beta, z, t) - g(Y, z, t) \in \mathfrak{m}^4$ and $g(\beta, z, t) - g(0, z, t) \in \mathfrak{m}^4$. Since y divides g_3, $g(0, z, t) \in \mathfrak{m}^4$. Thus

$$\sigma_1 Y + \sigma_2 Y^2 + \sigma_3 Y^3 + \sigma_4 Y^4 = g(Y + \beta, z, t) - g(\beta, z, t) \equiv g(Y, z, t) \mod \mathfrak{m}^4.$$

Comparing the parts of degree three, we obtain

$$g_3(Y, z, t) = \sigma_1^{(2)} Y + \sigma_2^{(1)} Y^2 + \sigma_3(0, 0) Y^3. \tag{C.2}$$

Since $g_3 \neq 0$, at least one of $\sigma_1^{(2)}$, $\sigma_2^{(1)}$, and $\sigma_3(0, 0)$ is non-zero.

Step 2. In this step we fix linear coordinates z', t' of $k[[z,t]]$ and a weight vector $\mathbf{w} = (w_X, w_Y, w_z, w_t)$ for X, Y, z', t', and we compute $\mathrm{in}_{\mathbf{w}} F$.

We use $\mathrm{wt}_{\mathbf{w}}$ and $\mathrm{in}_{\mathbf{w}}$ as in [Liu26, Definition 2.16], with k in place of $\mathbb{C}$. We distinguish three cases.

Case A: $\sigma_0^{(2)} \neq 0$ *or* $\sigma_1^{(2)} \neq 0$. Let $z' := z$, $t' := t$, and $\mathbf{w} := (2,2,1,1)$. Since $\sigma_0, \sigma_1 \in (z,t)^2$ and $\sigma_2 \in (z,t)$, the terms $\sigma_2 Y^2$, $\sigma_3 Y^3$, and $\sigma_4 Y^4$ of (C.1) have $\mathbf{w}$-weight at least 5, and

$$\mathrm{wt}_{\mathbf{w}}(F) = 4, \qquad \mathrm{in}_{\mathbf{w}} F = X^2 + 2\sigma_0^{(2)} X + \sigma_1^{(2)} Y.$$

Case B: $\sigma_0^{(2)} = \sigma_1^{(2)} = 0$ *and* $\sigma_1^{(3)} \neq 0$. Let $z' := z$, $t' := t$, and $\mathbf{w} := (3,3,1,1)$. Since $\sigma_0, \sigma_1 \in (z,t)^3$, the terms $\sigma_2 Y^2$, $\sigma_3 Y^3$, and $\sigma_4 Y^4$ have $\mathbf{w}$-weight at least 7, and

$$\mathrm{wt}_{\mathbf{w}}(F) = 6, \qquad \mathrm{in}_{\mathbf{w}} F = X^2 + 2\sigma_0^{(3)} X + \sigma_1^{(3)} Y.$$

Case C: $\sigma_0^{(2)} = \sigma_1^{(2)} = \sigma_1^{(3)} = 0$. By (C.2), $\sigma_2^{(1)} \neq 0$ or $\sigma_3(0,0) \neq 0$. If $\sigma_2^{(1)} \neq 0$, let $z' := \sigma_2^{(1)}$, and let t' be a linear form in z, t such that z', t' are linearly independent. In this case the coordinates z, t on D change by the invertible linear map $(z,t) \to (z', t')$. If $\sigma_2^{(1)} = 0$, let $z' := z$ and $t' := t$. We write $\sigma_0, \dots, \sigma_4$ as power series in z', t' (and Y), which keeps (C.1) and $D = (X = Y = 0)$. For $\sigma \in k[[z', t']]$ and an integer $n \geq 0$, we denote by $\sigma^{[n]}$ the part of weight n of σ for the weights $2, 1$ of z', t'. Let $\mathbf{w} := (3,2,2,1)$. Since $\sigma_0 \in (z', t')^3$, $\sigma_1 \in (z', t')^4$, and the linear part of σ_2 is z' or 0, the terms $2\sigma_0 X$, $\sigma_1 Y$, and $\sigma_2 Y^2$ have $\mathbf{w}$-weight at least 6. Since $\sigma_3 Y^3$ has $\mathbf{w}$-weight at least 6 and $\sigma_4 Y^4$ has $\mathbf{w}$-weight at least 8,

$$\mathrm{wt}_{\mathbf{w}}(F) = 6, \qquad \mathrm{in}_{\mathbf{w}} F = X^2 + 2\sigma_0^{[3]} X + \sigma_1^{[4]} Y + \sigma_2^{[2]} Y^2 + \sigma_3(0,0) Y^3,$$

where $\sigma_2^{[2]} - z' \in kt'^2$ if $\sigma_2^{(1)} \neq 0$, and $\sigma_2^{[2]} \in kt'^2$ if $\sigma_2^{(1)} = 0$.

In any case, any monomial of F has $\mathbf{w}$-weight at least $2w_X$, the coefficient of X^2 in F is 1, and

$$\mathrm{wt}_{\mathbf{w}}(F) = 2w_X, \qquad w_X + w_Y + w_z + w_t - \mathrm{wt}_{\mathbf{w}}(F) = 2, \qquad \frac{w_X w_Y}{\mathrm{wt}_{\mathbf{w}}(F)} = \frac{w_Y}{2} \geq 1. \tag{C.3}$$

Let $\mathrm{disc}_X(\mathrm{in}_{\mathbf{w}} F)$ be the discriminant of $\mathrm{in}_{\mathbf{w}} F$ as a monic quadratic polynomial in X. It is equal to

$$4\big((\sigma_0^{(2)})^2 - \sigma_1^{(2)} Y\big), \qquad 4\big((\sigma_0^{(3)})^2 - \sigma_1^{(3)} Y\big), \qquad 4\big((\sigma_0^{[3]})^2 - \sigma_1^{[4]} Y - \sigma_2^{[2]} Y^2 - \sigma_3(0,0) Y^3\big)$$

in Cases A, B, and C respectively. In Case A, $\mathrm{disc}_X(\mathrm{in}_{\mathbf{w}} F) \neq 0$, since the coefficient of Y is $-4\sigma_1^{(2)}$ and the coefficient of Y^0 is $4(\sigma_0^{(2)})^2$. In Case B, the coefficient of Y is $-4\sigma_1^{(3)} \neq 0$. In Case C, the coefficient of Y^2 is $-4\sigma_2^{[2]}$, which is non-zero if $\sigma_2^{(1)} \neq 0$, and the coefficient of Y^3 is $-4\sigma_3(0,0)$, which is non-zero if $\sigma_2^{(1)} = 0$. Thus

$$\mathrm{disc}_X(\mathrm{in}_{\mathbf{w}} F) \neq 0. \tag{C.4}$$

Step 3. In this step we describe the exceptional prime divisors of the weighted blow-up with weights $\mathbf{w}$, compute their log discrepancies, and describe the multiplicities of functions along them.

We use only the following properties, so **Step 3** holds for any algebraically closed field $\mathbf{k} \supset \mathbb{C}$, any $\mathbf{w} \in \{(2,2,1,1), (3,3,1,1), (3,2,2,1)\}$, and any power series $F \in \mathbf{k}[[X, Y, z', t']]$ such that

(i) $\mathbf{k}[[X, Y, z', t']]/(F)$ is a normal domain,
(ii) any monomial of F has $\mathbf{w}$-weight at least $2w_X$, and the coefficient of X^2 in F is 1, and
(iii) $\mathrm{disc}_X(\mathrm{in}_{\mathbf{w}} F) \neq 0$.

By **Steps 1 and 2**, the power series F of (C.1) satisfies (i)–(iii) with $\mathbf{k} = k$. For any $\mathbf{w} \in \{(2,2,1,1), (3,3,1,1), (3,2,2,1)\}$, $w_X + w_Y + w_z + w_t - 2w_X = 2$ and $w_t = 1$.

For any such F, we prove in this step that $\bar{f} = \theta_1 \cdots \theta_r$ for pairwise distinct monic irreducible polynomials θ_j, that $\mathrm{in}_{\mathbf{w}} F$ is reduced, that the π-exceptional prime divisors on W' are $E_1, \dots, E_r$, where E_j corresponds to θ_j and has center the closed point, that (C.6) and (C.7) hold, and that

Claim C.1 holds for any j. Here $\bar{f}$, θ_j, $\pi : W' \to \operatorname{Spec} \mathbf{k}[[X,Y,z',t']]/(F)$, and E_j are defined below.

The lowest part. Since X^3 has $\mathbf{w}$-weight $3w_X > 2w_X$, (ii) implies that $\operatorname{in}_{\mathbf{w}} F = X^2 + p_1 X + p_0$ for $\mathbf{w}$-weighted homogeneous polynomials $p_1, p_0 \in \mathbf{k}[Y,z',t']$. Let X_1, Y_1, z_1 be variables, and let

$$\bar{f} := \operatorname{in}_{\mathbf{w}} F(X_1, Y_1, z_1, 1) \in \mathbf{k}[X_1, Y_1, z_1].$$

Since $w_t = 1$, setting $t' = 1$ is injective on $\mathbf{w}$-weighted homogeneous polynomials of a fixed weight, so (iii) implies

$$\left(\frac{\partial \bar{f}}{\partial X_1}\right)^2 - 4\bar{f} = \operatorname{disc}_X(\operatorname{in}_{\mathbf{w}} F)(Y_1, z_1, 1) \neq 0. \tag{C.5}$$

Since $\bar{f}$ is monic in X_1, any irreducible factor of $\bar{f}$ has positive degree in X_1. If $\bar{f}$ is divisible by the square of an irreducible polynomial, then $\bar{f} = (X_1 + p)^2$ for some $p \in \mathbf{k}[Y_1, z_1]$, and the left-hand side of (C.5) is zero, a contradiction. Thus $\bar{f}$ is reduced, and

$$\bar{f} = \theta_1 \cdots \theta_r$$

for pairwise distinct monic irreducible polynomials $\theta_1, \dots, \theta_r$ of positive degree in X_1. If $\operatorname{in}_{\mathbf{w}} F$ is divisible by the square of an irreducible polynomial, then $\operatorname{in}_{\mathbf{w}} F = (X+q)^2$ for some $q \in \mathbf{k}[Y,z',t']$, and $\operatorname{disc}_X(\operatorname{in}_{\mathbf{w}} F) = 0$, contradicting (iii). Thus $\operatorname{in}_{\mathbf{w}} F$ is reduced, and t' does not divide $\operatorname{in}_{\mathbf{w}} F$ since the coefficient of X^2 is 1. Moreover, for any j, $\frac{\partial \bar{f}}{\partial X_1} \notin (\theta_j)$: otherwise, by (C.5), θ_j divides a non-zero element of $\mathbf{k}[Y_1, z_1]$, which is impossible since θ_j has positive degree in X_1.

The chart of weight one. Let $V \to \mathbb{A}^4 = \operatorname{Spec} \mathbf{k}[X,Y,z',t']$ be the weighted blow-up with weights $\mathbf{w}$ in X, Y, z', t'. Since the ideals of elements of $\mathbf{w}$-weight at least n are monomial ideals, the weighted blow-up of $\operatorname{Spec} \mathbf{k}[[X,Y,z',t']]$ with weights $\mathbf{w}$ is the base change of $V \to \mathbb{A}^4$ along the flat morphism $\operatorname{Spec} \mathbf{k}[[X,Y,z',t']] \to \mathbb{A}^4$ [Mat89, Theorem 8.8]. Since $w_t = 1$, the chart V_t of V where t' does not vanish is $\mathbb{A}^4$ with coordinates X_1, Y_1, z_1, ρ. Let $U := V_t \times_{\mathbb{A}^4} \operatorname{Spec} \mathbf{k}[[X,Y,z',t']]$, which is the chart of the weighted blow-up of $\operatorname{Spec} \mathbf{k}[[X,Y,z',t']]$ where t' does not vanish. Thus U is integral, since it is an open subscheme of the weighted blow-up of the integral scheme $\operatorname{Spec} \mathbf{k}[[X,Y,z',t']]$, and on V_t and on U, where we write X_1, Y_1, z_1, ρ also for their pullbacks to U,

$$X = \rho^{w_X} X_1, \qquad Y = \rho^{w_Y} Y_1, \qquad z' = \rho^{w_z} z_1, \qquad t' = \rho.$$

Hence $(X,Y,z',t')^n \subset (\rho^n)$ on V_t for any $n \geq 1$, and the subscheme $(\rho^n = 0)$ of V_t maps into $\operatorname{Spec} \mathbf{k}[X,Y,z',t']/(X,Y,z',t')^n$, over which the base change is an isomorphism. Thus the subscheme $(\rho^n = 0)$ of U is isomorphic to $\operatorname{Spec} \mathbf{k}[X_1, Y_1, z_1][\rho]/(\rho^n)$ for any $n \geq 1$. In particular the exceptional divisor of the weighted blow-up on U is $(\rho = 0) \cong \operatorname{Spec} \mathbf{k}[X_1, Y_1, z_1]$, and the completion of the coordinate ring of U along $(\rho = 0)$ is $\mathbf{k}[X_1, Y_1, z_1][[\rho]]$.

For $P \in \mathbf{k}[[X,Y,z',t']]$ and an integer $n \geq 0$, let $P^{[n]}$ be the $\mathbf{w}$-weighted homogeneous part of weight n of P, and let

$$\widetilde{P} := \sum_{n \geq 0} P^{[n]}(X_1, Y_1, z_1, 1)\rho^n \in \mathbf{k}[X_1, Y_1, z_1][[\rho]],$$

which is the image of the pullback of P to U. Since $(w_z, w_t) = (2,1)$ in Case C of **Step 2**, for $\sigma \in \mathbf{k}[[z',t']]$ the part $\sigma^{[n]}$ of **Step 2** is the part $P^{[n]}$ for $P = \sigma$. By (ii), $\widetilde{F} = \rho^{2w_X} F^\circ$, where

$$F^\circ := \sum_{n \geq 0} F^{[2w_X + n]}(X_1, Y_1, z_1, 1)\rho^n, \qquad F^\circ \equiv \bar{f} \mod \rho.$$

Moreover, by (ii), F belongs to the ideal of $\mathbf{k}[[X,Y,z',t']]$ generated by the monomials of $\mathbf{w}$-weight at least $2w_X$, which generates the ideal (ρ^{2w_X}) on U. Hence $F = \rho^{2w_X} F^\circ$ on U for a regular function on U, unique since U is integral, which we denote also by F°. The image of the regular function F° in $\mathbf{k}[X_1, Y_1, z_1][[\rho]]$ is the power series F°, since ρ is a non-zero-divisor of $\mathbf{k}[X_1, Y_1, z_1][[\rho]]$. Since F° is not divisible by ρ, the strict transform of $\operatorname{Spec} \mathbf{k}[[X,Y,z',t']]/(F)$ on U is $(F^\circ = 0)$.

The exceptional prime divisors. Let $\pi : W' \to \operatorname{Spec} \mathbf{k}[[X, Y, z', t']]/(F)$ be the composition of the normalization of the strict transform with the weighted blow-up. Since $\mathbf{k}[[X, Y, z', t']]/(F)$ is excellent, the normalization is finite and π is proper. Since $\mathbf{k}[[X, Y, z', t']]/(F)$ is normal, π is an isomorphism over the complement of the closed point. For any j, let $\mathfrak{p}_j$ be the generic point of $(\rho = \theta_j = 0) \subset U$, and let A_j be the local ring of U at $\mathfrak{p}_j$. Since $\rho \neq 0$ is a non-zero-divisor of the local domain A_j and $A_j/\rho A_j = \mathbf{k}[X_1, Y_1, z_1]_{(\theta_j)}$ is a discrete valuation ring with uniformizer θ_j,

$$\dim A_j = \dim A_j/\rho A_j + 1 = 2,$$

and A_j is a regular local ring with maximal ideal (ρ, θ_j). Let $\eta_j := \bar{f}/\theta_j$. Since $\bar{f}$ is reduced, $\eta_j \notin (\theta_j)$, so η_j is a unit of A_j. Since $F^\circ \equiv \eta_j \theta_j \mod \rho$, $F^\circ \notin (\rho, \theta_j)^2$. Thus $A_j/(F^\circ)$ is a discrete valuation ring, and since

$$\theta_j \equiv -\eta_j^{-1}(F^\circ - \bar{f}) \in \rho A_j \mod F^\circ,$$

ρ is a uniformizer of $A_j/(F^\circ)$. In particular the strict transform is regular at $\mathfrak{p}_j$, so the normalization of the strict transform is an isomorphism over a neighborhood of $\mathfrak{p}_j$. Let E_j be the prime divisor on W' whose generic point maps to $\mathfrak{p}_j$.

The fiber of the strict transform over the closed point is the hypersurface $(\operatorname{in}_{\mathbf{w}} F = 0)$ of the weighted projective space $\mathbb{P}(w_X, w_Y, w_z, w_t)$, since the initial ideal of (F) with respect to the $\mathbf{w}$-filtration is $(\operatorname{in}_{\mathbf{w}} F)$, as $\mathbf{k}[X, Y, z', t']$ is a domain. Since t' does not divide $\operatorname{in}_{\mathbf{w}} F$, any irreducible component of $(\operatorname{in}_{\mathbf{w}} F = 0)$ is the closure of $(\rho = \theta_j = 0)$ for some j. Since the normalization is finite, any π-exceptional prime divisor on W' maps onto such a component, so it is E_j for some j. Let E_{tot} be the pullback to W' of the exceptional $\mathbb{Q}$-Cartier divisor of the weighted blow-up. On U, the exceptional divisor is $\operatorname{div}(\rho)$, and ρ is a uniformizer at the generic point of E_j for any j. Thus

$$E_{\mathrm{tot}} = \sum_{j=1}^{r} E_j. \tag{C.6}$$

Since all the weights are positive, the center of E_j on $\operatorname{Spec} \mathbf{k}[[X, Y, z', t']]/(F)$ is the closed point for any j.

Log discrepancies. We use the canonical divisors of [dFEM11, Appendix A], with which log discrepancies on schemes of finite type over $\mathbf{k}[[X, Y, z', t']]$ are defined as in [dFEM11, Section 2]. For completions of local rings of varieties, they are the log discrepancies of [Liu26, Definition 3.2]. By the conormal sequence of special differentials [dFEM11, Lemma A.1], which is left exact at regular points by the comparison of ranks $4 = 1 + 3$ [dFEM11, Proposition A.8], taking determinants shows that the residue of $dX \wedge dY \wedge dz' \wedge dt'/F$ generates $\mathcal{O}(K_{\operatorname{Spec} \mathbf{k}[[X,Y,z',t']]/(F)})$ at the regular points of height one. On U, since $w_t = 1$,

$$\frac{dX \wedge dY \wedge dz' \wedge dt'}{F} = \rho^{w_X + w_Y + w_z - 2w_X} \frac{dX_1 \wedge dY_1 \wedge dz_1 \wedge d\rho}{F^\circ},$$

and, since the module of special differentials of U is free on $dX_1, dY_1, dz_1, d\rho$, the conormal sequence of special differentials [dFEM11, Lemma A.1] for the strict transform $(F^\circ = 0) \subset U$ shows that the residue of $dX_1 \wedge dY_1 \wedge dz_1 \wedge d\rho/F^\circ$ generates the sheaf $\mathcal{O}(K)$ of the strict transform at $\mathfrak{p}_j$, where the strict transform is regular. Thus $K_{W'} - \pi^* K_{\operatorname{Spec} \mathbf{k}[[X,Y,z',t']]/(F)}$ has coefficient $w_X + w_Y + w_z - 2w_X$ along E_j, and by [Rei87, Section 4, (4.8)],

$$a\big(E_j, \operatorname{Spec} \mathbf{k}[[X, Y, z', t']]/(F), 0\big) = w_X + w_Y + w_z + w_t - 2w_X = 2 \tag{C.7}$$

for any j.

Multiplicities. For any j, let L_j be the field of fractions of $\mathbf{k}[X_1, Y_1, z_1]/(\theta_j)$, and let $\mathcal{O}_j \subset L_j$ be the localization of $\mathbf{k}[X_1, Y_1, z_1]/(\theta_j)$ at the element $\frac{\partial \bar{f}}{\partial X_1}$, which is non-zero by the first part of this step. For $\xi \in \mathcal{O}_j[[\rho]]$ and

$$P = \sum_n c_n(X_1, Y_1, z_1)\rho^n \in \mathbf{k}[X_1, Y_1, z_1][[\rho]],$$

let

$$P(\xi) := \sum_n c_n(\xi, Y_1, z_1)\rho^n \in \mathcal{O}_j[[\rho]].$$

Claim C.1. *Let $j \in \{1, \dots, r\}$.*

(1) There exists a unique $\xi_j \in \mathcal{O}_j[[\rho]]$ such that $\xi_j - X_1 \in \rho\mathcal{O}_j[[\rho]]$ and $F^\circ(\xi_j) = 0$.
(2) Let n be a positive integer and $\Xi \in \mathcal{O}_j[[\rho]]$ such that $\Xi - X_1 \in \rho\mathcal{O}_j[[\rho]]$ and $F^\circ(\Xi) \in \rho^n\mathcal{O}_j[[\rho]]$. Then $\Xi - \xi_j \in \rho^n\mathcal{O}_j[[\rho]]$.
(3) For any $P \in \mathbf{k}[[X, Y, z', t']]$ and any integer $n \geq 0$, $\operatorname{mult}_{E_j} P \geq n$ if and only if $\widetilde{P}(\xi_j) \in \rho^n\mathcal{O}_j[[\rho]]$.

Proof. Let $\Xi, \Xi' \in \mathcal{O}_j[[\rho]]$ such that $\Xi - X_1, \Xi' - X_1 \in \rho\mathcal{O}_j[[\rho]]$. Since the coefficients of F° are polynomials in X_1, the Taylor expansion of the coefficients of F° yields

$$F^\circ(\Xi') - F^\circ(\Xi) = (\Xi' - \Xi)Q$$

for some $Q \in \mathcal{O}_j[[\rho]]$. Since

$$F^\circ \equiv \bar{f} = X_1^2 + p_1(Y_1, z_1, 1)X_1 + p_0(Y_1, z_1, 1) \mod \rho,$$

we have

$$Q \equiv \Xi' + \Xi + p_1(Y_1, z_1, 1) \equiv \frac{\partial \bar{f}}{\partial X_1} \mod \rho.$$

Since $\frac{\partial \bar{f}}{\partial X_1}$ is a unit of $\mathcal{O}_j$, Q is a unit of $\mathcal{O}_j[[\rho]]$. Thus $\Xi' - \Xi = Q^{-1}(F^\circ(\Xi') - F^\circ(\Xi))$, which implies the uniqueness in (1), and (2) once ξ_j exists.

For the existence in (1), let $\Xi_1 := X_1$. Since θ_j divides $\bar{f}$,

$$F^\circ(\Xi_1) \equiv \bar{f} \equiv 0 \mod \rho\mathcal{O}_j[[\rho]].$$

For $n \geq 1$, if $\Xi_n - X_1 \in \rho\mathcal{O}_j[[\rho]]$ and $F^\circ(\Xi_n) \in \rho^n\mathcal{O}_j[[\rho]]$, let

$$\Xi_{n+1} := \Xi_n - \left(\frac{\partial \bar{f}}{\partial X_1}\right)^{-1} F^\circ(\Xi_n).$$

By the first paragraph,

$$F^\circ(\Xi_{n+1}) = F^\circ(\Xi_n)\left(1 - \left(\frac{\partial \bar{f}}{\partial X_1}\right)^{-1} Q_n\right)$$

for some $Q_n \equiv \frac{\partial \bar{f}}{\partial X_1} \mod \rho$, so $F^\circ(\Xi_{n+1}) \in \rho^{n+1}\mathcal{O}_j[[\rho]]$. Since $\Xi_{n+1} - \Xi_n \in \rho^n\mathcal{O}_j[[\rho]]$, the sequence $(\Xi_n)_n$ converges to some $\xi_j \in \mathcal{O}_j[[\rho]]$ with $F^\circ(\xi_j) = 0$ (cf. [ZS60, Chapter VII, §1, Theorem 5 and Corollary 1]).

For (3), the local ring of W' at the generic point of E_j is $A_j/(F^\circ)$, which is a discrete valuation ring with uniformizer ρ and residue field L_j. The homomorphism $\mathbf{k}[X_1, Y_1, z_1][[\rho]] \to L_j[[\rho]]$ given by $P \to P(\xi_j)$ maps F° to 0. Any element of the coordinate ring of U outside $\mathfrak{p}_j$ maps to a power series whose constant term is non-zero in L_j, hence to a unit. Thus $P \to P(\xi_j)$ induces a local homomorphism $A_j/(F^\circ) \to L_j[[\rho]]$ which maps ρ to ρ. Hence $\operatorname{mult}_{E_j} P$ is the largest integer n such that $\widetilde{P}(\xi_j) \in \rho^n L_j[[\rho]]$. Since

$$\rho^n L_j[[\rho]] \cap \mathcal{O}_j[[\rho]] = \rho^n\mathcal{O}_j[[\rho]],$$

(3) follows. □

Proof of Lemma 9.2 continued.

Step 4. In this step we find a prime divisor E such that $a(E, \operatorname{Spec} R, 0) = 2$ and $\sum_l b_l \operatorname{mult}_E \phi_l \geq c_D$, and we prove the sentence of Lemma 9.2 beginning with 'In particular' and the first sentence of the paragraph of Lemma 9.2 beginning with 'Moreover'.

Let $\mathbf{k} = k$, let F be the power series of (C.1), and let $\pi : W' \to \operatorname{Spec} R$, $E_1, \dots, E_r$, and E_{tot} be as in **Step 3**. Let D' be the strict transform of D on W'. For any l and j, let $n_l := \operatorname{mult}_D \phi_l$ and $s_{lj} := \operatorname{mult}_{E_j} \phi_l$. Since π is an isomorphism over the generic point of D,

$$\operatorname{div}_{W'}(\phi_l) = D_l + \sum_{j=1}^{r} s_{lj} E_j$$

for a Weil divisor $D_l \geq 0$ on W' with no π-exceptional component such that $D_l - n_l D' \geq 0$.

Let $H := -E_{\mathrm{tot}}$, a $\mathbb{Q}$-Cartier divisor on W'. The restriction $H|_{E_{\mathrm{tot}}}$ is the pullback of $\mathcal{O}(1)$ under the finite morphism from E_{tot} onto the hypersurface $(\operatorname{in}_{\mathbf{w}} F = 0)$ of $\mathbb{P}(w_X, w_Y, w_z, w_t)$, so $H|_{E_{\mathrm{tot}}}$ is ample and $(H^2 \cdot E_j) > 0$ for any j. By **Step 3**, E_j maps birationally onto a component of $(\operatorname{in}_{\mathbf{w}} F = 0)$, and $\operatorname{in}_{\mathbf{w}} F$ is reduced. The Hilbert series of $k[X, Y, z', t']/(\operatorname{in}_{\mathbf{w}} F)$ is $(1 - s^{2w_X})/((1 - s^{w_X})(1 - s^{w_Y})(1 - s^{w_z})(1 - s^{w_t}))$, so the degree of $(\operatorname{in}_{\mathbf{w}} F = 0)$ with respect to $\mathcal{O}(1)$ implies

$$\sum_{j=1}^{r} (H^2 \cdot E_j) = \frac{\operatorname{wt}_{\mathbf{w}}(F)}{w_X w_Y w_z w_t} = \frac{2}{w_Y w_z w_t}. \tag{C.8}$$

The strict transform of $(X = Y = 0)$ in the weighted blow-up of $\operatorname{Spec} k[[X, Y, z', t']]$ is the weighted blow-up of $\operatorname{Spec} k[[z', t']]$ with weights w_z, w_t in z', t', which is a normal surface S whose exceptional curve C is isomorphic to $\mathbb{P}(w_z, w_t)$. Since $D = (X = Y = 0)$ and π is an isomorphism over the generic point of D, D' maps finitely and birationally onto S, hence isomorphically. On $U \cap D'$, $E_{\mathrm{tot}} = \operatorname{div}(\rho)$ and ρ is a local parameter of C, so $E_{\mathrm{tot}}|_{D'} = C$. Moreover $H|_C$ is $\mathcal{O}_{\mathbb{P}(w_z, w_t)}(1)$. By the Hilbert series of $k[z', t']$ with weights w_z, w_t,

$$(D' \cdot E_{\mathrm{tot}} \cdot H) = \frac{1}{w_z w_t}. \tag{C.9}$$

Since $\operatorname{div}_{W'}(\phi_l)$ is principal, $(\operatorname{div}_{W'}(\phi_l) \cdot E_{\mathrm{tot}} \cdot H) = 0$. By (C.6), $(E_j \cdot E_{\mathrm{tot}} \cdot H) = -(H^2 \cdot E_j)$ for any j. Since $D_l - n_l D' \geq 0$ has no component in $\operatorname{Supp} E_{\mathrm{tot}}$, the intersection of $D_l - n_l D'$ with E_{tot} is an effective 1-cycle supported in $\pi^{-1}(o)$, and H is ample on $\pi^{-1}(o)$. Thus, by (C.9),

$$\sum_{j=1}^{r} s_{lj} (H^2 \cdot E_j) = (D_l \cdot E_{\mathrm{tot}} \cdot H) \geq n_l (D' \cdot E_{\mathrm{tot}} \cdot H) = \frac{n_l}{w_z w_t}. \tag{C.10}$$

Multiplying (C.10) by $b_l \geq 0$ and summing over l, by (C.8) and (C.3) we obtain

$$\sum_{j=1}^{r} \Big(\sum_{l} b_l s_{lj} \Big) (H^2 \cdot E_j) \geq \frac{c_D}{w_z w_t} = \frac{c_D w_Y}{2} \sum_{j=1}^{r} (H^2 \cdot E_j) \geq c_D \sum_{j=1}^{r} (H^2 \cdot E_j),$$

where the last inequality holds because $c_D \geq 0$. Since $(H^2 \cdot E_j) > 0$ for any j, there exists j_0 such that $\sum_l b_l \operatorname{mult}_{E_{j_0}} \phi_l \geq c_D$. Let $E := E_{j_0}$. By **Step 3**, E has center o and $a(E, \operatorname{Spec} R, 0) = 2$.

Since $K_{\operatorname{Spec} R} + B$ is $\mathbb{R}$-Cartier,

$$\operatorname{mld}(\operatorname{Spec} R \ni o, B) \leq a(E, \operatorname{Spec} R, B) = 2 - \sum_{l} b_l \operatorname{mult}_E \phi_l \leq 2 - c_D.$$

If $h \in R \backslash \{0\}$ and $\operatorname{mult}_D h \geq m$, then applying the last inequality to $B = \frac{1}{m} \operatorname{div}(h)$, for which $c_D = \frac{1}{m} \operatorname{mult}_D h \geq 1$, we obtain $\operatorname{mld}(\operatorname{Spec} R \ni o, \frac{1}{m} \operatorname{div}(h)) \leq 1$. The coordinates X, Y, z', t' satisfy $D = (X = Y = 0)$, the linear forms z', t' in y, z, t are fixed in **Steps 1 and 2**, $\mathbf{w} \in \{(2, 2, 1, 1), (3, 3, 1, 1), (3, 2, 2, 1)\}$, $\operatorname{in}_{\mathbf{w}} F$ is reduced by **Step 3**, and E is an exceptional prime divisor of the weighted blow-up with weights $\mathbf{w}$ in X, Y, z', t'.

Step 5. In this step we prove the existence of the integer N for the component $E = E_{j_0}$ fixed in **Step 4**.

Let $n := \max_l \operatorname{mult}_E \phi_l$ and $N := 2w_X + n + 1$. The coordinate changes of **Steps 1 and 2** form a k-automorphism of $k[[x, y, z, t]]$ which preserves $\mathfrak{m}$, so the congruence modulo $\mathfrak{m}^N$ does not

depend on whether we use x, y, z, t or X, Y, z', t'. Let g' and ϕ'_l be as in the lemma, where we use representatives of ϕ_l and ϕ'_l in $k[[x, y, z, t]]$ which are congruent modulo $\mathfrak{m}^N$. Let $F' := x^2 + g'$, regarded as a power series in X, Y, z', t'. Then $F' - F \in \mathfrak{m}^N$. Since any monomial of degree at least N has $\mathbf{w}$-weight at least $N > 2w_X$, F' satisfies (ii), and $\mathrm{in}_{\mathbf{w}} F'$ is equal to $\mathrm{in}_{\mathbf{w}} F$. Thus F' satisfies (iii), and $\bar{f}$, $\theta_1, \dots, \theta_r$, and $\mathcal{O}_1, \dots, \mathcal{O}_r$ of **Step 3** for F' are equal to those for F. Since R' is normal, F' satisfies (i). By **Step 3** for F', the weighted blow-up of $\operatorname{Spec} R'$ with weights $\mathbf{w}$ in X, Y, z', t' has exceptional prime divisors $E'_1, \dots, E'_r$ corresponding to $\theta_1, \dots, \theta_r$, with $a(E'_j, \operatorname{Spec} R', 0) = 2$ for any j. Let $E' := E'_{j_0}$, and let ξ_{j_0} and ξ'_{j_0} be the power series of Claim C.1(1) for F and for F' respectively.

Since $F' - F \in \mathfrak{m}^N$, we have

$$F'^{\circ} - F^{\circ} \in \rho^{N-2w_X}\mathbf{k}[X_1, Y_1, z_1][[\rho]] = \rho^{n+1}\mathbf{k}[X_1, Y_1, z_1][[\rho]], \qquad \widetilde{\phi'_l} - \widetilde{\phi_l} \in \rho^N \mathbf{k}[X_1, Y_1, z_1][[\rho]].$$

Thus

$$F'^{\circ}(\xi_{j_0}) = F'^{\circ}(\xi_{j_0}) - F^{\circ}(\xi_{j_0}) \in \rho^{n+1}\mathcal{O}_{j_0}[[\rho]],$$

and by Claim C.1(2) for F', $\xi'_{j_0} - \xi_{j_0} \in \rho^{n+1}\mathcal{O}_{j_0}[[\rho]]$. Since the coefficients of $\widetilde{\phi_l}$ are polynomials in X_1, the difference of $\widetilde{\phi_l}(\xi'_{j_0})$ and $\widetilde{\phi_l}(\xi_{j_0})$ is divisible by $\xi'_{j_0} - \xi_{j_0}$. Hence

$$\widetilde{\phi'_l}(\xi'_{j_0}) \equiv \widetilde{\phi_l}(\xi'_{j_0}) \equiv \widetilde{\phi_l}(\xi_{j_0}) \mod \rho^{n+1}\mathcal{O}_{j_0}[[\rho]].$$

Since $\operatorname{mult}_E \phi_l \le n$, Claim C.1(3) for F and the last congruence imply

$$\widetilde{\phi_l}(\xi_{j_0}) \in \rho^{\operatorname{mult}_E \phi_l}\mathcal{O}_{j_0}[[\rho]], \qquad \widetilde{\phi'_l}(\xi'_{j_0}) \in \rho^{\operatorname{mult}_E \phi_l}\mathcal{O}_{j_0}[[\rho]],$$

and by Claim C.1(3) for F', $\operatorname{mult}_{E'} \phi'_l \ge \operatorname{mult}_E \phi_l$ for any l.

Step 6. In this step we prove the last paragraph of the lemma.

Generic limits are constructed in [dFEM11, Section 3], and compatible generic limits are defined in Construction-Proposition 4.1. By the definition of a compatible generic limit (Construction-Proposition 4.1(a)), there is a subsequence of the pairs (F_i, h_i) such that, for any positive integer n, the n-jet of (F_*, h_*) maps to the generic point of an irreducible closed subset Z_n of the affine space $\mathcal{J}_n$ of n-jets, and any non-empty open subset of Z_n contains the n-jets of (F_i, h_i) for all but finitely many i in the subsequence.

The data on the limit. Since $x^2 + g \in \mathfrak{m}^2$, the ring R has embedding dimension four, so $F_* \in \mathfrak{m}^2$. Thus a k-algebra isomorphism $R \to k[[x, y, z, t]]/(F_*)$ lifts to a k-algebra endomorphism γ of $k[[x, y, z, t]]$ which induces an isomorphism on $\mathfrak{m}/\mathfrak{m}^2$. Hence γ is an automorphism and $\gamma((x^2 + g)) = (F_*)$. We regard D as a prime divisor on $\operatorname{Spec} k[[x, y, z, t]]/(F_*)$ through the isomorphism induced by γ. Let X, Y, z', t', $\mathbf{w}$, and F be as in **Steps 1 and 2**. Then

$$F(\gamma(X), \gamma(Y), \gamma(z'), \gamma(t')) = \gamma(x^2 + g) = vF_*$$

for a unit $v \in k[[x, y, z, t]]$. Replacing X, Y, z', t' with $\gamma(X), \gamma(Y), \gamma(z'), \gamma(t')$, we may assume that $vF_* = F(X, Y, z', t')$ and $D = (X = Y = 0)$. Let $\mathbf{u} = (u_x, u_y, u_z, u_t)$ be the power series in X, Y, z', t' such that

$$x = u_x(X, Y, z', t'), \qquad y = u_y(X, Y, z', t'), \qquad z = u_z(X, Y, z', t'), \qquad t = u_t(X, Y, z', t').$$

Then $F_*(\mathbf{u}) = v(\mathbf{u})^{-1}F$. Let c be the coefficient of X^2 in $F_*(\mathbf{u})$, that is, $c = v(0)^{-1}$. The power series $c^{-1}F_*(\mathbf{u}) = v(0)v(\mathbf{u})^{-1}F$ satisfies (i)–(iii) of **Step 3**, $\mathrm{in}_{\mathbf{w}}(c^{-1}F_*(\mathbf{u}))$ is equal to $\mathrm{in}_{\mathbf{w}} F$, and W', $E_1, \dots, E_r$, and $\mathcal{O}_1, \dots, \mathcal{O}_r$ of **Step 3** for $c^{-1}F_*(\mathbf{u})$ are equal to those for F. If $h_* \in (F_*)$, let j_0 be any index. Otherwise, by **Step 4** for $B = \operatorname{div}(h_*)$, there exists j_0 such that $\operatorname{mult}_{E_{j_0}} h_* \ge \operatorname{mult}_D h_* \ge m$. Let $E := E_{j_0}$, let $\theta := \theta_{j_0}$, let $\eta := \bar{f}/\theta$, and let $e \in \{1, 2\}$ be the degree of θ in X_1. Let $\xi \in \mathcal{O}_{j_0}[[\rho]]$ be the power series of Claim C.1(1) for $c^{-1}F_*(\mathbf{u})$. If $h_* \notin (F_*)$, then by Claim C.1(3),

$$\widetilde{h_*(\mathbf{u})}(\xi) \in \rho^m \mathcal{O}_{j_0}[[\rho]].$$

If $h_* \in (F_*)$, then $h_*(\mathbf{u})$ is a multiple of $c^{-1}F_*(\mathbf{u})$, whose image in $k[X_1, Y_1, z_1][[\rho]]$ is $\rho^{2w_X}(c^{-1}F_*(\mathbf{u}))^\circ$. Since $P \to \widetilde{P}$ and $P \to P(\xi)$ are ring homomorphisms and $(c^{-1}F_*(\mathbf{u}))^\circ(\xi) = 0$, we obtain

$$\widetilde{h_*(\mathbf{u})}(\xi) = 0,$$

so the last membership holds in this case too.

Since $\mathcal{O}_{j_0}$ is the localization of $k[X_1, Y_1, z_1]/(\theta)$ at $\frac{\partial \bar{f}}{\partial X_1}$, there exist an integer $\epsilon \geq 0$ and polynomials $P_1, \dots, P_{m-1} \in k[X_1, Y_1, z_1]$ such that

$$\Xi := X_1 + \sum_{n=1}^{m-1} P_n \Big(\frac{\partial \bar{f}}{\partial X_1}\Big)^{-\epsilon} \rho^n \equiv \xi \mod \rho^m \mathcal{O}_{j_0}[[\rho]].$$

Let $N_m := 2w_X + m$ and $\epsilon_1 := \epsilon N_m$. Since the power series

$$(c^{-1}F_*(\mathbf{u}))^\circ(\Xi) = (c^{-1}F_*(\mathbf{u}))^\circ(\Xi) - (c^{-1}F_*(\mathbf{u}))^\circ(\xi), \qquad \widetilde{h_*(\mathbf{u})}(\Xi) - \widetilde{h_*(\mathbf{u})}(\xi)$$

are divisible by $\Xi - \xi$, both

$$(c^{-1}F_*(\mathbf{u}))^\circ(\Xi) \qquad \text{and} \qquad \widetilde{h_*(\mathbf{u})}(\Xi)$$

belong to $\rho^m \mathcal{O}_{j_0}[[\rho]]$. For $0 \leq n < m$, the coefficients of ρ^n in

$$\Big(\frac{\partial \bar{f}}{\partial X_1}\Big)^{\epsilon_1} (c^{-1}F_*(\mathbf{u}))^\circ(\Xi), \qquad \Big(\frac{\partial \bar{f}}{\partial X_1}\Big)^{\epsilon_1} \widetilde{h_*(\mathbf{u})}(\Xi),$$

computed in $k[X_1, Y_1, z_1]_{\partial \bar{f}/\partial X_1}$, are polynomials, because a monomial of $\mathbf{w}$-weight less than N_m has degree less than N_m in X. Since θ is irreducible and $\frac{\partial \bar{f}}{\partial X_1} \notin (\theta)$, a polynomial with zero image in $\mathcal{O}_{j_0}$ belongs to (θ). Thus the coefficients of ρ^n above are θa_n and θb_n for some polynomials $a_n, b_n \in k[X_1, Y_1, z_1]$.

The parts of $\mathbf{w}$-weight less than N_m of $F_*(\mathbf{u})$ and of $h_*(\mathbf{u})$ depend only on the parts of degree less than N_m of F_*, h_*, and $\mathbf{u}$, since $\mathbf{u}$ has no constant terms and any monomial of degree at least N_m has $\mathbf{w}$-weight at least N_m. The coefficients of ρ^n with $n < m$ in

$$(c^{-1}F_*(\mathbf{u}))^\circ \qquad \text{and} \qquad \widetilde{h_*(\mathbf{u})}$$

depend only on the parts of $\mathbf{w}$-weight less than N_m of $F_*(\mathbf{u})$ and $h_*(\mathbf{u})$.

The incidence space. Let M be an integer which bounds the degrees of θ, η, P_n, a_n, and b_n. Let $\mathcal{V}$ be the affine space over $\mathbb{C}$ of tuples

$$\big(\mathbf{u}^\sharp, c^\sharp, \theta^\sharp, \eta^\sharp, (P_n^\sharp)_{1 \leq n < m}, (a_n^\sharp, b_n^\sharp)_{0 \leq n < m}\big),$$

where $\mathbf{u}^\sharp$ is a 4-tuple of polynomials in X, Y, z', t' of degree less than N_m with zero constant terms, $c^\sharp$ is a number, $\theta^\sharp$ and $\eta^\sharp$ are polynomials in X_1, Y_1, z_1 of degree at most M which are monic in X_1 of degrees e and $2 - e$ respectively, and $P_n^\sharp$, $a_n^\sharp$, and $b_n^\sharp$ are polynomials in X_1, Y_1, z_1 of degree at most M. Let $\mathcal{I} \subset \mathcal{J}_{N_m} \times \mathcal{V}$ be the subset of the points $((F^\sharp, h^\sharp), \mathbf{u}^\sharp, c^\sharp, \dots)$ with the following properties.

(a) The Jacobian determinant of $\mathbf{u}^\sharp$ at 0 is non-zero, and $c^\sharp \neq 0$.
(b) Any monomial of $F^\sharp(\mathbf{u}^\sharp)$ of $\mathbf{w}$-weight less than $2w_X$ has coefficient zero, and the coefficient of X^2 in $F^\sharp(\mathbf{u}^\sharp)$ is $c^\sharp$.
(c) Let

$$\bar{f}^\sharp := (c^\sharp)^{-1}(F^\sharp(\mathbf{u}^\sharp))^{[2w_X]}(X_1, Y_1, z_1, 1).$$

Then the discriminant of $\bar{f}^\sharp$ as a polynomial in X_1 is non-zero, and $\bar{f}^\sharp = \theta^\sharp \eta^\sharp$.
(d) Let

$$F^{\sharp\circ} := (c^\sharp)^{-1} \sum_{n<m} (F^\sharp(\mathbf{u}^\sharp))^{[2w_X+n]}(X_1, Y_1, z_1, 1)\rho^n,$$

let

$$\widetilde{h^\sharp} := \sum_{n<m} (h^\sharp(\mathbf{u}^\sharp))^{[n]}(X_1, Y_1, z_1, 1)\rho^n,$$

and let

$$\Xi^\sharp := X_1 + \sum_{n=1}^{m-1} P_n^\sharp \Big(\frac{\partial \bar{f}^\sharp}{\partial X_1}\Big)^{-\epsilon} \rho^n.$$

Then, for $0 \le n < m$, the coefficients of ρ^n in

$$\Big(\frac{\partial \bar{f}^\sharp}{\partial X_1}\Big)^{\epsilon_1} F^{\sharp\circ}(\Xi^\sharp), \qquad \Big(\frac{\partial \bar{f}^\sharp}{\partial X_1}\Big)^{\epsilon_1} \widetilde{h^\sharp}(\Xi^\sharp)$$

are $\theta^\sharp a_n^\sharp$ and $\theta^\sharp b_n^\sharp$ respectively.

By the degree argument in *The data on the limit*, which uses only N_m and ϵ, the coefficients of ρ^n, $n < m$, in (d) are polynomials in X_1, Y_1, z_1 whose coefficients are polynomials in the coordinates of $\mathcal{J}_{N_m} \times \mathcal{V}$ and in $(c^\sharp)^{-1}$. After multiplying by powers of $c^\sharp$, the conditions in (b), (d), and the equality in (c) are finitely many polynomial identities in the coordinates of $\mathcal{J}_{N_m} \times \mathcal{V}$, and the conditions in (a) and the non-vanishing in (c) are open. Thus $\mathcal{I}$ is locally closed, and by [Har77, Exercise II.3.19] the image $\mathrm{pr}(\mathcal{I})$ of $\mathcal{I}$ under the projection pr of $\mathcal{J}_{N_m} \times \mathcal{V}$ onto $\mathcal{J}_{N_m}$ is constructible.

By the data on the limit, the N_m-jet of (F_*, h_*), together with the part of degree less than N_m of $\mathbf{u}$ and with c, θ, η, $(P_n)_{1\le n<m}$, and $(a_n, b_n)_{0\le n<m}$, is a k-point of $\mathcal{I}$. Thus $\mathrm{pr}(\mathcal{I})$ contains the generic point of Z_{N_m}, and by [Har77, Exercise II.3.18(b)], $\mathrm{pr}(\mathcal{I})$ contains a non-empty open subset of Z_{N_m}. Hence, for all but finitely many i in the subsequence, the N_m-jet of (F_i, h_i) belongs to $\mathrm{pr}(\mathcal{I})$. Fix such i. Since the fiber of $\mathrm{pr}\,|_{\mathcal{I}}$ over the N_m-jet of (F_i, h_i) is a non-empty scheme of finite type over $\mathbb{C}$, it has a $\mathbb{C}$-point

$$\big(\mathbf{u}_i, c_i, \theta_i, \eta_i, (P_{i,n})_n, (a_{i,n}, b_{i,n})_n\big).$$

The data (F_i, h_i). By (a), there exist formal coordinates X_i, Y_i, z_i', t_i' of $\mathbb{C}[[x, y, z, t]]$ such that $x = u_{i,x}(X_i, Y_i, z_i', t_i')$, $y = u_{i,y}(X_i, Y_i, z_i', t_i')$, $z = u_{i,z}(X_i, Y_i, z_i', t_i')$, and $t = u_{i,t}(X_i, Y_i, z_i', t_i')$. Since $\mathbf{u}_i$ has no constant terms and any monomial of degree at least N_m has $\mathbf{w}$-weight at least N_m, the parts of $\mathbf{w}$-weight less than N_m of $F_i(\mathbf{u}_i)$ and $h_i(\mathbf{u}_i)$ depend only on the parts of degree less than N_m of F_i, h_i, and $\mathbf{u}_i$. The local ring of W_i at 0 is a normal G-ring [Mat89, Corollary to Theorem 32.6], so the map to the completion $\mathbb{C}[[x, y, z, t]]/(F_i)$ is faithfully flat [Mat89, Theorem 8.14] and regular, and $\mathbb{C}[[x, y, z, t]]/(F_i)$ is a normal local ring [Mat89, Theorem 32.2(i)], hence a normal domain. Thus, by (b) and (c), the power series $c_i^{-1}F_i(\mathbf{u}_i)$ in X_i, Y_i, z_i', t_i' satisfies (i)–(iii) of **Step 3** with $\mathbf{k} = \mathbb{C}$, and the polynomial $\bar{f}$ of **Step 3** for $c_i^{-1}F_i(\mathbf{u}_i)$ is $\bar{f}_i := \theta_i\eta_i$.

Since θ_i is monic in X_1 of degree $e \ge 1$ (θ_i itself need not be irreducible), θ_i has a monic irreducible factor θ'', which is one of the irreducible factors of $\bar{f}_i$ in **Step 3**. Let E'' be the corresponding exceptional prime divisor of the weighted blow-up of $\operatorname{Spec}\mathbb{C}[[x, y, z, t]]/(F_i)$ with weights $\mathbf{w}$ in X_i, Y_i, z_i', t_i', let $\mathcal{O}''$ be the localization of $\mathbb{C}[X_1, Y_1, z_1]/(\theta'')$ at $\frac{\partial \bar{f}_i}{\partial X_1}$, and let $\xi'' \in \mathcal{O}''[[\rho]]$ be the power series of Claim C.1(1).

By (C.7), $a(E'', \operatorname{Spec}\mathbb{C}[[x, y, z, t]]/(F_i), 0) = 2$.

Let

$$\Xi_i := X_1 + \sum_{n=1}^{m-1} P_{i,n}\Big(\frac{\partial \bar{f}_i}{\partial X_1}\Big)^{-\epsilon} \rho^n \in \mathcal{O}''[[\rho]].$$

For $n < m$, the coefficients of ρ^n in

$$(c_i^{-1}F_i(\mathbf{u}_i))^\circ \qquad \text{and} \qquad \widetilde{h_i(\mathbf{u}_i)}$$

are equal to the coefficients of ρ^n in $F^{\sharp\circ}$ and in $\widetilde{h^\sharp}$ of (d) for the N_m-jet of (F_i, h_i), $\mathbf{u}_i$, and c_i. Since θ'' divides θ_i, (d) implies that

$$(c_i^{-1}F_i(\mathbf{u}_i))^\circ(\Xi_i),\ \widetilde{h_i(\mathbf{u}_i)}(\Xi_i) \in \rho^m\mathcal{O}''[[\rho]].$$

By Claim C.1(2), $\Xi_i - \xi'' \in \rho^m\mathcal{O}''[[\rho]]$. Since the difference

$$\widetilde{h_i(\mathbf{u}_i)}(\Xi_i) - \widetilde{h_i(\mathbf{u}_i)}(\xi'')$$

is divisible by $\Xi_i - \xi''$, we obtain

$$\widetilde{h_i(\mathbf{u}_i)}(\xi'') \in \rho^m\mathcal{O}''[[\rho]].$$

By Claim C.1(3), $\mathrm{mult}_{E''} h_i \geq m$.

The prime divisor E'' is a formal prime divisor over $\operatorname{Spec}\mathbb{C}[[x,y,z,t]]/(F_i)$ with center the closed point. The completion of the local ring of W_i at 0 is $\mathbb{C}[[x,y,z,t]]/(F_i)$. By [Liu26, Lemma 3.3(3)], the restriction of the valuation $\mathrm{mult}_{E''}$ to the function field of W_i is mult_{E_i} for a prime divisor E_i over W_i with center 0, and

$$a(E_i, W_i, 0) = a(E'', \operatorname{Spec}\mathbb{C}[[x,y,z,t]]/(F_i), 0) = 2.$$

Since h_i belongs to the function field of W_i, $\mathrm{mult}_{E_i} h_i = \mathrm{mult}_{E''} h_i \geq m$. Since i is any index outside a finite subset of the subsequence, there exist infinitely many such i. □

C.2. **The double-plane divisor.** In this subsection, we describe the double-plane divisor E_0 of Set-up 9.3 on the root charts.

Lemma C.2. *Notation as in Set-up 9.3. We write* wt *and* in *for* $\mathrm{wt}_{(3,2,2,2)}$ *and* $\mathrm{in}_{(3,2,2,2)}$ *in the parameters* x,y,z,t [Liu26, Definition 2.16]*, and we let* $\mathcal{R} := \mathbb{C}[x,y,z,t]/(x^2+g_3)$*, graded by* wt*. For* $g \in G$*, let* $\chi_x(g), \chi_y(g), \chi_z(g), \chi_t(g) \in \mathbb{C}^*$ *be the scalars with* $g^*x = \chi_x(g)x$*,* $g^*y = \chi_y(g)y$*,* $g^*z = \chi_z(g)z$*, and* $g^*t = \chi_t(g)t$*. Fix* $u \in \{y,z,t\}$*, let* $\{u_1,u_2\} = \{y,z,t\}\backslash\{u\}$*, and let* σ, X, u', u'' *be the coordinates of the* u*-root chart of Set-up 9.3(7), so that* $u = \sigma^2$*,* $x = \sigma^3 X$*,* $u_1 = \sigma^2 u'$*, and* $u_2 = \sigma^2 u''$*. For a polynomial* Q *in* x,y,z,t*, let* $\bar{Q} \in \mathbb{C}[X,u',u'']$ *be the polynomial obtained by substituting* $X, 1, u', u''$ *for* x, u, u_1, u_2*. Let* $\tilde{f} := \sigma^{-6}f$*, let* $W' = (\tilde{f}=0)$ *be the strict transform of* W *on the* u*-root chart,* $\pi : W' \to W$ *the induced morphism,* $S_0 := W' \cap (\sigma = 0)$ *the scheme-theoretic intersection, and* $Y_u \subset Y$ *the preimage of the* u*-chart of the weighted blow-up. Then the following hold.*

1. $\mathrm{in}(f) = x^2 + g_3$*, and* x^2+g_3 *is irreducible. The scheme* S_0 *is the reduced surface* $(X^2 + \bar{g}_3 = 0) \subset \mathbb{A}^3_{X,u',u''}$*, and* $\mathrm{div}(\sigma) = S_0$ *on* W'*. A point of* S_0 *is a singular point of* S_0 *if and only if* $X = 0$ *and* $(u',u'') \in \mathrm{Sing}(\bar{g}_3 = 0)$*. At any smooth point of* S_0*,* W' *is smooth. Moreover,* W' *is normal and irreducible.*
2. *Let* ι *be the involution* $(\sigma, X, u', u'') \to (-\sigma, -X, u', u'')$*. Then* $\iota(W') = W'$*,* $Y_u = W'/\langle\iota\rangle$ *with quotient morphism* $\psi : W' \to Y_u$*, and the morphism* $(u_1/u, u_2/u) : E_0 \cap Y_u \to \mathbb{A}^2$ *is an isomorphism. In particular,* E_0 *is the unique exceptional prime divisor of* $Y \to W$*,* E_0 *is* G*-stable, and the morphism* $E_0 \to \mathbb{P}^2$ *given by* $(y:z:t)$ *is an isomorphism.*
3. *Let* Φ *be a regular function on the ambient germ such that* $\mathrm{in}(\Phi) \notin (x^2+g_3)$*. Then* $\mathrm{mult}_{E_0}(\Phi|_W) = \mathrm{wt}(\Phi)$*, and the image of* $\Phi|_W$ *in the graded ring of* E_0 *is the class of* $\mathrm{in}(\Phi)$ *in* $\mathcal{R}$*. Any non-zero regular function* φ *on* W *near* o *is the restriction of such a function* Φ*, and if* φ *is* G*-semi-invariant, then there is such a* Φ *which is* G*-semi-invariant with the character of* φ*. In particular, the graded ring of* E_0 *is* $\mathcal{R}$*.*
4. $a(E_0, W, 0) = 3$*. Moreover,* $\pi^*h = \sigma^s h'$ *for a regular function* h' *on* W' *such that* $h'|_{S_0} = \bar{P}|_{S_0}$ *if the image of* h *in Set-up 9.3(6) is* P*,* $h'|_{S_0} = X\bar{P}|_{S_0}$ *if it is* xP*,* $h'|_{S_0}$ *does not vanish identically on any irreducible component of* S_0*, and*

$$K_{W'} + B_{W'} = \pi^*\Big(K_W + \frac{1}{m}\mathrm{div}(h)\Big), \qquad B_{W'} := \frac{1}{m}\mathrm{div}(h') + \Big(\frac{s}{m} - 2\Big)\mathrm{div}(\sigma).$$

(5) Let $\tilde{G}$ be the group of pairs $(g,\eta) \in G \times \mathbb{C}^*$ with $\eta^2 = \chi_u(g)$, acting on the u-root chart by

$$(g,\eta)^*\sigma = \eta\sigma, \quad (g,\eta)^*X = \chi_x(g)\eta^{-3}X, \quad (g,\eta)^*u' = \chi_{u_1}(g)\eta^{-2}u', \quad (g,\eta)^*u'' = \chi_{u_2}(g)\eta^{-2}u''.$$

Then $\tilde{G}$ is a finite abelian group which acts faithfully on W' and preserves $B_{W'}$, and the homomorphism $\tilde{G} \to G$, $(g,\eta) \to g$, is surjective with kernel $\langle\iota\rangle$, and π is equivariant with respect to it. For any prime divisor F' over W' whose center on W' is contained in S_0,

$$a(F', W', B_{W'}) \geq \alpha |I_{\tilde{G}}(F')|. \tag{C.11}$$

Let F be a prime divisor over W computing α, that is, the center of F on W is o and $a(F, W, \frac{1}{m}\operatorname{div}(h)) = \alpha|I_G(F)|$. If the center Z of F on Y intersects Y_u, then there exists a prime divisor F' over W' whose center on W' is contained in S_0 and maps onto $Z \cap Y_u$, such that $a(F', W', B_{W'}) = \alpha|I_{\tilde{G}}(F')|$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1), except for the irreducibility of W'.

The quadratic part of f is x^2, any cubic monomial divisible by x has weight at least 7, and any monomial of degree at least four has weight at least 8. Thus $\operatorname{in}(f) = x^2 + g_3$. Since $x^2 + g_3$ is monic in x over the factorial ring $\mathbb{C}[y,z,t]$, $x^2 + g_3$ is reducible if and only if $-g_3$ is a square in $\mathbb{C}[y,z,t]$, which is impossible because $g_3 \neq 0$ is homogeneous of degree three.

Any monomial $x^a y^b z^c t^d$ pulls back to $\sigma^{3a+2b+2c+2d}$ times a monomial in X, u', u''. Let Φ be a regular function on the ambient germ and $k := \operatorname{wt}(\Phi)$. Since, after shrinking the ambient germ, $\Phi - \operatorname{in}(\Phi)$ belongs to the ideal generated by the monomials of weight at least $k+1$ (the expansion of $\Phi - \operatorname{in}(\Phi)$ in $\mathbb{C}[[x,y,z,t]]$ has only such monomials, and the completion $\widehat{\mathcal{O}}$ of the local ring $\mathcal{O}$ of the ambient germ at o is faithfully flat over $\mathcal{O}$ [Mat89, Theorems 8.14 and 7.5(ii)]), the pullback of Φ to the u-root chart is

$$\sigma^k \overline{\operatorname{in}(\Phi)} + \sigma^{k+1}\Theta \tag{C.12}$$

for a regular function Θ. Applied to f, (C.12) shows that $\tilde{f} = X^2 + \bar{g}_3 + \sigma\Theta_f$ for a regular function Θ_f, so $S_0 = (\sigma = X^2 + \bar{g}_3 = 0)$. Since $g_3 \neq 0$ is homogeneous, $\bar{g}_3 \neq 0$. Hence $X^2 + \bar{g}_3$ is either irreducible or equal to $(X-b)(X+b)$ for a non-zero $b \in \mathbb{C}[u', u'']$. In both cases S_0 is a reduced surface, and $\operatorname{div}(\sigma) = S_0$. By the Jacobian criterion, a point of S_0 is singular on S_0 if and only if $X = 0$, $\bar{g}_3 = 0$ and both partial derivatives of $\bar{g}_3$ vanish there. If p is a smooth point of S_0, then $\mathcal{O}_{W',p}/(\sigma) = \mathcal{O}_{S_0,p}$ is regular of dimension 2 and $\dim \mathcal{O}_{W',p} = 3$, so W' is smooth at p.

The Jacobian determinant of the morphism from the u-root chart to the ambient germ is $\pm 2\sigma^8$, so the morphism from the u-root chart to the ambient germ is étale on $(\sigma \neq 0)$. Since W is smooth outside o, $W' \backslash S_0$ is smooth. Thus the singular locus of W' is contained in the singular locus of S_0, whose dimension is at most one. As a hypersurface in a smooth fourfold, W' is Cohen–Macaulay, so W' is normal by Serre's criterion. Since $\dim S_0 = 2$, no irreducible component of W' is contained in $(\sigma = 0)$.

Step 2. In this step we fix an irreducible component D of S_0, and we prove that $\operatorname{mult}_D(\pi^*(\Phi|_W)) = \operatorname{wt}(\Phi)$ for any Φ as in (3), and that any non-zero regular function φ on W near o is the restriction of such a Φ, the class of $\operatorname{in}(\Phi)$ in $\mathcal{R}$ depending only on φ, and that, if φ is G-semi-invariant, there is such a Φ which is G-semi-invariant with the character of φ.

Let $k := \operatorname{wt}(\Phi)$. Dividing $\operatorname{in}(\Phi)$ by the monic polynomial $x^2 + g_3$ in x, we write $\operatorname{in}(\Phi) = (x^2+g_3)Q + A + xB$ with $A, B \in \mathbb{C}[y,z,t]$ homogeneous and $\operatorname{wt}(A) = \operatorname{wt}(xB) = k$. Since non-zero homogeneous elements of $\mathbb{C}[y,z,t]$ have even weight, $A = 0$ or $B = 0$, and $A + xB \neq 0$ because $\operatorname{in}(\Phi) \notin (x^2 + g_3)$. On W' we have $X^2 + \bar{g}_3 = -\sigma\Theta_f$, so by (C.12),

$$\pi^*(\Phi|_W) = \sigma^k\big(\bar{A} + X\bar{B} + \sigma\Theta'\big) \tag{C.13}$$

for a regular function Θ' on W'. Any irreducible component of S_0 is defined by a factor of $X^2+\bar{g}_3$ monic in X, hence maps onto $\mathbb{A}^2_{u',u''}$. Thus a non-zero polynomial in u',u'' does not vanish identically on D. Moreover, X does not vanish identically on D, because $S_0\cap(X=0)=(X=\bar{g}_3=0)$ and $\dim(X=\bar{g}_3=0)\leq 1$. Hence $\bar{A}+X\bar{B}$ does not vanish identically on D. Since $\mathrm{mult}_D\,\sigma=1$, (C.13) implies that $\mathrm{mult}_D(\pi^*(\Phi|_W))=k$.

Let $\varphi\neq 0$ be a regular function on W near o, and W_1 the irreducible component of W' containing D. Since W_1 dominates W, $\pi^*\varphi$ does not vanish identically on W_1, and by (C.12), $\mathrm{wt}(\Phi)\leq\mathrm{mult}_D(\pi^*\varphi)$ for any regular function Φ on the ambient germ with $\Phi|_W=\varphi$. Let Φ be such a function of maximal weight k. If $\mathrm{in}(\Phi)=(x^2+g_3)\Psi$, with Ψ homogeneous of weight $k-6$, then $\Phi-\Psi f$ restricts to φ and has weight larger than k, because $\Phi-\mathrm{in}(\Phi)$ and $\Psi(f-\mathrm{in}(f))$ have weight larger than k. This is a contradiction, so $\mathrm{in}(\Phi)\notin(x^2+g_3)$. If Φ_1,Φ_2 are two functions restricting to φ with $\mathrm{in}(\Phi_j)\notin(x^2+g_3)$, then both have weight $k=\mathrm{mult}_D(\pi^*\varphi)$, and $\Phi_1-\Phi_2=\Theta f$ for a regular function Θ. Either $\mathrm{wt}(\Theta f)>k$ and $\mathrm{in}(\Phi_1)=\mathrm{in}(\Phi_2)$, or $\mathrm{wt}(\Theta f)=k$ and

$$\mathrm{in}(\Phi_1)-\mathrm{in}(\Phi_2)=(x^2+g_3)\,\mathrm{in}(\Theta).$$

Thus the class of $\mathrm{in}(\Phi)$ in $\mathcal{R}$ depends only on φ.

Assume moreover that φ is G-semi-invariant with character χ, that is, $g^*\varphi=\chi(g)\varphi$ for $g\in G$. Since W is G-stable, $g^*f=v_gf$ for a unit v_g, so $g^*(x^2+g_3)=v_g(o)(x^2+g_3)$ by $\mathrm{in}(f)=x^2+g_3$. As G acts diagonally on x,y,z,t, g^* preserves the weight of every monomial, so G acts on the graded ring $\mathcal{R}$ and $\mathrm{in}(g^*\Phi)=g^*\,\mathrm{in}(\Phi)$. Since $g^*\Phi$ restricts to $\chi(g)\varphi$, the class of $\mathrm{in}(\Phi)$ in $\mathcal{R}$ is χ-semi-invariant. Let

$$\Phi^\chi:=\frac{1}{|G|}\sum_{g\in G}\chi(g)^{-1}g^*\Phi.$$

Then $\Phi^\chi|_W=\varphi$, Φ^χ is G-semi-invariant with character χ, and the weight k part of Φ^χ is the χ-component

$$\mathrm{in}(\Phi)^\chi:=\frac{1}{|G|}\sum_{g\in G}\chi(g)^{-1}g^*\,\mathrm{in}(\Phi)$$

of $\mathrm{in}(\Phi)$, whose class in $\mathcal{R}$ is the non-zero class of $\mathrm{in}(\Phi)$. Hence $\mathrm{wt}(\Phi^\chi)=k$, $\mathrm{in}(\Phi^\chi)=\mathrm{in}(\Phi)^\chi$, and $\mathrm{in}(\Phi^\chi)\notin(x^2+g_3)$.

Step 3. In this step we prove that W' is irreducible, and we prove (2).

On $(\sigma\neq 0)$, the morphism from the u-root chart to $(u\neq 0)$ is the étale double cover given by $\sigma^2=u$, with deck transformation ι. Thus, by **Step 1**, $W'\backslash S_0\to W\backslash(u=0)$ is finite étale of degree two with deck transformation ι, and $W'\backslash S_0$ is dense in W'. Assume that W' is reducible. Since W' is normal, W' is the disjoint union of an irreducible component W_1 and $\iota(W_1)$, and $W_1\backslash S_0\to W\backslash(u=0)$ is an isomorphism. Thus $\sigma|_{W_1}=\pi^*\theta$ for a regular function θ on $W\backslash(u=0)$ with $\theta^2=u$, and θ is regular on W near o because $\theta^2=u$ and W is normal. For an irreducible component D of S_0 contained in W_1, $\mathrm{mult}_D(\pi^*\theta)=\mathrm{mult}_D\,\sigma=1$. By **Step 2**, $\mathrm{mult}_D(\pi^*\theta)$ is the weight of a regular function on the ambient germ, which is 0 or at least 2. This is a contradiction, so W' is irreducible.

The u-chart of the weighted blow-up of the ambient germ is the quotient of the u-root chart by $\langle\iota\rangle\cong\mu_2$. The u-chart is the affine toric variety of the cone τ spanned by $(3,2,2,2)$ and the standard basis vectors e_x,e_{u_1},e_{u_2} with respect to $\mathbb{Z}^4$. Moreover, σ,X,u',u'' are the characters dual to the basis $(3,2,2,2),e_x,e_{u_1},e_{u_2}$ of a sublattice $N'\subset\mathbb{Z}^4$ of index two. So the u-root chart is the affine toric variety of τ with respect to N', and $\mathbb{Z}^4/N'$, generated by the class of e_u, acts by ι [CLS11, Proposition 1.3.18]. Thus $W'/\langle\iota\rangle$ maps finitely onto the strict transform of W on the u-chart, and birationally because $W'\backslash S_0\to W\backslash(u=0)$ is the quotient by ι. Since $W'/\langle\iota\rangle$ is normal, $Y_u=W'/\langle\iota\rangle$.

If $\sigma=0$, then $x=u=u_1=u_2=0$, so $\pi^{-1}(o)=S_0$ as sets. Hence the preimage of o in Y_u, with the reduced structure, is $E':=\psi(S_0)$. The functions $u_1/u=u'$ and $u_2/u=u''$ are ι-invariant, hence regular on Y_u. Since ι acts on the free $\mathbb{C}[u',u'']$-module $\mathbb{C}[X,u',u'']/(X^2+\bar{g}_3)$ with basis $1,X$

by $X \to -X$, the quotient $S_0/\langle\iota\rangle$ is $\mathbb{A}^2_{u',u''}$. Thus $\psi|_{S_0}$ induces a surjective morphism $\mathbb{A}^2 \to E'$ whose composite with $(u_1/u, u_2/u) : E' \to \mathbb{A}^2$ is the identity, and $(u_1/u, u_2/u) : E' \to \mathbb{A}^2$ is an isomorphism.

A point of the exceptional divisor of the weighted blow-up which belongs to the strict transform of W is a zero of $\mathrm{in}(f) = x^2 + g_3$, and $x^2 + g_3$ has no zero with $y = z = t = 0$ other than the origin. Hence Y_y, Y_z, and Y_t cover the preimage E of o in Y. The morphism $E \to \mathbb{P}^2$ given by $(y : z : t)$ restricts on $E \cap Y_u = E'$ to the isomorphism onto $(u \neq 0)$ above, and $E \cap Y_u$ is the preimage of $(u \neq 0)$. Thus $E \to \mathbb{P}^2$ is an isomorphism. In particular, E is irreducible of dimension 2, so $E = E_0$ is the unique exceptional prime divisor of $Y \to W$. Since G acts diagonally on x, y, z, t, G acts on the weighted blow-up and on Y, and E_0 is G-stable.

Step 4. In this step we prove (3).

Let D be an irreducible component of S_0. Then $\psi(D) = E_0 \cap Y_u$. If $\iota(D) \neq D$, then ι does not stabilize D, so $\iota \notin I_{\langle\iota\rangle}(D)$. If $\iota(D) = D$, then ι acts non-trivially on D because X does not vanish identically on D (**Step 2**). In both cases $I_{\langle\iota\rangle}(D)$ is trivial, and by [Liu26, Lemma 2.6], $\mathrm{mult}_{E_0}\varphi = \mathrm{mult}_D(\pi^*\varphi)$ for any regular function φ on W near o. By the equality $\mathrm{mult}_{E_0}\varphi = \mathrm{mult}_D(\pi^*\varphi)$ and **Step 2**, $\mathrm{mult}_{E_0}(\Phi|_W) = \mathrm{wt}(\Phi)$ for any Φ as in (3). The third sentence of (3) is proved in **Step 2**. Moreover, $\mathrm{mult}_{E_0}\varphi \geq \mathrm{wt}(\Phi)$ for any Φ with $\Phi|_W = \varphi$. Hence, if $\mathrm{mult}_{E_0}\varphi_1 = \mathrm{mult}_{E_0}\varphi_2 = k$ and the classes of φ_1 and φ_2 in $\mathcal{R}$ are equal, then $\varphi_1 - \varphi_2$ is the restriction of a function of weight larger than k, so $\mathrm{mult}_{E_0}(\varphi_1 - \varphi_2) > k$. For functions Φ_1, Φ_2 as in (3) restricting to φ_1, φ_2, we have

$$\mathrm{in}(\Phi_1\Phi_2) = \mathrm{in}(\Phi_1)\,\mathrm{in}(\Phi_2) \notin (x^2 + g_3),$$

because $x^2 + g_3$ is irreducible. Hence the map sending φ to the class of $\mathrm{in}(\Phi)$ induces an injective homomorphism of graded rings from the graded ring of E_0 to $\mathcal{R}$. The homomorphism is surjective, because $\mathrm{in}(Q) = Q$ for any homogeneous polynomial $Q \notin (x^2 + g_3)$.

Step 5. In this step we prove (4).

The sheaves ω_W and $\omega_{W'}$ are invertible [Har77, Theorem III.7.11] (the proof there is local). On the smooth loci, adjunction [Har77, Proposition II.8.20] shows that they are generated by the residue forms $dy \wedge dz \wedge dt/\partial_x f$ and $du' \wedge du'' \wedge d\sigma/\partial_X \tilde{f}$. Since W and W' are normal, $dy \wedge dz \wedge dt/\partial_x f$ generates ω_W near o, and $du' \wedge du'' \wedge d\sigma/\partial_X \tilde{f}$ generates $\omega_{W'}$. Since

$$du_1 \wedge du_2 \wedge du = 2\sigma^5 du' \wedge du'' \wedge d\sigma \quad \text{and} \quad \pi^*(\partial_x f) = \sigma^{-3}\partial_X(\sigma^6\tilde{f}) = \sigma^3\partial_X\tilde{f},$$

we obtain

$$\pi^*\Big(\frac{dy \wedge dz \wedge dt}{\partial_x f}\Big) = \pm 2\sigma^2 \frac{du' \wedge du'' \wedge d\sigma}{\partial_X \tilde{f}}.$$

Thus $K_{W'} = \pi^*K_W + 2\,\mathrm{div}(\sigma)$. Let $(Y_u, B^0_{Y_u})$ be the crepant pullback of $(W, 0)$ to Y_u. Then $(Y_u, B^0_{Y_u})$ is the crepant quotient of $(W', -2\,\mathrm{div}(\sigma))$ by $\langle\iota\rangle$, because

$$\psi^*(K_{Y_u} + B^0_{Y_u}) = K_{W'} - 2\,\mathrm{div}(\sigma).$$

As $I_{\langle\iota\rangle}(D)$ is trivial (**Step 4**) and $\mathrm{mult}_D\,\mathrm{div}(\sigma) = 1$, [Liu26, Lemma 2.6] implies that

$$a(E_0, W, 0) = a(E_0, Y_u, B^0_{Y_u}) = a(D, W', -2\,\mathrm{div}(\sigma)) = 3.$$

By (3), $h = \Phi|_W$ for a regular function Φ on the ambient germ with $\mathrm{wt}(\Phi) = s$ and $\mathrm{in}(\Phi) \equiv P$, resp. xP, $\mathrm{mod}\,(x^2 + g_3)$. By (C.13) with $A + xB = P$, resp. xP, $\pi^*h = \sigma^s h'$ with $h' := \bar{P} + \sigma\Theta'$, resp. $h' := X\bar{P} + \sigma\Theta'$. So $h'|_{S_0} = \bar{P}|_{S_0}$, resp. $X\bar{P}|_{S_0}$, and $h'|_{S_0}$ does not vanish identically on any irreducible component of S_0 by **Step 2**. Finally,

$$\pi^*\Big(K_W + \frac{1}{m}\,\mathrm{div}(h)\Big) = K_{W'} - 2\,\mathrm{div}(\sigma) + \frac{1}{m}\big(s\,\mathrm{div}(\sigma) + \mathrm{div}(h')\big) = K_{W'} + B_{W'}.$$

Step 6. In this step we prove (5).

A direct calculation shows that the action of (g,η) on the u-root chart lifts the action of g on the ambient germ, that is, $(g,\eta)^*$ multiplies x,u,u_1,u_2 by $\chi_x(g),\chi_u(g),\chi_{u_1}(g),\chi_{u_2}(g)$. Since W is G-stable and W' is the closure of $\pi^{-1}(W\backslash(u=0))$, W' is $\tilde{G}$-stable and π is equivariant. As a subgroup of the finite abelian group $G\times\mu_{2|G|}$ (note that $\eta^{2|G|}=\chi_u(g)^{|G|}=1$), $\tilde{G}$ is finite and abelian. The homomorphism $\tilde{G}\to G$ is surjective because $\chi_u(g)$ has a square root, and the kernel consists of $(1,\pm1)$, where $(1,-1)$ acts as ι. The orders of x,u_1,u_2 at o are smaller than the order 2 of f, so none of x,u_1,u_2 vanishes on W. Thus σ,X,u',u'' restrict to non-zero functions on W'. If (g,η) acts trivially on W', then

$$\eta=\chi_x(g)\eta^{-3}=\chi_{u_1}(g)\eta^{-2}=\chi_{u_2}(g)\eta^{-2}=1,$$

so $\chi_x(g)=\chi_u(g)=\chi_{u_1}(g)=\chi_{u_2}(g)=1$. Hence g acts trivially on the ambient germ, so $g=1$ because G acts faithfully on W. Thus $\tilde{G}$ acts faithfully on W'. Since $K_W+\frac{1}{m}\operatorname{div}(h)$ is G-invariant and π is equivariant, $B_{W'}$ is $\tilde{G}$-invariant by (4).

Let $(W/G,B_{W/G})$ be the crepant quotient of $(W,\frac{1}{m}\operatorname{div}(h))$ by G, and (V,B_V) the crepant quotient of $(W',B_{W'})$ by $\tilde{G}$. Since $V=Y_u/G$ by (2), the morphism $Y_u\to W$ induces a birational morphism $\beta:V\to W/G$. The pullbacks of K_V+B_V and of $\beta^*(K_{W/G}+B_{W/G})$ to W' are both $K_{W'}+B_{W'}$ by (4). Since pullback of $\mathbb{R}$-divisors by the finite morphism $W'\to V$ is injective, $K_V+B_V=\beta^*(K_{W/G}+B_{W/G})$. Let F' be a prime divisor over W', and $\bar{F}$ the image of F' over V, a prime divisor over W/G. By [Liu26, Lemma 2.6],

$$a(F',W',B_{W'})=|I_{\tilde{G}}(F')|a(\bar{F},W/G,B_{W/G}). \tag{C.14}$$

If the center of F' on W' is contained in $S_0=\pi^{-1}(o)$, then the center of $\bar{F}$ on W/G is the image $\bar{o}$ of o, so $a(\bar{F},W/G,B_{W/G})\geq\operatorname{mld}(W/G\ni\bar{o},B_{W/G})=\alpha$ [Liu26, Lemma 2.6]. This proves (C.11).

Let F and Z be as in (5), and $\bar{F}$ the image of F over W/G. By [Liu26, Lemma 2.6],

$$\alpha|I_G(F)|=a\Big(F,W,\frac{1}{m}\operatorname{div}(h)\Big)=|I_G(F)|a(\bar{F},W/G,B_{W/G}),$$

so $a(\bar{F},W/G,B_{W/G})=\alpha$. Regarding $\bar{F}$ as a prime divisor over V, let F' be a prime divisor over W' whose image over V is $\bar{F}$. Then equality holds in (C.11) by (C.14). The center of $\bar{F}$ on V is the image of $Z\cap Y_u$, so the image under ψ of the center of F' is $g(Z\cap Y_u)$ for some $g\in G$. Possibly replacing F' with $\tilde{g}^{-1}(F')$ for an element $\tilde{g}\in\tilde{G}$ over g, which does not change $\bar{F}$, we may assume that the center of F' maps onto $Z\cap Y_u$. Since Z is contained in the preimage of o, the center of F' is contained in $\pi^{-1}(o)=S_0$. □

C.3. **A double line: a normal form.** In this subsection, we bring the equation of a source whose residual cubic is y^2z into a normal form (Lemma C.3), which is used in the proof of Lemma E.23 in Appendix E.

Lemma C.3. *Let $(W\ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one in a smooth fourfold germ, G a finite abelian group acting faithfully on W and fixing o, and x,y,z,t ambient G-eigenparameters in which the quadratic part of a local equation f of W is x^2 and the residual cubic is y^2z. Then there exists a G-equivariant automorphism of $\mathbb{C}[[x,y,z,t]]$ inducing the identity on the tangent space under which f becomes a unit times $x^2+y^2z+\lambda yt^\ell+q(z,t)$, where $\lambda\in\mathbb{C}$, $\ell\geq3$ is an integer, and $q\in\mathbb{C}[[z,t]]$ with $\operatorname{mult}_0 q\geq4$.*

Proof. Let $\mathfrak{n}$ be the maximal ideal of $\mathbb{C}[[y,z,t]]$, and for a G-semi-invariant series r let χ_r be the character of r. An automorphism of $\mathbb{C}[[x,y,z,t]]$ (or of $\mathbb{C}[[y,z,t]]$) sending each eigenparameter s to a G-semi-invariant series with character χ_s is G-equivariant. All automorphisms below send each eigenparameter s to a G-semi-invariant series with character χ_s, equal to s plus an element of the square of the maximal ideal, so they are G-equivariant and induce the identity on the tangent space.

First we remove the terms of f containing x. Since the quadratic part of f is x^2, the Weierstrass preparation theorem [GLS07, Chapter I, Theorem 1.6] implies

$$f = e \cdot (x^2 + p_1 x + p_0)$$

for a unit e and $p_0, p_1 \in \mathfrak{n}$. Let $\gamma \in G$. Since y, z, t are eigenparameters, γ^* preserves $\mathbb{C}[[y,z,t]]$, and

$$\chi_x(\gamma)^{-2}\gamma^*(x^2 + p_1 x + p_0) = x^2 + \chi_x(\gamma)^{-1}(\gamma^* p_1)x + \chi_x(\gamma)^{-2}\gamma^* p_0$$

is again a Weierstrass polynomial in x. Since W is G-stable, $\gamma^* f$ is a unit times f, so $\chi_x(\gamma)^{-2}\gamma^*(x^2 + p_1 x + p_0)$ is a unit times $x^2 + p_1 x + p_0$. The uniqueness in the Weierstrass preparation theorem implies $\gamma^* p_1 = \chi_x(\gamma) p_1$ and $\gamma^* p_0 = \chi_x(\gamma)^2 p_0$. Comparing quadratic parts, $e(0) = 1$, the linear part of p_1 is zero, and $p_0 \in \mathfrak{n}^3$. Let Φ_1 be the automorphism of $\mathbb{C}[[x,y,z,t]]$ with $\Phi_1(x) = x - \frac{1}{2}p_1$ fixing y, z, t. Then

$$\Phi_1(f) = \Phi_1(e) \cdot (x^2 + g), \qquad g := p_0 - \tfrac{1}{4}p_1^2 \in \mathfrak{n}^3,$$

and g is G-semi-invariant with $\chi_g = \chi_x^2$. Since $e(0) = 1$ and $p_1^2 \in \mathfrak{n}^4$, the cubic part of g is the cubic part of $f(0,y,z,t) = e(0,y,z,t)p_0$, which is the residual cubic $y^2 z$. In particular $\chi_g = \chi_y^2 \chi_z$.

Second, we remove degree by degree the monomials of g other than $y^2 z$ that are divisible by y^2 or by yz. Let $N \geq 4$ and assume that

$$g = y^2 z + \nu_N(t) y + b_N(z,t) + g_N + g',$$

where $\nu_N(t)y + b_N(z,t)$ is a polynomial of degree less than N with $\operatorname{mult}_0 \nu_N \geq 3$ and $\operatorname{mult}_0 b_N \geq 4$, g_N is homogeneous of degree N, and $g' \in \mathfrak{n}^{N+1}$. For $N = 4$ this holds with $\nu_4 = b_4 = 0$. A monomial of g_N is divisible by y^2, or divisible by yz but not by y^2, or a multiple of yt^{N-1}, or a monomial in z, t. Accordingly

$$g_N = y^2 P + yzQ + c_N y t^{N-1} + \bar{b}_N(z,t),$$

where $P \in \mathbb{C}[y,z,t]$ and $Q \in \mathbb{C}[z,t]$ are homogeneous of degree $N-2$, $c_N \in \mathbb{C}$, and $\bar{b}_N$ is homogeneous of degree N. Since any monomial of g has character $\chi_y^2 \chi_z$, P is semi-invariant with $\chi_P = \chi_z$, and Q is semi-invariant with $\chi_Q = \chi_y$. Let σ_N be the automorphism of $\mathbb{C}[[y,z,t]]$ with

$$\sigma_N(y) = y - \tfrac{1}{2}Q, \qquad \sigma_N(z) = z - P, \qquad \sigma_N(t) = t.$$

Then

$$\sigma_N(y^2 z) = y^2 z - y^2 P - yzQ + yPQ + \tfrac{1}{4}Q^2 z - \tfrac{1}{4}PQ^2,$$

and the last three terms belong to $\mathfrak{n}^{2N-3} \subset \mathfrak{n}^{N+1}$. Since σ_N moves y and z by elements of $\mathfrak{n}^{N-2}$, $\sigma_N(r) - r \in \mathfrak{n}^{k+N-3}$ for any $r \in \mathfrak{n}^k$, and $\nu_N(t)y + b_N(z,t) + g_N + g' \in \mathfrak{n}^4$. Thus

$$\sigma_N(g) = y^2 z + \big(\nu_N(t) + c_N t^{N-1}\big)y + b_N(z,t) + \bar{b}_N(z,t) + g''$$

with $g'' \in \mathfrak{n}^{N+1}$, which is the assumed form for $N+1$. Since σ_N moves the variables by elements of $\mathfrak{n}^{N-2}$, the composites $\sigma_N \circ \cdots \circ \sigma_4$ converge in the $\mathfrak{n}$-adic topology to a G-equivariant automorphism σ of $\mathbb{C}[[y,z,t]]$ inducing the identity on the tangent space, and

$$\sigma(g) = y^2 z + \nu(t) y + b(z,t), \qquad \nu \in \mathbb{C}[[t]],\ b \in \mathbb{C}[[z,t]],\ \operatorname{mult}_0 \nu \geq 3,\ \operatorname{mult}_0 b \geq 4.$$

Let Φ_2 be the automorphism of $\mathbb{C}[[x,y,z,t]]$ which is σ on $\mathbb{C}[[y,z,t]]$ and fixes x.

Third, we normalize ν. If $\nu = 0$, let $\lambda := 0$, $\ell := 3$, $q := b$, and let Φ_3 be the identity. Otherwise write $\nu = \lambda t^\ell U(t)$ with $\lambda \in \mathbb{C}^*$, $\ell \geq 3$, and $U(0) = 1$. Since any monomial $t^k y$ of $\nu(t)y$ has character $\chi_y^2 \chi_z$, any monomial t^k of ν has character $\chi_y \chi_z$, so U is G-invariant. Let Φ_3 be the automorphism of $\mathbb{C}[[x,y,z,t]]$ with $\Phi_3(y) = U(t)^{-1} y$ and $\Phi_3(z) = U(t)^2 z$ fixing x, t. Then

$$\Phi_3\big(x^2 + y^2 z + \nu(t)y + b(z,t)\big) = x^2 + y^2 z + \lambda y t^\ell + q(z,t), \qquad q(z,t) := b(U(t)^2 z, t),$$

and $\operatorname{mult}_0 q \geq 4$. The automorphism $\Phi := \Phi_3 \circ \Phi_2 \circ \Phi_1$ is G-equivariant, induces the identity on the tangent space, and

$$\Phi(f) = \Phi(e) \cdot \big(x^2 + y^2 z + \lambda y t^\ell + q(z,t)\big).$$

□

Appendix D. Proofs for sources of type cE

Here we prove Theorem 10.1, together with the results of Section 10 on which the proof depends. Subsections D.1 and D.2 prove Theorem 10.2 and Construction-Proposition 10.3, Subsections D.3 and D.4 treat circles on sources of type D and E, and Subsection D.5 proves a uniform margin. Subsections D.6 and D.7 then assemble the proof of Theorem 10.1.

D.1. **Log canonical thresholds of functions of order one.** In this subsection, we prove Theorem 10.2.

Proof of Theorem 10.2. The proof is long, so we divide it into two steps.

Step 1. In this step we prove that for any Du Val surface germ $(S \ni o)$ of type E_6, E_7, or E_8 and any regular function $f \in \mathfrak{m}_S \backslash \mathfrak{m}_S^2$, where $\mathfrak{m}_S$ is the maximal ideal of the local ring of S at o,

$$\left(S, \frac{1}{15}\operatorname{div}(f)\right) \text{ is lc near } o. \tag{D.1}$$

By [Roa04, Table 1], there exist a finite subgroup $\Pi \subset \mathrm{SL}_2(\mathbb{C})$, namely the binary tetrahedral, octahedral, or icosahedral group when S is of type E_6, E_7, or E_8 respectively, and an isomorphism $\widehat{\mathcal{O}}_{S,o} \cong \mathbb{C}[[u,v]]^\Pi$, where Π acts linearly on u, v. Moreover, $\mathbb{C}[u,v]^\Pi$ is generated by homogeneous polynomials X_1, X_2, X_3 of degrees $(6,8,12)$, $(8,12,18)$, or $(12,20,30)$ respectively, subject to one weighted homogeneous relation of degree 24, 36, or 60. No generator has the degree of the relation, so the relation has no linear term in X_1, X_2, X_3. Thus X_1, X_2, X_3 form a basis of $\widehat{\mathfrak{m}}_S/\widehat{\mathfrak{m}}_S^2$, where $\widehat{\mathfrak{m}}_S$ is the maximal ideal of $\widehat{\mathcal{O}}_{S,o}$. Since $\mathfrak{m}_S^2 = \widehat{\mathfrak{m}}_S^2 \cap \mathcal{O}_{S,o}$, we have $f \notin \widehat{\mathfrak{m}}_S^2$, and in $\widehat{\mathcal{O}}_{S,o}$ we may write

$$f = c_1X_1 + c_2X_2 + c_3X_3 + \varphi(X_1, X_2, X_3),$$

where $(c_1, c_2, c_3) \in \mathbb{C}^3 \backslash \{0\}$ and φ is a power series whose monomials have degree at least 2 in X_1, X_2, X_3. Let d be the least degree of a generator X_i with $c_i \neq 0$. Then $d \leq 30$.

Let $\tilde{f} \in \mathbb{C}[[u,v]]^\Pi$ be the image of f, and for any $j \geq 0$ let $\tilde{f}_j$ be the homogeneous part of degree j of $\tilde{f}$. The three generators have distinct degrees, and $\deg X_j > 0$ for $j = 1, 2, 3$, so a monomial of degree at least 2 in X_1, X_2, X_3 with weighted degree d contains only generators of degree less than d. Since $c_j = 0$ for any generator X_j of degree less than d, we obtain

$$\tilde{f}_d = c_iX_i + P, \qquad P \in \mathbb{C}[X_j \mid \deg X_j < d],$$

where the monomials of P have degree at least 2 in the generators. If $\tilde{f}_d = 0$, then $X_i = -c_i^{-1}P \in \widehat{\mathfrak{m}}_S^2$, which is impossible because X_1, X_2, X_3 form a basis of $\widehat{\mathfrak{m}}_S/\widehat{\mathfrak{m}}_S^2$. Thus $\tilde{f}_d \neq 0$, and

$$k := \operatorname{mult}_0 \tilde{f} \leq d \leq 30.$$

Since Π acts linearly on u, v and $\tilde{f}$ is Π-invariant, the leading form $\tilde{f}_k$ is a non-zero Π-invariant form of degree k. Write $\tilde{f}_k = \prod_p l_p^{m_p}$ up to a non-zero constant, where p runs over the distinct points of $\mathbb{P}^1$ defined by the linear factors l_p of $\tilde{f}_k$, and $\sum_p m_p = k$. Then Π permutes the points p and preserves the multiplicities m_p. The group Π has no fixed point on $\mathbb{P}^1$: a fixed point is a Π-invariant line in $\mathbb{C}^2$, which has a Π-invariant complement by Maschke's theorem, so Π would be diagonal in some basis, hence abelian, whereas the binary tetrahedral, octahedral and icosahedral groups are not abelian. Thus any Π-orbit on $\mathbb{P}^1$ has at least two points, and

$$m_p \leq \frac{k}{2} \quad \text{for any } p. \tag{D.2}$$

Fix an integer $N \geq 30$, and let $f_N := \sum_{j \leq N} \tilde{f}_j$. Since Π preserves degrees, f_N is a Π-invariant polynomial, hence a regular function on $S_0 := \mathbb{A}^2/\Pi$, and f_N has the leading form $\tilde{f}_k$ because

$k \le N$. Let $\mu : Y \to \mathbb{A}^2$ be the blow-up of the origin, E the exceptional curve, and D_N the strict transform of $\operatorname{div}(f_N)$. Since $K_Y = \mu^* K_{\mathbb{A}^2} + E$ and $\mu^* \operatorname{div}(f_N) = D_N + kE$,

$$K_Y + E + \frac{2}{k} D_N = \mu^* \Big(K_{\mathbb{A}^2} + \frac{2}{k} \operatorname{div}(f_N) \Big).$$

Under $E \cong \mathbb{P}^1$, $D_N|_E = \sum_p m_p p$, so by (D.2) the coefficients of $\frac{2}{k} D_N|_E$ are at most 1, and $(E, \frac{2}{k} D_N|_E)$ is lc. Since E is a smooth curve on the smooth surface Y and $E \not\subset \operatorname{Supp} D_N$, $K_E = (K_Y + E)|_E$ and the different of $\frac{2}{k} D_N$ on E is $\frac{2}{k} D_N|_E$. By inversion of adjunction [Kaw07, Theorem], $(Y, E + \frac{2}{k} D_N)$ is lc near E. Thus $(\mathbb{A}^2, \frac{2}{k} \operatorname{div}(f_N))$ is lc near 0.

Any non-identity element of finite order of $\mathrm{SL}_2(\mathbb{C})$ is diagonalizable with eigenvalues $\zeta, \zeta^{-1} \ne 1$, so Π contains no pseudo-reflection. Hence the quotient map $\mathbb{A}^2 \to S_0$ is étale in codimension one, so the crepant quotient of $(\mathbb{A}^2, \frac{2}{k} \operatorname{div}(f_N))$ by Π is $(S_0, \frac{2}{k} \operatorname{div}(f_N))$, with no branch divisor. By [Liu26, Lemma 2.6], which holds for any finite group acting faithfully on a normal quasi-projective variety, $(S_0, \frac{2}{k} \operatorname{div}(f_N))$ is lc near 0. Therefore $\operatorname{lct}(S_0 \ni 0; \operatorname{div}(f_N)) \ge \frac{2}{k}$ (Definition 2.11) for any $N \ge 30$.

It remains to pass from f_N to f. By [Roa04, Table 1],

$$\widehat{\mathcal{O}}_{S_0,0} = \mathbb{C}[[u,v]]^{\Pi} \cong \widehat{\mathcal{O}}_{S,o},$$

and the isomorphism maps f to $\tilde{f}$. The homogeneous parts of the Π-invariant series $\tilde{f} - f_N$ have degrees at least $N+1$, so $\tilde{f} - f_N$ is a power series in monomials in X_1, X_2, X_3 of weighted degree at least $N+1$, hence of degree at least $(N+1)/30$ in X_1, X_2, X_3. Fix $\epsilon > 0$, take $N \ge \max\{30, 60/\epsilon\}$, and set $l := \lceil (N+1)/30 \rceil \ge 2/\epsilon$. Then $\tilde{f} - f_N \in \widehat{\mathfrak{m}}_S^l$, so

$$(\tilde{f}) + \widehat{\mathfrak{m}}_S^l = (f_N) + \widehat{\mathfrak{m}}_S^l.$$

By [dFEM11, Proposition 2.11], $\operatorname{lct}(S \ni o; \operatorname{div}(f))$ and $\operatorname{lct}(S_0 \ni 0; \operatorname{div}(f_N))$ are the lc thresholds of $\tilde{f}$ and f_N on the completion $\mathbb{C}[[u,v]]^{\Pi}$ of the klt germ $(S_0 \ni 0)$. Thus [dFEM11, Proposition 2.19(ii)], applied with $n = 2$ and exponent 1, implies

$$\big| \operatorname{lct}(S \ni o; \operatorname{div}(f)) - \operatorname{lct}(S_0 \ni 0; \operatorname{div}(f_N)) \big| \le \epsilon,$$

so $\operatorname{lct}(S \ni o; \operatorname{div}(f)) \ge \frac{2}{k} - \epsilon$. Since $\epsilon > 0$ is arbitrary,

$$\operatorname{lct}(S \ni o; \operatorname{div}(f)) \ge \frac{2}{k} \ge \frac{2}{30} = \frac{1}{15},$$

which proves (D.1).

Step 2. In this step we cut W by a hyperplane section S of type E_6, E_7, or E_8 on which h restricts to an element of $\mathfrak{m}_S \backslash \mathfrak{m}_S^2$, and deduce the theorem from (D.1).

Let $x_1, \dots, x_4$ be regular parameters at o of a smooth fourfold germ containing W as a hypersurface. Since W is singular at o, the equation of W belongs to the square of the maximal ideal, so $x_1, \dots, x_4$ form a basis of $\mathfrak{m}_o / \mathfrak{m}_o^2$. Write $h \equiv \sum_j b_j x_j \pmod{\mathfrak{m}_o^2}$ with $(b_1, \dots, b_4) \ne 0$. Let ℓ_0^2 be the quadratic part of the equation of W, where ℓ_0 is a non-zero linear form in $x_1, \dots, x_4$, and let ℓ_1^3 be the residual cubic of W, where ℓ_1 is a non-zero linear form on the hyperplane $(\ell_0 = 0)$ of the tangent space.

By [Rei83, Theorem 1.1], $(W \ni o)$ is an isolated cDV point, and by [Kaw02, Definition 2.2 and Theorem 2.3] a general hyperplane section of W through o is a normal Du Val surface germ. Thus there exists a dense open subset $\mathcal{U} \subset \mathbb{C}^4$ such that for any $(a_1, \dots, a_4) \in \mathcal{U}$, $L := \sum_j a_j x_j$ defines a normal Du Val surface germ $S := W \cap (L = 0)$ at o. We take $(a_1, \dots, a_4) \in \mathcal{U}$ not proportional to $(b_1, \dots, b_4)$ and with L not proportional to ℓ_0.

Since

$$\mathfrak{m}_S / \mathfrak{m}_S^2 = \mathfrak{m}_o / (\mathfrak{m}_o^2 + \mathbb{C} L)$$

and L is not proportional to $\sum_j b_j x_j$, we have $h|_S \in \mathfrak{m}_S \backslash \mathfrak{m}_S^2$. The quadratic part of the equation of S is the restriction of ℓ_0^2 to $(L = 0)$, which is non-zero, so S has quadratic rank one. By [Liu26, Lemma 11.3(1)], applied with the trivial group to the isolated cDV germ $(W \ni o)$, there

exist formal parameters x, y, z, t at o in which $W = (x^2 + g(y, z, t) = 0)$ with $\operatorname{mult} g \geq 3$, and the linear part of x is a non-zero multiple of ℓ_0. Since L is not proportional to ℓ_0, the linear part of L in x, y, z, t has a non-zero coefficient of one of y, z, t. After permuting y, z, t, we may assume that it is the coefficient of t, and we solve $L = 0$ for t as a power series in x, y, z. Substituting into $x^2 + g$, we obtain $x^2 e + xA + B$, where e is a unit, $\operatorname{mult} A \geq 2$, and B is a power series in y, z. Completing the square in x replaces B by $B - A^2/(4e)$, which changes only terms of order at least 4 (see [KM98, 4.24(1)(2)] for the elimination of the terms involving x). Thus S is given by $x^2 + q(y, z) = 0$ with $\operatorname{mult} q \geq 3$, and the cubic part q_3 of q is the restriction of ℓ_1^3 to the plane $(\ell_0 = L = 0)$. If $\operatorname{mult} q \geq 4$, then S is not Du Val [KM98, 4.25, Step 9]. Hence $q_3 \neq 0$, and q_3 is the cube of a non-zero linear form. A Du Val germ $x^2 + q(y, z)$ whose cubic part is a non-zero cube is of type E_6, E_7, or E_8 by [KM98, 4.25, Steps 5–8]. Thus S is of type E_6, E_7, or E_8.

By (D.1) applied to $f = h|_S$, the pair $(S, \frac{1}{15}\operatorname{div}(h|_S))$ is lc near o. The surface $S = \operatorname{div}(L|_W)$ is a normal Cartier divisor on the Gorenstein threefold W, so $K_S = (K_W + S)|_S$. Since $h|_S \neq 0$, S is not a component of $\operatorname{div}(h)$, and the different of $\frac{1}{15}\operatorname{div}(h)$ on S is $\frac{1}{15}\operatorname{div}(h|_S)$. By inversion of adjunction [Kaw07, Theorem], $(W, S + \frac{1}{15}\operatorname{div}(h))$ is lc near S, in particular near o. Since $S \geq 0$, $(W, \frac{1}{15}\operatorname{div}(h))$ is lc near o. □

D.2. **Normal forms of sources of type cE.** In this subsection, we prove Construction-Proposition 10.3.

The following lemma is a G-equivariant normal form, along the parameter t, for deformations of a weighted homogeneous Du Val equation of type D or E. It is used in the proof of Construction-Proposition 10.3(2) and in Subsections D.4 and D.6.

Lemma D.1. *Fix one of the Du Val types of Theorem 10.4, with $f_0 = x^2 + g_0(y, z)$ of q-weighted degree d, and a basis $M_1, \dots, M_\mu$ of $\mathbb{C}[y, z]/(\partial_y g_0, \partial_z g_0)$ consisting of monomials. Let G be a finite abelian group acting on $\mathbb{C}[[x, y, z, t]]$ through characters of x, y, z, and t, and let $\hat{f} \in \mathbb{C}[[x, y, z, t]]$ be a G-semi-invariant power series such that any monomial of $\hat{f}(x, y, z, 0) - f_0$ has q-weighted degree larger than d. Then there exists a G-equivariant automorphism ψ of $\mathbb{C}[[x, y, z, t]]$ with $\psi(t) = t$ such that $\psi(\hat{f})$ is the product of a G-invariant unit and*

$$f_0 + \sum_{j=1}^{\mu} a_j(t) M_j(y, z), \qquad a_j \in t\mathbb{C}[[t]],$$

where each non-zero $a_j(t)M_j$ is G-semi-invariant with the character of $\hat{f}$.

Proof. The proof is long, so we divide it into several steps. Write $\mathfrak{R} := \mathbb{C}[[x, y, z, t]]$, $(x_1, x_2, x_3) := (x, y, z)$, and $(q_1, q_2, q_3) := (q_x, q_y, q_z)$. Since G acts through characters of x, y, z, and t, any monomial is G-semi-invariant. For a character θ of G and $\varphi \in \mathfrak{R}$, let $\pi_\theta(\varphi)$ be the sum of the terms of φ whose monomials have character θ. We say that $\pi_\theta(\varphi)$ is the projection of φ to the character θ. If $\varphi' \in \mathfrak{R}$ is G-semi-invariant with character θ', then

$$\pi_\theta(\varphi\varphi') = \pi_{\theta\theta'^{-1}}(\varphi)\varphi'.$$

Let χ be the character of $\hat{f}$, χ_i the character of x_i, and χ_t the character of t. Weighted degrees are q-weighted degrees in x, y, z.

Step 1. In this step we show that

$$\mathbb{C}[[x, y, z]] = \bigoplus_{j=1}^{\mu} \mathbb{C}M_j \oplus J, \qquad J := (\partial_x f_0, \partial_y f_0, \partial_z f_0)\mathbb{C}[[x, y, z]], \tag{D.3}$$

and that the ideal $(\partial_x f_0, \partial_y f_0, \partial_z f_0)$ of $\mathbb{C}[x, y, z]$ contains any polynomial whose monomials have weighted degree larger than $d - 2$.

Since $\partial_x f_0 = 2x$, we have

$$\mathbb{C}[x, y, z]/(\partial_x f_0, \partial_y f_0, \partial_z f_0) \cong Q := \mathbb{C}[y, z]/(\partial_y g_0, \partial_z g_0),$$

and Q is graded by the weighted degree. A change of the coordinates y, z preserving the weights q_y, q_z induces a graded isomorphism of the algebras Q, so we may assume that g_0 is one of $y^2z + z^{k-1}$, $y^3 + z^4$, $y^3 + yz^3$, and $y^3 + z^5$. A direct calculation shows that Q has the basis

$$1, z, \dots, z^{k-2}, y; \qquad y^i z^j \ (i \le 1,\ j \le 2); \qquad 1, y, z, yz, z^2, z^3, z^4; \qquad y^i z^j \ (i \le 1,\ j \le 3),$$

respectively, whose largest weighted degrees are $2k-4$, 10, 16, and 28. In each case the largest weighted degree is $d-2$. Hence $Q_e = 0$ for $e > d-2$, that is, the ideal $(\partial_x f_0, \partial_y f_0, \partial_z f_0)$ of $\mathbb{C}[x,y,z]$ contains any polynomial whose monomials have weighted degree larger than $d-2$. Since $\dim Q = \mu < +\infty$, the ideal $(\partial_x f_0, \partial_y f_0, \partial_z f_0)$ contains a power of (x,y,z). Hence $\mathbb{C}[[x,y,z]]/J \cong Q$, and the monomials $M_1, \dots, M_\mu$ map to a basis of Q. Thus (D.3) holds.

Step 2. In this step we construct a G-equivariant automorphism ψ_1 of $\mathfrak{R}$ with $\psi_1(t) = t$ and $\psi_1(\mathbb{C}[[x,y,z]]) \subset \mathbb{C}[[x,y,z]]$ such that $\psi_1(\hat{f})(x,y,z,0) = f_0$.

Let $\phi := \hat{f}(x,y,z,0)$. The monomials of ϕ are monomials of $\hat{f}$, so they have character χ. Since f_0 is the sum of the terms of ϕ of weighted degree d, f_0 is G-semi-invariant with character χ, and $\partial_i f_0 := \partial f_0/\partial x_i$ is G-semi-invariant with character $\chi\chi_i^{-1}$.

Let $\phi' \in \mathbb{C}[[x,y,z]]$ be G-semi-invariant with character χ such that the least weighted degree of a monomial of $\phi' - f_0$ is an integer $w > d$, and let ϕ'_w be the sum of the terms of $\phi' - f_0$ of weighted degree w. Since $w > d-2$, **Step 1** implies $\phi'_w = \sum_i \xi_i \partial_i f_0$ for weighted homogeneous polynomials ξ_i of weighted degree $w - d + q_i$. Projecting the equality $\phi'_w = \sum_i \xi_i \partial_i f_0$ to the character χ, we may replace ξ_i with $\pi_{\chi_i}(\xi_i)$, so we may assume that ξ_i has character χ_i. Let σ be the endomorphism of $\mathfrak{R}$ with $\sigma(x_i) = x_i - \xi_i$ and $\sigma(t) = t$. Since ξ_i has weighted degree larger than q_i, the linear part of σ is the identity plus a map raising the weights, so σ is an automorphism. Since ξ_i has character χ_i, σ is G-equivariant. By the Taylor formula, the terms of $f_0(x-\xi) - f_0 + \sum_i \xi_i \partial_i f_0$ have weighted degree at least

$$(d - q_i - q_j) + (w - d + q_i) + (w - d + q_j) = 2w - d > w,$$

and for any monomial x^β of $\phi' - f_0$ of weighted degree $e \ge w$, the terms of $\sigma(x^\beta) - x^\beta$ have weighted degree at least $e + w - d > e$. Hence the monomials of $\sigma(\phi') - f_0$ have weighted degree larger than w.

Starting from $\phi' = \phi$ and iterating, we obtain G-equivariant automorphisms $\sigma_1, \sigma_2, \dots$ such that the least weighted degree w_n of $\sigma_n \circ \cdots \circ \sigma_1(\phi) - f_0$ tends to $+\infty$, and $\sigma_{n+1}(x_i) - x_i$ has weighted degree larger than $w_n - d$. Thus $\sigma_n \circ \cdots \circ \sigma_1$ converges in the (x,y,z)-adic topology to a G-equivariant endomorphism ψ_1 of $\mathfrak{R}$ with $\psi_1(t) = t$, $\psi_1(\mathbb{C}[[x,y,z]]) \subset \mathbb{C}[[x,y,z]]$, and $\psi_1(\phi) = f_0$. The map induced by ψ_1 on $\mathfrak{m}/\mathfrak{m}^2$, where $\mathfrak{m}$ is the maximal ideal of $\mathfrak{R}$, is unipotent, so ψ_1 is an automorphism. Without the group G, ψ_1 is the weighted elimination of [Kol98, Construction 2.6]. Here each correction ξ_i is projected to the character χ_i, so that ψ_1 is G-equivariant. Since $\psi_1(t) = t$ and ψ_1 maps $\mathbb{C}[[x,y,z]]$ into itself, $\psi_1(\hat{f})(x,y,z,0) = \psi_1(\phi) = f_0$.

Step 3. In this step we perform the t-adic Jacobi preparation. Let $F := \psi_1(\hat{f})$, a G-semi-invariant power series with character χ and $F(x,y,z,0) = f_0$.

We construct by induction on $n \ge 1$ G-equivariant automorphisms $\tau_1, \tau_2, \dots$ of $\mathfrak{R}$ and polynomials $a_j^{(n)} \in t\mathbb{C}[t]$ of degree less than n with the following properties: $\tau_n(t) = t$, $\tau_n(x_i) - x_i \in t^n\mathfrak{R}$, each non-zero $a_j^{(n)} M_j$ is G-semi-invariant with character χ, and

$$F^{(n)} := \tau_{n-1} \circ \cdots \circ \tau_1(F) \equiv f_0 + \sum_{j=1}^{\mu} a_j^{(n)}(t) M_j \pmod{t^n\mathfrak{R}}. \tag{D.4}$$

For $n = 1$, take $a_j^{(1)} = 0$. Then (D.4) is $F(x,y,z,0) = f_0$. Assume (D.4) for some n and write

$$F^{(n)} = f_0 + \sum_j a_j^{(n)}(t) M_j + t^n R_n(x,y,z) + t^{n+1} R'_n, \qquad R_n \in \mathbb{C}[[x,y,z]],\ R'_n \in \mathfrak{R}.$$

All terms of $F^{(n)}$ have character χ, so R_n is G-semi-invariant with character $\chi\chi_t^{-n}$. By (D.3), $R_n = \sum_j c_j M_j + \sum_i \xi_i \partial_i f_0$ with $c_j \in \mathbb{C}$ and $\xi_i \in \mathbb{C}[[x,y,z]]$. Projecting the equality $R_n = \sum_j c_j M_j + \sum_i \xi_i \partial_i f_0$ to the character $\chi\chi_t^{-n}$ and using the uniqueness of the decomposition (D.3), we see that $c_j = 0$ unless $t^n M_j$ has character χ, and that we may assume that ξ_i has character $\chi_i\chi_t^{-n}$. Let τ_n be the endomorphism of $\mathfrak{R}$ with $\tau_n(x_i) = x_i - t^n\xi_i$ and $\tau_n(t) = t$. It is G-equivariant because $t^n\xi_i$ has character χ_i, and it is an automorphism because it induces an invertible map on $\mathfrak{m}/\mathfrak{m}^2$. By the Taylor formula in t,

$$\tau_n(F^{(n)}) \equiv F^{(n)} - t^n \sum_i \xi_i \partial_i F^{(n)} \equiv f_0 + \sum_j \big(a_j^{(n)}(t) + c_j t^n\big) M_j \pmod{t^{n+1}\mathfrak{R}},$$

where we use $\partial_i F^{(n)} \equiv \partial_i f_0 \pmod{t\mathfrak{R}}$ and $a_j^{(n)} \in t\mathbb{C}[t]$. Thus $a_j^{(n+1)} := a_j^{(n)} + c_j t^n$ satisfies (D.4) for $n+1$, and each non-zero $a_j^{(n+1)} M_j$ has character χ.

Since $\tau_n(\varphi) - \varphi \in t^n\mathfrak{R}$ for any $\varphi \in \mathfrak{R}$, the automorphisms $\tau_n \circ \cdots \circ \tau_1$ converge in the t-adic topology, hence in the topology of $\mathfrak{R}$, to a G-equivariant endomorphism τ of $\mathfrak{R}$ with $\tau(t) = t$, which induces an invertible map on $\mathfrak{m}/\mathfrak{m}^2$. Thus τ is an automorphism. The polynomials $a_j^{(n)}$ converge t-adically to $a_j \in t\mathbb{C}[[t]]$, and (D.4) implies

$$\tau(F) \equiv f_0 + \sum_j a_j(t) M_j \pmod{t^n\mathfrak{R}}$$

for any n. Hence $\tau(F) = f_0 + \sum_j a_j(t) M_j$, and any non-zero $a_j(t)M_j$ is a sum of terms with character χ. Thus $\psi := \tau \circ \psi_1$ satisfies the conclusion of Lemma D.1, with the G-invariant unit equal to 1. □

Proof of Construction-Proposition 10.3(1). By [Rei83, Theorem 1.1], $(W \ni o)$ is an isolated cDV threefold singularity. By [Liu26, Lemma 11.3(1)], W has a G-equivariant closed embedding into a smooth fourfold germ $(Y \ni o)$ with $W = (f = 0)$ for a regular G-semi-invariant function f on Y, and there exist G-semi-invariant elements x_1, y_1, z_1, t_1 of the maximal ideal of $\widehat{\mathcal{O}}_{Y,o}$ such that $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x_1, y_1, z_1, t_1]]$, G acts diagonally on x_1, y_1, z_1, t_1, and the expansion $\hat{f}$ of f in $\widehat{\mathcal{O}}_{Y,o}$ is

$$\hat{f} = x_1^2 + g(y_1, z_1, t_1),$$

where g is a G-semi-invariant power series with $\operatorname{mult} g \geq 3$ whose cubic part g_3 is the residual ternary cubic. Let χ be the character of g.

By assumption, $g_3 = \ell^3$ for a linear form ℓ in y_1, z_1, t_1, and $\ell \neq 0$, since a general hyperplane section of the cDV point $(W \ni o)$ is Du Val, so its residual cubic, the restriction of g_3 to a general plane, is non-zero [KM98, 4.25, Step 9]. For any $\gamma \in G$, $(\gamma \cdot \ell)^3 = \chi(\gamma)\ell^3$, so $\gamma \cdot \ell \in \mathbb{C}\ell$ by unique factorization. Choosing ℓ and an eigenbasis of a G-stable complement of $\mathbb{C}\ell$ in the span of y_1, z_1, t_1 as new linear coordinates, we may assume that $g_3 = y_1^3$. Then $g(y_1, 0, 0)$ has order three, and by the Weierstrass preparation theorem there exist a unique unit $u \in \mathbb{C}[[y_1, z_1, t_1]]$ and unique $a_0, a_1, a_2 \in (z_1, t_1)\mathbb{C}[[z_1, t_1]]$ with

$$g = u(y_1^3 + a_2 y_1^2 + a_1 y_1 + a_0). \tag{D.5}$$

Taking the parts of degree at most three in (D.5), we obtain $u(0) = 1$, $\operatorname{mult} a_2 \geq 2$, $\operatorname{mult} a_1 \geq 3$, and $\operatorname{mult} a_0 \geq 4$. Let η be the character of y_1. For $\gamma \in G$, applying γ to (D.5) implies

$$g = \chi(\gamma)^{-1}\eta(\gamma)^3(\gamma \cdot u)\big(y_1^3 + \eta(\gamma)^{-1}(\gamma \cdot a_2)y_1^2 + \eta(\gamma)^{-2}(\gamma \cdot a_1)y_1 + \eta(\gamma)^{-3}(\gamma \cdot a_0)\big),$$

and the uniqueness in (D.5) implies $\gamma \cdot u = \chi(\gamma)\eta(\gamma)^{-3}u$ and $\gamma \cdot a_k = \eta(\gamma)^{3-k}a_k$. Since $u(0) = 1$, we obtain $\chi = \eta^3$, and u is G-invariant. Let $u^{-1/2}$ be the square root of u^{-1} with constant term 1. For $\gamma \in G$, $\gamma \cdot u^{-1/2}$ is a square root of u^{-1} with constant term 1, so $u^{-1/2}$ is G-invariant. Let $x_2 := u^{-1/2}x_1$ and $y_2 := y_1 + \frac{1}{3}a_2$. Then x_2, y_2, z_1, t_1 are G-semi-invariant parameters with the

characters of x_1, y_1, z_1, t_1, and a direct calculation shows

$$u^{-1}(x_1^2+g) = x_2^2 + y_2^3 + y_2\hat{A}(z_1,t_1) + \hat{B}(z_1,t_1), \qquad \hat{A} := a_1 - \frac{a_2^2}{3}, \quad \hat{B} := a_0 - \frac{a_1a_2}{3} + \frac{2a_2^3}{27},$$

with mult $\hat{A} \geq 3$ and mult $\hat{B} \geq 4$.

Let x, y, z, t be G-eigenparameters of Y at o whose classes in $\mathfrak{m}_{Y,o}/\mathfrak{m}_{Y,o}^2$ are those of x_2, y_2, z_1, t_1, let $\mathfrak{R} := \widehat{\mathcal{O}}_{Y,o}$, so that $\mathfrak{R} = \mathbb{C}[[x,y,z,t]]$, with maximal ideal $\mathfrak{m}$, and let ψ be the automorphism of $\mathfrak{R}$ with $\psi(x_2) = x$, $\psi(y_2) = y$, $\psi(z_1) = z$, and $\psi(t_1) = t$. Then ψ is G-equivariant. Since $\hat{f} = x_1^2 + g$ and u is G-invariant, $v := \psi(u^{-1})$ is a G-invariant unit of $\mathfrak{R}$ with

$$v\psi(\hat{f}) = x^2 + y^3 + y\hat{A}(z,t) + \hat{B}(z,t).$$

Below, we apply Construction-Theorem 4.13 with ψ and v.

Since $h \in \mathfrak{m}_o^2$ and $\mathfrak{m}_o^2$ is the image of $\mathfrak{m}_{Y,o}^2$, we may take h to be a regular G-semi-invariant function on Y in $\mathfrak{m}_{Y,o}^2$, by projecting any lift in $\mathfrak{m}_{Y,o}^2$ to the character of h. Let $\hat{h}$ be the expansion of h in $\mathfrak{R}$.

Since $(W, \frac{1}{m}\operatorname{div}(h))$ is terminal, it is klt. Let N_1 be the integer of Construction-Theorem 4.13 for $n = 4$, the single mark h with coefficient $\frac{1}{m}$, no further vectors $c^{(j)}$ and no prime divisors P_i, and the data ψ, v, and $u_1 := 1$. Fix $N \geq \max(N_1, 3)$, let L_1 be the integer of Construction-Theorem 4.13 for N, and fix $L \geq \max(L_1, 6)$. Let A and B be the sums of the homogeneous parts of $\hat{A}$ and $\hat{B}$ of degree less than L, let h' be the sum of the homogeneous parts of $\psi(\hat{h})$ of degree less than N, and let

$$f' := x^2 + y^3 + yA(z,t) + B(z,t), \qquad W' := (f' = 0) \subset \mathbb{A}^4,$$

with the diagonal action of G in which x, y, z, t have their characters in $\mathfrak{R}$. Since G acts diagonally and ψ is G-equivariant, f' and h' are G-semi-invariant polynomials with the characters of f and h, and

$$h' \equiv \psi(\hat{h}) \pmod{(x,y,z,t)^N}, \qquad f' \equiv v\psi(\hat{f}) \pmod{(x,y,z,t)^L}.$$

By Construction-Theorem 4.13(1), G acts faithfully on W', the pair $(W', \frac{1}{m}\operatorname{div}(h'))$ is klt near 0, and

$$\operatorname{mld}((W' \ni 0, \tfrac{1}{m}\operatorname{div}(h'))/G') = \operatorname{mld}((W \ni o, \tfrac{1}{m}\operatorname{div}(h))/G')$$

for any subgroup $G' \subset G$. By Construction-Theorem 4.13(3), W' is an isolated Gorenstein terminal germ at 0, and the completions of the local rings of W at o and of W' at 0 are G-equivariantly isomorphic. Since $\psi(\hat{h}) \in \mathfrak{m}^2$ and $N \geq 3$, we have $h' \in \mathfrak{m}^2$. Since $L \geq 6$, mult $A \geq 3$, mult $B \geq 4$, and the homogeneous parts A_3, B_4, and B_5 of A and B are those of $\hat{A}$ and $\hat{B}$.

It remains to show that exactly one of (E_6), (E_7), and (E_8) holds. The three conditions exclude each other, so it suffices to show that B_4, A_3, and B_5 do not all vanish. Assume to the contrary that $B_4 = A_3 = B_5 = 0$, so mult $A \geq 4$ and mult $B \geq 6$. By [Rei83, Theorem 1.1], the Gorenstein terminal point $0 \in W'$ is a cDV point, and by [Kaw02, Definition 2.2] a general hyperplane section of W' through 0 is a Du Val surface germ. A general hyperplane through 0 is $(t = \ell')$ with $\ell' := \alpha x + \beta y + \gamma z$ for complex numbers α, β, γ, and the hyperplane section is the surface germ $S \subset \mathbb{A}^3_{x,y,z}$ defined by

$$F_S := x^2 + y^3 + yA(z,\ell') + B(z,\ell').$$

Let $w := (3,2,1)$ be the weights of x, y, z. For a homogeneous polynomial $P_k(z,t)$ of degree k, the terms of $P_k(z,\ell') - P_k(1,\gamma)z^k$ have w-weight larger than k. Hence

$$\operatorname{in}_w(F_S) = x^2 + y^3 + A_4(1,\gamma)yz^4 + B_6(1,\gamma)z^6.$$

If $\operatorname{in}_w(F_S)$ were reducible, then, being monic of degree two in x, it would be $(x+P)(x-P)$ with $-P^2 = y^3 + A_4(1,\gamma)yz^4 + B_6(1,\gamma)z^6$, which is impossible because the right-hand side has degree three in y. Thus $\operatorname{in}_w(F_S)$ is irreducible and reduced. Let E be the prime divisor over S given by the weighted blow-up of $\mathbb{A}^3$ with weights w: the strict transform of S intersects the exceptional

divisor $\mathbb{P}(3,2,1)$ along the irreducible reduced curve $(\mathrm{in}_w(F_S)=0)$, and Lemma 4.12, applied with $n=3$ to S, which is normal near 0 as a Du Val surface germ, and to $\mathrm{in}_w(F_S)$, whose zero set in $\mathbb{A}^3$ is integral and not contained in a coordinate hyperplane, implies

$$a(E,S,0)=3+2+1-6=0.$$

Since a Du Val surface germ is klt, $a(E,S,0)>0$, a contradiction. Thus exactly one of (E_6), (E_7), and (E_8) holds. The three cases are the normal forms cE_6, cE_7, and cE_8 of [Kol98, Theorem 2.10]. □

Proof of Construction-Proposition 10.3(2). The proof is long, so we divide it into several steps. Let $(W_i \ni o_i, G_i, h_i)$ be the data, with α_i strictly increasing in $(0,1)$. For each i, let $W_i'=(f_i'=0)$ with $f_i'=x^2+y^3+yA_i+B_i$ and h_i' be as in (1), and let $A_{i,k}$ and $B_{i,k}$ be the homogeneous parts of A_i and B_i of degree k. By Construction-Theorem 4.13(1),

$$\alpha_i=\mathrm{mld}\big((W_i'\ni 0,\tfrac{1}{m}\,\mathrm{div}(h_i'))/G_i\big). \tag{D.6}$$

The germ W_i' is normal and klt, G_i acts diagonally on $\mathbb{A}^4$ and faithfully on W_i', h_i' is a G_i-semi-invariant polynomial, and $(W_i',\frac{1}{m}\,\mathrm{div}(h_i'))$ is klt near 0. We regard G_i as a subgroup of $\mathbb{T}^4$, where $\theta=(\theta_x,\theta_y,\theta_z,\theta_t)\in\mathbb{T}^4$ acts on the monomial $x^{e_1}y^{e_2}z^{e_3}t^{e_4}$ by multiplication by

$$\exp(2\pi\sqrt{-1}(e_1\theta_x+e_2\theta_y+e_3\theta_z+e_4\theta_t)).$$

Step 1. In this step we show that $|G_i|\to+\infty$.

Assume that $|G_i|\le M'$ for a positive integer M' and infinitely many i. Since W_i is Gorenstein, $m(K_{W_i}+\frac{1}{m}\,\mathrm{div}(h_i))$ is Cartier near o_i. By Lemma 4.8, for these i,

$$\alpha_i\in\frac{1}{m\,\mathrm{lcm}(1,2,\dots,M')}\mathbb{Z}\cap(0,1),$$

which is a finite set. This contradicts the assumption that α_i is strictly increasing. Hence $|G_i|\to+\infty$. Moreover $\alpha_i\ge\alpha_1>0$ for any i.

Step 2. In this step we show that, after passing to a subsequence, the G_i converge in the Hausdorff distance to a closed subgroup $H\subset\mathbb{T}^4$ with $\dim H=1$.

There exist finitely many monomials z^at^b with $a+b\in\{3,4,5\}$. Hence, after passing to a subsequence, we may assume that (E_6) holds for any i, or (E_7) holds for any i, or (E_8) holds for any i, and that there exist non-negative integers a,b such that z^at^b has a non-zero coefficient in $B_{i,4}$ in case (E_6), in $A_{i,3}$ in case (E_7), and in $B_{i,5}$ in case (E_8), for any i. Since f_i' is G_i-semi-invariant, the monomials x^2, y^3, and z^at^b (resp. yz^at^b in case (E_7)) have the character of f_i' on G_i. Hence $G_i\subset S$, where $S\subset\mathbb{T}^4$ is the closed subgroup defined by

$$2\theta_x=3\theta_y=a\theta_z+b\theta_t \text{ in cases } (\mathrm{E}_6) \text{ and } (\mathrm{E}_8),\qquad 2\theta_x=3\theta_y,\ 2\theta_y=a\theta_z+b\theta_t \text{ in case } (\mathrm{E}_7).$$

The relations defining S are linearly independent, so S^0 is the image of the linear subspace $V\subset\mathbb{R}^4$ defined by $2\theta_x=3\theta_y=a\theta_z+b\theta_t$ in cases (E_6) and (E_8), and by $2\theta_x=3\theta_y$ and $2\theta_y=a\theta_z+b\theta_t$ in case (E_7), with $\theta\in\mathbb{R}^4$, and $\dim V=2$. Since $a+b=4$, 3, and 5 respectively, V contains the vector

$$w_6:=(6,4,3,3),\qquad w_7:=(9,6,4,4),\qquad w_8:=(15,10,6,6), \tag{D.7}$$

respectively, whose coordinates are positive.

By Lemma 4.6, after passing to a subsequence, the G_i converge in the Hausdorff distance to a closed subgroup $H\subset\mathbb{T}^4$ such that $G_i\subset H$ and G_i intersects any connected component of H for $i\gg 0$, and $\dim H\ge 1$ by **Step 1**. Since S is closed and $G_i\subset S$, we have $H\subset S$, so $H^0\subset S^0$ and $\dim H\le 2$. If $\dim H=2$, then $H^0=S^0$, and the Lie algebra V of H contains the positive vector of (D.7). Then Lemma 4.7, applied to W_i', G_i, and the boundary $\frac{1}{m}\,\mathrm{div}(h_i')$, implies $\alpha_i\to 0$ by (D.6). This contradicts $\alpha_i\ge\alpha_1>0$. Hence $\dim H=1$.

Step 3. In this step we construct the circle and show that the coordinates of w are neither all positive nor all negative, that $C \neq D$, and that the non-zero part $B_{i,4}$, $A_{i,3}$, or $B_{i,5}$ is $\lambda_i z^a t^b$ with $\lambda_i \in \mathbb{C}^*$.

By Lemma 4.6, H^0 is the image of $\mathbb{R}w$ for a primitive $w \in \mathbb{Z}^4$, $R_i := G_i \cap \mathbf{T}_{(w)}$ is cyclic of order $r_i \to +\infty$, and $[G_i : R_i] = |H/H^0| =: e_0$ for $i \gg 0$. Since $\mathbb{R}w \subset V$, the coordinates of w satisfy $2w_1 = 3w_2$, so w_2 is even and $w = (3s, 2s, C, D)$ for integers s, C, D, with $\gcd(s, C, D) = \gcd(3s, 2s, C, D) = 1$. The second equation of V implies

$$6s = aC + bD \text{ in cases } (\mathrm{E}_6) \text{ and } (\mathrm{E}_8), \qquad 4s = aC + bD \text{ in case } (\mathrm{E}_7).$$

If $3s, 2s, C, D$ are all positive or all negative, then w or $-w$ is a positive vector in the Lie algebra $\mathbb{R}w$ of H, and Lemma 4.7 implies $\alpha_i \to 0$ as in **Step 2**, a contradiction. Hence $3s, 2s, C, D$ are neither all positive nor all negative. If $C = D$, then $(a+b)C = 6s$, $4s$, or $6s$ with $a+b = 4$, 3, or 5, so

$$w = \frac{C}{3} w_6, \qquad w = \frac{C}{4} w_7, \qquad w = \frac{C}{6} w_8$$

respectively. Since $w \neq 0$, $C \neq 0$, and the coordinates of w are all positive or all negative, which we have just excluded. Hence $C \neq D$.

The group R_i is $\mu_{r_i}(3s, 2s, C, D)$, and a generator of R_i acts on $z^{a'} t^{b'}$ by $\zeta_i^{a'C + b'D}$ for a primitive r_i-th root of unity ζ_i. Let $z^{a'} t^{b'}$ be a monomial with a non-zero coefficient in $B_{i,4}$, $A_{i,3}$, or $B_{i,5}$ respectively. Then $a' + b' = a + b$, and since f_i' is G_i-semi-invariant, $a'C + b'D \equiv aC + bD \pmod{r_i}$, that is, r_i divides $(a' - a)(C - D)$. Since $C \neq D$ is fixed, $|a' - a| \leq 5$, and $r_i \to +\infty$, we obtain $a' = a$ for $i \gg 0$. Thus, after removing finitely many i, the non-zero part $B_{i,4}$, $A_{i,3}$, or $B_{i,5}$ is $\lambda_i z^a t^b$ with $\lambda_i \in \mathbb{C}^*$.

Step 4. In this step we show that $a, b \geq 1$, which finishes the proof.

Assume to the contrary that $ab = 0$. Since $a + b \geq 3$, exactly one of a, b is zero. Exchanging z and t, which exchanges C and D, we may assume that $b = 0$. Then $a = 4$, 3, or 5 in the three cases, since $a + b = 4$, 3, or 5. Let $f_0 = x^2 + g_0(y, z)$, $q = (q_x, q_y, q_z)$, and d be the polynomial, the weights, and the q-weighted degree of Theorem 10.4 for the type E_6, E_7, or E_8 respectively, so that $g_0 = y^3 + z^4$, $y^3 + yz^3$, or $y^3 + z^5$, $q = (6, 4, 3)$, $(9, 6, 4)$, or $(15, 10, 6)$, and $d = 12$, 18, or 30, and let $M_1, \dots, M_\mu$ be the monomial basis of $\mathbb{C}[y, z]/(\partial_y g_0, \partial_z g_0)$ of **Step 1** of the proof of Lemma D.1. Any M_j has q-weighted degree at most $d - 2$.

With $b = 0$, the relation of **Step 3** is $6s = 4C$, $4s = 3C$, or $6s = 5C$ in the three cases, and since $q_y/q_z = 4/3$, $3/2$, or $5/3$, it is equivalent to $2sq_z = Cq_y$, and $3sq_y = 2sq_x$ because $q_x/q_y = 3/2$. Hence $(3s, 2s, C)$ is an integral vector proportional to the primitive vector q, and there exists an integer c' with $(3s, 2s, C) = c'(q_x, q_y, q_z)$. Let $v := D$. Then

$$w = (c'q_x, c'q_y, c'q_z, v), \qquad \gcd(c', v) = 1.$$

Let $\omega_i \in \mathbb{C}^*$ with $\lambda_i \omega_i^a = 1$, and let σ_i be the G_i-equivariant automorphism of $\mathfrak{R} = \mathbb{C}[[x, y, z, t]]$ with $\sigma_i(z) = \omega_i z$ fixing x, y, t. Let $\mathfrak{m}$ be the maximal ideal of $\mathfrak{R}$. Then

$$\sigma_i(f_i')(x, y, z, 0) = x^2 + y^3 + yA_i(\omega_i z, 0) + B_i(\omega_i z, 0),$$

and any monomial of $\sigma_i(f_i')(x, y, z, 0) - f_0$ has q-weighted degree larger than d, because $\operatorname{mult} A_i \geq 3$, $\operatorname{mult} B_i \geq 4$, and the following degree bounds hold. In case (E_6), the monomials yz^k $(k \geq 3)$ and z^k $(k \geq 5)$ have q-weighted degrees at least 13 and 15. In case (E_7), $B_{i,4} = 0$, and the monomials yz^k $(k \geq 4)$ and z^k $(k \geq 5)$ have q-weighted degrees at least 22 and 20. In case (E_8), $A_{i,3} = B_{i,4} = 0$, and the monomials yz^k $(k \geq 4)$ and z^k $(k \geq 6)$ have q-weighted degrees at least 34 and 36. By Lemma D.1, there exists a G_i-equivariant automorphism ψ_i of $\mathfrak{R}$ with $\psi_i(t) = t$ such that $\psi_i \sigma_i(f_i')$ is the product of a G_i-invariant unit and

$$\hat{f}_i := f_0 + \sum_{j=1}^{\mu} a_{ij}(t) M_j, \qquad a_{ij} \in t\mathbb{C}[[t]],$$

where each non-zero $a_{ij}(t)M_j$ is G_i-semi-invariant with the character of f_i'. Thus $\psi_i\sigma_i$ induces a G_i-equivariant isomorphism $\widehat{\mathcal{O}}_{W_i',0} \cong \mathfrak{R}/(\hat{f}_i)$, where G_i acts on $\mathfrak{R}$ through the diagonal action of G_i on $\mathbb{A}^4$. In particular R_i acts on x, y, z, t with the weights w. Moreover $\hat{f}_i \in \mathfrak{m}^2$ and the quadratic part of $\hat{f}_i$ has rank one, because $\psi_i\sigma_i$ is an automorphism and the quadratic part of f_i' is x^2. The quadratic part of $\hat{f}_i$ is $x^2 + t\ell_i$ for a linear form ℓ_i in y, z, t, so $\ell_i = 0$ and the quadratic part of $\hat{f}_i$ is x^2. We now consider the signs of c' and v.

Assume that $c' = 0$. Then $v = \pm 1$ and $\mathbf{T}_{(w)} = \mathbf{T}_{(0,0,0,1)}$, so $R_i = G_i \cap \mathbf{T}_{(0,0,0,1)} = \mu_{r_i}(0,0,0,1)$. The germ W_i' is an isolated singular cDV germ at 0 by [Kol98, Theorem 2.4], G_i is a finite diagonal abelian group preserving f_i' up to a character and acting faithfully on W_i', $r_i \to +\infty$, $[G_i : R_i] \le e_0$, and h_i' does not vanish identically on W_i'. Theorem 8.5, with the coordinates ordered as t, x, y, z, the DCC set $\{\frac{1}{m}\}$, and the boundary $\frac{1}{m}\operatorname{div}(h_i')$, implies that the numbers α_i of (D.6) do not form a strictly increasing sequence with $r_i \to +\infty$, a contradiction.

Assume that $v = 0$. Then $c' = \pm 1$, and since $\mathbf{T}_{(w)} = \mathbf{T}_{(-w)}$ we may assume that $c' = 1$. A generator of R_i acts on $x^{e_1}y^{e_2}z^{e_3}t^k$ by ζ_i^e, where e is the q-weighted degree of $x^{e_1}y^{e_2}z^{e_3}$. If t^kM_j has a non-zero coefficient in $a_{ij}(t)M_j$, then the character of t^kM_j on R_i equals the character of f_0 on R_i, so r_i divides $d - e_j$, where e_j is the q-weighted degree of M_j. Since $2 \le d - e_j \le d$, this is impossible for $r_i > d$. Hence $\hat{f}_i = f_0$ for $i \gg 0$. The ideal of $\mathfrak{R}$ generated by f_0 and the partial derivatives of f_0 is contained in (x, y, z), so it is not primary to $\mathfrak{m}$. On the other hand, since W_i' has an isolated singularity at 0, the ideal generated by f_i' and the partial derivatives of f_i' is primary to $\mathfrak{m}$, and by the chain rule $\psi_i\sigma_i$ maps it onto the ideal generated by $\psi_i\sigma_i(f_i')$ and the partial derivatives of $\psi_i\sigma_i(f_i')$. Since $\psi_i\sigma_i(f_i')/\hat{f}_i$ is a unit, the last ideal is generated by $\hat{f}_i$ and the partial derivatives of $\hat{f}_i$, a contradiction.

Assume that $c'v > 0$. Then the coordinates of w are all positive or all negative, which is excluded in **Step 3**.

Assume that $c'v < 0$. Since $\mathbf{T}_{(w)} = \mathbf{T}_{(-w)}$, we may assume that $v > 0$, and we let $c := -c'$. Then c, v are coprime positive integers and $\mathbf{T}_{(w)} = \mathbf{T}_{(-cq_x,-cq_y,-cq_z,v)}$. The type, f_0, the basis $M_1, \dots, M_\mu$, the integers c, v, e_0, and the DCC set $\{\frac{1}{m}\}$ do not depend on i. For any i, W_i' is an isolated Gorenstein terminal germ with the G_i-equivariant isomorphism $\widehat{\mathcal{O}}_{W_i',0} \cong \mathfrak{R}/(\hat{f}_i)$, G_i acts diagonally on x, y, z, t and preserves f_0 and each non-zero $a_{ij}(t)M_j$ up to the character of f_i', the quadratic part of $\hat{f}_i$ is x^2, $\hat{f}_i(x, y, z, 0) = f_0$, $\mathbf{T}_{(w)}$ acts on x, y, z, t with the weights $(-cq_x, -cq_y, -cq_z, v)$, $[G_i : G_i \cap \mathbf{T}_{(w)}] \le e_0$, $|G_i \cap \mathbf{T}_{(w)}| = r_i \to +\infty$, and $(W_i', \frac{1}{m}\operatorname{div}(h_i'))$ is klt near 0, with the boundary $\frac{1}{m}\operatorname{div}(h_i')$ having labels in $\{\frac{1}{m}\}$. By Theorem 10.4, the numbers (D.6) are not positive and strictly increasing, a contradiction.

Hence $a, b \ge 1$. Together with **Steps 2 and 3**, after passing to a subsequence, the integers s, C, D and e_0 satisfy the conclusions of Construction-Proposition 10.3(2). □

D.3. **Circles $(-cq_x, -cq_y, -cq_z, v)$ on sources of type D and E: comparisons.** In this subsection, we prove Lemmas D.4 and D.9, the claims of Construction D.8, and Proposition D.11. They are used in the proof of Theorem 10.4 in Subsection D.4. Lemma D.3 is the lifting of diagonal automorphisms through the binary polyhedral quotient, and Lemma D.6 compares the log discrepancies on $W_0 = S \times \mathbb{A}^1_t$, where $S = (f_0 = 0) \subset \mathbb{A}^3_{x,y,z}$, with those on $V = S \times \mathbb{A}^1_\varpi$ through the covers given by $t = s^v$ and $\varpi = s^r$.

Definition D.2. Let c, v be coprime positive integers and $q = (q_x, q_y, q_z)$ the weights of one of the Du Val types of Theorem 10.4. For a finite group acting diagonally on an affine space and a coordinate u of the space, χ_u denotes the character of u, that is, $g^*u = \chi_u(g)u$ for any element g of the group, where g^* is the pullback of functions. For $\lambda \in \mathbb{C}^*$, $\mathbf{T}(\lambda)$ denotes the element of $\mathbf{T} = \mathbf{T}_{(-cq_x,-cq_y,-cq_z,v)}$ with

$$\mathbf{T}(\lambda)^*(x, y, z, t) = (\lambda^{-cq_x}x, \lambda^{-cq_y}y, \lambda^{-cq_z}z, \lambda^v t).$$

Since $\gcd(cq_x, cq_y, cq_z, v) = \gcd(c, v) = 1$, the vector $(-cq_x, -cq_y, -cq_z, v)$ is primitive and $\lambda \to \mathbf{T}(\lambda)$ is an isomorphism $\mathbb{C}^* \cong \mathbf{T}$. We write $\omega := (q_x, q_y, q_z, 1)$.

Lemma D.3. *Let $\Pi \subset \mathrm{SL}_2(\mathbb{C})$ be a finite subgroup, q_x, q_y, q_z positive integers, and $F_x, F_y, F_z \in \mathbb{C}[\xi_1, \xi_2]$ Π-invariant homogeneous polynomials of degrees $2q_x$, $2q_y$, $2q_z$ such that*

$$\pi := (F_x, F_y, F_z) : \mathbb{A}^2_{\xi_1,\xi_2} \to S := \pi(\mathbb{A}^2) \subset \mathbb{A}^3_{x,y,z}$$

is the quotient morphism by Π. Let Q be a finite group acting on $\mathbb{A}^3_{x,y,z}$ by diagonal matrices and preserving S, and for $\sigma \in Q$ let σ_S be the automorphism of S induced by σ. Then Π acts freely on $\mathbb{A}^2\backslash\{0\}$, and π is étale over $S\backslash\{0\}$. Moreover, the set $\widehat{Q}$ of pairs $(\hat{\sigma}, \sigma)$ with $\hat{\sigma} \in \mathrm{GL}_2(\mathbb{C})$, $\sigma \in Q$, and $\pi \circ \hat{\sigma} = \sigma_S \circ \pi$ is a finite group, the projection $\widehat{Q} \to Q$ is surjective with kernel $\Pi \times \{1\}$, the image $\widehat{Q}_1$ of $\widehat{Q}$ in $\mathrm{GL}_2(\mathbb{C})$ is the group of lifts through π of the automorphisms σ_S, Π is a normal subgroup of $\widehat{Q}_1$, and $\widehat{Q}_1/\Pi$ is isomorphic to the image of Q in $\mathrm{Aut}(S)$.

Proof. An element of Π other than the identity has finite order and determinant one, so both eigenvalues are different from 1. Thus Π acts freely on $\mathbb{A}^2\backslash\{0\}$, and π restricts to an étale Galois cover $\mathbb{A}^2\backslash\{0\} \to S\backslash\{0\}$ with group Π, which is the universal cover because $\mathbb{A}^2\backslash\{0\}$ is simply connected. Let $\sigma \in Q$. Since σ_S fixes 0, the map

$$\sigma_S \circ \pi : \mathbb{A}^2\backslash\{0\} \to S\backslash\{0\}$$

is a covering map, and it lifts through the universal cover π to a biholomorphic map $\hat{\sigma}$ of $\mathbb{A}^2\backslash\{0\}$ (the lift of $\sigma_S^{-1} \circ \pi$ is its inverse up to an element of Π). As π is finite and $\pi^{-1}(0) = \{0\}$, $\hat{\sigma}$ is bounded near 0, so it extends over 0 by the Riemann extension theorem to an automorphism of the analytic germ $(\mathbb{A}^2, 0)$ with $\pi \circ \hat{\sigma} = \sigma_S \circ \pi$. The automorphisms $\hat{\sigma}'$ of $(\mathbb{A}^2, 0)$ with $\pi \circ \hat{\sigma}' = \sigma_S \circ \pi$ form the coset $\Pi\hat{\sigma}$, because $\hat{\sigma}' \circ \hat{\sigma}^{-1}$ is a deck transformation of the local universal cover π, whose deck group is Π [Pri67, Proposition 1].

We show that $\hat{\sigma}$ is linear. For $\lambda \in \mathbb{C}^*$, let ρ_λ be the automorphism of $\mathbb{A}^3$ multiplying x, y, and z by λ^{2q_x}, λ^{2q_y}, and λ^{2q_z}. Since F_u is homogeneous of degree $2q_u$, $\pi(\lambda\xi) = \rho_\lambda(\pi(\xi))$, and ρ_λ commutes with σ because both are diagonal. For $0 < |\lambda| \leq 1$, the germ $\hat{\sigma}_\lambda(\xi) := \lambda^{-1}\hat{\sigma}(\lambda\xi)$ is an automorphism of $(\mathbb{A}^2, 0)$ with

$$\pi \circ \hat{\sigma}_\lambda = \rho_\lambda^{-1} \circ \sigma_S \circ \rho_\lambda \circ \pi = \sigma_S \circ \pi,$$

so $\hat{\sigma}_\lambda \in \Pi\hat{\sigma}$. Fix a point $\xi_0 \neq 0$ in the domain of $\hat{\sigma}$. Since $\hat{\sigma}(\xi_0) \neq 0$ and Π acts freely on $\mathbb{A}^2\backslash\{0\}$, the $|\Pi|$ elements of $\Pi\hat{\sigma}$ take pairwise distinct values at ξ_0. The value $\hat{\sigma}_\lambda(\xi_0)$ depends continuously on λ in the connected set $\{0 < |\lambda| \leq 1\}$, hence $\hat{\sigma}_\lambda = \hat{\sigma}$ for any such λ. Writing $\hat{\sigma} = \sum_{m \geq 1} \hat{\sigma}_m$ with $\hat{\sigma}_m$ homogeneous of degree m, we obtain $\lambda^{m-1}\hat{\sigma}_m = \hat{\sigma}_m$ for any m, so $\hat{\sigma} = \hat{\sigma}_1 \in \mathrm{GL}_2(\mathbb{C})$.

Thus, for any $\sigma \in Q$, the lifts of σ_S form a coset of Π in $\mathrm{GL}_2(\mathbb{C})$. Hence $\widehat{Q}$ is a group of order $|\Pi||Q|$ mapping onto Q, and the kernel consists of the pairs $(\hat{\sigma}, 1)$ with $\pi \circ \hat{\sigma} = \pi$, that is, of $\Pi \times \{1\}$. Since π is surjective, $\hat{\sigma} \in \widehat{Q}_1$ determines the automorphism σ_S with $\pi \circ \hat{\sigma} = \sigma_S \circ \pi$, and $\hat{\sigma} \to \sigma_S$ is a homomorphism from $\widehat{Q}_1$ onto the image of Q in $\mathrm{Aut}(S)$ with kernel Π. □

Lemma D.4. *Fix one of the Du Val types of Theorem 10.4, with $f_0 = x^2 + g_0(y, z)$ and weights $q = (q_x, q_y, q_z)$, and let $S := (f_0 = 0) \subset \mathbb{A}^3_{x,y,z}$. There exist a finite non-cyclic subgroup $\Pi \subset \mathrm{SL}_2(\mathbb{C})$ and Π-invariant homogeneous polynomials $F_x, F_y, F_z \in \mathbb{C}[\xi_1, \xi_2]$ of degrees $2q_x, 2q_y, 2q_z$ such that $\pi := (F_x, F_y, F_z) : \mathbb{A}^2_{\xi_1,\xi_2} \to S$ is the quotient morphism by Π and is étale over $S\backslash\{0\}$. No line of $\mathbb{C}^2$ is Π-invariant. Let c, v be coprime positive integers, $\mathbf{T} := \mathbf{T}_{(-cq_x,-cq_y,-cq_z,v)}$, and G a finite abelian group acting diagonally on $\mathbb{A}^4_{x,y,z,t}$ and preserving f_0 up to a character, with $R := G \cap \mathbf{T}$ and $r := |R|$. Let $\widehat{G}$ be the set of pairs $(\hat{g}, g) \in \mathrm{GL}_2(\mathbb{C}) \times G$ with $\pi \circ \hat{g} = g_S \circ \pi$, where g_S is the automorphism of S induced by g, acting on $\mathbb{A}^2_{\xi_1,\xi_2} \times \mathbb{A}^1_t$ by $\hat{g}$ on (ξ_1, ξ_2) and by g on t. Then $\widehat{G}$ is a finite group, the projection $\widehat{G} \to G$ is surjective with kernel $\Pi \times \{1\}$, and $\pi \times \mathrm{id}_{\mathbb{A}^1}$ induces an isomorphism $(\mathbb{A}^2 \times \mathbb{A}^1_t)/\widehat{G} \cong (S \times \mathbb{A}^1_t)/G$. Moreover, let $\bar{c} := c/\gcd(c, 2)$,*

$\bar{v} := 2v/\gcd(c,2)$, *and* $\widehat{\mathbf{T}} := \mathbf{T}_{(-\bar{c},-\bar{c},\bar{v})}$ *acting on* (ξ_1,ξ_2,t). *Then* $\widehat{G}\cap\widehat{\mathbf{T}}$ *is central in* $\widehat{G}$, *and*

$$|\widehat{G}\cap\widehat{\mathbf{T}}| = \frac{2r}{\gcd(c,2)}, \qquad [\widehat{G}:\widehat{G}\cap\widehat{\mathbf{T}}] = \frac{\gcd(c,2)|\Pi|}{2}[G:R].$$

Proof. Let Π be the binary dihedral group of order $4(k-2)$ if the type is D_k, and the binary tetrahedral, octahedral, or icosahedral group, of order 24, 48, or 120, if the type is E_6, E_7, or E_8. By [Roa04, Table 1], the ring $\mathbb{C}[\xi_1,\xi_2]^\Pi$ is generated by homogeneous invariants X, Y, Z of degrees 4, $2(k-2)$, $2(k-1)$ for D_k, 6, 8, 12 for E_6, 8, 12, 18 for E_7, and 12, 20, 30 for E_8, with the single relation $X^{k-1}-XY^2+Z^2$, $X^4+Y^3+Z^2$, $X^3Y+Y^3+Z^2$, and $X^5+Y^3+Z^2$, respectively. Let $F_x := Z$, $F_z := X$, and $F_y := \sqrt{-1}Y$ for D_k and $F_y := Y$ for E_6, E_7, E_8. Then $\deg F_u = 2q_u$ for $u\in\{x,y,z\}$, and the relation is $f_0(F_x,F_y,F_z)=0$. If f_0 is the image of one of the four polynomials listed in Theorem 10.4 under a fixed change of y,z preserving the weights, we apply that change of y,z to (F_x,F_y,F_z). Then π is the quotient morphism by Π. By Lemma D.3, applied with the trivial group Q, π is étale over $S\backslash\{0\}$. The group Π is not abelian, in particular not cyclic.

Assume that a line $\ell_1\subset\mathbb{C}^2$ is Π-invariant. Since Π is finite, ℓ_1 has a Π-invariant complement ℓ_2, and in a basis of $\ell_1\oplus\ell_2$ any element of Π is a diagonal matrix of determinant one. Then Π is abelian, a contradiction.

The group G acts on $\mathbb{A}^3_{x,y,z}$ by the diagonal matrices with entries $\chi_x(g)$, $\chi_y(g)$, $\chi_z(g)$ and preserves S, because $g^*f_0=\chi_{f_0}(g)f_0$ for a character χ_{f_0}. By Lemma D.3 for the action of G on $\mathbb{A}^3_{x,y,z}$, $\widehat{G}$ is a finite group of order $|\Pi||G|$, and $\widehat{G}\to G$ is surjective with kernel $\Pi\times\{1\}$. The normal subgroup $\Pi\times\{1\}$ acts trivially on t, and $\pi\times\mathrm{id}_{\mathbb{A}^1}$ is the quotient morphism of $\mathbb{A}^2\times\mathbb{A}^1_t$ by $\Pi\times\{1\}$, with the induced action of $\widehat{G}/(\Pi\times\{1\})\cong G$ on $S\times\mathbb{A}^1_t$ equal to the given one. Taking the quotient in stages, $(\mathbb{A}^2\times\mathbb{A}^1_t)/\widehat{G}\cong(S\times\mathbb{A}^1_t)/G$.

If $(\hat{g},g)\in\widehat{G}$ acts trivially on $\mathbb{A}^2\times\mathbb{A}^1_t$, then $\hat{g}=1$ and $\chi_t(g)=1$, so $g_S=\mathrm{id}_S$. Since S is not contained in a coordinate hyperplane, $\chi_x(g)=\chi_y(g)=\chi_z(g)=1$, and $g=1$ because G is a group of diagonal matrices. Thus the action of $\widehat{G}$ on $\mathbb{A}^2\times\mathbb{A}^1_t$ is faithful, and we regard $\widehat{G}$ as a subgroup of $\mathrm{Aut}(\mathbb{A}^2\times\mathbb{A}^1_t)$.

Recall that $\bar{c}=c/\gcd(c,2)$ and $\bar{v}=2v/\gcd(c,2)$. If $\gcd(c,2)=1$, then $\gcd(\bar{c},\bar{v})=\gcd(c,2v)=1$, and if $\gcd(c,2)=2$, then $\gcd(\bar{c},\bar{v})=\gcd(c/2,v)=1$. Thus $\eta\to\widehat{\mathbf{T}}(\eta)$, where $\widehat{\mathbf{T}}(\eta)$ multiplies ξ_1, ξ_2, and t by $\eta^{-\bar{c}}$, $\eta^{-\bar{c}}$, and $\eta^{\bar{v}}$, is an isomorphism $\mathbb{C}^*\cong\widehat{\mathbf{T}}$. Let $\eta\in\mathbb{C}^*$ and $\lambda := \eta^{2/\gcd(c,2)}$. Since F_u is homogeneous of degree $2q_u$,

$$\pi(\eta^{-\bar{c}}\xi) = \big(\eta^{-2\bar{c}q_x}F_x(\xi),\eta^{-2\bar{c}q_y}F_y(\xi),\eta^{-2\bar{c}q_z}F_z(\xi)\big) = \big(\lambda^{-cq_x}F_x(\xi),\lambda^{-cq_y}F_y(\xi),\lambda^{-cq_z}F_z(\xi)\big),$$

and $\eta^{\bar{v}}=\lambda^v$. Thus $\widehat{\mathbf{T}}(\eta)\in\widehat{G}$ if and only if there exists $g\in G$ acting on (x,y,z,t) as $\mathbf{T}(\lambda)$, that is, if and only if $\mathbf{T}(\lambda)\in R$. Since R is the subgroup of order r of $\mathbf{T}\cong\mathbb{C}^*$, $\mathbf{T}(\lambda)\in R$ if and only if $\lambda^r=1$. The map $\eta\to\eta^{2/\gcd(c,2)}$ is surjective of degree $2/\gcd(c,2)$, so

$$|\widehat{G}\cap\widehat{\mathbf{T}}| = \frac{2}{\gcd(c,2)}\cdot r = \frac{2r}{\gcd(c,2)}.$$

The elements of $\widehat{\mathbf{T}}$ act on (ξ_1,ξ_2) by scalars and on t by a scalar, and any element of $\widehat{G}$ acts on t by a scalar, so $\widehat{G}\cap\widehat{\mathbf{T}}$ is central in $\widehat{G}$. Finally,

$$[\widehat{G}:\widehat{G}\cap\widehat{\mathbf{T}}] = \frac{|\Pi|r[G:R]}{2r/\gcd(c,2)} = \frac{\gcd(c,2)|\Pi|}{2}[G:R].$$

□

Set-up D.5. In the situation of Theorem 10.4, let d be the q-weighted degree of f_0 and $S := (f_0=0)\subset\mathbb{A}^3_{x,y,z}$, and let $\Pi\subset\mathrm{SL}_2(\mathbb{C})$ and the quotient morphism $\pi=(F_x,F_y,F_z):\mathbb{A}^2_{\xi_1,\xi_2}\to S$ by Π be given by Lemma D.4. Let γ be the least positive element of Γ ($\gamma:=1$ if $\Gamma\subset\{0\}$), $M_{\deg} := \lceil 2/\gamma\rceil$, and $\kappa_{\max} := M_{\deg}\max\{cq_x,v\}$. Let $N_0 := \mathrm{lcm}(q_x,q_y,q_z)$, $I_0 := (x^{N_0/q_x},y^{N_0/q_y},z^{N_0/q_z})\mathcal{O}_S$, and $\mathfrak{m} := (\xi_1,\xi_2)\subset\mathbb{C}[\xi_1,\xi_2]$.

Lemma D.6. *Fix one of the Du Val types of Theorem 10.4, with $f_0 = x^2 + g_0(y,z)$ and weights $q = (q_x, q_y, q_z)$, and let $S := (f_0 = 0) \subset \mathbb{A}^3_{x,y,z}$ and $W_0 := S \times \mathbb{A}^1_t \subset \mathbb{A}^4$. Let c, v be coprime positive integers, $\mathbf{T} := \mathbf{T}_{(-cq_x,-cq_y,-cq_z,v)}$, and G a finite abelian group acting diagonally on $\mathbb{A}^4_{x,y,z,t}$ and preserving f_0 up to a character, with $R := G \cap \mathbf{T}$ and $r := |R|$. Let G_S be the group of pairs (g,ζ) with $g \in G$ and $\zeta^v = \chi_t(g)$, acting on $\mathbb{A}^3_{x,y,z}$ by $(g,\zeta)^* u = \chi_u(g)\zeta^{cq_u} u$ for $u \in \{x,y,z\}$ and on $\mathbb{A}^1_\varpi$ by $(g,\zeta)^*\varpi = \zeta^r \varpi$. Let $h_1, \dots, h_s$ be G-semi-invariant polynomials not divisible by f_0 such that h_l is $\mathbf{T}$-semi-invariant of $\mathbf{T}$-weight κ_l, let $b_1, \dots, b_s$ be real numbers, and let*

$$D_0 := \sum_l b_l \operatorname{div}(h_l|_{W_0}), \quad D_S := \sum_l b_l \operatorname{div}(h_l(x,y,z,1)|_S), \quad \tau := \sum_l b_l \kappa_l, \quad \delta := \tau + c - v.$$

Here $h_l(x,y,z,1)|_S \neq 0$, because the q-weighted homogeneous components of $h_l(x,y,z,1)$ are the coefficients of the powers of t in h_l, and f_0 is weighted homogeneous. Then the following hold.

1. *The kernel R_S of the action of G_S on S is $\{(\mathbf{T}(\lambda), \lambda) \mid \lambda^r = 1\}$, it has order r and acts trivially on $\mathbb{A}^1_\varpi$, and $\widetilde{H} := G_S/R_S$ has order $v[G:R]$, acts faithfully on S, and preserves D_S.*
2. *If (W_0, D_0) is lc near 0, then (S, D_S) is lc near 0, and $a(\nu, S, D_S) \geq \delta m_\nu$ for any prime divisor ν over S with center 0, where $m_\nu := \min_{u \in \{x,y,z\}} \operatorname{mult}_\nu u/(cq_u)$.*
3. *Let $V := S \times \mathbb{A}^1_\varpi$ with the induced action of $\widetilde{H}$, ν a prime divisor over S with center 0, k_0 a positive integer with $k_0 \leq r m_\nu$, and E a prime divisor over V such that mult_E is the Gauss extension of mult_ν with ϖ-value k_0. Then there exists a prime divisor F_0 over W_0 such that*

$$\frac{a(F_0, W_0, D_0)}{|I_G(F_0)|} = \frac{a\big(E, V, \operatorname{pr}_1^* D_S + (1 + \delta/r)\operatorname{div}(\varpi)\big)}{|I_{\widetilde{H}}(E)|},$$

and such that for any polynomial $\varphi \in \mathbb{C}[x,y,z,t]$ not divisible by f_0, with $\varphi = \sum_\kappa \varphi_\kappa$ the decomposition into $\mathbf{T}$-semi-invariant parts φ_κ of $\mathbf{T}$-weight κ,

$$\frac{\operatorname{mult}_{F_0} \varphi}{|I_G(F_0)|} = \frac{1}{|I_{\widetilde{H}}(E)|} \min_\kappa \Big(\operatorname{mult}_\nu \big(\varphi_\kappa(x,y,z,1)|_S\big) + \frac{\kappa k_0}{r} \Big),$$

where mult_ν of the zero function is $+\infty$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1). An element $(g,\zeta) \in G_S$ acts trivially on S if and only if $\chi_u(g)\zeta^{cq_u} = 1$ for $u \in \{x,y,z\}$, because S is not contained in a coordinate hyperplane. Since also $\chi_t(g) = \zeta^v$, the element (g,ζ) acts trivially on S if and only if g acts on (x,y,z,t) as $\mathbf{T}(\zeta)$, that is, $g = \mathbf{T}(\zeta) \in R$. As $R = \{\mathbf{T}(\lambda) \mid \lambda^r = 1\}$, we obtain $R_S = \{(\mathbf{T}(\lambda), \lambda) \mid \lambda^r = 1\}$, of order r, and $\lambda^r = 1$ means that R_S acts trivially on $\mathbb{A}^1_\varpi$. For any $g \in G$ there exist exactly v values of ζ with $\zeta^v = \chi_t(g)$, so $|G_S| = v|G|$ and $|\widetilde{H}| = v|G|/r = v[G:R]$.

Let h be one of the h_l, and $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ a monomial of h. The $\mathbf{T}$-weight of $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ is $\kappa_l = vn_4 - c(q_x n_1 + q_y n_2 + q_z n_3)$, and $(g,\zeta)^*$ multiplies $x^{n_1}y^{n_2}z^{n_3}$ by

$$\chi_x(g)^{n_1}\chi_y(g)^{n_2}\chi_z(g)^{n_3}\zeta^{c(q_x n_1 + q_y n_2 + q_z n_3)} = \chi_h(g)\chi_t(g)^{-n_4}\zeta^{vn_4 - \kappa_l} = \chi_h(g)\zeta^{-\kappa_l},$$

which does not depend on the monomial. Hence $h(x,y,z,1)$ is G_S-semi-invariant, and $\widetilde{H}$ preserves D_S.

Step 2. In this step we pass to the cover $t = s^v$ and to the morphism $\varrho_0 : \widetilde{W}_0 \to \widetilde{V}$. Let $\widetilde{W}_0 := (f_0(x,y,z) = 0) \subset \mathbb{A}^4_{x,y,z,s}$ and $p : \widetilde{W}_0 \to W_0$, $(x,y,z,s) \to (x,y,z,s^v)$.

Let G_S act on $\widetilde{W}_0$ by $(g,\zeta)^* u = \chi_u(g)u$ for $u \in \{x,y,z\}$ and $(g,\zeta)^* s = \zeta s$. The action of G_S on $\widetilde{W}_0$ is faithful, p is the quotient morphism by the normal subgroup $\{(1,\zeta) \mid \zeta^v = 1\} \cong \mu_v$, and

$(g,\zeta)\to g$ induces an isomorphism $G_S/\mu_v\cong G$ compatible with p. Since p is a cyclic cover of degree v totally ramified along $(s=0)$, the Hurwitz formula implies

$$K_{\widetilde{W}_0}+\widetilde{B}=p^*(K_{W_0}+D_0),\qquad \widetilde{B}:=(1-v)\operatorname{div}(s)+\sum_l b_l\operatorname{div}(p^*h_l).$$

Let $\widetilde{V}:=(f_0(X,Y,Z)=0)\subset\mathbb{A}^4_{X,Y,Z,s}$, so that $\widetilde{V}=S\times\mathbb{A}^1_s$ with the first projection $\mathrm{pr}_1:\widetilde{V}\to S$, and let

$$\varrho_0:\widetilde{W}_0\to\widetilde{V},\qquad (x,y,z,s)\to(xs^{cq_x},ys^{cq_y},zs^{cq_z},s).$$

Since f_0 is weighted homogeneous, ϱ_0 is well defined. It is an isomorphism over $(s\neq 0)$, and it is G_S-equivariant for the action of G_S on $\widetilde{V}$ given by the action of G_S on S in the statement and by $(g,\zeta)^*s=\zeta s$. For a monomial $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ of $\mathbf{T}$-weight κ, we have

$$x^{n_1}y^{n_2}z^{n_3}s^{vn_4}=s^\kappa\varrho_0^*(X^{n_1}Y^{n_2}Z^{n_3}).$$

Hence

$$p^*\varphi_\kappa=s^\kappa\varrho_0^*\big(\varphi_\kappa(X,Y,Z,1)\big)\tag{D.8}$$

for any $\mathbf{T}$-semi-invariant polynomial φ_κ of $\mathbf{T}$-weight κ. The canonical sheaves of $\widetilde{W}_0$ and of $\widetilde{V}$ are generated by $\widetilde{\Omega}:=dy\wedge dz\wedge ds/(2x)$ and $\Omega:=dY\wedge dZ\wedge ds/(2X)$, the residues of the volume forms divided by f_0 [KM98, Proposition 5.73]. Since the q-weighted degree $2q_x$ of f_0 equals $q_x+q_y+q_z-1$ for each type of Theorem 10.4, we have $q_y+q_z-q_x=1$, and a direct calculation shows $\varrho_0^*\Omega=s^c\widetilde{\Omega}$. For a prime divisor P over $\widetilde{W}_0$,

$$a(P,\widetilde{W}_0,\widetilde{B})=1+\operatorname{mult}_P\widetilde{\Omega}-\operatorname{mult}_P\widetilde{B},$$

where $\operatorname{mult}_P\widetilde{\Omega}$ is the order of $\widetilde{\Omega}$ along P, and similarly on $\widetilde{V}$. By (D.8) applied to the h_l,

$$a(P,\widetilde{W}_0,\widetilde{B})=a(P,\widetilde{V},B_s),\qquad B_s:=\mathrm{pr}_1^*D_S+(1+\delta)\operatorname{div}(s),\tag{D.9}$$

for any prime divisor P over $\widetilde{W}_0$, because $-c-(1-v)-\tau=-(1+\delta)$. Finally, R_S acts trivially on S and by $s\to\lambda s$, $\lambda^r=1$, on $\mathbb{A}^1_s$, so $\widetilde{V}\to V$, $\varpi=s^r$, is the quotient morphism by R_S, and the Hurwitz formula implies

$$K_{\widetilde{V}}+B_s=q_r^*\big(K_V+B_V\big),\qquad B_V:=\mathrm{pr}_1^*D_S+\Big(1+\frac{\delta}{r}\Big)\operatorname{div}(\varpi),\tag{D.10}$$

where q_r denotes the quotient morphism, because $r(1+\delta/r)-(r-1)=1+\delta$.

Step 3. In this step we prove (2). The torus $\mathbf{T}$ acts on $\mathbb{A}^4$ preserving W_0 and each $\operatorname{div}(h_l|_{W_0})$, so the closed subset Nlc of points near which (W_0,D_0) is not lc is $\mathbf{T}$-invariant. By assumption $0\notin\mathrm{Nlc}$. Since $\mathbf{T}(\lambda)(0,0,0,1)=(0,0,0,\lambda^v)$ tends to 0 as $\lambda\to 0$, the closure of the $\mathbf{T}$-orbit of $(0,0,0,1)$ contains 0, hence $(0,0,0,1)\notin\mathrm{Nlc}$. The morphism p is étale over $(t\neq 0)$, so $(\widetilde{W}_0,\widetilde{B})$ is lc near $(0,0,0,1)$ by [Liu26, Lemma 2.6(2)]. Over $(s\neq 0)$, ϱ_0 is an isomorphism mapping $(0,0,0,1)$ to $(0,0,0,1)$, and $B_s=\mathrm{pr}_1^*D_S$ there. Thus $(\widetilde{V}\cap(s\neq 0),\mathrm{pr}_1^*D_S)$ is lc near $(0,1)$. Taking the product of a log resolution of (S,D_S) with $\mathbb{A}^1_s\backslash\{0\}$, we see that (S,D_S) is lc near 0.

Let ν be a prime divisor over S with center 0, and a_1,a_2 coprime positive integers with $a_2/a_1<m_\nu$. Let $S'\to S$ be a higher model on which ν is a prime divisor E_ν. At the generic point of $E_\nu\times\{0\}\subset S'\times\mathbb{A}^1_s$, a local equation w_ν of $E_\nu\times\mathbb{A}^1_s$ and s are regular parameters, and the only components through the generic point of $E_\nu\times\{0\}$ of the crepant transform of B_s are $E_\nu\times\mathbb{A}^1_s$ and $(s=0)$, with coefficients $1-a(\nu,S,D_S)$ and $1+\delta$. Let P be the exceptional divisor of the weighted blow-up of the generic point of $E_\nu\times\{0\}$ with weights (a_1,a_2) on (w_ν,s). A direct calculation shows $a(P,S'\times\mathbb{A}^1_s,0)=a_1+a_2$, $\operatorname{mult}_P w_\nu=a_1$, and $\operatorname{mult}_P s=a_2$, so

$$a(P,\widetilde{V},B_s)=a_1+a_2-a_1\big(1-a(\nu,S,D_S)\big)-a_2(1+\delta)=a_1a(\nu,S,D_S)-a_2\delta.$$

Moreover $\operatorname{mult}_P\psi=a_1\operatorname{mult}_\nu\psi$ for any regular function ψ on S. Hence

$$\operatorname{mult}_P x=\operatorname{mult}_P(Xs^{-cq_x})=a_1\operatorname{mult}_\nu x-a_2cq_x>0,$$

and similarly for y and z, while $\operatorname{mult}_P s = a_2 > 0$. Thus the center of P on $\widetilde{W}_0$ is 0, and the center on W_0 of the prime divisor under P is 0. Since (W_0, D_0) is lc near 0, [Liu26, Lemma 2.6(2)] applied to p and (D.9) imply

$$a(P, \widetilde{V}, B_s) = a(P, \widetilde{W}_0, \widetilde{B}) \geq 0.$$

Thus $a(\nu, S, D_S) \geq \delta a_2/a_1$ for any rational number $a_2/a_1 \in (0, m_\nu)$, and so $a(\nu, S, D_S) \geq \delta m_\nu$.

Step 4. In this step we prove (3). Let $g_1 := \gcd(r, k_0)$. Let S', E_ν, and w_ν be as in **Step 3**, and let P be the exceptional divisor of the weighted blow-up of the generic point of $E_\nu \times \{0\} \subset S' \times \mathbb{A}^1_s$ with weights $(r/g_1, k_0/g_1)$ on (w_ν, s). Then P is the monomial valuation in (w_ν, s), so for rational functions ψ_j on S,

$$\operatorname{mult}_P \Big(\sum_j \psi_j s^j \Big) = \min_j \Big(\frac{r}{g_1} \operatorname{mult}_\nu \psi_j + \frac{jk_0}{g_1} \Big) = \frac{r}{g_1} \min_j \Big(\operatorname{mult}_\nu \psi_j + \frac{jk_0}{r} \Big).$$

With $s^r = \varpi$, the valuation mult_P restricts to $\frac{r}{g_1} \operatorname{mult}_E$ on the rational functions on V. Thus P is a prime divisor over E with ramification index r/g_1. Since $k_0 \leq r m_\nu$,

$$\operatorname{mult}_P u = \frac{r}{g_1} \Big(\operatorname{mult}_\nu u - \frac{c q_u k_0}{r} \Big) \geq 0 \quad \text{for } u \in \{x, y, z\},$$

and $\operatorname{mult}_P s = k_0/g_1 > 0$, so the valuation mult_P has a center on the affine variety $\widetilde{W}_0$. As ϱ_0 is birational and G_S-equivariant, P is also a prime divisor over $\widetilde{W}_0$, and the inertia group $I_{G_S}(P)$ is one subgroup of G_S whether P is regarded over $\widetilde{V}$ or over $\widetilde{W}_0$. Let F_0 be the prime divisor over W_0 under P.

By [Liu26, Lemma 2.6(1)], the order of the inertia group of a prime divisor is the ramification index over the quotient. Since ramification indices are multiplicative in the towers $\widetilde{W}_0 \to W_0 \to W_0/G$ and $\widetilde{V} \to V \to V/\widetilde{H}$, whose composites are both the quotient by G_S,

$$|I_{G_S}(P)| = e(P/F_0)|I_G(F_0)| = \frac{r}{g_1} |I_{\widetilde{H}}(E)|,$$

where $e(P/F_0)$ is the ramification index of P over F_0. By [Liu26, Lemma 2.6(2)(5)], the Hurwitz formula of **Step 2**, (D.9), and (D.10),

$$\frac{a(F_0, W_0, D_0)}{|I_G(F_0)|} = \frac{a(P, \widetilde{W}_0, \widetilde{B})}{|I_{G_S}(P)|} = \frac{a(P, \widetilde{V}, B_s)}{|I_{G_S}(P)|} = \frac{a(E, V, B_V)}{|I_{\widetilde{H}}(E)|}.$$

Let $\varphi = \sum_\kappa \varphi_\kappa$ be as in (3). Since $\operatorname{mult}_P(p^*\varphi) = e(P/F_0) \operatorname{mult}_{F_0} \varphi$, (D.8) and the formula for mult_P imply

$$\frac{\operatorname{mult}_{F_0} \varphi}{|I_G(F_0)|} = \frac{\operatorname{mult}_P \big(\sum_\kappa s^\kappa \varphi_\kappa(X, Y, Z, 1) \big)}{|I_{G_S}(P)|} = \frac{1}{|I_{\widetilde{H}}(E)|} \min_\kappa \Big(\operatorname{mult}_\nu \big(\varphi_\kappa(x, y, z, 1)|_S \big) + \frac{\kappa k_0}{r} \Big).$$

□

Lemma D.7. *Notation as in Set-up D.5. Write $f = f_0 + A$ with $A = \sum_j a_j(t) M_j$, let d_j be the q-weighted degree of M_j, and write $D = \sum_l b_l \operatorname{div}(h_l)$ with $b_l \in \Gamma$ and the h_l written with x-degree at most one. Then the following hold.*

(1) $0 \leq d_j \leq d - 2$ for any j, and any non-zero term $a_{jn} t^n M_j$ of A satisfies

$$vn + c(d - d_j) = \ell r \quad \text{for an integer } \ell \geq 1. \tag{D.11}$$

In particular, each monomial of $f - f_0$ has total degree at least $(r - cd)/v$.

(2) Assume $r > v(d+1) + cd$. Then $\operatorname{in}_\omega(f) = f_0$, the ω-weighted blow-up of $\mathbb{A}^4$ defines a prime divisor E_W over W with center 0 and $a(E_W, W, 0) = 2$, and $\sum_l b_l \operatorname{wt}_\omega(h_l) < 2$. In particular, fewer than $2/\gamma$ of the h_l satisfy $b_l > 0$ and $h_l(0) = 0$, and each such h_l has a monomial of total degree less than $M_{\deg}$.

(3) *Assume moreover* $r > 2M_{\deg}\max\{cq_x, v\}$. *Then for any* l *with* $b_l > 0$ *and* $h_l(0) = 0$, *the number* κ_l *and the block* h_l^0 *of Set-up 5.1 for* $\mathbf{T}$ *are well defined,*

$$|\kappa_l| \le \kappa_{\max}, \tag{D.12}$$

and any monomial of $h_l - h_l^0$ *has total degree at least* $(r - \kappa_{\max})/\max\{cq_x, v\}$.

(4) *For any prime divisor* P *over* $\mathbb{A}^2$,

$$\operatorname{mult}_P(\pi^* I_0) = 2N_0 \operatorname{mult}_P \mathfrak{m}.$$

Proof. We prove (1). The partial derivatives $\partial_y g_0$ and $\partial_z g_0$ are weighted homogeneous of degrees $d - q_y$ and $d - q_z$. The monomial bases of $\mathbb{C}[y,z]/(\partial_y g_0, \partial_z g_0)$ for the four polynomials listed in Theorem 10.4 are $1, z, \dots, z^{k-2}, y$ for D_k, $y^a z^b$ with $a \le 1$ and $b \le 2$ for E_6, $1, y, z, yz, z^2, z^3, z^4$ for E_7, and $y^a z^b$ with $a \le 1$ and $b \le 3$ for E_8. Their largest weighted degrees are $2(k-2) = d-2$, $4+6 = d-2$, $16 = d-2$, and $10+18 = d-2$, respectively. A weight-preserving change of y, z keeps the weighted degrees of a weighted homogeneous basis. Hence $0 \le d_j \le d-2$ for any j.

Let $a_{jn} t^n M_j$ be a non-zero term of A. It has the G-character of f_0, so any element of R, which acts on monomials through their $\mathbf{T}$-weights, multiplies $t^n M_j$ and f_0 by one scalar. The $\mathbf{T}$-weights of the monomials of $t^n M_j$ and of f_0 are $vn - cd_j$ and $-cd$, so r divides $vn + c(d - d_j)$, and (D.11) holds with $\ell \ge 1$ because $n \ge 1$ and $d_j \le d-2$. Thus $n \ge (r - cd)/v$, and each monomial of $f - f_0$ has total degree at least $(r-cd)/v$.

We prove (2) and (3). Assume $r > v(d+1) + cd$. Then the terms of A have ω-weight $n + d_j > d$ by (1), so $\operatorname{in}_\omega(f) = f_0$ and $\operatorname{wt}_\omega(f) = d$. The polynomial $f_0 = x^2 + g_0$ is irreducible because g_0 is not a square in $\mathbb{C}[y,z]$, so $(f_0 = 0) \subset \mathbb{A}^4_{x,y,z,t}$ is integral, and it is not contained in a coordinate hyperplane. By Lemma 4.12, the ω-weighted blow-up of $\mathbb{A}^4$ defines a prime divisor E_W over W with center 0 and

$$a(E_W, W, 0) = q_x + q_y + q_z + 1 - d = 2.$$

Each $\operatorname{in}_\omega(h_l)$ has x-degree at most one, so it is not divisible by f_0, and $\operatorname{mult}_{E_W} h_l = \operatorname{wt}_\omega(h_l)$ by Lemma 4.12. Since (W, D) is klt near 0,

$$0 < a(E_W, W, D) = 2 - \sum_l b_l \operatorname{wt}_\omega(h_l).$$

If $b_l > 0$ and $h_l(0) = 0$, then, as $\omega \in \mathbb{Z}^4_{\ge 1}$, $\operatorname{wt}_\omega(h_l) \ge 1$ and $b_l \ge \gamma$. Thus fewer than $2/\gamma$ of the h_l satisfy $b_l > 0$ and $h_l(0) = 0$, and for each such h_l, $\operatorname{wt}_\omega(h_l) < 2/\gamma$. Hence each such h_l has a monomial of total degree less than $2/\gamma \le M_{\deg}$, and f has the monomial x^2 of total degree $2 \le M_{\deg}$. By Lemma 5.3 with $M = M_{\deg}$, for $r > 2M_{\deg}\max\{cq_x, v\}$ the numbers κ_l and the blocks h_l^0 are well defined, $|\kappa_l| \le \kappa_{\max}$, and any monomial of $h_l - h_l^0$ has total degree at least $(r - \kappa_{\max})/\max\{cq_x, v\}$.

We prove (4). The ideal $\pi^* I_0$ is generated by $F_u^{N_0/q_u}$, $u \in \{x, y, z\}$, which are homogeneous of degree $2N_0$. They have no common zero on $\mathbb{A}^2 \backslash \{0\}$, since $\pi^{-1}(0) = \{0\}$. On the blow-up of the origin of $\mathbb{A}^2$ with exceptional curve E_1, the pullback of a homogeneous polynomial F of degree m is $\xi_1^m F(1, \xi_2/\xi_1)$ on the chart with coordinates $\xi_1, \xi_2/\xi_1$, and similarly on the other chart. Hence the pullbacks of $\pi^* I_0$ and of $\mathfrak{m}^{2N_0}$ are both $\mathcal{O}(-2N_0 E_1)$. This proves (4), since mult_P of an ideal is determined by the pullback of the ideal to a higher model on which P is a divisor. □

Construction D.8. In the situation of Theorem 10.4, write $D = \sum_l b_l \operatorname{div}(h_l)$ with $b_l \in \Gamma$ and the h_l written with x-degree at most one, let h_l^0, κ_l, $\tau = \sum_l b_l \kappa_l$, and δ be as in Set-up 5.1 for $\mathbf{T}$, let d, S, Π, π, γ, N_0, I_0, and $\mathfrak{m}$ be as in Set-up D.5, and let $D_S := \sum_l b_l \operatorname{div}(h_l^0(x, y, z, 1)|_S)$. Removing the h_l with $b_l = 0$ or $h_l(0) \ne 0$ changes neither D and D_S near 0 nor τ, and we assume that $b_l \ge \gamma$ and $h_l(0) = 0$ for any l. Let G_S be the group of pairs (g, ζ) with $g \in G$ and $\zeta^v = \chi_t(g)$, acting on $\mathbb{A}^3_{x,y,z}$ by multiplying x, y, and z by $\chi_x(g)\zeta^{cq_x}$, $\chi_y(g)\zeta^{cq_y}$, and $\chi_z(g)\zeta^{cq_z}$, which preserves S, and on $\mathbb{A}^1_\varpi$ by multiplying ϖ by ζ^r. Let R_S be the kernel of the action of G_S on S, $\widetilde{H} := G_S/R_S$, $V := S \times \mathbb{A}^1_\varpi$, and $\widehat{G}_S \subset \mathrm{GL}_2(\mathbb{C})$ the group of lifts through π of the

automorphisms of S given by G_S. Then there exists an integer r_0, depending only on the type, c, v, e_0, and Γ, such that for $r \geq r_0$ the following hold.

(1) $f^0 = f_0$, $\delta = \tau + c - v$, each monomial of $f - f_0$ has total degree at least $(r - cd)/v$, and $(S \times \mathbb{A}^1_t, \sum_l b_l \operatorname{div}(h_l^0))$ is lc near 0.

(2) The group R_S acts trivially on $\mathbb{A}^1_\varpi$, so $\widetilde{H}$ acts on V. The group $\widetilde{H}$ has order $v[G : R]$ and acts faithfully on S, the pair (S, D_S) is lc near 0, and

$$\beta_- := \operatorname{mld}((S \ni 0, D_S)/G_S) = \operatorname{mld}((\mathbb{A}^2 \ni 0, \pi^* D_S)/\widehat{G}_S).$$

(3) If $\delta \geq 0$, then $(S, D_S, I_0^{\delta/(cN_0)})$ is lc near 0, and

$$\beta_+ := \operatorname{mld}((S \ni 0, D_S, I_0^{\delta/(cN_0)})/G_S) = \operatorname{mld}((\mathbb{A}^2 \ni 0, \pi^* D_S, \mathfrak{m}^{2\delta/c})/\widehat{G}_S).$$

If $\delta = 0$, then $\beta_+ = \beta_-$.

(4) Let $\beta := \beta_-$ if $\delta \leq 0$ and $\beta := \beta_+$ if $\delta \geq 0$. Then $\beta \leq 2$, the numbers β belong to a set satisfying the ACC that depends only on the type, c, v, e_0, and Γ, and β is computed by a prime divisor over $\mathbb{A}^2$ with center 0 whose log discrepancy with respect to $\mathbb{A}^2$ is bounded in terms of the type, c, v, e_0, and Γ.

Proof. The proof is long, so we divide it into several steps. Let γ, $M_{\deg}$, and $\kappa_{\max}$ be as in Set-up D.5, and $w := (-cq_x, -cq_y, -cq_z, v)$, so that $\max_i |w_i| = \max\{cq_x, v\}$ because $q_x \geq q_y, q_z$ for each type.

Step 1. In this step we reduce the marks and compute f^0 and δ. The normalization in the statement is possible: if $b_l = 0$ or $h_l(0) \neq 0$, then $b_l \operatorname{div}(h_l)$ is zero near 0, $b_l \kappa_l = 0$, and $b_l \operatorname{div}(h_l^0(x, y, z, 1)|_S)$ is zero near 0. Let $r > v(d+1) + cd$ and $r > 2M_{\deg} \max\{cq_x, v\}$. By Lemma D.7(2)(3), there are fewer than $2/\gamma$ marks, and the κ_l and h_l^0 are well defined. The monomial of least degree of f is x^2, of $\mathbf{T}$-weight $-2cq_x = -cd$, and the terms of A have $\mathbf{T}$-weights $\ell r - cd \neq -cd$ by (D.11). Since f has no other monomial of total degree at most two, $\kappa(f) = -cd$ and $f^0 = f_0$, so $\epsilon_{\mathbf{T}} = -c(q_x + q_y + q_z) + v + cd = v - c$ and $\delta = \tau - \epsilon_{\mathbf{T}} = \tau + c - v$. By Lemma D.7(1), each monomial of $f - f_0$ has total degree at least $(r - cd)/v$.

Step 2. In this step we prove the lc claim of (1). Let $\Lambda := (\Gamma \setminus \{0\}) \cup \{1\}$, and let L_1 be the largest of the integers given by Lemma 4.14 for $n = 4$, Λ, and all numbers of functions at most $1 + 2/\gamma$. Let r_0 be an integer larger than $v(d+1) + cd$, than $2M_{\deg} \max\{cq_x, v\}$, than $vL_1 + cd$, and than $L_1 \max\{cq_x, v\} + \kappa_{\max}$. Then r_0 depends only on the type, c, v, and Γ. Let $r \geq r_0$. Then the conclusions of **Step 1** hold.

Since W is a normal Cartier divisor on $\mathbb{A}^4$ and (W, D) is klt near 0, [Kaw07, Theorem] implies that the pair

$$(\mathbb{A}^4, \operatorname{div}(f) + \sum_l b_l \operatorname{div}(h_l))$$

is lc near 0. By [dFEM11, Proposition 2.11(i)], the pair $(\operatorname{Spec} \mathbb{C}[[x, y, z, t]], f \cdot \prod_l h_l^{b_l})$ is lc. By Lemma D.7(1)(3), $f - f_0$ and $h_l - h_l^0$ belong to $(x, y, z, t)^{L_1}$, the number of factors is less than $1 + 2/\gamma$, and the exponents belong to Λ. By Lemma 4.14, $(\operatorname{Spec} \mathbb{C}[[x, y, z, t]], f_0 \cdot \prod_l (h_l^0)^{b_l})$ is lc, and so, by [dFEM11, Proposition 2.11(i)], the pair

$$(\mathbb{A}^4, \operatorname{div}(f_0) + \sum_l b_l \operatorname{div}(h_l^0))$$

is lc near 0. Each h_l^0 has x-degree at most one and is non-zero, so it is not divisible by f_0. Since $S \times \mathbb{A}^1_t = (f_0 = 0)$ is normal, [Kaw07, Theorem] implies that $(S \times \mathbb{A}^1_t, \sum_l b_l \operatorname{div}(h_l^0))$ is lc near 0. This proves (1).

Step 3. In this step we prove (2) and (3). The quotients by G_S in (2) and (3) are the quotients by $\widetilde{H}$, and $\widehat{G}_S$ is the group $\widehat{Q}_1$ of Lemma D.3 for the diagonal action of G_S on $\mathbb{A}^3_{x,y,z}$. Each h_l^0 is G-semi-invariant, because G is diagonal and h_l^0 is a sum of terms of h_l, and it is $\mathbf{T}$-semi-invariant

of $\mathbf{T}$-weight κ_l. Let $W_0 := S \times \mathbb{A}^1_t$ and $D_0 := \sum_l b_l \operatorname{div}(h_l^0|_{W_0})$, so that (W_0, D_0) is lc near 0 by (1). The groups G_S, R_S, and $\widetilde{H}$ of Lemma D.6 for the h_l^0 and the b_l are those of the statement, because they are defined by identical formulas. By Lemma D.6(1), R_S acts trivially on $\mathbb{A}^1_\varpi$, so $\widetilde{H}$ acts on V, and $\widetilde{H}$ has order $v[G:R]$ and acts faithfully on S. By Lemma D.6(2), (S, D_S) is lc near 0.

By Lemma D.3, Π is normal in $\widehat{G}_S$ and $\widehat{G}_S/\Pi \cong \widetilde{H}$. Since π is étale over $S\backslash\{0\}$, the crepant quotient of $(\mathbb{A}^2, \pi^* D_S)$ by Π is (S, D_S). Taking the quotient in stages [Liu26, Lemma 2.6(5)], the germs $(\mathbb{A}^2 \ni 0, \pi^* D_S)/\widehat{G}_S$ and $(S \ni 0, D_S)/\widetilde{H}$ coincide, and the equality in (2) follows.

Assume $\delta \geq 0$. Let ν be a prime divisor over S. Since I_0 is generated by monomials,

$$\operatorname{mult}_\nu I_0 = \min_{u \in \{x,y,z\}} \frac{N_0}{q_u} \operatorname{mult}_\nu u = cN_0 m_\nu,$$

with $m_\nu := \min_{u\in\{x,y,z\}} \operatorname{mult}_\nu u/(cq_u)$. If the center of ν is 0, then

$$a(\nu, S, D_S, I_0^{\delta/(cN_0)}) = a(\nu, S, D_S) - \delta m_\nu \geq 0$$

by Lemma D.6(2). If the center of ν is a curve through 0, then $\operatorname{mult}_\nu I_0 = 0$, because the cosupport of I_0 is $\{0\}$, and $a(\nu, S, D_S, I_0^{\delta/(cN_0)}) = a(\nu, S, D_S) \geq 0$ by (2). Thus $(S, D_S, I_0^{\delta/(cN_0)})$ is lc near 0.

Let P be a prime divisor over $\mathbb{A}^2$ with center 0 and ν the prime divisor over S under P, with ramification index $e_P = |I_\Pi(P)|$ [Liu26, Lemma 2.6(1)]. By Lemma D.7(4), $\operatorname{mult}_P(\pi^* I_0) = 2N_0 \operatorname{mult}_P \mathfrak{m}$, and $\operatorname{mult}_P(\pi^* I_0) = e_P \operatorname{mult}_\nu I_0$. By [Liu26, Lemma 2.6(2)],

$$a\big(P, \mathbb{A}^2, \pi^* D_S, \mathfrak{m}^{2\delta/c}\big) = a\big(P, \mathbb{A}^2, \pi^* D_S\big) - \frac{\delta}{cN_0} \operatorname{mult}_P(\pi^* I_0) = e_P a\big(\nu, S, D_S, I_0^{\delta/(cN_0)}\big).$$

Since $|I_{\widehat{G}_S}(P)| = e_P |I_{\widetilde{H}}(\nu)|$ by the multiplicativity of ramification indices,

$$\frac{a(P, \mathbb{A}^2, \pi^* D_S, \mathfrak{m}^{2\delta/c})}{|I_{\widehat{G}_S}(P)|} = \frac{a(\nu, S, D_S, I_0^{\delta/(cN_0)})}{|I_{\widetilde{H}}(\nu)|}.$$

Any prime divisor over S with center 0 is the prime divisor under some prime divisor over $\mathbb{A}^2$ with center 0 [Liu26, Lemma 2.6(3)], since $\pi^{-1}(0) = \{0\}$, so the infima defining both sides of the equality in (3) (Definition 2.4) coincide. If $\delta = 0$, then the ideal has exponent 0, so $\beta_+ = \beta_-$.

Step 4. In this step we describe β_+ on a surface and compute the coefficients. Let n_l be the largest integer with $t^{n_l} \mid h_l^0$, let $b_t := \sum_l b_l n_l$, and write $h_l^0 = \sum_n t^n h_{l,n}(x, y, z)$. Since $\operatorname{div}(h_l^0|_{W_0}) \geq n_l(S \times \{0\})$ and (W_0, D_0) is lc near 0, we have $b_t \leq 1$. A monomial $x^a y^b z^e t^n$ of h_l^0 has $c(q_x a + q_y b + q_z e) = vn - \kappa_l$, so $h_{l,n}(F_x, F_y, F_z)$ is homogeneous of degree $2(vn - \kappa_l)/c$. Moreover $h_{l,n_l}(F_x, F_y, F_z) \neq 0$, because h_{l,n_l} has x-degree at most one and is not divisible by f_0. Hence

$$\operatorname{mult}_0 \pi^* D_S = \sum_l b_l \frac{2(vn_l - \kappa_l)}{c} = \frac{2(vb_t - \tau)}{c}. \tag{D.13}$$

Let $\sigma_1 : Y_1 \to \mathbb{A}^2$ be the blow-up of the origin with exceptional curve E_1. By (D.13),

$$a\big(E_1, \mathbb{A}^2, \pi^* D_S, \mathfrak{m}^{2\delta/c}\big) = 2 - \frac{2(vb_t - \tau)}{c} - \frac{2(\tau + c - v)}{c} = \frac{2v(1 - b_t)}{c}.$$

Thus, for $\delta \geq 0$, the pair $(\mathbb{A}^2, \pi^* D_S, \mathfrak{m}^{2\delta/c})$ has the crepant transform (Y_1, B_1) with $B_1 := \sigma_{1*}^{-1} \pi^* D_S + (1 - 2v(1 - b_t)/c)E_1$, and β_+ is the minimum of $a(P, Y_1, B_1)/|I_{\widehat{G}_S}(P)|$ over the prime divisors P over Y_1 with center in E_1.

Let $\widehat{G}_S^{\mathrm{sc}}$ be the subgroup of scalar matrices of $\widehat{G}_S$. It is the inertia group of E_1, because an element of $\mathrm{GL}_2(\mathbb{C})$ acts trivially on $E_1 \cong \mathbb{P}^1$ if and only if it is a scalar. Let $\lambda^{2v} = 1$. As $\gcd(c, v) = 1$ and $\lambda^2 \in \mu_v$, there exists ζ with $\zeta^v = 1$ and $\zeta^c = \lambda^2$. Then $(1, \zeta) \in G_S$ multiplies $u \in \{x, y, z\}$ by $\zeta^{cq_u} = \lambda^{2q_u}$, which is the automorphism induced by the scalar λ through π. Thus

$\mu_{2v} \subset \widehat{G}_S^{\mathrm{sc}}$. Since -1 is the unique element of order 2 of Π, $\widehat{G}_S^{\mathrm{sc}} \cap \Pi = \{\pm 1\}$, and $\widehat{G}_S^{\mathrm{sc}}/\{\pm 1\}$ embeds in $\widehat{G}_S/\Pi \cong \widetilde{H}$. Hence $|\widehat{G}_S^{\mathrm{sc}}| = 2ve_1$ for a positive integer $e_1 \le [G : R] \le e_0$. Let $(Y, B_Y) := (Y_1, B_1)/\widehat{G}_S$ and E_Y the image of E_1. By [Liu26, Lemma 2.6(4)],

$$\mathrm{mult}_{E_Y} B_Y = 1 - \frac{2v(1-b_t)/c}{2ve_1} = 1 - \frac{1-b_t}{ce_1} \in \left[1 - \frac{1}{ce_1}, 1\right]. \tag{D.14}$$

Step 5. In this step we prove (4). Let $\Lambda_0 := \Psi(\Gamma_+ \cap [0,1])$, a DCC set by [Liu26, Lemma 2.10(1)]. We have $\pi^* D_S \in \Gamma_+$, and the coefficients of $\pi^* D_S$ are at most one because $(\mathbb{A}^2, \pi^* D_S)$ is lc near 0. By [Liu26, Lemma 2.6(4)] and [Liu26, Lemma 2.10(5)], the coefficients of the crepant quotient $(Y_-, B_-) := (\mathbb{A}^2, \pi^* D_S)/\widehat{G}_S$ near the image y_- of 0, and those of B_Y other than $\mathrm{mult}_{E_Y} B_Y$, belong to Λ_0. By (D.14) and $b_t \in \Gamma_+ \cap [0,1]$, also $\mathrm{mult}_{E_Y} B_Y \in \Lambda_0$. Thus $B_- \ge 0$, $B_Y \ge 0$, $B_- \in \Lambda_0$, and $B_Y \in \Lambda_0$.

Let $\mathcal{M}$ be the set of Theorem 2.9(1) for Λ_0 and l_0 the integer of Theorem 2.9(2) for Λ_0. The number $\beta_- = \mathrm{mld}(Y_- \ni y_-, B_-)$ belongs to $\mathcal{M}$. Let $\delta \ge 0$. The pair (Y, B_Y) is lc near E_Y by (3). At a closed point $y \in E_Y$ where Y and E_Y are smooth and E_Y is the only component of B_Y, [KM98, Corollary 2.31(3)] and the blow-up of y give $\mathrm{mld}(Y \ni y, B_Y) = 2 - \mathrm{mult}_{E_Y} B_Y > a(E_Y, Y, B_Y)$. Thus

$$\beta_+ = \min\left\{1 - \mathrm{mult}_{E_Y} B_Y,\ \min_{y \in \Theta_Y} \mathrm{mld}(Y \ni y, B_Y)\right\}$$

for a finite set $\Theta_Y \subset E_Y$, and $\beta_+ \in \mathcal{M} \cup \{1 - b \mid b \in \Lambda_0\}$. Both $\mathcal{M}$ and $\{1 - b \mid b \in \Lambda_0\}$ satisfy the ACC and depend only on Γ.

The curve E_1 is $\widehat{G}_S$-invariant, and $a(E_1, \mathbb{A}^2, \pi^* D_S) \le 2$ and $a(E_1, \mathbb{A}^2, \pi^* D_S, \mathfrak{m}^{2\delta/c}) \le 2$, so $\beta \le 2$.

Let $\beta = \beta_-$. By Theorem 2.9(2), there exists a prime divisor P_- over Y_- with center y_- computing β_- and $a(P_-, Y_-, 0) \le l_0$. Let P be a prime divisor over $\mathbb{A}^2$ over P_-. By [Liu26, Lemma 2.6(2)], P computes β_-, and

$$a(P, \mathbb{A}^2, 0) = |I_{\widehat{G}_S}(P)| a(P_-, Y_-, B_{\mathrm{br}}) \le |\widehat{G}_S| l_0,$$

where $B_{\mathrm{br}} \ge 0$ is the branch boundary of $\mathbb{A}^2 \to Y_-$. Let $\beta = \beta_+$. If E_Y computes β_+, then $P := E_1$ computes β_+ and $a(E_1, \mathbb{A}^2, 0) = 2$. Otherwise there exist $y \in \Theta_Y$ and a prime divisor P_y over Y with center y computing β_+ and $a(P_y, Y, 0) \le l_0$. Let P be a prime divisor over Y_1 over P_y. Then P computes β_+, the center of P on $\mathbb{A}^2$ is 0, and $a(P, Y_1, 0) \le |\widehat{G}_S| l_0$ by [Liu26, Lemma 2.6(2)] for the quotient morphism $Y_1 \to Y$, as in the case $\beta = \beta_-$. At the center of P on Y_1, a local equation w_1 of E_1 is part of a system of regular parameters, so $a(P, Y_1, 0) \ge \mathrm{mult}_P w_1 + 1$ [Liu26, Lemma 2.13]. Since $K_{Y_1} = \sigma_1^* K_{\mathbb{A}^2} + E_1$,

$$a(P, \mathbb{A}^2, 0) = a(P, Y_1, 0) + \mathrm{mult}_P E_1 \le 2a(P, Y_1, 0) \le 2|\widehat{G}_S| l_0.$$

Finally $|\widehat{G}_S| = |\Pi| v [G : R] \le 120ve_0$ for the E types and $|\widehat{G}_S| \le 4(k-2)ve_0$ for D_k, which proves (4). □

Lemma D.9. *In the situation of Theorem 10.4, write* $f_0 = x^2 + g_0(y,z)$ *and* $f = f_0 + A$ *with* $A := \sum_j a_j(t) M_j$, *let* $W_0 := (f_0 = 0) \subset \mathbb{A}^4$, *and let* $N \ge 0$ *be an integer. Then there exists a bijection* $F \to F_0$ *from the set of prime divisors* F *over* W *with center* 0 *and* $\mathrm{mult}_F A > \mathrm{mult}_F g_0 + N$ *onto the set of prime divisors* F_0 *over* W_0 *with center* 0 *and* $\mathrm{mult}_{F_0} A > \mathrm{mult}_{F_0} g_0 + N$ *such that the following hold for any such* F.

(1) $\mathrm{mult}_{F_0} \phi = \mathrm{mult}_F \phi$ *for any* $\phi \in \mathbb{C}[y,z,t]$, *and* $\mathrm{mult}_{F_0} x = \mathrm{mult}_F x$.
(2) $a(F_0, W_0, 0) = a(F, W, 0)$, $I_G(F_0) = I_G(F)$, $\mathrm{Stab}_G(F_0) = \mathrm{Stab}_G(F)$, *and the residue fields of* F *and* F_0 *are isomorphic.*
(3) $\mathrm{mult}_{F_0} h = \mathrm{mult}_F h$ *for any* $h \in \mathbb{C}[x,y,z,t]$ *with* $\min\{\mathrm{mult}_F h, \mathrm{mult}_{F_0} h\} \le N$.

Proof. Let $g := g_0 + A$, so that $W = (x^2 + g = 0)$ and $W_0 = (x^2 + g_0 = 0)$ are double covers of $\mathbb{A}^3_{y,z,t}$. The hypersurfaces W and W_0 are integral: g_0 is not a square in $\mathbb{C}[y,z]$, since $(f_0 = 0) \subset \mathbb{A}^3_{x,y,z}$ is a normal surface, and g is not a square in $\mathbb{C}[y,z,t]$, since $a_j \in t\mathbb{C}[t]$ and so the restriction of g to $(t = 0)$ is g_0. Since G preserves f_0 and each non-zero $a_j(t)M_j$ up to a common character, g_0, A, and g are G-semi-invariant of one character, and $u := g_0/g$ is a G-invariant rational function of y, z, t with $u - 1 = -A/g$.

Let F be a prime divisor over W with center 0 and $\operatorname{mult}_F A > \operatorname{mult}_F g_0 + N$. Then $\operatorname{mult}_F g = \operatorname{mult}_F g_0$ and $\operatorname{mult}_F(u-1) = \operatorname{mult}_F A - \operatorname{mult}_F g_0 > N$.

Claim D.10. *Let $g', g'' \in \mathbb{C}[y,z,t]$, let $W' := (x^2 + g' = 0)$ and $W'' := (x^2 + g'' = 0)$ be integral hypersurfaces in $\mathbb{A}^4$, let $u' := g''/g'$, and let F be a prime divisor over W' with center 0 and $\operatorname{mult}_F(u'-1) > N$. Let $\mathcal{O}_F$ be the local ring of a higher model of W' at the generic point of F, with maximal ideal $\mathfrak{m}_{\mathcal{O}_F}$ and residue field $\kappa(F)$, and let $\mathcal{O}'_F$ be the localization of $\mathcal{O}_F[U]/(U^2 - u')$ at the maximal ideal generated by $\mathfrak{m}_{\mathcal{O}_F}$ and $U - 1$. Then the following hold.*

(1) The ring $\mathcal{O}'_F$ is a discrete valuation ring étale over $\mathcal{O}_F$ with residue field $\kappa(F)$, and its valuation mult'_F extends mult_F and satisfies $\operatorname{mult}'_F(U-1) > N$.

(2) Let $x'' := Ux$. Then $(x,y,z,t) \to (x'',y,z,t)$ embeds the coordinate ring of W'' in the fraction field of $\mathcal{O}'_F$, and the restriction of mult'_F is $\operatorname{mult}_{F''}$ for a prime divisor F'' over W''. The ring $\mathcal{O}'_F$ is the localization of $\mathcal{O}_{F''}[U'']/(U''^2 - u'^{-1})$, $U'' := U^{-1}$, at the maximal ideal generated by $\mathfrak{m}_{\mathcal{O}_{F''}}$ and $U'' - 1$, and it is étale over $\mathcal{O}_{F''}$.

(3) $\operatorname{mult}_{F''} \phi = \operatorname{mult}_F \phi$ for any $\phi \in \mathbb{C}[y,z,t]$.

Proof. Since $2U$ is a unit at the maximal ideal of $\mathcal{O}'_F$, $\mathcal{O}_F \to \mathcal{O}'_F$ is étale with residue field $\kappa(F)$. Thus $\mathcal{O}'_F$ is a discrete valuation ring whose valuation mult'_F extends mult_F and has value group $\mathbb{Z}$ and residue field $\kappa(F)$. As $U+1$ is a unit,

$$\operatorname{mult}'_F(U-1) = \operatorname{mult}'_F(U^2-1) = \operatorname{mult}_F(u'-1) > N.$$

This proves (1).

Let K' be the fraction field of $\mathcal{O}'_F$. Since $x''^2 = u'x^2 = -g''$ in K', $(x,y,z,t) \to (x'',y,z,t)$ defines a homomorphism from the coordinate ring of W'' to K'. It is injective, because the image contains y, z, t, which are algebraically independent, and W'' is integral of dimension three. Let $\mathcal{O}_1$ be the valuation ring of the restriction of mult'_F to the rational functions on W''. The field K' is generated over the rational functions on W'' by U, and $U^2 - u'$ has coefficients in $\mathcal{O}_1$ and reduction $(U-1)(U+1)$ with two simple roots in the residue field of $\mathcal{O}_1$. By Hensel's lemma, $\mathcal{O}'_F$ is unramified over $\mathcal{O}_1$ with trivial residue field extension. Thus the restriction of mult'_F has value group $\mathbb{Z}$ and residue field $\kappa(F)$, of transcendence degree two over $\mathbb{C}$, so it is $\operatorname{mult}_{F''}$ for a prime divisor F'' over W'' [KM98, Lemma 2.45]. Moreover $\mathcal{O}'_F$ is the localization of $\mathcal{O}_{F''}[U'']/(U''^2 - u'^{-1})$ at the ideal generated by $\mathfrak{m}_{\mathcal{O}_{F''}}$ and $U''-1$, so $\mathcal{O}_{F''} \to \mathcal{O}'_F$ is étale as well. This proves (2). For $\phi \in \mathbb{C}[y,z,t]$, $\operatorname{mult}_{F''} \phi = \operatorname{mult}'_F \phi = \operatorname{mult}_F \phi$, which proves (3). □

Proof of Lemma D.9 continued. We apply Claim D.10 to $W' := W$, $W'' := W_0$, $g' := g$, $g'' := g_0$, and F, so that $u' = u$. Let F_0 be the prime divisor F'' over W_0 given by Claim D.10(2), $x_0 := Ux$, and K' the fraction field of $\mathcal{O}'_F$.

For $\phi \in \mathbb{C}[y,z,t]$, $\operatorname{mult}_{F_0} \phi = \operatorname{mult}_F \phi$ by Claim D.10(3), and $\operatorname{mult}_{F_0} x_0 = \operatorname{mult}'_F(U) + \operatorname{mult}_F x = \operatorname{mult}_F x$. This proves (1). In particular F_0 has center 0 and

$$\operatorname{mult}_{F_0} A - \operatorname{mult}_{F_0} g_0 = \operatorname{mult}_F A - \operatorname{mult}_F g_0 > N.$$

For $h \in \mathbb{C}[x,y,z,t]$, write $h = \sum_j h_j(y,z,t)x^j$. Then

$$h(x_0,y,z,t) - h(x,y,z,t) = \sum_{j \geq 1} h_j(y,z,t)x^j(U-1)(U^{j-1} + \cdots + 1),$$

and each term has mult'_F larger than N by Claim D.10(1). Thus $\operatorname{mult}_{F_0} h$ and $\operatorname{mult}_F h$ coincide if one of them is at most N, which proves (3).

Let F_0 be a prime divisor over W_0 with center 0 and $\operatorname{mult}_{F_0} A > \operatorname{mult}_{F_0} g_0 + N$. Then $\operatorname{mult}_{F_0}(u^{-1}-1) = \operatorname{mult}_{F_0} A - \operatorname{mult}_{F_0} g_0 > N$, and Claim D.10 for $W' := W_0$, $W'' := W$, $g' := g_0$, $g'' := g$, and F_0, so that $u' = u^{-1}$, and with U^{-1} in place of U, yields a discrete valuation ring $\mathcal{O}''_{F_0}$ étale over $\mathcal{O}_{F_0}$ and a prime divisor F' over W. If F_0 is the prime divisor F'' given by Claim D.10(2) for $W' := W$, $W'' := W_0$, and a prime divisor F over W, then $\mathcal{O}''_{F_0}$ is dominated by $\mathcal{O}'_F$ by Claim D.10(2), and both have fraction field K', so $\mathcal{O}''_{F_0} = \mathcal{O}'_F$ and $F' = F$. Thus the maps $F \to F_0$ and $F_0 \to F'$ are mutually inverse, and $F \to F_0$ is a bijection.

We prove (2). The residue fields of F and F_0 are both identified with $\kappa(F)$. The canonical sheaves of W and W_0 are generated near 0 by the residues $\Omega := dy\wedge dz\wedge dt/(2x)$ and $\Omega_0 := dy\wedge dz\wedge dt/(2x_0)$ [KM98, Proposition 5.73], so $a(F,W,0) = 1+\operatorname{mult}_F \Omega$ and $a(F_0,W_0,0) = 1+\operatorname{mult}_{F_0}\Omega_0$, where $\operatorname{mult}_F \Omega$ is the order of Ω along F, computed in the free module of differential 3-forms of $\mathcal{O}_F$. The modules of differential 3-forms of $\mathcal{O}_F$, $\mathcal{O}'_F$, and $\mathcal{O}_{F_0}$ are compatible with the étale homomorphisms $\mathcal{O}_F \to \mathcal{O}'_F$ and $\mathcal{O}_{F_0} \to \mathcal{O}'_F$. Since $\Omega_0 = U^{-1}\Omega$ and U is a unit of $\mathcal{O}'_F$, $\operatorname{mult}_{F_0}\Omega_0 = \operatorname{mult}_F \Omega$, and $a(F_0,W_0,0) = a(F,W,0)$.

Let $\sigma \in G$. Since G is diagonal, $\sigma^* x = \chi_x(\sigma)x$, and σ acts on W_0 with $\sigma^* x_0 = \chi_x(\sigma)x_0$. The set of prime divisors F in the statement is G-stable because A and g_0 are G-semi-invariant. As u is G-invariant, σ maps the branch $U \equiv 1$ of $\sqrt{u}$ along F to the branch $U \equiv 1$ along $\sigma(F)$, and the identity $x_0 = Ux$ is preserved by σ. Hence $\sigma(F)_0 = \sigma(F_0)$, so $\operatorname{Stab}_G(F_0) = \operatorname{Stab}_G(F)$. An element $\sigma \in \operatorname{Stab}_G(F)$ acts on $\mathcal{O}'_F$, fixing U, compatibly with the homomorphisms from $\mathcal{O}_F$ and $\mathcal{O}_{F_0}$. Thus σ acts trivially on the residue field of F if and only if it acts trivially on that of F_0, and $I_G(F_0) = I_G(F)$. □

Proposition D.11. *In the situation of Theorem 10.4 and Construction D.8, there exists a real number $C_0 \geq 0$, depending only on the type, c, v, e_0, and Γ, such that* $\operatorname{mld}((W \ni 0, D)/G) \leq \beta + C_0/r$ *for $r \geq r_0$.*

Proof. The proof is long, so we divide it into several steps. Let $\alpha := \operatorname{mld}((W \ni 0, D)/G)$, let γ, $M_{\deg}$, and $\kappa_{\max}$ be as in Set-up D.5, let $W_0 := S \times \mathbb{A}^1_t$ and $D_0 := \sum_l b_l \operatorname{div}(h_l^0|_{W_0})$, and let R_S, $\widetilde{H}$, and V, with the action of $\widetilde{H}$ on V, be as in Construction D.8. By Construction D.8, $b_l \geq \gamma$ and $h_l(0) = 0$ for any l. Then the number of marks is less than $2/\gamma$ by Lemma D.7(2), and

$$|\delta| \leq \delta_{\max} := \frac{2\kappa_{\max}}{\gamma} + c + v. \tag{D.15}$$

Step 1. In this step we fix a computing divisor on S and bound the orders of functions along it. By Construction D.8(4), there exist a prime divisor P over $\mathbb{A}^2$ with center 0 computing β and an integer l_1, depending only on the type, c, v, e_0, and Γ, with $a(P,\mathbb{A}^2,0) \leq l_1$. Let ν be the prime divisor over S under P, with ramification index $e_P = |I_\Pi(P)|$, and $e := |I_{\widetilde{H}}(\nu)|$. By [Liu26, Lemma 2.6(1)(2)] and Lemma D.7(4),

$$e\beta = \begin{cases} a(\nu,S,D_S) & \text{if } \delta \leq 0, \\ a(\nu,S,D_S) - \delta m_\nu & \text{if } \delta \geq 0, \end{cases} \tag{D.16}$$

with $m_\nu := \min_{u\in\{x,y,z\}} \operatorname{mult}_\nu u/(cq_u)$, and $e \leq |\widetilde{H}| = v[G:R] \leq ve_0$.

We bound the orders along ν. Since $\operatorname{mult}_P(\pi^*\psi) = e_P \operatorname{mult}_\nu \psi$ for any rational function ψ on S and $e_P \geq 1$, it suffices to bound the orders along P. For a linear form θ_1, choose a linear form θ_2 such that θ_1, θ_2 are regular parameters at 0. Both vanish at the center 0 of P, so $\operatorname{mult}_P \theta_1 + 1 \leq a(P,\mathbb{A}^2,0) \leq l_1$ by [Liu26, Lemma 2.13]. As F_u is a product of $2q_u$ linear forms,

$$1 \leq \operatorname{mult}_\nu u \leq \operatorname{mult}_P F_u \leq 2q_u l_1 \quad \text{for } u \in \{x,y,z\}.$$

In particular $1/(cq_x) \leq m_\nu \leq 2l_1/c$. Let $H_l := h_l^0(x,y,z,1)|_S$. By (D.16),

$$a(P,\mathbb{A}^2,\pi^* D_S) \geq e_P e\beta + \max\{\delta,0\}\cdot e_P m_\nu \geq 0,$$

so $\sum_l b_l \operatorname{mult}_P(\pi^* H_l) \le a(P, \mathbb{A}^2, 0) \le l_1$. As $b_l \ge \gamma$,

$$\operatorname{mult}_\nu H_l \le l_1/\gamma \quad \text{for any } l.$$

Let $C_2 := l_1/\gamma + 2\kappa_{\max} l_1/c$ and $C_3 := ve_0(C_2+1) + 4q_x l_1 + 2C_2 + 1$. We use C_2 and C_3 for the order estimates along the prime divisor F_0 constructed below: C_2 bounds $\operatorname{mult}_\nu H_l + \kappa_l k_0/r$ for any l, and C_3 is the margin of k_0 from 0 if $\delta \le 0$ and from rm_ν if $\delta > 0$: for r large, it makes $\operatorname{mult}_\nu \big(h_\kappa(x,y,z,1)|_S\big) + \kappa k_0/r$, the quantity of Lemma D.6(3) that C_2 bounds for h_l^0, larger than C_2 for the part h_κ of each h_l of any $\mathbf{T}$-weight $\kappa \ne \kappa_l$, and makes the deformation term A have larger order than g_0 along F_0.

Step 2. In this step we choose the ϖ-value k_0 of the Gauss extension. Let χ_ϖ be the character of $\widetilde{H}$ with $\sigma^*\varpi = \chi_\varpi(\sigma)\varpi$, which is well defined by Lemma D.6(1), and $q' \in \{1, \dots, e\}$ the integer of Lemma 3.1 for $V = S$, $H = \widetilde{H}$, $B = D_S$, the character χ_ϖ, and ν. The group $\widetilde{H}$ acts faithfully on S, fixes 0, and preserves D_S by Lemma D.6(1). If $\delta \le 0$, let k_0 be the integer in $[C_3, C_3 + e)$ with $k_0 \equiv q' \pmod e$. If $\delta > 0$, let k_0 be an integer in $(rm_\nu - C_3 - e - 1, rm_\nu - C_3]$ with $k_0 \equiv q' \pmod e$. Let

$$\sigma_\nu := m_\nu - \frac{k_0}{r}.$$

Let $r_1 \ge r_0$ be an integer larger than $2\kappa_{\max}$, than $2cq_x(C_3 + ve_0)$, than $4cq_xC_2 + 2\kappa_{\max}$, and than $cq_x(ve_0(C_2+1) + 4q_xl_1 + 2C_2 + C_3 + ve_0 + 1)$. Then r_1 depends only on the type, c, v, e_0, and Γ. Let $r \ge r_1$. If $\delta \le 0$, then $\sigma_\nu \ge 1/(cq_x) - (C_3 + ve_0)/r \ge 1/(2cq_x)$. If $\delta > 0$, then $C_3/r \le \sigma_\nu < (C_3 + ve_0 + 1)/r$ and $k_0 \ge r/(cq_x) - C_3 - ve_0 - 1 \ge ve_0(C_2+1) + 4q_xl_1 + 2C_2$. In both cases $\sigma_\nu > 0$ and $k_0 > 0$.

By Lemma 3.1, the Gauss extension of mult_ν with ϖ-value k_0 is mult_E for a prime divisor E over V with center $(0,0)$, $I_{\widetilde{H}}(E) = I_{\widetilde{H}}(\nu)$, and $a(E, V, \operatorname{pr}_1^* D_S) = a(\nu, S, D_S) + k_0$. Since $\sigma_\nu > 0$, we have $k_0 < rm_\nu$. Let F_0 be the prime divisor over W_0 given by Lemma D.6(3) for E, and for a polynomial φ not divisible by f_0 let $[\varphi] := \operatorname{mult}_{F_0} \varphi / |I_G(F_0)|$. Since $\operatorname{mult}_E \varpi = k_0$,

$$\frac{a(F_0, W_0, D_0)}{|I_G(F_0)|} = \frac{a(\nu, S, D_S) + k_0 - (1 + \delta/r)k_0}{e} = \frac{a(\nu, S, D_S) - \delta k_0/r}{e}. \tag{D.17}$$

Step 3. In this step we compute the orders along F_0. By Lemma D.6(3), for $u \in \{x, y, z\}$,

$$e[u] = \operatorname{mult}_\nu u - \frac{cq_u k_0}{r} \ge cq_u\sigma_\nu > 0, \qquad e[t] = \frac{vk_0}{r} > 0,$$

so the center of F_0 on W_0 is 0.

Let $h := h_l$ for some l. By Lemma 5.3, the $\mathbf{T}$-semi-invariant parts of h have $\mathbf{T}$-weights $\kappa = \kappa_l + jr$, $j \in \mathbb{Z}$, and the part with $j = 0$ is h_l^0. A monomial $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ of $\mathbf{T}$-weight κ has

$$c(q_xn_1 + q_yn_2 + q_zn_3) = vn_4 - \kappa \ge -\kappa,$$

so

$$\operatorname{mult}_\nu(x^{n_1}y^{n_2}z^{n_3}) \ge cm_\nu(q_xn_1 + q_yn_2 + q_zn_3) \ge -\kappa m_\nu.$$

Let h_κ be the part of $\mathbf{T}$-weight κ. If $j \le -1$, then $-\kappa \ge r - \kappa_{\max}$ by (D.12), and

$$\operatorname{mult}_\nu \big(h_\kappa(x,y,z,1)|_S\big) + \frac{\kappa k_0}{r} \ge -\kappa\Big(m_\nu - \frac{k_0}{r}\Big) \ge (r - \kappa_{\max})\sigma_\nu > C_2,$$

where the last inequality follows from $\sigma_\nu \ge 1/(2cq_x)$ and $r > 4cq_xC_2 + 2\kappa_{\max}$ if $\delta \le 0$, and from $\sigma_\nu \ge C_3/r$, $r > 2\kappa_{\max}$, and $C_3 > 2C_2$ if $\delta > 0$. If $j \ge 1$, then $\kappa \ge r - \kappa_{\max} \ge r/2$, and the left hand side is at least $\kappa k_0/r \ge k_0/2 > C_2$. For $j = 0$,

$$\operatorname{mult}_\nu H_l + \frac{\kappa_l k_0}{r} \le \frac{l_1}{\gamma} + \kappa_{\max} m_\nu \le C_2,$$

because $0 < k_0/r < m_\nu$. Thus, by Lemma D.6(3), the minimum is attained only at $j = 0$, and

$$[h_l] = [h_l^0] \le \frac{C_2}{e} \le C_2 \quad \text{for any } l. \tag{D.18}$$

The polynomial g_0 is $\mathbf{T}$-semi-invariant of $\mathbf{T}$-weight $-cd$ and $g_0|_S = -x^2|_S$. A non-zero term $a_{jn}t^nM_j$ of A is $\mathbf{T}$-semi-invariant of $\mathbf{T}$-weight $vn - cd_j = \ell r - cd$ with $\ell \geq 1$ by (D.11). Hence

$$e[g_0] = 2\operatorname{mult}_\nu x - \frac{cdk_0}{r}, \qquad e[t^nM_j] \geq \ell k_0 - \frac{cdk_0}{r},$$

and

$$e\big([A] - [g_0]\big) \geq k_0 - 2\operatorname{mult}_\nu x \geq k_0 - 4q_xl_1 > ve_0(C_2 + 1),$$

and $[A] - [g_0] > C_2 + 1$ since $e \leq ve_0$.

Step 4. In this step we pass to W and conclude. Let $N := \lceil |I_G(F_0)|C_2 \rceil$. By **Step 3**,

$$\operatorname{mult}_{F_0} A - \operatorname{mult}_{F_0} g_0 = |I_G(F_0)|\big([A] - [g_0]\big) > |I_G(F_0)|C_2 + 1 > N,$$

and $\operatorname{mult}_{F_0} h_l \leq |I_G(F_0)|C_2 \leq N$ by (D.18). Let F be the prime divisor over W with center 0 corresponding to F_0 by Lemma D.9. By Lemma D.9(2)(3) and (D.18),

$$a(F, W, D) = a(F, W, 0) - \sum_l b_l \operatorname{mult}_F h_l = a(F_0, W_0, 0) - \sum_l b_l \operatorname{mult}_{F_0} h_l^0 = a(F_0, W_0, D_0),$$

and $|I_G(F)| = |I_G(F_0)|$. By the formula at the end of [Liu26, Lemma 2.6] and (D.17),

$$\alpha \leq \frac{a(F, W, D)}{|I_G(F)|} = \frac{a(\nu, S, D_S) - \delta k_0/r}{e}.$$

If $\delta \leq 0$, then (D.16) implies

$$\alpha \leq \beta + \frac{|\delta|k_0}{re} \leq \beta + \frac{\delta_{\max}(C_3 + ve_0)}{r}.$$

If $\delta > 0$, then (D.16) implies

$$\alpha \leq \beta + \frac{\delta\sigma_\nu}{e} \leq \beta + \frac{\delta_{\max}(C_3 + ve_0 + 1)}{r}.$$

Let $r_0 \leq r < r_1$. The prime divisor E_W of Lemma D.7(2) has center 0 and $a(E_W, W, 0) = 2$, so $\alpha \leq a(E_W, W, D)/|I_G(E_W)| \leq 2 \leq 2r_1/r$, where $a(E_W, W, D) \geq 0$ because (W, D) is klt near 0. Since $\beta \geq 0$, the proposition holds with $C_0 := \max\{2r_1, \delta_{\max}(C_3 + ve_0 + 1)\}$. □

D.4. **Circles $(-cq_x, -cq_y, -cq_z, v)$ on sources of type D and E: limits.** In this subsection, we prove Theorem 10.4.

The lower bound for the minimal log discrepancy of the quotient depends on a real number θ larger than 1 such that $(\mathbb{A}^2, \theta\pi^*D_S, \mathfrak{m}^{2\delta_\theta^+/c})$ is lc near 0 (Claim D.12), where δ_θ^+ is the maximum of $\theta\tau + c - v$ and 0, τ is the number of Construction D.8, and $\pi : \mathbb{A}^2 \to S$ is the quotient morphism of Lemma D.4. We obtain such a θ from a compatible generic limit of the reference pairs, which is klt because no line of the plane is invariant under the binary polyhedral group (Lemma D.4).

Proof of Theorem 10.4. The proof is long, so we divide it into several steps.

Step 1. In this step we fix notation and two consequences of Construction D.8.

Let γ, $M_{\deg}$, $\kappa_{\max} = M_{\deg}\max\{cq_x, v\}$, N_0, I_0, and $\mathfrak{m}$ be as in Set-up D.5. Write $D = \sum_l b_l \operatorname{div}(h_l)$ as in Construction D.8, so that $b_l \geq \gamma$, $h_l(0) = 0$, and the h_l are G-semi-invariant polynomials of x-degree at most one. Write $f = f_0 + A$ with $A := \sum_j a_j(t)M_j$, let $W_0 := (f_0 = 0) \subset \mathbb{A}^4$, so that $W_0 = S \times \mathbb{A}^1_t$, and let $\alpha := \operatorname{mld}((W \ni 0, D)/G)$. Let h_l^0, κ_l, $\tau = \sum_l b_l\kappa_l$, δ, d, G_S, R_S, $\widetilde{H}$, $\widehat{G}_S$, D_S, V, β_-, β_+, β, and r_0 be as in Construction D.8. Let $\Pi \subset \mathrm{SL}_2(\mathbb{C})$ and the quotient morphism $\pi = (F_x, F_y, F_z) : \mathbb{A}^2_{\xi_1,\xi_2} \to S$ by Π be given by Lemma D.4. Let $\widehat{G}$, $\bar{c}$, $\bar{v}$, and $\widehat{\mathbf{T}}$, acting on $\mathbb{A}^3_{\xi_1,\xi_2,t}$, be given by Lemma D.4. Let C_0 be the constant of Proposition D.11. For a real number θ, let

$$\delta_\theta := \theta\tau + c - v \qquad \text{and} \qquad \delta_\theta^+ := \max\{\delta_\theta, 0\}.$$

By Construction D.8(1), $\delta_1 = \delta$. Let $H_l := h_l^0(x,y,z,1) \in \mathbb{C}[x,y,z]$ and $\widehat{H}_l := H_l(F_x, F_y, F_z) \in \mathbb{C}[\xi_1,\xi_2]$, so that $D_S = \sum_l b_l \operatorname{div}(H_l|_S)$ and $\pi^* D_S = \sum_l b_l \operatorname{div}(\widehat{H}_l)$. In **Steps 1–3** we assume that $r \geq r_0$.

By Lemma D.7(4), $\operatorname{mult}_P(\pi^* I_0) = 2N_0 \operatorname{mult}_P \mathfrak{m}$ for any prime divisor P over $\mathbb{A}^2$. Since π is étale over $S\backslash\{0\}$, the crepant quotient of $(\mathbb{A}^2, \theta\pi^* D_S)$ by Π is $(S, \theta D_S)$ for any $\theta \geq 0$. Hence, by [Liu26, Lemma 2.6(2)] and Lemma D.7(4), for any $\theta, \lambda \geq 0$, any prime divisor P over $\mathbb{A}^2$, and the prime divisor ν over S under P, whose center on S is the image of the center of P,

$$a(P, \mathbb{A}^2, \theta\pi^* D_S, \mathfrak{m}^{2\lambda/c}) = |I_\Pi(P)| \cdot a(\nu, S, \theta D_S, I_0^{\lambda/(cN_0)}),$$

and any prime divisor over S is the prime divisor under some prime divisor over $\mathbb{A}^2$ [Liu26, Lemma 2.6(3)]. In particular,

$$(S, \theta D_S, I_0^{\lambda/(cN_0)}) \text{ is lc near } 0 \iff (\mathbb{A}^2, \theta\pi^* D_S, \mathfrak{m}^{2\lambda/c}) \text{ is lc near } 0, \tag{D.19}$$

and, by Construction D.8(2)(3) (with $\mathfrak{m}^0 := \mathcal{O}_{\mathbb{A}^2}$),

$$\beta = \operatorname{mld}\big((\mathbb{A}^2 \ni 0, \pi^* D_S, \mathfrak{m}^{2\delta_1^+/c})/\widehat{G}_S\big), \tag{D.20}$$

where for $\delta = 0$ we use $\beta_+ = \beta_-$.

Enlarging r_0, we may assume that $r_0 > v(d+1) + cd$ and $r_0 > 2M_{\deg} \max\{cq_x, v\}$. By Lemma D.7(2)(3), fewer than $2/\gamma$ of the h_l occur, each h_l has a monomial of total degree less than $M_{\deg}$, and $|\kappa_l| \leq \kappa_{\max}$ for any l. By Lemma 5.3 and Lemma D.7(3), for any l we may write

$$h_l = \sum_{\ell \in \mathbb{Z}} h_{l,\ell},$$

where $h_{l,\ell}$ is the sum of the terms of h_l of $\mathbf{T}$-weight $\kappa_l + \ell r$, $h_{l,0} = h_l^0$, and any monomial of $h_{l,\ell}$ with $\ell \neq 0$ has total degree at least $(r - \kappa_{\max})/\max\{cq_x, v\}$. Here $q_x = \max\{q_x, q_y, q_z\}$.

Step 2. In this step we pass to the cover $t = s^v$ and to the morphism $\varrho : \widetilde{W} \to V_r$, and express the quotient values of (W, D) through V_r.

Let $a_j(t) = \sum_k a_{jk} t^k$. For any $a_{jk} \neq 0$, (D.11) of Lemma D.7(1) shows that

$$\ell_{jk} := \frac{kv + c(d - d_j)}{r}$$

is a positive integer, where d_j is the q-weighted degree of M_j. Let

$$F_r := f_0(X,Y,Z) + \sum_{j,k} a_{jk} \varpi^{\ell_{jk}} M_j(Y,Z) \in \mathbb{C}[X,Y,Z,\varpi],$$

and let $V_r := (F_r = 0) \subset \mathbb{A}^4_{X,Y,Z,\varpi}$. Then $V_r \cap (\varpi = 0) = S \times \{0\}$, which we identify with S.

Let $\widetilde{W}$ be the normalization of $W \times_{\mathbb{A}^1_t} \mathbb{A}^1_s$, where $\mathbb{A}^1_s \to \mathbb{A}^1_t$ is given by $t = s^v$, and let $p : \widetilde{W} \to W$ be the induced finite morphism. Near 0, $W \times_{\mathbb{A}^1_t} \mathbb{A}^1_s$ is the hypersurface $(f(x,y,z,s^v) = 0) \subset \mathbb{A}^4_{x,y,z,s}$. Since $\operatorname{div}(t|_W) = S$ is reduced and irreducible, the Cartier divisor $(s = 0)$ of $W \times_{\mathbb{A}^1_t} \mathbb{A}^1_s$ is the normal surface S, and $W \times_{\mathbb{A}^1_t} \mathbb{A}^1_s$ is étale over the normal variety W on $(s \neq 0)$. Thus $W \times_{\mathbb{A}^1_t} \mathbb{A}^1_s$ is normal near 0 (a Noetherian local ring is normal if its quotient by a non-zero-divisor in the maximal ideal is normal, by Serre's criterion [Mat89, Theorem 23.8]), hence irreducible near 0, and $\widetilde{W} = W \times_{\mathbb{A}^1_t} \mathbb{A}^1_s$ near 0. The group G_S acts on $\widetilde{W}$ through G on x, y, and z and by multiplying s by ζ for $(g, \zeta) \in G_S$. The morphism p is G_S-equivariant for the action of G_S on W through G, and it is the quotient morphism by $\{1\} \times \mu_v$. Since $G_S \to G$ is surjective with kernel $\{1\} \times \mu_v$, we obtain $W/G = \widetilde{W}/G_S$.

As for $W \times_{\mathbb{A}^1_t} \mathbb{A}^1_s$, the hypersurface V_r is normal near S, because $S = V_r \cap (\varpi = 0)$ is a normal Cartier divisor on V_r [Mat89, Theorem 23.8]. Let

$$\varrho : \widetilde{W} \to V_r, \qquad (x,y,z,s) \to (xs^{cq_x}, ys^{cq_y}, zs^{cq_z}, s^r).$$

Since f_0 and the M_j are weighted homogeneous, $F_r(xs^{cq_x}, ys^{cq_y}, zs^{cq_z}, s^r) = s^{cd}f(x,y,z,s^v)$, so ϱ is well defined. The group G_S acts diagonally on $\mathbb{A}^4_{X,Y,Z,\varpi}$ through its actions on $\mathbb{A}^3_{X,Y,Z}$ and on $\mathbb{A}^1_\varpi$ of Construction D.8. The group G_S preserves V_r, and ϱ is G_S-equivariant. By Construction D.8(2) and Lemma D.6(1), the kernel of the action of G_S on $\mathbb{A}^4_{X,Y,Z,\varpi}$ is R_S, so $\widetilde{H} = G_S/R_S$ acts on $\mathbb{A}^4_{X,Y,Z,\varpi}$, faithfully on S, and $|\widetilde{H}| = v[G : R]$. Since $x = Xs^{-cq_x}$, $y = Ys^{-cq_y}$, $z = Zs^{-cq_z}$, and $s^r = \varpi$, the restriction of ϱ to $\widetilde{W} \cap (s \neq 0)$ is the quotient morphism by R_S onto $V_r \cap (\varpi \neq 0)$. Hence $\widetilde{W}/G_S$, W/G, and $V_r/\widetilde{H}$ are birational.

Let $\widetilde{F}$ be a prime divisor over $\widetilde{W}$, and let F and F' be the prime divisors over W and over V_r dominated by $\widetilde{F}$. For a rational function ψ on $\widetilde{W}$, let

$$[\psi] := \frac{\operatorname{mult}_{\widetilde{F}} \psi}{|I_{G_S}(\widetilde{F})|},$$

and for rational functions ψ on W and ψ' on V_r write $[\psi] := [p^*\psi]$ and $[\psi'] := [\varrho^*\psi']$. By [Liu26, Lemma 2.6(1)] and the multiplicativity of ramification indices in $\widetilde{W} \to W \to W/G$ and in $\widetilde{W} \to V_r \to V_r/\widetilde{H}$,

$$[\psi] = \frac{\operatorname{mult}_F \psi}{|I_G(F)|}, \qquad [\psi'] = \frac{\operatorname{mult}_{F'} \psi'}{|I_{\widetilde{H}}(F')|}.$$

Let

$$A_W := \frac{a(F,W,0)}{|I_G(F)|}, \qquad A_{V_r} := \frac{a(F',V_r,0)}{|I_{\widetilde{H}}(F')|}.$$

The canonical sheaves of W and V_r are generated near 0 by the residue forms

$$\Omega_W := \frac{dy \wedge dz \wedge dt}{\partial_x f}, \qquad \Omega_{V_r} := \frac{dY \wedge dZ \wedge d\varpi}{\partial_X F_r}$$

[KM98, Proposition 5.73]. A direct calculation using $q_x + q_y + q_z = d + 1$ shows that $\varrho^*\Omega_{V_r} = \frac{r}{v} s^{c+r-v} p^*\Omega_W$ on $\widetilde{W}$. Hence, by the Hurwitz formula,

$$A_{V_r} = A_W + (c + r - v)[s]$$

whenever F and F' have center 0 on W and on V_r. For any l and ℓ, a monomial $x^{n_1}y^{n_2}z^{n_3}t^{n_4}$ of $\mathbf{T}$-weight $\kappa_l + \ell r$ satisfies

$$p^*(x^{n_1}y^{n_2}z^{n_3}t^{n_4}) = s^{\kappa_l + \ell r}\varrho^*(X^{n_1}Y^{n_2}Z^{n_3}),$$

so

$$s^{-\kappa_l} p^* h_{l,\ell} = \varrho^*\big(\varpi^\ell h_{l,\ell}(X,Y,Z,1)\big).$$

Let F be a prime divisor over W with center 0, let $\widetilde{F}$ be a prime divisor over $\widetilde{W}$ dominating F, and let F' be the prime divisor over V_r dominated by $\widetilde{F}$, with $[\cdot]$ defined by $\widetilde{F}$. Since $[x], [y], [z], [t] > 0$, we have $[\varpi] = r[t]/v > 0$ and $[X_i] = [x_i] + cq_i[\varpi]/r > 0$, where $(x_1, x_2, x_3) = (x, y, z)$, $(X_1, X_2, X_3) = (X, Y, Z)$, and $(q_1, q_2, q_3) = (q_x, q_y, q_z)$. Thus F' has center 0 on V_r, and

$$m_{\widetilde{F}} := \min_i \frac{[X_i]}{cq_i} > \frac{[\varpi]}{r}. \tag{D.21}$$

We have

$$a(F,W,D) = a(F,W,0) - \sum_l b_l \operatorname{mult}_F h_l, \qquad \frac{a(F,W,D)}{|I_G(F)|} = A_W - \sum_l b_l [h_l].$$

Since $[h_l] = \kappa_l[\varpi]/r + [s^{-\kappa_l}p^*h_l]$ and $\tau = \sum_l b_l\kappa_l$, we obtain

$$\frac{a(F,W,D)}{|I_G(F)|} = A_{V_r} - \Big(1 + \frac{\delta}{r}\Big)[\varpi] - \sum_l b_l [s^{-\kappa_l} p^* h_l]. \tag{D.22}$$

Replacing f by f_0, D by θD_0 with $D_0 := \sum_l b_l \operatorname{div}(h_l^0|_{W_0})$, $\widetilde{W}$ by $S \times \mathbb{A}^1_s$, and V_r by V in the computation above, we see that any prime divisor F_0 over W_0 with center 0 satisfies

$$\frac{a(F_0, W_0, \theta D_0)}{|I_G(F_0)|} = A_V - \Big(1 + \frac{\delta_\theta}{r}\Big)[\varpi] - \theta \sum_l b_l [H_l(X, Y, Z)], \tag{D.23}$$

because $s^{-\kappa_l} p^* h_l^0 = \varrho^* H_l(X, Y, Z)$.

Step 3. In this step we prove the lower bound $\alpha \geq \beta$ under the hypothesis of Claim D.12.

Claim D.12. *Let $\theta > 1$ be a real number. Then there exists an integer $r_1 \geq r_0$, depending only on the type, c, v, e_0, Γ, and θ, such that the following holds. Let $(W \ni 0, G, D)$ be data of kind (1) of Theorem 10.4, use the notation of **Step 1**, and assume that $r \geq r_1$ and that $(\mathbb{A}^2, \theta\pi^* D_S, \mathfrak{m}^{2\delta_\theta^+/c})$ is lc near 0. Then $(W, \theta D)$ and $(W_0, \theta \sum_l b_l \operatorname{div}(h_l|_{W_0}))$ are lc near 0, and $\alpha \geq \beta$.*

Proof. We first show that $(W_0, \theta D_0)$ is lc near 0. Let $W_0' \to W_0$ be a log resolution of $(W_0, \theta D_0)$ on which the preimage of 0 is a divisor and the union of the preimage of 0, the exceptional locus, and the strict transform of D_0 is a simple normal crossing divisor. By [KM98, Lemma 2.30 and Corollary 2.31(3)], $(W_0, \theta D_0)$ is lc near 0 if $a(E, W_0, \theta D_0) \geq 0$ for any prime divisor E on W_0' whose image in W_0 contains 0, and such an E whose image is $\{0\}$ has center 0. Let E be a prime divisor on W_0' whose image in W_0 contains 0 and is not $\{0\}$, and assume that $a(E, W_0, \theta D_0) < 0$. Blowing up a component of $E \cap P$ for a component P of the preimage of 0 meeting E, and then repeatedly the intersection of the last exceptional divisor with the strict transform of E, we obtain prime divisors E_k over W_0 with center 0 and

$$a(E_k, W_0, \theta D_0) = a(P, W_0, \theta D_0) + k a(E, W_0, \theta D_0), \qquad k = 1, 2, \dots,$$

by [KM98, Lemmas 2.29 and 2.30], since each center is a component of the intersection of exactly two components of the simple normal crossing divisor. Thus $a(E_k, W_0, \theta D_0) < 0$ for k large. Hence it suffices to show that $a(F_0, W_0, \theta D_0) \geq 0$ for any prime divisor F_0 over W_0 with center 0. Let $[\cdot]$ be as in **Step 2** for F_0. The surface $S = V \cap (\varpi = 0)$ is a normal Du Val complete intersection in $\mathbb{A}^4_{X,Y,Z,\varpi}$, so the hypotheses of Lemma 5.5 hold for V in place of X, the trivial group, $u = \varpi$, and $S_0 = S$. If $\delta_\theta \leq 0$, then $(S, \theta D_S)$ is lc near 0 by (D.19), and Lemma 5.5 with $\mathfrak{a} = \prod_l (H_l)^{\theta b_l}$ and $b = 1 + \delta_\theta/r \leq 1$ shows that the right-hand side of (D.23) is non-negative. If $\delta_\theta \geq 0$, then $(S, \theta D_S, I_0^{\delta_\theta/(cN_0)})$ is lc near 0 by (D.19). Since $[I_0] = cN_0 m_{\widetilde{F}}$, the right-hand side of (D.23) is equal to

$$\Big(A_V - [\varpi] - \theta \sum_l b_l [H_l] - \frac{\delta_\theta}{cN_0}[I_0]\Big) + \delta_\theta \Big(m_{\widetilde{F}} - \frac{[\varpi]}{r}\Big).$$

By Lemma 5.5 with $b = 1$ and

$$\mathfrak{a} = \prod_l (H_l)^{\theta b_l} \cdot I_0^{\delta_\theta/(cN_0)},$$

the first term is non-negative, and by (D.21), the second term is non-negative. Hence $(W_0, \theta D_0)$ is lc near 0.

By [Kaw07, Theorem], the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(f_0) + \theta \sum_l b_l \operatorname{div}(h_l^0)\Big)$$

is lc near 0. Let L be the maximum of the integers of Lemma 4.14 for $n = 4$, all numbers of functions at most $1 + 2/\gamma$, and the DCC set $\{1\} \cup \{\theta b : b \in \Gamma, b > 0\}$. Here lc near 0 is tested on $\operatorname{Spec} \mathbb{C}[[x, y, z, t]]$ [dFEM11, Proposition 2.11(i)]. By Construction D.8(1) and **Step 1**,

$f - f_0 \in (x,y,z,t)^L$ and $h_l - h_l^0 \in (x,y,z,t)^L$ once r is larger than a constant depending only on the type, c, v, Γ, and θ. Lemma 4.14 and [Kaw07, Theorem] imply that

$$(W, \theta D) \quad \text{and} \quad \Big(W_0, \theta \sum_l b_l \operatorname{div}(h_l|_{W_0})\Big) \quad \text{are lc near 0.} \tag{D.24}$$

Since W and W_0 are klt and $\theta > 1$, (W, D) and $(W_0, \sum_l b_l \operatorname{div}(h_l|_{W_0}))$ are klt near 0.

Let F be a prime divisor over W with center 0, and write $[\psi] := \operatorname{mult}_F \psi/|I_G(F)|$ for a regular function ψ near 0, which equals the bracket $[\psi]$ of **Step 2** for any prime divisor $\widetilde{F}$ over $\widetilde{W}$ dominating F. Since $\beta \le 2$ by Construction D.8(4), we may assume that $a(F,W,D)/|I_G(F)| \le 2$. By (D.24), $a(F,W,D) - (\theta-1)\operatorname{mult}_F D \ge 0$, so

$$[h_l] \le B_1 := \frac{2}{(\theta-1)\gamma} \quad \text{for any } l, \qquad \frac{a(F,W,0)}{|I_G(F)|} \le B_2 := \frac{2\theta}{\theta-1}. \tag{D.25}$$

Since S is klt, there exists a real number $\epsilon_0 > 0$, depending only on the type, such that $(S, \epsilon_0 \operatorname{div}(xyz|_S))$ is lc near 0. Since $W \cap (t = 0) = S$, [Kaw07, Theorem] implies that $(W, \epsilon_0 \operatorname{div}(xyz) + \operatorname{div}(t))$ is lc near 0, hence

$$\epsilon_0([x] + [y] + [z]) + [t] \le B_2. \tag{D.26}$$

For $i = 1,2,3$, the monomial $x_i^v t^{cq_i}$ is $\mathbf{T}$-invariant, so the character of $x_i^v t^{cq_i}$ is trivial on R, and, G being abelian, the G-invariant function $(x_i^v t^{cq_i})^{[G:R]}$ vanishes at the image of 0 in W/G, and mult_F of a G-invariant function is $|I_G(F)|$ times the order of the function along the prime divisor over W/G under F. Thus $[G:R](v[x_i] + cq_i[t]) \ge 1$, which gives, since $[G:R] \le e_0$,

$$v[x_i] + cq_i[t] \ge \frac{1}{e_0} \quad \text{for } i = 1,2,3. \tag{D.27}$$

Let $C_1 := B_1 + 1$. We distinguish two cases according to $[A] - [g_0]$.

Case 1: $[A] - [g_0] \le C_1$. Since $x^2 = -(g_0 + A)$ on W, either $[A] = 2[x]$, or $[g_0] = 2[x] < [A]$, or $[g_0] = [A] \le 2[x]$. In each case $[A] \le 2[x] + C_1$. Some term $a_{jk} t^k M_j$ of A satisfies $[t^k M_j] \le [A]$, and $vk = \ell_{jk} r - c(d - d_j) \ge r - cd$ by (D.11). By (D.26),

$$[t] \le \frac{v(2B_2/\epsilon_0 + C_1)}{r - cd}.$$

Hence, if r is larger than a constant depending only on the type, c, v, e_0, Γ, and θ, then $cq_i[t] \le 1/(2e_0)$, and by (D.27), $[x_i] \ge 1/(2ve_0)$ for $i = 1,2,3$.

Let $h_l^< := \sum_{\ell<0} h_{l,\ell}$. A monomial $x^{n_1} y^{n_2} z^{n_3} t^{n_4}$ of $h_{l,\ell}$ with $\ell < 0$ satisfies

$$c(q_x n_1 + q_y n_2 + q_z n_3) = v n_4 - \kappa_l - \ell r \ge r - |\kappa_l|,$$

so

$$[h_l^<] \ge \frac{r - |\kappa_l|}{cq_x} \cdot \frac{1}{2ve_0} > B_1 \ge [h_l]$$

for r larger than a constant depending only on the type, c, v, e_0, Γ, and θ. Thus $[h_l - h_l^<] = [h_l]$. By **Step 2**,

$$s^{-\kappa_l} p^*(h_l - h_l^<) = \varrho^* H_{r,l}, \qquad H_{r,l} := \sum_{\ell \ge 0} \varpi^\ell h_{l,\ell}(X,Y,Z,1) \in \mathbb{C}[X,Y,Z,\varpi],$$

where $H_{r,l}$ is $\widetilde{H}$-semi-invariant with $H_{r,l}(X,Y,Z,0) = H_l(X,Y,Z)$. With $[\cdot]$ as in **Step 2**, (D.22) becomes

$$\frac{a(F,W,D)}{|I_G(F)|} = A_{V_r} - \Big(1 + \frac{\delta}{r}\Big)[\varpi] - \sum_l b_l [H_{r,l}]. \tag{D.28}$$

By **Step 2**, V_r is normal near S, and (V_r, S) is plt near S by [KM98, Theorem 5.50(1)] with $B = 0$, so V_r is klt near 0. The group $\widetilde{H}$ acts diagonally on $\mathbb{A}^4_{X,Y,Z,\varpi}$ and faithfully on S. Thus

the hypotheses of Lemma 5.5 hold for V_r in place of X, the group $\widetilde{H}$, $H = \widetilde{H}$, $u = \varpi$, and $S_0 = S$. Let F' be as in **Step 2**, so that the right-hand side of (D.28) is

$$\frac{a\big(F', V_r, (1+\delta/r)S, \prod_l (H_{r,l})^{b_l}\big)}{|I_{\widetilde{H}}(F')|}.$$

If $\delta \le 0$, then (S, D_S) is lc near 0 by Construction D.8(2), and Lemma 5.5 with $\mathfrak{a} = \prod_l (H_{r,l})^{b_l}$ and $b = 1 + \delta/r \le 1$ shows that

$$\frac{a(F, W, D)}{|I_G(F)|} \ge \mathrm{mld}((S \ni 0, D_S)/\widetilde{H}) = \beta_- = \beta.$$

If $\delta \ge 0$, then $(S, D_S, I_0^{\delta/(cN_0)})$ is lc near 0 by Construction D.8(3), and, since $[I_0] = cN_0 m_{\widetilde{F}}$, by (D.28) we have

$$\frac{a(F, W, D)}{|I_G(F)|} = \frac{a\big(F', V_r, S, \prod_l (H_{r,l})^{b_l} \cdot I_0^{\delta/(cN_0)}\big)}{|I_{\widetilde{H}}(F')|} + \delta\Big(m_{\widetilde{F}} - \frac{[\varpi]}{r}\Big) \ge \beta_+ = \beta$$

by Lemma 5.5 with $b = 1$ and by (D.21). Here $\widetilde{H}$ is the image of G_S in $\mathrm{Aut}(S)$.

Case 2: $[A] - [g_0] > C_1$. Let $N := \lfloor B_1 |I_G(F)| \rfloor$. Then

$$\mathrm{mult}_F A - \mathrm{mult}_F g_0 > C_1 |I_G(F)| \ge B_1 |I_G(F)| + 1 > N.$$

Let F_0 be the prime divisor over W_0 with center 0 given by Lemma D.9 for N. By (D.25), $\mathrm{mult}_F h_l \le N$, so $\mathrm{mult}_{F_0} h_l = \mathrm{mult}_F h_l$ for any l. By Lemma D.9(2),

$$\frac{a(F, W, D)}{|I_G(F)|} = \frac{a(F_0, W_0, D')}{|I_G(F_0)|}, \qquad D' := \sum_l b_l \,\mathrm{div}(h_l|_{W_0}).$$

By Lemma D.4, $\pi \times \mathrm{id}_{\mathbb{A}^1} : \mathbb{A}^3_{\xi_1,\xi_2,t} \to W_0$ is étale in codimension one and induces an isomorphism $\mathbb{A}^3/\widehat{G} \cong W_0/G$. Let $\hat{h}_l := h_l(F_x, F_y, F_z, t)$ and $\widehat{B} := \sum_l b_l \,\mathrm{div}(\hat{h}_l)$. Then $(\mathbb{A}^3, \widehat{B})$ is klt near 0 [KM98, Proposition 5.20], and the crepant quotients of $(\mathbb{A}^3, \widehat{B})$ by $\widehat{G}$ and of (W_0, D') by G are equal. Let $\widehat{F}$ be a prime divisor over $\mathbb{A}^3$ with center 0 over F_0, so that

$$\frac{a(F_0, W_0, D')}{|I_G(F_0)|} = \frac{a(\widehat{F}, \mathbb{A}^3, \widehat{B})}{|I_{\widehat{G}}(\widehat{F})|}.$$

The group $I_{\widehat{G}}(\widehat{F})$ acts faithfully on the conormal space of $\widehat{F}$ at a general point, so it is cyclic. By Lemma D.4, $\widehat{G} \cap \widehat{\mathbf{T}}$ is central in $\widehat{G}$. Hence the subgroup $\widehat{G}_F$ generated by $I_{\widehat{G}}(\widehat{F})$ and $\widehat{G} \cap \widehat{\mathbf{T}}$ is abelian, and $I_{\widehat{G}_F}(\widehat{F}) = I_{\widehat{G}}(\widehat{F})$. After a linear change of the coordinates ξ_1, ξ_2, which does not change the action of $\widehat{\mathbf{T}}$, $\widehat{G}_F$ acts diagonally on $\mathbb{A}^3$. By Lemma D.4,

$$|\widehat{G}_F \cap \widehat{\mathbf{T}}| = \frac{2r}{\gcd(c,2)}, \qquad [\widehat{G}_F : \widehat{G}_F \cap \widehat{\mathbf{T}}] \le \frac{\gcd(c,2)|\Pi|}{2}[G : R] \le |\Pi| e_0.$$

The polynomial $\hat{h}_l^0 := h_l^0(F_x, F_y, F_z, t)$ is $\widehat{\mathbf{T}}$-homogeneous, and, for $\ell \ne 0$, the polynomial $h_{l,\ell}(F_x, F_y, F_z, t)$ is $\widehat{\mathbf{T}}$-homogeneous of a different weight, with all monomials of total degree at least $(r - \kappa_{\max})/\max\{cq_x, v\}$ by **Step 1**. Since h_l^0 has x-degree at most one, $h_l^0|_{W_0} \ne 0$. Let k_l be the t-degree of a monomial of least degree of h_l^0, so $k_l < M_{\deg}$. The coefficient of t^{k_l} in h_l^0 is a non-zero polynomial of x-degree at most one, hence non-zero on S, so $\hat{h}_l^0$ has a non-zero term of total degree at most $2(vM_{\deg} + |\kappa_l|)/c + M_{\deg}$. Thus, for r larger than a constant depending only on the type, c, v, and Γ, the principal block of $\hat{h}_l$ for $\widehat{\mathbf{T}}$ in the sense of Set-up 5.1 is $\hat{h}_l^0$.

The vectors $(\bar{c}, \bar{c}, -\bar{v})$ and $(-\bar{c}, -\bar{c}, \bar{v})$ both define the circle $\widehat{\mathbf{T}}$ of Lemma D.4. As in [Liu26, Theorem 16.21], we take all weights and adjusted masses below with respect to $(\bar{c}, \bar{c}, -\bar{v})$. By [Liu26, Theorem 16.21(2)], applied to $\mathbb{A}^3_{\xi_1,\xi_2,t}$, the circle $\mathbf{T}_{(\bar{c},\bar{c},-\bar{v})} = \widehat{\mathbf{T}}$ with $\bar{c} = c/\gcd(c,2)$ and $\bar{v} = 2v/\gcd(c,2)$, for which $\gcd(\bar{c}, \bar{v}) = 1$, the index bound $|\Pi|e_0$, the DCC set Γ, the finite

diagonal group $\widehat{G}_F$, and the klt pair $(\mathbb{A}^3, \widehat{B})$, for r larger than a constant depending only on the type, c, v, e_0, and Γ, we have

$$\mathrm{mld}((\mathbb{A}^3 \ni 0, \widehat{B})/\widehat{G}_F) \geq \hat{\beta},$$

where $\hat{\beta}$ is the number of [Liu26, Construction-Theorem 16.9] for the principal blocks $\hat{h}^0_l$ and the group $\widehat{G}_F$. We identify $\hat{\beta}$ by the final equality in [Liu26, Construction-Theorem 16.9], applied with $(\bar{c}, \bar{c}, \bar{v})$ in place of the integers (p, q, r) there and $H' = \widehat{G}_F/(\widehat{G}_F \cap \widehat{\mathbf{T}})$, the group denoted H there. Let τ' be the number denoted τ there, the adjusted mass of $\widehat{B}$ along $\mathbf{T}_{(\bar{c},\bar{c},-\bar{v})}$. Then

$$\tau' = \bar{v} - 2\bar{c} - \frac{2}{\gcd(c,2)}\tau = -\frac{2}{\gcd(c,2)}\delta,$$

so the number τ'^- denoted τ^- there is $\frac{2}{\gcd(c,2)}\delta^+$. Let σ be the variable of the cover there, $\sigma^{\bar{v}} = t$, and let $\xi'_1 := \xi_1\sigma^{\bar{c}}$ and $\xi'_2 := \xi_2\sigma^{\bar{c}}$, which are the coordinates denoted u and v there. Since F_x, F_y, and F_z are homogeneous of degrees $2q_x$, $2q_y$, and $2q_z$, and h^0_l has $\mathbf{T}$-weight κ_l,

$$\hat{h}^0_l(\xi_1, \xi_2, \sigma^{\bar{v}}) = \sigma^{2\kappa_l/\gcd(c,2)}\widehat{H}_l(\xi'_1, \xi'_2),$$

so the divisor denoted C there is π^*D_S in the coordinates (ξ'_1, ξ'_2). An element $(\hat{g}, \iota)$ of the group of the cover there, with $(\hat{g}, g) \in \widehat{G}_F$ and $\iota^{\bar{v}} = \chi_t(g)$, acts on (ξ'_1, ξ'_2) by $\iota^{\bar{c}}\hat{g}$. Let $\zeta := \iota^{2/\gcd(c,2)}$. Then $\zeta^v = \chi_t(g)$, so $(g, \zeta) \in G_S$, and

$$F_x\big(\iota^{\bar{c}}\hat{g}(\xi_1, \xi_2)\big) = \chi_x(g)\zeta^{cq_x}F_x(\xi_1, \xi_2),$$

and likewise for F_y and F_z. Thus $\iota^{\bar{c}}\hat{g}$ is the lift through π of an automorphism of S given by (g, ζ), and the image of the group of the cover there in $\mathrm{GL}_2(\mathbb{C})$ is the subgroup $\widehat{G}_{S,F} \subset \widehat{G}_S$ of the lifts $\iota^{\bar{c}}\hat{g}$ with $(\hat{g}, g) \in \widehat{G}_F$ and $\iota^{\bar{v}} = \chi_t(g)$. The ideal denoted $\mathfrak{b}$ there pulls back to $(\xi_1'^{\bar{c}\bar{v}|H'|}, \xi_2'^{\bar{c}\bar{v}|H'|})$, whose integral closure is $(\xi'_1, \xi'_2)^{\bar{c}\bar{v}|H'|}$. Since

$$(\xi'_1, \xi'_2)^{\tau'^-/\bar{c}} = (\xi'_1, \xi'_2)^{2\delta^+/c},$$

the $\mathbb{R}$-ideals $\mathfrak{b}^{\tau'^-/(\bar{c}^2\bar{v}|H'|)}$ and $(\xi'_1, \xi'_2)^{2\delta^+/c}$ give equal log discrepancies along any prime divisor. Since $\delta_1 = \delta$, $\hat{\beta}$ is the minimal log discrepancy of the crepant quotient of $(\mathbb{A}^2 \ni 0, \pi^*D_S, \mathfrak{m}^{2\delta_1^+/c})$ by $\widehat{G}_{S,F}$. For any prime divisor Q over $\mathbb{A}^2$ with center 0, $I_{\widehat{G}_{S,F}}(Q) \subset I_{\widehat{G}_S}(Q)$, so

$$\frac{a(Q, \mathbb{A}^2, \pi^*D_S, \mathfrak{m}^{2\delta_1^+/c})}{|I_{\widehat{G}_{S,F}}(Q)|} = [I_{\widehat{G}_S}(Q) : I_{\widehat{G}_{S,F}}(Q)] \cdot \frac{a(Q, \mathbb{A}^2, \pi^*D_S, \mathfrak{m}^{2\delta_1^+/c})}{|I_{\widehat{G}_S}(Q)|} \geq \beta$$

by (D.20). Hence $\hat{\beta} \geq \beta$, and

$$\frac{a(F, W, D)}{|I_G(F)|} = \frac{a(\widehat{F}, \mathbb{A}^3, \widehat{B})}{|I_{\widehat{G}_F}(\widehat{F})|} \geq \mathrm{mld}((\mathbb{A}^3 \ni 0, \widehat{B})/\widehat{G}_F) \geq \beta.$$

In both cases, $a(F, W, D)/|I_G(F)| \geq \beta$. Since F is arbitrary, $\alpha \geq \beta$. □

Proof of Theorem 10.4 continued.

Step 4. In this step we take a compatible generic limit of the reference pairs, and show that the limit is lc near 0 and has no lc place with center 0.

Suppose that there exists a sequence of data $(W_i \ni 0, G_i, D_i)$ of kind (1) of Theorem 10.4 with $r_i := |G_i \cap \mathbf{T}| \to +\infty$ such that the numbers $\alpha_i := \mathrm{mld}((W_i \ni 0, D_i)/G_i)$ satisfy $0 < \alpha_1 < \alpha_2 < \cdots$. We add the index i to the notation of **Steps 1**–**3**, write $D_i = \sum_l b_{i,l}\,\mathrm{div}(h_{i,l})$, and let $\alpha_\infty := \lim_i \alpha_i$. By **Step 1**, fewer than $2/\gamma$ of the $h_{i,l}$ occur and $|\kappa_{i,l}| \leq \kappa_{\max}$ (Lemma D.7). By Construction D.8(4), the numbers β_i belong to a set satisfying the ACC. Since Γ is a DCC set, passing to a subsequence, we may assume that $r_i \geq r_0$, that the number m of the $h_{i,l}$ and the numbers $\kappa_l := \kappa_{i,l}$ do not depend on i, that $b_{i,l}$ is non-decreasing with limit $b^*_l > 0$ for any l, and

that β_i is non-increasing with limit β_∞. Then $\tau_i \to \tau_* := \sum_l b_l^* \kappa_l$ and $\delta_i \to \delta_* := \tau_* + c - v$. By Proposition D.11, $\alpha_i \le \beta_i + C_0/r_i$, hence

$$\beta_i \ge \beta_\infty \ge \alpha_\infty > \alpha_i > 0 \quad \text{for any } i. \tag{D.29}$$

If $m = 0$, then $D_i = 0$ near 0, so $\delta_{i,\theta} = c - v$ for any θ, and the pair

$$(\mathbb{A}^2, \theta\pi^* D_{S,i}, \mathfrak{m}^{2\delta_{i,\theta}^+/c}) = (\mathbb{A}^2, \mathfrak{m}^{2(c-v)^+/c})$$

is lc near 0 for any θ, since $2(c-v)^+/c < 2$. Then Claim D.12 with $\theta = 2$ and (D.29) give

$$\alpha_i \ge \beta_i \ge \beta_\infty \ge \alpha_\infty > \alpha_i$$

for i large, a contradiction. Thus $m \ge 1$.

By Construction D.8(2) and Lemma D.3, applied to the diagonal action of $G_{S,i}$ on $\mathbb{A}^3_{x,y,z}$ (so that $\widehat{G}_{S,i}$ is the group $\widehat{Q}_1$ there), Π is normal in $\widehat{G}_{S,i}$, $\widehat{G}_{S,i}/\Pi$ is the faithful quotient of $G_{S,i}$ acting on S, $|\widehat{G}_{S,i}| = |\Pi| v [G_i : R_i] \le |\Pi| v e_0$, and Π is the kernel of the action of $\widehat{G}_{S,i}$ on S. Thus $\widehat{G}_{S,i}$ is contained in the normalizer of Π in $\mathrm{GL}_2(\mathbb{C})$, which is contained in $\mathbb{C}^* \cdot \Pi^+$, where $\Pi^+ \subset \mathrm{SL}_2(\mathbb{C})$ is the preimage of the normalizer of the image of Π in $\mathrm{PGL}_2(\mathbb{C})$. The image $\bar{\Pi}$ of Π in $\mathrm{PGL}_2(\mathbb{C})$ is finite and non-cyclic (dihedral of order $2(k-2) \ge 4$ for type D_k, and A_4, S_4, A_5 for E_6, E_7, E_8, by Klein's classification of the finite subgroups of $\mathrm{PGL}_2(\mathbb{C})$). Its normalizer N is a Zariski closed subgroup of $\mathrm{PGL}_2(\mathbb{C})$, and the identity component N^0 centralizes $\bar{\Pi}$ because $\mathrm{Aut}(\bar{\Pi})$ is finite. If $N^0 \ne \{1\}$, then N^0 contains a subgroup U isomorphic to $\mathbb{G}_m$ or to $\mathbb{G}_a$ (a Borel subgroup of N^0 is non-trivial by [Bor91, Theorem 11.10], and it contains such a subgroup by [Bor91, Theorems 10.6(2)(4) and 10.9]). The centralizer of U in $\mathrm{PGL}_2(\mathbb{C})$ is U (a maximal torus, resp. a maximal unipotent subgroup), so $\bar{\Pi} \subset U$ is cyclic, a contradiction. Thus N is finite, and so is its preimage Π^+ in $\mathrm{SL}_2(\mathbb{C})$. An element $\lambda g'$ of $\widehat{G}_{S,i}$ with $\lambda \in \mathbb{C}^*$ and $g' \in \Pi^+$ satisfies $\lambda^n g'^n = 1$ with $n := |\widehat{G}_{S,i}| \le |\Pi| v e_0$, so $\lambda^n \in \Pi^+$ and $\lambda^{n|\Pi^+|} = 1$. Thus $\widehat{G}_{S,i}$ is contained in the finite group generated by Π^+ and the roots of unity of order dividing $(|\Pi| v e_0)! |\Pi^+|$, which has finitely many subgroups. Hence $\widehat{G}_{S,i}$ ranges over finitely many subgroups of $\mathrm{GL}_2(\mathbb{C})$, and passing to a subsequence, we may assume that $\widehat{G}_S := \widehat{G}_{S,i}$ does not depend on i and that the character of the $\widehat{G}_S$-semi-invariant polynomial $\widehat{H}_{i,l}$ depends only on l. Since $(\mathbb{A}^2, \pi^* D_{S,i})$ is lc near 0, $\sum_l b_{i,l} \operatorname{mult}_0 \widehat{H}_{i,l} \le 2$, so $\operatorname{mult}_0 \widehat{H}_{i,l} \le 2/\gamma$.

We apply Construction-Proposition 4.1(a) to $\mathbb{A}^2_{\xi_1,\xi_2}$, the finite group $\widehat{G}_S \subset \mathrm{GL}_2(\mathbb{C})$, and the tuples $(\widehat{H}_{i,1}, \dots, \widehat{H}_{i,m})$. Passing to a subsequence, we obtain an algebraically closed field $k \supset \mathbb{C}$ and a compatible generic limit $(\widehat{H}_{\infty,l})_l$, where the $\widehat{H}_{\infty,l} \in k[[\xi_1, \xi_2]]$ are non-zero $\widehat{G}_S$-semi-invariant power series. A compatible generic limit of the sequence is also a compatible generic limit of any infinite subsequence. Let $\mathcal{A}_k := \operatorname{Spec} k[[\xi_1, \xi_2]]$, and let

$$\Delta_i := \sum_l b_l^* \operatorname{div}(\widehat{H}_{i,l}), \quad \Delta_\infty := \sum_l b_l^* \operatorname{div}(\widehat{H}_{\infty,l}), \quad \lambda_i := \frac{2\delta_i^+}{c}, \quad \lambda_* := \frac{2\delta_*^+}{c}.$$

Then Δ_i and Δ_∞ are the boundaries C_i and C_∞ of Theorem 4.2 with the coefficients b_l^*. Then $\lambda_i \to \lambda_*$, and $(\mathbb{A}^2, \pi^* D_{S,i}, \mathfrak{m}^{\lambda_i})$ is lc near 0 by Construction D.8(2)(3) and (D.19).

Let E_0 be the exceptional curve of the blow-up of $0 \in \mathbb{A}^2$. By (D.20) and (D.29),

$$2 - \lambda_i \ge a(E_0, \mathbb{A}^2, \pi^* D_{S,i}, \mathfrak{m}^{\lambda_i}) \ge |I_{\widehat{G}_S}(E_0)| \beta_i \ge \beta_\infty,$$

so $\lambda_* \le 2 - \beta_\infty < 2$. Thus $(\mathbb{A}^2, \mathfrak{m}^\lambda)$ is klt near 0 for any $\lambda \le \lambda_*$, and the hypotheses of Theorem 4.2 hold for $\mathfrak{c} = \mathfrak{m}^\lambda$.

We show that $(\mathcal{A}_k, \Delta_\infty, \mathfrak{m}^{\lambda_*})$ is lc near 0. Let $\epsilon \in (0,1)$. For i sufficiently large, $(1-\epsilon) b_l^* \le b_{i,l}$ for any l and $(1-\epsilon)\lambda_* \le \lambda_i$, so the pair

$$(\mathbb{A}^2, (1-\epsilon)\Delta_i, \mathfrak{m}^{(1-\epsilon)\lambda_*})$$

is lc near 0. By Theorem 4.2(1), applied to the tail of the sequence with $\mathfrak{c} = \mathfrak{m}^{(1-\epsilon)\lambda_*}$ and the real number $1-\epsilon$, the pair

$$(\mathcal{A}_k, (1-\epsilon)\Delta_\infty, \mathfrak{m}^{(1-\epsilon)\lambda_*})$$

is lc near 0. Since log discrepancies are affine functions of ϵ, letting $\epsilon \to 0$ we obtain the lc property of $(\mathcal{A}_k, \Delta_\infty, \mathfrak{m}^{\lambda_*})$ near 0.

We show that $a(E, \mathcal{A}_k, \Delta_\infty, \mathfrak{m}^{\lambda_*}) > 0$ for any prime divisor E over $\mathcal{A}_k$ with center 0. Suppose that $a(E, \mathcal{A}_k, \Delta_\infty, \mathfrak{m}^{\lambda_*}) = 0$, and let $a_E := a(E, \mathcal{A}_k, 0)$ and $m_E := \operatorname{mult}_E \mathfrak{m} \geq 1$. Then $b_l^* \operatorname{mult}_E \widehat{H}_{\infty,l} \leq a_E$ for any l. Let d' be an integer with $d' > a_E / \min_l b_l^*$ such that the conclusion of Theorem 4.2(3) with $\mathfrak{c} = \mathfrak{m}$ holds for d' in place of d (the pair $(\mathbb{A}^2, \mathfrak{m})$ is klt near 0), let $Z_{d'}^\circ \subset Z_{d'}$ be the open subset given there, and let $\varphi^{(d')}$ denote the d'-jet of a power series φ. Since $\varphi - \varphi^{(d')} \in \mathfrak{m}^{d'}$ and $\operatorname{mult}_E \widehat{H}_{\infty,l} < d' \leq d' m_E$, we have

$$\operatorname{mult}_E \widehat{H}_{\infty,l}^{(d')} = \operatorname{mult}_E \widehat{H}_{\infty,l},$$

so

$$a(E, \mathcal{A}_k, \Delta_\infty^{(d')}, \mathfrak{m}^{\lambda_*}) = 0 \qquad \text{for} \qquad \Delta_\infty^{(d')} := \sum_l b_l^* \operatorname{div}(\widehat{H}_{\infty,l}^{(d')}).$$

The d'-jet of $(\widehat{H}_{\infty,l})_l$ maps to the generic point of $Z_{d'}$, so it is a point of $Z_{d'}^\circ$. Let $Y_\infty \to \mathbb{A}_k^2$ be the fiber over it of the $\widehat{G}_S$-equivariant log resolution of Theorem 4.2(3), let Υ be the snc divisor on Y_∞ formed by the exceptional curves and the strict transforms of the zero loci of the $\widehat{H}_{\infty,l}^{(d')}$, and let o_E be the center of E on Y_∞. Since the ideals $\mathfrak{m}\mathcal{O}_{Y_\infty}$ and $\widehat{H}_{\infty,l}^{(d')}\mathcal{O}_{Y_\infty}$ are invertible and supported on Υ near the fiber over 0, there is an $\mathbb{R}$-divisor B_{Y_∞} supported on Υ with

$$a(P, \mathbb{A}_k^2, \Delta_\infty^{(d')}, \mathfrak{m}^{\lambda_*}) = a(P, Y_\infty, B_{Y_\infty})$$

for any prime divisor P over Y_∞ whose center maps to 0. The coefficient in B_{Y_∞} of a component Q of Υ is $1 - a(Q, \mathbb{A}_k^2, \Delta_\infty^{(d')}, \mathfrak{m}^{\lambda_*})$.

By [KM98, Lemma 2.45], there are an integer $n_E \geq 0$ and blow-ups $Y^{j+1} \to Y^j$ for $0 \leq j < n_E$, with $Y^0 := Y_\infty$, such that $Y^{j+1} \to Y^j$ is the blow-up of the center o^j of E on Y^j, o^j is a closed point, and the center of E on Y^{n_E} is a curve. For $0 \leq j \leq n_E$, let Υ^j be the reduced divisor on Y^j formed by the strict transforms of the components of Υ and the exceptional curves of $Y^j \to Y_\infty$. Let Θ be an exceptional curve of Y_∞ through o_E, or the exceptional curve of $Y^j \to Y^{j-1}$ for some $1 \leq j \leq n_E$. The center of E on Y^0 in the first case, and on Y^j in the second case, is contained in Θ, so $\operatorname{mult}_\Theta \psi \leq \operatorname{mult}_E \psi$ for any $\psi \in k[[\xi_1, \xi_2]]$, and Θ maps to 0, so $\operatorname{mult}_\Theta \mathfrak{m} \geq 1$. Hence

$$\operatorname{mult}_\Theta \widehat{H}_{\infty,l}^{(d')} \leq \operatorname{mult}_E \widehat{H}_{\infty,l}^{(d')} < d' \leq d' \operatorname{mult}_\Theta \mathfrak{m},$$

so

$$\operatorname{mult}_\Theta \widehat{H}_{\infty,l}^{(d')} = \operatorname{mult}_\Theta \widehat{H}_{\infty,l}$$

for any l, and

$$a(\Theta, \mathbb{A}_k^2, \Delta_\infty^{(d')}, \mathfrak{m}^{\lambda_*}) = a(\Theta, \mathcal{A}_k, \Delta_\infty, \mathfrak{m}^{\lambda_*}) \geq 0,$$

since $(\mathcal{A}_k, \Delta_\infty, \mathfrak{m}^{\lambda_*})$ is lc near 0. Thus the coefficient of Θ in the crepant transform of $(\mathbb{A}_k^2, \Delta_\infty^{(d')}, \mathfrak{m}^{\lambda_*})$ is at most one.

Let $\mathcal{B}$ be an $\mathbb{R}$-divisor with snc support on a smooth surface S', and $o' \in S'$ a closed point such that any component of $\mathcal{B}$ through o' has non-positive coefficient, except at most one component, whose coefficient is at most one. Then $a(P', S', \mathcal{B}) \geq 1$ for any prime divisor P' over S' with center o'. The exceptional curve Ξ' of the blow-up of o' has log discrepancy $2 - \operatorname{mult}_{o'} \mathcal{B} \geq 1$, hence non-positive coefficient in the crepant transform of $\mathcal{B}$, and any closed point of Ξ' is contained in at most one strict transform of a component of $\mathcal{B}$ through o', so the hypothesis holds at any closed point of Ξ'. Since P' is the exceptional curve of the last of finitely many blow-ups of the centers of P' [KM98, Lemma 2.45], induction on their number gives $a(P', S', \mathcal{B}) \geq 1$.

Since o_E maps to 0, it is contained in an exceptional curve of Y_∞. Suppose that $n_E \geq 1$ and that, for some $0 \leq j < n_E$, the point o^j is contained in only one component Θ^j of Υ^j. Then Θ^j is an exceptional curve of Y_∞ through o_E if $j = 0$, and the exceptional curve of $Y^j \to Y^{j-1}$ if $j \geq 1$, so the coefficient of Θ^j in the crepant transform of $(\mathbb{A}^2_k, \Delta^{(d')}_\infty, \mathfrak{m}^{\lambda_*})$ is at most one, and the previous paragraph, applied to the crepant transform of $(\mathbb{A}^2_k, \Delta^{(d')}_\infty, \mathfrak{m}^{\lambda_*})$ on Y^j at o^j, gives

$$a(E, \mathbb{A}^2_k, \Delta^{(d')}_\infty, \mathfrak{m}^{\lambda_*}) \geq 1,$$

a contradiction. Hence either $n_E = 0$ and E is a component of Υ over 0, or $n_E \geq 1$ and, for any $0 \leq j < n_E$, the point o^j is the intersection point of two components of Υ^j. In the second case, let Υ_1 and Υ_2 be the components of Υ through o_E, with local equations v_1 and v_2 at o_E. The blow-ups $Y^{j+1} \to Y^j$ are toric in v_1, v_2, and E is the monomial valuation with positive weights (w_1, w_2) in v_1, v_2. In both cases E is the monomial valuation with positive weights along a stratum of Υ over 0.

Since the log discrepancy over Y_∞ of such a monomial valuation is the sum of the weights [JM12, Proposition 5.1 and (5.1)], the relative canonical divisor of $Y_\infty \to \mathbb{A}^2_k$ is supported on Υ, and near o_E the ideals $\mathfrak{m}\mathcal{O}_{Y_\infty}$ and $\widehat{H}^{(d')}_{\infty,l}\mathcal{O}_{Y_\infty}$ are generated by monomials in local equations of the components of Υ, the numbers $a(E, \mathbb{A}^2_k, 0)$, $\operatorname{mult}_E \mathfrak{m}$, and $\operatorname{mult}_E \widehat{H}^{(d')}_{\infty,l}$ are the linear combinations with the weights of E of the corresponding numbers of the components of the stratum. By Theorem 4.2(3), the incidences of the components of the fibers of the log resolution, whether they map to 0, their log discrepancies over $\mathbb{A}^2$, and the multiplicities along them of $\mathfrak{m}$ and of the d'-jets, also along the strict transforms of the zero loci of the jets, do not depend on the point of $Z^\circ_{d'}$. Hence the weights of E define, for all but finitely many i, a prime divisor E_i over $\mathbb{A}^2$ with center 0 such that

$$a(E_i, \mathbb{A}^2, 0) = a_E, \qquad \operatorname{mult}_{E_i} \mathfrak{m} = m_E, \qquad \operatorname{mult}_{E_i} \widehat{H}^{(d')}_{i,l} = \operatorname{mult}_E \widehat{H}^{(d')}_{\infty,l} \text{ for any } l.$$

Since $\widehat{H}_{i,l} - \widehat{H}^{(d')}_{i,l} \in \mathfrak{m}^{d'}$ and $\operatorname{mult}_{E_i} \widehat{H}^{(d')}_{i,l} < d' \leq d' m_E$, we obtain

$$\operatorname{mult}_{E_i} \widehat{H}_{i,l} = \operatorname{mult}_{E_i} \widehat{H}^{(d')}_{i,l} = \operatorname{mult}_E \widehat{H}_{\infty,l}.$$

Therefore

$$\begin{aligned} a(E_i, \mathbb{A}^2, \pi^* D_{S,i}, \mathfrak{m}^{\lambda_i}) &= \sum_l (b^*_l - b_{i,l}) \operatorname{mult}_E \widehat{H}_{\infty,l} + (\lambda_* - \lambda_i) m_E \\ &\leq \frac{a_E}{\gamma} \sum_l (b^*_l - b_{i,l}) + |\lambda_* - \lambda_i| m_E, \end{aligned}$$

which tends to 0. On the other hand, by (D.20) and (D.29), the left-hand side is at least $|I_{\widehat{G}_S}(E_i)|\beta_i \geq \beta_\infty > 0$, a contradiction.

Step 5. In this step we show that $(\mathcal{A}_k, \Delta_\infty, \mathfrak{m}^{\lambda_*})$ is klt near 0.

Suppose not. By **Step 4**, $(\mathcal{A}_k, \Delta_\infty, \mathfrak{m}^{\lambda_*})$ is lc near 0 and has no lc place with center 0. Hence there exists a prime divisor Σ on $\mathcal{A}_k$ through 0 with coefficient one in Δ_∞. By **Step 4**, $a(E_0, \mathcal{A}_k, \Delta_\infty, \mathfrak{m}^{\lambda_*}) > 0$, that is,

$$\operatorname{mult}_0 \Delta_\infty + \lambda_* < 2. \tag{D.30}$$

Since $\lambda_* \geq 0$ and $\operatorname{mult}_0 \Sigma \geq 1$, (D.30) shows that Σ is the only prime divisor through 0 with coefficient one in Δ_∞, and that $\operatorname{mult}_0 \Sigma = 1$. Since the $\widehat{H}_{\infty,l}$ are $\widehat{G}_S$-semi-invariant, Δ_∞ is $\widehat{G}_S$-invariant, hence so is Σ. In particular, Σ is Π-invariant, because $\Pi \subset \widehat{G}_S$. Since Σ is smooth at 0, the tangent line of Σ at 0 is a Π-invariant line of k^2. The Π-invariant lines form a closed subset $\Lambda \subset \mathbb{P}^1$ defined over $\mathbb{C}$. Since Λ has a k-point, it has a $\mathbb{C}$-point. This contradicts Lemma D.4, by which no line of $\mathbb{C}^2$ is Π-invariant. Thus $(\mathcal{A}_k, \Delta_\infty, \mathfrak{m}^{\lambda_*})$ is klt near 0.

Step 6. In this step we obtain a margin and finish the proof of Theorem 10.4 for data of kind (1).

Klt near 0 is tested on a log resolution of $\mathcal{A}_k$, Δ_∞, and $\mathfrak{m}$, whose finitely many log discrepancies are affine functions of the coefficient of Δ_∞ and the exponent of $\mathfrak{m}$. By **Step 5**, there exist real numbers $\theta_1 > 1$ and $\eta > 0$ such that

$$(\mathcal{A}_k, \theta_1\Delta_\infty, \mathfrak{m}^{\lambda_*+\eta}) \text{ is klt near } 0.$$

In particular, $\lambda_* + \eta < 2$. By Theorem 4.2(1) with $\mathfrak{c} = \mathfrak{m}^{\lambda_*+\eta}$, after passing to a subsequence, $(\mathbb{A}^2, \theta_1\Delta_i, \mathfrak{m}^{\lambda_*+\eta})$ is lc near 0 for any i. Let $\theta \in (1, \theta_1]$ be a real number with $(\theta-1)|\tau_*| < c\eta/4$. Since

$$\delta_{i,\theta} = \theta\tau_i + c - v \to \delta_* + (\theta-1)\tau_* < \delta_*^+ + \frac{c\eta}{4},$$

for i sufficiently large we have $2\delta_{i,\theta}^+/c \le \lambda_* + \eta$. Moreover $\theta b_{i,l} \le \theta_1 b_l^*$ for any l. Hence, for i sufficiently large, the pair

$$(\mathbb{A}^2, \theta\pi^* D_{S,i}, \mathfrak{m}^{2\delta_{i,\theta}^+/c})$$

is lc near 0. Let r_1 be the integer of Claim D.12 for θ. For i sufficiently large, $r_i \ge r_1$, and Claim D.12 and (D.29) imply

$$\alpha_i \ge \beta_i \ge \beta_\infty \ge \alpha_\infty > \alpha_i,$$

a contradiction. This proves Theorem 10.4 for data of kind (1).

Step 7. In this step we reduce Theorem 10.4 for data of kind (2) to data of kind (1).

Let $(W \ni o, G, D)$, with $D = \sum_l b_l \operatorname{div}(h_l)$, be data of kind (2) of Theorem 10.4 with $r := |G \cap \mathbf{T}| > v + cd$, and with the G-equivariant isomorphism $\widehat{\mathcal{O}}_{W,o} \cong \mathfrak{R}/(\hat{f})$, where $\mathfrak{R} := \mathbb{C}[[x, y, z, t]]$. We construct data $(W_1 \ni 0, G, D_1)$ of kind (1) of Theorem 10.4, with the given action of G on x, y, z, t and the labels b_l, such that

$$\operatorname{mld}((W_1 \ni 0, D_1)/G) = \operatorname{mld}((W \ni o, D)/G). \tag{D.31}$$

Since $G \cap \mathbf{T}$ depends only on the characters of x, y, z, t, and a sequence of data of kind (2) with $|G \cap \mathbf{T}| \to +\infty$ satisfies $|G \cap \mathbf{T}| > v + cd$ for all but finitely many members, Theorem 10.4 for data of kind (1) then excludes a sequence of data of kind (2) as in the theorem.

Discarding the h_l with $b_l = 0$, which does not change D, we may assume that $b_l > 0$ for any l. Since the isomorphism $\widehat{\mathcal{O}}_{W,o} \cong \mathfrak{R}/(\hat{f})$ is G-equivariant for the diagonal action of G on $\mathfrak{R}$, G preserves the ideal $(\hat{f})$, so for any $g \in G$ we have $g^*\hat{f} = w_g\hat{f}$ for a unit w_g of $\mathfrak{R}$. Since $\hat{f}(x, y, z, 0) = f_0$, the series $\hat{f}$ has neither a constant term nor a term x, so comparing the coefficients of x^2 gives $w_g(0) = \chi_x(g)^2$. Hence the sum $\hat{f}'$ of the terms of $\hat{f}$ whose monomials have the character χ_x^2, which is $|G|^{-1}\sum_{g\in G}\chi_x(g)^{-2}g^*\hat{f}$, is the product of $\hat{f}$ and a unit with constant term 1. Moreover G preserves the ideal $(\hat{f}, t)/(t) = (f_0)$ of $\mathbb{C}[[x, y, z]]$, so for any $g \in G$, g^*f_0 is the product of f_0 and a unit, which is a constant because g^*f_0 is weighted homogeneous of the weighted degree of f_0. Comparing the coefficients of x^2, f_0 is G-semi-invariant with the character χ_x^2, so $\hat{f}'(x, y, z, 0) = f_0$. Replacing $\hat{f}$ by $\hat{f}'$, we may assume that $\hat{f}$ is G-semi-invariant with the character χ_x^2. The quadratic part of $\hat{f}$ is still x^2, and $\hat{f}(x, y, z, 0) = f_0$.

The ideal $(\partial_y g_0, \partial_z g_0)$ is weighted homogeneous, so $\mathbb{C}[y, z]/(\partial_y g_0, \partial_z g_0)$ has a basis $(M_j)_j$ consisting of monomials, and Theorem 10.4 for data of kind (1) holds with the basis $(M_j)_j$. Since $\hat{f}$ is G-semi-invariant and $\hat{f}(x, y, z, 0) = f_0$, Lemma D.1 yields a G-equivariant automorphism ψ_1 of $\mathfrak{R}$ with $\psi_1(t) = t$, a G-invariant unit u_1, and series $\hat{a}_j \in t\mathbb{C}[[t]]$ such that

$$\psi_1(\hat{f}) = u_1\Big(f_0 + \sum_j \hat{a}_j(t)M_j\Big),$$

where each non-zero $\hat{a}_j(t)M_j$ is G-semi-invariant with the character of $\hat{f}$.

Since $\hat{f}(x, y, z, 0) = f_0$, the only monomial of degree one that may occur in $\hat{f}$ is t. For a primitive r-th root of unity λ, the element $\mathbf{T}(\lambda)$ of $G \cap \mathbf{T}$ multiplies t and x^2 by λ^v and $\lambda^{-2cq_x} = \lambda^{-cd}$, which are different since $0 < v + cd < r$. Thus t does not have the character χ_x^2 of $\hat{f}$, so $\hat{f} \in (x, y, z, t)^2$, and the embedding dimension of W at o is four. Choose G-semi-invariant

regular functions $x_1, \dots, x_4$ near o with the characters of x, y, z, t whose classes in $\mathfrak{m}_o/\mathfrak{m}_o^2$ are the images of x, y, z, t. Possibly shrinking W to a neighborhood of o, there is a G-equivariant closed embedding of W into a smooth fourfold germ $(Y \ni o)$ with G-eigenparameters $x_1, \dots, x_4$ (embed W equivariantly into an affine space with a linear action of G, and cut out Y by semi-invariant equations whose differentials at o are linearly independent), and $W = (f = 0)$ for a G-semi-invariant regular function f on Y. Identify $\widehat{\mathcal{O}}_{Y,o}$ with $\mathfrak{R}$ through $x_1, \dots, x_4$, and let $\hat{f}_Y$ and $\hat{h}_l$ be the expansions of f and of G-semi-invariant regular functions on Y restricting to h_l. The composite isomorphism

$$\mathfrak{R}/(\hat{f}_Y) \cong \widehat{\mathcal{O}}_{W,o} \cong \mathfrak{R}/(\hat{f})$$

lifts to a G-equivariant automorphism ψ_2 of $\mathfrak{R}$, and $\psi_2(\hat{f}_Y) = u_2\hat{f}$ for a unit u_2, which is G-invariant because $\hat{f}_Y$ and $\hat{f}$ are G-semi-invariant and $u_2(0) \neq 0$. Let $\psi := \psi_1 \circ \psi_2$ and $v_0 := (\psi_1(u_2)u_1)^{-1}$, so that

$$v_0\psi(\hat{f}_Y) = f_0 + \sum_j \hat{a}_j(t)M_j.$$

We apply Construction-Theorem 4.13 with $n = 4$, the functions h_l on Y, the labels $c^{(0)} = (b_l)_l$, no further vectors and no prime divisors, the automorphism ψ, the unit v_0, and 1 in place of each unit u_k of Construction-Theorem 4.13. Let N_1 be the integer given there, let $N \geq N_1$, let L_1 be the integer given there for N, let $L \geq L_1$, let h_l' be the sum of the terms of $\psi(\hat{h}_l)$ of degree less than N (the polynomials $h_{k,0}$ of Construction-Theorem 4.13), and let $a_j \in t\mathbb{C}[t]$ be the sum of the terms of $\hat{a}_j$ of degree less than L. Let

$$f_1 := f_0 + \sum_j a_j(t)M_j, \qquad W_1 := (f_1 = 0) \subset \mathbb{A}^4, \qquad D_1 := \sum_l b_l \operatorname{div}(h_l'|_{W_1}).$$

Then $h_l' \equiv \psi(\hat{h}_l) \pmod{(x,y,z,t)^N}$ and $f_1 \equiv v_0\psi(\hat{f}_Y) \pmod{(x,y,z,t)^L}$. By Construction-Theorem 4.13(1)(3), G acts faithfully on W_1, W_1 is an isolated Gorenstein terminal germ at 0, (W_1, D_1) is klt near 0, and (D.31) holds. The polynomials h_l' are G-semi-invariant, and G preserves f_0 and each non-zero $a_j(t)M_j$ up to the character of $\hat{f}$. Thus $(W_1 \ni 0, G, D_1)$ is data of kind (1) of Theorem 10.4 with the basis $(M_j)_j$, with labels in Γ and $[G : G \cap \mathbf{T}] \leq e_0$. □

D.5. **A uniform margin.** In this subsection, we prove Lemmas D.13 and D.14 and Theorem 10.5.

The following lemma reads off log discrepancies, log canonicity, and the minimal log discrepancy at the closed point from one log resolution over a local base.

Lemma D.13. *Let V be the spectrum of an excellent normal local domain containing $\mathbb{Q}$, o the closed point of V, and $\mathfrak{c} = \prod_{j=1}^s \mathfrak{c}_j^{r_j}$ an $\mathbb{R}$-ideal on V with non-zero ideals $\mathfrak{c}_j$ and real numbers $r_j > 0$. Assume that K_V is $\mathbb{Q}$-Cartier. Let $g : Y \to V$ be a proper birational morphism from a regular scheme Y such that each $\mathfrak{c}_j\mathcal{O}_Y$ is invertible, and let $\sum_l E_l$ be a reduced simple normal crossing divisor on Y whose support contains* $\operatorname{Exc}(g)$ *and the supports of the divisors of the $\mathfrak{c}_j\mathcal{O}_Y$. Then the following hold.*

(1) For any prime divisor F over Y, $a(F, Y, \sum_l E_l) \geq 0$, $a(F, Y, \sum_l E_l) \geq 1$ if $\operatorname{mult}_F E_l = 0$ for all l, and

$$a(F, V, \mathfrak{c}) = a\Big(F, Y, \sum_l E_l\Big) + \sum_l a(E_l, V, \mathfrak{c}) \operatorname{mult}_F E_l.$$

(2) $(V, \mathfrak{c})$ is lc (resp. klt) if and only if $a(E_l, V, \mathfrak{c}) \geq 0$ (resp. $a(E_l, V, \mathfrak{c}) > 0$) for all l.

(3) If $\dim V \geq 2$ and $\operatorname{mld}(V \ni o, \mathfrak{c}) \geq 0$, then $(V, \mathfrak{c})$ is lc.

(4) Assume that $\dim V \geq 2$ and that $\mathfrak{m}_o\mathcal{O}_Y$ is invertible. If $a(E_l, V, \mathfrak{c}) < 0$ for some l with $E_l \cap g^{-1}(o) \neq \emptyset$, then $\operatorname{mld}(V \ni o, \mathfrak{c}) = -\infty$. Otherwise

$$\operatorname{mld}(V \ni o, \mathfrak{c}) = \min\{a(E_l, V, \mathfrak{c}) \mid g(E_l) = o\} \geq 0.$$

Proof. (1) Since Y is regular and $\sum_l E_l$ is a reduced simple normal crossing divisor, $(Y, \sum_l E_l)$ is lc [Kol13, Corollary 2.11], so $a(F, Y, \sum_l E_l) \geq 0$. If $\operatorname{mult}_F E_l = 0$ for all l, then $a(F, Y, \sum_l E_l) = a(F, Y, 0) \geq 1$ as Y is regular. Since $K_Y - g^*K_V$ is supported on $\operatorname{Exc}(g)$ and the divisors of the $\mathfrak{c}_j\mathcal{O}_Y$ are supported on $\sum_l E_l$,

$$K_Y - g^*K_V - \sum_j r_j \operatorname{div}(\mathfrak{c}_j\mathcal{O}_Y) = \sum_l \big(a(E_l, V, \mathfrak{c}) - 1\big)E_l,$$

where a component E_l which is neither g-exceptional nor contained in the support of the divisor of some $\mathfrak{c}_j\mathcal{O}_Y$ has $a(E_l, V, \mathfrak{c}) = 1$ and coefficient zero. Hence

$$a(F, V, \mathfrak{c}) = a(F, Y, 0) + \sum_l \big(a(E_l, V, \mathfrak{c}) - 1\big) \operatorname{mult}_F E_l,$$

which is the formula of (1).

(2) Any prime divisor over V is a prime divisor over Y. If $a(E_l, V, \mathfrak{c}) \geq 0$ for all l, then (1) implies $a(F, V, \mathfrak{c}) \geq 0$ for any F. If $a(E_l, V, \mathfrak{c}) > 0$ for all l, then (1) implies $a(F, V, \mathfrak{c}) > 0$ for any F, since either $\operatorname{mult}_F E_l > 0$ for some l or $a(F, Y, \sum_l E_l) \geq 1$. The converse implications hold since the E_l are prime divisors over V.

(4) Since $\dim V \geq 2$, $\mathfrak{m}_o$ is not principal, so g is not a local isomorphism at any point of $g^{-1}(o)$. Thus $g^{-1}(o)$, the support of the divisor defined by $\mathfrak{m}_o\mathcal{O}_Y$, is a divisor contained in $\operatorname{Exc}(g)$, and it is the union of the E_l with $g(E_l) = o$. Assume first that $a(E_1, V, \mathfrak{c}) < 0$ and $E_1 \cap g^{-1}(o) \neq \emptyset$, possibly renumbering the E_l. If $g(E_1) = o$, then E_1 is proper over the residue field of o, and $\dim E_1 = \dim V - 1 \geq 1$ (dimension formula [Mat89, Theorem 15.6], as excellent rings are universally catenary). In this case we let ξ be the generic point of a prime divisor on E_1 not contained in E_l for any $l \neq 1$. Otherwise there exists E_2 with $g(E_2) = o$ and $E_1 \cap E_2 \neq \emptyset$, and we let ξ be the generic point of a component of $E_1 \cap E_2$. In both cases there exist regular parameters u, v of $\mathcal{O}_{Y,\xi}$ such that $E_1 = (u = 0)$ and $\operatorname{Supp} \sum_l E_l \subset (uv = 0)$ near ξ. For a positive integer p, let F_p be the prime divisor over Y with center $\overline{\{\xi\}}$ given by the monomial valuation with weights $(p, 1)$ on (u, v). A direct calculation and (1) imply

$$a(F_p, V, \mathfrak{c}) = p \cdot a(E_1, V, \mathfrak{c}) + c_v,$$

where $c_v = a(E_{l_v}, V, \mathfrak{c})$ if $(v = 0)$ is a component E_{l_v} of $\sum_l E_l$ near ξ, and $c_v = 1$ otherwise. Since $\overline{\{\xi\}} \subset g^{-1}(o)$, the center of F_p on V is o. Letting $p \to +\infty$, we obtain $\operatorname{mld}(V \ni o, \mathfrak{c}) = -\infty$.

Assume next that $a(E_l, V, \mathfrak{c}) \geq 0$ whenever $E_l \cap g^{-1}(o) \neq \emptyset$, and let F be a prime divisor over V with center o. The center of F on Y is contained in $g^{-1}(o)$, so $\operatorname{mult}_F E_l > 0$ only if $E_l \cap g^{-1}(o) \neq \emptyset$, and $\operatorname{mult}_F E_l \geq 1$ for some l with $g(E_l) = o$. Thus (1) implies

$$a(F, V, \mathfrak{c}) \geq \min\{a(E_l, V, \mathfrak{c}) \mid g(E_l) = o\} \geq 0.$$

The E_l with $g(E_l) = o$ are prime divisors over V with center o, and there is at least one since $g^{-1}(o)$ is a non-empty divisor, so (4) holds.

(3) By [Tem18, Theorem 1.1.11], applied with Z defined by the product of the $\mathfrak{c}_j$, the ideal of the singular locus of V, and $\mathfrak{m}_o$, there exist a proper birational morphism $g_1 : Y_1 \to V$ from a regular scheme Y_1 and a reduced simple normal crossing divisor $\sum_l E'_l$ on Y_1 which satisfy the hypotheses of the lemma and such that $\mathfrak{m}_o\mathcal{O}_{Y_1}$ is invertible. Suppose that $(V, \mathfrak{c})$ is not lc. By (2) for g_1, $a(E'_1, V, \mathfrak{c}) < 0$ for some E'_1. Since $g_1(E'_1)$ is a non-empty closed subset of the local scheme V, $o \in g_1(E'_1)$, so $E'_1 \cap g_1^{-1}(o) \neq \emptyset$. Thus (4) for g_1 implies $\operatorname{mld}(V \ni o, \mathfrak{c}) = -\infty$, a contradiction. □

The generic limits in the proof of Theorem 10.5 live on formal power series rings over large fields, so we need the precise inversion of adjunction of [EM04, Theorem 1.1] and [EMY03, Theorem 0.1], which is proved there for varieties, for a hypersurface in $\operatorname{Spec} k[[x_1, \dots, x_n]]$ with $k \supset \mathbb{C}$ algebraically closed and for products of ideals with real exponents. The ring $R/(f)$ below is not assumed to be the completion of a local ring of a variety, so we give the jet argument of [EM04, Sections 2–3] for $R/(f)$. **Step 1** sets up the log resolution over R [Tem18, Theorem 1.1.11] and

the jets at o, which depend only on finite truncations of f and of the generators of the $\mathfrak{a}_j$, and **Steps 4 to 7** follow [EMY03, EM04].

Lemma D.14 (cf. [EM04, Theorem 1.1], [EMY03, Theorem 0.1])**.** *Let $k \supset \mathbb{C}$ be an algebraically closed field, $n \geq 3$ an integer, $R := k[[x_1, \dots, x_n]]$ with closed point o, $f \in R$ a non-zero element such that $R/(f)$ is a normal domain, and $\mathfrak{a} = \prod_{j=1}^{s} \mathfrak{a}_j^{r_j}$ with real numbers $r_j > 0$ and ideals $\mathfrak{a}_j \subset R$ not contained in (f). Let $W := \operatorname{Spec} R/(f)$. Then*

$$\operatorname{mld}(W \ni o, \mathfrak{a}\mathcal{O}_W) = \operatorname{mld}(\operatorname{Spec} R \ni o, (f) \cdot \mathfrak{a}),$$

where the ideal (f) has exponent one.

Proof. The proof is long, so we divide it into several steps. Let $X := \operatorname{Spec} R$ and $\mathfrak{m}$ the maximal ideal of R. Since $f \neq 0$ and $R/(f)$ is a domain, $f \in \mathfrak{m}$. Let

$$\mathfrak{j} := \Big(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\Big), \qquad \mathfrak{b} := f\mathfrak{j}\mathfrak{m}\prod_{j=1}^{s}\mathfrak{a}_j,$$
$$\mu_X := \operatorname{mld}(X \ni o, (f)\cdot\mathfrak{a}), \qquad \mu_W := \operatorname{mld}(W \ni o, \mathfrak{a}\mathcal{O}_W).$$

Step 1. In this step we fix a log resolution, introduce jets and arcs at o, compute Jacobian matrices in regular parameters, and prove Newton's method in the form of Claim D.15.

Let $d := \operatorname{mult}_0 f$ and f_{in} the homogeneous part of degree d of f. If $\mathfrak{j} \subset (f)$, then $\partial f/\partial x_l \in \mathfrak{m}^d$ for any l, so $\partial f_{\mathrm{in}}/\partial x_l = 0$, which contradicts Euler's formula $d \cdot f_{\mathrm{in}} = \sum_l x_l \partial f_{\mathrm{in}}/\partial x_l$. Thus $\mathfrak{j} \not\subset (f)$, that is, some $\partial f/\partial x_l$ does not vanish on W.

By [Tem18, Theorem 1.1.11], applied to X with Z the closed subscheme defined by $\mathfrak{b}$, there exists a sequence of blow-ups $g : Y \to X$ with centers over $V(\mathfrak{b})$ such that Y is regular and $\mathfrak{b}\mathcal{O}_Y$ is an invertible ideal whose support $g^{-1}(V(\mathfrak{b}))$ is a simple normal crossing divisor. Since $\mathfrak{b}\mathcal{O}_Y$ is invertible, so are $f\mathcal{O}_Y$, $\mathfrak{j}\mathcal{O}_Y$, $\mathfrak{m}\mathcal{O}_Y$, and the $\mathfrak{a}_j\mathcal{O}_Y$. In particular $g^{-1}(o)$ is the support of a divisor. The local ring of X at the generic point of W is a discrete valuation ring, so g is an isomorphism over a neighborhood of the generic point of W. Let W_Y be the strict transform of W, which is a component of $g^{-1}(V(\mathfrak{b}))$ of multiplicity one in $\operatorname{div}(g^*f)$, and $E_1, \dots, E_t$ the other components of $g^{-1}(V(\mathfrak{b}))$. Then W_Y is regular, and $g_W := g|_{W_Y} : W_Y \to W$ is projective and birational. Let $\mathcal{D}$ be the set of prime divisors on W_Y which are components of $E_i \cap W_Y$ for some i. Then $\sum_{D\in\mathcal{D}} D$ is a simple normal crossing divisor, and any $D \in \mathcal{D}$ is a component of $E_i \cap W_Y$ for exactly one i, which we denote by $i(D)$.

Any closed point y of Y belongs to $g^{-1}(o)$, which is proper over k, so the residue field of y is k, $\dim \mathcal{O}_{Y,y} = n$ (dimension formula [Mat89, Theorem 15.6], as excellent rings are universally catenary), and $\widehat{\mathcal{O}}_{Y,y} = k[[y_1, \dots, y_n]]$ for regular parameters $y_1, \dots, y_n$ of $\mathcal{O}_{Y,y}$. We take them such that any component of $g^{-1}(V(\mathfrak{b}))$ through y is defined by one of them near y, and $W_Y = (y_n = 0)$ if $y \in W_Y$. For i with $y \in E_i$, let u_i be the parameter defining E_i, and let $g_l := g^*x_l \in k[[y_1, \dots, y_n]]$.

For a prime divisor F over X (resp. W), $a(F, X, 0) - 1$ (resp. $a(F, W, 0) - 1$) is the order along F of the pullback of $dx_1 \wedge \cdots \wedge dx_n$ (resp. of the generator ω of the canonical module of W, the residue of $dx_1 \wedge \cdots \wedge dx_n/f$) [Kol13, Definitions 1.5, 1.6, and 2.4]. Here we take the base $B := X$, for which $\omega_{X/B} = \mathcal{O}_X$ and $\omega_{W/B} = \mathcal{O}_W(W) = \omega_{X/B}(W)|_W$ [Kol13, (1.6.2)], and for g the relative canonical divisor $K_{Y/X}$ is the divisor of the Jacobian determinant in regular parameters (cf. [dFEM11, Lemma A.11 and Remark A.12]). Moreover

$$\omega = (-1)^{l-1}\frac{dx_1 \wedge \cdots \wedge \widehat{dx_l} \wedge \cdots \wedge dx_n}{\partial f/\partial x_l} \quad \text{on the locus where } \frac{\partial f}{\partial x_l} \neq 0.$$

Thus there exists a unit ϑ of $k[[y_1, \dots, y_n]]$ such that

$$\det\Big(\frac{\partial g_l}{\partial y_{l'}}\Big)_{1\leq l,l'\leq n} = \vartheta \prod_{i: y\in E_i} u_i^{a(E_i,X,0)-1}. \tag{D.32}$$

Assume that $y \in W_Y$, so that $y_1, \dots, y_{n-1}$ restrict to regular parameters of W_Y at y and $\widehat{\mathcal{O}}_{W_Y,y} = k[[y_1, \dots, y_{n-1}]]$, and $g^*f = \vartheta' y_n \prod_{i: y \in E_i} u_i^{\operatorname{mult}_{E_i} f}$ for a unit ϑ'. For $1 \le l \le n$, let

$$f_l := \Big(\frac{\partial f}{\partial x_l}\Big)(g_1, \dots, g_n)\Big|_{y_n=0}, \qquad M_l := \det\Big(\frac{\partial g_{l'}}{\partial y_{l''}}\Big|_{y_n=0}\Big)_{l' \ne l,\ l'' \le n-1}.$$

Differentiating g^*f and restricting to $(y_n = 0)$, the row vector $(f_1, \dots, f_n)$ times the matrix $(\partial g_l/\partial y_{l'}|_{y_n=0})_{l,l'}$ is

$$\Big(0, \dots, 0, \vartheta'|_{y_n=0} \prod_i u_i^{\operatorname{mult}_{E_i} f}\Big),$$

where u_i also denotes the restriction of u_i to $(y_n = 0)$. Multiplying from the right by the adjugate matrix, whose last row is $((-1)^{n+l} M_l)_l$, and using (D.32), we obtain

$$M_l = (-1)^{n+l} \vartheta_y f_l \prod_{i: y \in E_i} u_i^{a(E_i,X,0)-1-\operatorname{mult}_{E_i} f} \tag{D.33}$$

for a unit ϑ_y of $k[[y_1, \dots, y_{n-1}]]$ which does not depend on l. Since $\mathfrak{j} \not\subset (f)$, W is integral, and g_W is birational, some f_l is non-zero, and

$$g_W^*(dx_1 \wedge \cdots \wedge \widehat{dx_l} \wedge \cdots \wedge dx_n) = M_l \cdot dy_1 \wedge \cdots \wedge dy_{n-1},$$

so (D.33) implies

$$g_W^* \omega = \pm \vartheta_y \prod_i u_i^{a(E_i,X,0)-1-\operatorname{mult}_{E_i} f} \cdot dy_1 \wedge \cdots \wedge dy_{n-1}.$$

As $\mathfrak{a}_j \mathcal{O}_{W_Y}$ is the restriction of $\mathfrak{a}_j \mathcal{O}_Y$ to W_Y, we obtain, for any $D \in \mathcal{D}$ and $i = i(D)$,

$$a(D, W, 0) = a(E_i, X, 0) - \operatorname{mult}_{E_i} f, \qquad a(D, W, \mathfrak{a}\mathcal{O}_W) = a(E_i, X, (f) \cdot \mathfrak{a}). \tag{D.34}$$

Moreover the pullbacks of ω, $\mathfrak{j}$, $\mathfrak{m}$, and the $\mathfrak{a}_j$ to W_Y have divisors supported on $\sum_{D \in \mathcal{D}} D$, and $g_W^{-1}(o) = W_Y \cap g^{-1}(o)$ is the union of some members of $\mathcal{D}$.

For a field $K \supset k$ and an integer $m \ge 1$, an m-jet (resp. an arc) of X at o over K is a homomorphism of k-algebras $\gamma : R \to K[\tau]/(\tau^{m+1})$ (resp. $K[[\tau]]$) with $\gamma(\mathfrak{m}) \subset (\tau)$. Since R is complete, γ is determined by $(\gamma(x_1), \dots, \gamma(x_n))$, which is an arbitrary element of $(\tau K[\tau]/(\tau^{m+1}))^n$ (resp. $(\tau K[[\tau]])^n$). Thus the m-jets of X at o are the K-points of the affine space

$$J_m := \big(\tau k[\tau]/(\tau^{m+1})\big)^{\oplus n} \cong \mathbb{A}_k^{nm}.$$

For an arc or a jet γ and $l \le m$, we write $\pi_l(\gamma) \in J_l$ for the truncation of γ, and $\gamma(c\tau)$ for the action of $c \in k^*$, which defines an action of $\mathbb{G}_m$ on J_m. For $\varphi \in R$, let $\operatorname{mult}_\gamma \varphi := \operatorname{mult}_0 \gamma(\varphi)$ be the order in τ of $\gamma(\varphi)$, where $\operatorname{mult}_\gamma \varphi \ge m+1$ means $\gamma(\varphi) = 0$ for an m-jet, and for an ideal $\mathfrak{c} \subset R$, let $\operatorname{mult}_\gamma \mathfrak{c}$ be the minimum of $\operatorname{mult}_\gamma \varphi$ for $\varphi \in \mathfrak{c}$. For an arc δ and an integer $e \ge 1$, we write $\delta(\tau^e)$ for the arc obtained by substituting τ^e for τ, so that $\operatorname{mult}_{\delta(\tau^e)} \varphi = e \operatorname{mult}_\delta \varphi$ for any $\varphi \in R$. The coefficient of τ^l in $\gamma(\varphi)$ is a polynomial in the coefficients of γ which depends only on the class of φ in $R/\mathfrak{m}^{l+1}$. Hence, for $p \le m+1$, the contact locus $\{\gamma \in J_m \mid \operatorname{mult}_\gamma \mathfrak{c} \ge p\}$ is closed in J_m, defined by polynomials which depend only on the classes modulo $\mathfrak{m}^p$ of generators of $\mathfrak{c}$. The m-jets of W at o form the closed subset $\{\gamma \in J_m \mid \operatorname{mult}_\gamma f \ge m+1\}$, and an arc of W at o is an arc γ of X at o with $\gamma(f) = 0$.

Similarly, for a closed point y of Y (resp. of W_Y), the m-jets and the arcs of Y (resp. W_Y) at y over K are determined by the images of $y_1, \dots, y_n$ (resp. $y_1, \dots, y_{n-1}$), which are arbitrary elements of $\tau K[\tau]/(\tau^{m+1})$ (resp. $\tau K[[\tau]]$), and composing with g^* we obtain jets and arcs of X at o. The m-jets of Y (resp. W_Y) at points of $g^{-1}(o)$ are the m-jets of the scheme $Y \times_X \operatorname{Spec} R/\mathfrak{m}^{m+1}$ (resp. $W_Y \times_X \operatorname{Spec} R/\mathfrak{m}^{m+1}$), which is of finite type over k, so they form a scheme of finite type over k whose closed points are the m-jets over k at closed points. For such a jet or arc γ' and a prime divisor P on Y (resp. W_Y), $\operatorname{mult}_{\gamma'} P$ denotes the order in τ of γ' applied to a local equation of P.

We use Newton's method in the following form.

Claim D.15. *Let $K \supset k$ be a field, N a positive integer, $\varphi \in K[[\tau, z_1, \dots, z_N]]^N$, $\gamma \in (\tau K[[\tau]])^N$, and $\mathbf{J}$ the matrix $(\partial\varphi_l/\partial z_{l'})_{l,l'}$ evaluated at $z = \gamma$, a matrix over $K[[\tau]]$. Assume that $\kappa := \operatorname{mult}_0 \det \mathbf{J}$ is finite, and let $m \geq 2\kappa$ be an integer. Then for any $\psi \in \varphi(\gamma) + \tau^{m+1}K[[\tau]]^N$, there exists $w \in K[[\tau]]^N$ with $\varphi(\gamma + \tau^{m+1-\kappa}w) = \psi$.*

Proof. Let $\sigma := m + 1 - \kappa$. Write $\det \mathbf{J} = \tau^\kappa \vartheta_0$ with a unit ϑ_0, $\psi = \varphi(\gamma) + \tau^{m+1}\psi'$, and, for $w \in K[[\tau]]^N$,

$$Q(w) := \varphi(\gamma + \tau^\sigma w) - \varphi(\gamma) - \tau^\sigma \mathbf{J} w, \qquad \Phi(w) := \vartheta_0^{-1} \operatorname{adj}(\mathbf{J})\big(\psi' - \tau^{-m-1}Q(w)\big).$$

By the Taylor formula, $Q(w) - Q(w') \in \tau^{2\sigma+r}K[[\tau]]^N$ whenever $w - w' \in \tau^r K[[\tau]]^N$, and $Q(0) = 0$. Since $2\sigma \geq m+1$, $\Phi(w) \in K[[\tau]]^N$. Since $\det \mathbf{J} \neq 0$ and $\operatorname{adj}(\mathbf{J})\mathbf{J}$ is $\tau^\kappa\vartheta_0$ times the identity matrix, $\varphi(\gamma + \tau^\sigma w) = \psi$ if and only if $\Phi(w) = w$. As $m \geq 2\kappa$, $\Phi(w) - \Phi(w') \in \tau^{r+1}K[[\tau]]^N$ whenever $w - w' \in \tau^r K[[\tau]]^N$, so the sequence $(\Phi^l(0))_l$ converges in the τ-adic topology to a fixed point of Φ. □

Step 2. In this step we read off μ_X and μ_W from g and g_W.

Let $(V, V', h, \mathfrak{c}, \mathcal{P})$ be either $(X, Y, g, (f)\cdot\mathfrak{a}, \{W_Y, E_1, \dots, E_t\})$ or $(W, W_Y, g_W, \mathfrak{a}\mathcal{O}_W, \mathcal{D})$, so that $h : V' \to V$. We show that, for any prime divisor F over V',

$$a(F, V, \mathfrak{c}) = a\Big(F, V', \sum_{P\in\mathcal{P}} P\Big) + \sum_{P\in\mathcal{P}} a(P, V, \mathfrak{c}) \operatorname{mult}_F P, \qquad a\Big(F, V', \sum_{P\in\mathcal{P}} P\Big) \geq 0, \tag{D.35}$$

and that the following hold.

(a) If $a(P, V, \mathfrak{c}) < 0$ for some $P \in \mathcal{P}$ with $P \cap h^{-1}(o) \neq \emptyset$, then $\operatorname{mld}(V \ni o, \mathfrak{c}) = -\infty$.
(b) Otherwise,

$$\operatorname{mld}(V \ni o, \mathfrak{c}) = \min\{a(P, V, \mathfrak{c}) \mid P \in \mathcal{P},\ h(P) = o\} \geq 0.$$

By **Step 1**, g with $\sum_{P\in\{W_Y, E_1, \dots, E_t\}} P$ and g_W with $\sum_{D\in\mathcal{D}} D$ satisfy the hypotheses of Lemma D.13 for $(X, (f)\cdot\mathfrak{a})$ and for $(W, \mathfrak{a}\mathcal{O}_W)$. The canonical divisors K_X and K_W are Cartier as X is regular and W is a hypersurface, and $\dim X > \dim W = n - 1 \geq 2$. The ideals $\mathfrak{m}\mathcal{O}_Y$ and $\mathfrak{m}\mathcal{O}_{W_Y}$ are invertible. Since g is a sequence of blow-ups with centers over $V(\mathfrak{b})$, $\operatorname{Exc}(g) \subset g^{-1}(V(\mathfrak{b})) = W_Y \cup \bigcup_i E_i$. Since X is regular, hence factorial [Mat89, Theorem 20.3], $\operatorname{Exc}(g)$ is of pure codimension one by the purity theorem of van der Waerden [Gro67, Théorème 21.12.12], and it does not contain W_Y, as g is an isomorphism over a neighborhood of the generic point of W (**Step 1**). Hence $\operatorname{Exc}(g) \subset \bigcup_i E_i$, so

$$\operatorname{Exc}(g_W) \subset W_Y \cap \operatorname{Exc}(g) \subset \bigcup_{D\in\mathcal{D}} D.$$

Thus (D.35) is Lemma D.13(1), and (a) and (b) are Lemma D.13(4). Moreover $h^{-1}(o)$, the support of the divisor defined by $\mathfrak{m}\mathcal{O}_{V'}$, is the union of the $P \in \mathcal{P}$ with $h(P) = o$.

Step 3. In this step we show $\mu_X \leq \mu_W$.

Let $D \in \mathcal{D}$ and $i := i(D)$. If $g_W(D) = o$, then $D \subset g^{-1}(o)$, which is the union of the $E_{i'}$ with $g(E_{i'}) = o$, so $D \subset E_i \cap E_{i'} \cap W_Y$ for such an i', and $i' = i$ since D has codimension two in Y. Thus $g(E_i) = o$ if $g_W(D) = o$. If $a(D, W, \mathfrak{a}\mathcal{O}_W) < 0$ for some $D \in \mathcal{D}$ with $D \cap g_W^{-1}(o) \neq \emptyset$, then $E_{i(D)} \cap g^{-1}(o) \neq \emptyset$ and $a(E_{i(D)}, X, (f)\cdot\mathfrak{a}) < 0$ by (D.34), so $\mu_X = -\infty$ by **Step 2**(a). Otherwise, by **Step 2**(b), $\mu_W = a(D, W, \mathfrak{a}\mathcal{O}_W)$ for some $D \in \mathcal{D}$ with $g_W(D) = o$. Since $E_{i(D)}$ has center o, (D.34) implies

$$\mu_X \leq a(E_{i(D)}, X, (f)\cdot\mathfrak{a}) = a(D, W, \mathfrak{a}\mathcal{O}_W) = \mu_W.$$

Step 4. In this step we show that the contact loci in J_m defined by a monomial valuation on Y over o are large.

Let $\nu = (\nu_1, \dots, \nu_t) \in \mathbb{Z}^t_{\geq 0}$ such that $\bigcap_{\nu_i>0} E_i \neq \emptyset$ and $g(E_{i_0}) = o$ for some i_0 with $\nu_{i_0} > 0$. Let

$$A_\nu := \sum_i \nu_i a(E_i, X, 0), \qquad p_0 := \sum_i \nu_i \operatorname{mult}_{E_i} f, \qquad p_j := \sum_i \nu_i \operatorname{mult}_{E_i} \mathfrak{a}_j \quad (1 \leq j \leq s),$$

and let $m \geq 2A_\nu + p_0 + \sum_j p_j$ be an integer. We show that there exists an irreducible closed subset $\Theta \subset J_m$, stable under the action of $\mathbb{G}_m$, such that

$$\text{(D.36)} \quad \dim \Theta \geq n(m+1) - A_\nu, \qquad \operatorname{mult}_\gamma f \geq p_0 \ \text{ and } \ \operatorname{mult}_\gamma \mathfrak{a}_j \geq p_j \ \text{ for any } \gamma \in \Theta \text{ and any } j.$$

Let E°_ν be the set of points of $\bigcap_{\nu_i>0} E_i$ which are contained neither in W_Y nor in E_i for any i with $\nu_i = 0$. Then E°_ν is a non-empty open subset of $\bigcap_{\nu_i>0} E_i \subset g^{-1}(o)$, and $\dim E^\circ_\nu = n - \#\{i \mid \nu_i > 0\}$. Let Λ be the set of m-jets γ' of Y at closed points of E°_ν with $\operatorname{mult}_{\gamma'} E_i = \nu_i$ for any i with $\nu_i > 0$. Since $\nu_i \leq A_\nu \leq m$, the jets in Λ at a fixed closed point form an open dense subset of a linear subspace of codimension $\sum_{\nu_i>0}(\nu_i - 1)$ of the affine space of m-jets of Y at the point, whose dimension is nm. Hence $\dim \Lambda \geq n(m+1) - \sum_i \nu_i$. Near a point of E°_ν, the components of $g^{-1}(V(\mathfrak{b}))$ are the E_i with $\nu_i > 0$, so $\operatorname{mult}_\gamma f = p_0$ and $\operatorname{mult}_\gamma \mathfrak{a}_j = p_j$ for the image $\gamma \in J_m$ of any $\gamma' \in \Lambda$.

We show that the fibers of the map $\Lambda \to J_m$ induced by g have dimension at most $\kappa := \sum_i \nu_i(a(E_i, X, 0) - 1)$ at closed points. Since $a(E_i, X, 0) \geq 1$ for any i, we have $\kappa \leq A_\nu$ and $\nu_i \leq A_\nu$, so the choice of m gives $m \geq 2\kappa$ and $m + 1 - \kappa \geq A_\nu + 1 > \nu_i$. Let $\gamma', \eta' \in \Lambda$ be closed points with equal images in J_m, y the base point of γ', and $\tilde\gamma'$, $\tilde\eta'$ the arcs of Y whose truncations are γ', η' and whose coefficients of τ^l vanish for $l > m$. Let $\mathbf{J}_{\gamma'}$ be the matrix $(\partial g_l/\partial y_{l'})_{l,l'}$ evaluated at $\tilde\gamma'$, whose determinant has order κ by (D.32). Since the arcs of X induced by $\tilde\gamma'$ and $\tilde\eta'$ coincide modulo τ^{m+1} and $m \geq 2\kappa$, Claim D.15 implies that there exists $w \in k[[\tau]]^n$ such that the arcs of X induced by $\tilde\gamma' + \tau^{m+1-\kappa} w$ and by $\tilde\eta'$ coincide. As $m + 1 - \kappa > \nu_i$ for any i, the orders of $\tilde\gamma' + \tau^{m+1-\kappa} w$ and of $\tilde\eta'$ along the E_i with $\nu_i > 0$ are finite, so the generic points of $\tilde\gamma' + \tau^{m+1-\kappa} w$ and of $\tilde\eta'$ are not contained in $g^{-1}(V(\mathfrak{b}))$, over the complement of which g is an isomorphism. Since Y is separated, $\tilde\eta' = \tilde\gamma' + \tau^{m+1-\kappa} w$.

Since $2(m + 1 - \kappa) \geq m + 1$, the Taylor formula implies $\mathbf{J}_{\gamma'} w \in \tau^\kappa k[[\tau]]^n$. By the elementary divisor theorem over $k[[\tau]]$, there exist invertible matrices U, U' over $k[[\tau]]$ and integers $c_l \geq 0$ with $\sum_l c_l = \kappa$ such that

$$\mathbf{J}_{\gamma'} = U \operatorname{diag}(\tau^{c_1}, \dots, \tau^{c_n}) U'.$$

Thus the classes modulo τ^κ of the $w \in k[[\tau]]^n$ with $\mathbf{J}_{\gamma'} w \in \tau^\kappa k[[\tau]]^n$ form a k-vector space of dimension $\sum_l \min\{c_l, \kappa\} = \kappa$. As $\eta' = \pi_m(\tilde\gamma' + \tau^{m+1-\kappa} w)$ is determined by the class of w modulo τ^κ, the fiber through γ' is of dimension at most κ. Therefore the dimension of the image of Λ in J_m is at least $\dim \Lambda - \kappa \geq n(m+1) - A_\nu$. Let Θ be the closure of an irreducible component of maximal dimension of the image of Λ. The set Λ and the map $\Lambda \to J_m$ are compatible with the actions of the connected group $\mathbb{G}_m$, so Θ is stable under $\mathbb{G}_m$, and (D.36) holds since the conditions on the orders are closed.

Step 5. In this step we bound from below the dimension of the jets of W at o whose truncations belong to the set Θ of **Step 4**.

First, let $K \supset k$ be a field, $0 \leq \lambda' \leq q$ integers, and ρ an arc of X at o over K with $\operatorname{mult}_\rho f \geq q + \lambda' + 1$ and $\operatorname{mult}_\rho \mathfrak{j} = \lambda'$. We show

$$\text{(D.37)} \qquad \text{there exists an arc } \delta \text{ of } W \text{ at } o \text{ over } K \text{ with } \pi_q(\delta) = \pi_q(\rho).$$

Let l with $\operatorname{mult}_\rho(\partial f/\partial x_l) = \lambda'$, and $\varphi \in K[[\tau, z]]$ the series given by substituting $\rho(x_{l'})$ for $x_{l'}$, $l' \neq l$, and z for x_l in f. By Claim D.15 with $N = 1$, $\gamma = \rho(x_l)$, λ' in place of κ, $q + \lambda'$ in place of m, and $\psi = 0$, there exists $z \in \rho(x_l) + \tau^{q+1} K[[\tau]]$ with $\varphi(z) = 0$, which defines δ.

Keep ν, m, and Θ as in **Step 4**, and let λ be the minimum of $\operatorname{mult}_\delta \mathfrak{j}$ over the arcs δ of W at o over k with $\pi_m(\delta) \in \Theta$. We show $\lambda < +\infty$. Let $D \in \mathcal{D}$ with $g_W(D) = o$, y a closed point of D not contained in other members of $\mathcal{D}$, and δ' the arc of W_Y at y given by $y_1 \to \tau$ and $y_l \to 0$ for $2 \leq l \leq n-1$, where $D = (y_1 = 0)$ near y. The generic point of δ' is not contained

in $\bigcup_{D'\in\mathcal{D}} D'$, so the arc δ_0 of W at o induced by δ' satisfies $\mathrm{mult}_{\delta_0}\mathfrak{j} < +\infty$ by **Step 1**. The truncation $\pi_m(\delta_0(\tau^{m+1}))$ of the arc $\delta_0(\tau^{m+1})$ of W at o is 0, which belongs to Θ since Θ is closed, stable under $\mathbb{G}_m$, and non-empty, and $\gamma(c\tau) \to 0$ as $c \to 0$ for any $\gamma \in J_m$. Moreover

$$\mathrm{mult}_{\delta_0(\tau^{m+1})}\mathfrak{j} = (m+1)\,\mathrm{mult}_{\delta_0}\mathfrak{j} < +\infty,$$

so $\lambda < +\infty$.

Let $q \geq \max\{m, \lambda\}$ be an integer, and

$$T := \{\gamma \in J_{q+\lambda} \mid \pi_m(\gamma) \in \Theta,\ \mathrm{mult}_\gamma f \geq q+\lambda+1\}, \qquad T^\circ := \{\gamma \in T \mid \mathrm{mult}_\gamma \mathfrak{j} \leq \lambda\}.$$

We show the following.

(i) T° is a non-empty open subset of T, and for any closed point $\gamma \in T^\circ$ there exists an arc δ of W at o over k with $\pi_q(\delta) = \pi_q(\gamma)$, $\pi_m(\delta) \in \Theta$, and $\mathrm{mult}_\delta \mathfrak{j} = \lambda$.

(ii) $\dim \pi_q(T^\circ) \geq (n-1)(q+1) - A_\nu + p_0 - \lambda$.

(i) The subset T° is open in T, and it contains $\pi_{q+\lambda}(\delta)$ for an arc δ attaining λ. Let $\gamma \in T^\circ$ be a closed point and $\lambda' := \mathrm{mult}_\gamma \mathfrak{j} \leq \lambda \leq q$. By (D.37), applied to the arc of X whose truncation is γ and whose coefficients of τ^l vanish for $l > q+\lambda$, there exists an arc δ of W at o over k with $\pi_q(\delta) = \pi_q(\gamma)$. Then $\pi_m(\delta) = \pi_m(\gamma) \in \Theta$ and $\mathrm{mult}_\delta \mathfrak{j} = \lambda'$, so $\lambda' \geq \lambda$ by the minimality of λ, that is, $\lambda' = \lambda$.

(ii) The inverse image $\Theta_{q+\lambda}$ of Θ in $J_{q+\lambda}$ is irreducible of dimension $\dim\Theta + n(q+\lambda-m)$. Since $\mathrm{mult}_\gamma f \geq p_0$ for $\gamma \in \Theta_{q+\lambda}$, the set T is defined in $\Theta_{q+\lambda}$ by the vanishing of the coefficients of $\tau^{p_0}, \dots, \tau^{q+\lambda}$ in $\gamma(f)$, which are $q+\lambda-p_0+1$ polynomials. Let T_1 be an irreducible component of T with $T_1 \cap T^\circ \neq \emptyset$. Then $\dim T_1 \geq \dim\Theta + n(q+\lambda-m) - (q+\lambda-p_0+1)$. The fibers of $\pi_q : J_{q+\lambda} \to J_q$ are of dimension $n\lambda$, so (D.36) implies

$$\dim \pi_q(T^\circ) \geq \dim T_1 - n\lambda \geq (n-1)(q+1) - A_\nu + p_0 - \lambda.$$

Step 6. In this step we bound $\dim \pi_q(T^\circ)$ from above when $\mu_W \geq 0$ and q is large.

Assume $\mu_W \geq 0$. By **Step 2** applied to W, $a(D, W, \mathfrak{a}\mathcal{O}_W) \geq 0$ for any $D \in \mathcal{D}$ with $D \cap g_W^{-1}(o) \neq \emptyset$, and $a(D, W, \mathfrak{a}\mathcal{O}_W) \geq \mu_W$ if $g_W(D) = o$. Keep ν, m, Θ, and λ as in **Step 5**, let $\Xi := \lambda + 1 + \sum_j p_j$, and assume

$$q \geq \max\{m, \lambda\} + \Xi + 2\lambda + 2\Xi \sum_{D\in\mathcal{D}} |a(D, W, 0) - 1|.$$

We show

$$\dim \pi_q(T^\circ) \leq (n-1)(q+1) - \lambda - \sum_{j=1}^{s} r_j p_j - \mu_W. \tag{D.38}$$

First, let δ be an arc of W at o over k with $\mathrm{mult}_\delta \mathfrak{j} = \lambda$. We show that there exist a field $K \supset k$ and an arc δ'' of W at o over K such that $\pi_q(\delta'') = \pi_q(\delta)$ and the homomorphism $R/(f) \to K[[\tau]]$ given by δ'' is injective. After a linear change of the coordinates $x_1, \dots, x_n$, used only to construct δ'', we may assume that $f(0, \dots, 0, x_n) \neq 0$ and $\mathrm{mult}_\delta(\partial f/\partial x_n) = \lambda$, since both are conditions on the last coordinate vector outside proper closed subsets of k^n. Let $K := k(\zeta_{i,l} \mid 1 \leq i \leq n-1,\ l \geq 0)$ be a purely transcendental extension of k, and

$$\delta''(x_i) := \delta(x_i) + \tau^{q+1} \sum_{l\geq 0} \zeta_{i,l}\tau^l \quad \text{for } i \leq n-1, \qquad \bar\delta := (\delta''(x_1), \dots, \delta''(x_{n-1})).$$

Substituting $\bar\delta$ for $(x_1, \dots, x_{n-1})$ in f, we obtain $\varphi \in K[[\tau, z]]$ with $\mathrm{mult}_0 \varphi(\delta(x_n)) \geq q+1+\lambda$ by the Taylor formula, and $\mathrm{mult}_0(\partial\varphi/\partial z)(\delta(x_n)) = \lambda$ as $q+1 > \lambda$. By Claim D.15 with $N = 1$, $\gamma = \delta(x_n)$, λ in place of κ, $q+\lambda$ in place of m, and $\psi = 0$, there exists $\delta''(x_n) \in \delta(x_n) + \tau^{q+1}K[[\tau]]$ with $\varphi(\delta''(x_n)) = 0$, which defines δ''. Suppose that a non-zero $\chi \in k[[x_1, \dots, x_{n-1}]]$ satisfies $\chi(\bar\delta) = 0$. Applying the derivation $\partial/\partial\zeta_{i,0}$ of K to the coefficients, we obtain $\tau^{q+1}(\partial\chi/\partial x_i)(\bar\delta) = 0$, since the $\delta(x_i)$ have coefficients in k. By induction, $(\partial^\beta\chi)(\bar\delta) = 0$ for any $\beta \in \mathbb{Z}_{\geq 0}^{n-1}$, which fails for β with $(\partial^\beta\chi)(0) \neq 0$. Thus $k[[x_1, \dots, x_{n-1}]] \to K[[\tau]]$ is injective. By the Weierstrass

preparation theorem, $R/(f)$ is finite over $k[[x_1,\dots,x_{n-1}]]$. If a non-zero $\varrho \in R/(f)$ maps to zero, let $\varrho^e + b_1\varrho^{e-1} + \cdots + b_e = 0$ with $b_l \in k[[x_1,\dots,x_{n-1}]]$ and e minimal. Then $b_e \neq 0$ as $R/(f)$ is a domain, and b_e maps to zero, a contradiction.

Let γ be a closed point of $\pi_q(T^\circ)$. By **Step 5**(i) and the last paragraph, $\gamma = \pi_q(\delta'')$ for an arc δ'' of W at o over a field $K \supset k$ such that $R/(f) \to K[[\tau]]$ is injective, $\pi_m(\delta'') \in \Theta$, and $\operatorname{mult}_{\delta''} \mathfrak{j} = \lambda$. The generic point of $\operatorname{Spec} K[[\tau]]$ maps to the generic point of W, over which g_W is an isomorphism, and g_W is proper, so δ'' lifts to an arc δ' of W_Y whose closed point is contained in $g_W^{-1}(o)$, and $\operatorname{mult}_{\delta'} D < +\infty$ for any $D \in \mathcal{D}$. Let $\mathbf{v}(\delta') \in \{0,\dots,\Xi\}^{\mathcal{D}}$ be the vector with $\mathbf{v}(\delta')_D := \min\{\operatorname{mult}_{\delta'} D, \Xi\}$. Consider the following conditions on a vector $\mathbf{v} = (v_D)_D \in \{0,\dots,\Xi\}^{\mathcal{D}}$:

(α) $v_D \geq 1$ for some D with $g_W(D) = o$, and any D with $v_D > 0$ intersects $g_W^{-1}(o)$;
(β) $\sum_D v_D \operatorname{mult}_D(\mathfrak{a}_j \mathcal{O}_{W_Y}) \geq p_j$ for any j.

The vector $\mathbf{v}(\delta')$ satisfies (α), since the closed point of δ' is contained in $g_W^{-1}(o)$, the union of the D with $g_W(D) = o$. It satisfies (β), since $\Xi > p_j$ and, by (D.36),

$$\sum_D \operatorname{mult}_{\delta'} D \cdot \operatorname{mult}_D(\mathfrak{a}_j \mathcal{O}_{W_Y}) = \operatorname{mult}_{\delta''} \mathfrak{a}_j \geq p_j.$$

For $\mathbf{v} \in \{0,\dots,\Xi\}^{\mathcal{D}}$, let $\Lambda_{\mathbf{v}}$ be the set of q-jets η' of W_Y at points of $g_W^{-1}(o)$ such that, for any $D \in \mathcal{D}$, $\operatorname{mult}_{\eta'} D = v_D$ if $v_D < \Xi$, and $\operatorname{mult}_{\eta'} D \geq \Xi$ if $v_D = \Xi$. Let $\Lambda_{\mathbf{v}}^T \subset \Lambda_{\mathbf{v}}$ be the set of $\eta' \in \Lambda_{\mathbf{v}}$ whose images in J_q belong to $\pi_q(T^\circ)$. Then $\pi_q(\delta') \in \Lambda_{\mathbf{v}(\delta')}^T$, so the closed points of the constructible set $\pi_q(T^\circ)$ belong to the union of the images of the $\Lambda_{\mathbf{v}}^T$ over the finitely many vectors $\mathbf{v}$ satisfying (α) and (β).

Fix a vector $\mathbf{v}$ satisfying (α) and (β). The base point of a jet in $\Lambda_{\mathbf{v}}$ is contained in $g_W^{-1}(o) \cap \bigcap_{v_D>0} D$, of dimension at most $n - 1 - \#\{D \mid v_D > 0\}$, and the jets in $\Lambda_{\mathbf{v}}$ at a fixed closed point form a subset of a linear subspace of codimension $\sum_{v_D>0}(v_D - 1)$ of the affine space of q-jets of W_Y at the point, whose dimension is $(n-1)q$. Hence

$$\dim \Lambda_{\mathbf{v}}^T \leq (n-1)(q+1) - \sum_D v_D.$$

Let $c := \lambda + \sum_D v_D(a(D,W,0) - 1)$. We show that the fibers of the map $\Lambda_{\mathbf{v}}^T \to J_q$ have dimension at least c at closed points. Let $\eta' \in \Lambda_{\mathbf{v}}^T$ be a closed point, y the base point of η', and $\tilde{\eta}'$ the arc of W_Y at y with $\tilde{\eta}'(y_l) := \eta'(y_l) + \tau^{q+1}$ for $1 \leq l \leq n-1$, where $\eta'(y_l)$ is viewed as a polynomial in τ of degree at most q, so that $\operatorname{mult}_{\tilde{\eta}'} D \leq q+1$ for any D, since any D through y is defined near y by one of $y_1,\dots,y_{n-1}$. Let $\mathbf{J}_{\eta'}$ be the $n \times (n-1)$ matrix of the derivatives $\partial g_l/\partial y_{l'}|_{y_n=0}$, $l' \leq n-1$, evaluated at $\tilde{\eta}'$. By (D.33) and (D.34), the minimum of the orders of the maximal minors $M_l(\tilde{\eta}')$ is

$$\operatorname{mult}_{\tilde{\eta}'} \mathfrak{j} + \sum_{D \ni y} (a(D,W,0) - 1) \operatorname{mult}_{\tilde{\eta}'} D, \qquad \text{where} \quad \operatorname{mult}_{\tilde{\eta}'} \mathfrak{j} = \sum_D \operatorname{mult}_{\tilde{\eta}'} D \cdot \operatorname{mult}_D(\mathfrak{j}\mathcal{O}_{W_Y}) = \lambda,$$

since the image of η' belongs to $\pi_q(T^\circ)$ and $\lambda \leq q$. Thus $\operatorname{mult}_{\tilde{\eta}'} D \leq \lambda < \Xi$ if $\operatorname{mult}_D(\mathfrak{j}\mathcal{O}_{W_Y}) > 0$. Moreover $a(D,W,0) - 1 \geq -\operatorname{mult}_D(\mathfrak{j}\mathcal{O}_{W_Y})$ for any D by (D.33), as the M_l are power series. Hence $a(D,W,0) - 1 \geq 0$ whenever $v_D = \Xi$, and the minimum of the orders of the $M_l(\tilde{\eta}')$ is at least c.

By the elementary divisor theorem over $k[[\tau]]$, $\mathbf{J}_{\eta'} = U\mathbf{J}_0U'$ with invertible matrices U, U' over $k[[\tau]]$, where $\mathbf{J}_0$ is the $n \times (n-1)$ matrix whose first $n-1$ rows form $\operatorname{diag}(\tau^{c_1},\dots,\tau^{c_{n-1}})$ and whose last row is zero, $c_l \in \mathbb{Z}_{\geq 0} \cup \{+\infty\}$, $\tau^{+\infty} := 0$, and $\sum_l c_l$ is the minimum of the orders of the maximal minors, so $\sum_l c_l \geq c$. Thus the classes modulo τ^c of the $w \in k[[\tau]]^{n-1}$ with $\mathbf{J}_{\eta'} w \in \tau^c k[[\tau]]^n$ form a k-vector space of dimension $\sum_l \min\{c_l, c\} \geq c$. For such w, let $\eta'_w := \pi_q(\tilde{\eta}' + \tau^{q+1-c} w)$. Since $c \leq \lambda + \Xi \sum_D |a(D,W,0) - 1|$, we have $2(q+1-c) \geq q+1$ and $q+1-c > \Xi$. Hence the images of η'_w and η' in J_q coincide by the Taylor formula, and for any D, either $\operatorname{mult}_{\eta'_w} D = \operatorname{mult}_{\eta'} D$ or $\operatorname{mult}_{\eta'_w} D, \operatorname{mult}_{\eta'} D \geq \Xi$. Thus $\eta'_w \in \Lambda_{\mathbf{v}}^T$, and distinct classes of w

yield distinct η'_w. Therefore the fiber through η' is of dimension at least c, and the dimension of the image of $\Lambda_{\mathbf{v}}^T$ in J_q is at most

$$(n-1)(q+1) - \sum_D v_D - c = (n-1)(q+1) - \lambda - \sum_D v_D a(D, W, 0).$$

Since

$$a(D, W, 0) = a(D, W, \mathfrak{a}\mathcal{O}_W) + \sum_j r_j \operatorname{mult}_D(\mathfrak{a}_j \mathcal{O}_{W_Y}),$$

the conditions (α) and (β) and the bounds on the $a(D, W, \mathfrak{a}\mathcal{O}_W)$ at the beginning of this step imply

$$\sum_D v_D a(D, W, 0) \geq \mu_W + \sum_j r_j p_j.$$

Thus (D.38) holds.

Step 7. In this step we show $\mu_W \leq \mu_X$, which together with **Step 3** proves the lemma.

Suppose $\mu_X < \mu_W$. Then $\mu_W > -\infty$, so $\mu_W \geq 0$ by **Step 2**. We choose ν as in **Step 4** with

$$\sum_i \nu_i a(E_i, X, (f) \cdot \mathfrak{a}) < \mu_W. \tag{D.39}$$

If $\mu_X > -\infty$, then the hypothesis of **Step 2**(a) fails, and **Step 2**(b) implies that there exists i_0 with $g(E_{i_0}) = o$ and $a(E_{i_0}, X, (f) \cdot \mathfrak{a}) = \mu_X$, as $g(W_Y) \neq o$. We let $\nu_{i_0} := 1$ and $\nu_i := 0$ for $i \neq i_0$. If $\mu_X = -\infty$, then the conclusion of **Step 2**(b) fails, so there exists i_1 with $E_{i_1} \cap g^{-1}(o) \neq \emptyset$ and $a(E_{i_1}, X, (f) \cdot \mathfrak{a}) < 0$, since $a(W_Y, X, (f) \cdot \mathfrak{a}) = 0$. Let N_1 be a positive integer. If $g(E_{i_1}) = o$, we let $\nu_{i_1} := N_1$ and $\nu_i := 0$ for $i \neq i_1$. Otherwise there exists i_2 with $g(E_{i_2}) = o$ and $E_{i_1} \cap E_{i_2} \neq \emptyset$, and we let $\nu_{i_1} := N_1$, $\nu_{i_2} := 1$, and $\nu_i := 0$ for $i \neq i_1, i_2$. For N_1 large, (D.39) holds.

Let m, Θ, λ, q, and T° be as in **Steps 4, 5, and 6** for ν. By **Step 5**(ii) and (D.38),

$$(n-1)(q+1) - A_\nu + p_0 - \lambda \leq (n-1)(q+1) - \lambda - \sum_j r_j p_j - \mu_W,$$

that is,

$$\sum_i \nu_i a(E_i, X, (f) \cdot \mathfrak{a}) = A_\nu - p_0 - \sum_j r_j p_j \geq \mu_W,$$

which contradicts (D.39). Thus $\mu_W \leq \mu_X$, and $\mu_W = \mu_X$ by **Step 3**. □

The next lemma deduces the klt property from a gap above one at the closed point when no component of the boundary has coefficient one.

Lemma D.16. *Let V be the spectrum of a normal complete local domain of dimension three containing $\mathbb{Q}$, o the closed point of V, and $B = \sum_{j=1}^s b_j \operatorname{div}(h_j)$ with real numbers $b_j > 0$ and non-zero elements h_j of the maximal ideal. Assume that K_V is $\mathbb{Q}$-Cartier, $\lfloor B \rfloor = 0$, and $\operatorname{mld}(V \ni o, B) > 1$. Then (V, B) is klt, and there exists a real number $\theta > 1$ such that $(V, \theta B)$ is klt.*

Proof. We first treat rational coefficients.

Claim D.17. *Let $B' = \sum_{j=1}^s b'_j \operatorname{div}(h_j)$ with rational numbers $b'_j > 0$. If (V, B') is lc and not klt and $\lfloor B' \rfloor = 0$, then $\operatorname{mld}(V \ni o, B') \leq 1$.*

Proof. Since K_V is $\mathbb{Q}$-Cartier and the $\operatorname{div}(h_j)$ are Cartier, $K_V + B'$ is $\mathbb{Q}$-Cartier. By [Fuj11, Theorem 10.4], whose construction runs a relative minimal model program with scaling for a klt pair over V that is big over V, and by [LM22, Theorem A(0)], which provides such a program over the excellent base V and the termination of such a program (the log resolution with a relatively anti-ample exceptional divisor, the choice of general members, and the negativity lemma in that construction are [Tem18, Theorem 1.1.11], [LM22, Corollary 10.4], and [LM22, Lemma 5.15] over V), there exists a projective birational morphism $g' : Y' \to V$ such that Y'

is $\mathbb{Q}$-factorial, $(Y', B'_{Y'})$ is dlt, where $K_{Y'} + B'_{Y'} = g'^*(K_V + B')$, and $a(E, V, B') = 0$ for any g'-exceptional prime divisor E. In particular $B'_{Y'} \geq 0$, and $B'_{Y'}$ is the sum of $g'^{-1}_* B'$ and the g'-exceptional prime divisors.

If g' has no exceptional prime divisor, then $\lfloor B'_{Y'} \rfloor = \lfloor g'^{-1}_* B' \rfloor = 0$, so the dlt pair $(Y', B'_{Y'})$ is klt, and (V, B') is klt ([Kol13, Notation 2.6], with the base a formal power series ring over a field of which the ring of V is a quotient by Cohen's structure theorem), a contradiction. Thus there exists a g'-exceptional prime divisor E. Since $g'(E)$ is a non-empty closed subset of the local scheme V, $o \in g'(E)$, and $\dim g'(E) \leq 1$. If $g'(E) = \{o\}$, then $\operatorname{mld}(V \ni o, B') \leq a(E, V, B') = 0$. Thus we may assume that $g'(E)$ is a curve. Since $E \to g'(E)$ is proper and $\dim E = 2 > \dim g'(E)$ [Mat89, Theorem 15.6], the fiber $E \cap g'^{-1}(o)$ is not finite: otherwise $E \to g'(E)$ would be quasi-finite over a neighborhood of o, that is, over $g'(E)$, hence finite, and $\dim E = \dim g'(E)$. Let $C \subset E \cap g'^{-1}(o)$ be an irreducible curve and ξ the generic point of C. Then $\dim \mathcal{O}_{Y',\xi} = 2$ [Mat89, Theorem 15.6]. We find a prime divisor F over Y' with center C such that $a(F, Y', B'_{Y'}) \leq 1$.

Let $S := \operatorname{Spec} \mathcal{O}_{Y',\xi}$, a normal excellent two-dimensional local scheme with closed point s, and E_S, B'_S the restrictions of E, $B'_{Y'}$ to S. Then E_S is reduced and passes through s, $B'_S \geq E_S$ since E has coefficient one in $B'_{Y'} \geq 0$, and $K_S + E_S$ is $\mathbb{Q}$-Cartier since Y' is $\mathbb{Q}$-factorial. By [Kol13, Theorem 2.25], there exists a proper birational morphism $h : S' \to S$ such that S' is regular and $K_{S'} + h^{-1}_* E_S$ is h-nef. If h is an isomorphism, then S is regular, and we let F_S be the exceptional prime divisor of the blow-up of s, so that $a(F_S, S, E_S) = 2 - \operatorname{mult}_s E_S \leq 1$. Otherwise, let $F_1, \dots, F_\ell$ be the h-exceptional prime divisors, where $\ell \geq 1$, and write

$$K_{S'} + h^{-1}_* E_S + N = h^*(K_S + E_S), \qquad N = \sum_{i=1}^{\ell} \bigl(1 - a(F_i, S, E_S)\bigr) F_i.$$

Then $(N \cdot F_i) = -((K_{S'} + h^{-1}_* E_S) \cdot F_i) \leq 0$ for any i. Write $N = N^+ - N^-$, where $N^+, N^- \geq 0$ have no common components. Then

$$((N^-)^2) = (N^+ \cdot N^-) - (N \cdot N^-) \geq 0,$$

and the intersection matrix of $F_1, \dots, F_\ell$ is negative definite [Kol13, Theorem 10.1], so $N^- = 0$. Thus $N \geq 0$, that is, $a(F_i, S, E_S) \leq 1$ for any i, and we let $F_S := F_1$. In both cases F_S has center s, and

$$a(F_S, S, B'_S) \leq a(F_S, S, E_S) \leq 1.$$

The prime divisor F over Y' with center C determined by F_S satisfies

$$a(F, Y', B'_{Y'}) = a(F_S, S, B'_S) \leq 1.$$

The center of F on V is $g'(C) = o$, so

$$\operatorname{mld}(V \ni o, B') \leq a(F, V, B') = a(F, Y', B'_{Y'}) \leq 1.$$

□

Proof of Lemma D.16 continued. For real numbers $u_j > 0$, Lemma D.13 holds for the pair $(V, \sum_j u_j \operatorname{div}(h_j))$, which has the log discrepancies of the $\mathbb{R}$-ideal $\prod_j (h_j)^{u_j}$. By Lemma D.13(3), (V, B) is lc. Let $g : Y \to V$ be a log resolution of $(V, \sum_j \operatorname{div}(h_j))$, which exists by [Tem18, Theorem 1.1.11] applied with Z defined by the product of $\prod_j h_j$ and the ideal of the singular locus of V, and let $\sum_l E_l$ be the reduced divisor on Y whose support is the union of $\operatorname{Exc}(g)$ and $g^{-1}(\operatorname{Supp} \sum_j \operatorname{div}(h_j))$. For $u = (u_1, \dots, u_s) \in \mathbb{R}^s$, let $B(u) := \sum_j u_j \operatorname{div}(h_j)$ and

$$a_l(u) := a(E_l, V, B(u)) = a(E_l, V, 0) - \sum_j u_j \operatorname{mult}_{E_l} h_j.$$

Since K_V is $\mathbb{Q}$-Cartier, $a(E_l, V, 0) \in \mathbb{Q}$, so each a_l is an affine function with rational coefficients. Let $b := (b_1, \dots, b_s)$, so that $B = B(b)$.

Suppose that (V, B) is not klt. By Lemma D.13(2), $a_l(b) \geq 0$ for all l, and the set L_0 of indices l with $a_l(b) = 0$ is non-empty. If E_l is the strict transform of a prime divisor P on

V, then $a_l(b) = 1 - \operatorname{mult}_P B > 0$ since $\lfloor B \rfloor = 0$. Fix a real number $t \in (0,1)$ such that $(1-t)\operatorname{mld}(V \ni o, B) > 1$. The affine subspace

$$\mathcal{U} := \{u \in \mathbb{R}^s \mid a_l(u) = 0 \text{ for any } l \in L_0\}$$

is defined by linear equations with rational coefficients and contains b, so $\mathcal{U}$ contains rational points arbitrarily close to b. Since $b_j > 0$ for any j and $a_l(b) > (1-t)a_l(b)$ for any $l \notin L_0$, there exists a rational point $b' \in \mathcal{U}$ such that $b'_j > 0$ for any j and

$$a_l(b') \geq (1-t)a_l(b) \geq 0 \quad \text{for all } l,$$

where equality holds for $l \in L_0$. Let F be a prime divisor over V. By Lemma D.13(1), applied to $B(b')$ and to B,

$$a(F, V, B(b')) = a\Big(F, Y, \sum_l E_l\Big) + \sum_l a_l(b') \operatorname{mult}_F E_l \geq (1-t)a(F, V, B).$$

Thus $\operatorname{mld}(V \ni o, B(b')) \geq (1-t)\operatorname{mld}(V \ni o, B) > 1$. By Lemma D.13(2), $(V, B(b'))$ is lc and not klt, as $a_l(b') = 0$ for $l \in L_0$. Moreover $\lfloor B(b') \rfloor = 0$, since for the strict transform E_l of a prime divisor P on V we have $l \notin L_0$ and $1 - \operatorname{mult}_P B(b') = a_l(b') > 0$. This contradicts Claim D.17. Thus (V, B) is klt.

By Lemma D.13(2), $a_l(b) > 0$ for all l. The functions $\theta \to a_l(\theta b)$ are affine and finitely many, so there exists a real number $\theta > 1$ such that $a_l(\theta b) > 0$ for all l. By Lemma D.13(2), $(V, \theta B)$ is klt. □

The next lemma computes the order of a function along E_0 and the multiplicity of the divisor of the function on a germ of quadratic rank one whose residual cubic is y^3.

Lemma D.18. *Let $k \supset \mathbb{C}$ be an algebraically closed field, and $g \in k[[y,z,t]]$ a power series such that $g - y^3 \in (y,z,t)^4$ and $R := k[[x,y,z,t]]/(x^2+g)$ is a normal domain. Let o be the closed point of* $\operatorname{Spec} R$. *Let $a, d \in k[[y,z,t]]$, not both zero, such that the image h of $a + xd$ in R belongs to the maximal ideal, and let $n := \min\{2\operatorname{mult}_0 a, 3 + 2\operatorname{mult}_0 d\}$, where mult_0 denotes the multiplicity at the origin and $\operatorname{mult}_0 0 := +\infty$. Then the following hold.*

(1) The strict transform of $\operatorname{Spec} R$ *under the weighted blow-up of* $\operatorname{Spec} k[[x,y,z,t]]$ *with weights $(3,2,2,2)$ in x, y, z, t has a unique exceptional prime divisor E_0. The center of E_0 on* $\operatorname{Spec} R$ *is o, $a(E_0, \operatorname{Spec} R, 0) = 3$, and $\operatorname{mult}_{E_0} h = n$.*

(2) $e(R/hR) = n$, where e denotes the Hilbert–Samuel multiplicity with respect to the maximal ideal. Moreover,

$$n = \sum_D \operatorname{mult}_D h \cdot e(R/I_D),$$

where D runs over the prime divisors on $\operatorname{Spec} R$ *and I_D is the ideal of D.*

Proof. For a power series φ, write φ_m for the homogeneous part of degree m. Since h belongs to the maximal ideal, $a(0) = 0$, so $n \geq 2$ is an integer. Since $2\operatorname{mult}_0 a$ is even and $3 + 2\operatorname{mult}_0 d$ is odd, exactly one of $2\operatorname{mult}_0 a = n$ and $3 + 2\operatorname{mult}_0 d = n$ holds.

(1) Since $g - y^3 \in (y,z,t)^4$, the part of lowest weight of $x^2 + g$ with respect to the weights $(3,2,2,2)$ is $x^2 + y^3$, of weight 6, and $x^2 + y^3$ is reduced and irreducible. The intersection of the strict transform of $\operatorname{Spec} R$ with the exceptional divisor $\mathbb{P}(3,2,2,2)$ of the weighted blow-up is $(x^2 + y^3 = 0)$, which is irreducible and not contained in $(t = 0)$, so we compute on the chart U_t of the weighted blow-up where t generates the exceptional ideal. On U_t, $x = x't'^3$, $y = y't'^2$, $z = z't'^2$, and $t = t'^2$, and U_t is the quotient by μ_2 acting by $(x', y', z', t') \to (-x', y', z', -t')$. Since $g - y^3 \in (y,z,t)^4$,

$$x^2 + g = t'^6(x'^2 + y'^3 + t'^2\chi_1)$$

for a power series χ_1, so the intersection of the strict transform $(x'^2 + y'^3 + t'^2\chi_1 = 0)$ with $(t' = 0)$ is the irreducible surface $(x'^2 + y'^3 = 0)$, whose image is the unique exceptional prime divisor E_0 of the strict transform of $\operatorname{Spec} R$, with center o. At a general point of $(x'^2 + y'^3 = 0)$

we have $x' \neq 0$, so μ_2 acts freely there, and the strict transform is smooth and intersects $(t' = 0)$ transversally. The exceptional divisor of the weighted blow-up has log discrepancy $3+2+2+2 = 9$ over $\operatorname{Spec} k[[x,y,z,t]]$, and the order of x^2+g along it is 6, so adjunction gives $a(E_0, \operatorname{Spec} R, 0) = 9-6 = 3$. For a power series φ of weight e whose part $\varphi_{(e)}$ of lowest weight is not divisible by x^2+y^3, $\varphi = t'^e(\varphi_{(e)}(x',y',z',1) + t'\chi_2)$ for a power series χ_2, and $\varphi_{(e)}(x',y',z',1)$ does not vanish on $(x'^2+y'^3=0)$, so $\operatorname{mult}_{E_0}\varphi = e$. The parts of lowest weight of a and xd are $a_{\operatorname{mult}_0 a}$ and $xd_{\operatorname{mult}_0 d}$, of weights $2\operatorname{mult}_0 a$ and $3+2\operatorname{mult}_0 d$. Hence the part of lowest weight of $a+xd$ is $a_{\operatorname{mult}_0 a}$ if $n = 2\operatorname{mult}_0 a$, and $xd_{\operatorname{mult}_0 d}$ if $n = 3+2\operatorname{mult}_0 d$. In both cases it has degree at most one in x, so it is not divisible by x^2+y^3, and $\operatorname{mult}_{E_0} h = n$.

(2) Let $N := a^2 + gd^2 \in k[[y,z,t]]$. In R,

$$(a+xd)(a-xd) = a^2 - x^2d^2 = a^2 + gd^2 = N,$$

so $N \in hR$. Since R is a domain, $h \neq 0$, and $a - xd \neq 0$ in R (by the uniqueness in the Weierstrass division theorem by the monic polynomial x^2+g of degree two in x, as a and d are not both zero), we have $N \neq 0$ in R, hence $N \neq 0$. Since $g_3 = y^3$, the lowest homogeneous part N_0 of N is $a^2_{\operatorname{mult}_0 a}$ if $n = 2\operatorname{mult}_0 a$, and $y^3 d^2_{\operatorname{mult}_0 d}$ if $n = 3+2\operatorname{mult}_0 d$. In both cases N_0 is a non-zero form in y,z,t of degree n.

Let $p = (p_1,p_2,p_3) \in k^3$ be a general point, and ℓ_1, ℓ_2 linear forms in y,z,t whose common zero locus in k^3 is the line kp. We may assume that $N_0(p) \neq 0$, $p_1 \neq 0$, $a_{\operatorname{mult}_0 a}(p) \neq 0$ if $a \neq 0$, and $d_{\operatorname{mult}_0 d}(p) \neq 0$ if $d \neq 0$. The tangent cone of R/hR is a quotient of $k[x,y,z,t]/(x^2, N_0)$, since x^2 and N_0 are the lowest homogeneous parts of x^2+g and of $N \in (x^2+g, a+xd)$. The common zero locus of x, N_0, ℓ_1, ℓ_2 in $\mathbb{P}^3$ is empty, since x, ℓ_1, ℓ_2 vanish only at the point $(0 : p_1 : p_2 : p_3)$ and $N_0(p) \neq 0$. Hence the images of ℓ_1, ℓ_2 generate an ideal of the tangent cone of R/hR whose radical is the irrelevant ideal. Let $\mathfrak{n}$ be the maximal ideal of R/hR. Since the irrelevant ideal is finitely generated, the ideal generated by the images of ℓ_1, ℓ_2 contains the part of degree r of the tangent cone for some $r \geq 1$, that is,

$$\mathfrak{n}^r = (\ell_1, \ell_2)\mathfrak{n}^{r-1} + \mathfrak{n}^{r+1},$$

and Nakayama's lemma implies $\mathfrak{n}^r = (\ell_1,\ell_2)\mathfrak{n}^{r-1}$. Thus (ℓ_1,ℓ_2) is a reduction of $\mathfrak{n}$. Since R/hR is a Cohen–Macaulay ring of dimension two,

$$e(R/hR) = \operatorname{length} R/(h, \ell_1, \ell_2)$$

[Mat89, Theorems 14.13 and 17.11]. Let $R' := R/(\ell_1,\ell_2)$. Substituting $(y,z,t) = (p_1 w, p_2 w, p_3 w)$, we obtain $R' \cong k[[x,w]]/(x^2 + p_1^3 w^3 + w^4\psi(w))$ for a power series ψ. The ring R' is a one-dimensional local domain whose normalization is $k[[\tau]]$, where w and x map to τ^2 and τ^3 times units (we omit the parametrization $w \to \tau^2$ and $x \to \epsilon\tau^3$ of the cusp R', where ϵ is a unit with $\epsilon^2 = -p_1^3 - \tau^2\psi(\tau^2)$). Since the normalization is finite over R' with cokernel of finite length, $\operatorname{length} R'/hR'$ is the exponent e such that the image of h in $k[[\tau]]$ is τ^e times a unit. In R', $h = a(pw) + xd(pw)$, and by the choice of p the images of $a(pw)$ and $xd(pw)$ in $k[[\tau]]$ are $\tau^{2\operatorname{mult}_0 a}$ and $\tau^{3+2\operatorname{mult}_0 d}$ times units (or zero if $a=0$, resp. $d=0$). Since $2\operatorname{mult}_0 a$ and $3+2\operatorname{mult}_0 d$ have different parity, the image of h is τ^n times a unit. Thus $e(R/hR) = n$.

Since R is normal, the localization of R at the ideal I_D of a prime divisor D is a discrete valuation ring, and the length of $(R/hR)_{I_D}$ is $\operatorname{mult}_D h$. The minimal primes of hR are the ideals I_D of the prime divisors D with $\operatorname{mult}_D h > 0$, and $\dim R/I_D = 2 = \dim R/hR$. By the associativity formula for multiplicities [Mat89, Theorem 14.7],

$$e(R/hR) = \sum_D \operatorname{mult}_D h \cdot e(R/I_D).$$

□

The proof of Theorem 10.5 is by contradiction: a sequence on which the margin fails has a generic limit whose minimal log discrepancy at the closed point is still at least $1+\epsilon$, the prime divisor E_0 bounds the number of terms of the boundary and the orders of the h_j along

E_0, Lemma D.18(2) implies that any component of the limit boundary with coefficient one is smooth, Lemma 9.2 excludes such a component, and log canonicity of the limit with slightly larger coefficients passes back to the sequence.

Proof of Theorem 10.5. The proof is long, so we divide it into several steps.

Step 1. In this step we fix formal coordinates for the data of Theorem 10.5 and bound the number s of terms of Δ and the orders of the h_j along E_0.

Let $\mathcal{A} := \operatorname{Spec} \mathbb{C}[[x,y,z,t]]$ with closed point o. Fix $\gamma \in (0,1]$ and $\epsilon \in (0,2)$. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form, and $\Delta = \sum_{j=1}^{s} c_j \operatorname{div}(h_j)$ with non-zero $h_j \in \mathfrak{m}_o$ and $\gamma \le c_j \le 1$. Assume that $\operatorname{mld}(W \ni o, \Delta) \ge 1+\epsilon$. The quadratic rank and the residual cubic of $(W \ni o)$ are as recalled in Section 2 [Liu26, Definition-Lemma 11.1]. Completing the square in x, applying the Weierstrass preparation theorem in y (the residual cubic is y^3), and removing the term in y^2, we obtain formal coordinates x, y, z, t such that

$$\widehat{\mathcal{O}}_{W,o} \cong \mathbb{C}[[x,y,z,t]]/(f), \qquad f = x^2 + y^3 + yA(z,t) + B(z,t), \qquad \operatorname{mult}_0 A \ge 3, \quad \operatorname{mult}_0 B \ge 4.$$

For an integer m, let A_m and B_m be the homogeneous parts of degree m of A and B. We show that one of the following holds:

- (E_6) $B_4 \neq 0$.
- (E_7) $B_4 = 0$ and $A_3 \neq 0$.
- (E_8) $B_4 = A_3 = 0$ and $B_5 \neq 0$.

The cases (E_6), (E_7), and (E_8) are those where a general hyperplane section of W through o is a Du Val point of type E_6, E_7, and E_8 respectively (cf. [KM98, Corollary 5.38]). Suppose that $B_4 = A_3 = B_5 = 0$. Then any monomial of f has weight at least 6 with respect to the weights $(3,2,1,1)$ on (x,y,z,t). Let E be the exceptional prime divisor of the weighted blow-up of $\mathcal{A}$ with weights $(3,2,1,1)$. Then $a(E,\mathcal{A},0) = 7$ and $\operatorname{mult}_E f \ge 6$. Since $\widehat{\mathcal{O}}_{W,o}$ is a normal domain, Lemma D.14 with $k = \mathbb{C}$ and [dFEM11, Lemma 2.21 and Proposition A.14] imply

$$\operatorname{mld}(W \ni o, 0) = \operatorname{mld}(\mathcal{A} \ni o, (f)) \le a(E,\mathcal{A},0) - \operatorname{mult}_E f \le 1.$$

This is impossible, since W is Gorenstein and terminal, so that $a(F,W,0) \ge 2$ for any prime divisor F over W with center o.

Since f is monic of degree two in x, the Weierstrass division theorem implies that there exist unique $a_j, d_j \in \mathbb{C}[[y,z,t]]$ such that h_j is the image of $a_j + xd_j$ in $\widehat{\mathcal{O}}_{W,o}$. Since $h_j \neq 0$, a_j and d_j are not both zero. Let

$$n_j := \min\{2 \operatorname{mult}_0 a_j, 3 + 2 \operatorname{mult}_0 d_j\}.$$

Since $h_j \in \mathfrak{m}_o$, $a_j(0) = 0$, so $n_j \ge 2$. By Lemma D.18(1) and [dFEM11, Lemma 2.21 and Proposition A.14], E_0 is a prime divisor over W with center o, $a(E_0,W,0) = 3$, and $\operatorname{mult}_{E_0} h_j = n_j$. The assumption $\operatorname{mld}(W \ni o, \Delta) \ge 1+\epsilon$ implies $3 - \sum_j c_j n_j = a(E_0,W,\Delta) \ge 1+\epsilon$, that is,

$$\sum_{j=1}^{s} c_j n_j \le 2 - \epsilon. \tag{D.40}$$

Since $c_j \ge \gamma$ and $n_j \ge 2$, we obtain $s < 1/\gamma$ and $n_j < 2/\gamma$ for any j.

Step 2. In this step we construct a generic limit of a sequence on which the conclusion fails, and show that the limit hypersurface is normal.

Suppose that for any real number $\theta > 1$ there exist $(W \ni o)$ and Δ satisfying the assumptions of Theorem 10.5 such that $(W, \theta\Delta)$ is not lc near o. Taking $\theta = 1 + \frac{1}{i}$, for any positive integer i there exist $(W_i \ni o)$ and $\Delta_i = \sum_{j=1}^{s_i} c_{i,j} \operatorname{div}(h_{i,j})$ satisfying the assumptions of Theorem 10.5 such that $(W_i, (1+\frac{1}{i})\Delta_i)$ is not lc near o. Since W_i is terminal, $\Delta_i \neq 0$, so $s_i \ge 1$. Let $f_i = x^2 + y^3 + yA_i + B_i$, $a_{i,j}$, $d_{i,j}$, and $n_{i,j}$ be as in **Step 1**. By **Step 1**, $s_i < 1/\gamma$ and $n_{i,j} < 2/\gamma$. Thus, after passing to a subsequence, we may assume that $s_i = s$ and $n_{i,j} = n_j$ do not depend

on i, that a single case among (E_6), (E_7), and (E_8) holds for any i, and that $c_{i,j} \to c_{*,j} \in [\gamma, 1]$ for any j. Since there are finitely many monomials of a fixed degree, after passing to a further subsequence we may also fix a monomial whose coefficient is non-zero for any i in $B_{i,4}$, $A_{i,3}$, or $B_{i,5}$ according to the case, and, for any j, a monomial of degree $n_j/2$ whose coefficient in $a_{i,j}$ is non-zero for any i if n_j is even, and a monomial of degree $(n_j - 3)/2$ whose coefficient in $d_{i,j}$ is non-zero for any i if n_j is odd.

The results of [Kaw21, Section 3] hold for the germ of a klt variety of any dimension, and [Kaw21, Remark 3.3] treats ideals in the completion. We use them for the germ $0 \in \mathbb{A}^4$, with the principal ideals (f_i) as one entry of the tuples. For an integer $l \geq 1$, let $\mathcal{T}_l$ be the affine space of $(s+1)$-tuples of complex polynomials in x, y, z, t of degree less than l, and $\varphi_{i,l} \in \mathcal{T}_l$ the tuple of the truncations in degree less than l of $(f_i, h_{i,1}, \dots, h_{i,s})$. After passing to a subsequence, there exist infinite sets N_l of indices with $N_{l+1} \subset N_l$ and irreducible locally closed subsets $Z_l \subset \mathcal{T}_l$ such that $\{\varphi_{i,l}\}_{i \in N_l}$ is a dense subset of Z_l and the truncation maps $Z_{l+1} \to Z_l$ are dominant: inductively, Z_{l+1} is the intersection of the inverse image of Z_l with an irreducible component of the closure of $\{\varphi_{i,l+1}\}_{i \in N_l}$ which dominates Z_l and contains $\varphi_{i,l+1}$ for infinitely many $i \in N_l$ (such a component exists by the pigeonhole principle if Z_l is a point, and if $\dim Z_l \geq 1$, any component C dominating Z_l has $\dim C \geq 1$ and contains the dense subset $C \cap \{\varphi_{i,l+1}\}_{i \in N_l}$ of C, which is infinite), and N_{l+1} is the set of $i \in N_l$ with $\varphi_{i,l+1} \in Z_{l+1}$. By generic flatness, after shrinking the Z_l to dense open subsets compatibly with the truncation maps and the N_l accordingly, the ideals on $\mathbb{A}^4 \times Z_l$ generated by $\mathfrak{m}_0^l$ and by an entry of the universal tuple are flat over Z_l, where $\mathfrak{m}_0$ is the ideal of the origin. This is a family of approximations of the sequence of tuples $((f_i), (h_{i,1}), \dots, (h_{i,s}))$ in the sense of [Kaw21, Definition 3.1 and Remark 3.3].

Let k_0 be the union of the function fields of the Z_l, and $(f_*, h_{*,1}, \dots, h_{*,s})$ the tuple of power series over k_0 whose truncation in degree less than l is the generic point of Z_l for any l. Let $\hat{\mathfrak{m}}$ be the maximal ideal of $k_0[[x, y, z, t]]$. The generic limit of $((f_i))_i$ [Kaw21, Definition 3.2] is $\varprojlim_l ((f_*) + \hat{\mathfrak{m}}^l) = (f_*)$, since (f_*) is closed in the $\hat{\mathfrak{m}}$-adic topology [Mat89, Theorem 8.10(i)], and similarly the generic limit of $((h_{i,j}))_i$ is $(h_{*,j})$. Let k be an algebraic closure of k_0. Log discrepancies and minimal log discrepancies at the closed point do not change under the base change from $k_0[[x, y, z, t]]$ to $k[[x, y, z, t]]$ [Kaw21, Remark 2.6]. The vanishing of a fixed coefficient is a closed condition, so it holds on Z_l if it holds for any $i \in N_l$. A fixed coefficient that is non-zero for any i does not vanish identically on Z_l, so it is non-zero at the generic point of Z_l. Thus

$$f_* = x^2 + y^3 + yA_*(z, t) + B_*(z, t), \qquad \operatorname{mult}_0 A_* \geq 3, \quad \operatorname{mult}_0 B_* \geq 4,$$

the case among (E_6), (E_7), and (E_8) that holds for all f_i holds for A_* and B_*, and $h_{*,j} = a_{*,j} + xd_{*,j}$ with $a_{*,j}, d_{*,j} \in k[[y, z, t]]$ and

$$\min\{2 \operatorname{mult}_0 a_{*,j}, 3 + 2 \operatorname{mult}_0 d_{*,j}\} = n_j. \tag{D.41}$$

Let $g_* := y^3 + yA_* + B_*$, $R_* := k[[x, y, z, t]]/(f_*)$, $W_* := \operatorname{Spec} R_*$, $\mathcal{A}_k := \operatorname{Spec} k[[x, y, z, t]]$, and

$$\Delta_* := \sum_{j=1}^{s} c_{*,j} \operatorname{div}(h_{*,j}).$$

We show that R_* is a normal domain. Suppose that g_* is divisible by the square of a non-unit of $k[[y, z, t]]$. Since g_* is a Weierstrass polynomial of degree three in y, the Weierstrass preparation theorem implies $g_* = (y + \alpha)^2(y + \beta)$ with $\alpha, \beta \in (z, t)k[[z, t]]$. As g_* has no term in y^2, $\beta = -2\alpha$, so $A_* = -3\alpha^2$ and $B_* = -2\alpha^3$. Since $\operatorname{mult}_0 A_* \geq 3$, $\operatorname{mult}_0 \alpha \geq 2$, hence $A_{*,3} = B_{*,4} = B_{*,5} = 0$, which contradicts each of (E_6), (E_7), and (E_8). Thus g_* is squarefree. Let $\mathfrak{p}$ be a height-one prime of R_*, and π a generator of the height-one prime $\mathfrak{p} \cap k[[y, z, t]]$. If $\pi \nmid g_*$, then R_* is étale over $k[[y, z, t]]$ at $\mathfrak{p}$, since $2g_*$ is a unit at (π). If $\pi \mid g_*$, then $\pi^2 \nmid g_*$, and $k[[y, z, t]]_{(\pi)}[x]/(x^2 + g_*)$ is a discrete valuation ring with uniformizer x by the Eisenstein criterion. In both cases $(R_*)_{\mathfrak{p}}$ is regular. Since R_* is a hypersurface, it is Cohen–Macaulay, so R_* is normal by Serre's criterion. A normal local ring is a domain.

Step 3. In this step we show

$$\text{(D.42)} \qquad \operatorname{mld}(W_* \ni o, \Delta_*) \geq 1+\epsilon \qquad \text{and} \qquad \sum_{j=1}^{s} c_{*,j} n_j \leq 2-\epsilon.$$

The second inequality follows from (D.40) for W_i by letting $i \to +\infty$. Let F be a prime divisor over $\mathcal{A}_{k_0} := \operatorname{Spec} k_0[[x,y,z,t]]$ with center o. By [Kaw21, Remark 3.8(i)], there exist infinitely many i and prime divisors F_i over $\mathcal{A}$ with center o such that $a(F_i, \mathcal{A}, 0) = a(F, \mathcal{A}_{k_0}, 0)$, $\operatorname{mult}_{F_i} f_i = \operatorname{mult}_F f_*$, and $\operatorname{mult}_{F_i} h_{i,j} = \operatorname{mult}_F h_{*,j}$ for any j (the center of F_i is o, as F_i is obtained by restricting to the fiber over $\varphi_{i,l}$ a prime divisor over $\mathbb{A}^4 \times Z_l$ with center $o \times Z_l$ in [Kaw21, Remark 3.8(i)]). Since $\widehat{\mathcal{O}}_{W_i,o} = \mathbb{C}[[x,y,z,t]]/(f_i)$ is a normal domain and $h_{i,j} \notin (f_i)$, Lemma D.14 with $k = \mathbb{C}$ and [dFEM11, Lemma 2.21 and Proposition A.14] imply

$$a\Big(F_i, \mathcal{A}, (f_i) \cdot \prod_j (h_{i,j})^{c_{i,j}}\Big) \geq \operatorname{mld}\Big(\mathcal{A} \ni o, (f_i) \cdot \prod_j (h_{i,j})^{c_{i,j}}\Big) = \operatorname{mld}(W_i \ni o, \Delta_i) \geq 1+\epsilon.$$

The left-hand side is

$$a(F, \mathcal{A}_{k_0}, 0) - \operatorname{mult}_F f_* - \sum_j c_{i,j} \operatorname{mult}_F h_{*,j}.$$

Letting $i \to +\infty$ along the infinite set of such indices, we obtain

$$a\Big(F, \mathcal{A}_{k_0}, (f_*) \cdot \prod_j (h_{*,j})^{c_{*,j}}\Big) \geq 1+\epsilon.$$

Since F is arbitrary, the minimal log discrepancy at o of $\mathcal{A}_{k_0}$ with the $\mathbb{R}$-ideal in the last display is at least $1+\epsilon$, and [Kaw21, Remark 2.6] implies

$$\operatorname{mld}\Big(\mathcal{A}_k \ni o, (f_*) \cdot \prod_j (h_{*,j})^{c_{*,j}}\Big) \geq 1+\epsilon.$$

By (D.41), $a_{*,j}$ and $d_{*,j}$ are not both zero, so $h_{*,j} \notin (f_*)$ by the uniqueness in the Weierstrass division theorem. Since R_* is a normal domain (**Step 2**), Lemma D.14 over k implies the first inequality of (D.42).

Step 4. In this step we show $\lfloor \Delta_* \rfloor = 0$.

Let D be a prime divisor on W_*, and $c_D := \sum_j c_{*,j} \operatorname{mult}_D h_{*,j}$ the coefficient of D in Δ_*. Suppose that $c_D \geq 1$. By (D.41), $\operatorname{mult}_0 a_{*,j} \geq 1$, so $h_{*,j}$ belongs to the maximal ideal of R_*. Since $g_* - y^3 \in (y,z,t)^4$, Lemma D.18(2) and (D.41) imply

$$\sum_j c_{*,j} n_j = \sum_j c_{*,j} \sum_{D'} \operatorname{mult}_{D'} h_{*,j} \cdot e(R_*/I_{D'}) = \sum_{D'} c_{D'} e(R_*/I_{D'}) \geq c_D e(R_*/I_D) \geq e(R_*/I_D),$$

where D' runs over the prime divisors on W_*. By (D.42), $e(R_*/I_D) \leq 2-\epsilon < 2$, so $e(R_*/I_D) = 1$. The ring R_*/I_D is a complete local domain of dimension two, hence unmixed, so it is regular by Nagata's criterion [Nag62, (40.6)]. Since $\operatorname{mult}_0 g_* = 3$ and $g_{*,3} = y^3 \neq 0$, Lemma 9.2 over k, for R_*, g_*, D, and Δ_*, implies

$$\operatorname{mld}(W_* \ni o, \Delta_*) \leq 2 - c_D \leq 1,$$

which contradicts (D.42). Thus $c_D < 1$ for any prime divisor D on W_*, that is, $\lfloor \Delta_* \rfloor = 0$.

Step 5. In this step we show that $(W_i, (1+\frac{1}{i})\Delta_i)$ is lc near o for infinitely many i, which contradicts the choice of the sequence.

The scheme W_* is the spectrum of a normal complete local domain of dimension three (**Step 2**), and K_{W_*} is Cartier since W_* is a hypersurface. By **Steps 3 and 4**, $\lfloor \Delta_* \rfloor = 0$ and $\operatorname{mld}(W_* \ni o, \Delta_*) \geq 1+\epsilon > 1$. By Lemma D.16, there exists a real number $\theta_0 > 1$ such that $(W_*, \theta_0 \Delta_*)$ is klt. In particular $\operatorname{mld}(W_* \ni o, \theta_0 \Delta_*) \geq 0$. By Lemma D.14 over k,

$$\operatorname{mld}\Big(\mathcal{A}_k \ni o, (f_*) \cdot \prod_j (h_{*,j})^{\theta_0 c_{*,j}}\Big) = \operatorname{mld}(W_* \ni o, \theta_0 \Delta_*) \geq 0,$$

and [Kaw21, Remark 2.6] implies

$$\operatorname{mld}\Big(\mathcal{A}_{k_0} \ni o, (f_*) \cdot \prod_j (h_{*,j})^{\theta_0 c_{*,j}}\Big) \geq 0.$$

Thus $\mathcal{A}_{k_0}$ with the $\mathbb{R}$-ideal in the last display is lc by Lemma D.13(3). The exponents $1, \theta_0 c_{*,1}, \dots, \theta_0 c_{*,s}$ are fixed positive real numbers, so [Kaw21, Theorem 3.10] implies that $(\mathcal{A}, (f_i) \cdot \prod_j (h_{i,j})^{\theta_0 c_{*,j}})$ is lc for infinitely many i. For any such i,

$$\operatorname{mld}\Big(\mathcal{A} \ni o, (f_i) \cdot \prod_j (h_{i,j})^{\theta_0 c_{*,j}}\Big) \geq 0.$$

On $\operatorname{Spec} \widehat{\mathcal{O}}_{W_i,o}$, the $\mathbb{R}$-divisor $\theta_0 \sum_j c_{*,j} \operatorname{div}(h_{i,j})$ has the log discrepancies of the $\mathbb{R}$-ideal $\prod_j (h_{i,j})^{\theta_0 c_{*,j}}$, so Lemma D.14 with $k = \mathbb{C}$, Lemma D.13(3) for this $\mathbb{R}$-ideal, and [dFEM11, Proposition 2.11(i)] imply that $(W_i, \theta_0 \sum_j c_{*,j} \operatorname{div}(h_{i,j}))$ is lc near o.

Let $\theta_1 := (1+\theta_0)/2$. Since $c_{i,j} \to c_{*,j} \geq \gamma > 0$ and $\theta_0/\theta_1 > 1$, for i large we have $\theta_1 c_{i,j} \leq \theta_0 c_{*,j}$ for any j. Since $\operatorname{div}(h_{i,j}) \geq 0$ near o for any j,

$$\Big(1 + \frac{1}{i}\Big)\Delta_i \leq \theta_1 \Delta_i \leq \theta_0 \sum_j c_{*,j} \operatorname{div}(h_{i,j})$$

near o for i large with $1 + \frac{1}{i} \leq \theta_1$. Thus $(W_i, (1 + \frac{1}{i})\Delta_i)$ is lc near o for infinitely many i, which contradicts the choice of $(W_i \ni o)$ and Δ_i in **Step 2**. Hence there exists a real number $\theta = \theta(\gamma, \epsilon) > 1$ such that $(W, \theta\Delta)$ is lc near o for any $(W \ni o)$ and Δ satisfying the assumptions of Theorem 10.5.

Step 6. In this step we prove the remaining inequalities.

Let $\eta := (\theta - 1)/\theta$, and let P be a prime divisor over W whose center contains o. Let $m_P := \sum_j c_j \operatorname{mult}_P h_j$. Since $(W, \theta\Delta)$ is lc near o and the center of P contains o, $a(P, W, \theta\Delta) = a(P, W, 0) - \theta m_P \geq 0$. Thus

$$a(P, W, \Delta) = a(P, W, 0) - m_P \geq \Big(1 - \frac{1}{\theta}\Big) a(P, W, 0) = \eta a(P, W, 0).$$

Let G be a finite group fixing o and preserving Δ, $L > 0$ a real number, and P a prime divisor over W with center o such that $0 < a(P, W, \Delta) \leq L |I_G(P)|$. Then

$$a(P, W, 0) \leq \frac{a(P, W, \Delta)}{\eta} \leq \frac{L}{\eta} |I_G(P)|,$$

and

$$\sum_j c_j \operatorname{mult}_P h_j = a(P, W, 0) - a(P, W, \Delta) \leq \frac{1-\eta}{\eta} a(P, W, \Delta) \leq \frac{(1-\eta)L}{\eta} |I_G(P)|.$$

□

D.6. **Proof of the class theorem: gaps, margins and the case $CD = 0$.** In this subsection, we prove Lemmas D.20 and D.21 and Propositions D.22 and D.23, which reduce Theorem 10.1 to the circles $(3s, 2s, C, D)$ with $C < 0 < D$. The finiteness of the set of labels enters through Theorem 2.13, in the following form.

Lemma D.19. *Fix a finite set $S \subset [0, 1]$ and a positive integer M. Then there exists a subset $\mathcal{D}_{S,M} \subset [0, +\infty)$ whose intersection with any bounded interval is finite, with the following property. Let $(W \ni o)$ be a normal Gorenstein threefold germ, G a finite group of order at most M acting faithfully on W and fixing o, and $B = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in S$ and non-zero G-semi-invariant regular functions h_j, such that (W, B) is lc near o. Then $\operatorname{mld}((W \ni o, B)/G) \in \mathcal{D}_{S,M}$, and there exists a prime divisor F over W with center o such that $a(F, W, B) = |I_G(F)| \operatorname{mld}((W \ni o, B)/G)$.*

Proof. Let $\mathcal{D}_0 \subset \mathbb{R}$ be the set of Theorem 2.13(2) for $d = 3$, $r = 1$, and $I = S$. By Theorem 2.13(2) it has no accumulation point in $\mathbb{R}$, hence it is finite in any bounded interval. Let

$$\mathcal{D}_{S,M} := \left\{ \frac{a}{e} \;\middle|\; a \in \mathcal{D}_0 \cap [0, +\infty),\ e \in \{1, \dots, M\} \right\},$$

which is finite in any bounded interval. Since K_W is Cartier and each $\operatorname{div}(h_j) \geq 0$ is a Cartier divisor, $a(F, W, B) \in \mathcal{D}_0 \cap [0, +\infty)$ for any prime divisor F over W. By [Liu26, Lemma 2.6],

$$\operatorname{mld}((W \ni o, B)/G) = \inf_F \frac{a(F, W, B)}{|I_G(F)|},$$

where F runs over the prime divisors over W with center o, and $|I_G(F)| \leq |G| \leq M$. Thus all the quotients in the infimum belong to $\mathcal{D}_{S,M}$. Since $\mathcal{D}_{S,M}$ has only finitely many elements below any given real number, the infimum is a minimum, and it belongs to $\mathcal{D}_{S,M}$. □

Lemma D.20. *Fix a finite set $S \subset [0, 1]$. There exist real numbers $\epsilon_S \in (0, 2)$ and $\theta_S > 1$ with the following property. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form, and $B = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in S$ and non-zero regular functions h_j vanishing at o, such that (W, B) is terminal near o. Let E_0 be the prime divisor over W of Lemma D.18(1), taken in coordinates of the completion of W at o in which the equation is $x^2 + g$ with $g - y^3 \in (y, z, t)^4$. Then $\operatorname{mld}(W \ni o, B) \geq 1 + \epsilon_S$, $\sum_j c_j \operatorname{mult}_{E_0} h_j \leq 2 - \epsilon_S$, and $(W, \theta_S B)$ is lc near o.*

Proof. Let $\mathcal{D}_{S,1}$ be the set of Lemma D.19 for $M = 1$. Since $\mathcal{D}_{S,1} \cap (1, 3]$ is finite, there exists $\epsilon_S \in (0, 1]$ such that $\mathcal{D}_{S,1} \cap (1, 1 + \epsilon_S) = \emptyset$. If $S \subset \{0\}$, let $\theta_S := 2$. Otherwise let $\gamma := \min(S \setminus \{0\})$, and let $\theta_S := \theta(\gamma, \epsilon_S) > 1$ be the number of Theorem 10.5.

Let $(W \ni o)$ and B be as in the lemma. Removing the terms with $c_j = 0$ does not change B, so we may assume that $c_j \geq \gamma$ for any j, and that $B = 0$ if $S \subset \{0\}$. Since (W, B) is terminal near o, it is lc near o. By Lemma D.19 with the trivial group, there exists a prime divisor F over W with center o such that $a(F, W, B) = \operatorname{mld}(W \ni o, B) \in \mathcal{D}_{S,1}$. Since F is exceptional over W and (W, B) is terminal near o, $a(F, W, B) > 1$. In the completed local ring of W at o, after completing the square in x and a linear change of y, z, t, the equation of W is $x^2 + g$ with $g - y^3 \in (y, z, t)^4$, since the residual cubic is the cube of a linear form. We work in the coordinates of the statement, in which E_0 is taken. Any element h of the maximal ideal is the image of $a + xd$ with $a, d \in \mathbb{C}[[y, z, t]]$. By Lemma D.18(1) and [dFEM11, Lemma 2.21 and Proposition A.14], E_0 is a prime divisor over W with center o, $a(E_0, W, 0) = 3$, and

$$\operatorname{mult}_{E_0} h = \min\{2 \operatorname{mult}_0 a, 3 + 2 \operatorname{mult}_0 d\} \geq 2.$$

Hence

$$1 < \operatorname{mld}(W \ni o, B) \leq a(E_0, W, B) = 3 - \sum_j c_j \operatorname{mult}_{E_0} h_j \leq 3.$$

Thus $\operatorname{mld}(W \ni o, B) \in \mathcal{D}_{S,1} \cap (1, 3]$, so $\operatorname{mld}(W \ni o, B) \geq 1 + \epsilon_S$, and the display implies $\sum_j c_j \operatorname{mult}_{E_0} h_j \leq 2 - \epsilon_S$. If $B = 0$, then $(W, \theta_S B) = (W, 0)$ is lc. Otherwise $\gamma \leq c_j \leq 1$, each h_j is a non-zero element of $\mathfrak{m}_o$, and $\operatorname{mld}(W \ni o, B) \geq 1 + \epsilon_S$, so $(W, \theta_S B)$ is lc near o by Theorem 10.5. □

Recall that in Construction-Theorem 4.13, $n \in \{3, 4\}$ is the dimension of a smooth germ $(Y \ni o)$ containing W, the data (ii) are a G-equivariant closed embedding $W \subset Y$ with $W = (f = 0)$ if $n = 4$ and G-eigenparameters $x_1, \dots, x_n$ of Y at o, (iii) are G-semi-invariant regular functions h_k on Y whose restrictions to W are marks, with positive labels c_k, such that $(W, \sum_k c_k \operatorname{div}(h_k|_W))$ is klt near o, (iv) are finitely many further label vectors with klt pairs, (v) are finitely many prime divisors over W with center o, and (vi) are a G-equivariant automorphism ψ of $\mathbb{C}[[x_1, \dots, x_n]]$, a G-invariant unit v, and units u_k with trivial character.

Lemma D.21. *Fix a finite set $S \subset [0,1]$ and real numbers $1 < \theta_1 < \theta_2 < \theta_3 \le \theta_S$, where θ_S is the number given by Lemma D.20. Let $(W \ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form, G a finite cyclic group acting faithfully on W and fixing o, and $B = \sum_j c_j \operatorname{div}(h_j)$ a labelled boundary (Definition 2.7) with $c_j \in S$ such that (W, B) is terminal near o. Fix data as in (ii), (v), and (vi) of Construction-Theorem 4.13 for $(W, (h_j)_j)$, together with G-semi-invariant lifts to Y of the marks h_j, and let c be the vector of the positive labels c_j. Then there exists an integer N_2 such that for any integer $N \ge N_2$ there exists an integer L_2 such that for any integer $L \ge L_2$ the following holds. Let $h_{j,0}$ and f_0 be G-semi-invariant polynomials satisfying the congruences of Construction-Theorem 4.13 with N and L for the fixed data (ii), (v), and (vi) and the fixed lifts of the marks, and let $W_0 := (f_0 = 0) \subset \mathbb{A}^4$ and $B_0 := \sum_j c_j \operatorname{div}(h_{j,0}|_{W_0})$. Then the conclusions of Construction-Theorem 4.13 hold for $(W_0, (h_{j,0})_j)$, with the fixed lifts of the marks of positive label and the labels c in (iii) and the vectors $\theta_1 c$ and $\theta_2 c$ in (iv). In particular, W_0 is isolated Gorenstein terminal, $(W_0, \theta_2 B_0)$ is lc near 0, and $(W_0, \theta_1 B_0)$ is klt near 0.*

Proof. Removing the marks with label zero changes neither B nor B_0, so we may assume that $c_j > 0$ for any j. The marks that do not vanish at o are units near o and do not change B near o, so Lemma D.20, applied to the marks with positive label that vanish at o, shows that $(W, \theta_S B)$ is lc near o. Let $\theta \in [0, \theta_S)$. For any prime divisor E over W whose center contains o,

$$a(E, W, \theta B) = \Big(1 - \frac{\theta}{\theta_S}\Big) a(E, W, 0) + \frac{\theta}{\theta_S} a(E, W, \theta_S B) > 0,$$

since $a(E, W, 0) > 0$ and $a(E, W, \theta_S B) \ge 0$. Hence $(W, \theta B)$ is klt near o. In particular, (W, B), $(W, \theta_1 B)$, and $(W, \theta_2 B)$ are klt near o, because $\theta_1 < \theta_2 < \theta_3 \le \theta_S$.

Since W has quadratic rank one, it is singular at o, so $n = 4$ in Construction-Theorem 4.13. We use Construction-Theorem 4.13 with the fixed data (ii), (v), (vi), with the fixed G-semi-invariant lifts to Y of the marks h_j and the labels $c := (c_j)_j$ in (iii), and with $c^{(1)} := \theta_1 c$ and $c^{(2)} := \theta_2 c$ in (iv). Let N_2 be the integer N_1 of Construction-Theorem 4.13 for the data (ii)–(vi) above and, for $N \ge N_2$, let L_2 be the integer L_1 of Construction-Theorem 4.13 for N. For $N \ge N_2$ and $L \ge L_2$, the conclusions of Construction-Theorem 4.13 hold for $(W_0, (h_{j,0})_j)$ with the data (ii)–(vi) above. In particular, conclusion (1) for $j = 1, 2$ implies that $(W_0, \theta_1 B_0)$ and $(W_0, \theta_2 B_0)$ are klt near 0, hence $(W_0, \theta_2 B_0)$ is lc near 0, and conclusion (3) implies that W_0 is an isolated Gorenstein terminal germ at 0. □

The next proposition reduces an increasing sequence of quotient values to one circle.

Proposition D.22. *Fix a finite set $S \subset [0,1]$ and real numbers $1 < \theta_1 < \theta_2 < \theta_3 \le \theta_S$, where θ_S is the number given by Lemma D.20. For any $i \ge 1$, let $(W_i \ni o_i)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form, let G_i be a finite cyclic group acting faithfully on W_i and fixing o_i, and let $\Delta_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$ be a labelled boundary (Definition 2.7) with $c_{ij} \in S$ such that (W_i, Δ_i) is terminal near o_i. Assume that the numbers $\alpha_i := \operatorname{mld}((W_i \ni o_i, \Delta_i)/G_i)$ form a strictly increasing sequence in $(0,1)$. Then $|G_i| \to +\infty$. Moreover, after passing to a subsequence, replacing the data by numerical admissions for the groups G_i (Construction-Theorem 4.13 and Lemma D.21), and exchanging z and t if necessary, the labels $c_j := c_{ij}$ and their number do not depend on i, each $W_i = (x^2 + y^3 + yA_i(z,t) + B_i(z,t) = 0)$ is an isolated Gorenstein terminal polynomial germ with G_i diagonal, $\operatorname{mult} A_i \ge 3$, and $\operatorname{mult} B_i \ge 4$, all in one row ($E_6$), ($E_7$), or ($E_8$) of Construction-Proposition 10.3(1). The numerical admissions do not change the numbers α_i, the marks h_{ij} of the new data are G_i-semi-invariant polynomials, $(W_i, \theta_1 \Delta_i)$ is klt near 0, and $(W_i, \theta_2 \Delta_i)$ is lc near 0. One of the following holds.*

(1) There is $e_0 \ge 1$ such that $|G_i \cap \mathbf{T}_{(3,2,6,0)}| \to +\infty$ and $[G_i : G_i \cap \mathbf{T}_{(3,2,6,0)}] \le e_0$, and either $B_{i,4}$ is a non-zero multiple of zt^3, or $B_{i,4} = A_{i,3} = 0$ and $B_{i,5}$ is a non-zero multiple of zt^4.

(2) *There are integers* $s \geq 0$, $C < 0 < D$ *with* $\gcd(s, C, D) = 1$, *and* $e_0 \geq 1$ *such that* $|G_i \cap \mathbf{T}| \to +\infty$ *and* $[G_i : G_i \cap \mathbf{T}] \leq e_0$ *for* $\mathbf{T} := \mathbf{T}_{(3s,2s,C,D)}$, *and the non-zero homogeneous part* $B_{i,4}$, $A_{i,3}$, *or* $B_{i,5}$ *of the row is* $b_i z^a t^b$ *with* $b_i \in \mathbb{C}^*$, $a, b \geq 1$, *and* $6s = aC + bD$, $4s = aC + bD$, *or* $6s = aC + bD$, *respectively.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the labels and show that $|G_i| \to +\infty$. Removing the marks with label zero and the marks that do not vanish at o_i does not change Δ_i near o_i, so we may assume that $c_{ij} \geq \gamma := \min(S \backslash \{0\})$ and $h_{ij}(o_i) = 0$ for any i, j (if $S \subset \{0\}$, then $\Delta_i = 0$, and the steps below hold without marks). Since $h_{ij}(o_i) = 0$, Lemma D.18(1) implies $\operatorname{mult}_{E_0} h_{ij} \geq 2$, in coordinates of the completion of W_i at o_i in which the equation is $x^2 + g$ with $g - y^3 \in (y, z, t)^4$. By Lemma D.20,

$$2\gamma \cdot \#\{j\} \leq \sum_j c_{ij} \operatorname{mult}_{E_0} h_{ij} \leq 2 - \epsilon_S < 2.$$

Thus there are fewer than $1/\gamma$ marks, and there are only finitely many possible vectors of labels. Passing to a subsequence, we may assume that the labels $c_j := c_{ij}$ and their number do not depend on i.

If there exists an integer M such that $|G_i| \leq M$ for infinitely many i, then for any such i, α_i belongs to the finite set $\mathcal{D}_{S,M} \cap [0, 1]$ of Lemma D.19, since W_i is Gorenstein and (W_i, Δ_i) is lc near o_i. This contradicts the strict increase of α_i. Hence $|G_i| \to +\infty$.

Step 2. In this step we bring W_i to the polynomial form of the proposition by numerical admissions. By [Liu26, Lemma 11.3(1)], there exist a G_i-equivariant closed embedding of W_i into a smooth fourfold germ $(Y_i \ni o_i)$, a G_i-semi-invariant regular function f_i on Y_i with $W_i = (f_i = 0)$, G_i-semi-invariant regular parameters of Y_i at o_i, and G_i-semi-invariant elements x, y, z, t of $\widehat{\mathcal{O}}_{Y_i,o_i}$, such that $\widehat{\mathcal{O}}_{Y_i,o_i} = \mathbb{C}[[x, y, z, t]]$, G_i acts diagonally on x, y, z, t, and the expansion of f_i in x, y, z, t is $x^2 + g_i(y, z, t)$, where $\operatorname{mult} g_i \geq 3$ and the cubic part $g_{i,3}$ of g_i is the residual cubic. Let $\hat{f}_i$ be the expansion of f_i in the regular parameters, let ψ_i' be the G_i-equivariant automorphism of $\mathbb{C}[[x, y, z, t]]$ which replaces the regular parameters by x, y, z, t, where the regular parameters are ordered so that the k-th regular parameter has the character of the k-th of x, y, z, t (possible since the regular parameters and x, y, z, t induce bases of eigenvectors of the cotangent space of Y_i at o_i, so their characters, counted with multiplicities, coincide). Then $\psi_i'(\hat{f}_i) = x^2 + g_i$. By assumption $g_{i,3} = l_i^3$ for a linear form l_i. Since $g_{i,3}$ is G_i-semi-invariant, l_i is a G_i-eigenvector. After a G_i-equivariant linear change of y, z, t (reordering the regular parameters accordingly), we may assume that $l_i = y$, with $\hat{f}_i$, ψ_i', and g_i taken for the new coordinates x, y, z, t and the reordered regular parameters.

By the Weierstrass preparation theorem in y, $g_i = U_i \cdot (y^3 + a_{i,2}y^2 + a_{i,1}y + a_{i,0})$ with a unit U_i, $U_i(0) = 1$, and $a_{i,k} \in \mathbb{C}[[z, t]]$. The uniqueness in the Weierstrass preparation theorem implies that U_i is G_i-invariant and that each $a_{i,k}y^k$ has the character of y^3. Since $g_{i,3} = y^3$, we have $\operatorname{mult} a_{i,2} \geq 2$, $\operatorname{mult} a_{i,1} \geq 3$, and $\operatorname{mult} a_{i,0} \geq 4$. Let ψ_i be the composite of ψ_i' and the G_i-equivariant automorphism which maps y to $y - a_{i,2}/3$, x to $\tilde{U}_i^{1/2}x$, and fixes z, t, where $\tilde{U}_i := U_i(y - a_{i,2}/3, z, t)$ and $\tilde{U}_i^{1/2}$ is the square root with value 1 at 0. Here $\tilde{U}_i$ is G_i-invariant, since U_i is G_i-invariant and $a_{i,2}$ has the character of y. Let $v_i := \tilde{U}_i^{-1}$. Then

$$v_i\psi_i(\hat{f}_i) = x^2 + y^3 + y\hat{A}_i(z, t) + \hat{B}_i(z, t), \qquad \hat{A}_i = a_{i,1} - \frac{a_{i,2}^2}{3}, \qquad \hat{B}_i = a_{i,0} - \frac{a_{i,1}a_{i,2}}{3} + \frac{2a_{i,2}^3}{27}.$$

Thus $\operatorname{mult} \hat{A}_i \geq 3$ and $\operatorname{mult} \hat{B}_i \geq 4$.

Let $\hat{h}_{ij}$ be the expansion in the regular parameters of a G_i-semi-invariant lift of h_{ij} to Y_i (it exists since G_i is finite: project any lift to the summand of the character of h_{ij}). We use Lemma D.21 with the embedding $W_i \subset Y_i$, the equation f_i, and the regular parameters above as the data (ii), the lifts of the marks h_{ij} with expansions $\hat{h}_{ij}$, the empty set of prime divisors in

(v), and ψ_i, v_i, $u_j = 1$ in (vi). Let N_2 be the integer of Lemma D.21, let $N \geq N_2$, let L_2 be the integer given there for N, and let $L \geq \max(6, L_2)$, so that the conclusions of Lemma D.21 hold for the numerical admission formed by the polynomials below. Let A_i, B_i, and $h_{ij}^\sharp$ be the sums of the terms of degree less than $L-1$, L, and N of $\hat{A}_i$, $\hat{B}_i$, and $\psi_i(\hat{h}_{ij})$, respectively. Since G_i acts diagonally, $x^2+y^3+yA_i+B_i$ and $h_{ij}^\sharp$ are G_i-semi-invariant with the characters of $\hat{f}_i$ and $\hat{h}_{ij}$, and

$$x^2+y^3+yA_i+B_i \equiv v_i\psi_i(\hat{f}_i) \pmod{(x,y,z,t)^L}, \qquad h_{ij}^\sharp \equiv \psi_i(\hat{h}_{ij}) \pmod{(x,y,z,t)^N}.$$

By Construction-Theorem 4.13(1) for $j=0$ and $G' = G_i$, and by Lemma D.21, the numerical admission $(W_i^\sharp, (h_{ij}^\sharp)_j)$ with $W_i^\sharp = (x^2+y^3+yA_i+B_i = 0)$ and

$$\Delta_i^\sharp := \sum_j c_j \operatorname{div}(h_{ij}^\sharp|_{W_i^\sharp})$$

satisfies $\operatorname{mld}((W_i^\sharp \ni 0, \Delta_i^\sharp)/G_i) = \alpha_i$, $W_i^\sharp$ is isolated Gorenstein terminal, $(W_i^\sharp, \theta_1\Delta_i^\sharp)$ is klt near 0, and $(W_i^\sharp, \theta_2\Delta_i^\sharp)$ is lc near 0. Possibly replacing $(W_i, (h_{ij})_j)$ with $(W_i^\sharp, (h_{ij}^\sharp)_j)$, we may assume that $W_i = (f_i = 0)$ with $f_i = x^2+y^3+yA_i+B_i$, that G_i acts diagonally, that the marks h_{ij} are G_i-semi-invariant polynomials, that $\operatorname{mld}((W_i \ni 0, \Delta_i)/G_i) = \alpha_i$, and that $(W_i, \theta_1\Delta_i)$ is klt near 0 and $(W_i, \theta_2\Delta_i)$ is lc near 0, with $\operatorname{mult} A_i \geq 3$, $\operatorname{mult} B_i \geq 4$. In particular, (W_i, Δ_i) is klt near 0, since $\theta_1 > 1$ and $\Delta_i \geq 0$.

Assume that none of the three rows holds for f_i. Then $\operatorname{mult} A_i \geq 4$ and $\operatorname{mult} B_i \geq 6$, so $\operatorname{wt}_{(3,2,1,1)}(f_i) = 6$, and Lemma 4.7 with the trivial group, $D=0$, and $w = (3,2,1,1)$ implies $\operatorname{mld}(W_i \ni 0) \leq 7-6 = 1$. Since W_i is Gorenstein, the log discrepancies $a(E, W_i, 0)$ are integers, and terminality implies $\operatorname{mld}(W_i \ni 0) \geq 2$, a contradiction. Thus exactly one row holds, since the rows are mutually exclusive. Passing to a subsequence, we may assume that all f_i are in one row, and we fix a monomial z^at^b with $a+b = 4$, 3, or 5 which has a non-zero coefficient in $B_{i,4}$, $A_{i,3}$, or $B_{i,5}$, respectively, for any i.

Step 3. In this step we find the circle. Regard G_i as a subgroup of $\mathbb{T}^4$. Since G_i preserves f_i up to a character and x^2, y^3, and z^at^b (resp. yz^at^b in row (E_7)) are monomials of f_i, G_i is contained in the closed subgroup

$$\mathbf{T}_0 := \{u \in \mathbb{T}^4 \mid 2u_1 - 3u_2 = 3u_2 - au_3 - bu_4 = 0\} \quad (\text{resp. } 2u_1 - 3u_2 = 2u_2 - au_3 - bu_4 = 0).$$

The Lie algebra $\mathfrak{t}_0 \subset \mathbb{R}^4$ of $\mathbf{T}_0$ is defined in $\mathbb{R}^4$ by the linear equations in the definition of $\mathbf{T}_0$, so $\dim \mathbf{T}_0 = 2$, and $\mathfrak{t}_0$ contains the positive vector $Q = (6,4,3,3)$, $(9,6,4,4)$, or $(15,10,6,6)$ in rows (E_6), (E_7), and (E_8), respectively. By Lemma 4.6, after passing to a subsequence, the G_i converge to a closed subgroup $H \subset \mathbb{T}^4$, and $\dim H \geq 1$ by **Step 1**. Since $\mathbf{T}_0$ is closed, $H \subset \mathbf{T}_0$, so $\dim H \leq 2$. Since $\alpha_i \geq \alpha_1 > 0$, Lemma 4.7 shows that the Lie algebra of H contains no vector with all coordinates positive. If $\dim H = 2$, then the Lie algebra of H is $\mathfrak{t}_0 \ni Q$, a contradiction. Thus $\dim H = 1$, and by Lemma 4.6 there exists a primitive $w \in \mathbb{Z}^4 \cap \mathfrak{t}_0$ such that $r_i := |G_i \cap \mathbf{T}_{(w)}| \to +\infty$ and $[G_i : G_i \cap \mathbf{T}_{(w)}] \leq e_0$ for $e_0 := |H/H^0|$ and $i \gg 0$. Neither w nor $-w$ has all coordinates positive.

Since $2w_1 = 3w_2$, w_2 is even. Write $w = (3s, 2s, C, D)$ with $s, C, D \in \mathbb{Z}$. Then $\gcd(s, C, D) = 1$, and

$$6s = aC + bD \text{ in rows } (E_6), (E_8), \qquad 4s = aC + bD \text{ in row } (E_7). \tag{D.43}$$

Possibly replacing w by $-w$, which does not change $\mathbf{T}_{(w)}$, we may assume $s \geq 0$. The group $G_i \cap \mathbf{T}_{(w)}$ is generated by the element acting by $(\zeta_i^{3s}, \zeta_i^{2s}, \zeta_i^C, \zeta_i^D)$ for a primitive r_i-th root of unity ζ_i, so all monomials $x^\alpha y^\beta z^p t^q$ of f_i satisfy $3s\alpha + 2s\beta + Cp + Dq \equiv 6s \pmod{r_i}$.

If $C = D$, then (D.43) and $a+b \in \{3,4,5\}$ imply that $C = D$ is a positive multiple of s, and $s \neq 0$ since $w \neq 0$. Then all coordinates of w are positive, which was excluded. Thus $C \neq D$.

Let z^pt^q and $z^{p'}t^{q'}$ be two monomials of $B_{i,4}$ (resp. let yz^pt^q and $yz^{p'}t^{q'}$ be two monomials of $yA_{i,3}$, or z^pt^q and $z^{p'}t^{q'}$ two monomials of $B_{i,5}$). Since $p+q=p'+q'\le 5$,

$$(p-p')(C-D)=(p-p')C+(q-q')D\equiv 0 \pmod{r_i}, \qquad |(p-p')(C-D)|\le 5|C-D|.$$

For $i\gg 0$ this implies $p=p'$. Since z^at^b has a non-zero coefficient for any i (**Step 2**), after discarding finitely many i the non-zero homogeneous part of the row is $b_iz^at^b$ (resp. $b_iyz^at^b$ for $yA_{i,3}$) with $b_i\in\mathbb{C}^*$.

Step 4. In this step we exclude $ab=0$. Exchanging z and t if necessary, we may assume $b=0$, so the monomial is z^4, yz^3, or z^5, and (D.43) reads $4C=6s$, $3C=4s$, or $5C=6s$. Let $q=(q_x,q_y,q_z)$ be $(6,4,3)$, $(9,6,4)$, or $(15,10,6)$, let f_0 be $x^2+y^3+z^4$, $x^2+y^3+yz^3$, or $x^2+y^3+z^5$, respectively, and let $d=12$, 18, or 30 be the q-weighted degree of f_0. Since q is primitive, $(3s,2s,C)=c'q$ for an integer c', and $\gcd(c',D)=1$ because w is primitive. After multiplying z by a constant, $b_i=1$. Any monomial of $f_i(x,y,z,0)-f_0$ has q-weighted degree larger than d. In row (E_6), the monomials of $f_i(x,y,z,0)-f_0$ are yz^k with $k\ge 3$ and z^k with $k\ge 5$, of degrees $4+3k\ge 13$ and $3k\ge 15$. In row (E_7), they are yz^k with $k\ge 4$ and z^k with $k\ge 5$, of degrees $6+4k\ge 22$ and $4k\ge 20$. In row (E_8), they are yz^k with $k\ge 4$ and z^k with $k\ge 6$, of degrees $10+6k\ge 34$ and $6k\ge 36$.

We use the monomial basis M_j of $\mathbb{C}[y,z]/(\partial_y g_0,\partial_z g_0)$, where $f_0=x^2+g_0$, given by $1,z,z^2,y,yz,yz^2$ in row (E_6), by $1,z,z^2,z^3,z^4,y,yz$ in row (E_7), and by $1,z,z^2,z^3,y,yz,yz^2,yz^3$ in row (E_8). Their q-weighted degrees d_j satisfy

$$d-d_j\in\{12,9,8,6,5,2\},\quad \{18,14,12,10,8,6,2\},\quad \{30,24,20,18,14,12,8,2\},$$

respectively. In particular, $2\le d-d_j\le d$. By Lemma D.1, there exists a G_i-equivariant automorphism ϕ_i of $\mathbb{C}[[x,y,z,t]]$ with $\phi_i(t)=t$ such that $\phi_i(f_i)$ is the product of a G_i-invariant unit and

$$\tilde{f}_i:=f_0+\sum_j a_{ij}(t)M_j, \qquad a_{ij}\in t\mathbb{C}[[t]],$$

where each non-zero $a_{ij}(t)M_j$ has the character of f_i. Thus there exists a G_i-equivariant isomorphism

$$\widehat{\mathcal{O}}_{W_i,0}\cong\mathbb{C}[[x,y,z,t]]/(\tilde{f}_i), \tag{D.44}$$

and $\tilde{f}_i(x,y,z,0)=f_0$. The quadratic part of $\tilde{f}_i$ is x^2+Q_i' with Q_i' in the span of t^2,ty,tz, and the rank of x^2+Q_i' is the quadratic rank one of W_i, so $Q_i'=0$. Moreover the section $(t=0)$ of W_i is formally isomorphic to the Du Val singularity $(f_0=0)$, so W_i is an isolated cDV germ.

Since $R_i:=G_i\cap\mathbf{T}_{(w)}$ acts on t^kM_j by $\zeta_i^{kD+c'd_j}$ and on f_i by $\zeta_i^{c'd}$, a non-zero summand $a_{ij}(t)M_j$ contains a monomial t^kM_j with $k\ge 1$ and

$$kD\equiv c'(d-d_j) \pmod{r_i}. \tag{D.45}$$

We consider the signs of c' and D.

- If $c'=0$, then $w=(0,0,0,\pm 1)$. We order the coordinates as (t,x,y,z). By Theorem 8.5 for W_i, G_i, $R_i=\mu_{r_i}(1,0,0,0)$, and Δ_i with $\Gamma=S$, where $r_0=r_0(S)$ is the integer of Theorem 8.5 and $r_i\ge r_0$ for $i\gg 0$ since $r_i\to+\infty$, the α_i do not form a strictly increasing sequence, a contradiction.
- If $D=0$ and $c'\ne 0$, then (D.45) reads $c'(d-d_j)\equiv 0 \pmod{r_i}$, which is impossible once $r_i>|c'|d$, since $0<|c'|(d-d_j)\le|c'|d$. Thus all a_{ij} vanish and $\tilde{f}_i=f_0(x,y,z)$, so $\mathbb{C}[[x,y,z,t]]/(\tilde{f}_i)$ is singular along the t-axis, which contradicts the isolatedness of the singularity of W_i at 0.
- If $c'D>0$, then all coordinates of w or of $-w$ are positive, which was excluded.
- If c' and D are non-zero of opposite signs, possibly replacing w by $-w$, we may assume $c'<0<D$. Let $c:=-c'$ and $v:=D$, coprime positive integers. Then $\mathbf{T}_{(w)}=\mathbf{T}_{(-cq_x,-cq_y,-cq_z,v)}$, and the part of Theorem 10.4 for germs with a G-equivariant

isomorphism of the completion, applied to W_i, $\tilde{f}_i$, G_i, and Δ_i with the isomorphism (D.44) and $\Gamma = S$, contradicts the strict increase of α_i with $r_i \to +\infty$.

Thus $a, b \geq 1$.

Step 5. In this step we conclude. Assume first that $CD = 0$. Exchanging z and t if necessary, we may assume $D = 0$. If $s = 0$, then (D.43) implies $aC = 0$, so $C = 0$ and $w = 0$, which is absurd. Thus $s > 0$, and (D.43) implies $C > 0$. All monomials t^k have $\mathbf{T}_{(w)}$-weight $0 \not\equiv 6s \pmod{r_i}$ for $r_i > 6s$, so $f_i(0,0,0,t) = 0$, and the t-axis is contained in W_i. Since the singularity of W_i at 0 is isolated, W_i is smooth at the points $(0,0,0,t)$ with $t \neq 0$ small, and $\partial_t f_i$ vanishes on the t-axis. Hence f_i contains a monomial xt^k, yt^k, or zt^k, of weight $3s$, $2s$, or C, congruent to $6s$ modulo r_i. For $r_i > 6s + C$, the weights $3s$ and $2s$ are not congruent to $6s$ modulo r_i, and $C \equiv 6s \pmod{r_i}$ implies $C = 6s$. Then $\gcd(s, C, D) = s = 1$ and $w = (3, 2, 6, 0)$. In row (E_7), (D.43) reads $4 = 6a$, which is impossible. In rows (E_6) and (E_8), it reads $6 = 6a$, so $a = 1$ and $b = 3$ or $b = 4$, respectively. Thus (1) holds.

Assume next that $CD > 0$. Since $a, b \geq 1$, (D.43) implies that $sC > 0$, so all coordinates of w or of $-w$ are positive, which was excluded. Finally, if C and D are non-zero of opposite signs, then possibly replacing w by $-w$ (if $s = 0$) and exchanging z and t, we may assume $s \geq 0$ and $C < 0 < D$, and (2) holds by **Steps 3 and 4**. **Steps 3 to 5** only pass to subsequences and exchange z and t, which keep the properties of the data obtained in **Step 2**. □

The circle $\mathbf{T}_{(3,2,6,0)}$ of Proposition D.22(1) is excluded by the weighted blow-up of $\mathbb{A}^4$ along the t-axis with weights $(3, 2, 6)$ on x, y, z: the fiber over the origin of the exceptional divisor of the strict transform of W is a union of curves, each lying on a surface with Du Val singularities.

Proposition D.23. *Fix a finite set $S \subset [0,1]$ and an integer $e_0 \geq 1$. Let W be an isolated Gorenstein terminal polynomial germ*

$$W = (x^2 + y^3 + yA(z,t) + B(z,t) = 0)$$

with $\operatorname{mult} A \geq 3$ *and* $\operatorname{mult} B \geq 4$*, such that either B_4 is a non-zero multiple of zt^3, or $B_4 = A_3 = 0$ and B_5 is a non-zero multiple of zt^4. Let G be a finite diagonal abelian group acting faithfully on W and preserving the equation of W up to a character, with $[G : G \cap \mathbf{T}_{(3,2,6,0)}] \leq e_0$, and let Δ be a labelled boundary $\Delta = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in S$ and polynomial marks h_j such that (W, Δ) is klt near 0. Then the numbers* $\operatorname{mld}((W \ni 0, \Delta)/G)$ *in $(0,1)$ for such data satisfy the ACC. The ACC also holds with z and t exchanged throughout, so that $\mathbf{T}_{(3,2,6,0)}$ becomes $\mathbf{T}_{(3,2,0,6)}$ in the condition on G.*

Proof. The proof is long, so we divide it into several steps. By the symmetry exchanging z and t throughout, it suffices to treat the data without the exchange. Assume that there exist data (W_i, G_i, Δ_i) $(i \geq 1)$ as in the proposition such that $\alpha_i := \operatorname{mld}((W_i \ni 0, \Delta_i)/G_i)$ form a strictly increasing sequence in $(0,1)$. Write $W_i = (f_i = 0)$ with $f_i = x^2 + y^3 + yA_i + B_i$, and $\Delta_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$. Let $w := (3, 2, 6, 0)$, $\mathbf{T} := \mathbf{T}_{(w)}$, $R_i := G_i \cap \mathbf{T}$, $r_i := |R_i|$, and $\gamma := \min(S \setminus \{0\})$ (if $S \subset \{0\}$, then $\Delta_i = 0$, and the steps below hold without marks).

Step 1. In this step we normalize the marks and show that $r_i \to +\infty$. Removing the marks with label zero and the marks that do not vanish at 0 does not change Δ_i near 0, so we may assume $c_{ij} \geq \gamma$ and $h_{ij}(0) = 0$. Since f_i is monic of degree two in x, we may replace h_{ij} by the remainder of the division of h_{ij} by f_i in $\mathbb{C}[y, z, t][x]$, which agrees with h_{ij} on W_i and has the character of h_{ij}. Thus we may assume that each h_{ij} has degree at most one in x. Let $w_0 := (3, 2, 2, 2)$. Since $\operatorname{mult} A_i \geq 3$ and $\operatorname{mult} B_i \geq 4$, $\operatorname{in}_{w_0}(f_i) = x^2 + y^3$, which defines an integral subvariety of $\mathbb{P}(w_0)$ not contained in any coordinate hyperplane. By Lemma 4.12, there exists a prime divisor E over W_i with center 0, $a(E, W_i, 0) = 9 - 6 = 3$, and $\operatorname{mult}_E h_{ij} \geq \operatorname{wt}_{w_0}(h_{ij}) \geq 2$. Since (W_i, Δ_i) is klt near 0, $0 < a(E, W_i, \Delta_i) \leq 3 - 2\gamma \cdot \#\{j\}$, so there are fewer than $3/(2\gamma)$ marks. Passing to a subsequence, we may assume that the labels $c_j := c_{ij}$ and their number do not depend on i.

If r_i is bounded on a subsequence, then so is $|G_i| \leq e_0 r_i$, and on the subsequence α_i belongs to a finite set by Lemma D.19, since W_i is Gorenstein. This contradicts the strict increase of α_i.

Hence $r_i \to +\infty$, and we may assume $r_i > 6$. The group R_i is generated by the element acting by $(\zeta_i^3, \zeta_i^2, \zeta_i^6, 1)$ for a primitive r_i-th root of unity ζ_i. All monomials of f_i have the character of x^2 on R_i, so their $\mathbf{T}$-weights are congruent to 6 modulo r_i. The monomials of $\mathbf{T}$-weight 6 in f_i are x^2, y^3, and the monomials zt^q of B_i, since yz^pt^q has weight $2+6p$ and z^pt^q has weight $6p$. Thus

$$f_i = \sum_{\ell \geq 0} f_{i,\ell}, \qquad f_{i,0} = \mathrm{in}_w(f_i) = x^2 + y^3 + zb_i(t),$$

where $f_{i,\ell}$ is the sum of the terms of f_i of $\mathbf{T}$-weight $6 + \ell r_i$, and $b_i \in \mathbb{C}[t]$ has order 3 at $t = 0$ in the first case of the proposition and order 4 in the second case. Similarly, with $m_{ij} := \mathrm{wt}_w(h_{ij})$,

$$h_{ij} = \sum_{\ell \geq 0} h_{ij,\ell}, \qquad h_{ij,0} = \mathrm{in}_w(h_{ij}),$$

where $h_{ij,\ell}$ is the sum of the terms of h_{ij} of $\mathbf{T}$-weight $m_{ij} + \ell r_i$.

Step 2. In this step we bound m_{ij}. Since the monomials t^q have $\mathbf{T}$-weight $0 \not\equiv 6 \pmod{r_i}$, $f_i(0,0,0,t) = 0$, and the t-axis $L := (x = y = z = 0)$ is contained in W_i. Let $k := \mathbb{C}(t)$ and let $W_{i,k} := (f_i = 0) \subset \mathbb{A}^3_k$ be the generic fiber of t on W_i, a normal hypersurface in the coordinates x, y, z through the point 0 corresponding to the generic point of L. For $w' := (3, 2, 6)$, $\mathrm{in}_{w'}(f_i) = x^2 + y^3 + b_i(t)z$ with $b_i(t) \in k^*$, which defines an integral subvariety of $\mathbb{P}_k(w')$ not contained in any coordinate hyperplane. By Lemma 4.12, there exists a prime divisor P_i over W_i with center L such that

$$a(P_i, W_i, 0) = 3 + 2 + 6 - 6 = 5, \qquad \mathrm{mult}_{P_i} h_{ij} \geq \mathrm{wt}_{w'}(h_{ij}) = m_{ij}.$$

Since $0 \in L$ and (W_i, Δ_i) is klt near 0, $0 < a(P_i, W_i, \Delta_i) \leq 5 - \sum_j c_j m_{ij}$. Thus $m_{ij} < 5/\gamma$, and passing to a subsequence we may assume that $m_j := m_{ij}$ does not depend on i. Let

$$d_0 := 5 - \sum_j c_j m_j \in (0, 5].$$

Step 3. In this step we write the weighted blow-up of $\mathbb{A}^4$ along L with weights w' on x, y, z on root charts. Let $w_x := 3$, $w_y := 2$, $w_z := 6$. For $\iota = x$, y, and z, let $\mathbb{A}^4_\iota$ be the affine space with coordinates (ξ, Y, Z, t), (ξ, X, Z, t), and (ξ, X, Y, t), respectively, whose coordinates other than ξ and t we call the *capital coordinates*, and let $p_\iota : \mathbb{A}^4_\iota \to \mathbb{A}^4$ be

$$p_x(\xi, Y, Z, t) = (\xi^3, \xi^2 Y, \xi^6 Z, t), \qquad p_y(\xi, X, Z, t) = (\xi^3 X, \xi^2, \xi^6 Z, t),$$
$$p_z(\xi, X, Y, t) = (\xi^3 X, \xi^2 Y, \xi^6, t).$$

Then p_ι is the composite of the quotient by μ_{w_ι}, an open immersion into the weighted blow-up, and the blow-up morphism, and the images of p_x, p_y, p_z cover the weighted blow-up. For $g \in G_i$ acting by

$$g^*(x, y, z, t) = (\chi_x(g)x, \chi_y(g)y, \chi_z(g)z, \chi_t(g)t),$$

let

$$G'_{i,\iota} := \{(g, \mu) \in G_i \times \mathbb{C}^* \mid \mu^{w_\iota} = \chi_\iota(g)\},$$

acting diagonally on $\mathbb{A}^4_\iota$ by multiplying ξ by μ, t by $\chi_t(g)$, X by $\chi_x(g)\mu^{-3}$, Y by $\chi_y(g)\mu^{-2}$, and Z by $\chi_z(g)\mu^{-6}$. Then p_ι is equivariant along the projection $G'_{i,\iota} \to G_i$, whose kernel is μ_{w_ι}. The subgroup

$$R'_i := \{(\zeta^w, \zeta) \mid \zeta \in \mu_{r_i}\} \subset G'_{i,\iota},$$

where ζ^w acts by $(\zeta^3, \zeta^2, \zeta^6, 1)$, acts only on ξ. Let $\varpi := \xi^{r_i}$, so that $\mathbb{A}^4_\iota / R'_i$ is the affine space with ξ replaced by ϖ, and let $\Upsilon_{i,\iota} := G'_{i,\iota}/R'_i$, a finite abelian group acting diagonally, with

$$|\Upsilon_{i,\iota}| = \frac{w_\iota |G_i|}{r_i} \leq 6e_0 =: N_G.$$

Let

$$F_{x,\ell} := f_{i,\ell}(1, Y, Z, t), \qquad F_{y,\ell} := f_{i,\ell}(X, 1, Z, t), \qquad F_{z,\ell} := f_{i,\ell}(X, Y, 1, t),$$

and define $H_{j,x,\ell} := h_{ij,\ell}(1,Y,Z,t)$, $H_{j,y,\ell} := h_{ij,\ell}(X,1,Z,t)$, and $H_{j,z,\ell} := h_{ij,\ell}(X,Y,1,t)$. Since $f_{i,\ell}$ and $h_{ij,\ell}$ are $\mathbf{T}$-homogeneous,

$$p_\iota^* f_i = \xi^6 F_\iota, \quad p_\iota^* h_{ij} = \xi^{m_j} H_{j,\iota}, \qquad F_\iota := \sum_{\ell\ge 0} \varpi^\ell F_{\iota,\ell}, \quad H_{j,\iota} := \sum_{\ell\ge 0} \varpi^\ell H_{j,\iota,\ell},$$

which are $\Upsilon_{i,\iota}$-semi-invariant polynomials in ϖ, t, and the capital coordinates. Let

$$V_\iota := (F_\iota = 0) \subset \mathbb{A}^4_\iota/R'_i, \qquad S_\iota := (F_{\iota,0} = 0) \subset \mathbb{A}^3 := (\varpi = 0), \qquad \Sigma_\iota := S_\iota \cap (t = 0),$$

so that $S_\iota = V_\iota \cap (\varpi = 0)$, and

$$F_{x,0} = 1 + Y^3 + Zb_i(t), \qquad F_{y,0} = X^2 + 1 + Zb_i(t), \qquad F_{z,0} = X^2 + Y^3 + b_i(t).$$

Each $F_{\iota,0}$ is irreducible and is not a multiple of a coordinate. On Σ_x, $Y^3 = -1$ and $\partial_Y F_{x,0} = 3Y^2 \neq 0$. On Σ_y, $X^2 = -1$ and $\partial_X F_{y,0} = 2X \neq 0$. On Σ_z, $\partial_X F_{z,0} = 2X$ and $\partial_Y F_{z,0} = 3Y^2$ vanish only at the origin, where S_z is a Du Val singularity of type D_4 or E_6 according to the order 3 or 4 of b_i. Thus each S_ι is normal with Du Val singularities near Σ_ι. The hypersurface V_ι is Gorenstein, hence Cohen–Macaulay [Mat89, Theorems 21.3 and 18.1], so it satisfies (S_2). Since S_ι is a Cartier divisor of V_ι, V_ι is regular at the regular points of S_ι, so $\operatorname{Sing} V_\iota \cap S_\iota \subset \operatorname{Sing} S_\iota$, which is finite near Σ_ι. Since any irreducible component of $\operatorname{Sing} V_\iota$ through a point of Σ_ι intersects the Cartier divisor S_ι in a finite set near Σ_ι, $\dim \operatorname{Sing} V_\iota \le 1$ near Σ_ι. Thus V_ι satisfies (R_1) near Σ_ι, and it is normal near Σ_ι [Mat89, Theorem 23.8].

An element of $G'_{i,\iota}$ acting trivially on S_ι acts trivially on t and on the capital coordinates, since S_ι is not contained in a coordinate hyperplane. Then it is (g,μ) with $g = \mu^w$, so $g \in R_i$, $g = \zeta^w$ with $\zeta \in \mu_{r_i}$, and $(\mu/\zeta)^3 = (\mu/\zeta)^2 = 1$ implies $(g,\mu) = (\zeta^w,\zeta) \in R'_i$. Thus $\Upsilon_{i,\iota}$ acts faithfully on S_ι and on V_ι. Since $f_{i,0}$ is irreducible of degree two in x, it does not divide the non-zero polynomial $h_{ij,0}$ of degree at most one in x, nor any monomial multiple of $h_{ij,0}$. Since $f_{i,0}$ and $h_{ij,0}$ are $\mathbf{T}$-homogeneous, $F_{\iota,0}$ does not divide $H_{j,\iota,0}$, so $H_{j,\iota,0}$ does not vanish identically on S_ι.

Step 4. In this step we compute the crepant boundaries on the charts. Let

$$C_\iota := \operatorname{div}(F_\iota) + \sum_j c_j \operatorname{div}(H_{j,\iota}) + \Big(1 - \frac{d_0}{r_i}\Big) \operatorname{div}(\varpi),$$

$$\Xi_\iota := \sum_j c_j \operatorname{div}(H_{j,\iota}|_{V_\iota}) + \Big(1 - \frac{d_0}{r_i}\Big) \operatorname{div}(\varpi|_{V_\iota}),$$

which satisfy $C_\iota \ge 0$ and $\Xi_\iota \ge 0$ since $r_i > 6 > d_0$. The pullback of $dx \wedge dy \wedge dz \wedge dt$ by p_ι is $w_\iota \xi^{3+2+6-1} d\xi \wedge dU_1 \wedge dU_2 \wedge dt$, where U_1, U_2 are the capital coordinates of $\mathbb{A}^4_\iota$. Since $-10 + 6 + \sum_j c_j m_j = 1 - d_0$,

$$p_\iota^* \Big(K_{\mathbb{A}^4} + \operatorname{div}(f_i) + \sum_j c_j \operatorname{div}(h_{ij}) \Big) = K_{\mathbb{A}^4_\iota} + \operatorname{div}(F_\iota) + \sum_j c_j \operatorname{div}(H_{j,\iota}) + (1 - d_0) \operatorname{div}(\xi),$$

and by the Hurwitz formula for $\varpi = \xi^{r_i}$ the right side is the pullback of $K_{\mathbb{A}^4_\iota/R'_i} + C_\iota$. Taking residues along $(f_i = 0)$ and $(F_\iota = 0)$, the pullbacks of $K_{W_i} + \Delta_i$ and of $K_{V_\iota} + \Xi_\iota$ to $(F_\iota = 0) \subset \mathbb{A}^4_\iota$ coincide. By [Liu26, Lemma 2.6(2)(5)], for prime divisors P over W_i and P_ι over V_ι inducing a common prime divisor over W_i/G_i, and for prime divisors P' over $\mathbb{A}^4$ and P'_ι over $\mathbb{A}^4_\iota/R'_i$ inducing a common prime divisor over $\mathbb{A}^4/G_i$,

$$\begin{aligned} \frac{a(P, W_i, \Delta_i)}{|I_{G_i}(P)|} &= \frac{a(P_\iota, V_\iota, \Xi_\iota)}{|I_{\Upsilon_{i,\iota}}(P_\iota)|}, \\ \frac{a(P', \mathbb{A}^4, \operatorname{div}(f_i) + \sum_j c_j \operatorname{div}(h_{ij}))}{|I_{G_i}(P')|} &= \frac{a(P'_\iota, \mathbb{A}^4_\iota/R'_i, C_\iota)}{|I_{\Upsilon_{i,\iota}}(P'_\iota)|}. \end{aligned} \tag{D.46}$$

Since p_ι maps $(\xi = t = 0)$ to 0, a prime divisor over V_ι whose center intersects a small neighborhood of Σ_ι induces a prime divisor over W_i whose center intersects a small neighborhood of 0. Thus (D.46) implies that (V_ι, Ξ_ι) is klt near Σ_ι, and so is V_ι, because $\Xi_\iota \ge 0$.

Step 5. In this step we show that, for $i \gg 0$ and any ι, the following pairs are lc near Σ_ι, where

$$M_{i,\iota} := \operatorname{div}(F_\iota) + \sum_j c_j \operatorname{div}(H_{j,\iota}) + \operatorname{div}(\varpi), \qquad M_{i,\iota,0} := \operatorname{div}(F_{\iota,0}) + \sum_j c_j \operatorname{div}(H_{j,\iota,0}),$$

$$\Xi_{S_\iota} := \sum_j c_j \operatorname{div}(H_{j,\iota,0}|_{S_\iota}).$$

(1) $(\mathbb{A}^4_\iota/R'_i, M_{i,\iota})$.
(2) $(\mathbb{A}^3, M_{i,\iota,0})$.
(3) (S_ι, Ξ_{S_ι}).

By **Step 4** and inversion of adjunction [Kaw07, Theorem] along the normal Cartier divisor V_ι, the pair $(\mathbb{A}^4_\iota/R'_i, C_\iota)$ is lc near Σ_ι. Since $(1-d_0/r_i)M_{i,\iota} \le C_\iota$, the lc threshold of $M_{i,\iota}$ at any point of Σ_ι is at least $1-d_0/r_i$. The coefficients of $M_{i,\iota}$ belong to the DCC set $(S\cup\{1\})_+$ [Liu26, Lemma 2.10(1)]. If for infinitely many i there exist ι and a point $\mathfrak{y}_i \in \Sigma_\iota$ at which the lc threshold of $M_{i,\iota}$ is less than one, then the lc thresholds at the $\mathfrak{y}_i$ tend to one, and a subsequence of them is strictly increasing. This contradicts Theorem 2.12(1) applied to small neighborhoods of the $\mathfrak{y}_i$ in $\mathbb{A}^4$ with the boundary 0. Thus the pair (1) is lc near Σ_ι for $i \gg 0$. Inversion of adjunction [Kaw07, Theorem] along $(\varpi = 0)$ implies that the pair (2) is lc near Σ_ι, and inversion of adjunction along the normal Cartier divisor S_ι of $\mathbb{A}^3$ implies that the pair (3) is lc near Σ_ι. We assume $i \gg 0$ from now on.

Step 6. In this step we define β_i and show that it belongs to a finite set. For a scheme point $\eta \in \Sigma_\iota$, let $\beta_i(\iota,\eta)$ be the minimal log discrepancy at the image of η of the crepant quotient of (S_ι, Ξ_{S_ι}) by $\Upsilon_{i,\iota}$. Let $\mathfrak{y}$ be a closed point of $\bar\eta$, and let $\bar G \subset \Upsilon_{i,\iota}$ be the subgroup of the elements fixing $\mathfrak{y}$ and preserving $\bar\eta$. Any element of the inertia group in $\Upsilon_{i,\iota}$ of a prime divisor over $\mathbb{A}^4_\iota/R'_i$, V_ι, $\mathbb{A}^3$, or S_ι with center $\bar\eta$ acts trivially on $\bar\eta$, so it belongs to $\bar G$. Hence, by [Liu26, Lemma 2.6], $\beta_i(\iota,\eta)$ is the quotient value at η for the group $\bar G$. Any coordinate v with $v(\mathfrak{y}) \ne 0$ has trivial $\bar G$-character, so after replacing any such v by $v - v(\mathfrak{y})$, the group $\bar G$ acts diagonally on a chart centered at $\mathfrak{y}$, and it acts faithfully on S_ι by **Step 3**. By Theorem 3.3(1) if $\eta = \mathfrak{y}$ and Theorem 3.3(3) if η is the generic point of a curve,

$$\beta_i(\iota,\eta) = \operatorname{mld}\Big(\Big(\mathbb{A}^3 \ni \eta, \operatorname{div}(F_{\iota,0}) + \sum_j c_j \operatorname{div}(H_{j,\iota,0})\Big)/\bar G\Big), \tag{D.47}$$

which is the minimal log discrepancy at the image of η of the crepant quotient $(\mathbb{A}^3/\bar G, \Theta)$ of the pair (2) of **Step 5**.

Let Γ' be a prime divisor on $\mathbb{A}^3$ through $\mathfrak{y}$ with ramification index $e' \le |\bar G| \le N_G$ over $\mathbb{A}^3/\bar G$. The coefficient of Γ' in $M_{i,\iota,0}$ is $n_0 + \sum_j c_j n_j$ with $n_0 \in \{0,1\}$, since $F_{\iota,0}$ is irreducible, and $n_j \in \mathbb{Z}_{\ge 0}$. It is at most one by **Step 5**, so $n_j \le 1/\gamma$, and the coefficient of the image of Γ' in Θ is $1-(1-n_0-\sum_j c_j n_j)/e'$ [Liu26, Lemma 2.6(4)]. Thus all coefficients of Θ near the image of $\mathfrak{y}$ belong to the finite set

$$\Lambda := \Big\{1 - \frac{1-n_0-\sum_j c_j n_j}{e'} \;\Big|\; n_0 \in \{0,1\},\ n_j \in \mathbb{Z}_{\ge 0},\ n_0 + \sum_j c_j n_j \le 1,\ 1 \le e' \le N_G\Big\}.$$

Let $\pi : \mathbb{A}^3 \to \mathbb{A}^3/\bar G$ be the quotient, and Θ' a prime divisor on $\mathbb{A}^3/\bar G$. Since $\mathbb{A}^3$ is factorial, $\pi^*\Theta' = \operatorname{div}(\varphi)$ for a regular function φ, and with $\operatorname{Nm}(\varphi) := \prod_{g\in\bar G} g^*\varphi$, a $\bar G$-invariant function,

$$|\bar G|\Theta' = \pi_*\pi^*\Theta' = \pi_* \operatorname{div}(\varphi) = \operatorname{div}(\operatorname{Nm}(\varphi)).$$

Thus $|\bar G|\Theta'$ is Cartier. The display holds for any Weil divisor on $\mathbb{A}^3/\bar G$ in place of Θ', so $|\bar G|K_{\mathbb{A}^3/\bar G}$ is Cartier as well. Let $\Theta_{\mathfrak{y}}$ be the sum of the components of Θ through $\pi(\mathfrak{y})$, with their coefficients in Θ. Then the coefficients of $\Theta_{\mathfrak{y}}$ belong to $\Lambda\backslash\{0\}$, and $\Theta_{\mathfrak{y}} = \Theta$ on an open neighborhood of $\pi(\mathfrak{y})$,

which contains $\pi(\eta)$. Since $|\bar{G}|$ divides $N_G!$, $N_G!K_{\mathbb{A}^3/\bar{G}}$ is Cartier, and

$$\Theta_{\mathfrak{y}} = \sum_k \frac{\lambda_k}{N_G!} \cdot N_G!\Theta_k$$

with $\lambda_k \in \Lambda$ and Cartier divisors $N_G!\Theta_k \geq 0$. Let $\mathcal{D}_1$ be the set of Theorem 2.13(2) for $d = 3$, $r = N_G!$, and $I = \{\lambda/N_G! \mid \lambda \in \Lambda\}$. By Theorem 2.13(2), it has no accumulation point in $\mathbb{R}$, hence it is finite in any bounded interval, and it depends only on S, the labels, and e_0. Since $(\mathbb{A}^3/\bar{G}, \Theta)$ is lc near $\pi(\mathfrak{y})$ and $\Theta_{\mathfrak{y}} = \Theta$ near $\pi(\mathfrak{y})$, the log discrepancies of prime divisors with center $\overline{\pi(\eta)}$ with respect to $(\mathbb{A}^3/\bar{G}, \Theta)$ and to $(\mathbb{A}^3/\bar{G}, \Theta_{\mathfrak{y}})$ are equal, and they are non-negative elements of $\mathcal{D}_1$, so $\beta_i(\iota, \eta) \in \mathcal{D}_1$, and $\beta_i(\iota, \eta)$ is computed by a prime divisor.

The fiber $\Sigma_x = (Y^3 = -1,\ t = 0)$ consists of three lines. At the generic point η of one of them, $\beta_i(x, \eta)$ is one minus the coefficient of the image of $\bar{\eta}$ in the quotient boundary, so $\beta_i(x, \eta) \leq 1$. Since $\mathcal{D}_1 \cap [0, 1]$ is finite and $\beta_i(x, \eta) \leq 1$, the minimum

$$\beta_i := \min\{\beta_i(\iota, \eta) \mid \iota \in \{x, y, z\},\ \eta \in \Sigma_\iota\}$$

exists, and $\beta_i \in \mathcal{D}_1 \cap [0, 1]$.

Step 7. In this step we show that $\alpha_i \geq \beta_i$. Let P be a prime divisor over W_i with center 0. Since the strict transform of W_i in the weighted blow-up of **Step 3** is proper over W_i, the center of P on it is contained in the fiber over 0, which is covered by the images of the p_ι. Thus there exist ι and a prime divisor P_ι over V_ι such that P_ι and P induce a common prime divisor over W_i/G_i and the center $\bar{\eta}$ of P_ι is contained in $(\varpi = t = 0) \cap V_\iota = \Sigma_\iota$. Let $\mathfrak{y}$ and $\bar{G}$ be as in **Step 6** for η, with coordinates centered at $\mathfrak{y}$. Then $I_{\Upsilon_{i,\iota}}(P_\iota) = I_{\bar{G}}(P_\iota)$. We use Lemma 5.5 with the group $\bar{G}$, $X = V_\iota$, $u = \varpi$, $S_0 = S_\iota$, the $\mathbb{R}$-ideal $\prod_j (H_{j,\iota})^{c_j}$, and $b = 1 - d_0/r_i \leq 1$. The hypotheses of Lemma 5.5 hold by **Steps 3, 4, and 5**. If η is the generic point of a curve, we use the case of the $\bar{G}$-invariant curve $\bar{\eta} \ni \mathfrak{y}$ in Lemma 5.5. By (D.46) and (D.47),

$$\frac{a(P, W_i, \Delta_i)}{|I_{G_i}(P)|} = \frac{a(P_\iota, V_\iota, \Xi_\iota)}{|I_{\bar{G}}(P_\iota)|} \geq \beta_i(\iota, \eta) + \frac{d_0 \operatorname{mult}_{P_\iota} \varpi}{r_i |I_{\bar{G}}(P_\iota)|} \geq \beta_i.$$

Taking the infimum over P, [Liu26, Lemma 2.6] implies $\alpha_i \geq \beta_i$.

Step 8. In this step we show that $\alpha_i \leq \beta_i + M_1/r_i$ for a constant M_1, and we conclude. The subgroups $\bar{G}$ of **Step 6** act diagonally on $\mathbb{A}^3$ with order at most N_G, so they belong to the finite set of subgroups of $(\mathbb{C}^*)^3$ of order at most N_G. Let l be the maximum of the integers of Theorem 2.10(3) for the klt threefolds $\mathbb{A}^3/\bar{G}$ for all such subgroups and the DCC set $\Lambda\backslash\{0\}$, and let $\kappa := N_G l/\gamma$ and $M_1 := 5(\kappa + N_G)$.

Choose ι and $\eta \in \Sigma_\iota$ with $\beta_i = \beta_i(\iota, \eta)$, and $\mathfrak{y}$, $\bar{G}$ as in **Step 6**. The coefficients of $\Theta_{\mathfrak{y}}$ of **Step 6** belong to $\Lambda\backslash\{0\}$, $\Theta_{\mathfrak{y}}$ is $\mathbb{R}$-Cartier since $|\bar{G}|$ times any Weil divisor on $\mathbb{A}^3/\bar{G}$ is Cartier (**Step 6**), and $\Theta_{\mathfrak{y}} = \Theta$ on an open neighborhood of $\pi(\mathfrak{y})$ containing $\pi(\eta)$. Thus $(\mathbb{A}^3/\bar{G}, \Theta_{\mathfrak{y}})$ is lc near $\pi(\eta)$, and by (D.47) its minimal log discrepancy at $\pi(\eta)$ is β_i, and any prime divisor computing it computes β_i for $(\mathbb{A}^3/\bar{G}, \Theta)$. Theorem 2.10(3), applied to $(\mathbb{A}^3/\bar{G}, \Theta_{\mathfrak{y}})$, implies that there exists a prime divisor over $\mathbb{A}^3/\bar{G}$ computing β_i whose log discrepancy with respect to $(\mathbb{A}^3/\bar{G}, 0)$ is at most l. Let ν be a prime divisor over $\mathbb{A}^3$ inducing it, and $e := |I_{\bar{G}}(\nu)| \leq N_G$. By [Liu26, Lemma 2.6(2)] and the Hurwitz formula, $a(\nu, \mathbb{A}^3, 0) \leq el \leq N_G l$. Since the pair (2) of **Step 5** is lc,

$$\operatorname{mult}_\nu F_{\iota,0} + \sum_j c_j \operatorname{mult}_\nu H_{j,\iota,0} \leq a(\nu, \mathbb{A}^3, 0) \leq N_G l,$$

so $\operatorname{mult}_\nu F_{\iota,0} \leq \kappa$ and $\operatorname{mult}_\nu H_{j,\iota,0} \leq \kappa$ for any j.

Assume first that $\eta = \mathfrak{y}$ is a closed point. We use Proposition 5.4 with $\mathbb{A}^3$ (centered at $\mathfrak{y}$) as S there, $s = \mathfrak{y}$, $\mathbb{A}^4_\iota/R'_i = \mathbb{A}^3 \times \mathbb{A}^1_\varpi$ as V there, $\tilde{H} = \bar{G}$, χ the character of ϖ, $b = 1 - d_0/r_i$, and the functions F_ι (with label 1) and $H_{j,\iota}$ (with labels c_j), so that $C_V = C_\iota$ and C_S is the boundary of the pair (2) of **Step 5**. Let q be the integer of Lemma 3.1 for ν and χ, and k_0 the integer with

$k_0 \equiv q \pmod e$ and $\kappa < k_0 \le \kappa + e$. The Gauss extension of mult_ν with ϖ-value k_0 is a prime divisor P'_ι over $\mathbb{A}^4_\iota/R'_i$ with center $(\mathfrak{y}, 0)$, $I_{\bar G}(P'_\iota) = I_{\bar G}(\nu)$, and

$$\frac{a(P'_\iota, \mathbb{A}^4_\iota/R'_i, C_\iota)}{|I_{\bar G}(P'_\iota)|} = \beta_i + \frac{d_0 k_0}{r_i e} \le \beta_i + \frac{M_1}{r_i}. \tag{D.48}$$

Assume next that η is the generic point of a curve. Let $\bar G_\eta \subset \bar G$ be the subgroup of the elements acting trivially on $\bar\eta$. By **Step 6**, the inertia group in $\bar G$ of any prime divisor with center $\bar\eta$ is contained in $\bar G_\eta$, so by [Liu26, Lemma 2.6] the quotient values at η for $\bar G$ and for $\bar G_\eta$ coincide, ν computes both, and $I_{\bar G_\eta}(\nu) = I_{\bar G}(\nu)$. Possibly replacing $\bar G$ with $\bar G_\eta$, we may assume that $\bar G = \bar G_\eta$. Let ϕ be a $\bar G$-invariant monomial which is not constant on $\bar\eta$ (a power of Z if $\iota \in \{x, y\}$, and a power of Y if $\iota = z$), so that ϕ is regular at η. Let Ω be an algebraic closure of $\mathbb{C}(\phi)$, $\mathfrak{c}$ a closed point over η of the base change to Ω of the generic fiber of ϕ on $\mathbb{A}^3$, and S' the connected component of the base change containing $\mathfrak{c}$. We use Lemma 3.22(2)(3) (cf. [Liu26, Lemma 3.6]) for the pair (2) of **Step 5**, the curve $\bar\eta$, $\bar G_\eta$ as the group H there, and $t = \phi$. Thus $\bar G_\eta$ fixes $\mathfrak{c}$, and a prime divisor ν' over S' with center $\mathfrak{c}$ whose valuation restricts to mult_ν computes β_i for the restriction of the pair (2) to S' and the group $\bar G_\eta$, with $I_{\bar G_\eta}(\nu') = I_{\bar G_\eta}(\nu)$. Such ν' exists: take a component over $\mathfrak{c}$ of the pullback of ν to the base change of a model on which ν is a divisor. Since the valuation of ν' restricts to mult_ν, $\mathrm{mult}_{\nu'} F_{\iota,0} = \mathrm{mult}_\nu F_{\iota,0}$ and $\mathrm{mult}_{\nu'} H_{j,\iota,0} = \mathrm{mult}_\nu H_{j,\iota,0}$ for any j. We use Proposition 5.4 over Ω with S' as S there, $s = \mathfrak{c}$, $\tilde H = \bar G_\eta$, ν' in place of ν, and the other data as in the case of a closed point, with q and k_0 defined for ν'. Here $\bar G_\eta$ acts faithfully on S' since $S' \to \mathbb{A}^3$ is dominant, and (S', C_S) is lc near $\mathfrak{c}$ by the comparisons of Lemma 3.22(2)(3). Let F' be the prime divisor over $S' \times \mathbb{A}^1_\varpi$ given there. The valuation of F', restricted to the function field of $\mathbb{A}^4_\iota/R'_i$, is the Gauss extension of mult_ν with ϖ-value k_0, which is a prime divisor P'_ι over $\mathbb{A}^4_\iota/R'_i$ with center $\bar\eta \times \{0\}$. Restricting to the generic fiber of ϕ on $\mathbb{A}^4_\iota/R'_i$ changes neither the log discrepancy with respect to C_ι nor the inertia group of a prime divisor with center $\bar\eta \times \{0\}$, since the local ring of the generic fiber at the point given by $\bar\eta \times \{0\}$ is $\mathcal{O}_{\mathbb{A}^4_\iota/R'_i, \bar\eta\times\{0\}}$. This is the comparison of Lemma 3.22(2), whose argument does not use that the dimension is three. The base change to Ω is a filtered union of finite separable base changes, so it changes neither the log discrepancy nor the inertia group of F', by the comparison of Lemma 3.22(3), whose argument again does not use the dimension. Thus the log discrepancy of P'_ι with respect to $(\mathbb{A}^4_\iota/R'_i, C_\iota)$ and $|I_{\bar G_\eta}(P'_\iota)|$ are equal to those of F' with respect to $(S' \times \mathbb{A}^1_\varpi, C_V)$, and P'_ι satisfies (D.48).

In both cases the inertia group of P'_ι in $\Upsilon_{i,\iota}$ acts trivially on the center of P'_ι, so it is $I_{\bar G}(P'_\iota)$. Let P' be a prime divisor over $\mathbb{A}^4$ such that P' and P'_ι induce a common prime divisor over $\mathbb{A}^4/G_i$. Since p_ι maps $(\xi = t = 0)$ to 0, the center of P' is 0. By Theorem 3.3(1), [Liu26, Lemma 2.6], and (D.46),

$$\alpha_i = \mathrm{mld}\Big(\Big(\mathbb{A}^4 \ni 0, \mathrm{div}(f_i), \prod_j h_{ij}^{c_j}\Big)/G_i\Big) \le \frac{a(P', \mathbb{A}^4, \mathrm{div}(f_i) + \sum_j c_j \mathrm{div}(h_{ij}))}{|I_{G_i}(P')|} \le \beta_i + \frac{M_1}{r_i}.$$

By **Steps 6, 7, and 8**, $\beta_i \le \alpha_i \le \beta_i + M_1/r_i$ with β_i in the finite set $\mathcal{D}_1 \cap [0,1]$ and $r_i \to +\infty$. Passing to a subsequence, $\beta_i = \beta$ does not depend on i. Then $\alpha_i \to \beta$ and $\alpha_i \ge \beta$, which is impossible for a strictly increasing sequence. Hence the numbers $\mathrm{mld}((W \ni 0, \Delta)/G)$ in $(0,1)$ satisfy the ACC. □

D.7. **Proof of the class theorem: the case $CD < 0$.** In this subsection, we prove Lemmas D.24 and D.25 and Theorem 10.1. In the proofs of this subsection, for the finite set $S \subset [0,1]$ of the statement being proved, $\gamma := \min(S\backslash\{0\})$ if $S \not\subset \{0\}$ and $\gamma := 1$ otherwise, so that every non-zero element of S is at least γ.

Lemma D.24. *Fix a finite set $S \subset [0,1]$. In case (2) of Proposition D.22, let $r_i := |G_i \cap \mathbf{T}|$, and let $Q = (Q_x, Q_y, Q_z, Q_t)$ be $(6,4,3,3)$, $(9,6,4,4)$, $(15,10,6,6)$, F_* be $x^2+y^3+z^at^b$, $x^2+y^3+yz^at^b$, $x^2+y^3+z^at^b$, and d be 12, 18, 30 in the rows (E_6), (E_7), (E_8), respectively. Then, after omitting the marks with label zero and passing to a subsequence, we may replace $(W_i, (h_{ij})_j)$ with a*

numerical admission, with the group G_i, of the terminal data of Theorem 10.1 from which it was obtained in Proposition D.22, given by Construction-Theorem 4.13 with integers $N \geq N_2$ and $L \geq \max(L_2, 6)$, where N_2 is the integer given by Lemma D.21 and L_2 is the integer given by that lemma for N. After the replacement, the equation f_i of W_i satisfies $f_i - x^2 \in \mathbb{C}[y,z,t]$, the principal block (Set-up 5.1) of f_i along $\mathbf{T}$ is F_*, and, with the marks h_{ij} written with x-degree at most one and h_{ij}^0 their principal blocks along $\mathbf{T}$, the following hold.

(1) The least Q-weighted degree n_j of a monomial of h_{ij}, and the $\mathbf{T}$-weight κ_j of the monomials of h_{ij} of Q-weighted degree n_j, do not depend on i, and

$$\sum_j c_j n_j \leq Q_x + Q_y + Q_z + Q_t - d - 1 - \epsilon_S,$$

where ϵ_S is the number given by Lemma D.20.

(2) Each monomial of $f_i - F_*$ has $\mathbf{T}$-weight $6s + lr_i$, and each monomial of $h_{ij} - h_{ij}^0$ has $\mathbf{T}$-weight $\kappa_j + lr_i$, for a non-zero integer l, and the least total degree of a monomial of $f_i - F_*$ or of $h_{ij} - h_{ij}^0$ tends to $+\infty$ with i.

Proof. The proof is long, so we divide it into several steps. The $\mathbf{T}$-weight of a monomial $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}t^{\alpha_4}$ is $3s\alpha_1 + 2s\alpha_2 + C\alpha_3 + D\alpha_4$, and the Q-weighted degree of $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}t^{\alpha_4}$ is $Q_x\alpha_1 + Q_y\alpha_2 + Q_z\alpha_3 + Q_t\alpha_4$. For $g \in G_i$, we write $g^*x = \chi_x(g)x$, $g^*y = \chi_y(g)y$, $g^*z = \chi_z(g)z$, and $g^*t = \chi_t(g)t$, where g^* denotes the pullback of functions. By Proposition D.22(2),

$$aC + bD = 6s \ \text{ in the rows } (\mathrm{E}_6) \text{ and } (\mathrm{E}_8), \qquad aC + bD = 4s \ \text{ in the row } (\mathrm{E}_7).$$

Hence any monomial of F_* has $\mathbf{T}$-weight $6s$ and Q-weighted degree d. Let $g_* := F_* - x^2 \in \mathbb{C}[y,z,t]$, and let $J \subset \mathbb{C}[y,z,t]$ be the ideal generated by $\partial g_*/\partial y$, $\partial g_*/\partial z$, and $\partial g_*/\partial t$.

Step 1. In this step we show that any monomial $m = y^k z^p t^q$ of $\mathbf{T}$-weight $6s$ and Q-weighted degree greater than d belongs to J.

Since $C < 0 < D$, two integers u, v with $uC + vD = 0$ satisfy either $u = v = 0$ or $uv > 0$. For $k \leq 1$ in the rows (E_6) and (E_8), and for $k \leq 2$ in the row (E_7), we show below that the condition on the $\mathbf{T}$-weight of m is equivalent to a relation $uC + vD = 0$ and the condition on the Q-weighted degree of m is equivalent to the inequality $u + v > 0$, for a pair of integers u, v. Thus, for such k, $u > 0$ and $v > 0$.

Consider the rows (E_6) and (E_8). Here $Q_z = Q_t$, $3Q_y = d = Q_z(a+b)$, $\partial g_*/\partial y = 3y^2$, and $z\partial g_*/\partial z = az^a t^b$, so $y^2 \in J$ and $z^a t^b \in J$. If $k \geq 2$, then $m \in (y^2) \subset J$. If $k = 1$, then the conditions $2s + pC + qD = 6s$ and $Q_y + Q_z(p+q) > d$ are equivalent to

$$(3p - 2a)C + (3q - 2b)D = 0 \quad \text{and} \quad (3p - 2a) + (3q - 2b) > 0.$$

Thus $3p > 2a$ and $3q > 2b$. If $a \leq 3$ and $b \leq 3$, which holds in the row (E_6) and in the row (E_8) with $\{a, b\} = \{2, 3\}$, then $p \geq a$ and $q \geq b$, and $m \in (z^a t^b) \subset J$. In the row (E_8) with $(a, b) = (4, 1)$, we have $p \geq 3$ and $q \geq 1$, so $m \in (z^3 t) = (\partial g_*/\partial z) \subset J$. The case $(a, b) = (1, 4)$ is symmetric, with $m \in (zt^3) = (\partial g_*/\partial t)$. If $k = 0$, then the conditions $pC + qD = 6s$ and $Q_z(p+q) > d$ are equivalent to

$$(p - a)C + (q - b)D = 0 \quad \text{and} \quad (p - a) + (q - b) > 0.$$

Thus $p > a$ and $q > b$, and $m \in (z^a t^b) \subset J$.

Consider the row (E_7). Let $M_* := z^a t^b$, so $g_* = y^3 + yM_*$, $a + b = 3$, $Q = (9, 6, 4, 4)$, and $d = 18$. Since

$$\frac{\partial g_*}{\partial y} = 3y^2 + M_* \quad \text{and} \quad z\frac{\partial g_*}{\partial z} = ayM_*,$$

we have $yM_* \in J$, $y^3 = \frac{1}{3}\big(y\partial g_*/\partial y - yM_*\big) \in J$, and $M_*^2 = M_*\partial g_*/\partial y - 3y \cdot yM_* \in J$. If $k \geq 3$, then $m \in (y^3) \subset J$. If $k = 2$, then the conditions $4s + pC + qD = 6s$ and $12 + 4(p+q) > 18$ are equivalent to

$$(2p - a)C + (2q - b)D = 0 \quad \text{and} \quad (2p - a) + (2q - b) > 0.$$

Thus $2p > a$ and $2q > b$. Since $a, b \le 2$, we obtain $p \ge a$ and $q \ge b$, so $m \in (y^2 M_*) \subset J$. If $k = 1$, then the conditions $2s + pC + qD = 6s$ and $6 + 4(p+q) > 18$ are equivalent to

$$(p-a)C + (q-b)D = 0 \quad \text{and} \quad (p-a) + (q-b) > 0.$$

Thus $p > a$ and $q > b$, and $m \in (yM_*) \subset J$. If $k = 0$, then the conditions $pC + qD = 6s$ and $4(p+q) > 18$ are equivalent to

$$(2p-3a)C + (2q-3b)D = 0 \quad \text{and} \quad (2p-3a) + (2q-3b) > 0.$$

Thus $2p > 3a$ and $2q > 3b$. Since $a, b \le 2$, we obtain $p \ge 2a$ and $q \ge 2b$, so $m \in (M_*^2) \subset J$.

Step 2. In this step we construct, for any i, a G_i-equivariant automorphism ψ_i of $\mathbb{C}[[x,y,z,t]]$ with the following properties: $\psi_i(x) = x$, the power series $\psi_i(y), \psi_i(z), \psi_i(t)$ belong to $\mathbb{C}[[y,z,t]]$, any monomial of $\psi_i(y)$ (resp. $\psi_i(z)$, $\psi_i(t)$) has the $\mathbf{T}$-weight of y (resp. z, t), and the sum of the terms of $\mathbf{T}$-weight $6s$ of $\psi_i(f_i)$ is F_*.

Recall that $f_i = x^2 + y^3 + yA_i(z,t) + B_i(z,t)$ with $\operatorname{mult} A_i \ge 3$ and $\operatorname{mult} B_i \ge 4$, and that the non-zero homogeneous part $B_{i,4}$, $A_{i,3}$, or $B_{i,5}$ of the row is $b_i z^a t^b$ (Proposition D.22). The monomials of f_i of Q-weighted degree at most d are x^2, y^3, and $z^a t^b$ (resp. $yz^a t^b$ in the row (E_7)), with coefficients 1, 1, and b_i, since the remaining monomials of yA_i and B_i are $yz^p t^q$ and $z^p t^q$ with $p+q \ge 3$ and $p+q \ge 5$ in the row (E_6), with $p+q \ge 4$ and $p+q \ge 5$ in the row (E_7), and with $p+q \ge 4$ and $p+q \ge 6$ in the row (E_8), and their Q-weighted degrees are at least $13 > 12$, $20 > 18$, and $34 > 30$, respectively. Let f_i^0 be the principal block of f_i, that is, the sum of the terms of f_i of $\mathbf{T}$-weight $6s$. Then $f_i^0 - x^2 \in \mathbb{C}[y,z,t]$, and $f_i^0 - x^2 - y^3 - b_i z^a t^b$ (resp. $f_i^0 - x^2 - y^3 - b_i y z^a t^b$) is a sum of monomials of Q-weighted degree greater than d. Let $\psi_{i,d+1}$ be the automorphism mapping z to $b_i^{-1/a} z$, where $b_i^{-1/a}$ is a fixed a-th root of b_i^{-1}, and fixing x, y, and t. It is G_i-equivariant since G_i is diagonal, and any monomial of $\psi_{i,d+1}(f_i^0) - F_*$ has Q-weighted degree at least $d+1$.

Let $e > d$ be an integer, and let $\psi_{i,e}$ be a G_i-equivariant automorphism of $\mathbb{C}[[x,y,z,t]]$ with the properties of ψ_i above except the last one, such that any monomial of $\psi_{i,e}(f_i^0) - F_*$ has Q-weighted degree at least e. Let Φ_e be the sum of the terms of $\psi_{i,e}(f_i^0) - F_*$ of Q-weighted degree e. Then $\Phi_e \in \mathbb{C}[y,z,t]$, any monomial of Φ_e has $\mathbf{T}$-weight $6s$, and Φ_e is G_i-semi-invariant with the character χ_f of f_i. By **Step 1**,

$$\Phi_e = \varphi_y \frac{\partial g_*}{\partial y} + \varphi_z \frac{\partial g_*}{\partial z} + \varphi_t \frac{\partial g_*}{\partial t}$$

for polynomials $\varphi_y, \varphi_z, \varphi_t \in \mathbb{C}[y,z,t]$. The polynomial $\partial g_*/\partial y$ has $\mathbf{T}$-weight $6s - 2s$, Q-weighted degree $d - Q_y$, and character $\chi_f \chi_y^{-1}$, and similarly for z and t. Since $\mathbf{T}$-weights, Q-weighted degrees, and characters are additive (resp. multiplicative) on products of monomials, we may replace φ_y by the sum of the terms of φ_y of $\mathbf{T}$-weight $2s$, Q-weighted degree $e - d + Q_y$, and character χ_y, and similarly for φ_z and φ_t. Let Ξ_e be the endomorphism of $\mathbb{C}[[x,y,z,t]]$ given by

$$x \to x, \quad y \to y - \varphi_y, \quad z \to z - \varphi_z, \quad t \to t - \varphi_t,$$

and let $\psi_{i,e+1} := \Xi_e \circ \psi_{i,e}$. The endomorphism Ξ_e is G_i-equivariant and preserves the $\mathbf{T}$-weights of monomials. Since φ_y has Q-weighted degree $e - d + Q_y > Q_y$, and similarly for z and t, the linear terms of φ_y, φ_z, and φ_t are multiples of coordinates of Q-weight larger than Q_y, Q_z, and Q_t, respectively, so the linear part of Ξ_e is the identity plus a nilpotent map, Ξ_e is an automorphism, and the Taylor expansion of F_* implies that any monomial of

$$\psi_{i,e+1}(f_i^0) - F_* = \Xi_e(F_*) - F_* + \Xi_e(\Phi_e) + \Xi_e\big(\psi_{i,e}(f_i^0) - F_* - \Phi_e\big)$$

has Q-weighted degree at least $e+1$, because $\Xi_e(F_*) - F_* + \Phi_e$ and $\Xi_e(\Phi_e) - \Phi_e$ are sums of monomials of Q-weighted degree at least $2e - d > e$.

The polynomials $\varphi_y, \varphi_z, \varphi_t$ defining Ξ_e have total degree at least $(e-d)/Q_x$, which tends to $+\infty$ with e. Thus the automorphisms $\psi_{i,e}$ converge in the (x,y,z,t)-adic topology to a G_i-equivariant endomorphism ψ_i with $\psi_i(f_i^0) = F_*$. For $e > d + Q_y$, the polynomials $\varphi_y, \varphi_z, \varphi_t$

have no linear terms, so the linear part of ψ_i is the linear part of $\psi_{i,e}$ for any $e > d + Q_y$, which is invertible, and ψ_i is an automorphism. Since ψ_i maps a monomial of $\mathbf{T}$-weight κ to a power series whose monomials have $\mathbf{T}$-weight κ, the sum of the terms of $\mathbf{T}$-weight $6s$ of $\psi_i(f_i)$ is $\psi_i(f_i^0) = F_*$. Moreover, $\psi_i(f_i) - x^2 \in \mathbb{C}[[y,z,t]]$.

Step 3. In this step we replace W_i and the marks with a numerical admission given by Construction-Theorem 4.13 with integers $N \geq N_2$ and $L \geq \max(L_2, 6)$.

The marks with label zero do not change Δ_i, and we omit them. Thus $c_j > 0$, and hence $c_j \geq \gamma$, for any j. If $S \subset \{0\}$, then there exists no mark and the statements on the marks are empty.

Let $(W_i' \ni o_i)$, G_i, and $\Delta_i' = \sum_j c_j \operatorname{div}(h_{ij}')$ be the terminal data of Theorem 10.1 from which $(W_i, (h_{ij})_j)$ was obtained in Proposition D.22, and let x', y', z', t' and f_i' be the eigenparameters and the equation in the data (ii) of Construction-Theorem 4.13 for the numerical admission of Proposition D.22. By Construction-Theorem 4.13(3), there exists a G_i-equivariant isomorphism Θ_i from the completion of the local ring of W_i' at o_i onto $\mathbb{C}[[x,y,z,t]]/(f_i)$. Lifting the images of x', y', z', t' under Θ_i to G_i-semi-invariant power series with the characters of x', y', z', t', respectively, we obtain a G_i-equivariant homomorphism

$$\tilde{\Theta}_i : \mathbb{C}[[x', y', z', t']] \to \mathbb{C}[[x,y,z,t]]$$

which induces Θ_i. Since f_i' and f_i belong to the squares of the maximal ideals, Θ_i induces an isomorphism between the cotangent spaces of $\mathbb{A}^4$ at the origins, so $\tilde{\Theta}_i$ is an isomorphism. Since Θ_i is an isomorphism, $\tilde{\Theta}_i$ maps the ideal (f_i') onto (f_i), so $\tilde{\Theta}_i(f_i') = u_i f_i$ for a unit u_i. The unit u_i is G_i-invariant, since f_i' and f_i both have the character χ_f (Construction-Theorem 4.13). We identify $\mathbb{C}[[x', y', z', t']]$ with $\mathbb{C}[[x,y,z,t]]$ by $x' \to x$, $y' \to y$, $z' \to z$, $t' \to t$, after exchanging z' and t' if z and t were exchanged in Proposition D.22. The identification is G_i-equivariant, since by Construction-Theorem 4.13 the coordinates x, y, z, t have the characters of x', y', z', t'. Then $\tilde{\Theta}_i$ is a G_i-equivariant automorphism of $\mathbb{C}[[x,y,z,t]]$, and we write $\hat{f}_i'$ and $\hat{h}_{ij}'$ for the expansions in $\mathbb{C}[[x,y,z,t]]$ of f_i' and of the regular functions h_{ij}' on the ambient germ given in the data (ii) and (iii) of Construction-Theorem 4.13. Then $\psi := \psi_i \circ \tilde{\Theta}_i$ is a G_i-equivariant automorphism of $\mathbb{C}[[x,y,z,t]]$. Let $v := \psi_i(u_i)^{-1}$, a G_i-invariant unit. Then

$$v\psi(\hat{f}_i') = \psi_i(u_i)^{-1}\psi_i(u_i f_i) = \psi_i(f_i).$$

Let N_2 be the integer of Lemma D.21 for $(W_i' \ni o_i)$, G_i, Δ_i', the data (ii) above, the regular functions h_{ij}' on the ambient germ of (iii) above as the fixed lifts of the marks, no prime divisors in (v), and the data (vi) given by ψ, v, and $u_j := 1$ for any j. Let $N \geq N_2$, let L_2 be the integer of Lemma D.21 for N, and let $L \geq \max(L_2, 6)$. Let $f_i^\sharp$ be the sum of the terms of $\psi_i(f_i)$ of total degree less than L, and let $h_{ij}^\sharp$ be the sum of the terms of $\psi(\hat{h}_{ij}')$ of total degree less than N. The polynomials $f_i^\sharp$ and $h_{ij}^\sharp$ are G_i-semi-invariant with the characters of f_i' and h_{ij}', and

$$f_i^\sharp \equiv v\psi(\hat{f}_i') \pmod{(x,y,z,t)^L}, \qquad h_{ij}^\sharp \equiv \psi(\hat{h}_{ij}') \pmod{(x,y,z,t)^N}.$$

Let $W_i^\sharp := (f_i^\sharp = 0) \subset \mathbb{A}^4$ and

$$\Delta_i^\sharp := \sum_j c_j \operatorname{div}(h_{ij}^\sharp|_{W_i^\sharp}).$$

By Lemma D.21, the following hold.

- $W_i^\sharp$ is an isolated Gorenstein terminal germ at 0, G_i acts faithfully on $W_i^\sharp$, and the coordinates x, y, z, t have the characters $\chi_x, \chi_y, \chi_z, \chi_t$. Thus the diagonal action of G_i on $\mathbb{A}^4$, and hence $\mathbf{T}$, r_i, and e_0, do not change when we replace W_i with $W_i^\sharp$ below.
- For any subgroup $G' \subset G_i$,

$$\operatorname{mld}((W_i^\sharp \ni 0, \Delta_i^\sharp)/G') = \operatorname{mld}((W_i' \ni o_i, \Delta_i')/G').$$

For $G' = G_i$, both sides are equal to α_i. For the trivial subgroup, Lemma D.20, applied after removing the marks that do not vanish at o_i, which does not change Δ_i' near o_i, implies

$$\mathrm{mld}(W_i^\sharp \ni 0, \Delta_i^\sharp) = \mathrm{mld}(W_i' \ni o_i, \Delta_i') \geq 1 + \epsilon_S.$$

- $(W_i^\sharp, \theta_2\Delta_i^\sharp)$ is lc near 0, and $(W_i^\sharp, \theta_1\Delta_i^\sharp)$ is klt near 0.

Moreover, $f_i^\sharp - x^2 \in \mathbb{C}[y,z,t]$, and, by **Step 2** and $L > 5$, the principal block of $f_i^\sharp$ along $\mathbf{T}$ is F_*. Possibly replacing W_i, f_i, h_{ij}, and Δ_i with $W_i^\sharp$, $f_i^\sharp$, $h_{ij}^\sharp$, and $\Delta_i^\sharp$, we may assume that $f_i = x^2 + g_i$ with $g_i \in \mathbb{C}[y,z,t]$ and that the principal block of f_i is F_*. Replacing x^2 with $-g_i$ in each h_{ij}, we write each h_{ij} with x-degree at most one without changing $h_{ij}|_{W_i}$, and h_{ij} remains G_i-semi-invariant, and the character of h_{ij} does not change, since x^2 and g_i have the character χ_f.

Step 4. In this step we prove the statement (2) for f_i.

Let $R_i := G_i \cap \mathbf{T}$, a cyclic group of order r_i. Since $\gcd(3s, 2s, C, D) = \gcd(s, C, D) = 1$, the group R_i is generated by the element acting on x, y, z, t by $\zeta^{3s}, \zeta^{2s}, \zeta^C, \zeta^D$ for a primitive r_i-th root of unity ζ, and the generator multiplies a monomial of $\mathbf{T}$-weight κ by ζ^κ. Since f_i and h_{ij} are G_i-semi-invariant, the $\mathbf{T}$-weights of any two monomials of f_i (resp. of h_{ij}) are congruent modulo r_i. Since x^2 is a monomial of f_i of $\mathbf{T}$-weight $6s$ and the principal block of f_i is F_*, any monomial of $f_i - F_*$ has $\mathbf{T}$-weight $6s + lr_i$ for a non-zero integer l.

Let $w_{\max} := \max\{3s, -C, D\} > 0$. The $\mathbf{T}$-weight of a monomial m has absolute value at most $w_{\max} \deg m$, where $\deg m$ is the total degree of m. Hence a monomial of $\mathbf{T}$-weight $6s + lr_i$ with $l \neq 0$ has total degree at least $(r_i - 6s)/w_{\max}$, which tends to $+\infty$ with i. In particular, after passing to a subsequence, any monomial of $f_i - F_*$ has total degree greater than $d/3$, hence Q-weighted degree greater than d, since $\min(Q_x, Q_y, Q_z, Q_t) \geq 3$. Thus $\mathrm{in}_Q(f_i) = F_*$.

Step 5. In this step we prove the statement (1).

The polynomial g_* is not a square in $\mathbb{C}[y,z,t]$, since $g_*(y,1,1)$ is y^3+1 or y^3+y. Thus $F_* = x^2 + g_*$ is irreducible, $(F_* = 0) \subset \mathbb{A}^4$ is integral, and $(F_* = 0)$ is not contained in any coordinate hyperplane, since otherwise F_* would divide a power of a coordinate. The vector Q is primitive, and W_i is normal near 0. By **Step 4** and Lemma 4.12, there exists a prime divisor E_i over W_i with center 0 such that

$$a(E_i, W_i, 0) = Q_x + Q_y + Q_z + Q_t - d,$$

and $\mathrm{mult}_{E_i}(h|_{W_i}) = \mathrm{wt}_Q(h)$ for any polynomial h such that F_* does not divide $\mathrm{in}_Q(h)$. Let n_{ij} be the least Q-weighted degree of a monomial of h_{ij}. Since $\mathrm{in}_Q(h_{ij})$ is non-zero with x-degree at most one and F_* is monic of degree two in x, the polynomial F_* does not divide $\mathrm{in}_Q(h_{ij})$, so $\mathrm{mult}_{E_i} h_{ij} = n_{ij}$. By **Step 3**,

$$Q_x + Q_y + Q_z + Q_t - d - \sum_j c_j n_{ij} = a(E_i, W_i, \Delta_i) \geq \mathrm{mld}(W_i \ni 0, \Delta_i) \geq 1 + \epsilon_S,$$

which is the inequality of (1) with n_{ij} in place of n_j.

Since $Q_x + Q_y + Q_z + Q_t - d \in \{4, 5, 7\}$ and $c_j \geq \gamma$, we obtain $n_{ij} < 7/\gamma$. A monomial of Q-weighted degree n_{ij} has total degree at most $n_{ij}/3 < 7/(3\gamma)$, and so does a monomial of h_{ij} of least total degree. Hence the $\mathbf{T}$-weights of the monomials of h_{ij} of Q-weighted degree n_{ij} or of least total degree have absolute value less than $7w_{\max}/(3\gamma)$. By **Step 4**, these $\mathbf{T}$-weights are congruent modulo r_i. Thus, for $r_i > 14w_{\max}/(3\gamma)$, all the monomials of h_{ij} of Q-weighted degree n_{ij} and all the monomials of h_{ij} of least total degree have $\mathbf{T}$-weight κ_{ij}, for an integer κ_{ij}. In particular, $\kappa_{ij} = \kappa(h_{ij})$ in the notation of Set-up 5.1, and h_{ij}^0 is the sum of the terms of h_{ij} of $\mathbf{T}$-weight κ_{ij}. Since $n_{ij} < 7/\gamma$ and $|\kappa_{ij}| < 7w_{\max}/(3\gamma)$ are integers, after passing to a subsequence, $n_j := n_{ij}$ and $\kappa_j := \kappa_{ij}$ do not depend on i. This proves (1).

Step 6. In this step we prove the statement (2) for the marks.

By **Steps 4 and 5**, any monomial of h_{ij} has $\mathbf{T}$-weight $\kappa_j + lr_i$ for an integer l, and the monomials with $l = 0$ are the monomials of h_{ij}^0. Thus any monomial of $h_{ij} - h_{ij}^0$ has $\mathbf{T}$-weight $\kappa_j + lr_i$ with $l \neq 0$, and the total degree of such a monomial is at least

$$\frac{r_i - |\kappa_j|}{w_{\max}} \geq \frac{r_i - 7w_{\max}/(3\gamma)}{w_{\max}},$$

which tends to $+\infty$ with i. Together with **Step 4**, this proves (2). □

Lemma D.25. *In the situation of Lemma D.24, let θ_2 be the second of the real numbers $\theta_1 < \theta_2 < \theta_3$ fixed in Proposition D.22. There exists a real number M_0, depending only on S and θ_2, with the following property. Let F be a prime divisor over W_i with center 0 such that $a(F, W_i, \Delta_i) \leq 3e$, where $e := |I_{G_i}(F)|$. Then there exists a prime divisor E over $\mathbb{A}^4$ with center 0 such that the following hold.*

(1) $I_{G_i}(E) = I_{G_i}(F)$, $\operatorname{mult}_E \phi = \operatorname{mult}_F \phi$ for any $\phi \in \mathbb{C}[y,z,t]$, $\operatorname{mult}_E x = \operatorname{mult}_F x$, and $\operatorname{mult}_E h_{ij} = \operatorname{mult}_F h_{ij}$ for any j.

(2) We have

$$a\Big(E, \mathbb{A}^4, \operatorname{div}(f_i) + \sum_j c_j \operatorname{div}(h_{ij})\Big) = a(F, W_i, \Delta_i).$$

(3) The numbers $a(E, \mathbb{A}^4, 0)$, $\operatorname{mult}_E f_i$, $\operatorname{mult}_E x$, and $\operatorname{mult}_E h_{ij}$ for any j are at most $M_0 e$.

(4) $\operatorname{mult}_E f_i > 2 \operatorname{mult}_E x$.

Proof. The proof is long, so we divide it into several steps. By Lemma D.24, $c_j > 0$ for any j, $f_i = x^2 + g_i$ with $g_i \in \mathbb{C}[y,z,t]$, and $h_{ij} = p_{ij} + xq_{ij}$ with $p_{ij}, q_{ij} \in \mathbb{C}[y,z,t]$. Then $c_j \geq \gamma$ for any j. Let

$$a_0 := \frac{3\theta_2}{\theta_2 - 1} \quad \text{and} \quad M_0 := 31a_0 + \frac{3}{\gamma(\theta_2 - 1)} + 1.$$

Fix i and F as in the statement, let $I := I_{G_i}(F)$, and let

$$a_F := a(F, W_i, 0), \qquad X := \operatorname{mult}_F x, \qquad H_j := \operatorname{mult}_F h_{ij}.$$

Since the center of F is 0, the integers X, $\operatorname{mult}_F y$, $\operatorname{mult}_F z$, and $\operatorname{mult}_F t$ are positive.

Step 1. In this step we show that

$$a_F \leq a_0 e, \qquad H_j \leq \frac{3e}{\gamma(\theta_2 - 1)} \text{ for any } j, \qquad X \leq 15 a_0 e.$$

Since $(W_i, \theta_2 \Delta_i)$ is lc near 0 (Lemma D.24 and Lemma D.21), we have $a_F - \theta_2 \sum_k c_k H_k \geq 0$. Together with $a(F, W_i, \Delta_i) = a_F - \sum_k c_k H_k \leq 3e$, we obtain

$$\Big(1 - \frac{1}{\theta_2}\Big) a_F \leq a_F - \sum_k c_k H_k \leq 3e \quad \text{and} \quad \gamma H_j \leq \sum_k c_k H_k \leq \frac{a_F}{\theta_2} \leq \frac{3e}{\theta_2 - 1}.$$

By Construction-Theorem 4.13(3), the germ $(W_i \ni 0)$ is an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form. Since $f_i \in (x,y,z,t)^2$, the coordinate x restricts to an element of $\mathfrak{m}_0 \backslash \mathfrak{m}_0^2$. By Theorem 10.2, $(W_i, \frac{1}{15}\operatorname{div}(x))$ is lc near 0, so $a_F - X/15 \geq 0$, that is, $X \leq 15 a_F \leq 15 a_0 e$.

Step 2. In this step we extend mult_F to a prime divisor F' over $W_i \times \mathbb{A}^1_w$.

Let χ_x be the character of G_i with $g^* x = \chi_x(g) x$ for $g \in G_i$, let $\chi := \chi_x^2$, and let $q_F \in \{1, \dots, e\}$ be the integer of Lemma 3.1 for F and χ. Let $m_F := \max(2X, X + \max_j H_j)$, where $\max_j H_j := 0$ if there is no mark, and let k_1 be the integer such that

$$k_1 \equiv q_F \pmod{e} \quad \text{and} \quad m_F < k_1 \leq m_F + e.$$

Possibly shrinking W_i to a G_i-invariant affine neighborhood of 0, we may assume that W_i is normal. Let G_i act on $W_i \times \mathbb{A}^1_w$ by $g \cdot (p, w) = (g \cdot p, \chi(g) w)$. By Lemma 3.1 with $B = 0$, the

Gauss extension of mult_F with w-value k_1 [Liu26, Definition-Proposition 2.11] is $\mathrm{mult}_{F'}$ for a prime divisor F' over $W_i \times \mathbb{A}^1_w$ with center $(0,0)$, $I_{G_i}(F') = I$, and

$$a(F', W_i \times \mathbb{A}^1_w, 0) = a_F + k_1.$$

By the definition of the Gauss extension, $\mathrm{mult}_{F'} \phi = \mathrm{mult}_F \phi$ for any non-zero rational function ϕ on W_i, and $\mathrm{mult}_{F'} w = k_1$.

Step 3. In this step we adjoin a square root of $x^2 + w$.

Let $\mathbf{k}_1$ be the function field of $W_i \times \mathbb{A}^1_w$, let $\widehat{\mathbf{k}}_1$ be the completion of $\mathbf{k}_1$ along $\mathrm{mult}_{F'}$, and let $\hat{\nu}$ be the extension of $\mathrm{mult}_{F'}$ to $\widehat{\mathbf{k}}_1$. Since $\mathrm{mult}_{F'}(w/x^2) = k_1 - 2X > 0$, the series

$$\xi := \sum_{l \geq 0} \binom{1/2}{l} \left(\frac{w}{x^2}\right)^l$$

converges in $\widehat{\mathbf{k}}_1$, $\xi^2 = 1 + w/x^2$, and $\hat{\nu}(\xi - 1) = k_1 - 2X$. Let $\tilde{x} := x\xi$, let $\mathbf{k} := \mathbf{k}_1(\tilde{x}) \subset \widehat{\mathbf{k}}_1$, and let $\nu := \hat{\nu}|_{\mathbf{k}}$. Then

$$\tilde{x}^2 = x^2 + w, \qquad \nu(\tilde{x}) = X, \qquad \nu(\tilde{x} - x) = X + (k_1 - 2X) = k_1 - X.$$

Since $x^2 + w$ has degree one in w, it is not a square in $\mathbf{k}_1$, so $[\mathbf{k} : \mathbf{k}_1] = 2$. Since $\mathbf{k}_1 \subset \mathbf{k} \subset \widehat{\mathbf{k}}_1$ and $\mathbf{k}_1$ is dense in $\widehat{\mathbf{k}}_1$, the valuation ν has value group $\mathbb{Z}$, and the residue field of ν is the residue field of $\mathrm{mult}_{F'}$.

Step 4. In this step we construct E.

Since $f_i = x^2 + g_i$ vanishes on W_i, we have $x^2 = -g_i(y,z,t)$ in $\mathbf{k}_1$, and

$$f_i(\tilde{x}, y, z, t) = \tilde{x}^2 + g_i(y,z,t) = x^2 + w + g_i(y,z,t) = w.$$

Let $\mathbf{k}_2 := \mathbb{C}(\tilde{x}, y, z, t) \subset \mathbf{k}$. Since $x^2 = -g_i(y,z,t) \in \mathbf{k}_2$ and $w = f_i(\tilde{x}, y, z, t) \in \mathbf{k}_2$, the field $\mathbf{k} = \mathbf{k}_2(x)$ is algebraic over $\mathbf{k}_2$. Thus $\tilde{x}, y, z, t$ are algebraically independent, and $\mathbf{k}_2$ is the function field of $\mathbb{A}^4$ with coordinates $\tilde{x}, y, z, t$. We use $\tilde{x}, y, z, t$ as the coordinates x, y, z, t of $\mathbb{A}^4$ in the statement. Since $\nu(w/\tilde{x}^2) = k_1 - 2X > 0$, the series

$$\xi' := \sum_{l \geq 0} \binom{1/2}{l} \left(-\frac{w}{\tilde{x}^2}\right)^l$$

converges in the closure $\widehat{\mathbf{k}}_2$ of $\mathbf{k}_2$ in $\widehat{\mathbf{k}}_1$. Since $\tilde{x}^2 = x^2\xi^2$,

$$(\xi\xi')^2 = \frac{\tilde{x}^2}{x^2} \cdot \frac{\tilde{x}^2 - w}{\tilde{x}^2} = 1,$$

and $\hat{\nu}(\xi\xi' - 1) > 0$. Hence $\xi\xi' = 1$, and $x = \tilde{x}\xi' \in \widehat{\mathbf{k}}_2$. Thus $\mathbf{k}_1$, which is generated over $\mathbb{C}$ by x, y, z, t, w, is contained in $\widehat{\mathbf{k}}_2$, and $\widehat{\mathbf{k}}_2 = \widehat{\mathbf{k}}_1$. Therefore $\nu|_{\mathbf{k}_2}$ has value group $\mathbb{Z}$, and the residue field of $\nu|_{\mathbf{k}_2}$ is the residue field of $\mathrm{mult}_{F'}$, which has transcendence degree three over $\mathbb{C}$. By [KM98, Lemma 2.45], $\nu|_{\mathbf{k}_2} = \mathrm{mult}_E$ for a prime divisor E over $\mathbb{A}^4$. Since $\mathrm{mult}_E \tilde{x} = X$, $\mathrm{mult}_E y = \mathrm{mult}_F y$, $\mathrm{mult}_E z = \mathrm{mult}_F z$, and $\mathrm{mult}_E t = \mathrm{mult}_F t$ are positive, the center of E is 0.

Step 5. In this step we compute the orders along E and $a(E, \mathbb{A}^4, 0)$.

For any $\phi \in \mathbb{C}[y,z,t]$, **Step 2** implies $\mathrm{mult}_E \phi = \nu(\phi) = \mathrm{mult}_{F'} \phi = \mathrm{mult}_F \phi$. Moreover, $\mathrm{mult}_E \tilde{x} = X = \mathrm{mult}_F x$, and $\mathrm{mult}_E f_i(\tilde{x}, y, z, t) = \nu(w) = k_1$. For any j,

$$h_{ij}(\tilde{x}, y, z, t) - h_{ij}(x, y, z, t) = (\tilde{x} - x) q_{ij}(y, z, t),$$

and $\nu((\tilde{x} - x)q_{ij}) \geq k_1 - X > H_j = \nu(h_{ij}(x,y,z,t))$ by **Step 3** and the choice of k_1. Thus $\mathrm{mult}_E h_{ij}(\tilde{x}, y, z, t) = H_j = \mathrm{mult}_F h_{ij}$.

By **Steps 3 and 4**, ν has ramification index one over $\mathrm{mult}_{F'}$ and over mult_E, and the three residue fields coincide. Thus the valuation ring of ν is étale over the valuation rings of $\mathrm{mult}_{F'}$ and of mult_E, and the order along ν of a rational 4-form ω on $W_i \times \mathbb{A}^1_w$ (resp. on $\mathbb{A}^4$) is the order of ω along F' (resp. E). Since $\partial f_i / \partial x = 2x$, the form $dy \wedge dz \wedge dt/(2x)$ generates the canonical sheaf of W_i near 0, so $a(F', W_i \times \mathbb{A}^1_w, 0) - 1$ is the order along F' of $dw \wedge dy \wedge dz \wedge dt/(2x)$.

Similarly, $a(E,\mathbb{A}^4,0)-1$ is the order along E of $d\tilde{x}\wedge dy\wedge dz\wedge dt$. In the module of 4-forms of $\mathbf{k}$, we have $2xdx=-dg_i(y,z,t)$, so $dx\wedge dy\wedge dz\wedge dt=0$, and $2\tilde{x}d\tilde{x}=2xdx+dw$. Hence

$$d\tilde{x}\wedge dy\wedge dz\wedge dt=\frac{x}{\tilde{x}}dw\wedge\frac{dy\wedge dz\wedge dt}{2x}.$$

Since $\nu(x/\tilde{x})=0$, **Step 2** implies

$$a(E,\mathbb{A}^4,0)=a(F',W_i\times\mathbb{A}^1_w,0)=a_F+k_1.$$

Step 6. In this step we show that $I_{G_i}(E)=I$.

Since $\mathbf{k}=\mathbf{k}_1(\tilde{x})$ with $\tilde{x}^2=x^2+w$, and $g^*(x^2+w)=\chi_x(g)^2(x^2+w)$ for any $g\in G_i$, the action of G_i on $\mathbf{k}_1$ extends to an action on $\mathbf{k}$ with $g^*\tilde{x}=\chi_x(g)\tilde{x}$. On $\mathbf{k}_2$, it is the diagonal action of G_i on $\mathbb{A}^4$, since $\tilde{x}$ has the character χ_x. For $g\in G_i$, let $g\cdot\nu:=\nu\circ g^*$.

The polynomial $T^2-(x^2+w)$ has the roots $\pm x\xi$ in $\widehat{\mathbf{k}}_1$. Hence $\mathrm{mult}_{F'}$ has exactly two extensions to $\mathbf{k}$ (the extensions correspond to the irreducible factors over the completion [Ser79, Chapter II, §3, Theorem 1(iii) and Corollary 1]), and ν is the extension with $\nu(\tilde{x}/x-1)>0$. Let $g\in\mathrm{Stab}_{G_i}(F')$. Then $g\cdot\nu$ extends $\mathrm{mult}_{F'}$, and $(g\cdot\nu)(\tilde{x}/x-1)=\nu(\tilde{x}/x-1)>0$, since $g^*(\tilde{x}/x)=\tilde{x}/x$. Thus $g\cdot\nu=\nu$. Conversely, $g\cdot\nu=\nu$ implies $g\in\mathrm{Stab}_{G_i}(F')$ by restriction to $\mathbf{k}_1$. Similarly, x is a root of $T^2+g_i(y,z,t)$, whose roots in $\widehat{\mathbf{k}}_2$ are $\pm\tilde{x}\xi'$. Hence mult_E has at most two extensions to $\mathbf{k}$, and ν is the extension with $\nu(x/\tilde{x}-1)>0$. Let $g\in\mathrm{Stab}_{G_i}(E)$. Then $g\cdot\nu$ extends mult_E, and $(g\cdot\nu)(x/\tilde{x}-1)=\nu(x/\tilde{x}-1)>0$, since $g^*(x/\tilde{x})=x/\tilde{x}$. Thus $g\cdot\nu=\nu$. Conversely, $g\cdot\nu=\nu$ implies $g\in\mathrm{Stab}_{G_i}(E)$ by restriction to $\mathbf{k}_2$. Thus

$$\mathrm{Stab}_{G_i}(E)=\mathrm{Stab}_{G_i}(\nu)=\mathrm{Stab}_{G_i}(F').$$

The residue fields of mult_E, ν, and $\mathrm{mult}_{F'}$ coincide by **Steps 3 and 4**. Hence an element of the common stabilizer acts trivially on the residue field of mult_E if and only if it acts trivially on the residue field of $\mathrm{mult}_{F'}$. Thus $I_{G_i}(E)=I_{G_i}(F')=I$ by **Step 2**.

Step 7. In this step we conclude the proof.

The statement (1) follows from **Steps 5 and 6**. By **Step 5**,

$$a\Big(E,\mathbb{A}^4,\mathrm{div}(f_i)+\sum_j c_j\,\mathrm{div}(h_{ij})\Big)=a_F+k_1-k_1-\sum_j c_jH_j=a(F,W_i,\Delta_i),$$

which is (2). By **Step 1**,

$$k_1\le m_F+e\le 2X+\max_j H_j+e\le\Big(30a_0+\frac{3}{\gamma(\theta_2-1)}+1\Big)e.$$

Thus $\mathrm{mult}_E f_i=k_1\le M_0e$ and $a(E,\mathbb{A}^4,0)=a_F+k_1\le M_0e$. Moreover, by **Step 1**,

$$\mathrm{mult}_E x=X\le 15a_0e\le M_0e\quad\text{and}\quad\mathrm{mult}_E h_{ij}=H_j\le M_0e.$$

This proves (3). Since $k_1>m_F\ge 2X$, we have $\mathrm{mult}_E f_i=k_1>2X=2\,\mathrm{mult}_E x$. This proves (4). □

We now prove Theorem 10.1.

Proof of Theorem 10.1. The proof is long, so we divide it into several steps.

Step 1. In this step we prove the inequality with η_S, reduce the ACC to case (2) of Proposition D.22, and fix notation.

Let $\theta_S>1$ be the number given by Lemma D.20 for S, and set $\eta_S:=(\theta_S-1)/\theta_S$. Let $(W\ni o)$ be an isolated Gorenstein terminal hypersurface threefold germ of quadratic rank one whose residual cubic is the cube of a linear form, G a finite cyclic group acting faithfully on W and fixing o, and Δ a labelled boundary (Definition 2.7) with labels in S such that (W,Δ) is terminal near o. After removing the marks that do not vanish at o, which does not change Δ near o,

Lemma D.20 implies that $(W, \theta_S\Delta)$ is lc near o. Thus any prime divisor P over W whose center contains o satisfies $a(P, W, \theta_S\Delta) \geq 0$, that is, $\theta_S \operatorname{mult}_P \Delta \leq a(P, W, 0)$, and hence

$$a(P, W, \Delta) = a(P, W, 0) - \operatorname{mult}_P \Delta \geq \Big(1 - \frac{1}{\theta_S}\Big) a(P, W, 0) = \eta_S a(P, W, 0).$$

Assume that the ACC fails. Then there exist data $(W_i \ni o_i)$, G_i, and $\Delta_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$ $(i \geq 1)$ with the properties of $(W \ni o)$, G, and Δ above such that the numbers $\alpha_i := \operatorname{mld}((W_i \ni o_i, \Delta_i)/G_i)$ form a strictly increasing sequence in $(0, 1)$. Fix real numbers $1 < \theta_1 < \theta_2 < \theta_3 := \theta_S$. By Proposition D.22, after passing to a subsequence and applying numerical admissions, case (1) or case (2) of Proposition D.22 holds. By Proposition D.22, the numerical admissions above do not change α_i. Assume that case (1) of Proposition D.22 holds. The group G_i is diagonal and preserves the equation of W_i up to a character, the marks h_{ij} are polynomials by Proposition D.22, and (W_i, Δ_i) is klt near 0, since $(W_i, \theta_1\Delta_i)$ is klt near 0 by Proposition D.22. Thus Proposition D.23 implies that the numbers α_i satisfy the ACC, a contradiction. Hence case (2) of Proposition D.22 holds. By Lemma D.24, after a further numerical admission and passing to a subsequence, the conclusions of Lemma D.24 hold. The further admission is one of the numerical admissions of Lemma D.21, so α_i does not change and $(W_i, \theta_2\Delta_i)$ is lc near 0.

From now on we use the following notation. The circle is $\mathbf{T} = \mathbf{T}_{(3s,2s,C,D)}$ with integers $s \geq 0$ and $C < 0 < D$ such that $\gcd(s, C, D) = 1$. We write $w_x := 3s$, $w_y := 2s$, $w_z := C$, $w_t := D$ for the $\mathbf{T}$-weights of the coordinates x, y, z, t of $\mathbb{A}^4$, and the letter q denotes any one of x, y, z, t. Let $\mathcal{P}$ be the set of coordinates q with $w_q > 0$. Thus $\mathcal{P} = \{x, y, t\}$ if $s > 0$, $\mathcal{P} = \{t\}$ if $s = 0$, and z is the only coordinate of negative weight. Let $w_{\max} := \max\{3s, -C, D\}$. Let $R_i := G_i \cap \mathbf{T}$ and $r_i := |R_i|$, so that $r_i \to +\infty$ and $[G_i : R_i] \leq e_0$. We have $W_i = (f_i = 0) \subset \mathbb{A}^4$, and the principal block of f_i along $\mathbf{T}$ is the polynomial F_* of Lemma D.24. The labels c_j do not depend on i, and after deleting the marks with label zero we may assume that $c_j > 0$ for any j, so $c_j \geq \gamma$ for any j, with γ as fixed at the beginning of Subsection D.7. The marks h_{ij} have x-degree at most one, and h^0_{ij} is the principal block of h_{ij}, of $\mathbf{T}$-weight κ_j. Let $\mathfrak{a}_i := \prod_j h_{ij}^{c_j}$, an $\mathbb{R}$-ideal on $\mathbb{A}^4$. For an integer l, let $f_{i,l}$ and $h_{ij,l}$ be the sums of the terms of f_i and h_{ij} of $\mathbf{T}$-weight $6s + lr_i$ and $\kappa_j + lr_i$, respectively. By Lemma D.24(2),

$$f_i = \sum_{l \in \mathbb{Z}} f_{i,l}, \quad h_{ij} = \sum_{l \in \mathbb{Z}} h_{ij,l}, \quad f_{i,0} = F_*, \quad h_{ij,0} = h^0_{ij},$$

and the least degree d_i of a monomial of $f_i - F_*$ or of some $h_{ij} - h^0_{ij}$ satisfies $d_i \to +\infty$. Since $\epsilon_{\mathbf{T}} = -s + C + D$, the adjusted mass of Δ_i along $\mathbf{T}$ (Set-up 5.1) is

$$\delta = \sum_j c_j\kappa_j + s - C - D,$$

which does not depend on i. Finally, let

$$K_1 := \max\{6s, |\kappa_j| \mid j\}, \quad N_G := w_{\max}e_0, \quad N_0 := e_0 w_{\max}^2, \quad \epsilon_0 := \frac{1}{2N_0}.$$

Step 2. In this step we define the root charts and the reference surfaces, and we prove that the reference pairs are lc.

Let $\iota \in \{1, -1\}$ and let u be a coordinate with $\iota w_u > 0$, so that (ι, u) is one of $(-1, z)$, $(1, t)$, and, if $s > 0$, $(1, x)$, $(1, y)$. Let $\mathbb{A}^3_{\iota,u}$ be the affine space with the three coordinates q', $q \neq u$, and set $u' := 1$. We use the root chart of Definition-Lemma 11.4(2) for the weight vector $\iota(3s, 2s, C, D)$ at the coordinate u, now with weights of both signs, and denote it by (ι, u): it is $Y^{\iota,u} := \mathbb{A}^1_\lambda \times \mathbb{A}^3_{\iota,u}$ with the rational map

$$\pi_{\iota,u} : Y^{\iota,u} \dashrightarrow \mathbb{A}^4, \qquad u = \lambda^{\iota w_u}, \quad q = \lambda^{\iota w_q} q' \ \ (q \neq u).$$

Over $\lambda \neq 0$, $\pi_{\iota,u}$ is the quotient of $\mathbb{G}_{m,\lambda} \times \mathbb{A}^3_{\iota,u}$ by the free action of $\mu_{\iota w_u}$ given by $\lambda \to \zeta\lambda$, $q' \to \zeta^{-\iota w_q} q'$, onto the open subset $(u \neq 0)$, so it is étale there. If g is a $\mathbf{T}$-semi-invariant

polynomial of weight κ, then $\pi^*_{\iota,u}g = \lambda^{\iota\kappa}g^{\iota,u}$, where $g^{\iota,u} := g(x', y', z', t')$ is a polynomial on $\mathbb{A}^3_{\iota,u}$. Let $\varpi := \lambda^{r_i}$. We set

$$f_i^{\iota,u} := \lambda^{-6s\iota}\pi^*_{\iota,u}f_i = \sum_l \varpi^{\iota l} f_{i,l}^{\iota,u}, \qquad h_{ij}^{\iota,u} := \lambda^{-\iota\kappa_j}\pi^*_{\iota,u}h_{ij} = \sum_l \varpi^{\iota l} h_{ij,l}^{\iota,u},$$

which are polynomials in ϖ, ϖ^{-1} and the coordinates of $\mathbb{A}^3_{\iota,u}$. In superscripts we also write $+$ and $-$ for $\iota = 1$ and $\iota = -1$.

Let $\widehat{G}_i^{\iota,u}$ be the group of the automorphisms $(\lambda, (q')) \to (\zeta\lambda, (\zeta_q q'))$ of $Y^{\iota,u}$ lifting some $g = (g_q) \in G_i$ through $\pi_{\iota,u}$, that is, with $\zeta^{\iota w_u} = g_u$ and $\zeta_q = g_q\zeta^{-\iota w_q}$ for $q \neq u$. It contains the lifts of the identity. Any g has exactly ιw_u lifts. Since $(3s, 2s, C, D)$ is primitive, $\mathbf{T} \cong \mathbb{G}_m$ through $c \to (c^{w_q})$, and $R_i \cong \mu_{r_i}$. The lifts $(\lambda, (q')) \to (c^\iota\lambda, (q'))$ of the elements $c \in \mu_{r_i} \cong R_i$ form a subgroup $\widehat{R}_i \cong \mu_{r_i}$ acting only on λ. Let

$$H_i^{\iota,u} := \widehat{G}_i^{\iota,u}/\widehat{R}_i, \qquad V_i^{\iota,u} := \mathbb{A}^1_\varpi \times \mathbb{A}^3_{\iota,u} = Y^{\iota,u}/\widehat{R}_i.$$

Then $H_i^{\iota,u}$ acts diagonally on $V_i^{\iota,u}$, and $|H_i^{\iota,u}| = \iota w_u|G_i|/r_i \le N_G$. If a lift acts trivially on $\mathbb{A}^3_{\iota,u}$, then $g_q = (\zeta^\iota)^{w_q}$ for any q, so $g \in R_i$ and the lift belongs to $\widehat{R}_i$. Thus $H_i^{\iota,u}$ acts faithfully on $\mathbb{A}^3_{\iota,u}$. Moreover, since $\pi_{\iota,u}$ is the quotient by the lifts of the identity over $(u \neq 0)$, the quotients $Y^{\iota,u}/\widehat{G}_i^{\iota,u}$, $\mathbb{A}^4/G_i$, and $V_i^{\iota,u}/H_i^{\iota,u}$ are birational to each other through $\pi_{\iota,u}$. All $f_i^{\iota,u}$ and $h_{ij}^{\iota,u}$ are $H_i^{\iota,u}$-semi-invariant, and so is each summand $\varpi^{\iota l}f_{i,l}^{\iota,u}$, $\varpi^{\iota l}h_{ij,l}^{\iota,u}$, since the action is diagonal.

The reference objects of the chart (ι, u) are

$$S^{\iota,u} := (F_*^{\iota,u} = 0) \subset \mathbb{A}^3_{\iota,u}, \qquad \Sigma^{\iota,u} := S^{\iota,u} \cap \bigcap_{q\neq u,\ \iota w_q \le 0} (q' = 0), \qquad B_i^{\iota,u} := \sum_j c_j \operatorname{div}\left(h_{ij}^{0,\iota,u}|_{S^{\iota,u}}\right),$$

where $h_{ij}^{0,\iota,u} := (h_{ij}^0)^{\iota,u}$. The surface $S^{\iota,u}$ and the closed subset $\Sigma^{\iota,u}$ do not depend on i, and $H_i^{\iota,u}$ preserves $S^{\iota,u}$, $\Sigma^{\iota,u}$, and $B_i^{\iota,u}$. Write $F_* = x^2 + y^3 + z^at^b$ in the rows (E_6), (E_8) and $F_* = x^2 + y^3 + yz^at^b$ in the row (E_7), with $a, b \ge 1$ and $a + b \in \{4, 5, 3\}$, respectively. We list the charts.

(1) If $s > 0$, then $\Sigma^{+,x}$ is $(z' = 1 + y'^3 = 0)$ and $\Sigma^{+,y}$ is $(z' = x'^2 + 1 = 0)$. Each is a disjoint union of lines on which only t' is non-constant, and $S^{+,x}$, $S^{+,y}$ are smooth along them, since $\partial F_*^{+,x}/\partial y' = 3y'^2 \neq 0$ and $\partial F_*^{+,y}/\partial x' = 2x' \neq 0$ there.

(2) $\Sigma^{+,t}$ is the curve $(z' = x'^2 + y'^3 = 0)$ if $s > 0$, and the origin if $s = 0$. The surface $S^{+,t}$ is smooth along $\Sigma^{+,t}$ outside the origin, and the equation of $S^{+,t}$ is $x'^2 + y'^3 + z'^a$ or $x'^2 + y'^3 + y'z'^a$.

(3) $\Sigma^{-,z}$ is the origin, and the equation of $S^{-,z}$ is $x'^2 + y'^3 + t'^b$ or $x'^2 + y'^3 + y't'^b$.

Since $a, b \le 4$, and $a, b \le 2$ in the row (E_7), the origin is a smooth point or a Du Val singularity of type A_1, A_2, D_4, or E_6 of $S^{+,t}$ and $S^{-,z}$. Thus $S^{\iota,u}$ is a normal surface with klt singularities near $\Sigma^{\iota,u}$. Since $F_*^{\iota,u}$ is not divisible by a coordinate, $H_i^{\iota,u}$ acts faithfully on $S^{\iota,u}$. Each one-dimensional irreducible component $\mathcal{C}$ of $\Sigma^{\iota,u}$ contains a point $p_\mathcal{C}$ at which any coordinate non-constant on $\mathcal{C}$ vanishes, namely the point with $t' = 0$ in (1) and the origin in (2), and $p_\mathcal{C}$ is fixed by the stabilizer of $\mathcal{C}$ in $H_i^{\iota,u}$.

The mark h_{ij}^0 does not vanish identically on $W^0 := (F_* = 0) \subset \mathbb{A}^4$, since h_{ij}^0 has x-degree at most one and F_* is monic of degree two in x. Thus $h_{ij}^{0,\iota,u}|_{S^{\iota,u}} \neq 0$. The threefold W^0 is normal by Serre's criterion [Mat89, Theorem 23.8], since it is a hypersurface, hence Cohen–Macaulay [Mat89, Theorems 21.3 and 18.1], and its singular locus is contained in the curve $(x = y = zt = 0)$. Let $\Delta_i^0 := \sum_j c_j \operatorname{div}(h_{ij}^0|_{W^0})$.

We show that $(W^0, \theta_2\Delta_i^0)$ is lc near 0 for any $i \gg 0$. Since $(W_i, \theta_2\Delta_i)$ is lc near 0, the pair $(\mathbb{A}^4, W_i + \theta_2\sum_j c_j \operatorname{div}(h_{ij}))$ is lc near 0 by [Kaw07, Theorem]. Thus the $\mathbb{R}$-ideals $\mathfrak{b}_i := f_i\prod_j h_{ij}^{\theta_2 c_j}$ and $\mathfrak{b}_i^0 := F_*\prod_j (h_{ij}^0)^{\theta_2 c_j}$ satisfy $\operatorname{lct}_0(\mathbb{A}^4; \mathfrak{b}_i) \ge 1$. Let $\epsilon > 0$. The factors of $\mathfrak{b}_i$ and $\mathfrak{b}_i^0$ are congruent modulo $(x, y, z, t)^{d_i}$, and the exponents belong to the finite set $\{1\} \cup \{\theta_2 c_j\}$. By Theorem 2.12(2)

with $n=4$, $\mathrm{lct}_0(\mathbb{A}^4;\mathfrak{b}_i^0)\ge 1-\epsilon$ once $d_i\ge 4/(\epsilon\min\{1,\theta_2\gamma\})$. If $\mathrm{lct}_0(\mathbb{A}^4;\mathfrak{b}_i^0)<1$ for infinitely many i, then the numbers $\mathrm{lct}_0(\mathbb{A}^4;\mathfrak{b}_i^0)$ converge to 1 from below, so they contain a strictly increasing sequence. They are lc thresholds of the $\mathbb{R}$-divisors

$$\mathrm{div}(F_*)+\theta_2\sum_j c_j\,\mathrm{div}(h_{ij}^0)$$

on neighborhoods of 0 in $\mathbb{A}^4$, whose coefficients are finite sums of elements of the finite set $\{1\}\cup\{\theta_2 c_j\}$ of positive real numbers. Only finitely many such sums are less than any given real number, so they form a DCC set, and Theorem 2.12(1) gives a contradiction. Hence $(\mathbb{A}^4, W^0+\theta_2\sum_j c_j\,\mathrm{div}(h_{ij}^0))$ is lc near 0 for $i\gg 0$, and $(W^0,\theta_2\Delta_i^0)$ is lc near 0 by [Kaw07, Theorem]. Removing finitely many i, we may assume that $(W^0,\theta_2\Delta_i^0)$ is lc near 0 for any i.

We show that $(S^{\iota,u},\theta_2 B_i^{\iota,u})$ is lc near $\Sigma^{\iota,u}$. The torus $\mathbf{T}$ acts on W^0 and multiplies each h_{ij}^0 by a character, so the non-lc locus N_i of $(W^0,\theta_2\Delta_i^0)$ is a closed $\mathbf{T}$-invariant subset with $0\notin N_i$. Let $\lambda_0\ne 0$, $p'\in\Sigma^{\iota,u}$, and $p:=\pi_{\iota,u}(\lambda_0,p')$. Any coordinate q with $\iota w_q\le 0$ vanishes at p, so the point $c\cdot p=(c^{w_q}q(p))$ tends to 0 as $c^\iota\to 0$. Thus 0 belongs to the closure of $\mathbf{T}\cdot p$, and $p\notin N_i$. The preimage of W^0 under $\pi_{\iota,u}$ over $\lambda\ne 0$ is $\mathbb{G}_{m,\lambda}\times S^{\iota,u}$, and the pullback of Δ_i^0 there is $\mathrm{pr}_2^*B_i^{\iota,u}$, since

$$\pi_{\iota,u}^*h_{ij}^0=\lambda^{\iota\kappa_j}h_{ij}^{0,\iota,u}.$$

Since $\pi_{\iota,u}$ is étale there, the pair

$$\big(\mathbb{G}_{m,\lambda}\times S^{\iota,u},\theta_2\,\mathrm{pr}_2^*B_i^{\iota,u}\big)$$

is lc near $\mathbb{G}_{m,\lambda}\times\Sigma^{\iota,u}$. Hence $(S^{\iota,u},\theta_2B_i^{\iota,u})$ is lc near $\Sigma^{\iota,u}$. In particular, $(S^{\iota,u},B_i^{\iota,u})$ is lc near $\Sigma^{\iota,u}$, and $(\mathbb{A}^3_{\iota,u},S^{\iota,u}+B_{i,\mathbb{A}}^{\iota,u})$ is lc near $\Sigma^{\iota,u}$ by [Kaw07, Theorem], where

$$B_{i,\mathbb{A}}^{\iota,u}:=\sum_j c_j\,\mathrm{div}(h_{ij}^{0,\iota,u})\quad\text{on }\mathbb{A}^3_{\iota,u}.$$

Step 3. In this step we define the reference values $\beta_i^\pm$ and show that they are attained, belong to a fixed discrete set, and are computed by prime divisors of bounded log discrepancy.

For $\iota\in\{1,-1\}$, let

$$\beta_i^\iota:=\inf\frac{a(\nu,S^{\iota,u},B_i^{\iota,u})}{|I_{H_i^{\iota,u}}(\nu)|},$$

where u runs over the coordinates with $\iota w_u>0$ and ν runs over the prime divisors over $S^{\iota,u}$ whose center is contained in $\Sigma^{\iota,u}$. The center Z of such a ν is either a closed point p of $\Sigma^{\iota,u}$, and we set $p_Z:=p$, or a one-dimensional component $\mathcal{C}$ of $\Sigma^{\iota,u}$, and we set $p_Z:=p_{\mathcal{C}}$. Let $H_Z\subset H_i^{\iota,u}$ be the stabilizer of Z. Then $I_{H_i^{\iota,u}}(\nu)\subset H_Z$, and H_Z fixes p_Z. The coordinates q' with $q'(p_Z)\ne 0$ are H_Z-invariant, since H_Z is diagonal. After translating them, H_Z acts diagonally on $\mathbb{A}^3_{\iota,u}$ and fixes p_Z. By Theorem 3.3(1) and (3), applied with $n=3$, $c=1$, $X=S^{\iota,u}$, and the group H_Z,

$$\mathrm{mld}\big((S^{\iota,u}\ni Z,B_i^{\iota,u})/H_Z\big)=\mathrm{mld}\big((\mathbb{A}^3_{\iota,u}\ni Z,S^{\iota,u}+B_{i,\mathbb{A}}^{\iota,u})/H_Z\big),\tag{D.49}$$

where both minimal log discrepancies are taken at the image of p, or of the generic point of $\mathcal{C}$. By [Liu26, Lemma 2.6], β_i^ι is the infimum of the left side of (D.49) over all u and Z, and $\beta_i^\iota\ge 0$.

Possibly shrinking $\mathbb{A}^3_{\iota,u}$ to an $H_i^{\iota,u}$-invariant neighborhood of $\Sigma^{\iota,u}$, we may assume that $(\mathbb{A}^3_{\iota,u},S^{\iota,u}+B_{i,\mathbb{A}}^{\iota,u})$ is lc. Let

$$(X_Z,\Delta_Z):=(\mathbb{A}^3_{\iota,u},S^{\iota,u}+B_{i,\mathbb{A}}^{\iota,u})/H_Z.$$

Any prime divisor P on $\mathbb{A}^3_{\iota,u}$ has coefficient $n_0+\sum_j c_jn_j\le 1$ in $S^{\iota,u}+B_{i,\mathbb{A}}^{\iota,u}$, with integers $n_0\in\{0,1\}$ and $0\le n_j\le 1/\gamma$. Thus the coefficients of $S^{\iota,u}+B_{i,\mathbb{A}}^{\iota,u}$ belong to a finite set Λ_0, and by [Liu26, Lemma 2.6(4)] the coefficients of Δ_Z belong to the finite set

$$\Lambda:=\{1-(1-b)/e\mid b\in\Lambda_0\cup\{0\},\ 1\le e\le N_G\}.$$

The group H_Z is a diagonal abelian group of order at most N_G, so X_Z is an open subset of one of the finitely many $\mathbb{Q}$-factorial klt threefolds $\mathbb{A}^3/H'$ [KM98, Lemma 5.16 and Proposition 5.20(4)], where H' runs over the diagonal subgroups of $\mathbb{G}_m^3$ of order at most N_G. For a prime divisor $\bar{P}$ on X_Z with ramification index $e_{\bar{P}}$, let g be a polynomial generating the ideal of the reduced preimage of $\bar{P}$ in the factorial ring of $\mathbb{A}^3_{\iota,u}$. Then the norm $\prod_{h\in H_Z} h^*g$ is H_Z-invariant and defines the Cartier divisor $(|H_Z|/e_{\bar{P}})\bar{P}$ on X_Z. Thus $N_G!\bar{P}$ is Cartier for any prime divisor $\bar{P}$, and $N_G!K_{X_Z}$ is Cartier. Hence $\Delta_Z = \sum_{\bar{P}}(\lambda_{\bar{P}}/N_G!)N_G!\bar{P}$ with $\lambda_{\bar{P}} \in \Lambda$. By Theorem 2.13(2) with $d = 3$, $r = N_G!$, and $I = \{\lambda/N_G! \mid \lambda \in \Lambda\}$, the log discrepancies of the pairs (X_Z, Δ_Z), which are lc by [KM98, Proposition 5.20(4)], valid for $\mathbb{R}$-divisors by [Liu26, Remark 2.17], form a subset $\mathcal{D}_1 \subset \mathbb{R}$ which does not depend on i and has finite intersection with any bounded interval. By Theorem 2.10(3), applied on the finitely many quotients $\mathbb{A}^3/H'$ to the closure of Δ_Z, which has coefficients in $\Lambda\backslash\{0\}$ and is $\mathbb{R}$-Cartier since $\mathbb{A}^3/H'$ is $\mathbb{Q}$-factorial, and to the image of p or of the generic point of $\mathcal{C}$, there exists a positive integer l_0, independent of i, u, and Z, such that the right side of (D.49) is computed by a prime divisor $\bar{E}$ over X_Z with $a(\bar{E}, X_Z, 0) \le l_0$. In particular, the right side of (D.49) is attained, so it is a non-negative element of $\mathcal{D}_1$. Since $\mathcal{D}_1 \cap [0, x]$ is finite for any real number x, β_i^ι belongs to $\mathcal{D}_1$ and is attained by some u, Z, and ν, since the left side of (D.49) is attained: for a closed point Z by [Nak16a, Remark 2.1] applied to the lc pair $(S^{\iota,u}, B_i^{\iota,u})/H_Z$ and [Liu26, Lemma 2.6(2) and (3)], and for $Z = \mathcal{C}$ by $\nu = \mathcal{C}$.

By [Liu26, Lemma 2.6(3)], there is a prime divisor E_Z over $\mathbb{A}^3_{\iota,u}$ over $\bar{E}$. By [Liu26, Lemma 2.6(2)], E_Z has center Z,

$$\frac{a(E_Z, \mathbb{A}^3_{\iota,u}, S^{\iota,u} + B^{\iota,u}_{i,\mathbb{A}})}{|I_{H_Z}(E_Z)|} = \mathrm{mld}\big((S^{\iota,u} \ni Z, B_i^{\iota,u})/H_Z\big), \tag{D.50}$$

and

$$a(E_Z, \mathbb{A}^3_{\iota,u}, 0) \le |I_{H_Z}(E_Z)| a(\bar{E}, X_Z, 0) \le N_G l_0,$$

since $K_{\mathbb{A}^3_{\iota,u}}$ is the pullback of $K_{X_Z} + \Delta_Z^0$ for a $\mathbb{Q}$-divisor $\Delta_Z^0 \ge 0$ on X_Z. Since $(\mathbb{A}^3_{\iota,u}, (q' = 0))$ and $(\mathbb{A}^3_{\iota,u}, S^{\iota,u} + B^{\iota,u}_{i,\mathbb{A}})$ are lc near Z, the numbers $\mathrm{mult}_{E_Z} q'$, $\mathrm{mult}_{E_Z} F_*^{\iota,u}$, and $\mathrm{mult}_{E_Z} h_{ij}^{0,\iota,u}$ are at most $M_2 := N_G l_0/\gamma$. Here a coordinate q' which does not vanish on Z has $\mathrm{mult}_{E_Z} q' = 0$.

Step 4. In this step we compare log discrepancies on $\mathbb{A}^4$ and on the root charts, and we prove the gap between β_i^+ and β_i^-.

For a prime divisor E over a normal variety Y, a real number $\varsigma > 0$, a normal variety X with a dominant generically finite rational map $Y \dashrightarrow X$, and an $\mathbb{R}$-divisor B on X, we set $a(\varsigma\,\mathrm{mult}_E, X, B) := \varsigma e_E a(E_X, X, B)$, where E_X is the prime divisor over X dominated by E and e_E is the ramification index of E over E_X. The number $a(\varsigma\,\mathrm{mult}_E, X, B)$ depends only on E_X and ςe_E, and we define $a(\varsigma\,\mathrm{mult}_E, X, B, \mathfrak{a})$ for an $\mathbb{R}$-ideal $\mathfrak{a}$ on X likewise. If a finite group H acts faithfully on X and F is a prime divisor over X, then $a(\mathrm{mult}_F/|I_H(F)|, X, B) = a(F, X, B)/|I_H(F)|$. By [Liu26, Lemma 2.6], the ramification index of F over the prime divisor $\bar{F}$ over X/H dominated by F is $|I_H(F)|$.

Claim D.26. *Let (ι, u) be a root chart, E a prime divisor over $Y^{\iota,u}$, and $v = \varsigma\,\mathrm{mult}_E$ with $\varsigma > 0$, such that v has a center on $Y^{\iota,u}$ and on $\mathbb{A}^4$. Then*

$$a(v, \mathbb{A}^4, \mathrm{div}(f_i), \mathfrak{a}_i) = a\Big(v, Y^{\iota,u}, \mathrm{div}(\lambda) + \mathrm{div}(f_i^{\iota,u}) + \sum_j c_j \,\mathrm{div}(h_{ij}^{\iota,u})\Big) - \iota\delta v(\lambda). \tag{D.51}$$

Let instead E be a prime divisor over $\mathbb{A}^1_\lambda \times S^{\iota,u}$ and $v = \varsigma\,\mathrm{mult}_E$ with $\varsigma > 0$ such that v has a center on W^0. Then for any real number $\theta > 0$,

$$a(v, W^0, \theta\Delta_i^0) = a\big(v, \mathbb{A}^1_\lambda \times S^{\iota,u}, \mathrm{div}(\lambda) + \theta\,\mathrm{pr}_2^* B_i^{\iota,u}\big) - \iota\Big(\theta \sum_j c_j\kappa_j + s - C - D\Big)v(\lambda). \tag{D.52}$$

Proof. Let X be $\mathbb{A}^4$ or W^0, and ω the generator $dx\wedge dy\wedge dz\wedge dt$ of $\omega_{\mathbb{A}^4}$ or the generator Ω of ω_{W^0} defined below. By the Hurwitz formula,

$$a(v,X,0)=\varsigma\big(1+\operatorname{mult}_E\operatorname{div}(\pi_{\iota,u}^*\omega)\big).$$

Let ω' be the wedge product of the forms dq', $q\neq u$. A direct calculation shows that

$$\pi_{\iota,u}^*(dx\wedge dy\wedge dz\wedge dt)=\pm\iota w_u\lambda^{\iota(5s+C+D)}\frac{d\lambda}{\lambda}\wedge\omega'.$$

Since

$$K_{Y^{\iota,u}}=\operatorname{div}(d\lambda\wedge\omega'),\qquad \pi_{\iota,u}^*f_i=\lambda^{6s\iota}f_i^{\iota,u},\qquad \pi_{\iota,u}^*h_{ij}=\lambda^{\iota\kappa_j}h_{ij}^{\iota,u},$$

the left side of (D.51) is

$$a\Big(v,Y^{\iota,u},\operatorname{div}(\lambda)+\operatorname{div}(f_i^{\iota,u})+\sum_j c_j\operatorname{div}(h_{ij}^{\iota,u})\Big)+\iota\Big(5s+C+D-6s-\sum_j c_j\kappa_j\Big)v(\lambda),$$

which is the right side. For (D.52), the residue Ω of $dx\wedge dy\wedge dz\wedge dt/F_*$ along W^0 generates ω_{W^0}. Since $\pi_{\iota,u}^*F_*=\lambda^{6s\iota}F_*^{\iota,u}$, we obtain

$$\pi_{\iota,u}^*\Omega=\pm\iota w_u\lambda^{\iota(-s+C+D)}\frac{d\lambda}{\lambda}\wedge\eta,$$

where the residue η of $\omega'/F_*^{\iota,u}$ along $S^{\iota,u}$ generates $\omega_{S^{\iota,u}}$ near $\Sigma^{\iota,u}$. Repeating the computation of the first part with Ω, η, the marks h_{ij}^0, and the labels θc_j, we obtain (D.52). □

Proof of Theorem 10.1 continued. We also use the Gauss extensions of [Liu26, Definition-Proposition 2.11], up to a positive rational multiple. Lemma 3.1 gives the following formula for $\varsigma=1$, a positive integer w-value in a fixed class modulo the order of the inertia group, and a prime divisor whose center is a closed point, together with the inertia group. In the proof of Claim D.28(1) we need it without a group, for rational ς and ξ and any center.

Claim D.27. *Let Y be a normal variety, B an $\mathbb{R}$-divisor on Y with K_Y+B $\mathbb{R}$-Cartier, ν a prime divisor over Y, and $\varsigma>0$ and $\xi\geq 0$ rational numbers. Then the Gauss extension v of $\varsigma\operatorname{mult}_\nu$ over $Y\times\mathbb{A}^1_w$ with w-value ξ, that is,*

$$v\Big(\sum_n a_nw^n\Big)=\min_n(\varsigma\operatorname{mult}_\nu a_n+n\xi)$$

for rational functions a_n on Y, is a positive multiple of mult_E for a prime divisor E over $Y\times\mathbb{A}^1_w$, and

$$a(v,Y\times\mathbb{A}^1_w,\operatorname{pr}_1^*B+\operatorname{div}(w))=\varsigma a(\nu,Y,B).\tag{D.53}$$

Proof. For $\xi=0$, $v=\varsigma\operatorname{mult}_{\nu\times\mathbb{A}^1}$ and (D.53) is immediate. Let $\xi>0$. Choose a positive integer N such that $N\varsigma$ and $N\xi$ are integers, let $N_1:=\gcd(N\varsigma,N\xi)$, and write $N\varsigma=N_1m_1$ and $N\xi=N_1m_2$ with coprime positive integers m_1 and m_2. By the formula for v, Nv is the Gauss extension of $N\varsigma\operatorname{mult}_\nu$ with w-value $N\xi$, and $Nv=N_1v'$, where v' is the Gauss extension of $m_1\operatorname{mult}_\nu$ with w-value m_2. By [Liu26, Definition-Proposition 2.11 and Lemma 18.1], applied to ν and the coprime positive integers m_1 and m_2, $v'=\operatorname{mult}_E$ for a prime divisor E over $Y\times\mathbb{A}^1_w$ and

$$a(E,Y\times\mathbb{A}^1_w,\operatorname{pr}_1^*B)=m_1a(\nu,Y,B)+m_2.$$

Since $\operatorname{mult}_E w=m_2$, we obtain

$$a\big(E,Y\times\mathbb{A}^1_w,\operatorname{pr}_1^*B+\operatorname{div}(w)\big)=m_1a(\nu,Y,B).$$

Thus $v=(N_1/N)\operatorname{mult}_E$, and by the definition of $a(\varsigma\operatorname{mult}_E,X,B)$ before Claim D.26 (with $X=Y\times\mathbb{A}^1_w$ and $e_E=1$), multiplying v by N multiplies both sides of (D.53) by N, and

$$a\big(v,Y\times\mathbb{A}^1_w,\operatorname{pr}_1^*B+\operatorname{div}(w)\big)=\frac{N_1}{N}a\big(E,Y\times\mathbb{A}^1_w,\operatorname{pr}_1^*B+\operatorname{div}(w)\big)=\frac{N_1m_1}{N}a(\nu,Y,B)=\varsigma a(\nu,Y,B).$$

□

Claim D.28. *The following hold.*

(1) Let (ι,u) be a root chart, ν a prime divisor over $S^{\iota,u}$ with center contained in $\Sigma^{\iota,u}$, and

$$l(\nu) := \min\left\{ \frac{\operatorname{mult}_\nu q'}{|w_q||I_{H_i^{\iota,u}}(\nu)|} \;\middle|\; q \neq u,\ \iota w_q < 0 \right\}.$$

Then $l(\nu) \geq 1/N_0$, and there exist a root chart $(-\iota,u_1)$ and a prime divisor ν_1 over $S^{-\iota,u_1}$ with center contained in $\Sigma^{-\iota,u_1}$ such that

$$\frac{a(\nu_1, S^{-\iota,u_1}, B_i^{-\iota,u_1})}{|I_{H_i^{-\iota,u_1}}(\nu_1)|} = \frac{a(\nu, S^{\iota,u}, B_i^{\iota,u})}{|I_{H_i^{\iota,u}}(\nu)|} - \iota\delta l(\nu). \tag{D.54}$$

(2) If $\delta<0$, then $\beta_i^- \geq \beta_i^+ + |\delta|/N_0$. If $\delta>0$, then $\beta_i^+ \geq \beta_i^- + \delta/N_0$. If $\delta=0$, then $\beta_i^+ = \beta_i^-$.

Proof. (1) Let $H := H_i^{\iota,u}$ and $v_\nu := \operatorname{mult}_\nu / |I_H(\nu)|$. The coordinates q' with $q \neq u$ and $\iota w_q < 0$ vanish on $\Sigma^{\iota,u}$, so $l := l(\nu) > 0$. Let u_1 be a coordinate with $\iota w_{u_1} < 0$ and $v_\nu(u_1') = |w_{u_1}| l$. The monomial $m := u^{|w_{u_1}|} u_1^{\iota w_u}$ has $\mathbf{T}$-weight zero and $\pi_{\iota,u}^* m = (u_1')^{\iota w_u}$. The group G_i acts on m by a character trivial on R_i, so $m^{[G_i:R_i]}$ is G_i-invariant. Thus $(u_1')^{\iota w_u [G_i:R_i]}|_{S^{\iota,u}}$ is an H-invariant regular function vanishing on the center of ν, and $\iota w_u [G_i : R_i] v_\nu(u_1')$ is a positive integer by [Liu26, Lemma 2.6]. Hence $l \geq 1/(w_{\max}^2 e_0) = 1/N_0$.

For a rational number $\xi \in [0,l]$, let v_ξ be the Gauss extension of v_ν over $\mathbb{A}^1_\lambda \times S^{\iota,u}$ with λ-value ξ. Then $v_\xi(\pi_{\iota,u}^* u) = \iota w_u \xi$ and $v_\xi(\pi_{\iota,u}^* q) = \iota w_q \xi + v_\nu(q')$ for $q \neq u$. Since $l > 0$ and the center of ν is contained in $\Sigma^{\iota,u}$, we obtain $v_l(\pi_{\iota,u}^* q) \geq 0$ for any q, with equality for $q = u_1$, and $v_l(\pi_{\iota,u}^* q) > 0$ if $\iota w_q \geq 0$. In particular, v_l has a center on W^0. By Claim D.27, v_l is a positive multiple of the order along a prime divisor over $\mathbb{A}^1_\lambda \times S^{\iota,u}$. By (D.52) with $\theta = 1$ and (D.53),

$$a(v_l, W^0, \Delta_i^0) = \frac{a(\nu, S^{\iota,u}, B_i^{\iota,u})}{|I_H(\nu)|} - \iota\delta l. \tag{D.55}$$

The map $\pi_{\iota,u}$ restricts to a dominant generically finite rational map $\mathbb{A}^1_\lambda \times S^{\iota,u} \dashrightarrow W^0$. Let $v_l^\natural$ be the valuation on W^0 given by $v_l^\natural(\varphi) := v_l(\pi_{\iota,u}^* \varphi)$ for rational functions φ on W^0. It is a positive multiple of the order along the prime divisor over W^0 dominated by the prime divisor of v_l. For $c \in \mathbb{C}^*$, the automorphism $\lambda \to c\lambda$ of $\mathbb{A}^1_\lambda \times S^{\iota,u}$ preserves v_l, and $\pi_{\iota,u}$ carries it to the action of $c^\iota \in \mathbf{T}$ on W^0, so $v_l^\natural$ is $\mathbf{T}$-invariant.

Consider the root chart $(-\iota,u_1)$ with radial coordinate λ_1 and coordinates q'', $q \neq u_1$, of $\mathbb{A}^3_{-\iota,u_1}$. The map $\pi_{-\iota,u_1}$ restricts to a dominant generically finite rational map $\mathbb{A}^1_{\lambda_1} \times S^{-\iota,u_1} \dashrightarrow W^0$. Let $\mathcal{V}$ be the set of valuations $\tilde v'$ on $\mathbb{A}^1_{\lambda_1} \times S^{-\iota,u_1}$ with $\tilde v'(\pi_{-\iota,u_1}^* \varphi) = v_l^\natural(\varphi)$ for rational functions φ on W^0. Each element of $\mathcal{V}$ is a positive multiple of the order along a prime divisor dominating the prime divisor of $v_l^\natural$, and $\mathcal{V}$ is finite, since $\pi_{-\iota,u_1}$ is dominant and generically finite. Let $\tilde v \in \mathcal{V}$. The automorphisms $\lambda_1 \to c\lambda_1$, $c \in \mathbb{C}^*$, are carried by $\pi_{-\iota,u_1}$ to the action of $\mathbf{T}$ on W^0, which preserves $v_l^\natural$, so they permute $\mathcal{V}$. Since a divisible group has no non-trivial finite quotient, they fix $\tilde v$. For $g = \sum_n g_n \lambda_1^n$ with rational functions g_n on $S^{-\iota,u_1}$, each $g_n\lambda_1^n$ is a linear combination of the functions $g(c_k\lambda_1)$ for finitely many distinct $c_k \in \mathbb{C}^*$, so $\tilde v(g_n\lambda_1^n) \geq \tilde v(g)$. Thus $\tilde v$ is the Gauss extension of the valuation $\nu_1'(\psi) := \tilde v(\mathrm{pr}_2^* \psi)$ on $S^{-\iota,u_1}$, with $\tilde v(\lambda_1) = v_l^\natural(u_1)/|w_{u_1}| = 0$. For $q \neq u_1$, $\nu_1'(q'') = v_l^\natural(q) \geq 0$, and $v_l^\natural(q) > 0$ if $-\iota w_q \leq 0$. The valuation $\tilde v$ is a positive multiple of the order along a prime divisor over the threefold $\mathbb{A}^1_{\lambda_1} \times S^{-\iota,u_1}$, so its value group is discrete and its residue field has transcendence degree two. The residue field of $\tilde v$ is generated over that of ν_1' by the residue of λ_1 [Bou72, Chapter VI, §10, no. 1, Proposition 2], so the residue field of ν_1' has transcendence degree at least one, hence exactly one by the inequality of Abhyankar [Bou72, Chapter VI, §10, no. 3, Corollary 1 to Theorem 1], since ν_1' is a non-trivial ($\nu_1'(u'') > 0$) valuation of the function field of the surface $S^{-\iota,u_1}$, trivial on $\mathbb{C}$. The value group of ν_1' is a non-zero subgroup of that of $\tilde v$, hence discrete. By [KM98, Lemma 2.45], $\nu_1' = c_\nu \operatorname{mult}_{\nu_1}$ for a

prime divisor ν_1 over $S^{-\iota,u_1}$ and a real number $c_\nu > 0$. Hence the center of ν_1 is contained in $\Sigma^{-\iota,u_1}$.

The subvariety $W^{0,\circ} := W^0 \cap (xyzt \neq 0)$ is closed in the torus $\mathbb{G}_m^4 = (xyzt \neq 0)$ and invariant under its closed subgroup $\mathbf{T} \cdot G_i$, which acts freely on $\mathbb{G}_m^4$ by translations. Hence the quotient $U_i := W^{0,\circ}/(\mathbf{T} \cdot G_i)$ exists, namely the image of $W^{0,\circ}$ in the torus $\mathbb{G}_m^4/(\mathbf{T} \cdot G_i)$, and U_i is normal, as a quotient of the normal variety $W^{0,\circ}$ (**Step 2**). Since

$$\pi_{\iota,u}(\lambda, p') = \lambda^\iota \cdot \pi_{\iota,u}(1, p')$$

for the action of $\mathbf{T}$, and each $\mathbf{T}$-orbit in $W^{0,\circ}$ intersects the slice $(u = 1)$ in one orbit of the lifts of the identity, the map $p' \to \pi_{\iota,u}(1, p')$ induces a bijective morphism of $(S^{\iota,u} \cap (\prod_{q \neq u} q' \neq 0))/H$ onto the normal variety U_i, which is birational, as is any bijective morphism between varieties, and hence an isomorphism by Zariski's main theorem. Likewise, $\pi_{-\iota,u_1}$ induces an isomorphism of $(S^{-\iota,u_1} \cap (\prod_{q \neq u_1} q'' \neq 0))/H_1$ onto U_i, where $H_1 := H_i^{-\iota,u_1}$. So $S^{\iota,u}/H$ and $S^{-\iota,u_1}/H_1$ are birational through U_i. A rational function φ on U_i pulls back to a $\mathbf{T}$-invariant rational function on W^0, whose pullbacks to $\mathbb{A}^1_\lambda \times S^{\iota,u}$ and $\mathbb{A}^1_{\lambda_1} \times S^{-\iota,u_1}$ are the pullbacks under pr_2 of the pullbacks φ_S and φ_{S_1} of φ to $S^{\iota,u}$ and $S^{-\iota,u_1}$. Since v_l and $\tilde{v}$ both induce the valuation $v_l^\natural$ on W^0, we obtain

$$v_\nu(\varphi_S) = v_l^\natural(\varphi) = \nu_1'(\varphi_{S_1}).$$

By [Liu26, Lemma 2.6], v_ν induces on U_i the order along the image $\bar{\nu}$ of ν, so $\bar{\nu}$ is also the image of ν_1, and $\nu_1' = c_\nu \operatorname{mult}_{\nu_1}$ induces $c_\nu |I_{H_1}(\nu_1)| \operatorname{mult}_{\bar{\nu}}$. Thus $c_\nu = 1/|I_{H_1}(\nu_1)|$. By (D.52) on the root chart $(-\iota, u_1)$ and (D.53) with $\xi = 0$,

$$a(v_l, W^0, \Delta_i^0) = a\big(\tilde{v}, \mathbb{A}^1_{\lambda_1} \times S^{-\iota,u_1}, \operatorname{div}(\lambda_1) + \mathrm{pr}_2^* B_i^{-\iota,u_1}\big) = \frac{a(\nu_1, S^{-\iota,u_1}, B_i^{-\iota,u_1})}{|I_{H_1}(\nu_1)|}.$$

Comparing the last display with (D.55), we obtain (D.54).

(2) Assume that $\delta < 0$. Let ν be a prime divisor attaining β_i^- (**Step 3**). By (1) with $\iota = -1$ applied to ν and by the definition of β_i^+,

$$\beta_i^+ \leq \beta_i^- + \delta l(\nu) \leq \beta_i^- - \frac{|\delta|}{N_0}.$$

Assume that $\delta > 0$. Applying (1) with $\iota = 1$ to a prime divisor attaining β_i^+, we obtain $\beta_i^- \leq \beta_i^+ - \delta/N_0$. If $\delta = 0$, (1) implies $\beta_i^- \leq \beta_i^+$ and $\beta_i^+ \leq \beta_i^-$. □

Proof of Theorem 10.1 continued. Set $\iota_0 := 1$ if $\delta \leq 0$ and $\iota_0 := -1$ if $\delta > 0$, and let $\beta_i := \beta_i^{\iota_0}$. Then $-\iota_0 \delta = |\delta|$, and by Claim D.28(2), $\beta_i^{-\iota_0} \geq \beta_i + |\delta|/N_0$.

Step 5. In this step we prove that $\alpha_i \geq \beta_i$ for $i \gg 0$.

Since (W_i, Δ_i) is klt near 0, there exists a prime divisor F over W_i with center 0 such that $a(F, W_i, \Delta_i)/|I_{G_i}(F)| = \alpha_i < 1$. Since $a(F, W_i, \Delta_i) \leq 3|I_{G_i}(F)|$, Lemma D.25 provides a prime divisor E over $\mathbb{A}^4$ with center 0 for F, and $\operatorname{mult}_E f_i > 2 \operatorname{mult}_E x$ by Lemma D.25(4). Let $e := |I_{G_i}(E)| = |I_{G_i}(F)|$ and $v := \operatorname{mult}_E /e$. By Lemma D.25, $\alpha_i = a(v, \mathbb{A}^4, \operatorname{div}(f_i), \mathfrak{a}_i)$, and $v(f_i)$, $v(h_{ij})$, and $v(q)$ are at most M_0 for any coordinate q, since $(\mathbb{A}^4, (q = 0))$ is lc. For $q \in \mathcal{P}$, the monomial $q^{-C} z^{w_q}$ has $\mathbf{T}$-weight zero, so $(q^{-C} z^{w_q})^{[G_i:R_i]}$ is a G_i-invariant polynomial vanishing at 0. By [Liu26, Lemma 2.6],

$$-Cv(q) + w_q v(z) \geq \frac{1}{e_0} \quad (q \in \mathcal{P}), \qquad v(q) \geq \frac{1}{e_0} \quad (w_q = 0). \tag{D.56}$$

Since $\epsilon_0 \leq 1/(2e_0 w_{\max})$, (D.56) implies the following: if $v(z) < \epsilon_0$, then $v(q) \geq \epsilon_0$ for any $q \in \mathcal{P}$, and if $v(q) < \epsilon_0$ for some $q \in \mathcal{P}$, then $v(z) \geq \epsilon_0$. Let i be so large that $\epsilon_0(r_i - K_1) > w_{\max} M_0$. Let H denote the group $H_i^{\iota,u}$ of the root chart under consideration.

Case A: $v(z) \geq \epsilon_0$. Let $f_i^{\geq 0} := \sum_{l \geq 0} f_{i,l}$ and $h_{ij}^{\geq 0} := \sum_{l \geq 0} h_{ij,l}$. A monomial $x^{\alpha_x} y^{\alpha_y} z^{\alpha_z} t^{\alpha_t}$ of some $f_{i,l}$ or $h_{ij,l}$ with $l \leq -1$ has $\mathbf{T}$-weight at most $K_1 - r_i$, so $-C\alpha_z \geq r_i - K_1$, and the value of

the monomial under v is at least $\alpha_z v(z) \geq \epsilon_0(r_i - K_1)/w_{\max} > M_0$. Hence $v(f_i^{\geq 0}) = v(f_i)$ and $v(h_{ij}^{\geq 0}) = v(h_{ij})$. Let

$$\sigma := \min_{q \in \mathcal{P}} \frac{v(q)}{w_q},$$

attained by $u \in \mathcal{P}$, and let v also denote a valuation on $Y^{+,u}$ with $v(\pi_{+,u}^* \varphi) = v(\varphi)$ for rational functions φ on $\mathbb{A}^4$, a positive multiple of the order along a prime divisor dominating E. Then $v(\lambda) = \sigma$, $v(q') = v(q) - w_q\sigma \geq 0$ for $q \in \mathcal{P}$, $v(z') = v(z) - C\sigma > 0$, and $v(q') = v(q) > 0$ if $w_q = 0$. The functions

$$f_i^{\geq 0,+,u} := \lambda^{-6s} \pi_{+,u}^* f_i^{\geq 0} = \sum_{l \geq 0} \varpi^l f_{i,l}^{+,u}, \qquad h_{ij}^{\geq 0,+,u} := \lambda^{-\kappa_j} \pi_{+,u}^* h_{ij}^{\geq 0}$$

are polynomials on $V := V_i^{+,u}$ whose restrictions to $(\varpi = 0)$ are $F_*^{+,u}$ and $h_{ij}^{0,+,u}$. The left side of (D.51) depends only on $v(f_i)$ and $v(h_{ij})$, and the cyclic cover $Y^{+,u} \to V$, $\lambda \to \varpi = \lambda^{r_i}$, is branched only along $(\lambda = 0)$. Thus Claim D.26 implies

$$\alpha_i = a\Big(v, V, \operatorname{div}(\varpi) + \operatorname{div}(f_i^{\geq 0,+,u}) + \sum_j c_j \operatorname{div}(h_{ij}^{\geq 0,+,u})\Big) - \delta\sigma. \tag{D.57}$$

Moreover $v(\varpi) = r_i\sigma > 0$ and $v(f_i^{\geq 0,+,u}) = v(f_i) - 6s\sigma > 0$, since $6s\sigma \leq 2v(x) < v(f_i)$ if $s > 0$. Hence the center of v on V is contained in

$$(\varpi = f_i^{\geq 0,+,u} = 0) \cap \bigcap_{w_q \leq 0} (q' = 0) = \{0\} \times \Sigma^{+,u},$$

so it is $\{0\} \times Z$ for Z as in **Step 3**. Since V/H and $\mathbb{A}^4/G_i$ are birational (**Step 2**), the valuation induced by v on V is $\operatorname{mult}_{E_V}/|I_H(E_V)|$ for a prime divisor E_V over V with center $\{0\} \times Z$, and $I_H(E_V) \subset H_Z$. After translating coordinates as in **Step 3**, $S^{+,u} = (f_i^{\geq 0,+,u} = \varpi = 0)$ is a complete intersection of codimension two in $V \cong \mathbb{A}^4$, and $(S^{+,u}, B_i^{+,u})$ is lc near $\Sigma^{+,u}$ (**Step 2**). By Theorem 3.3(1) and (3) with $n = 4$, $c = 2$, $X = S^{+,u}$, the group H_Z, and $\mathfrak{a} = \prod_j (h_{ij}^{\geq 0,+,u})^{c_j}$,

$$\begin{aligned} \frac{a\big(E_V, V, \operatorname{div}(\varpi) + \operatorname{div}(f_i^{\geq 0,+,u}), \mathfrak{a}\big)}{|I_H(E_V)|} &\geq \operatorname{mld}\Big(\big(V \ni \{0\} \times Z, \operatorname{div}(f_i^{\geq 0,+,u}) + \operatorname{div}(\varpi), \mathfrak{a}\big)/H_Z\Big) \\ &= \operatorname{mld}\big((S^{+,u} \ni Z, B_i^{+,u})/H_Z\big) \geq \beta_i^+. \end{aligned}$$

By (D.57), $\alpha_i \geq \beta_i^+ - \delta\sigma$.

Case B: $v(q) \geq \epsilon_0$ for any $q \in \mathcal{P}$. A monomial $x^{\alpha_x} y^{\alpha_y} z^{\alpha_z} t^{\alpha_t}$ of some $f_{i,l}$ or $h_{ij,l}$ with $l \geq 1$ has $\mathbf{T}$-weight at least $r_i - K_1$, so $\sum_{q \in \mathcal{P}} w_q \alpha_q \geq r_i - K_1$, and the value of the monomial under v is at least $\epsilon_0(r_i - K_1)/w_{\max} > M_0$. Let $f_i^{\leq 0} := \sum_{l \leq 0} f_{i,l}$ and $h_{ij}^{\leq 0} := \sum_{l \leq 0} h_{ij,l}$. On the root chart $(-1, z)$, let v also denote a valuation on $Y^{-,z}$ with $v(\pi_{-,z}^* \varphi) = v(\varphi)$ for rational functions φ on $\mathbb{A}^4$, a positive multiple of the order along a prime divisor dominating E, and let

$$\rho := v(\lambda) = \frac{v(z)}{-C}.$$

Then $v(q') = v(q) + w_q\rho > 0$ for $q \in \mathcal{P}$, and $v(q') = v(q) > 0$ if $w_q = 0$. The functions

$$f_i^{\leq 0,-,z} := \lambda^{6s} \pi_{-,z}^* f_i^{\leq 0} = \sum_{l \leq 0} \varpi^{-l} f_{i,l}^{-,z}, \qquad h_{ij}^{\leq 0,-,z} := \lambda^{\kappa_j} \pi_{-,z}^* h_{ij}^{\leq 0}$$

are polynomials on $V_i^{-,z}$, and $v(f_i^{\leq 0,-,z}) = v(f_i) + 6s\rho > 0$. The argument of Case A, with $(1, u)$, σ, $f_i^{\geq 0}$, $h_{ij}^{\geq 0}$ replaced by $(-1, z)$, ρ, $f_i^{\leq 0}$, $h_{ij}^{\leq 0}$, implies $\alpha_i \geq \beta_i^- + \delta\rho$.

By the first paragraph of this step, Case A or Case B holds. Assume that $\delta<0$, so $\beta_i=\beta_i^+$. In Case A, $\alpha_i\ge\beta_i^++|\delta|\sigma\ge\beta_i$. Otherwise $v(z)<\epsilon_0$, Case B holds, and $\rho<\epsilon_0\le 1/N_0$, so

$$\alpha_i\ge\beta_i^--|\delta|\rho\ge\beta_i+\frac{|\delta|}{N_0}-|\delta|\rho\ge\beta_i.$$

Assume that $\delta>0$, so $\beta_i=\beta_i^-$. In Case B, $\alpha_i\ge\beta_i^-+\delta\rho\ge\beta_i$. Otherwise $v(q)<\epsilon_0$ for some $q\in\mathcal{P}$, Case A holds, and $\sigma\le v(q)/w_q<\epsilon_0\le 1/N_0$, so

$$\alpha_i\ge\beta_i^+-\delta\sigma\ge\beta_i+\frac{\delta}{N_0}-\delta\sigma\ge\beta_i.$$

If $\delta=0$, then $\beta_i^+=\beta_i^-=\beta_i$ and either case implies $\alpha_i\ge\beta_i$. Hence $\alpha_i\ge\beta_i$.

Step 6. In this step we prove that $\alpha_i\le\beta_i+M_1/r_i$ for $i\gg 0$, where $M_1:=|\delta|(M_2+N_G)$.

Let $\iota:=\iota_0$. By **Step 3**, there exist a root chart (ι,u) and $Z\subset\Sigma^{\iota,u}$, a closed point or a one-dimensional component, such that $\mathrm{mld}((S^{\iota,u}\ni Z,B_i^{\iota,u})/H_Z)=\beta_i$, and the prime divisor E_Z over $\mathbb{A}^3_{\iota,u}$ of (D.50) has center Z, mult_{E_Z} at most M_2 on the coordinates, on $F_*^{\iota,u}$, and on the $h_{ij}^{0,\iota,u}$, and

$$\frac{a(E_Z,\mathbb{A}^3_{\iota,u},S^{\iota,u}+B_{i,\mathbb{A}}^{\iota,u})}{|I_{H_Z}(E_Z)|}=\beta_i.$$

Let $H:=H_i^{\iota,u}$, $I:=I_H(E_Z)=I_{H_Z}(E_Z)$, and $e:=|I|\le N_G$. By [Liu26, Lemma 2.6(1)], I is cyclic and acts faithfully on $\mathfrak{m}_{E_Z}/\mathfrak{m}_{E_Z}^2$ by a character χ_E. Let χ be the character of H with $h^*\varpi=\chi(h)\varpi$, and let $q_0\in\{1,\dots,e\}$ be the integer with $\chi|_I=\chi_E^{q_0}$. Let k_0 be the integer with $k_0\equiv q_0\pmod e$ and $M_2<k_0\le M_2+e$, and let $\hat F$ be the Gauss extension of mult_{E_Z} over $V:=V_i^{\iota,u}=\mathbb{A}^1_\varpi\times\mathbb{A}^3_{\iota,u}$ with ϖ-value k_0. Then $\hat F$ is a prime divisor over V with center $\{0\}\times Z$, $I_H(\hat F)=I$, and

$$a\big(\hat F,V,\mathrm{pr}_2^*(S^{\iota,u}+B_{i,\mathbb{A}}^{\iota,u})+\mathrm{div}(\varpi)\big)=a(E_Z,\mathbb{A}^3_{\iota,u},S^{\iota,u}+B_{i,\mathbb{A}}^{\iota,u}).\tag{D.58}$$

If Z is a closed point, this is Lemma 3.1, applied to the group H_Z and the restriction of χ to H_Z after translating coordinates as in **Step 3**, together with the equality $I_H(\hat F)=I_{H_Z}(\hat F)$, which holds since an element of $I_H(\hat F)$ stabilizes the center $\{0\}\times Z$ of $\hat F$ and hence belongs to H_Z. If Z is a curve, [Liu26, Definition-Proposition 2.11 and Lemma 18.1] implies that $\hat F$ is a prime divisor over V with center $\{0\}\times Z$ satisfying (D.58), and the residue field of $\hat F$ is generated over that of E_Z by the residue of ϖ/φ^{k_0}, where φ is a rational function on $\mathbb{A}^3_{\iota,u}$ with $\mathrm{mult}_{E_Z}\varphi=1$. An element of $I_H(\hat F)$ fixes E_Z and acts trivially on the residue field of E_Z, so $I_H(\hat F)\subset I$. Conversely, $g\in I$ multiplies the residue of ϖ/φ^{k_0} by $\chi(g)\chi_E(g)^{-k_0}=1$, so $I\subset I_H(\hat F)$.

We compute the orders along $\hat F$. For a polynomial g on $\mathbb{A}^3_{\iota,u}$, $\mathrm{mult}_{\hat F}g=\mathrm{mult}_{E_Z}g\ge 0$. A summand $\varpi^{\iota l}f_{i,l}^{\iota,u}$ with $\iota l\ge 1$ has order at least $k_0>M_2\ge\mathrm{mult}_{E_Z}F_*^{\iota,u}$. Let $\iota l\le -1$, and let $x^{\alpha_x}y^{\alpha_y}z^{\alpha_z}t^{\alpha_t}$ be a monomial of $f_{i,l}$. The $\mathbf{T}$-weight of the monomial is $6s+lr_i$, so $\sum_{\iota w_q<0}|w_q|\alpha_q\ge|l|r_i-6s$. The coordinates q' with $\iota w_q<0$ vanish on $Z\subset\Sigma^{\iota,u}$, so $\mathrm{mult}_{E_Z}q'\ge 1$, and the corresponding term $\varpi^{\iota l}\prod_{q\ne u}q'^{\alpha_q}$ of $f_i^{\iota,u}$ has order at least

$$-|l|k_0+\frac{|l|r_i-6s}{w_{\max}}\ge\frac{r_i-K_1}{w_{\max}}-k_0>M_2$$

if $r_i>w_{\max}(2M_2+N_G)+K_1$, which we assume. Replacing $f_{i,l}$ and $6s$ by $h_{ij,l}$ and κ_j above and using $|\kappa_j|\le K_1$, we obtain the corresponding bounds for the summands of $h_{ij}^{\iota,u}$. Hence

$$\mathrm{mult}_{\hat F}f_i^{\iota,u}=\mathrm{mult}_{E_Z}F_*^{\iota,u},\qquad\mathrm{mult}_{\hat F}h_{ij}^{\iota,u}=\mathrm{mult}_{E_Z}h_{ij}^{0,\iota,u}.$$

Let v be a valuation on $Y^{\iota,u}$ with $v(\varphi(\lambda^{r_i},\cdot))=\mathrm{mult}_{\hat F}\varphi/e$ for rational functions φ on V, so that $v(\lambda)=k_0/(er_i)$. For $q\ne u$, $v(\pi_{\iota,u}^*q)=\iota w_q v(\lambda)+v(q')$. Thus $v(\pi_{\iota,u}^*q)>0$ if $\iota w_q>0$, $v(\pi_{\iota,u}^*q)\ge 1/e$ if $w_q=0$, and $v(\pi_{\iota,u}^*q)\ge(1-w_{\max}k_0/r_i)/e>0$ if $\iota w_q<0$. Moreover

$v(\pi^*_{\iota,u}u) = \iota w_u v(\lambda) > 0$. Thus the valuation $v^\natural(\varphi) := v(\pi^*_{\iota,u}\varphi)$ on $\mathbb{A}^4$ has center 0. Since $\mathbb{A}^4/G_i$ and V/H are birational (**Step 2**), $v^\natural = \mathrm{mult}_{E^\natural}/|I_{G_i}(E^\natural)|$ for a prime divisor $E^\natural$ over $\mathbb{A}^4$ with center 0. By Theorem 3.3(1) with $n = 4$, $c = 1$, $X = W_i$, and the group G_i,

$$\alpha_i = \mathrm{mld}\big((\mathbb{A}^4 \ni 0, \mathrm{div}(f_i), \mathfrak{a}_i)/G_i\big) \le \frac{a(E^\natural, \mathbb{A}^4, \mathrm{div}(f_i), \mathfrak{a}_i)}{|I_{G_i}(E^\natural)|} = a(v^\natural, \mathbb{A}^4, \mathrm{div}(f_i), \mathfrak{a}_i).$$

By Claim D.26 (whose left side for v is $a(v^\natural, \mathbb{A}^4, \mathrm{div}(f_i), \mathfrak{a}_i)$ by definition), the cyclic cover $Y^{\iota,u} \to V$, the orders above, and (D.58),

$$\begin{aligned} a(v^\natural, \mathbb{A}^4, \mathrm{div}(f_i), \mathfrak{a}_i) &= \frac{1}{e}\Big(a(\hat{F}, V, \mathrm{div}(\varpi)) - \mathrm{mult}_{\hat{F}} f_i^{\iota,u} - \sum_j c_j \,\mathrm{mult}_{\hat{F}} h_{ij}^{\iota,u}\Big) - \iota\delta\frac{k_0}{er_i} \\ &= \frac{a(E_Z, \mathbb{A}^3_{\iota,u}, S^{\iota,u} + B^{\iota,u}_{i,\mathbb{A}})}{e} - \iota\delta\frac{k_0}{er_i} = \beta_i + |\delta|\frac{k_0}{er_i}. \end{aligned}$$

Since $k_0 \le M_2 + N_G$, we obtain $\alpha_i \le \beta_i + M_1/r_i$.

Step 7. In this step we conclude the proof.

By **Steps 5 and 6**, $\beta_i \le \alpha_i \le \beta_i + M_1/r_i$ for $i \gg 0$. By **Step 3**, β_i belongs to $\mathcal{D}_1 \cap [0,1]$, which is a finite set. Passing to a subsequence, we may assume that $\beta_i = \beta$ for any i. Since $r_i \to +\infty$, α_i converges to β. Since α_i is strictly increasing, $\alpha_i < \beta$ for any i, which contradicts $\alpha_i \ge \beta$. Hence the numbers in Theorem 10.1 satisfy the ACC. □

Appendix E. Proofs for several labels

In this section, we prove the propositions of Section 11. In the proofs below, scores are evaluated at one prime divisor P at a time, and $a_D(P) = \sum_k \mu_k a_k(P)$ (Definition-Lemma 11.4).

E.1. **Quadratic rank two, unbounded split radical order.** In this subsection, we prove Proposition 11.11. The lower bound (E.10) in the proof is proved by one augmentation of the plane traces of the marks: along the circle that remains, the plane trace of the k-th mark begins with a block $z^{p_k}t^{q_k}\varphi_{ik}$, where φ_{ik} is an invariant unit, and adding $\mathrm{div}(z)$ and $\mathrm{div}(t)$ with the coefficients $1 - \sum_k \mu_k p_k/m$ and $1 - \sum_k \mu_k q_k/m$ bounds $a(P, W, D)$ from below, linearly in the labels, at all prime divisors P at once.

Proof of Proposition 11.11. The proof is long, so we divide it into several steps. Assume that there exists a sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \ge 1$, as in Set-up 11.1, with W_i of quadratic rank two and split radical orders $n_i \to +\infty$, such that the numbers α_i form a strictly increasing sequence in $(0,1)$. We write $h_{ik} := h_{i,k}$, $c_k = \mu_k/m$, and

$$\Gamma := \{c_1, \dots, c_s\}.$$

Then $\Gamma \subset [0,1]$ is a finite set, in particular a DCC set. Passing to a subsequence, we may assume that $n_i \ge 2m$ for any i. Since $(\alpha_i)_i$ is strictly increasing, $\alpha_i \ge \alpha_1 > 0$ and $\alpha_i < \lim_{j\to+\infty}\alpha_j$ for any i.

Step 1. In this step we pass to polynomial equations and marks by a numerical admission, and fix the prime divisors which the admission protects.

Fix i, and fix a G_i-equivariant embedding of W_i into a smooth fourfold germ $(M_i \ni o_i)$ in which W_i is defined by a G_i-semi-invariant equation, given by ambient G_i-eigenparameters as in Definition-Lemma 11.4(2), which exists since W_i has embedding dimension four and G_i is abelian. By Definition 8.1, there exist formal coordinates x, y, z, t of M_i at o_i, on which G_i acts diagonally on z, t and monomially on the pair $\{x, y\}$, such that the completed local equation of W_i is $xy + \hat{g}_i(z,t)$ and $\mathrm{mult}_0 \hat{g}_i = n_i$. The formal coordinates x, y, z, t are used only to define E_i and to compare W_i with the numerical admission in this step. After the identification of $E_{i,0}$ below, the formal coordinates are not used again. Let E_i be the prime divisor over W_i with center o_i which corresponds [dFEM11, Lemma 2.21] to the exceptional prime divisor of the strict

transform of $\operatorname{Spec}\widehat{\mathcal{O}}_{W_i,o_i}$ in the weighted blow-up with weights $(\lfloor n_i/2\rfloor, \lceil n_i/2\rceil, 1, 1)$ in x, y, z, t (the strict transform intersects the exceptional divisor in a single prime divisor, since the lowest weighted part of $xy + \hat{g}_i$ is irreducible and not divisible by a coordinate, as shown after the definition of w_i below). Let F_i be a prime divisor over W_i with center o_i such that $a_D(F_i) = \alpha_i$, which exists by Proposition 11.7(0).

Since W_i is singular of quadratic rank two and G_i is cyclic, Theorem 8.11 and Proposition 11.8 provide a numerical admission $(W_{i,0}, (h_{ik,0})_k)$ of $(W_i, (h_{ik})_k)$ (Construction-Theorem 4.13) with group G_i, for the labels $c_1, \dots, c_s$, protecting F_i together with $I_{G_i}(F_i)$, and protecting E_i. When the admission is taken, we also fix an integer $L' > n_i$, so that $L' > \lceil n_i/2\rceil$, and take the admission as in the sentence of Theorem 8.11(1) on formal coordinates for this L'. If G_i acts diagonally on x, y, z, t, we apply that sentence to x, y, z, t (by the last sentence of Theorem 8.11(1) if x, y come from $x^2 + y^2 + \hat{g}_i$ with equal characters in Definition 8.1). Otherwise we apply it to the G_i-eigen formal coordinates $x' := (x + y)/2$, $y' := (x - y)/(2\sqrt{-1})$, z, t, in which the completed equation is $x'^2 + y'^2 + \hat{g}_i$ and $\chi_{x'} \neq \chi_{y'}$, since otherwise G_i would act diagonally on $x = x' + \sqrt{-1}y'$ and $y = x' - \sqrt{-1}y'$ (x', y', z, t are the coordinates of Definition 8.1 before the replacement by $x \pm \sqrt{-1}y$). Then the G_i-equivariant isomorphism of completions of Construction-Theorem 4.13(2) maps x, y, z, t to X, Y, z, t in case (i) below, and to $X + \sqrt{-1}Y$, $X - \sqrt{-1}Y$, z, t in case (ii) below, modulo the L'-th power of the maximal ideal. The first choice of coordinates leads to case (i) below and the second to case (ii), since the isomorphism maps x, y, z, t, resp. x', y', z, t, to X, Y, z, t modulo the square of the maximal ideal, so that the tangent cone of $W_{i,0}$ at 0 is the image ($XY = 0$), resp. ($X^2 + Y^2 = 0$), of the tangent cone ($xy = 0$), resp. ($x'^2 + y'^2 = 0$), of W_i at o_i, and XY and $X^2 + Y^2$ are not proportional. We use the following properties.

(1) $W_{i,0} \subset \mathbb{A}^4$ is an isolated Gorenstein terminal germ at 0, and either

$$W_{i,0} = (XY + g_{i,0}(z, t) = 0) \subset \mathbb{A}^4$$

and G_i acts diagonally on X, Y, z, t (case (i)), or

$$W_{i,0} = (X^2 + Y^2 + g_{i,0}(z, t) = 0) \subset \mathbb{A}^4$$

and G_i acts diagonally on X, Y, z, t with $\chi_X \neq \chi_Y$ (case (ii)). Here $g_{i,0} \in (z, t)^3$ is a polynomial, and $h_{i1,0}, \dots, h_{is,0}$ are G_i-semi-invariant polynomials.

(2) G_i acts faithfully on $W_{i,0}$. Let $D_{i,0} := \sum_k c_k \operatorname{div}(h_{ik,0})$. Then $(W_{i,0}, D_{i,0})$ is klt near 0, and $\alpha_i = \operatorname{mld}((W_{i,0} \ni 0, D_{i,0})/G_i)$.

(3) E_i corresponds to a prime divisor $E_{i,0}$ over $W_{i,0}$ with center 0 such that

$$a(E_{i,0}, W_{i,0}, 0) = a(E_i, W_i, 0), \qquad \operatorname{mult}_{E_{i,0}} h_{ik,0} = \operatorname{mult}_{E_i} h_{ik} \text{ for any } k.$$

In the first case let $x := X$ and $y := Y$, and in the second case let $x := X + \sqrt{-1}Y$ and $y := X - \sqrt{-1}Y$. Then $W_{i,0} = (xy + g_{i,0}(z, t) = 0)$. In the second case $\chi_Y = \pm\chi_X$, since $\chi_X^2 = \chi_Y^2$ is the character of the equation. An element $\sigma \in G_i$ with $\chi_Y(\sigma) = \chi_X(\sigma)$ acts on x and y by the scalar $\chi_X(\sigma)$, and an element σ with $\chi_Y(\sigma) = -\chi_X(\sigma)$ maps x to $\chi_X(\sigma)y$ and y to $\chi_X(\sigma)x$. Thus in both cases G_i acts linearly and monomially on the pair $\{x, y\}$ and diagonally on z, t, and preserves $xy + g_{i,0}$ up to a character χ. Let $G_i^{\mathrm{s}} \subset G_i$ be the subgroup of the elements preserving each of the lines $\mathbb{C}x$ and $\mathbb{C}y$. Then $G_i^{\mathrm{s}} = G_i$ in the first case. In the second case G_i^{s} consists of the elements σ with $\chi_Y(\sigma) = \chi_X(\sigma)$, so $[G_i : G_i^{\mathrm{s}}] = 2$, and any element of G_i^{s} acts on x and y by one scalar.

By the choice of the admission and the definition of x and y on $W_{i,0}$, in both cases the isomorphism of completions maps each of x, y, z, t to the corresponding coordinate of $\mathbb{A}^4$ modulo the L'-th power of the maximal ideal. Lifting the images of the coordinates, it is induced by an automorphism Φ of $\mathbb{C}[[x, y, z, t]]$ with $\Phi(x) - x, \Phi(y) - y, \Phi(z) - z, \Phi(t) - t \in (x, y, z, t)^{L'}$, and Φ maps the ideal $(xy + \hat{g}_i)$ onto the ideal $(xy + g_{i,0})$, since it induces an isomorphism of the quotients. Thus $\Phi(xy + \hat{g}_i) = u \cdot (xy + g_{i,0})$ for a unit u of $\mathbb{C}[[x, y, z, t]]$. Since $xy + \hat{g}_i \in (x, y, z, t)^2$, we have $\Phi(xy + \hat{g}_i) - (xy + \hat{g}_i) \in (x, y, z, t)^{L'+1}$. As the quadratic part of $xy + \hat{g}_i$ has rank two,

$\hat{g}_i \in (z,t)^3$, and $g_{i,0} \in (z,t)^3$. Comparing the quadratic parts, we obtain $u(0) = 1$, and setting $x = y = 0$, we obtain

$$u(0,0,z,t)g_{i,0} - \hat{g}_i \in (z,t)^{L'+1}.$$

Since $L' + 1 > n_i = \operatorname{mult}_0 \hat{g}_i$, it follows that $\operatorname{mult}_0 g_{i,0} = n_i$ and that $g_{i,0}$ and $\hat{g}_i$ coincide in degrees at most n_i.

Let $w_i := (\lfloor n_i/2 \rfloor, \lceil n_i/2 \rceil, 1, 1)$, and let $(\hat{g}_i)_{n_i} \neq 0$ be the homogeneous part of degree n_i of $\hat{g}_i$. The lowest weighted part $xy + (\hat{g}_i)_{n_i}$ for the weights w_i is irreducible, since it is linear in x and the coefficients y and $(\hat{g}_i)_{n_i}$ are coprime, and it is not divisible by a coordinate. Each of $\Phi(x) - x, \Phi(y) - y, \Phi(z) - z, \Phi(t) - t$ belongs to $(x,y,z,t)^{L'}$, so its w_i-weighted order is at least $L' > \lceil n_i/2 \rceil$, the largest entry of w_i. Hence Φ maps each ideal $I_j \subset \mathbb{C}[[x,y,z,t]]$ generated by the monomials of w_i-weight at least j onto itself and induces the identity on $\bigoplus_j I_j/I_{j+1}$, so Φ induces an automorphism of the w_i-weighted blow-up $\operatorname{Proj} \bigoplus_{j \geq 0} I_j$ of $\operatorname{Spec} \mathbb{C}[[x,y,z,t]]$ (cf. the last sentence of Lemma 4.12), under which the strict transforms of $(xy + \hat{g}_i = 0)$ and $(xy + g_{i,0} = 0)$ correspond. Since $xy + (\hat{g}_i)_{n_i}$ is irreducible and not divisible by a coordinate, the prime divisor $E_{i,0}$ corresponding to E_i is the exceptional prime divisor of the weighted blow-up of $W_{i,0}$ with weights w_i in x, y, z, t.

Any element of the coordinate ring of $W_{i,0}$ has a unique representative in $\mathbb{C}[x,y,z,t]$ without monomials divisible by xy, obtained by replacing xy with $-g_{i,0}$. The representative is unique since the homomorphism from the coordinate ring of $W_{i,0}$ to $\mathbb{C}[z,t][x,x^{-1}]$ which maps y to $-g_{i,0}/x$ is injective, as $W_{i,0}$ is integral ($xy + g_{i,0}$ is irreducible, being linear in x with coprime coefficients y and $g_{i,0} \neq 0$), and maps a representative

$$\psi(z,t) + \sum_{a \geq 1} x^a \psi_a(z,t) + \sum_{b \geq 1} y^b \psi'_b(z,t)$$

to the Laurent polynomial whose coefficients of x^0, x^a, and x^{-b} are ψ, ψ_a, and $(-g_{i,0})^b \psi'_b$. Any $\sigma \in G_i$ maps xy to a multiple of xy, so $\sigma(xy) = \chi(\sigma)xy$ and $\sigma(g_{i,0}) = \chi(\sigma)g_{i,0}$. Hence replacing xy with $-g_{i,0}$ commutes with G_i, and G_i maps representatives without monomials divisible by xy to such representatives and polynomials in z,t to polynomials in z,t. Possibly replacing each $h_{ik,0}$ with the representative of $h_{ik,0}|_{W_{i,0}}$, we may assume that

$$h_{ik,0} = \psi_{ik}(z,t) + x\xi_{ik}(x,z,t) + y\eta_{ik}(y,z,t) \tag{E.1}$$

for polynomials ψ_{ik}, ξ_{ik} and η_{ik}. By uniqueness, the plane trace ψ_{ik} of $h_{ik,0}$ is either zero or G_i-semi-invariant with the character of $h_{ik,0}$.

Step 2. In this step we show that $\psi_{ik} \neq 0$ and $\operatorname{mult}_0 \psi_{ik} \leq m - 1$ for any i and k.

On $W_{i,0} \backslash (x = 0)$ we have $y = -g_{i,0}/x$, so the projection to $\mathbb{A}^3_{x,z,t}$ restricts to an open immersion $W_{i,0} \backslash (x = 0) \to \mathbb{A}^3_{x,z,t}$ with image $\mathbb{A}^3_{x,z,t} \backslash (x = 0)$. Thus the projection $W_{i,0} \to \mathbb{A}^3_{x,z,t}$ is a birational morphism, and any prime divisor over $W_{i,0}$ is a prime divisor over $\mathbb{A}^3_{x,z,t}$. Conversely, a prime divisor P over $\mathbb{A}^3_{x,z,t}$ with $\operatorname{mult}_P(g_{i,0}/x) \geq 0$ is a prime divisor over $W_{i,0}$, since the coordinates x, y, z, t of $W_{i,0}$, where $y = -g_{i,0}/x$, then have non-negative multiplicities along P. The 3-form

$$\omega := \frac{dx}{x} \wedge dz \wedge dt$$

is the residue of $dx \wedge dy \wedge dz \wedge dt/(xy + g_{i,0})$ along $W_{i,0}$, so ω generates the canonical sheaf of $W_{i,0}$ near 0 by adjunction for the Cartier divisor $W_{i,0} \subset \mathbb{A}^4$ [KM98, Proposition 5.73]. On $\mathbb{A}^3_{x,z,t}$, ω has a simple pole along $(x = 0)$ and no other zero or pole. Thus, for any prime divisor P over $W_{i,0}$ with center 0,

$$a(P, W_{i,0}, 0) = a(P, \mathbb{A}^3_{x,z,t}, \operatorname{div}(x)). \tag{E.2}$$

For a prime divisor P over $W_{i,0}$ with center 0, let

$$\theta_P := a(P, W_{i,0}, 0) - \operatorname{mult}_P z - \operatorname{mult}_P t. \tag{E.3}$$

The integral hypersurface $W_{i,0} = (xy + g_{i,0} = 0) \subset \mathbb{A}^4$ is normal [Har77, Proposition II.8.23(b)], since by the Jacobian criterion it is regular outside the curve $(x = y = g_{i,0} = 0)$. By **Step 1**, $0 \neq g_{i,0} \in \mathbb{C}[z,t]$, and the finite cyclic group G_i acts faithfully on $W_{i,0}$, linearly on $\mathbb{A}^4$, monomially on $\{x,y\}$ and diagonally on z,t, and preserves $xy + g_{i,0}$ up to the character χ. The subgroup G_i^{s} of **Step 1** consists of the elements preserving each of x and y up to scalars. Thus, by Lemma B.16 for $W_{i,0}$ and G_i, for any prime divisor P over $W_{i,0}$ with center 0, $\theta_P \geq 0$, and if moreover $I_{G_i}(P) \not\subset G_i^{\mathrm{s}}$, then $|I_{G_i}(P)|$ is even and $\theta_P \geq |I_{G_i}(P)|/2$.

Since $g_{i,0}$ and $\hat{g}_i$ coincide in degrees at most n_i (**Step 1**),

$$f_{w_i} := \mathrm{in}_{w_i}(xy + g_{i,0}) = xy + (\hat{g}_i)_{n_i}, \qquad \mathrm{wt}_{w_i}(xy + g_{i,0}) = n_i.$$

By **Step 1**, f_{w_i} is irreducible, so $(f_{w_i} = 0) \subset \mathbb{A}^4$ is integral, and it is not contained in any coordinate hyperplane. The vector w_i is primitive. Thus $E_{i,0}$ is the prime divisor E_W of Lemma 4.12 for $n = 4$, $W = W_{i,0}$, and $w = w_i$, and by Lemma 4.12,

$$a(E_{i,0}, W_{i,0}, 0) = \left\lfloor \frac{n_i}{2} \right\rfloor + \left\lceil \frac{n_i}{2} \right\rceil + 1 + 1 - n_i = 2.$$

Thus $a(E_i, W_i, 0) = 2$ by **Step 1**(3). Since (W_i, C_k) is terminal near o_i for $C_k = \frac{1}{m}\operatorname{div}(h_{ik})$ (Set-up 11.1) and E_i is exceptional over W_i,

$$1 < a(E_i, W_i, C_k) = 2 - \frac{1}{m}\operatorname{mult}_{E_i} h_{ik}.$$

Hence $\operatorname{mult}_{E_i} h_{ik} \leq m - 1$, and $\operatorname{mult}_{E_{i,0}} h_{ik,0} \leq m - 1$ by **Step 1**(3).

Since $n_i \geq 2m$, $\lfloor n_i/2 \rfloor \geq m$ and $\lceil n_i/2 \rceil \geq m$. In (E.1), by the inequality of Lemma 4.12,

$$\operatorname{mult}_{E_{i,0}}(x\xi_{ik}) \geq \mathrm{wt}_{w_i}(x\xi_{ik}) \geq m, \qquad \operatorname{mult}_{E_{i,0}}(y\eta_{ik}) \geq \mathrm{wt}_{w_i}(y\eta_{ik}) \geq m.$$

If $\psi_{ik} \neq 0$, then $\mathrm{in}_{w_i}(\psi_{ik})$ is a non-zero form in z,t of degree $\operatorname{mult}_0 \psi_{ik}$, not divisible by f_{w_i}, so $\operatorname{mult}_{E_{i,0}} \psi_{ik} = \operatorname{mult}_0 \psi_{ik}$ by Lemma 4.12. If $\psi_{ik} = 0$ or $\operatorname{mult}_0 \psi_{ik} \geq m$, then $\operatorname{mult}_{E_{i,0}} h_{ik,0} \geq m$, a contradiction. Thus $\psi_{ik} \neq 0$ and $\operatorname{mult}_0 \psi_{ik} \leq m - 1$. The case where $h_{ik,0}$ is a unit near 0 is included, in which $\operatorname{mult}_0 \psi_{ik} = 0$. In particular, by (E.1), $h_{ik,0}$ is divisible neither by x nor by y.

Step 3. In this step we show that $|G_i| \to +\infty$ after passing to a subsequence, and take limits of the plane images of G_i.

Assume that there exists a positive integer B such that $|G_i| \leq B$ for infinitely many i. By Definition-Lemma 11.4(1), $\alpha_i = a_D(F_i) = \sum_k \mu_k a_k(F_i)$, where $a_k(F_i) > 0$ and $m|I_{G_i}(F_i)|a_k(F_i)$ is a positive integer. Since $\mu_k a_k(F_i) \leq \alpha_i < 1$,

$$a_k(F_i) \in \left\{ \frac{j}{me} \;\middle|\; 1 \leq e \leq B,\ j \in \mathbb{Z},\ 0 < j < \frac{me}{\mu_k} \right\}$$

for any k and any such i. The right hand side is a finite set depending only on m, B, and μ_k. Thus α_i belongs to a finite set for infinitely many i, which is impossible for a strictly increasing sequence. Hence, passing to a subsequence, we may assume that $|G_i| \to +\infty$.

Let Π_i and Π_i' be the images of G_i and G_i^{s} in the group of diagonal automorphisms of $\mathbb{A}^2_{z,t}$, regarded as finite subgroups of $\mathbb{T}^2$ [Liu26, Definition 2.15]. We call the image of $\sigma \in G_i$ in Π_i the *plane image* of σ. Then $\Pi_i' \subset \Pi_i$ and $[\Pi_i : \Pi_i'] \leq 2$. By Lemma 4.6, passing to a subsequence, there exist closed subgroups Π and Π' of $\mathbb{T}^2$ such that $\Pi_i \to \Pi$ and $\Pi_i' \to \Pi'$ in the Hausdorff distance, and $\Pi_i \subset \Pi$ and $\Pi_i' \subset \Pi'$ for $i \gg 0$. Passing to a subsequence, we may also assume that either $G_i^{\mathrm{s}} = G_i$ for any i or $G_i^{\mathrm{s}} \neq G_i$ for any i. Write $\Pi_i = \Pi_i' \cup \pi_i \Pi_i'$ with $\pi_i \in \Pi_i$. Passing to a subsequence, π_i converges to some $\pi \in \Pi$, and $\Pi = \Pi' \cup \pi\Pi'$, since Hausdorff limits are compatible with finite unions and $\pi_i \Pi_i' \to \pi\Pi'$. Thus Π' is a closed subgroup of Π of index at most two. In particular, Π' is open in Π, being a closed subgroup of finite index, and the identity components satisfy $\Pi^0 = (\Pi')^0$.

Step 4. In this step we show that Π is infinite.

Assume that Π is finite. Then $|\Pi_i| \leq |\Pi|$ for $i \gg 0$. Let $K_i \subset G_i$ be the kernel of $G_i \to \Pi_i$.

First assume that $G_i^{\mathrm{s}} = G_i$. An element of K_i acts on x, y, z, t by $(s, s', 1, 1)$ for some $s, s' \in \mathbb{C}^*$. Since $0 \neq g_{i,0}$ is invariant under K_i, $\chi|_{K_i} = 1$, and hence $ss' = 1$. Thus $K_i = R_i := G_i \cap \mathbf{T}_{(1,-1,0,0)}$. As a finite subgroup of a circle, $R_i = \mu_{r_i}(1, -1, 0, 0)$ with $r_i := |R_i|$, and $[G_i : R_i] = |\Pi_i| \leq |\Pi|$. Since $|G_i| \to +\infty$ (**Step 3**), $r_i \to +\infty$. Let $e_0 := |\Pi|$. The hypotheses of Theorem 8.3 for e_0, Γ, $W_{i,0} = (xy - (-g_{i,0}) = 0)$, G_i, and $D_{i,0}$ hold:

- $0 \neq -g_{i,0} \in (z, t)^3 \subset (z, t)^2$, and $W_{i,0}$ is normal and klt near 0, being isolated Gorenstein terminal (**Step 1**(1));
- G_i is a finite cyclic group acting diagonally on x, y, z, t, preserving the equation up to the character χ, and acting faithfully on $W_{i,0}$ (**Step 1**(2));
- $R_i = G_i \cap \mathbf{T}_{(1,-1,0,0)} = \mu_{r_i}(1, -1, 0, 0)$, $r_i \to +\infty$, and $[G_i : R_i] \leq e_0$;
- $D_{i,0} = \sum_k c_k \operatorname{div}(h_{ik,0})$ is a labelled boundary on $(W_{i,0} \ni 0)$ (Definition 2.7), since the $h_{ik,0}|_{W_{i,0}}$ are non-zero regular G_i-semi-invariant functions, possibly units, and the labels c_k belong to the DCC set $\Gamma \subset [0, 1]$;
- $(W_{i,0}, D_{i,0})$ is klt near 0 (**Step 1**(2)).

Thus, by Theorem 8.3, the numbers $\alpha_i = \operatorname{mld}((W_{i,0} \ni 0, D_{i,0})/G_i)$ do not form a strictly increasing sequence of positive numbers, a contradiction.

Next assume that $G_i^{\mathrm{s}} \neq G_i$. Then we are in the second case of **Step 1**(1), and any element of G_i^{s} acts on x and y by one scalar (**Step 1**). An element of $K_i \cap G_i^{\mathrm{s}}$ acts by $(a, a, 1, 1)$, and χ is trivial on K_i since $0 \neq g_{i,0}$ is invariant under K_i, so $a^2 = 1$. Hence $|K_i \cap G_i^{\mathrm{s}}| \leq 2$. If $\tau, \tau' \in K_i \backslash G_i^{\mathrm{s}}$, then $\tau'\tau^{-1} \in K_i \cap G_i^{\mathrm{s}}$, so $|K_i \backslash G_i^{\mathrm{s}}| \leq 2$. Thus $|K_i| \leq 4$ and $|G_i| \leq 4|\Pi|$ for $i \gg 0$, which contradicts **Step 3**.

Therefore $\dim \Pi \geq 1$.

Step 5. In this step we construct prime divisors over $W_{i,0}$ which bound α_i from above, and show that Π^0 is neither $\mathbb{T}^2$ nor a circle $\mathbf{T}_{(u,v)}$ with $uv > 0$.

Let $(Z, T) \in \mathbb{R}^2_{>0}$ be a vector whose class modulo $\mathbb{Z}^2$ belongs to Π'. Since $\Pi'_i \to \Pi'$ in the Hausdorff distance, there exist $(Z_i, T_i) \in \mathbb{Q}^2$ whose classes modulo $\mathbb{Z}^2$ belong to Π'_i such that $(Z_i, T_i) \to (Z, T)$. Let $\epsilon := \frac{1}{2}\min(Z, T)$. Possibly removing finitely many i, we may assume that $Z_i, T_i \geq \epsilon$ for any i. Let $\gamma_i \in G_i^{\mathrm{s}}$ be an element whose plane image is the class of (Z_i, T_i), and let d_i be the order of γ_i. Then γ_i acts on z, t by $e^{2\pi\sqrt{-1}Z_i}, e^{2\pi\sqrt{-1}T_i}$, and on x by $e^{2\pi\sqrt{-1}\zeta_i}$ for some $\zeta_i \in \frac{1}{d_i}\mathbb{Z}$. Let

$$\ell_i := \min\{aZ_i + bT_i \mid z^a t^b \text{ is a monomial of } g_{i,0}\}.$$

Then $\ell_i \geq n_i\epsilon \to +\infty$. Let $\nu_i \in \zeta_i + \mathbb{Z}$ with $|\nu_i - \ell_i/2| \leq \frac{1}{2}$, and let $\nu'_i := \ell_i - \nu_i$. Then $\nu_i, \nu'_i \to +\infty$, and possibly removing finitely many i, $\nu_i, \nu'_i > 0$.

Since $\gamma_i^{d_i} = 1$, the numbers $d_i\nu_i$, d_iZ_i, and d_iT_i are positive integers. Let k_i be their greatest common divisor, and let

$$(\hat{\nu}_i, \hat{Z}_i, \hat{T}_i) := \frac{d_i}{k_i}(\nu_i, Z_i, T_i) \in \mathbb{Z}^3_{>0},$$

a primitive vector. Let $\tilde{P}_i$ be the exceptional divisor of the weighted blow-up of $\mathbb{A}^3_{x,z,t}$ with weights $(\hat{\nu}_i, \hat{Z}_i, \hat{T}_i)$, a prime divisor over $\mathbb{A}^3_{x,z,t}$. That is, $\tilde{P}_i$ is the monomial prime divisor in x, z, t with weights $\hat{w}_i := (\hat{\nu}_i, \hat{Z}_i, \hat{T}_i)$ [Liu26, Definition 2.16] (cf. [CLS11, Proposition 4.1.1]). For a Laurent polynomial $f = \sum c_{abc}x^a z^b t^c \in \mathbb{C}[x, x^{-1}, z, t]$,

$$\operatorname{mult}_{\tilde{P}_i} f = \frac{d_i}{k_i}\min\{a\nu_i + bZ_i + cT_i \mid c_{abc} \neq 0\}. \tag{E.4}$$

For a polynomial f, (E.4) holds by definition, and for a Laurent polynomial f, it follows from the polynomial case applied to $x^j f$ for $j \gg 0$. We call $a\nu_i + bZ_i + cT_i$ the *weight* of the monomial $x^a z^b t^c$. Thus $\operatorname{mult}_{\tilde{P}_i} f$ is $\frac{d_i}{k_i}$ times the least weight of a monomial of f, and the $\hat{w}_i$-weight of a monomial is $\frac{d_i}{k_i}$ times its weight. Thus $\operatorname{mult}_{\tilde{P}_i} x > 0$, $\operatorname{mult}_{\tilde{P}_i} z > 0$, $\operatorname{mult}_{\tilde{P}_i} t > 0$, $\operatorname{mult}_{\tilde{P}_i} g_{i,0} = \frac{d_i}{k_i}\ell_i$,

and

$$\operatorname{mult}_{\tilde{P}_i} y = \frac{d_i}{k_i}(\ell_i - \nu_i) = \frac{d_i}{k_i}\nu_i' > 0.$$

Hence $\tilde{P}_i$ is a prime divisor over $W_{i,0}$ by **Step 2**, and the center of $\tilde{P}_i$ on $W_{i,0}$ is 0. By (E.2) and $a(\tilde{P}_i, \mathbb{A}^3_{x,z,t}, 0) = \hat{\nu}_i + \hat{Z}_i + \hat{T}_i$ [JM12, Proposition 5.1 and (5.1)],

$$a(\tilde{P}_i, W_{i,0}, 0) = (\hat{\nu}_i + \hat{Z}_i + \hat{T}_i) - \hat{\nu}_i = \frac{d_i}{k_i}(Z_i + T_i).$$

We show that $\gamma_i \in I_{G_i}(\tilde{P}_i)$. Since $\gamma_i \in G_i^{\mathrm{s}}$ preserves $(x = 0)$, the open immersion of **Step 2** is γ_i-equivariant, where γ_i acts on $\mathbb{A}^3_{x,z,t}$ by

$$\left(e^{2\pi\sqrt{-1}\nu_i}, e^{2\pi\sqrt{-1}Z_i}, e^{2\pi\sqrt{-1}T_i}\right) = \left(s_i^{\hat{\nu}_i}, s_i^{\hat{Z}_i}, s_i^{\hat{T}_i}\right), \qquad s_i := e^{2\pi\sqrt{-1}k_i/d_i},$$

that is, by the image of s_i under the one-parameter subgroup $s \to (s^{\hat{\nu}_i}, s^{\hat{Z}_i}, s^{\hat{T}_i})$ of $(\mathbb{C}^*)^3$. Thus γ_i multiplies any polynomial in x, z, t which is homogeneous of $\hat{w}_i$-weight j by s_i^j, so γ_i preserves $\operatorname{mult}_{\tilde{P}_i}$ by (E.4). Any element of the residue field of $\operatorname{mult}_{\tilde{P}_i}$ is the residue class of f_1/f_2 for non-zero polynomials f_1, f_2 in x, z, t with $\operatorname{mult}_{\tilde{P}_i} f_1 = \operatorname{mult}_{\tilde{P}_i} f_2$, and it is equal to the residue class of

$$\frac{\operatorname{in}_{\hat{w}_i}(f_1)}{\operatorname{in}_{\hat{w}_i}(f_2)}.$$

By (E.4), $\operatorname{in}_{\hat{w}_i}(f_1)$ and $\operatorname{in}_{\hat{w}_i}(f_2)$ are homogeneous of $\hat{w}_i$-weight $\operatorname{mult}_{\tilde{P}_i} f_1 = \operatorname{mult}_{\tilde{P}_i} f_2$, so γ_i multiplies both by $s_i^{\operatorname{mult}_{\tilde{P}_i} f_1}$ and fixes the quotient. Thus γ_i fixes $\tilde{P}_i$ and acts trivially on it, that is, $\gamma_i \in I_{G_i}(\tilde{P}_i)$. Since G_i acts faithfully on $W_{i,0}$, $\langle\gamma_i\rangle$ has order d_i, so $|I_{G_i}(\tilde{P}_i)| \geq d_i$. Since $(W_{i,0}, D_{i,0})$ is klt near 0, $a(\tilde{P}_i, W_{i,0}, D_{i,0}) > 0$. By **Step 1**(2) and [Liu26, Lemma 2.6],

$$\alpha_i \leq \frac{a(\tilde{P}_i, W_{i,0}, D_{i,0})}{|I_{G_i}(\tilde{P}_i)|} \leq \frac{a(\tilde{P}_i, W_{i,0}, D_{i,0})}{d_i}. \tag{E.5}$$

Since $D_{i,0} \geq 0$, $a(\tilde{P}_i, W_{i,0}, D_{i,0}) \leq a(\tilde{P}_i, W_{i,0}, 0)$, and (E.5) implies $\alpha_i \leq (Z_i + T_i)/k_i \leq Z_i + T_i$. Thus

$$\limsup_{i \to +\infty} \alpha_i \leq Z + T. \tag{E.6}$$

Now assume that $\Pi^0 = \mathbb{T}^2$, or that $\Pi^0 = \mathbf{T}_{(u,v)}$ for a primitive vector $(u, v) \in \mathbb{Z}^2$ with $uv > 0$. In the second case, possibly replacing (u, v) with $(-u, -v)$, we may assume that $u, v > 0$. Let $(u, v) := (1, 1)$ in the first case. For any $\delta > 0$, the class of $\delta(u, v)$ modulo $\mathbb{Z}^2$ belongs to $\Pi^0 \subset \Pi'$, so (E.6) implies $\limsup_i \alpha_i \leq \delta(u + v)$. Hence $\limsup_i \alpha_i = 0$, which contradicts $\alpha_i \geq \alpha_1 > 0$.

Step 6. In this step we fix the circle Π^0 and select a block of each plane trace along it.

By **Steps 4 and 5**, $\Pi^0 = \mathbf{T}_{(u',v')}$ for a primitive vector $(u', v') \in \mathbb{Z}^2$ with $u'v' \leq 0$. Possibly exchanging z and t, and replacing (u', v') with $(-u', -v')$, we may assume that $\Pi^0 = \mathbf{T}_{(u,-v)}$, where $u \geq 1$, $v \geq 0$, and $\gcd(u, v) = 1$. Exchanging z and t does not affect **Steps 1–5**. The case $v = 0$ is $\Pi^0 = \mathbf{T}_{(1,0)}$.

Since $\Pi_i \subset \Pi$, the components of Π other than Π^0 have positive distance from Π^0, and $\Pi_i \to \Pi$ in the Hausdorff distance, $\Pi_i \cap \Pi^0 \to \Pi^0$ in the Hausdorff distance. Thus $\rho_i := |\Pi_i \cap \Pi^0| \to +\infty$. The group $\Pi_i \cap \Pi^0$ is cyclic, generated by the class of $(u/\rho_i, -v/\rho_i)$, which acts on a monomial $z^a t^b$ by $e^{2\pi\sqrt{-1}(ua-vb)/\rho_i}$. Since G_i acts on z, t through Π_i and ψ_{ik} is G_i-semi-invariant (**Steps 1 and 2**), any two monomials $z^a t^b$ and $z^{a'} t^{b'}$ of ψ_{ik} satisfy

$$ua - vb \equiv ua' - vb' \pmod{\rho_i}.$$

By **Step 2**, ψ_{ik} has a monomial $z^{a_*} t^{b_*}$ with $a_* + b_* \leq m - 1$. If $z^a t^b$ is a monomial of ψ_{ik} with $ua - vb \neq ua_* - vb_*$, then $\rho_i \leq |ua - vb - (ua_* - vb_*)| \leq (u + v)(a + b + m - 1)$, so

$$a + b \geq N_i := \frac{\rho_i}{u + v} - m + 1 \to +\infty.$$

Since $\gcd(u,v)=1$, the pairs $(a,b)\in\mathbb{Z}^2_{\geq 0}$ with $ua-vb=ua_*-vb_*$ are (a_0+jv,b_0+ju), $j\in\mathbb{Z}_{\geq 0}$, for the least such pair (a_0,b_0), and they increase componentwise with j. Let (p_{ik},q_{ik}) be the least such pair for which $z^{p_{ik}}t^{q_{ik}}$ is a monomial of ψ_{ik}. Then $p_{ik}\leq a_*$ and $q_{ik}\leq b_*$, so $p_{ik}+q_{ik}\leq m-1$. Passing to a subsequence, we may assume that $(p_{ik},q_{ik})=(p_k,q_k)$ does not depend on i, for any k. Thus

$$\psi_{ik}=z^{p_k}t^{q_k}\varphi_{ik}(z^vt^u)+\psi'_{ik},\qquad \varphi_{ik}(0)\neq 0,\quad p_k+q_k\leq m-1,\quad \psi'_{ik}\in(z,t)^{\lceil N_i\rceil}, \tag{E.7}$$

where φ_{ik} is a polynomial in one variable. All the monomials of $z^{p_k}t^{q_k}\varphi_{ik}(z^vt^u)$ are monomials of ψ_{ik}, so they have the character of ψ_{ik} on G_i, and $\varphi_{ik}(z^vt^u)$ is a G_i-invariant unit near 0.

Let

$$\bar{p}:=\sum_k\frac{\mu_kp_k}{m},\qquad \bar{q}:=\sum_k\frac{\mu_kq_k}{m},\qquad \lambda_z:=1-\bar{p},\qquad \lambda_t:=1-\bar{q}. \tag{E.8}$$

Then $\sum_k c_kp_k=\bar{p}$, $\sum_k c_kq_k=\bar{q}$, and $\bar{p}+\bar{q}\leq\sum_k\mu_k\frac{m-1}{m}=1-\frac{1}{m}$, since $p_k+q_k\leq m-1$ by (E.7) and $\sum_k\mu_k=1$ (Set-up 11.1(1)). In particular $\lambda_z,\lambda_t\in(0,1]$ and $\min(\lambda_z,\lambda_t)=1-\max(\bar{p},\bar{q})$. For a closed subgroup $\Pi''\subset\mathbb{T}^2$, let

$$b(\Pi''):=\inf\{\lambda_zZ+\lambda_tT\mid Z,T\in\mathbb{R}_{>0},\ (Z,T)\bmod\mathbb{Z}^2\in\Pi''\}. \tag{E.9}$$

Since the class of $(1,1)$ is 0, $b(\Pi'')\leq\lambda_z+\lambda_t$. If $\Pi''\subset\Pi'''$ are closed subgroups, then $b(\Pi'')\geq b(\Pi''')$.

Step 7. In this step we show that, for $i\gg 0$ and any prime divisor P over $W_{i,0}$ with center 0,

$$a(P,W_{i,0},D_{i,0})\geq\lambda_z\operatorname{mult}_P z+\lambda_t\operatorname{mult}_P t+\min(\lambda_z,\lambda_t)\theta_P, \tag{E.10}$$

and that $\alpha_i\geq b(\Pi)$ for $i\gg 0$.

If $\max(\bar{p},\bar{q})=0$, then $p_k=q_k=0$ for any k. By (E.7), $\psi_{ik}(0)\neq 0$, so $h_{ik,0}$ is a unit near 0 by (E.1). Thus $D_{i,0}=0$ near 0, $\lambda_z=\lambda_t=1$, and (E.10) is the definition (E.3) of θ_P.

Assume that $\max(\bar{p},\bar{q})>0$. Let

$$\lambda'_z:=1-\frac{\bar{p}}{\max(\bar{p},\bar{q})},\qquad \lambda'_t:=1-\frac{\bar{q}}{\max(\bar{p},\bar{q})}.$$

Then $\lambda'_z,\lambda'_t\in[0,1]$. Since $\varphi_{ik}(z^vt^u)$ is a unit near 0,

$$\sum_k\frac{c_k}{\max(\bar{p},\bar{q})}\operatorname{div}\left(z^{p_k}t^{q_k}\varphi_{ik}(z^vt^u)\right)+\lambda'_z\operatorname{div}(z)+\lambda'_t\operatorname{div}(t)=\operatorname{div}(z)+\operatorname{div}(t)$$

near 0 on $\mathbb{A}^2_{z,t}$, and $(\mathbb{A}^2_{z,t},\operatorname{div}(z)+\operatorname{div}(t))$ is lc. Let $\Lambda\subset\mathbb{R}_{>0}$ be the finite set of the positive numbers among $c_k/\max(\bar{p},\bar{q})$, λ'_z, λ'_t, and 1. Let L_2 (resp. L_4) be the maximum of the integers of Lemma 4.14 for Λ, $n=2$ (resp. $n=4$) and s' elements, $1\leq s'\leq s+3$. Since log canonicity of a germ is preserved under completion [dFEM11, Proposition 2.11], we may use Lemma 4.14 for polynomial germs at 0. For $i\gg 0$, $N_i\geq L_2$ and $n_i\geq L_4$.

By (E.7), $\psi_{ik}-z^{p_k}t^{q_k}\varphi_{ik}(z^vt^u)\in(z,t)^{L_2}$ for $i\gg 0$. Thus, by Lemma 4.14 with $n=2$, the pair

$$\left(\mathbb{A}^2_{z,t},\sum_k\frac{c_k}{\max(\bar{p},\bar{q})}\operatorname{div}(\psi_{ik})+\lambda'_z\operatorname{div}(z)+\lambda'_t\operatorname{div}(t)\right)$$

is lc near 0, where terms with coefficient zero are omitted. By (E.1), the restrictions of $h_{ik,0}$ to $(x=0)\cong\mathbb{A}^3_{y,z,t}$ and then to $(x=y=0)\cong\mathbb{A}^2_{z,t}$ are $\psi_{ik}+y\eta_{ik}$ and ψ_{ik}. By **Step 2**, $h_{ik,0}$ is divisible neither by x nor by y, and $\psi_{ik}\neq 0$, so $\operatorname{div}(y)$ has no common component with $\operatorname{div}(\psi_{ik}+y\eta_{ik})$ on $\mathbb{A}^3_{y,z,t}$. Thus inversion of adjunction [Kaw07, Theorem], applied near 0, first along $(y=0)\subset\mathbb{A}^3_{y,z,t}$ and then along $(x=0)\subset\mathbb{A}^4$, where the different is the restriction of the boundary (cf. [KM98, Proposition 5.73]), implies that

$$\left(\mathbb{A}^4,\operatorname{div}(xy)+\sum_k\frac{c_k}{\max(\bar{p},\bar{q})}\operatorname{div}(h_{ik,0})+\lambda'_z\operatorname{div}(z)+\lambda'_t\operatorname{div}(t)\right)$$

is lc near 0. Since

$$(xy+g_{i,0})-xy=g_{i,0}\in(x,y,z,t)^{n_i}\subset(x,y,z,t)^{L_4},$$

Lemma 4.14 with $n = 4$, applied to the elements xy, $h_{i1,0}, \dots, h_{is,0}$, z, t and to $xy + g_{i,0}$, $h_{i1,0}, \dots, h_{is,0}$, z, t with positive coefficients (at most $s+3$ of them), implies that

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + g_{i,0}) + \sum_k \frac{c_k}{\max(\bar{p}, \bar{q})} \operatorname{div}(h_{ik,0}) + \lambda'_z \operatorname{div}(z) + \lambda'_t \operatorname{div}(t)\Big)$$

is lc near 0. Since $W_{i,0} = (xy + g_{i,0} = 0)$ is normal and is not contained in the zero locus of $h_{ik,0}$, z, or t, the adjunction direction of [Kaw07, Theorem], applied near 0, where the different is the restriction of the boundary (cf. [KM98, Proposition 5.73]), implies that

$$\Big(W_{i,0}, \frac{1}{\max(\bar{p}, \bar{q})} D_{i,0} + \lambda'_z \operatorname{div}(z) + \lambda'_t \operatorname{div}(t)\Big)$$

is lc near 0. Hence, for any prime divisor P over $W_{i,0}$ with center 0,

$$a(P, W_{i,0}, 0) - \frac{\operatorname{mult}_P D_{i,0}}{\max(\bar{p}, \bar{q})} \geq \lambda'_z \operatorname{mult}_P z + \lambda'_t \operatorname{mult}_P t.$$

We multiply both sides by $\max(\bar{p}, \bar{q})$, add

$$(1 - \max(\bar{p}, \bar{q}))a(P, W_{i,0}, 0) = (1 - \max(\bar{p}, \bar{q}))(\theta_P + \operatorname{mult}_P z + \operatorname{mult}_P t)$$

to both sides, and obtain

$$a(P, W_{i,0}, D_{i,0}) \geq (1 - \bar{p}) \operatorname{mult}_P z + (1 - \bar{q}) \operatorname{mult}_P t + (1 - \max(\bar{p}, \bar{q}))\theta_P,$$

which is (E.10).

Let P be a prime divisor over $W_{i,0}$ with center 0. By [Liu26, Lemma 2.6(1)], $I_{G_i}(P)$ is cyclic and acts faithfully on $\mathfrak{m}_P/\mathfrak{m}_P^2$ by a character χ_P. Let γ be the generator of $I_{G_i}(P)$ with $\chi_P(\gamma) = e^{2\pi\sqrt{-1}/|I_{G_i}(P)|}$. For any G_i-semi-invariant regular function f near 0 with character χ_f, $\chi_f(\gamma) = \chi_P(\gamma)^{\operatorname{mult}_P f}$ [Liu26, Lemma 9.2(1)]. In particular,

$$\text{(E.11)} \qquad \text{the plane image of } \gamma \text{ is the class of } \Big(\frac{\operatorname{mult}_P z}{|I_{G_i}(P)|}, \frac{\operatorname{mult}_P t}{|I_{G_i}(P)|}\Big) \text{ modulo } \mathbb{Z}^2,$$

and both coordinates are positive. By Lemma B.16 (**Step 2**), $\theta_P \geq 0$, so (E.10) and (E.11) imply that, for $i \gg 0$,

$$\frac{a(P, W_{i,0}, D_{i,0})}{|I_{G_i}(P)|} \geq \lambda_z \frac{\operatorname{mult}_P z}{|I_{G_i}(P)|} + \lambda_t \frac{\operatorname{mult}_P t}{|I_{G_i}(P)|} \geq b(\Pi_i) \geq b(\Pi).$$

Since P is arbitrary, **Step 1**(2) and [Liu26, Lemma 2.6] imply $\alpha_i \geq b(\Pi)$ for $i \gg 0$.

Step 8. In this step we show that $\limsup_i \alpha_i \leq b(\Pi')$, and conclude when $\Pi = \Pi'$.

Let $(Z, T) \in \mathbb{R}^2_{>0}$ be a vector whose class modulo $\mathbb{Z}^2$ belongs to Π', and let Z_i, T_i, ϵ, γ_i, d_i, ℓ_i, ν_i, ν'_i, k_i, and $\tilde{P}_i$ be as in **Step 5**. We compute $\operatorname{mult}_{\tilde{P}_i} h_{ik,0}$ by (E.4). Through the open immersion of **Step 2**, $y = -g_{i,0}/x$, and (E.1) becomes the Laurent polynomial

$$h_{ik,0} = \psi_{ik} + x\xi_{ik}(x, z, t) + \Big(-\frac{g_{i,0}}{x}\Big)\eta_{ik}\Big(-\frac{g_{i,0}}{x}, z, t\Big).$$

The monomials of the three terms have x-exponents 0, at least 1, and at most -1, so no cancellation occurs among the three terms. Every monomial of the second term has weight at least ν_i, and every monomial of the third term has weight at least $\min_{b \geq 1}(-b\nu_i + b\ell_i) = \nu'_i$. For the first term, by (E.7), a monomial $z^{p_k + jv} t^{q_k + ju}$ of $z^{p_k} t^{q_k} \varphi_{ik}(z^v t^u)$ has weight

$$p_k Z_i + q_k T_i + j(vZ_i + uT_i) \geq p_k Z_i + q_k T_i,$$

with equality for $j = 0$, and a monomial of ψ'_{ik} has weight at least ϵN_i. Here $p_k Z_i + q_k T_i$ is at most $(m-1)\max(Z_i, T_i)$, hence bounded, while ν_i, ν'_i and ϵN_i tend to $+\infty$. Thus, for $i \gg 0$ and any k,

$$\operatorname{mult}_{\tilde{P}_i} h_{ik,0} = \frac{d_i}{k_i}(p_k Z_i + q_k T_i),$$

and by **Step 5** and (E.8),

$$a(\tilde{P}_i, W_{i,0}, D_{i,0}) = \frac{d_i}{k_i}\Big(Z_i + T_i - \sum_k c_k(p_k Z_i + q_k T_i)\Big) = \frac{d_i}{k_i}(\lambda_z Z_i + \lambda_t T_i).$$

By (E.5), for $i \gg 0$,

$$\alpha_i \le \frac{\lambda_z Z_i + \lambda_t T_i}{k_i} \le \lambda_z Z_i + \lambda_t T_i.$$

Hence $\limsup_i \alpha_i \le \lambda_z Z + \lambda_t T$. Since (Z, T) is arbitrary,

$$\limsup_{i\to+\infty} \alpha_i \le b(\Pi'). \tag{E.12}$$

If $\Pi = \Pi'$, then **Step 7** and (E.12) imply that, for $i \gg 0$,

$$\alpha_i \ge b(\Pi) = b(\Pi') \ge \limsup_{j\to+\infty} \alpha_j = \lim_{j\to+\infty} \alpha_j > \alpha_i,$$

a contradiction. Thus $[\Pi : \Pi'] = 2$. In particular $G_i^{\mathrm{s}} \ne G_i$ for any i, since otherwise $\Pi_i = \Pi_i'$ for any i (**Step 3**) and $\Pi = \Pi'$, and we are in the second case of **Step 1**(1) for any i.

Step 9. In this step we show that an element of G_i belongs to G_i^{s} if and only if the plane image belongs to Π', for $i \gg 0$, and exclude the case $v \ge 1$.

If $\Pi_i \subset \Pi'$ for infinitely many i, then $\Pi \subset \Pi'$, which contradicts $[\Pi : \Pi'] = 2$. Thus, for $i \gg 0$, $\Pi_i' \subset \Pi_i \cap \Pi' \subsetneq \Pi_i$. Since $[\Pi_i : \Pi_i'] \le 2$, we obtain $[\Pi_i : \Pi_i'] = 2$ and $\Pi_i' = \Pi_i \cap \Pi'$. The surjection $G_i \to \Pi_i$ maps G_i^{s} onto Π_i', and $[G_i : G_i^{\mathrm{s}}] = 2 = [\Pi_i : \Pi_i']$. Hence an element of G_i belongs to G_i^{s} if and only if the plane image belongs to $\Pi_i' = \Pi_i \cap \Pi'$.

Assume that $v \ge 1$. Then $u, v \ge 1$. We bound $b(\mathbf{T}_{(u,-v)})$. Let $j \in \mathbb{Z}$ with $ju \equiv 1 \pmod v$. For $\delta > 0$, the class of $(j/v - \delta)(u, -v)$ modulo $\mathbb{Z}^2$, a point of $\mathbf{T}_{(u,-v)}$, is the class of

$$\Big(\frac{ju}{v} - \delta u, -j + \delta v\Big) \equiv \Big(\frac{1}{v} - \delta u, \delta v\Big) \pmod{\mathbb{Z}^2},$$

and both coordinates on the right hand side are positive for small δ. Thus

$$b(\mathbf{T}_{(u,-v)}) \le \lim_{\delta\to 0}\Big(\lambda_z\Big(\frac{1}{v} - \delta u\Big) + \lambda_t \delta v\Big) = \frac{\lambda_z}{v}.$$

Similarly, let $j' \in \mathbb{Z}$ with $j'v \equiv -1 \pmod u$. For $\delta > 0$, the class of $(j'/u + \delta)(u, -v)$ modulo $\mathbb{Z}^2$ is the class of $(\delta u, \frac{1}{u} - \delta v)$, so $b(\mathbf{T}_{(u,-v)}) \le \lambda_t/u$. Since $\mathbf{T}_{(u,-v)} = \Pi^0 \subset \Pi'$ and $u, v \ge 1$,

$$b(\Pi') \le b(\mathbf{T}_{(u,-v)}) \le \min\Big(\frac{\lambda_z}{v}, \frac{\lambda_t}{u}\Big) \le \min(\lambda_z, \lambda_t). \tag{E.13}$$

Let $i \gg 0$, let P be a prime divisor over $W_{i,0}$ with center 0, and let γ be as in **Step 7**. First assume that $I_{G_i}(P) \subset G_i^{\mathrm{s}}$. Then the plane image of γ belongs to $\Pi_i' \subset \Pi'$, and as in **Step 7**, (E.10) and (E.11) imply $a(P, W_{i,0}, D_{i,0})/|I_{G_i}(P)| \ge b(\Pi')$. Next assume that $I_{G_i}(P) \not\subset G_i^{\mathrm{s}}$. Then $\theta_P \ge |I_{G_i}(P)|/2$ by Lemma B.16 (**Step 2**). Moreover $\gamma^2 \in G_i^{\mathrm{s}}$, as $[G_i : G_i^{\mathrm{s}}] = 2$, so by (E.11) the class of $(2\,\mathrm{mult}_P z/|I_{G_i}(P)|, 2\,\mathrm{mult}_P t/|I_{G_i}(P)|)$ belongs to $\Pi_i' \subset \Pi'$. Thus (E.10) and (E.13) imply

$$\frac{a(P, W_{i,0}, D_{i,0})}{|I_{G_i}(P)|} \ge \frac{1}{2}\Big(\lambda_z \frac{2\,\mathrm{mult}_P z}{|I_{G_i}(P)|} + \lambda_t \frac{2\,\mathrm{mult}_P t}{|I_{G_i}(P)|}\Big) + \frac{\min(\lambda_z, \lambda_t)}{2} \ge \frac{b(\Pi')}{2} + \frac{b(\Pi')}{2} = b(\Pi').$$

Since P is arbitrary, $\alpha_i \ge b(\Pi') \ge \lim_j \alpha_j > \alpha_i$ for $i \gg 0$ by (E.12), a contradiction. Thus $v = 0$, that is, $\Pi^0 = \mathbf{T}_{(1,0)}$.

Step 10. In this step we exclude the remaining case $\Pi^0 = \mathbf{T}_{(1,0)}$ and $[\Pi : \Pi'] = 2$ by the circle theorems.

Let $S_i \subset G_i$ be the preimage of $\Pi_i \cap \Pi^0$ under $G_i \to \Pi_i$. Then

$$[G_i : S_i] = [\Pi_i : \Pi_i \cap \Pi^0] \le |\Pi/\Pi^0|$$

for $i \gg 0$, since $\Pi_i \subset \Pi$. Since $|G_i| \to +\infty$ (**Step 3**), $|S_i| \to +\infty$. Since $\Pi^0 \subset \Pi'$, **Step 9** implies $S_i \subset G_i^{\mathrm{s}}$, so by **Step 1** any element of S_i acts on x and y by one scalar. As $\Pi^0 = \mathbf{T}_{(1,0)}$ acts trivially on t, any $\sigma \in S_i$ acts on x, y, z, t by $(a, a, \beta, 1)$ for some $a, \beta \in \mathbb{C}^*$, and preserves $xy + g_{i,0}$ up to $\chi(\sigma) = a^2$. By **Step 8**, we are in the second case of **Step 1**(1), so G_i acts diagonally on the coordinates $X = (x + y)/2$, $Y = (x - y)/(2\sqrt{-1})$, z, t, and $W_{i,0} = (X^2 + Y^2 + g_{i,0}(z, t) = 0)$. Passing to a subsequence, we may assume that either z divides $g_{i,0}$ for any i or z divides $g_{i,0}$ for no i.

First assume that z does not divide $g_{i,0}$. Then $g_{i,0}(0, t) \neq 0$, and a monomial t^l of $g_{i,0}$ is invariant under S_i. Hence $a^2 = \chi(\sigma) = 1$ for any $\sigma \in S_i$. Let $R_i \subset S_i$ be the subgroup of the elements with $a = 1$. Then $[S_i : R_i] \leq 2$, and R_i consists of the elements of G_i acting trivially on X, Y, and t. In the coordinates (z, X, Y, t), $R_i = G_i \cap \mathbf{T}_{(1,0,0,0)}$, and as a finite subgroup of a circle, $R_i = \mu_{r_i}(1, 0, 0, 0)$ with $r_i := |R_i| \geq |S_i|/2 \to +\infty$. Let $e_0 := 2|\Pi/\Pi^0|$. Then $[G_i : R_i] \leq e_0$. Let $r_0(\Gamma)$ be the integer of Theorem 8.5 for e_0 and Γ. In the coordinates (z, X, Y, t), the hypotheses of Theorem 8.5 for e_0, Γ, $f = X^2 + Y^2 + g_{i,0}(z, t)$, G_i, and $D_{i,0}$ hold:

- $W_{i,0} = (f = 0)$ is an isolated singular cDV germ at 0: it is isolated Gorenstein terminal (**Step 1**(1)), and a general hyperplane section through 0 is a Du Val singularity of type A_{n_i-1} (Definition 8.1);
- G_i is a finite cyclic group acting diagonally on z, X, Y, t, preserving f up to the character χ, and acting faithfully on $W_{i,0}$ (**Step 1**(2));
- $R_i = G_i \cap \mathbf{T}_{(1,0,0,0)} = \mu_{r_i}(1, 0, 0, 0)$ with $r_i \geq r_0(\Gamma)$ for $i \gg 0$, since $r_i \to +\infty$, and $[G_i : R_i] \leq e_0$;
- the $h_{ik,0}$ are G_i-semi-invariant polynomials in z, X, Y, t not vanishing identically on $W_{i,0}$, possibly units, with labels $c_k \in \Gamma$, where $\Gamma \subset [0, 1]$ is a DCC set;
- $(W_{i,0}, D_{i,0})$ is klt near 0 (**Step 1**(2)).

Thus, by Theorem 8.5, the numbers $\alpha_i = \mathrm{mld}((W_{i,0} \ni 0, D_{i,0})/G_i)$ do not form a strictly increasing sequence with $r_i \to +\infty$, a contradiction.

Next assume that z divides $g_{i,0}$. If z^2 divides $g_{i,0}$, then the partial derivatives of $xy + g_{i,0}$ vanish along $(x = y = z = 0)$, which contradicts the isolatedness of the singularity of $W_{i,0}$. Thus $g_{i,0} = zf_i(z, t)$ with $f_i(0, t) \neq 0$, and $f_i(0, 0) = 0$ since $g_{i,0} \in (z, t)^3$. A monomial zt^l of $g_{i,0}$ has character β on $\sigma \in S_i$, so $a^2 = \chi(\sigma) = \beta$, and $\sigma = (a, a, a^2, 1) \in \mathbf{T}_{(1,1,2,0)}$. Conversely, the plane image of an element of $G_i \cap \mathbf{T}_{(1,1,2,0)}$ belongs to $\Pi_i \cap \Pi^0$. Thus

$$R_i := G_i \cap \mathbf{T}_{(1,1,2,0)} = S_i, \qquad r_i := |R_i| \to +\infty, \qquad [G_i : R_i] \leq |\Pi/\Pi^0|.$$

Let $e_0 := |\Pi/\Pi^0|$. The data e_0, Γ, $W_{i,0} = (xy + zf_i(z, t) = 0)$, G_i, and $D_{i,0}$ satisfy Set-up 8.2, where G_i may exchange x and y as Theorem 8.7 allows, and the hypotheses of Theorem 8.7:

- $\Gamma \subset [0, 1]$ is a DCC set, and $\mathbf{T} = \mathbf{T}_{(1,1,2,0)}$ is a primitive circle;
- $W_{i,0} = (xy + zf_i(z, t) = 0)$ with $0 \neq zf_i \in (z, t)^2$, isolated Gorenstein terminal of quadratic rank two, in particular normal and klt near 0, and $f_i(0, 0) = 0$, $f_i(0, t) \neq 0$;
- G_i is a finite abelian group acting faithfully on $W_{i,0}$, linearly and monomially on the pair $\{x, y\}$ and diagonally on z, t, and preserving the equation up to the character χ (**Step 1**);
- $R_i = G_i \cap \mathbf{T}$, $r_i = |R_i| \to +\infty$, and $[G_i : R_i] \leq e_0$;
- $D_{i,0} = \sum_k c_k \operatorname{div}(h_{ik,0})$ is a labelled boundary (Definition 2.7) with labels in Γ, and $(W_{i,0}, D_{i,0})$ is klt near 0, with $\alpha_i = \mathrm{mld}((W_{i,0} \ni 0, D_{i,0})/G_i)$ (**Step 1**(2)).

By Theorem 8.7, α cannot increase along $\mathbf{T}_{(1,1,2,0)}$ (Set-up 7.3). Since Γ, e_0 and $(1, 1, 2, 0)$ are fixed, $r_i \to +\infty$, and the α_i are positive and strictly increasing, we obtain a contradiction.

Steps 4, 5, 8, 9, and 10 exclude all the possibilities for Π and Π'. Thus no such sequence exists. □

E.2. **Quadratic rank two, fixed split radical order: circles and graded quotients.** In this subsection, we prove Lemma E.1 and Proposition E.3. Both are used in Subsection E.4, in the proof of Proposition 11.12.

Lemma E.1 reduces a strictly increasing sequence of quotient values with a fixed split radical order to one circle of one of two types.

Lemma E.1. *Notation as in Set-up 11.1. Let $(W_i \ni o_i, G_i, (h_{i,k})_k)$ be a sequence of such data, with s, m and the μ_k fixed and with W_i of quadratic rank two and split radical order n independent of i, let $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$ and $\alpha_i := \operatorname{mld}((W_i \ni o_i, D_i)/G_i)$, and assume that the α_i are strictly increasing in $(0,1)$. Then, after passing to a subsequence and replacing each $(W_i \ni o_i, (h_{i,k})_k)$ with a numerical admission with group G_i (Proposition 11.8 and Theorem 8.11), there exist integers $a, b \geq 0$ with $a+b=n$, a primitive vector $w=(w_x,w_y,w_z,w_t) \in \mathbb{Z}^4$ with $w_x+w_y = aw_z+bw_t$, and a positive integer e_0 with the following properties, where $\mathbf{T} := \mathbf{T}_{(w)}$.*

(1) $W_i = (xy + g_i(z,t) = 0)$ with $\operatorname{mult}_0 g_i = n$, the monomial $z^a t^b$ occurs in g_i, and G_i acts diagonally on x,y,z,t.

(2) $|G_i \cap \mathbf{T}| \to +\infty$ and $[G_i : G_i \cap \mathbf{T}] \leq e_0$.

(3) After possibly exchanging x and y, possibly exchanging z and t together with a and b, and possibly replacing w by $-w$, one of the following holds.

(a) The monomial $z^a t^b$ is the only monomial of degree n of g_i, and either $w_z < 0 < w_t$, or $w_t = 0 < w_z$, $a \leq 1$, and $w_x < 0 < w_y$.

(b) $w = (-p, N+p, w_z, w_t)$ with $p, w_z, w_t > 0$ and $N = aw_z + bw_t$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that $|G_i| \to +\infty$, and we replace the data with numerical admissions of the form of Theorem 8.11.

Let M be a positive integer. By Proposition 11.7(1), applied with $\lambda = 1$, the numbers α_i with $|G_i| \leq M$ belong to a finite set. Since the α_i are strictly increasing, only finitely many i satisfy $|G_i| \leq M$. Thus $|G_i| \to +\infty$.

Since W_i has quadratic rank two, $(W_i \ni o_i)$ is singular. By [Liu26, Lemma 5.4(2) and its last paragraph], $(W_i \ni o_i)$ is an isolated cDV hypersurface singularity, there exists a G_i-equivariant closed embedding of W_i into a smooth fourfold germ such that W_i is the zero locus of a regular G_i-semi-invariant function, and any regular G_i-semi-invariant function on W_i near o_i with character χ lifts to a regular G_i-semi-invariant function with character χ. The quadratic part of the equation of W_i has rank two, and G_i is cyclic (Set-up 11.1(3)). By Proposition 11.8 and Theorem 8.11, there exists a numerical admission $(W_i', (h_{i,k}')_k)$ of $(W_i, (h_{i,k})_k)$ with group G_i, for the vectors and the prime divisors of Proposition 11.8(i) and (ii), such that one of the following holds, where $g_i \in (z,t)^3$ is a polynomial.

(i) $W_i' = (xy + g_i(z,t) = 0) \subset \mathbb{A}^4$, and G_i acts diagonally on x,y,z,t.

(ii) $W_i' = (X^2 + Y^2 + g_i(z,t) = 0) \subset \mathbb{A}^4$, and G_i acts diagonally on X,Y,z,t with $\chi_X \neq \chi_Y$.

Let $D_i' := \sum_k c_k \operatorname{div}(h_{i,k}'|_{W_i'})$. By Construction-Theorem 4.13(1) and (3), G_i acts faithfully on W_i', the pair (W_i', D_i') is klt near 0,

$$\alpha_i = \operatorname{mld}((W_i' \ni 0, D_i')/G_i),$$

$(W_i' \ni 0)$ is an isolated Gorenstein terminal germ, and the completions of the local rings of W_i at o_i and of W_i' at 0 are G_i-equivariantly isomorphic.

In case (ii), let $x := X + \sqrt{-1}Y$ and $y := X - \sqrt{-1}Y$, so that $X^2 + Y^2 = xy$. An element $\gamma \in G_i$ multiplies X and Y by scalars λ_X and λ_Y with $\lambda_X^2 = \lambda_Y^2$, since X^2 and Y^2 occur in the G_i-semi-invariant equation. If $\lambda_Y = \lambda_X$, then γ multiplies x and y by λ_X. If $\lambda_Y = -\lambda_X$, then $\gamma(x) = \lambda_X y$ and $\gamma(y) = \lambda_X x$. Thus in both cases $W_i' = (xy + g_i(z,t) = 0)$, and G_i acts linearly and monomially on the pair $\{x,y\}$ and diagonally on z,t. In particular $\operatorname{mult}_0 g_i$ is the split radical order of W_i' (Definition 8.1). The split radical order depends only on the completed local ring, since the argument after Definition 8.1 shows, in the completed local ring, that a general hyperplane section through the base point is a Du Val singularity of type A whose index is the split radical order minus one. As the completions of the local rings of W_i at o_i and of W_i' at 0 are isomorphic, $\operatorname{mult}_0 g_i = n$. Moreover $n \geq 3$, since $g_i \in (z,t)^3$.

We replace $(W_i \ni o_i)$, $(h_{i,k})_k$, and D_i with $(W_i' \ni 0)$, $(h'_{i,k})_k$, and D_i', and we drop the primes. Passing to a subsequence, we may assume that (i) holds for any i or that (ii) holds for any i. Since z, t have exactly $n+1$ monomials of degree n and the homogeneous part of degree n of g_i is non-zero, after passing to a subsequence we may assume that there exist integers $a, b \geq 0$ with $a+b=n$ such that the monomial $z^a t^b$ occurs in g_i for any i.

Let $\Gamma := \{c_1, \dots, c_s\}$. Since $c_k = \mu_k/m$ (Set-up 11.1) with $0 < \mu_k \leq 1$ and $m \geq 1$, $\Gamma \subset (0,1]$ is a finite set, hence a DCC set. For any i, D_i is a labelled boundary on $(W_i \ni 0)$ with labels in Γ, where units and repeated factors among the marks are allowed by Definition 2.7. Moreover (W_i, D_i) is klt near 0, W_i is normal and klt near 0, G_i is a finite abelian group acting faithfully on W_i and preserving the equation $xy + g_i$ up to a character, and $\alpha_i \geq \alpha_1 > 0$. We use the properties above in any application below of Lemma 4.7, Proposition 8.10, and Theorems 8.3, 8.5, 8.6, 8.7, 8.8, 8.9, and 8.13, and at each application we name the form of the group, the circle, and the bound for the index.

Step 2. In this step we find the circle $\mathbf{T}$ and the integer e_0.

We regard G_i as a subgroup of $\mathbb{T}^4$ through the diagonal action on x, y, z, t in case (i) and on X, Y, z, t in case (ii). In case (i), the monomials xy and $z^a t^b$ occur in the G_i-semi-invariant equation, so the characters of xy and $z^a t^b$ coincide. In case (ii), the characters of the monomials X^2, Y^2, and $z^a t^b$ coincide. Thus $G_i \subset S$ for the closed subgroup

$$S := \{\theta \in \mathbb{T}^4 \mid \theta_1 + \theta_2 = a\theta_3 + b\theta_4\} \qquad \text{in case (i)},$$
$$S := \{\theta \in \mathbb{T}^4 \mid 2\theta_1 = 2\theta_2 = a\theta_3 + b\theta_4\} \qquad \text{in case (ii)}.$$

By Lemma 4.6, after passing to a subsequence there exists a closed subgroup $H \subset \mathbb{T}^4$ such that the G_i converge to H in the Hausdorff distance and $G_i \subset H$ for $i \gg 0$. Since $|G_i| \to +\infty$, $\dim H \geq 1$. Since S is closed and contains G_i for any i, $H \subset S$. Let $V_H \subset \mathbb{R}^4$ be the linear subspace whose image in $\mathbb{T}^4 = \mathbb{R}^4/\mathbb{Z}^4$ is H^0 [Liu26, Lemma 4.3(1)]. Then $V_H \subset V_S$, where

$$V_S := \{v \in \mathbb{R}^4 \mid v_1 + v_2 = av_3 + bv_4\} \qquad \text{in case (i)},$$
$$V_S := \{v \in \mathbb{R}^4 \mid v_1 = v_2,\ 2v_1 = av_3 + bv_4\} \qquad \text{in case (ii)}.$$

Thus $\dim V_S = 3$ in case (i) and $\dim V_S = 2$ in case (ii). Since $2n = 2a + 2b$, the vector $(n, n, 2, 2)$ belongs to V_S in both cases.

If V_H contains a vector with all coordinates positive, then Lemma 4.7, applied in the coordinates in which G_i is diagonal, implies that $\alpha_i \to 0$, which contradicts $\alpha_i \geq \alpha_1 > 0$. Hence

(E.14) $\qquad V_H$ contains no vector with all coordinates positive.

In particular $V_H \neq V_S$. In case (ii), this implies $\dim V_H = 1$. In case (i), $\dim V_H \in \{1, 2\}$. If $\dim V_H = 2$ in case (i), then Proposition 8.10, applied with the fixed integers a, b, the set Γ and $\epsilon := \alpha_1$ to the diagonal groups G_i converging to H with $\dim H^0 = 2$, implies that the α_i are not strictly increasing, a contradiction. Thus $\dim V_H = 1$ in both cases.

By Lemma 4.6, H^0 is the image of $\mathbb{R}w$ for a primitive vector $w \in \mathbb{Z}^4$, the group $R_i := G_i \cap \mathbf{T}_{(w)}$ is cyclic of order $r_i \to +\infty$, and $[G_i : R_i] = |H/H^0|$ for $i \gg 0$. Let $e_0 := |H/H^0|$, and let $\mathbf{T} := \mathbf{T}_{(w)}$. Write $w = (w_x, w_y, w_z, w_t)$, where in case (ii) the first two coordinates are the weights on X and Y. Since $w \in V_H \subset V_S$,

$$w_x + w_y = aw_z + bw_t,$$

and in case (ii) moreover $w_x = w_y$. In case (ii), an element λ of $\mathbf{T}$ multiplies both X and Y by λ^{w_x}, hence multiplies both x and y by λ^{w_x}. Thus in both cases $\mathbf{T}$ acts diagonally on x, y, z, t with weights w_x, w_y, w_z, w_t. By (E.14) applied to w and $-w$, neither w nor $-w$ has all four coordinates positive. Discarding finitely many terms, we may assume that $[G_i : R_i] \leq e_0$ and

$$r_i > n(|w_z| + |w_t|)$$

for any i.

The exchange of x and y, the exchange of z and t together with a and b, and the replacement of w by $-w$ do not change $\mathbf{T}$, R_i, e_0, the relation $w_x + w_y = aw_z + bw_t$, the occurrence of $z^a t^b$ in g_i, or the form of the equation and of the group in (i) and (ii).

In the steps below, whenever a theorem shows that α cannot increase along $\mathbf{T}$ (Set-up 7.3) for the data (W_i, G_i, D_i), we obtain a contradiction, since Γ, e_0 and w are fixed, $r_i \to +\infty$, and the α_i are positive and strictly increasing.

Step 3. In this step we show that any monomial $z^k t^l$ of g_i satisfies

$$kw_z + lw_t \equiv w_x + w_y \pmod{r_i}, \tag{E.15}$$

and we describe the monomials of g_i when $w_t = 0 < w_z$.

The cyclic group R_i of order r_i is generated by the element of $\mathbf{T}$ which multiplies x, y, z, t by $e^{2\pi\sqrt{-1}w_x/r_i}$, $e^{2\pi\sqrt{-1}w_y/r_i}$, $e^{2\pi\sqrt{-1}w_z/r_i}$, $e^{2\pi\sqrt{-1}w_t/r_i}$. Since the equation $xy+g_i$ is G_i-semi-invariant, the generator multiplies xy and any monomial of g_i by equal scalars. This is (E.15).

Assume that $w_t = 0 < w_z$. Since $z^a t^b$ occurs in g_i, (E.15) implies $w_x + w_y \equiv aw_z \pmod{r_i}$, so any monomial $z^k t^l$ of g_i satisfies

$$kw_z \equiv aw_z \pmod{r_i}.$$

If $a = 0$, then any monomial $z^k t^{n-k}$ of degree n of g_i satisfies $kw_z \equiv 0 \pmod{r_i}$ with $0 \le kw_z \le nw_z < r_i$. Hence $k = 0$, and t^n is the only monomial of degree n of g_i. If $a \ge 1$, then $0 < aw_z \le nw_z < r_i$, so no monomial t^l occurs in g_i. Thus z divides g_i. If z^2 divided g_i, then all partial derivatives of $xy + g_i$ would vanish along the curve $(x = y = z = 0) \subset W_i$, which contradicts the fact that W_i has an isolated singularity at 0. Thus $g_i = zf_i(z,t)$ for a polynomial f_i with $f_i(0,t) \ne 0$, and $f_i(0,0) = 0$ since $\operatorname{mult}_0 g_i = n \ge 3$. A monomial zt^l occurs in g_i, so $(a-1)w_z \equiv 0 \pmod{r_i}$ with $0 \le (a-1)w_z < r_i$. Hence $a = 1$. Any monomial $z^k t^{n-k}$ of degree n of g_i has $k \ge 1$ and $(k-1)w_z \equiv 0 \pmod{r_i}$ with $0 \le (k-1)w_z < r_i$, so $k = 1$. Thus zt^{n-1} is the only monomial of degree n of g_i.

In conclusion, if $w_t = 0 < w_z$, then one of the following holds.

- $a = 0$, $w_x + w_y = 0$, and t^n is the only monomial of degree n of g_i.
- $a = 1$, $w_x + w_y = w_z$, zt^{n-1} is the only monomial of degree n of g_i, and $g_i = zf_i(z,t)$ with $f_i(0,0) = 0$ and $f_i(0,t) \ne 0$.

Step 4. In this step we assume that $w_z < 0 < w_t$. We show that $z^a t^b$ is the only monomial of degree n of g_i, and that $w_x = w_y$ and $b = 0$ do not hold together.

Let $z^k t^{n-k}$ and $z^{k'} t^{n-k'}$ be monomials of degree n of g_i. By (E.15),

$$(k-k')(w_z - w_t) \equiv 0 \pmod{r_i}.$$

Since $|(k-k')(w_z - w_t)| \le n(|w_z| + |w_t|) < r_i$ and $w_z - w_t \ne 0$, we have $k = k'$. Thus $z^a t^b$ is the only monomial of degree n of g_i.

Assume that $w_x = w_y$ and $b = 0$. Then $a = n$, z^n is the only monomial of degree n of g_i, and

$$2w_x = nw_z < 0.$$

If n is odd, then w_z is even. Thus $c_* := -\gcd(n,2)w_z/2$ is a positive integer, and

$$-w_x = \frac{nc_*}{\gcd(n,2)}, \qquad -w_z = \frac{2c_*}{\gcd(n,2)}.$$

Since $\gcd(n/\gcd(n,2), 2/\gcd(n,2)) = 1$, we have $\gcd(w_x, w_y, w_z) = c_*$, and the primitivity of w implies $\gcd(c_*, w_t) = 1$. The germ $(W_i \ni 0)$ is isolated Gorenstein terminal of quadratic rank two, the completed local ring of W_i at 0 is G_i-equivariantly $\mathbb{C}[[x,y,z,t]]/(xy + g_i(z,t))$, the quadratic form xy is non-degenerate, G_i acts linearly on the span of x, y and diagonally on z, t, and $\operatorname{mult}_0 g_i(z,0) = n$ because z^n occurs in g_i and $\operatorname{mult}_0 g_i = n$. The circle $\mathbf{T}$ acts by λ^{w_x} on x and y, by λ^{w_z} on z, and by λ^{w_t} on t. Thus the last statement of Theorem 8.6, applied with the integers c and v of the theorem equal to c_* and w_t, with the group G_i (which may exchange

x and y), the fixed circle $\mathbf{T} = \mathbf{T}_{(w_x,w_x,w_z,w_t)}$ and the bound e_0, implies that α cannot increase along $\mathbf{T}$, a contradiction.

Step 5. In this step we show that case (ii) does not occur. Together with **Step 1** and **Step 2**, this proves (1) and (2).

Assume that (ii) holds. Then $w_x = w_y$ and $2w_x = aw_z + bw_t$. We consider the signs of w_z and w_t.

If $w_z = w_t = 0$, then $w_x = w_y = 0$, which contradicts $w \neq 0$.

If $w_z w_t > 0$, then w_z, w_t are both positive or both negative, and so is $2w_x = aw_z + bw_t$, since $a, b \geq 0$ and $a + b = n > 0$. Thus w or $-w$ has all four coordinates positive, since $w_y = w_x$, which contradicts **Step 2**.

If exactly one of w_z, w_t is zero, then after possibly exchanging z and t together with a and b and possibly replacing w by $-w$, we have $w_t = 0 < w_z$. We apply **Step 3**. If $a = 0$, then $2w_x = 0$, so $w = (0, 0, 1, 0)$, and $\mathbf{T}$ acts only on z. The germ $(W_i \ni 0)$ is an isolated singular cDV germ (a singular isolated Gorenstein terminal germ by **Step 1**, hence isolated cDV by [KM98, Corollary 5.38]), G_i is diagonal on X, Y, z, t and preserves $X^2 + Y^2 + g_i$ up to a character, and $R_i = G_i \cap \mathbf{T}_{(0,0,1,0)}$ has order $r_i \to +\infty$ and index at most e_0. Theorem 8.5, applied in the coordinates z, X, Y, t with z as the first coordinate, implies that the α_i do not form a strictly increasing sequence, a contradiction. If $a = 1$, then $2w_x = w_z$, and the primitivity of w implies $w = (1, 1, 2, 0)$. Moreover $W_i = (xy + zf_i(z, t) = 0)$ with $f_i(0, 0) = 0$ and $f_i(0, t) \neq 0$. Theorem 8.7, applied to the groups G_i (which may exchange x and y), the fixed circle $\mathbf{T}_{(1,1,2,0)}$ and the bound e_0, implies that α cannot increase along $\mathbf{T}$, a contradiction.

If $w_z w_t < 0$, then after replacing w by $-w$ we may assume that $w_x \leq 0$, and after exchanging z and t together with a and b we may assume that $w_z < 0 < w_t$. By **Step 4**, $z^a t^b$ is the only monomial of degree n of g_i and $b \geq 1$. Since $aw_z = 2w_x - bw_t < 0$, we have $a \geq 1$. Let $\kappa_i \neq 0$ be the coefficient of $z^a t^b$ in g_i. Replacing x with x/κ_i and g_i with g_i/κ_i, which does not change W_i, the weights of $\mathbf{T}$, or the property that G_i acts linearly and monomially on $\{x, y\}$, we may assume that the homogeneous part of degree n of g_i is $z^a t^b$. The integers $-w_x \geq 0$, $-w_z > 0$, and $w_t > 0$ satisfy $\gcd(-w_x, -w_z, w_t) = 1$ and

$$a(-w_z) - bw_t = -2w_x,$$

and $a, b \geq 1$ with $a + b = n \geq 3$. The germs $(W_i \ni 0)$ are isolated Gorenstein terminal of quadratic rank two, with G_i-equivariant isomorphisms of the completed local rings with $\mathbb{C}[[x, y, z, t]]/(xy + g_i(z, t))$, on which G_i acts linearly and monomially on $\{x, y\}$ and diagonally on z, t, and $\operatorname{mult}_0 g_i = n$. Moreover $|G_i \cap \mathbf{T}| = r_i \to +\infty$ and $[G_i : G_i \cap \mathbf{T}] \leq e_0$. Thus Theorem 8.13, applied with the integers called A, C, D in Theorem 8.13 equal to $-w_x$, $-w_z$, w_t, and with D_i in place of the boundaries B_i there, implies that the α_i do not form a strictly increasing sequence of positive numbers, a contradiction.

Thus (ii) does not occur. Hence $W_i = (xy + g_i(z, t) = 0)$ with $\operatorname{mult}_0 g_i = n$, the monomial $z^a t^b$ occurs in g_i, and G_i acts diagonally on x, y, z, t. This is (1). By **Step 2**, w is primitive with $w_x + w_y = aw_z + bw_t$, $|G_i \cap \mathbf{T}| = r_i \to +\infty$, and $[G_i : G_i \cap \mathbf{T}] \leq e_0$. This is (2).

Step 6. In this step we prove (3). By **Step 5**, G_i is diagonal on x, y, z, t. We consider the signs of w_z and w_t.

Assume that $w_z = w_t = 0$. Then $w_x + w_y = 0$, and the primitivity of w implies $w = \pm(1, -1, 0, 0)$. After exchanging x and y, $w = (1, -1, 0, 0)$ and $R_i = \mu_{r_i}(1, -1, 0, 0)$. Theorem 8.3, applied to $W_i = (xy - (-g_i(z, t)) = 0)$, the diagonal groups G_i, the circle $\mathbf{T}_{(1,-1,0,0)}$ and the bound e_0, implies that the α_i do not form a strictly increasing sequence of positive numbers, a contradiction.

Assume that exactly one of w_z, w_t is zero. After possibly exchanging z and t together with a and b and possibly replacing w by $-w$, we have $w_t = 0 < w_z$. We apply **Step 3**.

- Let $a = 0$. Then t^n is the only monomial of degree n of g_i and $w_x + w_y = 0$. If $w_x = 0$, then $w = (0, 0, 1, 0)$. In this case $(W_i \ni 0)$ is an isolated singular cDV germ (a

singular isolated Gorenstein terminal germ by **Step 1**, hence isolated cDV by [KM98, Corollary 5.38]), G_i is diagonal, and $R_i = G_i \cap \mathbf{T}_{(0,0,1,0)}$ has order $r_i \to +\infty$ and index at most e_0. Theorem 8.5, applied in the coordinates z, x, y, t with z as the first coordinate, implies that the α_i do not form a strictly increasing sequence, a contradiction. Thus $w_x \neq 0$, and after exchanging x and y we have $w_x < 0 < w_y$. Hence (a) holds with $a = 0$.

- Let $a = 1$. Then zt^{n-1} is the only monomial of degree n of g_i, $w_x + w_y = w_z > 0$, and $W_i = (xy + zf_i(z,t) = 0)$ with $f_i(0,0) = 0$ and $f_i(0,t) \neq 0$. Assume that $w_x, w_y \geq 0$. The primitivity of $w = (w_x, w_y, w_x + w_y, 0)$ implies $\gcd(w_x, w_y) = 1$. Theorem 8.8, applied with the integers A and B of the theorem equal to w_x and w_y, to the diagonal groups G_i, the circle $\mathbf{T}_{(w_x,w_y,w_x+w_y,0)}$ and the bound e_0, implies that α cannot increase along $\mathbf{T}$, a contradiction. Thus one of w_x, w_y is negative, and the other is positive since $w_x + w_y > 0$. After exchanging x and y we have $w_x < 0 < w_y$. Hence (a) holds with $a = 1$.

Assume that $w_z w_t > 0$. After replacing w by $-w$, we have $w_z, w_t > 0$. Let $N := aw_z + bw_t$. Then $N > 0$ and $w_x + w_y = N$. By **Step 2**, w_x and w_y are not both positive.

- Assume that one of w_x, w_y is zero. Then the other one is N, so $\mathbf{T} = \mathbf{T}_{(0,N,w_z,w_t)}$ or $\mathbf{T} = \mathbf{T}_{(N,0,w_z,w_t)}$, and $\gcd(N, w_z, w_t) = 1$ by the primitivity of w. The $\mathbf{T}$-weight of $z^a t^b$ is N, so the sum of the terms of g_i of $\mathbf{T}$-weight N contains the non-zero term of $z^a t^b$, hence is non-zero. Moreover $0 \neq g_i \in (z,t)^3$. Theorem 8.9, applied with the integers called N, v, s in Theorem 8.9 equal to N, w_z, and w_t, to the diagonal groups G_i, the circle $\mathbf{T}$ and the bound e_0, implies that α cannot increase along $\mathbf{T}$, a contradiction.
- Thus one of w_x, w_y is negative, and the other one is larger than N. After exchanging x and y, let $p := -w_x > 0$. Then $w_y = N + p$ and $w = (-p, N + p, w_z, w_t)$ with $p, w_z, w_t > 0$. Hence (b) holds.

Assume that $w_z w_t < 0$. After exchanging z and t together with a and b, we have $w_z < 0 < w_t$. By **Step 4**, $z^a t^b$ is the only monomial of degree n of g_i. Hence (a) holds.

Thus (3) holds in any case, which completes the proof. □

We now introduce the graded quotients of the principal tuple along the circle of Lemma E.1, and prove the lower bound for the circles of case (a) of Lemma E.1(3).

Construction E.2. Let the data be as in the conclusion of Lemma E.1, and let $r_i := |G_i \cap \mathbf{T}|$. In case (a) of Lemma E.1(3), possibly replacing the data with numerical admissions that keep the groups G_i, we assume moreover that

$$g_i = \phi_i(z,t)\psi_i(z,t), \qquad \phi_i = z^a + \sum_{j=0}^{a-1} \phi_{i,j}(t) z^j, \qquad \psi_i = t^b + \sum_{j=0}^{b-1} \psi_{i,j}(z) t^j, \tag{E.16}$$

where ϕ_i and ψ_i are G_i-semi-invariant polynomials with the characters of z^a and t^b, $\phi_{i,j}(0) = \psi_{i,j}(0) = 0$ for any j, and $\phi_i = z^a$ if $w_t = 0$. Write each $h_{i,k}$ without monomials divisible by xy, and pass to a subsequence on which $\kappa_k := \kappa(h_{i,k})$ is independent of i for each k (Set-up 5.1 for $\mathbf{T}$). Here $\kappa(h_{i,k})$ takes finitely many values, since $c_k \operatorname{mult}_0 h_{i,k} \leq c_k \operatorname{mult}_E h_{i,k} < a(E, W_i, 0) = 2$ by the klt property of (W_i, D_i), where E is an irreducible component of the exceptional divisor $(xy = 0) \subset \mathbb{P}^3$ of the blow-up of W_i at 0 (the maximal ideal of 0 generates the ideal of the exceptional divisor on the blow-up, so $\operatorname{mult}_E h_{i,k} \geq \operatorname{mult}_0 h_{i,k}$, and $a(E, W_i, 0) = 2$ by adjunction). Let $h^0_{i,k}$ be the principal block of $h_{i,k}$, and let $V_i := (xy + z^a t^b = 0) \subset \mathbb{A}^4$ in case (a) of Lemma E.1(3) and $V_i := (xy + g^0_i(z,t) = 0) \subset \mathbb{A}^4$ in case (b), where g^0_i is the principal block of g_i. In case (a), there is a sequence of real numbers $L_i \to +\infty$ with

$$g_i - z^a t^b \in (z,t)^{\lceil L_i \rceil} \quad \text{and} \quad h_{i,k} - h^0_{i,k} \in (x,y,z,t)^{\lceil L_i \rceil} \text{ for any } k. \tag{E.17}$$

Let δ be the adjusted mass of D_i along $\mathbf{T}$, that is,

$$\delta = \sum_k c_k \kappa_k - w_z - w_t.$$

For $c' = (c'_1, \dots, c'_s) \in \mathbb{R}^s_{\geq 0}$, let $(Y_i^+, B_i^+(c'))$ and $(Y_i^-, B_i^-(c'))$ be the positive and negative graded quotients of $(V_i, \sum_k c'_k \operatorname{div}(h^0_{i,k}|_{V_i}))$ by $\mathbf{T}$, followed by the quotient by $G_i/(G_i \cap \mathbf{T})$, with the induced boundaries, let $C_i^\pm \subset Y_i^\pm$ be the fibers over the image of the origin, and let $\beta_i^\pm(c')$ be the minimum of the log discrepancies of $(Y_i^\pm, B_i^\pm(c'))$ over the prime divisors with center in $C_i^\pm$. Let $c := (c_1, \dots, c_s)$, $B_i^\pm := B_i^\pm(c)$, $\beta_i^\pm := \beta_i^\pm(c)$, and $\beta_i := \min(\beta_i^+, \beta_i^-)$.

Explicitly, write $(x_1, x_2, x_3, x_4) := (x, y, z, t)$ and $w = (w_1, w_2, w_3, w_4)$. The surface Y_i^+ is covered by the open subsets $Y_{i,l}^+$, $w_l > 0$, where $Y_{i,l}^+$ is the quotient of the slice $S_{i,l} := V_i \cap (x_l = 1)$ by $\mu_{|w_l|} \subset \mathbf{T}$ and then by $G_i/(G_i \cap \mathbf{T})$, and $B_i^+(c')$ on $Y_{i,l}^+$ is the crepant image of $\sum_k c'_k \operatorname{div}(h^0_{i,k}|_{S_{i,l}})$, all branch coefficients included. Likewise Y_i^- is covered by the open subsets $Y_{i,l}^-$, $w_l < 0$, where $Y_{i,l}^-$ is the quotient of $S_{i,l}$ by $\mu_{|w_l|} \subset \mathbf{T}$ and then by $G_i/(G_i \cap \mathbf{T})$, and $B_i^-(c')$ on $Y_{i,l}^-$ is the crepant image of $\sum_k c'_k \operatorname{div}(h^0_{i,k}|_{S_{i,l}})$.

The normalization (E.16) is possible. Indeed, the terms of degree n of g_i form a non-zero multiple of the monomial $z^a t^b$, and z^a and t^b are coprime, so by Hensel's lemma and the Weierstrass preparation theorem $g_i = v_i \phi_i \psi_i$ in $\mathbb{C}[[z,t]]$, where v_i is a unit and ϕ_i, ψ_i are Weierstrass polynomials in z and in t as in (E.16) with coefficients in $\mathbb{C}[[t]]$ and $\mathbb{C}[[z]]$. The three factors are unique and G_i is diagonal, so ϕ_i and ψ_i are semi-invariant with the characters of z^a and t^b, and v_i is G_i-invariant. If $w_t = 0$ and $a = 1$, then $\phi_{i,0}(t)$ has $\mathbf{T}$-weight zero and the character of z on $G_i \cap \mathbf{T}$, so $\phi_{i,0} = 0$ once $r_i > w_z$. Construction-Theorem 4.13, applied with the automorphism $x \to v_i x$, the invariant unit v_i^{-1}, polynomial truncations of ϕ_i and ψ_i, and the vectors and the prime divisors of Proposition 11.8(i) and (ii), replaces the data with numerical admissions that keep the groups G_i and have the equations $xy + \phi_i \psi_i$. Here $x \to v_i x$ is a G_i-equivariant automorphism of $\mathbb{C}[[x,y,z,t]]$ and v_i^{-1} is a G_i-invariant unit, as in Construction-Theorem 4.13(vi). The automorphism maps $xy + g_i$ to $v_i(xy + \phi_i \psi_i)$. The data of Lemma E.1 satisfy (1)–(3) and (4') of Set-up 11.1 (Lemma 11.10(1)), and the proofs of Proposition 11.8 and Lemma 11.10(1) use (4) only through the klt property of the pairs $(W_i, \frac{1}{m} \operatorname{div}(h_{i,k}))$ near 0 and the inequality $\operatorname{mld}(W_i \ni 0, \frac{1}{m} \operatorname{div}(h_{i,k})) > 0$. The klt property holds by Construction-Theorem 4.13(1) for the admission of Lemma E.1, and the inequality holds by (4'), since K_{W_i} is Cartier, so that

$$a\Big(P, W_i, \frac{1}{m} \operatorname{div}(h_{i,k})\Big) \in \frac{1}{m}\mathbb{Z}$$

for any prime divisor P over W_i with center 0. Thus the conclusions of Proposition 11.8 and Lemma 11.10(1) hold for the new data, with no condition on the automorphism and the unit of Construction-Theorem 4.13(vi). The monomials of degree n of the admitted g_i are those of g_i, and the values α_i do not change, so the conclusion of Lemma E.1 holds for the admitted data.

In case (a), we take

$$|w| := \max_j |w_j|, \qquad \bar\kappa := \max(n|w|, |\kappa_1|, \dots, |\kappa_s|), \qquad L_i := \frac{r_i - \bar\kappa}{|w|}.$$

Then $L_i \to +\infty$, since $r_i \to +\infty$ by Lemma E.1(2), and we show (E.17) for L_i. Since $G_i \cap \mathbf{T} = \{\zeta^w \mid \zeta \in \mu_{r_i}\}$ and $h_{i,k}$, ϕ_i, ψ_i are G_i-semi-invariant, the $\mathbf{T}$-weights of the monomials of $h_{i,k}$ belong to $\kappa_k + r_i\mathbb{Z}$, those of ϕ_i to $aw_z + r_i\mathbb{Z}$, and those of ψ_i to $bw_t + r_i\mathbb{Z}$. For $\ell \in \mathbb{Z}$ let $h^{(\ell)}_{i,k}$ be the sum of the terms of $h_{i,k}$ of weight $\kappa_k + \ell r_i$, so that $h^{(0)}_{i,k} = h^0_{i,k}$. A monomial of weight $\kappa_k + \ell r_i$ with $\ell \neq 0$ has degree at least $(r_i - |\kappa_k|)/|w| \geq L_i$, so $h_{i,k} - h^0_{i,k} \in (x,y,z,t)^{\lceil L_i \rceil}$. By Lemma E.1(3), either $w_z < 0 < w_t$, or $w_t = 0 < w_z$ and $a \leq 1$. Assume that $w_z < 0 < w_t$. A monomial $z^j t^{j'}$ of ϕ_i with $j < a$ has $j' \geq 1$ and weight $-j|w_z| + j' w_t > -a|w_z|$, so the weight is at least $-a|w_z| + r_i$, and

$$j' \geq \frac{r_i - a|w_z|}{w_t}.$$

Likewise a monomial $z^{j'}t^j$ of ψ_i with $j < b$ has weight at most $bw_t - r_i$, and $j' \geq (r_i - bw_t)/|w_z|$. Assume that $w_t = 0 < w_z$. Then $\phi_i = z^a$, and a monomial $z^{j'}t^j$ of ψ_i with $j < b$ has $j' \geq 1$ and weight $j'w_z \in r_i\mathbb{Z}$, so $j' \geq r_i/w_z$. In each case $j' \geq L_i$, since $a|w_z|, bw_t \leq n|w| \leq \bar{\kappa}$ and $|w_z|, w_t \leq |w|$. Thus the monomials of $\phi_i - z^a$ and $\psi_i - t^b$ have degree at least L_i, and $g_i - z^at^b \in (z,t)^{\lceil L_i \rceil}$ by

$$g_i - z^at^b = z^a(\psi_i - t^b) + (\phi_i - z^a)t^b + (\phi_i - z^a)(\psi_i - t^b).$$

Proposition E.3. *Notation as in Construction E.2, in case (a) of Lemma E.1(3).*

(1) For i large, the pairs (Y_i^+, B_i^+) and (Y_i^-, B_i^-) are lc near C_i^+ and C_i^-.

(2) There exists a positive integer Q, independent of i, such that for i large, $\beta_i^- \geq \beta_i^+ + |\delta|/Q$ if $\delta < 0$, $\beta_i^+ \geq \beta_i^- + \delta/Q$ if $\delta > 0$, and $\beta_i^+ = \beta_i^-$ if $\delta = 0$.

(3) Let $\theta > 1$ and $\delta_\theta := \theta\sum_k c_k\kappa_k - w_z - w_t$. Assume that there exists $\sigma \in \{+,-\}$ with $\sigma\delta_\theta \leq 0$ such that the pair $(Y_i^\sigma, B_i^\sigma(\theta c))$ is lc near C_i^σ for any i. Then $\alpha_i \geq \beta_i$ for i large.

Proof. The proof is long, so we divide it into several steps. We write $(x_1,x_2,x_3,x_4) := (x,y,z,t)$, $w = (w_1,w_2,w_3,w_4)$, $N := w_x + w_y = aw_z + bw_t$, L_w for the least common multiple of the non-zero $|w_j|$, and $H_i := G_i/(G_i \cap \mathbf{T})$, so that $|H_i| \leq e_0$ by Lemma E.1(2). In case (a), $V := V_i = (xy + z^at^b = 0)$ does not depend on i, and either $w_z < 0 < w_t$, or $w_t = 0 < w_z$, $a \leq 1$, and $w_x < 0 < w_y$. In both alternatives w has coordinates of both signs. For $c' \in \mathbb{R}^s_{\geq 0}$ let $D_i^0(c') := \sum_k c'_k \operatorname{div}(h^0_{i,k}|_V)$, $D_i^0 := D_i^0(c)$, and $\delta(c') := \sum_k c'_k\kappa_k - w_z - w_t$, so that $\delta(c) = \delta$ and $\delta(\theta c) = \delta_\theta$. For a real-power form ω (a top form multiplied by real powers of functions) and a prime divisor E over a variety, $\operatorname{mult}_E\omega$ is the order of ω along E, computed in local coordinates on a higher model on which E is a divisor.

Step 1. In this step we compare $xy + g_i$ with $xy + z^at^b$, and we factor g_i in two ways, $g_i = u_i^+g_i^+ = u_i^-g_i^-$.

Since $(xy + g_i) - (xy + z^at^b) = g_i - z^at^b$, we have $(xy + g_i) - (xy + z^at^b) \in (x,y,z,t)^{\lceil L_i \rceil}$ by (E.17), where $L_i = (r_i - \bar{\kappa})/|w|$ as in Construction E.2.

If $w_z < 0 < w_t$, let

$$g_i^+ := \phi_it^b, \quad u_i^+ := \psi_i/t^b, \quad g_i^- := z^a\psi_i, \quad u_i^- := \phi_i/z^a,$$

and if $w_t = 0 < w_z$, let

$$g_i^+ := z^a\psi_i, \quad u_i^+ := 1, \quad g_i^- := z^at^b, \quad u_i^- := \psi_i/t^b.$$

Then $g_i = u_i^+g_i^+ = u_i^-g_i^-$, and $u_i^\pm$ are G_i-invariant. By the lower bounds $j' \geq L_i$ in the monomials $z^jt^{j'}$ of ϕ_i and $z^{j'}t^j$ of ψ_i and by the $\mathbf{T}$-weights of the monomials of ϕ_i and ψ_i (Construction E.2), for $\sigma \in \{+,-\}$ the following hold.

(i) The monomials of g_i^σ have weights $N + \ell r_i$ with $\sigma\ell \geq 0$, and the only monomial of weight N of g_i^σ is z^at^b.

(ii) $u_i^\sigma - 1$ is a finite sum of terms $\gamma z^jt^{j'}$ with $j \geq -a$ and $j' \geq -b$ such that either $j \geq L_i$ and $\sigma w_z < 0$, or $j' \geq L_i$ and $\sigma w_t < 0$.

Step 2. In this step we show that there exists i_0 such that (V, D_i^0) is lc near 0 for $i \geq i_0$, and that for a fixed $\theta > 1$ there exists i_1 such that $(W_i, \theta D_i)$ is lc near 0 for any $i \geq i_1$ for which $(V, D_i^0(\theta c))$ is lc near 0.

The hypersurfaces W_i and V are normal (V is a hypersurface singular in codimension two, hence normal by Serre's criterion [Har77, Proposition II.8.23]), and no $h_{i,k}$ or $h^0_{i,k}$ vanishes identically on them, since a non-zero polynomial without monomials divisible by xy does not vanish identically on $(xy + g = 0)$. By lc inversion of adjunction [Kaw07, Theorem], $(W_i, \theta' D_i)$ is lc near 0 if and only if the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + g_i) + \theta'\sum_k c_k \operatorname{div}(h_{i,k})\Big)$$

is lc near 0, and $(V, D_i^0(\theta' c))$ is lc near 0 if and only if the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + z^a t^b) + \theta' \sum_k c_k \operatorname{div}(h_{i,k}^0)\Big)$$

is lc near 0, for $\theta' \in \{1, \theta\}$. Being lc near 0 may be checked on $\operatorname{Spec} \mathbb{C}[[x, y, z, t]]$ [dFEM11, Proposition 2.11]. Let L' and L'' be the integers of Lemma 4.14 in dimension 4, for $s+1$ functions and the finite sets $\{1, c_1, \dots, c_s\}$ and $\{1, \theta c_1, \dots, \theta c_s\}$, respectively. If $\lceil L_i \rceil \geq L'$ (resp. $\lceil L_i \rceil \geq L''$), then (E.17), **Step 1**, and Lemma 4.14, applied in both directions, show that the ambient pair of W_i for $\theta' = 1$ (resp. $\theta' = \theta$) is lc near 0 if and only if the ambient pair of V for this θ' is lc near 0. Let i_0 and i_1 be integers such that $\lceil L_i \rceil \geq L'$ for $i \geq i_0$ and $\lceil L_i \rceil \geq L''$ for $i \geq i_1$. Since (W_i, D_i) is klt near 0, both statements of this step follow.

Step 3. In this step we express the log discrepancies on the graded quotients and on V through one function on the valuations of the quotient $Z := \operatorname{Spec} \mathbb{C}[V]^{\mathbf{T}}$.

Let $o_Z \in Z$ be the image of 0, and let $(\widetilde{Y}^{\pm}, \widetilde{B}_i^{\pm}(c'))$ be the graded quotients before the quotient by H_i, so that $(Y_i^{\pm}, B_i^{\pm}(c'))$ is the crepant quotient of $(\widetilde{Y}^{\pm}, \widetilde{B}_i^{\pm}(c'))$ by H_i, and $\widetilde{Y}^{\pm} \to Z$ are projective and birational (Lemma 8.12, since w has coordinates of both signs). Since w is primitive and $w_y = N - w_x$, the integers w_x, w_z, w_t are coprime, so there exists a Laurent monomial ϖ in x, z, t of $\mathbf{T}$-weight one. Write $x_j = \varpi^{w_j} M_j$ and $h_{i,k}^0|_V = \varpi^{\kappa_k} H_{i,k}$ with $\mathbf{T}$-invariant M_j and $H_{i,k}$. The residue form $\Omega := \frac{dx}{x} \wedge dz \wedge dt$ generates ω_V near 0. The form $\frac{dx}{x}$ is $\mathbf{T}$-invariant and $dz \wedge dt$ has $\mathbf{T}$-weight $w_z + w_t$, so $\Omega' := \varpi^{-w_z - w_t} \Omega$ is $\mathbf{T}$-invariant. Let ξ be the vector field of $\mathbf{T}$ and $\omega_0 := \iota_\xi \Omega'$. Then ω_0 is $\mathbf{T}$-invariant with $\iota_\xi \omega_0 = 0$, hence a 2-form pulled back from Z, and since $\iota_\xi \frac{d\varpi}{\varpi} = 1$, the 3-form $\Omega' - \frac{d\varpi}{\varpi} \wedge \omega_0$ is killed by ι_ξ and $\mathbf{T}$-invariant, hence pulled back from the surface Z, hence zero. Thus $\Omega = \varpi^{w_z + w_t} \frac{d\varpi}{\varpi} \wedge \omega_0$. Let $\omega_i := \omega_0 \prod_k H_{i,k}^{-c_k}$, so that

$$\Omega \prod_k (h_{i,k}^0)^{-c_k} = \varpi^{-\delta} \frac{d\varpi}{\varpi} \wedge \omega_i. \tag{E.18}$$

For a prime divisor ν over Z with center o_Z, let

$$\eta_i(\nu) := 1 + \operatorname{mult}_\nu \omega_i, \qquad \underline{\rho}(\nu) := \max_{w_j > 0} \Big(- \frac{\operatorname{mult}_\nu M_j}{w_j} \Big), \qquad \overline{\rho}(\nu) := \min_{w_j < 0} \frac{\operatorname{mult}_\nu M_j}{|w_j|}.$$

We show the following.

(a) $\underline{\rho}(\nu) < \overline{\rho}(\nu)$, both numbers belong to $\frac{1}{L_w}\mathbb{Z}$, and $\operatorname{mult}_\nu M_j > 0$ if $w_j = 0$.

(b) The log discrepancies of ν on the graded quotients are

$$a(\nu, \widetilde{Y}^+, \widetilde{B}_i^+) = \eta_i(\nu) - \delta \underline{\rho}(\nu), \qquad a(\nu, \widetilde{Y}^-, \widetilde{B}_i^-) = \eta_i(\nu) - \delta \overline{\rho}(\nu).$$

(c) Let $\varrho \geq 1$ and c_ν be coprime integers with $c_\nu / \varrho \in (\underline{\rho}(\nu), \overline{\rho}(\nu))$, and let v be the Gauss extension of $\varrho \operatorname{mult}_\nu$ with ϖ-value c_ν [Liu26, Definition-Proposition 2.11]. Then $v = \operatorname{mult}_E$ for a prime divisor E over V with center 0, and $a(E, V, D_i^0) = \varrho \eta_i(\nu) - \delta c_\nu$.

For (a), let $w_j > 0 > w_{j'}$. The function $x_j^{|w_{j'}|} x_{j'}^{w_j} = M_j^{|w_{j'}|} M_{j'}^{w_j}$ is $\mathbf{T}$-invariant, regular, and vanishes at 0, so it vanishes at o_Z and $|w_{j'}| \operatorname{mult}_\nu M_j + w_j \operatorname{mult}_\nu M_{j'} > 0$. Thus $-\operatorname{mult}_\nu M_j / w_j < \operatorname{mult}_\nu M_{j'} / |w_{j'}|$, which implies $\underline{\rho}(\nu) < \overline{\rho}(\nu)$. If $w_j = 0$, then $x_j = M_j$ is invariant and vanishes at 0. Since $\operatorname{mult}_\nu M_j \in \mathbb{Z}$, both numbers belong to $\frac{1}{L_w}\mathbb{Z}$.

For (b), choose l with $w_l > 0$ and $\underline{\rho}(\nu) = -\operatorname{mult}_\nu M_l / w_l$. By the choice of l, any $\mathbf{T}$-invariant monomial regular on $V \cap (x_l \neq 0)$ has non-negative order along ν, so the center of ν on $\widetilde{Y}^+$ is contained in the chart $S_{i,l}/\mu_{w_l}$ of x_l. On $\mathbb{G}_m \times S_{i,l}$, with coordinate ρ on $\mathbb{G}_m$, the map $(\rho, s) \to \rho \cdot s$ (the action of $\mathbf{T}$) is the quotient by μ_{w_l} onto $V \cap (x_l \neq 0)$, and $\varpi = \zeta_0 \rho M_l^{-1/w_l}$ for a root of unity ζ_0. By (E.18) and $dM_l \wedge \omega_i = 0$, substituting $\varpi = \zeta_0 \rho M_l^{-1/w_l}$, so that

$$\frac{d\varpi}{\varpi} \wedge \omega_i = \frac{d\rho}{\rho} \wedge \omega_i \qquad \text{and} \qquad \varpi^{-\delta} = \zeta_0^{-\delta} \rho^{-\delta} M_l^{\delta / w_l},$$

we obtain

$$\Omega\prod_k (h^0_{i,k})^{-c_k} = \lambda_0\rho^{-\delta}\frac{d\rho}{\rho}\wedge M_l^{\delta/w_l}\omega_i, \qquad \lambda_0\in\mathbb{C}^*,$$

and $M_l^{\delta/w_l}\omega_i$ is the real-power form of the pair $(S_{i,l}, \sum_k c_k \operatorname{div}(h^0_{i,k}|_{S_{i,l}}))$, whose crepant image is $\widetilde{B}_i^+$ on the chart. Hence

$$a(\nu,\widetilde{Y}^+,\widetilde{B}_i^+) = 1+\operatorname{mult}_\nu\left(M_l^{\delta/w_l}\omega_i\right) = \eta_i(\nu)+\frac{\delta}{w_l}\operatorname{mult}_\nu M_l = \eta_i(\nu)-\delta\underline{\rho}(\nu).$$

For the negative graded quotient, the proof of (b) for $\widetilde{Y}^+$ works with $w_l<0$, $\overline{\rho}(\nu)=\operatorname{mult}_\nu M_l/|w_l|$ in place of $\underline{\rho}(\nu) = -\operatorname{mult}_\nu M_l/w_l$, and the map $(\rho,s)\to\rho^{-1}\cdot s$.

For (c), $v(x_j) = c_\nu w_j + \varrho\operatorname{mult}_\nu M_j > 0$ for any j by (a) and $c_\nu/\varrho\in(\underline{\rho}(\nu),\overline{\rho}(\nu))$, so the center of v on V is 0. Let $Z'\to Z$ be a resolution of singularities on which ν is a divisor, and let $B_{Z'} := -\operatorname{div}(\omega_i)$ on Z', so that $a(\cdot,Z',B_{Z'}) = 1+\operatorname{mult}.\,\omega_i$. The function field of V is that of $Z'\times\mathbb{A}^1_\varpi$, and by (E.18) the pair (V,D_i^0) and the pair

$$\left(Z'\times\mathbb{A}^1_\varpi, \operatorname{pr}^* B_{Z'} + (1+\delta)\operatorname{div}(\varpi)\right)$$

have equal log discrepancies at any prime divisor over V. We distinguish three cases.

If $c_\nu>0$, then by [Liu26, Lemma 18.1], $v=\operatorname{mult}_E$ for a prime divisor E, and

$$a(E,V,D_i^0) = \varrho a(\nu,Z',B_{Z'}) + c_\nu - (1+\delta)c_\nu = \varrho\eta_i(\nu)-\delta c_\nu.$$

If $c_\nu<0$, let $\varpi' := \varpi^{-1}$. Then

$$\varpi^{-\delta}\frac{d\varpi}{\varpi} = -\varpi'^{\delta}\frac{d\varpi'}{\varpi'},$$

so by (E.18) the pair (V,D_i^0) and the pair

$$\left(Z'\times\mathbb{A}^1_{\varpi'}, \operatorname{pr}^* B_{Z'} + (1-\delta)\operatorname{div}(\varpi')\right)$$

have equal log discrepancies at any prime divisor over V. The valuation v is the Gauss extension of $\varrho\operatorname{mult}_\nu$ with ϖ'-value $-c_\nu>0$, so by [Liu26, Lemma 18.1], $v=\operatorname{mult}_E$ for a prime divisor E, and

$$a(E,V,D_i^0) = \varrho\eta_i(\nu) - c_\nu + (1-\delta)c_\nu = \varrho\eta_i(\nu)-\delta c_\nu.$$

If $c_\nu=0$, then $\varrho=1$, since ϱ and c_ν are coprime, and $v=\operatorname{mult}_E$ for the prime divisor $E := \nu\times\mathbb{A}^1_\varpi$ of $Z'\times\mathbb{A}^1_\varpi$. Since E is not a component of $\operatorname{div}(\varpi)$, $a(E,V,D_i^0) = a(\nu,Z',B_{Z'}) = \eta_i(\nu)$.

In all three cases the center of E on V is 0, since the center of v on V is 0.

Step 4. In this step we prove (1). Let $i\geq i_0$, so that (V,D_i^0) is lc near 0 by **Step 2**. Since (Y_i^+,B_i^+) is the crepant quotient of $(\widetilde{Y}^+,\widetilde{B}_i^+)$ by H_i, it suffices to show that $(\widetilde{Y}^+,\widetilde{B}_i^+)$ is lc near the fiber $\widetilde{C}^+$ over o_Z [Liu26, Lemma 2.6]. Suppose not. Then there exist a closed point $p\in\widetilde{C}^+$ and a prime divisor E' over $\widetilde{Y}^+$ whose center contains p, with $a(E',\widetilde{Y}^+,\widetilde{B}_i^+)<0$. Let G' be a prime divisor with center p on a log resolution on which E' and G' are divisors meeting transversally at a general point of $E'\cap G'$. The toric valuations with weights $(d,1)$ along E' and G' at that point have center p and log discrepancies

$$d\cdot a(E',\widetilde{Y}^+,\widetilde{B}_i^+) + a(G',\widetilde{Y}^+,\widetilde{B}_i^+)$$

[JM12, Proposition 5.1], which are negative for d large. Thus there exists a prime divisor ν over $\widetilde{Y}^+$ whose center is p and with $a(\nu,\widetilde{Y}^+,\widetilde{B}_i^+)<0$ (cf. [KM98, Corollary 2.31(1)]). The center of ν on Z is o_Z, so **Step 3**(b) implies $\eta_i(\nu)-\delta\underline{\rho}(\nu)<0$. Choose coprime integers $\varrho\geq 1$ and c_ν with $c_\nu/\varrho\in(\underline{\rho}(\nu),\overline{\rho}(\nu))$ and

$$\delta\left(\frac{c_\nu}{\varrho}-\underline{\rho}(\nu)\right) > \eta_i(\nu)-\delta\underline{\rho}(\nu).$$

By **Step 3**(c),

$$a(E,V,D_i^0) = \varrho\left(\eta_i(\nu)-\delta\underline{\rho}(\nu)-\delta\left(\frac{c_\nu}{\varrho}-\underline{\rho}(\nu)\right)\right) < 0$$

for a prime divisor E over V with center 0, a contradiction. For (Y_i^-, B_i^-), the proof given for (Y_i^+, B_i^+) works with $\widetilde{Y}^-$, $\widetilde{B}_i^-$, $\widetilde{C}^-$, and $\overline{\rho}(\nu)$ in place of $\widetilde{Y}^+$, $\widetilde{B}_i^+$, $\widetilde{C}^+$, and $\underline{\rho}(\nu)$, and with c_ν/ϱ close to $\overline{\rho}(\nu)$.

Step 5. In this step we prove (2) with $Q := e_0 L_w$. Let E be a prime divisor over Y_i^- with center in C_i^-, and let ν be a prime divisor over $\widetilde{Y}^-$ lying over E. The center of ν on Z is o_Z. Since $\widetilde{Y}^\pm \to Z$ are proper and birational and H_i acts compatibly on them, ν is also a prime divisor over $\widetilde{Y}^+$ with center in $\widetilde{C}^+$, E is a prime divisor over Y_i^+ with center in C_i^+, and the inertia group $I := I_{H_i}(\nu)$ does not depend on whether ν is regarded over $\widetilde{Y}^+$ or over $\widetilde{Y}^-$. Conversely, any prime divisor over Y_i^+ with center in C_i^+ is obtained in this way. By [Liu26, Lemma 2.6] and **Step 3**(b),

$$a(E, Y_i^-, B_i^-) - a(E, Y_i^+, B_i^+) = \frac{a(\nu, \widetilde{Y}^-, \widetilde{B}_i^-) - a(\nu, \widetilde{Y}^+, \widetilde{B}_i^+)}{|I|} = -\frac{\delta\big(\overline{\rho}(\nu) - \underline{\rho}(\nu)\big)}{|I|}.$$

By **Step 3**(a), $\overline{\rho}(\nu) - \underline{\rho}(\nu) \geq 1/L_w$, and $|I| \leq |H_i| \leq e_0$. If $\delta < 0$, then

$$a(E, Y_i^-, B_i^-) \geq a(E, Y_i^+, B_i^+) + \frac{|\delta|}{Q} \geq \beta_i^+ + \frac{|\delta|}{Q},$$

and taking the minimum over E we obtain $\beta_i^- \geq \beta_i^+ + |\delta|/Q$. The case $\delta > 0$ is symmetric, and if $\delta = 0$, then $a(E, Y_i^-, B_i^-) = a(E, Y_i^+, B_i^+)$ for any E, so $\beta_i^+ = \beta_i^-$.

Step 6. In this step we show the following. Let $\sigma \in \{+, -\}$ and $c' \in \mathbb{R}^s_{\geq 0}$ with $\sigma\delta(c') \leq 0$. If $(Y_i^\sigma, B_i^\sigma(c'))$ is lc near C_i^σ, then $(V, D_i^0(c'))$ is lc near 0.

The argument of **Step 4**, with V, $D_i^0(c')$, and the point 0 in place of $\widetilde{Y}^+$, $\widetilde{B}_i^+$, and $\widetilde{C}^+$, shows that it suffices to prove $a(F, V, D_i^0(c')) \geq 0$ for any prime divisor F over V with center 0. Fix such F, and choose l with $\sigma w_l > 0$ such that $\mathrm{mult}_F\, x_l/|w_l|$ is minimal among the j with $\sigma w_j > 0$. Let x_j'', $j \neq l$, be the coordinates of the slice $S_{i,l} = V \cap (x_l = 1)$, and consider $\mathbb{A}^1_\rho \times S_{i,l}$ with the map $(\rho, s) \to \rho^\sigma \cdot s$, that is, $x_l = \rho^{|w_l|}$ and $x_j = \rho^{\sigma w_j} x_j''$. Over $\rho \neq 0$ the map is the quotient by $\mu_{|w_l|}$ onto $V \cap (x_l \neq 0)$. By (E.18) with c' in place of c, that is, with $\delta(c')$ and $\omega_0 \prod_k H_{i,k}^{-c'_k}$ in place of δ and ω_i, and the substitution of **Step 3**(b) (with ρ^σ in place of ρ), the pair

$$\Big(\mathbb{A}^1_\rho \times S_{i,l},\ (1 + \sigma\delta(c'))S_0 + \mathrm{pr}^*\, B_S(c')\Big), \qquad S_0 := \{0\} \times S_{i,l}, \quad B_S(c') := \sum_k c'_k \,\mathrm{div}(h^0_{i,k}|_{S_{i,l}}),$$

is crepant to $(V, D_i^0(c'))$ through the map. Hence, for a prime divisor F' over $\mathbb{A}^1_\rho \times S_{i,l}$ lying over F, we have $a(F, V, D_i^0(c')) \geq 0$ if and only if

$$a\big(F', \mathbb{A}^1_\rho \times S_{i,l}, (1 + \sigma\delta(c'))S_0 + \mathrm{pr}^*\, B_S(c')\big) \geq 0$$

[Liu26, Lemma 2.6]. Moreover $\mathrm{mult}_{F'} \rho > 0$, $\mathrm{mult}_{F'} x_j'' \geq 0$ if $\sigma w_j > 0$ by the choice of l, and $\mathrm{mult}_{F'} x_j'' > 0$ if $\sigma w_j \leq 0$. Thus the center of F' is contained in $S_0 = \{0\} \times S_{i,l}$. Let $p \in S_{i,l}$ be a closed point with $(0, p)$ in the center of F'. Any $\mathbf{T}$-invariant monomial in x, y, z, t which vanishes at 0 is a monomial in the x_j'' with positive order along F', so the image of p in Y_i^σ belongs to C_i^σ. Since $(Y_i^\sigma, B_i^\sigma(c'))$ is lc near C_i^σ, the pair $(S_{i,l}, B_S(c'))$ is lc near p [Liu26, Lemma 2.6]. By [Kaw07, Theorem], $(\mathbb{A}^1_\rho \times S_{i,l}, S_0 + \mathrm{pr}^*\, B_S(c'))$ is lc near $\{0\} \times \{p\}$, and since the center of F' contains $(0, p)$ and $1 + \sigma\delta(c') \leq 1$,

$$a\big(F', \mathbb{A}^1_\rho \times S_{i,l}, (1 + \sigma\delta(c'))S_0 + \mathrm{pr}^*\, B_S(c')\big) \geq a\big(F', \mathbb{A}^1_\rho \times S_{i,l}, S_0 + \mathrm{pr}^*\, B_S(c')\big) \geq 0.$$

Hence $a(F, V, D_i^0(c')) \geq 0$.

Step 7. In this step we fix θ as in (3), and we bound the orders along a prime divisor computing α_i.

Let σ be the sign of the hypothesis of (3), so that $\sigma\delta_\theta \leq 0$ and $(Y_i^\sigma, B_i^\sigma(\theta c))$ is lc near C_i^σ for any i. By **Step 6** with $c' = \theta c$, $(V, D_i^0(\theta c))$ is lc near 0 for any i, so by **Step 2**, $(W_i, \theta D_i)$

is lc near 0 for $i \geq i_1$. Fix $i \geq \max(i_0, i_1)$ and a prime divisor F over W_i with center 0 such that $a(F, W_i, D_i) = |I_{G_i}(F)|\alpha_i$ (Proposition 11.7(0) with (4'), which holds by Construction E.2). Let $e := |I_{G_i}(F)|$, and for a regular function f near 0 let $[f] := \operatorname{mult}_F f/e$ be the normalized order of f. Then $a(F, W_i, D_i) = e\alpha_i < e$, since $\alpha_i < 1$ by the hypothesis of Lemma E.1, and the numerical admissions of Lemma E.1 and Construction E.2 do not change α_i. Moreover $a(F, W_i, 0) \geq \theta \sum_k c_k \operatorname{mult}_F h_{i,k}$. Thus $\sum_k c_k \operatorname{mult}_F h_{i,k} < e/(\theta - 1)$, and with $C_1 := \theta/(\theta - 1)$ and $C_2 := \max_k 1/(c_k(\theta - 1))$ we obtain

$$[h_{i,k}] < C_2 \text{ for any } k, \qquad \frac{a(F, W_i, 0)}{e} < C_1. \tag{E.19}$$

The residue form $\Omega_i := \frac{dx}{x} \wedge dz \wedge dt$ generates ω_{W_i} near 0. On a higher model on which F is a divisor, $\frac{dx}{x}$ is a logarithmic 1-form along F, and dz and dt are products of z and t with logarithmic 1-forms along F. Hence $\operatorname{mult}_F \Omega_i \geq \operatorname{mult}_F z + \operatorname{mult}_F t - 1$, and by (E.19),

$$[z] + [t] \leq \frac{a(F, W_i, 0)}{e} < C_1. \tag{E.20}$$

Let $w_j > 0 > w_{j'}$. The monomial $x_j^{|w_{j'}|} x_{j'}^{w_j}$ is $\mathbf{T}$-invariant, so the character of the monomial on G_i is trivial on $G_i \cap \mathbf{T}$, and the $|H_i|$-th power of the monomial is a G_i-invariant regular function vanishing at 0. The order of such a function along F is at least e [Liu26, Lemma 2.6], so

$$|w_{j'}|[x_j] + w_j[x_{j'}] \geq \frac{1}{e_0}, \qquad \text{and} \qquad [x_j] \geq \frac{1}{e_0} \text{ if } w_j = 0. \tag{E.21}$$

Let $\epsilon := \min\left(\frac{1}{2e_0|w|}, \frac{1}{Q}\right)$.

Step 8. In this step we show the following. Let $\sigma \in \{+, -\}$, assume that $[x_j] \geq \epsilon$ for any j with $\sigma w_j < 0$, and let $\rho_F := \min_{\sigma w_j > 0}[x_j]/|w_j|$. Then there exists i_2, depending only on the sequence of data and on θ, not on F or σ, such that for $i \geq \max(i_0, i_1, i_2)$,

$$\frac{a(F, W_i, D_i)}{e} \geq \beta_i^\sigma - \sigma\delta\rho_F. \tag{E.22}$$

Let i_2 be an integer with $\epsilon L_i - nC_1 > C_2$ for any $i \geq i_2$.

First we replace x. Let $x' := x/u_i^\sigma$, a function on a dense open subset of W_i. By **Step 1**(ii), each term $\gamma z^j t^{j'}$ of $u_i^\sigma - 1$ has an exponent at least L_i on a coordinate of weight of sign $-\sigma$, whose normalized order is at least ϵ, while the other exponent is at least $-a$ or $-b$. By (E.20), $\operatorname{mult}_F(u_i^\sigma - 1) \geq e(\epsilon L_i - nC_1) > eC_2$. Hence $\operatorname{mult}_F x' = \operatorname{mult}_F x$, $x'y + g_i^\sigma = 0$ on W_i, and $\Omega_i = \frac{dx'}{x'} \wedge dz \wedge dt$, since u_i^σ only depends on z and t. For any polynomial h, $h(x, y, z, t) - h(x', y, z, t)$ is $(u_i^\sigma - 1)x'$ times a polynomial in u_i^σ, x', y, z, t, so $\operatorname{mult}_F\big(h(x, y, z, t) - h(x', y, z, t)\big) > eC_2$.

Next we remove the blocks of the opposite sign. Let

$$h_{i,k}^\sigma := \sum_{\sigma\ell \geq 0} h_{i,k}^{(\ell)}(x', y, z, t).$$

A monomial $\prod_j x_j^{\gamma_j}$ of weight $\kappa_k + \ell r_i$ with $\sigma\ell < 0$ satisfies $\sum_{\sigma w_j < 0} \gamma_j |w_j| \geq r_i - |\kappa_k|$, so the normalized order of the monomial along F, with x' in place of x, is at least $\epsilon(r_i - |\kappa_k|)/|w| \geq \epsilon L_i > C_2$. By (E.19) and the previous paragraph, $\operatorname{mult}_F h_{i,k} = \operatorname{mult}_F h_{i,k}^\sigma$ for any k, and therefore

$$a(F, W_i, D_i) = 1 + \operatorname{mult}_F\left(\Omega_i \prod_k (h_{i,k}^\sigma)^{-c_k}\right). \tag{E.23}$$

Next we pass to a root chart. Write $(x_1', x_2', x_3', x_4') := (x', y, z, t)$, and choose l with $\sigma w_l > 0$ and $[x_l]/|w_l| = \rho_F$, with $x_l \in \{x, y\}$ if $[x_j]/|w_j| = \rho_F$ for some $j \in \{1, 2\}$ with $\sigma w_j > 0$. On a $\mu_{|w_l|}$-cover of W_i let ρ satisfy $\rho^{|w_l|} = x_l'$, and let $x_j'' := x_j' \rho^{-\sigma w_j}$ for $j \neq l$. By **Step 1**(i) and the definition of $h_{i,k}^\sigma$,

$$x'y + g_i^\sigma = \rho^{\sigma N} F_i(\rho^{r_i}, x''), \qquad h_{i,k}^\sigma = \rho^{\sigma\kappa_k} \widehat{H}_{i,k}(\rho^{r_i}, x''),$$

for polynomials F_i and $\widehat{H}_{i,k}$ such that $F_i(0,x'')$ and $\widehat{H}_{i,k}(0,x'')$ are the restrictions of $xy+z^at^b$ and $h^0_{i,k}$ to the slice $S_{i,l}$, in the coordinates x''_j.

Since u_i^σ is invariant, x' has the character of x. Let $\hat{G}_l$ be the group of diagonal automorphisms of $\mathbb{A}^4_{\rho,x''}$ generated by the lifts of G_i and by $\mu_{|w_l|}$, where $\zeta\in\mu_{|w_l|}$ acts by $\rho\to\zeta\rho$ and $x''_j\to\zeta^{-\sigma w_j}x''_j$. Let $\hat{R}\subset\hat{G}_l$ be the subgroup of the elements acting on ρ alone, $r':=|\hat{R}|$, $q:=\rho^{r'}$, and $\bar{G}_l:=\hat{G}_l/\hat{R}$. The polynomial $F_i(\rho^{r_i},x'')$ is $\hat{R}$-semi-invariant, and $F_i(0,x'')$ is a non-zero $\hat{R}$-invariant, so $F_i(\rho^{r_i},x'')$ is a polynomial in q and x''. Likewise $\widehat{H}_{i,k}(\rho^{r_i},x'')$ is a polynomial in q and x'' for any k. Let

$$X:=(F_i=0)\subset\mathbb{A}^4_{q,x''},\qquad S_0:=X\cap(q=0)\cong S_{i,l},$$
$$\Delta_X:=\sum_k c_k\operatorname{div}(\widehat{H}_{i,k}|_X),\qquad b_q:=1+\frac{\sigma\delta}{r'}.$$

The root chart has Jacobian $\pm|w_l|\rho^{\sigma(N+w_z+w_t)-1}$, so, taking residues, it pulls back $\Omega_i\prod_k(h^\sigma_{i,k})^{-c_k}$ to

$$\pm|w_l|\rho^{-1-\sigma\delta}\omega'\prod_k\widehat{H}_{i,k}^{-c_k},$$

where ω' is the residue form of the hypersurface $X'\subset\mathbb{A}^4_{\rho,x''}$ defined by $F_i(\rho^{r_i},x'')$. Thus, near $(\rho=0)$, the divisor of the pulled-back form is $-\Delta_{X'}$ for

$$\Delta_{X'}:=(1+\sigma\delta)\operatorname{div}(\rho)+\sum_k c_k\operatorname{div}(\widehat{H}_{i,k}|_{X'}).$$

The hypersurface X' is normal near $(\rho=0)$ by the argument given for X below, and $X=X'/\hat{R}$. Since $\hat{R}$ acts on ρ through $\mu_{r'}$ and fixes the $\widehat{H}_{i,k}$, and $1-(1-(1+\sigma\delta))/r'=b_q$, the crepant quotient of $(X',\Delta_{X'})$ by $\hat{R}$ is $(X,b_qS_0+\Delta_X)$ [Liu26, Lemma 2.6(4)]. Hence, by (E.23), the Hurwitz formula along $X'\dashrightarrow W_i$ (cf. [KM98, Proposition 5.20]), and [Liu26, Lemma 2.6(1)(2)(5)], the quotient $X/\bar{G}_l$ is birational to W_i/G_i, and for the prime divisor P over X lying over F,

$$\frac{a(F,W_i,D_i)}{e}=\frac{a(P,X,b_qS_0+\Delta_X)}{|I_{\bar{G}_l}(P)|}.\tag{E.24}$$

The surface S_0 is smooth if $x_l\in\{x,y\}$, isomorphic to $(xy+t^b=0)$ if $x_l=z$, and isomorphic to $(xy+z^a=0)$ if $x_l=t$. In particular S_0 is a normal klt complete intersection with isolated singularities, since $(xy+t^b=0)$ and $(xy+z^a=0)$ are smooth or have a singularity of type A, which is canonical [KM98, Theorem 4.20].

Since S_0 is a Cartier divisor of X, X is smooth at the smooth points of S_0, and X is normal near S_0 by Serre's criterion. Let E be a prime divisor over X whose center contains a point $p_0\in S_0$. If the center of E is contained in S_0, then $a(E,X,0)\geq a(E,X,S_0)+\operatorname{mult}_E S_0>0$, since (X,S_0) is lc near S_0 by [Kaw07, Theorem]. Otherwise choose a function φ vanishing on the center of E but not identically on S_0. For $\tau>0$ small, $(S_0,\tau\operatorname{div}(\varphi|_{S_0}))$ is lc near p_0, so $(X,S_0+\tau\operatorname{div}(\varphi))$ is lc near p_0 by [Kaw07, Theorem], and $a(E,X,0)\geq\tau\operatorname{mult}_E\varphi>0$. Thus X is klt near S_0. An element of $\bar{G}_l$ acting trivially on S_0 fixes the x''_j, since S_0 is not contained in a hyperplane $(x''_j=0)$, so it is the image of an element of $\hat{R}$. Hence $\bar{G}_l$ acts faithfully on S_0 and on X.

Finally we bound the right-hand side of (E.24) by Lemma 5.5. Since $\operatorname{mult}_F x_l>0$, we have $\operatorname{mult}_P q>0$, so the center of P on X is contained in S_0. Since w has coordinates of both signs, there is j_1 with $\sigma w_{j_1}<0$, and x''_{j_1} has positive order along P. As S_0 is not contained in $(x''_{j_1}=0)$, the center of P is a closed point or a curve. Any $\mathbf{T}$-invariant monomial in x,y,z,t which vanishes at 0 has positive order along P, and it is a power of u_i^σ times a monomial M'' in the x''_j. Since $\operatorname{mult}_F(u_i^\sigma-1)>0$ and u_i^σ is G_i-invariant, $\operatorname{mult}_P u_i^\sigma=0$, so M'' has positive order along P, and M'' vanishes on the center of P. Thus the image of the center of P in $S_0/\bar{G}_l=Y^\sigma_{i,l}$ is contained

in C_i^σ, and $(S_0, \sum_k c_k \operatorname{div}(\widehat{H}_{i,k}|_{S_0}))$ is lc near the center of P by (1) and [Liu26, Lemma 2.6]. Let $\bar{b} := \min(b_q, 1)$ and $\mathfrak{a} := \prod_k \widehat{H}_{i,k}^{c_k}$.

Assume that the center of P is a closed point p, and let $\bar{G}_{l,p}$ be the stabilizer of p in $\bar{G}_l$. Each x_j'' with $x_j''(p) \neq 0$ has trivial character on $\bar{G}_{l,p}$, so in the coordinates q and $x_j'' - x_j''(p)$ the group $\bar{G}_{l,p}$ acts diagonally, p is the origin, $X = (F_i = 0)$ and $S_0 = (F_i = q = 0)$ are complete intersections in the coordinates q and $x_j'' - x_j''(p)$, and $\bar{G}_{l,p}$ acts faithfully on S_0. The inertia group of P fixes p, so $I_{\bar{G}_l}(P) = I_{\bar{G}_{l,p}}(P)$. Let

$$\beta_0 := \operatorname{mld}\Big(\Big(S_0 \ni p, \sum_k c_k \operatorname{div}(\widehat{H}_{i,k}|_{S_0})\Big)/\bar{G}_{l,p}\Big).$$

After a translation by an element of $\bar{G}_l$, the prime divisors over Y_i^σ with center the image of p are the prime divisors under the prime divisors over S_0 with center p [Liu26, Lemma 2.6(2)(3)], and for a prime divisor Γ' over S_0 with center p, the inertia group $I_{\bar{G}_l}(\Gamma')$ fixes p, so $I_{\bar{G}_l}(\Gamma') = I_{\bar{G}_{l,p}}(\Gamma')$. By [Liu26, Lemma 2.6], β_0 is the minimal log discrepancy of (Y_i^σ, B_i^σ) at the image of p, so $\beta_0 \geq \beta_i^\sigma$. By Lemma 5.5 for the group $\bar{G}_{l,p}$, with $u = q$ and the $\mathbb{R}$-ideal $\mathfrak{a}$,

$$\frac{a(P, X, \bar{b}S_0 + \Delta_X)}{|I_{\bar{G}_l}(P)|} \geq \beta_0 + \frac{(1-\bar{b})\operatorname{mult}_P q}{|I_{\bar{G}_l}(P)|}. \tag{E.25}$$

Assume that the center of P is a curve C. We first show that $0 \in S_0$. If $x_l = z$ and $b = 0$, then $a = n \geq 3$, so $w_z < 0 < w_t$ and $\sigma = -$, and $g_i^- = z^n$ since $\psi_i = 1$. Thus $x'y = -z^n$ on W_i and $[x] + [y] = n[z] = n|w_z|\rho_F$, while $N = w_x + w_y = nw_z$. If $w_x \geq 0$, then $|w_y| \geq n|w_z|$ and $[y] \geq |w_y|\rho_F \geq n[z] = [x] + [y] > [y]$, which is absurd. Thus $w_x < 0$, and likewise $w_y < 0$. Then $[x] \geq |w_x|\rho_F$, $[y] \geq |w_y|\rho_F$, and $[x] + [y] = (|w_x| + |w_y|)\rho_F$, so $[x]/|w_x| = \rho_F$, against the choice of l. If $x_l = t$ and $a = 0$, then $w_t \neq 0$ since $\sigma w_t > 0$, so $w_z < 0 < w_t$ by Lemma E.1(3), $\sigma = +$, and $g_i^+ = t^n$ since $\phi_i = 1$. Thus $x'y = -t^n$ on W_i and $[x] + [y] = n[t] = nw_t\rho_F$, while $N = w_x + w_y = nw_t$. If $w_x \leq 0$, then $w_y \geq nw_t$ and $[y] \geq w_y\rho_F \geq n[t] = [x] + [y] > [y]$, which is absurd. Thus $w_x > 0$, and likewise $w_y > 0$. Then $[x] \geq w_x\rho_F$, $[y] \geq w_y\rho_F$, and $[x]+[y] = (w_x+w_y)\rho_F$, so $[x]/w_x = \rho_F$, against the choice of l. Hence S_0 is defined in $\mathbb{A}^3_{x''}$ by one of the binomials $x_2'' + x_3''^a x_4''^b$, $x_1'' + x_3''^a x_4''^b$, $x_1''x_2'' + x_4''^b$ with $b \geq 1$, and $x_1''x_2'' + x_3''^a$ with $a \geq 1$, so $0 \in S_0$. The curve C is an irreducible component of $S_0 \cap (x_{j_1}'' = 0)$, and since S_0 is defined by such a binomial, $C = S_0 \cap \bigcap_{j \in J}(x_j'' = 0)$ for a set J of indices. Thus C contains 0 and is $\bar{G}_l$-invariant. Every $\mathbf{T}$-invariant monomial in x, y, z, t which vanishes at 0 is a monomial of positive degree in the x_j'' times a power of u_i^σ, so the image of 0 in $Y_{i,l}^\sigma$ belongs to C_i^σ, so $(S_0, \sum_k c_k \operatorname{div}(\widehat{H}_{i,k}|_{S_0}))$ is lc near 0, hence near the generic point of C, by (1) and [Liu26, Lemma 2.6]. Let β_0 be the minimal log discrepancy of the quotient of $(S_0, \sum_k c_k \operatorname{div}(\widehat{H}_{i,k}|_{S_0}))$ by $\bar{G}_l$ at the image of the generic point of C. By [Liu26, Lemma 2.6], β_0 is the minimal log discrepancy of (Y_i^σ, B_i^σ) at the image of the generic point of C, which belongs to C_i^σ, so $\beta_0 \geq \beta_i^\sigma$. Since $\dim S_0 = 2$ and $\bar{G}_l$ acts diagonally on $\mathbb{A}^4_{q,x''}$ and faithfully on S_0 and on X, the second part of Lemma 5.5 for the group $\bar{G}_l$, with $u = q$ and the $\mathbb{R}$-ideal $\mathfrak{a}$, gives (E.25).

Since

$$a(P, X, b_q S_0 + \Delta_X) = a(P, X, \bar{b}S_0 + \Delta_X) - (b_q - \bar{b})\operatorname{mult}_P q,$$

(E.25) and $\beta_0 \geq \beta_i^\sigma$ give, whether $b_q \leq 1$ or $b_q > 1$,

$$\frac{a(P, X, b_q S_0 + \Delta_X)}{|I_{\bar{G}_l}(P)|} \geq \beta_i^\sigma + \frac{(1-b_q)\operatorname{mult}_P q}{|I_{\bar{G}_l}(P)|} = \beta_i^\sigma - \sigma\delta\frac{\operatorname{mult}_P q}{r'|I_{\bar{G}_l}(P)|}.$$

It remains to compute $\operatorname{mult}_P q$. Let $N' := |\hat{G}_l|$ and $\Xi := x_l'^{N'} = \rho^{|w_l|N'}$. Then Ξ is $\hat{G}_l$-invariant, hence a rational function on the common quotient of W_i by G_i and of X by $\bar{G}_l$, and r' divides N', so $\Xi = q^{|w_l|N'/r'}$. Let $\bar{F}$ be the prime divisor over the common quotient under F and P. By

[Liu26, Lemma 2.6],

$$N' \operatorname{mult}_F x_l = \operatorname{mult}_F \Xi = e \operatorname{mult}_{\bar F} \Xi \qquad \text{and} \qquad \frac{N'|w_l|}{r'} \operatorname{mult}_P q = \operatorname{mult}_P \Xi = |I_{\bar G_l}(P)| \operatorname{mult}_{\bar F} \Xi,$$

where the first equality uses $\operatorname{mult}_F x_l' = \operatorname{mult}_F x_l$. Dividing the second equality by the first, we obtain

$$\frac{\operatorname{mult}_P q}{r'|I_{\bar G_l}(P)|} = \frac{[x_l]}{|w_l|} = \rho_F.$$

Together with (E.24), this proves (E.22).

Step 9. In this step we prove (3). Let $i \geq \max(i_0, i_1, i_2)$, and let F, e, and $[\cdot]$ be as in **Step 7**. Let $\sigma_0 := +$ if $\delta \leq 0$ and $\sigma_0 := -$ if $\delta > 0$. Then $-\sigma_0\delta = |\delta|$, and $\beta_i = \beta_i^{\sigma_0}$ by (2).

Assume that $[x_j] \geq \epsilon$ for any j with $\sigma_0 w_j < 0$. Then (E.22) with $\sigma = \sigma_0$ shows

$$\alpha_i = \frac{a(F, W_i, D_i)}{e} \geq \beta_i^{\sigma_0} + |\delta|\rho_F \geq \beta_i.$$

Assume that $[x_{j_0}] < \epsilon$ for some j_0 with $\sigma_0 w_{j_0} < 0$. For any j with $\sigma_0 w_j > 0$, (E.21) shows

$$|w_{j_0}|[x_j] \geq \frac{1}{e_0} - |w_j|[x_{j_0}] \geq \frac{1}{e_0} - |w|\epsilon \geq \frac{1}{2e_0},$$

so $[x_j] \geq 1/(2e_0|w|) \geq \epsilon$. Thus the assumption of **Step 8** holds for $\sigma = -\sigma_0$, so (E.22) holds with $\sigma = -\sigma_0$, and $\rho_F \leq [x_{j_0}]/|w_{j_0}| < \epsilon$. By (2) and $\epsilon \leq 1/Q$,

$$\alpha_i \geq \beta_i^{-\sigma_0} + \sigma_0\delta\rho_F \geq \beta_i^{-\sigma_0} - |\delta|\epsilon \geq \beta_i^{\sigma_0} + \frac{|\delta|}{Q} - |\delta|\epsilon \geq \beta_i.$$

Since i_0, i_1 and i_2 do not depend on F, $\alpha_i \geq \beta_i$ for i large. □

E.3. **Quadratic rank two, fixed split radical order: the plane construction and upper bounds.** In this subsection, we prove Proposition E.4 and Lemmas E.6 and E.7.

Proposition E.4 is the counterpart of Proposition E.3 for the circles of Lemma E.1(3)(b). For the circles of type (b), the negative graded quotient is a quotient of the plane, and the references (Y_i^+, B_i^+) and (Y_i^-, B_i^-) are related by the crepant identity of Proposition E.4(1).

Proposition E.4. *Notation as in Construction E.2, in case (b) of Lemma E.1(3).*

(1) Y_i^- is the quotient of $\mathbb{A}^2_{\zeta_1,\zeta_2}$ by a finite group $\bar G_i$ of order $p[G_i : G_i \cap \mathbf{T}]$ acting faithfully and diagonally and containing the image of the central $\mu_p \subset \mathbf{T}$, where $\zeta_1 = \xi^{w_z} z$ and $\zeta_2 = \xi^{w_t} t$ on the root chart $x = \xi^p$, and C_i^- is the image of the origin. The crepant pullback of $(Y_i^-, B_i^-(c'))$ to $\mathbb{A}^2$ is $(\mathbb{A}^2, \sum_k c_k' \operatorname{div}(P_{i,k}))$ with

$$P_{i,k} := h_{i,k}^0(1, -g_i^0(\zeta_1, \zeta_2), \zeta_1, \zeta_2).$$

There are a projective birational morphism $\pi_i : Y_i^+ \to Y_i^-$ with $\pi_i^{-1}(C_i^-) = C_i^+$ and a $\mathbb{Q}$-divisor $E_i \geq 0$ on Y_i^+ with support C_i^+ such that, for every $c' \in \mathbb{R}^s_{\geq 0}$,

$$K_{Y_i^+} + B_i^+(c') = \pi_i^*\big(K_{Y_i^-} + B_i^-(c')\big) - \Big(\sum_k c_k'\kappa_k - w_z - w_t\Big)E_i.$$

Moreover, if $L := \operatorname{lcm}(w_z, w_t, N+p)|\bar G_i|$ and $\mathfrak{a}_i$ is the ideal on Y_i^- generated by the images of ζ_1^{L/w_z}, ζ_2^{L/w_t}, and $g_i^0(\zeta_1, \zeta_2)^{L/(N+p)}$, then $\mathfrak{a}_i \mathcal{O}_{Y_i^+} = \mathcal{O}_{Y_i^+}(-LE_i)$.

(2) For i large, the pairs (Y_i^+, B_i^+) and (Y_i^-, B_i^-) are lc near C_i^+ and C_i^-. There exists a positive integer Q, independent of i, such that for i large, $\beta_i^- \geq \beta_i^+ + |\delta|/Q$ if $\delta < 0$, and $\beta_i^+ \geq \beta_i^-$ if $\delta \geq 0$.

(3) Let $\theta > 1$ and $\delta_\theta := \theta \sum_k c_k\kappa_k - w_z - w_t$. Assume that there exists $\sigma \in \{+, -\}$ with $\sigma\delta_\theta \leq 0$ such that the pair $(Y_i^\sigma, B_i^\sigma(\theta c))$ is lc near C_i^σ for any i. Then $\alpha_i \geq \beta_i$ for i large.

Proof. The proof is long, so we divide it into several steps. For $c' \in \mathbb{R}^s_{\geq 0}$ let

$$\delta(c') := \sum_k c'_k \kappa_k - w_z - w_t,$$

so that $\delta = \delta(c)$ and $\delta_\theta = \delta(\theta c)$. Let $w_{\max} := \max(N+p, w_z, w_t)$.

Step 1. In this step we fix coordinates and groups, and we describe $Y_i^{\pm}$ by root charts.

By Lemma E.1, $W_i = (xy + g_i(z,t) = 0) \subset \mathbb{A}^4$, the group G_i acts diagonally on x, y, z, t, $w = (-p, N+p, w_z, w_t)$ with $p, w_z, w_t > 0$ and $N = aw_z + bw_t$, the group $R_i := G_i \cap \mathbf{T}$ has order $r_i \to +\infty$, and $[G_i : R_i] \leq e_0$. The circle $\mathbf{T}$ acts by

$$\lambda \cdot (x,y,z,t) = (\lambda^{-p}x, \lambda^{N+p}y, \lambda^{w_z}z, \lambda^{w_t}t), \quad \lambda \in \mathbb{C}^*,$$

and, since w is primitive, $\lambda \to \lambda \cdot (-)$ is injective, so R_i is the image of μ_{r_i}. For $\sigma \in G_i$ let $\chi_x(\sigma), \chi_y(\sigma), \chi_z(\sigma), \chi_t(\sigma)$ be the scalars by which σ multiplies x, y, z, t, and define $\chi_x, \chi_y, \chi_z, \chi_t$ on $\mathbf{T}$ likewise. Since $xy + g_i$ is G_i-semi-invariant, the monomial $z^a t^b$ occurs in g_i, and $-p + (N+p) = aw_z + bw_t$, we have

$$\chi_x \chi_y = \chi_z^a \chi_t^b \quad \text{on } G_i \text{ and on } \mathbf{T}. \tag{E.26}$$

For a polynomial f and an integer m, let $f_{[m]}$ be the sum of the terms of f of $\mathbf{T}$-weight m. Since $R_i = \mu_{r_i}$ acts on a monomial of $\mathbf{T}$-weight m by λ^m, the $\mathbf{T}$-weights of the monomials of a G_i-semi-invariant polynomial are congruent modulo r_i. Since g_i is a polynomial in z, t and $w_z, w_t > 0$, for $r_i > N$ we obtain

$$g_i = g_i^0 + \sum_{l \geq 1} (g_i)_{[N + l r_i]}, \qquad h_{i,k} = h_{i,k}^0 + \sum_{l \neq 0} (h_{i,k})_{[\kappa_k + l r_i]}.$$

A monomial of $\mathbf{T}$-weight m with no factor xy has degree at least $|m|/w_{\max}$. Thus, for $r_i \geq 2 \max_k |\kappa_k|$, each monomial of $g_i - g_i^0$ and of $h_{i,k} - h_{i,k}^0$ has degree at least $r_i/(2w_{\max})$. The polynomial $h_{i,k}^0$ has no monomial divisible by xy, and it is non-zero. Division by $xy + g_i^0$ with respect to the monomial xy shows that a non-zero polynomial with no monomial divisible by xy does not vanish on V_i. Thus $h_{i,k}^0|_{V_i} \neq 0$. The hypersurface V_i is normal, since its singular locus is contained in the closed subset $(x = y = 0) \cap (g_i^0 = \partial_z g_i^0 = \partial_t g_i^0 = 0)$ of $\mathbb{A}^4$, whose dimension is at most one as $g_i^0 \neq 0$ [Har77, Proposition II.8.23(b)].

The projection $(x,y,z,t) \to (x,z,t)$ restricts to G_i-equivariant birational maps from W_i and from V_i onto $\mathbb{A}^3_{x,z,t}$, where G_i acts on $\mathbb{A}^3_{x,z,t}$ through χ_x, χ_z, χ_t, and the inverse maps are given by $y = -g_i/x$ and $y = -g_i^0/x$ respectively. Let Ω be the rational 3-form on $\mathbb{A}^3_{x,z,t}$, and hence on W_i and V_i, given by

$$\Omega := \frac{dx \wedge dz \wedge dt}{x}.$$

Since $\partial(xy + g_i)/\partial y = \partial(xy + g_i^0)/\partial y = x$, the form Ω is the residue of $dx \wedge dy \wedge dz \wedge dt/(xy + g_i)$ on W_i and of $dx \wedge dy \wedge dz \wedge dt/(xy + g_i^0)$ on V_i. Thus Ω generates the canonical sheaves of the hypersurfaces W_i and V_i near 0.

The x-root chart. Let $\mathbb{A}^3$ have coordinates ξ, ζ_1, ζ_2, and let

$$\varphi_x : \mathbb{A}^3 \backslash (\xi = 0) \to V_i, \quad (\xi, \zeta_1, \zeta_2) \to \big(\xi^p, -\xi^{-N-p} g_i^0(\zeta_1, \zeta_2), \xi^{-w_z}\zeta_1, \xi^{-w_t}\zeta_2\big).$$

Since g_i^0 has $\mathbf{T}$-weight N,

$$g_i^0(\xi^{-w_z}\zeta_1, \xi^{-w_t}\zeta_2) = \xi^{-N} g_i^0(\zeta_1, \zeta_2),$$

so the image of φ_x is contained in V_i. On V_i we have $x = \xi^p$, $\zeta_1 = \xi^{w_z} z$, and $\zeta_2 = \xi^{w_t} t$, and φ_x is the quotient of $\mathbb{A}^3 \backslash (\xi = 0)$ onto $V_i \cap (x \neq 0)$ by μ_p acting by

$$\epsilon \cdot (\xi, \zeta_1, \zeta_2) = (\epsilon\xi, \epsilon^{w_z}\zeta_1, \epsilon^{w_t}\zeta_2).$$

Applying the homogeneity $f_{[m]}(\lambda\cdot v)=\lambda^m f_{[m]}(v)$ with $\lambda=\xi$ at $v=\varphi_x(\xi,\zeta_1,\zeta_2)$, we obtain, for any polynomial f and any m,

$$f_{[m]}\circ\varphi_x=\xi^{-m}f_{[m]}\big(1,-g_i^0(\zeta_1,\zeta_2),\zeta_1,\zeta_2\big),\quad \text{in particular}\quad h^0_{i,k}\circ\varphi_x=\xi^{-\kappa_k}P_{i,k}(\zeta_1,\zeta_2), \tag{E.27}$$

with $P_{i,k}$ as in (1). Since $h^0_{i,k}|_{V_i}\neq 0$, $P_{i,k}\neq 0$. Since $dx/x=pd\xi/\xi$, and $dz\wedge dt\equiv\xi^{-w_z-w_t}d\zeta_1\wedge d\zeta_2$ modulo $d\xi$,

$$\varphi_x^*\Omega=p\xi^{-w_z-w_t-1}d\xi\wedge d\zeta_1\wedge d\zeta_2. \tag{E.28}$$

Let $\hat G_x$ be the group of pairs $(\sigma,\epsilon)\in G_i\times\mathbb{C}^*$ with $\epsilon^p=\chi_x(\sigma)$, acting on $\mathbb{A}^3$ by

$$(\sigma,\epsilon)\cdot(\xi,\zeta_1,\zeta_2)=\big(\epsilon\xi,\epsilon^{w_z}\chi_z(\sigma)\zeta_1,\epsilon^{w_t}\chi_t(\sigma)\zeta_2\big).$$

By (E.26) and the homogeneity of g_i^0, φ_x is equivariant for $\hat G_x\to G_i$, $(\sigma,\epsilon)\to\sigma$, and $|\hat G_x|=p|G_i|$. The group $\hat G_x$ acts faithfully and diagonally on $\mathbb{A}^3$. An element (σ,ϵ) acts trivially on ζ_1,ζ_2 if and only if $\chi_x(\sigma)=\epsilon^p$, $\chi_z(\sigma)=\epsilon^{-w_z}$, and $\chi_t(\sigma)=\epsilon^{-w_t}$, that is, by (E.26), if and only if σ is the element ϵ^{-1} of $\mathbf{T}$. Thus the subgroup $\hat R_x\subset\hat G_x$ of elements acting on ξ alone is isomorphic to R_i, and it acts on ξ through μ_{r_i}. Let

$$\bar G_i:=\hat G_x/\hat R_x,\qquad |\bar G_i|=p|G_i|/r_i=p[G_i:R_i].$$

Then $\bar G_i$ acts faithfully and diagonally on $\mathbb{A}^2_{\zeta_1,\zeta_2}$. The elements $(1,\epsilon)$ of $\hat G_x$ form the group μ_p of the quotient φ_x, and they act on ζ_1,ζ_2 as the central $\mu_p\subset\mathbf{T}$, the kernel of χ_x on $\mathbf{T}$, acts on z,t. Since $(1,\epsilon)\in\hat R_x$ only for $\epsilon=1$, the image of the central $\mu_p\subset\mathbf{T}$ in $\bar G_i$ is a subgroup of order p, and it acts faithfully on the plane, that is, $\gcd(p,w_z,w_t)=1$. The equality $\gcd(p,w_z,w_t)=1$ also follows from the primitivity of w, since a common divisor of p,w_z,w_t divides $N=aw_z+bw_t$ and $N+p$.

The positive root charts. Let $u\in\{y,z,t\}$, and let $\mathbb{A}^4$ have coordinates ρ and x'_j for the three coordinates $x_j\in\{x,y,z,t\}\backslash\{u\}$. Let

$$\psi_u:\mathbb{A}^4\backslash(\rho=0)\to\mathbb{A}^4,\quad u=\rho^{w_u},\quad x_j=\rho^{w_j}x'_j,$$

where w_j is the entry of w at x_j, so that $x=\rho^{-p}x'$. For any polynomial f and any m,

$$f_{[m]}\circ\psi_u=\rho^m f_{[m]}|_{u=1}(x'), \tag{E.29}$$

where $f_{[m]}|_{u=1}(x')$ denotes the result of substituting 1 for u and x'_j for x_j in $f_{[m]}$. Let $S_u\subset\mathbb{A}^3_{x'}$ be the surface defined by $(xy+g_i^0)|_{u=1}$ (the slice $S_{i,u}$ of Construction E.2, written without the index i in this proof), that is,

$$S_y=(x'+g_i^0(z',t')=0),\qquad S_z=(x'y'+g_i^0(1,t')=0),\qquad S_t=(x'y'+g_i^0(z',1)=0).$$

The surface S_y is the graph of $-g_i^0$, hence smooth. Since $g_i^0(1,t')$ and $g_i^0(z',1)$ are non-zero, S_z and S_t are normal surfaces whose singular points are the points $x'=y'=0$ at the multiple roots of $g_i^0(1,t')$, resp. $g_i^0(z',1)$, where the singularities are of type A, hence canonical [KM98, Theorem 4.20], in particular klt. Thus S_u is a normal klt complete intersection in $\mathbb{A}^3$.

Let $\hat G_u$ be the group of pairs $(\sigma,\epsilon)\in G_i\times\mathbb{C}^*$ with $\epsilon^{w_u}=\chi_u(\sigma)$, acting on $\mathbb{A}^4$ by $\rho\to\epsilon\rho$ and $x'_j\to\epsilon^{-w_j}\chi_{x_j}(\sigma)x'_j$. Then ψ_u is equivariant for $\hat G_u\to G_i$, $|\hat G_u|=w_u|G_i|$, and $\hat G_u$ acts faithfully and diagonally. As for $\hat G_x$, an element (σ,ϵ) acts trivially on the x'_j if and only if σ is the element ϵ of $\mathbf{T}$, so the subgroup $\hat R_u\subset\hat G_u$ of elements acting on ρ alone is isomorphic to R_i and acts on ρ through μ_{r_i}. Let $\bar G_u:=\hat G_u/\hat R_u$. It acts faithfully and diagonally on $\mathbb{A}^3_{x'}$, preserves S_u, and acts faithfully on S_u, since S_u is not contained in a coordinate hyperplane.

The graded quotients. The coordinate x is the only coordinate of negative weight. Thus the negative graded quotient of V_i by $\mathbf{T}$ is covered by the root chart at x, and, by (E.27), the crepant boundary of $\sum_k c'_k\operatorname{div}(h^0_{i,k}|_{V_i})$ on the chart is $\sum_k c'_k\operatorname{div}(P_{i,k})$ on the plane $\mathbb{A}^2_{\zeta_1,\zeta_2}$. Dividing by

$\bar{G}_i = \hat{G}_x/\hat{R}_x$, we obtain that $Y_i^- = \mathbb{A}^2_{\zeta_1,\zeta_2}/\bar{G}_i$, that $B_i^-(c')$ is the boundary of the crepant quotient by $\bar{G}_i$ of the pair

$$(\mathbb{A}^2_{\zeta_1,\zeta_2}, \sum_k c'_k \operatorname{div}(P_{i,k})),$$

and that C_i^- is the image of the origin. Similarly, by (E.29) and dividing by $\bar{G}_u$, Y_i^+ is covered by the three open subsets $S_u/\bar{G}_u$, $u \in \{y,z,t\}$, and $B_i^+(c')$ restricted to $S_u/\bar{G}_u$ is the boundary of the crepant quotient of $(S_u, \sum_k c'_k \operatorname{div}(h^0_{i,k}|_{u=1}|_{S_u}))$ by $\bar{G}_u$. A $\mathbf{T}$-orbit in $V_i\backslash\{0\}$ with $(y,z,t) \neq 0$ has 0 in the closure of the orbit if and only if $x = 0$ on the orbit. Thus C_i^+ is covered by the images of the curves $S_u \cap (x' = 0)$.

Let A_i be the ring of $\mathbf{T}$-invariant regular functions on V_i. A μ_p-invariant monomial $\zeta_1^{\gamma_1}\zeta_2^{\gamma_2}$ satisfies $p \mid w_z\gamma_1 + w_t\gamma_2$, and it is the restriction to the slice $x = 1$ of

$$x^{(w_z\gamma_1+w_t\gamma_2)/p} z^{\gamma_1} t^{\gamma_2} \in A_i.$$

Restriction to the slice $x = 1$ is injective on A_i, since $\mathbf{T} \cdot (V_i \cap (x = 1)) = V_i \cap (x \neq 0)$ is dense in V_i and a $\mathbf{T}$-invariant function is determined by its values on the slice. Together with the previous display, it gives an isomorphism $A_i \cong \mathbb{C}[\zeta_1,\zeta_2]^{\mu_p}$, compatible with the actions of $\hat{G}_x$ through $\hat{G}_x \to G_i$ on A_i and through $\bar{G}_i = \hat{G}_x/\hat{R}_x$ on $\mathbb{C}[\zeta_1,\zeta_2]$, since restriction to the slice takes the action of $\sigma \in G_i$ to the action of $(\sigma,\epsilon) \in \hat{G}_x$. Taking invariants, $A_i^{G_i} \cong \mathbb{C}[\zeta_1,\zeta_2]^{\bar{G}_i}$, so $Y_i^- = \mathbb{A}^2/\bar{G}_i \cong \operatorname{Spec} A_i^{G_i}$. Since Y_i^+ is projective over $\operatorname{Spec} A_i^{G_i}$, we obtain a projective morphism $\pi_i : Y_i^+ \to Y_i^-$. A $\mathbf{T}$-orbit with $x \neq 0$ and $(y,z,t) \neq 0$ defines a point of both Y_i^+ and Y_i^-, so π_i is an isomorphism over $Y_i^-\backslash C_i^-$, and π_i is birational. Each non-constant monomial in A_i is divisible by x, since x is the only coordinate of negative weight. Thus $\pi_i(C_i^+) = C_i^-$. A point of $Y_i^+\backslash C_i^+$ is given by an orbit with $x \neq 0$ and $(y,z,t) \neq 0$, hence by a point of the slice $x = 1$ with $(\zeta_1,\zeta_2) \neq 0$, since $\zeta_1 = \zeta_2 = 0$ implies $y = -g_i^0(0,0) = 0$ there. Thus $\pi_i^{-1}(C_i^-) = C_i^+$.

Step 2. In this step we prove (1).

The first two sentences of (1) are proved in **Step 1**: the group $\bar{G}_i$, the order $|\bar{G}_i|$, and the image of the central $\mu_p \subset \mathbf{T}$ in $\bar{G}_i$ are in the paragraph on the x-root chart, and $Y_i^- = \mathbb{A}^2_{\zeta_1,\zeta_2}/\bar{G}_i$, C_i^-, and the crepant pullback of $(Y_i^-, B_i^-(c'))$ to $\mathbb{A}^2_{\zeta_1,\zeta_2}$ are in the paragraph on the graded quotients. The morphism π_i and the equality $\pi_i^{-1}(C_i^-) = C_i^+$ are also proved in **Step 1**. Let E_i be the $\mathbb{Q}$-divisor on Y_i^+ whose pullback to S_u is $\frac{1}{p}\operatorname{div}(x'|_{S_u})$ for any $u \in \{y,z,t\}$. It is well defined: $x'|_{S_u}$ is $\bar{G}_u$-semi-invariant, so a power of it is $\bar{G}_u$-invariant, and on the intersection of the charts at u and u' the function x' of the chart at u and the function x' of the chart at u' differ by the factor $u'^{p/w_{u'}}$, which is a unit there. Thus $E_i \geq 0$, and the support of E_i is C_i^+.

Fix $u \in \{y,z,t\}$ and $c' \in \mathbb{R}^s_{\geq 0}$. On the slice $V_i \cap (u = 1) = S_u$ we have $x' = x$. Let $z' = 1$ if $u = z$, $t' = 1$ if $u = t$, and $y' = 1$ if $u = y$. Let χ be the character of $\bar{G}_u$ with $g^*x' = \chi(g)x'$ for $g \in \bar{G}_u$. Let S' be an irreducible component of the surface $(\xi^p = x') \subset \mathbb{A}^1_\xi \times S_u$, let $\tilde{S}$ be the normalization of S', and let $\mathfrak{G}$ be the group of the pairs (g,η) with $g \in \bar{G}_u$ and $\eta^p = \chi(g)$ which preserve S', where (g,η) acts by $\xi \to \eta\xi$ and by g on S_u. The surface $(\xi^p = x')$ is finite and flat of degree p over S_u. Thus every irreducible component of $(\xi^p = x')$ maps onto S_u, and the elements $(1,\eta)$ with $\eta^p = 1$ permute the irreducible components of $(\xi^p = x')$ transitively. Thus, for $g \in \bar{G}_u$ and η_0 with $\eta_0^p = \chi(g)$, some $(1,\eta_1)$ maps $(g,\eta_0)S'$ back to S', so $\mathfrak{G} \to \bar{G}_u$ is surjective, with kernel the stabilizer $\mu_{S'} \subset \mu_p$ of S'. The number of components is $p/|\mu_{S'}|$, so $S' \to S_u$ has degree $|\mu_{S'}|$, and $\mu_{S'}$ acts faithfully on S'. Hence $\tilde{S}/\mu_{S'} \to S_u$ is finite and birational onto the normal surface S_u, hence an isomorphism, and $\tilde{S}/\mathfrak{G} \cong S_u/\bar{G}_u$. The morphism $\tilde{S} \to \mathbb{A}^2_{\zeta_1,\zeta_2}$ given by $\zeta_1 = \xi^{w_z}z'$ and $\zeta_2 = \xi^{w_t}t'$ is $\mathfrak{G}$-equivariant for the homomorphism $\mathfrak{G} \to \bar{G}_i$ which sends $((\sigma,\epsilon)\hat{R}_u, \eta)$ to $(\sigma, \eta\epsilon^{-1})\hat{R}_x$, and it induces a birational map $\tilde{S}/\mathfrak{G} \dashrightarrow Y_i^-$, since, by **Step 1**, both $S_u/\bar{G}_u$ and Y_i^- are birational to the quotient of $V_i \cap (x \neq 0) \cap (u \neq 0)$ by G_i and $\mathbf{T}$. Let $\tilde{\Pi}$ be the normalization of $\mathbb{A}^2_{\zeta_1,\zeta_2}$ in $\tilde{S}$. Let ω_u be the residue of $dx'_{j_1} \wedge dx'_{j_2} \wedge dx'_{j_3}/(xy + g_i^0)|_{u=1}$, a

generator of ω_{S_u}. On each of the three slices,

$$d\zeta_1 \wedge d\zeta_2 = \pm\frac{w_u}{p}\xi^{w_z+w_t}\omega_u \quad \text{on } \tilde{S}.$$

On S_y, we have $\omega_y = \pm dz'\wedge dt'$ and $d\xi/\xi = \frac{1}{p}dg_i^0/g_i^0$, so, by the Euler identity $w_z z\partial_z g_i^0 + w_t t\partial_t g_i^0 = Ng_i^0$,

$$d\zeta_1 \wedge d\zeta_2 = \xi^{w_z+w_t}\Big(1 + \frac{w_z z'\partial_z g_i^0 + w_t t'\partial_t g_i^0}{pg_i^0}\Big)dz' \wedge dt' = \frac{N+p}{p}\xi^{w_z+w_t}dz' \wedge dt'.$$

On S_z we have $z' = 1$, $\omega_z = \pm dx' \wedge dt'/x'$, since $\partial(x'y' + g_i^0(1,t'))/\partial y' = x'$, and $d\xi/\xi = \frac{1}{p}dx'/x'$, so

$$d\zeta_1 \wedge d\zeta_2 = w_z\xi^{w_z+w_t}\frac{d\xi}{\xi} \wedge dt' = \frac{w_z}{p}\xi^{w_z+w_t}\frac{dx' \wedge dt'}{x'} = \pm\frac{w_z}{p}\xi^{w_z+w_t}\omega_z.$$

On S_t, exchanging z and t in the calculation on S_z gives $d\zeta_1 \wedge d\zeta_2 = \pm\frac{w_t}{p}\xi^{w_z+w_t}\omega_t$. By (E.27) applied at the points of S_u with $x' \neq 0$, $P_{i,k}(\zeta_1,\zeta_2) = \xi^{\kappa_k}h_{i,k}^0|_{u=1}$. Let

$$\sigma^- := d\zeta_1 \wedge d\zeta_2 \cdot \prod_k P_{i,k}^{-c'_k}, \qquad \sigma^+ := \omega_u \cdot \prod_k \big(h_{i,k}^0|_{u=1}\big)^{-c'_k},$$

multivalued rational 2-forms on $\mathbb{A}^2_{\zeta_1,\zeta_2}$ and on S_u. Then, on $\tilde{S}$,

$$\sigma^- = \pm\frac{w_u}{p}\xi^{-\delta(c')}\sigma^+. \tag{E.30}$$

Then $\tilde{S}/\mathfrak{G}$ is the chart $S_u/\bar{G}_u$ of Y_i^+, and $\tilde{\Pi}/\mathfrak{G} = Y_i^-$, since $\tilde{\Pi}/\mathfrak{G} \to \mathbb{A}^2/\bar{G}_i$ is finite and birational onto a normal variety. The crepant pullbacks $B_{\tilde{S}}$ and $B_{\tilde{\Pi}}$ of $(Y_i^+, B_i^+(c'))$ and $(Y_i^-, B_i^-(c'))$ satisfy $B_{\tilde{S}} = -\operatorname{div}(\sigma^+)$ and $B_{\tilde{\Pi}} = -\operatorname{div}(\sigma^-)$, where σ^+ and σ^- are pulled back to $\tilde{S}$ and $\tilde{\Pi}$, by **Step 1**, since ω_u and $d\zeta_1 \wedge d\zeta_2$ generate ω_{S_u} and $\omega_{\mathbb{A}^2}$, and by the Hurwitz formula (cf. [KM98, Proposition 5.20]) for the finite morphisms $\tilde{S} \to S_u$ and $\tilde{\Pi} \to \mathbb{A}^2_{\zeta_1,\zeta_2}$. Let Γ be a prime divisor over Y_i^+ whose center intersects the chart $S_u/\bar{G}_u$, and let $\tilde{\Gamma}$ be a prime divisor over $\tilde{S}$, and hence over $\tilde{\Pi}$, over Γ. By the definition of log discrepancies and (E.30),

$$a(\tilde{\Gamma}, \tilde{\Pi}, B_{\tilde{\Pi}}) = 1 + \operatorname{mult}_{\tilde{\Gamma}}\sigma^- = 1 + \operatorname{mult}_{\tilde{\Gamma}}\sigma^+ - \delta(c')\operatorname{mult}_{\tilde{\Gamma}}\xi = a(\tilde{\Gamma}, \tilde{S}, B_{\tilde{S}}) - \delta(c')\operatorname{mult}_{\tilde{\Gamma}}\xi.$$

Dividing by $|I_{\mathfrak{G}}(\tilde{\Gamma})|$, [Liu26, Lemma 2.6(2)] implies

$$a(\Gamma, Y_i^-, B_i^-(c')) = a(\Gamma, Y_i^+, B_i^+(c')) - \delta(c')\operatorname{mult}_\Gamma E_i, \tag{E.31}$$

since $\operatorname{mult}_{\tilde{\Gamma}}\xi = \frac{1}{p}\operatorname{mult}_{\tilde{\Gamma}}x'$ and the pullback of E_i to $\tilde{S}$ is $\frac{1}{p}\operatorname{div}(x')$. We apply (E.31) to the prime divisors Γ on Y_i^+. The coefficient of Γ in

$$K_{Y_i^+} + B_i^+(c') - \pi_i^*\big(K_{Y_i^-} + B_i^-(c')\big) + \delta(c')E_i$$

is

$$a(\Gamma, Y_i^-, B_i^-(c')) - a(\Gamma, Y_i^+, B_i^+(c')) + \delta(c')\operatorname{mult}_\Gamma E_i = 0.$$

This proves the equality of (1). Since the three charts cover Y_i^+, (E.31) holds for any prime divisor Γ over Y_i^+ and any $c' \in \mathbb{R}^s_{\geq 0}$.

We prove the last sentence of (1). Let $M_1 := L/w_z$, $M_2 := L/w_t$, and $M_3 := L/(N+p)$. By the definition of $\hat{G}_x$ and the semi-invariance and the homogeneity of g_i^0, the group $\bar{G}_i$ multiplies ζ_1, ζ_2, and $g_i^0(\zeta_1,\zeta_2)$ by characters of $\bar{G}_i$. Their orders divide $|\bar{G}_i|$, and $|\bar{G}_i|$ divides M_1, M_2, and M_3. Thus $\zeta_1^{M_1}$, $\zeta_2^{M_2}$, and $g_i^0(\zeta_1,\zeta_2)^{M_3}$ are $\bar{G}_i$-invariant, and they induce regular functions on Y_i^-. Fix $u \in \{y,z,t\}$. The composition of $\tilde{S} \to \mathbb{A}^2_{\zeta_1,\zeta_2}$ and the quotient morphism to Y_i^- coincides with the composition of $\tilde{S} \to S_u/\bar{G}_u \subset Y_i^+$ and π_i, since, by **Step 1**, both send a point $v \in \tilde{S}$ with $\xi(v) \neq 0$ to the point of Y_i^- given by the $\mathbf{T}$-orbit of the image of v in $S_u \subset V_i$, $\tilde{S}$ is reduced, and Y_i^- is separated. Since p divides $|\bar{G}_i|$, L/p is an integer, and $\xi^L = x'^{L/p}$. By the

homogeneity of g_i^0, $g_i^0(\zeta_1,\zeta_2)=\xi^N g_i^0(z',t')$, and $g_i^0(z',t')=-x'y'$ on S_u. Thus the pullbacks of the three functions to $\tilde{S}$ are

$$\zeta_1^{M_1}=x'^{L/p}z'^{M_1},\qquad \zeta_2^{M_2}=x'^{L/p}t'^{M_2},\qquad g_i^0(\zeta_1,\zeta_2)^{M_3}=(-1)^{M_3}x'^{L/p}y'^{M_3}.$$

The order of χ divides $|\bar{G}_u|=w_u[G_i:R_i]$, which divides $L/p=\operatorname{lcm}(w_z,w_t,N+p)[G_i:R_i]$. Thus $\phi:=x'^{L/p}$ is a $\bar{G}_u$-invariant regular function on S_u, and $\operatorname{div}(\phi)=LE_i$ on $S_u/\bar{G}_u$, since the pullbacks to S_u of both divisors are $\frac{L}{p}\operatorname{div}(x'|_{S_u})$. The functions z'^{M_1}, t'^{M_2}, and $(-1)^{M_3}y'^{M_3}$ on S_u are regular, they are $\bar{G}_u$-invariant as quotients of $\bar{G}_u$-invariant functions by ϕ, and one of them is ±1. Thus, on $S_u/\bar{G}_u$,

$$\mathfrak{a}_i\mathcal{O}_{Y_i^+}=\phi\mathcal{O}_{Y_i^+}=\mathcal{O}_{Y_i^+}(-LE_i).$$

Since the three charts cover Y_i^+, $\mathfrak{a}_i\mathcal{O}_{Y_i^+}=\mathcal{O}_{Y_i^+}(-LE_i)$. This proves (1).

Step 3. In this step we prove (2).

By Lemma E.1 and Construction-Theorem 4.13(1), the pair (W_i,D_i) is klt near 0. Since W_i is a normal Cartier divisor on $\mathbb{A}^4$ and $h_{i,k}|_{W_i}\neq0$ for any k, [Kaw07, Theorem] implies that $(\mathbb{A}^4,W_i+\sum_k c_k\operatorname{div}(h_{i,k}))$ is lc near 0. Let L' be the integer of Lemma 4.14 for $n=4$, $s+1$ functions and the finite set $\{1,c_1,\dots,c_s\}$. By **Step 1**, $(xy+g_i)-(xy+g_i^0)$ and $h_{i,k}-h_{i,k}^0$ belong to $(x,y,z,t)^{L'}$ when $r_i\geq 2w_{\max}L'$. A pair on $\mathbb{A}^4$ whose boundary is a combination of divisors of polynomials is lc near 0 if and only if its pullback to $\operatorname{Spec}\mathbb{C}[[x,y,z,t]]$ is lc [dFEM11, Proposition 2.11]. Thus Lemma 4.14 implies that $(\mathbb{A}^4,V_i+\sum_k c_k\operatorname{div}(h_{i,k}^0))$ is lc near 0 for i large.

Fix such i. The pair $(\mathbb{A}^4,V_i+\sum_k c_k\operatorname{div}(h_{i,k}^0))$ is **T**-invariant, so the locus where it is not lc is a closed **T**-invariant subset of $\mathbb{A}^4$ not containing 0. Thus the non-lc locus contains no point whose **T**-orbit has 0 in the closure of the orbit, in particular no point of $(x=0)$ and no point of the x-axis. By [Kaw07, Theorem], $(V_i,\sum_k c_k\operatorname{div}(h_{i,k}^0|_{V_i}))$ is lc near $V_i\cap(x=0)$ and near the x-axis. The morphism φ_x is the quotient of $\mathbb{A}^3\backslash(\xi=0)$ by μ_p acting freely, and, by (E.27), the pullback of $\sum_k c_k\operatorname{div}(h_{i,k}^0|_{V_i})$ is $\sum_k c_k\operatorname{div}(P_{i,k})$ on $(\mathbb{A}^1_\xi\backslash\{0\})\times\mathbb{A}^2_{\zeta_1,\zeta_2}$. The x-axis corresponds to $\zeta_1=\zeta_2=0$. Thus the pair

$$(\mathbb{A}^2_{\zeta_1,\zeta_2},\sum_k c_k\operatorname{div}(P_{i,k}))$$

is lc near 0. Similarly, for $u\in\{y,z,t\}$, ψ_u maps $(\mathbb{A}^1_\rho\backslash\{0\})\times S_u$ onto $V_i\cap(u\neq0)$ as the quotient by the elements $(1,\epsilon)$ of $\hat{G}_u$, which act freely, and, by (E.29), the pullback of the boundary is the pullback of $\sum_k c_k\operatorname{div}(h_{i,k}^0|_{u=1}|_{S_u})$ by the projection to S_u. Thus $(S_u,\sum_k c_k\operatorname{div}(h_{i,k}^0|_{u=1}|_{S_u}))$ is lc near $S_u\cap(x'=0)$. By **Step 1** and [Liu26, Lemma 2.6(2)(3)], (Y_i^-,B_i^-) is lc near C_i^- and (Y_i^+,B_i^+) is lc near C_i^+. Thus both pairs are lc for i large, which proves the first sentence of (2).

Let

$$Q:=pe_0w_{\max}.$$

Let Γ be a prime divisor over Y_i^+ with center in C_i^+. Choose $u\in\{y,z,t\}$ such that the center of Γ intersects the chart $S_u/\bar{G}_u$, and let Γ' be a prime divisor over S_u over Γ. Since x' vanishes on the center of Γ', $\operatorname{mult}_{\Gamma'}x'\geq1$, and since $|\bar{G}_u|=w_u[G_i:R_i]\leq w_{\max}e_0$, [Liu26, Lemma 2.6(1)] implies

$$\operatorname{mult}_\Gamma E_i=\frac{\operatorname{mult}_{\Gamma'}x'}{p|I_{\bar{G}_u}(\Gamma')|}\geq\frac{1}{Q}.$$

Since $\pi_i^{-1}(C_i^-)=C_i^+$ and π_i is birational, the prime divisors over Y_i^- with center C_i^- are the prime divisors over Y_i^+ with center in C_i^+. If $\delta<0$, then (E.31) with $c'=c$ implies

$$a(\Gamma,Y_i^-,B_i^-)=a(\Gamma,Y_i^+,B_i^+)+|\delta|\operatorname{mult}_\Gamma E_i\geq\beta_i^++\frac{|\delta|}{Q}$$

for any such Γ, hence $\beta_i^-\geq\beta_i^++|\delta|/Q$. If $\delta\geq0$, then (E.31) implies

$$a(\Gamma,Y_i^+,B_i^+)=a(\Gamma,Y_i^-,B_i^-)+\delta\operatorname{mult}_\Gamma E_i\geq\beta_i^-,$$

hence $\beta_i^+ \geq \beta_i^-$. This proves (2).

In particular, for i large,

$$\beta_i = \beta_i^+ \text{ if } \delta < 0, \qquad \beta_i = \beta_i^- \text{ if } \delta \geq 0, \qquad \beta_i \leq \beta_i^- \leq 2. \tag{E.32}$$

For the last inequality, let F_1 be the exceptional divisor of the blow-up of $\mathbb{A}^2_{\zeta_1,\zeta_2}$ at 0. Then

$$a(F_1, \mathbb{A}^2, \sum_k c_k \operatorname{div}(P_{i,k})) \leq a(F_1, \mathbb{A}^2, 0) = 2,$$

so the prime divisor over Y_i^- under F_1 has center C_i^- and log discrepancy at most 2 by [Liu26, Lemma 2.6(2)].

Step 4. In this step we compute log discrepancies on the root charts.

Let $U \in \{W_i, V_i\}$, let F be a prime divisor over U with center 0, and let $e := |I_{G_i}(F)|$. For a G_i-semi-invariant rational function f on U, let

$$[f] := \frac{\operatorname{mult}_F f}{e}.$$

More generally, let X be a normal variety with a faithful action of a finite group H, and let $X \dashrightarrow U$ be an H-equivariant dominant rational map for a homomorphism $H \to G_i$ which induces a birational map $X/H \dashrightarrow U/G_i$. For a prime divisor P over X whose image on a higher model of U/G_i is the image of F, and for an H-semi-invariant rational function f on X, let $[f] := \operatorname{mult}_P f/|I_H(P)|$. By [Liu26, Lemma 2.6(1)], $[f]$ does not depend on the choice of P, and $[f] = \operatorname{mult}_F f/e$ if f is the pullback of a rational function on U. Moreover, if $H' \subset H$ is a normal subgroup and P' is the image of P over X/H', then $\operatorname{mult}_{P'} f/|I_{H/H'}(P')| = [f]$ for an H-semi-invariant rational function f on X/H' by [Liu26, Lemma 2.6(1)(5)].

We use the following Hurwitz formula. Let ϖ be a multivalued rational 3-form on U, let $\Delta_U := -\operatorname{div}(\varpi)$ and $\Delta_X := -\operatorname{div}(\varpi_X)$, where ϖ_X is the pullback of ϖ to X, so that $a(E, U, \Delta_U) = 1 + \operatorname{mult}_E \varpi$ and $a(E', X, \Delta_X) = 1 + \operatorname{mult}_{E'} \varpi_X$ for any prime divisors E over U and E' over X. If P is a prime divisor over X whose image over U is a prime divisor F_P over U, with ramification index e_P, then

$$a(P, X, \Delta_X) = e_P\big(1 + \operatorname{mult}_{F_P} \varpi\big) = e_P a(F_P, U, \Delta_U). \tag{E.33}$$

To prove (E.33), let $\bar{U}$ be a higher model of U on which F_P is a divisor and $\bar{X}$ the normalization of $\bar{U}$ in the function field of X. Then P is a divisor on $\bar{X}$, since $\bar{X} \to \bar{U}$ is finite and $\bar{X}$ is normal. At the generic point of P, let s be a local parameter along P, t a local parameter along F_P, and y_2, y_3 regular functions on $\bar{U}$ at the generic point of F_P whose restrictions form a separating transcendence basis of the residue field of F_P, hence of the residue field of P. Then $t = vs^{e_P}$ with v a unit, and

$$dt \wedge dy_2 \wedge dy_3 = s^{e_P - 1}(e_P v + sg) ds \wedge dy_2 \wedge dy_3$$

with g regular, and $ds \wedge dy_2 \wedge dy_3$ generates the canonical sheaf of $\bar{X}$ at the generic point of P, and likewise $dt \wedge dy_2 \wedge dy_3$ generates the canonical sheaf of $\bar{U}$ at the generic point of F_P. Thus mult_P of the pullback of ϖ is $e_P \operatorname{mult}_{F_P} \varpi + e_P - 1$, which gives (E.33).

Claim E.5. *Let $U \in \{W_i, V_i\}$, and let $f_U := xy + g_i$ if $U = W_i$ and $f_U := xy + g_i^0$ if $U = V_i$. Let $f_1, \dots, f_s$ be G_i-semi-invariant polynomials such that $f_k|_U \neq 0$, the $\mathbf{T}$-weights of the monomials of f_k belong to $\kappa_k + r_i\mathbb{Z}$, and $(f_k)_{[\kappa_k]} = h_{i,k}^0$. Let $c' \in \mathbb{R}^s_{\geq 0}$, and let F be a prime divisor over U with center 0 and $e := |I_{G_i}(F)|$. Assume that $r_i > N$.*

(1) Assume that no monomial of any f_k has weight less than κ_k. Let $u \in \{y, z, t\}$ with $[u]/w_u = \min([y]/(N+p), [z]/w_z, [t]/w_t)$, and let $q := \rho^{r_i}$. Then $f_U \circ \psi_u = \rho^N f'_U$ and $f_k \circ \psi_u = \rho^{\kappa_k} f'_k$ for polynomials f'_U, f'_k in q and x'. The hypersurface $X := (f'_U = 0) \subset \mathbb{A}^4_{q,x'}$ is normal and klt near $S_0 := X \cap (q = 0)$, and $S_0 = S_u$. The group $\bar{G}_u$ acts diagonally on $\mathbb{A}^4_{q,x'}$ and faithfully on X and on S_0, and $f'_k|_{S_0} = h_{i,k}^0|_{u=1}|_{S_u}$. There exists

a prime divisor P over X with center in $S_0 \cap (x' = 0)$, whose image on a higher model of U/G_i is the image of F, such that

$$\frac{a(F, U, \sum_k c'_k \operatorname{div}(f_k))}{e} = \frac{a\big(P, X, (1 + \delta(c')/r_i)S_0 + \sum_k c'_k \operatorname{div}(f'_k|_X)\big)}{|I_{\bar{G}_u}(P)|}.$$

Moreover, $[q] = r_i[u]/w_u$ and $[x'] = [x] + p[u]/w_u$.

(2) *Assume that $U = V_i$ and that no monomial of any f_k has weight greater than κ_k. Let $q := \xi^{r_i}$. Then $f_k \circ \varphi_x = \xi^{-\kappa_k} \tilde{f}_k$ for polynomials $\tilde{f}_k$ in q, ζ_1, ζ_2 with $\tilde{f}_k|_{q=0} = P_{i,k}$, and there exists a prime divisor P over $\mathbb{A}^3_{q,\zeta_1,\zeta_2}$ with center 0, whose image on a higher model of V_i/G_i is the image of F, such that*

$$\frac{a(F, V_i, \sum_k c'_k \operatorname{div}(f_k))}{e} = \frac{a\big(P, \mathbb{A}^3, (1 - \delta(c')/r_i)\operatorname{div}(q) + \sum_k c'_k \operatorname{div}(\tilde{f}_k)\big)}{|I_{\bar{G}_i}(P)|}.$$

Moreover, $[q] = r_i[x]/p$.

(3) *In the situation of (1), let $\mathfrak{a} = \prod_j \mathfrak{a}_j^{\tau_j}$ be an $\mathbb{R}$-ideal on $\mathbb{A}^4_{q,x'}$ with $\tau_j > 0$ and each $\mathfrak{a}_j$ generated by $\bar{G}_u$-semi-invariant polynomials, such that $\mathfrak{a}|_{S_0}$ is non-zero and $(S_0, \mathfrak{a}|_{S_0})$ is lc near $S_0 \cap (x' = 0)$. Let $\beta_{\mathfrak{a}}$ be the infimum of $a(\Gamma', S_0, \mathfrak{a}|_{S_0})/|I_{\bar{G}_u}(\Gamma')|$ over the prime divisors Γ' over S_0 with center in $S_0 \cap (x' = 0)$. Then for any real number $b \le 1$,*

$$\frac{a(P, X, bS_0, \mathfrak{a}|_X)}{|I_{\bar{G}_u}(P)|} \ge \beta_{\mathfrak{a}} + (1 - b)[q].$$

Proof. (1) Since xy has weight N and $g_i - g_i^0$ has only blocks of weights $N + lr_i$ with $l \ge 1$, (E.29) implies

$$f_U \circ \psi_u = \rho^N \Big((xy + g_i^0)|_{u=1} + \sum_{l \ge 1} q^l (g_i)_{[N+lr_i]}|_{u=1}\Big),$$

$$f_k \circ \psi_u = \rho^{\kappa_k} \Big(h_{i,k}^0|_{u=1} + \sum_{l \ge 1} q^l (f_k)_{[\kappa_k + lr_i]}|_{u=1}\Big),$$

where we regard $(g_i)_{[N+lr_i]}$ as zero if $U = V_i$. Let f'_U and f'_k be the factors in large parentheses. Then $f'_U|_{q=0} = (xy + g_i^0)|_{u=1}$, so $S_0 = S_u$ and $f'_k|_{S_0} = h_{i,k}^0|_{u=1}|_{S_u}$. The group $\hat{G}_u$ multiplies q by ϵ^{r_i}, so $\bar{G}_u$ acts diagonally on $\mathbb{A}^4_{q,x'}$. By **Step 1**, an element of $\hat{G}_u$ acting trivially on the x'_j belongs to $\hat{R}_u$, and $\hat{R}_u$ acts on ρ through μ_{r_i}, hence trivially on $q = \rho^{r_i}$. Thus the kernel of the action of $\hat{G}_u$ on $\mathbb{A}^4_{q,x'}$ is $\hat{R}_u$, and $\bar{G}_u$ acts faithfully on $\mathbb{A}^4_{q,x'}$, hence on X, which is not contained in a coordinate hyperplane. The group $\bar{G}_u$ acts faithfully on $S_0 = S_u$ by **Step 1**. Since f'_U is semi-invariant, $\bar{G}_u$ preserves X.

We show that X is normal and klt near S_0. The divisor $S_0 = \operatorname{div}(q|_X)$ is Cartier, and a Cartier divisor through a singular point of X is singular there. Since S_0 is a normal surface, the hypersurface X is smooth in codimension one near S_0, hence normal near S_0. Since S_0 is a normal Cartier divisor on X and S_0 is klt (**Step 1**), [KM98, Theorem 5.50(1)] with $B = 0$ implies that (X, S_0) is plt near S_0. Since $a(E', X, 0) \ge a(E', X, S_0)$ for any prime divisor E' over X, X is klt near S_0.

Let $X' \subset \mathbb{A}^4_{\rho,x'}$ be the hypersurface defined by $f'_U(\rho^{r_i}, x')$. As for X, X' is normal near $(\rho = 0)$. Then $X = X'/\hat{R}_u$, and the morphism $\psi_u : X' \backslash (\rho = 0) \to U$ is $\hat{G}_u$-equivariant for $\hat{G}_u \to G_i$ and induces a birational map $X'/\hat{G}_u \dashrightarrow U/G_i$. Let F' be a prime divisor over X' mapping to F. Then $[\rho] = [u]/w_u > 0$, $[x'_j] = [x_j] - w_j[u]/w_u \ge 0$ for $x_j \in \{y, z, t\} \backslash \{u\}$ by the choice of u, and $[x'] = [x] + p[u]/w_u > 0$. Thus F' has a center on X', contained in $(\rho = 0) \cap (x' = 0)$. Since $\sum_j w_j = N + w_z + w_t$,

$$\psi_u^*(dx \wedge dy \wedge dz \wedge dt) = \pm w_u \rho^{N + w_z + w_t - 1} d\rho \wedge dx'_{j_1} \wedge dx'_{j_2} \wedge dx'_{j_3},$$

and taking residues, $\psi_u^*\Omega = \pm w_u \rho^{w_z+w_t-1}\omega'$, where ω' is the residue of

$$\frac{d\rho \wedge dx'_{j_1} \wedge dx'_{j_2} \wedge dx'_{j_3}}{f'_U(\rho^{r_i}, x')},$$

a generator of $\omega_{X'}$ near $(\rho = 0)$. Let

$$\Delta' := (1 + \delta(c')) \operatorname{div}(\rho) + \sum_k c'_k \operatorname{div}(f'_k|_{X'}).$$

Then, near $(\rho = 0)$, $-\Delta'$ is the divisor of

$$\rho^{-1-\delta(c')}\omega' \prod_k f_k'^{-c'_k} = \pm w_u^{-1}\psi_u^*(\Omega \prod_k f_k^{-c'_k}),$$

and $-\sum_k c'_k \operatorname{div}(f_k)$ is the divisor of $\Omega \prod_k f_k^{-c'_k}$ near 0. Let e' be the ramification index of F' over F. By (E.33), applied to $\varpi = \Omega \prod_k f_k^{-c'_k}$ and $X' \dashrightarrow U$, $a(F', X', \Delta') = e'a(F, U, \sum_k c'_k \operatorname{div}(f_k))$. Moreover, by [Liu26, Lemma 2.6(1)], $|I_{\hat{G}_u}(F')| = e'e$. By [Liu26, Lemma 2.6(4)], the crepant quotient of (X', Δ') by $\hat{R}_u$ is

$$(X, (1 + \delta(c')/r_i)S_0 + \sum_k c'_k \operatorname{div}(f'_k|_X)),$$

since $\hat{R}_u$ acts on ρ through μ_{r_i} and fixes the f'_k. Let P be the image of F'. Then [Liu26, Lemma 2.6(2)(5)] implies

$$\frac{a(F, U, \sum_k c'_k \operatorname{div}(f_k))}{e} = \frac{a(F', X', \Delta')}{|I_{\hat{G}_u}(F')|} = \frac{a\big(P, X, (1 + \delta(c')/r_i)S_0 + \sum_k c'_k \operatorname{div}(f'_k|_X)\big)}{|I_{\bar{G}_u}(P)|}.$$

Finally, $q^{w_u} = u^{r_i}$ and $x'^{w_u} = u^p x^{w_u}$, so $[q] = r_i[u]/w_u$ and $[x'] = [x] + p[u]/w_u$.

(2) By (E.27),

$$f_k \circ \varphi_x = \xi^{-\kappa_k}\Big(P_{i,k} + \sum_{l\geq 1} q^l (f_k)_{[\kappa_k - lr_i]}\big(1, -g_i^0(\zeta_1, \zeta_2), \zeta_1, \zeta_2\big)\Big),$$

which defines $\tilde{f}_k$. The morphism $\varphi_x : \mathbb{A}^3 \backslash (\xi = 0) \to V_i$ is $\hat{G}_x$-equivariant for $\hat{G}_x \to G_i$ and induces a birational map $\mathbb{A}^3_{\xi,\zeta_1,\zeta_2}/\hat{G}_x \dashrightarrow V_i/G_i$. Let F' be a prime divisor over $\mathbb{A}^3$ mapping to F. Since $[\xi] = [x]/p > 0$, $[\zeta_1] = [z] + w_z[x]/p > 0$ and $[\zeta_2] = [t] + w_t[x]/p > 0$, the center of F' is 0. Let

$$\Delta'_x := (1 - \delta(c')) \operatorname{div}(\xi) + \sum_k c'_k \operatorname{div}(\tilde{f}_k).$$

By (E.28), $-\Delta'_x$ is the divisor of

$$\xi^{\delta(c')-1} d\xi \wedge d\zeta_1 \wedge d\zeta_2 \prod_k \tilde{f}_k^{-c'_k} = p^{-1}\varphi_x^*(\Omega \prod_k f_k^{-c'_k}).$$

Let e' be the ramification index of F' over F. By (E.33), applied to φ_x,

$$a(F', \mathbb{A}^3, \Delta'_x) = e'a(F, V_i, \sum_k c'_k \operatorname{div}(f_k)),$$

and $|I_{\hat{G}_x}(F')| = e'e$ by [Liu26, Lemma 2.6(1)]. By [Liu26, Lemma 2.6(4)], the crepant quotient of $(\mathbb{A}^3, \Delta'_x)$ by $\hat{R}_x$ is

$$(\mathbb{A}^3_{q,\zeta_1,\zeta_2}, (1 - \delta(c')/r_i) \operatorname{div}(q) + \sum_k c'_k \operatorname{div}(\tilde{f}_k)).$$

Let P be the image of F'. Then [Liu26, Lemma 2.6(2)(5)] implies the equality. Finally, $q^p = x^{r_i}$, so $[q] = r_i[x]/p$.

(3) Let Z be the center of P on X, a point or a curve in $S_0 \cap (x' = 0)$, and let o_P be a general closed point of Z. Let $H \subset \bar{G}_u$ be the stabilizer of o_P. Since $I_{\bar{G}_u}(P)$ acts trivially on the residue

field of P, it fixes Z pointwise, so $I_{\bar{G}_u}(P) = I_H(P)$. The group H acts trivially on the coordinates which do not vanish at o_P. After translating the coordinates which do not vanish at o_P, the point o_P is the origin, H acts diagonally, and the hypotheses of Lemma 5.5 hold for X, S_0, H, and $\mathfrak{a}$, with q in place of the function u of Lemma 5.5. If Z is a curve, then Z is H-invariant, since o_P is general. By Lemma 5.5,

$$\frac{a(P, X, bS_0, \mathfrak{a}|_X)}{|I_H(P)|} \geq \beta_0 + \frac{(1-b)\operatorname{mult}_P q}{|I_H(P)|} = \beta_0 + (1-b)[q],$$

where β_0 is the minimal log discrepancy of the quotient of $(S_0, \mathfrak{a}|_{S_0})$ by H at the image of o_P, or of the generic point of Z if Z is a curve. For a prime divisor Γ' over S_0 with center Z, the inertia group $I_{\bar{G}_u}(\Gamma')$ acts trivially on the residue field of Γ', hence fixes Z pointwise and in particular fixes o_P, so $I_{\bar{G}_u}(\Gamma') = I_H(\Gamma')$. Thus, by Definition 2.4 for the group H, β_0 is the infimum of $a(\Gamma', S_0, \mathfrak{a}|_{S_0})/|I_{\bar{G}_u}(\Gamma')|$ over the prime divisors Γ' over S_0 with center Z, and $\beta_0 \geq \beta_{\mathfrak{a}}$ since $Z \subset S_0 \cap (x' = 0)$. □

Proof of Proposition E.4 continued. We apply Claim E.5 to the references. Let $c' \in \mathbb{R}^s_{\geq 0}$ and $u \in \{y, z, t\}$. By **Step 1** and [Liu26, Lemma 2.6(2)], if $(Y_i^+, B_i^+(c'))$ is lc near C_i^+, then we may apply Claim E.5(3) to $\mathfrak{a} = \prod_k f_k'^{c_k'}$ (the product over the k with $c_k' > 0$), and

$$\beta_{\mathfrak{a}} \geq \beta_i^+(c'). \tag{E.34}$$

By (E.31), the pair

$$\Big(S_u, \sum_k c_k' \operatorname{div}(h_{i,k}^0|_{u=1}|_{S_u}) + \frac{\delta(c')}{p} \operatorname{div}(x'|_{S_u})\Big)$$

is the crepant pullback of $(Y_i^-, B_i^-(c'))$. Thus, if $\delta(c') \geq 0$ and $(Y_i^-, B_i^-(c'))$ is lc near C_i^-, then we may apply Claim E.5(3) to $\mathfrak{a} = \prod_k f_k'^{c_k'} \cdot (x')^{\delta(c')/p}$ (without the last factor if $\delta(c') = 0$), and

$$\beta_{\mathfrak{a}} \geq \beta_i^-(c'). \tag{E.35}$$

Step 5. In this step we assume the hypotheses of (3), and we show that $(W_i, \theta D_i)$ is lc near 0 for i large.

Fix i with $r_i > N$. First we show that $(Y_i^+, B_i^+(\theta c))$ is lc near C_i^+ and $(Y_i^-, B_i^-(\theta c))$ is lc near C_i^-. By the hypothesis of (3), either $\delta_\theta \leq 0$ and $(Y_i^+, B_i^+(\theta c))$ is lc near C_i^+ for any i, or $\delta_\theta \geq 0$ and $(Y_i^-, B_i^-(\theta c))$ is lc near C_i^- for any i. In the first case, let Γ be a prime divisor over Y_i^- whose center contains C_i^-. Since $\pi_i^{-1}(C_i^-) = C_i^+$, the center of Γ on Y_i^+ intersects C_i^+, and (E.31) with $c' = \theta c$ implies

$$a(\Gamma, Y_i^-, B_i^-(\theta c)) = a(\Gamma, Y_i^+, B_i^+(\theta c)) + |\delta_\theta| \operatorname{mult}_\Gamma E_i \geq 0.$$

If $(Y_i^-, B_i^-(\theta c))$ were not lc near C_i^-, then some prime divisor over Y_i^- whose center contains C_i^- would have negative log discrepancy. Thus $(Y_i^-, B_i^-(\theta c))$ is lc near C_i^-. In the second case, let Γ be a prime divisor over Y_i^+ whose center intersects C_i^+. Since the center of Γ on Y_i^- contains C_i^-, (E.31) implies

$$a(\Gamma, Y_i^+, B_i^+(\theta c)) = a(\Gamma, Y_i^-, B_i^-(\theta c)) + \delta_\theta \operatorname{mult}_\Gamma E_i \geq 0.$$

Thus $(Y_i^+, B_i^+(\theta c))$ is lc near C_i^+.

Let

$$\mathcal{D}_\theta := \theta \sum_k c_k \operatorname{div}(h_{i,k}^0|_{V_i}).$$

By **Step 1** and [Liu26, Lemma 2.6(2)], the pair

$$(\mathbb{A}^2_{\zeta_1,\zeta_2}, \theta \sum_k c_k \operatorname{div}(P_{i,k}))$$

is lc near 0, and, for $u \in \{y, z, t\}$, the pair

$$(S_u, \theta \sum_k c_k \operatorname{div}(h^0_{i,k}|_{u=1}|_{S_u}))$$

is lc near $S_u \cap (x' = 0)$. Through the quotient maps φ_x and ψ_u of **Step 3**, the pair $(V_i, \mathcal{D}_\theta)$ is lc near the points of V_i with $x \neq 0$ whose coordinates $(\zeta_1, \zeta_2) = (\xi^{w_z} z, \xi^{w_t} t)$ are close to 0, and near the points of $V_i \cap (x = 0)\backslash\{0\}$. For a point of V_i close to 0 with $x \neq 0$, the coordinates ζ_1, ζ_2 are close to 0. Thus there exists a neighborhood U of 0 such that $(V_i, \mathcal{D}_\theta)$ is lc on $U\backslash\{0\}$. If $(V_i, \mathcal{D}_\theta)$ were not lc near 0, then, by a log resolution, there would be a prime divisor F over V_i with center 0 such that $a(F, V_i, \mathcal{D}_\theta) < 0$. Let $e := |I_{G_i}(F)|$.

- If $\delta_\theta \geq 0$, apply Claim E.5(2) with $f_k = h^0_{i,k}$ and $c' = \theta c$. Then $\tilde{f}_k = P_{i,k}$. By Lemma 5.5, applied to $\mathbb{A}^3_{q,\zeta_1,\zeta_2}$, the function q, $S_0 = (q = 0) = \mathbb{A}^2_{\zeta_1,\zeta_2}$, $H = \bar{G}_i$, $\mathfrak{a} = \prod_k P^{\theta c_k}_{i,k}$, and $b = 1 - \delta_\theta / r_i \leq 1$,
$$\frac{a(F, V_i, \mathcal{D}_\theta)}{e} \geq \beta^-_i(\theta c) + \frac{\delta_\theta}{r_i}[q] \geq 0,$$
since
$$\beta^-_i(\theta c) = \operatorname{mld}((\mathbb{A}^2 \ni 0, \theta \sum_k c_k \operatorname{div}(P_{i,k}))/\bar{G}_i) \geq 0.$$
- If $\delta_\theta < 0$, apply Claim E.5(1) with $U = V_i$, $f_k = h^0_{i,k}$, and $c' = \theta c$, and Claim E.5(3) with $\mathfrak{a} = \prod_k f_k'^{\theta c_k}$ and $b = 1 + \delta_\theta / r_i < 1$. By (E.34),
$$\frac{a(F, V_i, \mathcal{D}_\theta)}{e} \geq \beta^+_i(\theta c) + \frac{|\delta_\theta|}{r_i}[q] \geq 0.$$

In both cases we obtain a contradiction. Thus $(V_i, \mathcal{D}_\theta)$ is lc near 0.

By [Kaw07, Theorem], $(\mathbb{A}^4, V_i + \theta \sum_k c_k \operatorname{div}(h^0_{i,k}))$ is lc near 0. Let L'_θ be the integer of Lemma 4.14 for $n = 4$, $s + 1$ functions and the finite set $\{1, \theta c_1, \dots, \theta c_s\}$. If $r_i \geq 2 w_{\max} L'_\theta$, then, by **Step 1**, [dFEM11, Proposition 2.11], and Lemma 4.14, $(\mathbb{A}^4, W_i + \theta \sum_k c_k \operatorname{div}(h_{i,k}))$ is lc near 0, and, by [Kaw07, Theorem], $(W_i, \theta D_i)$ is lc near 0.

Step 6. In this step we fix constants, and we prove the lower bound for the prime divisors F over W_i with $[x] > C_1 / r_i$, where C_1 is defined below.

By [Liu26, Lemma 2.6], $\alpha_i = \inf_F a(F, W_i, D_i)/|I_{G_i}(F)|$, where F runs over the prime divisors over W_i with center 0. By (E.32), it suffices to show that, for i large, $a_F := a(F, W_i, D_i)/e \geq \beta_i$ for any such F with $a_F \leq 2$, where $e := |I_{G_i}(F)|$. Fix i large, so that **Steps 3 and 5** hold, and fix such F. Let $[\cdot]$ be as in **Step 4**.

Since $(W_i, \theta D_i)$ is lc near 0,

$$0 \leq \frac{a(F, W_i, \theta D_i)}{e} = a_F - (\theta - 1) \sum_k c_k [h_{i,k}],$$

and since $a_F \leq 2$,

$$[h_{i,k}] \leq C_2 := \max_k \frac{2}{c_k(\theta - 1)} \quad \text{for any } k. \tag{E.36}$$

The monomials $x^{w_z} z^p$, $x^{w_t} t^p$, and $x^{N+p} y^p$ have $\mathbf{T}$-weight 0. Thus the characters of G_i on them are trivial on R_i, and their orders divide $[G_i : R_i] \leq e_0$. For a G_i-invariant regular function f on W_i vanishing at 0, [Liu26, Lemma 2.6(1)(2)] implies that $[f]$ is the multiplicity, along the prime divisor over W_i/G_i under F, of the function on W_i/G_i induced by f, so $[f] \geq 1$. Let m be one of the three monomials and d_m the order of the character of G_i on m, so $d_m \leq e_0$. Then m^{d_m} is a G_i-invariant regular function on W_i vanishing at 0, so $d_m[m] = [m^{d_m}] \geq 1$, and $[m] \geq 1/e_0$. Since $[x^{w_z} z^p] = w_z[x] + p[z]$, and likewise for the other two monomials, we obtain

$$w_z[x] + p[z] \geq \frac{1}{e_0}, \qquad w_t[x] + p[t] \geq \frac{1}{e_0}, \qquad (N + p)[x] + p[y] \geq \frac{1}{e_0}. \tag{E.37}$$

Let

$$C_1 := 2p(C_2+1),$$

and assume that $[x] > C_1/r_i$ and $r_i \geq 2\max_k |\kappa_k|$.

For $l \leq -1$, the block $(h_{i,k})_{[\kappa_k+lr_i]}$ has weight at most $\kappa_k - r_i < 0$, and the monomials of the block have no factor xy. Thus they have the form $x^\gamma z^{\gamma_1} t^{\gamma_2}$ with $p\gamma = w_z\gamma_1 + w_t\gamma_2 - \kappa_k - lr_i \geq r_i/2$, and

$$\left[x^\gamma z^{\gamma_1} t^{\gamma_2}\right] \geq \gamma[x] > \frac{r_i}{2p} \cdot \frac{C_1}{r_i} = C_2 + 1.$$

Let

$$h^{\geq}_{i,k} := h_{i,k} - \sum_{l \leq -1} (h_{i,k})_{[\kappa_k + lr_i]}.$$

Then $[h_{i,k} - h^{\geq}_{i,k}] > C_2 + 1$, and, by (E.36), $[h^{\geq}_{i,k}] = [h_{i,k}]$. In particular $h^{\geq}_{i,k}|_{W_i} \neq 0$, and $a_F = a(F, W_i, \sum_k c_k \operatorname{div}(h^{\geq}_{i,k}))/e$. Apply Claim E.5(1) with $U = W_i$, $f_k = h^{\geq}_{i,k}$, and $c' = c$, and let u, X, S_0, f'_k, and P be as there. Then

$$a_F = \frac{a\big(P, X, (1+\delta/r_i)S_0 + \sum_k c_k \operatorname{div}(f'_k|_X)\big)}{|I_{\bar{G}_u}(P)|}.$$

- Assume $\delta < 0$. By **Step 3**, (Y_i^+, B_i^+) is lc near C_i^+. Claim E.5(3) with $\mathfrak{a} = \prod_k f_k'^{c_k}$ and $b = 1 + \delta/r_i < 1$, together with (E.34) and (E.32), implies

$$a_F \geq \beta_i^+ + \frac{|\delta|}{r_i}[q] \geq \beta_i^+ = \beta_i.$$

- Assume $\delta \geq 0$. By **Step 3**, (Y_i^-, B_i^-) is lc near C_i^-. Let $\mathfrak{a} := \prod_k f_k'^{c_k} \cdot (x')^{\delta/p}$ (without the last factor if $\delta = 0$). Since

$$\Big(1 + \frac{\delta}{r_i}\Big) S_0 + \sum_k c_k \operatorname{div}(f'_k|_X) = S_0 + \sum_k c_k \operatorname{div}(f'_k|_X) + \frac{\delta}{p}\operatorname{div}(x'|_X) + \frac{\delta}{r_i} S_0 - \frac{\delta}{p}\operatorname{div}(x'|_X),$$

the relations $[q] = r_i[u]/w_u$ and $[x'] = [x] + p[u]/w_u$ of Claim E.5(1) imply

$$\begin{aligned} a_F &= \frac{a(P, X, S_0, \mathfrak{a}|_X)}{|I_{\bar{G}_u}(P)|} - \frac{\delta}{r_i}[q] + \frac{\delta}{p}[x'] \\ &= \frac{a(P, X, S_0, \mathfrak{a}|_X)}{|I_{\bar{G}_u}(P)|} + \frac{\delta}{p}[x]. \end{aligned}$$

By Claim E.5(3) with $b = 1$, (E.35) and (E.32),

$$a_F \geq \beta_i^- + \frac{\delta}{p}[x] \geq \beta_i^- = \beta_i.$$

Step 7. In this step we prove the lower bound for the prime divisors F with $[x] \leq C_1/r_i$, and we finish the proof.

Let F and e be as in **Step 6**, and assume $[x] \leq C_1/r_i$. Assume moreover that

$$r_i \geq 2\max_k |\kappa_k|, \qquad r_i \geq 2e_0 w_{\max} C_1, \qquad r_i \geq C_1, \qquad r_i \geq 4pe_0 w_{\max}(C_2+2), \qquad r_i \geq QC_1/p.$$

By (E.37), $p[z] \geq 1/e_0 - w_z C_1/r_i \geq 1/(2e_0)$, and similarly for t and y. Thus

$$[y], [z], [t] \geq \frac{1}{2pe_0}.$$

For $l \geq 1$, the block $(h_{i,k})_{[\kappa_k+lr_i]}$ has weight at least $r_i/2$, so each monomial of it is $x^\gamma z^{\gamma_1} t^{\gamma_2}$ with $w_z\gamma_1 + w_t\gamma_2 \geq r_i/2$ or $y^\gamma z^{\gamma_1} t^{\gamma_2}$ with $(N+p)\gamma + w_z\gamma_1 + w_t\gamma_2 \geq r_i/2$. In both cases the degree in y, z, t is at least $r_i/(2w_{\max})$. Similarly each monomial of $g_i - g_i^0$ has degree at least

$r_i/w_{\max}$ in z,t. On W_i we have $y=-g_i/x$. Let $y_0:=-g_i^0/x$, a rational function on $\mathbb{A}^3_{x,z,t}$. Then $y-y_0=-(g_i-g_i^0)/x$, so

$$[y-y_0]\ge \frac{r_i}{2pe_0w_{\max}}-[x]\ge 2(C_2+2)-1>C_2+1,$$

and $[y_0]\ge\min([y],[y-y_0])\ge 1/(2pe_0)$. Thus $[f]\ge r_i/(4pe_0w_{\max})\ge C_2+2$ for any monomial f of a block of weight κ_k+lr_i, $l\ge 1$, after substituting y or y_0 for y.

By **Step 1**, W_i and V_i are G_i-equivariantly birational to $\mathbb{A}^3_{x,z,t}$, with $y|_{V_i}=y_0$. Thus F is a prime divisor over V_i with inertia group $I_{G_i}(F)$ of order e, and since $[x],[y_0],[z],[t]>0$, the center of F on V_i is 0. Since Ω generates the canonical sheaves of W_i and V_i near 0, $a(F,W_i,0)=1+\operatorname{mult}_F\Omega=a(F,V_i,0)$. Let

$$\tilde h_{i,k}:=h^0_{i,k}+\sum_{l\le -1}(h_{i,k})_{[\kappa_k+lr_i]}.$$

We have $h_{i,k}(x,y,z,t)-h_{i,k}(x,y_0,z,t)=(y-y_0)\Phi_k$ for a polynomial Φ_k in x,y,y_0,z,t, so $[h_{i,k}(x,y,z,t)-h_{i,k}(x,y_0,z,t)]\ge[y-y_0]>C_2+1$. Moreover $h_{i,k}(x,y_0,z,t)-\tilde h_{i,k}(x,y_0,z,t)$ is the sum of the blocks of weights κ_k+lr_i, $l\ge1$, at y_0, of order at least C_2+2. Since $[h_{i,k}|_{W_i}]\le C_2$ by (E.36), we obtain $[\tilde h_{i,k}|_{V_i}]=[h_{i,k}|_{W_i}]$. Thus

$$a_F=\frac{a(F,V_i,\sum_k c_k\operatorname{div}(\tilde h_{i,k}|_{V_i}))}{e}.$$

Apply Claim E.5(2) with $f_k=\tilde h_{i,k}$ and $c'=c$, and let $\tilde f_k$ and P be as there. Then

$$a_F=\frac{a\big(P,\mathbb{A}^3,(1-\delta/r_i)\operatorname{div}(q)+\sum_k c_k\operatorname{div}(\tilde f_k)\big)}{|I_{\bar G_i}(P)|},\qquad \tilde f_k|_{q=0}=P_{i,k},\qquad [q]=\frac{r_i[x]}{p}.$$

By **Step 3**, (Y_i^-,B_i^-) is lc near C_i^-, so the pair

$$(\mathbb{A}^2_{\zeta_1,\zeta_2},\sum_k c_k\operatorname{div}(P_{i,k}))$$

is lc near 0, and, by **Step 1**,

$$\operatorname{mld}((\mathbb{A}^2\ni 0,\sum_k c_k\operatorname{div}(P_{i,k}))/\bar G_i)=\beta_i^-.$$

In both cases below, we use Lemma 5.5 for $\mathbb{A}^3_{q,\zeta_1,\zeta_2}$, the function q, $S_0=\mathbb{A}^2_{\zeta_1,\zeta_2}$, $H=\bar G_i$, and $\mathfrak{a}=\prod_k\tilde f_k^{c_k}$.

- If $\delta\ge0$, then Lemma 5.5 with $b=1-\delta/r_i\le1$ and (E.32) imply

$$a_F\ge\beta_i^-+\frac{\delta}{r_i}[q]=\beta_i^-+\frac{\delta}{p}[x]\ge\beta_i^-=\beta_i.$$

- If $\delta<0$, then

$$a(P,\mathbb{A}^3,(1+|\delta|/r_i)\operatorname{div}(q),\mathfrak{a})=a(P,\mathbb{A}^3,\operatorname{div}(q),\mathfrak{a})-\frac{|\delta|}{r_i}\operatorname{mult}_P q.$$

  Lemma 5.5 with $b=1$, (2) and (E.32) imply

$$a_F\ge\beta_i^--\frac{|\delta|}{r_i}[q]=\beta_i^--\frac{|\delta|}{p}[x]\ge\beta_i^++\frac{|\delta|}{Q}-\frac{|\delta|C_1}{pr_i}\ge\beta_i^+=\beta_i,$$

  since $r_i\ge QC_1/p$.

Thus, for i large, $a_F\ge\beta_i$ for any prime divisor F over W_i with center 0 and $a_F\le2$, by **Step 6** if $[x]>C_1/r_i$ and by **Step 7** otherwise. By (E.32), $a_F\ge\beta_i$ also holds if $a_F>2$. Hence $\alpha_i\ge\beta_i$ for i large. This proves (3). □

We now prove Lemmas E.6 and E.7: the reference values of Construction E.2 form a finite set, and the quotient values α_i exceed them by at most C_0/r_i.

The surfaces $Y_i^{\pm}$ are covered, near the fibers $C_i^{\pm}$, by quotients of surface germs of bounded type by groups of bounded order, and the boundaries have coefficients in a finite set. The following lemma shows that the corresponding quotient values form a finite set of real numbers and are computed by prime divisors of bounded log discrepancy. It replaces Lemma 4.8, which needs an integral multiple of the log canonical divisor to be Cartier and so does not apply when the coefficients are irrational.

Lemma E.6. *Let N be a positive integer and $\Lambda \subset [0,1]$ a finite set. Then there exist a positive integer l and a finite set $\mathcal{V} \subset \mathbb{R}$ with the following property. Let $(S \ni p)$ be a surface germ which is the quotient of a smooth surface germ by a finite group of order at most N, let H be a finite group of order at most N acting faithfully on S and fixing p, and let $B \geq 0$ be an H-invariant $\mathbb{R}$-divisor on S with coefficients in Λ such that (S,B) is lc near p. Then $\operatorname{mld}((S \ni p, B)/H) \in \mathcal{V}$, and there exists a prime divisor E over S with center p such that $a(E,S,0) \leq l$ and $a(E,S,B)/|I_H(E)| = \operatorname{mld}((S \ni p, B)/H)$.*

Proof. Let

$$\Lambda' := \left\{ 1 - \frac{1-\lambda}{e} \;\middle|\; \lambda \in \Lambda \cup \{0\},\ 1 \leq e \leq N \right\} \subset [0,1],$$

a finite set. By Theorem 2.9(2) applied to Λ', there exists a positive integer l' such that any lc surface germ $(X \ni x, B_X)$ with $B_X \in \Lambda'$ has a prime divisor E' over X with $\operatorname{center}_X E' = \{x\}$, $a(E',X,B_X) = \operatorname{mld}(X \ni x, B_X)$, and $a(E',X,0) \leq l'$. Let $M := (N^2)!$ and let $b_{\min}$ be the least positive element of Λ' (and $b_{\min} := 1$ if $\Lambda' \subset \{0\}$). We show that $l := Nl'$ and

$$\mathcal{V} := \left\{ a - \sum_{j=1}^{J} b_j m_j \;\middle|\; a, m_1, \dots, m_J \in \tfrac{1}{M}\mathbb{Z} \cap [0, \tfrac{l'}{b_{\min}}],\ b_1, \dots, b_J \in \Lambda',\ 0 \leq J \leq \tfrac{Ml'}{b_{\min}} \right\}$$

have the property of the lemma. The set $\mathcal{V}$ is finite.

Write $(S \ni p) = (T \ni 0)/G_0$, where $(T \ni 0)$ is a smooth surface germ and G_0 is a finite group of order at most N acting faithfully on T and fixing 0. Let $\pi : S \to S' := S/H$ be the quotient map, $p' := \pi(p)$, and let B' be the $\mathbb{R}$-divisor on S' with $K_S + B = \pi^*(K_{S'} + B')$, so that $\operatorname{mld}((S \ni p, B)/H) = \operatorname{mld}(S' \ni p', B')$ by definition. For a prime divisor P' on S' and a prime divisor P on S over P', the coefficient of B' along P' is $1 - (1 - \operatorname{mult}_P B)/|I_H(P)|$ [Liu26, Lemma 2.6]. Since $|I_H(P)| \leq |H| \leq N$, we have $B' \in \Lambda'$. Moreover, (S', B') is lc near p', and for any prime divisor E over S with center p, whose image on S' we denote by E',

$$a(E,S,B) = |I_H(E)|a(E',S',B') \quad \text{and} \quad a(E,S,0) = |I_H(E)|a(E',S',B'_0), \tag{E.38}$$

where $K_S = \pi^*(K_{S'} + B'_0)$ [Liu26, Lemma 2.6].

We show that MD' is Cartier near p' for any Weil divisor D' on S'. The composite $\varphi : T \to S \to S'$ is a finite morphism of degree $d := |G_0||H| \leq N^2$, and possibly shrinking S' to a neighborhood of p', we may assume that $\varphi^{-1}(p') = \{0\}$, since G_0 fixes 0 and H fixes p. Let φ^*D' be the Weil divisor on T which is the pullback of D' over the smooth locus of S'. Since T is smooth, $\varphi^*D' = \operatorname{div}(g)$ near 0 for a rational function g on T. Let $\operatorname{Nm}(g)$ be the norm of g along the finite morphism φ. Then

$$\operatorname{div}(\operatorname{Nm}(g)) = \varphi_* \operatorname{div}(g) = \varphi_*\varphi^* D' = dD'$$

near p' [Ful98, Proposition 1.4(b)]. Thus dD' is principal near p', and so is MD', since d divides M. In particular, $MK_{S'}$ and M times any component of B' are Cartier near p'.

By the choice of l', there exists a prime divisor E' over S' with center p' such that $a(E',S',B') = \operatorname{mld}(S' \ni p', B')$ and $a(E',S',0) \leq l'$. Write $B' = \sum_j b_j B'_j$ near p', where the B'_j are distinct prime divisors through p' and $b_j \in \Lambda' \backslash \{0\}$. By the previous paragraph, $a(E',S',0) \in \frac{1}{M}\mathbb{Z}$ and

$m_j := \operatorname{mult}_{E'} B'_j \in \frac{1}{M}\mathbb{N}^+$, the multiplicities being positive because $p' \in B'_j$. Since (S', B') is lc near p',

$$0 \le a(E', S', B') = a(E', S', 0) - \sum_j b_j m_j.$$

Hence $\sum_j b_j m_j \le a(E', S', 0) \le l'$, so $m_j \le l'/b_{\min}$ for any j, and the number of indices j is at most $Ml'/b_{\min}$. Thus $\operatorname{mld}((S \ni p, B)/H) = a(E', S', B') \in \mathcal{V}$.

Since π is finite and surjective, there exists a prime divisor E over S whose image on S' is E', and $\operatorname{center}_S E = \{p\}$, since $\pi^{-1}(p') = \{p\}$. By (E.38), $a(E, S, B)/|I_H(E)| = a(E', S', B') = \operatorname{mld}((S \ni p, B)/H)$. Since $B'_0 \ge 0$, (E.38) also implies $a(E, S, 0) \le |I_H(E)| a(E', S', 0) \le Nl' = l$. □

The next lemma shows that the reference values β_i of Construction E.2 take finitely many values, and bounds α_i from above by Gauss extensions of prime divisors computing β_i. It treats both classes (a) and (b) of Lemma E.1(3).

Lemma E.7. *Notation as in Construction E.2. Then, for i large, the numbers β_i belong to a finite set, and there exists a real number $C_0 \ge 0$, independent of i, such that $\alpha_i \le \beta_i + C_0/r_i$.*

Proof. The proof is long, so we divide it into several steps. In the proof, x_1, x_2, x_3, x_4 are the coordinates x, y, z, t of $\mathbb{A}^4$, $f_i := xy + g_i$, and $f_i^\circ := xy + z^a t^b$ in case (a) and $f_i^\circ := xy + g_i^0$ in case (b), so that $V_i = (f_i^\circ = 0)$ as in Construction E.2. Let $\kappa_f := w_x + w_y$ be the $\mathbf{T}$-weight of f_i°. Recall that $h_{i,k}$ has no monomial divisible by xy, that $h^0_{i,k}$ is the principal block of $h_{i,k}$, of $\mathbf{T}$-weight κ_k, and that $c_k = \mu_k/m > 0$.

Step 1. In this step we describe $Y_i^\pm$ near $C_i^\pm$ by root sections.

Fix a sign $\star \in \{+, -\}$, let $\varsigma := 1$ if $\star = +$ and $\varsigma := -1$ if $\star = -$, and let $\omega := \varsigma w = (\omega_1, \dots, \omega_4)$. Then $Y_i^\star$ is the quotient by $G_i/(G_i \cap \mathbf{T})$ of $\operatorname{Proj} \bigoplus_{d \ge 0} \mathbb{C}[V_i]_d$, where $\mathbb{C}[V_i]_d$ is the space of functions of ω-weight d. A monomial of positive ω-weight is divisible by some x_l with $\omega_l > 0$, so $\operatorname{Proj} \bigoplus_{d \ge 0} \mathbb{C}[V_i]_d$ is covered by the open subsets $D_+(x_l)$ with $\omega_l > 0$.

Fix l with $\omega_l > 0$, and let $\mathbb{A}^3_l$ be the affine space with coordinates $x' = (x'_j)_{j \ne l}$. For a polynomial h on $\mathbb{A}^4$ let $h|_{x_l=1}$ be the polynomial on $\mathbb{A}^3_l$ obtained by substituting $x_l = 1$ and $x_j = x'_j$ for $j \ne l$. Let

$$\bar f := f_i^\circ|_{x_l=1}, \qquad \bar h_k := h^0_{i,k}|_{x_l=1}, \qquad S_l := (\bar f = 0) \subset \mathbb{A}^3_l$$

(the slice $S_{i,l}$ of Construction E.2, with the index i dropped), and let $\bar B := \sum_k c_k \operatorname{div}(\bar h_k)$ on $\mathbb{A}^3_l$ and $B_{S_l} := \sum_k c_k \operatorname{div}(\bar h_k|_{S_l})$ on S_l. Let $\Phi_l : \mathbb{A}^3_l \times \mathbb{A}^1_\rho \dashrightarrow \mathbb{A}^4$ be the root chart with root parameter ρ, that is,

$$x_l = \rho^{\omega_l}, \qquad x_j = \rho^{\omega_j} x'_j \quad \text{for } j \ne l.$$

Let $\hat G_l$ be the group of pairs (g, θ) with $g \in G_i$ and $\theta^{\omega_l} = \chi_l(g)$, where $g^* x_j = \chi_j(g) x_j$. It acts diagonally on $\mathbb{A}^3_l \times \mathbb{A}^1_\rho$ by $\rho \to \theta\rho$ and $x'_j \to \chi_j(g)\theta^{-\omega_j} x'_j$, and Φ_l is equivariant for the surjection $\hat G_l \to G_i$, $(g, \theta) \to g$, whose kernel is the subgroup $\{(1, \theta)\} \cong \mu_{\omega_l}$ of $\hat G_l$. Over $(x_l \ne 0)$, Φ_l is the quotient by $\{(1, \theta)\}$. An element (g, θ) acts trivially on $\mathbb{A}^3_l$ if and only if $\chi_j(g) = \theta^{\omega_j}$ for any j, that is, g is the image of θ in $\mathbf{T}$. Such elements (g, θ) form the subgroup $\hat R \cong G_i \cap \mathbf{T} = \mu_{r_i}$, acting on ρ by scalars, so the quotient of $\mathbb{A}^3_l \times \mathbb{A}^1_\rho$ by $\hat R$ is $\mathbb{A}^3_l \times \mathbb{A}^1_q$ with $q := \rho^{r_i}$. Thus $\bar G_l := \hat G_l/\hat R$ acts on q by a character χ_q, and Φ_l induces a birational map

$$(\mathbb{A}^3_l \times \mathbb{A}^1_q)/\bar G_l \dashrightarrow \mathbb{A}^4/G_i. \tag{E.39}$$

The group $\bar G_l$ acts diagonally on $\mathbb{A}^3_l \times \mathbb{A}^1_q$, faithfully on $\mathbb{A}^3_l$ and on S_l, and $S_l/\bar G_l$ is the open subset of $Y_i^\star$ over $D_+(x_l)$. The action on S_l is faithful as shown at the end of this step, and $S_l/\bar G_l$ is the open subset over $D_+(x_l)$ since $D_+(x_l) = \operatorname{Spec} \mathbb{C}[S_l]^{\mu_{\omega_l}}$. Moreover, $|\bar G_l| = \omega_l |G_i|/r_i \le \omega_l e_0$.

The restriction of $B_i^\star$ to $S_l/\bar G_l$ is the crepant quotient of (S_l, B_{S_l}) by $\bar G_l$ (Construction E.2). The weight-zero monomials of positive degree generate the functions of ω-weight zero vanishing

at the origin, and any such monomial is divisible by some x_j with $\omega_j \le 0$, while $x_l^{|\omega_j|}x_j^{\omega_l}$ ($\omega_j < 0$) and x_j ($\omega_j = 0$) are such monomials. Thus the preimage of $C_i^\star$ in S_l is $S_l \cap L_l$, where

$$L_l := (x'_j = 0 \text{ for any } j \ne l \text{ with } \omega_j \le 0) \subset \mathbb{A}^3_l.$$

We describe S_l. Write $f_i^\circ = xy + g_i^\circ$, where $g_i^\circ = z^a t^b$ in case (a) and $g_i^\circ = g_i^0$ in case (b). If $x_l \in \{x, y\}$, then S_l is the graph of $-g_i^\circ$ over the plane of the coordinates x'_z, x'_t. If $x_l = z$, then $\bar f = x'y' + g_i^\circ(1, t')$, writing x', y', t' for x'_x, x'_y, x'_t, and $g_i^\circ(1, t')$ is a non-zero polynomial of degree at most n_1, where $n_1 := n$ in case (a) and $n_1 := \lfloor N/\min(w_z, w_t)\rfloor$ in case (b), since $z^a t^b$ occurs in g_i° and any monomial $z^c t^d$ of g_i° in case (b) satisfies $cw_z + dw_t = N$ (in particular S_l is irreducible, since $x'y' + g_i^\circ(1, t')$ has degree one in x' and $g_i^\circ(1, t') \ne 0$). At a point of S_l with $x' = y' = 0$ and $t' = t_0$, write $g_i^\circ(1, t') = (t' - t_0)^k u(t')$ with $u(t_0) \ne 0$ and $1 \le k \le n_1$. On the open subset $u \ne 0$, the substitution $x' \to x'/u(t')$ is an isomorphism from S_l onto $(x'y' + (t' - t_0)^k = 0)$, which is the quotient of $\mathbb{A}^2$ by μ_k acting with weights $(1, -1)$. At the other points of S_l, S_l is smooth. The case $x_l = t$ is symmetric. Thus, if

$$N_0 := \max\{n_1,\ e_0|w_1|, \dots, e_0|w_4|\},$$

then any germ of S_l at a closed point is the quotient of a smooth surface germ by a group of order at most N_0, S_l is normal and klt, and $|\bar G_l| \le N_0$. No coordinate divides $\bar f$, so S_l is not contained in a coordinate hyperplane. Hence a diagonal element acting trivially on S_l fixes a point with non-zero coordinates and is the identity. So $\bar G_l$ and any subgroup of $\bar G_l$ act faithfully on S_l. The polynomials $\bar h_k$ are $\bar G_l$-semi-invariant and non-zero on S_l, since $h^0_{i,k}$ is non-zero on V_i and $\mathbf{T}$-homogeneous.

Step 2. In this step we show that both (Y_i^+, B_i^+) and (Y_i^-, B_i^-) are lc near C_i^+ and C_i^- for i large, and that the β_i belong to a finite set.

In case (a), the lc property is Proposition E.3(1). In case (b), it is Proposition E.4(2).

Let

$$\Lambda := \Big\{ \sum_k c_k m_k \ \Big|\ m_1, \dots, m_s \in \mathbb{Z}_{\ge 0} \Big\} \cap [0, 1],$$

a finite set since $c_k > 0$ for any k. Fix $\star$ and i large. Any prime divisor over $Y_i^\star$ with center in $C_i^\star$ has as center either a closed point $y \in C_i^\star$ or an irreducible component Γ of $C_i^\star$ of dimension one. Thus

$$\beta_i^\star = \min\Big(\min_y \operatorname{mld}(Y_i^\star \ni y, B_i^\star),\ \min_\Gamma \big(1 - \operatorname{mult}_\Gamma B_i^\star\big)\Big).$$

Let $y \in C_i^\star$ be a closed point, fix l with $\omega_l > 0$ and y in the open subset $S_l/\bar G_l$, and let $\sigma \in S_l$ be a point over y, with stabilizer $\bar G_\sigma \subset \bar G_l$. Then

$$\operatorname{mld}(Y_i^\star \ni y, B_i^\star) = \operatorname{mld}((S_l \ni \sigma, B_{S_l})/\bar G_\sigma)$$

[Liu26, Lemma 2.6]: by [Liu26, Lemma 2.6(2)(3)], after a translation by an element of $\bar G_l$, the prime divisors over $Y_i^\star$ with center y are the prime divisors under the prime divisors over S_l with center σ, and the inertia group in $\bar G_l$ of a prime divisor over S_l with center σ fixes σ, hence is its inertia group in $\bar G_\sigma$. By **Step 1**, $(S_l \ni \sigma)$ is the quotient of a smooth surface germ by a group of order at most N_0, $\bar G_\sigma$ is a group of order at most N_0 acting faithfully on S_l and fixing σ, $B_{S_l} \ge 0$ is $\bar G_\sigma$-invariant, and $B_{S_l} \in \Lambda$, since (S_l, B_{S_l}) is lc near σ. Let $\mathcal{V}$ be the finite set of Lemma E.6 for N_0 and Λ. Then $\operatorname{mld}(Y_i^\star \ni y, B_i^\star) \in \mathcal{V}$. Let Γ be a component of $C_i^\star$ of dimension one, contained in $S_l/\bar G_l$ generically, and $Z \subset S_l$ a component of the preimage of Γ. Then

$$1 - \operatorname{mult}_\Gamma B_i^\star = \frac{1 - \operatorname{mult}_Z B_{S_l}}{|I_{\bar G_l}(Z)|}$$

[Liu26, Lemma 2.6], with $\operatorname{mult}_Z B_{S_l} \in \Lambda$ and $|I_{\bar G_l}(Z)| \le N_0$. Thus $\beta_i^\star$, and hence β_i, belongs to the finite set

$$\mathcal{V} \cup \Big\{ \frac{1 - \lambda}{e} \ \Big|\ \lambda \in \Lambda,\ 1 \le e \le N_0 \Big\}.$$

Step 3. In this step we fix, for i large, a prime divisor E over an affine space $\mathbb{A}^3_l$ computing β_i, and a Gauss extension F_q of E over $\mathbb{A}^3_l \times \mathbb{A}^1_q$, with bounded invariants.

Let $c_{\min} := \min_k c_k$. For a finite diagonal group G' acting on $\mathbb{A}^3$ with $|G'| \le N_0$, we have $G' \subset \mu^3_{N_0!}$, so there exist only finitely many such groups. Let

$$\Lambda_1 := \Big\{ m_0 + \sum_k c_k m_k \;\Big|\; m_0, \dots, m_s \in \mathbb{Z}_{\ge 0} \Big\} \cap [0,1] \quad \text{and}$$

$$\Lambda_2 := \Big\{ 1 - \frac{1-\lambda}{e} \;\Big|\; \lambda \in \Lambda_1,\ 1 \le e \le N_0 \Big\} \backslash \{0\},$$

finite subsets of $[0,1]$ and $(0,1]$. For any of the finitely many quotients $\mathbb{A}^3/G'$, which are klt threefolds, Theorem 2.10(3) with the DCC set Λ_2 provides a positive integer, and we let l_3 be the maximum of the integers so obtained. Let

$$C_1 := \max\{N_0 l_3/c_{\min},\ 1/c_{\min} + N_0 + 1\}, \qquad C_2 := C_1 + N_0 + 1.$$

Fix i large and a sign $\star$ with $\beta_i = \beta_i^\star$, and let ς and ω be as in **Step 1**. By **Step 2**, $\beta_i^\star = a(\Sigma_0, Y_i^\star, B_i^\star)$ for a prime divisor Σ_0 over $Y_i^\star$ whose center Θ is a closed point of $C_i^\star$ or a component of $C_i^\star$ of dimension one. Since any prime divisor over $Y_i^\star$ with center Θ has center in $C_i^\star$, we have $\beta_i = \mathrm{mld}(Y_i^\star \ni \eta_\Theta, B_i^\star)$, where η_Θ is the generic point of Θ. Fix l with $\omega_l > 0$ and $\Theta \cap (S_l/\bar{G}_l) \ne \emptyset$, and fix an irreducible component Z of the preimage of Θ in S_l. Thus Z is a closed point σ or a curve, and $Z \subset S_l \cap L_l$ by **Step 1**. We construct a prime divisor E over $\mathbb{A}^3_l$ with center Z, with $e := |I_{\bar{G}_l}(E)|$, a positive integer k_G, and a prime divisor F_q over $\mathbb{A}^3_l \times \mathbb{A}^1_q$ with center $Z \times \{0\}$ such that, with $\nu := \mathrm{mult}_E$,

(1) $a(E, \mathbb{A}^3_l, \mathrm{div}(\bar{f}) + \bar{B})/e = \beta_i$;
(2) $\nu(\bar{f}) \le C_1$ and $\nu(\bar{h}_k) \le C_1$ for any k;
(3) $C_1 < k_G \le C_2$, and mult_{F_q} is the Gauss extension of ν with q-value k_G [Liu26, Definition-Proposition 2.11], that is,
$$\mathrm{mult}_{F_q}\Big(\sum_j a_j q^j\Big) = \min_j \big(\nu(a_j) + j k_G\big) \quad \text{for rational functions } a_j \text{ on } \mathbb{A}^3_l;$$
(4) $I_{\bar{G}_l}(F_q) = I_{\bar{G}_l}(E)$, and $a(F_q, \mathbb{A}^3_l \times \mathbb{A}^1_q, 0) = a(E, \mathbb{A}^3_l, 0) + k_G$.

First assume that Z is a closed point σ. Let $\bar{G}_\sigma \subset \bar{G}_l$ be the stabilizer of σ. The coordinates x'_j with $x'_j(\sigma) \ne 0$ are $\bar{G}_\sigma$-invariant, so after replacing them with $x'_j - x'_j(\sigma)$, the group $\bar{G}_\sigma$ still acts diagonally, σ is the origin of $\mathbb{A}^3_l$, and $\bar{f}$ and the $\bar{h}_k$ are $\bar{G}_\sigma$-semi-invariant polynomials. By **Step 1**, S_l is, near σ, a normal integral hypersurface with klt singularities on which $\bar{G}_\sigma$ acts faithfully, the $\bar{h}_k$ are non-zero on S_l, and (S_l, B_{S_l}) is lc near σ by **Step 2**. Hence

$$\beta_i = \mathrm{mld}((S_l \ni \sigma, B_{S_l})/\bar{G}_\sigma) = \mathrm{mld}\big((\mathbb{A}^3_l \ni \sigma, \mathrm{div}(\bar{f}) + \bar{B})/\bar{G}_\sigma\big)$$

by **Step 2** and Theorem 3.3(1). Moreover, $(\mathbb{A}^3_l, S_l + \bar{B})$ is lc near σ by [Kaw07, Theorem].

Let $X := \mathbb{A}^3_l/\bar{G}_\sigma$, $\bar{\sigma}$ the image of σ, and Δ_X the $\mathbb{R}$-divisor on X with $K_{\mathbb{A}^3_l} + \mathrm{div}(\bar{f}) + \bar{B}$ equal to the pullback of $K_X + \Delta_X$. The coefficients of $\mathrm{div}(\bar{f}) + \bar{B}$ along the prime divisors through σ belong to Λ_1, so the coefficients of Δ_X along the prime divisors through $\bar{\sigma}$ belong to Λ_2 [Liu26, Lemma 2.6], and (X, Δ_X) is lc near $\bar{\sigma}$. Removing from Δ_X the components not containing $\bar{\sigma}$ changes neither $\mathrm{mld}(X \ni \bar{\sigma}, \Delta_X)$ nor the log discrepancies at prime divisors with center $\bar{\sigma}$, so we may assume that $\Delta_X \in \Lambda_2$. By Theorem 2.10(3) and the choice of l_3, there exists a prime divisor E' over X computing $\mathrm{mld}(X \ni \bar{\sigma}, \Delta_X) = \beta_i$ with $a(E', X, 0) \le l_3$. Let E be a prime divisor over $\mathbb{A}^3_l$ with image E' on X. Then $\mathrm{center}\, E = \{\sigma\}$, $I_{\bar{G}_l}(E) = I_{\bar{G}_\sigma}(E)$ since inertia elements fix the center, and [Liu26, Lemma 2.6] implies (1) and

$$a(E, \mathbb{A}^3_l, 0) \le |I_{\bar{G}_\sigma}(E)| a(E', X, 0) \le N_0 l_3.$$

Since $a(E, \mathbb{A}^3_l, \mathrm{div}(\bar{f}) + \bar{B}) \ge 0$, we obtain $\nu(\bar{f}) + \sum_k c_k \nu(\bar{h}_k) \le N_0 l_3$, which implies (2). Let $q_G \in \{1, \dots, e\}$ be the integer of Lemma 3.1 for $V = \mathbb{A}^3_l$, the point σ, $B = 0$, the group $\bar{G}_\sigma$ acting

faithfully on $\mathbb{A}^3_l$, the character $\chi_q|_{\bar{G}_\sigma}$, and E. Let k_G be the least integer with $k_G > C_1$ and $k_G \equiv q_G \pmod e$, so $C_1 < k_G \le C_1 + N_0 \le C_2$. By Lemma 3.1, the Gauss extension of ν with q-value k_G is mult_{F_q} for a prime divisor F_q over $\mathbb{A}^3_l \times \mathbb{A}^1_q$ with center $(\sigma, 0)$, $I_{\bar{G}_\sigma}(F_q) = I_{\bar{G}_\sigma}(E)$, and $a(F_q, \mathbb{A}^3_l \times \mathbb{A}^1_q, 0) = a(E, \mathbb{A}^3_l, 0) + k_G$. Since $I_{\bar{G}_l}(F_q) = I_{\bar{G}_\sigma}(F_q)$, (3) and (4) hold.

Next assume that Z is a curve, with generic point η. Let $I := I_{\bar{G}_l}(Z)$, the subgroup of $\bar{G}_l$ fixing Z pointwise, and $e_Z := |I|$. By **Step 2**,

$$\beta_i = \frac{1 - \mathrm{mult}_Z B_{S_l}}{e_Z}.$$

The ideal of Z in $\mathbb{C}[x']$ is stable under the stabilizer of Z in $\bar{G}_l$, a diagonal group, so it is generated by semi-invariant polynomials, and one of them, γ, restricts to a uniformizer of $\mathcal{O}_{S_l,\eta}$. Since S_l is regular at η, $(\gamma, \bar{f})$ is a regular system of parameters of $\mathcal{O}_{\mathbb{A}^3_l,\eta}$, and $(\gamma, \bar{f}, q)$ is one of the local ring of $\mathbb{A}^3_l \times \mathbb{A}^1_q$ at the generic point of $Z \times \{0\}$. Let $g \in I$ act on γ, $\bar{f}$, and q by $\chi_\gamma(g)$, $\chi_{\bar{f}}(g)$, and $\chi_q(g)$. If $\chi_\gamma(g) = 1$, then g acts trivially on the tangent space $T_{\sigma_1} S_l$ of S_l at a general closed point σ_1 of Z, where S_l is smooth (since g fixes Z pointwise and fixes γ, a local equation of Z on S_l at σ_1, it acts unipotently on $T_{\sigma_1} S_l$, hence trivially, as g has finite order). Let $\mathfrak{m}$ be the maximal ideal of $\widehat{\mathcal{O}}_{S_l,\sigma_1}$ and $u_1, u_2 \in \mathfrak{m}$ elements whose classes form a basis of $\mathfrak{m}/\mathfrak{m}^2$. Since g acts trivially on $\mathfrak{m}/\mathfrak{m}^2$, the g-invariant elements

$$u'_j := \frac{1}{|\langle g \rangle|} \sum_{g' \in \langle g \rangle} g'^* u_j$$

satisfy $u'_j - u_j \in \mathfrak{m}^2$, so $\widehat{\mathcal{O}}_{S_l,\sigma_1} = \mathbb{C}[[u'_1, u'_2]]$ (by the Cohen structure theorem [Har77, Theorem I.5.5A], as u'_1, u'_2 form a regular system of parameters of the complete regular local ring $\widehat{\mathcal{O}}_{S_l,\sigma_1}$) and g acts trivially on $\widehat{\mathcal{O}}_{S_l,\sigma_1}$, hence on its subring $\mathcal{O}_{S_l,\sigma_1}$. Thus g acts trivially near σ_1, hence on the irreducible surface S_l, and $g = 1$ by **Step 1**. Thus $\chi_\gamma|_I$ is injective, I is cyclic, and there exist integers $q_1, q_2 \in \{1, \dots, e_Z\}$ with $\chi_{\bar{f}}|_I = \chi_\gamma^{q_1}|_I$ and $\chi_q|_I = \chi_\gamma^{q_2}|_I$. Let M_1 be the least integer with $M_1 > 1/c_{\min}$ and $M_1 \equiv q_1 \pmod{e_Z}$, and k_G the least integer with $k_G > C_1$ and $k_G \equiv q_2 \pmod{e_Z}$. Let F_q be the prime divisor over $\mathbb{A}^3_l \times \mathbb{A}^1_q$ whose valuation is the monomial valuation in $(\gamma, \bar{f}, q)$ with weights $(1, M_1, k_G)$ at the generic point of $Z \times \{0\}$, and let E be the prime divisor over $\mathbb{A}^3_l$ whose valuation ν is the monomial valuation in $(\gamma, \bar{f})$ with weights $(1, M_1)$ at η. Then (3) holds, $a(E, \mathbb{A}^3_l, 0) = 1 + M_1$, and $a(F_q, \mathbb{A}^3_l \times \mathbb{A}^1_q, 0) = 1 + M_1 + k_G$, as E and F_q are the exceptional divisors of weighted blow-ups along regular parameters.

Let $m_k := \mathrm{mult}_Z(\bar{h}_k|_{S_l})$. Then $c_k m_k \le \mathrm{mult}_Z B_{S_l} \le 1$, so $m_k < M_1$. Since $\bar{h}_k|_{S_l}$ is a unit times γ^{m_k} in $\mathcal{O}_{S_l,\eta}$, we have $\bar{h}_k = u_k \gamma^{m_k} + \bar{f} h'_k$ with a unit u_k and $h'_k \in \mathcal{O}_{\mathbb{A}^3_l,\eta}$, so $\nu(\bar{h}_k) = m_k$. Together with $\nu(\bar{f}) = M_1 \le 1/c_{\min} + N_0 \le C_1$, we obtain (2) and

$$a(E, \mathbb{A}^3_l, \mathrm{div}(\bar{f}) + \bar{B}) = 1 + M_1 - M_1 - \sum_k c_k m_k = 1 - \mathrm{mult}_Z B_{S_l}.$$

An element of $I_{\bar{G}_l}(E)$ or of $I_{\bar{G}_l}(F_q)$ acts trivially on the residue field of η, hence belongs to I. The residue fields of ν and mult_{F_q} are generated over $\mathbb{C}(Z)$ by the classes of $\bar{f}/\gamma^{M_1}$, and of $\bar{f}/\gamma^{M_1}$ and q/γ^{k_G}, respectively, on which $g \in I$ acts by $\chi_{\bar{f}}(g)\chi_\gamma(g)^{-M_1} = 1$ and $\chi_q(g)\chi_\gamma(g)^{-k_G} = 1$. Hence $I_{\bar{G}_l}(E) = I = I_{\bar{G}_l}(F_q)$, $e = e_Z$, and (1) and (4) hold. Finally, $C_1 < k_G \le C_1 + e_Z \le C_2$.

Step 4. In this step we compute the orders of f_i and of the $h_{i,k}$ at the valuation v on $\mathbb{A}^4$ induced by the Gauss extension of ν with ρ-value k_G/r_i.

Let

$$\omega_{\max} := \max_j |w_j|, \qquad \kappa_{\max} := \max\{|\kappa_f|, |\kappa_1|, \dots, |\kappa_s|\},$$
$$C_3 := \omega_{\max}(C_1 + C_2 + a + b + 1) + \kappa_{\max}.$$

Assume that $r_i > C_3$, which holds for i large by Lemma E.1(2). Let $\tilde{v}$ be the Gauss extension of ν over $\mathbb{A}^3_l \times \mathbb{A}^1_\rho$ with ρ-value k_G/r_i (a valuation by [Bou72, Chapter VI, §10, no. 1, Lemma 1], equal to r_i^{-1} times the Gauss extension of $r_i\nu$ with ρ-value k_G [Liu26, Definition-Proposition 2.11]), that is,

$$\tilde{v}\Big(\sum_m a_m \rho^m\Big) = \min_m \Big(\nu(a_m) + \frac{mk_G}{r_i}\Big) \quad \text{for rational functions } a_m \text{ on } \mathbb{A}^3_l.$$

Let $p_q : \mathbb{A}^3_l \times \mathbb{A}^1_\rho \to \mathbb{A}^3_l \times \mathbb{A}^1_q$ be the morphism $p_q(x', \rho) = (x', \rho^{r_i})$. By **Step 3**(3), $\tilde{v}(p_q^*\phi) = \mathrm{mult}_{F_q}\phi$ for rational functions ϕ on $\mathbb{A}^3_l \times \mathbb{A}^1_q$. Let v be the valuation on $\mathbb{A}^4$ given by $v(\phi) := \tilde{v}(\Phi_l^*\phi)$.

Since $Z \subset L_l$, we have $\nu(x'_j) \geq 1$ for any $j \neq l$ with $\omega_j \leq 0$. Hence $v(x_l) = \omega_l k_G/r_i > 0$, $v(x_j) = \omega_j k_G/r_i + \nu(x'_j) > 0$ if $\omega_j > 0$, and $v(x_j) \geq 1 - \omega_{\max}k_G/r_i > 0$ if $\omega_j \leq 0$, since $r_i > \omega_{\max}C_2$. Thus the center of v on $\mathbb{A}^4$ is the origin.

Let P be one of $f_i, h_{i,1}, \dots, h_{i,s}$, let $\kappa := \varsigma\kappa_f$ and $\bar{P} := \bar{f}$ if $P = f_i$, and $\kappa := \varsigma\kappa_k$ and $\bar{P} := \bar{h}_k$ if $P = h_{i,k}$. We show that

$$v(P) = \frac{\kappa k_G}{r_i} + \nu(\bar{P}). \tag{E.40}$$

The polynomial P is G_i-semi-invariant, and $G_i \cap \mathbf{T} = \mu_{r_i}$ acts on a monomial x^α through $\langle\omega, \alpha\rangle$ modulo r_i. Thus the ω-weights of the monomials of P are congruent to κ modulo r_i, and $P = \sum_{m\in\mathbb{Z}} P_m$, where P_m is the sum of the terms of P of ω-weight $\kappa + mr_i$. If $P = h_{i,k}$, then $P_0 = h^0_{i,k}$. If $P = f_i$, write $P_0 = f_i^\circ + \epsilon_i$. Since

$$\Phi_l^* x^\alpha = \rho^{\langle\omega,\alpha\rangle} \prod_{j\neq l} x_j'^{\alpha_j},$$

we have

$$\Phi_l^* P = \sum_m \rho^{\kappa+mr_i} P_m|_{x_l=1}, \qquad \text{hence} \qquad v(P) = \min_m \Big(\frac{(\kappa + mr_i)k_G}{r_i} + \nu(P_m|_{x_l=1})\Big).$$

We show that $\nu(P_0|_{x_l=1}) = \nu(\bar{P})$. If $P = h_{i,k}$, then $P_0|_{x_l=1} = \bar{h}_k$. Let $P = f_i$. A non-leading monomial of ϕ_i (resp. ψ_i) is a monomial of $\phi_i - z^a$ (resp. $\psi_i - t^b$), with ϕ_i, ψ_i as in (E.16). In case (b), and in case (a) with $w_t = 0 < w_z$, we have $\epsilon_i = 0$: if $w_t = 0$, then $\phi_i = z^a$, and a non-leading monomial of ψ_i has positive z-degree, hence a $\mathbf{T}$-weight different from that of t^b. Let $w_z < 0 < w_t$ in case (a), so that $\kappa_f = aw_z + bw_t$. A non-leading monomial x^α of ϕ_i, with $\alpha_3 < a$ and $\alpha_4 \geq 1$, has the character of z^a on μ_{r_i} and $\mathbf{T}$-weight $\alpha_3 w_z + \alpha_4 w_t > aw_z$, hence $\mathbf{T}$-weight at least $aw_z + r_i$, and $\alpha_4 w_t \geq r_i - a|w_z|$. Likewise a non-leading monomial x^α of ψ_i, with $\alpha_4 < b$ and $\alpha_3 \geq 1$, has $\mathbf{T}$-weight at most $bw_t - r_i$, and $\alpha_3|w_z| \geq r_i - bw_t$. The product of z^a and a non-leading monomial of ψ_i has $\mathbf{T}$-weight less than κ_f, and the product of t^b and a non-leading monomial of ϕ_i has $\mathbf{T}$-weight greater than κ_f. Thus, by (E.16), every monomial x^α of ϵ_i is a product of non-leading monomials of ϕ_i and ψ_i, and

$$\alpha_3 \geq \frac{r_i - bw_t}{|w_z|} \qquad \text{and} \qquad \alpha_4 \geq \frac{r_i - a|w_z|}{w_t}.$$

Since $Z \subset L_l$, we have $\nu(x'_z) \geq 1$ if $\star = +$ and $\nu(x'_t) \geq 1$ if $\star = -$. Since $r_i > C_3 \geq \omega_{\max}(C_1 + a + b)$, it follows that $\nu(\epsilon_i|_{x_l=1}) > C_1 \geq \nu(\bar{f})$ by **Step 3**(2), so $\nu(P_0|_{x_l=1}) = \nu(\bar{f})$.

The term $m = 0$ is $\kappa k_G/r_i + \nu(\bar{P})$, and $\nu(\bar{P}) \leq C_1 < k_G$ by **Step 3**(2),(3). If $m \geq 1$, the term is at least $\kappa k_G/r_i + k_G$, since $\nu \geq 0$ on polynomials. Let $m \leq -1$ and let x^α be a monomial of P_m. Then

$$\sum_{\omega_j<0} |\omega_j|\alpha_j = \sum_{\omega_j>0} \omega_j\alpha_j - \kappa - mr_i \geq |m|r_i - \kappa_{\max},$$

so

$$\nu\Big(\prod_{j\neq l} x_j'^{\alpha_j}\Big) \geq \sum_{\omega_j<0} \alpha_j \geq \frac{|m|r_i - \kappa_{\max}}{\omega_{\max}},$$

and the term with index m is at least

$$\frac{\kappa k_G}{r_i} + |m|\Big(\frac{r_i}{\omega_{\max}} - k_G\Big) - \frac{\kappa_{\max}}{\omega_{\max}} \geq \frac{\kappa k_G}{r_i} + \frac{r_i - \kappa_{\max}}{\omega_{\max}} - k_G > \frac{\kappa k_G}{r_i} + C_1,$$

since $r_i > C_3$. Thus the minimum is attained only at $m = 0$, which proves (E.40).

Step 5. In this step we compute the log discrepancy at v and prove the upper bound.

For a smooth variety X and a positive real multiple $u = \tau \operatorname{mult}_\Sigma$ of the valuation of a prime divisor Σ over X, let $A_X(u) := \tau a(\Sigma, X, 0)$. Let $p : X' \dashrightarrow X$ be a dominant generically finite rational map of smooth varieties, let Σ' be a prime divisor over X', let Σ be the prime divisor over X dominated by Σ', with ramification index $e(\Sigma'/\Sigma)$, and let $dX = J dX'$, where dX and dX' are top-degree forms generating the canonical sheaves near the centers of Σ and Σ' and J is a rational function on X'. Then

$$e(\Sigma'/\Sigma) a(\Sigma, X, 0) = a(\Sigma', X', 0) + \operatorname{mult}_{\Sigma'} J$$

by the Hurwitz formula applied on higher models of X and X' on which Σ and Σ' are prime divisors: near the generic point of Σ', the form dX vanishes to order $e(\Sigma'/\Sigma)(a(\Sigma, X, 0) - 1) + (e(\Sigma'/\Sigma) - 1)$ along Σ', the form dX' vanishes to order $a(\Sigma', X', 0) - 1$ along Σ', and $dX = J dX'$. Hence, if $u = \tau \operatorname{mult}_{\Sigma'}$ for some $\tau > 0$, then $u(p^*\phi) = \tau e(\Sigma'/\Sigma) \operatorname{mult}_\Sigma \phi$ for rational functions ϕ on X, and

$$A_X\big(\tau e(\Sigma'/\Sigma) \operatorname{mult}_\Sigma\big) = A_{X'}(u) + u(J). \tag{E.41}$$

We have $\tilde{v} = \tau_\rho \operatorname{mult}_{F_\rho}$ for a prime divisor F_ρ over $\mathbb{A}^3_l \times \mathbb{A}^1_\rho$ dominating F_q and some $\tau_\rho > 0$, since $\tilde{v}(p_q^*\phi) = \operatorname{mult}_{F_q} \phi$ and p_q is finite. The valuation $\tilde{v}$ has a center on $\mathbb{A}^3_l \times \mathbb{A}^1_\rho$, since $\tilde{v}(x'_j) \geq 0$ and $\tilde{v}(\rho) > 0$. Since Φ_l is given by $x_l = \rho^{\omega_l}$ and $x_j = \rho^{\omega_j} x'_j$ for $j \neq l$, and p_q by $q = \rho^{r_i}$,

$$\Phi_l^*(dx \wedge dy \wedge dz \wedge dt) = \pm\omega_l \rho^{\sum_j \omega_j - 1} d\rho \wedge \bigwedge_{j \neq l} dx'_j, \qquad p_q^*\Big(dq \wedge \bigwedge_{j \neq l} dx'_j\Big) = r_i \rho^{r_i - 1} d\rho \wedge \bigwedge_{j \neq l} dx'_j.$$

Applying (E.41) with $X' = \mathbb{A}^3_l \times \mathbb{A}^1_\rho$ and $u = \tilde{v}$ to $p = \Phi_l$ and to $p = p_q$, whose induced valuations on $\mathbb{A}^4$ and on $\mathbb{A}^3_l \times \mathbb{A}^1_q$ are v and $\operatorname{mult}_{F_q}$, and using **Step 3**(4), we obtain

$$A_{\mathbb{A}^4}(v) = A_{\mathbb{A}^3_l \times \mathbb{A}^1_q}(\operatorname{mult}_{F_q}) + \Big(\sum_j \omega_j - r_i\Big)\frac{k_G}{r_i} = a(E, \mathbb{A}^3_l, 0) + \Big(\sum_j \omega_j\Big)\frac{k_G}{r_i}.$$

Let

$$\Delta_i := \operatorname{div}(f_i) + \sum_k c_k \operatorname{div}(h_{i,k}) \text{ on } \mathbb{A}^4 \qquad \text{and} \qquad A_{\mathbb{A}^4, \Delta_i}(v) := A_{\mathbb{A}^4}(v) - v(f_i) - \sum_k c_k v(h_{i,k}).$$

By (E.40) and **Step 3**(1),

$$A_{\mathbb{A}^4, \Delta_i}(v) = a(E, \mathbb{A}^3_l, \operatorname{div}(\bar{f}) + \bar{B}) + \frac{k_G}{r_i}\Big(\sum_j \omega_j - \varsigma\kappa_f - \varsigma \sum_k c_k \kappa_k\Big) = e\beta_i - \frac{\varsigma\delta k_G}{r_i},$$

since $\sum_j \omega_j - \varsigma\kappa_f = \varsigma(w_z + w_t)$ and $\delta = \sum_k c_k \kappa_k - w_z - w_t$.

Let $Q := \mathbb{A}^4/G_i$, $\pi_Q : \mathbb{A}^4 \to Q$ the quotient map, $\bar{0}$ the image of the origin, and Δ_Q the $\mathbb{R}$-divisor on Q such that $K_{\mathbb{A}^4} + \Delta_i$ is the pullback of $K_Q + \Delta_Q$. Let $\bar{F}$ be the image of F_q on $(\mathbb{A}^3_l \times \mathbb{A}^1_q)/\bar{G}_l$, a prime divisor over Q through the birational map (E.39). The ramification index of F_q over $\bar{F}$ is $|I_{\bar{G}_l}(F_q)| = e$ by **Step 3**(4), so $v(\pi_Q^*\bar{\phi}) = e \operatorname{mult}_{\bar{F}} \bar{\phi}$ for rational functions $\bar{\phi}$ on Q, and the center of $\bar{F}$ on Q is $\bar{0}$, since the center of v on $\mathbb{A}^4$ is the origin. Write $v = \tau_F \operatorname{mult}_F$ for a prime divisor F over $\mathbb{A}^4$ and $\tau_F > 0$. Then the image of F on Q is $\bar{F}$ and $\operatorname{mult}_F(\pi_Q^*\bar{\phi}) = |I_{G_i}(F)| \operatorname{mult}_{\bar{F}} \bar{\phi}$, so $\tau_F |I_{G_i}(F)| = e$, and $a(F, \mathbb{A}^4, \Delta_i) = |I_{G_i}(F)| a(\bar{F}, Q, \Delta_Q)$ by [Liu26, Lemma 2.6]. Hence

$$a(\bar{F}, Q, \Delta_Q) = \frac{\tau_F a(F, \mathbb{A}^4, \Delta_i)}{e} = \frac{A_{\mathbb{A}^4, \Delta_i}(v)}{e} = \beta_i - \frac{\varsigma\delta k_G}{e r_i}.$$

The hypersurface $W_i = (f_i = 0)$ is, near 0, normal, integral, and klt, G_i acts faithfully on it, the $h_{i,k}$ are G_i-semi-invariant and non-zero on W_i, and (W_i, D_i) is klt near 0 (Construction-Theorem 4.13(1) and (3)). By Theorem 3.3(1),

$$\alpha_i = \mathrm{mld}((W_i \ni 0, D_i)/G_i) = \mathrm{mld}((\mathbb{A}^4 \ni 0, \Delta_i)/G_i),$$

and

$$\mathrm{mld}((\mathbb{A}^4 \ni 0, \Delta_i)/G_i) = \mathrm{mld}(Q \ni \bar{0}, \Delta_Q) \le a(\bar{F}, Q, \Delta_Q).$$

Since $e \ge 1$ and $k_G \le C_2$, we conclude that

$$\alpha_i \le \beta_i + \frac{|\delta| k_G}{e r_i} \le \beta_i + \frac{C_0}{r_i}, \qquad C_0 := |\delta| C_2,$$

for i large. The constants N_0, l_3, C_1, C_2, and δ do not depend on i, so neither does C_0. □

E.4. **Quadratic rank two, fixed split radical order: closing the sequence.** In this subsection, we prove Proposition 11.12. The reference pairs of Construction E.2 are pairs with an ideal on a fixed plane. Before we take a compatible generic limit, we replace the ideal by a fixed number of general members, and the following lemma shows that the replacement does not change lc near the origin or the quotient value at the origin.

Lemma E.8. *Let $\bar{G} \subset GL_2(\mathbb{C})$ be a finite group acting linearly on $\mathbb{A}^2$, let $\mathfrak{m}$ be the maximal ideal of the origin $0 \in \mathbb{A}^2$, let Δ_0 be a $\bar{G}$-invariant reduced divisor on $\mathbb{A}^2$, and let $\gamma_1, \dots, \gamma_n$ be $\bar{G}$-invariant polynomials generating an ideal $\mathfrak{a}$ with $\mathfrak{m}^{N_0} \subset \mathfrak{a}$ for a positive integer N_0. Let M be a positive integer. Then there exists a non-empty Zariski open subset $\Omega \subset \mathbb{C}^{Mn}$ with the following property. Let $(t_{l,j})_{1 \le l \le M, 1 \le j \le n} \in \Omega$, let $\eta_l := \sum_{j=1}^n t_{l,j}\gamma_j$ for $1 \le l \le M$, let $\Delta \ge 0$ be a $\bar{G}$-invariant $\mathbb{R}$-divisor with $\operatorname{Supp}\Delta \subset \Delta_0$, let $\sigma \in [0, M)$ be a real number, and let*

$$\Delta_\eta := \Delta + \frac{\sigma}{M}\sum_{l=1}^{M} \operatorname{div}(\eta_l).$$

Then the following hold.

(1) The pair $(\mathbb{A}^2, \Delta_\eta)$ is lc near 0 if and only if $(\mathbb{A}^2, \Delta, \mathfrak{a}^\sigma)$ is lc near 0.

(2) If the pairs of (1) are lc near 0, then

$$\mathrm{mld}\big((\mathbb{A}^2 \ni 0, \Delta_\eta)/\bar{G}\big) = \mathrm{mld}\big((\mathbb{A}^2 \ni 0, \Delta, \mathfrak{a}^\sigma)/\bar{G}\big).$$

(3) Each η_l is non-zero, $\bar{G}$-invariant, and $\operatorname{mult}_0 \eta_l \le N_0$.

Proof. Let $f : Y \to \mathbb{A}^2$ be a $\bar{G}$-equivariant log resolution of $(\mathbb{A}^2, \Delta_0)$ and of $\mathfrak{a}$ near 0, obtained by blowing up $\bar{G}$-orbits of closed points, so that $\mathfrak{a}\mathcal{O}_Y = \mathcal{O}_Y(-F_\mathfrak{a})$ for a divisor $F_\mathfrak{a} \ge 0$ supported on $f^{-1}(0)$, and $f^{-1}(0) \cup \operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}\Delta_0$ is an snc divisor near $f^{-1}(0)$. Let $\Sigma \subset f^{-1}(0)$ be the finite set of the singular points of $f^{-1}(0) \cup \operatorname{Supp} f_*^{-1}\Delta_0$ together with the points y of $f^{-1}(0)$ whose stabilizer in $\bar{G}$ is not the inertia group $I_{\bar{G}}(\Gamma)$ of the component Γ of $f^{-1}(0)$ through y. The set Σ is finite since $\operatorname{Stab}_{\bar{G}}(\Gamma)/I_{\bar{G}}(\Gamma)$ acts faithfully on Γ.

The pullbacks $f^*\gamma_j$ generate $\mathcal{O}_Y(-F_\mathfrak{a})$, so the divisors $f^*\operatorname{div}(\eta) - F_\mathfrak{a}$, for η a linear combination of the γ_j, form a base point free linear system near $f^{-1}(0)$. By Bertini's theorem there exists a non-empty open subset Ω such that for $(t_{l,j}) \in \Omega$ the divisors $S_l := f^*\operatorname{div}(\eta_l) - F_\mathfrak{a}$ are smooth near $f^{-1}(0)$, intersect $f^{-1}(0) \cup \operatorname{Supp} f_*^{-1}\Delta_0$ transversally and only at points outside Σ, and satisfy $S_l \cap S_{l'} \cap f^{-1}(0) = \emptyset$ for $l \ne l'$. Moreover $\eta_l \ne 0$ and $\operatorname{mult}_0 \eta_l = \min_j \operatorname{mult}_0 \gamma_j \le N_0$, the last inequality because $\mathfrak{m}^{N_0} \subset \mathfrak{a}$. Each η_l is $\bar{G}$-invariant, so S_l is $\bar{G}$-invariant. This proves (3).

Define Δ_Y by $K_Y + \Delta_Y = f^*(K_{\mathbb{A}^2} + \Delta) + \sigma F_\mathfrak{a}$. Since $\mathfrak{a}\mathcal{O}_Y = \mathcal{O}_Y(-F_\mathfrak{a})$ and $f^*\operatorname{div}(\eta_l) = F_\mathfrak{a} + S_l$, for any prime divisor P over Y we have

$$a(P, \mathbb{A}^2, \Delta, \mathfrak{a}^\sigma) = a(P, Y, \Delta_Y),$$

$$a\Big(P, \mathbb{A}^2, \Delta + \frac{\sigma}{M}\sum_l \operatorname{div}(\eta_l)\Big) = a\Big(P, Y, \Delta_Y + \frac{\sigma}{M}\sum_l S_l\Big). \tag{E.42}$$

Near $f^{-1}(0)$ the divisor $\operatorname{Supp}\Delta_Y \cup \bigcup_l S_l$ is snc, and the coefficient of each S_l is $\sigma/M < 1$. Hence the pairs in (E.42) are lc near $f^{-1}(0)$ if and only if the coefficients of Δ_Y near $f^{-1}(0)$ are at most one [KM98, Corollary 2.31(3)]. Since f is proper, this proves (1).

For (2), recall that the quotient value at 0 is the minimum of $a(P)/|I_{\bar G}(P)|$ over the prime divisors P over $\mathbb{A}^2$ with center 0 [Liu26, Lemma 2.6]. If the center of P on Y is not contained in $\bigcup_l S_l$, then the values of P for the pairs of (E.42) coincide by (E.42). Otherwise the center of P is a point $y \in S_l \cap f^{-1}(0)$ for exactly one l, and $y \notin \Sigma$. Let Γ be the component of $f^{-1}(0)$ through y, let $d := \operatorname{mult}_\Gamma \Delta_Y \le 1$, and let $I := I_{\bar G}(\Gamma) = \operatorname{Stab}_{\bar G}(y)$, a cyclic group of order e_Γ that fixes Γ pointwise. Since $I_{\bar G}(P)$ fixes the center of P, we have $I_{\bar G}(P) \subset I$. We show that Y/I is smooth near the image of y. Let χ be the character by which I acts on the conormal line of Γ at y, let v be a local equation of Γ at y, and let u be a regular function near y whose restriction to Γ is a local coordinate at y. Since I fixes Γ pointwise, the functions $\tilde u := \frac{1}{e_\Gamma}\sum_{g\in I} g^*u$ and $\tilde v := \frac{1}{e_\Gamma}\sum_{g\in I}\chi(g)^{-1}g^*v$ satisfy $\tilde u|_\Gamma = u|_\Gamma$, $\tilde v \in v\mathcal{O}_{Y,y}$, and $\tilde v - v \in \mathfrak{m}_y^2$, so $\tilde u, \tilde v$ are local coordinates at y with $g^*\tilde u = \tilde u$ and $g^*\tilde v = \chi(g)\tilde v$ for any $g \in I$ (cf. [Car57]). An element of $\ker\chi$ fixes $\tilde u$ and $\tilde v$, so it acts trivially on $\widehat{\mathcal{O}}_{Y,y}$ and hence on Y. Thus χ is faithful, and the completion of the local ring of Y/I at the image of y is $\widehat{\mathcal{O}}^I_{Y,y}$ [Liu26, Lemma 3.3(1)], which is $\mathbb{C}[[\tilde u, \tilde v^{e_\Gamma}]]$. Hence Y/I is smooth near the image of y. The image Γ' of Γ has coefficient $d' := 1 - \frac{1-d}{e_\Gamma}$ in the crepant boundary. The curve S_l is the preimage of the smooth curve $S_l' := S_l/I$, which is transversal to Γ'. No non-identity element of I fixes S_l pointwise, since T_yS_l is the eigenline of I on which I acts by a faithful character, so the quotient map is unramified along S_l. Hence, for $\varsigma \in \{0, \sigma/M\}$, the crepant boundary of $(Y, \Delta_Y + \varsigma S_l)$ near the image of y is $d'\Gamma' + \varsigma S_l'$ [Liu26, Lemma 2.6(4)]. Let P' be the prime divisor over Y/I induced by P. By [Liu26, Lemmas 2.6(2) and 2.13], for $\varsigma \in \{0, \sigma/M\}$,

$$\begin{aligned}\frac{a(P, Y, \Delta_Y + \varsigma S_l)}{|I_{\bar G}(P)|} &= a(P', Y/I, d'\Gamma' + \varsigma S_l')\\ &\ge (1-d')\operatorname{mult}_{P'}\Gamma' + (1-\varsigma)\operatorname{mult}_{P'} S_l'\\ &\ge 1 - d' = \frac{a(\Gamma, Y, \Delta_Y)}{|I_{\bar G}(\Gamma)|}.\end{aligned}$$

The right hand side is the value of Γ for the pairs of (E.42), and Γ has center 0. Thus both quotient values are equal to the minimum over the prime divisors whose center on Y is not contained in $\bigcup_l S_l$, where the values coincide by (E.42). This proves (2). □

Proof of Proposition 11.12. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce to the statement $\alpha_i \ge \beta$ for one index i and one real number β, and we fix the side of the graded quotients.

Assume to the contrary that there exists a sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \ge 1$, as in Set-up 11.1, with W_i of quadratic rank two with a fixed split radical order, whose numbers

$$\alpha_i := \operatorname{mld}((W_i \ni o_i, \sum_k c_k \operatorname{div}(h_{i,k}))/G_i)$$

form a strictly increasing sequence in $(0,1)$, and let $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$. By Lemma E.1, after passing to a subsequence and replacing the data as there, we have integers $a, b \ge 0$, a primitive vector $w = (w_x, w_y, w_z, w_t)$ and a positive integer e_0 as in that lemma, and we use the objects V_i, κ_k, $h^0_{i,k}$, δ, $(Y_i^\pm, B_i^\pm(c'))$, $C_i^\pm$, $\beta_i^\pm$, and β_i of Construction E.2, with $r_i = |G_i \cap \mathbf{T}|$. After the replacements, $W_i = (xy + g_i(z,t) = 0) \subset \mathbb{A}^4$ by Lemma E.1(1), and $o_i = 0$. The replacements in Lemma E.1 and in Construction E.2 are by numerical admissions that keep the groups G_i, which do not change the numbers α_i by Proposition 11.8. Thus the new α_i are still strictly increasing in $(0,1)$, and a contradiction for the new data is a contradiction for the original ones. By Lemma E.7, for i large the numbers β_i belong to a finite set and $\alpha_i \le \beta_i + C_0/r_i$. Since

$r_i \to +\infty$ by Lemma E.1(2), possibly passing to a subsequence, we may assume that $\beta_i = \beta$ for any i, where β is a real number with

$$\beta \geq \lim_{i\to+\infty} \alpha_i > \alpha_j > 0 \quad \text{for any } j.$$

Thus it suffices to show that $\alpha_i \geq \beta$ for some i.

For $\theta \geq 1$ let $\delta_\theta := \theta \sum_k c_k\kappa_k - w_z - w_t$, so that $\delta_1 = \delta$ and $\delta_\theta = \theta\delta + (\theta-1)(w_z + w_t)$. We choose a sign $\epsilon \in \{+,-\}$ as follows. In case (a) of Lemma E.1(3), let $\epsilon := +$ if $\delta < 0$, or if $\delta = 0$ and $w_z + w_t \leq 0$, and $\epsilon := -$ otherwise. In case (b), let $\epsilon := +$ if $\delta < 0$ and $\epsilon := -$ if $\delta \geq 0$. Fix a real number $\theta_0 > 1$ such that δ_θ has the sign of δ for $\theta \in [1,\theta_0]$ when $\delta \neq 0$. Then the following hold for i large.

(i) $\beta_i = \beta_i^\epsilon$, and $(Y_i^\epsilon, B_i^\epsilon)$ is lc near C_i^ϵ.
(ii) For $\theta \in (1,\theta_0]$, $\epsilon\delta_\theta \leq 0$.

By Proposition E.3(1)(2) in case (a) and by Proposition E.4(2) in case (b), (i) holds. For (ii), let $\theta \in (1,\theta_0]$. If $\delta < 0$, then $\delta_\theta < 0$ by the choice of θ_0, and $\epsilon = +$ in both cases. If $\delta > 0$, then $\delta_\theta > 0$, and $\epsilon = -$ in both cases. If $\delta = 0$, then $\delta_\theta = (\theta-1)(w_z + w_t)$ has the sign of $w_z + w_t$. In case (a), $\epsilon = +$ if $w_z + w_t \leq 0$, and $\epsilon = -$ if $w_z + w_t > 0$. In case (b), $w_z + w_t > 0$ since $w_z, w_t > 0$, and $\epsilon = -$. Thus, by Propositions E.3(3) and E.4(3) with $\sigma = \epsilon$, if there exists $\theta \in (1,\theta_0]$ such that $(Y_i^\epsilon, B_i^\epsilon(\theta c))$ is lc near C_i^ϵ for any i, then $\alpha_i \geq \beta_i = \beta$ for i large.

Step 2. In this step we write the pairs $(Y_i^\epsilon, B_i^\epsilon(\theta c))$, $\theta \in [1,\theta_0]$, as pairs with an ideal on a fixed plane with a fixed group.

We show that, possibly passing to a subsequence, there exist a finite diagonal group $\bar G \subset GL_2(\mathbb{C})$ acting on $\mathbb{A}^2$ with coordinates ζ_1, ζ_2; affine functions $\omega_1, \omega_2 : [1,\theta_0] \to \mathbb{R}_{\geq 0}$ with $\omega_j(\theta) \leq \theta\omega_j(1)$; non-zero $\bar G$-semi-invariant polynomials $f_{i,k}$, $1 \leq k \leq s$, whose characters depend only on k, with $c_k \operatorname{mult}_0 f_{i,k} \leq 2$; a positive integer N_0 and $\bar G$-invariant polynomials $\gamma_{i,1}, \dots, \gamma_{i,n_1}$ of degree bounded independently of i generating an ideal $\mathfrak{a}_i$ with $\mathfrak{m}^{N_0} \subset \mathfrak{a}_i$; and a continuous function $\sigma : [1,\theta_0] \to \mathbb{R}_{\geq 0}$, such that, with

$$\Delta_i(\theta) := \sum_k \theta c_k \operatorname{div}(f_{i,k}) + \omega_1(\theta)\operatorname{div}(\zeta_1) + \omega_2(\theta)\operatorname{div}(\zeta_2),$$

for any $\theta \in [1,\theta_0]$ and i large,

$$\begin{gathered}(Y_i^\epsilon, B_i^\epsilon(\theta c)) \text{ is lc near } C_i^\epsilon \iff (\mathbb{A}^2, \Delta_i(\theta), \mathfrak{a}_i^{\sigma(\theta)}) \text{ is lc near } 0,\\ \beta = \beta_i = \operatorname{mld}\big((\mathbb{A}^2 \ni 0, \Delta_i(1), \mathfrak{a}_i^{\sigma(1)})/\bar G\big).\end{gathered} \tag{E.43}$$

Moreover, to any $\mathbf{T}$-invariant G_i-semi-invariant regular function U on V_i we attach below a polynomial u on $\mathbb{A}^2$ with the following property. For any real number $\lambda \geq 0$, the graded quotient $(Y_i^\epsilon, B_i^\epsilon + \lambda\operatorname{div}(U))$ on the side ϵ of

$$\Big(V_i, \sum_k c_k \operatorname{div}(h_{i,k}^0|_{V_i}) + \lambda\operatorname{div}(U|_{V_i})\Big),$$

followed by the quotient by $G_i/(G_i \cap \mathbf{T})$, is lc near C_i^ϵ if and only if

$$\big(\mathbb{A}^2, \Delta_i(1) + \lambda\operatorname{div}(u), \mathfrak{a}_i^{\sigma(1)}\big)$$

is lc near 0.

Case (a) of Lemma E.1(3). Here $V_i = V := (xy + z^a t^b = 0)$ does not depend on i. The line $\mathbb{R}w$ intersects the cone $\{X + Y = aZ + bT\} \cap \mathbb{R}^4_{\geq 0}$ only at 0, because w has a negative and a positive coordinate ($w_z < 0 < w_t$ or $w_x < 0 < w_y$). The group G_i is diagonal, so Lemma 8.12 can be used for V, w, G_i, and $D_V := \sum_k \theta c_k \operatorname{div}(h_{i,k}^0|_V)$, whose number δ is δ_θ. Let Q_0 be the toric surface of that lemma, and let $H_i := G_i/(G_i \cap \mathbf{T})$, which acts faithfully on Q_0 through the torus of Q_0. Let Y^ϵ be the graded quotient of V on the side ϵ in Lemma 8.12, which does not depend on i, and let $\tilde B_i^\epsilon(\theta c)$ be the $\mathbb{R}$-divisor B^ϵ of that lemma for D_V, so that $(Y_i^\epsilon, B_i^\epsilon(\theta c))$ is the crepant quotient of $(Y^\epsilon, \tilde B_i^\epsilon(\theta c))$ by H_i. By **Step 1**(ii) for $\theta > 1$, and by the choice of ϵ for

$\theta = 1$, where $\delta_1 = \delta$, we have $\epsilon\delta_\theta \le 0$, so ϵ is a sign allowed in Lemma 8.12 for δ_θ. Let $B_{0,i}(\theta)$ be the push-forward of $\tilde{B}_i^\epsilon(\theta c)$ to Q_0. By Lemma 8.12(3), there exist an H_i-invariant monomial ideal I on Q_0, equal to (1) or primary at the origin, and a positive integer N' (the integer N of that lemma), depending only on a, b, w, and the sign of δ_{θ_0}, such that for any prime divisor P over Q_0,

$$a\big(P, Y^\epsilon, \tilde{B}_i^\epsilon(\theta c)\big) = a\big(P, Q_0, B_{0,i}(\theta), I^{|\delta_\theta|/N'}\big). \tag{E.44}$$

By Lemma 8.12(3), I and N' depend on δ_θ only through its sign. If $\delta \ne 0$, then δ_θ has the sign of δ_{θ_0} for any $\theta \in [1, \theta_0]$ by the choice of θ_0. If $\delta = 0$, then $\delta_\theta = (\theta - 1)(w_z + w_t)$ has the sign of δ_{θ_0} for $\theta \in (1, \theta_0]$, and for $\theta = 1$ the exponent $|\delta_1|/N'$ is 0, so the ideal plays no role. Hence the ideal I and the integer N' given by Lemma 8.12(3) for δ_{θ_0} serve in (E.44) for any $\theta \in [1, \theta_0]$.

Since Q_0 is a normal affine toric surface, the sublattice generated by the primitive generators of the rays of Q_0 has finite index in the lattice of one-parameter subgroups, and the inclusion defines a finite toric morphism $\varphi : \mathbb{A}^2 \to Q_0$, the quotient by a finite cyclic diagonal group Ξ acting freely in codimension one (the affine toric variety of a simplicial cone is the quotient of an affine space by a finite abelian group, see [CLS11, Section 1.3]). Let ζ_1, ζ_2 be the coordinates of $\mathbb{A}^2$, with $\varphi^* T_1 = \operatorname{div}(\zeta_1)$ and $\varphi^* T_2 = \operatorname{div}(\zeta_2)$ for the toric prime divisors T_1, T_2 of Q_0. The image of G_i in the torus of Q_0 is a group of order at most e_0, and its preimage in the torus of $\mathbb{A}^2$ is a finite diagonal group of order at most $|\Xi| e_0$ which contains Ξ and whose quotient by Ξ is isomorphic to H_i. There are finitely many finite diagonal subgroups of $GL_2(\mathbb{C})$ of order at most $|\Xi| e_0$, so possibly passing to a subsequence, we may assume that the preimage of the image of G_i in the torus of $\mathbb{A}^2$ is a group $\bar{G}$ independent of i.

The boundary of the graded quotient is the sum of the images of the divisors $\operatorname{div}(h^0_{i,k}|_{V_v})$ on the sections $V_v = V \cap (v = 1)$, $v \in \{x, y, z, t\}$ with $\epsilon w_v > 0$ (the sections V_ℓ of Lemma 8.12), with coefficients θc_k, and of the branch divisors of the quotients by $\mu_{|w_v|}$. The branch divisors are torus-invariant. For $j = 1, 2$, let e_j be the ramification index of the quotient maps $V_v \to V_v/\mu_{|w_v|}$ along the strict transform of T_j on Y^ϵ, which depends only on a, b, w, and ϵ. By [Liu26, Lemma 2.6(4)],

$$B_{0,i}(\theta) = \sum_k \theta c_k \Gamma_{i,k} + \Big(1 - \frac{1}{e_1}\Big) T_1 + \Big(1 - \frac{1}{e_2}\Big) T_2$$

with $\mathbb{Q}$-divisors $\Gamma_{i,k} \ge 0$ whose coefficients are integers except along T_1 and T_2, where they belong to $\frac{1}{e_1}\mathbb{Z}$ and $\frac{1}{e_2}\mathbb{Z}$. Since $\varphi^* T_j = \operatorname{div}(\zeta_j)$, we may write

$$\varphi^* \Gamma_{i,k} = \operatorname{div}(f_{i,k}) + q_{i,k,1} \operatorname{div}(\zeta_1) + q_{i,k,2} \operatorname{div}(\zeta_2)$$

with $f_{i,k} \in \mathbb{C}[\zeta_1, \zeta_2]$ and $q_{i,k,j} \in [0, 1) \cap \frac{1}{e_j}\mathbb{Z}$. The polynomial $f_{i,k}$ is $\bar{G}$-semi-invariant since $\Gamma_{i,k}$ is H_i-invariant, and possibly passing to a subsequence, the character of $f_{i,k}$ depends only on k, and $q_{i,k,j} = q_{k,j}$ does not depend on i. Let

$$\omega_j(\theta) := 1 - \frac{1}{e_j} + \theta \sum_k c_k q_{k,j} \quad \text{for } j = 1, 2,$$

so that $\omega_j(\theta) \le \theta \omega_j(1)$ for $\theta \ge 1$. Let $m_1, \dots, m_{n_1}$ be monomial generators of the $\bar{G}$-invariant monomial ideal $I\mathcal{O}_{\mathbb{A}^2}$, and let $\gamma_{i,j} := m_j^{|\bar{G}|}$, which are $\bar{G}$-invariant and do not depend on i. They generate an ideal $\mathfrak{a}_i = \mathfrak{a}$ with $\operatorname{mult}_P \mathfrak{a} = |\bar{G}| \operatorname{mult}_P(I\mathcal{O}_{\mathbb{A}^2})$ for any prime divisor P, and $\mathfrak{m}^{N_0} \subset \mathfrak{a}$ for some N_0 since I is (1) or primary at the origin. Let $\sigma(\theta) := |\delta_\theta|/(N'|\bar{G}|)$. Since φ is quasi-étale,

$$\varphi^*(K_{Q_0} + B_{0,i}(\theta)) = K_{\mathbb{A}^2} + \Delta_i(\theta).$$

By (E.44), [Liu26, Lemma 2.6(2)(5)], and Definition 2.4, the log discrepancies over Y_i^ϵ with center in C_i^ϵ are, up to the positive factors $|I_{\bar{G}}(P)|$, those of $(\mathbb{A}^2, \Delta_i(\theta), \mathfrak{a}^{\sigma(\theta)})$ over prime divisors with center 0, and the divisors over Y_i^ϵ with center intersecting a neighborhood of C_i^ϵ correspond to the divisors over $\mathbb{A}^2$ with center intersecting a neighborhood of 0, since $Y^\epsilon \to Q_0$ is projective.

This proves (E.43). Applying (E.43) with $\theta = 1$, **Step 1**(i) implies that $(\mathbb{A}^2, \Delta_i(1))$ is lc near 0, so $\operatorname{mult}_0 \Delta_i(1) \le 2$ and $c_k \operatorname{mult}_0 f_{i,k} \le 2$.

If U is a $\mathbf{T}$-invariant G_i-semi-invariant regular function on V, then U is the pullback of a regular function u_0 on $Q_0 = \operatorname{Spec} \mathbb{C}[V]^{\mathbf{T}}$, and we let $u := \varphi^* u_0$. Since $\operatorname{div}(U)$ on Y^ϵ is the pullback of $\operatorname{div}(u_0)$, (E.44) implies

$$a\big(P, Y^\epsilon, \tilde{B}_i^\epsilon(c) + \lambda \operatorname{div}(U)\big) = a\big(P, Q_0, B_{0,i}(1) + \lambda \operatorname{div}(u_0), I^{|\delta|/N'}\big)$$

for any prime divisor P over Q_0. Since $u = \varphi^* u_0$, the argument proving (E.43) for $\theta = 1$, with $B_{0,i}(1) + \lambda \operatorname{div}(u_0)$ in place of $B_{0,i}(1)$ and $\Delta_i(1) + \lambda \operatorname{div}(u)$ in place of $\Delta_i(1)$, proves the last statement of the step.

Case (b) of Lemma E.1(3). Here $w = (-p, N+p, w_z, w_t)$ and $V_i = (xy + g_i^0(z,t) = 0)$. On the x-root chart $x = \xi^p$, with $\zeta_1 := \xi^{w_z} z$ and $\zeta_2 := \xi^{w_t} t$, the section $(x = 1)$ of V_i is the plane with coordinates ζ_1, ζ_2, on which $y = -g_i^0(\zeta_1, \zeta_2)$, and the graded part of a mark is

$$P_{i,k}(\zeta_1, \zeta_2) = h_{i,k}^0\big(1, -g_i^0(\zeta_1, \zeta_2), \zeta_1, \zeta_2\big).$$

By Proposition E.4(1), $Y_i^- = \mathbb{A}^2/\bar{G}_i$ for the plane group $\bar{G}_i$ of that proposition, a finite diagonal group of order $p[G_i : G_i \cap \mathbf{T}] \le pe_0$ which contains the image of the central $\mu_p \subset \mathbf{T}$, and the crepant pullback of $(Y_i^-, B_i^-(c'))$ to $\mathbb{A}^2$ is $(\mathbb{A}^2, \sum_k c'_k \operatorname{div}(P_{i,k}))$. Possibly passing to a subsequence, we may assume that $\bar{G}_i = \bar{G}$ and that the characters of the $P_{i,k}$ depend only on k. Let $f_{i,k} := P_{i,k}$ and $\omega_1 = \omega_2 := 0$.

If $\epsilon = -$, let $\mathfrak{a}_i := \mathcal{O}_{\mathbb{A}^2}$ (generated by $\gamma_{i,1} := 1$) and $\sigma := 0$. Then the first line of (E.43) follows from the crepant pullback of Proposition E.4(1) with $c' = \theta c$, since C_i^- is the image of the origin, and the second line follows from the definition of β_i^-, **Step 1**(i), and [Liu26, Lemma 2.6]. If $\epsilon = +$, then $\delta < 0$. Let $L := \operatorname{lcm}(w_z, w_t, N+p) \cdot |\bar{G}|$ and let $\mathfrak{a}_i$ be the ideal generated by

$$\gamma_{i,1} := \zeta_1^{L/w_z}, \quad \gamma_{i,2} := \zeta_2^{L/w_t}, \quad \gamma_{i,3} := g_i^0(\zeta_1, \zeta_2)^{L/(N+p)},$$

which are $\bar{G}$-invariant, since their characters are powers of exponent divisible by $|\bar{G}|$. The polynomial g_i^0 has $\mathbf{T}$-weight N, so $\deg g_i^0 \le N/\min(w_z, w_t)$. Hence the $\gamma_{i,j}$ have degree bounded independently of i. Since $\gamma_{i,1}, \gamma_{i,2} \in \mathfrak{a}_i$, we have $\mathfrak{m}^{N_0} \subset \mathfrak{a}_i$ with $N_0 := L/w_z + L/w_t$. By Proposition E.4(1), the ideal on Y_i^- generated by the images of the $\gamma_{i,j}$, which we also denote by $\mathfrak{a}_i$, satisfies $\mathfrak{a}_i \mathcal{O}_{Y_i^+} = \mathcal{O}_{Y_i^+}(-LE_i)$, where $\pi_i : Y_i^+ \to Y_i^-$ is the projective birational morphism and $E_i \ge 0$ the $\mathbb{Q}$-divisor with support C_i^+ of Proposition E.4(1). Let $\sigma(\theta) := -\delta_\theta / L$. Since $\pi_i^{-1}(C_i^-) = C_i^+$, the identity of Proposition E.4(1) with $c' = \theta c$ implies that

$$a\big(P, Y_i^+, B_i^+(\theta c)\big) = a\big(P, Y_i^-, B_i^-(\theta c), \mathfrak{a}_i^{\sigma(\theta)}\big)$$

for any prime divisor P over Y_i^-, and (E.43) follows from [Liu26, Lemma 2.6(2)] as in case (a). The bound $c_k \operatorname{mult}_0 f_{i,k} \le 2$ follows from **Step 1**(i) as in case (a).

If U is a $\mathbf{T}$-invariant G_i-semi-invariant regular function on V_i, let $u := U(1, -g_i^0(\zeta_1, \zeta_2), \zeta_1, \zeta_2)$ be the restriction of U to the section $(x = 1)$, and let q be the order of the character of U on G_i. Then U^q is $\mathbf{T}$-invariant and G_i-invariant, hence a regular function on Y_i^-, and we let

$$\operatorname{div}_{Y_i^\pm}(U) := \frac{1}{q} \operatorname{div}_{Y_i^\pm}(U^q).$$

Since U^q is a function on Y_i^-, $\operatorname{div}_{Y_i^+}(U)$ is the pullback of $\operatorname{div}_{Y_i^-}(U)$ by π_i, and its crepant pullback to $\mathbb{A}^2$ is $\operatorname{div}(u)$. Thus the identity of Proposition E.4(1) holds with $\lambda \operatorname{div}(U)$ added on both sides. This proves the last statement of the step in case (b).

Step 3. In this step we replace the ideals by general members and take a compatible generic limit on the fixed plane.

Fix an integer $M > \theta_0 \max_{[1,\theta_0]} \sigma$. For each i, let

$$\Delta_0 := \mathrm{Supp}\Big(\sum_k \mathrm{div}(f_{i,k}) + \mathrm{div}(\zeta_1\zeta_2)\Big),$$

and let Ω_i be the open subset given by Lemma E.8 for $\bar{G}$, Δ_0, $\gamma_{i,1}, \dots, \gamma_{i,n_1}$, and M. Since $\mathbb{C}$ is uncountable, there exists $(t_{l,j}) \in \bigcap_i \Omega_i$, which we fix. Let $\eta_{i,l} := \sum_j t_{l,j}\gamma_{i,j}$ and

$$C_i := \Delta_i(1) + \frac{\sigma(1)}{M}\sum_{l=1}^{M} \mathrm{div}(\eta_{i,l}).$$

Here C_i is a divisor on $\mathbb{A}^2$ (the letter of Theorem 4.2 and Definition-Proposition 5.7, which are applied to the sequence $(C_i)_i$ below). It is not the fiber C_i^ϵ of **Step 1**. By (E.43) and Lemma E.8(1), (2), for i large, $(\mathbb{A}^2, C_i)$ is lc near 0 and

$$\mathrm{mld}\big((\mathbb{A}^2 \ni 0, C_i)/\bar{G}\big) = \beta. \tag{E.45}$$

Consider the tuples

$$(f_{i,1}, \dots, f_{i,s}, \zeta_1, \zeta_2, \eta_{i,1}, \dots, \eta_{i,M}).$$

Their members are non-zero $\bar{G}$-semi-invariant polynomials whose characters depend only on the position in the tuple, and whose orders at 0 are bounded by $2/c_k$, 1, and N_0 (Lemma E.8(3)). By case (a) of Construction-Proposition 4.1 (the fixed affine space $\mathbb{A}^2$ with the fixed group $\bar{G}$), after passing to a subsequence there exist an algebraically closed field $\mathbf{k} \supset \mathbb{C}$ (the field k of Construction-Proposition 4.1, written $\mathbf{k}$ since k indexes the marks) and a compatible generic limit

$$(f_{\infty,1}, \dots, f_{\infty,s}, \zeta_{\infty,1}, \zeta_{\infty,2}, \eta_{\infty,1}, \dots, \eta_{\infty,M})$$

over $\mathbf{k}$. The entries ζ_1, ζ_2 do not depend on i. Hence, for every positive integer d, the points of $\mathcal{J}_d$ whose entries at the positions of ζ_1, ζ_2 are the d-jets of ζ_1 and ζ_2 form a closed subset $\mathcal{J}_d^\zeta \subset \mathcal{J}_d$ containing the d-jets of all the tuples. Since any non-empty open subset of the irreducible closed subset Z_d of Construction-Proposition 4.1 contains the d-jets of some of the tuples, $Z_d \subset \mathcal{J}_d^\zeta$. The d-jet of the compatible generic limit maps to the generic point of Z_d, so its entries at the positions of ζ_1, ζ_2 are the d-jets of ζ_1 and ζ_2 for every d, and $\zeta_{\infty,1} = \zeta_1$ and $\zeta_{\infty,2} = \zeta_2$. Let

$$C_\infty := \sum_k c_k \,\mathrm{div}(f_{\infty,k}) + \omega_1(1)\,\mathrm{div}(\zeta_1) + \omega_2(1)\,\mathrm{div}(\zeta_2) + \frac{\sigma(1)}{M}\sum_{l=1}^{M}\mathrm{div}(\eta_{\infty,l})$$

on $\mathcal{A}_\mathbf{k} = \mathrm{Spec}\,\mathbf{k}[[\zeta_1, \zeta_2]]$. Theorem 4.2 and Definition-Proposition 5.7 require positive coefficients. If some coefficient above is zero, we use the subtuple of the members with positive coefficients, which is a compatible generic limit of the subtuples by Construction-Proposition 4.1. Thus Theorem 4.2 and Definition-Proposition 5.7 apply to the sequence $(C_i)_i$, the limit C_∞, the group $\bar{G}$, and $\mathfrak{c} = \mathcal{O}$. Theorem 4.2 and Definition-Proposition 5.7 also apply to $C_i + \varsigma\,\mathrm{div}(\zeta_j)$ and $C_\infty + \varsigma\,\mathrm{div}(\zeta_j)$ for any $j \in \{1, 2\}$ and real number $\varsigma > 0$.

Step 4. In this step we treat the case when $(\mathcal{A}_\mathbf{k}, C_\infty)$ is klt near 0.

By Theorem 4.2(1), after passing to a subsequence, there exists a real number $\theta' > 1$ such that $(\mathbb{A}^2, \theta' C_i)$ is lc near 0 for any i. Possibly decreasing θ', we may assume that $\theta' \le \theta_0$. Since $\theta'\sigma(1) < M$, Lemma E.8(1) implies that $(\mathbb{A}^2, \theta'\Delta_i(1), \mathfrak{a}_i^{\theta'\sigma(1)})$ is lc near 0, that is, for each i there is an open neighborhood O_i of 0, which may depend on i, such that

$$a(P, \mathbb{A}^2, \theta'\Delta_i(1)) \ge \theta'\sigma(1)\,\mathrm{mult}_P\,\mathfrak{a}_i \tag{E.46}$$

for any prime divisor P over $\mathbb{A}^2$ whose center intersects O_i.

Claim E.9. *There exists $\theta \in (1, \theta']$ such that $(\mathbb{A}^2, \Delta_i(\theta), \mathfrak{a}_i^{\sigma(\theta)})$ is lc near 0 for any i.*

Proof. The blow-up of 0 is a log resolution of $\mathfrak{m}$, and $(\mathbb{A}^2, \mathfrak{m}^2)$ is lc, so $a(P, \mathbb{A}^2, 0) \geq 2\,\mathrm{mult}_P\,\mathfrak{m}$ for any prime divisor P over $\mathbb{A}^2$. Since $\mathfrak{m}^{N_0} \subset \mathfrak{a}_i$, we have

$$\mathrm{mult}_P\,\mathfrak{a}_i \leq N_0\,\mathrm{mult}_P\,\mathfrak{m} \leq \frac{N_0}{2}a(P, \mathbb{A}^2, 0)$$

for any prime divisor P over $\mathbb{A}^2$. Let $\theta \in (1, \theta']$. Since $\omega_j(\theta) \leq \theta\omega_j(1)$ for $j = 1, 2$, we have $\Delta_i(\theta) \leq \theta\Delta_i(1)$, and log discrepancies are affine in the boundary, so by (E.46), for any prime divisor P over $\mathbb{A}^2$ whose center intersects O_i,

$$\begin{aligned} a(P, \mathbb{A}^2, \Delta_i(\theta), \mathfrak{a}_i^{\sigma(\theta)}) &\geq \Big(1 - \frac{\theta}{\theta'}\Big)a(P, \mathbb{A}^2, 0) + \frac{\theta}{\theta'}a(P, \mathbb{A}^2, \theta'\Delta_i(1)) - \sigma(\theta)\,\mathrm{mult}_P\,\mathfrak{a}_i \\ &\geq \Big(1 - \frac{\theta}{\theta'}\Big)a(P, \mathbb{A}^2, 0) - \big(\sigma(\theta) - \theta\sigma(1)\big)\,\mathrm{mult}_P\,\mathfrak{a}_i \\ &\geq \Big(1 - \frac{\theta}{\theta'} - \frac{N_0}{2}\big|\sigma(\theta) - \theta\sigma(1)\big|\Big)a(P, \mathbb{A}^2, 0). \end{aligned}$$

The function $\theta \to \sigma(\theta) - \theta\sigma(1)$ is continuous and vanishes at $\theta = 1$, so the last factor is positive for $\theta > 1$ close to 1, independently of i. Hence, for such θ and these P,

$$a(P, \mathbb{A}^2, \Delta_i(\theta), \mathfrak{a}_i^{\sigma(\theta)}) \geq 0,$$

so $(\mathbb{A}^2, \Delta_i(\theta), \mathfrak{a}_i^{\sigma(\theta)})$ is lc near 0 for any i. □

Proof of Proposition 11.12 continued. By Claim E.9 and (E.43), $(Y_i^\epsilon, B_i^\epsilon(\theta c))$ is lc near C_i^ϵ for any i. By **Step 1**, $\alpha_i \geq \beta$ for i large, a contradiction.

Step 5. In this step we treat the case when $(\mathcal{A}_\mathbf{k}, C_\infty)$ is not klt near 0, and we construct a $\mathbf{T}$-invariant function U that bounds α_i from below by Lemma 5.8.

By (E.45), the numbers $\mathrm{mld}((\mathbb{A}^2 \ni 0, C_i)/\bar{G})$ are equal to $\beta > 0$. By Definition-Proposition 5.7 in case (a) of Construction-Proposition 4.1, $(\mathcal{A}_\mathbf{k}, C_\infty)$ is lc near 0, it has exactly one floor F, F is smooth at 0, $(\mathcal{A}_\mathbf{k}, C_\infty - F)$ is klt near 0, and

$$\beta = \frac{1 - b}{e},$$

where $b := \mathrm{mult}_0((C_\infty - F)|_F)$ and e is the order of the image of $\bar{G}$ in the automorphism group of T_0F (Lemma 5.6). In particular $b < 1$. Since F is the unique floor and $\bar{G}$ preserves C_∞, $\bar{G}$ preserves F, so T_0F is a $\bar{G}$-invariant line. Let χ_1, χ_2 be the characters of ζ_1, ζ_2. If $\chi_1 = \chi_2$, choose $j_0 \in \{1, 2\}$ with $\zeta_{j_0}|_{T_0F} \neq 0$. If $\chi_1 \neq \chi_2$, the only $\bar{G}$-invariant lines are $(\zeta_1 = 0)$ and $(\zeta_2 = 0)$, so $T_0F = (\zeta_j = 0)$ for some j, and let j_0 be the other index. In both cases $\zeta_{j_0}|_F$ is a parameter of F at 0, and $\bar{G}$ acts on the cotangent line of F at 0 through χ_{j_0}, so e is the order of χ_{j_0}. Let $C' := C_\infty - F + (1 - b)\,\mathrm{div}(\zeta_{j_0})$. Then $C' \geq 0$, and C' does not contain F, since F has coefficient one in C_∞ and $\zeta_{j_0}|_F$ is a parameter of F. Moreover,

$$\mathrm{mult}_0(C'|_F) = b + (1 - b) = 1.$$

By Lemma 2.14, applied to $S = \mathcal{A}_\mathbf{k}$, the curve $D = F$, which is regular since it is smooth at 0, and $B = C'$, the pair

$$\big(\mathcal{A}_\mathbf{k}, F + C'\big) = \big(\mathcal{A}_\mathbf{k}, C_\infty + (1 - b)\,\mathrm{div}(\zeta_{j_0})\big)$$

is lc near 0. By Theorem 4.2(1) and **Step 3**, after passing to a subsequence, the pair

$$\big(\mathbb{A}^2, C_i + (1 - b)\,\mathrm{div}(\zeta_{j_0})\big)$$

is lc near 0 for any i, and by Lemma E.8(1),

$$\big(\mathbb{A}^2, \Delta_i(1) + (1 - b)\,\mathrm{div}(\zeta_{j_0}), \mathfrak{a}_i^{\sigma(1)}\big) \text{ is lc near 0 for every } i. \tag{E.47}$$

We now define U, a positive integer ℓ, and the order q of the character of U on G_i.

Case (a) of Lemma E.1(3). Let ℓ be the order of $\chi_{j_0}|_\Xi$. Then $\zeta_{j_0}^\ell$ is Ξ-invariant, hence a monomial u_0 on $Q_0 = \mathbb{A}^2/\Xi$ vanishing at the origin. Through $\mathbb{C}[Q_0] = \mathbb{C}[V]^\mathbf{T}$, u_0 is a $\mathbf{T}$-invariant regular function U on V, the restriction of a monomial in x, y, z, t (the coordinate ring of V is

the semigroup ring of the characters non-negative on the cone of V). Since G_i is diagonal, U is G_i-semi-invariant, and $U(0) = 0$.

Case (b) of Lemma E.1(3). Here $\bar{G}$ contains the image of the central μ_p, which acts on ζ_1, ζ_2 with weights w_z, w_t modulo p. Let $n_0 := w_z$ and $L_0 := z$ if $j_0 = 1$, and $n_0 := w_t$ and $L_0 := t$ if $j_0 = 2$. Let $\ell := p/\gcd(n_0, p)$ and

$$U := L_0^{\ell} x^{n_0/\gcd(n_0,p)}.$$

The $\mathbf{T}$-weight of U is $\ell n_0 - \frac{n_0}{\gcd(n_0,p)} p = 0$, U is G_i-semi-invariant since G_i is diagonal, $U(0) = 0$, and on the x-root chart $U = (\xi^{n_0} L_0)^{\ell} = \zeta_{j_0}^{\ell}$. The integer ℓ is the order of χ_{j_0} on μ_p.

In both cases ℓ divides e, since ℓ is the order of the restriction of χ_{j_0} to a subgroup of $\bar{G}$. The function U is $\mathbf{T}$-invariant, so G_i acts on U through $G_i/(G_i \cap \mathbf{T})$, and the action of G_i on U is the action of $\bar{G}$ on $\zeta_{j_0}^{\ell}$. Hence the order of the character of U on G_i is the order of $\chi_{j_0}^{\ell}$, that is,

$$q = \frac{e}{\ell}.$$

Let $\lambda := (1 - b)/\ell$. Then $\lambda > 0$ and

$$\frac{\lambda}{q} = \frac{1 - b}{e} = \beta.$$

The function u of **Step 2** attached to U is $\zeta_{j_0}^{\ell}$ in both cases, so

$$\lambda \operatorname{div}(u) = (1 - b) \operatorname{div}(\zeta_{j_0}).$$

By (E.47) and the last statement of **Step 2**, $(Y_i^{\epsilon}, B_i^{\epsilon} + \lambda \operatorname{div}(U))$ is lc near C_i^{ϵ} for any i.

Claim E.10. *For i large, the pair $(W_i, D_i + \lambda \operatorname{div}(U))$ is lc near 0.*

Proof. Let

$$D_i' := \sum_k c_k \operatorname{div}(h_{i,k}^0|_{V_i}) + \lambda \operatorname{div}(U|_{V_i}).$$

First we show that (V_i, D_i') is lc near 0. The pair is $\mathbf{T}$-invariant, so it is lc near 0 if $a(P, V_i, D_i') \geq 0$ for the $\mathbf{T}$-invariant prime divisors P over V_i whose center contains 0 and which are prime divisors on a $\mathbf{T}$-equivariant log resolution $\tilde{V}_i$ of (V_i, D_i') (such a resolution exists [Kol07, Proposition 3.9.1 and Theorem 3.35], its exceptional prime divisors and the components of the strict transform of D_i' are $\mathbf{T}$-invariant since $\mathbf{T}$ is connected, and lc is checked on one log resolution [KM98, Corollary 2.31(3) and 2.32]). Let P be such a prime divisor. Among the coordinates $v \in \{x, y, z, t\}$ with $\epsilon w_v > 0$, choose v_0 minimizing $\operatorname{mult}_P v/|w_v|$, and let $V_{i,v_0} := V_i \cap (v_0 = 1)$. The morphism $\Phi : V_{i,v_0} \times \mathbb{G}_m \to V_i$, $(v, \tau) \to \tau^{\pm 1} \cdot v$, with the exponent $+1$ if $\epsilon = +$ and -1 if $\epsilon = -$, where τ acts through $\mathbf{T}$, is the quotient by $\mu_{|w_{v_0}|}$ onto $(v_0 \neq 0)$. We have $\Phi^* h_{i,k}^0 = \tau^{\pm \kappa_k} h_{i,k}^0|_{V_{i,v_0}}$, $\Phi^* U = U|_{V_{i,v_0}}$, and the canonical form Ω of V_i has $\mathbf{T}$-weight $w_z + w_t$, so, up to a non-zero constant,

$$\Phi^* \Omega = \tau^{\pm(w_z + w_t)} \Omega_{v_0} \wedge \frac{d\tau}{\tau},$$

where Ω_{v_0} is a generator of the canonical sheaf of V_{i,v_0}, and Ω generates the canonical sheaf of V_i near 0.

Let X' be the normalization of V_i in the function field of $V_{i,v_0} \times \mathbb{G}_m$. The group $\mu_{|w_{v_0}|}$ acts faithfully on X' with $X'/\mu_{|w_{v_0}|} = V_i$, and X' contains $V_{i,v_0} \times \mathbb{G}_m$ as the preimage of $(v_0 \neq 0)$. Let (X', B') be the crepant pullback of (V_i, D_i'). By [Liu26, Lemma 2.6(1)–(3)], there is a prime divisor P' over X' such that $\operatorname{mult}_{P'} = e_{P'} \operatorname{mult}_P$ on rational functions on V_i, where $e_{P'} := |I_{\mu_{|w_{v_0}|}}(P')|$, and

$$a(P', X', B') = e_{P'} a(P, V_i, D_i').$$

We show that $\operatorname{mult}_{P'}$ is a Gauss extension in τ. For $s \in \mathbb{C}^*$, let σ_s be the automorphism $(v, \tau) \to (v, s\tau)$ of $V_{i,v_0} \times \mathbb{G}_m$. Then $\Phi \circ \sigma_s = s^{\pm 1} \cdot \Phi$, so σ_s^* acts on rational functions on V_i through $\mathbf{T}$. Since P is $\mathbf{T}$-invariant, the valuation $\operatorname{mult}_{P'} \circ \sigma_s^*$ is again equal to $e_{P'} \operatorname{mult}_P$ on

rational functions on V_i. Hence it is the valuation of one of the finitely many prime divisors of the normalization of $\tilde{V}_i$ in the function field of X' lying over P, since its center there is finite over P. Thus the stabilizer of $\operatorname{mult}_{P'}$ in $\mathbb{C}^*$ has finite index, and it is $\mathbb{C}^*$, since $\mathbb{C}^*$ is divisible. Let $\bar{\nu}$ be the restriction of $\operatorname{mult}_{P'}$ to rational functions on V_{i,v_0}, and $\rho := \operatorname{mult}_{P'} \tau$. Let $f = \sum_{j=0}^{n} a_j \tau^j$ with rational functions a_j on V_{i,v_0}, and let $s_0, \dots, s_n \in \mathbb{C}^*$ be distinct. By the Vandermonde determinant, each $a_j\tau^j$ is a linear combination of $\sigma_{s_0}^* f, \dots, \sigma_{s_n}^* f$, so $\operatorname{mult}_{P'}(a_j\tau^j) \geq \operatorname{mult}_{P'} f$. Therefore

$$\operatorname{mult}_{P'} f = \min_j \big(\bar{\nu}(a_j) + j\rho\big), \tag{E.48}$$

that is, if $\bar{\nu}$ is non-trivial, then $\operatorname{mult}_{P'}$ is the Gauss extension of $\bar{\nu}$ with τ-value ρ [Liu26, Definition-Proposition 2.11].

Next we locate $\bar{\nu}$. Since $v_0 \circ \Phi = \tau^{|w_{v_0}|}$, we have $|w_{v_0}|\rho = e_{P'} \operatorname{mult}_P v_0 \geq 0$. For $u \in \{x, y, z, t\}$, we have $u \circ \Phi = \tau^{\pm w_u} u|_{V_{i,v_0}}$, so

$$\bar{\nu}(u|_{V_{i,v_0}}) = e_{P'} \operatorname{mult}_P u \mp w_u \rho.$$

If $\pm w_u \leq 0$, this is non-negative. If $\pm w_u > 0$, it is equal to

$$e_{P'}|w_u|\Big(\frac{\operatorname{mult}_P u}{|w_u|} - \frac{\operatorname{mult}_P v_0}{|w_{v_0}|}\Big) \geq 0,$$

where the inequality holds by the choice of v_0. Hence $\bar{\nu}$ is non-negative on the regular functions on V_{i,v_0}. If $\bar{\nu}$ is non-trivial, then $\bar{\nu} = \varrho \operatorname{mult}_{\bar{P}}$ for a positive integer ϱ and a prime divisor $\bar{P}$ over V_{i,v_0} [Liu26, Lemma 2.12]. The restriction of $\bar{\nu}$ to the $\mathbf{T}$-invariant functions is $e_{P'} \operatorname{mult}_P$, whose center on $\operatorname{Spec} \mathbb{C}[V_i]^{\mathbf{T}}$ is the image of the center of P and contains the image of 0. Since Y_i^ϵ is projective over the quotient of $\operatorname{Spec} \mathbb{C}[V_i]^{\mathbf{T}}$ by G_i, the center $Z_{\bar{P}}$ on Y_i^ϵ of the valuation induced by $\bar{P}$ intersects C_i^ϵ. Let $O \subset Y_i^\epsilon$ be an open neighborhood of C_i^ϵ on which $(Y_i^\epsilon, B_i^\epsilon + \lambda \operatorname{div}(U))$ is lc. Then $Z_{\bar{P}}$ intersects O, so the generic point of $Z_{\bar{P}}$ belongs to O, and it also belongs to the image of V_{i,v_0}, since it is the image of the generic point of the center of $\bar{P}$ on V_{i,v_0}. Hence the center of $\bar{P}$ intersects the preimage of O in V_{i,v_0}.

Now we compute the log discrepancy. Let $D_i'' := D_i'|_{V_{i,v_0}}$ and

$$\Theta := \operatorname{pr}^* D_i'' + (1 \pm \delta) \operatorname{div}(\tau) \quad \text{on } V_{i,v_0} \times \mathbb{A}^1_\tau.$$

By the formulas for $\Phi^* h_{i,k}^0$, $\Phi^* U$, and $\Phi^* \Omega$,

$$\Phi^* D_i' = \operatorname{pr}^* D_i'' \pm \sum_k c_k \kappa_k \operatorname{div}(\tau), \qquad \Phi^* \Omega = \tau^{\pm(w_z + w_t) - 1} \Omega_{v_0} \wedge d\tau.$$

Since $\pm \sum_k c_k \kappa_k \mp (w_z + w_t) + 1 = 1 \pm \delta$,

$$\Theta = \Phi^* D_i' - \operatorname{div}\Big(\frac{\Phi^* \Omega}{\Omega_{v_0} \wedge d\tau}\Big).$$

Since Ω and $\Omega_{v_0} \wedge d\tau$ generate the canonical sheaves of V_i near 0 and of $V_{i,v_0} \times \mathbb{A}^1_\tau$, both $a(P', X', B')$ and $a(P', V_{i,v_0} \times \mathbb{A}^1_\tau, \Theta)$ are equal to one plus the order of $\Phi^* \Omega$ along P' minus $\operatorname{mult}_{P'} \Phi^* D_i'$ [KM98, Definition 2.25]. Thus

$$e_{P'} a(P, V_i, D_i') = a\big(P', V_{i,v_0} \times \mathbb{A}^1_\tau, \operatorname{pr}^* D_i''\big) - (1 \pm \delta)\rho.$$

If $\bar{\nu}$ is trivial, then $\operatorname{mult}_{P'} = \rho \operatorname{mult}_{V_{i,v_0} \times \{0\}}$ by (E.48), so $\rho = 1$, P' is the prime divisor $V_{i,v_0} \times \{0\}$, and $e_{P'} a(P, V_i, D_i') = 1 - (1 \pm \delta) = \mp\delta$. Otherwise, by (E.48), the values of $\operatorname{mult}_{P'}$ on the polynomials in τ with coefficients rational functions on V_{i,v_0} belong to $\varrho\mathbb{Z} + \rho\mathbb{Z}$, and they include $\rho = \operatorname{mult}_{P'} \tau$ and ϱ, since $\operatorname{mult}_{\bar{P}}$ takes the value 1. Since any rational function on $V_{i,v_0} \times \mathbb{G}_m$ is a quotient of two such polynomials, the value group $\mathbb{Z}$ of $\operatorname{mult}_{P'}$ is $\varrho\mathbb{Z} + \rho\mathbb{Z} = \gcd(\varrho, \rho)\mathbb{Z}$, and $\gcd(\varrho, \rho) = 1$. If $\rho = 0$, then $\varrho = 1$, $\operatorname{mult}_{P'}$ is the valuation of the pullback of $\bar{P}$ to $V_{i,v_0} \times \mathbb{A}^1_\tau$, and the first term is $a(\bar{P}, V_{i,v_0}, D_i'')$. If $\rho > 0$, then [Liu26, Lemma 18.1], applied to V_{i,v_0}, D_i'', $\bar{P}$,

and the coprime positive integers ϱ and ρ, shows that the first term is $\varrho a(\bar{P}, V_{i,v_0}, D_i'') + \rho$. In the last two cases

$$e_{P'} a(P, V_i, D_i') = \varrho a(\bar{P}, V_{i,v_0}, D_i'') \mp \delta\rho,$$

with the sign $-$ if $\epsilon = +$ and $+$ if $\epsilon = -$. The quotient of V_{i,v_0} by $\mu_{|w_{v_0}|}$ followed by the quotient by $G_i/(G_i \cap \mathbf{T})$ is the root chart of Y_i^ϵ at v_0, and D_i'' is the crepant pullback of $B_i^\epsilon + \lambda \operatorname{div}(U)$. Since $(Y_i^\epsilon, B_i^\epsilon + \lambda \operatorname{div}(U))$ is lc on O, the pair (V_{i,v_0}, D_i'') is lc on the preimage of O [KM98, Proposition 5.20(4)], so $a(\bar{P}, V_{i,v_0}, D_i'') \geq 0$. Moreover $\rho \geq 0$, and $\delta \leq 0$ if $\epsilon = +$ and $\delta \geq 0$ if $\epsilon = -$ by the choice of ϵ, so $\mp\delta\rho \geq 0$ and $\mp\delta \geq 0$. Hence $a(P, V_i, D_i') \geq 0$ in all cases. Thus (V_i, D_i') is lc near 0.

Next, V_i is a normal hypersurface in $\mathbb{A}^4$, since the singular locus of V_i has codimension at least two, and neither $h_{i,k}^0$ nor U vanishes on V_i. By [Kaw07, Theorem] with $S = V_i$, applied after shrinking $\mathbb{A}^4$ to a neighborhood of 0, the pair

$$\Big(\mathbb{A}^4, V_i + \sum_k c_k \operatorname{div}(h_{i,k}^0) + \lambda \operatorname{div}(U)\Big)$$

is lc near 0. Recall that $W_i = (xy + g_i = 0)$. Each monomial of $h_{i,k}$ has the character of $h_{i,k}$ on $G_i \cap \mathbf{T} \cong \mu_{r_i}$, so the $\mathbf{T}$-weight of each monomial of $h_{i,k}$ is congruent to κ_k modulo r_i, and the monomials of $h_{i,k} - h_{i,k}^0$ have degree at least $(r_i - |\kappa_k|)/\max_v |w_v|$. In case (b), similarly, the monomials of $g_i - g_i^0$ have degree at least $(r_i - |\kappa(g_i)|)/\max_v |w_v|$, where $\kappa(g_i) = w_x + w_y$ is the $\mathbf{T}$-weight of g_i^0. In case (a), $V_i = (xy + z^a t^b = 0)$, and $\operatorname{mult}_0(g_i - z^a t^b) \to +\infty$ by Construction E.2. The exponents 1, $c_1, \dots, c_s$, and λ form a fixed finite set, and U does not depend on i. By Lemma 4.14 and [dFEM11, Proposition 2.11], for i large the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(xy + g_i) + \sum_k c_k \operatorname{div}(h_{i,k}) + \lambda \operatorname{div}(U)\Big)$$

is lc near 0. Since W_i is normal, [Kaw07, Theorem] with $S = W_i$ implies that $(W_i, D_i + \lambda \operatorname{div}(U))$ is lc near 0. □

Proof of Proposition 11.12 continued. The function U is G_i-semi-invariant, regular, vanishes at 0, and the character of U has order q. By Claim E.10 and Lemma 5.8 with $Z = \{0\}$, for i large,

$$\alpha_i = \operatorname{mld}\big((W_i \ni 0, D_i)/G_i\big) \geq \frac{\lambda}{q} = \beta,$$

a contradiction. **Steps 4 and 5** cover all cases, and the proof is complete. □

E.5. **Triangles and secant conics: circles and crepant models.** In this subsection, we prove Lemmas E.11 and E.13. They are used in Subsection E.6 to prove Proposition 11.13.

Part (2) of Lemma E.11 excludes limits of dimension two of the groups by one lc augmentation for all labels at once. In **Steps 4–6**, the characters of $\mathbf{S}$ trivial on the identity component of the limit form a group $\mathbb{Z}\nu$, the vector ν defines an invariant monomial Φ, and the number λ/q, where λ is the largest λ' with $(W_i, D_i + \lambda' \operatorname{div}(\Phi))$ lc for $i \gg 0$ (Claim E.12, computed on the toric cover of $(x^2 + yzt = 0)$) and q is the number of connected components of the limit of the groups, bounds the quotient values from below for $i \gg 0$ and their upper limit from above.

Lemma E.11. *Notation as in Set-up 11.1. Let $(W_i \ni o_i, G_i, (h_{i,k})_k)$ be a sequence of such data, with s, m and the μ_k fixed and with W_i of quadratic rank one whose residual cubic is a triangle or a secant conic (Definition-Lemma 6.4), let $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$ and $\alpha_i := \operatorname{mld}((W_i \ni o_i, D_i)/G_i)$, and assume that the α_i are strictly increasing in $(0, 1)$.*

(1) If F_i is a computing lift of α_i (Definition 2.7), then $\alpha_i = \operatorname{mld}((W_i \ni o_i, D_i)/I_{G_i}(F_i))$.

(2) Assume that there are ambient G_i-eigenparameters x, y, z, t in which the quadratic part of the equation of W_i is x^2 and the residual cubic is $yzt + \epsilon t^3$, with $\epsilon \in \{0, 1\}$ independent of i. Then, after passing to a subsequence and replacing the data with numerical admissions

with the groups G_i (Proposition 11.8), taken with the integers N and L of Construction-Theorem 4.13 larger than $\max_k \mathrm{mult}_{o_i} h_{i,k}$, we have $o_i = 0$ and $W_i = (f_i = 0) \subset \mathbb{A}^4$ for a G_i-semi-invariant polynomial $f_i \in x^2 + \mathbb{C}[y,z,t]$ with residual cubic $yzt + \epsilon t^3$, the $h_{i,k}$ are polynomials, and G_i acts diagonally in x,y,z,t. Moreover, there are a primitive vector $w \in \mathbb{Z}^4$ and a positive integer e_0 such that $|G_i \cap \mathbf{T}_{(w)}| \to +\infty$ and $[G_i : G_i \cap \mathbf{T}_{(w)}] \leq e_0$, and such that, after a permutation of y,z,t preserving $yzt + \epsilon t^3$ and replacing w by $-w$, either $w = (0,1,-1,0)$, or $w = (u,-A,-B,C)$ with $u \geq 0$, $A,B,C > 0$, and $C = 2u + A + B$, or $w = (u,-A,B,C)$ with $u,A,B,C > 0$ and $B + C = 2u + A$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1).

Fix i, and let $I := I_{G_i}(F_i)$. For a prime divisor P over W_i with center o_i, the inertia group of P in the subgroup I is $I \cap I_{G_i}(P)$, and $a(P,W_i,D_i) = |I_{G_i}(P)| a_{D_i}(P) > 0$ by Definition-Lemma 11.4(1). Hence

$$\frac{a(P,W_i,D_i)}{|I \cap I_{G_i}(P)|} \geq \frac{a(P,W_i,D_i)}{|I_{G_i}(P)|} = a_{D_i}(P) \geq \alpha_i, \tag{E.49}$$

and both inequalities are equalities for $P = F_i$, since $I \cap I_{G_i}(F_i) = I$ and F_i computes α_i. For any subgroup $G' \subset G_i$, the number $\mathrm{mld}((W_i \ni o_i, D_i)/G')$ is the infimum of $a(P,W_i,D_i)/|I_{G'}(P)|$ over the prime divisors P over W_i with center o_i [Liu26, Lemma 2.6]. Applied with $G' = I$ and with $G' = G_i$, (E.49) implies (1). The argument above uses only that $a(P,W_i,D_i) > 0$ for the prime divisors P over W_i with center o_i. For a numerical admission of the data, G_i acts faithfully on the admitted germ and the corresponding pair is klt near 0 (Construction-Theorem 4.13(1)), so (1) also holds for it.

Step 2. In this step we prove that $|G_i| \to +\infty$, and we replace the data with numerical admissions in a polynomial normal form.

If $|G_i|$ is bounded on an infinite set $\mathcal{I}$ of indices, then by Proposition 11.7(1) the numbers $\alpha_i \in (0,1)$, $i \in \mathcal{I}$, belong to a finite set, which is impossible for a strictly increasing sequence. Thus $|G_i| \to +\infty$.

Let $\mathfrak{R} := \mathbb{C}[[x,y,z,t]]$, and write $\mathfrak{m} := (x,y,z,t)$, both in $\mathbb{C}[x,y,z,t]$ and in $\mathfrak{R}$. Let $\hat{f}_i \in \mathfrak{R}$ be the expansion of a G_i-semi-invariant local equation of W_i in the eigenparameters of (2), and let $\hat{h}_{i,k} \in \mathfrak{R}$ be the expansion of a G_i-semi-invariant lift of $h_{i,k}$ to the ambient space, obtained by projecting any lift to the character of $h_{i,k}$, which is again a lift since $h_{i,k}$ is G_i-semi-invariant. By the Weierstrass preparation theorem in x, whose factorization is G_i-equivariant by uniqueness, followed by completion of the square in x, there exist a G_i-invariant unit $v_i \in \mathfrak{R}$ and a G_i-equivariant automorphism ψ_i of $\mathfrak{R}$ fixing y,z,t, with $\psi_i(x) - x \in \mathfrak{m}^2$, such that

$$v_i \psi_i(\hat{f}_i) = x^2 + \hat{g}_i(y,z,t),$$

where $\hat{g}_i \in \mathbb{C}[[y,z,t]]$ has no terms of degree at most two and has cubic part $yzt + \epsilon t^3$. Let F_i be a computing lift of α_i (Definition 2.7), and let E_0 be the prime divisor over W_i of the weighted blow-up with weights $(3,2,2,2)$ in x,y,z,t (Definition-Lemma 11.5(3)). Let $N \geq \max(N_1, \max_k \mathrm{mult}_{o_i} h_{i,k} + 1)$ be an integer, and after fixing N, let $L \geq \max(4, L_1, N)$ be an integer, where N_1 and L_1 are the integers of Construction-Theorem 4.13 for the data of index i. Let $g_{i,0}$ be the sum of the terms of $\hat{g}_i$ of degree less than L, and let $f_{i,0} := x^2 + g_{i,0}$. By Proposition 11.8, with ψ_i, v_i, and $u_1 = \cdots = u_s = 1$ in Construction-Theorem 4.13(vi), with the integers N and L, with F_i together with $I_{G_i}(F_i)$ and E_0 protected, and with $f_{i,0}$, the data of index i have a numerical admission $(W_{i,0} \ni 0, G_i, (h_{i,k,0})_k)$. Then $W_{i,0} = (x^2 + g_{i,0} = 0)$ has the residual cubic $yzt + \epsilon t^3$, and since $x^2 + g_{i,0}$ is a G_i-semi-invariant polynomial monic in x, replacing each $h_{i,k,0}$ by the remainder of the division by $x^2 + g_{i,0}$ changes neither the character of $h_{i,k,0}$ nor $h_{i,k,0}|_{W_{i,0}}$, so that each $h_{i,k,0}$ has x-degree at most one (as in Definition-Lemma 11.5(1)). For a positive integer j, $\mathrm{mult}_{o_i} h_{i,k} \geq j$ if and only if $\hat{h}_{i,k} \in \mathfrak{m}^j + (\hat{f}_i)$. The automorphism ψ_i

maps $\mathfrak{m}^j + (\hat{f}_i)$ onto $\mathfrak{m}^j + (x^2 + \hat{g}_i)$, as v_i is a unit, and

$$x^2 + \hat{g}_i \equiv f_{i,0} \pmod{\mathfrak{m}^L}, \qquad h_{i,k,0} \equiv \psi_i(\hat{h}_{i,k}) \pmod{\mathfrak{m}^N + (f_{i,0})}.$$

Thus, for $j \le \min(N, L)$, $\hat{h}_{i,k} \in \mathfrak{m}^j + (\hat{f}_i)$ if and only if $h_{i,k,0} \in \mathfrak{m}^j + (f_{i,0})$. Since $\min(N, L) > \max_k \operatorname{mult}_{o_i} h_{i,k}$ by the choice of N and L, $\operatorname{mult}_0 h_{i,k,0} = \operatorname{mult}_{o_i} h_{i,k}$. The germ $(W_i \ni o_i)$ is of type cD by Definition 9.1, since its residual cubic $yzt + \epsilon t^3$ is not the cube of a linear form, so $\sum_k c_k \operatorname{mult}_{o_i} h_{i,k} \le 1 - \frac{1}{m}$ by Definition-Lemma 11.5(3), and hence $\sum_k c_k \operatorname{mult}_0 h_{i,k,0} \le 1 - \frac{1}{m}$.

We replace the data $(W_i \ni o_i, G_i, (h_{i,k})_k)$ with $(W_{i,0} \ni 0, G_i, (h_{i,k,0})_k)$ and keep the notation W_i, $h_{i,k}$, D_i, with $o_i = 0$. Thus, for any i, the following hold.

(a) $W_i = (f_i = 0) \subset \mathbb{A}^4$ with $f_i = x^2 + g_i(y, z, t)$, where g_i is a polynomial without terms of degree at most two whose cubic part is $yzt + \epsilon t^3$, and $(W_i \ni 0)$ is an isolated Gorenstein terminal germ (Construction-Theorem 4.13(3)).

(b) G_i acts diagonally and faithfully on W_i in the coordinates x, y, z, t, and f_i is G_i-semi-invariant.

(c) The $h_{i,k}$ are G_i-semi-invariant polynomials of x-degree at most one, (W_i, D_i) is klt near 0, and $\alpha_i = \operatorname{mld}((W_i \ni 0, D_i)/G_i)$ (Proposition 11.8).

(d) $\sum_k c_k \operatorname{mult}_0 h_{i,k} \le 1 - \frac{1}{m}$.

The admitted data are not data of Set-up 11.1, and below we use only (a)–(d). By (a), the graded ring of $\mathcal{O}_{W_i,0}$ with respect to the maximal ideal is $\mathbb{C}[x, y, z, t]/(x^2)$, and the homogeneous part of least degree of $h_{i,k}$ is not divisible by x^2, having x-degree at most one. Thus $\operatorname{mult}_0 h_{i,k}$ is the least degree of a monomial of $h_{i,k}$. Since $c_k = \mu_k/m$, (d) implies that

(E.50) $\quad h_{i,k}$ has a monomial of degree at most $(m-1)/\mu_k$ for any i and k.

By (a)–(c), the replaced data satisfy the part of the conclusion of (2) that precedes the word "Moreover".

Step 3. In this step we take a limit H of the groups and prove that $\dim H = 1$, or $\dim H = 2$ and $\epsilon = 0$.

Let $\mathbb{T}^4 := \mathbb{R}^4/\mathbb{Z}^4$, and let $\rho : \mathbb{R}^4 \to \mathbb{T}^4$ be the quotient map. By (b) we regard G_i as a subgroup of $\mathbb{T}^4$, the element $\rho(\theta)$ acting by

$$(x, y, z, t) \to (e^{2\pi\sqrt{-1}\theta_1}x, e^{2\pi\sqrt{-1}\theta_2}y, e^{2\pi\sqrt{-1}\theta_3}z, e^{2\pi\sqrt{-1}\theta_4}t),$$

so that $\rho(\theta)$ multiplies the monomial $x^{n_0}y^{n_1}z^{n_2}t^{n_3}$ by $e^{2\pi\sqrt{-1}\langle n,\theta\rangle}$, where $n = (n_0, n_1, n_2, n_3)$. By Lemma 4.6 and **Step 2**, after passing to a subsequence there exists a closed subgroup $H \subset \mathbb{T}^4$ such that the G_i converge to H in the Hausdorff distance, $G_i \subset H$, G_i intersects any connected component of H for $i \gg 0$, and $\dim H \ge 1$. Let $V \subset \mathbb{R}^4$ be the linear subspace with $\rho(V) = H^0$.

By (a) and (b), x^2 and yzt are monomials of the G_i-semi-invariant polynomial f_i, and so is t^3 if $\epsilon = 1$. Hence G_i is contained in the closed subgroup

$$\mathbf{S} := \{\rho(\theta) \mid 2\theta_1 - \theta_2 - \theta_3 - \theta_4 \in \mathbb{Z}\},$$

and, if $\epsilon = 1$, also in

$$\mathbf{S}' := \{\rho(\theta) \mid 2\theta_1 - 3\theta_4 \in \mathbb{Z}\}.$$

Thus $H \subset \mathbf{S}$, $H \subset \mathbf{S}'$ if $\epsilon = 1$, and

(E.51) $\quad V \subset V_{\mathbf{S}} := \{w \in \mathbb{R}^4 \mid 2w_1 = w_2 + w_3 + w_4\}, \quad$ and $\quad V \subset \{w \in \mathbb{R}^4 \mid 2w_1 = 3w_4\}$ if $\epsilon = 1$.

If V contained a vector with all coordinates positive, then $\alpha_i \to 0$ by Lemma 4.7, whose hypotheses hold by (a)–(c), whereas $\alpha_i \ge \alpha_1 > 0$. Thus V contains no vector with all coordinates positive. The vector $(3, 2, 2, 2)$ has all coordinates positive and belongs to $V_{\mathbf{S}}$ and to $V_{\mathbf{S}} \cap \{2w_1 = 3w_4\}$, where $\dim V_{\mathbf{S}} = 3$ and $\dim(V_{\mathbf{S}} \cap \{2w_1 = 3w_4\}) = 2$. Hence, by (E.51), V is a proper subspace of $V_{\mathbf{S}}$, so $\dim H \le 2$, and if $\epsilon = 1$, V is a proper subspace of $V_{\mathbf{S}} \cap \{2w_1 = 3w_4\}$, so $\dim H = 1$ ($\dim H \ge 1$ by the choice of H).

We now show that $\dim H = 2$ is impossible. Assume that $\dim H = 2$, so $\epsilon = 0$.

Step 4. In this step we define the invariant monomial Φ, the blocks of the equation and of the marks along H^0, and the exponent vectors ξ_k.

Let

$$M_0 := \{(e_1, e_2, e_3) \in \mathbb{Z}^3 \mid e_1 \equiv e_2 \equiv e_3 \pmod 2\},$$

and for a monomial $N = x^{n_0} y^{n_1} z^{n_2} t^{n_3}$ let

$$\gamma(N) := n_0(1,1,1) + 2(n_1, n_2, n_3) \in M_0.$$

The subgroup $\mathbf{S}$ is the kernel of the primitive character $(2,-1,-1,-1)$ of $\mathbb{T}^4$, hence a three-dimensional subtorus, and the map $\mathbb{Z}^4 \to M_0$, $n \to n_0(1,1,1) + 2(n_1, n_2, n_3)$, is surjective with kernel $\mathbb{Z}(2,-1,-1,-1)$. Thus the character group of $\mathbf{S}$ is $\mathbb{Z}^4/\mathbb{Z}(2,-1,-1,-1) \cong M_0$, under which the restriction to $\mathbf{S}$ of the character of a monomial N corresponds to $\gamma(N)$, and we write characters of $\mathbf{S}$ as elements of M_0. The linear map

$$\varpi : \mathbb{R}^3 \to V_{\mathbf{S}}, \qquad v \to (v_1 + v_2 + v_3, 2v_1, 2v_2, 2v_3),$$

is an isomorphism, and $\langle n, \varpi(v)\rangle = \langle \gamma(N), v\rangle$ for the exponent vector n of N. In particular, $\varpi(v)$ has all coordinates positive if and only if $v_1, v_2, v_3 > 0$.

The characters of $\mathbf{S}$ which are trivial on H^0 form a subgroup of M_0 of rank one with torsion-free quotient, the character group of the torus $\mathbf{S}/H^0$, so they form $\mathbb{Z}\nu$ for a primitive vector $\nu \in M_0$. Then $V = \varpi(\{v \mid \langle \nu, v\rangle = 0\})$. If ν had a positive and a negative coordinate, there would exist $v_1, v_2, v_3 > 0$ with $\langle \nu, v\rangle = 0$, and $\varpi(v) \in V$ would have all coordinates positive, contrary to **Step 3**. Replacing ν by $-\nu$, we may assume that $\nu \in \mathbb{Z}^3_{\geq 0}\backslash\{0\}$. Since $\nu \in M_0$, we have $\nu_1 \equiv \nu_2 \equiv \nu_3 \pmod 2$, so $\nu = \iota(1,1,1) + 2\beta$ with $\iota \in \{0,1\}$ and $\beta \in \mathbb{Z}^3_{\geq 0}$. Let

$$\Phi := x^{\iota} y^{\beta_1} z^{\beta_2} t^{\beta_3},$$

a non-constant monomial with $\gamma(\Phi) = \nu$. Since ν is primitive in M_0, the kernel of ν on $\mathbf{S}$ is connected, hence equal to H^0. Therefore ν maps H/H^0 injectively onto the group μ_q of q-th roots of unity, where $q := |H/H^0|$, and

$$H = \{g \in \mathbf{S} \mid \nu(g)^q = 1\}, \tag{E.52}$$

both sides being unions of q cosets of H^0. As G_i intersects any connected component of H for $i \gg 0$, the character of Φ on G_i has order exactly q for $i \gg 0$.

For any k, by (E.50) we may pass to a subsequence and fix a monomial M_k of degree at most $(m-1)/\mu_k$ which is a monomial of $h_{i,k}$ for any i. Let f_i^ν be the sum of the terms of f_i whose monomials N satisfy $\gamma(N) - (2,2,2) \in \mathbb{Z}\nu$, and let $h_{i,k}^\nu$ be the sum of the terms of $h_{i,k}$ whose monomials N satisfy $\gamma(N) - \gamma(M_k) \in \mathbb{Z}\nu$. Two monomials have equal characters on H^0 if and only if the difference of their images under γ belongs to $\mathbb{Z}\nu$.

We prove that there exist integers $d_i \to +\infty$ such that

$$f_i - f_i^\nu \in \mathfrak{m}^{d_i} \quad \text{and} \quad h_{i,k} - h_{i,k}^\nu \in \mathfrak{m}^{d_i} \text{ for any } k. \tag{E.53}$$

Fix an integer d, and let Ξ_d be the finite set of characters of $\mathbb{T}^4$ of the form $\chi_N - \chi_{N'}$, where χ_N is the character of a monomial N of degree at most d, $N' \in \{x^2, M_1, \dots, M_s\}$, and $\chi_N - \chi_{N'}$ is non-trivial on H^0. If $\chi \in \Xi_d$ were trivial on G_i for infinitely many i, then χ would be trivial on H, since any point of H is a limit of points of the G_i with such i, hence trivial on H^0, contrary to the definition of Ξ_d. Hence there exists i_d such that any $\chi \in \Xi_d$ is non-trivial on G_i for $i \geq i_d$. As f_i and $h_{i,k}$ are G_i-semi-invariant, any two monomials of f_i (resp. $h_{i,k}$) have equal characters on G_i, so for $i \geq i_d$ any monomial of $f_i - f_i^\nu$ and of $h_{i,k} - h_{i,k}^\nu$ has degree larger than d. We may take $i_1 < i_2 < \cdots$, and (E.53) holds with $d_i := \max\{d \mid i_d \leq i\}$ for $i \geq i_1$.

We compute f_i^ν. Let $\nu' \in \mathbb{Z}^3_{\geq 0}$ be the primitive vector with $\nu \in \mathbb{N}^+\nu'$. Since $2\nu' \in M_0 \cap \mathbb{Q}\nu = \mathbb{Z}\nu$, for $n' \in \mathbb{Z}^3$ we have $2(n' - (1,1,1)) \in \mathbb{Z}\nu$ if and only if $n' - (1,1,1) \in \mathbb{Z}\nu'$. By (a), the monomials of f_i are x^2 and monomials $y^{n_1} z^{n_2} t^{n_3}$ of degree at least three, yzt having coefficient 1. Thus a

monomial $y^{n_1}z^{n_2}t^{n_3}$ of f_i^ν has $(n_1,n_2,n_3)=(1,1,1)+l\nu'$ with $l\in\mathbb{Z}$, and $l\geq 0$ since the degree is at least three. Hence

$$f_i^\nu = x^2+yzt\Upsilon_i(\Psi),\qquad \Psi:=y^{\nu_1'}z^{\nu_2'}t^{\nu_3'}, \tag{E.54}$$

for a polynomial Υ_i in one variable with $\Upsilon_i(0)=1$.

We compute the exponent vectors of the marks. On monomials of x-degree at most one, γ is injective, since $n_0\in\{0,1\}$ is the common parity of the entries of $\gamma(N)$. The images under γ of the monomials of $h_{i,k}^\nu$ belong to $(\gamma(M_k)+\mathbb{Z}\nu)\cap\mathbb{Z}^3_{\geq 0}$, which is totally ordered componentwise since $\nu\geq 0$. Let $\xi_{i,k}$ be the least vector $\gamma(N)$ for N a monomial of $h_{i,k}^\nu$. Then $\xi_{i,k}\leq\gamma(M_k)$ componentwise, so $\xi_{i,k}$ takes finitely many values, and we may pass to a subsequence with $\xi_{i,k}=:\xi_k=(\xi_{k1},\xi_{k2},\xi_{k3})$ independent of i. Let N_k be the unique monomial of x-degree at most one with $\gamma(N_k)=\xi_k$, a monomial of $h_{i,k}^\nu$ for any i, and let

$$\ell_j:=1-\sum_k c_k\xi_{kj}\ (j=1,2,3),\qquad \ell:=(\ell_1,\ell_2,\ell_3),\qquad \lambda:=\min_{j:\nu_j>0}\frac{\ell_j}{\nu_j}.$$

If $h_{i,k}$ is a unit for some $i\geq i_1$, then $d_i\geq 1$, so by (E.53) the constant term of $h_{i,k}$ is a term of $h_{i,k}^\nu$, and since $\gamma(1)=0$, we have $\xi_k=0$ and $N_k=1$.

Step 5. In this step we compute on the toric cover of $W^\natural:=(x^2+yzt=0)\subset\mathbb{A}^4$ which pairs $(W_i,D_i+\lambda'\operatorname{div}(\Phi))$ are lc.

Let $\varphi:\mathbb{A}^3\to W^\natural$ be the morphism given by $x=\sqrt{-1}abc$, $y=a^2$, $z=b^2$, $t=c^2$, where a,b,c are the coordinates of $\mathbb{A}^3$, also written a_1,a_2,a_3. For $e\in\mathbb{Z}^3_{\geq 0}$ let $\mathbf{a}^e:=a^{e_1}b^{e_2}c^{e_3}$, so that $\varphi^*N=(\sqrt{-1})^{n_0}\mathbf{a}^{\gamma(N)}$ for a monomial $N=x^{n_0}y^{n_1}z^{n_2}t^{n_3}$. The morphism φ is the quotient by the group of sign changes $(a,b,c)\to(\pm a,\pm b,\pm c)$ with an even number of minus signs, whose non-trivial elements fix only coordinate axes: a monomial $\mathbf{a}^e$ is invariant under the sign changes if and only if $e_1\equiv e_2\equiv e_3 \pmod 2$, so the ring of invariants is generated by a^2,b^2,c^2,abc, with the only relation $(abc)^2=a^2b^2c^2$. Thus φ is finite and étale in codimension one, $W^\natural$ is normal, and $K_{\mathbb{A}^3}=\varphi^*K_{W^\natural}$.

Claim E.12. *Let $\lambda'\geq 0$ be a real number. Then the following holds for any sufficiently large i: the pair $(W_i,D_i+\lambda'\operatorname{div}(\Phi))$ is lc near 0 if and only if*

$$\sum_k c_k\xi_{kj}+\lambda'\nu_j\leq 1\quad \text{for } j=1,2,3.$$

Proof. If $\lambda'=0$, the factor Φ is omitted below. Let $\Lambda:=\{1,c_1,\dots,c_s,\lambda'\}\backslash\{0\}$, a finite subset of $\mathbb{R}_{>0}$, and let L' be the integer of Lemma 4.14 for $n=4$, the number $s+2$ of functions ($s+1$ if $\lambda'=0$), and Λ. Let $\sqrt{\Upsilon_i(\Psi)}\in\mathbb{C}[[y,z,t]]$ be the square root with constant term 1, let U_i be the sum of the terms of $\sqrt{\Upsilon_i(\Psi)}$ of degree less than L', let $\mathcal{U}_i:=(U_i\neq 0)\subset\mathbb{A}^4$, an open neighborhood of 0, and let ω_i be the automorphism $(x,y,z,t)\to(U_ix,y,z,t)$ of $\mathcal{U}_i$. By (E.54),

$$\begin{aligned}\omega_i^*f_i^\nu &= U_i^2(x^2+yzt)+yzt\big(\Upsilon_i(\Psi)-U_i^2\big),\\ \Upsilon_i(\Psi)-U_i^2 &= \Big(\sqrt{\Upsilon_i(\Psi)}-U_i\Big)\Big(\sqrt{\Upsilon_i(\Psi)}+U_i\Big)\in\mathfrak{m}^{L'}.\end{aligned} \tag{E.55}$$

For regular functions $\phi_1,\dots,\phi_l$ near $0\in\mathbb{A}^4$ and real numbers $\sigma_1,\dots,\sigma_l\geq 0$, the pair $(\mathbb{A}^4,\sum_j\sigma_j\operatorname{div}(\phi_j))$ is lc near 0 if and only if $(\operatorname{Spec}\mathfrak{R},\prod_j(\phi_j)^{\sigma_j})$ is lc, since a log resolution near 0 induces a log resolution of $\operatorname{Spec}\mathfrak{R}$ with equal discrepancies [dFEM11, Proposition 2.11(i)], and we use the equivalence below without further mention.

Assume that $d_i\geq L'$. On $\mathbb{A}^4$ let

$$\Delta_i:=\sum_k c_k\operatorname{div}(h_{i,k})+\lambda'\operatorname{div}(\Phi),\qquad \Delta_i^\nu:=\sum_k c_k\operatorname{div}(h_{i,k}^\nu)+\lambda'\operatorname{div}(\Phi),$$

and let $\sigma_j:=\sum_k c_k\xi_{kj}+\lambda'\nu_j$ for $j=1,2,3$. We show that the following conditions are equivalent, where lc means lc near 0:

(P1) $(W_i, D_i + \lambda' \operatorname{div}(\Phi))$ is lc;
(P2) $(\mathbb{A}^4, \operatorname{div}(f_i) + \Delta_i)$ is lc;
(P3) $(\mathbb{A}^4, \operatorname{div}(f_i^\nu) + \Delta_i^\nu)$ is lc;
(P4) $(\mathcal{U}_i, \omega_i^* \Gamma_i^\nu)$ is lc, where $\Gamma_i^\nu := \operatorname{div}(f_i^\nu) + \Delta_i^\nu$;
(P5) $(\mathcal{U}_i, W^\natural + \omega_i^* \Delta_i^\nu)$ is lc;
(P6) $(W^\natural \cap \mathcal{U}_i, \Delta_i^\natural)$ is lc, where $\Delta_i^\natural := \omega_i^* \Delta_i^\nu|_{W^\natural}$;
(P7) $(\mathbb{A}^3, \sum_{j=1}^3 \sigma_j \operatorname{div}(a_j))$ is lc.

By (a) and (c), W_i is normal and is not contained in the zero locus of $h_{i,k}$ or of Φ, so (P1)⇔(P2) by inversion of adjunction [Kaw07, Theorem]. By (E.53) and $d_i \geq L'$, Lemma 4.14 applied to the tuples $(f_i, h_{i,1}, \dots, h_{i,s}, \Phi)$ and $(f_i^\nu, h_{i,1}^\nu, \dots, h_{i,s}^\nu, \Phi)$ in both directions implies (P2)⇔(P3). Since ω_i is an automorphism of $\mathcal{U}_i$ fixing 0, (P3)⇔(P4). By (E.55), $\omega_i^* f_i^\nu - U_i^2(x^2 + yzt) \in \mathfrak{m}^{L'}$, and U_i^2 is a unit near 0, so Lemma 4.14 implies (P4)⇔(P5). The polynomial $h_{i,k}^\nu$ has x-degree at most one, so $\omega_i^* h_{i,k}^\nu = Q_0 + xU_iQ_1$ for polynomials Q_0, Q_1 in y, z, t, not both zero. As U_i is a polynomial in y, z, t, the polynomials $\omega_i^* h_{i,k}^\nu$ and $\omega_i^* \Phi$ are non-zero of x-degree at most one, hence not divisible by the irreducible polynomial $x^2 + yzt$, which is monic of degree two in x. Thus $\omega_i^* h_{i,k}^\nu|_{W^\natural} \neq 0$ and $\omega_i^* \Phi|_{W^\natural} \neq 0$, and (P5)⇔(P6) by [Kaw07, Theorem].

For (P6)⇔(P7), we compute the pullbacks under φ. Since $\omega_i^* N = U_i^{n_0} N$ for a monomial $N = x^{n_0} y^{n_1} z^{n_2} t^{n_3}$,

$$\varphi^* \omega_i^* h_{i,k}^\nu = \mathbf{a}^{\xi_k} v_{i,k}, \qquad \varphi^* \omega_i^* \Phi = (\sqrt{-1})^\iota (\varphi^* U_i)^\iota \mathbf{a}^\nu,$$

where

$$v_{i,k} := \sum_N b_N (\sqrt{-1})^{n_0} (\varphi^* U_i)^{n_0} \mathbf{a}^{\gamma(N) - \xi_k},$$

the sum running over the monomials $N = x^{n_0} y^{n_1} z^{n_2} t^{n_3}$ of $h_{i,k}^\nu$, and b_N being the coefficient of N in $h_{i,k}^\nu$. For $N \neq N_k$ we have $\gamma(N) - \xi_k \in \mathbb{N}^+ \nu$ by the injectivity of γ, so $\mathbf{a}^{\gamma(N) - \xi_k}$ vanishes at 0. As $(\varphi^* U_i)(0) = 1$ and $b_{N_k} \neq 0$, $v_{i,k}$ and $\varphi^* U_i$ are units near 0. Since $\varphi^{-1}(0) = \{0\}$ and $K_{\mathbb{A}^3} = \varphi^* K_{W^\natural}$, the pullback of the boundary of (P6) near 0 is the boundary of (P7), and (P6)⇔(P7) by [KM98, Proposition 5.20(4)].

Finally, the boundary of (P7) has simple normal crossing support. If the coefficients of the boundary are at most 1, then (P7) holds [KM98, Corollary 2.31(3)] ([KM98, Corollary 2.31(3)] assumes $\mathbb{Q}$-coefficients, and its proof is valid without change for $\mathbb{R}$-coefficients), and if a coefficient is larger than 1, then the corresponding coordinate hyperplane has negative log discrepancy. This proves the claim. □

Proof of Lemma E.11 continued. Since (W_i, D_i) is klt near 0 by (c), Claim E.12 with $\lambda' = 0$ implies that

$$\ell_j \geq 0 \quad \text{for } j = 1, 2, 3, \quad \text{hence} \quad \lambda \geq 0. \tag{E.56}$$

Step 6. In this step we prove that $\limsup_i \alpha_i \leq \lambda/q$ and that $\alpha_i \geq \lambda/q$ for $i \gg 0$, which contradicts the strict increase of the α_i.

Choose j_0 with $\nu_{j_0} > 0$ and $\lambda = \ell_{j_0}/\nu_{j_0}$. For a real number η with $0 < \eta < 1/(q(1+\nu_1+\nu_2+\nu_3))$, let $v(\eta) \in \mathbb{R}^3_{>0}$ be the vector with $v(\eta)_j = \eta$ for $j \neq j_0$ and

$$v(\eta)_{j_0} = \frac{1}{\nu_{j_0}} \Big(\frac{1}{q} - \eta \sum_{j \neq j_0} \nu_j \Big).$$

Then $\langle \nu, v(\eta) \rangle = 1/q$, so $\rho(\varpi(v(\eta))) \in \mathbf{S}$ satisfies $\nu(\rho(\varpi(v(\eta)))) = e^{2\pi\sqrt{-1}/q}$, and $\rho(\varpi(v(\eta))) \in H$ by (E.52). Moreover,

$$\langle \ell, v(\eta) \rangle = \frac{\lambda}{q} + \eta \sum_{j \neq j_0} (\ell_j - \lambda \nu_j). \tag{E.57}$$

Fix such an η. As the G_i converge to H in the Hausdorff distance, there exist $\varsigma_i \in G_i$ converging to $\rho(\varpi(v(\eta)))$, and since ρ is a covering map, there exist $w_i \in \rho^{-1}(\varsigma_i)$ converging to $\varpi(v(\eta))$. For $i \gg 0$, $w_i \in N_{G_i} \cap \mathbb{R}^4_{>0}$, where $N_{G_i} = \rho^{-1}(G_i)$ [Liu26, Definition 2.15], G_i being regarded as a subgroup of $\mathbb{T}^4$ by (b). Since $\varsigma_i \in \mathbf{S}$, the number $2w_{i,1} - w_{i,2} - w_{i,3} - w_{i,4}$ is an integer converging to 0, hence equal to 0 for $i \gg 0$. Thus $w_i = \varpi(\bar{v}_i)$ with $\bar{v}_i \to v(\eta)$, and a monomial N has w_i-weight $\langle \gamma(N), \bar{v}_i \rangle$. The coordinates of w_i are bounded below by a positive constant for $i \gg 0$, so by (E.53) the w_i-weights of the monomials of $f_i - f_i^\nu$ and $h_{i,k} - h_{i,k}^\nu$ tend to $+\infty$. By (E.54), the monomials of f_i^ν are x^2 and $yzt\Psi^l$ with $l \geq 0$, of w_i-weights

$$2(\bar{v}_{i,1} + \bar{v}_{i,2} + \bar{v}_{i,3}) \quad \text{and} \quad 2(\bar{v}_{i,1} + \bar{v}_{i,2} + \bar{v}_{i,3}) + 2l\langle \nu', \bar{v}_i \rangle.$$

The images under γ of the monomials of $h_{i,k}^\nu$ belong to $\xi_k + \mathbb{Z}_{\geq 0}\nu$. Hence, for $i \gg 0$,

$$\mathrm{wt}_{w_i}(f_i) \geq 2(\bar{v}_{i,1} + \bar{v}_{i,2} + \bar{v}_{i,3}), \qquad \mathrm{wt}_{w_i}(h_{i,k}) \geq \langle \xi_k, \bar{v}_i \rangle \text{ for any } k.$$

By Lemma 4.7, whose hypotheses hold by (a)–(c),

$$\alpha_i \leq \sum_{j=1}^4 w_{i,j} - \mathrm{wt}_{w_i}(f_i) - \sum_k c_k \,\mathrm{wt}_{w_i}(h_{i,k}) \leq (\bar{v}_{i,1} + \bar{v}_{i,2} + \bar{v}_{i,3}) - \sum_k c_k \langle \xi_k, \bar{v}_i \rangle = \langle \ell, \bar{v}_i \rangle$$

for $i \gg 0$. Thus $\limsup_i \alpha_i \leq \langle \ell, v(\eta) \rangle$ for any such η, and letting $\eta \to 0$ in (E.57),

$$\limsup_i \alpha_i \leq \frac{\lambda}{q}. \tag{E.58}$$

Since $\alpha_i \geq \alpha_1 > 0$, (E.58) implies that $\lambda > 0$. By the definition of λ and (E.56), $\sum_k c_k \xi_{kj} + \lambda \nu_j \leq 1$ for $j = 1, 2, 3$: for $\nu_j > 0$ the inequality is equivalent to $\lambda \leq \ell_j / \nu_j$, and for $\nu_j = 0$ to $\ell_j \geq 0$. By Claim E.12 with $\lambda' = \lambda$, the pair $(W_i, D_i + \lambda \,\mathrm{div}(\Phi))$ is lc near 0 for $i \gg 0$. The monomial Φ is G_i-semi-invariant, vanishes at 0, and by **Step 4** has a character of order q on G_i for $i \gg 0$. By Lemma 5.8 with $Z = \{0\}$,

$$\alpha_i \geq \frac{\lambda}{q} \quad \text{for } i \gg 0.$$

As the α_i are strictly increasing, $\alpha_i < \alpha_{i+1} \leq \lim_i \alpha_i \leq \lambda/q$ by (E.58), a contradiction with $\alpha_i \geq \lambda/q$. Therefore $\dim H = 2$ is impossible.

Step 7. In this step we assume that $\dim H = 1$, which holds by **Steps 3**–**6**, and determine the circle.

By Lemma 4.6, H^0 is the image of $\mathbb{R}w$ for a primitive $w \in \mathbb{Z}^4$, $R_i := G_i \cap \mathbf{T}_{(w)}$ is cyclic of order $r_i \to +\infty$, and $[G_i : R_i] = |H/H^0|$ for $i \gg 0$. Let $e_0 := |H/H^0|$. Since $\mathbf{T}_{(-w)} = \mathbf{T}_{(w)}$, we may replace w by $-w$ and assume that $u := w_1 \geq 0$. As $V = \mathbb{R}w$, **Step 3** and (E.51) imply that

$$w_2 + w_3 + w_4 = 2u, \qquad 3w_4 = 2u \text{ if } \epsilon = 1, \tag{E.59}$$

and that neither w nor $-w$ has all coordinates positive. The element $\rho(w/r_i)$ generates R_i and multiplies a monomial with exponent vector n by $e^{2\pi\sqrt{-1}\langle n, w \rangle / r_i}$. Since f_i is G_i-semi-invariant and x^2 is a monomial of f_i,

$$\langle n, w \rangle \equiv 2u \pmod{r_i} \quad \text{for the exponent vector } n \text{ of any monomial of } f_i. \tag{E.60}$$

Assume first that $u > 0$. Suppose that exactly one of w_2, w_3, w_4 is zero, and write $\zeta_1, \zeta_2, \zeta_3$ for y, z, t in an order in which ζ_1 has weight 0 and ζ_2, ζ_3 have the non-zero weights w_2', w_3', so that $w_2' + w_3' = 2u$. Let i be so large that $r_i > 2u + |w_2'| + |w_3'|$. A monomial ζ_1^l, $\zeta_2\zeta_1^l$, or $\zeta_3\zeta_1^l$ has weight 0, $w_2' = 2u - w_3'$, or $w_3' = 2u - w_2'$, none of which is congruent to $2u$ modulo r_i. By (E.60), $g_i \in (\zeta_2, \zeta_3)^2$, so $f_i \in (x, \zeta_2, \zeta_3)^2$ and W_i is singular along the line $(x = \zeta_2 = \zeta_3 = 0) \subset W_i$, contradicting (a).

Suppose that exactly two of w_2, w_3, w_4 are zero, say the weights of ζ_1, ζ_2, so that ζ_3 has weight $2u$. A monomial in ζ_1, ζ_2 has weight $0 \not\equiv 2u \pmod{r_i}$ for $r_i > 2u$. By (E.60) and (a),

$$f_i = x^2 + \zeta_3 Q(\zeta_1, \zeta_2) + \zeta_3^2 Q'(\zeta_1, \zeta_2, \zeta_3)$$

for polynomials $Q \in (\zeta_1, \zeta_2)^2$ and Q'. On the plane $\Pi := (x = \zeta_3 = 0) \subset W_i$, the partial derivatives of f_i with respect to x, ζ_1, ζ_2 vanish and the one with respect to ζ_3 is Q. Thus W_i is singular along $\Pi \cap (Q = 0)$, and $\dim_0(\Pi \cap (Q = 0)) \geq 1$ as $Q(0,0) = 0$, contradicting (a). If $w_2 = w_3 = w_4 = 0$, then $u = 0$ by (E.59), contrary to $u > 0$. Therefore $w_2, w_3, w_4 \neq 0$. They are not all positive, as w does not have all coordinates positive, and not all negative, as $w_2 + w_3 + w_4 = 2u > 0$.

- If two of them are negative, then $\epsilon = 0$, since for $\epsilon = 1$ we have $w_4 = 2u/3 > 0$ and $w_2 + w_3 = 4u/3 > 0$ by (E.59), so at most one of w_2, w_3, w_4 is negative. After a permutation of y, z, t, which preserves yzt, $w = (u, -A, -B, C)$ with $A, B, C > 0$, and $C = 2u + A + B$ by (E.59).
- If one of them is negative, then $w = (u, -A, B, C)$ with $A, B, C > 0$ after a permutation of y, z, t preserving $yzt + \epsilon t^3$ which moves the negative weight to y, and $B + C = 2u + A$ by (E.59). Such a permutation exists: for $\epsilon = 0$ any permutation of y, z, t preserves yzt, and for $\epsilon = 1$, $w_4 = 2u/3 > 0$ by (E.59), so the negative weight is w_2 or w_3, and the exchange of y and z preserves $yzt + t^3$.

Assume next that $u = 0$. If $\epsilon = 1$, then $w_4 = 0$ and $w_2 = -w_3$ by (E.59), so $w = \pm(0, 1, -1, 0)$ as w is primitive, and we replace w by $-w$ if necessary. Let $\epsilon = 0$. If one of w_2, w_3, w_4 is zero, then exactly one is, since $w \neq 0$ and $w_2 + w_3 + w_4 = 0$. After a permutation of y, z, t moving the zero weight to t, $w = \pm(0, 1, -1, 0)$, and we replace w by $-w$ if necessary. If none of w_2, w_3, w_4 is zero, then they have both signs, and after replacing w by $-w$ if necessary, two of them are negative. After a permutation of y, z, t, $w = (0, -A, -B, C)$ with $A, B, C > 0$ and $C = A + B$.

Thus w is one of the vectors in (2), $|G_i \cap \mathbf{T}_{(w)}| = r_i \to +\infty$, and $[G_i : G_i \cap \mathbf{T}_{(w)}] = e_0$ for $i \gg 0$. This proves (2). □

We now prove Lemma E.13, which constructs the crepant models along the circles that remain in Lemma E.11(2). For all labels at once, Lemma E.13 fixes one weighted blow-up along a ray close to the circle, so that the coefficients of the exceptional divisor on the quotient charts form a non-decreasing sequence whose terms belong to $(0, 1)$ unless $w_1 = 0$ and all marks are units (Lemma E.13(3)).

Lemma E.13. *Notation as in Set-up 11.1. Let $(W_i \ni o_i, G_i, (h_{i,k})_k)$ be a sequence of such data with s, m and the μ_k fixed, or of numerical admissions with the groups G_i (Proposition 11.8) of such data $(W_i' \ni o_i', G_i, (h_{i,k}')_k)$, taken with the integers N and L of Construction-Theorem 4.13 larger than $\max_k \operatorname{mult}_{o_i'} h_{i,k}'$, such that $o_i = 0$ and $W_i = (f_i = 0) \subset \mathbb{A}^4$ for a polynomial $f_i \in x^2 + \mathbb{C}[y, z, t]$ in coordinates x, y, z, t on which G_i acts diagonally, with residual cubic $yzt + \epsilon t^3$, $\epsilon \in \{0, 1\}$, and such that the $h_{i,k}$ are polynomials. Let $w \in \mathbb{Z}^4$ be a primitive vector of one of the forms listed in Lemma E.11(2) such that, for $\mathbf{T} := \mathbf{T}_{(w)}$, $r_i := |G_i \cap \mathbf{T}| \to +\infty$ and $[G_i : G_i \cap \mathbf{T}]$ is bounded. Write each $h_{i,k}$ in normal form (Definition-Lemma 11.5), pass to a subsequence on which the weights κ_k of the principal blocks $h_{i,k}^0$ along $\mathbf{T}$ are independent of i, and let δ be the adjusted mass of $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$ along $\mathbf{T}$ (Set-up 5.1). Then, after passing to a further subsequence, there exist a non-empty open interval $I_0 \subset \mathbb{R}_{>0}$ and a finite set $\mathcal{G}$ of positive integers with the following property. For any $\theta_* \in I_0$, after passing to a further subsequence, there exist positive integers L_i coprime to r_i with L_i/r_i converging monotonically to θ_*, primitive vectors $v_i \in \mathbb{Z}^4_{>0}$, and G_i-equivariant changes of coordinates, each of which commutes with $\mathbf{T}$ and adds at most one monomial to one of y, z, t, such that in the new coordinates the following hold.*

(1) $v_i = (r_i v_0 + \sigma L_i w)/g$ for an integer $g \in \mathcal{G}$ and a pair (v_0, σ) independent of i, as follows. If $w = (u, -A, -B, C)$ with $u \geq 0$, $A, B, C > 0$, and $C = 2u + A + B$, then (v_0, σ) is $((1,1,1,0), 1)$ or $((1,0,0,2), -1)$. If $w = (u, -A, B, C)$ with $u, A, B, C > 0$ and $B + C = 2u + A$, then (v_0, σ) is $(v_0^+, 1)$ or $((1,0,1,1), -1)$, where $v_0^+ := (1,1,1,0)$ if $B \leq C$ and $v_0^+ := (1,1,0,1)$ if $B > C$. If $w = (0, 1, -1, 0)$, then $v_0 = (1,1,0,1)$, $\sigma = -1$, and $g = 1$, that is, $v_i = (r_i, r_i - L_i, L_i, r_i)$.

(2) *The lowest v_i-weighted part of f_i is x^2+yzt. If $w=(u,-A,-B,C)$ with $A=B$, then the changes of coordinates are the identity.*

(3) *Let j_k be the least v_0-weighted degree of a monomial of $h^0_{i,k}$, which is independent of i, and let $J:=\sum_k c_k j_k$. Then the coefficients b_i of $\mathrm{div}(q)$ on the quotient charts of Definition-Lemma 11.4(2) for v_i, with $G_i\cap\mathbf{T}$ acting on the radial parameter, are*

$$b_i=1-\frac{1}{g}+\frac{1}{g}\Big(J+\sigma\delta\frac{L_i}{r_i}\Big),$$

and they are non-decreasing in i. If $w_1=0$ and all $h_{i,k}$ are units, then $b_i=0$ for any i. Otherwise $b_i\in(0,1)$ for any i, and $\lim_i b_i\in(0,1)$.

Proof. The proof is long, so we divide it into several steps. If the data are numerical admissions as in the statement, then Definition-Lemmas 11.4 and 11.5(1)(2), used below and for the normal forms in the statement, apply to them by Lemma 11.10(1), and the bound of Definition-Lemma 11.5(3) holds for them by Lemma 11.10(2), since $N,L>\max_k \mathrm{mult}_{o'_i} h'_{i,k}$ and $(W'_i\ni o'_i)$ is of type cD (Definition 9.1): by Construction-Theorem 4.13(3), the completion of the local ring of W'_i at o'_i is isomorphic to that of W_i at o_i, so $(W'_i\ni o'_i)$ has quadratic rank one and, up to a linear change of coordinates and a non-zero scalar, the residual cubic $yzt+\epsilon t^3$ [Liu26, Definition-Lemma 11.1], which is not the cube of a linear form. Let $(x_1,x_2,x_3,x_4):=(x,y,z,t)$, $u:=w_1$, and $M:=\max_j|w_j|$. By Lemma E.11(2), one of the following holds:

(i) $w=(u,-A,-B,C)$ with $u\geq 0$, $A,B,C>0$, and $C=2u+A+B$;
(ii) $w=(u,-A,B,C)$ with $u,A,B,C>0$ and $B+C=2u+A$;
(iii) $w=(0,1,-1,0)$.

For a monomial x^α in x,y,z,t we write $|\alpha|$ for the total degree of x^α and $\langle w,\alpha\rangle$ for the $\mathbf{T}$-weight of x^α, and, for $v\in\mathbb{Z}^4_{\geq 0}$, we call $\langle v,\alpha\rangle$ the *v-degree* of x^α. Let $\tau:=\sum_k c_k\kappa_k$ be the mass of D_i along $\mathbf{T}$. In each of (i)–(iii) the sum of the coordinates of w is $3u$, and x^2 is the only monomial of least degree of f_i. Thus $\epsilon_{\mathbf{T}}=\sum_j w_j-\kappa(f_i)=3u-2u=u$ (Set-up 5.1), and

$$\delta=\tau-u.$$

Step 1. In this step we collect congruences and bounds for the weights and degrees of monomials, and we fix the degrees that enter J on a subsequence.

Let $\zeta_i:=\exp(2\pi\sqrt{-1}/r_i)$, and let $\eta_i\in\mathbf{T}$ be the element which multiplies x,y,z,t by $\zeta_i^{w_1},\zeta_i^{w_2},\zeta_i^{w_3},\zeta_i^{w_4}$ respectively. Since w is primitive, η_i generates the cyclic group $R_i:=G_i\cap\mathbf{T}$ of order r_i, and η_i multiplies a monomial x^α by $\zeta_i^{\langle w,\alpha\rangle}$. Since f_i and the normal forms of the $h_{i,k}$ are G_i-semi-invariant, the $\mathbf{T}$-weights of the monomials of f_i belong to $2u+r_i\mathbb{Z}$, and the $\mathbf{T}$-weights of the monomials of the normal form of $h_{i,k}$ belong to $\kappa_k+r_i\mathbb{Z}$. Hence, for $i\gg 0$, any monomial x^α of f_i with $\langle w,\alpha\rangle\neq 2u$ satisfies $|\langle w,\alpha\rangle|\geq r_i-2u$, and so

$$|\alpha|\geq\frac{r_i-2u}{M},\tag{E.61}$$

and any monomial x^α of the normal form of $h_{i,k}$ with $\langle w,\alpha\rangle\neq\kappa_k$ satisfies

$$|\alpha|\geq\frac{r_i-|\kappa_k|}{M}\tag{E.62}$$

(Definition-Lemma 11.5(2)). The inequalities (E.61) and (E.62) only depend on the $\mathbf{T}$-weights, so they hold again after any change of coordinates which commutes with $\mathbf{T}$.

If $\epsilon=1$, then t^3 is a monomial of f_i, and so $3w_4-2u\in r_i\mathbb{Z}$. In (i), $3C-2u=4u+3A+3B>0$, hence $\epsilon=0$ for $i\gg 0$. In (ii), $3C=2u$ for $i\gg 0$ if $\epsilon=1$, and then $B=2u+A-C>C$. After discarding finitely many indices, we may assume that $\epsilon=0$ in (i) and that $3C=2u$ in (ii) if $\epsilon=1$.

The residual cubic $yzt + \epsilon t^3$ is not the cube of a linear form, so $(W_i \ni o_i)$ is of type cD (Definition 9.1), and Definition-Lemma 11.5(3) yields

$$\sum_k c_k \operatorname{mult}_{o_i} h_{i,k} \le 1 - \frac{1}{m}. \tag{E.63}$$

Let $d_{i,k}$ be the least total degree of a monomial of the normal form of $h_{i,k}$. Then $\operatorname{mult}_{o_i} h_{i,k} \ge d_{i,k}$, and we show $d_{i,k} = \operatorname{mult}_{o_i} h_{i,k}$. If $h_{i,k}|_{W_i} \in \mathfrak{m}_{o_i}^{d_{i,k}+1}$, then the normal form H of $h_{i,k}$ satisfies $H \equiv Qf_i$ modulo $(x,y,z,t)^{d_{i,k}+1}$ for an element Q of the local ring of $\mathbb{A}^4$ at 0. Comparing homogeneous parts of least degree, $Q \ne 0$ and the homogeneous part of degree $d_{i,k}$ of H, a non-zero polynomial of x-degree at most one, is x^2 times the homogeneous part of least degree of Q, a contradiction. By (E.63), $d_{i,k} \le (1 - \frac{1}{m})/c_k$. By the definition of κ_k (Set-up 5.1), $h^0_{i,k}$ contains a monomial of total degree $d_{i,k}$. If $h_{i,k}$ is a unit, then $d_{i,k} = 0$, $\kappa_k = 0$, and $h^0_{i,k}$ contains the constant monomial 1. Otherwise $h^0_{i,k}$ has no constant term.

We now define the base vectors. In (i), let $v_0^+ := (1,1,1,0)$ and $v_0^- := (1,0,0,2)$. In (ii), let $(z_1, z_2, B_1, B_2) := (z, t, B, C)$ if $B \le C$ and $(z_1, z_2, B_1, B_2) := (t, z, C, B)$ if $B > C$. Thus $B_1 \le B_2$, the $\mathbf{T}$-weights of z_1 and z_2 are B_1 and B_2, $B_1 + B_2 = 2u + A$, so $2B_2 \ge 2u + A > 2u$, and $z_1 = t$ if $\epsilon = 1$. Let v_0^+ be the vector with coordinate 1 at x, y, z_1 and coordinate 0 at z_2, and let $v_0^- := (1,0,1,1)$. Thus $v_0^+ = (1,1,1,0)$ if $B \le C$ and $v_0^+ = (1,1,0,1)$ if $B > C$. Let $j^{\pm}_{i,k}$ be the least $v_0^{\pm}$-degree of a monomial of $h^0_{i,k}$, and in (iii) let $j_{i,k}$ be the least $(1,1,0,1)$-degree of a monomial of $h^0_{i,k}$. All coordinates of $v_0^{\pm}$ and $(1,1,0,1)$ are at most 2, so $j^{\pm}_{i,k} \le 2d_{i,k}$ in (i) and (ii), and $j_{i,k} \le 2d_{i,k}$ in (iii). Passing to a subsequence, we may assume that, for any k, the integers $j^{\pm}_{i,k}$ in (i) and (ii), the integers $j_{i,k}$ in (iii), and whether $h_{i,k}$ is a unit are independent of i. We write $j^{\pm}_k$ and j_k for them, and $J_{\pm} := \sum_k c_k j^{\pm}_k$. Each of $(1,1,1,0)$, $(1,1,0,1)$, and $(1,0,1,1)$ is at most $(1,1,1,1)$ coordinatewise, so the corresponding least degree is at most $d_{i,k}$, and (E.63) implies

$$J_+ \le 1 - \frac{1}{m}, \qquad J_- \le 1 - \frac{1}{m} \text{ in (ii)}, \qquad \sum_k c_k j_k \le 1 - \frac{1}{m} \text{ in (iii)}. \tag{E.64}$$

Step 2. In this step we fix the base vector v_0 and the sign σ, common to all labels, and we prove (a)–(c) below.

In (i) and (ii), let $(v_0, \sigma) := (v_0^+, 1)$ if $J_+ > 0$, and $(v_0, \sigma) := (v_0^-, -1)$ if $J_+ = 0$. In (iii), let $(v_0, \sigma) := ((1,1,0,1), -1)$. Let j_k be the least v_0-degree of a monomial of $h^0_{i,k}$, and $J := \sum_k c_k j_k$. We show the following.

(a) If $u > 0$, or if some $h_{i,k}$ is not a unit, then either $J > 0$, or $J = 0$ and $\sigma\delta > 0$.
(b) If $u = 0$ and all $h_{i,k}$ are units, then $J = \delta = 0$.
(c) In (i) with $\sigma = -1$, $\delta = \frac{C}{2}J - u$.

If $\sigma = 1$, then $J = J_+ > 0$.

Consider (i) with $\sigma = -1$. Since $J_+ = 0$, for any k the block $h^0_{i,k}$ contains a monomial of v_0^+-degree 0, that is, t^{l_k} for an integer $l_k \ge 0$, and $l_k = 0$ if and only if $h_{i,k}$ is a unit. Thus $\kappa_k = Cl_k$. A monomial $x^e y^p z^q t^l$ of $h^0_{i,k}$, with $e \in \{0,1\}$, satisfies $ue - Ap - Bq + Cl = Cl_k$, hence $Cl \ge Cl_k - ue > C(l_k - 1)$ as $u < C$. So $l \ge l_k$, and the v_0^--degree $e + 2l$ of the monomial is at least $2l_k$, with equality for t^{l_k}. Therefore $j_k = 2l_k$, $J = 2\sum_k c_k l_k$, and

$$\tau = \sum_k c_k C l_k = \frac{C}{2} J,$$

which implies (c). If some $h_{i,k}$ is not a unit, then $J \ge 2c_k > 0$. If all $h_{i,k}$ are units, then $J = \tau = 0$ and $\delta = -u$, so $\sigma\delta = u$, which is positive if $u > 0$ and zero if $u = 0$.

Consider (ii) with $\sigma = -1$. Since $J_+ = 0$, for any k the block $h^0_{i,k}$ contains $z_2^{l_k}$ for an integer $l_k \ge 0$, and $l_k = 0$ if and only if $h_{i,k}$ is a unit. Thus $\kappa_k = B_2 l_k$. A monomial of v_0^--degree 0 is a

power y^p, whose $\mathbf{T}$-weight $-Ap$ is at most 0. If $h_{i,k}$ is not a unit, then $\kappa_k > 0$, so $h^0_{i,k}$ has no monomial of v_0^--degree 0, $j_k \geq 1$, and $J \geq c_k > 0$. If all $h_{i,k}$ are units, then $J = 0$ and $\delta = -u$, so $\sigma\delta = u > 0$.

Consider (iii), where $u = 0$ and $\delta = \tau$. If all $h_{i,k}$ are units, then all κ_k and j_k are 0, so $J = \delta = 0$. Assume that some $h_{i,k}$ is not a unit and that $J = 0$. Then for any k the block $h^0_{i,k}$ contains a monomial of $(1,1,0,1)$-degree 0, that is, z^{q_k} for an integer $q_k \geq 0$, and $q_k = 0$ if and only if $h_{i,k}$ is a unit. Thus $\kappa_k = -q_k$, $\delta = -\sum_k c_k q_k < 0$, and $\sigma\delta > 0$. This proves (a), (b), and (c). Thus v_0, σ, and w are paired as in (1).

Step 3. In this step we change coordinates so that the part of least v_0-degree of the principal block of f_i is $x^2 + yzt$.

Let P_i be the sum of the terms of f_i of $\mathbf{T}$-weight $2u$. Then $P_i - x^2 \in \mathbb{C}[y,z,t]$ has no monomial of total degree at most two, and the cubic monomials of $P_i - x^2$ are among those of $yzt + \epsilon t^3$. A monomial $y^p z^q t^l$ of $P_i - x^2$ has $\mathbf{T}$-weight $2u$. We go through the five choices of (v_0, σ).

(i) with $\sigma = 1$. The v_0-degree of $y^p z^q t^l$ is $p + q$, and $-Ap - Bq + Cl = 2u$. If $p + q = 0$, then $Cl = 2u < C$, so $l = 0$, which is excluded since $f_i(0) = 0$. If $p + q = 1$, then Cl is $2u + A$ or $2u + B$, which belong to $(0, C)$, a contradiction. If $p + q = 2$, then yzt^l requires $l = 1$, and y^2t^l requires $Cl = 2u + 2A$, that is, $l = 1 + (A - B)/C \in (0, 2)$. Thus $l = 1$ and $A = B$, and y^2t is a cubic monomial which is not a monomial of yzt. Similarly, z^2t^l requires $l = 1$ and $A = B$, and z^2t is a cubic monomial which is not a monomial of yzt. No change of coordinates is made.

(i) with $\sigma = -1$. The v_0-degree of $y^p z^q t^l$ is $2l$. If $l = 0$, then $-Ap - Bq = 2u \geq 0$ forces $p = q = 0$, which is excluded. If $l = 1$, then $Ap + Bq = C - 2u = A + B$. Write $A = A' \gcd(A, B)$ and $B = B' \gcd(A, B)$. The integer solutions are $(p, q) = (1 + B'n, 1 - A'n)$, $n \in \mathbb{Z}$, and they are non-negative only for $n = 0$, for $n = 1$ if $A' = 1$, and for $n = -1$ if $B' = 1$. If $A' = B' = 1$, the extra monomials y^2t and z^2t are cubic monomials which are not monomials of yzt, and no change of coordinates is made. If $A' = 1 < B'$, the part of v_0-degree two of P_i is $x^2 + yzt + \beta_i y^{B'+1}t$ for some $\beta_i \in \mathbb{C}$, and we replace z by $z + \beta_i y^{B'}$. If $B' = 1 < A'$, the part of v_0-degree two of P_i is $x^2 + yzt + \beta_i z^{A'+1}t$ for some $\beta_i \in \mathbb{C}$, and we replace y by $y + \beta_i z^{A'}$. If $A = B$, then $A' = B' = 1$, so in (i) no change of coordinates is made for either choice of σ, which proves the last sentence of (2).

(ii) with $\sigma = 1$. Write the monomial as $y^p z_1^q z_2^l$. The v_0-degree of $y^p z_1^q z_2^l$ is $p + q$, and $-Ap + B_1 q + B_2 l = 2u = B_1 + B_2 - A$. If $p + q = 0$, then $B_2 l = 2u < 2B_2$, so $l \leq 1$, and both $l = 0$ (as $u > 0$) and the linear monomial z_2 are excluded. If $p + q = 1$, then yz_2^l requires $l = 1 + B_1/B_2 \in (1, 2]$, so $l = 2$, $B_1 = B_2$, and yz_2^2 is a cubic monomial which is not a monomial of $yzt + \epsilon t^3$, while $z_1 z_2^l$ requires $l = 1 - A/B_2 < 1$, so $l = 0$ and z_1 is linear, which is excluded. If $p + q = 2$, then $yz_1 z_2^l$ requires $l = 1$, the monomial $z_1^2 z_2^l$ requires $l = 1 - (A + B_1)/B_2 < 1$, so $l = 0$ and z_1^2 is quadratic, and $y^2 z_2^l$ requires $l = 1 + (A + B_1)/B_2 > 1$. In the last case, if l is an integer, then the part of v_0-degree two of P_i is $x^2 + yzt + \beta_i y^2 z_2^l$, and we replace z_1 by $z_1 + \beta_i y z_2^{l-1}$.

(ii) with $\sigma = -1$. The v_0-degree of $y^p z_1^q z_2^l$ is $q + l$. If $q + l = 0$, then $-Ap = 2u > 0$, a contradiction. If $q + l = 1$, then $y^p z_1$ requires $p = 1 - B_2/A < 1$ and $y^p z_2$ requires $p = 1 - B_1/A < 1$, so $p = 0$ and the monomial is linear. If $q + l = 2$, then $y^p z_1 z_2$ requires $p = 1$, the monomial $y^p z_1^2$ requires $p = 1 + (B_1 - B_2)/A \leq 1$, so it is quadratic or the cubic monomial yz_1^2 with $B_1 = B_2$, and $y^p z_2^2$ requires $p = 1 + (B_2 - B_1)/A \geq 1$. Here $p = 1$ is the cubic monomial yz_2^2, and if $p \geq 2$ is an integer, then the part of v_0-degree two of P_i is $x^2 + yzt + \beta_i y^p z_2^2$, and we replace z_1 by $z_1 + \beta_i y^{p-1} z_2$. None of the cubic monomials yz_1^2, yz_2^2 is a monomial of $yzt + \epsilon t^3$, since $z_1 = t$ if $\epsilon = 1$.

(iii). Here $\langle w, \alpha \rangle = 0$ means $p = q$, and the v_0-degree of $y^p z^p t^l$ is $p + l$. The monomials with $p + l \leq 2$ are 1, yz, t, t^2, yzt, and y^2z^2, and only the last two can occur in $P_i - x^2$. Thus the part of v_0-degree two of P_i is $x^2 + yzt + \beta_i y^2 z^2$, and we replace t by $t + \beta_i yz$.

In each case the change replaces one coordinate x_{j_0}, $j_0 \in \{2, 3, 4\}$, by $x_{j_0} + \beta_i \mathfrak{n}$, where $\mathfrak{n}$ is a monomial of total degree at least two in the coordinates x_j with $j \in \{2, 3, 4\} \backslash \{j_0\}$, whose

$\mathbf{T}$-weight and v_0-degree are equal to those of x_{j_0}. If $\beta_i \neq 0$, then $\mathfrak{n}yzt/x_{j_0}$ and yzt are monomials of the G_i-semi-invariant polynomial f_i, so they have equal G_i-characters, and hence x_{j_0} and $\mathfrak{n}$ have equal G_i-characters. Thus the change is G_i-equivariant and commutes with $\mathbf{T}$. It is an automorphism of $\mathbb{C}[x,y,z,t]$ which fixes x and preserves the $\mathbf{T}$-weights and the v_0-degrees of homogeneous polynomials. Since $\mathfrak{n}$ has total degree at least two, the change does not alter the homogeneous part of least degree of any polynomial. Therefore the new equation belongs to $x^2+\mathbb{C}[y,z,t]$ whose quadratic part and residual cubic are equal to the old ones, the new normal forms are the transforms of the old ones, the weights κ_k do not change, the principal blocks are the transforms of the old ones, and the j_k and J of **Step 2** do not change. We keep the names x,y,z,t, f_i, $h_{i,k}$, and P_i for the new coordinates and polynomials. We have proved that

$$P_i = x^2 + yzt + (\text{terms of } v_0\text{-degree at least } 3). \tag{E.65}$$

Step 4. In this step we compute the log discrepancy and the orders of the marks along the weighted blow-ups of rays close to $v_0+\sigma\theta w$.

Let I_1 be the set of $\theta>0$ such that all coordinates of $v_0+\sigma\theta w$ are positive. For any j with $(v_0)_j=0$ we have $\sigma w_j>0$. Namely, $\sigma w_4 = C$ in (i) with $\sigma=1$, $(\sigma w_2,\sigma w_3)=(A,B)$ in (i) with $\sigma=-1$, $\sigma w_j = B_2$ for the coordinate z_2 in (ii) with $\sigma=1$, $\sigma w_2 = A$ in (ii) with $\sigma=-1$, and $\sigma w_3 = 1$ in (iii). Thus $I_1=(0,\theta_1)$ for some $\theta_1\in(0,+\infty]$. We show the following. Let $\theta\in I_1$, let L_i be positive integers with $L_i/r_i\to\theta$, and let

$$v_i' := \frac{r_i v_0 + \sigma L_i w}{g_i'},$$

where g_i' is the greatest common divisor of the coordinates of $r_iv_0+\sigma L_i w$. Then for $i\gg 0$, v_i' is a primitive vector in $\mathbb{Z}^4_{>0}$, the lowest v_i'-weighted part of f_i is x^2+yzt, and the prime divisor E_i over W_i of Lemma 4.12 for v_i' has center o_i and satisfies

$$a(E_i,W_i,0) = \frac{r_i+\sigma u L_i}{g_i'}, \qquad \operatorname{mult}_{E_i} h_{i,k} = \frac{r_i j_k + \sigma L_i \kappa_k}{g_i'} \quad \text{for any } k, \tag{E.66}$$

and

$$a(E_i,W_i,D_i) = \frac{r_i}{g_i'}\Big(1-J-\sigma\delta\frac{L_i}{r_i}\Big) > 0. \tag{E.67}$$

Choose a closed interval $I'\subset I_1$ whose interior contains θ, and a real number $c_0>0$ such that all coordinates of $v_0+\sigma\theta' w$ are at least c_0 for any $\theta'\in I'$. For $i\gg 0$, $L_i/r_i\in I'$, so all coordinates of $r_iv_0+\sigma L_i w$ are at least $c_0 r_i$, and $v_i'\in\mathbb{Z}^4_{>0}$ is primitive. For any monomial x^α,

$$\langle r_iv_0+\sigma L_i w,\alpha\rangle = r_i\langle v_0,\alpha\rangle + \sigma L_i\langle w,\alpha\rangle \geq c_0 r_i|\alpha|. \tag{E.68}$$

By (E.65), the monomials of P_i of least $(r_iv_0+\sigma L_i w)$-degree are x^2 and yzt, of degree $2r_i+2\sigma u L_i$, and any other monomial of P_i has degree at least $3r_i+2\sigma u L_i$. By (E.61) and (E.68), any monomial of f_i-P_i has degree at least $c_0r_i(r_i-2u)/M$, which exceeds $3r_i+2\sigma uL_i$ for $i\gg 0$ since L_i/r_i is bounded. Thus the lowest v_i'-weighted part of f_i is x^2+yzt, of weight $(2r_i+2\sigma uL_i)/g_i'$. Similarly, the monomials of $h^0_{i,k}$ of least $(r_iv_0+\sigma L_iw)$-degree are those of v_0-degree j_k, of degree $r_ij_k+\sigma L_i\kappa_k$, any other monomial of $h^0_{i,k}$ has degree at least $r_i(j_k+1)+\sigma L_i\kappa_k$, and, by (E.62) and (E.68), any monomial of the normal form of $h_{i,k}$ outside $h^0_{i,k}$ has degree at least $c_0r_i(r_i-|\kappa_k|)/M$, which is larger than $r_i(j_k+1)+\sigma L_i\kappa_k$ for $i\gg 0$, since j_k and κ_k are fixed and L_i/r_i is bounded. Thus the lowest v_i'-weighted part of the normal form of $h_{i,k}$ is the sum of the terms of $h^0_{i,k}$ of v_0-degree j_k. It is a non-zero polynomial of x-degree at most one, so it is not divisible by x^2+yzt.

The polynomial x^2+yzt is irreducible, being x^2 plus a polynomial in y,z,t which is not a square, and no coordinate divides it. Thus $(x^2+yzt=0)\subset\mathbb{A}^4$ is integral and not contained in any coordinate hyperplane. By Lemma 4.12, $\operatorname{mult}_{E_i} h_{i,k}$ is the v_i'-weight of the normal form of

$h_{i,k}$, which is the second formula of (E.66), and, as the sum of the coordinates of v_0 is 3 and that of w is $3u$,

$$a(E_i, W_i, 0) = \frac{3r_i + 3\sigma u L_i}{g_i'} - \frac{2r_i + 2\sigma u L_i}{g_i'} = \frac{r_i + \sigma u L_i}{g_i'}.$$

Since v_i' has positive coordinates, the center of E_i on W_i is o_i. Hence, by (E.66) and $\delta = \tau - u$,

$$\begin{aligned} a(E_i, W_i, D_i) &= a(E_i, W_i, 0) - \sum_k c_k \operatorname{mult}_{E_i} h_{i,k} \\ &= \frac{r_i(1-J) - \sigma L_i(\tau - u)}{g_i'} = \frac{r_i}{g_i'}\Big(1 - J - \sigma\delta\frac{L_i}{r_i}\Big), \end{aligned}$$

and $a(E_i, W_i, D_i) = |I_{G_i}(E_i)| a_D(E_i) > 0$ by Definition-Lemma 11.4(1). This proves (E.67).

Step 5. In this step we construct the interval I_0 and the finite set $\mathcal{G}$.

Let $\theta \in I_1$. Applying **Step 4** with $L_i := \lceil \theta r_i \rceil$ and letting $i \to +\infty$ in (E.67), we obtain $J + \sigma\delta\theta \le 1$. Letting $\theta \to 0$, we obtain $J \le 1$. In (i) with $\sigma = -1$, if $J = 1$, then $\delta = \frac{C}{2} - u = \frac{A+B}{2} > 0$ by **Step 2**(c), so

$$J + \sigma\delta\theta = 1 - \frac{A+B}{2}\theta < 1 \quad \text{for any } \theta > 0.$$

For the other choices of (v_0, σ), $J \le 1 - \frac{1}{m}$ by (E.64). Together with **Step 2**(a), this implies the following. Unless $u = 0$ and all $h_{i,k}$ are units, there exists $\theta_0 \in (0, \theta_1)$ such that

$$0 < J + \sigma\delta\theta < 1 \quad \text{for any } \theta \in (0, \theta_0). \tag{E.69}$$

In this case let $I_0 := (0, \theta_0)$. If $u = 0$ and all $h_{i,k}$ are units, let $I_0 := I_1$. Then $J + \sigma\delta\theta = 0$ for any θ by **Step 2**(b). Let $\mathcal{G} := \{1, 2, \dots, M\}$. The interval I_0 and the set $\mathcal{G}$ only depend on w and on the integers fixed in **Steps 1 and 2**.

Step 6. In this step we fix $\theta_* \in I_0$, and we construct the integers L_i and the vectors v_i.

For real numbers $0 \le a < b$ and an integer $r \ge \max\{r_0, (b-a)^{-2}\}$, where r_0 is the integer of Lemma 4.9, the interval $(ar, br]$ has length at least $\sqrt{r}$, so it contains an integer coprime to r by Lemma 4.9.

If $\sigma\delta \ge 0$, we construct L_i with L_i/r_i strictly increasing to θ_*, and if $\sigma\delta < 0$, with L_i/r_i strictly decreasing to θ_*. We treat the first case, the second being symmetric. Fix $\vartheta_0 \in I_0$ with $\vartheta_0 < \theta_*$. Let $i_0 := 0$. Recursively, for $n \ge 1$, given i_{n-1} and ϑ_{n-1}, choose an index $i_n > i_{n-1}$ with r_{i_n} so large that the interval

$$\Big(\max\Big\{\vartheta_{n-1}, \theta_* - \frac{1}{n}\Big\}, \frac{1}{2}\Big(\theta_* + \max\Big\{\vartheta_{n-1}, \theta_* - \frac{1}{n}\Big\}\Big)\Big]$$

contains L/r_{i_n} for an integer L coprime to r_{i_n}, and let $L_{i_n} := L$ and $\vartheta_n := L/r_{i_n}$. Passing to the subsequence $(i_n)_n$, the L_i are coprime to r_i, $L_i/r_i \in I_0$ is strictly increasing with limit θ_*, and $L_i \to +\infty$.

Let g_i be the greatest common divisor of the coordinates of $r_i v_0 + \sigma L_i w$. If a prime p divides g_i and r_i, then p divides $L_i w_j$ for any j, hence p divides all w_j as $\gcd(L_i, r_i) = 1$, which contradicts the primitivity of w. If a prime p divides g_i and L_i, then p divides $r_i(v_0)_1 = r_i$, a contradiction. Thus $\gcd(g_i, r_i L_i) = 1$. For any j with $(v_0)_j = 0$, g_i divides $L_i w_j$, hence g_i divides w_j, and $w_j \ne 0$ by **Step 4**. Thus $g_i \le M$ and $g_i \in \mathcal{G}$. Passing to a subsequence, we may assume that $g_i = g$ is independent of i. Let $v_i := (r_i v_0 + \sigma L_i w)/g$. If $u = 0$, the first coordinate of $r_i v_0 + \sigma L_i w$ is r_i, so g divides r_i and $g = 1$. In (iii), g divides $w_3 = -1$, and

$$v_i = r_i(1, 1, 0, 1) - L_i(0, 1, -1, 0) = (r_i, r_i - L_i, L_i, r_i).$$

This proves (1). By **Step 4**, after discarding finitely many indices, $v_i \in \mathbb{Z}^4_{>0}$ is primitive and the lowest v_i-weighted part of f_i is $x^2 + yzt$, which proves the first sentence of (2), and (E.66) and (E.67) hold with $g_i' = g$ and $E_i = E_{v_i}$.

Step 7. In this step we compute the coefficients b_i and prove (3).

Fix i and an index l, and consider the root chart of v_i at x_l in Definition-Lemma 11.4(2), with coordinates ρ and x'_j, $j \neq l$. Since $\gcd(g, r_i) = 1$, there exists an integer e_i with $ge_i \equiv \sigma L_i \pmod{r_i}$, and $\gcd(e_i, r_i) = 1$. As $g(v_i)_j = r_i(v_0)_j + \sigma L_i w_j \equiv ge_i w_j \pmod{r_i}$, we have $(v_i)_j \equiv e_i w_j \pmod{r_i}$, so $\eta_i^{e_i}$ multiplies x_j by $\zeta_i^{(v_i)_j}$ for any j. Hence the automorphism $\rho \to \zeta_i \rho$, $x'_j \to x'_j$ of the root chart is a lift of $\eta_i^{e_i}$, since $x_j = \rho^{(v_i)_j} x'_j$ is multiplied by $\zeta_i^{(v_i)_j}$. It acts on ρ alone and has order r_i, and it maps onto the generator $\eta_i^{e_i}$ of $R_i = G_i \cap \mathbf{T}$. Let R'_i be the cyclic group generated by the automorphism $\rho \to \zeta_i \rho$, $x'_j \to x'_j$, and $q := \rho^{r_i}$. By Definition-Lemma 11.4(2) (where $d = 1$ by Definition-Lemma 11.4(2)(a), as $x^2 + yzt$ is not a linear monomial), with $s_k = \mathrm{mult}_{E_{v_i}} h_{i,k}$ and by (E.66) and (E.67),

$$\begin{aligned} b_i &= 1 - \frac{a(E_{v_i}, W_i, 0) - \sum_k c_k s_k}{r_i} = 1 - \frac{a(E_{v_i}, W_i, D_i)}{r_i} \\ &= 1 - \frac{1}{g}\Big(1 - J - \sigma\delta \frac{L_i}{r_i}\Big) = 1 - \frac{1}{g} + \frac{1}{g}\Big(J + \sigma\delta \frac{L_i}{r_i}\Big). \end{aligned}$$

The numbers g, J, σ, and δ are independent of i. The function $\theta \to \sigma\delta\theta$ is non-decreasing if $\sigma\delta \geq 0$ and decreasing if $\sigma\delta < 0$, and L_i/r_i is increasing in the first case and decreasing in the second. Thus b_i is non-decreasing in i.

If $u = 0$ and all $h_{i,k}$ are units, then $J = \delta = 0$ by **Step 2**(b) and $g = 1$ by **Step 6**, so $b_i = 0$ for any i. Otherwise, $b_i < 1$ by (E.67), and $J + \sigma\delta L_i/r_i > 0$ by (E.69) since $L_i/r_i \in I_0$, so $b_i > 1 - \frac{1}{g} \geq 0$. Moreover,

$$\lim_i b_i = 1 - \frac{1}{g} + \frac{1}{g}(J + \sigma\delta\theta_*) \in \Big(1 - \frac{1}{g}, 1\Big) \subset (0, 1)$$

by (E.69). This proves (3). □

E.6. **Triangles and secant conics: permutations and vertices.** In this subsection, we prove Proposition 11.13.

Proof of Proposition 11.13. The proof is long, so we divide it into several steps.

Assume the contrary: there exists a sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with W_i of quadratic rank one whose residual cubic defines a triangle or a secant conic (Definition-Lemma 6.4), whose numbers $\alpha_i := \mathrm{mld}((W_i \ni o_i, D_i)/G_i)$, where $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$, satisfy $0 < \alpha_1 < \alpha_2 < \cdots < 1$. We pass to subsequences without further mention.

We may assume that for each k whether $h_{i,k}$ is a unit does not depend on i. If all $h_{i,k}$ are units, then $D_i = 0$ near o_i, and the numbers $\alpha_i = \mathrm{mld}((W_i \ni o_i, 0)/G_i)$ satisfy the ACC by Remark 11.6 ([Liu26, Theorem 1.3]), which contradicts the strict increase of the α_i. Thus some $h_{i,k}$ is not a unit, and this is kept by the numerical admissions below, which map units to units and non-units to non-units.

Step 1. In this step we reduce to the case when G_i is the inertia group of a computing lift, and we write the action of G_i in coordinates adapted to the factors of the residual cubic.

Let F_i be a computing lift of α_i (Definition 2.7). By Lemma E.11(1),

$$\alpha_i = \mathrm{mld}((W_i \ni o_i, D_i)/I_{G_i}(F_i)).$$

Since the inertia group of F_i in $I_{G_i}(F_i)$ is $I_{G_i}(F_i)$, the prime divisor F_i is also a computing lift of $\mathrm{mld}((W_i \ni o_i, D_i)/I_{G_i}(F_i))$. Possibly replacing G_i with $I_{G_i}(F_i)$, we may assume that $G_i = I_{G_i}(F_i)$. Then G_i acts trivially on the residue field of F_i [Liu26, Definition 2.5]. Thus, for any normal variety Y with a G_i-action and any G_i-equivariant proper birational morphism $Y \to W_i$, the center Z of F_i on Y is G_i-invariant and G_i acts trivially on the residue field of the generic point of Z. In particular, if Z is a closed point, then Z is fixed by G_i.

By [Liu26, Lemma 11.3(1)] there exist G_i-semi-invariant elements x, y, z, t of the completion $\widehat{\mathcal{O}}_{o_i}$ of the local ring at o_i of the ambient smooth fourfold germ, forming a regular system of parameters, on which G_i acts diagonally, such that the completed equation of W_i is $x^2 + g_i(y, z, t)$ with $g_i \in (y, z, t)^3$, and the cubic part $g_{i,3}$ of g_i is the residual cubic. By [Liu26, Lemma 11.3(2)], the set of linear factors of $g_{i,3}$ is G_i-stable, and a linear factor of $g_{i,3}$ whose multiplicity is different from the multiplicities of the other linear factors is G_i-semi-invariant.

If the residual cubic defines a triangle, we replace y, z, t by the three linear factors of $g_{i,3}$, which are linearly independent since the three lines are not concurrent, so that $g_{i,3} = yzt$ after rescaling. Since G_i permutes the lines spanned by y, z, and t, it acts on the span of y, z, t by monomial matrices. If the residual cubic defines a secant conic, then $g_{i,3}$ has a unique linear factor t, which is G_i-semi-invariant, and we let y and z be the tangent forms of the conic at its two points on the line $(t = 0)$, which are the nodes of the cubic. The forms y, z, and t are linearly independent, since the tangent lines $(y = 0)$ and $(z = 0)$ meet at the pole of the secant line $(t = 0)$, which is not on $(t = 0)$. After rescaling, $g_{i,3} = t(yz + t^2)$, and G_i permutes the lines spanned by y and z, since it preserves the conic and the line $(t = 0)$. Thus the residual cubic is $yzt + \epsilon t^3$ with $\epsilon \in \{0, 1\}$ ($\epsilon = 0$ for a triangle and $\epsilon = 1$ for a secant conic), and G_i acts on the span of y, z, t by monomial matrices, permuting the lines spanned by y, z, and t. The elements x, y, z, t are only formal. In the next paragraph we pass to numerical admissions, on which they are coordinates of $\mathbb{A}^4$ and the equation and the marks are polynomials.

By Proposition 11.8, possibly replacing the data with numerical admissions with the groups G_i, protecting F_i and $I_{G_i}(F_i)$ and, for each subgroup $H \subset G_i$, a computing lift of $\mathrm{mld}((W_i \ni o_i, D_i)/H)$ (Proposition 11.7(0)), and with the integers N and L of Construction-Theorem 4.13 larger than $\max_k \mathrm{mult}_{o_i} h_{i,k}$ and at least the integers N_1 and L_1 of Construction-Theorem 4.13 for each subgroup of G_i in place of G_i (in Construction-Theorem 4.13, we choose the ambient G_i-eigenparameters of (ii) with the characters of the elements x, y, z, t of [Liu26, Lemma 11.3(1)], take for ψ in (vi) the G_i-equivariant automorphism mapping x, y, z, t to the corresponding eigenparameters, and for v the G_i-invariant unit such that $v\psi(\hat{f})$ is $x^2 + g_i(y, z, t)$ in the eigenparameters, so that the admitted equation and marks are truncations of $v\psi(\hat{f})$ and of the $\psi(\hat{h}_{i,k})$, and then we make the linear change of coordinates of the previous paragraph), we may assume that $W_i = (f_i = 0) \subset \mathbb{A}^4$, where $f_i = x^2 + g_i(y, z, t)$ is a polynomial, the cubic part of g_i is $yzt + \epsilon t^3$, the $h_{i,k}$ are polynomials, and G_i acts linearly on $\mathbb{A}^4$, with x a G_i-eigenvector, permuting the lines spanned by y, z, and t. We may assume that ϵ is independent of i. For any subgroup $H \subset G_i$, the new data are numerical admissions with the groups H (Proposition 11.8) of the data with the groups H before the admission, which are data as in Set-up 11.1.

Let $G_{i,0} \subset G_i$ be the kernel of the permutation action of G_i on the three lines spanned by y, z, and t. Then $G_{i,0}$ acts diagonally in x, y, z, t, and $[G_i : G_{i,0}] \leq 3$. Since G_i is cyclic, the image $\bar{G}_i$ of G_i in the permutations of the factors is trivial, generated by a transposition, or generated by a 3-cycle, and if $\epsilon = 1$, then $\bar{G}_i$ is trivial or generated by the exchange of y and z, that is, of the nodes of the cubic. We may assume that the type of $\bar{G}_i$ is independent of i. Let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of $G_{i,0}$ on x, y, z, t. Since f_i is G_i-semi-invariant,

$$\chi_x^2 = \chi_y\chi_z\chi_t, \quad \text{and} \quad \chi_x^2 = \chi_t^3 \ \text{ if } \epsilon = 1.$$

If $\gamma \in G_i$ maps the line spanned by y onto the line spanned by z, then $\gamma\eta\gamma^{-1} = \eta$ for any $\eta \in G_{i,0}$, hence $\chi_y(\eta) = \chi_z(\eta)$.

For a prime divisor P over W_i with center o_i, $I_{G_{i,0}}(P) = G_{i,0} \cap I_{G_i}(P) \subset I_{G_i}(P)$ and $a(P, W_i, D_i) > 0$. By the formula at the end of [Liu26, Lemma 2.6],

$$\mathrm{mld}((W_i \ni o_i, D_i)/G_{i,0}) \geq \mathrm{mld}((W_i \ni o_i, D_i)/G_i) = \alpha_i. \tag{E.70}$$

If $|G_{i,0}|$ is bounded, then $|G_i| \leq 3|G_{i,0}|$ is bounded, and Proposition 11.7(1) with $\lambda = 1$ implies that the α_i belong to a finite set, which is not possible. Hence we may assume that $|G_{i,0}| \to +\infty$.

Step 2. In this step we exclude the 3-cycle and the exchange of the conic, and we find the circle in the case of a transposition.

Assume that $\bar{G}_i$ is non-trivial. By Lemma 4.6 and [Liu26, Lemma 4.4], the groups $G_{i,0} \subset \mathbb{T}^4$ converge in the Hausdorff distance to a closed subgroup H with $\dim H \geq 1$, $G_{i,0} \subset H$ for $i \gg 0$, and $G_{i,0}$ intersects any connected component of H. Let $V \subset \mathbb{R}^4$ be the linear subspace whose image under the exponential map $\mathbb{R}^4 \to \mathbb{T}^4$ of [Liu26, Lemma 4.3(1)] is H^0. If V contains a vector with all coordinates positive, then Lemma 4.7, applied to W_i, D_i, and $G_{i,0}$, implies that $\mathrm{mld}((W_i \ni o_i, D_i)/G_{i,0}) \to 0$, which contradicts (E.70) and $\alpha_i \geq \alpha_1 > 0$. Hence V contains no vector with all coordinates positive.

The relations between the characters in **Step 1** define a closed subgroup $H' \subset \mathbb{T}^4$ containing all $G_{i,0}$. Thus $H \subset H'$, and V is contained in the Lie algebra of H'. In the coordinates (a_x, a_y, a_z, a_t) of $\mathbb{R}^4$, the Lie algebra of H' is contained in $\{2a_x = a_y + a_z + a_t\}$, and in $\{2a_x = 3a_t\}$ if $\epsilon = 1$.

- If $\bar{G}_i$ is generated by a 3-cycle, then $a_y = a_z = a_t$ on V, so $V \subset \mathbb{R}(3,2,2,2)$. Since $\dim V \geq 1$, $V = \mathbb{R}(3,2,2,2)$, which contains a vector with all coordinates positive.
- If $\epsilon = 1$ and $\bar{G}_i$ is generated by the exchange of y and z, then $a_y = a_z$ and $2a_x = 3a_t = 2a_y + a_t$ on V. Hence $a_t = a_y$ and again $V = \mathbb{R}(3,2,2,2)$.

Hence $\bar{G}_i$ is generated by a transposition and $\epsilon = 0$. Name the exchanged coordinates y and z for the moment. Then $V \subset \{(a_x, a, a, a_t) \mid 2a_x = 2a + a_t\}$. If $\dim V = 2$, then $(3,2,2,2) \in V$, which is excluded. Hence $V = \mathbb{R}w$ for a primitive vector $w = (u, a, a, c) \in \mathbb{Z}^4$ with $2u = 2a + c$. By Lemma 4.6, $G_{i,0} \cap \mathbf{T}_{(w)}$ is cyclic of order $r_i \to +\infty$ and $[G_{i,0} : G_{i,0} \cap \mathbf{T}_{(w)}] = |H/H^0|$. The elements of $\mathbf{T}_{(w)}$ act diagonally, while the elements of $G_i \backslash G_{i,0}$ do not. Thus $R_i := G_i \cap \mathbf{T}_{(w)} = G_{i,0} \cap \mathbf{T}_{(w)}$, and $[G_i : R_i] \leq e_0 := 2|H/H^0|$.

Possibly replacing w with $-w$, we may assume that $u \geq 0$, and that $a < 0$ if $u = 0$. Let $\mathbf{T} := \mathbf{T}_{(w)}$. The polynomial f_i is R_i-semi-invariant with the character of x^2, so any monomial of f_i has $\mathbf{T}$-weight in $2u + r_i\mathbb{Z}$. Assume that $u > 0$.

- If $c = 0$, then $w = (u, u, u, 0)$. For $r_i > 2u$, no monomial t^l and no monomial $x_j t^l$ with $x_j \in \{x, y, z\}$ has $\mathbf{T}$-weight in $2u + r_i\mathbb{Z}$. Hence f_i and all partial derivatives of f_i vanish along the t-axis, and W_i is singular along the t-axis.
- If $a = 0$, then $w = (u, 0, 0, 2u)$. For $r_i > 2u$, no monomial $y^p z^q$ and no monomial $xy^p z^q$ has $\mathbf{T}$-weight in $2u + r_i\mathbb{Z}$. Hence f_i, $\partial f_i/\partial x$, $\partial f_i/\partial y$, and $\partial f_i/\partial z$ vanish on the plane $(x = t = 0)$, and W_i is singular along the curve $(x = t = \partial f_i/\partial t = 0)$, which passes through 0 since $f_i \in (x, y, z, t)^2$.
- If $a > 0$ and $c > 0$, then w has all coordinates positive, which is excluded.

Since W_i has an isolated singularity at 0, if $u > 0$, then a and c are non-zero and not both positive. Since $c = 2u - 2a$, after a permutation of y, z, t preserving yzt exactly one of the following holds:

(N) $w = (u, -A, -A, C)$ with $u \geq 0$, $A > 0$, and $C = 2u + 2A$, and G_i exchanges the lines spanned by y and z;

(P) $w = (u, -A, B, B)$ with $u > 0$, $B > u$, and $A = 2(B - u)$, and G_i exchanges the lines spanned by z and t.

Here (N) is the case $a < 0$ (which includes $u = 0$), with $A = -a$ and $C = c$, and (P) is the case $a > 0 > c$, where $a > u$, after renaming y, z, t as z, t, y, with $B = a$ and $A = -c$. In the cases (N) and (P), w is one of the vectors listed in Lemma E.11(2): in the case (N), $w = (u, -A, -B, C)$ with $B = A$, and in the case (P), $w = (u, -A, B, C)$ with $C = B$.

Step 3. In this step we fix the circle $\mathbf{T}$, the blocks of the marks along $\mathbf{T}$, and the bases of the extractions, before the choice of the ray.

If $\bar{G}_i$ is trivial, then $G_i = G_{i,0}$ acts diagonally in x, y, z, t. The hypotheses of Lemma E.11(2) hold for the data before the admission in **Step 1**, which are data as in Set-up 11.1, with ambient G_i-eigenparameters whose linear parts are those of the elements x, y, z, t of **Step 1**, since in them the quadratic part of the equation of W_i is x^2 and the residual cubic is $yzt + \epsilon t^3$. We

replace the data with the numerical admissions with the groups G_i of that lemma, taken with $N, L > \max_k \operatorname{mult}_{o_i} h_{i,k}$, in place of those of **Step 1**. The groups G_i and the numbers α_i do not change (Proposition 11.8(1)), and the new data have the form obtained in **Step 1**. Let F_i be a computing lift of α_i for the new data. The equality $G_i = I_{G_i}(F_i)$ is used below only in the cases (N) and (P). By that lemma, there exist a primitive vector w of one of the forms $(0, 1, -1, 0)$, $(u, -A, -B, C)$ with $C = 2u + A + B$ (type (i)), or $(u, -A, B, C)$ with $B + C = 2u + A$ (type (ii)), and a positive integer e_0, such that $r_i := |G_i \cap \mathbf{T}_{(w)}| \to +\infty$ and $[G_i : G_i \cap \mathbf{T}_{(w)}] \le e_0$. In this case let $\mathbf{T} := \mathbf{T}_{(w)}$ and $R_i := G_i \cap \mathbf{T}$. In the cases (N) and (P), $\mathbf{T}$, R_i, r_i, and e_0 are defined in **Step 2**. In all cases $R_i = G_{i,0} \cap \mathbf{T}$ and $[G_{i,0} : R_i] \le e_0$.

Write each $h_{i,k}$ in normal form (Definition-Lemma 11.5(1) for the group $G_{i,0}$, which holds for the admitted data by Lemma 11.10(1)). By Definition-Lemma 11.5(2), the weight κ_k of the principal block $h^0_{i,k}$ along $\mathbf{T}$ is determined for $i \gg 0$, and any monomial of $h_{i,k} - h^0_{i,k}$ has degree at least $(r_i - |\kappa_k|)/M$, where $M := \max_j |w_j|$. By Definition-Lemma 11.5(3), which holds for the admitted data by Lemma 11.10(2), since, before the admission, $(W_i \ni o_i)$ is of type cD (Definition 9.1) because its residual cubic defines a triangle or a secant conic, and the admission in **Step 1** is taken with $N, L > \max_k \operatorname{mult}_{o_i} h_{i,k}$ (in the trivial case the admission is the one of Lemma E.11(2), which is also taken with $N, L > \max_k \operatorname{mult}_{o_i} h_{i,k}$), we have

$$\sum_k c_k \operatorname{mult}_{o_i} h_{i,k} \le 1 - \frac{1}{m}.$$

The normal form of $h_{i,k}$ restricts to $h_{i,k}$ on W_i, so it has a monomial of degree at most $\operatorname{mult}_{o_i} h_{i,k}$, and

$$c_k \operatorname{mult}_{o_i} h_{i,k} \le \sum_{k'} c_{k'} \operatorname{mult}_{o_i} h_{i,k'} \le 1 - \frac{1}{m},$$

so $\operatorname{mult}_{o_i} h_{i,k} < 1/c_k$. For $i \gg 0$, such a monomial belongs to $h^0_{i,k}$, since $|\kappa_k| < r_i/2$ and any monomial of $h_{i,k} - h^0_{i,k}$ has degree at least $(r_i - |\kappa_k|)/M > r_i/(2M)$ (Definition-Lemma 11.5(2)). Thus $h^0_{i,k}$ contains a monomial of degree at most $\operatorname{mult}_{o_i} h_{i,k}$, the integers κ_k are bounded, and we may assume that they are independent of i. Let $\tau := \sum_k c_k \kappa_k$ and let δ be the adjusted mass of D_i along $\mathbf{T}$ (Set-up 5.1). For w of type (i) or (ii), the least monomial of f_i is x^2, so $\epsilon_{\mathbf{T}} = \sum_j w_j - \kappa(f_i) = \sum_j w_j - 2u = u$ and $\delta = \tau - u$. In the cases (N) and (P), G_i commutes with $\mathbf{T}$, because $w_y = w_z$ in the case (N) and $w_z = w_t$ in the case (P). Hence G_i preserves the decomposition of a polynomial according to $\mathbf{T}$-weights. The normal forms are G_i-semi-invariant by the uniqueness in Definition-Lemma 11.5(1) for $G_{i,0}$, since G_i is abelian and preserves the line spanned by x and the span of y, z, t, so the blocks $h^0_{i,k}$ are G_i-semi-invariant.

For a vector $v_0 \in \mathbb{Z}^4_{\ge 0}$ let $j_k(v_0)$ be the least v_0-weighted degree of a monomial of $h^0_{i,k}$. Then $0 \le j_k(v_0) \le 2 \operatorname{mult}_{o_i} h_{i,k}$ for the vectors v_0 below, and we may assume that $j_k(v_0)$ is independent of i for any of the finitely many vectors v_0 listed below. Let $v_0^{\pm}$ be the base vectors of Lemma E.13(1): $v_0^+ := (1, 1, 1, 0)$ and $v_0^- := (1, 0, 0, 2)$ for w of type (i), and $v_0^+ := (1, 1, 1, 0)$ if $B \le C$, $v_0^+ := (1, 1, 0, 1)$ if $B > C$, $v_0^- := (1, 0, 1, 1)$ for w of type (ii). Let

$$J_{\pm} := \sum_k c_k j_k(v_0^{\pm}).$$

The trivial case and the case (N). By Lemma E.13 for the data $(W_i \ni o_i, G_{i,0}, (h_{i,k})_k)$, which satisfy the hypotheses of that lemma: they are numerical admissions with the groups $G_{i,0}$ (Proposition 11.8) of data as in Set-up 11.1, taken with $N, L > \max_k \operatorname{mult}_{o_i} h_{i,k}$ (**Step 1** in the case (N), and Lemma E.11(2) in the trivial case, where $G_{i,0} = G_i$), $o_i = 0$, $W_i = (f_i = 0)$ for a polynomial $f_i \in x^2 + \mathbb{C}[y, z, t]$ with residual cubic $yzt + \epsilon t^3$, the $h_{i,k}$ are polynomials, and $G_{i,0}$ acts diagonally in x, y, z, t (by **Step 1** in the case (N), restricted to the subgroup $G_{i,0}$, and by Lemma E.11(2) in the trivial case), and for w, we obtain a non-empty open interval $I_0 \subset \mathbb{R}_{>0}$ and a finite set $\mathcal{G}$ of positive integers such that, for any $\theta_* \in I_0$, the integers L_i,

the vectors v_i, and the changes of coordinates of that lemma exist. If $w \neq (0,1,-1,0)$, then the vectors of Lemma E.13(1) are $v_i = (r_i v_0 + \sigma L_i w)/g$ with $g \in \mathcal{G}$ and, by Lemma E.13(1), $(v_0, \sigma) \in \{(v_0^+, 1), (v_0^-, -1)\}$, and the coefficient of Lemma E.13(3) is $b_i = \beta(L_i/r_i)$ for one of the affine functions

$$\beta_{g,\pm}(\theta) := 1 - \frac{1}{g} + \frac{1}{g}\big(J_\pm \pm \delta\theta\big), \qquad g \in \mathcal{G}. \tag{E.71}$$

If $w = (0,1,-1,0)$, then $v_i = (r_i, r_i - L_i, L_i, r_i) = r_i v_0^{\mathrm{h}} - L_i w$ with $v_0^{\mathrm{h}} := (1,1,0,1)$, and we let $\sigma := -1$ and $g := 1$. In this case $\epsilon_{\mathbf{T}} = 0$ and $\delta = \tau$. Let $J_{\mathrm{h}} := \sum_k c_k j_k(v_0^{\mathrm{h}})$ and $\beta_{\mathrm{h}}(\theta) := J_{\mathrm{h}} - \delta\theta$. By Lemma E.13(1) and (3), for any integers L_i that the lemma provides, $v_0 = v_0^{\mathrm{h}}$, $\sigma = -1$, $g = 1$, and $b_i = J_{\mathrm{h}} - \delta L_i/r_i = \beta_{\mathrm{h}}(L_i/r_i)$.

In the case (N), $w = (u, -A, -A, C)$ is of type (i) with $B = A$, and Lemma E.13(1) uses $v_0 \in \{(1,1,1,0), (1,0,0,2)\}$. Both bases have equal entries on y and z, and so does w. Hence v_i has equal entries on y and z, and the v_i-weight filtration of $\mathbb{C}[x,y,z,t]$ is G_i-invariant. By Lemma E.13(2), with $A = B$, the changes of coordinates are the identity, so the lowest v_i-weighted part of f_i is $x^2 + yzt$ in the coordinates x, y, z, t, and it is G_i-semi-invariant.

The case (P). Here $w = (u, -A, B, B)$ and G_i exchanges the lines spanned by z and t. Lemma E.13 is not used here: since $B = C$, it may choose $v_0^+ = (1,1,1,0)$, whose entries on z and t differ, and then the v_i-weight filtration is not G_i-invariant. We use only the base $v_0^- = (1,0,1,1)$ with $\sigma = -1$. For coprime positive integers L and r, let

$$v_{\mathrm{raw}} := r v_0^- - Lw = (r - uL, AL, r - BL, r - BL).$$

(P1) The vector v_{raw} has equal entries on z and t, so the v_{raw}-weight filtration of $\mathbb{C}[x,y,z,t]$ is G_i-invariant. If $0 < L/r < 1/B$, then all entries of v_{raw} are positive, since $u < B$. Let g be the greatest common divisor of the entries of v_{raw}. A prime dividing g and L divides r, which is not possible, so $\gcd(g, L) = 1$ and $g \mid A$. A prime p dividing g and r divides uL and BL, hence u and B, which is not possible since w is primitive and $A = 2(B - u)$. Thus g is coprime to r and L, and g belongs to the set $\mathcal{G}$ of positive divisors of A.

(P2) A monomial $x^e y^p z^q t^l$ of $\mathbf{T}$-weight $2u$ satisfies $ue - Ap + B(q + l) = 2u$. If the v_0^--weighted degree $e + q + l$ is at most two and $e + p + q + l \geq 2$, then it is x^2 or $e = 0$, since $f_i \in x^2 + \mathbb{C}[y,z,t]$. Let $e = 0$, so that $Ap = B(q + l) - 2u$. If $q + l = 0$, then $Ap = -2u < 0$, which is not possible. If $q + l = 1$, then $Ap = B - 2u < 2(B - u) = A$, so $p = 0$ and the monomial is z or t, of degree one. If $q + l = 2$, then $Ap = 2B - 2u = A$, so $p = 1$. Thus the monomial is x^2 or $yz^q t^l$ with $q + l = 2$. The monomials yz^2 and yt^2 do not occur in f_i, since the cubic part of g_i is yzt. Hence the monomials of f_i of $\mathbf{T}$-weight $2u$ and of v_0^--weighted degree at most two are x^2 and yzt, and no triangular change of coordinates occurs.

(P3) For any k, $0 \leq j_k(v_0^-) \leq \mathrm{mult}_{o_i} h_{i,k}$, so $J_- \in [0, 1 - \frac{1}{m}]$ by Definition-Lemma 11.5(3). If $J_- > 0$, then $0 < J_- - \delta\theta < 1$ for any sufficiently small $\theta > 0$. If $J_- = 0$, then for any k the block $h_{i,k}^0$ contains a monomial of v_0^--weighted degree zero, which is a pure power y^{p_k} with $p_k \geq 0$. Hence $\kappa_k = -Ap_k \leq 0$, $\tau \leq 0$, and $\delta = \tau - u \leq -u < 0$. Thus $J_- - \delta\theta = |\delta|\theta \in (0,1)$ for $0 < \theta < 1/|\delta|$. In either case there exists $\theta_0 \in (0, 1/B)$ such that $0 < J_- - \delta\theta < 1$ for any $\theta \in I_0 := (0, \theta_0)$. In particular, $\delta = 0$ is possible only if $J_- > 0$.

Step 4. In this step we choose the limit θ_* of the rays and construct the extractions.

Let

$$I_{\mathrm{fin}} := \{\sum_k c_k n_k \mid n_k \in \mathbb{Z}_{\geq 0}\} \cap [0,1],$$

which is a finite set since all c_k are positive. Let $\mathcal{T}$ be the set of log canonical thresholds $\mathrm{lct}(X, \Delta; Q)$, where (X, Δ) is an lc pair of dimension three with $\Delta \in I_{\mathrm{fin}}$, and Q is a Cartier divisor with coefficients in $\mathbb{N}^+$. By Theorem 2.12(1), $\mathcal{T}$ satisfies the ACC. The closure $\overline{\mathcal{T}}$ of $\mathcal{T}$ in $\mathbb{R}$ has empty interior: if $\overline{\mathcal{T}}$ contains a non-empty open interval U, then $\mathcal{T} \cap U$ is dense

in U, and we may choose $t_1 < t_2 < \cdots$ in $\mathcal{T} \cap U$, which contradicts the ACC. Moreover, $1 = \mathrm{lct}(\mathbb{A}^3, 0; \mathrm{div}(x)) \in \mathcal{T}$.

If $\delta \neq 0$, then the functions $\beta_{g,\pm}$ of (E.71), for g in the finite set $\mathcal{G}$ and both signs, and the function β_{h}, are non-constant affine functions. The inverse image of $\overline{\mathcal{T}}$ under any $\beta_{g,\pm}$ or β_{h} is closed with empty interior, and so is the union of the finitely many inverse images. We choose $\theta_* \in I_0$ such that

$$\beta_{g,\pm}(\theta_*) \notin \overline{\mathcal{T}} \quad \text{for any } g \in \mathcal{G} \text{ and both signs, and } \beta_{\mathrm{h}}(\theta_*) \notin \overline{\mathcal{T}}. \tag{E.72}$$

If $\delta = 0$, let $\theta_* \in I_0$ be arbitrary.

In the trivial case and the case (N), Lemma E.13, applied with this θ_*, provides positive integers L_i coprime to r_i such that $\theta_i := L_i/r_i$ converges monotonically to θ_*, and primitive vectors v_i as in **Step 3**. We may assume that v_0, σ, and g are independent of i. Then $b_i = \beta(\theta_i)$ with $\beta := \beta_{g,+}$ if $\sigma = 1$, $\beta := \beta_{g,-}$ if $\sigma = -1$, and $\beta := \beta_{\mathrm{h}}$ if $w = (0, 1, -1, 0)$. By Lemma E.13(3), the $b_i \in (0, 1)$ are non-decreasing, since not all $h_{i,k}$ are units.

In the case (P) we construct L_i and v_i as follows.

(P4) Let r_0 be the integer of Lemma 4.9. For real numbers $0 \le a < b$ and $r \ge \max\{r_0, (b-a)^{-2}\}$, the interval $(ar, br]$ has length at least $\sqrt{r}$, so it contains an integer coprime to r by Lemma 4.9. Assume that $\delta < 0$ (the case $\delta > 0$ is symmetric, with the intervals above θ_*, and if $\delta = 0$, any choice of L_i coprime to r_i with $L_i/r_i \in I_0$ converging monotonically to θ_* works). Fix $\theta'_0 \in I_0$ with $\theta'_0 < \theta_*$, and let $i_0 := 0$. Recursively, for $n \ge 1$, choose an index $i_n > i_{n-1}$ with r_{i_n} so large that the interval $(a_n r_{i_n}, b_n r_{i_n}]$, where $a_n := \max\{\theta'_{n-1}, \theta_* - \frac{1}{n}\}$ and $b_n := \frac{1}{2}(\theta_* + a_n)$, contains an integer L coprime to r_{i_n}, and let $L_{i_n} := L$ and $\theta'_n := L/r_{i_n}$. Passing to the subsequence $(i_n)_n$, we obtain positive integers L_i coprime to r_i such that $\theta_i := L_i/r_i \in I_0$ converges monotonically to θ_* and $J_- - \delta\theta_i$ is non-decreasing. Let g'_i be the greatest common divisor of $v_{\mathrm{raw},i} := r_i v_0^- - L_i w$, and $v_i := v_{\mathrm{raw},i}/g'_i$. By (P1), $g'_i \in \mathcal{G}$, and we may assume that $g := g'_i$ is independent of i.

For $i \gg 0$ all entries of v_i are at least $c_0 r_i$ for a constant $c_0 > 0$, since $\theta_* \in (0, 1/B)$. A monomial of f_i whose $\mathbf{T}$-weight is not $2u$ has degree at least $(r_i - 2u)/M$, and a monomial of $h_{i,k} - h^0_{i,k}$ has degree at least $(r_i - |\kappa_k|)/M$. Hence their v_i-weights are at least $c_0 r_i(r_i - 2u)/M$ and $c_0 r_i(r_i - |\kappa_k|)/M$, which exceed $2r_i$ and $r_i(j_k(v_0^-) + |\kappa_k|)$ for $i \gg 0$. A monomial x^α of $\mathbf{T}$-weight κ has v_i-weight $(r_i\langle v_0^-, \alpha\rangle - L_i\kappa)/g$. By (P2), the lowest v_i-weighted part of f_i is $x^2 + yzt$, and the lowest v_i-weighted part of the normal form of $h_{i,k}$ consists of the monomials of $h^0_{i,k}$ of v_0^--weighted degree $j_k(v_0^-)$. The lowest v_i-weighted part of the normal form of $h_{i,k}$ has x-degree at most one, so it is not divisible by $x^2 + yzt$. By Lemma 4.12, for the prime divisor E_i over W_i given by v_i,

$$a(E_i, W_i, 0) = \sum_j v_{i,j} - 2v_{i,x} = \frac{r_i - uL_i}{g}, \qquad s_{i,k} := \mathrm{mult}_{E_i} h_{i,k} = \frac{r_i j_k(v_0^-) - L_i\kappa_k}{g},$$

where we use $\sum_j w_j = 3u$.

(P5) Since $gv_i \equiv -L_i w \pmod{r_i}$ and $\gcd(gL_i, r_i) = 1$, any element $\lambda^w \in R_i$ with $\lambda \in \mu_{r_i}$ is equal to ν^{v_i} for $\nu := \lambda^{-gL'_i}$, where $L'_i L_i \equiv 1 \pmod{r_i}$. Thus $R_i \subset \mathbf{T}_{(v_i)}$. Let z', t' be G_i-eigen-linear forms spanning the span of z and t, which exist since G_i is cyclic. Since $v_{i,z} = v_{i,t}$, the v_i-weighted blow-up in the G_i-eigenparameters x, y, z', t' is the one in x, y, z, t, and the lowest v_i-weighted part $x^2 + yzt$ of f_i is reduced and irreducible. On any root chart of v_i (Definition-Lemma 11.4(2)) the lifts of R_i which act on the radial parameter alone form a subgroup R'_i of order r_i. By Definition-Lemma 11.4(2), the coefficient of $\mathrm{div}(q)$ on the quotient charts is

$$b_i = 1 - \frac{a(E_i, W_i, 0) - \sum_k c_k s_{i,k}}{r_i} = 1 - \frac{1}{g} + \frac{1}{g}\big(J_- - \delta\theta_i\big) = \beta_{g,-}(\theta_i),$$

where we use $\tau = \sum_k c_k\kappa_k$ and $\delta = \tau - u$. By (P3), $b_i \in (1-\frac{1}{g}, 1) \subset [0,1)$, and $b_i > 0$. By (P4), the b_i are non-decreasing.

In all cases $b_i = \beta(\theta_i)$ for an affine function β independent of i, with $\beta := \beta_{g,-}$ in the case (P). If $\delta = 0$, then $b_i = b$ is constant. If $\delta \neq 0$, then $b_* := \lim_i b_i = \beta(\theta_*)$ belongs to $(0,1]$ and is not in $\overline{\mathcal{T}}$ by (E.72). Since $1 \in \mathcal{T}$, $b_* \in (0,1)$.

Step 5. In this step we return to the extraction of $E_i := E_{v_i}$ and show that, for $i \gg 0$, the center of F_i is the origin of a quotient chart.

In the cases (N) and (P), let x_1, x_2, x_3, x_4 be x, y', z', t, respectively x, y, z', t', where y', z' (respectively z', t') are G_i-eigen-linear forms spanning the span of y, z (respectively of z, t). In the trivial case let $x_1, \dots, x_4$ be the coordinates x, y, z, t after the change of coordinates of Lemma E.13(2). In all cases $x_1, \dots, x_4$ are ambient G_i-eigenparameters, the v_i-weighted blow-up in $x_1, \dots, x_4$ is the one in x, y, z, t (in the cases (N) and (P) because $v_{i,y} = v_{i,z}$, respectively $v_{i,z} = v_{i,t}$), and the lowest v_i-weighted part of f_i is reduced and irreducible. Let $Y_i' \to W_i$ be the normalized strict transform of W_i in the v_i-weighted blow-up, $Y_i := Y_i'/G_i$, $\phi_i : Y_i \to W_i/G_i$ the induced morphism, $\bar{o}_i$ the image of o_i in W_i/G_i, and $K_{Y_i} + \Delta_{Y_i} := \phi_i^*(K_{W_i/G_i} + D_{W_i/G_i})$.

By **Step 4** (and Lemma E.13(3) in the trivial case and the case (N)), R_i lifts to a subgroup R_i' of order r_i acting on the radial parameter alone of any root chart, so $R_i \subset I_{G_i}(E_i)$. Let $e_i := [I_{G_i}(E_i) : R_i] \leq [G_i : R_i] \leq e_0$. We may assume that $e_i = e$ is independent of i. By Proposition 11.9(0) and Definition-Lemma 11.4(2), the coefficient of the image of E_i in Δ_{Y_i} is

$$1 - \frac{a(E_i, W_i, 0) - \sum_k c_k \operatorname{mult}_{E_i} h_{i,k}}{er_i} = 1 - \frac{1-b_i}{e},$$

which belongs to $[0,1)$ and is non-decreasing in i. By Proposition 11.9(0), α_i is the minimum of $\operatorname{mld}(Y_i \ni \xi, \Delta_{Y_i})$ over the points $\xi \in \phi_i^{-1}(\bar{o}_i)$, attained at the generic point ξ_i of the center of F_i on Y_i. By Proposition 11.9(1), the values α_i for which ξ_i is not a closed point satisfy the ACC, and so do the values α_i for which ξ_i is a closed point and a quotient chart as in that proposition has a closed point over ξ_i at which it is a smooth germ, a hypersurface germ of quadratic rank four, or a non-isolated hypersurface germ of quadratic rank three. Since the α_i are strictly increasing, we may assume that neither occurs for any i.

We determine the closed points of the exceptional divisor E_i of $Y_i' \to W_i$ over which a quotient chart is an isolated singularity of quadratic rank three. Let $P_y, P_z, P_t \in E_i$ be the points at which only y, only z, respectively only t is non-zero, and write x_y, x_z, x_t for x_2, x_3, x_4, that is, for y', z', t in the case (N), for y, z', t' in the case (P), and for y, z, t in the trivial case. Since $(x^2 + yzt = 0) \subset \mathbb{P}(v_i)$ does not meet $(y = z = t = 0) = (x_y = x_z = x_t = 0)$, E_i is covered by the root charts of v_i at x_y, x_z, and x_t (Definition-Lemma 11.4(2)). Let $l \in \{y, z, t\}$, let ρ be the radial parameter of the root chart at x_l, and let X and $x_j', x_{j'}'$, $\{j, j'\} = \{y, z, t\} \backslash \{l\}$, be the other coordinates of the root chart. Let W_i' be the strict transform of W_i in the root chart, defined by $f_i' := \rho^{-2v_{i,x}} f_i$.

Let K_i be the group acting on the root chart generated by the lifts of G_i and by the deck group μ_{n_i}, $n_i := v_{i,l}$ (Definition-Lemma 11.4(2)), and $R_i'' \subset K_i$ the subgroup of the elements acting on ρ alone, so that the quotient chart of Proposition 11.9(1) is W_i'/R_i''. The group R_i'' acts on ρ by roots of unity and contains R_i', so it is cyclic of order $d_i r_i$ for a positive integer d_i. Let $q := \rho^{r_i}$ and $\bar{q} := q^{d_i}$. Write $x^2 + yzt$ in $x_1, \dots, x_4$, and let $\bar{f}_i' \in \mathbb{C}[X, x_j', x_{j'}']$ be $(x^2 + yzt)|_{x=X, x_l=1, x_j=x_j', x_{j'}=x_{j'}'}$. The polynomial f_i' is K_i-semi-invariant and congruent to $\bar{f}_i'$ modulo ρ, and R_i'' acts trivially on $\bar{f}_i'$, so f_i' is invariant under R_i''. Thus $f_i' = F(q, X, x_j', x_{j'}') = F''(\bar{q}, X, x_j', x_{j'}')$ for polynomials F and F'', the quotient chart $Y_i^\circ := W_i'/R_i'$ is the hypersurface $(F = 0) \subset \mathbb{A}^4_{q,X,x_j',x_{j'}'}$, and $W_i'/R_i'' = Y_i^\circ/\mu_{d_i}$, with μ_{d_i} acting on q alone, is the hypersurface $(F'' = 0) \subset \mathbb{A}^4_{\bar{q},X,x_j',x_{j'}'}$. Moreover, $F \equiv \bar{f}_i' \pmod{q}$ and $F'' \equiv \bar{f}_i' \pmod{\bar{q}}$. Hence Y_i°, respectively W_i'/R_i'', is smooth at

a point of $(q=0)$, respectively of $(\bar{q}=0)$, whose coordinates $X, x'_j, x'_{j'}$ give a smooth point of the surface $(\bar{f}'_i=0)\subset\mathbb{A}^3$.

We describe the singular points of the surfaces $(\bar{f}'_i=0)$. In the trivial case, $\bar{f}'_i=X^2+x'_jx'_{j'}$, and $(\bar{f}'_i=0)$ is singular only at the origin, which maps to P_l. In the cases (N) and (P), let $l_0:=t$, respectively $l_0:=y$, so that x_{l_0} is the coordinate t, respectively y, and $x^2+yzt=x^2+x_{l_0}\Theta$, where Θ is the product yz, respectively zt, of two linearly independent linear forms, written in the eigen-linear forms y',z', respectively z',t'. Thus Θ is a non-degenerate binary quadratic form. On the root chart at x_{l_0}, $\bar{f}'_i=X^2+\Theta(x'_j,x'_{j'})$, and $(\bar{f}'_i=0)$ is singular only at the origin, which maps to P_{l_0}. On the root chart at x_l with $l\neq l_0$, $\bar{f}'_i=X^2+x'_{l_0}\Theta_l$, where $\Theta_l:=\Theta|_{x_l=1}$ is a polynomial in one variable, which is non-zero and has only simple zeros since Θ is non-degenerate. Thus $(\bar{f}'_i=0)$ is singular exactly at the points with $X=x'_{l_0}=\Theta_l=0$, which map to the points of E_i at which x, x_{l_0}, and one of the linear factors y,z, respectively z,t, of Θ vanish, that is, to P_y and P_z, respectively to P_z and P_t. Therefore, on each of the three root charts, $(\bar{f}'_i=0)$ is smooth at the points which do not map to P_y, P_z, or P_t.

Since ξ_i is a closed point and $\phi_i(\xi_i)=\bar{o}_i$, the center Z_i of F_i on Y'_i is a closed point of E_i. If $Z_i\notin\{P_y,P_z,P_t\}$, then a root chart U has a point of $(\rho=0)\cap W'_i$ over Z_i at which $(\bar{f}'_i=0)$ is smooth, so the quotient chart W'_i/R''_i of U has a smooth closed point over ξ_i, which is excluded by our choice of i. In the cases (N) and (P), Z_i is fixed by G_i (**Step 1**), while an element of $G_i\backslash G_{i,0}$ exchanges P_y and P_z, respectively P_z and P_t. Hence $Z_i=P_l$ for some $l\in\{y,z,t\}$, with $l=t$ in the case (N) and $l=y$ in the case (P), and we may assume that l is independent of i. From now on we use the root chart at x_l, which is one of the coordinates y,z,t. On it, $\bar{f}'_i=X^2+\Theta(x'_j,x'_{j'})$ for a non-degenerate binary quadratic form Θ, where $\Theta:=x'_jx'_{j'}$ in the trivial case. The origin of the root chart maps to P_l, so the origin of W'_i/R''_i maps to ξ_i. Any closed point of W'_i/R''_i over ξ_i is on $(\bar{q}=0)$, and it is not a smooth point by our choice of i, so it is the origin. Since the origin of W'_i/R''_i is not a smooth point, F'' has no linear term. In particular, no monomial of F'' is equal to $\bar{q}$, and the quadratic part of F'', which is congruent to $X^2+\Theta(x'_j,x'_{j'})$ modulo $\bar{q}$, has rank at least three. Thus, by our choice of i, the origin of W'_i/R''_i is an isolated singularity of quadratic rank three. The only point of Y_i° over the origin of W'_i/R''_i is the origin y_i° of Y_i°, which we call a *vertex*. Hence the center of F_i on the quotient chart Y_i° is y_i°. If $d_i=1$, then $Y_i^\circ=W'_i/R''_i$. If $d_i\geq 2$, then any monomial of F divisible by q is divisible by q^{d_i} and is not q^{d_i}, so it has degree at least $d_i+1\geq 3$, and the quadratic part of F is $X^2+\Theta(x'_j,x'_{j'})$, of rank three. Moreover, the morphism

$$Y_i^\circ\backslash(q=0)\to(W'_i/R''_i)\backslash(\bar{q}=0)$$

is the quotient by a free action of μ_{d_i}, hence étale, and W'_i/R''_i is smooth on a punctured neighborhood of the origin. Thus in all cases $(Y_i^\circ\ni y_i^\circ)$ is an isolated hypersurface singularity of quadratic rank three.

For this l, any element of G_i has exactly n_i lifts to K_i, so $|K_i|=n_i|G_i|$. Let $\Pi_i:=K_i/R'_i$, which acts faithfully on Y_i°, fixes y_i°, and has order $n_i[G_i:R_i]$. Let $H_{i,k}$ be the weak transforms of $h_{i,k}$ and

$$\Delta_i^\circ:=\sum_k c_k\operatorname{div}(H_{i,k})+b_i\operatorname{div}(q)$$

on Y_i°. By Definition-Lemma 11.4(2) and [Liu26, Lemma 2.6(2)(5)],

$$\alpha_i=\operatorname{mld}((Y_i^\circ\ni y_i^\circ,\Delta_i^\circ)/\Pi_i). \tag{E.73}$$

The pair $(Y_i^\circ,\Delta_i^\circ)$ is klt near y_i°. By [Liu26, Lemma 2.6(2)], it suffices to show that (Y_i,Δ_{Y_i}) is klt near $\phi_i^{-1}(\bar{o}_i)$. A prime divisor over Y_i whose center is contained in $\phi_i^{-1}(\bar{o}_i)$ has log discrepancy at least $\alpha_i>0$ by Proposition 11.9(0), and the generic point of any other center is outside $\phi_i^{-1}(\bar{o}_i)$, where ϕ_i is an isomorphism and $(W_i/G_i,D_{W_i/G_i})$ is klt, since (W_i,D_i) is klt.

Step 6. In this step we find a growing cyclic subgroup of Π_i in a fixed circle. Recall from **Steps 3 and 4** that $v_i=(r_iv_0+\sigma L_iw)/g$ with v_0, σ, g independent of i, where

$v_0 \in \{(1,1,1,0),(1,0,0,2),(1,0,1,1),(1,1,0,1)\}$, L_i is coprime to r_i, and $w_l \neq 0$ unless $w = (0,1,-1,0)$ and $l = t$. Let $v_{0,l}$ and w_l be the x_l-entries of v_0 and w, and define the vector $\varpi = (\varpi_X, \varpi_j, \varpi_{j'}, \varpi_q)$ of $\mathbb{Z}^4$ by

$$\varpi_X := v_{0,l}w_x - w_l, \quad \varpi_j := v_{0,l}w_j - v_{0,j}w_l, \quad \varpi_{j'} := v_{0,l}w_{j'} - v_{0,j'}w_l, \quad \varpi_q := gw_l. \tag{E.74}$$

Claim E.14. *There exist a positive integer e_1 and cyclic subgroups $\Sigma_i \subset \Pi_i$ such that $|\Sigma_i| \to +\infty$, $[\Pi_i : \Sigma_i] \leq e_1$, and Σ_i is contained in $\mathbf{T}_{(\varpi)}$, acting diagonally on the coordinates $X, x'_j, x'_{j'}, q$ of Y_i°. Moreover, at least two entries of ϖ are non-zero, and $\varpi_j + \varpi_{j'} = 2\varpi_X$.*

Proof. Let ζ be a generator of μ_{n_i}. It acts on ρ by ζ and on the coordinate x'_m of weight $v_{i,m}$ by $\zeta^{-v_{i,m}}$, hence on $q = \rho^{r_i}$ by ζ^{r_i}. An element ζ^a acting trivially on $X, x'_j, x'_{j'}$ satisfies $n_i \mid a\gcd(v_{i,x}, v_{i,j}, v_{i,j'})$, and $\gcd(n_i, v_{i,x}, v_{i,j}, v_{i,j'}) = 1$ since v_i is primitive. Hence $\mu_{n_i} \cap R'_i = \{1\}$, and μ_{n_i} injects into Π_i. We write $x'_x := X$. In the cases (N) and (P), $x'_j, x'_{j'}$ are the coordinates of the root chart coming from the eigen-linear forms y', z', respectively z', t', of **Step 5**. Since v_0, w, and v_i have equal entries on y and z, respectively on z and t, the entries of v_0, w, and v_i at x'_j and $x'_{j'}$ are well defined, and the computation below holds for x'_j and $x'_{j'}$.

Assume that $v_{0,l} \geq 1$. Then $n_i = (r_i v_{0,l} + \sigma L_i w_l)/g \to +\infty$, and $r_i v_{0,l} \equiv -\sigma L_i w_l \pmod{n_i}$. For $m \in \{x, j, j'\}$,

$$v_{0,l} g v_{i,m} = v_{0,m} r_i v_{0,l} + \sigma L_i v_{0,l} w_m \equiv \sigma L_i (v_{0,l} w_m - v_{0,m} w_l) \pmod{n_i},$$
$$v_{0,l} g r_i \equiv -\sigma L_i g w_l \pmod{n_i}.$$

Since $v_{0,x} = 1$, the element $\zeta^{v_{0,l}g}$ acts on $X, x'_j, x'_{j'}, q$ by λ^{ϖ_X}, λ^{ϖ_j}, $\lambda^{\varpi_{j'}}$, λ^{ϖ_q} with $\lambda := \zeta^{-\sigma L_i}$. Let Σ_i be the subgroup generated by $\zeta^{v_{0,l}g}$. Then $|\Sigma_i| \geq n_i/(v_{0,l}g)$.

Assume that $v_{0,l} = 0$. Then $n_i = \sigma L_i w_l / g$, so $\sigma w_l > 0$. Since $v_{0,x} = 1$, g divides $r_i + \sigma L_i w_x$, and a prime dividing g and L_i divides r_i. Thus $\gcd(g, L_i) = 1$, $g \mid w_l$, and $n_i = (\sigma w_l/g) L_i$. Let $\eta := \zeta^{\sigma w_l/g}$, a generator of $\mu_{L_i} \subset \mu_{n_i}$, and let Σ_i be the subgroup generated by η. Since $g v_{i,m} \equiv r_i v_{0,m} \pmod{L_i}$ and $g r_i$ is invertible modulo L_i, the element $\eta^{g r_i^{-1}}$, where r_i^{-1} is an inverse of r_i modulo L_i, generates Σ_i and acts on $X, x'_j, x'_{j'}, q$ by $\eta^{-1}, \eta^{-v_{0,j}}, \eta^{-v_{0,j'}}, \eta^g$. Since $\varpi = w_l(-1, -v_{0,j}, -v_{0,j'}, g)$, $\Sigma_i \subset \mathbf{T}_{(\varpi)}$. Moreover $|\Sigma_i| = L_i \to +\infty$ and $[\mu_{n_i} : \Sigma_i] = \sigma w_l / g$.

In both cases

$$[\Pi_i : \Sigma_i] \leq [G_i : R_i] \cdot [\mu_{n_i} : \Sigma_i] \leq e_1 := e_0 \max\{2g, |w_l|\}.$$

For the four possible v_0, $v_{0,l} + v_{0,j} + v_{0,j'} = 2 = 2v_{0,x}$, and $w_y + w_z + w_t = 2w_x$ since $x^2 + yzt$ is $\mathbf{T}$-semi-invariant. Hence

$$\varpi_j + \varpi_{j'} = v_{0,l}(2w_x - w_l) - (2 - v_{0,l})w_l = 2\varpi_X.$$

If $w = (0,1,-1,0)$, then $\varpi = (-1,-1,-1,1)$ for $l = y$, $\varpi = (1,1,1,-1)$ for $l = z$, and $\varpi = (0,1,-1,0)$ for $l = t$. Otherwise $w_l \neq 0$, so $\varpi_q \neq 0$. If moreover $\varpi_X = \varpi_j = \varpi_{j'} = 0$, then $v_{0,l} \neq 0$, since otherwise $\varpi_X = -w_l \neq 0$, and $w = (w_l/v_{0,l})v_0$. Since each of the four possible v_0 has a zero entry at y, z, or t, while w has no zero entry at y, z, and t, this is not possible. □

Proof of Proposition 11.13 continued. In the cases (N) and (P), the claim reads as follows. In the case (N) with $v_0 = (1,1,1,0)$ the vertex is at t, $v_{0,t} = 0$, $n_i = CL_i/g$, and $\Sigma_i = \mu_{L_i}$ acts by the weights $(-1,-1,-1,g)$ on X, y', z', q, with $[\Pi_i : \Sigma_i] \leq (C/g)e_0$. In the case (N) with $v_0 = (1,0,0,2)$, $\varpi = (-2A, -2A, -2A, gC)$. In the case (P) the vertex is at y, $n_i = AL_i/g$, and $\Sigma_i = \mu_{L_i}$ acts by the weights $(-1,-1,-1,g)$ on X, z', t', q, with $[\Pi_i : \Sigma_i] \leq (A/g)e_0$.

Step 7. In this step we treat the case $\delta = 0$, and the case when the inertia group of a computing lift at the vertex has unbounded order.

Assume that $\delta = 0$. By **Step 4**, $b_i = b$ is independent of i, so

$$\Delta_i^\circ = \sum_k c_k \operatorname{div}(H_{i,k}) + b \operatorname{div}(q).$$

The germ $(Y_i^\circ \ni y_i^\circ)$ is a hypersurface threefold germ of quadratic rank three, Π_i is a finite group acting faithfully on Y_i° and fixing y_i°, and $(Y_i^\circ, \Delta_i^\circ)$ is klt near y_i° (**Step 5**). By Theorem 7.1 with the finite set $\{c_1, \dots, c_s, b\}$, the positive numbers α_i of (E.73) satisfy the ACC, which is not possible. Hence $\delta \neq 0$, $b_i \to b_* \in (0,1)$, and $b_* \notin \overline{\mathcal{T}}$ (**Step 4**).

Let F_i° be a computing lift of α_i in (E.73) (Definition 2.7), and $I_i^\circ := I_{\Pi_i}(F_i^\circ)$, which is cyclic [Liu26, Lemma 2.6(1)]. For any prime divisor P over Y_i° with center y_i°, $I_{I_i^\circ}(P) \subset I_{\Pi_i}(P)$, and $I_{I_i^\circ}(F_i^\circ) = I_{\Pi_i}(F_i^\circ)$. By the formula at the end of [Liu26, Lemma 2.6],

$$\alpha_i = \mathrm{mld}((Y_i^\circ \ni y_i^\circ, \Delta_i^\circ)/I_i^\circ). \tag{E.75}$$

We may assume that either $|I_i^\circ| \to +\infty$, or $|I_i^\circ|$ is bounded.

Assume that $|I_i^\circ| \to +\infty$. By Claim E.14,

$$[I_i^\circ : I_i^\circ \cap \Sigma_i] \leq [\Pi_i : \Sigma_i] \leq e_1,$$

so $|I_i^\circ \cap \Sigma_i| \to +\infty$. The germ $(Y_i^\circ \ni y_i^\circ)$ is an isolated compound Du Val threefold singularity, since a general hyperplane section through y_i° has a quadratic part of rank three and thus is a Du Val point of type A_1 [KM98, Definition 5.32], hence an isolated Gorenstein terminal threefold germ [KM98, Corollary 5.38], of quadratic rank three. By Proposition 7.31, applied to I_i°, the marks $H_{i,k}$ and q with the labels c_k and b_i, and the prime divisor F_i°, there exist an integer $N_i \geq 3$, a faithful diagonal action of I_i° on $W_{N_i} = (x^\circ y^\circ + (z^\circ)^2 + (t^\circ)^{N_i} = 0)$ whose tangent representation at 0 is isomorphic to the tangent representation of I_i° at y_i°, and I_i°-semi-invariant polynomials $p_{i,k}$ and $p_{i,q}$ with the characters of $H_{i,k}$ and q, such that, for

$$D_i^\circ := \sum_k c_k \operatorname{div}(p_{i,k}) + b_i \operatorname{div}(p_{i,q}),$$

the pair (W_{N_i}, D_i°) is klt near 0 and $\alpha_i = \mathrm{mld}((W_{N_i} \ni 0, D_i^\circ)/I_i^\circ)$ by (E.75). Replacing each $p_{i,k}$ and $p_{i,q}$ by the polynomial obtained by substituting $-(z^\circ)^2 - (t^\circ)^{N_i}$ for $x^\circ y^\circ$, which has the character of $p_{i,k}$, respectively $p_{i,q}$, and restricts to W_{N_i} as $p_{i,k}$, respectively $p_{i,q}$, we may assume that no monomial of $p_{i,k}$ or $p_{i,q}$ is divisible by $x^\circ y^\circ$.

The coordinates $x^\circ, y^\circ, z^\circ, t^\circ$ of W_{N_i} correspond to I_i°-eigenvectors of the tangent representation. The group $I_i^\circ \cap \Sigma_i$ is cyclic, generated by an element γ_i acting by λ^{ϖ_X}, λ^{ϖ_j}, $\lambda^{\varpi_{j'}}$, λ^{ϖ_q} on $X, x'_j, x'_{j'}, q$ (Claim E.14), where λ is a root of unity whose order is a multiple of $|I_i^\circ \cap \Sigma_i|$. For $i \gg 0$, distinct entries of ϖ are distinct modulo $|I_i^\circ \cap \Sigma_i|$, so any eigenvector of γ_i belongs to the span of the coordinates of Y_i° whose weights are equal to one entry of ϖ. Comparing dimensions of eigenspaces, there exists a permutation ϖ' of ϖ such that γ_i acts on $x^\circ, y^\circ, z^\circ, t^\circ$ by $\lambda^{\varpi'_x}$, $\lambda^{\varpi'_y}$, $\lambda^{\varpi'_z}$, $\lambda^{\varpi'_t}$. Here the entry of ϖ' with index x is the weight on x°, and so on. Since $x^\circ y^\circ$ and $(z^\circ)^2$ both have the character of the I_i°-semi-invariant equation $x^\circ y^\circ + (z^\circ)^2 + (t^\circ)^{N_i}$ and $|I_i^\circ \cap \Sigma_i| \to +\infty$, $\varpi'_x + \varpi'_y = 2\varpi'_z$ for $i \gg 0$. We may assume that ϖ' is independent of i. Let w° be the primitive vector in $\mathbb{Q}_{>0}\varpi'$, $\mathbf{T}^\circ := \mathbf{T}_{(w^\circ)}$, and $R_i^\circ := I_i^\circ \cap \mathbf{T}^\circ$. Then $I_i^\circ \cap \Sigma_i \subset R_i^\circ$, so $r_i^\circ := |R_i^\circ| \to +\infty$ and $[I_i^\circ : R_i^\circ] \leq e_1$. Since $(z^\circ)^2$ and $(t^\circ)^{N_i}$ both have the character of $x^\circ y^\circ + (z^\circ)^2 + (t^\circ)^{N_i}$ on R_i°, r_i° divides $N_i w_t^\circ - 2w_z^\circ$.

Let $\Gamma^\circ := \{c_1, \dots, c_s\} \cup \{b_i \mid i \geq 1\}$, which is a DCC subset of $[0,1]$ since the b_i are non-decreasing. Then $(W_{N_i} \ni 0)$ with the action of I_i°, the circle $\mathbf{T}^\circ$, e_1, Γ°, and D_i° are data as in Set-up 7.3, with $\alpha = \alpha_i$. We write the vertex data with the superscript $\circ$: the circle $\mathbf{T}^\circ$ is not $\mathbf{T}$, and the mass and the adjusted mass of D_i° along $\mathbf{T}^\circ$, which we do not need, are not τ and δ. By Claim E.14, at least two entries of w° are non-zero. We show that $\mathbf{T}^\circ$ is a circle of one of Theorems 7.21, 7.22, 7.23, 8.3, and Corollary 7.29. Since W_{N_i} is symmetric in x° and y°, we may exchange x° and y°, and we may replace w° by $-w^\circ$.

- If $w_t^\circ = 0$, then r_i° divides $2w_z^\circ$, so $w_z^\circ = 0$ for $i \gg 0$, and $w^\circ = \pm(1,-1,0,0)$. Then $W_{N_i} = (x^\circ y^\circ - g_i^\circ(z^\circ, t^\circ) = 0)$ with

$$g_i^\circ := -(z^\circ)^2 - (t^\circ)^{N_i} \in (z^\circ, t^\circ)^2,$$

$R_i^\circ = \mu_{r_i^\circ}(1,-1,0,0)$, and $\mathbf{T}^\circ$ is a circle of Theorem 8.3.

- Assume that $w_t^\circ > 0$, and let $v^\circ := w_t^\circ$. If $w_z^\circ \geq 0$, then $w^\circ = (u^\circ, 2s^\circ - u^\circ, s^\circ, v^\circ)$ with $s^\circ := w_z^\circ \geq 0$, $u^\circ := w_x^\circ$, $\gcd(u^\circ, s^\circ, v^\circ) = 1$, and $(u^\circ, s^\circ) \neq (0,0)$ since two entries of w° are non-zero. Then $\mathbf{T}^\circ$ is a circle of Corollary 7.29.
- Assume that $w_t^\circ > 0$ and $w_z^\circ < 0$. Let $c^\circ := -w_z^\circ$, so $w_x^\circ + w_y^\circ = -2c^\circ < 0$. If $w_x^\circ, w_y^\circ < 0$, then $w^\circ = (-a^\circ, -b^\circ, -c^\circ, v^\circ)$ with $a^\circ, b^\circ > 0$, $a^\circ + b^\circ = 2c^\circ$, $\gcd(a^\circ, c^\circ, v^\circ) = 1$, and $\mathbf{T}^\circ$ is a circle of Theorem 7.21. If one of w_x°, w_y° is zero, then after exchanging x° and y°, $w^\circ = (0, -2c^\circ, -c^\circ, v^\circ)$ with $\gcd(c^\circ, v^\circ) = 1$, and $\mathbf{T}^\circ$ is a circle of Theorem 7.23. Otherwise, after exchanging x° and y°, $w^\circ = (a^\circ, -b^\circ, -c^\circ, v^\circ)$ with $a^\circ > 0$, $b^\circ = a^\circ + 2c^\circ$, and $\gcd(a^\circ, c^\circ, v^\circ) = 1$, and $\mathbf{T}^\circ$ is a circle of Theorem 7.22.

None of Theorems 7.21, 7.22, 7.23, 8.3, and Corollary 7.29 has a hypothesis on the sign of the adjusted mass of D_i° along $\mathbf{T}^\circ$, so they imply that α_i cannot increase along $\mathbf{T}^\circ$, which contradicts the assumption that the α_i are strictly increasing.

Step 8. In this step we treat the case when $|I_i^\circ|$ is bounded, and conclude the proof.

Assume that $|I_i^\circ|$ is bounded. We check the hypotheses of Proposition 11.9(2) for the germs $(Y_i^\circ \ni y_i^\circ)$, the groups I_i°, the functions $H_{i,k}$ and q, and the coefficients b_i. The germ $(Y_i^\circ \ni y_i^\circ)$ is a hypersurface threefold germ of quadratic rank three, and I_i° is a cyclic group of bounded order acting faithfully on Y_i° and fixing y_i°. The functions $H_{i,k}$ and q are I_i°-semi-invariant, and $q(y_i^\circ) = 0$. The $b_i \in [0,1)$ are non-decreasing with limit $b_* \in (0,1)$ (**Step 7**). The pair $(Y_i^\circ, \Delta_i^\circ)$ is klt near y_i° (**Step 5**), and $\alpha_i > 0$ is given by (E.75). It remains to show that b_* is not in the closure of the set of log canonical thresholds of $\operatorname{div}(q)$ with respect to $(Y_i^\circ, \sum_k c_k \operatorname{div}(H_{i,k}))$ near y_i°, over all i. Let U_i be an affine open neighborhood of y_i°, small enough that $(Y_i^\circ, \Delta_i^\circ)$ is klt on U_i and the log canonical threshold of $\operatorname{div}(q)|_{U_i}$ with respect to $(U_i, \sum_k c_k \operatorname{div}(H_{i,k})|_{U_i})$ is the one near y_i°. The coefficients of $\sum_k c_k \operatorname{div}(H_{i,k})|_{U_i}$ have the form $\sum_k c_k n_k \leq 1$ with $n_k \in \mathbb{Z}_{\geq 0}$, so they belong to I_{fin}, and $\operatorname{div}(q)|_{U_i}$ is a Cartier divisor with coefficients in $\mathbb{N}^+$, since q is regular. Thus the log canonical thresholds in question belong to the set $\mathcal{T}$ of **Step 4**, and $b_* \notin \overline{\mathcal{T}}$ by (E.72).

By Proposition 11.9(2), there exists $\eta > 0$ such that the positive numbers

$$\ell_i := \operatorname{mld}\Big(\Big(Y_i^\circ \ni y_i^\circ, \sum_k c_k \operatorname{div}(H_{i,k}) + b_* \operatorname{div}(q)\Big)/I_i^\circ\Big)$$

satisfy the ACC and $\ell_i \leq \alpha_i \leq \ell_i + (b_* - b_i)\ell_i/\eta$ for $i \gg 0$. We may assume that the ℓ_i are non-increasing. Let $\alpha_\infty := \lim_i \alpha_i$. Since $\ell_i \leq \alpha_i < 1$ and $b_i \to b_*$, $\alpha_i - \ell_i \to 0$, so $\ell_i \to \alpha_\infty$. Since the ℓ_i are non-increasing, $\ell_i \geq \alpha_\infty > \alpha_i \geq \ell_i$, which is not possible.

Thus the case when $|I_i^\circ|$ is bounded is also excluded. By **Steps 5–8**, the α_i cannot be strictly increasing, a contradiction. We finish the proof of Proposition 11.13. □

E.7. **Concurrent lines, and the circles of cuspidal cubics and tangent conics.** In this subsection, we prove Proposition 11.14 and Lemma E.16.

Lemma E.15 brings the equation and the marks of a source whose residual cubic defines three concurrent lines into polynomial normal form, and keeps the quotient values of the labelled boundary and of each proxy for any subgroup.

Lemma E.15. *Notation as in Set-up 11.1, with $(W \ni o)$ of quadratic rank one whose residual cubic defines three concurrent lines (Definition-Lemma 6.4). Let $P_1, \dots, P_l$ be prime divisors over W with center o. Then there exist a faithful diagonal action of G on $\mathbb{A}^4$ with coordinates x, y, z, t, a G-semi-invariant polynomial f', G-semi-invariant polynomials $h'_1, \dots, h'_s$ such that h'_k has the character of h_k for any k, a real number $\eta > 0$, and a G-equivariant isomorphism*

$$\Phi : \widehat{\mathcal{O}}_{W,o} \to \widehat{\mathcal{O}}_{W',0}, \qquad W' := (f' = 0) \subset \mathbb{A}^4,$$

with the following properties, where $C'_k := \frac{1}{m} \operatorname{div}(h'_k|_{W'})$ and $D' := \sum_k \mu_k C'_k = \sum_k c_k \operatorname{div}(h'_k|_{W'})$.

(1) The polynomial f' is one of

$$\text{(I)}\quad x^2+y^2z+z^3+a_1(t)z^2+a_2(t)z+a_3(t)y+a_4(t),$$
$$\text{(II)}\quad x^2+y^3+z^3+a_1(t)yz+a_2(t)y+a_3(t)z+a_4(t),$$

where $a_1,\dots,a_4\in\mathbb{C}[t]$, $\operatorname{mult} a_1\geq 2$, $\operatorname{mult} a_2\geq 3$, $\operatorname{mult} a_3\geq 3$, and $\operatorname{mult} a_4\geq 4$. The characters χ_x,χ_y,χ_z of x,y,z satisfy $\chi_y^2=\chi_z^2$ and $\chi_x^2=\chi_z^3$ in (I), and $\chi_x^2=\chi_y^3=\chi_z^3$ with $\chi_y\chi_z^{-1}$ of order three in (II).

(2) $(W'\ni 0)$ is an isolated Gorenstein terminal threefold germ, and G acts faithfully on W'.

(3) For any k, $h'_k|_{W'}$ is a mark on $(W'\ni 0)$ with $\operatorname{mult}_0(h'_k|_{W'})=\operatorname{mult}_o h_k$, and $(W',(1+\eta)C'_k)$ is klt near 0. In particular, (W',D') is klt near 0, and

$$a(P,W',0)-\frac{1}{m}\operatorname{mult}_P h'_k>0$$

for any prime divisor P over W' with center 0 and any k.

(4) For any subgroup $H\subset G$ and any k,

$$\operatorname{mld}((W'\ni 0,D')/H)=\operatorname{mld}((W\ni o,D)/H),$$
$$\operatorname{mld}((W'\ni 0,C'_k)/H)=\operatorname{mld}((W\ni o,C_k)/H).$$

In particular, $\alpha=\operatorname{mld}((W'\ni 0,D')/G)$.

(5) For any i, there exists a prime divisor P'_i over W' with center 0 such that

$$a(P'_i,W',0)=a(P_i,W,0),\qquad \operatorname{mult}_{P'_i} h'_k=\operatorname{mult}_{P_i} h_k,$$
$$I_H(P'_i)=I_H(P_i),\qquad \operatorname{Stab}_H(P'_i)=\operatorname{Stab}_H(P_i)$$

for any k and any subgroup $H\subset G$.

Proof. The proof is long, so we divide it into several steps. Let $\mathfrak{R}:=\mathbb{C}[[x,y,z,t]]$ and let $\mathfrak{n}$ be the maximal ideal of $\mathfrak{R}$. For $F\in\mathfrak{R}$, let $J(F)$ be the ideal of $\mathfrak{R}$ generated by the four partial derivatives of F.

Step 1. In this step we construct a diagonal action of G on $\mathfrak{R}$ and a G-semi-invariant element $F\in\mathfrak{R}$ of the shape of (1), with formal power series in place of the polynomials $a_1,\dots,a_4$, such that $\widehat{\mathcal{O}}_{W,o}\cong\mathfrak{R}/(F)$ G-equivariantly.

Since $(W\ni o)$ is an isolated Gorenstein terminal threefold germ, it is an isolated cDV germ [KM98, Corollary 5.38], of quadratic rank one in the sense of [Liu26, Definition-Lemma 11.1]. Thus [Liu26, Lemma 11.3(1)] provides a smooth fourfold germ $(Y\ni o)$ with an action of G containing W as $(f=0)$ for a G-semi-invariant regular function f on Y, and G-semi-invariant generators x,y,z,t of the maximal ideal of $\widehat{\mathcal{O}}_{Y,o}$ such that the expansion of f in $\widehat{\mathcal{O}}_{Y,o}$ is x^2+g, where $g\in\mathbb{C}[[y,z,t]]$ has order at least three and cubic part g_3 equal to the residual cubic, and x^2+g is G-semi-invariant of the character $\chi:=\chi_x^2$, where χ_x is the character of x. We identify $\widehat{\mathcal{O}}_{Y,o}$ with $\mathfrak{R}$, on which G acts diagonally. An element of G which acts trivially on x,y,z,t acts trivially on $\widehat{\mathcal{O}}_{W,o}=\mathfrak{R}/(f)$, hence on W. Since G acts faithfully on W, the characters $\chi_x,\chi_y,\chi_z,\chi_t$ of x,y,z,t define a faithful diagonal action of G on $\mathbb{A}^4$.

Since the residual cubic defines three concurrent lines, $g_3=l_1l_2l_3$ for linear forms l_1,l_2,l_3 in y,z,t which define three distinct lines through one point of $\mathbb{P}^2$. Thus l_1,l_2,l_3 span a two-dimensional subspace V of $\mathbb{C}y+\mathbb{C}z+\mathbb{C}t$. Since g_3 is G-semi-invariant, G permutes the lines $(l_j=0)$, and V is G-stable. Since G acts diagonally on $\mathbb{C}y+\mathbb{C}z+\mathbb{C}t$, after a G-equivariant linear change of y,z,t we may assume that y,z span V and that y,z,t are G-semi-invariant.

Then $g_3\in\mathbb{C}[y,z]$ is a binary cubic form with three distinct zeros on $\mathbb{P}^1$ with coordinates $[y:z]$. The group G acts on $\mathbb{P}^1$ through the character $\zeta:=\chi_y\chi_z^{-1}$ and permutes the three zeros. An automorphism of $\mathbb{P}^1$ fixing three distinct points is the identity, so the image $\zeta(G)$ is isomorphic to an abelian subgroup of the symmetric group on three letters. Thus $\zeta(G)$ has order one, two, or three.

- If ζ is trivial, then any linear change of y, z is G-equivariant. Since the automorphism group of $\mathbb{P}^1$ acts transitively on triples of distinct points and
$$y^2z + z^3 = z(y + \sqrt{-1}z)(y - \sqrt{-1}z)$$
has three distinct zeros, after a linear change of y, z we may assume that $g_3 = y^2z + z^3$.
- If $\zeta(G) = \{\pm 1\}$, then the non-trivial element of $\zeta(G)$ acts by $[y : z] \to [-y : z]$. The fixed points of $[y : z] \to [-y : z]$ are $[1 : 0]$ and $[0 : 1]$, so $[y : z] \to [-y : z]$ fixes exactly one zero of g_3 and exchanges the other two. Possibly exchanging y and z, the fixed zero is $[1 : 0]$, and the other two zeros are $[\pm p : 1]$ with $p \neq 0$. Thus $g_3 = \kappa z(y^2 - p^2z^2)$ with $\kappa \neq 0$, and after rescaling y and z we may assume that $g_3 = y^2z + z^3$.
- If $\zeta(G)$ has order three, then no zero of g_3 is fixed by $\zeta(G)$, so g_3 vanishes neither at $[1 : 0]$ nor at $[0 : 1]$. The characters of y^3, y^2z, yz^2, z^3 are $\zeta^3\chi_z^3 = \chi_z^3$, $\zeta^2\chi_z^3$, $\zeta\chi_z^3$, and χ_z^3, and only the first and the last coincide. Since g_3 is G-semi-invariant and has three distinct zeros, $g_3 = \kappa_1 y^3 + \kappa_2 z^3$ with $\kappa_1\kappa_2 \neq 0$, and after rescaling y and z we may assume that $g_3 = y^3 + z^3$.

Let $g_0 := y^2z + z^3$ and $(M_1, M_2, M_3, M_4) := (z^2, z, y, 1)$ in the first two cases, which we call *type (I)*, and let $g_0 := y^3 + z^3$ and $(M_1, M_2, M_3, M_4) := (yz, y, z, 1)$ in the last case, which we call *type (II)*. Since $x^2 + g_0$ is the cubic truncation of the G-semi-invariant $x^2 + g$, the monomials of g_0 have the character $\chi = \chi_x^2$. In type (I), $\chi_y^2\chi_z = \chi_z^3 = \chi_x^2$, so $\chi_y^2 = \chi_z^2$ and $\chi_x^2 = \chi_z^3$. In type (II), $\chi_x^2 = \chi_y^3 = \chi_z^3$ and $\chi_y\chi_z^{-1} = \zeta$ has order three. A direct calculation shows that $M_1, \dots, M_4$ is a basis of $\mathbb{C}[y, z]/(\partial_y g_0, \partial_z g_0)$, where $(\partial_y g_0, \partial_z g_0) = (2yz, y^2 + 3z^2)$ in type (I) and $(3y^2, 3z^2)$ in type (II). Since $\partial_y g_0$ and $\partial_z g_0$ are quadratic forms, any form $p \in \mathbb{C}[y, z]$ of degree e can be written as
$$p = \alpha\partial_y g_0 + \beta\partial_z g_0 + \sum_{\deg M_j = e} \gamma_j M_j$$
with forms $\alpha, \beta \in \mathbb{C}[y, z]$ of degree $e - 2$ (zero if $e \leq 1$) and $\gamma_j \in \mathbb{C}$.

We remove the terms of $g - g_0$ degree by degree. Let $n \geq 4$ be an integer, and assume that, after a G-equivariant automorphism of $\mathfrak{R}$ fixing x and t, the element $x^2 + g$ is equal to
$$x^2 + g_0 + \sum_{j=1}^{4} a_j(t)M_j + Q,$$
where $a_j \in \mathbb{C}[[t]]$, $\operatorname{mult} a_1 \geq 2$, $\operatorname{mult} a_2 \geq 3$, $\operatorname{mult} a_3 \geq 3$, $\operatorname{mult} a_4 \geq 4$, for any j either $a_j(t)M_j = 0$ or $a_j(t)M_j$ is G-semi-invariant of the character χ, and $Q \in \mathbb{C}[[y, z, t]] \cap \mathfrak{n}^n$ is G-semi-invariant of the character χ. For $n = 4$ the equality holds with $a_j = 0$ and $Q = g - g_0$. Let Q_n be the homogeneous part of degree n of Q. Applying the decomposition above to the coefficients of the powers of t in Q_n, we obtain
$$Q_n = \alpha\partial_y g_0 + \beta\partial_z g_0 + \sum_{j=1}^{4} \gamma_j t^{n - \deg M_j} M_j,$$
where $\alpha, \beta \in \mathbb{C}[y, z, t]$ are homogeneous of degree $n - 2$ and $\gamma_j \in \mathbb{C}$. Replacing α, β, and each term $\gamma_j t^{n-\deg M_j} M_j$ by their components of the characters χ_y, χ_z, and χ respectively, which keeps the equality because Q_n, $\partial_y g_0$, and $\partial_z g_0$ are G-semi-invariant of the characters χ, $\chi\chi_y^{-1}$, and $\chi\chi_z^{-1}$, we may assume that α and β are G-semi-invariant of the characters χ_y and χ_z, and that each non-zero $\gamma_j t^{n-\deg M_j} M_j$ has the character χ. The substitution $y \to y - \alpha$, $z \to z - \beta$, fixing x and t, is a G-equivariant automorphism of $\mathfrak{R}$. By the Taylor formula,
$$g_0(y - \alpha, z - \beta) - g_0 + \alpha\partial_y g_0 + \beta\partial_z g_0 \in \mathfrak{n}^{2n-3}.$$

Since each non-zero $a_j(t)M_j$ has order at least four, $a_j(t)M_j(y-\alpha,z-\beta)-a_j(t)M_j\in\mathfrak{n}^{n+1}$, and $Q(y-\alpha,z-\beta,t)-Q\in\mathfrak{n}^{2n-3}$. Since $2n-3\geq n+1$, the substitution transforms x^2+g into

$$x^2+g_0+\sum_{j=1}^{4}\bigl(a_j(t)+\gamma_j t^{n-\deg M_j}\bigr)M_j+Q'$$

with $Q'\in\mathbb{C}[[y,z,t]]\cap\mathfrak{n}^{n+1}$ G-semi-invariant of the character χ. Since $n-\deg M_j\geq 4-\deg M_j$, $\operatorname{mult}(a_j+\gamma_j t^{n-\deg M_j})$ is at least 2, 3, 3, 4 for $j=1,2,3,4$. The substitutions for $n=4,5,\dots$ differ from the identity by elements of $\mathfrak{n}^{n-2}$, so their composite converges $\mathfrak{n}$-adically to a G-equivariant automorphism of $\mathfrak{R}$, which transforms x^2+g into

$$F:=x^2+g_0+\sum_{j=1}^{4}a_j(t)M_j,\qquad a_j\in\mathbb{C}[[t]],$$

with the bounds on $\operatorname{mult}a_j$ above. Thus F is G-semi-invariant of the character $\chi=\chi_x^2$, and $\widehat{\mathcal{O}}_{W,o}\cong\mathfrak{R}/(F)$ G-equivariantly.

Step 2. In this step we replace F by a polynomial f' of the shape of (1) and prove (2).

Since W has an isolated singularity at o, the ideal $(F)+J(F)$ is $\mathfrak{n}$-primary, by the contact invariance of the Tjurina algebra [GLS07, Chapter I, Lemma 2.10(2)] and [GLS07, Chapter I, Lemma 2.3] applied to a local equation of W. Thus there exists a positive integer ℓ_0 such that $\mathfrak{n}^{\ell_0}\subset(F)+\mathfrak{n}J(F)$. If $F^\flat\in\mathfrak{R}$ satisfies $F^\flat-F\in\mathfrak{n}^{\ell_0+1}$, then

$$(F^\flat)+\mathfrak{n}J(F^\flat)+\mathfrak{n}^{\ell_0+1}=(F)+\mathfrak{n}J(F)+\mathfrak{n}^{\ell_0+1}\supset\mathfrak{n}^{\ell_0},$$

so $\mathfrak{n}^{\ell_0}\subset(F^\flat)+\mathfrak{n}J(F^\flat)$ by Nakayama's lemma.

Fix an integer $n_0>2\ell_0+3$, and let f' be the sum of the terms of F of degree less than n_0. Then f' is G-semi-invariant of the character χ, $F-f'\in\mathfrak{n}^{n_0}$, and f' is of the shape of (1), with the coefficients a_j replaced by their truncations.

We show that there exist a G-equivariant automorphism φ of $\mathfrak{R}$ and a G-invariant unit $v\in\mathfrak{R}$ such that $\varphi(f')=vF$. Let $F_0:=f'$. Suppose that $F_p\in\mathfrak{R}$ is G-semi-invariant of the character χ with $F_p-F\in\mathfrak{n}^{n_p}$ for an integer $n_p\geq n_0$. By the previous paragraph, $\mathfrak{n}^{\ell_0}\subset(F_p)+\mathfrak{n}J(F_p)$, so

$$F_p-F\in\mathfrak{n}^{n_p-\ell_0}\mathfrak{n}^{\ell_0}\subset\mathfrak{n}^{n_p-\ell_0}F_p+\mathfrak{n}^{n_p-\ell_0+1}J(F_p).$$

Thus $F_p-F=\sigma F_p+\sum_{j=1}^{4}\delta_j\partial_jF_p$ with $\sigma\in\mathfrak{n}^{n_p-\ell_0}$ and $\delta_j\in\mathfrak{n}^{n_p-\ell_0+1}$, where $\partial_1,\dots,\partial_4$ are the partial derivatives with respect to $x_1,\dots,x_4:=x,y,z,t$. Since F_p-F and F_p have the character χ and ∂_jF_p has the character $\chi\chi_{x_j}^{-1}$, after replacing σ by the G-invariant component of σ and δ_j by the component of δ_j of the character χ_{x_j}, we may assume that σ is G-invariant and that δ_j is G-semi-invariant of the character χ_{x_j}. Let

$$F_{p+1}:=(1-\sigma)F_p(x_1-\delta_1,\dots,x_4-\delta_4).$$

By the Taylor formula,

$$F_p(x_1-\delta_1,\dots,x_4-\delta_4)-F_p+\sum_j\delta_j\partial_jF_p\in\mathfrak{n}^{2(n_p-\ell_0+1)},$$

and

$$\sigma\sum_j\delta_j\partial_jF_p\in\mathfrak{n}^{2n_p-2\ell_0+1}.$$

Thus

$$F_{p+1}-F=F_p-\sigma F_p-\sum_j\delta_j\partial_jF_p-F+(\text{an element of }\mathfrak{n}^{2n_p-2\ell_0+1})\in\mathfrak{n}^{2n_p-2\ell_0+1},$$

and F_{p+1} is G-semi-invariant of the character χ. Let $n_{p+1}:=2n_p-2\ell_0+1$. Since $n_p>2\ell_0$, $n_{p+1}>n_p$, so $n_p\to+\infty$. The substitutions $x_j\to x_j-\delta_j$ differ from the identity by elements of $\mathfrak{n}^{n_p-\ell_0+1}$ and the factors $1-\sigma$ differ from 1 by elements of $\mathfrak{n}^{n_p-\ell_0}$. Hence their composites and

products converge $\mathfrak{n}$-adically to a G-equivariant automorphism φ of $\mathfrak{R}$ and a G-invariant unit v^{-1} with $v^{-1}\varphi(f') = F$.

Since $\widehat{\mathcal{O}}_{W',0} = \mathfrak{R}/(f')$, we obtain a G-equivariant isomorphism

$$\Phi : \widehat{\mathcal{O}}_{W,o} \cong \mathfrak{R}/(F) \cong \widehat{\mathcal{O}}_{W',0}.$$

Since

$$\mathfrak{n}^{\ell_0} \subset (f') + \mathfrak{n}J(f') \subset (f') + J(f')$$

and the completion of the local ring of $\mathbb{A}^4$ at 0 is faithfully flat over it, the ideal generated by f' and the partial derivatives of f' is primary to the maximal ideal of 0 in the local ring of $\mathbb{A}^4$ at 0. Thus W' is smooth outside 0 near 0. Being a hypersurface, W' is Gorenstein, and it is normal near 0 by Serre's criterion. By [dFEM11, Lemma 2.21, Proposition A.14, and Remark A.15], prime divisors over W with center o and prime divisors over W' with center 0 correspond to prime divisors over the spectra of the completions with closed center, with equal log discrepancies, equal orders of elements of the completions, and isomorphic residue fields. Through the G-equivariant isomorphism Φ, we obtain a bijection between prime divisors P over W' with center 0 and prime divisors $P^\natural$ over W with center o such that

$$a(P, W', 0) = a(P^\natural, W, 0), \qquad \operatorname{mult}_P \Phi(\vartheta) = \operatorname{mult}_{P^\natural} \vartheta \ \text{ for any } \vartheta \in \widehat{\mathcal{O}}_{W,o},$$

and $I_H(P) = I_H(P^\natural)$ and $\operatorname{Stab}_H(P) = \operatorname{Stab}_H(P^\natural)$ for any subgroup $H \subset G$. Since W is terminal near o and W' is smooth outside 0 near 0, W' is terminal near 0. Finally, $f' \in \mathfrak{n}^2$, so the cotangent space of W' at 0 is spanned by x, y, z, t. An element of G acting trivially on W' near 0 acts trivially on the cotangent space of W' at 0, so it is the identity by **Step 1**. This proves (1) and (2).

Step 3. In this step we truncate the marks and prove (3), (4), and (5).

Since each (W, C_k) is terminal, hence klt, near o [KM98, Corollary 2.31(1) and (3)], there exists a real number $\eta \in (0,1)$ such that $(W, (1+2\eta)C_k)$ is klt near o for any k. Fix a prime divisor P_0 over W with center o, and fix a real number Θ such that

$$\Theta > a(P_0, W, D) \quad \text{and} \quad \Theta > a(P_0, W, C_k) \ \text{ for any } k.$$

For any k, let χ_k be the character of h_k, and let $\hat{h}_k \in \mathfrak{R}$ be a G-semi-invariant element of the character χ_k whose image in $\mathfrak{R}/(f') = \widehat{\mathcal{O}}_{W',0}$ is $\Phi(h_k)$. Such an element exists: the component of the character χ_k of any preimage of $\Phi(h_k)$ is again a preimage, since $\Phi(h_k)$ has the character χ_k. Fix an integer L such that

- $L \geq 3m/\eta$,
- $L > \operatorname{mult}_o h_k$, $L > \operatorname{mult}_{P_0} h_k$, and $L > \operatorname{mult}_{P_i} h_k$ for any i and k, and
- $(\min_k \mu_k)\eta L/m > \Theta|G|$.

Let h'_k be the sum of the terms of $\hat{h}_k$ of degree less than L. Then h'_k is a G-semi-invariant polynomial of the character χ_k, and $h'_k - \hat{h}_k \in \mathfrak{n}^L$. Since $\Phi(h_k) \in \mathfrak{m}_0^j \backslash \mathfrak{m}_0^{j+1}$ for $j := \operatorname{mult}_o h_k < L$, where $\mathfrak{m}_0$ is the maximal ideal of $\widehat{\mathcal{O}}_{W',0}$, the image of h'_k has order j as well. Thus $h'_k|_{W'}$ is a mark on $(W' \ni 0)$ with $\operatorname{mult}_0(h'_k|_{W'}) = \operatorname{mult}_o h_k$.

If h_k is a unit, then so is h'_k, because $L \geq 1$. Assume that h_k is not a unit. By [dFEM11, Proposition 2.11] and the isomorphism Φ, the log canonical threshold at 0 of the ideal generated by the image of $\hat{h}_k$ on the spectrum of $\widehat{\mathcal{O}}_{W',0}$ is equal to $\operatorname{lct}(W \ni o; (h_k)) > (1+2\eta)/m$. By [dFEM11, Proposition 2.19(ii)] with $\epsilon = \eta/m$ (see Theorem 2.12(2)), since $L \geq 3/\epsilon$, the log canonical threshold of h'_k differs from it by at most η/m. By [dFEM11, Proposition 2.11] again, $\operatorname{lct}(W' \ni 0; (h'_k)) > (1+\eta)/m$. In both cases, since W' is klt near 0, $(W', (1+\eta)C'_k)$ is klt near 0. Since $D' = \sum_k \mu_k C'_k$ with $\sum_k \mu_k = 1$, (W', D') is klt near 0. In particular, for any prime divisor P over W' with center 0 and any k,

$$a(P, W', 0) > \frac{1+\eta}{m} \operatorname{mult}_P h'_k \quad \text{and} \quad a(P, W', 0) > \frac{1+\eta}{m} \operatorname{mult}_P \hat{h}_k, \tag{E.76}$$

where the second inequality follows from the klt pair $(W,(1+2\eta)C_k)$ and the bijection of **Step 2**, since $\mathrm{mult}_P \hat{h}_k = \mathrm{mult}_{P^\natural} h_k$. This proves (3).

We prove (4). Let P be a prime divisor over W' with center 0. If $\mathrm{mult}_P h'_k = \mathrm{mult}_P \hat{h}_k$ for $k=1,\dots,s$, then $a(P,W',D') = a(P^\natural,W,D)$ and $a(P,W',C'_k) = a(P^\natural,W,C_k)$ for any k. Assume that $\mathrm{mult}_P h'_k \neq \mathrm{mult}_P \hat{h}_k$ for some k. Since $h'_k - \hat{h}_k \in \mathfrak{n}^L$, both $\mathrm{mult}_P h'_k$ and $\mathrm{mult}_P \hat{h}_k$ are at least $\mathrm{mult}_P(\mathfrak{n}^L) \geq L$. By (E.76),

$$a(P,W',C'_k) > \frac{\eta}{m}\mathrm{mult}_P h'_k \geq \frac{\eta L}{m} \quad \text{and} \quad a(P^\natural,W,C_k) > \frac{\eta}{m}\mathrm{mult}_P \hat{h}_k \geq \frac{\eta L}{m}.$$

Since $a(P,W',C'_j)>0$ and $a(P^\natural,W,C_j)>0$ for any j, we obtain

$$a(P,W',D') = \sum_j \mu_j a(P,W',C'_j) > (\min_j \mu_j)\frac{\eta L}{m} > \Theta|G|,$$

and, since $a(P^\natural,W,D) = \sum_j \mu_j a(P^\natural,W,C_j)$, also $a(P^\natural,W,D) > \Theta|G|$. Since $\min_j \mu_j \leq 1$, also $a(P,W',C'_k) > \Theta|G|$ and $a(P^\natural,W,C_k) > \Theta|G|$. Thus, for any subgroup $H \subset G$, the numbers

$$\frac{a(P,W',D')}{|I_H(P)|}, \quad \frac{a(P^\natural,W,D)}{|I_H(P^\natural)|}, \quad \frac{a(P,W',C'_k)}{|I_H(P)|}, \quad \frac{a(P^\natural,W,C_k)}{|I_H(P^\natural)|}$$

are larger than Θ whenever the order of the corresponding mark at P changes, namely whenever $\mathrm{mult}_P h'_j \neq \mathrm{mult}_P \hat{h}_j$ for some j for the first two numbers, and whenever $\mathrm{mult}_P h'_k \neq \mathrm{mult}_P \hat{h}_k$ for the last two numbers. Let P'_0 be the prime divisor over W' with $(P'_0)^\natural = P_0$. Since $L > \mathrm{mult}_{P_0} h_k$ for any k, the orders of all marks at P'_0 are equal to those at P_0, and

$$\frac{a(P'_0,W',D')}{|I_H(P'_0)|} = \frac{a(P_0,W,D)}{|I_H(P_0)|} \leq a(P_0,W,D) < \Theta,$$

and similarly for C'_k and C_k. By [Liu26, Lemma 2.6], a quotient value is the infimum of the log discrepancies divided by the orders of the inertia groups over the prime divisors with center the closed point. Thus each of the four quotient values $\mathrm{mld}((W' \ni 0, D')/H)$, $\mathrm{mld}((W \ni o, D)/H)$, $\mathrm{mld}((W' \ni 0, C'_k)/H)$, and $\mathrm{mld}((W \ni o, C_k)/H)$ is the infimum over the prime divisors at which the orders of the corresponding marks do not change, where the corresponding numbers for W' and W are equal. This proves (4).

For any i, let P'_i be the prime divisor over W' with $(P'_i)^\natural = P_i$. Since $L > \mathrm{mult}_{P_i} h_k$, $\mathrm{mult}_{P'_i} h'_k = \mathrm{mult}_{P'_i} \hat{h}_k = \mathrm{mult}_{P_i} h_k$ for any k, and the remaining equalities of (5) follow from **Step 2**. □

Proof of Proposition 11.14. The proof is long, so we divide it into several steps. Suppose that there exists a sequence of data $(W_i \ni o_i, G_i, h_{i,1},\dots,h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with W_i of quadratic rank one whose residual cubic defines three concurrent lines, whose numbers

$$\alpha_i := \mathrm{mld}((W_i \ni o_i, \sum_k c_k \,\mathrm{div}(h_{i,k}))/G_i)$$

form a strictly increasing sequence in $(0,1)$. Let $\Gamma := \{c_1,\dots,c_s\}$, a finite subset of $(0,1]$.

Step 1. In this step we pass to polynomial germs with diagonal groups of unbounded order.

By Proposition 11.7(1) with $\lambda = 1$, for any positive integer M, the numbers α_i with $|G_i| \leq M$ belong to a finite set. Since the α_i are pairwise distinct, after passing to a subsequence we may assume that $|G_i| \to +\infty$.

For any i, by Lemma E.15 applied to $(W_i \ni o_i, G_i, h_{i,1},\dots,h_{i,s})$ with $l=0$, there exist a faithful diagonal action of G_i on $\mathbb{A}^4$, a polynomial f'_i of type (I) or (II) of Lemma E.15(1), and G_i-semi-invariant polynomials $h'_{i,1},\dots,h'_{i,s}$ whose restrictions to $W'_i := (f'_i = 0)$ are marks. Let

$D_i' := \sum_k c_k \operatorname{div}(h_{i,k}'|_{W_i'})$. By Lemma E.15(2)–(4), $(W_i' \ni 0)$ is an isolated Gorenstein terminal threefold germ on which G_i acts faithfully, (W_i', D_i') is klt near 0, and

$$\alpha_i = \operatorname{mld}((W_i' \ni 0, D_i')/G_i).$$

After passing to a subsequence, we may assume that all f_i' are of type (I), or that all f_i' are of type (II). Let $g_0 := y^2z + z^3$ and $(M_1, M_2, M_3, M_4) := (z^2, z, y, 1)$ in type (I), and $g_0 := y^3 + z^3$ and $(M_1, M_2, M_3, M_4) := (yz, y, z, 1)$ in type (II), so that

$$f_i' = x^2 + g_0(y,z) + \sum_{j=1}^{4} a_{i,j}(t) M_j, \qquad a_{i,j} \in t\mathbb{C}[t].$$

We regard G_i as a subgroup of $\mathbb{T}^4$, the class of (u_x, u_y, u_z, u_t) corresponding to the element acting on x, y, z, t by $\exp(2\pi\sqrt{-1}u_x), \dots, \exp(2\pi\sqrt{-1}u_t)$. By the relations between the characters in Lemma E.15(1), $G_i \subset \mathbf{S}$, where $\mathbf{S} \subset \mathbb{T}^4$ is the closed subgroup of the classes of the vectors u with

$$\begin{aligned} &2u_y - 2u_z \in \mathbb{Z} \text{ and } 2u_x - 3u_z \in \mathbb{Z} && \text{in type (I)},\\ &2u_x - 3u_y \in \mathbb{Z} \text{ and } 3u_y - 3u_z \in \mathbb{Z} && \text{in type (II)}. \end{aligned}$$

In both types, $\dim \mathbf{S} = 2$, and the identity component $\mathbf{S}^0$ of $\mathbf{S}$ is the image of the plane $\mathbb{R}(3,2,2,0) + \mathbb{R}(0,0,0,1) \subset \mathbb{R}^4$.

Step 2. In this step we show that the groups G_i grow along one circle $\mathbf{T}_{(w)}$ with $w = (3d, 2d, 2d, v)$.

By Lemma 4.6, after passing to a subsequence, there exists a closed subgroup $H_\infty \subset \mathbb{T}^4$ such that the G_i converge to H_∞ in the Hausdorff distance, $G_i \subset H_\infty$ for $i \gg 0$, and $\dim H_\infty \geq 1$. Since $\mathbf{S}$ is closed and contains all G_i, $H_\infty \subset \mathbf{S}$, so $H_\infty^0 \subset \mathbf{S}^0$ and $\dim H_\infty \leq 2$.

Assume that $\dim H_\infty = 2$. Then $H_\infty^0 = \mathbf{S}^0$, so the Lie algebra of H_∞ is the plane $\mathbb{R}(3,2,2,0) + \mathbb{R}(0,0,0,1)$, which contains the vector $(3,2,2,1)$ with all coordinates positive. For any i, $W_i' = (f_i' = 0)$ is a normal terminal, hence klt, hypersurface near 0, G_i is a finite diagonal abelian group acting faithfully on W_i', the $h_{i,k}'$ are G_i-semi-invariant polynomials, and (W_i', D_i') is klt near 0 (Lemma E.15(2)–(3)). By Lemma 4.7 with $n = 4$,

$$\alpha_i = \operatorname{mld}((W_i' \ni 0, D_i')/G_i) \to 0,$$

which contradicts $\alpha_i > \alpha_1 > 0$ for $i > 1$.

Thus $\dim H_\infty = 1$. By Lemma 4.6, $H_\infty^0 = \mathbf{T}_{(w)}$ for a primitive vector $w \in \mathbb{Z}^4$, $R_i := G_i \cap \mathbf{T}_{(w)}$ is cyclic of order $r_i \to +\infty$, and $[G_i : R_i] = |H_\infty / H_\infty^0| =: e_0$ for $i \gg 0$. Since $\mathbf{T}_{(w)} \subset \mathbf{S}^0$, $w = (3d, 2d, 2d, v)$ for real numbers d, v. Since $w \in \mathbb{Z}^4$, $d = 3d - 2d \in \mathbb{Z}$ and $v \in \mathbb{Z}$, and since w is primitive, $\gcd(d, v) = 1$. Since $\mathbf{T}_{(w)} = \mathbf{T}_{(-w)}$, we may assume that either $v > 0$, or $v = 0$ and $d = 1$. The group R_i is generated by the class of w/r_i.

Step 3. In this step we exclude the circles $\mathbf{T}_{(w)}$ with $v = 0$, with $d = 0$, and with $d > 0$.

Assume that $v = 0$ and $d = 1$. The generator of R_i multiplies x, y, z, t by $\omega^3, \omega^2, \omega^2, 1$, where $\omega := \exp(2\pi\sqrt{-1}/r_i)$. Since f_i' is G_i-semi-invariant, all monomials of f_i' have the character of x^2, on which the generator acts by ω^6. The generator acts on a monomial $t^n M_j$ by ω^4, ω^2, ω^2, ω^0 for $j = 1, 2, 3, 4$. For $r_i > 6$, $\omega^6 \notin \{\omega^4, \omega^2, 1\}$, so $a_{i,j} = 0$ for any j and $f_i' = x^2 + g_0(y,z)$. Then W_i' is singular along the t-axis, which contradicts the isolated singularity of W_i' at 0.

Assume that $d = 0$. Then $v = 1$, $\mathbf{T}_{(w)} = \mathbf{T}_{(0,0,0,1)}$, and $R_i = G_i \cap \mathbf{T}_{(0,0,0,1)} = \mu_{r_i}(0,0,0,1)$. The germ $(W_i' \ni 0)$ is an isolated singular cDV germ by Lemma E.15(2) and [KM98, Corollary 5.38], since f_i' has no term of degree less than two. The group G_i is a finite diagonal abelian group preserving f_i' up to a character and acting faithfully on W_i', $[G_i : R_i] = e_0$, $r_i \to +\infty$, $c_k \in \Gamma$, the $h_{i,k}'$ are G_i-semi-invariant polynomials not vanishing identically on W_i', and (W_i', D_i') is klt near 0. Since $r_i \geq r_0(\Gamma)$ for $i \gg 0$, Theorem 8.5, applied with the coordinate t in the role of x, the coordinates x, y, z in the role of u_1, u_2, u_3, the finite set $\Gamma = \{c_1, \dots, c_s\}$, and the integer $e_0 = [G_i : R_i]$, implies that the numbers $\alpha_i = \operatorname{mld}((W_i' \ni 0, D_i')/G_i)$ do not form a strictly increasing sequence, a contradiction.

Assume that $d > 0$. By the choice of the sign of w in **Step 2**, either $v > 0$, or $v = 0$ and $d = 1$, and the case $v = 0$, $d = 1$ has been excluded above. Hence $v > 0$, so $w/r_i \in \mathbb{R}^4_{>0}$, and $w/r_i \in N_{G_i}$ because its class generates $R_i \subset G_i$. Since f_i' and the $h_{i,k}'$ are polynomials, $\mathrm{wt}_{w/r_i}(f_i') \geq 0$ and $\mathrm{wt}_{w/r_i}(h_{i,k}') \geq 0$. By Lemma 4.7 with the vector $w/r_i \in N_{G_i} \cap \mathbb{R}^4_{>0}$,

$$\alpha_i \leq \frac{3d + 2d + 2d + v}{r_i} = \frac{7d + v}{r_i} \to 0,$$

which contradicts $\alpha_i > \alpha_1 > 0$.

Step 4. In this step we exclude the remaining circles, with $d < 0$ and $v > 0$.

Let $c := -d$. Then c and v are coprime positive integers and $\mathbf{T}_{(w)} = \mathbf{T}_{(-3c,-2c,-2c,v)}$. Let $f_0 := x^2 + g_0(y,z)$ and $q := (3,2,2)$, so that $\mathbf{T}_{(w)} = \mathbf{T}_{(-cq_x,-cq_y,-cq_z,v)}$. In type (I), $f_0 = x^2 + y^2 z + z^3$ is the standard form of type D_4, with the weighted homogeneous basis $M_1, \dots, M_4 = z^2, z, y, 1$ of $\mathbb{C}[y,z]/(\partial_y g_0, \partial_z g_0)$. In type (II), $f_0 = x^2 + y^3 + z^3$ is the form of type D_4 with weights $(3,2,2)$ allowed in Theorem 10.4, with the basis $M_1, \dots, M_4 = yz, y, z, 1$. We check the hypotheses of Theorem 10.4 for the Du Val type D_4, the weights q, and the data $(W_i' \ni 0, G_i, D_i')$:

- $W_i' = (f_0 + \sum_j a_{i,j}(t) M_j = 0)$ with $a_{i,j} \in t\mathbb{C}[t]$ is an isolated Gorenstein terminal germ (Lemma E.15(2));
- G_i is a finite diagonal abelian group acting faithfully on W_i', and f_0 and each non-zero $a_{i,j}(t)M_j$ are G_i-semi-invariant of the character of f_i', since all monomials of f_i' have the character of f_i';
- $[G_i : G_i \cap \mathbf{T}_{(w)}] = e_0$;
- D_i' is a labelled boundary with labels in the finite set $\Gamma \subset [0,1]$, and (W_i', D_i') is klt near 0.

Since $|G_i \cap \mathbf{T}_{(w)}| = r_i \to +\infty$ and the numbers $\alpha_i = \mathrm{mld}((W_i' \ni 0, D_i')/G_i)$ are positive and strictly increasing, we obtain a contradiction with Theorem 10.4.

All circles have been excluded, which proves Proposition 11.14. □

We now prove Lemma E.16, which reduces Proposition 11.15 to three circles.

Lemma E.16. *Notation as in Set-up 11.1. Let $(W_i \ni o_i, G_i, (h_{i,k})_k)$ be a sequence of such data, with s, m and the μ_k fixed and with W_i of quadratic rank one whose residual cubic is a cuspidal cubic or a tangent conic (Definition-Lemma 6.4), let $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$ and $\alpha_i := \mathrm{mld}((W_i \ni o_i, D_i)/G_i)$, and assume that the α_i are strictly increasing in $(0,1)$. Then, after passing to a subsequence and replacing the data with numerical admissions that keep the groups G_i (Proposition 11.8), $W_i = (x^2 + g_i(y,z,t) = 0)$ with G_i diagonal on x, y, z, t and the cubic part of g_i equal to $y^2 z + t^3$ or to $y^2 z + y t^2$, and there exist a circle $\mathbf{T}$, equal to $\mathbf{T}_{(0,1,-2,0)}$ or $\mathbf{T}_{(3,4,-2,2)}$ in the first case and to $\mathbf{T}_{(1,2,-2,0)}$ in the second case, and a positive integer e_0 such that $|G_i \cap \mathbf{T}| \to +\infty$ and $[G_i : G_i \cap \mathbf{T}] \leq e_0$.*

Proof. The proof is long, so we divide it into several steps. Possibly passing to a subsequence, we may assume that all the residual cubics are cuspidal cubics (the *cuspidal case*), or that all of them are tangent conics (the *tangent case*).

Step 1. In this step we replace the data with polynomial numerical admissions in normal form and pass to groups of unbounded order.

By Proposition 11.7(1) with $\lambda = 1$, for any positive integer M, the numbers α_i with $|G_i| \leq M$ belong to a finite set. Since the α_i are pairwise distinct, after passing to a subsequence we may assume that $|G_i| \to +\infty$.

Fix i. An isolated Gorenstein terminal threefold germ is an isolated cDV germ [KM98, Corollary 5.38]. By [Liu26, Lemma 11.3(1)], there exist a smooth fourfold germ $(Y_i \ni o_i)$ with an action of G_i containing W_i as $(f_i = 0)$ for a G_i-semi-invariant regular function f_i, and G_i-semi-invariant generators x, y, z, t of the maximal ideal of $\widehat{\mathcal{O}}_{Y_i,o_i}$, such that f_i is a unit

times $x^2+g(y,z,t)$ with $g\in(y,z,t)^3$ and the cubic part g_3 of g is the residual cubic. By Proposition 3.17(1), in the cuspidal case we may assume, after a G_i-equivariant linear change of y,z,t, that $g_3=y^2z+t^3$. By Proposition 3.17(2), in the tangent case we may assume, after a G_i-equivariant automorphism of $\widehat{\mathcal{O}}_{Y_i,o_i}$, that $g=y^2z+yt^2+B(z,t)$ with $\operatorname{mult}B\geq 4$. Let $x_1,\dots,x_4$ be regular G_i-eigenparameters of Y_i at o_i whose classes in $\mathfrak{m}_{o_i}/\mathfrak{m}_{o_i}^2$ are those of x,y,z,t. They exist since the Reynolds projections onto the eigenspaces of G_i are surjective on the maximal ideal of $\mathcal{O}_{Y_i,o_i}$. Let $\hat{f}_i$ and $\hat{h}_{i,k}$ be the images in $\widehat{\mathcal{O}}_{Y_i,o_i}$ of f_i and of G_i-semi-invariant lifts of the $h_{i,k}$ to Y_i. Let ψ be the G_i-equivariant automorphism of $\mathbb{C}[[x_1,\dots,x_4]]$ with $\psi(x)=x_1$, $\psi(y)=x_2$, $\psi(z)=x_3$, and $\psi(t)=x_4$. Then $\psi(\hat{f}_i)=\mathfrak{u}\cdot(x_1^2+g(x_2,x_3,x_4))$ for a unit $\mathfrak{u}$. Since $\psi(\hat{f}_i)$ and $x_1^2+g(x_2,x_3,x_4)$ are semi-invariants (x^2+g is G_i-semi-invariant by [Liu26, Lemma 11.3(1)] in the cuspidal case and by the last sentence of Proposition 3.17 in the tangent case) and $\mathfrak{u}(0)\neq 0$, $\mathfrak{u}$ is G_i-invariant.

By Proposition 11.8, there exists a numerical admission with the following data. The automorphism is ψ, the invariant unit is $\mathfrak{u}^{-1}$, and the units are $u_k:=1$. The protected prime divisors are E_0 and a prime divisor computing α_i, where E_0 is the prime divisor over W_i given by the weights $(3,2,2,2)$ in x_1,x_2,x_3,x_4 (Definition-Lemma 11.5(3) and the paragraph after Set-up 9.3). The polynomial $f_{i,0}$ is the sum of the terms of $x_1^2+g(x_2,x_3,x_4)$ of degree less than L, and the admitted marks are the sums of the terms of the $\psi(\hat{h}_{i,k})$ of degree less than N, where $L\geq 4$ and N are integers large enough for Proposition 11.8. After renaming x_1,x_2,x_3,x_4 as x,y,z,t, we obtain a polynomial germ $(x^2+g_i(y,z,t)=0)\subset\mathbb{A}^4$ with $g_i\in(y,z,t)^3$, on which G_i acts diagonally and faithfully. Since $L\geq 4$, the cubic part of g_i is y^2z+t^3 in the cuspidal case, and $g_i=y^2z+yt^2+B_i(z,t)$ for a polynomial B_i with $\operatorname{mult}B_i\geq 4$ in the tangent case. By Construction-Theorem 4.13(3), the new germ is isolated Gorenstein terminal.

From now on, W_i, $o_i=0$, $h_{i,k}$, and D_i denote the admitted germ, the origin, the admitted marks, and the labelled boundary of the admitted marks. By Proposition 11.8, the quotient value of the admitted marks with the labels c_k is α_i, and, for any k and any prime divisor P over W_i with center o_i,

$$a(P,W_i,0)-\frac{1}{m}\operatorname{mult}_P h_{i,k}>0. \tag{E.77}$$

We replace each $h_{i,k}$ by the normal form of $h_{i,k}$ (Definition-Lemma 11.5(1), which holds for the admitted data by Lemma 11.10(1)), which does not change D_i, so that each $h_{i,k}$ is a polynomial of x-degree at most one.

Before the admission, $(W_i,\frac{1}{m}\operatorname{div}(h_{i,k}))$ is terminal and $a(E_0,W_i,0)=3$ (the standard computation for the weighted blow-up with weights $(3,2,2,2)$, Lemma C.2(4)), so $\operatorname{mult}_{E_0}h_{i,k}\leq 2m-1$. The orders $\operatorname{mult}_{E_0}h_{i,k}$ are protected by the admission. In the new coordinates x,y,z,t, the prime divisor over the admitted germ corresponding to E_0 is still given by the weights $(3,2,2,2)$ (it is the prime divisor E_0 of the admitted germ, since E_0 depends only on the completed local ring by the paragraph after Set-up 9.3 and Theorem 6.2(2)). The lowest $(3,2,2,2)$-weighted part of $h_{i,k}$ has x-degree at most one, so it is not divisible by $x^2+g_{i,3}$. Hence $\operatorname{mult}_{E_0}h_{i,k}$ is the $(3,2,2,2)$-weighted order of $h_{i,k}$ (Lemma C.2(3)). Since the weighted order of a monomial is at least twice the total degree, $h_{i,k}$ has a monomial of total degree at most $m-1$.

Step 2. In this step we show that the groups G_i approach one circle, and we list the possible circles.

Regard G_i as a subgroup of $\mathbb{T}^4=\mathbb{R}^4/\mathbb{Z}^4$, the class of u acting on x,y,z,t by $\exp(2\pi\sqrt{-1}u_x)$, $\dots,\exp(2\pi\sqrt{-1}u_t)$. Since x^2, y^2z, and t^3 (resp. yt^2) have non-zero coefficients in the equation of W_i, and G_i preserves x^2+g_i up to a character, $G_i\subset\mathbf{S}$, the closed subgroup of the classes of u with $2u_x-2u_y-u_z\in\mathbb{Z}$ and $2u_x-3u_t\in\mathbb{Z}$ in the cuspidal case, and with $2u_x-2u_y-u_z\in\mathbb{Z}$ and $2u_x-u_y-2u_t\in\mathbb{Z}$ in the tangent case. The Lie algebra of $\mathbf{S}$ is the plane $2w_x=2w_y+w_z=3w_t$, resp. $2w_x=2w_y+w_z=w_y+2w_t$, whose integral points are the vectors $(3v,a,6v-2a,2v)$, resp. $(2a+b,2a,2b,a+b)$, with integers v,a, resp. a,b. Hence the circles $\mathbf{T}_{(w)}$ with w in the Lie

algebra of $\mathbf{S}$ are the

$$\begin{aligned} &\mathbf{T}_{(3v,a,6v-2a,2v)},\ \gcd(v,a)=1 \quad \text{(cuspidal case)},\\ &\mathbf{T}_{(2a+b,2a,2b,a+b)},\ \gcd(a,b)=1 \quad \text{(tangent case)}, \end{aligned} \tag{E.78}$$

and the weight vectors in (E.78) are primitive, since a common divisor of $3v$, a, $6v-2a$, and $2v$ divides a and v, and a common divisor of $2a+b$, $2a$, $2b$, and $a+b$ divides $a=(2a+b)-(a+b)$ and b.

Possibly passing to a subsequence, by Lemma 4.6 the G_i converge in the Hausdorff distance to a closed subgroup H_∞ of $\mathbb{T}^4$ such that, for $i \gg 0$, $G_i \subset H_\infty$ and G_i intersects any connected component of H_∞. Since $\mathbf{S}$ is closed and contains all G_i, $H_\infty \subset \mathbf{S}$. Since $|G_i| \to +\infty$, $\dim H_\infty \geq 1$.

We claim that the Lie algebra of H_∞ contains no vector of $\mathbb{R}^4_{>0}$. The admitted data of **Step 1** are polynomial, G_i is a finite diagonal group acting faithfully on W_i, the $h_{i,k}$ are G_i-semi-invariant polynomials, and (W_i, D_i) is klt near 0 (Construction-Theorem 4.13(1)). Thus, if the Lie algebra of H_∞ contained a vector with all coordinates positive, then $\alpha_i \to 0$ by Lemma 4.7, which contradicts $\alpha_i \geq \alpha_1 > 0$. This proves the claim.

In both cases $(3,2,2,2)$ belongs to the Lie algebra of $\mathbf{S}$ ($v=1$, $a=2$, resp. $a=b=1$). Hence $\dim H_\infty = 2$ is impossible by the claim, and $\dim H_\infty = 1$. By Lemma 4.6, the identity component H^0_∞ is the image of $\mathbb{R}w$ for a primitive vector $w \in \mathbb{Z}^4$, and for $\mathbf{T} := \mathbf{T}_{(w)}$, the group $R_i := G_i \cap \mathbf{T}$ is cyclic of order $r_i \to +\infty$ and $[G_i : R_i] = |H_\infty / H^0_\infty| =: e_0$ for $i \gg 0$. Since $H^0_\infty \subset \mathbf{S}$, w belongs to the Lie algebra of $\mathbf{S}$, so $\mathbf{T} = \mathbf{T}_{(w)}$ is one of the circles (E.78), and by the claim neither w nor $-w$ belongs to $\mathbb{R}^4_{>0}$.

In the cuspidal case, replacing (v,a) with $(-v,-a)$, we may assume that $v \geq 0$. If $v=0$, then $a=\pm 1$, and $\mathbf{T} = \mathbf{T}_{(0,1,-2,0)}$. If $v>0$ and $0<a<3v$, then $w \in \mathbb{R}^4_{>0}$, which is excluded. If $a=3v$, then $(v,a)=(1,3)$ and $w=(3,3,0,2)$. The remaining circles have $v>0$ and either $a \leq 0$ or $a>3v$. They include $\mathbf{T}_{(3,4,-2,2)}$ ($v=1$, $a=4$), one of the circles of the lemma, and $\mathbf{T}_{(3,0,6,2)}$ ($v=1$, $a=0$), whose weights are non-negative but not all positive, so that the claim does not exclude it. It belongs to the set $\mathcal{M}$ below.

In the tangent case, $ab \leq 0$, since otherwise w or $-w$ belongs to $\mathbb{R}^4_{>0}$. If $a=0$, then $b=\pm 1$ and $\mathbf{T} = \mathbf{T}_{(1,0,2,1)}$. If $b=0$, then $\mathbf{T} = \mathbf{T}_{(2,2,0,1)}$. Otherwise, replacing (a,b) with $(-a,-b)$, we may assume that $a>0>b$. Among the remaining circles is $\mathbf{T}_{(1,2,-2,0)}$ ($a=1$, $b=-1$).

Let $\mathcal{M}$ be the set of the circles with $v>0$, $a \in (-\infty, 0] \cup (3v, +\infty)$, and $(v,a) \neq (1,4)$ in the cuspidal case, and with $a>0>b$ and $(a,b) \neq (1,-1)$ in the tangent case. It remains to exclude $\mathbf{T}_{(3,3,0,2)}$, $\mathbf{T}_{(2,2,0,1)}$, the circles in $\mathcal{M}$, and $\mathbf{T}_{(1,0,2,1)}$.

For any k, let κ_k be the $\mathbf{T}$-weight of a monomial of least total degree of $h_{i,k}$. By **Step 1**, $|\kappa_k| \leq (m-1)\max_j |w_j|$. Possibly passing to a subsequence, we may assume that each κ_k does not depend on i. Since R_i is generated by an element λ^w with λ a primitive r_i-th root of unity, all monomials of a G_i-semi-invariant polynomial have $\mathbf{T}$-weights in one class modulo r_i. By Definition-Lemma 11.5(2), for $i \gg 0$ the monomials of $h_{i,k}$ of $\mathbf{T}$-weight different from κ_k have total degree at least $(r_i - |\kappa_k|)/\max_j |w_j|$. Likewise, the monomials of x^2+g_i of $\mathbf{T}$-weight different from $2w_1$ have total degree at least $(r_i - 2|w_1|)/\max_j |w_j|$, since their $\mathbf{T}$-weights differ from $2w_1$ by non-zero multiples of r_i.

Step 3. In this step we exclude $\mathbf{T}_{(3,3,0,2)}$ and $\mathbf{T}_{(2,2,0,1)}$.

Suppose that $w=(3,3,0,2)$ (resp. $w=(2,2,0,1)$). The monomials of x^2+g_i have $\mathbf{T}$-weights congruent to 6 (resp. 4) modulo r_i. The monomials not in $(x,y,t)^2$ are z^p, xz^p, yz^p, and tz^p, with $\mathbf{T}$-weights 0, 3, 3, and 2 (resp. 0, 2, 2, and 1). For $r_i > 6$, none of them is congruent to 6 (resp. 4) modulo r_i. Hence $x^2+g_i \in (x,y,t)^2$. Thus the z-axis is contained in W_i, and x^2+g_i and all partial derivatives of x^2+g_i vanish along the z-axis. Hence W_i is singular along the z-axis, which contradicts the isolatedness of W_i at 0.

Step 4. In this step, for a circle $\mathbf{T} \in \mathcal{M}$, we fix primitive weight vectors $w_i \in \mathbb{Z}^4_{>0}$ with $w_i \equiv L_i w \pmod{r_i}$ for integers L_i prime to r_i.

In the cuspidal case, for real numbers σ_1, σ_2 let

$$w(\sigma_1, \sigma_2) := (3\sigma_1, \sigma_2, 6\sigma_1 - 2\sigma_2, 2\sigma_1),$$

so that $w = w(v, a)$. Let $\epsilon_N := -1$ if $a \leq 0$ and $\epsilon_N := 1$ if $a > 3v$, and fix integers n_0, n_1 with $an_0 - vn_1 = \epsilon_N$. For $\theta \in \mathbb{R}$ let $\sigma_1(\theta) := n_0 + v\theta$ and $\sigma_2(\theta) := n_1 + a\theta$, so that $a\sigma_1(\theta) - v\sigma_2(\theta) = \epsilon_N$. Let

$$\Omega := \{(\sigma_1, \sigma_2) \mid \sigma_1 > 0,\ 0 < \sigma_2 < 3\sigma_1,\ 5\sigma_1 - \sigma_2 < 1\},$$

the open triangle with vertices $(0, 0)$, $(\frac{1}{5}, 0)$, and $(\frac{1}{2}, \frac{3}{2})$. The linear function $a\sigma_1 - v\sigma_2$ takes the values 0, $\frac{a}{5}$, and $\frac{a-3v}{2}$ at the vertices of Ω, so on Ω it takes any value strictly between the least and the largest of them. If $a \leq 0$, then $\frac{a-3v}{2} < -1 < 0$. If $a > 3v$ and $(v, a) \neq (1, 5)$, then $a \geq 6$, since $\gcd(v, a) = 1$ and $(v, a) \neq (1, 4)$, so $\frac{a}{5} > 1$. In both cases the set I of θ with $(\sigma_1(\theta), \sigma_2(\theta)) \in \Omega$ is a non-empty open interval. For instance, if $a = 0$, then $v = 1$, and one may take $n_0 = 0$, $n_1 = 1$, and $I = (\frac{1}{3}, \frac{2}{5})$. If $(v, a) = (1, 5)$, let $n_0 := 0$, $n_1 := -1$, and $I := (\frac{1}{5}, \frac{1}{2})$. Then $0 < \sigma_2(\theta) < 3\sigma_1(\theta)$ and $5\sigma_1(\theta) - \sigma_2(\theta) = 1$ for $\theta \in I$.

In the tangent case, for real numbers U, V let $w(U, V) := (2U + V, 2U, 2V, U + V)$, so that $w = w(a, b)$. Fix integers u_0, v_0 with $-bu_0 + av_0 = 1$, and for $\theta \in \mathbb{R}$ let $U(\theta) := u_0 + a\theta$ and $V(\theta) := v_0 + b\theta$, so that $-bU(\theta) + aV(\theta) = 1$. The linear function $-bU + aV$ takes the values 0, $-b$, and $\frac{a}{2}$ at the vertices $(0, 0)$, $(1, 0)$, and $(0, \frac{1}{2})$ of the open triangle $\{U > 0,\ V > 0,\ U + 2V < 1\}$. If $-b > 1$ or $a > 2$, then the set I of θ with $(U(\theta), V(\theta))$ in the triangle is a non-empty open interval. Otherwise $(a, b) = (2, -1)$, since $(a, b) \neq (1, -1)$. In this case $-bU + aV = U + 2V$, and we let I be the set of θ with $U(\theta) > 0$ and $V(\theta) > 0$. Then $U(\theta) + 2V(\theta) = 1$ for $\theta \in I$.

Replacing (n_0, n_1) with $(n_0 - Mv, n_1 - Ma)$ (resp. (u_0, v_0) with $(u_0 - Ma, v_0 - Mb)$) and I with $I + M$ for a large integer M, we may assume that $I \subset (0, +\infty)$. Fix $\theta_* \in I$ and a compact interval $I' \subset I$ whose interior contains θ_*. Let

$$\omega := \sum_k c_k\kappa_k - (5v - a) \quad \text{(cuspidal case)}, \qquad \omega := \sum_k c_k\kappa_k - (a + 2b) \quad \text{(tangent case)}.$$

The number ω depends only on $\mathbf{T}$ and the κ_k, which were fixed in **Step 2**.

We claim that, after passing to a subsequence, there exist positive integers L_i prime to r_i such that $\theta_i := L_i/r_i \in I'$, and the θ_i strictly increase to θ_* if $\omega \geq 0$ and strictly decrease to θ_* if $\omega < 0$. Let $\varpi(r)$ be the number of distinct prime factors of a positive integer r. By inclusion-exclusion over the divisors of $\prod_{p|r} p$, an interval of length l contains at least

$$l \prod_{p|r} \left(1 - \frac{1}{p}\right) - 2^{\varpi(r)} \geq l2^{-\varpi(r)} - 2^{\varpi(r)}$$

integers prime to r, which is positive if $l > 4^{\varpi(r)}$. Since $r \geq \varpi(r)!$, $4^{\varpi(r)}/r \to 0$ as $r \to +\infty$. Hence for any open interval $J \subset I'$, for $i \gg 0$ the interval r_iJ contains an integer prime to r_i. Choosing such integers for a sequence of pairwise disjoint open intervals $J_1, J_2, \cdots \subset I'$ converging to θ_* from the left (resp. right), we obtain the claim.

In the cuspidal case, let

$$\sigma_{1,i} := r_in_0 + vL_i, \qquad \sigma_{2,i} := r_in_1 + aL_i, \qquad w_i := w(\sigma_{1,i}, \sigma_{2,i}), \qquad A_i := 5\sigma_{1,i} - \sigma_{2,i},$$

and in the tangent case let

$$U_i := r_iu_0 + aL_i, \qquad V_i := r_iv_0 + bL_i, \qquad w_i := w(U_i, V_i), \qquad A_i := U_i + 2V_i.$$

Then $a\sigma_{1,i} - v\sigma_{2,i} = \epsilon_N r_i$ (resp. $-bU_i + aV_i = r_i$). The integers $\sigma_{1,i}, \sigma_{2,i}$ (resp. U_i, V_i) are coprime: a common prime factor p divides r_i, and then $\sigma_{1,i} \equiv vL_i$ and $\sigma_{2,i} \equiv aL_i$ (resp. $U_i \equiv aL_i$ and $V_i \equiv bL_i$) modulo p, which contradicts $\gcd(L_i, r_i) = 1$ and $\gcd(v, a) = 1$ (resp. $\gcd(a, b) = 1$). The argument of **Step 2** shows that w_i is primitive. Since $\theta_i \in I'$, $w_i \in \mathbb{Z}^4_{>0}$, and there exists $C_1 > 1$ such that $r_i/C_1 \leq w_{i,j} \leq C_1r_i$ for any i and j. The integer A_i satisfies

$$0 < A_i \leq r_i,$$

with $A_i = r_i$ exactly when $(v,a) = (1,5)$ (resp. $(a,b) = (2,-1)$). Moreover $w_i = r_i w^{(0)} + L_i w$ for $w^{(0)} := w(n_0, n_1)$ (resp. $w^{(0)} := w(u_0, v_0)$), so that

$$\langle w_i, e\rangle = r_i \langle w^{(0)}, e\rangle + L_i \langle w, e\rangle \quad \text{for any } e \in \mathbb{Z}^4, \tag{E.79}$$

where $\langle w, e\rangle$ is the $\mathbf{T}$-weight of the monomial $x^{e_1} y^{e_2} z^{e_3} t^{e_4}$.

Step 5. In this step we compute the lowest w_i-weighted parts of the equation and of the marks, and the coefficient of the quotient chart of Definition-Lemma 11.4(2).

Let $e \in \mathbb{Z}^4_{\geq 0}$ be the exponent vector of a monomial of $x^2 + g_i$, and let $|e| := e_1 + e_2 + e_3 + e_4$. If $\langle w, e\rangle \neq 2w_1$, then $|e| \geq (r_i - 2|w_1|)/\max_j |w_j|$ by **Step 2**, so

$$\langle w_i, e\rangle \geq \frac{r_i(r_i - 2|w_1|)}{C_1 \max_j |w_j|},$$

which is larger than $2w_{i,1} \leq 2C_1 r_i$ for $i \gg 0$. If $\langle w, e\rangle = 2w_1$, then by (E.79),

$$\langle w_i, e\rangle - 2w_{i,1} = r_i\big(\langle w^{(0)}, e\rangle - 2w_1^{(0)}\big).$$

In the cuspidal case, $e = (2,0,0,0)$, or $e_1 = 0$ and $e_2 + e_3 + e_4 \geq 3$. If $e_1 = 0$ and $e_2 + e_3 + e_4 \geq 3$, then the condition $\langle w, e\rangle = 6v$ reads

$$a(e_2 - 2e_3) = 2v(3 - 3e_3 - e_4).$$

Since $w^{(0)} = (3n_0, n_1, 6n_0 - 2n_1, 2n_0)$ and $e_1 = 0$,

$$\begin{aligned}\langle w^{(0)}, e\rangle - 6n_0 &= n_1(e_2 - 2e_3) - 2n_0(3 - 3e_3 - e_4) = n_1(e_2 - 2e_3) - \frac{an_0}{v}(e_2 - 2e_3)\\ &= \frac{(vn_1 - an_0)(e_2 - 2e_3)}{v} = -\frac{\epsilon_N}{v}(e_2 - 2e_3),\end{aligned}$$

where the second equality uses $a(e_2 - 2e_3) = 2v(3 - 3e_3 - e_4)$ and the last one uses $an_0 - vn_1 = \epsilon_N$. If $\epsilon_N = -1$, then $a \leq 0$, and we show that $e_2 - 2e_3 \geq 0$. If $e_3 = 0$, the inequality is clear. If $e_3 \geq 1$ and $a < 0$, then $3 - 3e_3 - e_4 \leq 0$, so $a(e_2 - 2e_3) \leq 0$. If $e_3 \geq 1$ and $a = 0$, then $3e_3 + e_4 = 3$, so $e_3 = 1$, $e_4 = 0$, and $e_2 \geq 2$. If $\epsilon_N = 1$, then $a > 3v$, and we show that $2e_3 - e_2 \geq 0$. If $e_2 = e_3 = 0$, the inequality is clear. If $e_3 \geq 1$, then $a(e_2 - 2e_3) \leq 0$. If $e_3 = 0$ and $e_2 \geq 1$, then $3ve_2 < ae_2 = 2v(3 - e_4)$, so $3e_2 + 2e_4 < 6$, which is impossible since $e_2 + e_3 + e_4 \geq 3$. In both cases

$$\langle w_i, e\rangle - 2w_{i,1} = r_i\big(\langle w^{(0)}, e\rangle - 6n_0\big) = -\frac{\epsilon_N r_i}{v}(e_2 - 2e_3) \geq 0,$$

with equality only for $y^2 z$ and t^3 among the monomials with $e_1 = 0$.

In the tangent case, the monomials of $x^2 + g_i$ are x^2, $y^2 z$, yt^2, and the monomials $z^{e_3} t^{e_4}$ of B_i, with $e_3 + e_4 \geq 4$. If $z^{e_3} t^{e_4}$ has $\mathbf{T}$-weight $2w_1 = 4a + 2b$, then $a(e_4 - 4) = -b(2e_3 + e_4 - 2)$, so $e_4 - 4 = -b\ell$ and $2e_3 + e_4 - 2 = a\ell$ for an integer ℓ, and the difference of the w_i-weights of $z^{e_3} t^{e_4}$ and x^2 is

$$U_i(e_4 - 4) + V_i(2e_3 + e_4 - 2) = \ell(-bU_i + aV_i) = \ell r_i.$$

If $\ell < 0$, then $e_4 \leq 3$ and $2e_3 + e_4 < 2$, which contradicts $e_3 + e_4 \geq 4$. If $\ell = 0$, then $e_4 = 4$ and $2e_3 = -2$, which is impossible. Hence $\ell \geq 1$.

Thus, for $i \gg 0$, the lowest w_i-weighted part of $x^2 + g_i$ is $x^2 + y^2 z + t^3$ (resp. $x^2 + y^2 z + yt^2$), which is reduced and irreducible and not divisible by any coordinate. Hence, by Lemma 4.12, the prime divisor E_{w_i} over W_i given by w_i satisfies $a(E_{w_i}, W_i, 0) = \sum_j w_{i,j} - 2w_{i,1}$, which is $5\sigma_{1,i} - \sigma_{2,i} = A_i$ (resp. $U_i + 2V_i = A_i$).

Fix k. A monomial of least total degree of $h_{i,k}$ has $\mathbf{T}$-weight κ_k and w_i-weight at most $(m-1)C_1 r_i$. If e is the exponent vector of a monomial of $h_{i,k}$ with $\langle w, e\rangle \neq \kappa_k$, then $|e| \geq (r_i - |\kappa_k|)/\max_j |w_j|$ by **Step 2**, so $\langle w_i, e\rangle > (m-1)C_1 r_i$ for $i \gg 0$. Hence the least w_i-weight of a monomial of $h_{i,k}$ is attained only at monomials of $\mathbf{T}$-weight κ_k, and by (E.79) it is $r_i j_{i,k} + L_i \kappa_k$, where $j_{i,k}$ is the least value of $\langle w^{(0)}, e\rangle$ over the monomials $x^{e_1} y^{e_2} z^{e_3} t^{e_4}$ of $h_{i,k}$ of $\mathbf{T}$-weight κ_k. The integer $j_{i,k}$ is at most $(m-1)\max_j |w_j^{(0)}|$, and it is at least $-\theta_i|\kappa_k| \geq -|\kappa_k| \max I'$ since

w_i-weights are non-negative. Possibly passing to a subsequence, which keeps the monotonicity of the θ_i, we may assume that $j_{i,k} =: j_k$ does not depend on i. The lowest w_i-weighted part of $h_{i,k}$ is non-zero of x-degree at most one, so it is not divisible by the lowest w_i-weighted part of $x^2 + g_i$, which is monic of degree two in x. Hence, by Lemma 4.12,

$$s_{i,k} := \operatorname{mult}_{E_{w_i}} h_{i,k} = r_i j_k + L_i \kappa_k.$$

We use the root charts of w_i (Definition-Lemma 11.4(2)) with coordinates ρ, X, Y, Z, T (one of X, Y, Z, T omitted), $x = \rho^{w_{i,1}} X$, $y = \rho^{w_{i,2}} Y$, $z = \rho^{w_{i,3}} Z$, and $t = \rho^{w_{i,4}} T$ (the omitted coordinate set equal to 1). We work on any one of the root charts of w_i, and omit the chart from the notation of W_i', R_i', K_i, and F_i. The formula for b_i below does not depend on the chart. Since $w_i \equiv L_i w \pmod{r_i}$ and $\gcd(L_i, r_i) = 1$, $R_i = \{\zeta^{w_i} \mid \zeta^{r_i} = 1\}$, and the lift of ζ^{w_i} that maps ρ to $\zeta\rho$ fixes X, Y, Z, T. The lifts that map ρ to $\zeta\rho$ and fix X, Y, Z, T constitute a subgroup R_i' of order r_i of the group K_i of the chart, and R_i' acts on ρ alone. Let $q := \rho^{r_i}$. A monomial of $h_{i,k}$ with exponent vector e has $\mathbf{T}$-weight $\kappa_k + \ell r_i$ for an integer ℓ, so by (E.79),

$$\langle w_i, e\rangle - s_{i,k} = r_i(\langle w^{(0)}, e\rangle - j_k + \ell L_i) \in r_i \mathbb{Z}_{\geq 0}.$$

Thus $h_{i,k} = \rho^{s_{i,k}} H_{i,k}$ with $H_{i,k}$ a polynomial in q and the coordinates X, Y, Z, T of the chart. The monomials of $x^2 + g_i$ have $\mathbf{T}$-weights in $2w_1 + r_i\mathbb{Z}$ (**Step 2**), and their w_i-weights are at least $2w_{i,1}$, since the lowest w_i-weighted part of $x^2 + g_i$ is $x^2 + y^2 z + t^3$ (resp. $x^2 + y^2 z + y t^2$). Hence, by (E.79), their w_i-weights belong to $2w_{i,1} + r_i\mathbb{Z}_{\geq 0}$, and the equation $F_i := \rho^{-2w_{i,1}}(x^2 + g_i)$ of the strict transform W_i' of W_i is a polynomial in q and X, Y, Z, T. By Definition-Lemma 11.4(2), the boundary of the quotient chart W_i'/R_i' is

$$\Delta_i' = \sum_k c_k \operatorname{div}(H_{i,k}) + b_i \operatorname{div}(q), \qquad b_i = 1 - \frac{A_i - \sum_k c_k s_{i,k}}{r_i}.$$

Since $A_i/r_i = 5\sigma_1(\theta_i) - \sigma_2(\theta_i)$ (resp. $U(\theta_i) + 2V(\theta_i)$),

$$b_i = 1 - (5n_0 - n_1) + \sum_k c_k j_k + \Big[\sum_k c_k \kappa_k - (5v - a)\Big]\theta_i \quad \text{(cuspidal case)}, \tag{E.80}$$

$$b_i = 1 - (u_0 + 2v_0) + \sum_k c_k j_k + \Big[\sum_k c_k \kappa_k - (a + 2b)\Big]\theta_i \quad \text{(tangent case)}. \tag{E.81}$$

We have $b_i \geq 1 - A_i/r_i \geq 0$, since $A_i \leq r_i$ and $s_{i,k} \geq 0$. Since E_{w_i} has center 0, (E.77) and $\sum_k \mu_k = 1$ imply

$$A_i - \sum_k c_k s_{i,k} = \sum_k \mu_k \Big(A_i - \frac{1}{m} s_{i,k}\Big) > 0,$$

so $b_i < 1$. The coefficient of θ_i in (E.80) and (E.81) is ω, and θ_i is increasing if $\omega \geq 0$ and decreasing if $\omega < 0$. Hence the $b_i \in [0, 1)$ are non-decreasing.

Step 6. In this step we exclude the circles in $\mathcal{M}$.

Let $Y_i' \to W_i$ be the normalized strict transform of W_i in the weighted blow-up with weights w_i, and let $Y_i := Y_i'/G_i$, with the boundary Δ_{Y_i} of Proposition 11.9(0). Since R_i' acts trivially on $(\rho = 0)$, $R_i \subset I_{G_i}(E_{w_i})$, and $e_i' := [I_{G_i}(E_{w_i}) : R_i] \leq e_0$. By Proposition 11.9(0), the coefficient of the image of E_{w_i} in Δ_{Y_i} is

$$1 - \frac{A_i - \sum_k c_k s_{i,k}}{e_i' r_i} = 1 - \frac{1 - b_i}{e_i'}.$$

Possibly passing to a subsequence, we may assume that e_i' does not depend on i. Then the coefficients of the images of the E_{w_i} in Δ_{Y_i} belong to $[0, 1)$ and are non-decreasing by **Step 5**.

The surface $(\rho = 0)$ on the root chart at x, y, z, and t is defined by

$$1 + Y^2 Z + T^3, \quad X^2 + Z + T^3, \quad X^2 + Y^2 + T^3, \quad X^2 + Y^2 Z + 1 \quad \text{(cuspidal case)},$$
$$1 + Y^2 Z + Y T^2, \quad X^2 + Z + T^2, \quad X^2 + Y^2 + Y T^2, \quad X^2 + Y^2 Z + Y \quad \text{(tangent case)},$$

respectively. A direct calculation shows that each of them is smooth, except the surface on the root chart at z, which is singular only at $X=Y=T=0$. The surface $(X^2+Y^2+T^3=0)$ (resp. $(X^2+Y^2+YT^2=0)$) has an A_2 singularity (resp. an A_3 singularity, since $X^2+Y^2+YT^2 = X^2+(Y+\frac{1}{2}T^2)^2-\frac{1}{4}T^4$) at $X=Y=T=0$. Since $(\rho=0)$ is a Cartier divisor on W_i', W_i' is smooth at any point of $(\rho=0)$ other than the vertex $\rho=X=Y=T=0$ of the root chart at z, and ρ is part of a system of local parameters of W_i' there. Since R_i' acts on ρ alone, W_i'/R_i' is smooth at the images of the points of $(\rho=0)$ other than the vertex, with the parameter $q=\rho^{r_i}$. The quotient chart of Proposition 11.9(1) is W_i'/R_i'', where $R_i''\supset R_i'$ is the subgroup of the elements of K_i acting on ρ alone, that is, the quotient of W_i'/R_i' by the cyclic group R_i''/R_i', which fixes X,Y,Z,T and acts on q alone. At the images of the points of $(\rho=0)$ other than the vertex, q and two of the coordinates X,Y,Z,T of the chart, minus their values, form a system of local parameters of W_i'/R_i', so W_i'/R_i'' is smooth at the images of the points of $(\rho=0)$ other than the vertex as well. By Proposition 11.9(1), the values α_i whose computing lifts have a divisor or a curve as center on Y_i, or a closed point as center with a smooth preimage on the quotient chart W_i'/R_i'' of one of the root charts, satisfy the ACC. Since the α_i are strictly increasing, after passing to a subsequence, the center on Y_i of a computing lift of α_i is the image of the vertex 0 of the root chart at z for any i.

From now on, W_i', R_i', K_i, and F_i denote the objects on the root chart at z. Let $\overline{W}_i := W_i'/R_i' \subset \mathbb{A}^4_{X,Y,T,q}$ be the quotient chart at z, with the group $\overline{K}_i := K_i/R_i'$ and the boundary Δ_i' of **Step 5**. It is defined by F_i. An element of K_i acting trivially on $\overline{W}_i$ acts trivially on X,Y,T,q, since $\overline{W}_i$ is contained in no coordinate hyperplane, so it acts on ρ by an r_i-th root of unity and fixes X,Y,T. Such elements form a group of order r_i containing R_i', so they form R_i'. Hence $\overline{K}_i$ acts faithfully on $\overline{W}_i$. The vertex 0 is fixed by $\overline{K}_i$. Since the vertex of the root chart maps to $0\in W_i$, the vertex $0\in\overline{W}_i$ maps to the image of 0 in W_i/G_i, whose preimage in W_i is $\{0\}$. Hence, for any prime divisor P' over $\overline{W}_i$ with center 0 and any prime divisor P over W_i such that the prime divisors over W_i/G_i under P' and under P coincide, P has center 0, and by Definition-Lemma 11.4(2)(c),

$$\frac{a(P',\overline{W}_i,\Delta_i')}{|I_{\overline{K}_i}(P')|}=a_D(P)\geq\alpha_i.$$

For any computing lift P of α_i, the prime divisor over W_i/G_i under P is the prime divisor under such a P' [Liu26, Lemma 2.6(3)], since the preimage in $\overline{W}_i$ of the image of the vertex is $\{0\}$. Therefore

$$\alpha_i=\mathrm{mld}((\overline{W}_i\ni 0,\Delta_i')/\overline{K}_i).$$

The pair $(\overline{W}_i,\Delta_i')$ is klt near 0: $\Delta_i'\geq 0$, and $K_{\overline{W}_i}+\Delta_i'$ is the pullback of $K_{W_i/G_i}+D_{W_i/G_i}$ under a finite morphism onto an open subset of Y_i followed by $Y_i\to W_i/G_i$, so any prime divisor over $\overline{W}_i$ whose center intersects a small neighborhood of 0 has positive log discrepancy by [Liu26, Lemma 2.6(2)] and Construction-Theorem 4.13(1).

Note that $w_{i,3}=2d_i$ for $d_i:=3\sigma_{1,i}-\sigma_{2,i}$ (resp. $d_i:=V_i$). The group $\mu_{w_{i,3}}\subset K_i$ acts by $\rho\to\zeta\rho$, $X\to\zeta^{-w_{i,1}}X$, $Y\to\zeta^{-w_{i,2}}Y$, and $T\to\zeta^{-w_{i,4}}T$. The integers $w_{i,1}$, $w_{i,2}$, and $w_{i,4}$ are coprime, since they are $3\sigma_{1,i},\sigma_{2,i},2\sigma_{1,i}$ (resp. $2U_i+V_i,2U_i,U_i+V_i$). Hence $\mu_{w_{i,3}}\cap R_i'=\{1\}$, and $\mu_{w_{i,3}}$ injects into $\overline{K}_i$. The subgroup of $\mu_{w_{i,3}}$ of order d_i acts on (X,Y,T,q) with weights

$$(-3\sigma_{1,i},-\sigma_{2,i},-2\sigma_{1,i},r_i)\pmod{d_i}\qquad(\text{resp. }(-2U_i-V_i,-2U_i,-U_i-V_i,r_i)\pmod{d_i}).$$

In the cuspidal case, $\sigma_{2,i}\equiv 3\sigma_{1,i}$ and

$$r_i=\epsilon_N(a\sigma_{1,i}-v\sigma_{2,i})\equiv\epsilon_N(a-3v)\sigma_{1,i}\pmod{d_i},$$

and $\gcd(\sigma_{1,i},d_i)=\gcd(\sigma_{1,i},\sigma_{2,i})=1$. In the tangent case, $r_i=-bU_i+aV_i\equiv -bU_i$ modulo d_i, and $\gcd(U_i,d_i)=1$. Changing the generator, we see that the subgroup of order d_i is $\mu_{d_i}(-3,-3,-2,v')$ with $v':=\epsilon_N(a-3v)$ in the cuspidal case, and $\mu_{d_i}(-2,-2,-1,v')$ with $v':=-b>0$ in the tangent case. In the cuspidal case, $v'=3v-a>0$ if $a\leq 0$ and $v'=a-3v>0$ if $a>3v$. Let

$\mathbf{T}'' := \mathbf{T}_{(-3,-3,-2,v')}$ (resp. $\mathbf{T}'' := \mathbf{T}_{(-2,-2,-1,v')}$). Then $|\overline{K}_i \cap \mathbf{T}''| \geq d_i \to +\infty$. The group K_i is an extension of G_i by $\mu_{w_{i,3}}$, so $|\overline{K}_i| = w_{i,3}[G_i : R_i]$ and

$$[\overline{K}_i : \overline{K}_i \cap \mathbf{T}''] \leq \frac{w_{i,3}e_0}{d_i} = 2e_0.$$

All monomials of F_i have the weight of X^2, namely -6 (resp. -4), modulo d_i. The monomials q, q^2, qX, qY, and qT have weights $v', 2v', v'-3, v'-3, v'-2$ (resp. $v', 2v', v'-2, v'-2, v'-1$), none of which is congruent to -6 (resp. -4) modulo d_i once $d_i > 2v'+6$ (resp. $d_i > 2v'+4$). Since $F_i(X, Y, T, 0) = X^2+Y^2+T^3$ (resp. $X^2+Y^2+YT^2$), the quadratic part of F_i is X^2+Y^2 for $i \gg 0$. Away from $(q = 0)$, $\overline{W}_i$ is the quotient of $W_i'\backslash(\rho = 0)$ by the free action of R_i', and $W_i'\backslash(\rho = 0)$ is étale over $W_i \cap (z \neq 0) \subset W_i\backslash\{0\}$, so $\overline{W}_i$ is smooth there. On $(q = 0)$, $\overline{W}_i$ is smooth away from 0. Hence 0 is an isolated singular point of quadratic rank two, and the hyperplane section $(q = 0)$ through it is a Du Val singularity. Thus $(\overline{W}_i \ni 0)$ is an isolated cDV germ (a general hyperplane section through 0 is then also Du Val, being an isolated hypersurface singularity of quadratic rank two, hence of type A, cf. [KM98, Definition 5.32]), hence an isolated Gorenstein terminal germ [KM98, Corollary 5.38]. By Proposition 3.15(4), applied with $(u_1, u_2, z, t) = (X, Y, T, q)$, there exist $\overline{K}_i$-semi-invariant X', Y' with the characters of X, Y and a $\overline{K}_i$-equivariant isomorphism, under which T and q correspond to T and q,

$$\widehat{\mathcal{O}}_{\overline{W}_i,0} \cong \mathbb{C}[[X', Y', T, q]]/(Q_2(X', Y') + \hat{g}_i(T, q)),$$

where $Q_2 \in \{xy, x^2 + y^2\}$ is the quadratic form of Proposition 3.15, so that $Q_2(X', Y')$ is a non-degenerate quadratic form in X', Y'. Since the quotient by q is an A_2 (resp. A_3) singularity, $\operatorname{mult}_0 \hat{g}_i(T, 0) = 3$ (resp. 4).

Let $\Gamma := \{c_1, \dots, c_s\} \cup \{b_i \mid i \geq 1\}$, a DCC subset of $[0, 1]$ since the $b_i \in [0, 1)$ are non-decreasing. The boundary Δ_i' of **Step 5** is a labelled boundary on $(\overline{W}_i \ni 0)$ with labels in Γ. Labels 0 are allowed, so $b_i \geq 0$ suffices. We apply the last two sentences of Theorem 8.6 with $n = 3$, $c = 1$, v' in place of v, the bound $2e_0$ in place of e_0, and Γ in the cuspidal case, and with $n = 4$, $c = 1$, v' in place of v, the bound $2e_0$, and Γ in the tangent case. The integers $A = nc/\gcd(n, 2)$ and $B = 2c/\gcd(n, 2)$ of Theorem 8.6 are $A = 3$, $B = 2$ for $n = 3$ and $A = 2$, $B = 1$ for $n = 4$, so in both cases $\mathbf{T}''$ acts by λ^{-A} on X' and Y', by λ^{-B} on T, and by $\lambda^{v'}$ on q, $\overline{K}_i$ is diagonal, and $\operatorname{mult}_0 \hat{g}_i(T, 0) = n$. Since $|\overline{K}_i \cap \mathbf{T}''| \to +\infty$ and $[\overline{K}_i : \overline{K}_i \cap \mathbf{T}''] \leq 2e_0$, Theorem 8.6 shows that the values $\operatorname{mld}((\overline{W}_i \ni 0, \Delta_i')/\overline{K}_i) = \alpha_i$ do not form a strictly increasing sequence, a contradiction. Hence $\mathbf{T} \notin \mathcal{M}$.

Step 7. In this step, for $\mathbf{T} = \mathbf{T}_{(1,0,2,1)}$ in the tangent case, we construct a crepant model over the y-axis and compute the boundaries of the quotient charts.

In **Steps 7–9** several charts are used at once, so we index the objects by the chart. Let $w = (1, 0, 2, 1)$, and write $B_i = \sum_e \lambda_e z^{e_3} t^{e_4}$. The monomials of $x^2 + g_i$ have $\mathbf{T}$-weights congruent to 2 modulo r_i. A monomial $z^{e_3}t^{e_4}$ of B_i has $\mathbf{T}$-weight $2e_3 + e_4 \geq e_3 + e_4 \geq 4$, so for $r_i > 2$,

$$2e_3 + e_4 = 2 + d_e r_i \quad \text{for a positive integer } d_e.$$

All $\mathbf{T}$-weights are non-negative, and $0 \leq \kappa_k \leq 2m - 2$ by **Step 1**. For $r_i > 2m - 2$, κ_k is the least $\mathbf{T}$-weight of a monomial of $h_{i,k}$, since all $\mathbf{T}$-weights of monomials of $h_{i,k}$ are congruent to κ_k modulo r_i, and any other monomial of $h_{i,k}$ has $\mathbf{T}$-weight at least $\kappa_k + r_i$. The sum $h^0_{i,k}$ of the terms of $h_{i,k}$ of $\mathbf{T}$-weight κ_k is non-zero of x-degree at most one, so it is not divisible by $x^2 + y^2z + yt^2$.

Let $\widetilde{W}_i \to W_i$ be the normalized strict transform of W_i in the weighted blow-up of $\mathbb{A}^4$ with weights w, which is $\mathbb{A}^1_y$ times the weighted blow-up of $\mathbb{A}^3_{x,z,t}$ with weights $(1, 2, 1)$. It is G_i-equivariant, proper, and an isomorphism away from the y-axis, which is contained in W_i. The lowest w-weighted part of $x^2 + g_i$ is $x^2 + y^2z + yt^2$, which is irreducible, so the exceptional locus is a prime divisor $E_{w,i}$ with

$$a(E_{w,i}, W_i, 0) = 1 + 0 + 2 + 1 - 2 = 2 \quad \text{and} \quad \operatorname{mult}_{E_{w,i}} h_{i,k} = \kappa_k.$$

Here $a(E_{w,i}, W_i, 0) = 2$, since near any point of the y-axis other than 0, W_i is smooth with local coordinates x, y, t, as $\partial_z(x^2 + g_i) = y^2 + \partial_z B_i$ does not vanish there, and on the root chart at t below, the morphism to W_i is $(x, y, t) = (\rho X, y, \rho)$ in the coordinates x, y, t, which is a chart of the blow-up of the curve $(x = t = 0)$ [KM98, Lemma 2.29]. The root charts at x, t, and z have coordinates (ρ, y, Z, T) with $(x, z, t) = (\rho, \rho^2 Z, \rho T)$, (ρ, X, y, Z) with $(x, z, t) = (\rho X, \rho^2 Z, \rho)$, and (ρ, X, y, T) with $(x, z, t) = (\rho X, \rho^2, \rho T)$, respectively. The strict transforms of W_i are defined by

$$\begin{aligned} &1 + y^2 Z + y T^2 + \sum_e \lambda_e \rho^{d_e r_i} Z^{e_3} T^{e_4} \quad \text{(chart at } x\text{)}, \\ &X^2 + y^2 Z + y + \sum_e \lambda_e \rho^{d_e r_i} Z^{e_3} \quad \text{(chart at } t\text{)}, \\ &X^2 + y^2 + y T^2 + \sum_e \lambda_e \rho^{d_e r_i} T^{e_4} \quad \text{(chart at } z\text{)}. \end{aligned} \tag{E.82}$$

Let $K_{i,\star}$ be the group generated by the lifts of G_i to the root chart at $\star \in \{x, t, z\}$ and by $\mu_{w_\star}$, as in Definition-Lemma 11.4(2), where $w_x = w_t = 1$ and $w_z = 2$. Since $R_i = \{\zeta^w \mid \zeta^{r_i} = 1\}$, the lifts of R_i mapping ρ to $\zeta\rho$ fix the other coordinates and form a subgroup $R'_{i,\star}$ of order r_i. Let $q := \rho^{r_i}$, let

$$\overline{W}_{i,\star} := W'_{i,\star}/R'_{i,\star} \subset \mathbb{A}^4$$

be the quotient of the strict transform $W'_{i,\star}$, defined by the polynomial $F_{i,\star}$ given by the equation of the chart at $\star$ in (E.82) with ρ^{r_i} replaced by q, and let $\overline{K}_{i,\star} := K_{i,\star}/R'_{i,\star}$. Then $|\overline{K}_{i,\star}| = w_\star[G_i : R_i] \le 2e_0$. A monomial of $h_{i,k}$ of $\mathbf{T}$-weight $\kappa_k + \ell r_i$, $\ell \ge 0$, is $\rho^{\kappa_k + \ell r_i}$ times a monomial in the other coordinates. Thus $h_{i,k} = \rho^{\kappa_k} H_{i,k}$ with $H_{i,k}$ a polynomial in q and the other coordinates, and $H_{i,k}|_{q=0}$ is $h^0_{i,k}$ written in the coordinates of the chart.

Let $\Delta'_{i,\star}$ be the $\mathbb{R}$-divisor on $\overline{W}_{i,\star}$ such that $K_{\overline{W}_{i,\star}} + \Delta'_{i,\star}$ is the pullback of $K_{W_i/G_i} + D_{W_i/G_i}$. The coefficient of $E_{w,i}$ in the crepant pullback of $K_{W_i} + D_i$ to $\widetilde{W}_i$ is $1 - (2 - \sum_k c_k\kappa_k)$. The morphism from $W'_{i,\star}$ to the image of $W'_{i,\star}$ in $\widetilde{W}_i$ is the quotient by $\mu_{w_\star}$, which is étale in codimension one since $-1 \in \mu_2$ acts on X by -1, and $(\rho = 0)$ is unramified over $E_{w,i}$, since the order along $(\rho = 0)$ of each coordinate x_j of $\mathbb{A}^4$ is $w_j = \mathrm{mult}_{E_{w,i}} x_j$. The quotient by $R'_{i,\star}$ is ramified along $(\rho = 0)$ with index r_i. By the Hurwitz formula,

$$\Delta'_{i,\star} = \sum_k c_k \operatorname{div}(H_{i,k}) + b_i \operatorname{div}(q), \qquad b_i := 1 - \frac{2 - \sum_k c_k\kappa_k}{r_i}. \tag{E.83}$$

Here b_i plays the role of the b_i of **Step 5**, which was defined only for the circles in $\mathcal{M}$. Since $\kappa_k \le 2m - 2$,

$$2 - \sum_k c_k\kappa_k = \sum_k \mu_k\Big(2 - \frac{1}{m}\kappa_k\Big) \in \Big[\frac{2}{m}, 2\Big].$$

Hence $b_i \in [0, 1)$ for $r_i \ge 2$, and $b_i \to 1$. Since $\overline{W}_{i,\star}/\overline{K}_{i,\star}$ is an open subset of $\widetilde{W}_i/G_i$ and $\widetilde{W}_i/G_i \to W_i/G_i$ is birational, [Liu26, Lemma 2.6(2)] implies that

$$\frac{a(P', \overline{W}_{i,\star}, \Delta'_{i,\star})}{|I_{\overline{K}_{i,\star}}(P')|} = a_D(P) \tag{E.84}$$

for any prime divisor P' over $\overline{W}_{i,\star}$ and any prime divisor P over W_i with center 0 lying over a common prime divisor over W_i/G_i. The argument of **Step 6** shows that $(\overline{W}_{i,\star}, \Delta'_{i,\star})$ is klt over a neighborhood of the image of 0 in W_i/G_i.

The fiber of $\widetilde{W}_i$ over 0 is $E_{w,i} \cap (y = 0)$. On the root chart at x, $\rho = y = 0$ implies $1 = 0$, so the root chart at x does not intersect the fiber. On the root chart at t (resp. z), the fiber is the curve C_t (resp. C_z) defined by $q = y = X = 0$, with coordinate Z (resp. T). The fiber is the

union of the images of C_t and C_z. Let $S_\star := \overline{W}_{i,\star} \cap (q = 0)$ for $\star \in \{t, z\}$:

$$S_t = (X^2 + y^2 Z + y = 0), \qquad S_z = (X^2 + y^2 + yT^2 = 0).$$

The surfaces S_t and S_z do not depend on i. Near C_t, S_t is smooth with local coordinates X, Z, since $\partial_y(X^2 + y^2 Z + y) = 1$ at $y = 0$. The surface S_z has an A_3 singularity at 0, and away from 0 it is smooth near C_z with local coordinates X, T, since $y + T^2$ is a unit there. In both cases $C_\star = (X = 0)$ near any smooth point of $S_\star$ on $C_\star$. Since $(q = 0)$ is a Cartier divisor, $\overline{W}_{i,t}$ is smooth near C_t, and $\overline{W}_{i,z}$ is normal near C_z and smooth near $C_z \backslash \{0\}$. Any prime divisor P over W_i with center 0 has center in the fiber of the proper morphism $\widetilde{W}_i \to W_i$ over 0, so there exists a prime divisor P' over some $\overline{W}_{i,\star}$, $\star \in \{t, z\}$, lying over the prime divisor over W_i/G_i under P [Liu26, Lemma 2.6(3)], whose center is a closed point of $C_\star$ or the curve $C_\star$.

We claim that $\overline{K}_{i,\star}$ acts faithfully on $S_\star$. Let $g' \in K_{i,\star}$ act trivially on $S_\star$. Since $S_\star$ is contained in no coordinate hyperplane and g' acts diagonally, g' fixes X, y, and Z (resp. T), and maps ρ to $\zeta\rho$ for some $\zeta \in \mathbb{C}^*$. The image of g' in G_i acts on W_i by $(\zeta, 1, \zeta^2, \zeta) \in \mathbf{T}$, hence belongs to R_i. Thus $\zeta^{r_i} = 1$ and $g' \in R'_{i,\star}$, which proves the claim.

Step 8. In this step we show that the sections $(S_\star, B_{i,\star})$ are lc near $C_\star$ for $i \gg 0$, where

$$B_{i,\star} := \sum_k c_k \operatorname{div}(H_{i,k}|_{S_\star}),$$

and we bound α_i from below.

The restriction $H_{i,k}|_{S_\star}$ is non-zero, since $h^0_{i,k}$ has X-degree at most one in the coordinates of the chart and $S_\star$ is defined by an irreducible polynomial monic of degree two in X. Let η be a closed point of $C_\star$. Since $(\overline{W}_{i,\star}, \Delta'_{i,\star})$ is klt near η, [Kaw07, Theorem] applied to the normal hypersurface $\overline{W}_{i,\star} \subset \mathbb{A}^4$ shows that the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(F_{i,\star}) + \sum_k c_k \operatorname{div}(H_{i,k}) + b_i \operatorname{div}(q)\Big)$$

is lc near η. Let

$$M_{i,\star} := \operatorname{div}(F_{i,\star}) + \sum_k c_k \operatorname{div}(H_{i,k}) + \operatorname{div}(q)$$

on $\mathbb{A}^4$. Since $b_i \le 1$, $(\mathbb{A}^4, b_i M_{i,\star})$ is lc near η, that is, $\operatorname{lct}_\eta(\mathbb{A}^4; M_{i,\star}) \ge b_i$. The coefficients of $M_{i,\star}$ belong to

$$\mathcal{J} := \Big(\mathbb{Z}_{\ge 0} + \sum_k c_k \mathbb{Z}_{\ge 0}\Big) \backslash \{0\},$$

which is a DCC set, since it has finitely many elements in any bounded interval. By Theorem 2.12(1) with $n = 4$, $I = \{0\}$, and $J = \mathcal{J}$, applied to small neighborhoods of the points η, the numbers $\operatorname{lct}_\eta(\mathbb{A}^4; M_{i,\star})$ for all i, $\star$ and η satisfy the ACC. We show that $(\mathbb{A}^4, M_{i,\star})$ is lc near $C_\star$ for $\star \in \{t, z\}$ and $i \gg 0$. Otherwise, for infinitely many i there exist $\star_i \in \{t, z\}$ and a closed point $\eta_i \in C_{\star_i}$ with $b_i \le \operatorname{lct}_{\eta_i}(\mathbb{A}^4; M_{i,\star_i}) < 1$. Since $b_i \to 1$, a subsequence of the thresholds $\operatorname{lct}_{\eta_i}(\mathbb{A}^4; M_{i,\star_i})$ is strictly increasing, which contradicts the ACC. By [Kaw07, Theorem] for $\overline{W}_{i,\star} \subset \mathbb{A}^4$ and then for the normal Cartier divisor $S_\star \subset \overline{W}_{i,\star}$, the pairs

$$\Big(\overline{W}_{i,\star}, \sum_k c_k \operatorname{div}(H_{i,k}) + S_\star\Big) \quad \text{and} \quad (S_\star, B_{i,\star})$$

are lc near $C_\star$.

For a closed point $\eta \in C_\star$, let $\overline{K}_{i,\star,\eta} \subset \overline{K}_{i,\star}$ be the stabilizer of η, and let

$$\beta_{i,\star}(\eta) := \operatorname{mld}((S_\star \ni \eta, B_{i,\star})/\overline{K}_{i,\star,\eta}).$$

Let $\xi_\star$ be the generic point of $C_\star$, and let $\beta_{i,\star}(\xi_\star)$ be the minimal log discrepancy of the quotient of $(S_\star, B_{i,\star})$ by $\overline{K}_{i,\star}$ at the image of $\xi_\star$, that is, by [Liu26, Lemma 2.6(2)],

$$\beta_{i,\star}(\xi_\star) = \frac{a(C_\star, S_\star, B_{i,\star})}{|I_{\overline{K}_{i,\star}}(C_\star)|}.$$

Let β_i be the infimum of the $\beta_{i,\star}(\eta)$ over $\star \in \{t, z\}$ and over the closed points and the generic point η of $C_\star$.

We claim that $\alpha_i \geq \beta_i$. Let P be a prime divisor over W_i with center 0, and let P' be as at the end of **Step 7**, with center η. Assume first that η is a closed point. The stabilizer $\overline{K}_{i,\star,\eta}$ acts diagonally on $\mathbb{A}^4$ after replacing the coordinate Z (resp. T) with $Z - Z(\eta)$ (resp. $T - T(\eta)$), since $\overline{K}_{i,\star,\eta}$ fixes $Z(\eta)$ (resp. $T(\eta)$). In Lemma 5.5 at η, we take the group and the subgroup of the lemma both equal to $\overline{K}_{i,\star,\eta}$, $\overline{W}_{i,\star}$ in place of X, which is klt near η since $\Delta'_{i,\star} \geq 0$, the polynomial q in place of the function u of the lemma, so that $S_0 = S_\star$, $\mathfrak{a} := \prod_k (H_{i,k})^{c_k}$, and $b := b_i$. By **Step 7**, $\overline{K}_{i,\star,\eta}$ acts faithfully on $S_\star$, and $(S_\star, B_{i,\star})$ is lc near η. Since

$$I_{\overline{K}_{i,\star}}(P') \subset \overline{K}_{i,\star,\eta},$$

(E.84) and Lemma 5.5 imply

$$a_D(P) = \frac{a(P', \overline{W}_{i,\star}, \Delta'_{i,\star})}{|I_{\overline{K}_{i,\star,\eta}}(P')|} \geq \beta_{i,\star}(\eta) + \frac{(1 - b_i)\operatorname{mult}_{P'} q}{|I_{\overline{K}_{i,\star,\eta}}(P')|} \geq \beta_i.$$

If $\eta = \xi_\star$, the inequality $a_D(P) \geq \beta_i$ follows from the last sentence of Lemma 5.5, applied at the origin of the chart with the group $\overline{K}_{i,\star}$ and the $\overline{K}_{i,\star}$-invariant curve $C_\star \subset S_\star$. Since P is arbitrary, $\alpha_i \geq \beta_i$.

Step 9. In this step we show that the β_i take finitely many values and that $\alpha_i \leq \beta_i + C_0/r_i$ for a constant C_0, which excludes $\mathbf{T}_{(1,0,2,1)}$.

Let

$$\Lambda := \Big(\sum_k c_k \mathbb{Z}_{\geq 0}\Big) \cap [0, 1],$$

a finite set since $c_k > 0$. For a closed point $\eta \in C_\star$, $(S_\star \ni \eta)$ is smooth or the A_3 singularity $\mathbb{A}^2/\mu_4$, $|\overline{K}_{i,\star,\eta}| \leq 2e_0$, and near η, after removing the components of $B_{i,\star}$ not passing through η, $B_{i,\star} \in \Lambda$, since $(S_\star, B_{i,\star})$ is lc near η. Let l_0 and $\mathcal{V}$ be the integer and the finite set of Lemma E.6 with $\max(4, 2e_0)$ in place of N and with Λ. Then $\beta_{i,\star}(\eta) \in \mathcal{V}$, and there exists a prime divisor ν over $S_\star$ with center η such that $a(\nu, S_\star, 0) \leq l_0$ and

$$\frac{a(\nu, S_\star, B_{i,\star})}{|I_{\overline{K}_{i,\star,\eta}}(\nu)|} = \beta_{i,\star}(\eta).$$

For the generic point, $\beta_{i,\star}(\xi_\star) = (1 - \gamma)/e$ with $\gamma = \operatorname{mult}_{C_\star} B_{i,\star} \in \Lambda$ and $1 \leq e \leq 2e_0$. Hence all β_i belong to the finite set

$$\mathcal{V} \cup \{(1 - \gamma)/e \mid \gamma \in \Lambda,\ 1 \leq e \leq 2e_0\},$$

and β_i is attained.

Fix $i \gg 0$, and fix $\star$ and η with $\beta_{i,\star}(\eta) = \beta_i$. Let $\overline{K}_\eta := \overline{K}_{i,\star,\eta}$ if η is closed and $\overline{K}_\eta := \overline{K}_{i,\star}$ if $\eta = \xi_\star$. If η is closed, let ν be a prime divisor over $S_\star$ with center η, $a(\nu, S_\star, 0) \leq l_0$, and

$$a(\nu, S_\star, B_{i,\star}) = \beta_i |I_{\overline{K}_\eta}(\nu)|,$$

and if $\eta = \xi_\star$, let $\nu := C_\star$. Let $e_\nu := |I_{\overline{K}_\eta}(\nu)| \leq 2e_0$, let χ be the character of q under $\overline{K}_\eta$, and let q_G be the integer of Lemma 3.1 for ν and χ, that is, $q_G \in \{1, \dots, e_\nu\}$ with $\chi|_{I_{\overline{K}_\eta}(\nu)} = \lambda^{q_G}$, where λ is the character of $I_{\overline{K}_\eta}(\nu)$ on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$.

We bound the orders along ν. Since $(S_\star, B_{i,\star})$ is lc near η,

$$c_k \operatorname{mult}_\nu(H_{i,k}|_{S_\star}) \leq a(\nu, S_\star, 0) \leq l_0.$$

There exists $c_X \in (0,1]$ such that $(S_\star, c_X \operatorname{div}(X))$ is lc near $C_\star$ for $\star \in \{t,z\}$: X is a local parameter of $S_\star$ at any smooth point of $S_\star$ on $C_\star$ by **Step 7**, and the only other point is the vertex of S_z. Thus $c_X \operatorname{mult}_\nu X \le l_0$. If $\eta = \xi_\star$, then $\operatorname{mult}_\nu X = 1$ and $\operatorname{mult}_\nu(H_{i,k}|_{S_\star}) \le 1/c_k$. Let $C_2 := \max(l_0,1)\cdot(1/c_X + \max_k 1/c_k)$, so that

$$\operatorname{mult}_\nu X \le C_2 \quad \text{and} \quad \operatorname{mult}_\nu(H_{i,k}|_{S_\star}) \le C_2 \quad \text{for any } k.$$

Fix the Gauss value k_G, a positive integer with $k_G \equiv q_G \pmod{e_\nu}$ and $3C_2 < k_G \le 3C_2 + 2e_0$.

Let $V_0 := S_\star \times \mathbb{A}^1_q$, defined in $\mathbb{A}^4$ by $F_{i,\star}|_{q=0}$, with the diagonal action of $\overline{K}_\eta$ by the characters of the coordinates, and let $H^0_{i,k}$ be $H_{i,k}|_{S_\star}$ regarded as a function on V_0 independent of q. If η is closed, the hypotheses of Proposition 5.4, with the S, s, u, $\tilde{H}$, χ, b, and Φ_j of the proposition equal to $S_\star$, η, q, $\overline{K}_\eta$, χ, b_i, and $H^0_{i,k}$, hold by **Steps 7 and 8** and since $k_G > \operatorname{mult}_\nu(H_{i,k}|_{S_\star})$. By Proposition 5.4, there exists a prime divisor F_0 over V_0 with center $(\eta,0)$ such that $I_{\overline{K}_\eta}(F_0) = I_{\overline{K}_\eta}(\nu)$, $\operatorname{mult}_{F_0} H^0_{i,k} = \operatorname{mult}_\nu(H_{i,k}|_{S_\star})$, $\operatorname{mult}_{F_0} q = k_G$, $\operatorname{mult}_{F_0}$ restricts to mult_ν on functions on $S_\star$, and

$$\frac{a(F_0, V_0, b_i \operatorname{div}(q) + \sum_k c_k \operatorname{div}(H^0_{i,k}))}{e_\nu} = \beta_i + \frac{(1-b_i)k_G}{e_\nu}. \tag{E.85}$$

If $\eta = \xi_\star$, let F_0 be the prime divisor over V_0 with center $C_\star \times \{0\}$ given by the monomial valuation with values 1 on X and k_G on q at the generic point of $C_\star \times \{0\}$, where X and q are local parameters. Then $a(F_0, V_0, 0) = 1 + k_G$, $\operatorname{mult}_{F_0} H^0_{i,k} = \operatorname{mult}_{C_\star}(H_{i,k}|_{S_\star})$, and $\operatorname{mult}_{F_0} q = k_G$, so

$$a\Big(F_0, V_0, b_i \operatorname{div}(q) + \sum_k c_k \operatorname{div}(H^0_{i,k})\Big) = a(C_\star, S_\star, B_{i,\star}) + (1-b_i)k_G.$$

The residue field of F_0 is generated over the function field of $C_\star$ by the residue of q/X^{k_G}, on which $I_{\overline{K}_\eta}(C_\star)$ acts by $\chi\lambda^{-k_G} = 1$, since λ has order e_ν [Liu26, Lemma 2.6(1)]. Elements of $\overline{K}_\eta$ outside $I_{\overline{K}_\eta}(C_\star)$ act non-trivially on the function field of $C_\star$. Hence $I_{\overline{K}_\eta}(F_0) = I_{\overline{K}_\eta}(C_\star)$, and (E.85) holds in this case as well.

We now pass from V_0 to $\overline{W}_{i,\star}$. Write $F_{i,\star} = X^2 + \Upsilon$ with Υ a polynomial in y, q, and Z (resp. T), by (E.82), and let $\Upsilon_0 := \Upsilon|_{q=0}$, so that $V_0 = (X_0^2 + \Upsilon_0 = 0)$, where X_0 denotes the coordinate X on V_0. Both $\overline{W}_{i,\star}$ and V_0 are $\overline{K}_\eta$-equivariant double covers of $\mathbb{A}^3$ with coordinates y,q,Z (resp. y,q,T), the quotients by $X \to -X$. Let ϑ be the restriction of $\operatorname{mult}_{F_0}$ to the function field of $\mathbb{A}^3$, and let $\epsilon' := (\Upsilon - \Upsilon_0)/\Upsilon_0$. Any monomial of $\Upsilon - \Upsilon_0$ is divisible by q, and y, Z (resp. T) have non-negative values, so $\vartheta(\Upsilon - \Upsilon_0) \ge k_G$. Moreover

$$\vartheta(\Upsilon_0) = \operatorname{mult}_{F_0}(X_0^2) = 2\operatorname{mult}_\nu X \le 2C_2.$$

Hence

$$\vartheta(\epsilon') \ge k_G - 2C_2 > C_2.$$

The valuation ϑ is a discrete valuation of rank one, since it is the restriction of $\operatorname{mult}_{F_0}$ to a subfield over which the function field of V_0 is finite. Let $\widehat{\mathcal{K}}$ be the completion of the function field of $\mathbb{A}^3$ with respect to ϑ. Since $\vartheta(\epsilon') > 0$, the binomial series

$$\varphi := \sum_{p\ge 0} \binom{1/2}{p} \epsilon'^p$$

converges in $\widehat{\mathcal{K}}$, $\varphi^2 = 1 + \epsilon'$, and φ is the unique square root of $1+\epsilon'$ with residue 1. The map $X \to \varphi X_0$ is an isomorphism of $\widehat{\mathcal{K}}$-algebras

$$\widehat{\mathcal{K}}[X]/(X^2 + \Upsilon) \cong \widehat{\mathcal{K}}[X_0]/(X_0^2 + \Upsilon_0).$$

In characteristic zero, for a finite extension $\mathcal{L}$ of the function field of $\mathbb{A}^3$, the Jacobson radical of $\widehat{\mathcal{K}} \otimes \mathcal{L}$ is zero. Thus the extensions of ϑ to $\mathcal{L}$ correspond to the factors of $\widehat{\mathcal{K}} \otimes \mathcal{L}$, each factor being the completion of $\mathcal{L}$ with respect to the corresponding extension, with value group and residue

field equal to those of the extension [Bou72, Chapter VI, §8, no. 2, Proposition 2 and Corollary 2]. Taking for $\mathcal{L}$ the function fields of $\overline{W}_{i,\star}$ and of V_0, the extensions of ϑ to them correspond to the factors of $\widehat{\mathcal{K}}[X]/(X^2+\Upsilon)$ and of $\widehat{\mathcal{K}}[X_0]/(X_0^2+\Upsilon_0)$, respectively. Thus mult_{F_0} corresponds to an extension ϑ' of ϑ to the function field of $\overline{W}_{i,\star}$ with $\vartheta'(X)=\mathrm{mult}_{F_0} X_0$, whose value group and ramification index over ϑ equal those of mult_{F_0} and whose residue field is isomorphic to that of mult_{F_0}. In particular $\vartheta'=\mathrm{mult}_{P'}$ for a prime divisor P' over $\overline{W}_{i,\star}$ [KM98, Lemma 2.45], whose center is η, since $\mathrm{mult}_{P'} X=\mathrm{mult}_{F_0} X_0$ and $\mathrm{mult}_{P'} f=\vartheta(f)=\mathrm{mult}_{F_0} f$ for any rational function f on $\mathbb{A}^3$, in particular for $f=y$, $f=q$, and $f=Z-c$ (resp. $f=T-c$) with $c\in\mathbb{C}$.

We compare the data of P' and F_0. Let $\overline{K}_\vartheta\subset\overline{K}_\eta$ be the subgroup preserving ϑ. It acts on $\widehat{\mathcal{K}}$, by extending the action on the function field of $\mathbb{A}^3$ continuously. Since Υ and Υ_0 have the character of X^2, ϵ' is invariant, so φ is fixed by $\overline{K}_\vartheta$ by uniqueness. Since X_0 has the character of X, the isomorphism $X\to\varphi X_0$ is $\overline{K}_\vartheta$-equivariant. The decomposition and inertia groups of P' and of F_0 in $\overline{K}_\eta$ are contained in $\overline{K}_\vartheta$, and the correspondence between the extensions of ϑ and between their residue fields is $\overline{K}_\vartheta$-equivariant. Hence the decomposition groups of P' and F_0 coincide, and so do the inertia groups. Since inertia elements fix the center,

$$|I_{\overline{K}_{i,\star}}(P')|=|I_{\overline{K}_\eta}(P')|=e_\nu.$$

Since $\overline{W}_{i,\star}$ is normal near η, Υ has no multiple factor near the image of η. By [Liu26, Lemma 2.6(2)] for the involution $X\to -X$, with the crepant boundaries $\frac{1}{2}\operatorname{div}(\Upsilon)$ and $\frac{1}{2}\operatorname{div}(\Upsilon_0)$ on $\mathbb{A}^3$, and since $\vartheta(\Upsilon)=\vartheta(\Upsilon_0)$, which holds because $\Upsilon=\Upsilon_0(1+\epsilon')$ and $\vartheta(\epsilon')>0$, and the ramification indices over ϑ are equal, $a(P',\overline{W}_{i,\star},0)=a(F_0,V_0,0)$. Moreover $\mathrm{mult}_{P'} q=\vartheta(q)=k_G$. Under the isomorphism $X\to\varphi X_0$, $H_{i,k}$ corresponds to $H_{i,k}(\varphi X_0,\cdot,q)$, and

$$H_{i,k}(\varphi X_0,\cdot,q)-H^0_{i,k}=\big(H_{i,k}(\varphi X_0,\cdot,q)-H_{i,k}(\varphi X_0,\cdot,0)\big)+\big(H_{i,k}(\varphi X_0,\cdot,0)-H_{i,k}(X_0,\cdot,0)\big).$$

The first difference is divisible by q, so it has value at least k_G. The second is divisible by $(\varphi-1)X_0$, and $\varphi-1=\epsilon'/(\varphi+1)$ has value $\vartheta(\epsilon')>C_2$. Both values are larger than $C_2\geq\mathrm{mult}_{F_0} H^0_{i,k}$. Hence $\mathrm{mult}_{P'} H_{i,k}=\mathrm{mult}_{F_0} H^0_{i,k}$ for any k, and

$$a(P',\overline{W}_{i,\star},\Delta'_{i,\star})=a\Big(F_0,V_0,b_i\operatorname{div}(q)+\sum_k c_k\operatorname{div}(H^0_{i,k})\Big).$$

Let P be a prime divisor over W_i lying over the prime divisor over W_i/G_i under P' [Liu26, Lemma 2.6(3)]. The center of P is 0, since η maps to the image of 0 in W_i/G_i, whose preimage in W_i is $\{0\}$. By (E.84), (E.85), and (E.83),

$$\alpha_i\leq a_D(P)=\beta_i+\frac{(1-b_i)k_G}{e_\nu}=\beta_i+\frac{(2-\sum_k c_k\kappa_k)k_G}{r_i e_\nu}\leq\beta_i+\frac{C_0}{r_i},\qquad C_0:=2(3C_2+2e_0).$$

Possibly passing to a subsequence, we may assume that **Steps 8 and 9** apply to any i, and that $\beta_i=\beta$ does not depend on i. By **Step 8**, $\alpha_1\geq\beta$, and $\lim_i\alpha_i\leq\beta$ since $r_i\to+\infty$. Thus $\alpha_2>\alpha_1\geq\beta\geq\lim_i\alpha_i\geq\alpha_2$, a contradiction. Hence $\mathbf{T}\neq\mathbf{T}_{(1,0,2,1)}$.

Step 10. In this step we conclude the proof.

By **Steps 3–9**, the circle $\mathbf{T}$ of **Step 2** is $\mathbf{T}_{(0,1,-2,0)}$ or $\mathbf{T}_{(3,4,-2,2)}$ in the cuspidal case, and $\mathbf{T}_{(1,2,-2,0)}$ in the tangent case. By **Steps 1 and 2**, after passing to a subsequence and replacing the data with numerical admissions that keep the groups G_i, we may assume that $W_i=(x^2+g_i(y,z,t)=0)$, that G_i is diagonal, that the cubic part of g_i is y^2z+t^3 (resp. y^2z+yt^2), that $|G_i\cap\mathbf{T}|=r_i\to+\infty$, and that $[G_i:G_i\cap\mathbf{T}]=e_0$ for $i\gg 0$. □

E.8. **Cuspidal cubics and tangent conics: reference pairs and upper bounds.** In this subsection, we prove Lemmas E.18 and E.21. For each of the three circles, Lemma E.18 bounds α_i above by β_i+C_0/r_i, where β_i is the minimal log discrepancy of a quotient pair on $\mathbb{A}^2$ (Construction E.17). Since $c_k>0$ for any k, a prime divisor of bounded log discrepancy computing β_i bounds the order of every mark along it, so one Gauss value k_G is used for all marks.

Construction E.17. Let the data be as in the conclusion of Lemma E.16, write $W_i = (f_i = 0)$ with $f_i = x^2 + g_i(y,z,t)$, and let $r_i := |G_i \cap \mathbf{T}|$. Possibly passing to a subsequence and replacing each $(W_i \ni o_i, G_i, (h_{i,k})_k)$ with a numerical admission with group G_i (Construction-Theorem 4.13 and Proposition 11.8), we may assume that

- $f_i = x^2 + y^2 z + t^3 + B_i(z,t)$, where any monomial $z^p t^j$ of B_i satisfies $p \geq 1$ and $2p \equiv 0 \pmod{r_i}$, if $\mathbf{T} = \mathbf{T}_{(0,1,-2,0)}$;
- $f_i = x^2 + y^2 z + t^3 + B_i(y,z,t)$, where any monomial of B_i has $\mathbf{T}$-weight $6 + nr_i$ for an integer $n \neq 0$, and any monomial of B_i with $n < 0$ does not involve y, if $\mathbf{T} = \mathbf{T}_{(3,4,-2,2)}$;
- $f_i = x^2 + y^2 z + yt^2 + B_i(z,t)$, where any monomial $z^p t^j$ of B_i satisfies $2p + 2 \equiv 0 \pmod{r_i}$, if $\mathbf{T} = \mathbf{T}_{(1,2,-2,0)}$.

Let $g^0 := y^2 z + t^3$ if $\mathbf{T}$ is $\mathbf{T}_{(0,1,-2,0)}$ or $\mathbf{T}_{(3,4,-2,2)}$, and $g^0 := y^2 z + yt^2$ if $\mathbf{T} = \mathbf{T}_{(1,2,-2,0)}$, and let $f^0 := x^2 + g^0$ and $W^0 := (f^0 = 0) \subset \mathbb{A}^4$. Then W^0 is a normal hypersurface, and for i large, f^0 is the principal block of f_i along $\mathbf{T}$ (Set-up 5.1) and any monomial of $f_i - f^0$ has degree at least $(r_i - 6)/4$. Write each $h_{i,k}$ in normal form (Definition-Lemma 11.5), and pass to a subsequence on which the weight κ_k of the principal block $h^0_{i,k}$ of $h_{i,k}$ along $\mathbf{T}$ (Set-up 5.1) is independent of i for each k. For $c' \in \mathbb{R}^s_{\geq 0}$, let $\delta(c')$ be the adjusted mass of $\sum_k c'_k \operatorname{div}(h_{i,k})$ along $\mathbf{T}$ and $D^0_i(c') := \sum_k c'_k \operatorname{div}(h^0_{i,k}|_{W^0})$, and let $c := (c_1, \dots, c_s)$, $\delta := \delta(c)$, and $D^0_i := D^0_i(c)$. In each case below we define two $\mathbf{T}$-invariant functions ζ_1, ζ_2, an $\mathbb{R}$-divisor $\Delta_i(c')$ and an $\mathbb{R}$-ideal $\mathfrak{c}(c')$ on $\mathbb{A}^2_{\zeta_1,\zeta_2}$. Let H_i be the image of G_i in the automorphism group of $\mathbb{A}^2_{\zeta_1,\zeta_2}$, $\Delta_i := \Delta_i(c)$, $\mathfrak{c} := \mathfrak{c}(c)$, and

$$\beta_i := \operatorname{mld}\big((\mathbb{A}^2 \ni 0, \Delta_i, \mathfrak{c})/H_i\big).$$

(a) If $\mathbf{T} = \mathbf{T}_{(0,1,-2,0)}$, then $\delta(c') = 1 + \sum_k c'_k \kappa_k$. Let $(\zeta_1, \zeta_2) := (x,t)$ and $\Phi := x^2 + t^3$. Modulo $x^2 + y^2 z + t^3$, write $h^0_{i,k} = y^{\kappa_k} \phi_{i,k}(x,t)$ if $\kappa_k \geq 0$, and if $\kappa_k < 0$ write $h^0_{i,k} = y^{\epsilon_k} z^{l_k} \psi_{i,k}(x,t)$ with $l_k := \lceil -\kappa_k/2 \rceil$ and $\epsilon_k := \kappa_k + 2l_k$, and let $\phi_{i,k} := (-\Phi)^{l_k} \psi_{i,k}$. Let

$$\Delta_i(c') := \sum_k c'_k \operatorname{div}(\phi_{i,k}) + \frac{\max(\delta(c'), 0)}{2} \operatorname{div}(\Phi), \qquad \mathfrak{c}(c') := \mathcal{O}.$$

(b) If $\mathbf{T} = \mathbf{T}_{(3,4,-2,2)}$, then $\delta(c') = \sum_k c'_k \kappa_k - 1$. Let $(\zeta_1, \zeta_2) := (yz^2, tz)$ and $Q := \zeta_1^2 + \zeta_2^3$. Write $h^0_{i,k} = x^{\epsilon_k} z^{-\kappa'_k} A_{i,k}(yz^2, tz)$ with $\epsilon_k \in \{0,1\}$ and $\kappa'_k := (\kappa_k - 3\epsilon_k)/2$, and let

$$\Delta_i(c') := \frac{1 + \sum_k c'_k \epsilon_k}{2} \operatorname{div}(Q) + \sum_k c'_k \operatorname{div}(A_{i,k}), \qquad \mathfrak{c}(c') := (Q^2, \zeta_1^3, \zeta_2^6)^{\max(-\delta(c'),0)/12}.$$

(c) If $\mathbf{T} = \mathbf{T}_{(1,2,-2,0)}$, then $\delta(c') = 1 + \sum_k c'_k \kappa_k$. Let $(\zeta_1, \zeta_2) := (yz, t)$. Write $\kappa_k = \epsilon_k + 2l_k$ with $\epsilon_k \in \{0,1\}$, and $h^0_{i,k} = x^{\epsilon_k} y^{l_k} \psi_{i,k}(yz,t)$ if $l_k \geq 0$ and $h^0_{i,k} = x^{\epsilon_k} z^{-l_k} \psi_{i,k}(yz,t)$ if $l_k < 0$. Let $\mathfrak{c}(c') := \mathcal{O}$ and

$$\Delta_i(c') := \frac{1 + \sum_k c'_k \epsilon_k}{2} \operatorname{div}(\zeta_1 + \zeta_2^2) + \Big(\sum_k c'_k \max(-l_k, 0) + \frac{\max(\delta(c'), 0)}{2}\Big) \operatorname{div}(\zeta_1) + \sum_k c'_k \operatorname{div}(\psi_{i,k}).$$

Proof. We first justify the normal form of f_i. By [KM98, Corollary 5.38], W_i is an isolated cDV germ of quadratic rank one, so it satisfies the hypotheses of Proposition 3.17 in the eigencoordinates x, y, z, t of $\mathbb{A}^4$. Since $R_i := G_i \cap \mathbf{T}$ is cyclic of order r_i and acts on a monomial of $\mathbf{T}$-weight ω through the ω-th power of a primitive character, the $\mathbf{T}$-weights of the monomials of a G_i-semi-invariant power series are congruent modulo r_i.

Let $\mathbf{T} = \mathbf{T}_{(0,1,-2,0)}$. Proposition 3.17(1) transforms f_i into a unit times $x^2 + y^2 z + \lambda y t^\ell + b(z,t)$, where the unit is G_i-invariant and the series is G_i-semi-invariant with the character of f_i (the last sentence of the proposition). The weight of x^2 is 0 and that of yt^ℓ is 1, so $\lambda = 0$ once $r_i > 1$. The part of b independent of z is $t^3 \varrho(t)$ with a G_i-invariant unit ϱ and $\varrho(0) = 1$, because the

cubic part of g_i is y^2z+t^3. The change $t \to t\varrho(t)^{1/3}$ is G_i-equivariant and preserves $\mathbf{T}$-weights, since t has weight 0. After it, $b-t^3 \in (z)$, and a monomial z^pt^j with $p \geq 1$ has weight $-2p \equiv 0 \pmod{r_i}$. Let $\mathbf{T} = \mathbf{T}_{(3,4,-2,2)}$. Here f_i becomes a unit times $x^2+y^2z+\lambda yt^\ell+b(z,t)$ again. The monomials of weight 6 in b are the z^pt^{p+3}, so the part of b of weight 6 is $t^3F(zt)$ with $F(0)=1$, and F is G_i-invariant because t^3 and $t^3F(zt)$ have the character of f_i. The change $t \to tF(zt)^{1/3}$ is G_i-equivariant and preserves $\mathbf{T}$-weights, since zt has weight 0. After it, the monomials of $f_i - x^2-y^2z-t^3$ have weights $6+nr_i$ with $n \neq 0$, those coming from yt^ℓ have weight at least $4+2\ell > 6$, and those coming from b do not involve y. Let $\mathbf{T} = \mathbf{T}_{(1,2,-2,0)}$. Proposition 3.17(2) transforms f_i into a unit times $x^2+y^2z+yt^2+B(z,t)$, and z^pt^j has weight $-2p \equiv 2 \pmod{r_i}$.

In each case let Ξ_i be the composite G_i-equivariant automorphism of $\mathbb{C}[[x,y,z,t]]$. The unit is G_i-invariant, since both f_i and the bracket have the character of x^2. Truncating the bracket in degrees below a large integer keeps the form above. Let $\hat{h}_{i,k}$ be the expansion of $h_{i,k}$ in $\mathbb{C}[[x,y,z,t]]$. We take the integers N and L of Construction-Theorem 4.13 larger than $\max_k \operatorname{mult}_{o_i} h_{i,k}$, and we truncate $\Xi_i(\hat{h}_{i,k})$ below degree N and the bracket below degree L. Then Construction-Theorem 4.13, with $\psi := \Xi_i$, with v the inverse of the unit, with $u_k = 1$, and with the truncations of $\Xi_i(\hat{h}_{i,k})$ below degree N as the polynomials $h_{k,0}$ (notation of Construction-Theorem 4.13), together with Proposition 11.8, replaces each $(W_i \ni o_i, G_i, (h_{i,k})_k)$ by data with group G_i, number α_i, and integer r_i, which again satisfy the conclusion of Lemma E.16. Proposition 11.8 holds for the present data, which are themselves admitted. Its proof uses Set-up 11.1(4) only through the klt property of $(W_i, \frac{1}{m}\operatorname{div}(h_{i,k}))$ near o_i and the inequality $\operatorname{mld}(W_i \ni o_i, \frac{1}{m}\operatorname{div}(h_{i,k})) \geq 1$ for any k. Both hold for the data of Lemma E.16 by Construction-Theorem 4.13(1), with the trivial subgroup, for the vectors of Proposition 11.8(i), since the pairs (W, C_k) of Set-up 11.1 are terminal near o. Moreover (W_i, D_i) is klt near o_i by Construction-Theorem 4.13(1) for the vector c. Hence the new data satisfy Set-up 11.1(1)–(3) and (4'), by Proposition 11.8(2) and Construction-Theorem 4.13(1)(3), so Definition-Lemma 11.5(1)(2) holds for them.

We prove the sentence on W^0 and f^0. The partial derivatives of f^0 vanish simultaneously only on the z-axis, so W^0 is singular only along the z-axis, a curve in the threefold W^0. Hence W^0 is normal by Serre's criterion, since a hypersurface is S_2 and a singular locus of codimension two gives R_1. The monomials of $f_i - f^0 = B_i$ have $\mathbf{T}$-weights different from the weight of x^2: in case (a), z^pt^j with $p \geq 1$ has weight $-2p \neq 0$, in case (b), the weight is $6+nr_i$ with $n \neq 0$, and in case (c), z^pt^j has weight $-2p \neq 2$. Any monomial of B_i has degree at least $(r_i-6)/4$: in cases (a) and (c), $2p \equiv 0$, resp. $2p+2 \equiv 0 \pmod{r_i}$, forces $p \geq r_i/2-1$, and in case (b), the weight $6+nr_i$ has absolute value at least r_i-6, while the weights of x,y,z,t are at most 4 in absolute value. Thus for $r_i > 14$ the monomial x^2 has least degree in f_i, and f^0 is the principal block of f_i along $\mathbf{T}$.

We show that the weights κ_k are bounded independently of i, so that the subsequence on which they do not depend on i exists. For $r_i > 18$, the lowest $(3,2,2,2)$-weighted part of f_i is f^0, since any monomial of $f_i - f^0$ has degree at least $(r_i-6)/4$, and f^0 is not divisible by any coordinate and is irreducible, as g^0 has degree one in z. By Lemma 4.12, the prime divisor E_0 over W_i given by the weights $(3,2,2,2)$ satisfies $a(E_0, W_i, 0) = 9-6 = 3$, and $\operatorname{mult}_{E_0} h_{i,k}$ is the $(3,2,2,2)$-weighted order of the normal form $h_{i,k}$, since the lowest weighted part of $h_{i,k}$ has x-degree at most one, so it is not divisible by f^0. By Proposition 11.8(2), $\operatorname{mult}_{E_0} h_{i,k} < 3m$. As all the weights are at least 2, a monomial of least degree of $h_{i,k}$ has total degree less than $\frac{3m}{2}$, so its $\mathbf{T}$-weight κ_k satisfies $|\kappa_k| < 6m$, since the weights of x,y,z,t along $\mathbf{T}$ are at most 4 in absolute value.

Next we verify the formulas. The numbers $\epsilon_{\mathbf{T}} = \sum_j w_j - \kappa(f_i)$ are $-1-0=-1$ for $(0,1,-2,0)$, $7-6=1$ for $(3,4,-2,2)$, and $1-2=-1$ for $(1,2,-2,0)$, since x^2 has weight 0, 6, and 2, respectively. Thus $\delta(c') = \sum_k c'_k\kappa_k \pm 1$, with the signs in (a), (b), and (c) of Construction E.17.

In case (a), a monomial $x^ay^bz^pt^j$ of $h^0_{i,k}$ has $b-2p = \kappa_k$. If $\kappa_k \geq 0$, then modulo $x^2+y^2z+t^3$ the monomial equals

$$y^{\kappa_k}(y^2z)^px^at^j \equiv y^{\kappa_k}(-\Phi)^px^at^j.$$

If $\kappa_k < 0$, then $2p \geq -\kappa_k$, so $p \geq l_k$, $b = 2(p-l_k)+\epsilon_k$, and the monomial equals $y^{\epsilon_k} z^{l_k} (y^2 z)^{p-l_k} x^a t^j$. Thus $\phi_{i,k}$ and $\psi_{i,k}$ exist, and they are non-zero because $h^0_{i,k}$ has x-degree at most one. In case (b), a monomial $x^a y^b z^p t^j$ of $h^0_{i,k}$ with $a \in \{0,1\}$ has $3a + 4b - 2p + 2j = \kappa_k$. Thus $a \equiv \kappa_k \pmod 2$, hence $a = \epsilon_k$ for any monomial of $h^0_{i,k}$, $p - 2b - j = -\kappa'_k$, and the monomial equals $x^{\epsilon_k} z^{-\kappa'_k} (yz^2)^b (tz)^j$. In case (c), a monomial $x^a y^b z^p t^j$ of $h^0_{i,k}$ with $a \in \{0,1\}$ has $a + 2b - 2p = \kappa_k$. Thus $a = \epsilon_k$, $b - p = l_k$, and the monomial is $x^{\epsilon_k} y^{l_k} (yz)^p t^j$ or $x^{\epsilon_k} z^{-l_k} (yz)^b t^j$.

Since G_i is diagonal and ζ_1, ζ_2 are monomials, H_i is a diagonal group. The group R_i acts trivially on ζ_1, ζ_2, so $|H_i| \leq [G_i : R_i] \leq e_0$. The polynomials Φ, Q, $\zeta_1 + \zeta_2^2$, and ζ_1 are H_i-semi-invariant, because x^2, $y^2 z$, t^3, and yt^2 have one character. The polynomials $\phi_{i,k}$, $A_{i,k}$, and $\psi_{i,k}$ are H_i-semi-invariant, because each of their monomials, multiplied by a monomial in x, y, z depending only on i and k, has the character of $h_{i,k}$, and the substitution $y^2 z \to -\Phi$ preserves characters. Thus $\Delta_i(c')$ is H_i-invariant, and $\mathfrak{c}(c')$ is an $\mathbb{R}$-ideal with H_i-invariant factors (Definition 2.4), since $(Q^2, \zeta_1^3, \zeta_2^6)$ is generated by H_i-semi-invariant polynomials. □

Lemma E.18. *Notation as in Construction E.17. Then, for i large, $(\mathbb{A}^2, \Delta_i, \mathfrak{c})$ is lc near 0, the numbers β_i belong to a finite set, and there exists a real number $C_0 \geq 0$, independent of i, such that $\alpha_i \leq \beta_i + C_0/r_i$.*

Proof. The proof is long, so we divide it into several steps. Let f^0, W^0, and D^0_i be as in Construction E.17, and let i be large enough for the sentence on f^0 there. The origin 0 of $\mathbb{A}^4$ is a point of W^0. Let ω_{W^0} and ω_{W_i} be the Poincaré residues of $dx \wedge dy \wedge dz \wedge dt / f^0$ and $dx \wedge dy \wedge dz \wedge dt / f_i$, generators of the dualizing sheaves of the hypersurfaces W^0 and W_i.

Step 1. In this step we describe one chart of W^0 for each case, the radial quotient of the chart, and the reference surface on it. Let $s_{\mathbf{T}} := 1$ if $\mathbf{T} = \mathbf{T}_{(0,1,-2,0)}$ and $\delta < 0$, or if $\mathbf{T} = \mathbf{T}_{(1,2,-2,0)}$. Let $s_{\mathbf{T}} := -1$ if $\mathbf{T} = \mathbf{T}_{(0,1,-2,0)}$ and $\delta \geq 0$, or if $\mathbf{T} = \mathbf{T}_{(3,4,-2,2)}$. In each case we define a normal surface $S \ni 0$, a dominant rational map $U := S \times \mathbb{A}^1_\rho \dashrightarrow W^0$, regular functions $H_{i,k}$ on S, and a reflection σ of S as follows.

- (a_y) Case (a) with $\delta < 0$: $S := \mathbb{A}^2_{x,t}$, $(x, t, \rho) \to (x, \rho, -\Phi/\rho^2, t)$, which is birational, $H_{i,k} := \phi_{i,k}$, and no reflection. Here $\omega_{W^0} = \pm\rho^{-2} dx \wedge d\rho \wedge dt$.
- (a_z) Case (a) with $\delta \geq 0$: $S := (x^2 + Y^2 + t^3 = 0) \subset \mathbb{A}^3_{x,Y,t}$, $(x, Y, t, \rho) \to (x, Y/\rho, \rho^2, t)$, which is the quotient by $(x, Y, t, \rho) \to (x, -Y, t, -\rho)$, $H_{i,k} := h^0_{i,k}(x, Y, 1, t)$, and $\sigma(x, Y, t) = (x, -Y, t)$. Here f^0 pulls back to $x^2 + Y^2 + t^3$ and $\omega_{W^0} = \pm 2\omega_S \wedge d\rho$.
- (b) $S := (X^2 + Y^2 + T^3 = 0) \subset \mathbb{A}^3_{X,Y,T}$,
$$(X, Y, T, \rho) \to (X\rho^{-3}, Y\rho^{-4}, \rho^2, T\rho^{-2}),$$
which is the quotient by $(X, Y, T, \rho) \to (-X, Y, T, -\rho)$, $H_{i,k} := h^0_{i,k}(X, Y, 1, T)$, and $\sigma(X, Y, T) = (-X, Y, T)$. Here f^0 pulls back to $\rho^{-6}(X^2 + Y^2 + T^3)$, and
$$\omega_{W^0} = \pm 2\rho^{-2} \omega_S \wedge d\rho.$$
- (c) $S := \mathbb{A}^2_{A,t}$ and, in this case, $\Phi := A^2 + t^2$, $(A, t, \rho) \to (\rho A, \rho^2, -\Phi/\rho^2, t)$, which is the quotient by $(A, t, \rho) \to (-A, t, -\rho)$,
$$H_{i,k} := A^{\epsilon_k} (-\Phi)^{\max(-l_k, 0)} \psi_{i,k}(-\Phi, t),$$
and $\sigma(A, t) = (-A, t)$. Here $\omega_{W^0} = \pm 2\rho^{-2} dA \wedge d\rho \wedge dt$.

We compute ω_{W^0} in (a_y), (a_z), (b), and (c) as follows. In (a_y) and (c), $\omega_{W^0} = \pm dx \wedge dy \wedge dt / y^2$, and $dx \wedge dy \wedge dt$ pulls back to $dx \wedge d\rho \wedge dt$, resp. $2\rho^2 dA \wedge d\rho \wedge dt$. In ($\mathrm{a}_z$) and (b), $dx \wedge dy \wedge dz \wedge dt$ pulls back to $\pm 2dx \wedge dY \wedge d\rho \wedge dt$, resp. $\pm 2\rho^{-8} dX \wedge dY \wedge d\rho \wedge dT$. Since $h^0_{i,k}$ is a sum of monomials of $\mathbf{T}$-weight κ_k, substitution of the chart gives

$$h^0_{i,k} = \rho^{s_{\mathbf{T}} \kappa_k} H_{i,k} \quad \text{on } U. \tag{E.86}$$

For instance in (b), $x^a y^b z^p t^j = \rho^{-(3a+4b-2p+2j)} X^a Y^b T^j$, and in (c),

$$x^{\epsilon_k} y^{l_k} (yz)^p t^j = \rho^{\kappa_k} A^{\epsilon_k} (-\Phi)^p t^j, \qquad x^{\epsilon_k} z^{-l_k} (yz)^b t^j = \rho^{\kappa_k} A^{\epsilon_k} (-\Phi)^{-l_k} (-\Phi)^b t^j.$$

In (a_z), (b), and (c) the ring of σ-invariants of S is a polynomial ring in two variables whose generators are the ζ_1, ζ_2 of Construction E.17. Let $\pi : S \to \mathbb{A}^2_{\zeta_1,\zeta_2}$ be the quotient by σ (the identity in (a_y)). In (a_z), the ring of σ-invariants is $\mathbb{C}[x,t]$ and π is branched along $\Phi = x^2 + t^3 = 0$, since $Y^2 = -\Phi$. In (b), the ring of σ-invariants is $\mathbb{C}[Y,T]$, where $yz^2 = Y$ and $tz = T$ on U, and π is branched along $Q = 0$, since $X^2 = -Q$. In (c), the ring of σ-invariants is $\mathbb{C}[-\Phi, t]$ with $\Phi = A^2 + t^2$, where $yz = -\Phi$ on U, and π is branched along $\zeta_1 + \zeta_2^2 = 0$, since $A^2 = -(\zeta_1 + \zeta_2^2)$.

Let $\tilde{G}_i$ be the group of diagonal automorphisms of U lifting the elements of G_i. In (a_z), (b), and (c), it contains the deck involution and $|\tilde{G}_i| = 2|G_i|$. In (a_y), the map $U \dashrightarrow W^0$ is birational and $|\tilde{G}_i| = |G_i|$. Let $\tilde{R}_i \subset \tilde{G}_i$ be the subgroup acting trivially on S. We claim that $\tilde{R}_i$ is cyclic of order r_i and acts faithfully on ρ. Let $g = (g_x, g_y, g_z, g_t) \in G_i$ have a lift acting by $\rho \to \mu\rho$. In (a_z) and (b), $\mu^2 = g_z$, and the lift acts trivially on S if and only if $g = (1, \mu^{-1}, \mu^2, 1)$, resp. $g = (\mu^{-3}, \mu^{-4}, \mu^2, \mu^{-2})$. In (c), $\mu^2 = g_y$, and the lift acts trivially on S if and only if $g_x = \mu$ and $g_t = 1$, and then $g_z = g_x^2 g_y^{-2} = \mu^{-2}$ because x^2 and $y^2 z$ have one character. In (a_y), g acts trivially on x, t if and only if $g = (1, g_y, g_y^{-2}, 1)$, because x^2 and $y^2 z$ have one character. Thus in all cases such a g belongs to $R_i = G_i \cap \mathbf{T}$, and any element of R_i has exactly one lift acting trivially on S, which proves the claim. Let $q := \rho^{r_i}$ and $J_i := \tilde{G}_i / \tilde{R}_i$, acting on $S \times \mathbb{A}^1_q$. Then J_i acts faithfully on S, $|J_i| \le 2e_0$, $\sigma \in J_i$ is the image of the deck involution in (a_z), (b), and (c), and the image of J_i in the automorphism group of $\mathbb{A}^2_{\zeta_1,\zeta_2}$ is H_i, with kernel generated by σ (in (a_y), $J_i = H_i$).

The crepant boundary of (W^0, D_i^0) on U is determined by the forms above. By (E.86) it is

$$B_U := (1 + s_{\mathbf{T}}\delta)\operatorname{div}(\rho) + \sum_k c_k \operatorname{div}(H_{i,k}) :$$

in (a_y) and (c) the coefficient of $\operatorname{div}(\rho)$ is $2 + \sum_k c_k\kappa_k = 1 + \delta$, in ($a_z$) it is $-\sum_k c_k\kappa_k = 1 - \delta$, and in (b) it is $2 - \sum_k c_k\kappa_k = 1 - \delta$. By [Liu26, Lemma 2.6(4)], the crepant quotient of (U, B_U) by $\tilde{R}_i$ is

$$\text{(E.87)} \qquad \Big(S \times \mathbb{A}^1_q,\ b_i \operatorname{div}(q) + \sum_k c_k \operatorname{div}(H_{i,k})\Big), \qquad b_i := 1 - \frac{1 - (1 + s_{\mathbf{T}}\delta)}{r_i} = 1 + \frac{s_{\mathbf{T}}\delta}{r_i}.$$

Let $B'_S := \sum_k c_k \operatorname{div}(H_{i,k})$ on S, and let

$$B_S := \begin{cases} B'_S + \frac{\max(\delta,0)}{2}\operatorname{div}(\Phi) & \text{in } (a_y) \text{ and (c)}, \\ B'_S + \max(-\delta, 0)\operatorname{div}(Y) & \text{in } (a_z), \\ B'_S & \text{in (b)}, \end{cases}$$

In case (b), let $\mathfrak{I} := (X^4, Y^3, T^6)$ and $\mathfrak{J} := \mathfrak{I}^{\max(-\delta,0)/12}$, and in the other cases let $\mathfrak{J} := \mathcal{O}_S$. In particular, if $s_{\mathbf{T}}\delta \le 0$, then $B_S = B'_S$ and $\mathfrak{J} = \mathcal{O}_S$.

We prove that $(S, B_S, \mathfrak{J})$ is the crepant pullback of $(\mathbb{A}^2, \Delta_i, \mathfrak{c})$ under π, that is, $K_S + B_S = \pi^*(K_{\mathbb{A}^2} + \Delta_i)$ and $\mathfrak{J} = \mathfrak{c}\mathcal{O}_S$. In ($a_y$), $B_S = \Delta_i$ by definition. In (a_z), $K_S = \pi^* K_{\mathbb{A}^2} + \operatorname{div}(Y)$ by the Hurwitz formula [Kol13, 2.41], since the ramification divisor of π is the reduced curve $\operatorname{div}(Y)$, fixed pointwise by σ, and $\pi^* \operatorname{div}(\Phi) = 2\operatorname{div}(Y)$. On S we have

$$H_{i,k} = \begin{cases} Y^{\kappa_k}\phi_{i,k} & \text{if } \kappa_k \ge 0, \\ Y^{\epsilon_k}\psi_{i,k} = Y^{\epsilon_k}(-\Phi)^{-l_k}\phi_{i,k} & \text{if } \kappa_k < 0. \end{cases}$$

Since $\epsilon_k - 2l_k = \kappa_k$, in both cases

$$\operatorname{div}(H_{i,k}) = \kappa_k \operatorname{div}(Y) + \pi^* \operatorname{div}(\phi_{i,k}).$$

Thus

$$B_S = (\delta - 1 + \max(-\delta, 0)) \operatorname{div}(Y) + \pi^* \sum_k c_k \operatorname{div}(\phi_{i,k}) = \pi^* \Delta_i - \operatorname{div}(Y).$$

In (b), by the Hurwitz formula [Kol13, 2.41], $K_S = \pi^* K_{\mathbb{A}^2} + \operatorname{div}(X)$, $\pi^* \operatorname{div}(Q) = 2 \operatorname{div}(X)$, and $H_{i,k} = X^{\epsilon_k} A_{i,k}(Y, T)$, so

$$B_S = \sum_k c_k \epsilon_k \operatorname{div}(X) + \pi^* \sum_k c_k \operatorname{div}(A_{i,k}) = \pi^* \Delta_i - \operatorname{div}(X),$$

and

$$\mathfrak{c} \mathcal{O}_S = (Q^2, Y^3, T^6)^{\max(-\delta, 0)/12} = \mathfrak{J}$$

because $Q^2 = X^4$ on S. In (c), by the Hurwitz formula [Kol13, 2.41],

$$K_S = \pi^* K_{\mathbb{A}^2} + \operatorname{div}(A), \qquad \pi^* \operatorname{div}(\zeta_1 + \zeta_2^2) = 2 \operatorname{div}(A), \qquad \pi^* \operatorname{div}(\zeta_1) = \operatorname{div}(\Phi),$$

so

$$\pi^* \Delta_i - \operatorname{div}(A) = \sum_k c_k \big(\epsilon_k \operatorname{div}(A) + \max(-l_k, 0) \operatorname{div}(\Phi) + \operatorname{div}(\psi_{i,k}(-\Phi, t)) \big) + \frac{\max(\delta, 0)}{2} \operatorname{div}(\Phi) = B_S.$$

Thus, by [KM98, Proposition 5.20], which holds for $\mathbb{R}$-divisors [Liu26, Remark 2.17], and [Liu26, Lemma 2.6(2)(5)], $(S, B_S, \mathfrak{J})$ is lc near 0 if and only if $(\mathbb{A}^2, \Delta_i, \mathfrak{c})$ is lc near 0, and in this case

$$\beta_i = \operatorname{mld}\big((S \ni 0, B_S, \mathfrak{J})/J_i\big), \tag{E.88}$$

where for a prime divisor ν over $\mathbb{A}^2$ with center 0 and a prime divisor ν_S over S lying over it,

$$\frac{a(\nu_S, S, B_S, \mathfrak{J})}{|I_{J_i}(\nu_S)|} = \frac{a(\nu, \mathbb{A}^2, \Delta_i, \mathfrak{c})}{|I_{H_i}(\nu)|}.$$

Here $a(\nu_S, S, B_S, \mathfrak{J})$ is the ramification index of ν_S over ν times $a(\nu, \mathbb{A}^2, \Delta_i, \mathfrak{c})$: for the pairs this is the Hurwitz formula, and $\operatorname{mult}_{\nu_S}(\mathfrak{c} \mathcal{O}_S)$ is the ramification index times $\operatorname{mult}_\nu \mathfrak{c}$.

Finally, a prime divisor F over W^0 has center 0 if and only if mult_F is positive on x, y, z, t. On U the functions x, y, z, t are $x, \rho, -\Phi\rho^{-2}, t$ in (a_y), $x, Y\rho^{-1}, \rho^2, t$ in (a_z), $X\rho^{-3}, Y\rho^{-4}, \rho^2, T\rho^{-2}$ in (b), and $\rho A, \rho^2, -\Phi\rho^{-2}, t$ in (c).

Step 2. In this step we construct prime divisors over W^0 from prime divisors over S by Gauss extensions. Let ν be a prime divisor over S with center 0, let $e := |I_{J_i}(\nu)|$, and let q_G be the integer of Lemma 3.1 for ν, the group J_i and the character of q. We write $\nu(g) := \operatorname{mult}_\nu g$. Since ν has center 0, $\nu(g) \geq 1$ for any coordinate g of S, and $\nu(\Phi) \geq 2$ in (a_y) and (c). In (b) let $\theta_\nu := \min(\nu(X)/3, \nu(Y)/4, \nu(T)/2) \geq \frac{1}{4}$, so that $\operatorname{mult}_\nu \mathfrak{J} = 12\theta_\nu$, and in (c) let $\theta_\nu := \nu(\Phi)/2 \geq 1$.

Let k_G be a positive integer with $k_G \equiv q_G \pmod e$. By Lemma 3.1, the Gauss extension of mult_ν with q-value k_G is $\operatorname{mult}_{F_q}$ for a prime divisor F_q over $S \times \mathbb{A}^1_q$ with center $(0, 0)$ and $I_{J_i}(F_q) = I_{J_i}(\nu)$, and

$$a(F_q, S \times \mathbb{A}^1_q, \operatorname{pr}_1^* B'_S) = a(\nu, S, B'_S) + k_G,$$
$$a(F_q, S \times \mathbb{A}^1_q, \operatorname{pr}_1^* B'_S + \operatorname{div}(q)) = a(\nu, S, B'_S).$$

Since a log discrepancy is an affine function of the coefficient of $\operatorname{div}(q)$, the boundary (E.87) has log discrepancy $a(\nu, S, B'_S) - s_{\mathbf{T}} \delta k_G / r_i$ at F_q. Let F_U be a prime divisor over U lying over F_q, let F^0 be the prime divisor over W^0 given by the restriction of $\operatorname{mult}_{F_U}$ to the function field of W^0, and let v be the positive multiple of $\operatorname{mult}_{F_U}$ which restricts to $\operatorname{mult}_{F_q}$. The crepant quotient of (U, B_U) by $\tilde{G}_i$ and the crepant quotient of (W^0, D_i^0) by G_i are crepant birational, and the quotient by $\tilde{G}_i$ factors through the quotient by $\tilde{R}_i$. Thus [Liu26, Lemma 2.6(2)(5)] implies

$$\frac{a(F^0, W^0, D_i^0)}{|I_{G_i}(F^0)|} = \frac{1}{e} \Big(a(\nu, S, B'_S) - \frac{s_{\mathbf{T}} \delta k_G}{r_i} \Big). \tag{E.89}$$

Moreover $v(q) = k_G$, $v(\rho) = k_G/r_i$, $v(g) = \nu(g)$ for g on S, and $v(h^0_{i,k}) = s_{\mathbf{T}}\kappa_k k_G/r_i + \nu(H_{i,k})$ by (E.86). By the end of **Step 1**, the values of x, y, z, t are

$$\begin{aligned}
&(\mathrm{a}_y): \quad \nu(x),\ \tfrac{k_G}{r_i},\ \nu(\Phi) - \tfrac{2k_G}{r_i},\ \nu(t);\\
&(\mathrm{a}_z): \quad \nu(x),\ \nu(Y) - \tfrac{k_G}{r_i},\ \tfrac{2k_G}{r_i},\ \nu(t);\\
&(\mathrm{b}): \quad \nu(X) - \tfrac{3k_G}{r_i},\ \nu(Y) - \tfrac{4k_G}{r_i},\ \tfrac{2k_G}{r_i},\ \nu(T) - \tfrac{2k_G}{r_i};\\
&(\mathrm{c}): \quad \nu(A) + \tfrac{k_G}{r_i},\ \tfrac{2k_G}{r_i},\ \nu(\Phi) - \tfrac{2k_G}{r_i},\ \nu(t).
\end{aligned} \tag{E.90}$$

Thus F^0 has center 0 if $k_G/r_i < \frac{1}{4}$ in the cases with $s_{\mathbf{T}}\delta \le 0$, and if $k_G/r_i < \theta_\nu$ in the cases with $s_{\mathbf{T}}\delta > 0$, namely (b) with $\delta < 0$ and (c) with $\delta > 0$. In (b) with $\delta < 0$ and (c) with $\delta > 0$, $a(\nu, S, B_S, \mathfrak{J}) = a(\nu, S, B'_S) - |\delta|\theta_\nu$, because $\mathfrak{J} = \mathfrak{J}^{|\delta|/12}$ in (b) and $B_S = B'_S + \frac{\delta}{2}\operatorname{div}(\Phi)$ in (c). Hence (E.89) reads

$$\frac{ea(F^0, W^0, D^0_i)}{|I_{G_i}(F^0)|} = \begin{cases} a(\nu, S, B_S, \mathfrak{J}) + |\delta|\dfrac{k_G}{r_i} & \text{if } s_{\mathbf{T}}\delta \le 0,\\[2ex] a(\nu, S, B_S, \mathfrak{J}) + |\delta|\Big(\theta_\nu - \dfrac{k_G}{r_i}\Big) & \text{if } s_{\mathbf{T}}\delta > 0. \end{cases} \tag{E.91}$$

Step 3. In this step we prove that $(\mathbb{A}^2, \Delta_i, \mathfrak{c})$ is lc near 0 for i large. We first show that (W^0, D^0_i) is lc near 0 for i large. The pair (W_i, D_i) is klt near o_i, by Construction-Theorem 4.13(1) for the vector c, applied to both admissions (the admission of Lemma E.16 and that of Construction E.17). For the first one, the pair of Set-up 11.1 is klt because D is a convex combination of the terminal proxies C_k, and the normal forms do not change D_i. Here we regard the normal forms $h_{i,k}$ as polynomials on $\mathbb{A}^4$. By [Kaw07, Theorem], the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(f_i) + \sum_k c_k \operatorname{div}(h_{i,k})\Big)$$

is lc near 0, hence so is the corresponding pair on $\operatorname{Spec}\mathbb{C}[[x, y, z, t]]$ [dFEM11, Proposition 2.11]. The differences $f_i - f^0$ and $h_{i,k} - h^0_{i,k}$ have order tending to infinity, by Construction E.17 and Definition-Lemma 11.5(2). Thus Lemma 4.14, applied with $\Lambda := \{1, c_1, \dots, c_s\}$, shows that $(\operatorname{Spec}\mathbb{C}[[x, y, z, t]], (f^0)\prod_k (h^0_{i,k})^{c_k})$ is lc for i large. By [dFEM11, Proposition 2.11] and [Kaw07, Theorem] again, (W^0, D^0_i) is lc near 0.

Fix such an i with moreover $r_i > 8e_0$, and suppose that $(\mathbb{A}^2, \Delta_i, \mathfrak{c})$ is not lc near 0. Let $\varphi : Y' \to \mathbb{A}^2$ be a log resolution of $(\mathbb{A}^2, \Delta_i)$, of the ideal $(Q^2, \zeta_1^3, \zeta_2^6)$ in case (b), and of the maximal ideal of 0, and let $(Y', \Delta_{Y'})$ be the crepant pullback, on which the ideal becomes a divisor. Then some component E of $\Delta_{Y'}$ meeting $\varphi^{-1}(0)$ has coefficient larger than one [KM98, Corollary 2.31(3) and Lemma 2.30]. Let $P \in E \cap \varphi^{-1}(0)$ be a point which is on at most one further component E' of $\operatorname{Supp}\Delta_{Y'} \cup \varphi^{-1}(0)$. If there is no such component, let E' be a smooth curve on a neighborhood of P in Y', transverse to E at P, so that $a(E', Y', \Delta_{Y'}) = 1$. Let E_n, $n \ge 1$, be the prime divisors over $\mathbb{A}^2$ obtained by blowing up P, and then repeatedly the point of the last exceptional curve on the strict transform of E. They have center 0. Blowing up a point of a smooth surface which is on exactly two components of the support of a crepant pullback, meeting transversally there and with log discrepancies a_1, a_2, gives an exceptional curve of log discrepancy $a_1 + a_2$ [KM98, Lemmas 2.29 and 2.30]. Each later center is only on the last exceptional curve and on the strict transform of E, so induction on n gives

$$a(E_n, \mathbb{A}^2, \Delta_i, \mathfrak{c}) = na(E, Y', \Delta_{Y'}) + a(E', Y', \Delta_{Y'}),$$

which tends to $-\infty$. By **Step 1**, for a prime divisor ν over S lying over E_n, $a(\nu, S, B_S, \mathfrak{J})$ is a positive integer multiple of $a(E_n, \mathbb{A}^2, \Delta_i, \mathfrak{c})$. Fix such a ν with $a(\nu, S, B_S, \mathfrak{J}) < -2|\delta|$. If $s_{\mathbf{T}}\delta \le 0$, let $k_G := q_G$. Then $k_G/r_i \le 2e_0/r_i < \frac{1}{4}$, and (E.91) implies

$$\frac{ea(F^0, W^0, D^0_i)}{|I_{G_i}(F^0)|} < -2|\delta| + \frac{|\delta|}{4} \le 0.$$

If $s_{\mathbf{T}}\delta > 0$, let k_G be the largest integer with $k_G \equiv q_G \pmod e$ and $k_G < r_i\theta_\nu$. Since $r_i\theta_\nu \geq r_i/4 > e$, we have $k_G > 0$ and $0 < \theta_\nu - k_G/r_i \leq e/r_i < 1$, so (E.91) implies $a(F^0, W^0, D_i^0) < 0$. In both cases F^0 has center 0 by **Step 2**. This contradicts the lc property of (W^0, D_i^0) near 0. Thus $(\mathbb{A}^2, \Delta_i, \mathfrak{c})$ is lc near 0, and so is $(S, B_S, \mathfrak{J})$ by **Step 1**.

Step 4. In this step we prove that the numbers β_i belong to a finite set, and we find prime divisors of bounded log discrepancy computing them. Let i be as in **Step 3**. The polynomials Φ, Q, $\zeta_1 + \zeta_2^2$, and ζ_1 of Construction E.17 are irreducible and reduced. Thus the coefficient of a prime divisor $\Gamma \ni 0$ in Δ_i is $\sum_k c_k \operatorname{mult}_\Gamma g_{i,k} + \lambda_\Gamma$, where $g_{i,k}$ is $\phi_{i,k}$, $A_{i,k}$, or $\psi_{i,k}$, and λ_Γ belongs to the finite set

$$\Big\{0,\ \frac{\max(\delta,0)}{2},\ \frac{1+\sum_k c_k\epsilon_k}{2},\ \sum_k c_k \max(-l_k,0) + \frac{\max(\delta,0)}{2}\Big\},$$

which does not depend on i. By **Step 3** the coefficient of Γ is at most one, so $\operatorname{mult}_\Gamma g_{i,k} \leq 1/c_k$. Hence the coefficients of Δ_i near 0 belong to a finite set Λ_1 independent of i. If $\mathfrak{c} = \mathcal{O}$, Lemma E.6, applied with $N := e_0$ and $\Lambda := \Lambda_1$, implies that β_i belongs to a finite set $\mathcal{V}$ and that there exists a prime divisor ν over $\mathbb{A}^2$ with center 0 computing β_i and with $a(\nu, \mathbb{A}^2, 0) \leq l$, where l and $\mathcal{V}$ depend only on e_0 and Λ_1. In case (b) with $\delta < 0$ we first replace the $\mathbb{R}$-ideal by general members.

Claim E.19. *Let H be a finite diagonal group acting on $\mathbb{A}^2$, $N_H := |H|$, $\Delta \geq 0$ an H-invariant $\mathbb{R}$-divisor on $\mathbb{A}^2$, $\mathfrak{a}$ an ideal of $\mathbb{C}[\zeta_1, \zeta_2]$ with cosupport $\{0\}$ generated by H-semi-invariant polynomials, and $\gamma > 0$ a real number such that $(\mathbb{A}^2, \Delta, \mathfrak{a}^\gamma)$ is lc near 0. Let $n := \lfloor \gamma \rfloor + 1$ and $\eta := \gamma/(nN_H)$. Then there exist H-invariant polynomials $g_1, \dots, g_n$ such that $\Delta' := \Delta + \eta \sum_j \operatorname{div}(g_j)$ satisfies the following: $(\mathbb{A}^2, \Delta')$ is lc near 0, the coefficients of Δ' near 0 belong to the set of coefficients of Δ together with η,*

$$\operatorname{mld}((\mathbb{A}^2 \ni 0, \Delta')/H) = \operatorname{mld}((\mathbb{A}^2 \ni 0, \Delta, \mathfrak{a}^\gamma)/H),$$

and for any prime divisor E over $\mathbb{A}^2$ with center 0 computing the left-hand side there exists a prime divisor E_1 with center 0 computing both sides with $a(E_1, \mathbb{A}^2, 0) \leq a(E, \mathbb{A}^2, 0)$.

Proof. Let $\mathfrak{m}_0$ be the maximal ideal of 0, and let $\varphi : Y' \to \mathbb{A}^2$ be the functorial principalization of the H-invariant ideal $\mathfrak{a} \cdot \mathfrak{m}_0 \cdot I_{\operatorname{Supp}\Delta}$ [Kol07, Theorem 3.35]. It is H-equivariant because it commutes with the automorphisms in H (part (4) of loc. cit.), its centers are H-invariant finite sets of closed points, since blowing up a smooth curve on a surface is an isomorphism, and the pullback of the ideal is the ideal of a simple normal crossing divisor containing the exceptional curves and the strict transform of $\operatorname{Supp}\Delta$ near $\varphi^{-1}(0)$. Thus φ is an H-equivariant log resolution of $(\mathbb{A}^2, \Delta)$, of $\mathfrak{a}$, and of $\mathfrak{m}_0$. Write $\mathfrak{a}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-A_{Y'})$, and let V be the linear system spanned by a set of H-semi-invariant generators of $\mathfrak{a}$. Then $\varphi^*V - A_{Y'}$ is base point free, $A_{Y'}$ is H-invariant, and $h^*V = V$ for any $h \in H$. For $g \in V$ let $g^H := \prod_{h\in H} h^*g$, which is H-invariant, and let V^H be the linear system spanned by the g^H, $g \in V$, whose members are H-invariant and vanish to order at least $N_H \operatorname{mult}_E \mathfrak{a}$ along any prime divisor E over $\mathbb{A}^2$. For $g \in V$ let $M_g := \varphi^* \operatorname{div}(g) - A_{Y'}$. Then $\varphi^* \operatorname{div}(h^*g) = A_{Y'} + h^{-1}(M_g)$ for any $h \in H$, and for a point $P' \in \varphi^{-1}(0)$ a general $g \in V$ has M_g avoiding the finite set $H \cdot P'$, so that $\varphi^* \operatorname{div}(g^H) = N_H A_{Y'}$ near P'. Thus $\varphi^*V^H - N_H A_{Y'}$ is base point free near $\varphi^{-1}(0)$. Let $g_1, \dots, g_n$ be general members of V^H, each general with respect to the previous ones. By Bertini's theorem [Har77, Corollary III.10.9 and Remark III.10.9.2], the strict transforms of the curves $\operatorname{div}(g_j)$ are smooth near $\varphi^{-1}(0)$, they are pairwise disjoint near $\varphi^{-1}(0)$, since each g_j avoids the finitely many points where the strict transforms for the earlier $g_{j'}$ meet $\varphi^{-1}(0)$, and they meet $\varphi^{-1}(0) \cup \operatorname{Supp}\varphi_*^{-1}\Delta$ transversally at general points of exceptional curves. Moreover, for any exceptional curve E'' of φ, the finitely many points of E'' whose stabilizer in H is larger than $I_H(E'')$ are avoided, because $\operatorname{Stab}_H(E'')/I_H(E'')$ acts faithfully on E''. In particular the coefficients of Δ' near 0 are those of Δ and η.

For any prime divisor E over $\mathbb{A}^2$ with center 0, $\operatorname{mult}_E g_j \geq N_H \operatorname{mult}_E \mathfrak{a}$, so

$$a(E, \mathbb{A}^2, \Delta') \leq a(E, \mathbb{A}^2, \Delta) - n\eta N_H \operatorname{mult}_E \mathfrak{a} = a(E, \mathbb{A}^2, \Delta, \mathfrak{a}^\gamma),$$

with equality if the center P_E of E on Y' is not on the strict transforms of the curves $\operatorname{div}(g_j)$, since then $\operatorname{mult}_E g_j = N_H \operatorname{mult}_E \mathfrak{a}$. If P_E is on such a strict transform G, then P_E is on exactly one exceptional curve E_1 and on no further component of $\varphi^{-1}(0) \cup \operatorname{Supp} \varphi_*^{-1}\Delta$, and $\operatorname{Stab}_H(P_E) = I_H(E_1)$. Write

$$a_1 := a(E_1, \mathbb{A}^2, \Delta') = a(E_1, \mathbb{A}^2, \Delta, \mathfrak{a}^\gamma) \geq 0.$$

Near P_E the crepant pullback of Δ' is $(1 - a_1)E_1 + \eta G$, so [Liu26, Lemma 2.13] implies

$$a(E, \mathbb{A}^2, \Delta') \geq a_1 \operatorname{mult}_E e_1 + (1 - \eta) \operatorname{mult}_E g \geq a_1,$$

where e_1 and g are local equations of E_1 and G. Since $I_H(E) \subset \operatorname{Stab}_H(P_E) = I_H(E_1)$,

$$\frac{a(E, \mathbb{A}^2, \Delta')}{|I_H(E)|} \geq \frac{a_1}{|I_H(E_1)|} = \frac{a(E_1, \mathbb{A}^2, \Delta, \mathfrak{a}^\gamma)}{|I_H(E_1)|}. \tag{E.92}$$

By [Liu26, Lemma 2.6], both minimal log discrepancies are the infima of $a(E, \cdot)/|I_H(E)|$ over E with center 0, so they are equal. The crepant pullback of Δ' to Y' has simple normal crossing support near $\varphi^{-1}(0)$ and coefficients at most one there, because the coefficients of Δ near 0 are at most one, $\eta < 1$, and

$$a(E'', \mathbb{A}^2, \Delta') = a(E'', \mathbb{A}^2, \Delta, \mathfrak{a}^\gamma) \geq 0$$

for any exceptional curve E'' of φ over 0. Thus $(\mathbb{A}^2, \Delta')$ is lc near 0 [KM98, Corollary 2.31(3) and Lemma 2.30].

Let E compute the left-hand side. If the center P_E of E on Y' is not on the curves G, then E computes the right-hand side, and we let $E_1 := E$. Otherwise the inequality (E.92) is an equality, and E_1 computes both sides. In this case [Liu26, Lemma 2.13] implies

$$a(E, \mathbb{A}^2, 0) = a\Big(E, Y', \sum_j (1 - a(E'_j, \mathbb{A}^2, 0))E'_j\Big) \geq \sum_j a(E'_j, \mathbb{A}^2, 0) \operatorname{mult}_E e'_j \geq a(E_1, \mathbb{A}^2, 0),$$

where the E'_j are the exceptional curves through P_E and e'_j their local equations. □

Proof of Lemma E.18 continued. In case (b) with $\delta < 0$, apply the Claim with $H := H_i$, $\Delta := \Delta_i$, $\mathfrak{a} := (Q^2, \zeta_1^3, \zeta_2^6)$, and $\gamma := -\delta/12$. Since $N_{H_i} \leq e_0$, the number η belongs to a finite set. Lemma E.6, applied to Δ' with $N := e_0$ and $\Lambda := \Lambda_1 \cup \{\gamma/(nN') \mid 1 \leq N' \leq e_0\}$, together with the Claim, again implies that β_i belongs to a finite set and that a prime divisor ν over $\mathbb{A}^2$ with center 0 and $a(\nu, \mathbb{A}^2, 0) \leq l$ computes β_i.

In all cases let ν_S be a prime divisor over S lying over ν. By (E.88), ν_S computes β_i on S. Since K_S is the pullback of $K_{\mathbb{A}^2}$ plus half the branch curve and the ramification index of ν_S over ν is at most two, $a(\nu_S, S, 0) \leq 2a(\nu, \mathbb{A}^2, 0) \leq 2l$. Since $(S, B_S, \mathfrak{J})$ is lc near 0 and $B_S - c_k \operatorname{div}(H_{i,k}) \geq 0$, we have $c_k \nu_S(H_{i,k}) \leq a(\nu_S, S, 0) \leq 2l$. By [Liu26, Lemma 2.13], $\nu(x) + \nu(t) \leq l$ in (a_y) and $\nu_S(A) + \nu_S(t) \leq 2l$ in (c). In (a_z) and (b), the σ-invariant coordinates x, t, resp. Y, T, have ν_S-values at most $2l$, and $2\nu_S(X) = \nu_S(Q) \leq 2\nu(Q) \leq 4l$ in (b), because Q has coefficient at least $\frac{1}{2}$ in Δ_i. In (c) with $\delta > 0$, $\frac{\delta}{2}\nu_S(\Phi) \leq a(\nu_S, S, 0) \leq 2l$. Hence there exists a constant $C_1 \geq 1$, depending only on l, e_0, the c_k, and δ, such that $|J_i| \leq C_1$, $\nu_S(H_{i,k}) \leq C_1$ for any k, the ν_S-values of x, t in (a_y) and (a_z), of X, Y, T in (b), and of A, t in (c) are at most C_1, and $\theta_{\nu_S} \leq C_1$ in the cases with $s_{\mathbf{T}}\delta > 0$.

Step 5. In this step we pass from a prime divisor over W^0 to a prime divisor over W_i of equal score, through a simple branch of residue one. We say that we are in the x-case in case (a) with $\delta \geq 0$ and in case (b), and in the z-case in case (a) with $\delta < 0$ and in case (c). Let $s_0 := y^2z + t^3$, and let $p_0 := f^0 - y^2z$, that is, $p_0 = x^2 + t^3$ in (a) and $p_0 = x^2 + yt^2$ in (c). Recall that $f_i = x^2 + s_0 + B_i$ in the x-case and $f_i = y^2z + p_0 + B_i(z, t)$ in the z-case.

Claim E.20. *Let i be as in **Step 3**, F^0 a prime divisor over W^0, and v a positive multiple of mult_{F^0} with $v(x), v(y), v(z), v(t) > 0$, where functions on $\mathbb{A}^4$ are evaluated on W^0. Let $\tau > \max_k v(h^0_{i,k})$ be a real number. Assume that $v(m) - 2v(x) \geq \tau$ for any monomial m of B_i in the x-case, and that $v(z^{j_1-1}t^{j_2}) - 2v(y) \geq \tau$ for any monomial $z^{j_1}t^{j_2}$ of B_i in the z-case. Assume moreover that $v(m) > v(h^0_{i,k})$ for any k and any monomial m of $h_{i,k} - h^0_{i,k}$. Then there exists a prime divisor F over W_i with center o_i such that*

$$\frac{a(F, W_i, D_i)}{|I_{G_i}(F)|} = \frac{a(F^0, W^0, D^0_i)}{|I_{G_i}(F^0)|}.$$

Proof. Let $\mathcal{K}^0$ and $\mathcal{K}$ be the function fields of W^0 and W_i, $\widehat{\mathcal{K}^0}$ the completion of $\mathcal{K}^0$ with respect to v, and $\hat{v}$ the extension of v to $\widehat{\mathcal{K}^0}$. In the x-case, let $\mathcal{L} := \mathbb{C}(y, z, t)$, so that $\mathcal{K}^0 = \mathcal{L}(x)$ with $x^2 = -s_0$ and $\mathcal{K} = \mathcal{L}(x)$ with $x^2 = -s_0 - B_i$. Then $\gamma_B := B_i/s_0 \in \mathcal{L}$ has $v(\gamma_B) \geq \tau > 0$, because $v(s_0) = 2v(x)$ on W^0. By Hensel's lemma [Ser79, Chapter II, §4, Proposition 7] there exists a unique ξ_+ in the closure $\widehat{\mathcal{L}}$ of $\mathcal{L}$ in $\widehat{\mathcal{K}^0}$ with $\xi_+^2 = 1 + \gamma_B$ and $v(\xi_+ - 1) > 0$, and $v(\xi_+ - 1) = v(\gamma_B) \geq \tau$. Since $(x\xi_+)^2 = -s_0 - B_i$, there exists an embedding $\iota : \mathcal{K} \to \widehat{\mathcal{K}^0}$ over $\mathcal{L}$ with $\iota(x) = x\xi_+$. In the z-case, let $\mathcal{L} := \mathcal{K}^0 = \mathbb{C}(x, y, t)$, in which $z = z_0 := -p_0/y^2$, so that $\mathcal{K} = \mathcal{L}(z)$ with $y^2 z + p_0 + B_i(z, t) = 0$. The polynomial

$$P(Z) := \frac{y^2 z_0 Z + p_0 + B_i(z_0 Z, t)}{y^2 z_0} = Z - 1 + \sum b_{j_1 j_2} \frac{z_0^{j_1 - 1} t^{j_2}}{y^2} Z^{j_1},$$

where $B_i = \sum b_{j_1 j_2} z^{j_1} t^{j_2}$, has coefficients of value at least 0, and the terms coming from B_i have coefficients of value at least $\tau > 0$. By Hensel's lemma there exists a unique $Z_+ \in \widehat{\mathcal{K}^0}$ with $P(Z_+) = 0$ and $v(Z_+ - 1) > 0$, and $v(Z_+ - 1) \geq \tau$. Let $\iota : \mathcal{K} \to \widehat{\mathcal{K}^0}$ be the embedding over $\mathcal{L}$ with $\iota(z) = z_0 Z_+$.

In both cases let $v' := \hat{v} \circ \iota$. The closure of $\iota(\mathcal{K})$ contains $\widehat{\mathcal{L}}$ and $\iota(x)\xi_+^{-1} = x$ in the x-case, and it contains $\widehat{\mathcal{L}} = \widehat{\mathcal{K}^0}$ in the z-case. Thus ι extends to an isomorphism from the completion of $\mathcal{K}$ with respect to v' onto $\widehat{\mathcal{K}^0}$. In particular v' and v have equal value groups and residue fields identified by ι, so $v' = \ell\,\mathrm{mult}_F$ for a prime divisor F over W_i [KM98, Lemma 2.45], where $\ell > 0$ is the real number with $v = \ell\,\mathrm{mult}_{F^0}$. The values of x, y, z, t under v' and under v are equal, since ξ_+ and Z_+ have value 0, so F has center o_i.

We compare the groups. The element γ_B and the coefficients of P are G_i-invariant, since s_0, B_i, p_0, $y^2 z_0$ and each monomial of B_i have the character of f_i. Let $g \in \mathrm{Stab}_{G_i}(F^0)$. Then g acts continuously on $\widehat{\mathcal{K}^0}$ and preserves $\widehat{\mathcal{L}}$, and $g^*\xi_+$, resp. g^*Z_+, is again a root of residue one of $\xi^2 - (1 + \gamma_B)$, resp. P, so it is equal to ξ_+, resp. Z_+. Thus $g^* \circ \iota = \iota \circ g^*$ on the generators of $\mathcal{K}$, hence on $\mathcal{K}$, and $g \in \mathrm{Stab}_{G_i}(F)$. Conversely, let $\widehat{\mathcal{K}}$ be the completion of $\mathcal{K}$ with respect to v', $\hat{v}'$ the extension of v' to $\widehat{\mathcal{K}}$, $\hat{\iota}$ the isomorphism from $\widehat{\mathcal{K}}$ onto $\widehat{\mathcal{K}^0}$ extending ι, and

$$\iota' := \hat{\iota}^{-1}|_{\mathcal{K}^0} : \mathcal{K}^0 \to \widehat{\mathcal{K}},$$

so that $v = \hat{v}' \circ \iota'$. In the z-case, ι' is the inclusion of $\mathcal{K}^0 = \mathcal{L}$ into $\mathcal{K} \subset \widehat{\mathcal{K}}$. In the x-case, ι' is the identity on $\mathcal{L}$ and $\iota'(x) = x\xi_-$, where $\xi_- := \hat{\iota}^{-1}(\xi_+^{-1})$ is in the closure of $\mathcal{L}$ in $\widehat{\mathcal{K}}$, satisfies $\xi_-^2 = (1 + \gamma_B)^{-1}$ because $x^2 = -s_0(1 + \gamma_B)$ in $\mathcal{K}$, and has $\hat{v}'(\xi_- - 1) = \hat{v}(\xi_+^{-1} - 1) > 0$. Let $g \in \mathrm{Stab}_{G_i}(F)$. Then g acts continuously on $\widehat{\mathcal{K}}$, preserves $\hat{v}'$ and the closure of $\mathcal{L}$ in $\widehat{\mathcal{K}}$, and fixes $(1 + \gamma_B)^{-1}$. Thus $g^*\xi_-$ is a root of $\xi^2 - (1 + \gamma_B)^{-1}$ with $\hat{v}'(g^*\xi_- - 1) > 0$, so $g^*\xi_- = \xi_-$, because the roots are $\pm\xi_-$ and $\hat{v}'(-\xi_- - 1) = 0$. Since g^*x is a constant multiple of x, we get $g^*\iota'(x) = g^*(x)\xi_- = \iota'(g^*x)$. Hence in both cases $g^* \circ \iota' = \iota' \circ g^*$ on $\mathcal{K}^0$, and $v \circ g^* = \hat{v}' \circ g^* \circ \iota' = \hat{v}' \circ \iota' = v$ on $\mathcal{K}^0$. Thus

$$\mathrm{Stab}_{G_i}(F) \subset \mathrm{Stab}_{G_i}(F^0).$$

The stabilizers of F and F^0 act compatibly on the common residue field, so $I_{G_i}(F) = I_{G_i}(F^0)$.

We compare the log discrepancies. Let $\mathcal{N} \subset \widehat{\mathcal{K}^0}$ be the field generated by $\mathcal{K}^0$ and $\iota(\mathcal{K})$. It is a finite extension of both, and the restriction of $\hat{v}$ to $\mathcal{N}$ has ramification index one and residue degree one over v and over v'. Let $F_{\mathcal{N}}$ be the corresponding prime divisor. Since the ramification index of $\hat{v}|_{\mathcal{N}}$ over v and over v' is one, the Hurwitz formula (for a finite morphism between normal varieties, the log discrepancy of a prime divisor over the source with respect to the crepant pullback is the ramification index times the log discrepancy of the prime divisor under it, cf. [KM98, Proposition 5.20] and [Liu26, Remark 2.17]) shows that $a(F^0, W^0, D_i^0)$ and $a(F, W_i, D_i)$ are equal to the log discrepancies of $F_{\mathcal{N}}$ with respect to the crepant pullbacks of (W^0, D_i^0) and of (W_i, D_i) to the normalizations of W^0 and W_i in $\mathcal{N}$. Computed on one model of $\mathcal{N}$ on which $F_{\mathcal{N}}$ is a divisor, the log discrepancies of $F_{\mathcal{N}}$ with respect to the crepant pullbacks of (W^0, D_i^0) and of (W_i, D_i) differ by $\operatorname{mult}_{F_{\mathcal{N}}} \vartheta$, where

$$\vartheta := \frac{\iota(\omega_{W_i})}{\omega_{W^0}} \prod_k \Big(\frac{h_{i,k}^0}{\iota(h_{i,k})} \Big)^{c_k}.$$

In the x-case, $\omega_{W^0} = \pm dy \wedge dz \wedge dt/(2x)$ and $\iota(\omega_{W_i}) = \pm dy \wedge dz \wedge dt/(2x\xi_+)$, so the ratio of forms is $\pm\xi_+^{-1}$, of value 0. In the z-case,

$$\omega_{W^0} = \pm \frac{dx \wedge dy \wedge dt}{y^2}, \qquad \iota(\omega_{W_i}) = \pm \frac{dx \wedge dy \wedge dt}{y^2 + \partial_z B_i(z_0 Z_+, t)},$$

and $v(\partial_z B_i(z_0 Z_+, t)/y^2) \geq \tau > 0$ by the assumption, so the ratio has value 0. Finally, $\iota(h_{i,k})$ is $h_{i,k}^0$ evaluated on W^0, plus the terms $m(\xi_+^a - 1)$, resp. $m(Z_+^a - 1)$, for the monomials m of $h_{i,k}^0$ with x-degree a, resp. z-degree a, plus the monomials of $h_{i,k} - h_{i,k}^0$ multiplied by powers of ξ_+, resp. Z_+. The first kind of terms have value at least $v(m) + \tau \geq \tau > v(h_{i,k}^0)$, since all coordinates have positive value, and the second kind have value larger than $v(h_{i,k}^0)$ by the assumption. Thus $v(\iota(h_{i,k})) = v(h_{i,k}^0)$, $\operatorname{mult}_{F_{\mathcal{N}}} \vartheta = 0$, and $a(F, W_i, D_i) = a(F^0, W^0, D_i^0)$. □

Proof of Lemma E.18 continued.

Step 6. In this step we prove the upper bound. Let i be as in **Step 3**, let ν_S, $e := |I_{J_i}(\nu_S)|$, and C_1 be as in **Step 4**, and write $\nu := \nu_S$ in the formulas of **Step 2**. Let

$$C_2 := (\max_k |\kappa_k| + 1)C_1 + 1, \qquad C_3 := 2C_2 + 8C_1 + 2,$$

and $C_4 := 16(C_3 + C_1 + C_2 + \max_k |\kappa_k| + 6)$. Assume $r_i \geq C_4$. If $s_{\mathbf{T}}\delta \leq 0$, let k_G be the least integer with $k_G \geq C_3$ and $k_G \equiv q_G \pmod e$, so that $C_3 \leq k_G < C_3 + C_1$ and $k_G/r_i < \frac{1}{8}$. If $s_{\mathbf{T}}\delta > 0$, let k_G be the largest integer with $k_G \leq r_i\theta_\nu - C_3$ and $k_G \equiv q_G \pmod e$, so that

$$\frac{C_3}{r_i} \leq \theta_\nu - \frac{k_G}{r_i} < \frac{C_3 + C_1}{r_i} \leq \frac{1}{8},$$

and $k_G > 0$ because $\theta_\nu \geq \frac{1}{4}$. Let F^0 and v be as in **Step 2**. In all cases $k_G/r_i \leq C_1$, so $v(h_{i,k}^0) = s_{\mathbf{T}}\kappa_k k_G/r_i + \nu(H_{i,k}) < C_2$ for any k. We verify the assumptions of the Claim of **Step 5** with $\tau := C_2$, using (E.90). Write $m = x^a y^b z^p t^j$ for a monomial of B_i or of $h_{i,k} - h_{i,k}^0$, so that $a = 0$ for B_i and $a \leq 1$ for $h_{i,k}$, and the $\mathbf{T}$-weight of m is $\kappa(f_i) + nr_i$, resp. $\kappa_k + nr_i$, with $n \neq 0$.

Case (a) with $\delta \geq 0$. Here $v(x), v(t) \in [1, C_1]$, $v(y) \geq 1 - k_G/r_i \geq \frac{1}{2}$, and $v(z) = 2k_G/r_i \geq 2C_3/r_i$. A monomial $z^p t^j$ of B_i has $p \geq r_i/2$, so $v(m) - 2v(x) \geq C_3 - 2C_1 > C_2$. For a monomial of $h_{i,k} - h_{i,k}^0$ we have $b - 2p = \kappa_k + nr_i$. If $n > 0$, then $b \geq r_i/2$ and $v(m) \geq b/2 \geq r_i/4 > C_2$. If $n < 0$, then $2p \geq r_i/2$ and $v(m) \geq p \cdot 2C_3/r_i \geq C_3/2 > C_2$.

Case (a) with $\delta < 0$. Here $v(x), v(t) \geq 1$, $v(y) = k_G/r_i \geq C_3/r_i$, and $v(z) = \nu(\Phi) - 2k_G/r_i \geq 1$. A monomial $z^p t^j$ of B_i has $v(z^{p-1}t^j) - 2v(y) \geq p - 1 - \frac{1}{4} \geq r_i/2 - 2 > C_2$. For a monomial of $h_{i,k} - h_{i,k}^0$, if $n > 0$ then $b \geq r_i/2$ and $v(m) \geq bC_3/r_i \geq C_3/2 > C_2$, and if $n < 0$ then $2p \geq r_i/2$ and $v(m) \geq p \geq r_i/4 > C_2$.

Case (b) with $\delta \geq 0$. Here $k_G/r_i < \frac{1}{8}$ implies $v(x), v(y), v(t) \geq \frac{1}{2}$, moreover $v(x) \leq C_1$ and $v(z) = 2k_G/r_i \geq 2C_3/r_i$. A monomial of B_i with $n < 0$ is $z^p t^j$ with $2p \geq r_i - 6$, so $v(m) \geq p \cdot 2C_3/r_i \geq C_3/2$. A monomial of B_i with $n > 0$ has $4b + 2j \geq r_i$, so $v(m) \geq (b+j)/2 \geq r_i/8$. In both cases $v(m) - 2v(x) > C_2$. For a monomial of $h_{i,k} - h^0_{i,k}$ we have $3a + 4b - 2p + 2j = \kappa_k + nr_i$. If $n < 0$, then $2p \geq r_i/2$ and $v(m) \geq C_3/2 > C_2$. If $n > 0$, then $3a + 4b + 2j \geq r_i/2$ and $v(m) \geq (a + b + j)/2 \geq r_i/16 > C_2$.

Case (b) with $\delta < 0$. Here $\theta_\nu - k_G/r_i \geq C_3/r_i$ implies $v(x) \geq 3C_3/r_i$, $v(y) \geq 4C_3/r_i$, and $v(t) \geq 2C_3/r_i$, moreover $v(x) \leq C_1$ and $v(z) = 2k_G/r_i \geq 2\theta_\nu - \frac{1}{4} \geq \frac{1}{4}$. A monomial of B_i with $n < 0$ is $z^p t^j$ with $p \geq r_i/4$, so $v(m) \geq r_i/16$. A monomial of B_i with $n > 0$ has $v(m) \geq (4b + 2j)C_3/r_i \geq C_3$. In both cases $v(m) - 2v(x) \geq C_3 - 2C_1 > C_2$. For a monomial of $h_{i,k} - h^0_{i,k}$, if $n < 0$ then $p \geq r_i/4$ and $v(m) \geq r_i/16 > C_2$, and if $n > 0$ then $v(m) \geq (3a + 4b + 2j)C_3/r_i \geq C_3/2 > C_2$.

Case (c) with $\delta \leq 0$. Here $v(x), v(t) \geq 1$, $v(y) = 2k_G/r_i \geq 2C_3/r_i$, and $v(z) = \nu(\Phi) - 2k_G/r_i \geq 1$. A monomial $z^p t^j$ of B_i has $p \geq r_i/2 - 1$, so $v(z^{p-1}t^j) - 2v(y) \geq r_i/2 - 3 > C_2$. For a monomial of $h_{i,k} - h^0_{i,k}$ we have $a + 2b - 2p = \kappa_k + nr_i$. If $n > 0$, then $2b \geq r_i/2$ and $v(m) \geq b \cdot 2C_3/r_i \geq C_3/2 > C_2$. If $n < 0$, then $2p \geq r_i/2$ and $v(m) \geq p \geq r_i/4 > C_2$.

Case (c) with $\delta > 0$. Here $v(x), v(t) \geq 1$, $v(z) = 2(\theta_\nu - k_G/r_i) \geq 2C_3/r_i$, and $v(y) = 2k_G/r_i \in [1, 2C_1]$. A monomial $z^p t^j$ of B_i has $p \geq r_i/2 - 1$, so

$$v(z^{p-1}t^j) - 2v(y) \geq (r_i/2 - 2)2C_3/r_i - 4C_1 \geq C_3/2 - 4C_1 > C_2.$$

For a monomial of $h_{i,k} - h^0_{i,k}$, if $n > 0$ then $2b \geq r_i/2$ and $v(m) \geq b \geq r_i/4 > C_2$, and if $n < 0$ then $2p \geq r_i/2$ and $v(m) \geq p \cdot 2C_3/r_i \geq C_3/2 > C_2$.

In all cases F^0 has center 0 by **Step 2**, and the Claim of **Step 5** yields a prime divisor F over W_i with center o_i and with score equal to that of F^0. Since ν_S computes β_i on S, $a(\nu_S, S, B_S, \mathfrak{J}) = e\beta_i$. Thus [Liu26, Lemma 2.6] and (E.91) imply

$$\alpha_i \leq \frac{a(F, W_i, D_i)}{|I_{G_i}(F)|} = \frac{a(F^0, W^0, D^0_i)}{|I_{G_i}(F^0)|} \leq \beta_i + \frac{|\delta|(C_3 + C_1)}{er_i} \leq \beta_i + \frac{C_0}{r_i},$$

where $C_0 := |\delta|(C_3 + C_1)$. The constant C_0 does not depend on i, which finishes the proof. □

We now show that, after passing to a subsequence, β_i is constant and positive, and find a real number $\theta > 1$ for which the three pairs in the following lemma are lc.

Lemma E.21. *Notation as in Construction E.17. After passing to a subsequence, the number $\beta := \beta_i$ is independent of i and positive, and there exists a real number $\theta > 1$ such that, for i large, $(\mathbb{A}^2, \Delta_i(\theta c), \mathfrak{c}(\theta c))$ is lc near 0, and the pairs $(W_i, \theta D_i)$ and $(W_i, \theta \sum_k c_k \operatorname{div}(h^0_{i,k}))$ are lc near o_i.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the hypersurface of principal blocks, bound the orders of the marks, and show that, after passing to a subsequence, β_i is a positive constant.

By Lemma E.16, $W_i = (f_i = 0) \subset \mathbb{A}^4$ with $f_i := x^2 + g_i$, the group G_i is diagonal on x, y, z, t, and o_i is the origin. For a regular function h near o_i, we write $\operatorname{mult}_{o_i} h$ for the order of $h|_{W_i}$ at o_i. Let g^0, f^0, and W^0 be as in Construction E.17. The relations of Construction E.17 between $h^0_{i,k}$ and the plane functions hold on W^0. By Construction E.17, W^0 is normal, f^0 is the principal block of f_i along $\mathbf{T}$ for i large, and any monomial of $f_i - f^0$ has degree at least $(r_i - 6)/4$. Together with Definition-Lemma 11.5(2), we obtain

$$\lim_{i \to +\infty} \operatorname{mult}_0(f_i - f^0) = +\infty \quad \text{and} \quad \lim_{i \to +\infty} \operatorname{mult}_0(h_{i,k} - h^0_{i,k}) = +\infty \quad \text{for any } k, \tag{E.93}$$

where mult_0 is the order at $0 \in \mathbb{A}^4$ and the order of 0 is $+\infty$.

Since $g_i, g^0 \in (y, z, t)^3$, the completions of the local rings of W_i and of W^0 at 0 are equal to $\mathbb{C}[[y, z, t]] \oplus x\mathbb{C}[[y, z, t]]$ (Weierstrass division by f_i, resp. f^0, which is monic of degree two in x, [GLS07, Chapter I, Theorem 1.8] with the trivial valuation on the field, for which $K\langle x \rangle = K[[x]]$

by [GLS07, Chapter I, Remark 1.1.1(1)]), and the n-th powers of their maximal ideals are $(y,z,t)^n \oplus x(y,z,t)^{n-1}$. For any field $\mathbf{k} \supset \mathbb{C}$, we also have $\mathbf{k}[[x,y,z,t]]/(f^0) = \mathbf{k}[[y,z,t]] \oplus x\mathbf{k}[[y,z,t]]$, with n-th power of the maximal ideal equal to $(y,z,t)^n \oplus x(y,z,t)^{n-1}$. Hence, for ϕ and ξ independent of x, the order of $\phi + x\xi$ at 0 on W_i, on W^0, or on $(f^0 = 0) \subset \operatorname{Spec} \mathbf{k}[[x,y,z,t]]$ is equal to the order of $\phi + x\xi$ at 0 in the ambient space. The principal block $h^0_{i,k}$ of the normal form $h_{i,k}$ contains a monomial of least degree of $h_{i,k}$ (Set-up 5.1), so

$$\operatorname{mult}_0\bigl(h^0_{i,k}|_{W^0}\bigr) = \operatorname{mult}_0 h^0_{i,k} = \operatorname{mult}_0 h_{i,k} = \operatorname{mult}_{o_i} h_{i,k}. \tag{E.94}$$

The data to which Lemma E.16 is applied are data of Set-up 11.1 whose residual cubic is a cuspidal cubic or a tangent conic, not the cube of a linear form, so they are of type cD (Definition 9.1). Since Construction-Theorem 4.13 allows any $N \geq N_1$ and $L \geq L_1$, we may assume that the admissions of Lemma E.16 and of Construction E.17 are taken with N and L larger than the orders at the origin of all the marks before the admissions. By Definition-Lemma 11.5(3), where m is the integer of Set-up 11.1, and Lemma 11.10(2), the marks of Lemma E.16 satisfy the inequality (E.95) below. We show that the admission of Construction E.17 does not change the orders of the marks at the origin. Let $e := \min(N, L)$ and $\mathfrak{m} := (x,y,z,t)$. In the notation of Construction-Theorem 4.13, we have $f_0 \equiv v\psi(\hat{f})$ and $h_{k,0} \equiv u_k\psi(\hat{h}_k)$ modulo $\mathfrak{m}^e$, so ψ induces an isomorphism

$$\mathfrak{R}/\bigl(\mathfrak{m}^e + (\hat{f})\bigr) \to \mathfrak{R}/\bigl(\mathfrak{m}^e + (f_0)\bigr)$$

which maps the image of $\mathfrak{m}^j$ onto the image of $\mathfrak{m}^j$ for any j and the image of $\hat{h}_k$ to the image of $u_k^{-1}h_{k,0}$. Since the order at the origin of a function on $(\hat{f} = 0)$, resp. $(f_0 = 0)$, is the largest j with the function in $\mathfrak{m}^j + (\hat{f})$, resp. $\mathfrak{m}^j + (f_0)$, the orders smaller than e are preserved. Replacing $h_{i,k}$ with its normal form does not change $\operatorname{mult}_{o_i} h_{i,k}$ either. Thus, for the marks of Construction E.17,

$$\sum_k c_k \operatorname{mult}_{o_i} h_{i,k} \leq 1 - \frac{1}{m}. \tag{E.95}$$

By Lemma E.18, after passing to a subsequence, $(\mathbb{A}^2, \Delta_i, \mathfrak{c})$ is lc near 0 for any i, the numbers β_i belong to a finite set, and $\alpha_i \leq \beta_i + C_0/r_i$. Passing to a further subsequence, $\beta_i = \beta$ is independent of i. Since the α_i are strictly increasing in $(0,1)$ and $r_i \to +\infty$ (Lemma E.16), we have

$$0 < \alpha_1 < \lim_{i \to +\infty} \alpha_i \leq \beta.$$

Step 2. In this step we take a compatible generic limit of the plane references and show that β is the quotient value of the limit pair.

The functions ζ_1, ζ_2 are $\mathbf{T}$-invariant monomials, so $G_i \cap \mathbf{T}$ acts trivially on $\mathbb{A}^2_{\zeta_1,\zeta_2}$ and H_i is a diagonal subgroup of $GL_2(\mathbb{C})$ of order at most $[G_i : G_i \cap \mathbf{T}] \leq e_0$ (Lemma E.16). Such a subgroup is contained in $\mu_{e_0!} \times \mu_{e_0!}$, so after passing to a subsequence, $H := H_i$ is independent of i.

We write the boundaries Δ_i as follows. In case (a), if $\kappa_k \geq 0$, let $l_k := 0$ and $\psi_{i,k} := \phi_{i,k}$, so that $\phi_{i,k} = (-\Phi)^{l_k}\psi_{i,k}$ for any k, and let $b_2 := \sum_k c_k l_k + \max(\delta, 0)/2$. Then

$$\Delta_i = \sum_k c_k \operatorname{div}(\psi_{i,k}) + b_2 \operatorname{div}(\Phi). \tag{E.96}$$

In cases (b) and (c), let

$$b_1 := \frac{1 + \sum_k c_k \epsilon_k}{2}, \qquad \text{and in case (c) let} \qquad b_2 := \sum_k c_k \max(-l_k, 0) + \frac{\max(\delta, 0)}{2}.$$

Then

$$\begin{aligned} &\Delta_i = b_1 \operatorname{div}(Q) + \sum_k c_k \operatorname{div}(A_{i,k}) && \text{in case (b)},\\ &\Delta_i = b_1 \operatorname{div}(\zeta_1 + \zeta_2^2) + b_2 \operatorname{div}(\zeta_1) + \sum_k c_k \operatorname{div}(\psi_{i,k}) && \text{in case (c)}. \end{aligned} \tag{E.97}$$

In case (b), let $\mathfrak{a} := (Q^2, \zeta_1^3, \zeta_2^6)$ and $\eta := \max(-\delta, 0)/12$, so that $\mathfrak{c} = \mathfrak{a}^\eta$.

Since G_i is diagonal and f^0 is the principal block of f_i, the monomials of f^0 have one G_i-character. Hence Φ, Q, and $\zeta_1 + \zeta_2^2$ are semi-invariant. By Construction E.17, the functions $\psi_{i,k}$ and $A_{i,k}$ are equal to the G_i-semi-invariant polynomial $h^0_{i,k}$ divided by a monomial, after replacing y^2z with $-\Phi$ in case (a), so they are H-semi-invariant. Passing to a subsequence, their characters are independent of i. In cases (a) and (c), $h^0_{i,k}|_{W^0}$ is a monomial times $\psi_{i,k}(x,t)$, resp. $\psi_{i,k}(yz,t)$, so $\operatorname{mult}_0 \psi_{i,k} \le \operatorname{mult}_{o_i} h_{i,k}$ by (E.94). In case (b), a monomial $\zeta_1^p \zeta_2^{p'}$ of $A_{i,k}$ yields the monomial $x^{\epsilon_k} y^p t^{p'} z^{2p+p'-\kappa'_k}$ of $h^0_{i,k}$, of degree at least $2(p+p') - \kappa'_k$, so $2\operatorname{mult}_0 A_{i,k} \le \operatorname{mult}_{o_i} h_{i,k} + \kappa'_k$. By (E.95), $\operatorname{mult}_{o_i} h_{i,k} \le 1/c_k$, so $\operatorname{mult}_0 \psi_{i,k}$ and $\operatorname{mult}_0 A_{i,k}$ are bounded independently of i.

In case (b), the ideal $\mathfrak{a}$ is generated by semi-invariant polynomials. For any prime divisor F over $\mathbb{A}^2$, $a(F, \mathbb{A}^2, 0) \ge \operatorname{mult}_F \zeta_1 + \operatorname{mult}_F \zeta_2$ since $(\mathbb{A}^2, \operatorname{div}(\zeta_1\zeta_2))$ is lc, while

$$\operatorname{mult}_F \mathfrak{a} \le \min(3\operatorname{mult}_F \zeta_1, 6\operatorname{mult}_F \zeta_2) \le 2(\operatorname{mult}_F \zeta_1 + \operatorname{mult}_F \zeta_2),$$

so $(\mathbb{A}^2, \mathfrak{a}^{1/2})$ is lc. Moreover, κ_k is the $\mathbf{T}$-weight of a monomial of degree $\operatorname{mult}_{o_i} h_{i,k}$, and the weights $(3, 4, -2, 2)$ are at least -2, so $\sum_k c_k\kappa_k \ge -2\sum_k c_k \operatorname{mult}_{o_i} h_{i,k} > -2$ by (E.95). Thus $-\delta = 1 - \sum_k c_k\kappa_k < 3$ and $\eta < \frac{1}{4}$. Hence $a(F, \mathbb{A}^2, \mathfrak{c}) \ge (1-2\eta)a(F, \mathbb{A}^2, 0) > 0$ for any F, and $(\mathbb{A}^2, \mathfrak{c})$ is klt.

Apply Construction-Proposition 4.1, with the ambient data (a), $n = 2$, and the group H, to the tuples $(\psi_{i,1}, \dots, \psi_{i,s}, \Phi)$ in case (a), $(A_{i,1}, \dots, A_{i,s}, Q)$ in case (b), and $(\psi_{i,1}, \dots, \psi_{i,s}, \zeta_1 + \zeta_2^2, \zeta_1)$ in case (c). After passing to a subsequence, we obtain an algebraically closed field $\mathbf{k} \supset \mathbb{C}$ and a compatible generic limit, whose entries are H-semi-invariant power series $\psi_{\infty,k}$, resp. $A_{\infty,k}$, in $\mathbf{k}[[\zeta_1, \zeta_2]]$, and Φ, Q, $\zeta_1 + \zeta_2^2$, and ζ_1 themselves, since, for any positive integer d, the d-jets of Φ, Q, $\zeta_1 + \zeta_2^2$, and ζ_1 do not depend on i and any non-empty open subset of the set Z_d of Construction-Proposition 4.1 contains the d-jets of the tuples for all but finitely many i. Let $\mathcal{A}_\mathbf{k} := \operatorname{Spec} \mathbf{k}[[\zeta_1, \zeta_2]]$, and let Δ_∞ be the $\mathbb{R}$-divisor on $\mathcal{A}_\mathbf{k}$ given by (E.96), resp. (E.97), with $\psi_{\infty,k}$, $A_{\infty,k}$ in place of $\psi_{i,k}$, $A_{i,k}$. In case (a), let $\phi_{\infty,k} := (-\Phi)^{l_k}\psi_{\infty,k}$. The labels c_k, $b_1 \ge \frac{1}{2}$, and b_2 are positive: if every term $c_k l_k$, resp. $c_k \max(-l_k, 0)$, of b_2 vanishes, then $\kappa_k \ge 0$ for any k, since $l_k = \lceil -\kappa_k/2 \rceil \ge 1$ if $\kappa_k < 0$ in case (a), resp. $\kappa_k = \epsilon_k + 2l_k$ in case (c), so $\delta = 1 + \sum_k c_k\kappa_k \ge 1$ and $b_2 \ge \frac{1}{2}$. The group H is diagonal, and in case (b), $(\mathbb{A}^2, \mathfrak{c})$ is klt. Moreover, in case (b), $\mathfrak{c} = \mathfrak{a}^\eta$ does not depend on i, since δ does not. Following Theorem 4.2, we write $\mathfrak{c}$ also for its pullback to $\mathcal{A}_\mathbf{k}$. By Theorem 4.2(1), after passing to a subsequence, $(\mathcal{A}_\mathbf{k}, \Delta_\infty, \mathfrak{c})$ is lc near 0, since $(\mathbb{A}^2, \Delta_i, \mathfrak{c})$ is lc near 0 for any i. By Theorem 4.2(2), applied to the ambient data (a) of Construction-Proposition 4.1 with $n = 2$, after passing to a subsequence,

$$\beta = \operatorname{mld}\big((\mathcal{A}_\mathbf{k} \ni 0, \Delta_\infty, \mathfrak{c})/H\big). \tag{E.98}$$

Step 3. In this step we show that $(\mathcal{A}_\mathbf{k}, \Delta_\infty, \mathfrak{c})$ is klt near 0, that is, the limit has no floor.

By Construction-Proposition 4.1, for any positive integer d, the d-jet of the limit tuple maps to the generic point of an irreducible closed subset Z_d of the space $\mathcal{J}_d$ of d-jets, and any non-empty open subset of Z_d contains the d-jets of the tuples for all but finitely many i. Let $\mathcal{C} \subset \mathcal{J}_d$ be a closed subset. If $Z_d \not\subset \mathcal{C}$, then $Z_d \backslash \mathcal{C}$ contains the d-jets for all but finitely many i. If $Z_d \subset \mathcal{C}$, then $\mathcal{C}$ contains the d-jets for all but finitely many i. Moreover, $Z_d \subset \mathcal{C}$ if and only if the d-jet of the limit tuple belongs to $\mathcal{C}$. Hence:

(∗) if the d-jets of the tuples belong to $\mathcal{C}$ for infinitely many i, then the d-jet of the limit tuple belongs to $\mathcal{C}$, and if the d-jet of the limit tuple belongs to $\mathcal{C}$, then the d-jets of the tuples belong to $\mathcal{C}$ for all but finitely many i.

For any k, we define a power series $h^0_{\infty,k} \in \mathbf{k}[[x, y, z, t]]$ as follows. In case (a), $h^0_{\infty,k} := y^{\kappa_k}\psi_{\infty,k}(x,t)$ if $\kappa_k \ge 0$, and $h^0_{\infty,k} := y^{\epsilon_k} z^{l_k} \psi_{\infty,k}(x,t)$ if $\kappa_k < 0$. In case (b),

$$h^0_{\infty,k} := x^{\epsilon_k} z^{-\kappa'_k} A_{\infty,k}(yz^2, tz).$$

In case (c), $h^0_{\infty,k} := x^{\epsilon_k} y^{l_k} \psi_{\infty,k}(yz,t)$ if $l_k \geq 0$, and $h^0_{\infty,k} := x^{\epsilon_k} z^{-l_k} \psi_{\infty,k}(yz,t)$ if $l_k < 0$. By Construction E.17, the formulas defining $h^0_{\infty,k}$ with $\psi_{i,k}$, $A_{i,k}$ in place of $\psi_{\infty,k}$, $A_{\infty,k}$ define functions equal to $h^0_{i,k}$ on W^0. In case (b), $h^0_{\infty,k}$ is a power series: since $h^0_{i,k}$ is a polynomial, the coefficient of $\zeta_1^p \zeta_2^{p'}$ in $A_{i,k}$ vanishes for any i whenever $2p + p' < \kappa'_k$, which is a closed condition on d-jets, so, by $(*)$, the coefficient of $\zeta_1^p \zeta_2^{p'}$ in $A_{\infty,k}$ vanishes whenever $2p + p' < \kappa'_k$.

Let $W^0_{\mathbf{k}} := (f^0 = 0) \subset \operatorname{Spec} \mathbf{k}[[x,y,z,t]]$, and let σ_k be the order of $h^0_{\infty,k}$ at 0 on $W^0_{\mathbf{k}}$. We claim that

$$\sigma_k \leq \operatorname{mult}_{o_i} h_{i,k} \quad \text{for any } k \text{ and } i \text{ large.} \tag{E.99}$$

Let σ be a non-negative integer with $\sigma \leq \sigma_k$, and let d be a positive integer with $d \geq \sigma + |\kappa'_k|$ in case (b) and $d \geq \sigma$ otherwise. Changing $\psi_{\infty,k}$, resp. $A_{\infty,k}$, by terms of degree at least d changes the series given by the formula defining $h^0_{\infty,k}$ by elements of the σ-th power of the maximal ideal. Thus the condition that the series defined from a d-jet by the formula defining $h^0_{\infty,k}$ has order at least σ on $W^0_{\mathbf{k}}$, resp. on W^0, is a linear condition on the d-jet, and defines a closed subset of $\mathcal{J}_d$. By $(*)$, $h^0_{i,k}|_{W^0}$ has order at least σ for i large, so $\sigma \leq \operatorname{mult}_{o_i} h_{i,k} \leq 1/c_k$ by (E.94) and (E.95). Hence σ_k is finite, and $\sigma = \sigma_k$ gives (E.99).

Assume that $(\mathcal{A}_{\mathbf{k}}, \Delta_\infty, \mathfrak{c})$ is not klt near 0. We first show that there exists an integral curve $\Gamma \subset \mathcal{A}_{\mathbf{k}}$ with $\operatorname{mult}_\Gamma \Delta_\infty = 1$. In cases (a) and (c), $\mathfrak{c} = \mathcal{O}$, the pairs $(\mathbb{A}^2, \Delta_i)$ are lc near 0, and $\beta_i = \beta > 0$, so Definition-Proposition 5.7 implies that $(\mathcal{A}_{\mathbf{k}}, \Delta_\infty)$ has a floor Γ, a curve with coefficient one in Δ_∞. In case (b), the hypothesis $\mathfrak{c} = \mathcal{O}$ of Definition-Proposition 5.7 may fail, and we argue directly. Since $\mathcal{A}_{\mathbf{k}}$ is the spectrum of a local ring, there exists a prime divisor F over $\mathcal{A}_{\mathbf{k}}$ with $a(F, \mathcal{A}_{\mathbf{k}}, \Delta_\infty, \mathfrak{c}) \leq 0$. By (E.98) and Definition 2.4, for any prime divisor F' over $\mathcal{A}_{\mathbf{k}}$ with center 0,

$$a(F', \mathcal{A}_{\mathbf{k}}, \Delta_\infty, \mathfrak{c}) \geq |I_H(F')|\beta > 0,$$

so the center of F is an integral curve Γ, and $F = \Gamma$. Since $\mathfrak{a} \supset (\zeta_1^3, \zeta_2^6)$, we have $\operatorname{mult}_\Gamma \mathfrak{c} = 0$, so $\operatorname{mult}_\Gamma \Delta_\infty \geq 1$, and $\operatorname{mult}_\Gamma \Delta_\infty = 1$ since the pair is lc.

Let $\gamma \in \mathbf{k}[[\zeta_1, \zeta_2]]$ be a prime element defining Γ. We show that, in each case, there exist numbers q_k with

$$\sum_k c_k q_k = 1 \quad \text{and} \quad q_k \leq \sigma_k \quad \text{for any } k, \tag{E.100}$$

or Γ is excluded directly.

Case (a). Let $\Gamma \neq (\Phi = 0)$ and $j_k := \operatorname{mult}_\Gamma \psi_{\infty,k}$. Since Φ is irreducible, $\operatorname{mult}_\Gamma \phi_{\infty,k} = j_k$ and the coefficient of Γ in Δ_∞ is $\sum_k c_k j_k$. Since $\gamma \in (x,t)$, we have $\psi_{\infty,k} \in (x,t)^{j_k}$, so $j_k \leq \sigma_k$, and $q_k := j_k$ satisfies (E.100). Let $\Gamma = (\Phi = 0)$, and write $\psi_{\infty,k} = \Phi^{j_k} \psi'_k$ with $\Phi \nmid \psi'_k$. On $W^0_{\mathbf{k}}$, $\Phi = -y^2 z$, so

$$h^0_{\infty,k} = (-1)^{j_k} y^{\kappa_k + 2j_k} z^{j_k} \psi'_k \text{ if } \kappa_k \geq 0, \qquad h^0_{\infty,k} = (-1)^{j_k} y^{\epsilon_k + 2j_k} z^{l_k + j_k} \psi'_k \text{ if } \kappa_k < 0.$$

Since $l_k = 0$ if $\kappa_k \geq 0$ and $\kappa_k + 2l_k = \epsilon_k$ if $\kappa_k < 0$, we obtain

$$\sigma_k \geq \max(\kappa_k + 2l_k + 2j_k, l_k + j_k).$$

The coefficient of Γ in Δ_∞ is $\sum_k c_k(l_k + j_k) + \max(\delta, 0)/2$. If $\delta \geq 0$, then $\delta = 1 + \sum_k c_k \kappa_k$, and coefficient one means $\sum_k c_k(\kappa_k + 2l_k + 2j_k) = 1$, so $q_k := \kappa_k + 2l_k + 2j_k$ satisfies (E.100). If $\delta < 0$, coefficient one means $\sum_k c_k(l_k + j_k) = 1$, and $q_k := l_k + j_k$ satisfies (E.100).

Case (b). Let $j_k := \operatorname{mult}_\Gamma A_{\infty,k}$. The ring $\mathbf{k}[[y,z,t]]$ is factorial [Mat89, Theorem 20.3], and we use the following remark: if $z \nmid P$ and $z^{-\kappa'_k} A_{\infty,k}(yz^2, tz) = P^j R$ with $R \in \mathbf{k}[[y,z,t]][z^{-1}]$, then $R \in \mathbf{k}[[y,z,t]]$, since the left hand side is a power series. Let $\Gamma = (Q = 0)$, and write $A_{\infty,k} = Q^{j_k} A'_k$. Since $Q(yz^2, tz) = z^3 g^0$ and $z \nmid g^0$, we obtain $z^{-\kappa'_k} A_{\infty,k}(yz^2, tz) = (g^0)^{j_k} R_k$ with $R_k \in \mathbf{k}[[y,z,t]]$. On $W^0_{\mathbf{k}}$, $g^0 = -x^2$, so $h^0_{\infty,k} = (-1)^{j_k} x^{\epsilon_k + 2j_k} R_k$ and $\sigma_k \geq \epsilon_k + 2j_k$. The coefficient of Γ in Δ_∞ is $b_1 + \sum_k c_k j_k$, and coefficient one means $\sum_k c_k(\epsilon_k + 2j_k) = 1$. Thus $q_k := \epsilon_k + 2j_k$ satisfies (E.100). Let $\Gamma \neq (Q = 0)$. Write $\gamma(yz^2, tz) = z^{a_\gamma} \bar{\gamma}$ with $z \nmid \bar{\gamma}$. A monomial

$\zeta_1^p\zeta_2^{p'}$ of γ yields $y^p t^{p'} z^{2p+p'}$, and $\gamma(0)=0$, so the part of $\bar\gamma$ independent of z is a non-zero series without constant term, and $\bar\gamma(0)=0$. Thus $z^{-\kappa'_k}A_{\infty,k}(yz^2,tz)=\bar\gamma^{j_k}R_k$ with $R_k\in\mathbf{k}[[y,z,t]]$, and $\sigma_k\ge j_k$. The coefficient of Γ in Δ_∞ is $\sum_k c_k j_k$, so $q_k:=j_k$ satisfies (E.100).

Case (c). Let $\Gamma=(\zeta_1=0)$, and let E be the prime divisor over $\mathcal{A}_\mathbf{k}$ with center 0 given by the weights $(2,1)$ in ζ_1,ζ_2. Then $a(E,\mathcal{A}_\mathbf{k},0)=3$, $\operatorname{mult}_E(\zeta_1+\zeta_2^2)=2$, and $\operatorname{mult}_E\psi_{\infty,k}\ge 2\operatorname{mult}_\Gamma\psi_{\infty,k}$. Since the coefficient $b_2+\sum_k c_k\operatorname{mult}_\Gamma\psi_{\infty,k}$ of Γ in Δ_∞ is one and $b_1\ge\frac12$,

$$a(E,\mathcal{A}_\mathbf{k},\Delta_\infty)\le 3-2b_1-2\Big(b_2+\sum_k c_k\operatorname{mult}_\Gamma\psi_{\infty,k}\Big)=1-2b_1\le 0,$$

which contradicts (E.98), Definition 2.4 (with $\mathfrak{c}=\mathcal{O}$), and $\beta>0$. Let $\Gamma=(\zeta_1+\zeta_2^2=0)$, and write $\psi_{\infty,k}=(\zeta_1+\zeta_2^2)^{j_k}\psi'_k$. Then $\psi_{\infty,k}(yz,t)=(yz+t^2)^{j_k}\psi'_k(yz,t)$ and $\operatorname{mult}_0(yz+t^2)=2$, so $\sigma_k\ge\epsilon_k+2j_k$. Coefficient one means $b_1+\sum_k c_k j_k=1$, that is, $\sum_k c_k(\epsilon_k+2j_k)=1$, and $q_k:=\epsilon_k+2j_k$ satisfies (E.100). For any other Γ, let $j_k:=\operatorname{mult}_\Gamma\psi_{\infty,k}$. Since $\gamma(yz,t)\in(y,z,t)$, we have $\sigma_k\ge j_k$, and the coefficient of Γ in Δ_∞ is $\sum_k c_k j_k$, so $q_k:=j_k$ satisfies (E.100).

In all cases, (E.100), (E.99), and (E.95) imply, for i large,

$$1=\sum_k c_k q_k\le\sum_k c_k\sigma_k\le\sum_k c_k\operatorname{mult}_{o_i}h_{i,k}\le 1-\frac{1}{m},$$

a contradiction. Thus $(\mathcal{A}_\mathbf{k},\Delta_\infty,\mathfrak{c})$ is klt near 0.

Step 4. In this step we find $\theta>1$ such that $(\mathbb{A}^2,\Delta_i(\theta c),\mathfrak{c}(\theta c))$ is lc near 0 for any i.

By **Step 3** and Theorem 4.2(1), after passing to a subsequence, there exists $\theta'>1$ such that $(\mathbb{A}^2,\theta'\Delta_i,\mathfrak{c})$ is lc near 0 for any i. Let $\tau:=\sum_k c_k\kappa_k$ be the mass. Scaling the labels by $\theta\ge1$ replaces τ with $\theta\tau$, so $1+\tau$ becomes $\theta(1+\tau)-(\theta-1)\le\theta(1+\tau)$, and $\tau-1$ becomes $\theta(\tau-1)+(\theta-1)\ge\theta(\tau-1)$. Thus, whatever the sign of δ,

$$\max(\delta(\theta c),0)\le\theta\max(\delta,0)\text{ in cases (a) and (c)},\qquad \max(-\delta(\theta c),0)\le\theta\max(-\delta,0)\text{ in case (b)},$$

and the sign of δ need not be preserved. Since moreover

$$\frac{1+\theta\sum_k c_k\epsilon_k}{2}\le\theta\frac{1+\sum_k c_k\epsilon_k}{2},$$

the formulas of Construction E.17 imply that

$$\Delta_i(\theta c)\le\theta\Delta_i,\qquad \mathfrak{c}(\theta c)=\mathcal{O}\text{ in cases (a) and (c)},\qquad \mathfrak{c}(\theta c)=\mathfrak{a}^{\eta_\theta}\text{ with }\eta_\theta\le\theta\eta\text{ in case (b)}.$$

In cases (a) and (c), let $\theta\in(1,\theta']$. Then $(\mathbb{A}^2,\Delta_i(\theta c))$ is lc near 0 for any i, since $\Delta_i(\theta c)\le\theta\Delta_i\le\theta'\Delta_i$. In case (b), since $2\eta<1$, we may take $\theta\in(1,\theta']$ with

$$\frac12\Big(1-\frac{\theta}{\theta'}\Big)\ge\theta\eta\Big(1-\frac{1}{\theta'}\Big).$$

Let $\vartheta:=\theta/\theta'\in(0,1]$. Then $\theta\Delta_i=\vartheta\theta'\Delta_i$ and $\eta_\theta\le\theta\eta\le\vartheta\eta+(1-\vartheta)/2$. Since the log discrepancy is affine in the boundary and in the exponent of $\mathfrak{a}$, for any prime divisor F over $\mathbb{A}^2$ whose center intersects a neighborhood of 0 on which $(\mathbb{A}^2,\theta'\Delta_i,\mathfrak{a}^\eta)$ is lc, **Step 2** implies

$$\begin{aligned}a\big(F,\mathbb{A}^2,\Delta_i(\theta c),\mathfrak{a}^{\eta_\theta}\big)&\ge a\big(F,\mathbb{A}^2,\vartheta\theta'\Delta_i,\mathfrak{a}^{\vartheta\eta+(1-\vartheta)/2}\big)\\&=\vartheta a\big(F,\mathbb{A}^2,\theta'\Delta_i,\mathfrak{a}^{\eta}\big)+(1-\vartheta)a\big(F,\mathbb{A}^2,\mathfrak{a}^{1/2}\big)\ge 0.\end{aligned}$$

Thus $(\mathbb{A}^2,\Delta_i(\theta c),\mathfrak{c}(\theta c))$ is lc near 0 for any i.

Step 5. In this step we show the following: for any $c'\in\mathbb{R}^s_{\ge0}$ and any i, if $(\mathbb{A}^2,\Delta_i(c'),\mathfrak{c}(c'))$ is lc near 0, then $(W^0,D_i^0(c'))$ is lc near 0, where $D_i^0(c')$ is as in Construction E.17.

Let $\delta':=\delta(c')$. Let $u:=y$ in cases (a) and (c) and $u:=z$ in case (b), and let pr be the projection of $\mathbb{A}^2_{\zeta_1,\zeta_2}\times\mathbb{A}^1_u$ onto $\mathbb{A}^2_{\zeta_1,\zeta_2}$. Let $U_2\ni0$ be an open subset of $\mathbb{A}^2_{\zeta_1,\zeta_2}$ on which $(\mathbb{A}^2,\Delta_i(c'),\mathfrak{c}(c'))$ is lc. Then

$$\big(U_2\times\mathbb{A}^1_u,\operatorname{pr}^*\Delta_i(c')+U_2\times\{0\},\operatorname{pr}^*\mathfrak{c}(c')\big)$$

is lc, because for a log resolution $\mu : U_2' \to U_2$ of $(U_2, \Delta_i(c'), \mathfrak{c}(c'))$, the morphism $\mu \times \mathrm{id}_{\mathbb{A}^1_u}$ is a log resolution of the product pair, and the prime divisors on $U_2' \times \mathbb{A}^1_u$ that are exceptional or contained in the support of the boundary or of the ideal are the strict transform of $U_2 \times \{0\}$, with coefficient one, and the prime divisors $P \times \mathbb{A}^1_u$, for prime divisors P on U_2', with

$$a\big(P \times \mathbb{A}^1_u, U_2 \times \mathbb{A}^1_u, \mathrm{pr}^*\, \Delta_i(c'), \mathrm{pr}^*\, \mathfrak{c}(c')\big) = a\big(P, U_2, \Delta_i(c'), \mathfrak{c}(c')\big) \geq 0.$$

Let $U_0 := W^0 \cap (\zeta_1, \zeta_2)^{-1}(U_2)$, an open neighborhood of 0 in W^0. We show that for any prime divisor F over W^0 whose center on W^0 intersects U_0, there exist a positive integer e_F and a prime divisor E_F over $\mathbb{A}^2_{\zeta_1,\zeta_2} \times \mathbb{A}^1_u$ whose center intersects $U_2 \times \mathbb{A}^1_u$, such that

$$a\big(F, W^0, D_i^0(c')\big) \geq e_F a\big(E_F, \mathbb{A}^2_{\zeta_1,\zeta_2} \times \mathbb{A}^1_u, \mathrm{pr}^*\, \Delta_i(c') + \mathrm{div}(u), \mathrm{pr}^*\, \mathfrak{c}(c')\big). \tag{E.101}$$

The right hand side is non-negative since the product pair above is lc, so $(W^0, D_i^0(c'))$ is lc on U_0. For such F, the functions x, y, z, t are regular at a point of the center of F, so $\mathrm{mult}_F\, x, \mathrm{mult}_F\, y, \mathrm{mult}_F\, z, \mathrm{mult}_F\, t \geq 0$.

Case (a). Here $(\zeta_1, \zeta_2, u) = (x, t, y)$, and $\delta' = 1 + \sum_k c_k' \kappa_k$. The projection $p : W^0 \to \mathbb{A}^3_{x,t,y}$ is birational and maps U_0 into $U_2 \times \mathbb{A}^1_u$. The Poincaré residue ω of $dx \wedge dy \wedge dz \wedge dt / f^0$ generates the canonical sheaf of W^0, and $p^*(dx \wedge dt \wedge dy) = \pm y^2 \omega$ since $\partial f^0 / \partial z = y^2$. Hence $a(F, W^0, 0) = a(F, \mathbb{A}^3, 2\,\mathrm{div}(y))$ for any prime divisor F over W^0. On W^0, $h^0_{i,k} = y^{\kappa_k} \phi_{i,k}(x, t)$ as rational functions, where for $\kappa_k < 0$ we use $z = -\Phi / y^2$. Thus

$$a\big(F, W^0, D_i^0(c')\big) = a\Big(F, \mathbb{A}^3, \mathrm{pr}^* \sum_k c_k' \,\mathrm{div}(\phi_{i,k}) + (1 + \delta')\,\mathrm{div}(y)\Big).$$

On W^0, $y^2 z = -\Phi$, so $2\,\mathrm{mult}_F\, y + \mathrm{mult}_F\, z = \mathrm{mult}_F\, \Phi$. If $\delta' \geq 0$, we replace $\delta'\,\mathrm{mult}_F\, y$ with $\frac{\delta'}{2}(\mathrm{mult}_F\, \Phi - \mathrm{mult}_F\, z)$. We obtain

$$a\big(F, W^0, D_i^0(c')\big) = a\big(F, \mathbb{A}^3, \mathrm{pr}^*\, \Delta_i(c') + \mathrm{div}(y)\big) + \max(-\delta', 0)\,\mathrm{mult}_F\, y + \frac{\max(\delta', 0)}{2}\,\mathrm{mult}_F\, z,$$

and (E.101) holds with $E_F = F$ and $e_F = 1$.

Case (b). Here $\delta' = \sum_k c_k' \kappa_k - 1$. The involution $(x, y, z, t) \to (-x, y, z, t)$ of W^0 generates a group $\langle \iota \rangle$ of order two, and the quotient is $\pi : W^0 \to \mathbb{A}^3_{y,z,t}$, ramified with index two along $(x = 0) = \pi^{-1}(g^0 = 0)$. Since $h^0_{i,k} = x^{\epsilon_k} z^{-\kappa_k'} A_{i,k}(yz^2, tz)$ and $2\,\mathrm{div}(x) = \pi^*\,\mathrm{div}(g^0)$, the $\mathbb{R}$-divisor $D_i^0(c')$ is $\langle \iota \rangle$-invariant, and the crepant quotient of $(W^0, D_i^0(c'))$ by $\langle \iota \rangle$ is $(\mathbb{A}^3_{y,z,t}, B')$ with

$$B' := b_1' \,\mathrm{div}(g^0) + \sum_k c_k' \,\mathrm{div}\big(z^{-\kappa_k'} A_{i,k}(yz^2, tz)\big), \qquad b_1' := \frac{1 + \sum_k c_k' \epsilon_k}{2}.$$

By [Liu26, Lemma 2.6(2)],

$$a(F, W^0, D_i^0(c')) = |I_{\langle \iota \rangle}(F)| a(E, \mathbb{A}^3_{y,z,t}, B'),$$

where E is the prime divisor over $\mathbb{A}^3_{y,z,t}$ under F, whose center intersects $\pi(U_0)$. The morphism $q : \mathbb{A}^3_{y,z,t} \to \mathbb{A}^2 \times \mathbb{A}^1_u$, $(y, z, t) \to (yz^2, tz, z)$, is birational, maps $\pi(U_0)$ into $U_2 \times \mathbb{A}^1_u$, and satisfies

$$q^*(d\zeta_1 \wedge d\zeta_2 \wedge du) = z^3 dy \wedge dt \wedge dz, \qquad q^* Q = z^3 g^0.$$

Since $3 - 3b_1' - \sum_k c_k' \kappa_k' = 1 - \frac{\delta'}{2}$, we obtain

$$a(E, \mathbb{A}^3_{y,z,t}, B') = a\Big(E, \mathbb{A}^2_{\zeta_1,\zeta_2} \times \mathbb{A}^1_u, \mathrm{pr}^*\, \Delta_i(c') + \Big(1 - \frac{\delta'}{2}\Big)\,\mathrm{div}(u)\Big).$$

We have $\mathrm{mult}_E\, y, \mathrm{mult}_E\, z, \mathrm{mult}_E\, t \geq 0$, so $\mathrm{mult}_E\, \zeta_1 \geq 2\,\mathrm{mult}_E\, u$, $\mathrm{mult}_E\, \zeta_2 \geq \mathrm{mult}_E\, u$, $\mathrm{mult}_E\, Q \geq 3\,\mathrm{mult}_E\, u$, and $\mathrm{mult}_E\, \mathfrak{a} \geq 6\,\mathrm{mult}_E\, u$. If $\delta' \geq 0$, then $\mathfrak{c}(c') = \mathcal{O}$ and the right hand side is

$$a\big(E, \mathbb{A}^2 \times \mathbb{A}^1_u, \mathrm{pr}^*\, \Delta_i(c') + \mathrm{div}(u)\big) + \frac{\delta'}{2}\,\mathrm{mult}_E\, u.$$

If $\delta' < 0$, then

$$\frac{-\delta'}{2}\operatorname{mult}_E u \le \frac{\max(-\delta', 0)}{12}\operatorname{mult}_E \mathfrak{a},$$

and the right hand side is at least

$$a\big(E, \mathbb{A}^2 \times \mathbb{A}^1_u, \operatorname{pr}^* \Delta_i(c') + \operatorname{div}(u), \operatorname{pr}^* \mathfrak{c}(c')\big).$$

Thus (E.101) holds with E_F the prime divisor E over $\mathbb{A}^2 \times \mathbb{A}^1_u$ and $e_F = |I_{\langle\iota\rangle}(F)|$.

Case (c). Here $(\zeta_1, \zeta_2) = (yz, t)$ and $\delta' = 1 + \sum_k c'_k \kappa_k$. Let ι and $\pi : W^0 \to \mathbb{A}^3_{y,z,t}$ be as in case (b), now ramified along $(x = 0) = \pi^{-1}(y(\zeta_1 + \zeta_2^2) = 0)$. As rational functions,

$$h^0_{i,k} = x^{\epsilon_k} y^{l_k} \zeta_1^{\max(-l_k, 0)} \psi_{i,k}(\zeta_1, \zeta_2),$$

and $2\operatorname{div}(x) = \pi^* \operatorname{div}(g^0)$ with $g^0 = y(\zeta_1 + \zeta_2^2)$. Since $b'_1 + \sum_k c'_k l_k = (1 + \sum_k c'_k \kappa_k)/2 = \delta'/2$ for $b'_1 := (1 + \sum_k c'_k \epsilon_k)/2$, the crepant quotient of $(W^0, D^0_i(c'))$ by $\langle\iota\rangle$ is $(\mathbb{A}^3_{y,z,t}, B')$ with

$$B' := b'_1 \operatorname{div}(\zeta_1 + \zeta_2^2) + \frac{\delta'}{2}\operatorname{div}(y) + \sum_k c'_k \max(-l_k, 0)\operatorname{div}(\zeta_1) + \sum_k c'_k \operatorname{div}(\psi_{i,k}(\zeta_1, \zeta_2)),$$

and, by [Liu26, Lemma 2.6(2)],

$$a(F, W^0, D^0_i(c')) = |I_{\langle\iota\rangle}(F)| a(E, \mathbb{A}^3_{y,z,t}, B').$$

The morphism $q : \mathbb{A}^3_{y,z,t} \to \mathbb{A}^2 \times \mathbb{A}^1_u$, $(y, z, t) \to (yz, t, y)$, is birational, maps $\pi(U_0)$ into $U_2 \times \mathbb{A}^1_u$, and satisfies

$$q^*(d\zeta_1 \wedge d\zeta_2 \wedge du) = \pm y dy \wedge dz \wedge dt.$$

Hence

$$a(E, \mathbb{A}^3_{y,z,t}, B') = a\Big(E, \mathbb{A}^2_{\zeta_1,\zeta_2} \times \mathbb{A}^1_u, \operatorname{pr}^*\Big(\Delta_i(c') - \frac{\max(\delta', 0)}{2}\operatorname{div}(\zeta_1)\Big) + \Big(1 + \frac{\delta'}{2}\Big)\operatorname{div}(u)\Big).$$

Since $\operatorname{mult}_E \zeta_1 = \operatorname{mult}_E u + \operatorname{mult}_E z$ and $\operatorname{mult}_E u, \operatorname{mult}_E z \ge 0$, we obtain

$$a(E, \mathbb{A}^3_{y,z,t}, B') = a\big(E, \mathbb{A}^2 \times \mathbb{A}^1_u, \operatorname{pr}^* \Delta_i(c') + \operatorname{div}(u)\big) + \frac{\max(-\delta', 0)}{2}\operatorname{mult}_E u + \frac{\max(\delta', 0)}{2}\operatorname{mult}_E z,$$

and (E.101) holds with $E_F = E$ and $e_F = |I_{\langle\iota\rangle}(F)|$.

Step 6. In this step we show that $(W_i, \theta D_i)$ and $(W_i, \theta \sum_k c_k \operatorname{div}(h^0_{i,k}))$ are lc near o_i for any i, with θ as in **Step 4**.

Consider on $\mathbb{A}^4$, with the trivial group, the three sequences of tuples

$$\mathcal{S}_1 : (f^0, h^0_{i,1}, \dots, h^0_{i,s}), \qquad \mathcal{S}_2 : (f_i, h_{i,1}, \dots, h_{i,s}), \qquad \mathcal{S}_3 : (f_i, h^0_{i,1}, \dots, h^0_{i,s}),$$

with the labels $(1, \theta c_1, \dots, \theta c_s)$. Their orders at 0 are bounded by (E.94) and (E.95). By Construction-Proposition 4.1, with the ambient data (a), after passing to a subsequence, $\mathcal{S}_2$ has a compatible generic limit $(f_\infty, h_{\infty,1}, \dots, h_{\infty,s})$ over an algebraically closed field $\mathbf{k}' \supset \mathbb{C}$. By (E.93), for any d, the d-jets of $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$ coincide for i large. Hence $(f_\infty, h_{\infty,1}, \dots, h_{\infty,s})$ is also a compatible generic limit of $\mathcal{S}_1$ and of $\mathcal{S}_3$. By Theorem 4.2(1) with $\mathfrak{c} = \mathcal{O}$, applied with the real number 1 to $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$, after passing to a subsequence, the following holds for $j = 1, 2, 3$: the pair

$$\Big(\operatorname{Spec} \mathbf{k}'[[x, y, z, t]], \operatorname{div}(f_\infty) + \theta \sum_k c_k \operatorname{div}(h_{\infty,k})\Big)$$

is lc near 0 if and only if the pair on $\mathbb{A}^4$ defined by $\mathcal{S}_j$ is lc near 0 for any i.

By **Steps 4 and 5** with $c' = \theta c$, the pair $(W^0, D^0_i(\theta c))$ is lc near 0 for any i. The hypersurface W^0 is normal, and $h^0_{i,k}$ is a non-zero polynomial of x-degree at most one, hence not divisible by f^0. By [Kaw07, Theorem] (the different is the restriction since W^0 is Cartier, [KM98, Remark 5.47]),

$(\mathbb{A}^4, W^0 + \theta \sum_k c_k \operatorname{div}(h^0_{i,k}))$ is lc near 0 for any i, which is the pair defined by $\mathcal{S}_1$. Thus the limit pair above is lc near 0, and the pairs defined by $\mathcal{S}_2$ and $\mathcal{S}_3$,

$$\Big(\mathbb{A}^4, W_i + \theta \sum_k c_k \operatorname{div}(h_{i,k})\Big) \quad \text{and} \quad \Big(\mathbb{A}^4, W_i + \theta \sum_k c_k \operatorname{div}(h^0_{i,k})\Big),$$

are lc near 0 for any i. The hypersurface W_i is normal, and $h_{i,k}$ and $h^0_{i,k}$ are non-zero polynomials of x-degree at most one, hence not divisible by f_i. By [Kaw07, Theorem] (the different is the restriction since W_i is Cartier, [KM98, Remark 5.47]), $(W_i, \theta D_i)$ and $(W_i, \theta \sum_k c_k \operatorname{div}(h^0_{i,k}))$ are lc near o_i for any i.

Steps 1, 4, and 6 prove the lemma. □

E.9. **Cuspidal cubics and tangent conics: lower bounds.** In this subsection, we prove Proposition 11.15. The lower bounds are proved on charts which carry a section with coefficient one, through the following form of Lemma 5.5 at points away from the origin.

Lemma E.22. *Let $n \geq 3$ be an integer, G a finite abelian group acting diagonally on $\mathbb{A}^n$, $X \subset \mathbb{A}^n$ either $\mathbb{A}^n$ or the common zero scheme of G-semi-invariant polynomials $f_1, \dots, f_c$, u a G-semi-invariant polynomial, $S_0 := X \cap (u = 0)$, and $\mathfrak{a}$ an $\mathbb{R}$-ideal on $\mathbb{A}^n$ whose factors are generated by G-semi-invariant polynomials. Let $p \in S_0$ be a closed point, and let $V \subset S_0$ be either $\{p\}$ or an integral curve through p. Assume that near p, X is a normal integral complete intersection with klt singularities, S_0 is a normal integral complete intersection of dimension at least 2 with klt singularities, G acts faithfully on X and on S_0, $\mathfrak{a}|_{S_0}$ is non-zero, and $(S_0, \mathfrak{a}|_{S_0})$ is lc. Let β_V be the infimum of $a(E, S_0, \mathfrak{a}|_{S_0})/|I_G(E)|$ over all prime divisors E over S_0 with center V. Then any prime divisor P over X with center V satisfies*

$$\frac{a(P, X, S_0, \mathfrak{a}|_X)}{|I_G(P)|} \geq \beta_V.$$

Proof. Let $H := I_G(P)$. Since H acts trivially on the residue field of P, which contains the function field of V, H fixes V pointwise. In particular H fixes $p = (p_1, \dots, p_n)$. If $h \in H$ acts by λ_j on the coordinate x_j, then $\lambda_j p_j = p_j$, so $\lambda_j = 1$ whenever $p_j \neq 0$, and hence $h(x_j - p_j) = \lambda_j(x_j - p_j)$ for any j. Thus H acts diagonally on the coordinates $x_j - p_j$, and $f_1, \dots, f_c$, u, and the generators of the factors of $\mathfrak{a}$ are H-semi-invariant polynomials in the coordinates $x_j - p_j$, where $f_1, \dots, f_c$ and u vanish at p. The hypotheses on X, S_0, and $\mathfrak{a}$ are local at p, and H acts faithfully on X and on S_0 as a subgroup of G. Hence Lemma 5.5, applied in the coordinates $x_j - p_j$ with the group H in place of both G and H, with $b = 1$, and with the curve V when V is a curve, implies

$$\frac{a(P, X, S_0, \mathfrak{a}|_X)}{|I_H(P)|} \geq \beta'_V,$$

where β'_V is the minimal log discrepancy of the quotient of $(S_0, \mathfrak{a}|_{S_0})$ by H at the image of the generic point of V. By [Liu26, Lemma 2.6], β'_V is the infimum of $a(E, S_0, \mathfrak{a}|_{S_0})/|I_H(E)|$ over all prime divisors E over S_0 with center V. Since $I_H(P) = I_G(P)$ and $I_H(E) = H \cap I_G(E) \subset I_G(E)$ for any such E, we have $\beta'_V \geq \beta_V$ (see also [Liu26, Lemma 3.1]), and Lemma E.22 follows. □

Proof of Proposition 11.15. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the data and reduce the proposition to a lower bound for the quotient values.

Suppose that there exists a sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with W_i of quadratic rank one whose residual cubic defines a cuspidal cubic or a tangent conic, whose numbers

$$\alpha_i := \operatorname{mld}((W_i \ni o_i, \textstyle\sum_k c_k \operatorname{div}(h_{i,k}))/G_i)$$

form a strictly increasing sequence in $(0, 1)$. By Lemma E.16, after passing to a subsequence and replacing the data with numerical admissions whose groups are again the G_i, which does

not change the numbers α_i, we have $W_i = (f_i = 0)$ with $f_i = x^2 + g_i(y,z,t)$, the group G_i is diagonal on x,y,z,t, and there exist a circle $\mathbf{T} = \mathbf{T}_{(w)}$ in the list of Lemma E.16 and a positive integer e_0 such that $r_i := |R_i| \to +\infty$ and $[G_i : R_i] \le e_0$, where $R_i := G_i \cap \mathbf{T}$. We use the notation of Construction E.17, and we call the cases $\mathbf{T} = \mathbf{T}_{(0,1,-2,0)}$, $\mathbf{T}_{(3,4,-2,2)}$, and $\mathbf{T}_{(1,2,-2,0)}$ the *cases (a), (b), and (c)*. Let $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$. By Lemmas E.18 and E.21, after passing to a subsequence, $\beta := \beta_i$ is a positive real number independent of i, and there exist real numbers $C_0 \ge 0$ and $\theta > 1$ such that for i large, $\alpha_i \le \beta + C_0/r_i$, and the pairs $(W_i, \theta D_i)$ and $(W_i, \theta \sum_k c_k \operatorname{div}(h^0_{i,k}))$ are lc near o_i. Since the α_i are strictly increasing and $r_i \to +\infty$,

$$\alpha_i < \lim_{j\to+\infty} \alpha_j \le \beta \quad \text{for any } i. \tag{E.102}$$

We show that $\alpha_i \ge \beta$ for i large, which contradicts (E.102).

In each case δ is the adjusted mass of Set-up 5.1. Since the principal block of f_i is the sum of the terms of f_i of $\mathbf{T}$-weight $\kappa(f_i) = \kappa(x^2)$, we have $\epsilon_{\mathbf{T}} = (0+1-2+0) - 0 = -1$ in case (a), $\epsilon_{\mathbf{T}} = (3+4-2+2) - 6 = 1$ in case (b), and $\epsilon_{\mathbf{T}} = (1+2-2+0) - 2 = -1$ in case (c). Thus $\delta = 1 + \sum_k c_k\kappa_k$ in cases (a) and (c), and $\delta = \sum_k c_k\kappa_k - 1$ in case (b), as in Construction E.17.

By Construction E.17, after the replacement made there, the equations f_i have the following form. For i large, the principal block of f_i along $\mathbf{T}$ is the polynomial f^0 of Construction E.17, that is, $x^2 + y^2z + t^3$ in cases (a) and (b) and $x^2 + y^2z + yt^2$ in case (c). In cases (a) and (c), $f_i = f^0 + B_i(z,t)$. In case (b), $f_i = f^0 + B_i(y,z,t)$, where the monomials of B_i of $\mathbf{T}$-weight less than 6 do not involve y. Since f_i is R_i-semi-invariant with the character of x^2, any monomial of B_i has $\mathbf{T}$-weight in $\kappa(f^0) + r_i(\mathbb{Z}\backslash\{0\})$, where $\kappa(f^0) = 0, 6, 2$ in cases (a), (b), (c). Let W^0 be as in Construction E.17.

From now on we drop the index i from the marks, and write $h_k := h_{i,k}$ and $h^0_k := h^0_{i,k}$, each h_k in normal form (Definition-Lemma 11.5(1)). By Definition-Lemma 11.5(2), any monomial M of $h_k - h^0_k$ has $\mathbf{T}$-weight $\kappa_k + nr_i$ for an integer $n \ne 0$. We call M *positive* if $n > 0$ and *negative* if $n < 0$, and we let h^+_k (resp. h^-_k) be the sum of the terms of h_k which are not negative (resp. not positive). The marks $h^\pm_k$ have the character of h_k. A unit h_k has $\kappa_k = 0$, and it adds 0 to any of the sums of orders below. For a prime divisor F over W_i with center o_i we write

$$e := |I_{G_i}(F)|, \qquad [f] := \frac{\operatorname{mult}_F f}{e}, \qquad a_{D_i}(F) := \frac{a(F, W_i, D_i)}{e}.$$

By [Liu26, Lemma 2.6], α_i is the infimum of $a_{D_i}(F)$ over all prime divisors F over W_i with center o_i.

Step 2. In this step we bound the orders at a prime divisor of small score, and we compute the coefficients of the radial parameters on the charts.

Let F be a prime divisor over W_i with center o_i and $a_{D_i}(F) \le 3$. Since $(W_i, \theta D_i)$ is lc near o_i, $a(F, W_i, 0) \ge \theta \sum_k c_k \operatorname{mult}_F h_k$. Since $a(F, W_i, 0) - \sum_k c_k \operatorname{mult}_F h_k = e a_{D_i}(F) \le 3e$, we obtain $a(F, W_i, 0) \le 3e + a(F, W_i, 0)/\theta$. Hence

$$\frac{a(F, W_i, 0)}{e} \le C_1 := \frac{3\theta}{\theta - 1}, \qquad [h_k] \le \frac{a(F, W_i, 0)}{c_k \theta e} \le \frac{3}{c_k(\theta - 1)} \le C_2 \tag{E.103}$$

for any k, where $C_2 := \max_k 3/(c_k(\theta-1))$. Since $(W_i, \theta \sum_k c_k \operatorname{div}(h^0_k))$ is lc near o_i, $a(F, W_i, 0) \ge \theta \sum_k c_k \operatorname{mult}_F h^0_k$, so the first inequality of (E.103) gives $[h^0_k] \le a(F, W_i, 0)/(c_k\theta e) \le C_2$ for any k. On the other hand, $\beta \le 2$: the exceptional divisor E_1 of the blow-up of the origin of $\mathbb{A}^2_{\zeta_1,\zeta_2}$ is invariant under the diagonal group H_i, and $a(E_1, \mathbb{A}^2, \Delta_i, \mathfrak{c}) \le a(E_1, \mathbb{A}^2, 0) = 2$. Thus any prime divisor F over W_i with center o_i and $a_{D_i}(F) > 3$ satisfies $a_{D_i}(F) > \beta$, and it suffices to prove that for i large,

$$a_{D_i}(F) \ge \beta \quad \text{for any prime divisor } F \text{ over } W_i \text{ with center } o_i \text{ and } a_{D_i}(F) \le 3. \tag{E.104}$$

Next we bound the orders of the coordinates. Near 0, the singular locus of W^0 is the z-axis, and at the points of the z-axis other than 0, W^0 is locally the product of a curve and a Du Val singularity of type A_2 in cases (a) and (b), and of type A_3 in case (c) (the partial derivatives of

f^0 vanish together only on the z-axis, and at $z = a \neq 0$ one completes the square in y). A general hyperplane section H_0 of W^0 through 0 is a Du Val surface of type D_4: for $z = a_1x + a_2y + a_3t$ with general a_j, completing the square in x leaves a binary cubic form in (y,t) with three distinct linear factors. By [Kaw07, Theorem] (the different is the restriction since H_0 is Cartier on W^0, [KM98, Remark 5.47]), (W^0, H_0) is lc near 0, so $a(E, W^0, 0) \geq \mathrm{mult}_E H_0 > 0$ for any prime divisor E over W^0 with center 0. Thus W^0 is klt near 0, and there exists a rational number $\eta > 0$ such that $(W^0, \eta \operatorname{div}(xyzt))$ is klt near 0 [KM98, Corollary 2.35(2)]. By [Kaw07, Theorem] (the different is the restriction since W^0 is Cartier, [KM98, Remark 5.47]), the pair

$$(\mathbb{A}^4, \operatorname{div}(f^0) + \eta \operatorname{div}(xyzt))$$

is lc near 0. Any monomial of $f_i - f^0$ has $\mathbf{T}$-weight $\kappa(f^0) + nr_i$ with $n \neq 0$, hence degree at least $(r_i - |\kappa(f^0)|)/\max_j |w_j|$. By Lemma 4.14 with $\Lambda = \{1, \eta\}$, applied on the completion of the local ring of $\mathbb{A}^4$ at 0 (see [dFEM11, Proposition 2.11(i)]), the pair

$$(\mathbb{A}^4, \operatorname{div}(f_i) + \eta \operatorname{div}(xyzt))$$

is lc near 0 for i large, and by [Kaw07, Theorem] (the different is the restriction since W_i is Cartier, [KM98, Remark 5.47]), $(W_i, \eta \operatorname{div}(xyzt))$ is lc near o_i. Hence $a(F, W_i, 0) \geq \eta \,\mathrm{mult}_F(xyzt)$, and with (E.103),

$$[x] + [y] + [z] + [t] \leq C_3 := C_1/\eta \tag{E.105}$$

for any F as in (E.104).

Let f be a G_i-semi-invariant regular function near o_i, fixed by R_i, with $f(o_i) = 0$. Then

$$[f] \geq \frac{1}{e_0}. \tag{E.106}$$

The group $I := I_{G_i}(F)$ is cyclic and acts faithfully on $\mathfrak{m}_F/\mathfrak{m}_F^2$ by a character χ_F [Liu26, Lemma 2.6(1)], and the character of f restricted to I is $\chi_F^{\mathrm{mult}_F f}$ [Liu26, Lemma 9.2(1)]. Since $I \cap R_i$ fixes f, $|I \cap R_i|$ divides $\mathrm{mult}_F f > 0$, so $\mathrm{mult}_F f \geq |I \cap R_i| \geq |I|/[G_i : R_i] \geq e/e_0$.

Finally we compute the coefficients of the radial parameters. Let $w' \in \{w, -w\}$ and let l be an index with $w'_l > 0$. On the root chart of w' at x_l, with coordinates ρ and x'_j, $j \neq l$, given by $x_l = \rho^{w'_l}$ and $x_j = \rho^{w'_j} x'_j$, write $f_i = \rho^{w'(f^0)} f'$, where $w'(f^0) = \pm\kappa(f^0)$ is the w'-weight of f^0, and $h_k = \rho^{w'(h_k^0)} \tilde{h}_k$ with $w'(h_k^0) = \pm\kappa_k$. A direct calculation shows that the pullback of $dx \wedge dy \wedge dz \wedge dt$ is

$$\pm w'_l \rho^{\sum_j w'_j - 1} d\rho \wedge \bigwedge_{j \neq l} dx'_j,$$

so taking residues, the pullback of the generator ω_{W_i} of the canonical sheaf of W_i given by the residue of $dx \wedge dy \wedge dz \wedge dt/f_i$ is $\pm w'_l \rho^{\sum_j w'_j - w'(f^0) - 1}$ times the residue generator on $(f' = 0)$. Hence the crepant boundary of (W_i, D_i) on $(f' = 0)$ is

$$\lambda_{w'} \operatorname{div}(\rho) + \sum_k c_k \operatorname{div}(\tilde{h}_k)$$

with

$$\lambda_{w'} = 1 - \sum_j w'_j + w'(f^0) + \sum_k c_k w'(h_k^0) = \begin{cases} 1 + \delta & \text{if } w' = w, \\ 1 - \delta & \text{if } w' = -w. \end{cases} \tag{E.107}$$

In the charts below, R_i lifts to a subgroup R' of order r_i acting on ρ alone, and on the quotient by R' with the parameter $q := \rho^{r_i}$, the coefficient of $\operatorname{div}(q)$ is $1 - (1 - \lambda_{w'})/r_i$ [Liu26, Lemma 2.6(4)], that is, $1 + \delta/r_i$ on the charts of w and $1 - \delta/r_i$ on the charts of $-w$. Let G' be the group generated by the lifts of G_i and by the group $\mu_{w'_l}$ of the root chart, and $G'' := G'/R'$. By the residue computation above and [Liu26, Lemma 2.6(0)], the quotient of the chart pair by G' and the quotient of (W_i, D_i) by G_i are crepant birational, with both log canonical divisors computed from ω_{W_i} and the h_k. Since a log discrepancy depends only on the valuation and the log canonical

divisor [KM98, Definition 2.25], a prime divisor with center on both quotients has equal log discrepancies on both. Thus by [Liu26, Lemma 2.6(2)(5)], if V'' is the quotient chart, B'' the crepant boundary on V'', and a prime divisor F' over V'' and a prime divisor F over W_i have equal images in the common quotient, then

$$\frac{a(F', V'', B'')}{|I_{G''}(F')|} = \frac{a(F, W_i, D_i)}{|I_{G_i}(F)|}. \tag{E.108}$$

Equality (E.108) also holds with D_i replaced by $\sum_k c_k \operatorname{div}(h_k^{\pm})$ or $\sum_k c_k \operatorname{div}(h_k^0)$, and with f_i replaced by an equation with principal block f^0.

Step 3. In this step we prove (E.104) in case (a) when either $\operatorname{mult}_F h_k = \operatorname{mult}_F h_k^0$ for any k, or $[z] \le \varpi_i$ for the number ϖ_i defined below.

In case (a) the weights of x, y, z, t are $0, 1, -2, 0$. A monomial $x^a y^b z^c t^d$ of $\mathbf{T}$-weight $\kappa_k + nr_i$ satisfies $b - 2c = \kappa_k + nr_i$, so $b \ge r_i - \kappa_{\max}$ if $n > 0$ and $c \ge (r_i - \kappa_{\max})/2$ if $n < 0$, where $\kappa_{\max} := \max_k |\kappa_k|$. The monomials of B_i are $z^c t^d$ with $c \ge r_i/2$. Let F be as in (E.104). If $\operatorname{mult}_F h_k \ne \operatorname{mult}_F h_k^0$ for some k, then $h_k - h_k^0$ has a monomial of order at most $\max(\operatorname{mult}_F h_k, \operatorname{mult}_F h_k^0) \le C_2 e$, hence

$$\min([y], [z]) \le \varpi_i := \frac{2C_2}{r_i - \kappa_{\max}}.$$

The function $y^2 z$ is fixed by R_i, so (E.106) implies

$$2[y] + [z] \ge \frac{1}{e_0}. \tag{E.109}$$

Assume first that $[z] \le \varpi_i$. Then $[y] \ge 1/(4e_0)$ for i large by (E.109), so any positive monomial of any h_k has order at least $(r_i - \kappa_{\max})e/(4e_0) > C_2 e$ for i large. Since $\operatorname{mult}_F h_k \le C_2 e$, we obtain $\operatorname{mult}_F h_k^- = \operatorname{mult}_F h_k$ for any k, and we let $h_k^\sharp := h_k^-$. If instead $\operatorname{mult}_F h_k = \operatorname{mult}_F h_k^0$ for any k, we let $h_k^\sharp := h_k^0$. In both cases the $h_k^\sharp$ have no positive monomials, and

$$a\Big(F, W_i, \sum_k c_k \operatorname{div}(h_k^\sharp)\Big) = a(F, W_i, D_i),$$

since a log discrepancy with respect to $\sum_k c_k \operatorname{div}(h_k^\sharp)$ depends only on the orders of the $h_k^\sharp$.

Consider the root chart of $-w = (0, -1, 2, 0)$ at z: $z = u^2$, $y = u^{-1}Y$, with coordinates u, x, Y, t. The element $(1, \lambda, \lambda^{-2}, 1)$ of R_i lifts to $u \to \lambda^{-1}u$, which fixes x, Y, t, and the subgroup R' of **Step 2** consists of such lifts. Let $q := u^{r_i}$. Since $z^c = u^{2c}$ and $2c \in r_i\mathbb{N}^+$ for any monomial $z^c t^d$ of B_i, we have $f_i = x^2 + Y^2 + t^3 + B_i'(q, t)$ with a polynomial $B_i' \in (q)$. A monomial $x^a y^b z^c t^d$ of $h_k^\sharp$ of $\mathbf{T}$-weight $\kappa_k + nr_i$, $n \le 0$, equals $u^{-\kappa_k} q^{-n} x^a Y^b t^d$. Thus $h_k^\sharp = u^{-\kappa_k} \tilde{h}_k(q, x, Y, t)$, where $\tilde{h}_k$ is regular and $\tilde{h}_k(0, x, Y, t) = \hat{h}_k := h_k^0(x, Y, 1, t)$. Let

$$V'' := (x^2 + Y^2 + t^3 + B_i'(q, t) = 0) \subset \mathbb{A}^4_{q,x,Y,t}, \qquad S_0 := V'' \cap (q = 0) = (x^2 + Y^2 + t^3 = 0).$$

By (E.107), the crepant boundary of $(W_i, \sum_k c_k \operatorname{div}(h_k^\sharp))$ on V'' is

$$B'' := \Big(1 - \frac{\delta}{r_i}\Big) \operatorname{div}(q) + \sum_k c_k \operatorname{div}(\tilde{h}_k).$$

Since $x^2 + Y^2 = (x + \sqrt{-1}Y)(x - \sqrt{-1}Y)$ and $t^3 + B_i'(q, t)$ is a non-zero polynomial, V'' is near 0 a normal hypersurface of type cA, so it has canonical, in particular klt, singularities near 0 (see [KM98, Definition 5.32 and Theorem 5.34]). The surface S_0 is a Du Val surface of type A_2. The group G'' of **Step 2** acts diagonally on $\mathbb{A}^4_{q,x,Y,t}$. It acts faithfully on S_0: an element of G' is a pair of $g \in G_i$ and μ with $\mu^2 = g_z$, acting on x, Y, t by $g_x, g_y\mu, g_t$, and if it acts trivially on x, Y, t, then $g_x = g_t = 1$ and $g_z = \mu^2 = g_y^{-2}$, so $g \in R_i$ and the pair belongs to R'.

Let ι be the involution $u \to -u$ of the root chart, which belongs to G' and acts on S_0 by $Y \to -Y$. Let $\pi : S_0 \to \mathbb{A}^2_{x,t}$ be the quotient by ι, branched along $\Phi = x^2 + t^3 = -Y^2$. By the

formulas of Construction E.17(a), $\hat{h}_k = Y^{\kappa_k}\pi^*\phi_{i,k}$ as rational functions in both cases $\kappa_k \geq 0$ and $\kappa_k < 0$, because $\phi_{i,k} = (-\Phi)^{l_k}\psi_{i,k} = Y^{2l_k}\psi_{i,k}$ when $\kappa_k < 0$. Since

$$\operatorname{div}_{S_0}(Y) = \frac{1}{2}\pi^*\operatorname{div}(\Phi), \qquad K_{S_0} = \pi^*\Big(K_{\mathbb{A}^2} + \frac{1}{2}\operatorname{div}(\Phi)\Big),$$

and since $\sum_k c_k\kappa_k + \max(-\delta, 0) = \max(\delta, 0) - 1$, the pair

$$\Big(S_0, B_{S_0} := \sum_k c_k \operatorname{div}(\hat{h}_k) + \max(-\delta, 0)\operatorname{div}(Y)\Big)$$

satisfies $K_{S_0} + B_{S_0} = \pi^*(K_{\mathbb{A}^2} + \Delta_i)$. The involution ι may change the signs of the $\tilde{h}_k$, and only the invariance of the divisors $\operatorname{div}(\tilde{h}_k)$ is used. The group $G''/\langle\iota\rangle$ acts on $\mathbb{A}^2_{x,t}$ through the action of G_i on x, t, that is, as H_i. Thus by [Liu26, Lemma 2.6(5)], the infimum of $a(E, S_0, B_{S_0})/|I_{G''}(E)|$ over the prime divisors E over S_0 with center 0 is $\beta_i = \beta$, and (S_0, B_{S_0}) is lc near 0 because $(\mathbb{A}^2, \Delta_i)$ is lc near 0 (Lemma E.18 and [KM98, Proposition 5.20]).

Let F' be a prime divisor over the root chart whose valuation extends mult_F up to a positive multiple, and F'' the prime divisor over V'' given by the restriction of $\operatorname{mult}_{F'}$ (both valuations are divisorial by [KM98, Lemma 2.45], see also [Liu26, Lemma 2.6]). Then $\operatorname{mult}_{F'} u = \frac{1}{2}\operatorname{mult}_{F'} z > 0$, $\operatorname{mult}_{F'} Y = \operatorname{mult}_{F'} u + \operatorname{mult}_{F'} y > \operatorname{mult}_{F'} u$, and $\operatorname{mult}_{F'} x, \operatorname{mult}_{F'} t > 0$, so the center of F'' on V'' is $0 \in S_0$. Let

$$B^\natural := \operatorname{div}(q) + \sum_k c_k \operatorname{div}(\tilde{h}_k) + \max(-\delta, 0)\operatorname{div}(Y),$$

whose restriction to S_0 is B_{S_0}. Lemma E.22, applied with $V'' \subset \mathbb{A}^4$ in the role of X, the group G'', q in the role of u,

$$\mathfrak{a} = \prod_k (\tilde{h}_k)^{c_k} \cdot (Y)^{\max(-\delta,0)},$$

and $V = \{0\}$, implies $a(F'', V'', B^\natural) \geq \beta|I_{G''}(F'')|$. The difference $a(F'', V'', B'') - a(F'', V'', B^\natural)$ is a positive multiple of

$$\frac{\delta}{r_i}\operatorname{mult}_{F'} q + \max(-\delta, 0)\operatorname{mult}_{F'} Y = \delta\operatorname{mult}_{F'} u + \max(-\delta, 0)\operatorname{mult}_{F'} Y,$$

which is non-negative: it is at least 0 if $\delta \geq 0$, and it equals $-\delta\operatorname{mult}_{F'} y > 0$ if $\delta < 0$. By (E.108), $a_{D_i}(F) = a(F'', V'', B'')/|I_{G''}(F'')| \geq \beta$.

Step 4. In this step we prove (E.104) in case (a) when $[y] \leq \varpi_i$.

By (E.109), $[z] \geq 1/(2e_0)$ for i large, so any negative monomial of any h_k has order at least $(r_i - \kappa_{\max})e/(4e_0) > C_2 e$, and $\operatorname{mult}_F h_k^+ = \operatorname{mult}_F h_k$ for any k. By (E.105), $[y], [z] \leq C_3$.

Let $\mathcal{L}_0 := \mathbb{C}(x, y, t)$, a subfield of the function field $\mathcal{K} := \mathbb{C}(W_i)$ of W_i with $\mathcal{K} = \mathcal{L}_0(z)$ finite over $\mathcal{L}_0$, let $F^\flat$ be the prime divisor over $\mathbb{A}^3_{x,y,t}$ under F for the projection $\operatorname{pr} : W_i \to \mathbb{A}^3_{x,y,t}$, and $v_0 := \operatorname{mult}_F \circ \operatorname{pr}^*$ on $\mathcal{L}_0$, a positive multiple of $\operatorname{mult}_{F^\flat}$. Let $z_0 := -\Phi/y^2 \in \mathcal{L}_0$ and $Z := z/z_0 \in \mathcal{K}$. Since $y^2 z + \Phi + B_i(z, t) = 0$ on W_i, Z is a root of

$$\mathcal{P}(Z) := Z - 1 + \frac{B_i(z_0 Z, t)}{y^2 z_0} \in \mathcal{L}_0[Z].$$

Any monomial $z^c t^d$ of B_i has $c \geq r_i/2$, so $\operatorname{mult}_F B_i \geq r_i e/(4e_0)$, while $\operatorname{mult}_F(y^2 z) \leq 3C_3 e$. Hence $\operatorname{mult}_F \Phi = \operatorname{mult}_F(y^2 z + B_i) = \operatorname{mult}_F(y^2 z)$ and $\operatorname{mult}_F z_0 = \operatorname{mult}_F z$. Each coefficient of $\mathcal{P}(Z) - (Z - 1)$ has v_0-value at least

$$N_i e := \Big(\frac{r_i}{2} - 1\Big)\operatorname{mult}_F z - 2\operatorname{mult}_F y \geq \Big(\frac{r_i - 2}{4e_0} - 2C_3\Big)e,$$

and $N_i \to +\infty$. Thus the reduction of $\mathcal{P}$ modulo the maximal ideal of the valuation ring of the completion $\widehat{\mathcal{L}_0}$ of $\mathcal{L}_0$ at v_0 is $Z - 1$, and by Hensel's lemma [Ser79, Chapter II, §4, Proposition 7] $\mathcal{P}$ has a unique root $Z_1 \in \widehat{\mathcal{L}_0}$ with residue 1. In the completion $\widehat{\mathcal{K}}$ of $\mathcal{K}$ at mult_F, which contains $\widehat{\mathcal{L}_0}$, Z is a root of $\mathcal{P}$ with $\operatorname{mult}_F(Z - 1) = \operatorname{mult}_F B_i - \operatorname{mult}_F \Phi > 0$. Since the derivative of $\mathcal{P}$

has residue 1, $\mathcal{P}$ has only one root with residue 1 in $\widehat{\mathcal{K}}$, so $Z = Z_1 \in \widehat{\mathcal{L}_0}$, and the closure of $\mathcal{K} = \mathcal{L}_0(Z)$ in $\widehat{\mathcal{K}}$ is $\widehat{\mathcal{L}_0}$, that is, $\widehat{\mathcal{K}} = \widehat{\mathcal{L}_0}$. Thus the value groups of mult_F and v_0 are equal, and so are their residue fields, and the extension $\mathcal{K}/\mathcal{L}_0$ is unramified along F. In particular, since the value group of mult_F is $\mathbb{Z}$, $v_0 = \mathrm{mult}_{F^\flat}$, that is, $\mathrm{mult}_F(\mathrm{pr}^*\,\phi) = \mathrm{mult}_{F^\flat}\,\phi$ for any $\phi \in \mathcal{L}_0$.

The group G_i acts faithfully on $\mathcal{L}_0$: an element acting trivially on x, y, t fixes the rational function Φ/y^2 of x, y, t, and z has the character of Φ/y^2, so the element fixes z. The function Z is G_i-invariant, because z_0 has the character of z. If $g \in G_i$ preserves v_0, then g maps mult_F to an extension of v_0 to $\mathcal{K}$ in which Z has residue 1, and such an extension is unique since Z_1 is the unique root of $\mathcal{P}$ with residue 1. Hence the decomposition groups of F and $F^\flat$ coincide, and since their residue fields coincide, $I_{G_i}(F) = I_{G_i}(F^\flat)$.

The residue ω_{W_i} of $dx \wedge dy \wedge dz \wedge dt/f_i$ equals $\pm dx \wedge dy \wedge dt/(y^2 + \partial B_i/\partial z)$, and

$$\mathrm{mult}_F(\partial B_i/\partial z) \geq N_i e + 2\,\mathrm{mult}_F\, y > 2\,\mathrm{mult}_F\, y,$$

so $y^2/(y^2 + \partial B_i/\partial z)$ is a unit with residue 1 along F. Since $\mathcal{K}/\mathcal{L}_0$ is unramified along F and the residue fields of mult_F and $\mathrm{mult}_{F^\flat}$ coincide, the valuation ring of mult_F is étale over that of $\mathrm{mult}_{F^\flat}$, and by the Hurwitz formula with ramification index 1, the order of $dx \wedge dy \wedge dt$ along F equals that along $F^\flat$. Hence $a(F, W_i, 0) = a(F^\flat, \mathbb{A}^3, 0) - 2\,\mathrm{mult}_{F^\flat}\, y$. For any k let $h_k^\flat := h_k^+(x, y, z_0, t) \in \mathcal{L}_0$. A monomial Mz^c of h_k^+, with M free of z, satisfies $Mz^c - Mz_0^c = Mz^c(1 - Z^{-c})$, of order at least $\mathrm{mult}_F(Z - 1) \geq N_i e > C_2 e$ for i large. Thus $\mathrm{mult}_{F^\flat}\, h_k^\flat = \mathrm{mult}_F\, h_k^+ = \mathrm{mult}_F\, h_k$, independently of the number of monomials, and

$$a_{D_i}(F) = \frac{a\big(F^\flat, \mathbb{A}^3_{x,y,t}, 2\,\mathrm{div}(y) + \sum_k c_k\,\mathrm{div}(h_k^\flat)\big)}{|I_{G_i}(F^\flat)|}. \tag{E.110}$$

A monomial $x^a y^b z^c t^d$ of h_k^+ of $\mathbf{T}$-weight $\kappa_k + nr_i$, $n \geq 0$, becomes $y^{\kappa_k}(y^{r_i})^n x^a(-\Phi)^c t^d$ after the substitution $z = z_0$. Thus $h_k^\flat = y^{\kappa_k}\tilde{h}_k(y^{r_i}, x, t)$, where $\tilde{h}_k$ is regular and $\tilde{h}_k(0, x, t) = \phi_{i,k}$ by Construction E.17(a). The group R_i acts on y alone. Let $q := y^{r_i}$. The coefficient of $\mathrm{div}(y)$ in the boundary on the right-hand side of (E.110) is $2 + \sum_k c_k\kappa_k = 1 + \delta$, so by [Liu26, Lemma 2.6(1)(2)(4)(5)] the right-hand side of (E.110) equals $a(F'', \mathbb{A}^3_{q,x,t}, B'')/|I_{\bar{G}_i}(F'')|$, where F'' is the prime divisor over $\mathbb{A}^3_{q,x,t}$ given by the restriction of v_0 along the quotient map $\mathbb{A}^3_{x,y,t} \to \mathbb{A}^3_{q,x,t}$, $\bar{G}_i := G_i/R_i$, and

$$B'' := \Big(1 + \frac{\delta}{r_i}\Big)\,\mathrm{div}(q) + \sum_k c_k\,\mathrm{div}(\tilde{h}_k).$$

An element of G_i acting trivially on x, t satisfies $g_y^2 g_z = 1$, because $y^2 z$ and t^3 have the character of x^2, so it belongs to R_i. Thus $\bar{G}_i$ acts faithfully on the plane $(q = 0) = \mathbb{A}^2_{x,t}$, as H_i, and the section pair of

$$B^\natural := \mathrm{div}(q) + \sum_k c_k\,\mathrm{div}(\tilde{h}_k) + \frac{\max(\delta, 0)}{2}\,\mathrm{div}(\Phi)$$

is $(\mathbb{A}^2_{x,t}, \Delta_i)$, with quotient minimum β at 0. Since $\mathrm{mult}_{F^\flat} x, \mathrm{mult}_{F^\flat} t, \mathrm{mult}_{F^\flat} y > 0$, the center of F'' is 0. Lemma E.22, applied with $X = \mathbb{A}^3_{q,x,t}$, the group $\bar{G}_i$, $u = q$,

$$\mathfrak{a} = \prod_k (\tilde{h}_k)^{c_k} \cdot (\Phi)^{\max(\delta,0)/2},$$

and $V = \{0\}$, implies $a(F'', \mathbb{A}^3, B^\natural) \geq \beta |I_{\bar{G}_i}(F'')|$. Since

$$\mathrm{mult}_{F^\flat}\,\Phi = \mathrm{mult}_F(y^2 z) = 2\,\mathrm{mult}_{F^\flat}\, y + \mathrm{mult}_F\, z,$$

the difference $a(F'', \mathbb{A}^3, B'') - a(F'', \mathbb{A}^3, B^\natural)$ is a positive multiple of

$$-\delta\,\mathrm{mult}_{F^\flat}\, y + \frac{\max(\delta, 0)}{2}\big(2\,\mathrm{mult}_{F^\flat}\, y + \mathrm{mult}_F\, z\big),$$

which equals $\delta\,\mathrm{mult}_F\, z/2 \geq 0$ if $\delta \geq 0$ and $-\delta\,\mathrm{mult}_{F^\flat}\, y > 0$ if $\delta < 0$. Hence $a_{D_i}(F) \geq \beta$.

If $\mathrm{mult}_F h_k \neq \mathrm{mult}_F h_k^0$ for some k and $[z] > \varpi_i$, then $[y] \leq \varpi_i$, since $\min([y],[z]) \leq \varpi_i$. Thus by **Steps 3 and 4**, (E.104) holds in case (a) for i large.

Step 5. In this step we describe the reference surface S of case (b) and the weighted blow-up of S with weights $(3,4,2)$, and we show that β is the infimum of the local quotient minima over the fiber of the blow-up.

In case (b) the weights of x,y,z,t are $3,4,-2,2$. Consider the root chart of $-w = (-3,-4,2,-2)$ at z: $z=u^2$, $x=u^{-3}X$, $y=u^{-4}Y$, $t=u^{-2}T$. The element $(\lambda^3,\lambda^4,\lambda^{-2},\lambda^2)$ of R_i lifts to $u \to \lambda^{-1}u$, which fixes X,Y,T, and R' consists of such lifts. Let $J := G'' = G'/R'$, which acts diagonally on $\mathbb{A}^4_{q,X,Y,T}$ with $q := u^{r_i}$. An element of G' is a pair of $g \in G_i$ and μ with $\mu^2 = g_z$, acting on X,Y,T by $g_x\mu^3, g_y\mu^4, g_t\mu^2$. If it acts trivially on X,Y,T, then $g = (\lambda^3,\lambda^4,\lambda^{-2},\lambda^2)$ with $\lambda = \mu^{-1}$, so $g \in R_i$ and the pair belongs to R'. Hence J acts faithfully on $\mathbb{A}^3_{X,Y,T}$ and on

$$S := (X^2+Y^2+T^3=0) \subset \mathbb{A}^3_{X,Y,T},$$

since $f^0 = u^{-6}(X^2+Y^2+T^3)$. The involution $\iota : u \to -u$ belongs to G' and acts by $X \to -X$, $Y \to Y$, $T \to T$. By Construction E.17(b),

$$h_k^0 = x^{\epsilon_k} z^{-\kappa'_k} A_{i,k}(yz^2, tz) = u^{-\kappa_k}\hat{h}_k, \qquad \hat{h}_k := h_k^0(X,Y,1,T) = X^{\epsilon_k}A_{i,k}(Y,T),$$

because $yz^2 = Y$, $tz = T$, and $\kappa_k = 3\epsilon_k + 2\kappa'_k$. Let $\delta^- := \max(-\delta, 0)$, $\mathfrak{I} := (X^4, Y^3, T^6)$, and $B_S := \sum_k c_k \operatorname{div}(\hat{h}_k)$ on S. The quotient $\pi : S \to \mathbb{A}^2_{Y,T} = \mathbb{A}^2_{\zeta_1,\zeta_2}$ by ι is branched along $Q = Y^2+T^3 = -X^2$. Since $\operatorname{div}_S(X) = \frac{1}{2}\pi^*\operatorname{div}(Q)$, $K_S = \pi^*(K_{\mathbb{A}^2} + \frac{1}{2}\operatorname{div}(Q))$, and $\mathfrak{I}\mathcal{O}_S = \pi^*(Q^2, \zeta_1^3, \zeta_2^6)$, we have

$$K_S + B_S = \pi^*\Big(K_{\mathbb{A}^2} + \frac{1+\sum_k c_k\epsilon_k}{2}\operatorname{div}(Q) + \sum_k c_k \operatorname{div}(A_{i,k})\Big) = \pi^*(K_{\mathbb{A}^2} + \Delta_i),$$

and $\mathfrak{I}^{\delta^-/12}\mathcal{O}_S$ is the pullback of $\mathfrak{c}$. The group $J/\langle\iota\rangle$ acts on $(\zeta_1,\zeta_2) = (yz^2, tz)$ through G_i, that is, as H_i. Thus by [Liu26, Lemma 2.6(5)], the quotient minimum of $(S \ni 0, B_S, \mathfrak{I}^{\delta^-/12})$ by J is β, and $(S, B_S, \mathfrak{I}^{\delta^-/12})$ is lc near 0 since $(\mathbb{A}^2, \Delta_i, \mathfrak{c})$ is lc near 0 (Lemma E.18).

Let ς be the radial parameter of the weighted blow-up of $\mathbb{A}^3_{X,Y,T}$ with weights $(3,4,2)$. The three root charts of the blow-up are

$$X = \varsigma^3,\ Y = \varsigma^4 Y_1,\ T = \varsigma^2 T_1;$$
$$X = \varsigma^3 X_1,\ Y = \varsigma^4,\ T = \varsigma^2 T_1;$$
$$X = \varsigma^3 X_1,\ Y = \varsigma^4 Y_1,\ T = \varsigma^2,$$

and the strict transforms S'_X, S'_Y, S'_T of S are

$$1 + \varsigma^2 Y_1^2 + T_1^3 = 0, \qquad X_1^2 + \varsigma^2 + T_1^3 = 0, \qquad X_1^2 + \varsigma^2 Y_1^2 + 1 = 0.$$

A monomial $x^\epsilon y^a z^c t^b$ of h_k^0 has $\mathbf{T}$-weight $3\epsilon + 4a - 2c + 2b = \kappa_k$ and $c \geq 0$, and the corresponding monomial $X^\epsilon Y^a T^b$ of $\hat{h}_k$ becomes $\varsigma^{\kappa_k}(\varsigma^2)^c$ times a monomial in the coordinates of the chart other than ς. Thus

$$\hat{h}_k = \varsigma^{\kappa_k}\check{h}_k(\varsigma^2, \cdot),$$

where $\check{h}_k$ is regular and is the polynomial h_k^0 written in the coordinates $X_1, Y_1, Z := \varsigma^2, T_1$, one of X_1, Y_1, T_1 being 1. The ideal $\mathfrak{I}$ becomes ς^{12} times the unit ideal on each chart. The pullback of $dX \wedge dY \wedge dT$ is $\pm w_l \varsigma^8$ times a coordinate 3-form, where $w_l \in \{3,4,2\}$, and the equation of S is ς^6 times that of S'_l, so the residue forms differ by $\pm w_l \varsigma^2$. Hence the crepant boundary of $(S, B_S, \mathfrak{I}^{\delta^-/12})$ on S'_l is

$$\Big(-2 + \sum_k c_k\kappa_k + \delta^-\Big)\operatorname{div}(\varsigma) + \sum_k c_k \operatorname{div}(\check{h}_k) = (-1 + \max(\delta, 0))\operatorname{div}(\varsigma) + \sum_k c_k \operatorname{div}(\check{h}_k),$$

since $\sum_k c_k\kappa_k = \delta + 1$.

Let J_l be the group generated by the lifts of J and by the group μ_{w_l} of the root chart. The involution $\varsigma \to -\varsigma$ fixing the other two coordinates belongs to J_l: in each chart it maps X to $-X$ and fixes Y and T, so it is a lift of ι. Let $S_l := S'_l/\langle \varsigma \to -\varsigma \rangle$, with the coordinate $Z = \varsigma^2$, and $J'_l := J_l/\langle \varsigma \to -\varsigma \rangle$. Then S_X, S_Y, S_T are the surfaces

$$1 + Y_1^2 Z + T_1^3 = 0, \qquad X_1^2 + Z + T_1^3 = 0, \qquad X_1^2 + Y_1^2 Z + 1 = 0, \tag{E.111}$$

which are smooth: the partial derivatives of the first equation with respect to Z and T_1, and the partial derivatives of the third with respect to X_1 and Z, vanish simultaneously only where the equation is 1, and the second is a graph. By [Liu26, Lemma 2.6(4)] the coefficient of $\operatorname{div}(Z)$ on S_l is $1 - (1 - (-1 + \max(\delta, 0)))/2$, so the crepant boundary on S_l is

$$B_l := v \operatorname{div}(Z) + \sum_k c_k \operatorname{div}(\check{h}_k), \qquad v := \frac{\max(\delta, 0)}{2},$$

and $B_l \geq 0$. The functions $\check{h}_k$ are invariant up to sign under $\varsigma \to -\varsigma$, and only the invariance of the divisors is used.

We claim that for any scheme point ξ of the curve $(Z = 0) \subset S_l$, any prime divisor E over S_l with center $\overline{\{\xi\}}$ satisfies $a(E, S_l, B_l)/|I_{J'_l}(E)| \geq \beta$, and that β is the infimum of the local quotient minima at the scheme points of the curves $(Z = 0)$. Let $S^w \to S$ be the weighted blow-up, with crepant boundary B^w (the ideal $\mathfrak{I}$ becomes the divisor 12 times the exceptional curve). By [KM98, Lemma 2.30], the log discrepancies of a prime divisor over S computed on S and on (S^w, B^w) are equal. A prime divisor over S with center 0 has center in the exceptional curve of S^w, and a prime divisor with center in the exceptional curve has center 0 on S because $S^w \to S$ is proper. The surface S'_l/μ_{w_l} is a chart of S^w, the group J_l is generated by μ_{w_l} and lifts of J, so S'_l/J_l is the corresponding chart of S^w/J, and $S_l/J'_l = S'_l/J_l$ since $J'_l = J_l/\langle \varsigma \to -\varsigma \rangle$. The group J_l acts faithfully on S'_l: the morphism $S'_l \to S$ is equivariant and dominant, so an element of J_l acting trivially on S'_l maps to the identity of J, which acts faithfully on S, and hence belongs to μ_{w_l}, which acts freely on the dense open subset $(\varsigma \neq 0)$ of S'_l. Hence J'_l acts faithfully on S_l [Liu26, Lemma 2.6(5)]. Thus by [Liu26, Lemma 2.6(2)(5)],

$$\frac{a(E, S_l, B_l)}{|I_{J'_l}(E)|} = \frac{a(E_S, S, B_S, \mathfrak{I}^{\delta^-/12})}{|I_J(E_S)|}$$

for the prime divisor E_S over S whose image in the common quotient is the image of E. Conversely, any prime divisor E_S over S with center 0 has center on S^w the closure of a scheme point of the exceptional curve, so the infimum of the local quotient minima over the scheme points of the curves $(Z = 0)$ is at most the infimum of $a(E_S, S, B_S, \mathfrak{I}^{\delta^-/12})/|I_J(E_S)|$, which is β. The claim follows.

Step 6. In this step we prove (E.104) in case (b) when $[z] \geq c_{\mathrm{b}}$, where c_{b} is a fixed real number with $0 < c_{\mathrm{b}} < 1/(3e_0)$.

The functions x^2z^3, yz^2, tz are fixed by R_i, so by (E.106),

$$2[x] + 3[z] \geq \frac{1}{e_0}, \qquad [y] + 2[z] \geq \frac{1}{e_0}, \qquad [t] + [z] \geq \frac{1}{e_0}. \tag{E.112}$$

A monomial $x^a y^b z^c t^d$ of $\mathbf{T}$-weight $\kappa + nr_i$ satisfies $3a + 4b - 2c + 2d = \kappa + nr_i$. Let $\kappa^+_{\max} := \max(\kappa_{\max}, 6)$. If $n < 0$, then $c \geq (r_i - \kappa^+_{\max})/2$, so any negative monomial of any h_k or of B_i has order at least $N'_i e$ with

$$N'_i := c_{\mathrm{b}}(r_i - \kappa^+_{\max})/2 \to +\infty.$$

Thus $\operatorname{mult}_F h^+_k = \operatorname{mult}_F h_k$ for any k and i large. Let B^-_i be the sum of the negative monomials of B_i, $g^+ := g_i - B^-_i$, and $W^+ := (x^2 + g^+ = 0)$, which is integral since $y^2z + t^3$ is not a square, and normal near 0 by Serre's criterion: the lowest-degree part $y^2z + t^3$ of g^+ is irreducible, so g^+ has no repeated factor in $\mathbb{C}[[y, z, t]]$, and near 0, $\operatorname{Sing}(W^+)$ is contained in $\{0\} \times \operatorname{Sing}(g^+ = 0)$, so $\dim \operatorname{Sing}(W^+) \leq 1$.

We replace W_i with W^+ by multiplying the coordinate x by a square root of residue one, taken in a completion along F. Let $\mathcal{L}_1 := \mathbb{C}(y,z,t)$ and $\epsilon_F := -B_i^-/g_i \in \mathcal{L}_1$, so that $1+\epsilon_F = g^+/g_i$ and $\operatorname{mult}_F \epsilon_F = \operatorname{mult}_F B_i^- - 2\operatorname{mult}_F x \geq (N_i'-2C_3)e$ by (E.105). Let $v_1 := \operatorname{mult}_F \circ \mathrm{pr}^*$ on $\mathcal{L}_1$ for the projection $\mathrm{pr} : W_i \to \mathbb{A}^3_{y,z,t}$. By Hensel's lemma [Ser79, Chapter II, §4, Proposition 7], $1+\epsilon_F$ has a square root ξ_0 with residue 1 in the completion $\widehat{\mathcal{L}_1}$ of $\mathcal{L}_1$ at v_1. In the completion $\widehat{\mathcal{K}}$ of the function field $\mathcal{K} := \mathbb{C}(W_i)$ of W_i at mult_F, the element $x^+ := \xi_0 x$ satisfies $(x^+)^2 + g^+ = 0$. Thus $\mathcal{K}^+ := \mathcal{L}_1(x^+) \subset \widehat{\mathcal{K}}$ is the function field of W^+, and the restriction of the valuation of $\widehat{\mathcal{K}}$ to $\mathcal{K}^+$ defines a prime divisor F^+ over W^+. The closures of $\mathcal{K}$ and $\mathcal{K}^+$ in $\widehat{\mathcal{K}}$ are both $\widehat{\mathcal{L}_1}(x) = \widehat{\mathcal{L}_1}(x^+)$, so the value groups of mult_F and $\operatorname{mult}_{F^+}$ are equal, and so are their residue fields. The element ϵ_F is G_i-invariant, so for g in the decomposition group of F, $g(\xi_0)$ is a square root of $1+\epsilon_F$ with residue 1, that is, $g(\xi_0) = \xi_0$, and $g(x^+)$ is x^+ times the character of x. Hence the inclusion $\mathcal{K}^+ \subset \widehat{\mathcal{K}}$ is equivariant for the action of G_i on W^+, and since $x = \xi_0^{-1}x^+$, the decomposition and inertia groups of F and F^+ coincide. The residue forms $dy\wedge dz\wedge dt/(2x)$ and $dy\wedge dz\wedge dt/(2x^+)$ differ by the unit ξ_0 with residue 1, so $a(F,W_i,0) = a(F^+,W^+,0)$. Each h_k^+ is $A+xB$ in normal form, and $h_k^+(x^+) - h_k^+(x) = (\xi_0-1)xB$ has order at least $\operatorname{mult}_F \epsilon_F > C_2 e$, so the orders of the marks are kept. Thus

$$a_{D_i}(F) = \frac{a\big(F^+, W^+, \sum_k c_k \operatorname{div}(h_k^+)\big)}{|I_{G_i}(F^+)|}.$$

Choose $l \in \{x,y,t\}$ for which $\operatorname{mult}_{F^+} x_l/w_l$ is the least of $\operatorname{mult}_{F^+} x^+/3$, $\operatorname{mult}_{F^+} y/4$, $\operatorname{mult}_{F^+} t/2$, and consider the root chart of w at x_l: $x_l = \rho^{w_l}$ and $x_j = \rho^{w_j}x_j'$ for $j \neq l$, where x stands for x^+. In particular $z = \rho^{-2}Z'$. The element $(\lambda^3,\lambda^4,\lambda^{-2},\lambda^2)$ of R_i lifts to $\rho \to \lambda\rho$, which fixes the x_j', and R' consists of such lifts. Let $q := \rho^{r_i}$. Any monomial of x^2+g^+ has $\mathbf{T}$-weight $6+nr_i$ with $n \geq 0$, so $x^2+g^+ = \rho^6 f'$ with f' regular in q and the x_j', and $f'|_{q=0}$ is f^0 with $x_l = 1$. Likewise $h_k^+ = \rho^{\kappa_k}\tilde{h}_k$ with $\tilde{h}_k$ regular and $\tilde{h}_k|_{q=0} = \check{h}_k$. Let $V'' := (f'=0) \subset \mathbb{A}^4$ be the quotient chart. By (E.107), the crepant boundary on V'' is

$$B'' := \Big(1+\frac{\delta}{r_i}\Big)\operatorname{div}(q) + \sum_k c_k \operatorname{div}(\tilde{h}_k).$$

The section $V'' \cap (q=0)$ is the surface S_l of (E.111), with $Z = Z'$ and with Y_1, T_1, X_1 the coordinates x_j'. It is smooth, so V'' is smooth near it. The groups G'' and J_l' induce equal groups of automorphisms of S_l: an element of G' is a pair of $g \in G_i$ and μ with $\mu^{w_l} = g_l$, acting on x_j' by $g_j\mu^{-w_j}$, and an element of J_l is given by $g \in G_i$, ν with $\nu^2 = g_z$, and ζ with $\zeta^{w_l} = g_l\nu^{w_l}$, acting on the coordinates of S_l' other than ς by $g_j\nu^{w_j}\zeta^{-w_j}$ and on $Z = \varsigma^2$ by ζ^2. With $\mu := \zeta/\nu$, the characters are $g_j\mu^{-w_j}$ for $j \neq z$ and $g_z\mu^2 = g_z\mu^{-w_z}$. If an element (g,μ) of G' acts trivially on S_l, then $g_j = \mu^{w_j}$ for $j \in \{x,y,t\}\backslash\{l\}$, $g_l = \mu^{w_l}$ by the definition of G', and $g_z = \mu^{-2} = \mu^{w_z}$ from the action on Z, that is, $g = (\lambda^3,\lambda^4,\lambda^{-2},\lambda^2)$ with $\lambda = \mu$, so $g \in R_i$ and (g,μ) is the lift $\rho \to \lambda\rho$, which belongs to R'. Hence G'' acts faithfully on S_l.

Let F' be a prime divisor over the root chart whose valuation extends $\operatorname{mult}_{F^+}$ up to a positive multiple, and F'' the prime divisor over V'' given by the restriction of $\operatorname{mult}_{F'}$. By the choice of l,

$$\operatorname{mult}_{F'} x_j' = w_j(\operatorname{mult}_{F'} x_j/w_j - \operatorname{mult}_{F'} x_l/w_l) \geq 0 \quad \text{for } j \in \{x,y,t\},$$

and $\operatorname{mult}_{F'} Z' = \operatorname{mult}_{F'} z + 2\operatorname{mult}_{F'}\rho > 0$, $\operatorname{mult}_{F'} q > 0$. Thus the center of F'' is a closed point or an irreducible component of the curve $(Z=0) \subset S_l$. Let

$$B^\natural := \operatorname{div}(q) + \sum_k c_k \operatorname{div}(\tilde{h}_k) + v\operatorname{div}(Z'),$$

whose restriction to S_l is B_l. The pair (S_l, B_l) is lc near the curve $(Z=0)$, since by **Step 5** its crepant boundary on S_l' is the crepant boundary of $(S, B_S, \mathfrak{I}^{\delta^-/12})$, which is lc near 0, and lc is preserved under crepant pullback by the weighted blow-up and by finite morphisms, and descends along the finite morphism $S_l' \to S_l$ [KM98, Lemma 2.30 and Proposition 5.20(4)]. By Lemma E.22,

applied with V'' in the role of X, the group G'', q in the role of u, $\mathfrak{a} = \prod_k (\tilde{h}_k)^{c_k} \cdot (Z')^v$, a closed point p of the center of F'', and V that center, together with **Step 5**, $a(F'', V'', B^\natural) \geq \beta |I_{G''}(F'')|$. The difference $a(F'', V'', B'') - a(F'', V'', B^\natural)$ is a positive multiple of

$$-\delta \operatorname{mult}_{F'} \rho + \frac{\max(\delta, 0)}{2} \big(\operatorname{mult}_{F'} z + 2 \operatorname{mult}_{F'} \rho \big),$$

which equals $\delta \operatorname{mult}_{F'} z/2 \geq 0$ if $\delta \geq 0$ and $-\delta \operatorname{mult}_{F'} \rho > 0$ if $\delta < 0$. By (E.108) for W^+, $a_{D_i}(F) \geq \beta$.

Step 7. In this step we prove (E.104) in case (b) when $[z] < c_{\mathrm{b}}$.

By (E.112), $[x], [y], [t] \geq \gamma_* := (1/e_0 - 3c_{\mathrm{b}})/2 > 0$. If $n > 0$ in the relation $3a + 4b - 2c + 2d = \kappa + nr_i$, then $a + b + d \geq (r_i - \kappa^+_{\max})/4$, so any positive monomial of any h_k or of B_i has order at least $\gamma_*(r_i - \kappa^+_{\max})e/4$, which tends to infinity. Thus $\operatorname{mult}_F h_k^- = \operatorname{mult}_F h_k$ for any k and i large. Let B_i^+ be the sum of the positive monomials of B_i and $W^- := (x^2 + g_i - B_i^+ = 0)$, which is integral and normal near 0 by the argument for W^+ applied to $g_i - B_i^+ = y^2 z + t^3 + B_i^-$. Possibly replacing B_i^- with B_i^+ and h_k^+ with h_k^-, the argument of **Step 6** gives a prime divisor F^- over W^- whose decomposition and inertia groups are those of F, and

$$a_{D_i}(F) = \frac{a\big(F^-, W^-, \sum_k c_k \operatorname{div}(h_k^-)\big)}{|I_{G_i}(F^-)|}.$$

We use the root chart of $-w$ at z of **Step 5**, with x standing for the coordinate of W^-. The monomials of $B_i - B_i^+$ have $\mathbf{T}$-weight less than 6, so they do not involve y, nor x. A monomial $z^c t^b$ of $\mathbf{T}$-weight $6 + nr_i$, $n < 0$, equals $u^{-6} q^{-n} T^b$. Thus $x^2 + g_i - B_i^+ = u^{-6}(X^2 + Y^2 + T^3 + C_i(q, T))$ with a polynomial $C_i \in (q)$, and $h_k^- = u^{-\kappa_k} \tilde{h}_k(q, X, Y, T)$ with $\tilde{h}_k$ regular and $\tilde{h}_k|_{q=0} = \hat{h}_k$. Let

$$V'' := (X^2 + Y^2 + T^3 + C_i(q, T) = 0) \subset \mathbb{A}^4_{q,X,Y,T}.$$

By (E.107), the crepant boundary on V'' is

$$B'' := \Big(1 - \frac{\delta}{r_i}\Big) \operatorname{div}(q) + \sum_k c_k \operatorname{div}(\tilde{h}_k).$$

Since $X^2 + Y^2 = (X + \sqrt{-1}Y)(X - \sqrt{-1}Y)$ and $T^3 + C_i(q, T) \neq 0$, V'' is near 0 a normal hypersurface of type cA, hence klt near 0 (see [KM98, Definition 5.32 and Theorem 5.34]). The section $V'' \cap (q = 0)$ is S, on which $J = G''$ acts faithfully by **Step 5**. Let F' be a prime divisor over the root chart whose valuation extends $\operatorname{mult}_{F^-}$ up to a positive multiple, and F'' the prime divisor over V'' given by the restriction of $\operatorname{mult}_{F'}$. Then

$$\operatorname{mult}_{F'} u > 0, \quad \operatorname{mult}_{F'} X > 3 \operatorname{mult}_{F'} u, \quad \operatorname{mult}_{F'} Y > 4 \operatorname{mult}_{F'} u, \quad \operatorname{mult}_{F'} T > 2 \operatorname{mult}_{F'} u,$$

so the center of F'' is 0 and

$$\operatorname{mult}_{F'} \mathfrak{I} = \min(4 \operatorname{mult}_{F'} X, 3 \operatorname{mult}_{F'} Y, 6 \operatorname{mult}_{F'} T) > 12 \operatorname{mult}_{F'} u.$$

By Lemma E.22, applied with V'' in the role of X, the group J, q in the role of u,

$$\mathfrak{a} = \prod_k (\tilde{h}_k)^{c_k} \cdot \mathfrak{I}^{\delta^-/12},$$

and $V = \{0\}$, together with **Step 5**,

$$a\big(F'', V'', \operatorname{div}(q) + \sum_k c_k \operatorname{div}(\tilde{h}_k), \mathfrak{I}^{\delta^-/12}\big) \geq \beta |I_J(F'')|.$$

The difference between $a(F'', V'', B'')$ and the left-hand side is a positive multiple of

$$\delta \operatorname{mult}_{F'} u + \frac{\delta^-}{12} \operatorname{mult}_{F'} \mathfrak{I},$$

which is $\delta \operatorname{mult}_{F'} u \geq 0$ if $\delta \geq 0$, and $-\delta(\operatorname{mult}_{F'} \mathfrak{I}/12 - \operatorname{mult}_{F'} u) > 0$ if $\delta < 0$. By (E.108) for W^-, $a_{D_i}(F) \geq \beta$.

By **Steps 6 and 7**, (E.104) holds in case (b) for i large.

Step 8. In this step we prove (E.104) in case (c).

In case (c) the weights of x, y, z, t are $1, 2, -2, 0$, and a monomial $x^a y^b z^c t^d$ of $\mathbf{T}$-weight $\kappa + nr_i$ satisfies $a + 2b - 2c = \kappa + nr_i$. The monomials of B_i are $z^c t^d$ with $-2c = 2 + nr_i$, so $c \geq (r_i - 2)/2$. The functions yz, $x^2 z$, t are fixed by R_i, so by (E.106),

$$[y] + [z] \geq \frac{1}{e_0}, \qquad 2[x] + [z] \geq \frac{1}{e_0}, \qquad [t] \geq \frac{1}{e_0}.$$

Fix a real number c_* with $0 < c_* < 1/e_0$, and let F be as in (E.104).

Assume first that $[z] \geq c_*$. Any negative monomial of any h_k and any monomial of B_i has $c \geq (r_i - \kappa^+_{\max})/2$, hence order at least $c_*(r_i - \kappa^+_{\max})e/2$, and $\operatorname{mult}_F h_k^+ = \operatorname{mult}_F h_k$ for any k and i large. Let $z_0 := -(x^2 + yt^2)/y^2$. Possibly replacing the function $x^2 + t^3$ of **Step 4** with $x^2 + yt^2$ and the bound $[z] \geq 1/(2e_0)$ with $[z] \geq c_*$, the argument of **Step 4** gives the prime divisor $F^\flat$ over $\mathbb{A}^3_{x,y,t}$ given by the restriction of mult_F, with $I_{G_i}(F^\flat) = I_{G_i}(F)$, $\operatorname{mult}_{F^\flat}(x^2 + yt^2) = 2\operatorname{mult}_F y + \operatorname{mult}_F z$, and

$$a_{D_i}(F) = \frac{a\big(F^\flat, \mathbb{A}^3_{x,y,t}, 2\operatorname{div}(y) + \sum_k c_k \operatorname{div}(h_k^\flat)\big)}{|I_{G_i}(F^\flat)|}, \qquad h_k^\flat := h_k^+(x, y, z_0, t).$$

Consider the root chart of w at y: $y = u^2$, $x = uA$, with coordinates u, A, t, and let $\Phi := A^2 + t^2$, so that $z_0 = -\Phi/u^2$ and $yz_0 = -\Phi$. The pullback of $dx \wedge dy \wedge dt$ is $\pm 2u^2 du \wedge dA \wedge dt$. The element $(\lambda, \lambda^2, \lambda^{-2}, 1)$ of R_i lifts to $u \to \lambda u$, which fixes A, t, and R' consists of such lifts. Let $q := u^{r_i}$. By the formulas of Construction E.17(c), a monomial of h_k^+ of $\mathbf{T}$-weight $\kappa_k + nr_i$, $n \geq 0$, becomes $u^{\kappa_k} q^n$ times a polynomial in A, t after the substitution $z = z_0$, so $h_k^\flat = u^{\kappa_k} \tilde{h}_k(q, A, t)$ with $\tilde{h}_k$ regular and

$$\tilde{h}_k(0, A, t) = \phi_k := A^{\epsilon_k}(-\Phi)^{\max(-l_k, 0)} \psi_{i,k}(-\Phi, t).$$

The crepant boundary on the chart has coefficient $4 + \sum_k c_k \kappa_k - 2 = 1 + \delta$ on $\operatorname{div}(u)$, so on the quotient $\mathbb{A}^3_{q,A,t}$ the crepant boundary is

$$B'' := \Big(1 + \frac{\delta}{r_i}\Big) \operatorname{div}(q) + \sum_k c_k \operatorname{div}(\tilde{h}_k).$$

An element of G' is a pair of $g \in G_i$ and μ with $\mu^2 = g_y$, acting on A, t by $g_x \mu^{-1}, g_t$. If it acts trivially on A, t, then $g_t = 1$, $g_y = g_x^2$, and $g_z = g_x^{-2}$ because $y^2 z$ has the character of x^2, so $g \in R_i$ and the pair belongs to R'. Hence G'' acts faithfully on the plane $\mathbb{A}^2_{A,t}$. The involution $\iota : u \to -u$ acts by $A \to -A$, and the quotient $\pi : \mathbb{A}^2_{A,t} \to \mathbb{A}^2_{\zeta_1,\zeta_2}$, $(\zeta_1, \zeta_2) = (yz, t) = (-\Phi, t)$, is branched along $A = 0$, that is, along $\zeta_1 + \zeta_2^2 = 0$. Since

$$\operatorname{div}(A) = \frac{1}{2}\pi^* \operatorname{div}(\zeta_1 + \zeta_2^2), \qquad -\Phi = \pi^* \zeta_1,$$

the pair

$$\Big(\mathbb{A}^2_{A,t}, \sum_k c_k \operatorname{div}(\phi_k) + \frac{\max(\delta, 0)}{2} \operatorname{div}(\Phi)\Big)$$

is the crepant pullback of $(\mathbb{A}^2_{\zeta_1,\zeta_2}, \Delta_i)$, and $G'''/\langle\iota\rangle$ acts on (ζ_1, ζ_2) as H_i. Thus the pair on $\mathbb{A}^2_{A,t}$ above is lc near 0, and the quotient minimum of the pair at 0 by G'' is β by [Liu26, Lemma 2.6(5)]. Let F'' be the prime divisor over $\mathbb{A}^3_{q,A,t}$ given by the restriction of an extension of $\operatorname{mult}_{F^\flat}$, with an extension F' over the root chart. Then $\operatorname{mult}_{F'} u > 0$, $\operatorname{mult}_{F'} t > 0$, and $\operatorname{mult}_{F'} \Phi = 2\operatorname{mult}_{F'} u + \operatorname{mult}_{F'} z_0 > 0$, so $\operatorname{mult}_{F'} A > 0$: otherwise $A^2 + t^2$ would have negative order, or order 0 with non-zero residue. Thus the center of F'' is 0. Let

$$B^\natural := \operatorname{div}(q) + \sum_k c_k \operatorname{div}(\tilde{h}_k) + \frac{\max(\delta, 0)}{2} \operatorname{div}(\Phi).$$

By Lemma E.22 with $\mathbb{A}^3_{q,A,t}$ in the role of X, the group G'', q in the role of u,

$$\mathfrak{a} = \prod_k (\tilde{h}_k)^{c_k} \cdot (\Phi)^{\max(\delta,0)/2},$$

and $V = \{0\}$, $a(F'', \mathbb{A}^3, B^\natural) \geq \beta |I_{G''}(F'')|$, and the difference $a(F'', \mathbb{A}^3, B'') - a(F'', \mathbb{A}^3, B^\natural)$ is a positive multiple of

$$-\delta \operatorname{mult}_{F'} u + \frac{\max(\delta,0)}{2}\big(2 \operatorname{mult}_{F'} u + \operatorname{mult}_{F'} z_0\big) \geq 0.$$

By [Liu26, Lemma 2.6(2)(4)(5)] and [KM98, Lemma 2.30], applied to the G_i-equivariant weighted blow-up of $\mathbb{A}^3_{x,y,t}$ with weights $(1,2,0)$, whose y-chart is $\mathbb{A}^3_{u,A,t}/\mu_2$, the right-hand side of the equality for $a_{D_i}(F)$ above equals $a(F'', \mathbb{A}^3_{q,A,t}, B'')/|I_{G''}(F'')|$. Hence $a_{D_i}(F) \geq \beta$.

Assume next that $[z] < c_*$. Then $[y] \geq 1/e_0 - c_*$ and $[x] \geq (1/e_0 - c_*)/2$. A positive monomial of any h_k has $a + 2b \geq r_i - \kappa_{\max}$, hence order at least $(1/e_0 - c_*)(r_i - \kappa_{\max})e/2$, and $\operatorname{mult}_F h_k^- = \operatorname{mult}_F h_k$ for any k and i large. All monomials of B_i are negative and are kept.

Consider the root chart of $-w = (-1,-2,2,0)$ at z: $z = u^2$, $x = u^{-1}X$, $y = u^{-2}Y$, with coordinates u, X, Y, t. The element $(\lambda, \lambda^2, \lambda^{-2}, 1)$ of R_i lifts to $u \to \lambda^{-1}u$, which fixes X, Y, t, and R' consists of such lifts. Let $q := u^{r_i}$. A monomial $z^c t^d$ of B_i with $-2c = 2 + nr_i$ equals $u^{-2}q^{-n}t^d$, so $f_i = u^{-2}(X^2 + Y^2 + Yt^2 + C_i(q,t))$ with a polynomial $C_i \in (q)$, and $h_k^- = u^{-\kappa_k}\tilde{h}_k(q, X, Y, t)$ with $\tilde{h}_k$ regular and

$$\tilde{h}_k|_{q=0} = \hat{h}_k := X^{\epsilon_k} Y^{\max(l_k,0)} \psi_{i,k}(Y,t).$$

Let

$$V'' := (X^2 + Y^2 + Yt^2 + C_i(q,t) = 0) \subset \mathbb{A}^4_{q,X,Y,t}, \qquad S_3 := V'' \cap (q = 0) = (X^2 + Y^2 + Yt^2 = 0).$$

Since $X^2 + Y^2 + Yt^2 = X^2 + (Y + t^2/2)^2 - t^4/4$ and $C_i(q,t) - t^4/4 \neq 0$, V'' is near 0 a normal hypersurface of type cA, hence klt near 0 (see [KM98, Definition 5.32 and Theorem 5.34]), and S_3 is a Du Val surface of type A_3. By (E.107), the crepant boundary on V'' is

$$B'' := \Big(1 - \frac{\delta}{r_i}\Big) \operatorname{div}(q) + \sum_k c_k \operatorname{div}(\tilde{h}_k).$$

An element of G' is a pair of $g \in G_i$ and μ with $\mu^2 = g_z$, acting on X, Y, t by $g_x\mu, g_y g_z, g_t$. If it acts trivially, then $g_t = 1$, $g_y = g_z^{-1} = \mu^{-2}$, and $g_x = \mu^{-1}$, so $g \in R_i$ and the pair belongs to R'. Hence G'' acts faithfully on S_3. The involution $\iota : u \to -u$ acts by $X \to -X$, and the quotient $\pi : S_3 \to \mathbb{A}^2_{Y,t} = \mathbb{A}^2_{\zeta_1,\zeta_2}$ is branched along $Y(Y + t^2) = -X^2$. Let $C_Y := (X = Y = 0)$ and $C' := (X = Y + t^2 = 0)$ on S_3. Then $\operatorname{div}_{S_3}(X) = C_Y + C'$,

$$\operatorname{div}_{S_3}(Y) = 2C_Y = \pi^* \operatorname{div}(\zeta_1), \qquad 2C' = \pi^* \operatorname{div}(\zeta_1 + \zeta_2^2),$$

and

$$K_{S_3} = \pi^*\Big(K_{\mathbb{A}^2} + \frac{1}{2}\operatorname{div}(\zeta_1) + \frac{1}{2}\operatorname{div}(\zeta_1 + \zeta_2^2)\Big).$$

Thus the pair

$$\Big(S_3, \sum_k c_k \operatorname{div}(\hat{h}_k) + \frac{\max(-\delta,0)}{2}\operatorname{div}(Y)\Big)$$

is the crepant pullback of the plane pair with coefficient $\frac{1+\sum_k c_k\epsilon_k}{2}$ on $\operatorname{div}(\zeta_1 + \zeta_2^2)$, the $\mathbb{R}$-divisor $\sum_k c_k \operatorname{div}(\psi_{i,k})$, and coefficient

$$\frac{1 + \sum_k c_k\epsilon_k}{2} + \sum_k c_k \max(l_k, 0) + \frac{\max(-\delta,0)}{2} = \sum_k c_k \max(-l_k, 0) + \frac{\max(\delta,0)}{2}$$

on $\operatorname{div}(\zeta_1)$, where the equality follows from

$$\max(l_k,0) - \max(-l_k,0) = l_k, \qquad \max(-\delta,0) - \max(\delta,0) = -\delta, \qquad 1 + \sum_k c_k(\epsilon_k + 2l_k) = \delta.$$

The plane pair is $(\mathbb{A}^2, \Delta_i)$, and $G''/\langle\iota\rangle$ acts on $(\zeta_1, \zeta_2) = (yz, t)$ as H_i, so the section pair is lc near 0 and has quotient minimum β at 0 by [Liu26, Lemma 2.6(5)]. Let F' be a prime divisor over the root chart extending mult_F up to a positive multiple, and F'' the prime divisor over V'' given by the restriction of $\operatorname{mult}_{F'}$. Then

$$\operatorname{mult}_{F'} u = \frac{1}{2}\operatorname{mult}_{F'} z > 0, \qquad \operatorname{mult}_{F'} X = \operatorname{mult}_{F'} x + \operatorname{mult}_{F'} u > 0,$$
$$\operatorname{mult}_{F'} Y = \operatorname{mult}_{F'} y + 2\operatorname{mult}_{F'} u > 2\operatorname{mult}_{F'} u, \qquad \operatorname{mult}_{F'} t > 0,$$

so the center of F'' is 0. Let

$$B^\natural := \operatorname{div}(q) + \sum_k c_k \operatorname{div}(\tilde{h}_k) + \frac{\max(-\delta, 0)}{2}\operatorname{div}(Y).$$

By Lemma E.22 with V'' in the role of X, the group G'', q in the role of u,

$$\mathfrak{a} = \prod_k (\tilde{h}_k)^{c_k} \cdot (Y)^{\max(-\delta,0)/2},$$

and $V = \{0\}$, $a(F'', V'', B^\natural) \geq \beta |I_{G''}(F'')|$. The difference $a(F'', V'', B'') - a(F'', V'', B^\natural)$ is a positive multiple of

$$\delta \operatorname{mult}_{F'} u + \frac{\max(-\delta, 0)}{2}\operatorname{mult}_{F'} Y,$$

which is at least 0 if $\delta \geq 0$, and equals $-\delta(\operatorname{mult}_{F'} Y/2 - \operatorname{mult}_{F'} u) > 0$ if $\delta < 0$. By (E.108), $a_{D_i}(F) \geq \beta$.

Step 9. In this step we conclude the proof.

By **Steps 3, 4, 6, 7, and 8**, (E.104) holds for i large in each case. By **Step 2**, $a_{D_i}(F) \geq \beta$ for any prime divisor F over W_i with center o_i, so $\alpha_i \geq \beta$ for i large by [Liu26, Lemma 2.6]. This contradicts (E.102). □

E.10. **A double line: reduction to one circle.** In this subsection, we prove Lemma E.23.

The proof first shows that the orders $\operatorname{mult}_0 S_i$ of the sums S_i of the terms of degree at least four of the equations $x^2 + y^2 z + S_i$ are bounded (**Steps 2–5**): otherwise the degeneration to the $(2m+2, 2m+1, 2, 1)$-initial forms gives $\alpha_i \geq \gamma$ for $i \gg 0$ (**Step 4**), and Lemma 4.7 gives $\limsup_i \alpha_i \leq \gamma$ (**Step 5**) for one positive number γ, contrary to the strict increase. Then the Hausdorff limit of the groups G_i gives the circle $\mathbf{T}$ (**Step 6**).

Lemma E.23. *Notation as in Set-up 11.1. Let $(W_i \ni o_i, G_i, (h_{i,k})_k)$ be a sequence of such data, with s, m and the μ_k fixed and with W_i of quadratic rank one whose residual cubic is a double line (Definition-Lemma 6.4), let $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$ and $\alpha_i := \operatorname{mld}((W_i \ni o_i, D_i)/G_i)$, and assume that the α_i are strictly increasing in $(0, 1)$. Then, after passing to a subsequence and replacing the data with numerical admissions with the groups G_i (Proposition 11.8), the following hold.*

(1) The G_i act diagonally on ambient eigenparameters x, y, z, t in which

$$W_i = (x^2 + y^2 z + S_i(y, z, t) = 0),$$

where $S_i - S_i(0, z, t)$ is y times a polynomial in t of order at least 3 and $S_i(0, z, t)$ has order at least 4.

(2) The numbers $\operatorname{mult}_0 S_i$ are bounded.

(3) There exist integers u, v, w with $(u + v, v, 2u, w)$ primitive and a positive integer e_0 such that $|G_i \cap \mathbf{T}| \to +\infty$ and $[G_i : G_i \cap \mathbf{T}] \leq e_0$ for $\mathbf{T} := \mathbf{T}_{(u+v,v,2u,w)}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prepare the equations, replace the data with numerical admissions, and describe the groups.

If the orders $|G_i|$ were bounded on an infinite subsequence, then by Proposition 11.7(1), applied with the upper bound 1 for the quotient values, the α_i of the subsequence would belong to a

finite set, which is impossible for a strictly increasing sequence. Thus, possibly passing to a subsequence, we may assume that $|G_i| \to +\infty$.

Fix i. Since $(W_i \ni o_i)$ is an isolated Gorenstein terminal singularity, [Liu26, Lemma 5.4(2)] gives a G_i-equivariant closed embedding of $(W_i \ni o_i)$ into a smooth fourfold germ, a regular G_i-semi-invariant local equation f_i of W_i, and G_i-eigenparameters x, y, z, t of the fourfold germ at o_i. Since W_i has quadratic rank one, the quadratic part of f_i is x'^2 for a linear form x' in x, y, z, t. For $\sigma \in G_i$, $\sigma(x')^2 \in \mathbb{C}^* x'^2$, so $\sigma(x') \in \mathbb{C}^* x'$. Thus, after a G_i-equivariant linear change of the eigenparameters, we may assume that $x' = x$, so that the quadratic part of f_i is x^2. By [Liu26, Definition-Lemma 11.1] (Definition 6.1), the residual cubic $g_{i,3}(y, z, t)$ of f_i is the restriction of the cubic part of f_i to the kernel $(x = 0)$ of x^2, that is, the cubic part of $f_i(0, y, z, t)$. It is G_i-semi-invariant, since G_i acts diagonally and f_i is G_i-semi-invariant. By hypothesis, $g_{i,3} = y'^2 z'$ for two linearly independent linear forms y', z' in y, z, t. Any $\sigma \in G_i$ acts linearly on the span V_0 of y, z, t and maps $y'^2 z'$ to a non-zero multiple of $y'^2 z'$. Thus, by unique factorization in $\mathbb{C}[y, z, t]$, $\sigma(y') \in \mathbb{C}^* y'$ and $\sigma(z') \in \mathbb{C}^* z'$. Since G_i acts diagonally on V_0, the G_i-stable plane spanned by y' and z' has a G_i-stable complement in V_0 spanned by an eigenvector t'. Possibly replacing (y, z, t) with (y', z', t'), we may assume that $g_{i,3} = y^2 z$.

By Lemma C.3, there exists a G_i-equivariant automorphism ψ_i of $\mathfrak{R} := \mathbb{C}[[x, y, z, t]]$ inducing the identity on the tangent space such that

$$\psi_i(\hat{f}_i) = U_i \hat{F}_i, \qquad \hat{F}_i := x^2 + y^2 z + \lambda_i y t^{\ell_i} + \hat{q}_i(z, t),$$

where $\hat{f}_i$ is the expansion of f_i, U_i is a unit of $\mathfrak{R}$, $\lambda_i \in \mathbb{C}$, $\ell_i \geq 3$ is an integer, and $\hat{q}_i \in \mathbb{C}[[z, t]]$ with $\operatorname{mult}_0 \hat{q}_i \geq 4$. For $\sigma \in G_i$, let $\sigma_x, \sigma_y, \sigma_z, \sigma_t \in \mathbb{C}^*$ be the scalars by which σ multiplies x, y, z, t. Since f_i is G_i-semi-invariant with quadratic part x^2, σ multiplies $\hat{f}_i$, and hence $\psi_i(\hat{f}_i)$, by σ_x^2. Thus

$$\sigma_x^{-2} \sigma(\hat{F}_i) = \frac{U_i}{\sigma(U_i)} \hat{F}_i.$$

Both $\hat{F}_i$ and $\sigma_x^{-2}\sigma(\hat{F}_i)$ are monic polynomials of degree two in x whose other coefficients belong to the maximal ideal of $\mathbb{C}[[y, z, t]]$. Since $\hat{F}_i(x, 0, 0, 0) = x^2$, the power series $\hat{F}_i$ is x-general of order 2. Thus, by the uniqueness of the unit and of the coefficients in the Weierstrass preparation theorem [GLS07, Chapter I, Theorem 1.6] (with the trivial valuation on $\mathbb{C}$, for which $K\langle x \rangle = K[[x]]$ by [GLS07, Chapter I, Remark 1.1.1(1)]), $\sigma(U_i) = U_i$. Thus U_i is G_i-invariant, and $\hat{F}_i$ is G_i-semi-invariant with the character of x^2.

For any k, let $\hat{h}_{i,k} \in \mathfrak{R}$ be the expansion of a G_i-semi-invariant lift of $h_{i,k}$ to the ambient germ, and let $c^{(k)} \in \mathbb{R}^s$ be the vector whose k-th coordinate is $\frac{1}{m}$ and whose other coordinates are 0. We apply Construction-Theorem 4.13 to W_i, f_i, x, y, z, t, the lifts of the $h_{i,k}$ and the labels $c_1, \ldots, c_s$, with the further label vectors $c^{(1)}, \ldots, c^{(s)}$ in (iv), with a prime divisor computing α_i and the prime divisor E_0 over W_i of Set-up 9.3(5) and the paragraph after it in (v), and with $\psi = \psi_i$, $v = U_i^{-1}$, and $u_1 = \cdots = u_s = 1$ in (vi). The prime divisor E_0 is protected for its use in Subsections E.11 and E.12. Hypothesis (iii) holds since $D_i = \sum_k \mu_k C_{i,k}$, so that $a(P, W_i, D_i) = \sum_k \mu_k a(P, W_i, C_{i,k})$ for any prime divisor P over W_i, and each $(W_i, C_{i,k})$ is terminal near o_i. Hypothesis (iv) holds since terminal pairs are klt [KM98, Corollary 2.31(1) and (3)].

Let $N \geq N_1$ and $L \geq L_1$ be integers as in Construction-Theorem 4.13. Let q_i be the sum of the terms of $\hat{q}_i$ of degree less than L, and let $h_{i,k,0}$ be the sum of the terms of $\psi_i(\hat{h}_{i,k})$ of degree less than N. Since G_i acts diagonally, q_i and $h_{i,k,0}$ are G_i-semi-invariant with the characters of $\hat{q}_i$ and $\hat{h}_{i,k}$. Let

$$f_{i,0} := x^2 + y^2 z + \lambda_i y t^{\ell_i} + q_i(z, t),$$

so that $f_{i,0} \equiv U_i^{-1} \psi_i(\hat{f}_i) \pmod{(x, y, z, t)^L}$, and let $W_{i,0} := (f_{i,0} = 0) \subset \mathbb{A}^4$. By Construction-Theorem 4.13(1) and (3), applied with the subgroups G_i and $\{1\}$, the following hold: $(W_{i,0} \ni 0)$ is an isolated Gorenstein terminal germ on which G_i acts faithfully, the pair

$(W_{i,0}, \sum_k c_k \operatorname{div}(h_{i,k,0}|_{W_{i,0}}))$ is klt near 0 with

$$\alpha_i = \operatorname{mld}\Big(\Big(W_{i,0} \ni 0, \sum_k c_k \operatorname{div}(h_{i,k,0}|_{W_{i,0}})\Big)/G_i\Big),$$

and for any k the pair $(W_{i,0}, \frac{1}{m}\operatorname{div}(h_{i,k,0}|_{W_{i,0}}))$ is klt near 0 with

$$\operatorname{mld}\Big(W_{i,0} \ni 0, \frac{1}{m}\operatorname{div}(h_{i,k,0}|_{W_{i,0}})\Big) = \operatorname{mld}(W_i \ni o_i, C_{i,k}) > 1,$$

where the inequality holds since $(W_i, C_{i,k})$ is terminal near o_i and its log discrepancies at the prime divisors over W_i with center o_i belong to $\frac{1}{m}\mathbb{Z}$ (Definition-Lemma 11.4(1)), hence are at least $1 + \frac{1}{m}$. In particular,

$$a(P, W_{i,0}, 0) - \frac{1}{m}\operatorname{mult}_P h_{i,k,0} > 0$$

for any k and any prime divisor P over $W_{i,0}$ with center 0. Thus $(W_{i,0}, (h_{i,k,0})_k)$ is a numerical admission of $(W_i, (h_{i,k})_k)$ (Construction-Theorem 4.13) with the group G_i, and it satisfies (i) and (ii) of Proposition 11.8.

Since $f_{i,0}$ is monic of degree two in x, division by $f_{i,0}$ writes $h_{i,k,0} = Q_{i,k}f_{i,0} + r_{i,k}$ with polynomials $Q_{i,k}$ and $r_{i,k}$, where $r_{i,k}$ has degree at most one in x. By the uniqueness of the division, $r_{i,k}$ is G_i-semi-invariant with the character of $h_{i,k,0}$, and $r_{i,k}|_{W_{i,0}} = h_{i,k,0}|_{W_{i,0}}$. Possibly replacing $(W_i \ni o_i, (h_{i,k})_k)$ with $(W_{i,0} \ni 0, (r_{i,k})_k)$, we may assume that the following hold for any i.

(a) $W_i = (f_i = 0) \subset \mathbb{A}^4$, where $f_i = x^2 + y^2 z + S_i$, $S_i = \lambda_i y t^{\ell_i} + q_i(z,t)$, $\lambda_i \in \mathbb{C}$, $\ell_i \geq 3$ is an integer, and q_i is a polynomial with $\operatorname{mult}_0 q_i \geq 4$. The germ $(W_i \ni 0)$ is isolated Gorenstein terminal. The group G_i is finite cyclic, acts diagonally on $\mathbb{A}^4$ and faithfully on W_i, and f_i is G_i-semi-invariant.

(b) The $h_{i,k}$ are G_i-semi-invariant polynomials of degree at most one in x which are non-zero on W_i. For $D_i := \sum_k c_k \operatorname{div}(h_{i,k}|_{W_i})$, the pair (W_i, D_i) is klt near 0 and $\alpha_i = \operatorname{mld}((W_i \ni 0, D_i)/G_i)$.

(c) For any k, the pair $(W_i, \frac{1}{m}\operatorname{div}(h_{i,k}|_{W_i}))$ is klt near 0 and $\operatorname{mld}(W_i \ni 0, \frac{1}{m}\operatorname{div}(h_{i,k}|_{W_i})) > 1$.

If $S_i = 0$, then all partial derivatives of $f_i = x^2 + y^2 z$ vanish along the surface $(x = y = 0)$, contrary to (a). Thus $\nu_i := \operatorname{mult}_0 S_i$ is an integer with $\nu_i \geq 4$.

We now describe the groups. For $\sigma \in G_i$, since f_i is G_i-semi-invariant and contains the monomials x^2 and $y^2 z$, $\sigma_x^2 = \sigma_y^2 \sigma_z$. Let $p(\sigma) := \sigma_x/\sigma_y$, $\beta(\sigma) := \sigma_y$, and $\theta(\sigma) := \sigma_t$. Then

$$(\sigma_x, \sigma_y, \sigma_z, \sigma_t) = (p(\sigma)\beta(\sigma), \beta(\sigma), p(\sigma)^2, \theta(\sigma)), \tag{E.113}$$

and $\sigma \to (p(\sigma), \beta(\sigma), \theta(\sigma))$ is injective since G_i acts faithfully on $\mathbb{A}^4$. Let $\mathbb{T}^4 = \mathbb{R}^4/\mathbb{Z}^4$, $\rho : \mathbb{R}^4 \to \mathbb{T}^4$, and N_{G_i} be as in [Liu26, Definition 2.15], where G_i is regarded as a subgroup of $\mathbb{T}^4$ by identifying σ with $\rho(\omega)$ for $\omega = (\omega_x, \omega_y, \omega_z, \omega_t)$ such that

$$(\sigma_x, \sigma_y, \sigma_z, \sigma_t) = (e^{2\pi\sqrt{-1}\omega_x}, e^{2\pi\sqrt{-1}\omega_y}, e^{2\pi\sqrt{-1}\omega_z}, e^{2\pi\sqrt{-1}\omega_t}),$$

so that $N_{G_i} = \rho^{-1}(G_i)$. If $\omega \in N_{G_i}$ and $d := \omega_x - \omega_y$, then $\sigma := \rho(\omega)$ satisfies $p(\sigma) = e^{2\pi\sqrt{-1}d}$, and (E.113) implies

$$l := \omega_z - 2d \in \mathbb{Z}. \tag{E.114}$$

Conversely, if $\sigma \in G_i$ and $P, B, T \in \mathbb{R}$ satisfy $p(\sigma) = e^{2\pi\sqrt{-1}P}$, $\beta(\sigma) = e^{2\pi\sqrt{-1}B}$, and $\theta(\sigma) = e^{2\pi\sqrt{-1}T}$, then (E.113) implies

$$(P + B, B, 2P, T) \in N_{G_i}. \tag{E.115}$$

Let $\bar{G}_i \subset (\mathbb{C}^*)^2$ be the image of G_i under $\sigma \to (p(\sigma), \theta(\sigma))$, regarded as a subgroup of $\mathbb{T}^2 := \mathbb{R}^2/\mathbb{Z}^2$ by identifying $(e^{2\pi\sqrt{-1}P}, e^{2\pi\sqrt{-1}T})$ with $\rho_2(P, T)$, where $\rho_2 : \mathbb{R}^2 \to \mathbb{T}^2$ is the quotient map, and let K_i be the kernel of $G_i \to \bar{G}_i$. Any $\sigma \in K_i$ multiplies x and y by $\beta(\sigma)$ and fixes z and t. Thus σ multiplies f_i by $\beta(\sigma)^2$, the monomial yt^{ℓ_i} by $\beta(\sigma)$, and any monomial of q_i by 1.

If $\beta(\sigma)^2 \neq 1$, then $\beta(\sigma) \neq \beta(\sigma)^2$, so $\lambda_i = 0$ and $q_i = 0$, contrary to $S_i \neq 0$. Thus $\beta(\sigma) \in \{\pm 1\}$ for any $\sigma \in K_i$. Since $\sigma \to \beta(\sigma)$ is injective on K_i, $|K_i| \leq 2$. In particular, $|\bar{G}_i| \geq |G_i|/2 \to +\infty$.

Step 2. In **Steps 2–5** we assume that $\nu_i \to +\infty$ after passing to a subsequence, and we derive a contradiction. In this step we describe the Hausdorff limit of the groups $\bar{G}_i$.

By Lemma 4.6, after passing to a subsequence, there exists a closed subgroup $H \subset \mathbb{T}^2$ such that $\bar{G}_i \to H$ in the Hausdorff distance and $\bar{G}_i \subset H$ for $i \gg 0$. If H were finite, then $|G_i| \leq 2|\bar{G}_i| \leq 2|H|$ for $i \gg 0$, contrary to **Step 1**. Thus $\dim H \geq 1$. By [Liu26, Lemma 4.3(1)], H/H^0 is finite and $H^0 = \rho_2(V)$ for a linear subspace $V \subset \mathbb{R}^2$ spanned by $V \cap \mathbb{Z}^2$.

We show that V contains no vector with both coordinates positive. Suppose that $(P_0, T_0) \in V$ with $P_0, T_0 > 0$. Possibly replacing (P_0, T_0) with a positive multiple, we may assume that $P_0 + T_0 < \alpha_1$. Since $\rho_2(P_0, T_0) \in H$, there exist $\sigma_i \in G_i$ and real numbers P_i, T_i with $p(\sigma_i) = e^{2\pi\sqrt{-1}P_i}$, $\theta(\sigma_i) = e^{2\pi\sqrt{-1}T_i}$, and $(P_i, T_i) \to (P_0, T_0)$. Let $B_i \in [1, 2)$ with $\beta(\sigma_i) = e^{2\pi\sqrt{-1}B_i}$. By (E.115),

$$\omega_i := (P_i + B_i, B_i, 2P_i, T_i) \in N_{G_i},$$

and for $i \gg 0$ all coordinates of ω_i are at least $\frac{1}{2}\min(P_0, T_0) > 0$. The monomials x^2 and $y^2 z$ have ω_i-weight $2(P_i + B_i) \leq 2(P_0 + 3)$, whereas

$$\mathrm{wt}_{\omega_i}(S_i) \geq \frac{1}{2}\nu_i \min(P_0, T_0) \to +\infty.$$

Thus $\mathrm{wt}_{\omega_i}(f_i) = 2(P_i + B_i)$ for $i \gg 0$, and since $\mathrm{wt}_{\omega_i}(h_{i,k}) \geq 0$, Lemma 4.7 implies

$$\alpha_i \leq (P_i + B_i) + B_i + 2P_i + T_i - 2(P_i + B_i) - \sum_k c_k \,\mathrm{wt}_{\omega_i}(h_{i,k}) \leq P_i + T_i.$$

Since $P_i + T_i \to P_0 + T_0 < \alpha_1 \leq \alpha_i$, this is a contradiction for $i \gg 0$.

If $\dim V = 2$, then $(1, 1) \in V$, which is impossible. Thus $\dim V = 1$, and $V = \mathbb{R}(v_1, v_2)$ with $v_1 v_2 \leq 0$, since otherwise (v_1, v_2) or $(-v_1, -v_2)$ would have both coordinates positive. Let (a', c') be the primitive vector of $\mathbb{Z}^2_{\geq 0}$ with $a'v_1 + c'v_2 = 0$, which exists since $v_1 v_2 \leq 0$. Then $V = \{(P, T) \mid a'P + c'T = 0\}$, and $a'\mathbb{Z} + c'\mathbb{Z} = \mathbb{Z}$ since $\gcd(a', c') = 1$. The subgroup $N_H := \rho_2^{-1}(H)$ of $\mathbb{R}^2$ contains $V + \mathbb{Z}^2$, and $N_H/(V + \mathbb{Z}^2) \cong H/H^0$ is finite [Liu26, Lemma 4.3(1)]. Hence the image of N_H under $(P, T) \to a'P + c'T$ is a subgroup of $\mathbb{R}$ containing $\mathbb{Z}$ with finite index, so it is $\frac{1}{n}\mathbb{Z}$ for a positive integer n, and $N_H = \{(P, T) \mid n(a'P + c'T) \in \mathbb{Z}\}$ since $N_H \supset V$. Thus

$$H = \{\rho_2(P, T) \mid n(a'P + c'T) \in \mathbb{Z}\}. \tag{E.116}$$

In particular, for $i \gg 0$, any $\sigma \in G_i$ and any real numbers P, T with $p(\sigma) = e^{2\pi\sqrt{-1}P}$ and $\theta(\sigma) = e^{2\pi\sqrt{-1}T}$ satisfy $n(a'P + c'T) \in \mathbb{Z}$.

Step 3. In this step we find, for any k, a monomial $z^{g_k} t^{j_k}$ of $h_{i,k}$ with $2g_k + j_k \leq m - 1$ such that, for $i \gg 0$, any other monomial $z^g t^j$ of $h_{i,k}$ of bounded degree satisfies $(2g - 2g_k, j - j_k) \in n\mathbb{Z}_{\geq 1}(a', c')$ (see (E.119) for the precise statement).

For $i \gg 0$ we have $\nu_i \geq 4m + 5$. The monomials x^2 and $y^2 z$ have $(2m + 2, 2m + 1, 2, 1)$-weight $4m + 4$, while S_i has weight at least $\nu_i \geq 4m + 5$. Thus

$$\mathrm{in}_{(2m+2,2m+1,2,1)}(f_i) = x^2 + y^2 z. \tag{E.117}$$

The polynomial $x^2 + y^2 z$ is irreducible, since z has exponent one in $y^2 z$, so that $-y^2 z$ is not a square in $\mathbb{C}[y, z, t]$. Thus $(x^2 + y^2 z = 0) \subset \mathbb{A}^4$ is integral, and it is not contained in any coordinate hyperplane. By Lemma 4.12, the weighted blow-up with weights $(2m + 2, 2m + 1, 2, 1)$ yields a prime divisor E_i' over W_i with center 0 and

$$a(E_i', W_i, 0) = (2m + 2) + (2m + 1) + 2 + 1 - (4m + 4) = 2.$$

For any k, the polynomial $\mathrm{in}_{(2m+2,2m+1,2,1)}(h_{i,k})$ is non-zero of degree at most one in x, so it is not divisible by x^2+y^2z, and Lemma 4.12 implies

$$\mathrm{mult}_{E_i'} h_{i,k} = \mathrm{wt}_{(2m+2,2m+1,2,1)}(h_{i,k}).$$

By **Step 1**(c),

$$2-\frac{1}{m}\,\mathrm{wt}_{(2m+2,2m+1,2,1)}(h_{i,k}) = a\Big(E_i', W_i, \frac{1}{m}\,\mathrm{div}(h_{i,k}|_{W_i})\Big) \geq \mathrm{mld}\Big(W_i \ni 0, \frac{1}{m}\,\mathrm{div}(h_{i,k}|_{W_i})\Big) > 1.$$

Since the weight is an integer, $\mathrm{wt}_{(2m+2,2m+1,2,1)}(h_{i,k}) \leq m-1$. A monomial $x^{\epsilon}y^{b_1}z^{b_2}t^{b_3}$ with $\epsilon+b_1 \geq 1$ has weight at least $2m+1$, and a monomial $z^g t^j$ has weight $2g+j$. Thus, for $i \gg 0$ and any k, $h_{i,k}$ contains a monomial $z^g t^j$ with $2g+j \leq m-1$. Units are allowed, so $g=j=0$ may occur. Since there are only finitely many such pairs (g,j), after passing to a subsequence there exist pairs (g_k', j_k'), $k=1,\dots,s$, with $2g_k'+j_k' \leq m-1$ such that $z^{g_k'}t^{j_k'}$ occurs in $h_{i,k}$ for any i and k.

Let $z^g t^j$ and $z^{\bar g}t^{\bar j}$ be monomials which occur together in $h_{i,k}$ for infinitely many i. Since $h_{i,k}$ is G_i-semi-invariant, $\sigma_z^{g-\bar g}\sigma_t^{j-\bar j}=1$ for any $\sigma \in G_i$, so by (E.113) the character

$$\rho_2(P,T) \to e^{2\pi\sqrt{-1}(2(g-\bar g)P+(j-\bar j)T)}$$

of $\mathbb{T}^2$ is trivial on $\bar G_i$ for infinitely many i. Since $\bar G_i \to H$, the character is trivial on H, and (E.116) implies

$$(2(g-\bar g), j-\bar j) \in n\mathbb{Z}(a',c'). \tag{E.118}$$

Fix k, and let Π_k be the set of pairs $(g,j) \in \mathbb{Z}^2_{\geq 0}$ with $(2(g-g_k'), j-j_k') \in n\mathbb{Z}(a',c')$. Since $a',c' \geq 0$ and $(a',c') \neq (0,0)$, the set Π_k is totally ordered by the componentwise order, and only finitely many elements of Π_k are at most (g_k', j_k'). Let (g_k, j_k) be the least element of Π_k such that $z^{g_k}t^{j_k}$ occurs in $h_{i,k}$ for infinitely many i. Passing to a subsequence for $k=1,\dots,s$ in turn, we may assume that $z^{g_k}t^{j_k}$ occurs in any $h_{i,k}$ and that no $z^g t^j$ with $(g,j) \in \Pi_k$ smaller than (g_k,j_k) occurs in any $h_{i,k}$. In particular, $2g_k+j_k \leq 2g_k'+j_k' \leq m-1$, and Π_k is also the set of pairs $(g,j) \in \mathbb{Z}^2_{\geq 0}$ with $(2(g-g_k), j-j_k) \in n\mathbb{Z}(a',c')$. By (E.118), applied with $(\bar g, \bar j) = (g_k, j_k)$, a monomial $z^g t^j$ with $(g,j) \notin \Pi_k$ occurs in only finitely many $h_{i,k}$. Since only finitely many pairs (g,j) satisfy $2g+j \leq C$ for a real number C, we obtain the following: for any real number $C>0$ there exists i_C such that for any $i \geq i_C$, any k, and any monomial $z^g t^j \neq z^{g_k}t^{j_k}$ of $h_{i,k}$ with $2g+j \leq C$,

$$(2g-2g_k, j-j_k) = en(a',c') \quad \text{for an integer } e \geq 1. \tag{E.119}$$

Applying (E.119) with $C=m-1$, any monomial $z^g t^j \neq z^{g_k}t^{j_k}$ of $h_{i,k}$ of weight $2g+j \leq m-1$ has weight $2g_k+j_k+en(a'+c') > 2g_k+j_k$, and monomials divisible by x or y have weight at least $2m+1$. Thus, for $i \gg 0$ and any k,

$$\mathrm{in}_{(2m+2,2m+1,2,1)}(h_{i,k}) = \vartheta_{i,k} z^{g_k}t^{j_k}, \qquad \vartheta_{i,k} \in \mathbb{C}^*. \tag{E.120}$$

Let

$$A_* := 1-2\sum_k c_k g_k, \qquad B_* := 1-\sum_k c_k j_k.$$

Since $2g_k+j_k \leq m-1$ for any k and $\sum_k c_k = \frac{1}{m}$,

$$2\sum_k c_k g_k + \sum_k c_k j_k \leq \frac{m-1}{m} = 1-\frac{1}{m}.$$

Thus $A_*, B_* \in [\frac{1}{m}, 1]$. In particular, $0 < A_* \leq 1$ and $0 < B_* \leq 1$.

Step 4. In this step we prove that $\alpha_i \geq \gamma$ for $i \gg 0$, where

$$\gamma := \min\Big(\Big\{\frac{A_*}{na'} \;\Big|\; a'>0\Big\} \cup \Big\{\frac{B_*}{nc'} \;\Big|\; c'>0\Big\}\Big) > 0.$$

Let

$$\Theta := \operatorname{div}(x^2 + y^2 z) + \sum_k c_k \operatorname{div}(z^{g_k} t^{j_k})$$

on $\mathbb{A}^4$, and for $\omega = (\omega_x, \omega_y, \omega_z, \omega_t) \in \mathbb{R}^4$ let

$$\phi_*(\omega) := \frac{A_*\omega_z}{2} + B_*\omega_t + \left|\omega_x - \omega_y - \frac{\omega_z}{2}\right|.$$

A direct calculation shows that

$$\omega_x + \omega_y + \omega_z - \min(2\omega_x, 2\omega_y + \omega_z) = \frac{\omega_z}{2} + \left|\omega_x - \omega_y - \frac{\omega_z}{2}\right|,$$

hence

$$\omega_x + \omega_y + \omega_z + \omega_t - \operatorname{wt}_\omega(x^2 + y^2 z) - \Big(\sum_k c_k g_k\Big)\omega_z - \Big(\sum_k c_k j_k\Big)\omega_t = \phi_*(\omega) \tag{E.121}$$

for any $\omega \in \mathbb{R}^4_{\geq 0}$.

Claim E.24. *For $i \gg 0$ and any $\omega \in N_{G_i} \cap \mathbb{R}^4_{>0}$, $\phi_*(\omega) \geq \gamma$.*

Proof. Let i be large enough that $\bar{G}_i \subset H$, let $\omega \in N_{G_i} \cap \mathbb{R}^4_{>0}$, and let $d := \omega_x - \omega_y$. By (E.114), $l = \omega_z - 2d \in \mathbb{Z}$. The element $\sigma := \rho(\omega)$ of G_i satisfies $p(\sigma) = e^{2\pi\sqrt{-1}d}$ and $\theta(\sigma) = e^{2\pi\sqrt{-1}\omega_t}$, so **Step 2** implies $n(a'd + c'\omega_t) \in \mathbb{Z}$. If $d \geq 0$, then

$$\frac{A_*\omega_z}{2} + \left|d - \frac{\omega_z}{2}\right| \geq A_* d, \tag{E.122}$$

since the left-hand side satisfies

$$\frac{A_*\omega_z}{2} + \left|d - \frac{\omega_z}{2}\right| \begin{cases} \geq \dfrac{A_*\omega_z}{2} \geq A_* d & \text{if } \dfrac{\omega_z}{2} \geq d, \\ = d - (1 - A_*)\dfrac{\omega_z}{2} \geq d - (1 - A_*)d = A_* d & \text{if } \dfrac{\omega_z}{2} < d, \end{cases}$$

where the second line uses $A_* \leq 1$.

Case 1: $a' = 0$. Then $c' = 1$ and $\gamma = \frac{B_*}{n}$. Since $n\omega_t \in \mathbb{Z}$ and $\omega_t > 0$, $\omega_t \geq \frac{1}{n}$, so $\phi_*(\omega) \geq B_*\omega_t \geq \frac{B_*}{n} = \gamma$.

Case 2: $c' = 0$. Then $a' = 1$, $\gamma = \frac{A_*}{n}$, and $nd \in \mathbb{Z}$. If $d > 0$, then $d \geq \frac{1}{n}$, and (E.122) implies $\phi_*(\omega) \geq A_* d \geq \frac{A_*}{n} = \gamma$. If $d < 0$, then $-d \geq \frac{1}{n}$ and

$$\phi_*(\omega) \geq \frac{\omega_z}{2} - d > -d \geq \frac{1}{n} \geq \frac{A_*}{n} = \gamma.$$

If $d = 0$, then $\omega_z = l$ is a positive integer, and

$$\phi_*(\omega) \geq \frac{(1 + A_*)\omega_z}{2} \geq \frac{1 + A_*}{2} \geq A_* \geq \gamma.$$

Case 3: $a', c' > 0$. Then $A_* \geq \gamma n a'$ and $B_* \geq \gamma n c'$. If $d \geq 0$, then (E.122) implies

$$\phi_*(\omega) \geq A_* d + B_*\omega_t \geq \gamma n(a'd + c'\omega_t) \geq \gamma,$$

since $n(a'd + c'\omega_t)$ is an integer and $n(a'd + c'\omega_t) \geq nc'\omega_t > 0$. If $d < 0$, then $l = \omega_z + 2(-d)$ is a positive integer, and

$$\phi_*(\omega) \geq \frac{\omega_z}{2} - d = \frac{l}{2} \geq \frac{1}{2}.$$

This proves the claim if $\gamma \leq \frac{1}{2}$. If $\gamma > \frac{1}{2}$, then $\frac{A_*}{na'} > \frac{1}{2}$ and $A_* \leq 1$ imply $na' = 1$, and similarly $nc' = 1$. Thus $n = a' = c' = 1$, $\gamma = \min(A_*, B_*)$, and $l' := d + \omega_t = n(a'd + c'\omega_t) \in \mathbb{Z}$. If $l' \leq -1$, then $-d \geq 1 + \omega_t > 1$, and

$$\phi_*(\omega) \geq \frac{\omega_z}{2} - d > 1 \geq A_* \geq \gamma.$$

If $l' \geq 0$, then $\omega_t \geq -d$, and

$$\phi_*(\omega) \geq \frac{(1+A_*)\omega_z}{2} + (1+B_*)(-d) \geq (1+\gamma)\Big(\frac{\omega_z}{2} - d\Big) = \frac{(1+\gamma)l}{2} \geq \frac{1+\gamma}{2} \geq \gamma.$$

Cases 1–3 exhaust all possibilities, since $(a', c') \in \mathbb{Z}^2_{\geq 0}$ is primitive. □

Proof of Lemma E.23 continued. We compute the quotient minimal log discrepancy of $(\mathbb{A}^4 \ni 0, \Theta)$ by G_i. We use [Liu26, Set-up 8.4] in dimension four, with the group G_i, $f_1 = x^2 + y^2 z$, $b_1 = 1$, $f_2 = z$, $b_2 = \sum_k c_k g_k$, $f_3 = t$, and $b_3 = \sum_k c_k j_k$, where f_2 is omitted if $\sum_k c_k g_k = 0$ and f_3 is omitted if $\sum_k c_k j_k = 0$. Recall from [Liu26, Set-up 8.4] that

$$\psi(\omega) = \sum \omega_\iota - \sum_j b_j \operatorname{wt}_\omega(f_j),$$

that the weighted leading boundary at $\omega \in \mathbb{R}^4_{>0}$ is

$$B_\omega = \sum_j b_j \operatorname{div}(\operatorname{in}_\omega(f_j))$$

on $(\mathbb{C}^*)^4$, that Σ is a regular subdivision of $\mathbb{R}^4_{\geq 0}$ with respect to N_{G_i} refining the Newton fans of the f_j, with toric morphism $\mu_\Sigma : V_\Sigma \to \mathbb{A}^4/G_i$, and that Δ_Σ is the reduced toric boundary of V_Σ and B'_Σ the strict transform of the closure of the restriction of the quotient boundary to $(\mathbb{C}^*)^4/G_i$. The polynomials f_j are G_i-semi-invariant by (E.113), and by (E.121) the function ψ of [Liu26, Set-up 8.4] is ϕ_*. For any $\omega \in \mathbb{R}^4_{>0}$, the polynomial $\operatorname{in}_\omega(x^2 + y^2 z)$ is x^2, $y^2 z$, or $x^2 + y^2 z$, so the zero locus of $\operatorname{in}_\omega(x^2 + y^2 z)$ on $(\mathbb{C}^*)^4$ is empty or smooth, since the derivative of $x^2 + y^2 z$ with respect to z is y^2. Thus the pair $((\mathbb{C}^*)^4, B_\omega)$ of [Liu26, Set-up 8.4] is lc for any $\omega \in \mathbb{R}^4_{>0}$. Let $\bar{0}$ be the image of 0 in $\mathbb{A}^4/G_i$. By [Liu26, Lemma 8.6(1)], $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$. Since $A_*, B_* \geq 0$, the function ϕ_* is non-negative on $\mathbb{R}^4_{\geq 0}$. Let $(\mathbb{A}^4/G_i, \Theta_{\mathbb{A}^4/G_i})$ be the crepant quotient of $(\mathbb{A}^4, \Theta)$ by G_i. By [Liu26, Lemma 8.5], the crepant pullback of $K_{\mathbb{A}^4/G_i} + \Theta_{\mathbb{A}^4/G_i}$ to V_Σ has boundary

$$\Delta_\Sigma + B'_\Sigma - \sum_v \phi_*(v) E_v \leq \Delta_\Sigma + B'_\Sigma,$$

where v runs over the primitive generators of the rays of Σ and E_v is the toric prime divisor of V_Σ corresponding to v, and the inequality holds since $\phi_* \geq 0$ on $\mathbb{R}^4_{\geq 0}$. Since $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$, the pair $(\mathbb{A}^4/G_i, \Theta_{\mathbb{A}^4/G_i})$ is lc near $\bar{0}$. Thus $(\mathbb{A}^4, \Theta)$ is lc near 0 by [Liu26, Lemma 2.6(2)]. By [Liu26, Lemma 8.6(2)] and Claim E.24, for $i \gg 0$,

$$\operatorname{mld}((\mathbb{A}^4 \ni 0, \Theta)/G_i) = \min\{\phi_*(\omega) \mid \omega \in N_{G_i} \cap \mathbb{R}^4_{>0}\} \geq \gamma. \tag{E.123}$$

We now compare α_i with (E.123). Let $X_i := \mathbb{A}^4/G_i$ with quotient map π_i, and let $\Delta_i \geq 0$ be the $\mathbb{Q}$-divisor on X_i with $K_{\mathbb{A}^4} = \pi_i^*(K_{X_i} + \Delta_i)$. We first verify the hypothesis of [Nak16b, Theorem 1.2] that X_i has a crepant resolution in the category of Deligne–Mumford stacks. The variety X_i is normal and $\mathbb{Q}$-factorial [KM98, Lemma 5.16], and Δ_i is $\mathbb{Q}$-Cartier. Let $G'_i \subset G_i$ be the subgroup generated by the elements which act non-trivially on exactly one of the coordinates. For $\iota \in \{x, y, z, t\}$, let $G_{i,\iota} \subset G_i$ be the subgroup of elements acting trivially on the other three coordinates, and $e_\iota := |G_{i,\iota}|$. Then $G'_i = \prod_\iota G_{i,\iota}$, the quotient $\mathbb{A}^4/G'_i$ is the affine space with coordinates $x^{e_x}, y^{e_y}, z^{e_z}, t^{e_t}$, and G_i/G'_i acts on it diagonally. If $\sigma \in G_i$ fixes y^{e_y}, z^{e_z} and t^{e_t}, then $\sigma_\iota^{e_\iota} = 1$ for $\iota \in \{y, z, t\}$. Since $G_{i,\iota}$ acts faithfully on ι through the group of e_ι-th roots of unity, there exist $\tau_\iota \in G_{i,\iota}$ multiplying the coordinate ι by σ_ι, so $\sigma\tau_y^{-1}\tau_z^{-1}\tau_t^{-1} \in G_{i,x}$ and $\sigma \in G'_i$. Replacing x by y, z, or t in the last two sentences shows that any $\sigma \in G_i$ fixing three of the coordinates $x^{e_x}, y^{e_y}, z^{e_z}, t^{e_t}$ belongs to G'_i, so G_i/G'_i contains no pseudoreflection of $\mathbb{A}^4/G'_i$. Since the fixed locus of a diagonal element is the linear subspace on which the coordinates with non-trivial eigenvalue vanish, G_i/G'_i acts on $\mathbb{A}^4/G'_i$ faithfully and freely in codimension one. By [Nak16b, the paragraph after Theorem 1.2], the quotient stack $[(\mathbb{A}^4/G'_i)/(G_i/G'_i)]$ is a crepant resolution of X_i by a smooth Deligne–Mumford stack.

Let $\tilde{f}_i, \tilde{h}_{i,k} \in \mathbb{C}[x,y,z,t,\varsigma]$ be the polynomials such that for $\varsigma \neq 0$

$$\tilde{f}_i = \varsigma^{-4m-4} f_i(\varsigma^{2m+2}x, \varsigma^{2m+1}y, \varsigma^2 z, \varsigma t), \qquad \tilde{h}_{i,k} = \varsigma^{-2g_k - j_k} h_{i,k}(\varsigma^{2m+2}x, \varsigma^{2m+1}y, \varsigma^2 z, \varsigma t).$$

They exist since any monomial of f_i has $(2m+2, 2m+1, 2, 1)$-weight at least $4m+4$, and any monomial of $h_{i,k}$ has weight at least $2g_k + j_k$ by (E.120). By (E.117) and (E.120), for $i \gg 0$ their values at $\varsigma = 0$ are $x^2 + y^2 z$ and $\vartheta_{i,k} z^{g_k} t^{j_k}$, and their values at $\varsigma = 1$ are f_i and $h_{i,k}$. The diagonal scaling commutes with G_i, so $\tilde{f}_i$ and $\tilde{h}_{i,k}$ are G_i-semi-invariant with the characters of f_i and $h_{i,k}$, and their $|G_i|$-th powers are G_i-invariant. Thus

$$\mathfrak{a}_i := (\tilde{f}_i^{|G_i|})^{1/|G_i|} \prod_k (\tilde{h}_{i,k}^{|G_i|})^{c_k/|G_i|}$$

is an $\mathbb{R}$-ideal on $X_i \times \mathbb{A}^1_\varsigma$ with positive exponents. Let $\mathfrak{a}_{i,\varsigma}$ be the restriction of $\mathfrak{a}_i$ to $X_i \times \{\varsigma\}$, and recall that $\bar{0} \in X_i$ is the image of 0. Let

$$\eta_i(\varsigma) := \mathrm{mld}(X_i \ni \bar{0}, \Delta_i, \mathfrak{a}_{i,\varsigma}).$$

For any polynomial f and any prime divisor P over $\mathbb{A}^4$, the order of the $\mathbb{R}$-ideal $(f^{|G_i|})^{1/|G_i|}$ along P is $\mathrm{mult}_P f$. Thus $(X_i, \Delta_i, \mathfrak{a}_{i,\varsigma})$ is the crepant quotient by G_i of the pair on $\mathbb{A}^4$ with boundary

$$\mathrm{div}(\tilde{f}_i|_\varsigma) + \sum_k c_k \,\mathrm{div}(\tilde{h}_{i,k}|_\varsigma)$$

[Liu26, Definition 2.5 and Lemma 2.6(2)], where $|_\varsigma$ denotes the value at ς. For $\varsigma = 1$, Theorem 3.3(1), applied to $\mathbb{A}^4$ with the single equation f_i, $X = W_i$, $\mathfrak{a} = \prod_k (h_{i,k})^{c_k}$ and the subgroup G_i, implies

$$\eta_i(1) = \mathrm{mld}((W_i \ni 0, D_i)/G_i) = \alpha_i,$$

where we use that for any prime divisor P over W_i, the order of the $\mathbb{R}$-ideal $\prod_k (h_{i,k}|_{W_i})^{c_k}$ along P is $\mathrm{mult}_P D_i$. The hypotheses of Theorem 3.3(1) hold by **Step 1**(a) and (b): G_i is finite abelian and diagonal, f_i and the $h_{i,k}$ are G_i-semi-invariant, W_i is near 0 a normal integral hypersurface with klt singularities on which G_i acts faithfully, the $h_{i,k}$ are non-zero on W_i, $c_k > 0$, and (W_i, D_i) is klt near 0. For $\varsigma \neq 0$, the scaling

$$(x,y,z,t) \to (\varsigma^{2m+2}x, \varsigma^{2m+1}y, \varsigma^2 z, \varsigma t)$$

is a G_i-equivariant automorphism of $\mathbb{A}^4$ fixing 0, so it induces an automorphism of X_i fixing $\bar{0}$ and preserving Δ_i. Since $\tilde{f}_i|_\varsigma$ and $\tilde{h}_{i,k}|_\varsigma$ are non-zero constant multiples of the pullbacks of f_i and $h_{i,k}$ under the automorphism of $\mathbb{A}^4$, $\eta_i(\varsigma) = \alpha_i$ for any $\varsigma \neq 0$. For $\varsigma = 0$, $(X_i, \Delta_i, \mathfrak{a}_{i,0})$ is the crepant quotient of $(\mathbb{A}^4, \Theta)$ by G_i, so

$$\eta_i(0) = \mathrm{mld}((\mathbb{A}^4 \ni 0, \Theta)/G_i).$$

By [Nak16b, Theorem 1.2(ii)], whose hypotheses on X_i were verified above, applied to X_i, the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $\Delta_i \geq 0$, the point $\bar{0}$, and the family $\mathfrak{a}_i$ over $\mathbb{A}^1_\varsigma$, the function $\varsigma \to \eta_i(\varsigma)$ is lower semi-continuous. Since $\eta_i(\varsigma) = \alpha_i$ for $\varsigma \neq 0$, we obtain $\eta_i(0) \leq \alpha_i$. Thus, by (E.123),

$$\alpha_i \geq \mathrm{mld}((\mathbb{A}^4 \ni 0, \Theta)/G_i) \geq \gamma$$

for $i \gg 0$.

Step 5. In this step we prove that $\limsup_i \alpha_i \leq \gamma$, which contradicts **Step 4**.

Let $\xi_0, \chi_0 > 0$ be real numbers with $n(a'\xi_0 + c'\chi_0) \in \mathbb{Z}$. By (E.116), $\rho_2(\xi_0, \chi_0) \in H$. Thus there exist $\sigma_i \in G_i$ and real numbers ξ_i, χ_i with $p(\sigma_i) = e^{2\pi\sqrt{-1}\xi_i}$, $\theta(\sigma_i) = e^{2\pi\sqrt{-1}\chi_i}$, and $(\xi_i, \chi_i) \to (\xi_0, \chi_0)$. Let

$$C_1 := 1 + \max_k (2g_k\xi_0 + j_k\chi_0),$$

and let $\zeta_i \in [C_1, C_1 + 1)$ with $\beta(\sigma_i) = e^{2\pi\sqrt{-1}\zeta_i}$. By (E.115),

$$\omega_i := (\xi_i + \zeta_i, \zeta_i, 2\xi_i, \chi_i) \in N_{G_i}.$$

Let $c_\circ := \frac{1}{2}\min(\xi_0, \chi_0, 1)$. For $i \gg 0$, $\xi_i \geq c_\circ$ and $\chi_i \geq c_\circ$, so all coordinates of ω_i are at least $c_\circ$.

We compute the weights of f_i and $h_{i,k}$. The monomials x^2 and y^2z have ω_i-weight $2(\xi_i+\zeta_i) \le 2(\xi_0+C_1+2)$, whereas

$$\mathrm{wt}_{\omega_i}(S_i) \ge c_\circ \nu_i \to +\infty.$$

Thus $\mathrm{wt}_{\omega_i}(f_i) = 2(\xi_i+\zeta_i)$ for $i \gg 0$. Fix k. A monomial of $h_{i,k}$ divisible by x or y has ω_i-weight at least $\zeta_i \ge C_1$, whereas $2g_k\xi_i + j_k\chi_i < C_1$ for $i \gg 0$. A monomial $z^g t^j$ of $h_{i,k}$ with $2g\xi_i + j\chi_i < C_1$ satisfies $2g + j \le C_2 := C_1/c_\circ$. By (E.119) with $C = C_2$, for $i \gg 0$ any such monomial other than $z^{g_k}t^{j_k}$ has ω_i-weight

$$2g\xi_i + j\chi_i = 2g_k\xi_i + j_k\chi_i + en(a'\xi_i + c'\chi_i) > 2g_k\xi_i + j_k\chi_i.$$

Since $z^{g_k}t^{j_k}$ occurs in $h_{i,k}$, we obtain $\mathrm{wt}_{\omega_i}(h_{i,k}) = 2g_k\xi_i + j_k\chi_i$ for $i \gg 0$. By Lemma 4.7, for $i \gg 0$,

$$\alpha_i \le (\xi_i+\zeta_i) + \zeta_i + 2\xi_i + \chi_i - 2(\xi_i+\zeta_i) - \sum_k c_k(2g_k\xi_i + j_k\chi_i) = A_*\xi_i + B_*\chi_i.$$

Thus $\limsup_i \alpha_i \le A_*\xi_0 + B_*\chi_0$.

We now let (ξ_0, χ_0) vary. If $a', c' > 0$, then the pairs of positive real numbers with $a'\xi_0 + c'\chi_0 = \frac{1}{n}$ form the open segment between $(\frac{1}{na'}, 0)$ and $(0, \frac{1}{nc'})$, on which the infimum of the linear function $A_*\xi_0 + B_*\chi_0$ is $\min(\frac{A_*}{na'}, \frac{B_*}{nc'}) = \gamma$. If $c' = 0$, then $a' = 1$, and we take $\xi_0 = \frac{1}{n}$ and $\chi_0 \to 0$, so that $A_*\xi_0 + B_*\chi_0 \to \frac{A_*}{n} = \gamma$. If $a' = 0$, then $c' = 1$, and we take $\chi_0 = \frac{1}{n}$ and $\xi_0 \to 0$, so that $A_*\xi_0 + B_*\chi_0 \to \frac{B_*}{n} = \gamma$. In each case $n(a'\xi_0 + c'\chi_0) \in \mathbb{Z}$. Thus $\limsup_i \alpha_i \le \gamma$.

By **Step 4**, $\alpha_i \ge \gamma$ for $i \gg 0$. Since the α_i are strictly increasing, $\alpha_i < \lim_{i'} \alpha_{i'} \le \gamma$ for any i, a contradiction. Thus no subsequence of (ν_i) tends to $+\infty$, that is, the numbers $\nu_i = \mathrm{mult}_0 S_i$ are bounded.

Step 6. In this step we show that $\dim \widetilde{H} = 1$ for the Hausdorff limit $\widetilde{H}$ of the groups G_i in $\mathbb{T}^4$, and we conclude the proof.

Let

$$\Lambda := \{\omega \in \mathbb{R}^4 \mid 2\omega_x - 2\omega_y - \omega_z = 0\},$$

and let $\Phi : \mathbb{R}^3 \to \Lambda$ be the linear isomorphism $\Phi(P,B,T) := (P+B, B, 2P, T)$. For $\omega \in N_{G_i}$, (E.114) implies $\omega - (0,0,l,0) \in \Lambda$, so $G_i \subset \rho(\Lambda)$, a closed subgroup of $\mathbb{T}^4$. By Lemma 4.6, after passing to a subsequence, there exists a closed subgroup $\widetilde{H} \subset \mathbb{T}^4$ such that $G_i \to \widetilde{H}$ in the Hausdorff distance and $G_i \subset \widetilde{H}$ for $i \gg 0$. Thus $\widetilde{H} \subset \rho(\Lambda)$, and since $|G_i| \to +\infty$, $\dim \widetilde{H} \ge 1$. Let $\widetilde{V} \subset \mathbb{R}^4$ be the linear subspace with $\rho(\widetilde{V}) = \widetilde{H}^0$ [Liu26, Lemma 4.3(1)]. Then $\widetilde{V} \subset \Lambda$, since $\widetilde{V}$ contains 0, is connected, and is contained in $\rho^{-1}(\rho(\Lambda)) = \Lambda + \mathbb{Z}^4$, the disjoint union of the hyperplanes $2\omega_x - 2\omega_y - \omega_z = k$, $k \in \mathbb{Z}$.

Assume that $\dim \widetilde{H} \ge 2$. Suppose that $\widetilde{V}$ contains a vector ω_0 with all coordinates positive. Possibly replacing ω_0 with a positive multiple, we may assume that the sum $|\omega_0|$ of the coordinates of ω_0 is less than α_1. Since $\rho(\omega_0) \in \widetilde{H}$ and $G_i \to \widetilde{H}$, there exist $\omega_i \in N_{G_i}$ with $\omega_i \to \omega_0$, and all coordinates of ω_i are positive for $i \gg 0$. Since $\mathrm{wt}_{\omega_i}(f_i) \ge 0$ and $\mathrm{wt}_{\omega_i}(h_{i,k}) \ge 0$, Lemma 4.7 gives $\alpha_i \le |\omega_i|$ for $i \gg 0$, as in **Step 2**. Since $|\omega_i| \to |\omega_0| < \alpha_1 \le \alpha_i$, this is a contradiction. Thus $\widetilde{V} \cap \mathbb{R}^4_{>0} = \emptyset$. Let $\widetilde{V}' := \Phi^{-1}(\widetilde{V})$, so that $\dim \widetilde{V}' \ge 2$. For $(P,B,T) \in \mathbb{R}^3$, $\Phi(P,B,T) \in \mathbb{R}^4_{>0}$ if and only if $(P,B,T) \in \mathbb{R}^3_{>0}$. Thus $\widetilde{V}' \cap \mathbb{R}^3_{>0} = \emptyset$. Since $(1,1,1) \notin \widetilde{V}'$, $\dim \widetilde{V}' = 2$, and $\widetilde{V}'$ is the kernel of the linear form $\langle \varphi, \cdot \rangle$ for a non-zero $\varphi \in \mathbb{R}^3$. The linear form $\langle \varphi, \cdot \rangle$ does not vanish on the convex set $\mathbb{R}^3_{>0}$, so it has one sign on $\mathbb{R}^3_{>0}$, and possibly replacing φ with $-\varphi$, we may assume that $\varphi \in \mathbb{R}^3_{\ge 0}$.

For a monomial $M = x^{\epsilon_1}y^{\epsilon_2}z^{\epsilon_3}t^{\epsilon_4}$, the Laurent monomial Mx^{-2} defines the character

$$\rho(\omega) \to e^{2\pi\sqrt{-1}\langle(\epsilon_1-2,\epsilon_2,\epsilon_3,\epsilon_4),\omega\rangle}$$

of $\mathbb{T}^4$, and on $\rho(\Phi(P,B,T))$ it takes the value

$$e^{2\pi\sqrt{-1}((\epsilon_1-2+2\epsilon_3)P + (\epsilon_1-2+\epsilon_2)B + \epsilon_4 T)}.$$

Let

$$\epsilon_M := (\epsilon_1 - 2 + 2\epsilon_3, \epsilon_1 - 2 + \epsilon_2, \epsilon_4) \in \mathbb{Z}^3.$$

The character is trivial on $\widetilde{H}^0 = \rho(\Phi(\widetilde{V}'))$ if and only if $\epsilon_M \in \mathbb{R}\varphi$. For $M = yt^\ell$ with $\ell \geq 3$, $\epsilon_M = (-2, -1, \ell)$, and for $M = z^a t^b$ with $a+b \geq 4$, $\epsilon_M = (2a-2, -2, b)$. If either vector belonged to $\mathbb{R}\varphi$, then, since the second coordinate of the vector is negative and $\varphi \in \mathbb{R}^3_{\geq 0}$, the vector would be a negative multiple of φ, so all coordinates of the vector would be non-positive. This is impossible for $(-2, -1, \ell)$ since $\ell \geq 3$, and for $(2a-2, -2, b)$ it forces $a \leq 1$ and $b = 0$, contrary to $a + b \geq 4$. Thus the character of Mx^{-2} is non-trivial on $\widetilde{H}$ for any such M. If such a monomial M occurred in f_i for infinitely many i, then, since f_i is G_i-semi-invariant and contains x^2, the character of Mx^{-2} would be trivial on G_i for infinitely many i, hence trivial on $\widetilde{H}$ since $G_i \to \widetilde{H}$. Thus any monomial of the form yt^ℓ or $z^a t^b$ occurs in only finitely many S_i. Since only finitely many such monomials have degree at most a given integer, $\nu_i \to +\infty$, contrary to **Step 5**. Thus $\dim \widetilde{H} = 1$.

By Lemma 4.6, $\widetilde{H}^0 = \rho(\mathbb{R}\tilde{w})$ for a primitive vector $\tilde{w} \in \mathbb{Z}^4$ with coordinates $\tilde{w}_x, \tilde{w}_y, \tilde{w}_z, \tilde{w}_t$, and for $i \gg 0$ the group $G_i \cap \mathbf{T}_{(\tilde{w})}$ is cyclic of order tending to $+\infty$ and has index $e_0 := |\widetilde{H}/\widetilde{H}^0|$ in G_i. Since $\tilde{w} \in \widetilde{V} \subset \Lambda$, $\tilde{w}_z = 2(\tilde{w}_x - \tilde{w}_y)$. Let $u := \tilde{w}_x - \tilde{w}_y$, $v := \tilde{w}_y$, and $w := \tilde{w}_t$. Then $\tilde{w} = (u+v, v, 2u, w)$ is primitive, $|G_i \cap \mathbf{T}| \to +\infty$, and $[G_i : G_i \cap \mathbf{T}] \leq e_0$ for $\mathbf{T} := \mathbf{T}_{(u+v,v,2u,w)}$. By **Step 1**(a), $W_i = (x^2 + y^2 z + S_i = 0)$ with $S_i - S_i(0, z, t) = \lambda_i y t^{\ell_i}$, $\ell_i \geq 3$, and $\operatorname{mult}_0 S_i(0, z, t) = \operatorname{mult}_0 q_i \geq 4$, and the G_i act diagonally on x, y, z, t. By **Step 5**, the numbers $\operatorname{mult}_0 S_i$ are bounded. This proves the lemma. □

E.11. **A double line: the circles.** In this subsection, we bound α_i along the circle $\mathbf{T}$ of Lemma E.23 from above and from below by the minima of two reference surfaces (Lemmas E.29 and E.32 below).

We first determine the lowest weighted part of the equation and the principal block of the equation along the circle of Lemma E.23.

Lemma E.25. *Notation as in the conclusion of Lemma E.23. Write $x_1, x_2, x_3, x_4 := x, y, z, t$, let $\mathbf{p} = (p_1, p_2, p_3, p_4) := (u+v, v, 2u, w)$ be the $\mathbf{T}$-weights of $x_1, \dots, x_4$, let $M := \max_j |p_j|$, let $r_i := |G_i \cap \mathbf{T}|$, and write $S_i = y\sigma_i(t) + S_i(0, z, t)$. Then, after passing to a subsequence, replacing (u, v, w) by $(-u, -v, -w)$ if necessary, and rescaling the coordinates by non-zero constants, the following hold for i large.*

(1) At least one of $p_1, \dots, p_4$ is not positive, at least one is not negative, and $u \neq w$.
(2) Any monomial of f_i whose $\mathbf{T}$-weight is not $2(u+v)$ has degree at least $(r_i - 2M)/M$.
(3) There exist a number $\eta \in \frac{1}{2}\mathbb{Z}$ with $\eta \geq 1$ and a polynomial f_*, both independent of i, such that $\operatorname{in}_{\mathbf{q}_\eta}(f_i) = f_*$ for $\mathbf{q}_\eta := 2(\eta+1, \eta, 2, 1)$ and f_* is one of

$$\begin{aligned} &I\colon\ x^2 + y^2 z + z^a t^b, \quad a, b \geq 0,\ a+b \geq 4,\ 2a+b = 2\eta+2,\ 2au + bw = 2(u+v); \\ &II\colon\ x^2 + y^2 z + y t^\ell, \quad \ell \geq 3,\ \ell = \eta + 2,\ \ell w = 2u + v. \end{aligned} \tag{E.124}$$

(4) $u > 0$, or $u = 0$ and $w \geq 0$, and exactly one of the following holds.
  (a) $u > 0 > w$.
  (b) $u > 0$, $w > 0$, $v < 0$, and either f_* is in row I with $a = 0$ and $2u > bw$, or f_* is in row II and $2u > \ell w$.
  (c) $(u, v, w) = (1, -1, 0)$, and f_* is in row I with $a = 0$.
  (d) $(u, v, w) = (1, -2, 0)$, and f_* is in row II.
  (e) $p_j \geq 0$ for any j, and either $(u, v, w) = (0, 0, 1)$ and f_* is in row I with $b = 0$, or $(u, v, w) = (0, 1, 2)$ and f_* is in row I with $b = 1$, or $v = 0$ and (u, w) is one of

$$\left(\frac{b}{2}, 1\right) \text{ or } (b, 2) \text{ with } f_* \text{ in row I and } a = 0, \qquad \left(\frac{\ell}{2}, 1\right) \text{ or } (\ell, 2) \text{ with } f_* \text{ in row II.}$$

(5) *The principal block* f_i^0 *of* f_i *along* $\mathbf{T}$ *is* $x^2+y^2z+y\sigma_i^0(t)+S_i^0(z,t)$, *where* $y\sigma_i^0(t)$ *and* $S_i^0(z,t)$ *are the sums of the terms of* $y\sigma_i(t)$ *and of* $S_i(0,z,t)$ *of* $\mathbf{T}$*-weight* $2(u+v)$, *and* $V_i:=(f_i^0=0)\subset\mathbb{A}^4$ *is a normal integral hypersurface.*

(6) *Let* $N_i\subset V_i$ *be the set of points at which all coordinates of non-positive* $\mathbf{T}$*-weight vanish or all coordinates of non-negative* $\mathbf{T}$*-weight vanish. Assume that* $(u,v,w)\neq(0,0,1)$. *Then for any index* l *with* $p_l\neq 0$ *and any point* $\tilde{y}\in N_i$ *with* $x_l(\tilde{y})=1$, *the germ at* $\tilde{y}$ *of the surface* $V_{i,l}:=V_i\cap(x_l=1)\subset\mathbb{A}^3$, *with coordinates* x_j, $j\neq l$, *is either smooth, or isomorphic to the germ at* 0 *of* $\mathbb{C}^2/\Gamma$ *for a finite subgroup* $\Gamma\subset\mathrm{SL}_2(\mathbb{C})$ *with* $|\Gamma|\leq N_1$, *where* $N_1:=4\max(a+1,b,3)$ *if* f_* *is in row I and* $N_1:=8\ell$ *if* f_* *is in row II.*

(7) *Assume that* $(u,v,w)\neq(0,0,1)$. *Then for any* i *there exist an open neighborhood* Ω_i *of* N_i *in* $\mathbb{A}^4$ *in the Euclidean topology, invariant under* $\mathbf{T}$ *and* G_i, *a holomorphic map* $\psi_i:\Omega_i\to\mathbb{A}^4$ *commuting with* $\mathbf{T}$ *and* G_i, *and a nowhere vanishing holomorphic function* v_i *on* Ω_i, *invariant under* $\mathbf{T}$ *and* G_i, *such that* $f_*\circ\psi_i=v_if_i^0$, $\psi_i(0)=0$, *the tangent map of* ψ_i *at* 0 *is the identity, and* ψ_i *is a local isomorphism at any point of* N_i. *In particular,* ψ_i *induces a* G_i*-equivariant and* $\mathbf{T}$*-equivariant automorphism of* $\mathbb{C}[[x,y,z,t]]$ *with identity tangent map which maps* f_* *to* $v_if_i^0$.

Proof of Lemma E.25. The proof is long, so we divide it into several steps.

Step 1. In this step we prove the first two statements of (1). Since α_i is strictly increasing in $(0,1)$, $\alpha_i\geq\alpha_1>0$ for any i. Suppose that $p_j>0$ for any j on an infinite subsequence. The case $p_j<0$ for any j follows after replacing (u,v,w) by $(-u,-v,-w)$. Let $E_{\mathbf{p}}$ be the toric prime divisor over $\mathbb{A}^4$ given by the monomial valuation with weights $\mathbf{p}$. Then $a(E_{\mathbf{p}},\mathbb{A}^4,0)=\sum_j p_j\leq 4M$, and $\mathrm{mult}_{E_{\mathbf{p}}}f_i>0$ and $\mathrm{mult}_{E_{\mathbf{p}}}h_{i,k}\geq 0$ for any k. The group $\mathbf{T}$ preserves $E_{\mathbf{p}}$ and acts trivially on the residue field of $E_{\mathbf{p}}$, which is the function field of the weighted projective space $\mathbb{P}(\mathbf{p})$. Thus $G_i\cap\mathbf{T}\subset I_{G_i}(E_{\mathbf{p}})$, and $|I_{G_i}(E_{\mathbf{p}})|\geq r_i$. The variety W_i is a normal hypersurface with klt singularities near 0 on which G_i acts faithfully, and (W_i,D_i) is klt near 0. By Theorem 3.3(1) with $n=4$, $c=1$, $H=G_i$, and $\mathfrak{a}=\prod_k h_{i,k}^{c_k}$, and by [Liu26, Lemma 2.6],

$$\alpha_i=\mathrm{mld}\Big(\Big(\mathbb{A}^4\ni 0,\mathrm{div}(f_i),\prod_k h_{i,k}^{c_k}\Big)/G_i\Big)\leq\frac{a(E_{\mathbf{p}},\mathbb{A}^4,\mathrm{div}(f_i)+\sum_k c_k\,\mathrm{div}(h_{i,k}))}{|I_{G_i}(E_{\mathbf{p}})|}\leq\frac{4M}{r_i}.$$

Since $r_i\to+\infty$, this contradicts $\alpha_i\geq\alpha_1>0$.

Step 2. In this step we prove (2) and (3), and that $u\neq w$. The group $G_i\cap\mathbf{T}$ is cyclic of order r_i and acts on a monomial of $\mathbf{T}$-weight d by a primitive r_i-th root of unity raised to the power d. Since f_i is G_i-semi-invariant and contains x^2, the $\mathbf{T}$-weight of any monomial of f_i is congruent to $2(u+v)$ modulo r_i. A monomial of degree d has $\mathbf{T}$-weight of absolute value at most Md. Thus a monomial of f_i whose $\mathbf{T}$-weight is not $2(u+v)$ satisfies $Md+2M\geq r_i$, which is (2).

Let

$$\eta_i:=\min\Big(\{\ell-2\mid yt^\ell\text{ is a monomial of }y\sigma_i\}\cup\Big\{a+\frac{b}{2}-1\;\Big|\;z^at^b\text{ is a monomial of }S_i(0,z,t)\Big\}\Big).$$

The set is not empty, since otherwise $f_i=x^2+y^2z$, whose singular locus $(x=y=0)$ is a plane. Since σ_i has order at least 3 and $S_i(0,z,t)$ has order at least 4, we have $\eta_i\in\frac{1}{2}\mathbb{Z}$ and $\eta_i\geq 1$. The value of a monomial of S_i in the set above is at most the degree of the monomial, so $\eta_i\leq\mathrm{mult}_0 S_i$. Since the numbers $\mathrm{mult}_0 S_i$ are bounded (Lemma E.23), possibly passing to a subsequence, we may assume that $\eta_i=\eta$ is independent of i. The $\mathbf{q}_\eta$-weights of x^2, y^2z, yt^ℓ, and z^at^b are $4\eta+4$, $4\eta+4$, $2\eta+2\ell$, and $4a+2b$. Thus $\mathrm{in}_{\mathbf{q}_\eta}(f_i)$ is x^2+y^2z plus the terms yt^ℓ with $\ell=\eta+2$ and z^at^b with $2a+b=2\eta+2$, and at least one such term occurs by the definition of η. The monomials of $\mathrm{in}_{\mathbf{q}_\eta}(f_i)$ have degree at most $2\eta+2$. By (2), for i large they have $\mathbf{T}$-weight $2(u+v)$, that is,

$$2au+bw=2(u+v)\quad\text{for }z^at^b,\qquad v+\ell w=2(u+v)\quad\text{for }yt^\ell.\tag{E.125}$$

If $u=w$, then each equation of (E.125), together with $2a+b=2\eta+2$ or $\ell=\eta+2$, implies $v=\eta u$, so $\mathbf{p}=u(\eta+1,\eta,2,1)$. Since $\mathbf{p}$ is primitive, $u\neq 0$, and all p_j have the sign of u, which

contradicts **Step 1**. Thus $u \neq w$. If $z^{a_1}t^{b_1}$ and $z^{a_2}t^{b_2}$ both occur, then $b_1 - b_2 = -2(a_1 - a_2)$, and (E.125) implies $2(u-w)(a_1-a_2) = 0$, so $a_1 = a_2$ and $b_1 = b_2$. If $z^a t^b$ and yt^ℓ both occur, then $b = 2\ell - 2 - 2a$ and $2\ell w = 4u + 2v$, so (E.125) implies

$$2au + (2\ell - 2 - 2a)w = 2u + 2v \quad \text{and hence} \quad 2(u-w)(a+1) = 0,$$

which is impossible. Thus

$$\mathrm{in}_{\mathbf{q}_\eta}(f_i) = x^2 + y^2 z + \lambda_i z^a t^b \quad \text{or} \quad \mathrm{in}_{\mathbf{q}_\eta}(f_i) = x^2 + y^2 z + \lambda_i y t^\ell, \qquad \lambda_i \neq 0.$$

Since only finitely many exponents satisfy $2a + b = 2\eta + 2$, possibly passing to a subsequence, they are independent of i. Rescaling t, or z and y if $b = 0$, by non-zero constants, we may assume that $\lambda_i = 1$. The diagonal rescaling commutes with G_i and keeps the form of f_i in Lemma E.23. This proves (3).

Step 3. In this step we prove (4). By **Step 2**, the $\mathbf{T}$-weight of any monomial of f_i is congruent to $2(u+v)$ modulo r_i. Since $r_i \to +\infty$ (Lemma E.23), we assume that $r_i > 3M$. Hence a monomial of f_i whose $\mathbf{T}$-weight d satisfies $|d - 2(u+v)| \leq 3M$ has $\mathbf{T}$-weight $2(u+v)$. If $w = 0$, the monomials t^n, yt^n, and zt^n have $\mathbf{T}$-weights 0, v, and $2u$, and if $u = 0$, the monomials z^n and $z^n t$ have $\mathbf{T}$-weights 0 and w, for any n, and $|d - 2(u+v)| \leq 3M$ for $d \in \{0, v, 2u, w\}$. Replacing (u, v, w) by $(-u, -v, -w)$ if necessary, we may assume that $u > 0$, or $u = 0$ and $w \geq 0$. By **Step 2**, $u = w = 0$ is impossible, and in the case $u = 0$ we have $w > 0$. If $u > 0$ and $w < 0$, then (a) holds.

Assume that $u > 0$ and $w > 0$. By **Step 1**, $v \leq 0$. In row I, $v = (a-1)u + \frac{bw}{2}$ by (E.125). If $a = 1$, then $b \geq 3$ and $v = \frac{bw}{2} > 0$, and if $a \geq 2$, then $v \geq u > 0$. Thus $a = 0$ and $2u \geq bw$, with equality if and only if $v = 0$. In row II, $v = \ell w - 2u$, so $2u \geq \ell w$, with equality if and only if $v = 0$. If $v < 0$, then (b) holds. If $v = 0$, then $\gcd(u, w) = 1$ since $\mathbf{p} = (u, 0, 2u, w)$ is primitive. In row I, $2u = bw$ implies $w \mid 2$, so $(u, w) = (\frac{b}{2}, 1)$ or $(b, 2)$. In row II, $2u = \ell w$ implies $(u, w) = (\frac{\ell}{2}, 1)$ or $(\ell, 2)$. Thus (e) holds.

Assume that $u > 0$ and $w = 0$. Then $\mathbf{p} = (u + v, v, 2u, 0)$. A monomial t^n has $\mathbf{T}$-weight 0. If $f_i(0, 0, 0, t) \neq 0$, then $0 = 2(u+v)$ by the congruence modulo r_i, so $u + v = 0$, $\mathbf{p} = (0, -u, 2u, 0)$, $u = 1$, and $(u, v, w) = (1, -1, 0)$. Otherwise the t-axis is contained in W_i. Since W_i has an isolated singularity at 0, some partial derivative of f_i does not vanish identically on the t-axis. On the t-axis,

$$\frac{\partial f_i}{\partial x} = 0, \quad \frac{\partial f_i}{\partial y} = \sigma_i(t), \quad \frac{\partial f_i}{\partial z} = \frac{\partial S_i}{\partial z}(0, 0, t), \quad \frac{\partial f_i}{\partial t} = \frac{\partial S_i}{\partial t}(0, 0, t) = 0.$$

If $\sigma_i \neq 0$, then a monomial yt^n of weight v occurs, so $v = 2(u+v)$, $\mathbf{p} = (-u, -2u, 2u, 0)$, $u = 1$, and $(u, v, w) = (1, -2, 0)$. If a monomial zt^n occurs, then $2u = 2(u+v)$, so $v = 0$ and $\mathbf{p} = (1, 0, 2, 0)$. In the last case the monomials yt^n have weight $0 \neq 2$, and the monomials $z^a t^b$ have weight $2a$, so $\sigma_i = 0$ and $S_i(0, z, t) = zP_i(z, t)$ with $P_i(0, 0) = 0$ for i large. Then $f_i = x^2 + z(y^2 + P_i)$, and all partial derivatives of f_i vanish on the curve $(x = z = y^2 + P_i(0, t) = 0)$, which passes through 0. This contradicts the isolated singularity of W_i. For $(u, v, w) = (1, -1, 0)$, (E.125) implies $a = 0$ in row I and excludes row II, since $\ell w = 0 \neq 1 = 2u + v$. For $(u, v, w) = (1, -2, 0)$, it excludes row I, since $2a = -2$, and allows row II. Thus (c) or (d) holds.

Assume that $u = 0$ and $w > 0$. Then $\mathbf{p} = (v, v, 0, w)$. By (E.125), $bw = 2v$ in row I and $\ell w = v$ in row II, so $v \geq 0$. If $v = 0$, then $\mathbf{p} = (0, 0, 0, w)$, $w = 1$, and f_* is in row I with $b = 0$. Assume that $v > 0$. A monomial z^n has weight $0 \neq 2v$, so no monomial z^n occurs in f_i, and the z-axis is contained in W_i. On the z-axis, $\partial f_i/\partial x$, $\partial f_i/\partial y = \sigma_i(0)$, and $\partial f_i/\partial z$ vanish, and $\partial f_i/\partial t$ is the sum of the terms of $S_i(0, z, t)$ of the form $\lambda z^n t$ divided by t. Since W_i has an isolated singularity at 0, a monomial $z^n t$ occurs, so $w = 2v$, $\mathbf{p} = (1, 1, 0, 2)$, and $(u, v, w) = (0, 1, 2)$. Then $bw = 2v$ implies $b = 1$ in row I, and $\ell w = v$ is impossible in row II. Thus (e) holds. The cases (a)–(e) are mutually exclusive by the signs of u, v, and w.

Step 4. In this step we prove (5). The polynomial f_i is the sum of $x^2 + y^2 z$, $y\sigma_i(t)$, and $S_i(0, z, t)$, so $f_i^0 = x^2 + y^2 z + y\sigma_i^0 + S_i^0$. Let $g := y^2 z + y\sigma_i^0(t) + S_i^0(z, t)$. By (3), $f_* - x^2 - y^2 z$ is a monomial

of $\mathbf{T}$-weight $2(u+v)$, so $(\sigma_i^0, S_i^0) \neq (0,0)$. A repeated factor of g independent of y would divide the coefficient z of y^2 twice, which is impossible. A repeated factor of positive degree in y would make the discriminant $\sigma_i^0(t)^2 - 4zS_i^0(z,t)$ vanish. If $\sigma_i^0 \neq 0$, then z does not divide $\sigma_i^0(t)^2$, and if $\sigma_i^0 = 0$, then $S_i^0 \neq 0$. Thus g is reduced. The coefficient z of y^2 is not a square, so $-g$ is not a square and $f_i^0 = x^2 + g$ is irreducible. The hypersurface V_i satisfies Serre's condition S_2 (a hypersurface is Cohen–Macaulay). The singular locus of V_i is contained in $(x=0) \times \mathrm{Sing}(g=0)$, and $\dim \mathrm{Sing}(g=0) \leq 1$ since g is reduced. Thus V_i satisfies R_1, and V_i is normal.

Step 5. In this step we compute f_i^0 in the cases (a)–(e) other than $(u,v,w) = (0,0,1)$. In each case we determine the monomials yt^n and $z^A t^B$ of $\mathbf{T}$-weight $2(u+v)$ by (E.125), using that the monomial $f_* - x^2 - y^2 z$ of (3) has the least $\mathbf{q}_\eta$-weight among the monomials of S_i.

- Case (a), row I. If $z^A t^B$ has weight $2(u+v)$, then $2u(A-a) = |w|(B-b)$, so $(A,B) = (a,b) + n(\frac{|w|}{g_0}, \frac{2u}{g_0})$ with $g_0 := \gcd(2u, |w|)$ and $n \in \mathbb{Z}$, and $n \geq 0$ by the minimality of $2a+b$. A monomial yt^n of weight $2(u+v)$ would satisfy $n = \frac{b}{2} + \frac{(a+1)u}{w} < \frac{b}{2} < \eta + 2$, which contradicts the minimality of η. Thus
$$f_i^0 = x^2 + y^2 z + z^a t^b U_i(\mu), \qquad \mu := z^{|w|/g_0} t^{2u/g_0},$$
where U_i is a polynomial with $U_i(0) = 1$.
- Case (a), row II. A monomial yt^n of weight $2(u+v)$ has $n = \ell$. If $z^A t^B$ has weight $2(u+v) = 2\ell w - 2u$, then $2u(A+1) = |w|(B - 2\ell)$, so $B \geq 2\ell + 1$. Thus $f_i^0 = x^2 + y^2 z + yt^\ell + t^{2\ell+1} Q_i(z,t)$ for a polynomial Q_i.
- Case (b), row I. A monomial $z^A t^B$ of weight $2(u+v) = bw$ satisfies $2uA + wB = bw$ and $2A + B \geq b$, so $2A(1 - \frac{u}{w}) \geq 0$, and $A = 0$, since $a = 0$ gives $b \geq 4$, so $2u > bw \geq 4w$ and $u > w$. A monomial yt^n of weight $2(u+v)$ has $n = n_* := \frac{u}{w} + \frac{b}{2}$, and $n_* > b$ since $2u > bw$. Thus $f_i^0 = x^2 + y^2 z + t^b(1 + \lambda_i y t^{n_* - b})$ with $\lambda_i \in \mathbb{C}$, where $\lambda_i = 0$ if $n_* \notin \mathbb{Z}$.
- Case (b), row II. A monomial $z^A t^B$ of weight $2(u+v) = 2\ell w - 2u$ satisfies $2u(A+1) = w(2\ell - B)$ and $2A + B \geq 2\ell - 2$. Since $2u > \ell w$, $(A+1)\ell < 2\ell - B$. If $A = 0$, then $2\ell - 2 \leq B < \ell$, which is impossible for $\ell \geq 3$, and if $A \geq 1$, then $B < 0$. Thus $f_i^0 = f_*$.
- Case (c). The monomials yt^n have weight $-1 \neq 0$, and $z^A t^B$ has weight $2A$. Thus $f_i^0 = x^2 + y^2 z + t^b U_i(t)$ with $U_i(0) = 1$.
- Case (d). All monomials yt^n have weight $-2 = 2(u+v)$, and no $z^A t^B$ has weight -2. Thus $f_i^0 = x^2 + y^2 z + yt^\ell U_i(t)$ with $U_i(0) = 1$.
- Case (e), $(u,v,w) = (0,1,2)$. The monomials yt^n have weight $1 + 2n \neq 2$, and $z^A t^B$ has weight $2B$. Thus $f_i^0 = x^2 + y^2 z + t z^a U_i(z)$ with $U_i(0) = 1$.
- Case (e), $v = 0$, row I. A monomial $z^A t^B$ of weight $2u = bw$ has $bA + B = b$, so it is t^b or z, and z does not occur since $S_i(0,z,t)$ has order at least 4. A monomial yt^n of weight $2u$ has $n = b$. Thus $f_i^0 = x^2 + y^2 z + t^b(1 + \lambda_i y)$ with $\lambda_i \in \mathbb{C}$.
- Case (e), $v = 0$, row II. A monomial yt^n of weight $2u = \ell w$ has $n = \ell$. A monomial $z^A t^B$ of weight ℓw has $\ell A + B = \ell$, so it is z or t^ℓ. The monomial z does not occur since $S_i(0,z,t)$ has order at least 4, and t^ℓ does not occur since $\frac{\ell}{2} - 1 < \ell - 2 = \eta$. Thus $f_i^0 = f_*$.

Step 6. In this step we prove (7) and (6). By **Step 5**, in each case there exist a $\mathbf{T}$-invariant monomial μ, which is G_i-semi-invariant and vanishes on N_i, and a polynomial U_i with $U_i(0) = 1$, as listed in **Step 5** ($\mu = z^{|w|/g_0} t^{2u/g_0}$, $yt^{n_* - b}$, t, t, z, y in the cases (a) row I, (b) row I, (c), (d), (e) with $u = 0$, and (e) with $v = 0$ row I, where $U_i(\mu) = 1 + \lambda_i \mu$ in the second and the last cases). The function $U_i(\mu)$ is G_i-invariant, since f_i^0 and f_* are G_i-semi-invariant with the character of x^2. Let Ω_i be the open subset of $\mathbb{A}^4$ on which $|\mu| < \epsilon_i$, where $\epsilon_i > 0$ is less than the absolute value of any root of U_i, so that the branches of the rational powers of $U_i(\mu)$ with value 1 at $\mu = 0$ are holomorphic on Ω_i. Let χ_μ be the character of G_i on μ. Its values are roots of unity, so $|\mu|$ and Ω_i are G_i-invariant. Since $U_i(\mu)$ is G_i-invariant and μ is a non-constant monomial,

$U_i(\chi_\mu(g)\xi) = U_i(\xi)$ as polynomials in ξ for any $g \in G_i$. Thus, for $\theta \in \mathbb{Q}$, the branch U_i^θ on the disc ($|\xi| < \epsilon_i$) with value 1 at 0 satisfies $U_i^\theta(\chi_\mu(g)\xi) = U_i^\theta(\xi)$, and the rational powers of $U_i(\mu)$ above are $\mathbf{T}$-invariant and G_i-invariant. Consider the holomorphic maps $\psi_i : \Omega_i \to \mathbb{A}^4$ given by

$$t \to tU_i(\mu)^{1/b} \text{ if } b \geq 1, \quad (x,z) \to (xU_i(\mu)^{\frac{1}{2(a-1)}}, zU_i(\mu)^{\frac{1}{a-1}}) \text{ if } b = 0,$$

in the case (a) row I, $t \to tU_i(\mu)^{1/b}$ in the cases (b) row I, (c) and (e) with $v = 0$ row I, $(y,z) \to (yU_i, zU_i^{-2})$ in the case (d), and $t \to tU_i(z)$ in the case (e) with $u = 0$, the other coordinates being fixed. A direct calculation shows that $f_* \circ \psi_i$ is f_i^0 times a power of $U_i(\mu)$. Each ψ_i multiplies the coordinates x_j by $U_i(\mu)^{\gamma_j}$ for some $\gamma \in \mathbb{Q}^4$, so the Jacobian determinant of ψ_i is

$$U_i(\mu)^{\sum_j \gamma_j}\Big(1 + \frac{U_i'(\mu)\mu}{U_i(\mu)} \sum_j \gamma_j \deg_{x_j} \mu\Big),$$

where $\deg_{x_j} \mu$ is the exponent of x_j in the monomial μ. The Jacobian determinant of ψ_i is 1 on $N_i \subset (\mu = 0)$. In the case (a) row II, let Ω_i be the open subset of $\mathbb{A}^4$ on which $|ztQ_i| < \frac{1}{4}$, let $U := (1 - 4ztQ_i)^{1/2}$ on Ω_i with value 1 on $(zt = 0)$, and let ψ_i be

$$(x,y,z,t) \to \Big(x,\ Uy + \frac{t^\ell U(1-U)}{2z},\ \frac{z}{U^2},\ t\Big).$$

Since $t^{2\ell+1}Q_i = t^{2\ell}(1-U^2)/(4z)$ and $1 - U$ is divisible by ztQ_i, ψ_i is holomorphic and a direct calculation shows $f_* \circ \psi_i = f_i^0$. The Jacobian determinant of ψ_i is $U \cdot \partial(zU^{-2})/\partial z$, which is 1 on $(z = 0)$ and on $(t = 0)$, so on N_i. In the cases (b) row II and (e) with $v = 0$ row II, let $\Omega_i := \mathbb{A}^4$ and let ψ_i be the identity.

All maps ψ_i commute with $\mathbf{T}$ and G_i, and are local isomorphisms at any point of N_i. Let v_i be the power of $U_i(\mu)$ with $f_* \circ \psi_i = v_i f_i^0$ above, and let $v_i := 1$ in the case (a) row II and in the cases where ψ_i is the identity. We show that ψ_i and v_i satisfy (7). The function v_i does not vanish on Ω_i by the choice of ϵ_i. The set Ω_i contains N_i, and it is invariant under $\mathbf{T}$ and G_i, since μ is $\mathbf{T}$-invariant and $|\mu|$ is G_i-invariant (resp. ztQ_i is $\mathbf{T}$-invariant and G_i-invariant, since $ztQ_i = \frac{y^2 z}{(yt^\ell)^2} \cdot t^{2\ell+1}Q_i$, and y^2z, yt^ℓ and the terms of $t^{2\ell+1}Q_i$ have the $\mathbf{T}$-weight and the character of x^2), and μ (resp. ztQ_i) vanishes on N_i. In the cases other than (a) row II where ψ_i is not the identity, $\psi_i(0) = 0$ and the tangent map of ψ_i at 0 is the identity, since $U_i(0) = 1$ and μ vanishes at 0. In the case (a) row II, $\psi_i(0) = 0$ and the tangent map of ψ_i at 0 is the identity, since $U - 1$ and $t^\ell(1-U)/z$ have order at least two at 0. This proves (7).

We prove (6). Let $\tilde{y} \in N_i$ with $x_l(\tilde{y}) = 1$. Since $\tilde{y}$ is not fixed by $\mathbf{T}$ and ψ_i is a $\mathbf{T}$-equivariant local isomorphism at $\tilde{y}$, the point $\psi_i(\tilde{y}) \in V_* := (f_* = 0)$ is not fixed by $\mathbf{T}$, so $x_{l'}(\psi_i(\tilde{y})) \neq 0$ for some index l' with $p_{l'} \neq 0$. Rescaling by an element of $\mathbf{T}$, we may assume that $x_{l'}(\psi_i(\tilde{y})) = 1$. The stabilizer of the slice $(x_l = 1)$ in $\mathbf{T}$ is $\mu_{|p_l|}$. It acts freely on $V_{i,l} \times \mathbb{G}_m$, through $\mathbf{T}$ on $V_{i,l}$ and by multiplication by ζ^{-1} on $\mathbb{G}_m$ for $\zeta \in \mu_{|p_l|}$, and the map $V_{i,l} \times \mathbb{G}_m \to V_i \cap (x_l \neq 0)$ given by the action is the quotient morphism by the free action of $\mu_{|p_l|}$, so it is étale. Thus the germ $(V_i \ni \tilde{y})$ is analytically isomorphic to $(V_{i,l} \ni \tilde{y}) \times (\mathbb{A}^1 \ni 0)$, and similarly for V_* at $\tilde{y}_* := \psi_i(\tilde{y})$ and the index l'. Since ψ_i is a local isomorphism at $\tilde{y}$,

$$(V_{i,l} \ni \tilde{y}) \times (\mathbb{A}^1 \ni 0) \cong (V_{*,l'} \ni \tilde{y}_*) \times (\mathbb{A}^1 \ni 0), \qquad V_{*,l'} := V_* \cap (x_{l'} = 1).$$

The sections of V_* by the hyperplanes $(x_j = 1)$ are as follows, at all their points. In row I, the x-section $1 + y^2z + z^a t^b$ is smooth, the y-section $x^2 + z + z^a t^b$ is smooth except at A_1 points, the z-section $x^2 + y^2 + t^b$ is smooth or has an A_{b-1} point, and the t-section $x^2 + y^2 z + z^a$ is smooth for $a = 0$, has A_1 points for $a = 1$, an A_3 point for $a = 2$, and a D_{a+1} point for $a \geq 3$. In row II, the x-section $1 + y^2 z + yt^\ell$, the y-section $x^2 + z + t^\ell$ and the t-section $x^2 + y^2 z + y$ are smooth, and the z-section $x^2 + y^2 + yt^\ell$ has an $A_{2\ell-1}$ point. Thus $(V_{*,l'} \ni \tilde{y}_*)$ is smooth or $\mathbb{C}^2/\Gamma_*$ for $\Gamma_* \subset \mathrm{SL}_2(\mathbb{C})$ of order at most N_1 (an A_n point is the quotient by a cyclic group of order $n+1$ and a D_n point by a binary dihedral group of order $4(n-2)$, [KM98, Theorem 4.20]

and [Roa04, Table 1]). If the product is smooth, then $(V_{i,l} \ni \tilde{y})$ is smooth. Otherwise the product is a canonical Gorenstein threefold germ, so $(V_{i,l} \ni \tilde{y})$ is a normal Gorenstein surface germ with canonical singularities (a resolution of the product may be taken to be a resolution of $(V_{i,l} \ni \tilde{y})$ times $\mathbb{A}^1$). Thus $(V_{i,l} \ni \tilde{y})$ is $\mathbb{C}^2/\Gamma$ with $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$ finite [KM98, Theorem 4.20 and Remark 4.21(1)]. The germs $(V_{i,l} \ni \tilde{y}) \times (\mathbb{A}^1 \ni 0)$ and $(V_{*,l'} \ni \tilde{y}_*) \times (\mathbb{A}^1 \ni 0)$ are the germs at 0 of $\mathbb{C}^3/(\Gamma \times \{1\})$ and $\mathbb{C}^3/(\Gamma_* \times \{1\})$. For $g \neq 1$ in a finite subgroup of $\mathrm{SL}_2(\mathbb{C})$, the eigenvalue 1 of $g \times 1$ has multiplicity one, so $\Gamma \times \{1\}$ and $\Gamma_* \times \{1\}$ are small subgroups of $\mathrm{GL}_3(\mathbb{C})$ in the sense of [Pri67, Definition 2]. By [Pri67, Theorem 2], they are linearly equivalent, so $|\Gamma| = |\Gamma_*| \leq N_1$. □

The principal blocks along $\mathbf{T}$ define the following reference surfaces.

Construction E.26. Let the data be as in the conclusion of Lemma E.23, oriented so that $u > 0$, or $u = 0$ and $w \geq 0$ (replace (u, v, w) by $(-u, -v, -w)$ if necessary), and let $r_i := |G_i \cap \mathbf{T}|$. Write each $h_{i,k}$ in normal form (Definition-Lemma 11.5(1)(2), which holds for the admitted data by Lemma 11.10(1)), and pass to a subsequence on which the weight κ_k of the principal block $h^0_{i,k}$ of $h_{i,k}$ along $\mathbf{T}$ (Set-up 5.1) is independent of i for each k. Let f^0_i be the principal block of the equation of W_i, $V_i := (f^0_i = 0) \subset \mathbb{A}^4$, and let δ be the adjusted mass of D_i along $\mathbf{T}$, that is, $\delta = \sum_k c_k\kappa_k - u - w$. For $c' \in \mathbb{R}^s_{\geq 0}$, let $(Y_i^+, B_i^+(c'))$ and $(Y_i^-, B_i^-(c'))$ be the positive and negative graded quotients Proj of $(V_i, \sum_k c'_k \operatorname{div}(h^0_{i,k}|_{V_i}))$ by $\mathbf{T}$, followed by the quotient by $G_i/(G_i \cap \mathbf{T})$, with the induced boundaries, where Y_i^- is empty if no coordinate has negative $\mathbf{T}$-weight. Let $C_i^{\pm} \subset Y_i^{\pm}$ be the fibers over the image of the origin, and let $\beta_i^{\pm}(c')$ be the minimum of the log discrepancies of $(Y_i^{\pm}, B_i^{\pm}(c'))$ over the prime divisors with center in $C_i^{\pm}$. Let $c := (c_1, \dots, c_s)$, $B_i^{\pm} := B_i^{\pm}(c)$, $\beta_i^{\pm} := \beta_i^{\pm}(c)$, and $\beta_i := \min(\beta_i^+, \beta_i^-)$, with $\beta_i := \beta_i^+$ if Y_i^- is empty.

The subsequence exists. By Proposition 11.8 the admitted data satisfy

$$a(E_0, W_i, 0) - \frac{1}{m} \operatorname{mult}_{E_0} h_{i,k} > 0$$

for the prime divisor E_0 of the weights $(3, 2, 2, 2)$. The lowest $(3, 2, 2, 2)$-weighted part of f_i is $x^2 + y^2 z$, so $a(E_0, W_i, 0) = 3$ and $\operatorname{mult}_{E_0} h_{i,k} \geq 2 \operatorname{mult}_0 h_{i,k}$ by Lemma 4.12. Thus $\operatorname{mult}_0 h_{i,k} < \frac{3m}{2}$, and the weights κ_k, which are the $\mathbf{T}$-weights of monomials of $h_{i,k}$ of least degree, are bounded. By Lemma E.25(5), V_i is a normal integral hypersurface. Moreover, $h^0_{i,k}|_{V_i} \neq 0$, since $h^0_{i,k}$ has degree at most one in x and f^0_i is monic of degree two in x.

We describe $(Y_i^{\pm}, B_i^{\pm}(c'))$ by charts.

Construction E.27. Notation as in Construction E.26. Write $x_1, x_2, x_3, x_4 := x, y, z, t$, let $\mathbf{p} = (p_1, p_2, p_3, p_4) := (u + v, v, 2u, w)$ be the $\mathbf{T}$-weights of $x_1, \dots, x_4$, let $M := \max_j |p_j|$, let χ_j be the character of G_i on x_j, and let $M_0 := \max(2M, |\kappa_1|, \dots, |\kappa_s|)$. For an index l with $p_l \neq 0$, let $\epsilon_l \in \{1, -1\}$ be the sign of p_l, $\mathbf{p}^{(l)} := \epsilon_l \mathbf{p}$, and $\delta_l := \epsilon_l \delta$. Let $\mathbb{A}^3$ be the affine space with coordinates x'_j, $j \neq l$, write $x'_l := 1$, and let

$$\Phi_l : \mathbb{G}_{m,\rho} \times \mathbb{A}^3 \to \mathbb{A}^4, \qquad (\rho, x') \to \left(\rho^{p_j^{(l)}} x'_j\right)_{j=1,\dots,4}.$$

Let $\widetilde{G}_{i,l}$ be the group of pairs $(g, \lambda) \in G_i \times \mathbb{C}^*$ with $\lambda^{|p_l|}\chi_l(g) = 1$, acting on $\mathbb{G}_m \times \mathbb{A}^3$ by $\rho \to \lambda^{-1}\rho$ and $x'_j \to \lambda^{p_j^{(l)}} \chi_j(g) x'_j$. Let $q := \rho^{r_i}$, let $R_{i,l} \subset \widetilde{G}_{i,l}$ be the subgroup of elements acting trivially on $\mathbb{A}^3$, and let $H_{i,l} := \widetilde{G}_{i,l}/R_{i,l}$, acting diagonally on $\mathbb{A}^1_q \times \mathbb{A}^3$. Recall $V_{i,l} = V_i \cap (x_l = 1) \subset \mathbb{A}^3$ from Lemma E.25(6), and for any k let $H^0_{i,k,l} := h^0_{i,k}|_{x_l=1}$, a polynomial on $\mathbb{A}^3$. Let

$$F_{i,l} := \rho^{-2\epsilon_l(u+v)} f_i \circ \Phi_l, \qquad H_{i,k,l} := \rho^{-\epsilon_l\kappa_k} h_{i,k} \circ \Phi_l.$$

Lemma E.28. *Notation as in Construction E.27, and let i be large. Then the following hold.*

(1) *Φ_l is equivariant for $\widetilde{G}_{i,l} \to G_i$, $(g,\lambda) \to g$, the group $R_{i,l}$ is cyclic of order r_i and acts on ρ alone, $|H_{i,l}| = |p_l|[G_i : G_i \cap \mathbf{T}] \le Me_0$, and Φ_l induces an isomorphism $\bar{\Phi}_l$ from $(\mathbb{G}_{m,q} \times \mathbb{A}^3)/H_{i,l}$ onto the open subset $(x_l \neq 0)/G_i$ of $\mathbb{A}^4/G_i$. The group $H_{i,l}$ acts faithfully on $\mathbb{A}^3$ and on $V_{i,l}$.*

(2) *$F_{i,l}$ and $H_{i,k,l}$ belong to $\mathbb{C}[x'][q,q^{-1}]$, and*

$$F_{i,l} = f_i^0|_{x_l=1} + \sum_{L\neq 0} q^L F_{i,l,L}, \qquad H_{i,k,l} = H_{i,k,l}^0 + \sum_{L\neq 0} q^L H_{i,k,l,L},$$

where for $L < 0$ any monomial of $F_{i,l,L}$ and of $H_{i,k,l,L}$ has degree at least $(|L|r_i - M_0)/M$ in the coordinates x'_j with $p_j^{(l)} < 0$.

(3) *Let P be a prime divisor over $\mathbb{A}^4$ and P' a prime divisor over $\mathbb{A}^1_q \times \mathbb{A}^3$ such that the prime divisors over $\mathbb{A}^4/G_i$ and over $(\mathbb{A}^1_q \times \mathbb{A}^3)/H_{i,l}$ induced by P and P' correspond under the birational map $\bar{\Phi}_l$. Then*

$$\frac{a(P,\mathbb{A}^4,\operatorname{div}(f_i) + \sum_k c_k \operatorname{div}(h_{i,k}))}{|I_{G_i}(P)|} = \frac{a\big(P',\mathbb{A}^1_q \times \mathbb{A}^3, \operatorname{div}(F_{i,l}) + \sum_k c_k \operatorname{div}(H_{i,k,l}) + (1+\frac{\delta_l}{r_i})\operatorname{div}(q)\big)}{|I_{H_{i,l}}(P')|},$$

where the divisors of the Laurent polynomials $F_{i,l}$ and $H_{i,k,l}$ are taken on $\mathbb{A}^1_q \times \mathbb{A}^3$ and may have negative coefficients along $(q=0)$.

(4) *Assume that $F_{i,l}$ and all $H_{i,k,l}$ are polynomials, and let $X_{i,l} := (F_{i,l} = 0) \subset \mathbb{A}^1_q \times \mathbb{A}^3$. Let P be a prime divisor over W_i and P' a prime divisor over $X_{i,l}$, with $X_{i,l}$ normal at the generic point of the center of P', such that the prime divisors over W_i/G_i and over $X_{i,l}/H_{i,l}$ induced by P and P' correspond under the birational map $X_{i,l}/H_{i,l} \dashrightarrow W_i/G_i$ induced by $\bar{\Phi}_l$. Then*

$$\frac{a(P,W_i,D_i)}{|I_{G_i}(P)|} = \frac{a\big(P',X_{i,l},\sum_k c_k \operatorname{div}(H_{i,k,l}|_{X_{i,l}}) + (1+\frac{\delta_l}{r_i})\operatorname{div}(q|_{X_{i,l}})\big)}{|I_{H_{i,l}}(P')|}.$$

(5) *The open subset $(x_l \neq 0)$ of $Y_i^{\epsilon_l}$ is $V_{i,l}/H_{i,l}$, and the restriction of $B_i^{\epsilon_l}(c')$ to it is the boundary of the crepant quotient by $H_{i,l}$ of the pair*

$$\Big(V_{i,l}, \sum_k c'_k \operatorname{div}(H^0_{i,k,l}|_{V_{i,l}})\Big).$$

A point of $V_{i,l}$ maps into $C_i^{\epsilon_l}$ if and only if the coordinates x'_j with $p_j^{(l)} \le 0$ vanish at it. The open subsets $(x_l \neq 0)$, for the indices l with $\epsilon_l = \pm 1$, cover $Y_i^{\pm}$.

Proof. (1) For $(g,\lambda) \in \widetilde{G}_{i,l}$, the j-th coordinate of

$$\Phi_l\big(\lambda^{-1}\rho, (\lambda^{p_j^{(l)}}\chi_j(g)x'_j)_j\big)$$

is $\chi_j(g)\rho^{p_j^{(l)}}x'_j$, also for $j = l$ since $\lambda^{-|p_l|} = \chi_l(g)$. Thus Φ_l is equivariant. An element (g,λ) acts trivially on $\mathbb{A}^3$ if and only if $g = (\lambda^{-p_j^{(l)}})_j$, that is, $g \in G_i \cap \mathbf{T}$ and λ is determined by g, since $\mathbf{p}$ is primitive. Thus $R_{i,l} \cong G_i \cap \mathbf{T}$ is cyclic of order r_i, it acts freely on $\mathbb{G}_{m,\rho}$ by r_i-th roots of unity, and the quotient of $\mathbb{G}_{m,\rho} \times \mathbb{A}^3$ by $R_{i,l}$ is $\mathbb{G}_{m,q} \times \mathbb{A}^3$. For each $g \in G_i$, exactly $|p_l|$ elements λ satisfy $\lambda^{|p_l|}\chi_l(g) = 1$, so $|\widetilde{G}_{i,l}| = |p_l||G_i|$ and $|H_{i,l}| = |p_l||G_i|/r_i \le Me_0$. The subgroup $\{(1,\lambda)\} \cong \mu_{|p_l|}$ of $\widetilde{G}_{i,l}$ acts freely on $\mathbb{G}_{m,\rho} \times \mathbb{A}^3$, and Φ_l is the quotient morphism by $\{(1,\lambda)\}$ onto $(x_l \neq 0)$, since $\rho^{|p_l|} = x_l$ and $x'_j = x_j/\rho^{p_j^{(l)}}$. The homomorphism $(g,\lambda) \to g$ from $\widetilde{G}_{i,l}$ to G_i is surjective with kernel $\{(1,\lambda)\}$. Thus Φ_l induces isomorphisms

$$(\mathbb{G}_{m,q} \times \mathbb{A}^3)/H_{i,l} \cong (\mathbb{G}_{m,\rho} \times \mathbb{A}^3)/\widetilde{G}_{i,l} \cong (x_l \neq 0)/G_i,$$

whose composite is $\bar{\Phi}_l$. By construction $H_{i,l}$ acts faithfully on $\mathbb{A}^3$. The polynomial $f_i^0|_{x_l=1}$ contains the restrictions of the monomials x^2 and y^2z to $(x_l=1)$, which are distinct monomials, so it is not a constant times a power of a coordinate, and $V_{i,l}$ is not contained in any coordinate hyperplane. A diagonal element acting trivially on $V_{i,l}$ thus fixes any coordinate, and $H_{i,l}$ acts faithfully on $V_{i,l}$.

(2) A monomial x^γ of $\mathbf{T}$-weight d satisfies $x^\gamma \circ \Phi_l = \rho^{\epsilon_l d} x'^\gamma$. By the first statement of Lemma 5.3, since f_i contains x^2, and by Definition-Lemma 11.5(2), any monomial of f_i (resp. $h_{i,k}$) has $\mathbf{T}$-weight $2(u+v)+L'r_i$ (resp. $\kappa_k + L'r_i$) with $L' \in \mathbb{Z}$, and $L'=0$ exactly for the monomials of the principal block. Thus $F_{i,l}$ and $H_{i,k,l}$ are as in (2) with $L = \epsilon_l L'$. If $L<0$, then $\sum_j p_j^{(l)}\gamma_j \le M_0 - |L|r_i$, so $\sum_{p_j^{(l)}<0} |p_j|\gamma_j \ge |L|r_i - M_0$, and the degree of x'^γ in the coordinates with $p_j^{(l)}<0$ is at least $(|L|r_i - M_0)/M$.

(3) For a rational function φ and a prime divisor E over a smooth variety X, we write

$$a(E, X, c\,\mathrm{div}(\varphi)) := a(E, X, 0) - c\,\mathrm{mult}_E\,\varphi,$$

and similarly for several functions. Let E be a prime divisor over $\mathbb{A}^4$, and $\pi : X' \to \mathbb{A}^4$ a higher model on which E is a divisor. We write $\mathrm{mult}_E(dx_1 \wedge \cdots \wedge dx_4)$ for the coefficient of E in $K_{X'/\mathbb{A}^4}$, the divisor of $\pi^*(dx_1 \wedge \cdots \wedge dx_4)$. Then

$$a\Big(E, \mathbb{A}^4, \sum_k c_k\,\mathrm{div}(\varphi_k)\Big) = 1 + \mathrm{mult}_E(dx_1 \wedge \cdots \wedge dx_4) - \sum_k c_k\,\mathrm{mult}_E\,\varphi_k,$$

and similarly on $\mathbb{A}^1_q \times \mathbb{A}^3$ with $dq \wedge dx'$. Up to a non-zero constant,

$$\Phi_l^*(dx_1 \wedge \cdots \wedge dx_4) = \rho^{\sum_j p_j^{(l)} - 1} d\rho \wedge dx' = \frac{1}{r_i}\rho^{\sum_j p_j^{(l)} - r_i} dq \wedge dx'.$$

Since $\rho^{r_i} = q$ and

$$\sum_j p_j^{(l)} - 2\epsilon_l(u+v) - \sum_k c_k\epsilon_l\kappa_k = \epsilon_l\Big(u + w - \sum_k c_k\kappa_k\Big) = -\delta_l,$$

we obtain, up to a non-zero constant,

$$\Phi_l^*\Big(\frac{dx_1 \wedge \cdots \wedge dx_4}{f_i \prod_k h_{i,k}^{c_k}}\Big) = \frac{q^{-1-\delta_l/r_i} dq \wedge dx'}{F_{i,l} \prod_k H_{i,k,l}^{c_k}}, \tag{E.126}$$

where both sides are formal products of rational forms and real powers of rational functions, so that their orders along a prime divisor are defined. Let P'' be a prime divisor over $\mathbb{A}^1_\rho \times \mathbb{A}^3$ such that $\mathrm{mult}_{P''}(\varphi(\rho^{r_i}, x'))$ is a positive multiple of $\mathrm{mult}_{P'}\,\varphi$ for the rational functions φ on $\mathbb{A}^1_q \times \mathbb{A}^3$. Such a prime divisor P'' exists by [Liu26, Lemma 2.6(3)], applied to the quotient of $\mathbb{A}^1_\rho \times \mathbb{A}^3$ by $R_{i,l}$. Then $\mathrm{mult}_{P''}(\varphi \circ \Phi_l)$ is a positive multiple of $\mathrm{mult}_{gP}\,\varphi$ for the rational functions φ on $\mathbb{A}^4$, for some $g \in G_i$, since $\mathrm{mult}_{P''}(\cdot \circ \Phi_l)$ and mult_P restrict on the pullbacks of the rational functions on $\mathbb{A}^4/G_i$ to positive multiples of the valuation of the prime divisor over $\mathbb{A}^4/G_i$ induced by P (by the choice of P'' and the hypothesis of (3)), and the prime divisors over $\mathbb{A}^4$ inducing it form one G_i-orbit [Ser79, Chapter I, §7, Proposition 19]. Since the divisors of f_i and of the $h_{i,k}$ are G_i-invariant and $I_{G_i}(gP) = I_{G_i}(P)$, the left side of (3) for gP equals that for P, so we may replace P by gP and assume that $g=1$. Let e and e' be the ramification indices of P'' over P and over P'. The morphisms Φ_l and $(\rho, x') \to (\rho^{r_i}, x')$ are the quotient morphisms by $\{(1,\lambda)\}$ and by $R_{i,l}$ (proof of (1)). Let π be a finite morphism between normal varieties, tamely ramified with index e along a prime divisor P'' over a prime divisor P, and η a form as in (E.126). Then $1 + \mathrm{mult}_{P''}\,\pi^*\eta = e(1 + \mathrm{mult}_P\,\eta)$ by the Hurwitz formula [Kol13, 2.41]. We apply it to the finite morphism from the normalization in $\mathbb{G}_m \times \mathbb{A}^3$ of a model of $\mathbb{A}^4$ on which P is a divisor, with the left form of (E.126), and to the finite morphism from the normalization in $\mathbb{G}_m \times \mathbb{A}^3$ of a model of $\mathbb{A}^1_q \times \mathbb{A}^3$ on which P' is a divisor, with the right form. Thus the left side of the formula of (3) multiplied by $e|I_{G_i}(P)|$ and the right side multiplied by $e'|I_{H_{i,l}}(P')|$ are both equal to 1 plus the

order of (E.126) along P''. By [Liu26, Lemma 2.6(1)], $|I_{\widetilde{G}_{i,l}}(P'')|$ is the ramification index of P'' over the prime divisor over $(\mathbb{G}_{m,\rho}\times\mathbb{A}^3)/\widetilde{G}_{i,l}$ induced by P'', so

$$e|I_{G_i}(P)| = |I_{\widetilde{G}_{i,l}}(P'')| = e'|I_{H_{i,l}}(P')|.$$

This proves (3).

(4) Near 0, ω_{W_i} is generated by $\mathrm{Res}_{W_i}(dx_1\wedge\cdots\wedge dx_4/f_i)$. Near the generic point of the center of P', where $X_{i,l}$ is normal by hypothesis, $\omega_{X_{i,l}}$ is generated by $\mathrm{Res}_{X_{i,l}}(dq\wedge dx'/F_{i,l})$. Since $\Phi_l^* f_i = \rho^{2\epsilon_l(u+v)}F_{i,l}$, $\Phi_l^{-1}(W_i)$ is the hypersurface $(F_{i,l}=0)$ of $\mathbb{G}_{m,\rho}\times\mathbb{A}^3$, which is étale over W_i and over $X_{i,l}\cap(q\neq 0)$, hence normal. Let $\varpi(\rho,x') := (\rho^{r_i},x')$. Taking residues of (E.126) along $(f_i=0)$ and along $(F_{i,l}=0)$, we obtain, up to a non-zero constant, on $\Phi_l^{-1}(W_i)$,

$$\Phi_l^*\Big(\mathrm{Res}_{W_i}\Big(\frac{dx_1\wedge\cdots\wedge dx_4}{f_i}\Big)\prod_k h_{i,k}^{-c_k}\Big) = \varpi^*\Big(q^{-1-\delta_l/r_i}\,\mathrm{Res}_{X_{i,l}}\Big(\frac{dq\wedge dx'}{F_{i,l}}\Big)\prod_k H_{i,k,l}^{-c_k}\Big).$$

The order along P of the form in the left parentheses is $a(P,W_i,D_i)-1$, and the order along P' of the form in the right parentheses is

$$a\Big(P',X_{i,l},\sum_k c_k\,\mathrm{div}(H_{i,k,l}|_{X_{i,l}}) + \Big(1+\frac{\delta_l}{r_i}\Big)\,\mathrm{div}(q|_{X_{i,l}})\Big) - 1,$$

where the factor $q^{-1-\delta_l/r_i}$ gives the coefficient $1+\delta_l/r_i$ of $\mathrm{div}(q|_{X_{i,l}})$. Let $\overline{W}_{i,l}$ be the closure of $\Phi_l^{-1}(W_i)$ in $\mathbb{A}^1_\rho\times\mathbb{A}^3$, and let P'' be a prime divisor over $\overline{W}_{i,l}$ over P and P', after replacing P by a G_i-translate as in (3), and let e and e' be its ramification indices over P and P'. By the Hurwitz formula [Kol13, 2.41], applied on the normalization of $\overline{W}_{i,l}$ as in (3), e times $a(P,W_i,D_i)$ and e' times the log discrepancy on the right side of (4) are both 1 plus the order along P'' of either side of the residue identity above. The argument of (3) with [Liu26, Lemma 2.6(1)], applied to the action of $\widetilde{G}_{i,l}$ on the normalization of $\overline{W}_{i,l}$, gives $e|I_{G_i}(P)| = e'|I_{H_{i,l}}(P')|$. Dividing proves (4).

(5) The open subset $(x_l\neq 0)$ of $\mathrm{Proj}\bigoplus_{d\geq 0}\mathbb{C}[V_i]_{\epsilon_l d}$ is the spectrum of the degree zero part of $\mathbb{C}[V_i][x_l^{-1}]$, which is the ring of invariants of $\mathbb{C}[V_{i,l}]$ under the stabilizer $\mu_{|p_l|}\subset\mathbf{T}$ of the slice $(x_l=1)$. The group $G_i/(G_i\cap\mathbf{T})$ acts on it through the image of $\widetilde{G}_{i,l}$ in the automorphisms of $\mathbb{A}^3$, which is $H_{i,l}$, and the image of $\{(1,\lambda)\}$ is $\mu_{|p_l|}$. Since crepant quotients may be taken in stages [Liu26, Lemma 2.6(5)], the first statement of (5) follows. The image of the origin in $\mathrm{Spec}\,\mathbb{C}[V_i]^{\mathbf{T}}$ is defined by the non-constant $\mathbf{T}$-invariant monomials x^γ, whose restrictions to the slice are x'^γ. A non-constant invariant monomial contains a coordinate x_j with $p_j^{(l)}\leq 0$, and conversely x_j with $p_j=0$ and $x_j^{|p_l|}x_l^{|p_j|}$ with $p_j^{(l)}<0$ are invariant monomials. Thus a point of $V_{i,l}$ maps to the image of the origin if and only if the coordinates x'_j with $p_j^{(l)}\leq 0$ vanish at it. A point of $\mathrm{Proj}\bigoplus_{d\geq 0}\mathbb{C}[V_i]_{\epsilon_l d}$ has a non-zero coordinate of positive weight $\epsilon_l p_l$, which proves the last statement of (5). □

Lemma E.29. *Notation as in Construction E.26. Then $(u,v)\neq(0,0)$, and the following hold for i large.*

(1) The pairs $(Y_i^\pm,B_i^\pm)$ are lc near $C_i^\pm$, the numbers β_i belong to a finite set, and there exists a real number $C_0\geq 0$, independent of i, such that $\alpha_i\leq\beta_i+C_0/r_i$.

(2) If no coordinate has negative $\mathbf{T}$-weight, then $\delta\leq 0$ and $\alpha_i\geq\beta_i$.

Proof. The proof is long, so we divide it into several steps. We keep the notation of Lemmas E.25 and E.28, and we write

$$B_{i,l} := \sum_k c_k\,\mathrm{div}(H^0_{i,k,l}|_{V_{i,l}}),\qquad F^0_{i,l} := f_i^0|_{x_l=1}.$$

Step 1. In this step we prove that $(u,v)\neq(0,0)$. Suppose that $u=v=0$. Then $w=1$ by Lemma E.25(4), and $\mathbf{T}=\mathbf{T}_{(0,0,0,1)}$. We verify the hypotheses of Theorem 8.5 with t as the first

coordinate, the finite set $\Gamma_0 := \{c_1, \dots, c_s\} \subset [0,1]$, which is a DCC set, taken as the set Γ of Theorem 8.5, and the bound e_0 of Lemma E.23. The admitted germ $(W_i \ni 0)$ is an isolated Gorenstein terminal threefold singularity (Construction-Theorem 4.13(3)), hence an isolated cDV hypersurface singularity [KM98, Corollary 5.38]. The group G_i is diagonal, preserves f_i up to a character, acts faithfully on W_i, and $G_i \cap \mathbf{T}_{(0,0,0,1)}$ is cyclic of order $r_i \geq r_0(\Gamma_0)$ for i large, with $[G_i : G_i \cap \mathbf{T}] \leq e_0$. The boundary $D_i = \sum_k c_k \operatorname{div}(h_{i,k})$ has labels in Γ_0 and G_i-semi-invariant polynomial marks not vanishing identically on W_i. The pair (W_i, D_i) is klt near 0 by Construction-Theorem 4.13(1), since the pair given by the i-th datum before the replacement by numerical admissions in Lemma E.23 is klt near its base point: in the notation of Set-up 11.1 for that datum, $K_W + D = \sum_k \mu_k(K_W + C_k)$ with $C_k = \frac{1}{m}\operatorname{div}(h_k)$, $\sum_k \mu_k = 1$, and each (W, C_k) terminal near o. By the last sentence of Theorem 8.5, the numbers α_i do not form a strictly increasing sequence with $r_i \to +\infty$, a contradiction.

Step 2. In this step we prove that $(V_i, \sum_k c_k \operatorname{div}(h^0_{i,k}|_{V_i}))$ is lc near N_i and that $(Y_i^{\pm}, B_i^{\pm})$ is lc near $C_i^{\pm}$. Since W_i is normal and (W_i, D_i) is klt near 0, the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(f_i) + \sum_k c_k \operatorname{div}(h_{i,k})\Big)$$

is lc near 0 by inversion of adjunction [Kaw07, Theorem], and so is the corresponding pair on $\operatorname{Spec} \mathbb{C}[[x_1, \dots, x_4]]$ [dFEM11, Proposition 2.11]. Let L_0 be the integer L of Lemma 4.14 for $n = 4$, $s+1$ functions and $\Lambda := \{1, c_1, \dots, c_s\}$. By Lemma E.25(2) and Definition-Lemma 11.5(2), $f_i - f_i^0$ and $h_{i,k} - h^0_{i,k}$ belong to $(x_1, \dots, x_4)^{L_0}$ for i large, and f_i^0 and $h^0_{i,k}$ are not zero. By Lemma 4.14 and [dFEM11, Proposition 2.11], the pair

$$\Big(\mathbb{A}^4, \operatorname{div}(f_i^0) + \sum_k c_k \operatorname{div}(h^0_{i,k})\Big) \tag{E.127}$$

is lc near 0. Since V_i is normal (Lemma E.25(5)), inversion of adjunction [Kaw07, Theorem] implies that $(V_i, \sum_k c_k \operatorname{div}(h^0_{i,k}|_{V_i}))$ is lc near 0.

The polynomials f_i^0 and $h^0_{i,k}$ are $\mathbf{T}$-semi-invariant, so the non-lc locus of the last pair is closed and $\mathbf{T}$-invariant. For $\tilde{y} \in N_i$, the point $\lambda \cdot \tilde{y}$ tends to 0 as $\lambda \to 0$ or as $\lambda \to +\infty$, since the non-zero coordinates of $\tilde{y}$ have $\mathbf{T}$-weights of one sign. Thus the pair is lc near N_i. The morphism $V_{i,l} \times \mathbb{G}_m \to V_i \cap (x_l \neq 0)$ given by the action of $\mathbf{T}$ is étale, and it pulls back $h^0_{i,k}$ to $\lambda^{\kappa_k} H^0_{i,k,l}$ for the coordinate λ of $\mathbb{G}_m$. Thus $(V_{i,l}, B_{i,l})$ is lc near the points of N_i. By [Liu26, Lemma 2.6] and Lemma E.28(5), $(Y_i^{\pm}, B_i^{\pm})$ is lc near $C_i^{\pm}$.

Step 3. In this step we prove that the β_i belong to a finite set. Let $N' := \max(N_1, Me_0)$, where $N_1 = 4\max(a+1, b, 3)$ if f_* is in row I and $N_1 = 8\ell$ if f_* is in row II, and let Λ_0 be the finite set of numbers $\sum_k c_k n_k \leq 1$ with $n_k \in \mathbb{Z}_{\geq 0}$. By Lemma E.6 applied to N' and Λ_0, there exists a finite set $\mathcal{V} \subset \mathbb{R}$, independent of i, which contains $\operatorname{mld}((S \ni p, B)/H)$ for any surface germ $(S \ni p)$, group H and boundary B satisfying the hypotheses of that lemma. Let $\mathcal{V}'$ be the finite set of numbers $\frac{1-c}{e}$ with $c \in \Lambda_0$ and $1 \leq e \leq Me_0$.

Fix a sign ϵ with Y_i^{ϵ} not empty. A prime divisor over Y_i^{ϵ} with center in C_i^{ϵ} either is a component Σ of C_i^{ϵ} which is a curve, or has a closed point $y \in C_i^{\epsilon}$ as center. Thus

$$\beta_i^{\epsilon} = \min\Big(\min_{\Sigma} a(\Sigma, Y_i^{\epsilon}, B_i^{\epsilon}),\ \inf_{y \in C_i^{\epsilon}} \operatorname{mld}(Y_i^{\epsilon} \ni y, B_i^{\epsilon})\Big).$$

By Lemma E.28(5), y and Σ meet an open subset $V_{i,l}/H_{i,l}$ with $\epsilon_l = \epsilon$. Let $\tilde{y} \in V_{i,l}$ be a point over y, so $\tilde{y} \in N_i$. By [Liu26, Lemma 2.6],

$$\operatorname{mld}(Y_i^{\epsilon} \ni y, B_i^{\epsilon}) = \operatorname{mld}((V_{i,l} \ni \tilde{y}, B_{i,l})/\operatorname{Stab}_{H_{i,l}}(\tilde{y})).$$

By Lemma E.25(6), $(V_{i,l} \ni \tilde{y})$ is the quotient of a smooth surface germ by a group of order at most N_1. The group $\operatorname{Stab}_{H_{i,l}}(\tilde{y})$ has order at most Me_0 and acts faithfully on $V_{i,l}$ (Lemma E.28(1)), and also on the germ at $\tilde{y}$, since the diagonal group $\mu_{|p_l|}$ permutes the components of $V_{i,l}$

transitively and none of them is contained in a coordinate hyperplane. The coefficient of a prime divisor Σ' of $V_{i,l}$ in $B_{i,l}$ is $\sum_k c_k \operatorname{mult}_{\Sigma'}(H^0_{i,k,l}|_{V_{i,l}})$, which is at most 1 near $\tilde{y}$ by **Step 2**, so it belongs to Λ_0. By **Step 2** and Lemma E.6, $\operatorname{mld}(Y_i^\epsilon \ni y, B_i^\epsilon) \in \mathcal{V}$. Let $\widetilde{\Sigma} \subset V_{i,l}$ be a curve over a component Σ. By [Liu26, Lemma 2.6(4)], $a(\Sigma, Y_i^\epsilon, B_i^\epsilon) = \frac{1-c}{e}$, where $c \in \Lambda_0$ is the coefficient of $\widetilde{\Sigma}$ in $B_{i,l}$ and $e \le |H_{i,l}| \le Me_0$ is the ramification index of $\widetilde{\Sigma}$, so $a(\Sigma, Y_i^\epsilon, B_i^\epsilon) \in \mathcal{V}'$. Thus β_i^ϵ is the least element of a subset of the finite set $\mathcal{V} \cup \mathcal{V}'$, and it is attained by a prime divisor with center in C_i^ϵ. Thus $\beta_i^\pm$ and β_i belong to $\mathcal{V} \cup \mathcal{V}'$.

Step 4. In this step we construct, for a sign ϵ with Y_i^ϵ not empty, a prime divisor over the ambient germ which bounds α_i from above by $\beta_i^\epsilon + C_0/r_i$. Since $\beta_i = \min(\beta_i^+, \beta_i^-)$, this proves the last statement of (1).

By **Step 3**, there exist an index l with $\epsilon_l = \epsilon$ and an irreducible closed subset $\widetilde{Z} \subset V_{i,l}$, a point or a curve, whose image in Y_i^ϵ is contained in C_i^ϵ, such that β_i^ϵ is the minimal log discrepancy at the image of the generic point of $\widetilde{Z}$. Let $H' \subset H_{i,l}$ be the subgroup of elements fixing $\widetilde{Z}$ pointwise, and let $\tilde{y} \in \widetilde{Z}$ be a closed point, general if $\widetilde{Z}$ is a curve. In the coordinates $x'_j - x'_j(\tilde{y})$ the group H' acts diagonally, since a coordinate which does not vanish at $\tilde{y}$ has trivial character on H'. By [Liu26, Lemma 2.6], β_i^ϵ is the minimal log discrepancy of the quotient of $(V_{i,l}, B_{i,l})$ by H' at the image of the generic point of $\widetilde{Z}$. Near $\tilde{y}$, $V_{i,l} = (F^0_{i,l} = 0)$ is a normal integral hypersurface with klt singularities (Lemma E.25(6)), H' acts faithfully on its germ at $\tilde{y}$ (Lemma E.28(1) and the argument of **Step 3**, since $H' \subset \operatorname{Stab}_{H_{i,l}}(\tilde{y})$), and $(V_{i,l}, B_{i,l})$ is lc (**Step 2**). By Theorem 3.3(1) if $\widetilde{Z}$ is a point, and by Theorem 3.3(3) if $\widetilde{Z}$ is a curve, both with $n = 3$, $c = 1$, $G = H = H'$, and $\mathfrak{a} = \prod_k (H^0_{i,k,l})^{c_k}$, β_i^ϵ is the minimal log discrepancy at the image of the generic point of $\widetilde{Z}$ of the crepant quotient $(\mathbb{A}^3/H', \Delta')$ by H' of the pair

$$\Big(\mathbb{A}^3, \operatorname{div}(F^0_{i,l}) + \sum_k c_k \operatorname{div}(H^0_{i,k,l})\Big). \tag{E.128}$$

We bound a computing divisor. The diagonal group H' has order at most Me_0, so it is contained in $\mu^3_{(Me_0)!}$, and $\mathbb{A}^3/H'$ is one of finitely many log terminal threefolds. Near $\tilde{y}$, the pair (E.128) is lc by inversion of adjunction [Kaw07, Theorem], $V_{i,l}$ is locally irreducible and is not a component of any $\operatorname{div}(H^0_{i,k,l})$, so the coefficients near $\tilde{y}$ of the pair belong to $\{1\} \cup \Lambda_0$. Let $\bar{y}$ be the image of $\tilde{y}$ in $\mathbb{A}^3/H'$, and let $\Delta'_{\tilde{y}}$ be the sum of the components of Δ' through $\bar{y}$, with their coefficients in Δ'. Since H' fixes $\tilde{y}$, the preimage of each such component passes through $\tilde{y}$, so by [Liu26, Lemma 2.6(4)] the coefficients of $\Delta'_{\tilde{y}}$ belong to the finite set I_0 of numbers $1 - \frac{1-c}{e}$ with $c \in \{1\} \cup \Lambda_0$ and $1 \le e \le Me_0$, after omitting zero coefficients. A component of Δ' not through $\bar{y}$ does not contain the image of $\widetilde{Z}$, so replacing Δ' by $\Delta'_{\tilde{y}}$ changes neither the minimal log discrepancy at the image of the generic point of $\widetilde{Z}$ nor the log discrepancies of the prime divisors with that center. The divisor $K_{\mathbb{A}^3/H'} + \Delta'_{\tilde{y}}$ is $\mathbb{R}$-Cartier since $\mathbb{A}^3/H'$ is $\mathbb{Q}$-factorial. By [Kaw24, Theorem 1.4], applied to each of the finitely many threefolds $\mathbb{A}^3/H'$ and the set I_0, there exist a positive integer l_1, independent of i, and a prime divisor E' over $\mathbb{A}^3/H'$ computing the minimal log discrepancy β_i^ϵ of $(\mathbb{A}^3/H', \Delta'_{\tilde{y}})$, hence of $(\mathbb{A}^3/H', \Delta')$, at the image of the generic point of $\widetilde{Z}$, with $a(E', \mathbb{A}^3/H', 0) \le l_1$. Let E be a prime divisor over $\mathbb{A}^3$ over E' with center $\widetilde{Z}$, and let $e := |I_{H'}(E)|$. Since an element of $I_{H_{i,l}}(E)$ preserves the center of E and acts trivially on the residue field of E, $I_{H_{i,l}}(E) = I_{H'}(E)$. By [Liu26, Lemma 2.6] and the effectivity of the branch divisor,

$$a\Big(E, \mathbb{A}^3, \operatorname{div}(F^0_{i,l}) + \sum_k c_k \operatorname{div}(H^0_{i,k,l})\Big) = e\beta_i^\epsilon, \qquad a(E, \mathbb{A}^3, 0) \le e a(E', \mathbb{A}^3/H', 0) \le C_1 := Me_0 l_1.$$

Since the first log discrepancy is non-negative,

$$n_F := \operatorname{mult}_E F^0_{i,l} \le C_1, \qquad n_k := \operatorname{mult}_E H^0_{i,k,l} \le C_1/c_k.$$

Let $C_2 := C_1/\min_k c_k$. Since $\widetilde{Z}$ maps into C_i^ϵ, any coordinate x'_j with $p_j^{(l)} \le 0$ vanishes on $\widetilde{Z}$ (Lemma E.28(5)), so $\operatorname{mult}_E x'_j \ge 1$ for such j.

We take a Gauss extension. Let χ_q be the character of $H_{i,l}$ on q, and let $I := I_{H_{i,l}}(E)$. By [Liu26, Lemma 2.6(1)], I is cyclic and acts on $\mathfrak{m}_E/\mathfrak{m}_E^2$ by a faithful character λ_E. Let $q_G \in \{1,\dots,e\}$ be the integer with $\chi_q|_I = \lambda_E^{q_G}$, and let k_G be the integer with $k_G \equiv q_G \pmod e$ and $C_2 < k_G \le C_2 + e$. Let F' be the Gauss extension of mult_E to $\mathbb{A}^1_q \times \mathbb{A}^3$ with q-value k_G [Liu26, Definition-Proposition 2.11]. It is $\operatorname{mult}_{F'}$ for a prime divisor F' over $\mathbb{A}^1_q \times \mathbb{A}^3$ with center $\{0\} \times \widetilde{Z}$, $\operatorname{mult}_{F'}(\varphi q^L) = \operatorname{mult}_E \varphi + Lk_G$ for any non-zero rational function φ on $\mathbb{A}^3$ and $L \in \mathbb{Z}$, and

$$a(F', \mathbb{A}^1_q \times \mathbb{A}^3, 0) = a(E, \mathbb{A}^3, 0) + k_G$$

by [Liu26, Lemma 18.1]. We show that $I_{H_{i,l}}(F') = I$. If $\widetilde{Z}$ is a point, this is Lemma 3.1, applied with $V = \mathbb{A}^3$, $0 = \tilde{y}$, the group H' and $\chi = \chi_q|_{H'}$, since $I_{H_{i,l}}(F')$ fixes the center of F'. In general, an element of $I_{H_{i,l}}(F')$ preserves the restriction mult_E of $\operatorname{mult}_{F'}$ and acts trivially on the residue field of E, which is contained in that of F', so it belongs to I. Conversely, let τ be a rational function on $\mathbb{A}^3$ with $\operatorname{mult}_E \tau = 1$ on which I acts by λ_E, obtained by projecting a uniformizer to the λ_E-eigenspace. The residue field of F' is generated over that of E by the residue of q/τ^{k_G} [Liu26, Lemma 18.1]. An element $g \in I$ preserves mult_E and the value of q, so it preserves F', and

$$g^*\Big(\frac{q}{\tau^{k_G}}\Big) = \chi_q(g)\lambda_E(g)^{-k_G}\frac{q}{\tau^{k_G}} = \frac{q}{\tau^{k_G}}.$$

Thus g acts trivially on the residue field of F', and $g \in I_{H_{i,l}}(F')$.

We compare orders. Let $r_* := M(C_1 + 2C_2 + Me_0) + M_0$, and let i be large, so that $r_i > r_*$. By Lemma E.28(2), $F_{i,l} - F^0_{i,l} = \sum_{L\ne 0} q^L F_{i,l,L}$. If $L \ge 1$, then $\operatorname{mult}_{F'}(q^L F_{i,l,L}) \ge k_G > C_1 \ge n_F$, since $k_G > C_2$, and $C_1 \le C_2$, since $c_k = \mu_k/m \le 1$. If $L \le -1$, then each monomial of $F_{i,l,L}$ has degree at least $(|L|r_i - M_0)/M$ in the coordinates x'_j with $p_j^{(l)} < 0$, so

$$\operatorname{mult}_{F'}(q^L F_{i,l,L}) \ge \frac{|L|r_i - M_0}{M} - |L|k_G \ge \frac{r_i - M_0}{M} - k_G > C_1 \ge n_F,$$

since the function $|L| \to \frac{|L|r_i - M_0}{M} - |L|k_G$ is increasing, as $r_i > Mk_G$, and $k_G \le C_2 + Me_0$. Thus $\operatorname{mult}_{F'} F_{i,l} \ge n_F$. For $H_{i,k,l}$, the bounds $k_G > C_2 \ge n_k$ for $L \ge 1$ and $\frac{r_i - M_0}{M} - k_G > C_1 + C_2 \ge n_k$ for $L \le -1$ imply $\operatorname{mult}_{F'} H_{i,k,l} \ge n_k$ for any k. Therefore

$$\begin{aligned} &a\Big(F', \mathbb{A}^1_q \times \mathbb{A}^3, \operatorname{div}(F_{i,l}) + \sum_k c_k \operatorname{div}(H_{i,k,l}) + \Big(1 + \frac{\delta_l}{r_i}\Big)\operatorname{div}(q)\Big) \\ &\quad \le a(E, \mathbb{A}^3, 0) + k_G - n_F - \sum_k c_k n_k - \Big(1 + \frac{\delta_l}{r_i}\Big)k_G \\ &\quad = e\beta_i^\epsilon - \frac{\delta_l k_G}{r_i}. \end{aligned}$$

We pass to $\mathbb{A}^4$. Let P be a prime divisor over $\mathbb{A}^4$ such that the prime divisors over $\mathbb{A}^4/G_i$ and over $(\mathbb{A}^1_q \times \mathbb{A}^3)/H_{i,l}$ induced by P and F' correspond under $\bar{\Phi}_l$ (Lemma E.28(1)), and let P'' be a prime divisor over $\mathbb{A}^1_\rho \times \mathbb{A}^3$ whose valuation restricts to positive multiples $e_1 \operatorname{mult}_P$ and $e_2 \operatorname{mult}_{F'}$. Then $\operatorname{mult}_{P''} \rho = e_2 k_G/r_i$ and, for any j, $\operatorname{mult}_{P''} x_j = e_2(p_j^{(l)} k_G/r_i + \operatorname{mult}_E x'_j)$. If $p_j^{(l)} > 0$, this is at least $e_2 k_G/r_i > 0$. If $p_j^{(l)} \le 0$, then $\operatorname{mult}_E x'_j \ge 1$, and it is at least $e_2(1 - Mk_G/r_i) > 0$ since $Mk_G < r_i$. Thus $\operatorname{mult}_P x_j > 0$ for any j, and the center of P on $\mathbb{A}^4$ is 0. By Theorem 3.3(1), [Liu26, Lemma 2.6], Lemma E.28(3) and $I_{H_{i,l}}(F') = I$,

$$\alpha_i \le \frac{a(P, \mathbb{A}^4, \operatorname{div}(f_i) + \sum_k c_k \operatorname{div}(h_{i,k}))}{|I_{G_i}(P)|} \le \beta_i^\epsilon - \frac{\delta_l k_G}{e r_i} \le \beta_i^\epsilon + \frac{C_0}{r_i}, \qquad C_0 := |\delta|(C_2 + Me_0).$$

The constant C_0 does not depend on i. This proves (1).

Step 5. In this step we prove (2). Assume that $p_j \geq 0$ for any j. Then Y_i^- is empty and $\beta_i = \beta_i^+$. Let $E_{\mathbf{p}}$ be the toric prime divisor over $\mathbb{A}^4$ given by the monomial valuation with weights $\mathbf{p}$. The center of $E_{\mathbf{p}}$ is the linear subspace $(x_j = 0 \mid p_j > 0)$, which contains 0, and $a(E_{\mathbf{p}}, \mathbb{A}^4, 0) = \sum_j p_j$. The polynomials f_i^0 and $h_{i,k}^0$ are $\mathbf{T}$-homogeneous of weights $2(u+v)$ and κ_k, so $\operatorname{mult}_{E_{\mathbf{p}}} f_i^0 = 2(u+v)$ and $\operatorname{mult}_{E_{\mathbf{p}}} h_{i,k}^0 = \kappa_k$. Since the pair (E.127) is lc near 0 (**Step 2**),

$$0 \leq a\Big(E_{\mathbf{p}}, \mathbb{A}^4, \operatorname{div}(f_i^0) + \sum_k c_k \operatorname{div}(h_{i,k}^0)\Big) = (u+v) + v + 2u + w - 2(u+v) - \sum_k c_k \kappa_k = -\delta.$$

Thus $\delta \leq 0$.

Let P be a prime divisor over W_i with center 0. Choose an index l with $p_l > 0$ such that $\operatorname{mult}_P x_l/p_l \leq \operatorname{mult}_P x_j/p_j$ for any j with $p_j > 0$. Then $\epsilon_l = 1$ and $\delta_l = \delta$. For i large, the $\mathbf{T}$-weights of the monomials of f_i and $h_{i,k}$ are non-negative, while $0 \leq 2(u+v) < r_i$ and $0 \leq \kappa_k < r_i$, so the integers L of Lemma E.28(2) are non-negative and $F_{i,l}$ and $H_{i,k,l}$ are polynomials. Let $X := X_{i,l} = (F_{i,l} = 0) \subset \mathbb{A}^1_q \times \mathbb{A}^3$. Then $X \cap (q = 0) = V_{i,l}$. Let P' be a prime divisor over X such that the prime divisors over W_i/G_i and over $X/H_{i,l}$ induced by P and P' correspond under the birational map induced by $\bar{\Phi}_l$, and let P'' be a prime divisor over the closure of $\Phi_l^{-1}(W_i)$ in $\mathbb{A}^1_\rho \times \mathbb{A}^3$ restricting to positive multiples $e_1 \operatorname{mult}_P$ and $e_2 \operatorname{mult}_{P'}$. Then

$$\operatorname{mult}_{P''} q = \frac{e_1 r_i \operatorname{mult}_P x_l}{p_l} > 0, \qquad \operatorname{mult}_{P''} x_j' = e_1\Big(\operatorname{mult}_P x_j - \frac{p_j \operatorname{mult}_P x_l}{p_l}\Big) \geq 0,$$

with strict inequality if $p_j = 0$. Thus the center $\widetilde{Z}$ of P' on X is contained in $V_{i,l}$ and in the locus where the coordinates x_j' with $p_j = 0$ vanish, that is, $\widetilde{Z}$ maps into C_i^+ (Lemma E.28(5)). Moreover, $\widetilde{Z}$ is a point or a curve, since $p_j = 0$ for some j by Lemma E.25(1) and no component of $V_{i,l}$ is contained in $(x_j' = 0)$.

Let $\tilde{y} \in \widetilde{Z}$ be a closed point, general if $\widetilde{Z}$ is a curve, and let $H' \subset H_{i,l}$ be the subgroup of elements fixing $\widetilde{Z}$ pointwise. Then $I_{H_{i,l}}(P') \subset H'$, and H' acts diagonally in the coordinates q and $x_j' - x_j'(\tilde{y})$. Near $\tilde{y}$, the surface $V_{i,l}$ is normal, integral and klt (Lemma E.25(6)), and it is the Cartier divisor $(q = 0)$ of the hypersurface X. Thus X is smooth at the smooth points of $V_{i,l}$, the singular locus of X has codimension at least two near $\tilde{y}$, and X is normal near $\tilde{y}$ by Serre's criterion. By inversion of adjunction [KM98, Theorem 5.50(1)], $(X, V_{i,l})$ is plt near $\tilde{y}$, so X is klt near $\tilde{y}$. The group H' fixes $\tilde{y}$, so it acts faithfully on the germ of $V_{i,l}$ at $\tilde{y}$ by the argument of **Step 3**, hence on the germ of X at $(0, \tilde{y})$. The hypotheses of Lemma 5.5 hold with $n = 4$, the group $G = H = H'$ on $\mathbb{A}^1_q \times \mathbb{A}^3$ centered at $(0, \tilde{y})$, the hypersurface X, the polynomial u there taken to be q, with $S_0 = V_{i,l}$, the $\mathbb{R}$-ideal $\mathfrak{a} = \prod_k H_{i,k,l}^{c_k}$, whose restriction to S_0 is $\prod_k (H_{i,k,l}^0|_{V_{i,l}})^{c_k}$, and the number b there taken to be $1 + \frac{\delta}{r_i}$, which is at most 1 since $\delta \leq 0$. The pair $(S_0, \mathfrak{a}|_{S_0}) = (V_{i,l}, B_{i,l})$ is lc near $\tilde{y}$ by **Step 2**. If $\widetilde{Z}$ is a point, Lemma 5.5 and [Liu26, Lemma 2.6] imply

$$\frac{a\big(P', X, (1 + \frac{\delta}{r_i})V_{i,l} + \mathfrak{a}|_X\big)}{|I_{H'}(P')|} \geq \operatorname{mld}((V_{i,l} \ni \tilde{y}, B_{i,l})/H') \geq \beta_i^+, \tag{E.129}$$

since the last minimal log discrepancy is that of (Y_i^+, B_i^+) at a point of C_i^+. If $\widetilde{Z}$ is a curve, the second statement of Lemma 5.5 with $C = \widetilde{Z}$ implies that the left side of (E.129) is at least the minimal log discrepancy of the quotient of $(V_{i,l}, B_{i,l})$ by H' at the image of the generic point of $\widetilde{Z}$, which is the log discrepancy of the image of $\widetilde{Z}$, a component of C_i^+, with respect to (Y_i^+, B_i^+), and hence at least β_i^+. Since $I_{H'}(P') = I_{H_{i,l}}(P')$, Lemma E.28(4) implies

$$\frac{a(P, W_i, D_i)}{|I_{G_i}(P)|} = \frac{a\big(P', X, \sum_k c_k \operatorname{div}(H_{i,k,l}|_X) + (1 + \frac{\delta}{r_i})V_{i,l}\big)}{|I_{H_{i,l}}(P')|} \geq \beta_i^+ = \beta_i.$$

Since P is an arbitrary prime divisor over W_i with center 0, $\alpha_i \geq \beta_i$ by [Liu26, Lemma 2.6]. This proves (2). □

We first write the equation of W_i in a form adapted to the signs of the weights. We use it in the proof of Lemma E.32. Until the end of that proof, the notation is that of Construction E.26, and we assume that $\mathbf{T}$ has weights of both signs.

Notation E.30. Let $x_1, \dots, x_4$, $\mathbf{p} = (p_1, \dots, p_4) = (u+v, v, 2u, w)$, and M be as in Construction E.27, so that p_j is the $\mathbf{T}$-weight of x_j, and $\gcd(p_1, \dots, p_4) = 1$. Let

$$\mathcal{P} := \{j \mid p_j > 0\}, \qquad \mathcal{N} := \{j \mid p_j < 0\},$$

which are non-empty by assumption. Let $R_i := G_i \cap \mathbf{T}$, so that $r_i = |R_i|$ and $[G_i : R_i] \leq e_0$ (Lemma E.23). The group R_i acts on a monomial of $\mathbf{T}$-weight κ through the character $\zeta \to \zeta^\kappa$ of μ_{r_i}, so two monomials with equal characters of G_i have $\mathbf{T}$-weights congruent modulo r_i.

For monomials m_1 and m_2 whose $\mathbf{T}$-weights are congruent modulo r_i, we say that m_2 has *shift* L with respect to m_1 if the $\mathbf{T}$-weight of m_2 is the $\mathbf{T}$-weight of m_1 plus Lr_i. For a G_i-semi-invariant polynomial or power series g and a monomial m with the character of g, a term of g has shift L with respect to m if its monomial has shift L with respect to m. The $\mathbf{T}$-weights of the monomials of f_i and of $h_{i,k}$ are congruent modulo r_i to $2(u+v)$ and to κ_k, respectively, and

$$f_i = \sum_{L \in \mathbb{Z}} f_{i,L}, \qquad h_{i,k} = \sum_{L \in \mathbb{Z}} h_{i,k,L},$$

where $f_{i,L}$ and $h_{i,k,L}$ are the sums of the terms of $\mathbf{T}$-weight $2(u+v)+Lr_i$ and κ_k+Lr_i, respectively, so that $f_{i,0} = f_i^0$ and $h_{i,k,0} = h_{i,k}^0$. We call $f_{i,L}$ and $h_{i,k,L}$ the *blocks* of shift L, and L the shift of their terms. For the terms of f_i, L is the shift with respect to x^2. Likewise, for a G_i-semi-invariant polynomial g with the character of f_i, respectively of $h_{i,k}$, the block g_L of shift L is the sum of the terms of g of $\mathbf{T}$-weight $2(u+v)+Lr_i$, respectively $\kappa_k + Lr_i$. The decompositions of f_i, of $h_{i,k}$, and of g into blocks are taken in the coordinates in force, so they apply as well after the replacement of the data in Lemma E.31.

If a monomial $\prod_j x_j^{n_j}$ has $\mathbf{T}$-weight $\kappa + Lr_i$ with $\kappa \in \mathbb{Z}$ and $L \geq 1$, then $Lr_i + \kappa = \sum_j p_j n_j \leq M \sum_{j \in \mathcal{P}} n_j$, and symmetrically for $L \leq -1$. Hence

$$\sum_{j \in \mathcal{P}} n_j \geq \frac{Lr_i - |\kappa|}{M} \text{ if } L \geq 1, \qquad \sum_{j \in \mathcal{N}} n_j \geq \frac{|L| r_i - |\kappa|}{M} \text{ if } L \leq -1. \tag{E.130}$$

In particular, a monomial of non-zero shift with respect to a monomial of $\mathbf{T}$-weight κ has degree at least $(r_i - |\kappa|)/M$, and the orders at 0 of $f_i - f_i^0$ and of $h_{i,k} - h_{i,k}^0$ tend to $+\infty$.

For $j \in \mathcal{P}$ and $l \in \mathcal{N}$, the polynomial $g_{jl} := x_j^{|p_l|} x_l^{p_j}$ is G_i-semi-invariant of $\mathbf{T}$-weight zero and vanishes at 0. Thus R_i acts trivially on g_{jl}, and the order n_{jl} of the character of g_{jl} divides $[G_i : R_i]$, so

$$g_{jl}^{n_{jl}} \in \mathbb{C}[x_1, \dots, x_4]^{G_i}, \qquad n_{jl} \leq e_0. \tag{E.131}$$

Lemma E.31. *Notation as in Construction E.26, and assume that $\mathbf{T}$ has weights of both signs. Let $\Lambda' \subset \mathbb{R}^s_{>0}$ be a finite set, and let the shifts be as in the notation above. Then, after passing to a subsequence and replacing the data with numerical admissions for the groups G_i (Proposition 11.8), composed with G_i-equivariant automorphisms of $\mathbb{C}[[x,y,z,t]]$ with diagonal tangent maps at 0, the following hold for i large.*

(1) The conclusion of Lemma E.23 holds with the old u, v, w, and e_0, and the numbers α_i, $\kappa_1, \dots, \kappa_s$, δ, β_i^+, and β_i^- do not change. For any $c' \in \Lambda'$ and any sign $\pm$, if the pair $(Y_i^\pm, B_i^\pm(c'))$ is lc near $C_i^\pm$ before the replacement, then it is lc near $C_i^\pm$ after the replacement.

(2) *Assume that* $w < 0$. *If* f_* *is in row I of* (E.124), *then there exist polynomials* $\sigma_i(t)$, $\phi_i(z,t) = z^{a+1} + \sum_{m\le a} \phi_{i,m}(t)z^m$, *and* $\psi_i(z,t) = t^b + \sum_{m<b} \psi_{i,m}(z)t^m$ *with*

$$f_i = x^2 + zy^2 + \sigma_i(t)y + \frac{\phi_i\psi_i + \sigma_i^2/4}{z}, \qquad \phi_i(0,t)t^b = -\frac{\sigma_i(t)^2}{4},$$

where the terms of $\phi_i - z^{a+1}$ *and of* $\sigma_i(t)y$ *have negative shift with respect to* z^{a+1} *and to* x^2, *respectively, and the terms of* $\psi_i - t^b$ *have positive shift with respect to* t^b. *Moreover* $f_i^0 = x^2 + zy^2 + z^a t^b(1+\varsigma_i)$, *where* $z^{a+1}t^b\varsigma_i$ *is the sum of the terms of* $(\phi_i - z^{a+1})(\psi_i - t^b)$ *of shift zero with respect to* $z^{a+1}t^b$. *The polynomial* ς_i *is* G_i*-invariant of* $\mathbf{T}$*-weight zero, and any monomial* $z^m t^n$ *of* ς_i *satisfies*

$$m \ge \frac{r_i - |w|b}{2u} - a - 1, \qquad n \ge \frac{r_i - 2u(a+1)}{|w|} - b.$$

If f_* *is in row II, then* $f_i^0 = x^2 + zy^2 + yt^\ell$, *and there exists a polynomial* $\psi_i = t^{2\ell} + \sum_{m<2\ell} \psi_{i,m}(z)t^m$, *whose terms other than* $t^{2\ell}$ *have positive shift with respect to* $t^{2\ell}$, *with*

$$f_i = x^2 + zy^2 + t^\ell y + \frac{t^{2\ell} - \psi_i}{4z}.$$

(3) *Assume that* $w \ge 0$. *Then* $f_i^0 = x^2 + zy^2 + \sigma^0(t)y + \tau^0(t)$ *for polynomials* σ^0 *and* τ^0 *independent of* i, *and* $f_i - f_i^0 = e_i(z,t) + e_i'(t)y$ *for polynomials* e_i *and* e_i', *where all the terms of* e_i *and of* $e_i'(t)y$ *have positive shift with respect to* x^2.

Proof. By Lemma E.25(4), after the subsequence and the rescaling of the coordinates given there, $u > 0$ and one of the cases (a)–(d) of Lemma E.25(4) holds, since in case (e) no coordinate has negative $\mathbf{T}$-weight. Thus $w < 0$ in case (a), $w > 0$ in case (b), and $w = 0$ in cases (c) and (d). Let $\mathbf{q}_\eta := 2(\eta+1, \eta, 2, 1)$. By Lemma E.25(3), any term of f_i other than the terms of f_* has $\mathbf{q}_\eta$-weight larger than $4\eta + 4$, the $\mathbf{q}_\eta$-weight of f_*. By Lemma E.23, $f_i = x^2 + zy^2 + \sigma_i(t)y + \rho_i(z,t)$, where σ_i has order at least 3 and $\rho_i := S_i(0,z,t)$ has order at least 4.

Case $w < 0$*: the terms of* $\sigma_i(t)y$. A term yt^n of $\sigma_i(t)y$ has $\mathbf{T}$-weight $2(u+v) + (nw - 2u - v)$. In row I, $2au + bw = 2(u+v)$, so $nw - 2u - v = (n - \frac{b}{2})w - (a+1)u$, and $2\eta + 2n > 4\eta + 4$, that is, $n > a + \frac{b}{2} + 1$. Hence $nw - 2u - v < 0$, and all the terms of $\sigma_i(t)y$ have negative shift with respect to x^2. In row II, $\ell w = 2u + v$, so $nw - 2u - v = (n - \ell)w$, and $n \ge \ell$, with equality only for the term yt^ℓ of f_*. Hence $\sigma_i = t^\ell s_i(t)$ with $s_i(0) = 1$, and all the terms of $(\sigma_i - t^\ell)y$ have negative shift with respect to x^2.

Case $w < 0$, *row I.* Let $\Delta_i := z\rho_i - \sigma_i^2/4$, so that

$$f_i = x^2 + z\Big(y + \frac{\sigma_i}{2z}\Big)^2 + \frac{\Delta_i}{z}.$$

Any monomial $z^m t^n$ of Δ_i other than $z^{a+1}t^b$ satisfies $2m + n > 2(a+1) + b$, and $z^{a+1}t^b$ has coefficient one: for the terms of $z\rho_i$ this is the bound on the $\mathbf{q}_\eta$-weights, and the terms of σ_i^2 are $t^{n+n'}$ with $n, n' > \eta + 2$. Hence $\Delta_i(t^2 z_1, t) = t^{2(a+1)+b}\Delta_i'(z_1, t)$ for a polynomial Δ_i' with $\Delta_i'(z_1, 0) = z_1^{a+1}$. Since f_i is a polynomial, ρ_i, σ_i, Δ_i, and Δ_i' are polynomials. The Weierstrass preparation and division theorems hold in the rings $\mathbb{C}\{z_1, t\}$ and $\mathbb{C}\{z, t\}$ of convergent power series [GLS07, Chapter I, Theorems 1.6 and 1.8], and, by the uniqueness, the factorizations below in the formal rings are the factorizations in the convergent rings. Hence P_i, ϕ_i, ψ_i, and υ_i below are convergent. The group G_i acts diagonally on $\mathbb{C}[[z_1, t]]$, with z_1 of the character of z/t^2, and Δ_i' is semi-invariant. By the Weierstrass preparation theorem, Δ_i' is a unit times a unique polynomial $P_i = z_1^{a+1} + \sum_{m\le a} P_{i,m}(t)z_1^m$ with $P_{i,m}(0) = 0$, and P_i is semi-invariant by the uniqueness. Let

$$\phi_i := t^{2(a+1)}P_i\Big(\frac{z}{t^2}, t\Big) = z^{a+1} + \sum_{m\le a} P_{i,m}(t)t^{2(a+1-m)}z^m.$$

By the Weierstrass division theorem, $\Delta_i = \Pi\phi_i + \Pi'$, where $\Pi \in \mathbb{C}[[z,t]]$ and $\Pi' \in \mathbb{C}[[t]][z]$ has degree at most a in z. Then $\Pi'(t^2 z_1, t)$ has degree at most a in z_1 and is divisible by P_i in $\mathbb{C}[[z_1,t]]$, so $\Pi' = 0$ by the uniqueness in the Weierstrass division theorem. For the weights $(2,1)$ of (z,t), the part of least weight of Δ_i/ϕ_i is t^b, the quotient of those of Δ_i and ϕ_i. Thus $(\Delta_i/\phi_i)(0,t)$ has order b, and the Weierstrass preparation theorem gives

$$\Delta_i = \upsilon_i\phi_i\psi_i, \qquad \psi_i = t^b + \sum_{m<b} \psi_{i,m}(z)t^m, \quad \psi_{i,m}(0) = 0,$$

with a unit υ_i of $\mathbb{C}[[z,t]]$. By the uniqueness, ϕ_i and ψ_i are semi-invariant with the characters of z^{a+1} and t^b, and υ_i is invariant, since Δ_i has the character of $z^{a+1}t^b$. A term $z^m t^n$ of $\phi_i - z^{a+1}$ has $m \le a$, so its $\mathbf{T}$-weight $2um + wn$ is smaller than $2u(a+1)$, and a term $z^m t^n$ of $\psi_i - t^b$ has $n < b$, so its $\mathbf{T}$-weight is larger than wb. Hence the terms of $\phi_i - z^{a+1}$ have negative shift with respect to z^{a+1}, and the terms of $\psi_i - t^b$ have positive shift with respect to t^b. Let $\sigma_i' := \upsilon_i(0,t)^{-1/2}\sigma_i$ and

$$x' := \frac{x}{\sqrt{\upsilon_i}}, \qquad y' := \frac{y + \sigma_i/(2z)}{\sqrt{\upsilon_i}} - \frac{\sigma_i'}{2z},$$

with the square roots equal to 1 at 0. Then y' is a power series, since

$$y' = \upsilon_i^{-1/2} y + \frac{\sigma_i}{2z}\big(\upsilon_i^{-1/2} - \upsilon_i(0,t)^{-1/2}\big)$$

and $\upsilon_i^{-1/2} - \upsilon_i(0,t)^{-1/2}$ is divisible by z. Moreover $(x,y,z,t) \to (x',y',z,t)$ is a G_i-equivariant automorphism of $\mathbb{C}[[x,y,z,t]]$ with identity tangent map, since σ_i has order at least 3. We obtain

$$\upsilon_i^{-1} f_i = x'^2 + zy'^2 + \sigma_i' y' + \frac{\phi_i\psi_i + \sigma_i'^2/4}{z},$$

and the restriction of $\Delta_i = \upsilon_i\phi_i\psi_i$ to $(z=0)$ implies $\phi_i(0,t)t^b = -\sigma_i'(t)^2/4$. The non-constant terms of the invariant unit $\upsilon_i(0,t)$ of $\mathbb{C}[[t]]$ have negative $\mathbf{T}$-weight, hence negative shift with respect to 1, so the terms of $\sigma_i' y'$ have negative shift with respect to x'^2.

Case $w < 0$, row II. Let $\tilde\Psi_i := \sigma_i^2 - 4z\rho_i$, so that $f_i = x^2 + z(y + \sigma_i/(2z))^2 - \tilde\Psi_i/(4z)$. Since $\tilde\Psi_i(0,t) = t^{2\ell}s_i(t)^2$, the Weierstrass preparation theorem gives $\tilde\Psi_i = \upsilon_i\psi_i$ with a unit υ_i and $\psi_i = t^{2\ell} + \sum_{m<2\ell}\psi_{i,m}(z)t^m$, $\psi_{i,m}(0) = 0$. Since $\tilde\Psi_i$ is a polynomial, υ_i and ψ_i are convergent, as in row I. By the uniqueness, ψ_i is semi-invariant with the character of $t^{2\ell}$ and υ_i is invariant, and $\upsilon_i(0,t) = s_i(t)^2$. As in row I, the terms of $\psi_i - t^{2\ell}$ have positive shift with respect to $t^{2\ell}$. Let

$$x' := \frac{x}{\sqrt{\upsilon_i}}, \qquad y' := \frac{y + \sigma_i/(2z)}{\sqrt{\upsilon_i}} - \frac{t^\ell}{2z},$$

a power series since $\sigma_i/s_i = t^\ell$. Then $(x,y,z,t) \to (x',y',z,t)$ is a G_i-equivariant automorphism of $\mathbb{C}[[x,y,z,t]]$ with identity tangent map, and

$$\upsilon_i^{-1} f_i = x'^2 + zy'^2 + t^\ell y' + \frac{t^{2\ell} - \psi_i}{4z}.$$

Case $w > 0$. By Lemma E.25(4)(b), $v < 0$, and either f_* is in row I with $a = 0$ and $2u > bw$, or f_* is in row II and $2u > \ell w$. The $\mathbf{T}$-weights of x^2 and zy^2 are $2(u+v)$, those of the terms of $\sigma_i(t)y$ are at least v, and those of the terms of ρ_i are non-negative. Since $2(u+v) - r_i < \min(v,0)$ for i large, no term of f_i has negative shift with respect to x^2. In row I, a term $z^m t^n$ of shift zero satisfies $2um + wn = bw < 2u$, so $m = 0$ and $n = b$, and there is at most one term yt^d of shift zero, namely the one with $v + dw = bw$. Thus $f_i^0 = x^2 + zy^2 + t^b + \lambda_i yt^d$ with $\lambda_i \in \mathbb{C}$, the coefficient of t^b being one by Lemma E.25(3). After passing to a subsequence on which either all λ_i vanish or none does, and taking the coordinates $\lambda_i y$ and $\lambda_i^{-2}z$ in place of y and z if $\lambda_i \ne 0$, we get $f_i^0 = x^2 + zy^2 + \sigma^0(t)y + \tau^0(t)$ with $\sigma^0 \in \{0, t^d\}$ and $\tau^0 := t^b$. In row II, a term yt^n of shift zero satisfies $v + nw = 2(u+v)$, so $n = \ell$, and a term $z^m t^n$ of shift zero satisfies $2u(m+1) = w(2\ell - n)$. Since $w < 2u/\ell$, the equation $2u(m+1) = w(2\ell - n)$ forces $m = 0$ and

$n = 2\ell - 2u/w < \ell$, and then t^n has $\mathbf{q}_\eta$-weight $2n < 4\eta + 4$, which is impossible. Thus $f_i^0 = f_*$, that is, $\sigma^0 = t^\ell$ and $\tau^0 = 0$. In both rows let $e_i' := \sigma_i - \sigma^0$ and $e_i := \rho_i - \tau^0$.

Case $w = 0$. By Lemma E.25(4)(c),(d), either $(u, v) = (1, -1)$ and f_* is in row I with $a = 0$, or $(u, v) = (1, -2)$ and f_* is in row II. The coordinate t has $\mathbf{T}$-weight zero, and x^2, y, and z have $\mathbf{T}$-weights $2(u + v)$, v, and 2. In the first case $2(u + v) = 0$, the terms of $\sigma_i(t)y$ have $\mathbf{T}$-weight $-1 \not\equiv 0 \pmod{r_i}$, so $\sigma_i = 0$, and a term $z^m t^n$ of ρ_i has shift $2m/r_i$. Thus $f_i^0 = x^2 + zy^2 + \tau_i(t)$, where τ_i is the sum of the terms of ρ_i not divisible by z, and $\tau_i = t^b s_i(t)$ with $s_i(0) = 1$. In the second case $2(u + v) = -2$, all the terms of $\sigma_i(t)y$ have shift zero, and a term $z^m t^n$ of ρ_i has $\mathbf{T}$-weight $2m \neq -2$, hence positive shift. Thus $f_i^0 = x^2 + zy^2 + \sigma_i(t)y$ and $\sigma_i = t^\ell s_i(t)$ with $s_i(0) = 1$. In both cases the terms of τ_i, respectively σ_i, have the character of t^b, respectively t^ℓ, so s_i is G_i-invariant. Hence $t' := ts_i(t)^{1/b}$, respectively $t' := ts_i(t)^{1/\ell}$, defines a G_i-equivariant and $\mathbf{T}$-equivariant automorphism of $\mathbb{C}[[x, y, z, t]]$ with identity tangent map, and in the coordinates x, y, z, t' we have $f_i^0 = x^2 + zy^2 + t'^b$, respectively $f_i^0 = x^2 + zy^2 + t'^\ell y$. We let $\sigma^0 := 0$, $\tau^0 := t^b$, $e_i := \rho_i - \tau_i$ in the first case, $\sigma^0 := t^\ell$, $\tau^0 := 0$, $e_i := \rho_i$ in the second case, and $e_i' := 0$, with t' in place of t.

Truncation. In each case we have found a G_i-equivariant automorphism of $\mathbb{C}[[x, y, z, t]]$ with diagonal tangent map at 0, which we write as a change of coordinates, and a G_i-invariant unit, namely v_i^{-1} if $w < 0$ and 1 if $w \geq 0$, whose product with f_i in the new coordinates has the form of (2) or (3), with power series in place of the polynomials σ_i, $\phi_{i,m}$, $\psi_{i,m}$, e_i, and e_i'. If $w > 0$, the series σ_i, e_i, and e_i' are polynomials, the automorphism is a rescaling of the coordinates, and we do not truncate.

Let $w \leq 0$. Let $J \subset \mathbb{C}[x, y, z, t]$ be the ideal generated by the $\mathbf{T}$-invariant monomials of positive degree. By Gordan's lemma, the monomials of $\mathbf{T}$-weight κ_k are the products of finitely many of them with $\mathbf{T}$-invariant monomials, and the $\mathbf{T}$-invariant monomials are the products of finitely many of them. Hence there is a positive integer C_J, independent of i, such that any monomial of $\mathbf{T}$-weight in $\{\kappa_1, \dots, \kappa_s\}$ and degree at least $C_J D$ belongs to J^D. We apply Construction-Theorem 4.13 to the old data, with the change of coordinates and the unit above, the units $u_k := 1$, and the vectors and the prime divisors listed in Proposition 11.8. For each i fix an integer N with $N \geq C_J i$, $N \geq r_i$, and $N \geq N_0$, where N_0 is the bound of Construction-Theorem 4.13 for the data of the previous sentence, and let $N' \geq \max(L_1 + b + 2, N)$, where L_1 is the integer given there for N, and $b := 0$ if f_* is in row II. Replace each of the power series σ_i, $\phi_{i,m}$ with $m \geq 1$, $\psi_{i,m}$, e_i, and e_i' by the sum of the terms of degree less than N' of that series, and, in row I of the case $w < 0$, let $\phi_{i,0} := -\sigma_i^2/(4t^b)$ for the new σ_i. It is a polynomial, since σ_i^2 was divisible by t^b before the replacement and $N' \geq b$. The replacement keeps the forms of (2) and (3), the identity $\phi_i(0, t)t^b = -\sigma_i^2/4$, and the signs of the shifts, and it changes the product of the unit above and f_i by terms of degree at least $N' - b - 2$. Replace each $h_{i,k}$, written in the new coordinates, by the sum of the terms of degree less than N of $h_{i,k}$. By the choice of N and N', Construction-Theorem 4.13 implies that the new data are a numerical admission of the old ones for the group G_i, and α_i does not change by Construction-Theorem 4.13(1), applied with $c^{(0)}$ and the group G_i.

Proposition 11.8 assumes the conditions of Set-up 11.1, in which the proxies are terminal near o, while the old pairs $(W_i, \frac{1}{m}\operatorname{div}(h_{i,k}))$ are only known to be klt near o. Its conclusions hold for the new data nevertheless. The pairs $(W_i, \frac{1}{m}\operatorname{div}(h_{i,k}))$ are klt near o by Construction-Theorem 4.13(1) for the numerical admissions of Lemma E.23, whose vectors in Construction-Theorem 4.13(iv) include those of Proposition 11.8(i). Hence the vectors of Proposition 11.8(i) are allowed in Construction-Theorem 4.13(iv) here, and Construction-Theorem 4.13(1) for them gives Proposition 11.8(2) for the new data. For the prime divisor F of Proposition 11.8 we take the prime divisor over the old W_i which computes α_i and corresponds to the prime divisor F of the numerical admission of Lemma E.23 (Proposition 11.8(1)), and Construction-Theorem 4.13(1) and (2) give Proposition 11.8(1) for the new data. Finally the old W_i is singular at o, since its equation $x^2 + zy^2 + S_i$ has order two.

In row I of the case $w<0$, a term z^mt^n of $\phi_i - z^{a+1}$ has $m \le a$ and negative shift, so $|w|n \ge r_i - 2u(a+1)$, and a term z^mt^n of $\psi_i - t^b$ has $n<b$ and positive shift, so $2um \ge r_i - |w|b$. The bounds on the monomials of ς_i follow. The terms of shift zero with respect to x^2 of the new f_i are x^2, zy^2, and the terms of shift zero of $\phi_i\psi_i/z$, since the terms of $\sigma_i(t)y$ and of $\sigma_i^2/(4z)$ have negative shift. Hence f_i^0 is given by the formula in (2).

The group G_i acts diagonally on the new coordinates with the old characters, and the new σ_i has order at least 3. The new $S_i(0,z,t)$ is equal to z^at^b in row I and to 0 in row II of the case $w<0$, and to the old $S_i(0,z,t)$ in the new coordinates if $w=0$, up to terms of degree at least $r_i/M - C$ for a constant C independent of i: the other terms have non-zero shift with respect to x^2 (see (E.130)), are terms of $z^at^b\varsigma_i$, whose monomials satisfy the bounds in (2), or have degree at least $N'-b-2$. Hence the conclusion of Lemma E.23 holds for the new data with the old u, v, w, and e_0. The terms of least degree of each mark do not change up to the diagonal tangent map, since N is larger than their degree, so $\kappa_1,\dots,\kappa_s$ and δ do not change. This proves (2), (3), and the first sentence of (1), except for $\beta_i^{\pm}$.

The references. We compare $\beta_i^{\pm}$ and the lc property near $C_i^{\pm}$ for the old and the new data, and we write $\tilde V_i$, $\tilde f_i^0$, $\tilde h_{i,k}^0$, $\tilde C_i^{\pm}$ for the new data. If $w>0$, the new references are the images of the old ones under the rescaling, and there is nothing to prove. Let $w\le 0$. The set N_i of Lemma E.25(6) is defined by the vanishing of coordinates, both for the old and for the new data. By $\mathbf{T}$-equivariance, a holomorphic function of $\mathbf{T}$-weight κ_k on a $\mathbf{T}$-invariant neighborhood of N_i whose expansion at 0 consists of terms of degree at least C_JD belongs to the ideal generated by J^D near any point of N_i. At 0 the expansion belongs to $J^D\mathbb{C}[[x,y,z,t]]$, and

$$J^D\mathbb{C}[[x,y,z,t]] \cap \mathbb{C}\{x,y,z,t\} = J^D\mathbb{C}\{x,y,z,t\}$$

by the faithful flatness of the completion [Mat89, Theorems 8.14 and 7.5(ii)]. The points of the neighborhood at which the membership fails form a closed $\mathbf{T}$-invariant analytic subset not containing 0, while the closure of the $\mathbf{T}$-orbit of any point of N_i contains 0. We show that there are a $\mathbf{T}$-equivariant and G_i-equivariant holomorphic map Θ_i, defined on a $\mathbf{T}$-invariant neighborhood of N_i, a local isomorphism at any point of N_i, and with $\tilde f_i^0$ equal to a unit times $f_i^0\circ\Theta_i$, and integers D_i with $D_i\to+\infty$, such that $\tilde h_{i,k}^0 - h_{i,k}^0\circ\Theta_i$ belongs, near any point of N_i, to the ideal generated by J^{D_i} for any k, and such that Θ_i is the identity on N_i, except in row II of the case $w<0$, where Θ_i is the identity on $(x=y=t=0)$ and is given on $(z=0)$ by $(x,y,0,t)\to(x,y+\xi_i(t),0,t)$ for a holomorphic function ξ_i with $\xi_i(0)=0$.

If $w=0$, let Θ_i be the inverse of the substitution $t\to t'$, which is defined near N_i and is the identity on N_i, since t vanishes on N_i. It maps blocks to blocks, so $\tilde f_i^0 = f_i^0\circ\Theta_i$, and $\tilde h_{i,k}^0 - h_{i,k}^0\circ\Theta_i$ consists of terms of degree at least N. Let $D_i := \lfloor N/C_J\rfloor$, which is at least i by the choice of N.

Let $w<0$. The $\mathbf{T}$-invariant monomials in z,t are the powers of $\iota := z^{|w|/g}t^{2u/g}$, $g := \gcd(|w|,2u)$, and ι vanishes on N_i. Let υ_i^0 be the sum of the terms of $\mathbf{T}$-weight zero of υ_i, a power series in ι with constant term $\upsilon_i(0)=1$ (compare, in row I, the coefficients of t^b in $(\Delta_i/\phi_i)(0,t) = \upsilon_i(0,t)t^b$, whose part of least weight is t^b, and use $\upsilon_i(0,t)=s_i(t)^2$ in row II), convergent since υ_i is (the terms of $\mathbf{T}$-weight zero of a convergent series form a convergent series), and invariant under G_i. Let

$$\Theta_i'(x,y,z,t) := \begin{cases} \left(\sqrt{\upsilon_i^0}x, \sqrt{\upsilon_i^0}y, z, t\right) & \text{in row I,} \\ \left(\sqrt{\upsilon_i^0}x, \sqrt{\upsilon_i^0}y + (\sqrt{\upsilon_i^0}-1)\frac{t^\ell}{2z}, z, t\right) & \text{in row II.} \end{cases}$$

The map Θ_i' is $\mathbf{T}$-equivariant and G_i-equivariant, and it is holomorphic where $|\iota|$ is small, since $\sqrt{\upsilon_i^0}-1$ is divisible by ι and ι is divisible by z. In row I, Θ_i' is the identity on N_i, since ι vanishes there. In row II, all the coordinates other than z have negative $\mathbf{T}$-weight, since $v=\ell w-2u$, so N_i is contained in $(x=y=t=0)\cup(z=0)$. The map Θ_i' is the identity on $(x=y=t=0)$, and it is given on $(z=0)$ by $(x,y,0,t)\to(x,y+\xi_i'(t),0,t)$, where $\xi_i'(t)$ is the restriction of

$(\sqrt{\upsilon_i^0}-1)t^\ell/(2z)$ to $(z=0)$ and $\xi_i'(0)=0$. By (E.130), the terms of non-zero shift of υ_i, of $\sigma_i - t^\ell$ in row II, and of σ_i and σ_i' in row I have degree at least $r_i/M - C$ for a constant C independent of i. Hence the inverse of the change of coordinates expresses x and y as the corresponding components of Θ_i' plus terms of non-zero shift with respect to x and y and terms of degree at least $2r_i/M - 2C$. The blocks of non-zero shift of $h_{i,k}$ have degree at least $(r_i - |\kappa_k|)/M$ by (E.130), so, after enlarging C, the terms of shift zero of $h_{i,k}$ in the new coordinates are the terms of $h_{i,k}^0 \circ \Theta_i'$ plus terms of degree at least $r_i/M - C$.

In row I, $f_i^0 = x^2 + zy^2 + z^a t^b \Upsilon_i$, where $z^{a+1}t^b\Upsilon_i$ is the sum of the terms of shift zero of Δ_i, and Υ_i is a polynomial in ι with constant term 1, since the terms of $\sigma_i(t)y$ and of σ_i^2 have non-zero shift and the other monomials of Δ_i satisfy $2m+n > 2(a+1)+b$. Here Δ_i, υ_i, ϕ_i, ψ_i are the series before the truncation, Υ_i is defined from the old f_i^0, and ς_i is the polynomial of (2), defined from the truncated ϕ_i and ψ_i. Let $\varpi_i := \Upsilon_i/(\upsilon_i^0(1+\varsigma_i))$, a convergent power series in ι with constant term 1. The terms of shift zero of $\upsilon_i\phi_i\psi_i/(z^{a+1}t^b)$ are υ_i^0 times the terms of shift zero of $\phi_i\psi_i/(z^{a+1}t^b)$, plus products of terms of non-zero shifts summing to zero, which have degree at least $2r_i/M - 2C$ by (E.130). The terms of shift zero of $\phi_i\psi_i/(z^{a+1}t^b)$ are 1 and the terms of shift zero of $(\phi_i - z^{a+1})(\psi_i - t^b)/(z^{a+1}t^b)$, and the terms of shift zero of $(\phi_i - z^{a+1})(\psi_i - t^b)/(z^{a+1}t^b)$ for the series before and after the truncation differ by terms of degree at least $N' - C$. Hence, after enlarging C, $\varpi_i - 1$ is a power series in ι whose terms have degree at least $\min(2r_i/M, N') - C$. Thus

$$f_i^0 \circ \Theta_i' = \upsilon_i^0\big(x^2 + zy^2 + z^a t^b(1+\varsigma_i)\varpi_i\big).$$

If $b \geq 1$, let $\varkappa_i$ be the power series in ι with $\varkappa_i(0) = 1$ and

$$\varkappa_i^b \cdot (1+\varsigma_i)(\iota\varkappa_i^{2u/g}) \cdot \varpi_i(\iota\varkappa_i^{2u/g}) = (1+\varsigma_i)(\iota),$$

given by the implicit function theorem, convergent by the holomorphic implicit function theorem since ς_i and ϖ_i are convergent, and G_i-invariant by the uniqueness of $\varkappa_i$, and let Θ_i'' be the map $t \to t\varkappa_i$. If $b = 0$, let $\varkappa_i$ be the convergent power series in ι with $\varkappa_i(0) = 1$ and

$$\varkappa_i^a \cdot (1+\varsigma_i)(\iota\varkappa_i^{|w|/g}) \cdot \varpi_i(\iota\varkappa_i^{|w|/g}) = (1+\varsigma_i)(\iota),$$

and let Θ_i'' be the map $(y,z) \to (y\varkappa_i^{-1/2}, z\varkappa_i)$. In row II, $f_i^0 = x^2 + z(y + t^\ell/(2z))^2 - t^{2\ell}\Upsilon_i/(4z)$, where $t^{2\ell}\Upsilon_i$ is the sum of the terms of shift zero of $\tilde\Psi_i = \upsilon_i\psi_i$, with $\tilde\Psi_i$, υ_i, ψ_i before the truncation. Let $\varpi_i := \Upsilon_i/\upsilon_i^0$, a convergent power series in ι with constant term 1. As in row I, the terms of shift zero of $\upsilon_i\psi_i/t^{2\ell}$ are υ_i^0 plus products of terms of non-zero shifts summing to zero, which have degree at least $2r_i/M - 2C$, since the terms of $\psi_i - t^{2\ell}$ have positive shift. Hence $\varpi_i - 1$ is a power series in ι whose terms have degree at least $\min(2r_i/M, N') - C$. In row II let

$$\Theta_i''(x,y,z,t) := \Big(\sqrt{\varpi_i}x, \sqrt{\varpi_i}y + (\sqrt{\varpi_i} - 1)\frac{t^\ell}{2z}, z, t\Big).$$

In both rows, for $\Theta_i := \Theta_i' \circ \Theta_i''$, the function $f_i^0 \circ \Theta_i$ is a unit times $\tilde f_i^0$, and the map Θ_i'' differs from the identity by terms divisible by a power of ι whose exponent tends to $+\infty$. In row I, Θ_i'' is the identity on N_i, since $\varkappa_i(0) = 1$ and ι vanishes on N_i, so Θ_i is the identity on N_i. In row II, Θ_i'', like Θ_i' with ϖ_i in place of υ_i^0, is the identity on $(x = y = t = 0)$ and is given on $(z = 0)$ by $(x,y,0,t) \to (x, y + \xi_i''(t), 0, t)$ with $\xi_i''(0) = 0$, so Θ_i is the identity on $(x = y = t = 0)$ and is given on $(z = 0)$ by $(x,y,0,t) \to (x, y + \xi_i(t), 0, t)$ with $\xi_i(0) = 0$. The Jacobian determinant of Θ_i' is υ_i^0, and that of Θ_i'' is

$$\varkappa_i + \frac{2u}{g}\iota\varkappa_i', \qquad \varkappa_i^{-1/2}\Big(\varkappa_i + \frac{|w|}{g}\iota\varkappa_i'\Big), \qquad \varpi_i$$

in row I with $b \geq 1$, in row I with $b = 0$, and in row II, respectively, where $\varkappa_i'$ is the derivative of $\varkappa_i$ in ι. All of them are 1 on $(\iota = 0)$, which contains N_i and is preserved by Θ_i'', so Θ_i is a local isomorphism at any point of N_i. Hence we may take for D_i the minimum of

$\lfloor(\min(r_i/M,N)-C)/C_J\rfloor$ and of the exponents of ι above, which tends to $+\infty$, since $N'\geq N\geq r_i$ by the choice of N and N'.

Conclusion. Let Λ'' be the set consisting of c and the vectors of Λ', let γ be the least entry of the vectors of Λ'', and let $\Lambda_0\subset[0,1]$ be the finite set of the numbers $\sum_k n_k c'_k\leq 1$ with $n_k\in\mathbb{Z}_{\geq 0}$ and $c'\in\Lambda''$. Let $N_1:=4\max(a+1,b,3)$ in row I and $N_1:=8\ell$ in row II, which bounds the orders of the groups in Lemma E.25(6). Let l_0 be the integer of Lemma E.6 for the bound $\max(N_1,Me_0)$ on the orders of the groups and the set Λ_0, and let L_0 be the integer of Lemma 4.14 for $n=3$, $s+1$ functions, and the finite set $\{1\}\cup\{c'_k\mid c'\in\Lambda'',\ 1\leq k\leq s\}$. Let i be so large that $D_i>\max(L_0,l_0/\gamma)$.

Fix an index l with $p_l\neq 0$, and let $\tilde{y}$ be a point of $\tilde{V}_{i,l}$ which maps into $\tilde{C}_i^{\epsilon_l}$ (Construction E.27). As a point of $\tilde{V}_i$, $\tilde{y}$ belongs to N_i (Lemma E.28(5)). In row II of the case $w<0$ with $\epsilon_l=-1$, we assume moreover that l is the index of t if $t(\tilde{y})\neq 0$, the index of x if $t(\tilde{y})=0$ and $x(\tilde{y})\neq 0$, and that $\tilde{y}=(0,1,0,0)$ otherwise. Every point of $\tilde{C}_i^-$ is the image of such a point $\tilde{y}$, and every point of C_i^- is the image of the analogous point of $V_{i,l}$, since the preimages of $\tilde{C}_i^-$ and of C_i^- are contained in $(z=0)$. Let $\hat{y}:=\Theta_i(\tilde{y})$. Then $x_l(\hat{y})=1$, since Θ_i is the identity on N_i, except in row II of the case $w<0$, where it preserves x and t on $(z=0)$ and fixes $(0,1,0,0)$, as $\xi_i(0)=0$. Since Θ_i preserves $(z=0)$ and $(x=y=t=0)$ in row II, the coordinates x_j with $\epsilon_l p_j\leq 0$ vanish at $\hat{y}$, so $\hat{y}$ belongs to N_i and maps into $C_i^{\epsilon_l}$ (Lemma E.28(5)).

Composing Θ_i with the element of $\mathbf{T}$ which restores $x_l=1$ (given near $\tilde{y}$ by a p_l-th root of $1/(x_l\circ\Theta_i)$ equal to 1 at $\tilde{y}$) gives an isomorphism from the germ of $\tilde{V}_{i,l}$ at $\tilde{y}$ onto the germ of $V_{i,l}$ at $\hat{y}$, equivariant for the stabilizer $H_{\tilde{y}}$ of $\tilde{y}$ in $H_{i,l}$, which is the stabilizer $H_{\hat{y}}$ of $\hat{y}$, since Θ_i commutes with $\mathbf{T}$ and G_i and is injective on N_i (it is the identity on N_i, except in row II of the case $w<0$, where it is the identity on $(x=y=t=0)$ and the bijection $(x,y,0,t)\to(x,y+\xi_i(t),0,t)$ of $(z=0)$). Under the isomorphism of germs, $\tilde{H}_k:=\tilde{h}^0_{i,k}|_{x_l=1}$ corresponds to a unit times H_k+E_k, where $H_k:=h^0_{i,k}|_{x_l=1}$ and E_k belongs to $\mathfrak{n}^{D_i}$, for the ideal $\mathfrak{n}$ generated by the x'_j with $\epsilon_l p_j\leq 0$, since $\Theta_i^*J=J$ near N_i, the element of $\mathbf{T}$ restoring $x_l=1$ preserves J, and a $\mathbf{T}$-invariant monomial of positive degree contains a coordinate x_j with $j\neq l$ and $\epsilon_l p_j\leq 0$. Here $\Theta_i^*J=J$ near N_i as for J^D above: both ideals are $\mathbf{T}$-invariant, and they are equal at 0, since near 0 the ideal J is generated by the $\mathbf{T}$-invariant holomorphic functions vanishing at 0, and Θ_i and its local inverse at 0 are $\mathbf{T}$-equivariant and fix 0. Hence the locus where they differ is closed, $\mathbf{T}$-invariant, and does not contain 0, so it misses N_i. By Lemma E.28(5), the coordinates x'_j with $\epsilon_l p_j\leq 0$ vanish at $\tilde{y}$ and at $\hat{y}$, and on the preimages of $\tilde{C}_i^{\epsilon_l}$ and of $C_i^{\epsilon_l}$.

Let $c'\in\Lambda''$, and assume that the quotient of $(V_{i,l},\sum_k c'_k\operatorname{div}(H_k))$ by $H_{\hat{y}}$ is lc near $\hat{y}$, which holds for $c'=c$ by Lemma E.29(1). By [Kaw07, Theorem], since $V_{i,l}$ is normal, the pair $(\mathbb{A}^3,V_{i,l}+\sum_k c'_k\operatorname{div}(H_k))$ is lc near $\hat{y}$, and Lemma 4.14, applied in the completion at $\hat{y}$, [dFEM11, Proposition 2.11], [Kaw07, Theorem], and [Liu26, Lemma 2.6] imply that the quotient of $(V_{i,l},\sum_k c'_k\operatorname{div}(H_k+E_k))$ by $H_{\hat{y}}$ is lc near $\hat{y}$. By Lemma E.25(6) and Lemma E.28(1), the germ of $V_{i,l}$ at $\hat{y}$ is the quotient of a smooth germ by a group of order at most N_1, and $|H_{\hat{y}}|\leq Me_0$. The coefficients of both boundaries belong to Λ_0, since the pairs are lc.

By Lemma E.6, both minimal log discrepancies at $\hat{y}$ of the quotients by $H_{\hat{y}}$ are computed by prime divisors E over $V_{i,l}$ with center $\hat{y}$ and $a(E,V_{i,l},0)\leq l_0$. For such E, the lc property implies $c'_k\operatorname{mult}_E H_k\leq l_0$ and $c'_k\operatorname{mult}_E(H_k+E_k)\leq l_0$, while $\operatorname{mult}_E E_k\geq D_i>l_0/\gamma$, so $\operatorname{mult}_E H_k=\operatorname{mult}_E(H_k+E_k)$ for any k. Hence the minimal log discrepancy at $\hat{y}$ of the old quotient pair and the minimal log discrepancy at $\tilde{y}$ of the new quotient pair are equal. Similarly, along a curve contained in the preimage of $C_i^{\epsilon_l}$, the coefficients of both boundaries are at most one, so the multiplicities of H_k and of H_k+E_k along it are less than D_i, and they are equal.

By [Liu26, Lemma 2.6], $\beta_i^{\pm}(c')$ is the minimum, over the points of $C_i^{\pm}$, of the minimal log discrepancies of the quotients by the stabilizers at a preimage of the point in any chart $V_{i,l}$ with $\epsilon_l=\pm 1$ containing it, and of the log discrepancies along the images of the curves contained in the preimages of $C_i^{\pm}$, and similarly for the new data. Since Θ_i maps $\tilde{V}_i\cap N_i$ bijectively onto

$V_i \cap N_i$ and commutes with $\mathbf{T}$ and G_i, and since, by the choice of $\tilde{y}$ and l, every point of $\tilde{C}_i^{\pm}$ is the image of a point $\tilde{y}$ of a chart $\tilde{V}_{i,l}$ and every point of $C_i^{\pm}$ is the image of the point $\Theta_i(\tilde{y})$ of $V_{i,l}$, $\beta_i^{\pm}$ does not change, and, for $c' \in \Lambda'$, the lc property of $(Y_i^{\pm}, B_i^{\pm}(c'))$ near $C_i^{\pm}$ is preserved. This proves (1). □

We now prove Lemma E.32, which treats the circles $\mathbf{T}$ with weights of both signs: for i large, the minima β_i^+ and β_i^- differ by at least $|\delta|/Q$, the smaller one being β_i^+ if $\delta < 0$ and β_i^- if $\delta > 0$, and, if for some $\theta > 1$ the boundaries of the reference pairs on a side fixed by the sign of δ_θ can be multiplied by θ keeping the reference pairs lc, then $\alpha_i \geq \beta_i$ for i large.

Lemma E.32. *Notation as in Construction E.26, and assume that* $\mathbf{T}$ *has weights of both signs.*

(1) There exists a positive integer Q, independent of i, such that, for i large, $\beta_i^- \geq \beta_i^+ + |\delta|/Q$ if $\delta < 0$, $\beta_i^+ \geq \beta_i^- + \delta/Q$ if $\delta > 0$, and $\beta_i^+ = \beta_i^-$ if $\delta = 0$.

(2) Let $\theta > 1$ and $\delta_\theta := \theta \sum_k c_k \kappa_k - u - w$. Assume that there exists a sign $\pm$ with $\pm\delta_\theta \leq 0$ such that the pair $(Y_i^{\pm}, B_i^{\pm}(\theta c))$ is lc near $C_i^{\pm}$ for any i. Then $\alpha_i \geq \beta_i$ for i large.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix notation and collect the properties of the data that we use.

We use Lemma E.31 with $\Lambda' := \{\theta c\}$ when we prove (2), where θ is the number in (2), and with $\Lambda' := \emptyset$ when we prove (1). The replacement of the data there changes neither α_i, κ_k, δ, δ_θ, nor $\beta_i^{\pm}$, and it preserves the hypotheses of (2). The conclusions of (1) and (2) hold or fail for each i separately, and the integer Q below does not depend on the subsequence, so it suffices to prove them for a subsequence of an arbitrary subsequence. Hence we may pass to subsequences in the proof. Thus we may assume that the conclusions of Lemma E.31 hold for the data. We use the notation x_j, $\mathbf{p}$, M, $\mathcal{P}$, $\mathcal{N}$, R_i, g_{jl}, n_{jl}, the shifts, and the blocks introduced before Lemma E.31.

We use the following properties of the data.

(a) By Lemma E.25(5) and (6) and Lemma E.28(5), $V_i = (f_i^0 = 0)$ is a normal integral hypersurface, and, for any j with $p_j \neq 0$, the surface $V_{i,j} = V_i \cap (x_j = 1) \subset \mathbb{A}^3$ of Construction E.27 is smooth or has a Du Val singularity at any point which maps into C_i^+ or C_i^-, since such a point belongs to the set N_i of Lemma E.25(6).

(b) By Lemma E.25(4), since $\mathbf{T}$ has weights of both signs, case (e) there does not occur, $u > 0$, and one of the following holds:
  (A) $w < 0$ (case (a) there);
  (B) $w > 0 > v$ (case (b) there);
  (C) $w = 0$ and $\mathbf{p} \in \{(0, -1, 2, 0), (-1, -2, 2, 0)\}$ (cases (c) and (d) there).

(c) In case (A), f_i has the presentation of Lemma E.31(2), with the polynomials σ_i, ϕ_i, ψ_i, and, in row I, ς_i there.

(d) In cases (B) and (C), $f_i^0 = x^2 + zy^2 + \sigma^0(t)y + \tau^0(t)$ and $f_i - f_i^0 = e_i(z,t) + e_i'(t)y$ (Lemma E.31(3)).

Moreover, the $h_{i,k}$ are polynomials (Lemma E.23), and W_i is normal and Gorenstein, so that

$$\Omega_{W_i} := \operatorname{Res} \frac{dx \wedge dy \wedge dz \wedge dt}{f_i}$$

generates ω_{W_i} near o. For a prime divisor F over a normal variety X and a non-zero rational section Ω of ω_X, let $\operatorname{mult}_F \Omega$ be the coefficient of F in the divisor of the pullback of Ω to a higher model of X on which F is a divisor. If Ω generates ω_X near the center of F and $B = \sum_k b_k \operatorname{div}(g_k)$, then

$$a(F, X, B) = 1 + \operatorname{mult}_F \Omega - \sum_k b_k \operatorname{mult}_F g_k. \tag{E.132}$$

If $X' \to X$ is a generically finite dominant rational map of normal varieties with $\dim X' = \dim X$ and F' is a prime divisor over X' lying over F with ramification index e, then

$$1 + \operatorname{mult}_{F'} \Omega = e(1 + \operatorname{mult}_F \Omega).$$

For the proof, let $\mathfrak{t}$ be a uniformizer of the valuation ring of F, and $\mathfrak{w}_2, \dots, \mathfrak{w}_n$ elements of the valuation ring of F whose residues form a transcendence basis of the residue field of F, where $n = \dim X$. Then $d\mathfrak{t} \wedge \mathfrak{W}$, where $\mathfrak{W} := d\mathfrak{w}_2 \wedge \cdots \wedge d\mathfrak{w}_n$, generates the n-forms at the generic point of F, so

$$\Omega = \mathfrak{t}^{\mathfrak{d}}\mathfrak{e} \cdot d\mathfrak{t} \wedge \mathfrak{W}$$

with $\mathfrak{d} = \operatorname{mult}_F \Omega$ and $\mathfrak{e}$ a unit. Write $\mathfrak{t} = \mathfrak{t}'^e\mathfrak{e}'$ with $\mathfrak{t}'$ a uniformizer of the valuation ring of F' and $\mathfrak{e}'$ a unit. Since the residue field of F' is a finite extension of the residue field of F, the residues of $\mathfrak{w}_2, \dots, \mathfrak{w}_n$ also form a transcendence basis of the residue field of F', so $d\mathfrak{t}' \wedge \mathfrak{W}$ generates the n-forms at the generic point of F'. As

$$d\mathfrak{t} = \mathfrak{t}'^{e-1}(e\mathfrak{e}'d\mathfrak{t}' + \mathfrak{t}'d\mathfrak{e}'),$$

the pullback of Ω is $\mathfrak{t}'^{e\mathfrak{d}+e-1}$ times a unit times $d\mathfrak{t}' \wedge \mathfrak{W}$, so $\operatorname{mult}_{F'} \Omega = e\mathfrak{d} + e - 1$.

Step 2. In this step we write the log discrepancies of $(Y_i^{\pm}, B_i^{\pm})$ through one function on the prime divisors over the base, and we prove (1).

Let $\mathbb{C}[V_i]_\kappa$ be the space of regular functions on V_i of $\mathbf{T}$-weight κ, let $Z_i' := \operatorname{Spec} \mathbb{C}[V_i]_0$ and $Z_i := Z_i'/G_i = \operatorname{Spec} \mathbb{C}[V_i]_0^{G_i}$, where G_i acts on Z_i' through G_i/R_i, and let $\bar{0} \in Z_i$ be the image of the origin. The surfaces $Y_i^{\pm}$ are projective over Z_i, and $C_i^{\pm}$ is the fiber over $\bar{0}$. Since $\mathbf{T}$ has weights of both signs, any rational function of $\mathbf{T}$-weight zero on V_i is a quotient of two regular functions of weight zero (multiply the numerator and the denominator by a monomial of the opposite weight). Hence $Y_i^+ \to Z_i$ and $Y_i^- \to Z_i$ are birational. In particular, the prime divisors over Y_i^+ with center in C_i^+ and the prime divisors over Y_i^- with center in C_i^- are both exactly the prime divisors P over Z_i with center $\bar{0}$.

Since $\gcd(p_1, \dots, p_4) = 1$, there exists a Laurent monomial λ in $x_1, \dots, x_4$ of $\mathbf{T}$-weight one. Let $\pi_i : V_i \dashrightarrow Z_i'$ be the rational map given by the regular functions of weight zero. The rational map $(\pi_i, \lambda) : V_i \dashrightarrow Z_i' \times \mathbb{G}_m$ is birational: a rational function of weight κ on V_i times $\lambda^{-\kappa}$ has weight zero, hence is a quotient of two regular functions of weight zero. Omitting pullbacks by (π_i, λ), write

$$x_j = \lambda^{p_j} N_j, \qquad h_{i,k}^0 = \lambda^{\kappa_k} \Psi_k$$

with rational functions N_j and Ψ_k on Z_i'. Let

$$\Omega_{V_i} := \operatorname{Res} \frac{dx \wedge dy \wedge dz \wedge dt}{f_i^0},$$

a generator of ω_{V_i}. Since f_i^0 is $\mathbf{T}$-homogeneous of weight $2(u+v)$, Ω_{V_i} has $\mathbf{T}$-weight $\sum_j p_j - 2(u+v) = u + w$. The rational 3-forms on $Z_i' \times \mathbb{G}_m$ are the forms $\varphi d\log\lambda \wedge \omega'$ with φ a rational function, for a fixed non-zero rational 2-form ω' on Z_i'. Thus $\Omega_{V_i} = \varphi d\log\lambda \wedge \omega'$, where φ has weight $u + w$, so $\varphi = \lambda^{u+w}\varphi_0$ for a rational function φ_0 on Z_i', and

$$\Omega_{V_i} = \lambda^{u+w} d\log\lambda \wedge \omega_0, \qquad \omega_0 := \varphi_0\omega'. \tag{E.133}$$

The functions N_j, Ψ_k and the form ω_0 are semi-invariant under G_i.

Let P be a prime divisor over Z_i with center $\bar{0}$, and take a prime divisor P_1 over Z_i' lying over P, with ramification index e_P. Let

$$\mathcal{B}(P) := \frac{1 + \operatorname{mult}_{P_1} \omega_0}{e_P} - \sum_k c_k \frac{\operatorname{mult}_{P_1} \Psi_k}{e_P},$$

$$\varrho_-(P) := \max_{j \in \mathcal{P}} \frac{-\operatorname{mult}_{P_1} N_j}{e_P p_j}, \qquad \varrho_+(P) := \min_{j \in \mathcal{N}} \frac{\operatorname{mult}_{P_1} N_j}{e_P |p_j|}.$$

The numbers $\mathcal{B}(P)$ and $\varrho_\pm(P)$ do not depend on the choice of P_1, since G_i permutes the prime divisors over Z_i' lying over P transitively, and N_j, Ψ_k, and ω_0 are G_i-semi-invariant.

Claim E.33. *For any prime divisor P over Z_i with center $\bar{0}$,*

$$a(P, Y_i^+, B_i^+) = \mathcal{B}(P) - \delta\varrho_-(P), \qquad a(P, Y_i^-, B_i^-) = \mathcal{B}(P) - \delta\varrho_+(P).$$

Proof. We prove the first equality. The second one is the first one for the circle with weights $-\mathbf{p}$, which exchanges Y_i^+ and Y_i^-, replaces δ by $-\delta$ and λ by λ^{-1}, and exchanges ϱ_- and $-\varrho_+$.

By Lemma E.28(5), for $j \in \mathcal{P}$ the open subset $(x_j \neq 0)$ of Y_i^+ is $V_{i,j}/H_{i,j}$, with the group $H_{i,j}$ of Construction E.27, and B_i^+ restricts there to the boundary of the crepant quotient of $(V_{i,j}, B_{i,j})$, where $B_{i,j} := \sum_k c_k \operatorname{div}(h^0_{i,k}|_{V_{i,j}})$. Moreover, the open subsets $(x_j \neq 0)$, $j \in \mathcal{P}$, cover Y_i^+. Choose $j \in \mathcal{P}$ such that the generic point of the center of P on Y_i^+ belongs to $(x_j \neq 0)$. The composite of the inclusion $V_{i,j} \subset V_i$ and π_i is dominant and generically finite, since $\mathbf{T} \cdot V_{i,j}$ is dense in V_i. Take a prime divisor P' over $V_{i,j}$ lying over P_1, and let e be its ramification index over P, so that its ramification index over P_1 is e/e_P. On $V_{i,j}$ we have $\lambda^{p_j} N_j = 1$, hence

$$\operatorname{mult}_{P'} \lambda = -\frac{e \operatorname{mult}_{P_1} N_j}{e_P p_j}.$$

For $l \in \mathcal{P}$, the function $x_l|_{V_{i,j}}$ is regular near the center of P', so

$$0 \leq \operatorname{mult}_{P'} x_l = p_l \operatorname{mult}_{P'} \lambda + \frac{e}{e_P} \operatorname{mult}_{P_1} N_l = e p_l \Big(\frac{-\operatorname{mult}_{P_1} N_j}{e_P p_j} - \frac{-\operatorname{mult}_{P_1} N_l}{e_P p_l} \Big).$$

Hence j attains the maximum defining $\varrho_-(P)$, and $\operatorname{mult}_{P'} \lambda = e\varrho_-(P)$.

Write $\Omega_{V_i} = d(x_j - 1) \wedge \omega_1$ with $\omega_1 := p_j^{-1} \lambda^{u+w} x_j^{-1} \omega_0$. Indeed $d \log x_j = p_j d \log \lambda + d \log N_j$, and $d \log N_j \wedge \omega_0 = 0$, as the pullback of a rational 3-form on the surface Z_i', so (E.133) implies $\Omega_{V_i} = p_j^{-1} \lambda^{u+w} d \log x_j \wedge \omega_0$. By adjunction to the Cartier divisor $V_{i,j}$ of V_i, the form

$$\Omega_{V_{i,j}} := \omega_1|_{V_{i,j}} = p_j^{-1} \lambda^{u+w} \omega_0|_{V_{i,j}}$$

generates $\omega_{V_{i,j}}$. By the Hurwitz formula, $1 + \operatorname{mult}_{P'} \omega_0 = e(1 + \operatorname{mult}_{P_1} \omega_0)/e_P$, and $\operatorname{mult}_{P'} \Psi_k = e \operatorname{mult}_{P_1} \Psi_k / e_P$. By (E.132) and $h^0_{i,k} = \lambda^{\kappa_k} \Psi_k$,

$$\begin{aligned} a(P', V_{i,j}, B_{i,j}) &= 1 + \operatorname{mult}_{P'} \omega_0 + (u+w) \operatorname{mult}_{P'} \lambda \\ &\quad - \sum_k c_k \big(\kappa_k \operatorname{mult}_{P'} \lambda + \operatorname{mult}_{P'} \Psi_k \big) \\ &= e\big(\mathcal{B}(P) - \delta \varrho_-(P) \big). \end{aligned}$$

Finally $a(P', V_{i,j}, B_{i,j}) = e a(P, Y_i^+, B_i^+)$ by [Liu26, Lemma 2.6], applied to $H_{i,j}$, which acts faithfully on $V_{i,j}$ (Lemma E.28(1)). □

Proof of Lemma E.32 continued. Let $Q := e_0 M^2$.

Claim E.34. *For any prime divisor P over Z_i with center $\bar{0}$, $\varrho_+(P) - \varrho_-(P) \geq 1/Q$.*

Proof. Let $j \in \mathcal{P}$ and $l \in \mathcal{N}$. On V_i, $g_{jl} = N_j^{|p_l|} N_l^{p_j}$, and g_{jl} is a regular function of $\mathbf{T}$-weight zero on V_i, not identically zero on V_i and vanishing at 0. By (E.131), $g_{jl}^{n_{jl}}$ is a regular function on Z_i vanishing at $\bar{0}$, and $n_{jl} \operatorname{mult}_{P_1} g_{jl} = e_P \operatorname{mult}_P(g_{jl}^{n_{jl}}) \geq e_P$. Hence, since $n_{jl} \leq e_0$,

$$\frac{\operatorname{mult}_{P_1} N_l}{e_P |p_l|} - \frac{-\operatorname{mult}_{P_1} N_j}{e_P p_j} = \frac{\operatorname{mult}_{P_1} g_{jl}}{e_P p_j |p_l|} \geq \frac{1}{e_0 p_j |p_l|} \geq \frac{1}{Q}.$$

Taking the minimum over $j \in \mathcal{P}$ and $l \in \mathcal{N}$ proves the claim. □

Proof of Lemma E.32 continued. We prove (1) with $Q = e_0 M^2$, which is independent of i. If $\delta = 0$, Claim E.33 implies $a(P, Y_i^+, B_i^+) = a(P, Y_i^-, B_i^-)$ for any prime divisor P over Z_i with center $\bar{0}$, so $\beta_i^+ = \beta_i^-$. If $\delta < 0$, Claims E.33 and E.34 imply

$$\begin{aligned} a(P, Y_i^-, B_i^-) &= \mathcal{B}(P) + |\delta| \varrho_+(P) \geq \mathcal{B}(P) + |\delta| \varrho_-(P) + \frac{|\delta|}{Q} \\ &= a(P, Y_i^+, B_i^+) + \frac{|\delta|}{Q} \geq \beta_i^+ + \frac{|\delta|}{Q} \end{aligned}$$

for any prime divisor P over Z_i with center $\bar{0}$. Taking the minimum over P implies $\beta_i^- \geq \beta_i^+ + |\delta|/Q$. The case $\delta > 0$ is symmetric. This proves (1). In particular, for i large, $\beta_i = \beta_i^+$ if $\delta < 0$, and $\beta_i = \beta_i^-$ if $\delta > 0$.

Step 3. In this step we bound from below, on a root chart, the scores of the prime divisors over a hypersurface whose blocks have shifts of one sign.

For $c' \in \mathbb{R}^s_{>0}$ let $\delta(c') := \sum_k c'_k \kappa_k - u - w$, so that $\delta(c) = \delta$ and $\delta(\theta c) = \delta_\theta$. Let $\mathcal{P}^+ := \mathcal{P}$ and $\mathcal{P}^- := \mathcal{N}$. In the following claim $\pm$ denotes one fixed sign, and $\mp$ the opposite one.

Claim E.35. *Let $c' \in \mathbb{R}^s_{>0}$, and assume that $(Y_i^\pm, B_i^\pm(c'))$ is lc near $C_i^\pm$. Let g be a G_i-semi-invariant polynomial with the character of f_i, and $h_1^\sharp, \dots, h_s^\sharp$ polynomials such that $h_k^\sharp$ is G_i-semi-invariant with the character of $h_{i,k}$, and such that all the terms of $g - f_i^0$ and of $h_k^\sharp - h_{i,k}^0$ have shift L with $\pm L \geq 1$. Let $W^\sharp$ be a G_i-invariant irreducible component of $(g = 0)$ which is not contained in any coordinate hyperplane, and $F^\sharp$ a prime divisor over $W^\sharp$ with center 0. Then the following hold.*

(1) g is reduced along $W^\sharp$.

(2) Let Ω_g be the restriction of $\mathrm{Res}(dx \wedge dy \wedge dz \wedge dt/g)$ to $W^\sharp$, which is defined by (1), let $[\varphi] := \mathrm{mult}_{F^\sharp} \varphi / |I_{G_i}(F^\sharp)|$, and let $\varrho^\pm := \min_{j \in \mathcal{P}^\pm} [x_j]/|p_j|$. Then

$$\frac{1 + \mathrm{mult}_{F^\sharp} \Omega_g - \sum_k c'_k \, \mathrm{mult}_{F^\sharp} h_k^\sharp}{|I_{G_i}(F^\sharp)|} \geq \beta_i^\pm(c') \mp \delta(c') \varrho^\pm.$$

Proof. The case of the sign $-$ is the case of the sign $+$ for the circle with weights $-\mathbf{p}$, which exchanges Y_i^+ and Y_i^-, the signs of the shifts, and $\delta(c')$ and $-\delta(c')$. We assume that the sign is $+$, and we drop the index i from G_i and r_i in this proof.

Let $j \in \mathcal{P}$ attain the minimum $\varrho := \varrho^+ = [x_j]/p_j$. We use the root chart $\Phi_j : \mathbb{G}_{m,\rho} \times \mathbb{A}^3 \to \mathbb{A}^4$ of Construction E.27, where $\epsilon_j = 1$, with the action of $\widetilde{G}_{i,j}$, and $q = \rho^r$. If $\mathcal{A}$ is a polynomial which is $\mathbf{T}$-homogeneous of weight κ, then $\mathcal{A} \circ \Phi_j = \rho^\kappa \mathcal{A}(x')$, where $\mathcal{A}(x')$ is $\mathcal{A}$ with x_j replaced by 1 and x_l by x'_l. Hence

$$g \circ \Phi_j = \rho^{2(u+v)} g', \qquad h'_k := \rho^{-\kappa_k} h_k^\sharp \circ \Phi_j,$$

where $g' := \sum_{L \geq 0} \rho^{Lr} g_L(x')$ and $h'_k = \sum_{L \geq 0} \rho^{Lr} h_{k,L}^\sharp(x')$, with g_L and $h_{k,L}^\sharp$ the blocks of shift L. Since all shifts are non-negative, g' and h'_k are polynomials in q and the x'_l. By Lemma E.28(1), the subgroup $R_{i,j}$ is cyclic of order r and acts on ρ alone, and $H_{i,j} = \widetilde{G}_{i,j}/R_{i,j}$ acts faithfully on $\mathbb{A}^3$ and on $V_{i,j}$. Let $W'' \subset \mathbb{A}^1_q \times \mathbb{A}^3$ be the hypersurface $(g'(q, x') = 0)$, with the diagonal action of $H_{i,j}$, and let

$$S_0 := W'' \cap (q = 0) = (g_0(x') = 0) = (f_i^0(x') = 0) = V_{i,j}.$$

Thus $H_{i,j}$ acts faithfully on S_0 and on W''. Since $h'_k|_{S_0} = h_{i,k}^0(x')$, Lemma E.28(5) implies that the crepant quotient of $(S_0, \sum_k c'_k \, \mathrm{div}(h'_k|_{S_0}))$ by $H_{i,j}$ is the open subset $(x_j \neq 0)$ of $(Y_i^+, B_i^+(c'))$. Let W' be an irreducible component of $(g'(\rho^r, x') = 0)$ which maps onto $W^\sharp \cap (x_j \neq 0)$, let $\widetilde{G}' \subset \widetilde{G}_{i,j}$ be its stabilizer, let W''_0 be the closure of the image of W' in W'', an irreducible component of W'', and let $H' \subset H_{i,j}$ be the stabilizer of W''_0. Since Φ_j is the quotient morphism onto $(x_j \neq 0)$ by the subgroup $\{(1, \lambda) \mid \lambda^{p_j} = 1\}$ of $\widetilde{G}_{i,j}$, which is the kernel of the surjective projection $\widetilde{G}_{i,j} \to G$ (proof of Lemma E.28(1)), and $W^\sharp \cap (x_j \neq 0)$ is irreducible, $\{(1, \lambda)\}$ permutes the irreducible components of the preimage of $W^\sharp$ transitively. Hence $\widetilde{G}' \to G$ is surjective, and the rational map $W'/\widetilde{G}' \dashrightarrow W^\sharp/G$ induced by Φ_j is birational. Similarly, $(\rho, x') \to (\rho^r, x')$ is the quotient morphism by $R_{i,j}$ over $(q \neq 0)$, so $R_{i,j}$ permutes the irreducible components of $(g'(\rho^r, x') = 0)$ over W''_0 transitively. Hence H' is the image of $\widetilde{G}'$ in $H_{i,j}$, and the rational map $W'/\widetilde{G}' \dashrightarrow W''_0/H'$ is birational. An element of $\widetilde{G}'$ acting trivially on W' acts trivially on $W^\sharp$, so its image in G is trivial, since G acts diagonally and $W^\sharp$ is not contained in any coordinate hyperplane. Then it is $(1, \xi)$ with $\xi^{p_j} = 1$, which acts on ρ by ξ^{-1}, and $\xi = 1$ since ρ does not

vanish on W'. Thus $\widetilde{G}'$ acts faithfully on W'. Moreover q and the x'_l do not vanish identically on W', so W''_0 is not contained in any coordinate hyperplane, and H', which acts diagonally, acts faithfully on W''_0.

Let F' be a prime divisor over W' lying over $F^\sharp$, with ramification index e_1, and F'' the prime divisor over W'' under F', with ramification index e' of F' over F''. Then $\operatorname{mult}_{F'} \rho = \operatorname{mult}_{F'} x_j/p_j > 0$, $\operatorname{mult}_{F'} x'_l = \operatorname{mult}_{F'} x_l - p_l \operatorname{mult}_{F'} \rho \geq 0$ for $l \in \mathcal{P}$ by the choice of j, and $\operatorname{mult}_{F'} x'_l > 0$ for $l \notin \mathcal{P}$. Thus F'' has a center Z'' on W'' contained in $S_0 \cap (x'_l = 0 \text{ for } l \notin \mathcal{P})$, and the image of Z'' in Y_i^+ is contained in C_i^+ by Lemma E.28(5).

By **Step 1**(a), S_0 is normal with at worst Du Val singularities near Z''. Hence W'' is normal near Z'', being a hypersurface with a normal Cartier divisor S_0, and (W'', S_0) is plt near Z'' by [KM98, Theorem 5.50(1)], since S_0 is klt near Z''. For an exceptional prime divisor E over W'' whose center intersects Z'', $a(E, W'', 0) \geq a(E, W'', S_0) > 0$, so W'' is klt near Z''. Thus W'' is normal, integral, and klt near Z''. In particular $g'(q, x')$ is reduced along the image of W', which is the component of W'' through Z''. Since $(\rho, x') \to (\rho^r, x')$ is étale over $(q \neq 0)$ and Φ_j is étale, $g'(\rho^r, x')$ is reduced along W', and g is reduced along $W^\sharp$. This proves (1).

We prove (2) by comparing the scores. Let $\Omega_{W''} := \operatorname{Res}(dq \wedge dx'/g')$, a generator of $\omega_{W''}$ near Z'', where dx' is the wedge product of the dx'_l. A direct computation of $\Phi_j^*(dx_1 \wedge \cdots \wedge dx_4)$ and of $dq = r\rho^{r-1} d\rho$ shows that, up to a non-zero constant,

$$\Phi_j^* \Omega_g = \rho^{\sum_l p_l - 2(u+v) - r} \Omega_{W''}|_{W'} = \rho^{u+w-r} \Omega_{W''}|_{W'}. \tag{E.134}$$

By the Hurwitz formula,

$$1 + \operatorname{mult}_{F'} \Omega_g = e_1(1 + \operatorname{mult}_{F^\sharp} \Omega_g), \qquad 1 + \operatorname{mult}_{F'} \Omega_{W''} = e'(1 + \operatorname{mult}_{F''} \Omega_{W''}). \tag{E.135}$$

Moreover

$$r \operatorname{mult}_{F'} \rho = \operatorname{mult}_{F'} q = e' \operatorname{mult}_{F''} q, \qquad \operatorname{mult}_{F'} h_k^\sharp = \kappa_k \operatorname{mult}_{F'} \rho + e' \operatorname{mult}_{F''} h'_k. \tag{E.136}$$

Combining (E.134), (E.135), and (E.136) with $u + w - \sum_k c'_k \kappa_k = -\delta(c')$, we obtain

$$e_1 \Big(1 + \operatorname{mult}_{F^\sharp} \Omega_g - \sum_k c'_k \operatorname{mult}_{F^\sharp} h_k^\sharp\Big) = e' \Big(a\big(F'', W'', S_0 + \sum_k c'_k \operatorname{div}(h'_k)\big) - \frac{\delta(c')}{r} \operatorname{mult}_{F''} q\Big).$$

The rational maps $W'/\widetilde{G}' \dashrightarrow W^\sharp/G$ and $W'/\widetilde{G}' \dashrightarrow W''_0/H'$ are birational, and $\widetilde{G}'$, G, and H' act faithfully on W', $W^\sharp$, and W''_0, so by [Liu26, Lemma 2.6] the ramification index of F' over $W^\sharp/G$ is

$$|I_{\widetilde{G}'}(F')| = e_1 |I_G(F^\sharp)| = e' |I_{H'}(F'')| = e' |I_{H_{i,j}}(F'')|,$$

where the last equality holds since an element of $H_{i,j}$ fixing F'' preserves W''_0. Hence the left-hand side of (2) is

$$\frac{a(F'', W'', S_0 + \sum_k c'_k \operatorname{div}(h'_k))}{|I_{H_{i,j}}(F'')|} - \delta(c') \frac{\operatorname{mult}_{F''} q}{r |I_{H_{i,j}}(F'')|},$$

and

$$\frac{\operatorname{mult}_{F''} q}{r |I_{H_{i,j}}(F'')|} = \frac{\operatorname{mult}_{F'} \rho}{|I_{\widetilde{G}'}(F')|} = \frac{[x_j]}{p_j} = \varrho.$$

It remains to bound the first term by $\beta_i^+(c')$. Let o'' be the closed point Z'', or a general closed point of the curve Z'', and let $H_{o''} \subset H_{i,j}$ be the subgroup fixing Z'' pointwise. For o'' general, $H_{o''}$ is the stabilizer of o''. If $x'_l(o'') \neq 0$, then $H_{o''}$ acts trivially on x'_l (an element of $H_{o''}$ fixes o'' and multiplies the semi-invariant x'_l by its character, so the character is trivial on $H_{o''}$), so $H_{o''}$ acts diagonally in the coordinates q and $x'_l - x'_l(o'')$, in which o'' is the origin. The hypotheses of Lemma 5.5 hold with $n = 4$, the group $H_{o''}$ in place of G and of H, the hypersurface W'' in place of X (a normal integral complete intersection with klt singularities near o'' on which $H_{o''}$ acts

faithfully), the section $S_0 = W'' \cap (q = 0)$ (a normal integral complete intersection of dimension two with klt singularities near o'', on which $H_{o''}$ acts faithfully) in place of $X \cap (u = 0)$, the $\mathbb{R}$-ideal $\mathfrak{a} := \prod_k (h'_k)^{c'_k}$, and $b = 1$. The pair $(S_0, \mathfrak{a}|_{S_0})$ is lc near o'', since the crepant quotient of $(S_0, \mathfrak{a}|_{S_0})$ by $H_{i,j}$ is $(Y_i^+, B_i^+(c'))$ near the image of o'' in C_i^+ [Liu26, Lemma 2.6]. Near o'' the quotient by $H_{i,j}$ is étale locally the quotient by $H_{o''}$, so the minimal log discrepancy of the lemma at o'', or at the generic point of Z'', is a minimal log discrepancy of $(Y_i^+, B_i^+(c'))$ at a point or at the generic point of a curve contained in C_i^+, and it is at least $\beta_i^+(c')$. Since the inertia group of F'' acts trivially on the residue field of F'', it fixes Z'' pointwise, so $I_{H_{o''}}(F'') = I_{H_{i,j}}(F'')$, and Lemma 5.5 implies

$$\frac{a(F'', W'', S_0 + \sum_k c'_k \operatorname{div}(h'_k))}{|I_{H_{i,j}}(F'')|} \geq \beta_i^+(c').$$

This proves (2). □

Step 4. In this step we fix θ as in (2), and we show that the margin θ passes from the reference to $(W_i, \theta D_i)$. We then bound the orders of the coordinates and of the marks at the prime divisors of small score.

Let $\pm$ be the sign of the hypothesis of (2), so that $\pm\delta_\theta \leq 0$ and $(Y_i^\pm, B_i^\pm(\theta c))$ is lc near $C_i^\pm$ for any i. We use the sign $\pm$ in Claim E.36.

Claim E.36. *For i large, the pair $(W_i, \theta D_i)$ is lc near o.*

Proof. Let $B_i^0 := \sum_k c_k \operatorname{div}(h^0_{i,k}|_{V_i})$, and let $F^\sharp$ be a prime divisor over V_i with center 0. Claim E.35(2), applied with the sign fixed above, $c' = \theta c$, $g = f_i^0$, $h_k^\sharp = h^0_{i,k}$, and $W^\sharp = V_i$, together with (E.132) for the generator Ω_{V_i} of ω_{V_i}, implies

$$\frac{a(F^\sharp, V_i, \theta B_i^0)}{|I_{G_i}(F^\sharp)|} \geq \beta_i^\pm(\theta c) \mp \delta_\theta \varrho^\pm \geq 0,$$

since $\beta_i^\pm(\theta c) \geq 0$ by the lc assumption, $\mp\delta_\theta \geq 0$, and $\varrho^\pm \geq 0$. If $(V_i, \theta B_i^0)$ were not lc near 0, then $\operatorname{mld}(V_i \ni 0, \theta B_i^0) = -\infty$ [Amb99, Section 1, the second paragraph after Definition 1.1], which is impossible. Thus $(V_i, \theta B_i^0)$ is lc near 0, and $(\mathbb{A}^4, V_i + \theta \sum_k c_k \operatorname{div}(h^0_{i,k}))$ is lc near 0 by [Kaw07, Theorem], since V_i is normal.

Passing to a subsequence, the set $\mathcal{K}$ of the indices k with $h_{i,k}(0) = 0$ does not depend on i. If $k \notin \mathcal{K}$, then $\kappa_k = 0$, the constant term of $h_{i,k}$ is a term of $h^0_{i,k}$, and both $h_{i,k}$ and $h^0_{i,k}$ are units near 0. Since $h_{i,k}$ and $h^0_{i,k}$ are units near 0 for $k \notin \mathcal{K}$, and units do not change lc near 0, it suffices to treat the indices $k \in \mathcal{K}$, and Lemma 4.14 is used below only for $k \in \mathcal{K}$. Let L_0 be the integer of Lemma 4.14 for $n = 4$, $|\mathcal{K}| + 1$ functions, and the finite set $\Lambda := \{1\} \cup \{\theta c_k \mid k \in \mathcal{K}\}$. By (E.130), for i large all the monomials of $f_i - f_i^0$ and of $h_{i,k} - h^0_{i,k}$ have degree at least L_0. Since lc is preserved by completion [dFEM11, Proposition 2.11], Lemma 4.14 applied to $(f_i^0, h^0_{i,k})_{k\in\mathcal{K}}$ and $(f_i, h_{i,k})_{k\in\mathcal{K}}$ implies that $(\mathbb{A}^4, W_i + \theta \sum_k c_k \operatorname{div}(h_{i,k}))$ is lc near 0. Since W_i is normal, $(W_i, \theta D_i)$ is lc near o by [Kaw07, Theorem]. □

Claim E.37. *The pair $(W_i, \frac{1}{2}\operatorname{div}(z) + \operatorname{div}(t))$ is lc near o. In particular, for any prime divisor F over W_i with center o,*

$$a(F, W_i, 0) \geq \tfrac{1}{2} \operatorname{mult}_F z + \operatorname{mult}_F t.$$

Proof. By Lemma E.23, $f_i = x^2 + zy^2 + y\sigma_i(t) + S_i(0, z, t)$, where σ_i has order at least 3 and $S_i(0, z, t)$ has order at least 4. Thus $f_i|_{t=0} = x^2 + zy^2 + S_i(0, z, 0)$, and either $S_i(0, z, 0) = 0$, or $S_i(0, z, 0) = z^N \upsilon(z)$ with $N \geq 4$ and $\upsilon(0) \neq 0$. By [Kaw07, Theorem], applied twice, first along the normal hypersurface W_i and then along the hyperplane $(t = 0)$, it suffices to prove that the pair

$$\left(\mathbb{A}^3, \operatorname{div}(g_0) + \frac{1}{2}\operatorname{div}(z)\right)$$

is lc near 0, where $g_0 := x^2 + zy^2 + S_i(0, z, 0)$ in the coordinates x, y, z. Since lc is preserved by completion [dFEM11, Proposition 2.11], we may replace z and y by $zv^{1/N}$ and $yv^{-1/(2N)}$, which preserves the divisor $(z = 0)$, and assume that $g_0 = x^2 + zy^2 + z^N$ or $g_0 = x^2 + zy^2$.

Any face polynomial of the Newton polyhedron of g_0 which contains x^2 has the non-zero derivative $2x$ on the torus, and the remaining face polynomials zy^2, z^N, and zy^2+z^N have the non-zero derivatives $2yz$, Nz^{N-1}, and $2yz$ there. Thus g_0 is non-degenerate, and the non-degeneracy holds for every face of the Newton polyhedron, compact or not, as [IP01, Definition 2.6] requires. Hence a toric log resolution of $(\mathbb{A}^3, (g_0 = 0) + (z = 0))$ near 0 exists [IP01, Proposition 2.7]. Explicitly, let π be the toric modification of [Var76, §10] for the support K of g_0, whose elements have degree at least 2 (the condition of [Var76, §9]). By [Var76, Lemma 10.3(2) and (3)], near $\pi^{-1}(0)$ the strict transform of $(g_0 = 0)$ is smooth and intersects every torus orbit contained in $\pi^{-1}(0)$ transversally, so near $\pi^{-1}(0)$ it forms a simple normal crossing divisor with the toric boundary, which contains the strict transform of $(z = 0)$. On the resolution the strict transform of $(g_0 = 0)$ has coefficient one, the strict transform of $(z = 0)$ has coefficient $\frac{1}{2}$, and the exceptional divisor attached to a primitive vector $(X_0, Y_0, Z_0) \in \mathbb{Z}^3_{\geq 0}$ has log discrepancy

$$\begin{aligned} &X_0 + Y_0 + Z_0 - \min(2X_0, 2Y_0 + Z_0, NZ_0) - \frac{Z_0}{2} \\ &\geq X_0 + Y_0 + \frac{Z_0}{2} - \frac{2X_0 + (2Y_0 + Z_0)}{2} = 0, \end{aligned}$$

where the term NZ_0 is omitted if $g_0 = x^2 + zy^2$. Hence the pair is lc. The second statement follows from (E.132). □

Proof of Lemma E.32 continued. By Lemma E.29(1), the numbers β_i belong to a finite set. Let C_1 be the maximum of the numbers β_i, and

$$C_2 := \frac{\theta C_1}{\theta - 1}, \qquad C_3 := \frac{C_2}{\theta \min_k c_k}.$$

For a prime divisor F over W_i with center o, write $e_F := |I_{G_i}(F)|$, $[\varphi] := \operatorname{mult}_F \varphi / e_F$ for a rational function φ, and let $a_D(F)$ be the score of F of Set-up 11.1 for the i-th data. Then α_i is the infimum of $a_D(F)$ over such F, and it suffices to prove that $a_D(F) \geq \beta_i$ for i large and any F with $a_D(F) \leq C_1$.

Fix such an F, and let $A := a(F, W_i, 0)$ and $\Sigma := \sum_k c_k \operatorname{mult}_F h_{i,k}$. For i large, Claim E.36 implies $A - \theta\Sigma \geq 0$, and $A - \Sigma \leq C_1 e_F$. Hence $(1 - \theta^{-1})A \leq C_1 e_F$, and, together with Claim E.37,

$$\frac{A}{e_F} \leq C_2, \qquad [h_{i,k}] \leq C_3, \qquad [z] \leq 2C_2, \qquad [t] \leq C_2. \tag{E.137}$$

For $j \in \mathcal{P}$ and $l \in \mathcal{N}$, the restriction to W_i of the invariant function $g_{jl}^{n_{jl}}$ of (E.131) descends to a regular function on W_i/G_i vanishing at the image of o, so $n_{jl}(|p_l|[x_j] + p_j[x_l]) \geq 1$ by [Liu26, Lemma 2.6]. Since $n_{jl} \leq e_0$,

$$|p_l|[x_j] + p_j[x_l] \geq \frac{1}{e_0} \qquad \text{for any } j \in \mathcal{P} \text{ and } l \in \mathcal{N}. \tag{E.138}$$

Fix a real number ϵ with $0 < \epsilon < \min(\frac{1}{2Q}, \frac{1}{4e_0M})$, where $Q = e_0M^2$ is the integer of **Step 2**, and let $c_0 := \frac{3}{4e_0M}$, so that $c_0 > \epsilon$. If $[x_l] < \epsilon$ for some $l \in \mathcal{N}$, then (E.138) implies, for any $j \in \mathcal{P}$,

$$[x_j] \geq \frac{1/e_0 - p_j\epsilon}{|p_l|} \geq \frac{1}{e_0M} - \epsilon > c_0 > \epsilon, \tag{E.139}$$

and symmetrically with $\mathcal{P}$ and $\mathcal{N}$ exchanged. We say that F belongs to $\mathcal{R}_+$ if $[x_l] \geq \epsilon$ for any $l \in \mathcal{N}$, and to $\mathcal{R}_-$ if $[x_j] \geq \epsilon$ for any $j \in \mathcal{P}$. By (E.139), F belongs to $\mathcal{R}_+$ or to $\mathcal{R}_-$.

Finally, let $\Xi_i := \epsilon r_i/(2M)$, so that $\Xi_i \to +\infty$. By (E.130), for any fixed $\kappa \in \mathbb{Z}$ and $C \geq 0$, if $F \in \mathcal{R}_+$, then any monomial of $\mathbf{T}$-weight $\kappa + Lr_i$ with $L \leq -1$ satisfies $[\cdot] \geq \epsilon(r_i - |\kappa|)/M \geq \Xi_i + C$ for i large, uniformly in F. Symmetrically, if $F \in \mathcal{R}_-$, then any monomial of $\mathbf{T}$-weight $\kappa + Lr_i$

with $L \geq 1$ satisfies $[\cdot] \geq \Xi_i + C$ for i large, uniformly in F. Below, a lower bound of the form $\epsilon(r_i - C)/M - C'$, with C and C' independent of i and F, is at least Ξ_i for i large, and we use the inequality $\epsilon(r_i - C)/M - C' \geq \Xi_i$ without further mention.

Step 5. In this step we replace, at a fixed prime divisor F with $a_D(F) \leq C_1$, the equation f_i by a polynomial whose blocks have shifts of one sign, without changing the score of F and the orders of the marks.

We use the following form of Newton's lemma. Let $\hat{\mathfrak{K}}$ be a field complete with respect to a discrete valuation ν, $\mathcal{F}$ a polynomial with coefficients in the valuation ring, and a an element of the valuation ring with $\nu(\mathcal{F}(a)) > 2\nu(\mathcal{F}'(a))$. Then $\mathcal{F}$ has a unique root ζ with $\nu(\zeta - a) > \nu(\mathcal{F}'(a))$ in any complete extension of $\hat{\mathfrak{K}}$, $\zeta \in \hat{\mathfrak{K}}$, and $\nu(\zeta - a) = \nu(\mathcal{F}(a)) - \nu(\mathcal{F}'(a))$. By Hensel's lemma [Eis95, Theorem 7.3], applied to the valuation ring of $\hat{\mathfrak{K}}$, there is a root $\zeta \in \hat{\mathfrak{K}}$ of $\mathcal{F}$ with $\nu(\zeta - a) > \nu(\mathcal{F}'(a))$. Since $\mathcal{F}'(\zeta) - \mathcal{F}'(a)$ is divisible by $\zeta - a$, $\nu(\mathcal{F}'(\zeta)) = \nu(\mathcal{F}'(a))$. If ζ' is a root with $\nu(\zeta' - a) > \nu(\mathcal{F}'(a))$ in a complete extension, then

$$0 = \mathcal{F}(\zeta') - \mathcal{F}(\zeta) = (\zeta' - \zeta)\big(\mathcal{F}'(\zeta) + (\zeta' - \zeta)c\big)$$

for an element c of the valuation ring, and the second factor has value $\nu(\mathcal{F}'(a)) < +\infty$, so $\zeta' = \zeta$. Finally $\mathcal{F}(a) = (a - \zeta)(\mathcal{F}'(\zeta) + (a - \zeta)c')$ for an element c' of the valuation ring, so $\nu(\zeta - a) = \nu(\mathcal{F}(a)) - \nu(\mathcal{F}'(a))$.

Let b be the exponent of t in f_* if f_* is in row I and ℓ the exponent of t in f_* if f_* is in row II, the other one being 0, and let

$$C_4 := 1 + \max(C_3, bC_2, \ell C_2), \qquad C_5 := 2C_2 + 2C_4.$$

The constants $C_2, \dots, C_5$ depend neither on i nor on F.

Claim E.38. *Let F be a prime divisor over W_i with center o and $a_D(F) \leq C_1$. Let $\mathfrak{L} \supset \mathbb{C}(W_i)$ be a finite field extension with an action of G_i extending the action on $\mathbb{C}(W_i)$, $\nu_{\mathfrak{L}}$ a valuation of $\mathfrak{L}$ extending* mult_F*, and $X_1, \dots, X_4 \in \mathfrak{L}$, where X_l is G_i-semi-invariant with the character of x_l. Let $\mathfrak{K}^\sharp := \mathbb{C}(X_1, \dots, X_4)$, and g a G_i-semi-invariant polynomial with the character of f_i and $g(X_1, \dots, X_4) = 0$. Assume the following.*

- *(i) The field $\mathfrak{L}$ is contained in the closures of $\mathbb{C}(W_i)$ and of $\mathfrak{K}^\sharp$ in the completion of $\mathfrak{L}$ with respect to $\nu_{\mathfrak{L}}$, and an element of G_i preserves $\nu_{\mathfrak{L}}$ if and only if it preserves* mult_F*, if and only if it preserves $\nu_{\mathfrak{L}}|_{\mathfrak{K}^\sharp}$.*
- *(ii) $\nu_{\mathfrak{L}}(X_l - x_l) \geq e_F(\Xi_i - C_5)$ for $l = 1, \dots, 4$.*
- *(iii) $\Omega_{W_i} = v\Omega_g$ as rational 3-forms of $\mathfrak{L}$, where $\nu_{\mathfrak{L}}(v) = 0$ and Ω_g is the residue form of $\operatorname{Res}(dX_1 \wedge \dots \wedge dX_4/g)$ on the component $W^\sharp := \operatorname{Spec} \mathbb{C}[X_1, \dots, X_4]$ of $(g = 0)$.*

Then, for any i with $\Xi_i - C_5 > \max(C_3, \epsilon)$ (a condition which depends neither on F nor on $\mathfrak{L}$ and X), $F^\sharp := \nu_{\mathfrak{L}}|_{\mathfrak{K}^\sharp}$ is a prime divisor over $W^\sharp$ with center 0, $I_{G_i}(F^\sharp) = I_{G_i}(F)$, $\operatorname{mult}_{F^\sharp} \Omega_g = \operatorname{mult}_F \Omega_{W_i}$, $\operatorname{mult}_{F^\sharp} h_{i,k}(X) = \operatorname{mult}_F h_{i,k}$ for any k, and, with $[X_l] := \operatorname{mult}_{F^\sharp} X_l/e_F$, $[X_l] \geq \min([x_l], \Xi_i - C_5)$, with equality if $[x_l] < \Xi_i - C_5$.

Proof. By (i), $\mathbb{C}(W_i)$ and $\mathfrak{K}^\sharp$ are dense in the completion of $\mathfrak{L}$ with respect to $\nu_{\mathfrak{L}}$. Hence the three valuations mult_F, $\nu_{\mathfrak{L}}$, and $F^\sharp$ have the value group and the residue field of that completion. In particular $\nu_{\mathfrak{L}}$ has ramification index one over mult_F and over $F^\sharp$, and $F^\sharp$ is a divisorial valuation of $\mathfrak{K}^\sharp$, since its value group is $\mathbb{Z}$ up to scaling and its residue field has transcendence degree two. Hence $\operatorname{mult}_{F^\sharp} \Omega_g = \operatorname{mult}_F \Omega_{W_i}$ by (iii) and the Hurwitz formula, applied to $\nu_{\mathfrak{L}}$ over $F^\sharp$ and over mult_F. By (i), the stabilizer of mult_F in G_i equals the stabilizer of $F^\sharp$, and it acts on their common residue field, so $I_{G_i}(F^\sharp) = I_{G_i}(F)$. By (ii), $[X_l] \geq \min([x_l], \Xi_i - C_5) > 0$, with equality if $[x_l] < \Xi_i - C_5$, so the center of $F^\sharp$ on $W^\sharp$ is 0. Finally $h_{i,k}(X) - h_{i,k}(x)$ belongs to the ideal generated by the $X_l - x_l$ in the ring $\mathbb{C}[x_1, \dots, x_4, X_1, \dots, X_4]$, on which $\nu_{\mathfrak{L}}$ is non-negative. Thus, since $\Xi_i - C_5 > C_3$, (E.137) implies

$$\nu_{\mathfrak{L}}\big(h_{i,k}(X) - h_{i,k}(x)\big) \geq e_F(\Xi_i - C_5) > e_F C_3 \geq \operatorname{mult}_F h_{i,k},$$

and $\operatorname{mult}_{F^\sharp} h_{i,k}(X) = \operatorname{mult}_F h_{i,k}$. □

Proof of Lemma E.32 continued. In each case below with $X \neq (x,y,z,t)$, $\mathfrak{L}$ is generated over $\mathbb{C}(W_i)$, and also over $\mathfrak{K}^\sharp$, by one semi-invariant element ϑ. The element ϑ is the root, given by Newton's lemma, of a semi-invariant polynomial near a semi-invariant approximant in the smaller field (1 for $\sqrt{U}$, $\sqrt{U'}$, and $\sqrt{U''}$, $\tilde{z}$ for z, and y for Y_0), and we let G_i act on ϑ through the character of the approximant.

If $\vartheta \in \mathbb{C}(W_i)$, its given action has the character of the approximant. For $\vartheta = z$ this holds as $\tilde{z}$ has the character of z. For $\vartheta = \sqrt{U}$ and $\sqrt{U'}$, resp. $\vartheta = \sqrt{U''}$, let $\mathfrak{L}_0 := \mathbb{C}(z,t)$, resp. $\mathfrak{L}_0 := \mathbb{C}(y,z,t)$, which contains the G_i-invariant element ϑ^2. Then $\vartheta \in \mathfrak{L}_0$. In case (A), the generic fiber of $W_i \to \mathbb{A}^2_{z,t}$ is a smooth conic by the forms of f_i below, so $\mathbb{C}(z,t)$ is algebraically closed in $\mathbb{C}(W_i)$. For $\sqrt{U''}$, $\mathbb{C}(W_i) = \mathfrak{L}_0(x)$ with $x^2 = -(\mathcal{G} + \tilde{e}_i)$, and $\vartheta \in \mathfrak{L}_0 x$ would make $-\mathcal{G} = -(\mathcal{G}+\tilde{e}_i)U''$ a square in $\mathfrak{L}_0$, which it is not, as $\mathcal{G}$ has degree two in y with leading coefficient z. Let ν_0 be the monomial valuation on $\mathfrak{L}_0$ with the weights 1 of z and N of t for $\sqrt{U}$, where $N > a+1$, with the weights N of z and 1 of t for $\sqrt{U'}$, where $N > \max(b, 2\ell)$, and with the $\mathbf{T}$-weights of y,z,t as weights for $\sqrt{U''}$. It is G_i-invariant, and $\nu_0(\vartheta^2 - 1) > 0$ by the descriptions of the terms of $\phi_i - z^{a+1}$, $\psi_i - t^b$, $\psi_i - t^{2\ell}$, ς_i, and $\tilde{e}_i$ below. Hence any $\gamma \in G_i$ fixes the one of $\pm\vartheta$ which is close to 1 for ν_0, and $\gamma\vartheta = \vartheta$.

For $\vartheta = Y_0$, since $t^{2\ell} - 4zx^2$ is irreducible and $\mathbb{C}(W_i) = \mathbb{C}(x,z,t)(y)$, $Y_0 \in \mathbb{C}(W_i)$ implies that the discriminants $t^{2\ell} - 4zx^2$ and $(t^\ell + e_i')^2 - 4z(x^2 + e_i)$ of $\mathcal{F}$ and of $\mathcal{F} + \tilde{e}_i$ in ζ differ by a square factor in $\mathbb{C}(x,z,t)$. They are then equal, as both have the coefficient $-4z$ of x^2, so $2t^\ell e_i' + e_i'^2 = 4ze_i$, and both sides vanish, as the left side does not depend on z. Thus $e_i = 0$, and $e_i' = 0$ since the terms of $e_i'(t)y$ have positive shift, so $\tilde{e}_i = 0$ and $Y_0 = y$.

Then (i) holds: ϑ belongs to the completion of the smaller field, and an element $\gamma \in G_i$ which preserves the valuation of the smaller field maps ϑ to the unique root close to γ applied to the approximant, which is $\gamma\vartheta$, so γ preserves $\nu_{\mathfrak{L}}$. We write only ϑ, the equations, (ii), and (iii). In the cases with $X = (x,y,z,t)$, we have $\mathfrak{K}^\sharp = \mathfrak{L}$ and $\nu_{\mathfrak{L}} = \mathrm{mult}_F$, there is no ϑ, and (i) holds since the three valuations in it are equal.

Case (A), row I, $F \in \mathcal{R}_+$. Here t is the only coordinate among z,t of negative weight, so by **Step 1**(c) and (E.130) any term $z^m t^n$ of $\phi_i - z^{a+1}$ has $n \geq (r_i - 2u(a+1))/M$ and $m \leq a$. Since $[t] \geq \epsilon$ and $[z] \leq 2C_2$,

$$[z^{m-a-1}t^n] \geq \frac{\epsilon(r_i - 2u(a+1))}{M} - 2(a+1)C_2,$$

so $[\phi_i/z^{a+1} - 1] \geq \Xi_i$. By Lemma E.31(2), any monomial $z^m t^n$ of ς_i has $n \geq (r_i - 2u(a+1))/M - b$, so $[\varsigma_i] \geq \Xi_i$. Let $U := \phi_i/(z^{a+1}(1+\varsigma_i))$, a rational function of z and t with $[U-1] \geq \Xi_i$. Similarly any term $t^n y$ of $\sigma_i(t)y$ has $n+1 \geq (r_i - 2|u+v|)/M$, so $[\sigma_i/z] \geq \Xi_i$. The element U is G_i-invariant. Let $\vartheta := \sqrt{U}$ be the square root of U near 1 given by the Newton lemma, and

$$X := \Big(\frac{x}{\sqrt{U}},\ \frac{y + \sigma_i/(2z)}{\sqrt{U}},\ z,\ t\Big), \qquad g := x^2 + zy^2 + z^a(1+\varsigma_i)\psi_i(z,t).$$

Then

$$f_i = x^2 + z\Big(y + \frac{\sigma_i}{2z}\Big)^2 + \frac{\phi_i\psi_i}{z} = Ug(X), \qquad \mathfrak{L} = \mathbb{C}(W_i)(\sqrt{U}) = \mathfrak{K}^\sharp(\sqrt{U}),$$

and X_2 has the character of y, since the characters of $\sigma_i y$ and zy^2 are equal. All the terms of $g - f_i^0 = z^a(1+\varsigma_i)(\psi_i - t^b)$ have positive shift. For (ii), $X_1 - x = (U^{-1/2} - 1)x$ and $X_2 - y = (U^{-1/2} - 1)y + U^{-1/2}\sigma_i/(2z)$, so $[X_l - x_l] \geq \Xi_i \geq \Xi_i - C_5$. For (iii), with $\Omega_{W_i} = dy \wedge dz \wedge dt/(\partial f_i/\partial x)$, we have $\partial f_i/\partial x = \sqrt{U}(\partial g/\partial x)(X)$ and $dy \wedge dz \wedge dt = \sqrt{U} dX_2 \wedge dz \wedge dt$, so $\Omega_{W_i} = \Omega_g$.

Case (A), row I, $F \in \mathcal{R}_-$. Any term $z^m t^n$ of $\psi_i - t^b$ has $n < b$ and $m \geq (r_i - |w|b)/M$, since z is the only coordinate among z,t of positive weight. Since $[z] \geq \epsilon$ and $[t] \leq C_2$, $[z^m t^{n-b}] \geq \epsilon(r_i - |w|b)/M - bC_2$, so $[\psi_i/t^b - 1] \geq \Xi_i$. By Lemma E.31(2), any monomial $z^m t^n$ of ς_i has $m \geq (r_i - |w|b)/M - a - 1$, so $[\varsigma_i] \geq \Xi_i$. Let $U' := \psi_i/(t^b(1+\varsigma_i))$, so that $[U'-1] \geq \Xi_i$,

let $\vartheta := \sqrt{U'}$ be the root near 1, and

$$X := \Big(\frac{x}{\sqrt{U'}},\ \frac{y+\sigma_i/(2z)}{\sqrt{U'}} - \frac{\sigma_i}{2z},\ z,\ t\Big),$$

$$g := x^2 + zy^2 + \sigma_i(t)y + \frac{\phi_i t^b(1+\varsigma_i) + \sigma_i^2/4}{z}.$$

By the identity $\phi_i(0,t)t^b = -\sigma_i^2/4$ of Lemma E.31(2), and since ς_i is divisible by z for i large (any monomial $z^m t^n$ of ς_i has $m \geq (r_i - |w|b)/(2u) - a - 1 \geq 1$ by Lemma E.31(2), as $r_i \to +\infty$), g is a polynomial, and $f_i = U'g(X)$. Moreover

$$g - f_i^0 = \sigma_i(t)y + (1+\varsigma_i)\sum_{1\leq m\leq a}\phi_{i,m}(t)z^{m-1}t^b + \frac{\varsigma_i}{z}\phi_{i,0}(t)t^b,$$

and all the terms of $g - f_i^0$ have negative shift. For (ii), $X_1 - x = (U'^{-1/2} - 1)x$ and $X_2 - y = (U'^{-1/2} - 1)(y + \sigma_i/(2z))$, where $[y + \sigma_i/(2z)] \geq -[z] \geq -2C_2$, so $[X_l - x_l] \geq \Xi_i - 2C_2 \geq \Xi_i - C_5$. For (iii), $\partial f_i/\partial x = \sqrt{U'}(\partial g/\partial x)(X)$ and $dy \wedge dz \wedge dt = \sqrt{U'}dX_2 \wedge dz \wedge dt$, so $\Omega_{W_i} = \Omega_g$.

Case (A), row II. If $F \in \mathcal{R}_+$, let $\mathfrak{L} := \mathbb{C}(W_i)$, $X := (x,y,z,t)$, and $g := f_i$, whose blocks other than f_i^0 have positive shift by **Step 1**(c). If $F \in \mathcal{R}_-$, let $U' := \psi_i/t^{2\ell}$, so that $[U' - 1] \geq \Xi_i$ as in row I, and define ϑ and X as in row I with $\sigma_i = t^\ell$. Since $f_i = x^2 + z(y + \frac{t^\ell}{2z})^2 - \frac{\psi_i}{4z}$, we have $f_i = U'g(X)$ with $g := x^2 + zy^2 + t^\ell y = f_i^0$, and (ii) and (iii) follow as in row I.

Cases (B) and (C), $F \in \mathcal{R}_+$. Let $\mathfrak{L} := \mathbb{C}(W_i)$, $X := (x,y,z,t)$, and $g := f_i$. By **Step 1**(d), the blocks of g other than f_i^0 have positive shift.

Cases (B) and (C), $F \in \mathcal{R}_-$. Write $f_i = f_i^0 + \tilde{e}_i$ with $\tilde{e}_i := e_i(z,t) + e_i'(t)y$ as in **Step 1**(d). All the terms of e_i, $e_i'y$, and $z\partial e_i/\partial z$ have positive shift, and all the coordinates of positive weight have $[\cdot] \geq \epsilon$, so $[\tilde{e}_i] \geq \Xi_i$ and $[\partial e_i/\partial z] \geq \Xi_i - 2C_2$. We distinguish three subcases.

If $[y] \leq C_4$, let $\mathfrak{L} := \mathbb{C}(W_i)$, $\tilde{z} := z + \tilde{e}_i/y^2$, $X := (x,y,\tilde{z},t)$, and $g := f_i^0$. As $\Xi_i > C_5 = 2C_2 + 2C_4$ for i large, $[\tilde{e}_i/y^2] \geq \Xi_i - 2C_4 > 2C_2 \geq [z]$, so $[\tilde{z}] = [z]$, and the bounds for e_i and $\partial e_i/\partial z$ above also hold at $(\tilde{z},t)$. Since $f_i^0 = x^2 + zy^2 + \sigma^0(t)y + \tau^0(t)$, $f_i = g(X)$. Here $\vartheta = z$ is the root near $\tilde{z}$ of

$$\mathcal{F}(\zeta) := \zeta y^2 + e_i(\zeta,t) + e_i'(t)y - \tilde{z}y^2,$$

since

$$\frac{\nu_{\mathfrak{L}}(\mathcal{F}(\tilde{z}))}{e_F} = [e_i(\tilde{z},t) + e_i'y] \geq \Xi_i, \qquad \frac{\nu_{\mathfrak{L}}(\mathcal{F}'(\tilde{z}))}{e_F} = 2[y] \leq 2C_4.$$

For (ii), $[\tilde{z} - z] \geq \Xi_i - 2C_4 \geq \Xi_i - C_5$. For (iii),

$$\Omega_g = \frac{dy \wedge d\tilde{z} \wedge dt}{2x} = \Big(1 + y^{-2}\frac{\partial e_i}{\partial z}\Big)\Omega_{W_i},$$

and $[y^{-2}\partial e_i/\partial z] > 0$.

If $[y] > C_4$ in row I, then $f_i^0 = x^2 + zy^2 + \sigma^0(t)y + t^b$ and

$$[zy^2] > 2C_4, \qquad [\sigma^0(t)y] > C_4, \qquad [t^b] \leq bC_2 < C_4,$$

so $\mathcal{G} := zy^2 + \sigma^0(t)y + t^b$ has $[\mathcal{G}] = b[t] \leq bC_2$. Let $U'' := \mathcal{G}/(\mathcal{G} + \tilde{e}_i)$, a rational function of y, z, and t which is G_i-invariant with $[U'' - 1] \geq \Xi_i - bC_2$, $\vartheta := \sqrt{U''}$ the root near 1, $X := (x\sqrt{U''}, y, z, t)$, and $g := f_i^0$. Then

$$g(X) = U''x^2 + \mathcal{G} = U''(x^2 + \mathcal{G} + \tilde{e}_i) = U''f_i.$$

For (ii), $X_1 - x = (\sqrt{U''} - 1)x$, so $[X_1 - x] \geq \Xi_i - bC_2 \geq \Xi_i - C_5$. For (iii),

$$\Omega_g = \frac{dy \wedge dz \wedge dt}{2x\sqrt{U''}} = U''^{-1/2}\Omega_{W_i}.$$

If $[y] > C_4$ in row II, then $f_i^0 = x^2 + zy^2 + t^\ell y$. Let $\mathcal{F}(\zeta) := x^2 + z\zeta^2 + t^\ell\zeta$. Then $\mathcal{F}(y) = -\tilde{e}_i$ and $\mathcal{F}'(y) = 2zy + t^\ell$ with $[2zy] > C_4 > \ell C_2 \geq [t^\ell]$, so $[\mathcal{F}'(y)] = \ell[t] \leq \ell C_2$ and $[\mathcal{F}(y)] \geq \Xi_i$. Let $\vartheta := Y_0$ be the root of $\mathcal{F}$ near y, $X := (x, Y_0, z, t)$, and $g := f_i^0$. Conversely y is the root near Y_0 of $\mathcal{F} + \tilde{e}_i$, by Newton's lemma with the bounds $[(\mathcal{F} + \tilde{e}_i)(Y_0)] \geq \Xi_i - \ell C_2$ and $[(\mathcal{F} + \tilde{e}_i)'(Y_0)] = \ell[t]$ in

place of the bounds for $\mathcal{F}$ at y, so $\mathfrak{L} := \mathbb{C}(W_i)(Y_0) = \mathfrak{K}^\sharp(y)$. For (ii), $[Y_0 - y] \geq \Xi_i - \ell C_2 \geq \Xi_i - C_5$. For (iii), with the forms $dx\wedge dz\wedge dt/(\partial f_i/\partial y)$ and $dx\wedge dz\wedge dt/(\partial g/\partial y)(X)$, we have $\Omega_g = \upsilon^{-1}\Omega_{W_i}$ with $\upsilon^{-1} = (2zy + t^\ell + e_i')/(2zY_0 + t^\ell)$, and the numerator and the denominator of υ^{-1} have $[\cdot] = \ell[t]$ and difference of order at least $\Xi_i - \ell C_2$, so $\nu_{\mathfrak{L}}(\upsilon) = 0$.

In all the cases with $F \in \mathcal{R}_+$, all the blocks of g other than f_i^0 have positive shift, and in all the cases with $F \in \mathcal{R}_-$ they have negative shift. In all cases g is G_i-semi-invariant with the character of f_i. Moreover $g - x^2 \in \mathbb{C}[y, z, t]$ has degree two in y with leading coefficient z, so $x^2 - g$ is not a square and g is irreducible. Hence the component $W^\sharp$ of $(g = 0)$ in Claim E.38(iii) is $(g = 0)$, which is G_i-invariant and is not contained in any coordinate hyperplane. The lower bounds for $[\cdot]$ in the cases of **Step 5** hold for any i above a threshold which does not depend on F, since the constants C_2, C_3, C_4, C_5, and those in the convention at the end of **Step 4** do not depend on F. Thus, for such i with moreover $\Xi_i - C_5 > \max(C_3, \epsilon)$, we may apply Claim E.38 to $(\mathfrak{L}, X, g)$.

Step 6. In this step we delete the blocks of the marks of the irrelevant sign, and we conclude the proof of (2).

Let F be a prime divisor over W_i with center o and $a_D(F) \leq C_1$, and let i be so large that $\Xi_i - C_5 > \max(C_3, \epsilon)$ and that the lower bounds of **Step 5** hold. The condition on i does not depend on F. Assume first that $F \in \mathcal{R}_+$, and let $(\mathfrak{L}, X, g)$ and $F^\sharp$ be as in **Step 5** for $\mathcal{R}_+$. Let $h_k^\sharp := \sum_{L\geq 0} h_{i,k,L}$. For $l \in \mathcal{N}$, Claim E.38 implies $[X_l] \geq \min([x_l], \Xi_i - C_5) \geq \epsilon$. Hence, by (E.130), any term of $h_{i,k} - h_k^\sharp$, evaluated at X, has $[\cdot] \geq \Xi_i > C_3$, while $[h_{i,k}(X)] = [h_{i,k}] \leq C_3$ by Claim E.38 and (E.137). Thus $\operatorname{mult}_{F^\sharp} h_k^\sharp(X) = \operatorname{mult}_F h_{i,k}$. By (E.132) and Claim E.38,

$$\begin{aligned} a_D(F) &= \frac{1 + \operatorname{mult}_F \Omega_{W_i} - \sum_k c_k \operatorname{mult}_F h_{i,k}}{e_F} \\ &= \frac{1 + \operatorname{mult}_{F^\sharp} \Omega_g - \sum_k c_k \operatorname{mult}_{F^\sharp} h_k^\sharp(X)}{|I_{G_i}(F^\sharp)|}. \end{aligned}$$

The pair (Y_i^+, B_i^+) is lc near C_i^+ by Lemma E.29(1). Claim E.35(2), applied with the sign $+$, $c' = c$, the coordinates $X_1, \dots, X_4$ in place of $x_1, \dots, x_4$, the polynomials g and $h_k^\sharp$, the component $W^\sharp$ and $F^\sharp$, implies

$$a_D(F) \geq \beta_i^+ - \delta\varrho^+, \qquad \varrho^+ := \min_{j\in\mathcal{P}} \frac{[X_j]}{p_j} \geq 0. \tag{E.140}$$

Symmetrically, if $F \in \mathcal{R}_-$, let $h_k^\sharp := \sum_{L\leq 0} h_{i,k,L}$ and use the data of **Step 5** for $\mathcal{R}_-$. Then

$$a_D(F) \geq \beta_i^- + \delta\varrho^-, \qquad \varrho^- := \min_{l\in\mathcal{N}} \frac{[X_l]}{|p_l|} \geq 0. \tag{E.141}$$

Recall from **Step 4** that F belongs to $\mathcal{R}_+$ or to $\mathcal{R}_-$.

Suppose that $\delta < 0$, so that $\beta_i = \beta_i^+$. If $F \in \mathcal{R}_+$, then (E.140) implies $a_D(F) \geq \beta_i^+ + |\delta|\varrho^+ \geq \beta_i$. Otherwise $[x_l] < \epsilon$ for some $l \in \mathcal{N}$, and $F \in \mathcal{R}_-$. Since $\epsilon < \Xi_i - C_5$, Claim E.38 implies $[X_l] = [x_l] < \epsilon$, so $\varrho^- \leq [X_l]/|p_l| < \epsilon$, as $|p_l| \geq 1$. By (E.141) and (1),

$$a_D(F) \geq \beta_i^- - |\delta|\epsilon \geq \beta_i^+ + |\delta|\Big(\frac{1}{Q} - \epsilon\Big) \geq \beta_i^+ = \beta_i,$$

since $\epsilon < \frac{1}{2Q}$.

Suppose that $\delta > 0$, so that $\beta_i = \beta_i^-$. If $F \in \mathcal{R}_-$, then (E.141) implies $a_D(F) \geq \beta_i^-$. Otherwise $[x_j] < \epsilon$ for some $j \in \mathcal{P}$, and $F \in \mathcal{R}_+$. Since $\epsilon < \Xi_i - C_5$, Claim E.38 implies $[X_j] = [x_j] < \epsilon$, so $\varrho^+ \leq [X_j]/p_j < \epsilon$, as $p_j \geq 1$. By (E.140) and (1),

$$a_D(F) \geq \beta_i^+ - \delta\epsilon \geq \beta_i^- + \delta\Big(\frac{1}{Q} - \epsilon\Big) \geq \beta_i^- = \beta_i.$$

Suppose that $\delta = 0$. Then $\beta_i^+ = \beta_i^- = \beta_i$ by (1), and (E.140) or (E.141) implies $a_D(F) \geq \beta_i$.

Thus, for i large, $a_D(F) \geq \beta_i$ for any prime divisor F over W_i with center o and $a_D(F) \leq C_1$. Since $\beta_i \leq C_1$, this implies $\alpha_i \geq \beta_i$ for i large, which proves (2). □

E.12. **A double line: the floor.** In this subsection, we prove Proposition 11.16 by comparing the quotient values α_i with the reference minima β_i of Construction E.26. If a compatible generic limit of the reference pairs is klt, Lemmas E.29 and E.32 squeeze α_i. Otherwise the limit has a floor, and by Lemmas E.39, 9.2, and E.40, infinitely many sources carry prime divisors of log discrepancy two that contradict the terminality of the proxies.

Lemma E.39. *Notation as in Construction E.26, assume that* $\mathbf{T}$ *has weights of both signs, and let* $\varsigma \in \{+,-\}$. *Let* $x_1,\dots,x_4$ *be the coordinates* x,y,z,t, *let* $\mathbf{p} = (p_1,\dots,p_4)$ *be their* $\mathbf{T}$*-weights, let* $M := \max_j |p_j|$, *and let* f_i *be the equation of* W_i, *so that* f_i^0 *is the principal block of* f_i. *Then, after passing to a subsequence and rescaling the coordinates* x,y,z,t *by the non-zero constants of Lemma E.25, the following hold, where* f_i *and* $h_{i,k}$ *are written in the rescaled coordinates.*

(1) For any k *and* i, *any monomial of* $h_{i,k} - h_{i,k}^0$ *has degree at least* $(r_i - |\kappa_k|)/M$, *any monomial of* $f_i - f_i^0$ *has degree at least* $(r_i - 2|u+v|)/M$, *and*

$$\sum_k c_k \operatorname{mult}_{E_0} h_{i,k} \leq 2 - \frac{1}{m},$$

where E_0 *is the exceptional prime divisor over* W_i *of the weighted blow-up with weights* $(3,2,2,2)$ *in eigenparameters* x,y,z,t *with quadratic part* x^2 *(the paragraph after Set-up 9.3).*

(2) Let $f_* = x^2 + g_*(y,z,t)$ *be the polynomial of Lemma E.25(3). Then* f_* *is* $\mathbf{T}$*-homogeneous of weight* $2(u+v)$, g_* *is squarefree,* $\operatorname{mult}_0 g_* = 3$, *and the cubic part of* g_* *is* $y^2 z$. *Moreover there exist a finite subgroup* $\bar{H}$ *of* $(\mathbb{C}^*)^4/\mathbf{T}$ *and, for each* i, *a* G_i*-equivariant and* $\mathbf{T}$*-equivariant automorphism* Φ_i *of* $\mathbb{C}[[x,y,z,t]]$ *with identity tangent map and a* $\mathbf{T}$*-invariant unit* $\mathfrak{u}_i$ *of* $\mathbb{C}[[x,y,z,t]]$, *such that* $\mathfrak{u}_i \Phi_i(f_i^0) = f_*$ *and the image of* G_i *in* $(\mathbb{C}^*)^4/\mathbf{T}$ *is* $\bar{H}$. *Moreover, for any* $c' \in \mathbb{R}^s_{\geq 0}$, Φ_i *induces an isomorphism of* $(Y_i^\varsigma, B_i^\varsigma(c'))$ *near* C_i^ς *with the pair* $(Y^\varsigma, B_{i,\Phi}^\varsigma(c'))$ *near the fiber* C^ς *of* Y^ς *over the image of the origin, where* Y^ς *is the graded quotient of the sign* ς *of* $V_* := (f_* = 0) \subset \mathbb{A}^4$ *by* $\mathbf{T}$ *followed by the quotient by* $\bar{H}$, *and* $B_{i,\Phi}^\varsigma(c')$ *is the boundary induced by* $\sum_k c'_k \operatorname{div}(\Phi_i(h_{i,k}^0)|_{V_*})$ *as in Construction E.26.*

(3) For a power series ϕ *and a positive integer* N, *let* $\phi_{<N}$ *be the sum of the terms of* ϕ *of degree less than* N. *There exist an algebraically closed field* $\mathbf{k} \supset \mathbb{C}$ *and a compatible generic limit* $(f_\infty, h_{\infty,1},\dots,h_{\infty,s})$ *over* $\mathbf{k}$, *in case (a) of Construction-Proposition 4.1 with* $n = 4$ *and trivial group, of the tuples*

$$\mathbf{t}_i := \big((\mathfrak{u}_i \Phi_i(f_i))_{<i}, \Phi_i(h_{i,1})_{<i}, \dots, \Phi_i(h_{i,s})_{<i}\big).$$

The tuple $(f_\infty, h_{\infty,1},\dots,h_{\infty,s})$ *is also a compatible generic limit of the principal tuples*

$$\mathbf{t}_i^0 := \big((\mathfrak{u}_i \Phi_i(f_i^0))_{<i}, \Phi_i(h_{i,1}^0)_{<i}, \dots, \Phi_i(h_{i,s}^0)_{<i}\big).$$

Moreover $f_\infty = f_*$, *and* $h_{\infty,k}$ *is* $\mathbf{T}$*-homogeneous of weight* κ_k *for any* k.

(4) Let J_-^ς *be the set of the coordinates of weight of sign* $-\varsigma$ *or of weight zero. For any index* ι *such that* x_ι *has weight of sign* ς, *let* $S_\iota := V_* \cap (x_\iota = 1) \subset \mathbb{A}^3$, *with the coordinates* x_j, $j \neq \iota$, *let* K_ι *be the group generated by* $\mu_{|p_\iota|} \subset \mathbf{T}$ *and the lifts of* $\bar{H}$ *to* S_ι *(as in Definition-Lemma 11.4(2)), that is, the group of the transformations* $x_j \to \tau_j x_j$, $j \neq \iota$, *given by the elements* $(\tau_1,\dots,\tau_4)$ *of the preimage of* $\bar{H}$ *in* $(\mathbb{C}^*)^4$ *with* $\tau_\iota = 1$, *let* $\mathfrak{n}_\iota \subset \mathbb{C}[S_\iota]$ *be the ideal generated by the coordinates in* J_-^ς, *whose zero locus is the preimage of* C^ς. *Then there exists a* K_ι*-invariant open neighborhood* $U_\iota \subset S_\iota$ *of the zero locus of* $\mathfrak{n}_\iota$ *with at worst quotient singularities; fix one. The charts* $U_\iota \to U_\iota/K_\iota \subset Y^\varsigma$, *with the fiber* $E := C^\varsigma$ *(a curve or a point) and the trivial group, are data* $\mathcal{A}^\varsigma$ *as in case (b) of Construction-Proposition 4.1. For any* i, k, *and* ι, *the restriction* $f_{i,k,\iota}$ *of* $\Phi_i(h_{i,k}^0)$ *to* S_ι

is an element of the $\mathfrak{n}_\iota$-adic completion of $\mathbb{C}[S_\iota]$, and the restriction $f_{\infty,k,\iota}$ of $h_{\infty,k}$ to $S_\iota \times_{\mathbb{C}} \operatorname{Spec}\mathbf{k}$ is an element of the $\mathfrak{n}_\iota$-adic completion of $\mathbf{k}[S_\iota]$. Moreover $(f_{\infty,k,\iota})_{k,\iota}$ is a compatible generic limit over $\mathbf{k}$ of the tuples $(f_{i,k,\iota})_{k,\iota}$ in case (b), and the boundary of the pair

$$\big(\mathcal{A}^\varsigma, \sum_k c_k \operatorname{div}(f_{i,k,\iota})\big)$$

in the sense of Theorem 4.2 is $B^\varsigma_{i,\Phi}(c)$.

(5) *For any d, all but finitely many monomials of a $\mathbf{T}$-homogeneous power series in $\mathbb{C}[[x,y,z,t]]$ or $\mathbf{k}[[x,y,z,t]]$ belong to the d-th power of the ideal generated by the coordinates in J^ς_-.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1). Since $R_i := G_i \cap \mathbf{T}$ has order r_i and $h_{i,k}$ is G_i-semi-invariant, the $\mathbf{T}$-weights of the monomials of $h_{i,k}$ belong to one class modulo r_i, and so do the $\mathbf{T}$-weights of the monomials of f_i. By Definition-Lemma 11.5(2) (Definition-Lemmas 11.4 and 11.5(1)(2) hold for the admitted data by Lemma 11.10(1)), for r_i large the class of $h_{i,k}$ contains κ_k, and any monomial of $h_{i,k} - h^0_{i,k}$ has weight $\kappa' \neq \kappa_k$ with $\kappa' \equiv \kappa_k \pmod{r_i}$, hence $|\kappa'| \geq r_i - |\kappa_k|$. A monomial of weight κ' has degree at least $|\kappa'|/M$. The principal block f^0_i is the sum of the terms of f_i of weight $\kappa(x^2) = 2(u+v)$, so any monomial of $f_i - f^0_i$ has weight $\kappa' \neq 2(u+v)$ with $\kappa' \equiv 2(u+v) \pmod{r_i}$, hence $|\kappa'| \geq r_i - 2|u+v|$ and the degree of the monomial is at least $(r_i - 2|u+v|)/M$.

The residual cubic of W_i defines the double line L^2M of Definition-Lemma 6.4, which is not the cube of a linear form, so W_i is of type cD (Definition 9.1). Hence Definition-Lemma 11.5(3) applied to the data before the numerical admission of Lemma E.23 implies $\sum_k c_k \operatorname{mult}_{E_0} h_k \leq 2 - \frac{1}{m}$ for the marks h_k of the data before admission. By Construction-Theorem 4.13(2), the correspondence of the protected prime divisors is induced by a G_i-equivariant isomorphism of the completions, and since E_0 depends only on the completion (the paragraph after Set-up 9.3), the prime divisor corresponding to E_0 is the prime divisor E_0 of the admitted germ. Hence the orders along E_0 are preserved, and the inequality of (1) follows.

Step 2. In this step we construct Φ_i and $\mathfrak{u}_i$, and prove (2). Since $\mathbf{T}$ has weights of both signs, $(u,v,w) \neq (0,0,1)$, so the hypothesis of Lemma E.25(6) and (7) holds. We pass to the subsequence of Lemma E.25 and rescale the coordinates as there. The orientation of Lemma E.25(4) is that of Construction E.26. Let $f_* = x^2 + g_*(y,z,t)$ be the polynomial of Lemma E.25(3), which is independent of i and $\mathbf{T}$-homogeneous of weight $2(u+v)$. Let ψ_i be the holomorphic map of Lemma E.25(7), and let υ_i be the nowhere vanishing function there (written v_i there), so that $f_* \circ \psi_i = \upsilon_i f^0_i$. Let Φ_i be the inverse of the automorphism of $\mathbb{C}[[x,y,z,t]]$ induced by ψ_i, and let $\mathfrak{u}_i := \Phi_i(\upsilon_i)$. Then Φ_i is G_i-equivariant and $\mathbf{T}$-equivariant with identity tangent map, $\mathfrak{u}_i$ is a $\mathbf{T}$-invariant unit, and applying Φ_i to $f_* \circ \psi_i = \upsilon_i f^0_i$ gives $\mathfrak{u}_i \Phi_i(f^0_i) = f_*$.

By Lemma E.25(3), the polynomial g_* is $y^2z + z^a t^b$ with $a+b \geq 4$, or $y(yz + t^\ell)$ with $\ell \geq 3$. In each case it is squarefree, since $y^2z + z^a t^b = z(y^2 + z^{a-1}t^b)$ for $a \geq 1$ with $z \nmid y^2 + z^{a-1}t^b$, the polynomials $y^2 + z^{a-1}t^b$ and $y^2z + t^b$ are squarefree, and $y \nmid yz + t^\ell$. It has order 3 with cubic part y^2z, as $a+b \geq 4$ and $\ell \geq 3$. The image of G_i in the torus $(\mathbb{C}^*)^4/\mathbf{T}$ has order $[G_i : R_i] \leq e_0$, so it belongs to the finite set of subgroups of the e_0!-torsion of $(\mathbb{C}^*)^4/\mathbf{T}$. After passing to a subsequence it is a fixed group $\bar{H}$.

By Lemma E.25(7), ψ_i is a $\mathbf{T}$-equivariant and G_i-equivariant local isomorphism at every point of N_i, and it maps $V_i \cap \Omega_i$ into V_*, since υ_i is nowhere vanishing. The set N_i consists of the origin and the points of V_i over the fibers $C^\pm_i$ (Lemma E.25(6) and Lemma E.28(5)). Under ψ_i, the function $h^0_{i,k}$ corresponds to $\Phi_i(h^0_{i,k})$. Taking graded quotients by $\mathbf{T}$ and quotients by $\bar{H}$ gives the isomorphism of $(Y^\varsigma_i, B^\varsigma_i(c'))$ near C^ς_i with $(Y^\varsigma, B^\varsigma_{i,\Phi}(c'))$ near C^ς, which proves the last part of (2). We omit the verification of the isomorphism of pairs above, which only uses that ψ_i is $\mathbf{T}$-equivariant and G_i-equivariant, is a local isomorphism along N_i, and maps the points of N_i

over C_i^ς bijectively onto the points of V_* over C^ς. The last property follows from Lemma E.25(7). Let $\mathcal{Z}^\varsigma \subset \mathbb{A}^4$ be the zero locus of the coordinates in J_-^ς. As in Lemma E.28(5), the points of V_i over C_i^ς, which belong to N_i, and the points of V_* over C^ς are the points other than 0 of $V_i \cap \mathcal{Z}^\varsigma$ and of $V_* \cap \mathcal{Z}^\varsigma$, respectively. The coordinates not in J_-^ς have $\mathbf{T}$-weights of sign ς, so for any $\eta \in \mathcal{Z}^\varsigma$ the closure of $\mathbf{T}\cdot\eta$ contains 0. As Ω_i is open, $\mathbf{T}$-invariant and contains 0, it contains $\mathcal{Z}^\varsigma$. As $\lambda\cdot\psi_i(\eta) = \psi_i(\lambda\cdot\eta)$ tends to $\psi_i(0) = 0$ when $\lambda\cdot\eta$ tends to 0, the coordinates in J_-^ς vanish at $\psi_i(\eta)$, so ψ_i maps $\mathcal{Z}^\varsigma$ into $\mathcal{Z}^\varsigma$. On the affine space $\mathcal{Z}^\varsigma$ the coordinate functions of ψ_i are holomorphic everywhere and $\mathbf{T}$-homogeneous, hence polynomials, and, as the tangent map of ψ_i at 0 is the identity, the one of index j is x_j plus a polynomial in the coordinates whose weights have absolute value less than $|p_j|$. Thus ψ_i restricts to an automorphism of $\mathcal{Z}^\varsigma$ fixing 0, which maps $V_i \cap \mathcal{Z}^\varsigma$ onto $V_* \cap \mathcal{Z}^\varsigma$, since $f_* \circ \psi_i = \upsilon_i f_i^0$ with υ_i nowhere vanishing.

Step 3. In this step we prove (3). Since Φ_i is an automorphism of the local ring $\mathbb{C}[[x,y,z,t]]$, it preserves the order at 0. By (1), the orders of $\Phi_i(f_i) - \Phi_i(f_i^0)$ and $\Phi_i(h_{i,k}) - \Phi_i(h_{i,k}^0)$ tend to infinity, and so do the orders of $\mathfrak{u}_i\Phi_i(f_i) - \mathfrak{u}_i\Phi_i(f_i^0)$. Hence, for any N, the N-jets of $\mathbf{t}_i$ and $\mathbf{t}_i^0$ coincide for all but finitely many i.

The component $(\mathfrak{u}_i\Phi_i(f_i))_{<i}$ of $\mathbf{t}_i$ has order 2 for $i \geq 3$. The order of $\Phi_i(h_{i,k})_{<i}$ is the order $o_{i,k}$ of $h_{i,k}$ for $i > o_{i,k}$. Since $\operatorname{mult}_{E_0} h_{i,k}$ is at least the $(3,2,2,2)$-weighted order of $h_{i,k}$ and the weights are at least 2, $2o_{i,k} \leq \operatorname{mult}_{E_0} h_{i,k} \leq (2 - \frac{1}{m})/c_k$ by (1), so $o_{i,k}$ is bounded. Thus the tuples $\mathbf{t}_i$ satisfy the hypotheses of case (a) of Construction-Proposition 4.1 with $n = 4$ and trivial group, and Construction-Proposition 4.1(a) implies, after passing to a subsequence, a compatible generic limit $(f_\infty, h_{\infty,1}, \dots, h_{\infty,s})$ over an algebraically closed field $\mathbf{k} \supset \mathbb{C}$, with closed subsets $Z_N \subset \mathcal{J}_N$ as there. As the N-jets of $\mathbf{t}_i$ and $\mathbf{t}_i^0$ coincide for all but finitely many i, the limit $(f_\infty, h_{\infty,1}, \dots, h_{\infty,s})$ with the closed subsets Z_N is also a compatible generic limit of the principal tuples $\mathbf{t}_i^0$.

Let $L_N \subset \mathcal{J}_N$ be the linear subspace of tuples of N-jets whose components are $\mathbf{T}$-homogeneous of weights $2(u+v), \kappa_1, \dots, \kappa_s$. The N-jets of the principal tuples belong to the closed subset L_N. If $Z_N \not\subset L_N$, then $Z_N \backslash L_N$ is a non-empty open subset of Z_N, so it contains the N-jets of all but finitely many principal tuples, which is absurd. Thus $Z_N \subset L_N$, and the N-jet of the limit, which maps to the generic point of Z_N, is $\mathbf{T}$-homogeneous. Since N is arbitrary, f_∞ and $h_{\infty,k}$ are $\mathbf{T}$-homogeneous. The first component of $\mathbf{t}_i^0$ is $(\mathfrak{u}_i\Phi_i(f_i^0))_{<i} = (f_*)_{<i}$, so for $i \geq N$ its N-jet is the N-jet of f_*. Let $L'_N \subset L_N$ be the closed subset of tuples whose first component is the N-jet of f_*. If $Z_N \not\subset L'_N$, then $Z_N \backslash L'_N$ is a non-empty open subset of Z_N, so it contains the N-jets of all but finitely many principal tuples, which is absurd. Thus $Z_N \subset L'_N$ for any N, and $f_\infty = f_*$.

Step 4. In this step we prove (4) and (5). Fix an index ι such that x_ι has weight of sign ς. By Lemma E.25(6), the points of $V_i \cap (x_\iota = 1)$ over C_i^ς are smooth points or quotient singularities of $V_i \cap (x_\iota = 1)$. As ψ_i is a $\mathbf{T}$-equivariant local isomorphism at the points of $V_i \cap (x_\iota = 1)$ over C_i^ς, it induces local isomorphisms of the root sections $V_i \cap (x_\iota = 1)$ and S_ι at corresponding points, and by the last part of (2) any point of S_ι over C^ς corresponds to such a point. Thus S_ι has at worst quotient singularities near the zero locus of $\mathfrak{n}_\iota$, and U_ι exists. The surface S_ι is normal: V_* is normal by Serre's criterion [Mat89, Theorem 23.8], since g_* is squarefree and the singular locus of V_* is contained in $(x = g_* = 0) \cap \operatorname{Sing}(g_* = 0)$, the morphism $S_\iota \times \mathbb{G}_m \to V_* \cap (x_\iota \neq 0)$, $(\tilde{y}, \lambda) \to \lambda\cdot\tilde{y}$, is the quotient by a free action of $\mu_{|p_\iota|}$, hence étale, and normality descends along the faithfully flat projection $S_\iota \times \mathbb{G}_m \to S_\iota$ [Mat89, Corollary to Theorem 23.9]. The group K_ι acts faithfully on S_ι, since S_ι is not contained in a coordinate hyperplane. As in Lemma E.28(5), the open subset $(x_\iota \neq 0)$ of Y^ς is S_ι/K_ι, a point of S_ι maps into C^ς if and only if the coordinates in J_-^ς vanish at it, and the open subsets S_ι/K_ι cover Y^ς. The fiber C^ς is proper and connected by Zariski's main theorem [Har77, Corollary III.11.4], as the image of a fiber of the graded quotient of the sign ς of V_* by $\mathbf{T}$, which is projective and birational over the normal variety $\operatorname{Spec}\mathbb{C}[V_*]^{\mathbf{T}}$.

Let $x^{\mathbf{n}}$ be a monomial of weight κ with $\mathbf{n} = (n_1, \dots, n_4) \in \mathbb{Z}^4_{\geq 0}$. Then

$$\sum_{\varsigma p_j > 0} |p_j| n_j = \varsigma\kappa + \sum_{\varsigma p_j < 0} |p_j| n_j \leq |\kappa| + M \sum_{x_j \in J^\varsigma_-} n_j.$$

Hence, if $\sum_{x_j \in J^\varsigma_-} n_j < d$, the degree of $x^{\mathbf{n}}$ is less than $|\kappa| + (M+1)d$. Thus, for any d, all but finitely many monomials of a $\mathbf{T}$-homogeneous power series belong to the d-th power of the ideal generated by the coordinates in J^ς_-, which proves (5). Monomials with $\sum_{x_j \in J^\varsigma_-} n_j \geq d$ restrict to S_ι as elements of $\mathfrak{n}_\iota^d$, so the restriction of the power series to S_ι is an element of the $\mathfrak{n}_\iota$-adic completion. The power series $\Phi_i(h^0_{i,k})$ is $\mathbf{T}$-homogeneous of weight κ_k since Φ_i is $\mathbf{T}$-equivariant, and so is $h_{\infty,k}$ by (3). Hence $f_{i,k,\iota}$ and $f_{\infty,k,\iota}$ are well defined. Moreover, for $N \geq \max_k|\kappa_k| + (M+1)d$, the class of $f_{i,k,\iota}$ in $\mathbb{C}[S_\iota]/\mathfrak{n}_\iota^d$ is the image of the N-jet of $\Phi_i(h^0_{i,k})$ under a morphism $\pi_{N,d} : L_N \to \mathcal{J}'_d$ given by polynomials, the restriction to $(x_\iota = 1)$ followed by the reduction modulo $\mathfrak{n}_\iota^d$, where $\mathcal{J}'_d$ is the affine space of the d-jets in case (b). In particular the classes of $f_{i,k,\iota}$ modulo $\mathfrak{n}_\iota^d$ are represented by polynomials of degree bounded independently of i. The morphism $\pi_{N,d}$ also maps the N-jet of $h_{\infty,k}$ to the class of $f_{\infty,k,\iota}$.

We check the hypotheses of Construction-Proposition 4.1(b) for the tuples $(f_{i,k,\iota})$.

(i) After passing to a subsequence, each $f_{i,k,\iota}$ is a non-zero K_ι-semi-invariant element of the $\mathfrak{n}_\iota$-adic completion of $\mathbb{C}[S_\iota]$ whose character depends only on k and ι. The group K_ι acts through the preimage of $\bar{H}$ in $(\mathbb{C}^*)^4$, which is $G_i\mathbf{T}$. The power series $\Phi_i(h^0_{i,k})$ is $\mathbf{T}$-homogeneous of weight κ_k and G_i-semi-invariant, and the characters of the fixed group $G_i\mathbf{T}$ restricting to the weight κ_k on $\mathbf{T}$ form a finite set, so after passing to a subsequence the character χ_k of $\Phi_i(h^0_{i,k})$ under $G_i\mathbf{T}$ is independent of i.

The element $f_{i,k,\iota}$ is non-zero: $h^0_{i,k}$ has degree at most one in x, so $h^0_{i,k} \notin (f^0_i)$, and as V_i is integral and normal at the points over C^ς_i, $h^0_{i,k}$ is non-zero in the completion of V_i at any such point, which corresponds under ψ_i to the completion of $S_\iota \times \mathbb{G}_m$ at a point over the zero locus of $\mathfrak{n}_\iota$.

(ii) The divisors $\operatorname{div}(f_{i,k,\iota})$ descend to the boundary of Construction E.26 on Y^ς (the group H is trivial).

(iii) On two charts, the restrictions of a $\mathbf{T}$-homogeneous function of weight κ_k with character χ_k to two root sections differ by a power of the ratio of the anchors times a value of χ_k, a unit independent of i by (i).

(iv) The orders of $f_{i,k,\iota}$ along the one-dimensional irreducible components of the zero locus of $\mathfrak{n}_\iota$ are at most m/μ_k: the pairs $(Y^\varsigma_i, B^\varsigma_i)$ are lc near C^ς_i by Lemma E.29(1), so their coefficients are at most one, and by [Liu26, Lemma 2.6(4)] this implies $c_k \operatorname{mult} f_{i,k,\iota} \leq 1$, where $c_k = \mu_k/m$ (Set-up 11.1).

If the zero locus of $\mathfrak{n}_\iota$ has a component which is a point $\tilde{y}$, then the pair $(U_\iota, \sum_k c_k \operatorname{div}(f_{i,k,\iota}))$ is lc at $\tilde{y}$ [Liu26, Lemma 2.6], and so is its pullback (U', Δ') to a smooth local cover U' of U_ι at $\tilde{y}$ [KM98, Proposition 5.20]. For the point $y' \in U'$ over $\tilde{y}$, the exceptional divisor of the blow-up of y' has log discrepancy $2 - \operatorname{mult}_{y'} \Delta' \geq 0$ with respect to (U', Δ'), so $\operatorname{mult}_{y'} \Delta' \leq 2$, and the order of $f_{i,k,\iota}$ at $\tilde{y}$ is at most $2/c_k = 2m/\mu_k$.

(v) The classes of $f_{i,k,\iota}$ modulo $\mathfrak{n}_\iota^d$ are represented by polynomials of degree bounded independently of i, as shown above.

By construction, the boundary $B_{\mathrm{br}} + \sum_k c_k \operatorname{div}(f_{i,k,\iota})$ descends to $B^\varsigma_{i,\Phi}(c)$.

Let Z'_d be the closure of $\pi_{N,d}(Z_N)$. It is irreducible, and $\pi_{N,d}$ maps the generic point of Z_N to the generic point of Z'_d, so the d-jet of $(f_{\infty,k,\iota})$ maps to the generic point of Z'_d. Let $O \subset Z'_d$ be a non-empty open subset. Then $\pi^{-1}_{N,d}(O) \cap Z_N$ is open in Z_N, and it is non-empty since it contains the generic point of Z_N. Hence it contains the N-jets of the principal tuples of (3) for all but finitely many i, so O contains the d-jets of $(f_{i,k,\iota})$ for all but finitely many i. The jets for different d are compatible because the morphisms $\pi_{N,d}$ commute with truncation.

The limit satisfies (i)–(iii). Let $d > 2m/\min_k \mu_k$. Taking $O = Z'_d$, the set Z'_d contains the d-jets of $(f_{i,k,\iota})$ for all but finitely many i, and their components are non-zero by (iv), since the elements of $\mathfrak{n}_\iota^d$ have order at least d along any irreducible component of the zero locus of $\mathfrak{n}_\iota$. Hence the closed subset of the d-jets with a zero component does not contain Z'_d, so it does not contain the generic point of Z'_d, and $f_{\infty,k,\iota} \neq 0$. Let $L''_N \subset L_N$ be the linear subspace of tuples whose component of index $k+1$ is semi-invariant with character χ_k under $G_i\mathbf{T}$ for any k. It contains the N-jets of the principal tuples, so the argument of **Step 3** implies $Z_N \subset L''_N$, and $h_{\infty,k}$ is semi-invariant with character χ_k. Hence $f_{\infty,k,\iota}$ is K_ι-semi-invariant with the character of $f_{i,k,\iota}$, and the restrictions of $h_{\infty,k}$ to two root sections differ by the unit of (iii). This proves (4). □

Lemma E.40. *Let $\mathbf{k} \supset \mathbb{C}$ be an algebraically closed field. For $i \geq 1$, let $F_i, h_{i,1}, \dots, h_{i,s}$ be polynomials in $\mathbb{C}[x,y,z,t]$ such that $0 \in W_i := (F_i = 0) \subset \mathbb{A}^4$, W_i is normal near 0, and no $h_{i,k}$ vanishes identically on W_i, let Φ_i be an automorphism of $\mathbb{C}[[x,y,z,t]]$, and let $\mathfrak{u}_i$ be a unit of $\mathbb{C}[[x,y,z,t]]$. Let $(F_\infty, h_{\infty,1}, \dots, h_{\infty,s})$ be a compatible generic limit over $\mathbf{k}$, in case (a) of Construction-Proposition 4.1 with $n = 4$ and trivial group, of the tuples*

$$\big((\mathfrak{u}_i\Phi_i(F_i))_{<i}, \Phi_i(h_{i,1})_{<i}, \dots, \Phi_i(h_{i,s})_{<i}\big),$$

where $\phi_{<N}$ is the sum of the terms of ϕ of degree less than N, and assume that the ring

$$R_\infty := \mathbf{k}[[x,y,z,t]]/(F_\infty)$$

is a normal domain and that $h_{\infty,k} \neq 0$ in R_∞ for any k. Let $\xi_1, \dots, \xi_4$ be formal coordinates of $\mathbf{k}[[x,y,z,t]]$ and $\mathbf{w} \in \mathbb{Z}^4_{>0}$ a primitive vector such that the lowest $\mathbf{w}$-weighted part of F_∞ in $\xi_1, \dots, \xi_4$ is reduced, let P_* be an exceptional prime divisor of the normalized weighted blow-up of $\operatorname{Spec} R_\infty$ with weights $\mathbf{w}$ in $\xi_1, \dots, \xi_4$, and let $n_k := \operatorname{mult}_{P_*} h_{\infty,k}$. Then for all but finitely many i there exists a prime divisor P_i over W_i with center 0 such that

$$a(P_i, W_i, 0) = a(P_*, \operatorname{Spec} R_\infty, 0), \qquad \operatorname{mult}_{P_i} h_{i,k} \geq n_k \quad \text{for } k = 1, \dots, s.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe P_* on a root chart. For a power series ϕ in $\xi_1, \dots, \xi_4$, let $\phi^{[e]}$ be the part of ϕ of $\mathbf{w}$-weight e, and let d be the weight of the lowest $\mathbf{w}$-weighted part $F_\infty^{[d]}$ of F_∞. The exceptional divisor of the weighted blow-up of the ambient space is the weighted projective space $\mathbb{P}(\mathbf{w})$, and P_* corresponds to an irreducible component Σ_* of $(F_\infty^{[d]} = 0) \subset \mathbb{P}(\mathbf{w})$. Choose an index l with $\Sigma_* \not\subset (\xi_l = 0)$. The root chart at ξ_l of the weighted blow-up of $\operatorname{Spec} \mathbf{k}[[\xi_1, \dots, \xi_4]]$ is $\operatorname{Spec} \mathcal{B}$, where

$$\mathcal{B} := \mathbf{k}[[\xi_1, \dots, \xi_4]] \otimes_{\mathbf{k}[\xi_1, \dots, \xi_4]} \mathbf{k}[\rho, \xi'_j \mid j \neq l], \qquad \xi_l \to \rho^{w_l}, \quad \xi_j \to \rho^{w_j}\xi'_j \ (j \neq l),$$

with the action of μ_{w_l} by $\rho \to \zeta\rho$ and $\xi'_j \to \zeta^{-w_j}\xi'_j$, whose quotient is a chart of the weighted blow-up. Since $\mathbf{k}[\rho, \xi']/(\rho^n)$ is a module over $\mathbf{k}[\xi_1, \dots, \xi_4]/(\xi_1, \dots, \xi_4)^n$, we have $\mathcal{B}/\rho^n\mathcal{B} = \mathbf{k}[\xi'][\rho]/(\rho^n)$ for any n, and ρ is a non-zero divisor of $\mathcal{B}$, which is flat over $\mathbf{k}[\rho, \xi']$. Write $\phi^{[e]}(\xi')$ for $\phi^{[e]}$ evaluated at $\xi_l = 1$, $\xi_j = \xi'_j$, and $\phi(\rho^{\mathbf{w}}\xi')$ for the image of ϕ in $\mathcal{B}$. The monomials of F_∞ have weight at least d, and they are multiples of finitely many of them (Dickson's lemma), so $F_\infty(\rho^{\mathbf{w}}\xi') = \rho^d F'_\infty$ for a unique $F'_\infty \in \mathcal{B}$. Modulo ρ^n,

$$\phi(\rho^{\mathbf{w}}\xi') \equiv \sum_{e<n} \rho^e \phi^{[e]}(\xi'), \qquad F'_\infty \equiv \sum_{d \leq e < d+n} \rho^{e-d} F_\infty^{[e]}(\xi').$$

Let $q \in \mathbf{k}[\xi']$ be an irreducible factor of $F_\infty^{[d]}(\xi')$ whose zero locus $\Sigma'_* := (q = 0) \subset \mathbb{A}^3_{\xi'}$ is an irreducible component of the preimage of Σ_* on the chart $\xi_l = 1$, and let Q' be the prime divisor $(\rho = q = 0)$ of the strict transform $(F'_\infty = 0)$. The morphism $\{\xi_l = 1\} \times \mathbb{G}_m \to \{\xi_l \neq 0\}$, $(\xi', \lambda) \to \lambda \cdot \xi'$, given by the $\mathbf{w}$-weighted action, is the quotient by the free action of μ_{w_l} by $\xi'_j \to \zeta^{-w_j}\xi'_j$ $(j \neq l)$ and $\lambda \to \zeta\lambda$, hence étale, and the pullback of $F_\infty^{[d]}$ under it is $\lambda^d F_\infty^{[d]}(\xi')$, so

$F_\infty^{[d]}(\xi')$ is reduced, as $F_\infty^{[d]}$ is. Hence $F_\infty^{[d]}(\xi') = qr'$ with $r' \notin (q)$. Let J be the set of indices $j \neq l$ with $\xi'_j \notin (q)$, and let

$$I_{Q'} := \{\zeta \in \mu_{w_l} \mid \zeta^{w_j} = 1 \text{ for } j \in J\},$$

the subgroup of μ_{w_l} fixing Q' pointwise.

We claim the following.

(a) The local ring $\mathcal{O}$ of $(F'_\infty = 0)$ at the generic point of Q' is a discrete valuation ring with uniformizer ρ.

(b) Let ν be the valuation of $\mathcal{O}$. Then

$$\operatorname{mult}_{P_*} \phi = \nu(\phi(\rho^{\mathbf{w}}\xi'))/|I_{Q'}|$$

for any ϕ, and $a(P_*, \operatorname{Spec} R_\infty, 0) = (\sum_j w_j - d)/|I_{Q'}|$.

(c) For a positive integer n', $\nu(\phi(\rho^{\mathbf{w}}\xi')) \geq n'$ if and only if there exist polynomials $A, B, \vartheta \in \mathbf{k}[\rho, \xi']$ with $\vartheta \notin (\rho, q)$ and

$$\vartheta \cdot \sum_{e<n'} \rho^e \phi^{[e]}(\xi') = A\rho^{n'} + B \sum_{d \leq e < d+n'} \rho^{e-d} F_\infty^{[e]}(\xi').$$

For (a), let $\mathcal{O}_2 := \mathcal{B}_{(\rho,q)\mathcal{B}}$, a regular local ring of dimension two, since ρ is a non-zero divisor and $\mathcal{O}_2/\rho\mathcal{O}_2 = \mathbf{k}[\xi']_{(q)}$ is a discrete valuation ring. Then $\mathcal{O} = \mathcal{O}_2/(F'_\infty)$ and $F'_\infty \neq 0$, as $F'_\infty \equiv qr' \not\equiv 0$ modulo ρ, so $\dim \mathcal{O} = 1$. Moreover, since r' is a unit of $\mathcal{O}_2$,

$$\mathcal{O}/\rho\mathcal{O} = \mathcal{O}_2/(\rho, qr') = \mathcal{O}_2/(\rho, q).$$

Thus the maximal ideal of $\mathcal{O}$ is generated by ρ, and (a) follows. For (b), the quotient map of the root chart by μ_{w_l} has ramification index $|I_{Q'}|$ along Q', and Q' maps onto the center of P_* on the weighted blow-up, which implies the first formula. The pullback of $d\xi_1 \wedge \cdots \wedge d\xi_4$ to the root chart is $w_l \rho^{\sum_j w_j - 1} d\rho \wedge \bigwedge_{j \neq l} d\xi'_j$, and the equation F_∞ pulls back to $\rho^d F'_\infty$. The canonical module of the hypersurface ring R_∞ is generated by the residue ω of $d\xi_1 \wedge \cdots \wedge d\xi_4/F_\infty$, and that of the strict transform $(F'_\infty = 0)$ by the residue ω' of $d\rho \wedge \bigwedge_{j \neq l} d\xi'_j/F'_\infty$ (cf. [KM98, Proposition 5.73]). Since $d\xi_1 \wedge \cdots \wedge d\xi_4$ and F_∞ pull back to $w_l \rho^{\sum_j w_j - 1} d\rho \wedge \bigwedge_{j \neq l} d\xi'_j$ and $\rho^d F'_\infty$, the pullback of ω is $w_l \rho^{\sum_j w_j - 1 - d} \omega'$, and ω' generates the canonical module at the generic point of Q', where the strict transform is regular by (a). Hence the log discrepancy of ν with respect to $(\operatorname{Spec} R_\infty, 0)$ is $\sum_j w_j - d$. Near the generic point of Q' the quotient map by μ_{w_l} is finite with ramification index $|I_{Q'}|$ along Q', and $\rho^{|I_{Q'}|}$ is a uniformizer of the quotient by $I_{Q'}$ at the image of Q'. As $I_{Q'}$ acts trivially on the residue field of Q' and

$$d(\rho^{|I_{Q'}|}) = |I_{Q'}| \rho^{|I_{Q'}|-1} d\rho,$$

a generator of the canonical module of the quotient at the image of Q' pulls back to $\rho^{|I_{Q'}|-1}$ times a generator at Q', that is, the canonical divisors differ by $(|I_{Q'}| - 1)Q'$ there. Comparing the orders of the pullback of ω along Q' and along P_* gives $\sum_j w_j - d = |I_{Q'}| a(P_*, \operatorname{Spec} R_\infty, 0)$. For (c), both sides of the equation only depend on $\phi(\rho^{\mathbf{w}}\xi')$ and F'_∞ modulo $\rho^{n'}$,

$$\mathcal{O}_2/\rho^{n'}\mathcal{O}_2 = (\mathbf{k}[\rho, \xi']/(\rho^{n'}))_{(\rho,q)},$$

and $\nu(\phi(\rho^{\mathbf{w}}\xi')) \geq n'$ means that the image of $\phi(\rho^{\mathbf{w}}\xi')$ belongs to $\rho^{n'}\mathcal{O}$. Clearing the denominators of the localization proves (c).

Step 2. In this step we fix finitely many data over $\mathbf{k}$. Let $n'_k := |I_{Q'}| n_k$, so that $\nu(h_{\infty,k}(\rho^{\mathbf{w}}\xi')) = n'_k$ by (b), finite since $h_{\infty,k} \neq 0$ in R_∞. Let N be an integer larger than $d + \max_k n'_k$. Since all weights are positive integers, any power series in $(\xi_1, \ldots, \xi_4)^N$ has only parts of weight at least N, so the parts of weight less than N of any power series in $\xi_1, \ldots, \xi_4$ only depend on the N-jet of the power series in x, y, z, t and on the N-jets of $\xi_1, \ldots, \xi_4$. Let $\bar{\xi}_j := (\xi_j)_{<N}$. The polynomials $\bar{\xi}_1, \ldots, \bar{\xi}_4$ are regular parameters of $\mathbf{k}[[x, y, z, t]]$, and the weighted parts of weight less than N of any power series in $\bar{\xi}_1, \ldots, \bar{\xi}_4$ and in $\xi_1, \ldots, \xi_4$ coincide. Choose a closed point p of $(q = 0) \subset \mathbb{A}^3_{\xi'}$

such that $r'(p) \neq 0$, some partial derivative of q does not vanish at p, $\xi'_j(p) \neq 0$ for $j \in J$, and $\vartheta_k(0,p) \neq 0$ for the polynomials ϑ_k of (c) for $\phi = h_{\infty,k}$ and $n' = n'_k$. Such a point exists, since $(q=0)$ is irreducible, none of r', ξ'_j ($j \in J$), and $\vartheta_k(0,\cdot)$ belongs to (q), and some $\partial q/\partial \xi'_j$ does not belong to (q), as q is irreducible and $\mathbf{k}$ has characteristic zero. For $j \notin J$, write $\xi'_j = qs_j$. Let N' be an integer larger than the degrees of $q, r', s_j, A_k, B_k, \vartheta_k$.

Step 3. In this step we construct an incidence space. Let $\mathcal{J}_N$ be the affine space of the N-jets of $(s+1)$-tuples, and let $\mathcal{T}$ be the affine space of the tuples $(\bar\xi, q, r', (s_j)_{j\notin J}, (A_k, B_k, \vartheta_k)_k, p)$, where $\bar\xi = (\bar\xi_1, \dots, \bar\xi_4)$ consists of polynomials of degree less than N without constant term, the other polynomials have degree at most N', and $p \in \mathbb{A}^3$. Let $\mathcal{I} \subset \mathcal{J}_N \times \mathcal{T}$ be the set of the points $((F, h_1, \dots, h_s), (\bar\xi, q, r', \dots, p))$ with the following properties, where the weighted parts are those of F and h_k expressed in the parameters $\bar\xi_1, \dots, \bar\xi_4$ and truncated at weight N:

(i) the Jacobian determinant of $\bar\xi$ at 0 is non-zero;
(ii) $F^{[e]} = 0$ for $e < d$, and $F^{[d]}(\xi') = qr'$;
(iii) $q(p) = 0$, $r'(p) \neq 0$, and some partial derivative of q does not vanish at p;
(iv) $\xi'_j = qs_j$ for $j \notin J$, and $\xi'_j(p) \neq 0$ for $j \in J$;
(v) for any k, $\vartheta_k(0,p) \neq 0$ and

$$\vartheta_k \cdot \sum_{e<n'_k} \rho^e h_k^{[e]}(\xi') = A_k \rho^{n'_k} + B_k \sum_{d\le e<d+n'_k} \rho^{e-d} F^{[e]}(\xi').$$

The coefficients of $F^{[e]}$ and $h_k^{[e]}$ for $e < N$ are polynomials in the coefficients of the jets, of $\bar\xi$, and of the inverse of the Jacobian determinant of $\bar\xi$ at 0, since the expression of an N-jet in the parameters $\bar\xi_1, \dots, \bar\xi_4$ is obtained by composing with the inverse of $\bar\xi$ modulo $(x,y,z,t)^N$. Thus (i)–(v) are finitely many polynomial equations and non-vanishing conditions with coefficients in $\mathbb{Q}$, and $\mathcal{I}$ is a locally closed subset of finite type. By Chevalley's theorem [Har77, Exercise II.3.19], the projection $\mathrm{pr}(\mathcal{I}) \subset \mathcal{J}_N$ is constructible.

By **Steps 1 and 2**, the point of $\mathcal{J}_N$ given by the N-jet of $(F_\infty, h_{\infty,1}, \dots, h_{\infty,s})$ belongs to $\mathrm{pr}(\mathcal{I})$. The N-jet of $(F_\infty, h_{\infty,1}, \dots, h_{\infty,s})$ maps to the generic point of the irreducible closed subset $Z_N \subset \mathcal{J}_N$ of the compatible generic limit. Thus $\mathrm{pr}(\mathcal{I}) \cap Z_N$ is a constructible subset of Z_N containing the generic point of Z_N, hence contains a non-empty open subset of Z_N. By the definition of a compatible generic limit, the open subset contains the N-jets of the tuples of the lemma for all but finitely many i. Hence, for such i with $i > N$, the N-jet of

$$(\mathfrak{u}_i \Phi_i(F_i), \Phi_i(h_{i,1}), \dots, \Phi_i(h_{i,s}))$$

belongs to $\mathrm{pr}(\mathcal{I})$. As $\mathbb{C}$ is algebraically closed, the fiber of $\mathcal{I}$ over such a jet has a $\mathbb{C}$-point

$$(\bar\xi_i, q_i, r'_i, (s_{i,j})_{j\notin J}, (A_{i,k}, B_{i,k}, \vartheta_{i,k})_k, p_i).$$

Step 4. In this step we construct P_i for such an index i. Let $F'_i := \Phi_i^{-1}(\mathfrak{u}_i) F_i$, so that $\Phi_i(F'_i) = \mathfrak{u}_i \Phi_i(F_i)$ and F'_i generates the ideal of W_i in $\mathbb{C}[[x,y,z,t]]$. Let $\xi_{i,j} := \Phi_i^{-1}(\bar\xi_{i,j})_{<N}$ for $j = 1, \dots, 4$, polynomials with invertible Jacobian matrix at 0. If $\Phi_i(\phi) = \Psi(\bar\xi_{i,1}, \dots, \bar\xi_{i,4})$ for a power series Ψ, then

$$\phi = \Psi(\Phi_i^{-1}(\bar\xi_{i,1}), \dots, \Phi_i^{-1}(\bar\xi_{i,4})),$$

and the parameters $\Phi_i^{-1}(\bar\xi_{i,j})$ and $\xi_{i,j}$ are congruent modulo $(x,y,z,t)^N$. Hence the weighted parts of weight less than N of F'_i and $h_{i,k}$ in $\xi_{i,1}, \dots, \xi_{i,4}$ are those of $\mathfrak{u}_i \Phi_i(F_i)$ and $\Phi_i(h_{i,k})$ in $\bar\xi_{i,1}, \dots, \bar\xi_{i,4}$, and (i)–(v) hold for F'_i, $h_{i,k}$ in the parameters $\xi_{i,j}$.

Consider the weighted blow-up of $\operatorname{Spec} \widehat{\mathcal{O}}_{W_i,0}$ with weights $\mathbf{w}$ in $\xi_{i,1}, \dots, \xi_{i,4}$ and the root chart $\operatorname{Spec} \mathcal{B}_i$ at $\xi_{i,l}$ of the ambient weighted blow-up, where, as in **Step 1**,

$$\mathcal{B}_i := \mathbb{C}[[\xi_{i,1}, \dots, \xi_{i,4}]] \otimes_{\mathbb{C}[\xi_{i,1},\dots,\xi_{i,4}]} \mathbb{C}[\rho, \xi'_j \mid j \neq l].$$

By (ii), the weighted parts of F'_i of weight less than d vanish, so $F'_i \in \rho^d \mathcal{B}_i$ as in **Step 1**. By (iii), p_i is a smooth point of $(q_i = 0)$, so it belongs to a unique irreducible component Σ'_i of $(q_i = 0) \subset \mathbb{A}^3_{\xi'}$,

the counterpart of Σ'_*, along which q_i vanishes to order one. Let $Q'_i := (\rho = 0) \times \Sigma'_i$. By (ii) and $r'_i(p_i) \neq 0$, the equation of Σ'_i divides $F_i'^{[d]}(\xi')$ exactly once. The argument of **Step 1**(a) shows that the local ring of the strict transform at the generic point of Q'_i is a discrete valuation ring with uniformizer ρ. Let ν_i be the corresponding valuation. By (iv), the coordinates ξ'_j with $j \notin J$ vanish on Σ'_i and the others do not, so the subgroup of μ_{w_l} fixing Q'_i pointwise is $I_{Q'}$. Let P_i be the prime divisor over $\operatorname{Spec} \widehat{\mathcal{O}}_{W_i,0}$ given by the image of Q'_i. The center of P_i on $\operatorname{Spec} \widehat{\mathcal{O}}_{W_i,0}$ is the closed point, since all weights are positive. By the computation of **Step 1**(b),

$$a(P_i, \operatorname{Spec} \widehat{\mathcal{O}}_{W_i,0}, 0) = (\sum_j w_j - d)/|I_{Q'}| = a(P_*, \operatorname{Spec} R_\infty, 0).$$

By (v) and $\vartheta_{i,k}(0, p_i) \neq 0$, $\vartheta_{i,k}$ is a unit at the generic point of Q'_i, so $\nu_i(h_{i,k}(\rho^{\mathbf{w}}\xi')) \geq n'_k$ and $\operatorname{mult}_{P_i} h_{i,k} \geq n'_k/|I_{Q'}| = n_k$. By [Liu26, Lemma 3.3(3)], P_i restricts to a prime divisor over W_i with center 0, still denoted P_i, such that $a(P_i, W_i, 0) = a(P_*, \operatorname{Spec} R_\infty, 0)$ and $\operatorname{mult}_{P_i} h_{i,k} \geq n_k$ for $k = 1, \dots, s$. □

Proof of Proposition 11.16. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce to the data of Construction E.26 and bound the unquotiented minimum from below. Suppose that there exists a sequence of data $(W_i \ni o_i, G_i, h_{i,1}, \dots, h_{i,s})$, $i \geq 1$, as in Set-up 11.1, with W_i of quadratic rank one whose residual cubic defines a double line (Definition-Lemma 6.4), and with α_i strictly increasing in $(0, 1)$. Let $D_i := \sum_k c_k \operatorname{div}(h_{i,k})$. Let P be a prime divisor over W_i with center o_i, and let $k \in \{1, \dots, s\}$. Since $(W_i, C_{i,k})$ is terminal near o_i for $C_{i,k} := \frac{1}{m} \operatorname{div}(h_{i,k})$ (Set-up 11.1),

$$a(P, W_i, C_{i,k}) = a(P, W_i, 0) - \frac{1}{m} \operatorname{mult}_P h_{i,k} > 1.$$

Since $a_k(P)|I_{G_i}(P)| = a(P, W_i, C_{i,k})$, the integrality in Definition-Lemma 11.4(1), which uses that W_i is Gorenstein and $\operatorname{mult}_P h_{i,k} \in \mathbb{Z}$, shows that $ma(P, W_i, C_{i,k})$ is an integer. Hence $a(P, W_i, C_{i,k}) \geq 1 + \frac{1}{m}$. As $\sum_k \mu_k = 1$ and $D_i = \sum_k \mu_k C_{i,k}$,

$$a(P, W_i, D_i) = \sum_k \mu_k a(P, W_i, C_{i,k}) \geq 1 + \frac{1}{m}.$$

Thus $\operatorname{mld}(W_i \ni o_i, D_i) \geq 1 + \frac{1}{m}$. This is the minimal log discrepancy of $(W_i \ni o_i, D_i)$ itself, without the quotient by G_i, which is the quantity that the numerical admission below preserves (as the minimal log discrepancy for the trivial subgroup).

By Lemma E.23, after passing to a subsequence and replacing the data with numerical admissions that keep the groups G_i, we may assume that the data are as in the conclusion of Lemma E.23. A numerical admission preserves the minimal log discrepancy of the real labelled boundary for any subgroup (Construction-Theorem 4.13(1) with $j = 0$), in particular for the trivial subgroup, and it preserves α_i. Hence, writing $W_i = (f_i = 0) \subset \mathbb{A}^4$, $o_i = 0$, and $D_i = \sum_k c_k \operatorname{div}(h_{i,k})$ for the admitted data,

$$\operatorname{mld}(W_i \ni 0, D_i) \geq 1 + \frac{1}{m} \quad \text{for any } i. \tag{E.142}$$

We use Construction E.26 with the orientation fixed there, and if $u = w = 0$, we choose the orientation with $v > 0$, which the rule of Construction E.26 allows. Then some coordinate has positive $\mathbf{T}$-weight: $2u > 0$ if $u > 0$, $w > 0$ if $u = 0 < w$, and $v > 0$ if $u = w = 0$. Thus either no coordinate has negative $\mathbf{T}$-weight, or $\mathbf{T}$ has weights of both signs. By Lemma E.29(1), the numbers β_i belong to a finite set, so after passing to a subsequence $\beta_i = \beta$ is independent of i, and, with the constant C_0 of Lemma E.29(1),

$$\alpha_i \leq \beta + \frac{C_0}{r_i} \quad \text{for any } i. \tag{E.143}$$

Since $r_i = |G_i \cap \mathbf{T}| \to +\infty$ (Lemma E.23(3)) and the α_i are strictly increasing, (E.143) implies $\alpha_i < \lim_j \alpha_j \le \beta$ for any i. In particular $\beta \ge \alpha_1 > 0$.

Step 2. In this step we exclude the case that no coordinate has negative $\mathbf{T}$-weight. In this case Lemma E.29(2) implies $\alpha_i \ge \beta_i = \beta$ for i large, which contradicts $\alpha_i < \beta$. From now on $\mathbf{T}$ has weights of both signs.

Step 3. In this step we choose a reference and exclude a klt limit. Let $\varsigma := +$ if $\delta < 0$, or if $\delta = 0$ and $u + w \le 0$, and let $\varsigma := -$ otherwise. By Lemma E.32(1), $\beta_i = \beta_i^\varsigma$ for i large. We use Lemma E.39 for ς, pass to the subsequence given there, and use the notation of Lemma E.39: f_*, Φ_i, $\mathfrak{u}_i$, Y^ς, C^ς, the field $\mathbf{k}$, the compatible generic limit $(f_\infty, h_{\infty,1}, \dots, h_{\infty,s})$ with $f_\infty = f_*$, and the data $\mathcal{A}^\varsigma$ of case (b) with the chart functions $f_{i,k,\iota}$ and $f_{\infty,k,\iota}$. Theorem 4.2 and Definition-Proposition 5.7 are used below in case (b) for $\mathcal{A}^\varsigma$, with $\mathfrak{c} = \mathcal{O}$, the labels $c_1, \dots, c_s$, and the trivial group. In their notation, C_i and C_∞ are written B_i^ς and B_∞^ς here, so that the pairs are

$$(\mathcal{A}^\varsigma, B_{\mathrm{br}} + \theta B_i^\varsigma) \quad \text{and} \quad (\mathcal{A}_{\mathbf{k}}^\varsigma, B_{\mathrm{br}} + \theta B_\infty^\varsigma).$$

By Lemma E.39(2)(4), $(\mathcal{A}^\varsigma, B_{\mathrm{br}} + \theta B_i^\varsigma)$ is isomorphic near the base to $(Y_i^\varsigma, B_i^\varsigma(\theta c))$ near C_i^ς for any $\theta \ge 0$, and the minimum of the log discrepancies of $(\mathcal{A}^\varsigma, B_{\mathrm{br}} + B_i^\varsigma)$ over the prime divisors with center contained in the base is β_i^ς, which equals β for i large.

Assume that $(\mathcal{A}_{\mathbf{k}}^\varsigma, B_{\mathrm{br}} + B_\infty^\varsigma)$ is klt near the base. By Theorem 4.2(1) with $\theta = 1$, there exists $\theta' > 1$ such that $(Y_i^\varsigma, B_i^\varsigma(\theta' c))$ is lc near C_i^ς for any i. Choose $\theta \in (1, \theta']$ so close to 1 that $\delta_\theta = \delta + (\theta - 1)\sum_k c_k \kappa_k$ has the sign of δ if $\delta \ne 0$. Since $\delta = \sum_k c_k \kappa_k - u - w$ (Construction E.26), if $\delta = 0$ then $\sum_k c_k \kappa_k = u + w$ and $\delta_\theta = (\theta - 1)(u + w)$, which has the sign of $u + w$. The coefficients $1 - (1 - b)/e$ of a crepant quotient increase with b [Liu26, Lemma 2.6(4)], so $(Y_i^\varsigma, B_i^\varsigma(\theta c))$ is lc near C_i^ς for any i. By the choice of ς and θ, $\varsigma\delta_\theta \le 0$, so Lemma E.32(2), used with the sign ς, implies $\alpha_i \ge \beta_i = \beta$ for i large, which contradicts $\alpha_i < \beta$. Hence $(\mathcal{A}_{\mathbf{k}}^\varsigma, B_{\mathrm{br}} + B_\infty^\varsigma)$ is not klt near the base.

Step 4. In this step we define the limit germ W_∞ and obtain a prime divisor S of W_∞ of coefficient one. Let

$$W_\infty := \operatorname{Spec} R_\infty, \qquad R_\infty := \mathbf{k}[[x, y, z, t]]/(f_*).$$

Let $V_{*,\mathbf{k}} := (f_* = 0) \subset \mathbb{A}^4_{\mathbf{k}}$, where $f_* = x^2 + g_*$. Since g_* is squarefree, the singular locus of $V_{*,\mathbf{k}}$ is contained in

$$(x = g_* = 0) \cap \operatorname{Sing}(g_* = 0),$$

so it has codimension at least two, and $V_{*,\mathbf{k}}$ is normal by Serre's criterion [Mat89, Theorem 23.8], and so is R_∞, the completion of an excellent normal local ring [Mat89, Theorem 32.2(i)]. Thus R_∞ is a normal complete local domain of dimension three.

The hypotheses of Definition-Proposition 5.7 hold for $\mathcal{A}^\varsigma$: the pairs $(Y_i^\varsigma, B_i^\varsigma)$ are lc near C_i^ς by Lemma E.29(1), the numbers $\beta_i^\varsigma = \beta$ converge to $\beta > 0$, and the limit is not klt near the base by **Step 3**. By Definition-Proposition 5.7, the limit is lc near the base. As it is not klt near the base, it has an lc center meeting the base, and any such lc center is a floor of the limit. Let S_Y be such a floor, that is, a curve with coefficient one in $B_{\mathrm{br}} + B_\infty^\varsigma$ which intersects the base $C_{\mathbf{k}}^\varsigma$, and whose preimage in the smooth formal cover $\mathcal{S}_{\tilde{y}}$ of Definition-Proposition 5.7 at any closed point $\tilde{y}$ over $C_{\mathbf{k}}^\varsigma$ of a chart cover is smooth.

We show that no component of $C_{\mathbf{k}}^\varsigma$ is a floor. If C^ς is a point, there is nothing to show. Let E' be an irreducible component of the zero locus of $\mathfrak{n}_\iota$ on a chart U_ι and n a non-negative integer. The tuples whose k-th component on U_ι has order at least n along E' form a closed subset $\mathcal{K}_n$ of the space of the n-jets of case (b): the elements of $\mathfrak{n}_\iota^n$ have order at least n along E', so the condition means that the class modulo $\mathfrak{n}_\iota^n$ belongs to a fixed linear subspace. If $\operatorname{mult}_{E'} f_{\infty,k,\iota} \ge n$, then $\mathcal{K}_n$ contains the generic point of the irreducible closed subset Z_n' attached by Construction-Proposition 4.1 to the compatible generic limit of Lemma E.39(4), hence $Z_n' \subset \mathcal{K}_n$. As Z_n' itself contains the n-jets of the tuples $(f_{i,k,\iota})$ for all but finitely many i, we obtain $\operatorname{mult}_{E'} f_{i,k,\iota} \ge n$ for all but finitely many i. Hence the coefficient of the image of E' in $B_{\mathrm{br}} + B_\infty^\varsigma$ is at most the

coefficient of the image of E' in $B_{\mathrm{br}} + B_i^\varsigma$ for all but finitely many i. If it were one, then for i large the log discrepancy of $(Y_i^\varsigma, B_i^\varsigma)$ along a component of C_i^ς would be zero, contrary to $\beta_i^\varsigma = \beta > 0$.

The branch coefficients of B_{br} are less than one, and by Lemma 3.7 a prime divisor of coefficient b on a chart cover with ramification index e has image of coefficient $1 - (1-b)/e$, which is one if and only if $b = 1$. Thus a component Γ of the preimage of S_Y in the completion of a chart cover $U_\iota \times_{\mathbb{C}} \operatorname{Spec} \mathbf{k}$ along the zero locus of $\mathfrak{n}_\iota$ satisfies $\sum_k c_k \operatorname{mult}_\Gamma f_{\infty,k,\iota} = 1$, intersects the preimage of $C_{\mathbf{k}}^\varsigma$, and is not contained in the zero locus of $\mathfrak{n}_\iota$. As $f_{\infty,k,\iota}$ is the restriction of $h_{\infty,k}$ to $S_\iota = V_* \cap (x_\iota = 1)$ (Lemma E.39(4)),

$$\sum_k c_k \operatorname{mult}_\Gamma(h_{\infty,k}|_{S_\iota}) = 1. \tag{E.144}$$

Let $\mathcal{Z}^\varsigma \subset \mathbb{A}^4_{\mathbf{k}}$ be the zero locus of the coordinates in J_-^ς, that is, of the coordinates of weight of sign $-\varsigma$ or of weight zero, and let $\mathcal{W}$ be the formal completion of $V_{*,\mathbf{k}}$ along $V_{*,\mathbf{k}} \cap \mathcal{Z}^\varsigma$. The morphism $S_\iota \times \mathbb{G}_m \to V_* \cap (x_\iota \neq 0)$, $(\tilde{y}, \lambda) \to \lambda \cdot \tilde{y}$, is the quotient by a free action of $\mu_{|p_\iota|}$, hence étale, and it is $\mathbf{T}$-equivariant, $\mathbf{T}$ acting on $\mathbb{G}_m$ by multiplication. It maps $(S_\iota \cap \mathcal{Z}^\varsigma) \times \mathbb{G}_m$ onto $V_* \cap \mathcal{Z}^\varsigma \cap (x_\iota \neq 0)$, so it induces an étale morphism from the completion of $S_{\iota,\mathbf{k}} \times \mathbb{G}_m$ along $(S_{\iota,\mathbf{k}} \cap \mathcal{Z}^\varsigma) \times \mathbb{G}_m$ to $\mathcal{W} \cap (x_\iota \neq 0)$. Let S° be the image of $\Gamma \times \mathbb{G}_m$, a $\mathbf{T}$-invariant prime divisor of $\mathcal{W} \cap (x_\iota \neq 0)$. For a $\mathbf{T}$-homogeneous function ϕ of weight κ on $\mathcal{W}$, the pullback of ϕ is $\lambda^\kappa \cdot \phi|_{S_\iota}$ with λ a unit, so $\operatorname{mult}_{S^\circ} \phi = \operatorname{mult}_\Gamma(\phi|_{S_\iota})$.

Let $\bar{S}$ be the closure of S° in $\mathcal{W}$. The coordinates in J_-^ς vanish at a point of Γ over $C_{\mathbf{k}}^\varsigma$, and the coordinates not in J_-^ς have $\mathbf{T}$-weight of sign ς, so the closure of the $\mathbf{T}$-orbit of such a point contains the origin. Thus $0 \in \bar{S}$.

Let $I_{\mathcal{Z}}$ be the ideal of $\mathcal{Z}^\varsigma$, let

$$\mathcal{R} := \Gamma(\mathcal{W}, \mathcal{O}_{\mathcal{W}}) = \mathbf{k}[x_j \mid x_j \notin J_-^\varsigma][[x_j \mid x_j \in J_-^\varsigma]]/(f_*),$$

let $\mathcal{R}_\iota$ be the ring of global functions on $\mathcal{W} \cap (x_\iota \neq 0)$, the $I_{\mathcal{Z}}$-adic completion of $\mathcal{R}[x_\iota^{-1}]$, let $\mathfrak{P}_\iota \subset \mathcal{R}_\iota$ be the prime ideal of S°, let $\mathfrak{P} := \mathfrak{P}_\iota \cap \mathcal{R}$, the prime ideal of $\bar{S}$, and let $\mathfrak{m}_0 \subset \mathcal{R}$ be the maximal ideal of the origin, so that $\mathfrak{P} \subset \mathfrak{m}_0$ and the completion of $\mathcal{R}_{\mathfrak{m}_0}$ is R_∞. By a theorem of Rotthaus [Mat89, §32, Remark 2 after the Corollary to Theorem 32.6], $\mathcal{R}$ is excellent. For an excellent ring D and an ideal $I \subset D$, the I-adic completion $D \to \widehat{D}$ is a regular homomorphism: it is flat [Mat89, Theorem 8.8], any maximal ideal $\mathfrak{M}$ of $\widehat{D}$ contains $I\widehat{D}$ [Mat89, Theorem 8.2], and the composite of $D_{\mathfrak{M}\cap D} \to \widehat{D}_{\mathfrak{M}}$ with the completion of $\widehat{D}_{\mathfrak{M}}$ is the completion of the excellent local ring $D_{\mathfrak{M}\cap D}$, so $D_{\mathfrak{M}\cap D} \to \widehat{D}_{\mathfrak{M}}$ is regular [Mat89, Theorem 32.1(ii)]. Thus $\mathbf{k}[V_{*,\mathbf{k}}] \to \mathcal{R}$, $\mathcal{R}[x_\iota^{-1}] \to \mathcal{R}_\iota$, and $\mathcal{R}_{\mathfrak{m}_0} \to R_\infty$ are regular homomorphisms, and $\mathcal{R}$ is normal [Mat89, Corollary to Theorem 23.9]. The set $V_{*,\mathbf{k}} \cap \mathcal{Z}^\varsigma$ is connected, since it is $\mathbf{T}$-stable and the closure of any $\mathbf{T}$-orbit in it contains the origin, and $I_{\mathcal{Z}}\mathcal{R}$ is contained in the Jacobson radical of $\mathcal{R}$, so $\operatorname{Spec} \mathcal{R}$ is connected and the normal ring $\mathcal{R}$ is a domain.

A $\mathbf{T}$-homogeneous element of R_∞ belongs to $\mathcal{R}$ by Lemma E.39(5). By (E.144) and the equality $\operatorname{mult}_{S^\circ} \phi = \operatorname{mult}_\Gamma(\phi|_{S_\iota})$ for $\mathbf{T}$-homogeneous ϕ, applied to the $h_{\infty,k}$ (Lemma E.39(3)), $\sum_k c_k \operatorname{mult}_{S^\circ} h_{\infty,k} = 1$, so some $h_{\infty,k}$ is a non-zero element of $\mathfrak{P}$. As $\mathcal{R}[x_\iota^{-1}] \to \mathcal{R}_\iota$ is flat, the height of $\mathfrak{P}$ is at most the height one of $\mathfrak{P}_\iota$ by going down [Mat89, Theorem 9.5], so $\mathfrak{P}$ has height one. The homomorphism

$$\mathcal{R}[x_\iota^{-1}]/\mathfrak{P} \to \mathcal{R}_\iota/\mathfrak{P}\mathcal{R}_\iota$$

is the base change of the regular homomorphism $\mathcal{R}[x_\iota^{-1}] \to \mathcal{R}_\iota$, so it is flat with regular fibers, and since $\mathcal{R}[x_\iota^{-1}]/\mathfrak{P}$ is a domain, the ring $\mathcal{R}_\iota/\mathfrak{P}\mathcal{R}_\iota$ is reduced by [Mat89, Corollary to Theorem 23.9(ii)], applied to the localizations at its maximal ideals, and $\mathfrak{P}_\iota$ is a minimal prime of $\mathfrak{P}\mathcal{R}_\iota$ [Mat89, Theorem 15.1], so $\mathfrak{P}$ generates the maximal ideal of $(\mathcal{R}_\iota)_{\mathfrak{P}_\iota}$, and $\operatorname{mult}_{S^\circ} \phi = v_{\mathfrak{P}}(\phi)$ for $\phi \in \mathcal{R}$, where $v_{\mathfrak{P}}$ is the valuation of the discrete valuation ring $\mathcal{R}_{\mathfrak{P}}$. Let $\mathfrak{p} := \mathfrak{P}\mathcal{R}_{\mathfrak{m}_0}$, let I_S be a minimal prime of $\mathfrak{p}R_\infty$, and let S be the prime divisor of W_∞ defined by I_S. Possibly replacing $\mathcal{R}[x_\iota^{-1}] \to \mathcal{R}_\iota$ with the regular homomorphism $\mathcal{R}_{\mathfrak{m}_0} \to R_\infty$, the argument above implies that I_S has height one, $I_S \cap \mathcal{R}_{\mathfrak{m}_0} = \mathfrak{p}$, and $\mathfrak{p}$ generates the maximal ideal of $(R_\infty)_{I_S}$, so $\operatorname{mult}_S \phi = v_{\mathfrak{P}}(\phi)$

for $\phi \in \mathcal{R}$. Hence, for any $\mathbf{T}$-homogeneous $\phi \in R_\infty$,

$$\operatorname{mult}_S \phi = \operatorname{mult}_{S^\circ} \phi = \operatorname{mult}_\Gamma(\phi|_{S_\iota}),$$

and, as the $h_{\infty,k}$ are $\mathbf{T}$-homogeneous (Lemma E.39(3)),

$$c_S := \sum_k c_k \operatorname{mult}_S h_{\infty,k} = 1. \tag{E.145}$$

Step 5. In this step we show that R_∞/I_S is regular. Let E_0 be the prime divisor over W_∞ defined in Lemma E.39(1) with W_∞ in place of W_i, that is, the exceptional prime divisor of the weighted blow-up of W_∞ with weights $(3,2,2,2)$ in x,y,z,t. The lowest $(3,2,2,2)$-weighted part of f_* is x^2+y^2z, which is irreducible, so the graded ring $\mathbf{k}[x,y,z,t]/(x^2+y^2z)$ is a domain, and $\operatorname{mult}_{E_0}\phi$ is the largest $(3,2,2,2)$-weighted order of a representative of $\phi \in R_\infty$ in $\mathbf{k}[[x,y,z,t]]$. Since R_∞ is a free $\mathbf{k}[[y,z,t]]$-module with basis $1,x$, any $\phi \in R_\infty$ is uniquely $\phi_0(y,z,t) + x\phi_1(y,z,t)$. The lowest weighted part of $\phi_0 + x\phi_1$ has degree at most one in x, so it is non-zero in the graded ring if $\phi \neq 0$, and

$$\operatorname{mult}_{E_0}\phi = \min\bigl(2\operatorname{mult}_0\phi_0, 3 + 2\operatorname{mult}_0\phi_1\bigr). \tag{E.146}$$

For i large, on $\operatorname{Spec}\mathbb{C}[[x,y,z,t]]/(\mathfrak{u}_i\Phi_i(f_i))$ the order along the exceptional prime divisor E_0 of the weighted blow-up with weights $(3,2,2,2)$ in x,y,z,t is again the largest $(3,2,2,2)$-weighted order of a representative and satisfies the analogue of (E.146), since the lowest $(3,2,2,2)$-weighted part of $\mathfrak{u}_i\Phi_i(f_i)$ is x^2+y^2z by Lemma E.39(1)(2). As $\Phi_i(x)-x,\dots,\Phi_i(t)-t$ belong to $(x,y,z,t)^2$, whose elements have weighted order at least 4, the automorphism Φ_i preserves the $(3,2,2,2)$-weighted filtration. Hence Φ_i maps E_0 over W_i, through the completion [dFEM11, Lemma 2.21], to E_0 over $\operatorname{Spec}\mathbb{C}[[x,y,z,t]]/(\mathfrak{u}_i\Phi_i(f_i))$, and $\operatorname{mult}_{E_0}\Phi_i(h_{i,k}) = \operatorname{mult}_{E_0} h_{i,k}$.

We claim that

$$\sum_k c_k \operatorname{mult}_{E_0} h_{\infty,k} \leq 2 - \frac{1}{m}. \tag{E.147}$$

Let $j_1,\dots,j_s$ be non-negative integers with $j_k \leq \operatorname{mult}_{E_0} h_{\infty,k}$, let $N > \max_k j_k$, and let $\mathfrak{f}_n \subset \mathbf{k}[[x,y,z,t]]$ be the ideal of the power series of $(3,2,2,2)$-weighted order at least n, so that $(x,y,z,t)^N \subset \mathfrak{f}_{j_k}$. The set $\mathcal{K}'_N \subset \mathcal{J}_N$ of the N-jets of tuples $(F,h_1,\dots,h_s)$ with

$$h_k \in (F) + \mathfrak{f}_{j_k} + (x,y,z,t)^N \quad \text{for any } k$$

is the image of the morphism

$$(F,(A_k)_k,(B_k)_k) \to \bigl(F,((A_kF + B_k)_{<N})_k\bigr),$$

where B_k runs through the N-jets in $\mathfrak{f}_{j_k}$, so it is constructible. It contains the N-jet of $(f_\infty, h_{\infty,1},\dots,h_{\infty,s})$, which maps to the generic point of Z_N, so $\mathcal{K}'_N \cap Z_N$ contains a non-empty open subset of Z_N, which contains the N-jets of the tuples of Lemma E.39(3) for all but finitely many i. For such $i > N$,

$$\Phi_i(h_{i,k}) \in (\mathfrak{u}_i\Phi_i(f_i)) + \mathfrak{f}_{j_k},$$

so $\operatorname{mult}_{E_0} h_{i,k} = \operatorname{mult}_{E_0}\Phi_i(h_{i,k}) \geq j_k$ for any k, and Lemma E.39(1) implies $\sum_k c_k j_k \leq 2 - \frac{1}{m}$. Taking j_k arbitrarily large shows that $\operatorname{mult}_{E_0} h_{\infty,k}$ is finite, in particular $h_{\infty,k} \neq 0$ in R_∞, and taking $j_k = \operatorname{mult}_{E_0} h_{\infty,k}$ proves (E.147).

For $0 \neq \phi \in R_\infty$, let $e(R_\infty/(\phi))$ be the Hilbert–Samuel multiplicity of the two-dimensional local ring $R_\infty/(\phi)$ with respect to the maximal ideal. We claim that

$$e(R_\infty/(\phi)) = \operatorname{mult}_{E_0}\phi. \tag{E.148}$$

Since $x^2 = -g_* \in (y,z,t)^3$ in R_∞, the ideal (y,z,t) is a reduction of the maximal ideal. The residue field is infinite, so for two general linear forms l_1,l_2 in y,z,t the ideal (l_1,l_2) is a minimal reduction of the maximal ideal of $R_\infty/(\phi)$, generated by a system of parameters [Mat89, Theorems 14.13 and 14.14]. The ring $R_\infty/(\phi)$ is Cohen–Macaulay, being a quotient of the

hypersurface ring R_∞ by a non-zero divisor, hence $e(R_\infty/(\phi))$ is the length of $R_\infty/(\phi, l_1, l_2)$ [Mat89, Theorem 17.11]. Let ζ be a linear form in y, z, t independent of l_1, l_2. Then

$$R_\infty/(l_1, l_2) \cong \mathbf{k}[[x, \zeta]]/(x^2 + \lambda_0\zeta^3 + \cdots),$$

where $\lambda_0 \neq 0$ is the value of the cubic part y^2z of g_* on the line $(l_1 = l_2 = 0)$, which is non-zero for general l_1, l_2. The cusp $R_\infty/(l_1, l_2)$ is unibranch, with normalization $\mathbf{k}[[\tau]]$ and $\zeta = \tau^2 \cdot (\text{unit})$, $x = \tau^3 \cdot (\text{unit})$. Thus the length of $R_\infty/(\phi, l_1, l_2)$ is the order in τ of $\phi_0|_{\text{line}} + x\phi_1|_{\text{line}}$. For general l_1, l_2, the lowest homogeneous parts of ϕ_0 and ϕ_1 do not vanish on the line, so $\phi_0|_{\text{line}}$ and $x\phi_1|_{\text{line}}$ have orders $2\operatorname{mult}_0\phi_0$ and $3 + 2\operatorname{mult}_0\phi_1$ in τ. The integers $2\operatorname{mult}_0\phi_0$ and $3 + 2\operatorname{mult}_0\phi_1$ have different parity, so no cancellation occurs, and (E.146) implies (E.148).

By (E.147) and (E.148),

$$\sum_k c_k e(R_\infty/(h_{\infty,k})) = \sum_k c_k \operatorname{mult}_{E_0} h_{\infty,k} \leq 2 - \frac{1}{m}. \tag{E.149}$$

Since R_∞ is normal, the localization of R_∞ at the ideal $I_{S'}$ of a prime divisor S' of W_∞ through 0 is a discrete valuation ring, and the associativity formula for multiplicities [Mat89, Theorem 14.7] rewrites the left side of (E.149) as

$$\sum_{S'} \Big(\sum_k c_k \operatorname{mult}_{S'} h_{\infty,k} \Big) e(R_\infty/I_{S'}) \leq 2 - \frac{1}{m} < 2,$$

where S' runs through the prime divisors of W_∞ through 0. All terms are non-negative and $e(R_\infty/I_{S'}) \geq 1$. By (E.145), the term of S is $e(R_\infty/I_S) \geq 1$. Hence $e(R_\infty/I_S) < 2$, that is, $e(R_\infty/I_S) = 1$. The ring R_∞/I_S is a complete local domain of dimension two with multiplicity one, hence regular by [Nag62, (40.6)], since a complete local domain is unmixed.

Step 6. In this step we find a prime divisor of log discrepancy two over infinitely many W_i and conclude. Consider Lemma 9.2 over $\mathbf{k}$ with $R := R_\infty$, $g := g_*$, with $D := S$, and with $B := \sum_k c_k \operatorname{div}(h_{\infty,k})$, that is, $b_k := c_k$ and $\phi_k := h_{\infty,k}$. The hypotheses of Lemma 9.2 hold: R_∞ is a normal complete local domain by **Step 4**, $\operatorname{mult}_0 g_* = 3$ and the cubic part y^2z of g_* is non-zero by Lemma E.39(2), R_∞/I_S is regular of dimension two by **Step 5**, and $h_{\infty,k} \neq 0$ in R_∞ by **Step 5**. By (E.145), $c_D = c_S = 1$. By Lemma 9.2, there exist a prime divisor P_* over W_∞ with center 0, formal coordinates X, Y, z', t' of $\mathbf{k}[[x, y, z, t]]$, and a weight vector $\mathbf{w} \in \{(2,2,1,1), (3,3,1,1), (3,2,2,1)\}$ such that

$$a(P_*, W_\infty, 0) = 2, \qquad \sum_k c_k \operatorname{mult}_{P_*} h_{\infty,k} \geq 1,$$

the lowest $\mathbf{w}$-weighted part of f_* in X, Y, z', t' is reduced, and P_* is an exceptional prime divisor of the weighted blow-up with weights $\mathbf{w}$ in X, Y, z', t'. Let $n_k := \operatorname{mult}_{P_*} h_{\infty,k}$, a non-negative integer, so that $\sum_k c_k n_k \geq 1$.

Consider Lemma E.40 with $F_i := f_i$, the marks $h_{i,k}$, the automorphisms Φ_i and the units $\mathbf{u}_i$ of Lemma E.39(2), the compatible generic limit $(f_\infty, h_{\infty,1}, \ldots, h_{\infty,s})$ of Lemma E.39(3), the coordinates $(\xi_1, \ldots, \xi_4) := (X, Y, z', t')$, the vector $\mathbf{w}$, and P_*. The hypotheses of Lemma E.40 hold: W_i is an isolated Gorenstein terminal threefold germ by Construction-Theorem 4.13(3), so it is normal near 0, no $h_{i,k}$ vanishes identically on W_i since $h_{i,k}$ is a mark, R_∞ is a normal domain by **Step 4**, and $h_{\infty,k} \neq 0$ in R_∞ for any k by **Step 5**. By Lemma E.40, for all but finitely many i there exists a prime divisor P_i over W_i with center 0 such that $a(P_i, W_i, 0) = 2$ and $\operatorname{mult}_{P_i} h_{i,k} \geq n_k$ for any k. For such i,

$$a(P_i, W_i, D_i) = a(P_i, W_i, 0) - \sum_k c_k \operatorname{mult}_{P_i} h_{i,k} \leq 2 - \sum_k c_k n_k \leq 1,$$

so $\operatorname{mld}(W_i \ni 0, D_i) \leq 1$. This contradicts (E.142). Therefore Proposition 11.16 holds. □

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